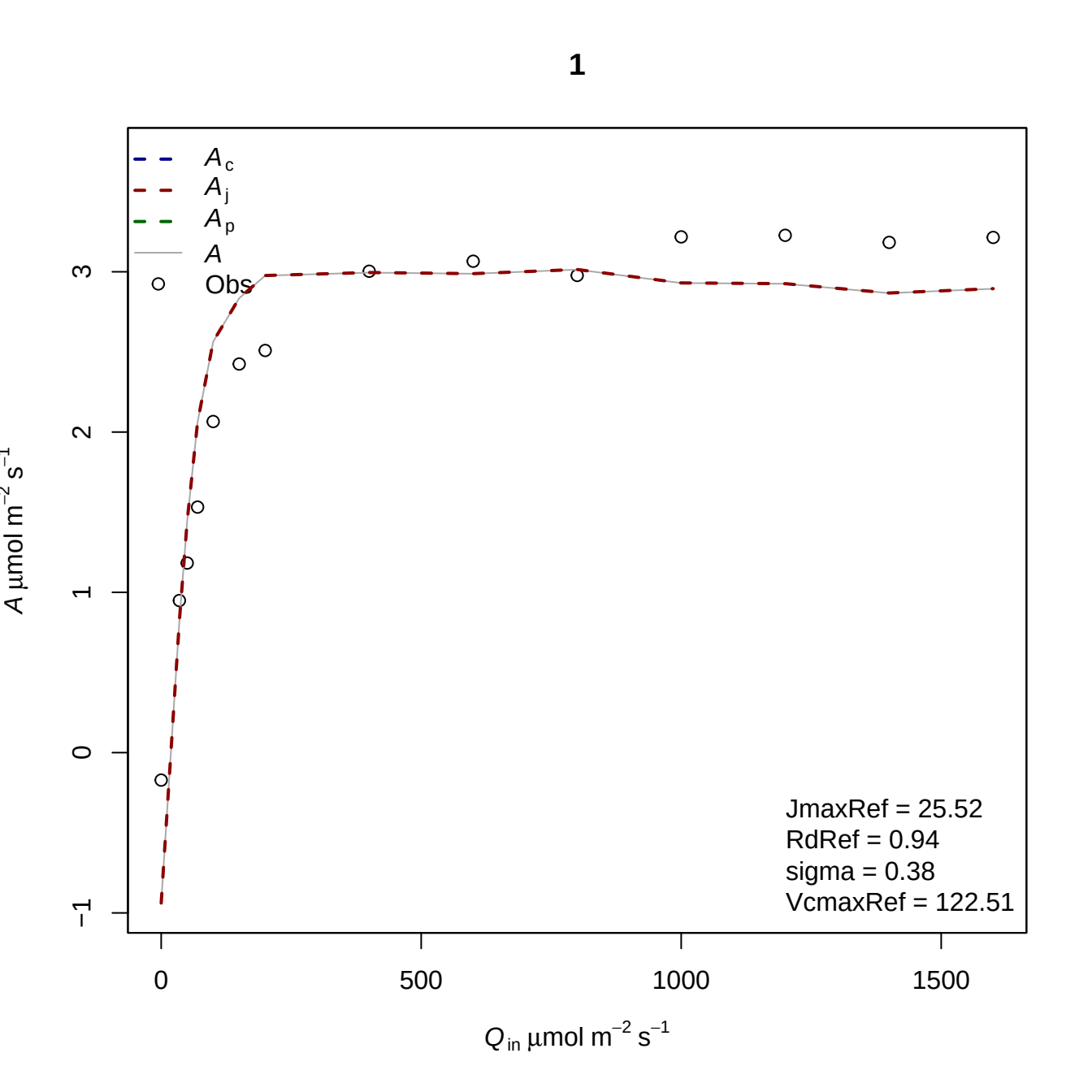
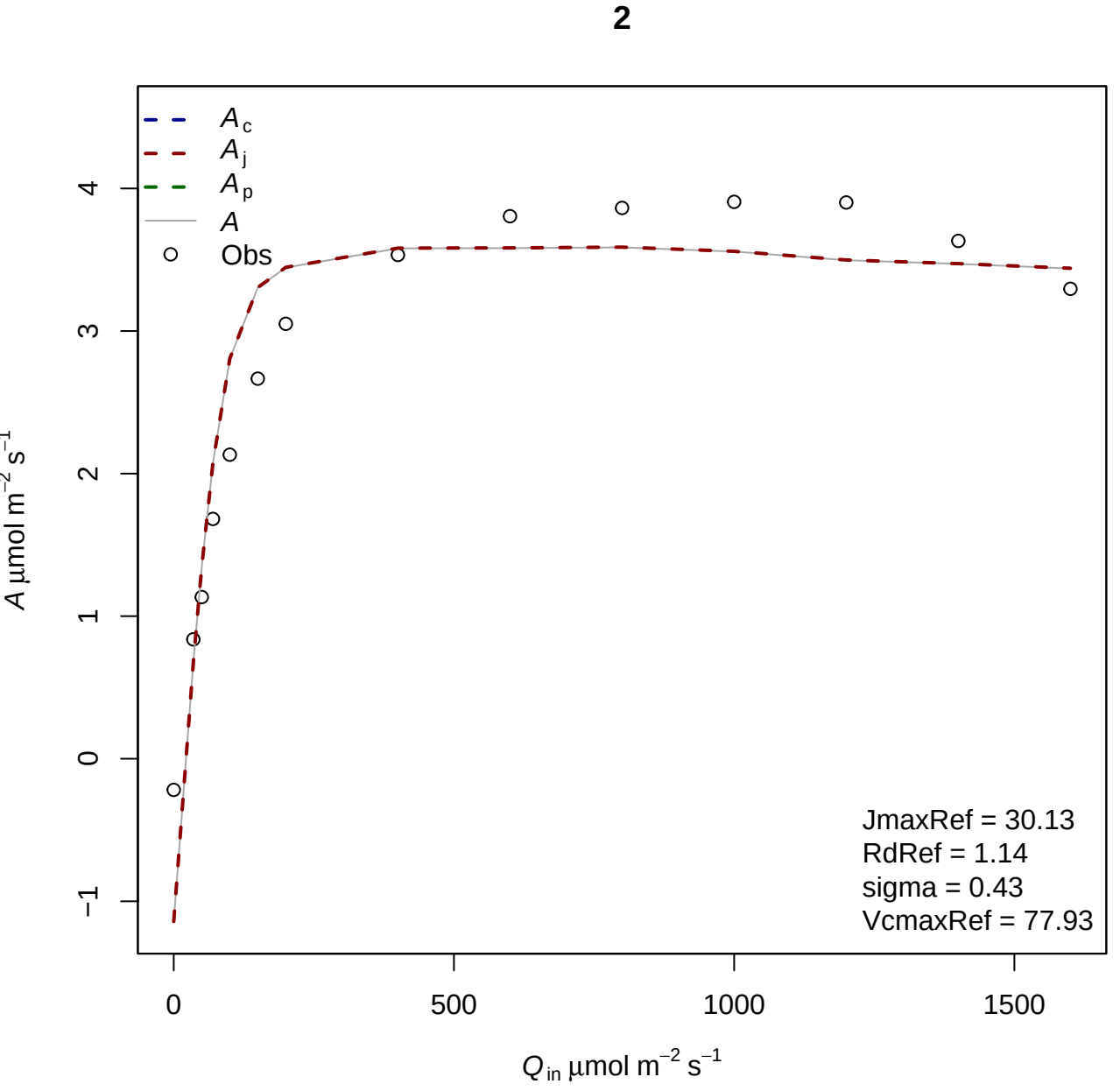
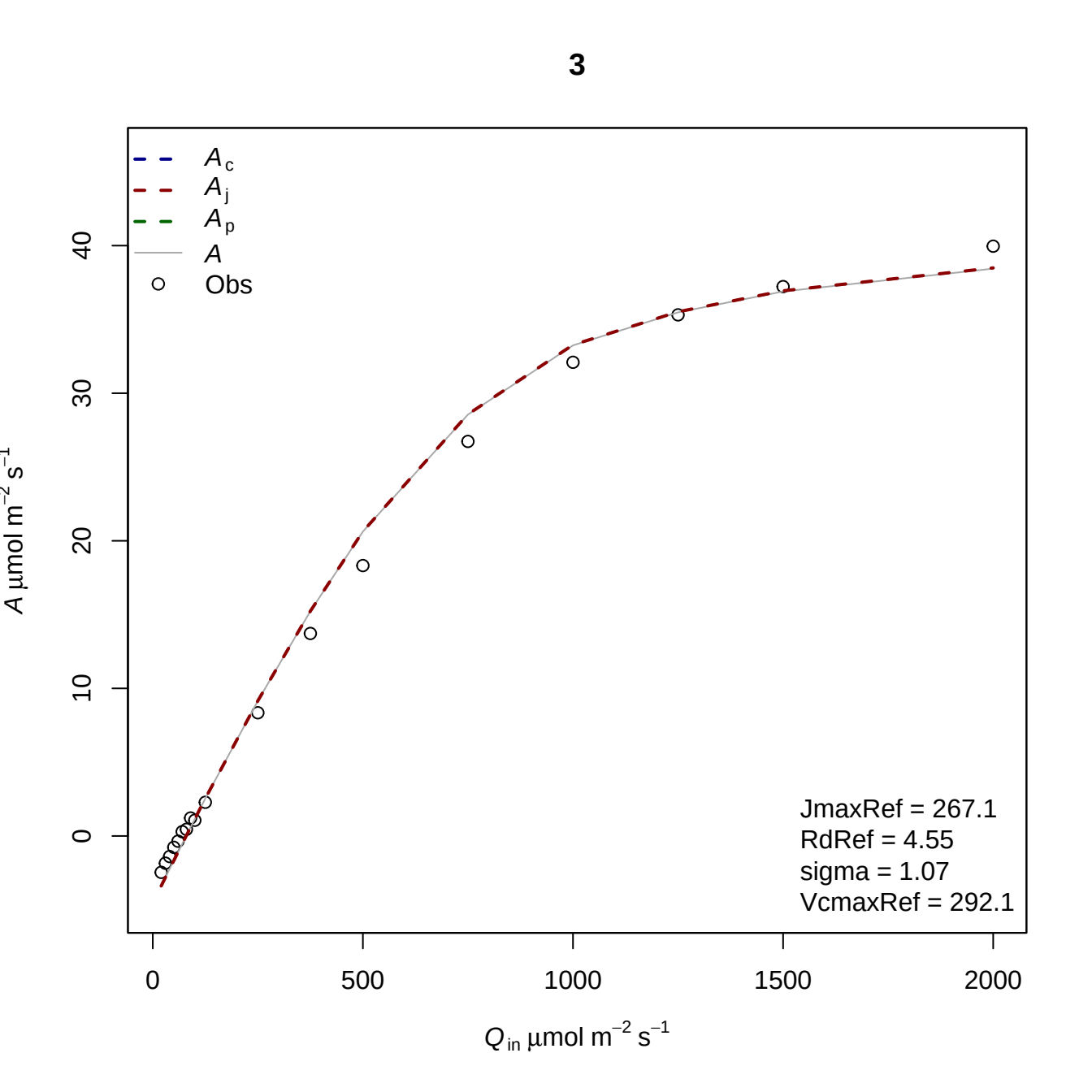
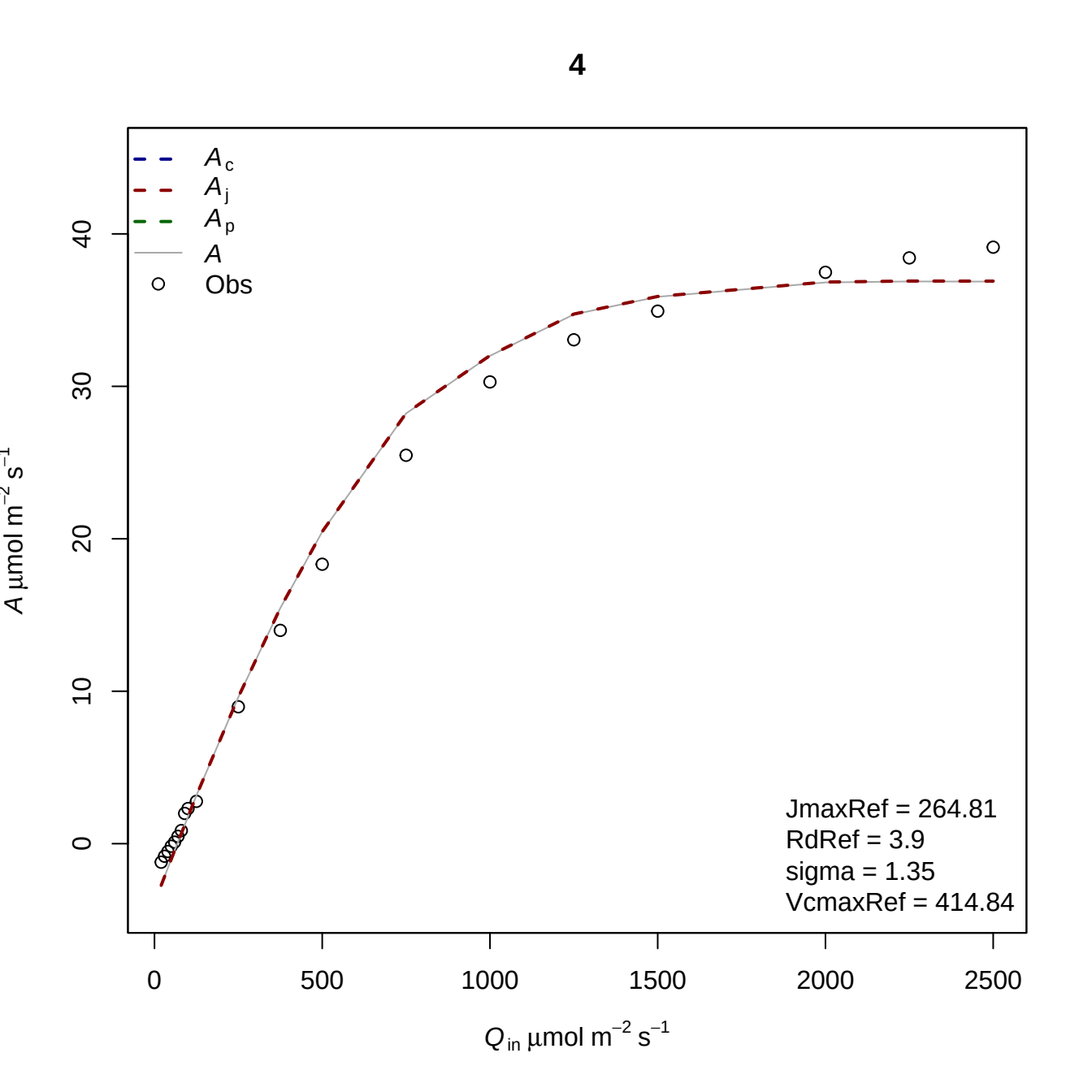
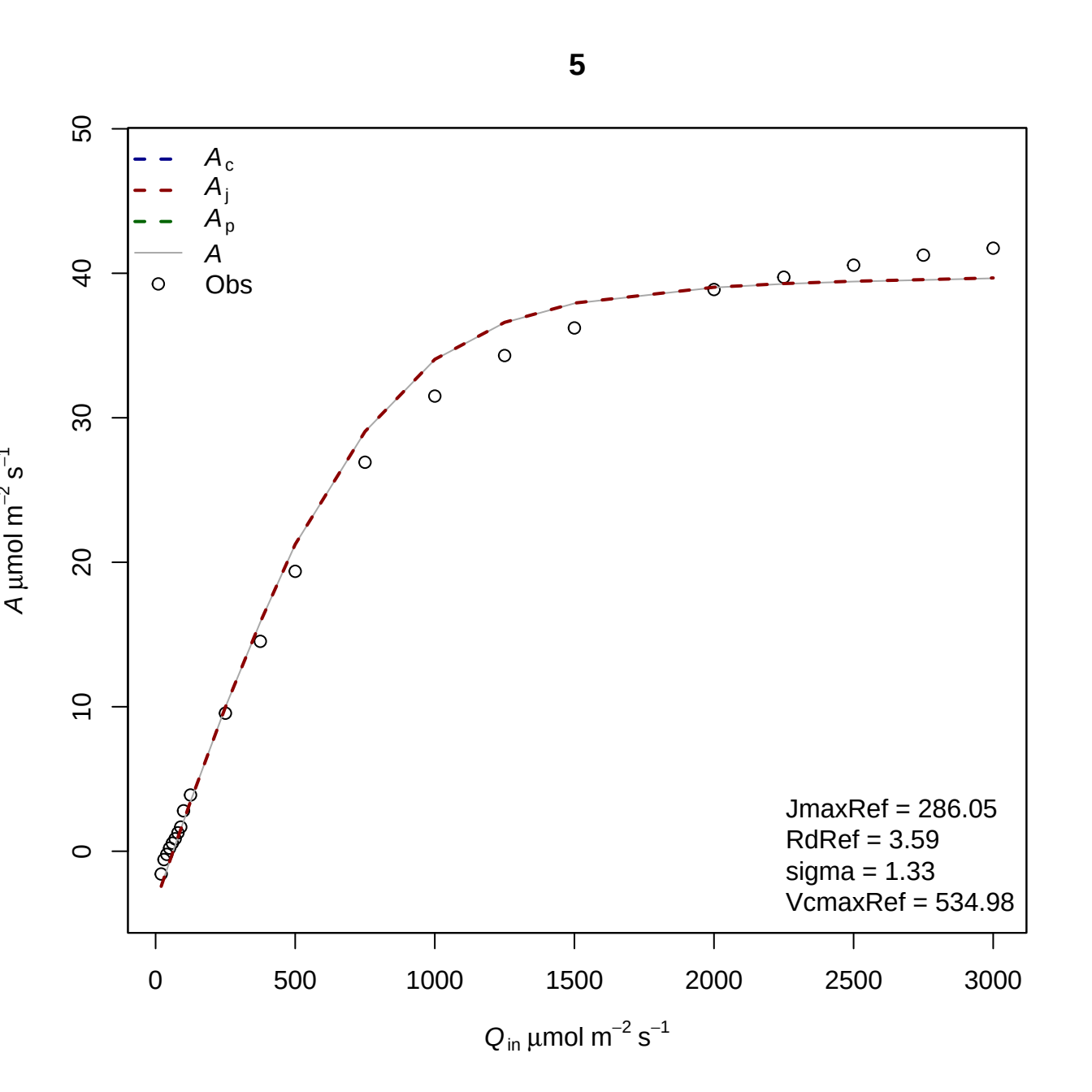
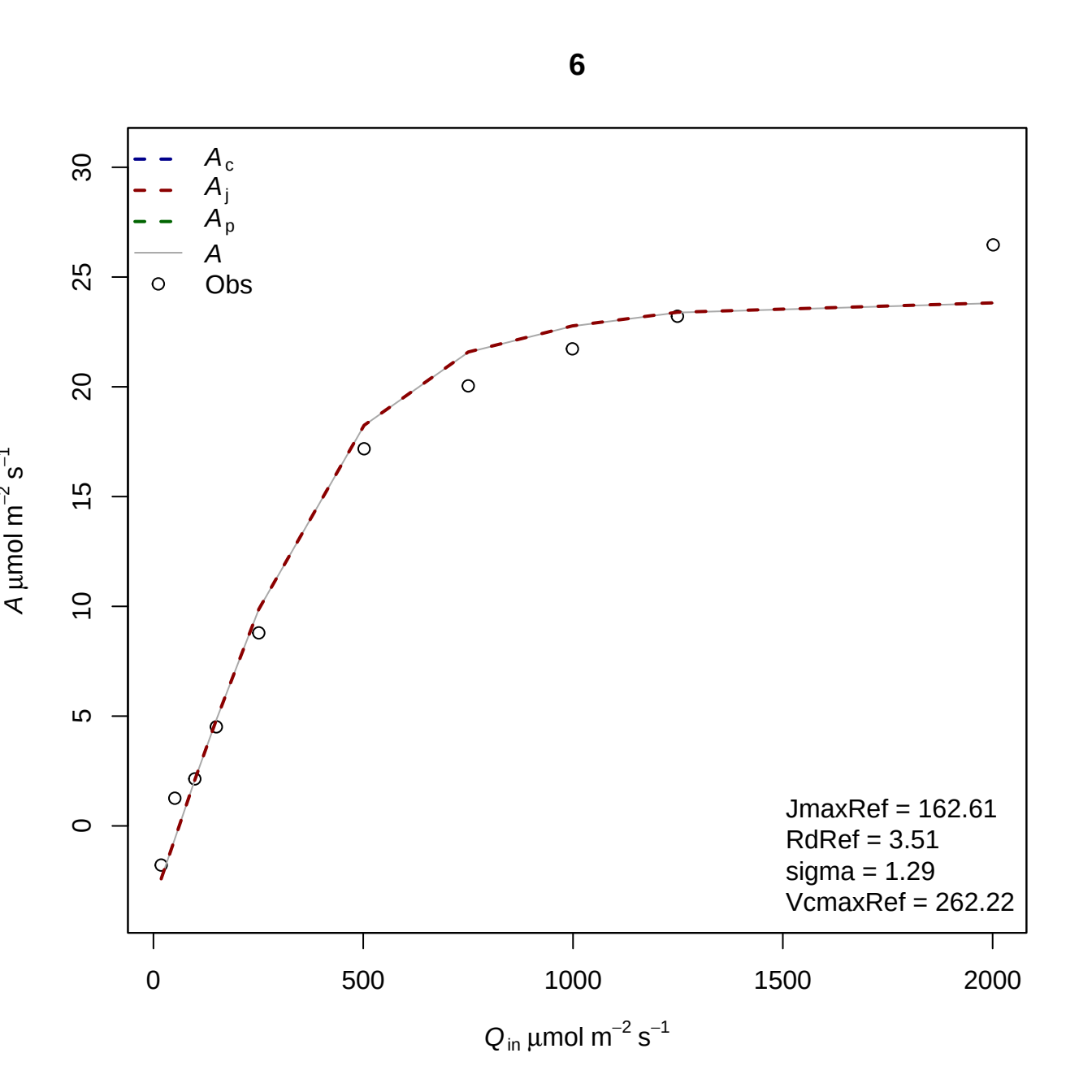

1
A_c
A_j
A_p
A
Obs
JmaxRef = 25.52
RdRef = 0.94
sigma = 0.38
VcmaxRef = 122.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

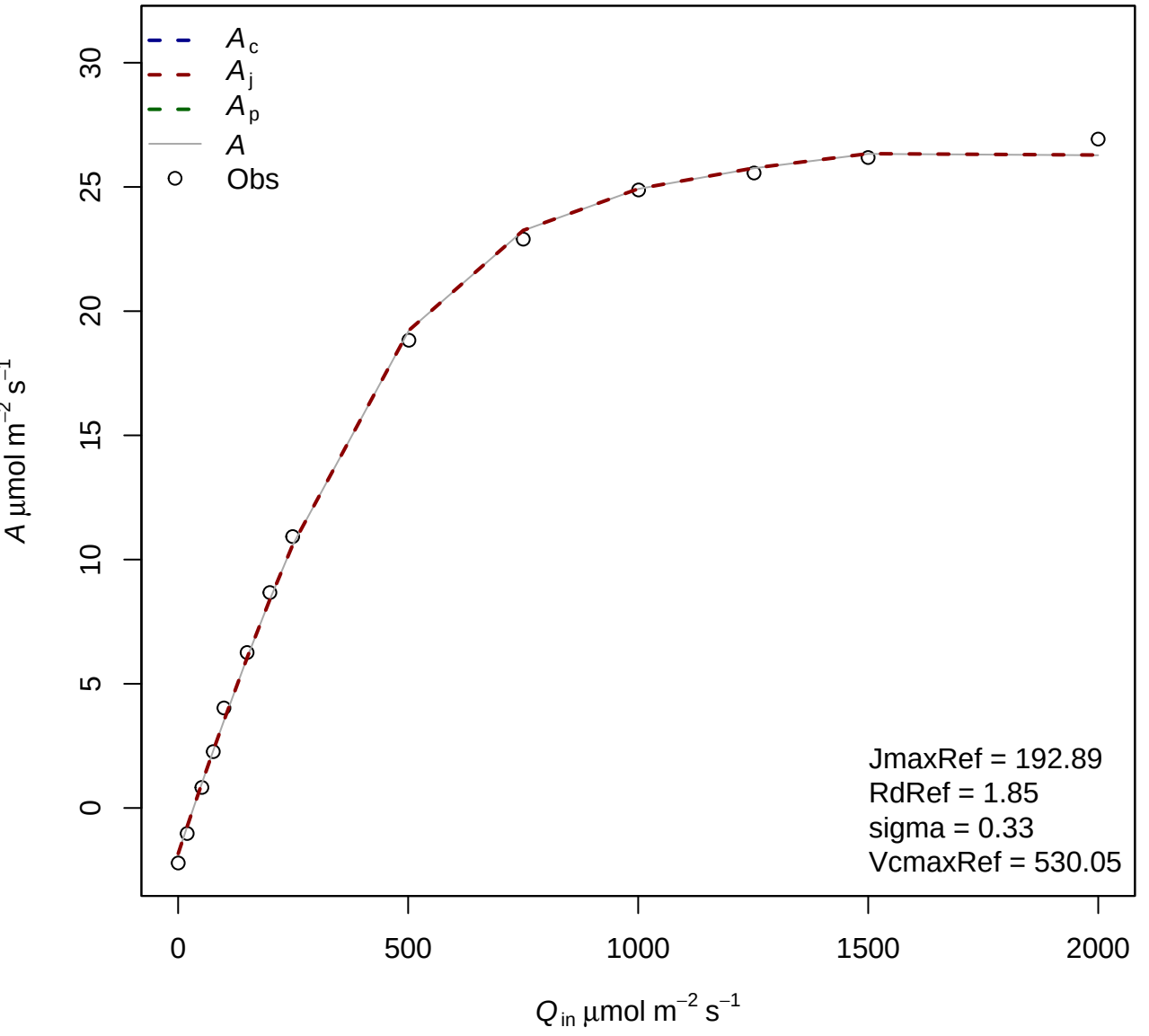

2
A_c
A_j
A_p
A
Obs
JmaxRef = 30.13
RdRef = 1.14
sigma = 0.43
VcmaxRef = 77.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

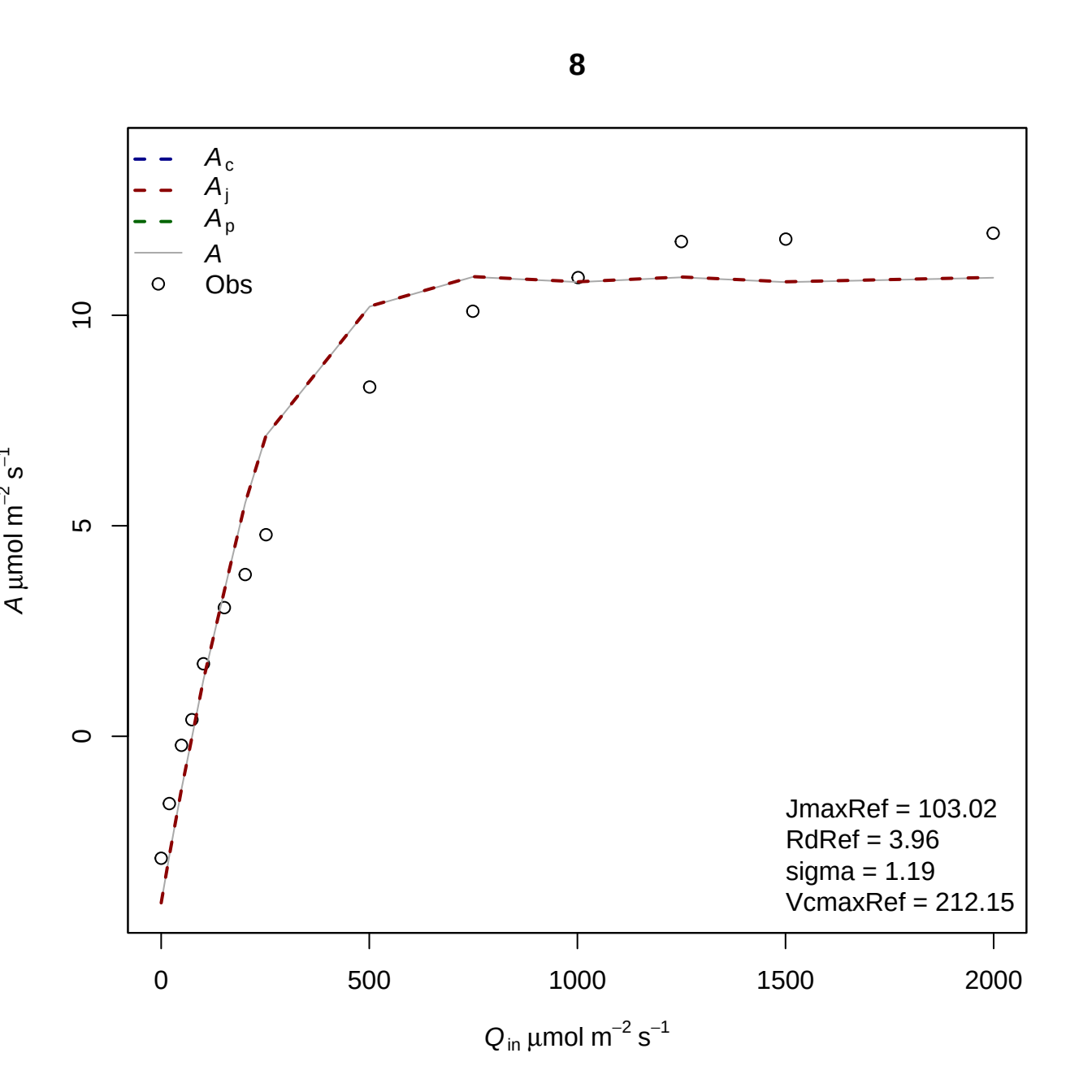

3
A_c
A_j
A_p
A
Obs
JmaxRef = 267.1
RdRef = 4.55
sigma = 1.07
VcmaxRef = 292.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

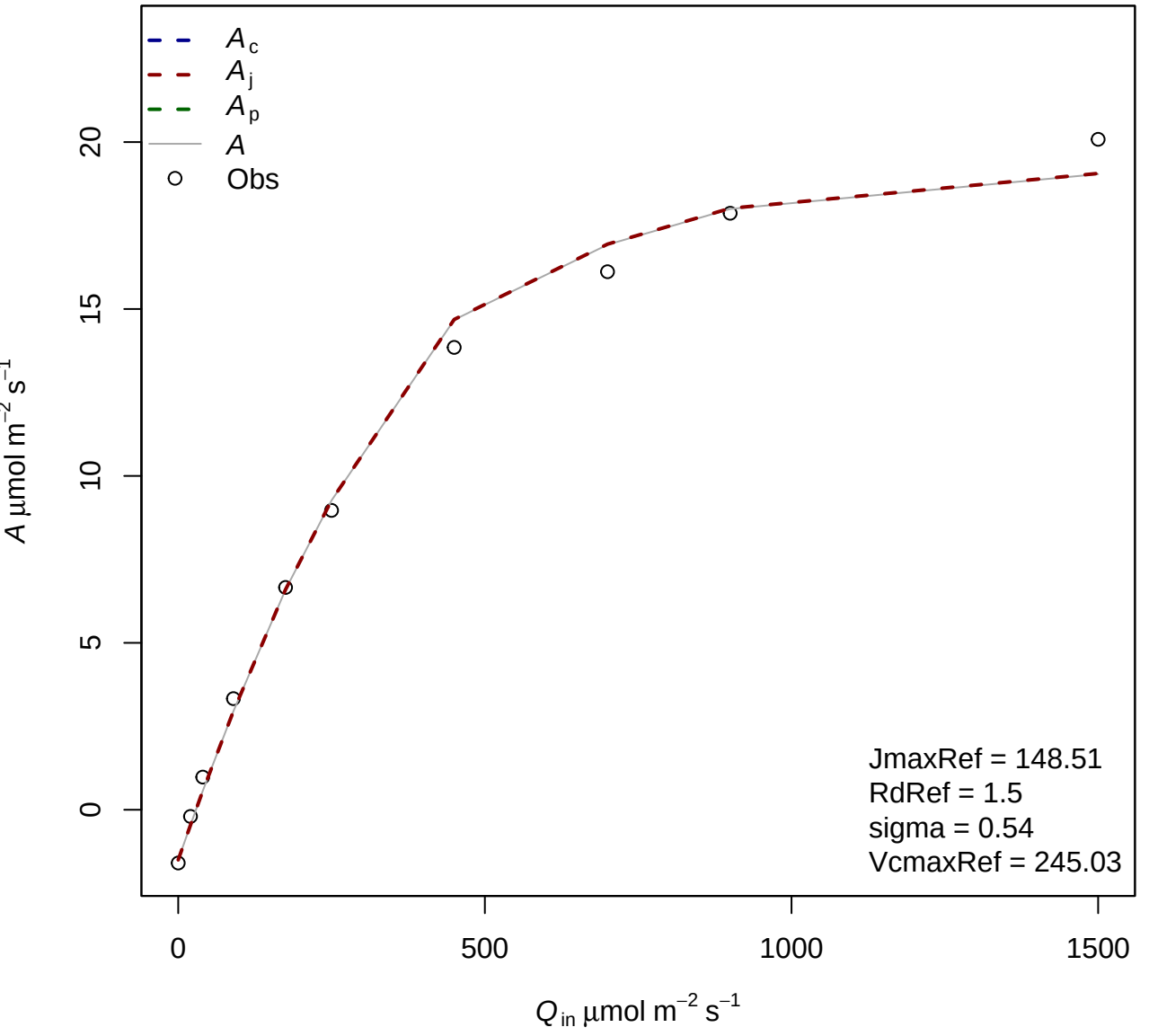

4
A_c
A_j
A_p
A
Obs
JmaxRef = 264.81
RdRef = 3.9
sigma = 1.35
VcmaxRef = 414.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000
2500

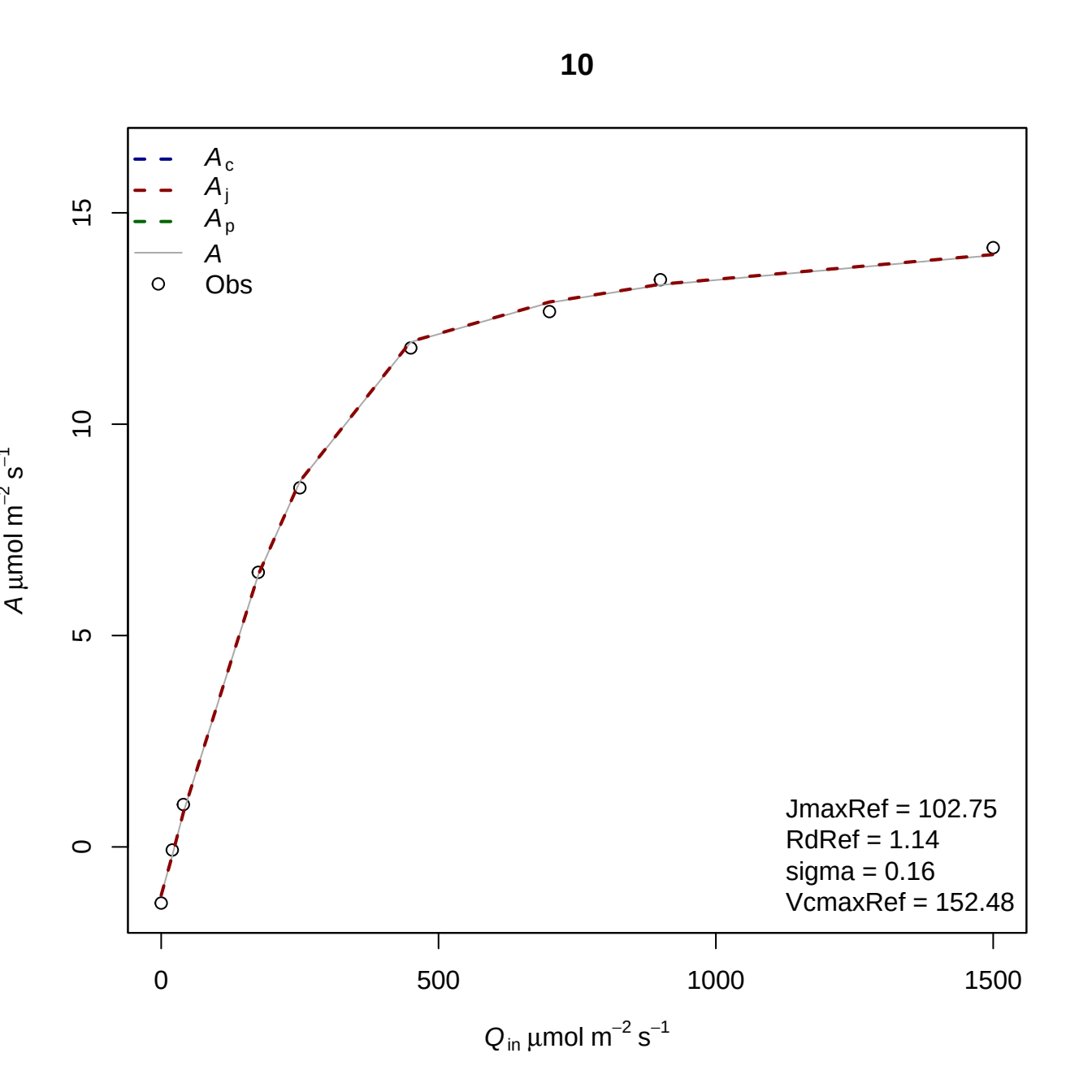

5
A_c
A_j
A_p
A
Obs
JmaxRef = 286.05
RdRef = 3.59
sigma = 1.33
VcmaxRef = 534.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
50
0
500
1000
1500
2000
2500
3000

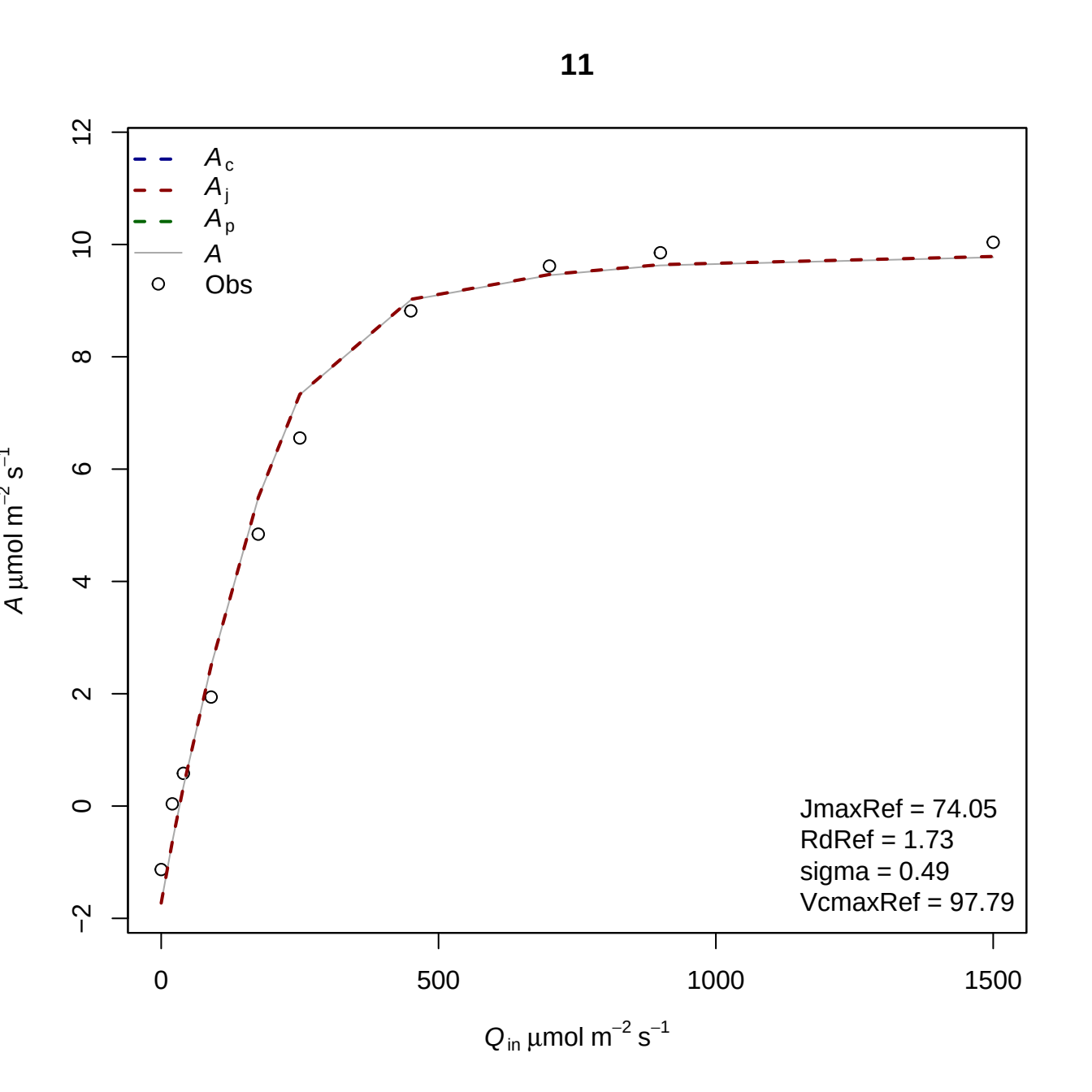

6
A_c
A_j
A_p
A
Obs
JmaxRef = 162.61
RdRef = 3.51
sigma = 1.29
VcmaxRef = 262.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

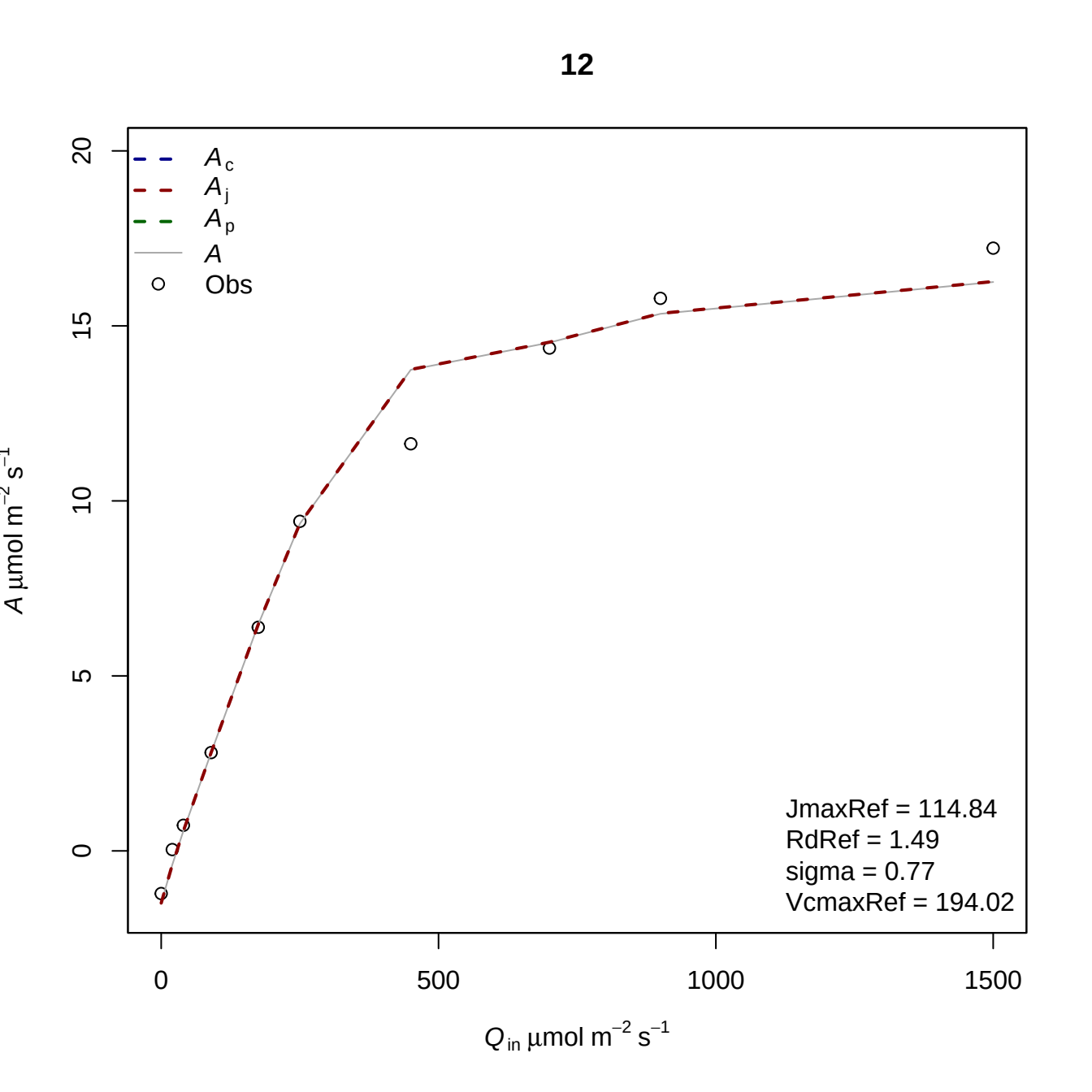

A_c
A_j
A_p
A
Obs
JmaxRef = 192.89
RdRef = 1.85
sigma = 0.33
VcmaxRef = 530.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

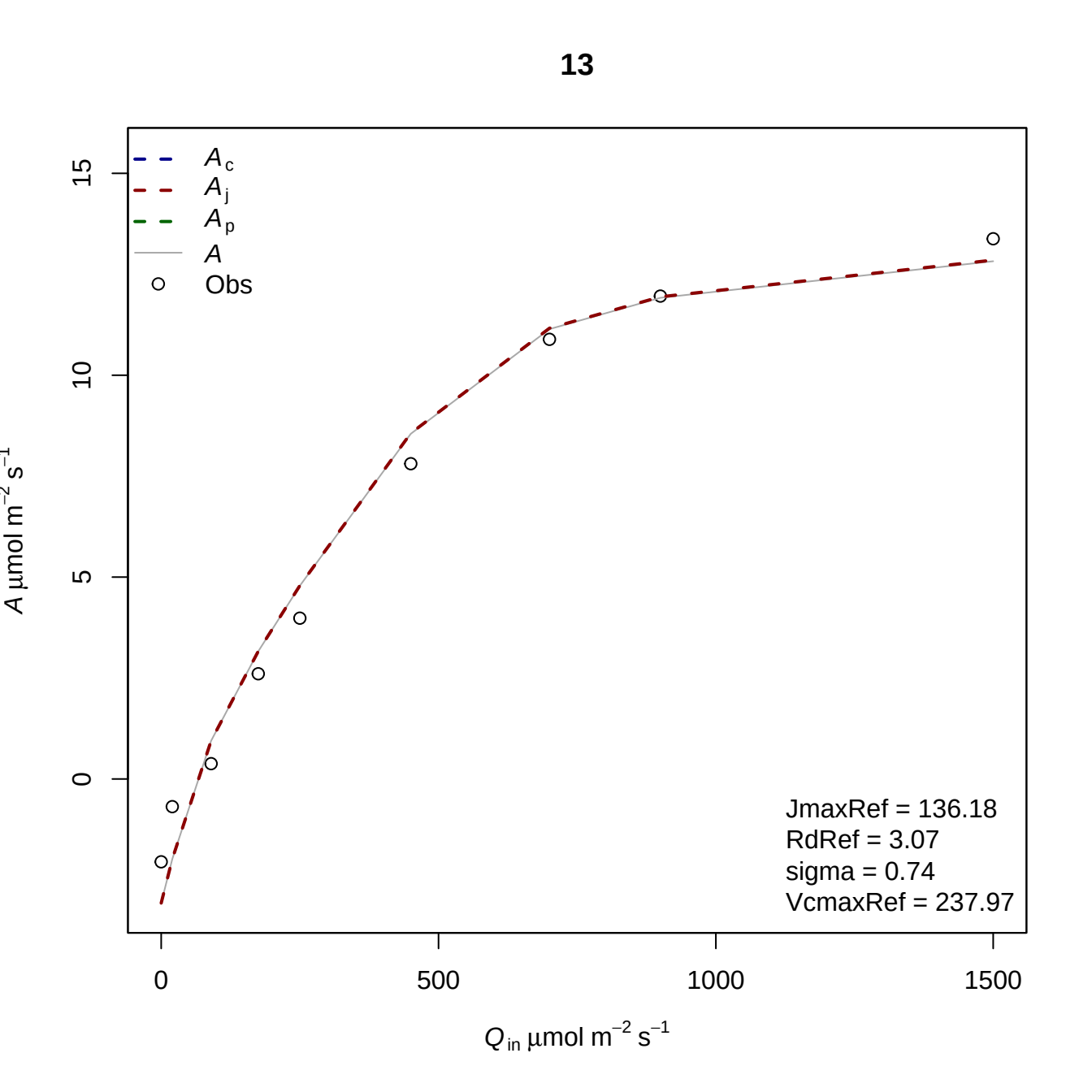

8
A_c
A_j
A_p
A
Obs
JmaxRef = 103.02
RdRef = 3.96
sigma = 1.19
VcmaxRef = 212.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

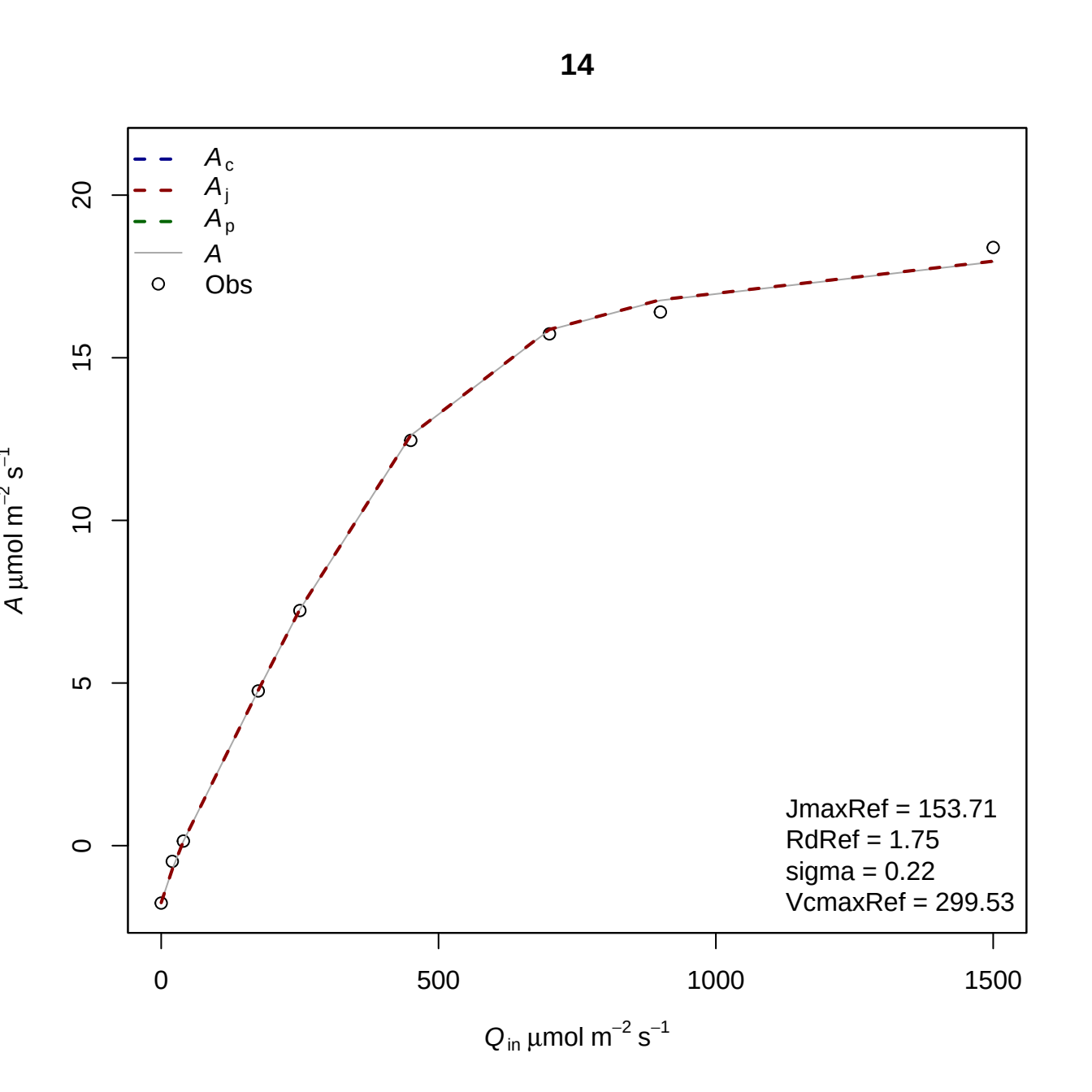

A_c
A_j
A_p
A
Obs
JmaxRef = 148.51
RdRef = 1.5
sigma = 0.54
VcmaxRef = 245.03
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

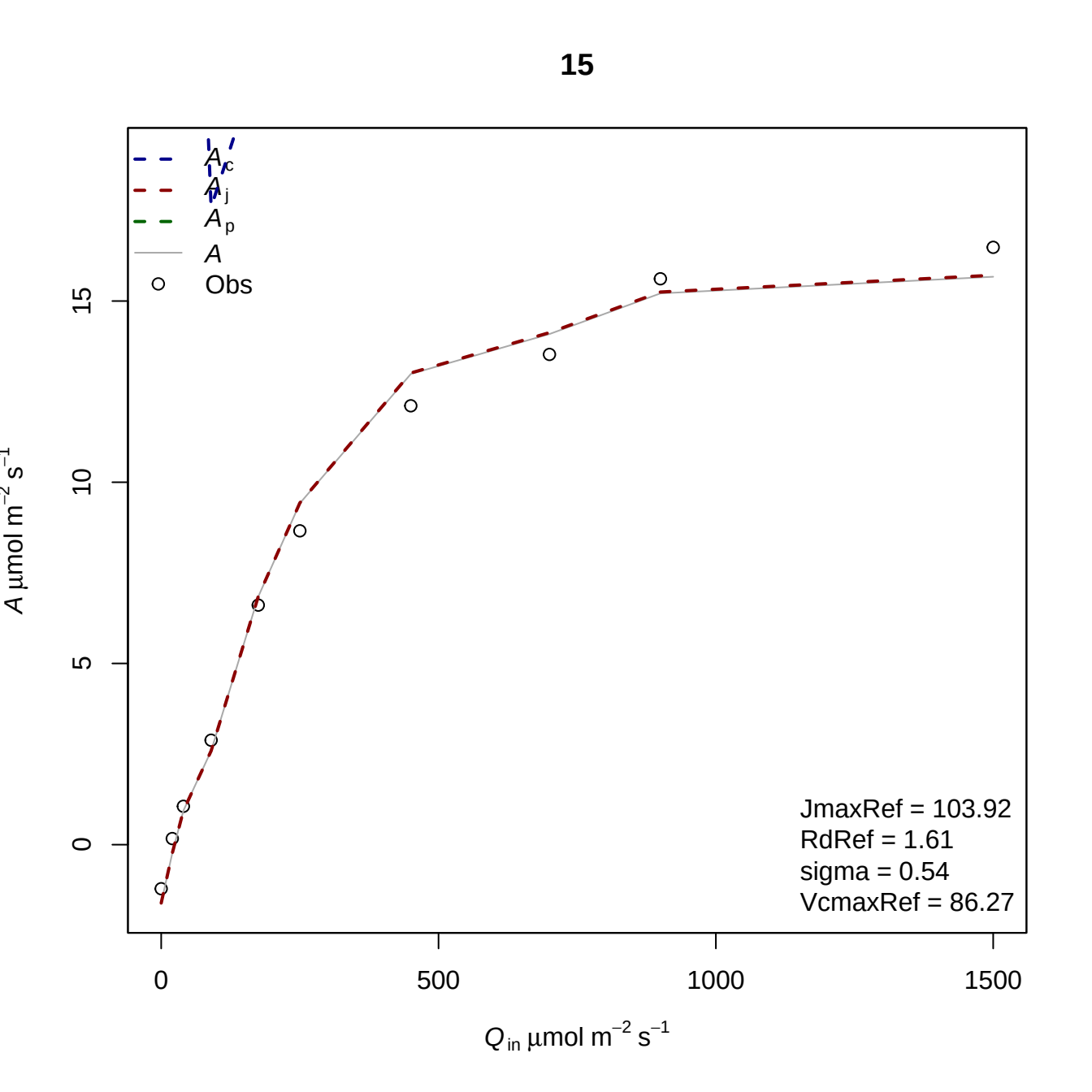

10
A_c
A_j
A_p
A
Obs
JmaxRef = 102.75
RdRef = 1.14
sigma = 0.16
VcmaxRef = 152.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

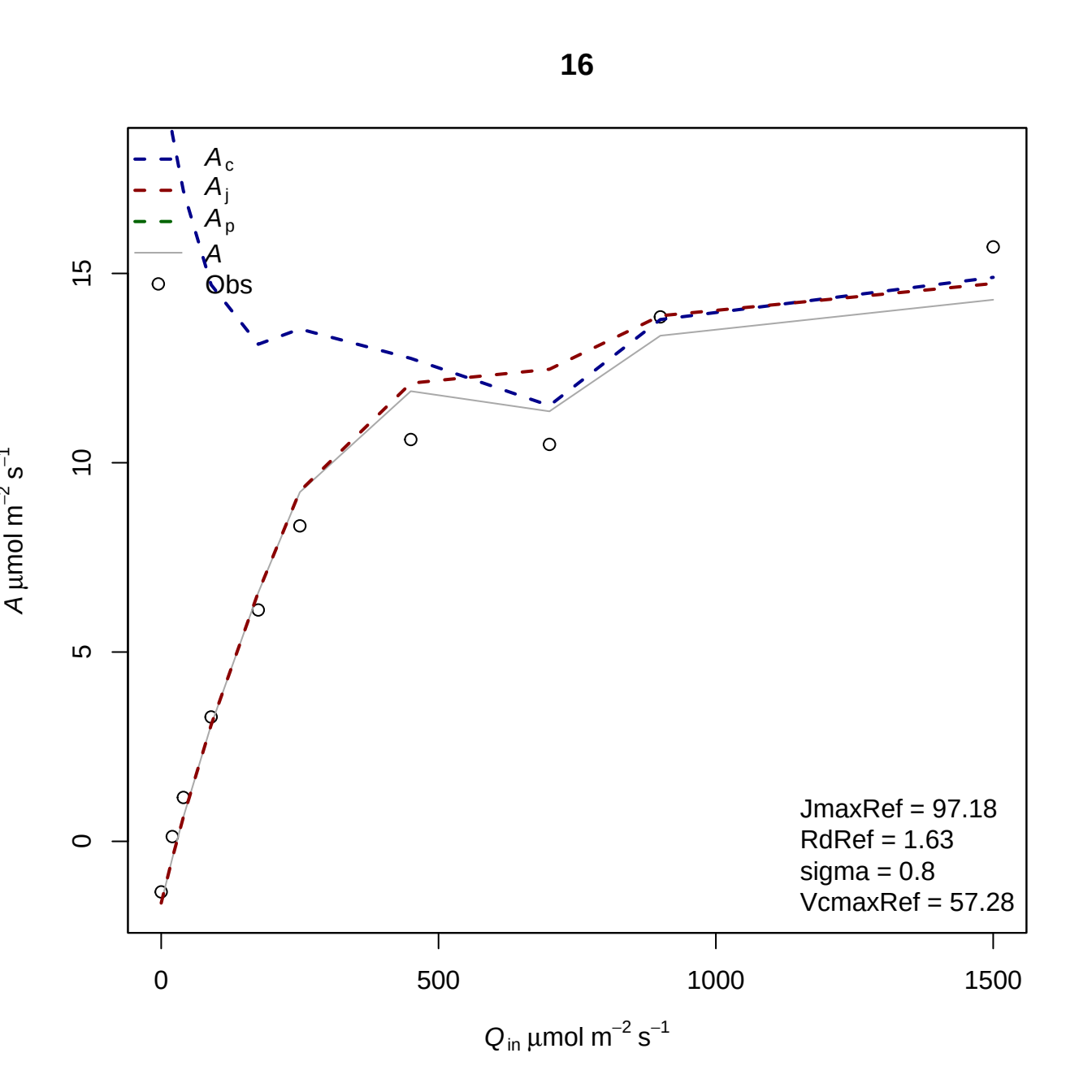

11
A_c
A_j
A_p
A
Obs
JmaxRef = 74.05
RdRef = 1.73
sigma = 0.49
VcmaxRef = 97.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

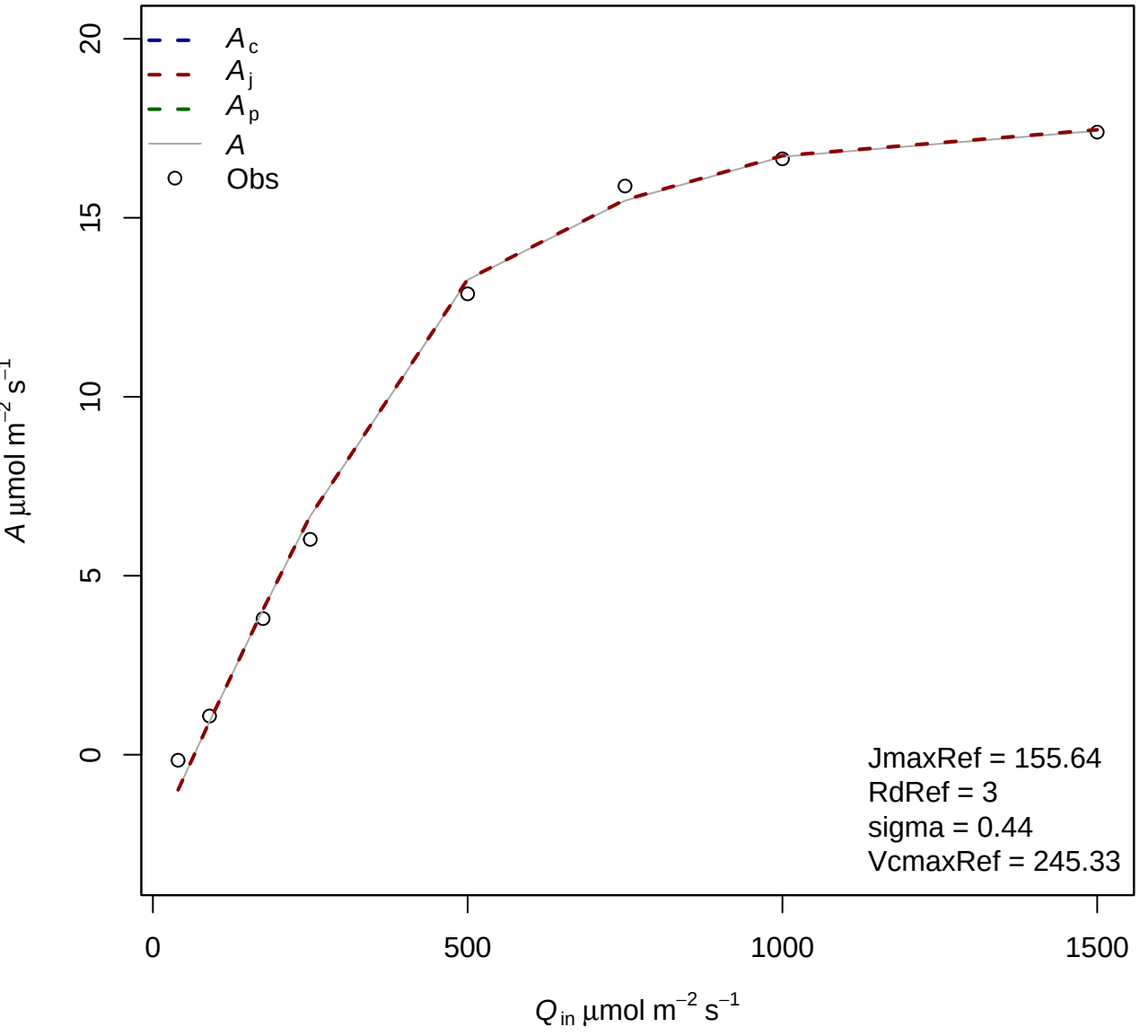

12
A_c
A_j
A_p
A
Obs
JmaxRef = 114.84
RdRef = 1.49
sigma = 0.77
VcmaxRef = 194.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

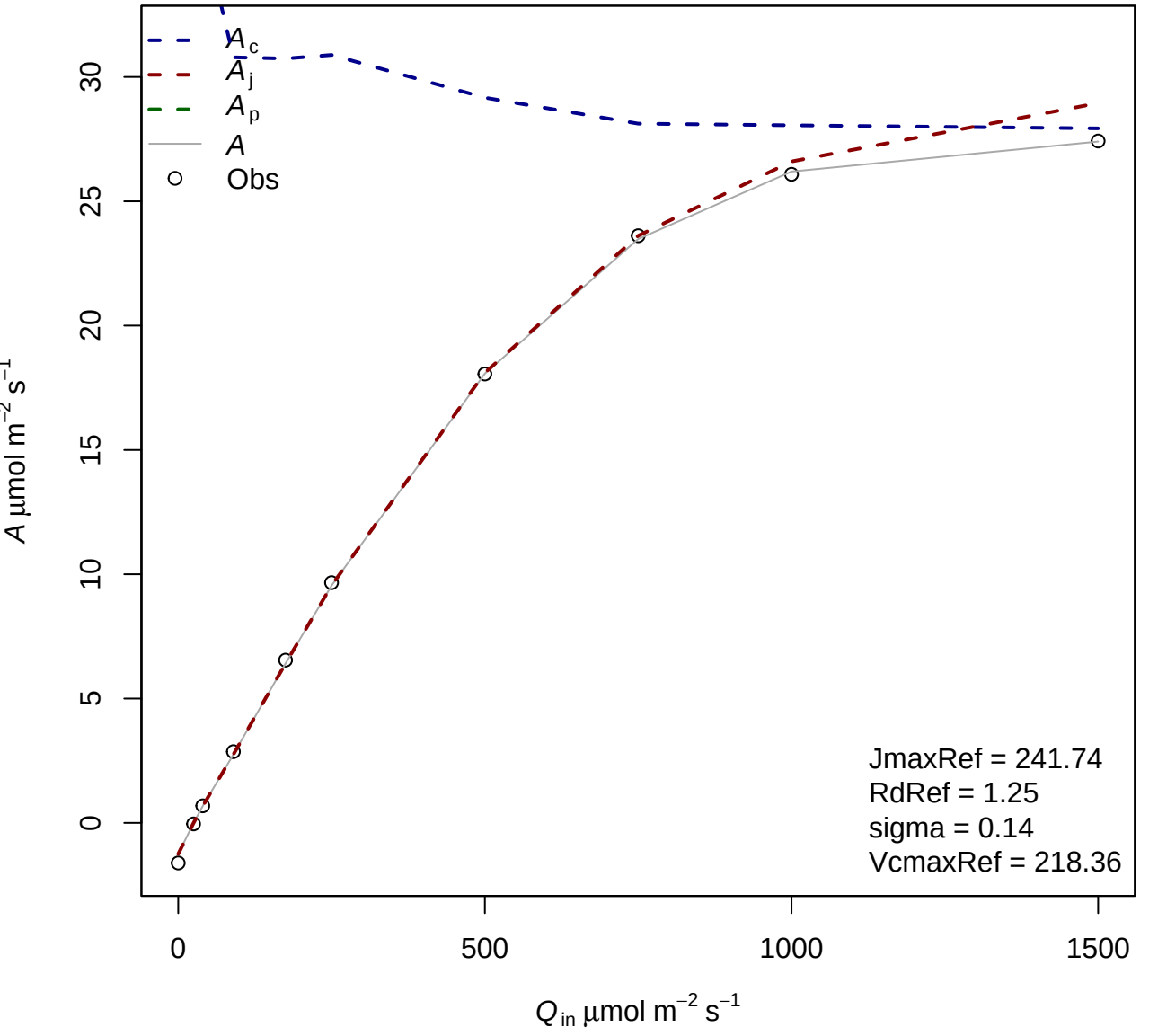

13
A_c
A_j
A_p
A
Obs
JmaxRef = 136.18
RdRef = 3.07
sigma = 0.74
VcmaxRef = 237.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

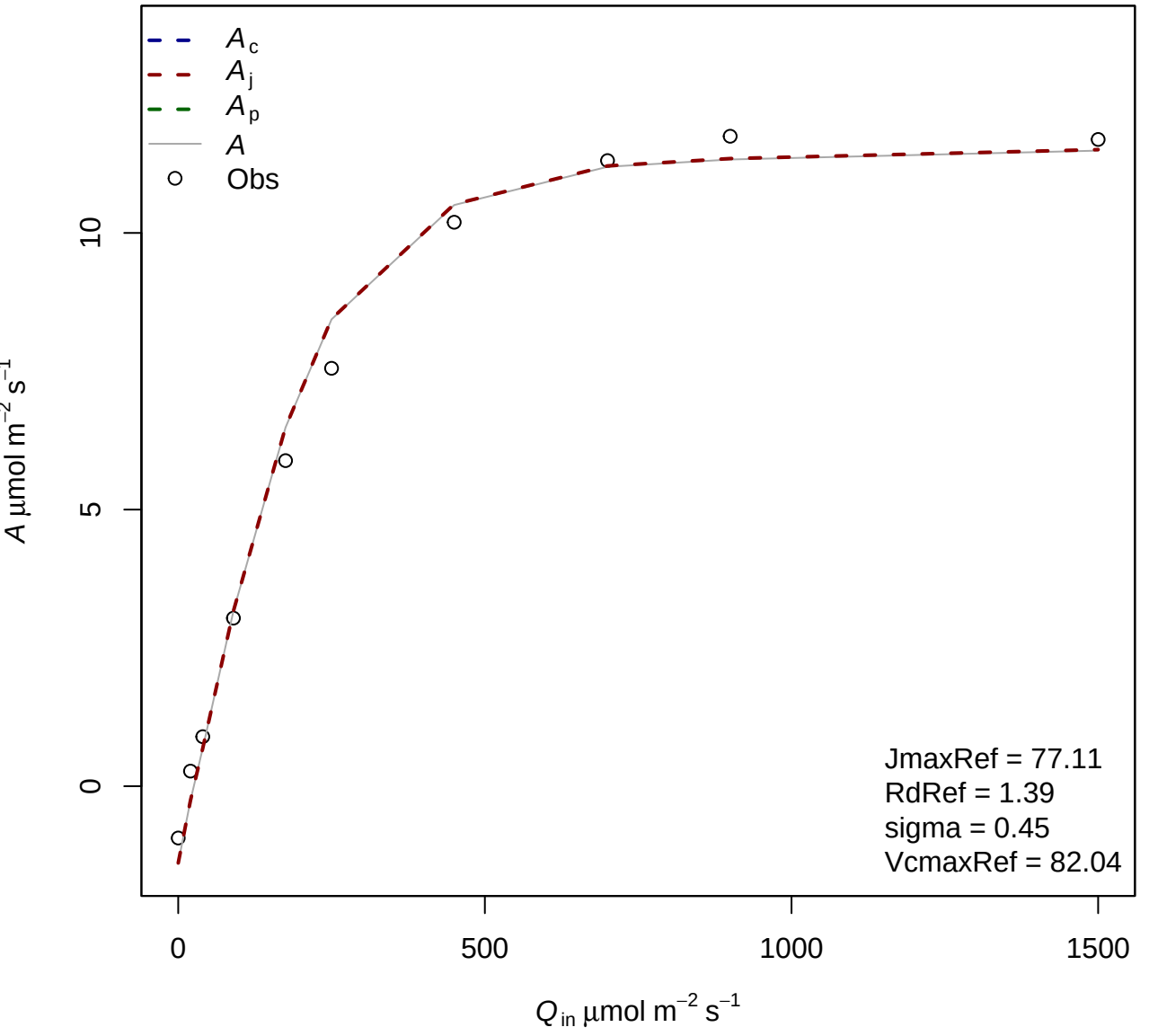

14
A_c
A_j
A_p
A
Obs
JmaxRef = 153.71
RdRef = 1.75
sigma = 0.22
VcmaxRef = 299.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

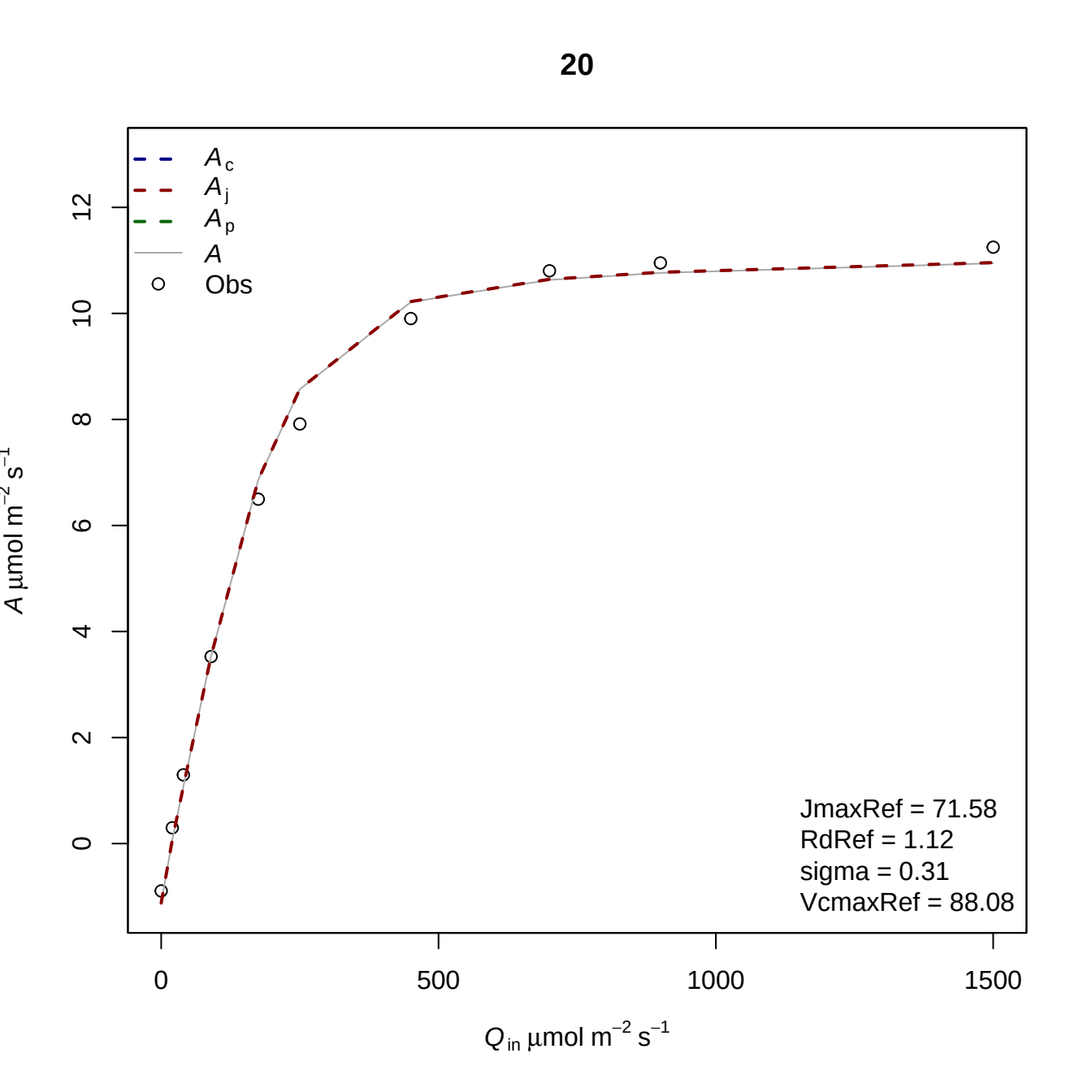

15
A_c
A_j
A_p
A
Obs
JmaxRef = 103.92
RdRef = 1.61
sigma = 0.54
VcmaxRef = 86.27
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

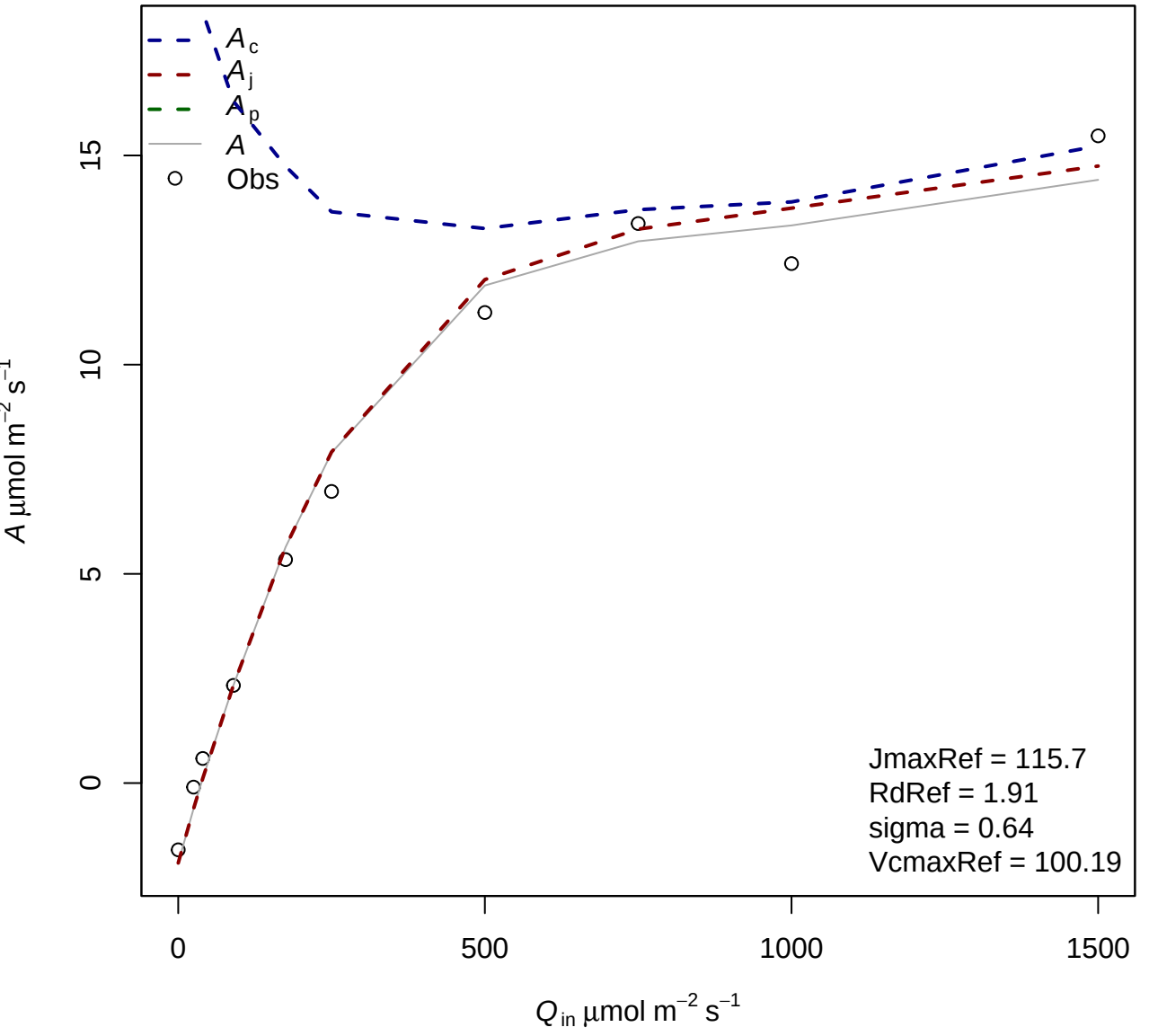

16
A_c
A_j
A_p
A
Obs
JmaxRef = 97.18
RdRef = 1.63
sigma = 0.8
VcmaxRef = 57.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

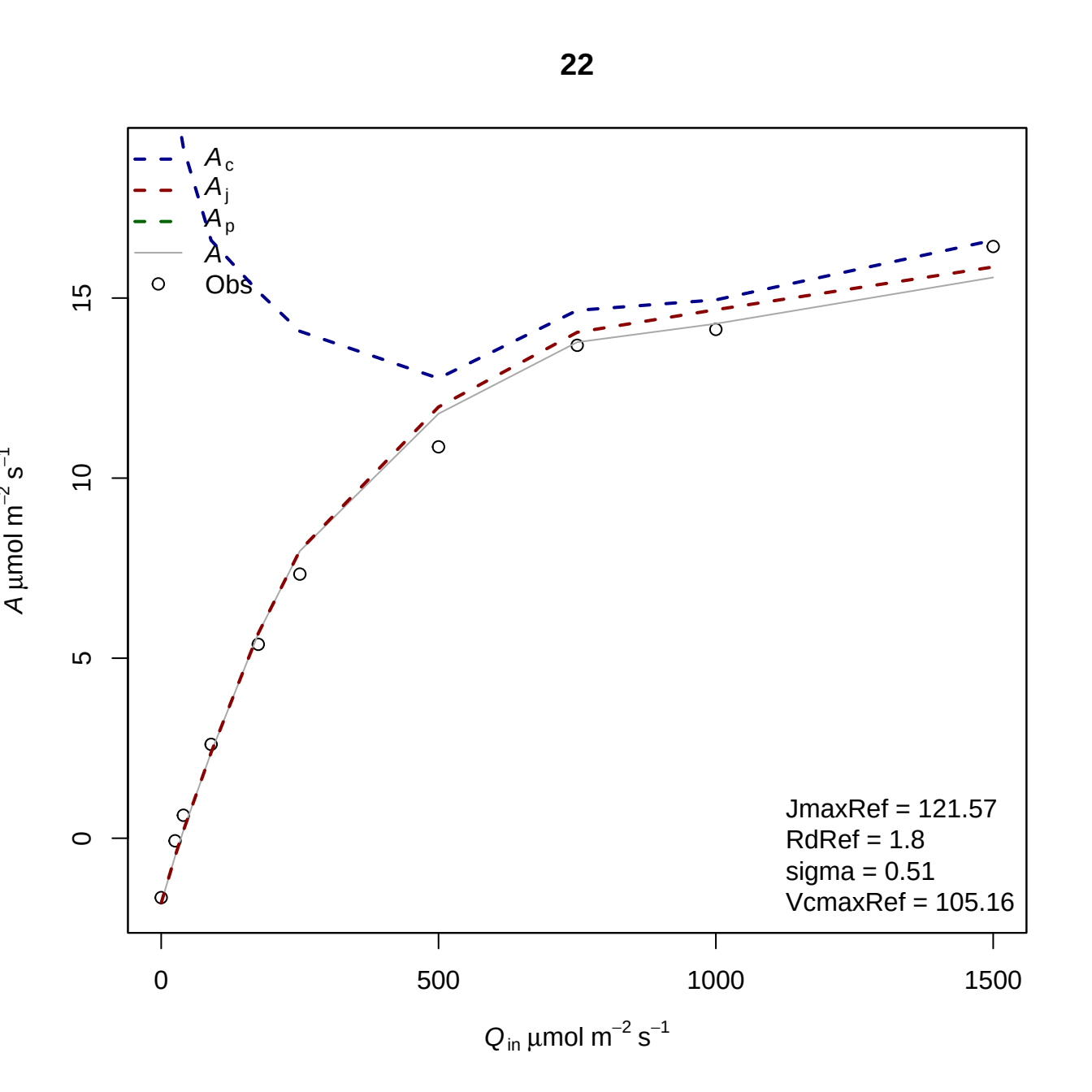

A_c
A_j
A_p
A
Obs
JmaxRef = 155.64
RdRef = 3
sigma = 0.44
VcmaxRef = 245.33
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

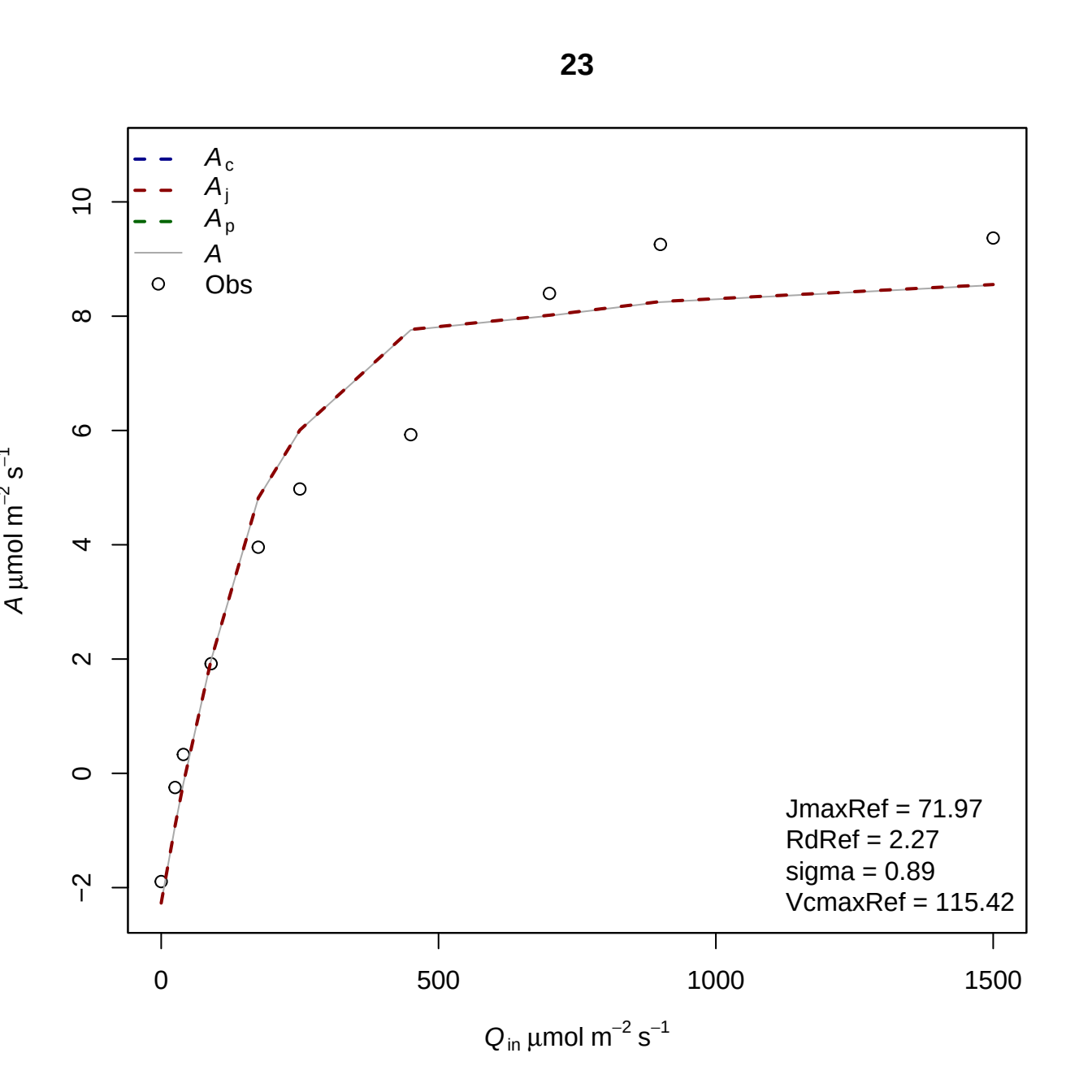

A_c
A_j
A_p
A
Obs
JmaxRef = 241.74
RdRef = 1.25
sigma = 0.14
VcmaxRef = 218.36
0
5
10
15
20
25
30
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

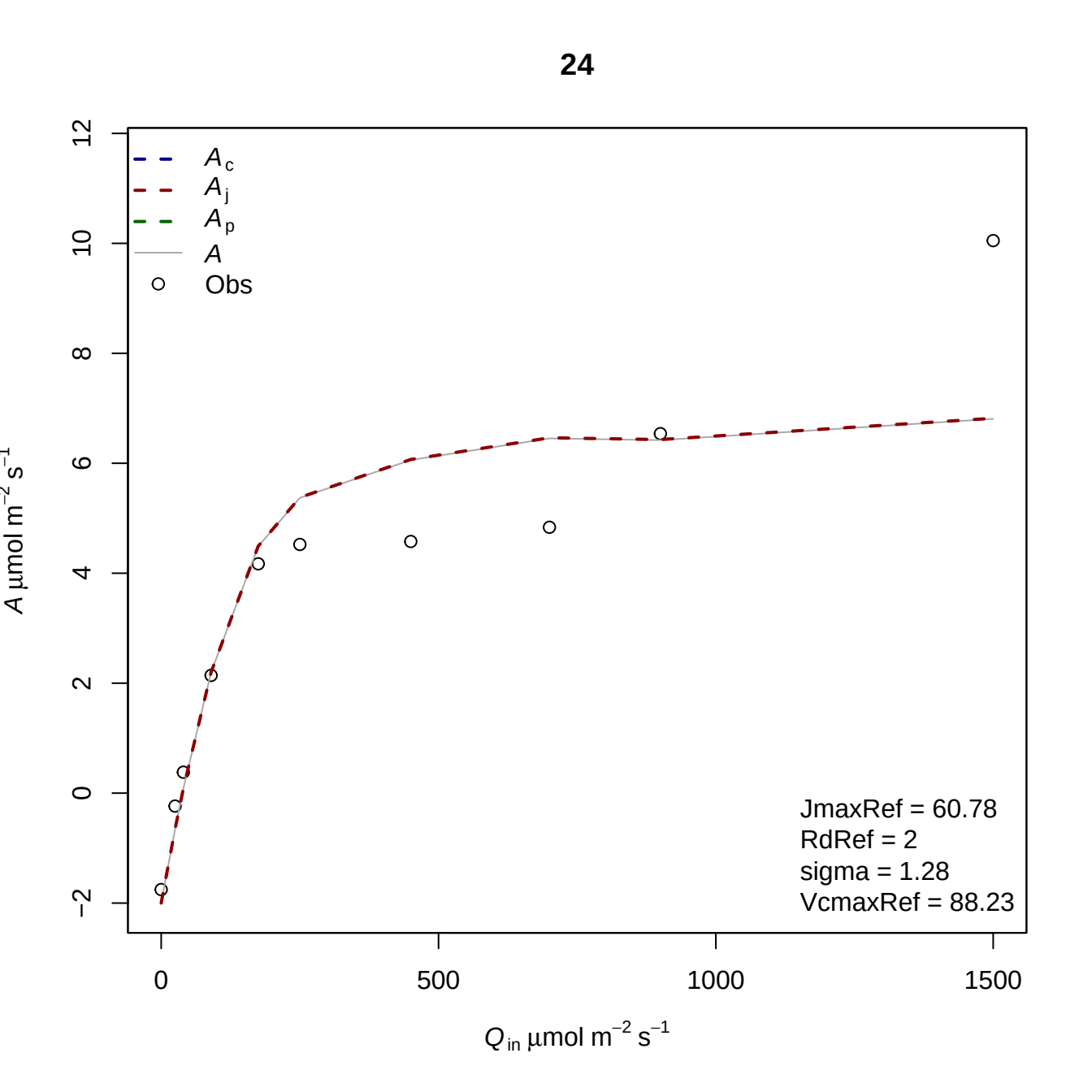

A_c
A_j
A_p
A
Obs
JmaxRef = 77.11
RdRef = 1.39
sigma = 0.45
VcmaxRef = 82.04
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

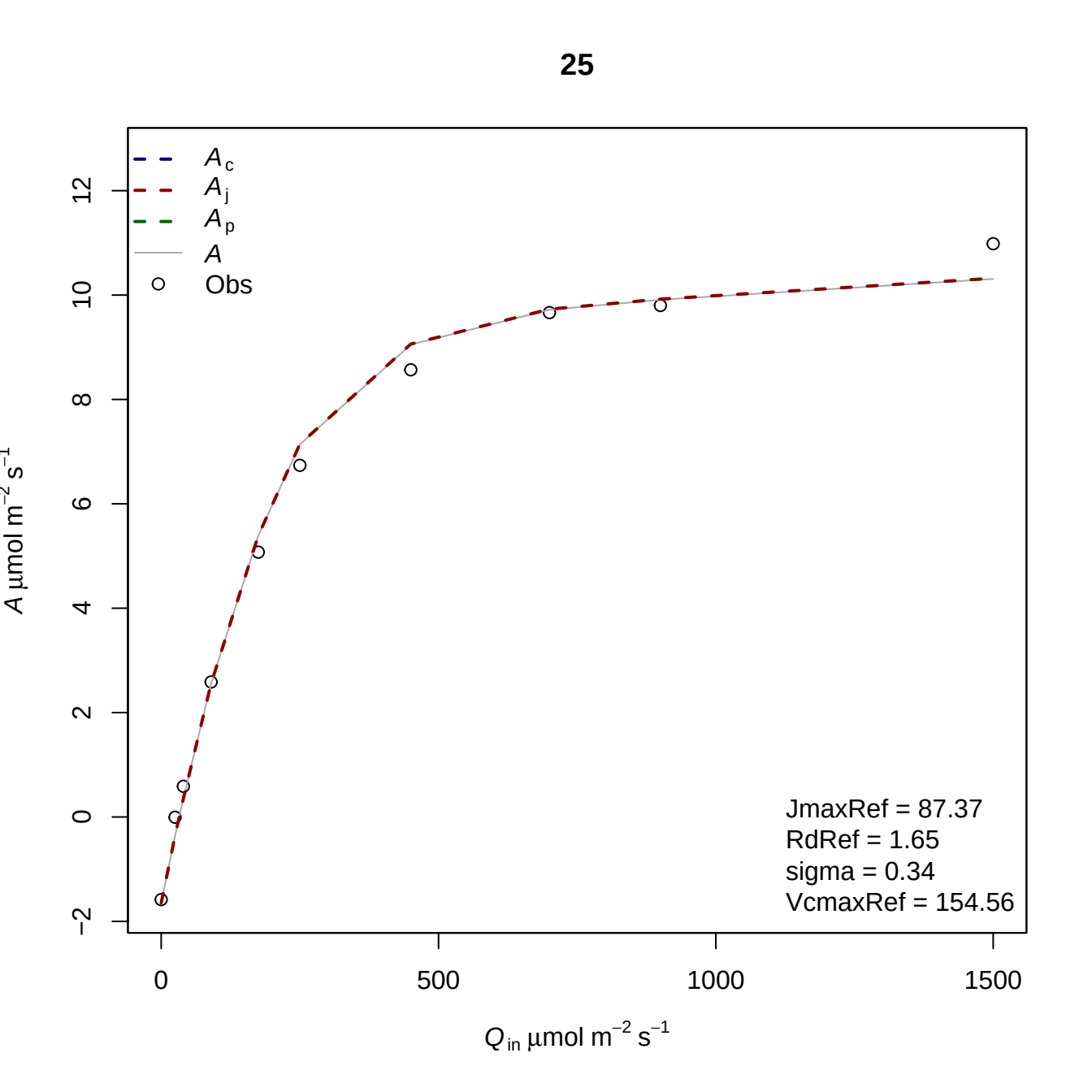

20
A_c
A_j
A_p
A
Obs
JmaxRef = 71.58
RdRef = 1.12
sigma = 0.31
VcmaxRef = 88.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

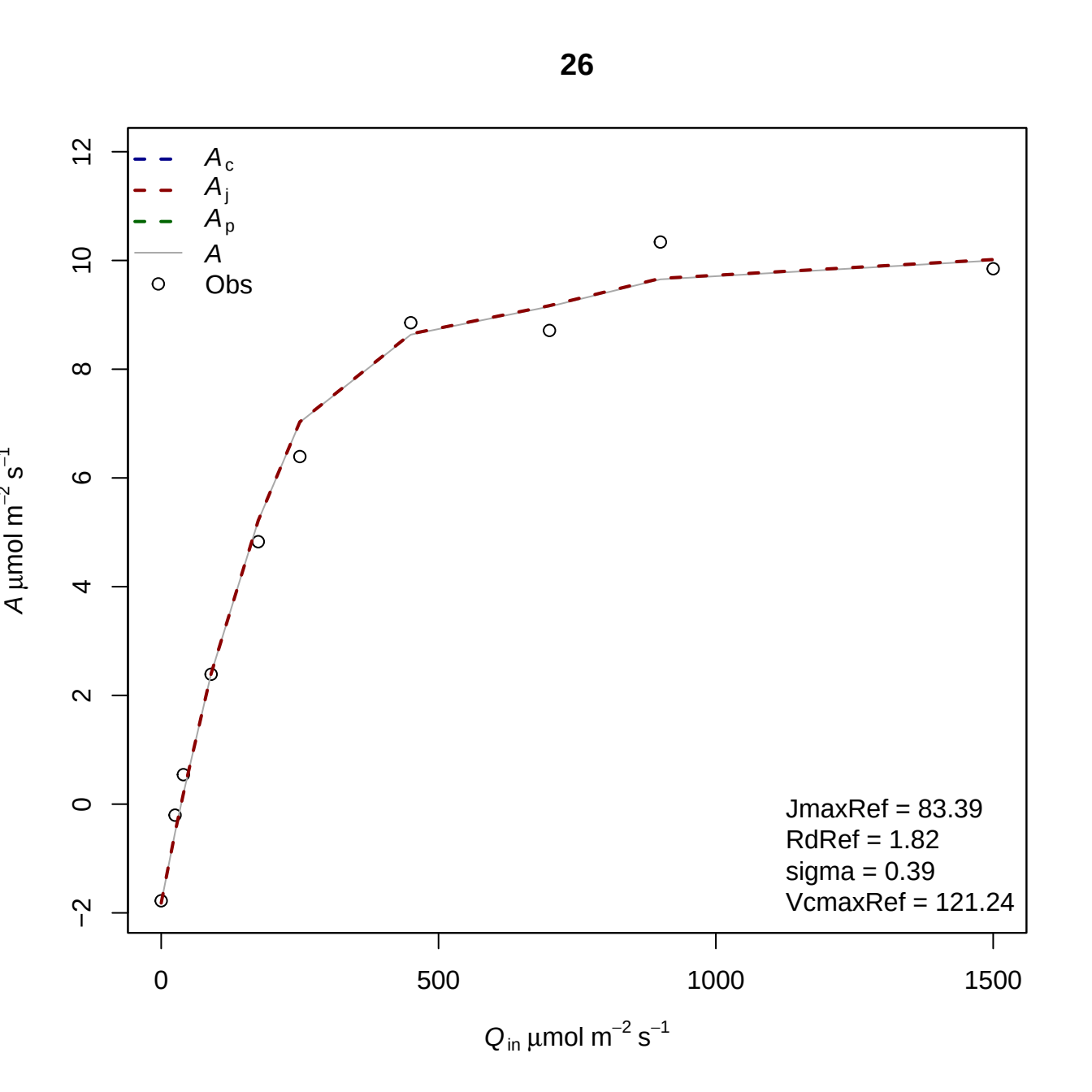

A_c
A_j
A_p
A
Obs
JmaxRef = 115.7
RdRef = 1.91
sigma = 0.64
VcmaxRef = 100.19
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

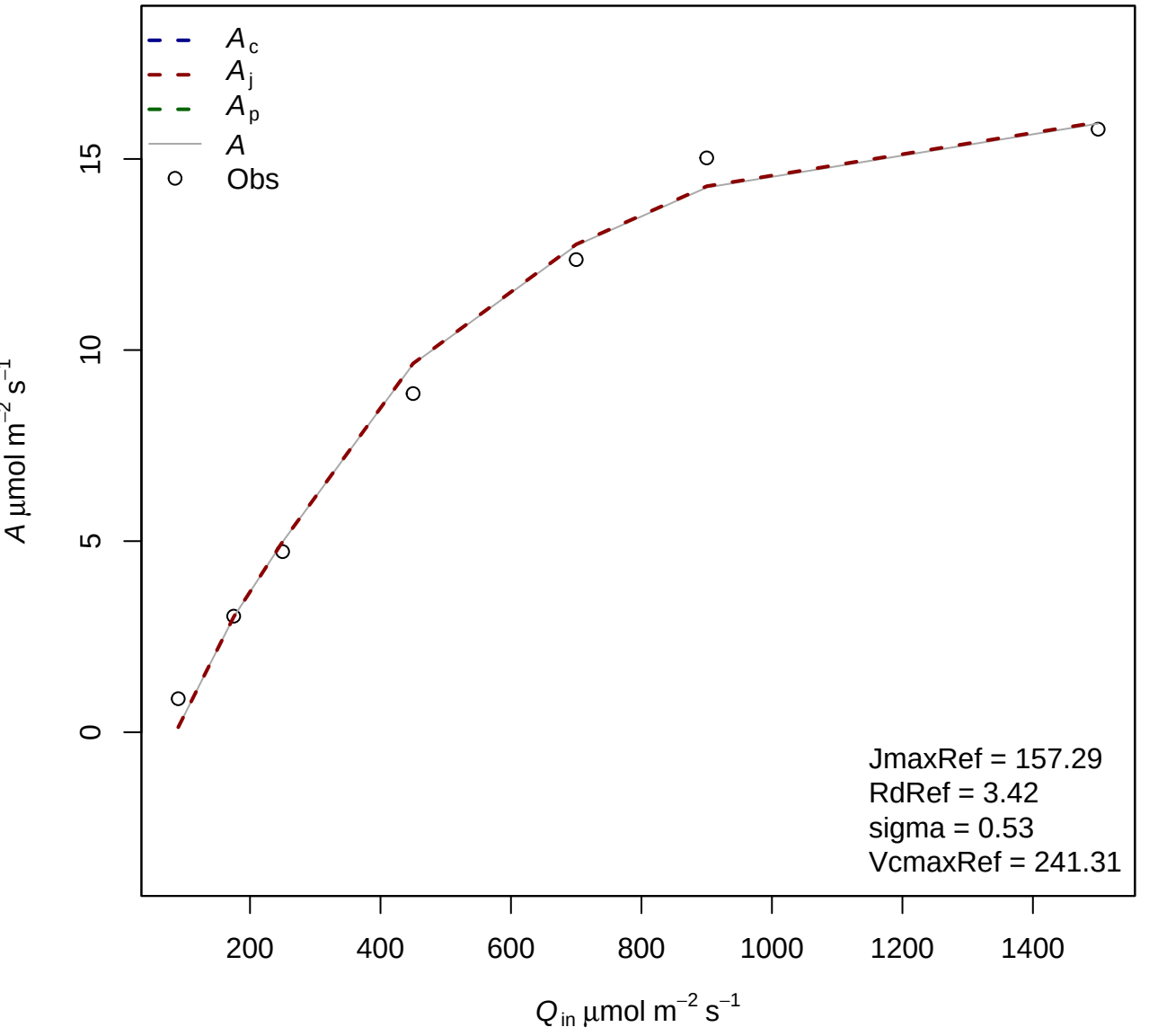

22
A_c
A_j
A_p
A
Obs
JmaxRef = 121.57
RdRef = 1.8
sigma = 0.51
VcmaxRef = 105.16
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

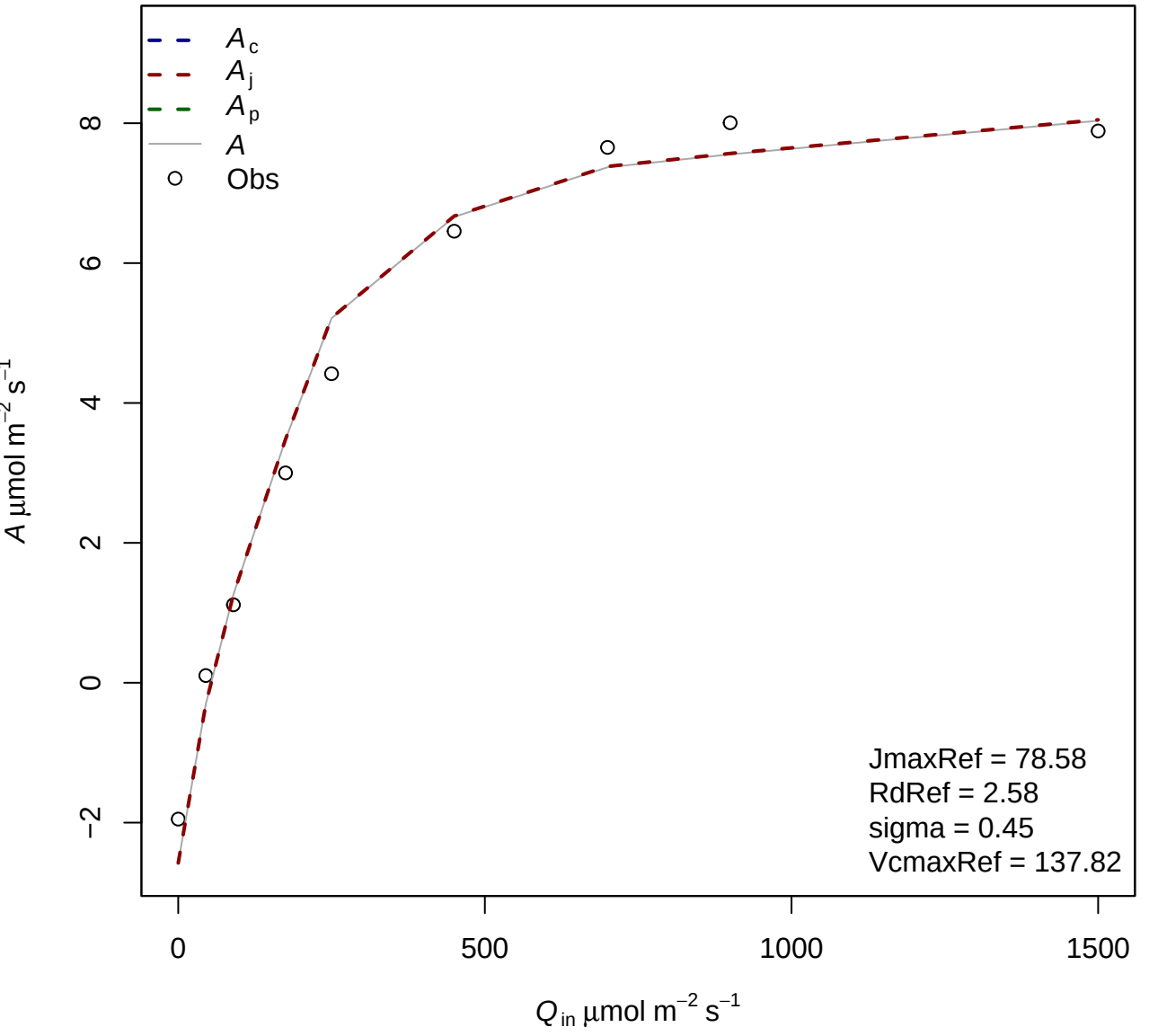

23
A_c
A_j
A_p
A
Obs
JmaxRef = 71.97
RdRef = 2.27
sigma = 0.89
VcmaxRef = 115.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

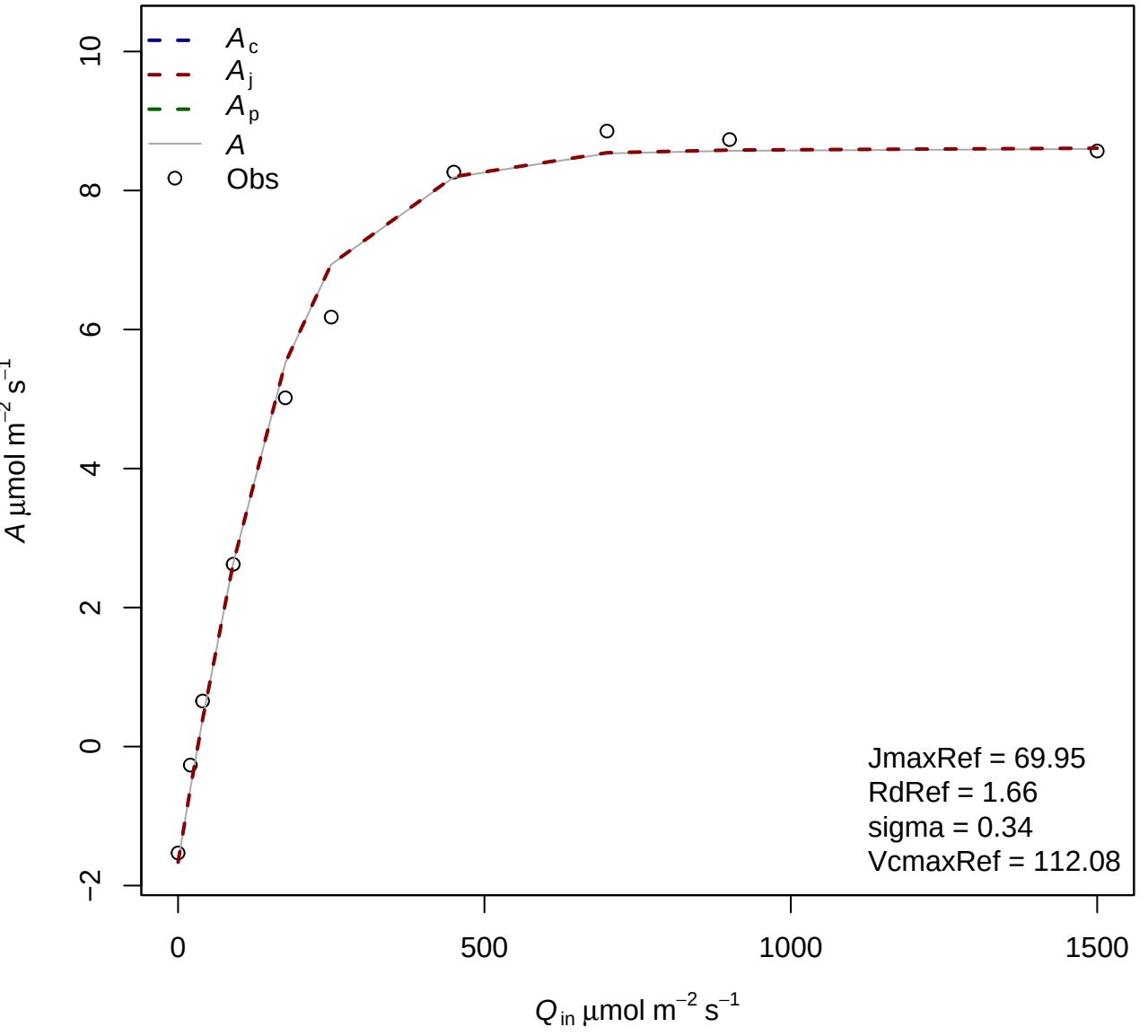

24
A_c
A_j
A_p
A
Obs
JmaxRef = 60.78
RdRef = 2
sigma = 1.28
VcmaxRef = 88.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

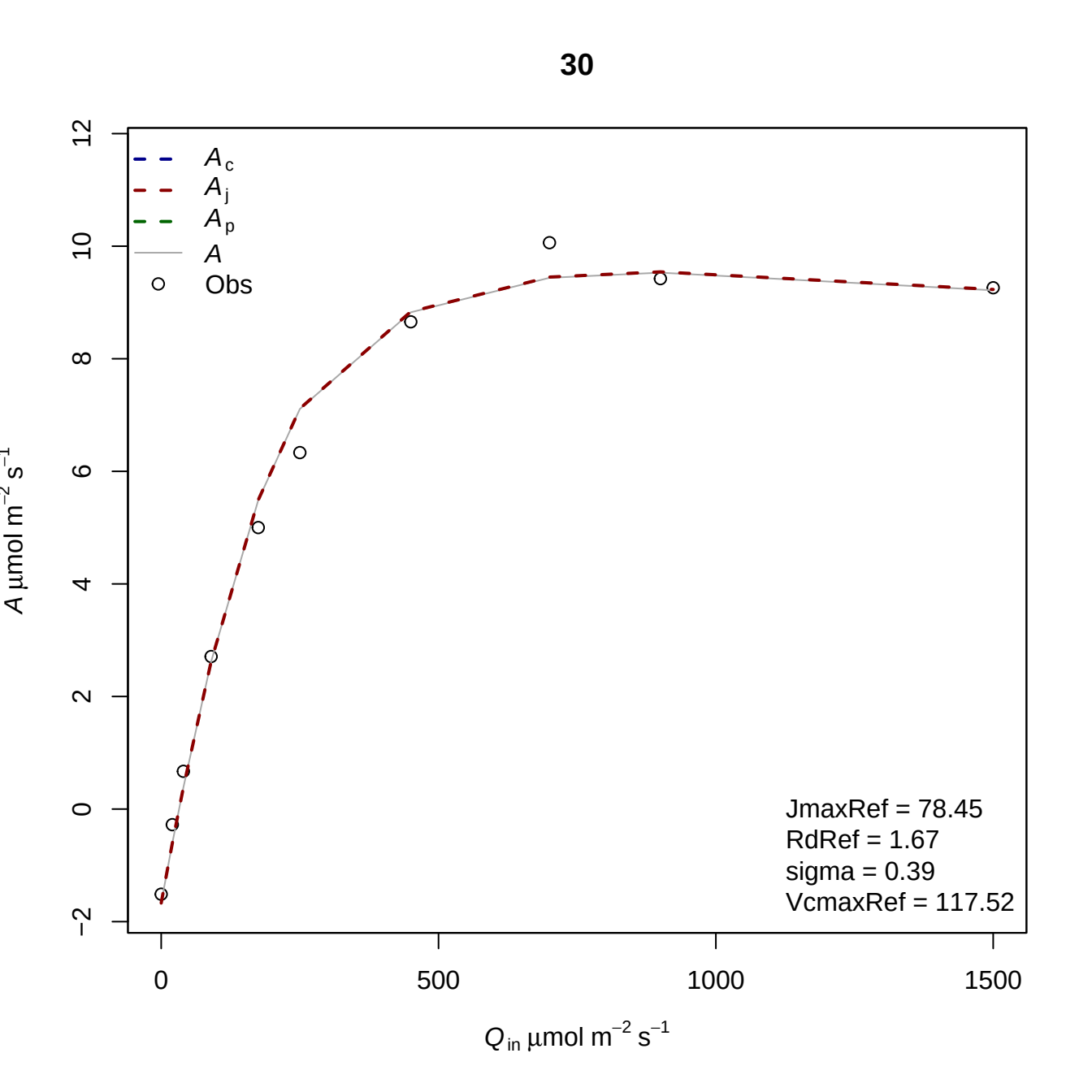

25
A_c
A_j
A_p
A
Obs
JmaxRef = 87.37
RdRef = 1.65
sigma = 0.34
VcmaxRef = 154.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

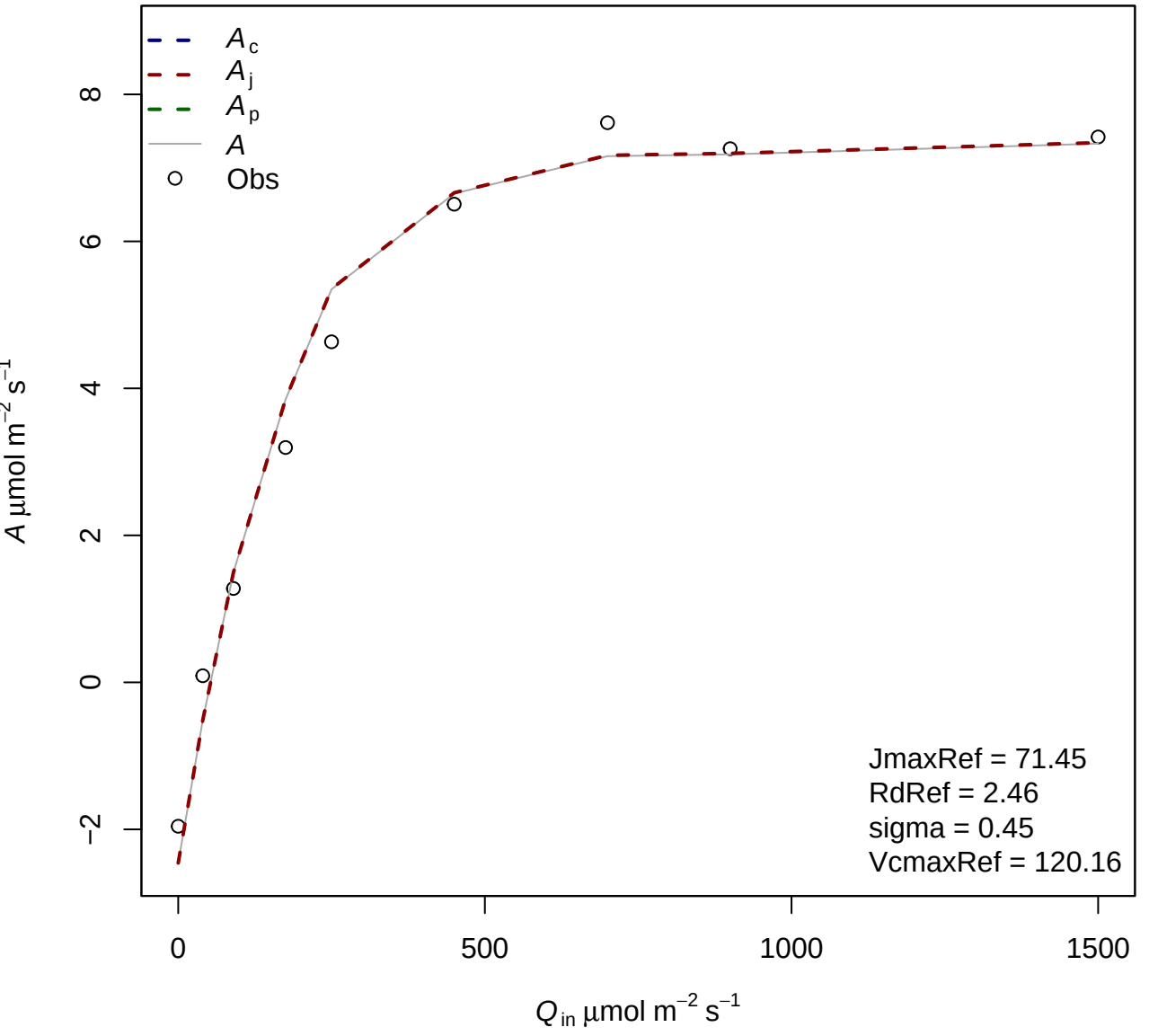

26
A_c
A_j
A_p
A
Obs
JmaxRef = 83.39
RdRef = 1.82
sigma = 0.39
VcmaxRef = 121.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

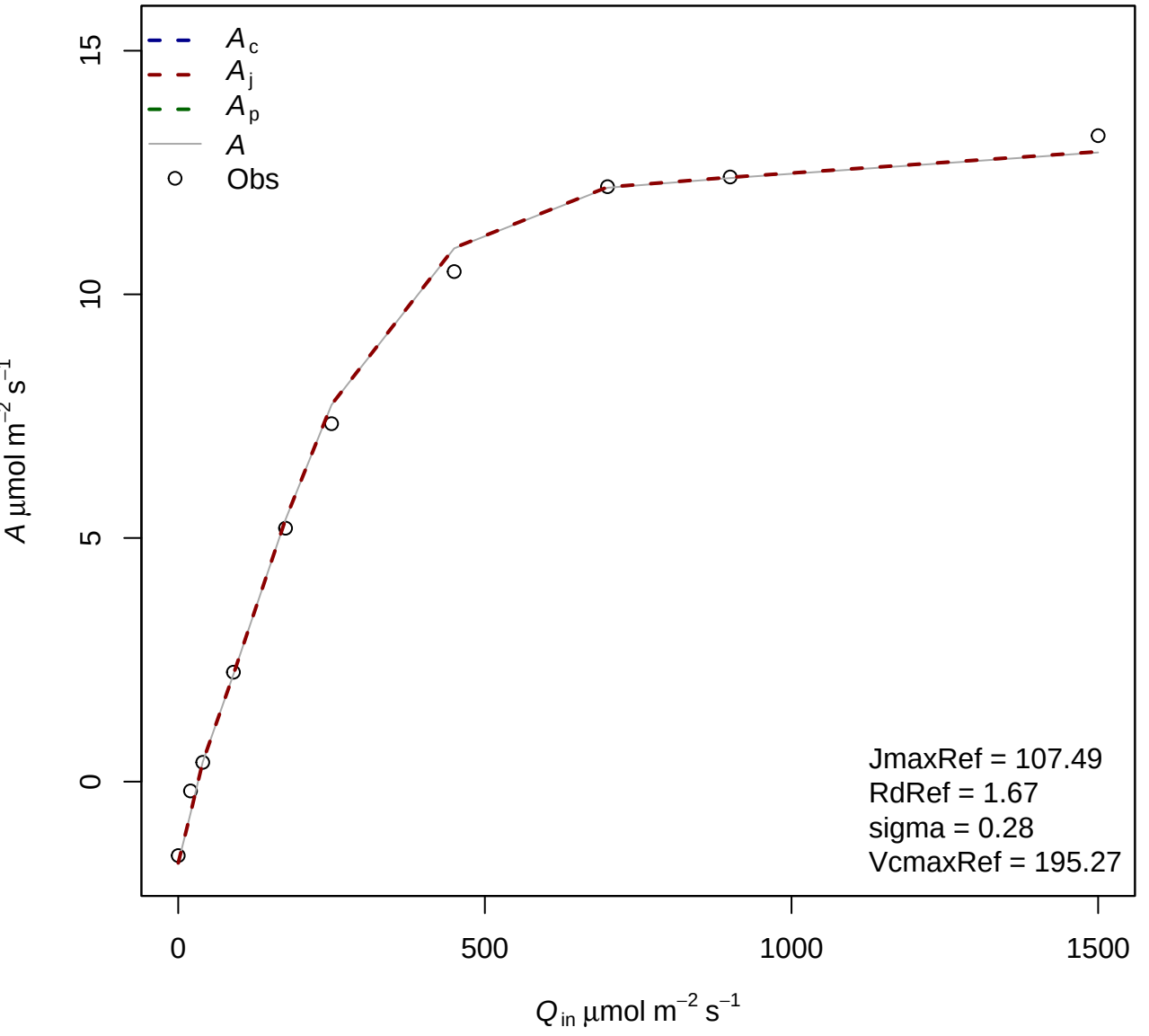

A_c
A_j
A_p
A
Obs
JmaxRef = 157.29
RdRef = 3.42
sigma = 0.53
VcmaxRef = 241.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
200
400
600
800
1000
1200
1400

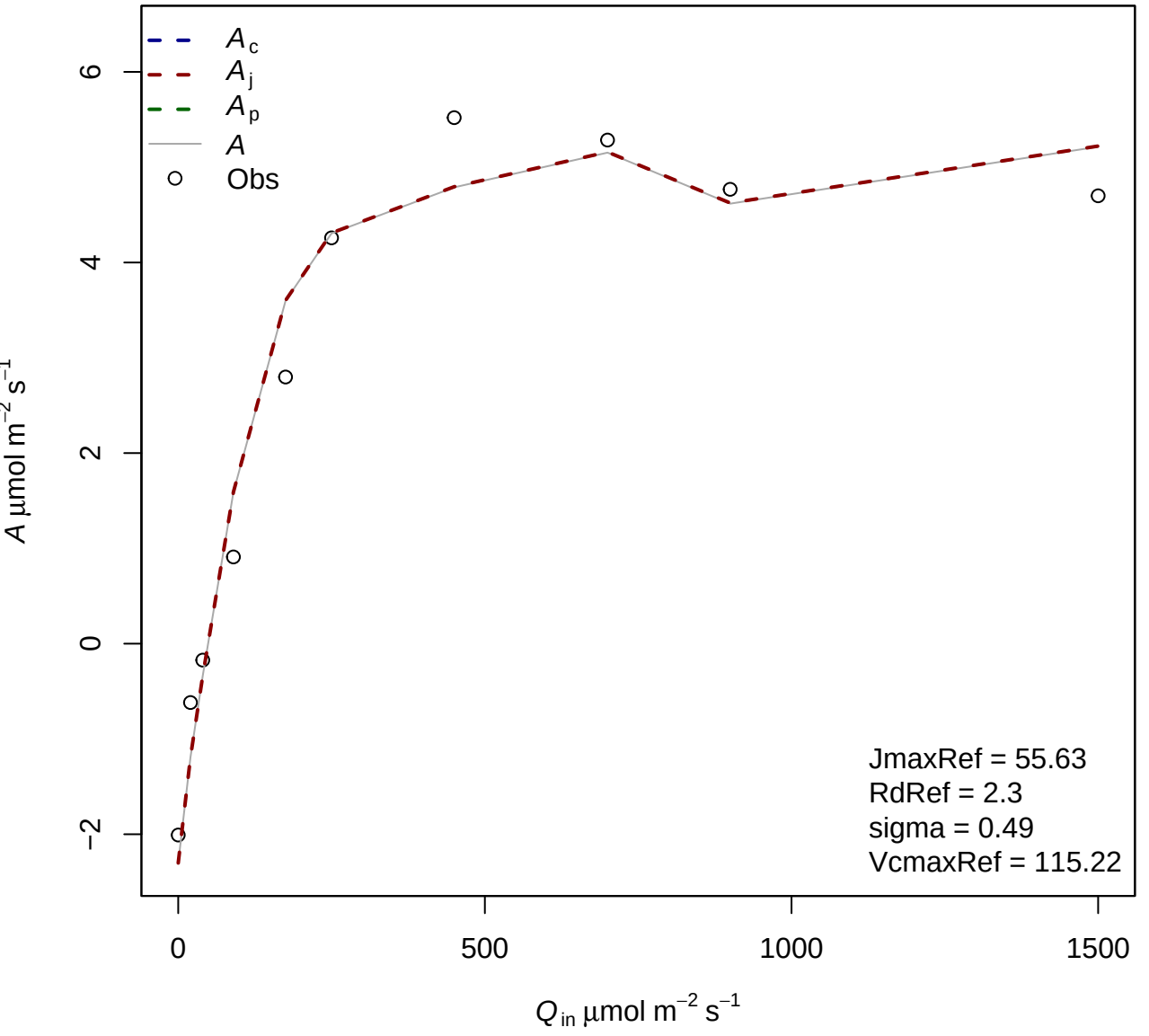

A_c
A_j
A_p
A
Obs
JmaxRef = 78.58
RdRef = 2.58
sigma = 0.45
VcmaxRef = 137.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

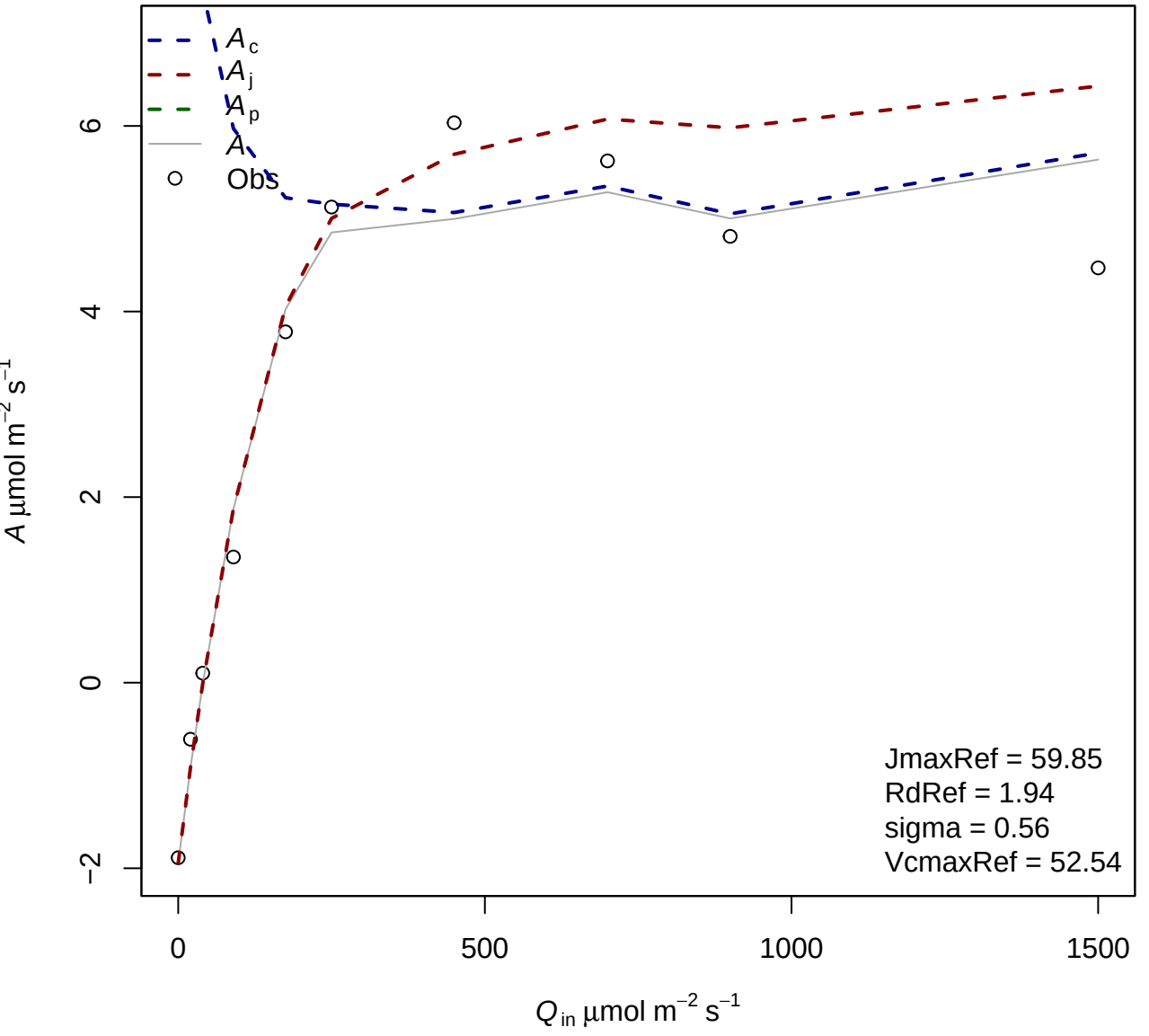

A_c
A_j
A_p
A
Obs
JmaxRef = 69.95
RdRef = 1.66
sigma = 0.34
VcmaxRef = 112.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

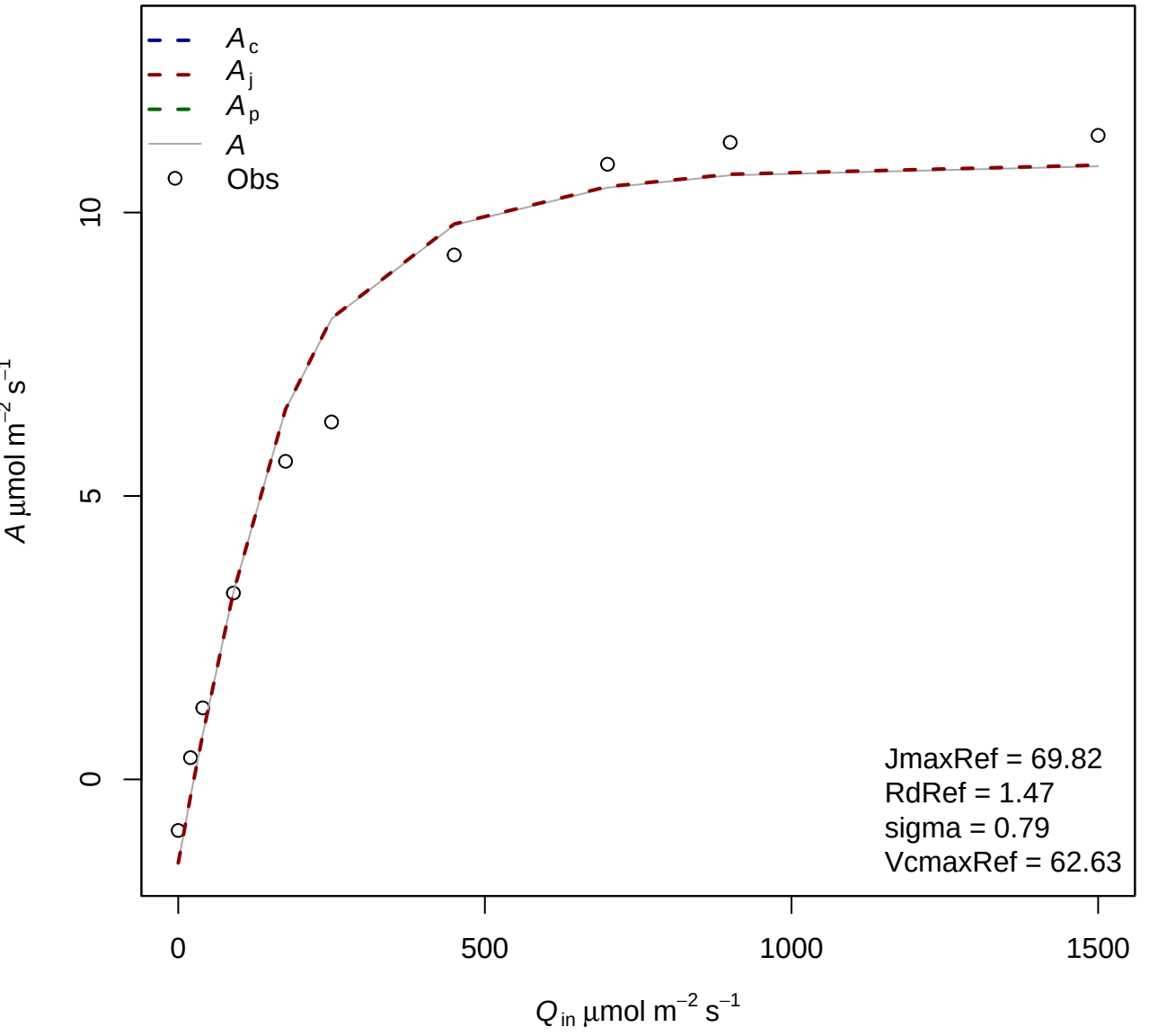

30
A_c
A_j
A_p
A
Obs
JmaxRef = 78.45
RdRef = 1.67
sigma = 0.39
VcmaxRef = 117.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

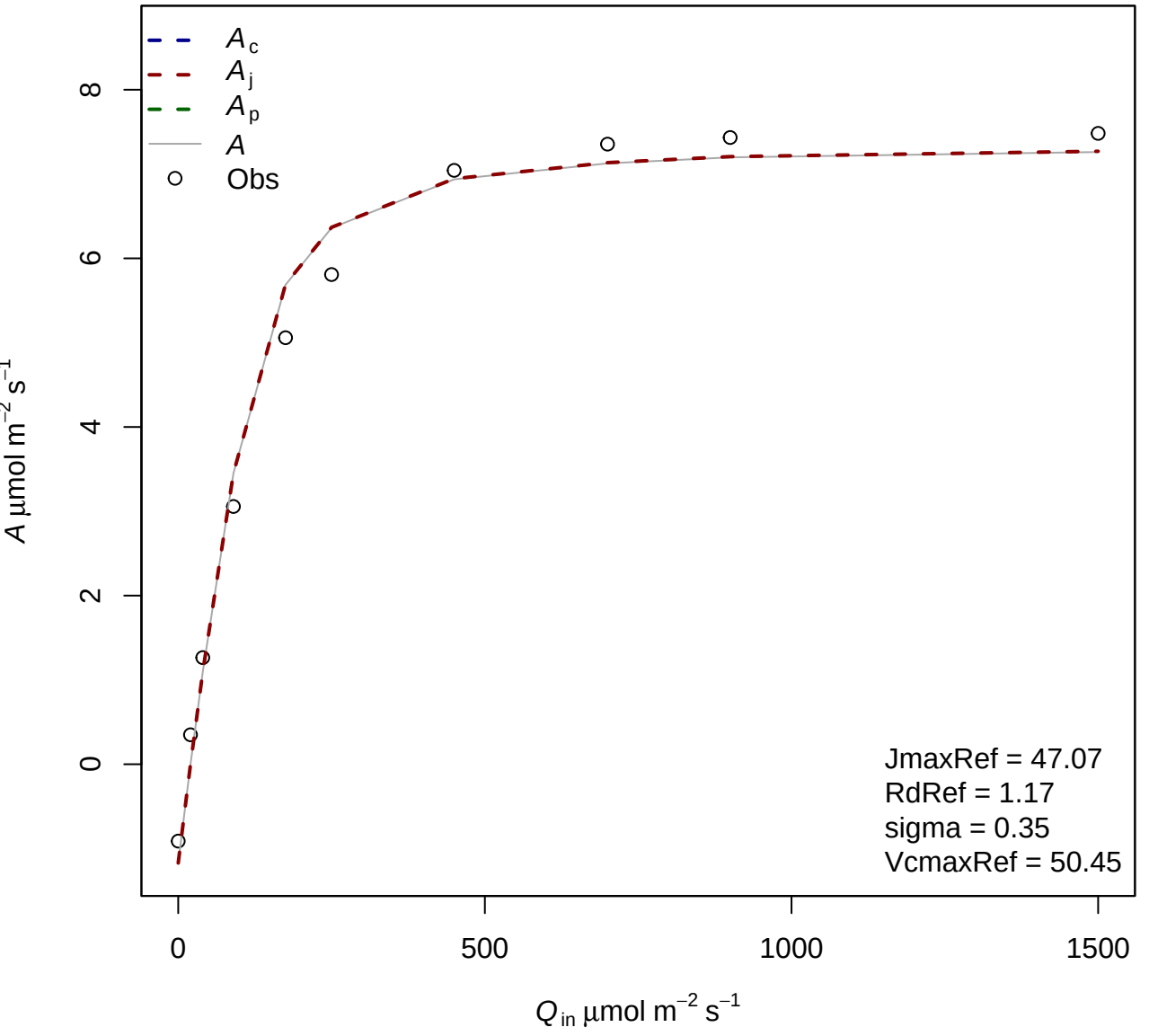

A_c
A_j
A_p
A
Obs
JmaxRef = 71.45
RdRef = 2.46
sigma = 0.45
VcmaxRef = 120.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

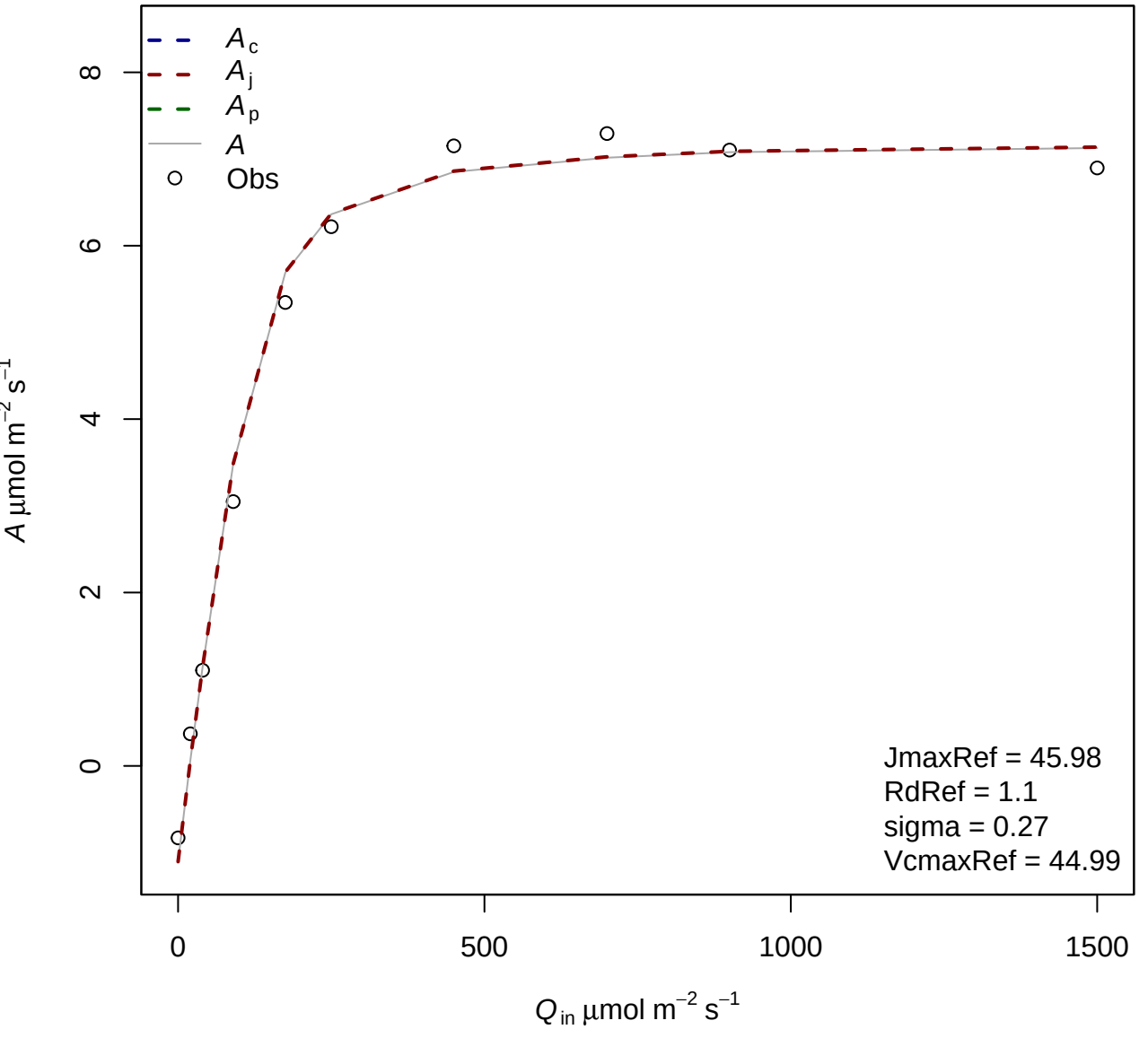

A_c
A_j
A_p
A
Obs
JmaxRef = 107.49
RdRef = 1.67
sigma = 0.28
VcmaxRef = 195.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

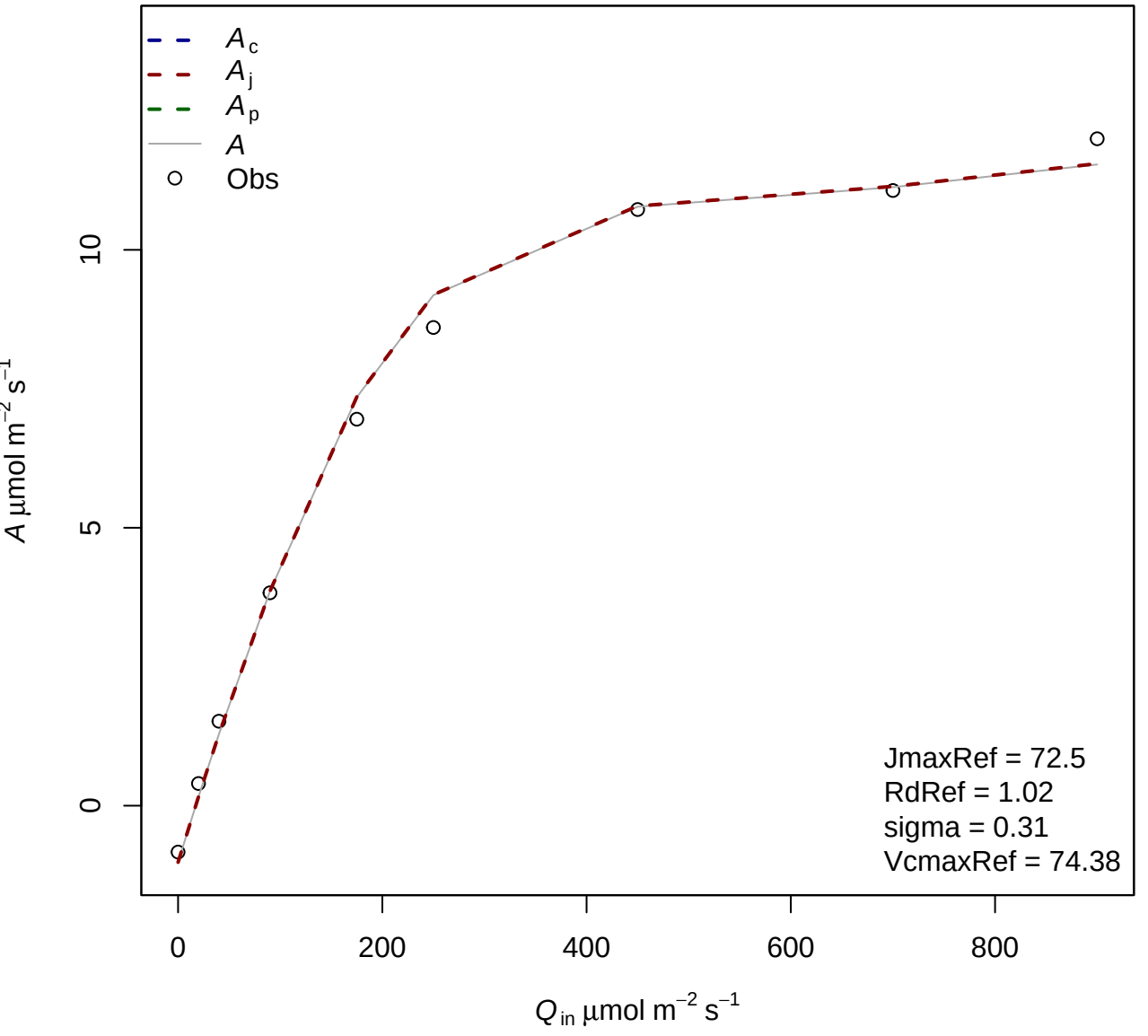

A_c
A_j
A_p
A
Obs
JmaxRef = 55.63
RdRef = 2.3
sigma = 0.49
VcmaxRef = 115.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

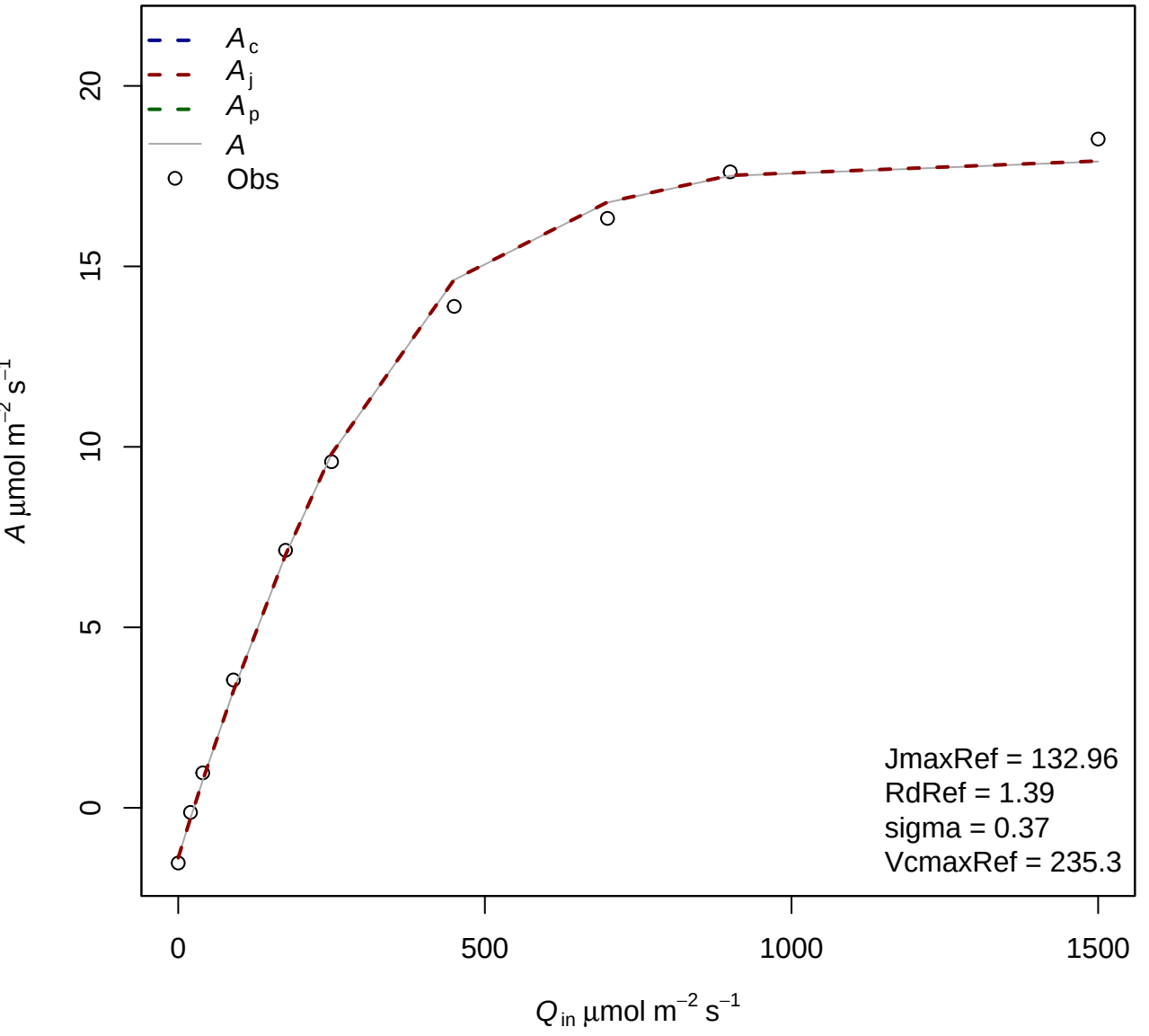

A_c
A_j
A_p
A
Obs
JmaxRef = 59.85
RdRef = 1.94
sigma = 0.56
VcmaxRef = 52.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

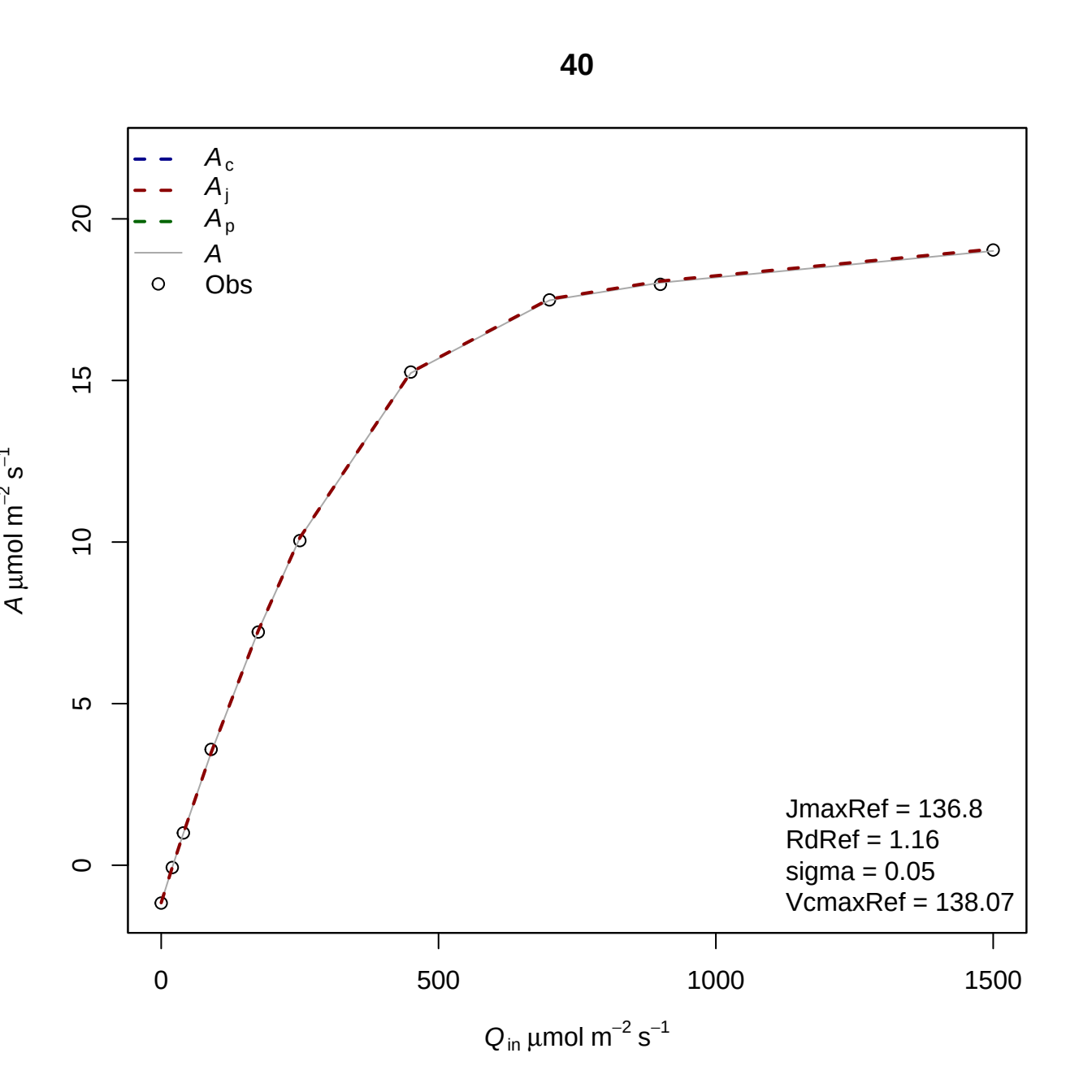

A_c
A_j
A_p
A
Obs
JmaxRef = 69.82
RdRef = 1.47
sigma = 0.79
VcmaxRef = 62.63
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

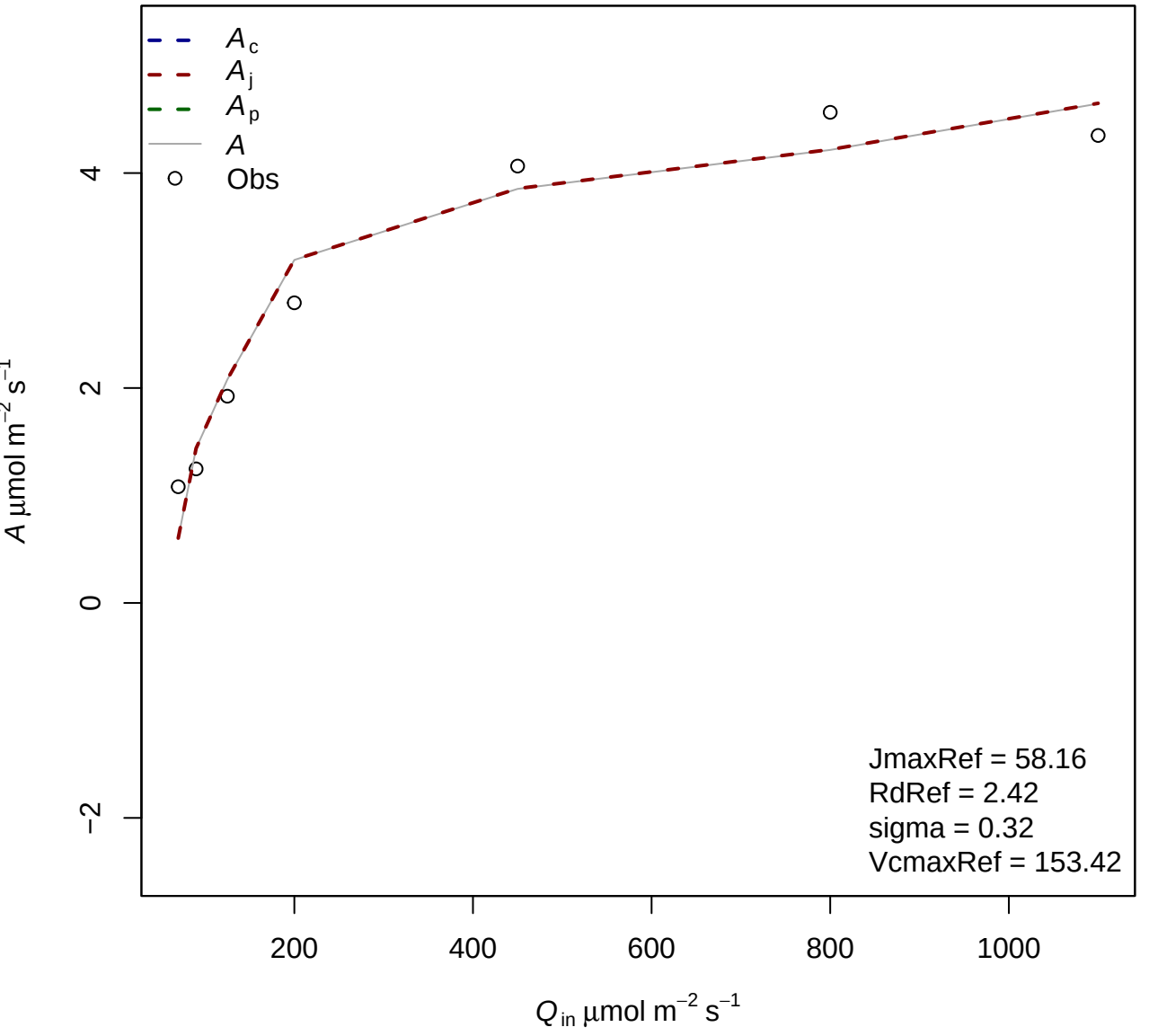

A_c
A_j
A_p
A
Obs
JmaxRef = 47.07
RdRef = 1.17
sigma = 0.35
VcmaxRef = 50.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

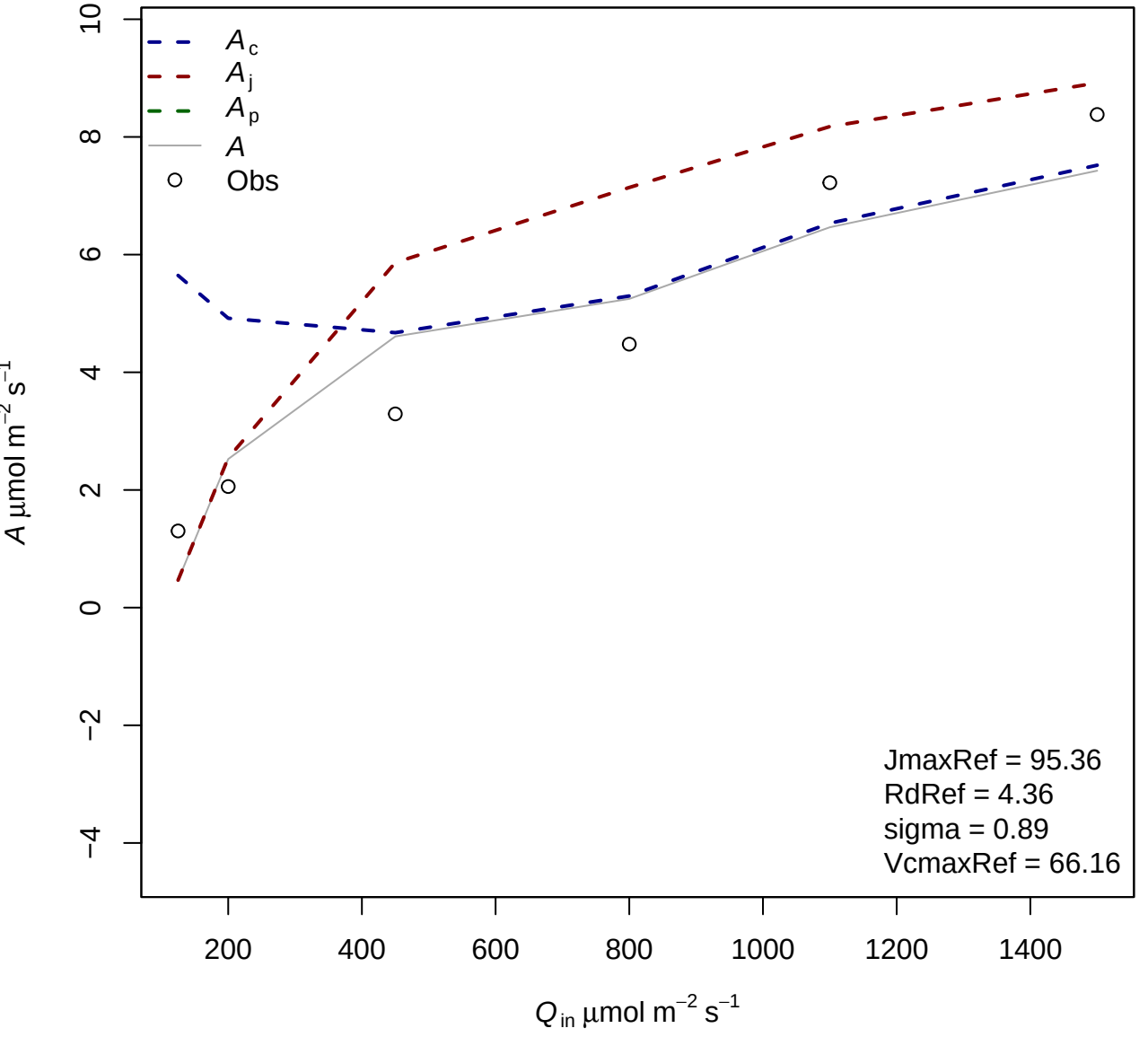

A_c
A_j
A_p
A
Obs
JmaxRef = 45.98
RdRef = 1.1
sigma = 0.27
VcmaxRef = 44.99
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

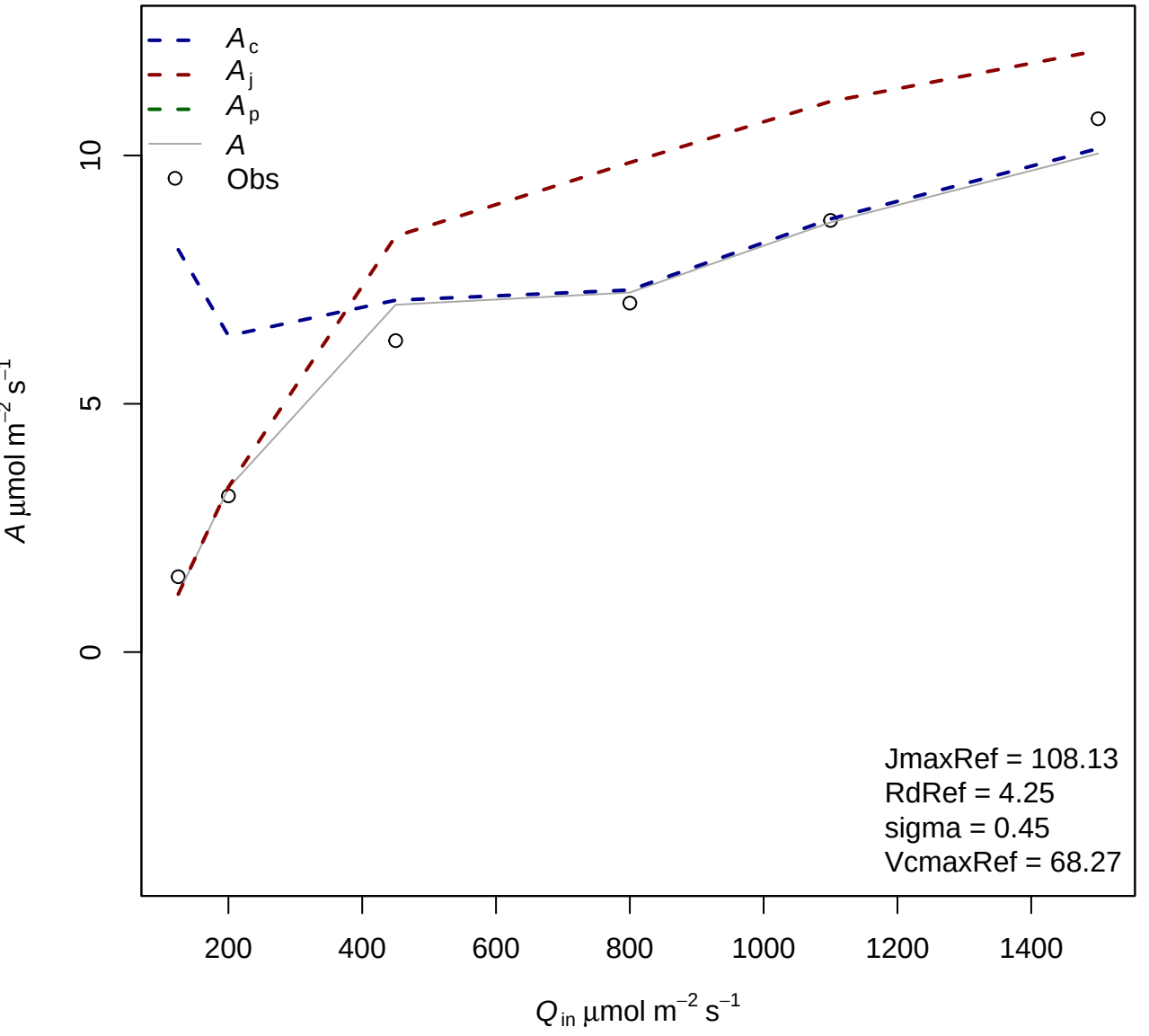

A_c
A_j
A_p
A
Obs
JmaxRef = 72.5
RdRef = 1.02
sigma = 0.31
VcmaxRef = 74.38
A µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
Q_{in} µmol m^{-2} s^{-1}

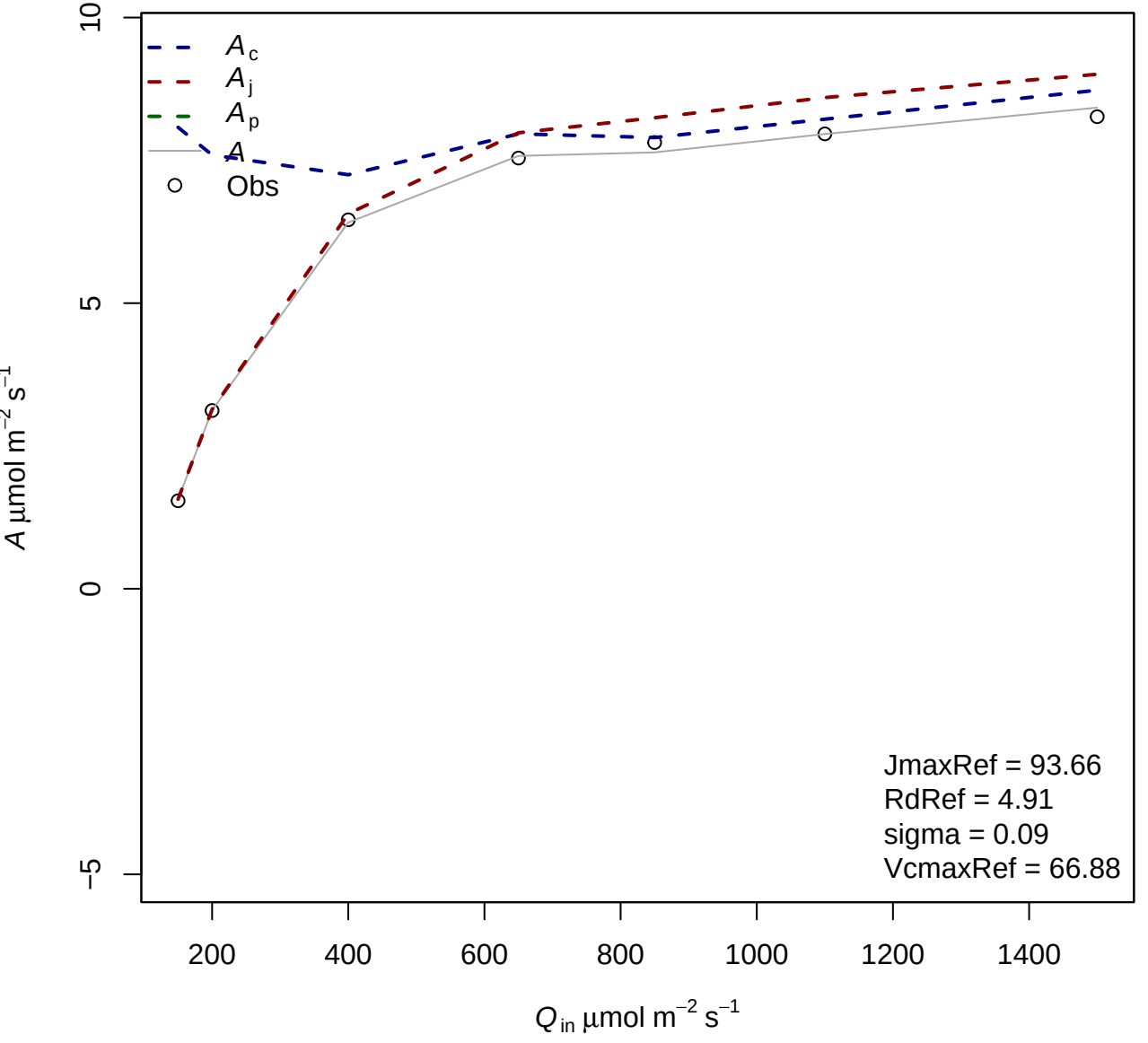

A_c
A_j
A_p
A
Obs
JmaxRef = 132.96
RdRef = 1.39
sigma = 0.37
VcmaxRef = 235.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

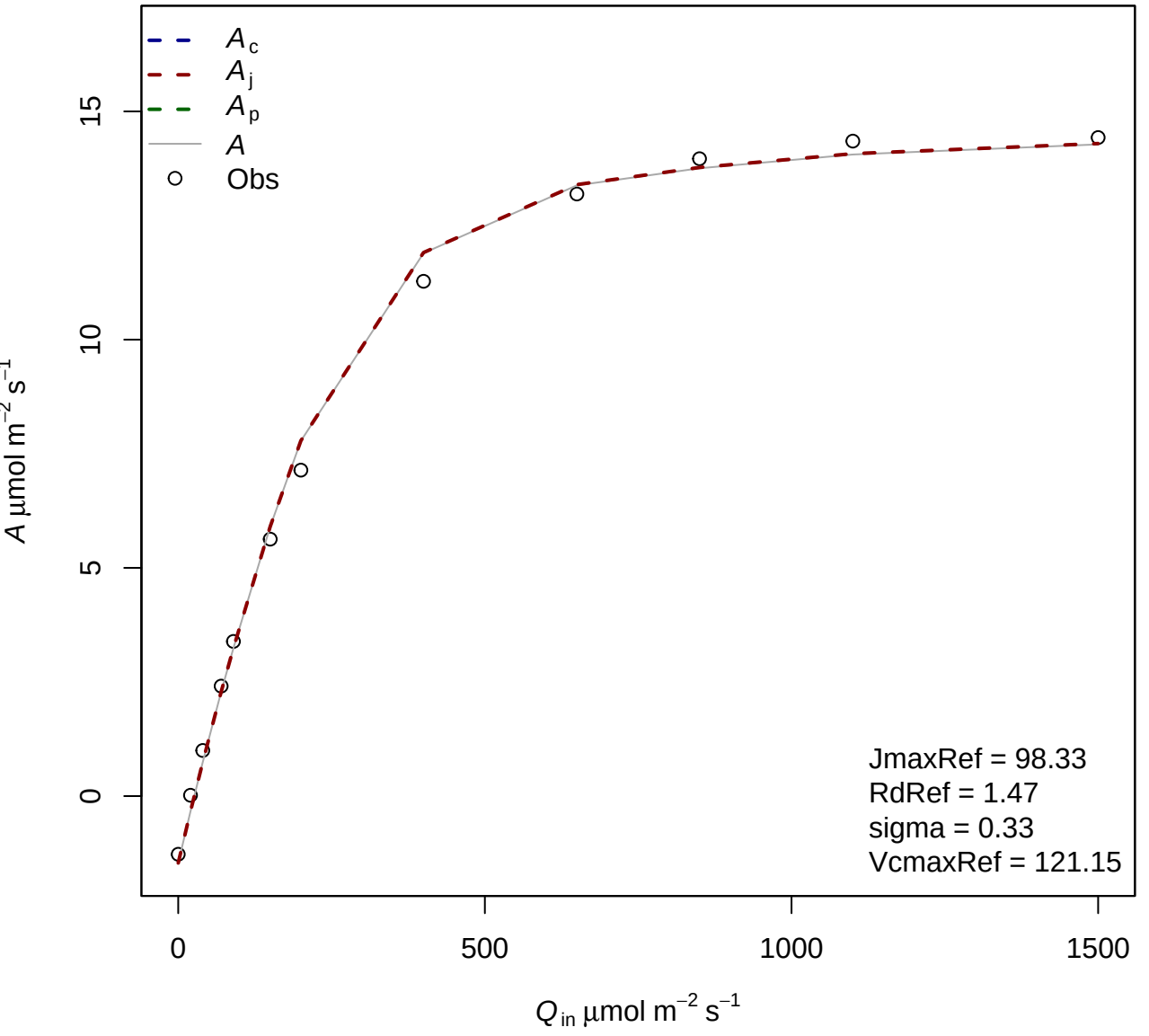

40
A_c
A_j
A_p
A
Obs
JmaxRef = 136.8
RdRef = 1.16
sigma = 0.05
VcmaxRef = 138.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

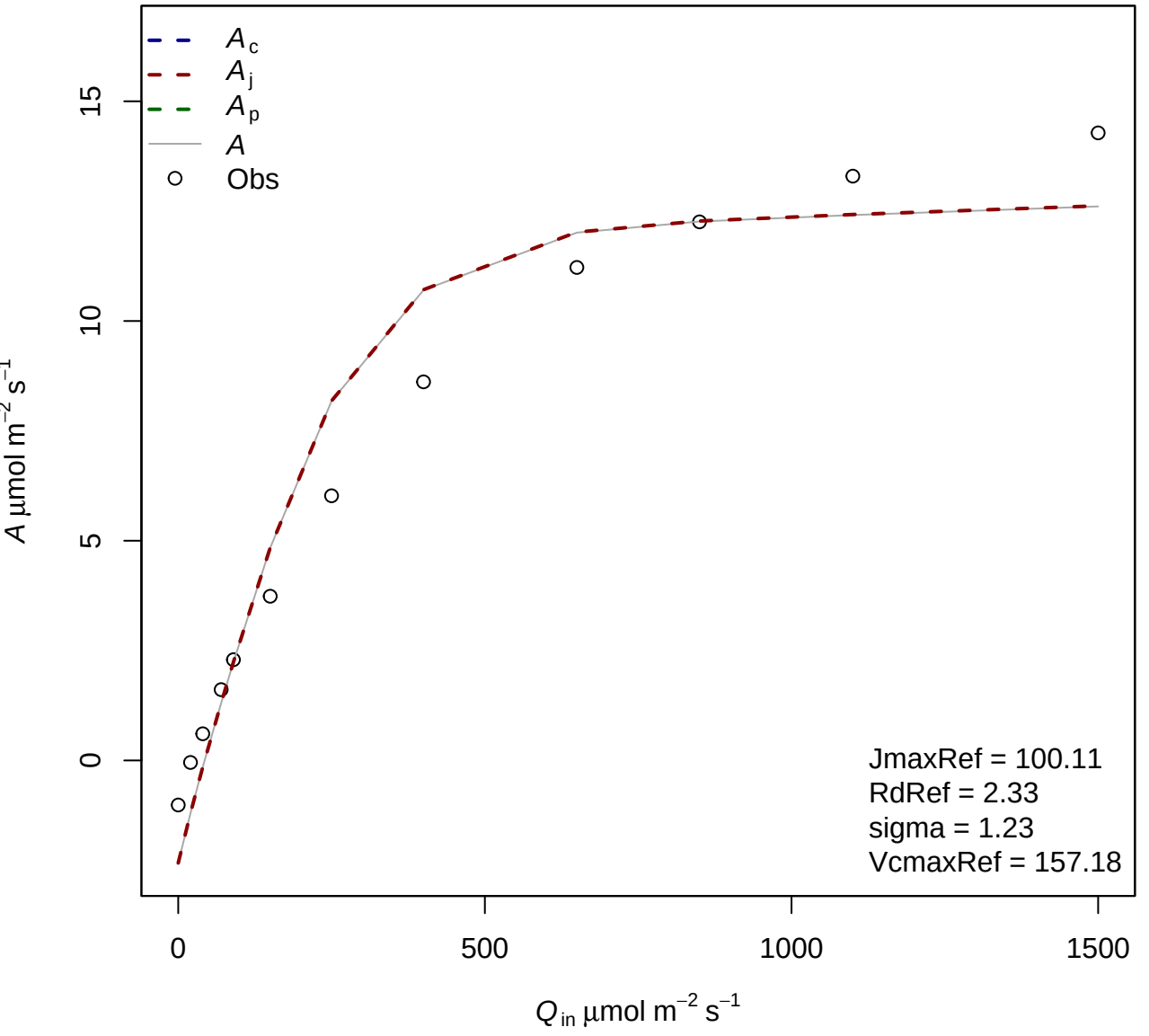

A_c
A_j
A_p
A
Obs
JmaxRef = 58.16
RdRef = 2.42
sigma = 0.32
VcmaxRef = 153.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
200
400
600
800
1000

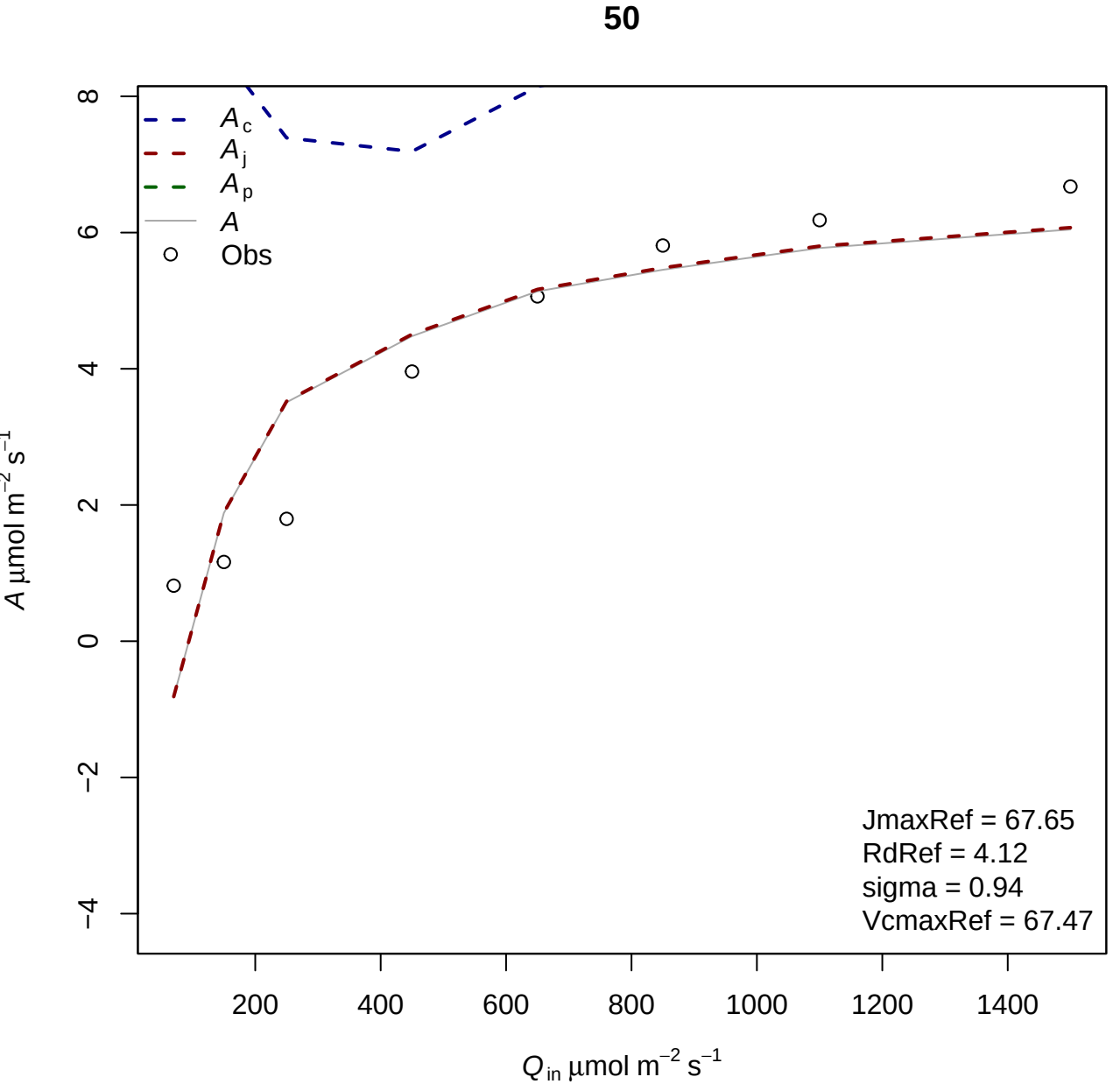

A_c
A_j
A_p
A
Obs
JmaxRef = 95.36
RdRef = 4.36
sigma = 0.89
VcmaxRef = 66.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

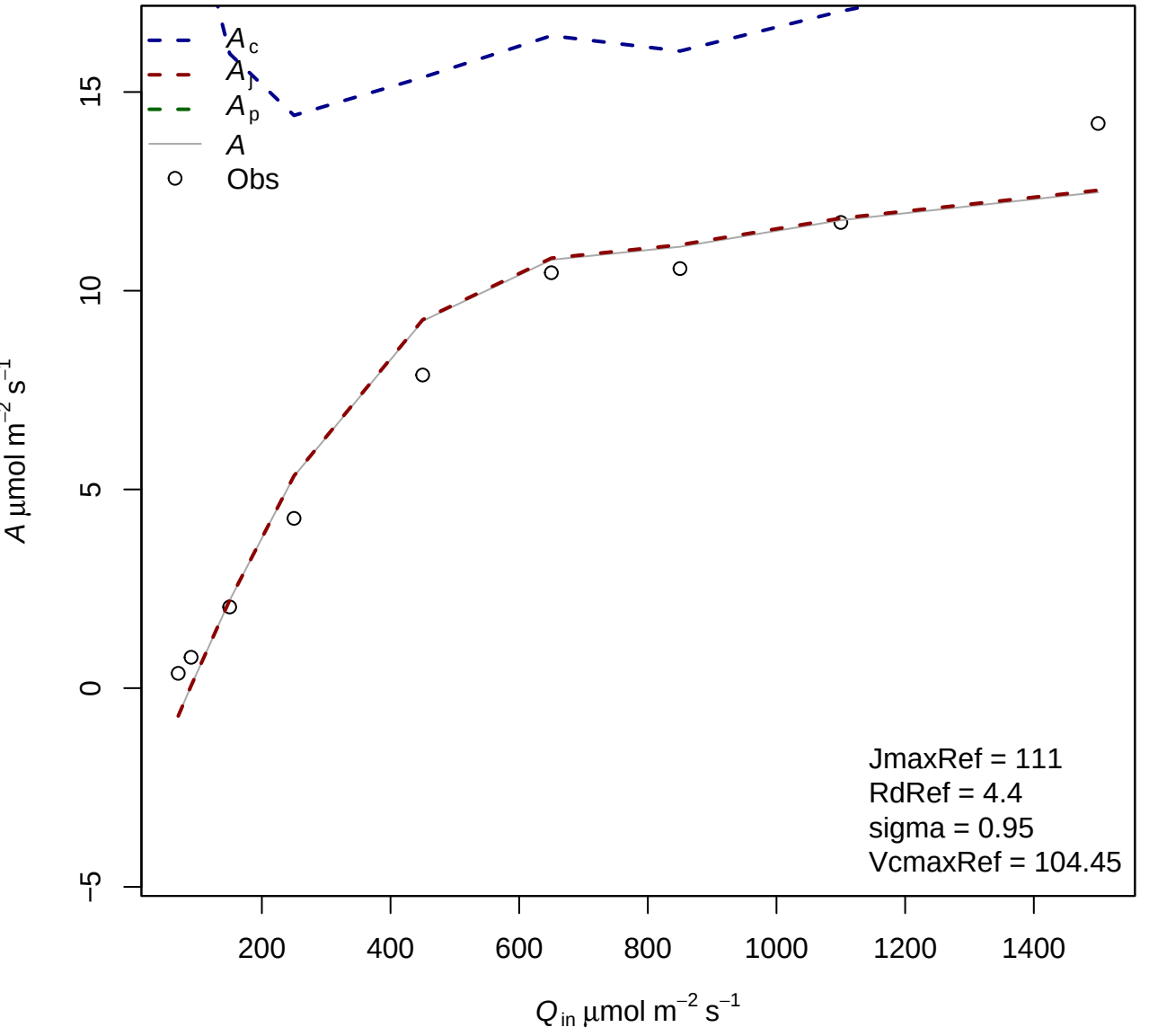

A_c
A_j
A_p
A
Obs
0
5
10
A µmol m^{-2} s^{-1}
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 108.13
RdRef = 4.25
sigma = 0.45
VcmaxRef = 68.27

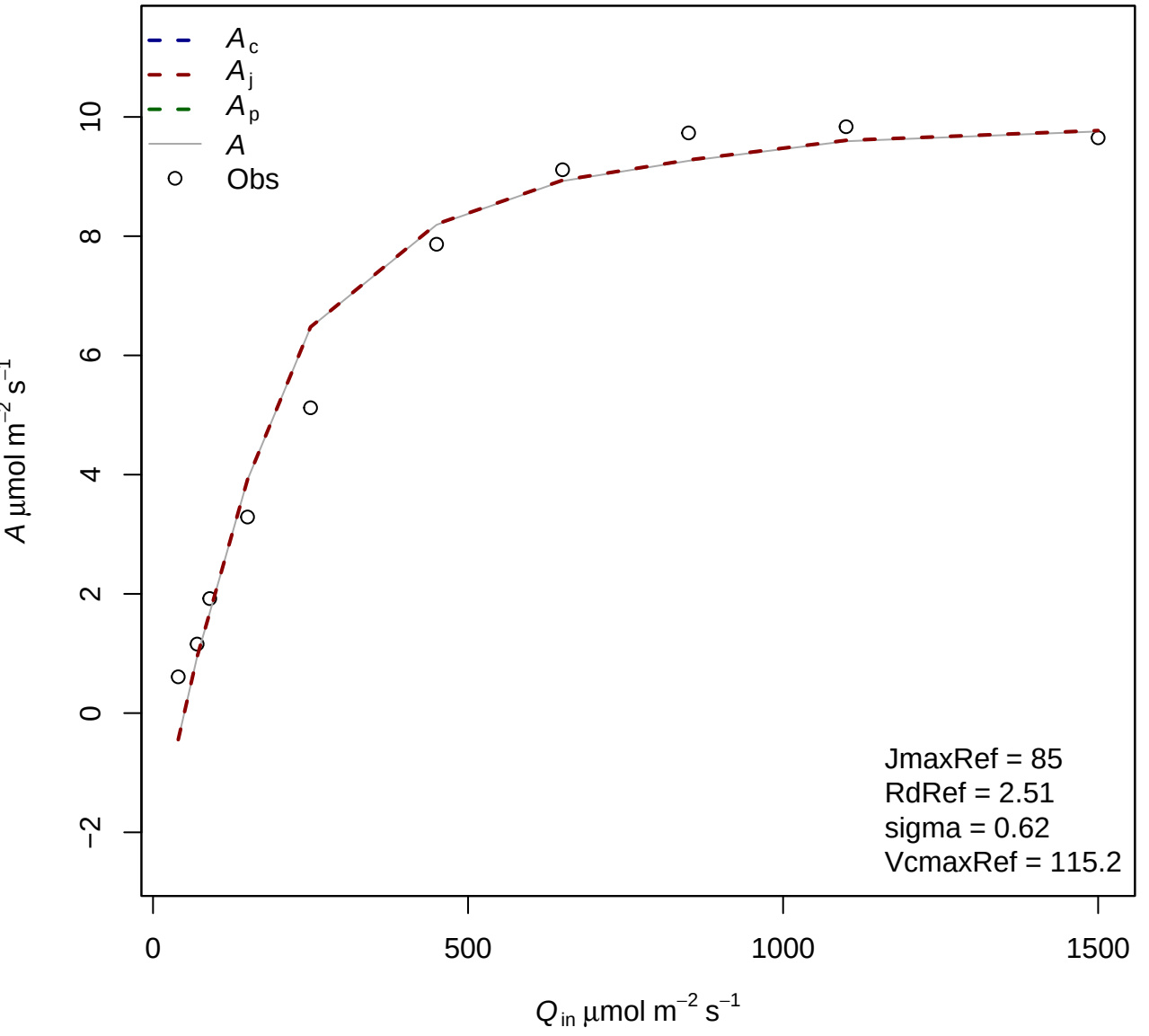

A_c
A_j
A_p
A
Obs
JmaxRef = 93.66
RdRef = 4.91
sigma = 0.09
VcmaxRef = 66.88
10
5
0
−5
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

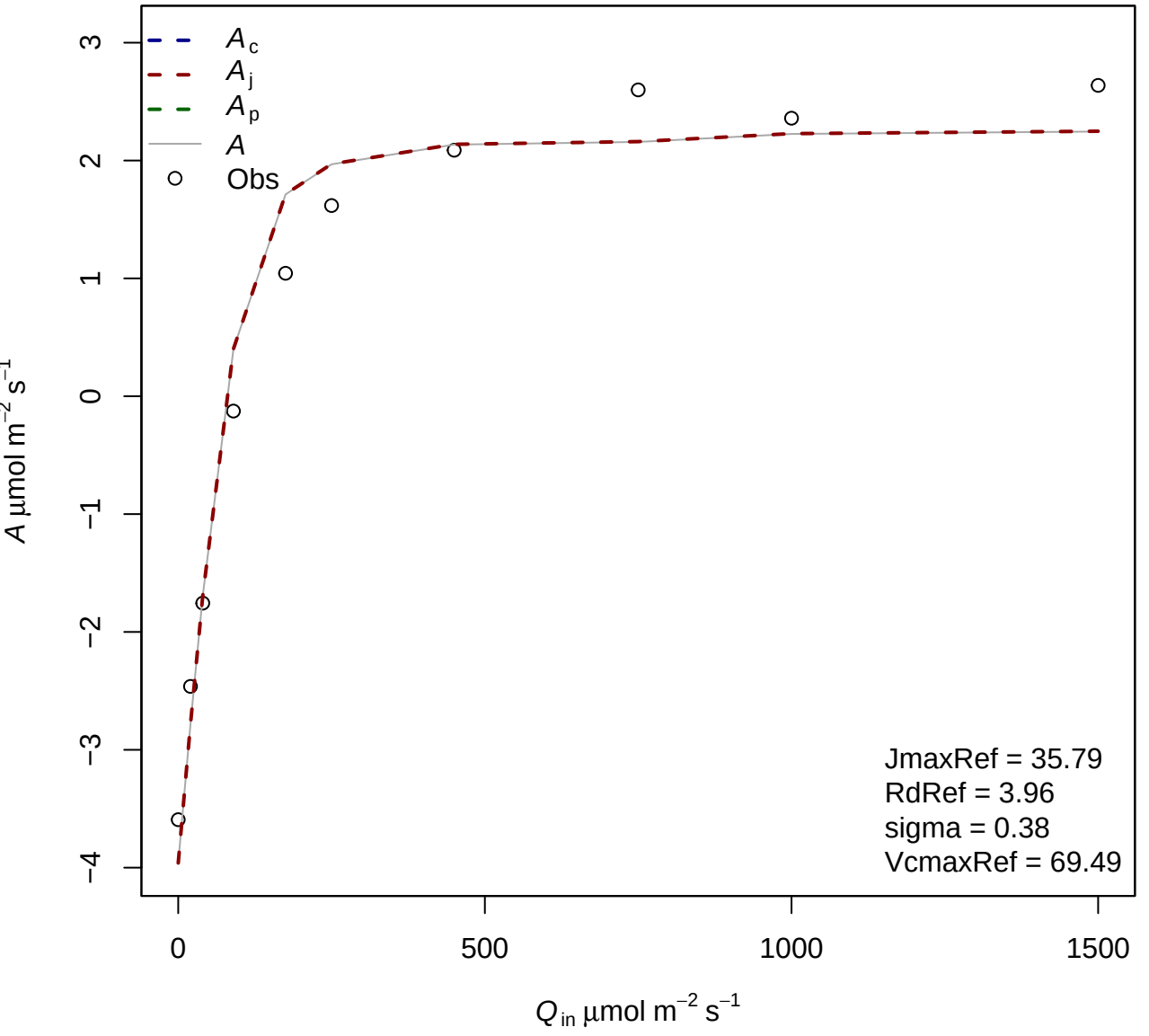

A_c
A_j
A_p
A
Obs
JmaxRef = 98.33
RdRef = 1.47
sigma = 0.33
VcmaxRef = 121.15
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

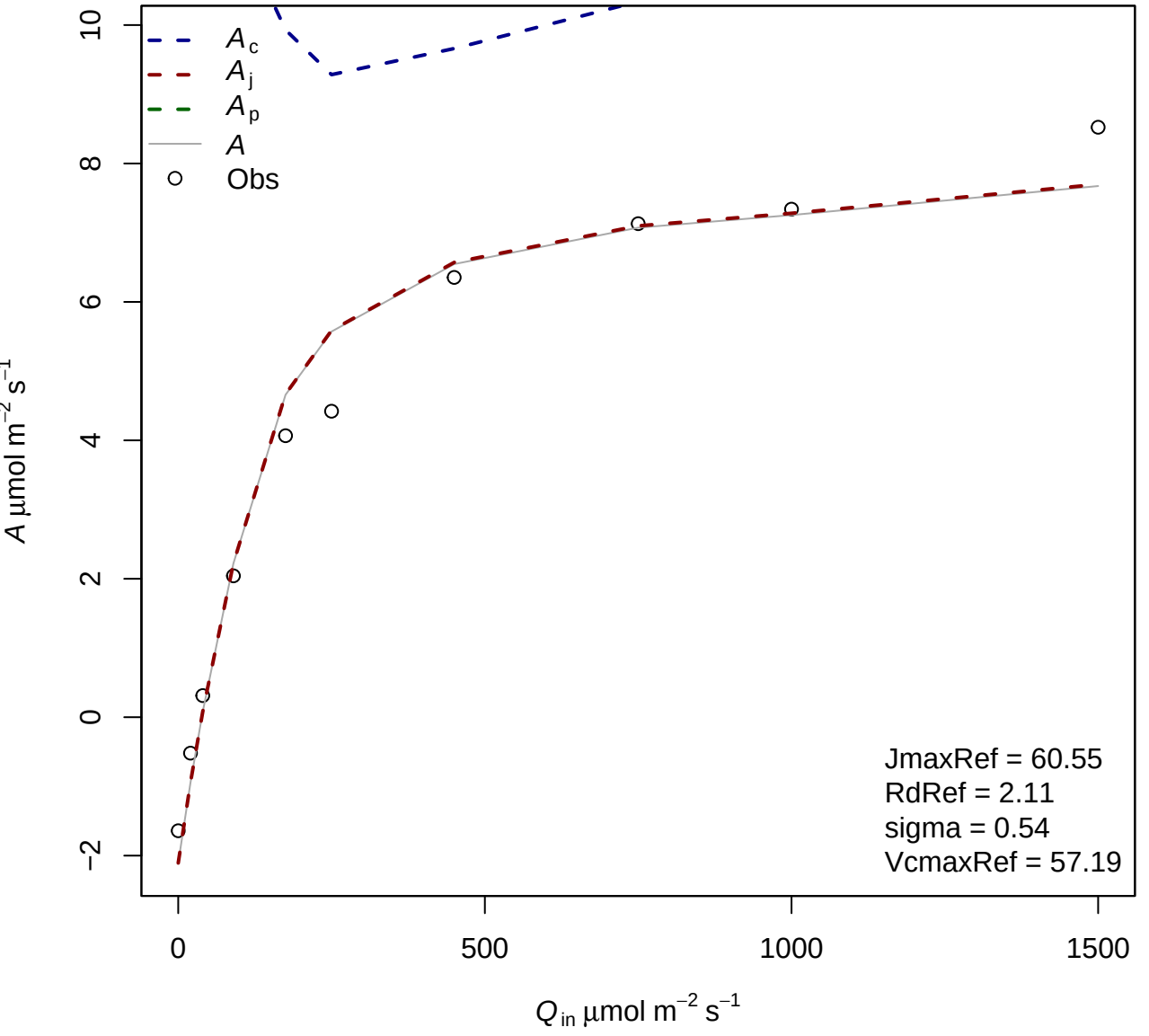

A_c
A_j
A_p
A
Obs
JmaxRef = 100.11
RdRef = 2.33
sigma = 1.23
VcmaxRef = 157.18
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

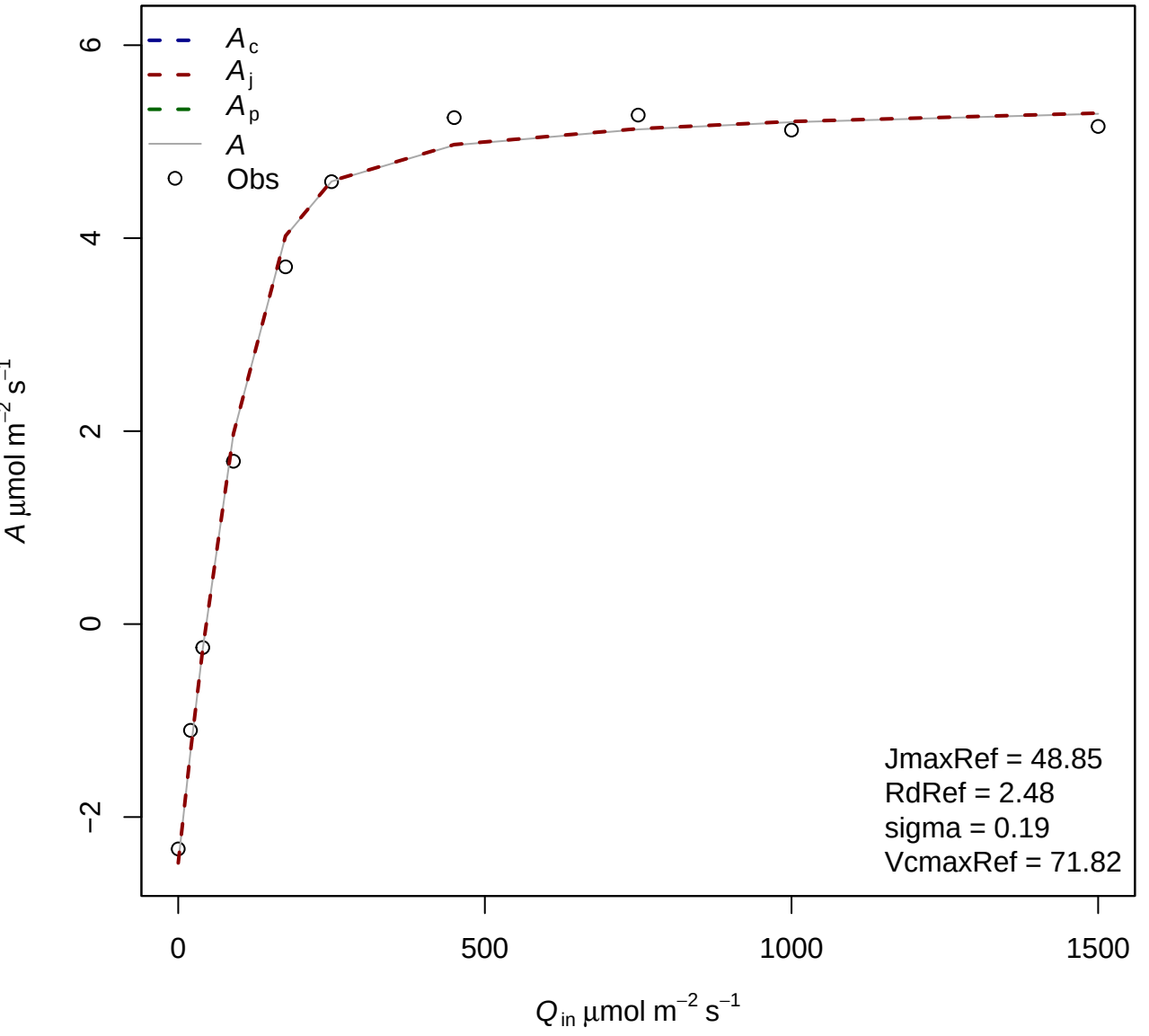

50
A_c
A_j
A_p
A
Obs
JmaxRef = 67.65
RdRef = 4.12
sigma = 0.94
VcmaxRef = 67.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

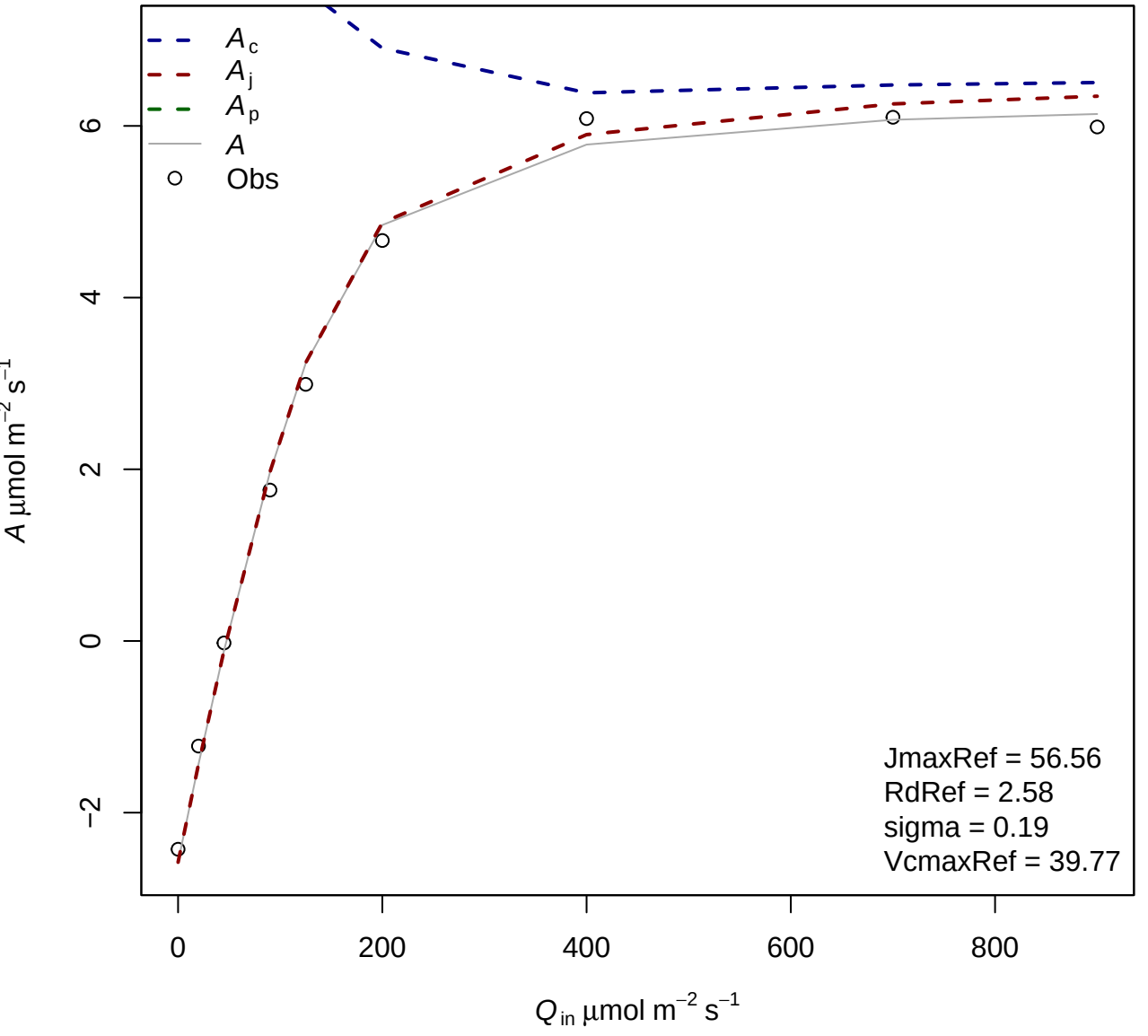

A_c
A_j
A_p
A
Obs
JmaxRef = 111
RdRef = 4.4
sigma = 0.95
VcmaxRef = 104.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
200
400
600
800
1000
1200
1400

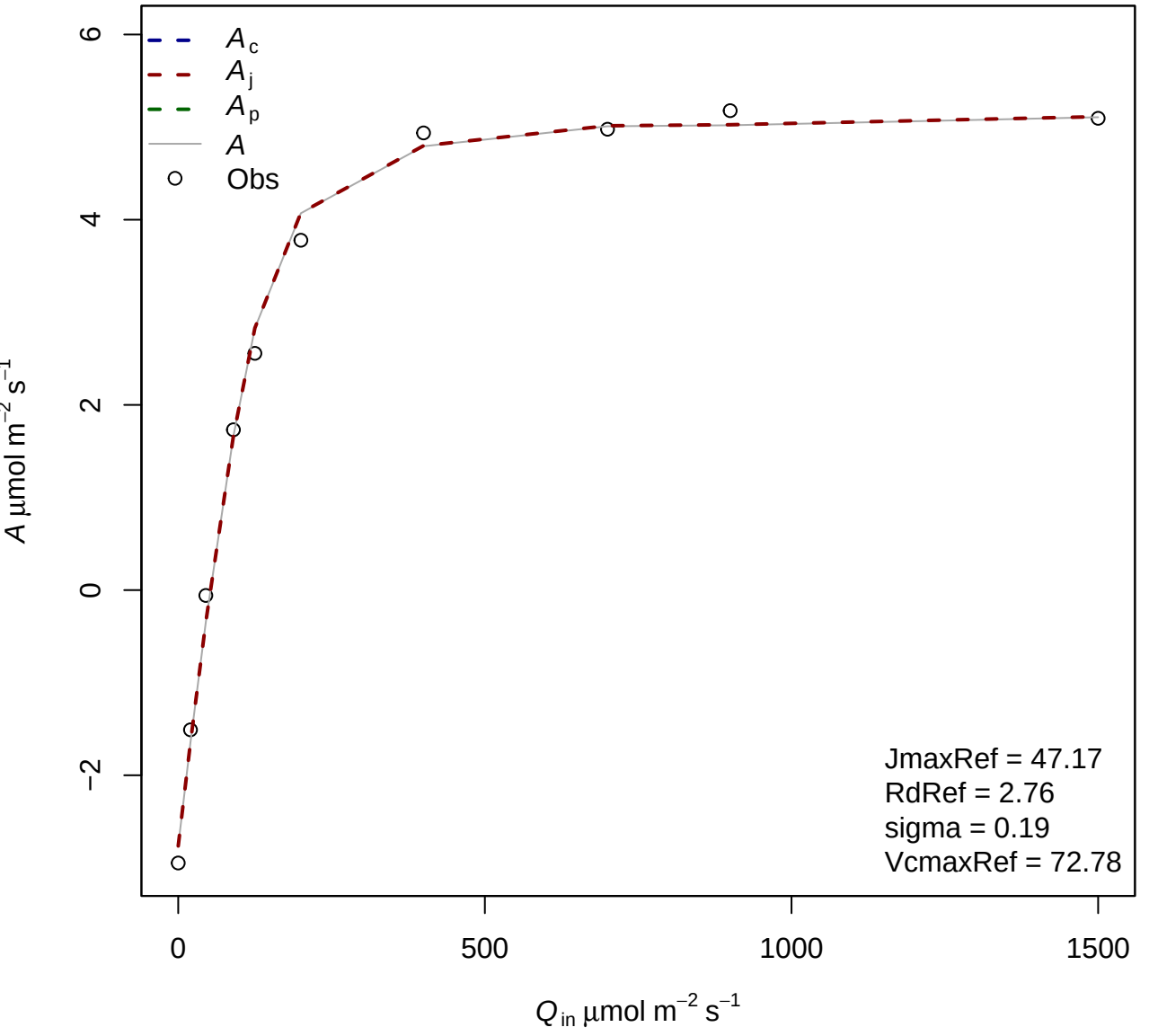

A_c
A_j
A_p
A
Obs
JmaxRef = 85
RdRef = 2.51
sigma = 0.62
VcmaxRef = 115.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

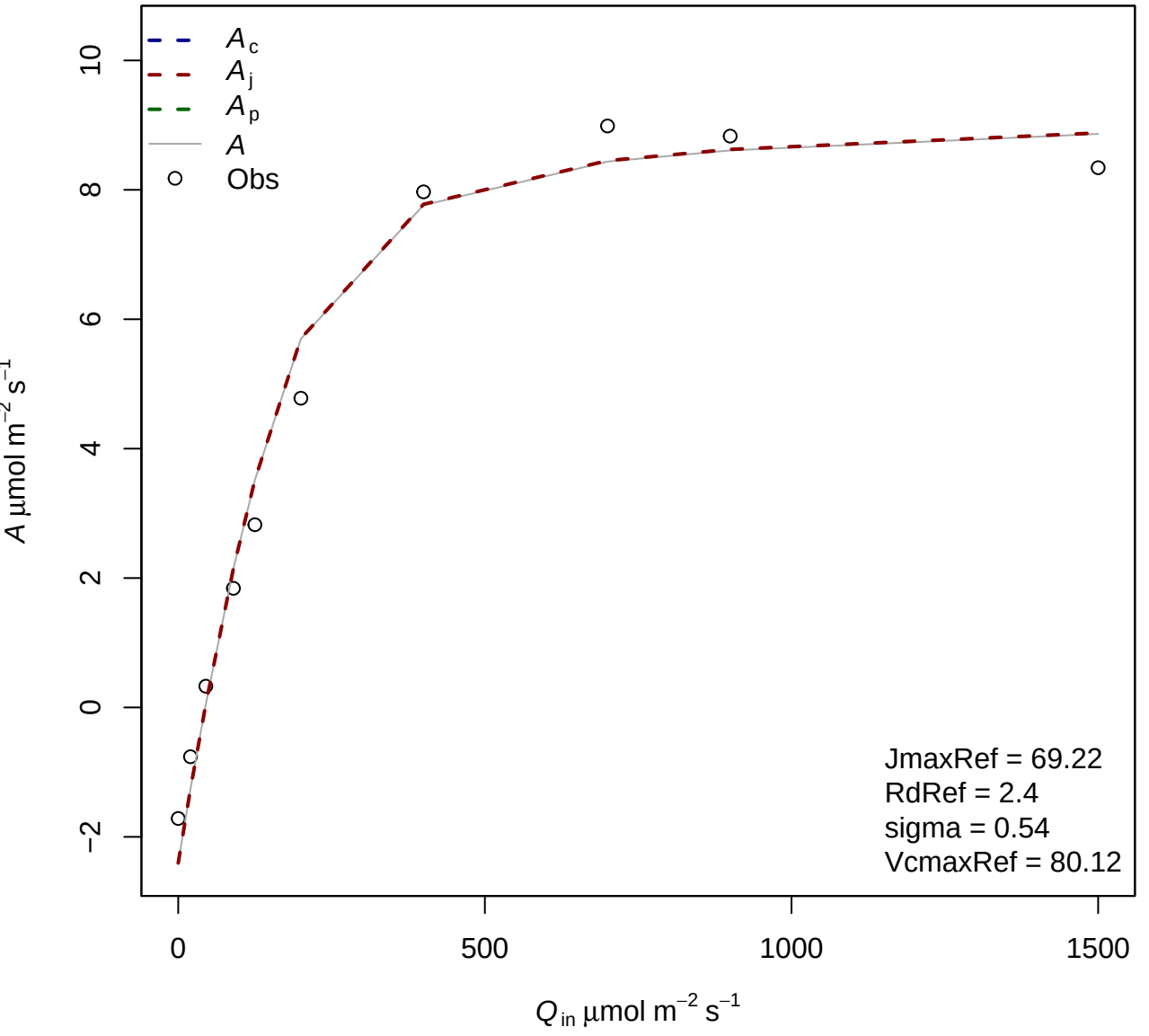

A_c
A_j
A_p
A
Obs
JmaxRef = 35.79
RdRef = 3.96
sigma = 0.38
VcmaxRef = 69.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
−4
0
500
1000
1500

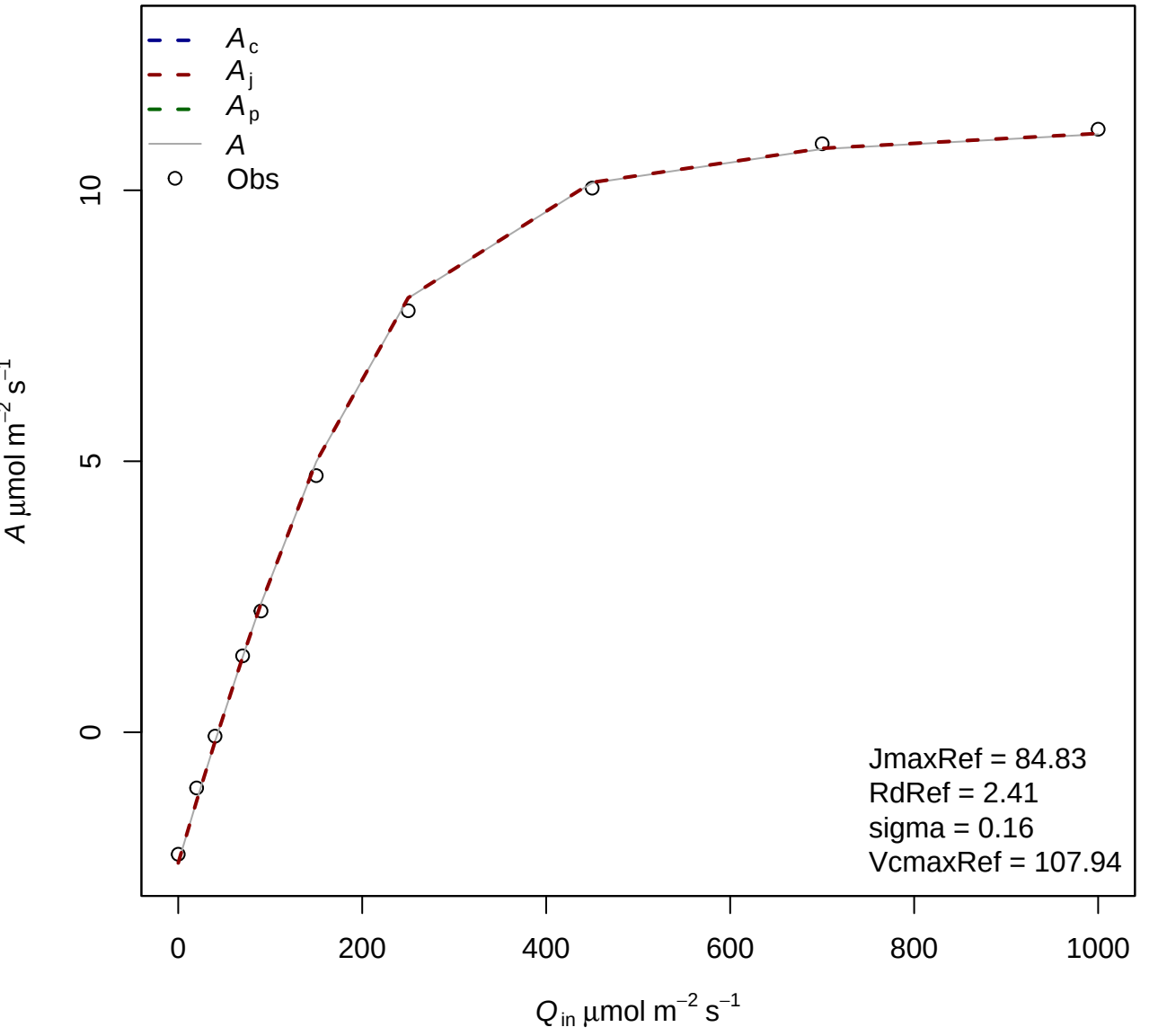

A_c
A_j
A_p
A
Obs
JmaxRef = 60.55
RdRef = 2.11
sigma = 0.54
VcmaxRef = 57.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

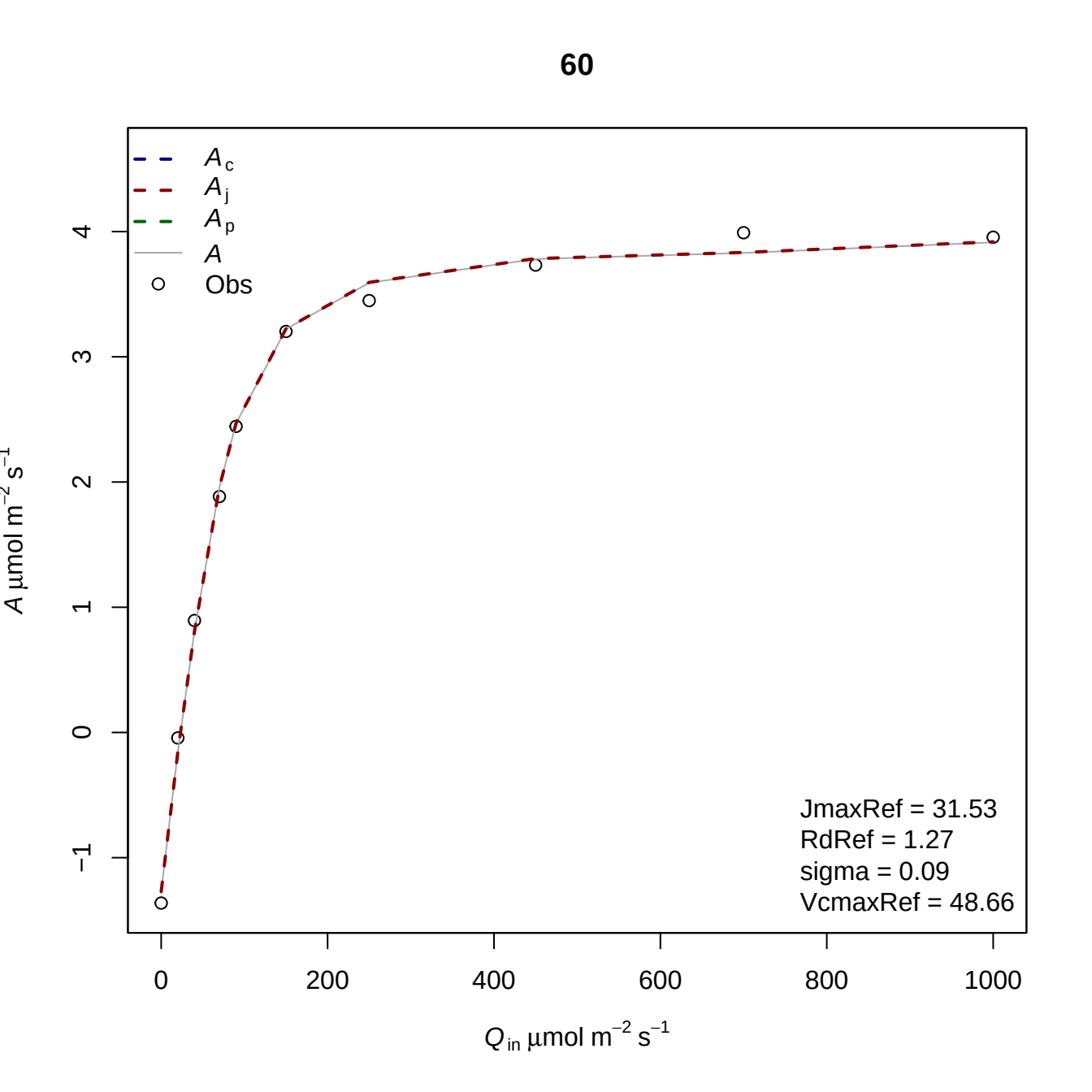

A_c
A_j
A_p
A
Obs
JmaxRef = 48.85
RdRef = 2.48
sigma = 0.19
VcmaxRef = 71.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

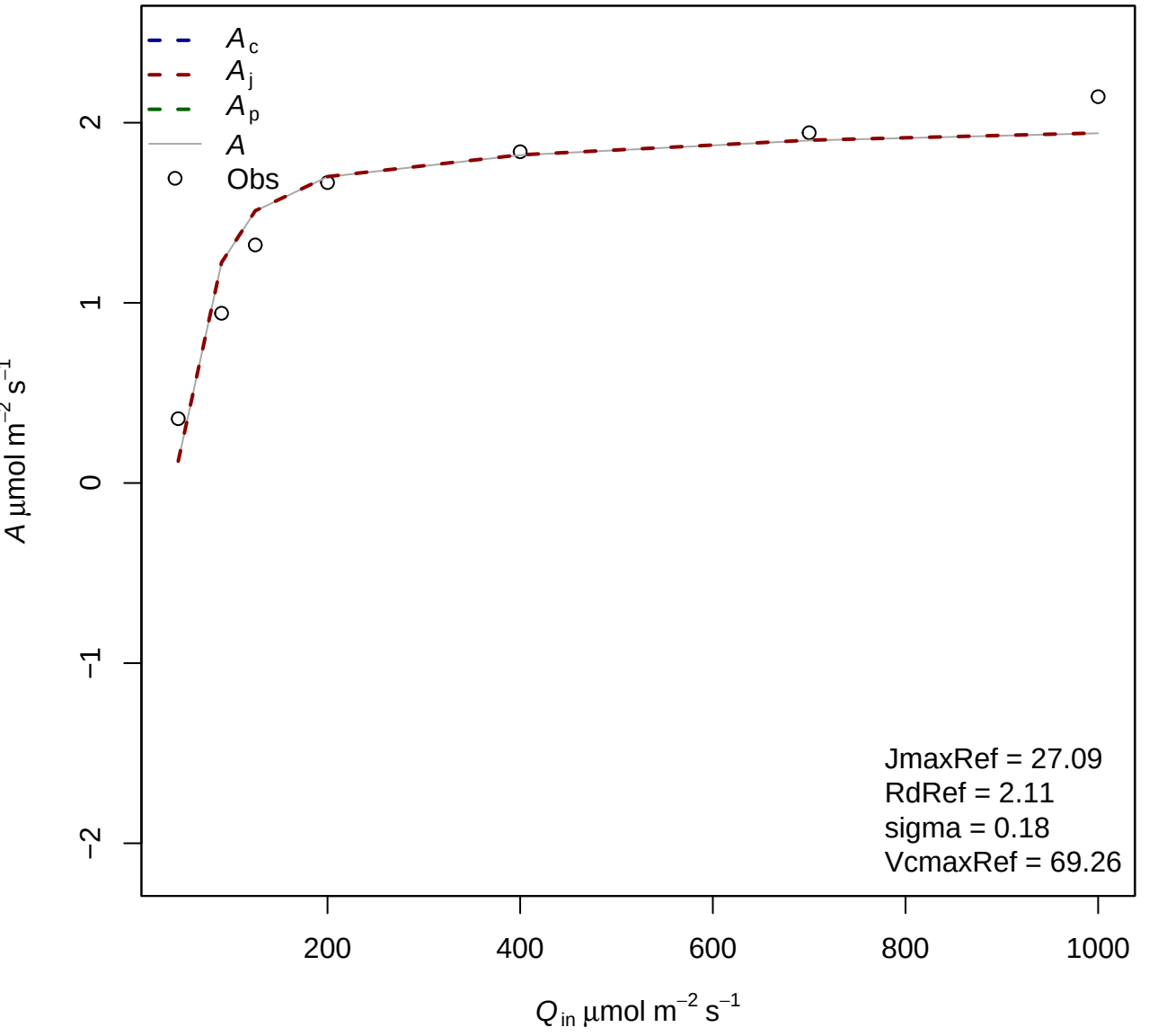

A_c
A_j
A_p
A
Obs
JmaxRef = 56.56
RdRef = 2.58
sigma = 0.19
VcmaxRef = 39.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

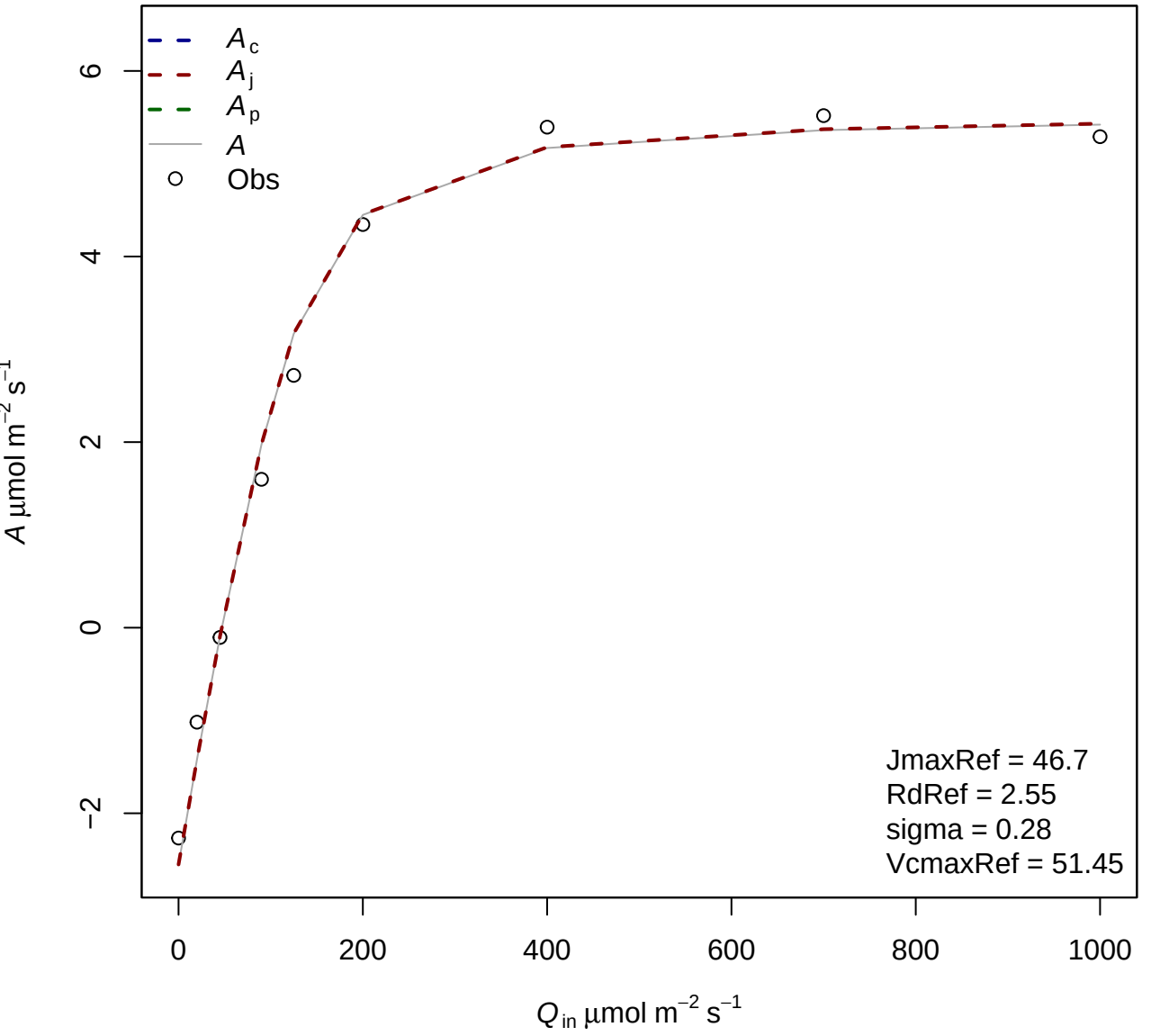

A_c
A_j
A_p
A
Obs
JmaxRef = 47.17
RdRef = 2.76
sigma = 0.19
VcmaxRef = 72.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

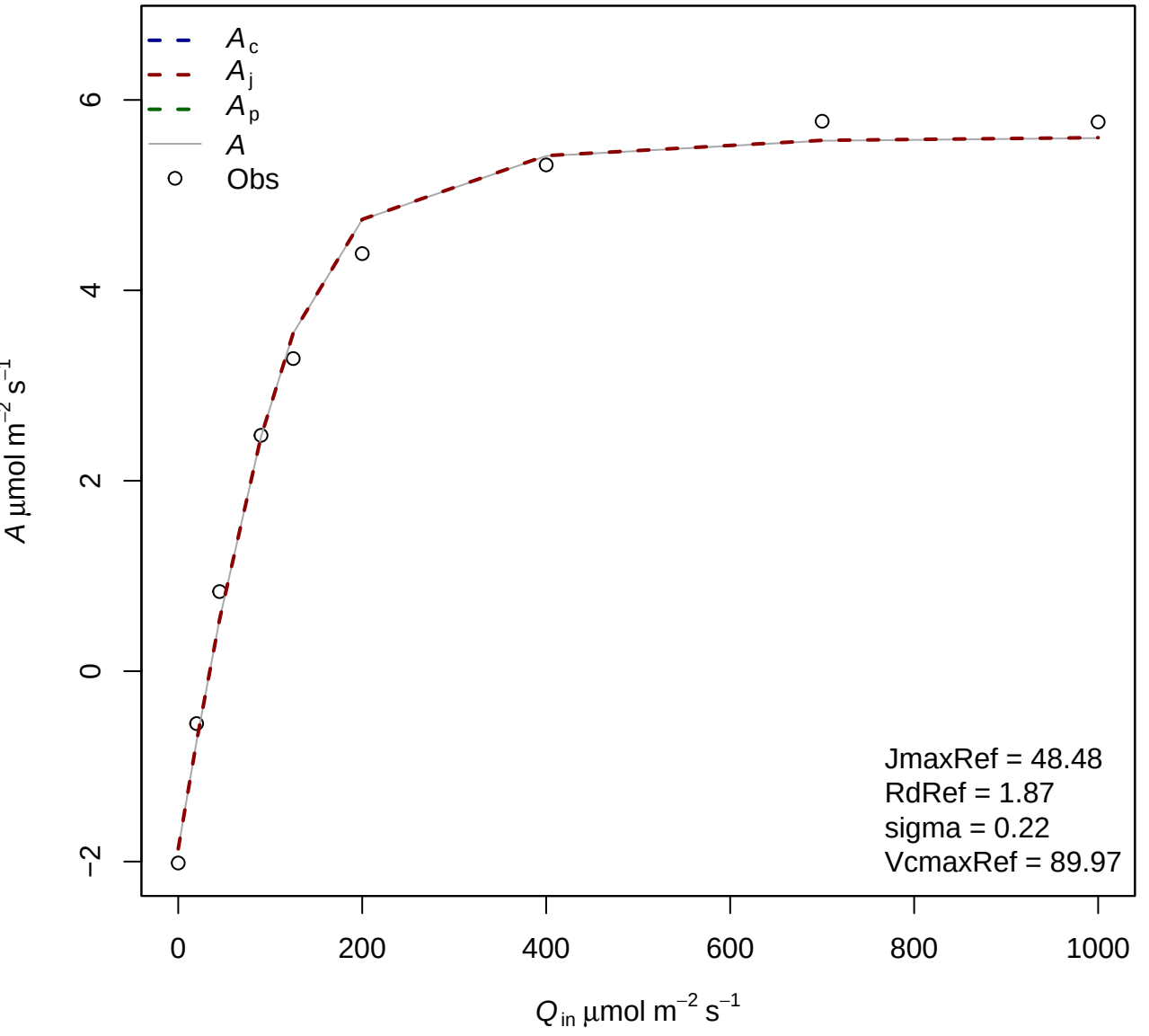

A_c
A_j
A_p
A
Obs
JmaxRef = 69.22
RdRef = 2.4
sigma = 0.54
VcmaxRef = 80.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

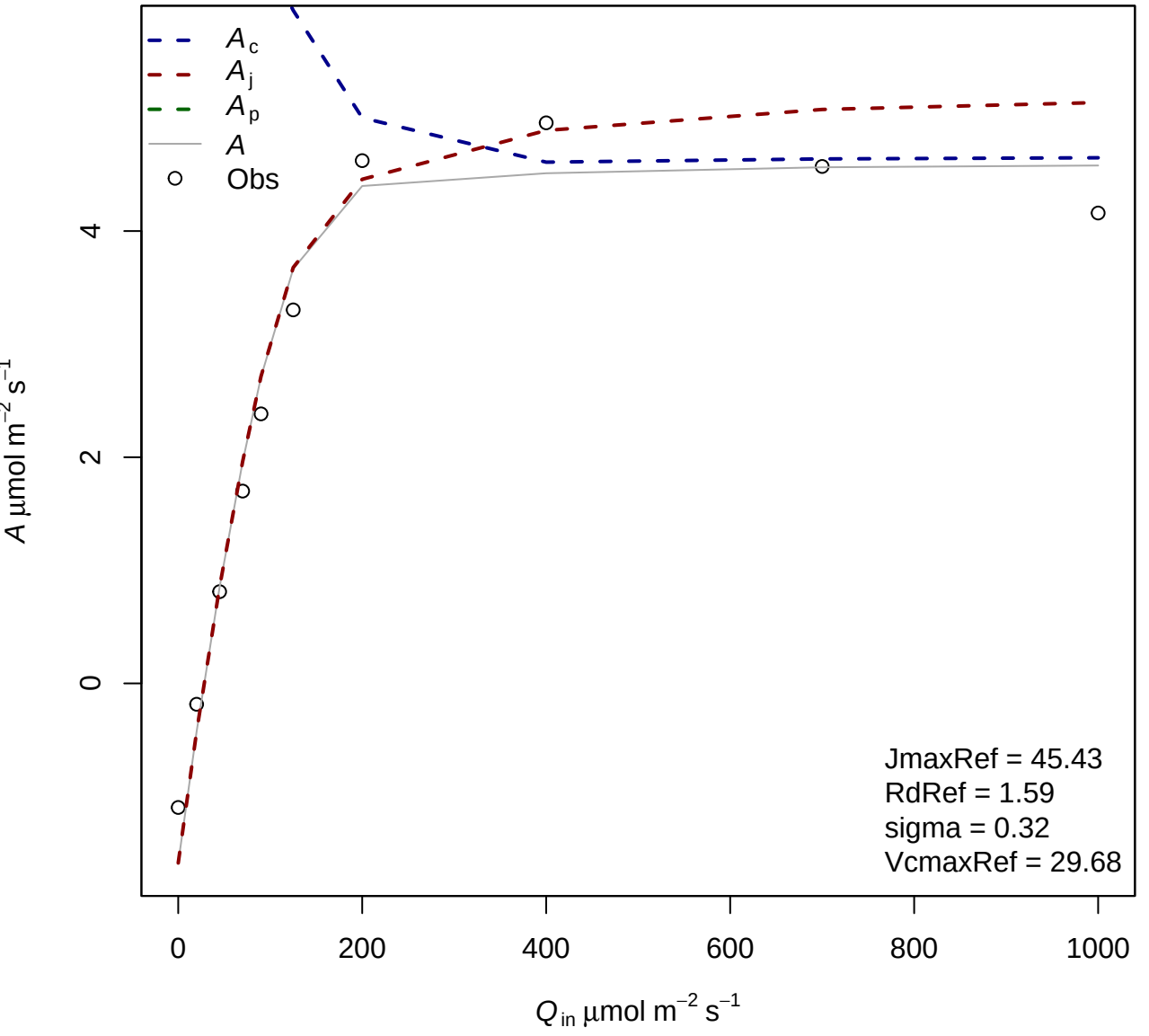

A_c
A_j
A_p
A
Obs
JmaxRef = 84.83
RdRef = 2.41
sigma = 0.16
VcmaxRef = 107.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

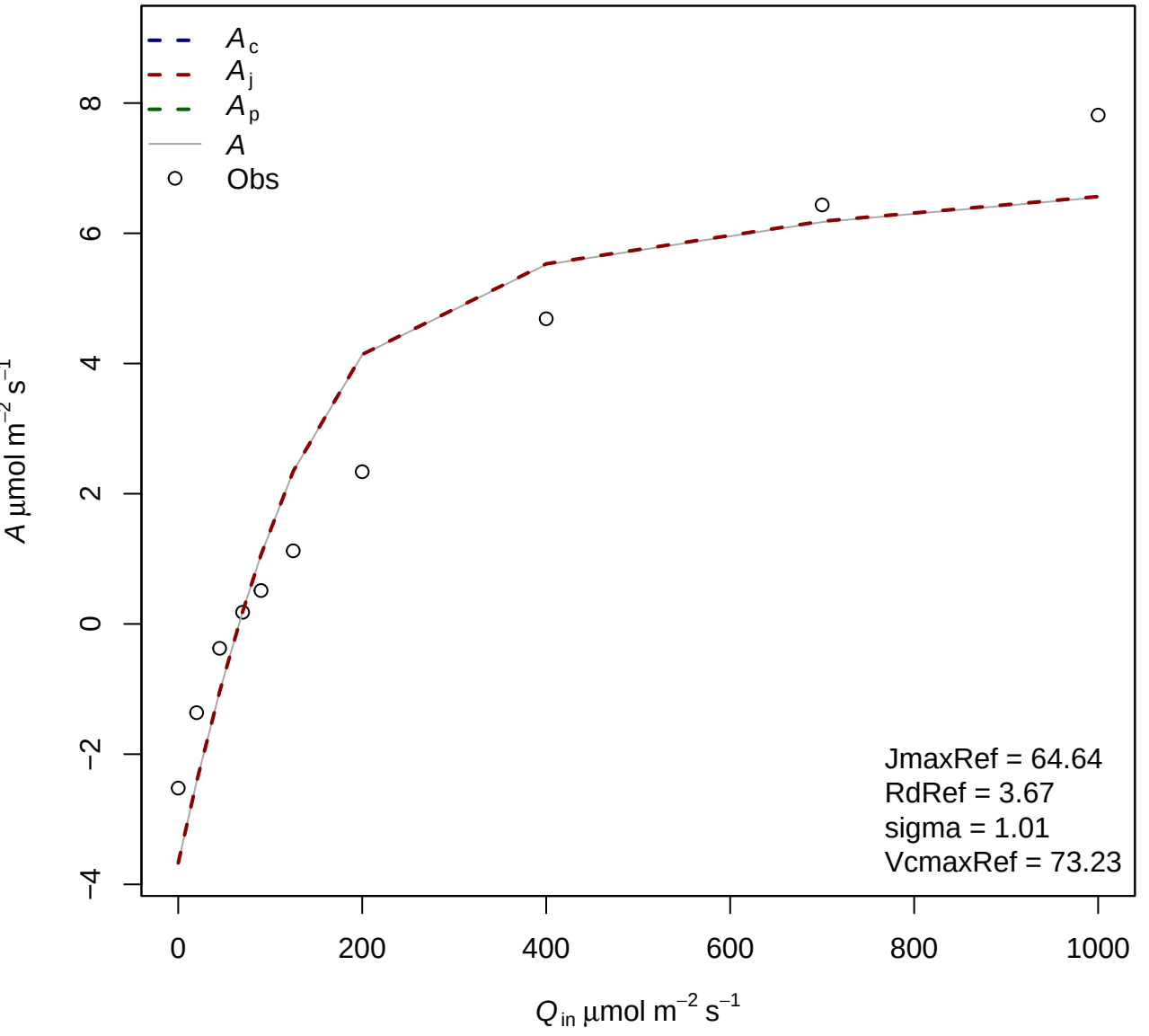

60
A_c
A_j
A_p
A
Obs
JmaxRef = 31.53
RdRef = 1.27
sigma = 0.09
VcmaxRef = 48.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

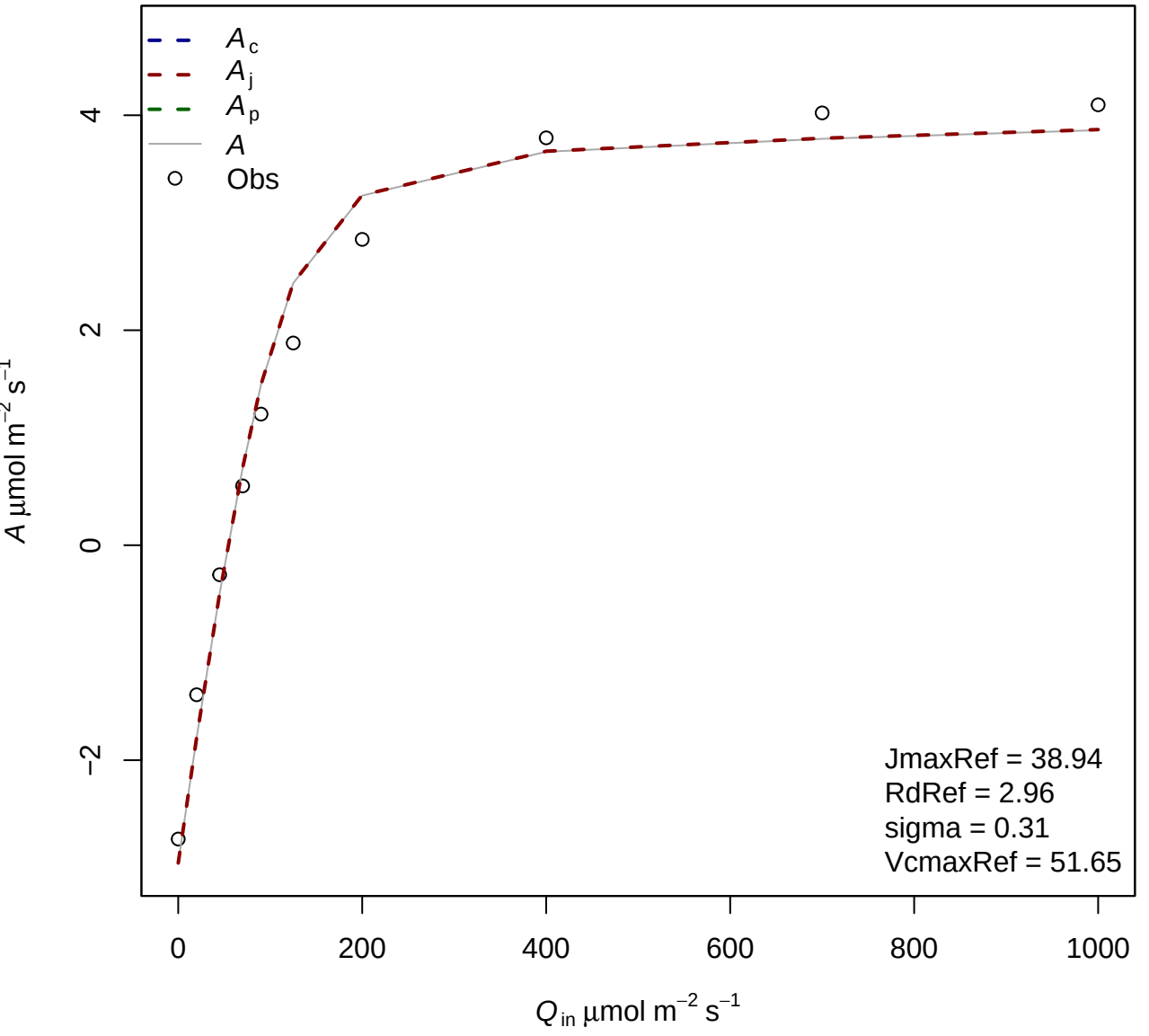

A_c
A_j
A_p
A
Obs
JmaxRef = 27.09
RdRef = 2.11
sigma = 0.18
VcmaxRef = 69.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
200
400
600
800
1000

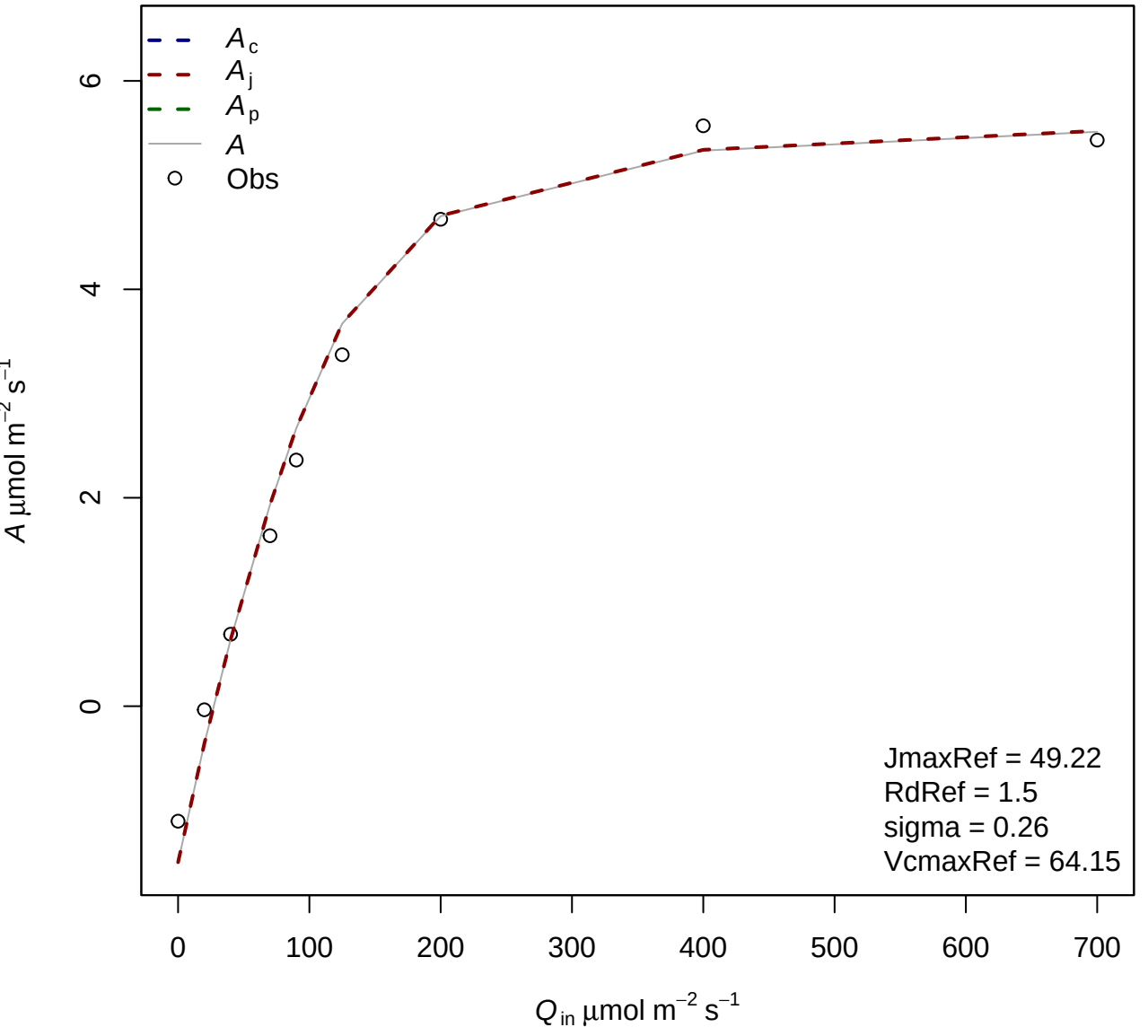

A_c
A_j
A_p
A
Obs
JmaxRef = 46.7
RdRef = 2.55
sigma = 0.28
VcmaxRef = 51.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

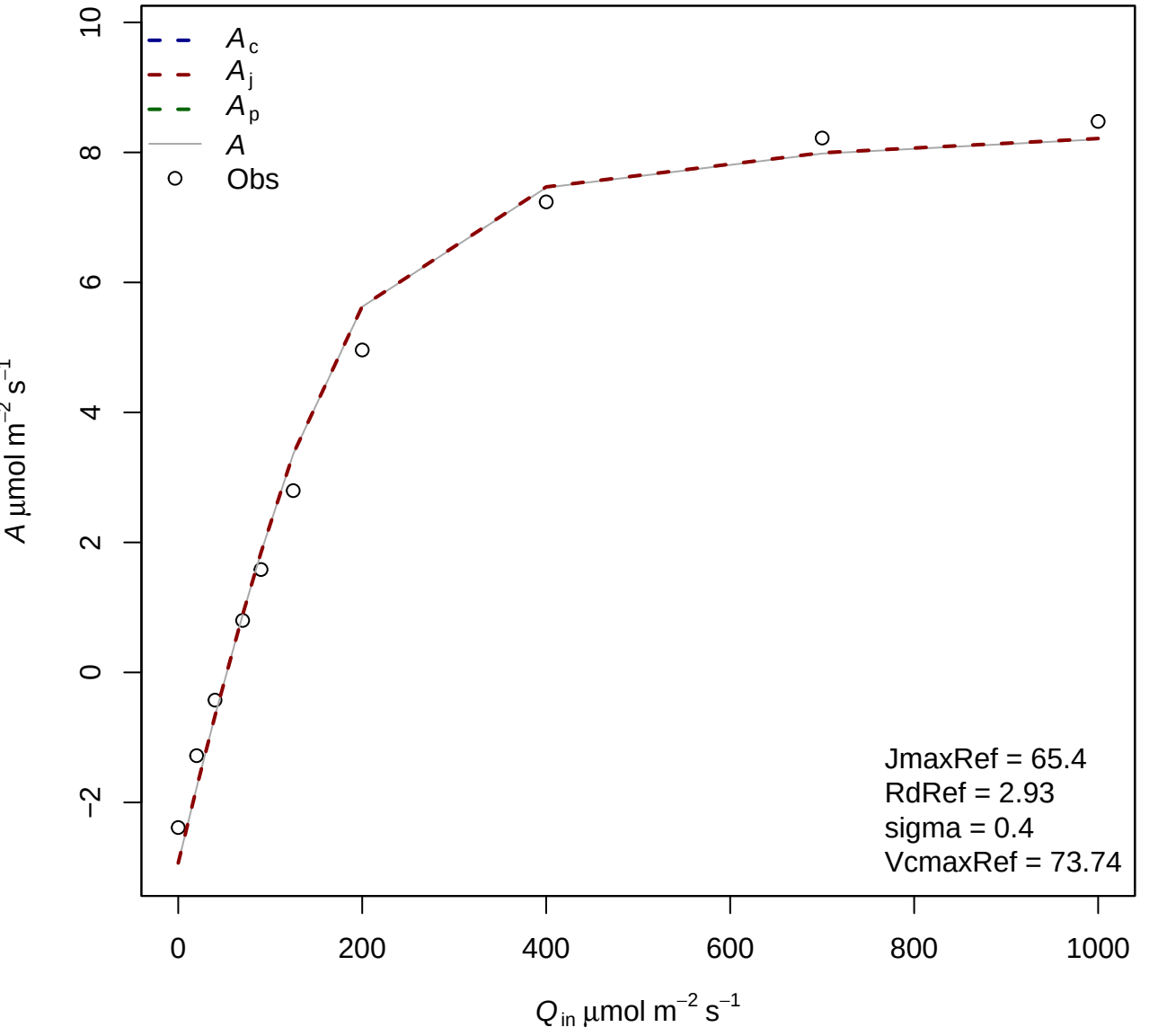

A_c
A_j
A_p
A
Obs
JmaxRef = 48.48
RdRef = 1.87
sigma = 0.22
VcmaxRef = 89.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

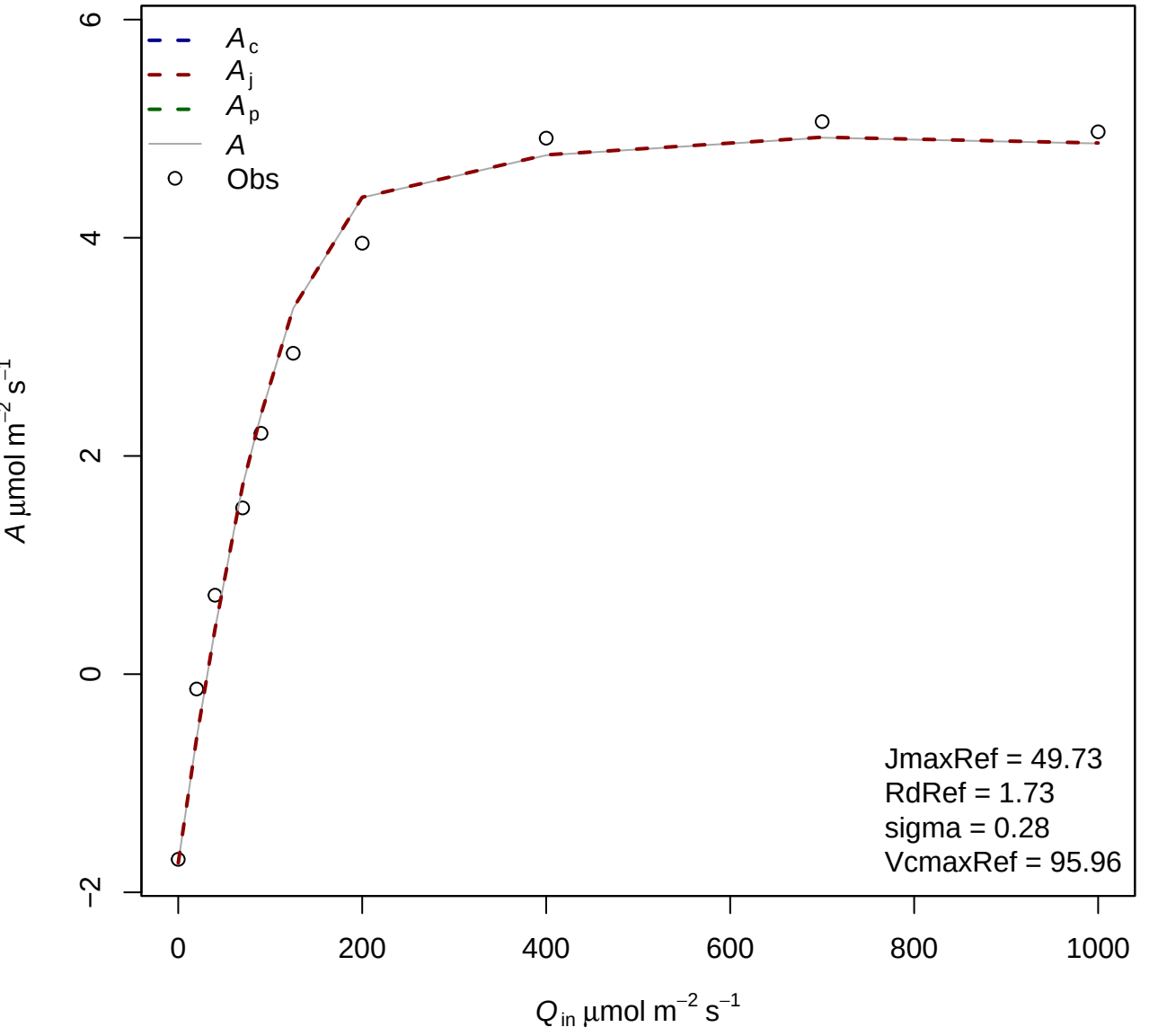

A_c
A_j
A_p
A
Obs
JmaxRef = 45.43
RdRef = 1.59
sigma = 0.32
VcmaxRef = 29.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
0
200
400
600
800
1000

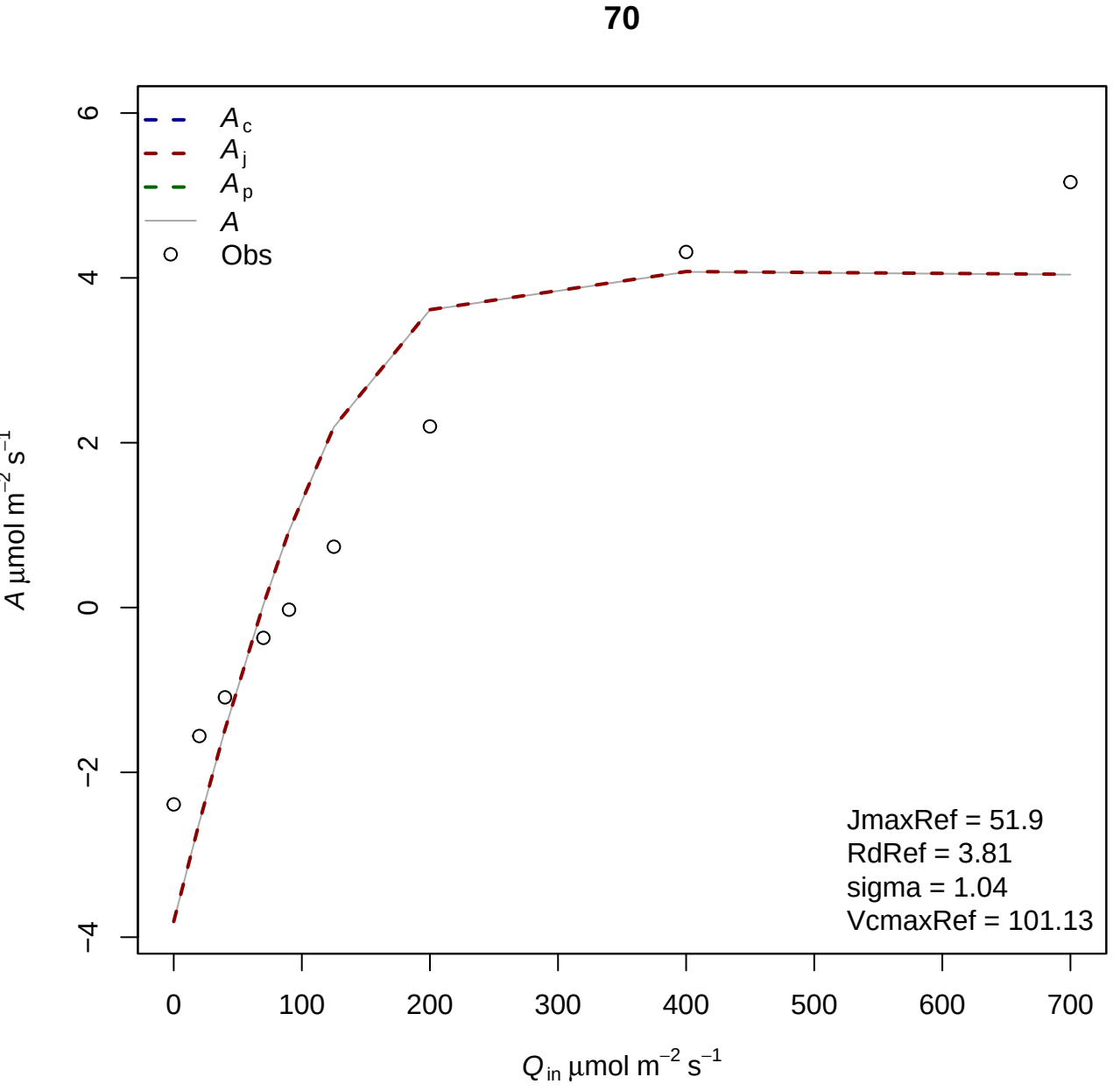

A_c
A_j
A_p
A
Obs
JmaxRef = 64.64
RdRef = 3.67
sigma = 1.01
VcmaxRef = 73.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
200
400
600
800
1000

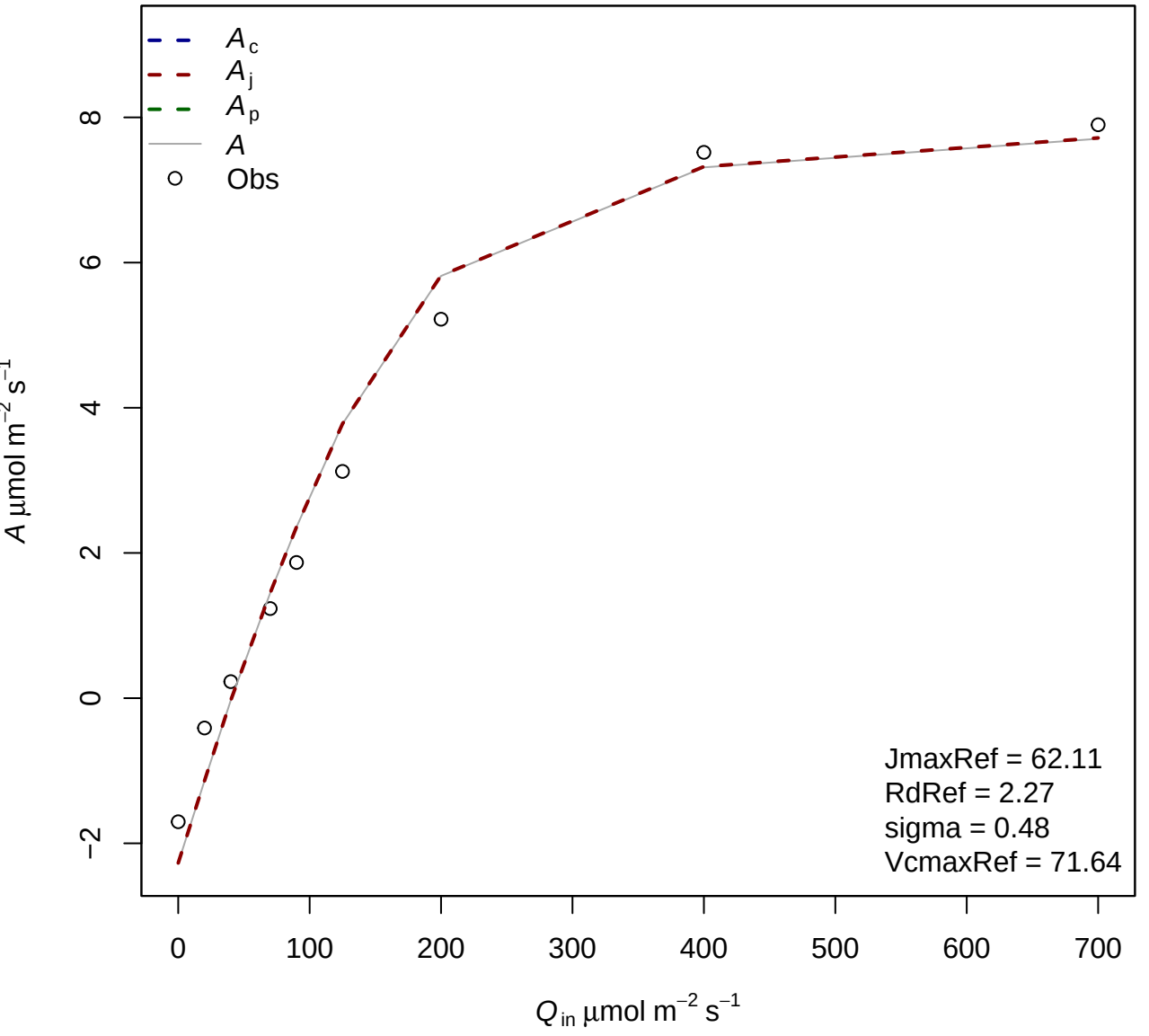

A_c
A_j
A_p
A
Obs
JmaxRef = 38.94
RdRef = 2.96
sigma = 0.31
VcmaxRef = 51.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

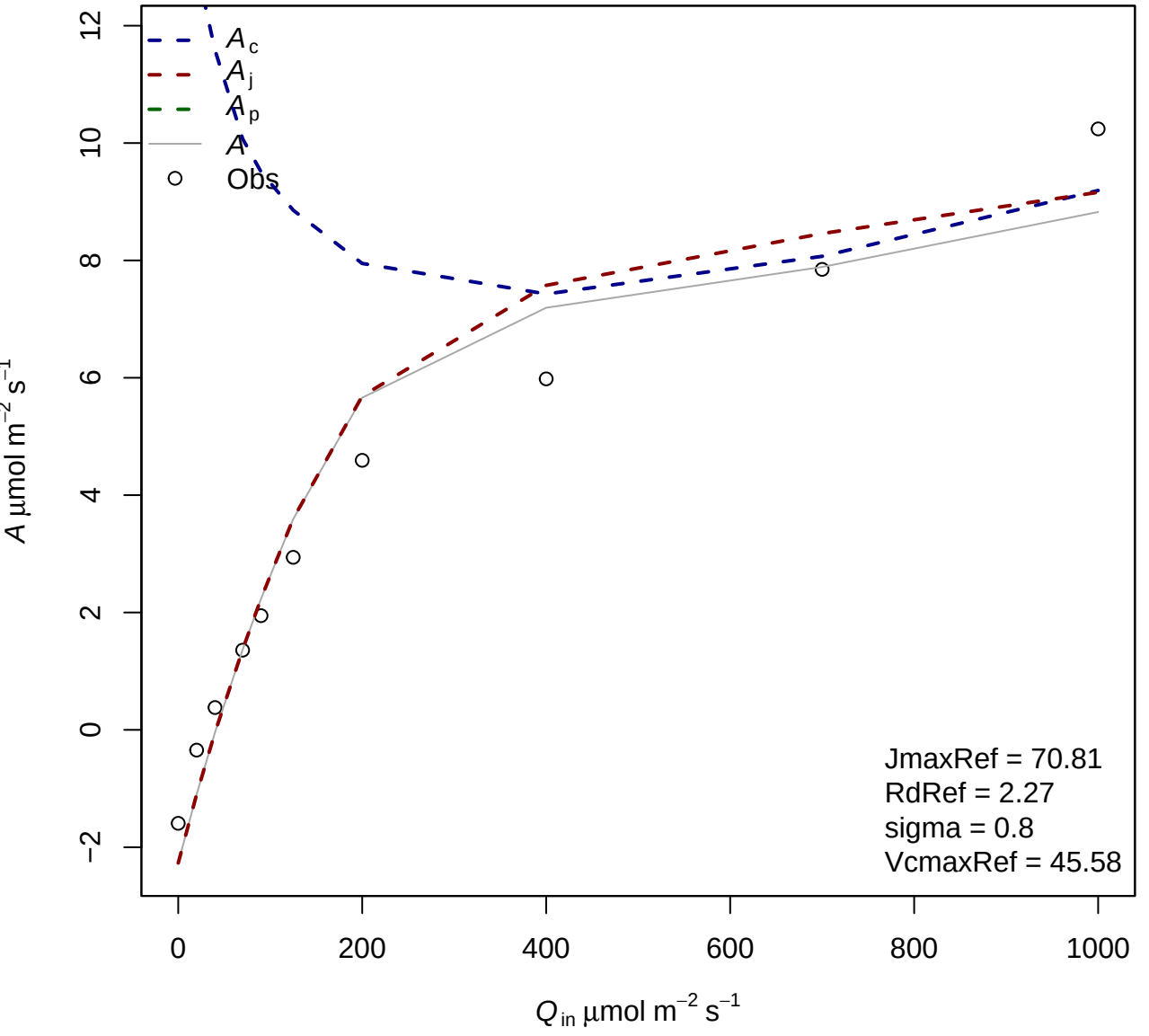

A_c
A_j
A_p
A
Obs
JmaxRef = 49.22
RdRef = 1.5
sigma = 0.26
VcmaxRef = 64.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
100
200
300
400
500
600
700

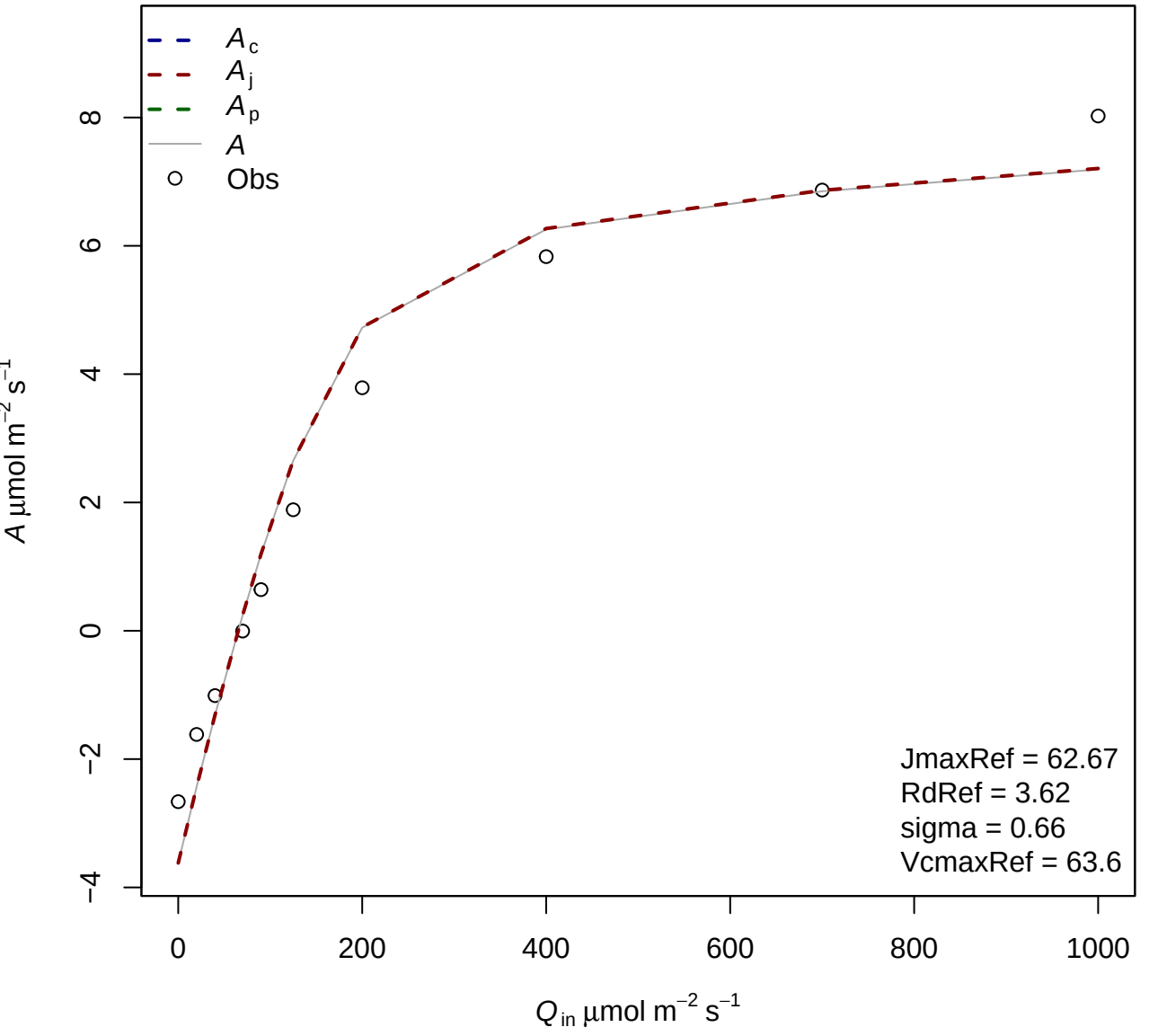

A_c
A_j
A_p
A
Obs
JmaxRef = 65.4
RdRef = 2.93
sigma = 0.4
VcmaxRef = 73.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000

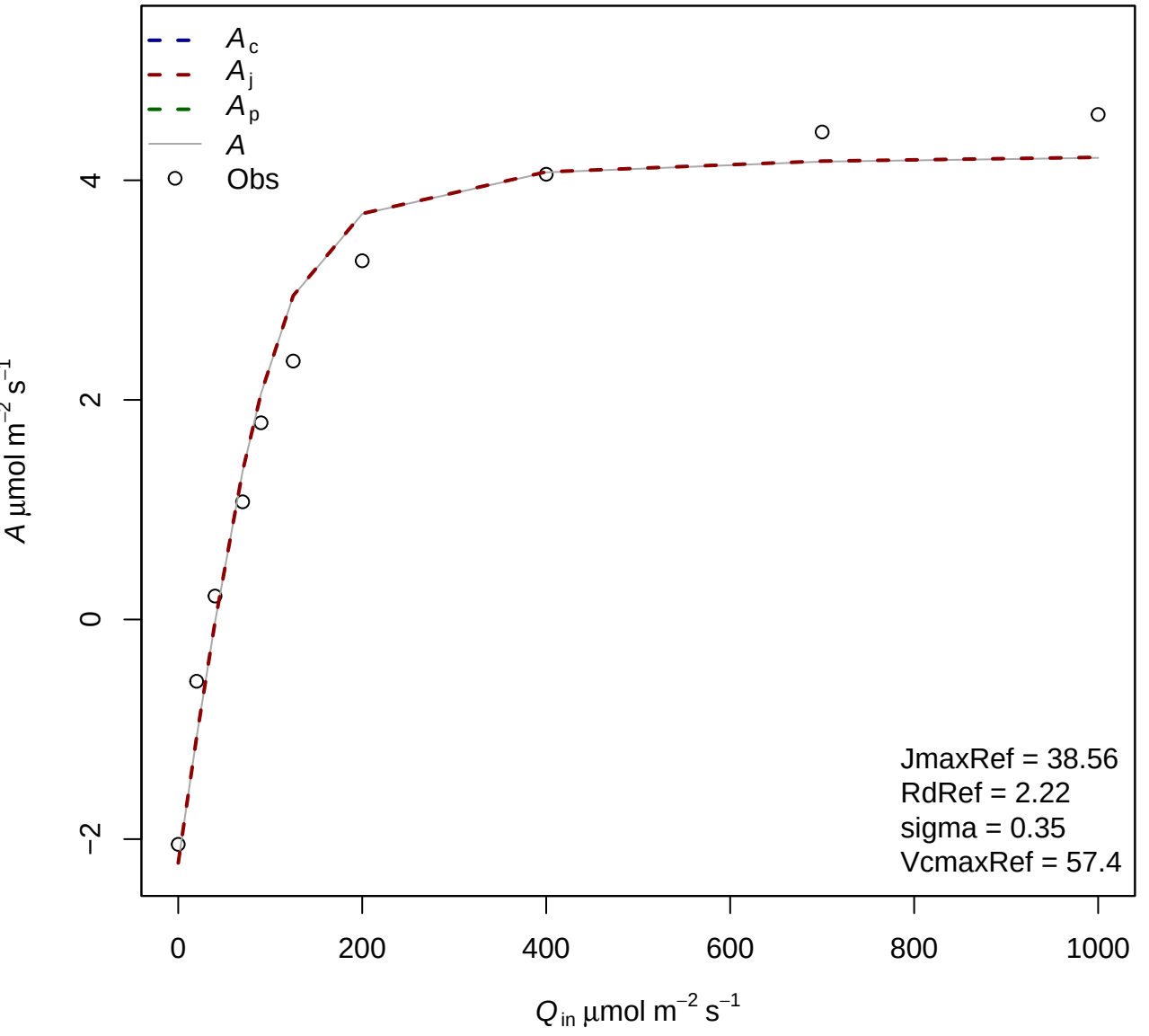

A_c
A_j
A_p
A
Obs
JmaxRef = 49.73
RdRef = 1.73
sigma = 0.28
VcmaxRef = 95.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

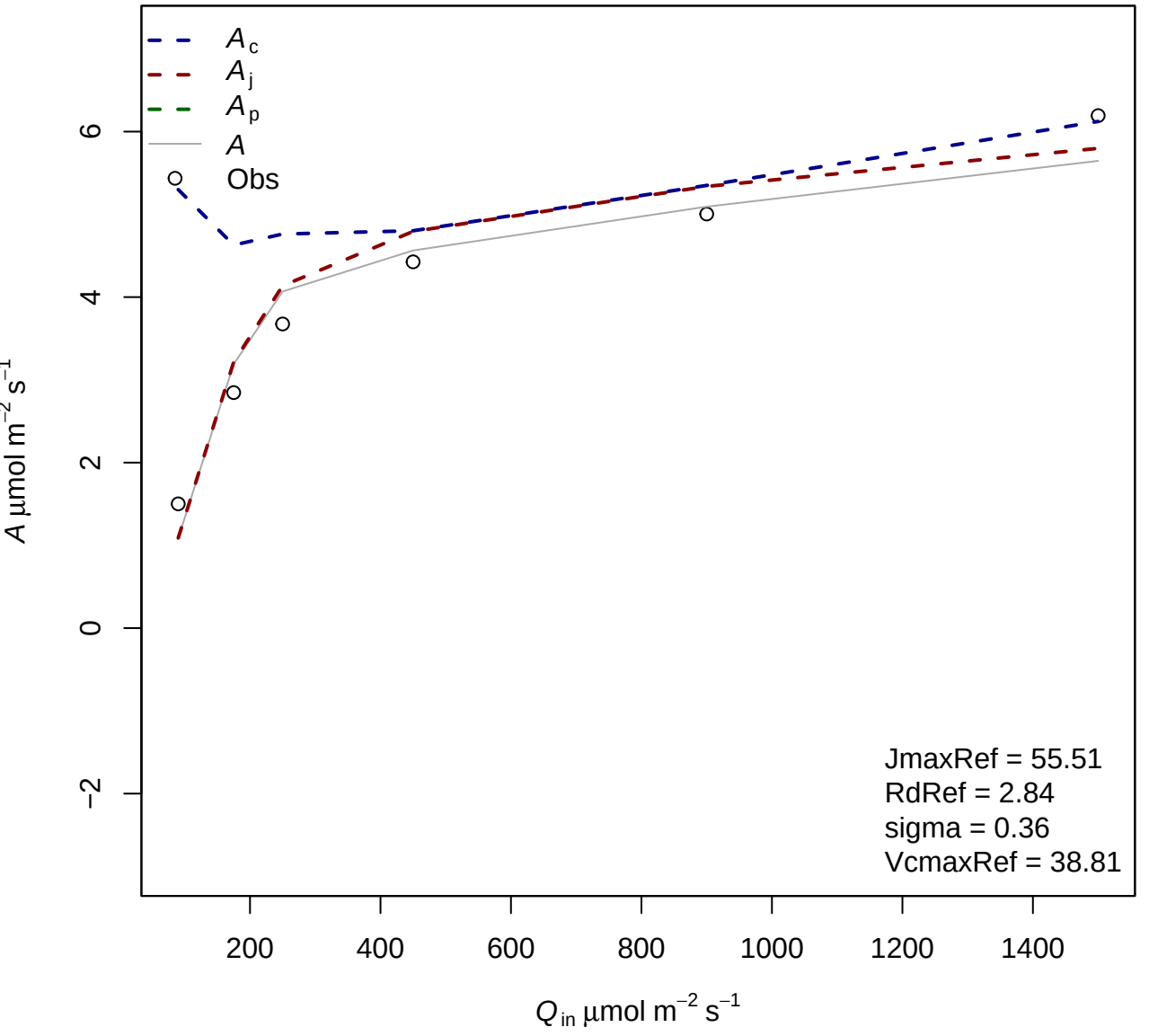

70
A_c
A_j
A_p
A
Obs
JmaxRef = 51.9
RdRef = 3.81
sigma = 1.04
VcmaxRef = 101.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
100
200
300
400
500
600
700

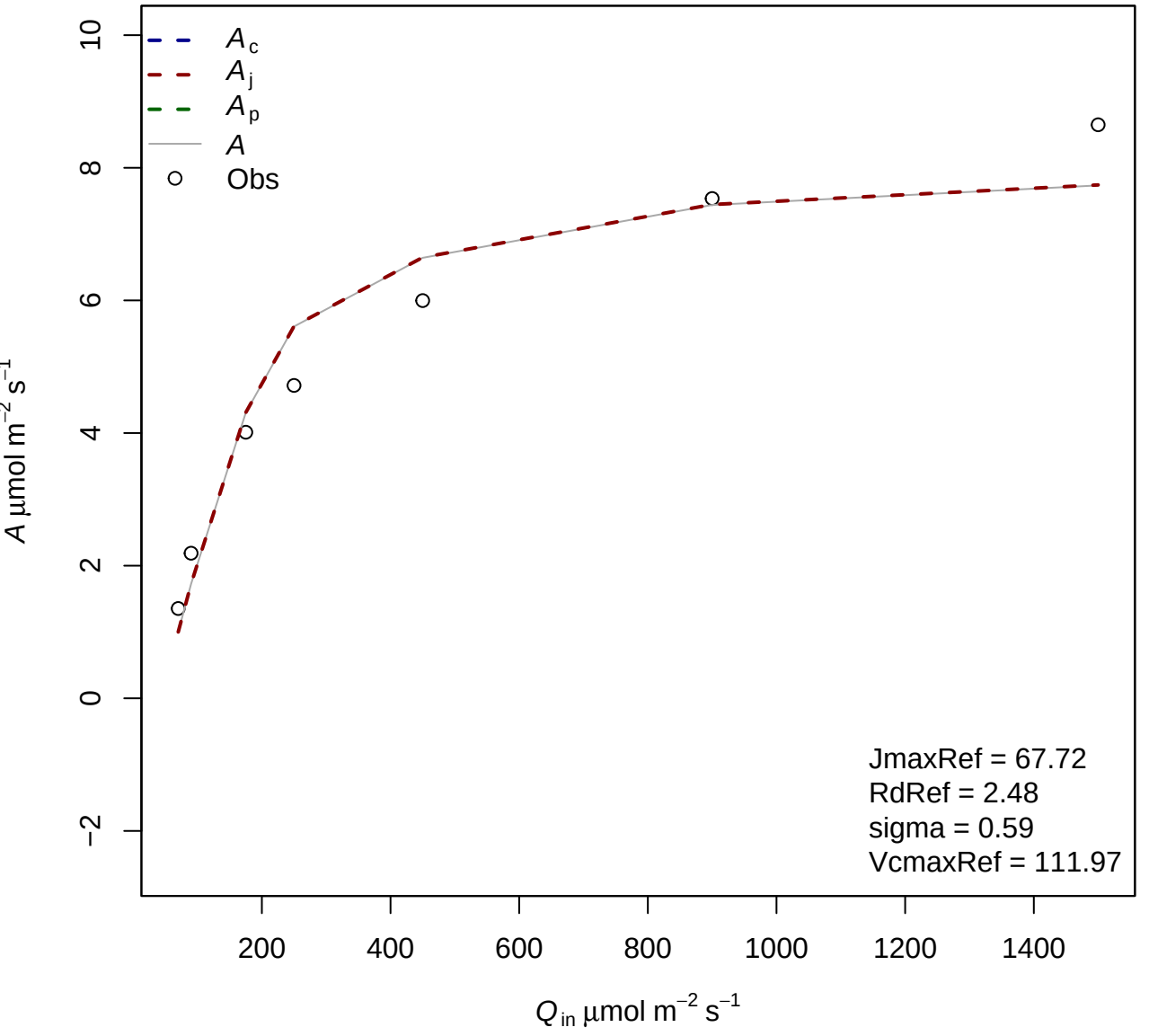

A_c
A_j
A_p
A
Obs
JmaxRef = 62.11
RdRef = 2.27
sigma = 0.48
VcmaxRef = 71.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
100
200
300
400
500
600
700

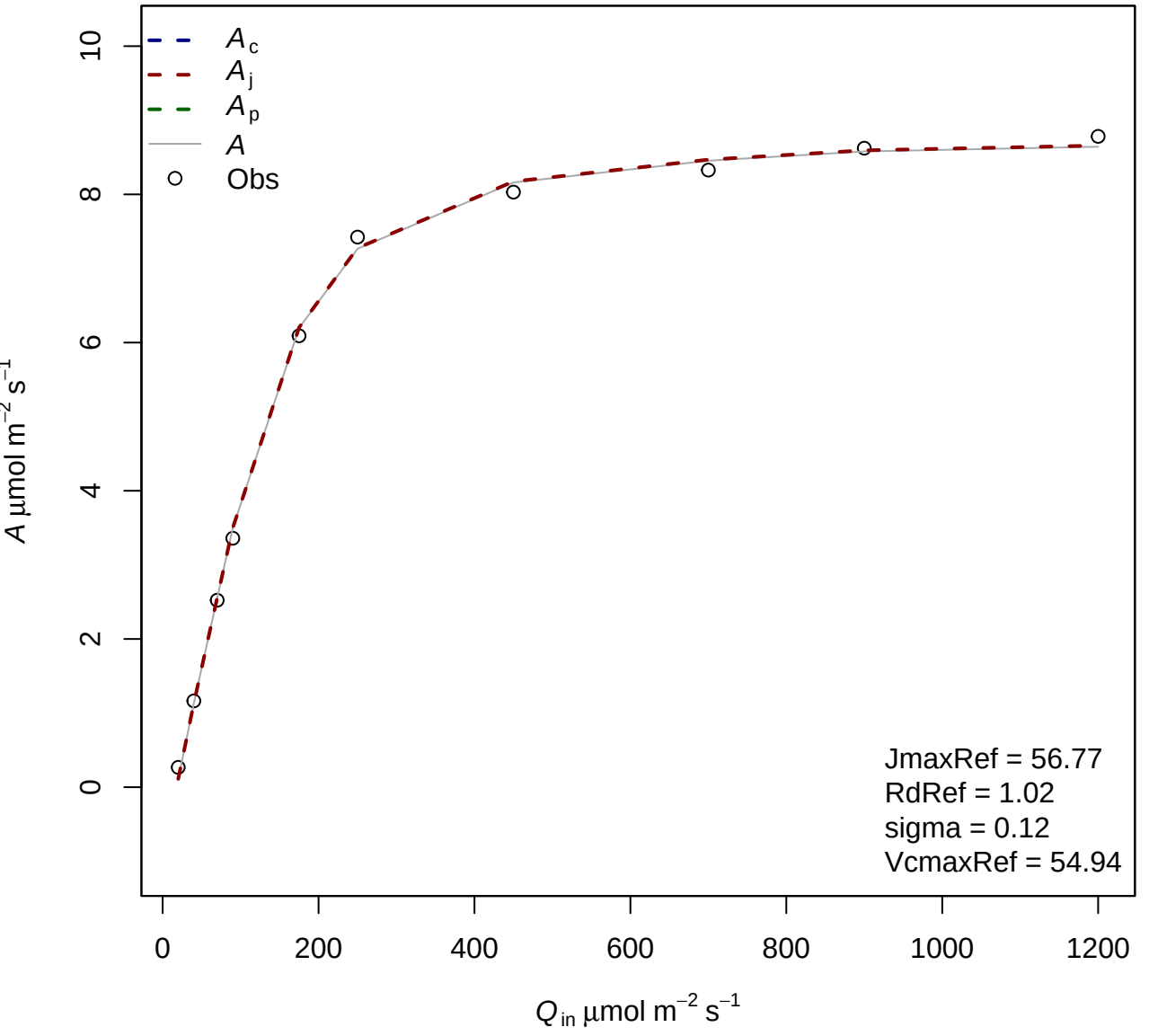

A_c
A_j
A_p
A
Obs
JmaxRef = 70.81
RdRef = 2.27
sigma = 0.8
VcmaxRef = 45.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
200
400
600
800
1000

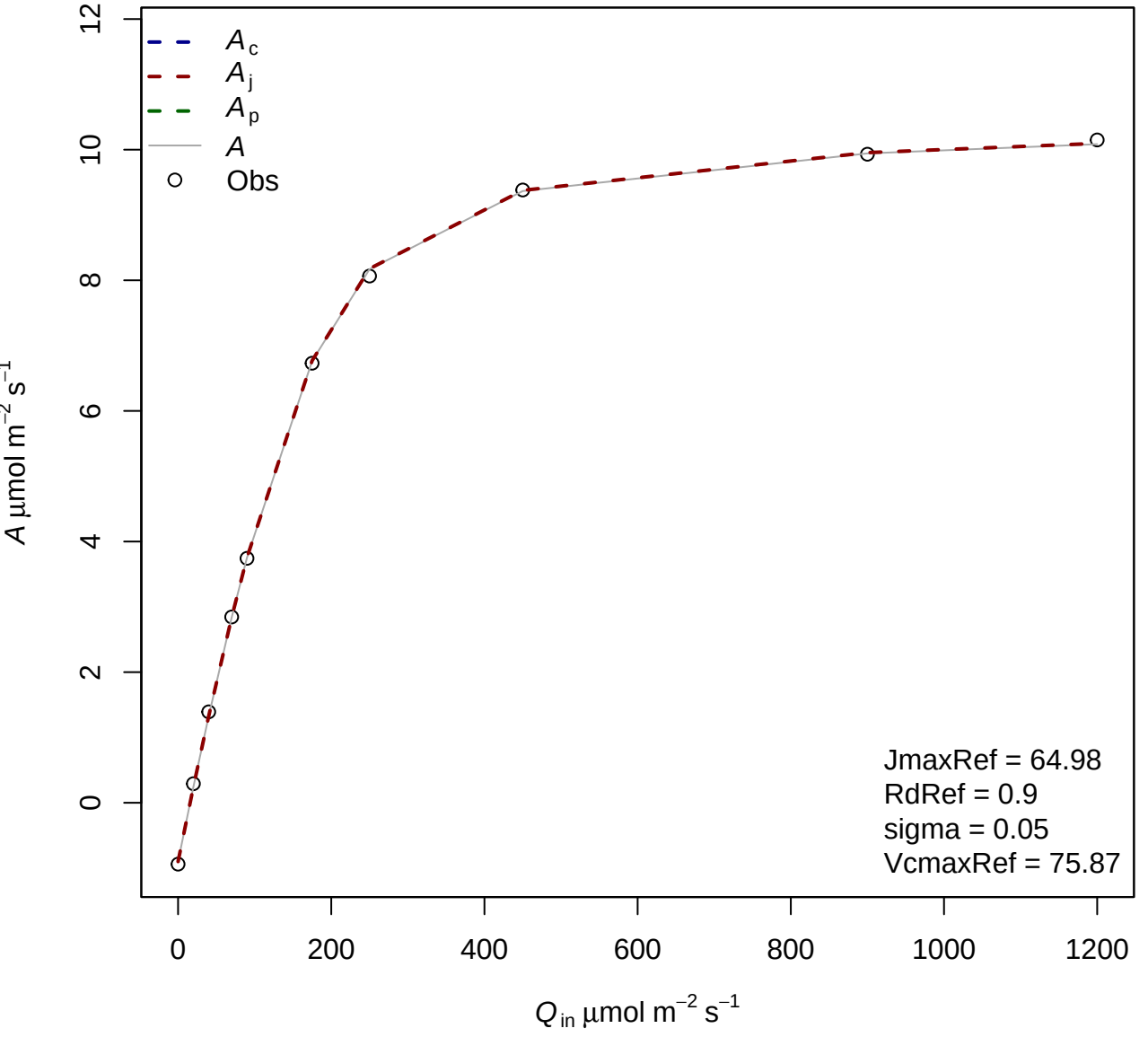

A_c
A_j
A_p
A
Obs
JmaxRef = 62.67
RdRef = 3.62
sigma = 0.66
VcmaxRef = 63.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
200
400
600
800
1000

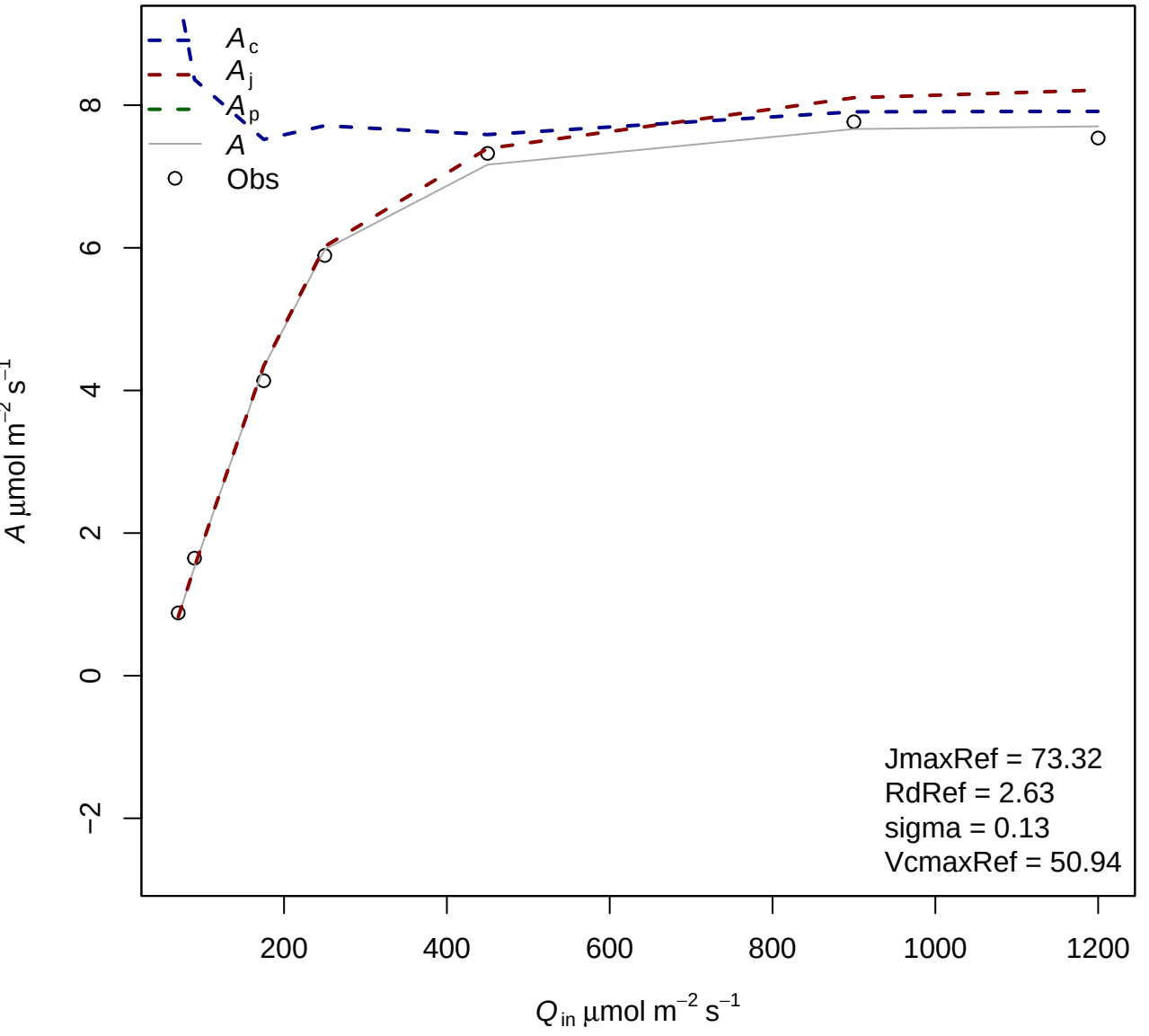

A_c
A_j
A_p
A
Obs
JmaxRef = 38.56
RdRef = 2.22
sigma = 0.35
VcmaxRef = 57.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

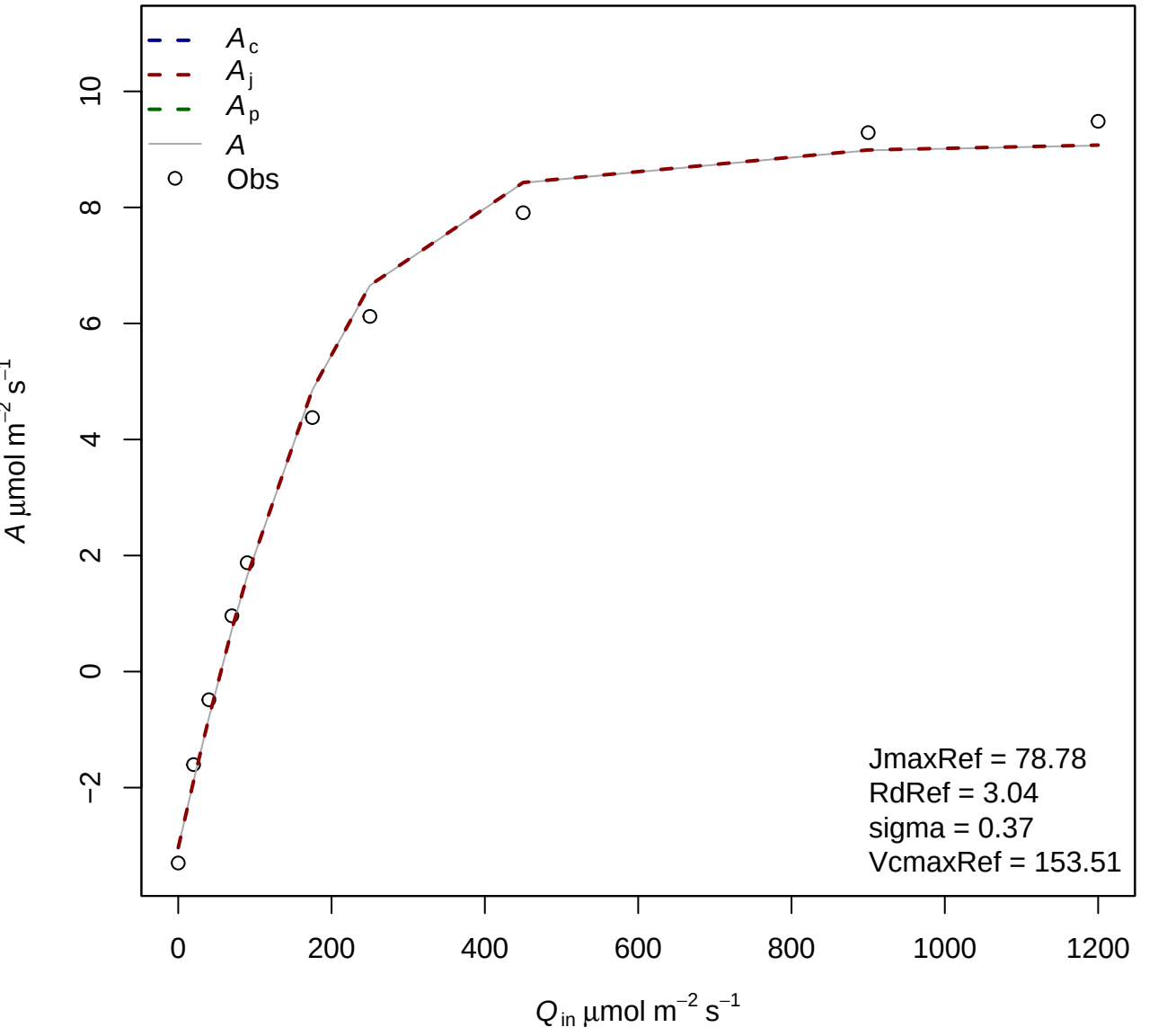

A_c
A_j
A_p
A
Obs
JmaxRef = 55.51
RdRef = 2.84
sigma = 0.36
VcmaxRef = 38.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000
1200
1400

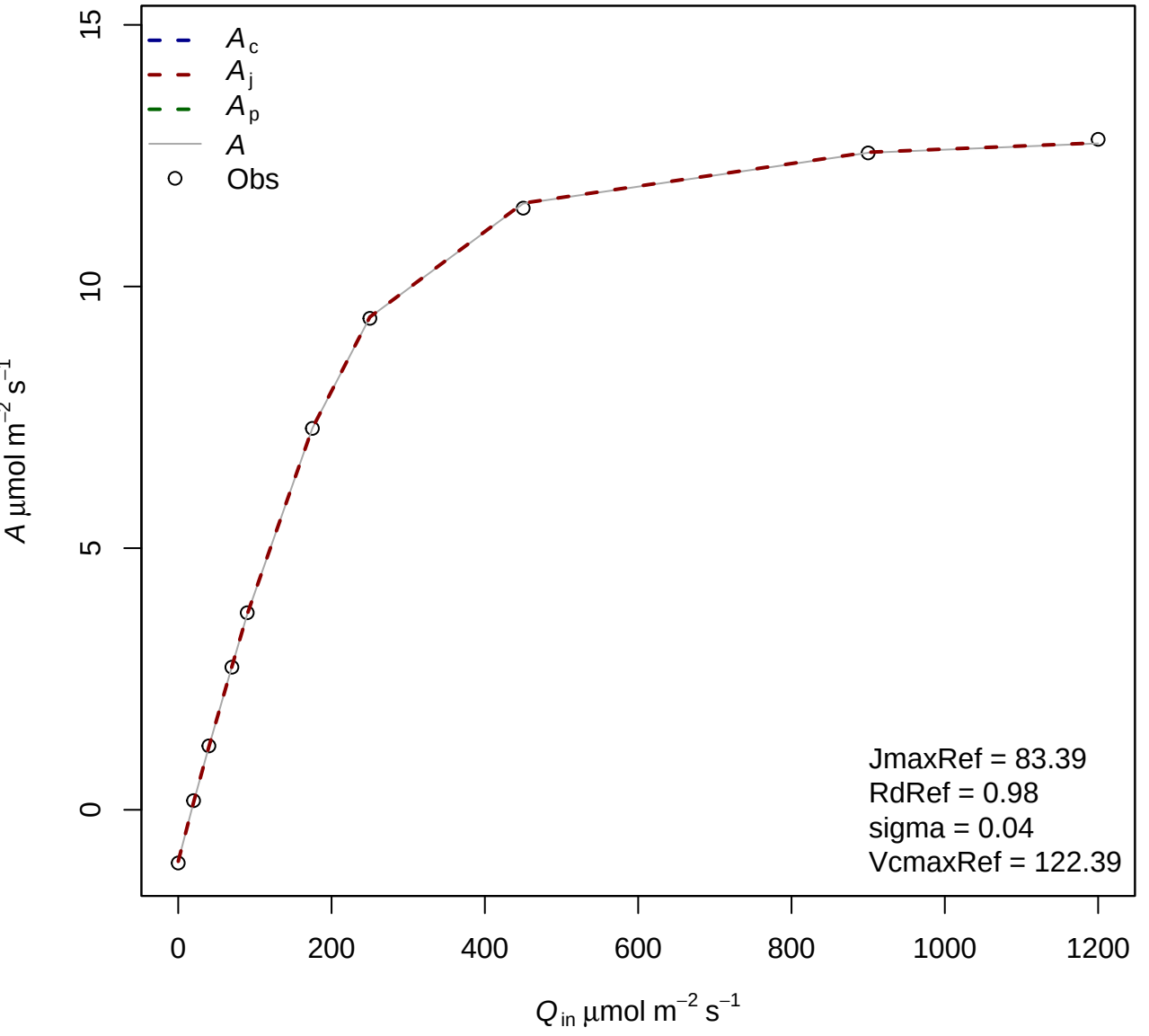

A_c
A_j
A_p
A
Obs
JmaxRef = 67.72
RdRef = 2.48
sigma = 0.59
VcmaxRef = 111.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

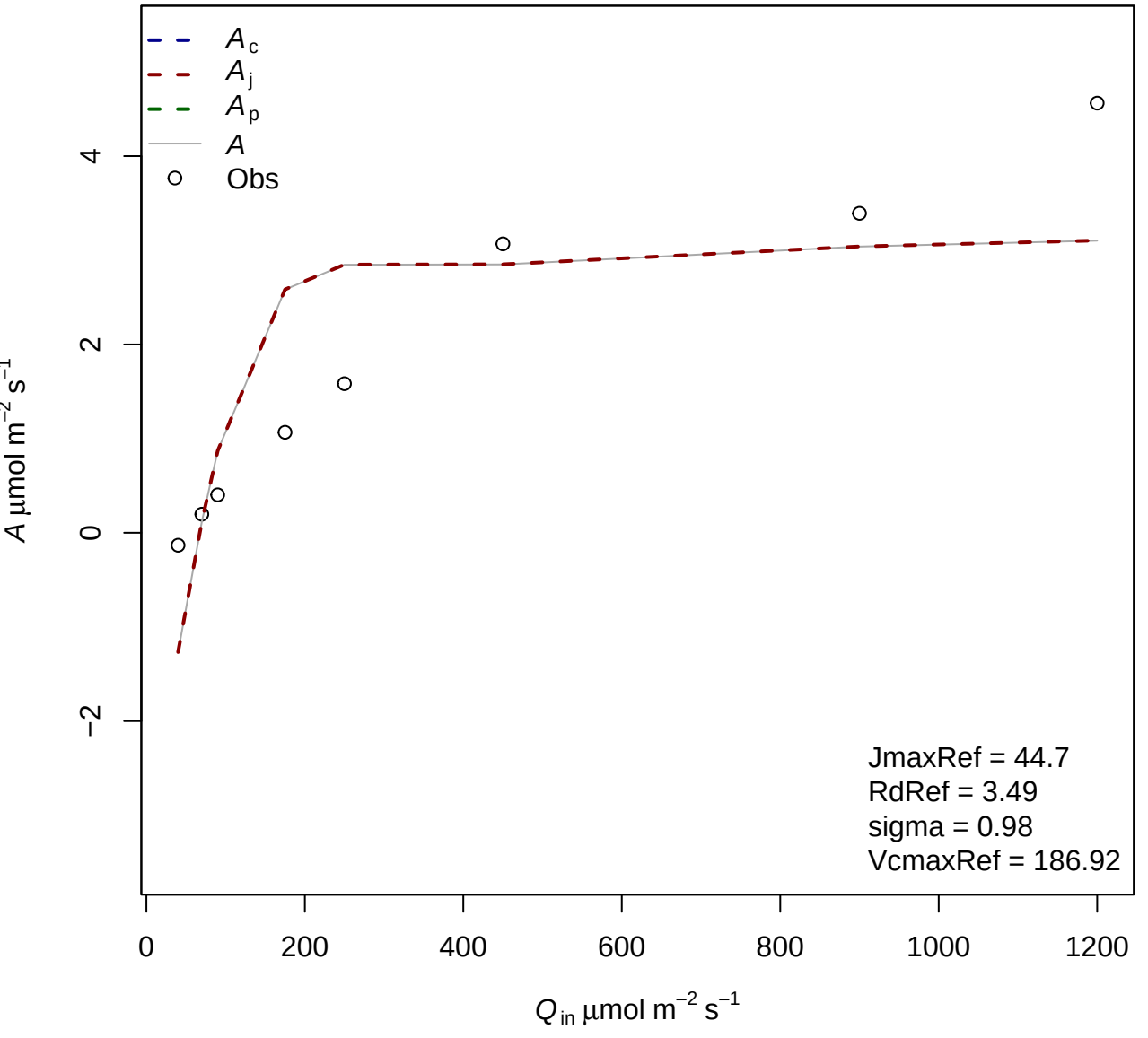

A_c
A_j
A_p
A
Obs
JmaxRef = 56.77
RdRef = 1.02
sigma = 0.12
VcmaxRef = 54.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

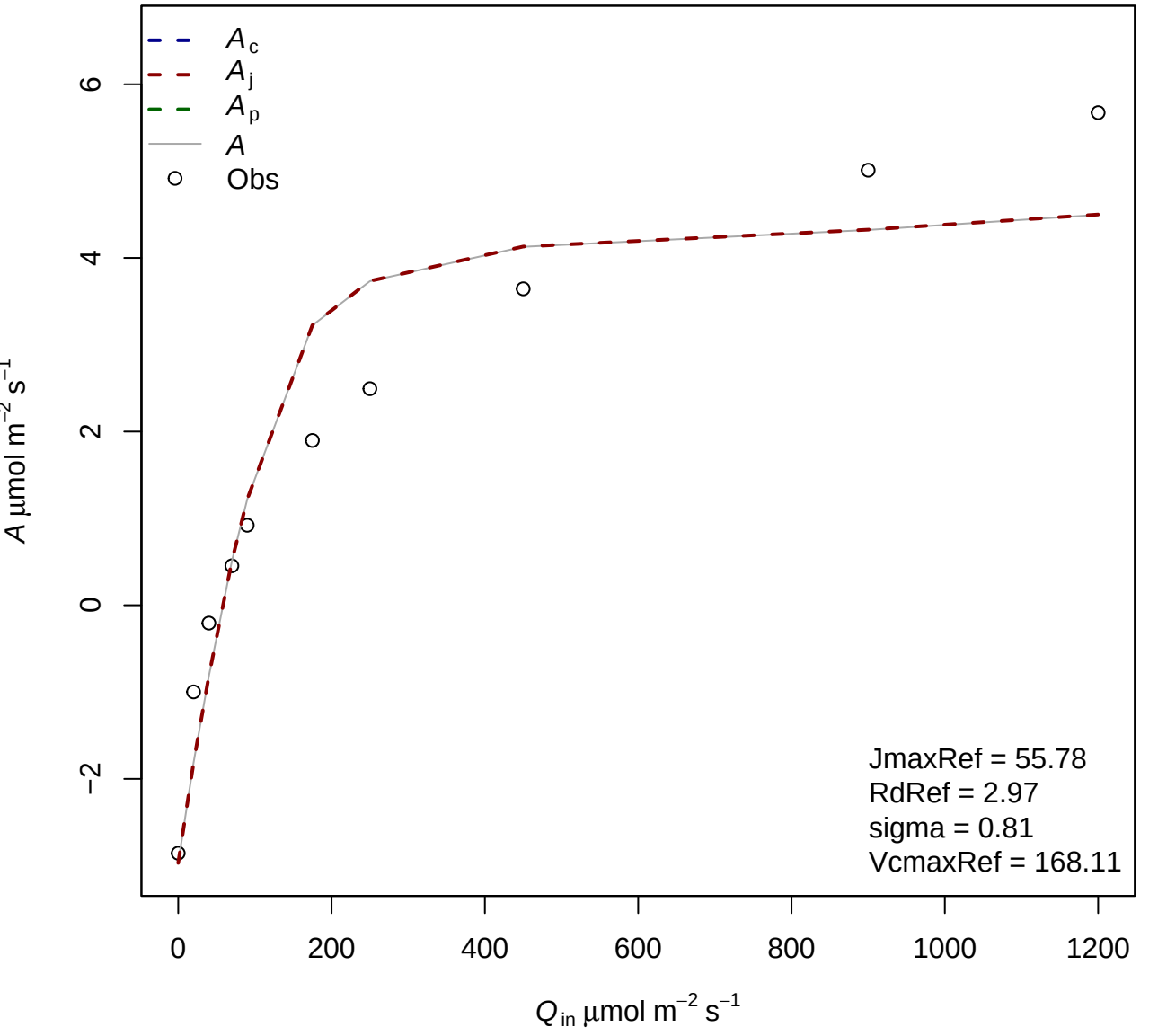

A_c
A_j
A_p
A
Obs
JmaxRef = 64.98
RdRef = 0.9
sigma = 0.05
VcmaxRef = 75.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

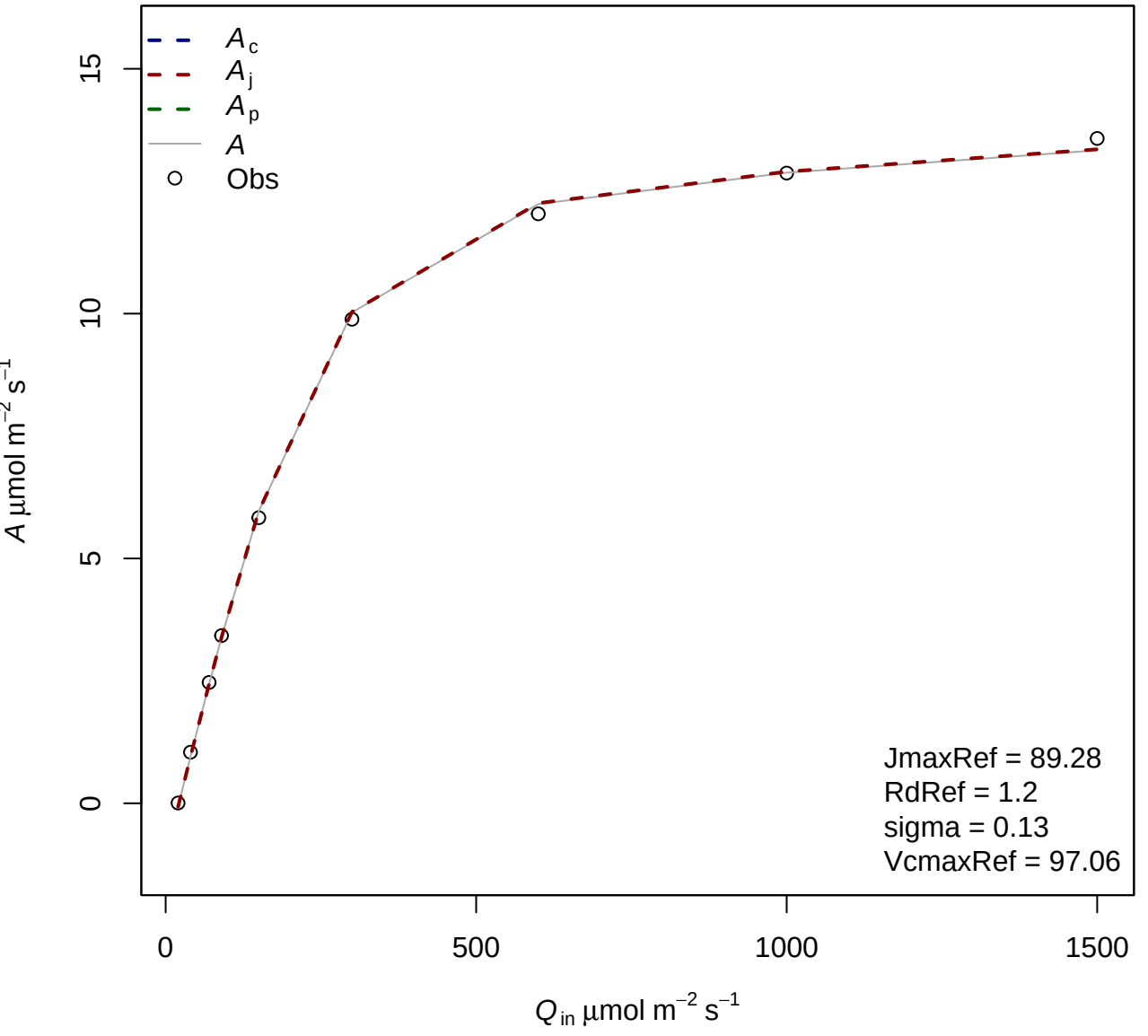

A_c
A_j
A_p
A
Obs
JmaxRef = 73.32
RdRef = 2.63
sigma = 0.13
VcmaxRef = 50.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200

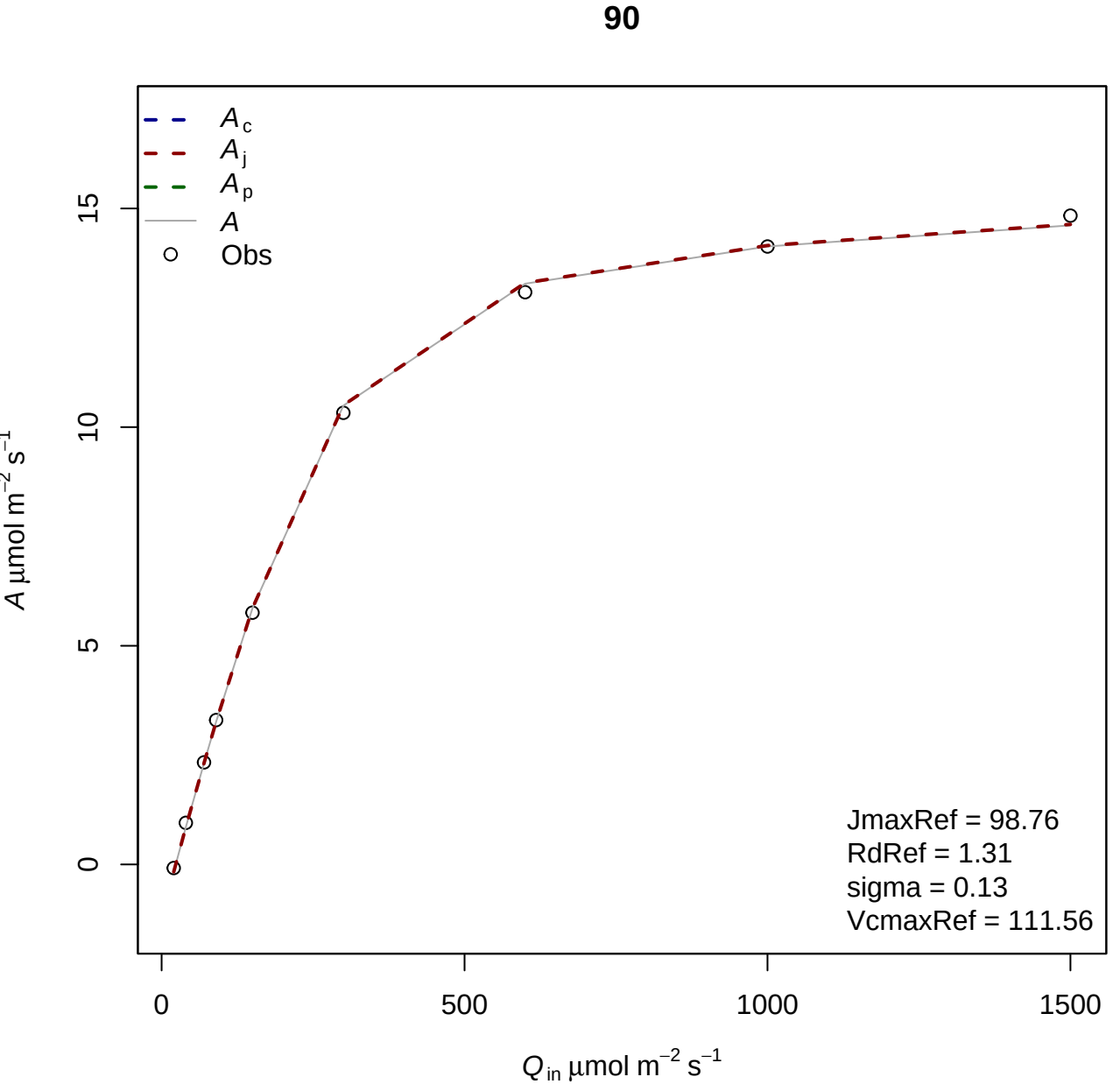

A_c
A_j
A_p
A
Obs
JmaxRef = 78.78
RdRef = 3.04
sigma = 0.37
VcmaxRef = 153.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

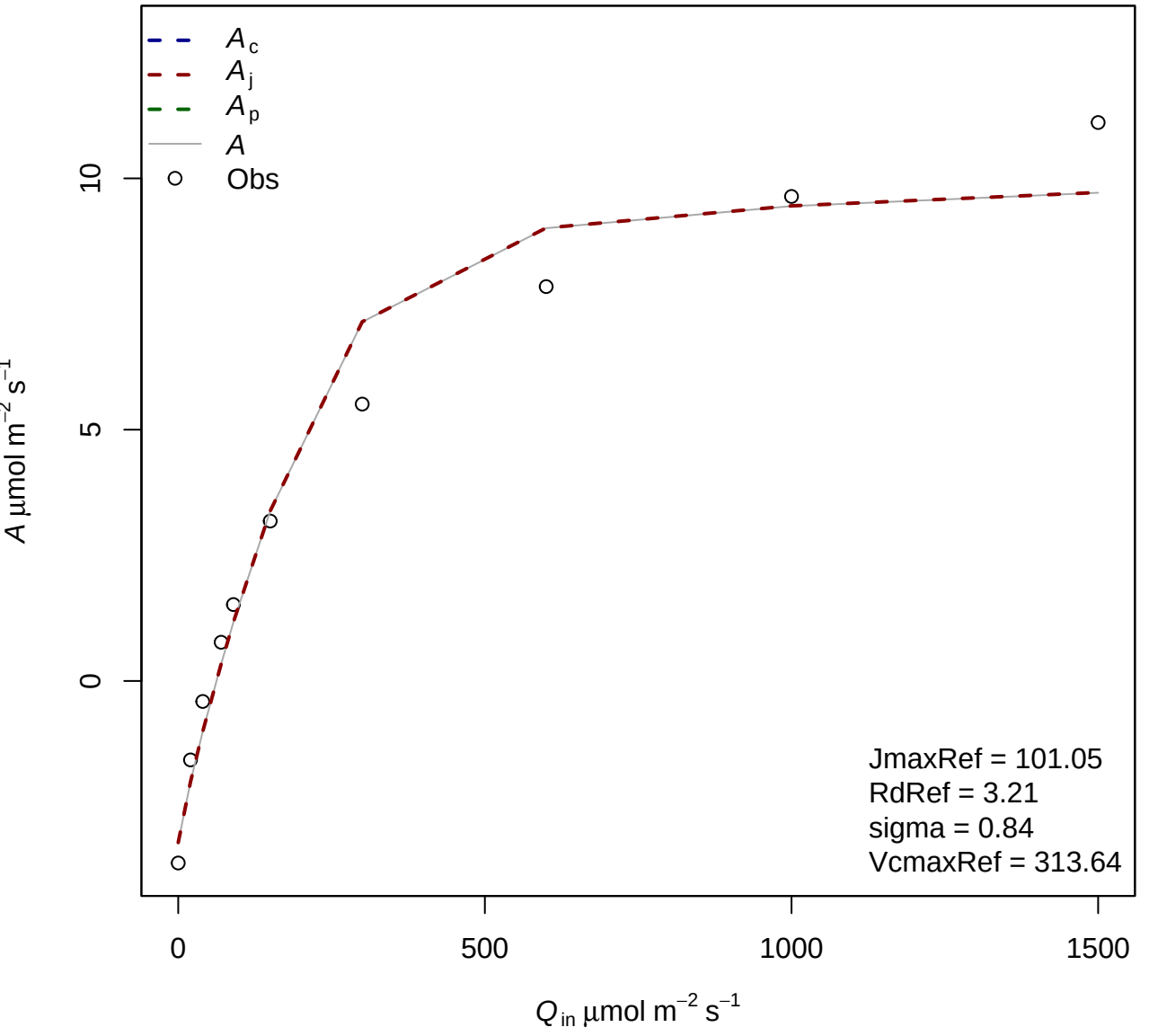

A_c
A_j
A_p
A
Obs
JmaxRef = 83.39
RdRef = 0.98
sigma = 0.04
VcmaxRef = 122.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

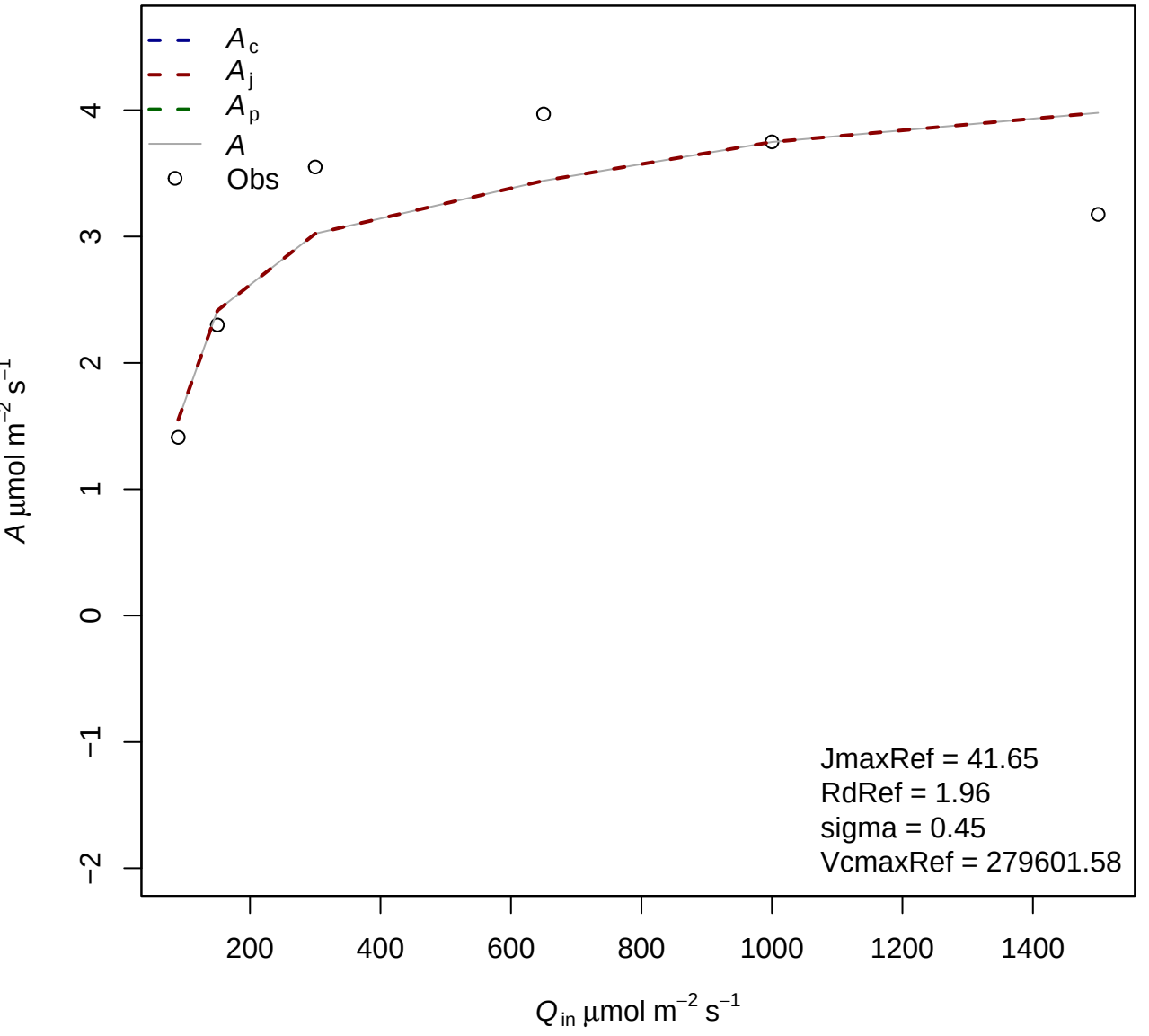

A_c
A_j
A_p
A
Obs
JmaxRef = 44.7
RdRef = 3.49
sigma = 0.98
VcmaxRef = 186.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200

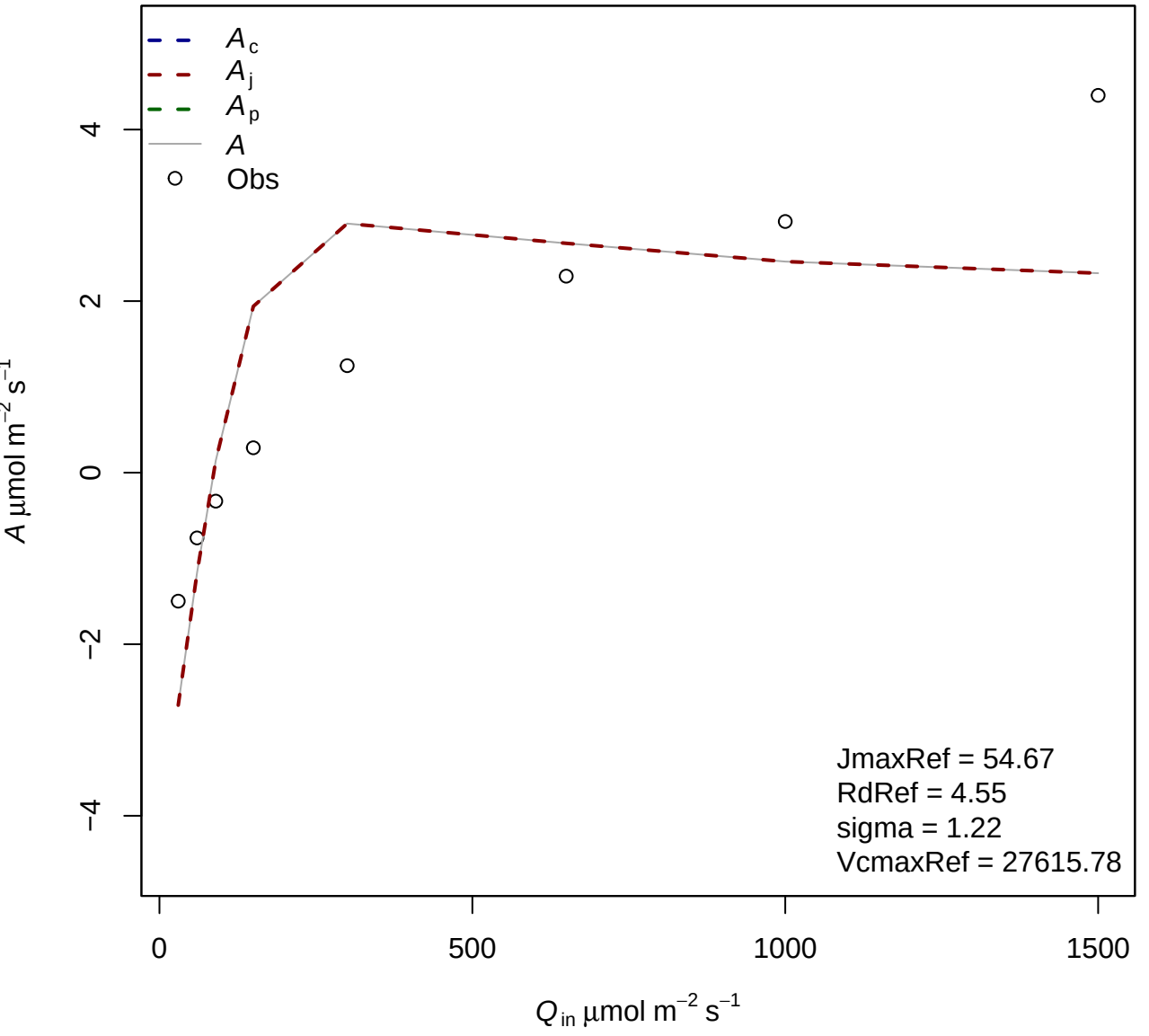

A_c
A_j
A_p
A
Obs
JmaxRef = 55.78
RdRef = 2.97
sigma = 0.81
VcmaxRef = 168.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

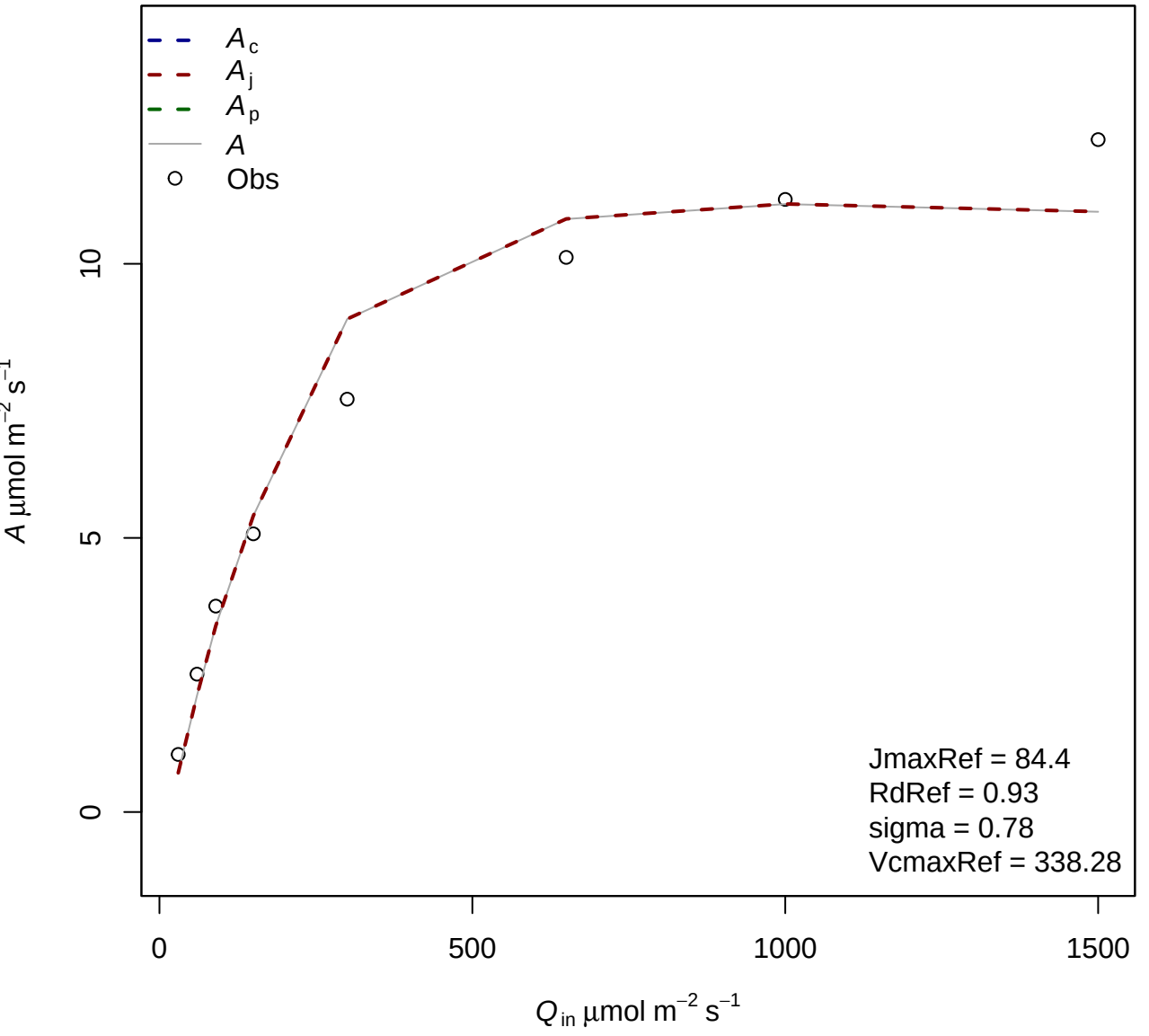

A_c
A_j
A_p
A
Obs
JmaxRef = 89.28
RdRef = 1.2
sigma = 0.13
VcmaxRef = 97.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

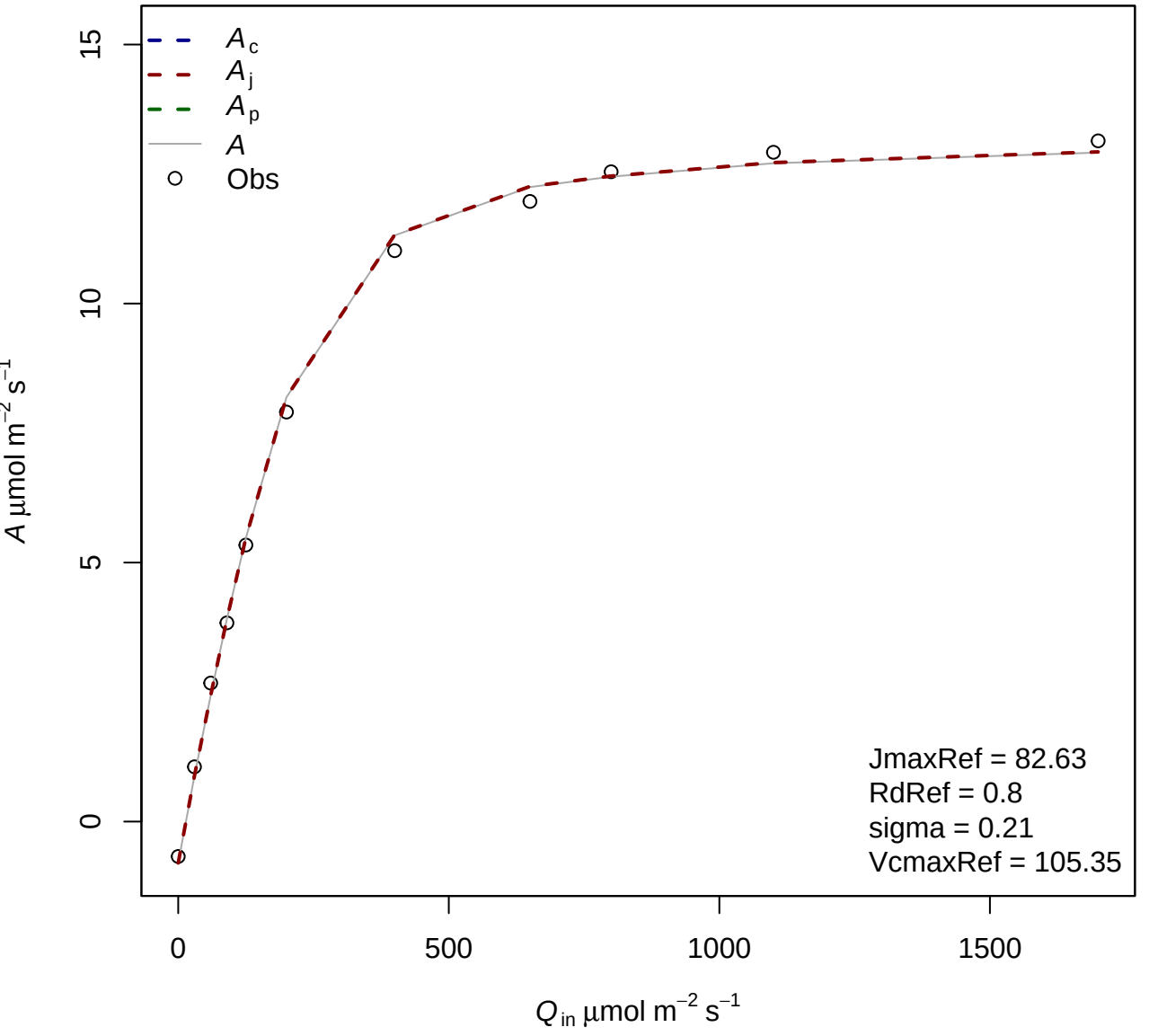

90
A_c
A_j
A_p
A
Obs
JmaxRef = 98.76
RdRef = 1.31
sigma = 0.13
VcmaxRef = 111.56
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

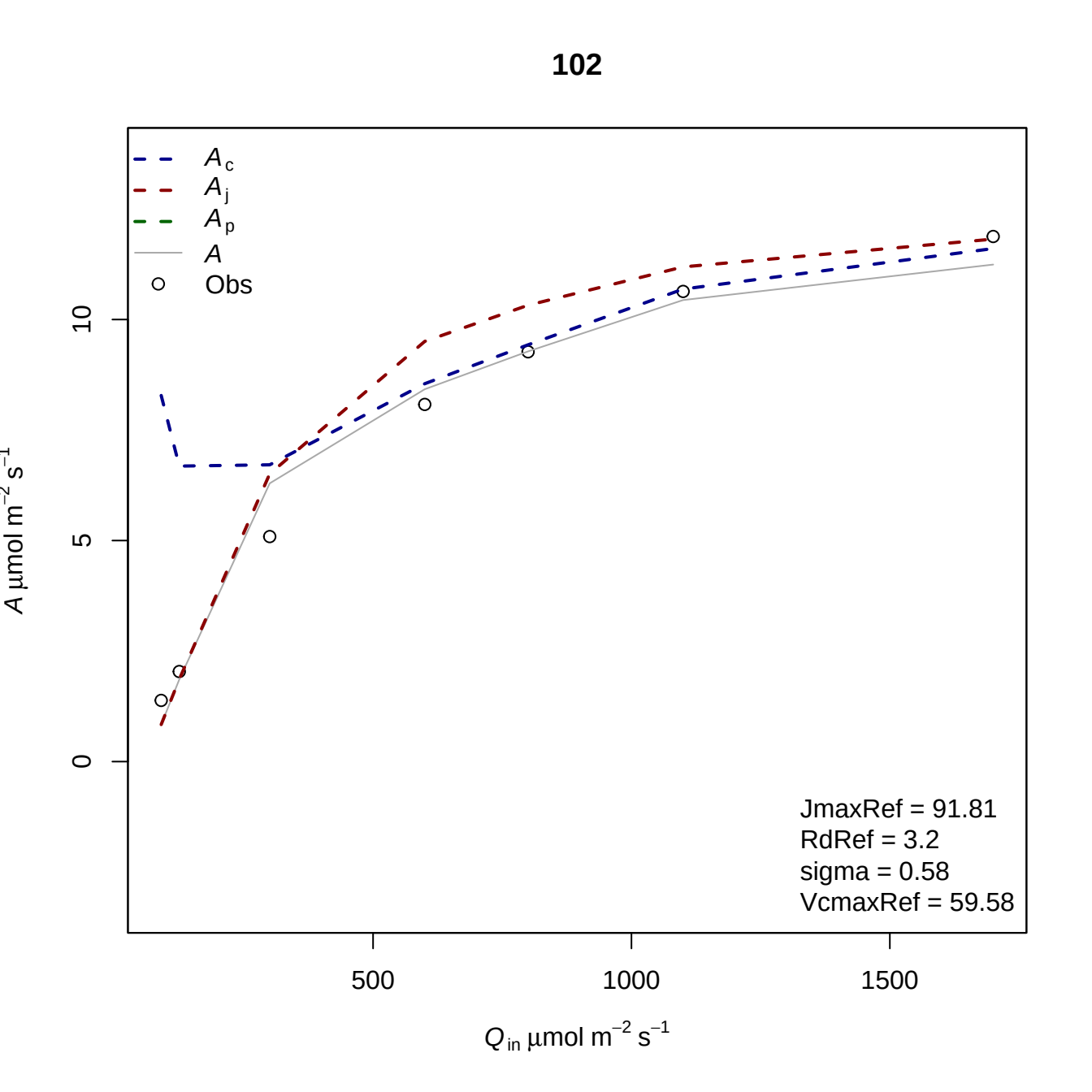

A_c
A_j
A_p
A
Obs
JmaxRef = 101.05
RdRef = 3.21
sigma = 0.84
VcmaxRef = 313.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

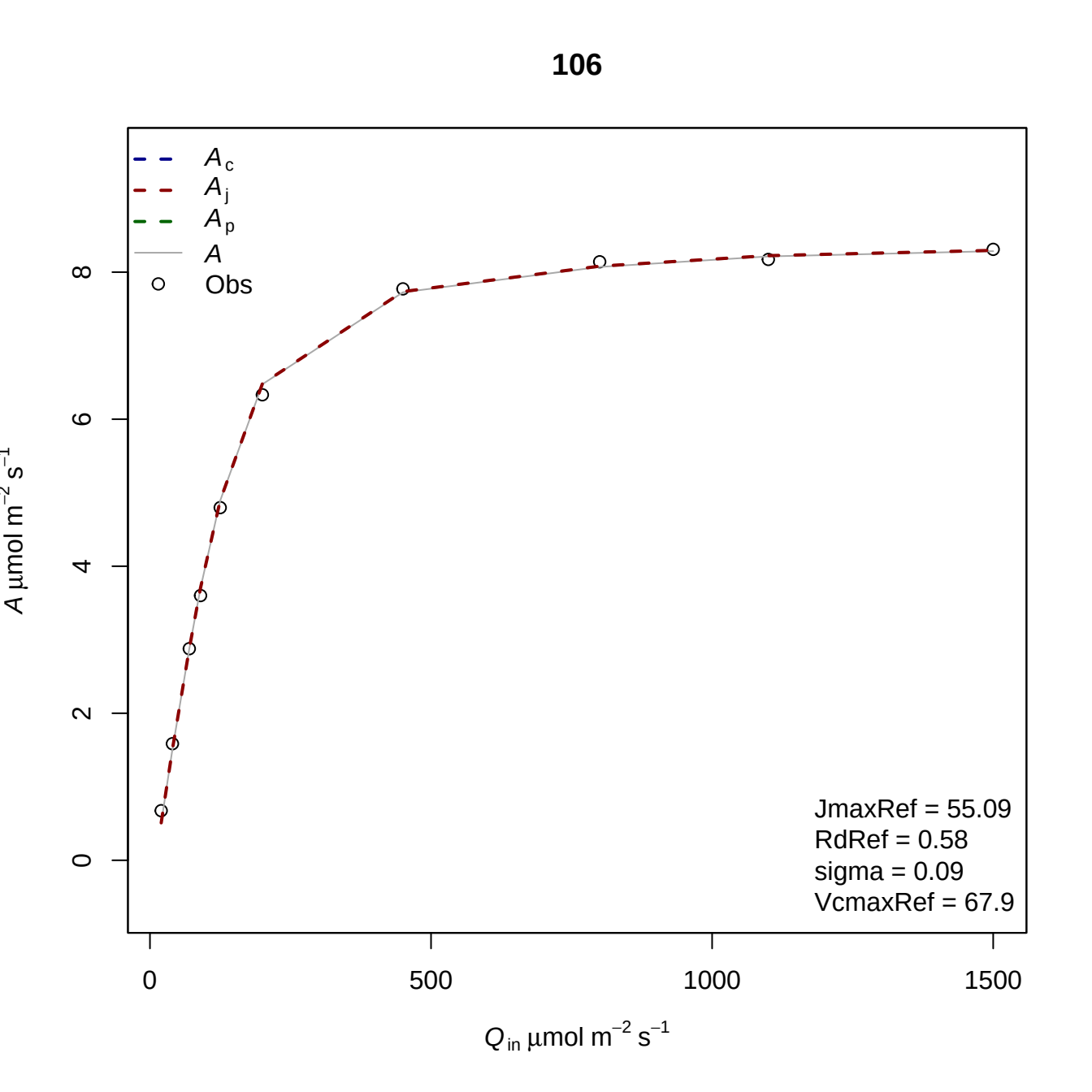

A_c
A_j
A_p
A
Obs
JmaxRef = 41.65
RdRef = 1.96
sigma = 0.45
VcmaxRef = 279601.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

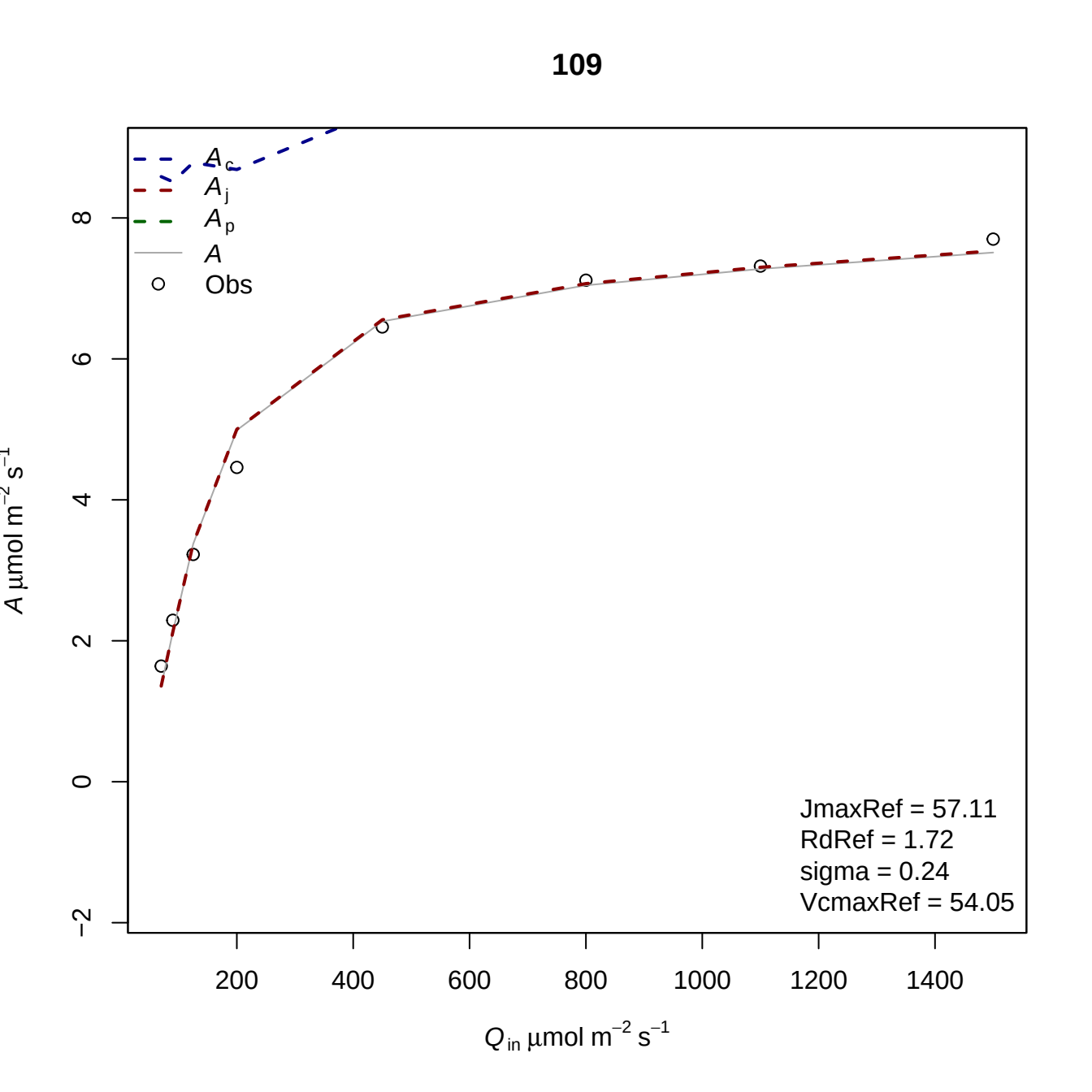

A_c
A_j
A_p
A
Obs
JmaxRef = 54.67
RdRef = 4.55
sigma = 1.22
VcmaxRef = 27615.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

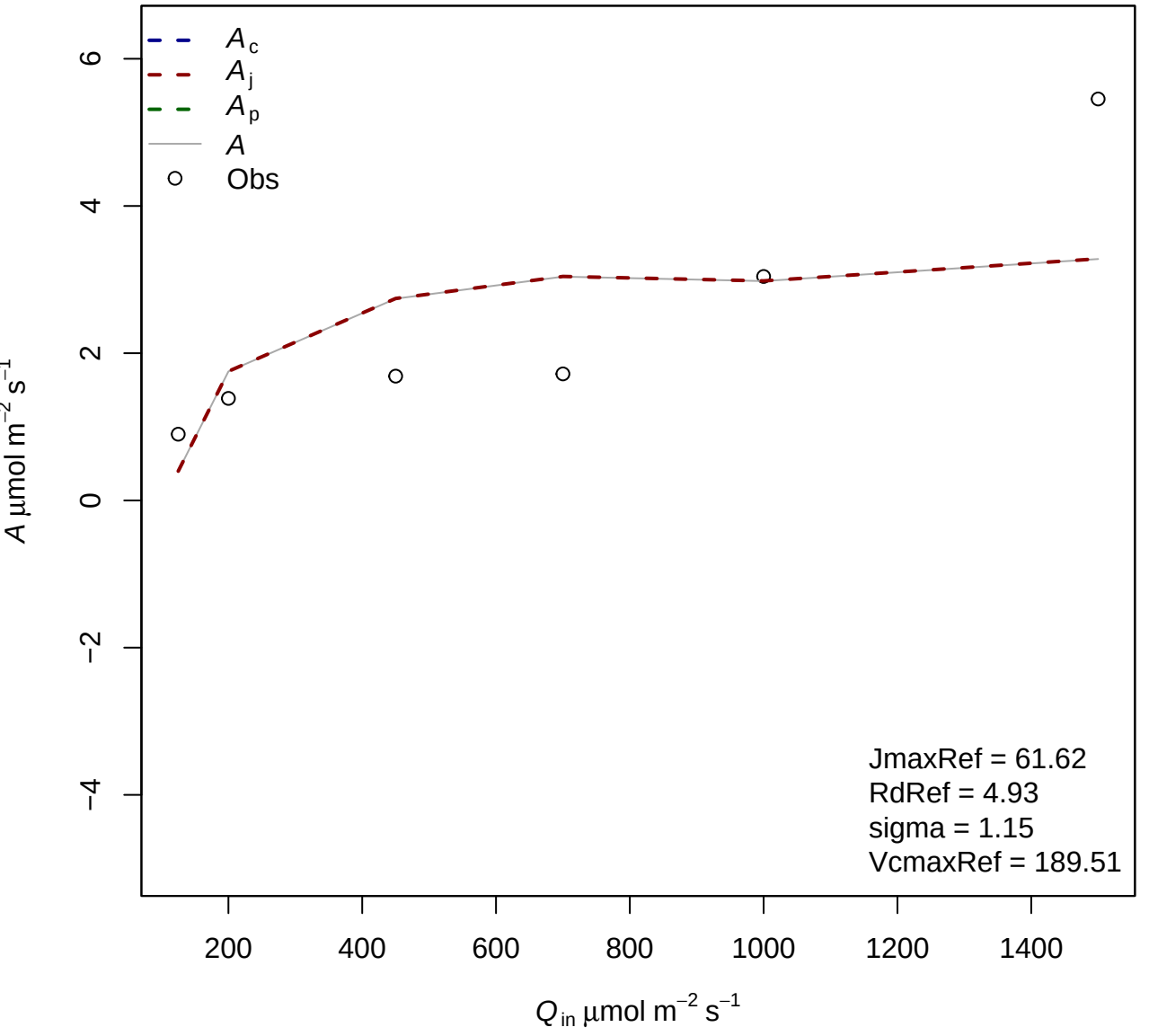

A_c
A_j
A_p
A
Obs
JmaxRef = 84.4
RdRef = 0.93
sigma = 0.78
VcmaxRef = 338.28
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

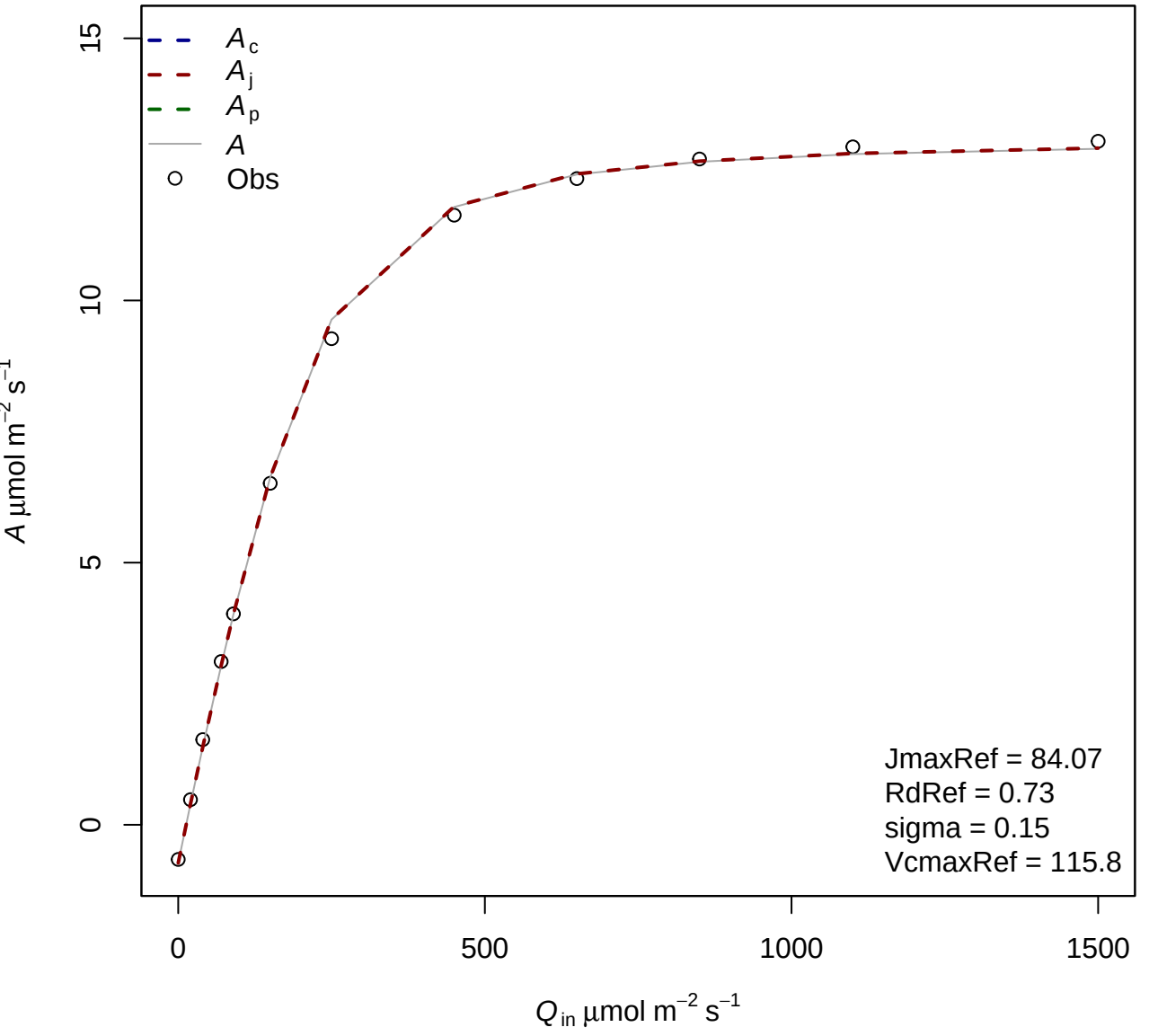

A_c
A_j
A_p
A
Obs
JmaxRef = 82.63
RdRef = 0.8
sigma = 0.21
VcmaxRef = 105.35
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

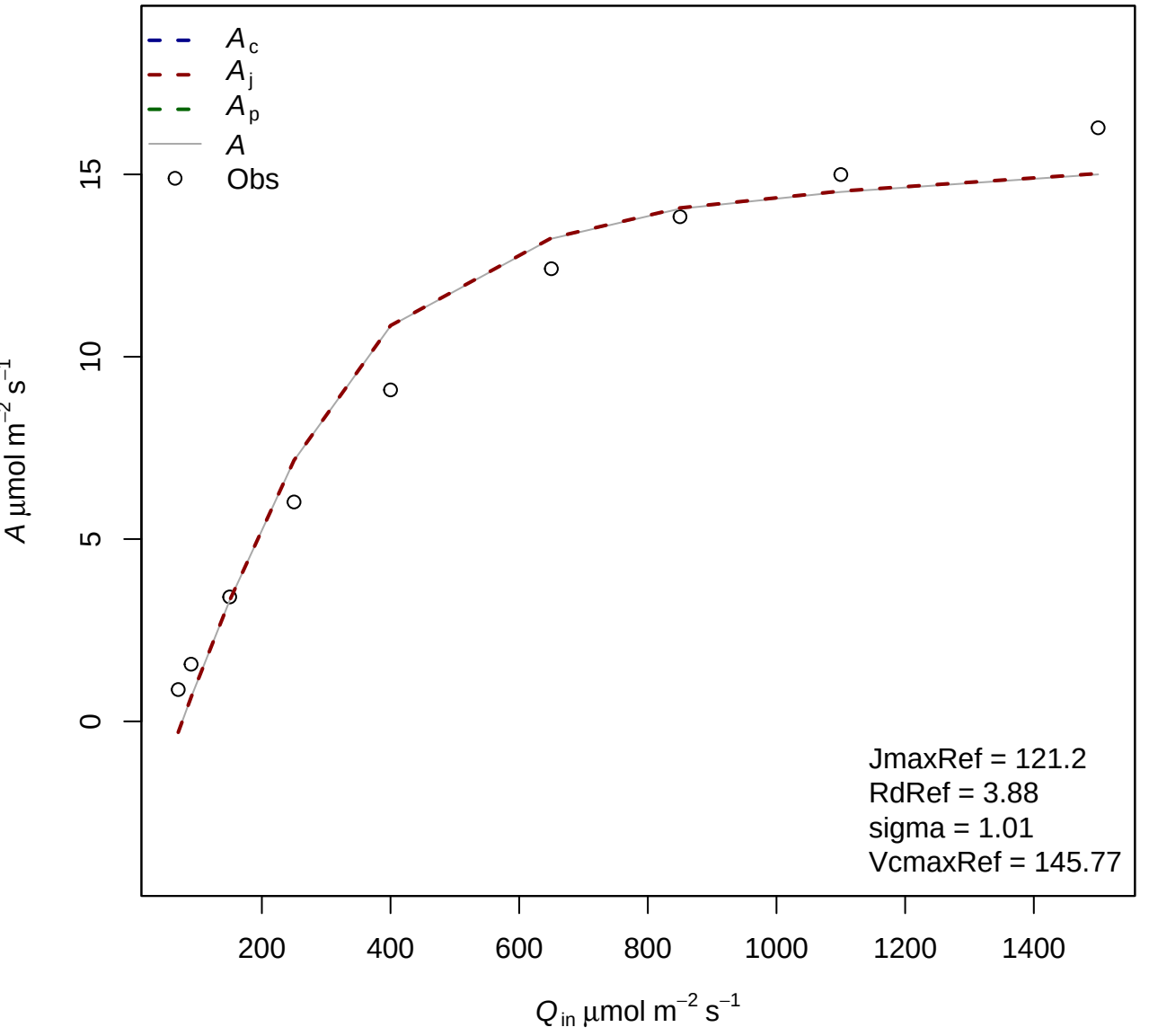

102
A_c
A_j
A_p
A
Obs
JmaxRef = 91.81
RdRef = 3.2
sigma = 0.58
VcmaxRef = 59.58
0
5
10
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

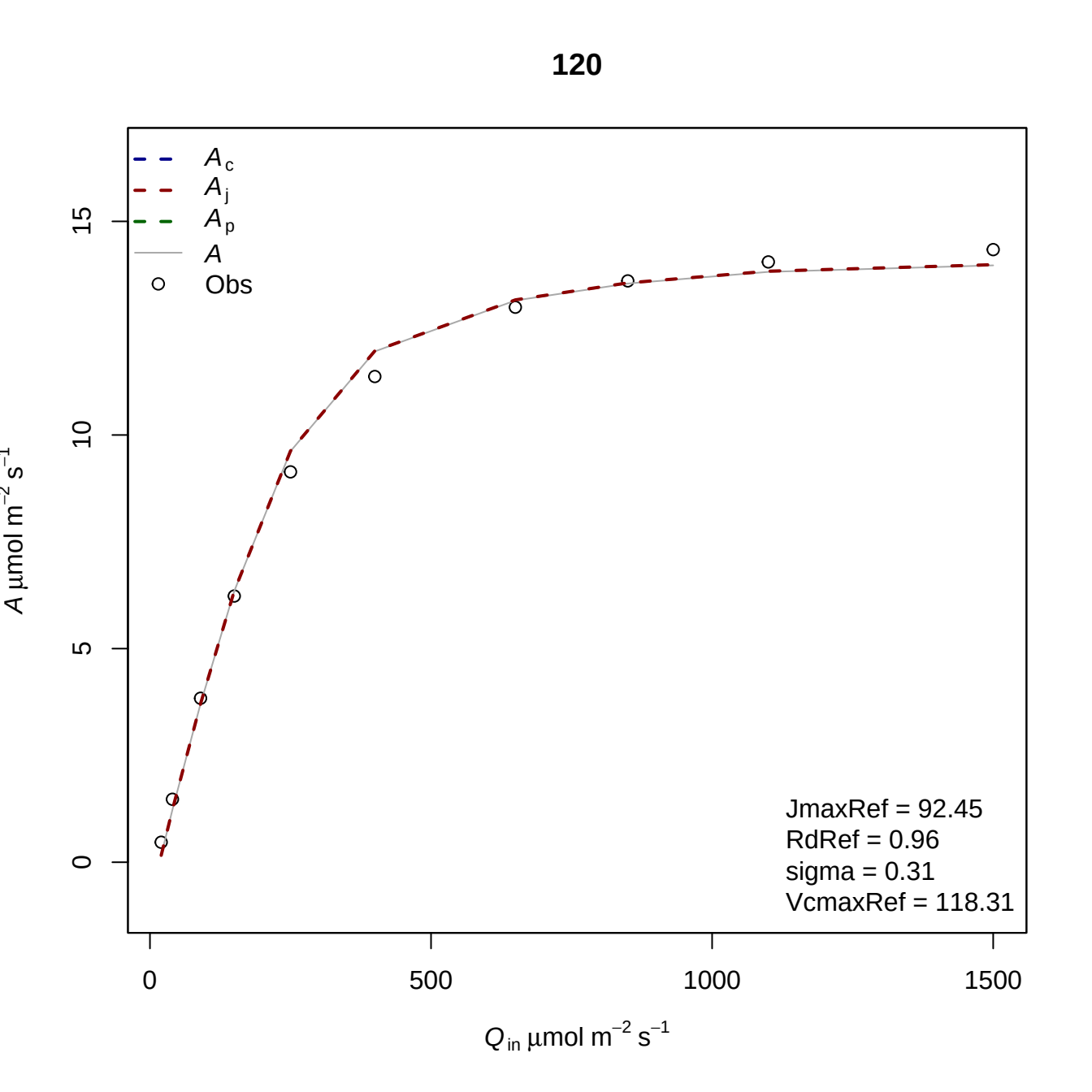

106
A_c
A_j
A_p
A
Obs
JmaxRef = 55.09
RdRef = 0.58
sigma = 0.09
VcmaxRef = 67.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

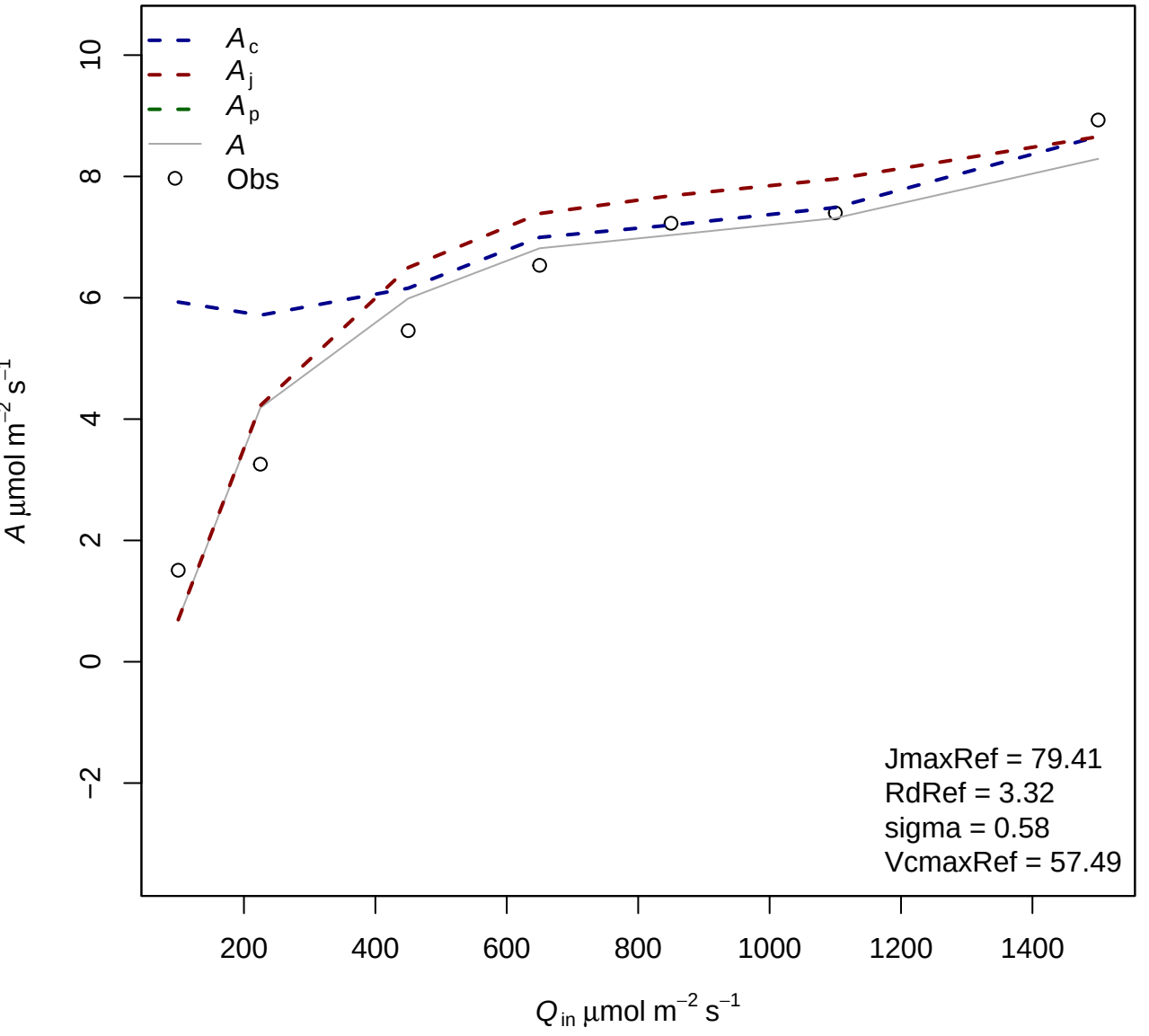

109
A_c
A_j
A_p
A
Obs
JmaxRef = 57.11
RdRef = 1.72
sigma = 0.24
VcmaxRef = 54.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

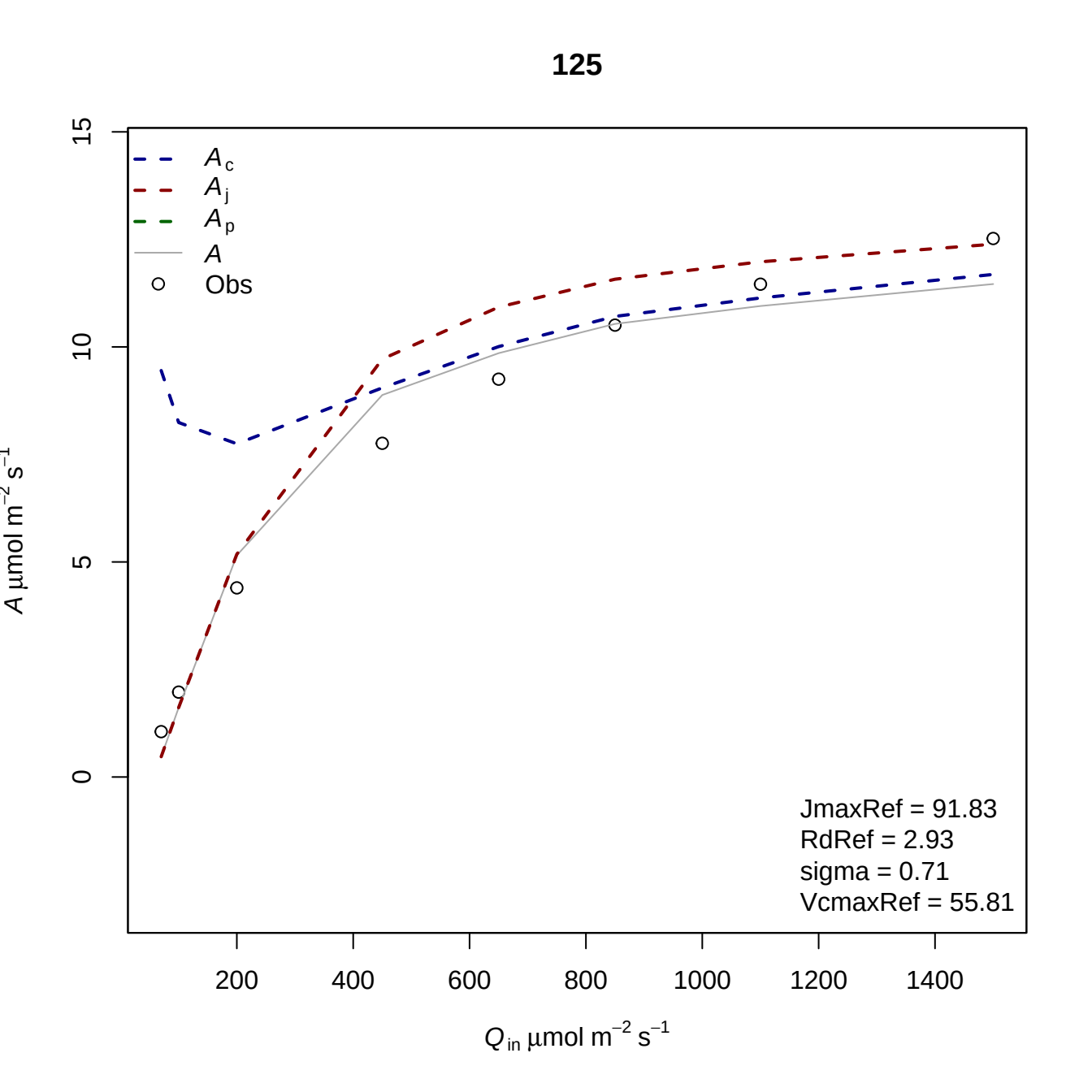

A_c
A_j
A_p
A
Obs
JmaxRef = 61.62
RdRef = 4.93
sigma = 1.15
VcmaxRef = 189.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400


A_c
A_j
A_p
A
Obs
JmaxRef = 84.07
RdRef = 0.73
sigma = 0.15
VcmaxRef = 115.8
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

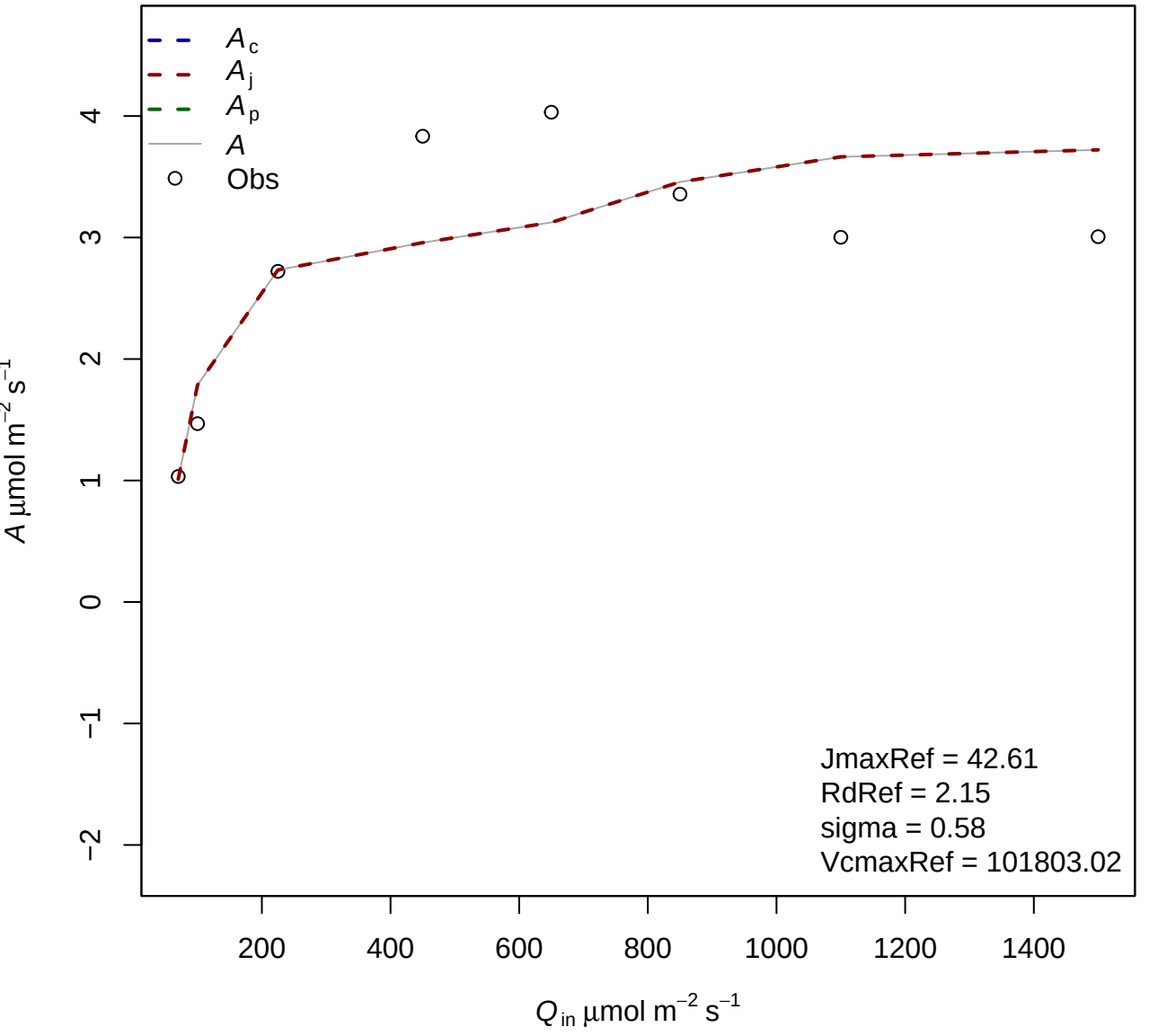

A_c
A_j
A_p
A
Obs
JmaxRef = 121.2
RdRef = 3.88
sigma = 1.01
VcmaxRef = 145.77
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

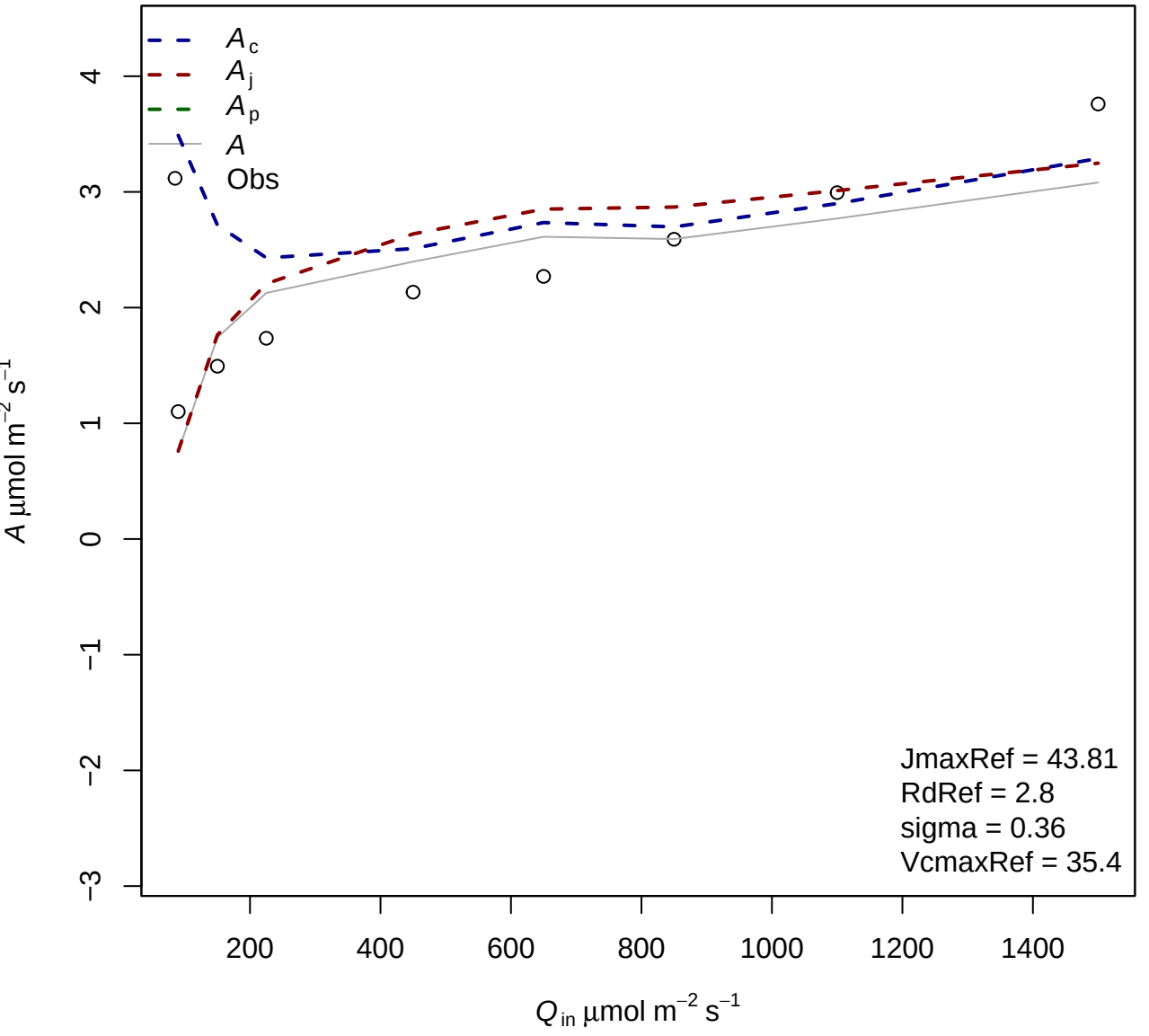

120
A_c
A_j
A_p
A
Obs
JmaxRef = 92.45
RdRef = 0.96
sigma = 0.31
VcmaxRef = 118.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

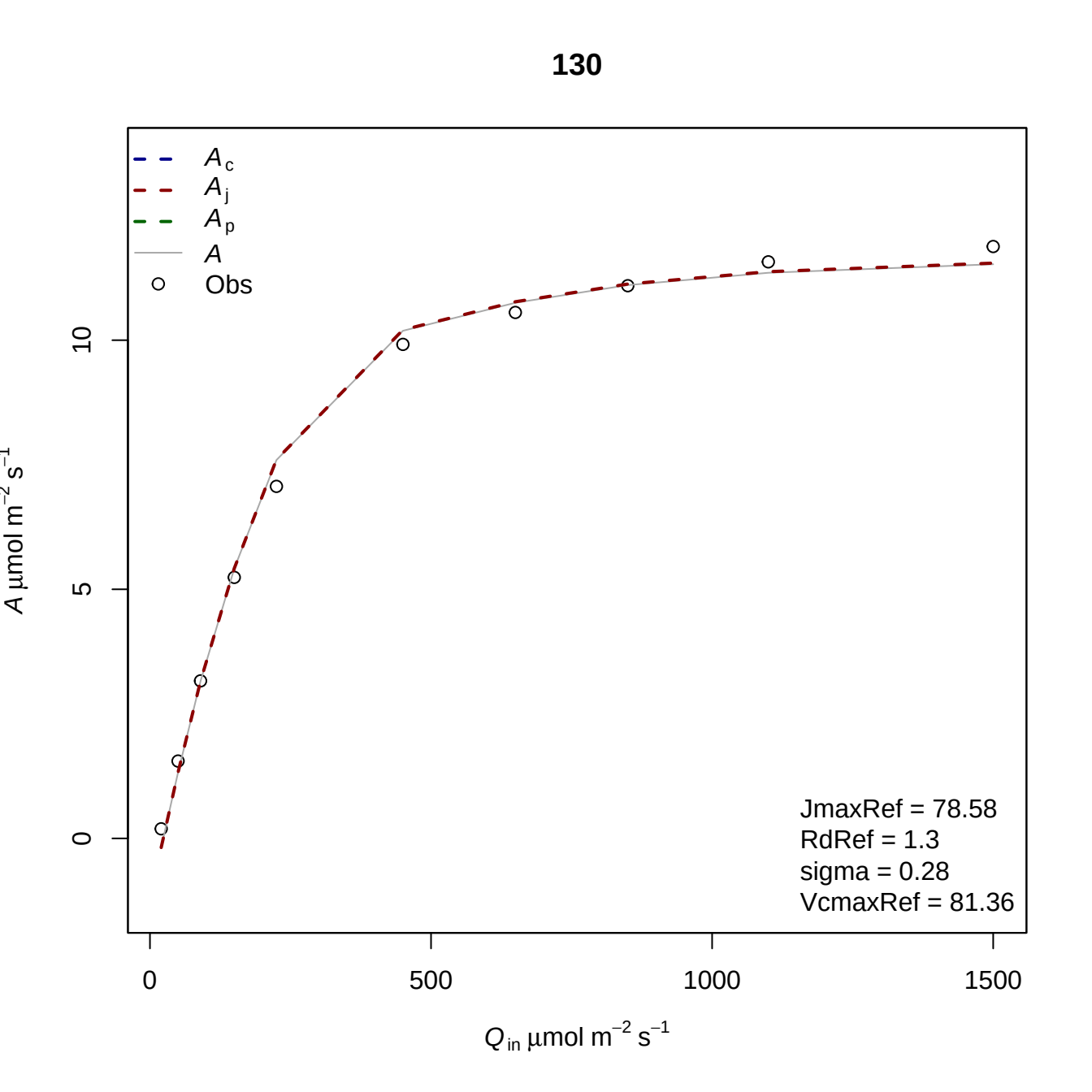

A_c
A_j
A_p
A
Obs
JmaxRef = 79.41
RdRef = 3.32
sigma = 0.58
VcmaxRef = 57.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

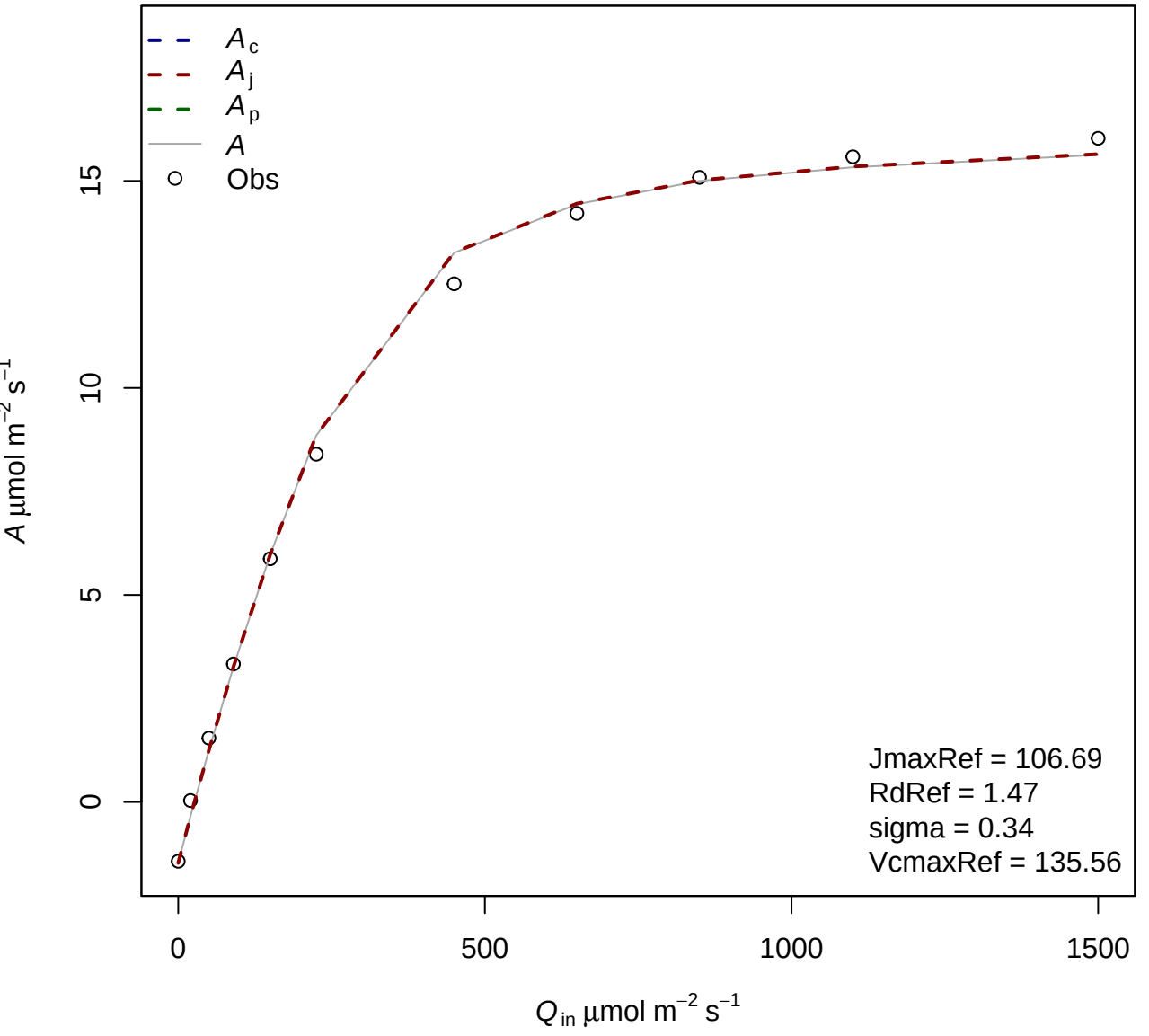

125
A_c
A_j
A_p
A
Obs
JmaxRef = 91.83
RdRef = 2.93
sigma = 0.71
VcmaxRef = 55.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
200
400
600
800
1000
1200
1400

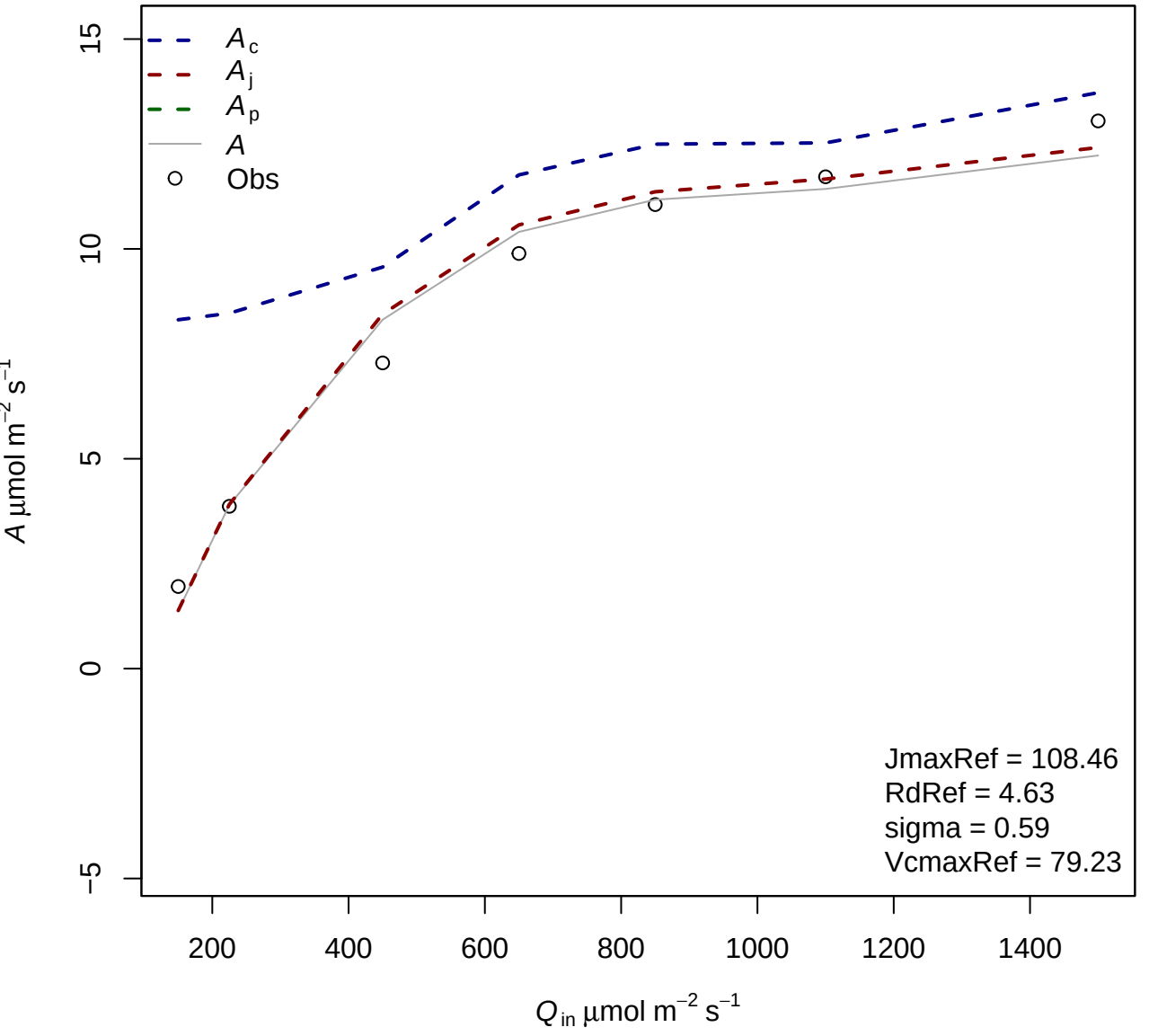
126

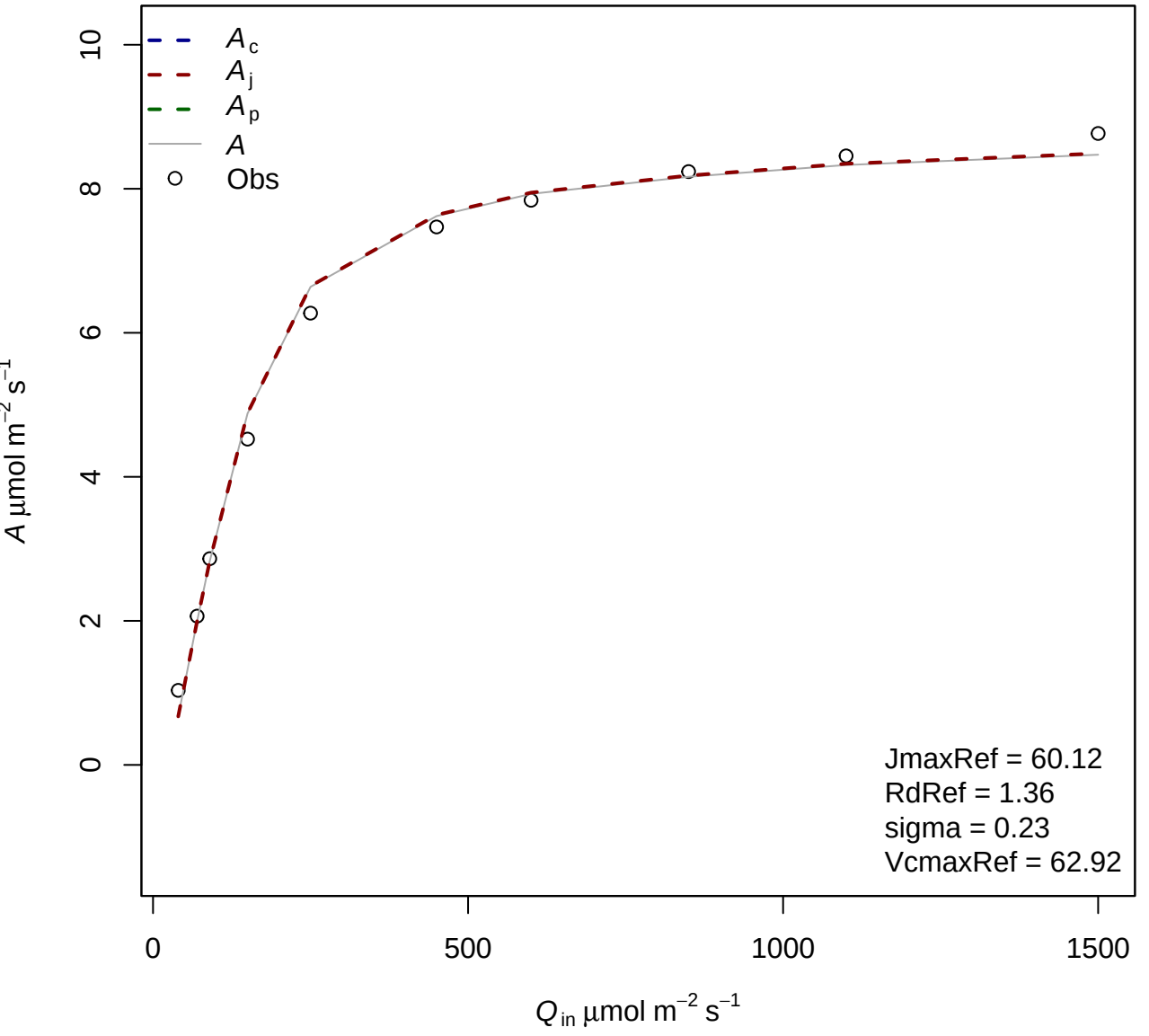

A_c
A_j
A_p
A
Obs
JmaxRef = 42.61
RdRef = 2.15
sigma = 0.58
VcmaxRef = 101803.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

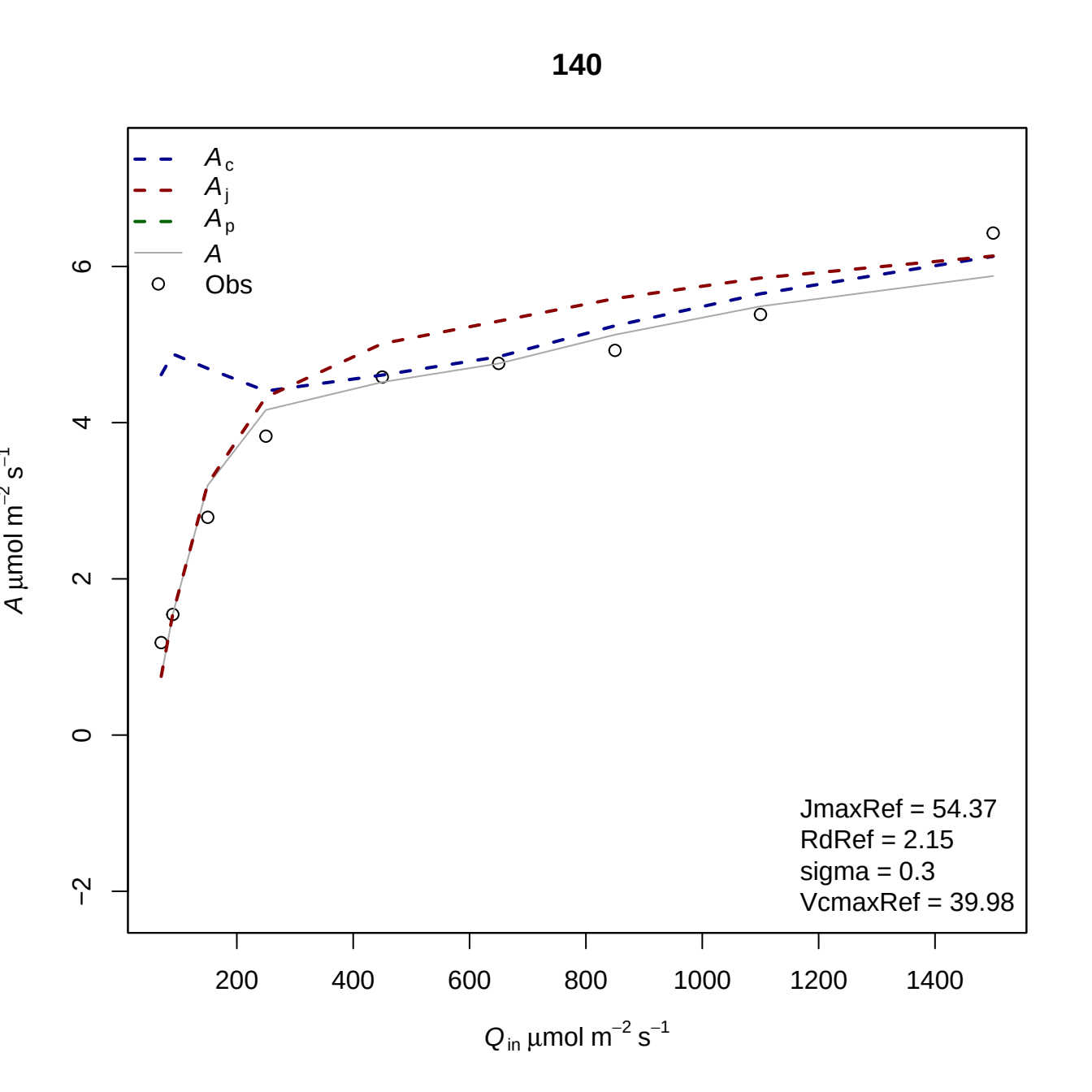

A_c
A_j
A_p
A
Obs
JmaxRef = 43.81
RdRef = 2.8
sigma = 0.36
VcmaxRef = 35.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

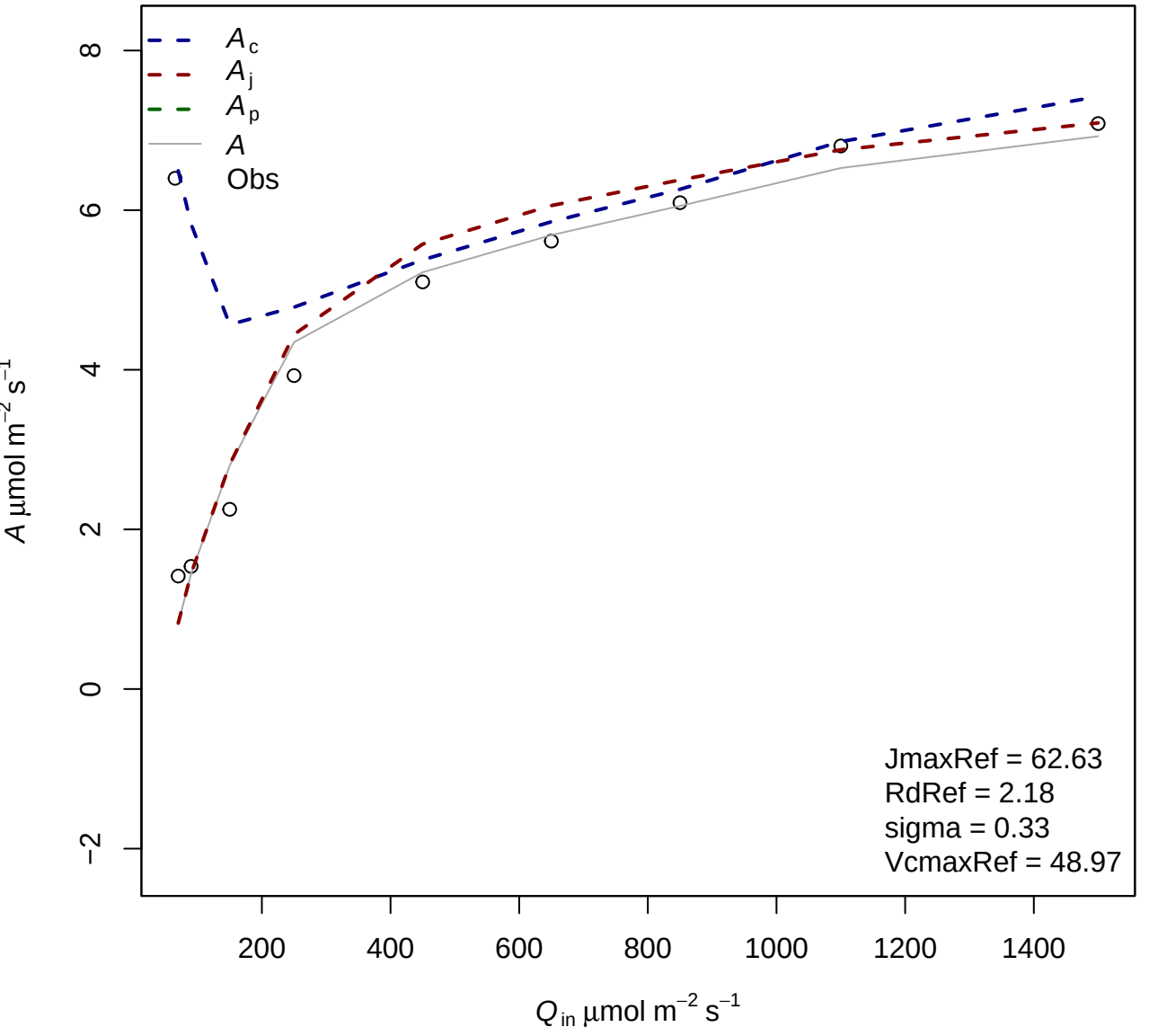

130
A_c
A_j
A_p
A
Obs
JmaxRef = 78.58
RdRef = 1.3
sigma = 0.28
VcmaxRef = 81.36
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

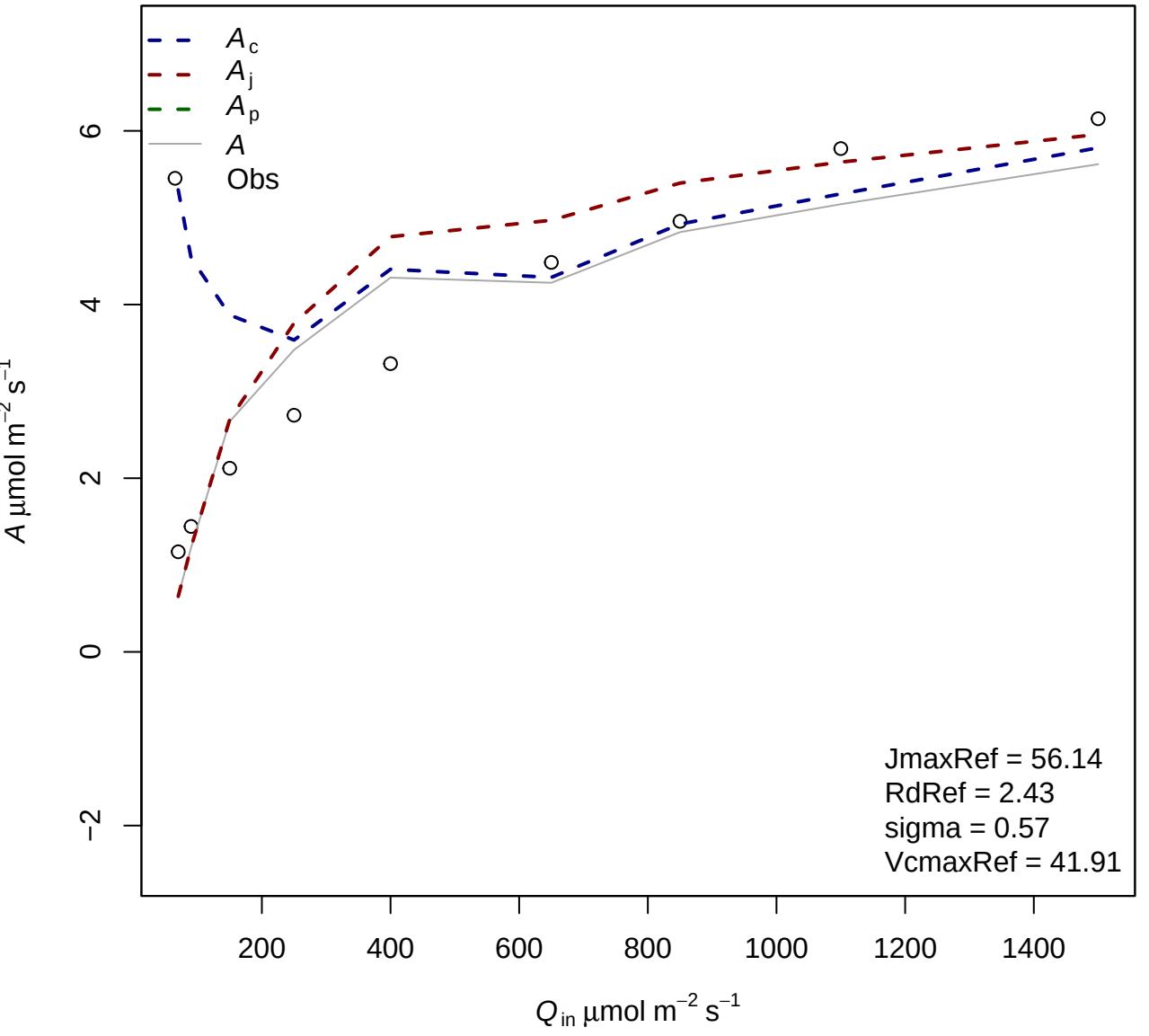

A_c
A_j
A_p
A
Obs
JmaxRef = 106.69
RdRef = 1.47
sigma = 0.34
VcmaxRef = 135.56
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

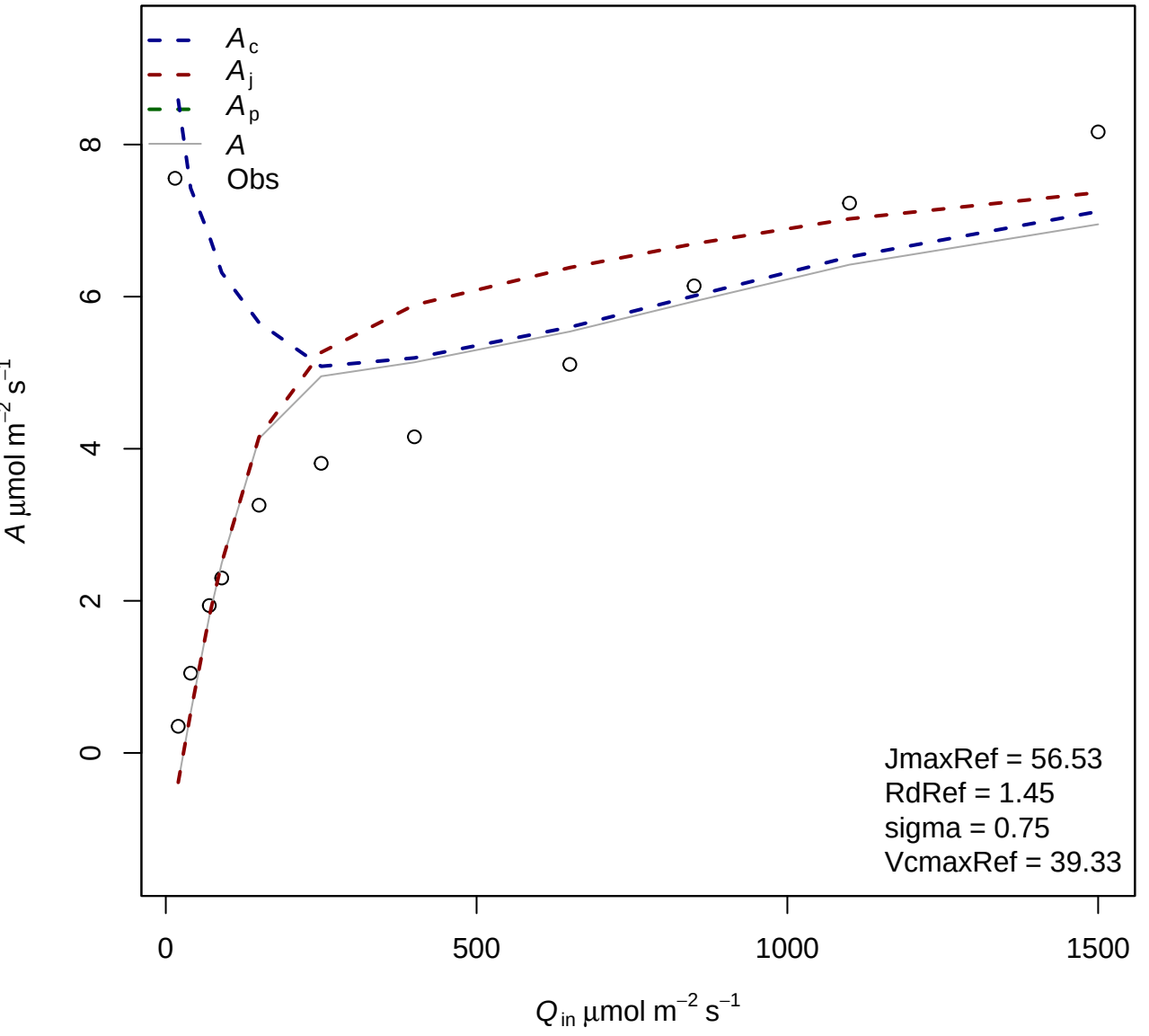

A_c
A_j
A_p
A
Obs
JmaxRef = 108.46
RdRef = 4.63
sigma = 0.59
VcmaxRef = 79.23
15
10
5
0
−5
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

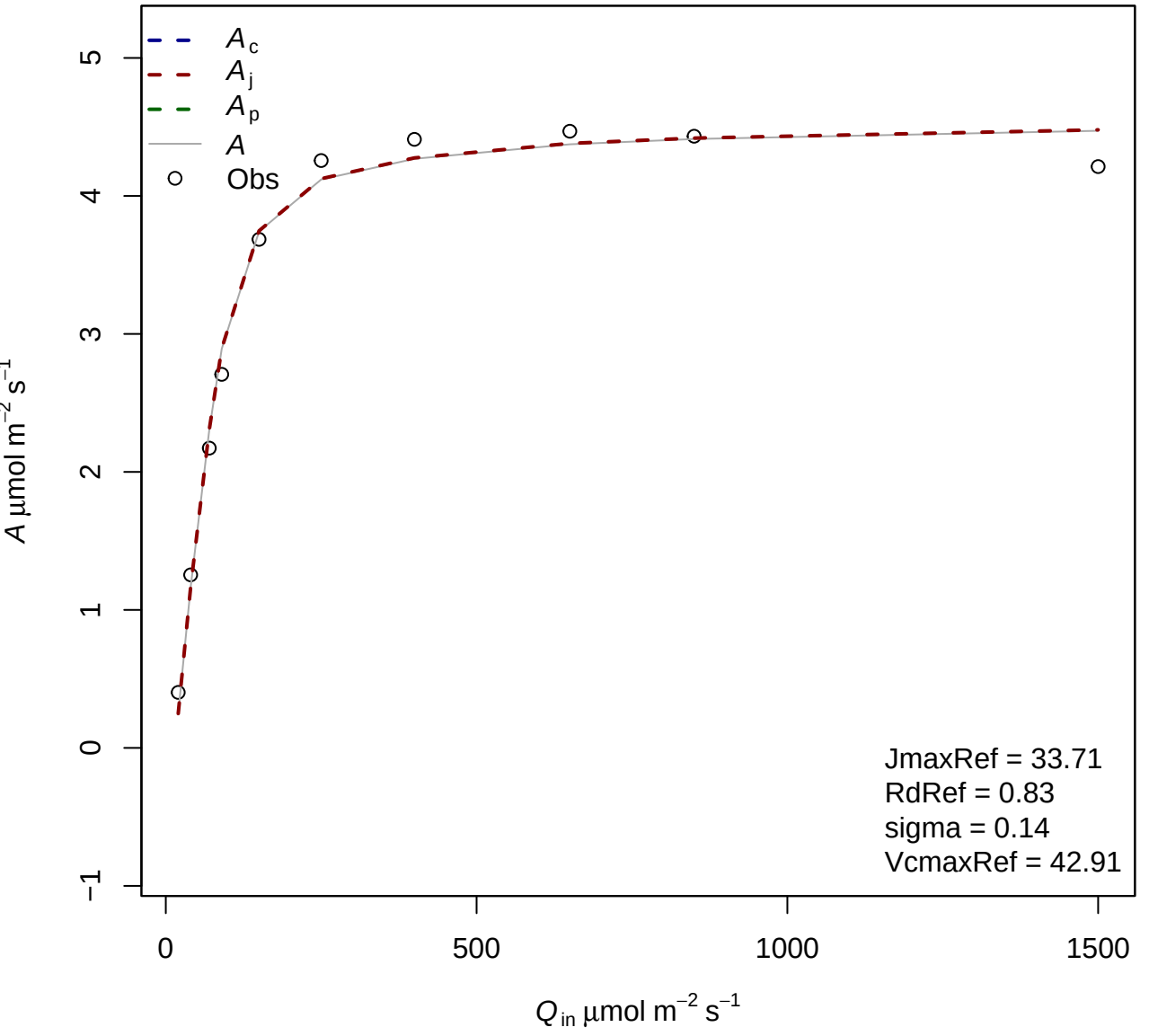

A_c
A_j
A_p
A
Obs
JmaxRef = 60.12
RdRef = 1.36
sigma = 0.23
VcmaxRef = 62.92
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

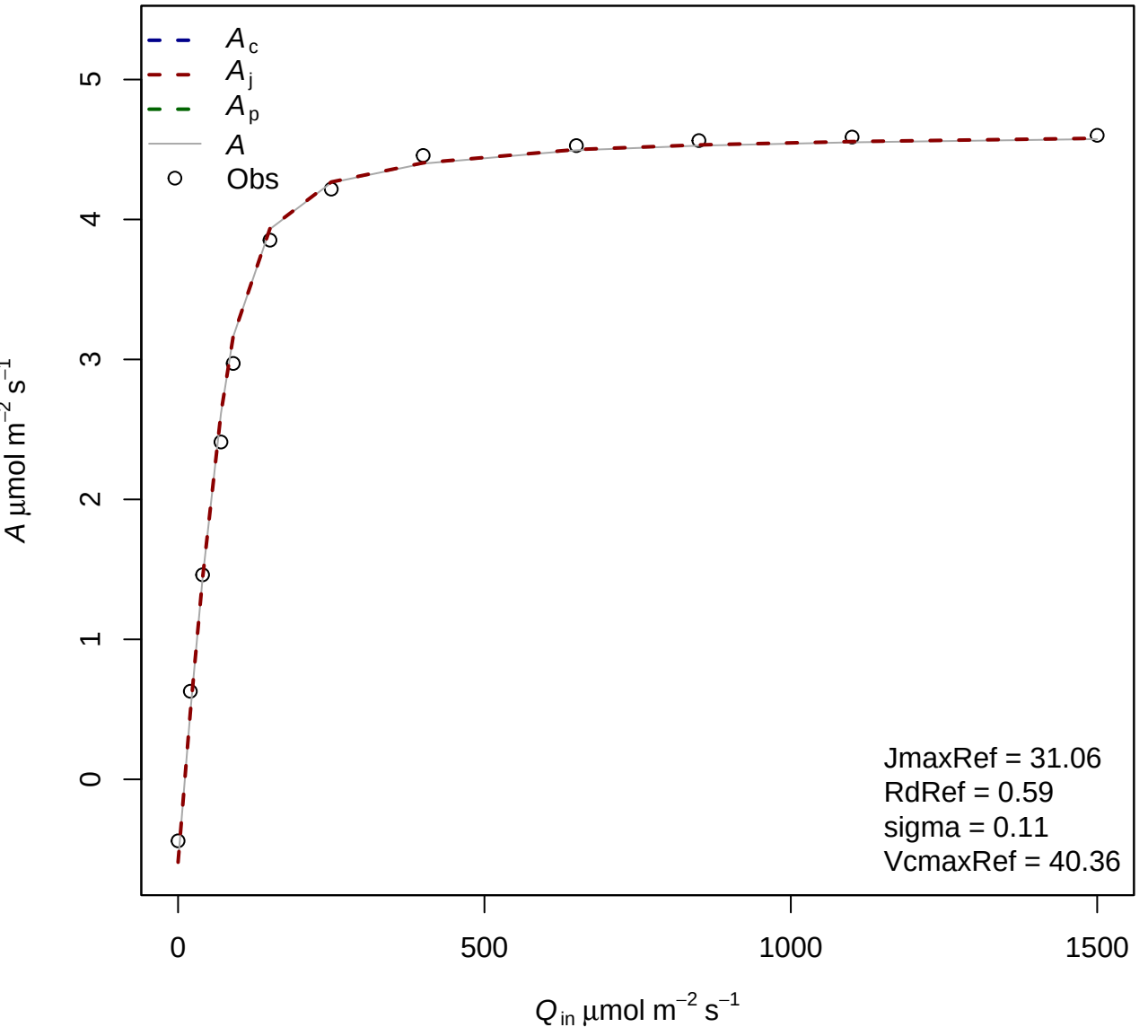

140
A_c
A_j
A_p
A
Obs
JmaxRef = 54.37
RdRef = 2.15
sigma = 0.3
VcmaxRef = 39.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000
1200
1400

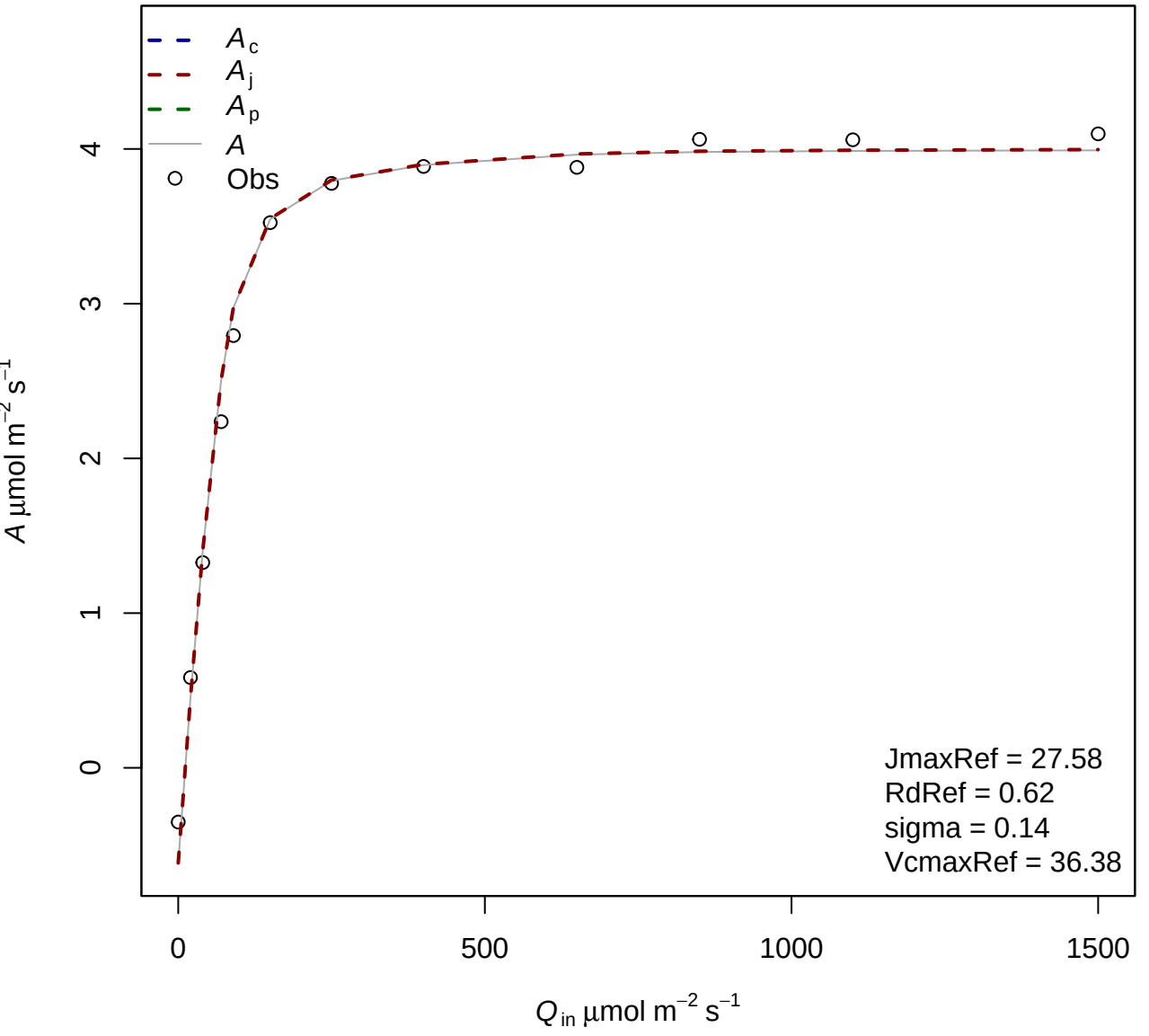

A_c
A_j
A_p
A
Obs
JmaxRef = 62.63
RdRef = 2.18
sigma = 0.33
VcmaxRef = 48.97
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

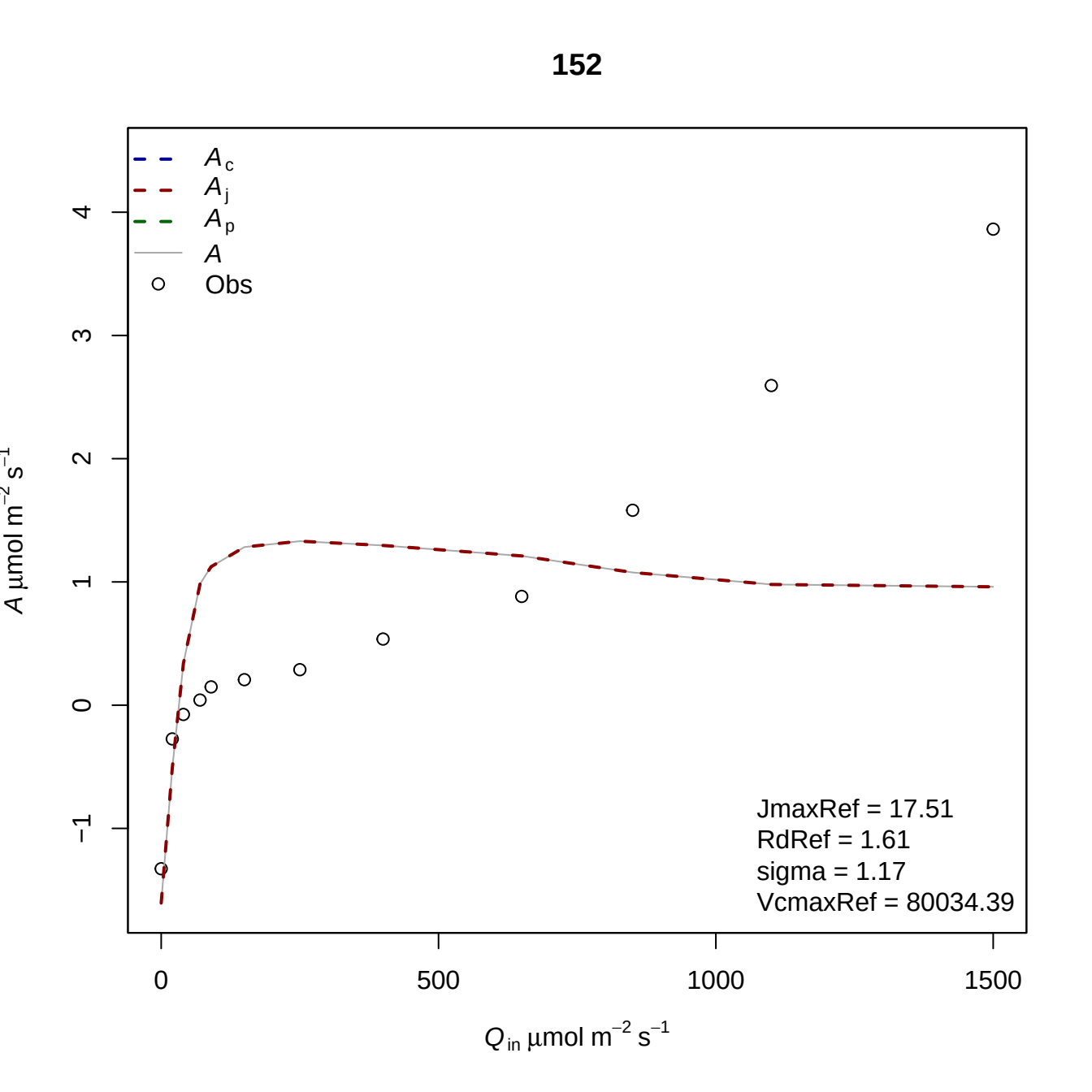

A_c
A_j
A_p
A
Obs
JmaxRef = 56.14
RdRef = 2.43
sigma = 0.57
VcmaxRef = 41.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000
1200
1400

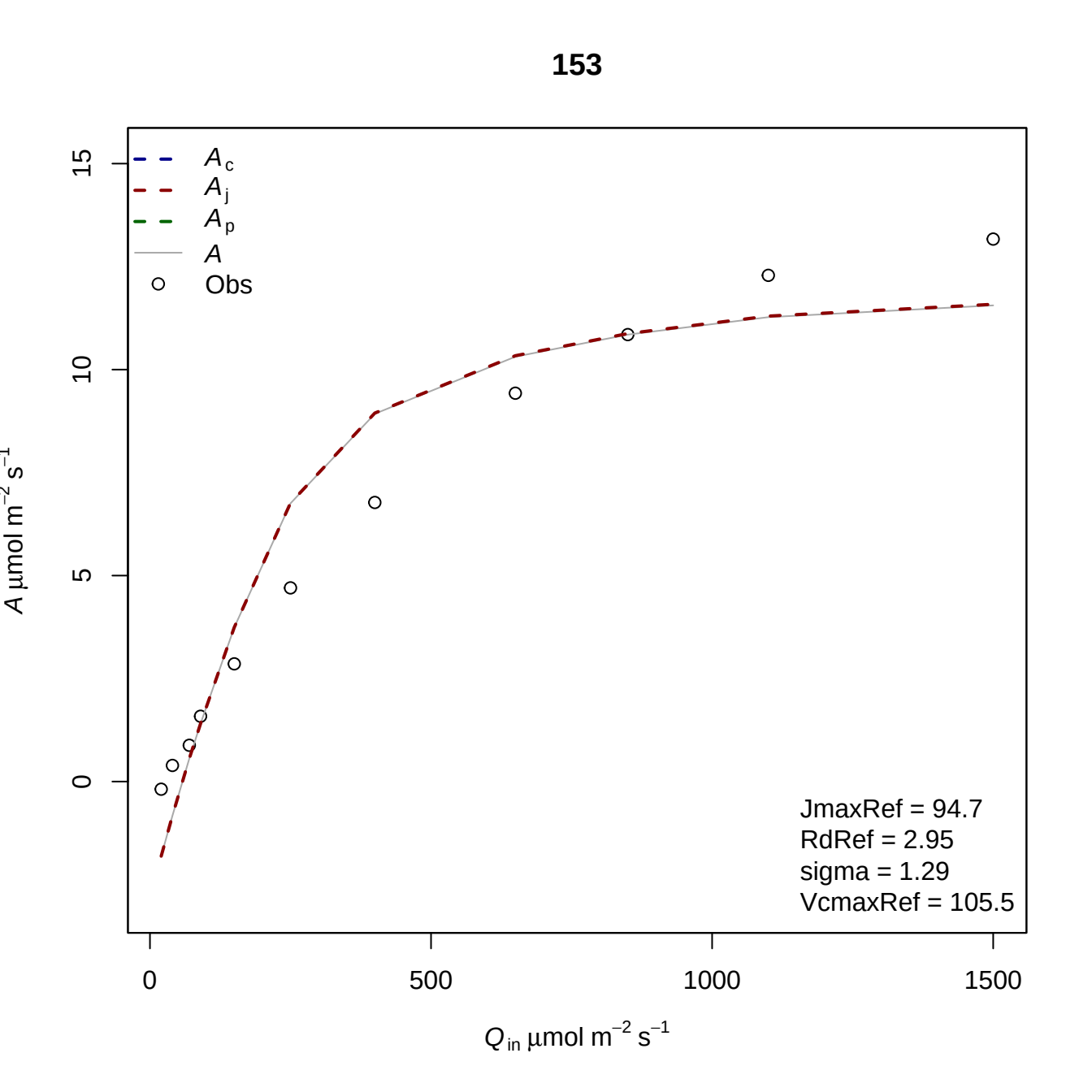

A_c
A_j
A_p
A
Obs
JmaxRef = 56.53
RdRef = 1.45
sigma = 0.75
VcmaxRef = 39.33
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

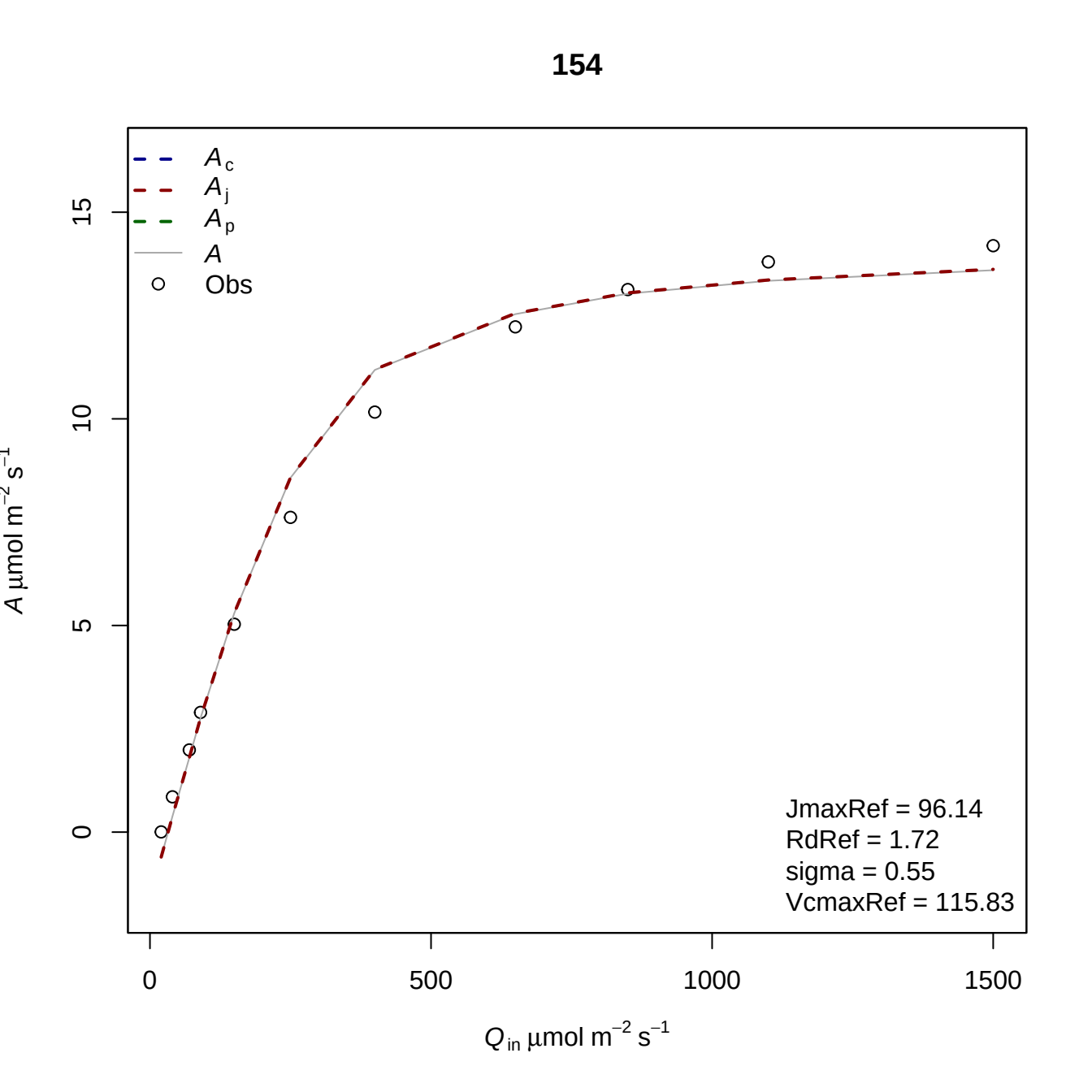

A_c
A_j
A_p
A
Obs
JmaxRef = 33.71
RdRef = 0.83
sigma = 0.14
VcmaxRef = 42.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

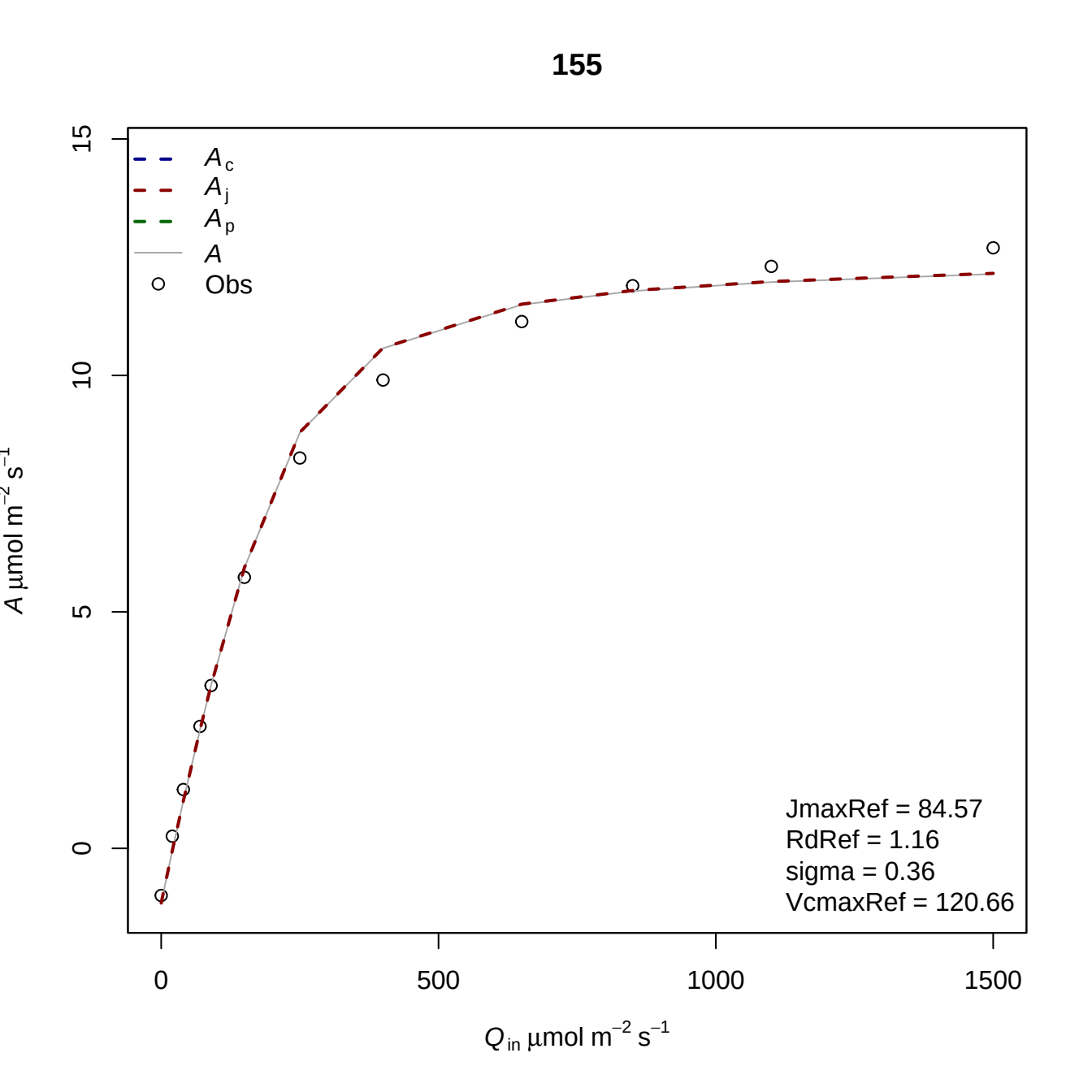

A_c
A_j
A_p
A
Obs
JmaxRef = 31.06
RdRef = 0.59
sigma = 0.11
VcmaxRef = 40.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

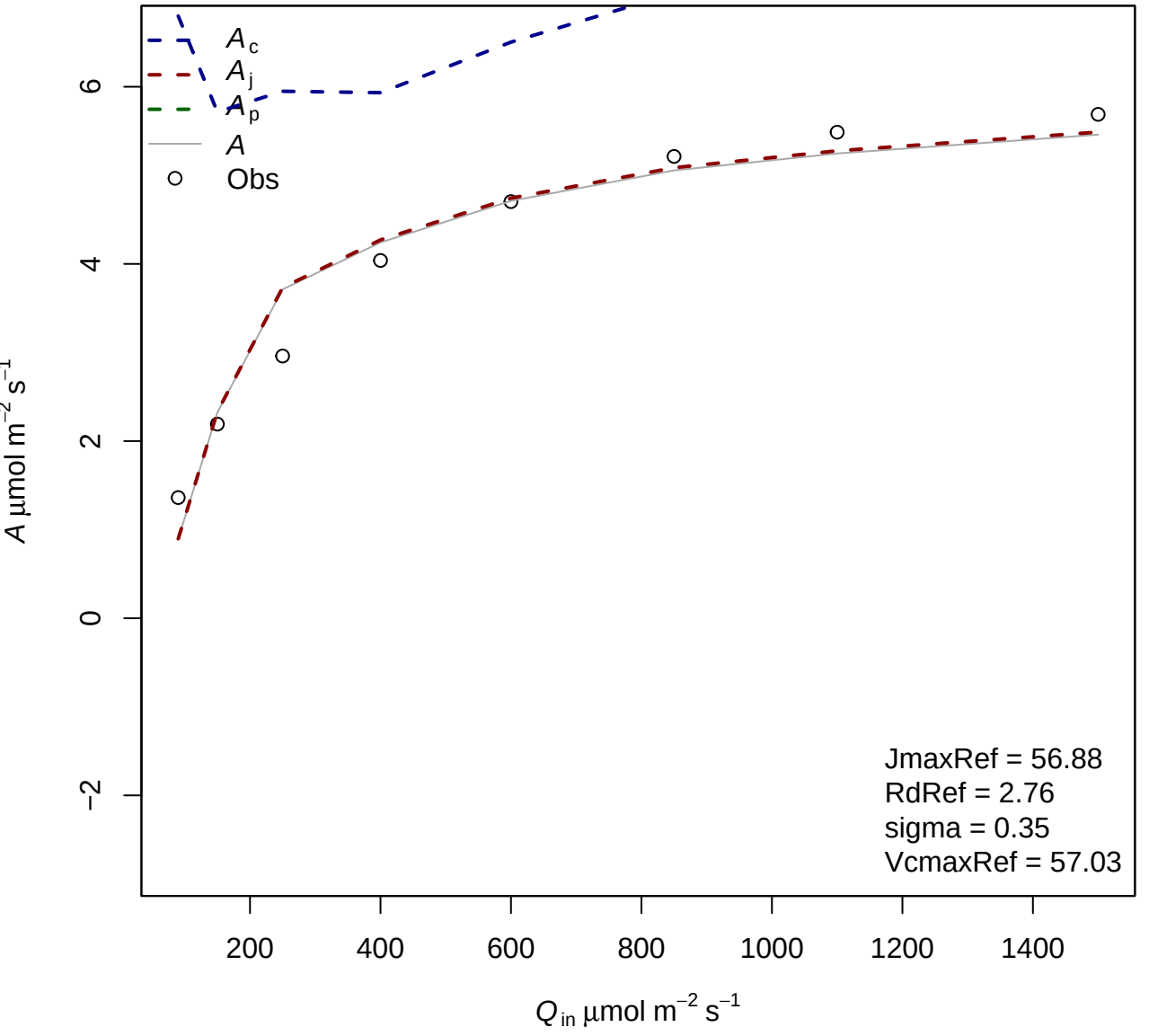

A_c
A_j
A_p
A
Obs
JmaxRef = 27.58
RdRef = 0.62
sigma = 0.14
VcmaxRef = 36.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

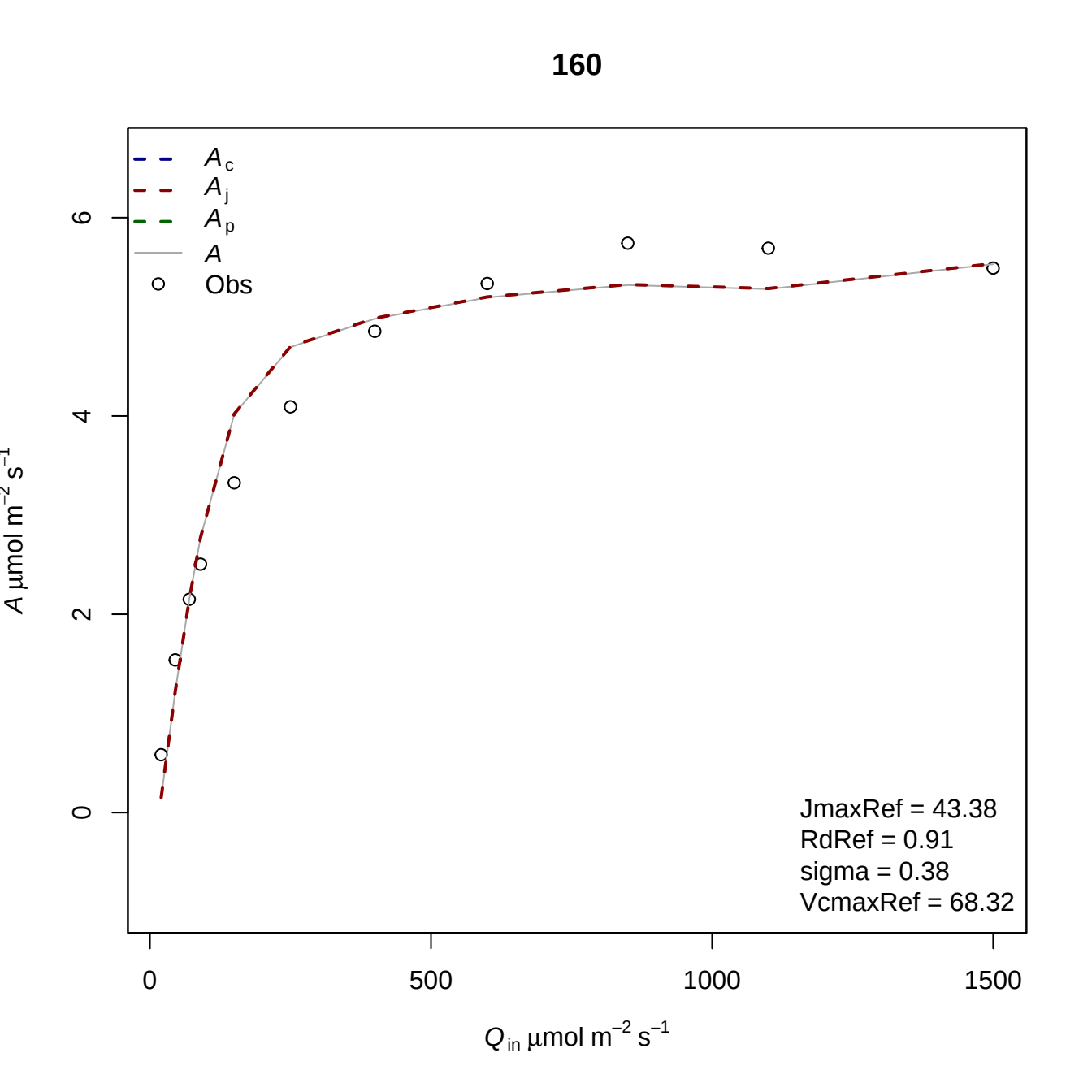

152
A_c
A_j
A_p
A
Obs
JmaxRef = 17.51
RdRef = 1.61
sigma = 1.17
VcmaxRef = 80034.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

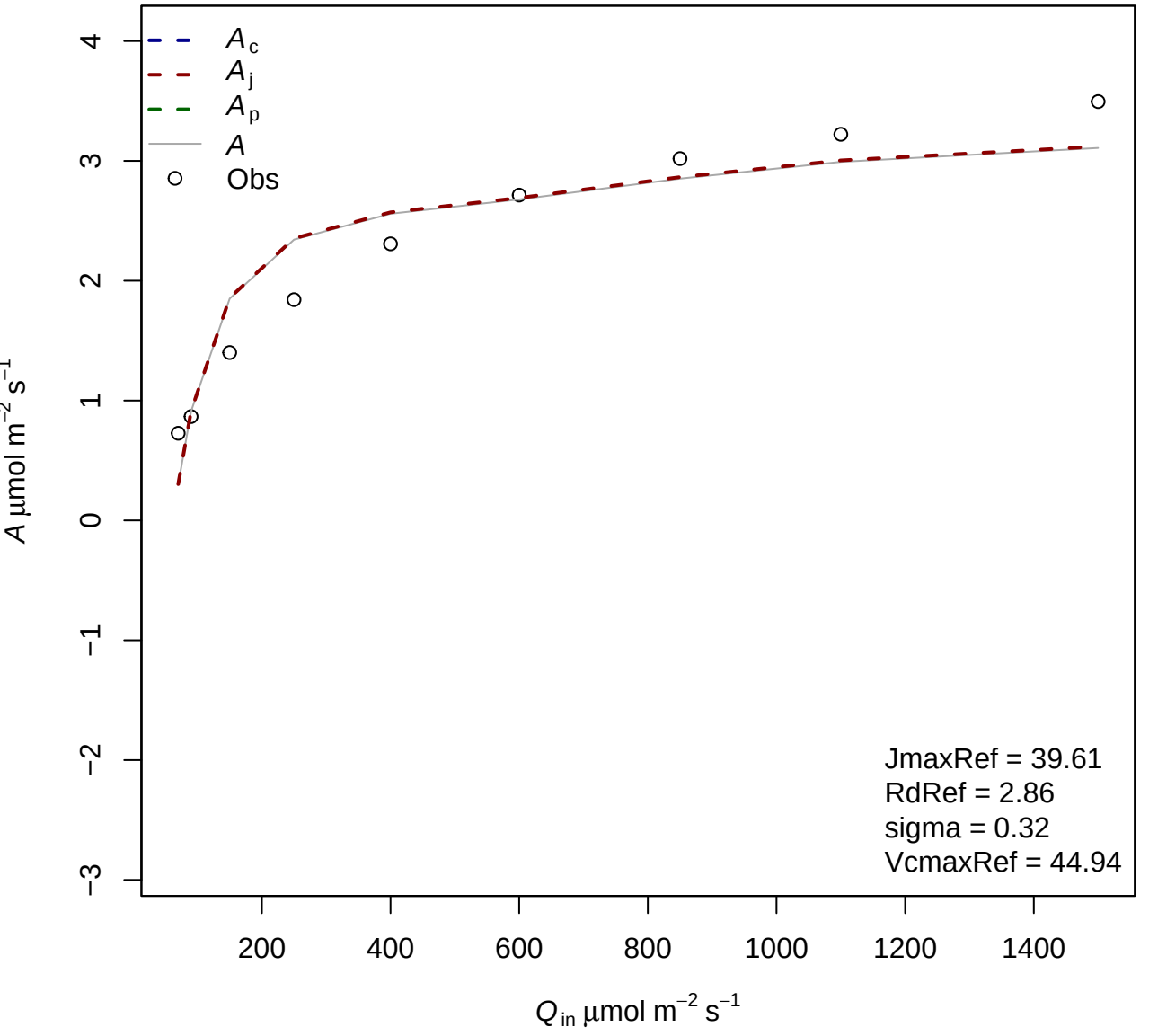

153
A_c
A_j
A_p
A
Obs
JmaxRef = 94.7
RdRef = 2.95
sigma = 1.29
VcmaxRef = 105.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

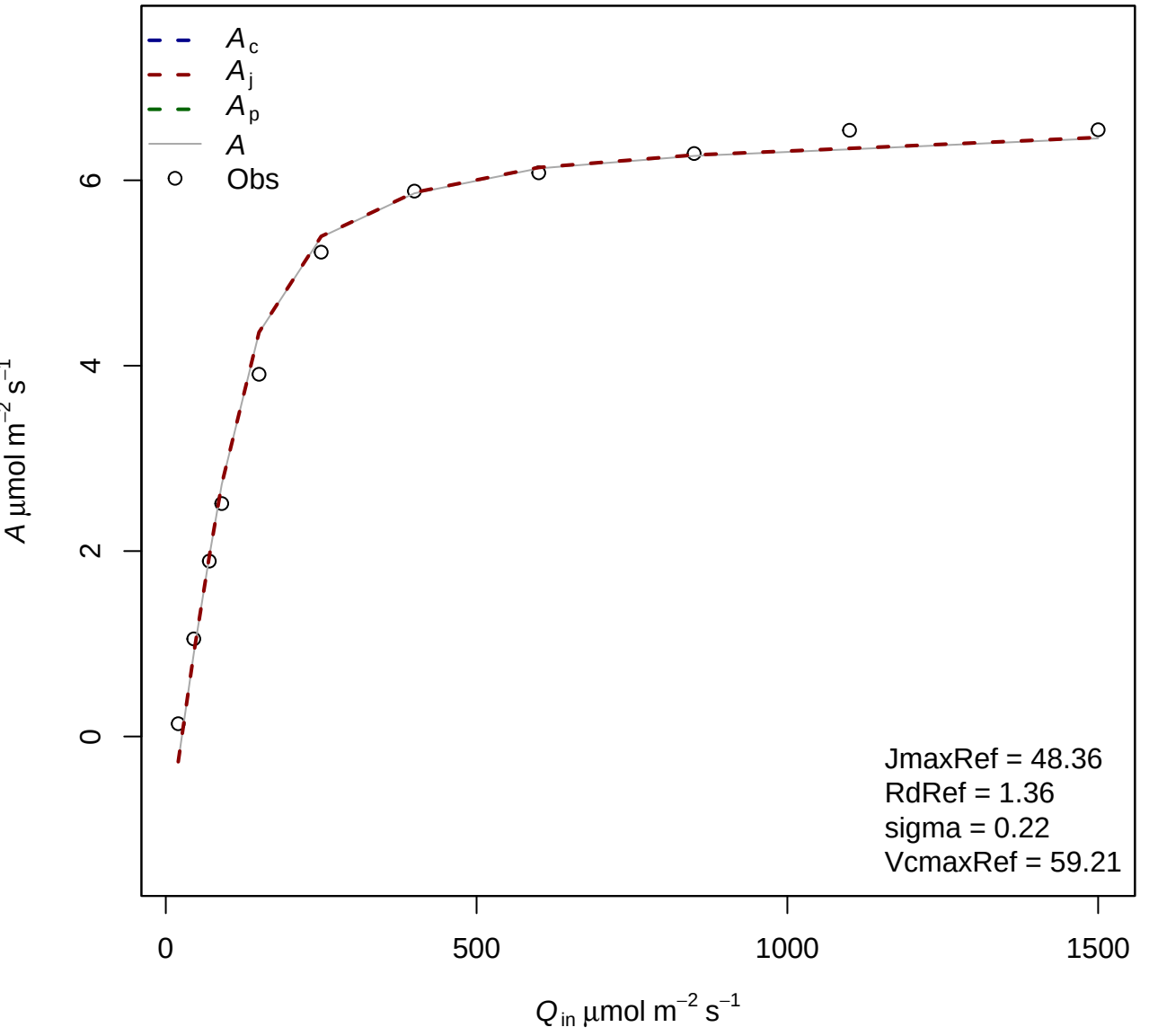

154
A_c
A_j
A_p
A
Obs
JmaxRef = 96.14
RdRef = 1.72
sigma = 0.55
VcmaxRef = 115.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

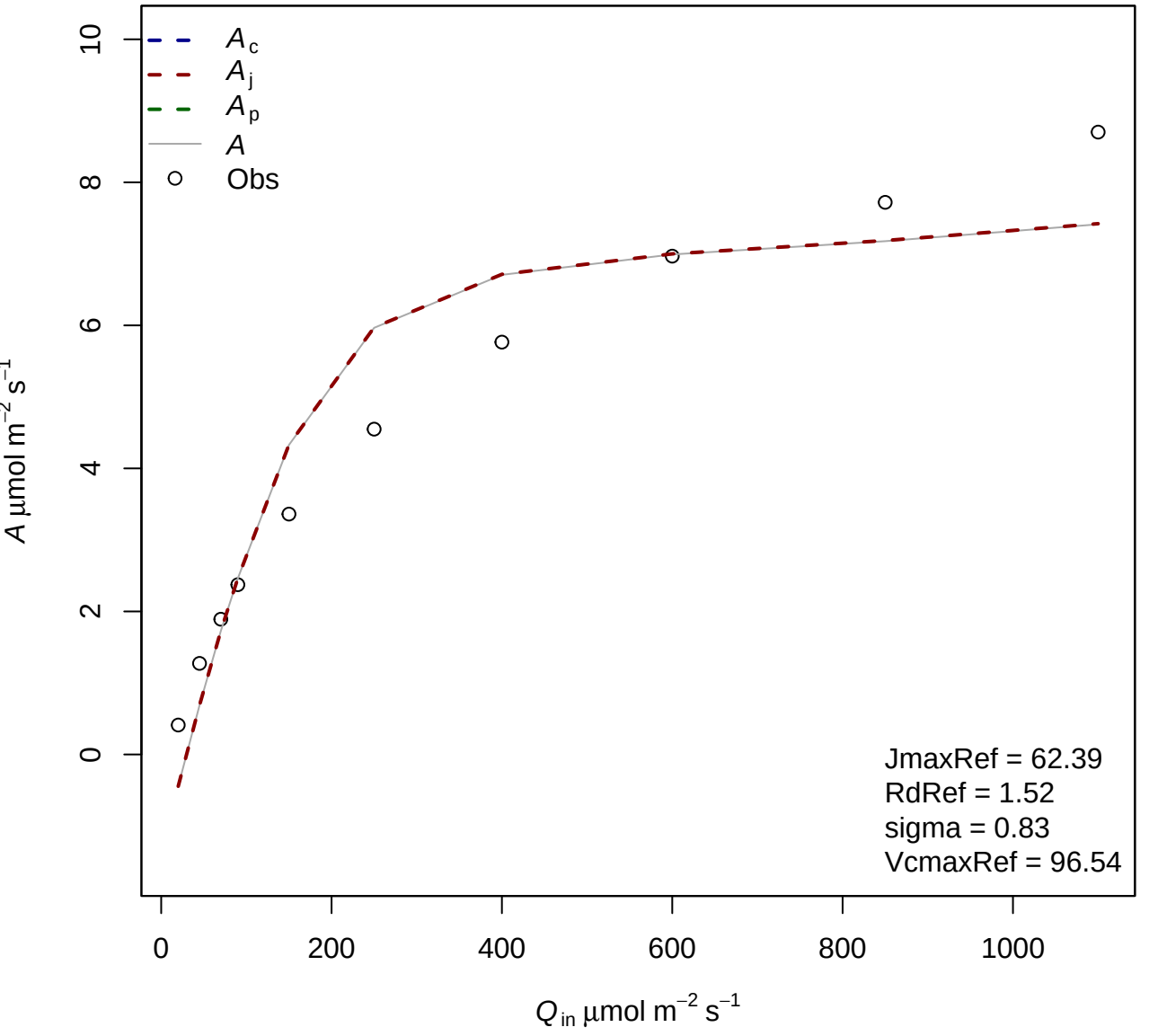

155
A_c
A_j
A_p
A
Obs
JmaxRef = 84.57
RdRef = 1.16
sigma = 0.36
VcmaxRef = 120.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

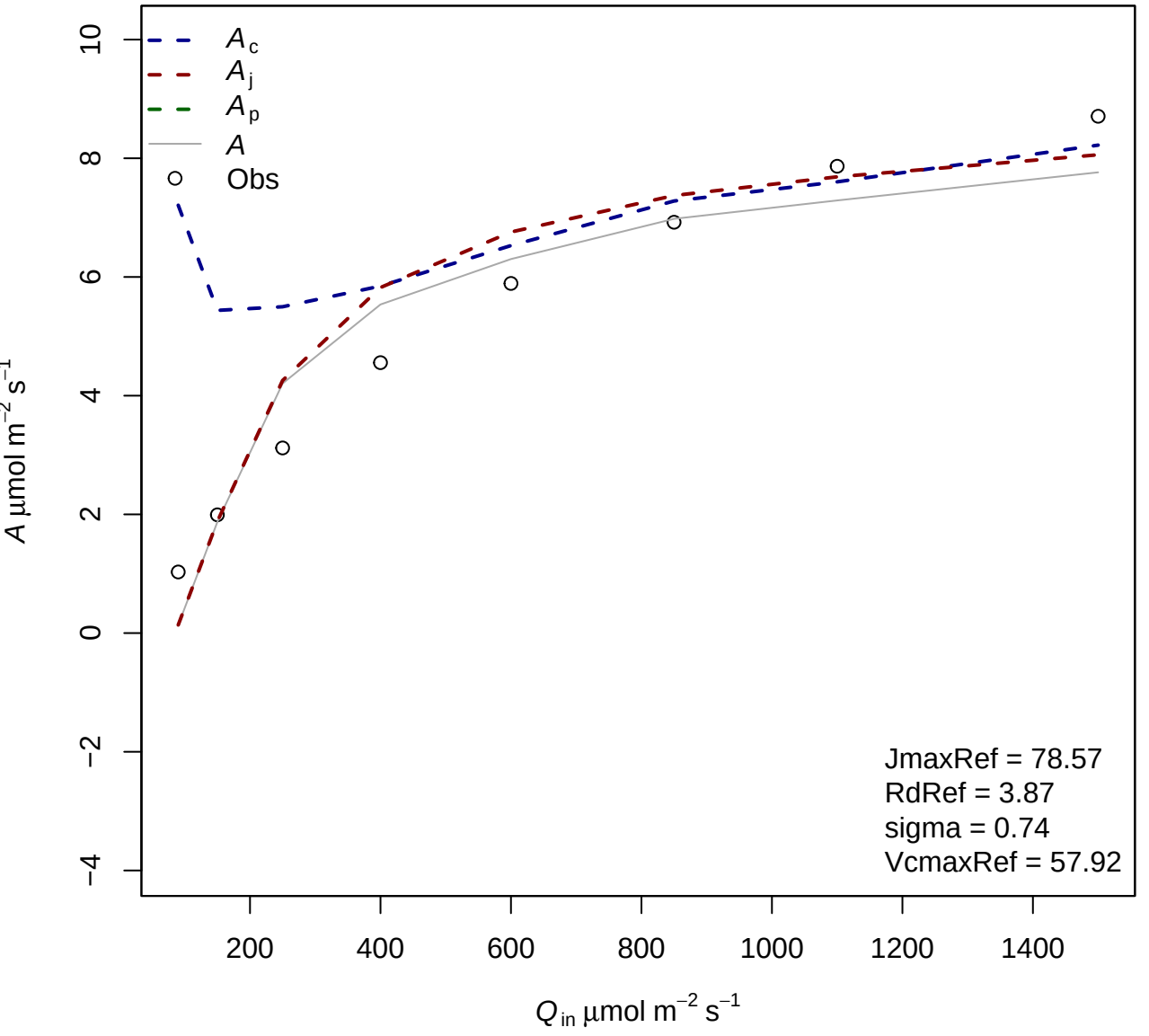

A_c
A_j
A_p
A
Obs
JmaxRef = 56.88
RdRef = 2.76
sigma = 0.35
VcmaxRef = 57.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000
1200
1400

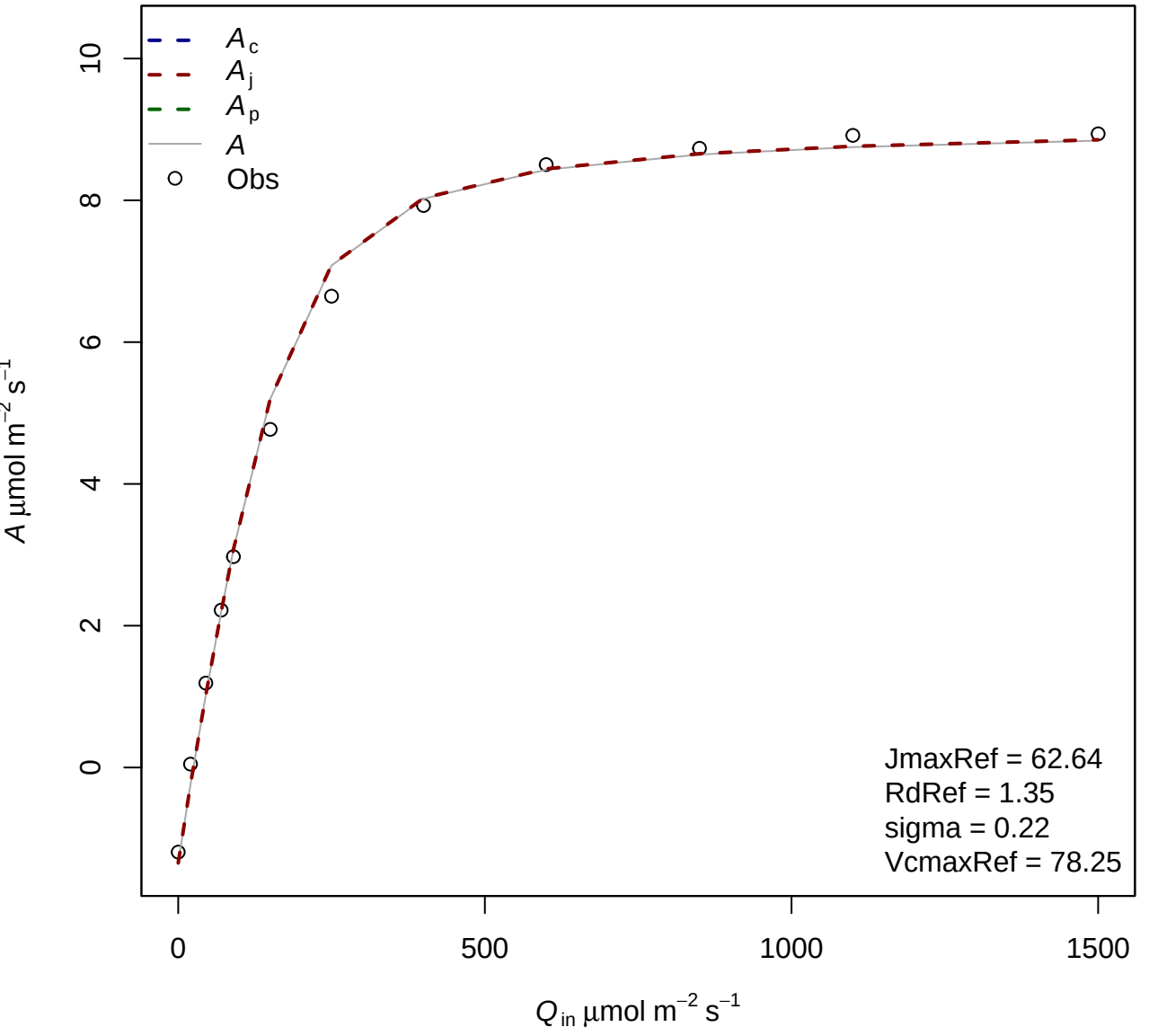

160
A_c
A_j
A_p
A
Obs
JmaxRef = 43.38
RdRef = 0.91
sigma = 0.38
VcmaxRef = 68.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

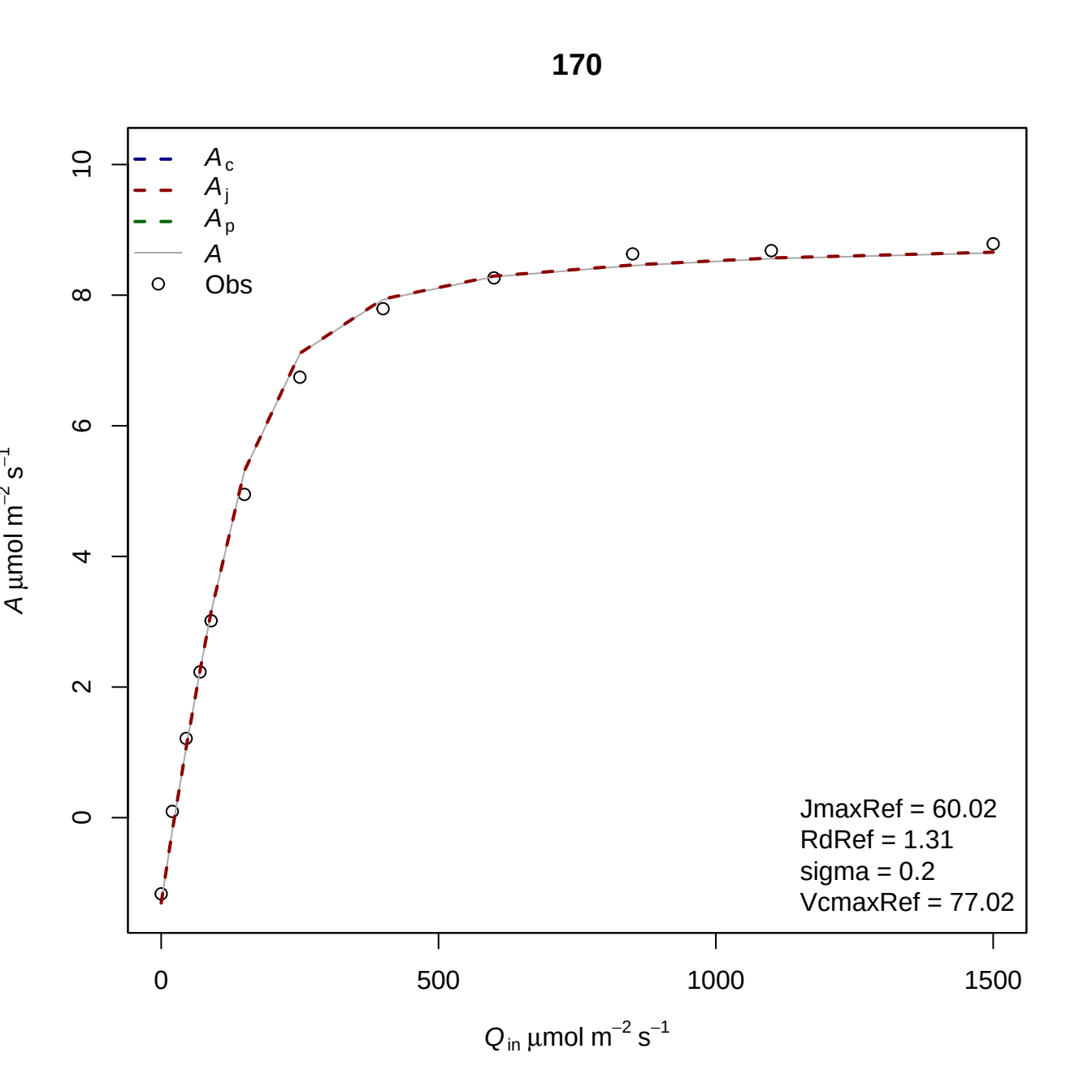

A_c
A_j
A_p
A
Obs
JmaxRef = 39.61
RdRef = 2.86
sigma = 0.32
VcmaxRef = 44.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
200
400
600
800
1000
1200
1400

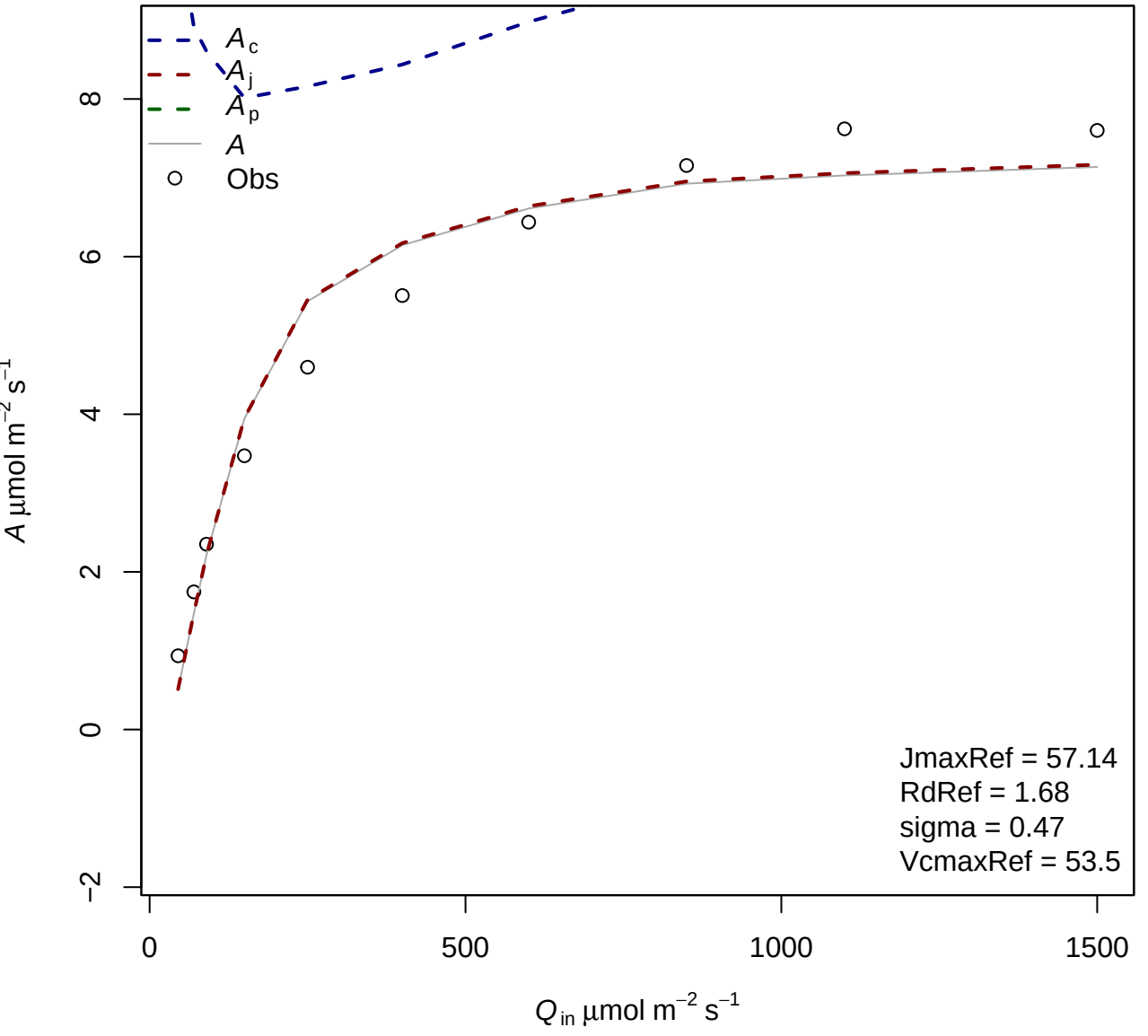

A_c
A_j
A_p
A
Obs
JmaxRef = 48.36
RdRef = 1.36
sigma = 0.22
VcmaxRef = 59.21
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

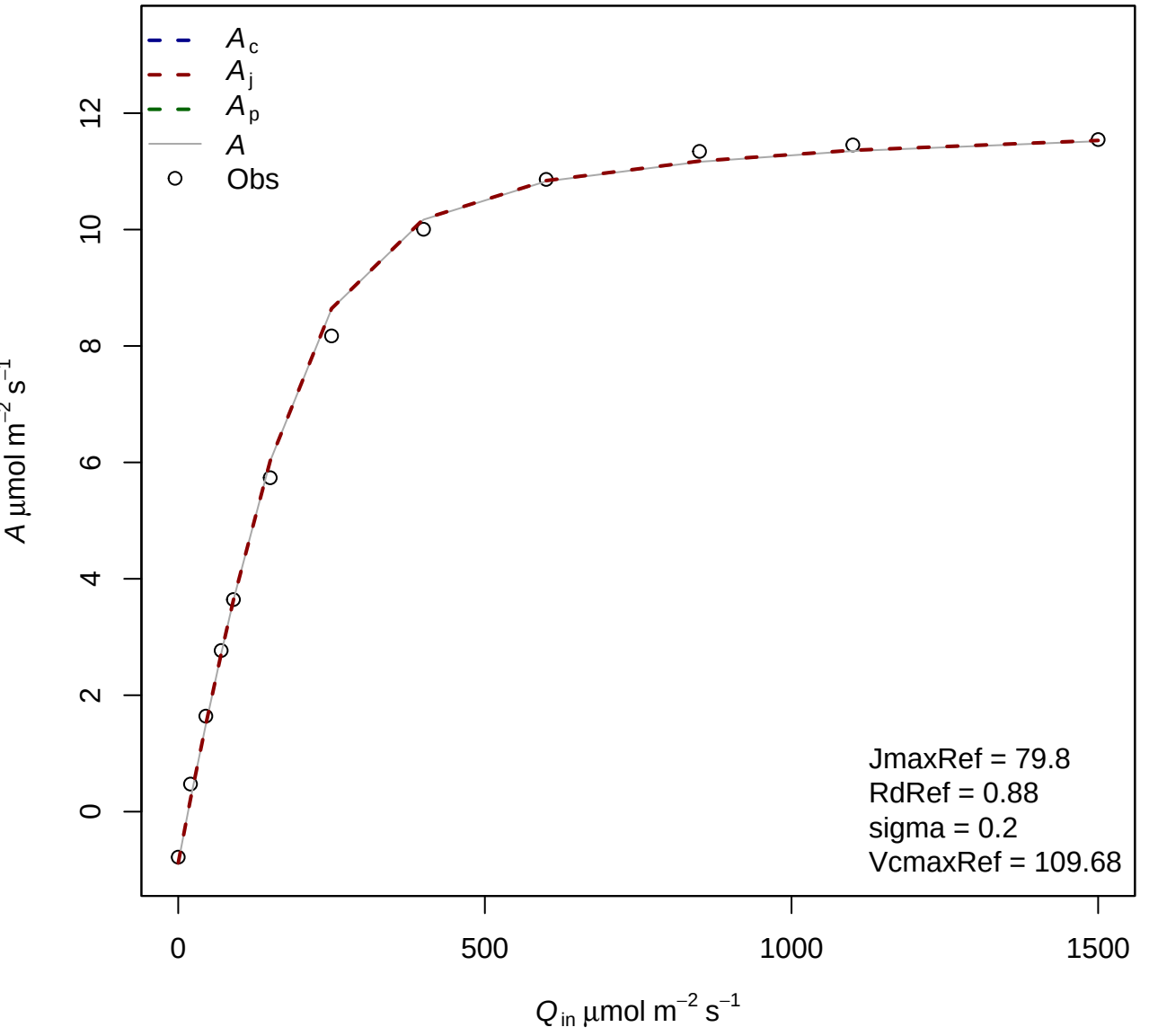

A_c
A_j
A_p
A
Obs
JmaxRef = 62.39
RdRef = 1.52
sigma = 0.83
VcmaxRef = 96.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000

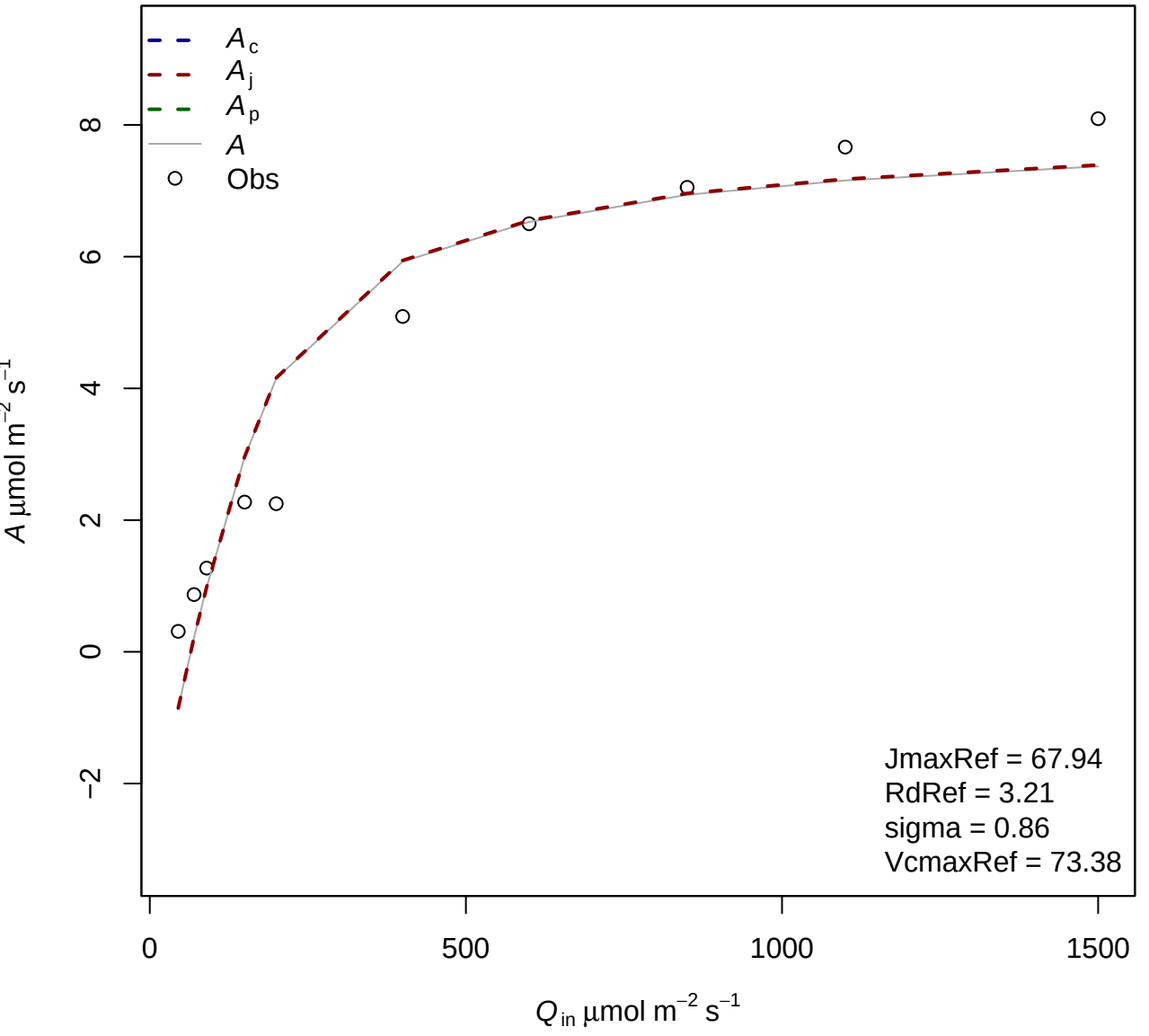

A_c
A_j
A_p
A
Obs
JmaxRef = 78.57
RdRef = 3.87
sigma = 0.74
VcmaxRef = 57.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

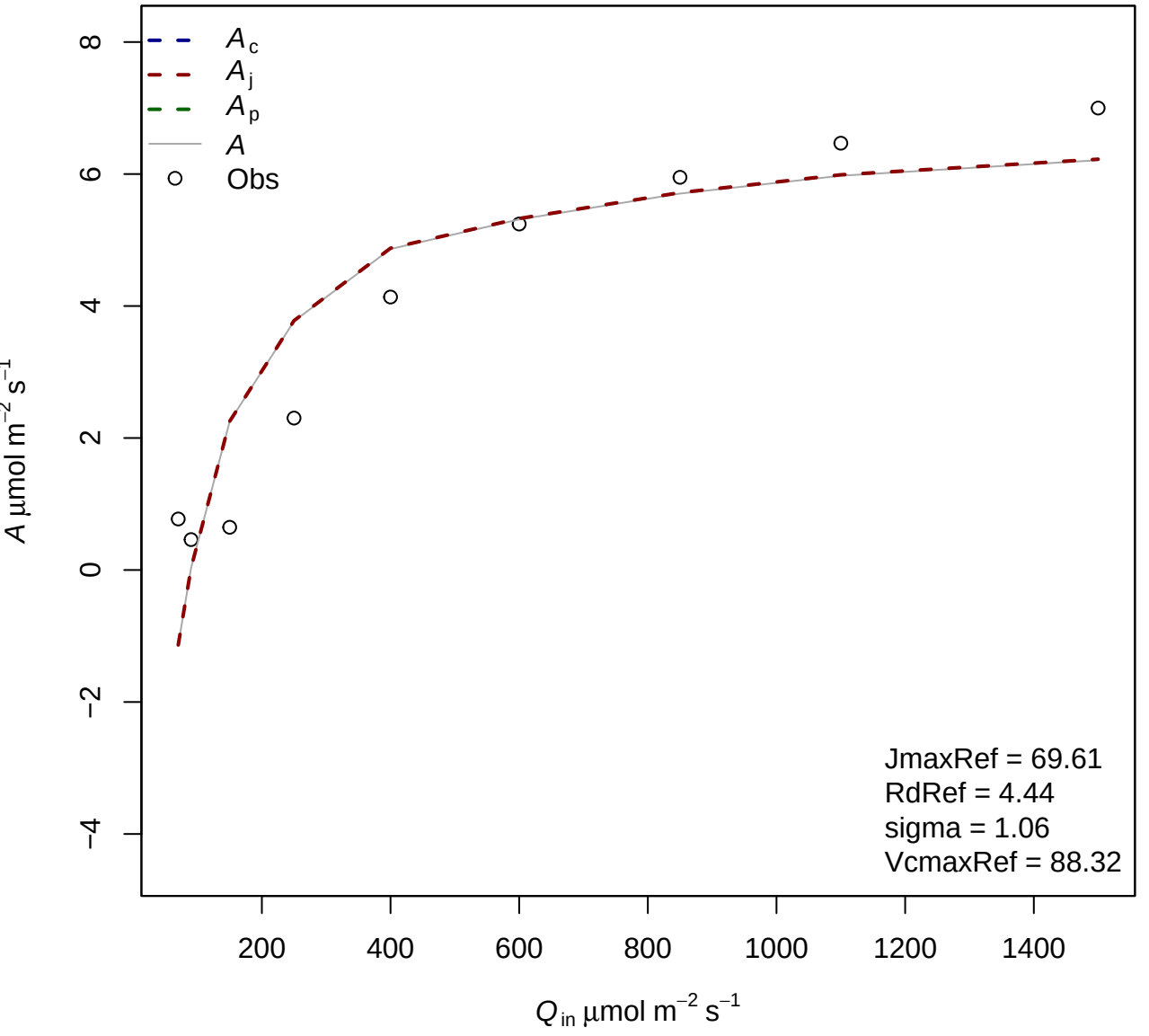

A_c
A_j
A_p
A
Obs
JmaxRef = 62.64
RdRef = 1.35
sigma = 0.22
VcmaxRef = 78.25
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

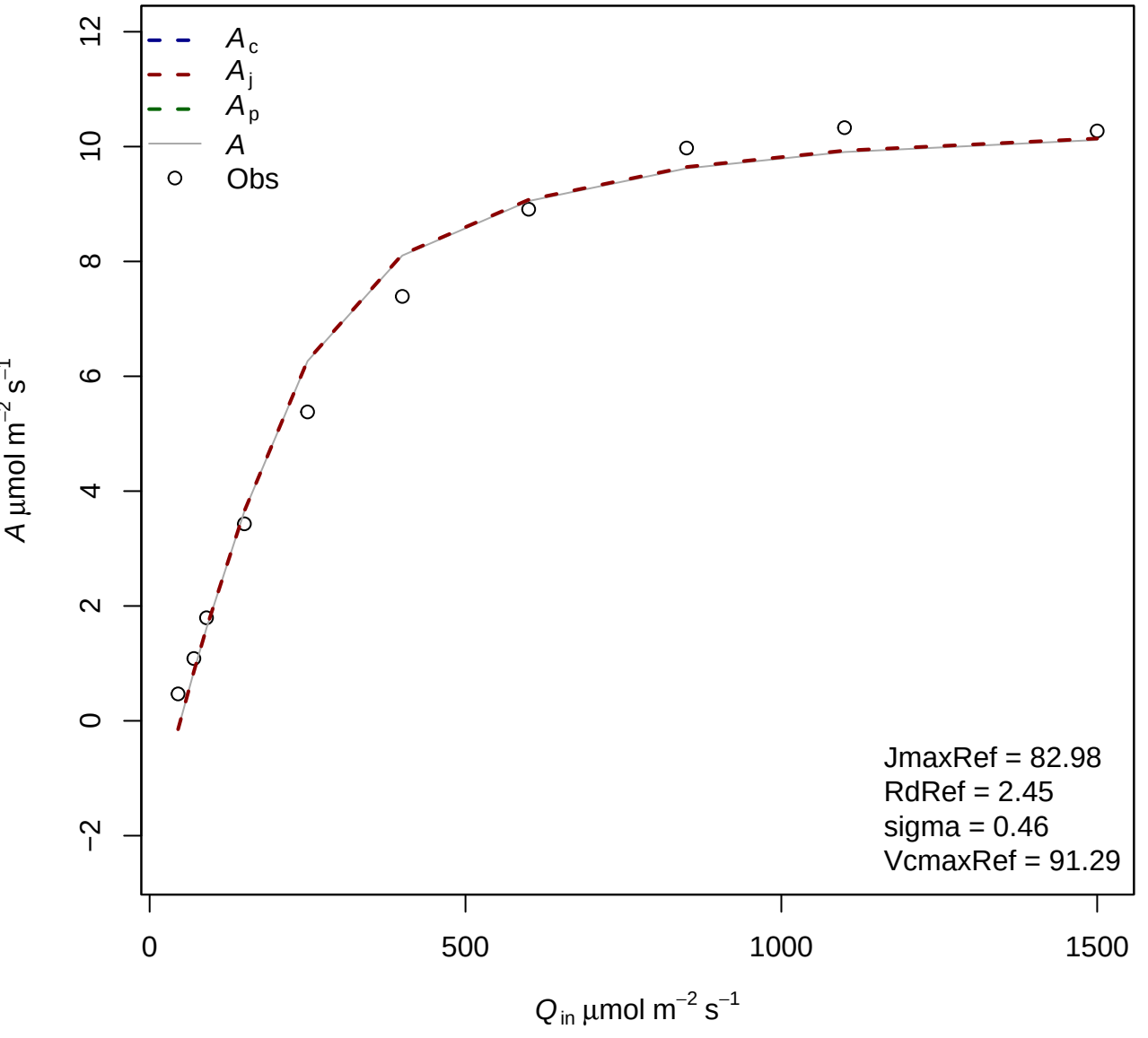

170
A_c
A_j
A_p
A
Obs
JmaxRef = 60.02
RdRef = 1.31
sigma = 0.2
VcmaxRef = 77.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

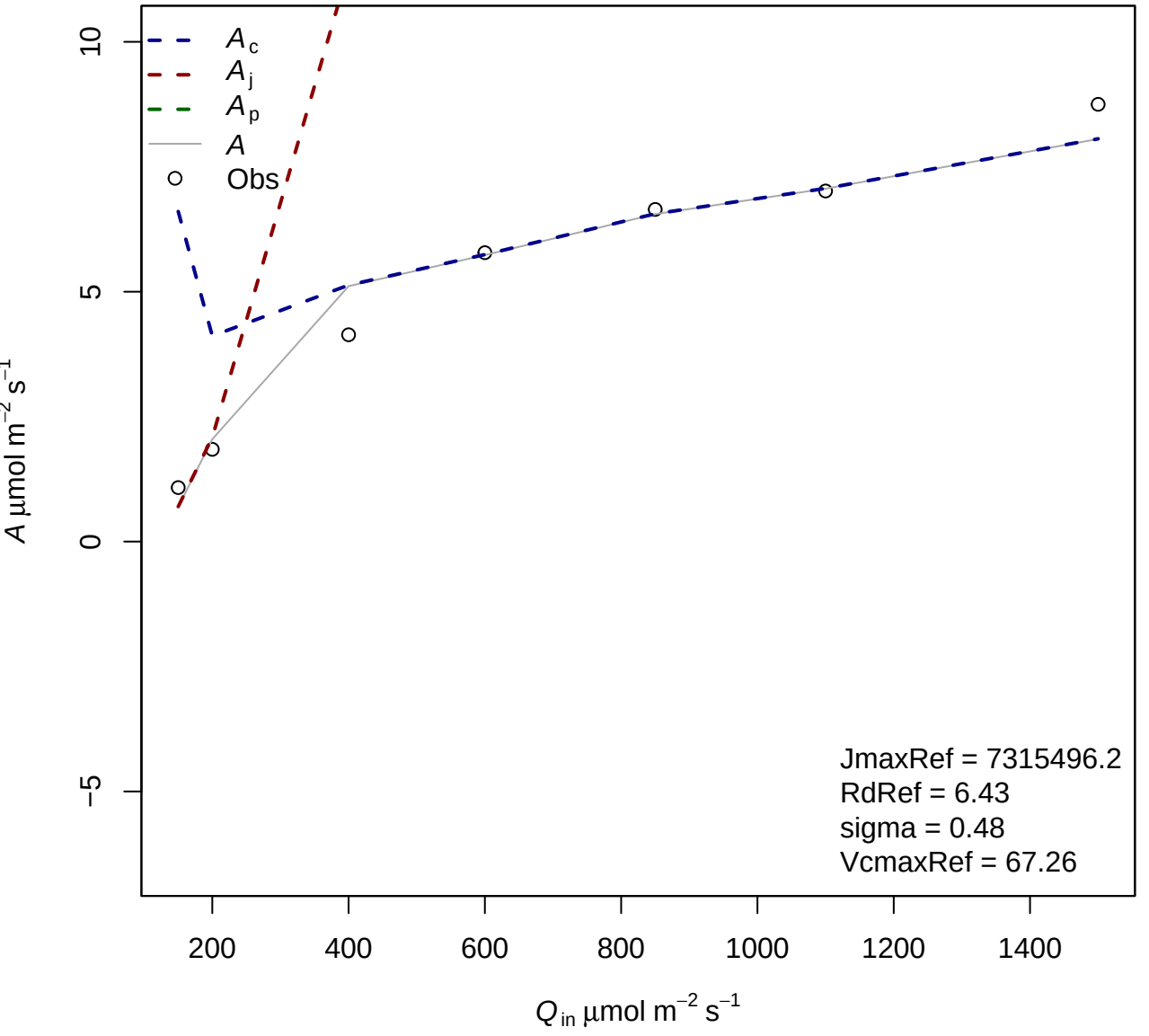

A_c
A_j
A_p
A
Obs
JmaxRef = 57.14
RdRef = 1.68
sigma = 0.47
VcmaxRef = 53.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

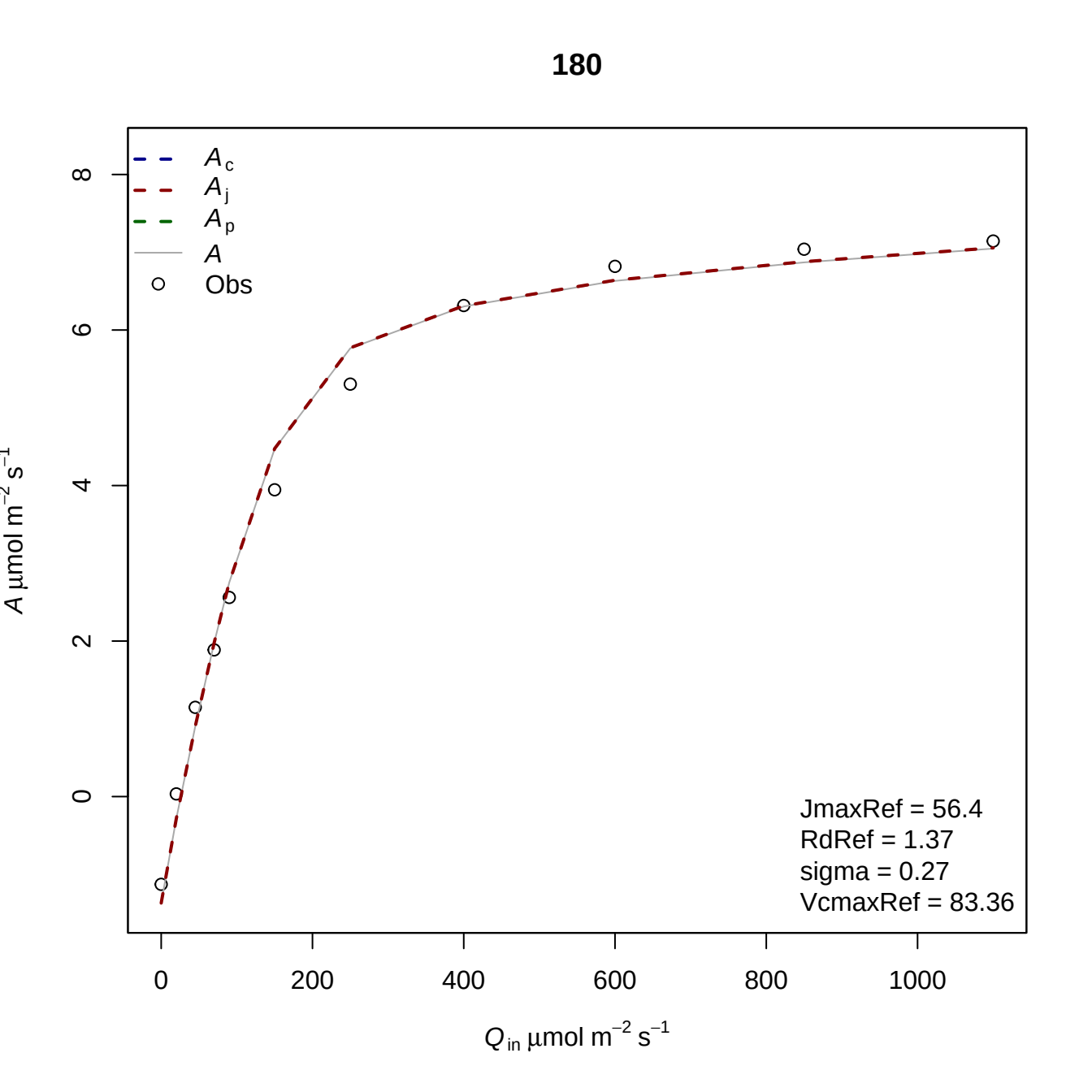

A_c
A_j
A_p
A
Obs
JmaxRef = 79.8
RdRef = 0.88
sigma = 0.2
VcmaxRef = 109.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

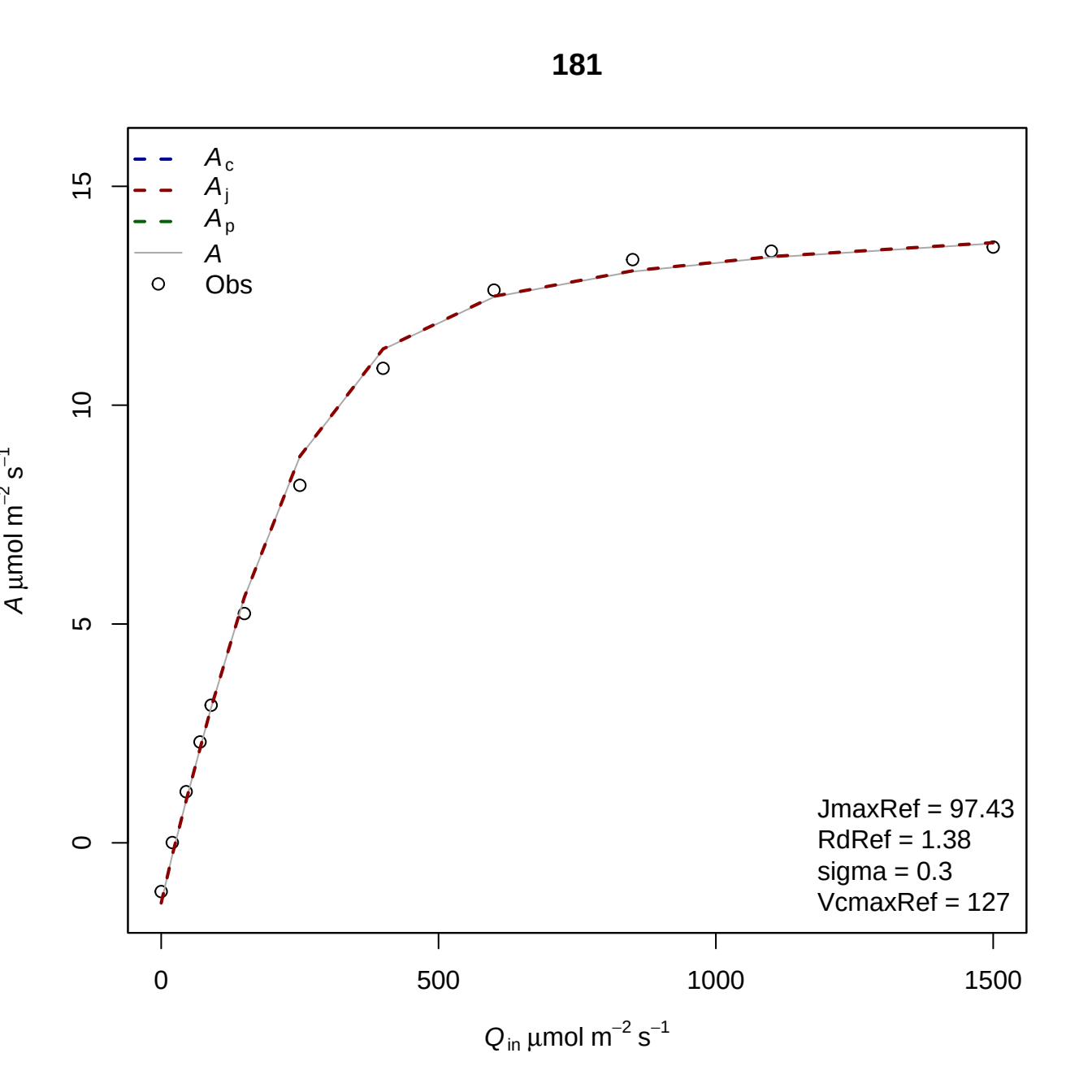

A_c
A_j
A_p
A
Obs
JmaxRef = 67.94
RdRef = 3.21
sigma = 0.86
VcmaxRef = 73.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

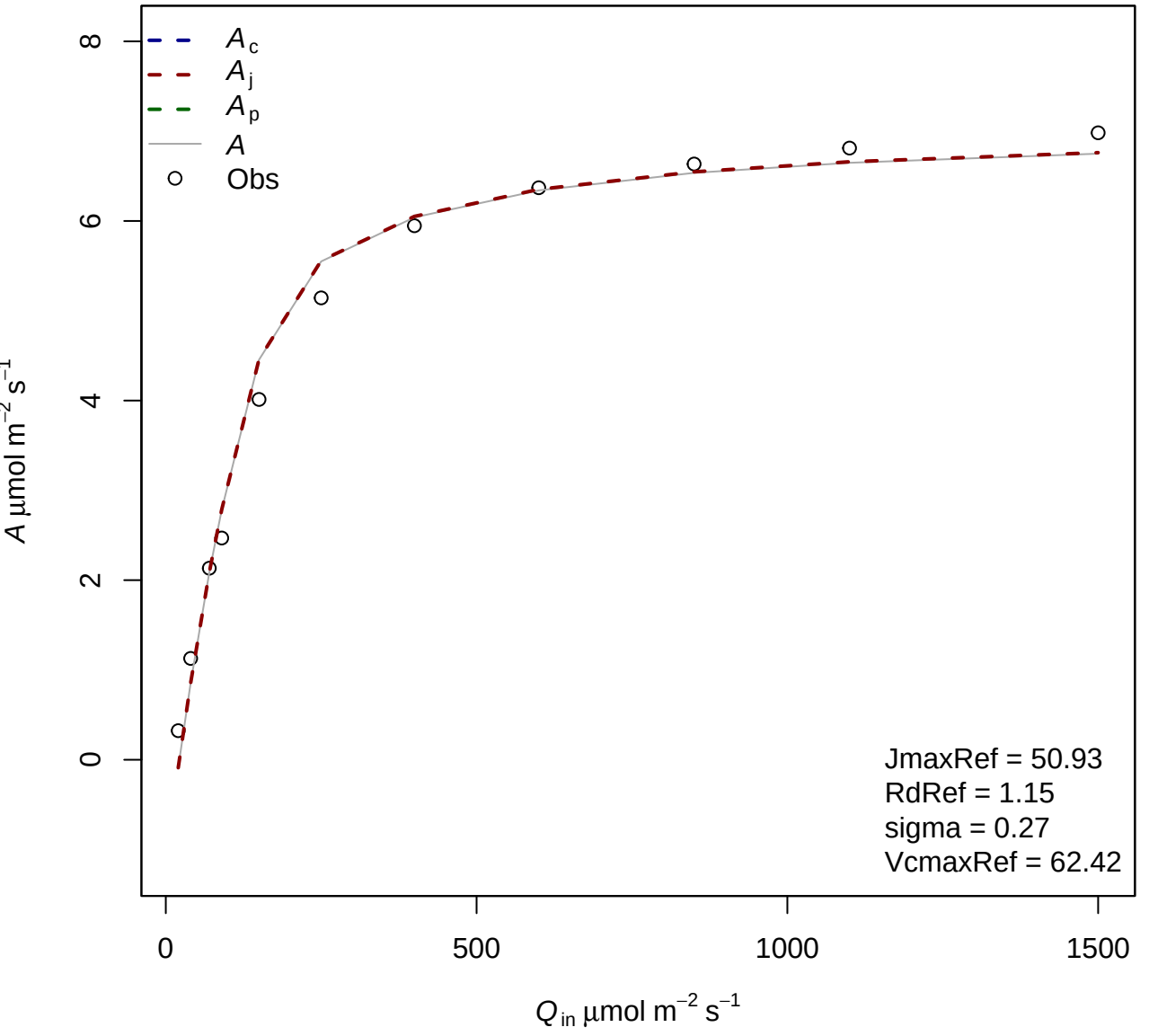

A_c
A_j
A_p
A
Obs
JmaxRef = 69.61
RdRef = 4.44
sigma = 1.06
VcmaxRef = 88.32
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

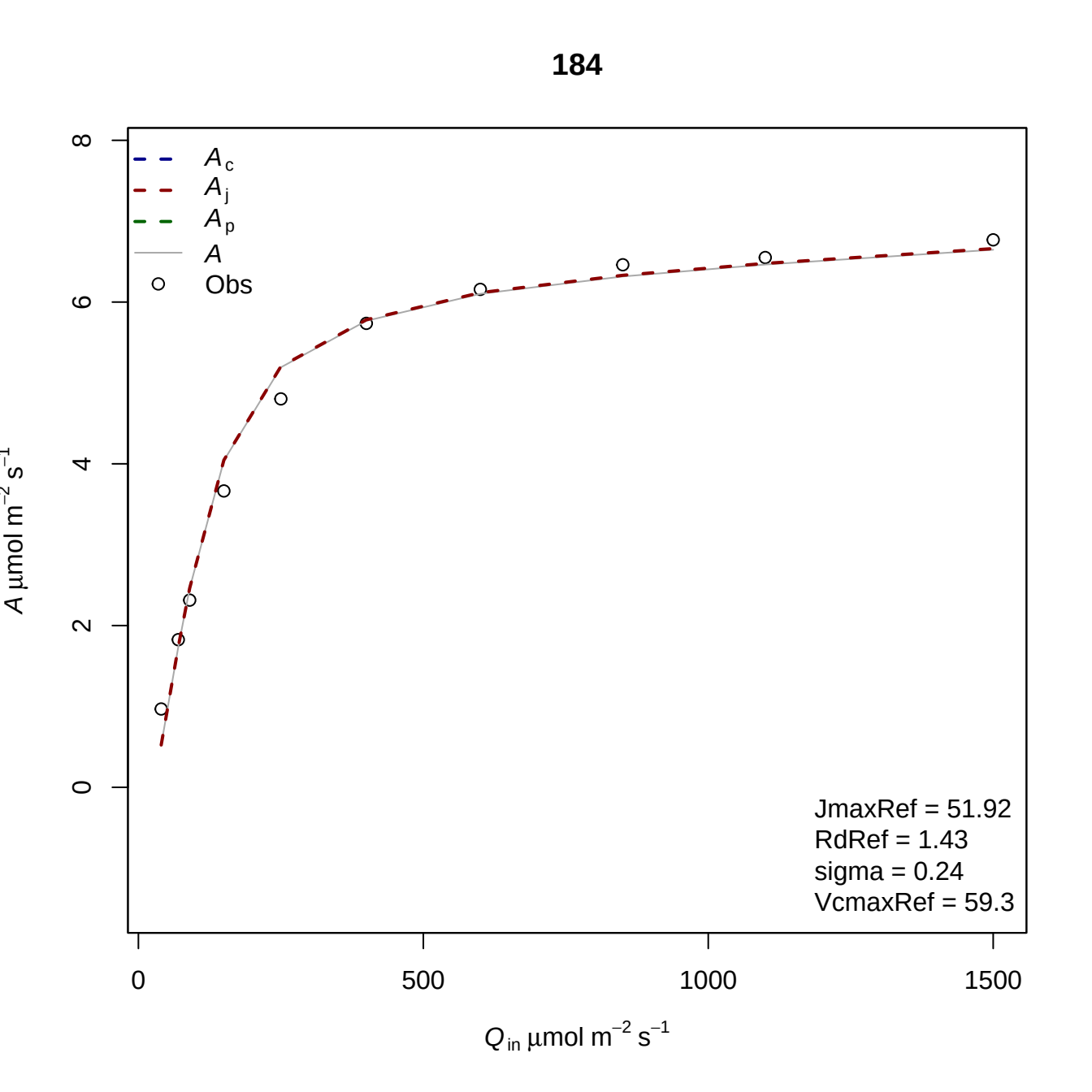

A_c
A_j
A_p
A
Obs
JmaxRef = 82.98
RdRef = 2.45
sigma = 0.46
VcmaxRef = 91.29
12
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

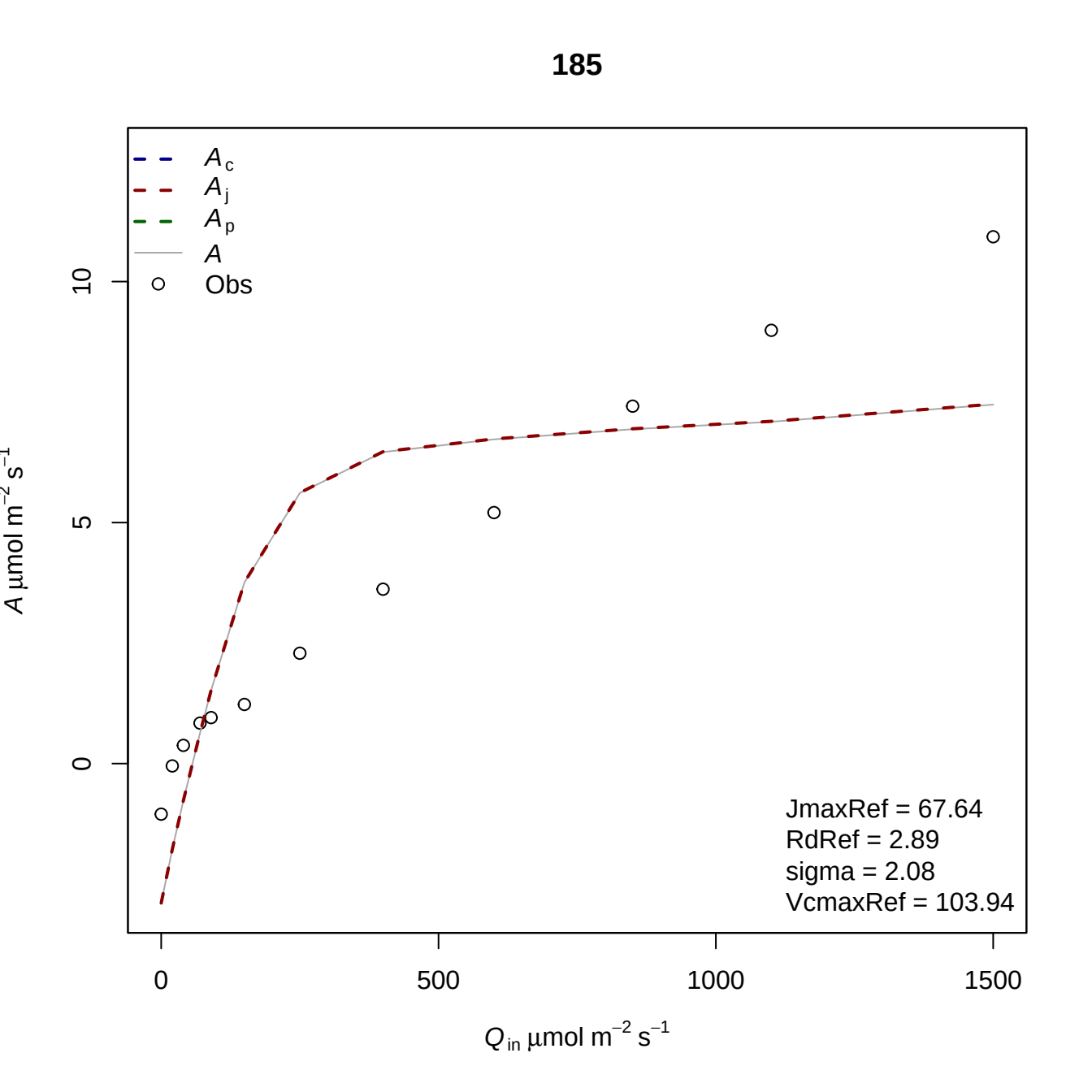

A_c
A_j
A_p
A
Obs
JmaxRef = 7315496.2
RdRef = 6.43
sigma = 0.48
VcmaxRef = 67.26
10
5
0
−5
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

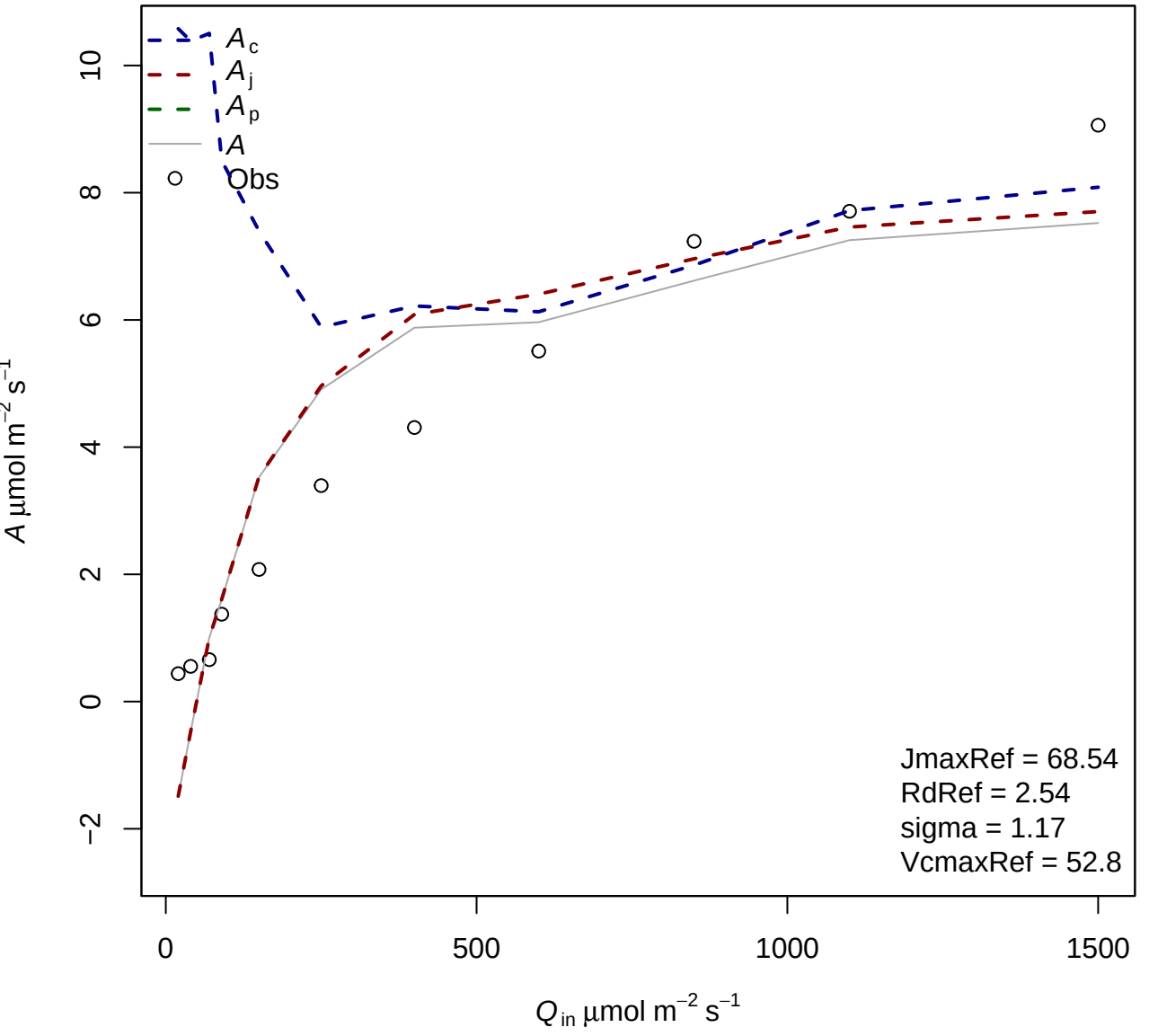

180
A_c
A_j
A_p
A
Obs
JmaxRef = 56.4
RdRef = 1.37
sigma = 0.27
VcmaxRef = 83.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000

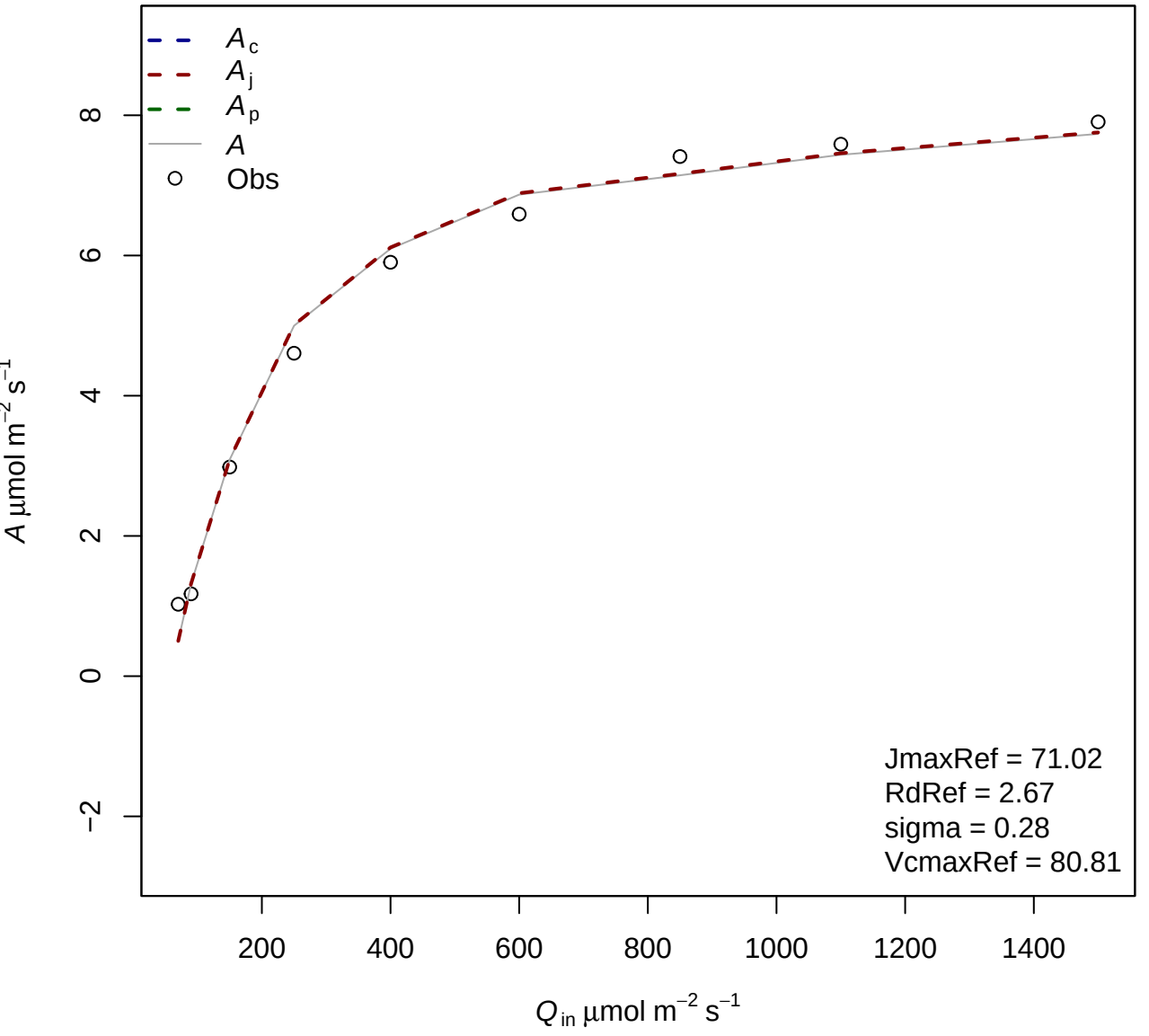

181
A_c
A_j
A_p
A
Obs
JmaxRef = 97.43
RdRef = 1.38
sigma = 0.3
VcmaxRef = 127
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

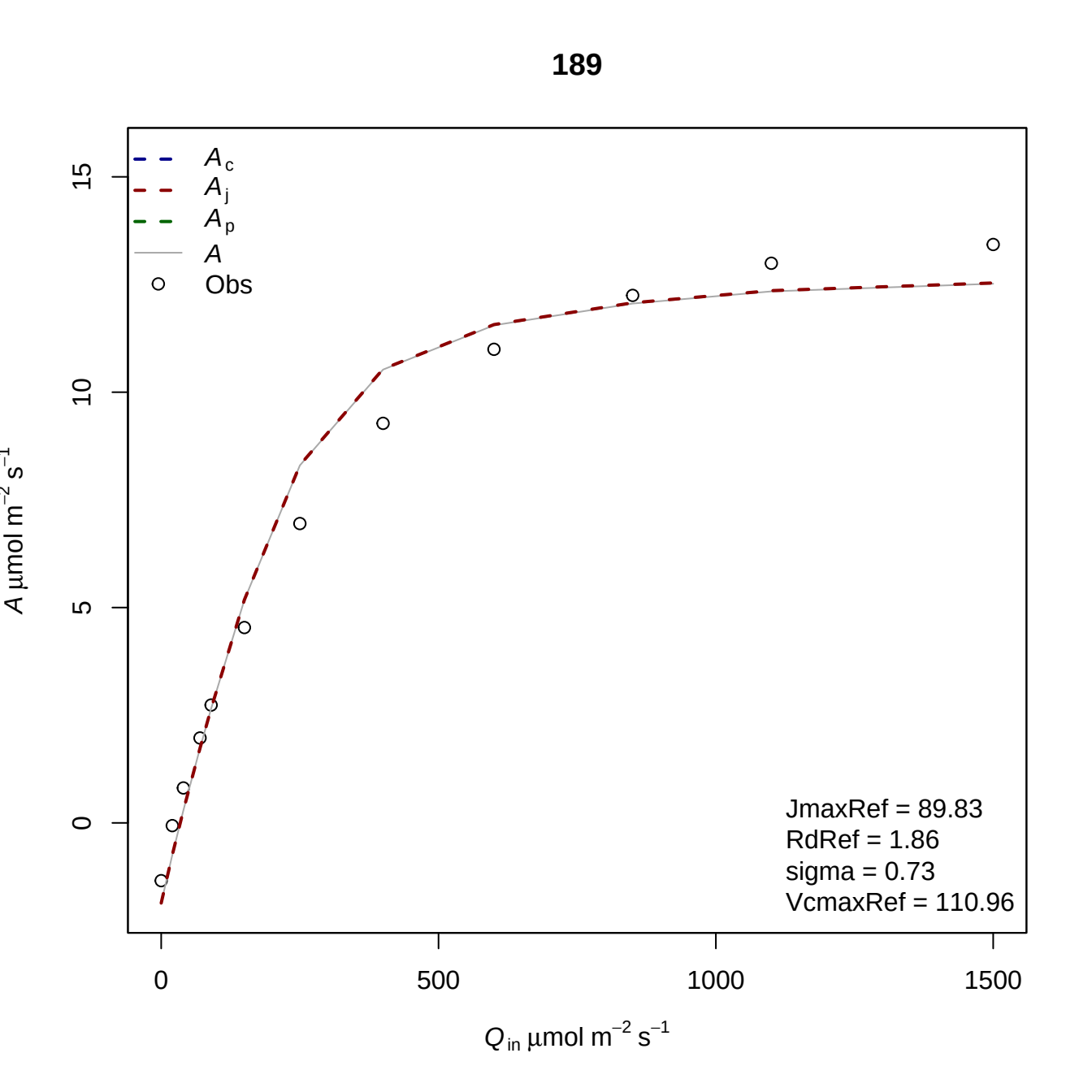

A_c
A_j
A_p
A
Obs
JmaxRef = 50.93
RdRef = 1.15
sigma = 0.27
VcmaxRef = 62.42
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

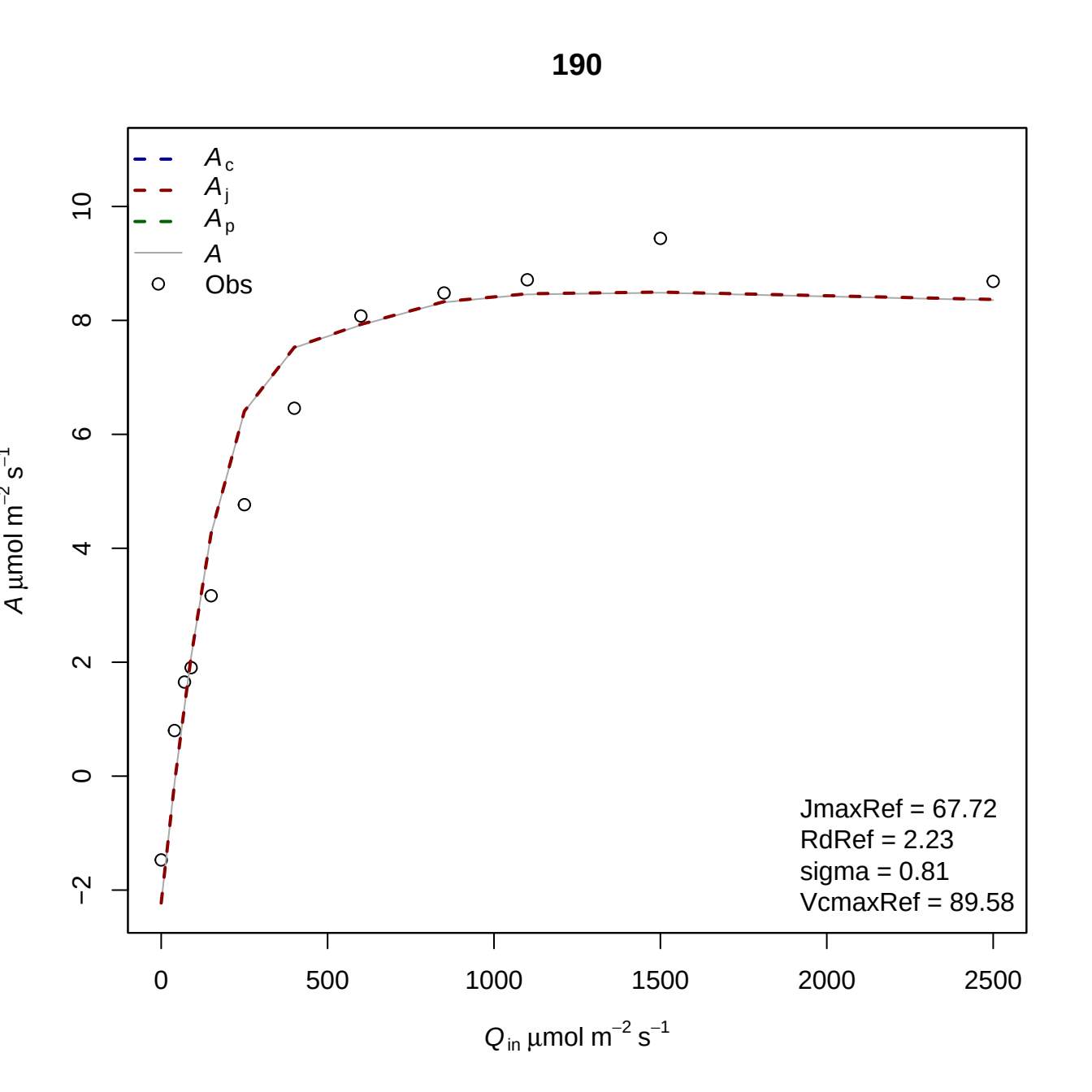

184
A_c
A_j
A_p
A
Obs
JmaxRef = 51.92
RdRef = 1.43
sigma = 0.24
VcmaxRef = 59.3
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

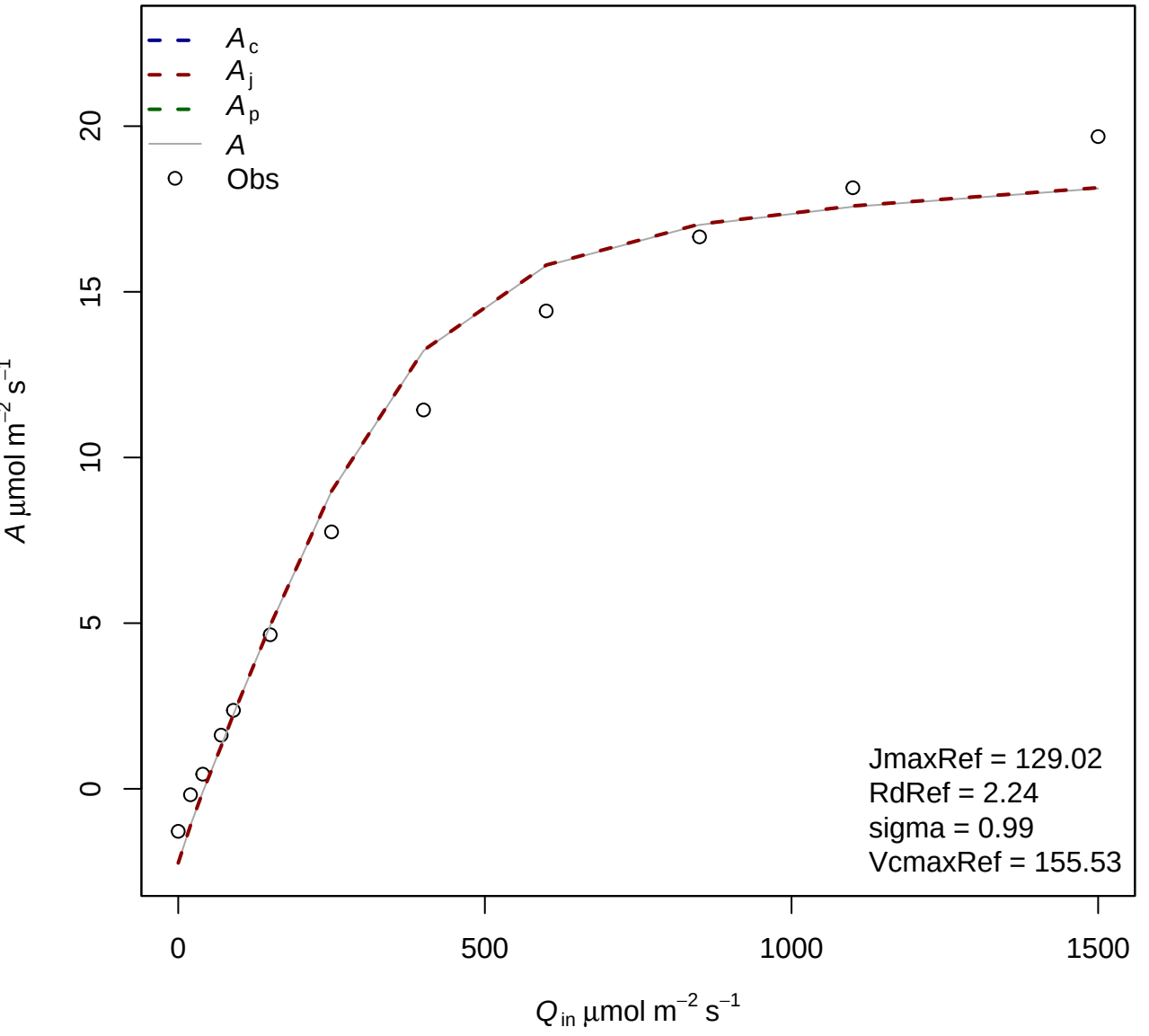

185
A_c
A_j
A_p
A
Obs
JmaxRef = 67.64
RdRef = 2.89
sigma = 2.08
VcmaxRef = 103.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

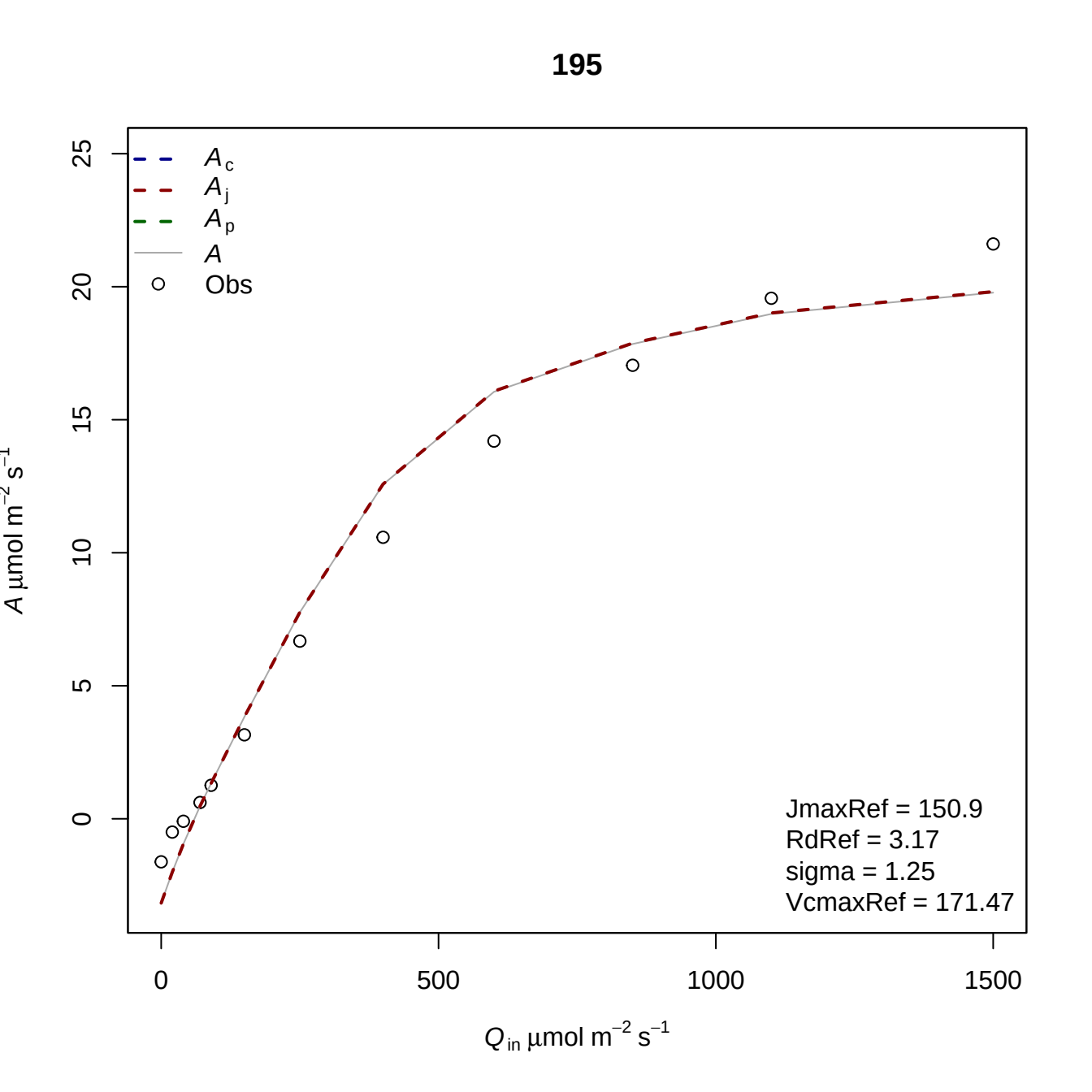

A_c
A_j
A_p
A
Obs
JmaxRef = 68.54
RdRef = 2.54
sigma = 1.17
VcmaxRef = 52.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

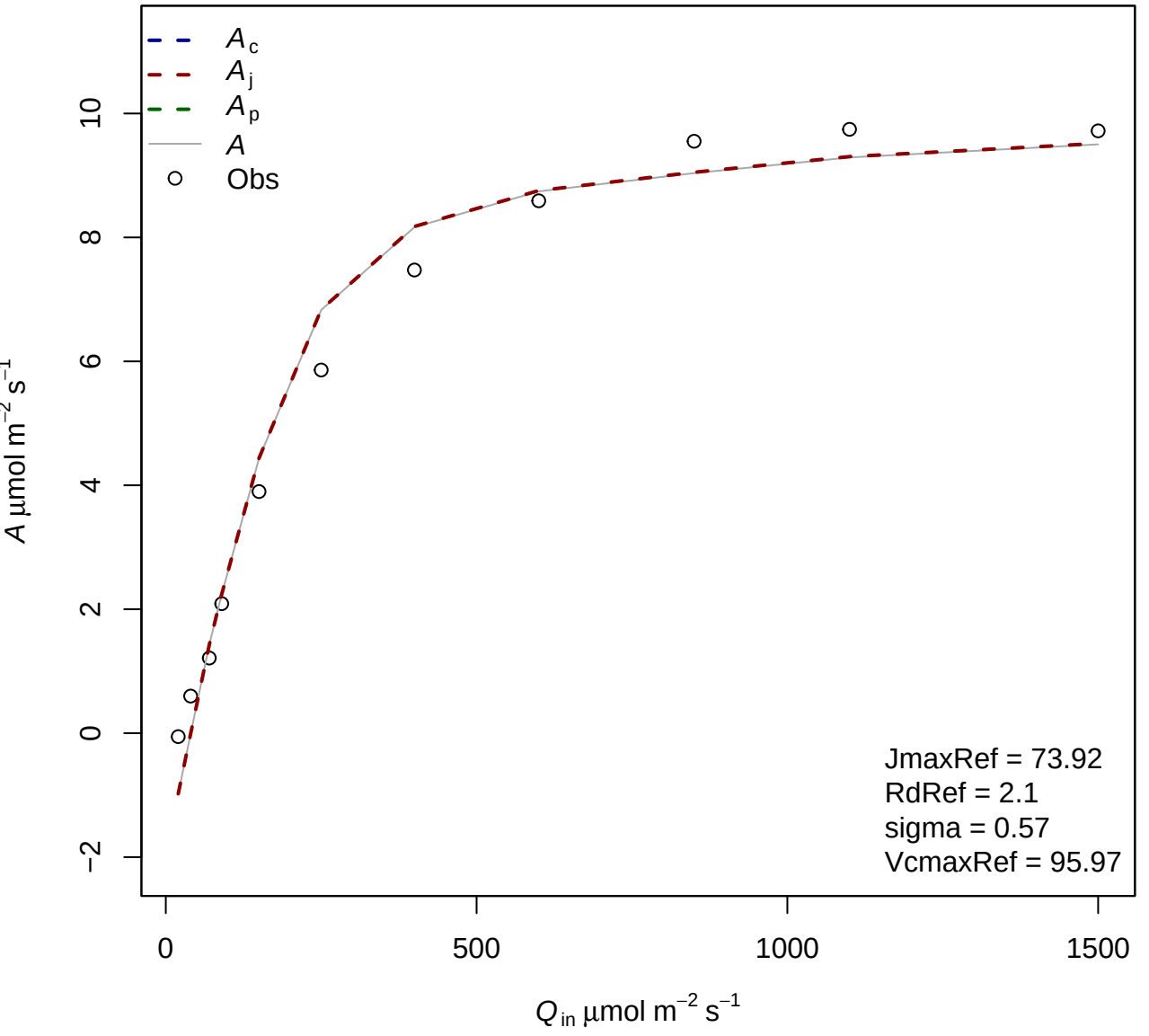

A_c
A_j
A_p
A
Obs
JmaxRef = 71.02
RdRef = 2.67
sigma = 0.28
VcmaxRef = 80.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

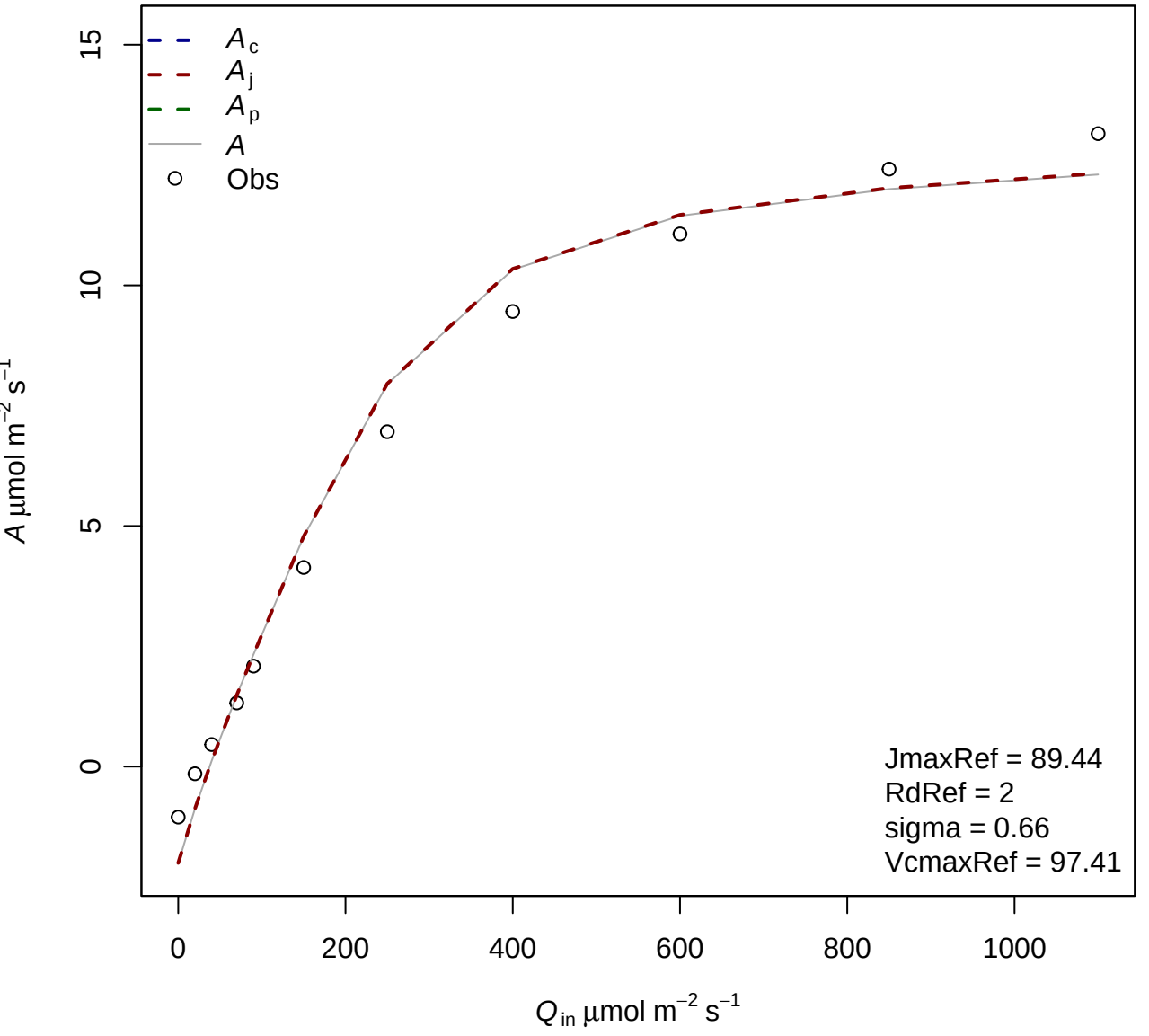

189
A_c
A_j
A_p
A
Obs
JmaxRef = 89.83
RdRef = 1.86
sigma = 0.73
VcmaxRef = 110.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

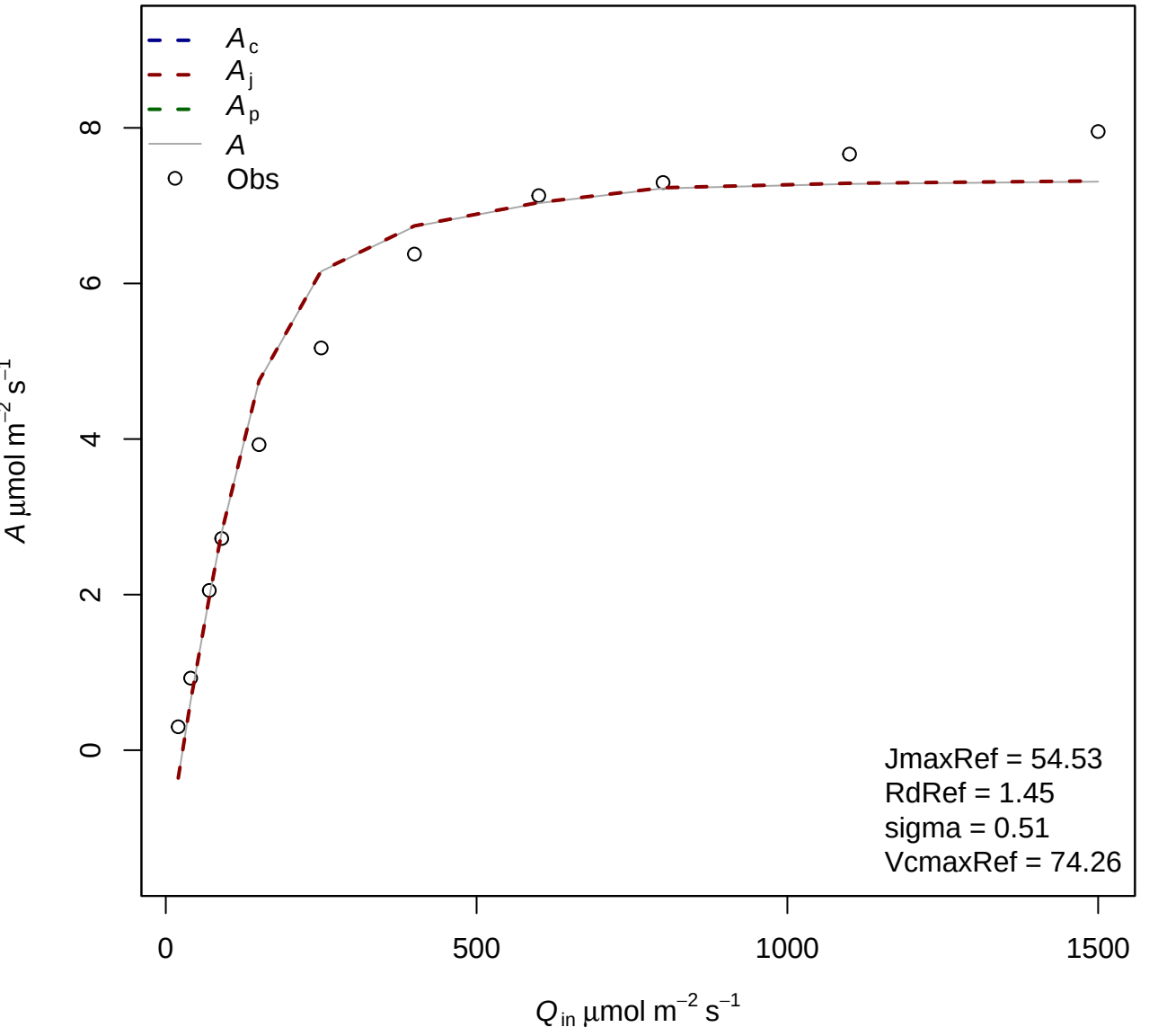

190
A_c
A_j
A_p
A
Obs
JmaxRef = 67.72
RdRef = 2.23
sigma = 0.81
VcmaxRef = 89.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000
2500

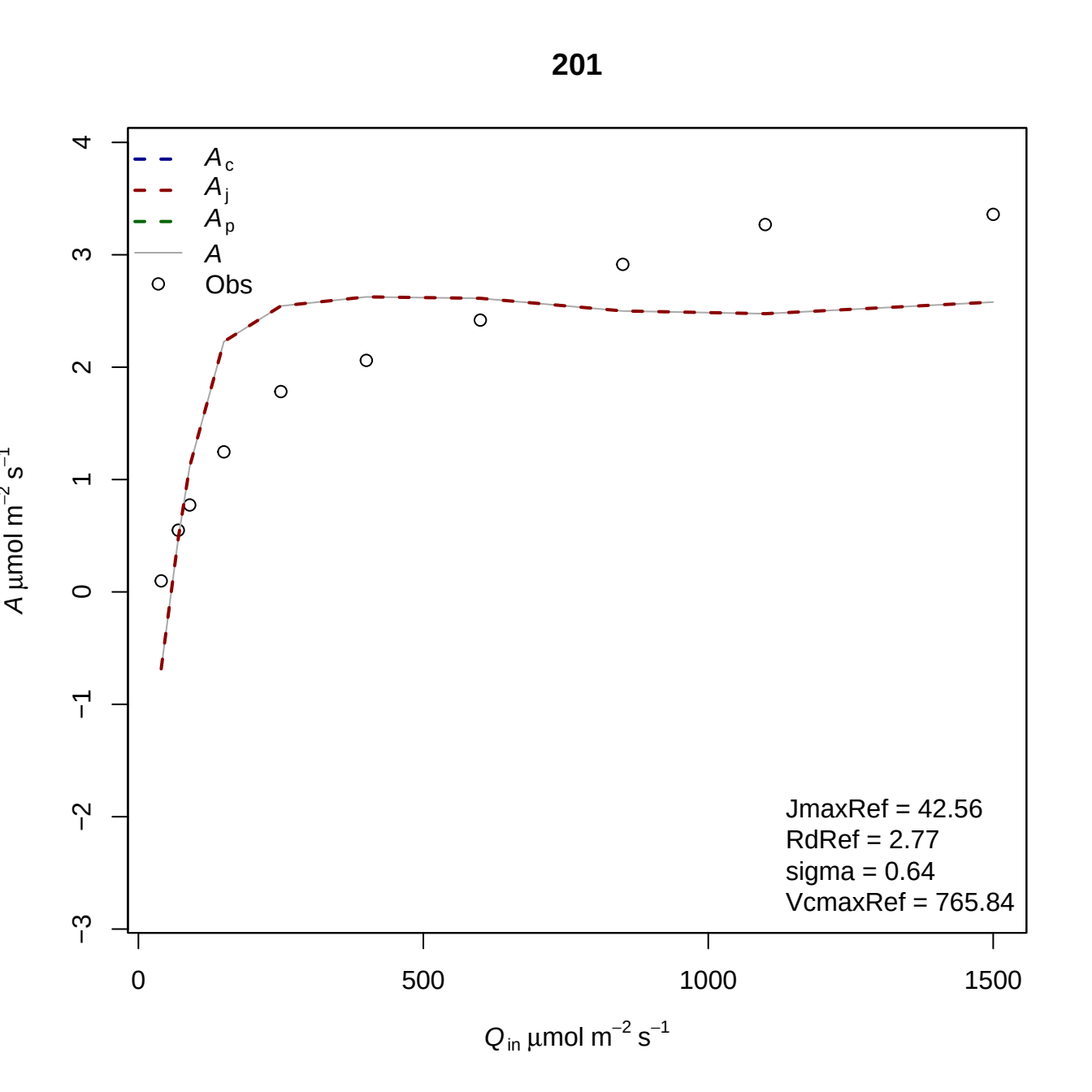

A_c
A_j
A_p
A
Obs
JmaxRef = 129.02
RdRef = 2.24
sigma = 0.99
VcmaxRef = 155.53
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

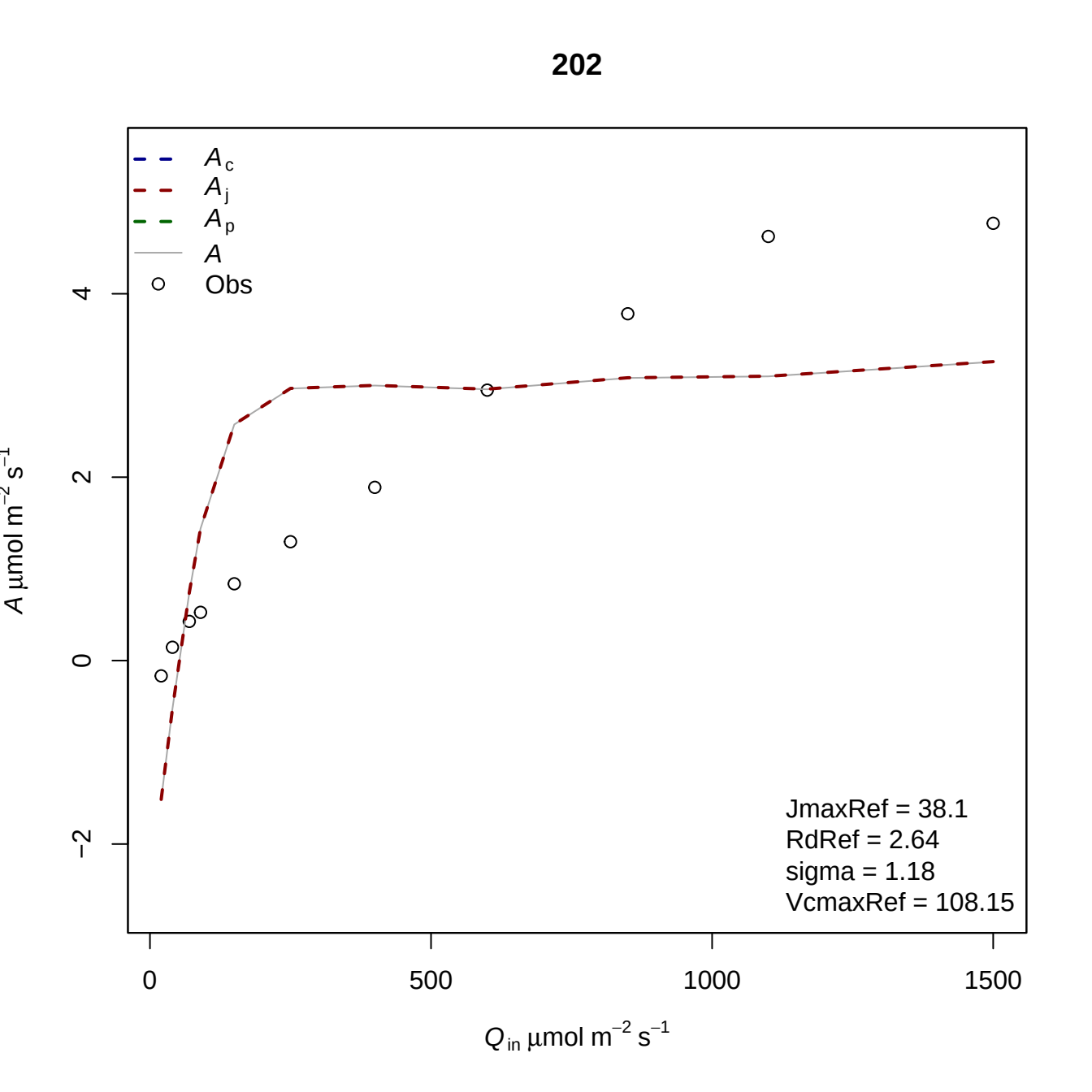

195
A_c
A_j
A_p
A
Obs
JmaxRef = 150.9
RdRef = 3.17
sigma = 1.25
VcmaxRef = 171.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

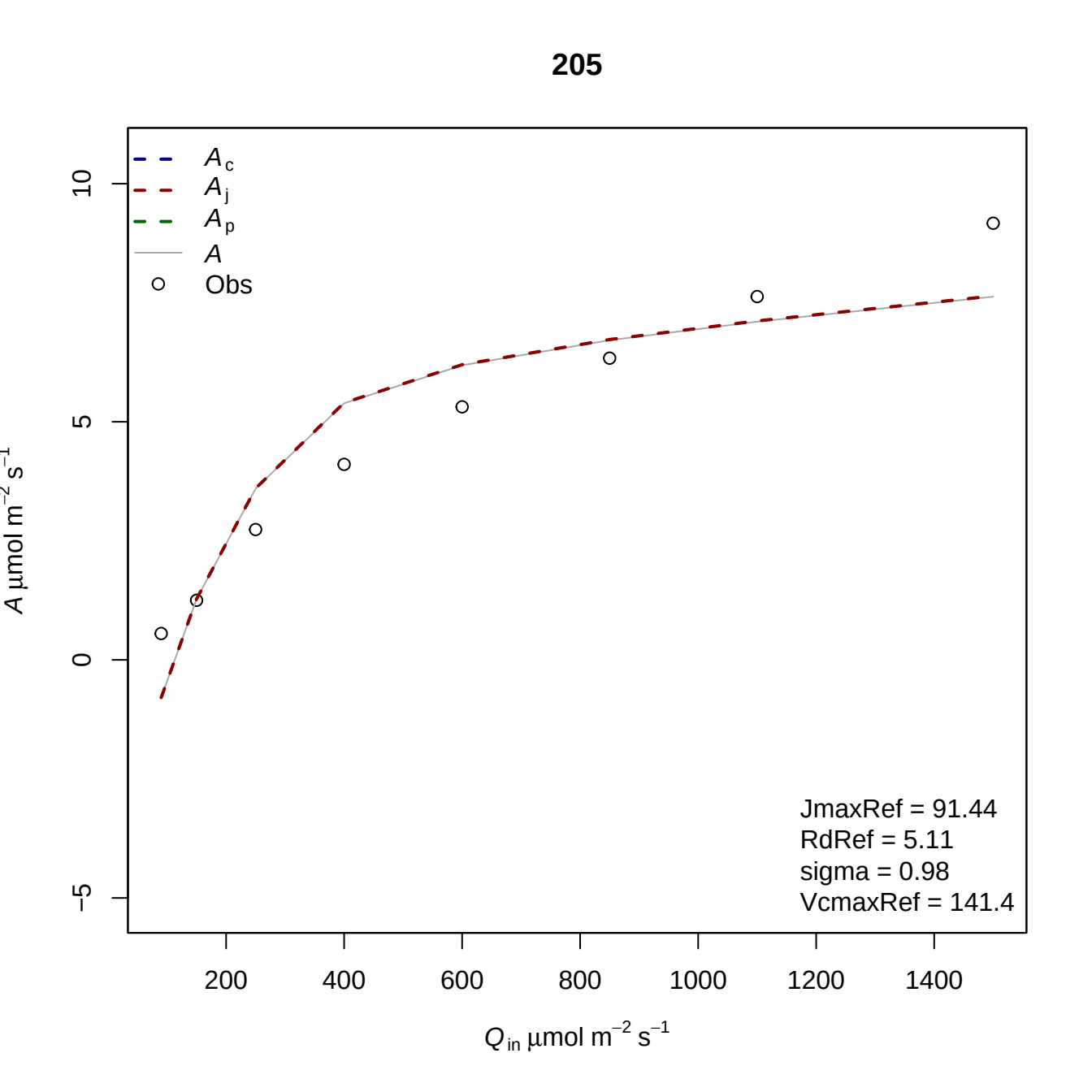

A_c
A_j
A_p
A
Obs
JmaxRef = 73.92
RdRef = 2.1
sigma = 0.57
VcmaxRef = 95.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

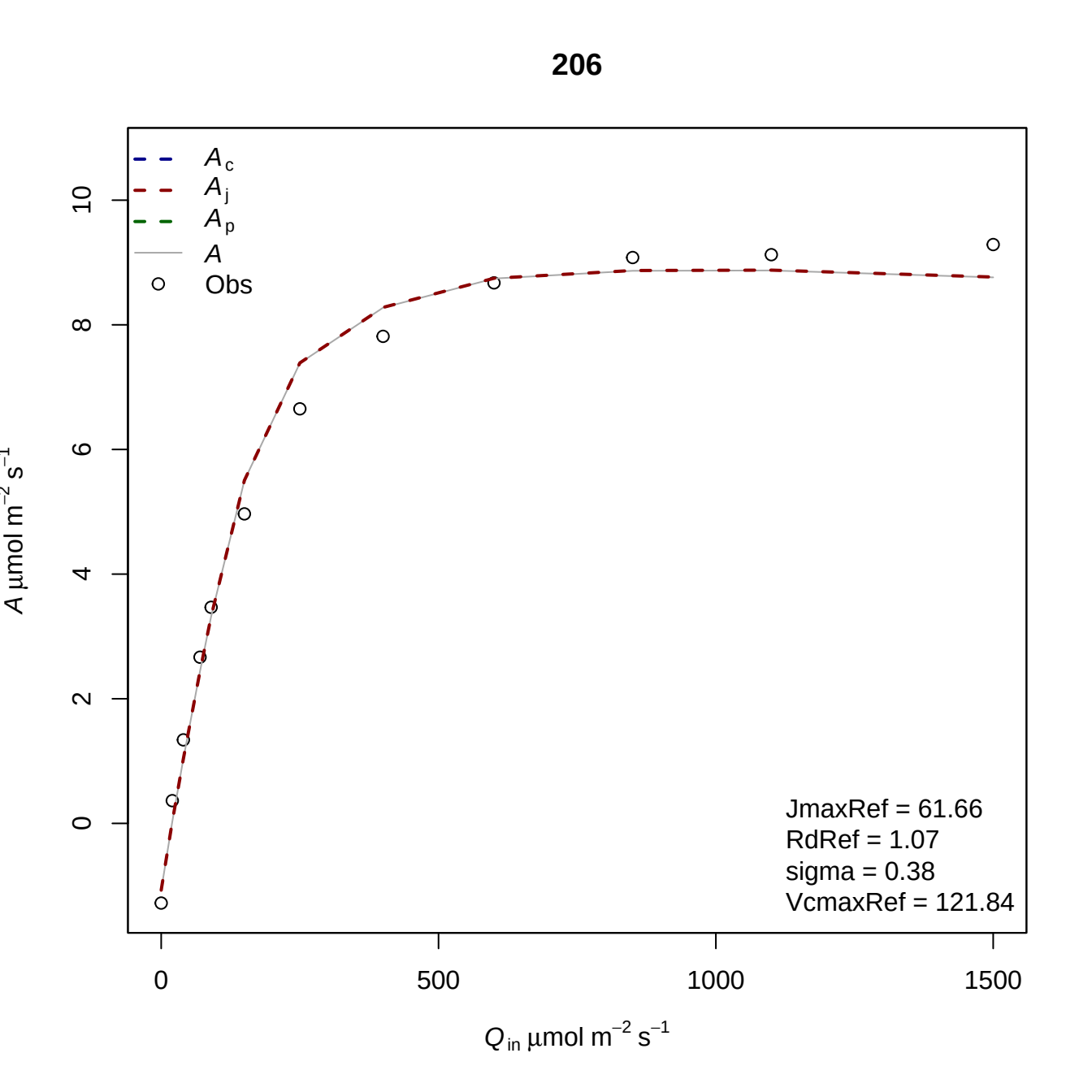

A_c
A_j
A_p
A
Obs
JmaxRef = 89.44
RdRef = 2
sigma = 0.66
VcmaxRef = 97.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

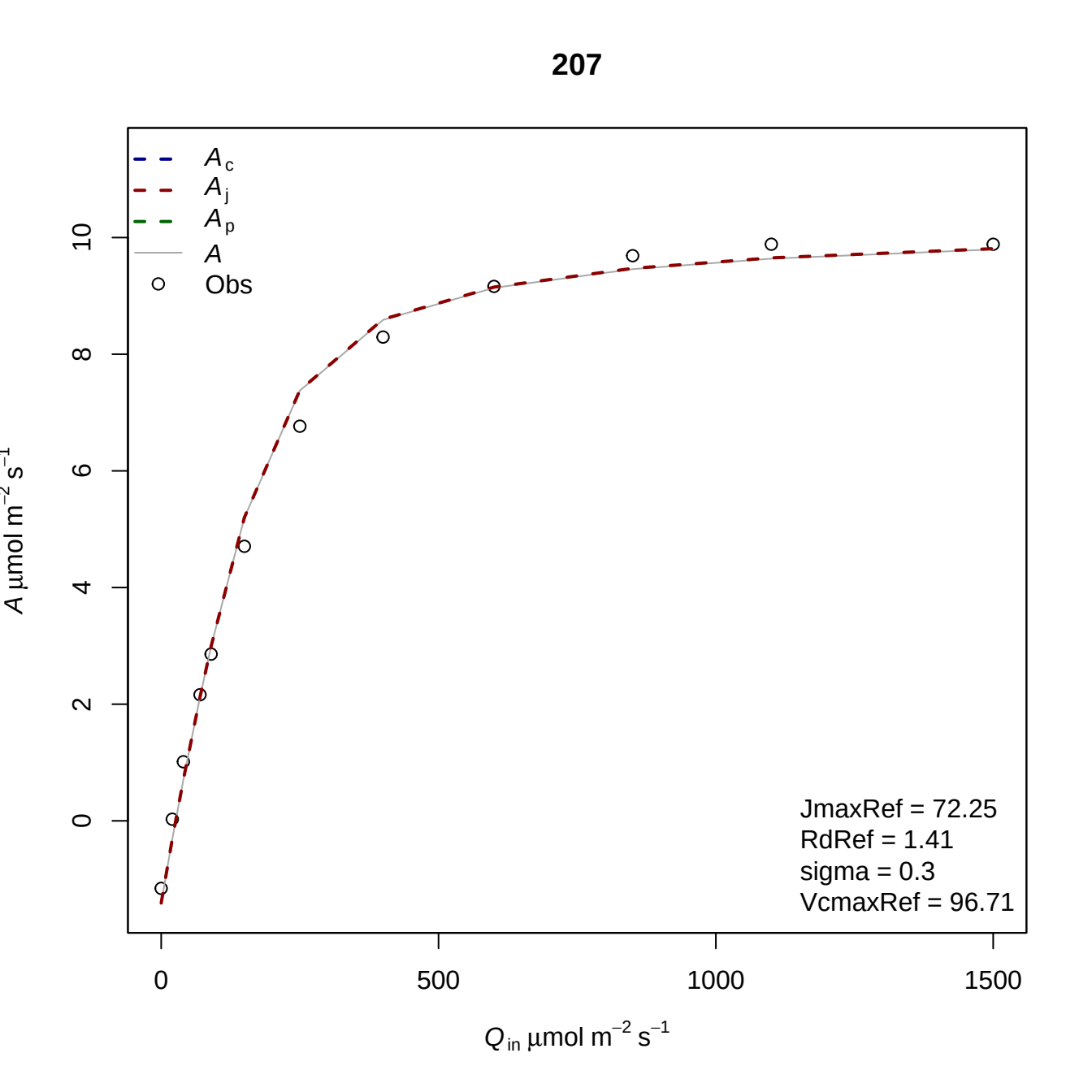

A_c
A_j
A_p
A
Obs
JmaxRef = 54.53
RdRef = 1.45
sigma = 0.51
VcmaxRef = 74.26
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

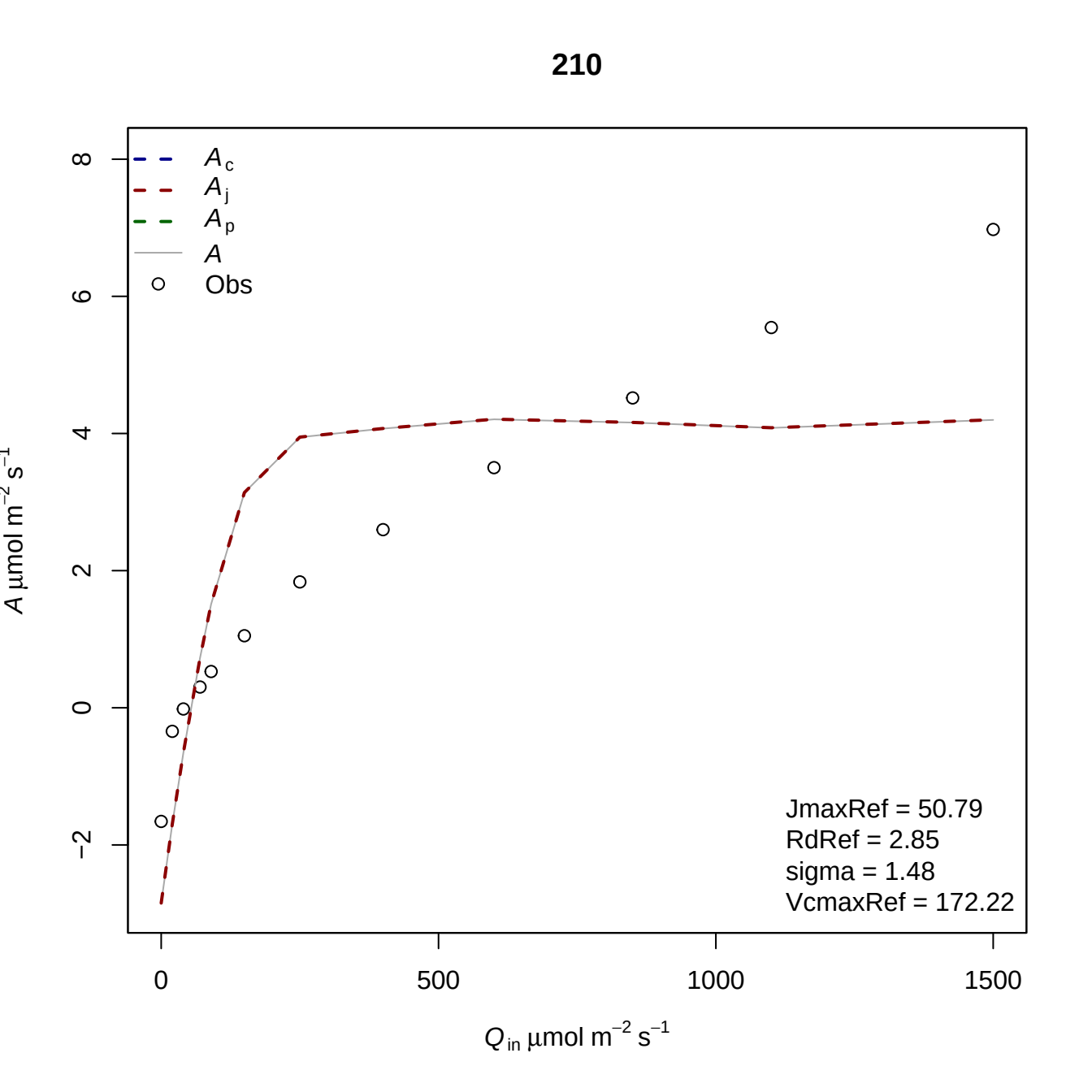

201
A_c
A_j
A_p
A
Obs
JmaxRef = 42.56
RdRef = 2.77
sigma = 0.64
VcmaxRef = 765.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

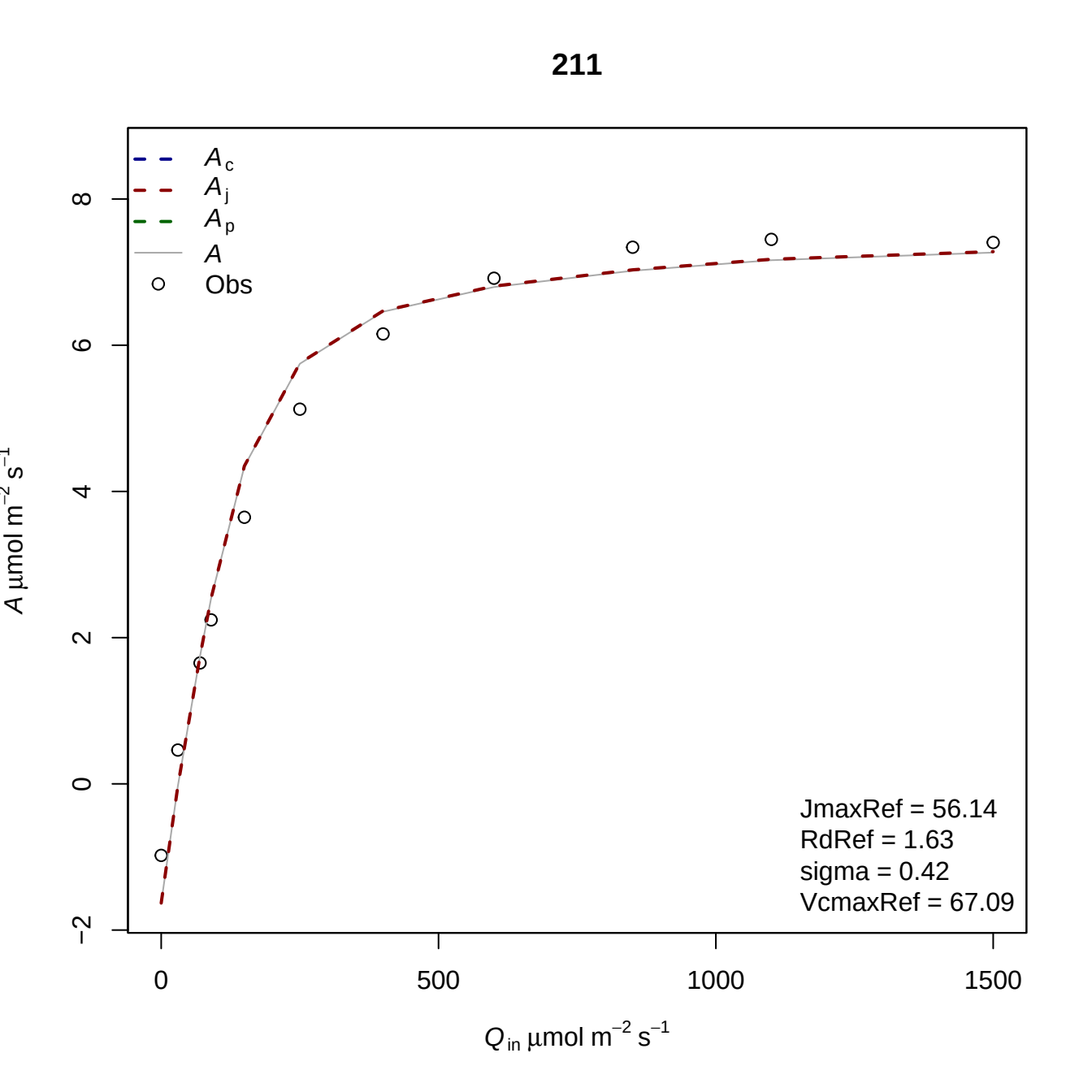

202
A_c
A_j
A_p
A
Obs
JmaxRef = 38.1
RdRef = 2.64
sigma = 1.18
VcmaxRef = 108.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

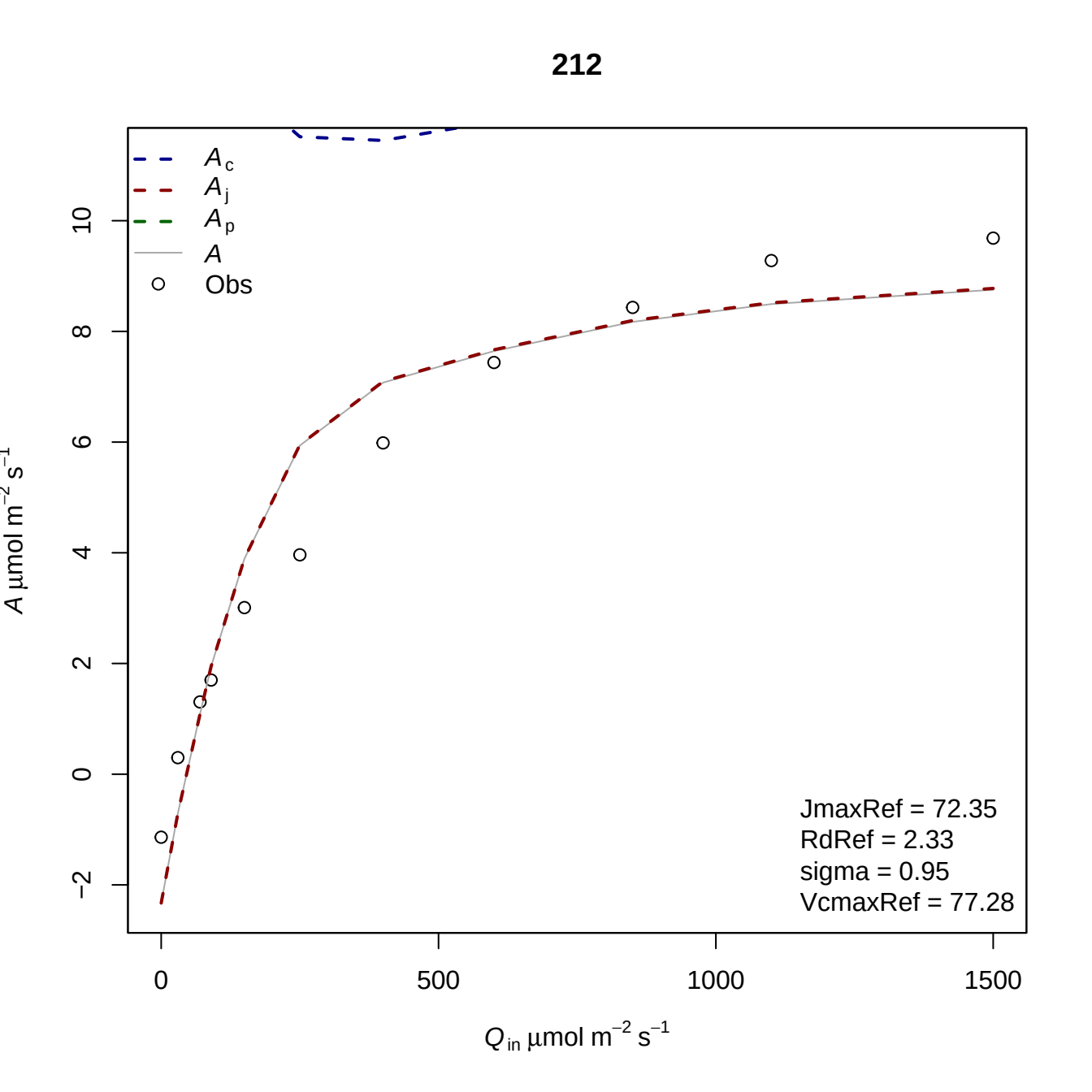

205
A_c
A_j
A_p
A
Obs
JmaxRef = 91.44
RdRef = 5.11
sigma = 0.98
VcmaxRef = 141.4
10
5
0
−5
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

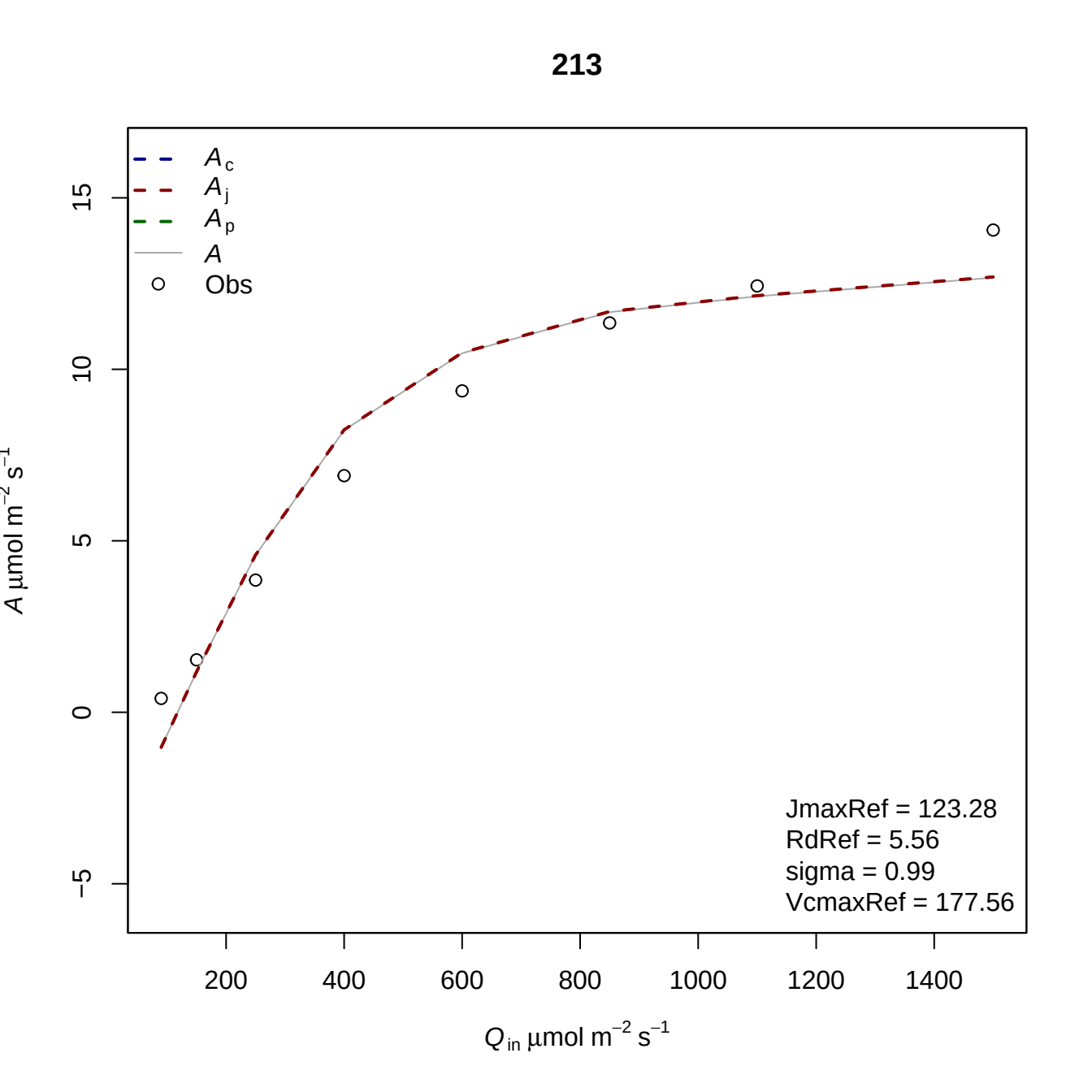

206
A_c
A_j
A_p
A
Obs
JmaxRef = 61.66
RdRef = 1.07
sigma = 0.38
VcmaxRef = 121.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

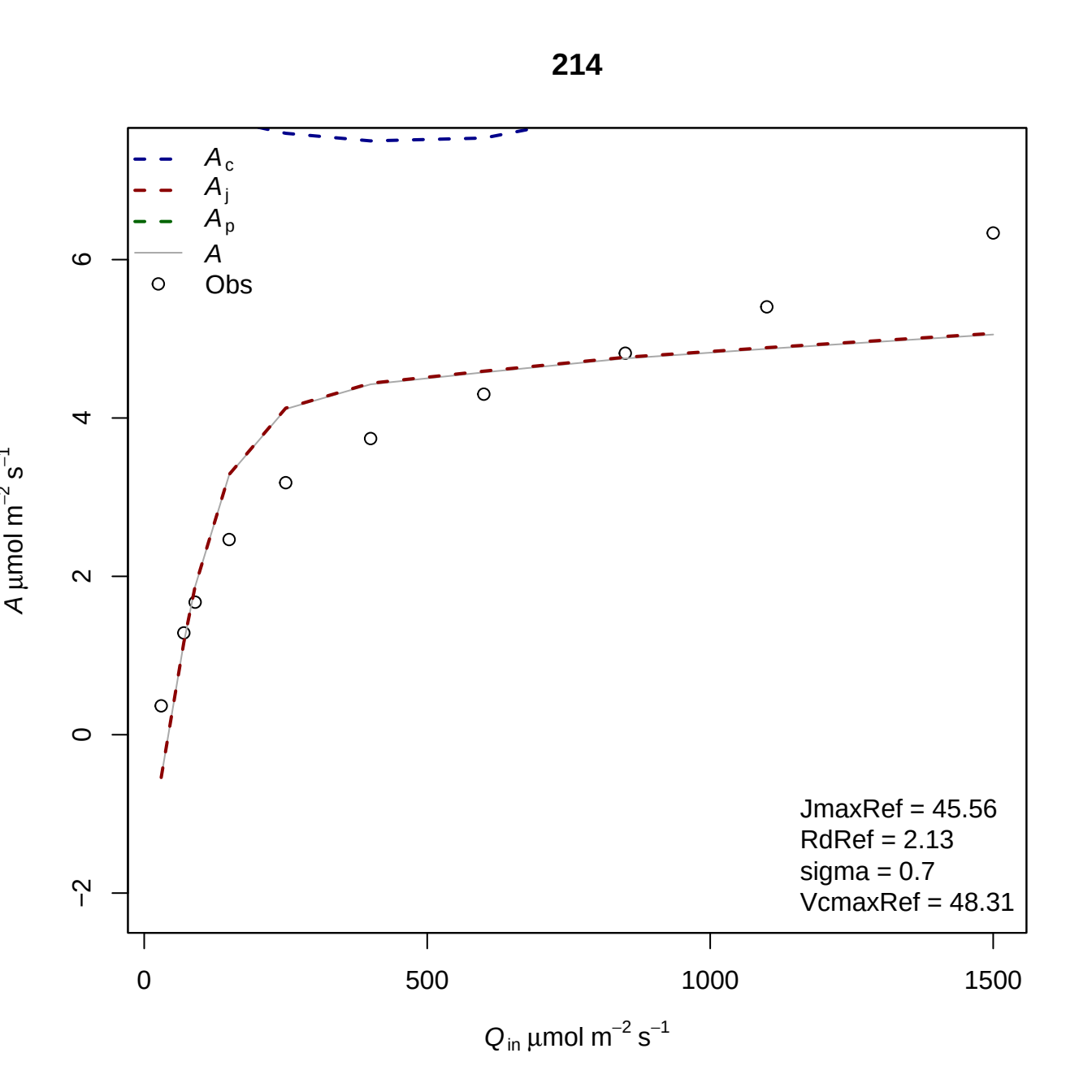

207
A_c
A_j
A_p
A
Obs
JmaxRef = 72.25
RdRef = 1.41
sigma = 0.3
VcmaxRef = 96.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

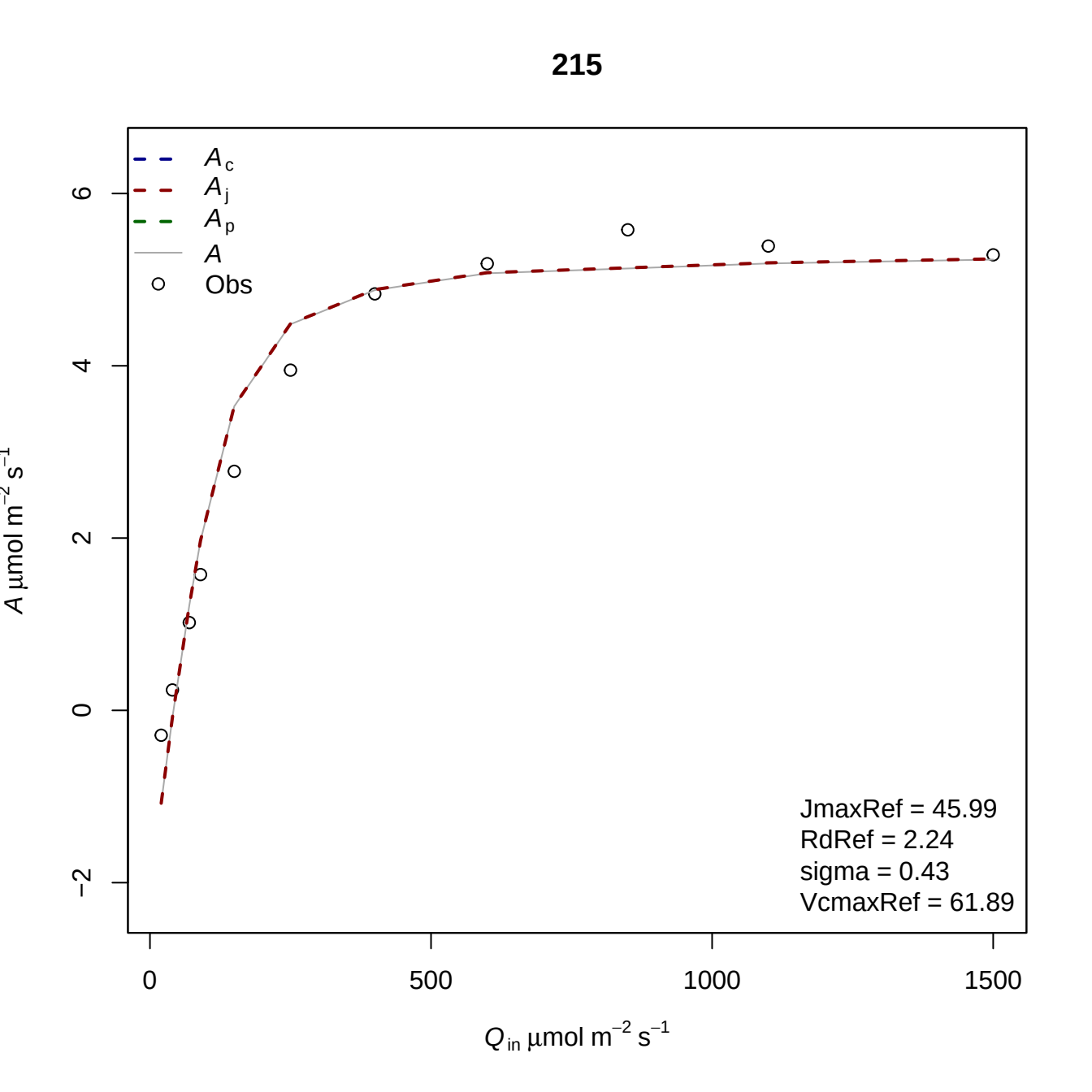

210
A_c
A_j
A_p
A
Obs
JmaxRef = 50.79
RdRef = 2.85
sigma = 1.48
VcmaxRef = 172.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

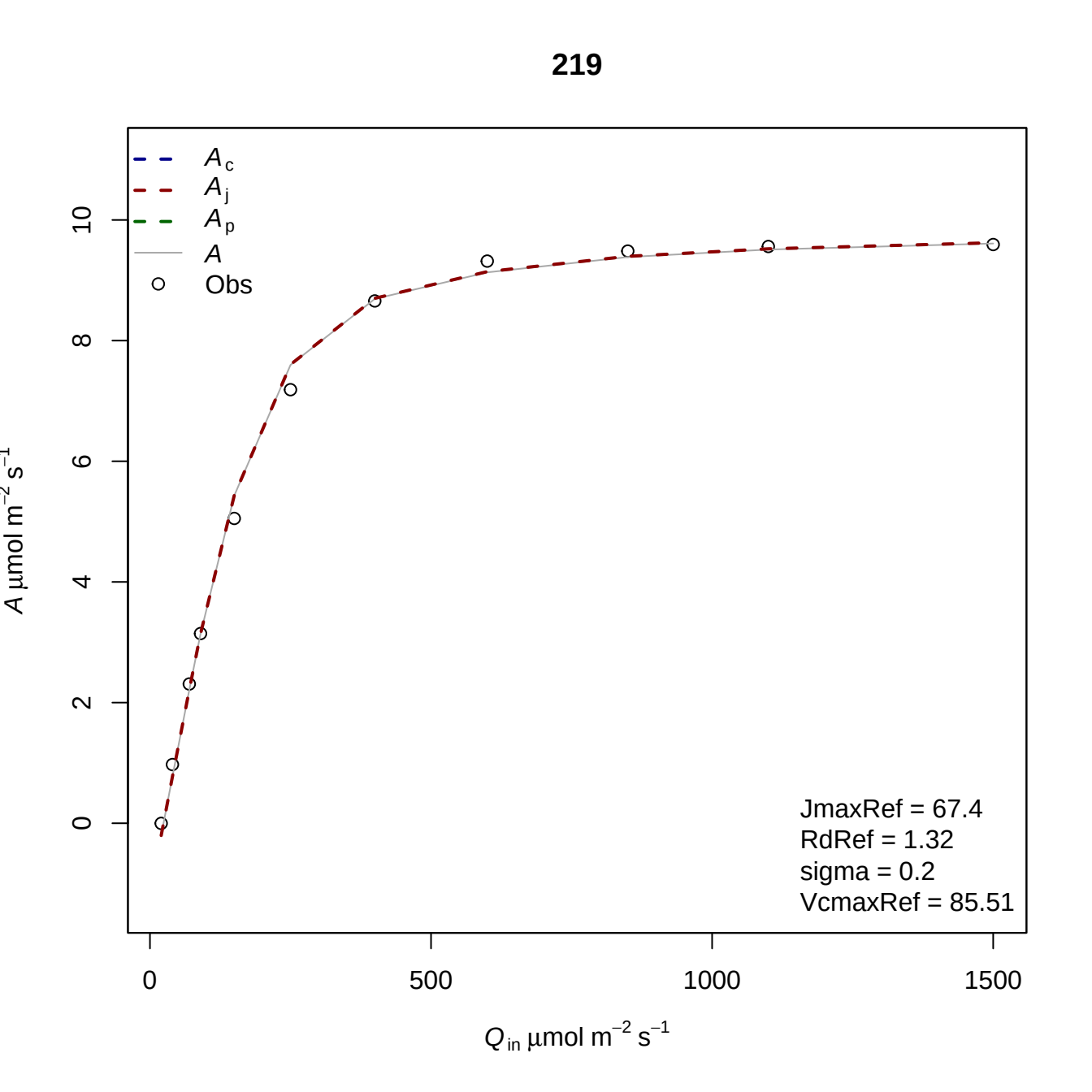

211
A_c
A_j
A_p
A
Obs
JmaxRef = 56.14
RdRef = 1.63
sigma = 0.42
VcmaxRef = 67.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

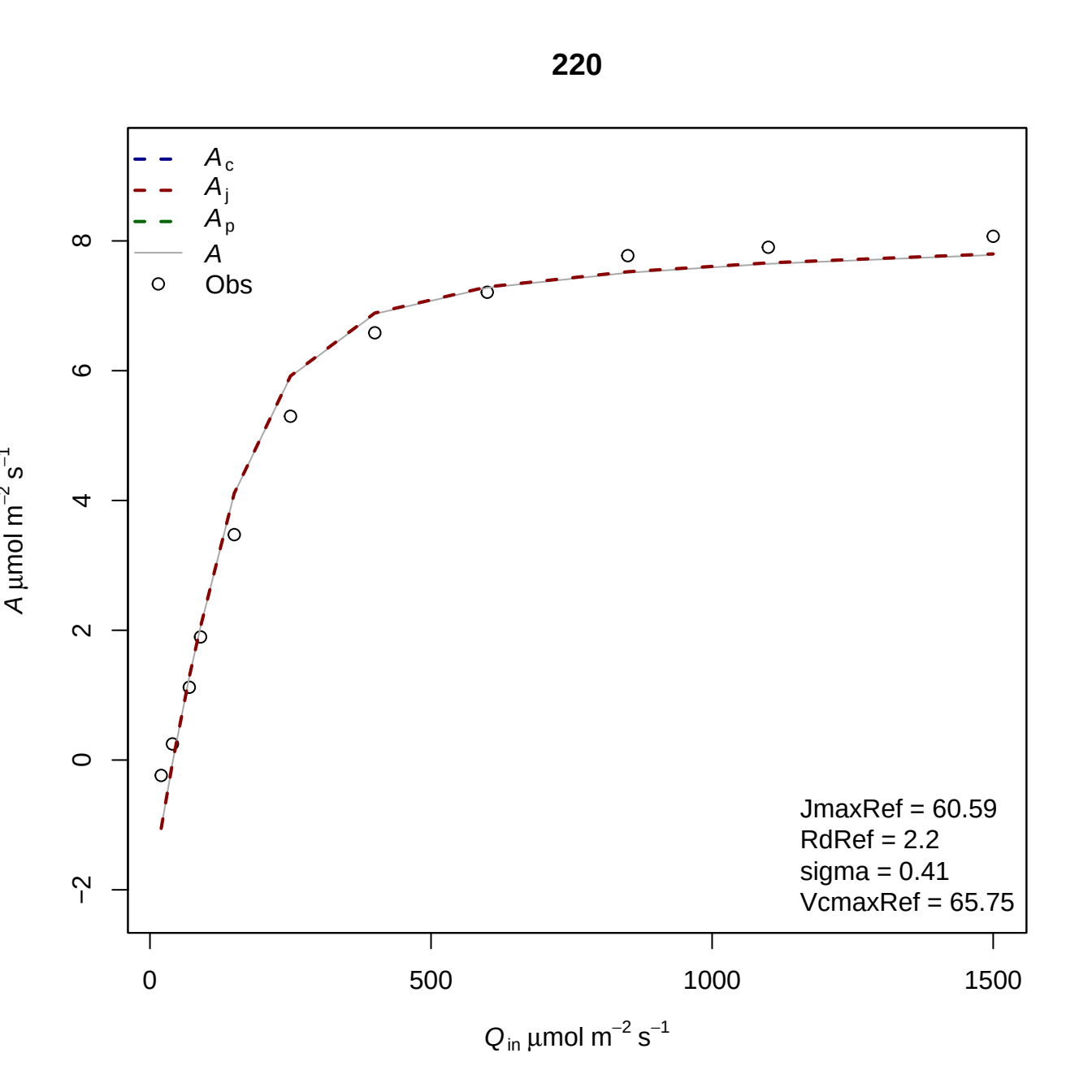

212
A_c
A_j
A_p
A
Obs
JmaxRef = 72.35
RdRef = 2.33
sigma = 0.95
VcmaxRef = 77.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

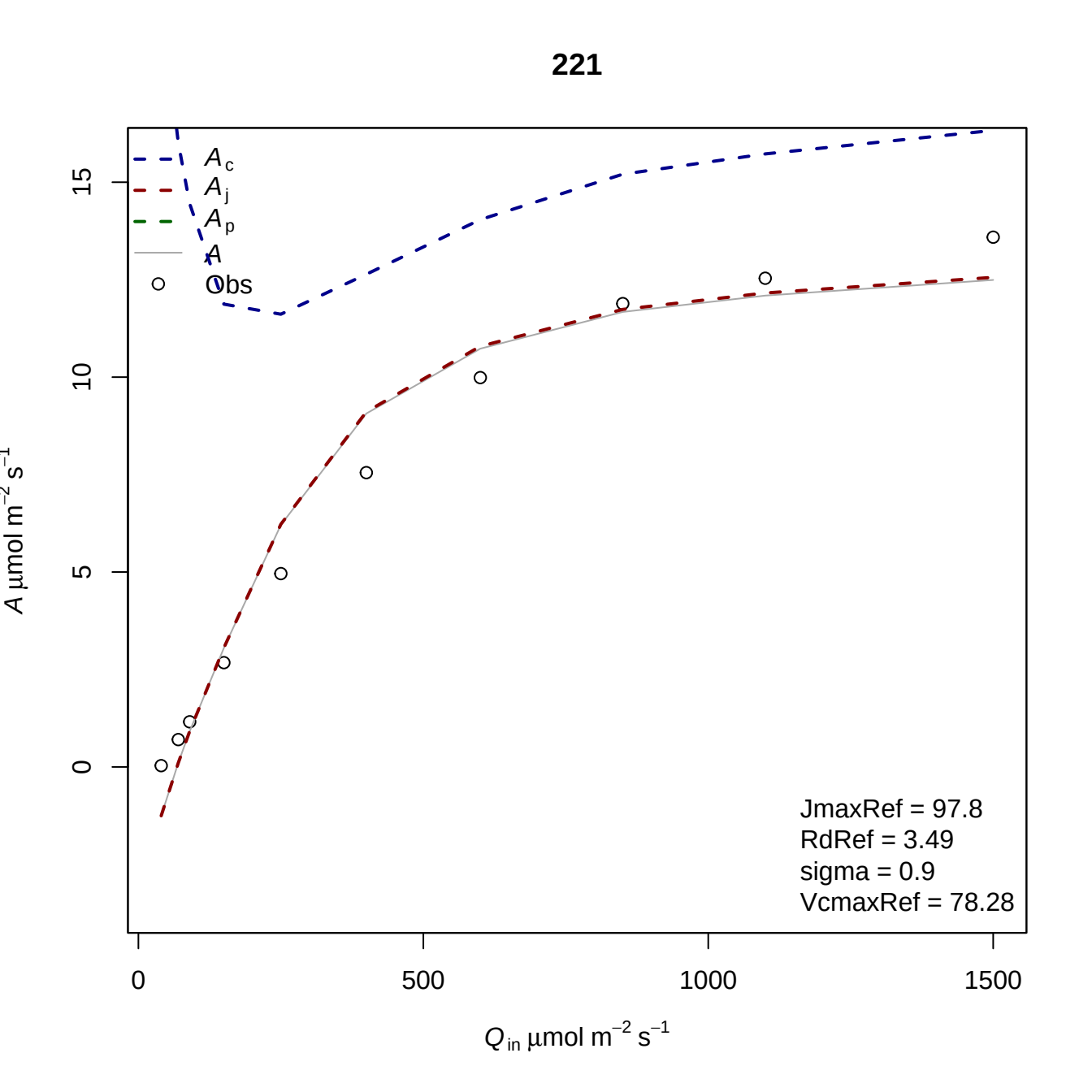

213
A_c
A_j
A_p
A
Obs
JmaxRef = 123.28
RdRef = 5.56
sigma = 0.99
VcmaxRef = 177.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
200
400
600
800
1000
1200
1400

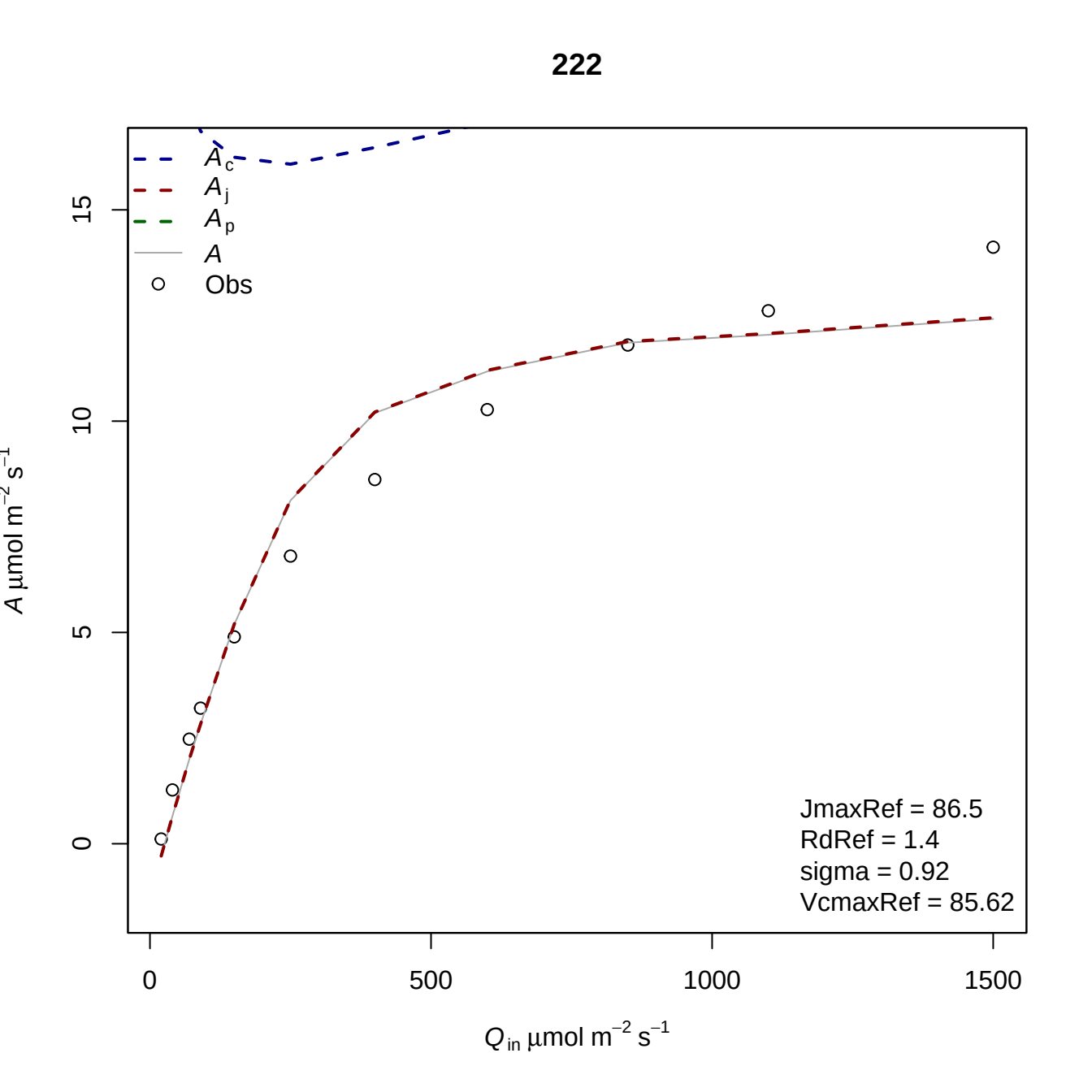

214
A_c
A_j
A_p
A
Obs
JmaxRef = 45.56
RdRef = 2.13
sigma = 0.7
VcmaxRef = 48.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

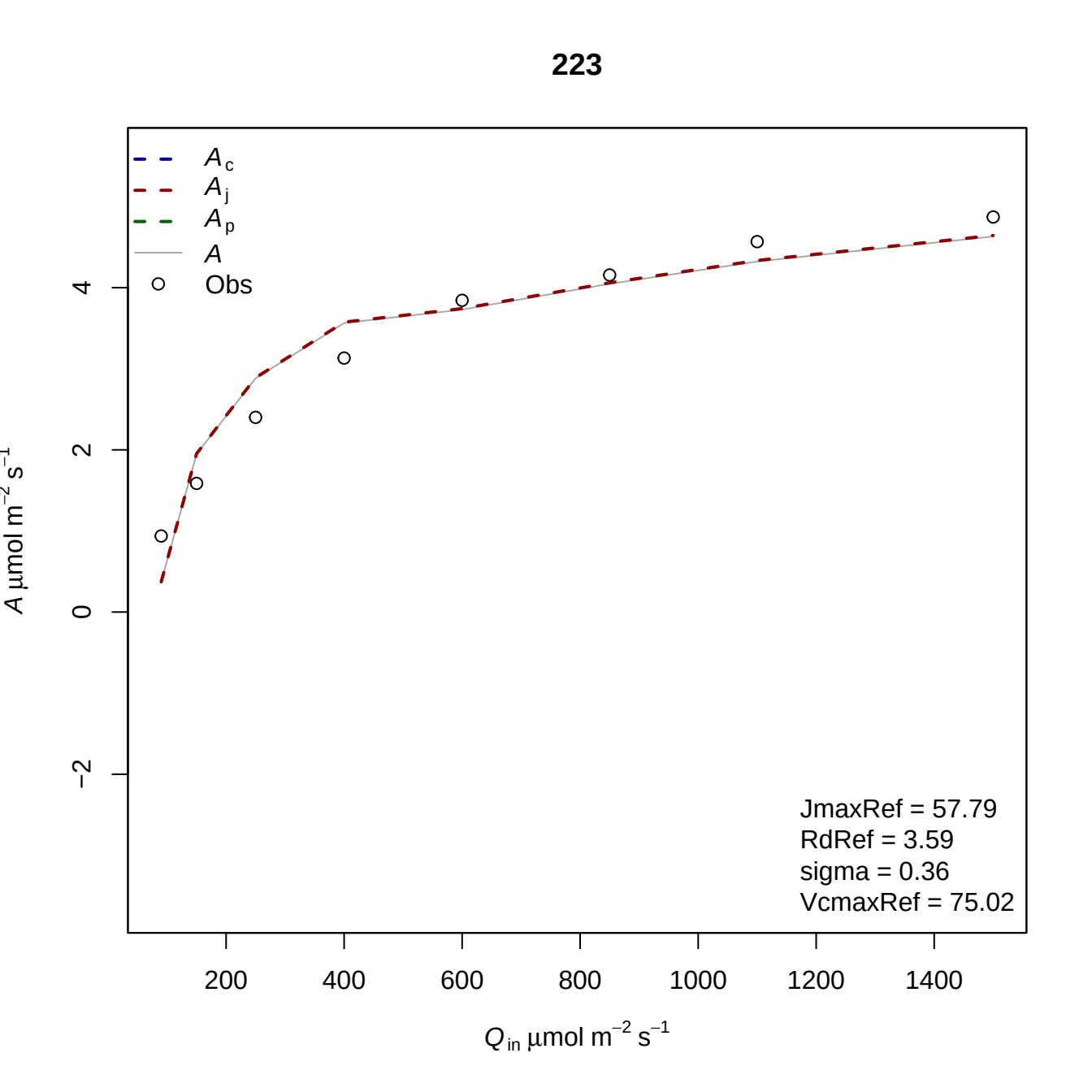

215
A_c
A_j
A_p
A
Obs
JmaxRef = 45.99
RdRef = 2.24
sigma = 0.43
VcmaxRef = 61.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

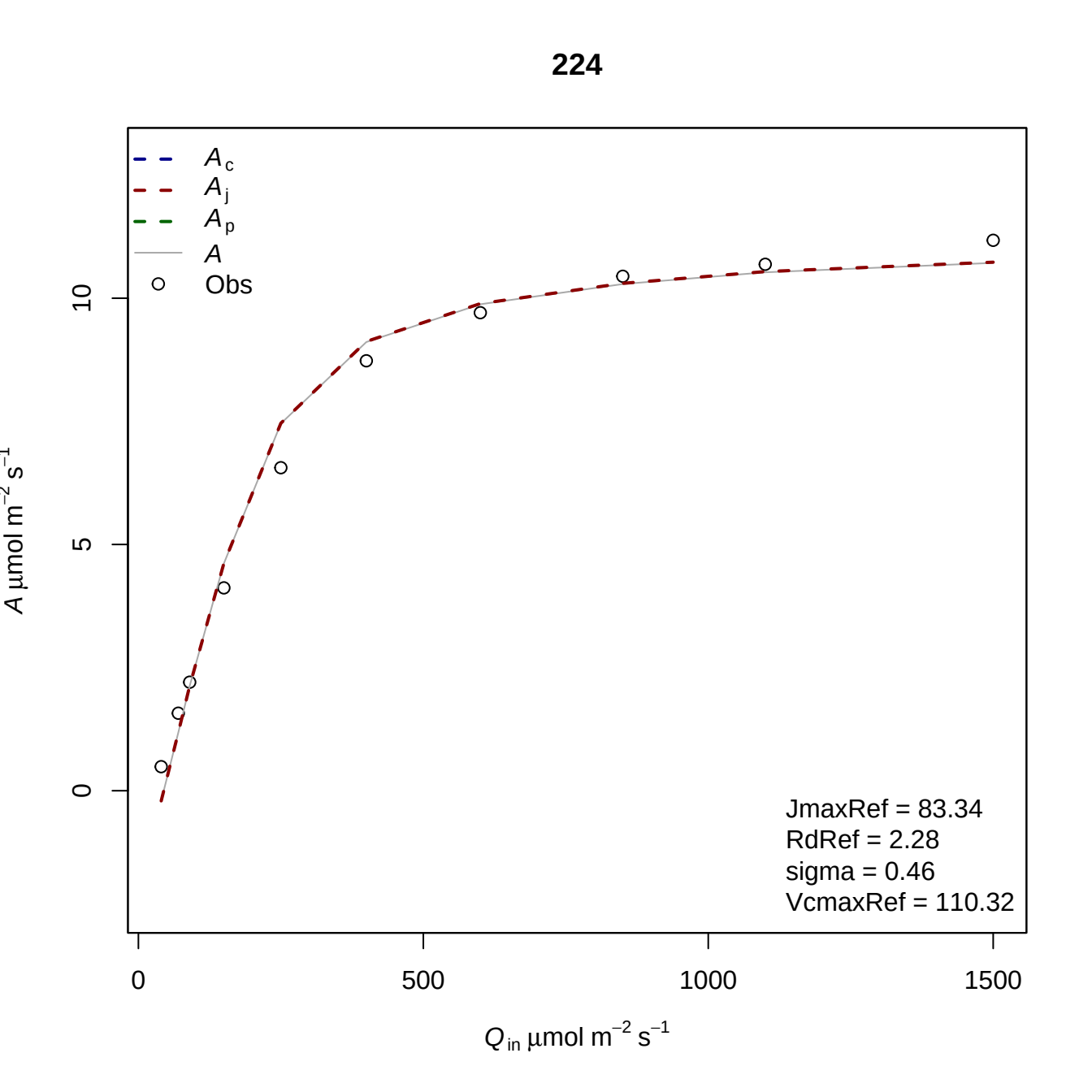

219
A_c
A_j
A_p
A
Obs
JmaxRef = 67.4
RdRef = 1.32
sigma = 0.2
VcmaxRef = 85.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

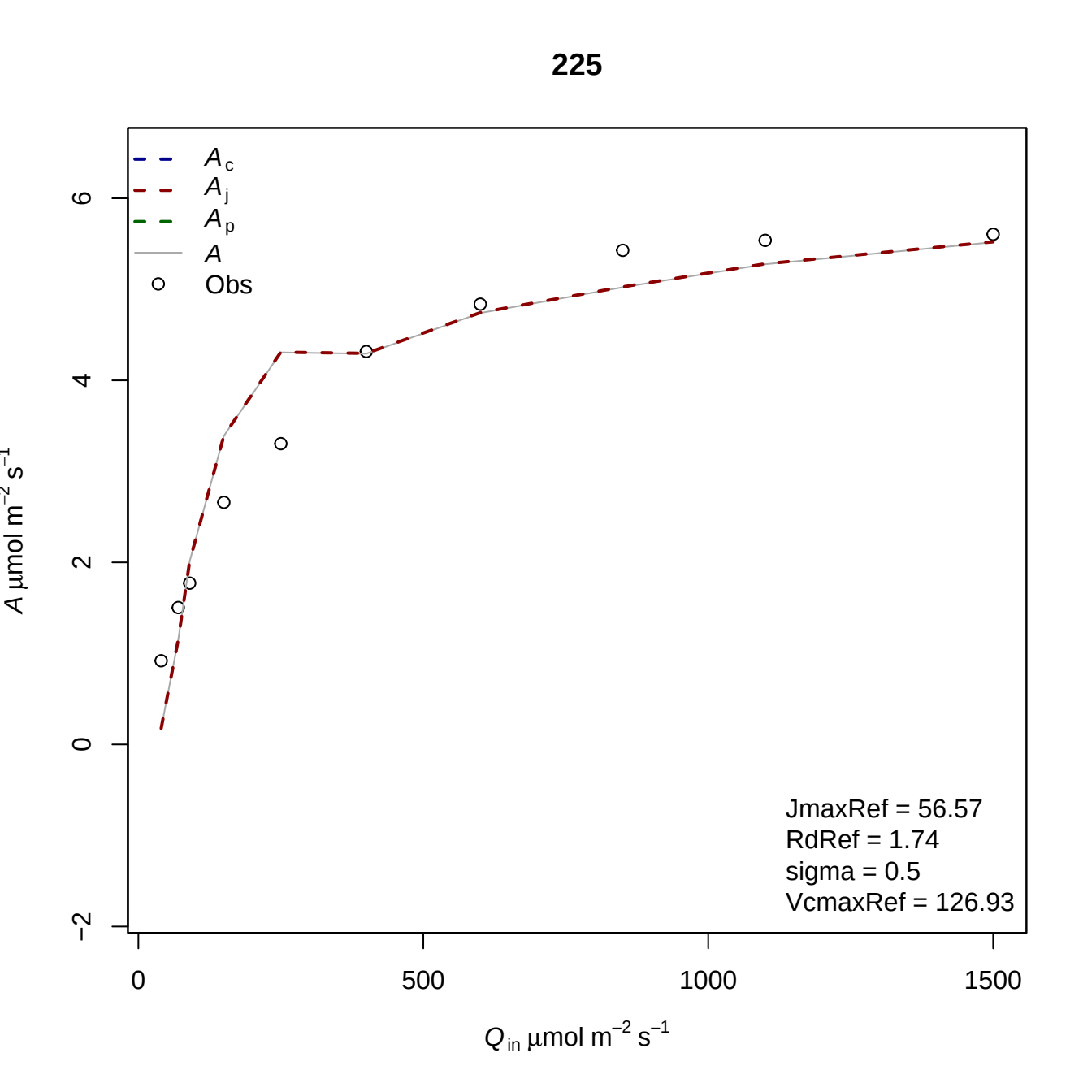

220
A_c
A_j
A_p
A
Obs
JmaxRef = 60.59
RdRef = 2.2
sigma = 0.41
VcmaxRef = 65.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

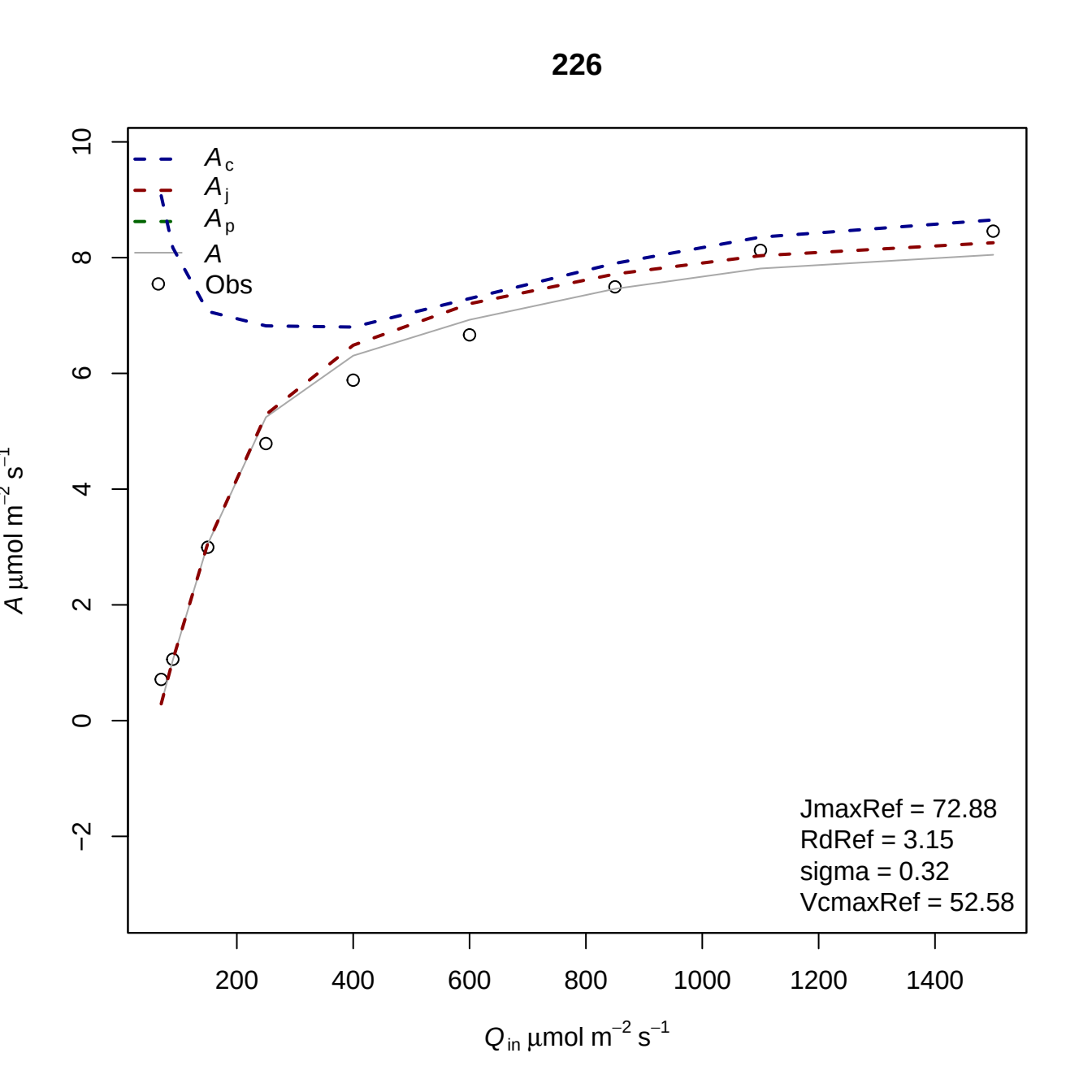

221
A_c
A_j
A_p
A
Obs
JmaxRef = 97.8
RdRef = 3.49
sigma = 0.9
VcmaxRef = 78.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

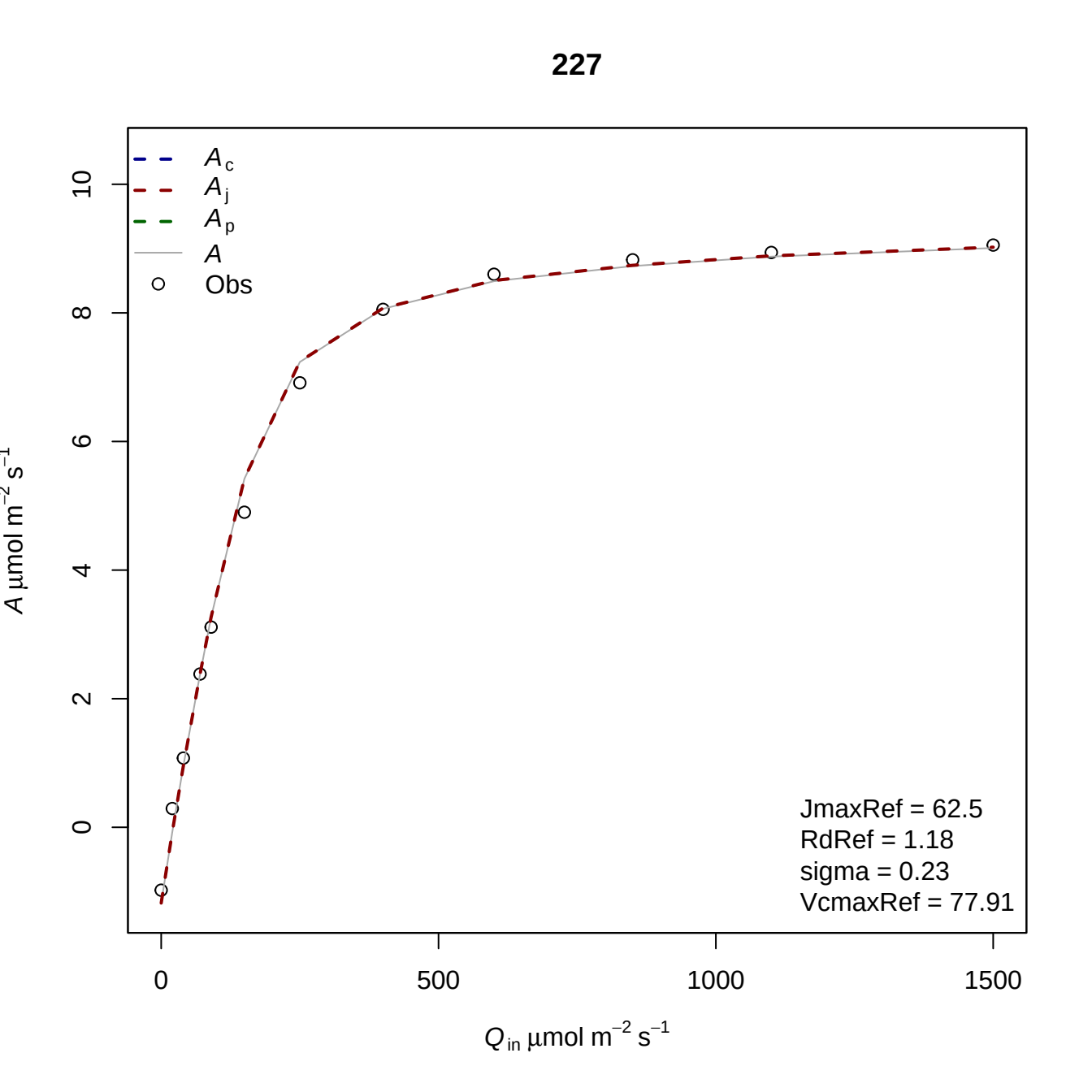

222
A_c
A_j
A_p
A
Obs
JmaxRef = 86.5
RdRef = 1.4
sigma = 0.92
VcmaxRef = 85.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

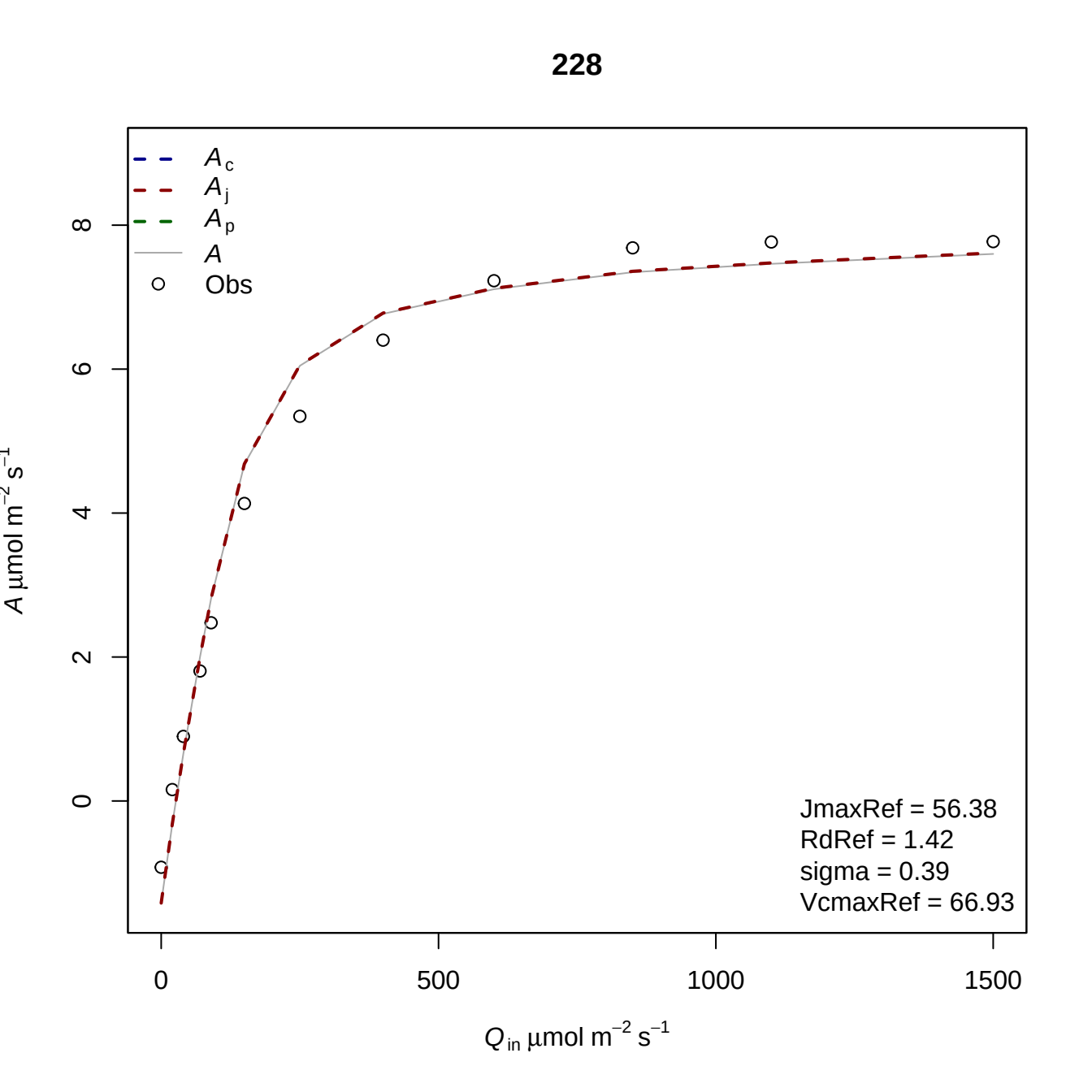

223
A_c
A_j
A_p
A
Obs
JmaxRef = 57.79
RdRef = 3.59
sigma = 0.36
VcmaxRef = 75.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
200
400
600
800
1000
1200
1400

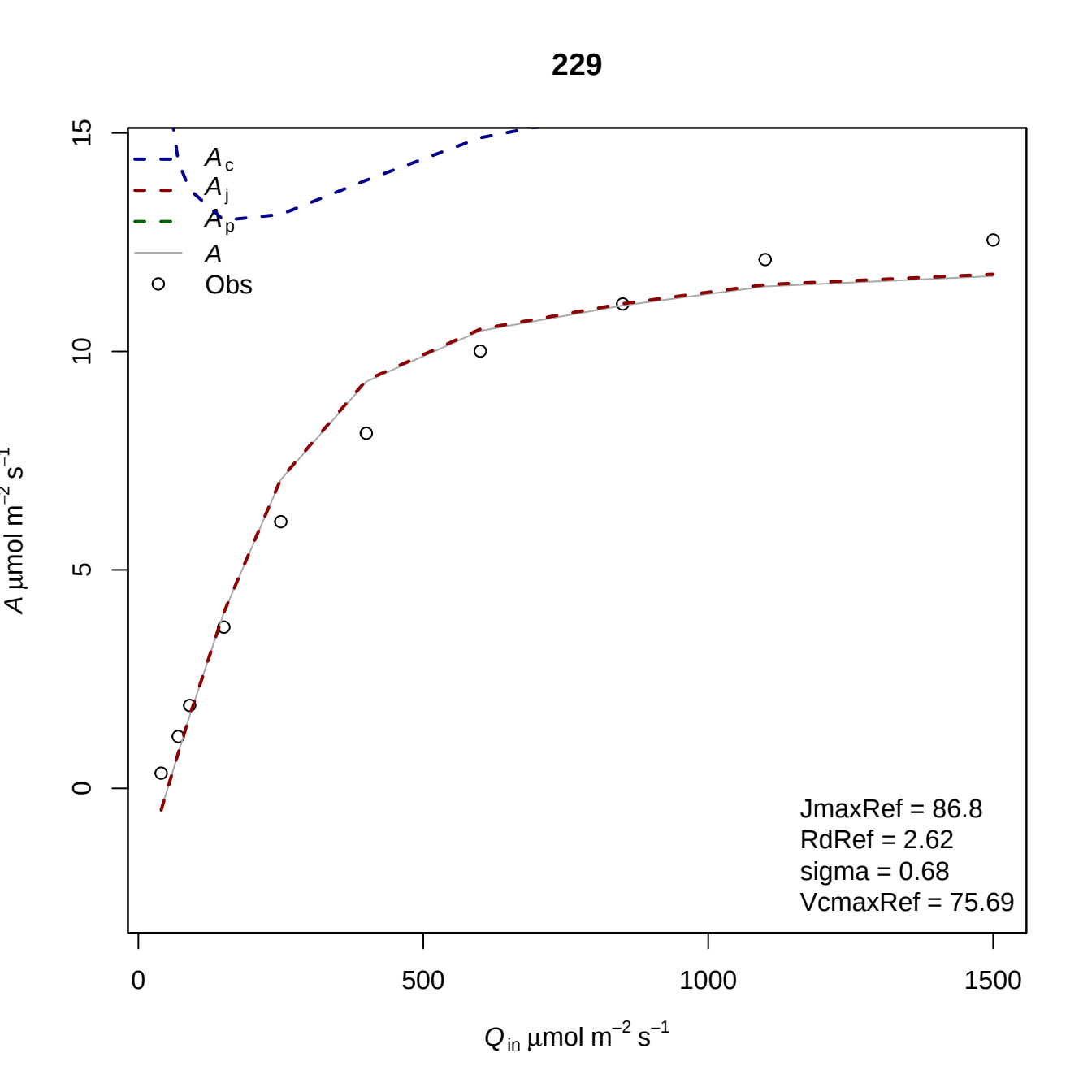

224
A_c
A_j
A_p
A
Obs
JmaxRef = 83.34
RdRef = 2.28
sigma = 0.46
VcmaxRef = 110.32
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

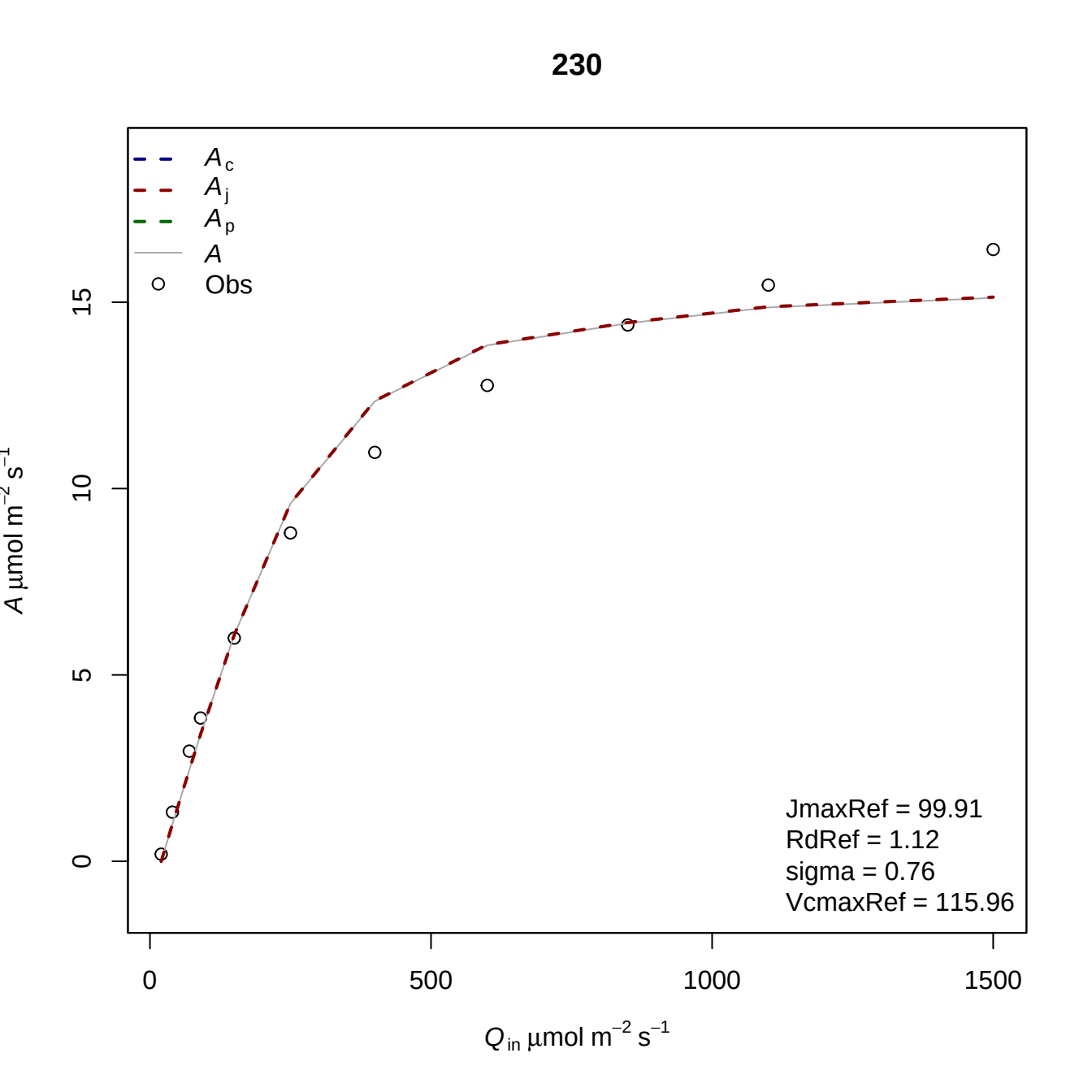

225
A_c
A_j
A_p
A
Obs
JmaxRef = 56.57
RdRef = 1.74
sigma = 0.5
VcmaxRef = 126.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

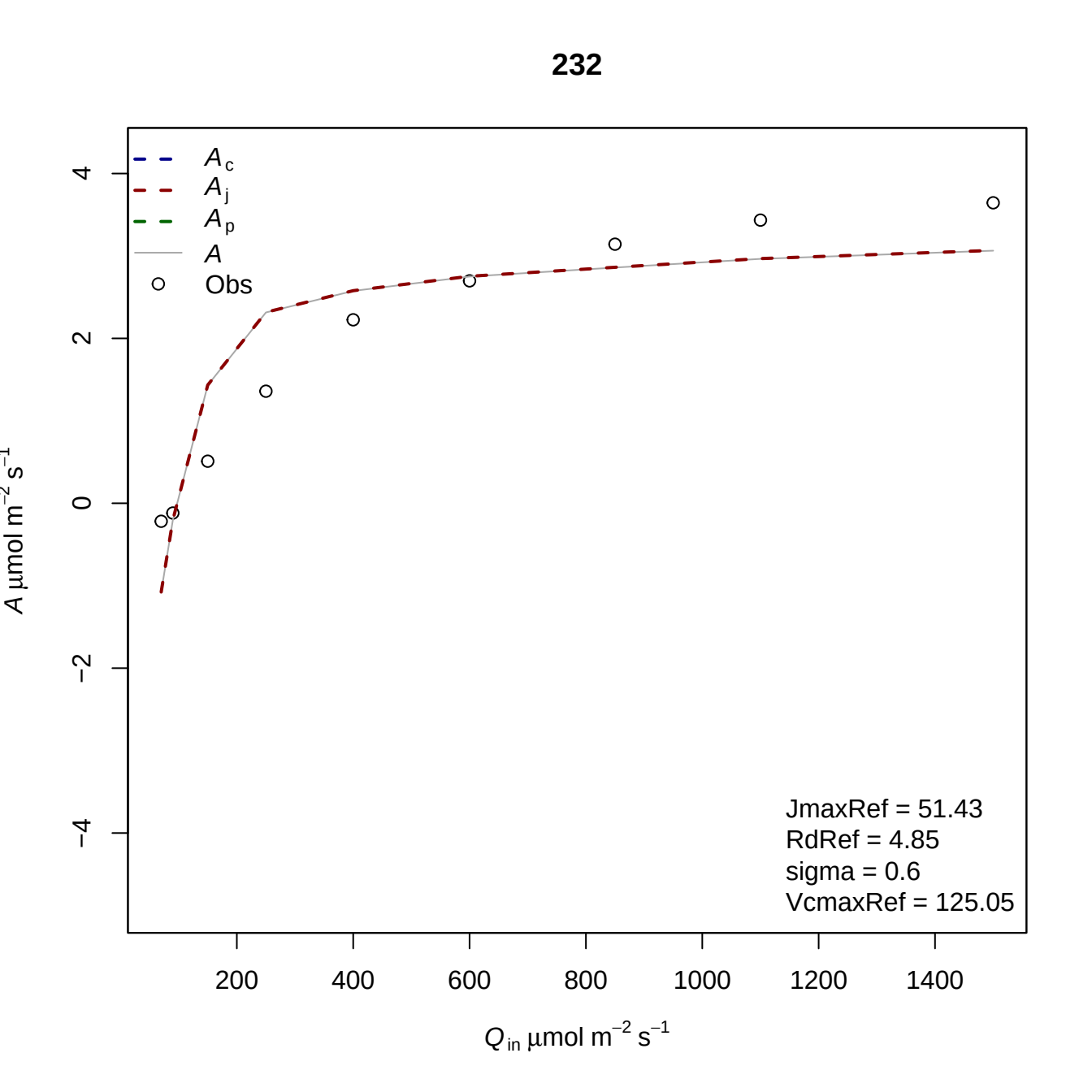

226
A_c
A_j
A_p
A
Obs
JmaxRef = 72.88
RdRef = 3.15
sigma = 0.32
VcmaxRef = 52.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

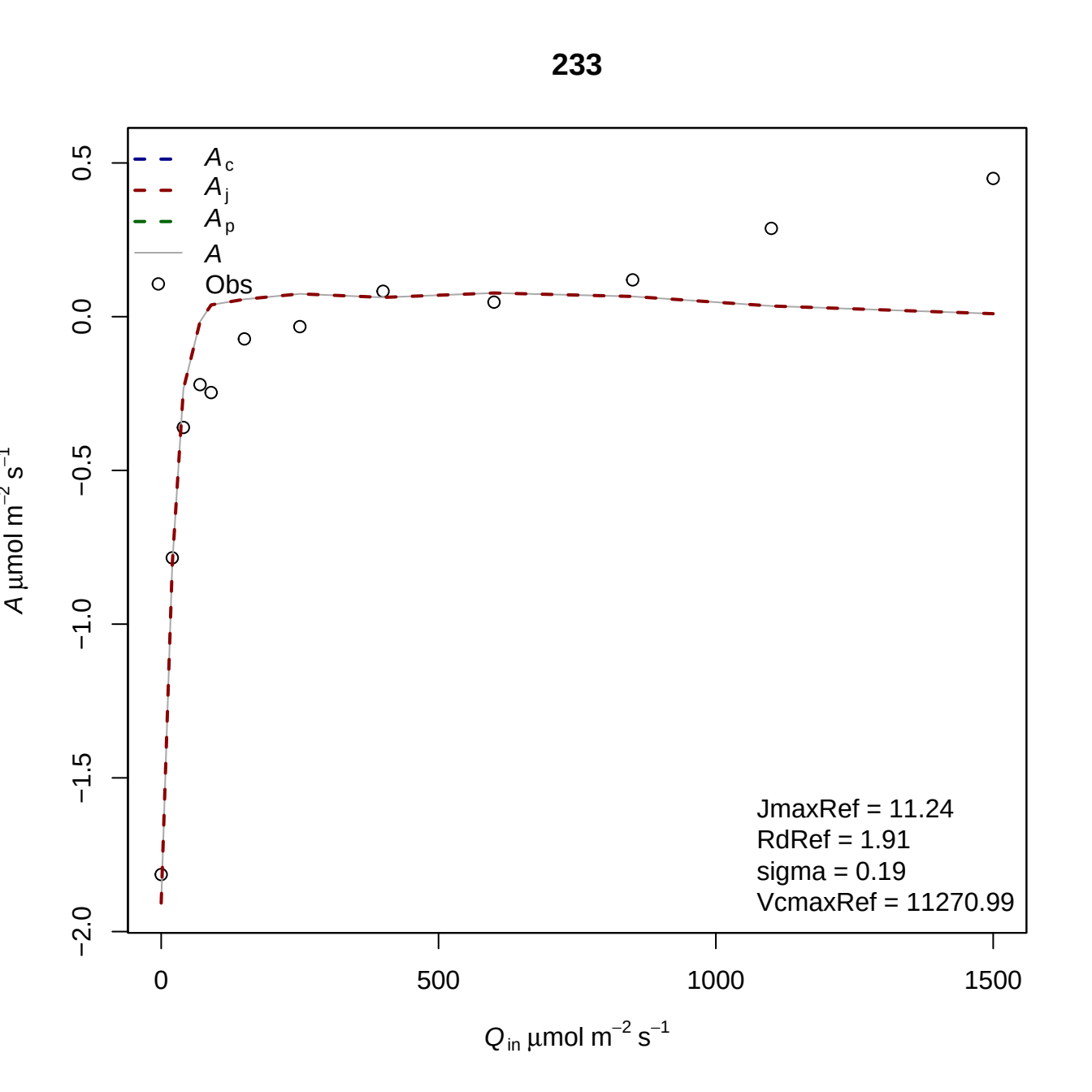

227
A_c
A_j
A_p
A
Obs
JmaxRef = 62.5
RdRef = 1.18
sigma = 0.23
VcmaxRef = 77.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

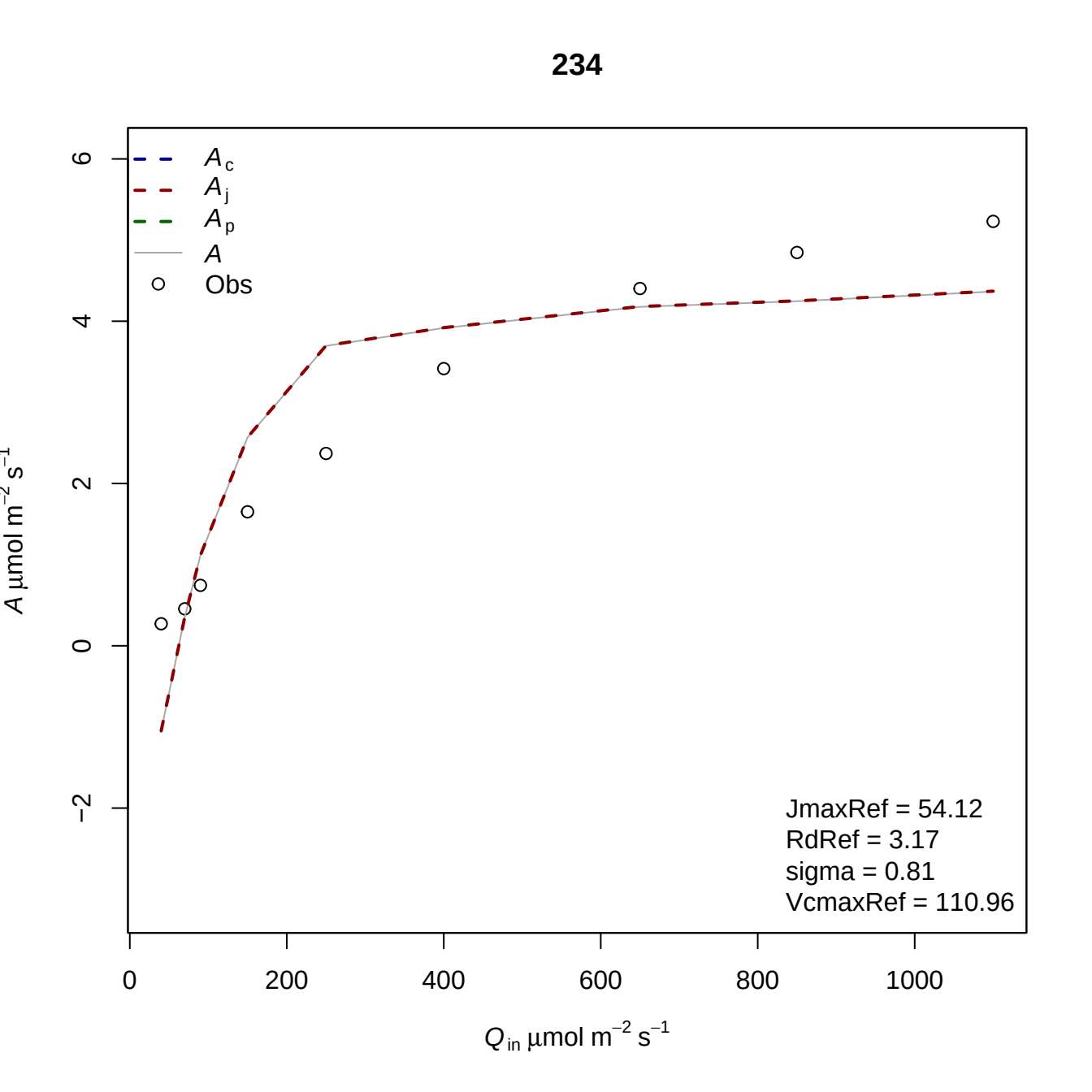

228
A_c
A_j
A_p
A
Obs
JmaxRef = 56.38
RdRef = 1.42
sigma = 0.39
VcmaxRef = 66.93
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

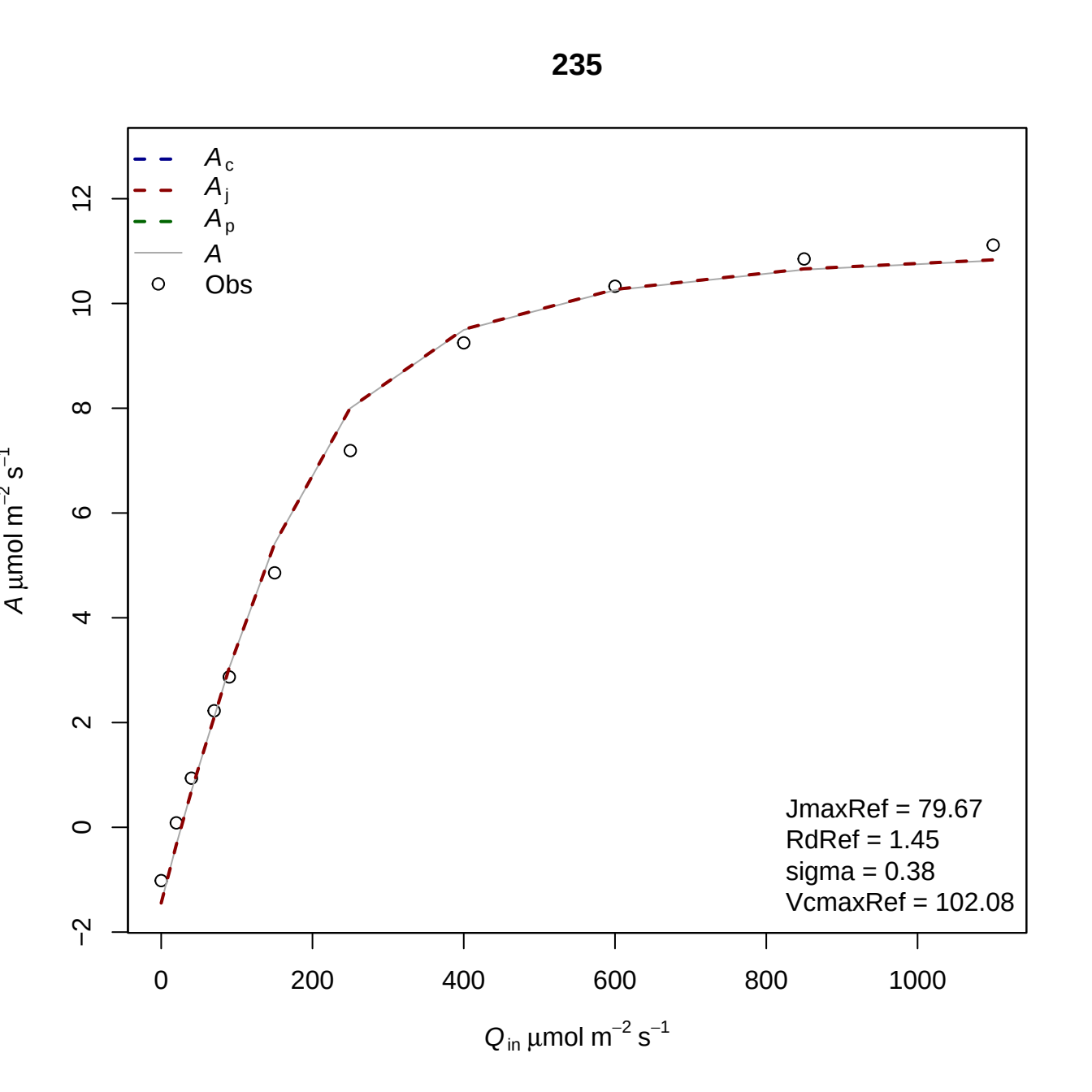

229
A_c
A_j
A_p
A
Obs
JmaxRef = 86.8
RdRef = 2.62
sigma = 0.68
VcmaxRef = 75.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

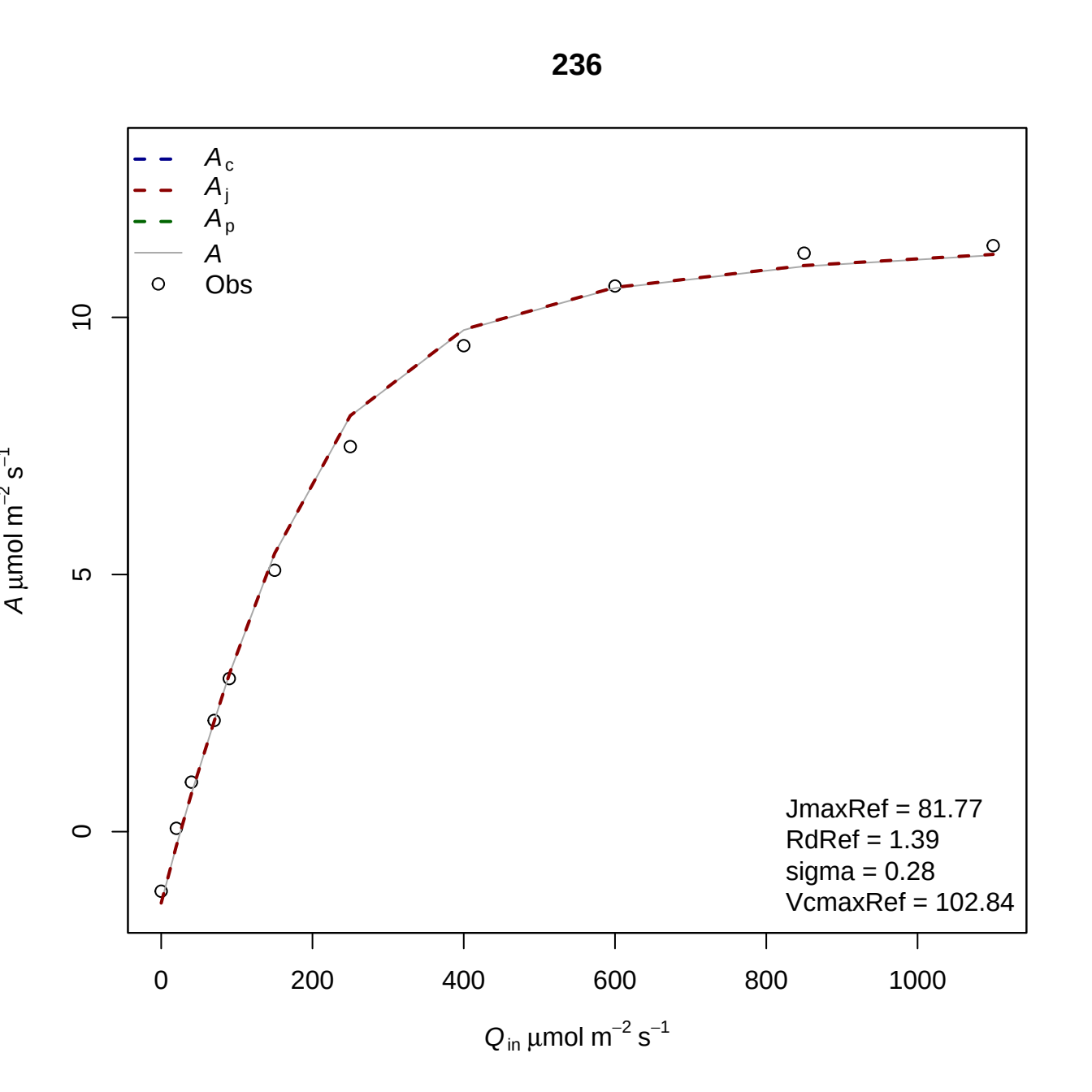

230
A_c
A_j
A_p
A
Obs
JmaxRef = 99.91
RdRef = 1.12
sigma = 0.76
VcmaxRef = 115.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

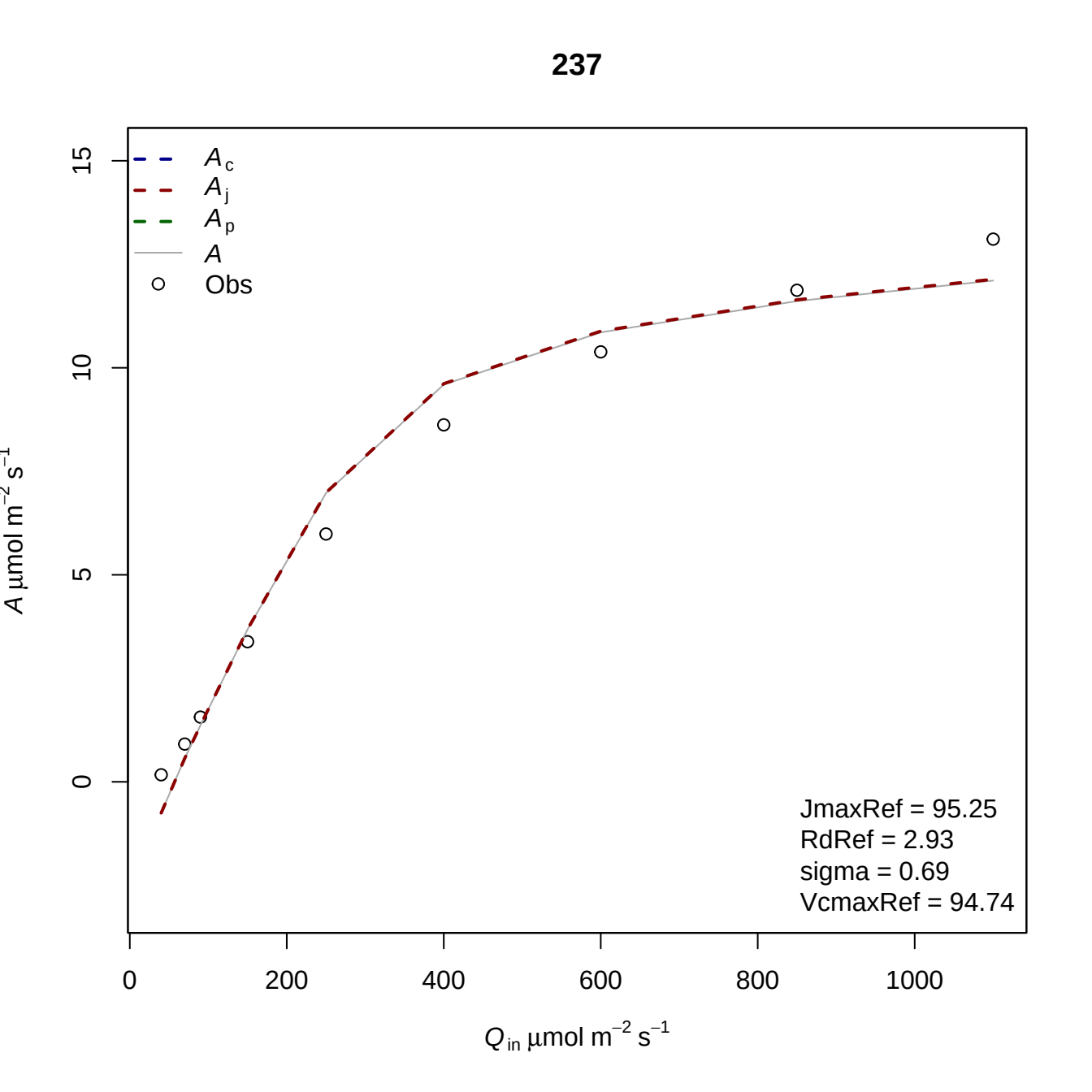

232
A_c
A_j
A_p
A
Obs
JmaxRef = 51.43
RdRef = 4.85
sigma = 0.6
VcmaxRef = 125.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

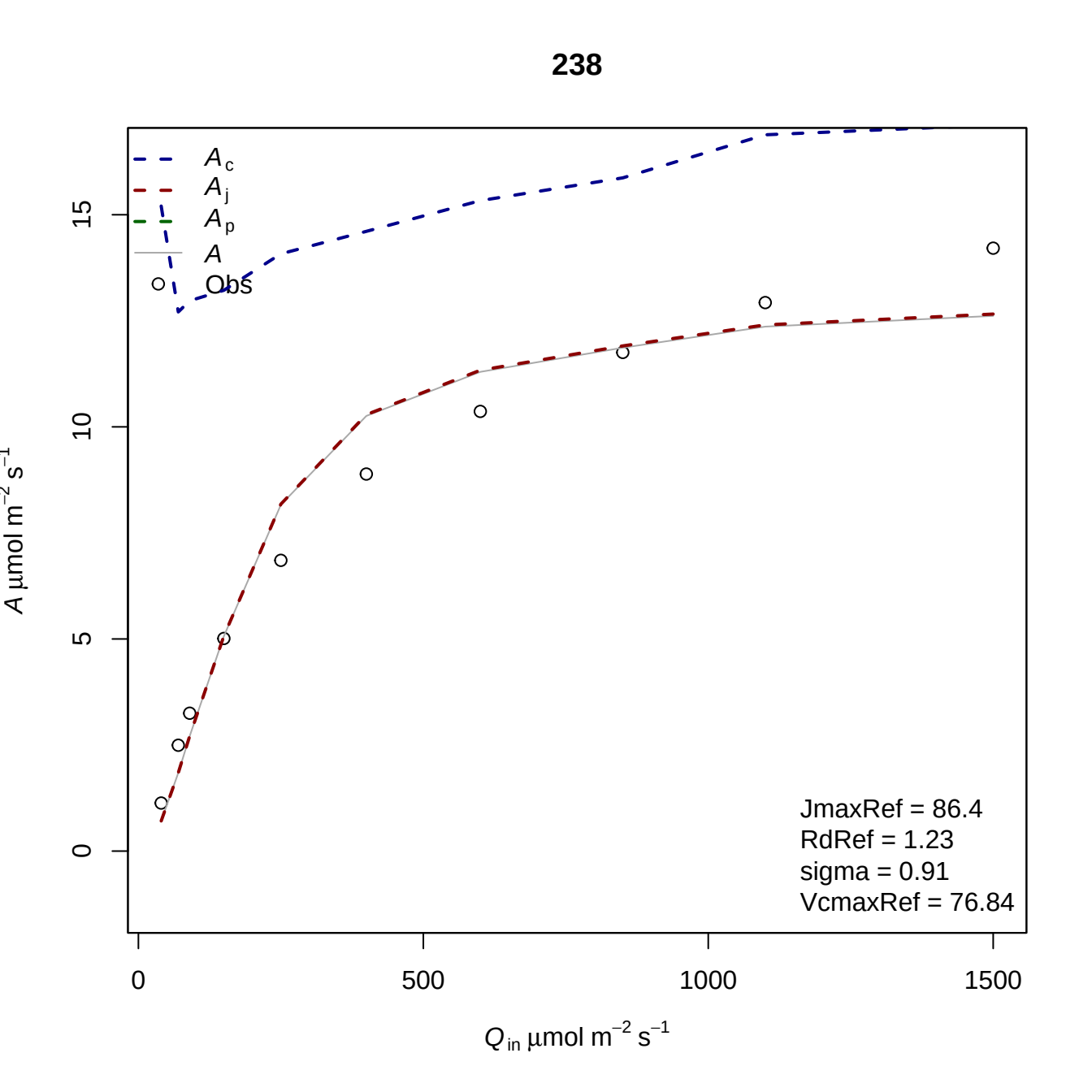

233
A_c
A_j
A_p
A
Obs
JmaxRef = 11.24
RdRef = 1.91
sigma = 0.19
VcmaxRef = 11270.99
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

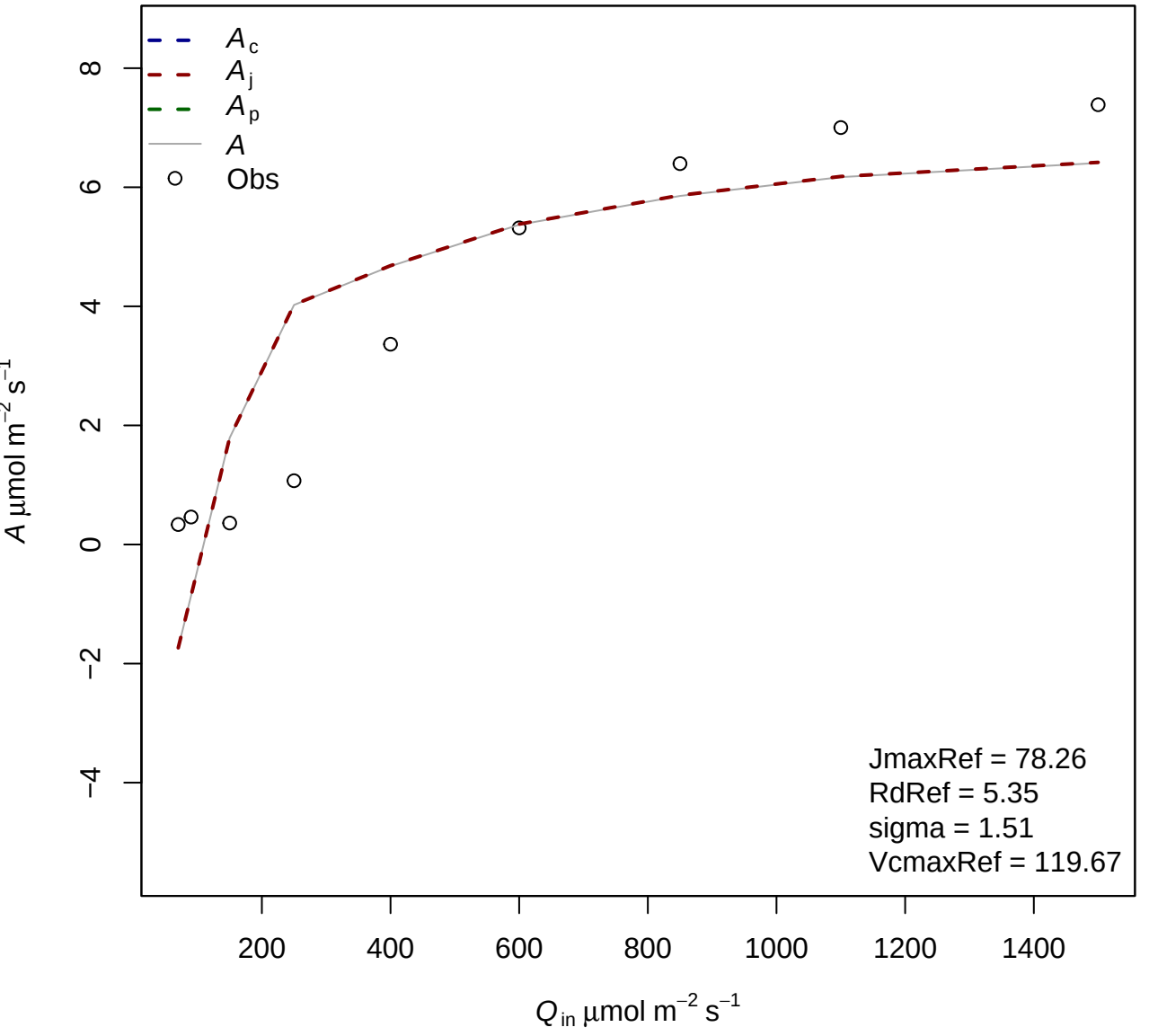

234
A_c
A_j
A_p
A
Obs
JmaxRef = 54.12
RdRef = 3.17
sigma = 0.81
VcmaxRef = 110.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

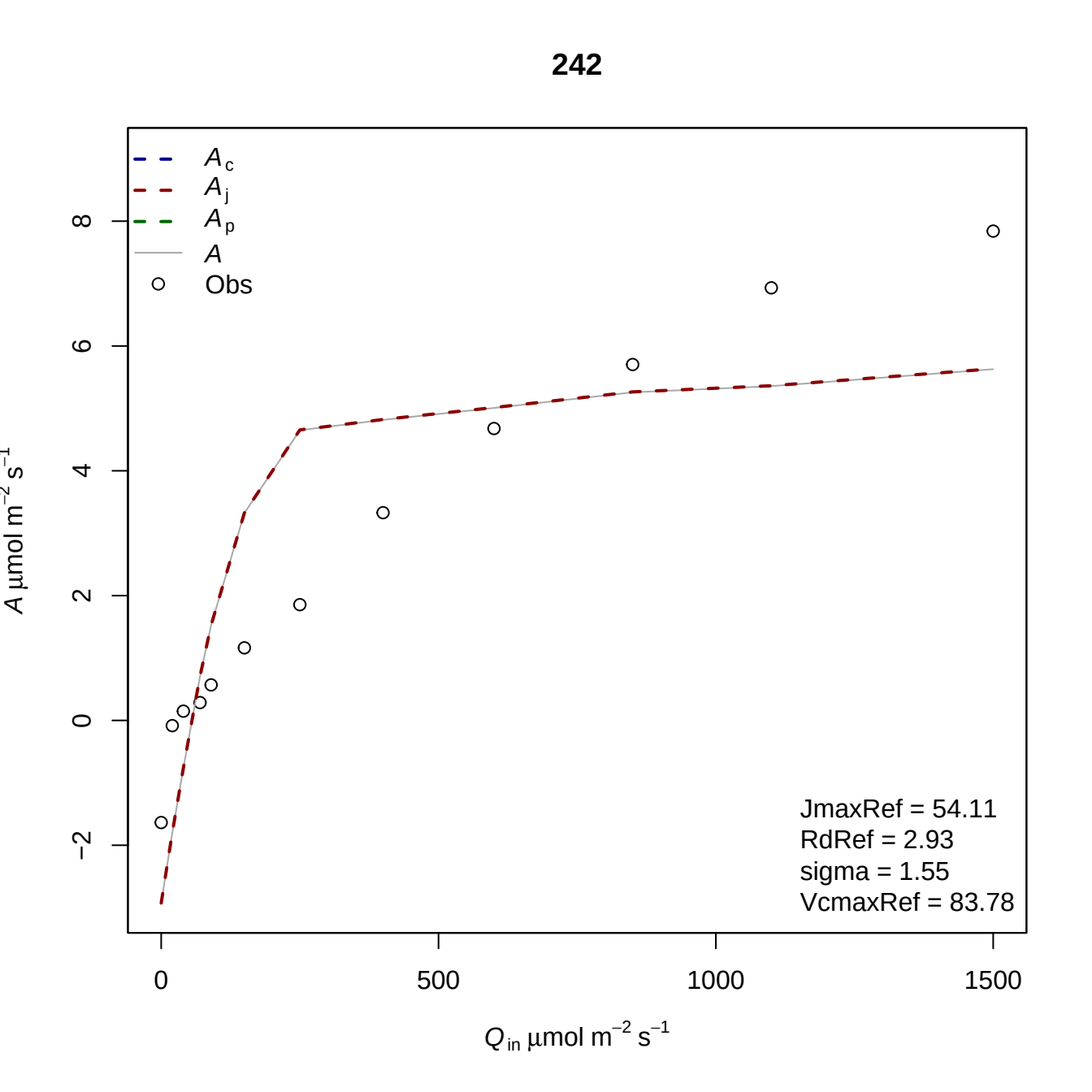

235
A_c
A_j
A_p
A
Obs
JmaxRef = 79.67
RdRef = 1.45
sigma = 0.38
VcmaxRef = 102.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000

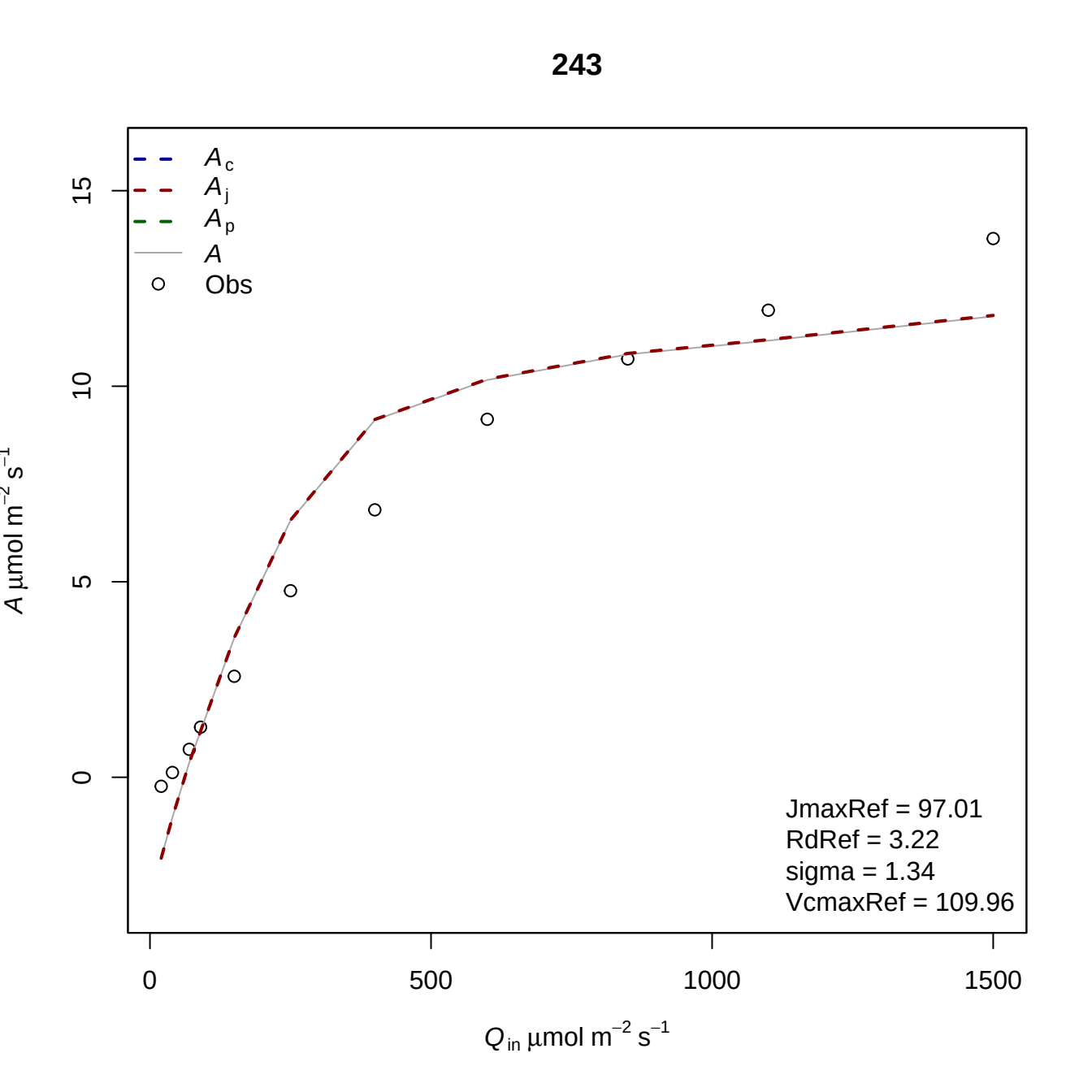

236
A_c
A_j
A_p
A
Obs
JmaxRef = 81.77
RdRef = 1.39
sigma = 0.28
VcmaxRef = 102.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

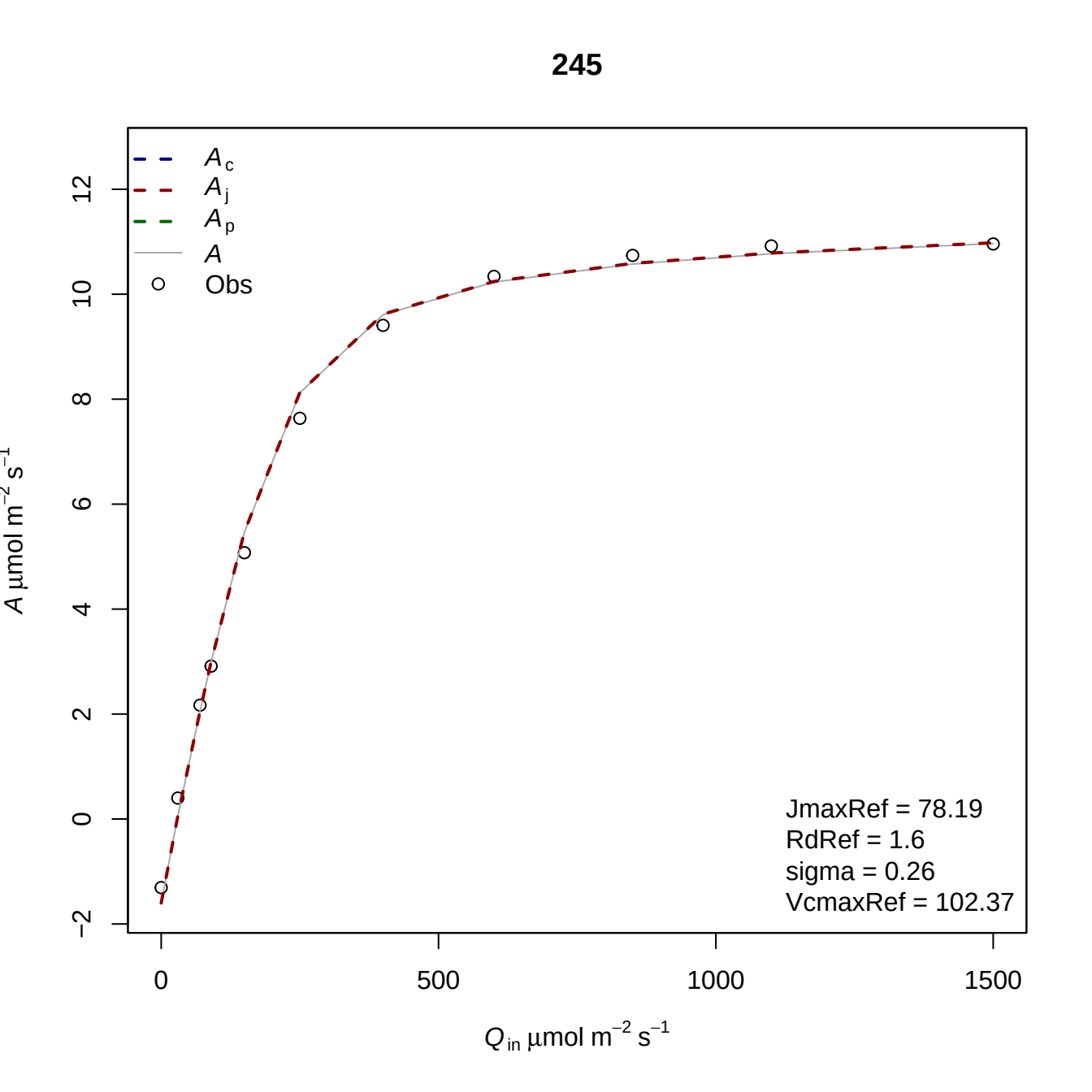

237
A_c
A_j
A_p
A
Obs
JmaxRef = 95.25
RdRef = 2.93
sigma = 0.69
VcmaxRef = 94.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

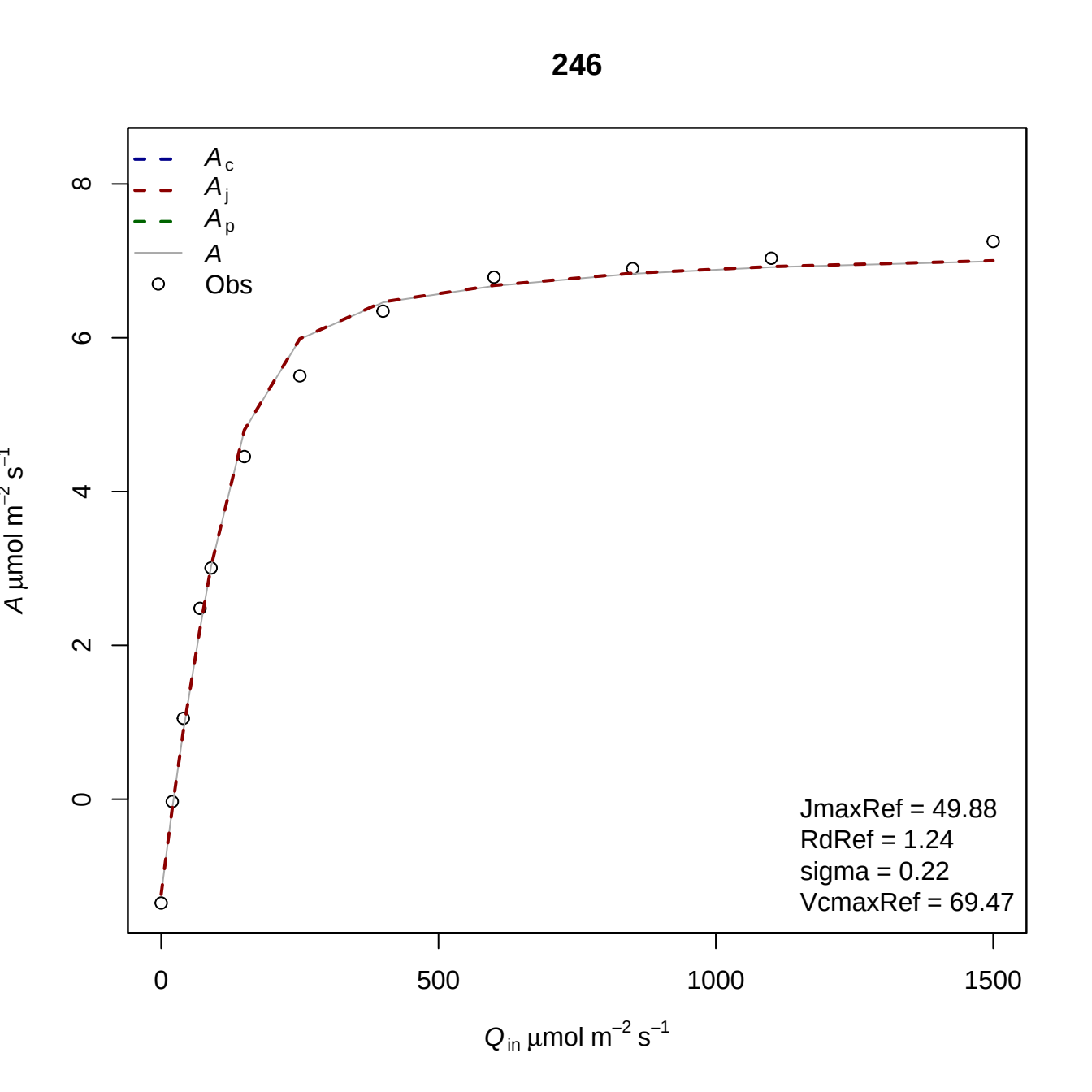

238
A_c
A_j
A_p
A
Obs
JmaxRef = 86.4
RdRef = 1.23
sigma = 0.91
VcmaxRef = 76.84
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

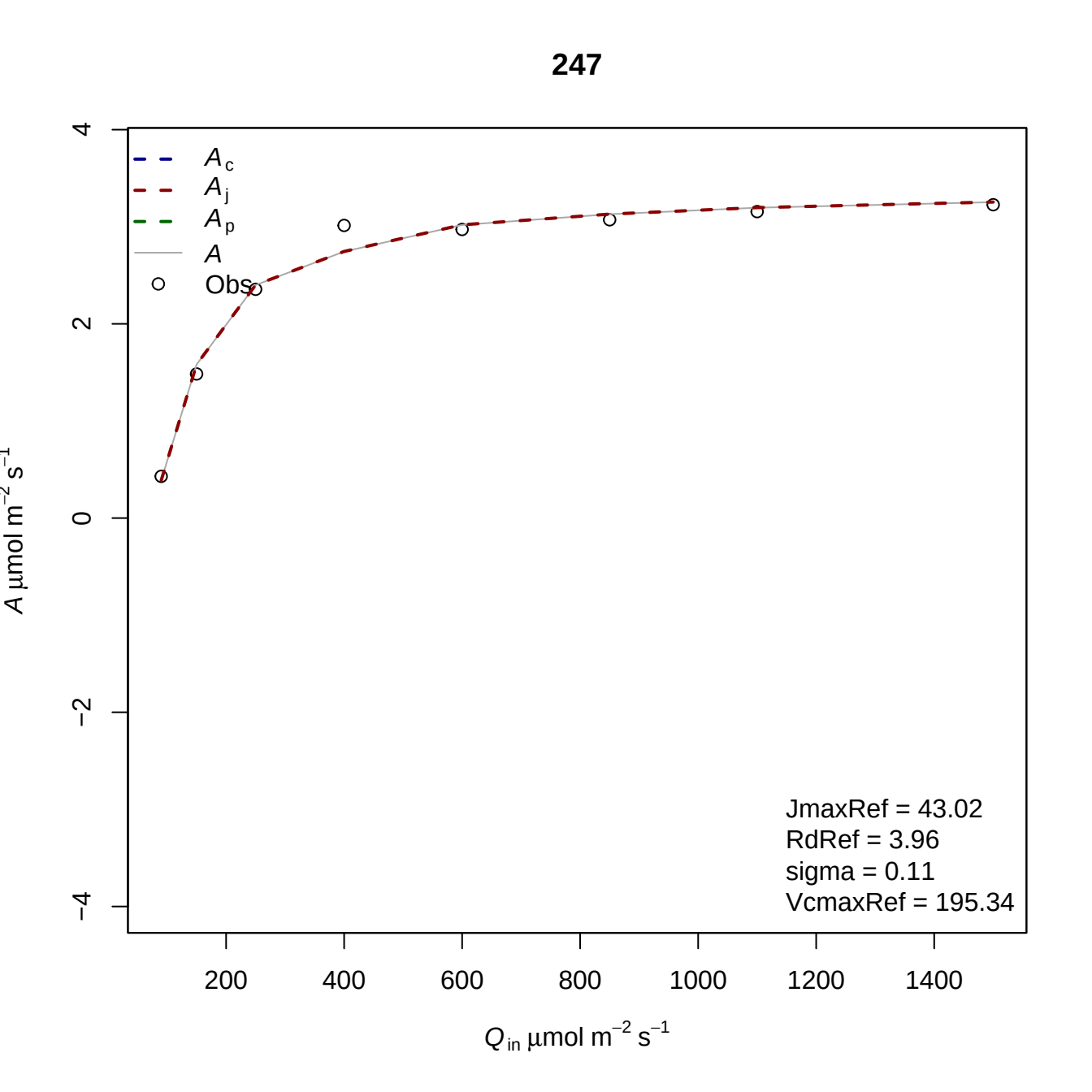

A_c
A_j
A_p
A
Obs
JmaxRef = 78.26
RdRef = 5.35
sigma = 1.51
VcmaxRef = 119.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

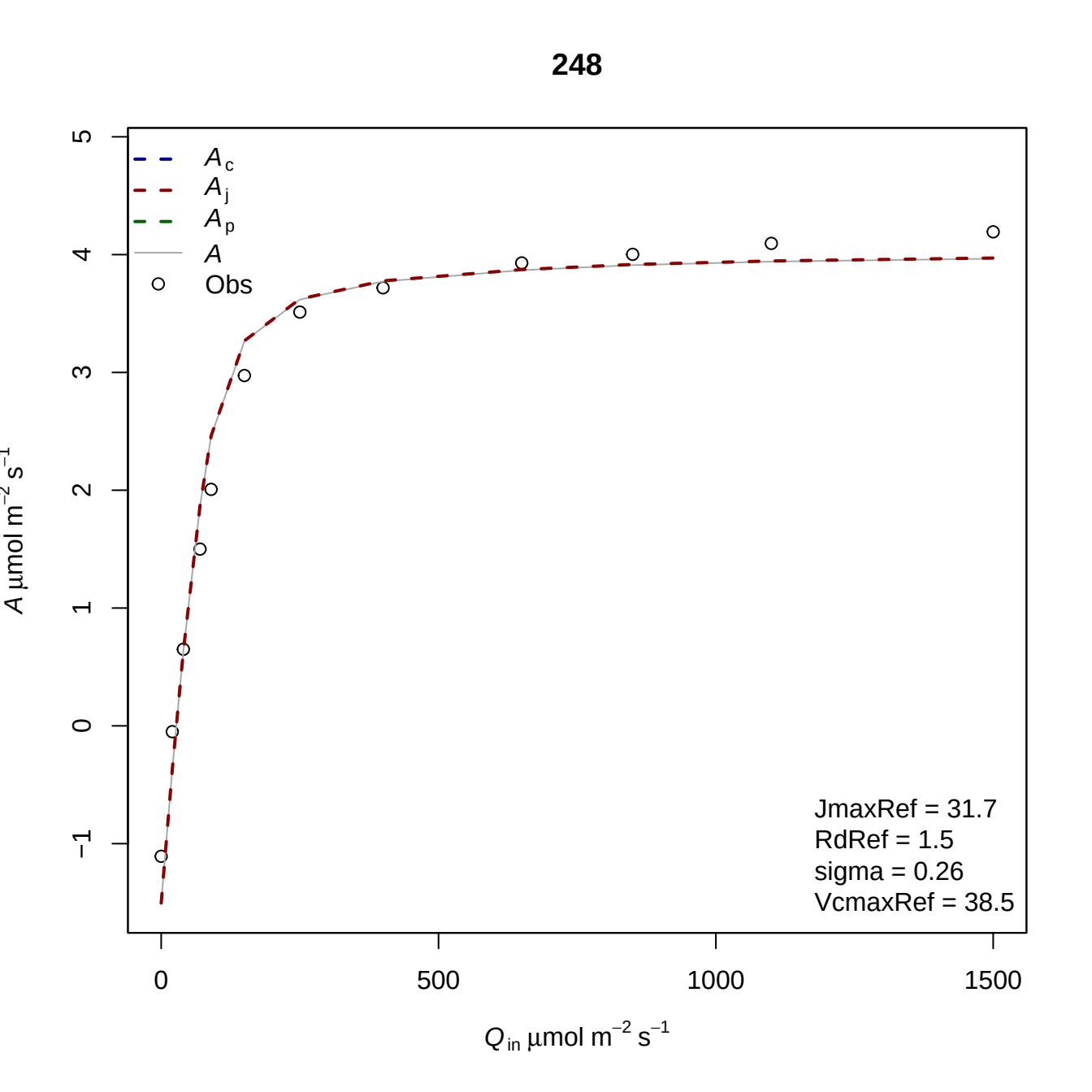

242
A_c
A_j
A_p
A
Obs
JmaxRef = 54.11
RdRef = 2.93
sigma = 1.55
VcmaxRef = 83.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

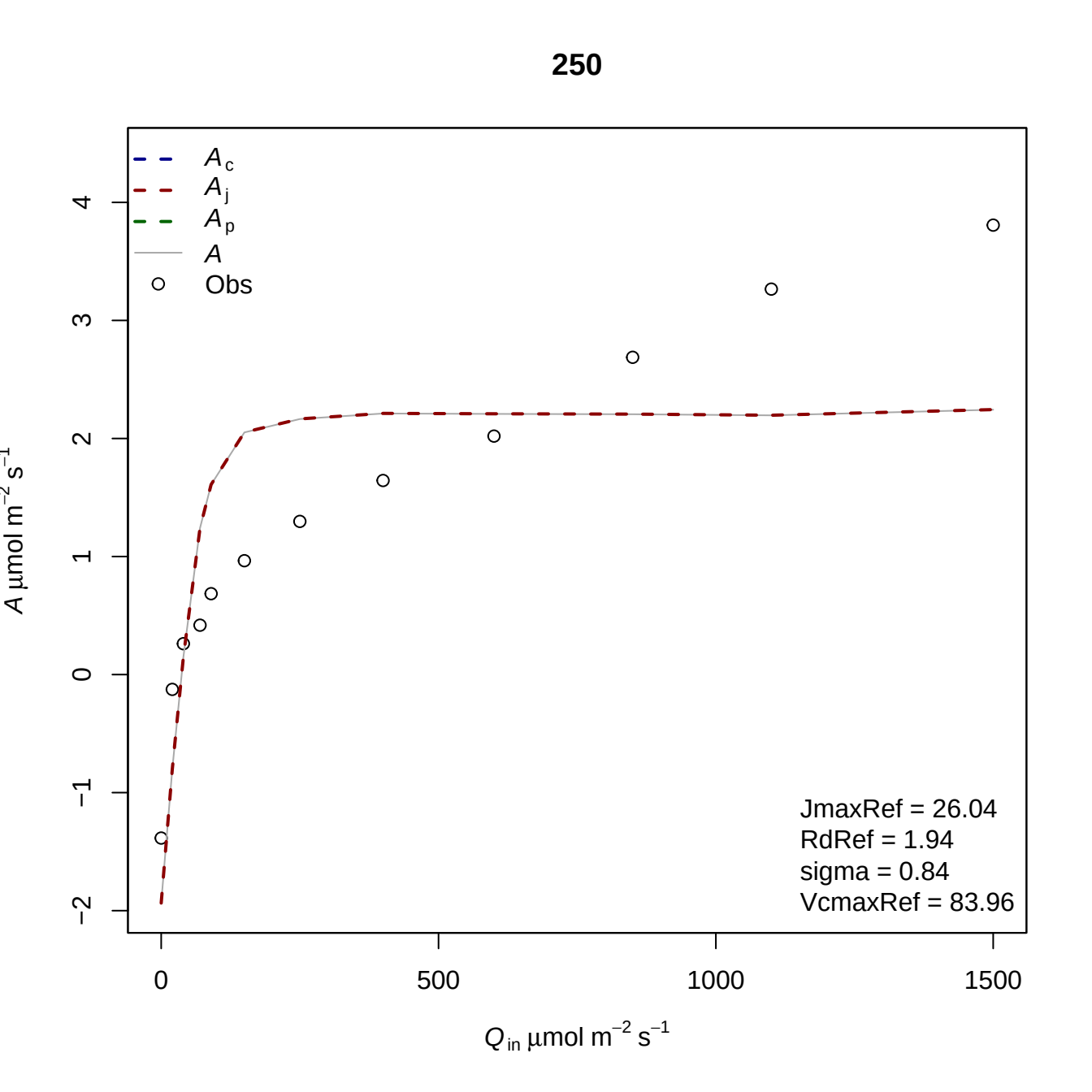

243
A_c
A_j
A_p
A
Obs
JmaxRef = 97.01
RdRef = 3.22
sigma = 1.34
VcmaxRef = 109.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

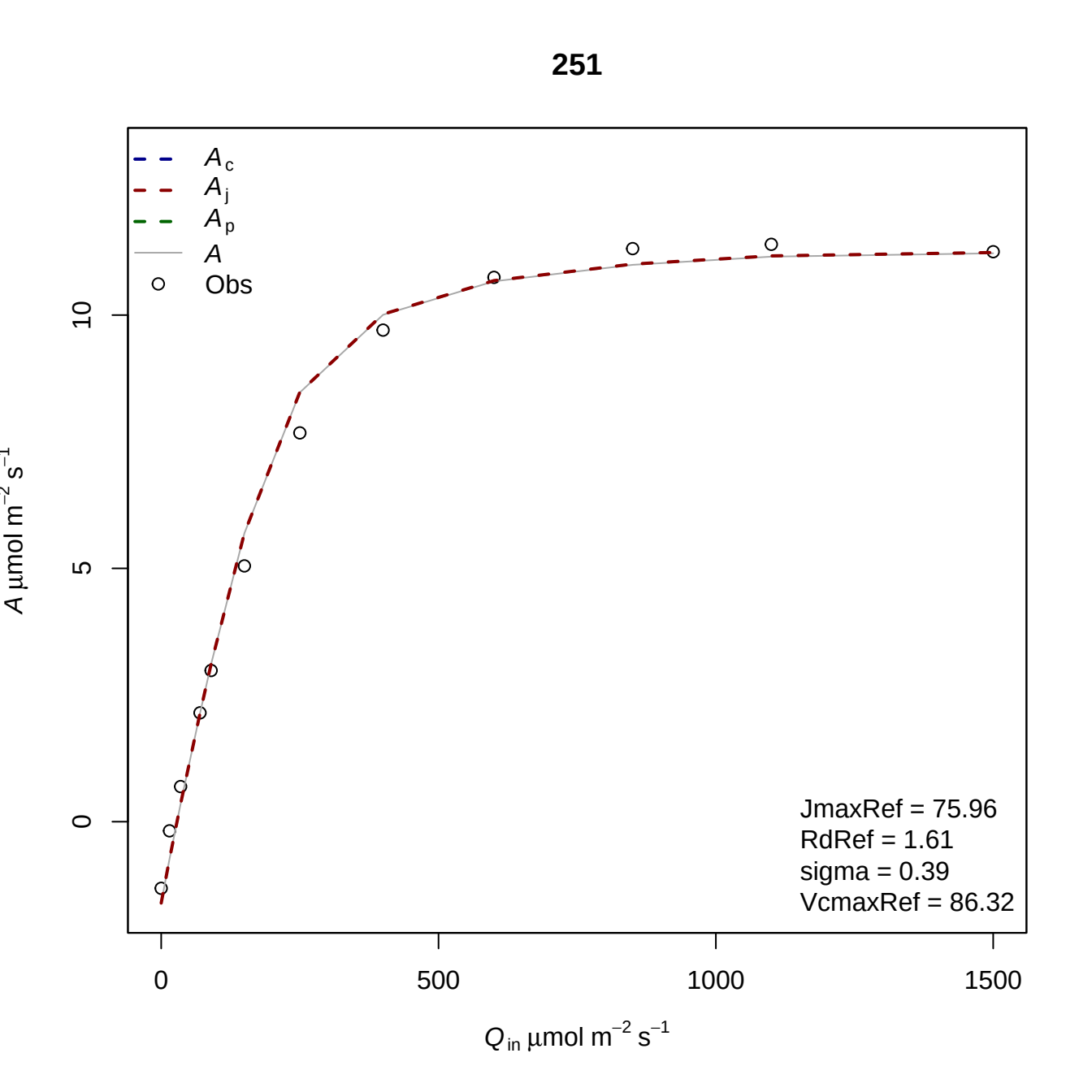

245
A_c
A_j
A_p
A
Obs
JmaxRef = 78.19
RdRef = 1.6
sigma = 0.26
VcmaxRef = 102.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

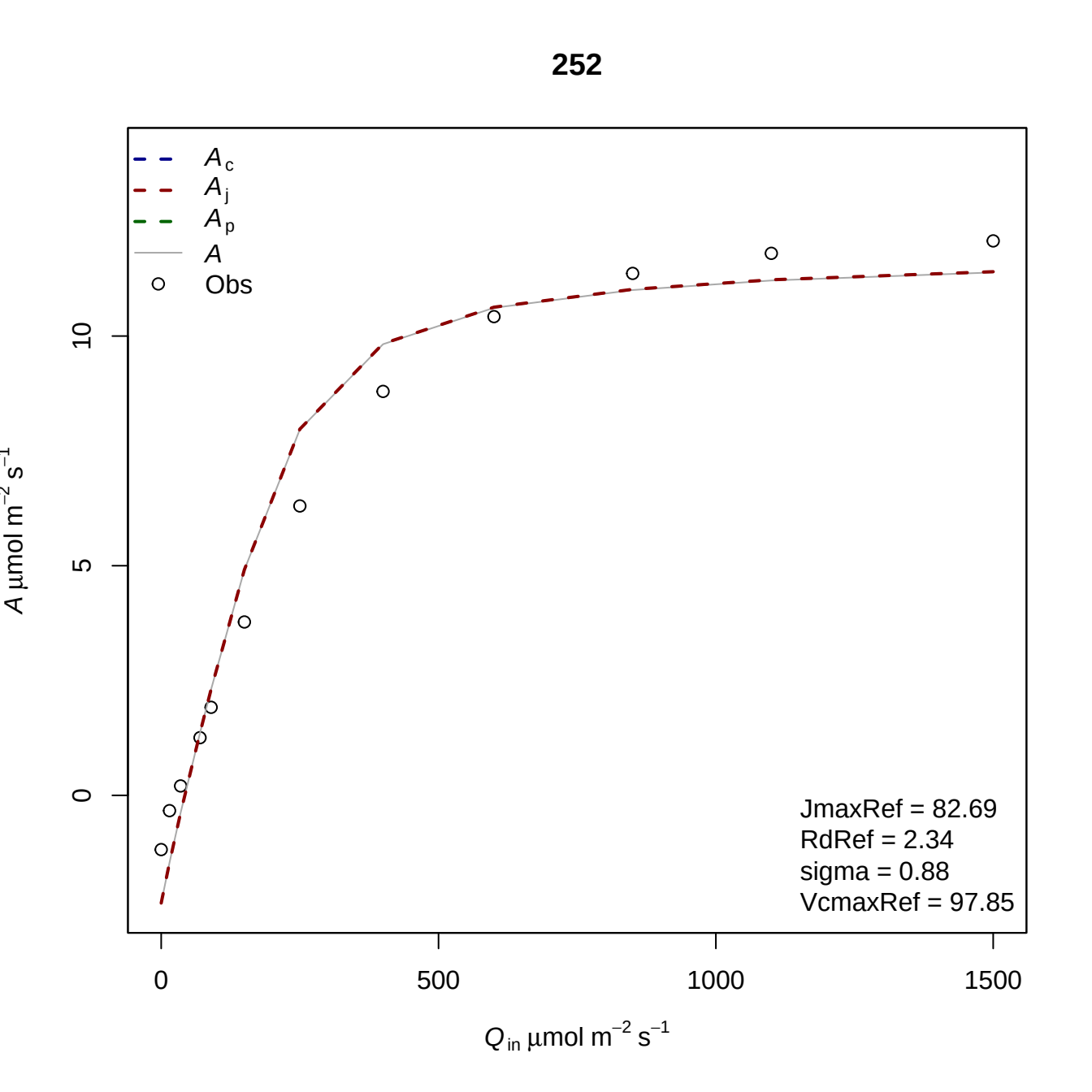

246
A_c
A_j
A_p
A
Obs
JmaxRef = 49.88
RdRef = 1.24
sigma = 0.22
VcmaxRef = 69.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

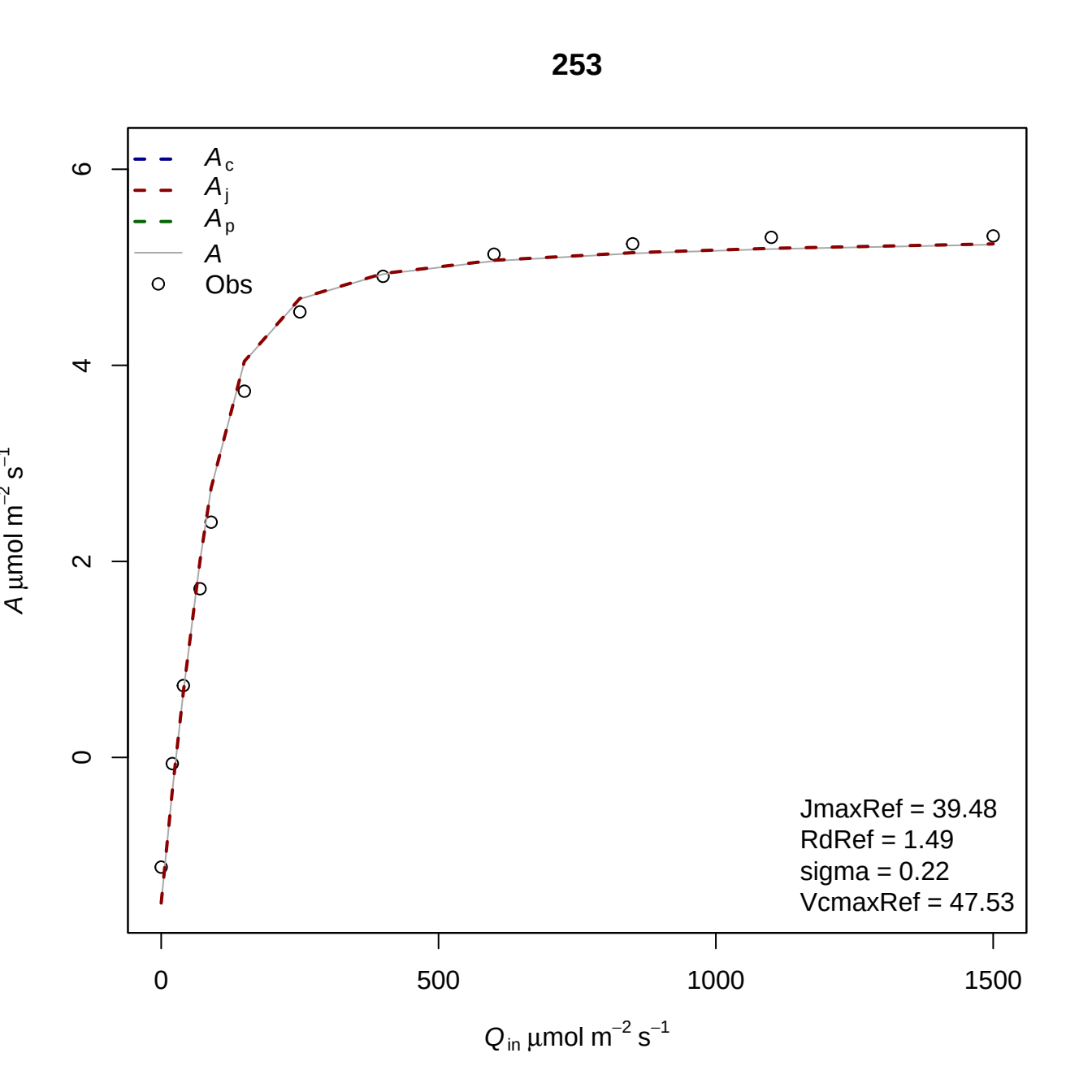

247
A_c
A_j
A_p
A
Obs
JmaxRef = 43.02
RdRef = 3.96
sigma = 0.11
VcmaxRef = 195.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

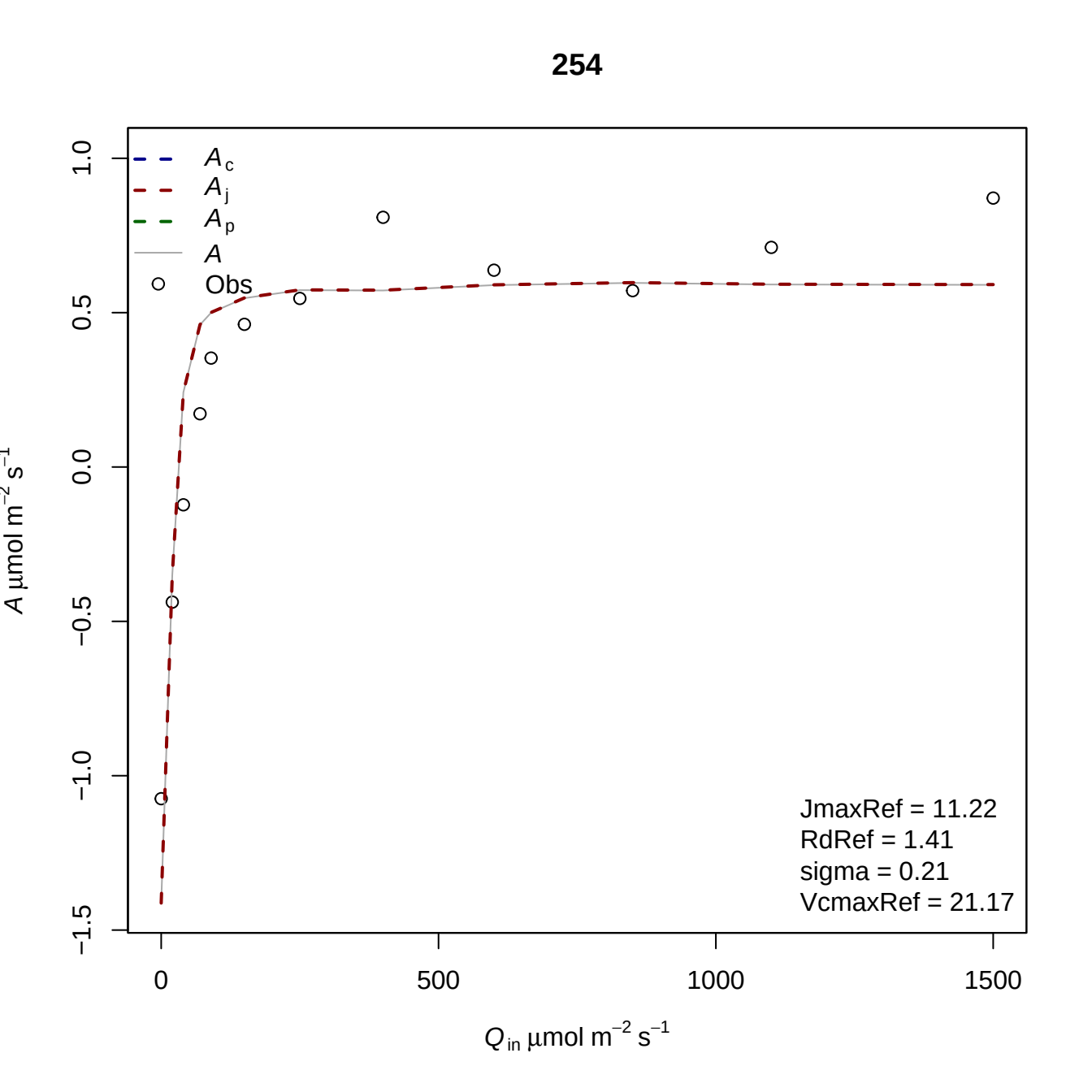

248
A_c
A_j
A_p
A
Obs
JmaxRef = 31.7
RdRef = 1.5
sigma = 0.26
VcmaxRef = 38.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500

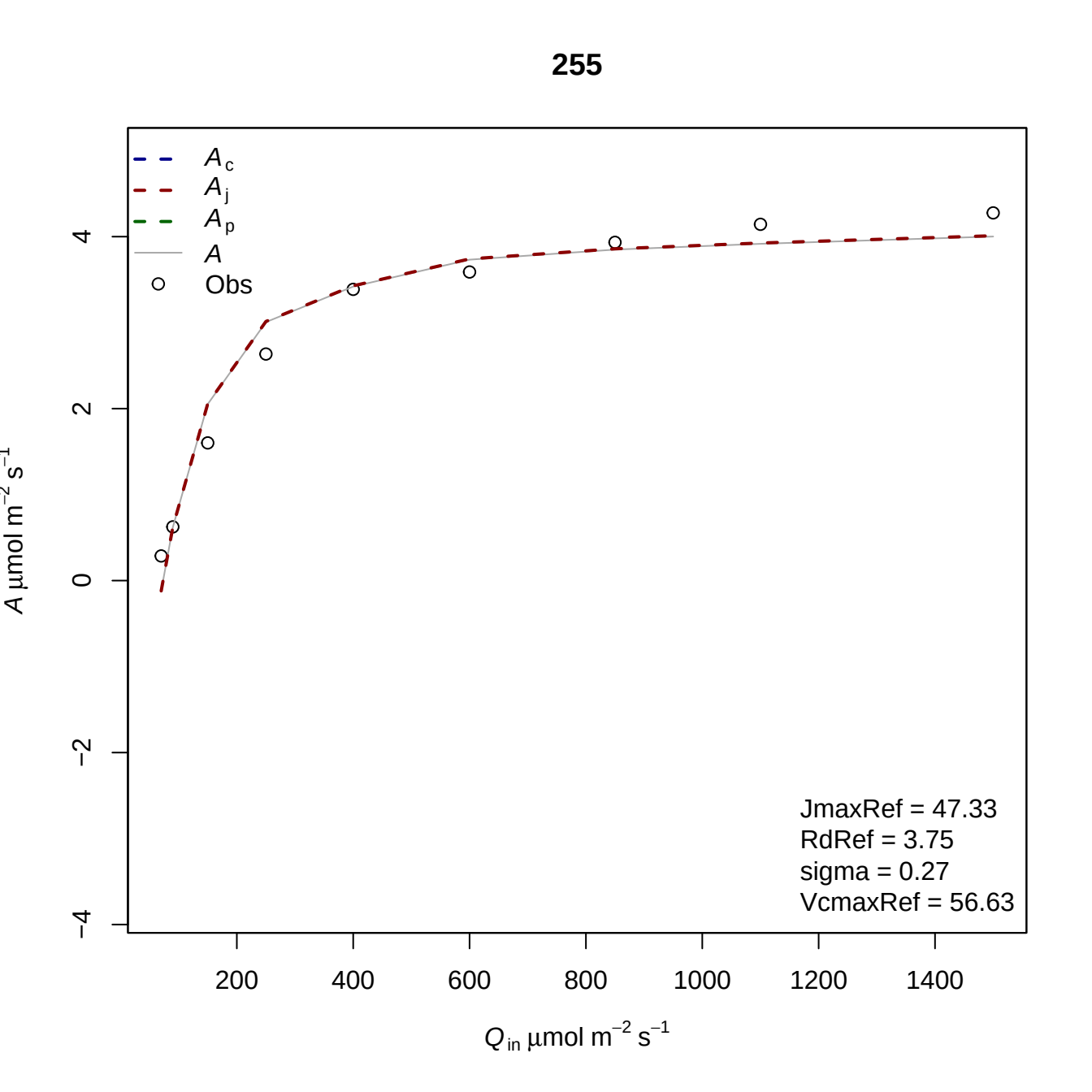

250
A_c
A_j
A_p
A
Obs
JmaxRef = 26.04
RdRef = 1.94
sigma = 0.84
VcmaxRef = 83.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

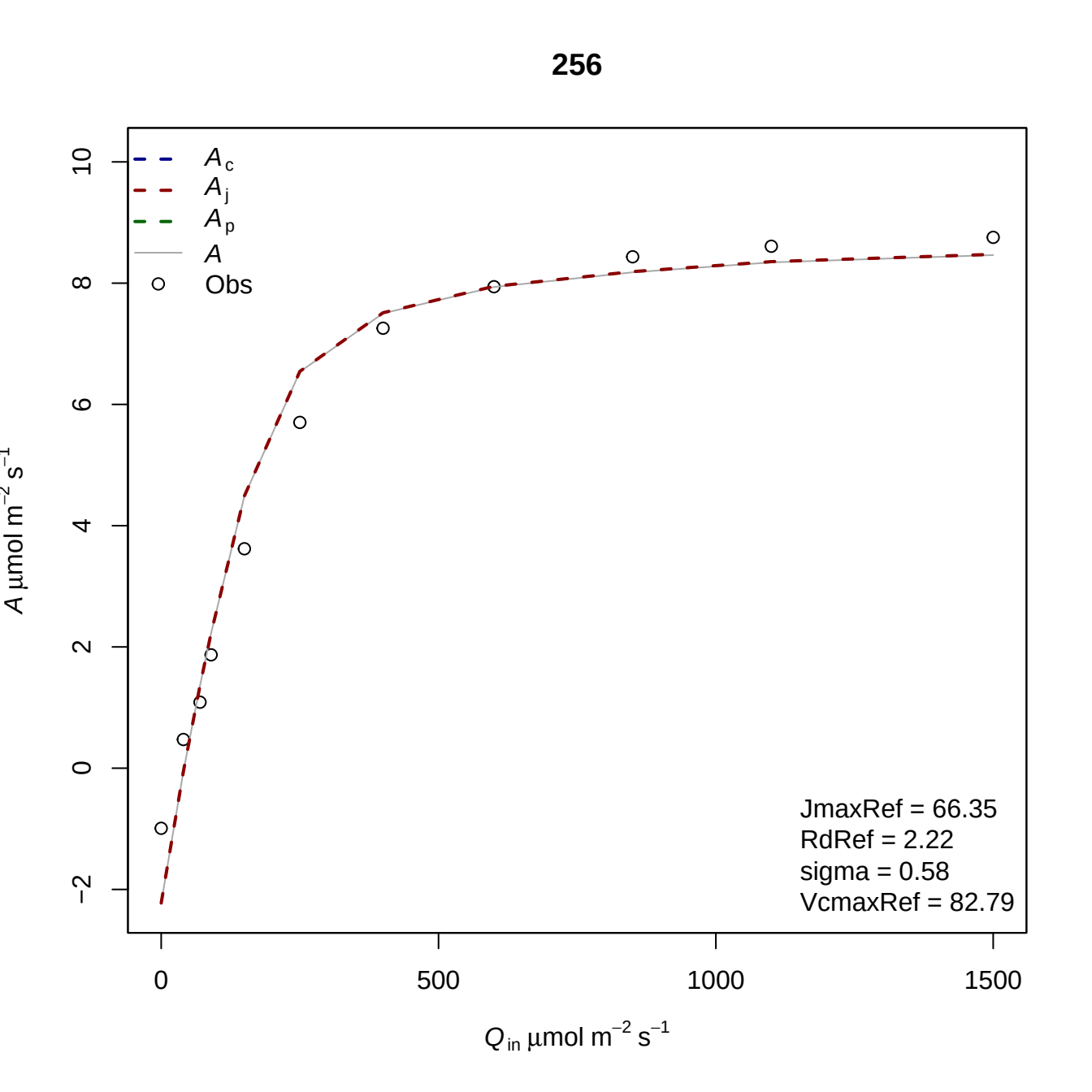

251
A_c
A_j
A_p
A
Obs
JmaxRef = 75.96
RdRef = 1.61
sigma = 0.39
VcmaxRef = 86.32
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

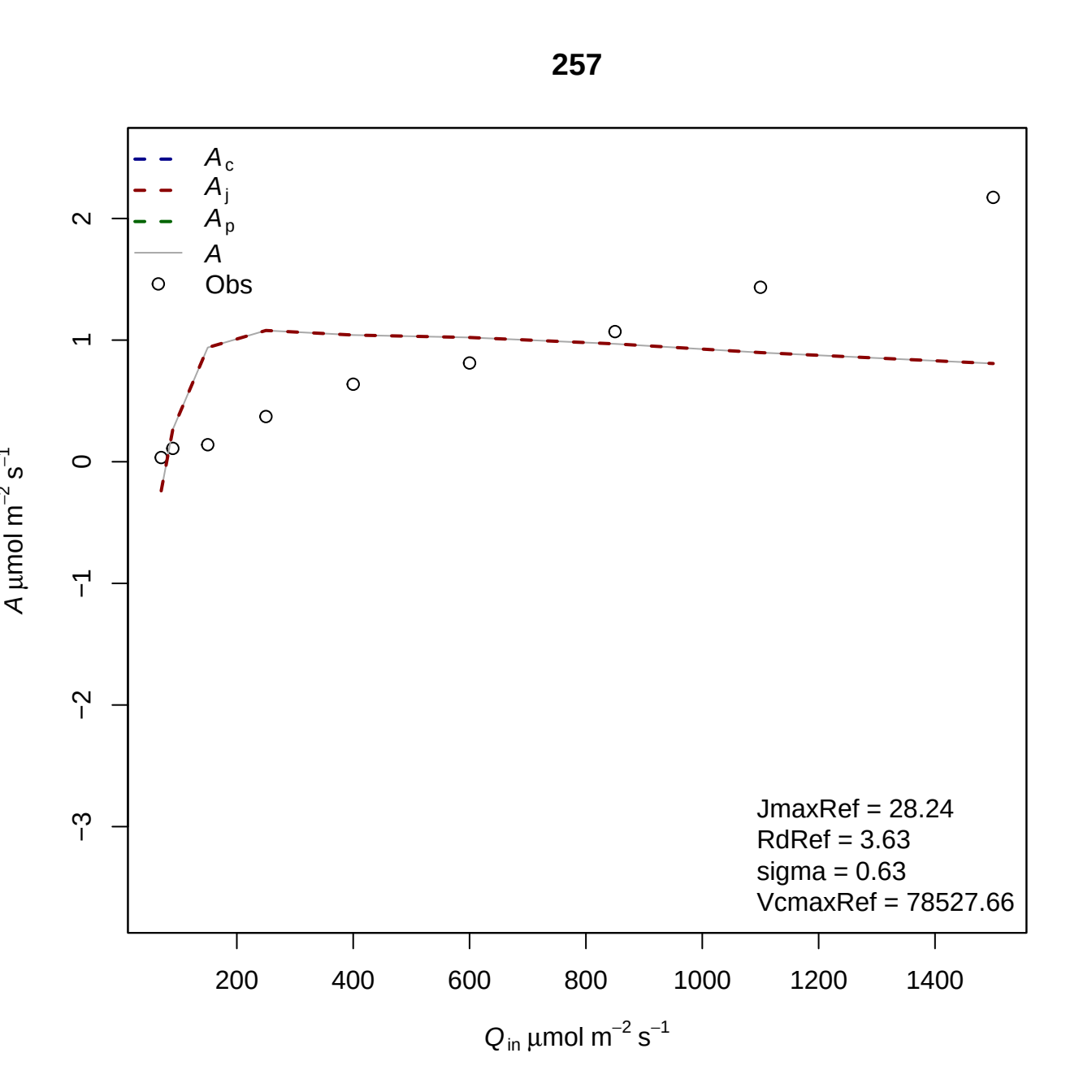

252
A_c
A_j
A_p
A
Obs
JmaxRef = 82.69
RdRef = 2.34
sigma = 0.88
VcmaxRef = 97.85
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

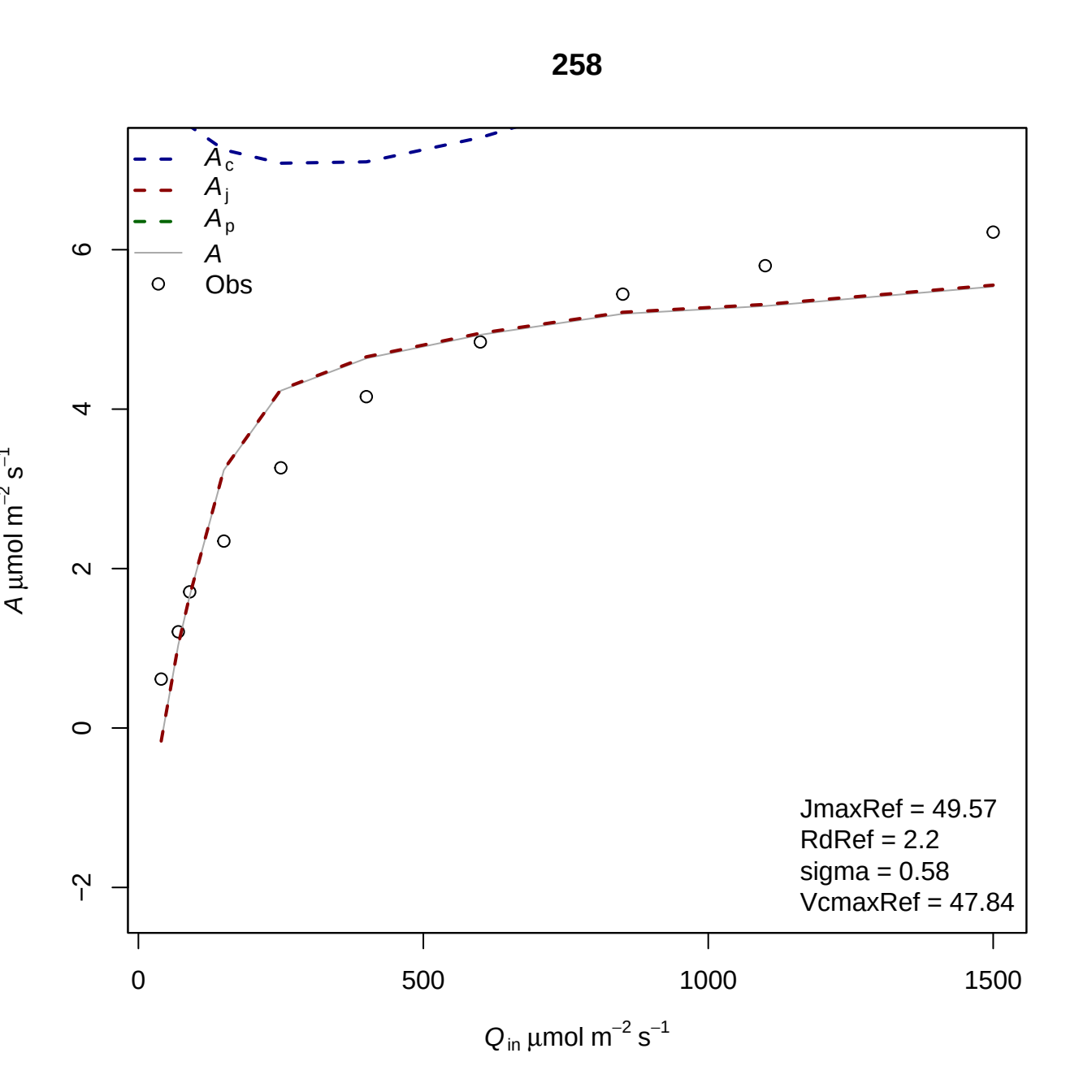

253
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 39.48
RdRef = 1.49
sigma = 0.22
VcmaxRef = 47.53

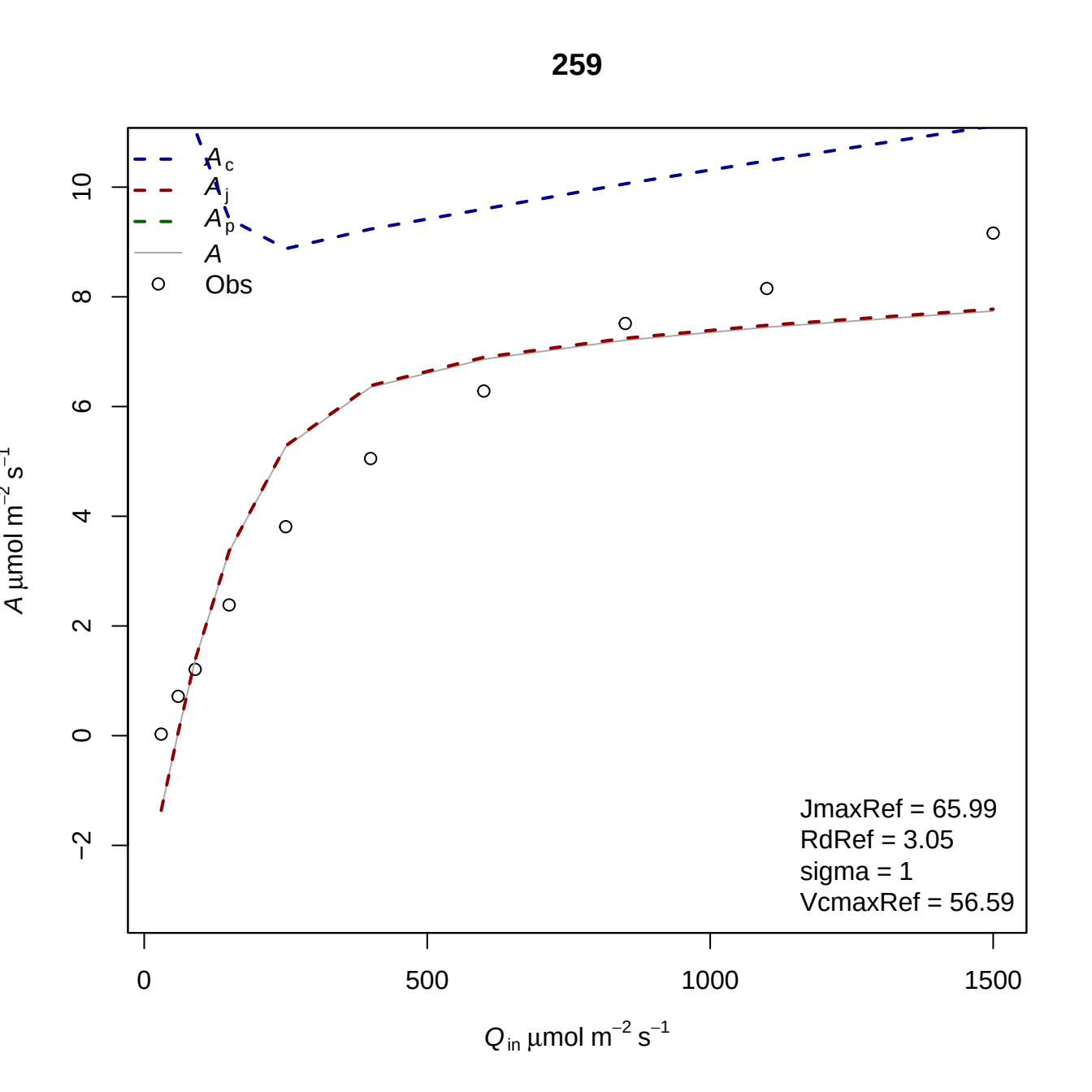

254
A_c
A_j
A_p
A
Obs
JmaxRef = 11.22
RdRef = 1.41
sigma = 0.21
VcmaxRef = 21.17
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

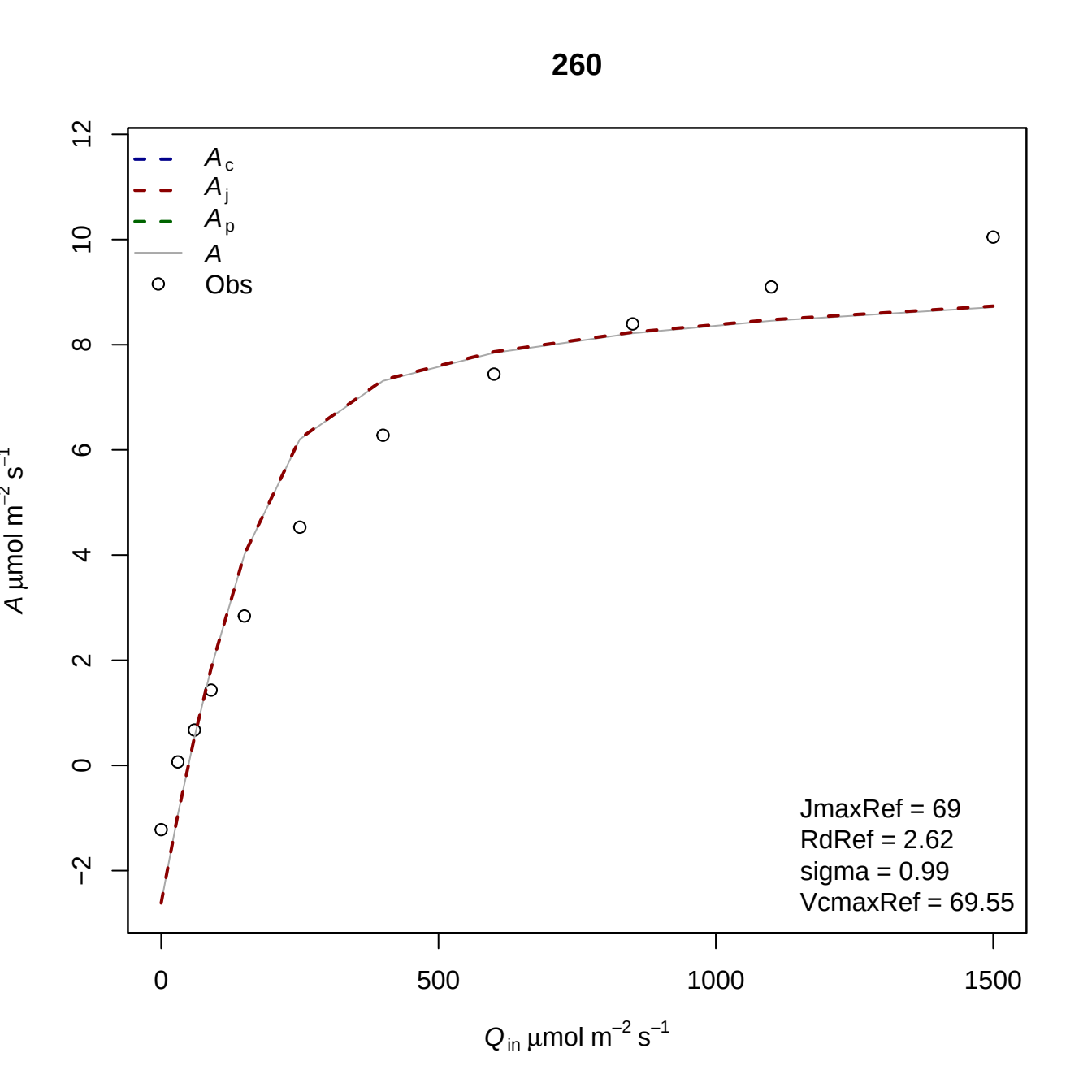

255
A_c
A_j
A_p
A
Obs
JmaxRef = 47.33
RdRef = 3.75
sigma = 0.27
VcmaxRef = 56.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

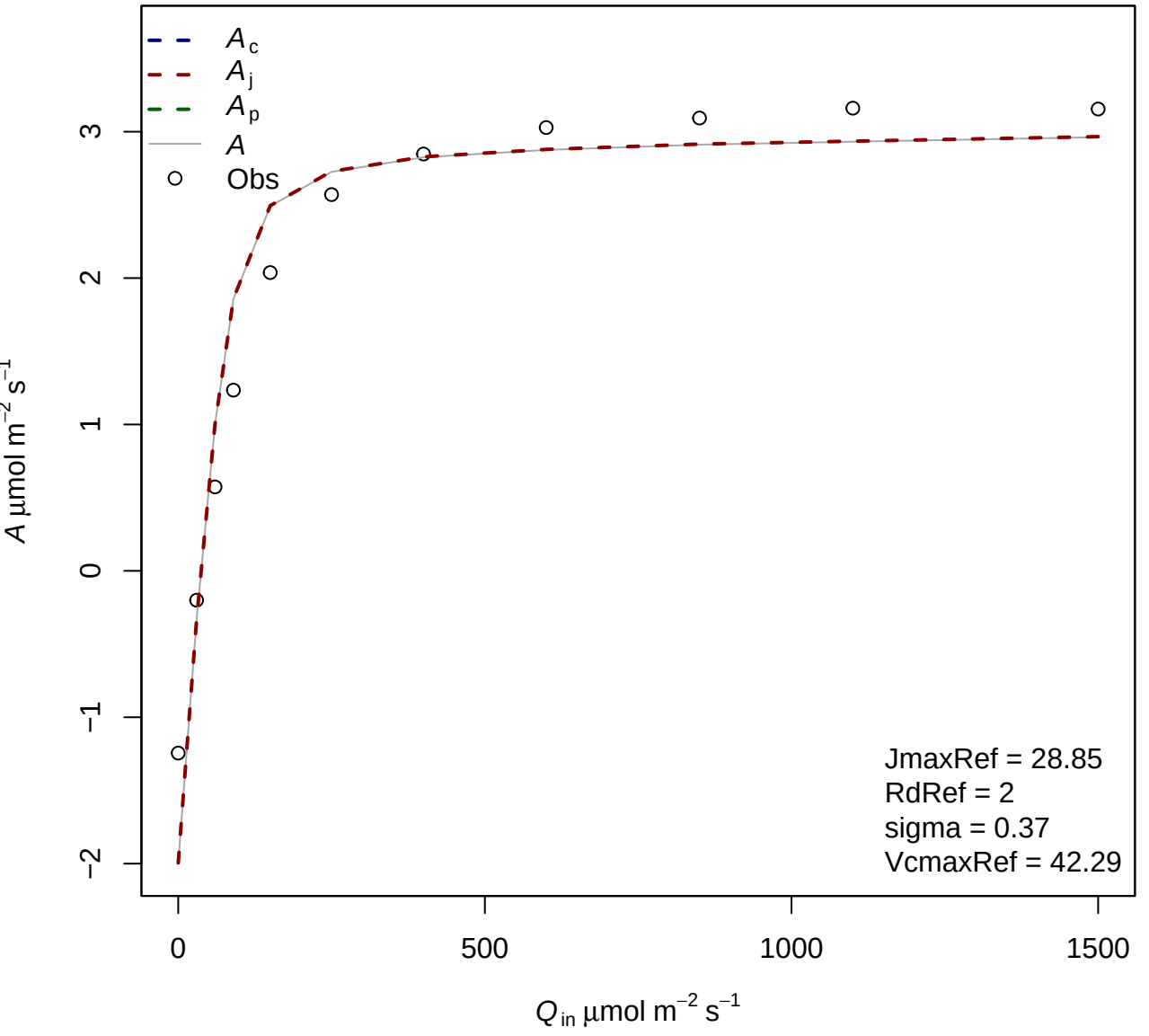

256
A_c
A_j
A_p
A
Obs
JmaxRef = 66.35
RdRef = 2.22
sigma = 0.58
VcmaxRef = 82.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

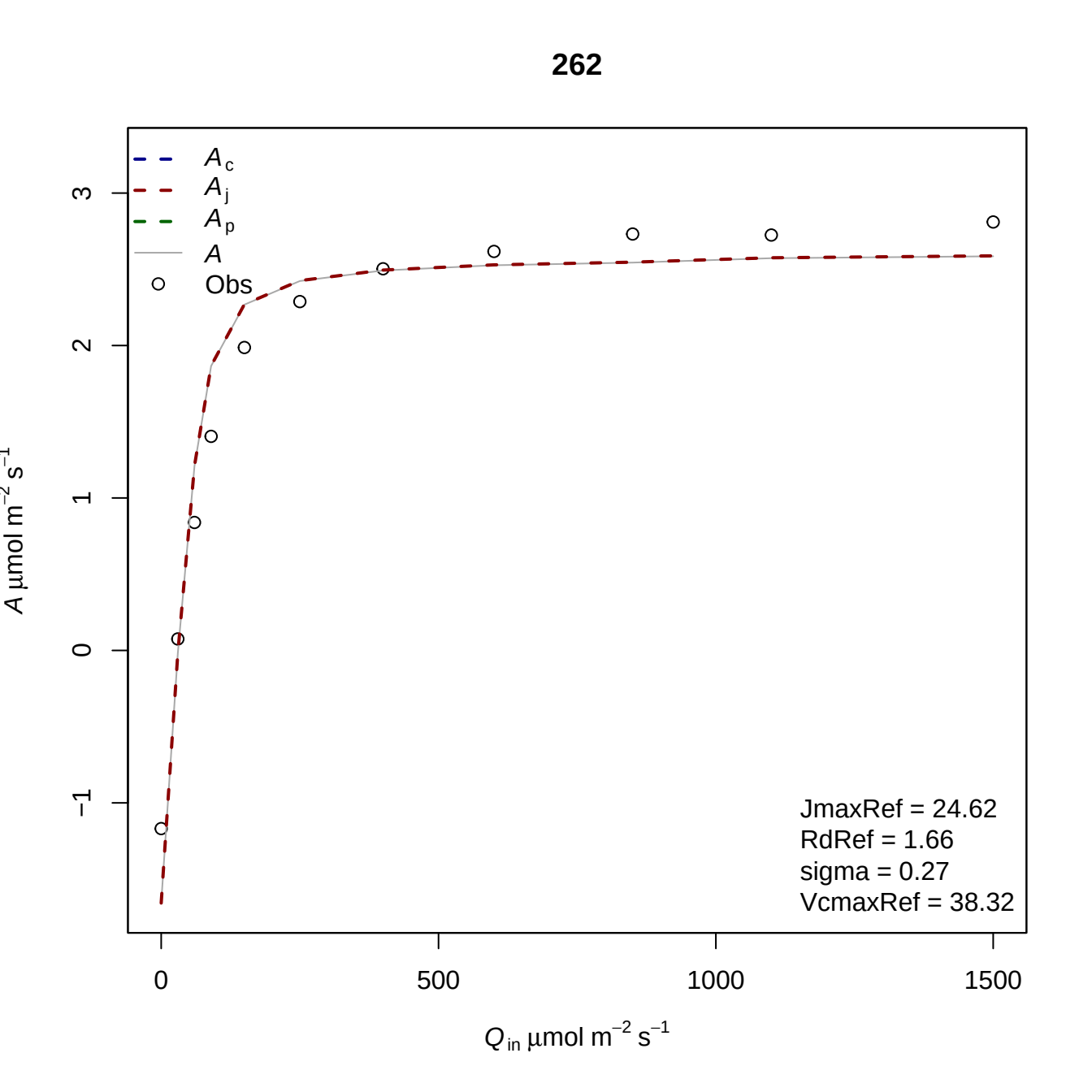

257
A_c
A_j
A_p
A
Obs
JmaxRef = 28.24
RdRef = 3.63
sigma = 0.63
VcmaxRef = 78527.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
200
400
600
800
1000
1200
1400

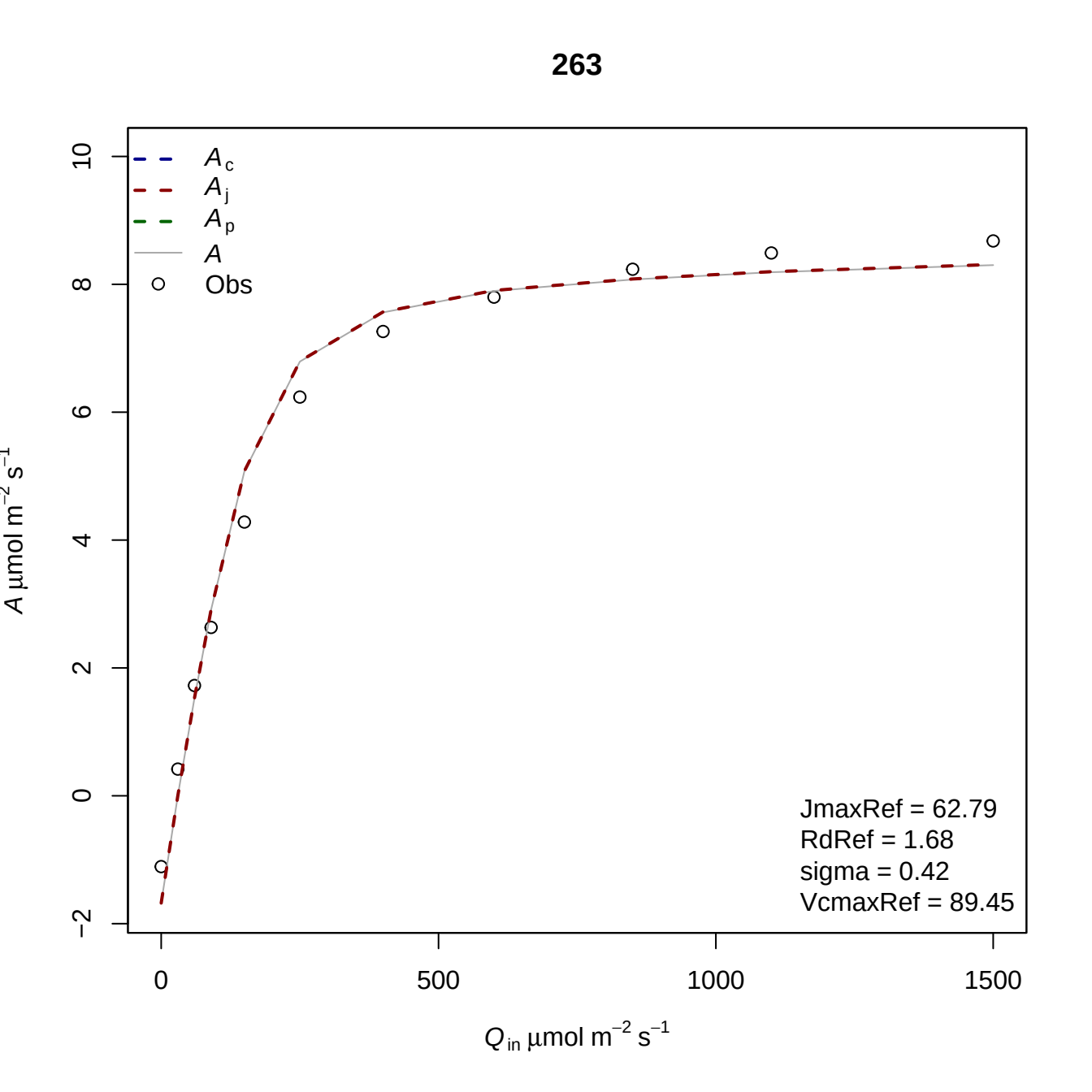

258
A_c
A_j
A_p
A
Obs
JmaxRef = 49.57
RdRef = 2.2
sigma = 0.58
VcmaxRef = 47.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

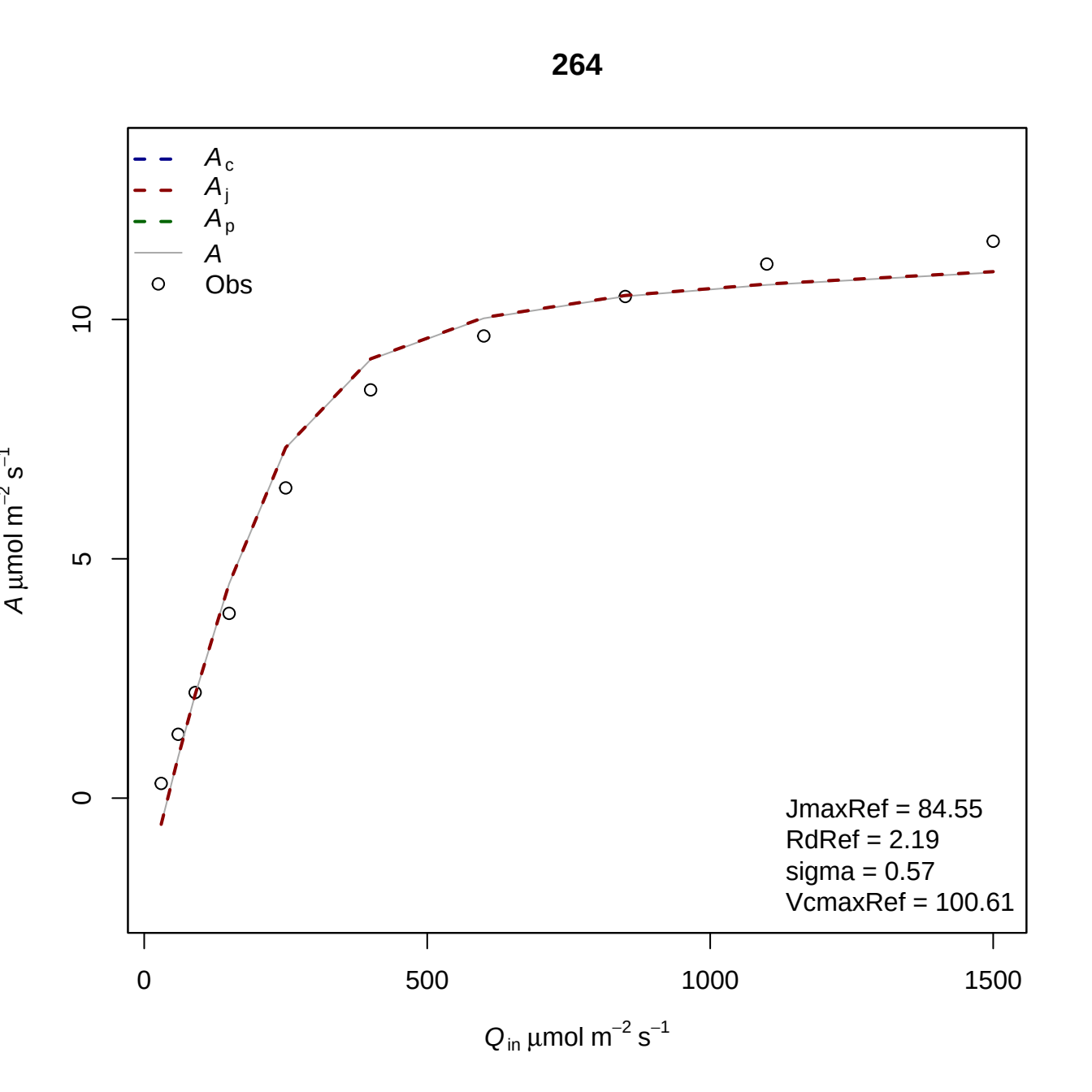

259
A_c
A_j
A_p
A
Obs
JmaxRef = 65.99
RdRef = 3.05
sigma = 1
VcmaxRef = 56.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

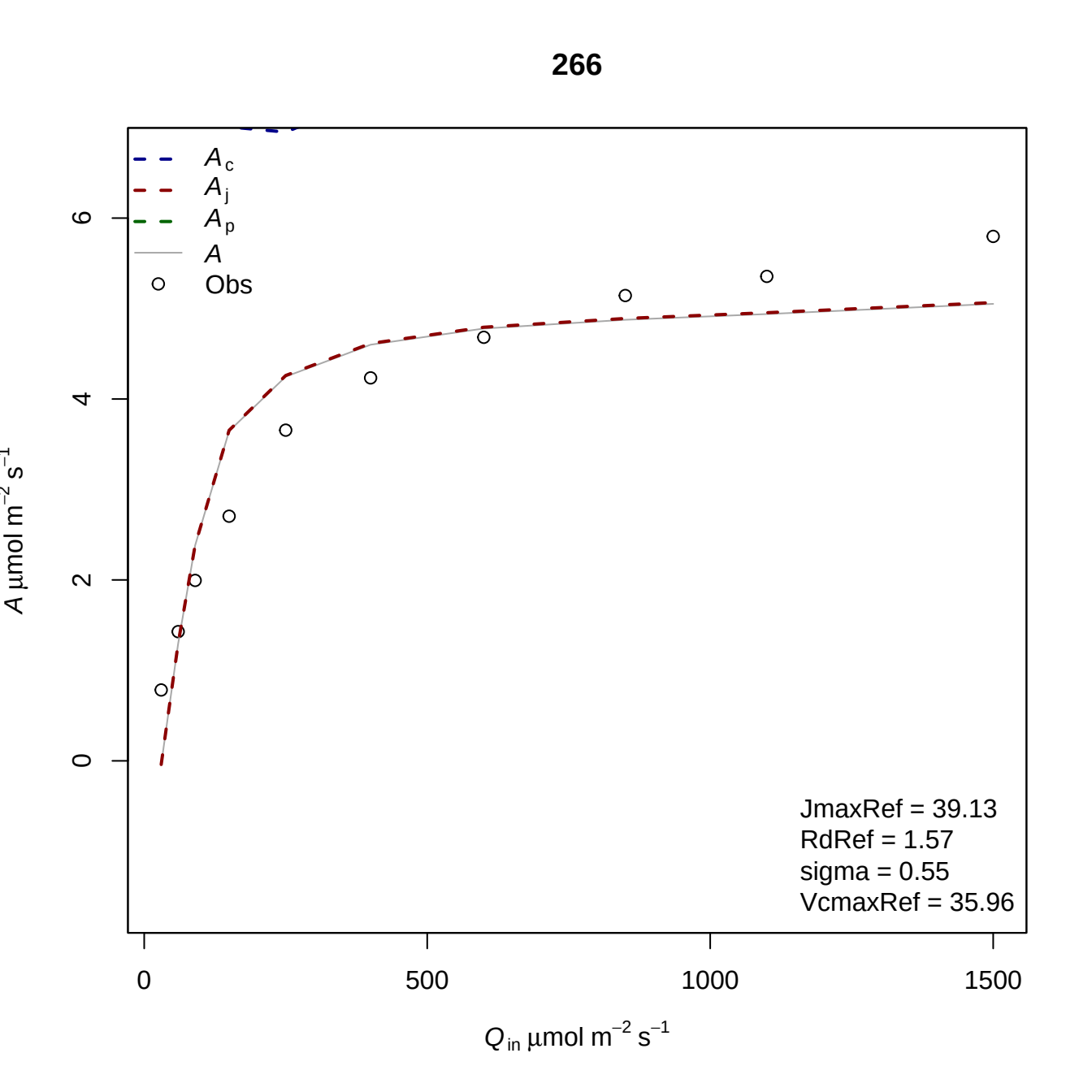

260
A_c
A_j
A_p
A
Obs
JmaxRef = 69
RdRef = 2.62
sigma = 0.99
VcmaxRef = 69.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

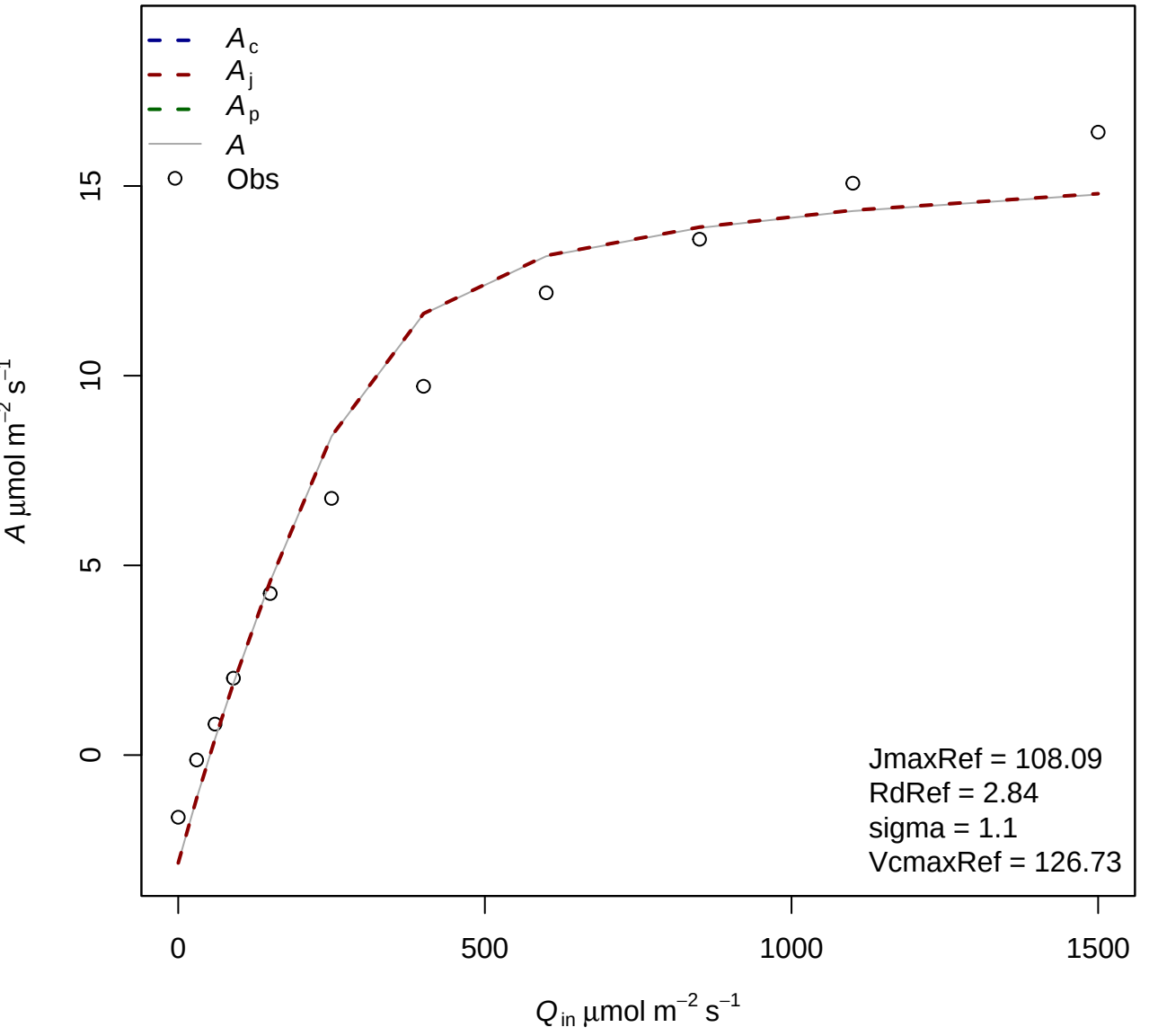

A_c
A_j
A_p
A
Obs
JmaxRef = 28.85
RdRef = 2
sigma = 0.37
VcmaxRef = 42.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
500
1000
1500

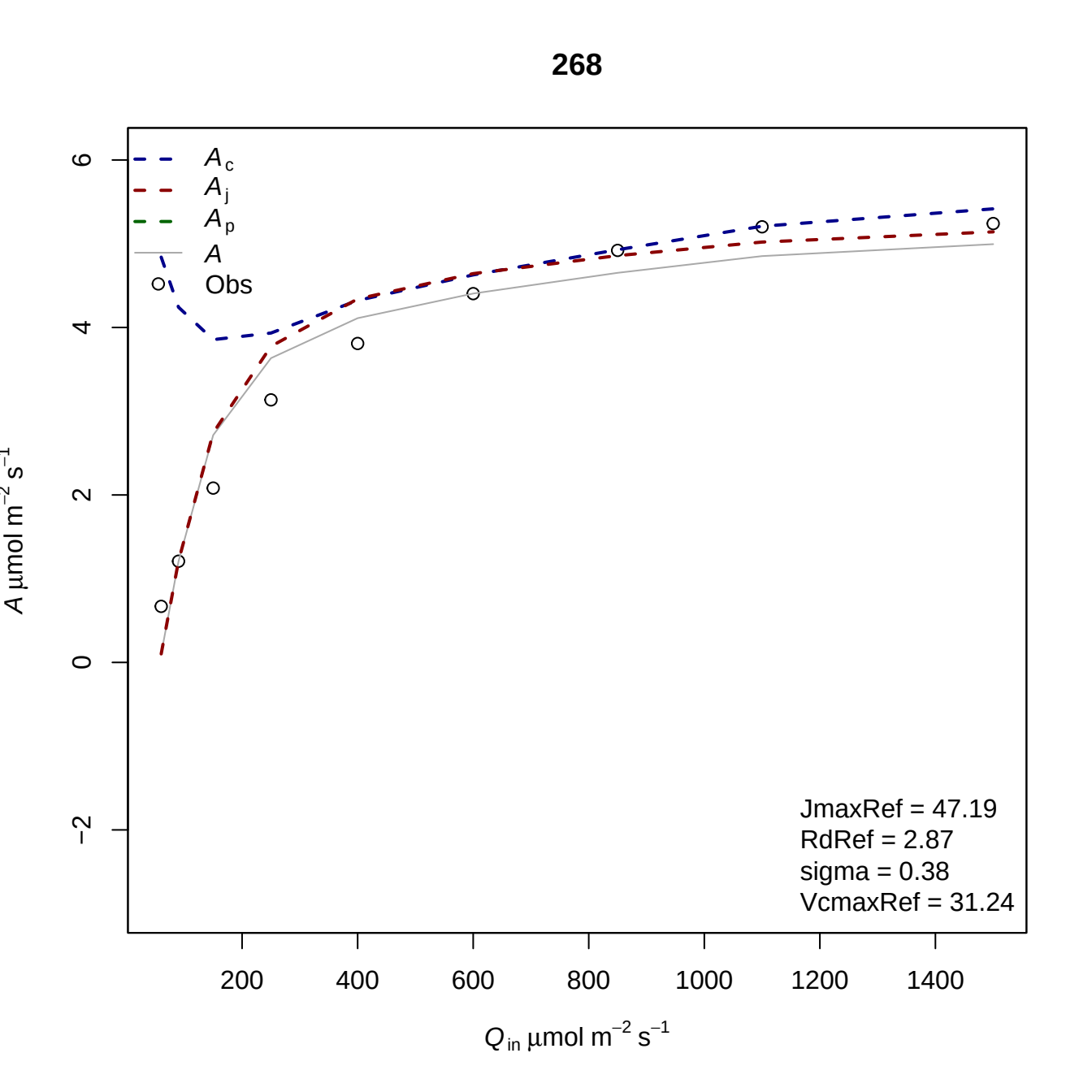

262
A_c
A_j
A_p
A
Obs
JmaxRef = 24.62
RdRef = 1.66
sigma = 0.27
VcmaxRef = 38.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

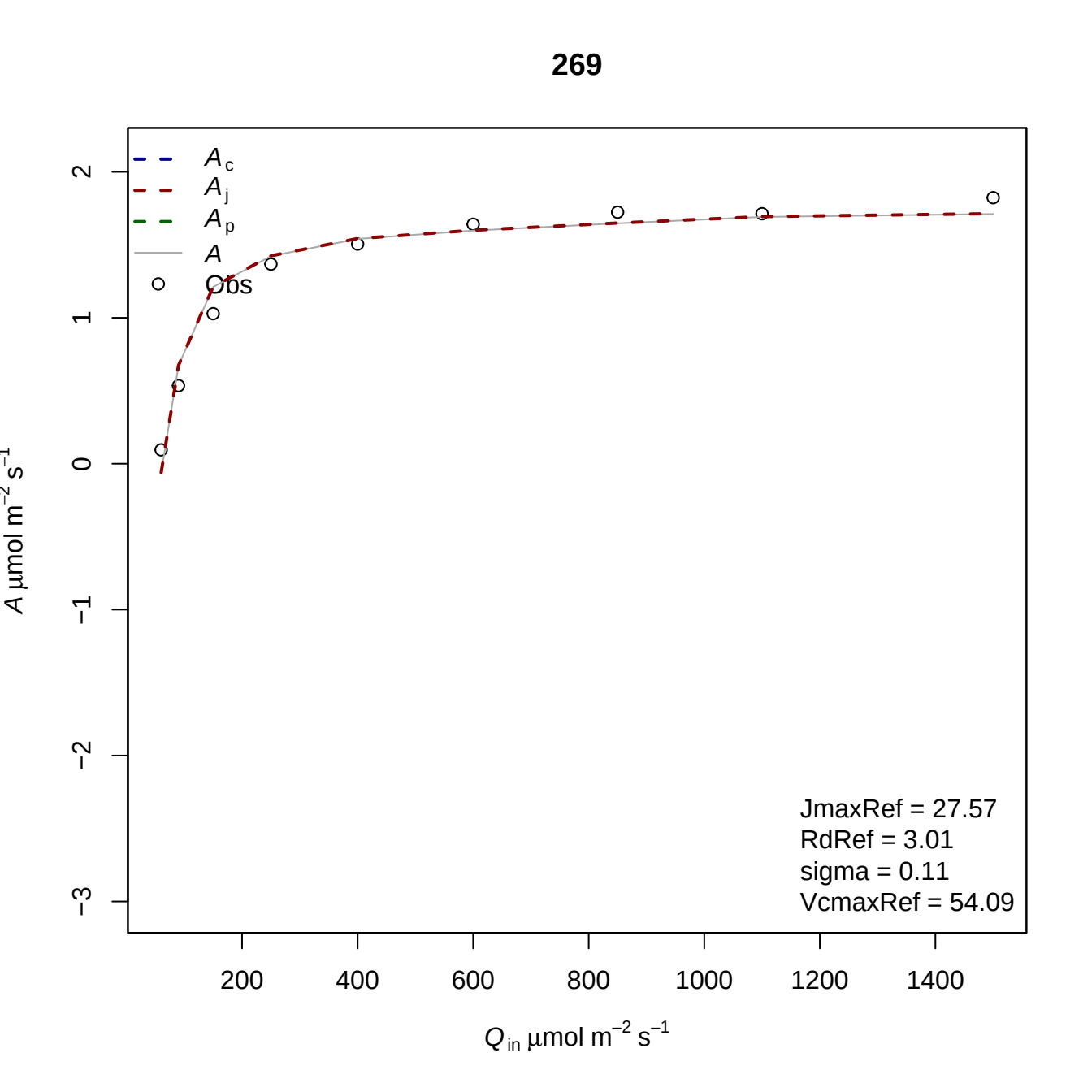

263
A_c
A_j
A_p
A
Obs
JmaxRef = 62.79
RdRef = 1.68
sigma = 0.42
VcmaxRef = 89.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

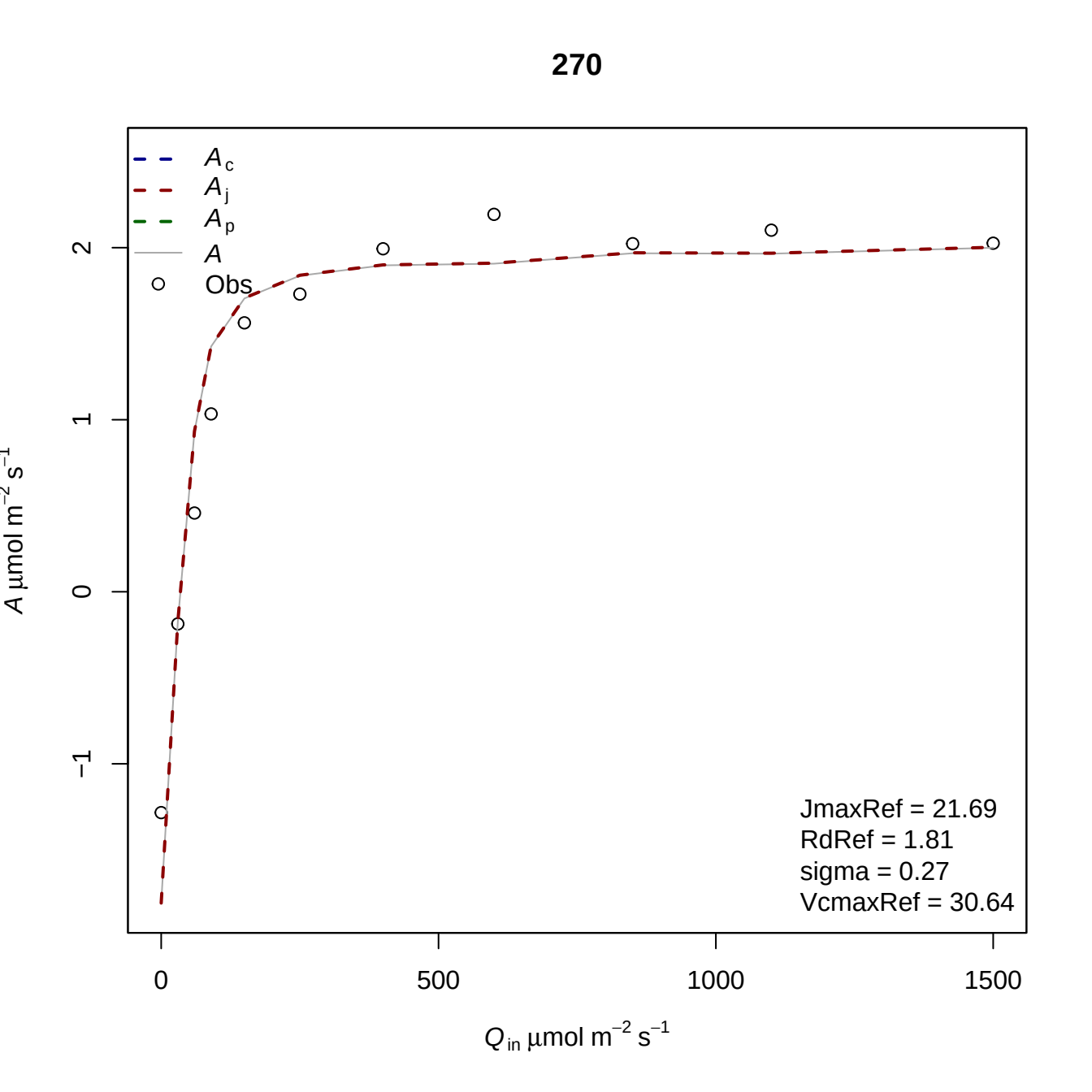

264
A_c
A_j
A_p
A
Obs
JmaxRef = 84.55
RdRef = 2.19
sigma = 0.57
VcmaxRef = 100.61
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

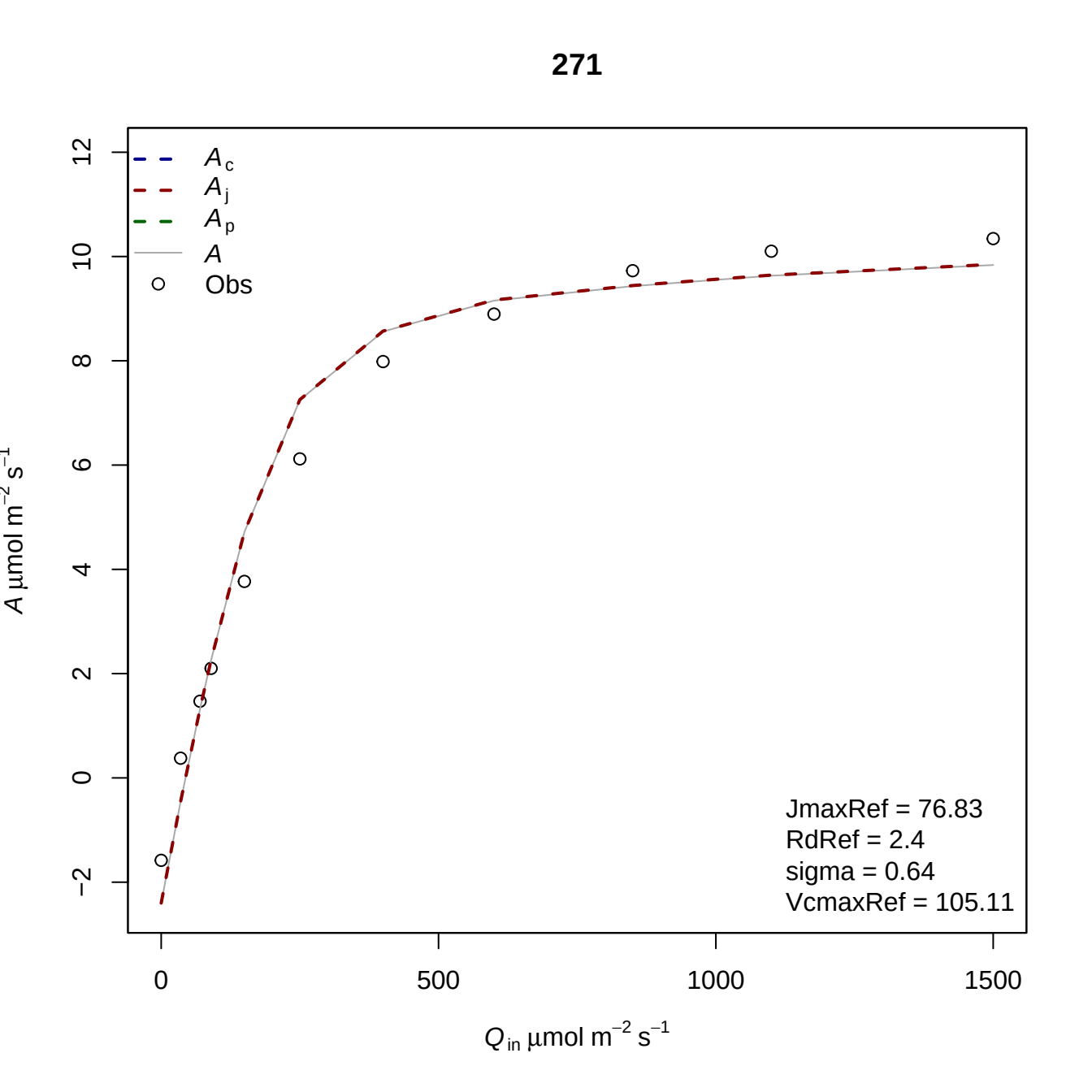

266
A_c
A_j
A_p
A
Obs
JmaxRef = 39.13
RdRef = 1.57
sigma = 0.55
VcmaxRef = 35.96
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

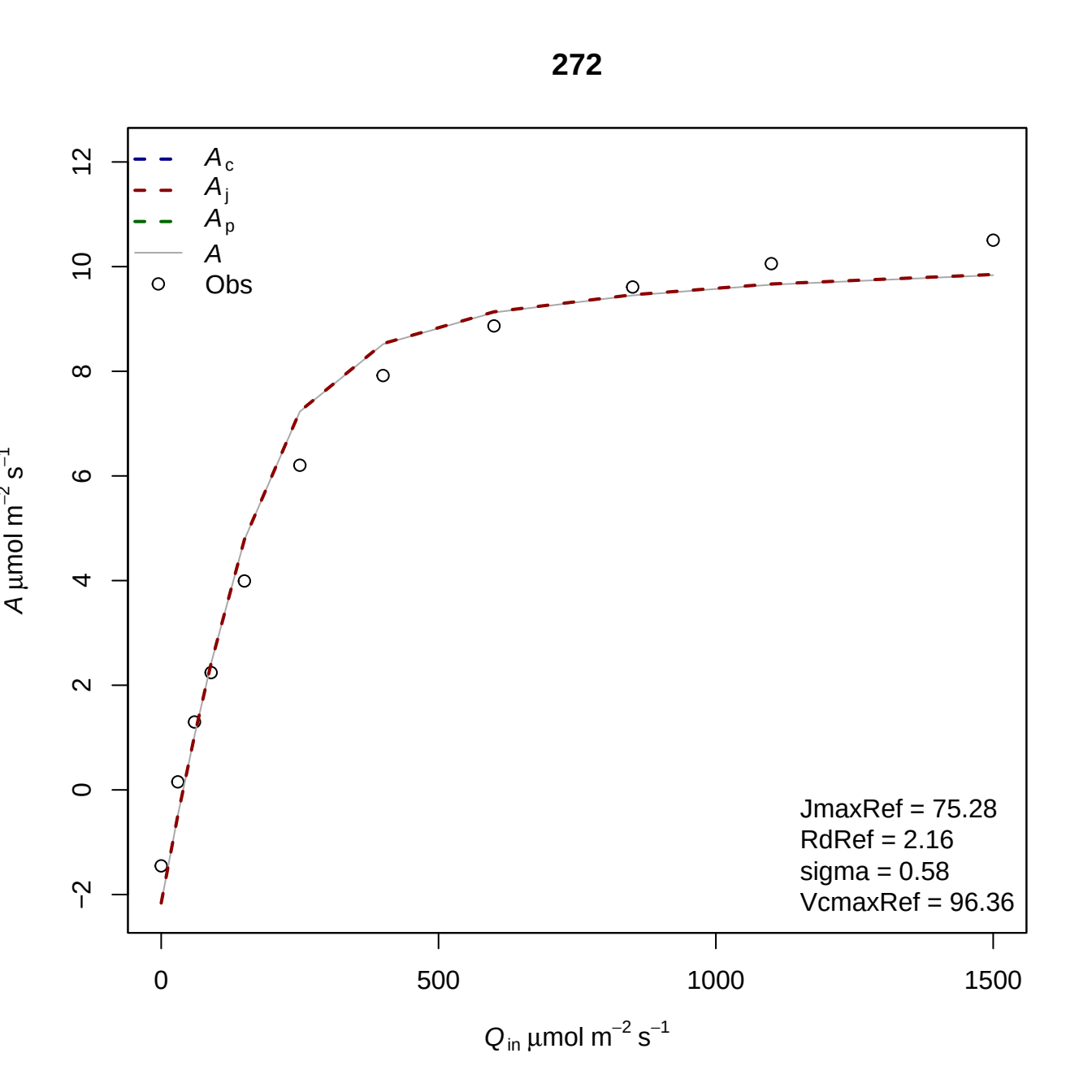

A_c
A_j
A_p
A
Obs
JmaxRef = 108.09
RdRef = 2.84
sigma = 1.1
VcmaxRef = 126.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

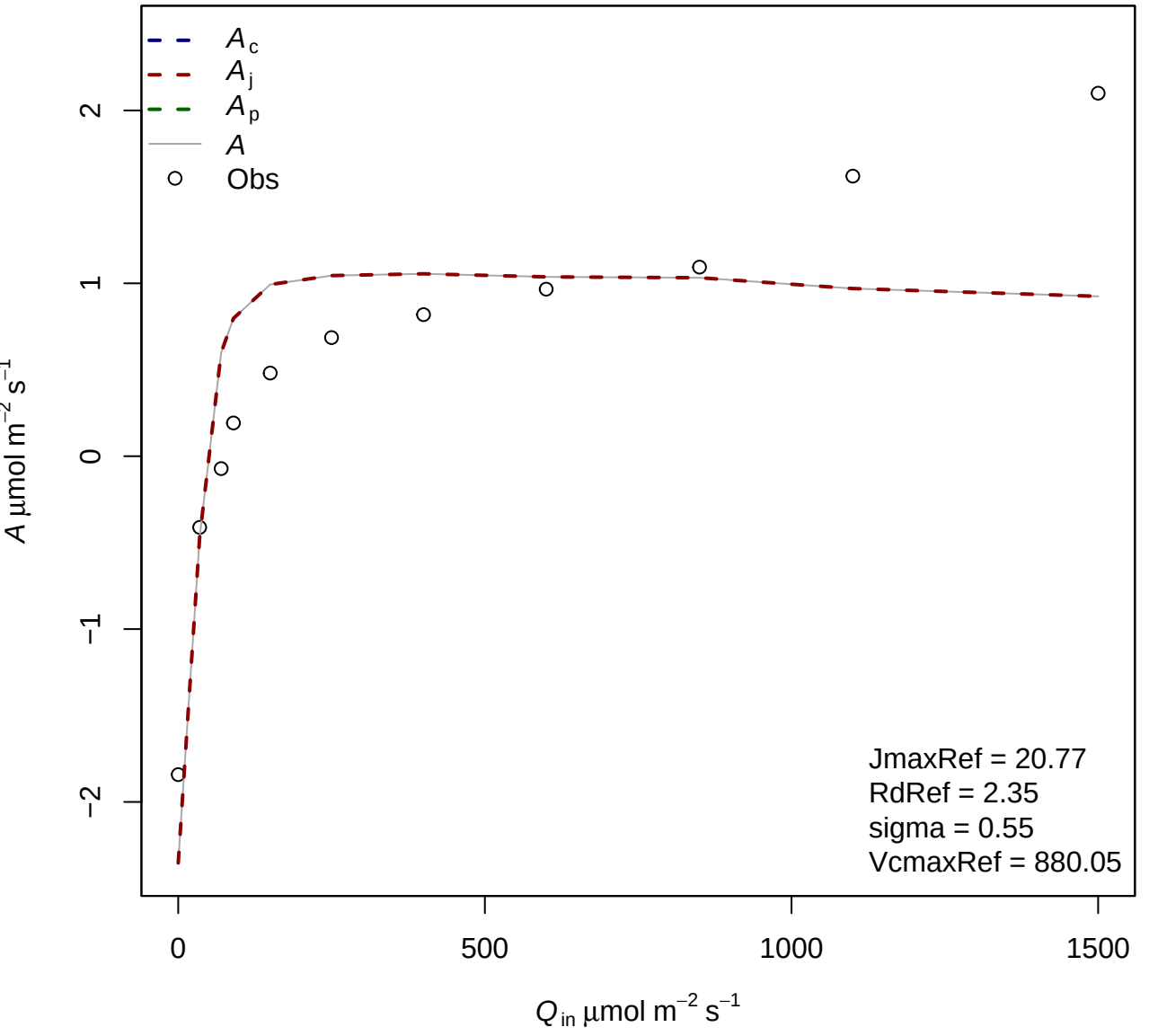

268
A_c
A_j
A_p
A
Obs
JmaxRef = 47.19
RdRef = 2.87
sigma = 0.38
VcmaxRef = 31.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
200
400
600
800
1000
1200
1400

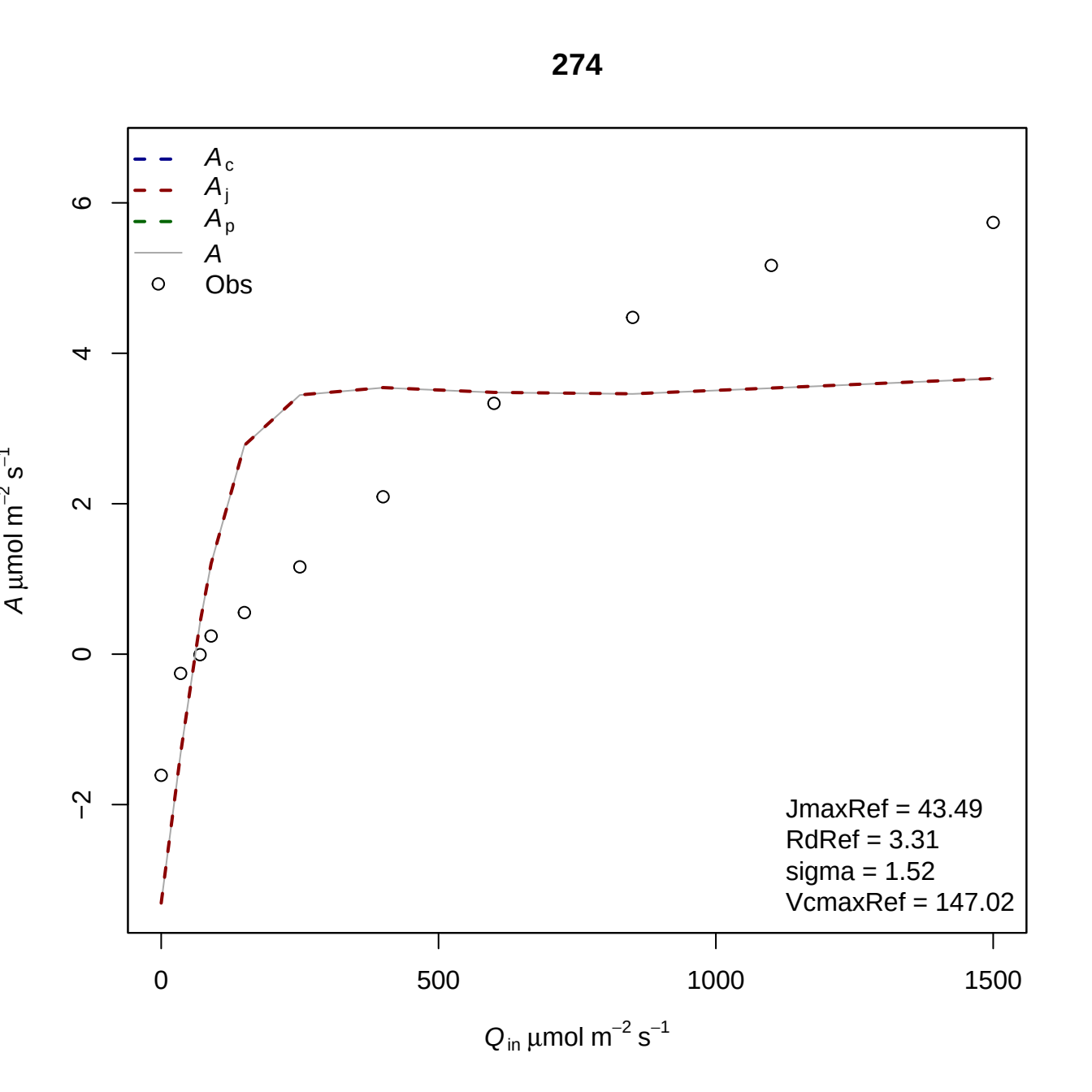

269
A_c
A_j
A_p
A
Obs
JmaxRef = 27.57
RdRef = 3.01
sigma = 0.11
VcmaxRef = 54.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
200
400
600
800
1000
1200
1400

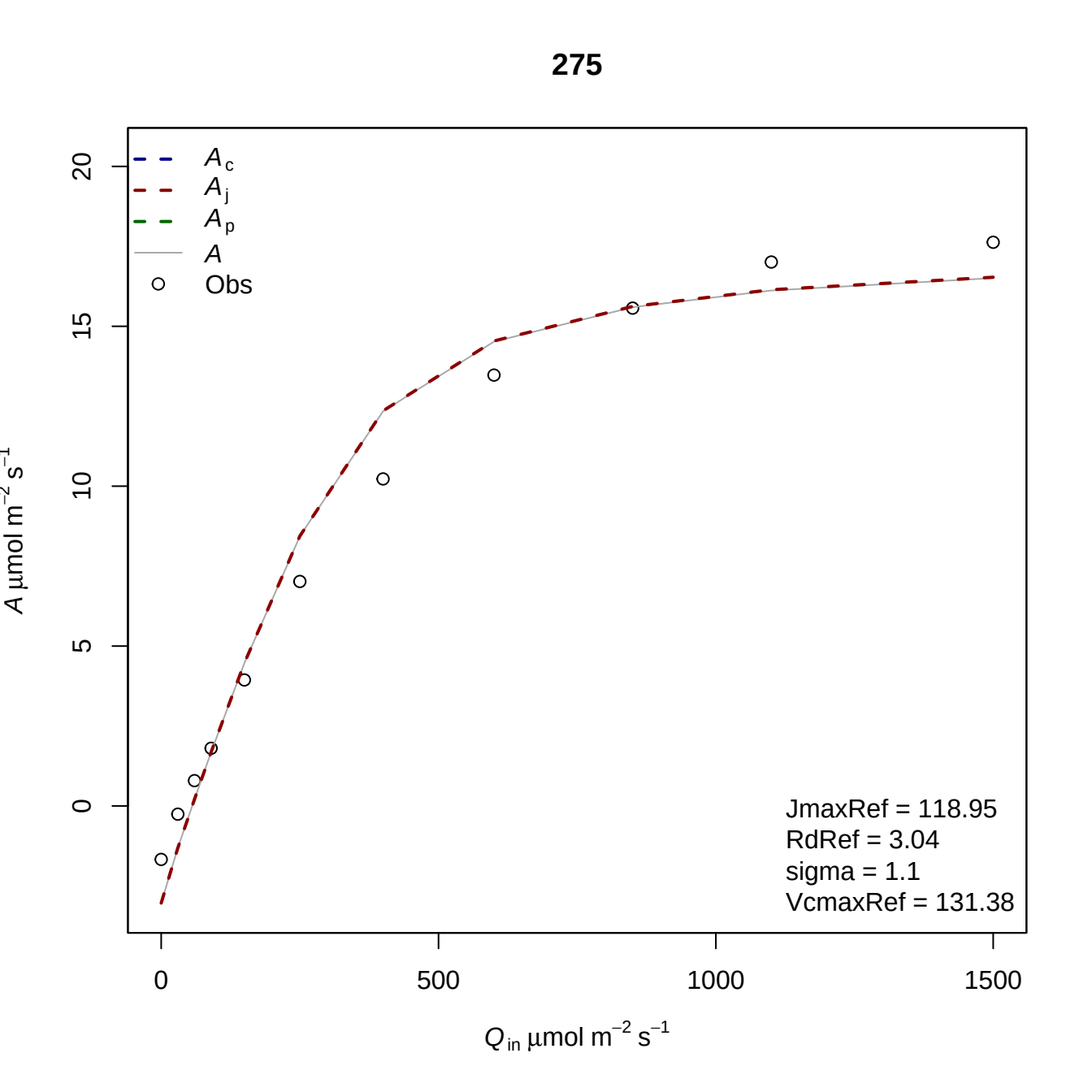

270
A_c
A_j
A_p
A
Obs
JmaxRef = 21.69
RdRef = 1.81
sigma = 0.27
VcmaxRef = 30.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
0
500
1000
1500

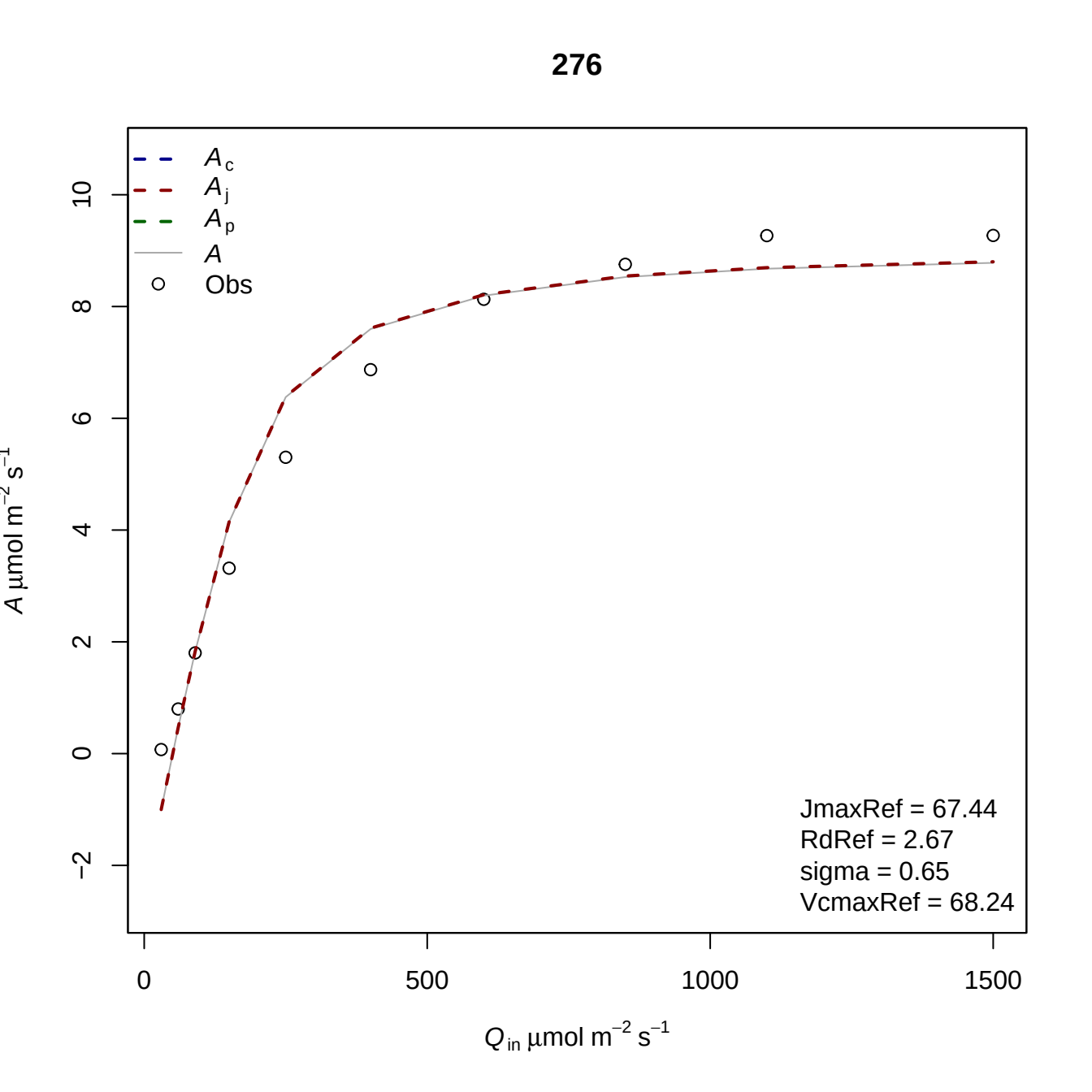

271
A_c
A_j
A_p
A
Obs
JmaxRef = 76.83
RdRef = 2.4
sigma = 0.64
VcmaxRef = 105.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

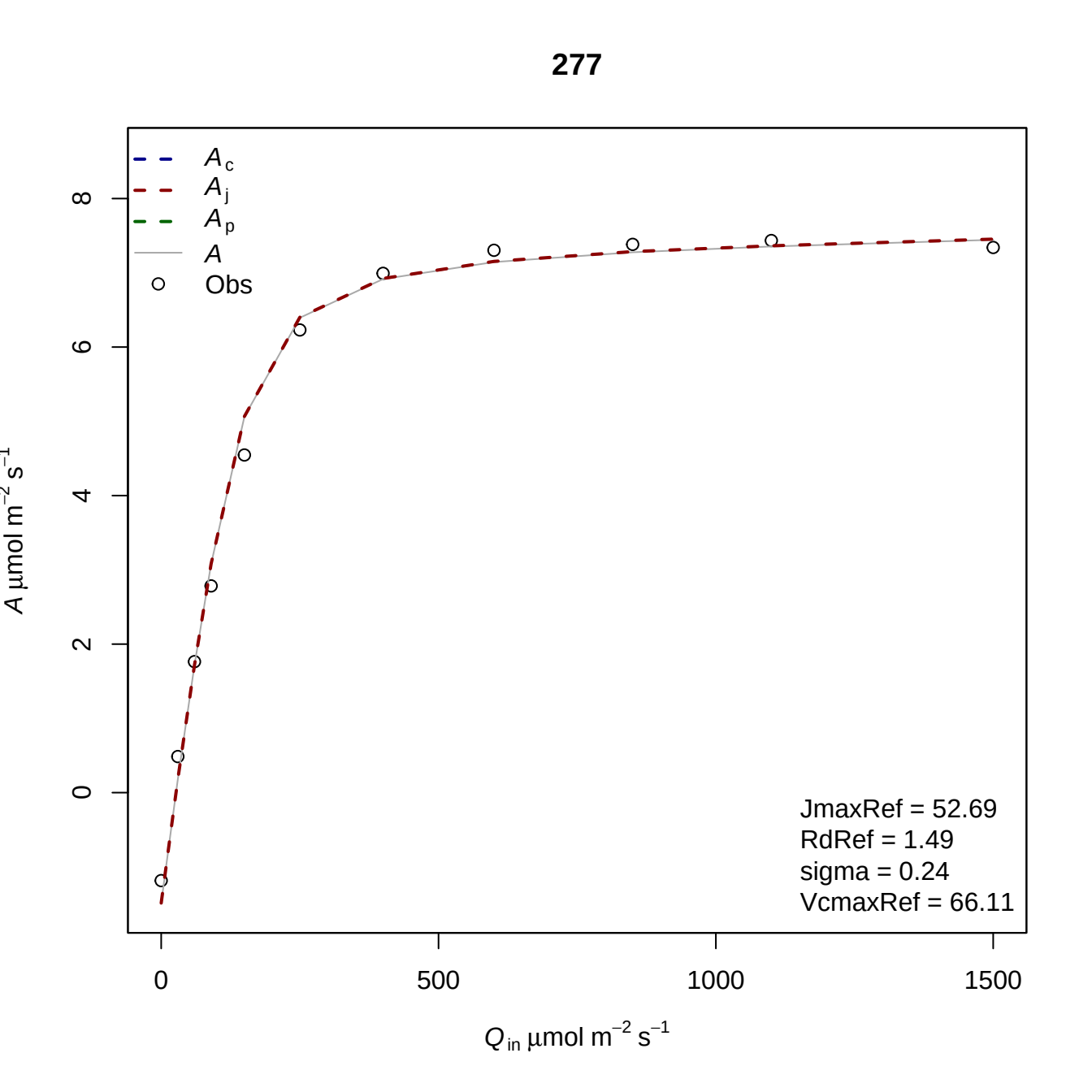

272
A_c
A_j
A_p
A
Obs
JmaxRef = 75.28
RdRef = 2.16
sigma = 0.58
VcmaxRef = 96.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

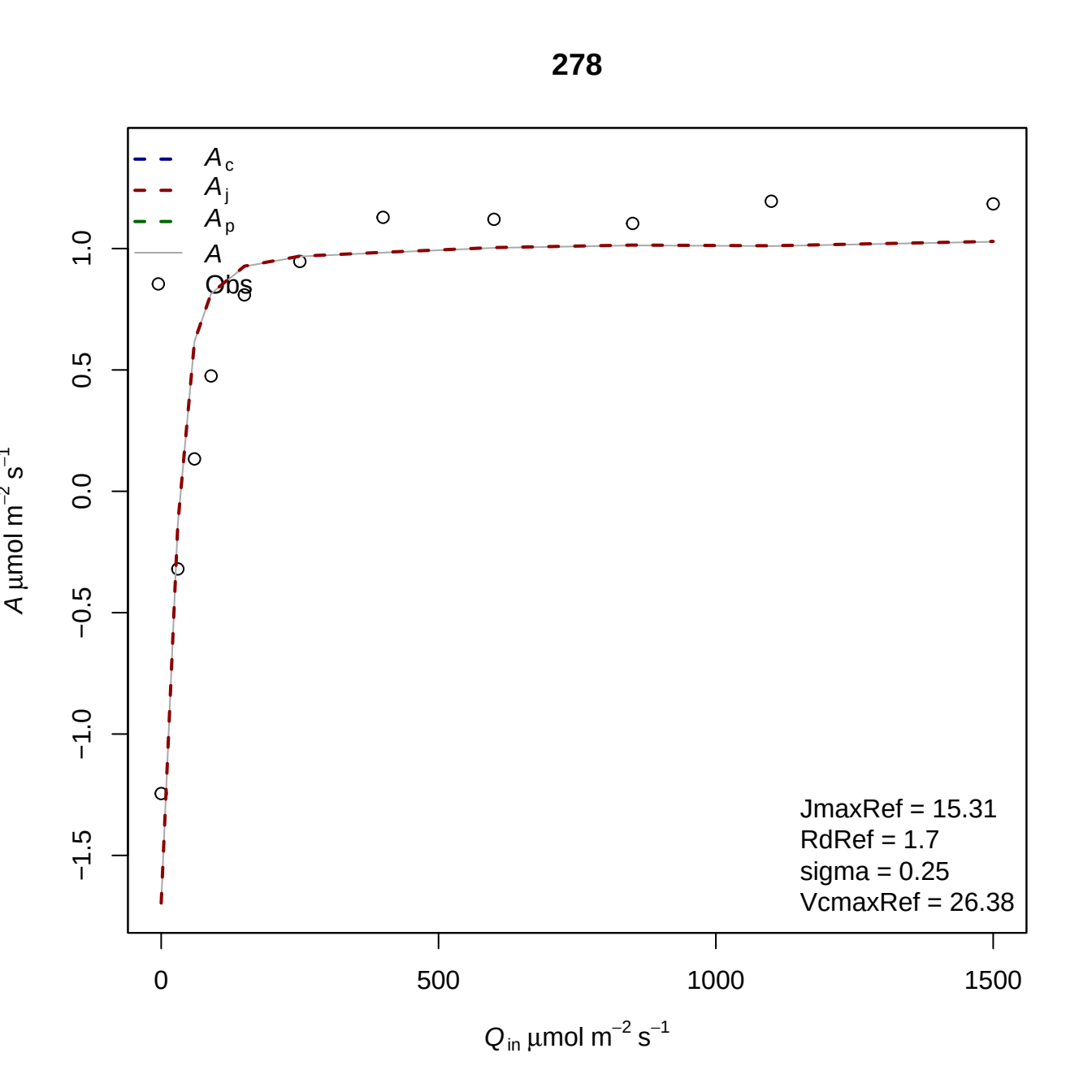

A_c
A_j
A_p
A
Obs
JmaxRef = 20.77
RdRef = 2.35
sigma = 0.55
VcmaxRef = 880.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500

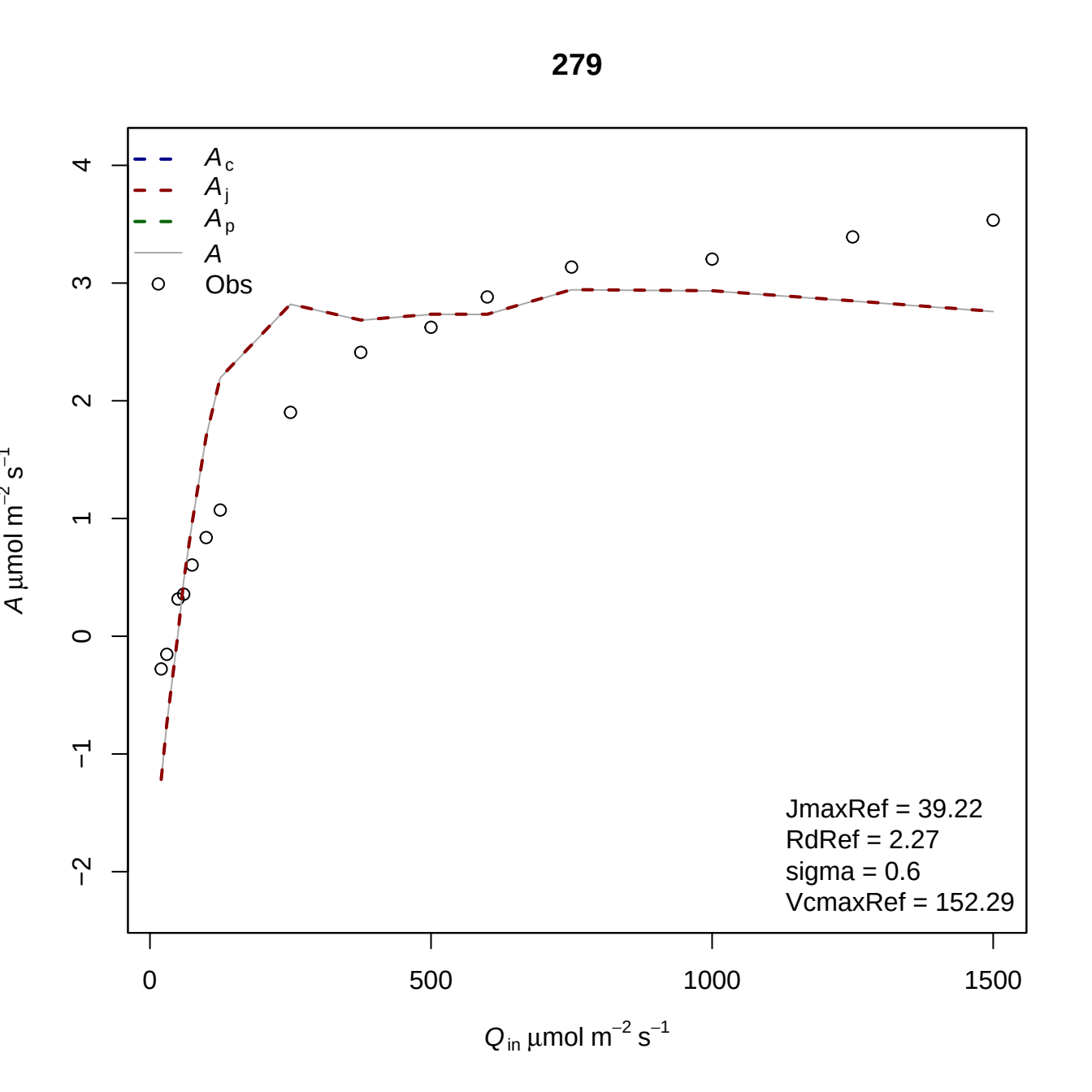

274
A_c
A_j
A_p
A
Obs
JmaxRef = 43.49
RdRef = 3.31
sigma = 1.52
VcmaxRef = 147.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

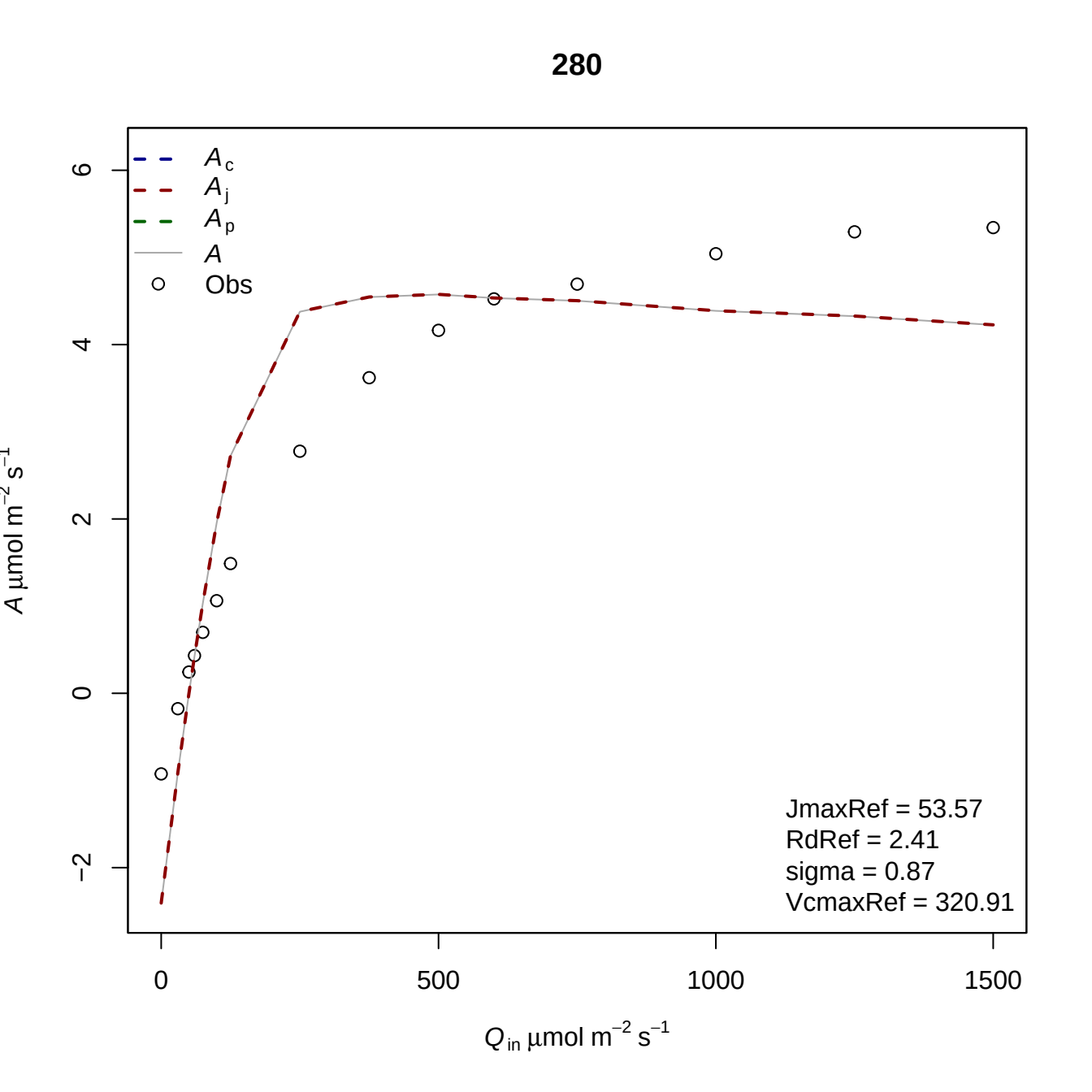

275
A_c
A_j
A_p
A
Obs
JmaxRef = 118.95
RdRef = 3.04
sigma = 1.1
VcmaxRef = 131.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

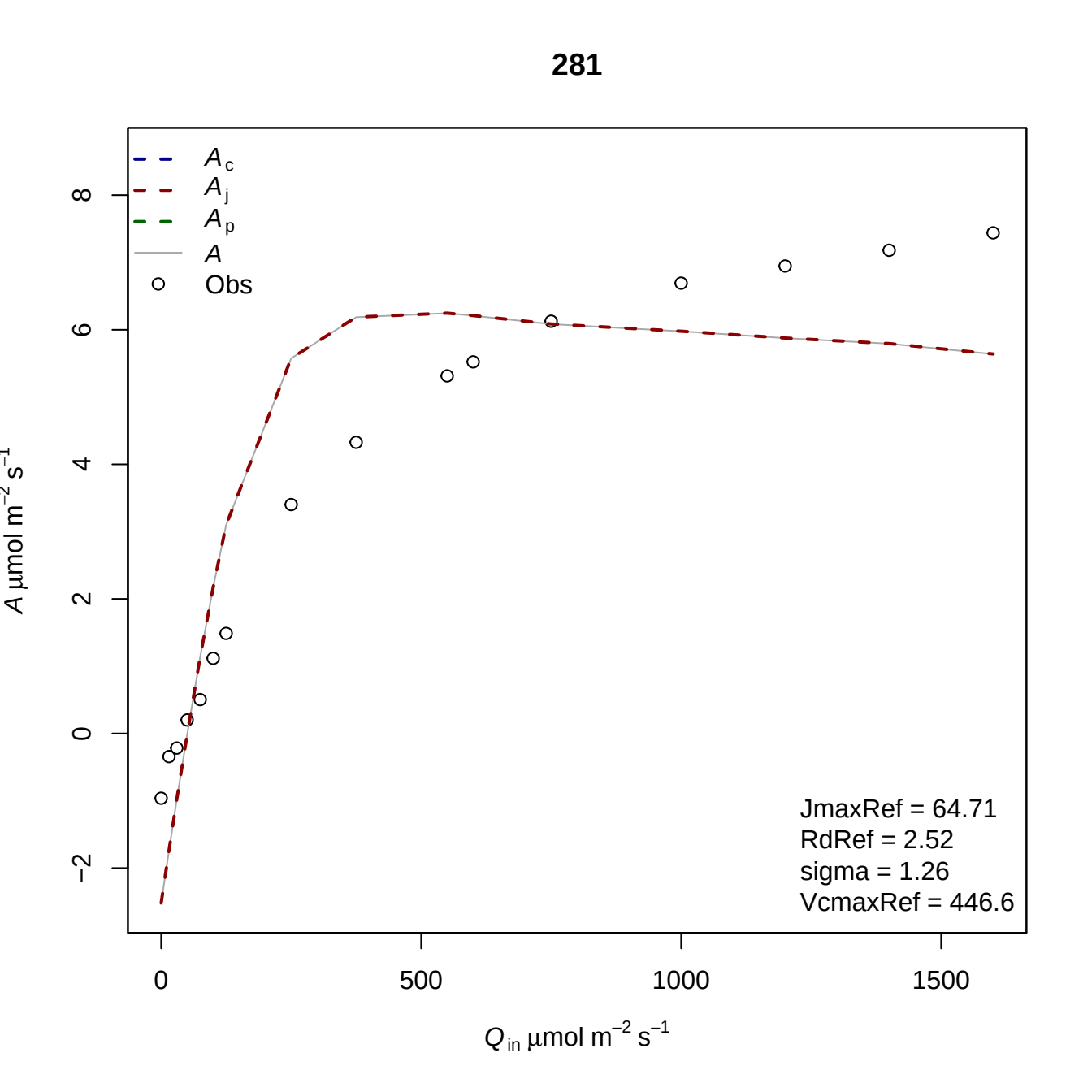

276
A_c
A_j
A_p
A
Obs
JmaxRef = 67.44
RdRef = 2.67
sigma = 0.65
VcmaxRef = 68.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

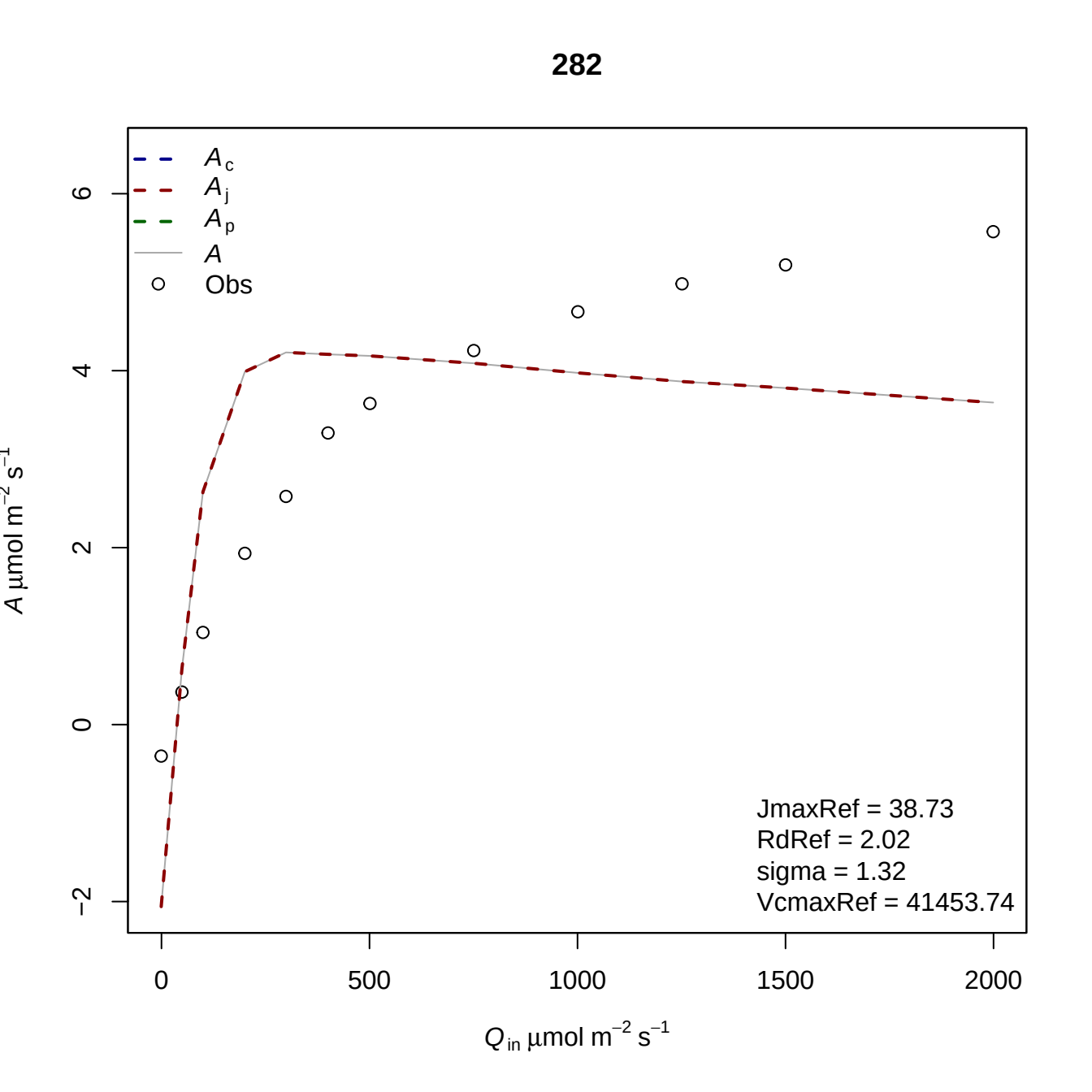

277
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 52.69
RdRef = 1.49
sigma = 0.24
VcmaxRef = 66.11

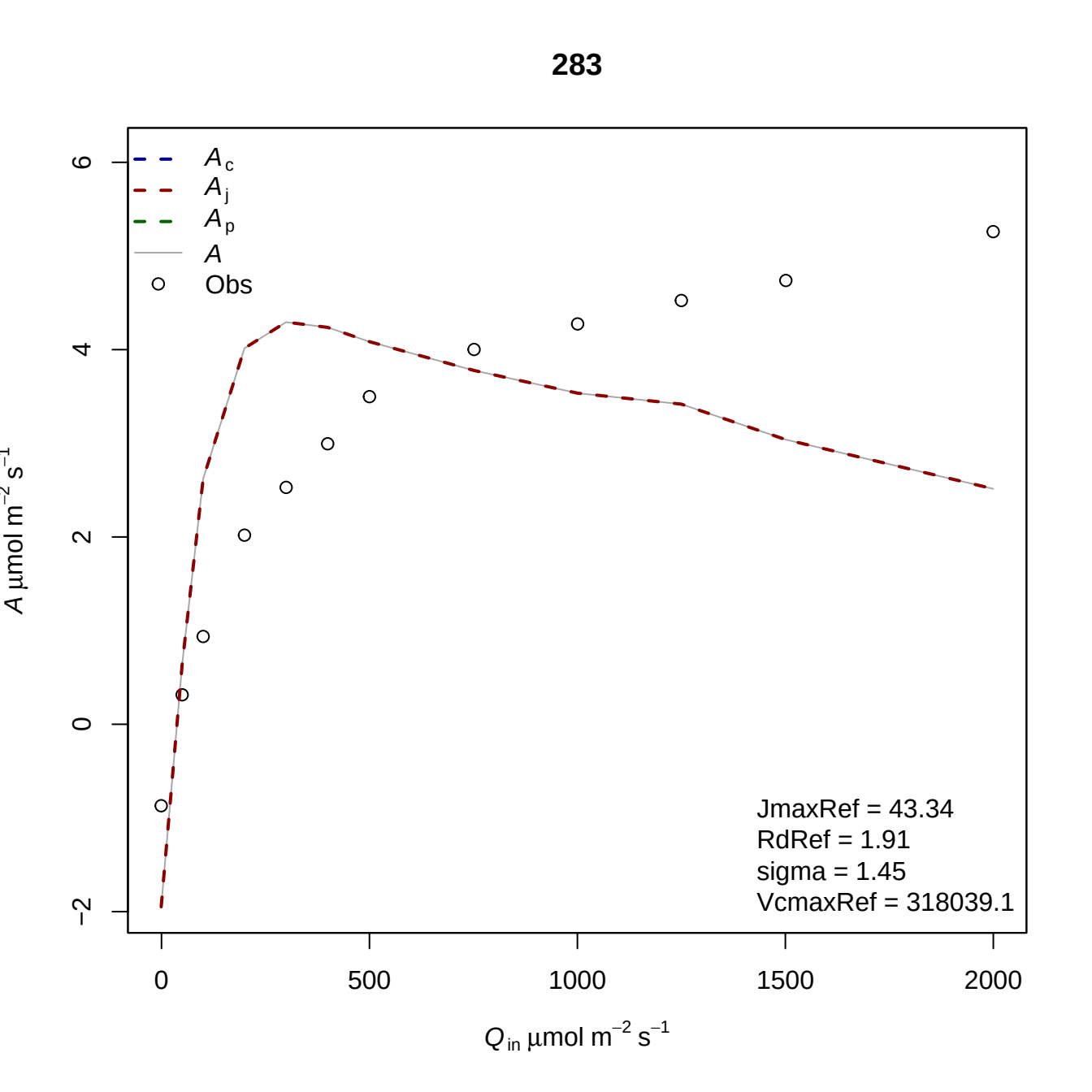

278
A_c
A_j
A_p
A
Obs
JmaxRef = 15.31
RdRef = 1.7
sigma = 0.25
VcmaxRef = 26.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500

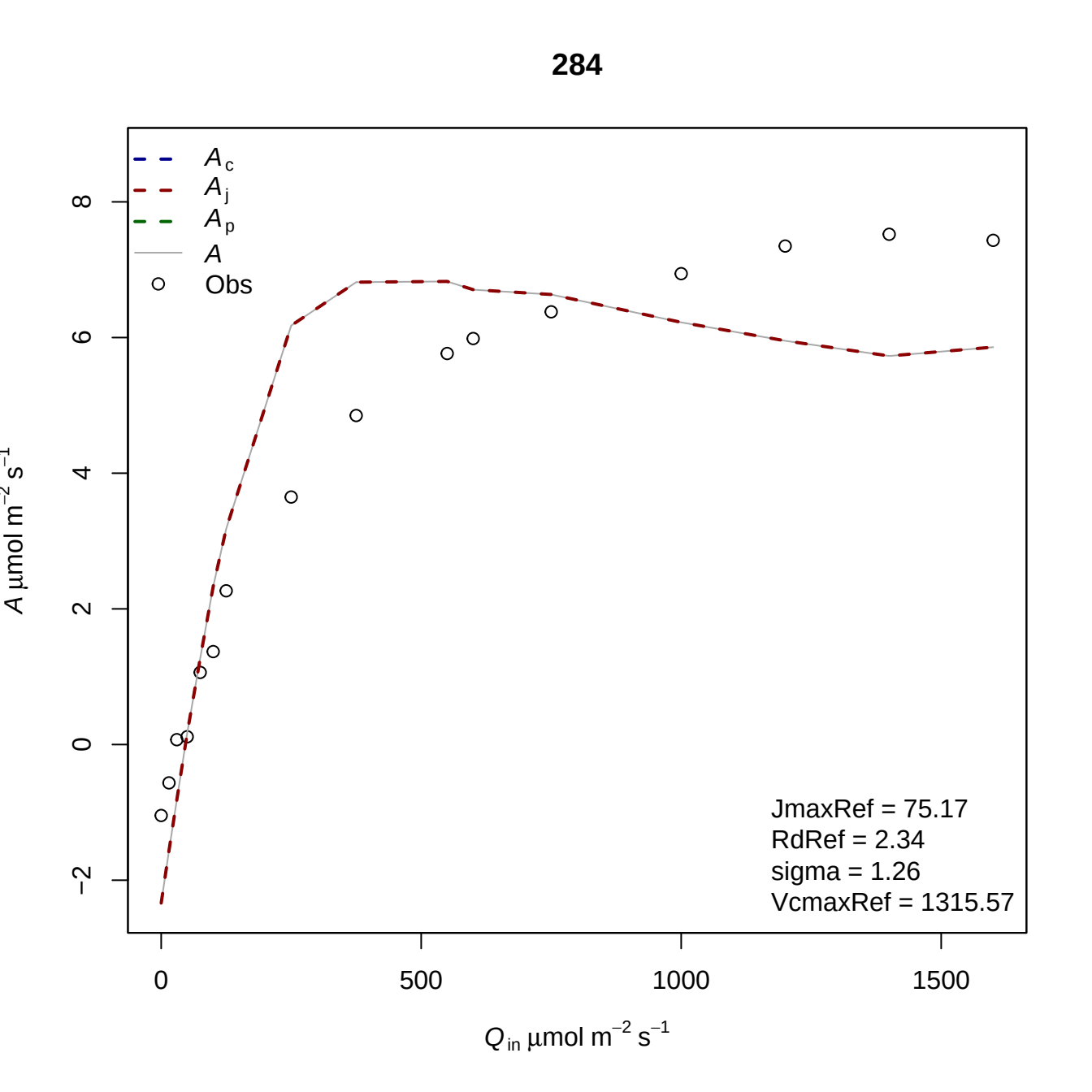

279
A_c
A_j
A_p
A
Obs
JmaxRef = 39.22
RdRef = 2.27
sigma = 0.6
VcmaxRef = 152.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

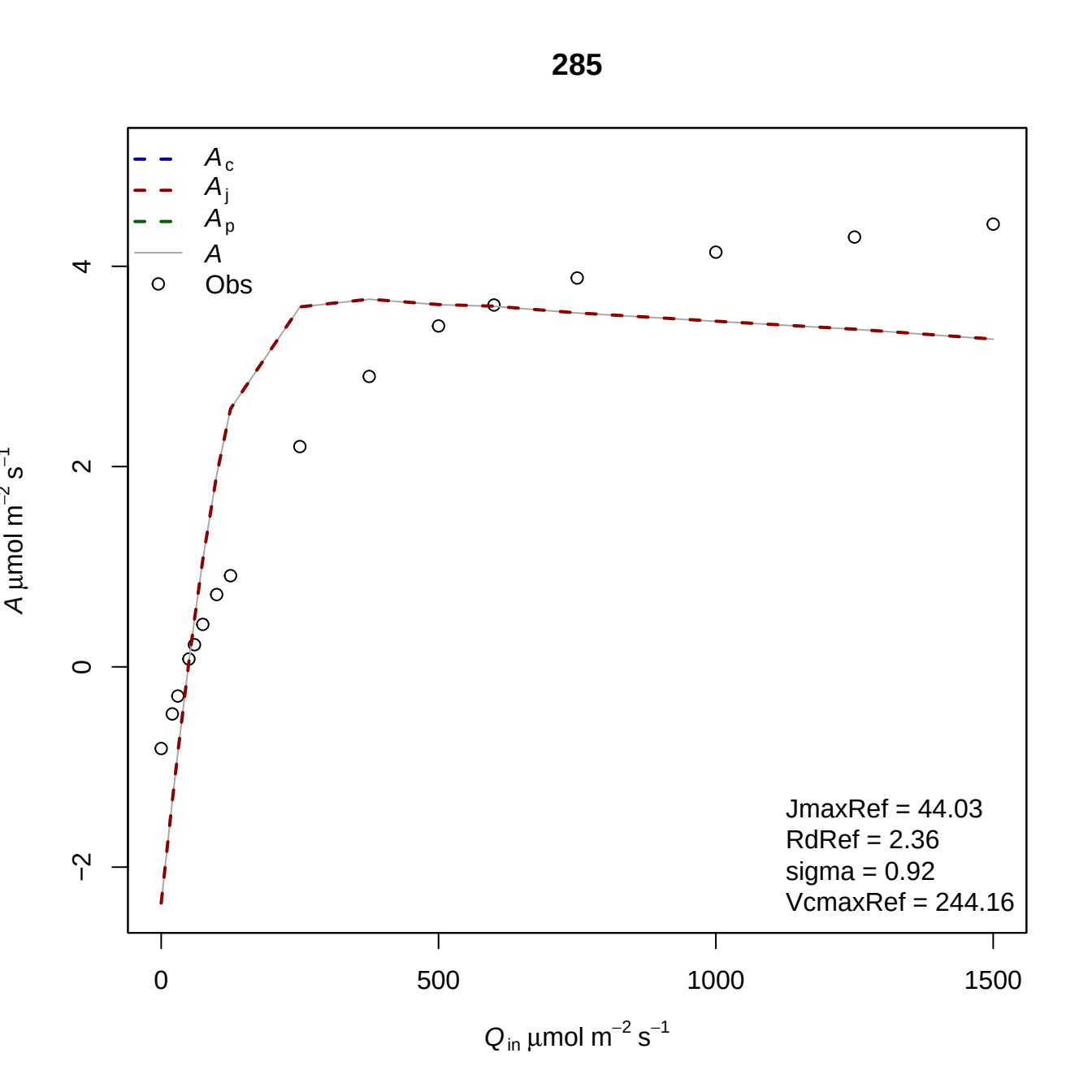

280
A_c
A_j
A_p
A
Obs
JmaxRef = 53.57
RdRef = 2.41
sigma = 0.87
VcmaxRef = 320.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

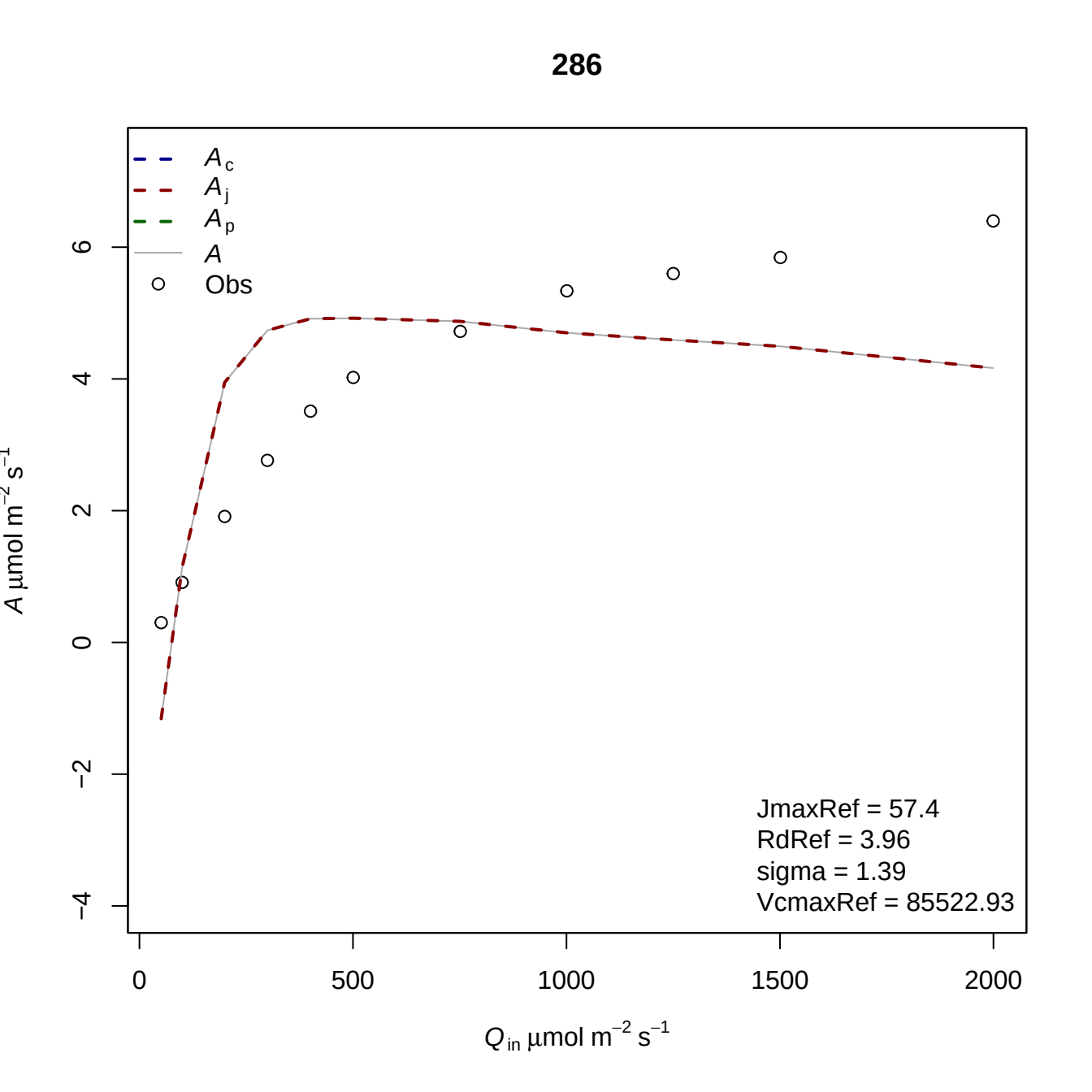

281
A_c
A_j
A_p
A
Obs
JmaxRef = 64.71
RdRef = 2.52
sigma = 1.26
VcmaxRef = 446.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

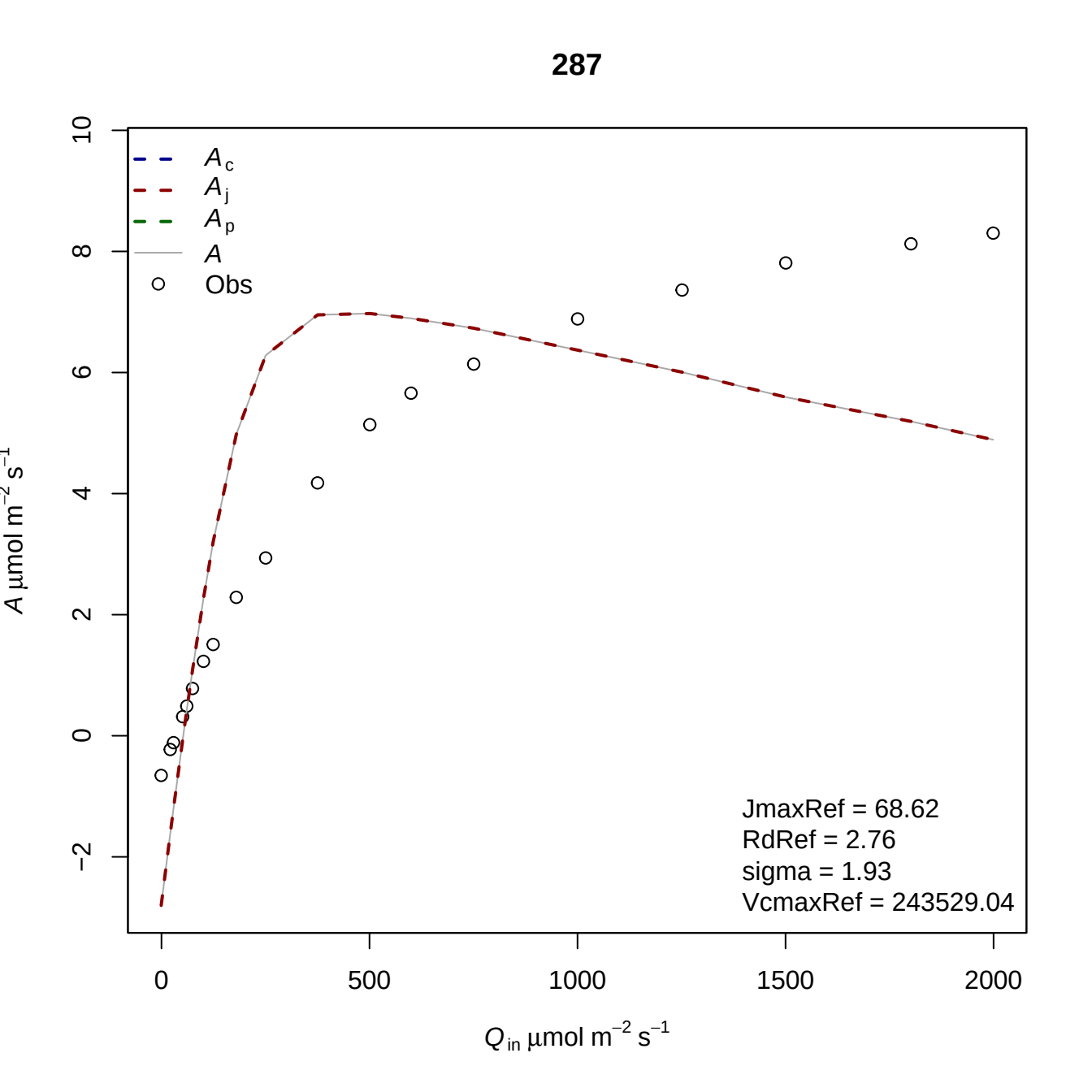

282
A_c
A_j
A_p
A
Obs
JmaxRef = 38.73
RdRef = 2.02
sigma = 1.32
VcmaxRef = 41453.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

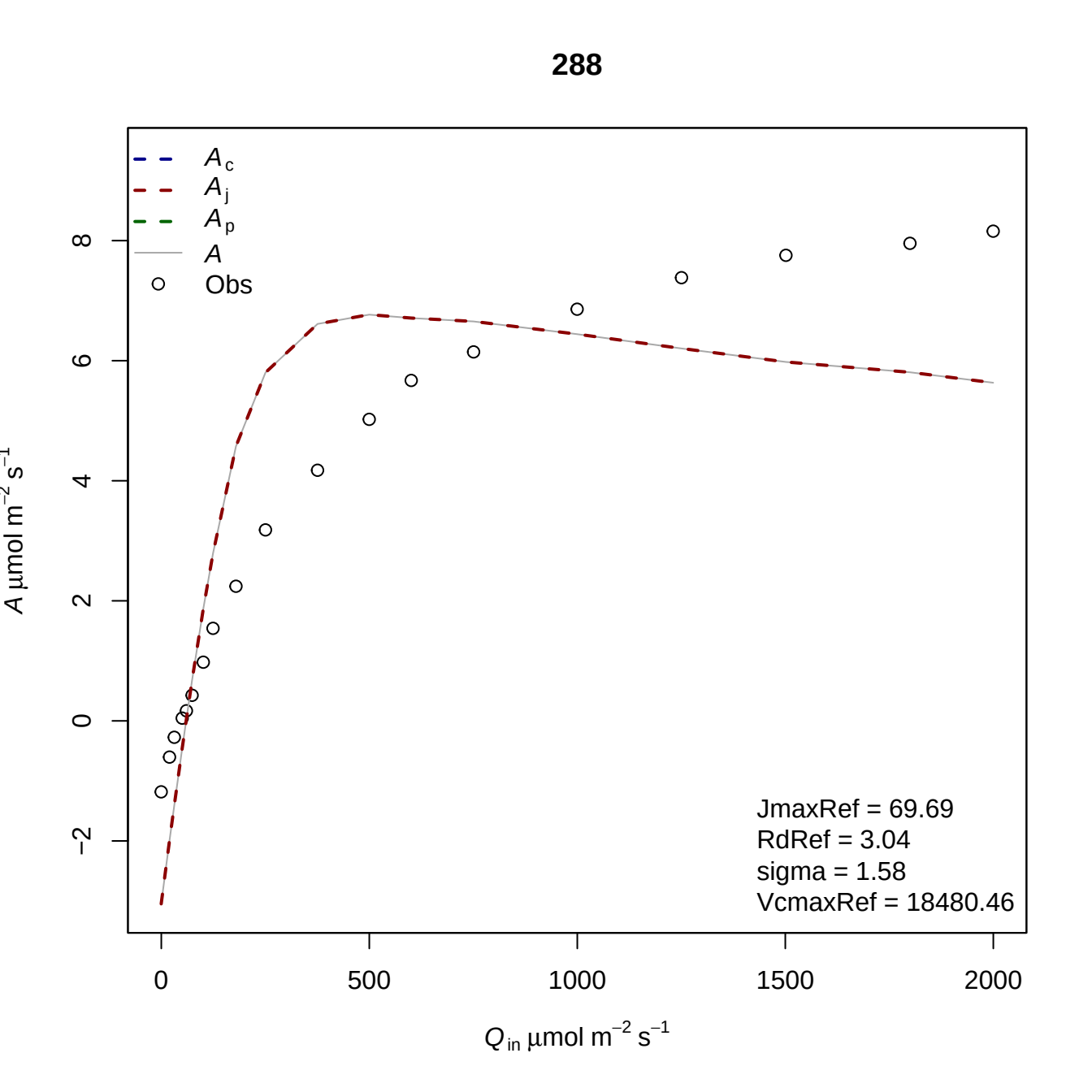

283
A_c
A_j
A_p
A
Obs
JmaxRef = 43.34
RdRef = 1.91
sigma = 1.45
VcmaxRef = 318039.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

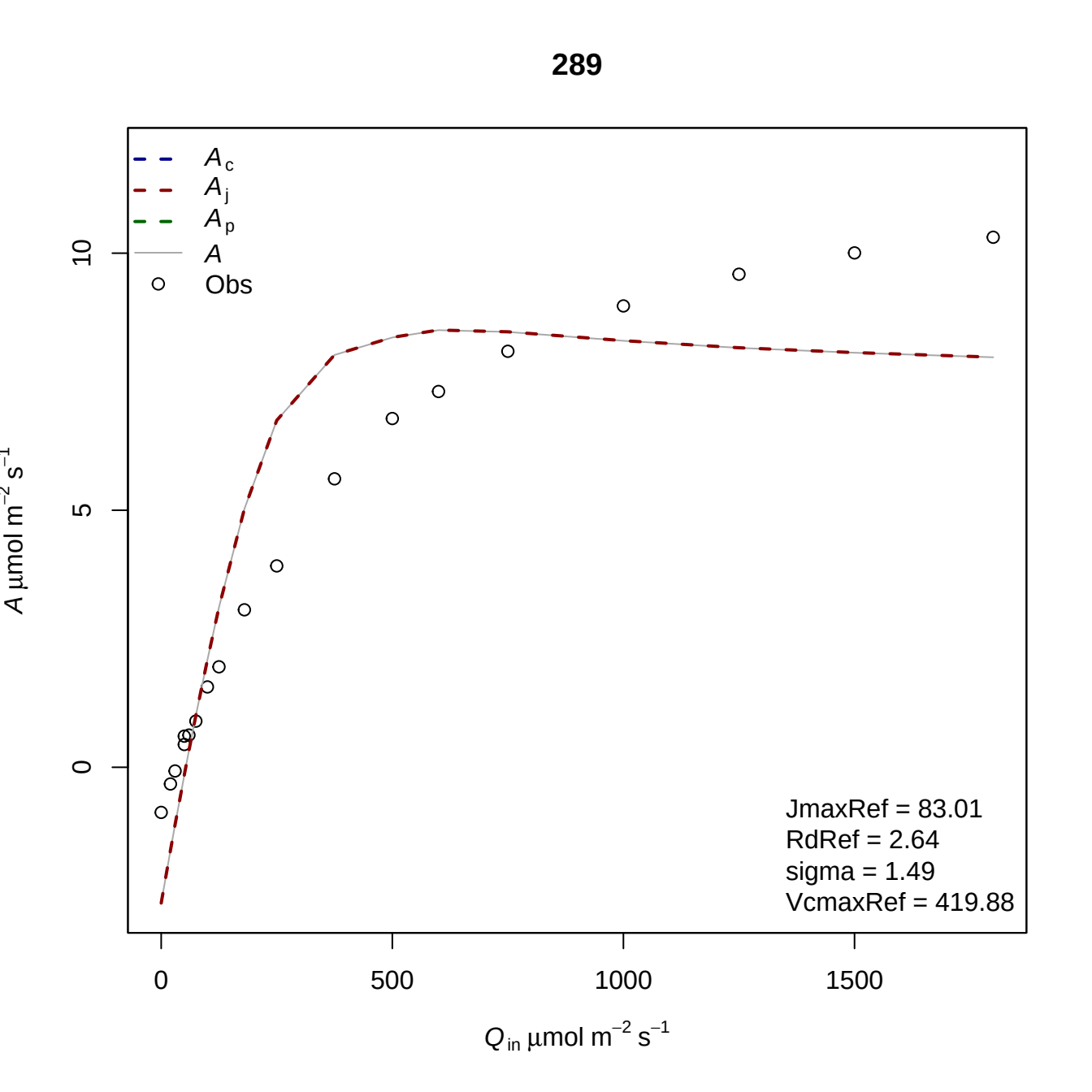

284
A_c
A_j
A_p
A
Obs
JmaxRef = 75.17
RdRef = 2.34
sigma = 1.26
VcmaxRef = 1315.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

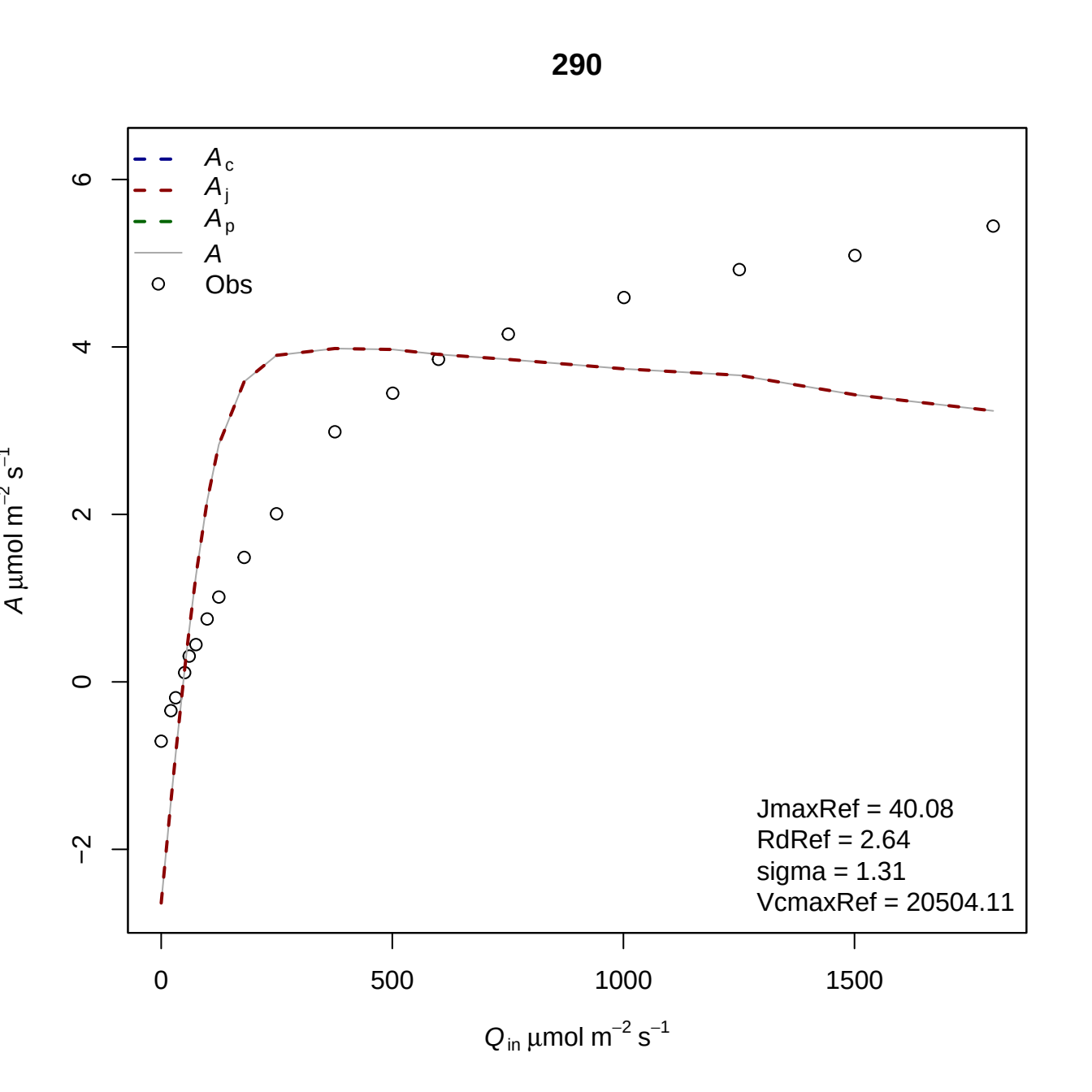

285
A_c
A_j
A_p
A
Obs
JmaxRef = 44.03
RdRef = 2.36
sigma = 0.92
VcmaxRef = 244.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

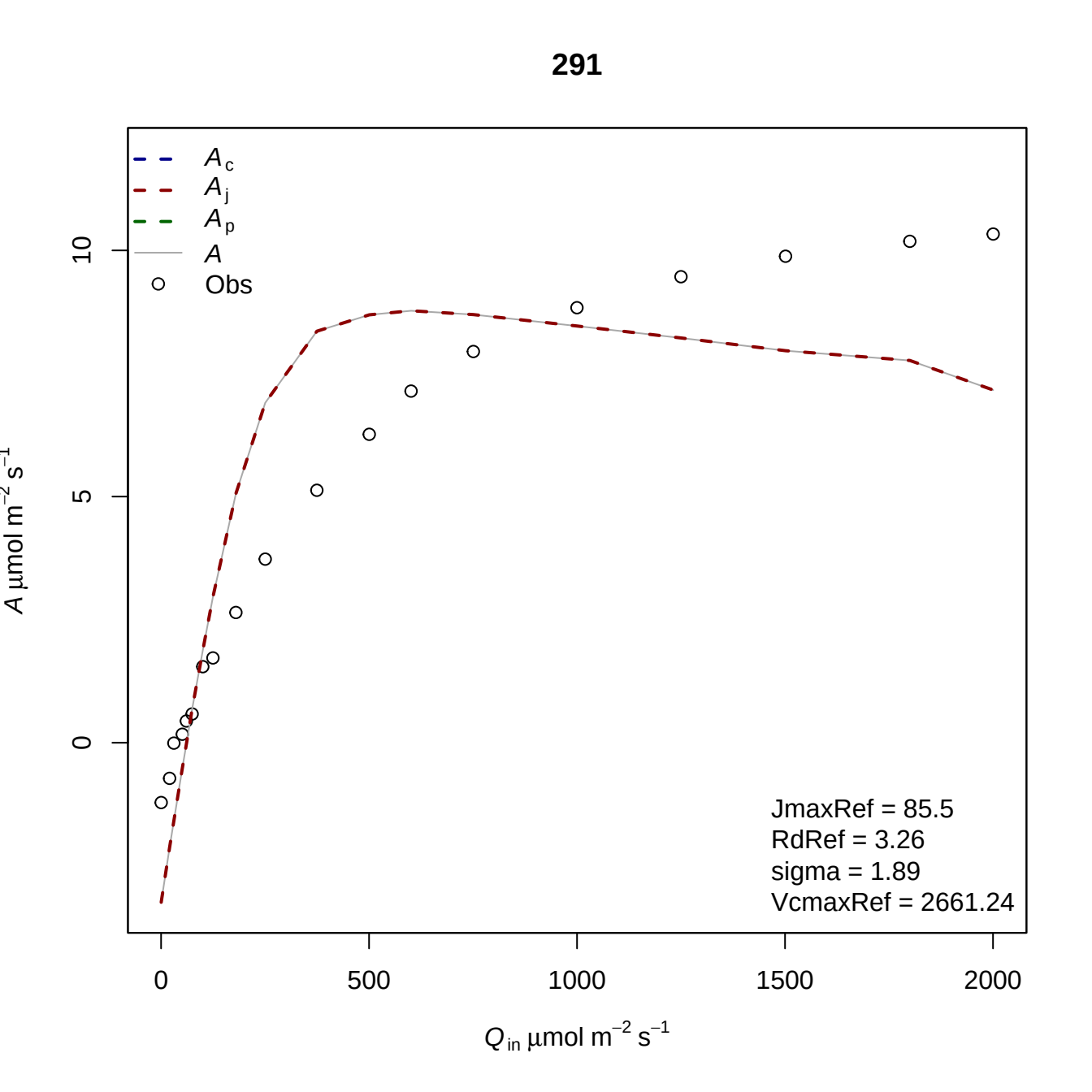

286
A_c
A_j
A_p
A
Obs
JmaxRef = 57.4
RdRef = 3.96
sigma = 1.39
VcmaxRef = 85522.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

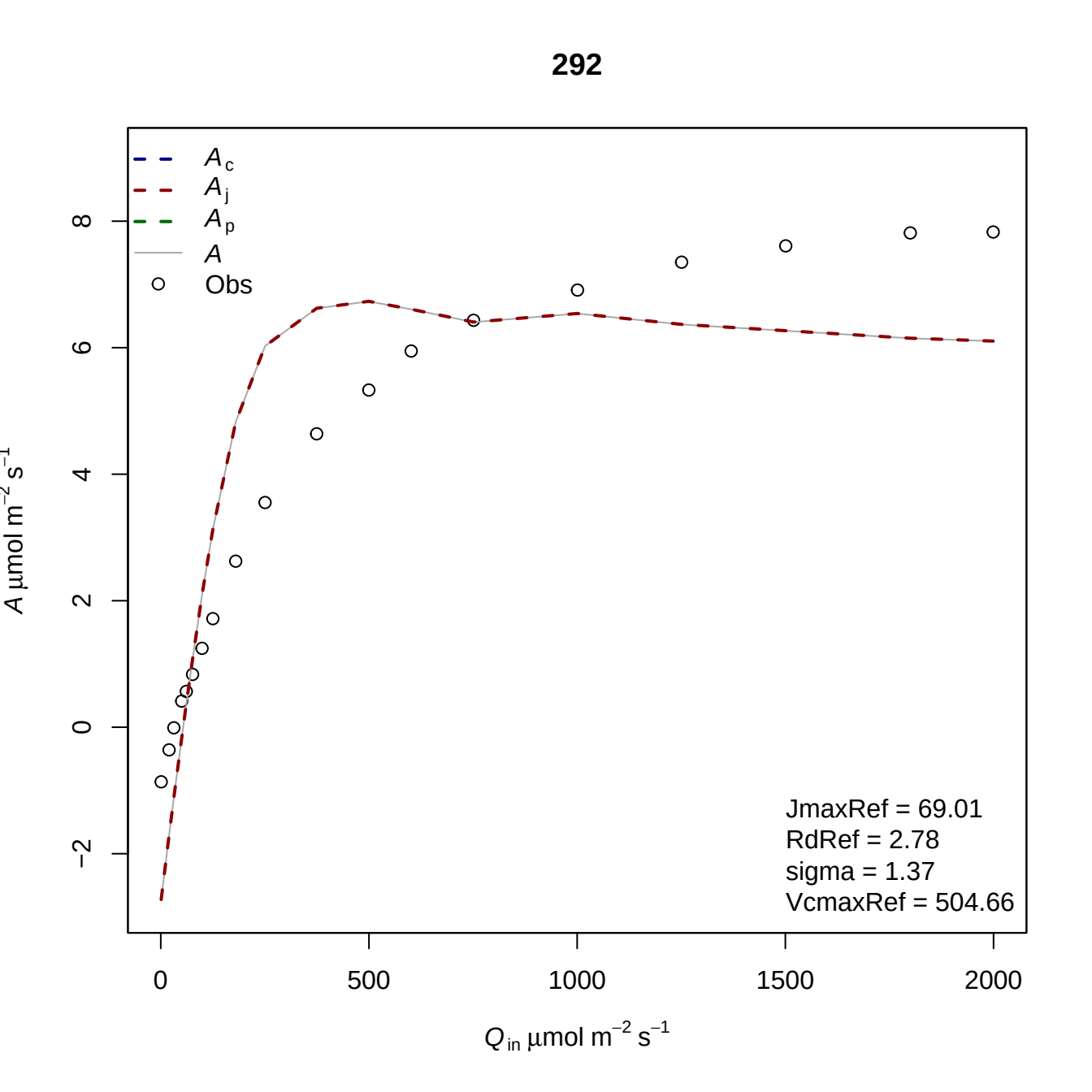

287
A_c
A_j
A_p
A
Obs
JmaxRef = 68.62
RdRef = 2.76
sigma = 1.93
VcmaxRef = 243529.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

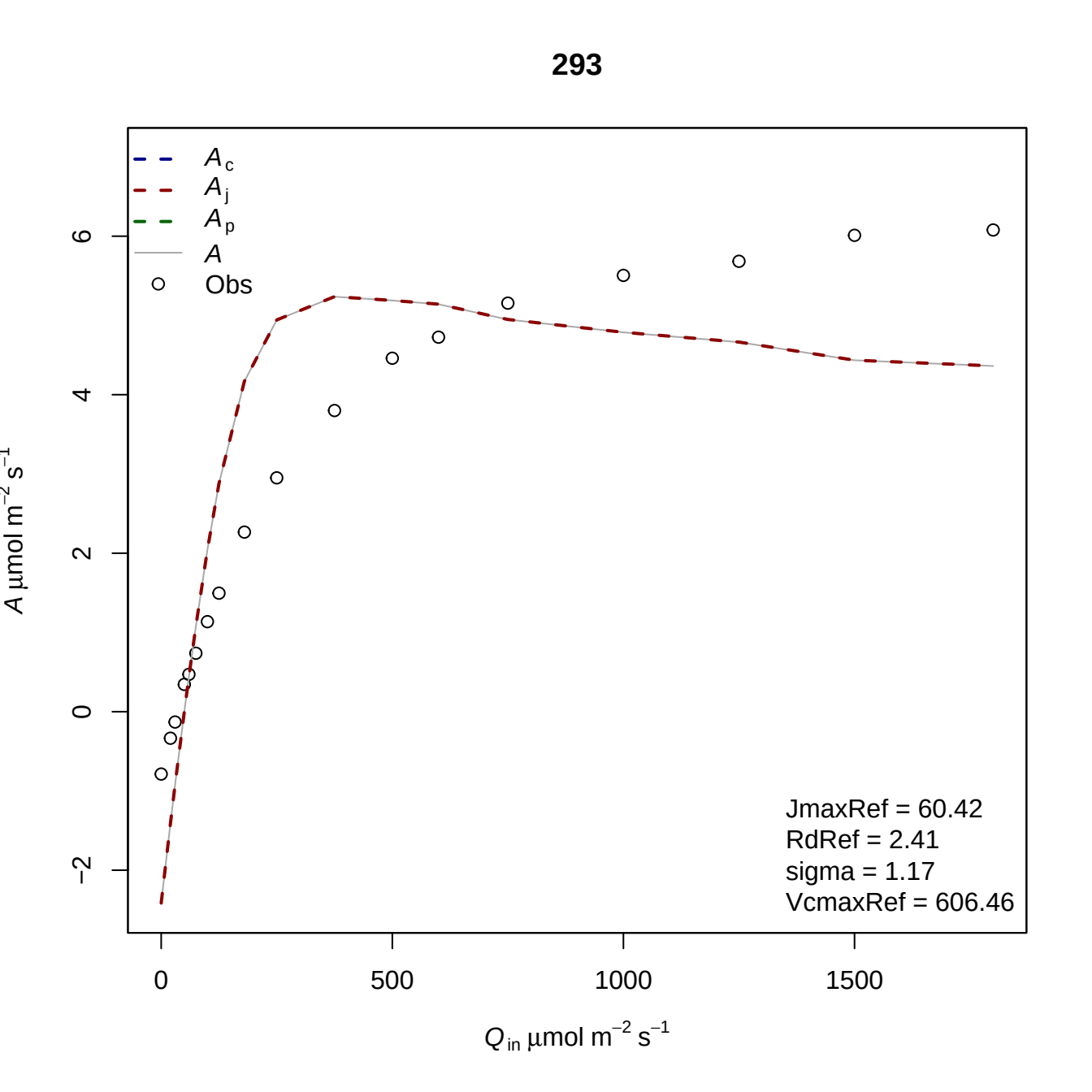

288
A_c
A_j
A_p
A
Obs
JmaxRef = 69.69
RdRef = 3.04
sigma = 1.58
VcmaxRef = 18480.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

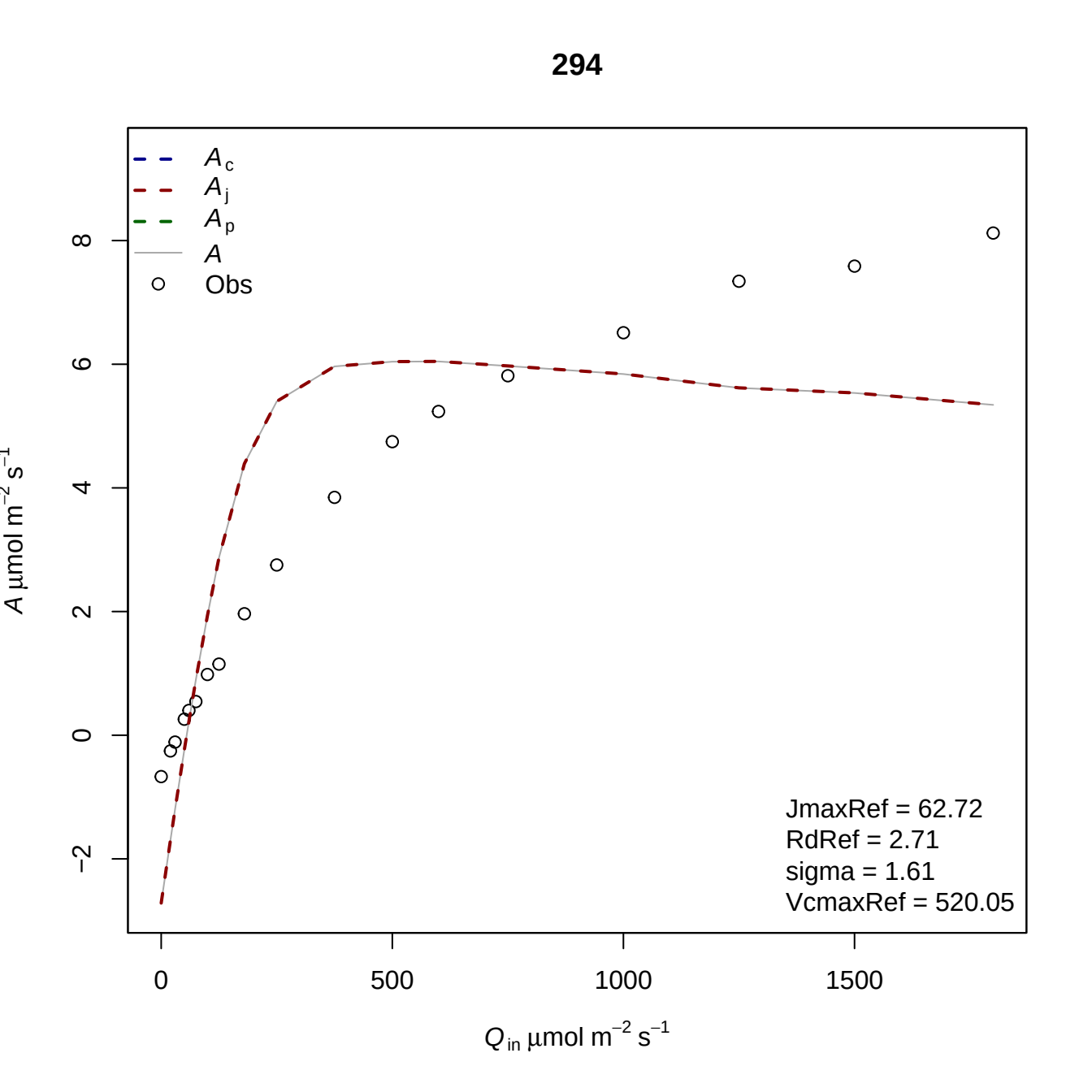

289
A_c
A_j
A_p
A
Obs
JmaxRef = 83.01
RdRef = 2.64
sigma = 1.49
VcmaxRef = 419.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

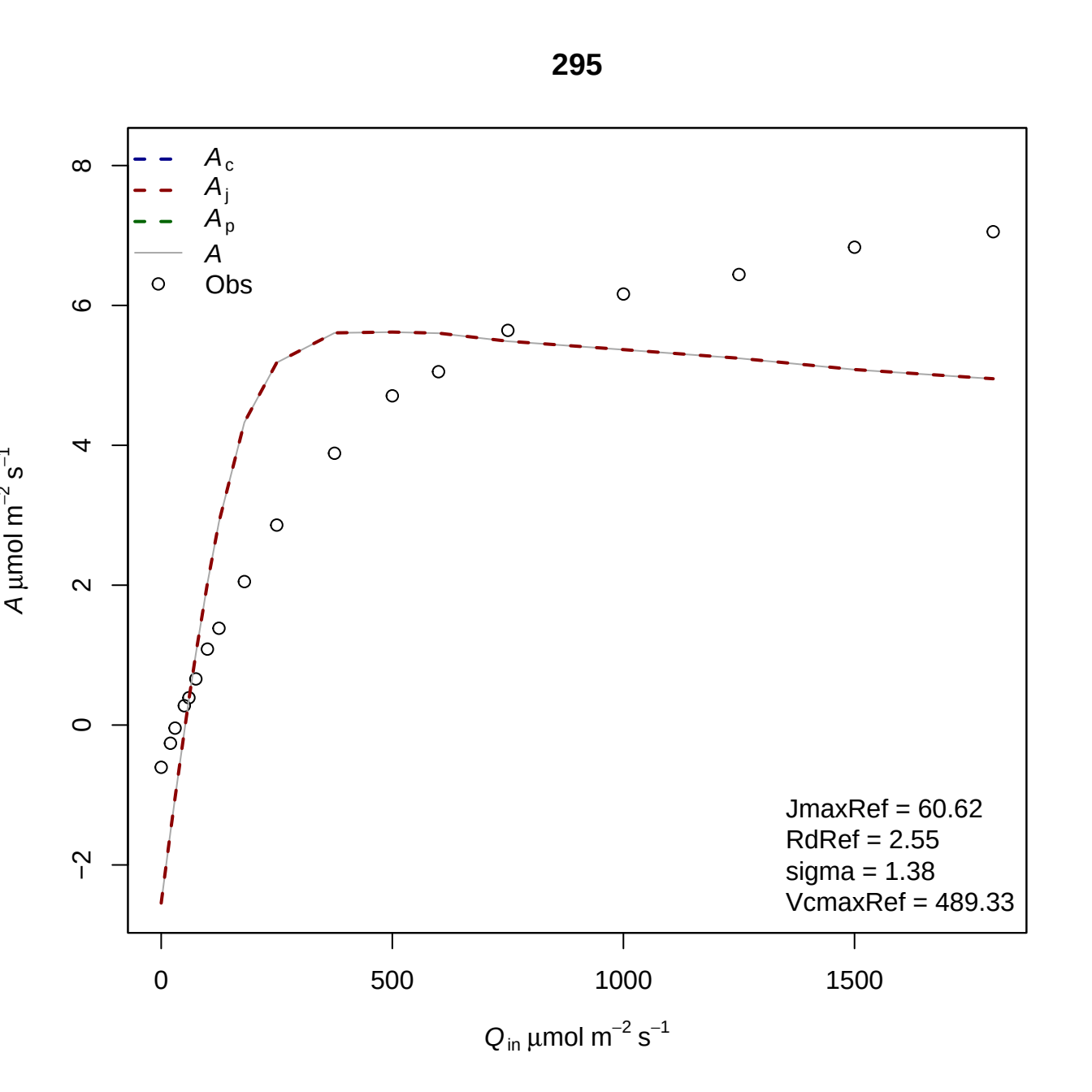

290
A_c
A_j
A_p
A
Obs
JmaxRef = 40.08
RdRef = 2.64
sigma = 1.31
VcmaxRef = 20504.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

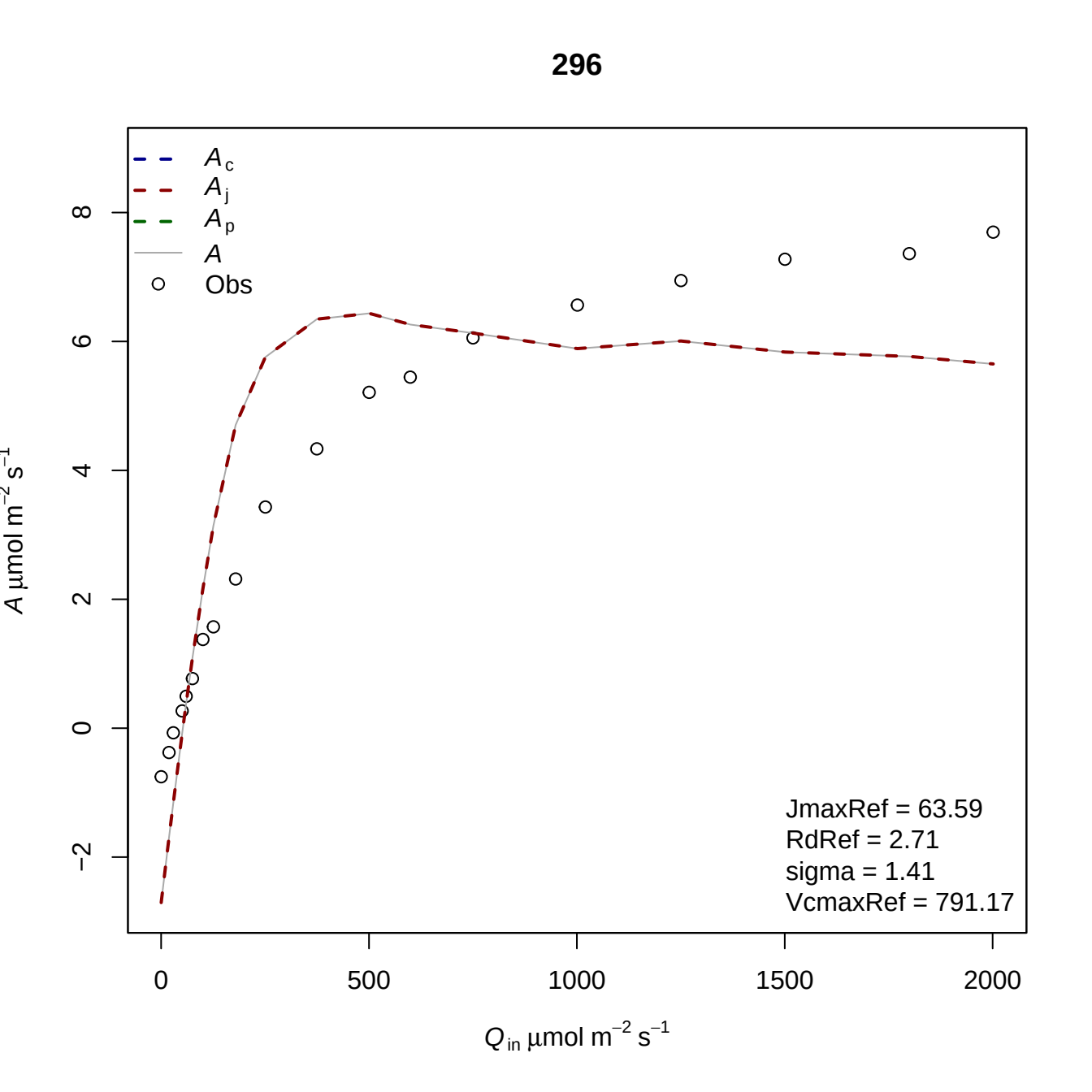

291
A_c
A_j
A_p
A
Obs
JmaxRef = 85.5
RdRef = 3.26
sigma = 1.89
VcmaxRef = 2661.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

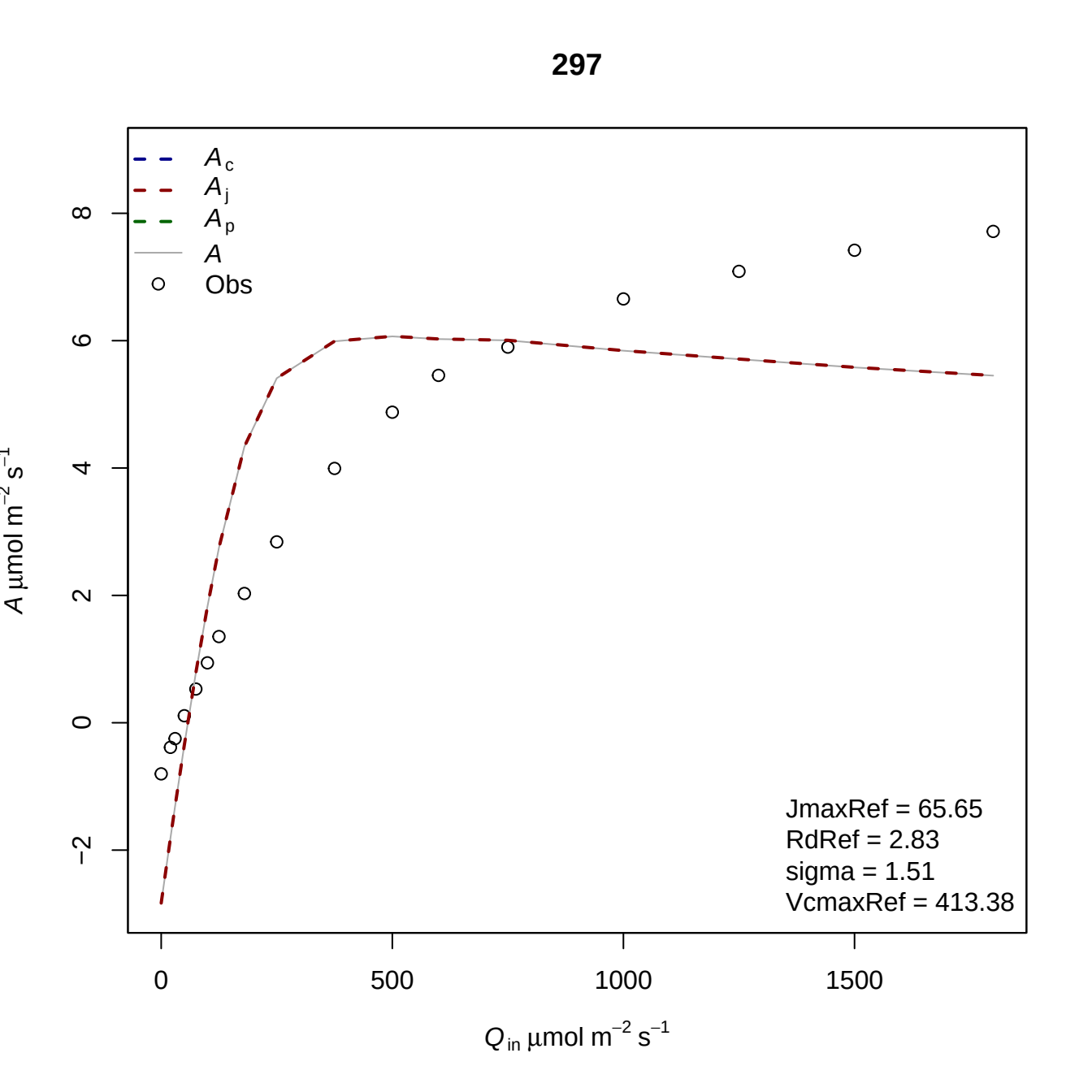

292
A_c
A_j
A_p
A
Obs
JmaxRef = 69.01
RdRef = 2.78
sigma = 1.37
VcmaxRef = 504.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

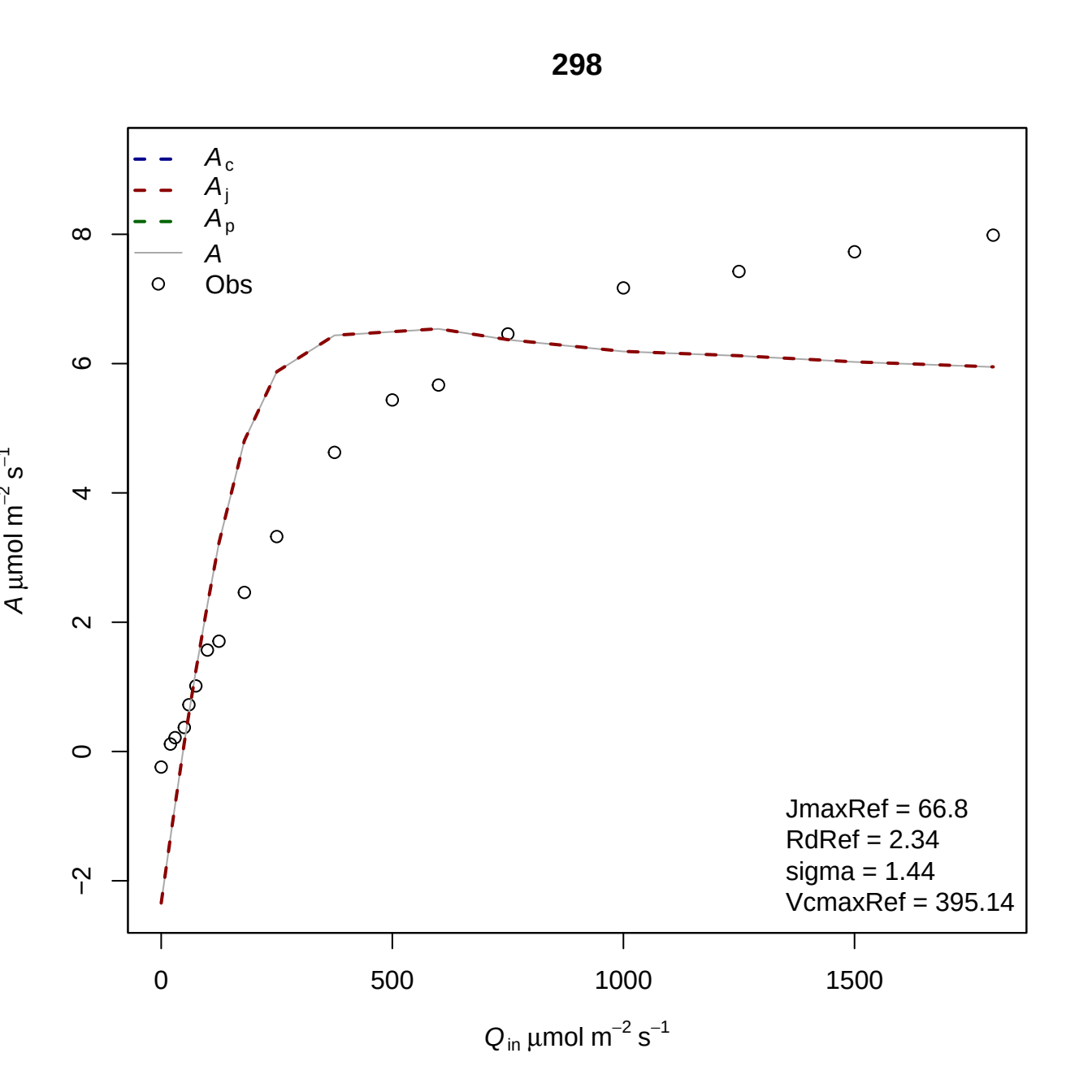

293
A_c
A_j
A_p
A
Obs
JmaxRef = 60.42
RdRef = 2.41
sigma = 1.17
VcmaxRef = 606.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

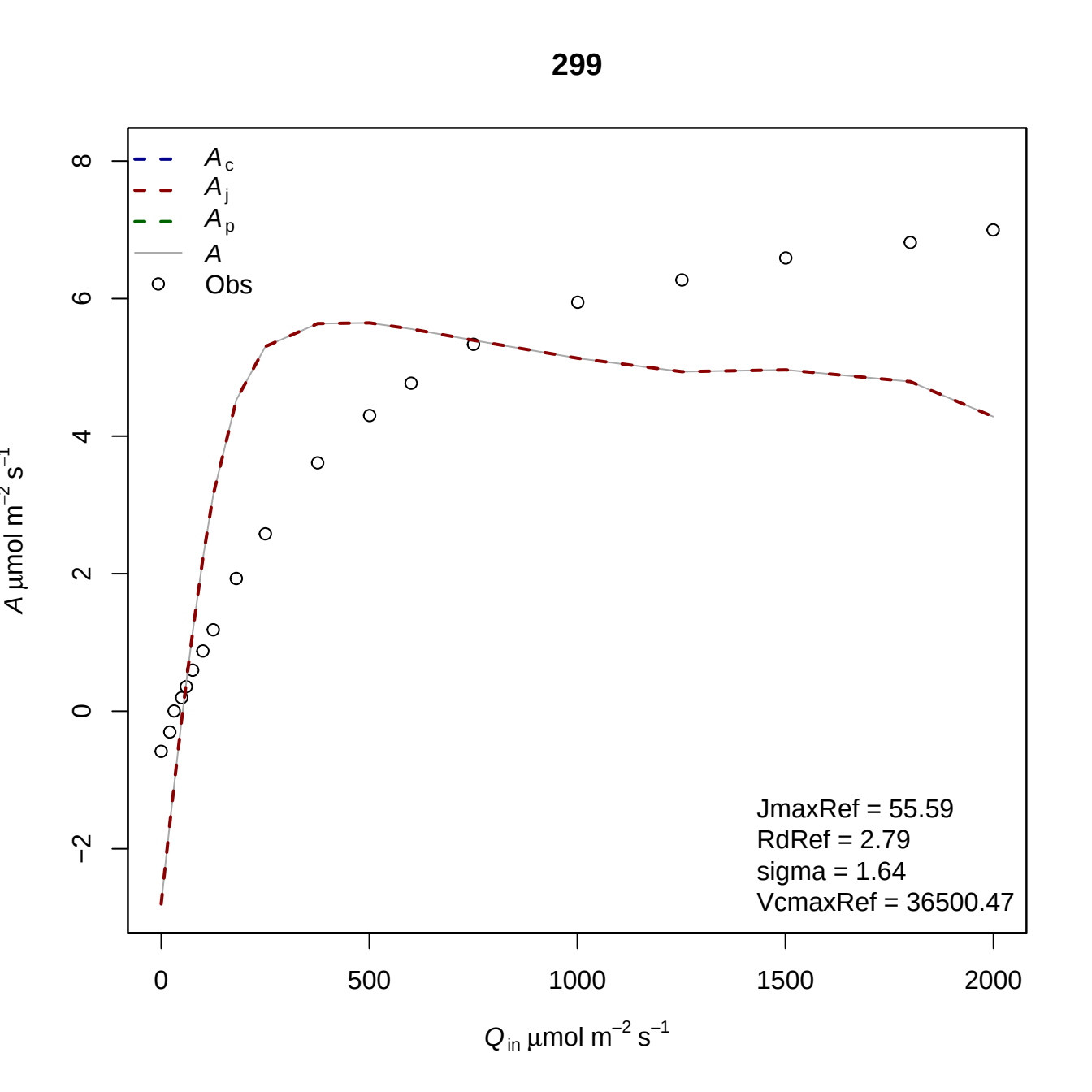

294
A_c
A_j
A_p
A
Obs
JmaxRef = 62.72
RdRef = 2.71
sigma = 1.61
VcmaxRef = 520.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

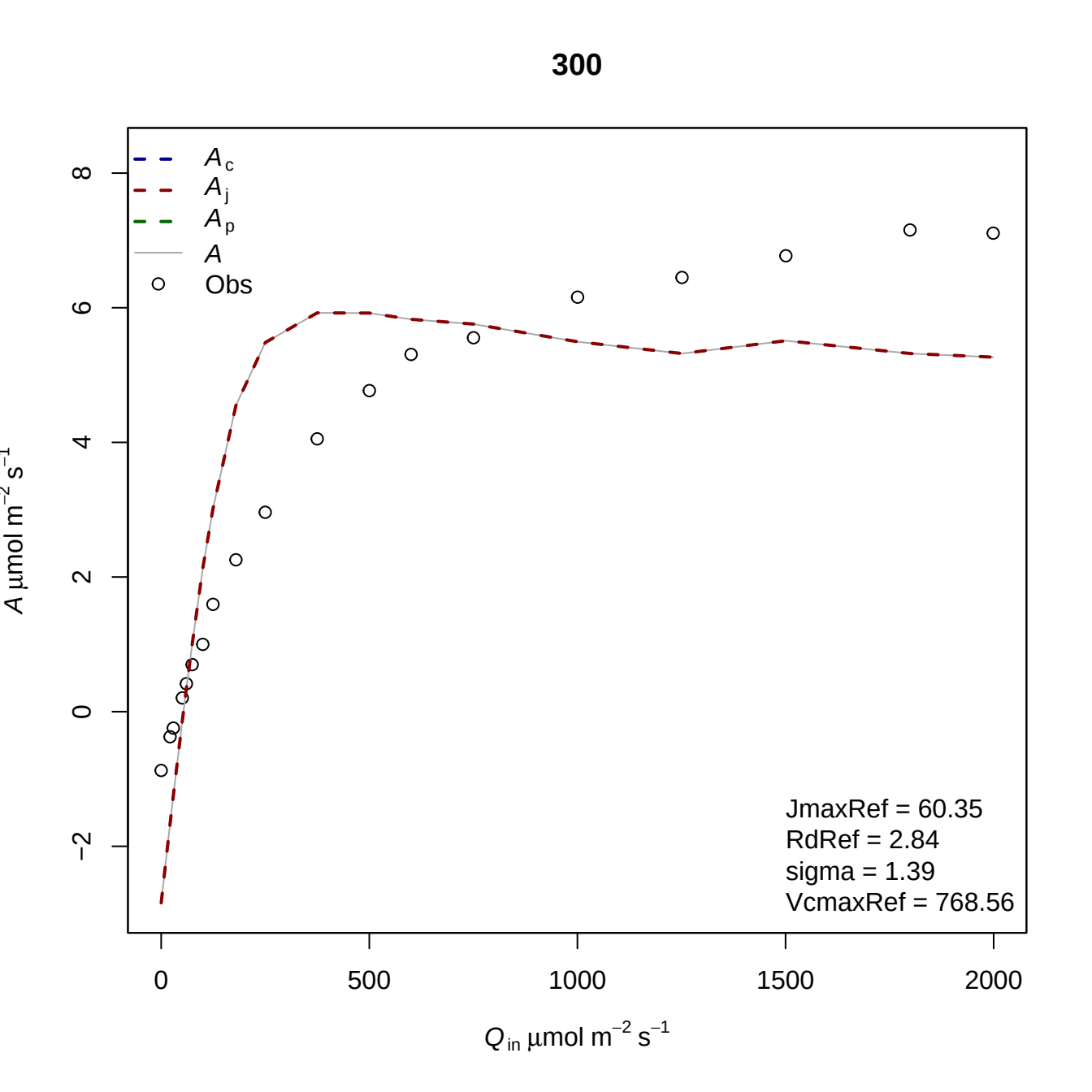

295
A_c
A_j
A_p
A
Obs
JmaxRef = 60.62
RdRef = 2.55
sigma = 1.38
VcmaxRef = 489.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

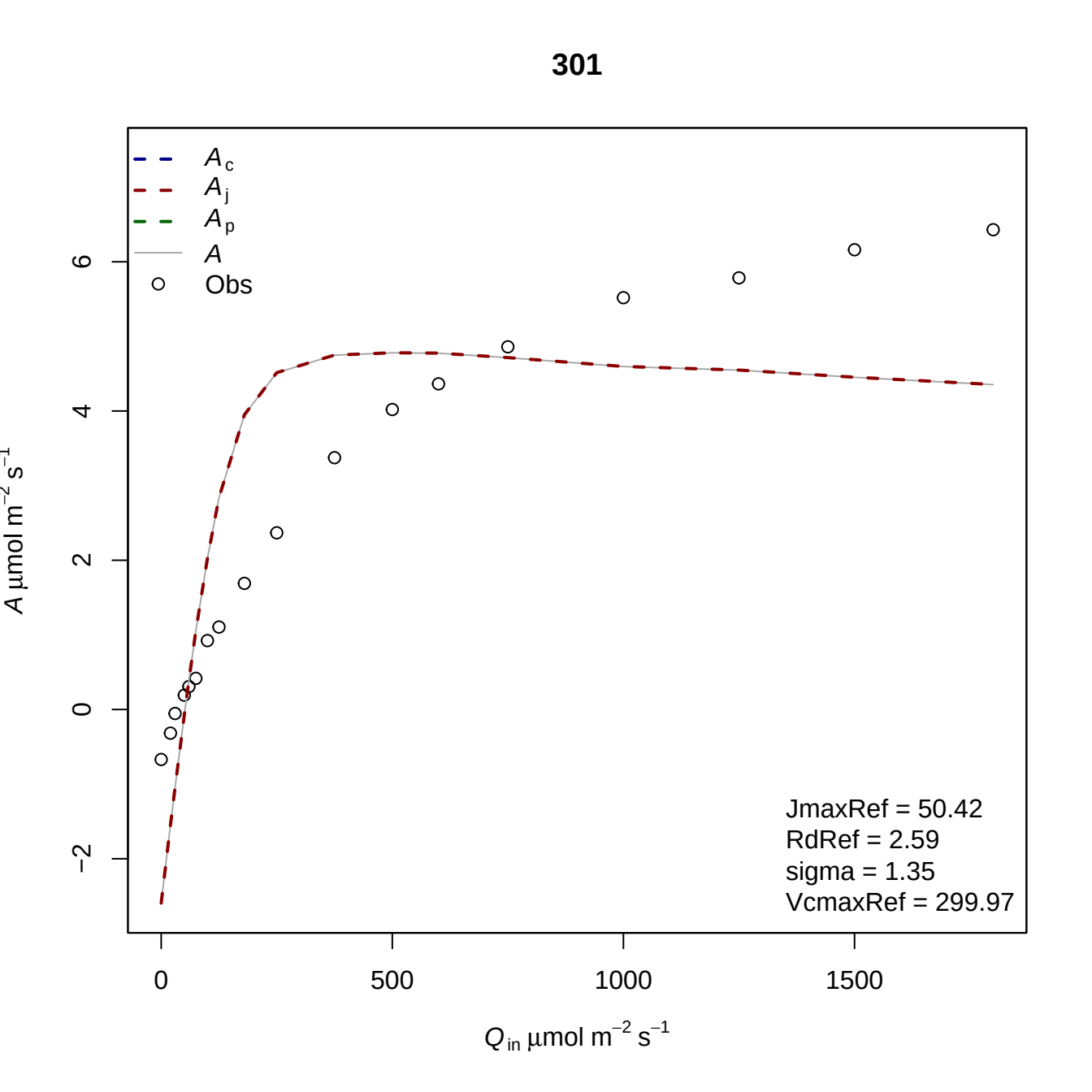

296
A_c
A_j
A_p
A
Obs
JmaxRef = 63.59
RdRef = 2.71
sigma = 1.41
VcmaxRef = 791.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

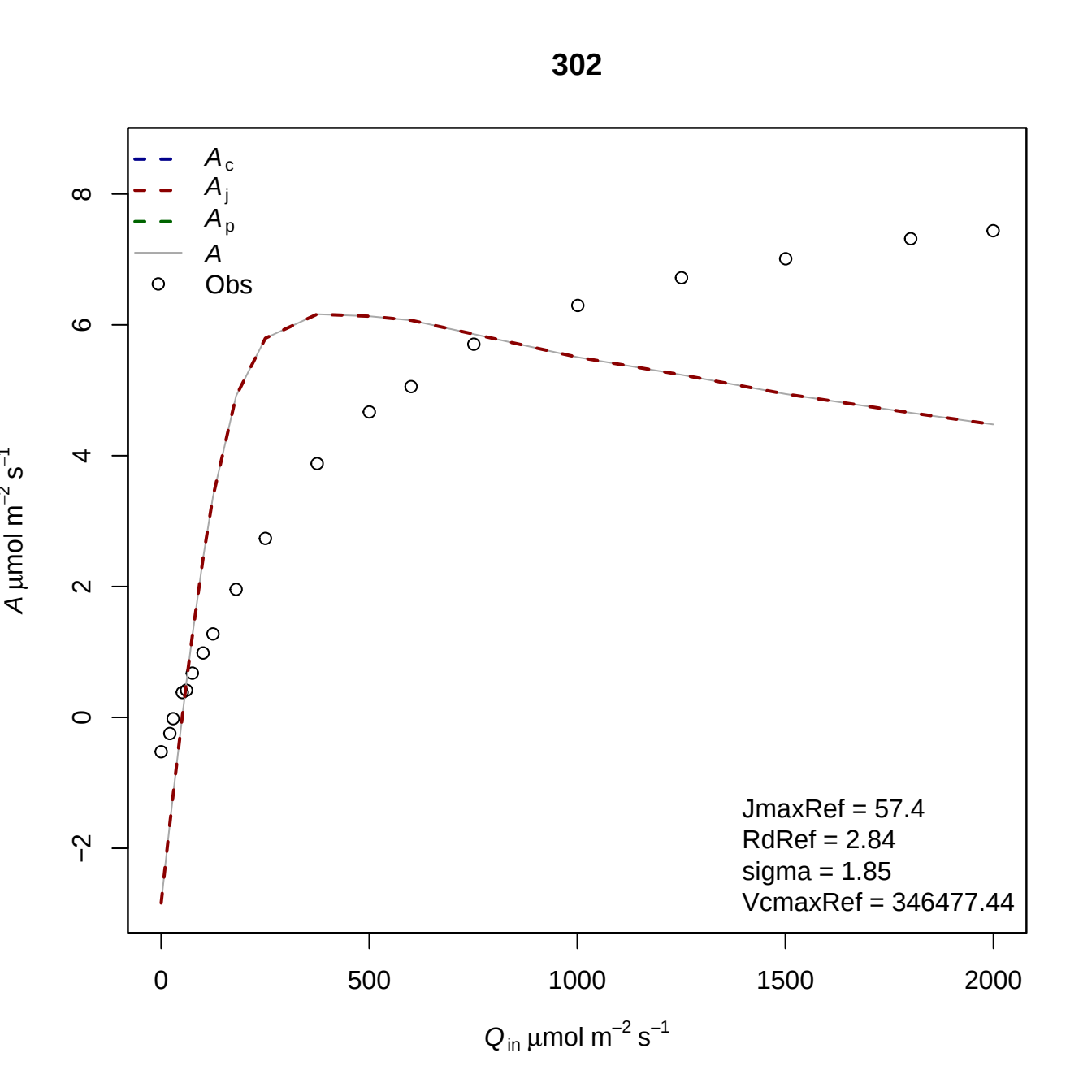

297
A_c
A_j
A_p
A
Obs
JmaxRef = 65.65
RdRef = 2.83
sigma = 1.51
VcmaxRef = 413.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

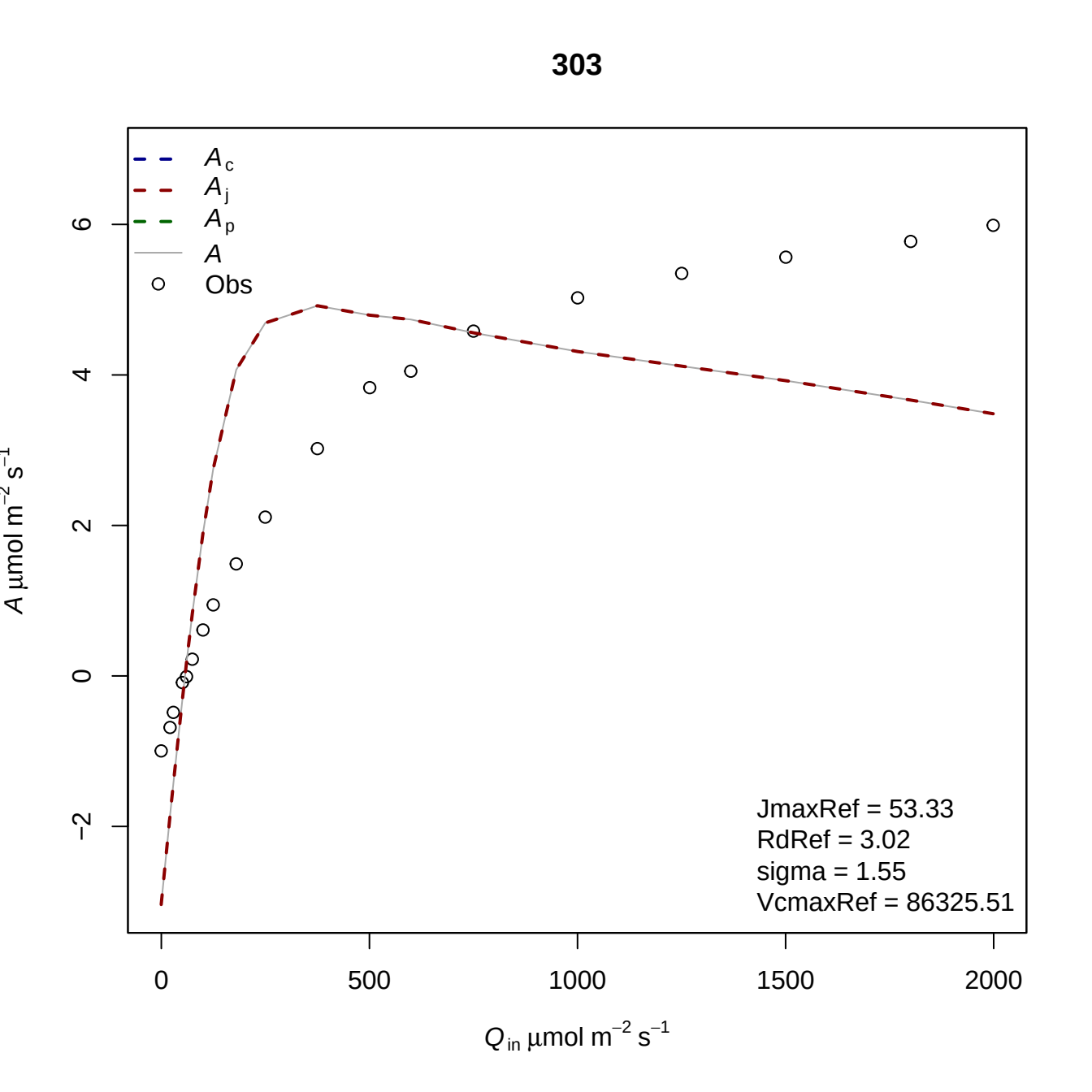

298
A_c
A_j
A_p
A
Obs
JmaxRef = 66.8
RdRef = 2.34
sigma = 1.44
VcmaxRef = 395.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

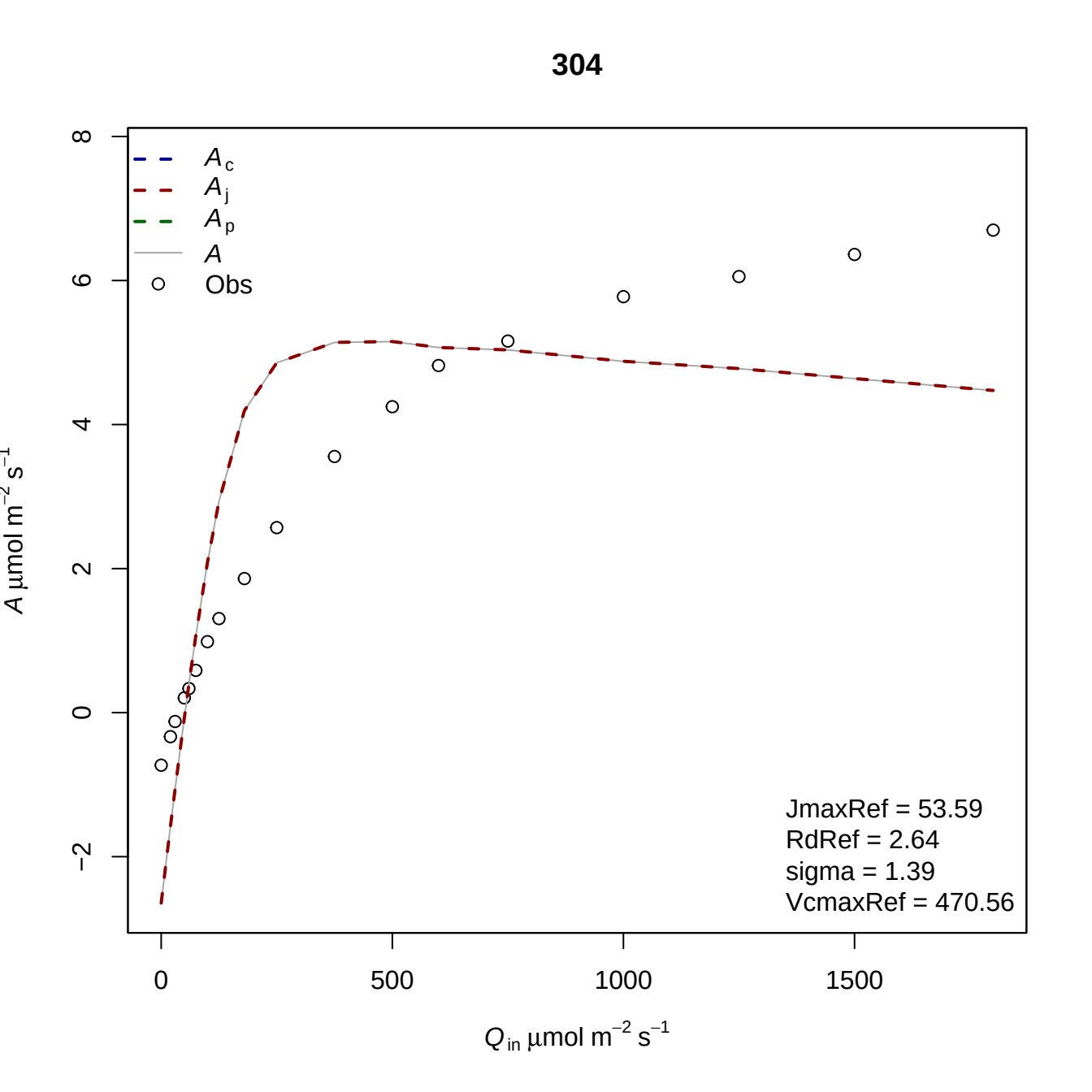

299
A_c
A_j
A_p
A
Obs
JmaxRef = 55.59
RdRef = 2.79
sigma = 1.64
VcmaxRef = 36500.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

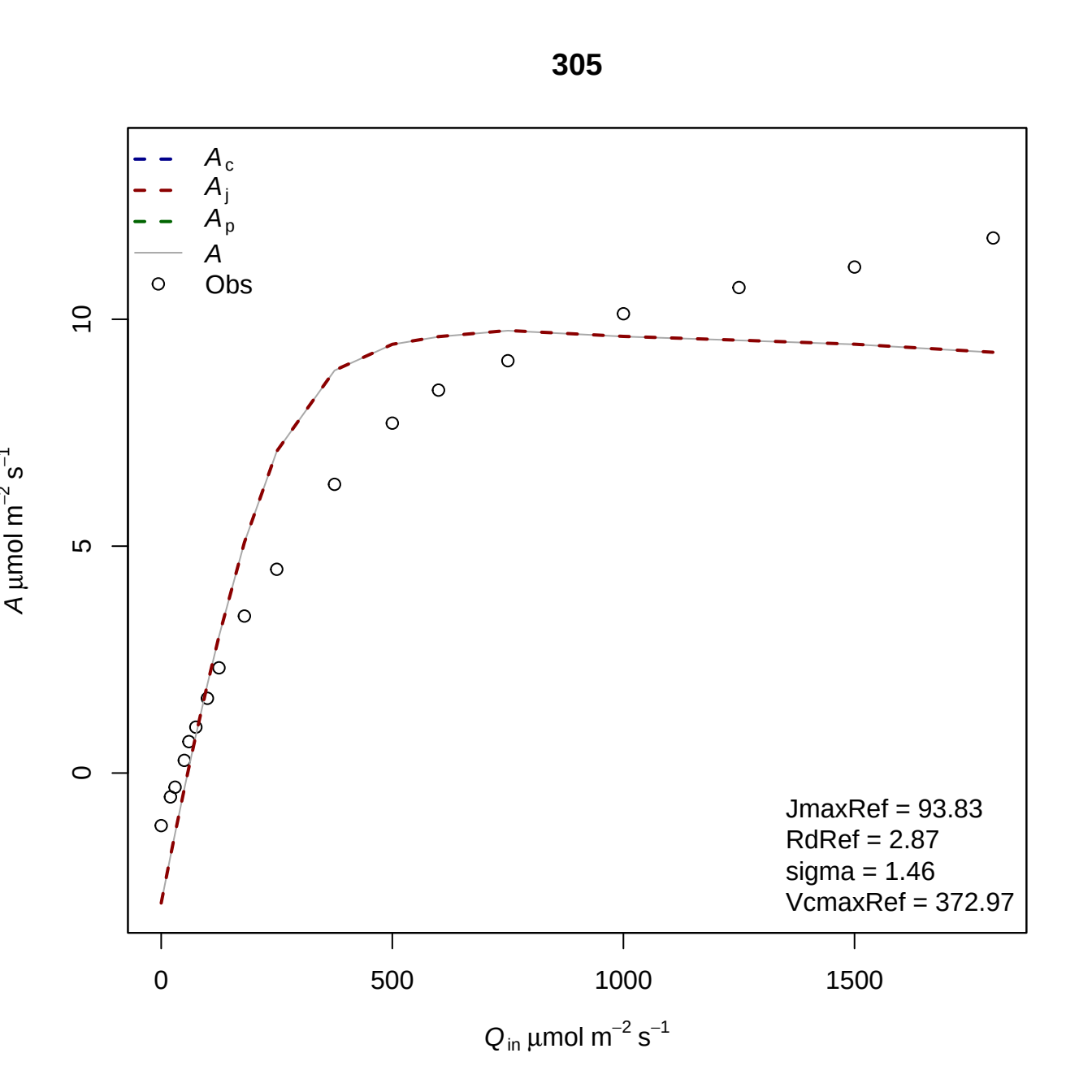

300
A_c
A_j
A_p
A
Obs
JmaxRef = 60.35
RdRef = 2.84
sigma = 1.39
VcmaxRef = 768.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

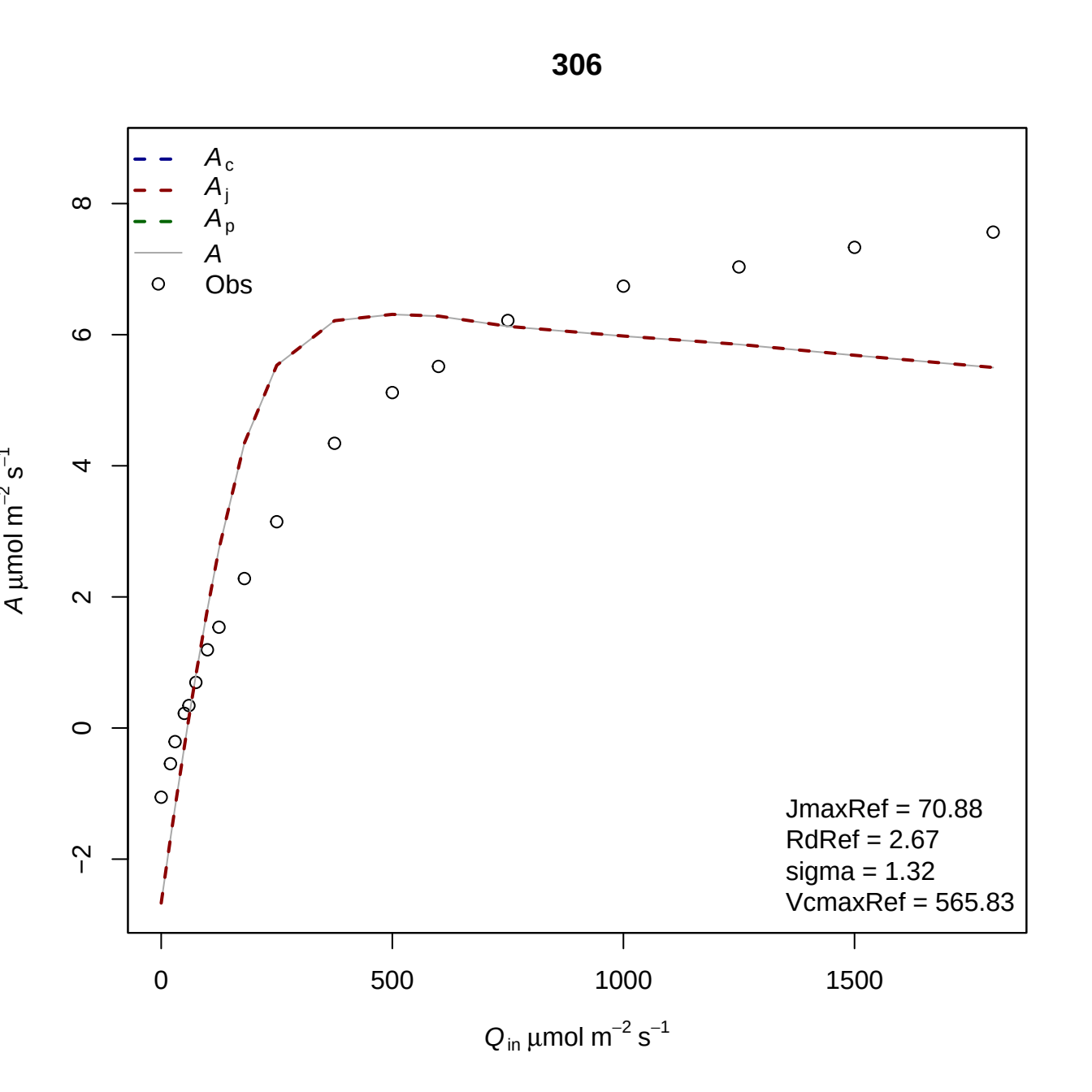

301
A_c
A_j
A_p
A
Obs
JmaxRef = 50.42
RdRef = 2.59
sigma = 1.35
VcmaxRef = 299.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

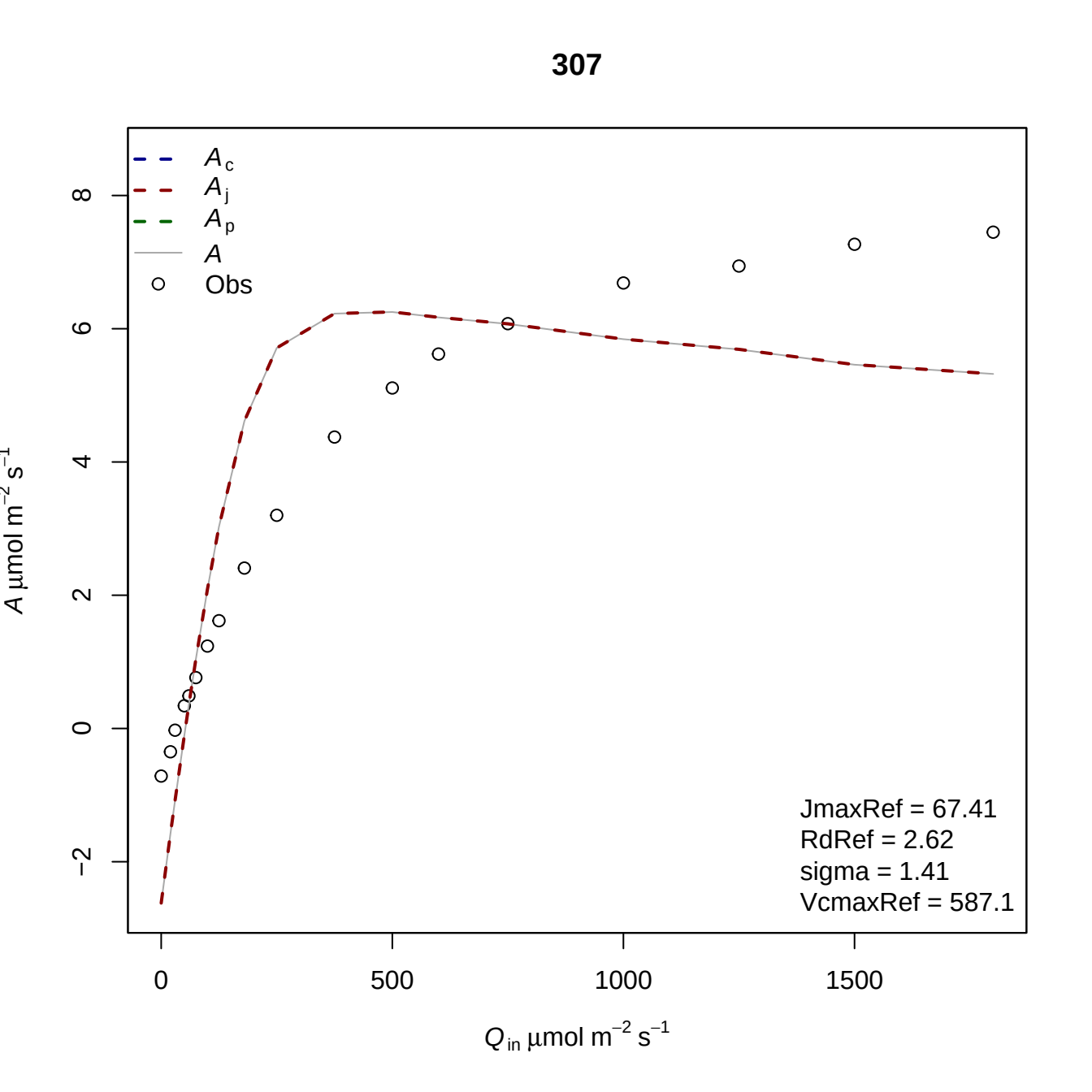

302
A_c
A_j
A_p
A
Obs
JmaxRef = 57.4
RdRef = 2.84
sigma = 1.85
VcmaxRef = 346477.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

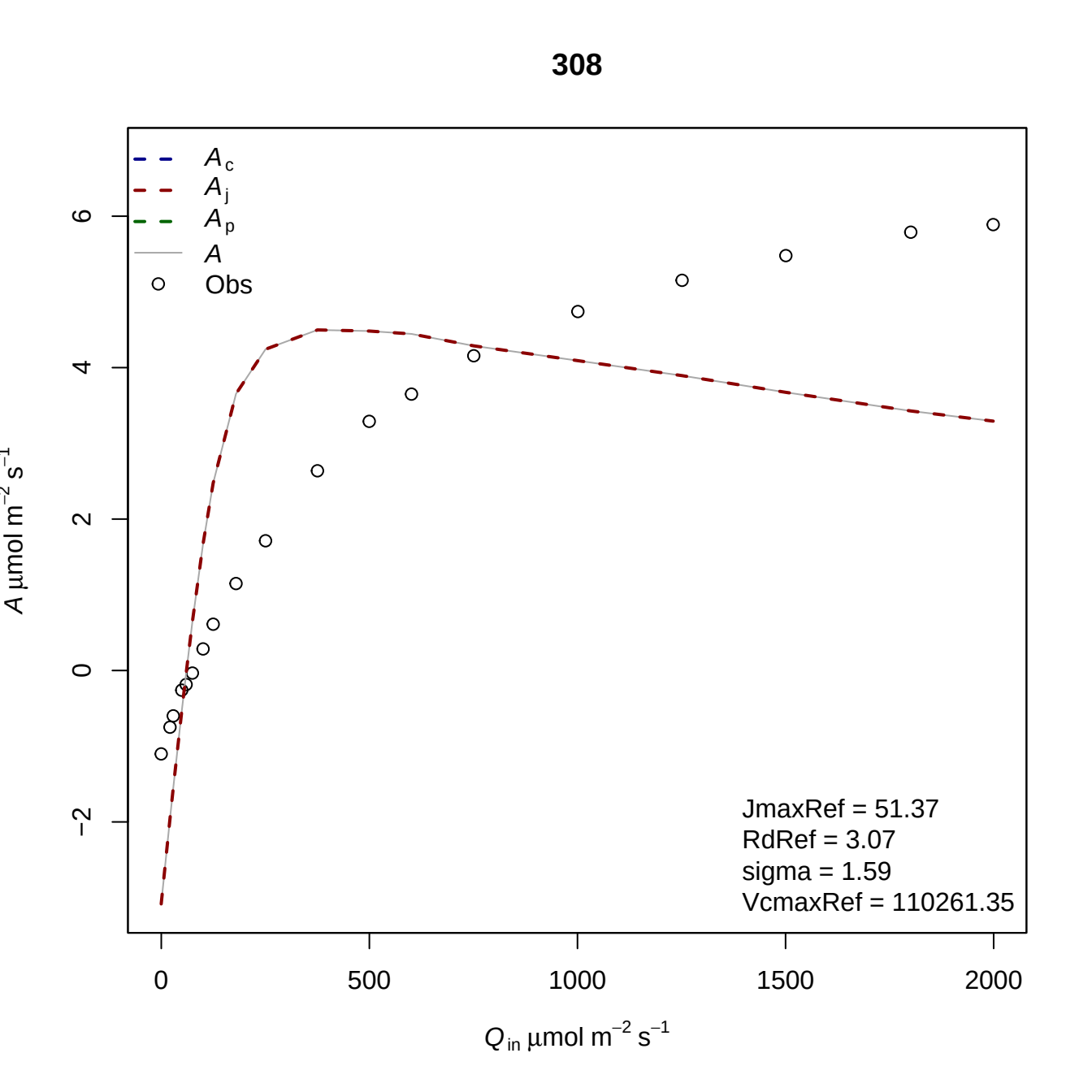

303
A_c
A_j
A_p
A
Obs
JmaxRef = 53.33
RdRef = 3.02
sigma = 1.55
VcmaxRef = 86325.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

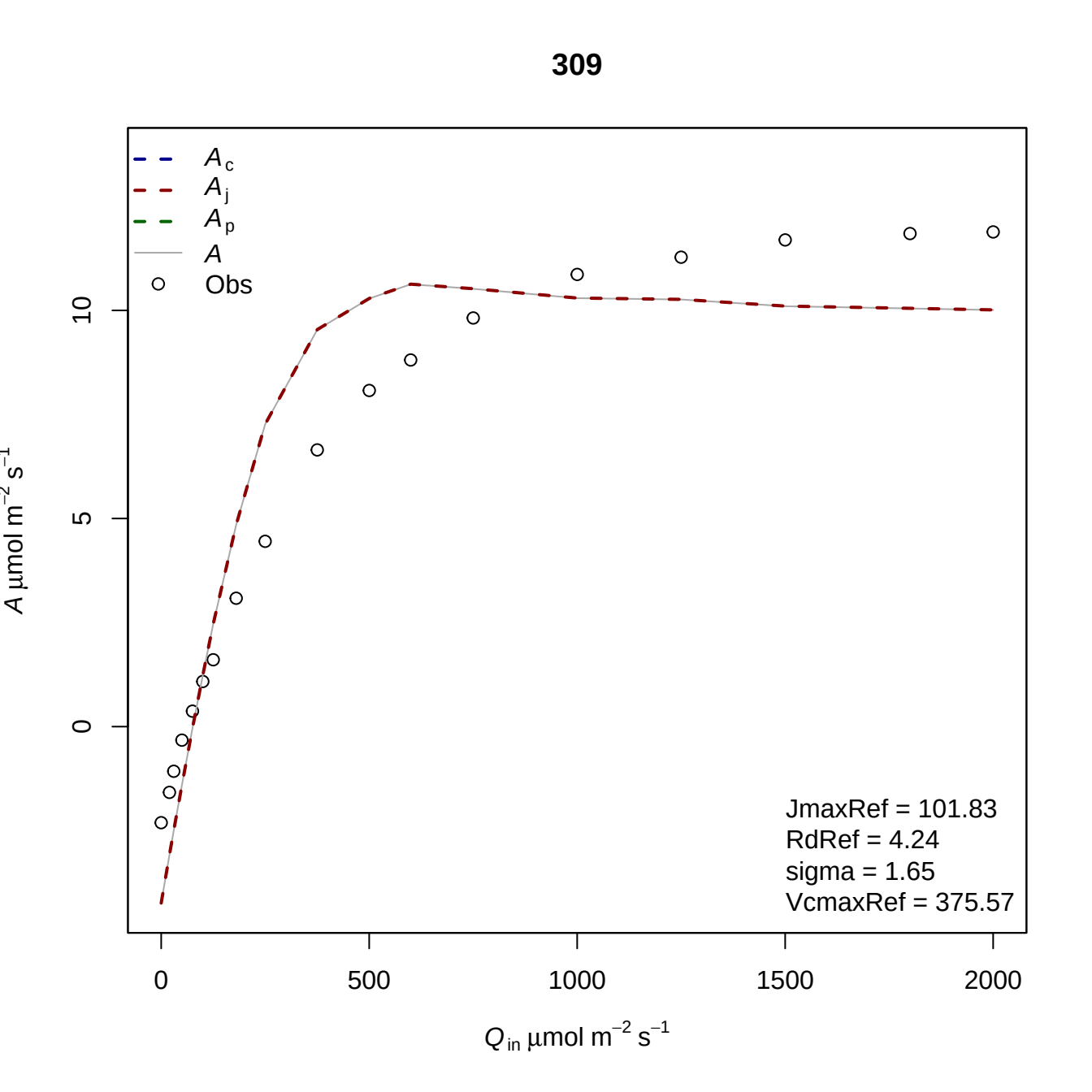

304
A_c
A_j
A_p
A
Obs
JmaxRef = 53.59
RdRef = 2.64
sigma = 1.39
VcmaxRef = 470.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

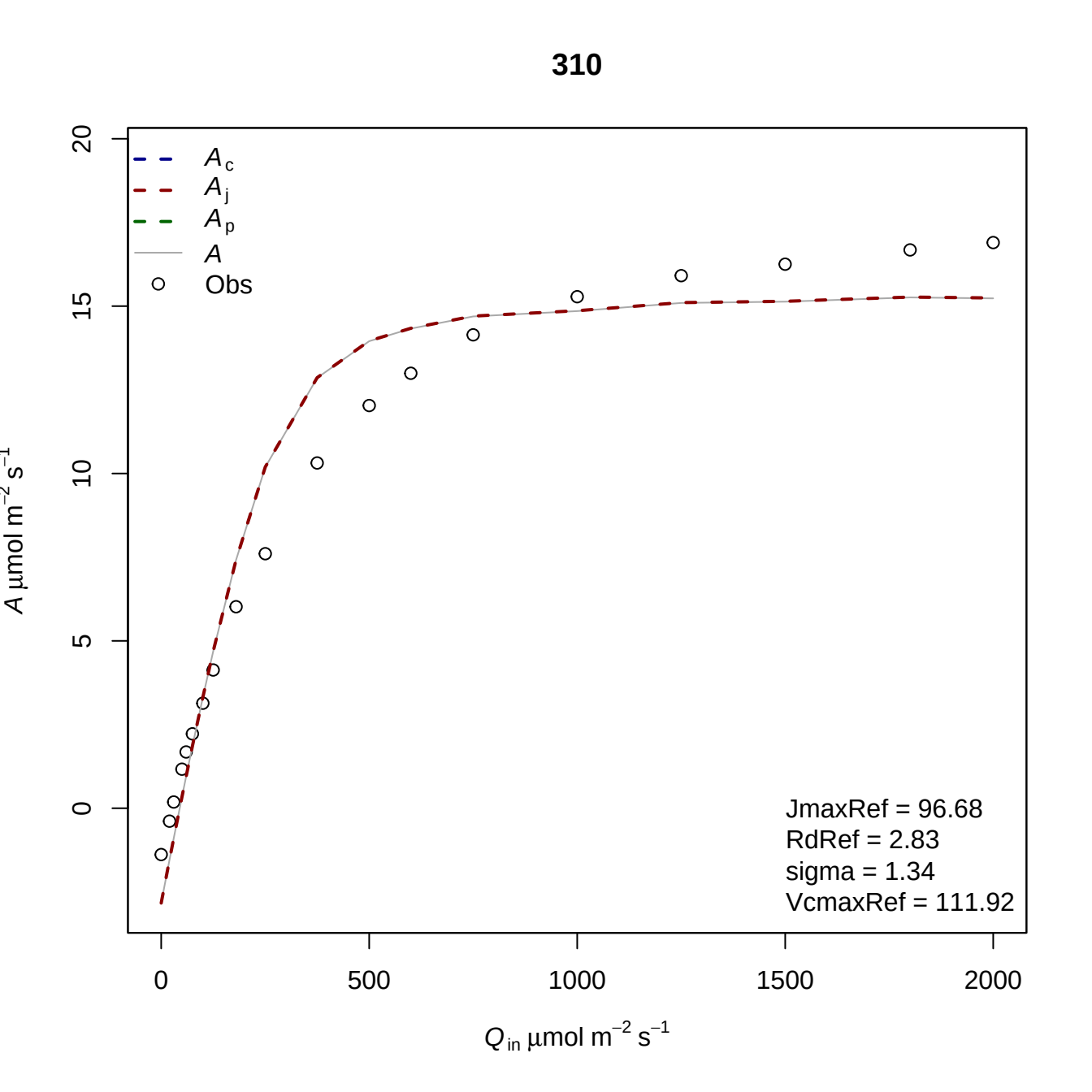

305
A_c
A_j
A_p
A
Obs
JmaxRef = 93.83
RdRef = 2.87
sigma = 1.46
VcmaxRef = 372.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

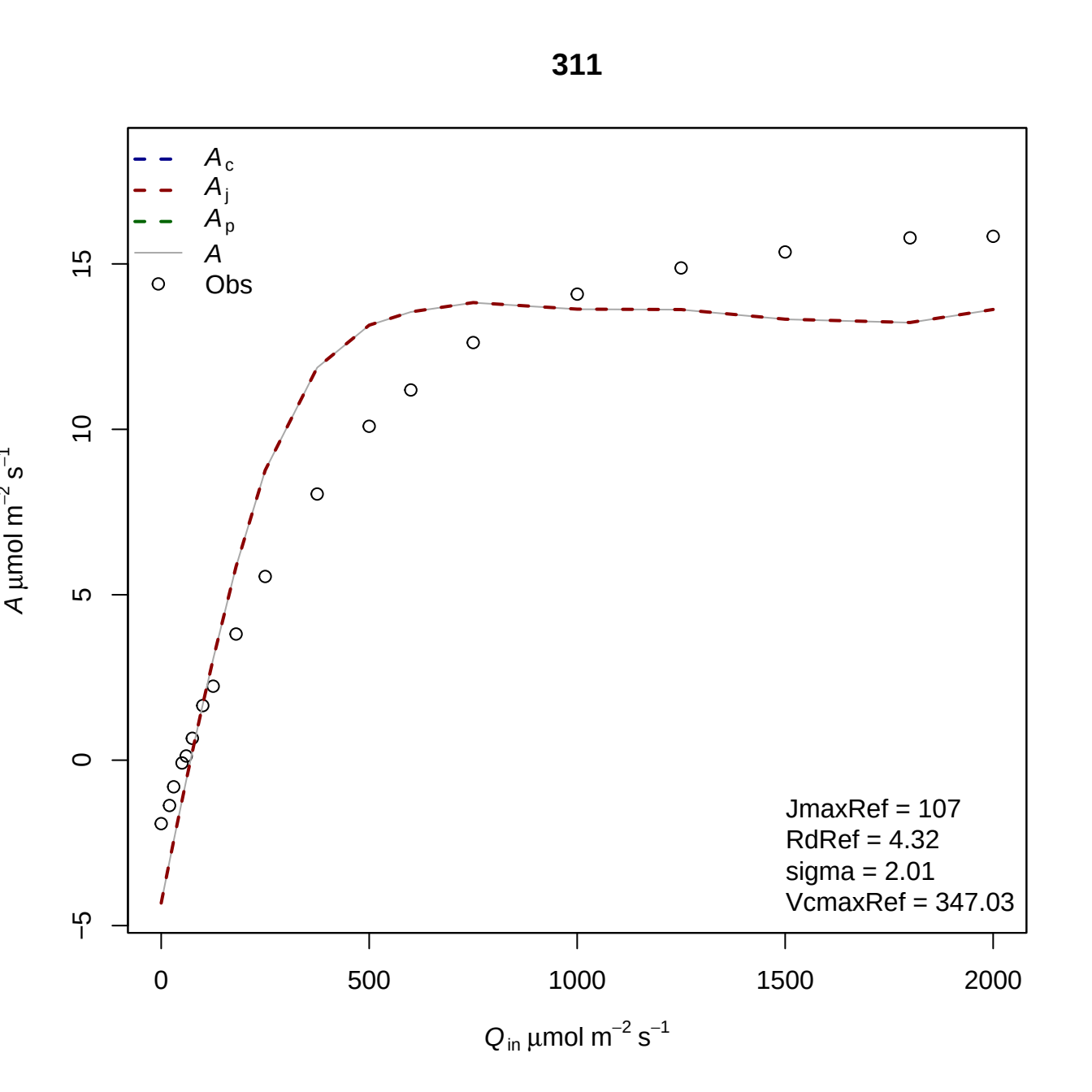

306
A_c
A_j
A_p
A
Obs
JmaxRef = 70.88
RdRef = 2.67
sigma = 1.32
VcmaxRef = 565.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

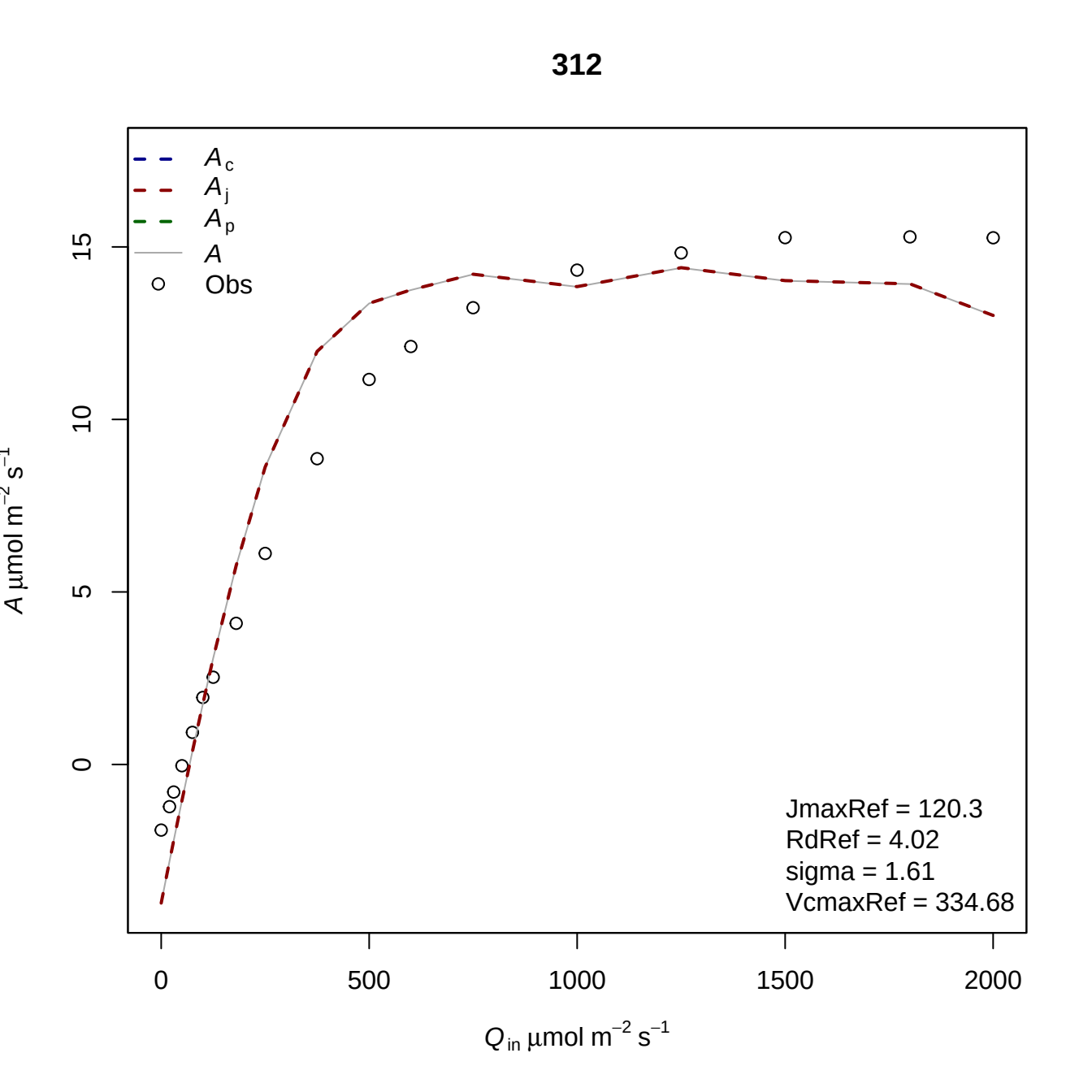

307
A_c
A_j
A_p
A
Obs
JmaxRef = 67.41
RdRef = 2.62
sigma = 1.41
VcmaxRef = 587.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

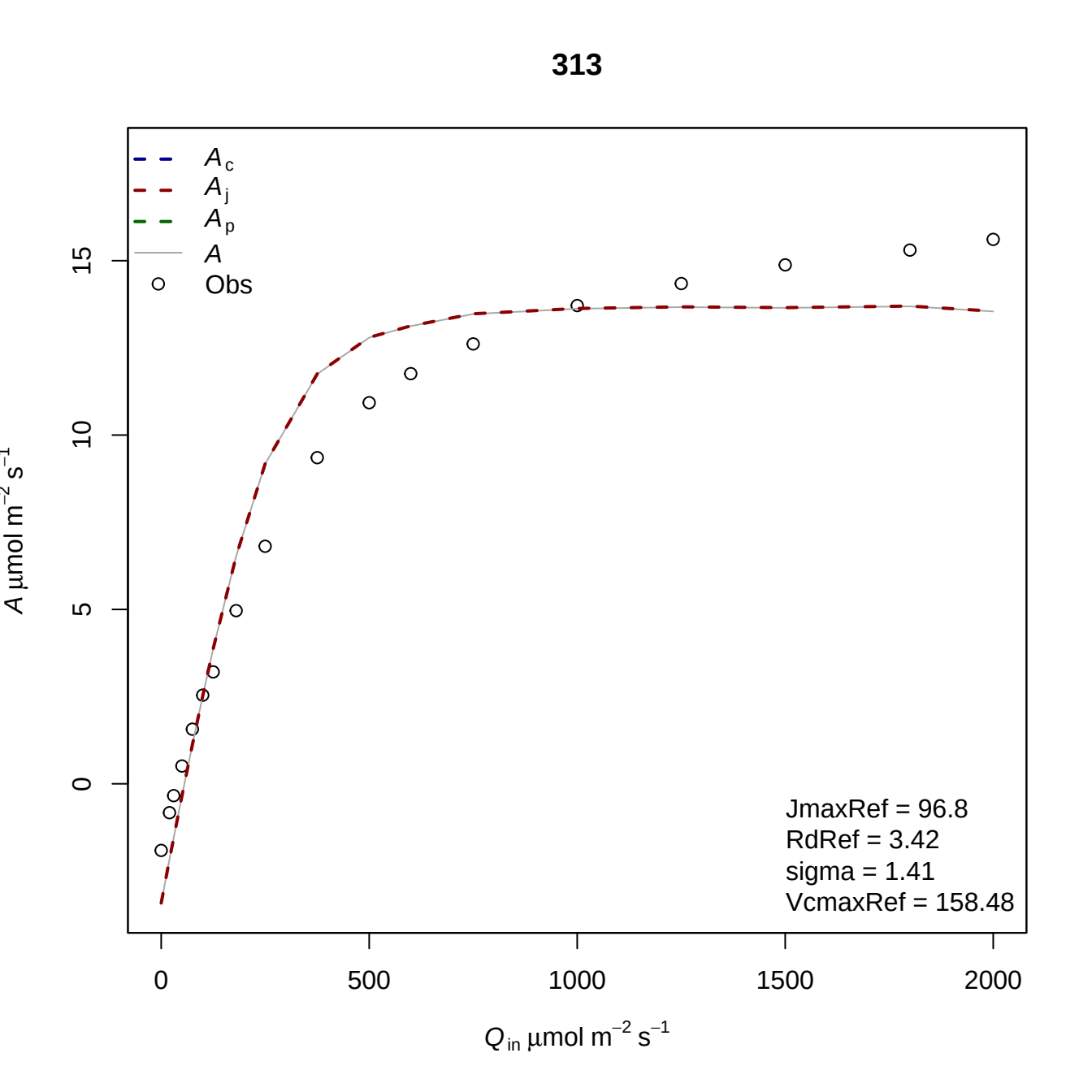

308
A_c
A_j
A_p
A
Obs
JmaxRef = 51.37
RdRef = 3.07
sigma = 1.59
VcmaxRef = 110261.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

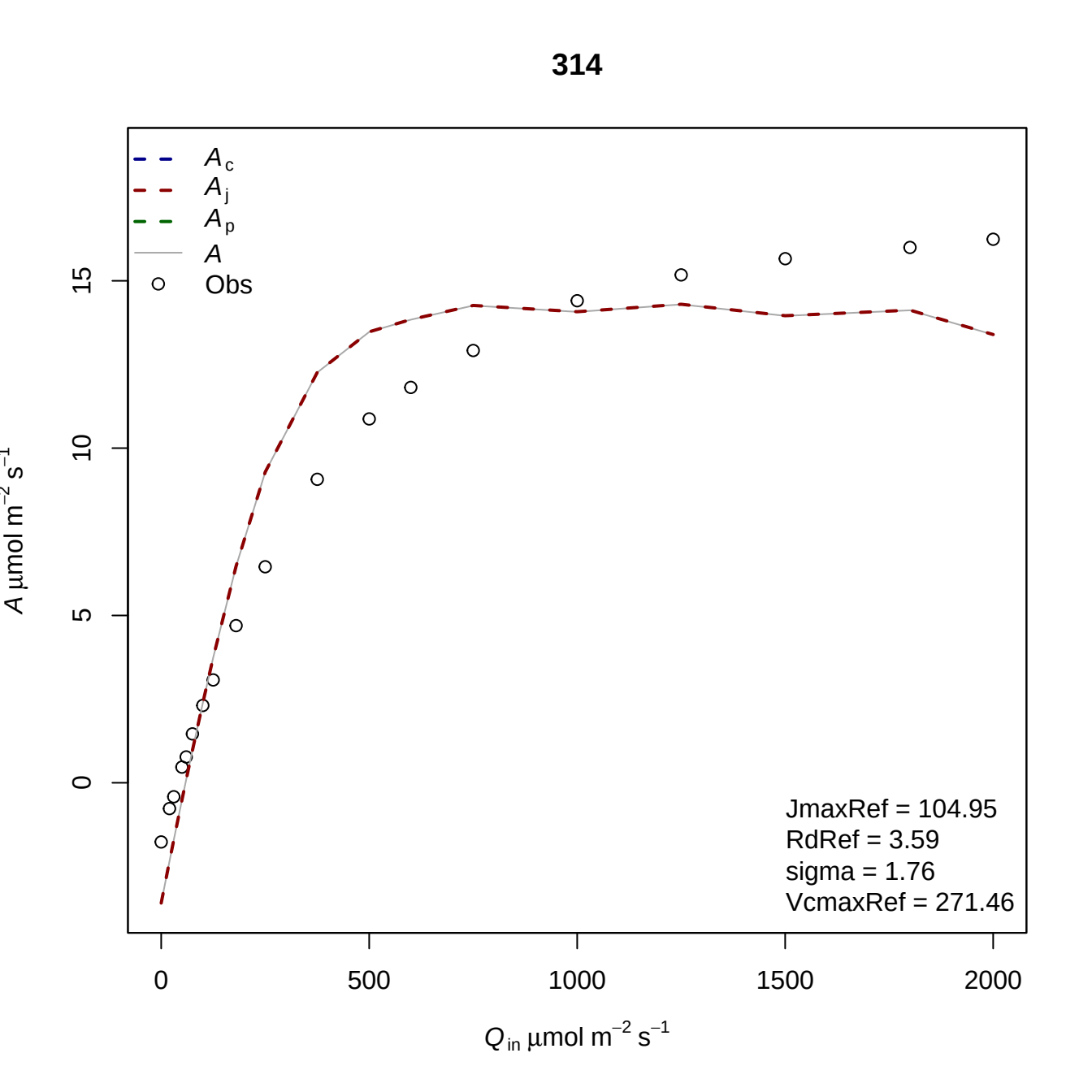

309
A_c
A_j
A_p
A
Obs
JmaxRef = 101.83
RdRef = 4.24
sigma = 1.65
VcmaxRef = 375.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

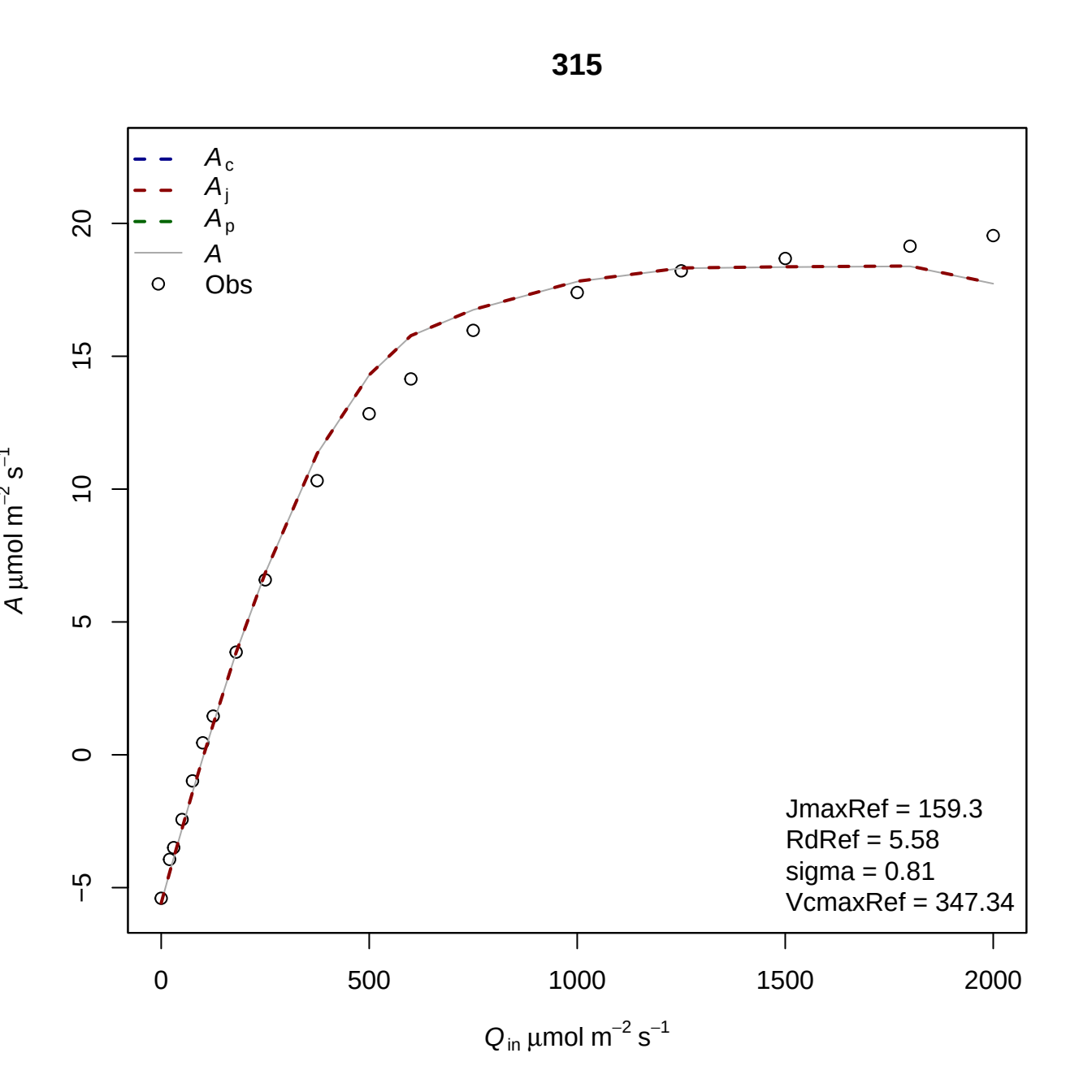

310
A_c
A_j
A_p
A
Obs
JmaxRef = 96.68
RdRef = 2.83
sigma = 1.34
VcmaxRef = 111.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

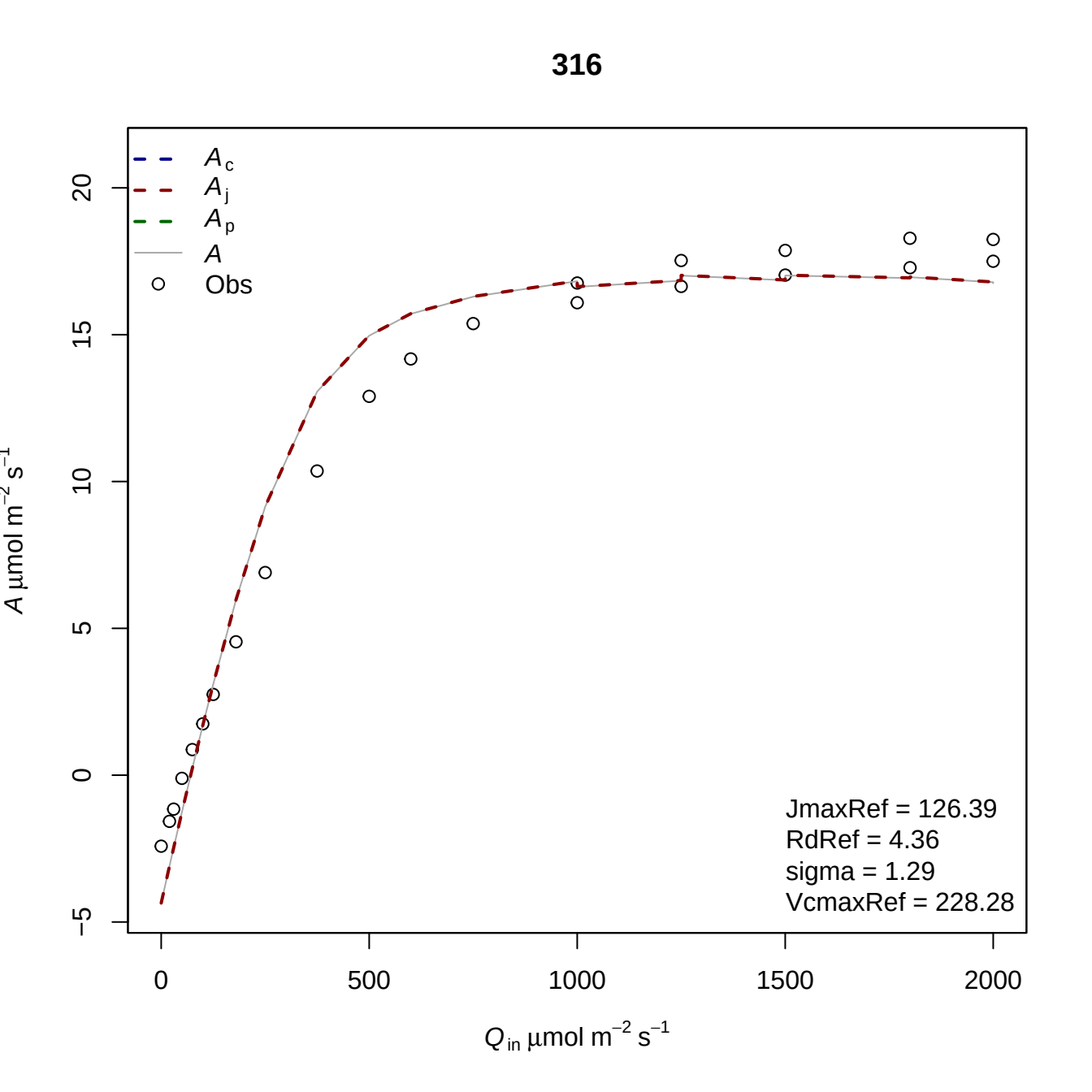

311
A_c
A_j
A_p
A
Obs
JmaxRef = 107
RdRef = 4.32
sigma = 2.01
VcmaxRef = 347.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

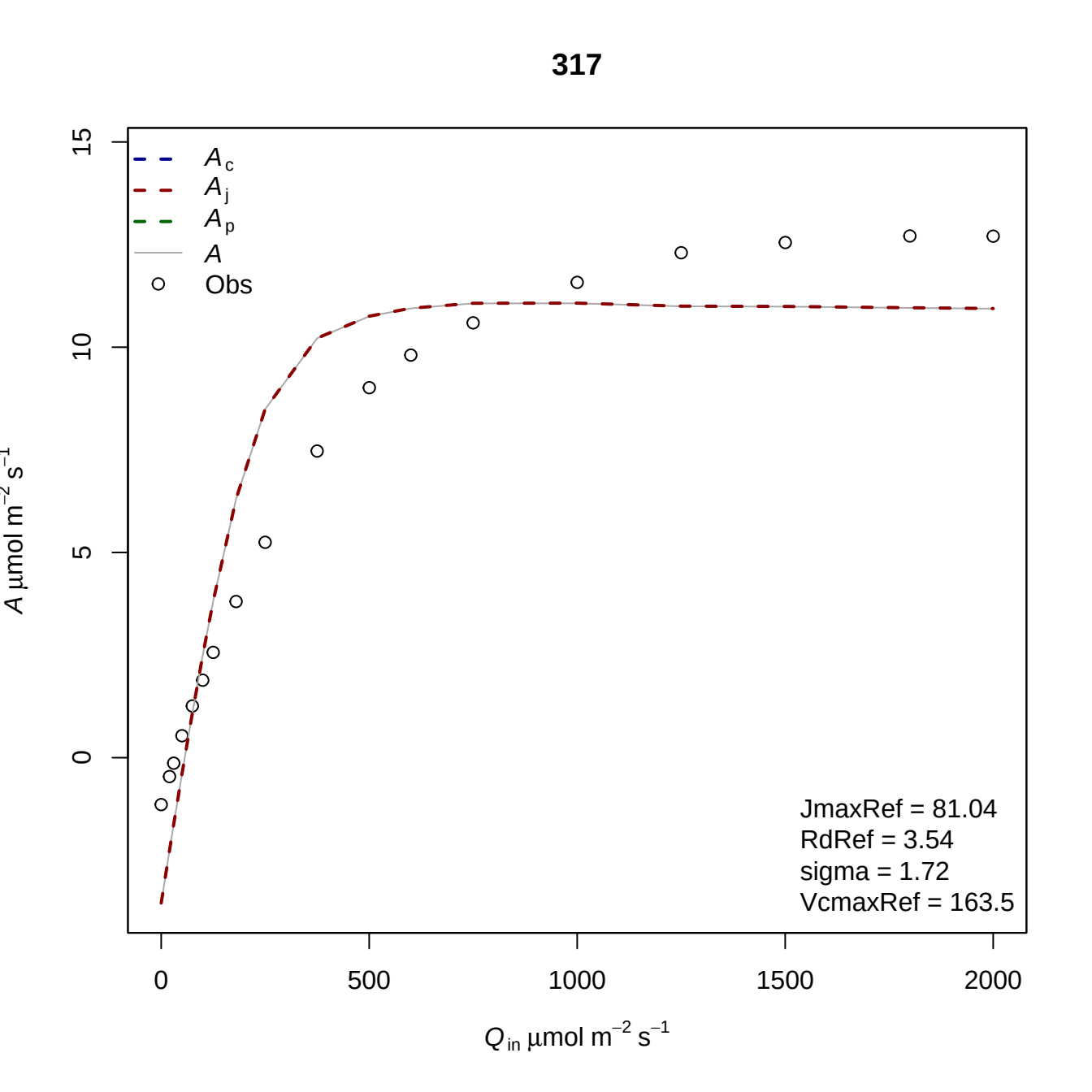

312
A_c
A_j
A_p
A
Obs
JmaxRef = 120.3
RdRef = 4.02
sigma = 1.61
VcmaxRef = 334.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

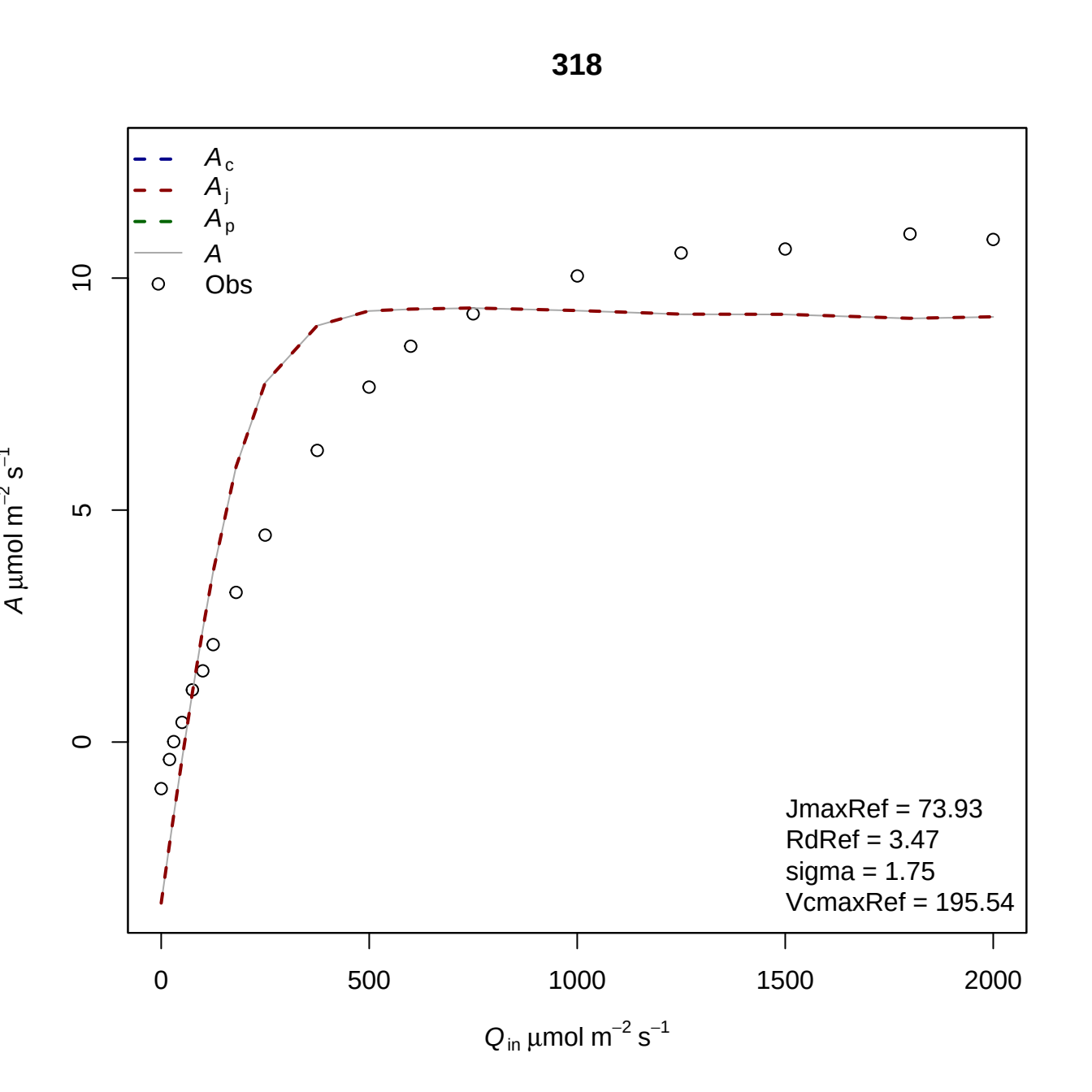

313
A_c
A_j
A_p
A
Obs
JmaxRef = 96.8
RdRef = 3.42
sigma = 1.41
VcmaxRef = 158.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

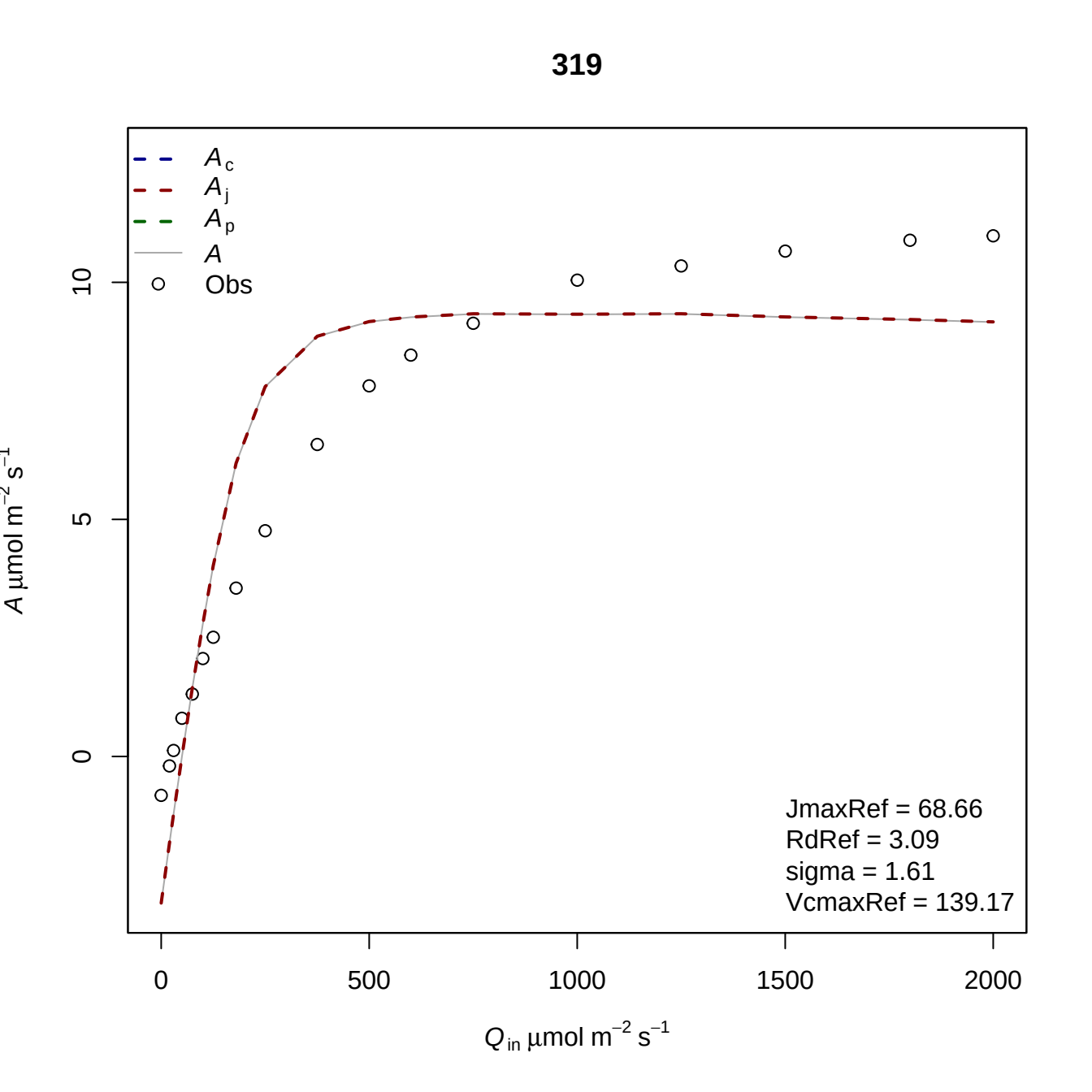

314
A_c
A_j
A_p
A
Obs
JmaxRef = 104.95
RdRef = 3.59
sigma = 1.76
VcmaxRef = 271.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

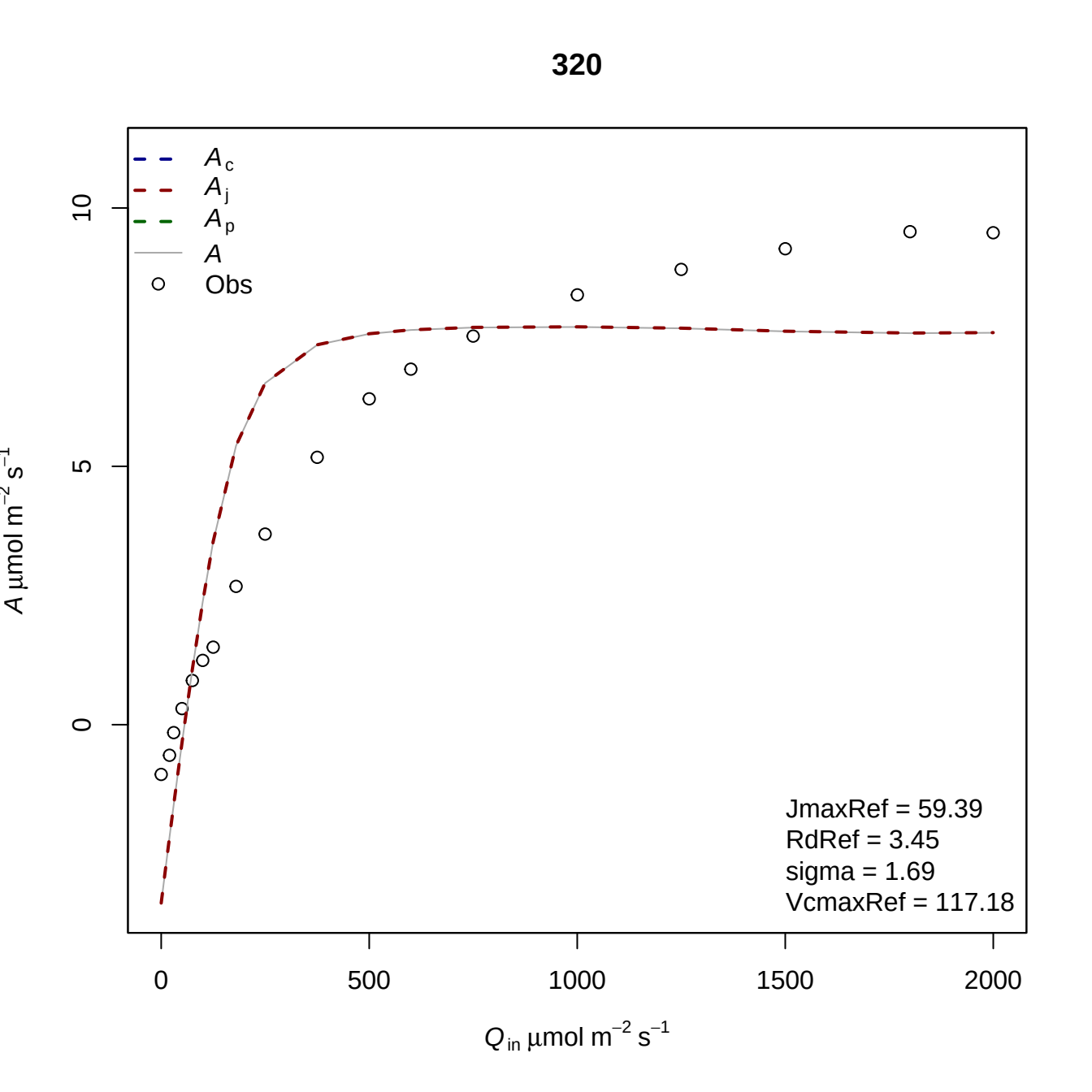

315
A_c
A_j
A_p
A
Obs
JmaxRef = 159.3
RdRef = 5.58
sigma = 0.81
VcmaxRef = 347.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

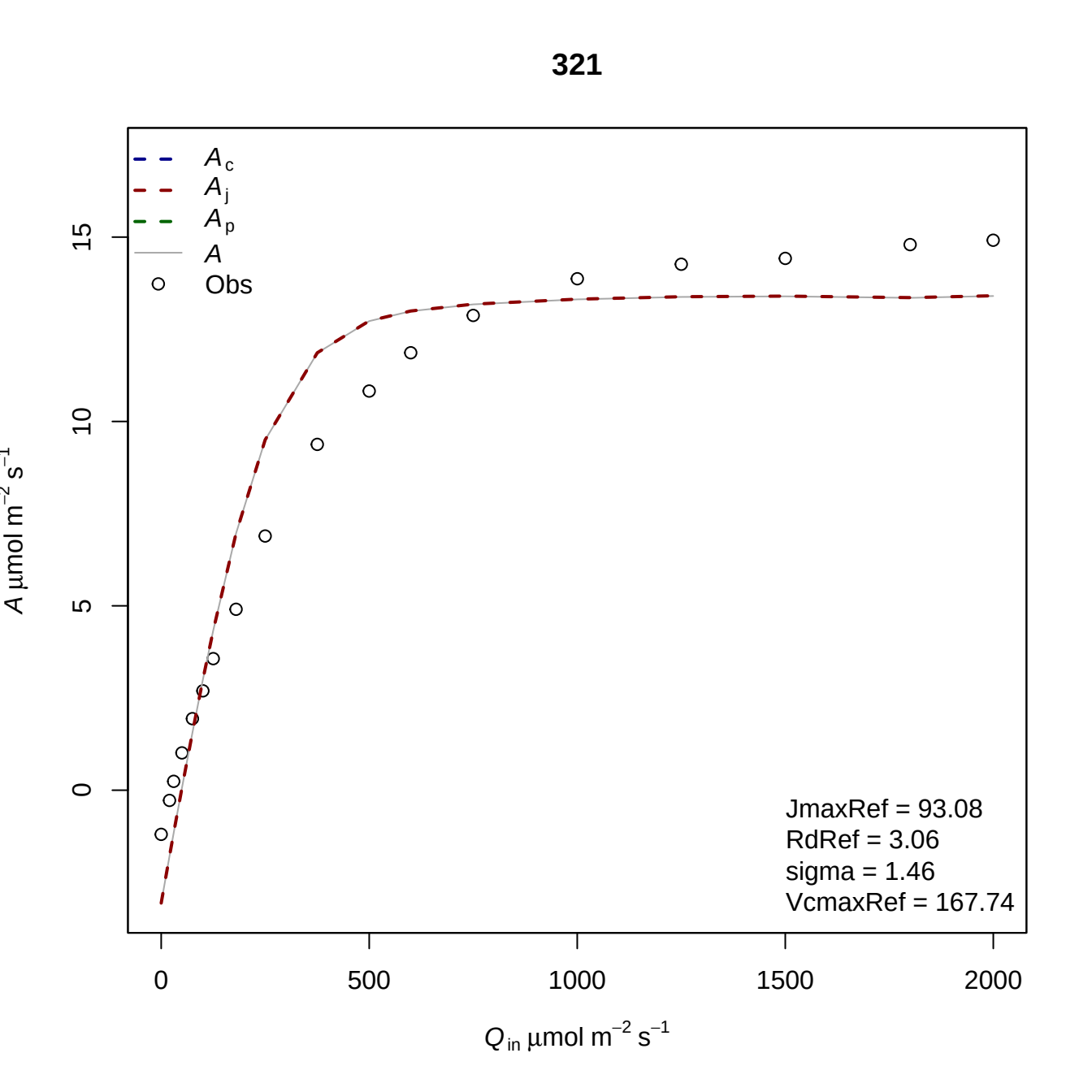

316
A_c
A_j
A_p
A
Obs
JmaxRef = 126.39
RdRef = 4.36
sigma = 1.29
VcmaxRef = 228.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

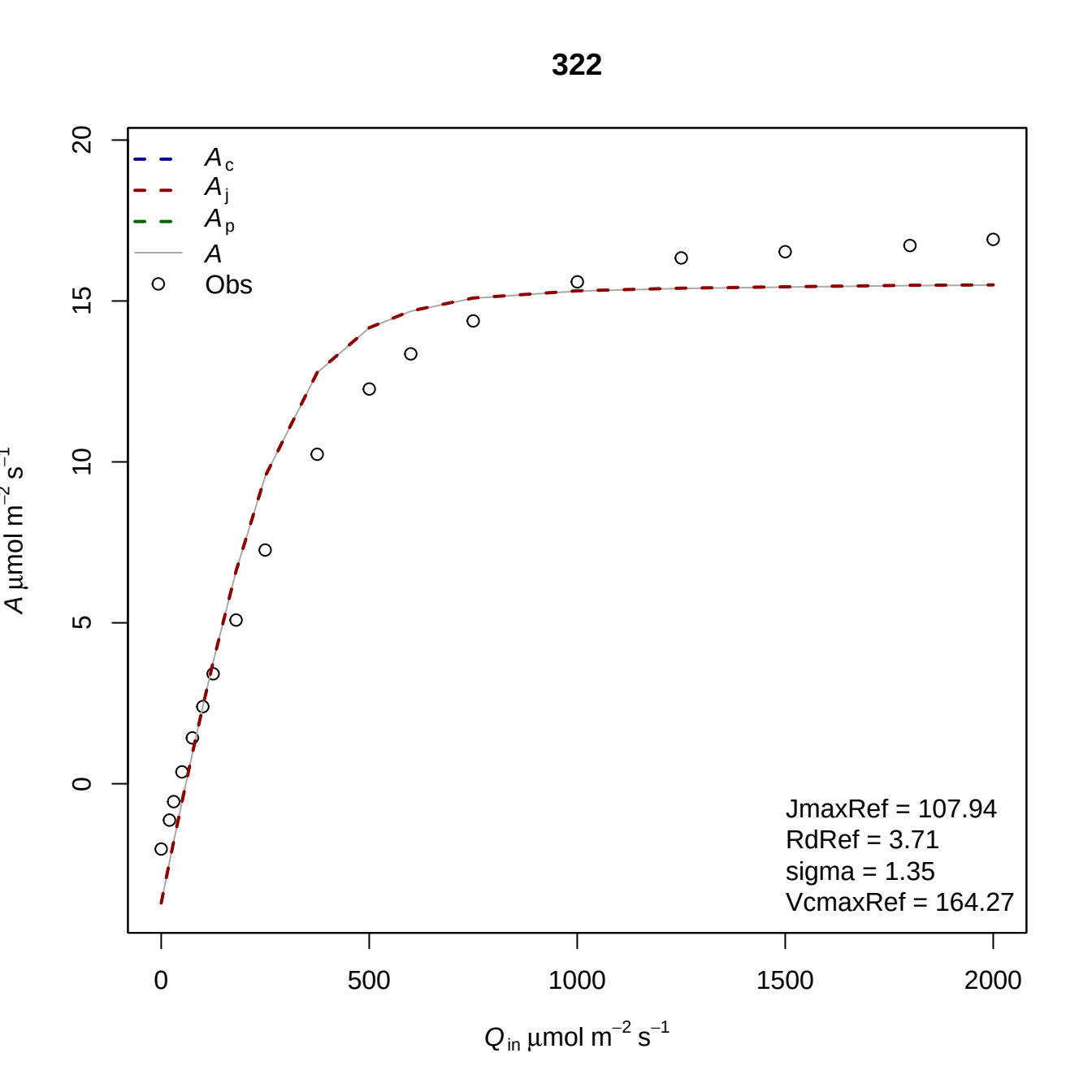

317
A_c
A_j
A_p
A
Obs
JmaxRef = 81.04
RdRef = 3.54
sigma = 1.72
VcmaxRef = 163.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

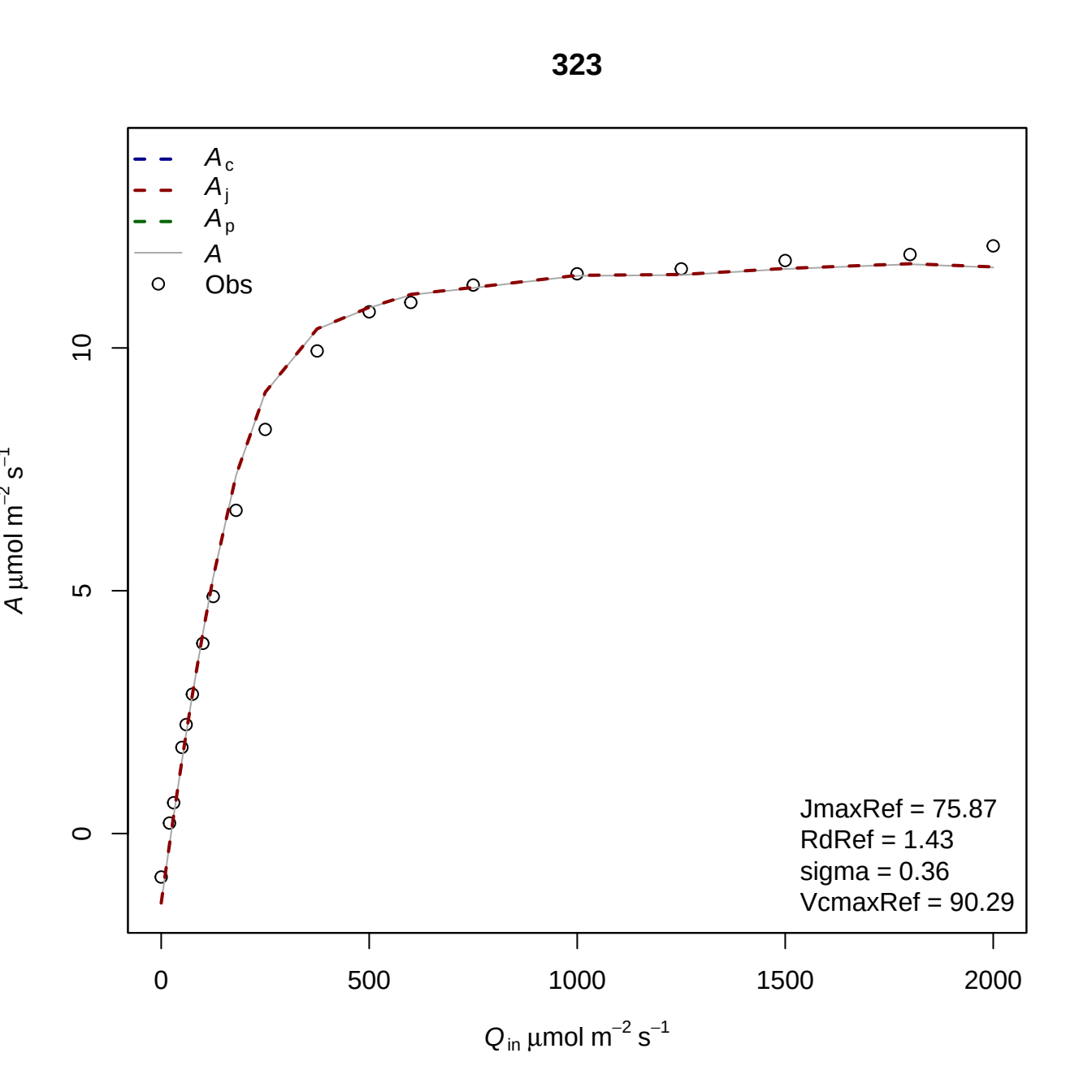

318
A_c
A_j
A_p
A
Obs
JmaxRef = 73.93
RdRef = 3.47
sigma = 1.75
VcmaxRef = 195.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

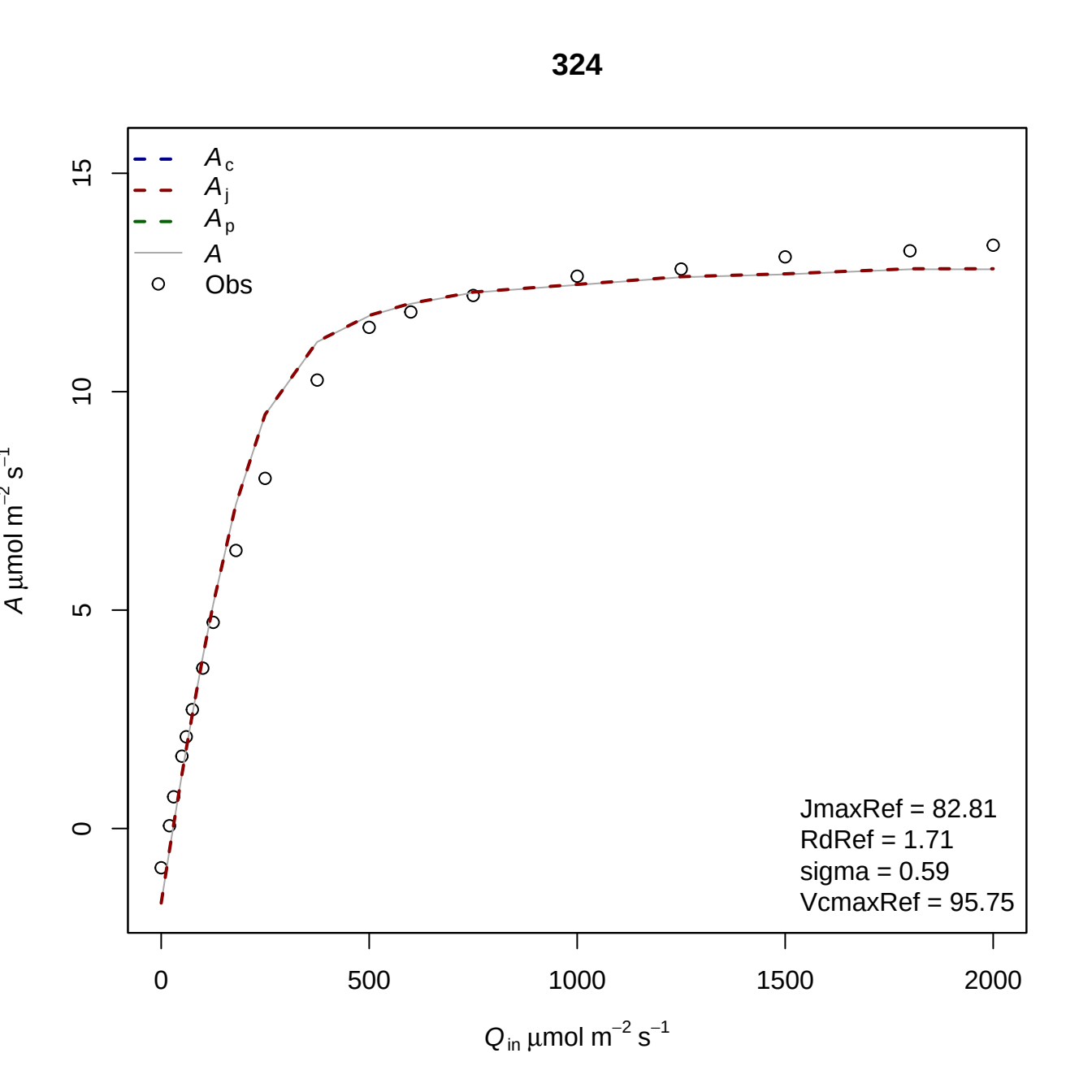

319
A_c
A_j
A_p
A
Obs
JmaxRef = 68.66
RdRef = 3.09
sigma = 1.61
VcmaxRef = 139.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

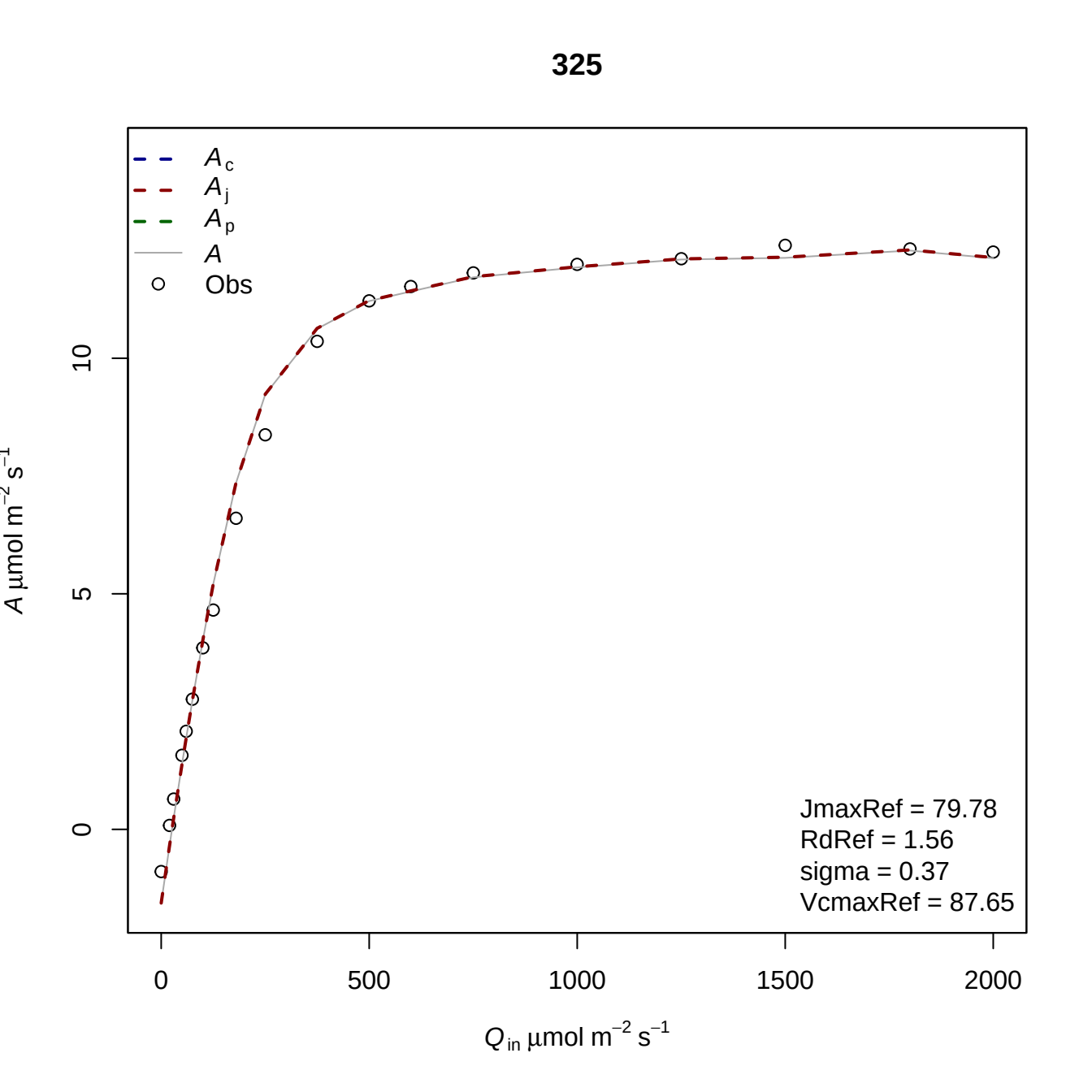

320
A_c
A_j
A_p
A
Obs
JmaxRef = 59.39
RdRef = 3.45
sigma = 1.69
VcmaxRef = 117.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

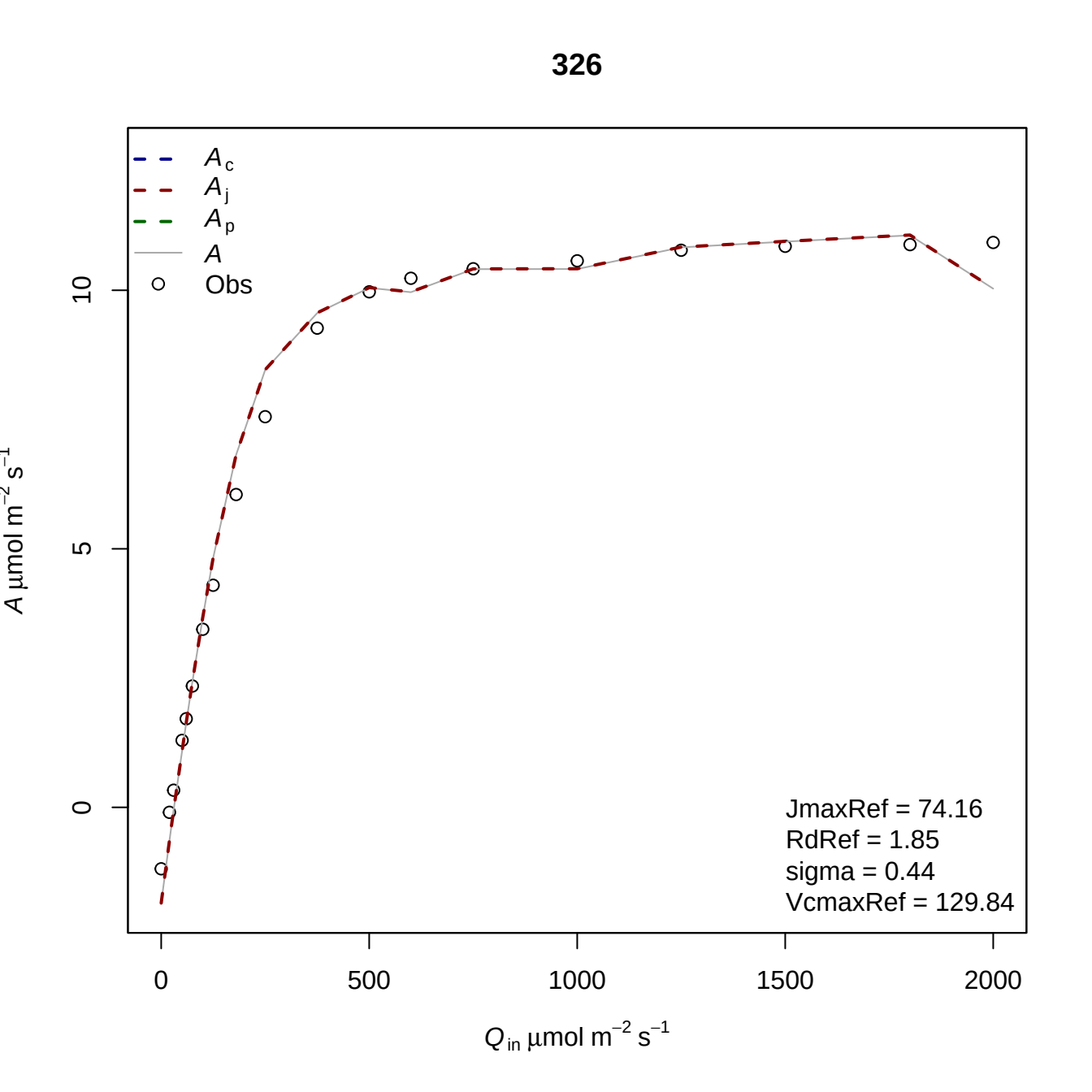

321
A_c
A_j
A_p
A
Obs
JmaxRef = 93.08
RdRef = 3.06
sigma = 1.46
VcmaxRef = 167.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

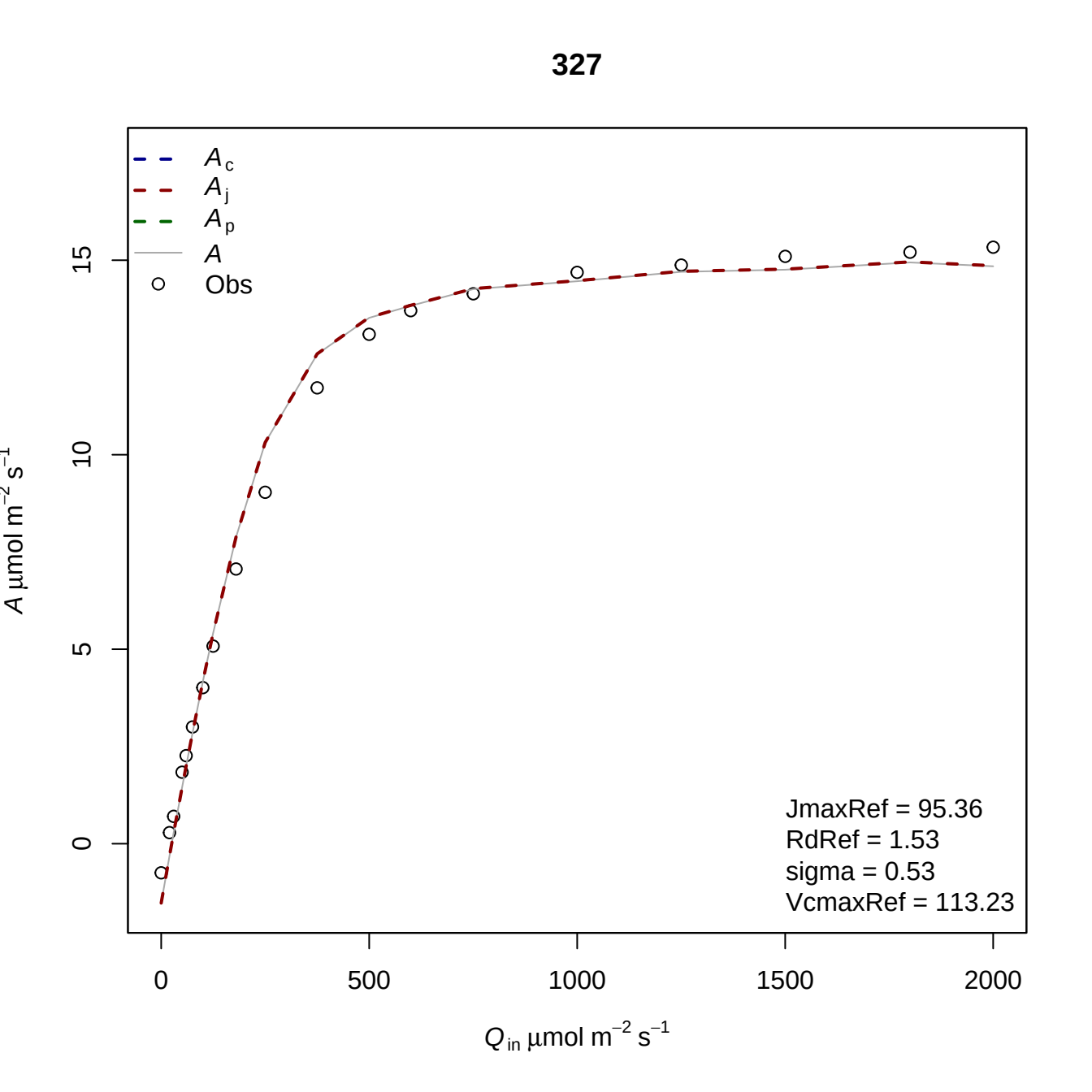

322
A_c
A_j
A_p
A
Obs
JmaxRef = 107.94
RdRef = 3.71
sigma = 1.35
VcmaxRef = 164.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

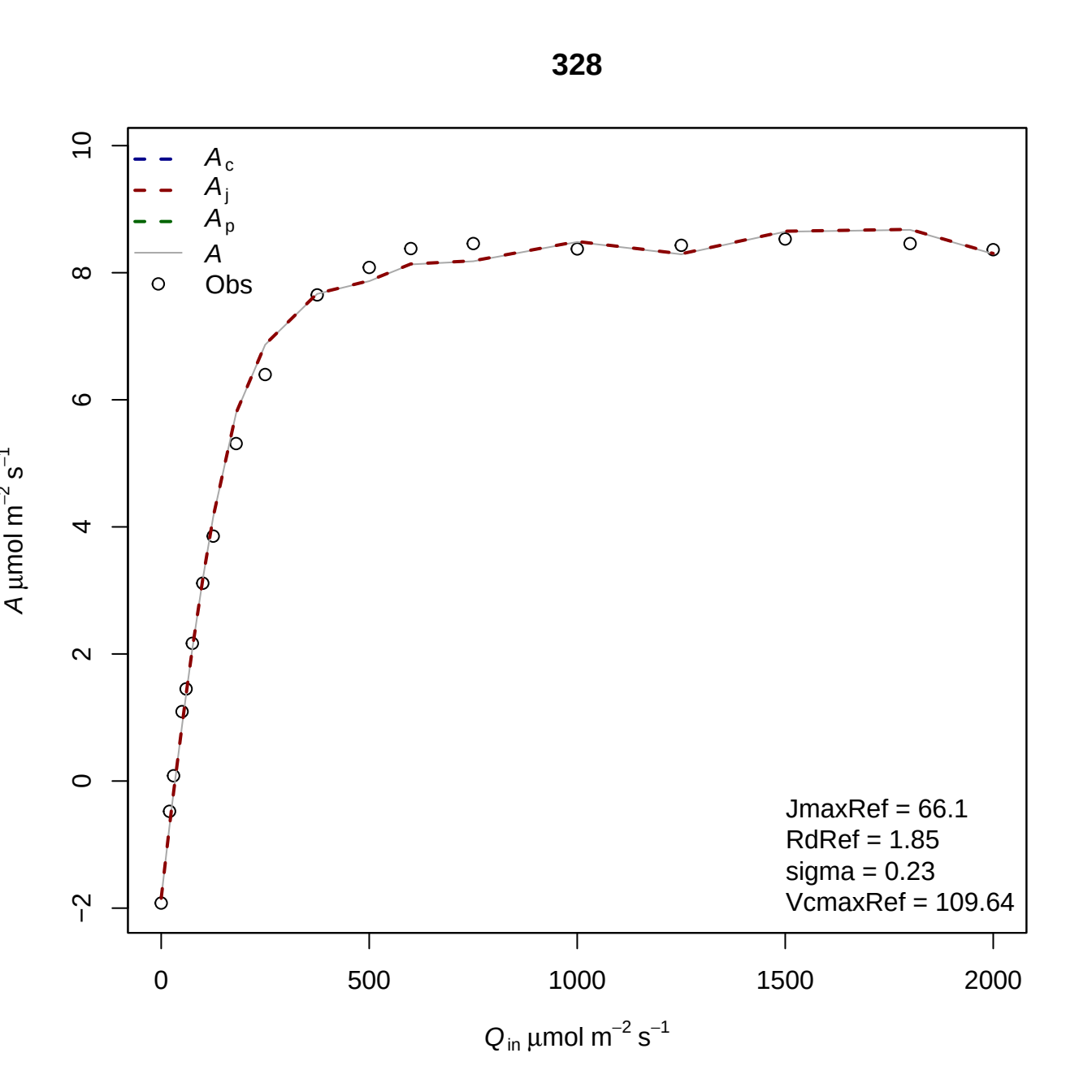

323
A_c
A_j
A_p
A
Obs
JmaxRef = 75.87
RdRef = 1.43
sigma = 0.36
VcmaxRef = 90.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

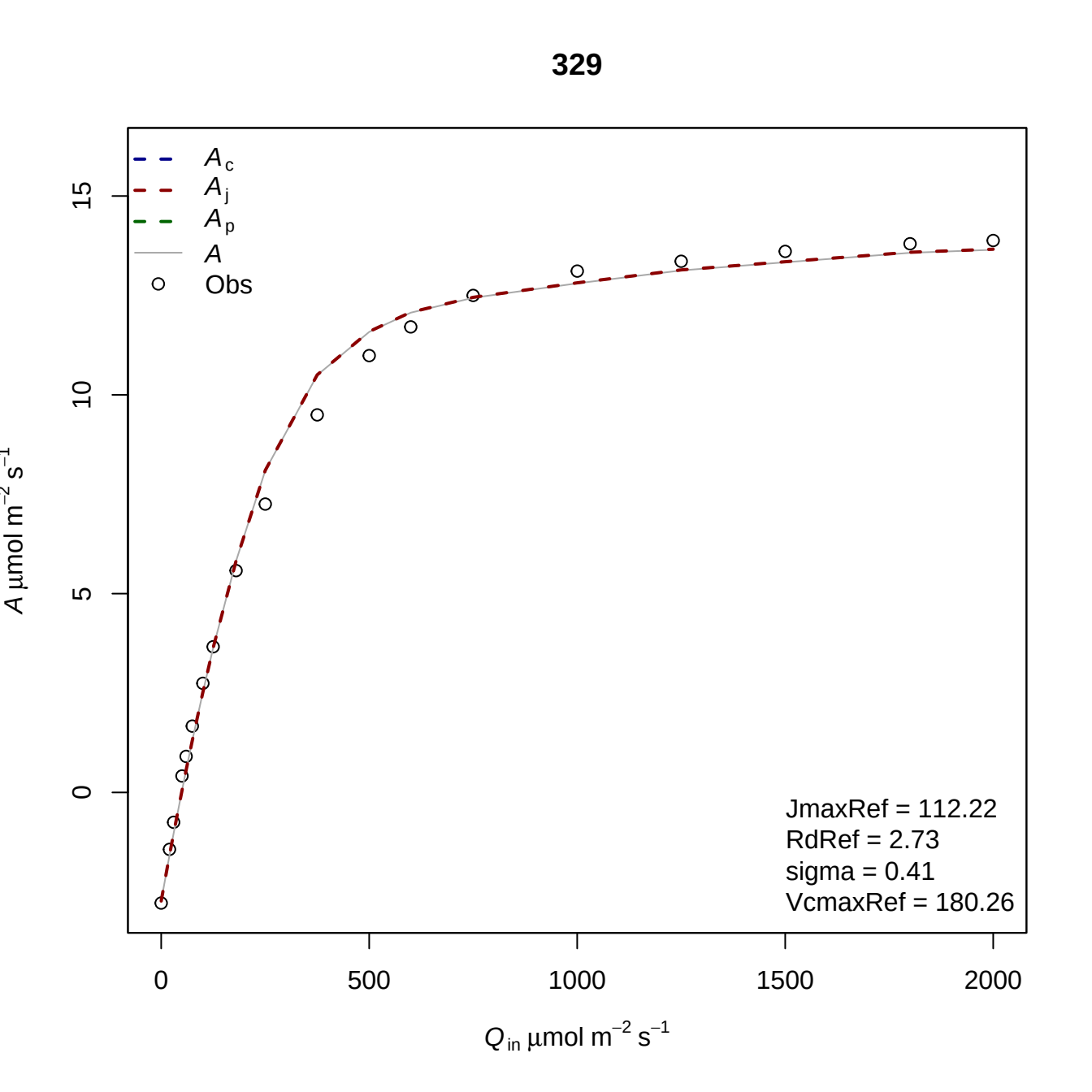

324
A_c
A_j
A_p
A
Obs
JmaxRef = 82.81
RdRef = 1.71
sigma = 0.59
VcmaxRef = 95.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

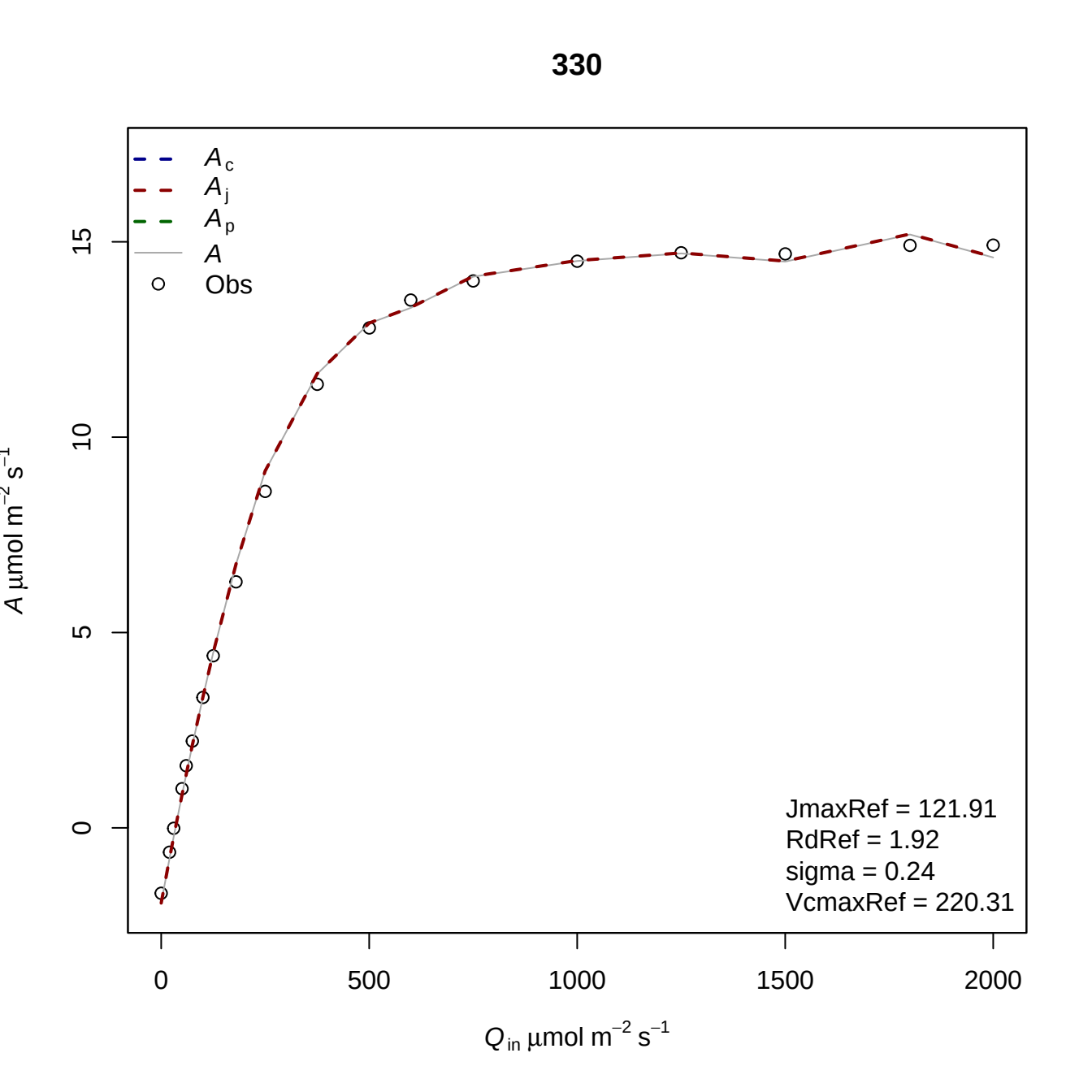

325
A_c
A_j
A_p
A
Obs
JmaxRef = 79.78
RdRef = 1.56
sigma = 0.37
VcmaxRef = 87.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

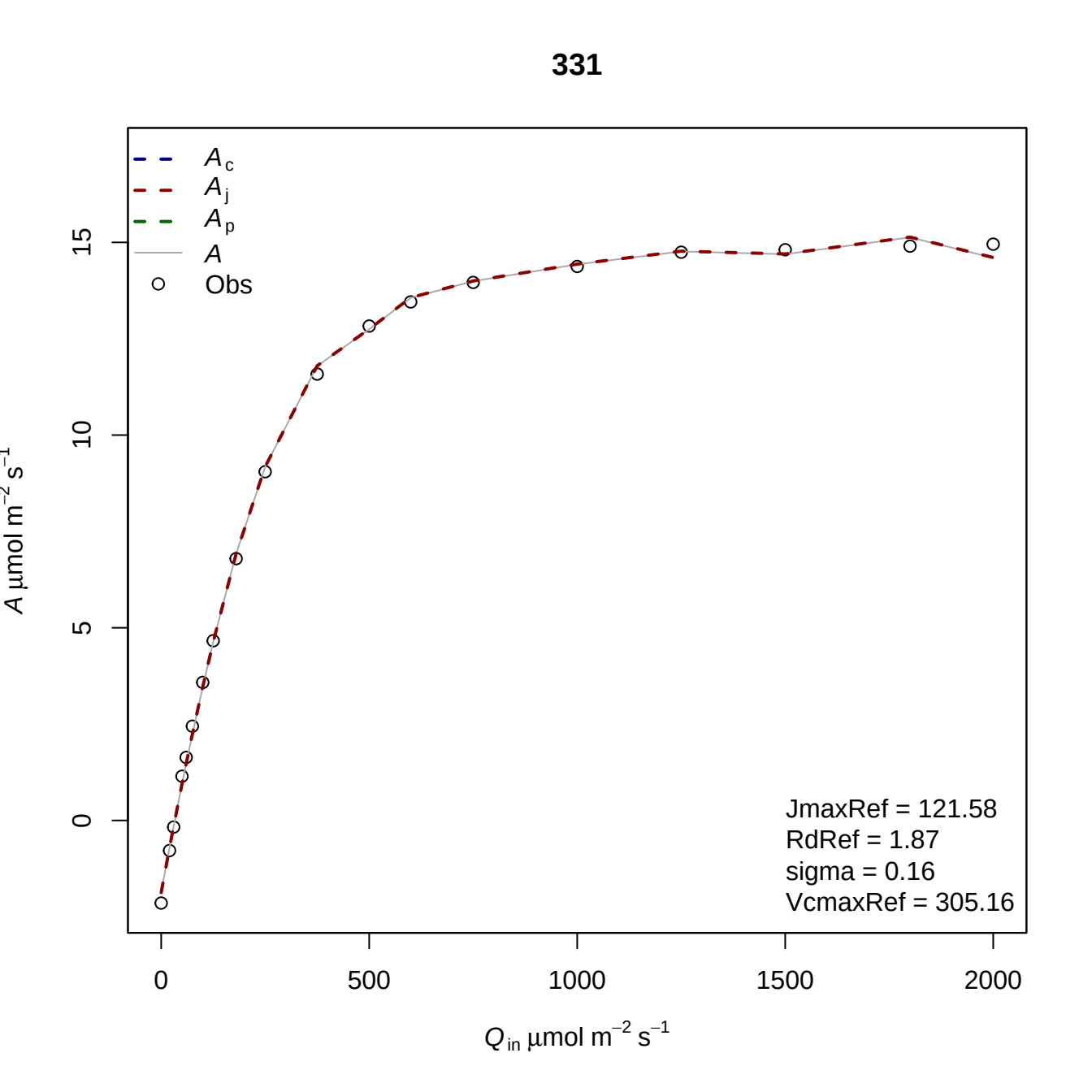

326
A_c
A_j
A_p
A
Obs
JmaxRef = 74.16
RdRef = 1.85
sigma = 0.44
VcmaxRef = 129.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

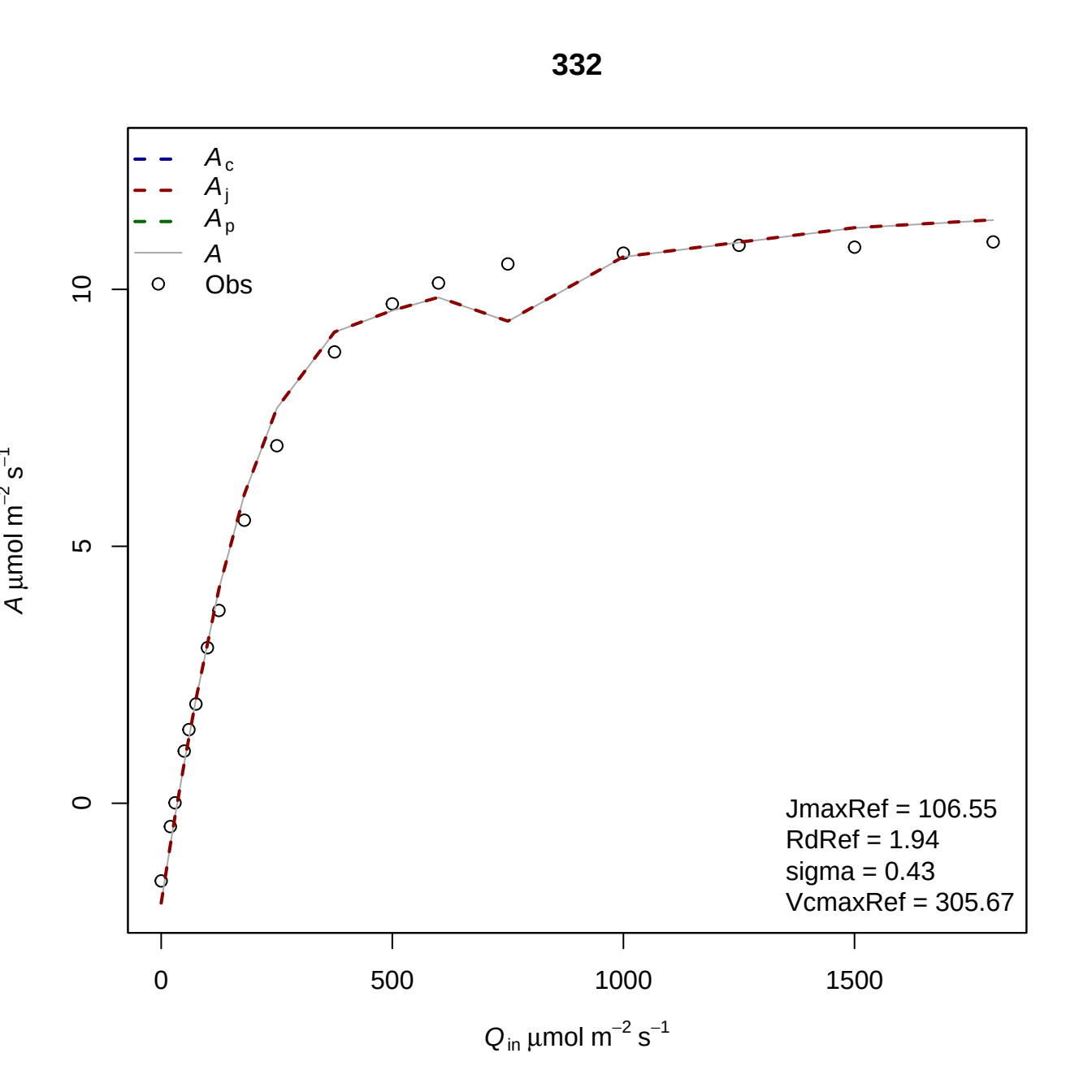

327
A_c
A_j
A_p
A
Obs
JmaxRef = 95.36
RdRef = 1.53
sigma = 0.53
VcmaxRef = 113.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

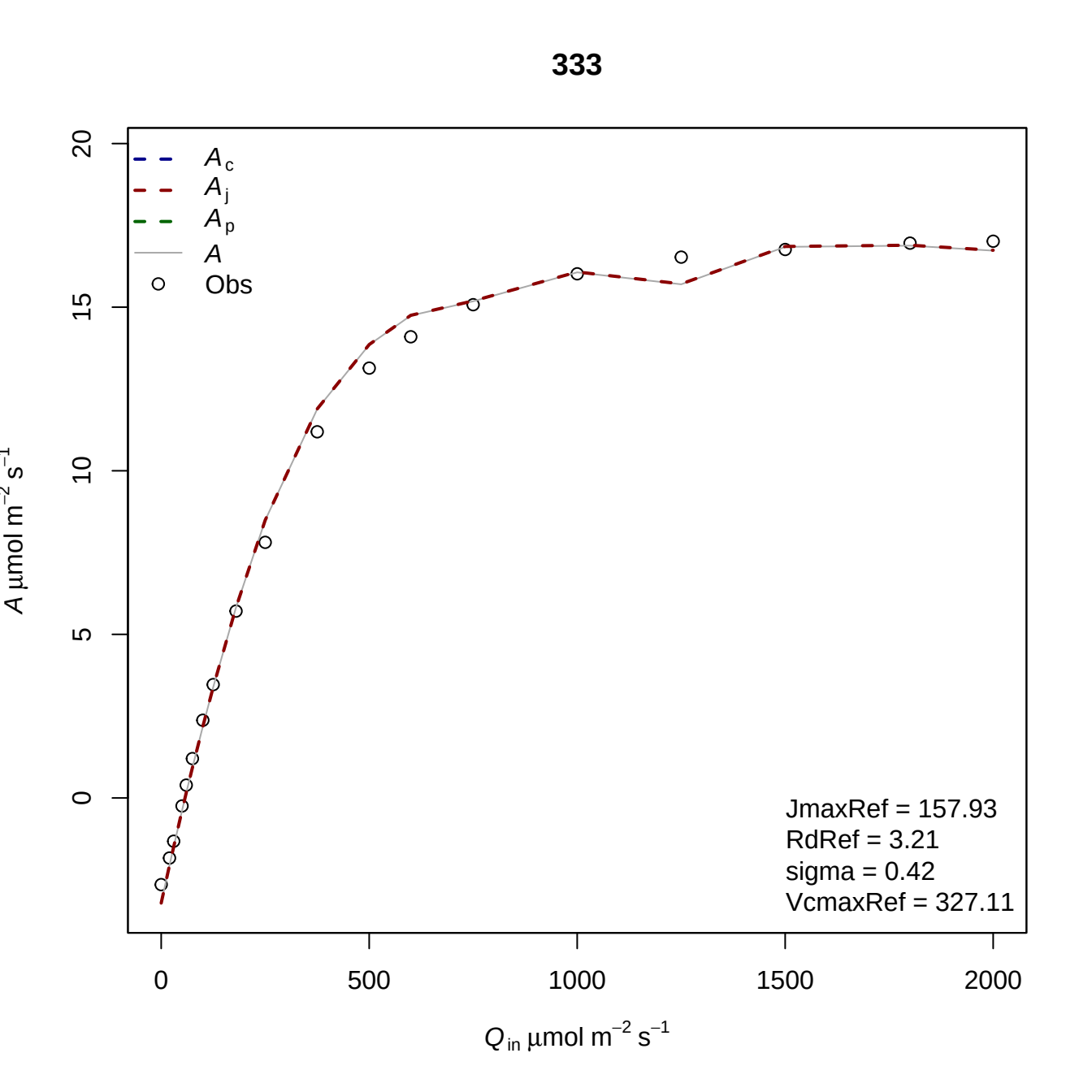

328
A_c
A_j
A_p
A
Obs
JmaxRef = 66.1
RdRef = 1.85
sigma = 0.23
VcmaxRef = 109.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

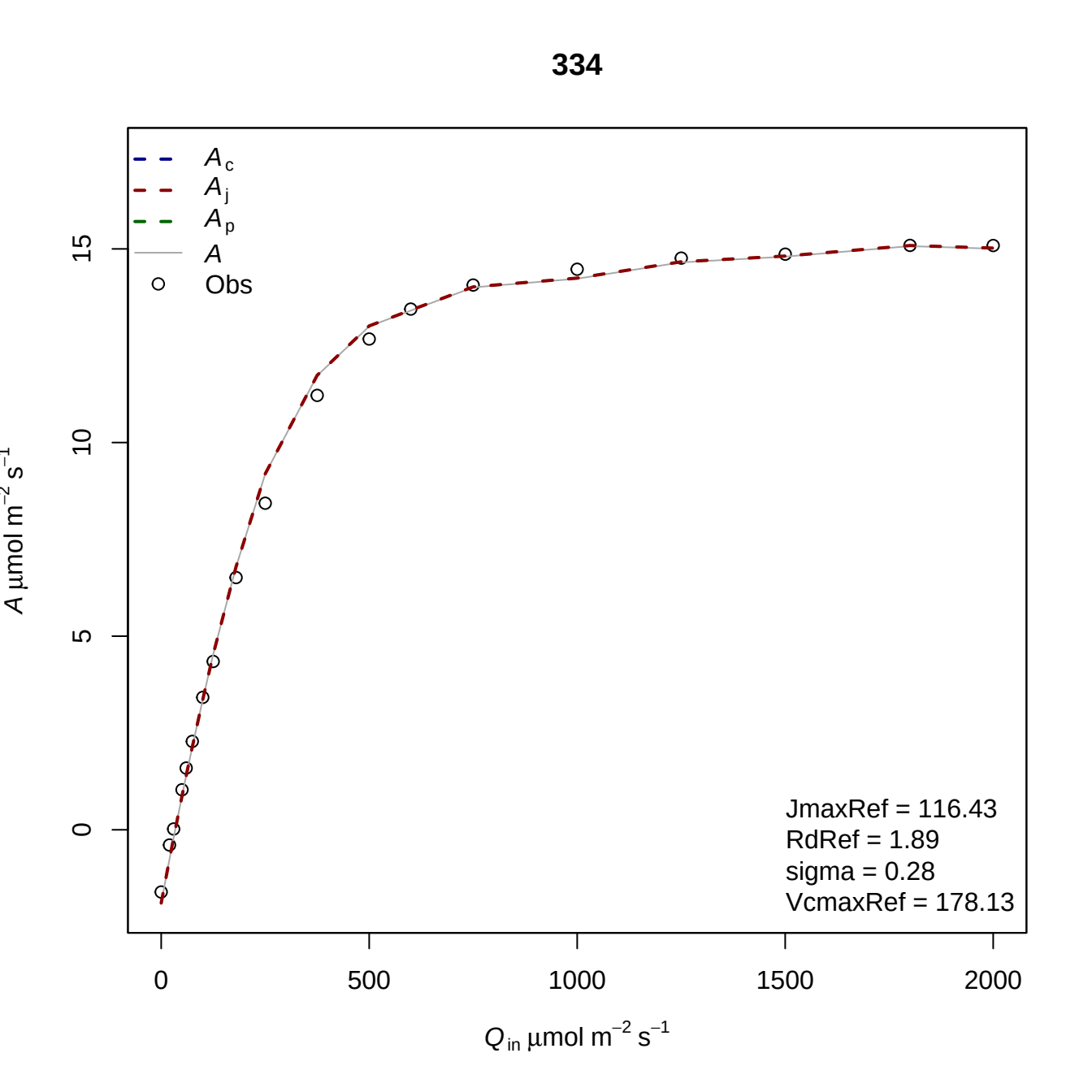

329
A_c
A_j
A_p
A
Obs
JmaxRef = 112.22
RdRef = 2.73
sigma = 0.41
VcmaxRef = 180.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

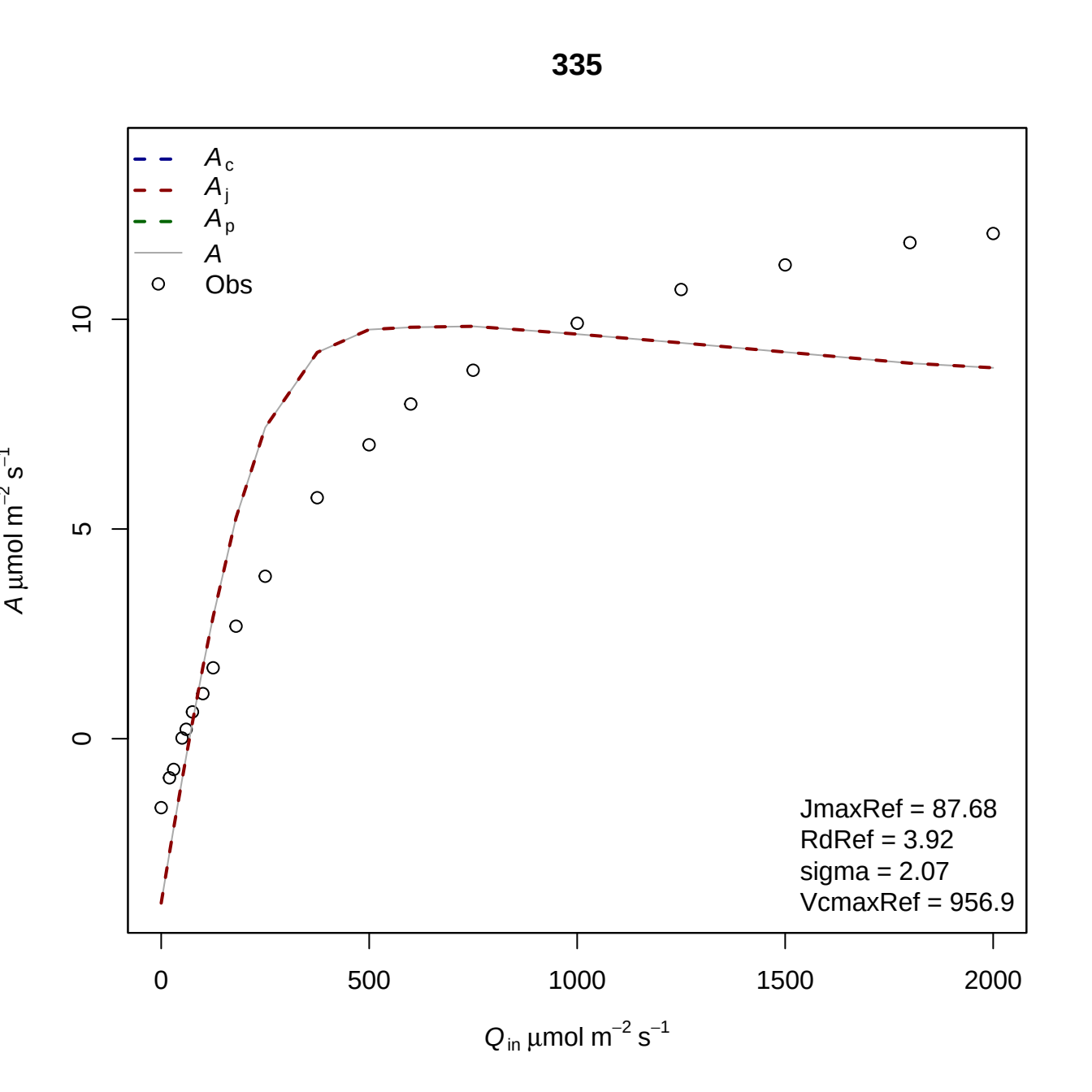

330
A_c
A_j
A_p
A
Obs
JmaxRef = 121.91
RdRef = 1.92
sigma = 0.24
VcmaxRef = 220.31
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

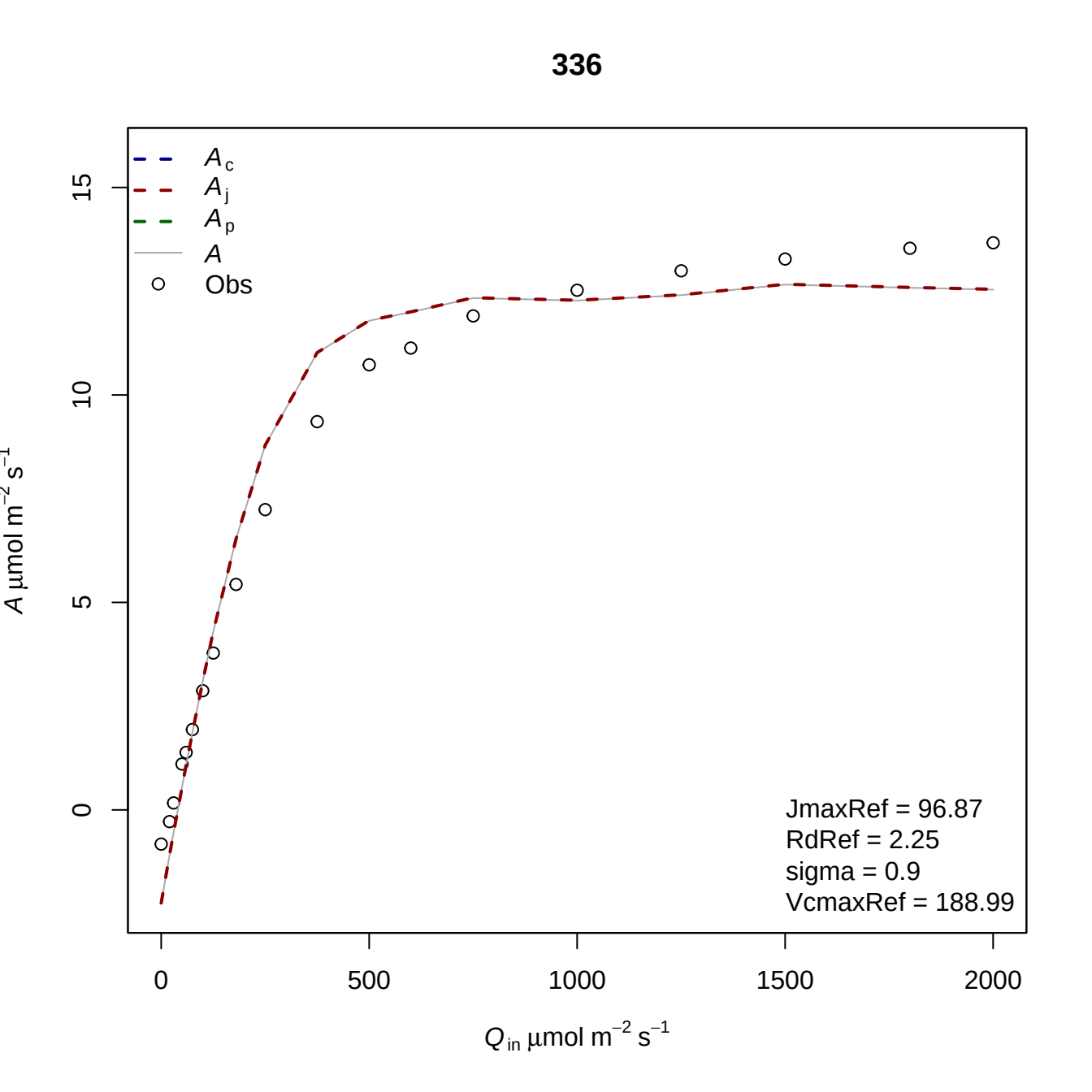

331
A_c
A_j
A_p
A
Obs
JmaxRef = 121.58
RdRef = 1.87
sigma = 0.16
VcmaxRef = 305.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

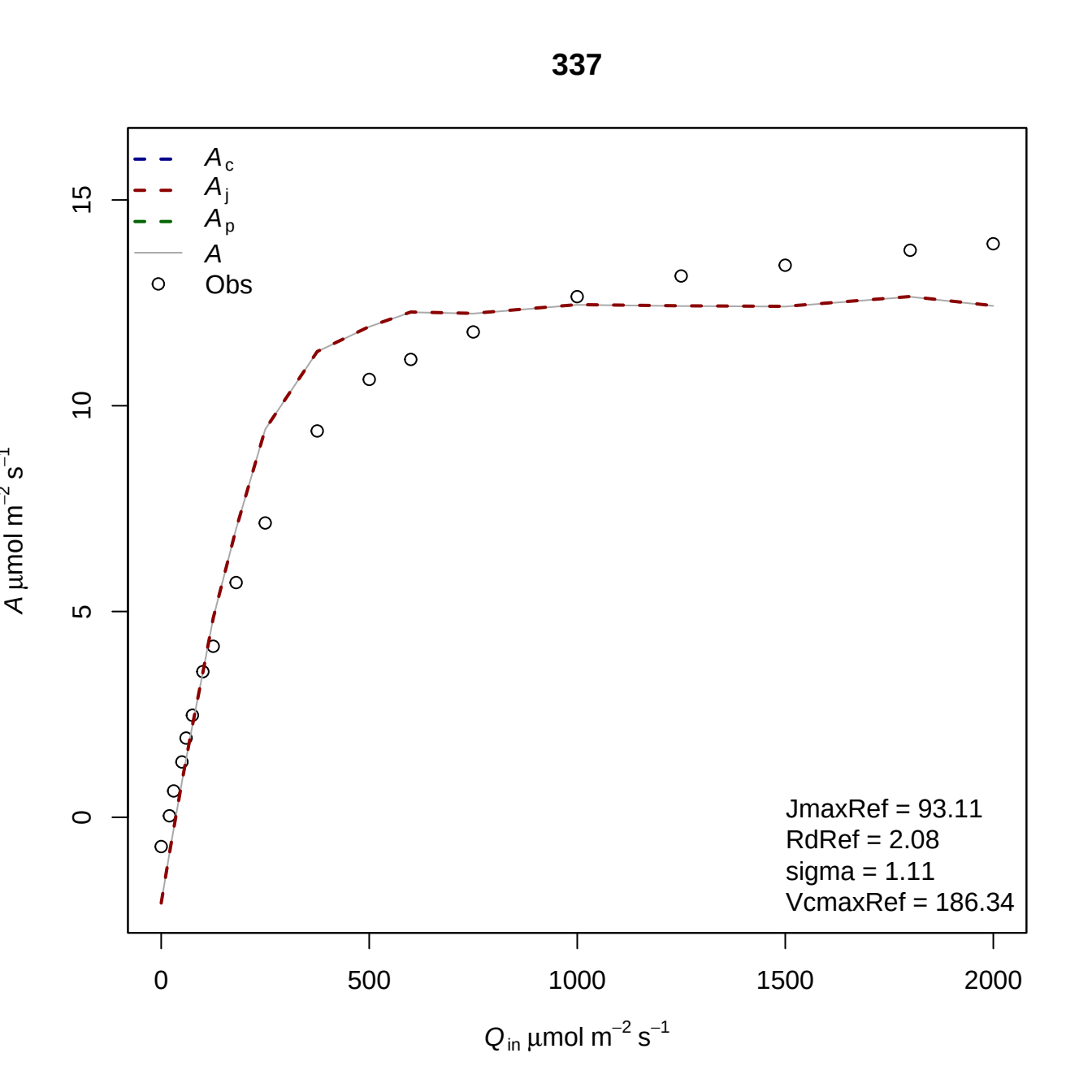

332
A_c
A_j
A_p
A
Obs
JmaxRef = 106.55
RdRef = 1.94
sigma = 0.43
VcmaxRef = 305.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

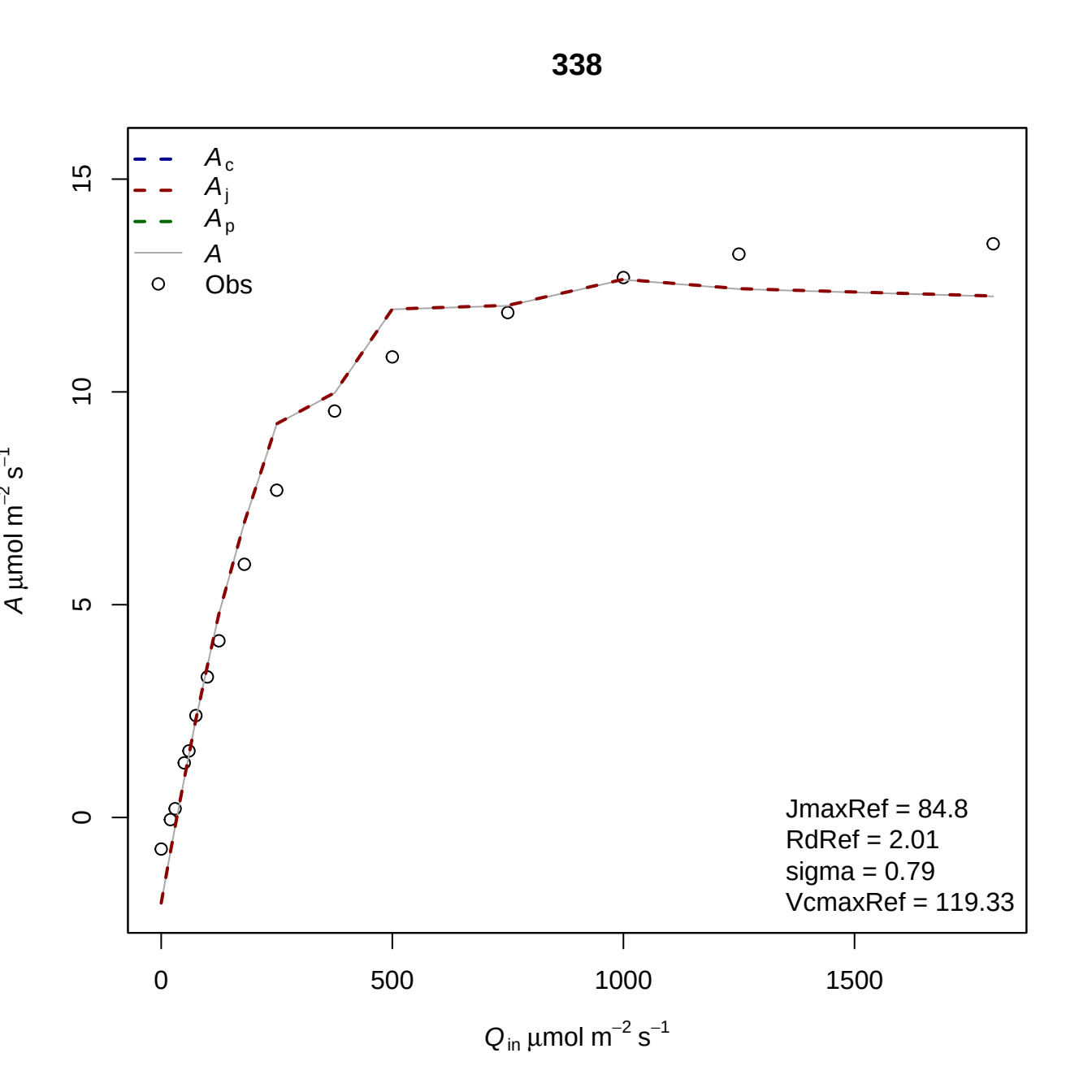

333
A_c
A_j
A_p
A
Obs
JmaxRef = 157.93
RdRef = 3.21
sigma = 0.42
VcmaxRef = 327.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

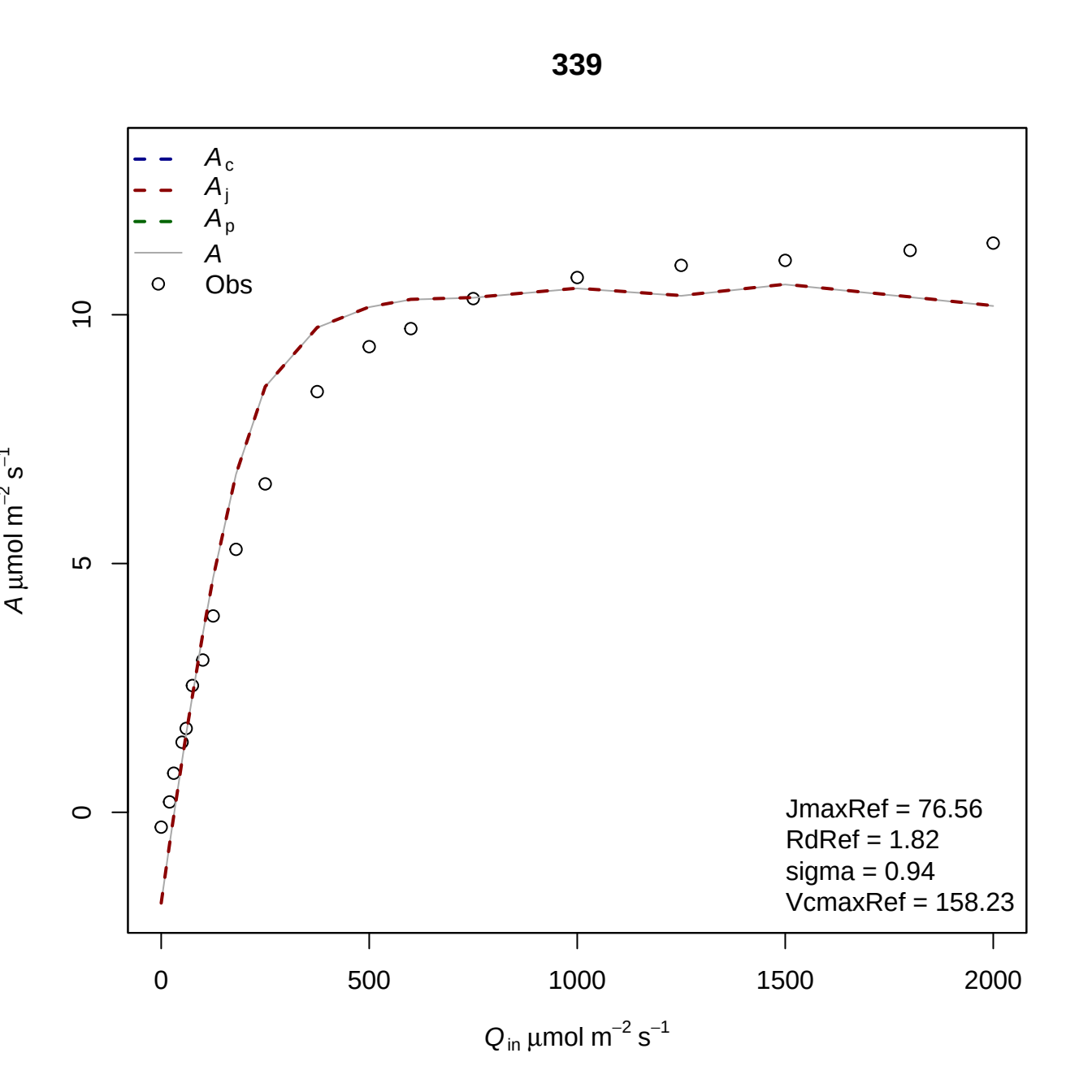

334
A_c
A_j
A_p
A
Obs
JmaxRef = 116.43
RdRef = 1.89
sigma = 0.28
VcmaxRef = 178.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

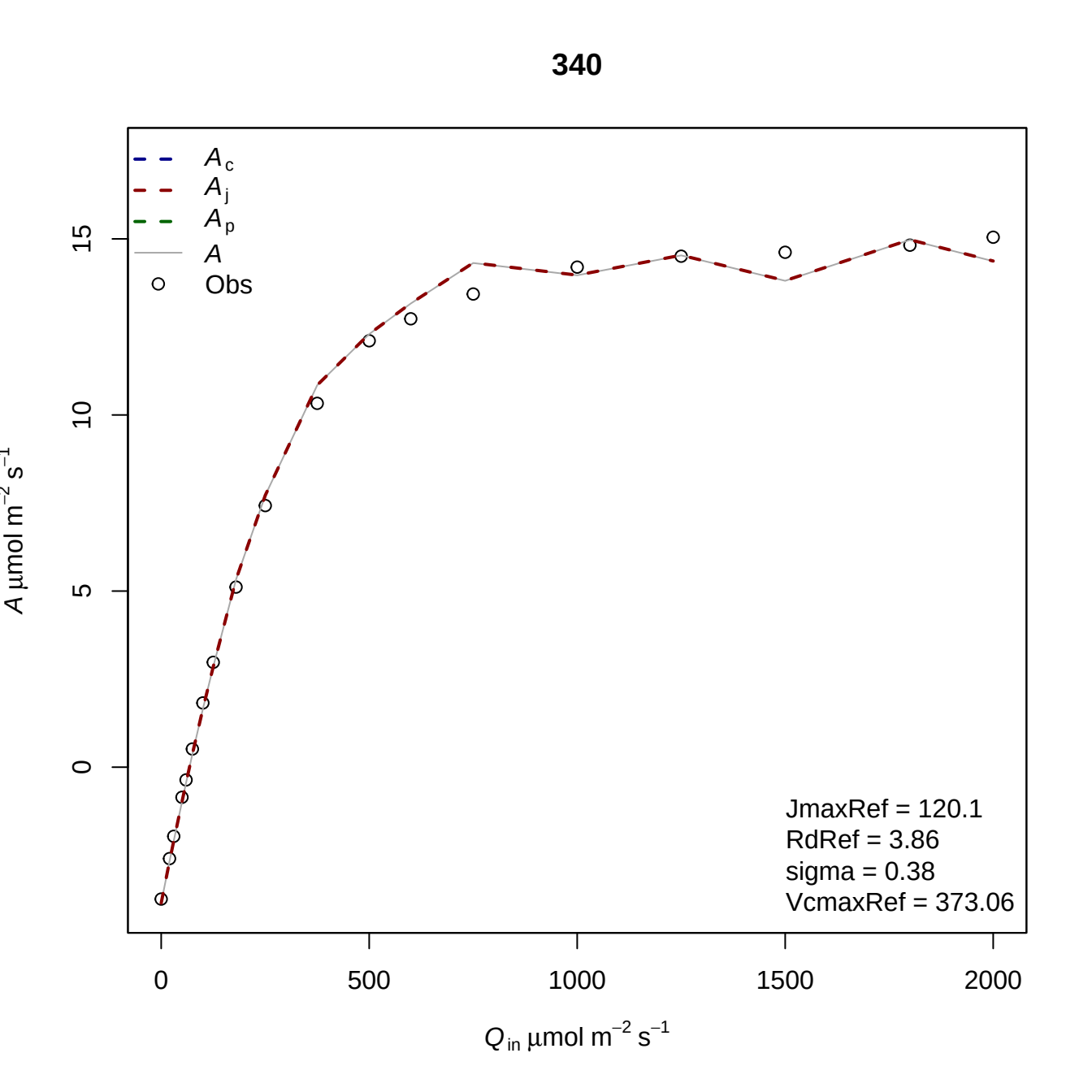

335
A_c
A_j
A_p
A
Obs
JmaxRef = 87.68
RdRef = 3.92
sigma = 2.07
VcmaxRef = 956.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

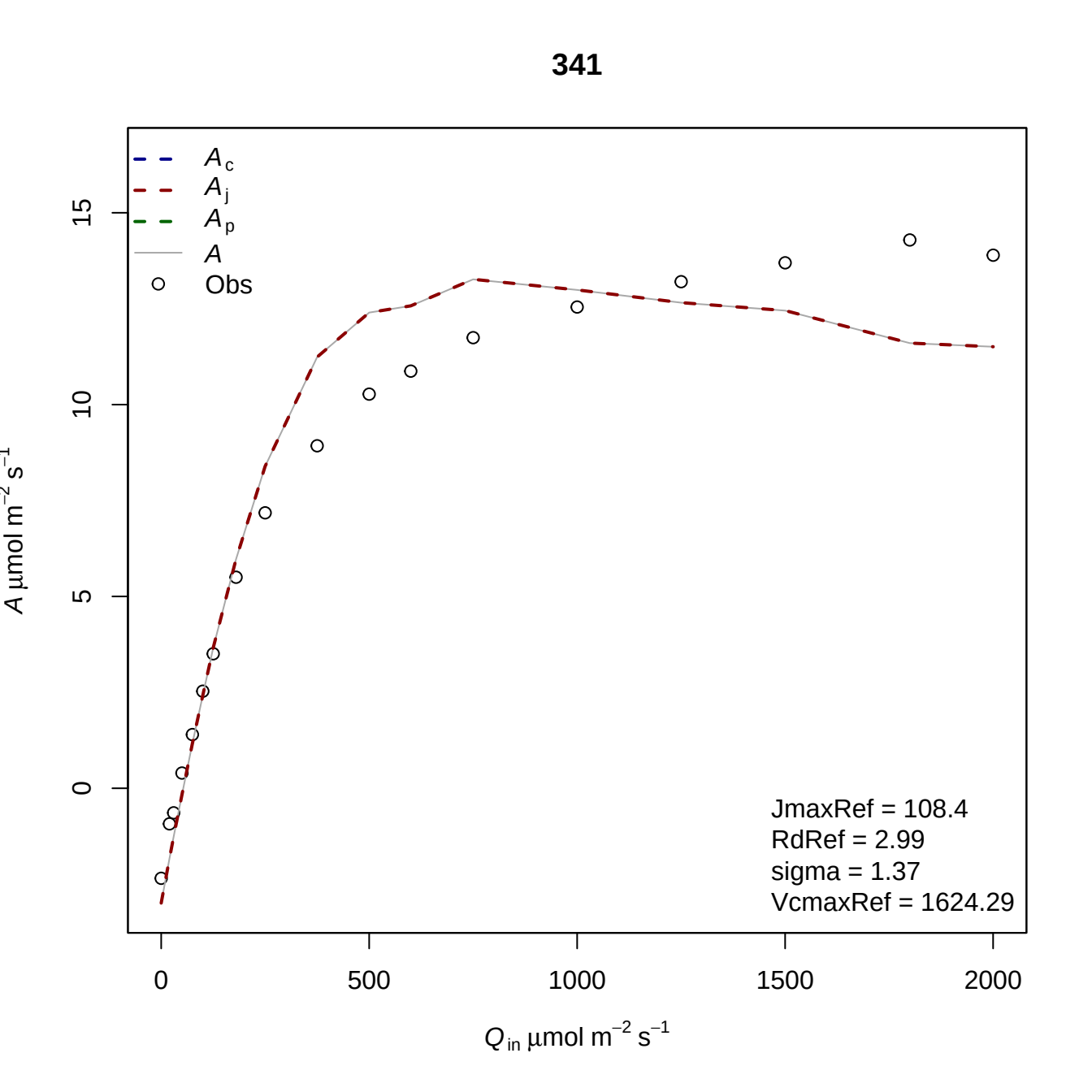

336
A_c
A_j
A_p
A
Obs
JmaxRef = 96.87
RdRef = 2.25
sigma = 0.9
VcmaxRef = 188.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

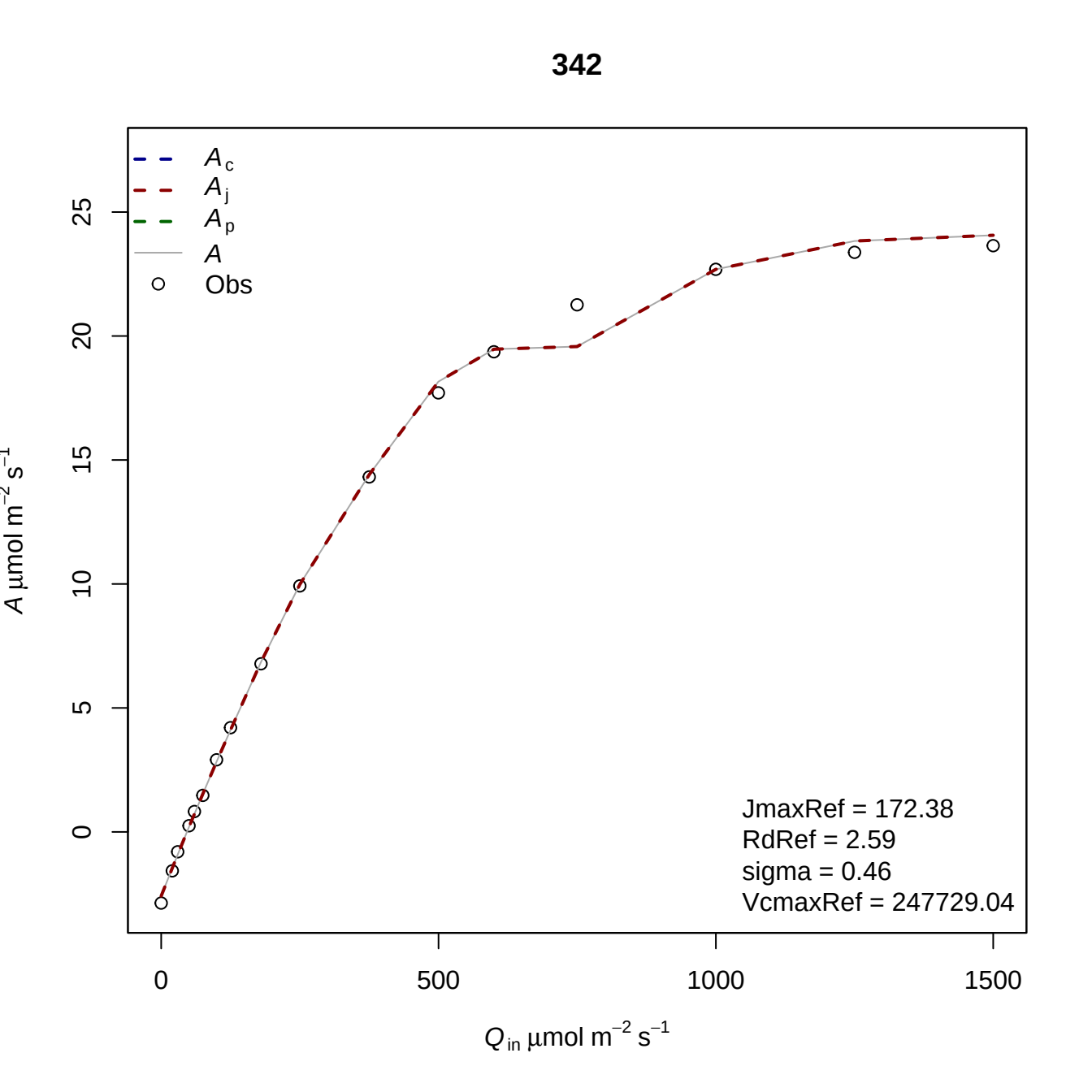

337
A_c
A_j
A_p
A
Obs
JmaxRef = 93.11
RdRef = 2.08
sigma = 1.11
VcmaxRef = 186.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

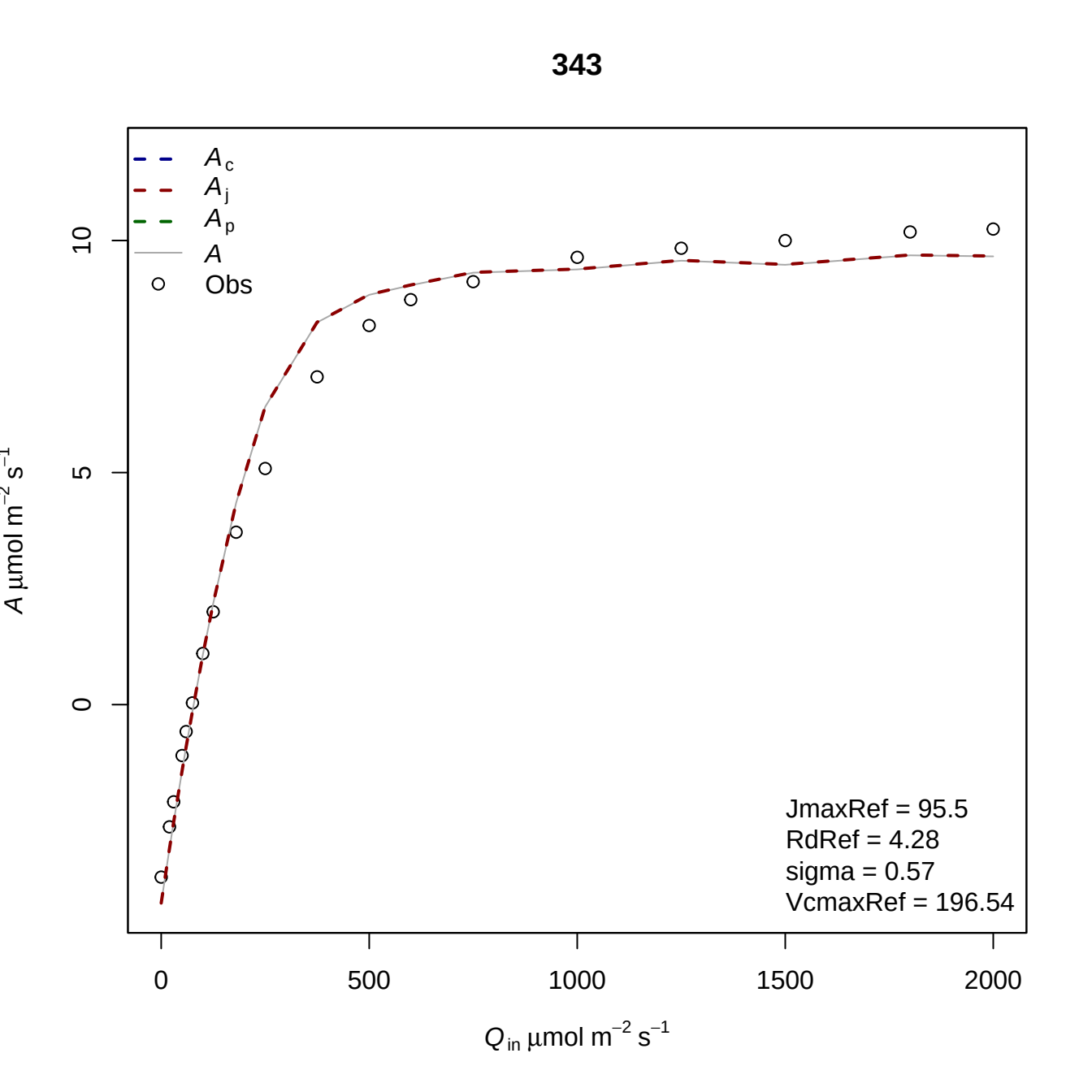

338
A_c
A_j
A_p
A
Obs
JmaxRef = 84.8
RdRef = 2.01
sigma = 0.79
VcmaxRef = 119.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

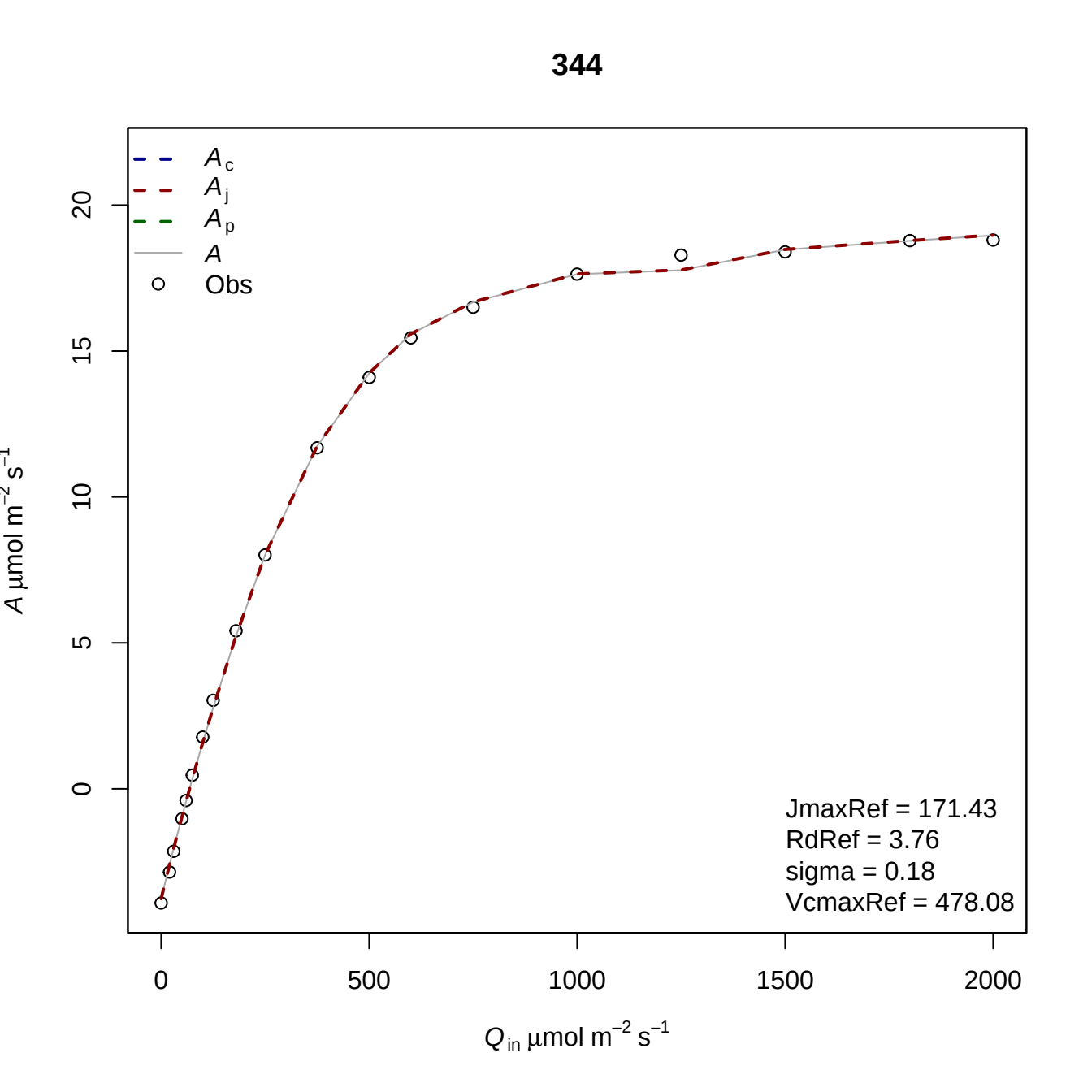

339
A_c
A_j
A_p
A
Obs
JmaxRef = 76.56
RdRef = 1.82
sigma = 0.94
VcmaxRef = 158.23
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

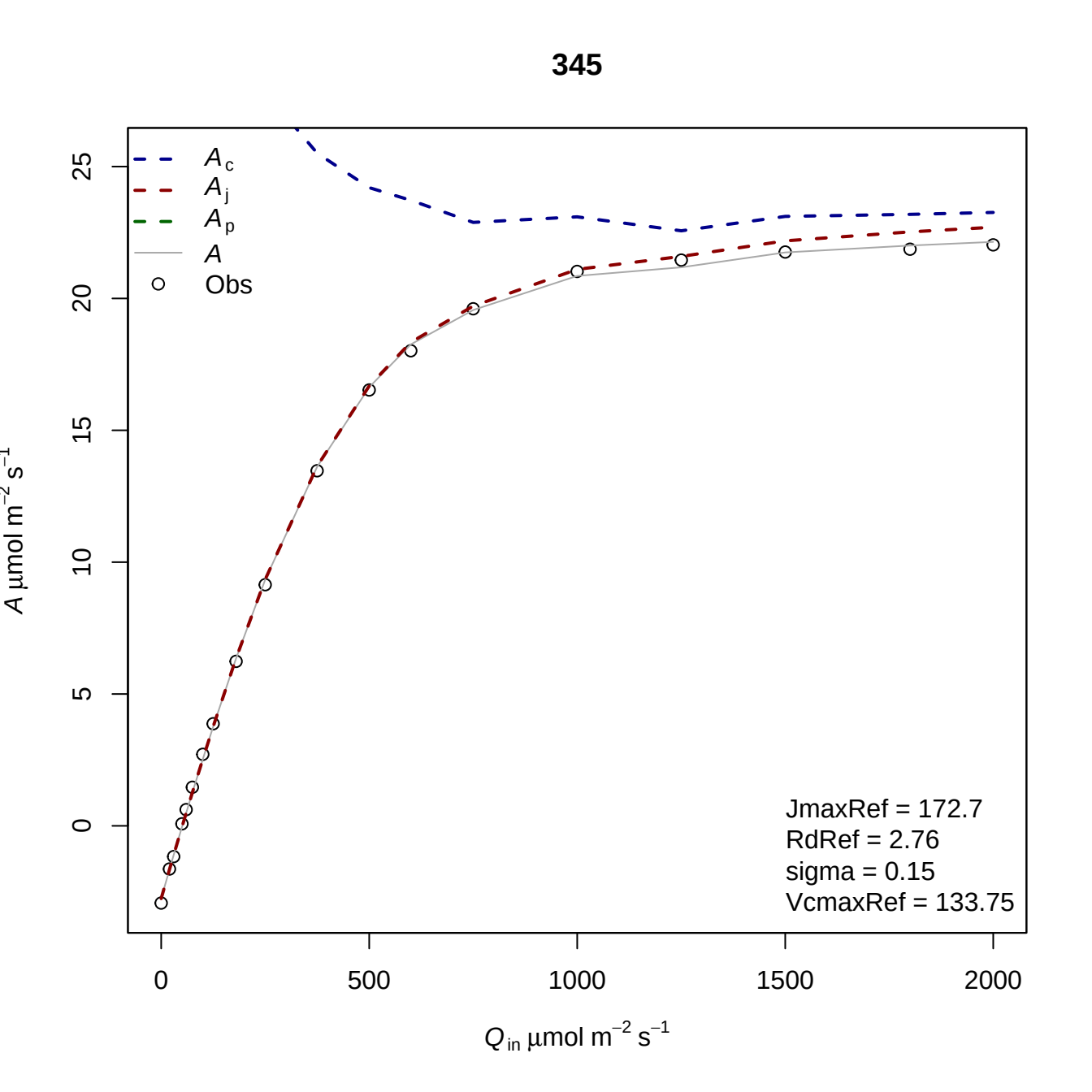

340
A_c
A_j
A_p
A
Obs
JmaxRef = 120.1
RdRef = 3.86
sigma = 0.38
VcmaxRef = 373.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

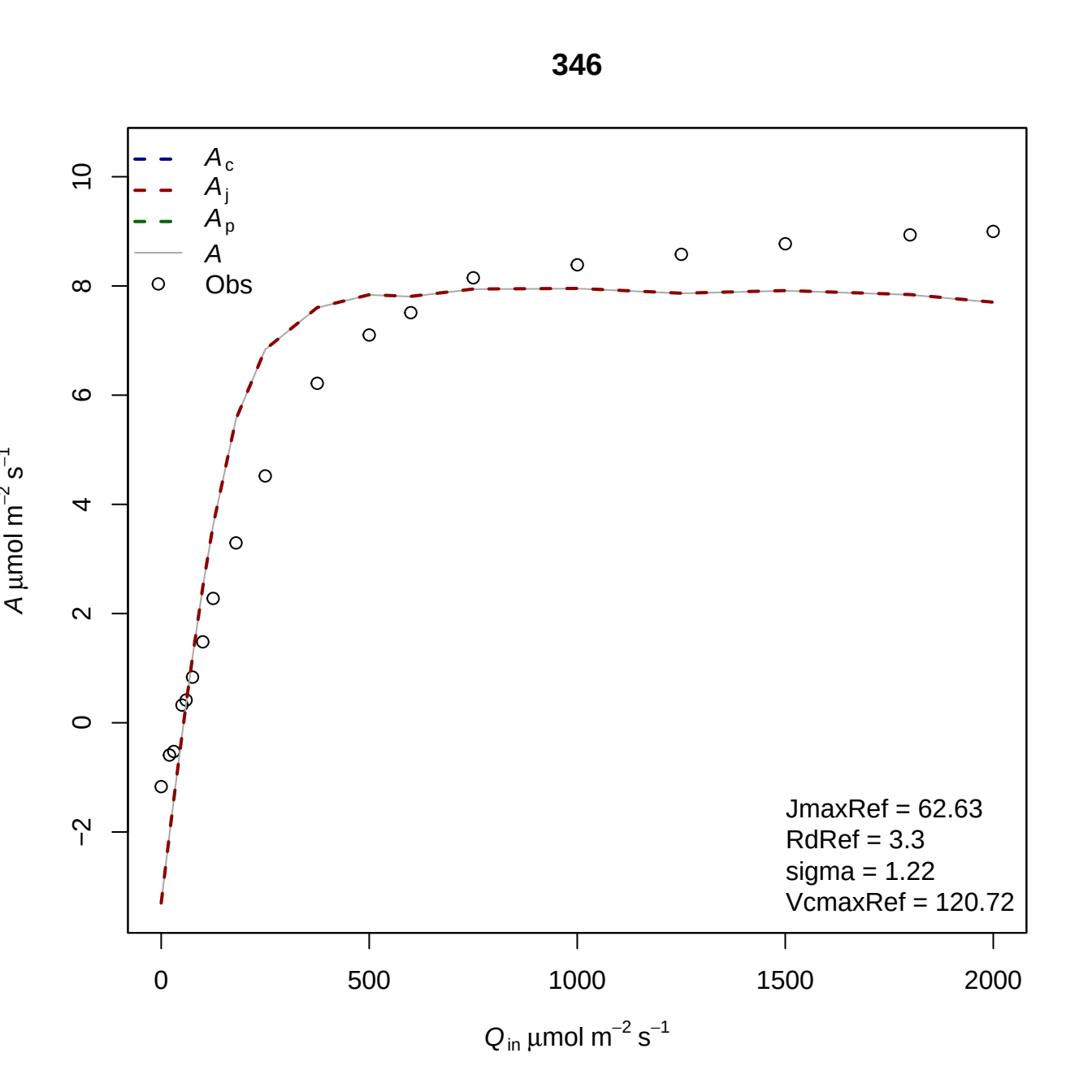

341
A_c
A_j
A_p
A
Obs
JmaxRef = 108.4
RdRef = 2.99
sigma = 1.37
VcmaxRef = 1624.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

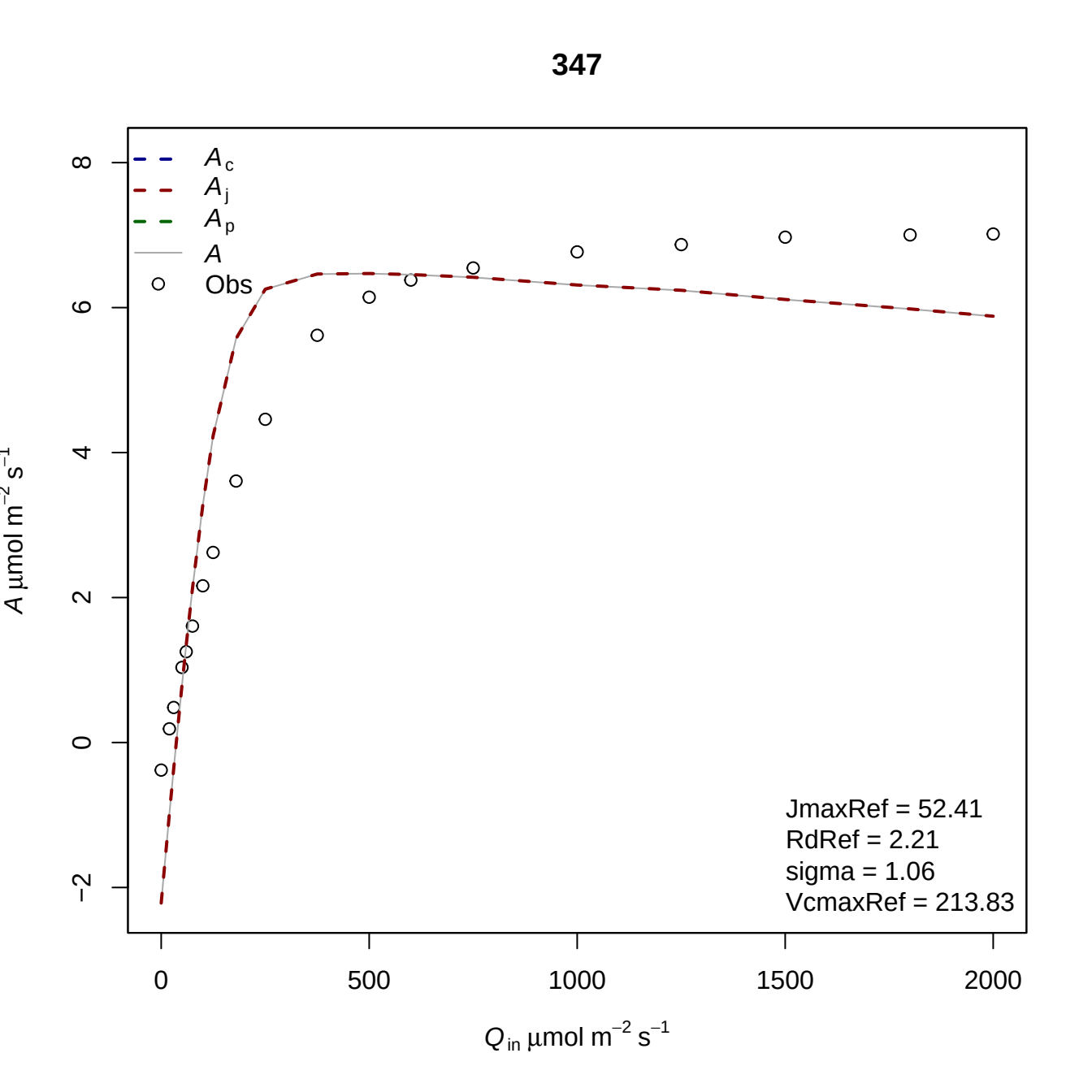

342
A_c
A_j
A_p
A
Obs
JmaxRef = 172.38
RdRef = 2.59
sigma = 0.46
VcmaxRef = 247729.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

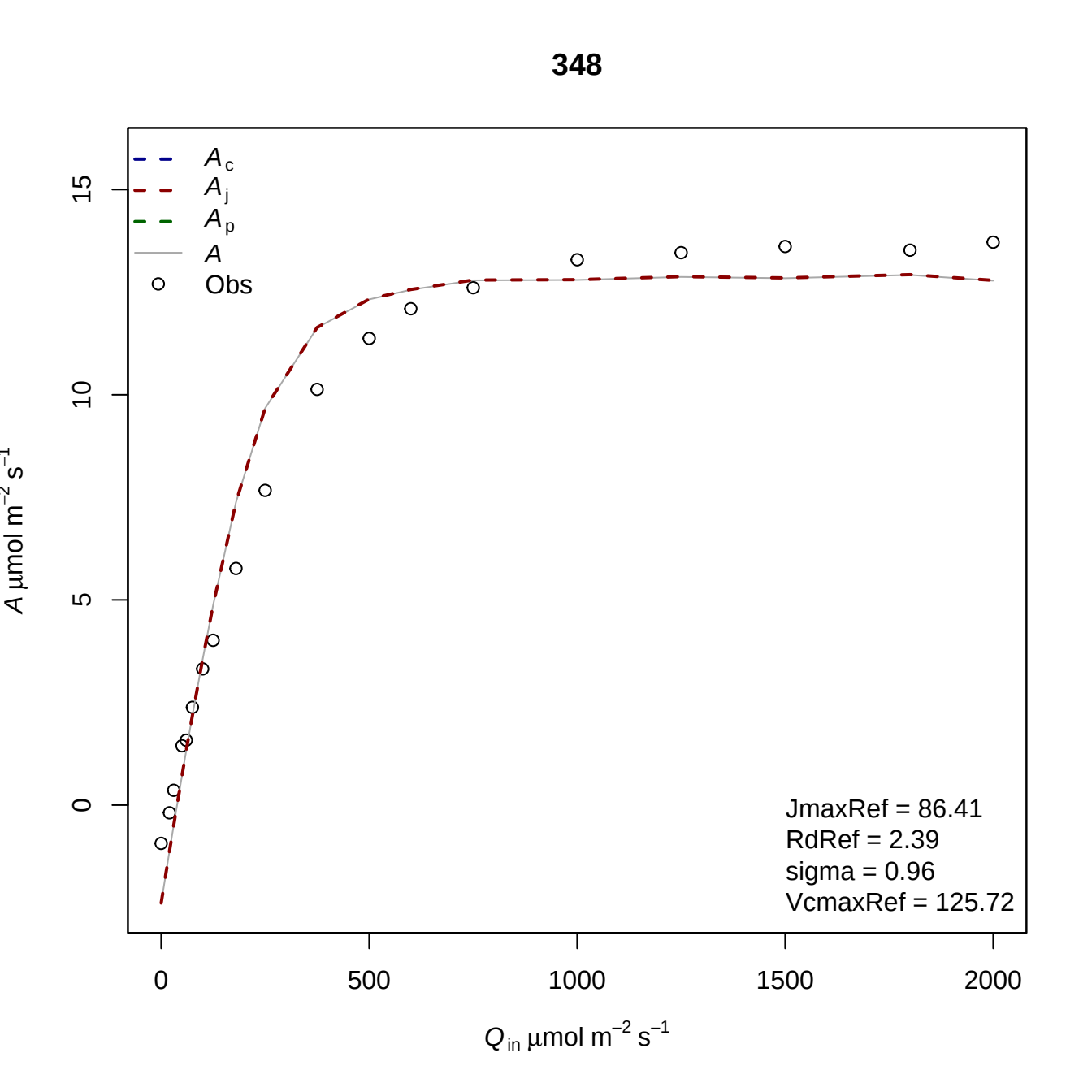

343
A_c
A_j
A_p
A
Obs
JmaxRef = 95.5
RdRef = 4.28
sigma = 0.57
VcmaxRef = 196.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

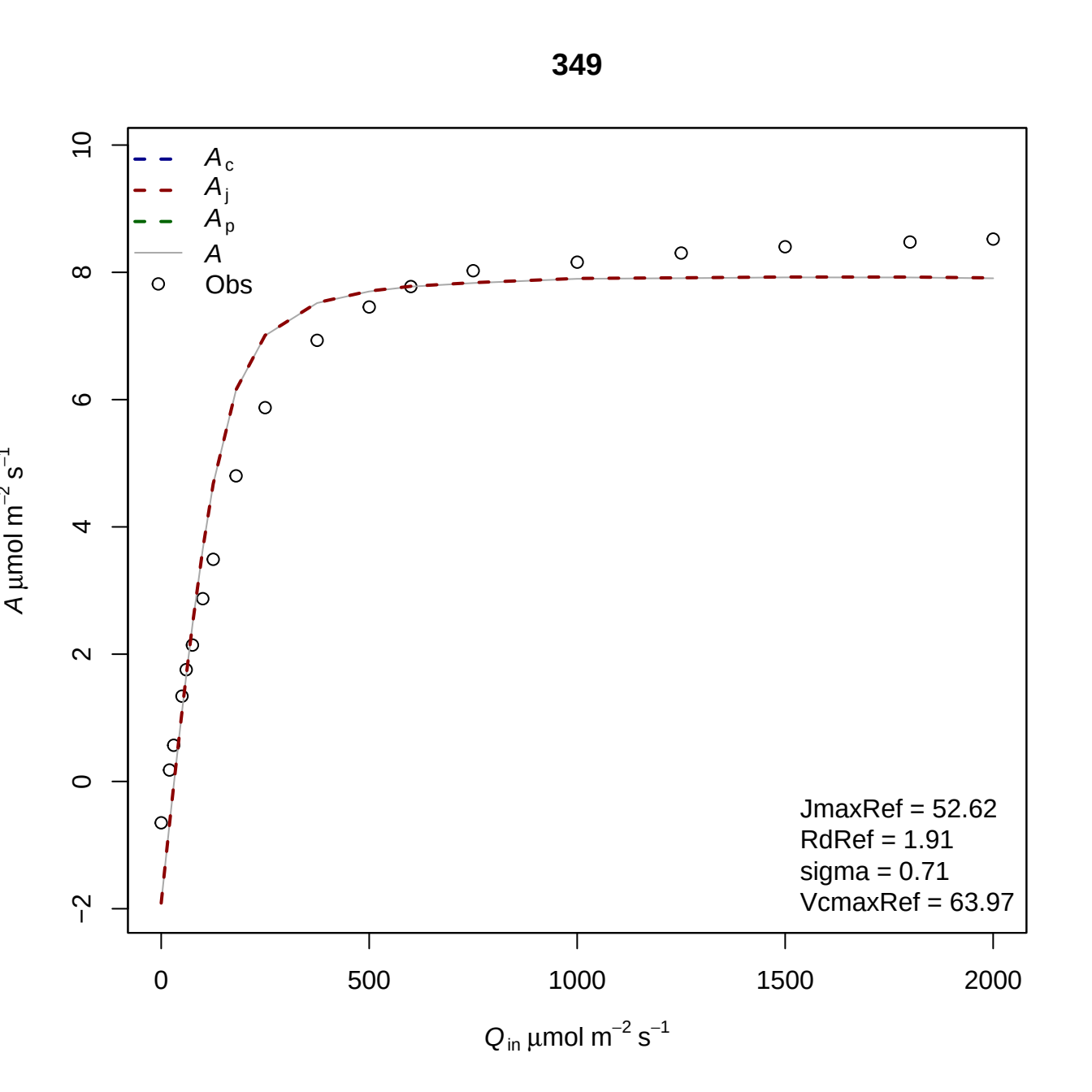

344
A_c
A_j
A_p
A
Obs
JmaxRef = 171.43
RdRef = 3.76
sigma = 0.18
VcmaxRef = 478.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

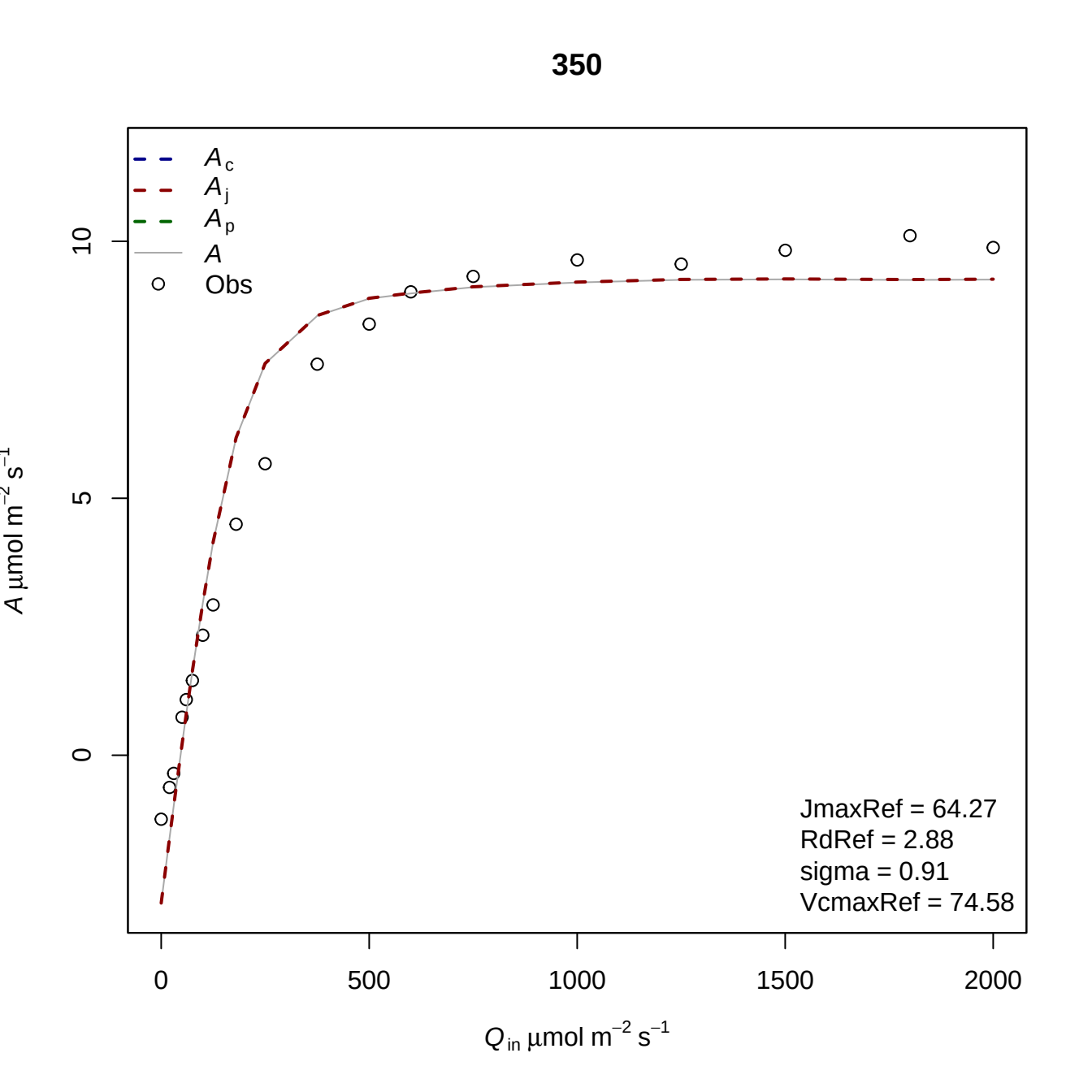

345
A_c
A_j
A_p
A
Obs
JmaxRef = 172.7
RdRef = 2.76
sigma = 0.15
VcmaxRef = 133.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

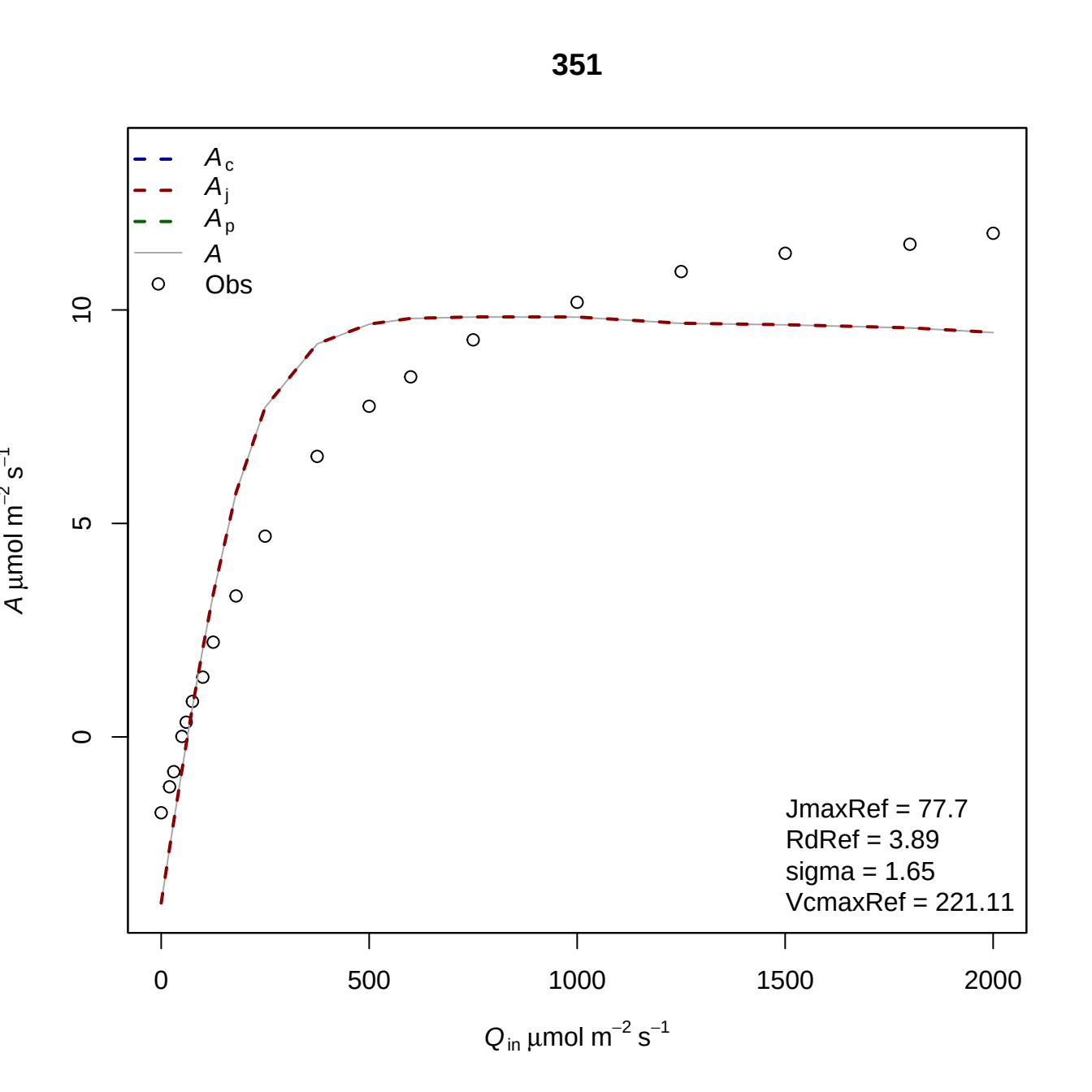

346
A_c
A_j
A_p
A
Obs
JmaxRef = 62.63
RdRef = 3.3
sigma = 1.22
VcmaxRef = 120.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

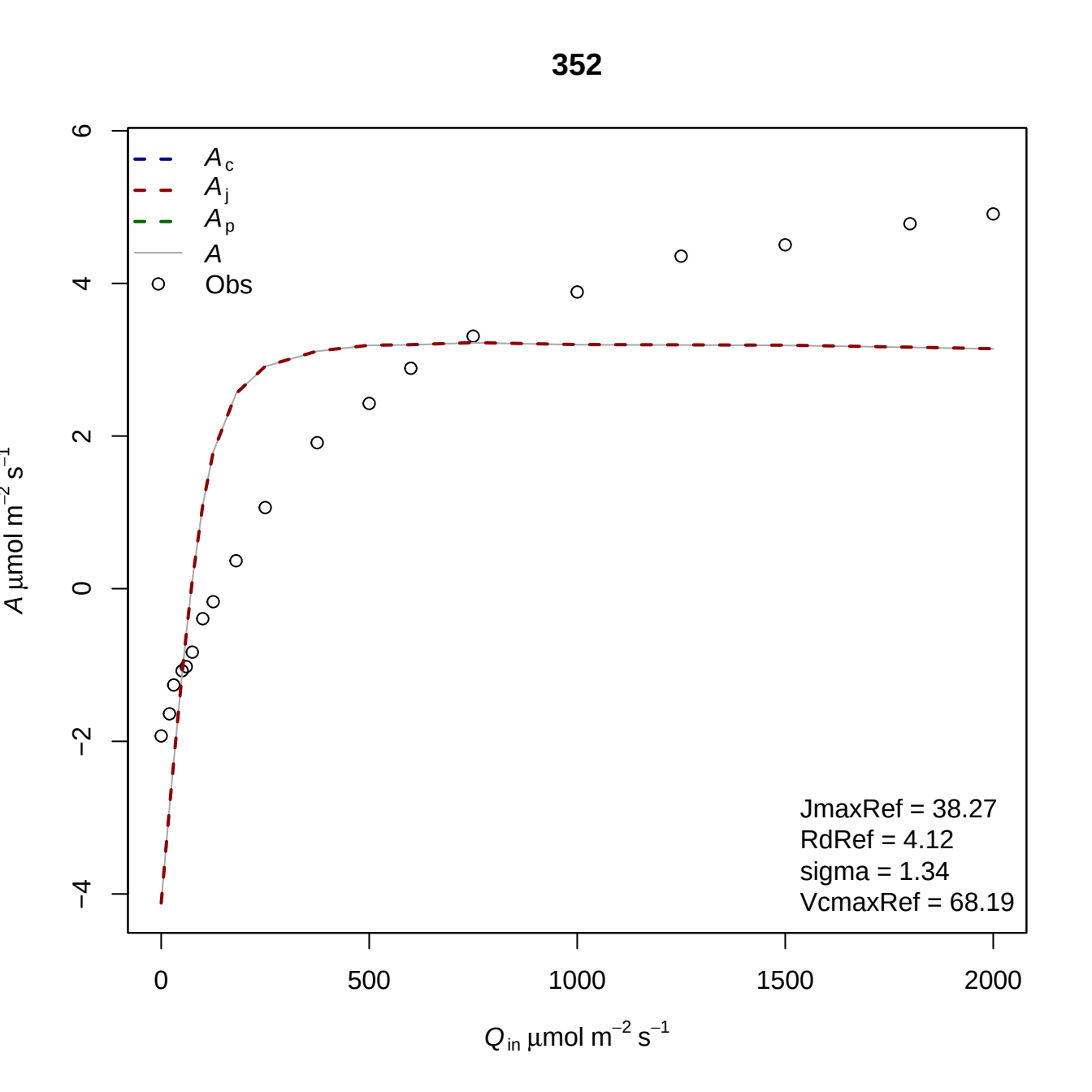

347
A_c
A_j
A_p
A
Obs
JmaxRef = 52.41
RdRef = 2.21
sigma = 1.06
VcmaxRef = 213.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

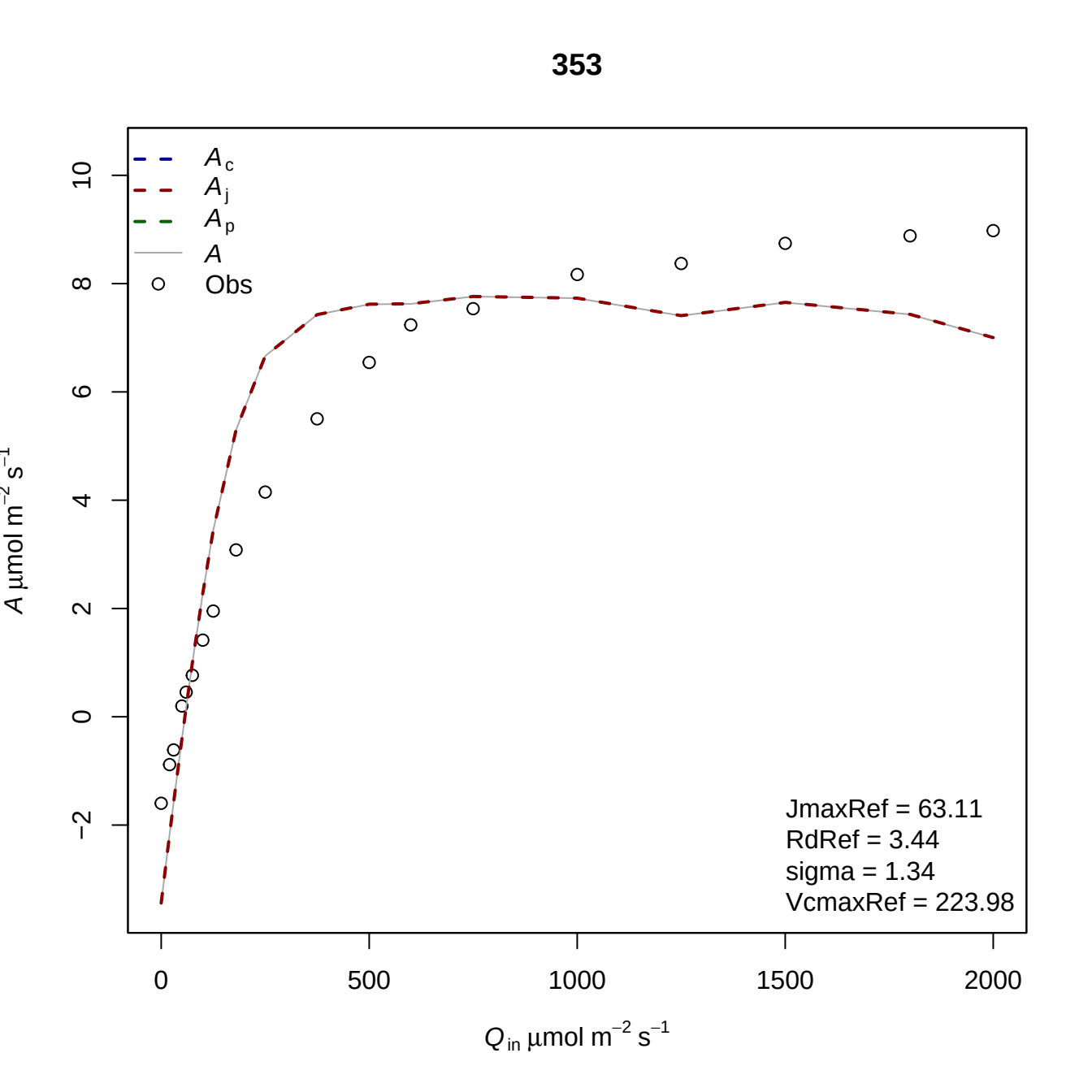

348
A_c
A_j
A_p
A
Obs
JmaxRef = 86.41
RdRef = 2.39
sigma = 0.96
VcmaxRef = 125.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

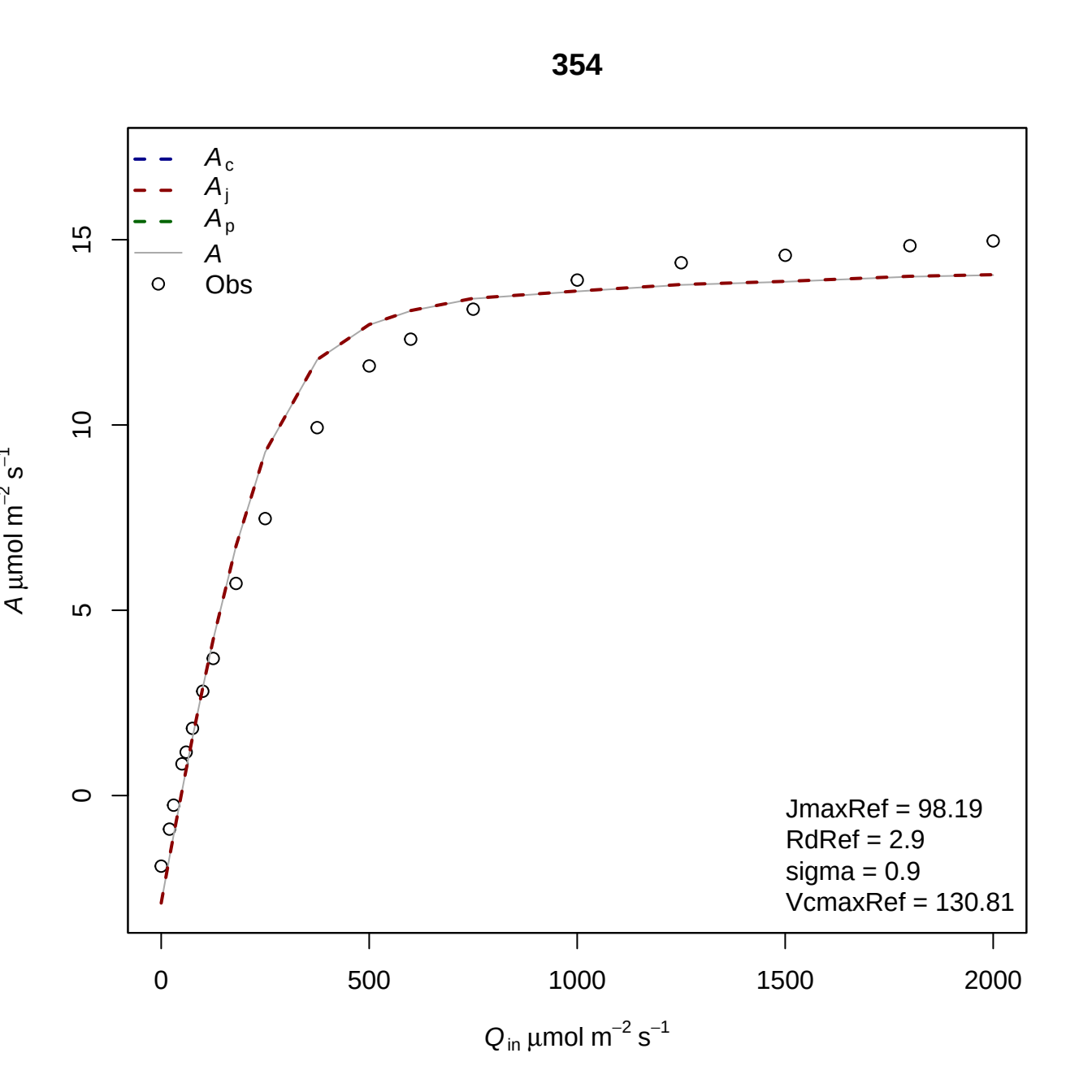

349
A_c
A_j
A_p
A
Obs
JmaxRef = 52.62
RdRef = 1.91
sigma = 0.71
VcmaxRef = 63.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

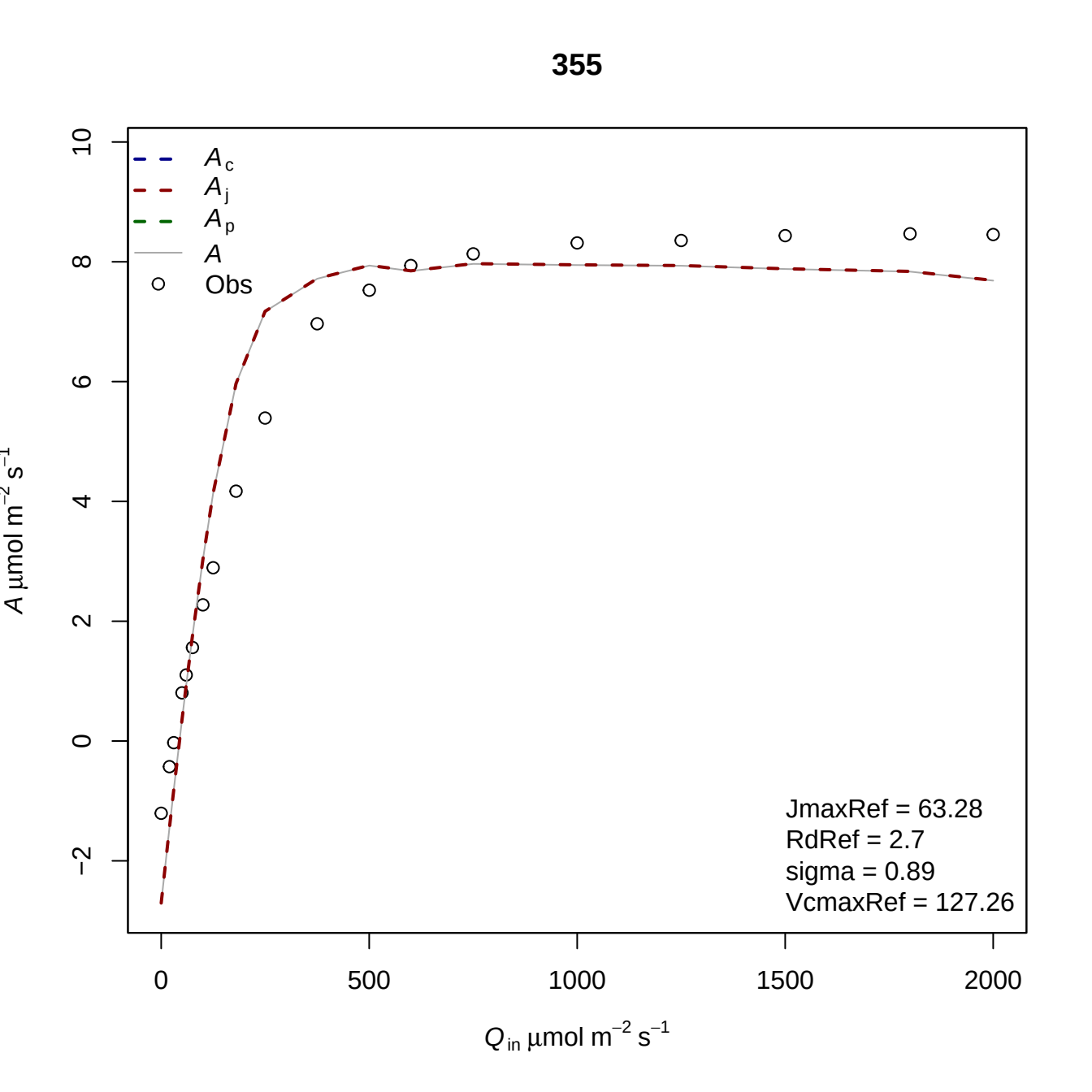

350
A_c
A_j
A_p
A
Obs
JmaxRef = 64.27
RdRef = 2.88
sigma = 0.91
VcmaxRef = 74.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

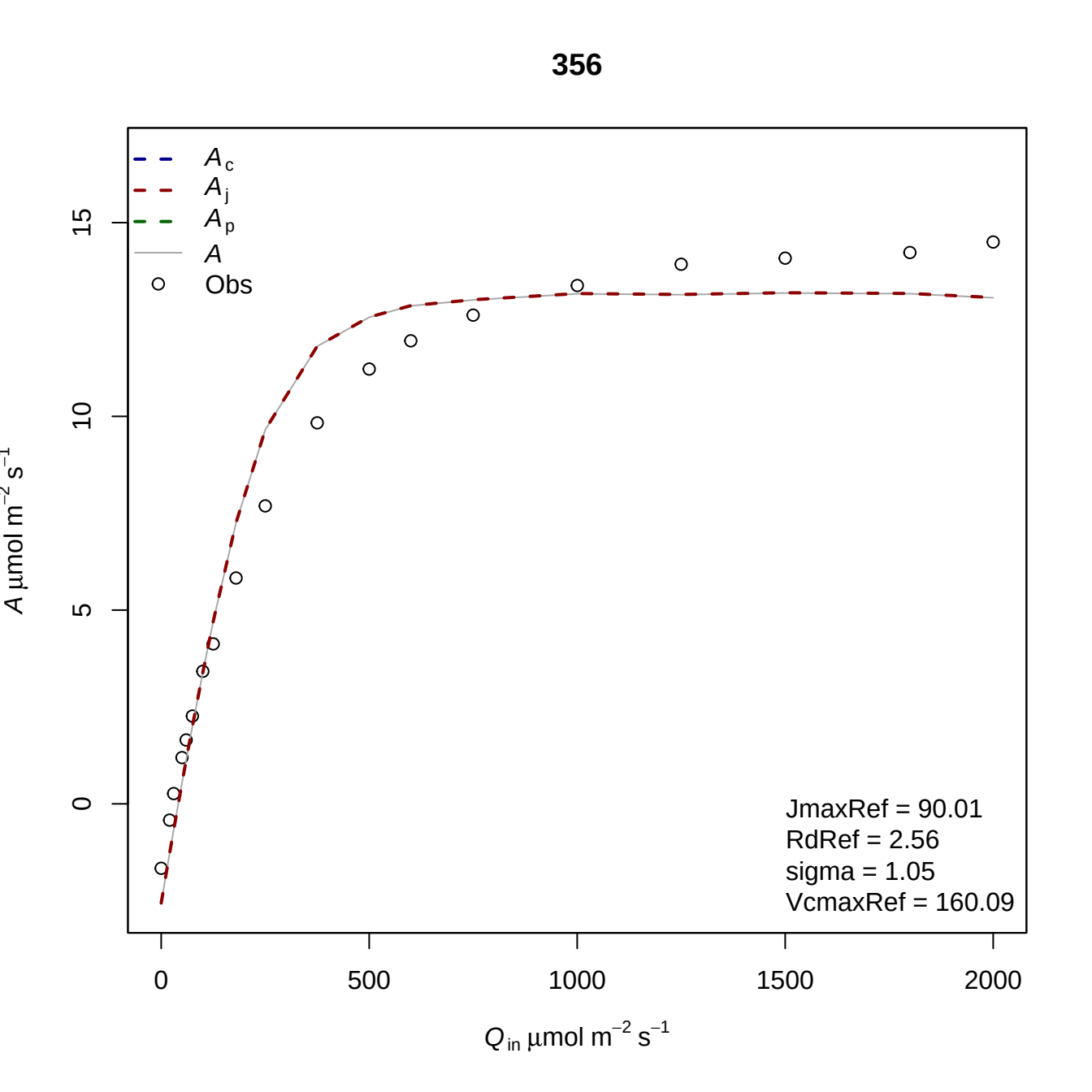

351
A_c
A_j
A_p
A
Obs
JmaxRef = 77.7
RdRef = 3.89
sigma = 1.65
VcmaxRef = 221.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

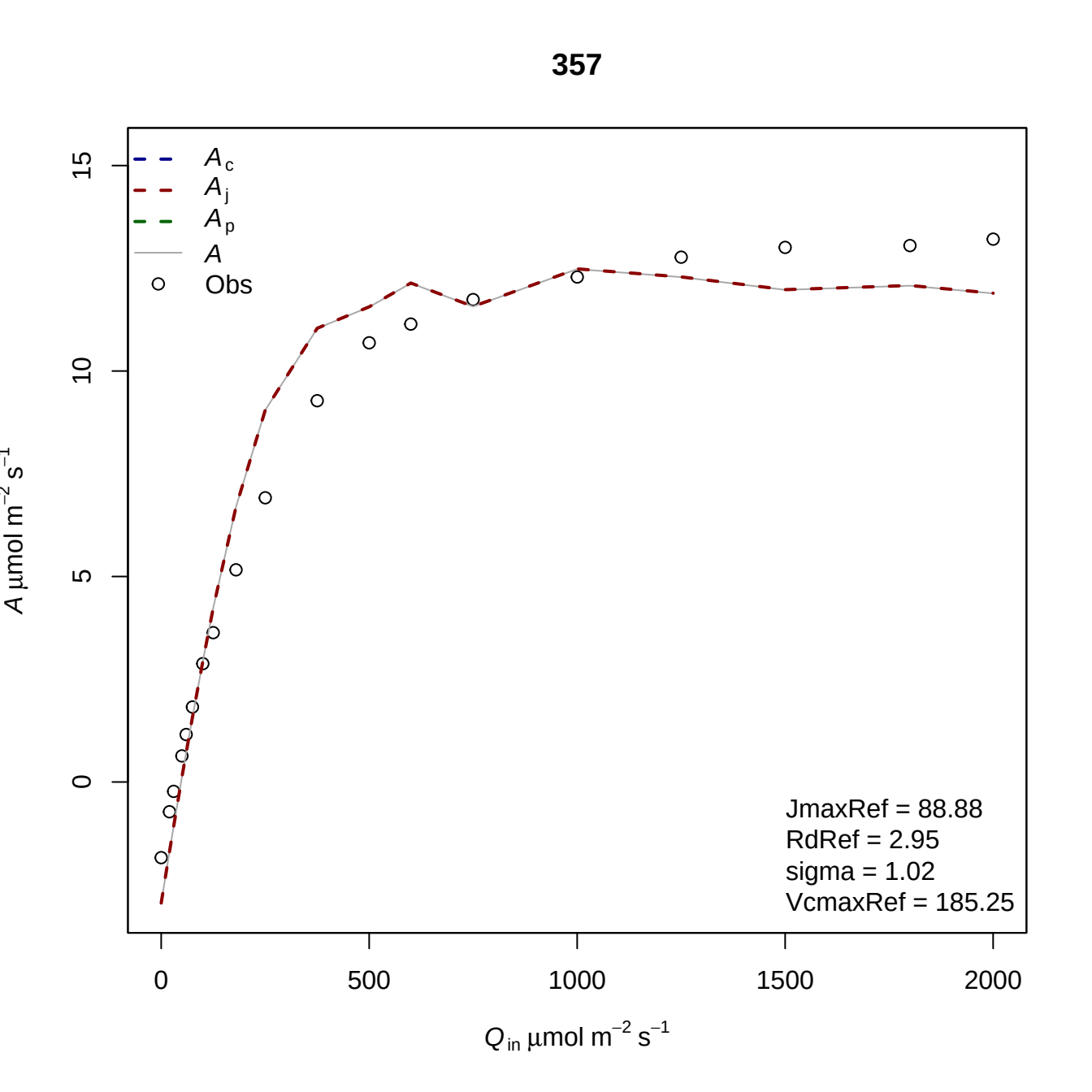

352
A_c
A_j
A_p
A
Obs
JmaxRef = 38.27
RdRef = 4.12
sigma = 1.34
VcmaxRef = 68.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

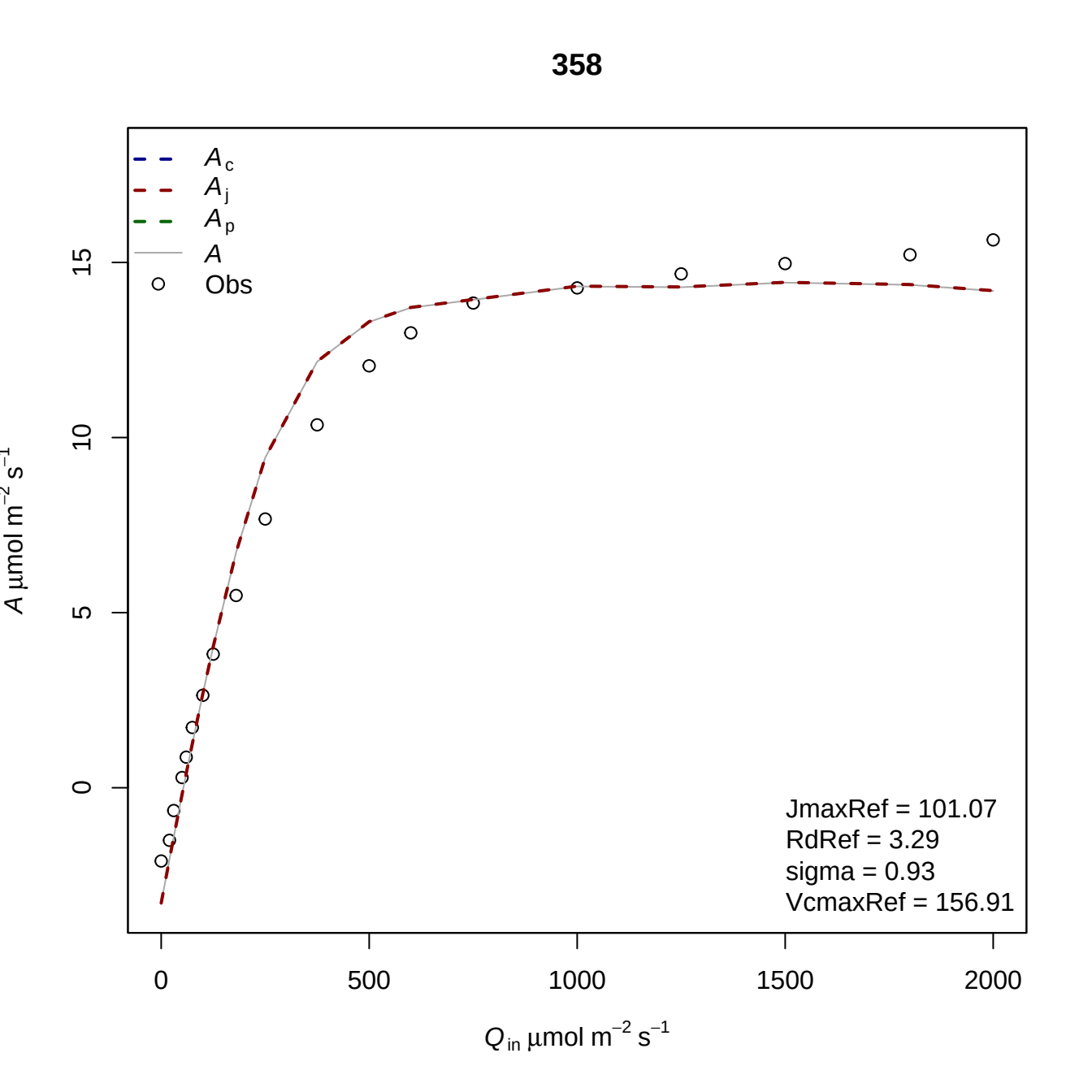

353
A_c
A_j
A_p
A
Obs
JmaxRef = 63.11
RdRef = 3.44
sigma = 1.34
VcmaxRef = 223.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

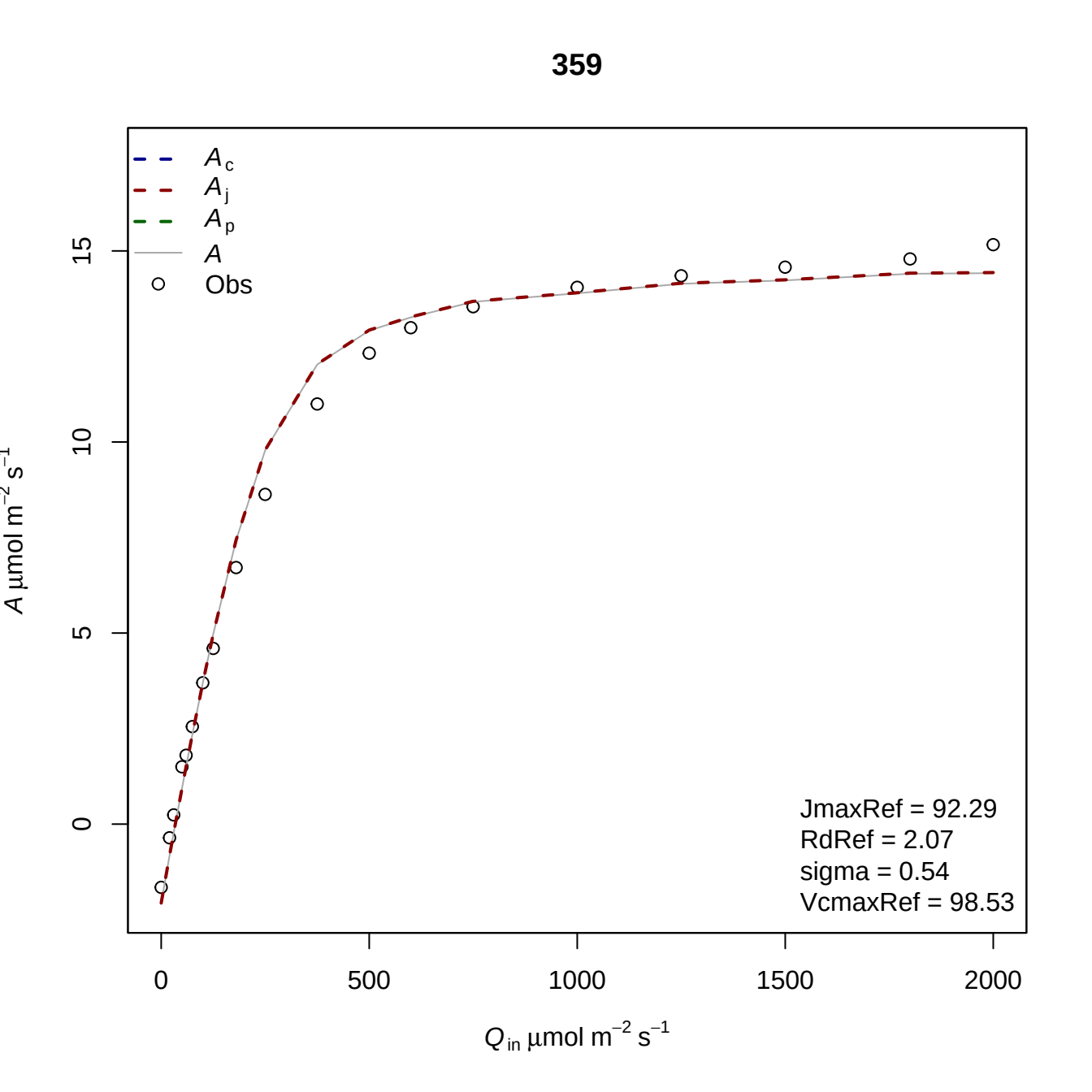

354
A_c
A_j
A_p
A
Obs
JmaxRef = 98.19
RdRef = 2.9
sigma = 0.9
VcmaxRef = 130.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

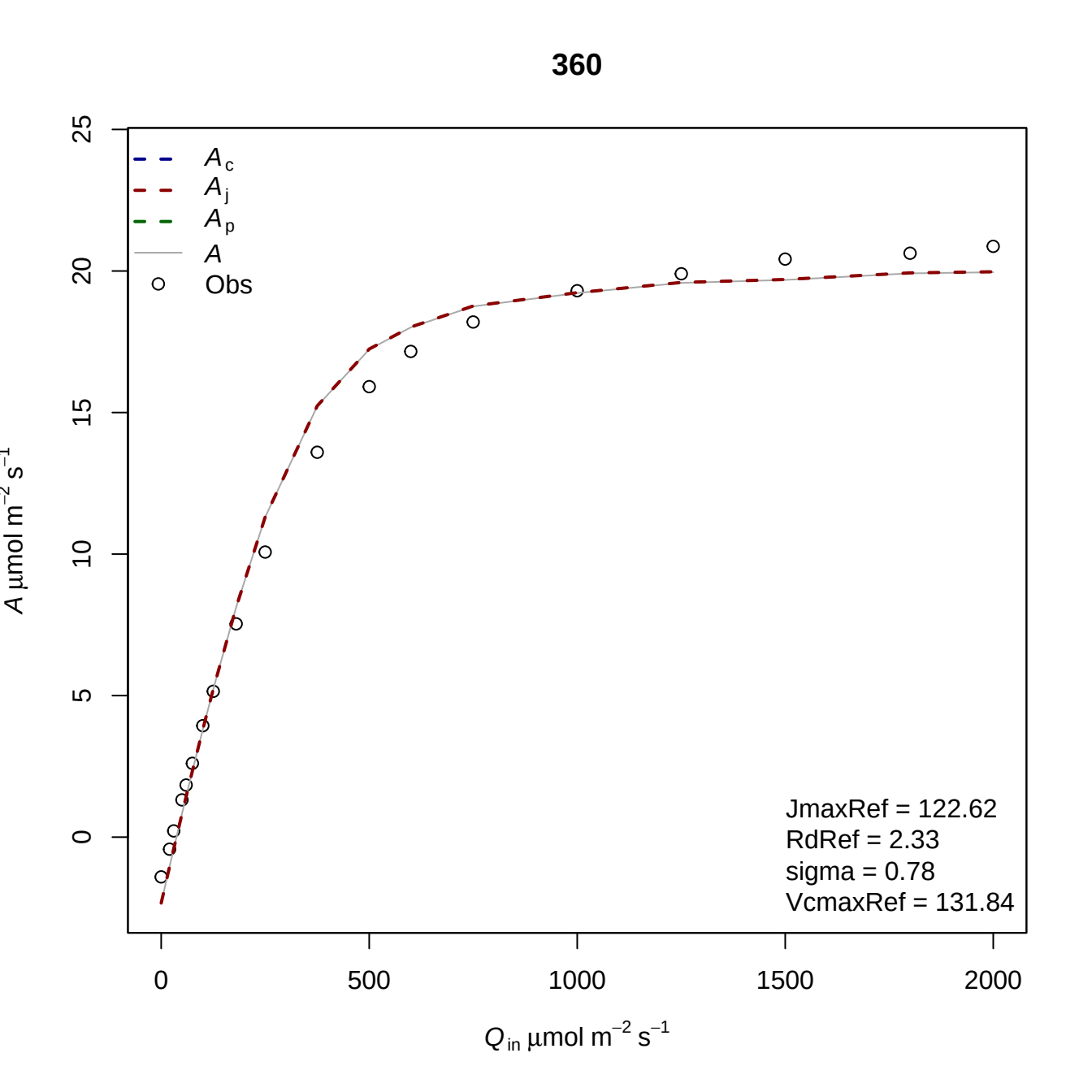

355
A_c
A_j
A_p
A
Obs
JmaxRef = 63.28
RdRef = 2.7
sigma = 0.89
VcmaxRef = 127.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

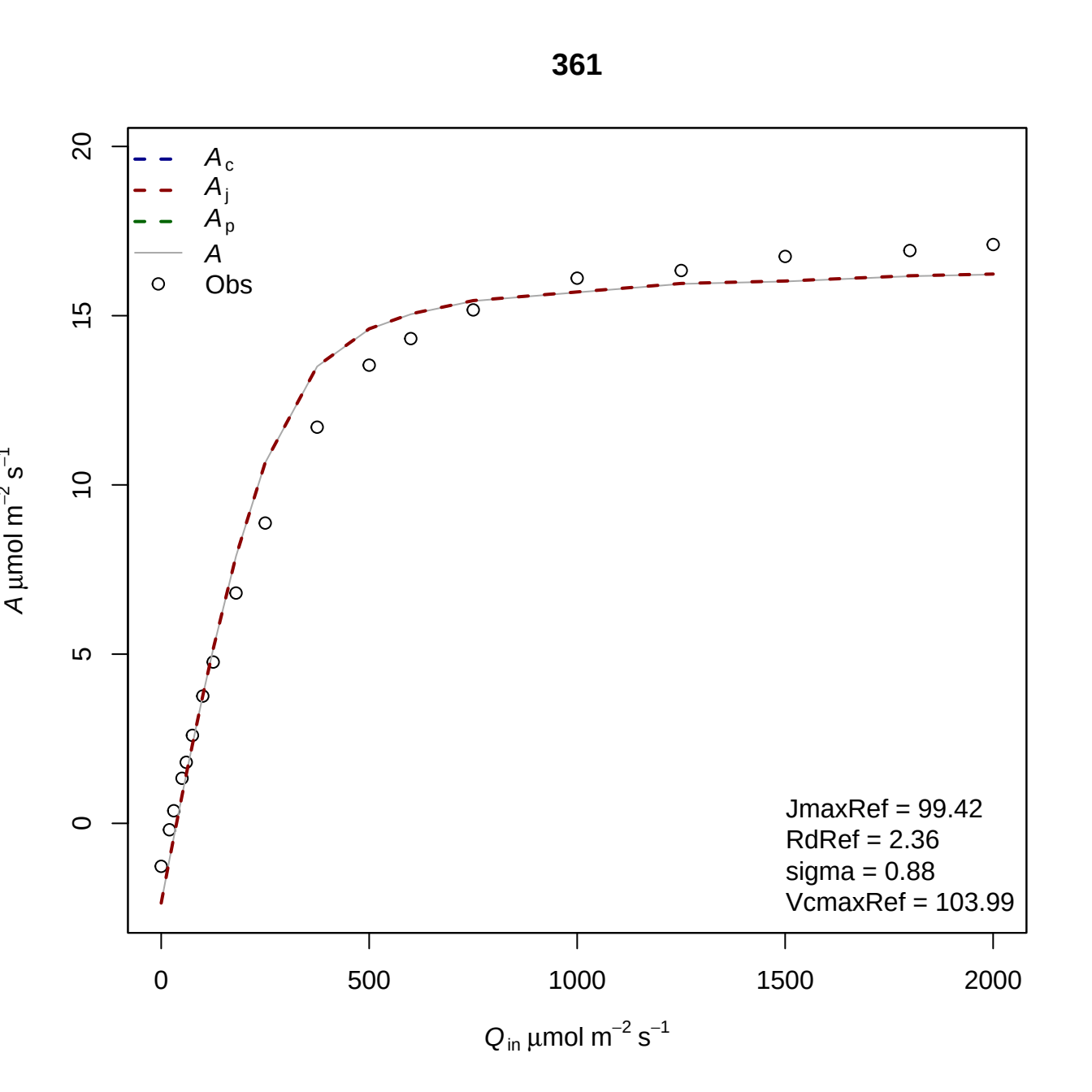

356
A_c
A_j
A_p
A
Obs
JmaxRef = 90.01
RdRef = 2.56
sigma = 1.05
VcmaxRef = 160.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

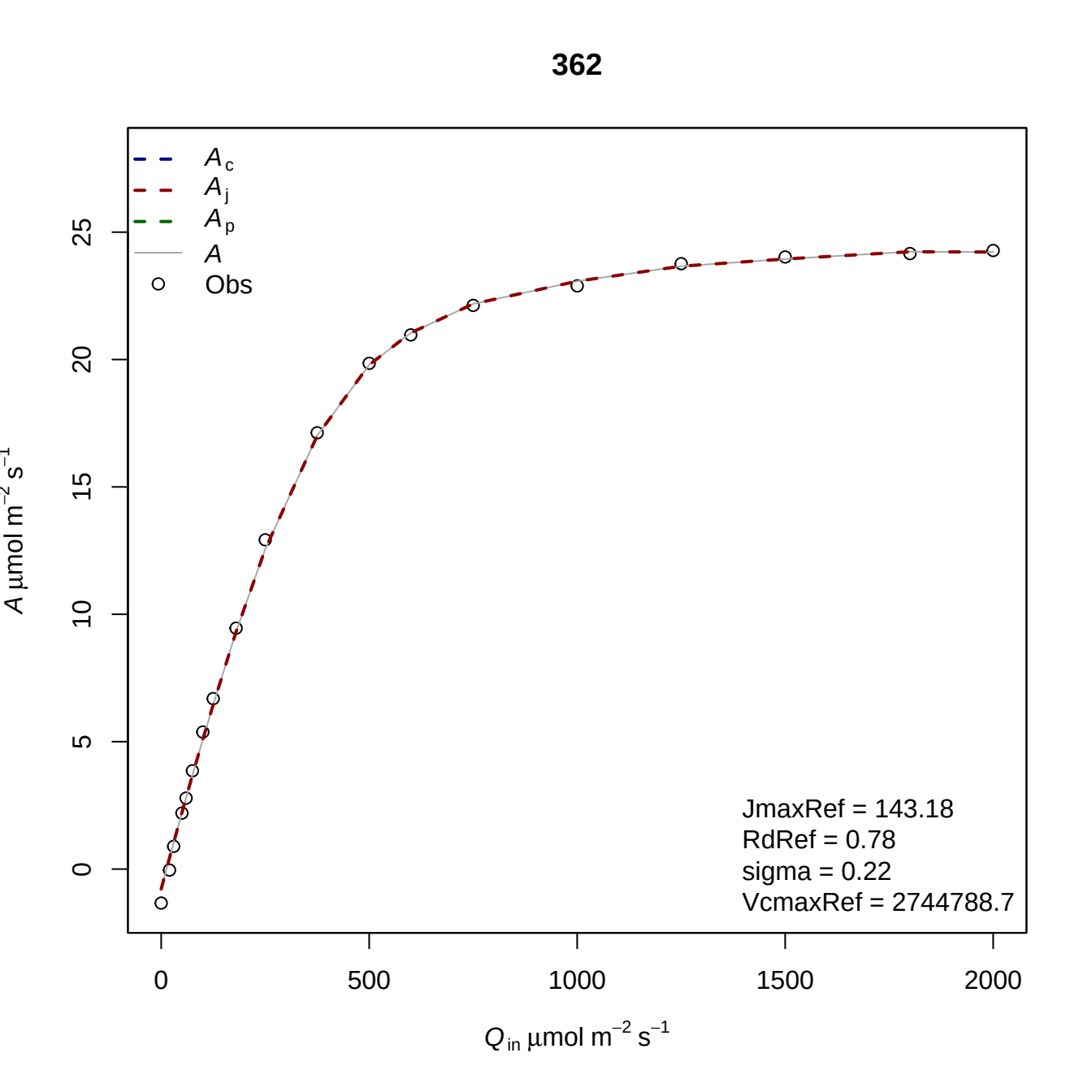

357
A_c
A_j
A_p
A
Obs
JmaxRef = 88.88
RdRef = 2.95
sigma = 1.02
VcmaxRef = 185.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

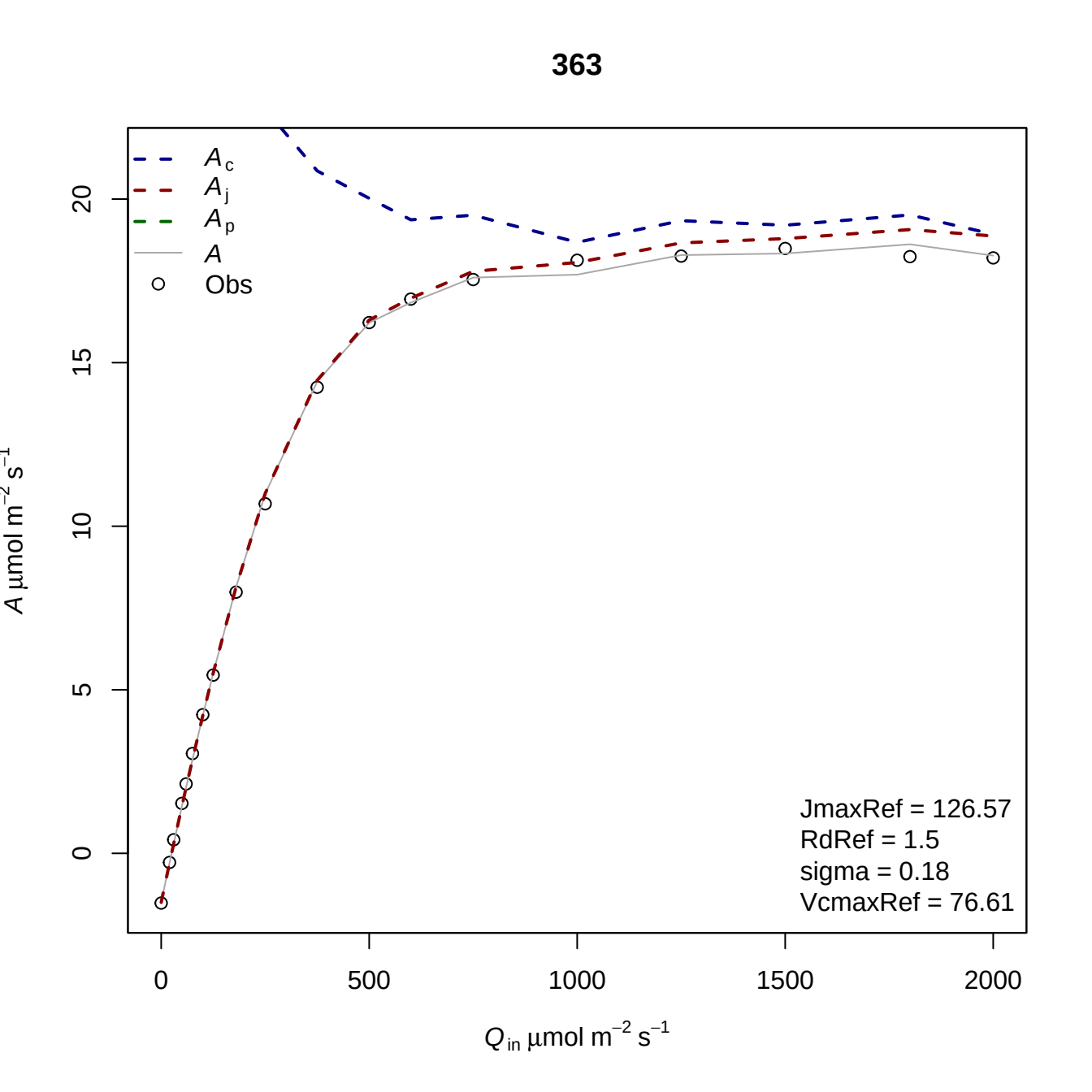

358
A_c
A_j
A_p
A
Obs
JmaxRef = 101.07
RdRef = 3.29
sigma = 0.93
VcmaxRef = 156.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

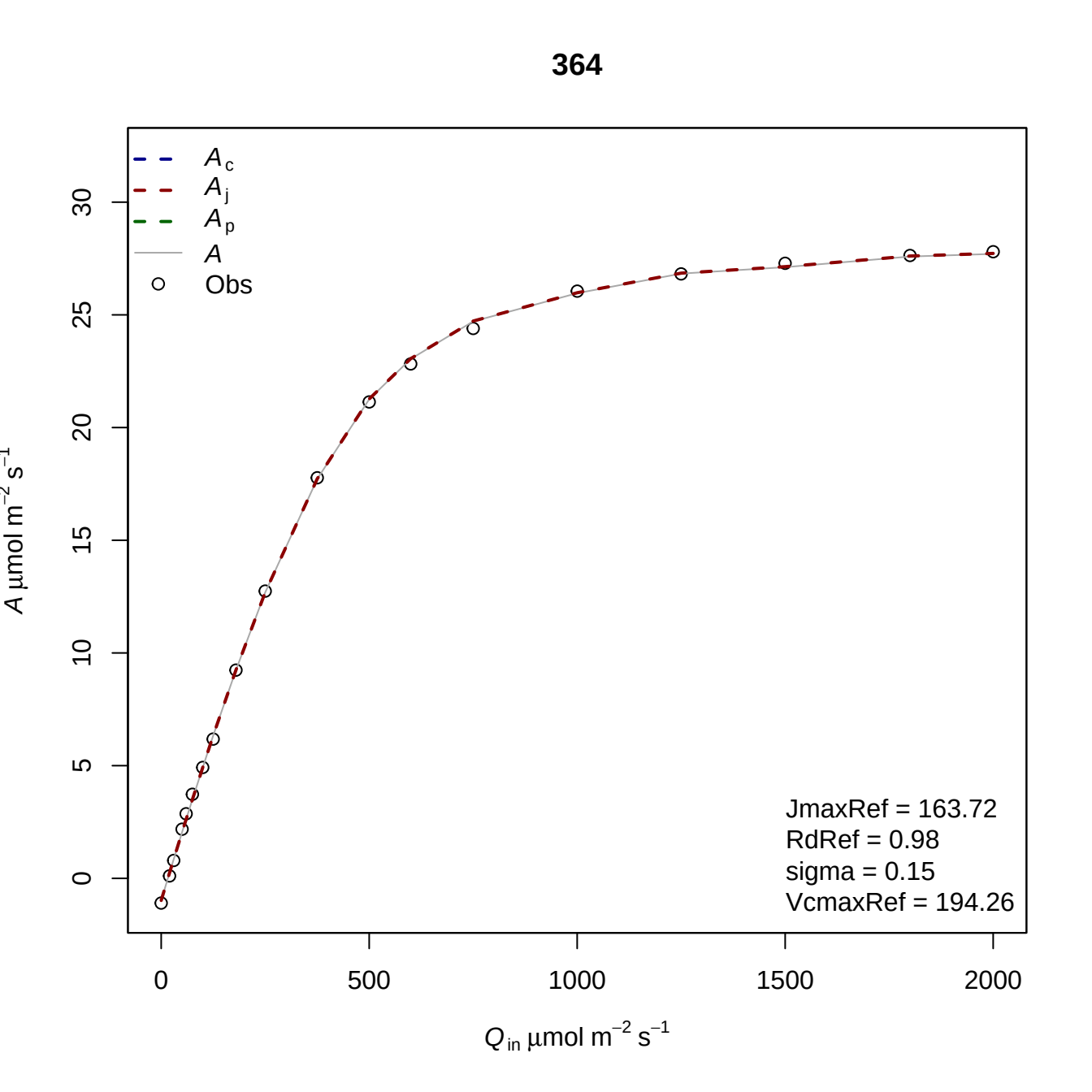

359
A_c
A_j
A_p
A
Obs
JmaxRef = 92.29
RdRef = 2.07
sigma = 0.54
VcmaxRef = 98.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

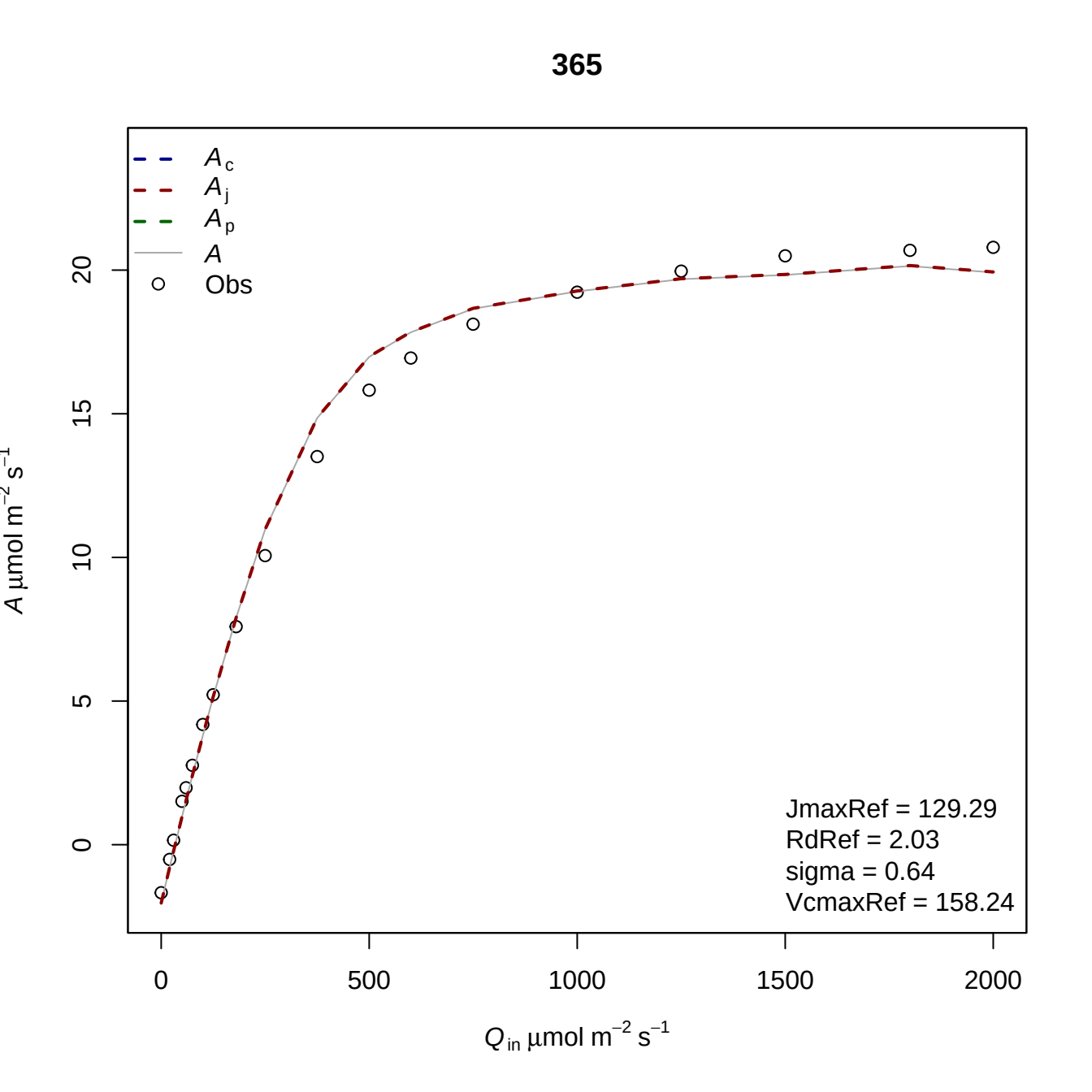

360
A_c
A_j
A_p
A
Obs
JmaxRef = 122.62
RdRef = 2.33
sigma = 0.78
VcmaxRef = 131.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

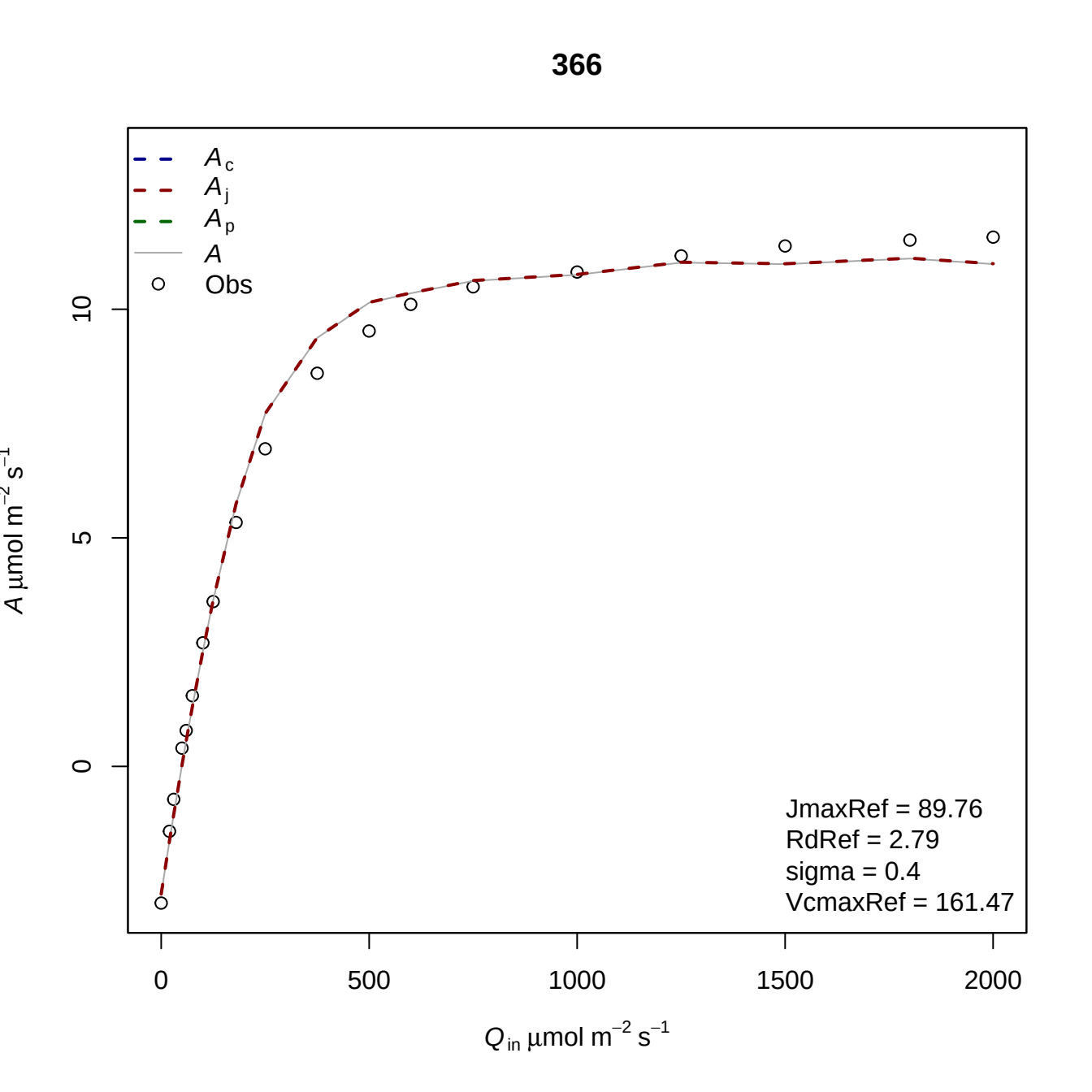

361
A_c
A_j
A_p
A
Obs
JmaxRef = 99.42
RdRef = 2.36
sigma = 0.88
VcmaxRef = 103.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

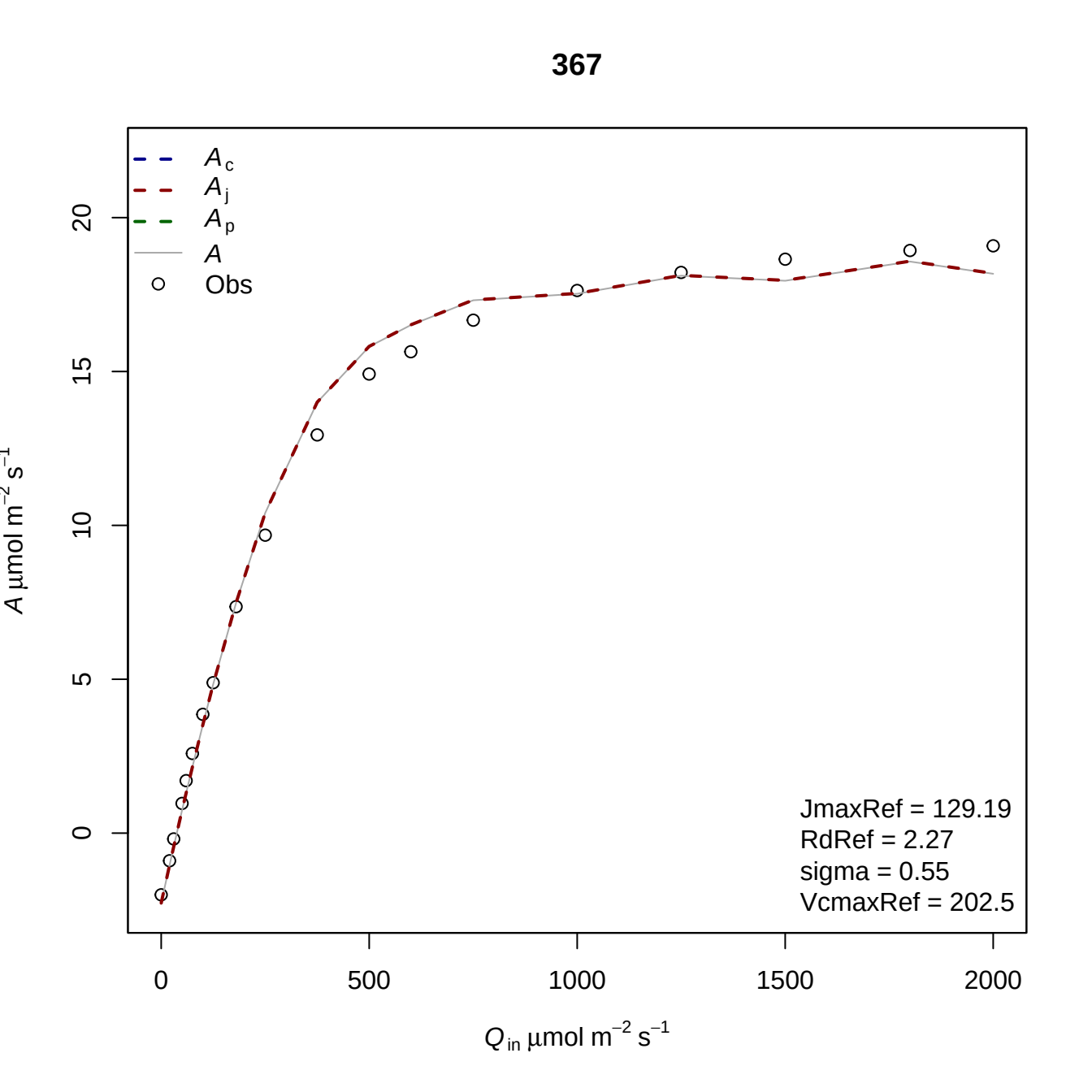

362
A_c
A_j
A_p
A
Obs
JmaxRef = 143.18
RdRef = 0.78
sigma = 0.22
VcmaxRef = 2744788.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

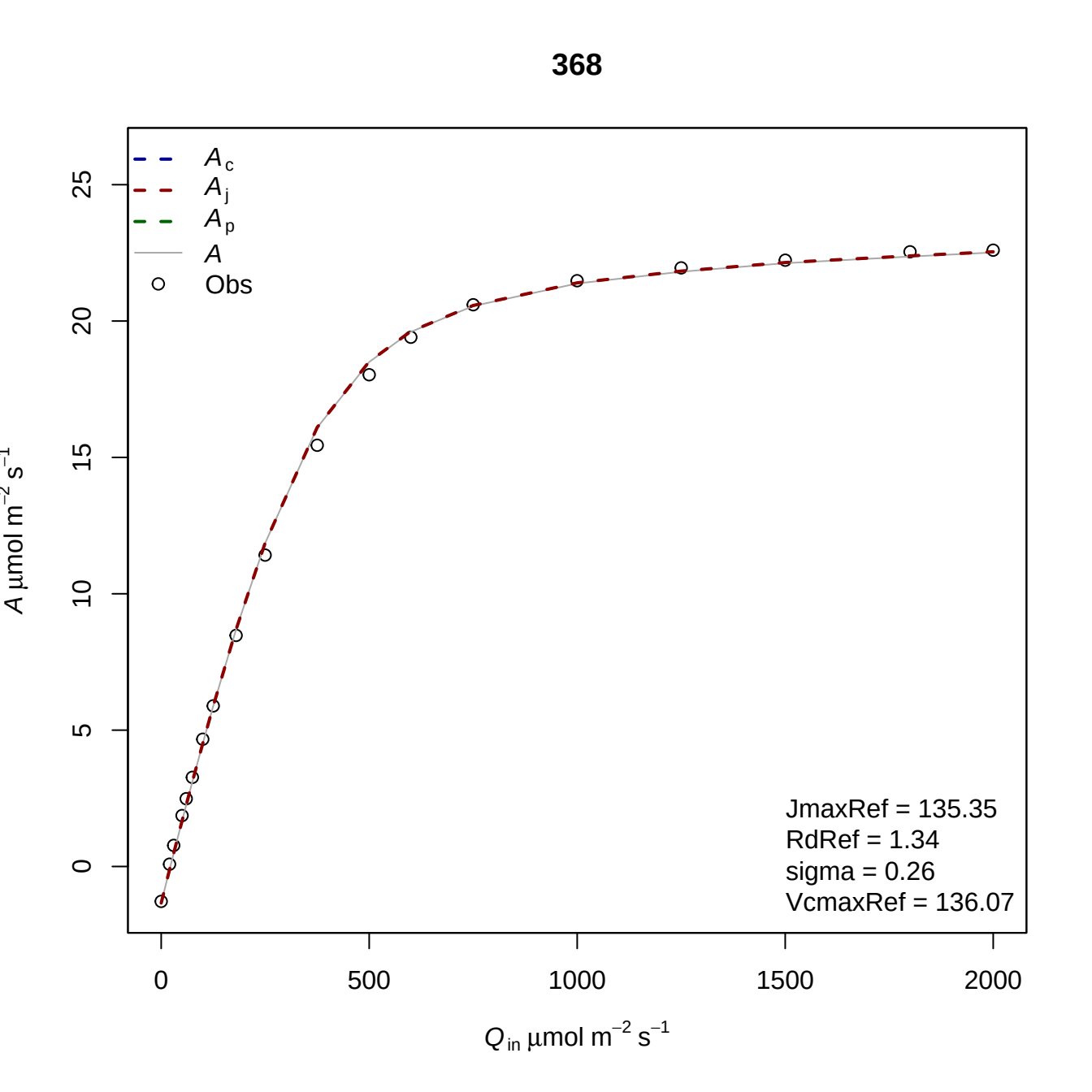

363
A_c
A_j
A_p
A
Obs
JmaxRef = 126.57
RdRef = 1.5
sigma = 0.18
VcmaxRef = 76.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

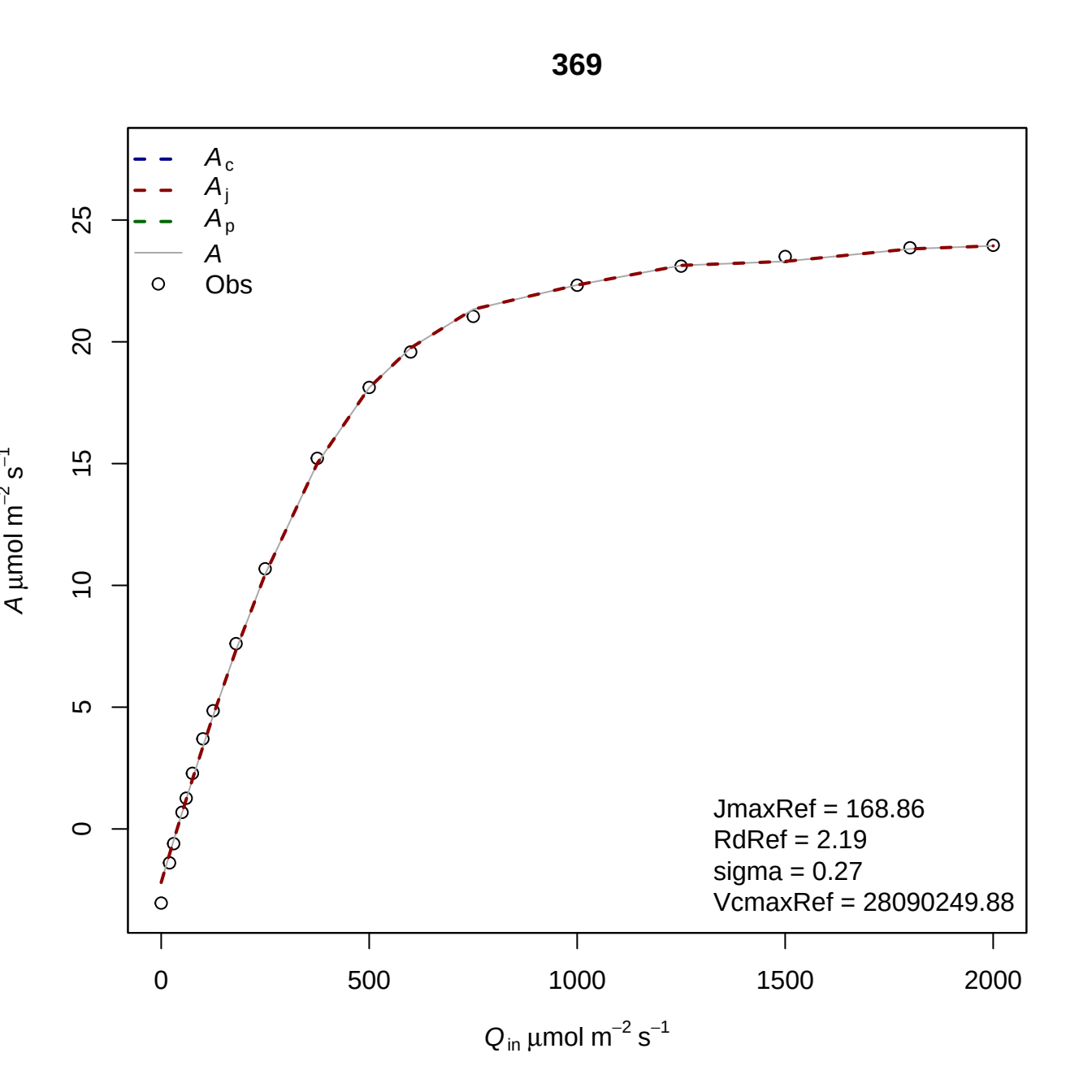

364
A_c
A_j
A_p
A
Obs
JmaxRef = 163.72
RdRef = 0.98
sigma = 0.15
VcmaxRef = 194.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

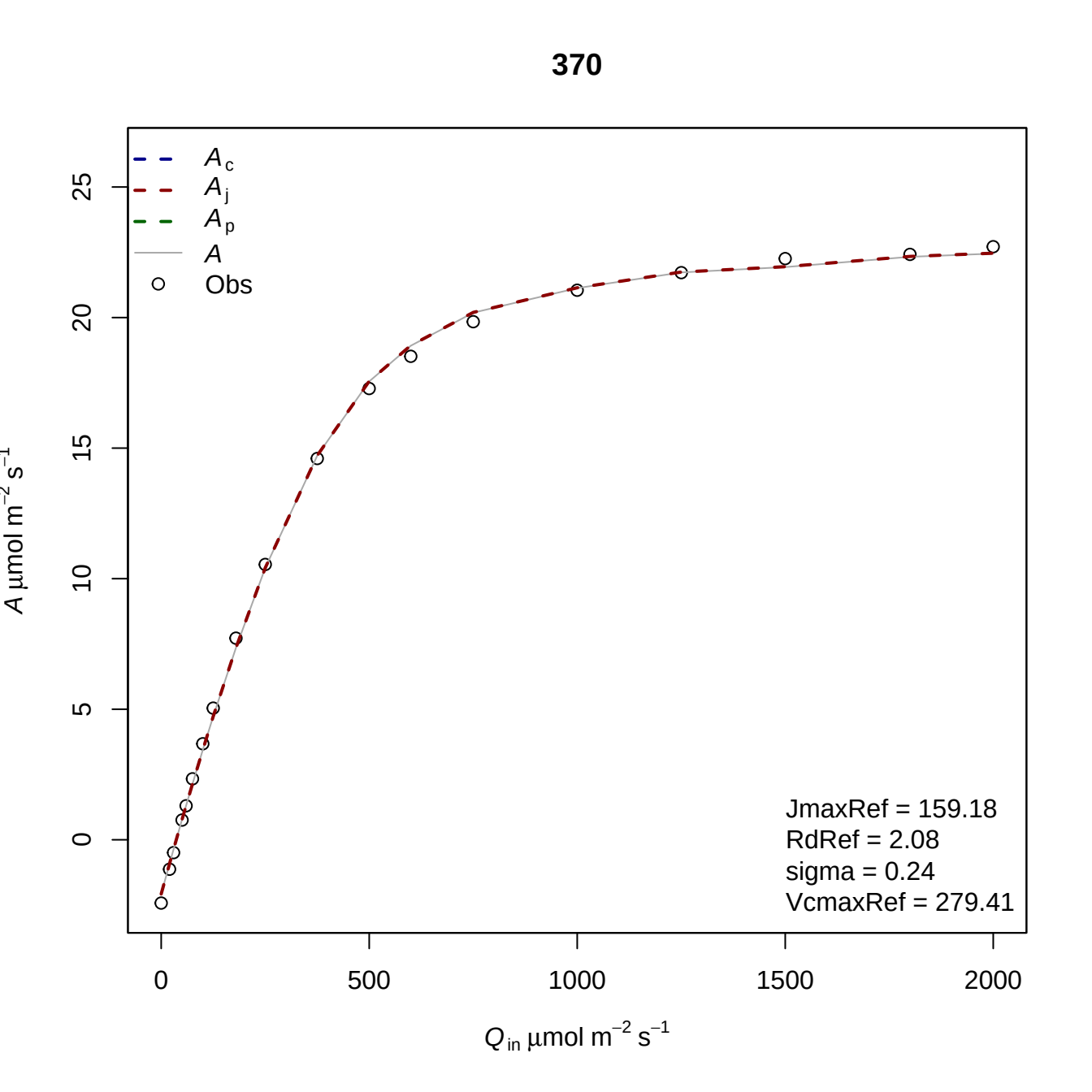

365
A_c
A_j
A_p
A
Obs
JmaxRef = 129.29
RdRef = 2.03
sigma = 0.64
VcmaxRef = 158.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

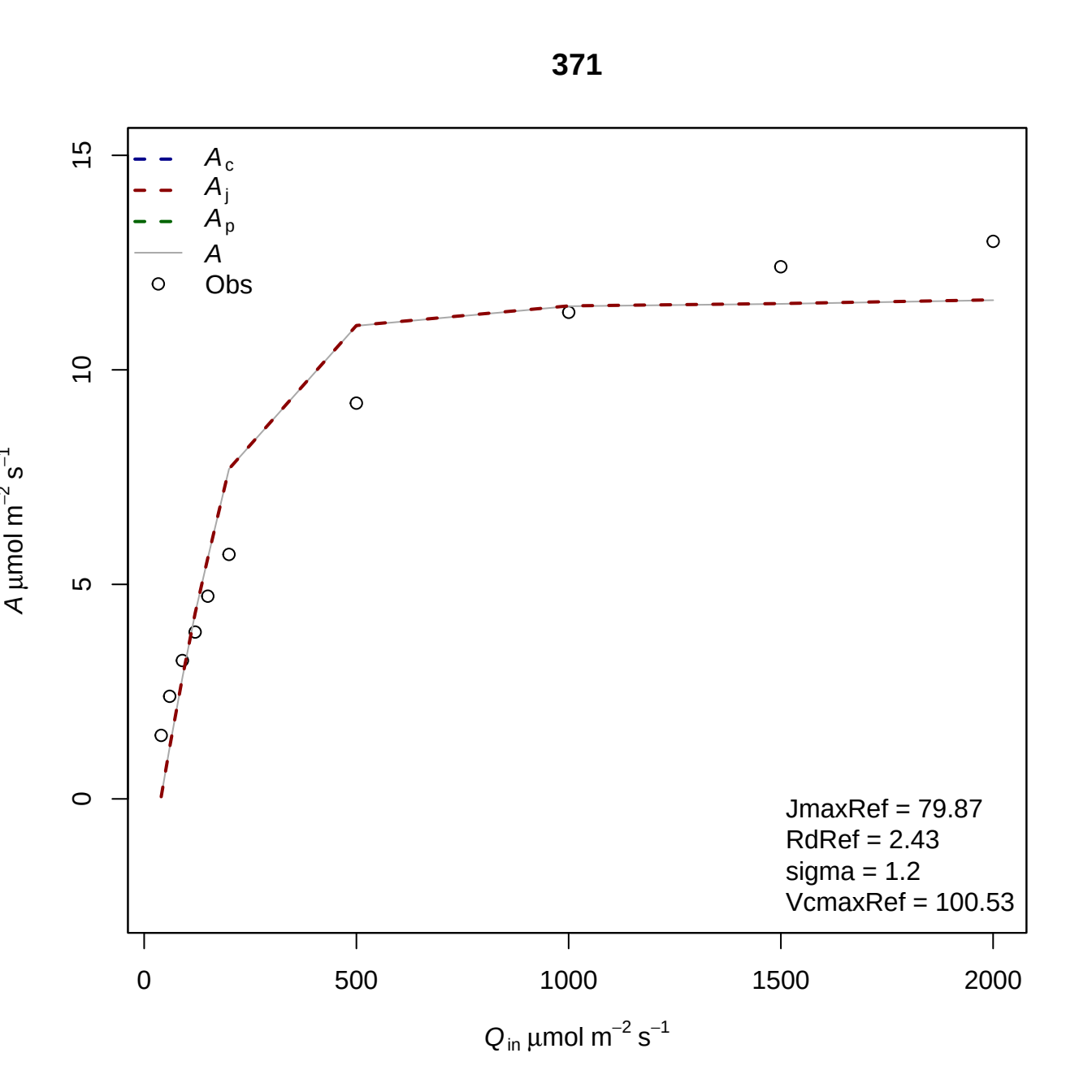

366
A_c
A_j
A_p
A
Obs
JmaxRef = 89.76
RdRef = 2.79
sigma = 0.4
VcmaxRef = 161.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

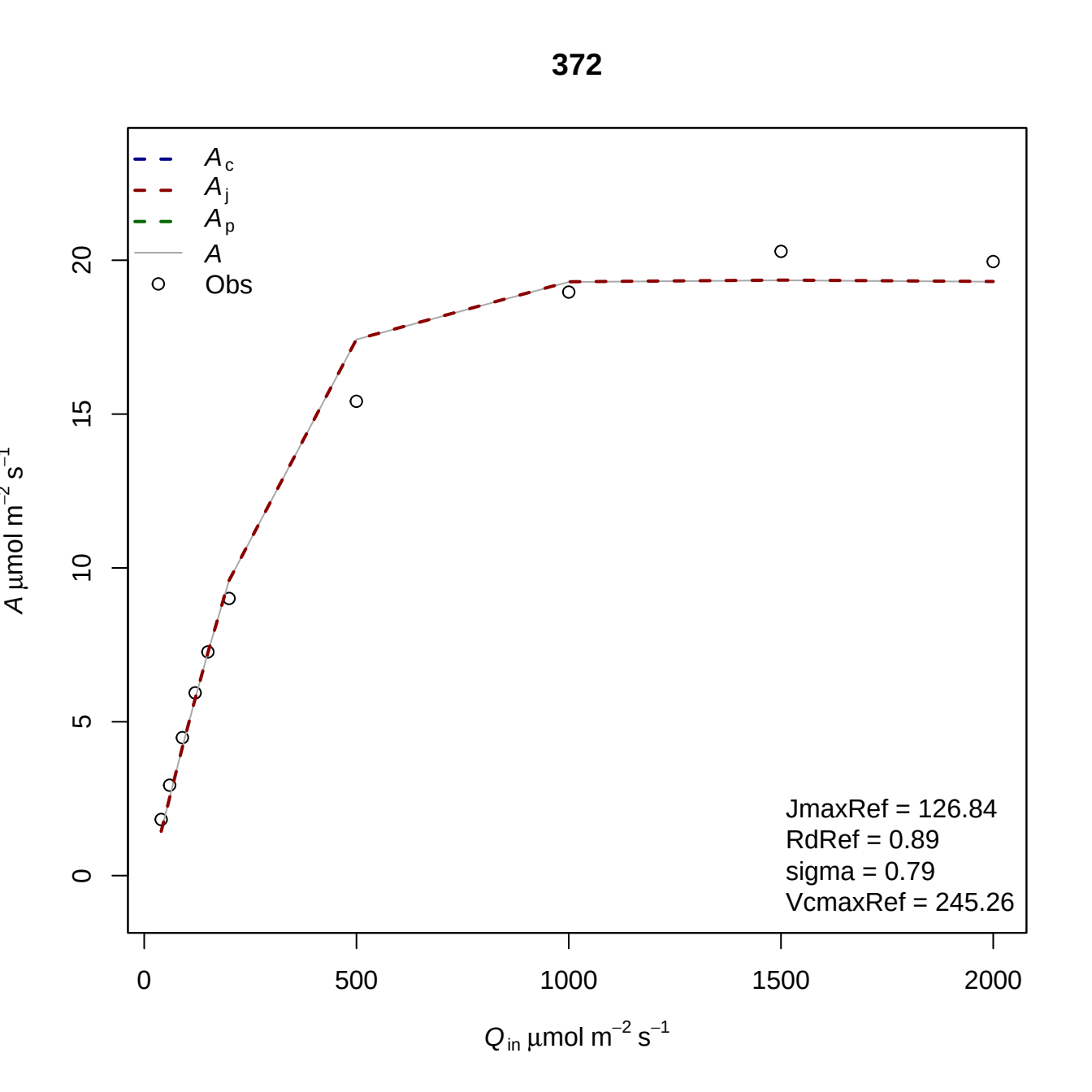

367
A_c
A_j
A_p
A
Obs
JmaxRef = 129.19
RdRef = 2.27
sigma = 0.55
VcmaxRef = 202.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

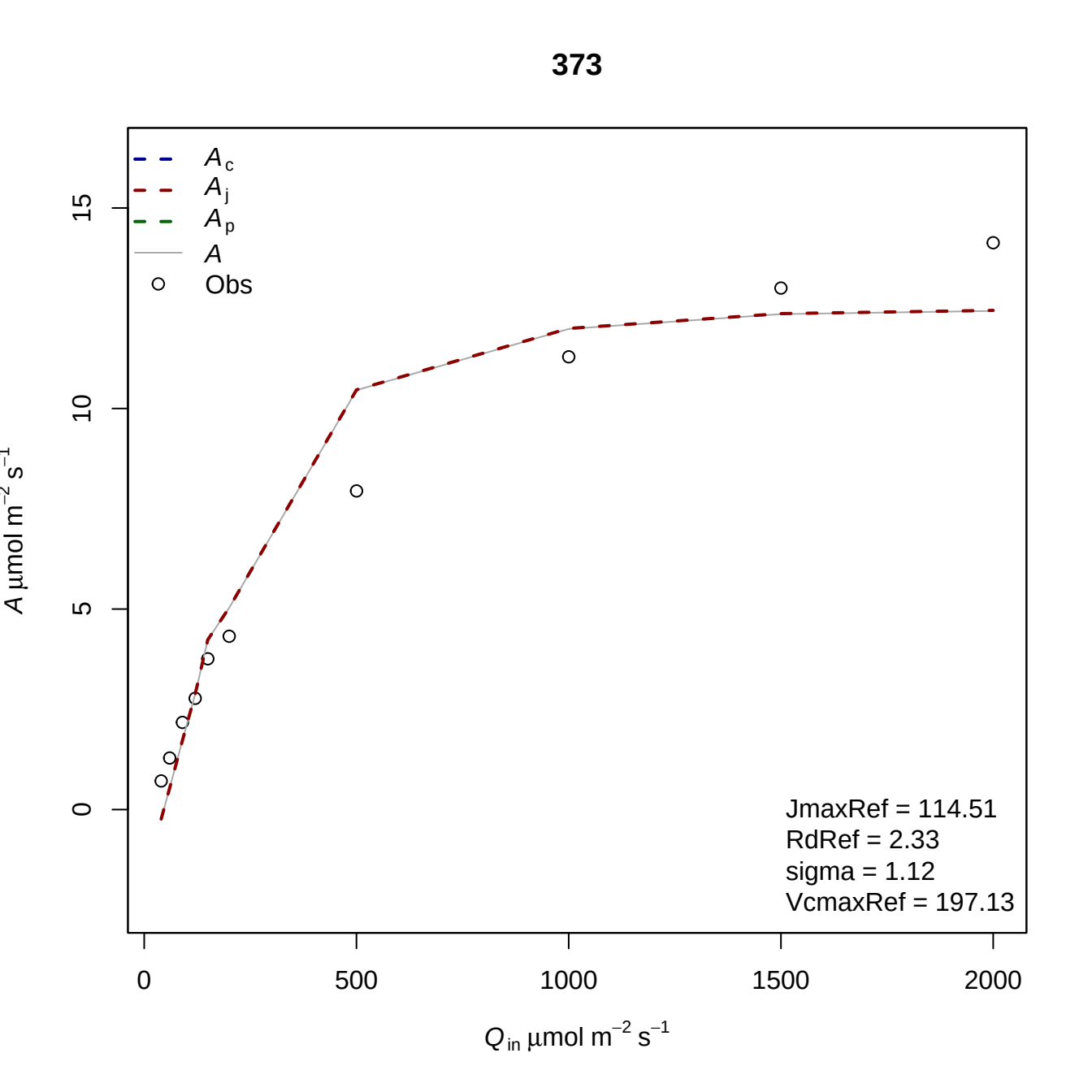

368
A_c
A_j
A_p
A
Obs
JmaxRef = 135.35
RdRef = 1.34
sigma = 0.26
VcmaxRef = 136.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

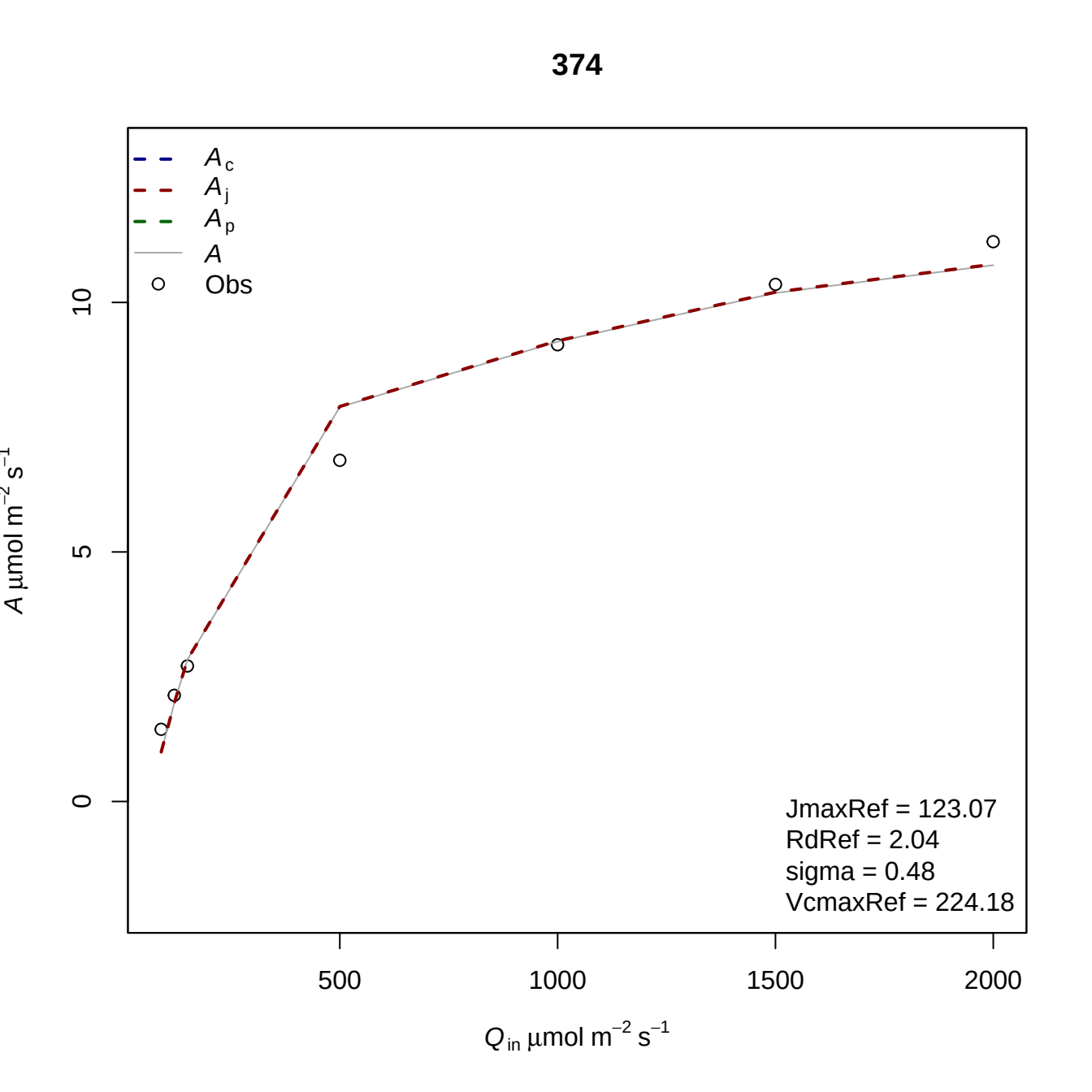

369
A_c
A_j
A_p
A
Obs
JmaxRef = 168.86
RdRef = 2.19
sigma = 0.27
VcmaxRef = 28090249.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

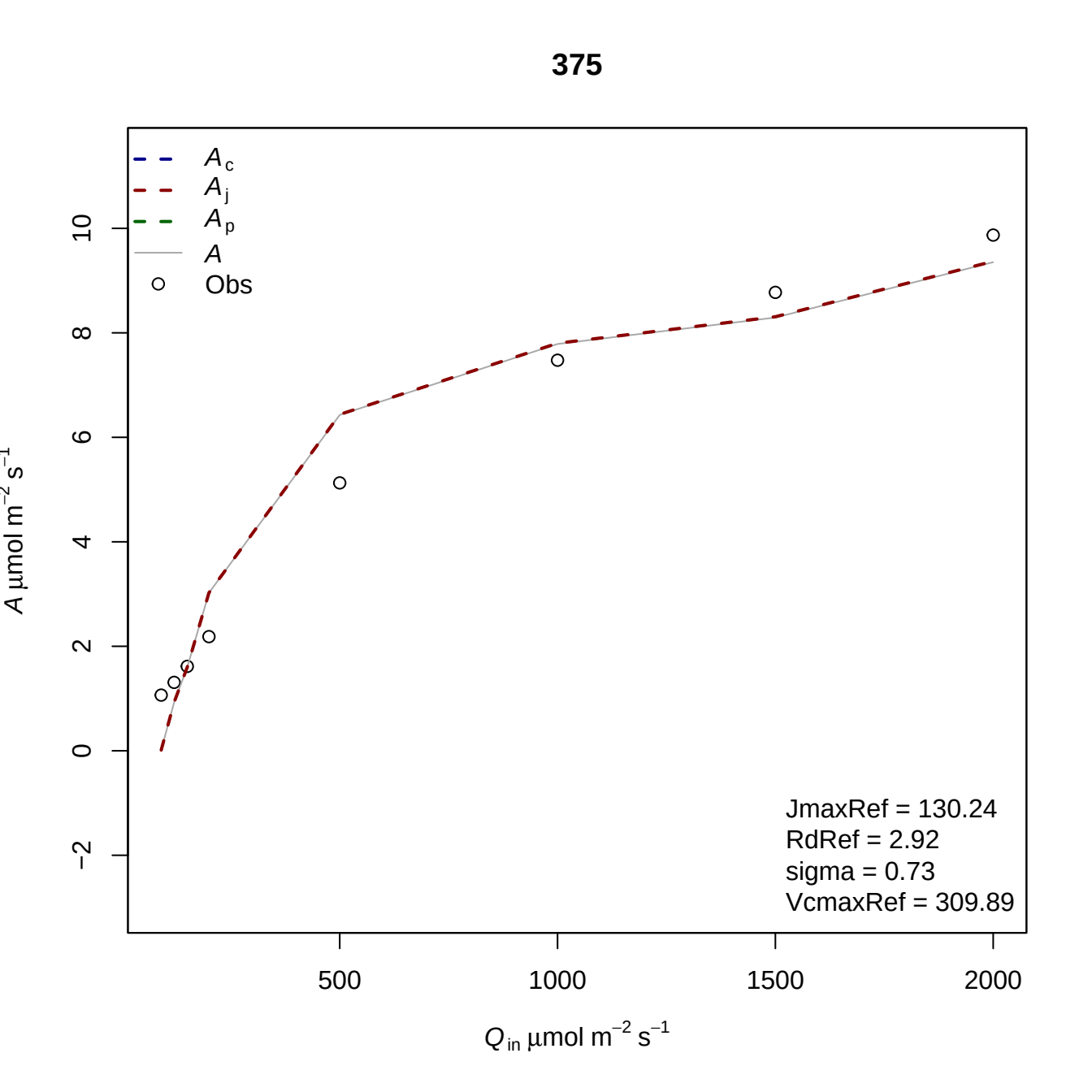

370
A_c
A_j
A_p
A
Obs
JmaxRef = 159.18
RdRef = 2.08
sigma = 0.24
VcmaxRef = 279.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

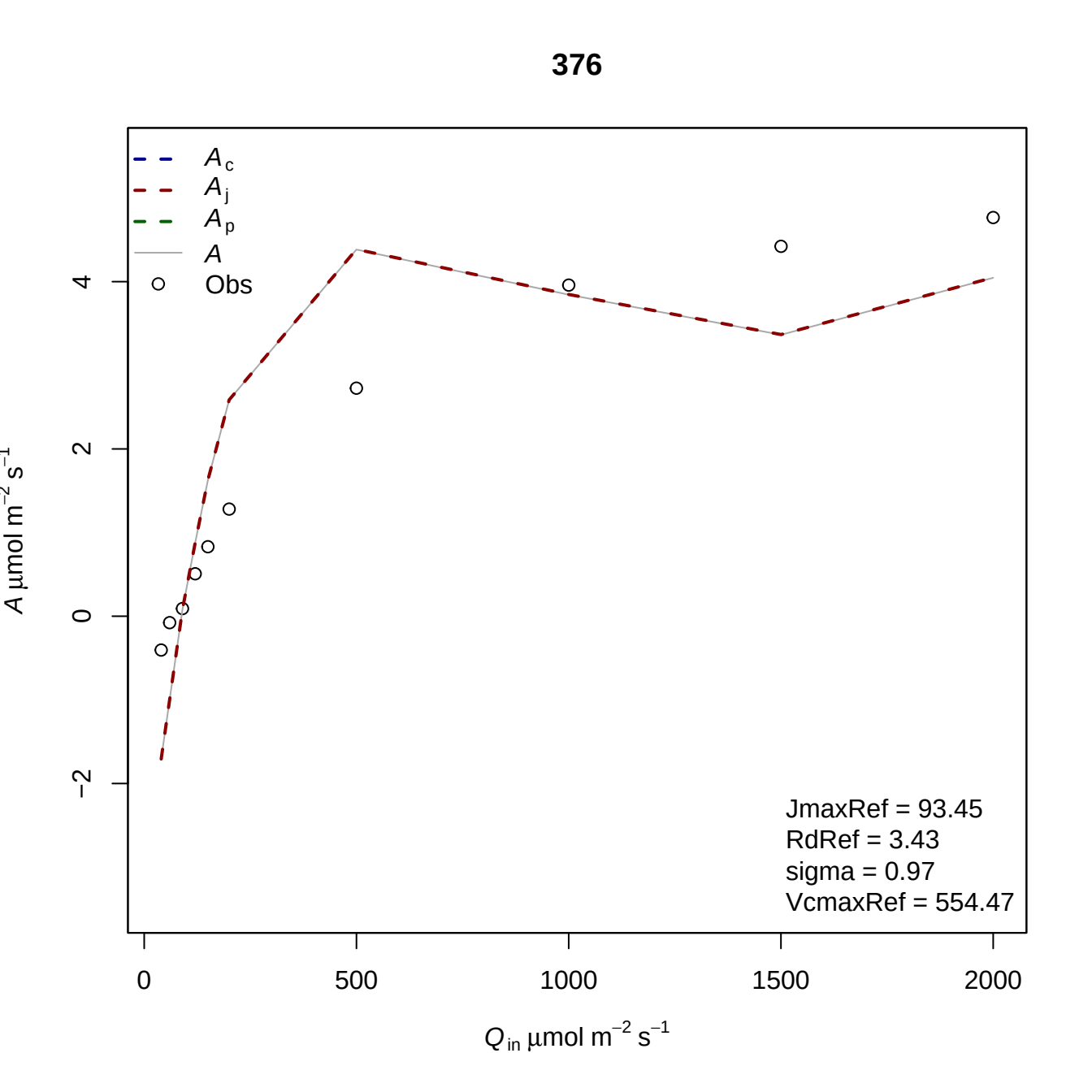

371
A_c
A_j
A_p
A
Obs
JmaxRef = 79.87
RdRef = 2.43
sigma = 1.2
VcmaxRef = 100.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

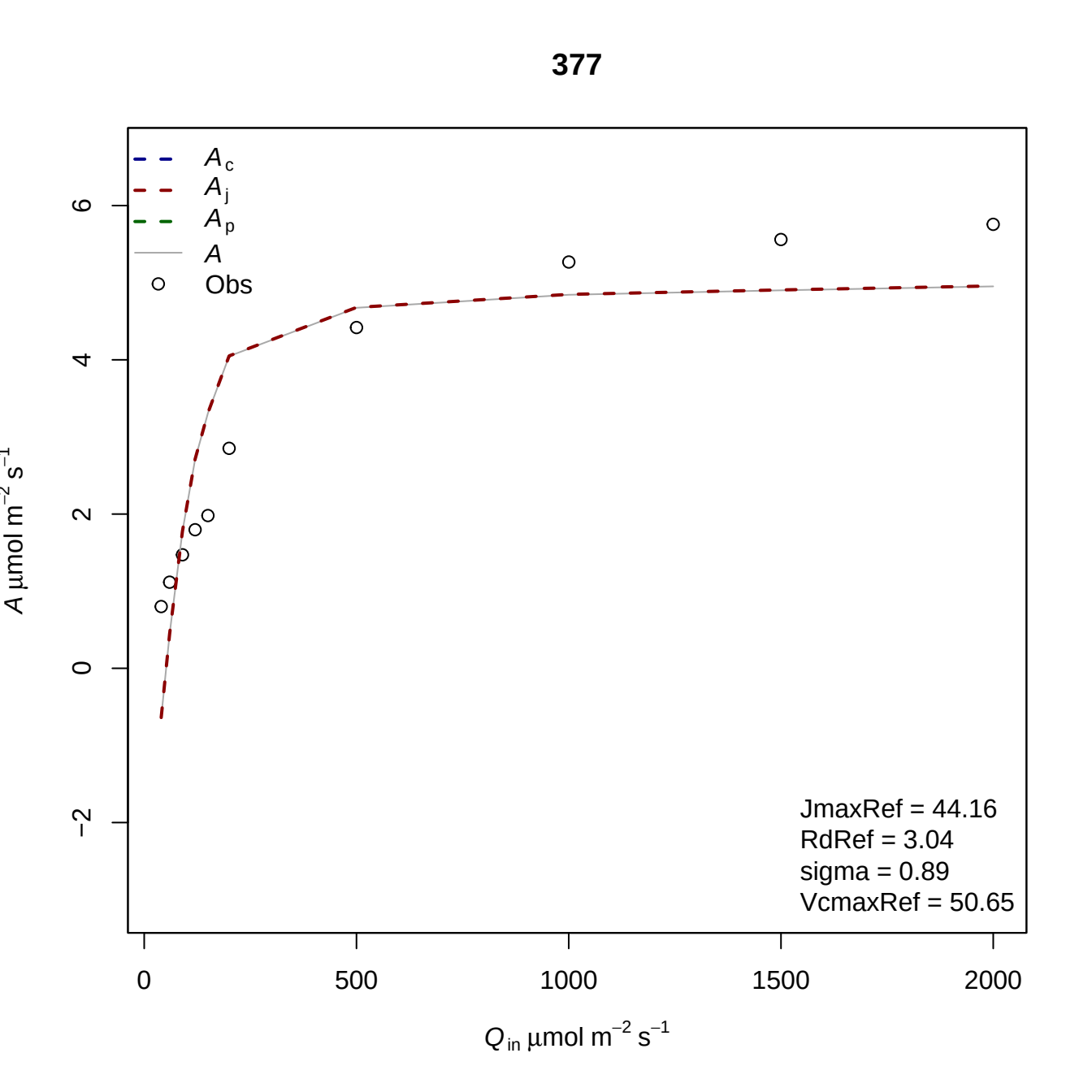

372
A_c
A_j
A_p
A
Obs
JmaxRef = 126.84
RdRef = 0.89
sigma = 0.79
VcmaxRef = 245.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

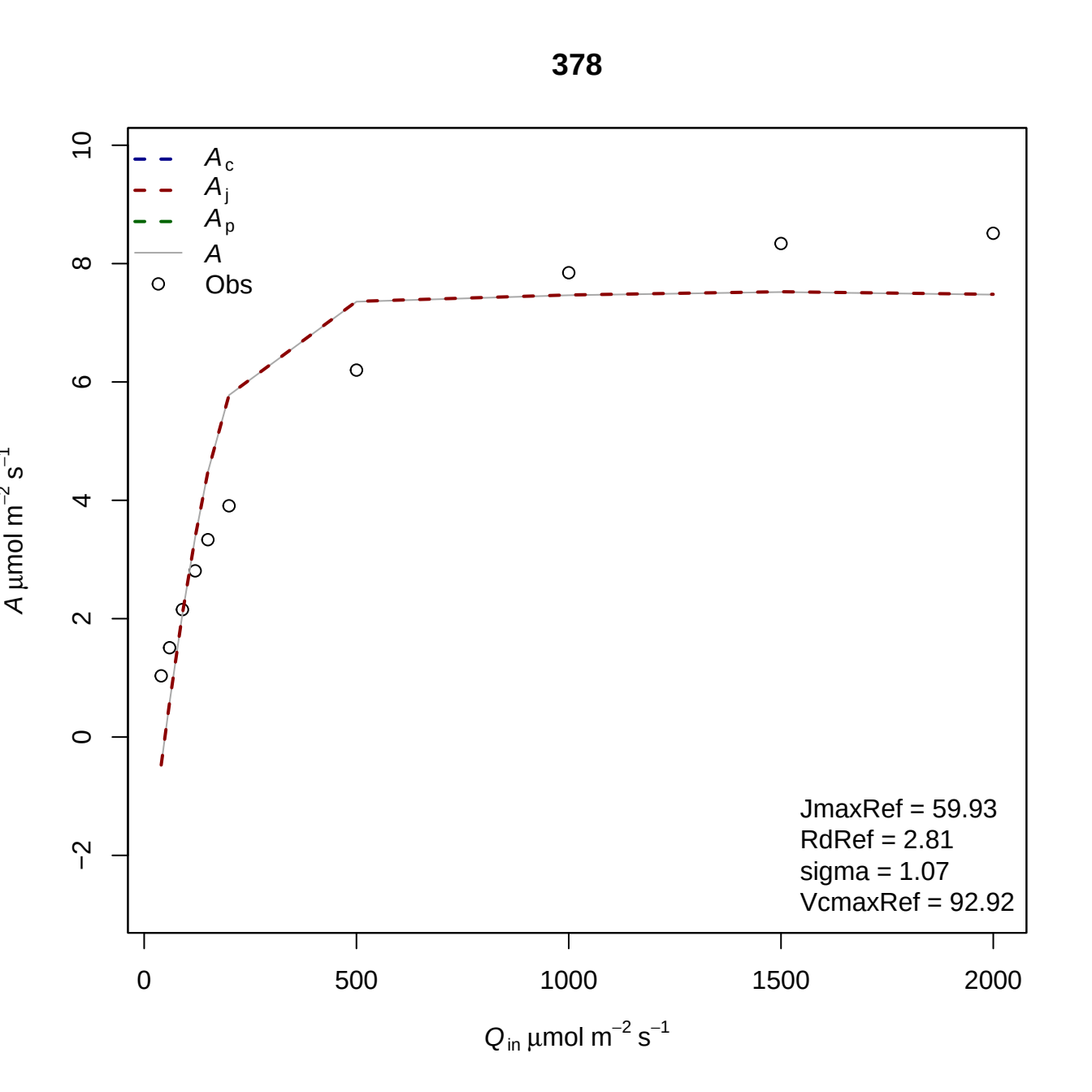

373
A_c
A_j
A_p
A
Obs
JmaxRef = 114.51
RdRef = 2.33
sigma = 1.12
VcmaxRef = 197.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

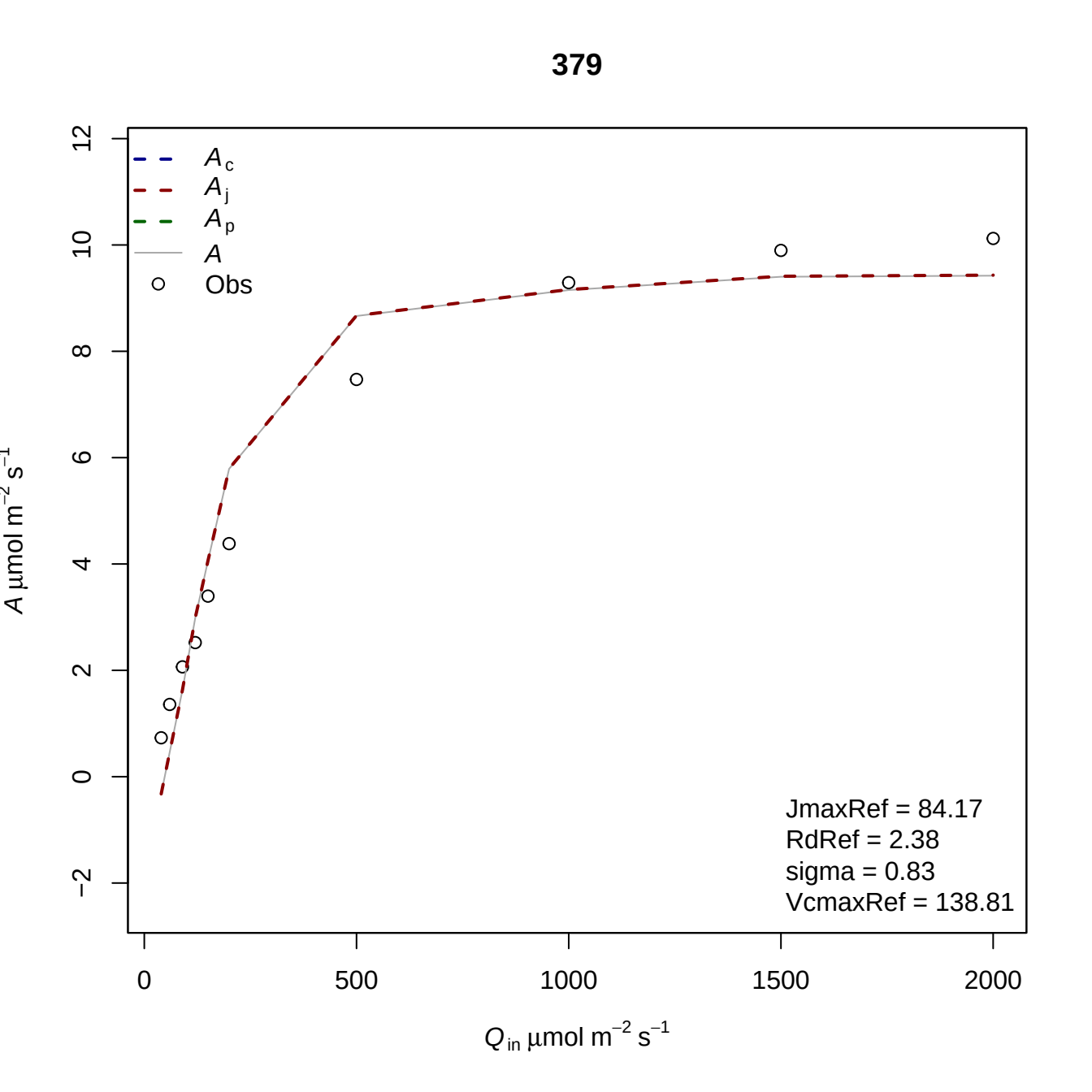

374
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 123.07
RdRef = 2.04
sigma = 0.48
VcmaxRef = 224.18

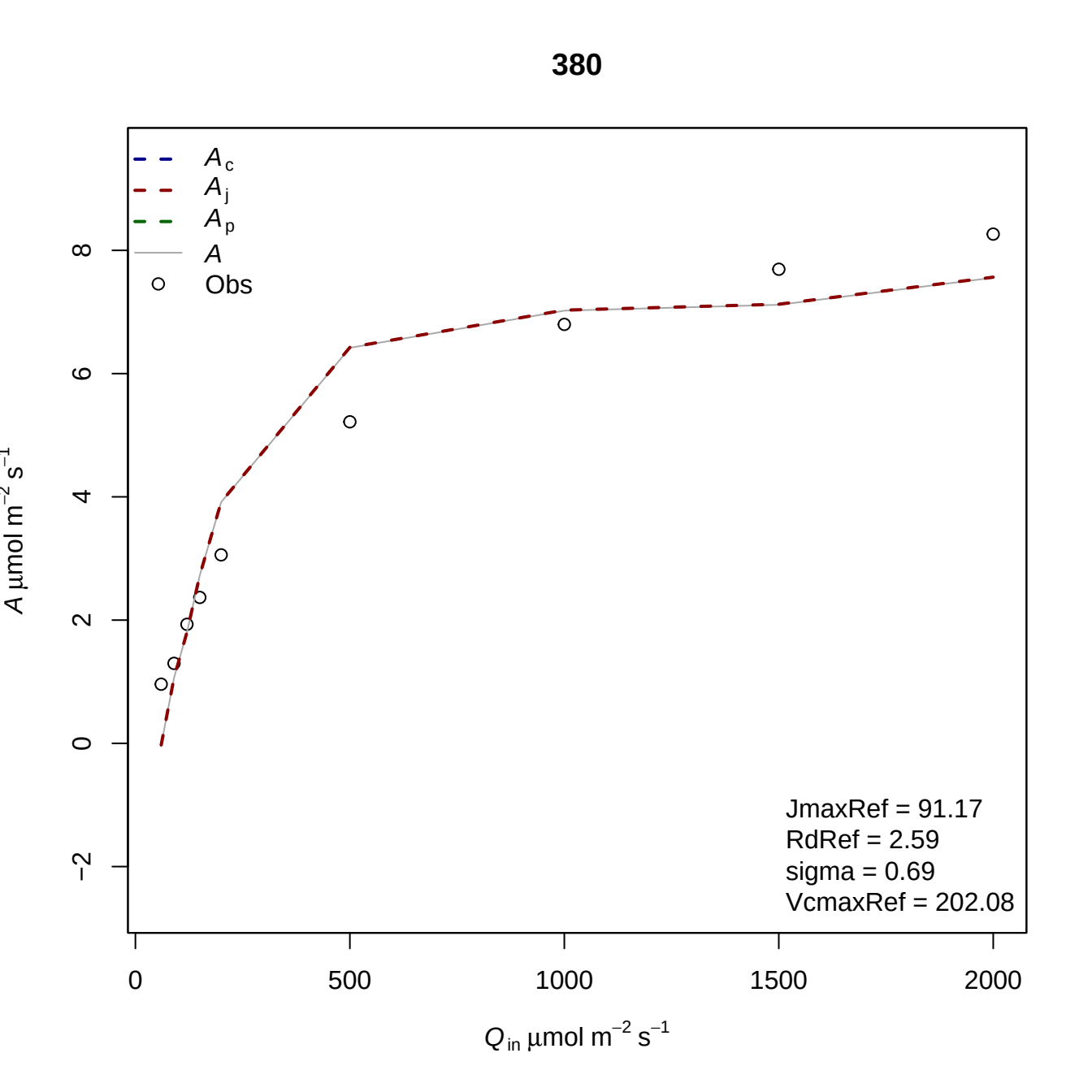

375
A_c
A_j
A_p
A
Obs
JmaxRef = 130.24
RdRef = 2.92
sigma = 0.73
VcmaxRef = 309.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

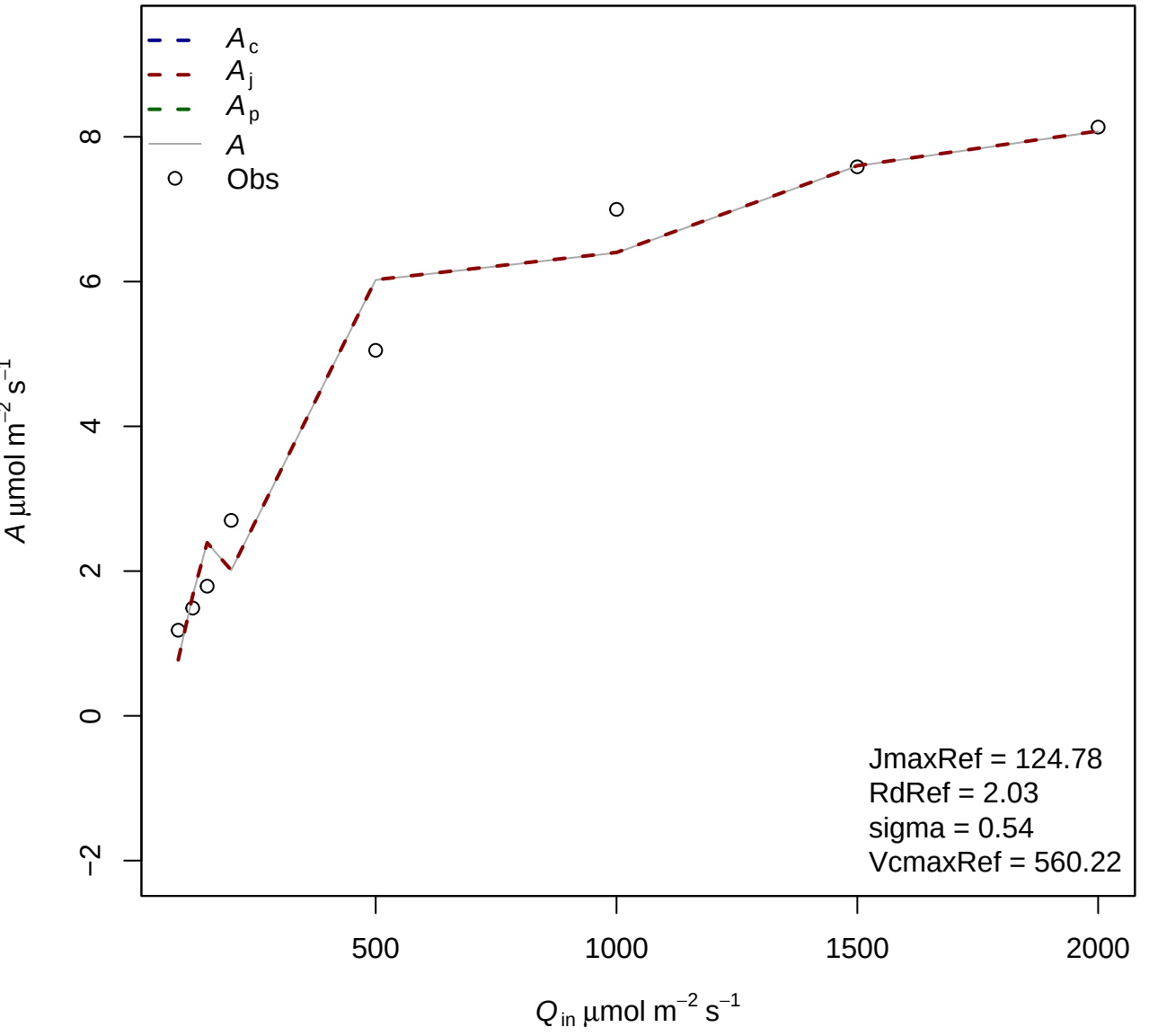

376
A_c
A_j
A_p
A
Obs
JmaxRef = 93.45
RdRef = 3.43
sigma = 0.97
VcmaxRef = 554.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

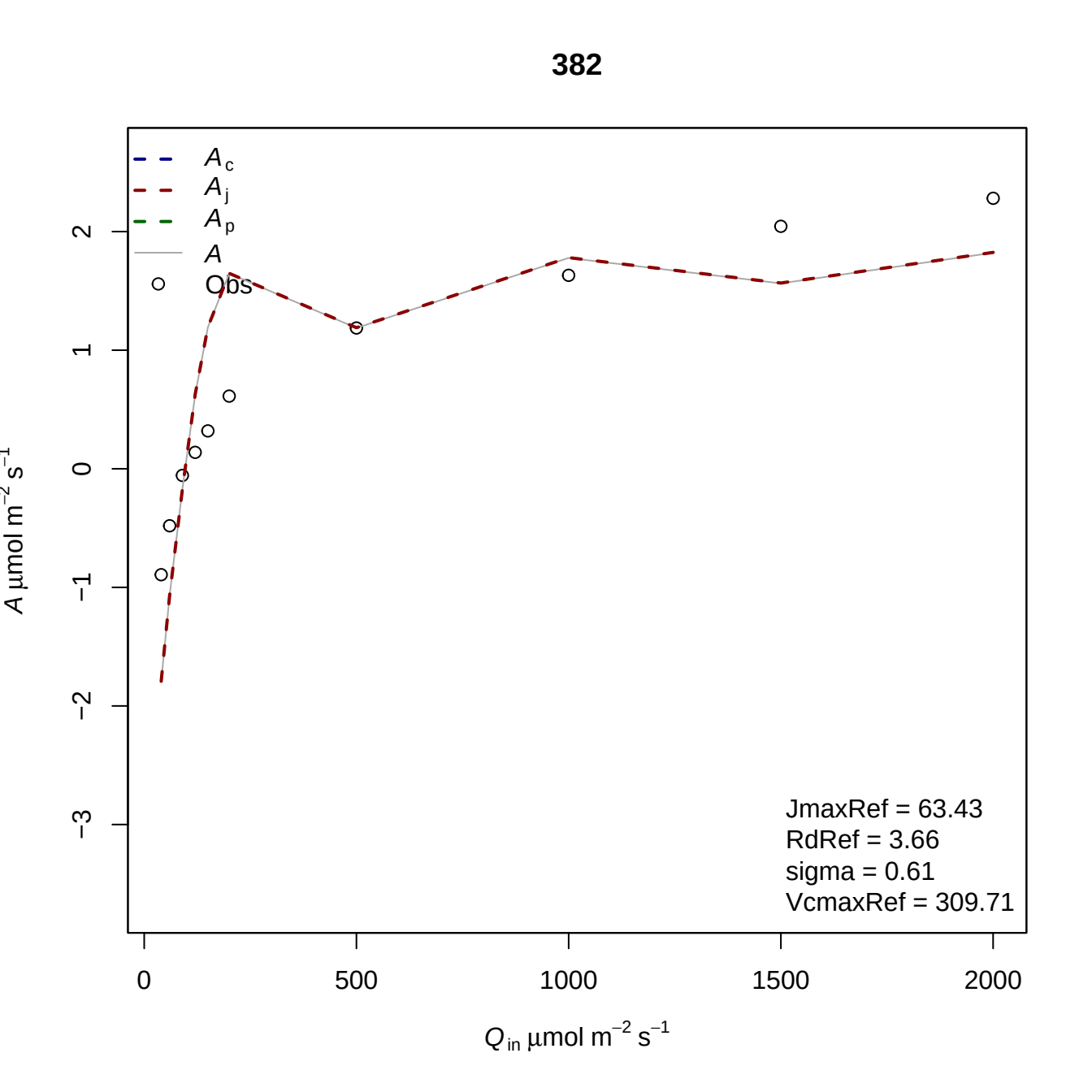

377
A_c
A_j
A_p
A
Obs
JmaxRef = 44.16
RdRef = 3.04
sigma = 0.89
VcmaxRef = 50.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

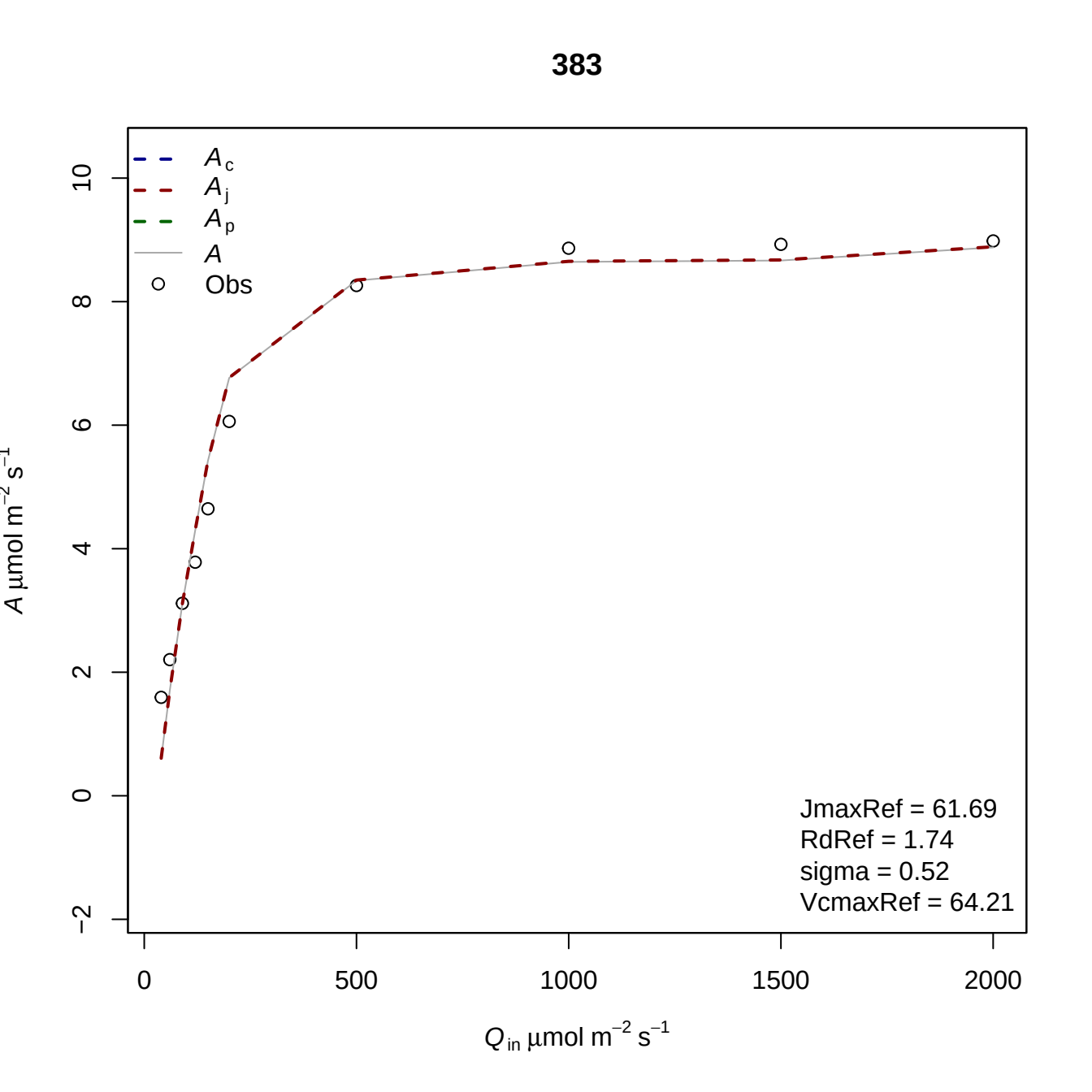

378
A_c
A_j
A_p
A
Obs
JmaxRef = 59.93
RdRef = 2.81
sigma = 1.07
VcmaxRef = 92.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

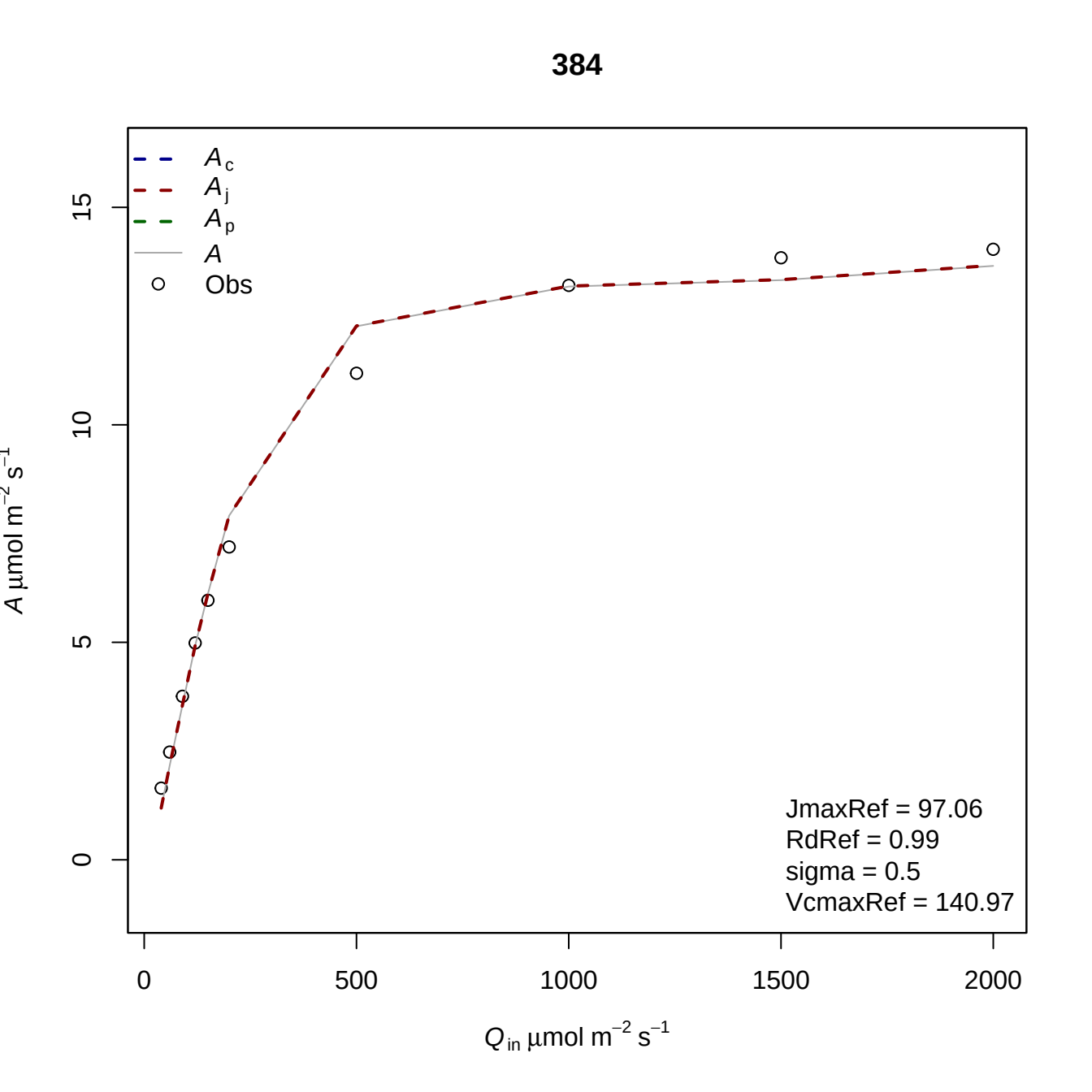

379
A_c
A_j
A_p
A
Obs
JmaxRef = 84.17
RdRef = 2.38
sigma = 0.83
VcmaxRef = 138.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

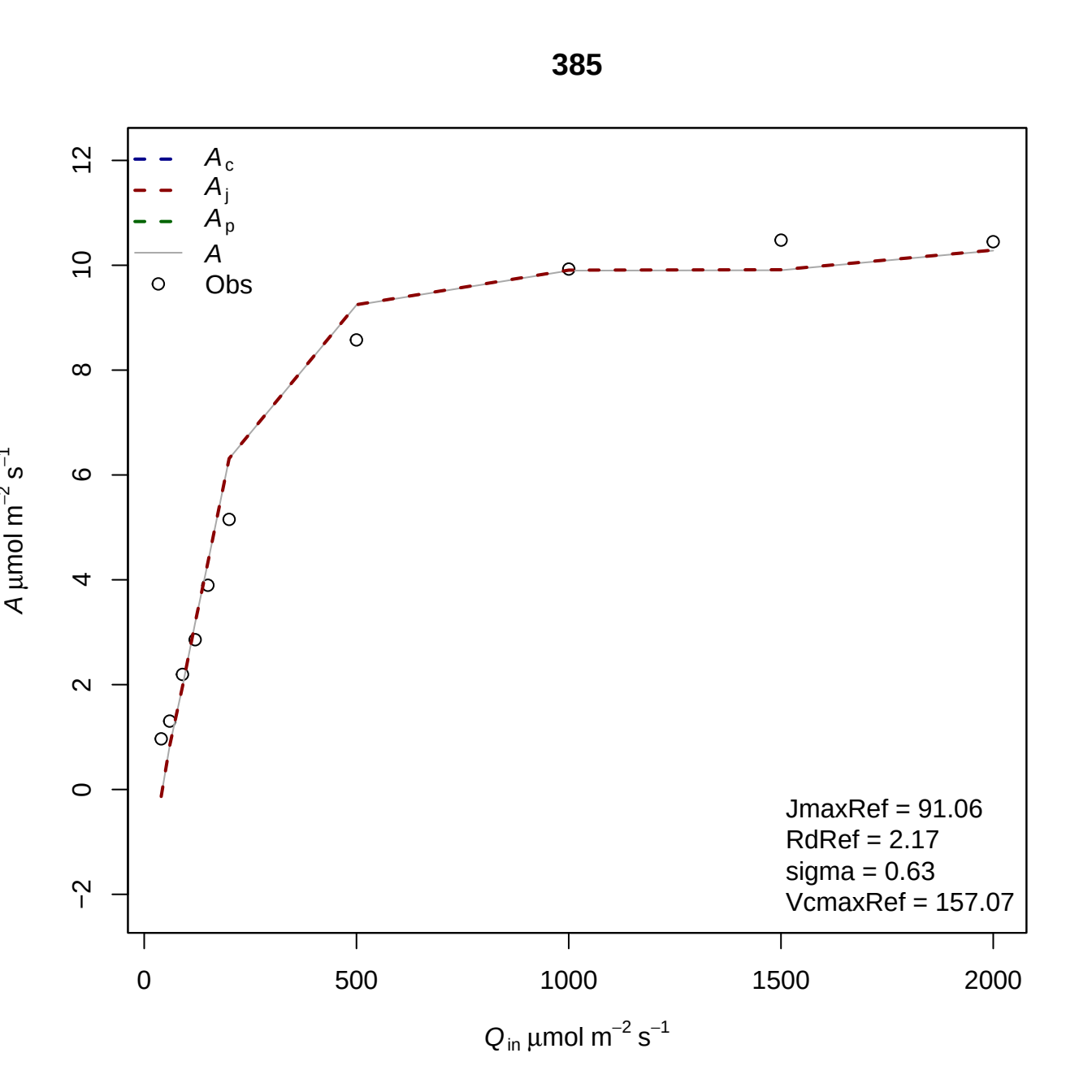

380
A_c
A_j
A_p
A
Obs
JmaxRef = 91.17
RdRef = 2.59
sigma = 0.69
VcmaxRef = 202.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

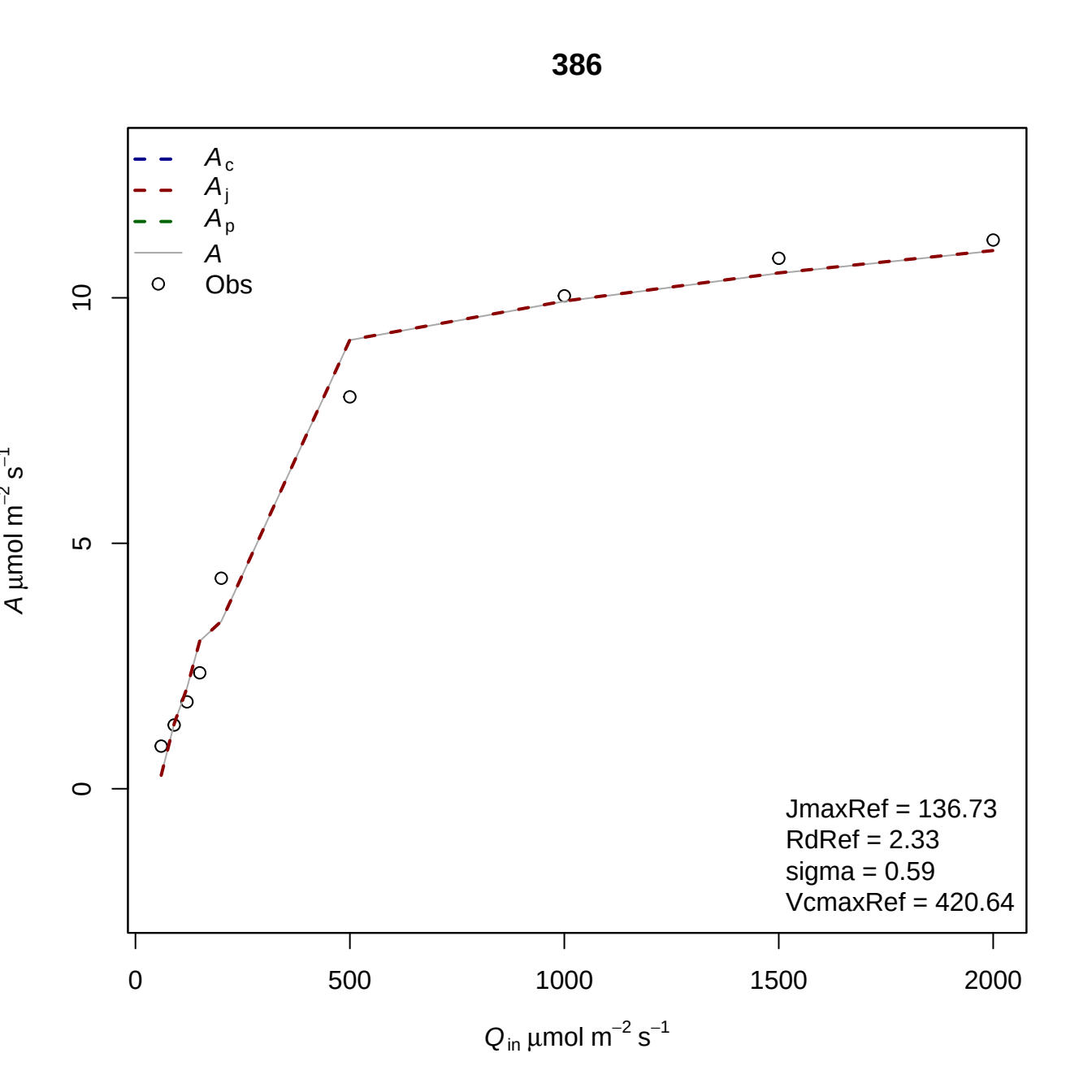

A_c
A_j
A_p
A
Obs
JmaxRef = 124.78
RdRef = 2.03
sigma = 0.54
VcmaxRef = 560.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
500
1000
1500
2000

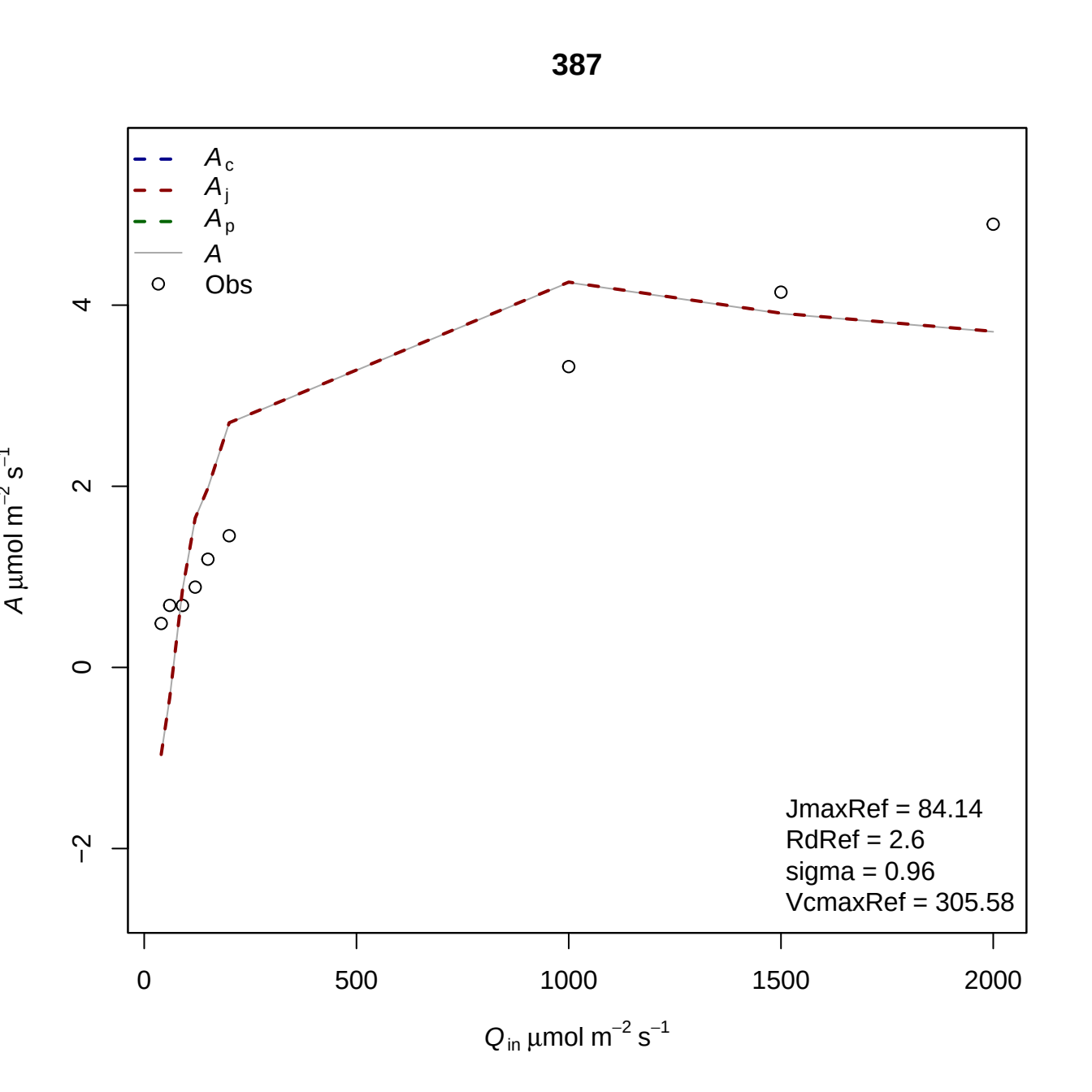

382
A_c
A_j
A_p
A
Obs
JmaxRef = 63.43
RdRef = 3.66
sigma = 0.61
VcmaxRef = 309.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
0
500
1000
1500
2000

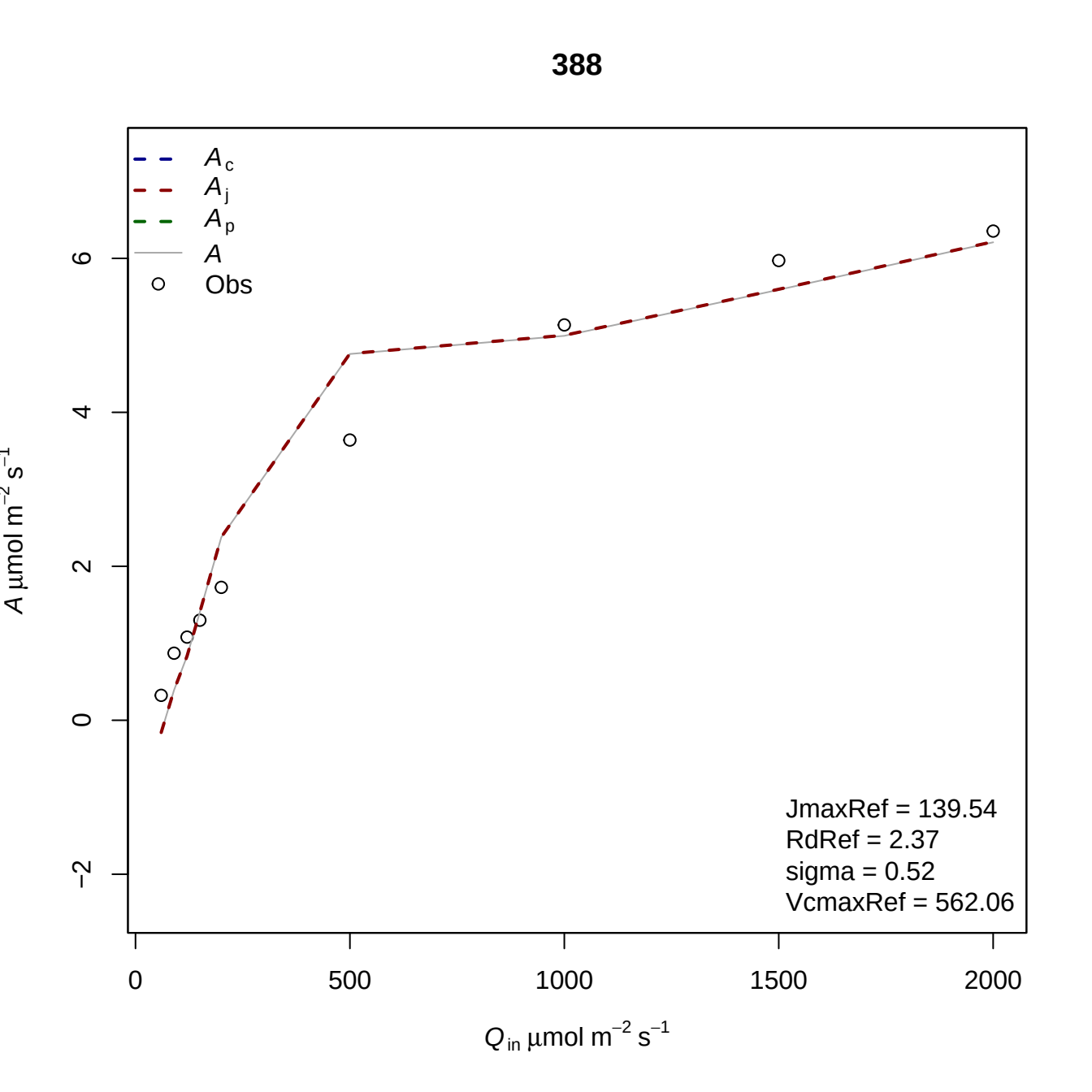

383
A_c
A_j
A_p
A
Obs
JmaxRef = 61.69
RdRef = 1.74
sigma = 0.52
VcmaxRef = 64.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

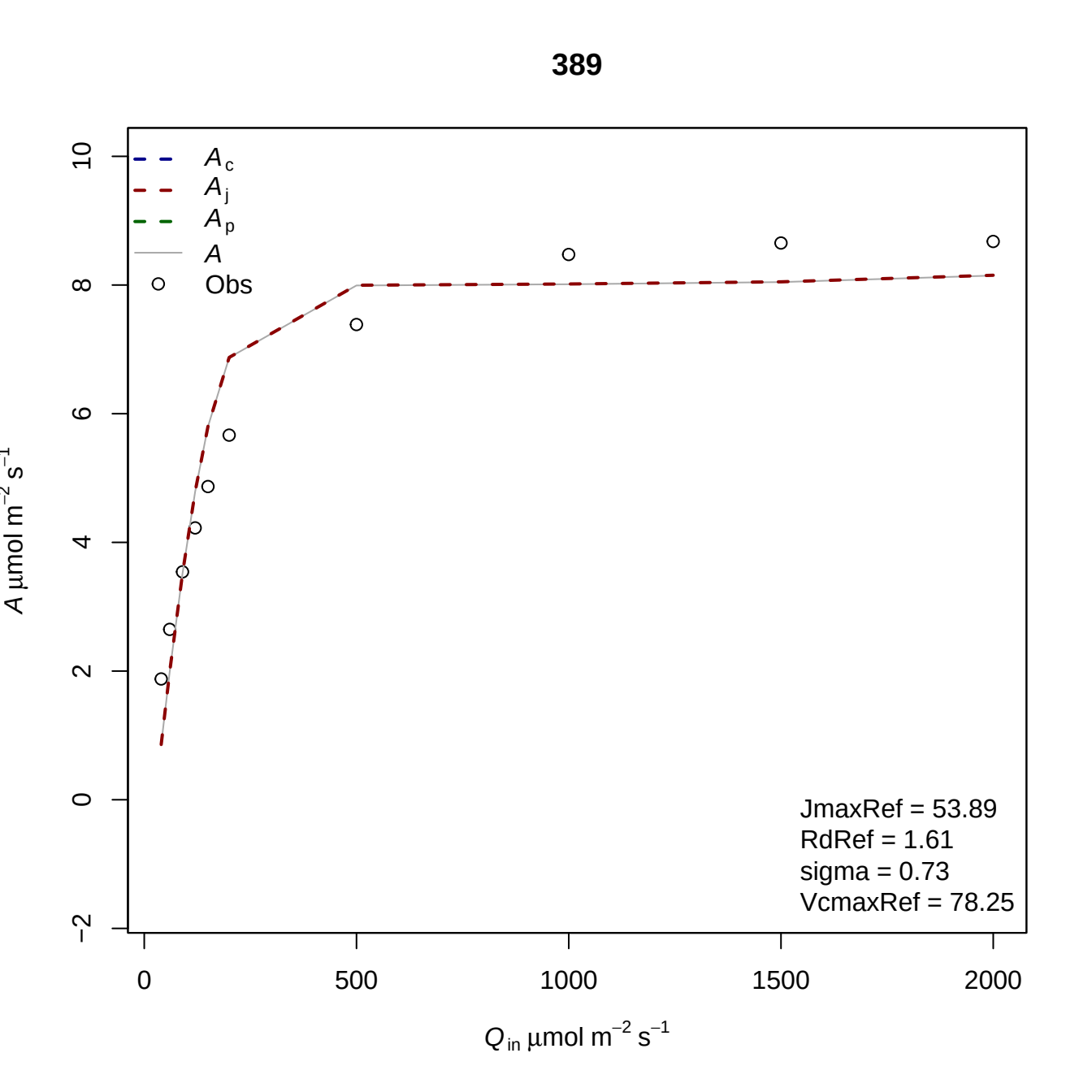

384
A_c
A_j
A_p
A
Obs
JmaxRef = 97.06
RdRef = 0.99
sigma = 0.5
VcmaxRef = 140.97
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

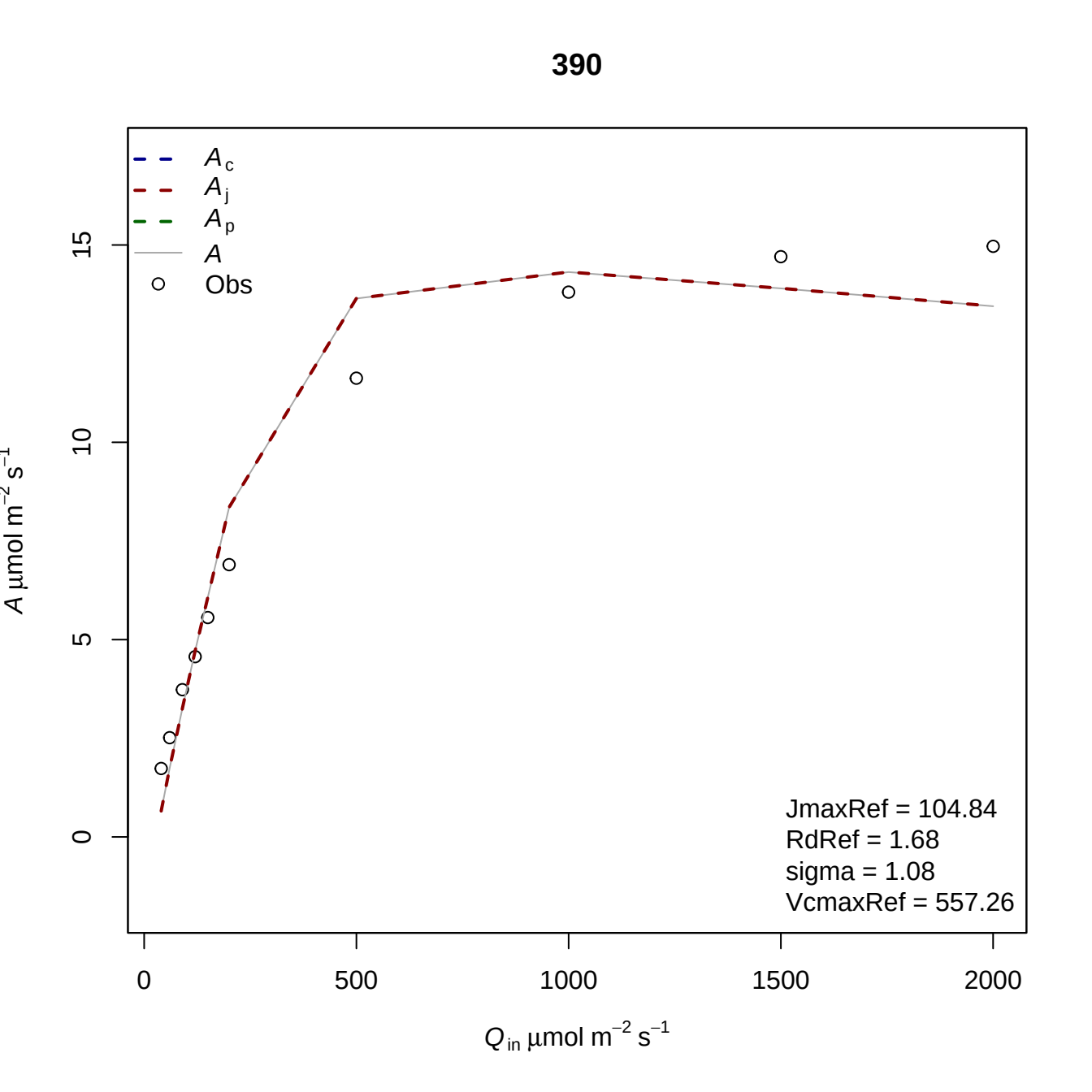

385
A_c
A_j
A_p
A
Obs
JmaxRef = 91.06
RdRef = 2.17
sigma = 0.63
VcmaxRef = 157.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

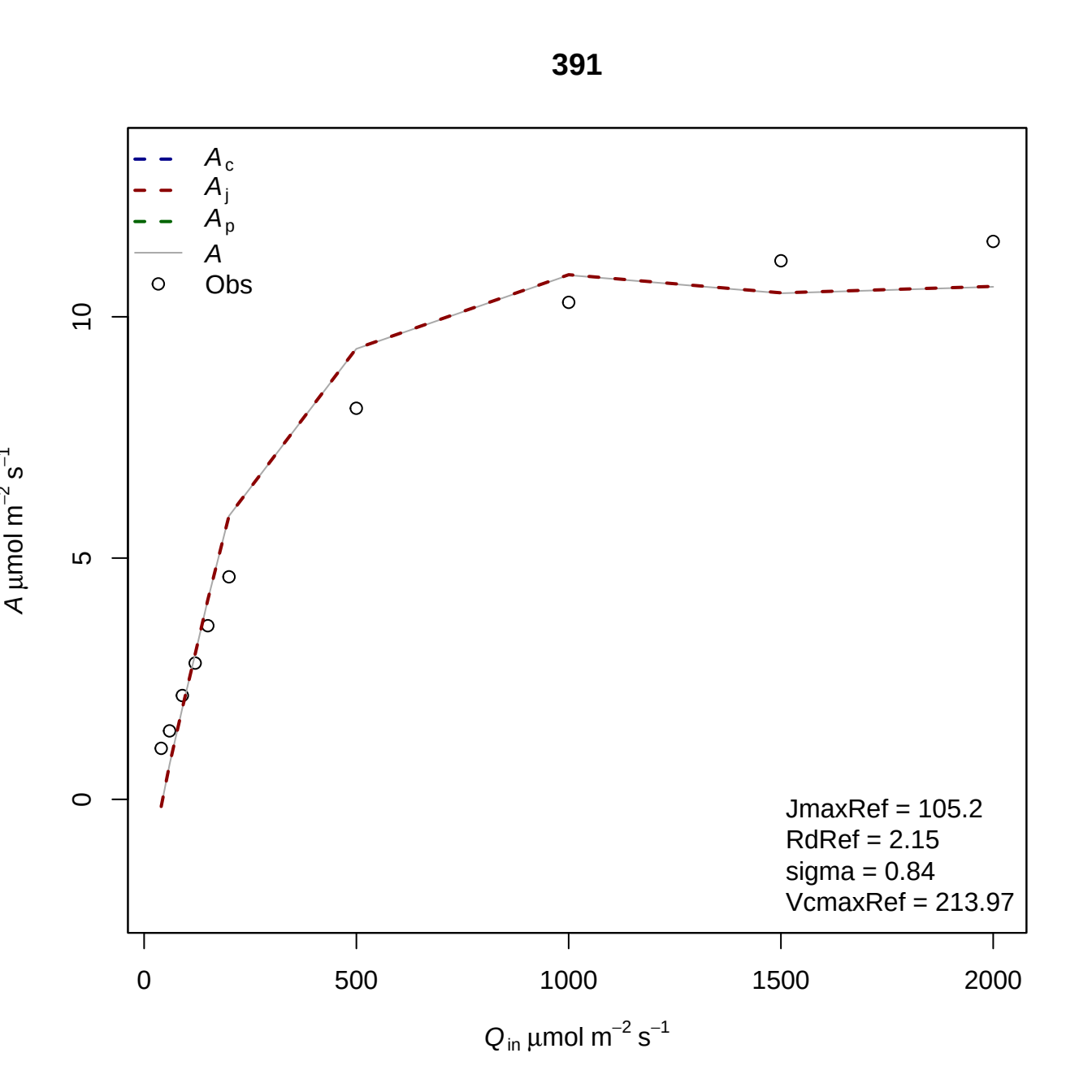

386
A_c
A_j
A_p
A
Obs
JmaxRef = 136.73
RdRef = 2.33
sigma = 0.59
VcmaxRef = 420.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

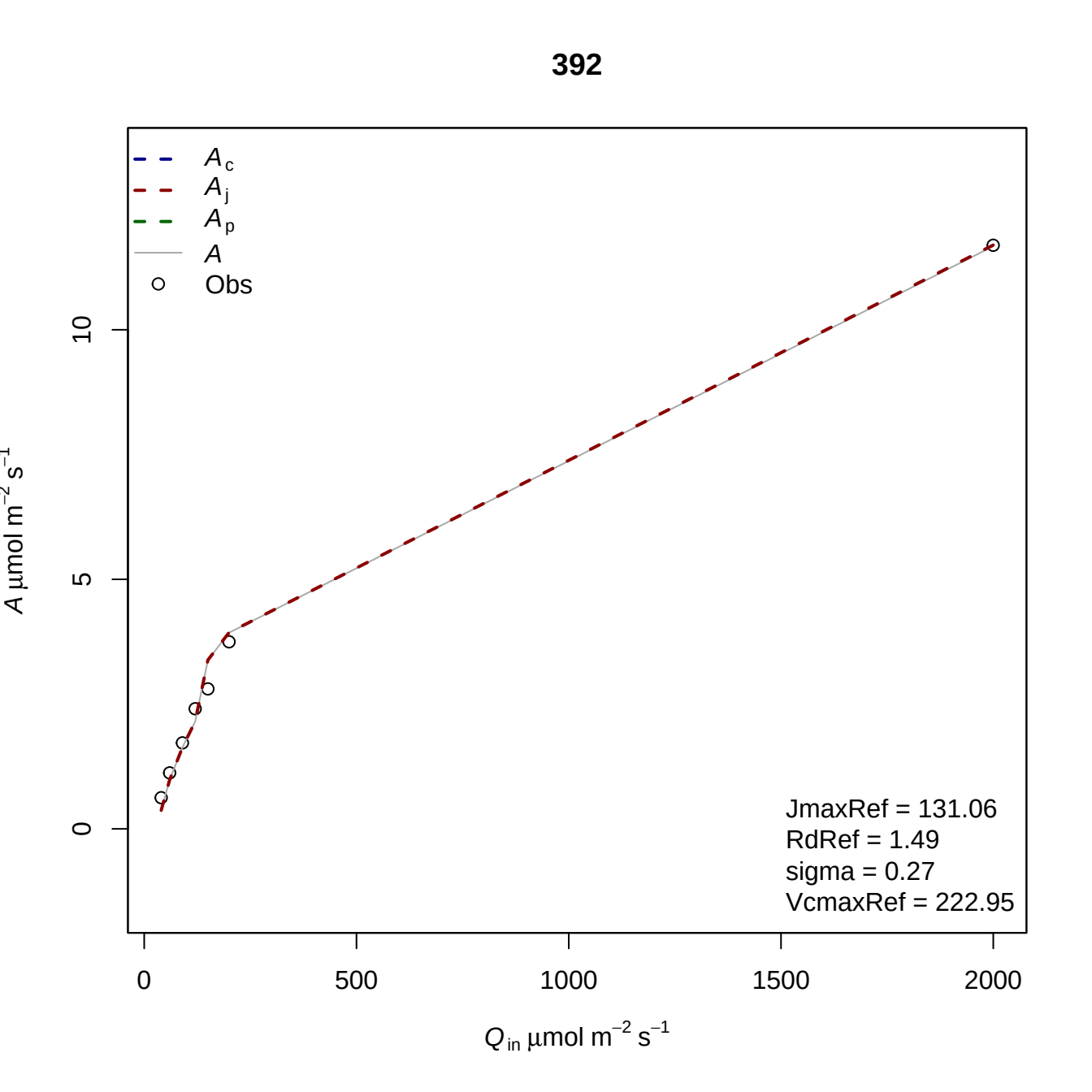

387
A_c
A_j
A_p
A
Obs
JmaxRef = 84.14
RdRef = 2.6
sigma = 0.96
VcmaxRef = 305.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

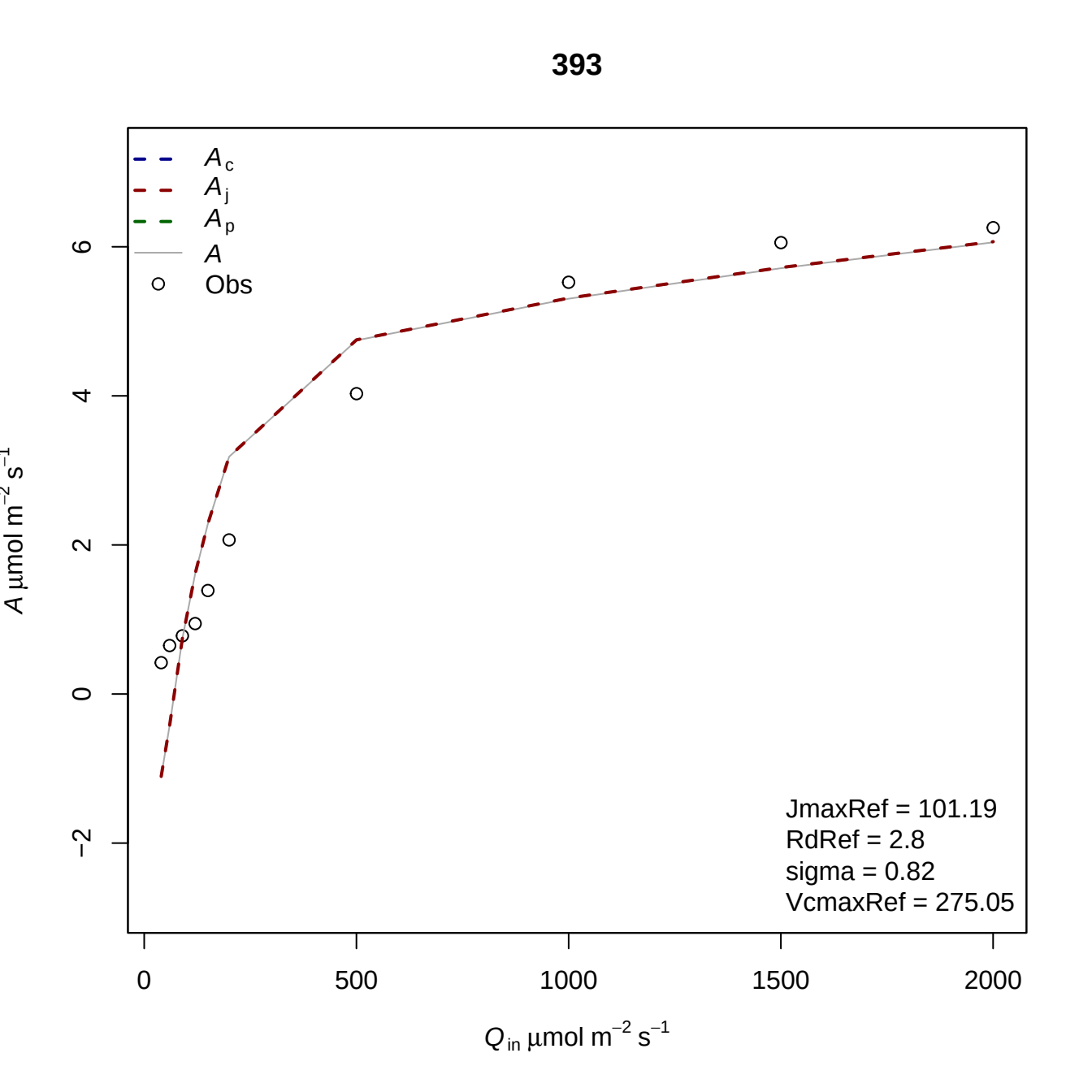

388
A_c
A_j
A_p
A
Obs
JmaxRef = 139.54
RdRef = 2.37
sigma = 0.52
VcmaxRef = 562.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

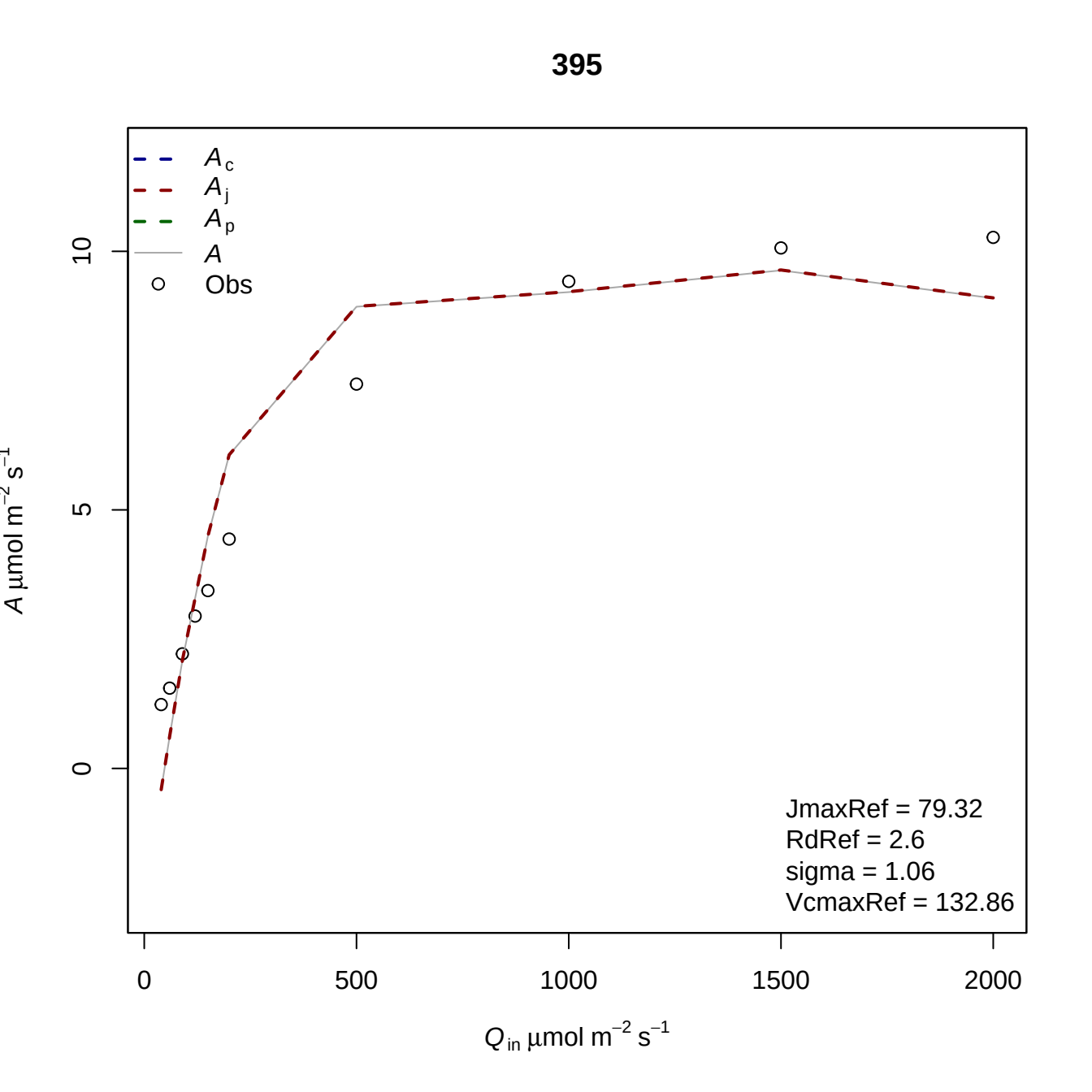

389
A_c
A_j
A_p
A
Obs
JmaxRef = 53.89
RdRef = 1.61
sigma = 0.73
VcmaxRef = 78.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

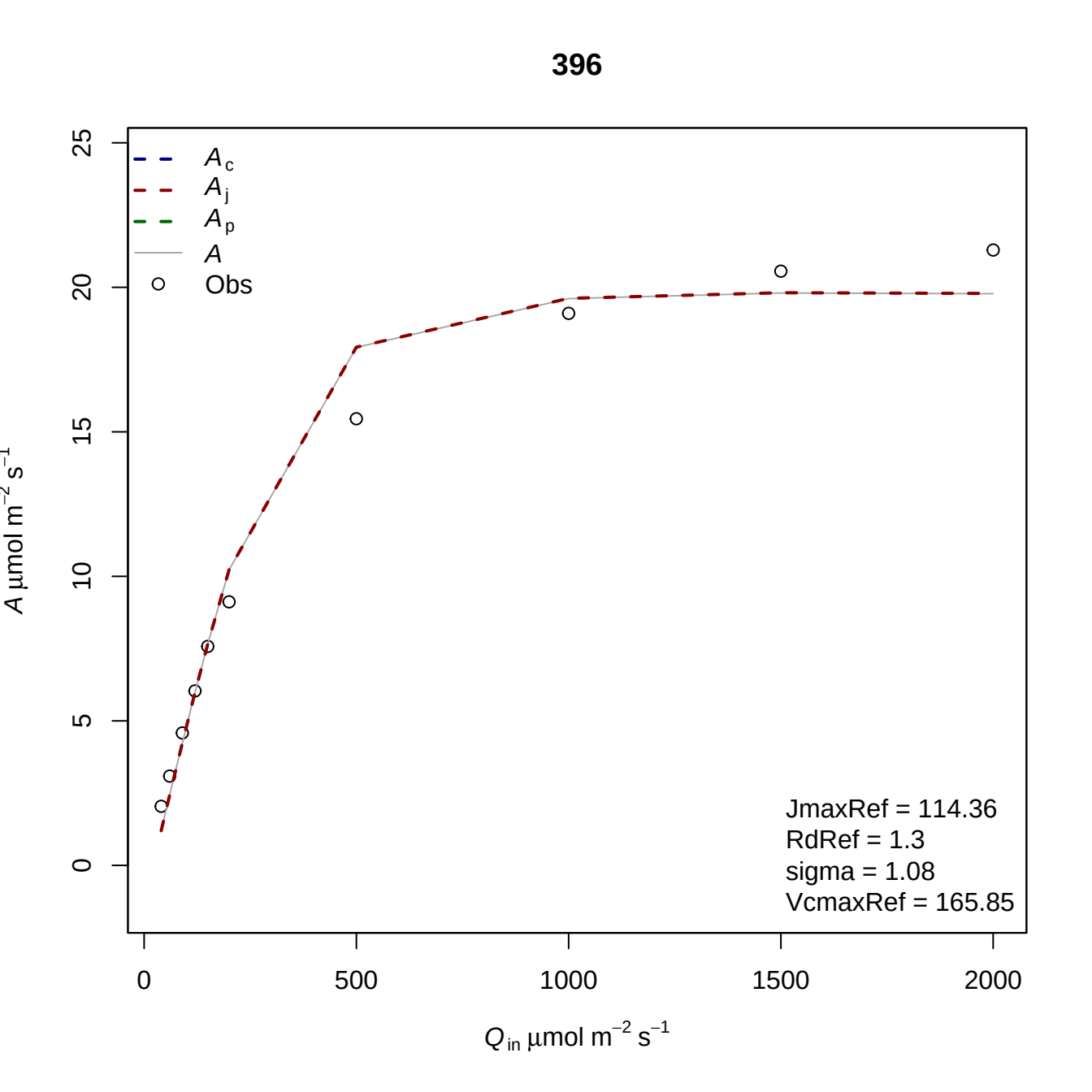

390
A_c
A_j
A_p
A
Obs
JmaxRef = 104.84
RdRef = 1.68
sigma = 1.08
VcmaxRef = 557.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

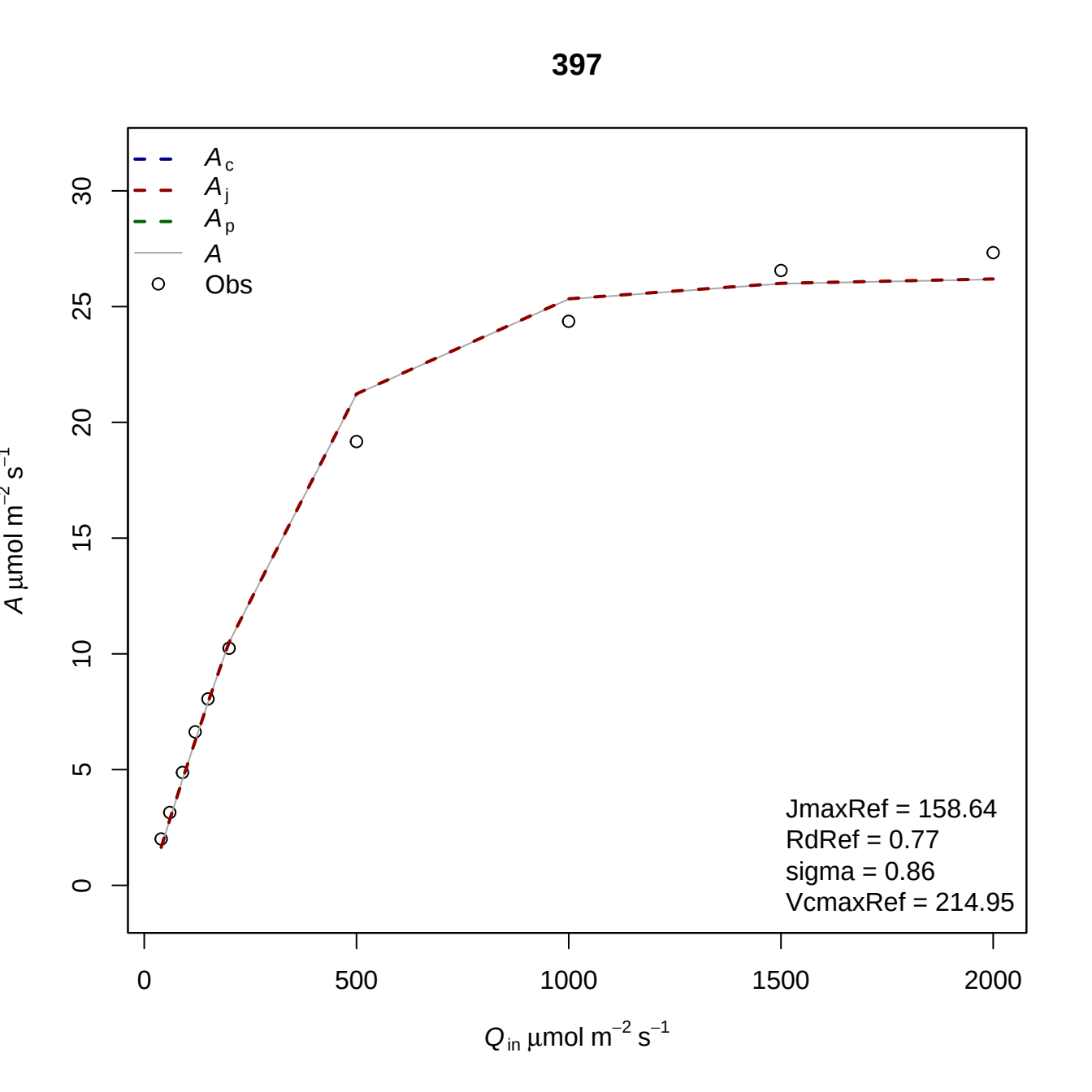

391
A_c
A_j
A_p
A
Obs
JmaxRef = 105.2
RdRef = 2.15
sigma = 0.84
VcmaxRef = 213.97
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

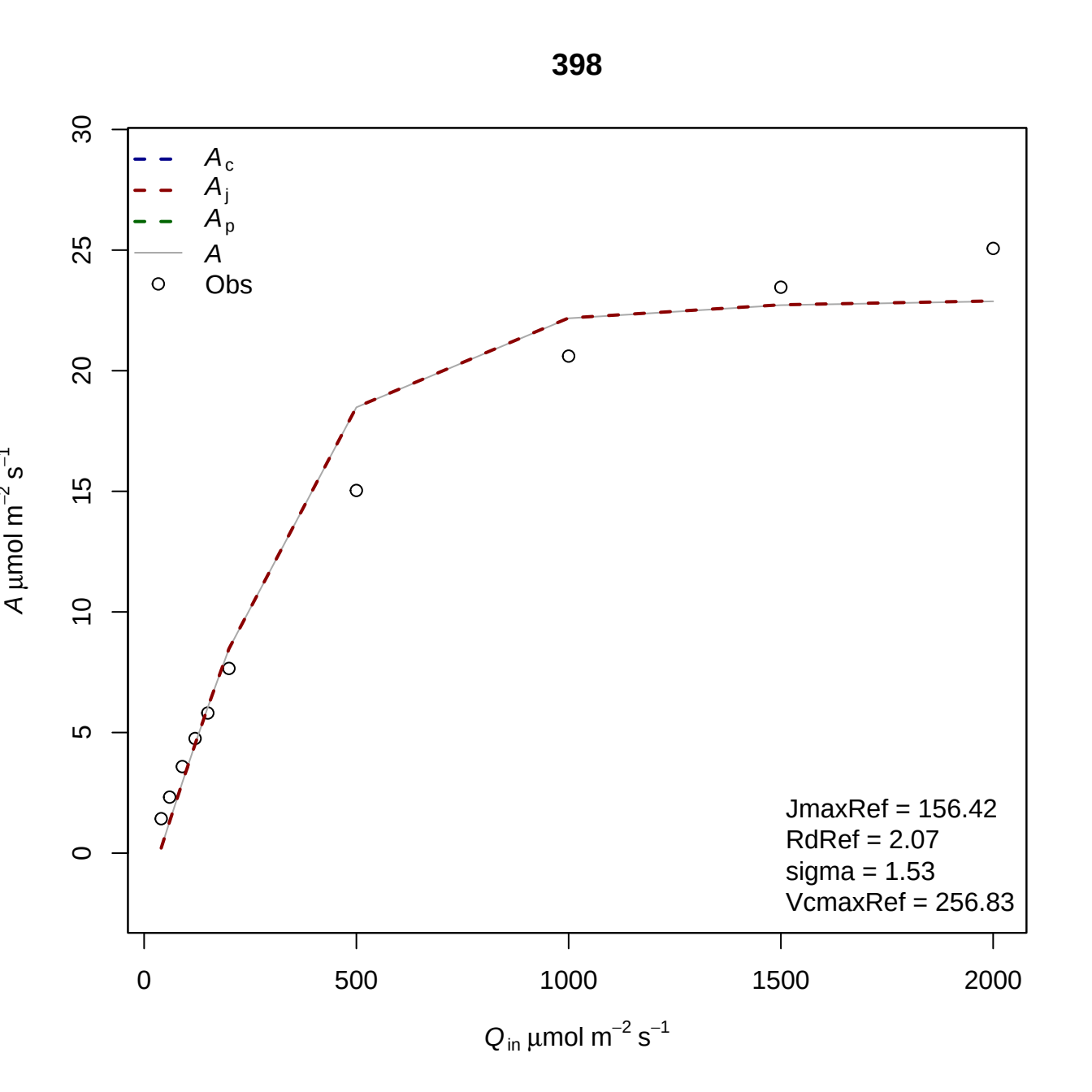

392
A_c
A_j
A_p
A
Obs
JmaxRef = 131.06
RdRef = 1.49
sigma = 0.27
VcmaxRef = 222.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

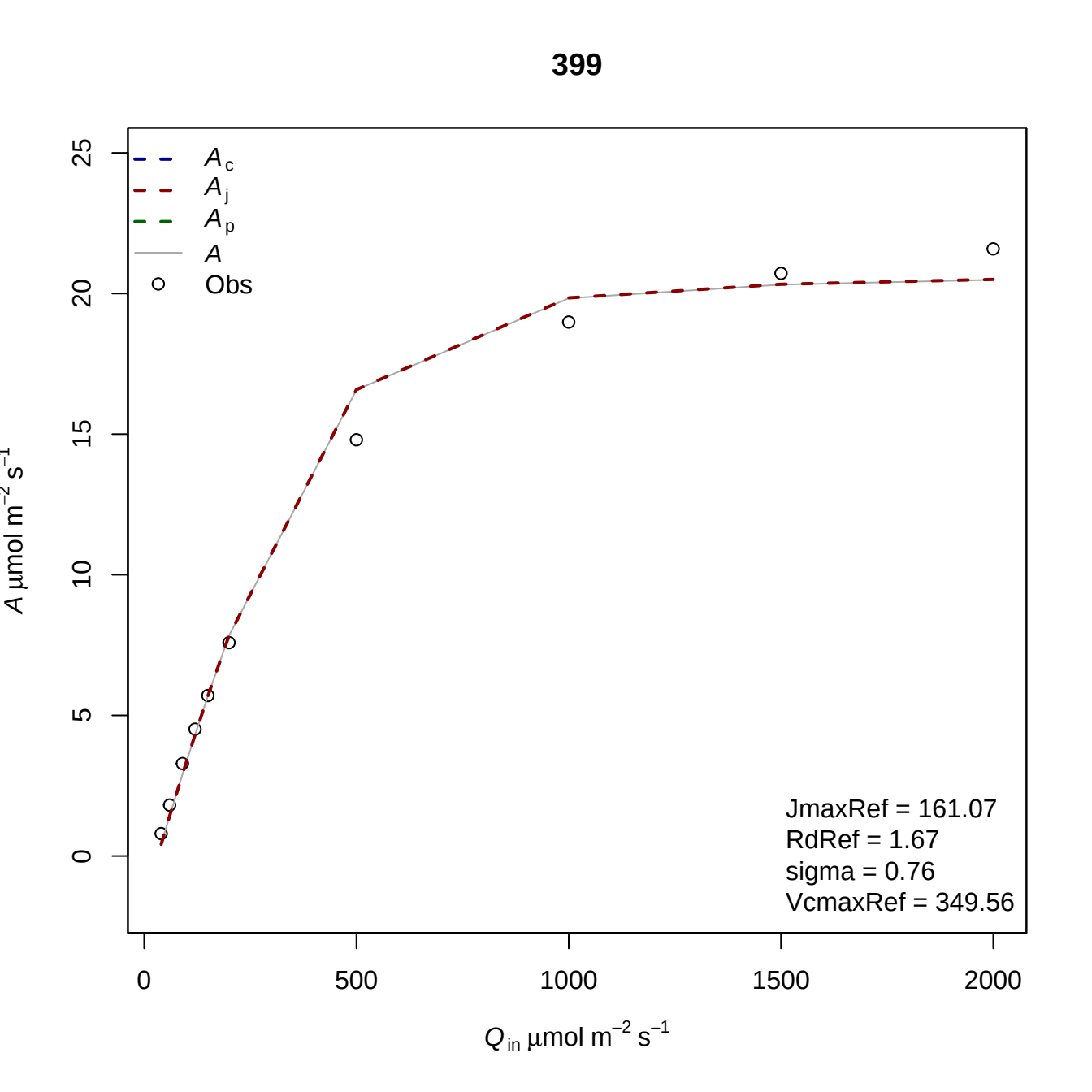

393
A_c
A_j
A_p
A
Obs
JmaxRef = 101.19
RdRef = 2.8
sigma = 0.82
VcmaxRef = 275.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

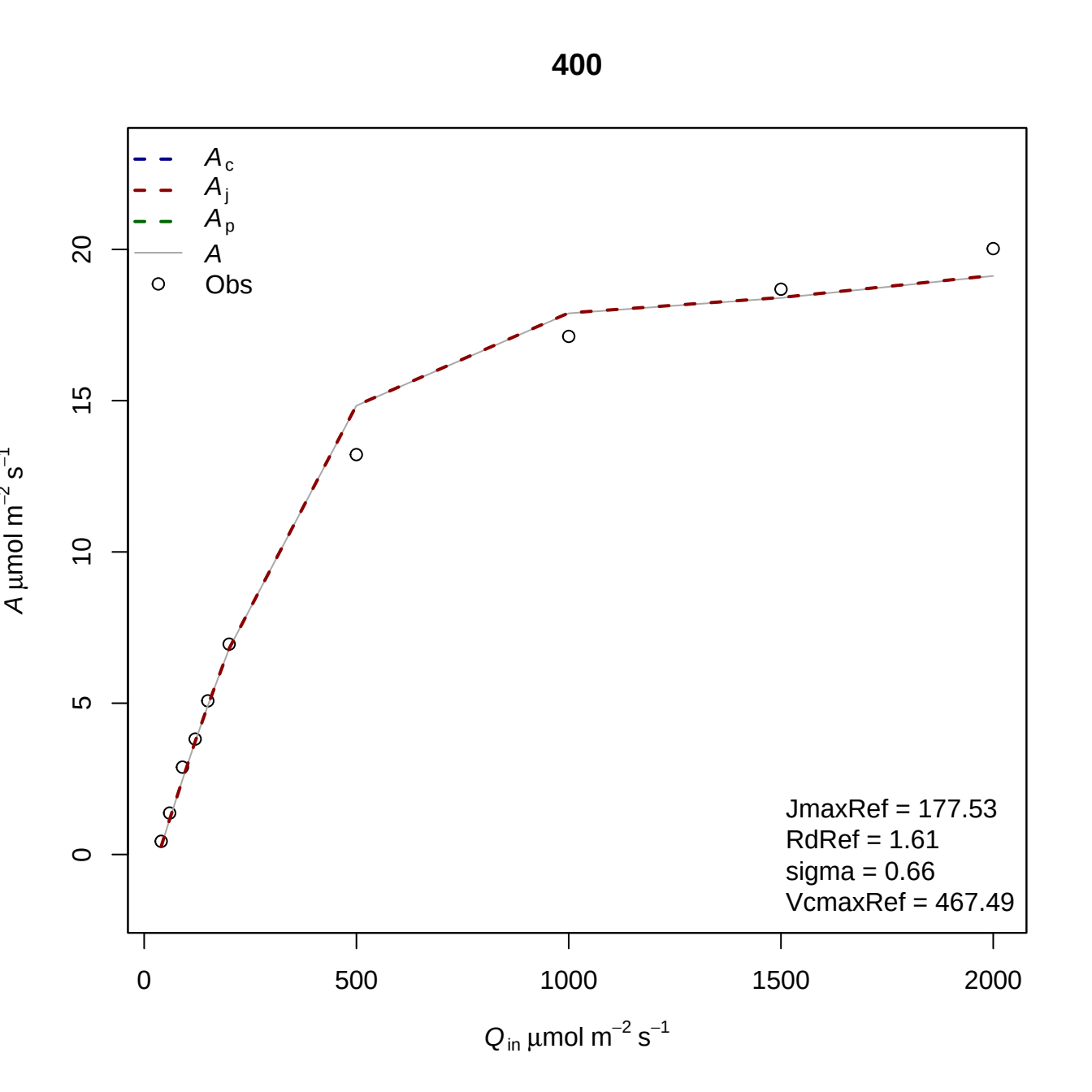

395
A_c
A_j
A_p
A
Obs
JmaxRef = 79.32
RdRef = 2.6
sigma = 1.06
VcmaxRef = 132.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

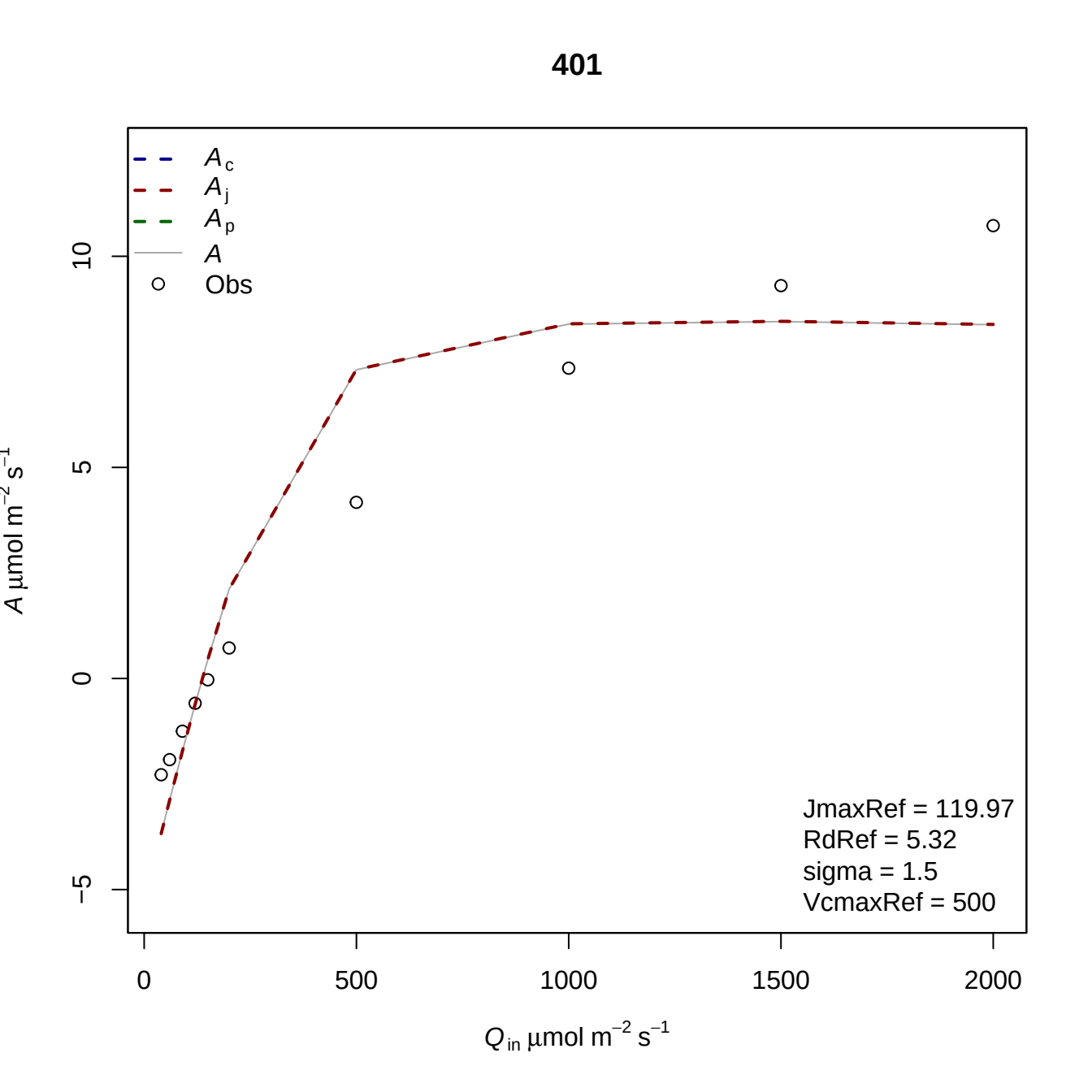

396
A_c
A_j
A_p
A
Obs
JmaxRef = 114.36
RdRef = 1.3
sigma = 1.08
VcmaxRef = 165.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

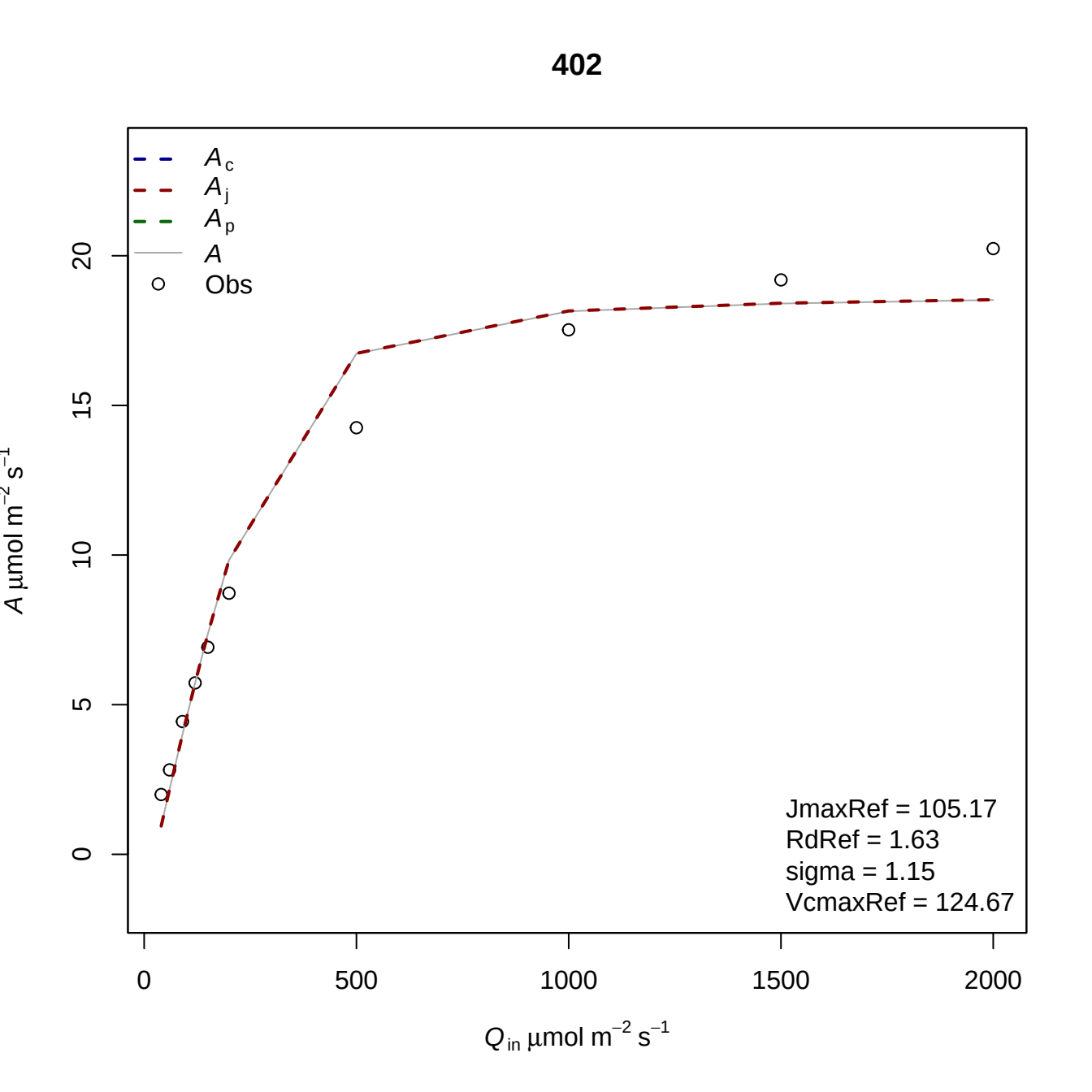

397
A_c
A_j
A_p
A
Obs
JmaxRef = 158.64
RdRef = 0.77
sigma = 0.86
VcmaxRef = 214.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

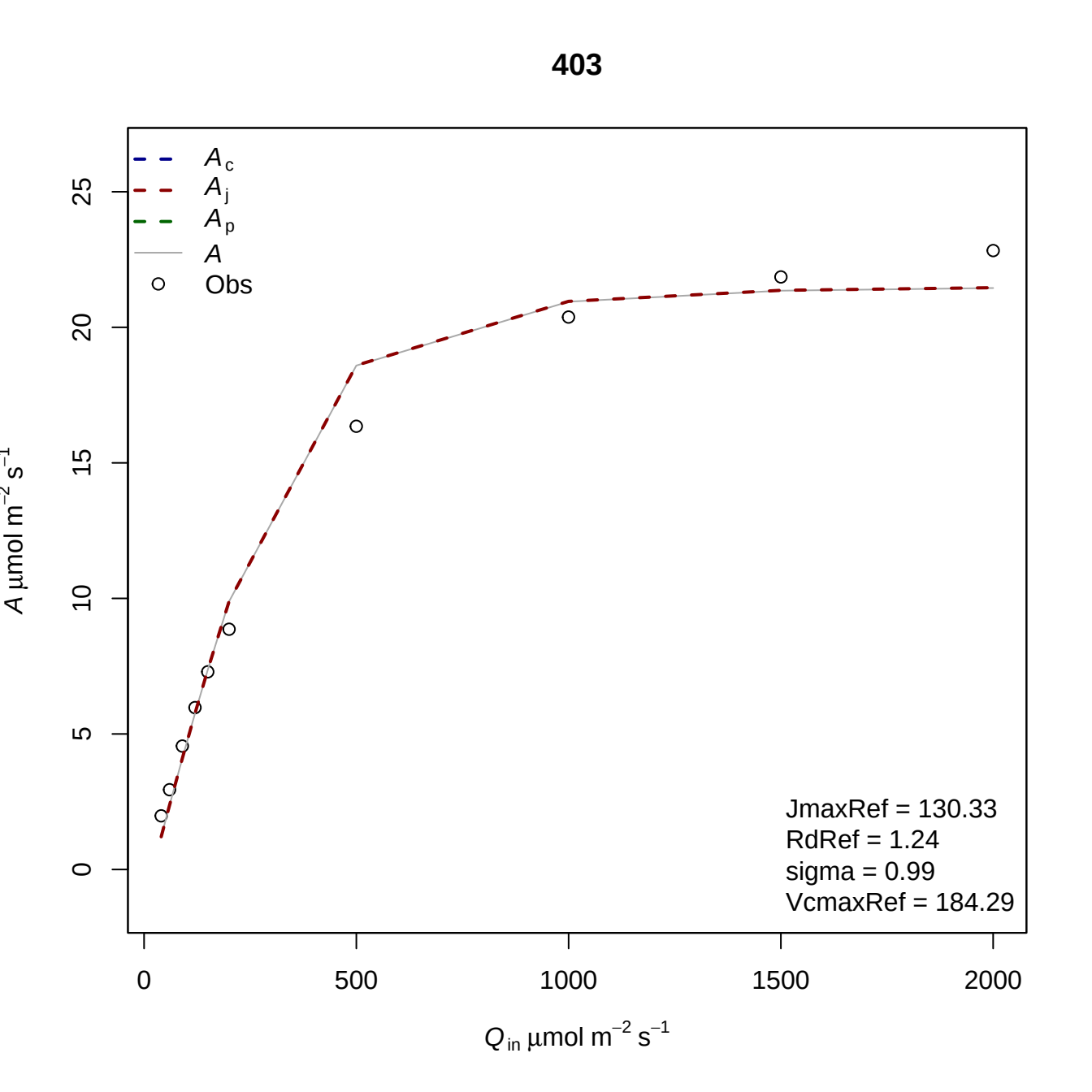

398
A_c
A_j
A_p
A
Obs
JmaxRef = 156.42
RdRef = 2.07
sigma = 1.53
VcmaxRef = 256.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

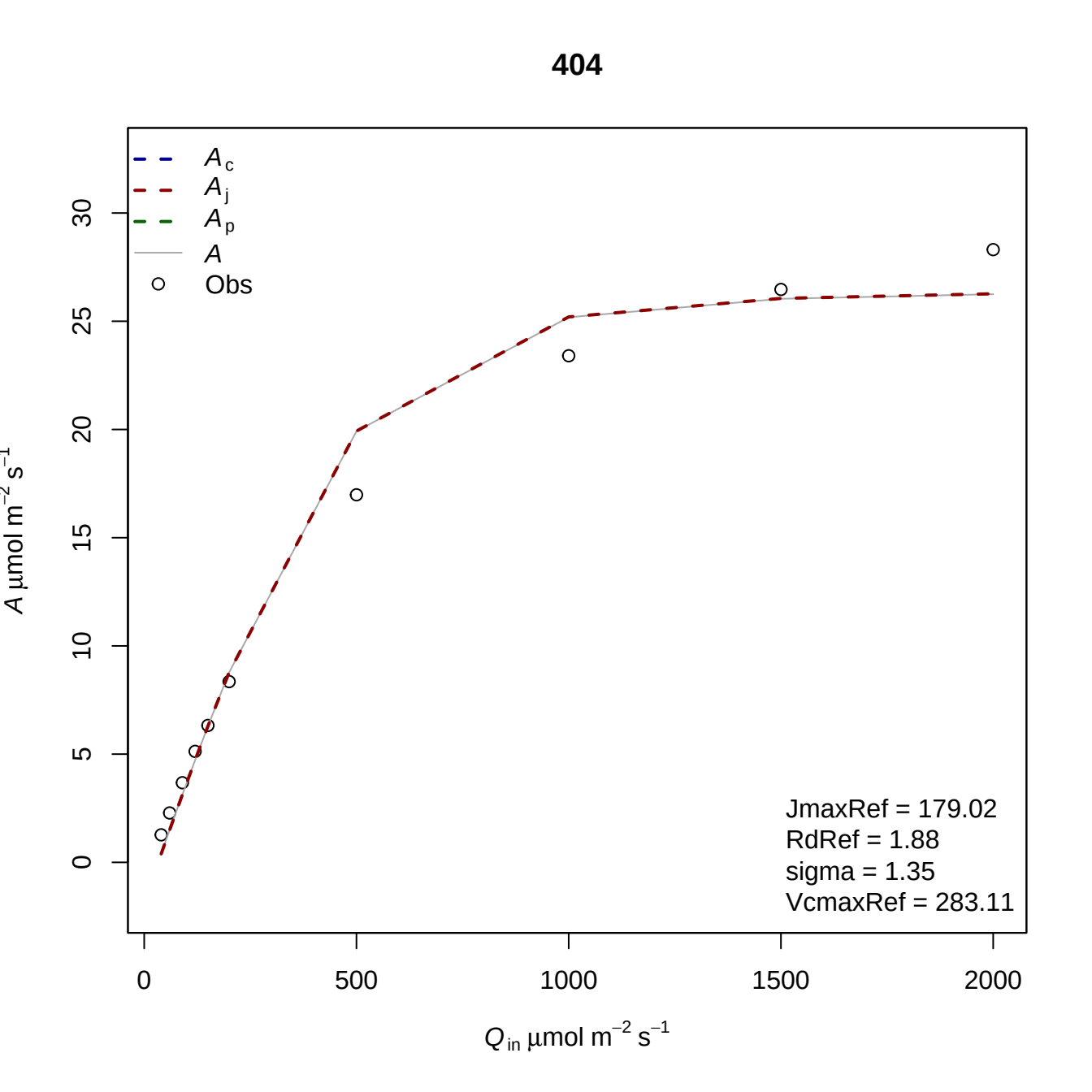

399
A_c
A_j
A_p
A
Obs
JmaxRef = 161.07
RdRef = 1.67
sigma = 0.76
VcmaxRef = 349.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

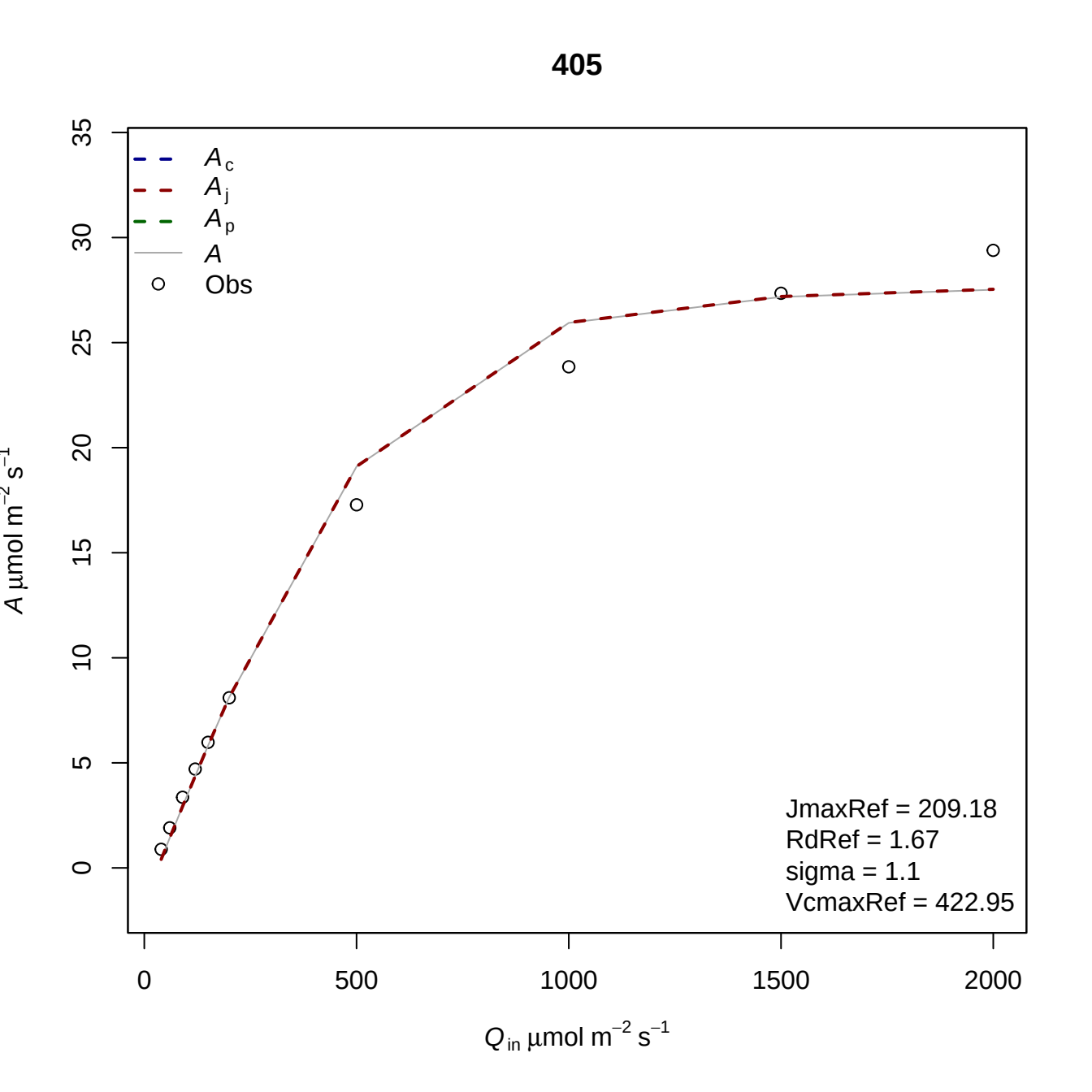

400
A_c
A_j
A_p
A
Obs
JmaxRef = 177.53
RdRef = 1.61
sigma = 0.66
VcmaxRef = 467.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

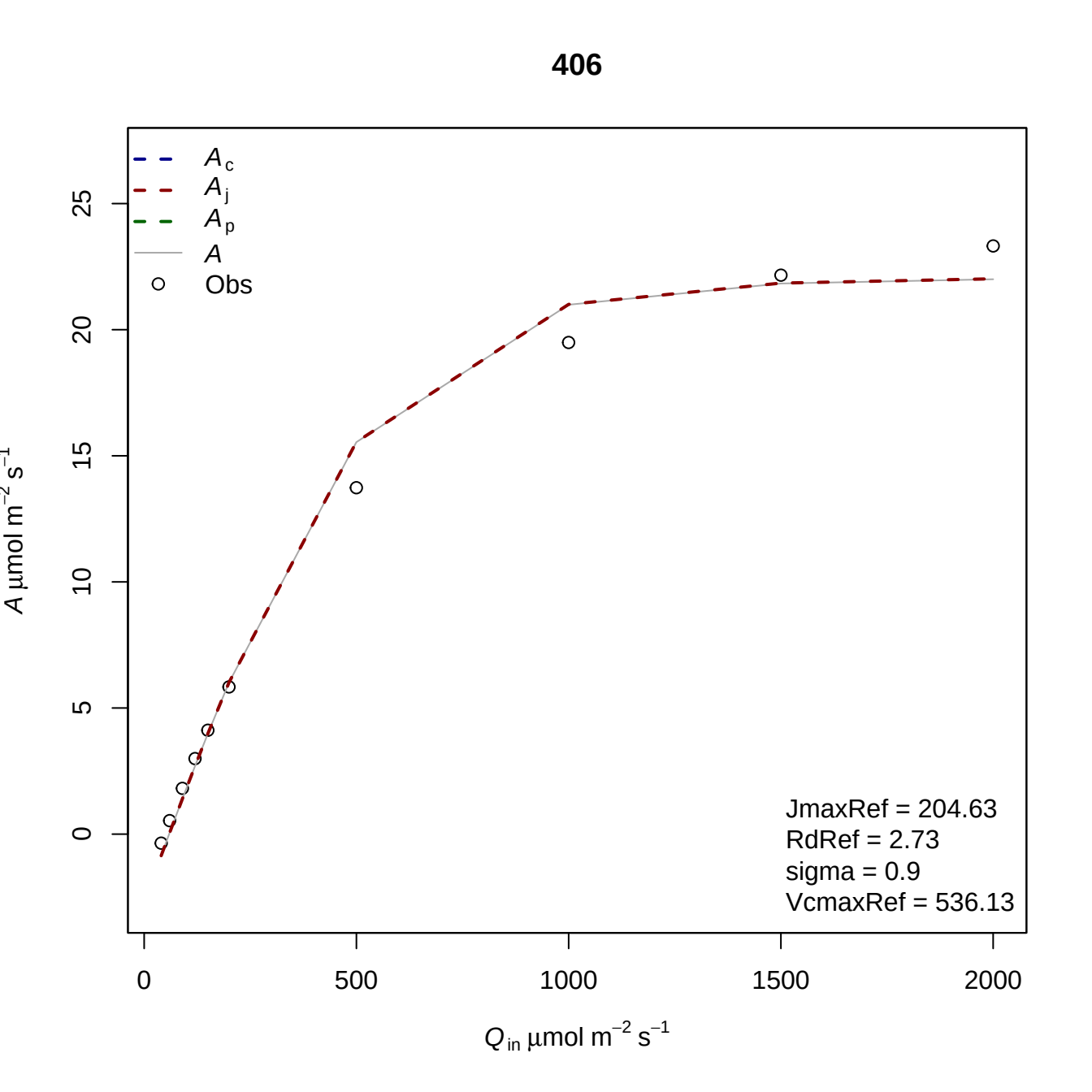

401
A_c
A_j
A_p
A
Obs
JmaxRef = 119.97
RdRef = 5.32
sigma = 1.5
VcmaxRef = 500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

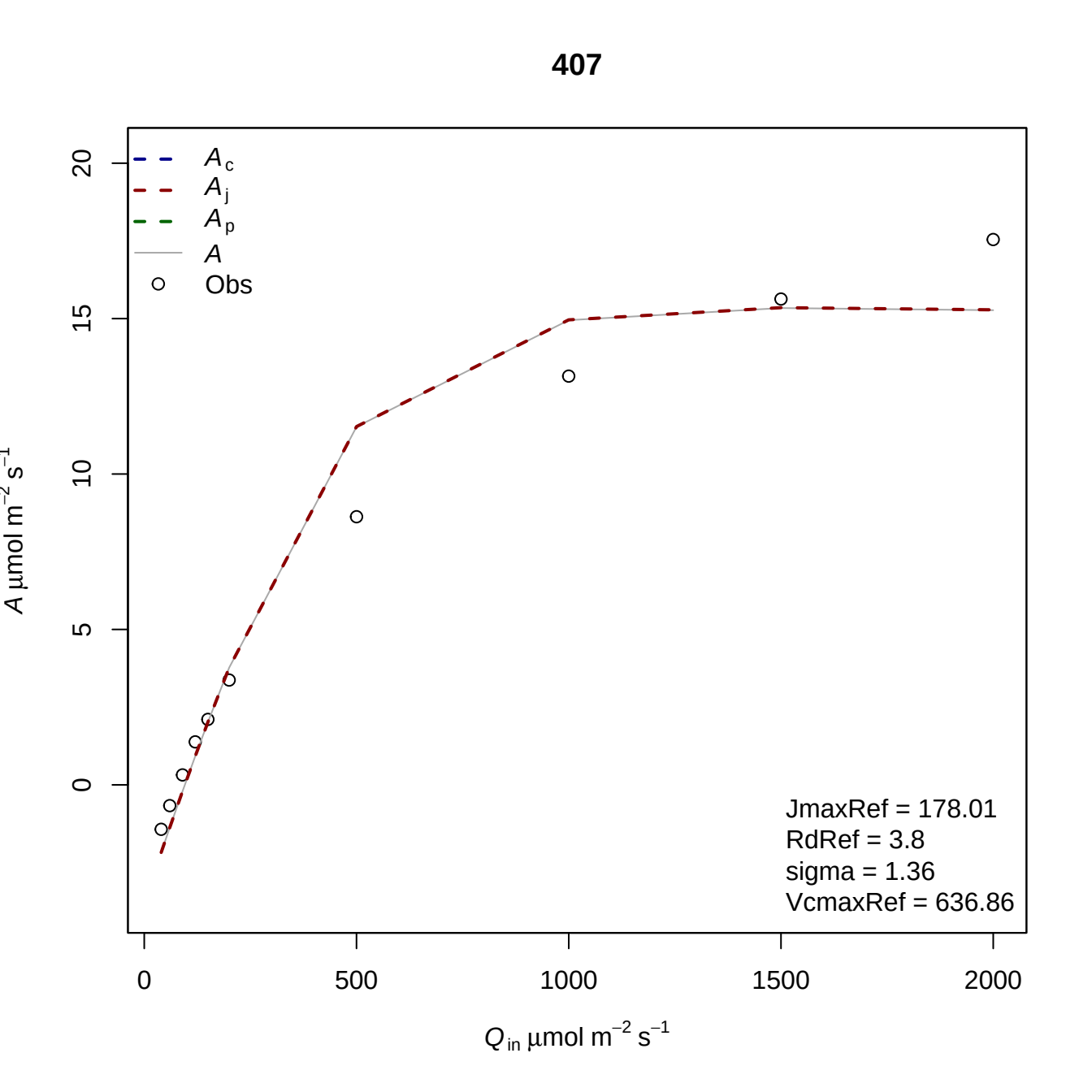

402
A_c
A_j
A_p
A
Obs
JmaxRef = 105.17
RdRef = 1.63
sigma = 1.15
VcmaxRef = 124.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

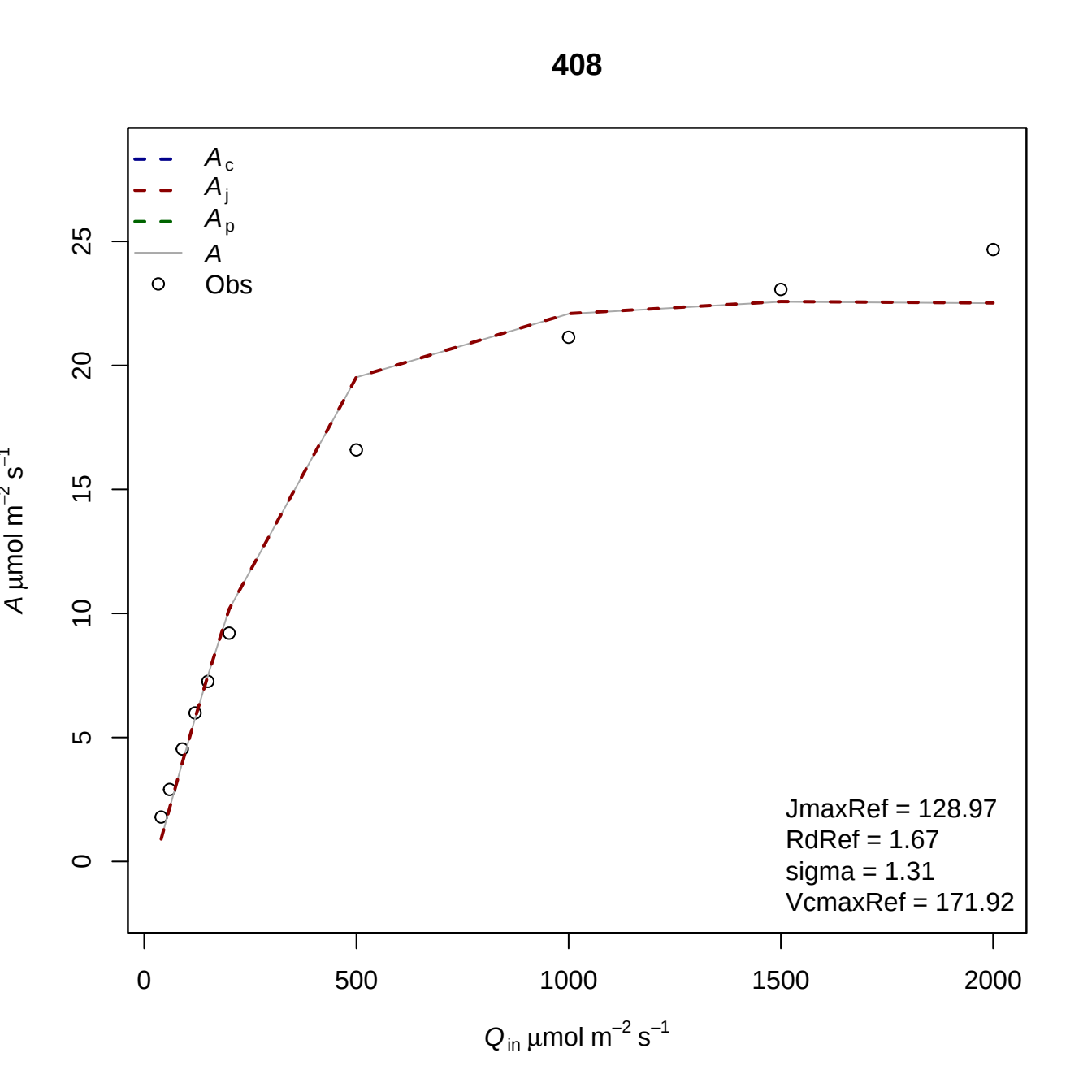

403
A_c
A_j
A_p
A
Obs
JmaxRef = 130.33
RdRef = 1.24
sigma = 0.99
VcmaxRef = 184.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

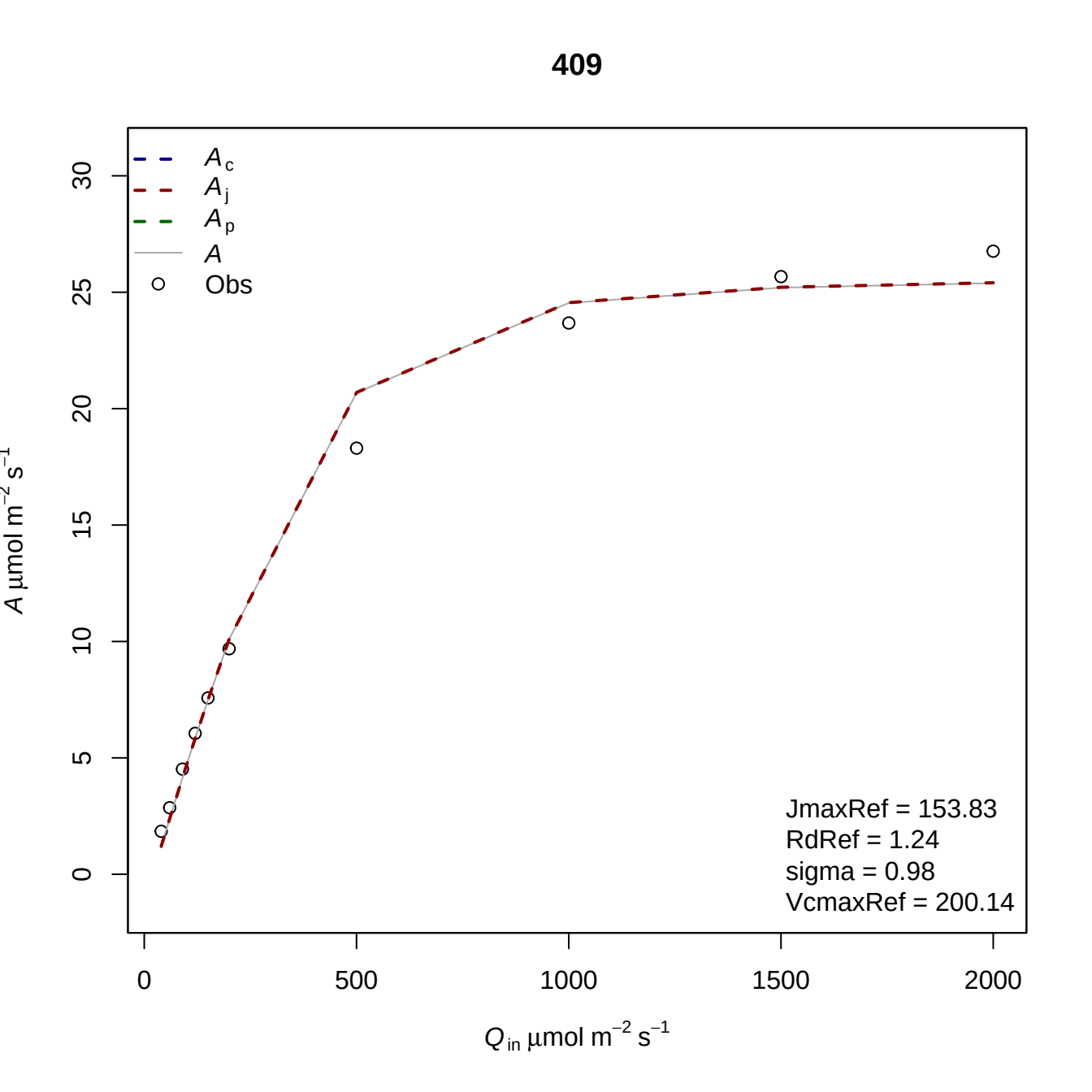

404
A_c
A_j
A_p
A
Obs
JmaxRef = 179.02
RdRef = 1.88
sigma = 1.35
VcmaxRef = 283.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

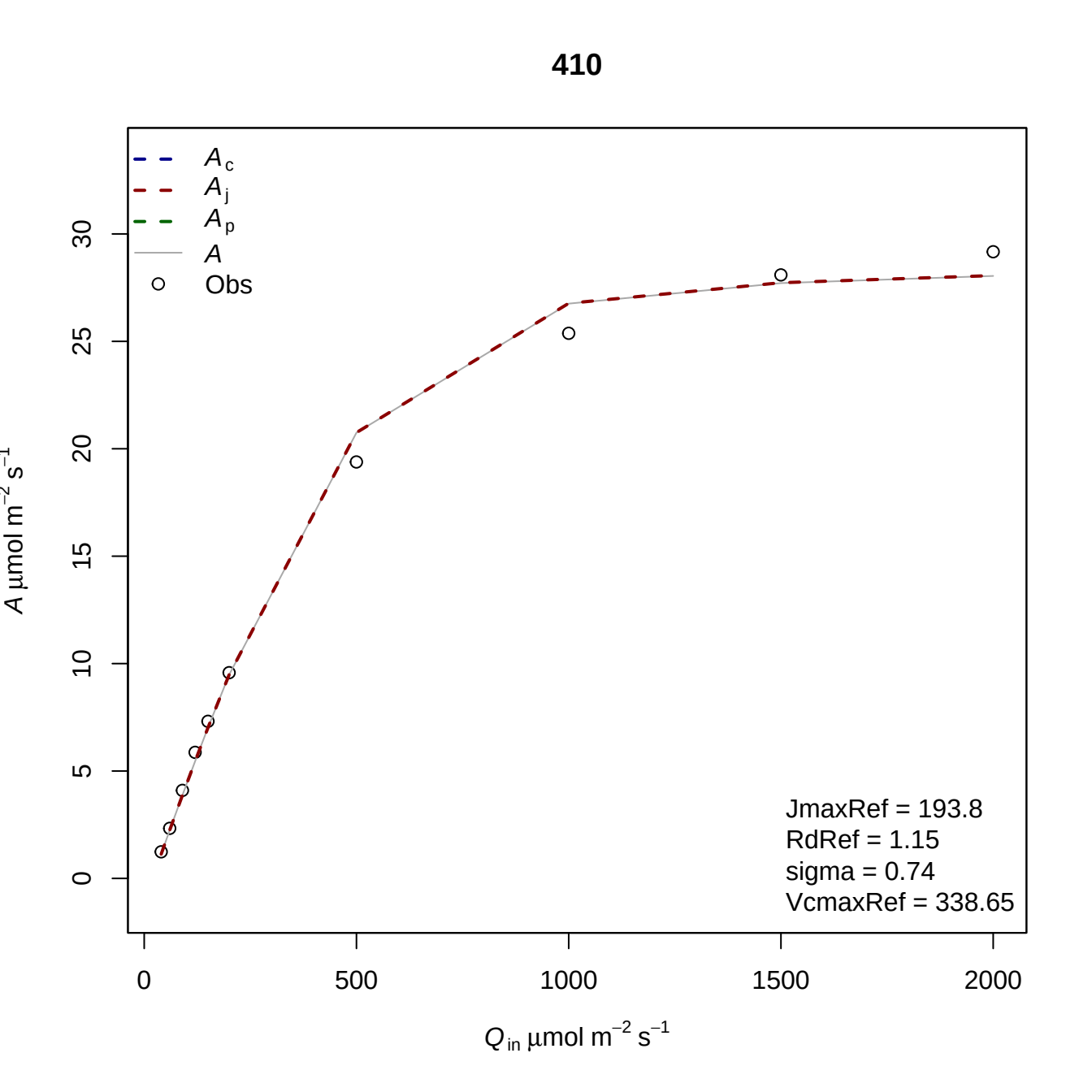

405
A_c
A_j
A_p
A
Obs
JmaxRef = 209.18
RdRef = 1.67
sigma = 1.1
VcmaxRef = 422.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

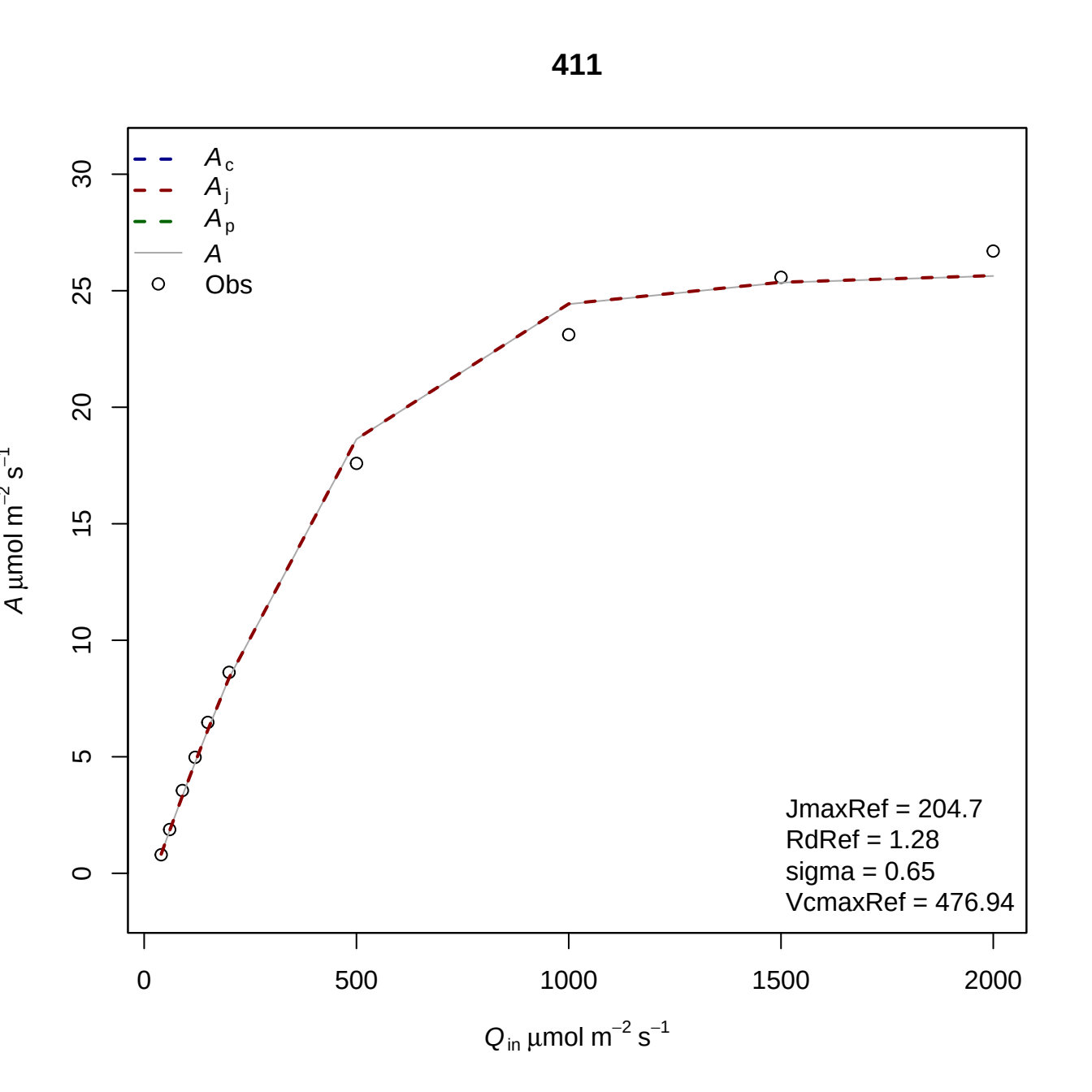

406
A_c
A_j
A_p
A
Obs
JmaxRef = 204.63
RdRef = 2.73
sigma = 0.9
VcmaxRef = 536.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

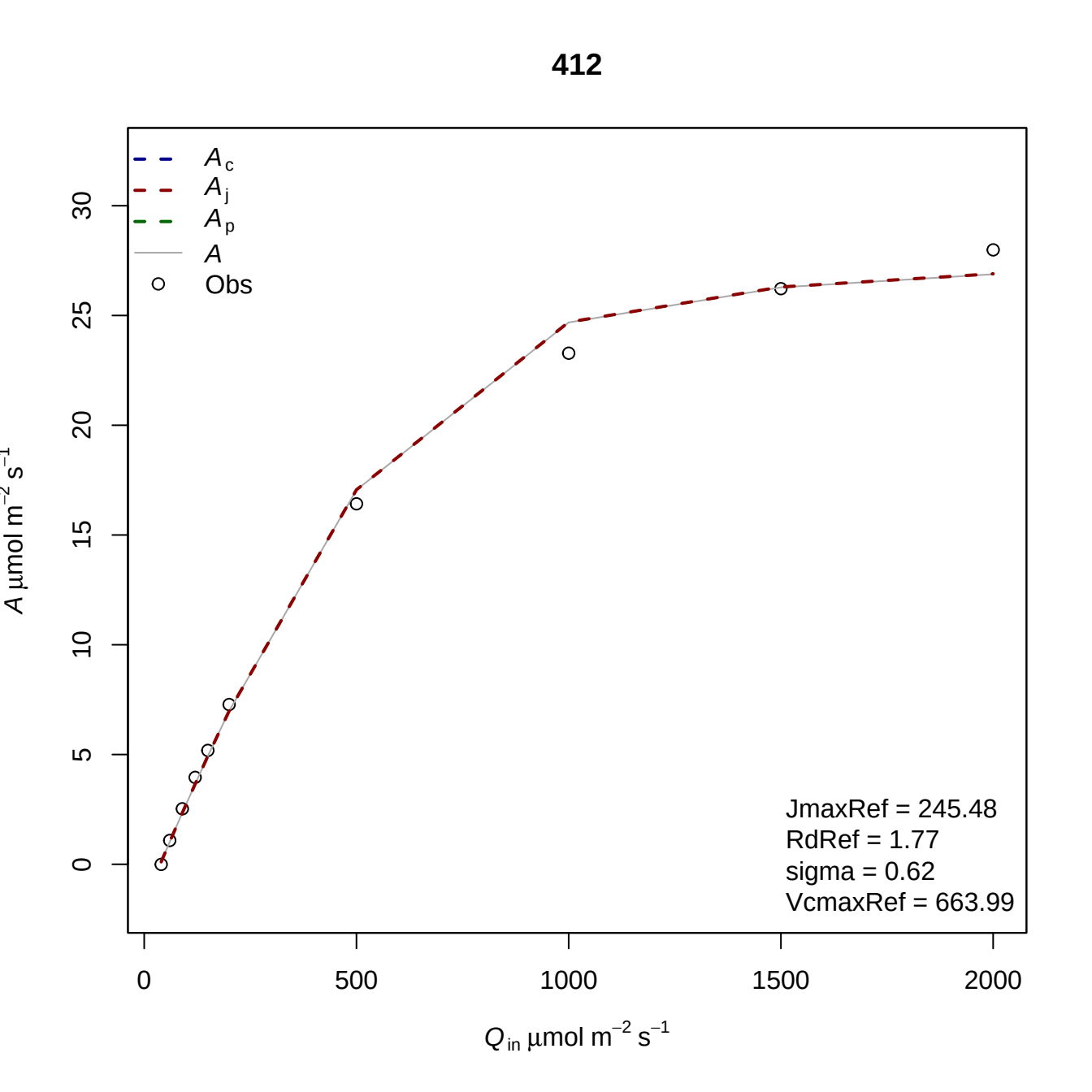

407
A_c
A_j
A_p
A
Obs
JmaxRef = 178.01
RdRef = 3.8
sigma = 1.36
VcmaxRef = 636.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

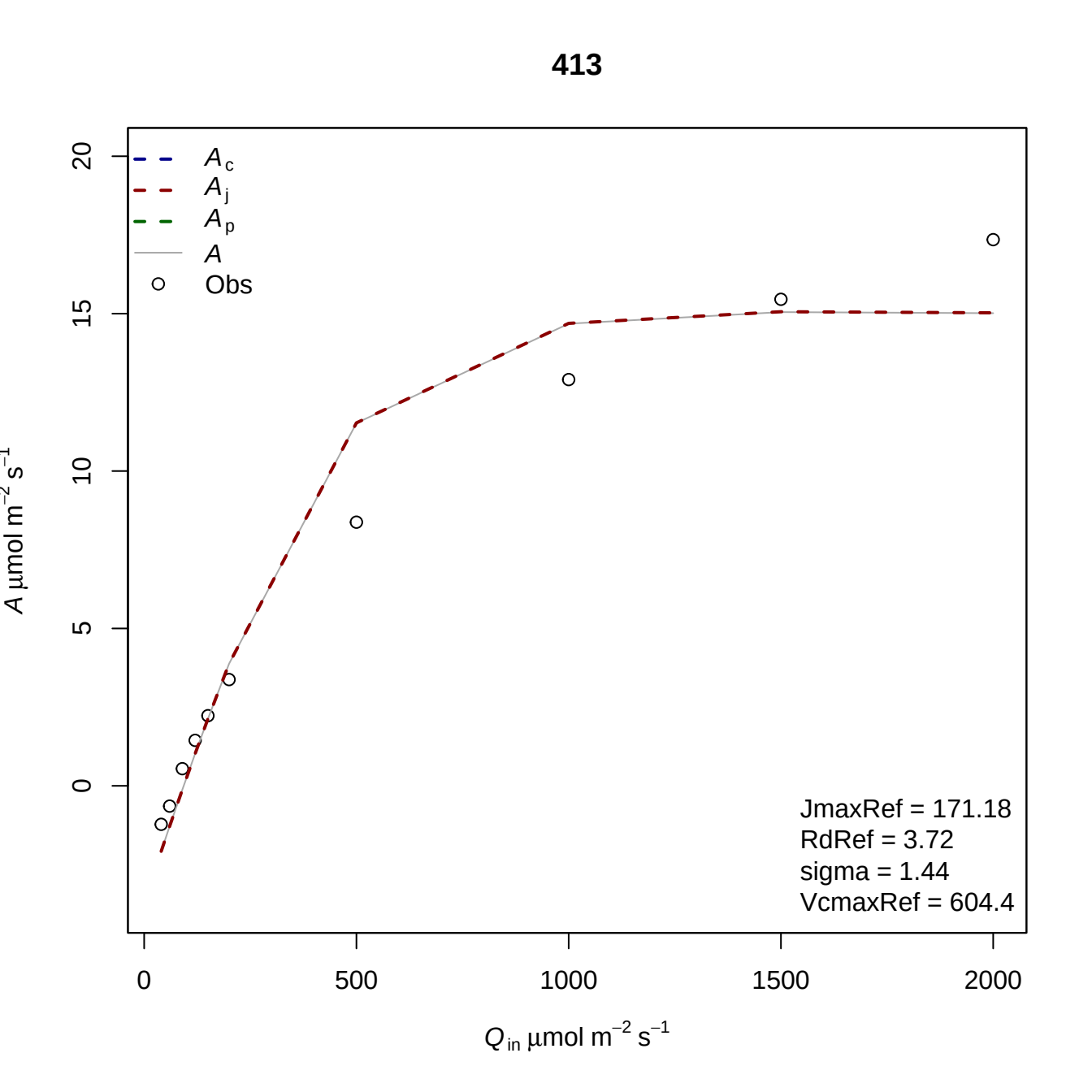

408
A_c
A_j
A_p
A
Obs
JmaxRef = 128.97
RdRef = 1.67
sigma = 1.31
VcmaxRef = 171.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

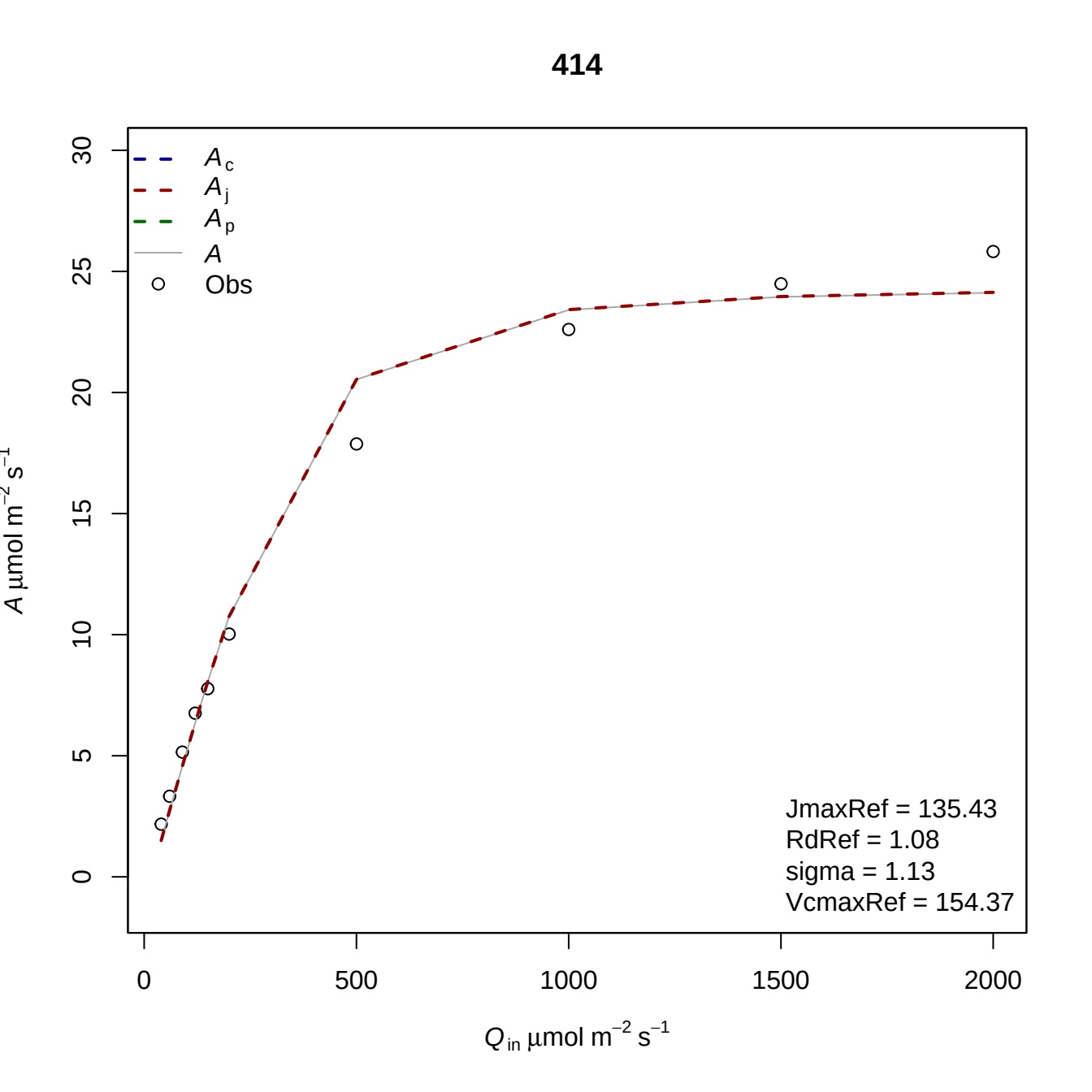

409
A_c
A_j
A_p
A
Obs
JmaxRef = 153.83
RdRef = 1.24
sigma = 0.98
VcmaxRef = 200.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

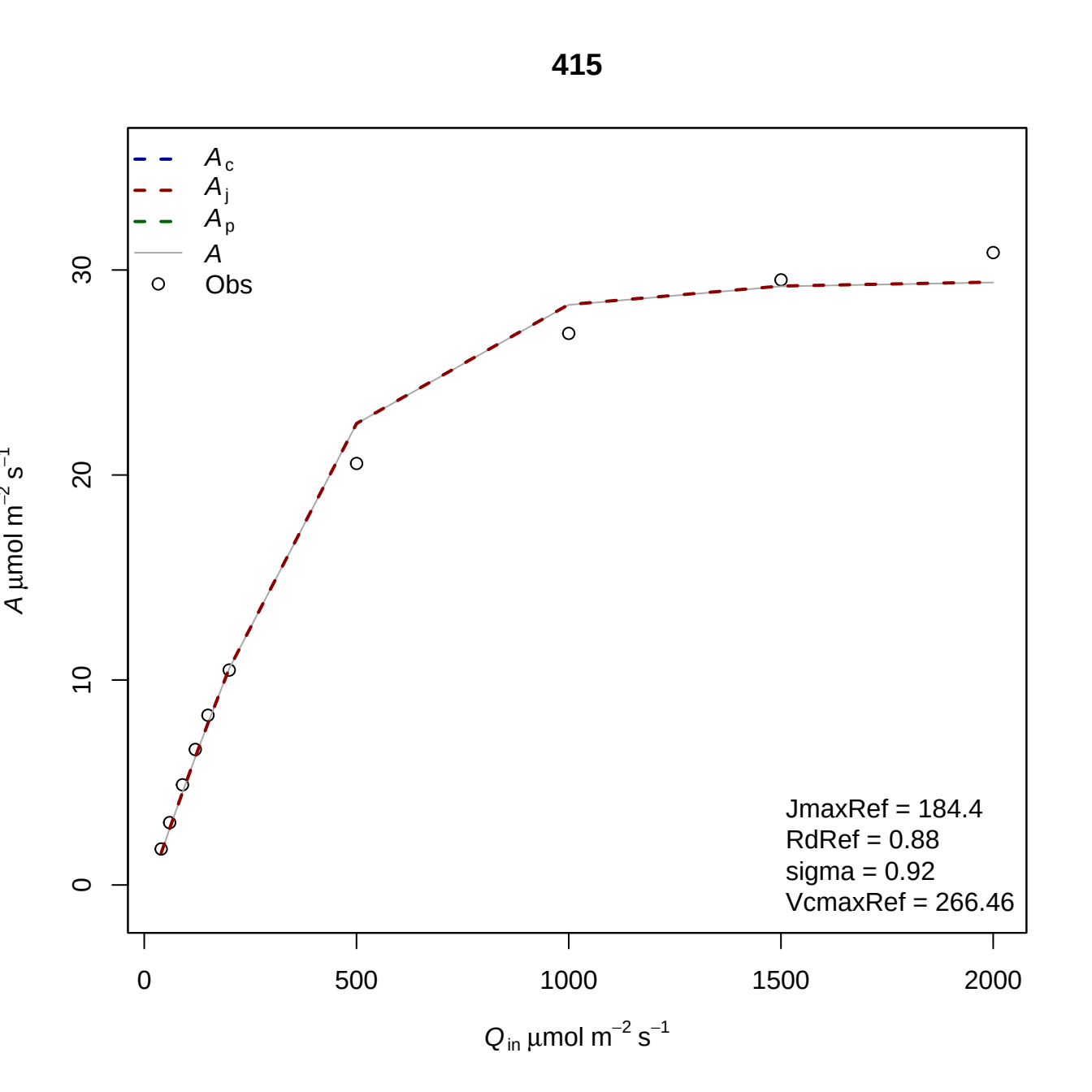

410
A_c
A_j
A_p
A
Obs
JmaxRef = 193.8
RdRef = 1.15
sigma = 0.74
VcmaxRef = 338.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

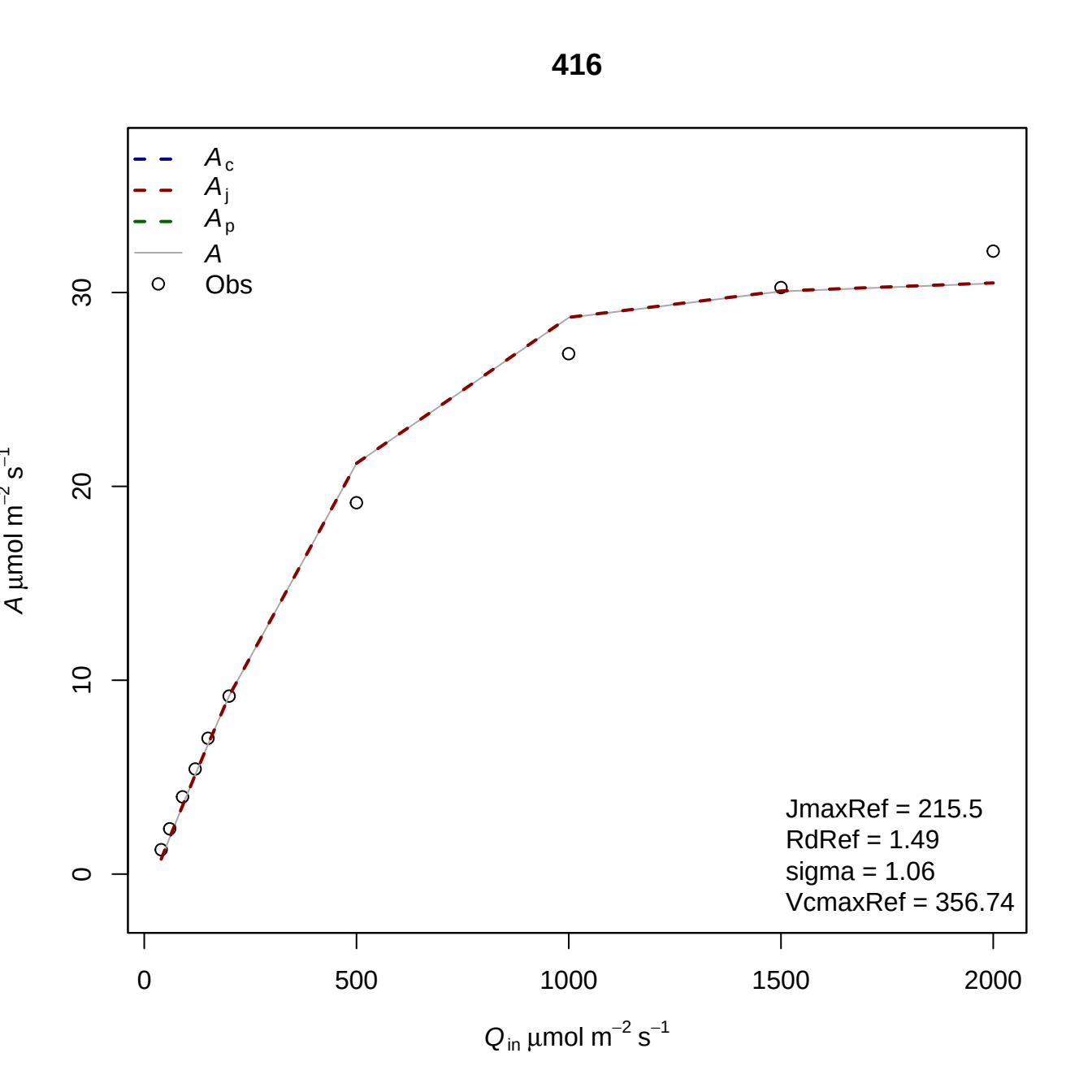

411
A_c
A_j
A_p
A
Obs
JmaxRef = 204.7
RdRef = 1.28
sigma = 0.65
VcmaxRef = 476.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

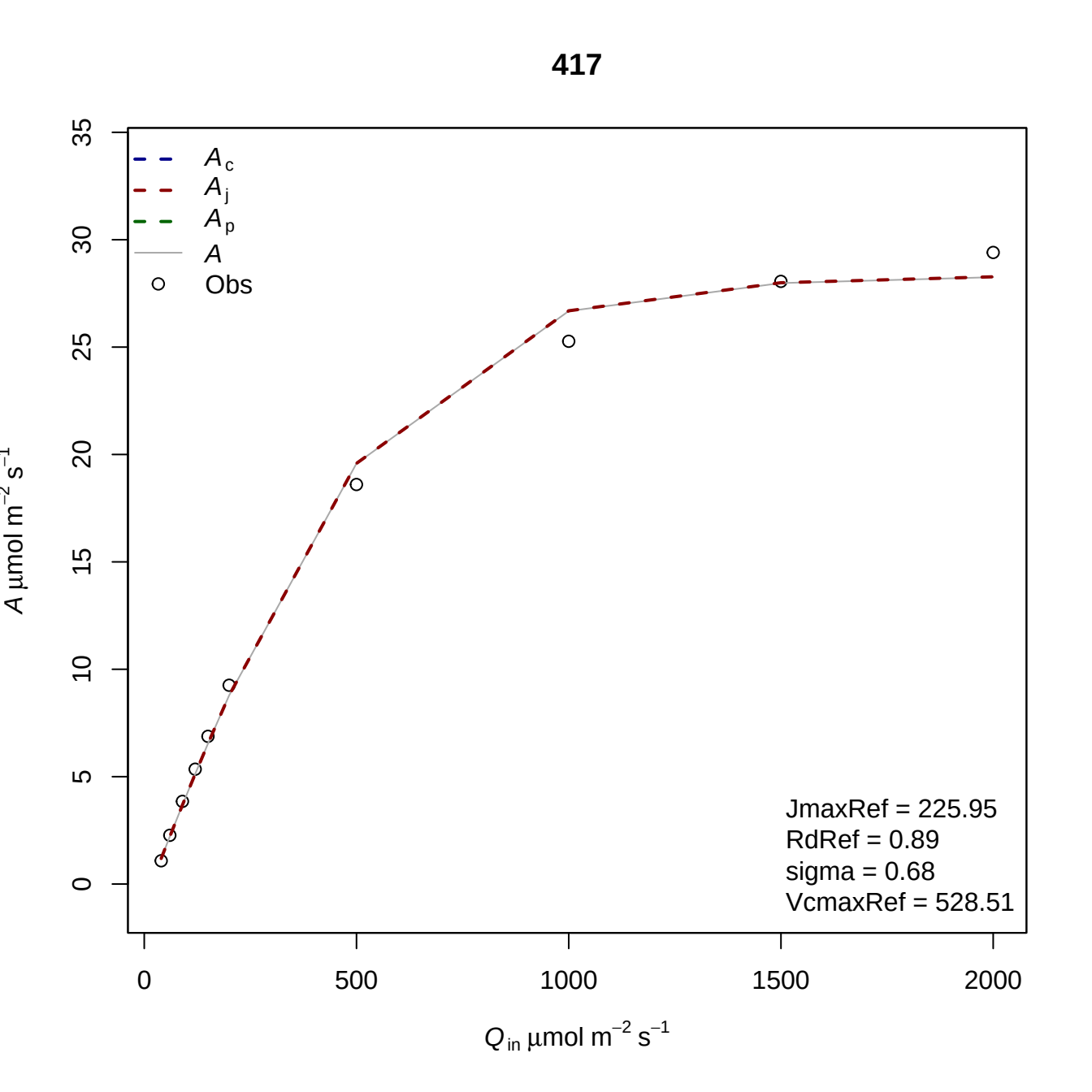

412
A_c
A_j
A_p
A
Obs
JmaxRef = 245.48
RdRef = 1.77
sigma = 0.62
VcmaxRef = 663.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

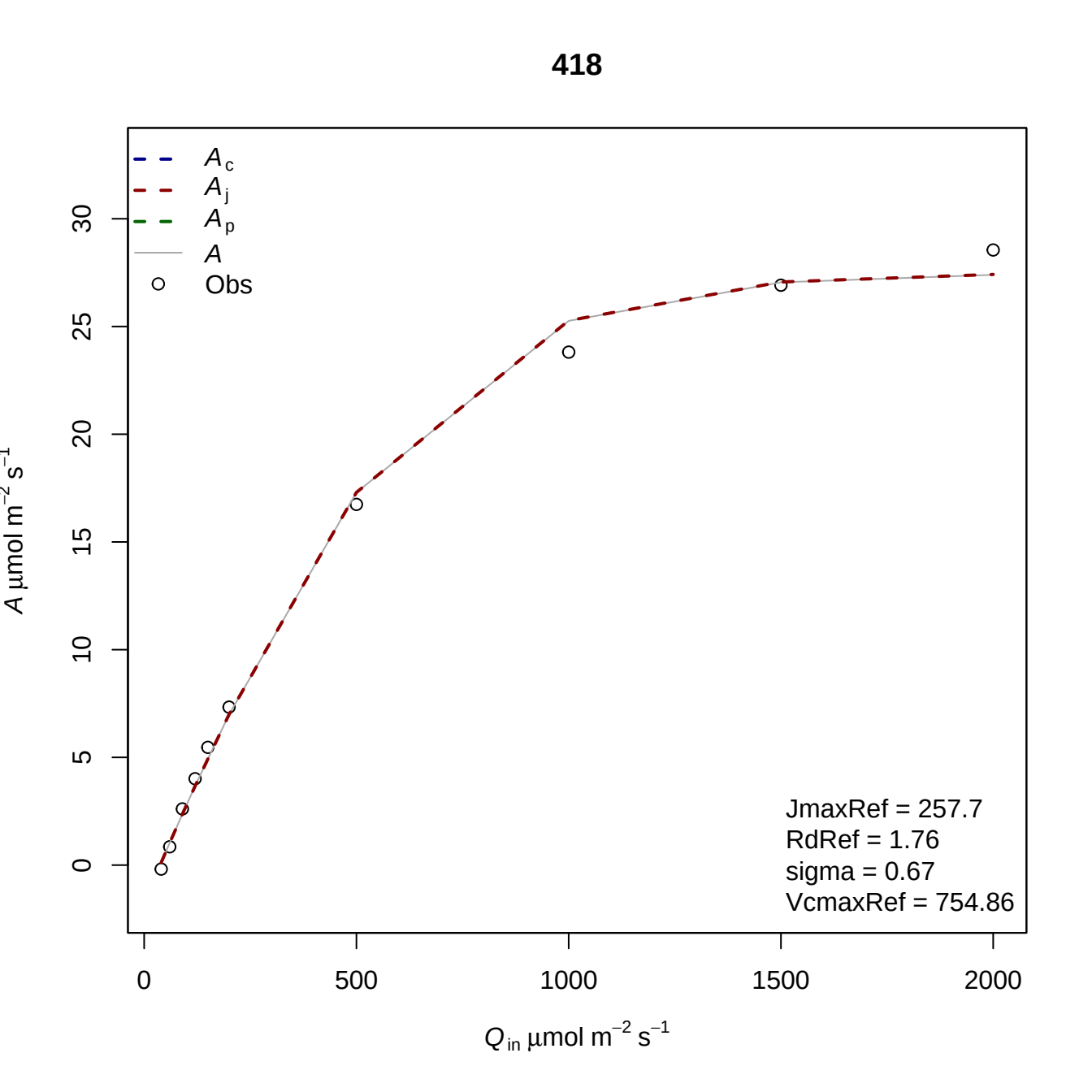

413
A_c
A_j
A_p
A
Obs
JmaxRef = 171.18
RdRef = 3.72
sigma = 1.44
VcmaxRef = 604.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

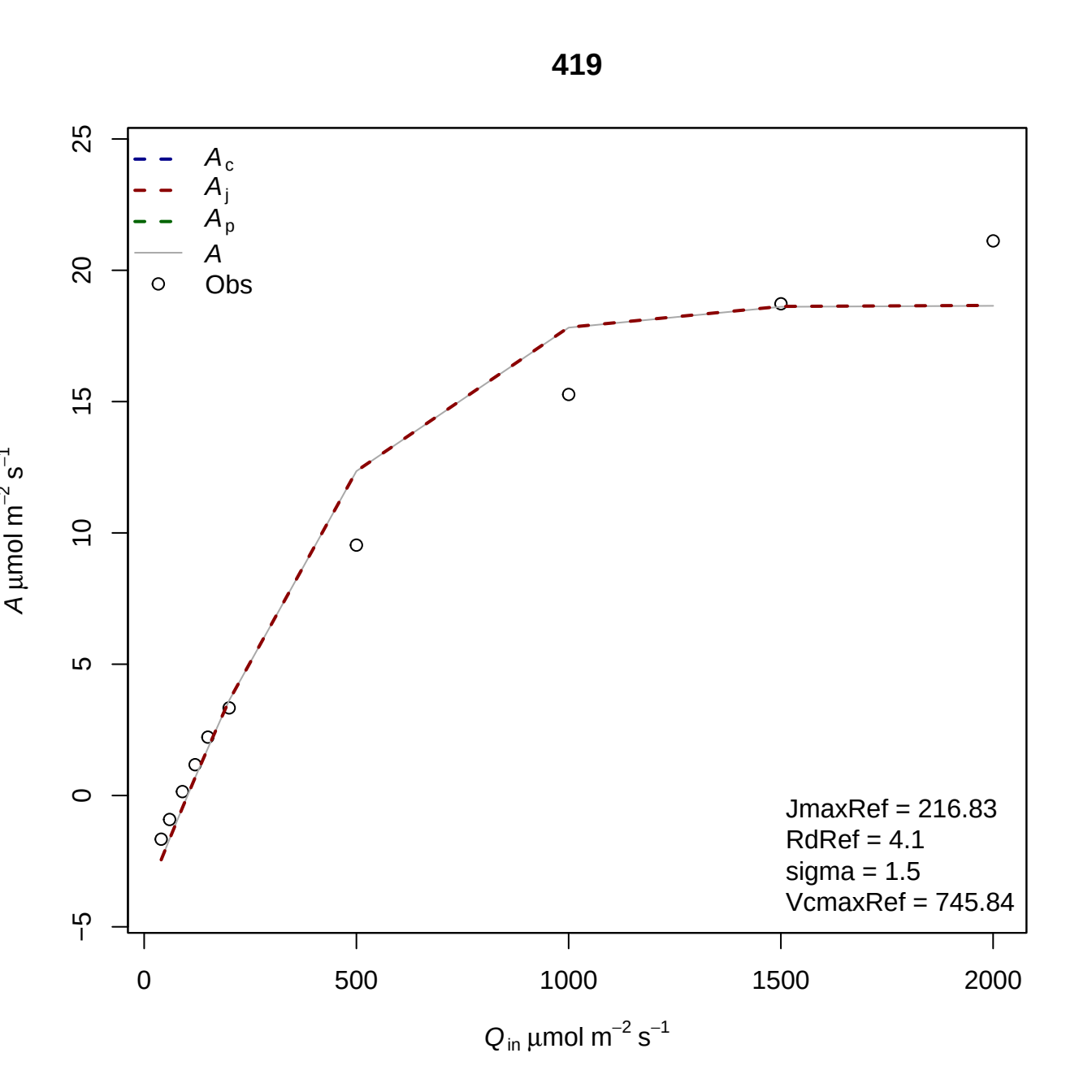

414
A_c
A_j
A_p
A
Obs
JmaxRef = 135.43
RdRef = 1.08
sigma = 1.13
VcmaxRef = 154.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

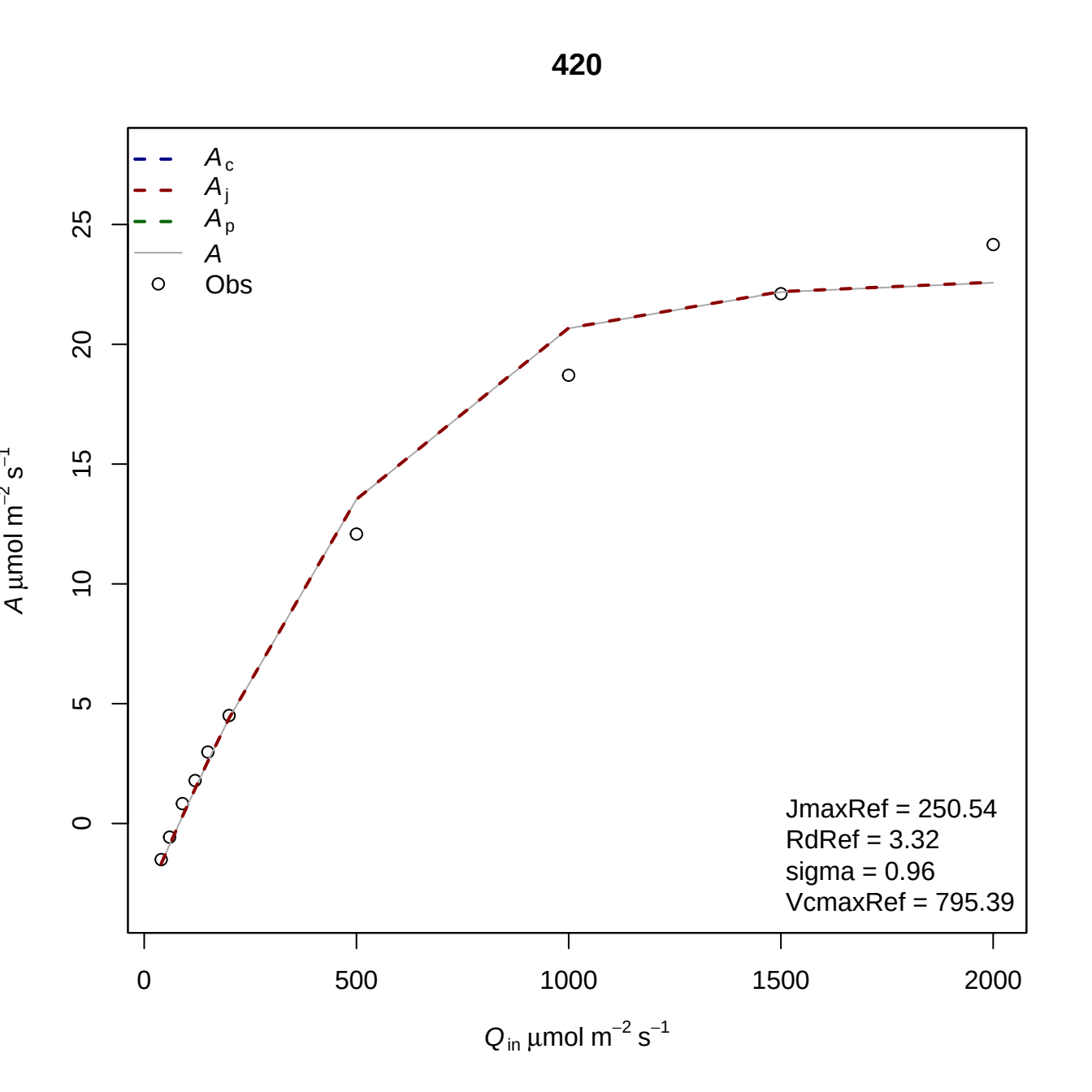

415
A_c
A_j
A_p
A
Obs
JmaxRef = 184.4
RdRef = 0.88
sigma = 0.92
VcmaxRef = 266.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

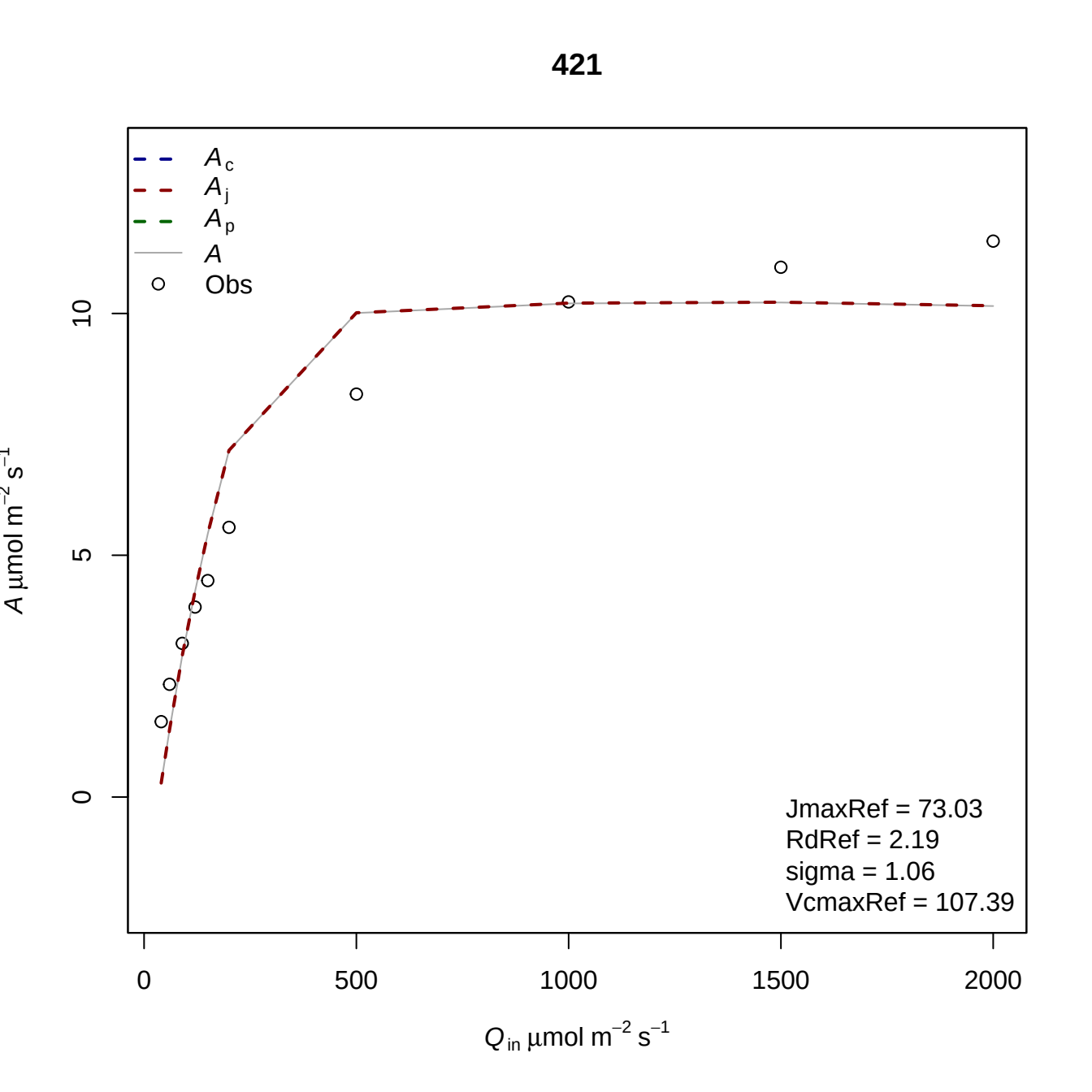

416
A_c
A_j
A_p
A
Obs
JmaxRef = 215.5
RdRef = 1.49
sigma = 1.06
VcmaxRef = 356.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

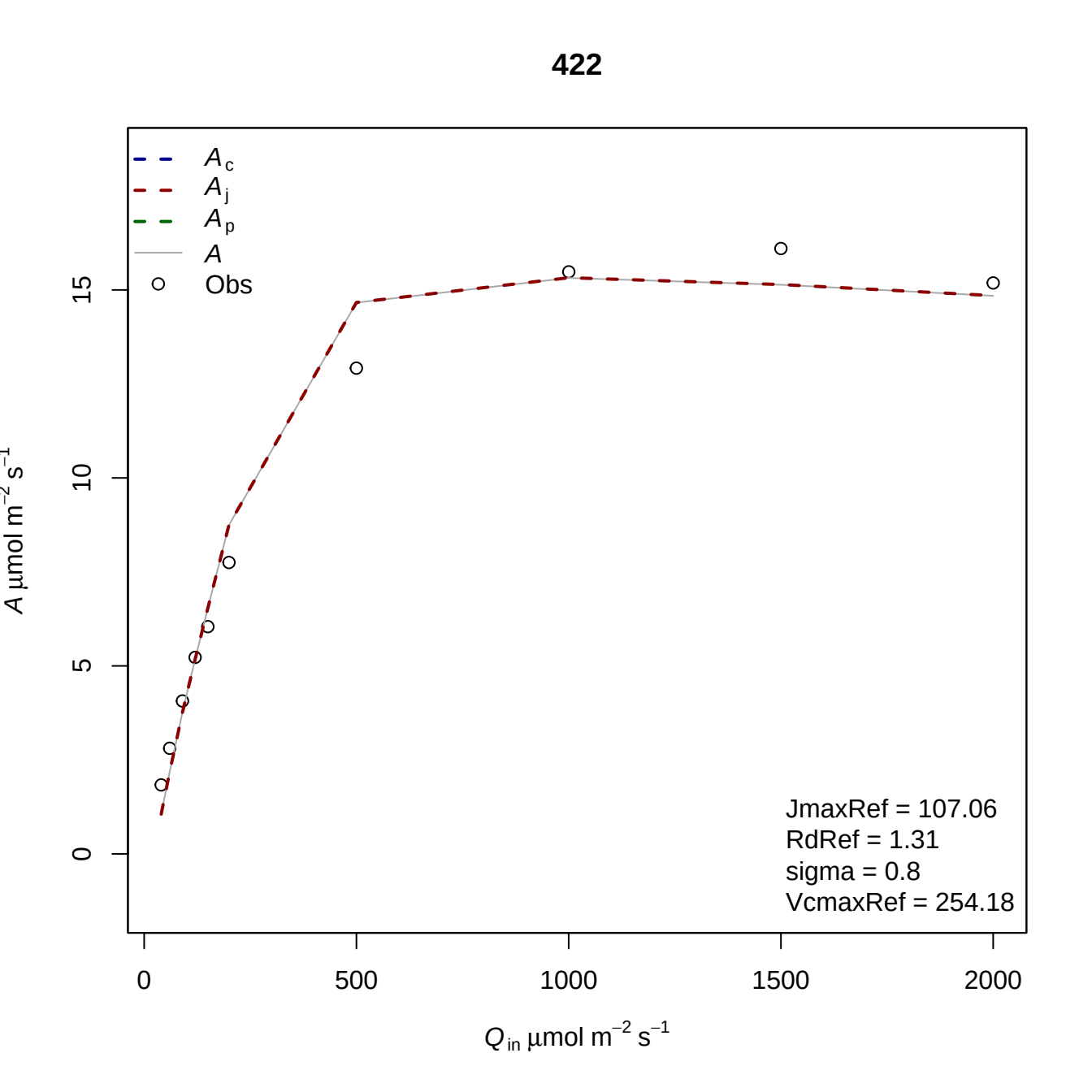

417
A_c
A_j
A_p
A
Obs
JmaxRef = 225.95
RdRef = 0.89
sigma = 0.68
VcmaxRef = 528.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

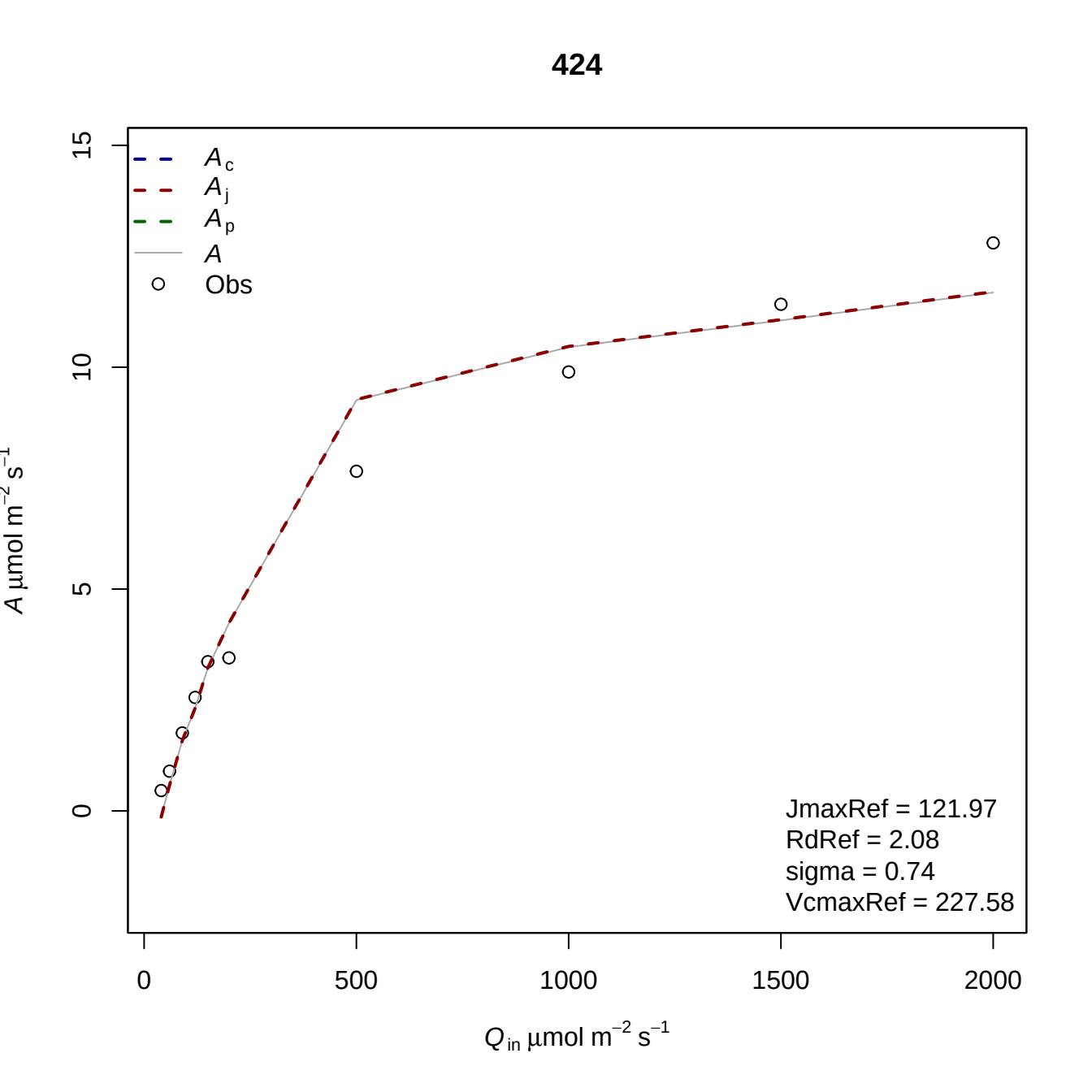

418
A_c
A_j
A_p
A
Obs
JmaxRef = 257.7
RdRef = 1.76
sigma = 0.67
VcmaxRef = 754.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

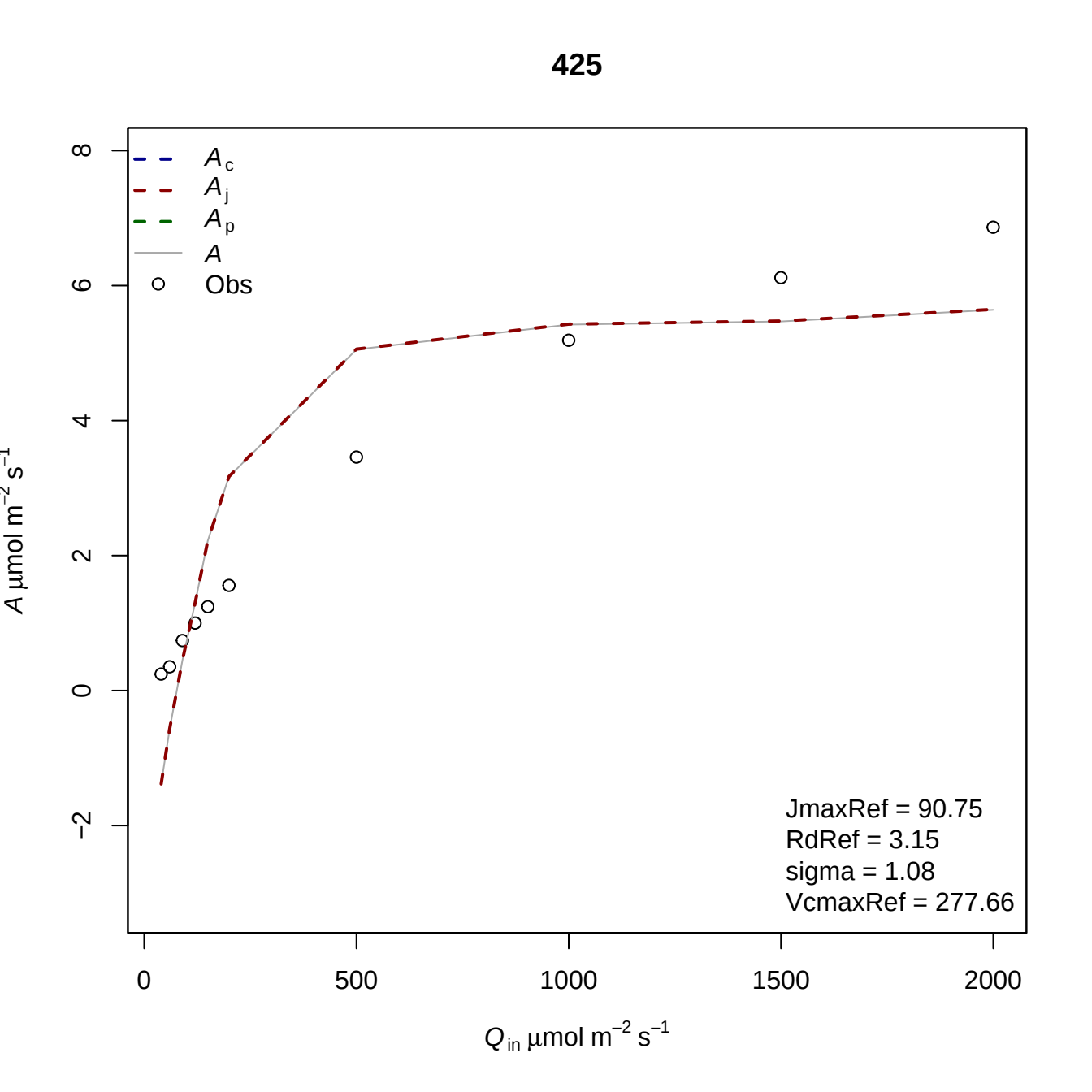

419
A_c
A_j
A_p
A
Obs
JmaxRef = 216.83
RdRef = 4.1
sigma = 1.5
VcmaxRef = 745.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

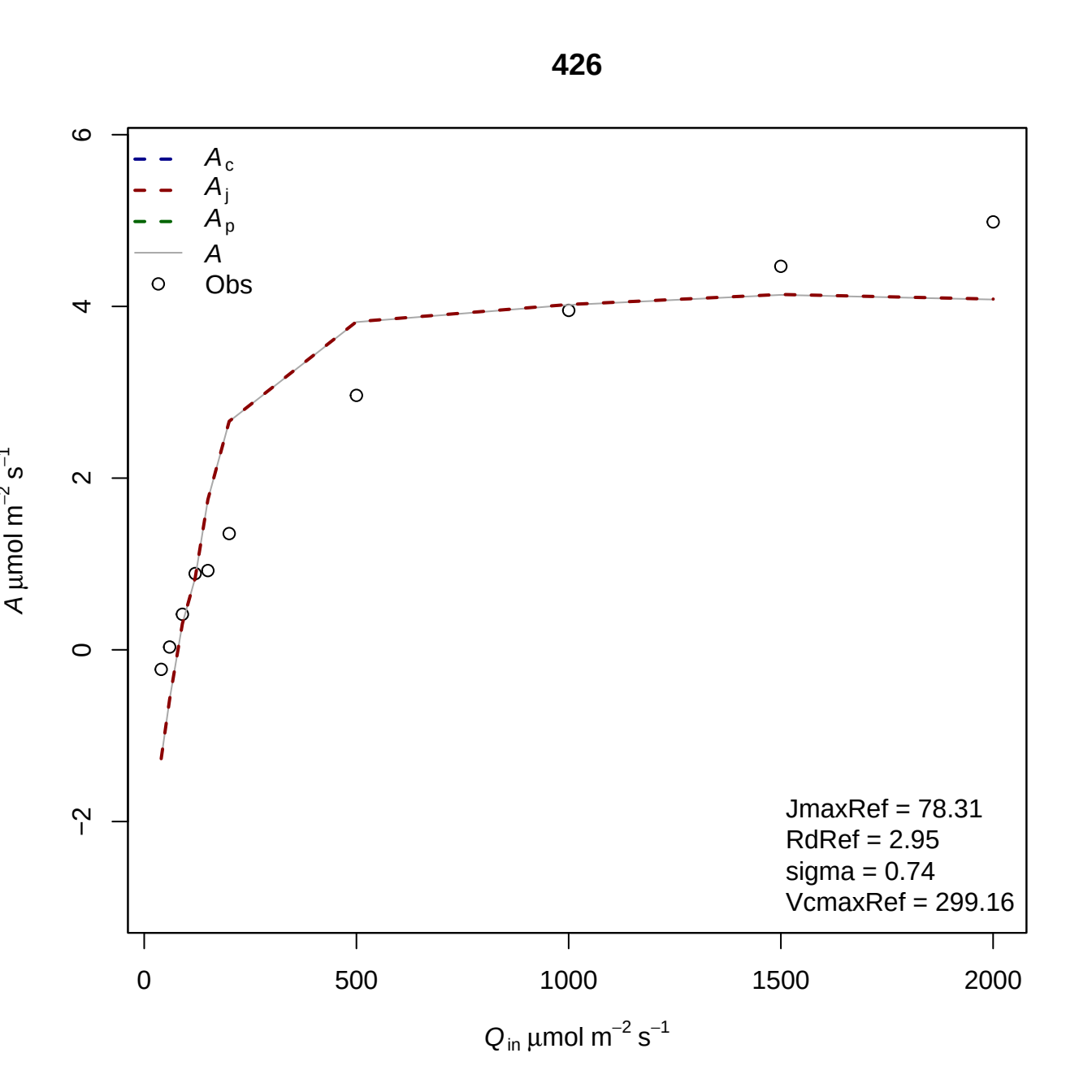

420
A_c
A_j
A_p
A
Obs
JmaxRef = 250.54
RdRef = 3.32
sigma = 0.96
VcmaxRef = 795.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

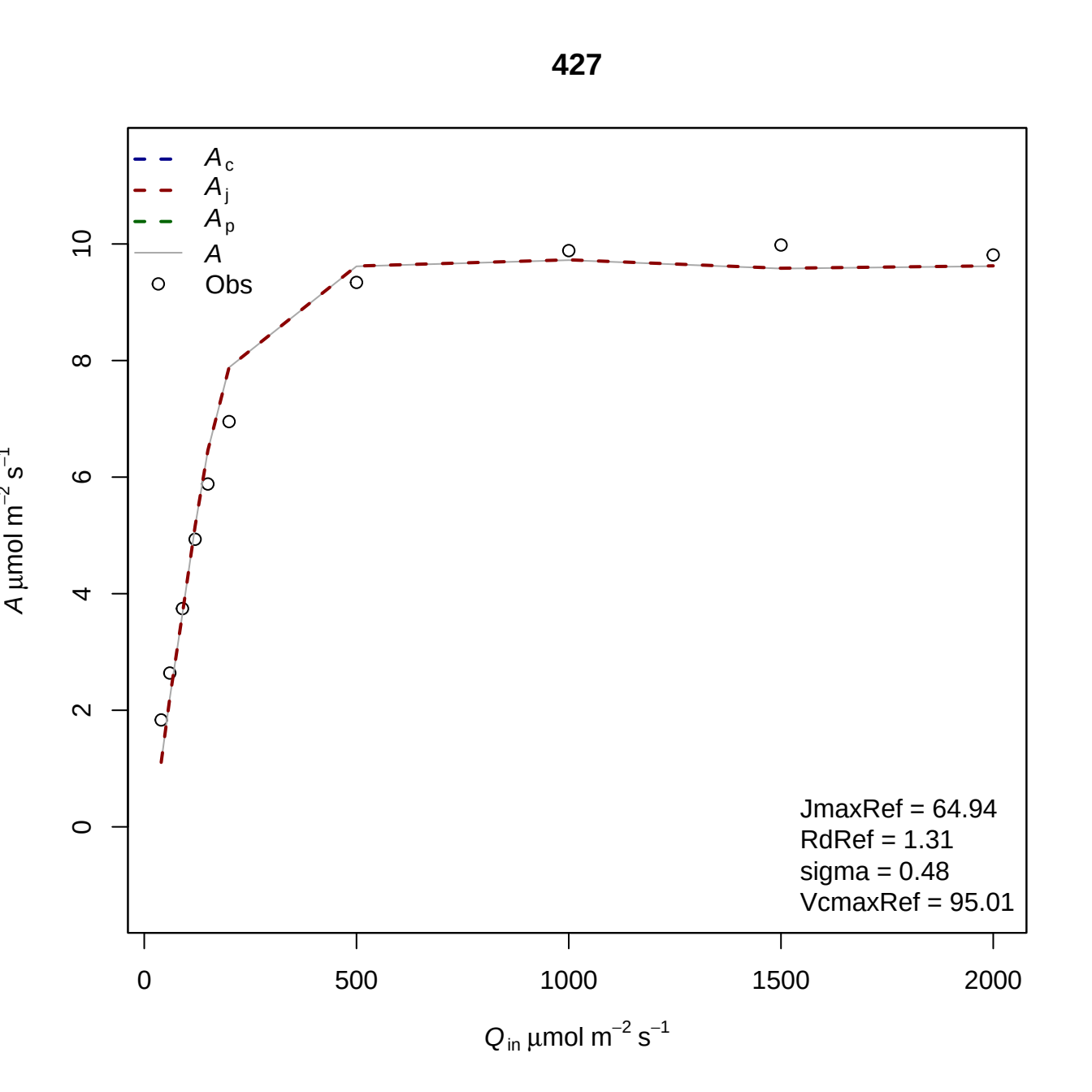

421
A_c
A_j
A_p
A
Obs
JmaxRef = 73.03
RdRef = 2.19
sigma = 1.06
VcmaxRef = 107.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

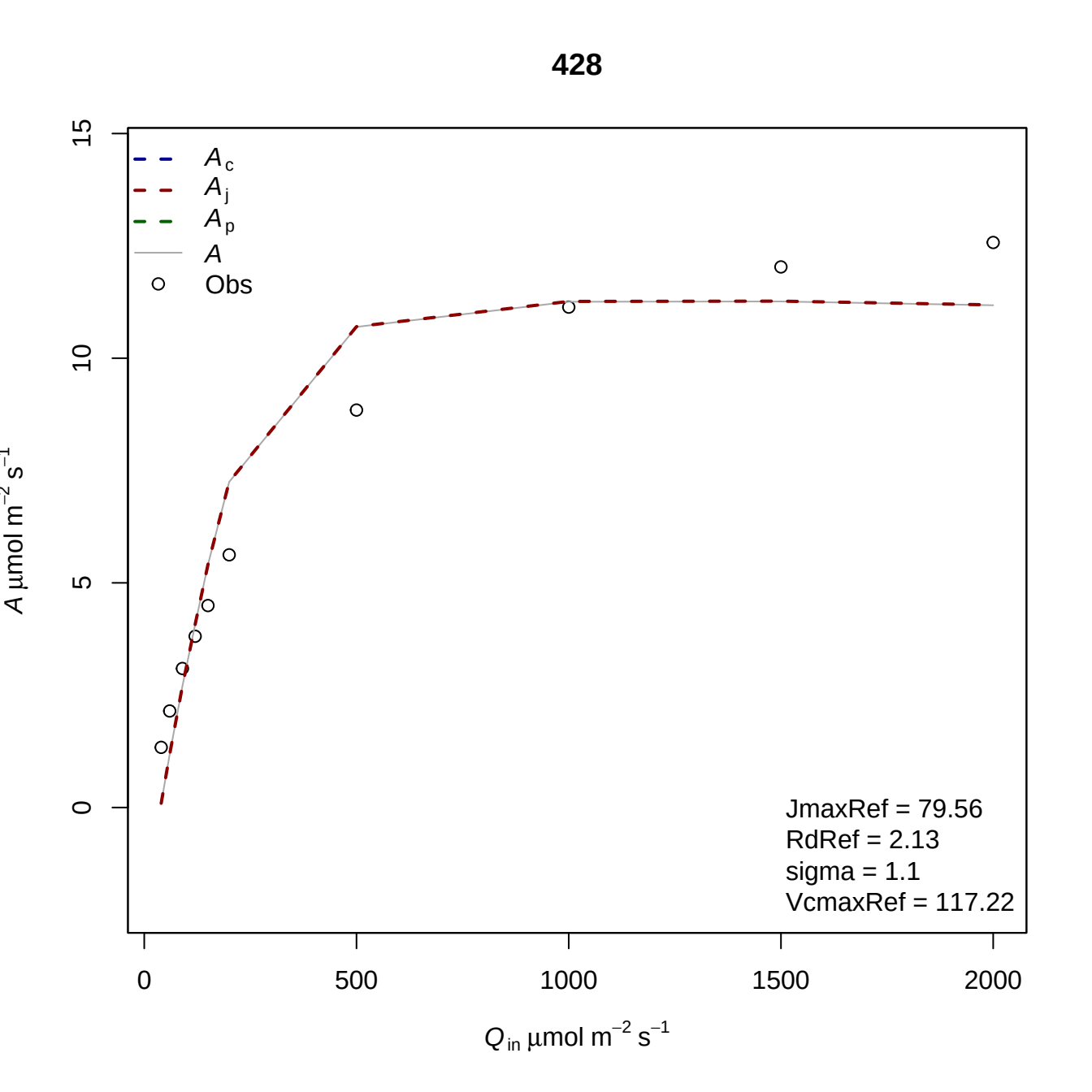

422
A_c
A_j
A_p
A
Obs
JmaxRef = 107.06
RdRef = 1.31
sigma = 0.8
VcmaxRef = 254.18
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

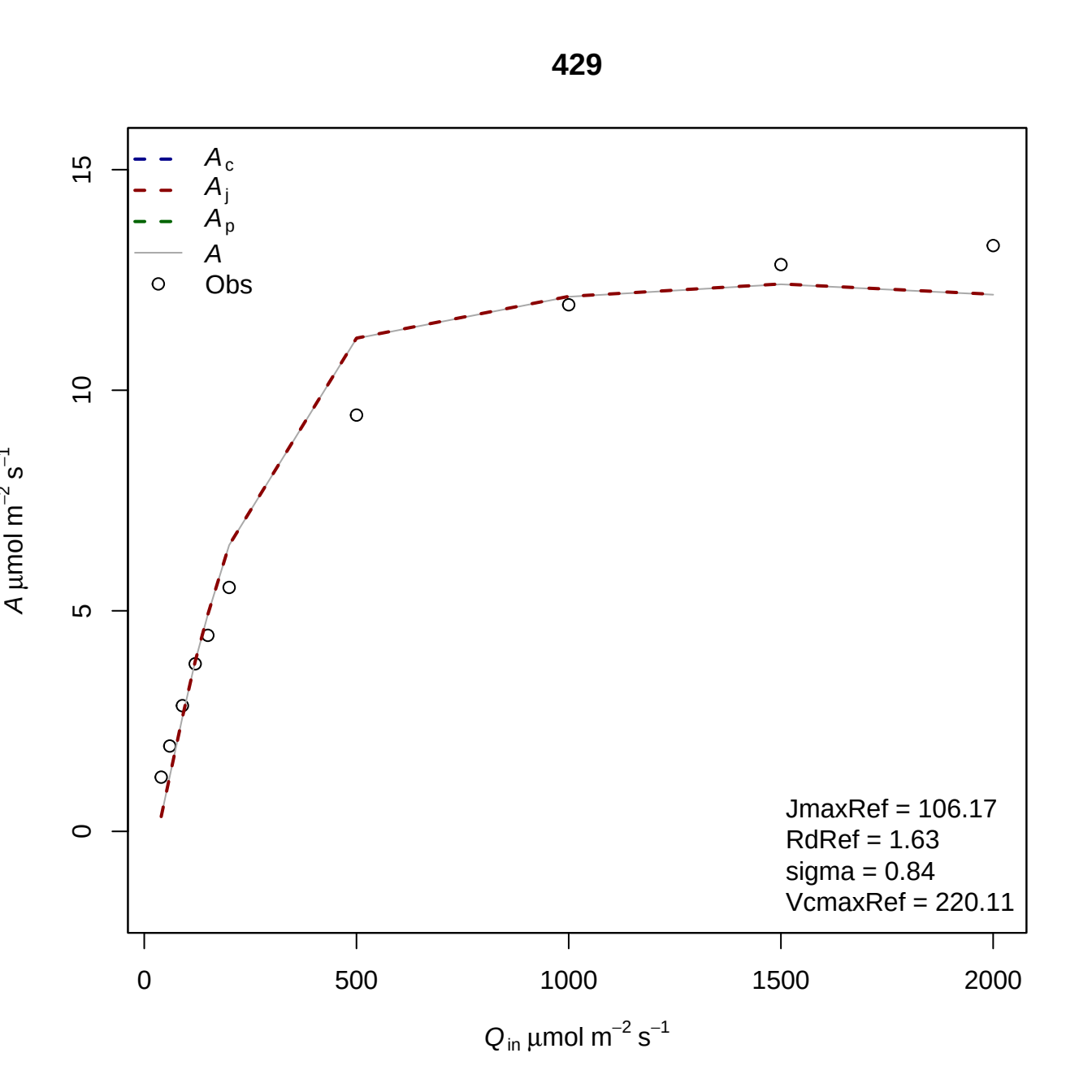

424
A_c
A_j
A_p
A
Obs
JmaxRef = 121.97
RdRef = 2.08
sigma = 0.74
VcmaxRef = 227.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

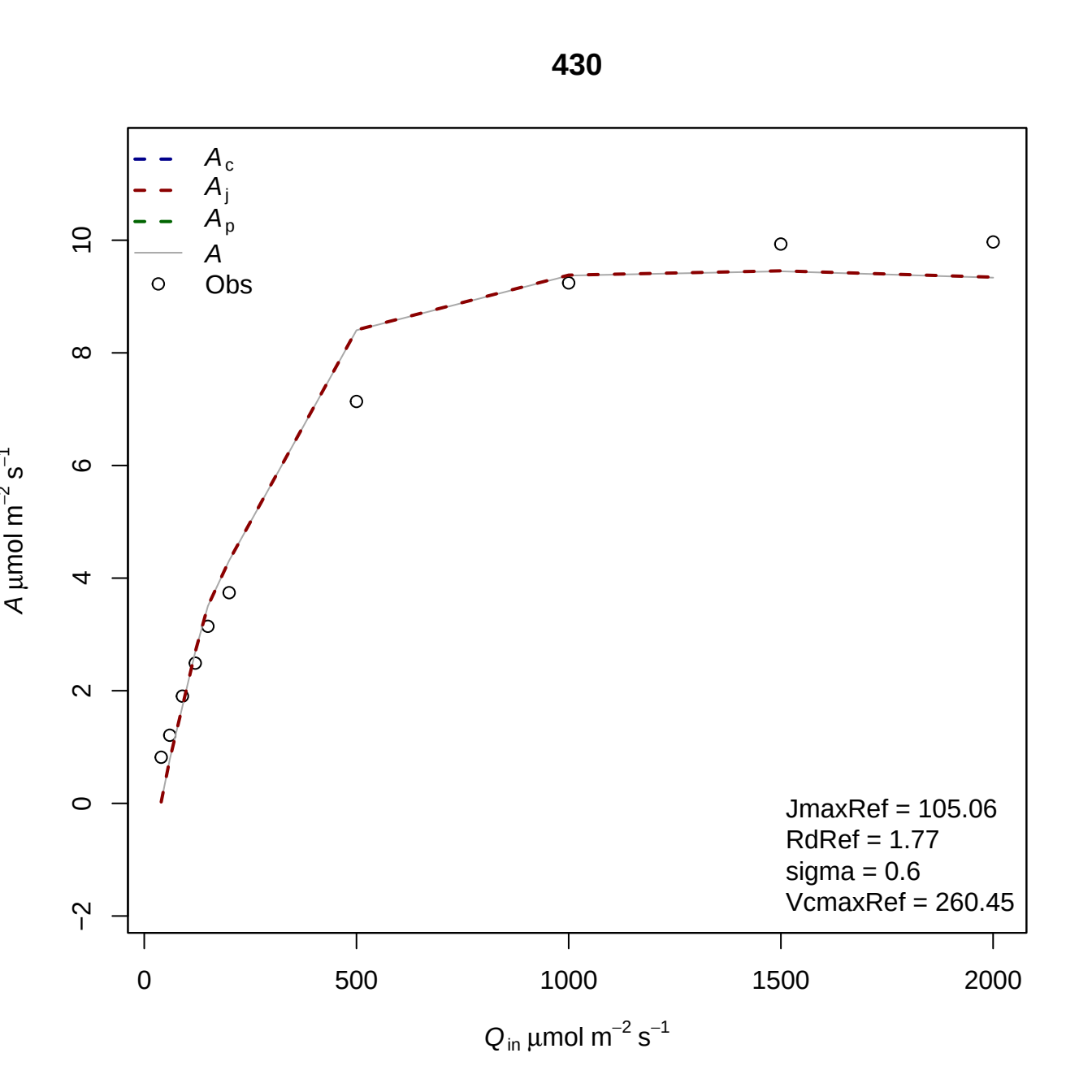

425
A_c
A_j
A_p
A
Obs
JmaxRef = 90.75
RdRef = 3.15
sigma = 1.08
VcmaxRef = 277.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

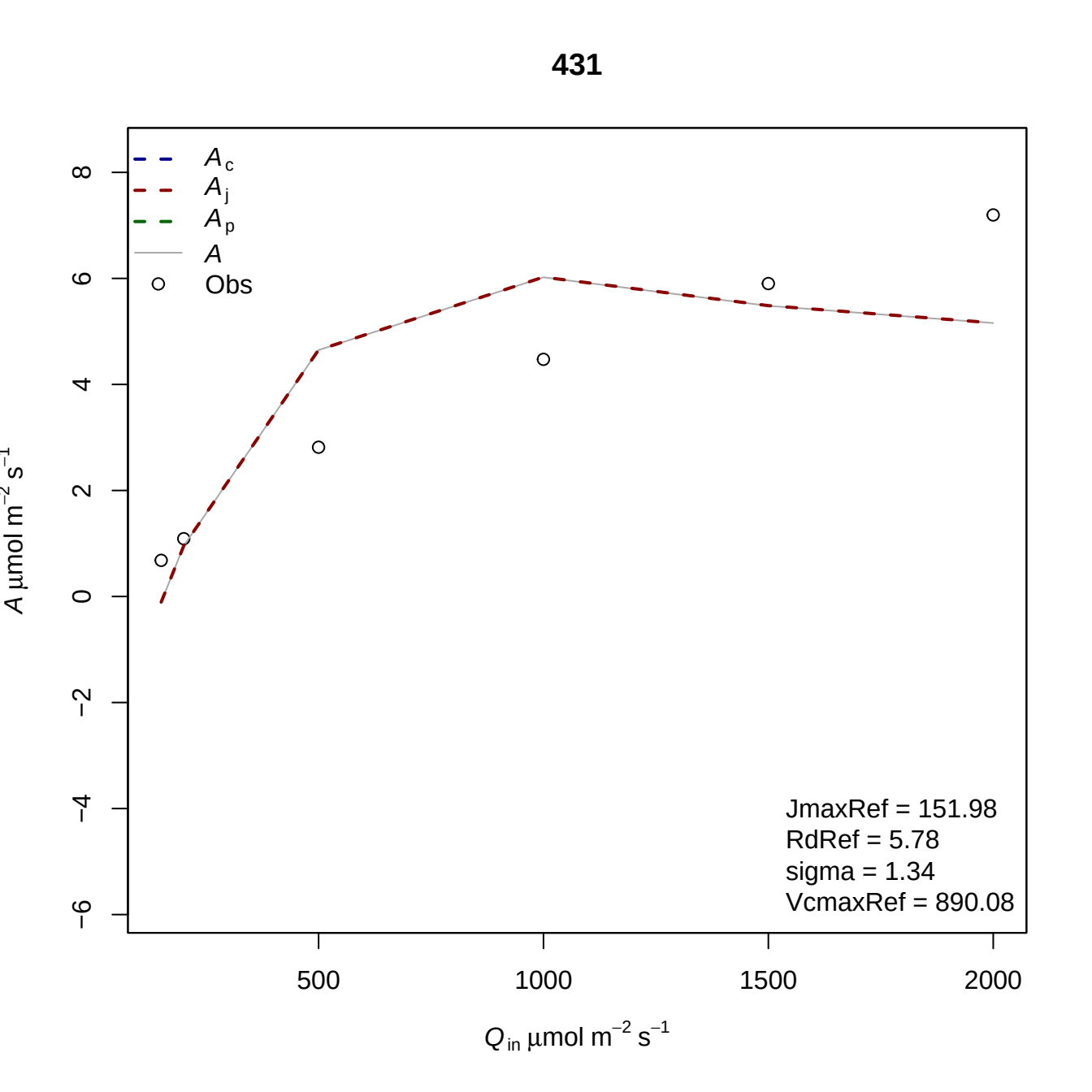

426
A_c
A_j
A_p
A
Obs
JmaxRef = 78.31
RdRef = 2.95
sigma = 0.74
VcmaxRef = 299.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

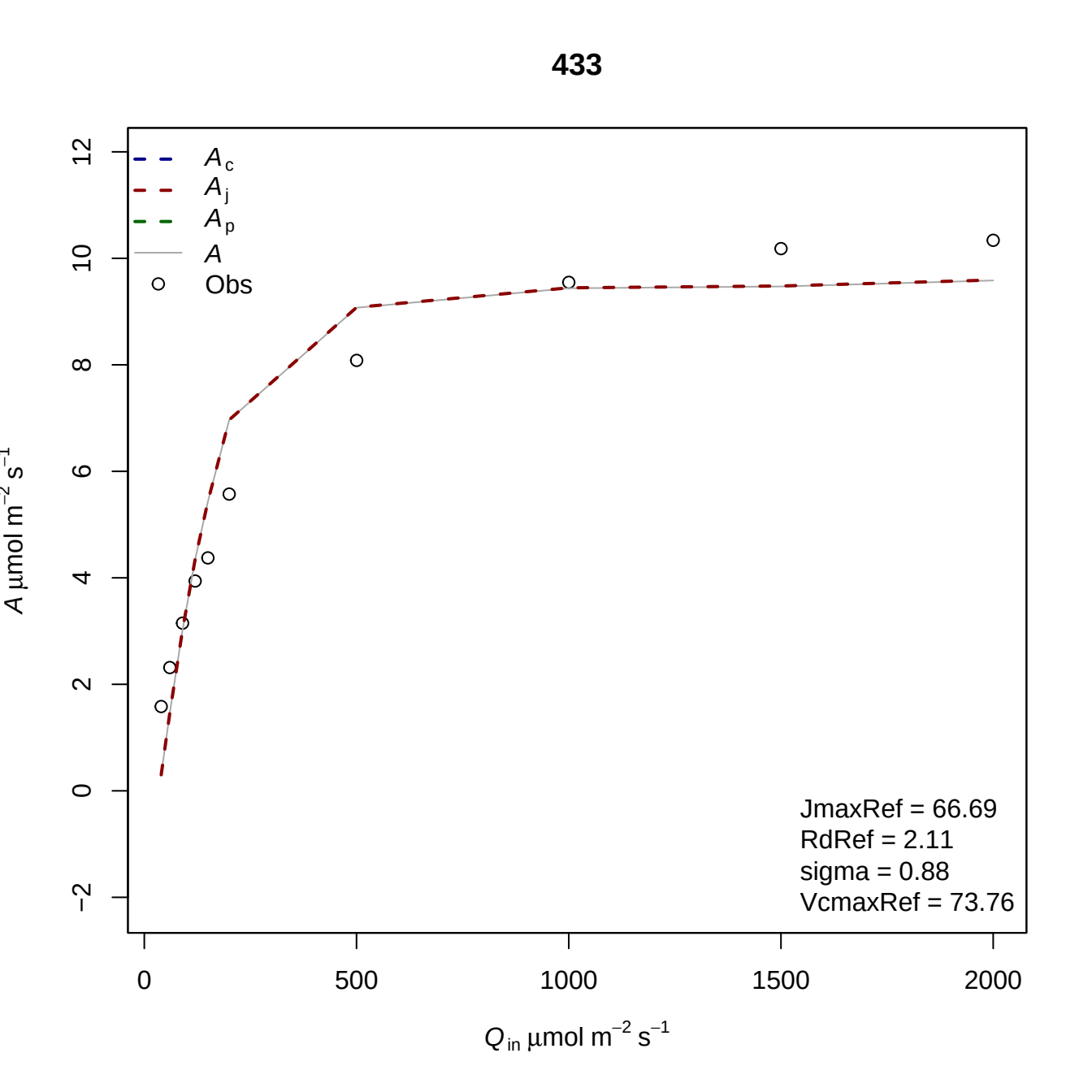

427
A_c
A_j
A_p
A
Obs
JmaxRef = 64.94
RdRef = 1.31
sigma = 0.48
VcmaxRef = 95.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

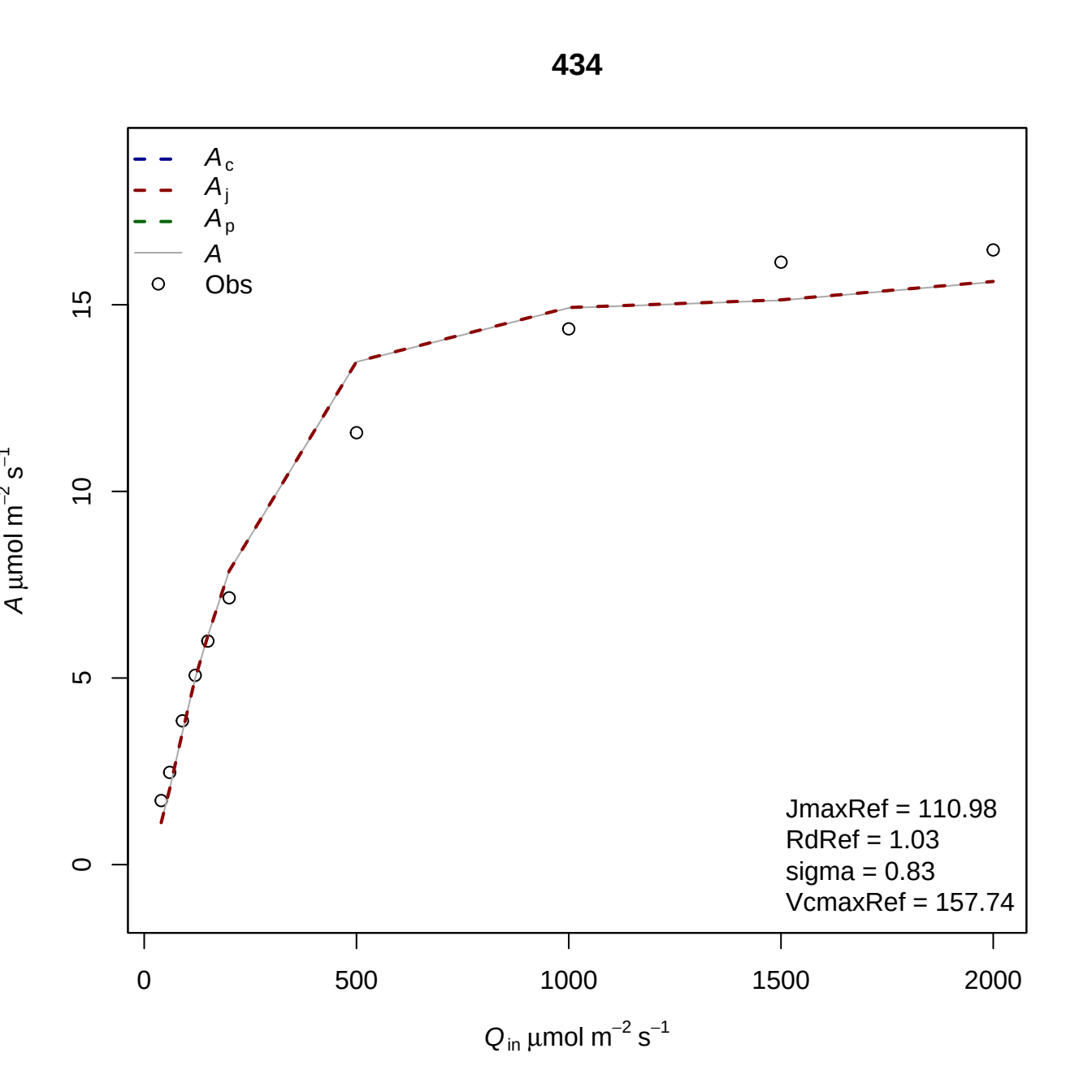

428
A_c
A_j
A_p
A
Obs
JmaxRef = 79.56
RdRef = 2.13
sigma = 1.1
VcmaxRef = 117.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

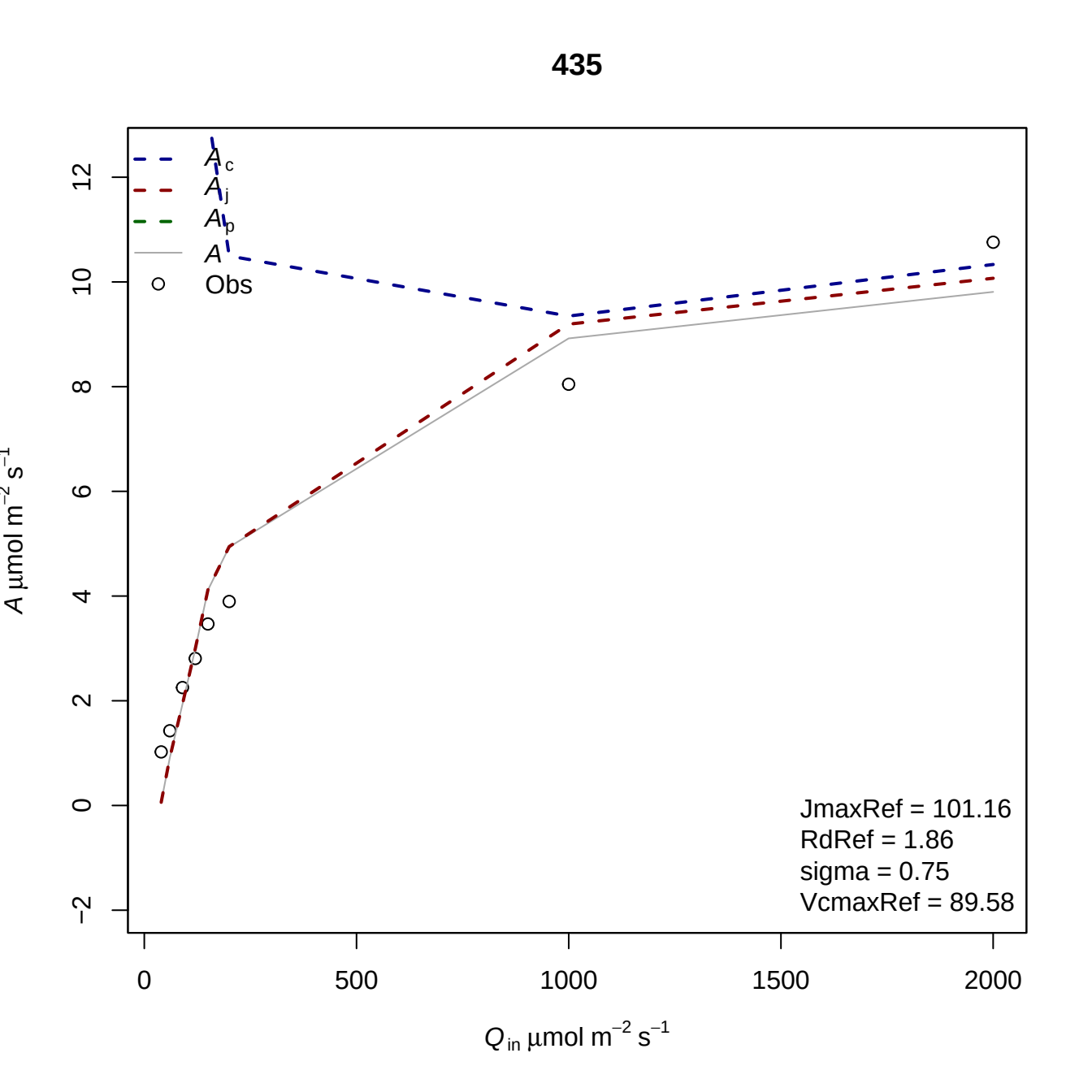

429
A_c
A_j
A_p
A
Obs
JmaxRef = 106.17
RdRef = 1.63
sigma = 0.84
VcmaxRef = 220.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

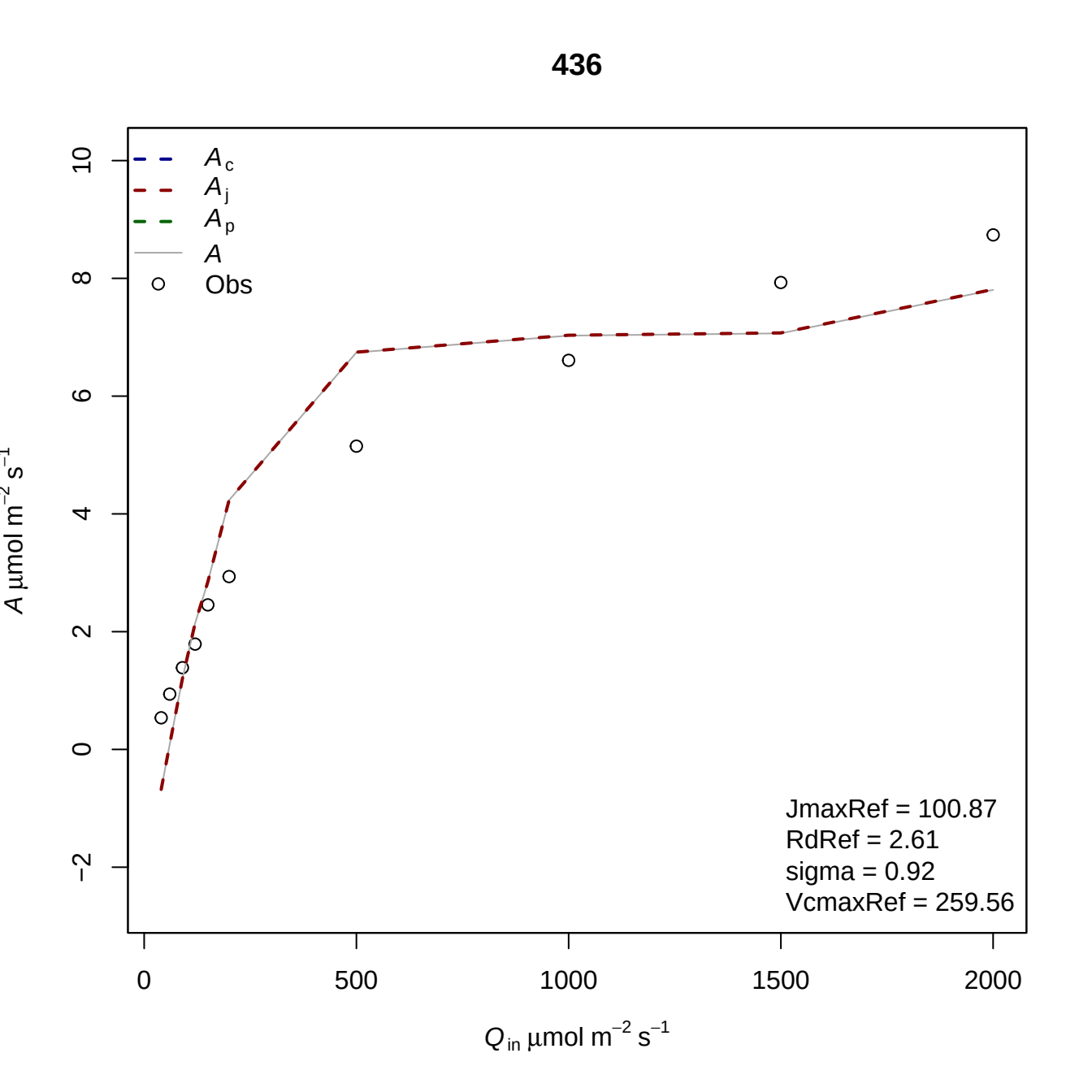

430
A_c
A_j
A_p
A
Obs
JmaxRef = 105.06
RdRef = 1.77
sigma = 0.6
VcmaxRef = 260.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

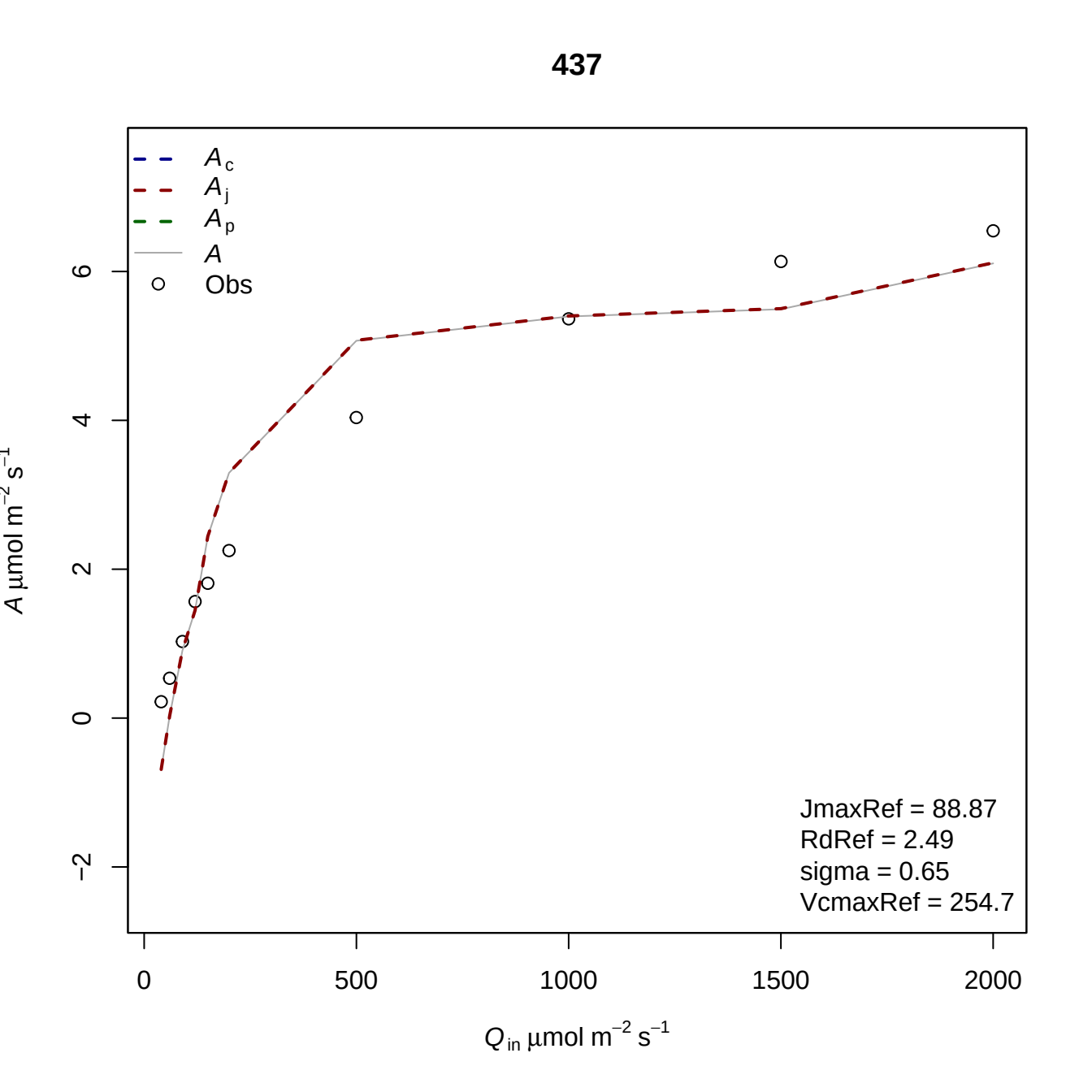

431
A_c
A_j
A_p
A
Obs
JmaxRef = 151.98
RdRef = 5.78
sigma = 1.34
VcmaxRef = 890.08
8
6
4
2
0
−2
−4
−6
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

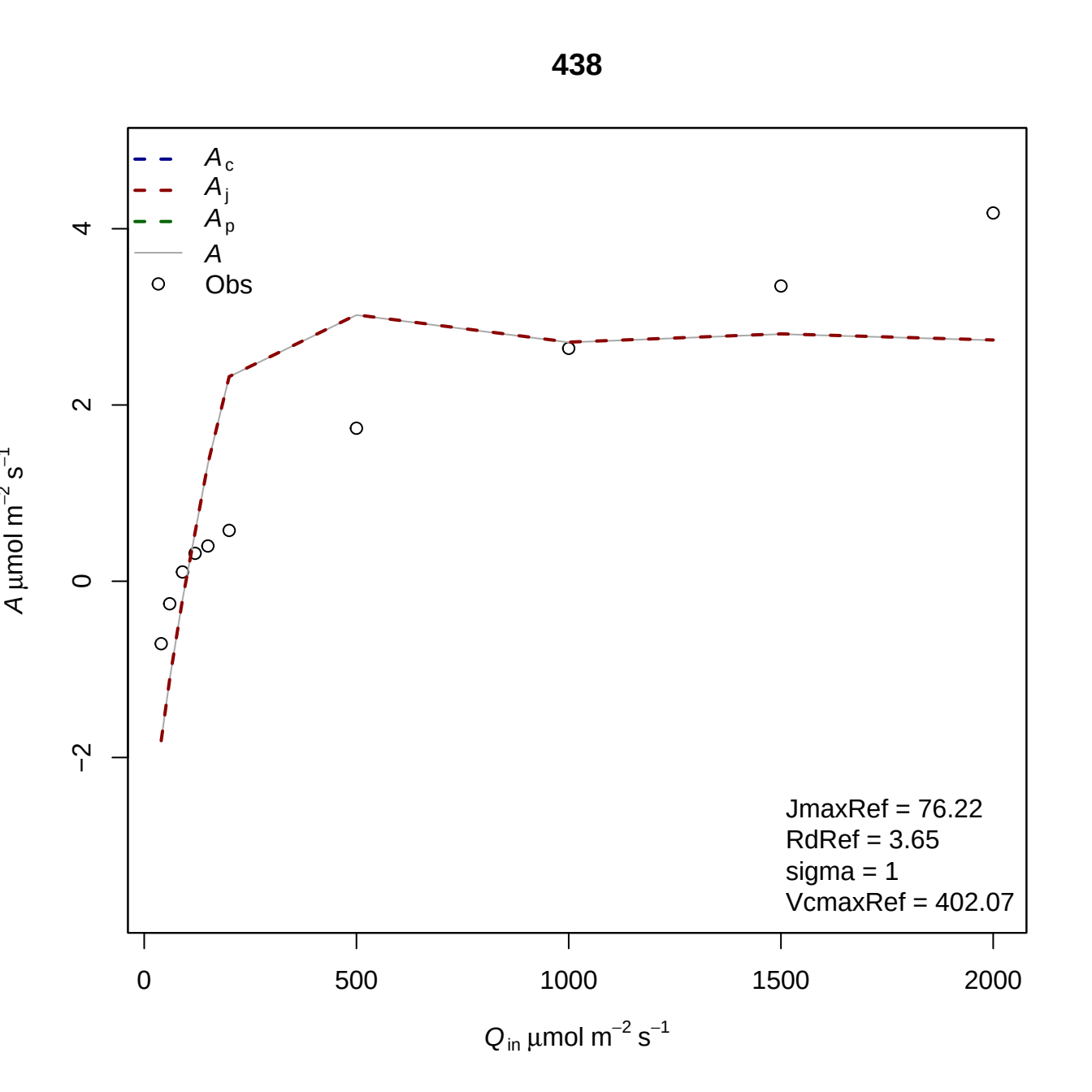

433
A_c
A_j
A_p
A
Obs
JmaxRef = 66.69
RdRef = 2.11
sigma = 0.88
VcmaxRef = 73.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

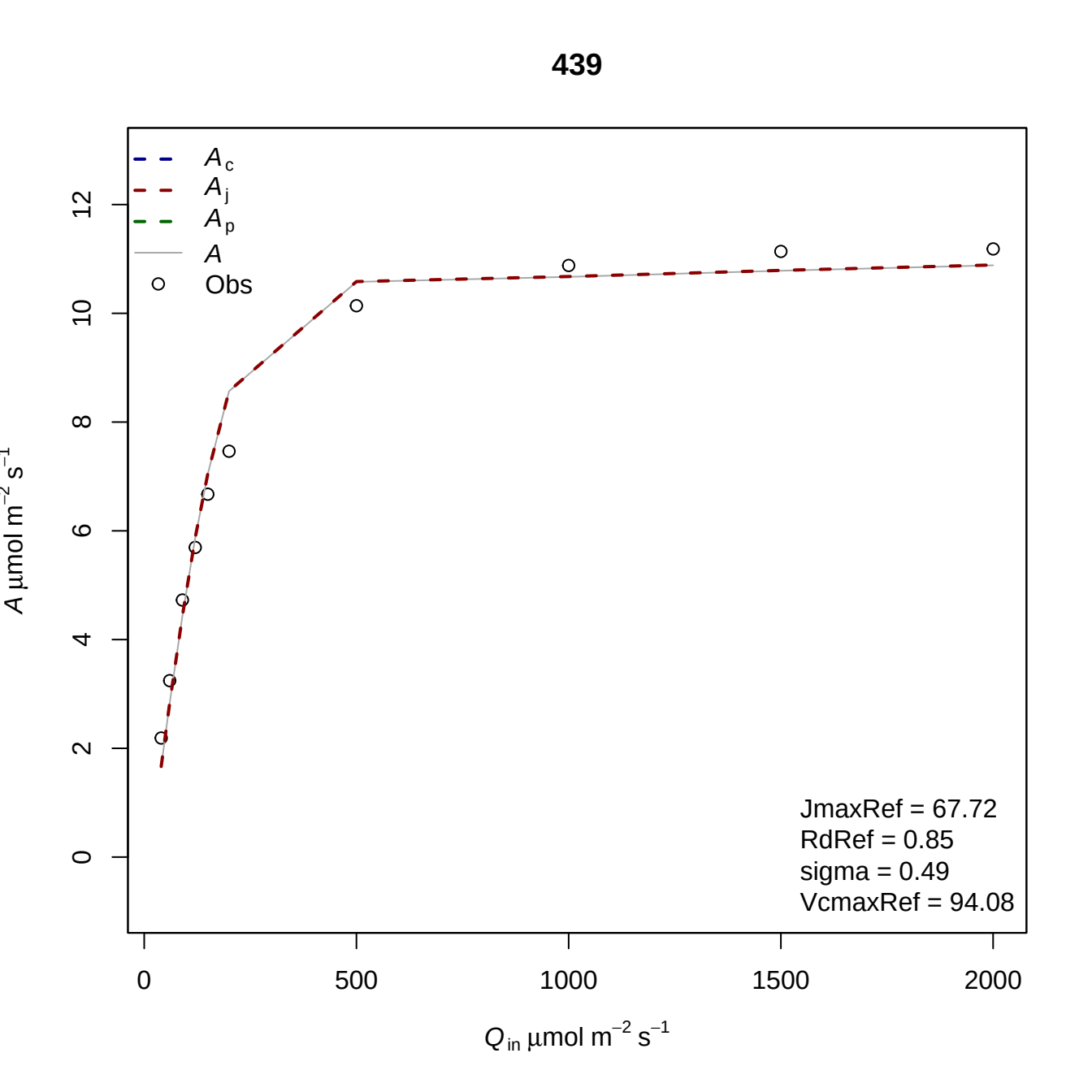

434
A_c
A_j
A_p
A
Obs
JmaxRef = 110.98
RdRef = 1.03
sigma = 0.83
VcmaxRef = 157.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

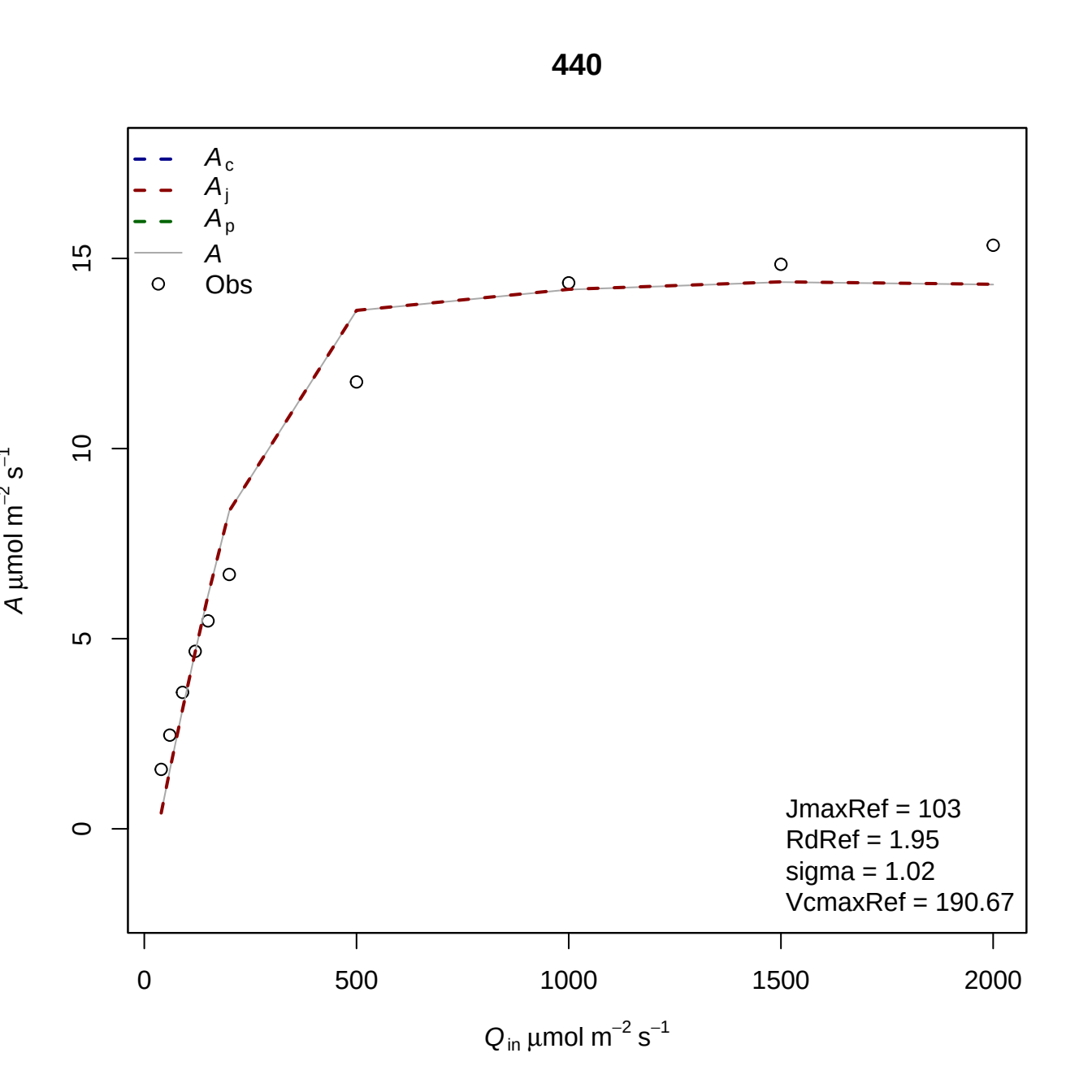

435
A_c
A_j
A_p
A
Obs
JmaxRef = 101.16
RdRef = 1.86
sigma = 0.75
VcmaxRef = 89.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

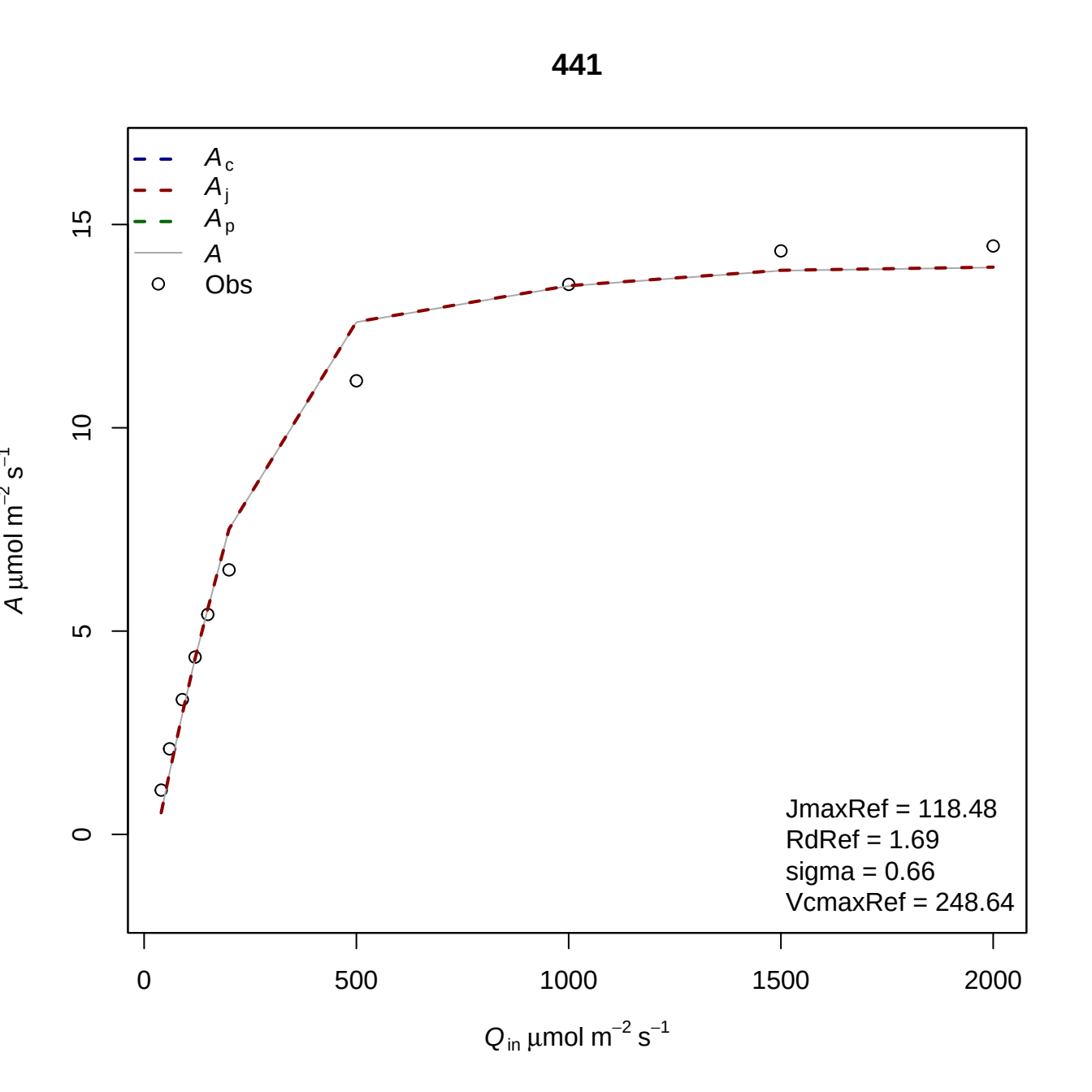

436
A_c
A_j
A_p
A
Obs
JmaxRef = 100.87
RdRef = 2.61
sigma = 0.92
VcmaxRef = 259.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

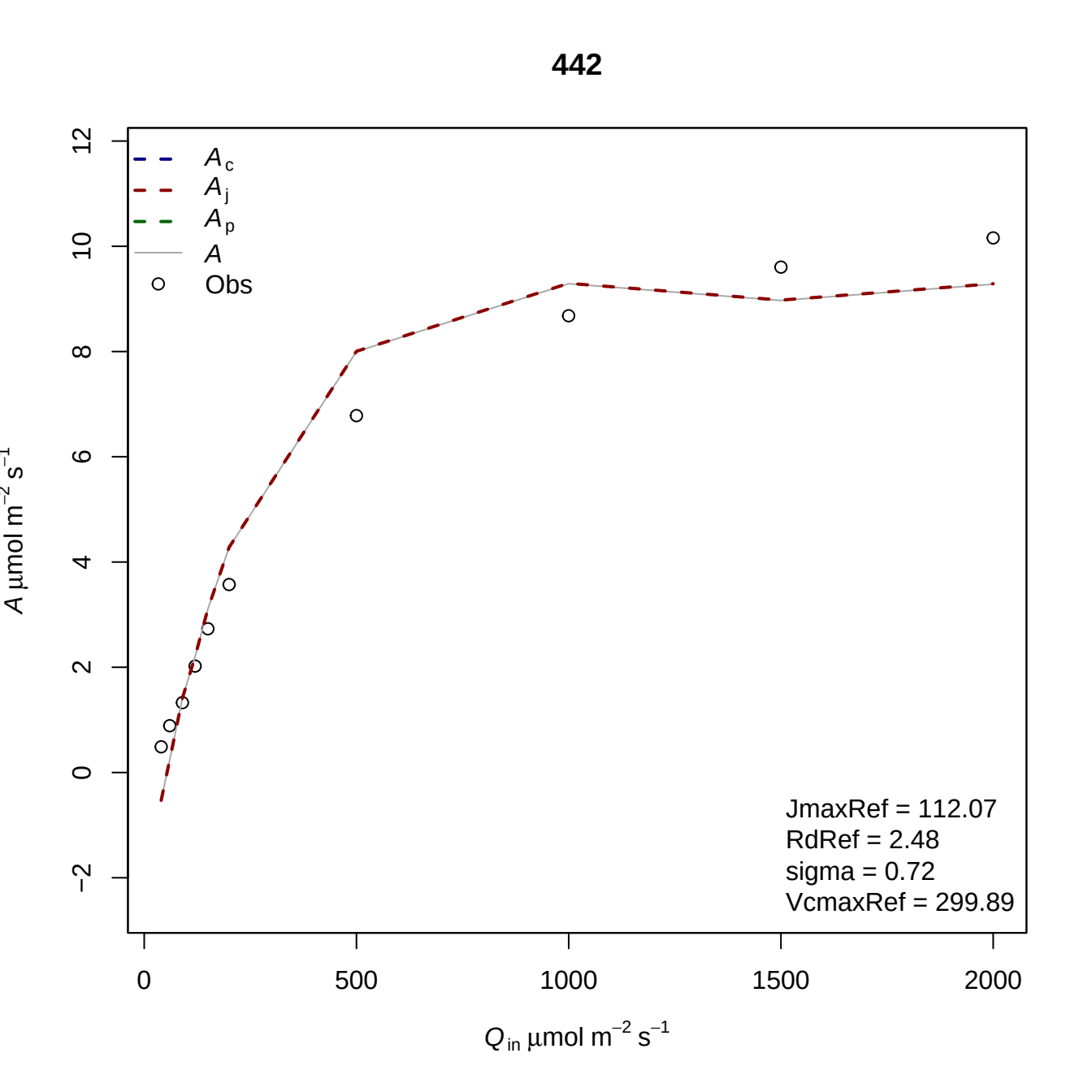

437
A_c
A_j
A_p
A
Obs
JmaxRef = 88.87
RdRef = 2.49
sigma = 0.65
VcmaxRef = 254.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

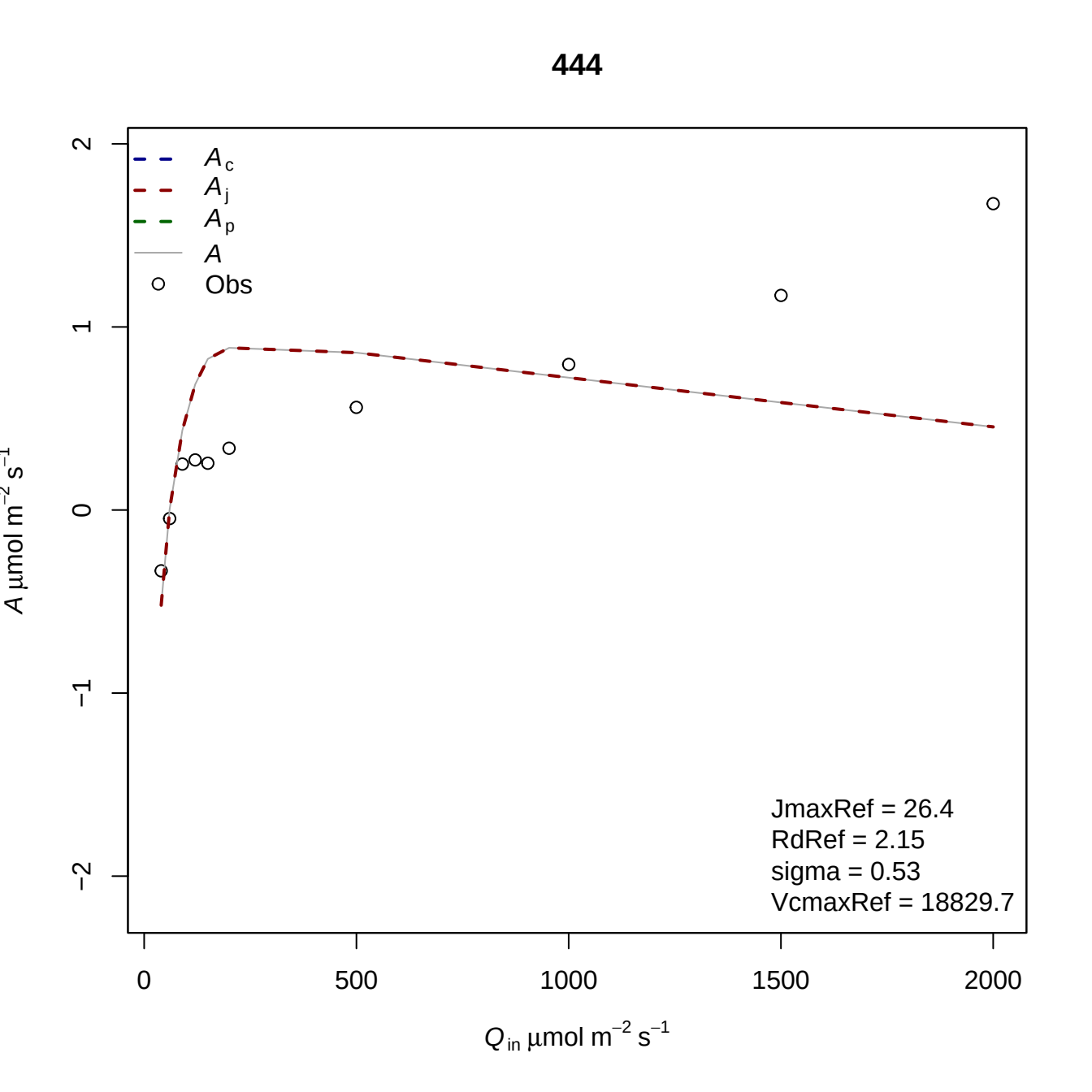

438
A_c
A_j
A_p
A
Obs
JmaxRef = 76.22
RdRef = 3.65
sigma = 1
VcmaxRef = 402.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

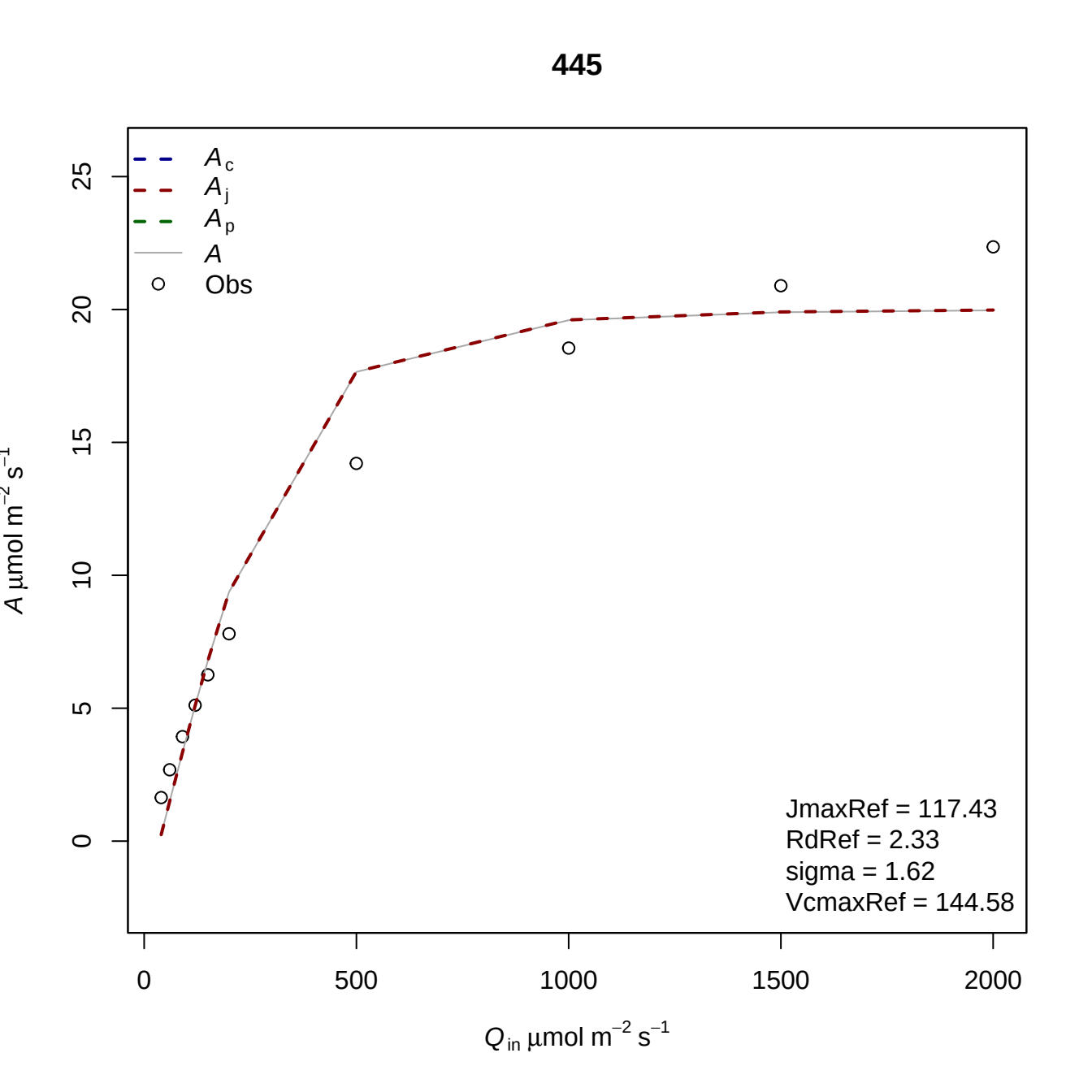

439
A_c
A_j
A_p
A
Obs
JmaxRef = 67.72
RdRef = 0.85
sigma = 0.49
VcmaxRef = 94.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

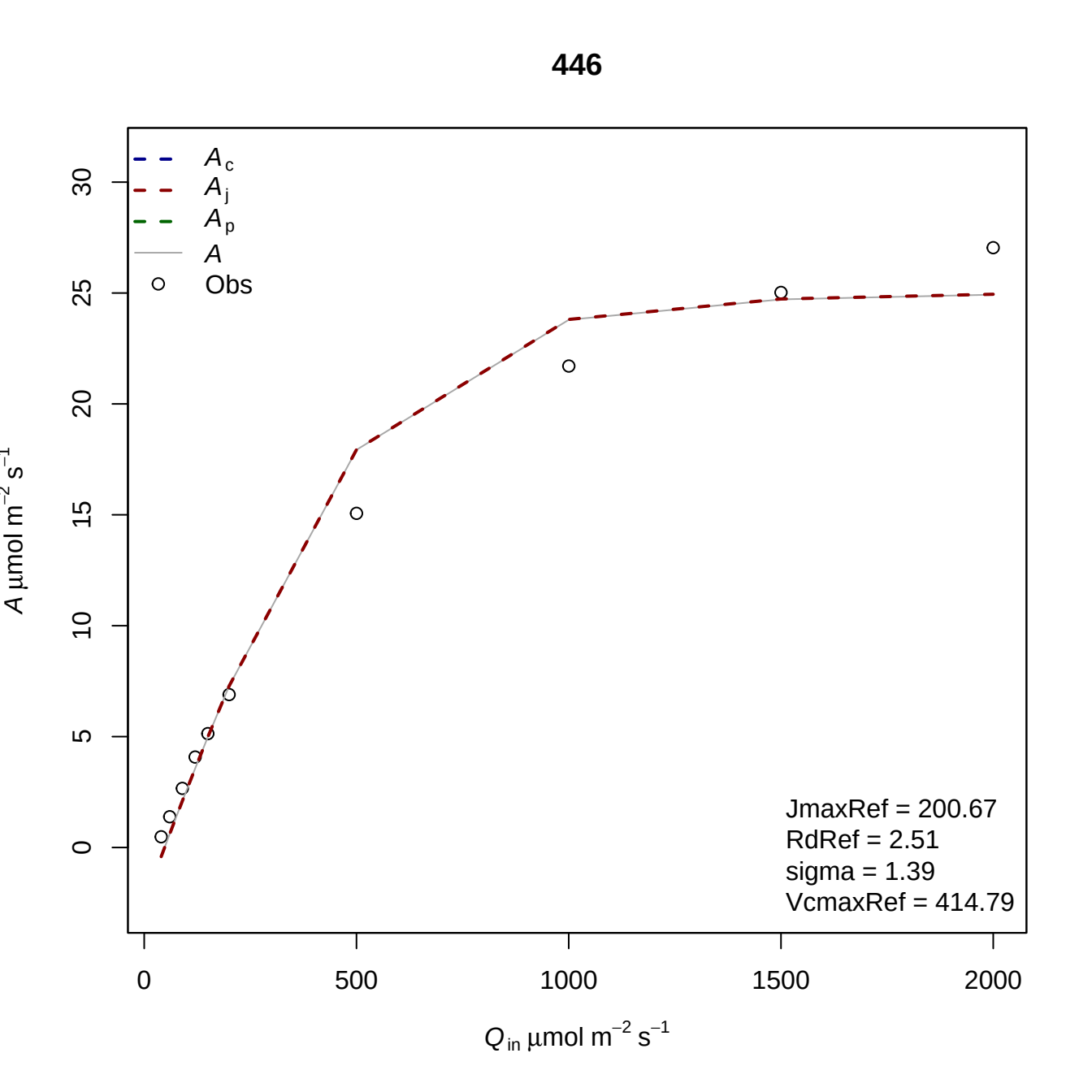

440
A_c
A_j
A_p
A
Obs
JmaxRef = 103
RdRef = 1.95
sigma = 1.02
VcmaxRef = 190.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

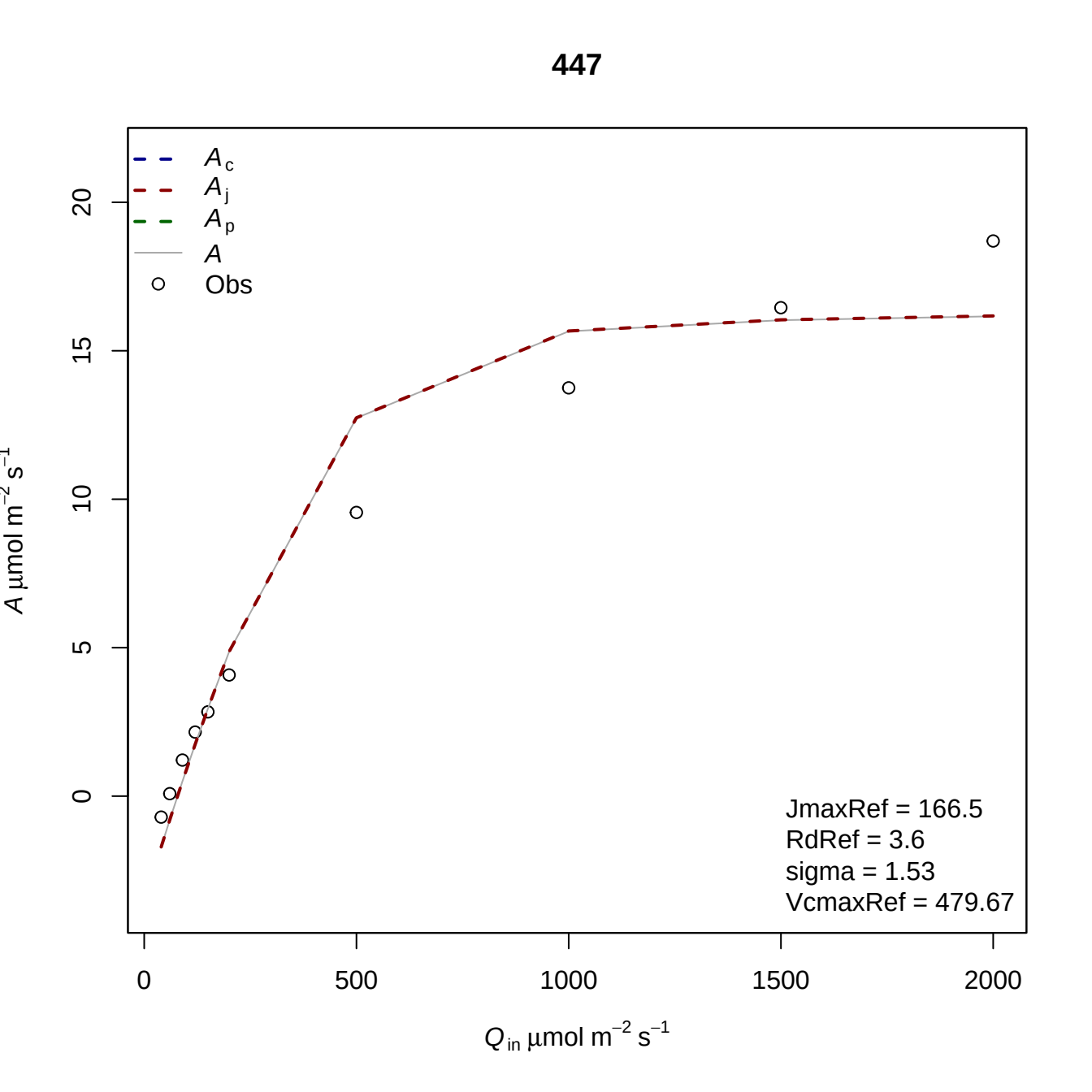

441
A_c
A_j
A_p
A
Obs
JmaxRef = 118.48
RdRef = 1.69
sigma = 0.66
VcmaxRef = 248.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

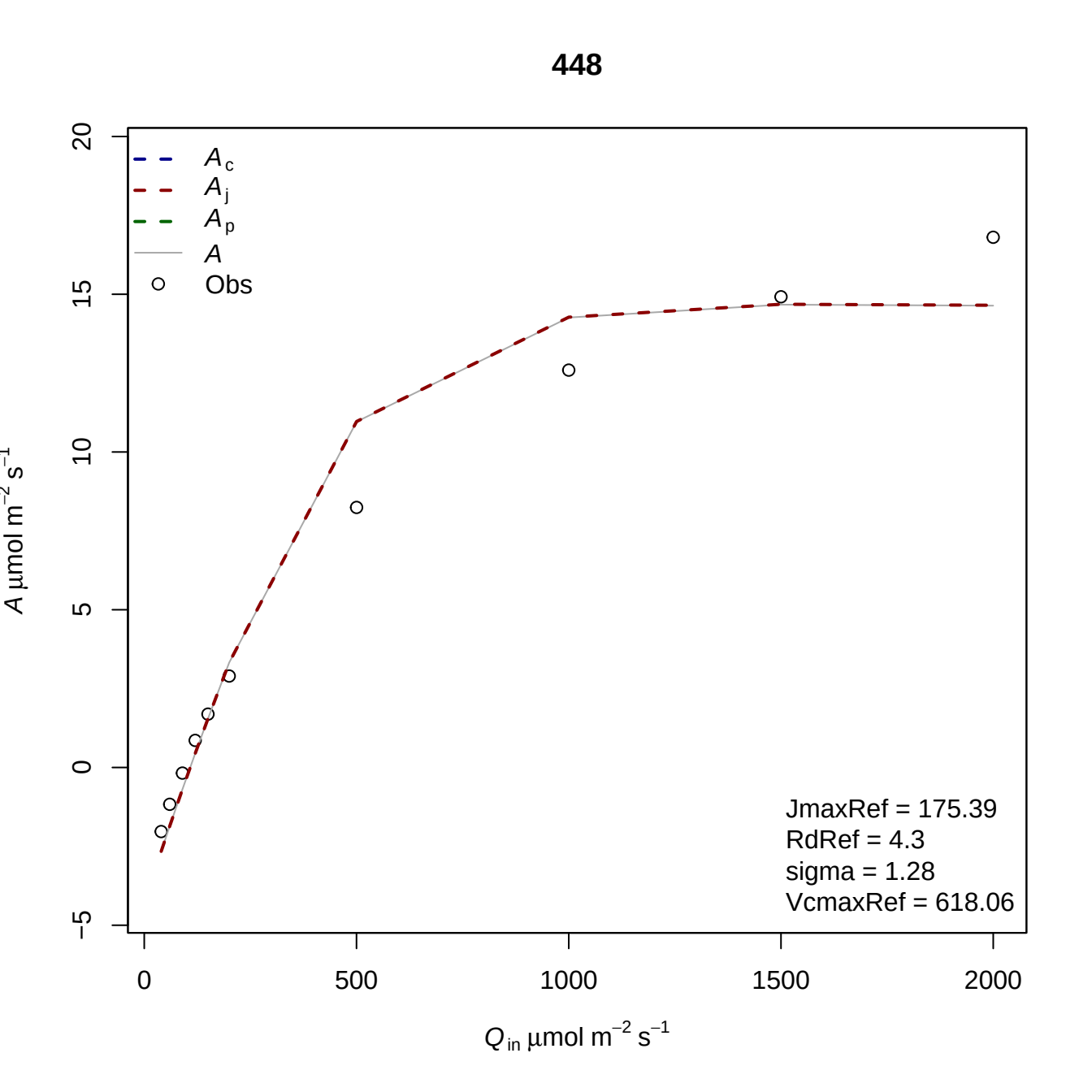

442
A_c
A_j
A_p
A
Obs
JmaxRef = 112.07
RdRef = 2.48
sigma = 0.72
VcmaxRef = 299.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

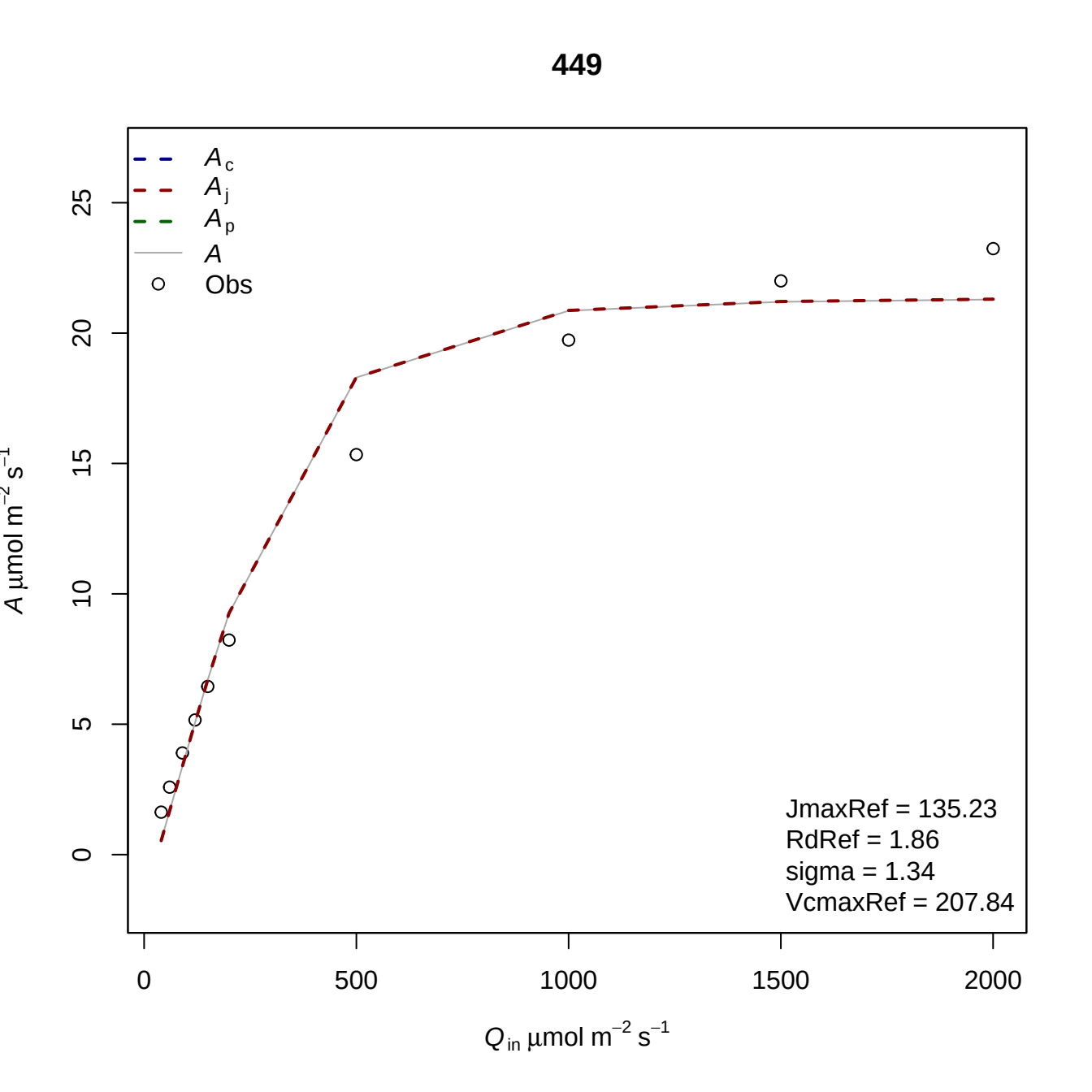

444
A_c
A_j
A_p
A
Obs
JmaxRef = 26.4
RdRef = 2.15
sigma = 0.53
VcmaxRef = 18829.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500
2000

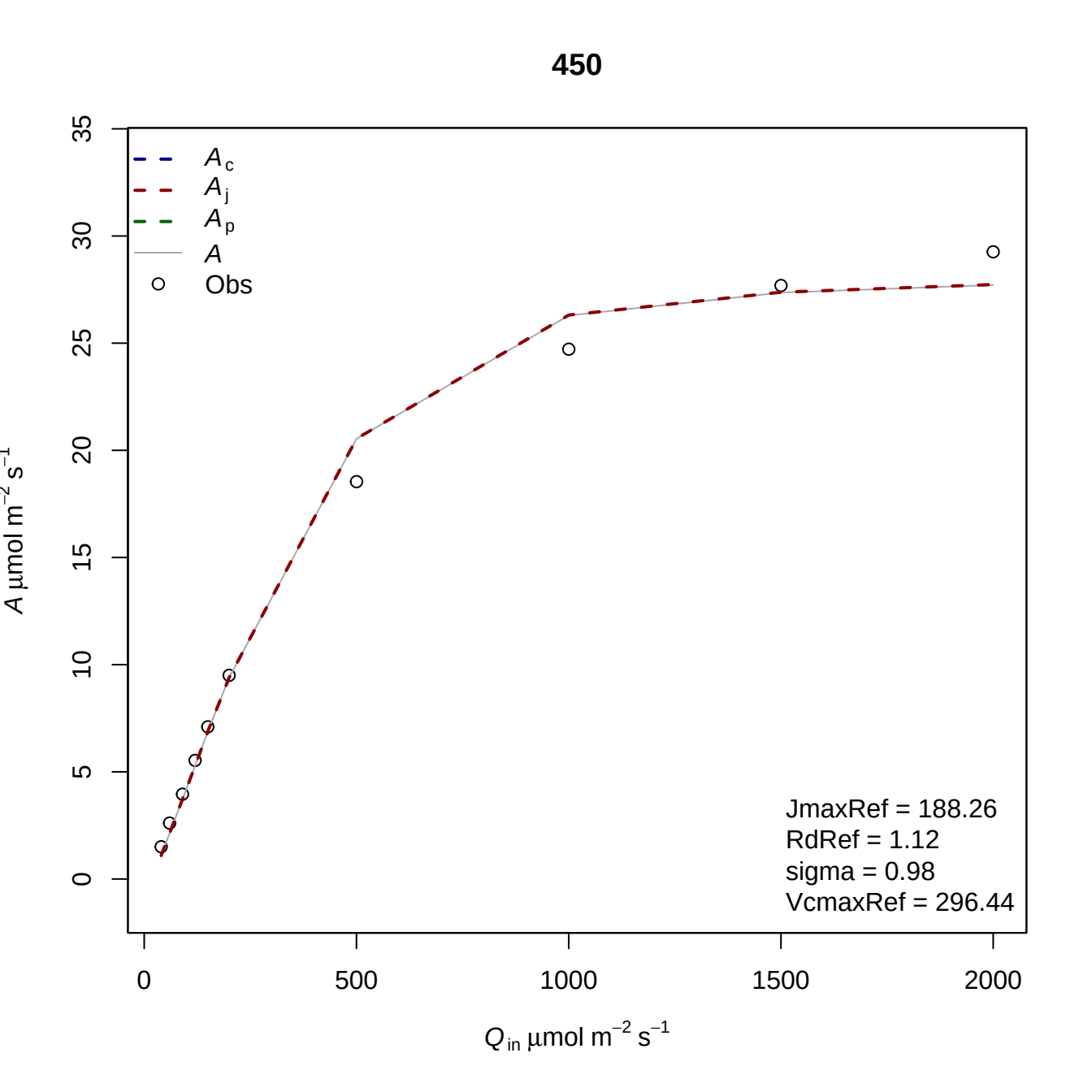

445
A_c
A_j
A_p
A
Obs
JmaxRef = 117.43
RdRef = 2.33
sigma = 1.62
VcmaxRef = 144.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

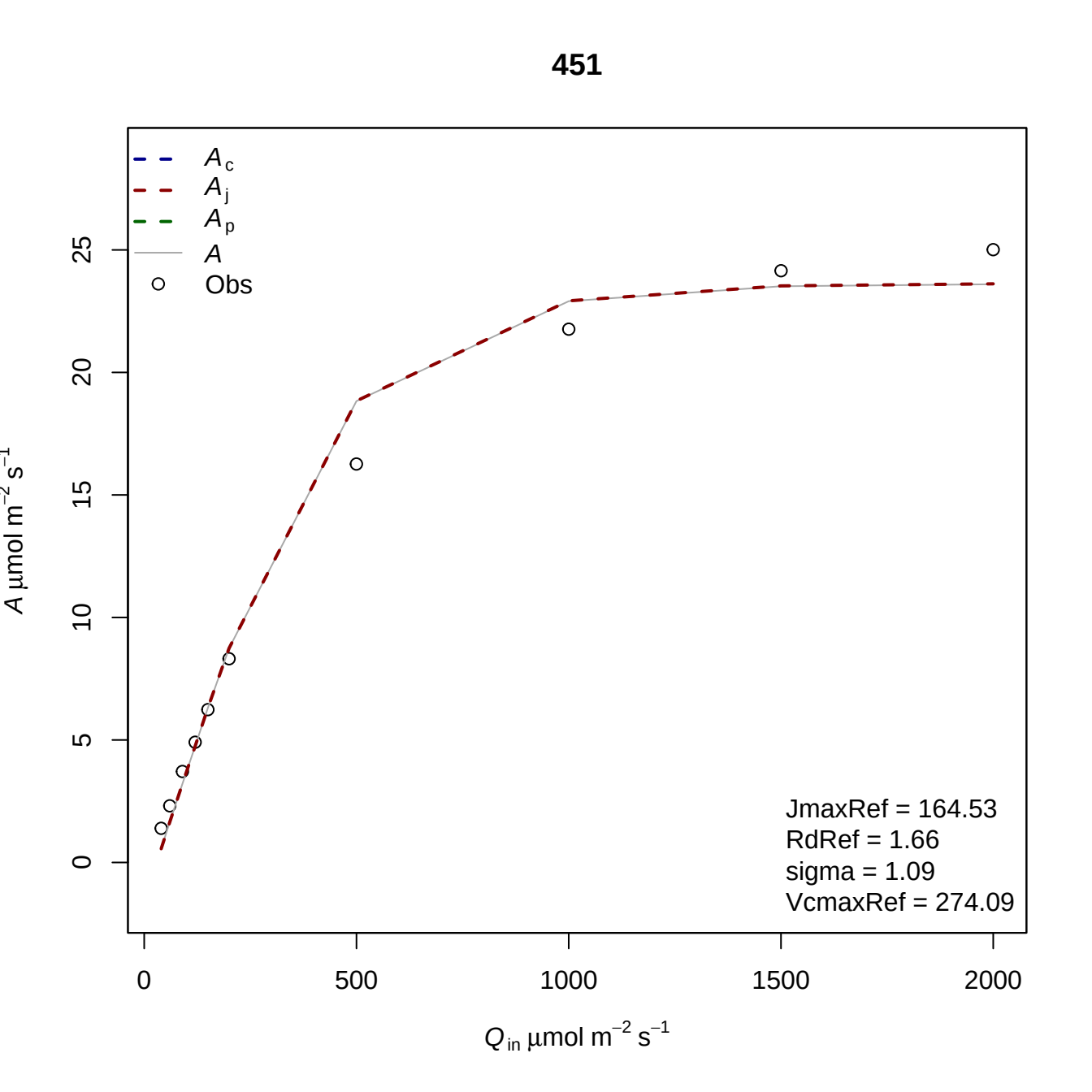

446
A_c
A_j
A_p
A
Obs
JmaxRef = 200.67
RdRef = 2.51
sigma = 1.39
VcmaxRef = 414.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

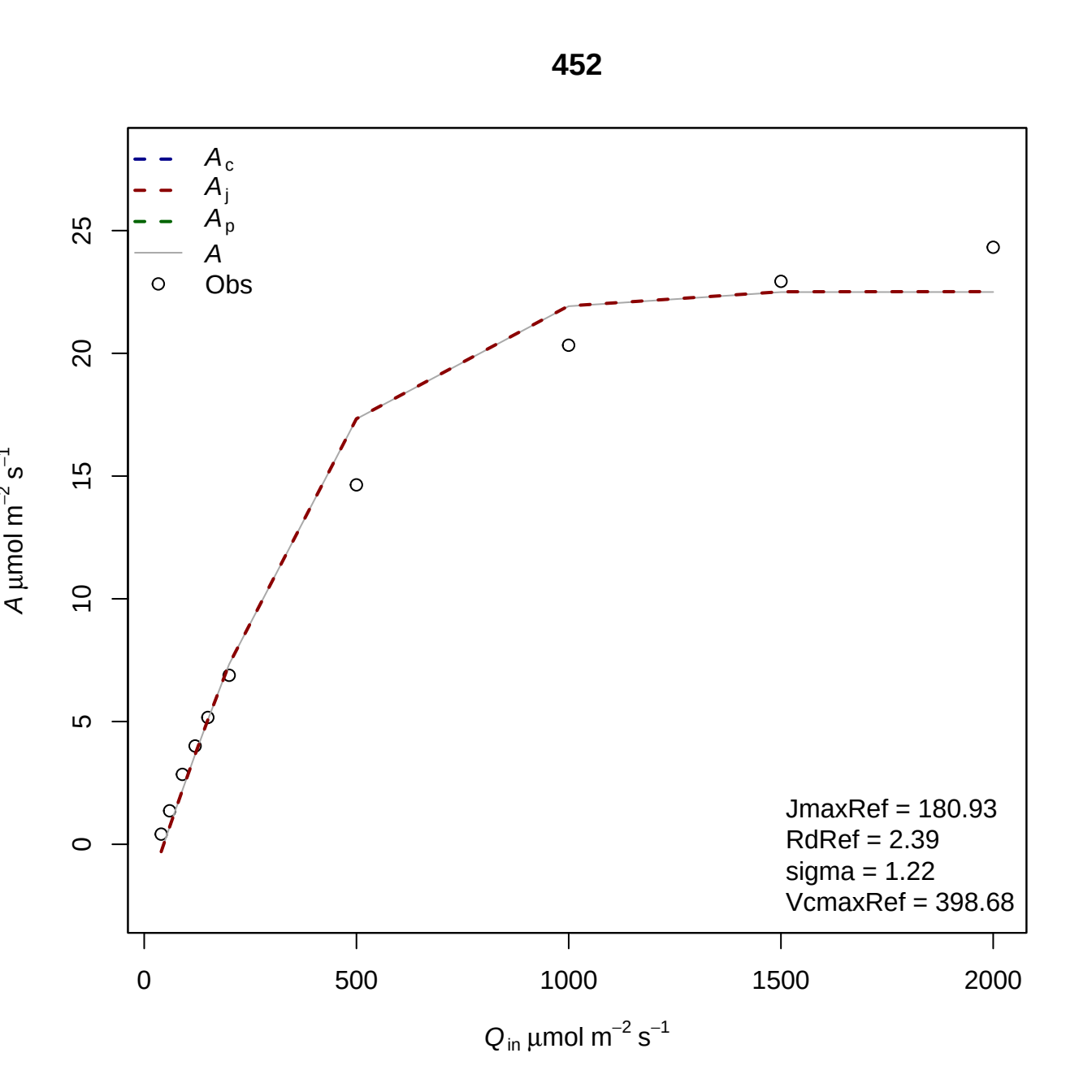

447
A_c
A_j
A_p
A
Obs
JmaxRef = 166.5
RdRef = 3.6
sigma = 1.53
VcmaxRef = 479.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

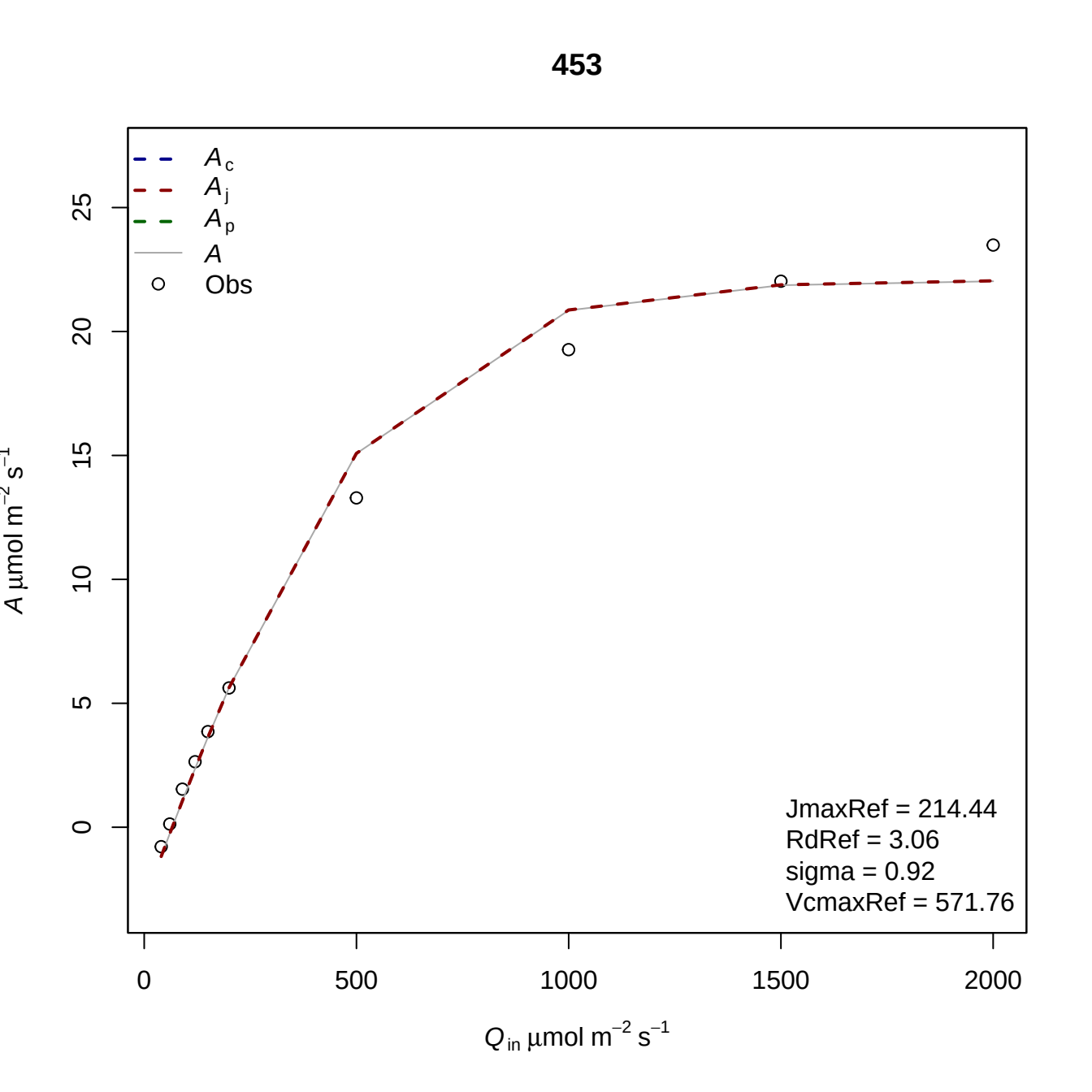

448
A_c
A_j
A_p
A
Obs
JmaxRef = 175.39
RdRef = 4.3
sigma = 1.28
VcmaxRef = 618.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

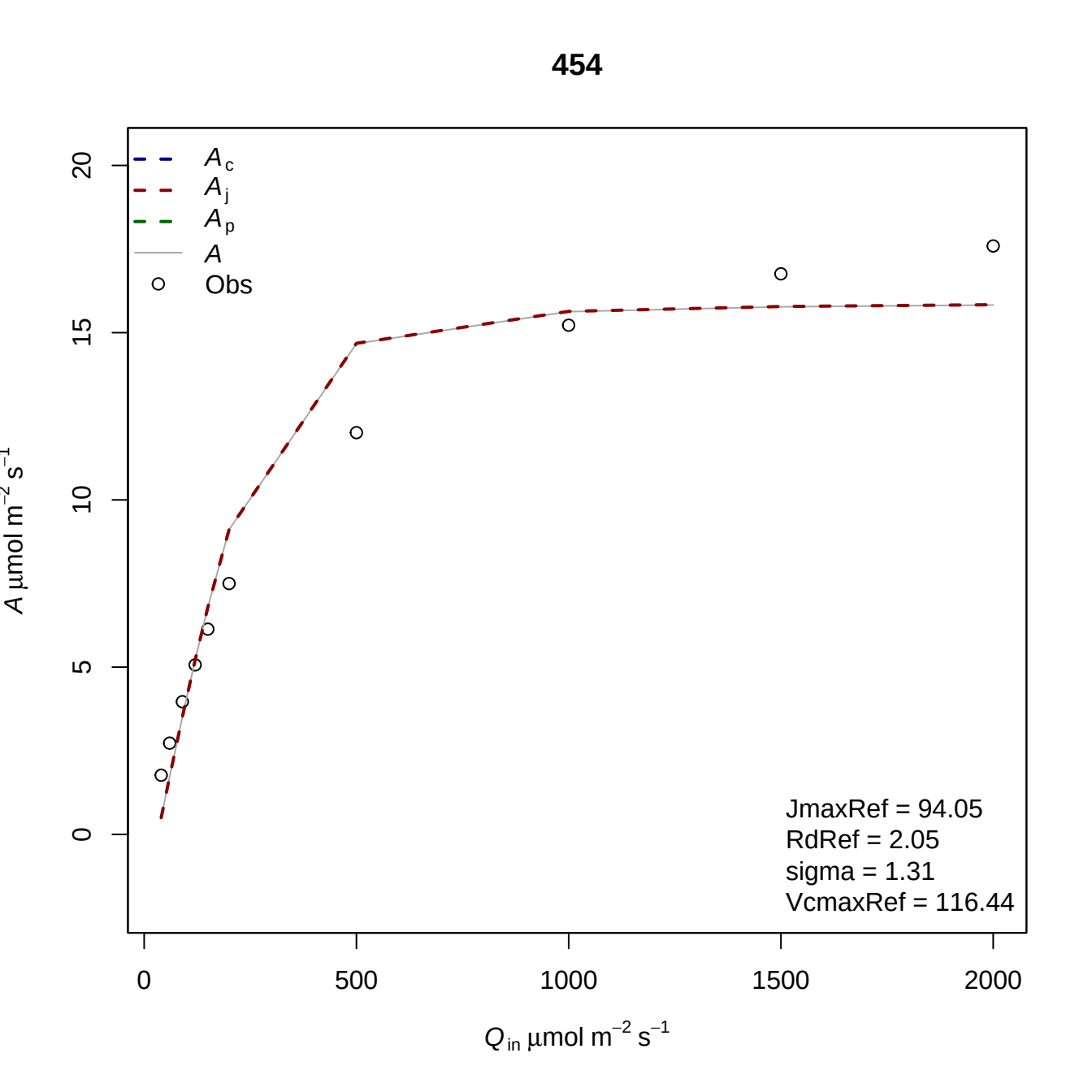

449
A_c
A_j
A_p
A
Obs
JmaxRef = 135.23
RdRef = 1.86
sigma = 1.34
VcmaxRef = 207.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

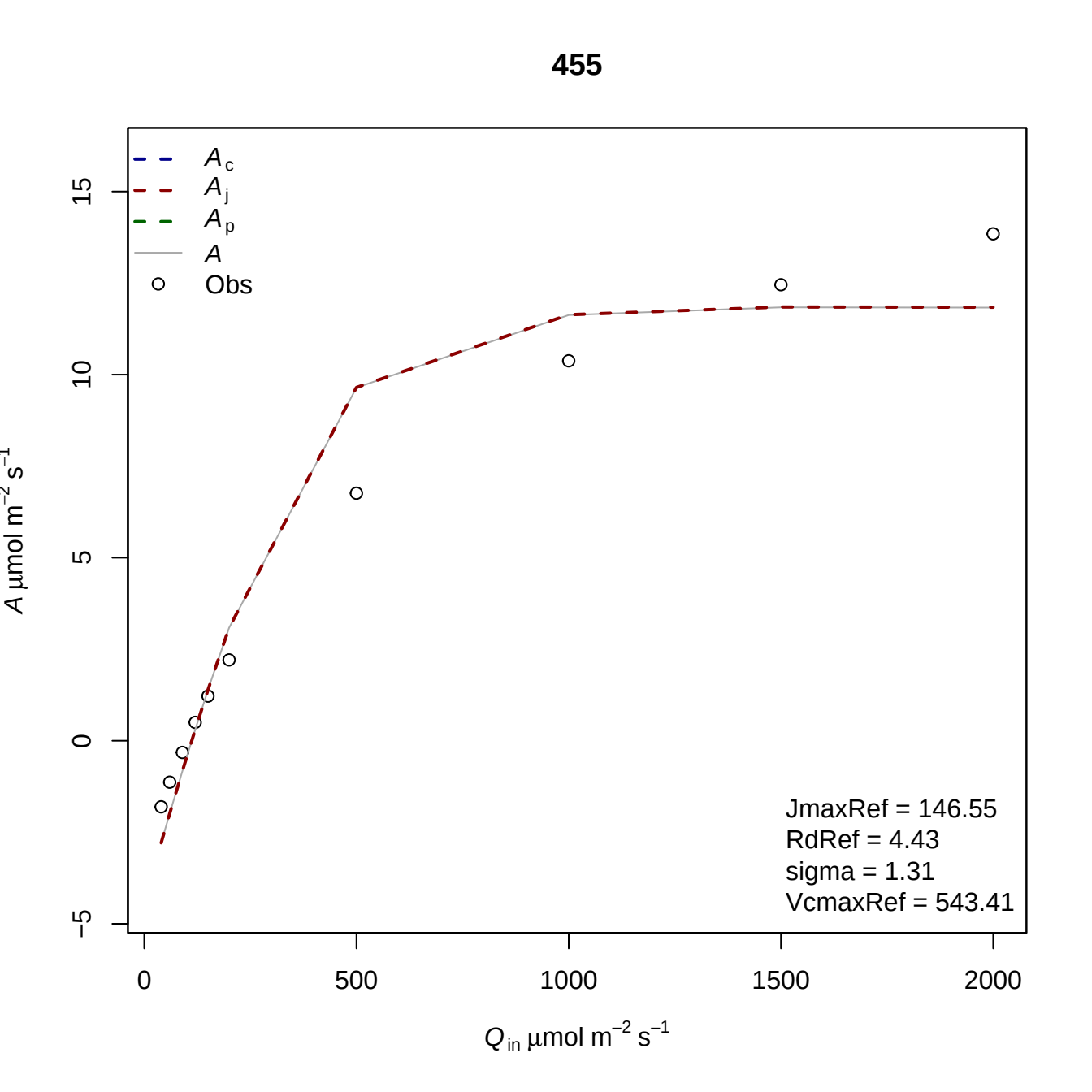

450
A_c
A_j
A_p
A
Obs
JmaxRef = 188.26
RdRef = 1.12
sigma = 0.98
VcmaxRef = 296.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

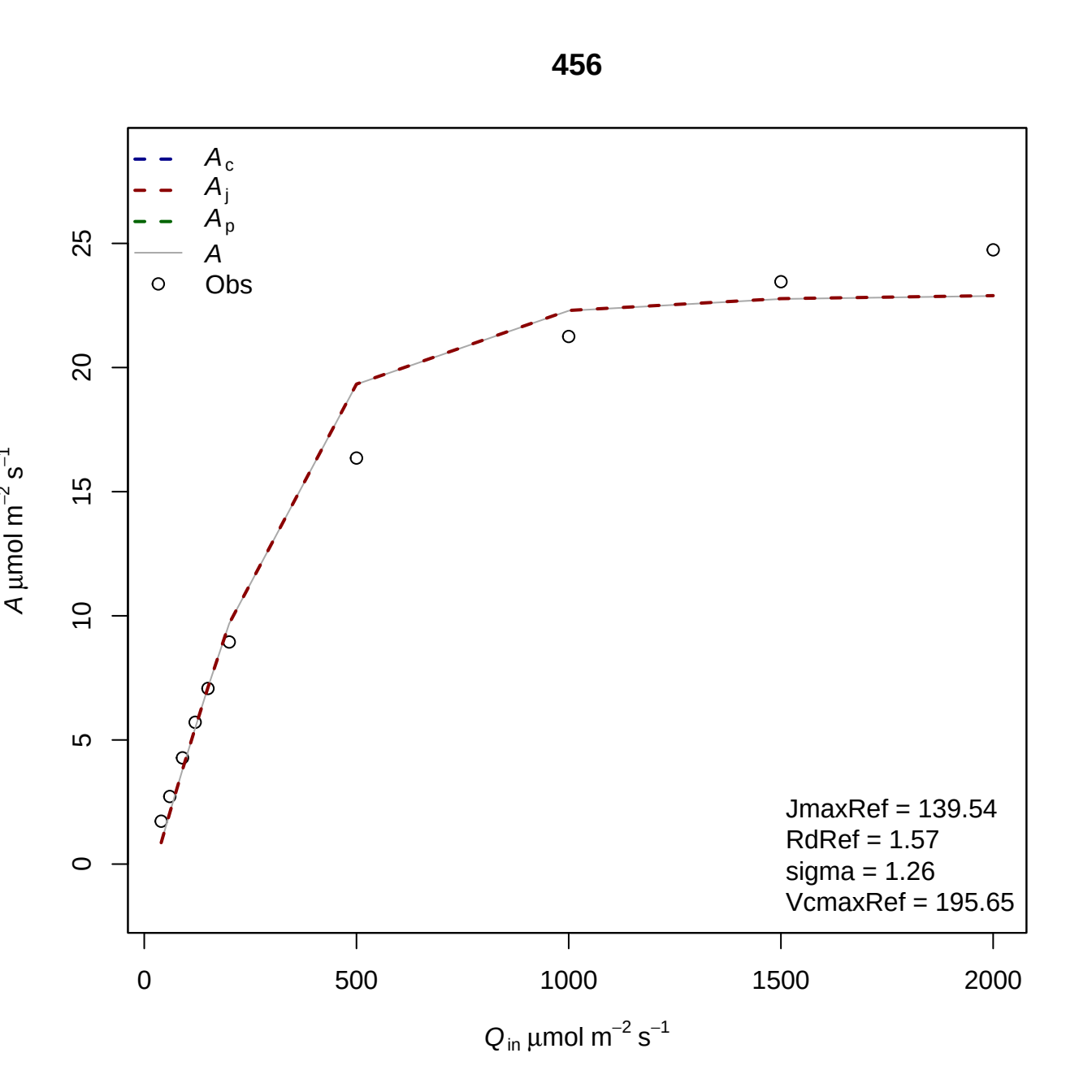

451
A_c
A_j
A_p
A
Obs
JmaxRef = 164.53
RdRef = 1.66
sigma = 1.09
VcmaxRef = 274.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

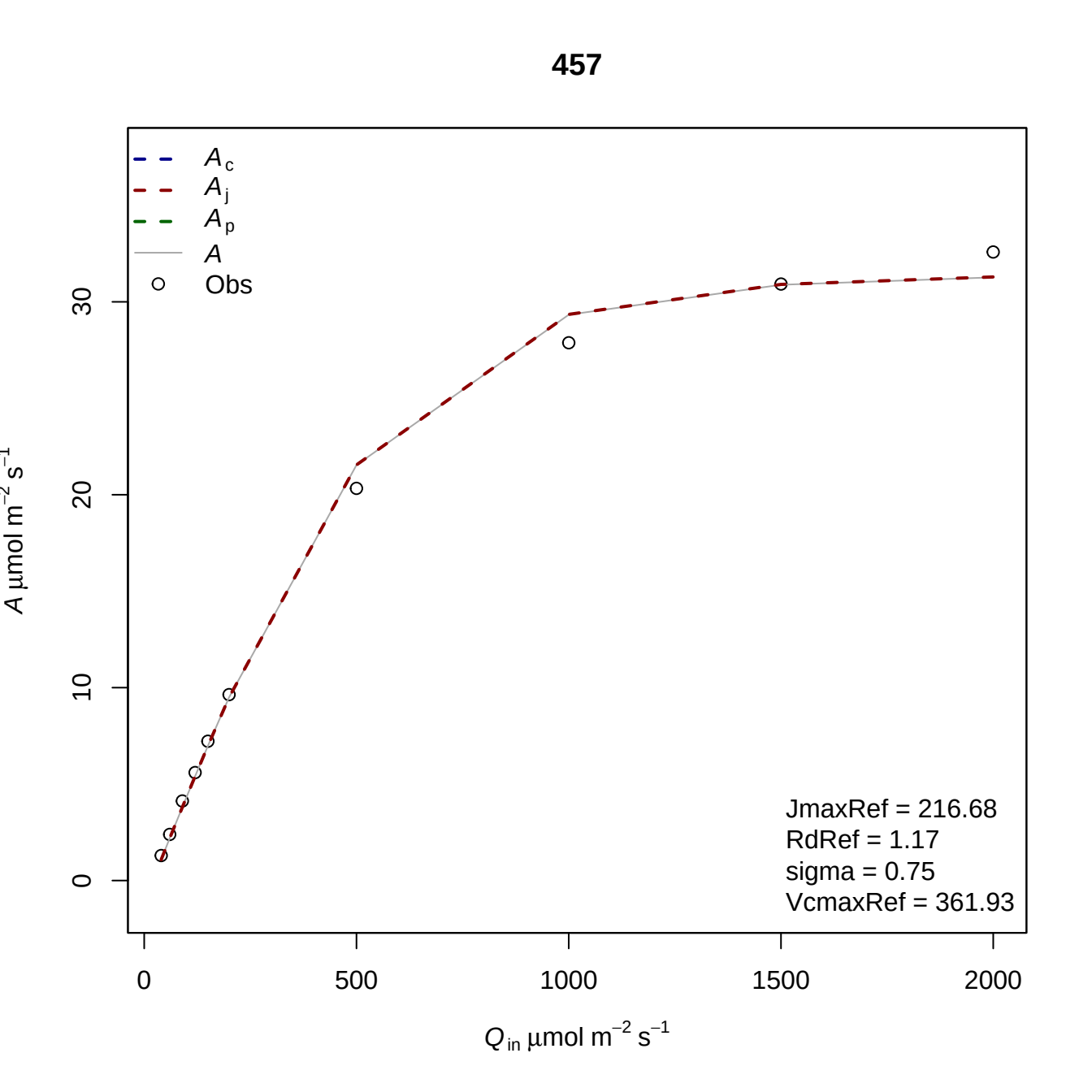

452
A_c
A_j
A_p
A
Obs
JmaxRef = 180.93
RdRef = 2.39
sigma = 1.22
VcmaxRef = 398.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

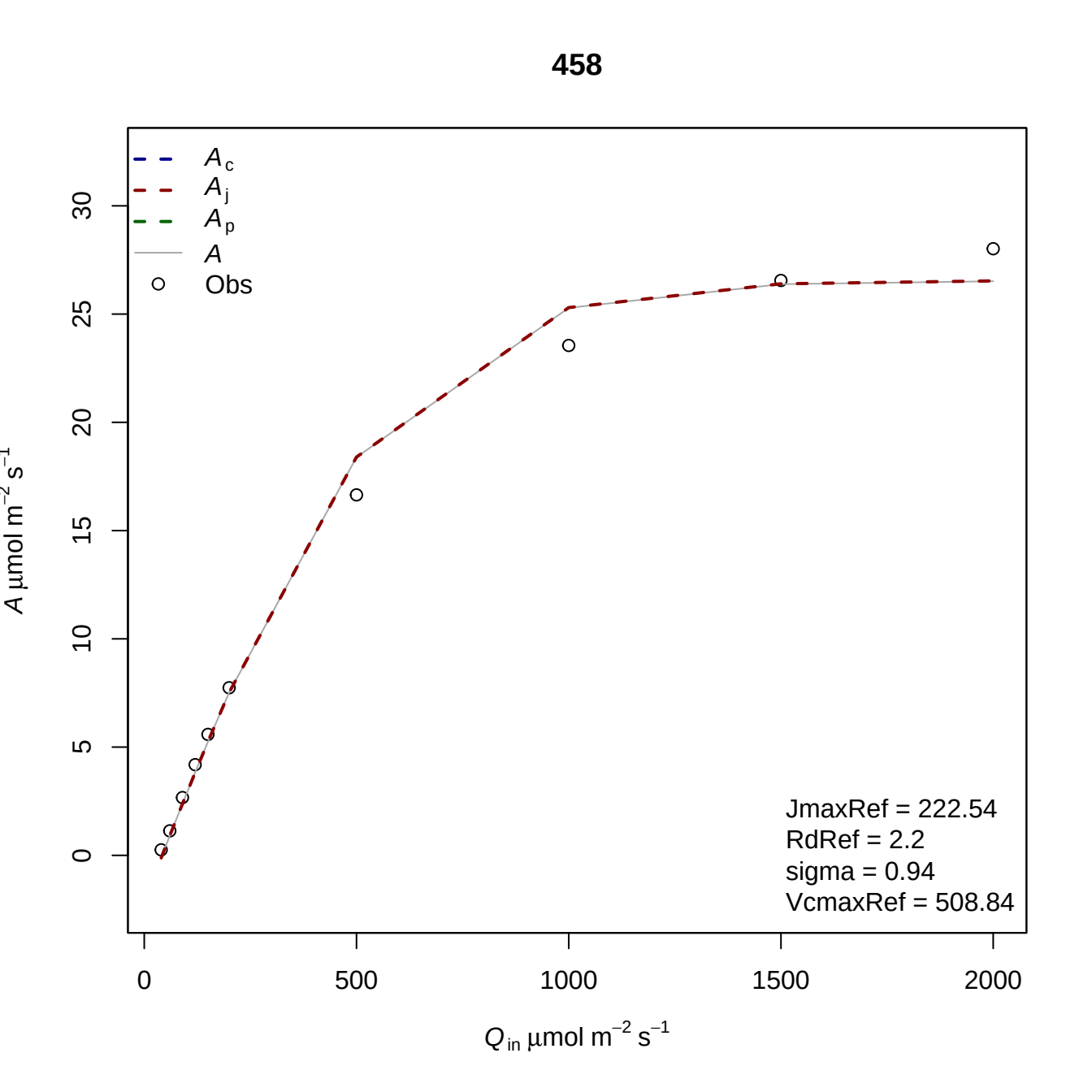

453
A_c
A_j
A_p
A
Obs
JmaxRef = 214.44
RdRef = 3.06
sigma = 0.92
VcmaxRef = 571.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

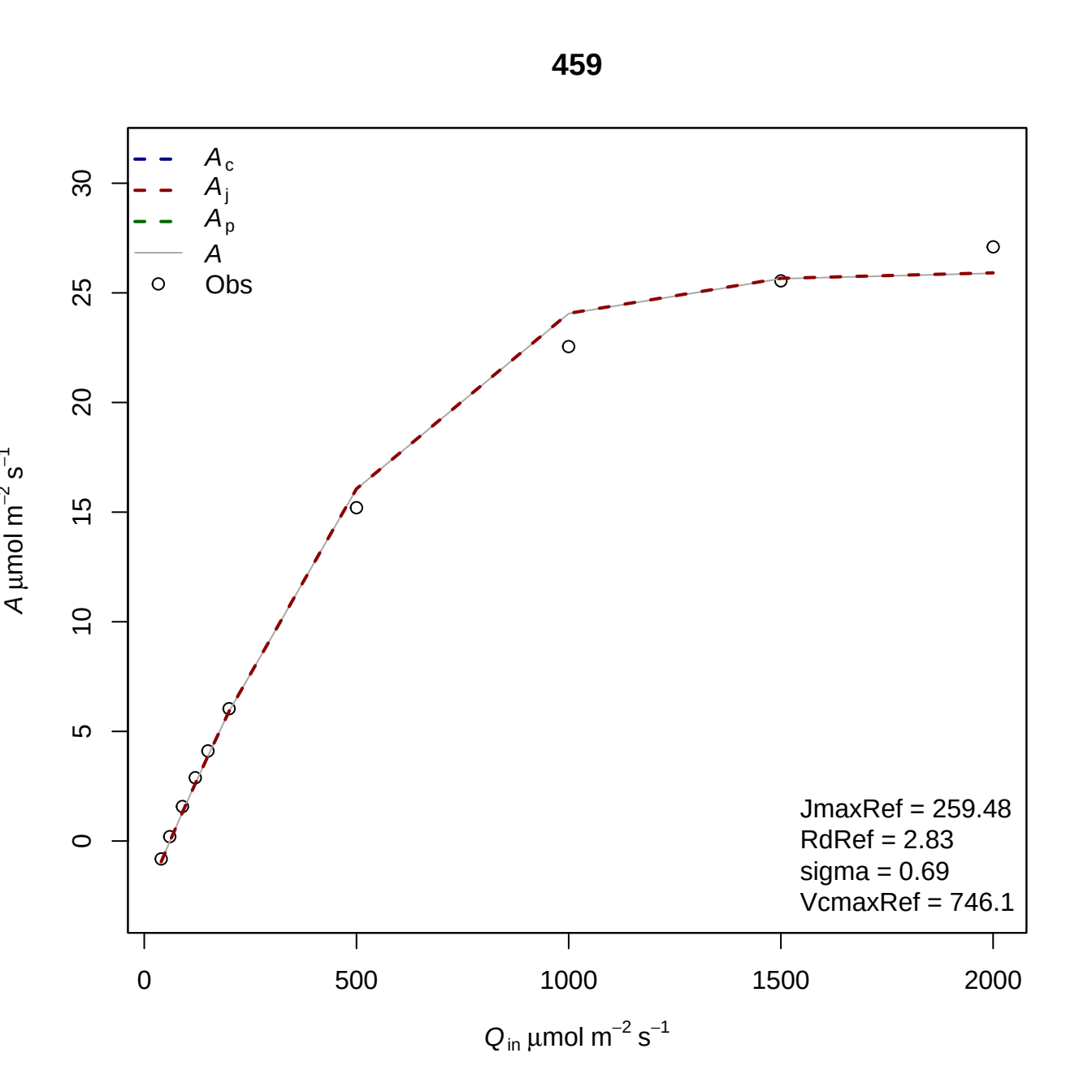

454
A_c
A_j
A_p
A
Obs
JmaxRef = 94.05
RdRef = 2.05
sigma = 1.31
VcmaxRef = 116.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

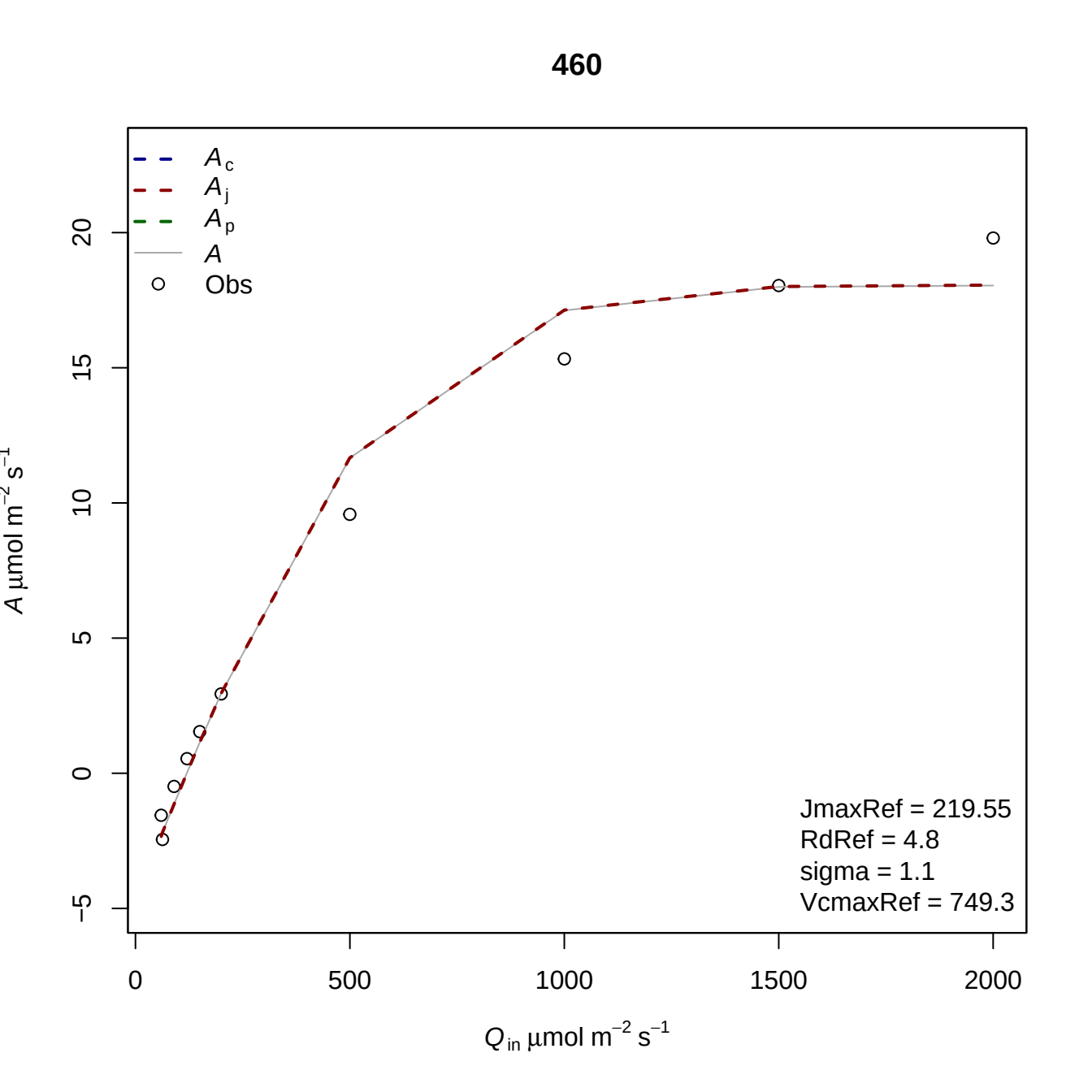

455
A_c
A_j
A_p
A
Obs
JmaxRef = 146.55
RdRef = 4.43
sigma = 1.31
VcmaxRef = 543.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

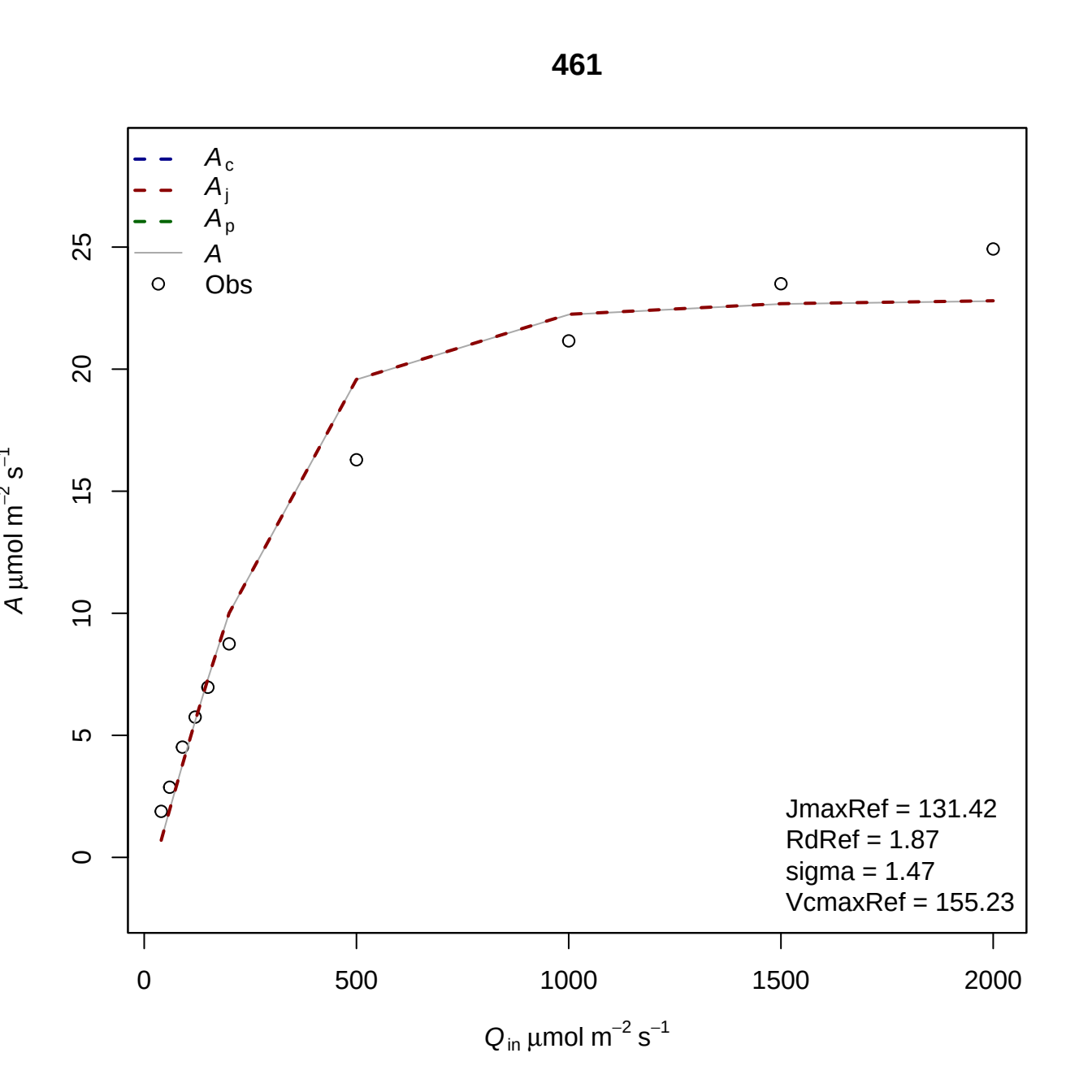

456
A_c
A_j
A_p
A
Obs
JmaxRef = 139.54
RdRef = 1.57
sigma = 1.26
VcmaxRef = 195.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

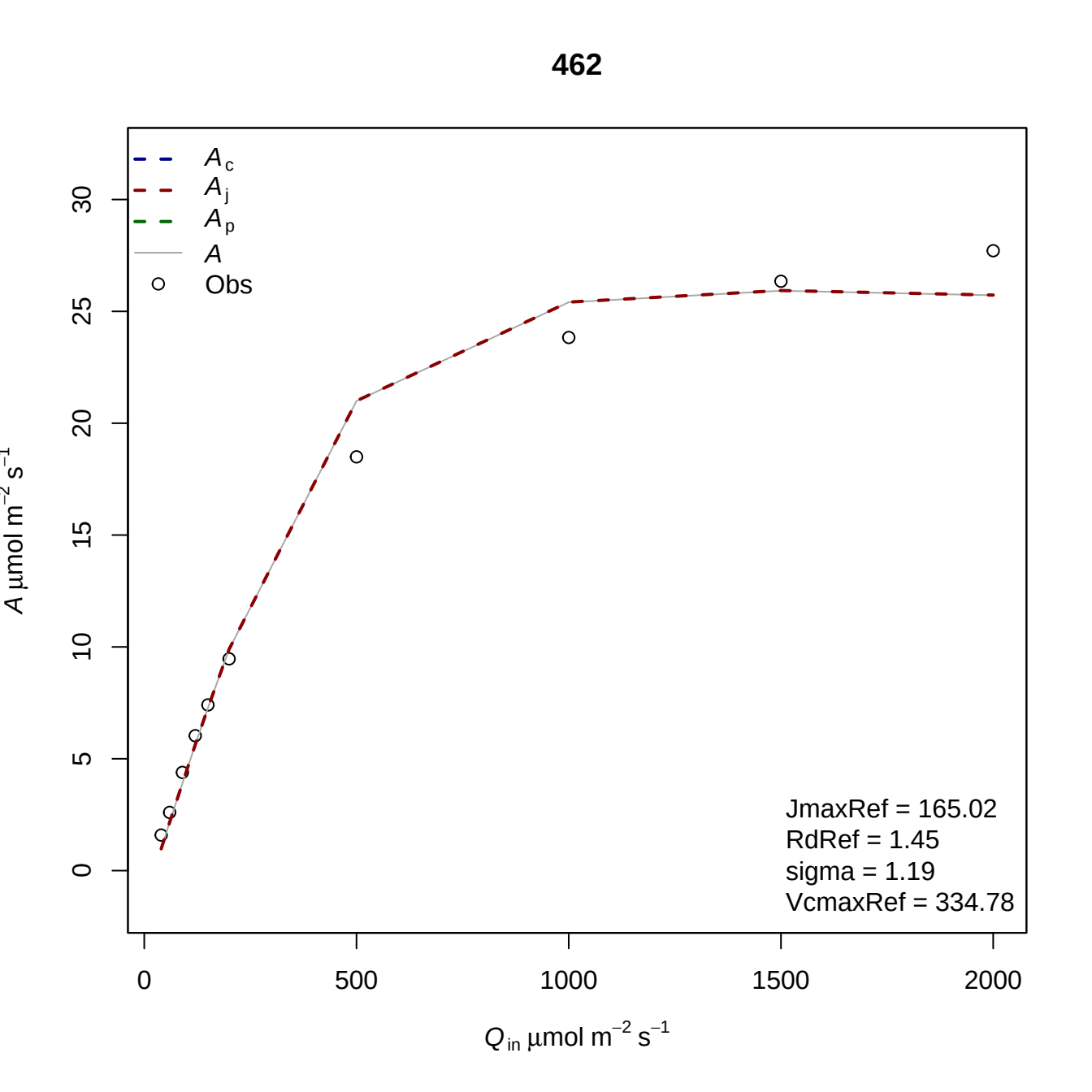

457
A_c
A_j
A_p
A
Obs
JmaxRef = 216.68
RdRef = 1.17
sigma = 0.75
VcmaxRef = 361.93
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

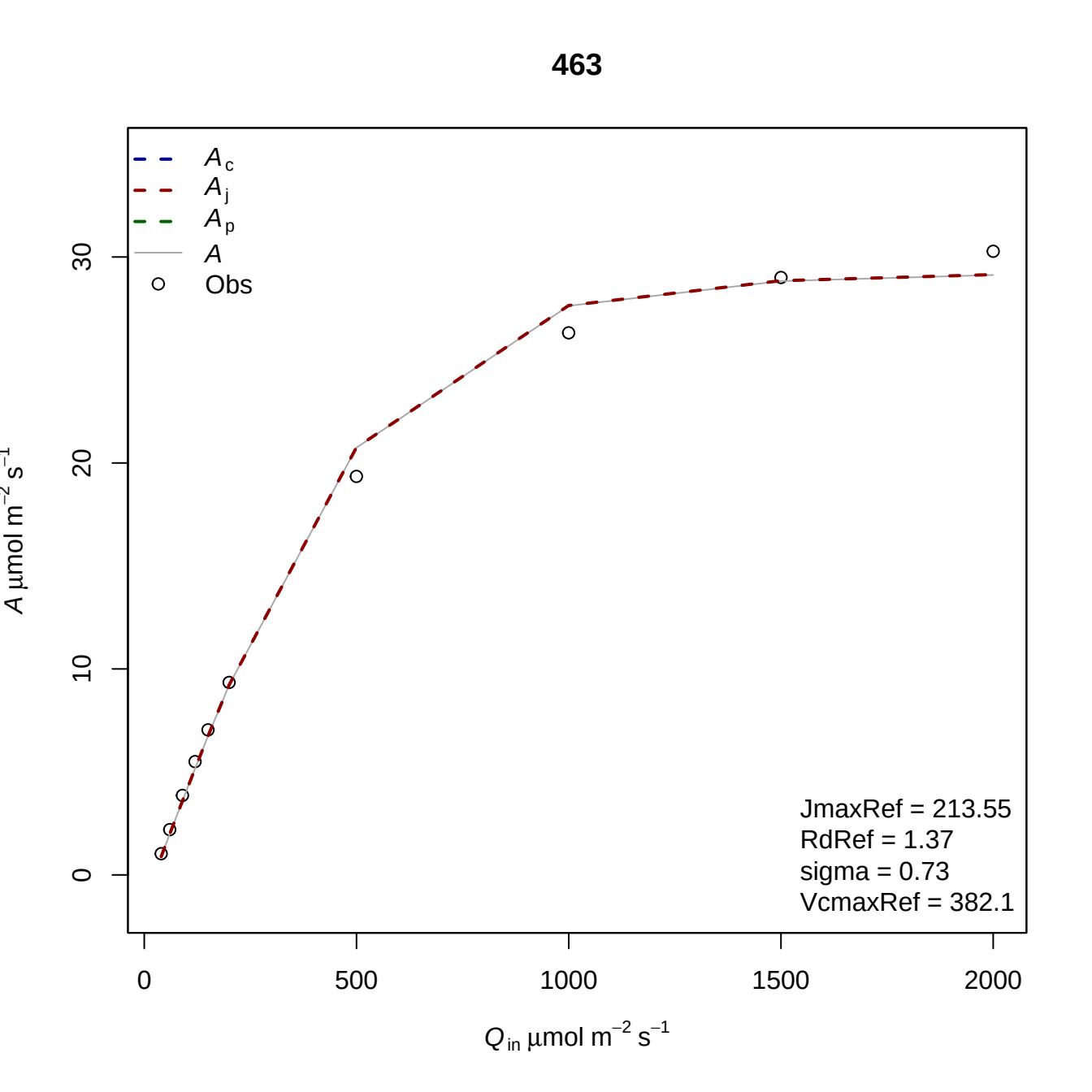

458
A_c
A_j
A_p
A
Obs
JmaxRef = 222.54
RdRef = 2.2
sigma = 0.94
VcmaxRef = 508.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

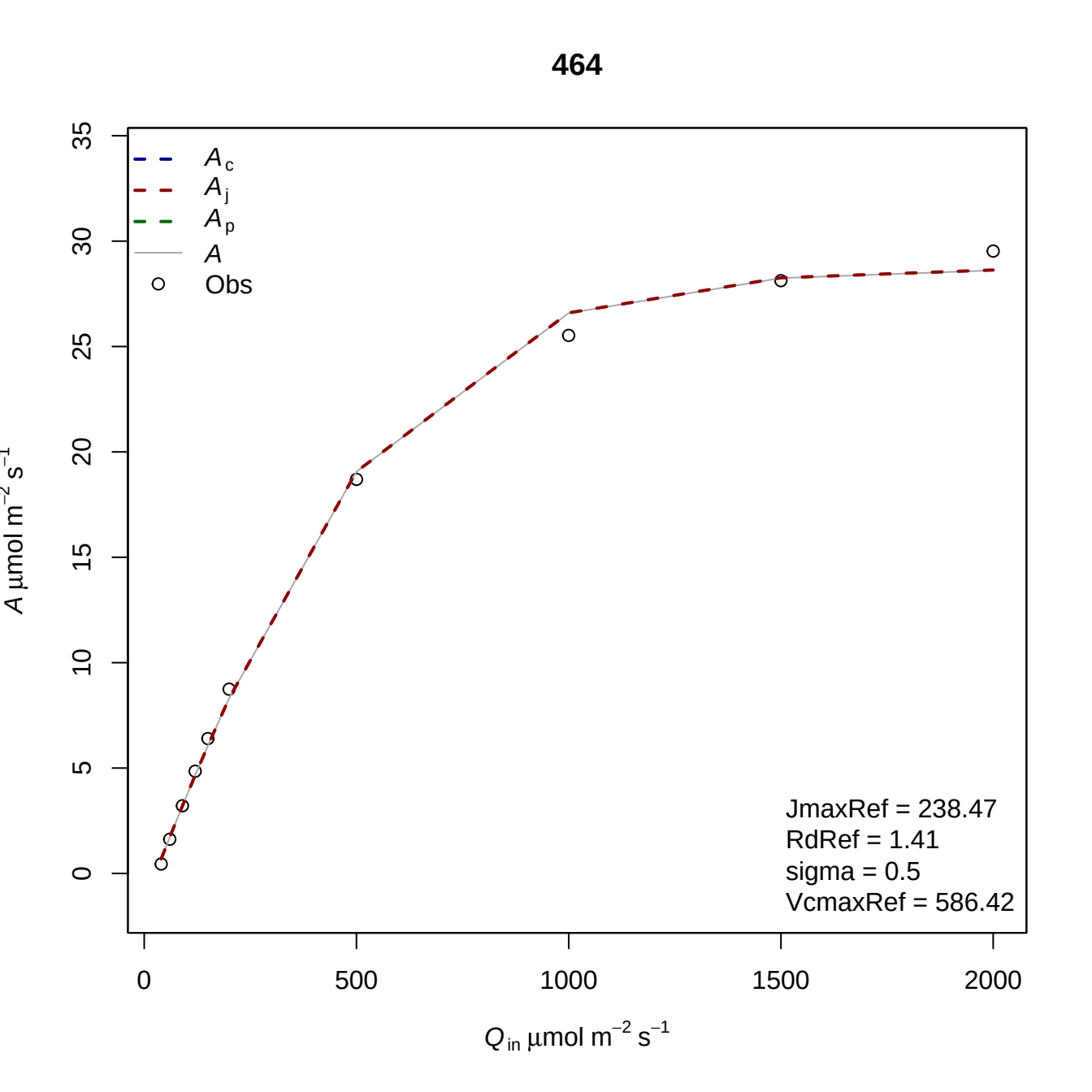

459
A_c
A_j
A_p
A
Obs
JmaxRef = 259.48
RdRef = 2.83
sigma = 0.69
VcmaxRef = 746.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

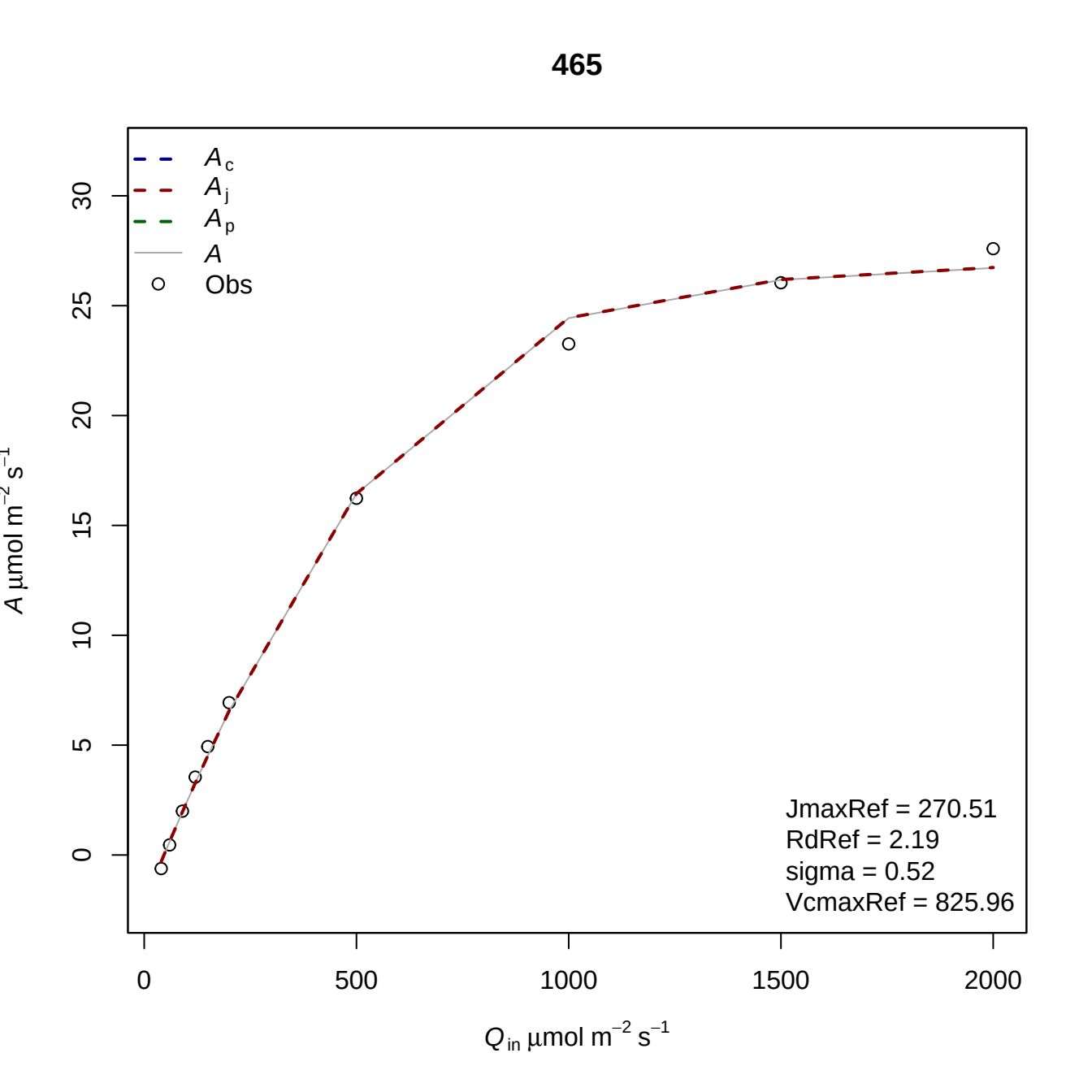

460
A_c
A_j
A_p
A
Obs
JmaxRef = 219.55
RdRef = 4.8
sigma = 1.1
VcmaxRef = 749.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

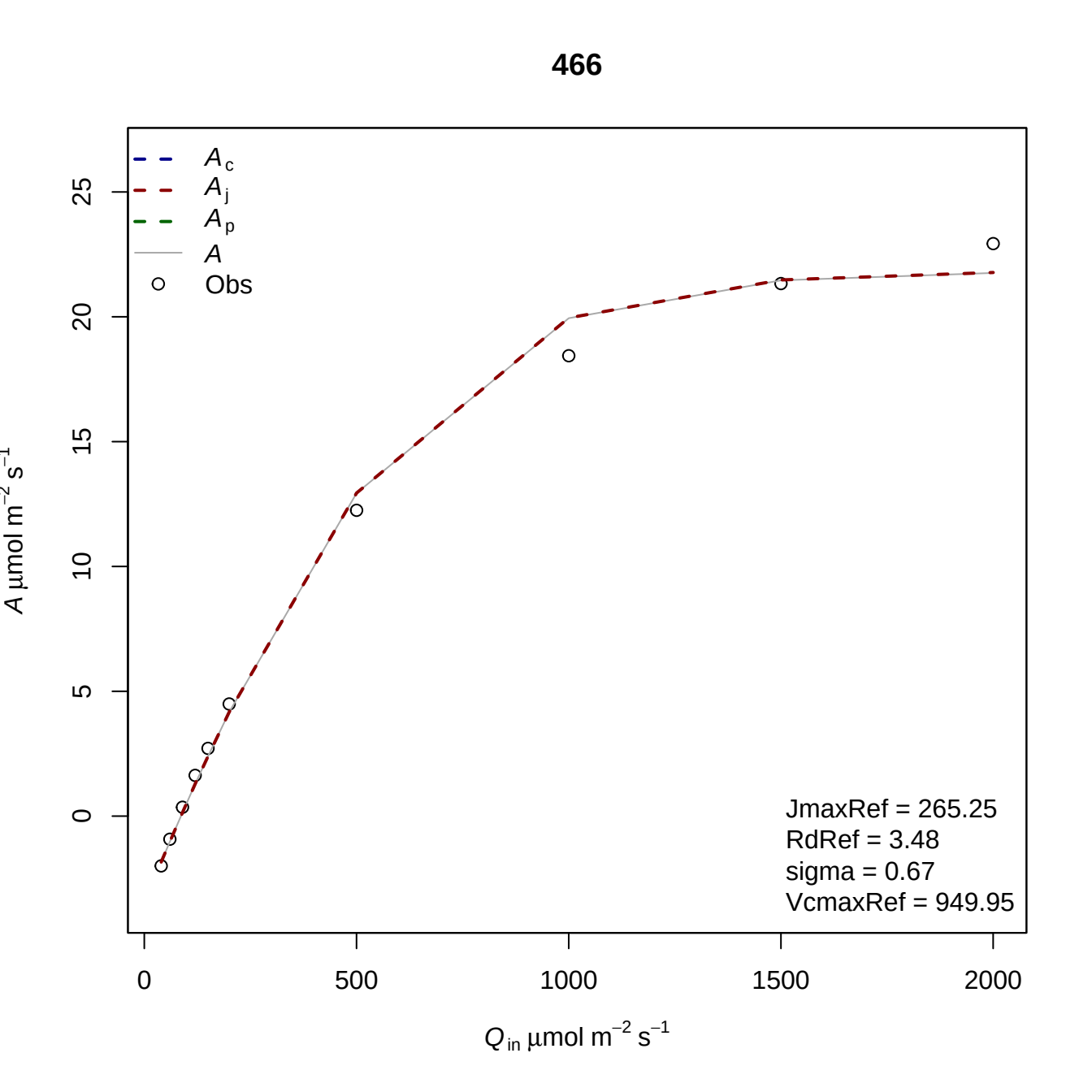

461
A_c
A_j
A_p
A
Obs
JmaxRef = 131.42
RdRef = 1.87
sigma = 1.47
VcmaxRef = 155.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

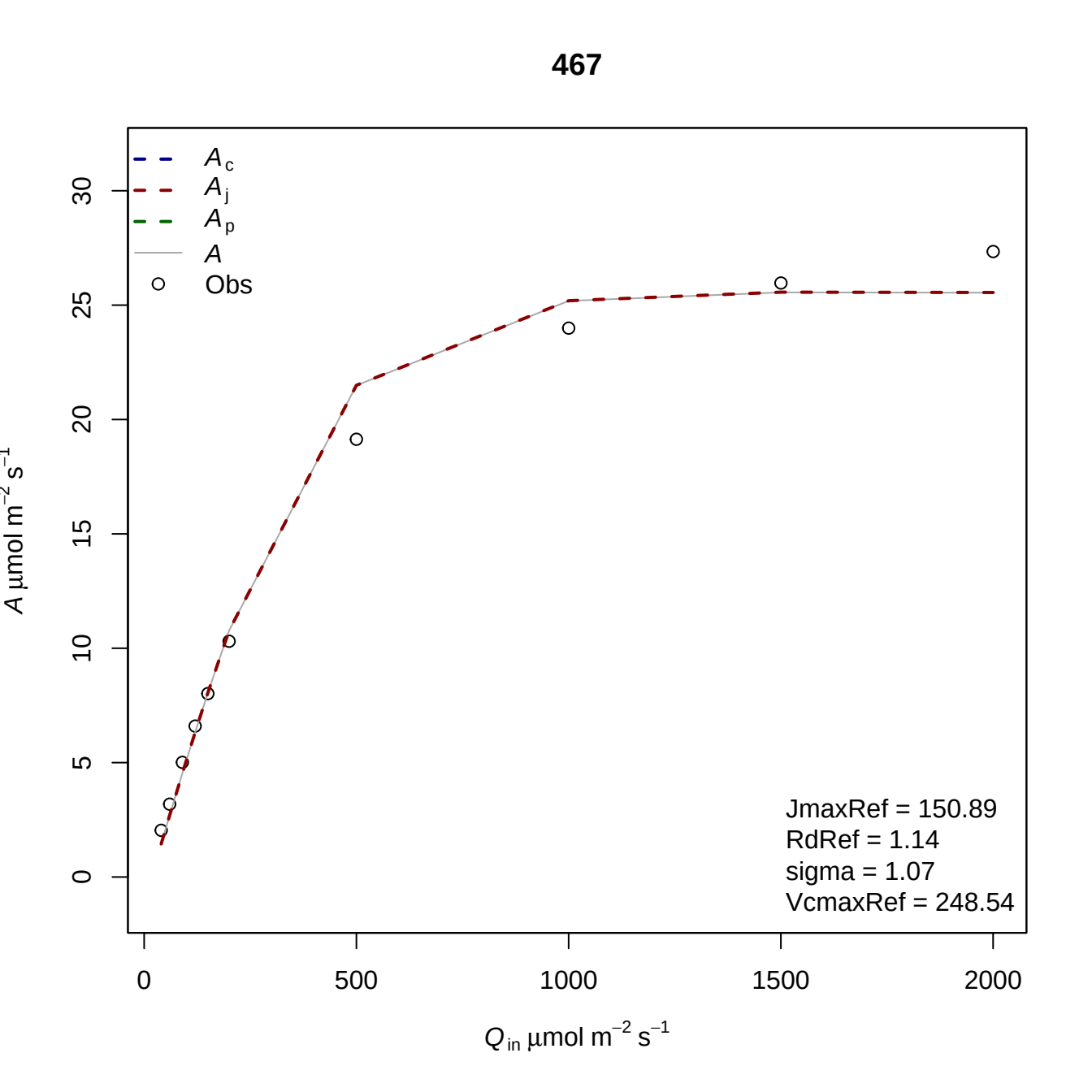

462
A_c
A_j
A_p
A
Obs
JmaxRef = 165.02
RdRef = 1.45
sigma = 1.19
VcmaxRef = 334.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

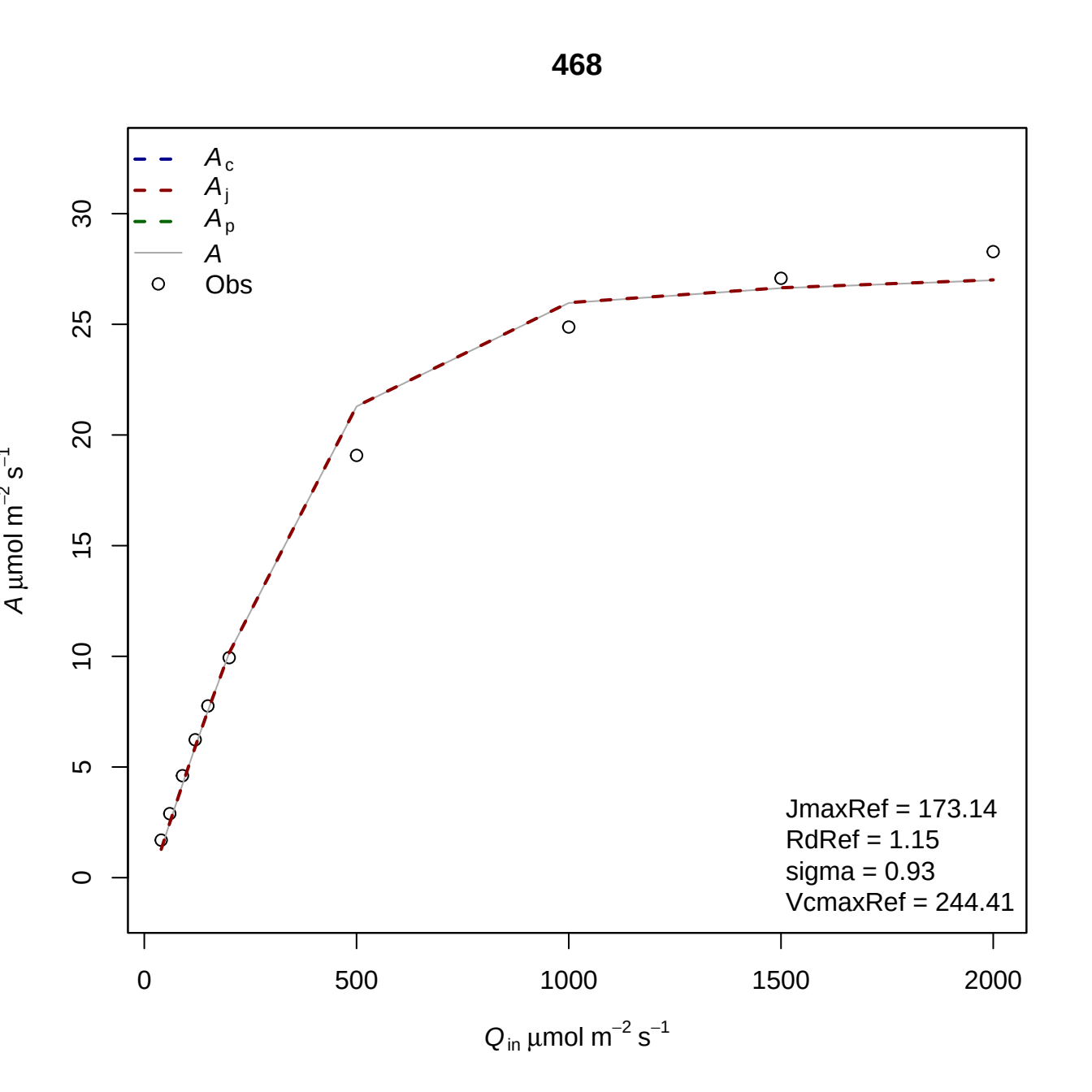

463
A_c
A_j
A_p
A
Obs
JmaxRef = 213.55
RdRef = 1.37
sigma = 0.73
VcmaxRef = 382.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

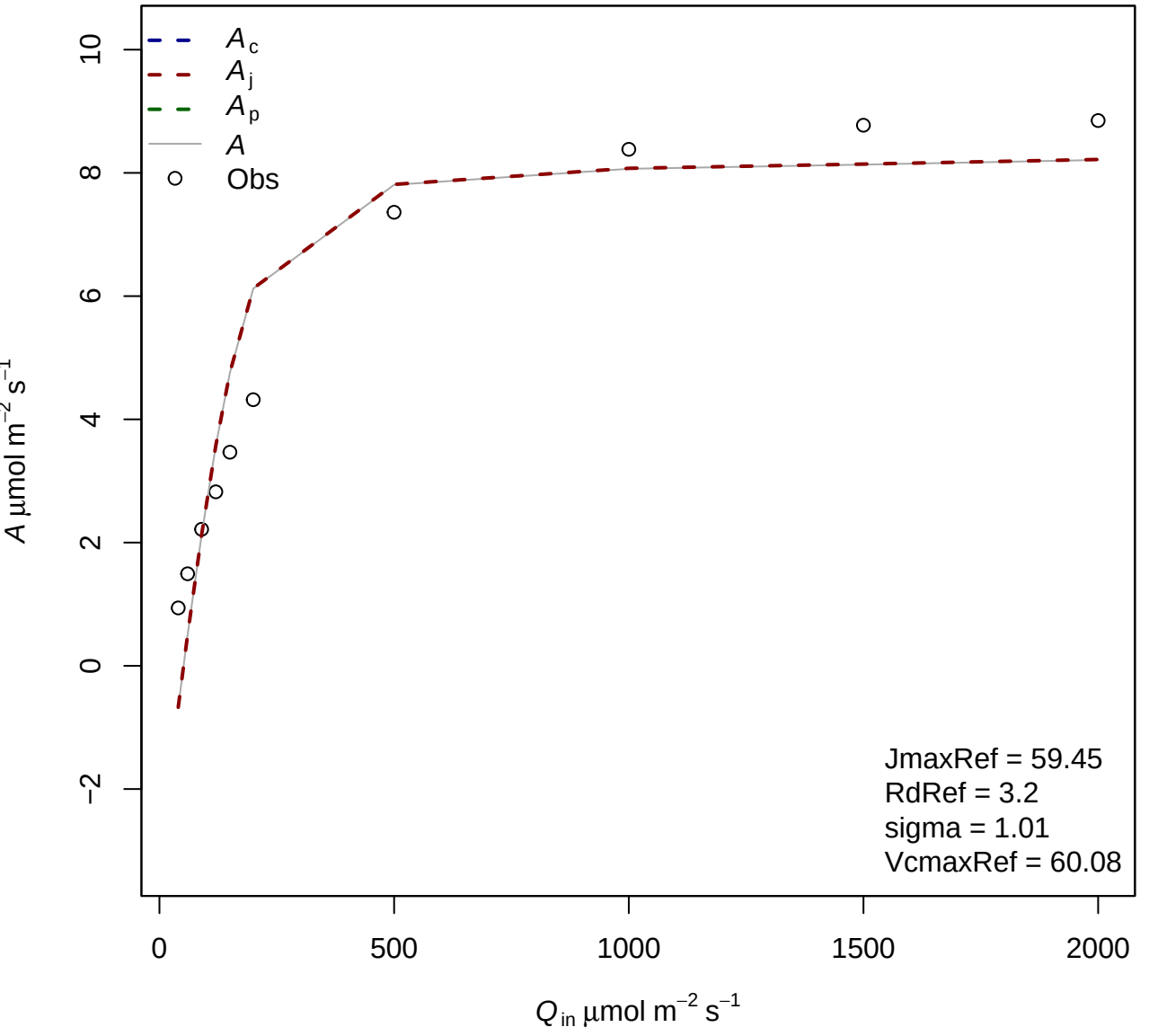

464
A_c
A_j
A_p
A
Obs
JmaxRef = 238.47
RdRef = 1.41
sigma = 0.5
VcmaxRef = 586.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

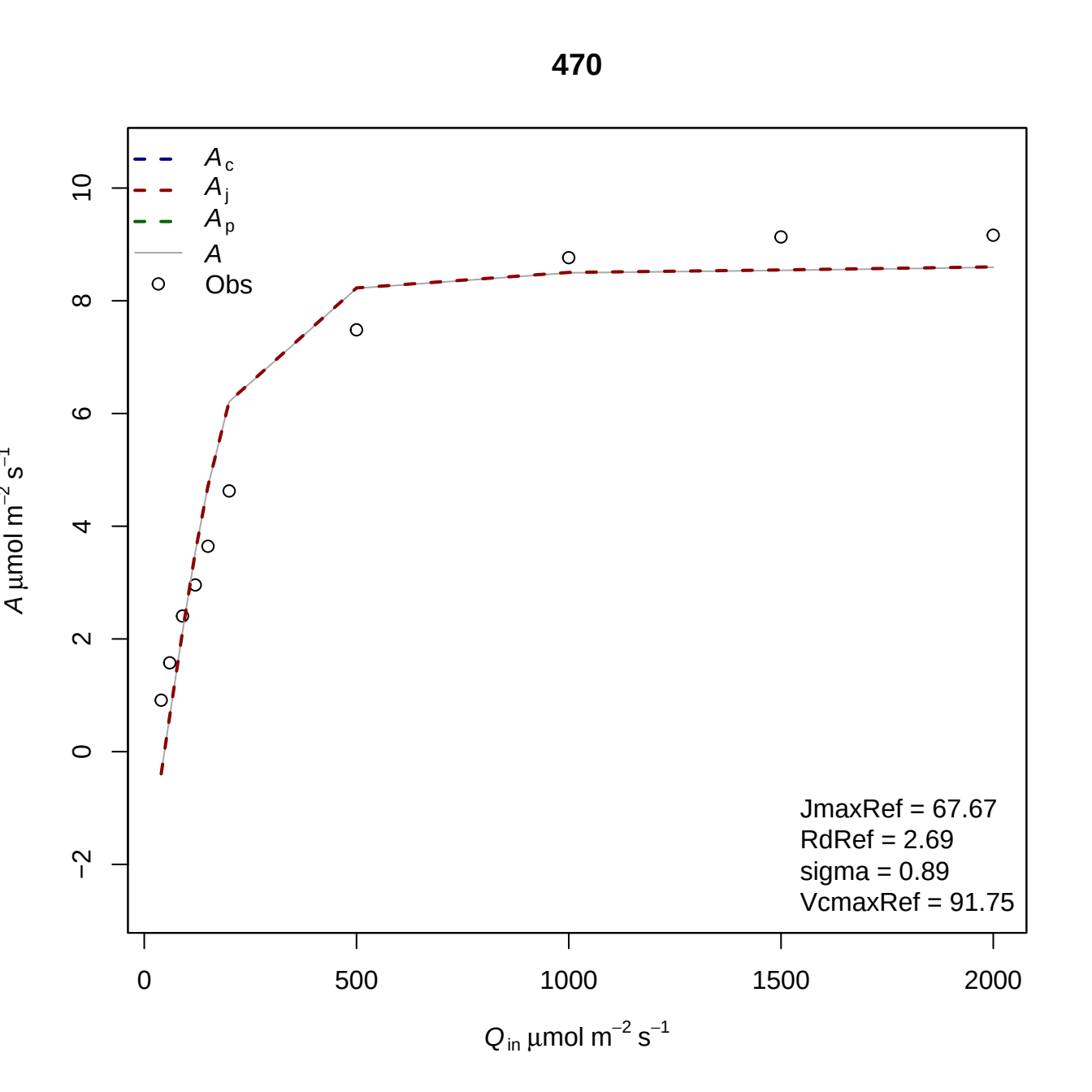

465
A_c
A_j
A_p
A
Obs
JmaxRef = 270.51
RdRef = 2.19
sigma = 0.52
VcmaxRef = 825.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

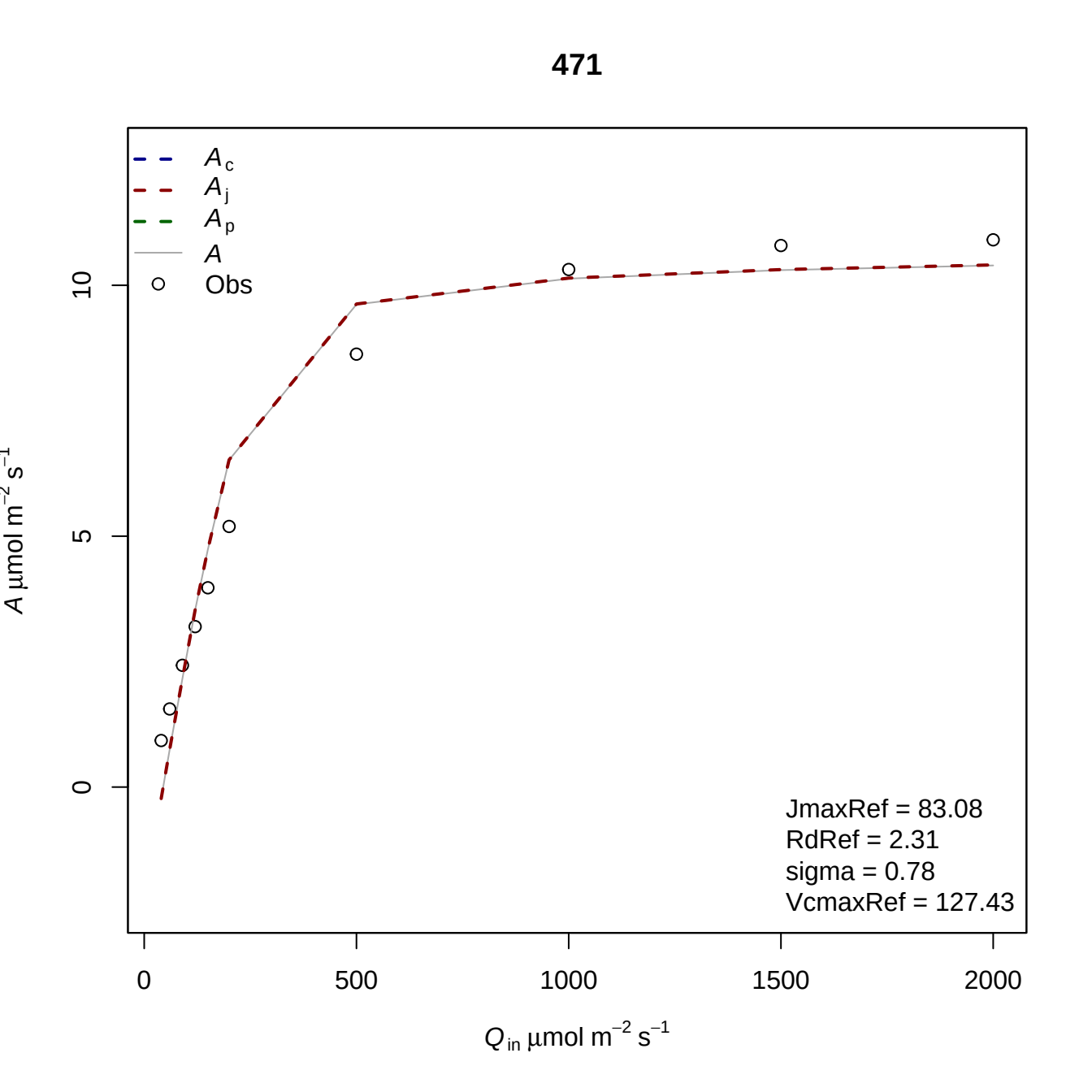

466
A_c
A_j
A_p
A
Obs
JmaxRef = 265.25
RdRef = 3.48
sigma = 0.67
VcmaxRef = 949.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

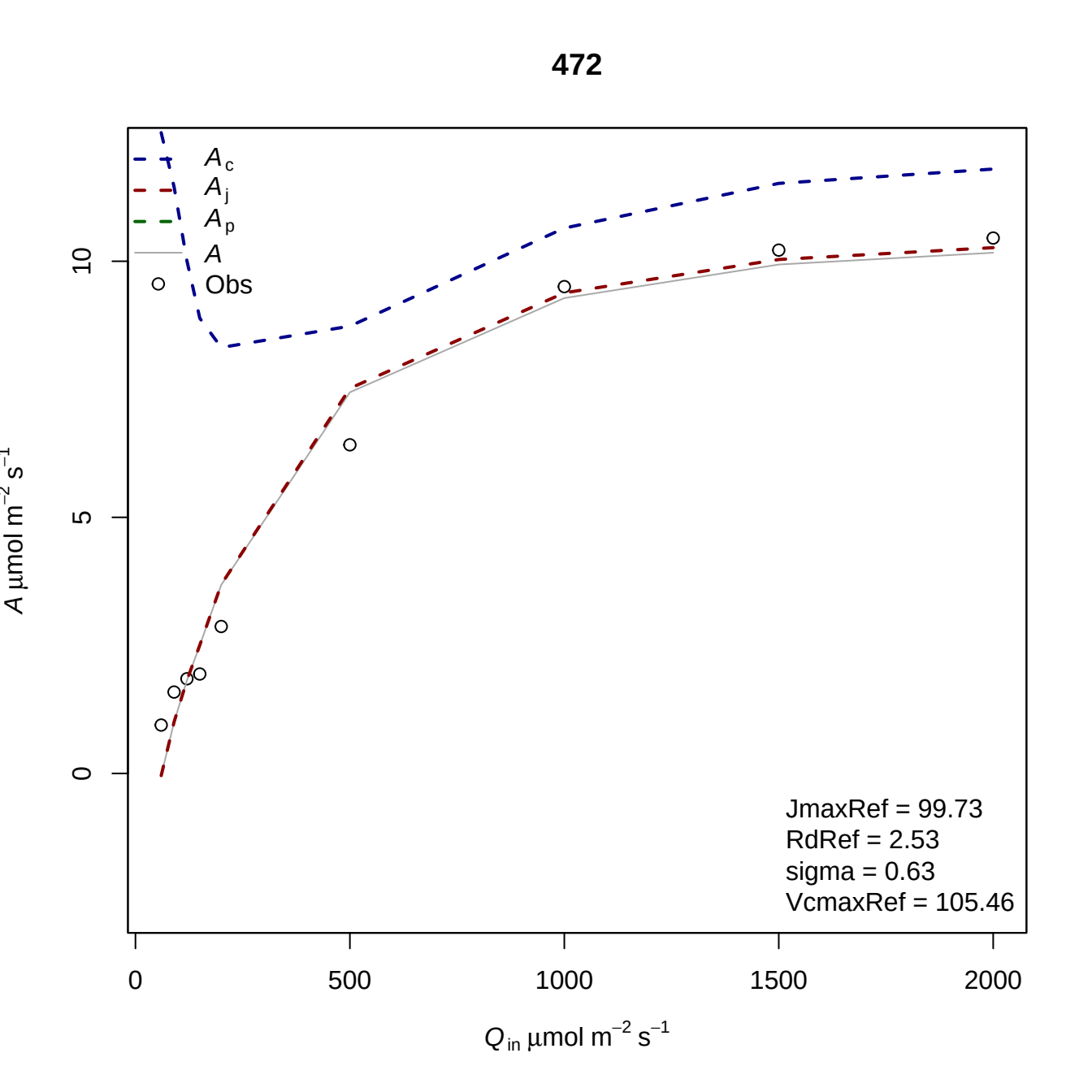

467
A_c
A_j
A_p
A
Obs
JmaxRef = 150.89
RdRef = 1.14
sigma = 1.07
VcmaxRef = 248.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

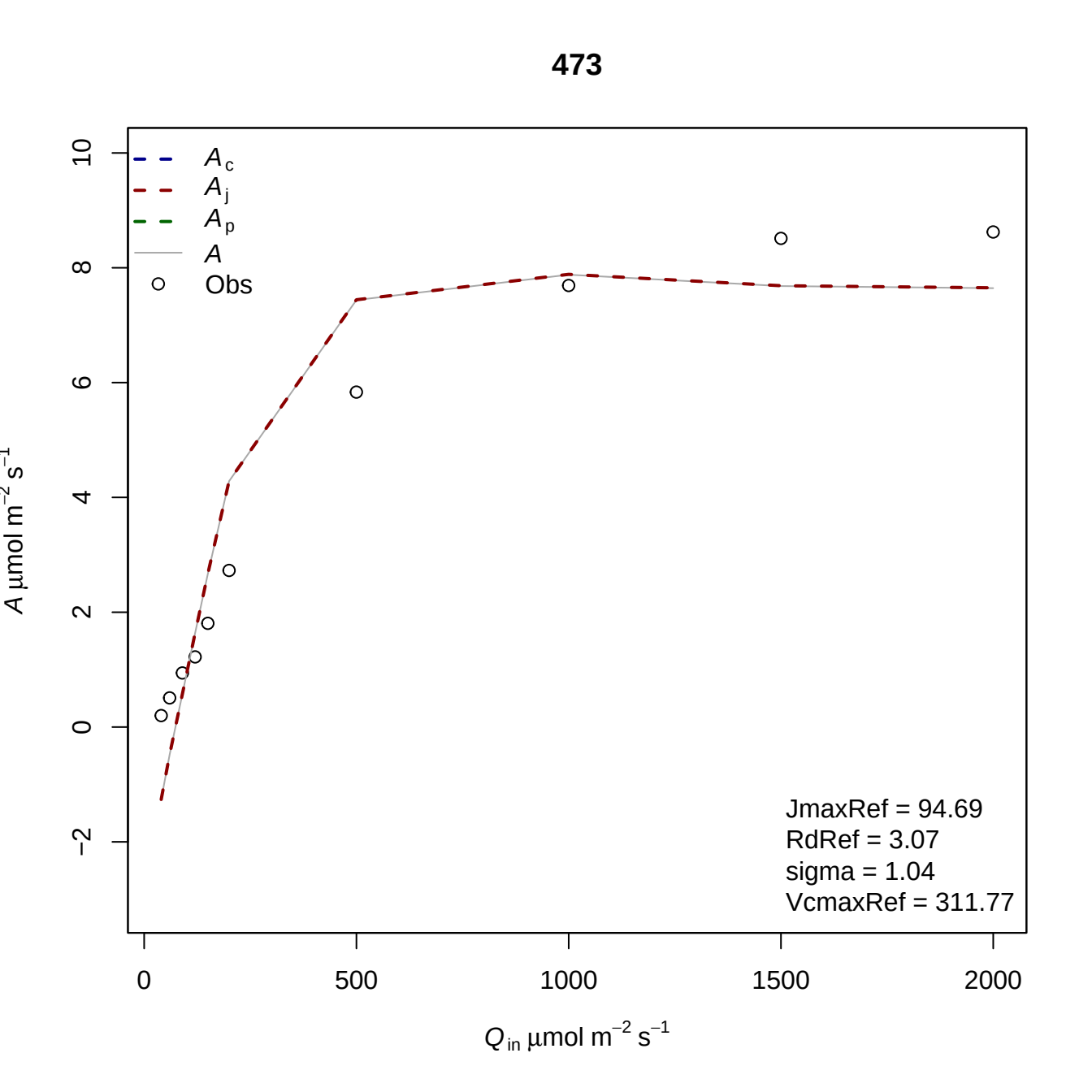

468
A_c
A_j
A_p
A
Obs
JmaxRef = 173.14
RdRef = 1.15
sigma = 0.93
VcmaxRef = 244.41
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

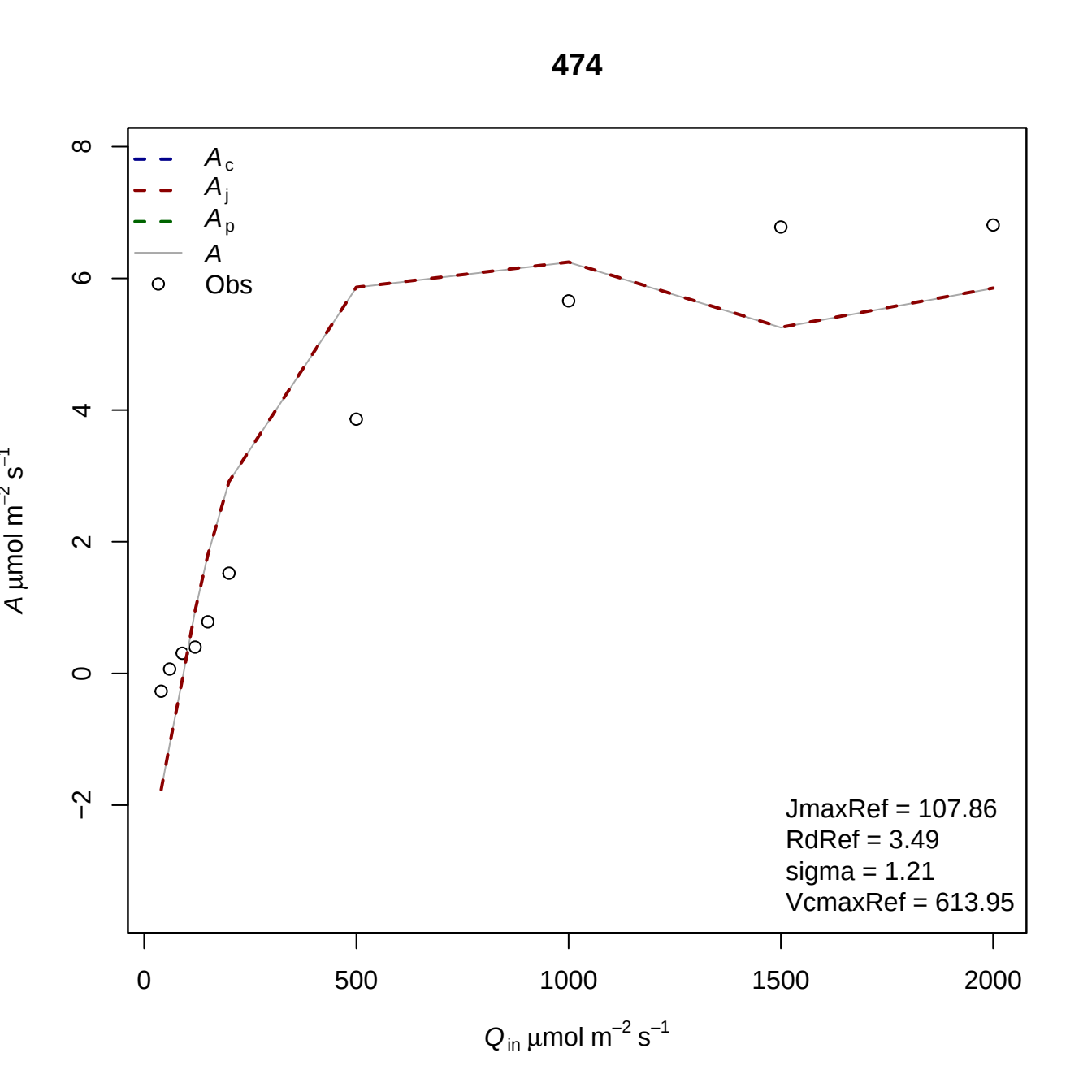

A_c
A_j
A_p
A
Obs
JmaxRef = 59.45
RdRef = 3.2
sigma = 1.01
VcmaxRef = 60.08
10
8
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

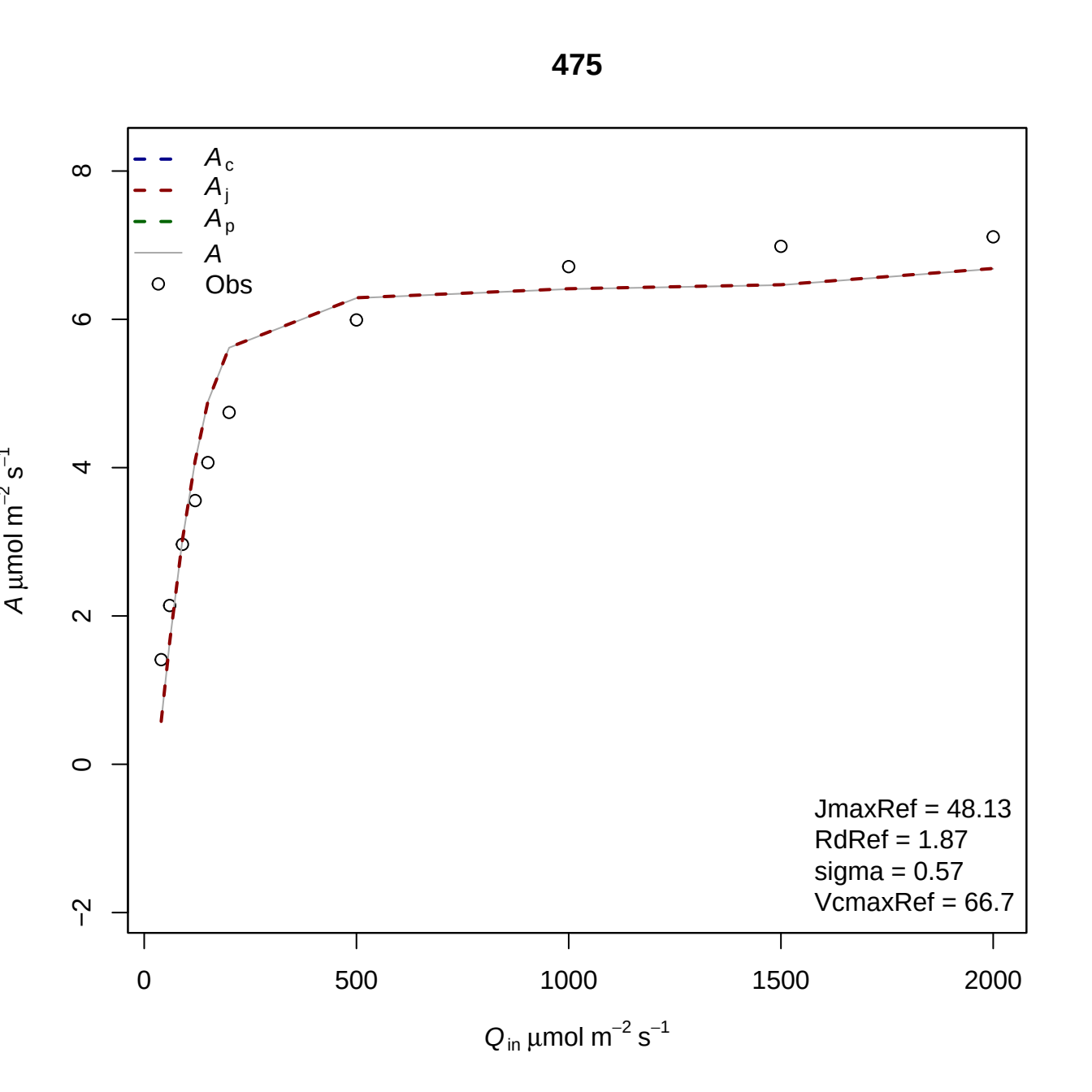

470
A_c
A_j
A_p
A
Obs
JmaxRef = 67.67
RdRef = 2.69
sigma = 0.89
VcmaxRef = 91.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

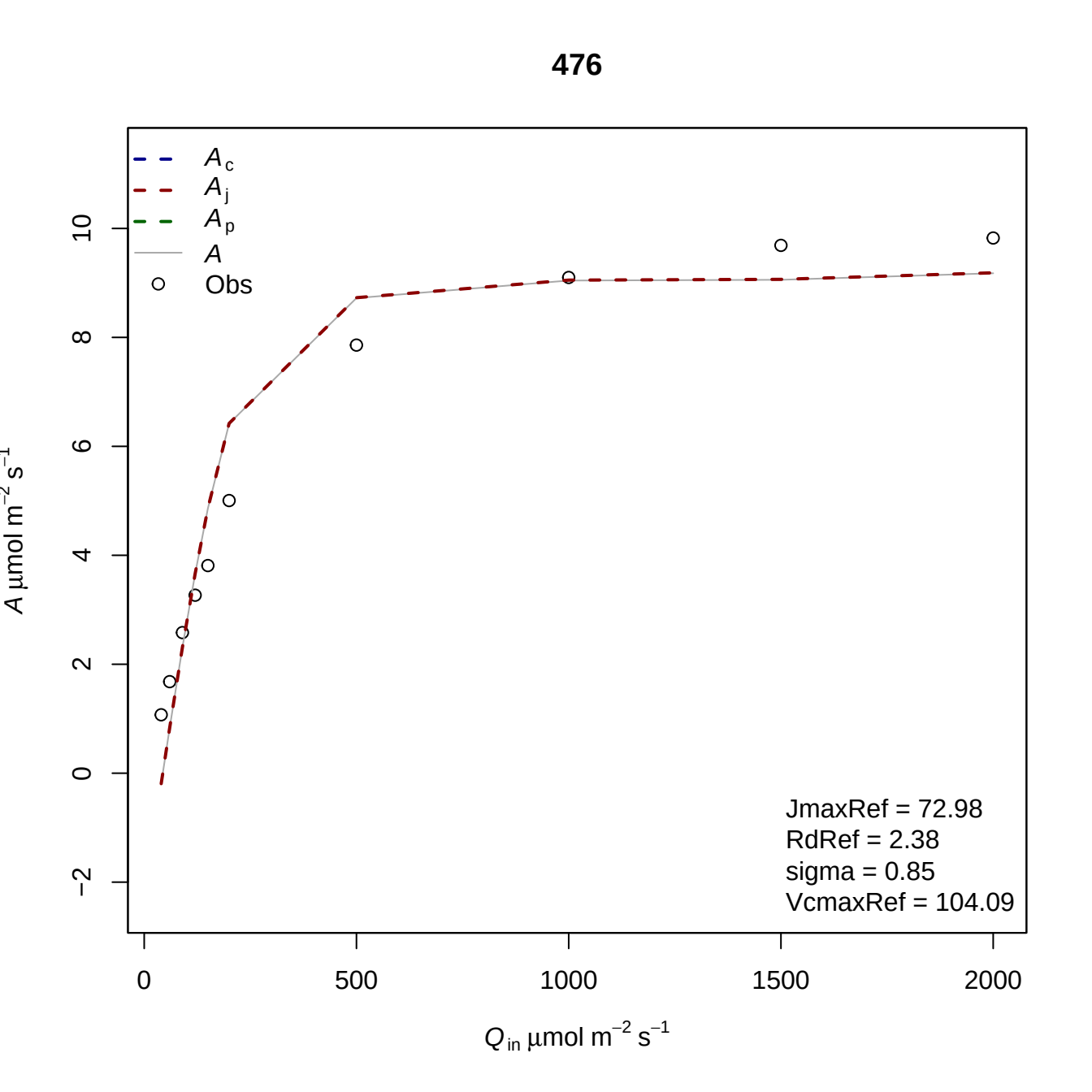

471
A_c
A_j
A_p
A
Obs
JmaxRef = 83.08
RdRef = 2.31
sigma = 0.78
VcmaxRef = 127.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

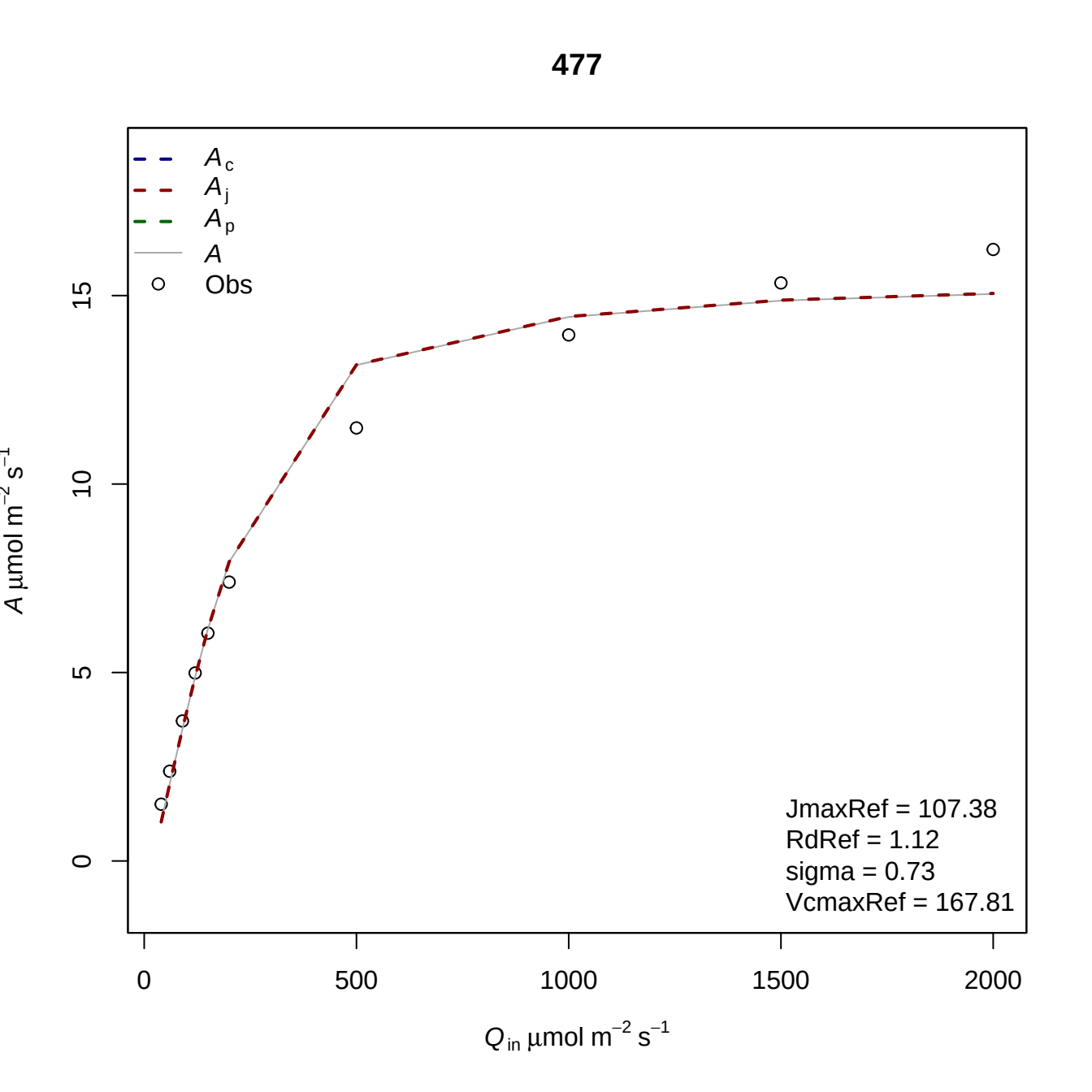

472
A_c
A_j
A_p
A
Obs
JmaxRef = 99.73
RdRef = 2.53
sigma = 0.63
VcmaxRef = 105.46
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

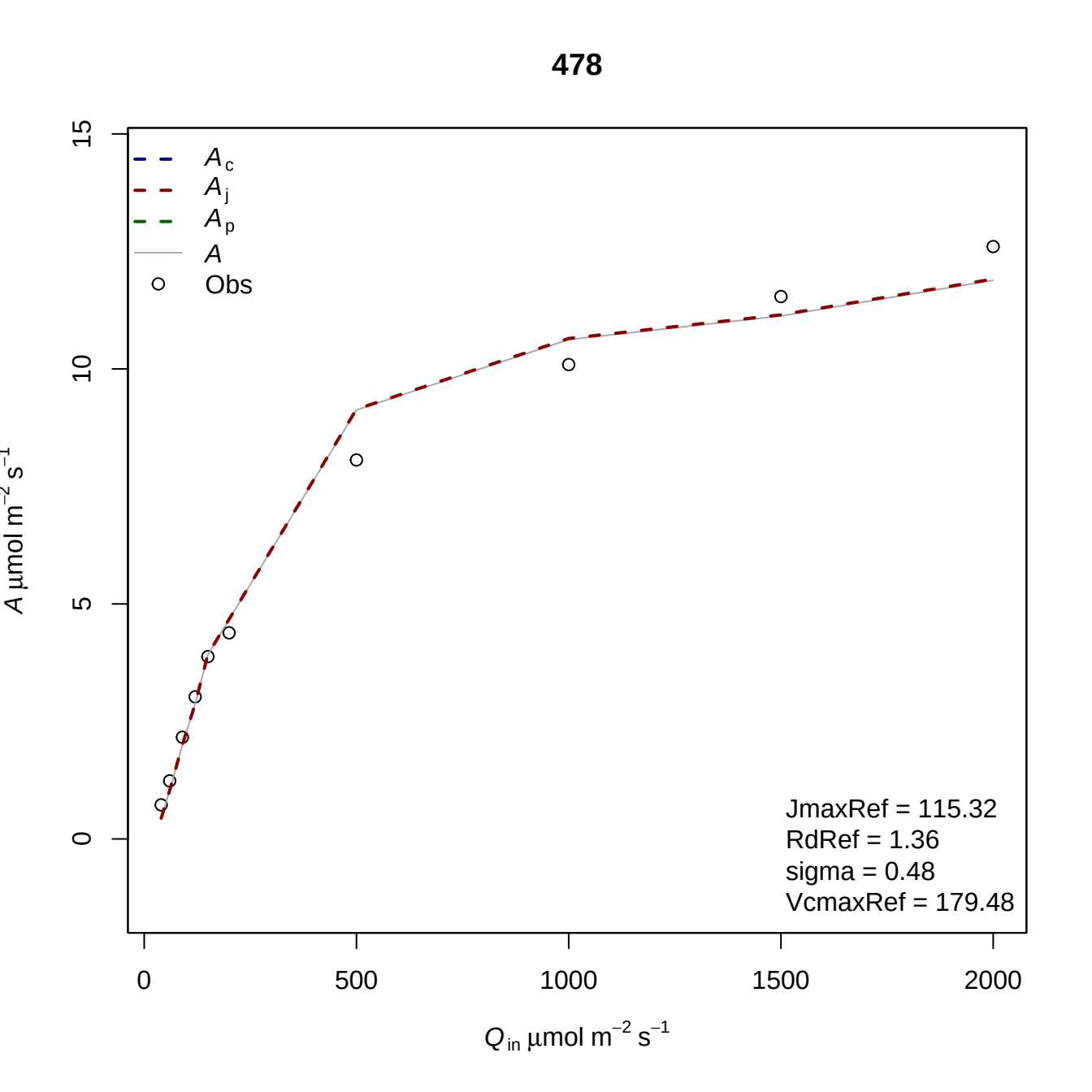

473
A_c
A_j
A_p
A
Obs
JmaxRef = 94.69
RdRef = 3.07
sigma = 1.04
VcmaxRef = 311.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

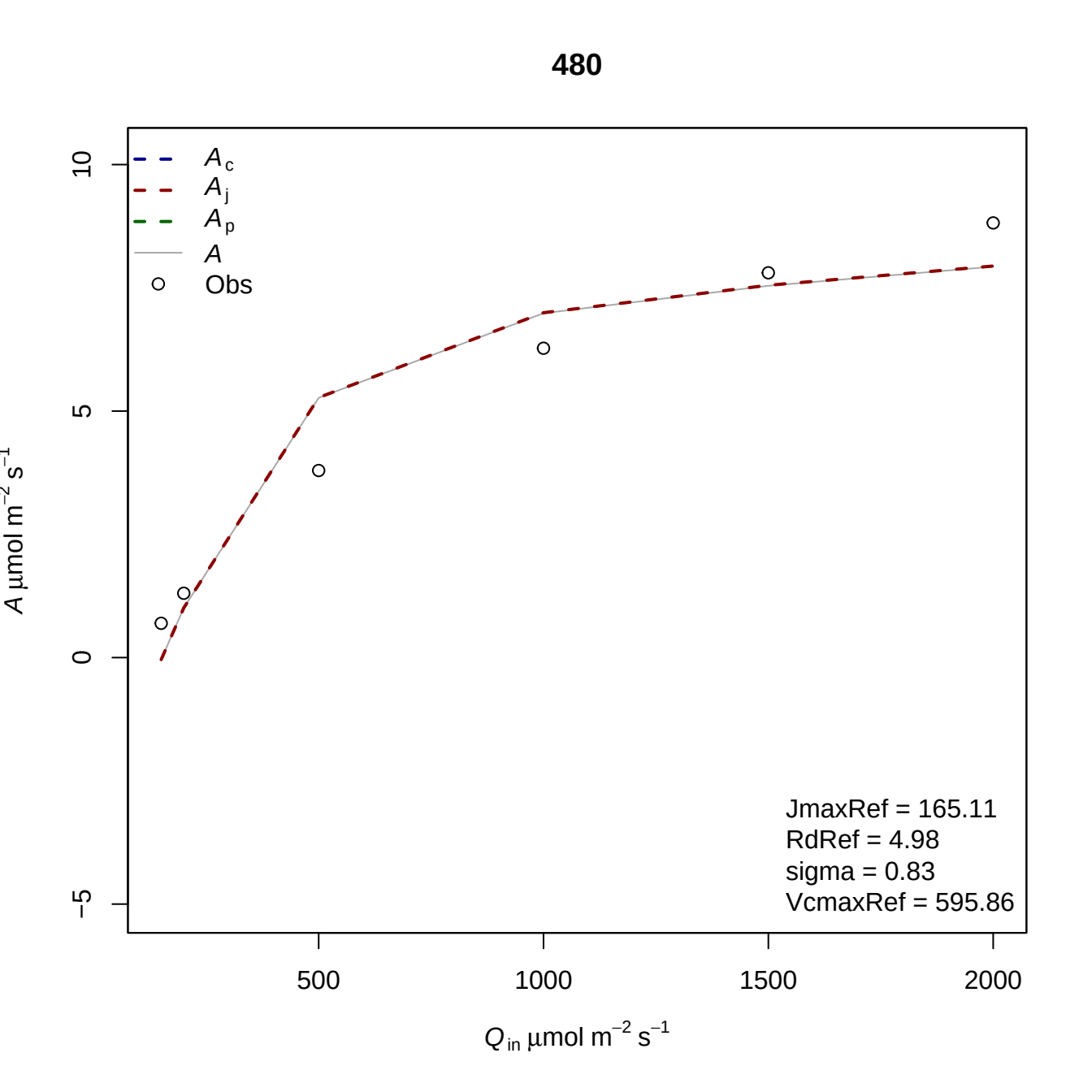

474
A_c
A_j
A_p
A
Obs
JmaxRef = 107.86
RdRef = 3.49
sigma = 1.21
VcmaxRef = 613.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

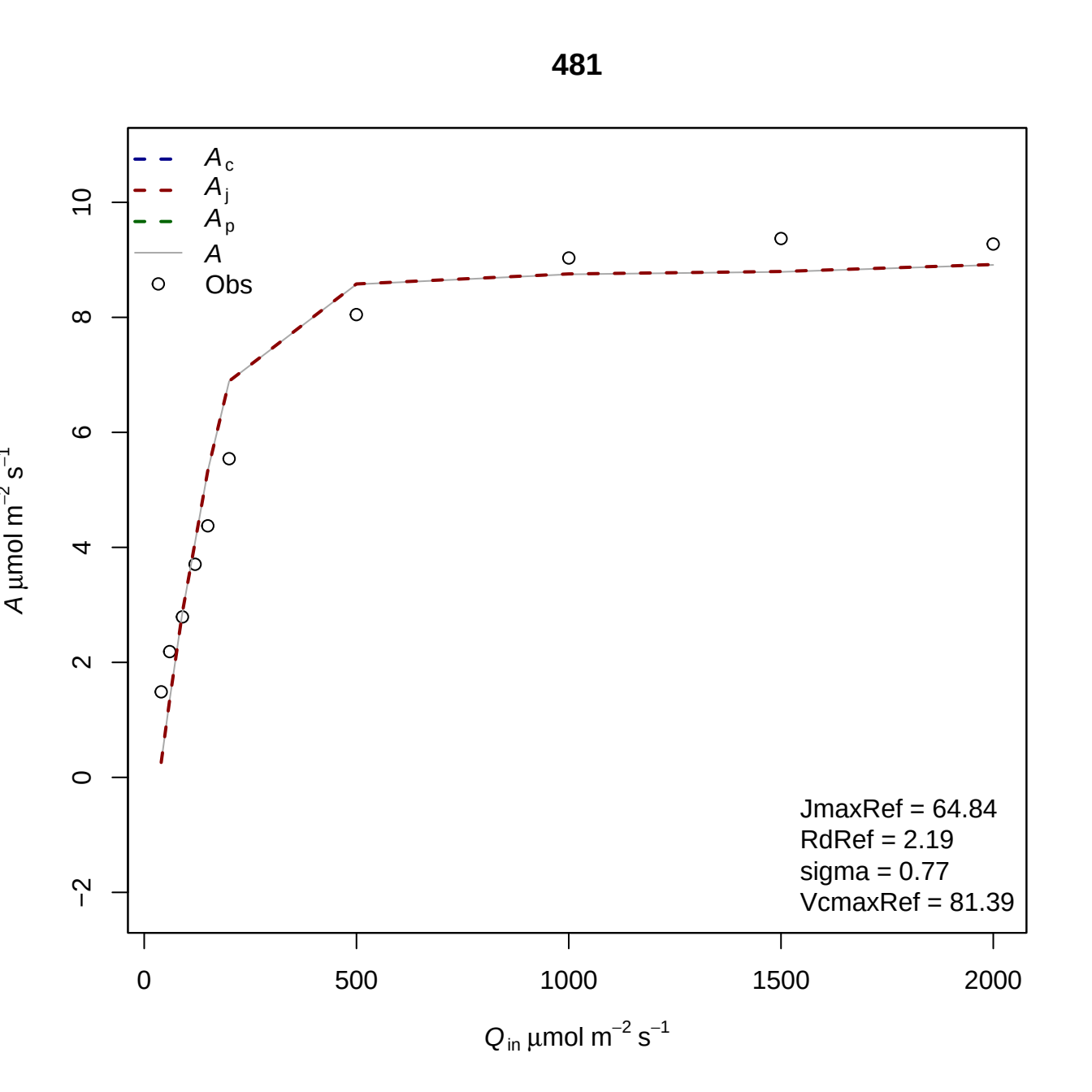

475
A_c
A_j
A_p
A
Obs
JmaxRef = 48.13
RdRef = 1.87
sigma = 0.57
VcmaxRef = 66.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

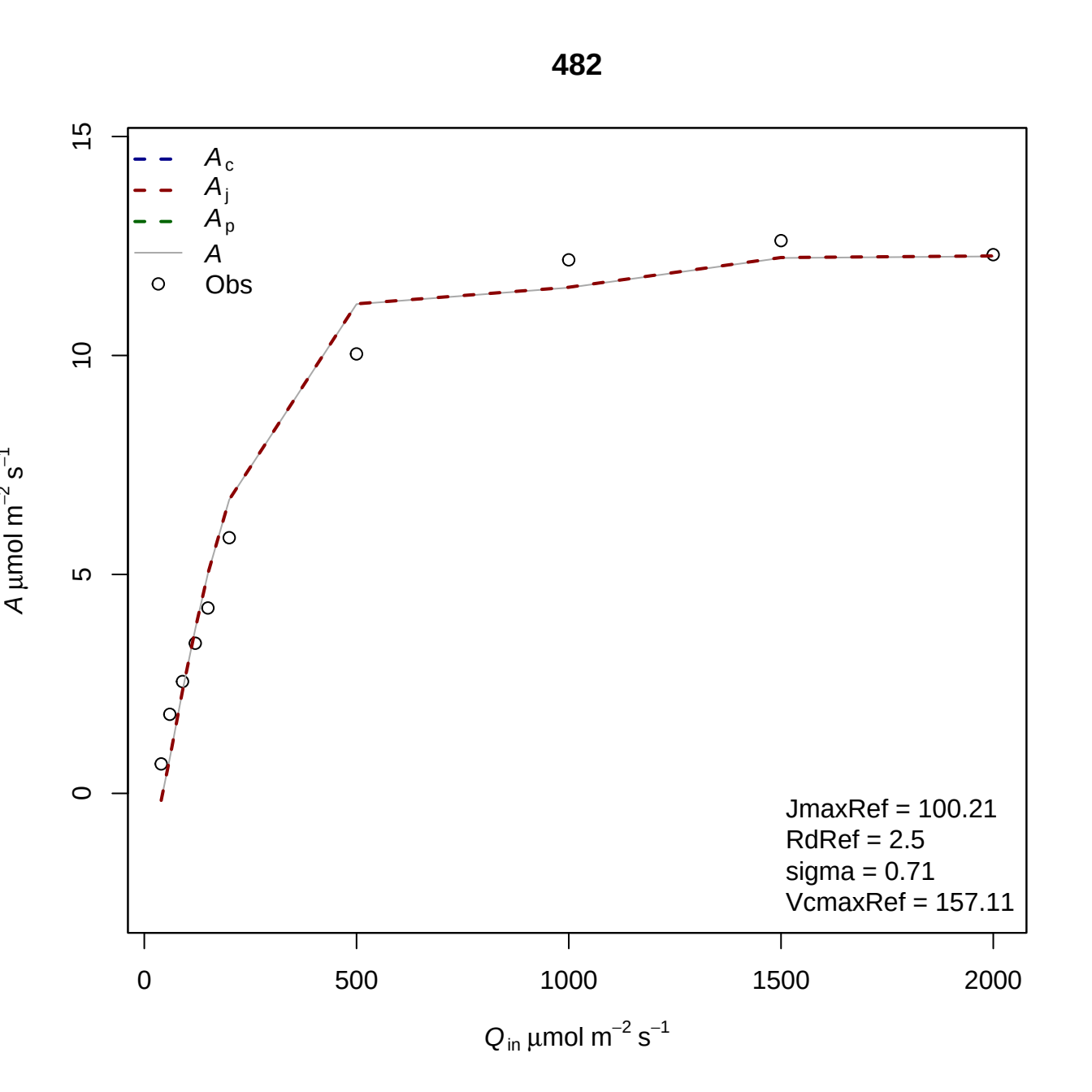

476
A_c
A_j
A_p
A
Obs
JmaxRef = 72.98
RdRef = 2.38
sigma = 0.85
VcmaxRef = 104.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

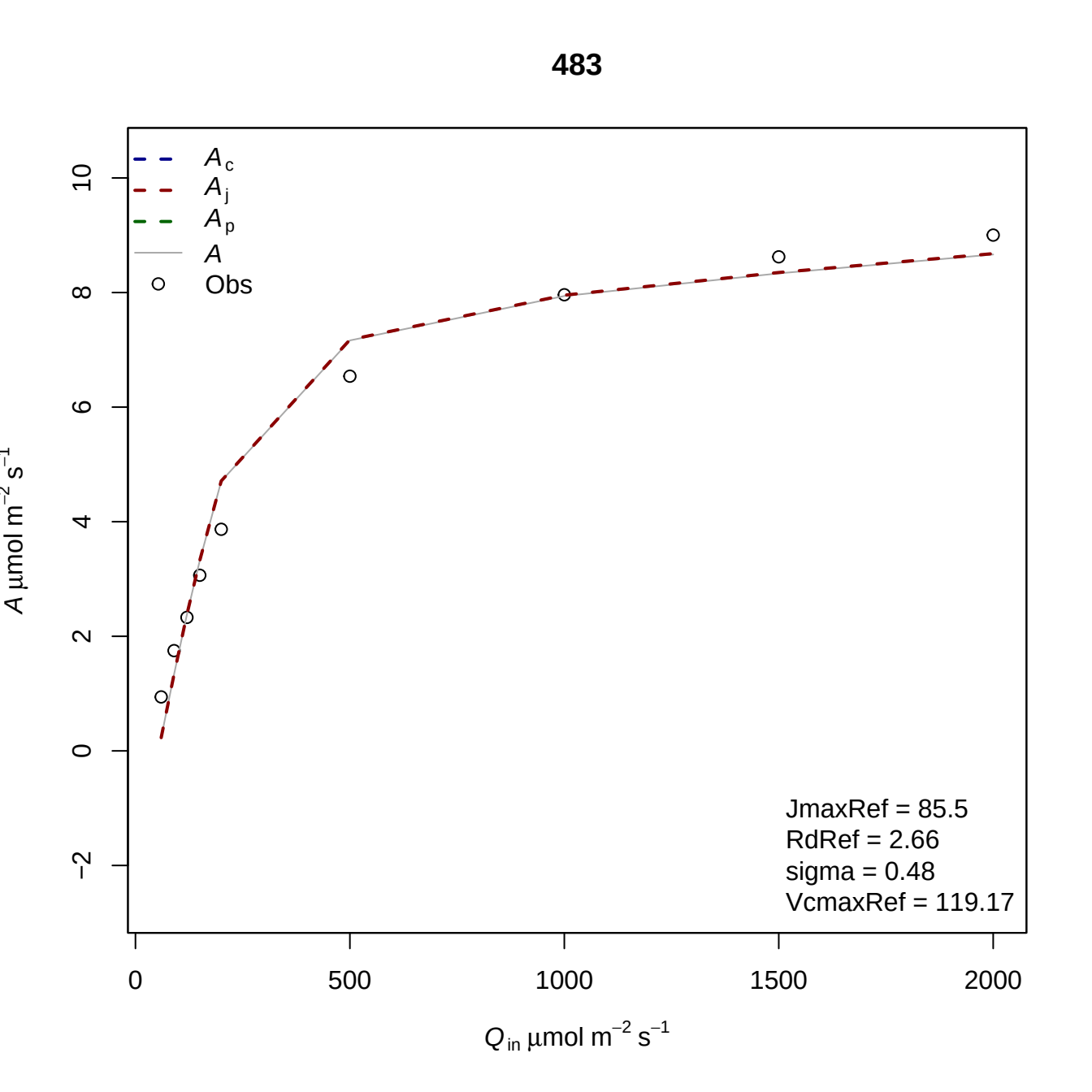

477
A_c
A_j
A_p
A
Obs
JmaxRef = 107.38
RdRef = 1.12
sigma = 0.73
VcmaxRef = 167.81
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

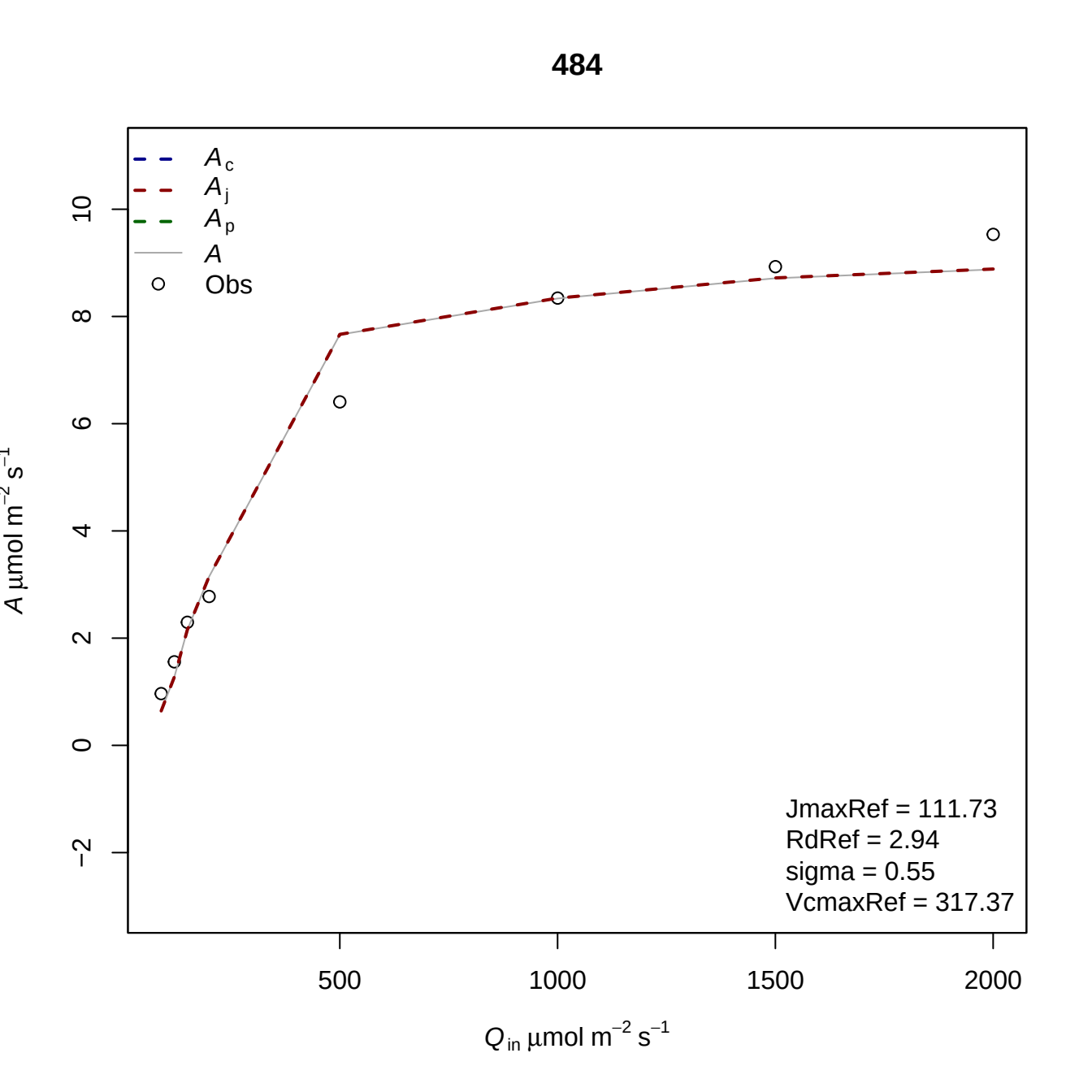

478
A_c
A_j
A_p
A
Obs
JmaxRef = 115.32
RdRef = 1.36
sigma = 0.48
VcmaxRef = 179.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

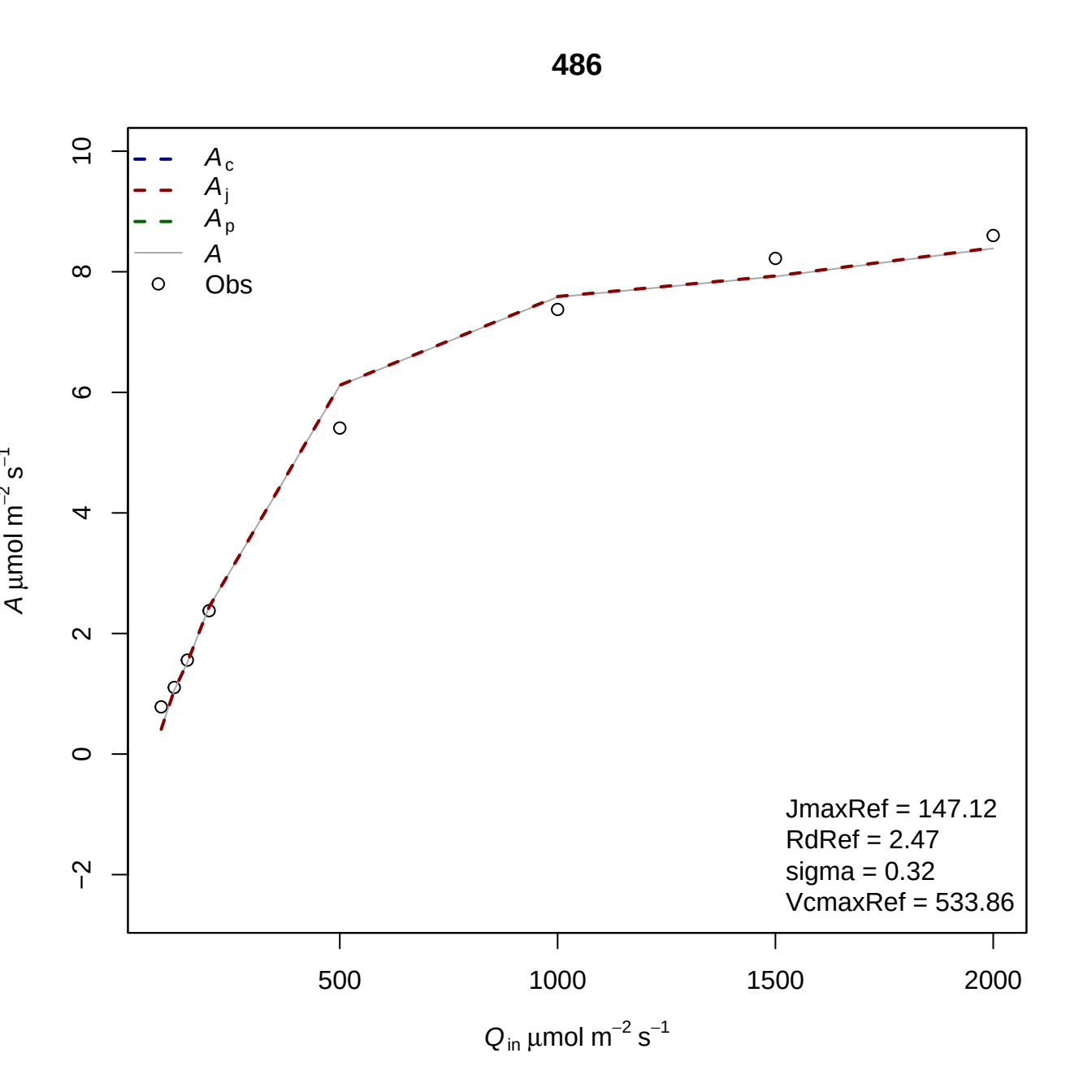

480
A_c
A_j
A_p
A
Obs
10
5
0
−5
A µmol m^{-2} s^{-1}
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 165.11
RdRef = 4.98
sigma = 0.83
VcmaxRef = 595.86

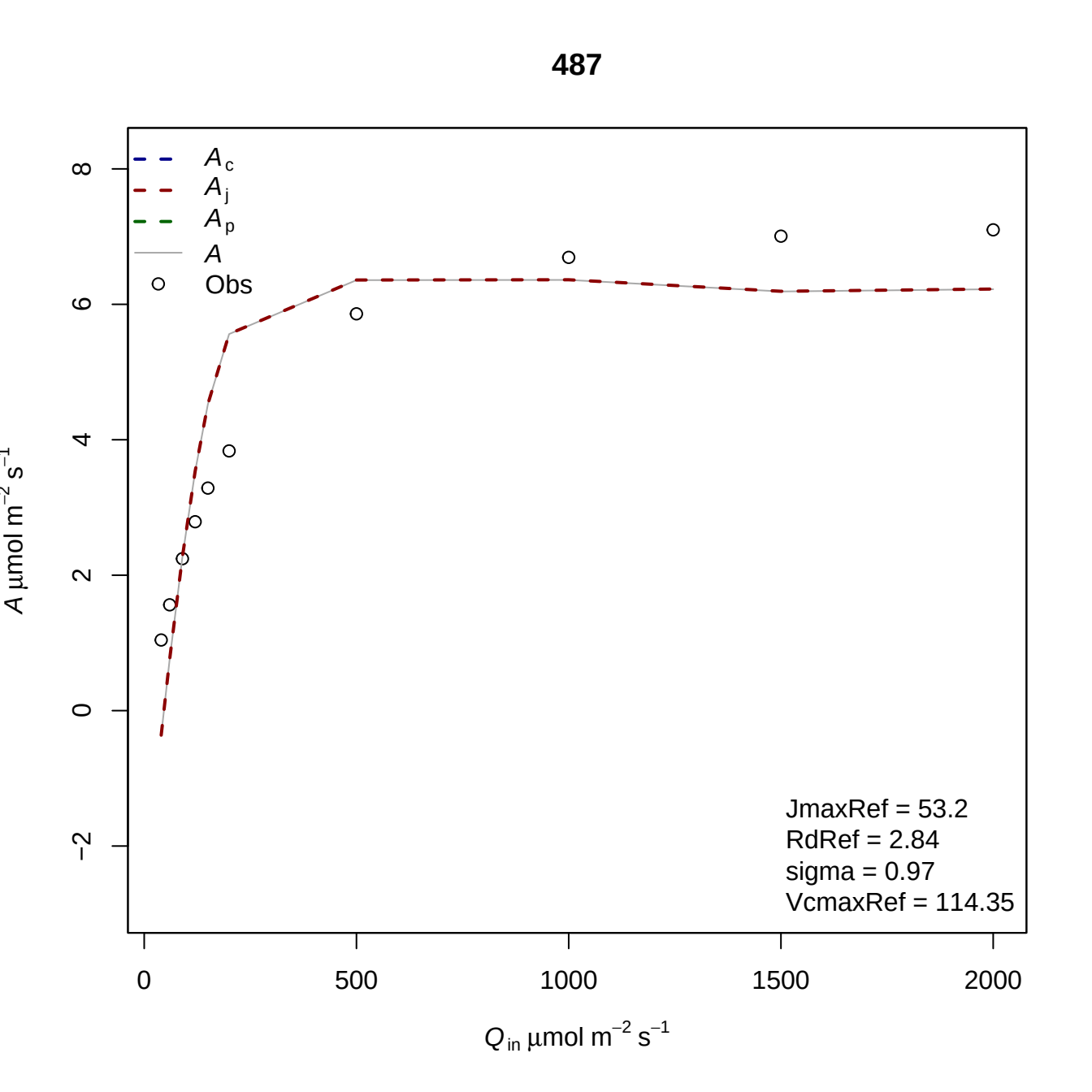

481
A_c
A_j
A_p
A
Obs
JmaxRef = 64.84
RdRef = 2.19
sigma = 0.77
VcmaxRef = 81.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

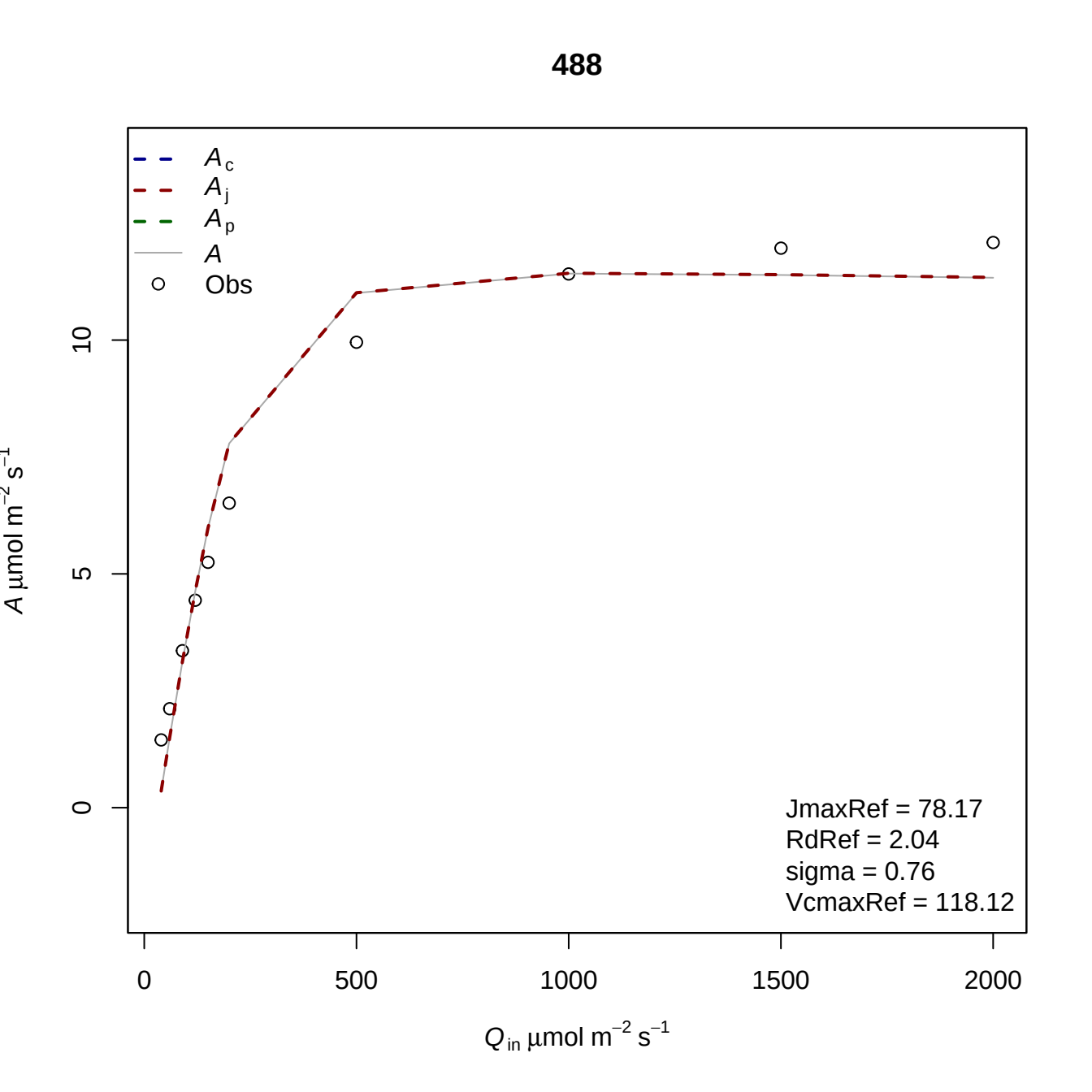

482
A_c
A_j
A_p
A
Obs
JmaxRef = 100.21
RdRef = 2.5
sigma = 0.71
VcmaxRef = 157.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

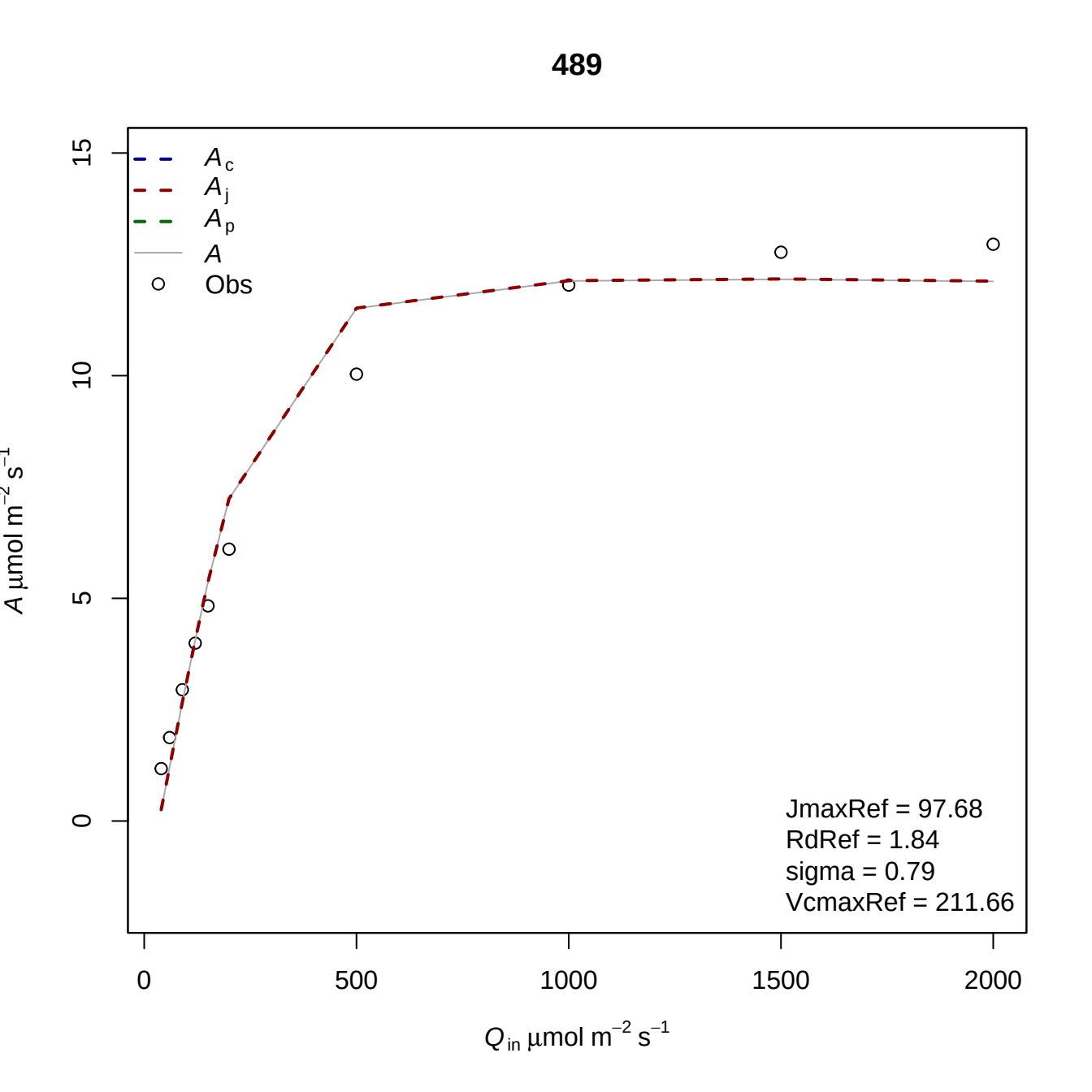

483
A_c
A_j
A_p
A
Obs
JmaxRef = 85.5
RdRef = 2.66
sigma = 0.48
VcmaxRef = 119.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

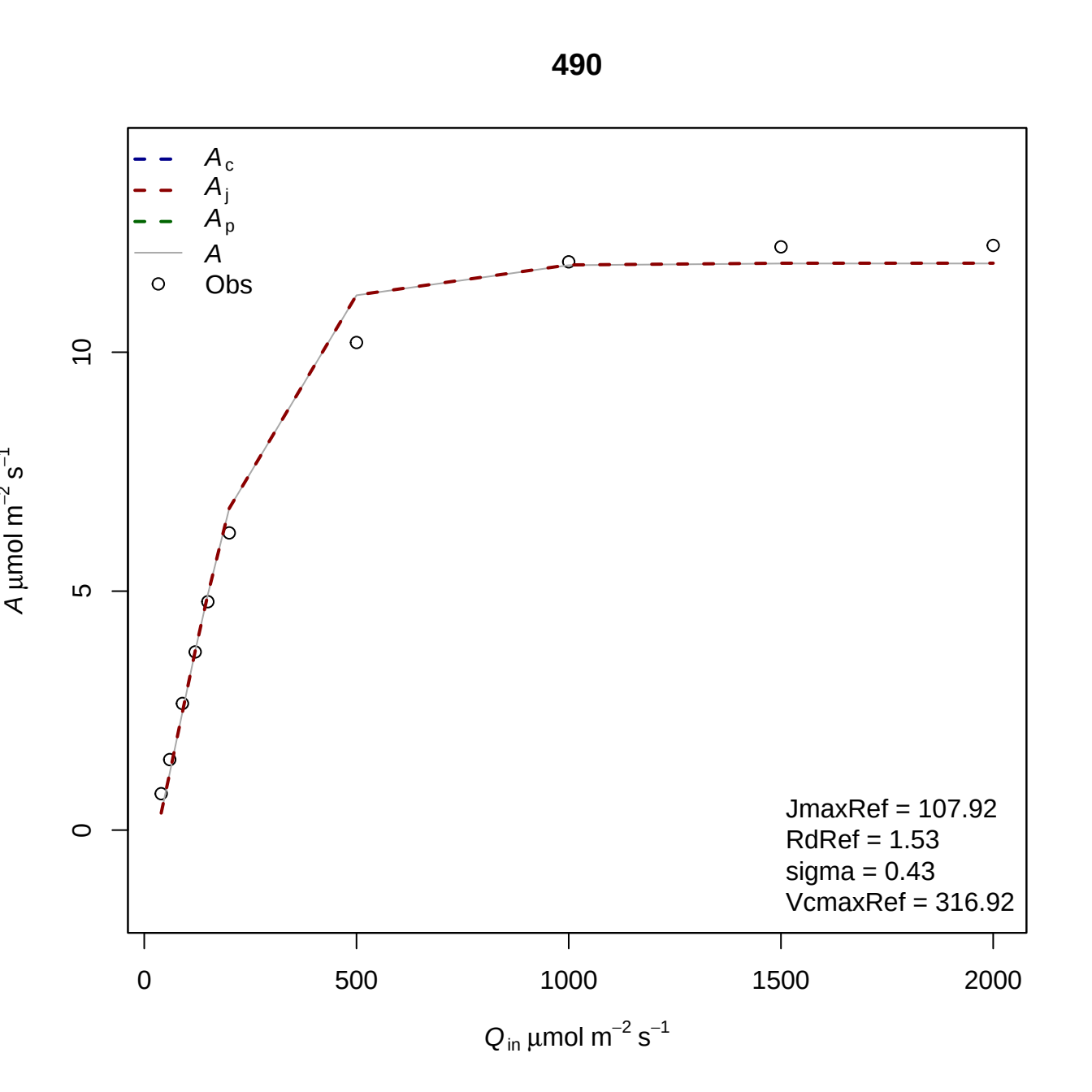

484
A_c
A_j
A_p
A
Obs
JmaxRef = 111.73
RdRef = 2.94
sigma = 0.55
VcmaxRef = 317.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

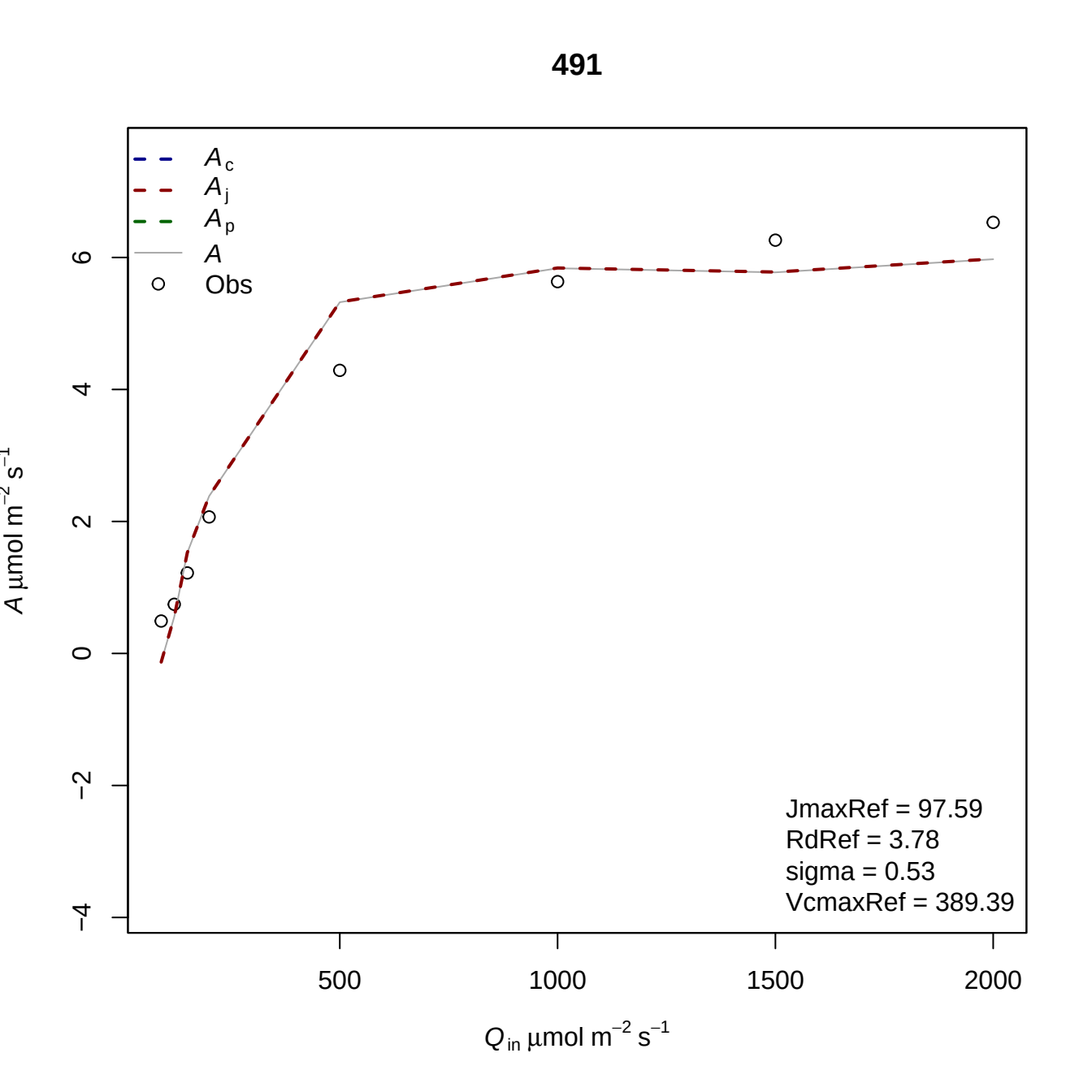

486
A_c
A_j
A_p
A
Obs
JmaxRef = 147.12
RdRef = 2.47
sigma = 0.32
VcmaxRef = 533.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
500
1000
1500
2000

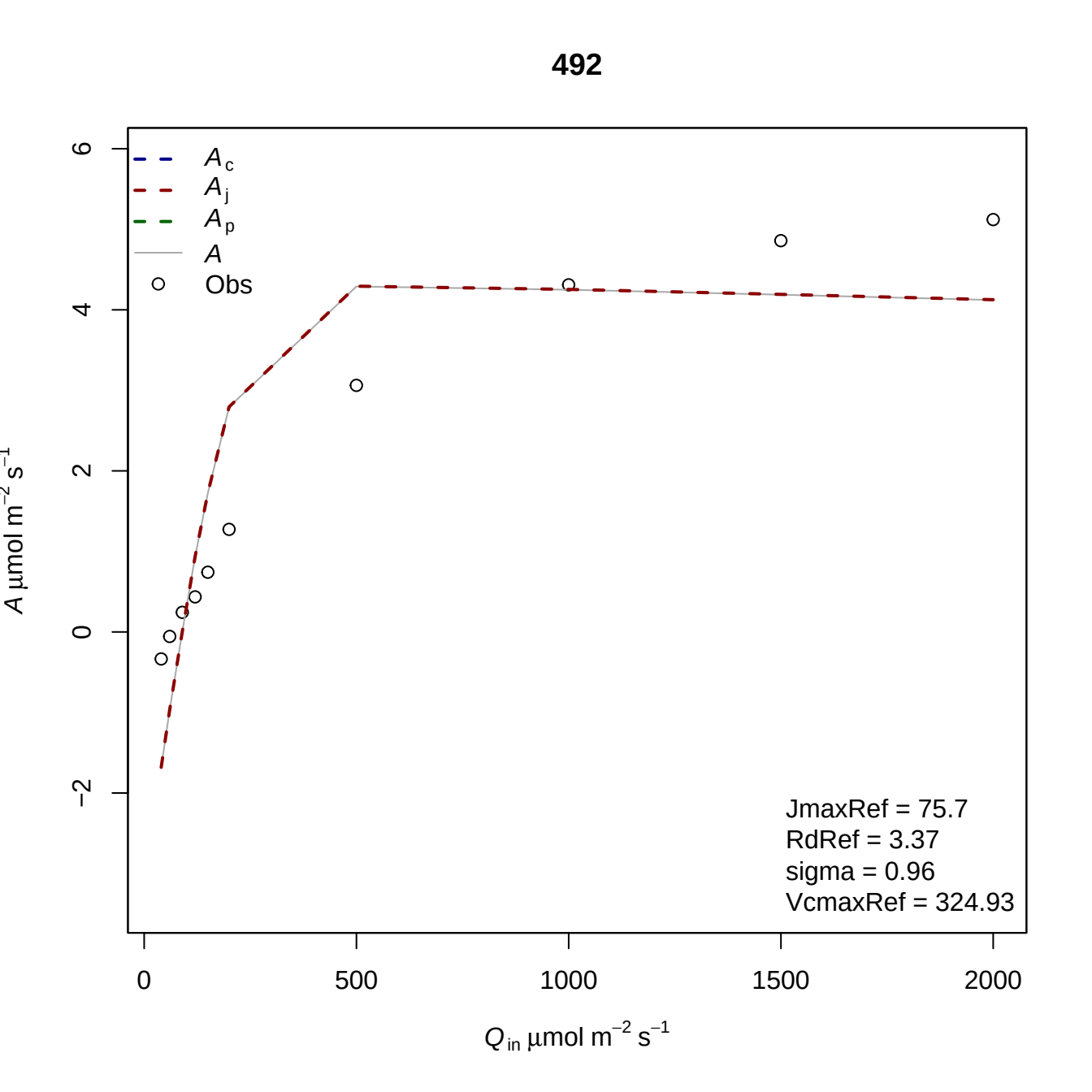

487
A_c
A_j
A_p
A
Obs
JmaxRef = 53.2
RdRef = 2.84
sigma = 0.97
VcmaxRef = 114.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

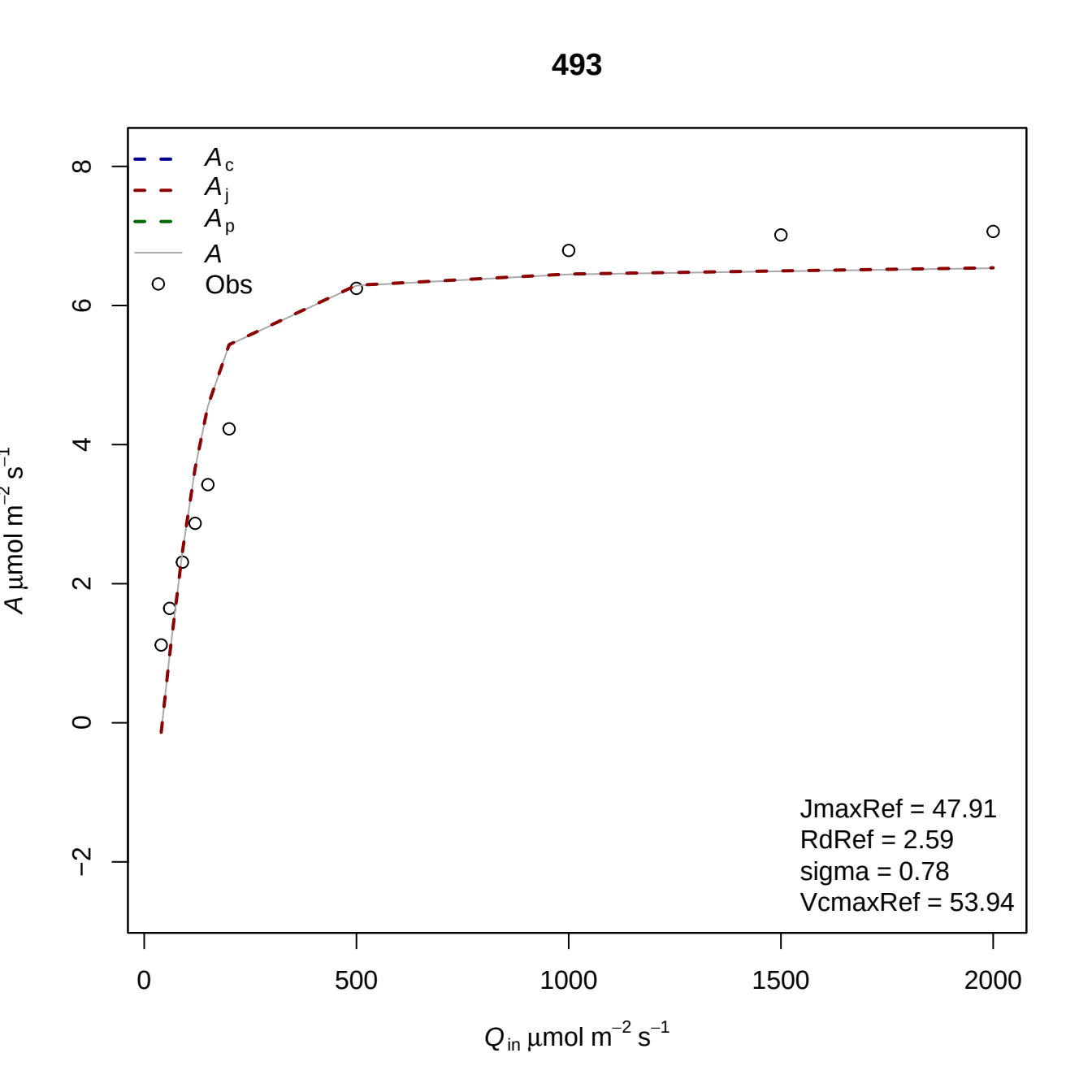

488
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 78.17
RdRef = 2.04
sigma = 0.76
VcmaxRef = 118.12

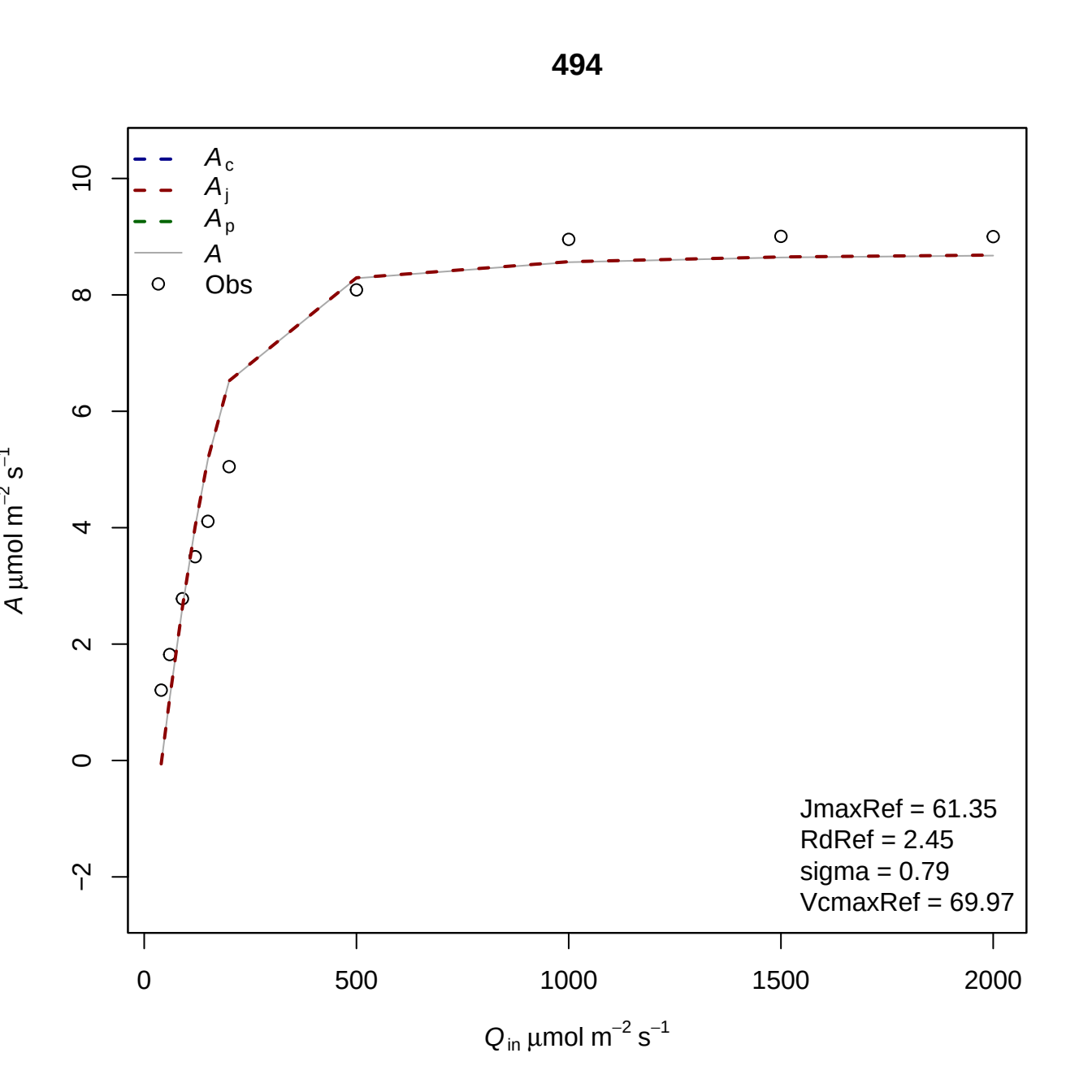

489
A_c
A_j
A_p
A
Obs
JmaxRef = 97.68
RdRef = 1.84
sigma = 0.79
VcmaxRef = 211.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

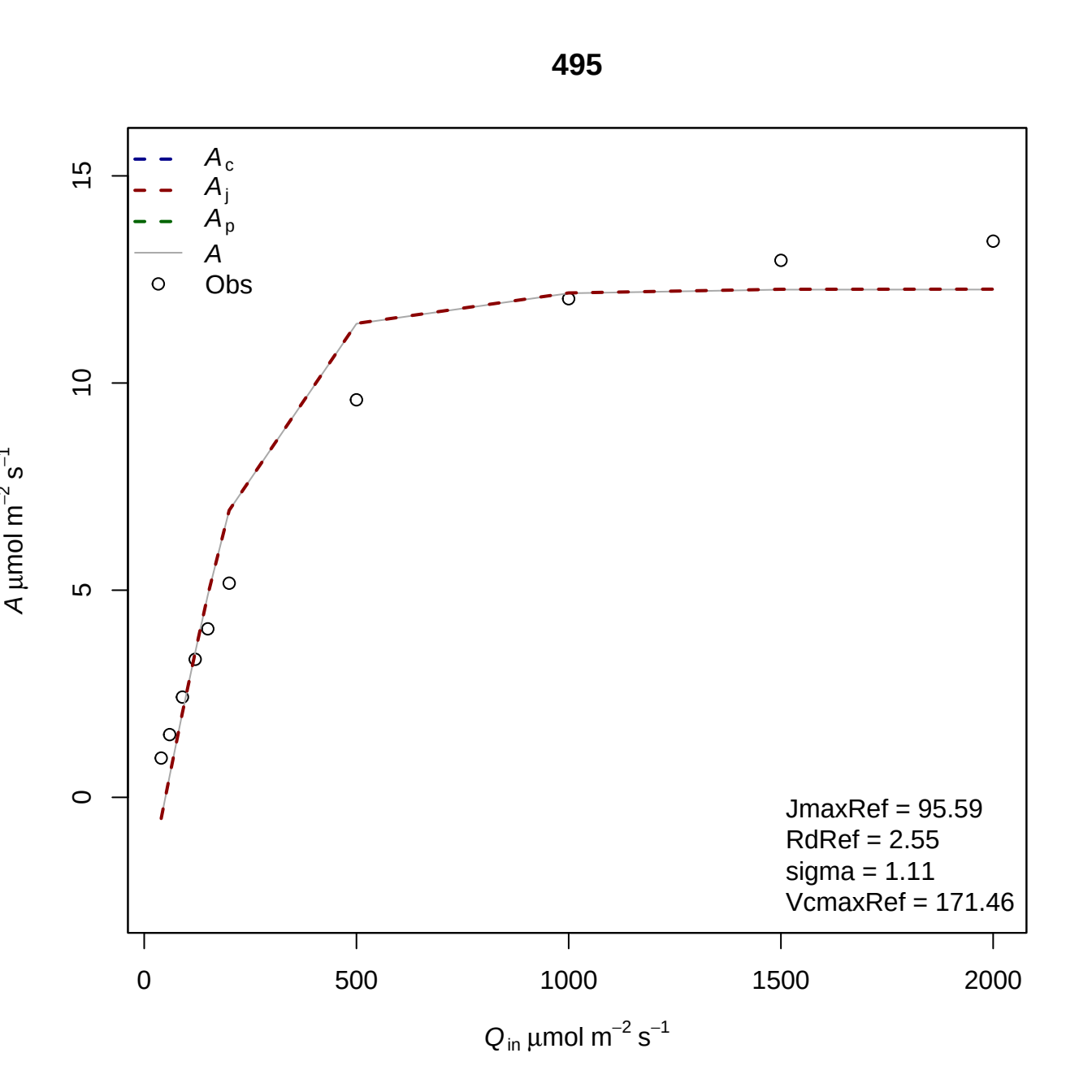

490
A_c
A_j
A_p
A
Obs
JmaxRef = 107.92
RdRef = 1.53
sigma = 0.43
VcmaxRef = 316.92
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

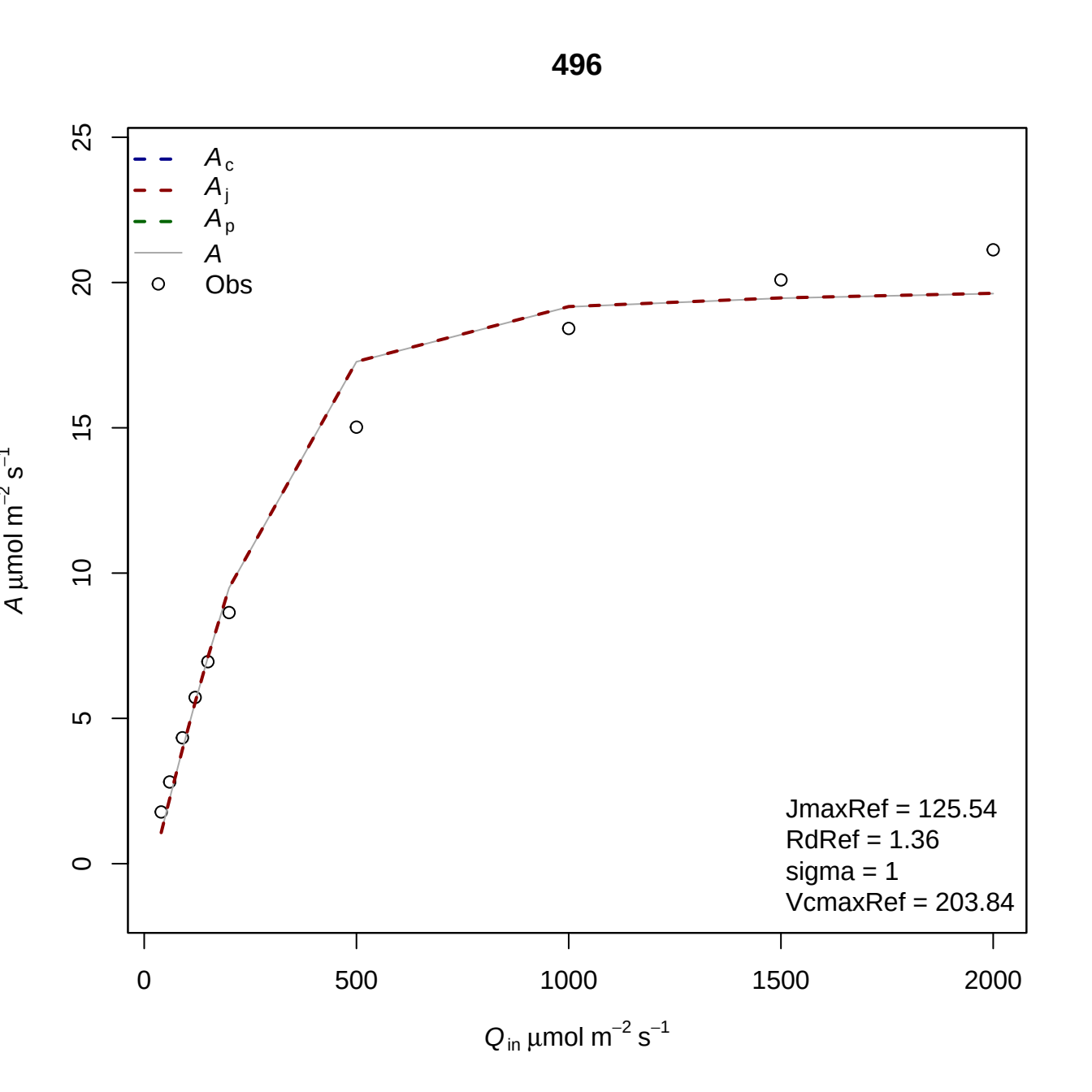

491
A_c
A_j
A_p
A
Obs
JmaxRef = 97.59
RdRef = 3.78
sigma = 0.53
VcmaxRef = 389.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
500
1000
1500
2000

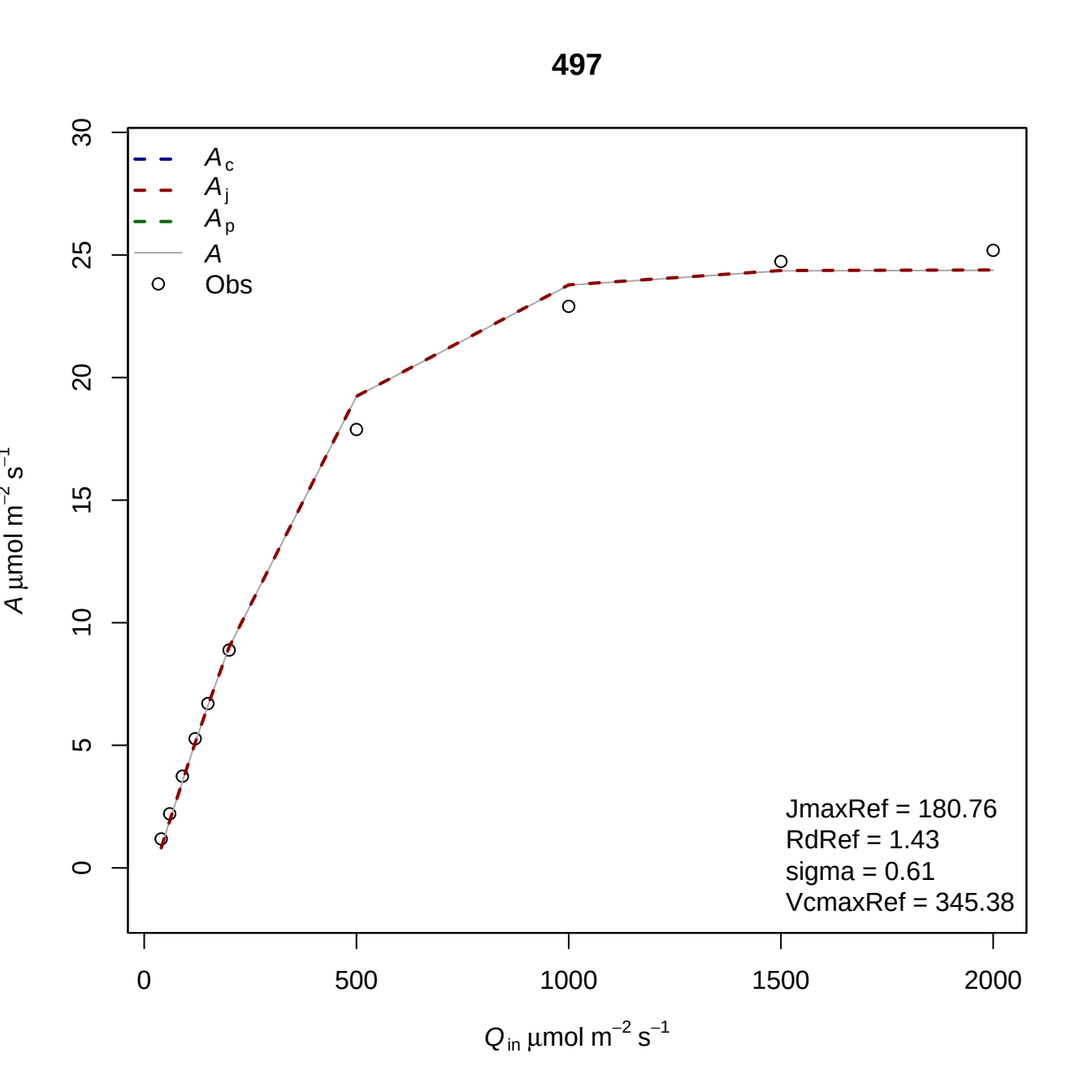

492
A_c
A_j
A_p
A
Obs
JmaxRef = 75.7
RdRef = 3.37
sigma = 0.96
VcmaxRef = 324.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

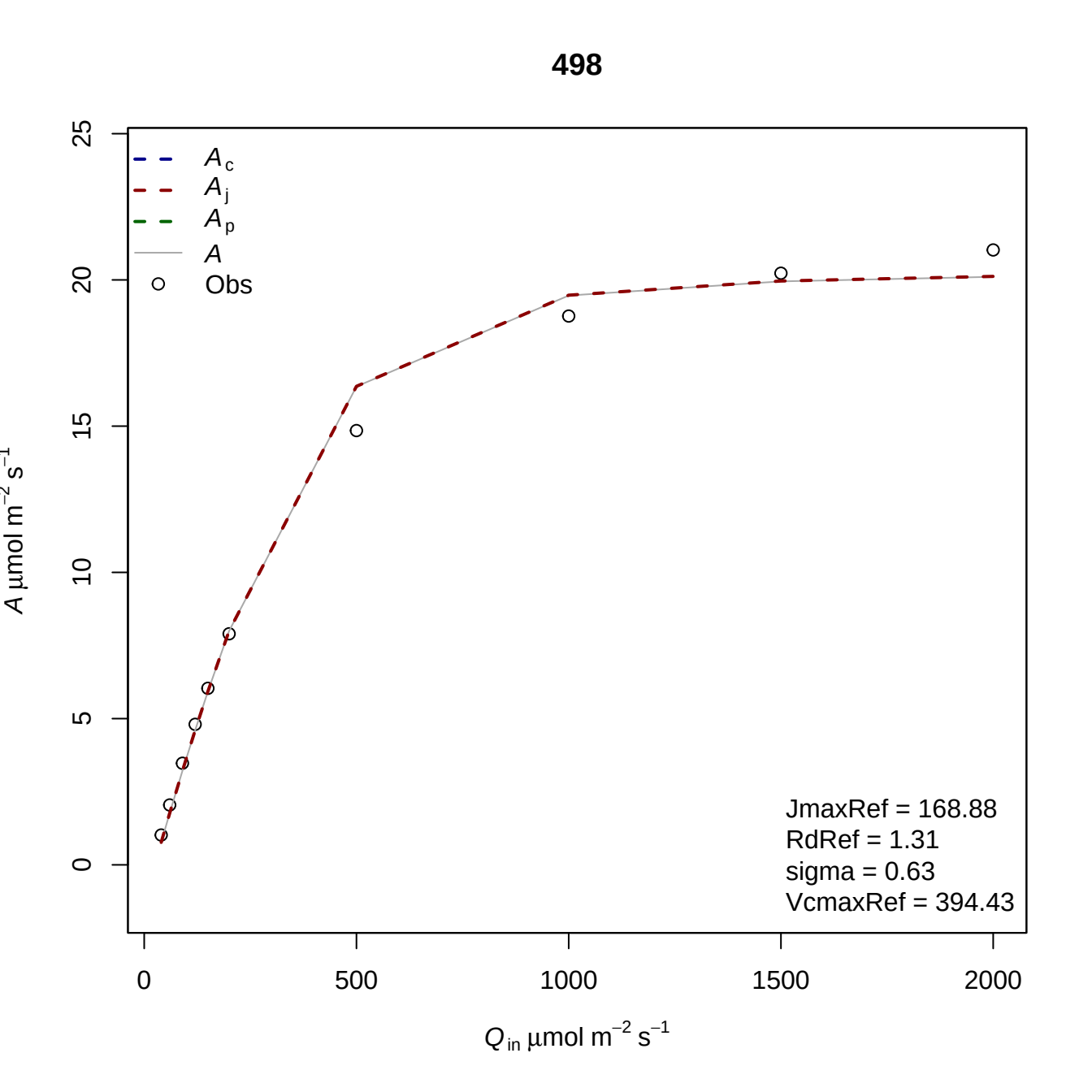

493
A_c
A_j
A_p
A
Obs
JmaxRef = 47.91
RdRef = 2.59
sigma = 0.78
VcmaxRef = 53.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

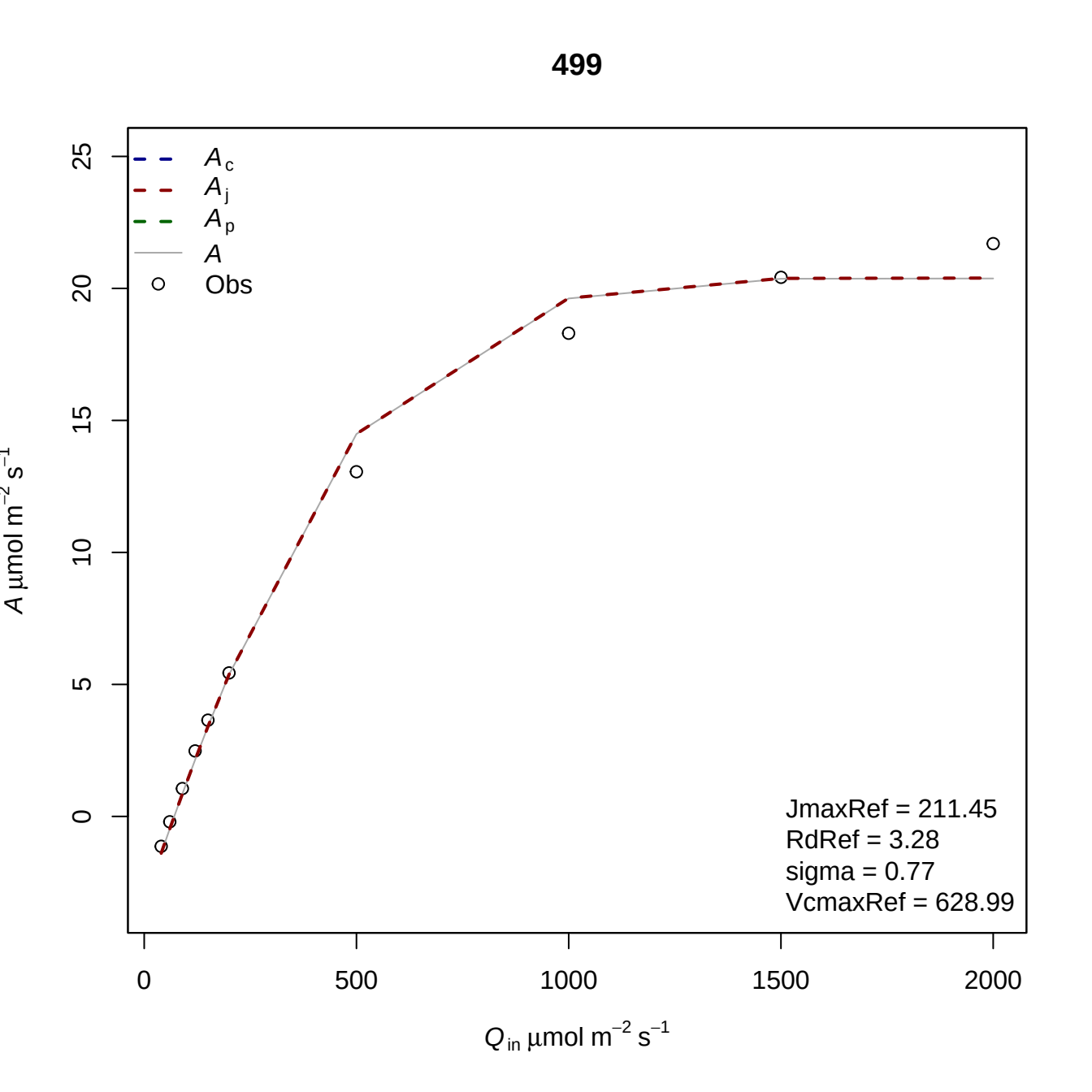

494
A_c
A_j
A_p
A
Obs
JmaxRef = 61.35
RdRef = 2.45
sigma = 0.79
VcmaxRef = 69.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

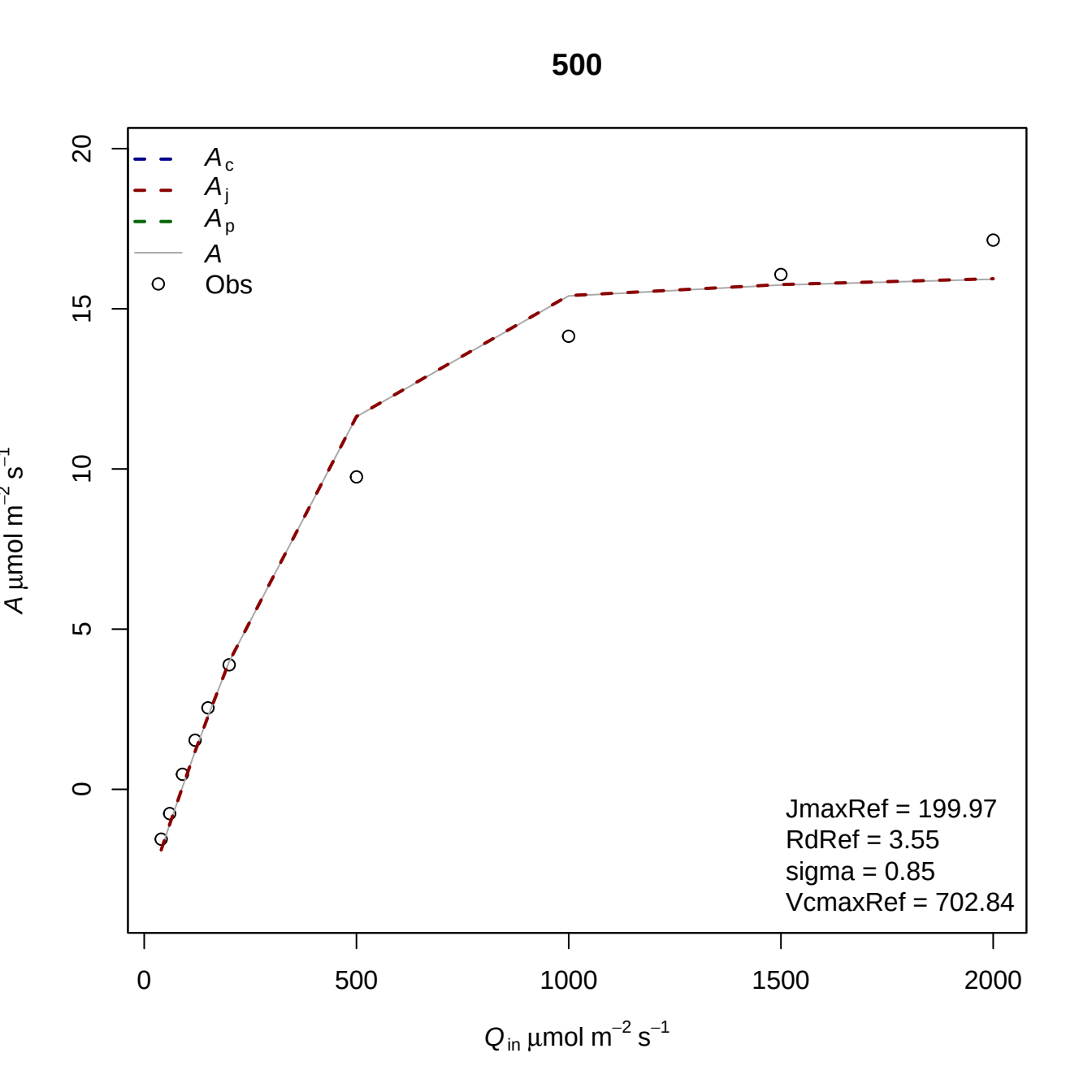

495
A_c
A_j
A_p
A
Obs
JmaxRef = 95.59
RdRef = 2.55
sigma = 1.11
VcmaxRef = 171.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

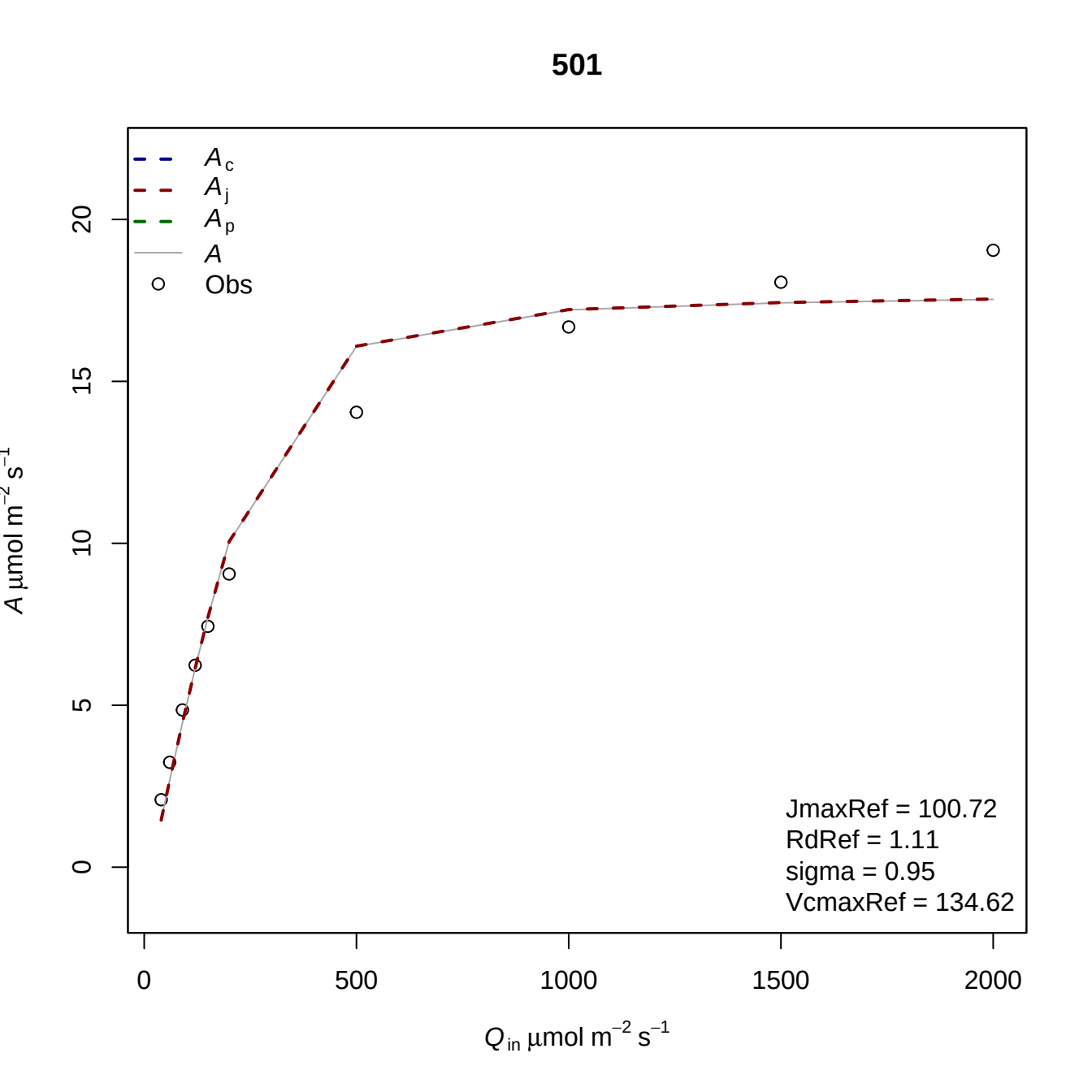

496
A_c
A_j
A_p
A
Obs
JmaxRef = 125.54
RdRef = 1.36
sigma = 1
VcmaxRef = 203.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

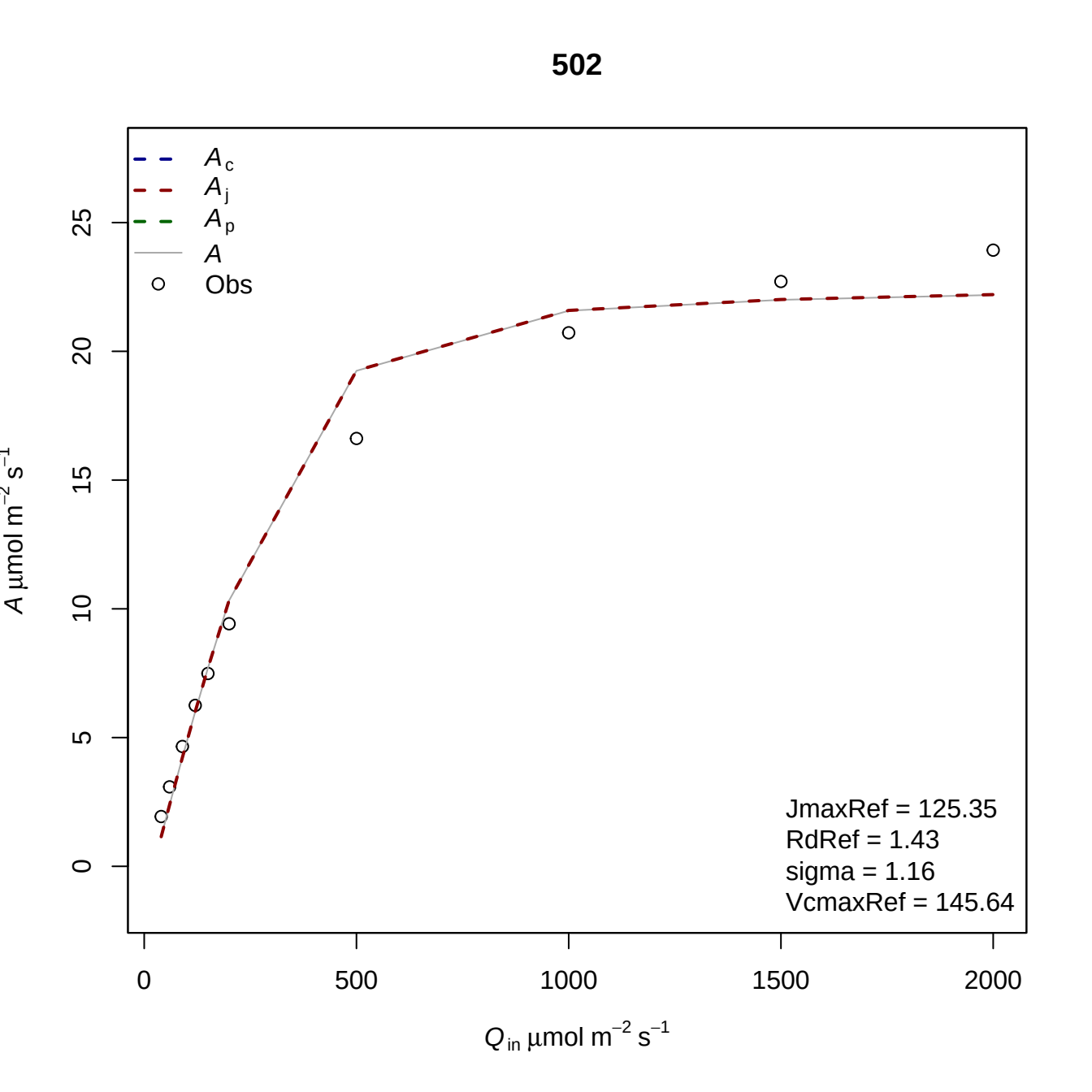

497
A_c
A_j
A_p
A
Obs
JmaxRef = 180.76
RdRef = 1.43
sigma = 0.61
VcmaxRef = 345.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

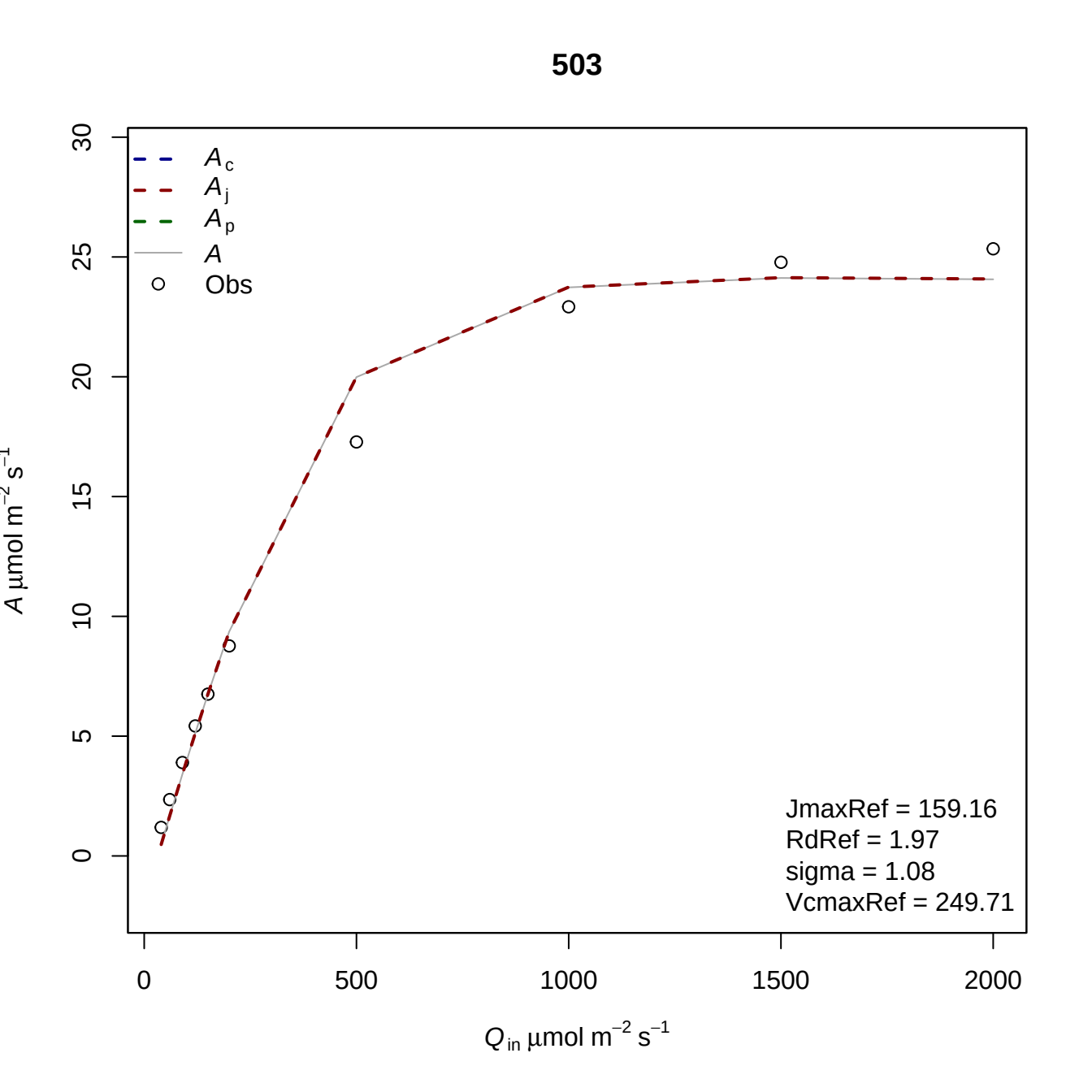

498
A_c
A_j
A_p
A
Obs
JmaxRef = 168.88
RdRef = 1.31
sigma = 0.63
VcmaxRef = 394.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

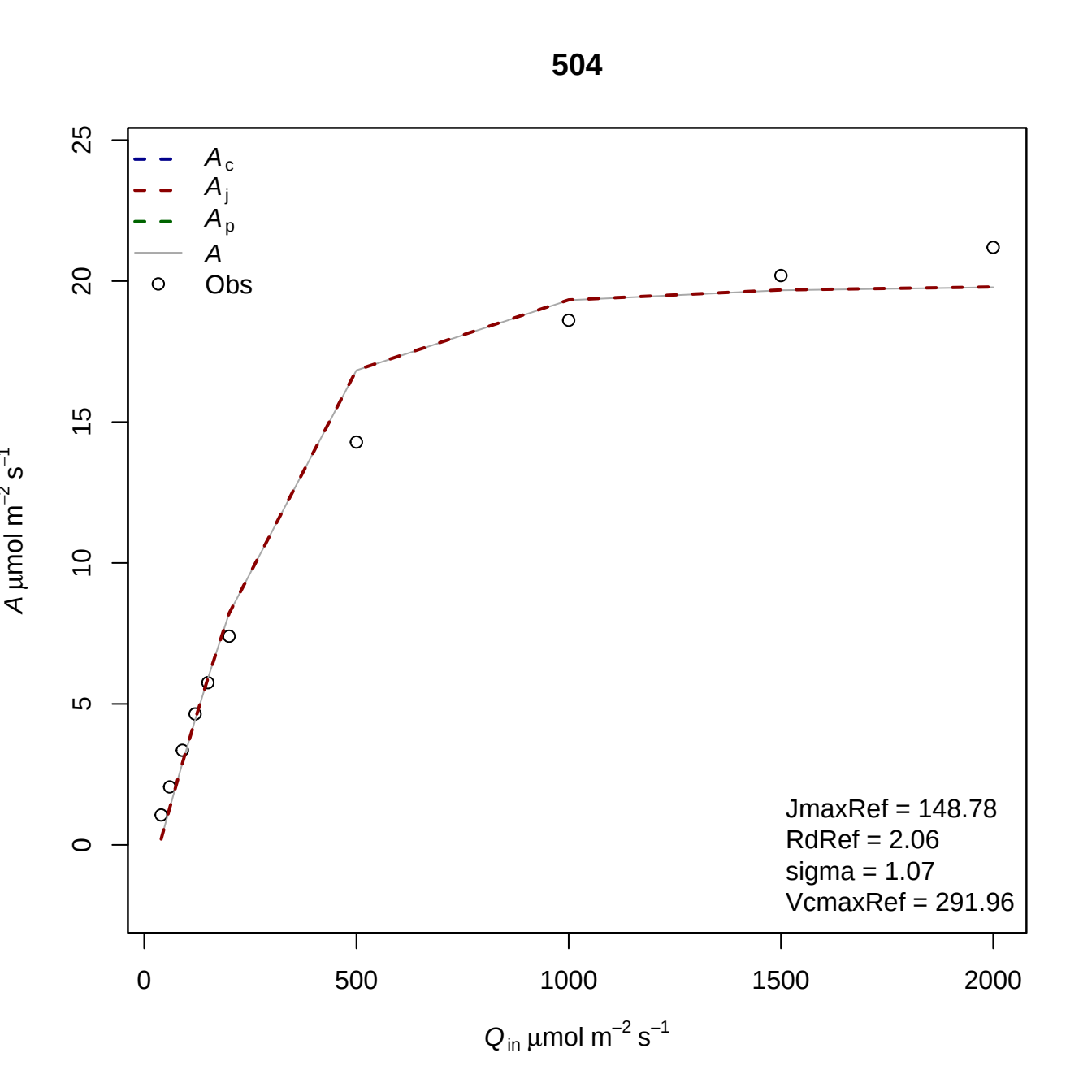

499
A_c
A_j
A_p
A
Obs
JmaxRef = 211.45
RdRef = 3.28
sigma = 0.77
VcmaxRef = 628.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

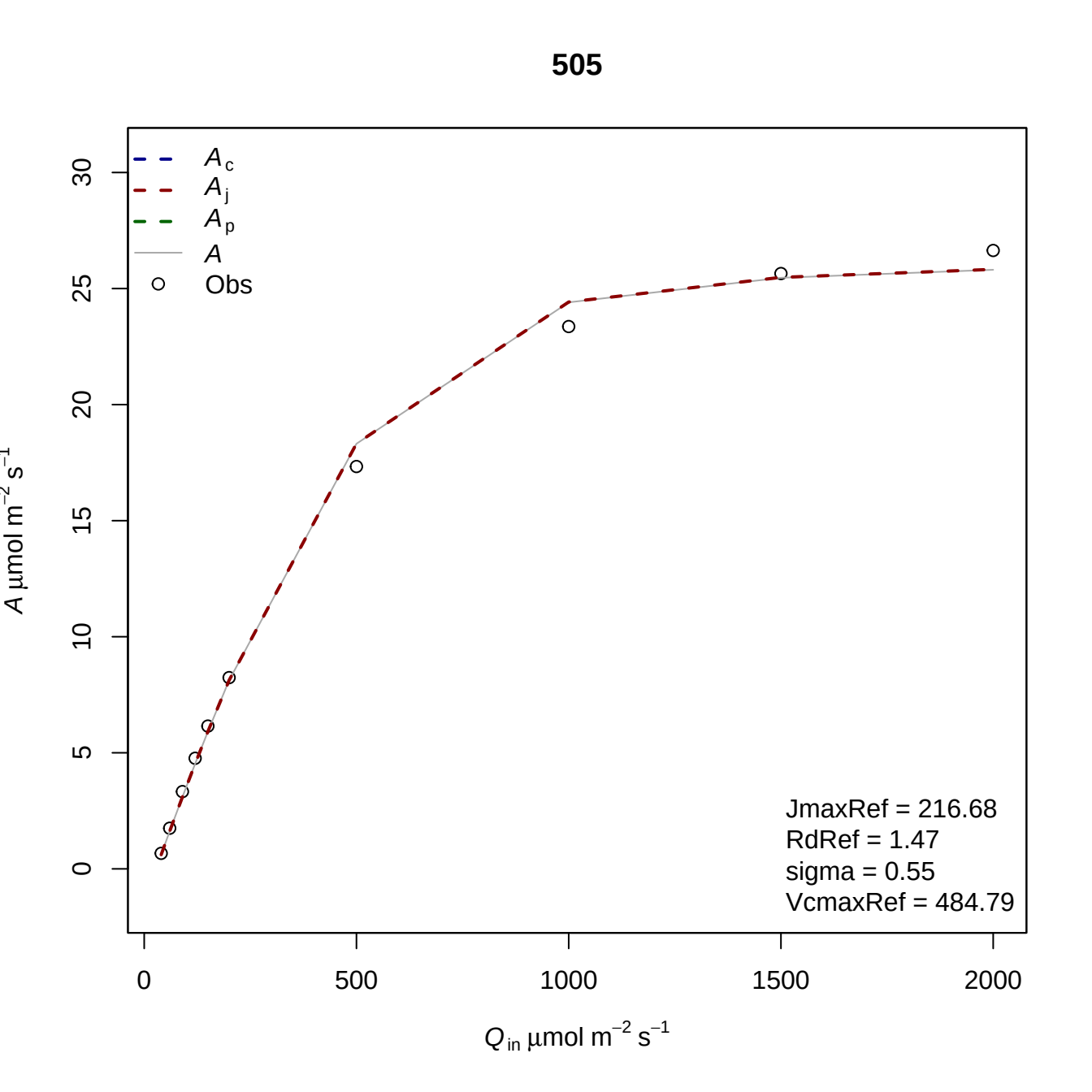

500
A_c
A_j
A_p
A
Obs
JmaxRef = 199.97
RdRef = 3.55
sigma = 0.85
VcmaxRef = 702.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

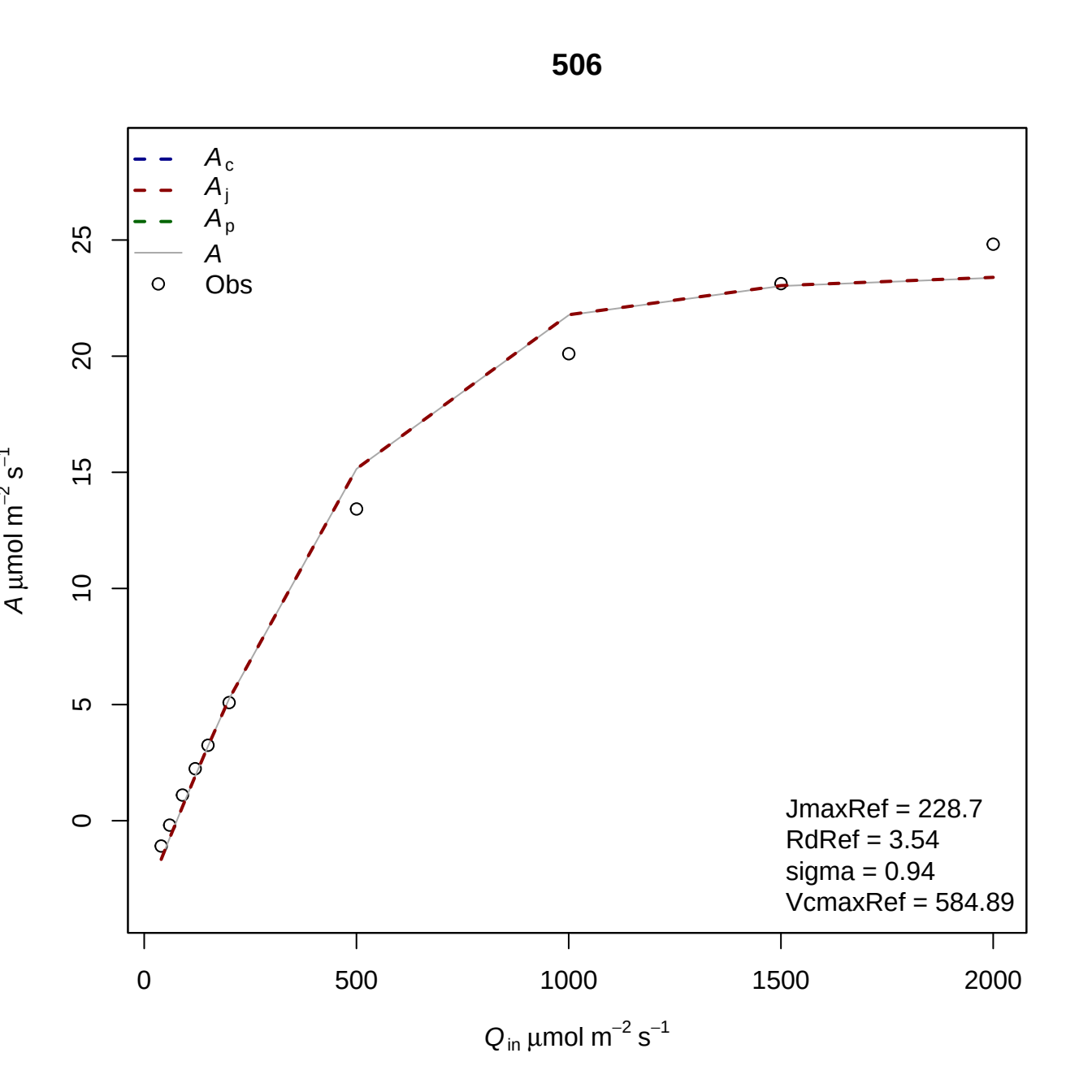

501
A_c
A_j
A_p
A
Obs
JmaxRef = 100.72
RdRef = 1.11
sigma = 0.95
VcmaxRef = 134.62
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

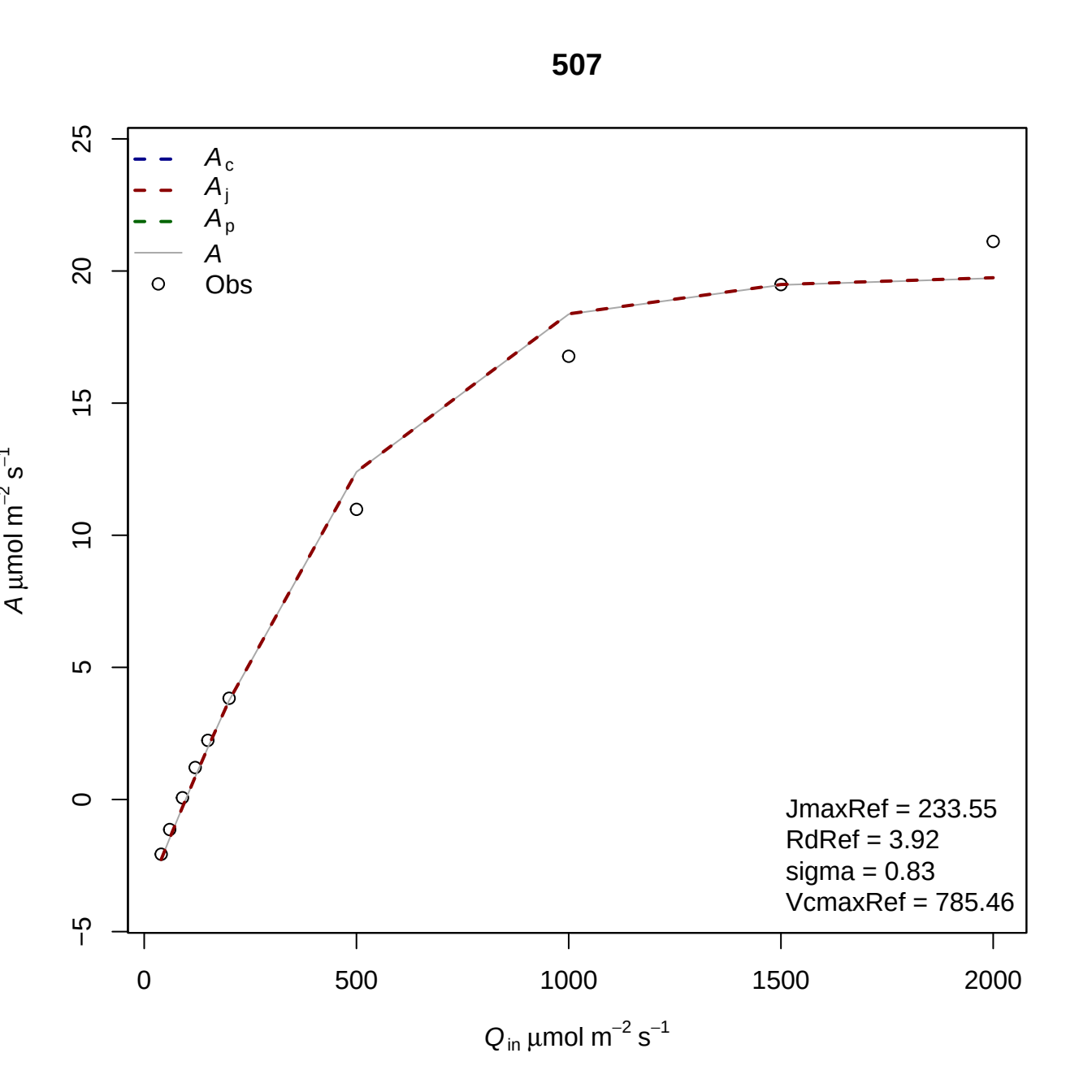

502
A_c
A_j
A_p
A
Obs
JmaxRef = 125.35
RdRef = 1.43
sigma = 1.16
VcmaxRef = 145.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

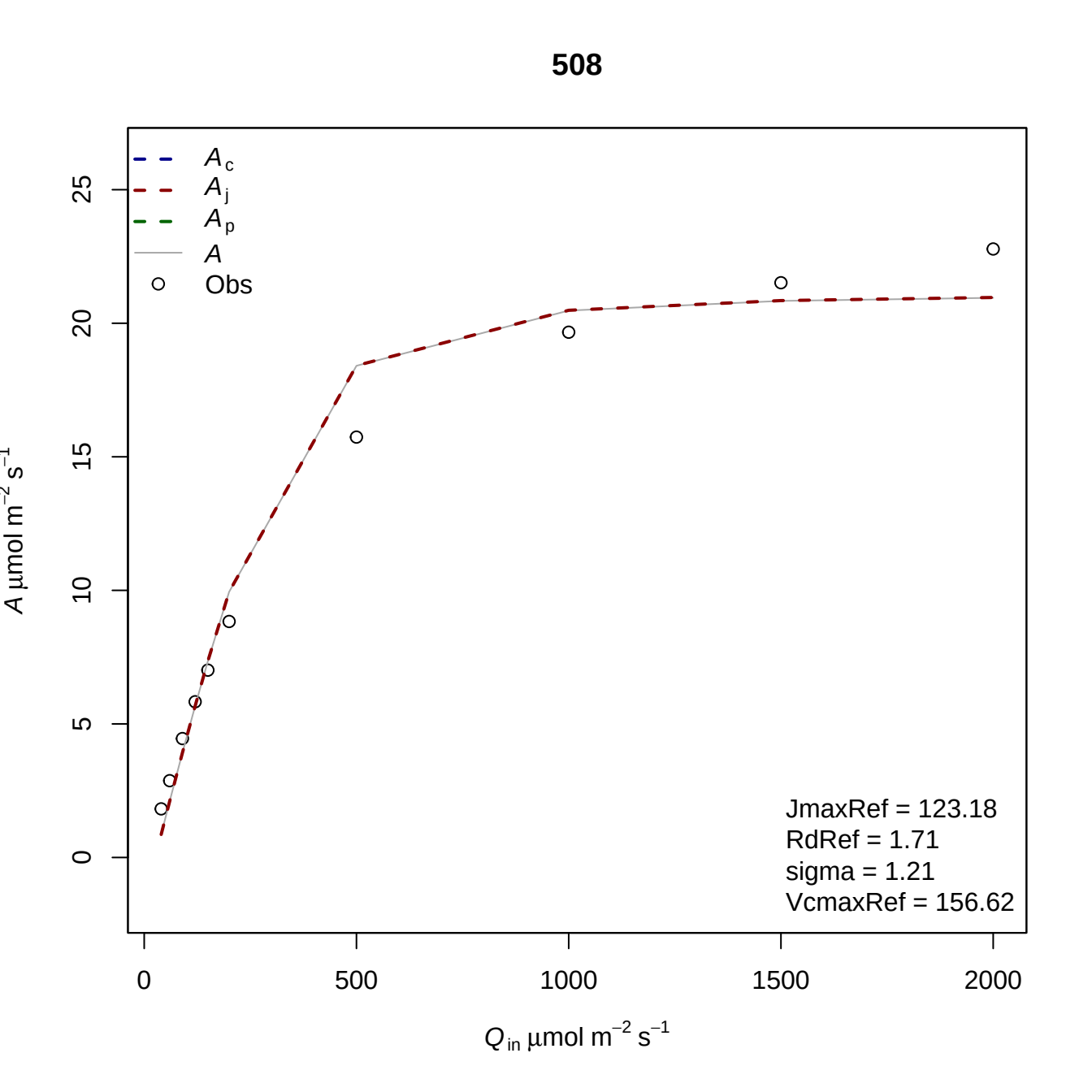

503
A_c
A_j
A_p
A
Obs
JmaxRef = 159.16
RdRef = 1.97
sigma = 1.08
VcmaxRef = 249.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

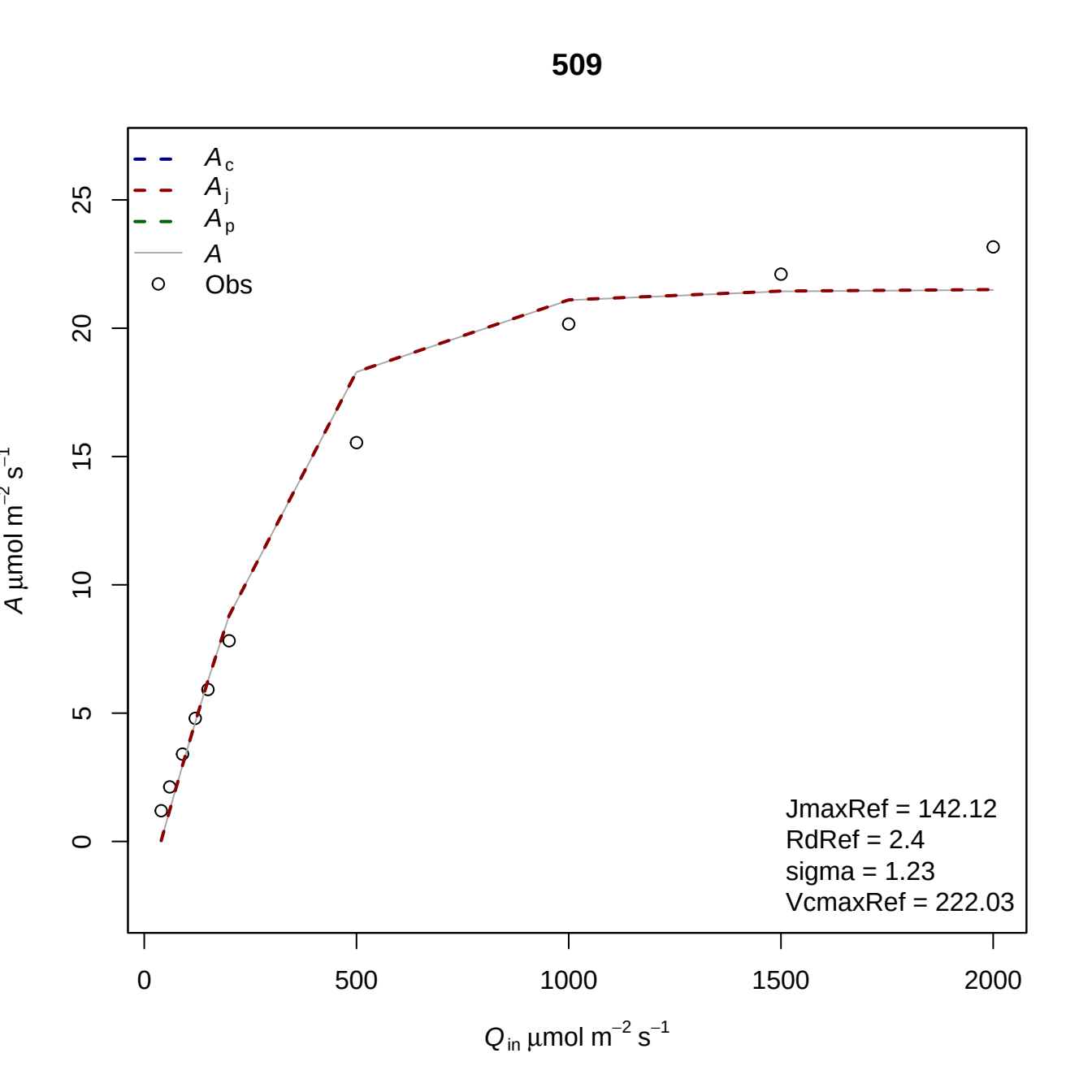

504
A_c
A_j
A_p
A
Obs
JmaxRef = 148.78
RdRef = 2.06
sigma = 1.07
VcmaxRef = 291.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

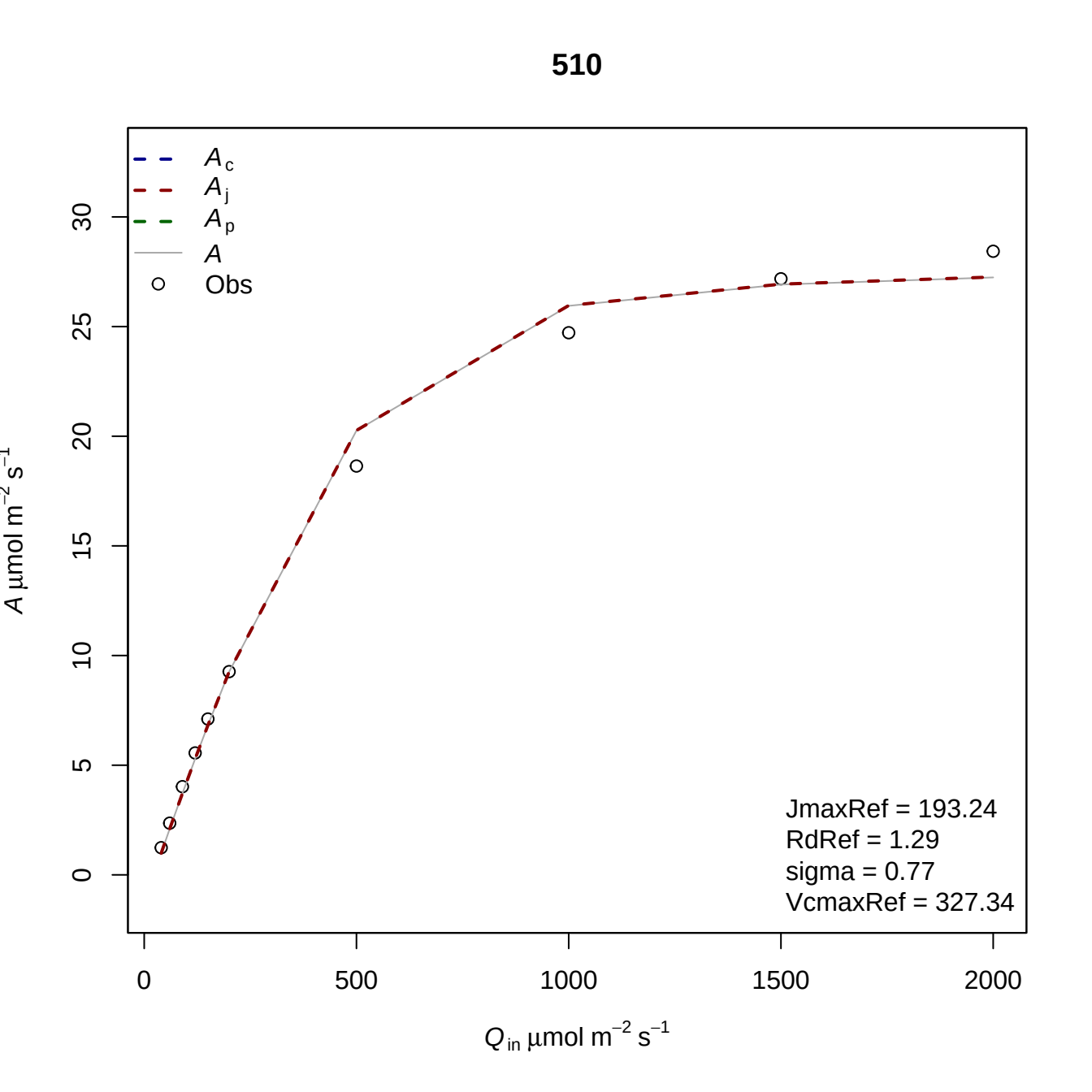

505
A_c
A_j
A_p
A
Obs
JmaxRef = 216.68
RdRef = 1.47
sigma = 0.55
VcmaxRef = 484.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

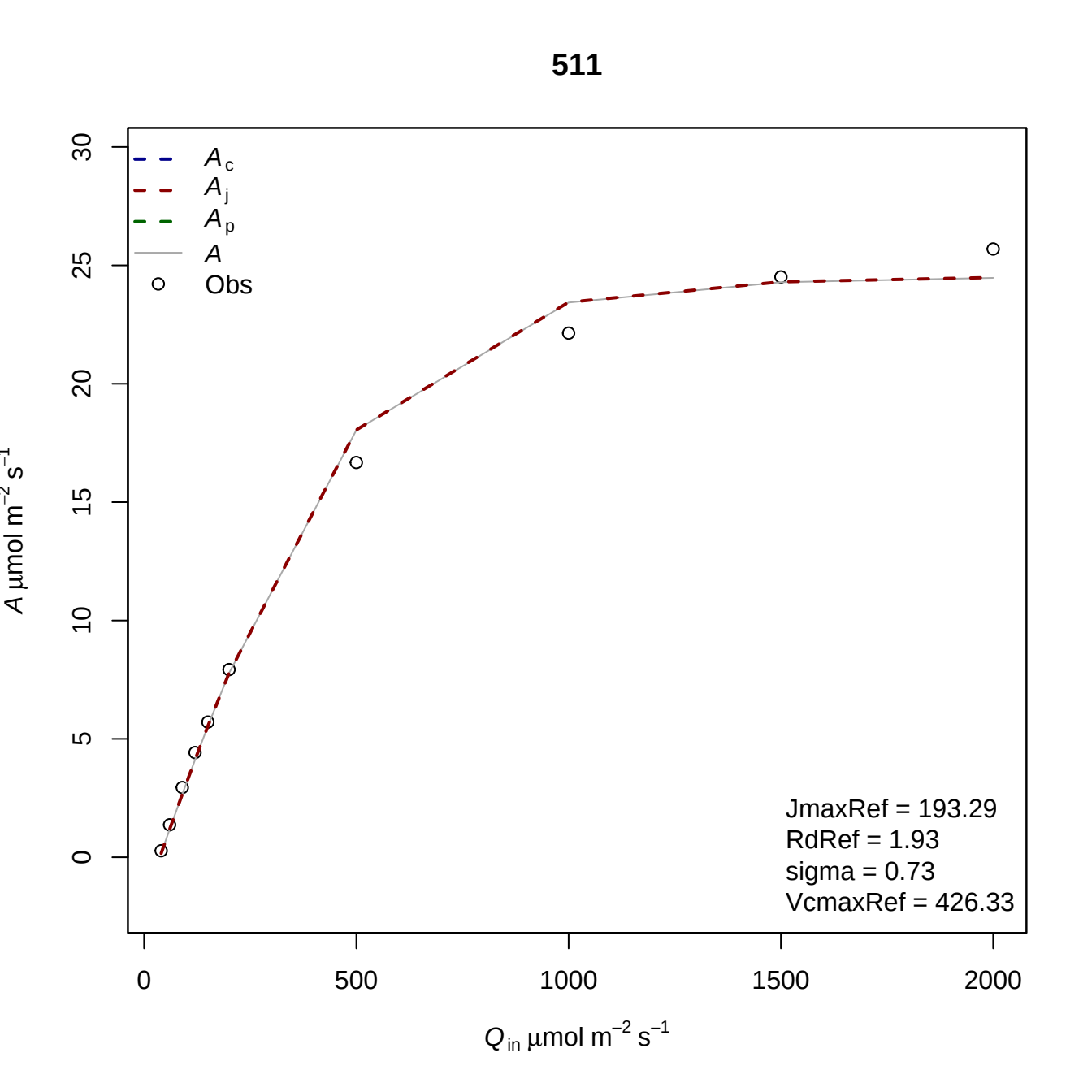

506
A_c
A_j
A_p
A
Obs
JmaxRef = 228.7
RdRef = 3.54
sigma = 0.94
VcmaxRef = 584.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

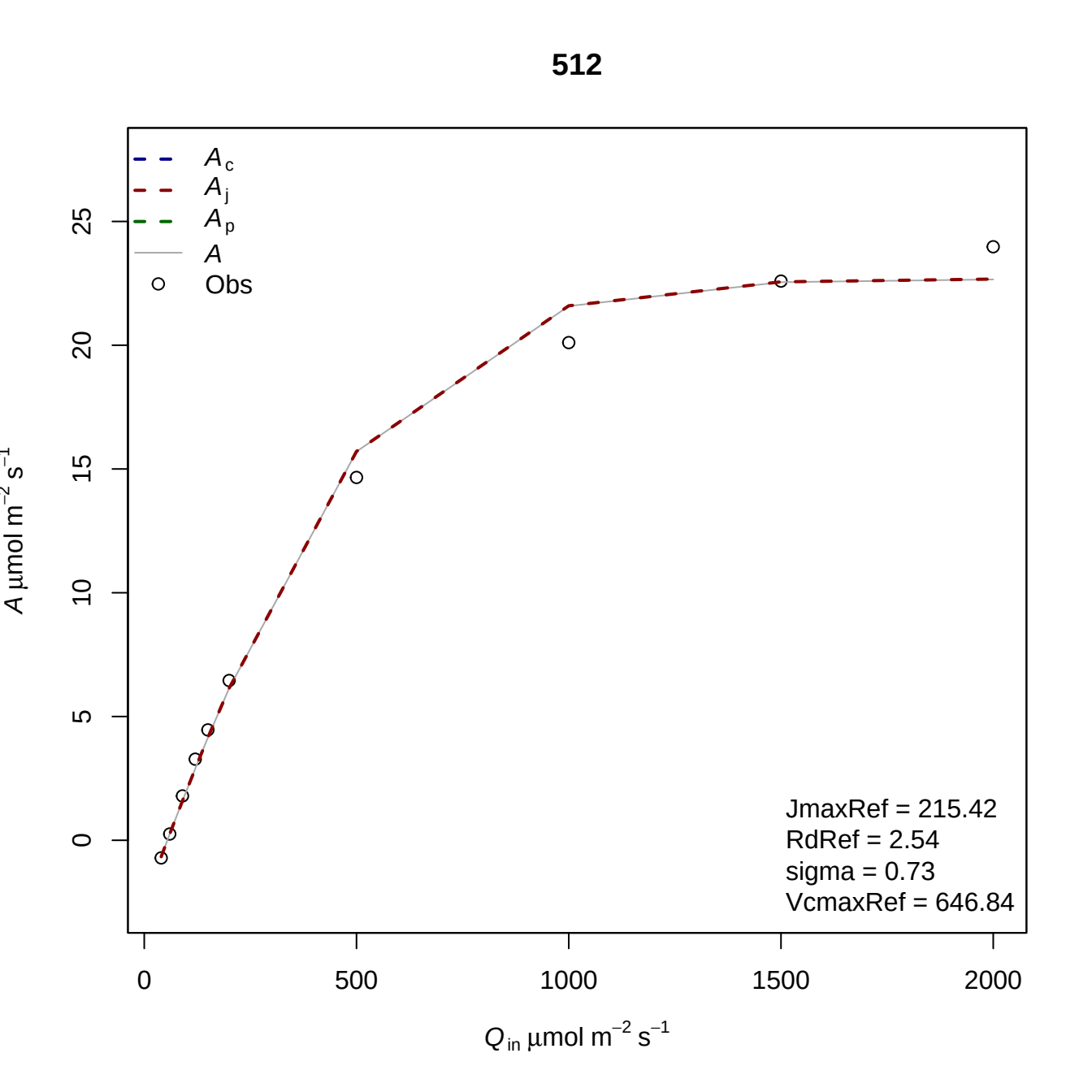

507
A_c
A_j
A_p
A
Obs
JmaxRef = 233.55
RdRef = 3.92
sigma = 0.83
VcmaxRef = 785.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

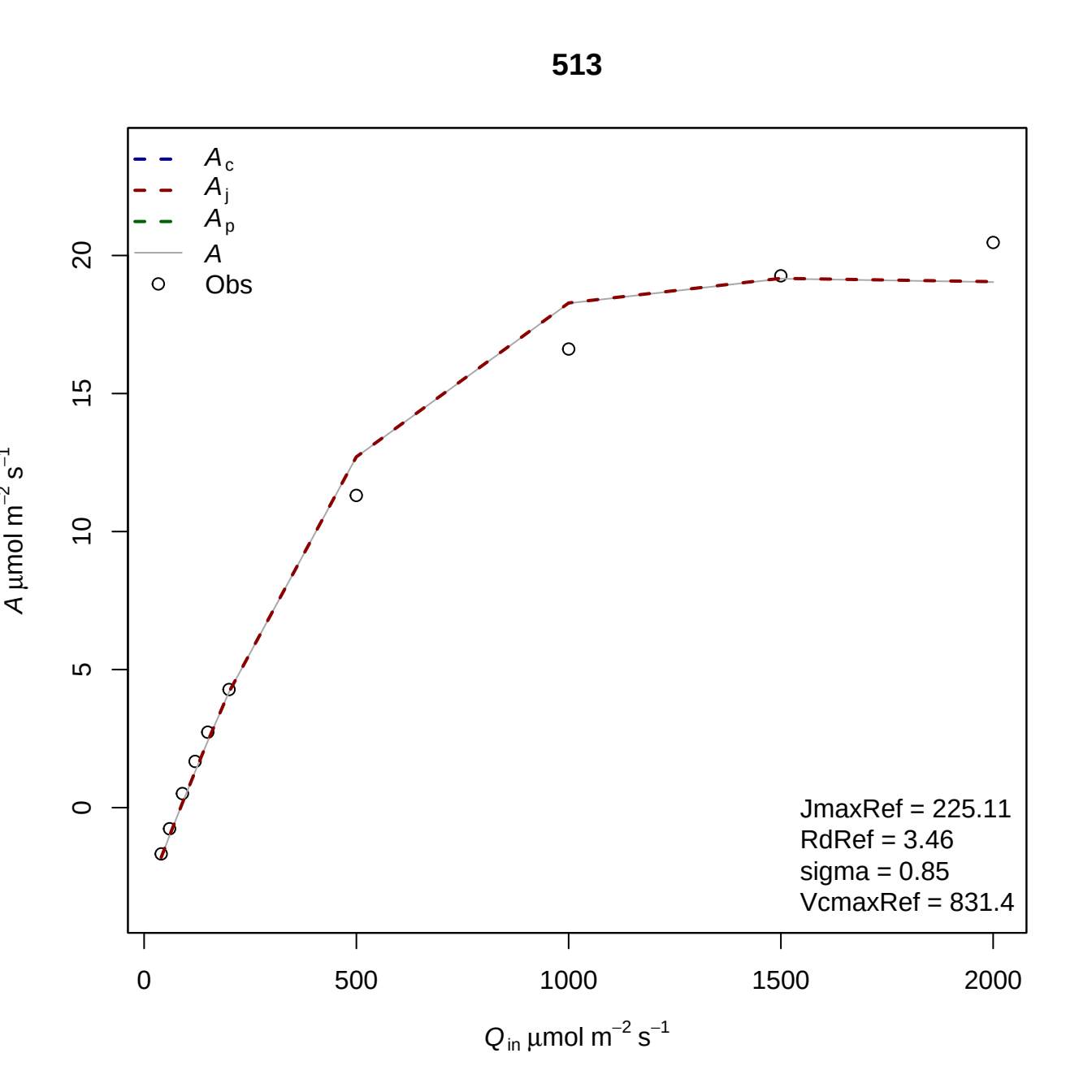

508
A_c
A_j
A_p
A
Obs
JmaxRef = 123.18
RdRef = 1.71
sigma = 1.21
VcmaxRef = 156.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

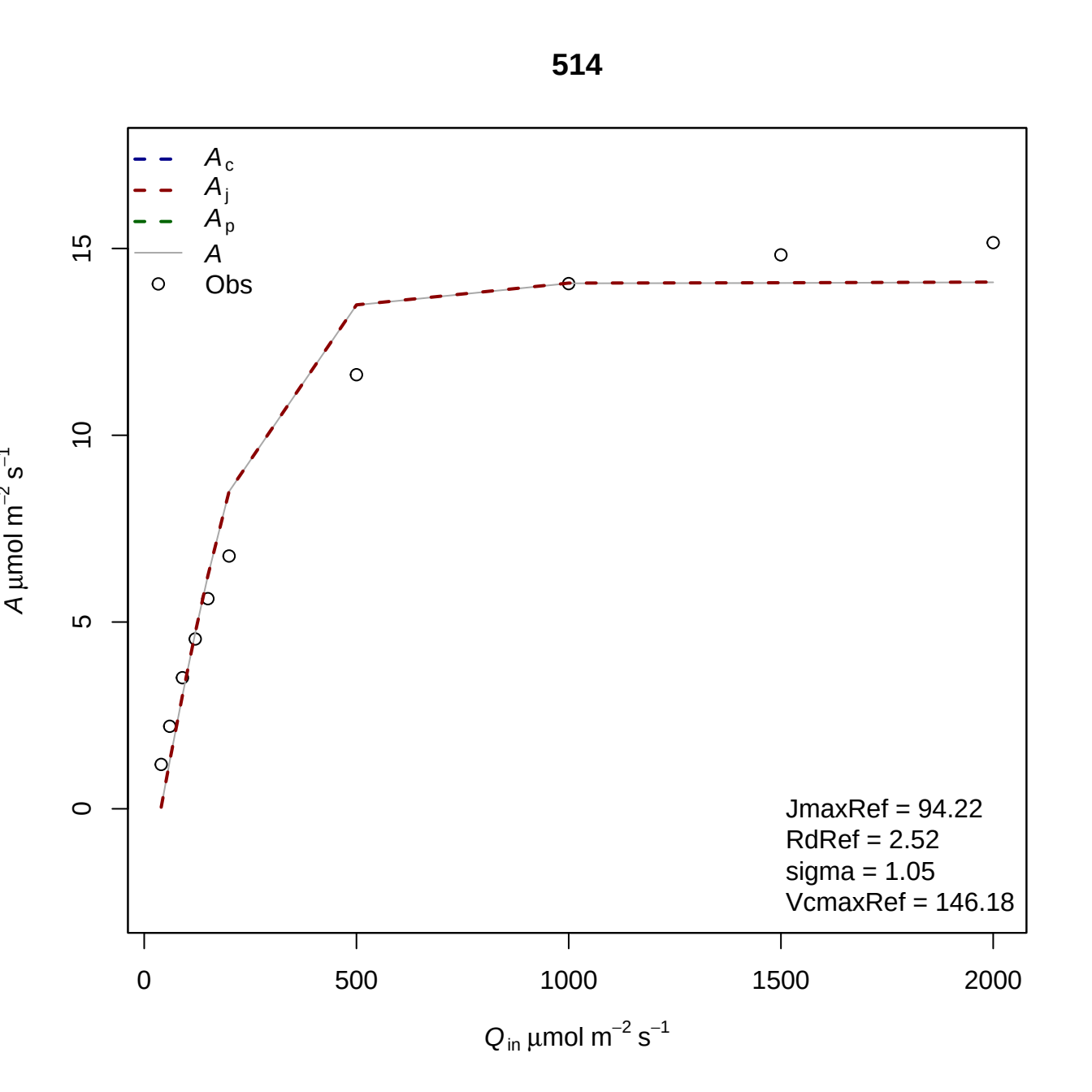

509
A_c
A_j
A_p
A
Obs
JmaxRef = 142.12
RdRef = 2.4
sigma = 1.23
VcmaxRef = 222.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

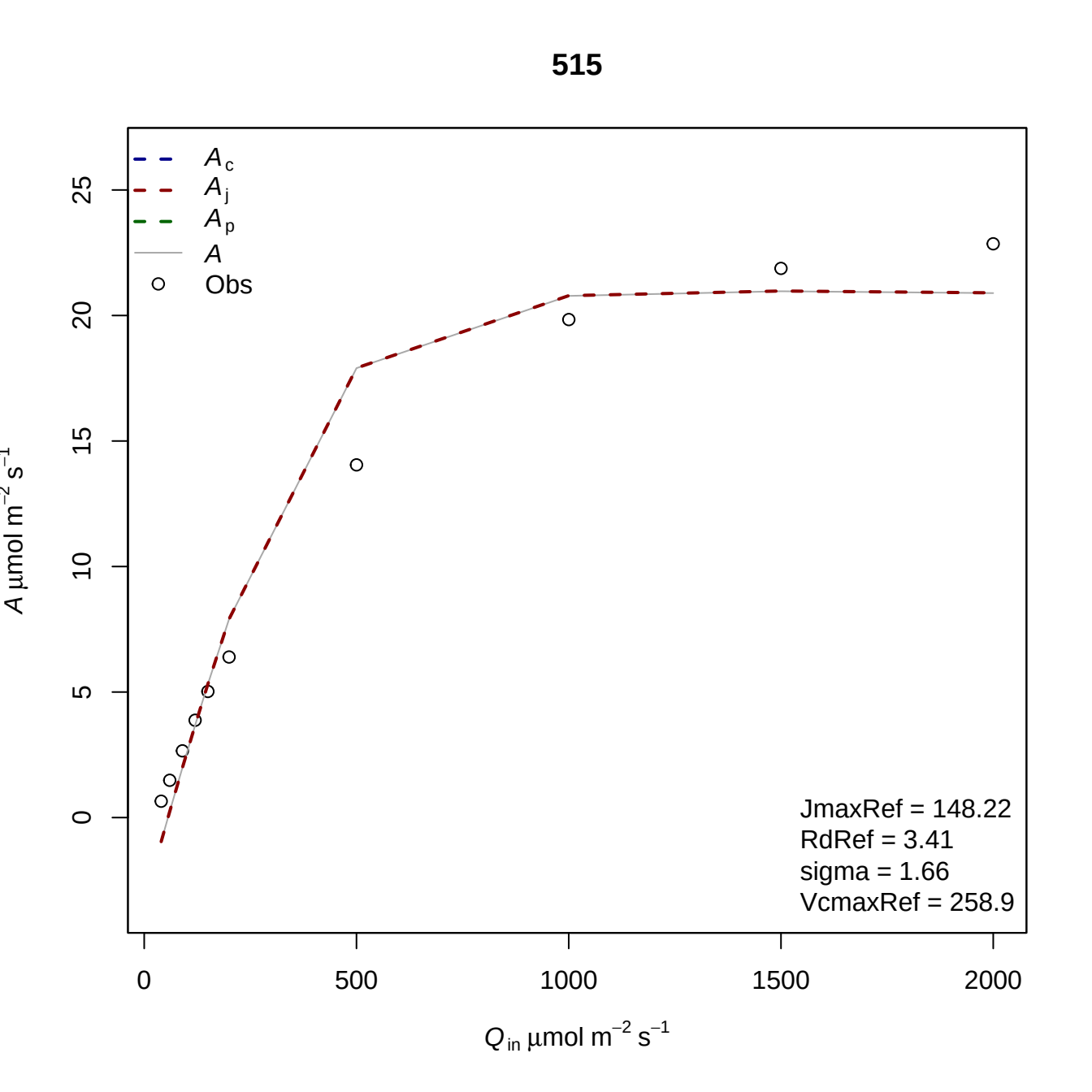

510
A_c
A_j
A_p
A
Obs
JmaxRef = 193.24
RdRef = 1.29
sigma = 0.77
VcmaxRef = 327.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

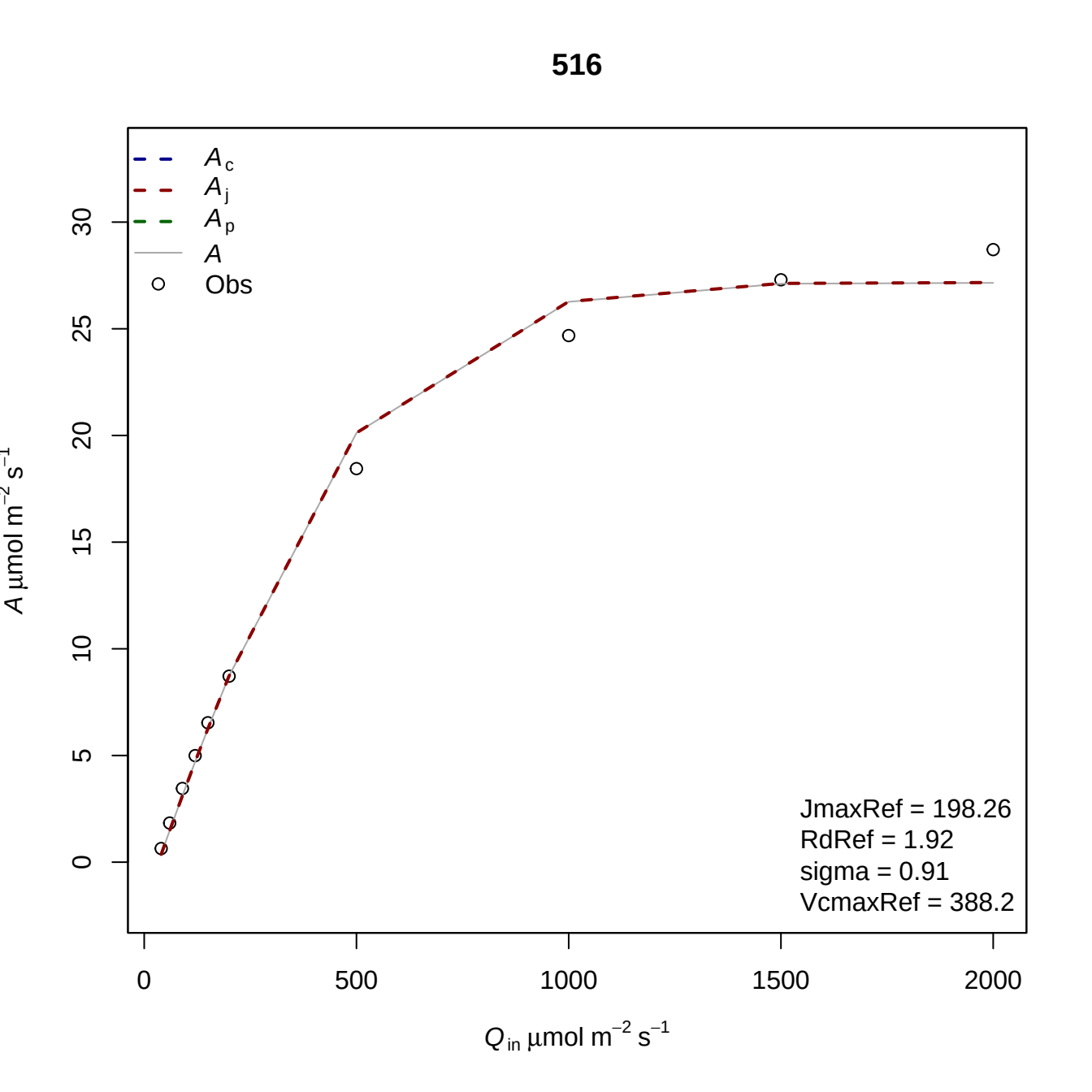

511
A_c
A_j
A_p
A
Obs
JmaxRef = 193.29
RdRef = 1.93
sigma = 0.73
VcmaxRef = 426.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

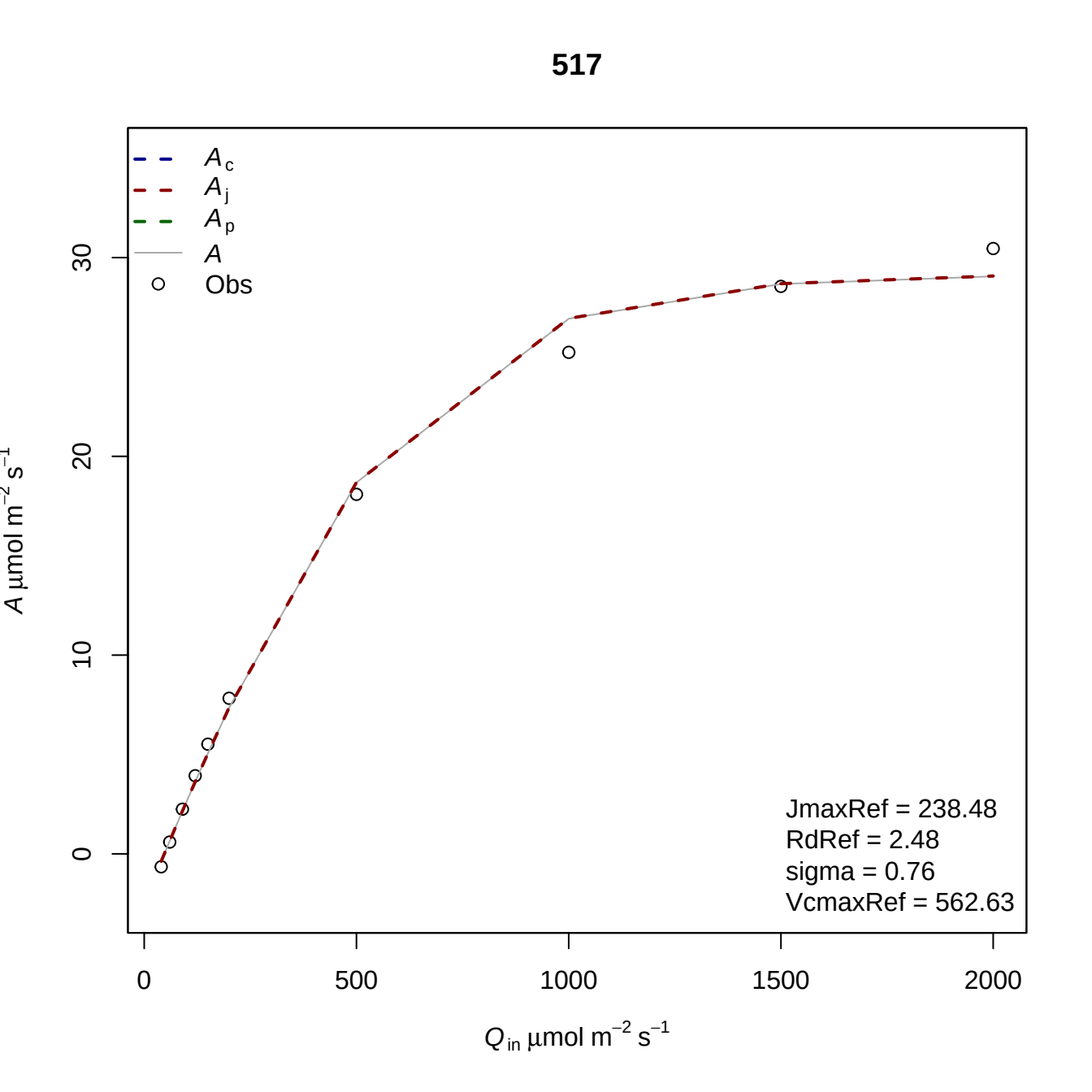

512
A_c
A_j
A_p
A
Obs
JmaxRef = 215.42
RdRef = 2.54
sigma = 0.73
VcmaxRef = 646.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

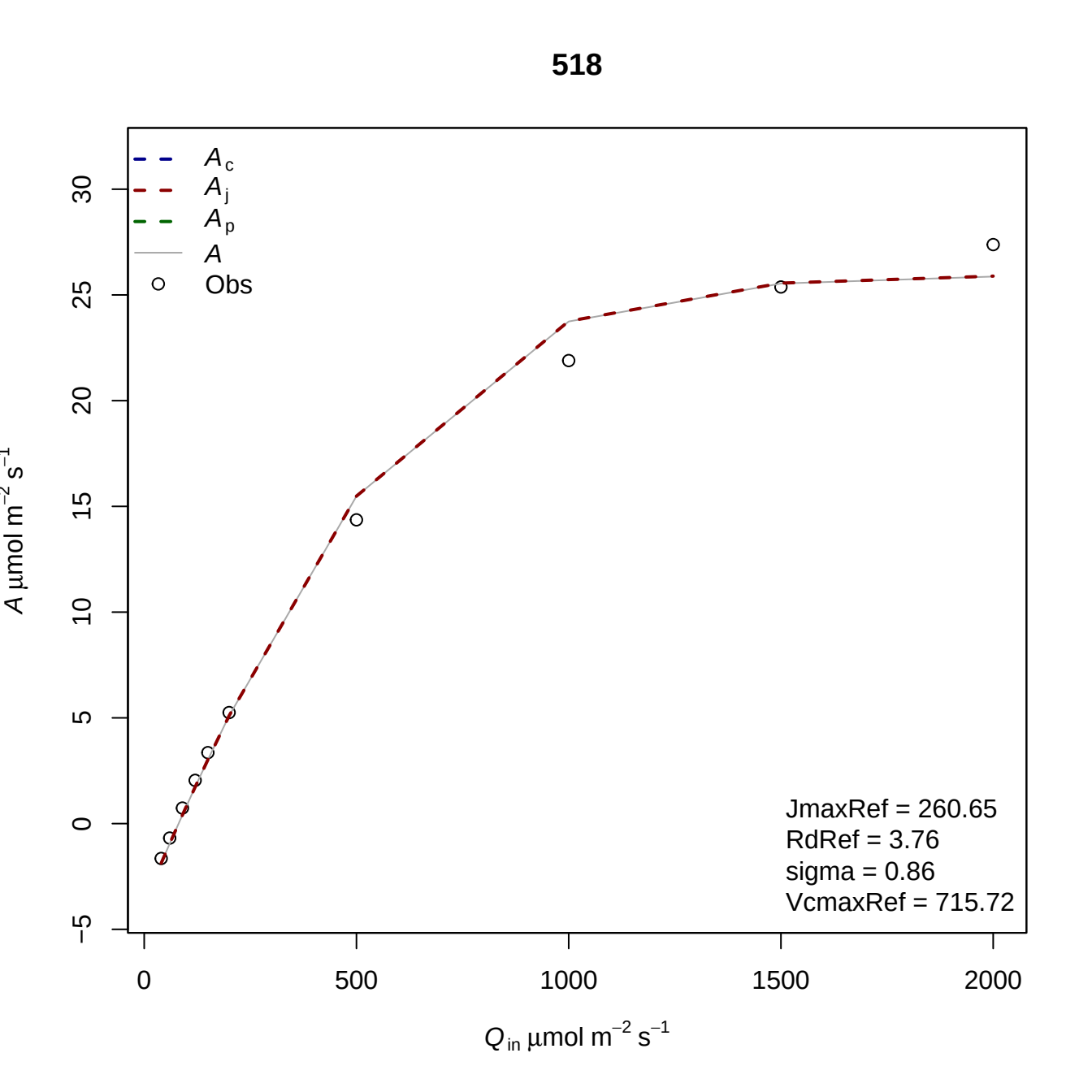

513
A_c
A_j
A_p
A
Obs
JmaxRef = 225.11
RdRef = 3.46
sigma = 0.85
VcmaxRef = 831.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

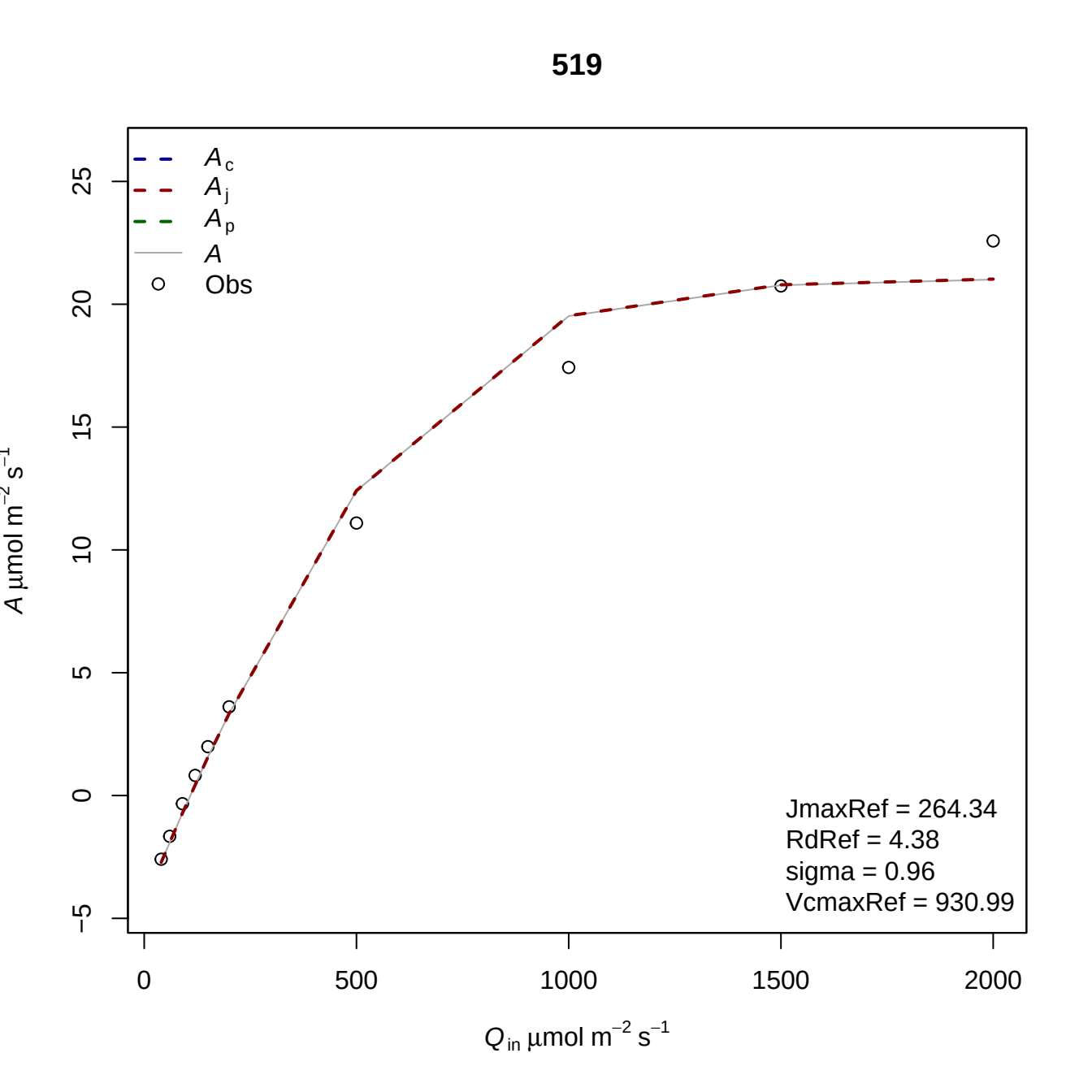

514
A_c
A_j
A_p
A
Obs
JmaxRef = 94.22
RdRef = 2.52
sigma = 1.05
VcmaxRef = 146.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

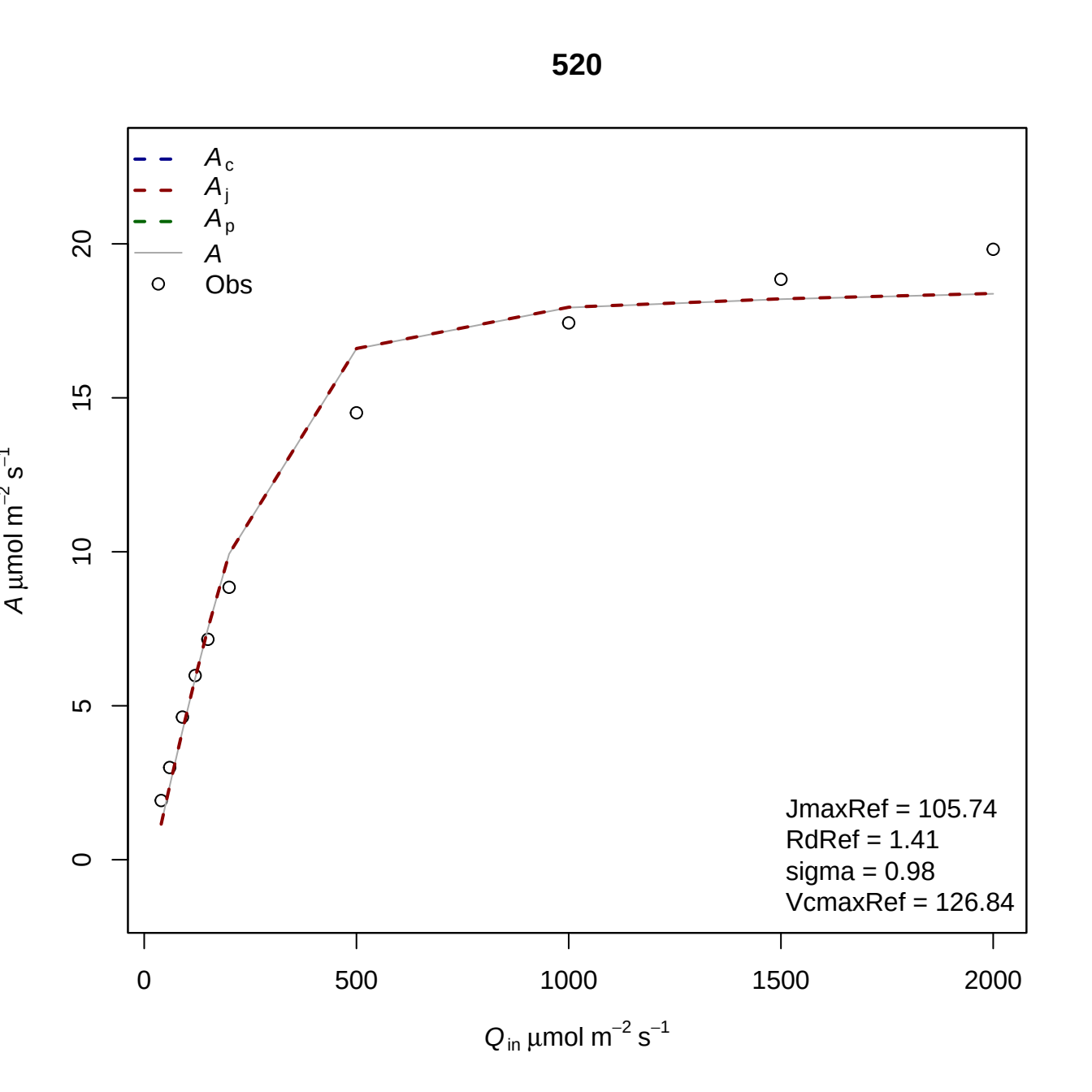

515
A_c
A_j
A_p
A
Obs
JmaxRef = 148.22
RdRef = 3.41
sigma = 1.66
VcmaxRef = 258.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

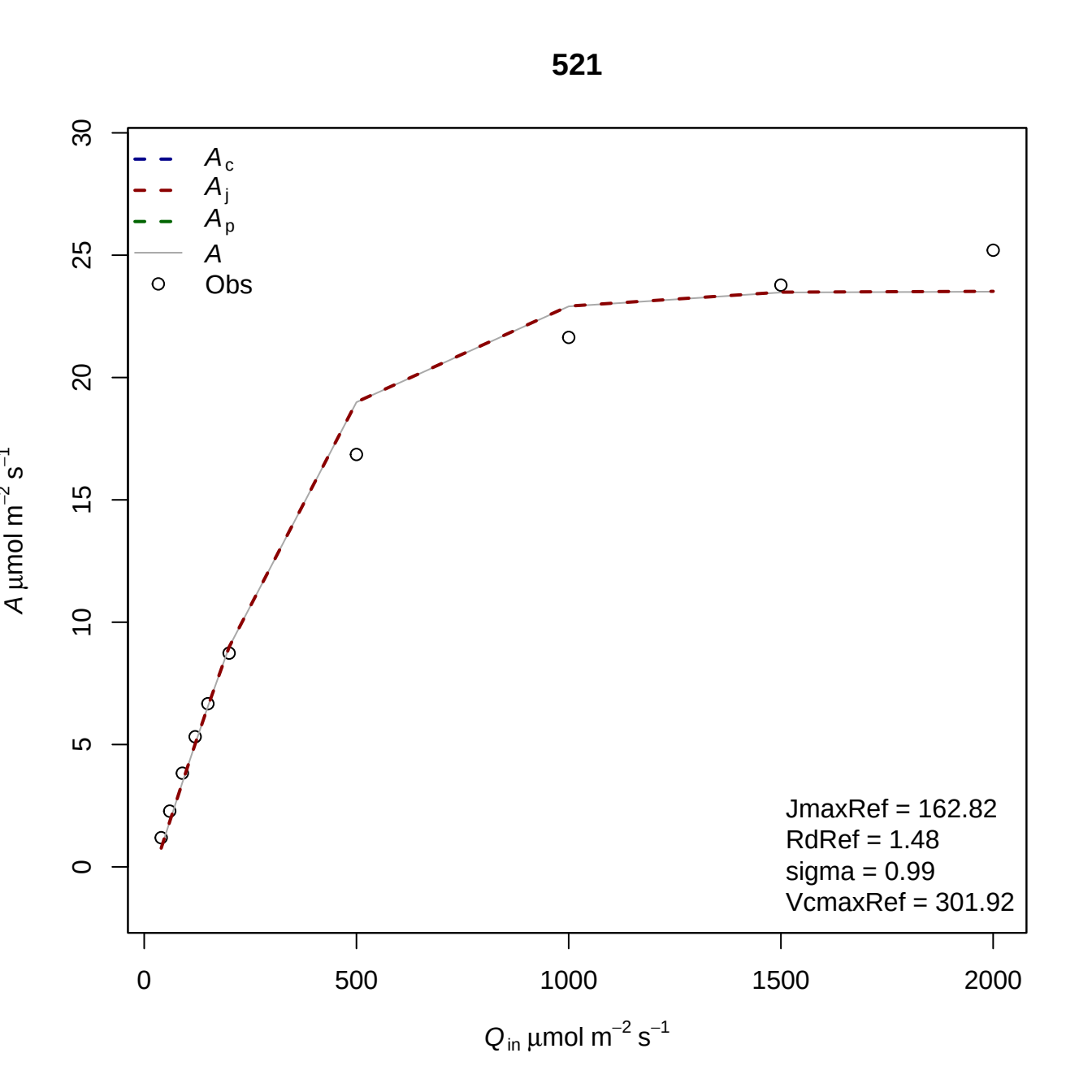

516
A_c
A_j
A_p
A
Obs
JmaxRef = 198.26
RdRef = 1.92
sigma = 0.91
VcmaxRef = 388.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

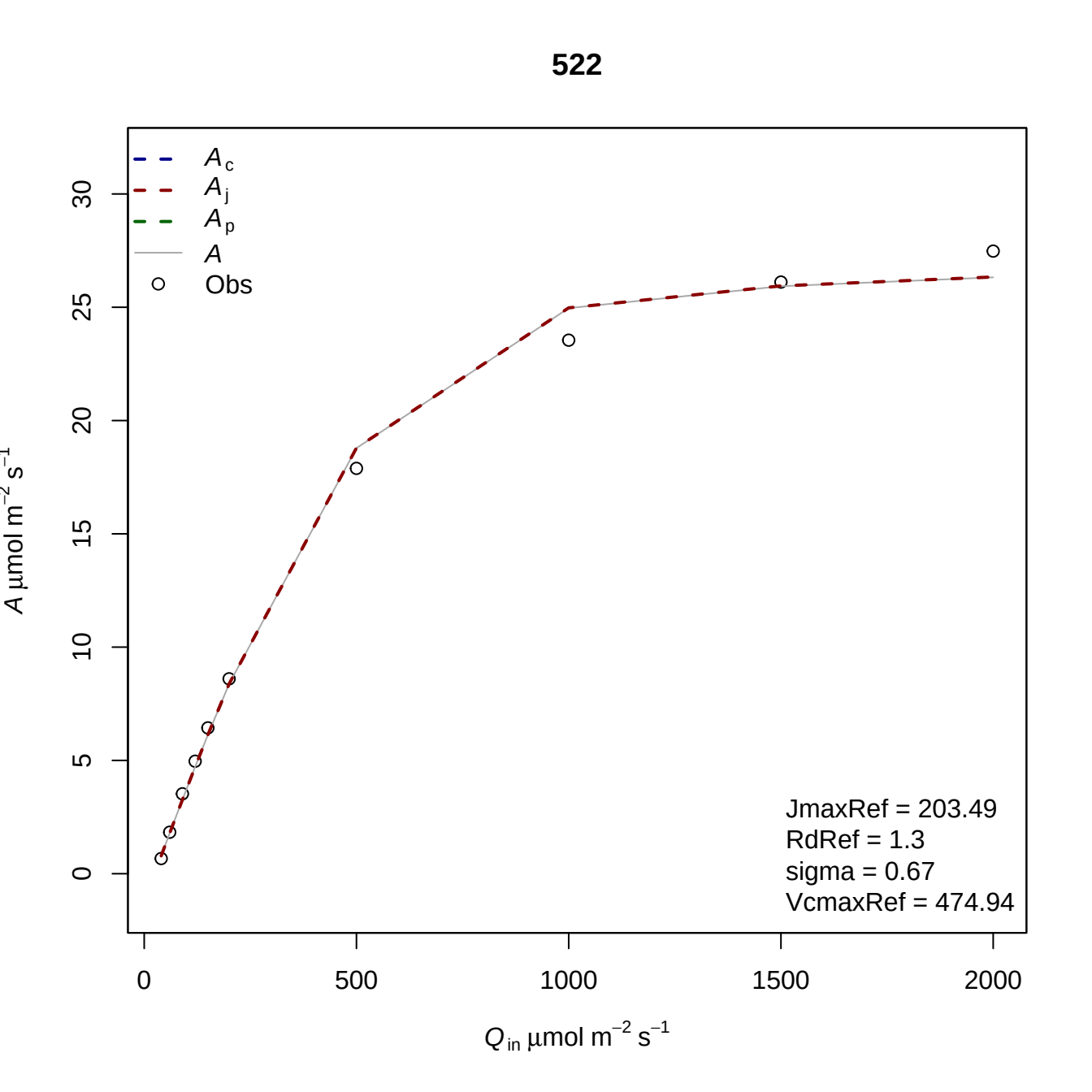

517
A_c
A_j
A_p
A
Obs
JmaxRef = 238.48
RdRef = 2.48
sigma = 0.76
VcmaxRef = 562.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

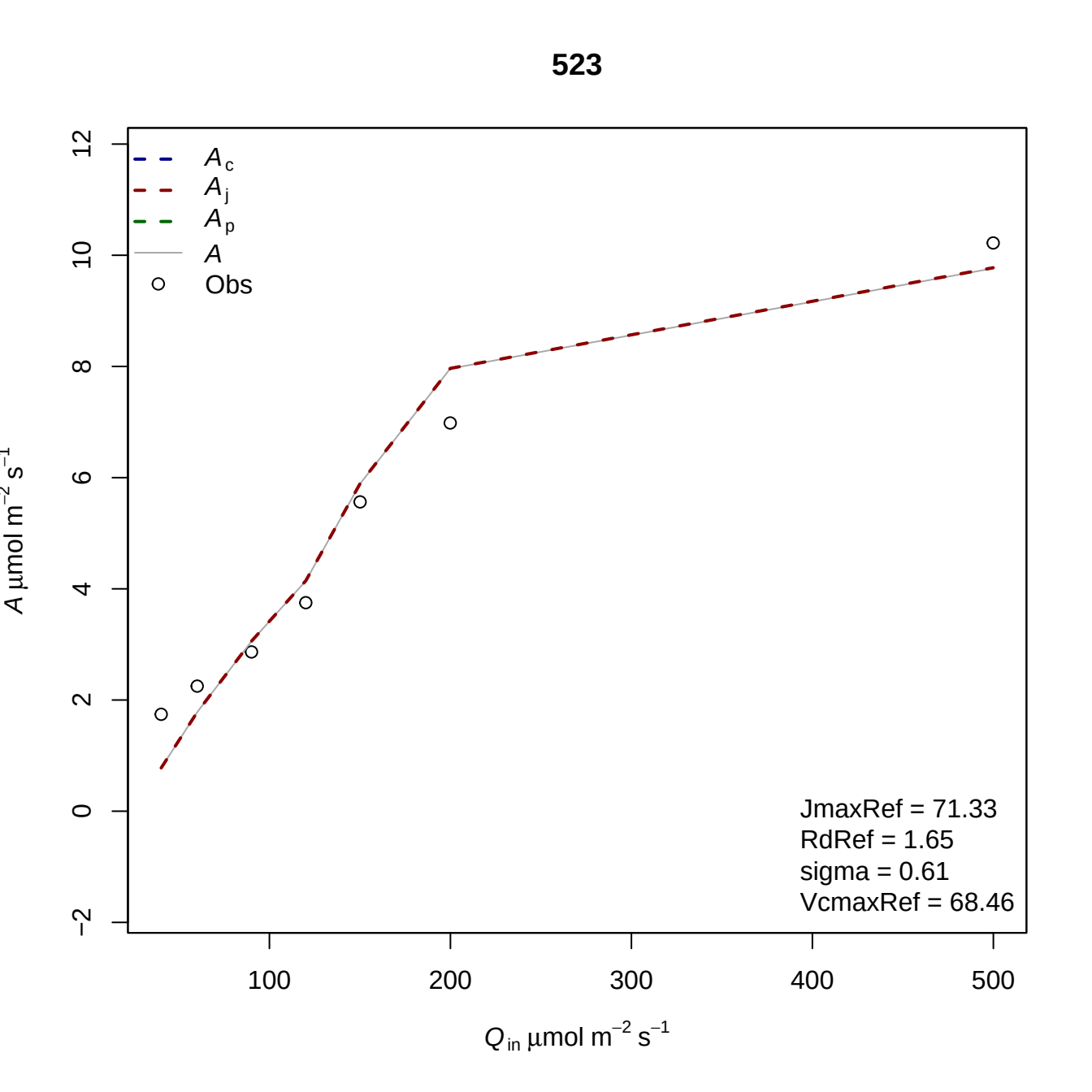

518
A_c
A_j
A_p
A
Obs
JmaxRef = 260.65
RdRef = 3.76
sigma = 0.86
VcmaxRef = 715.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
30
0
500
1000
1500
2000

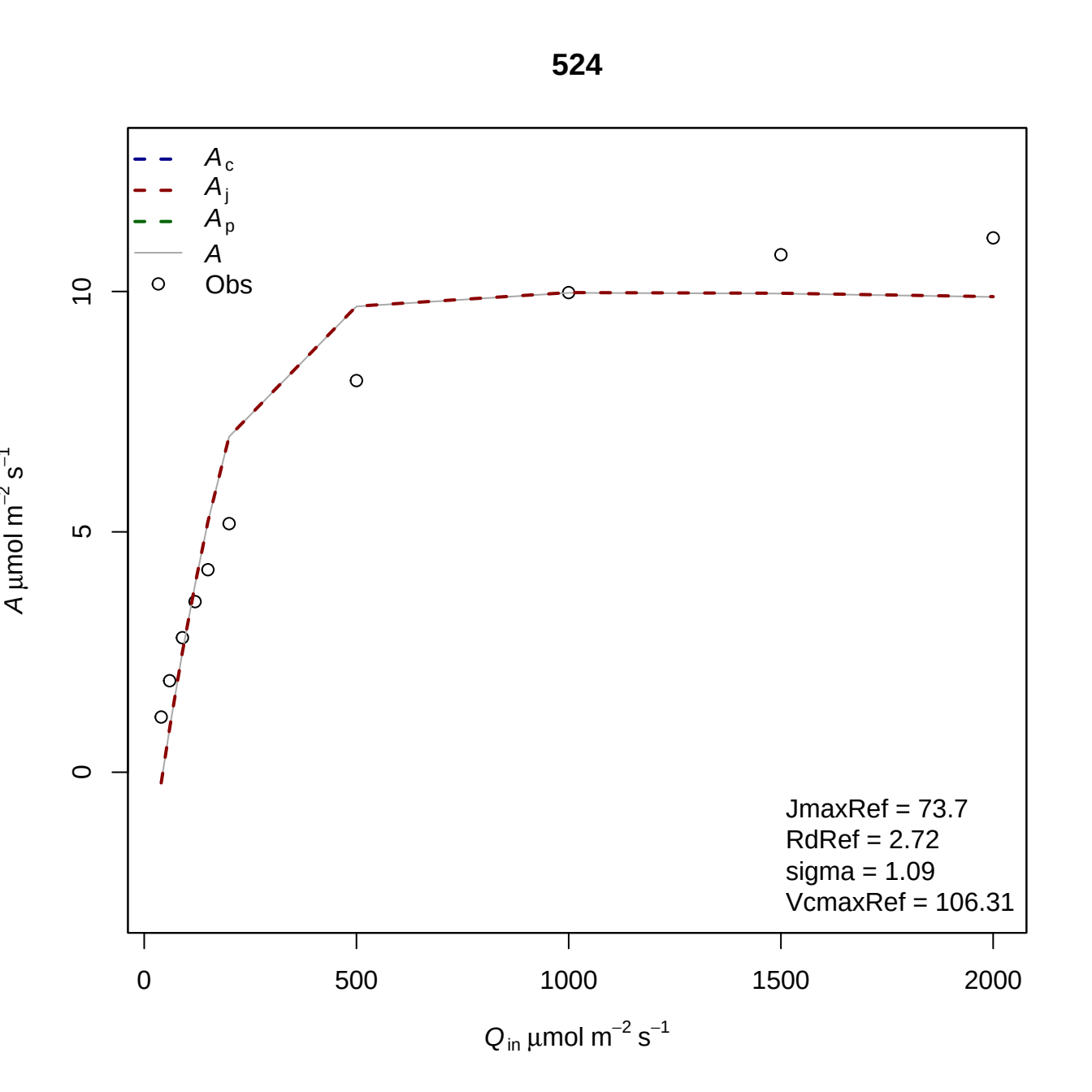

519
A_c
A_j
A_p
A
Obs
JmaxRef = 264.34
RdRef = 4.38
sigma = 0.96
VcmaxRef = 930.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

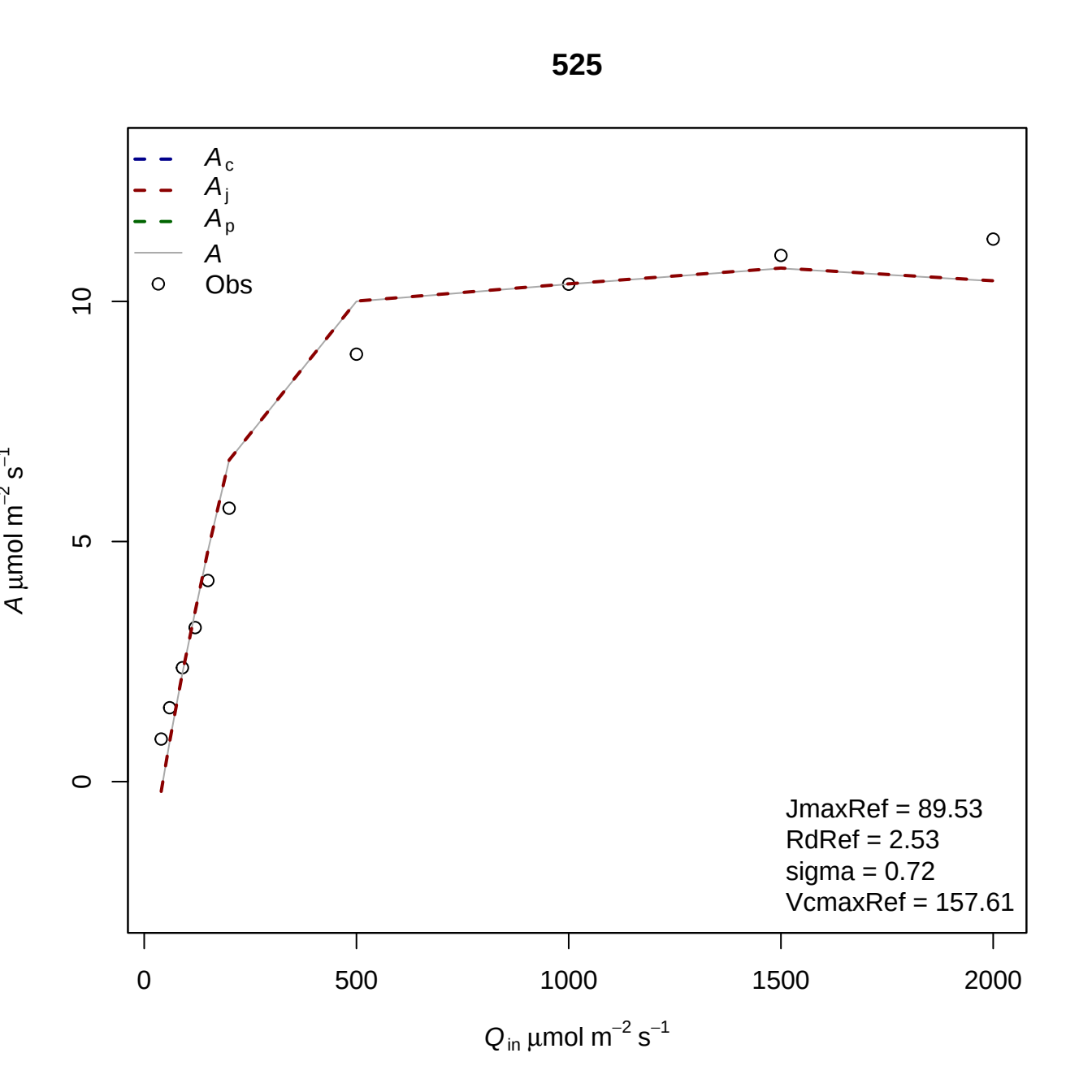

520
A_c
A_j
A_p
A
Obs
JmaxRef = 105.74
RdRef = 1.41
sigma = 0.98
VcmaxRef = 126.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

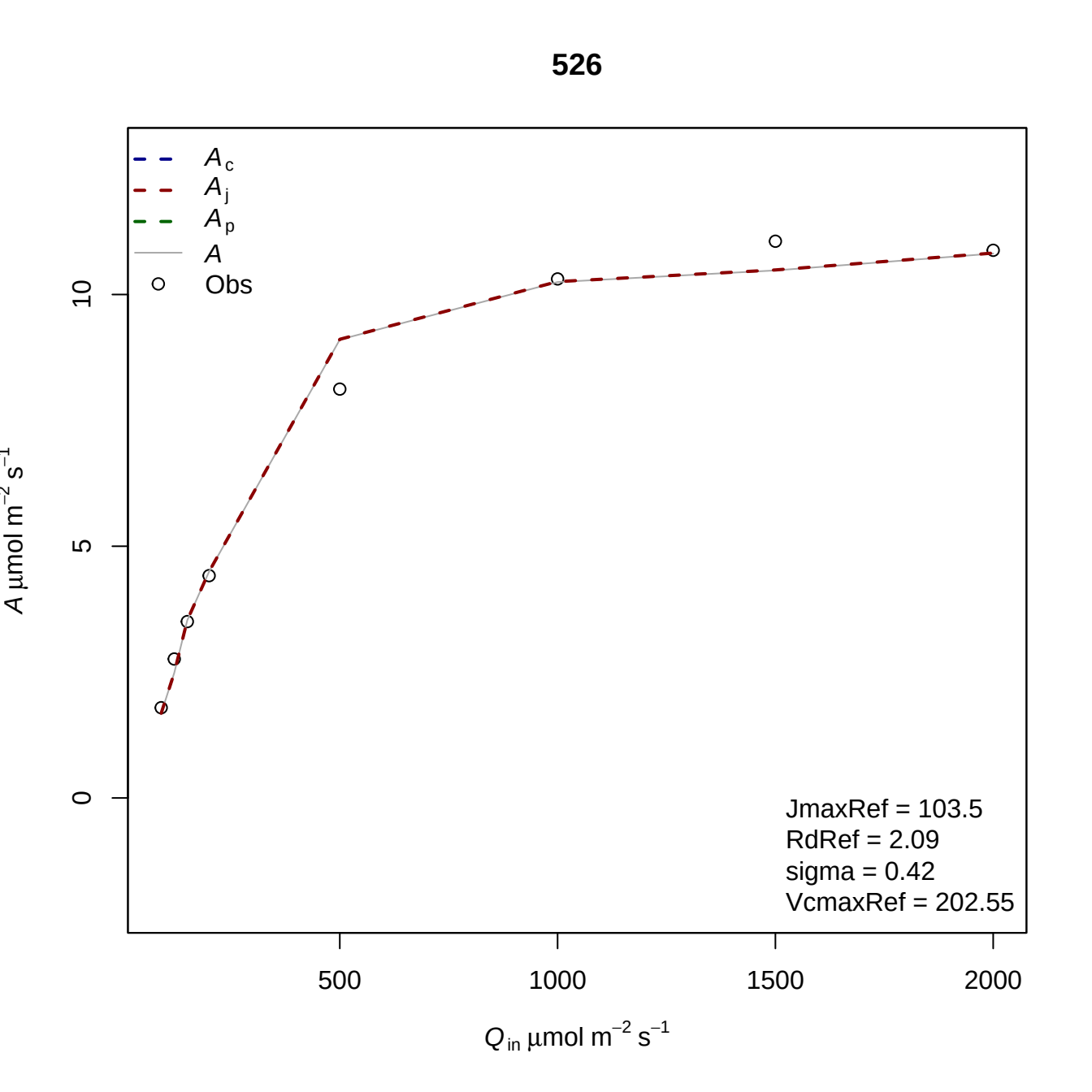

521
A_c
A_j
A_p
A
Obs
JmaxRef = 162.82
RdRef = 1.48
sigma = 0.99
VcmaxRef = 301.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

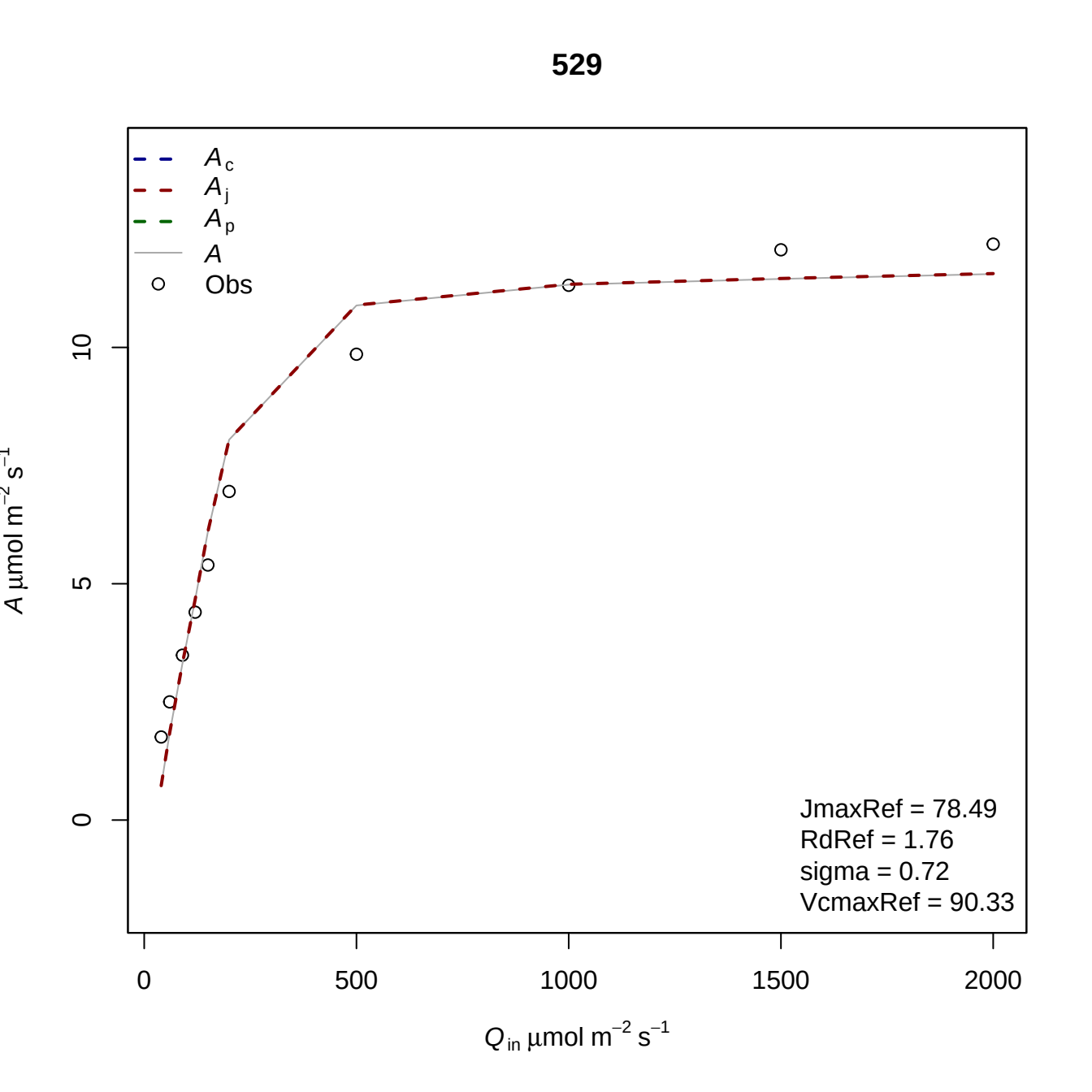

522
A_c
A_j
A_p
A
Obs
JmaxRef = 203.49
RdRef = 1.3
sigma = 0.67
VcmaxRef = 474.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

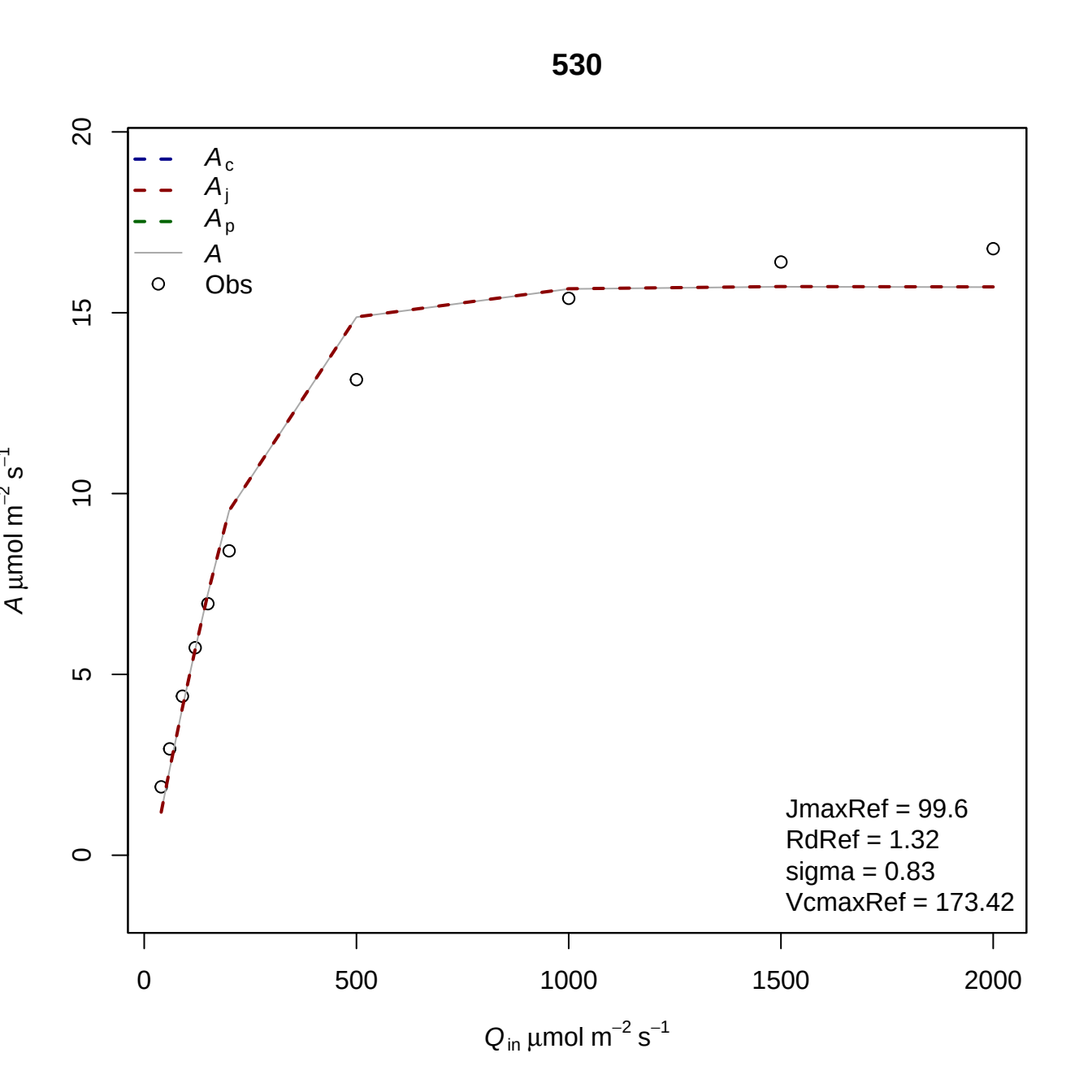

523
A_c
A_j
A_p
A
Obs
JmaxRef = 71.33
RdRef = 1.65
sigma = 0.61
VcmaxRef = 68.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
100
200
300
400
500

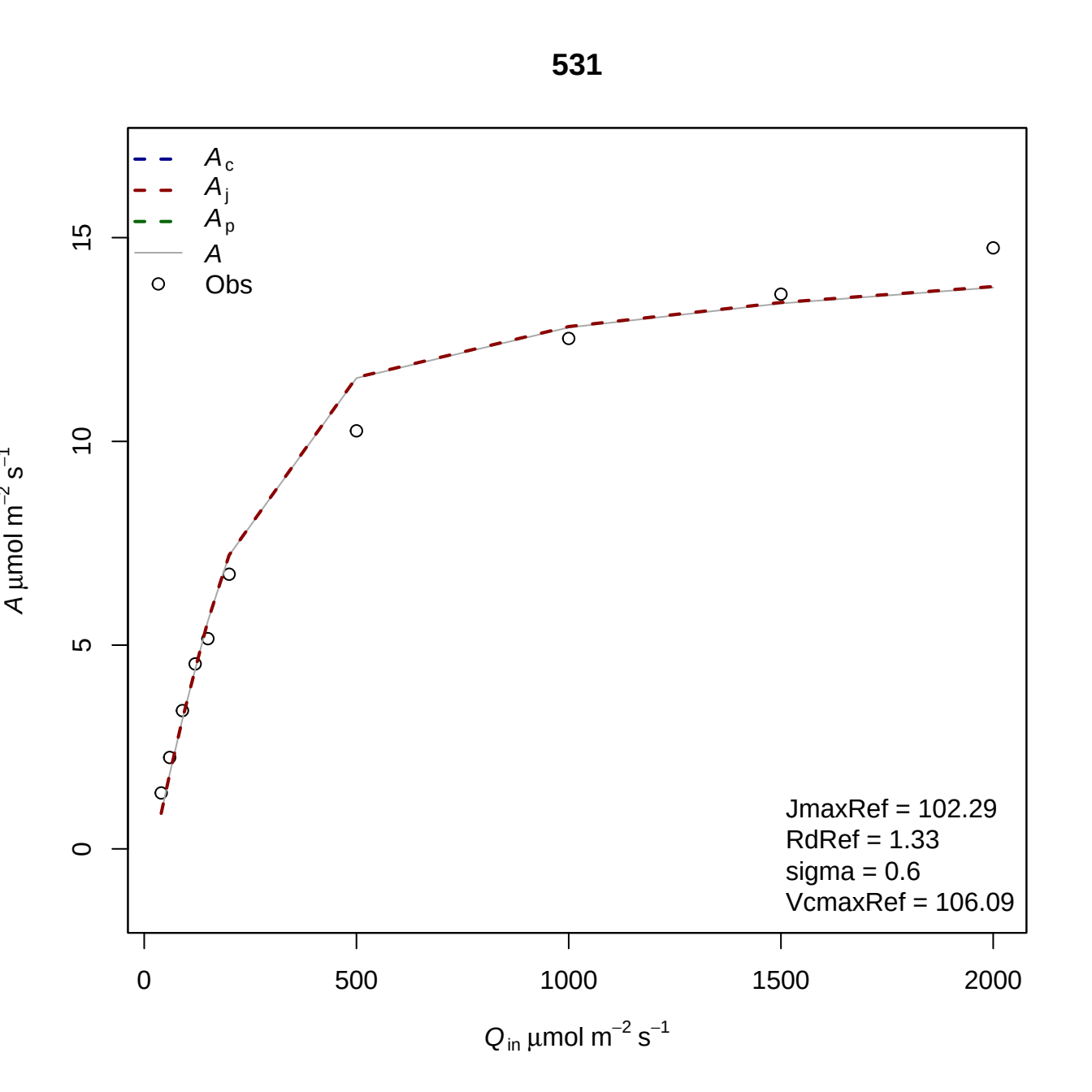

524
A_c
A_j
A_p
A
Obs
JmaxRef = 73.7
RdRef = 2.72
sigma = 1.09
VcmaxRef = 106.31
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

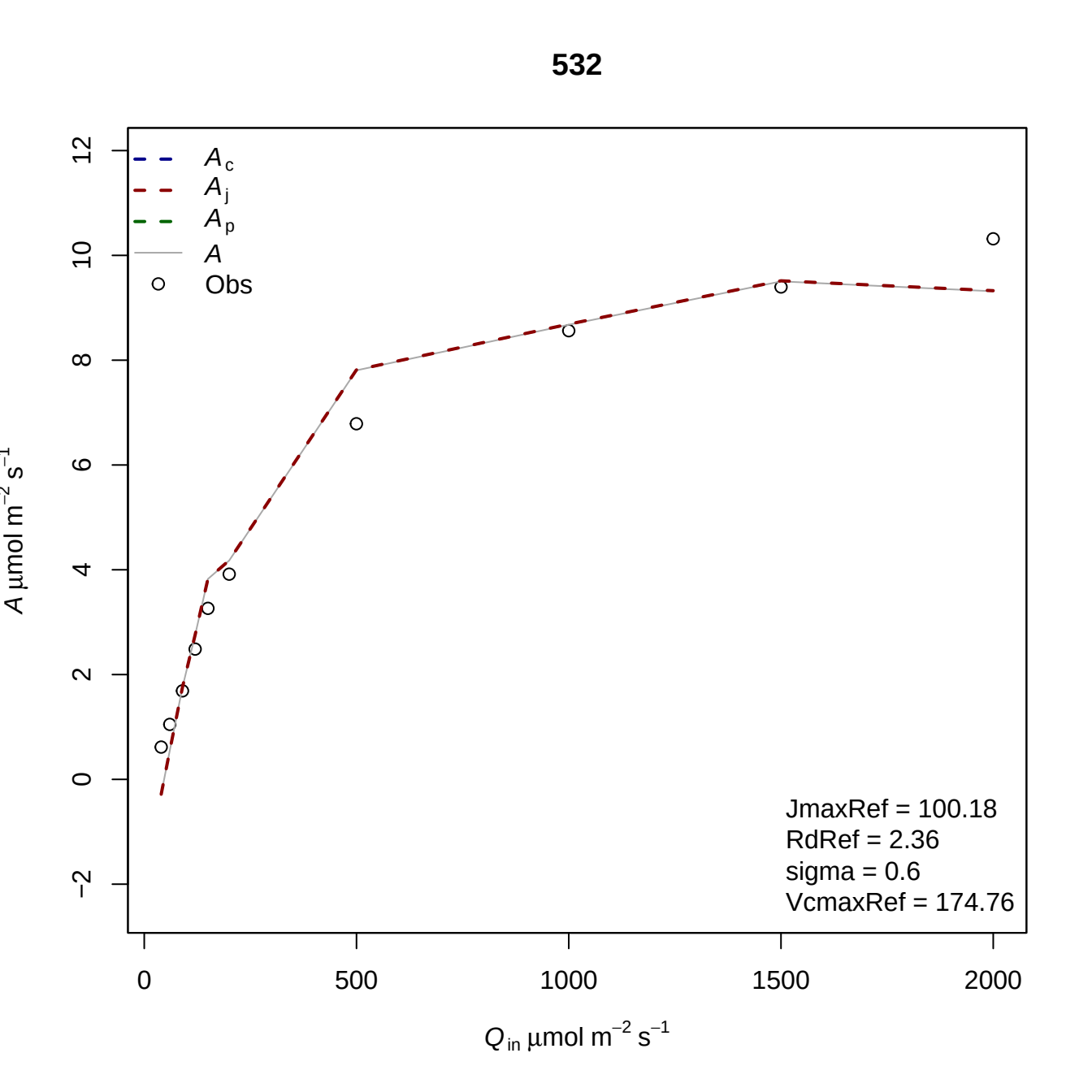

525
A_c
A_j
A_p
A
Obs
JmaxRef = 89.53
RdRef = 2.53
sigma = 0.72
VcmaxRef = 157.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

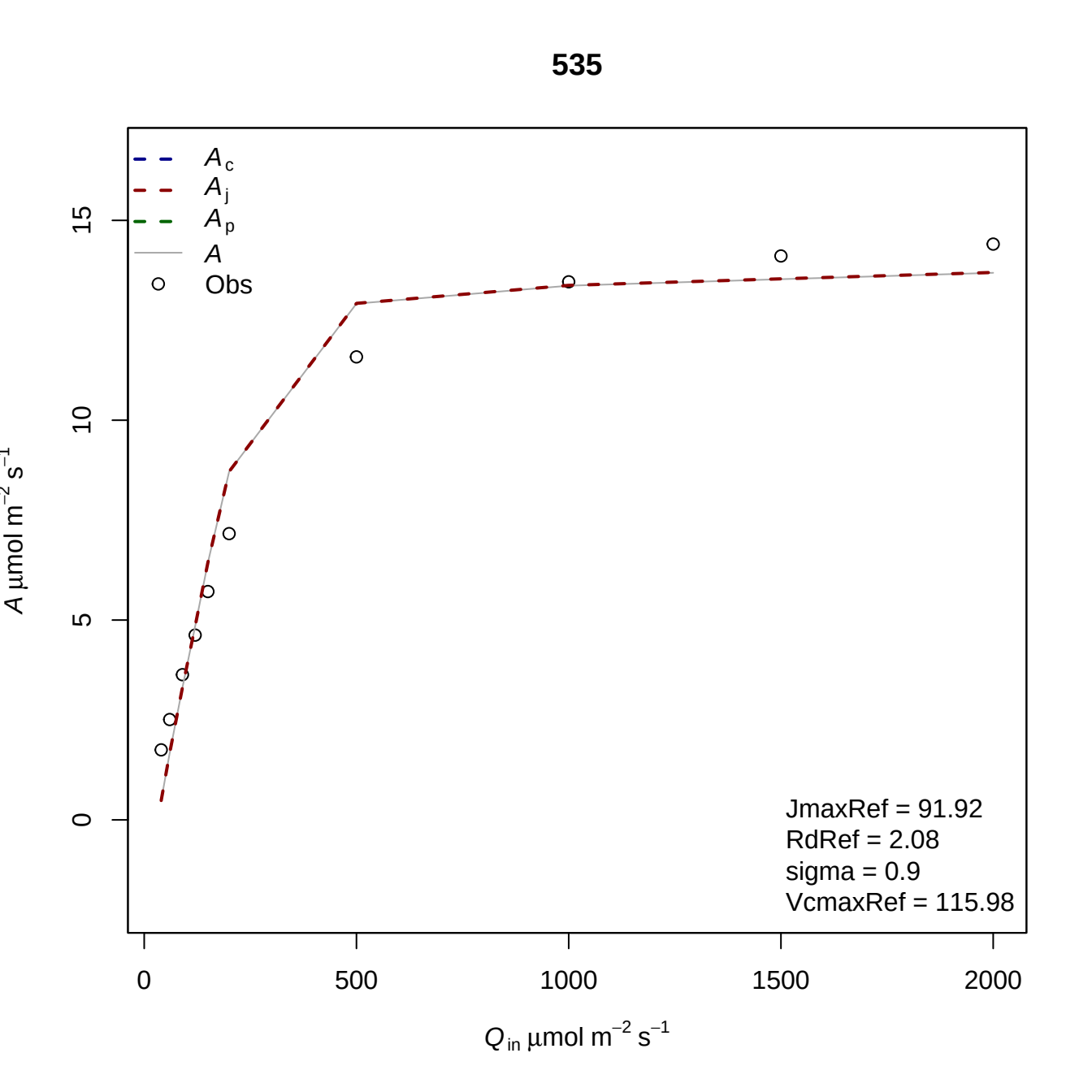

526
A_c
A_j
A_p
A
Obs
JmaxRef = 103.5
RdRef = 2.09
sigma = 0.42
VcmaxRef = 202.55
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

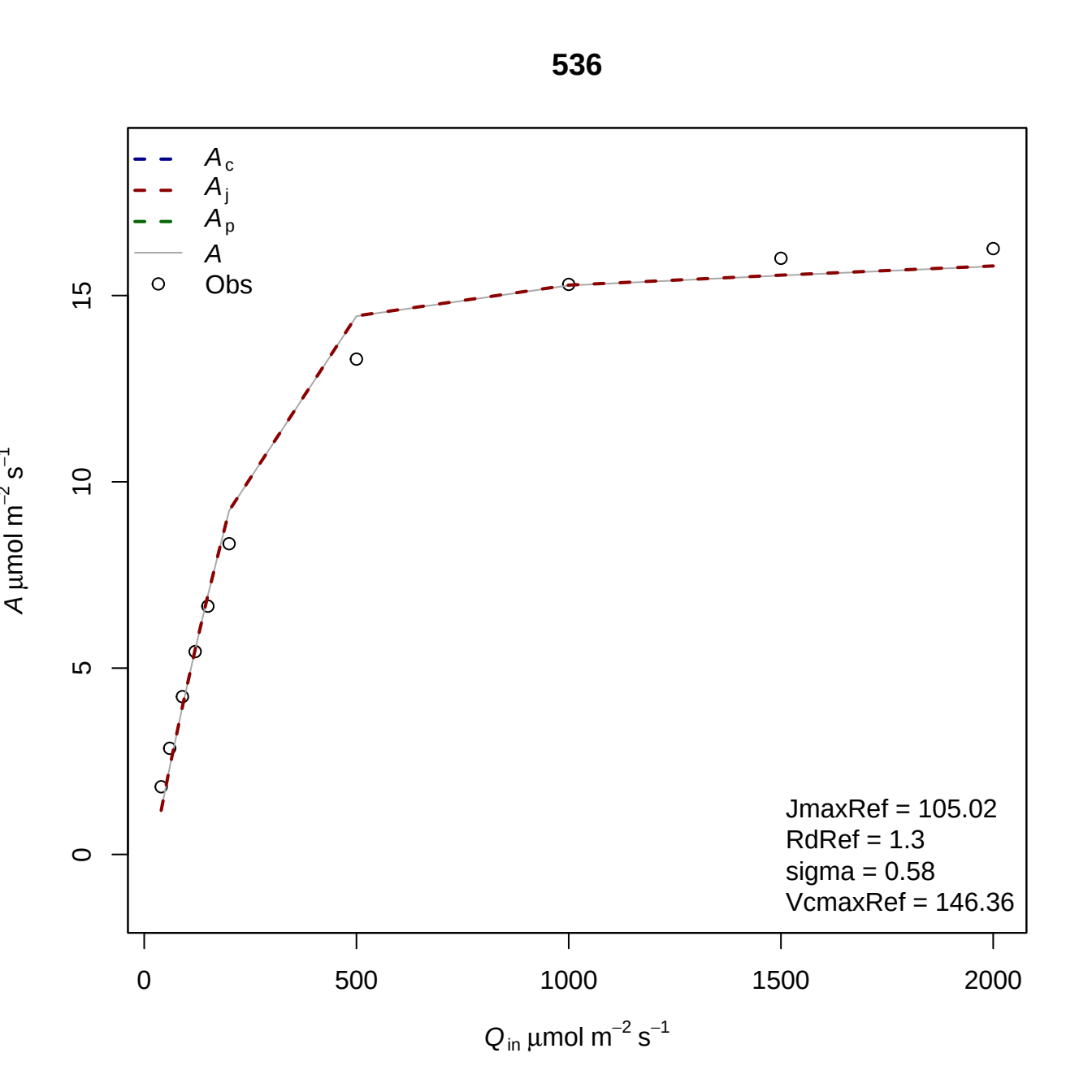

529
A_c
A_j
A_p
A
Obs
JmaxRef = 78.49
RdRef = 1.76
sigma = 0.72
VcmaxRef = 90.33
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

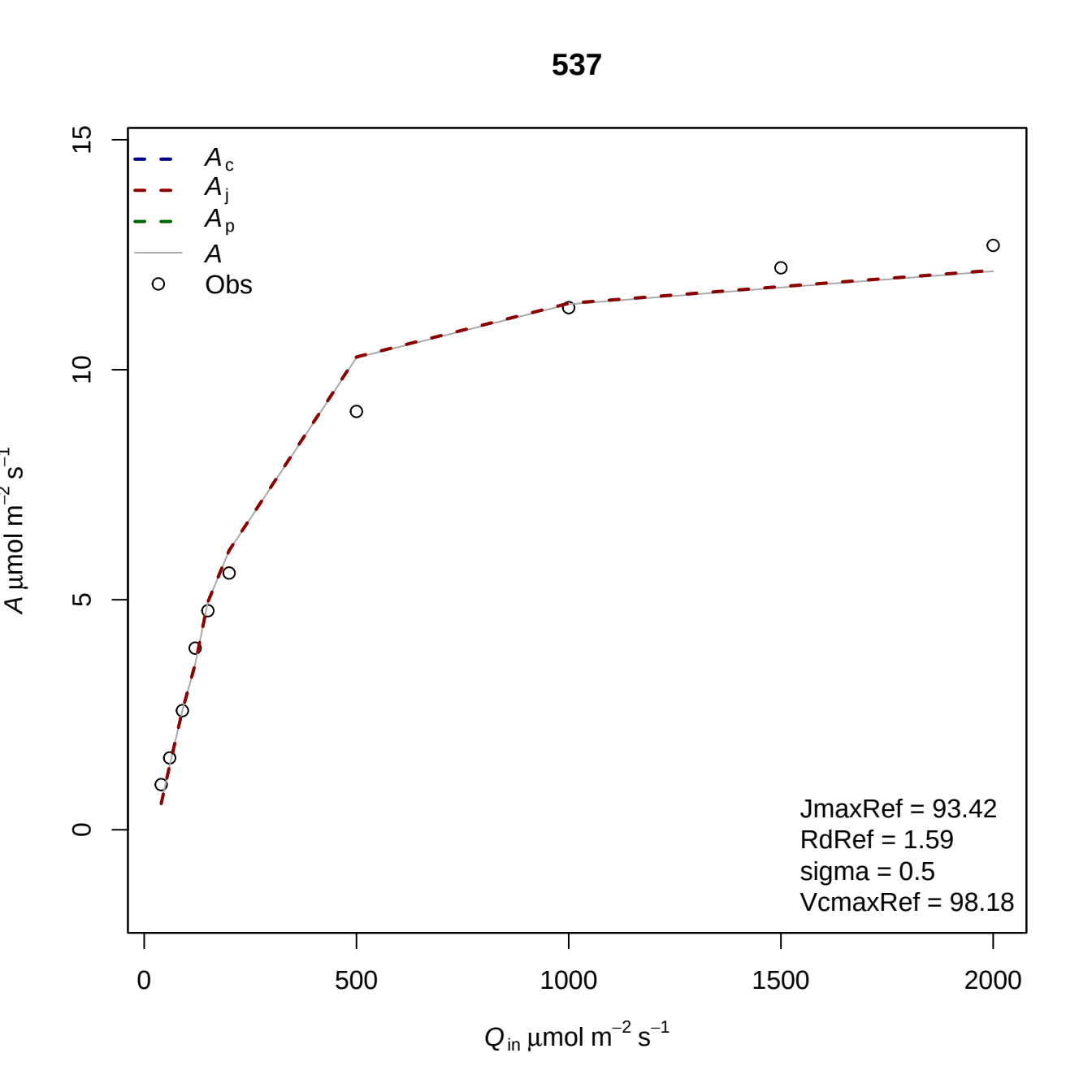

530
A_c
A_j
A_p
A
Obs
JmaxRef = 99.6
RdRef = 1.32
sigma = 0.83
VcmaxRef = 173.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

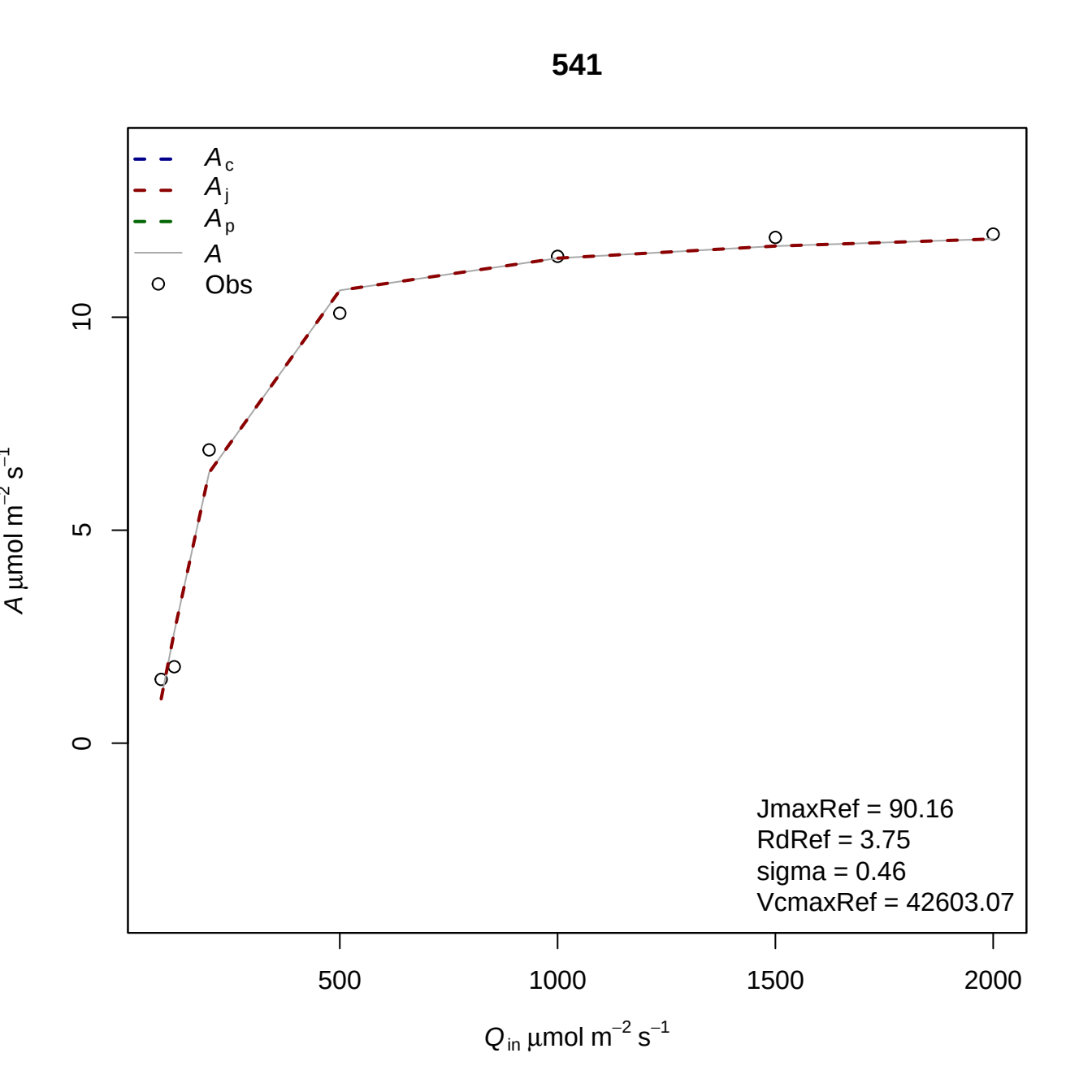

531
A_c
A_j
A_p
A
Obs
JmaxRef = 102.29
RdRef = 1.33
sigma = 0.6
VcmaxRef = 106.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

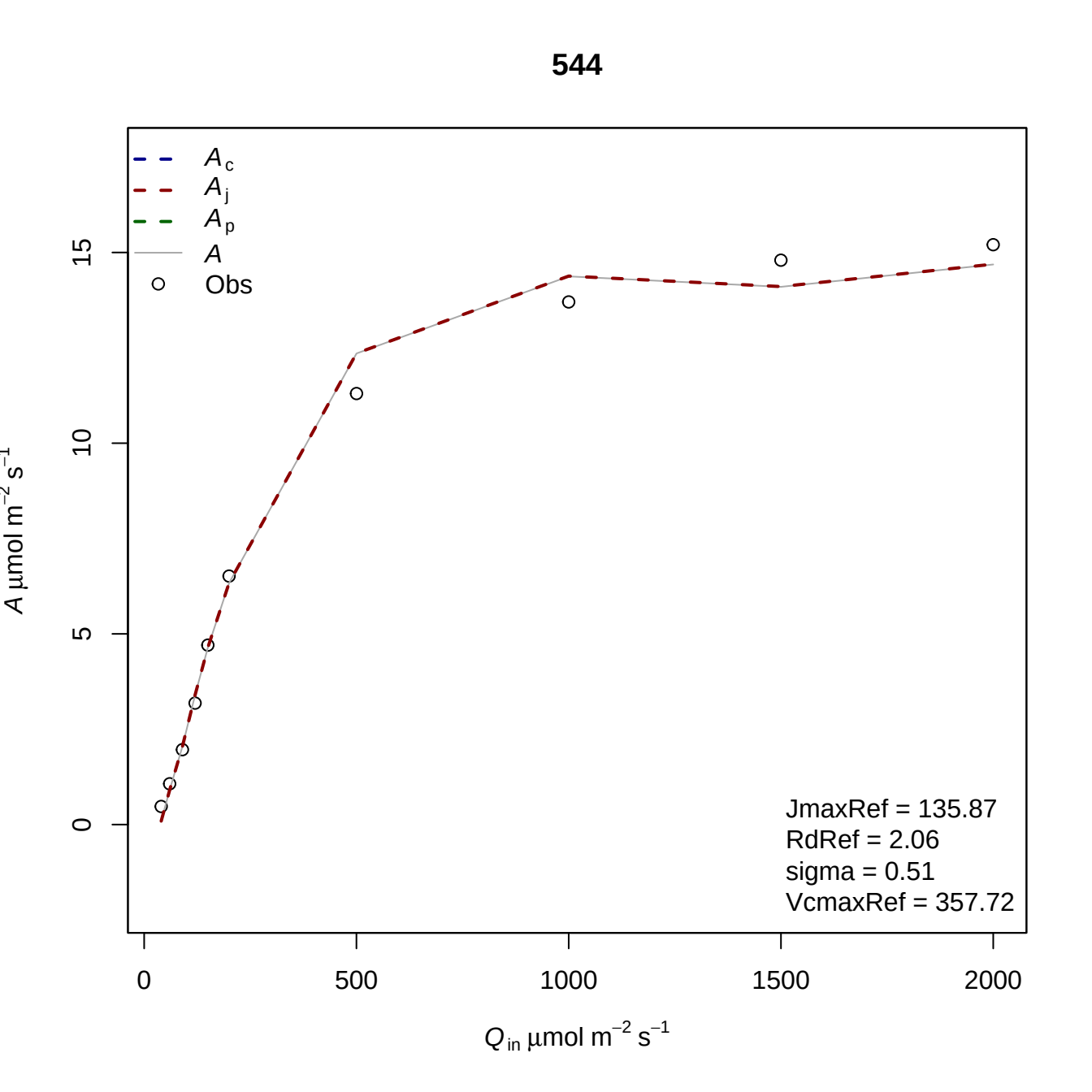

532
A_c
A_j
A_p
A
Obs
JmaxRef = 100.18
RdRef = 2.36
sigma = 0.6
VcmaxRef = 174.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

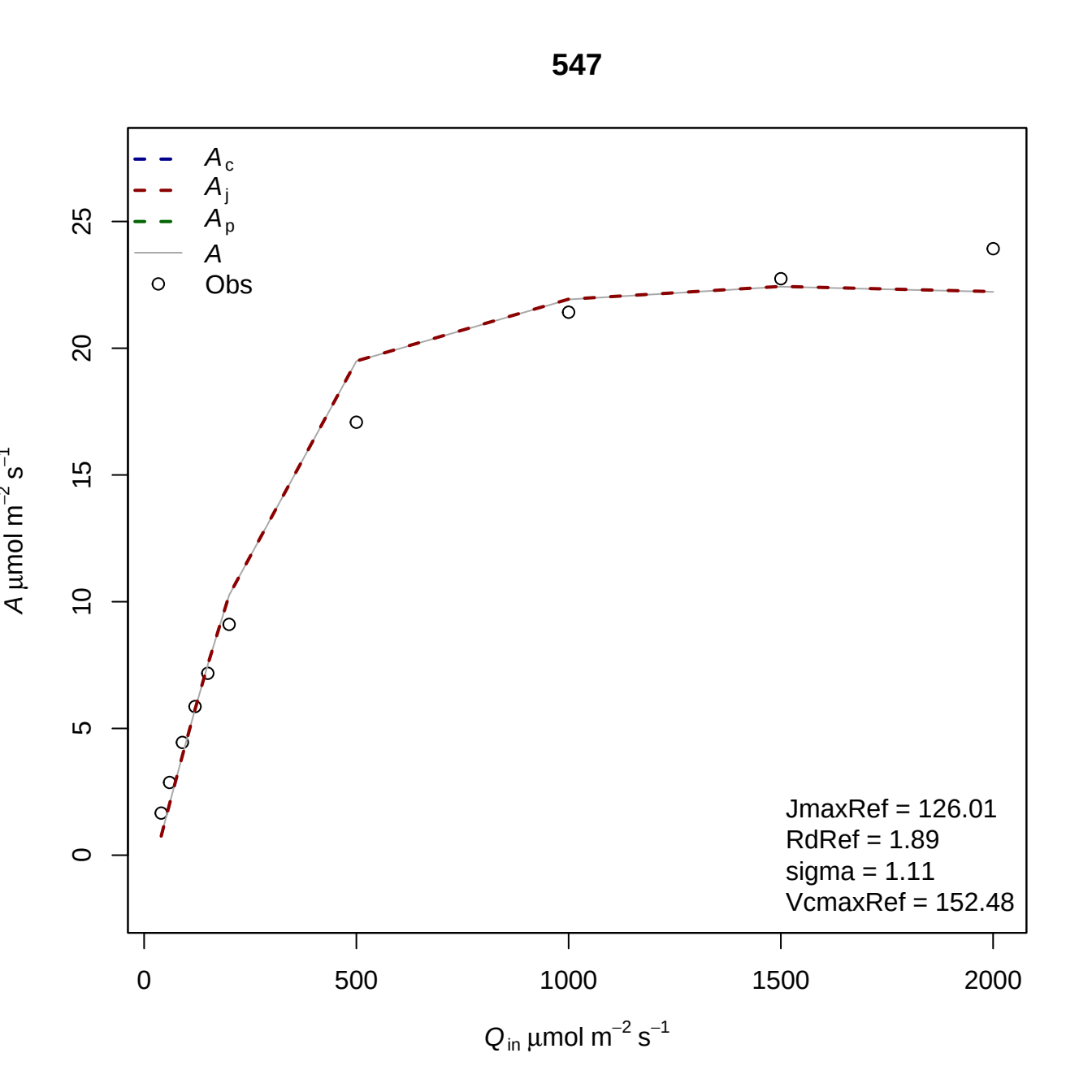

535
A_c
A_j
A_p
A
Obs
JmaxRef = 91.92
RdRef = 2.08
sigma = 0.9
VcmaxRef = 115.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

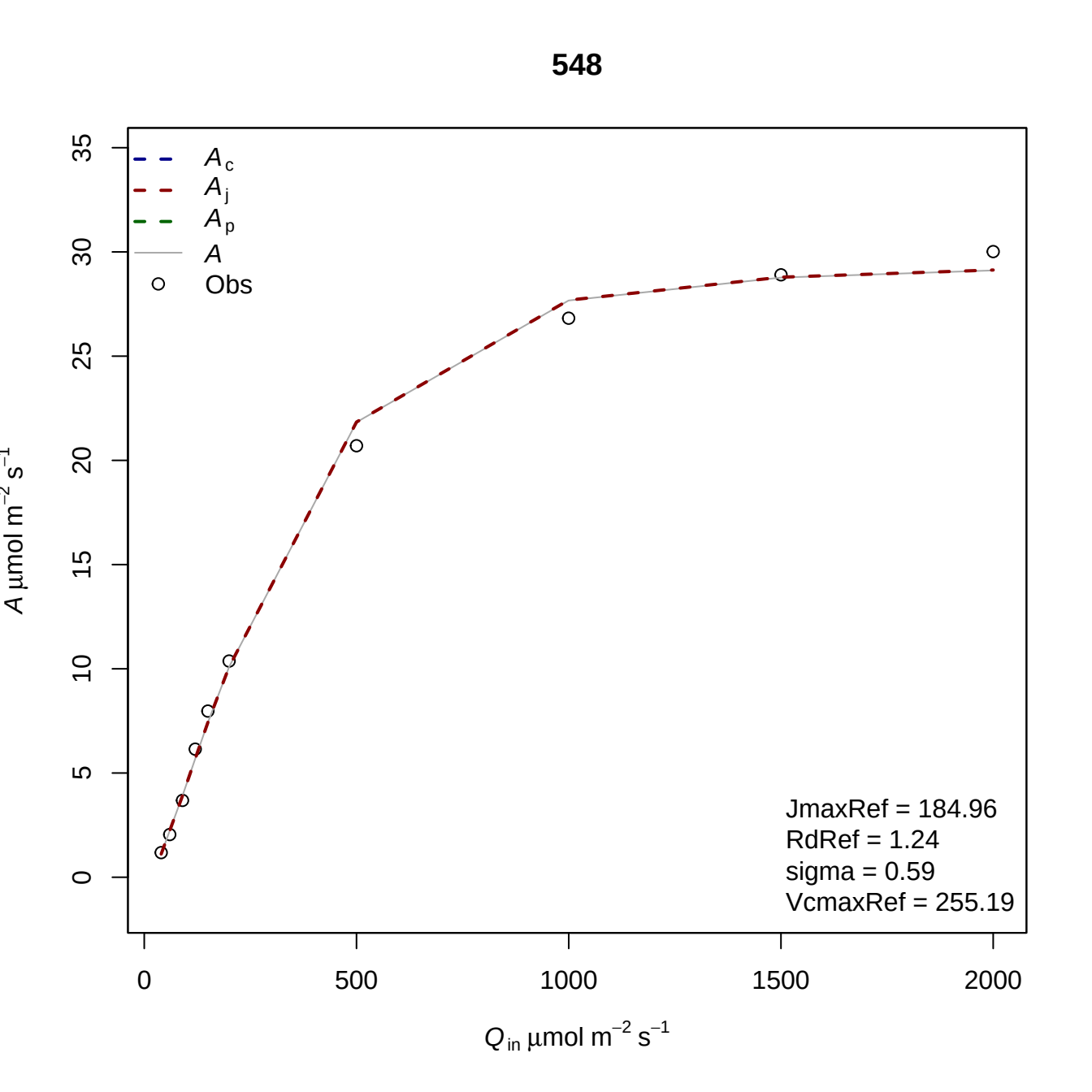

536
A_c
A_j
A_p
A
Obs
JmaxRef = 105.02
RdRef = 1.3
sigma = 0.58
VcmaxRef = 146.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

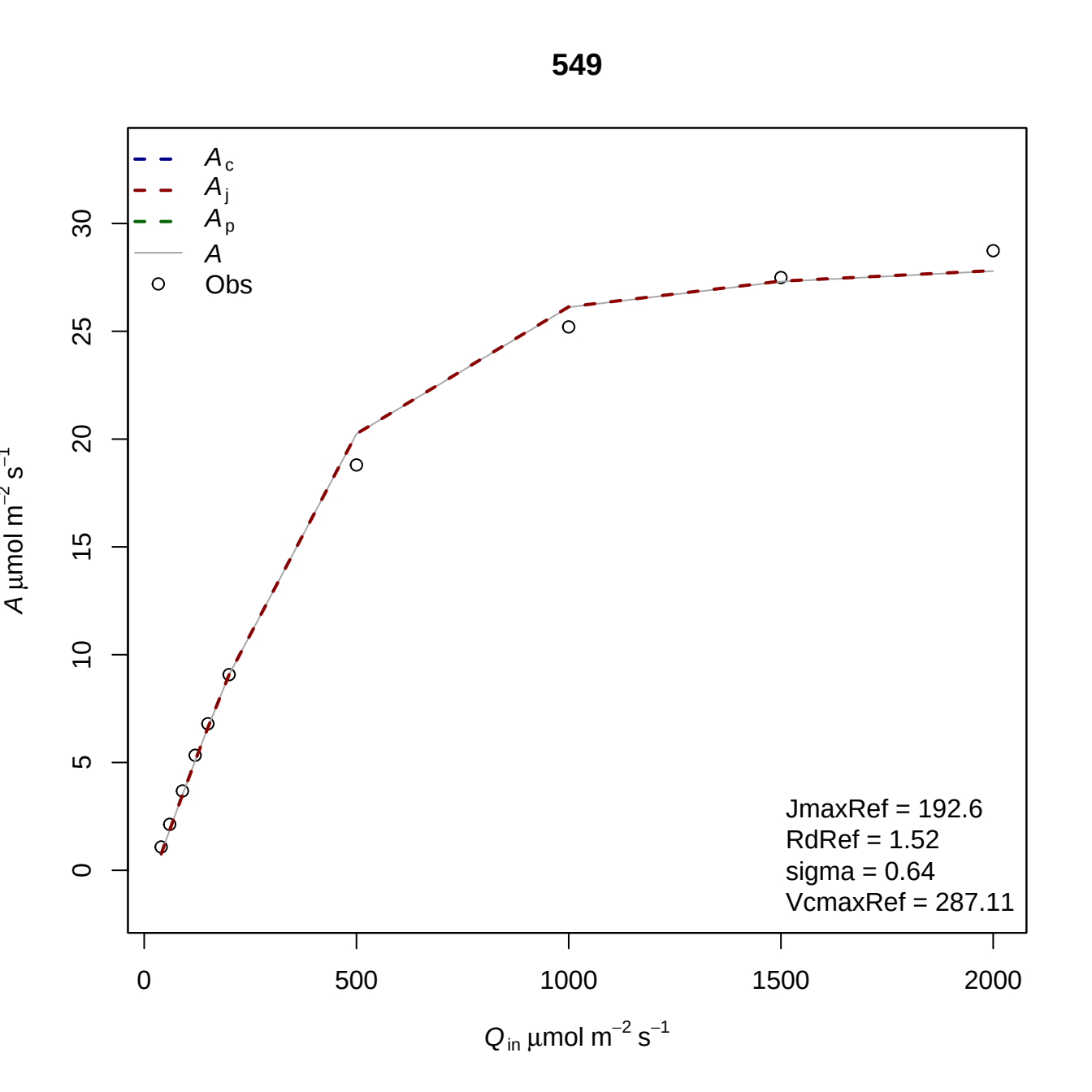

537
A_c
A_j
A_p
A
Obs
JmaxRef = 93.42
RdRef = 1.59
sigma = 0.5
VcmaxRef = 98.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

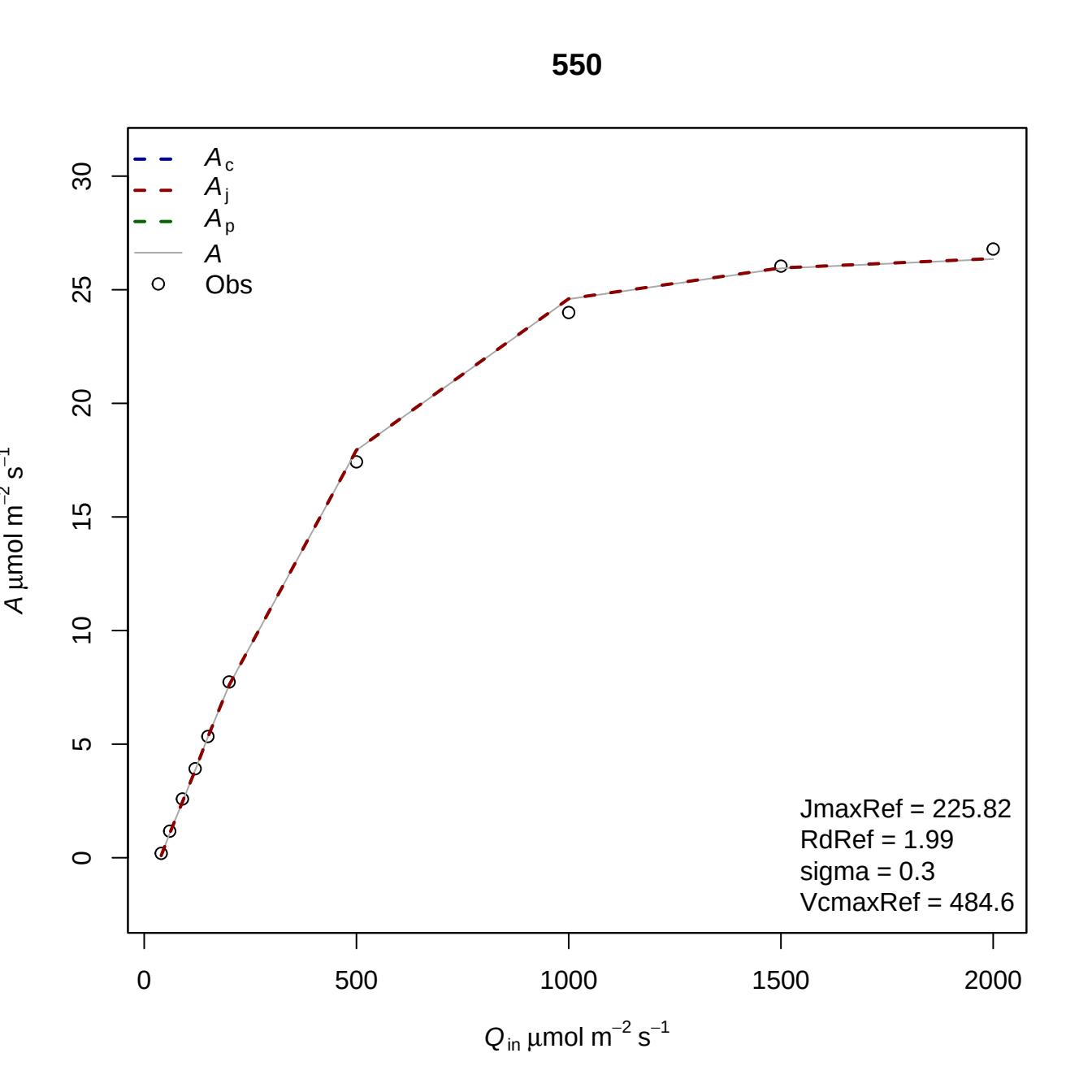

541
A_c
A_j
A_p
A
Obs
JmaxRef = 90.16
RdRef = 3.75
sigma = 0.46
VcmaxRef = 42603.07
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

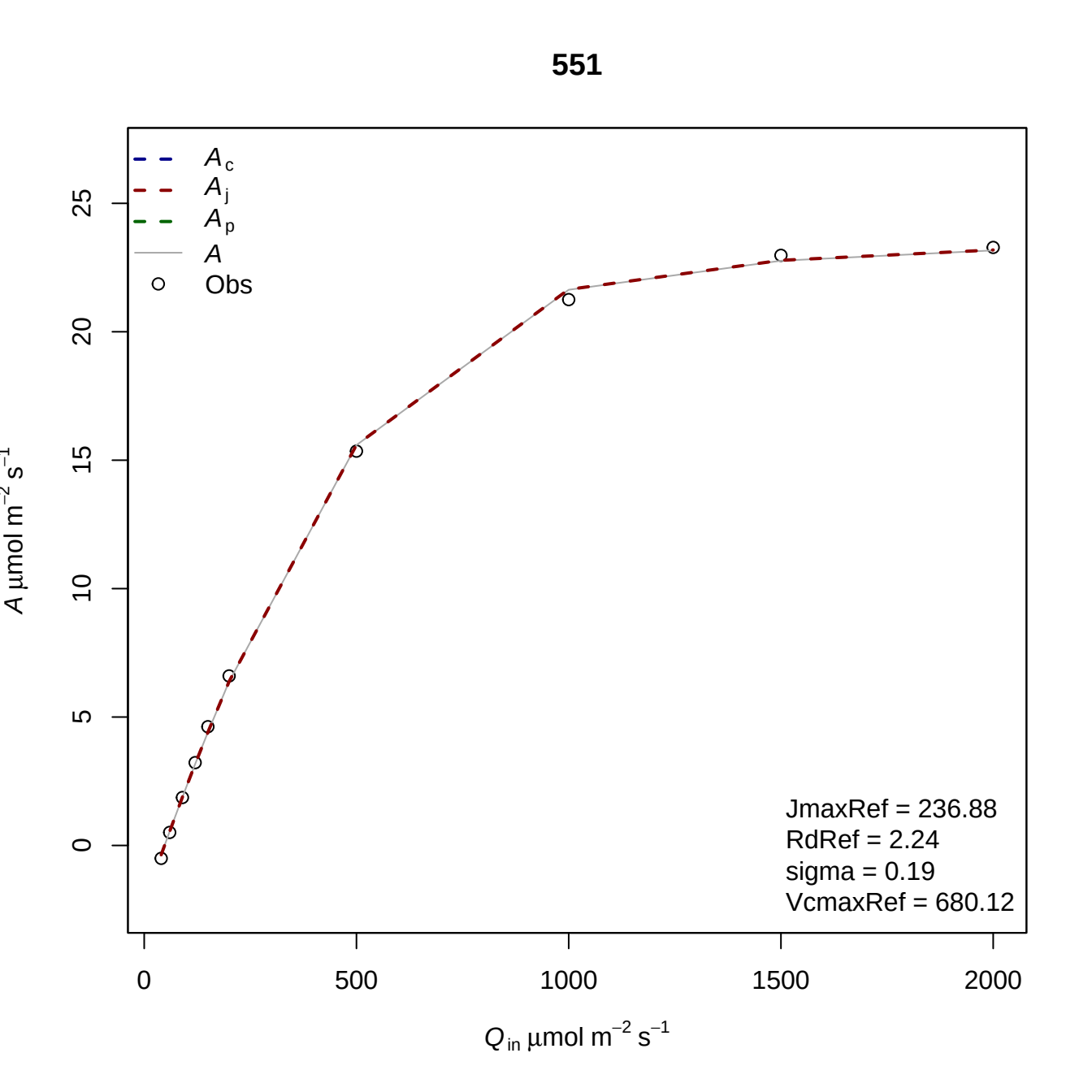

544
A_c
A_j
A_p
A
Obs
JmaxRef = 135.87
RdRef = 2.06
sigma = 0.51
VcmaxRef = 357.72
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

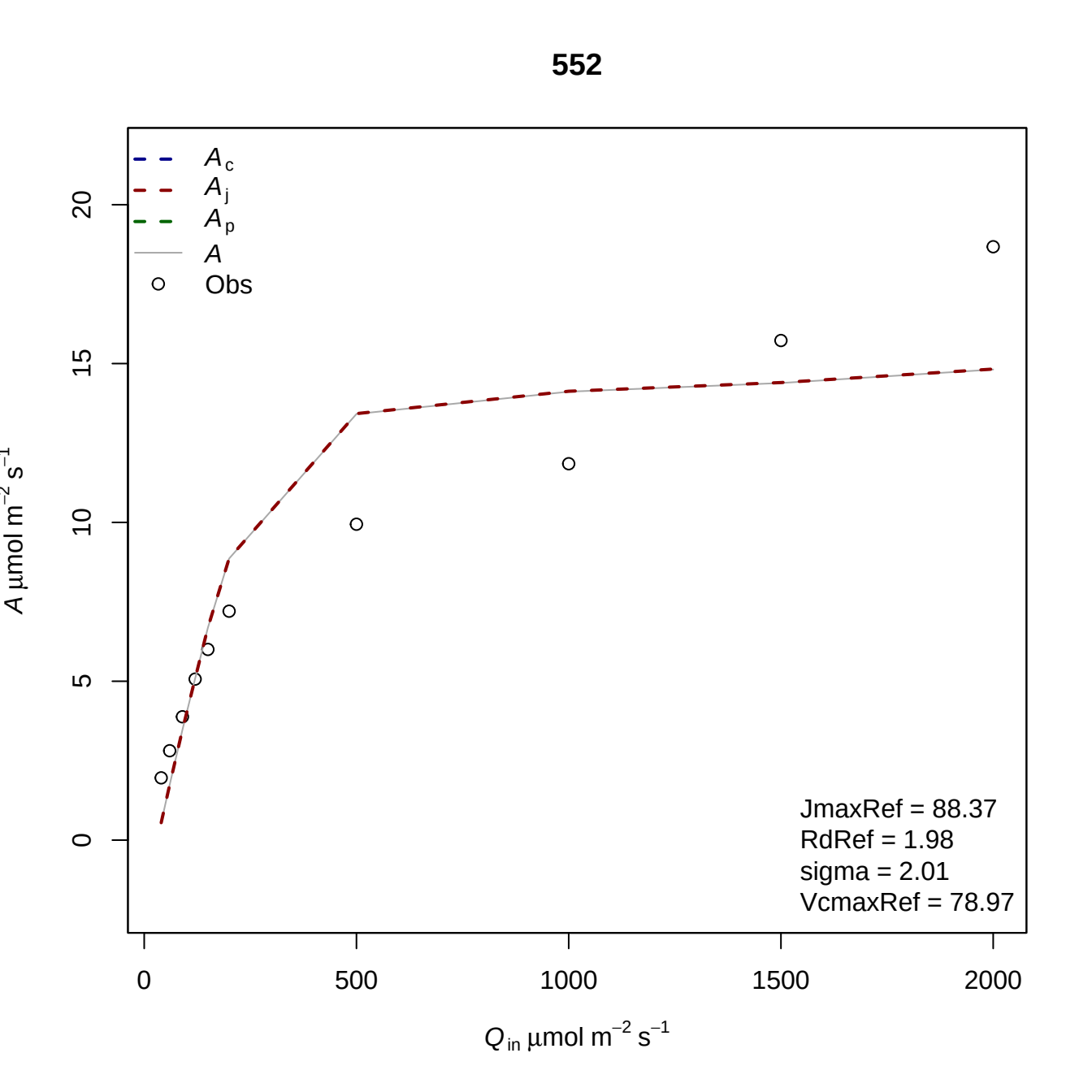

547
A_c
A_j
A_p
A
Obs
JmaxRef = 126.01
RdRef = 1.89
sigma = 1.11
VcmaxRef = 152.48
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

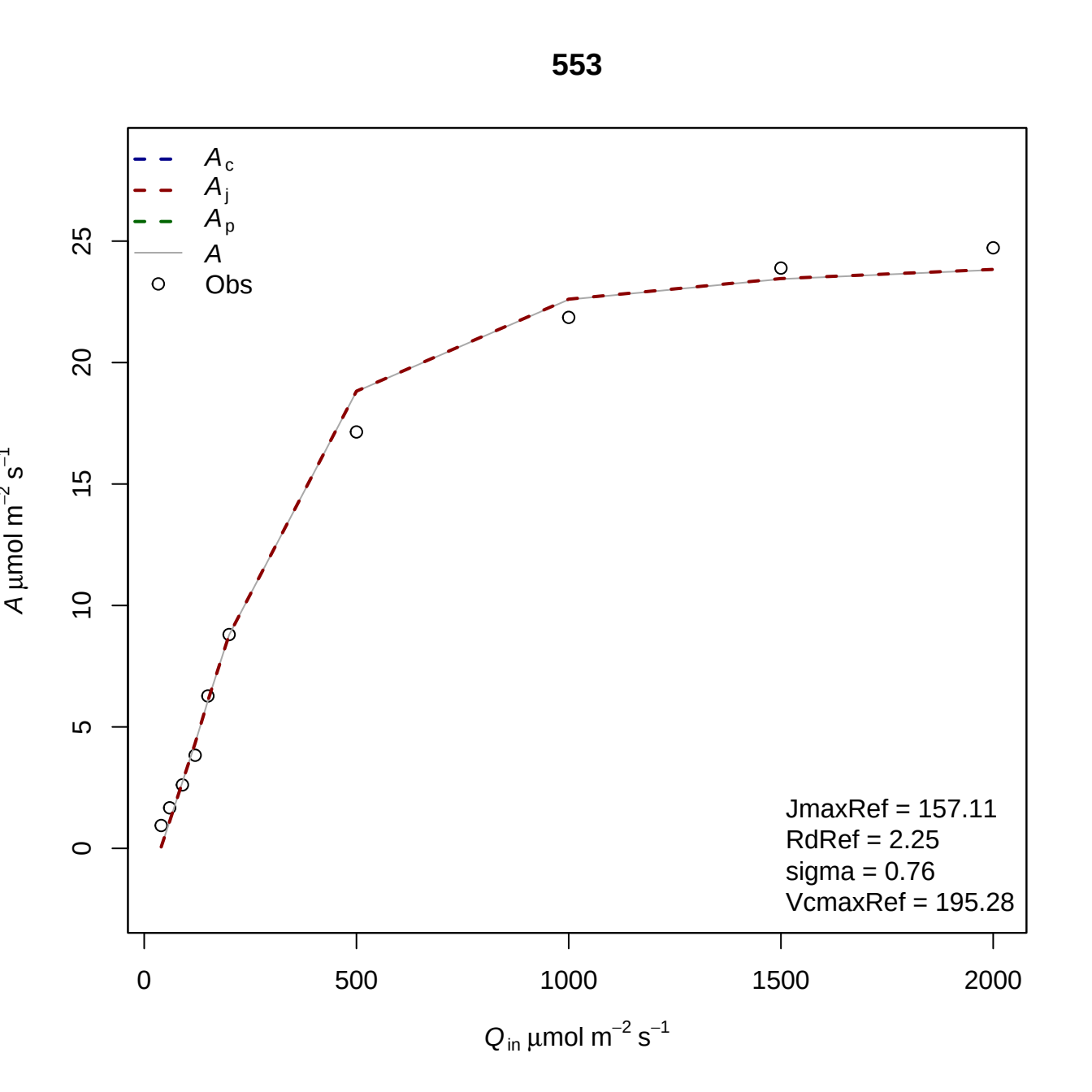

548
A_c
A_j
A_p
A
Obs
JmaxRef = 184.96
RdRef = 1.24
sigma = 0.59
VcmaxRef = 255.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

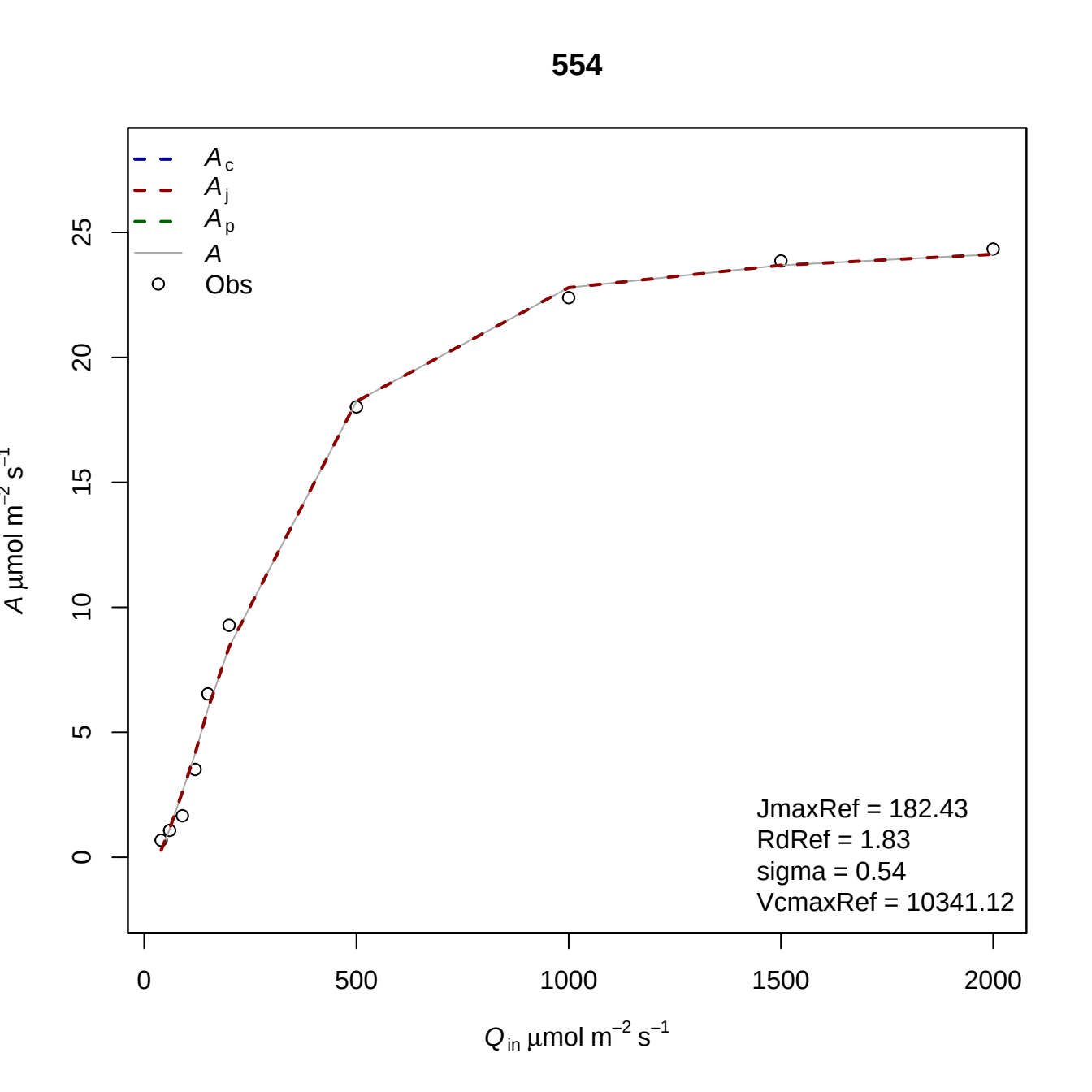

549
A_c
A_j
A_p
A
Obs
JmaxRef = 192.6
RdRef = 1.52
sigma = 0.64
VcmaxRef = 287.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

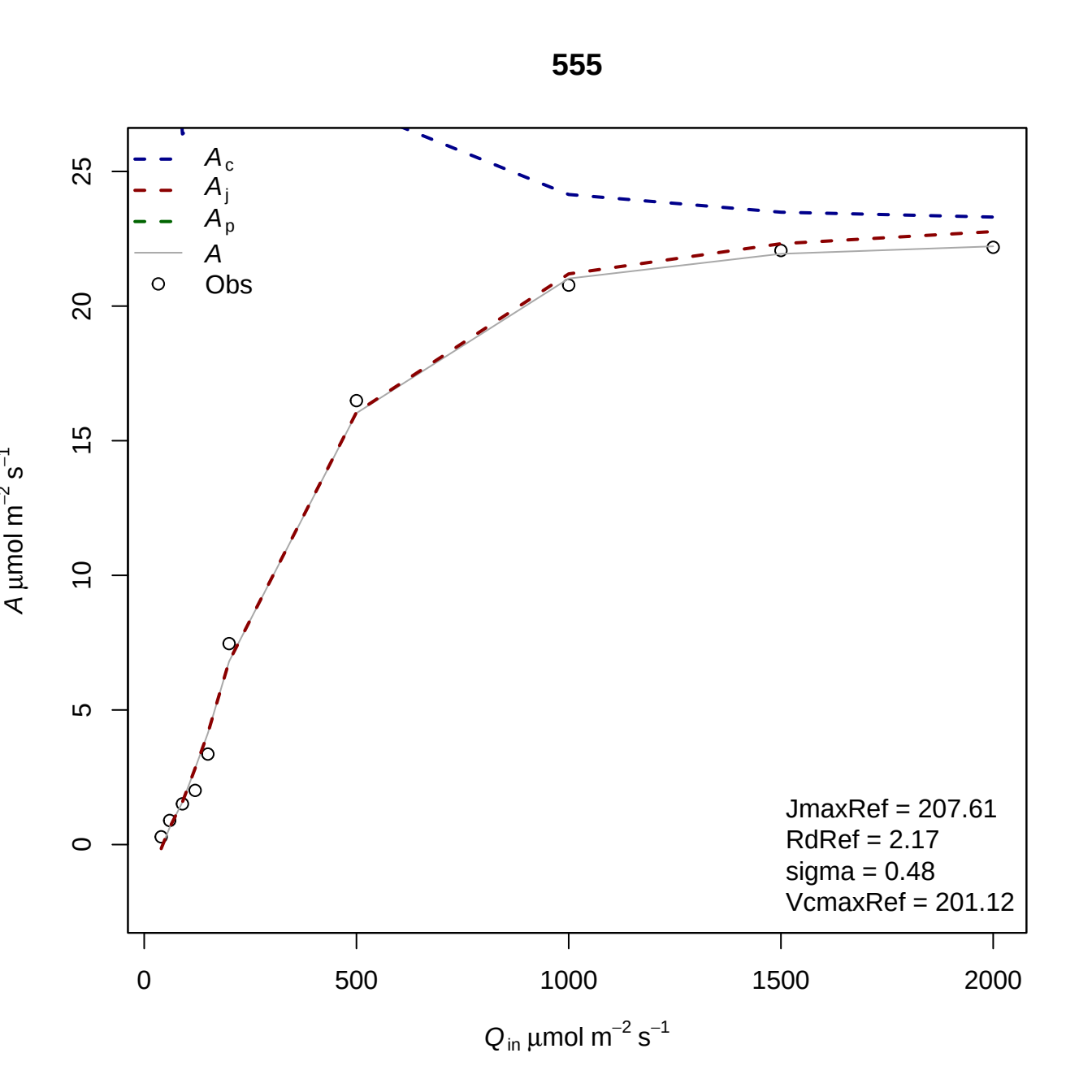

550
A_c
A_j
A_p
A
Obs
JmaxRef = 225.82
RdRef = 1.99
sigma = 0.3
VcmaxRef = 484.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

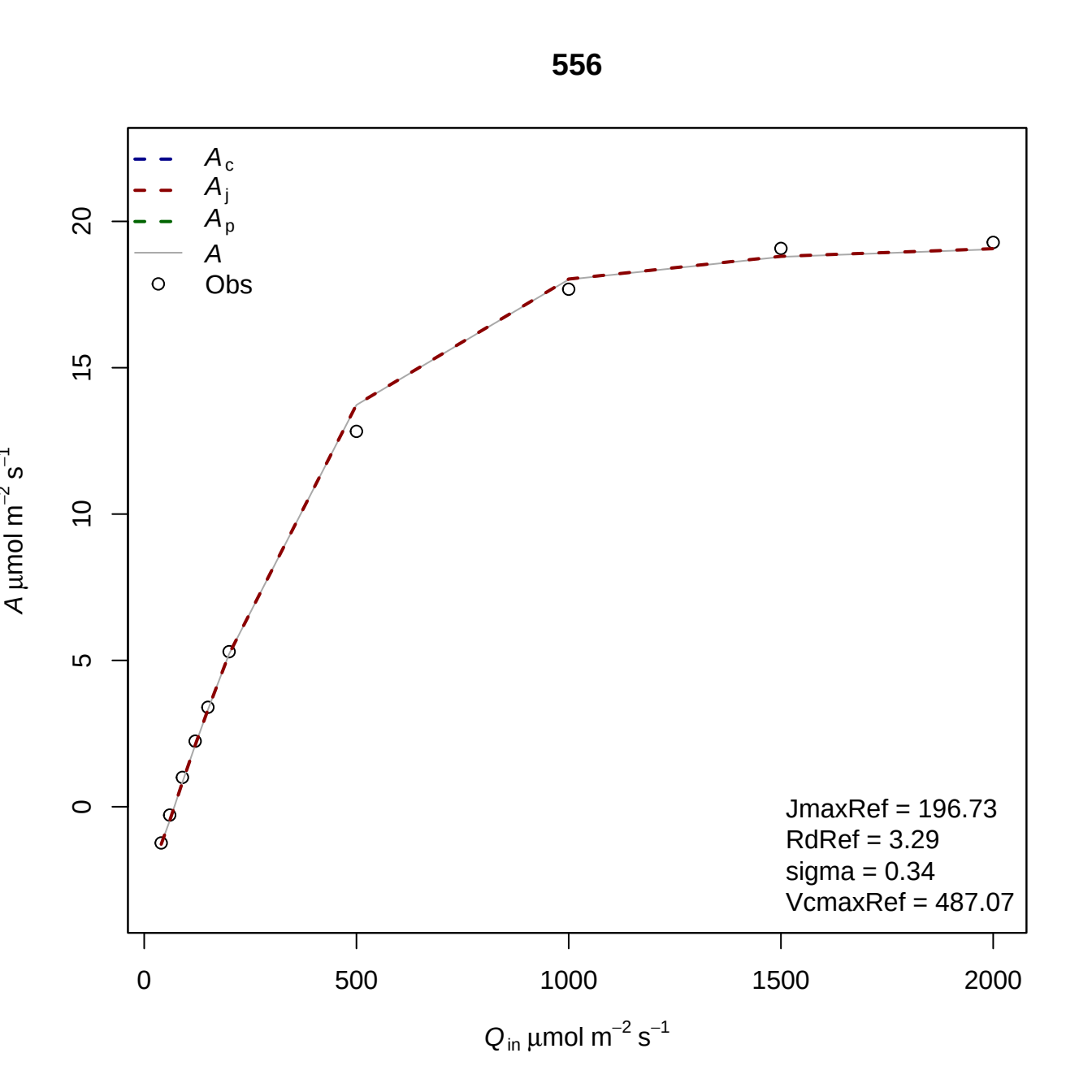

551
A_c
A_j
A_p
A
Obs
JmaxRef = 236.88
RdRef = 2.24
sigma = 0.19
VcmaxRef = 680.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

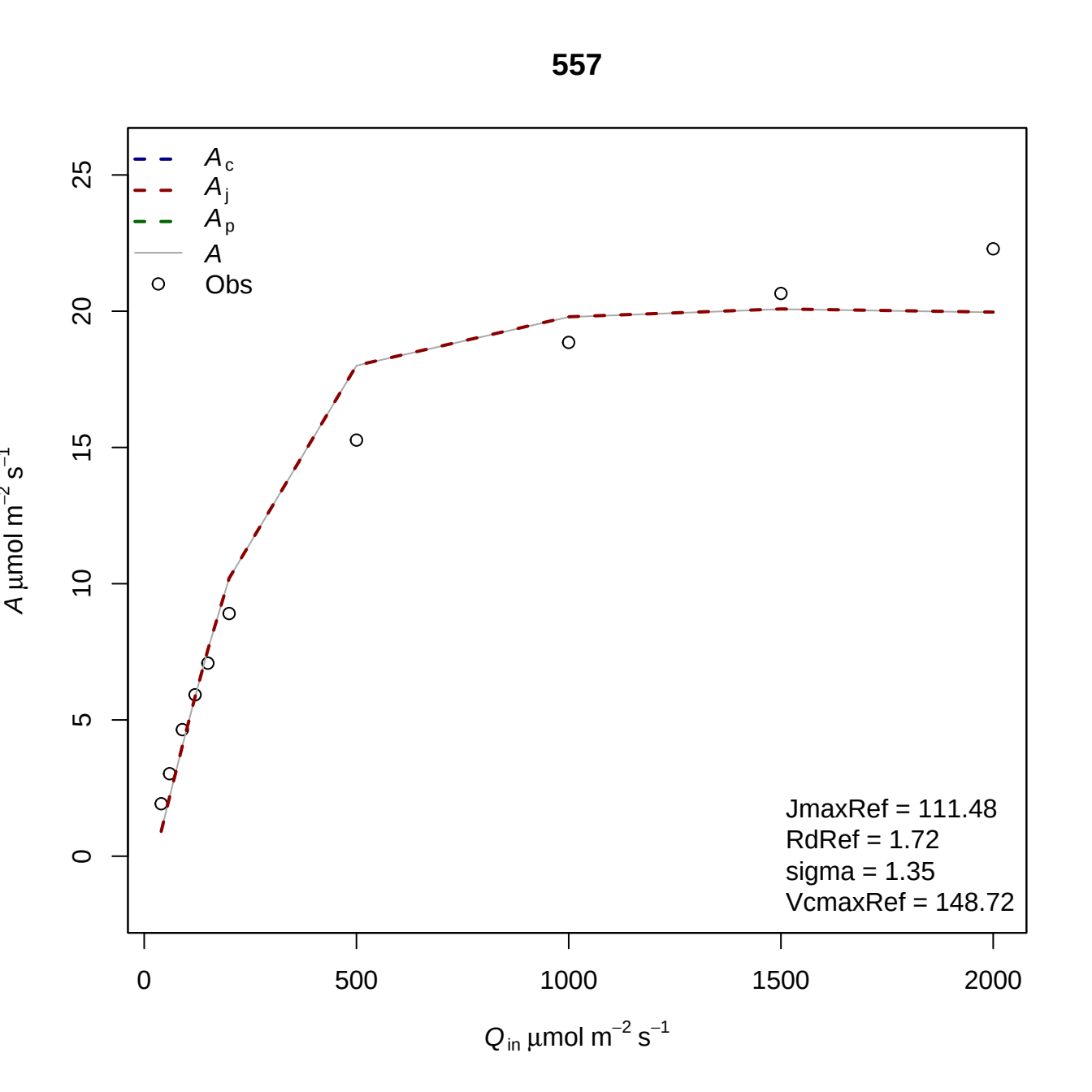

552
A_c
A_j
A_p
A
Obs
JmaxRef = 88.37
RdRef = 1.98
sigma = 2.01
VcmaxRef = 78.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

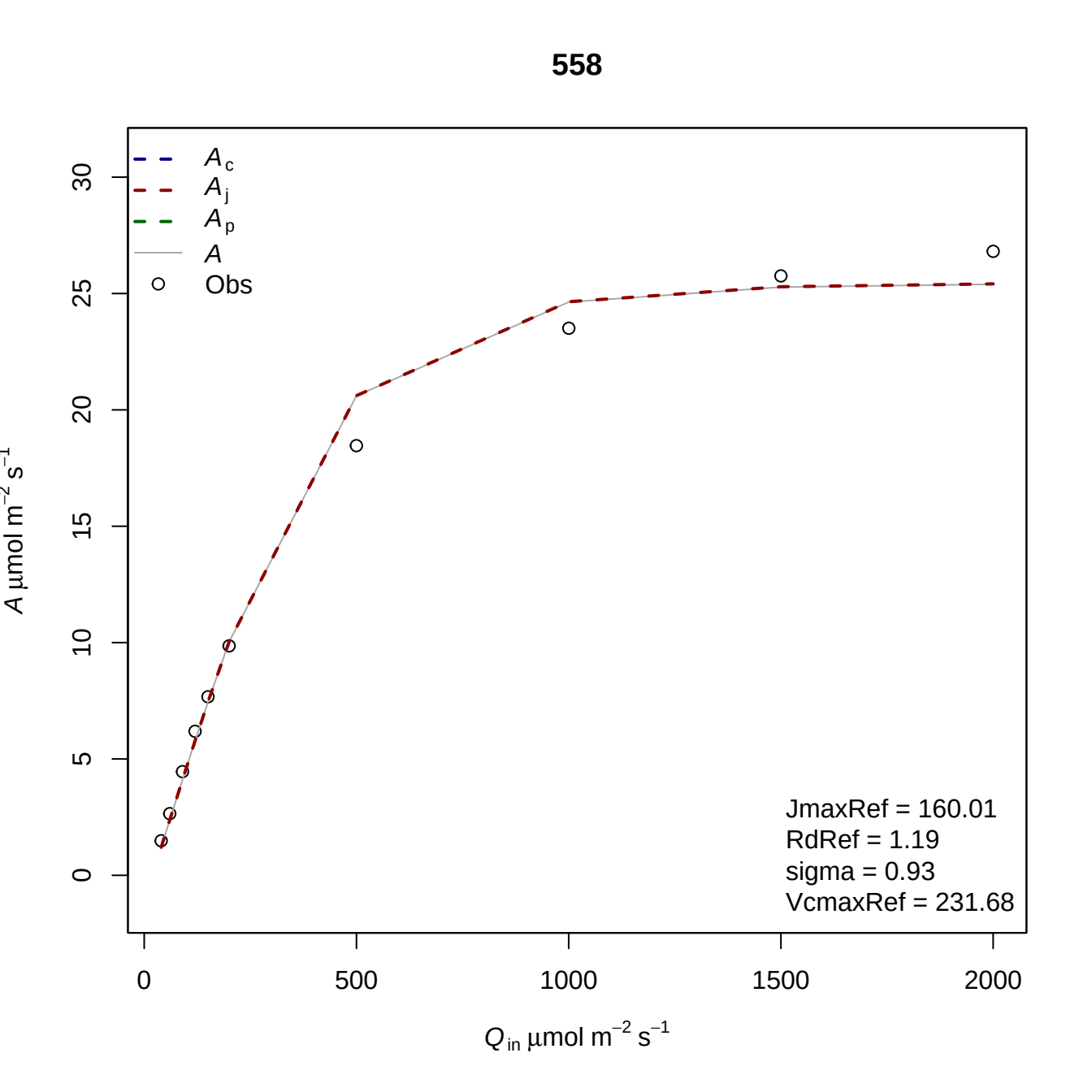

553
A_c
A_j
A_p
A
Obs
JmaxRef = 157.11
RdRef = 2.25
sigma = 0.76
VcmaxRef = 195.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

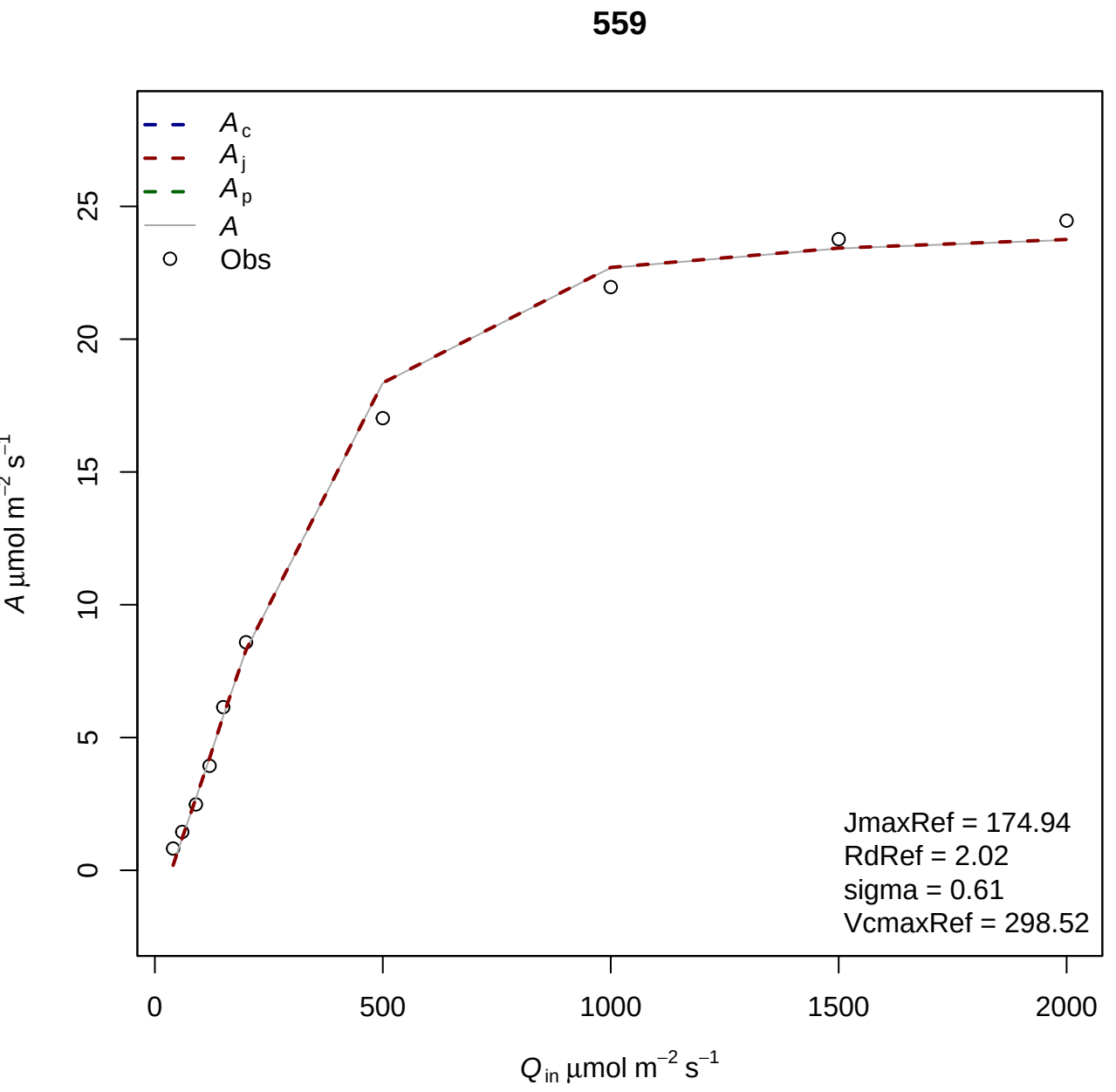

554
A_c
A_j
A_p
A
Obs
JmaxRef = 182.43
RdRef = 1.83
sigma = 0.54
VcmaxRef = 10341.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

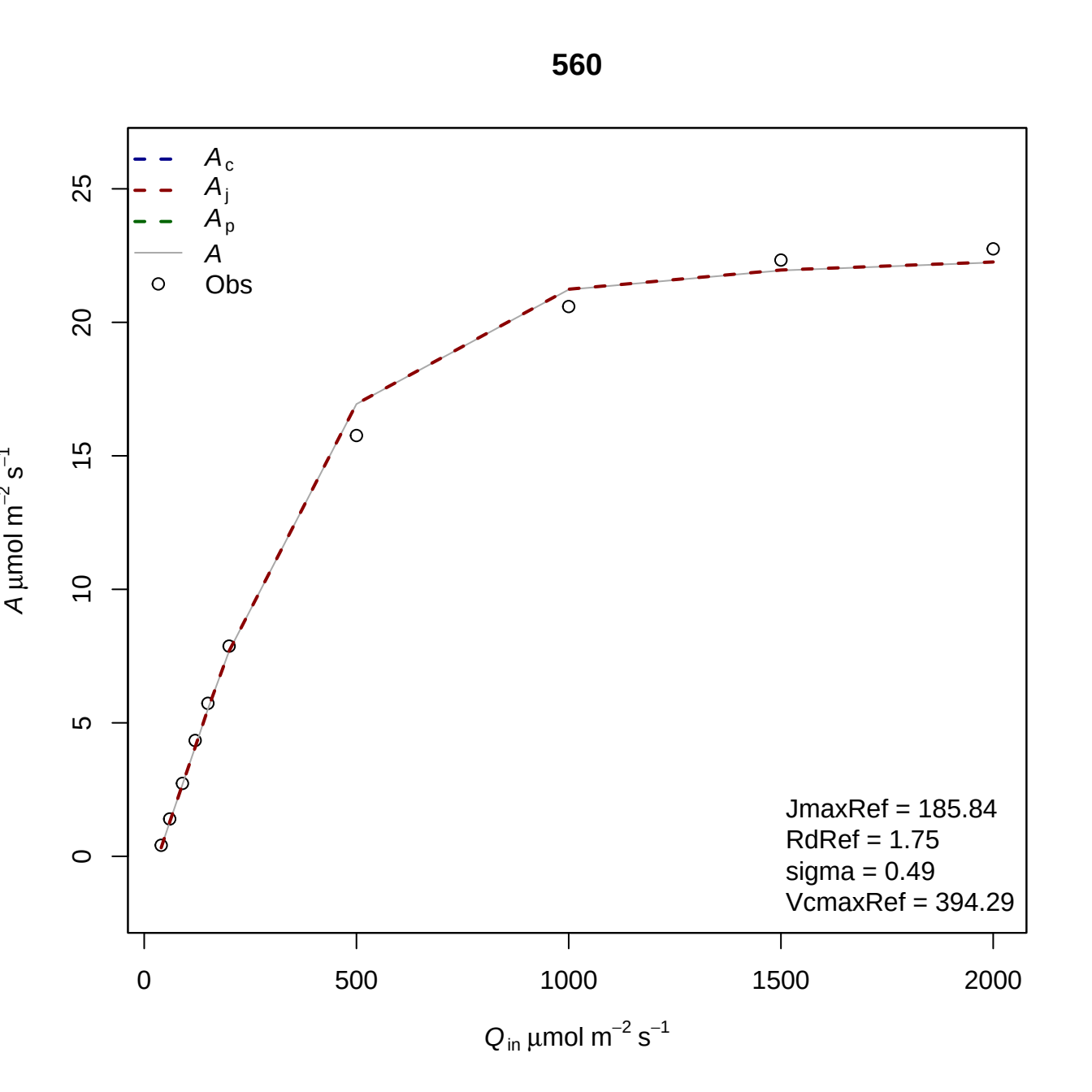

555
A_c
A_j
A_p
A
Obs
JmaxRef = 207.61
RdRef = 2.17
sigma = 0.48
VcmaxRef = 201.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

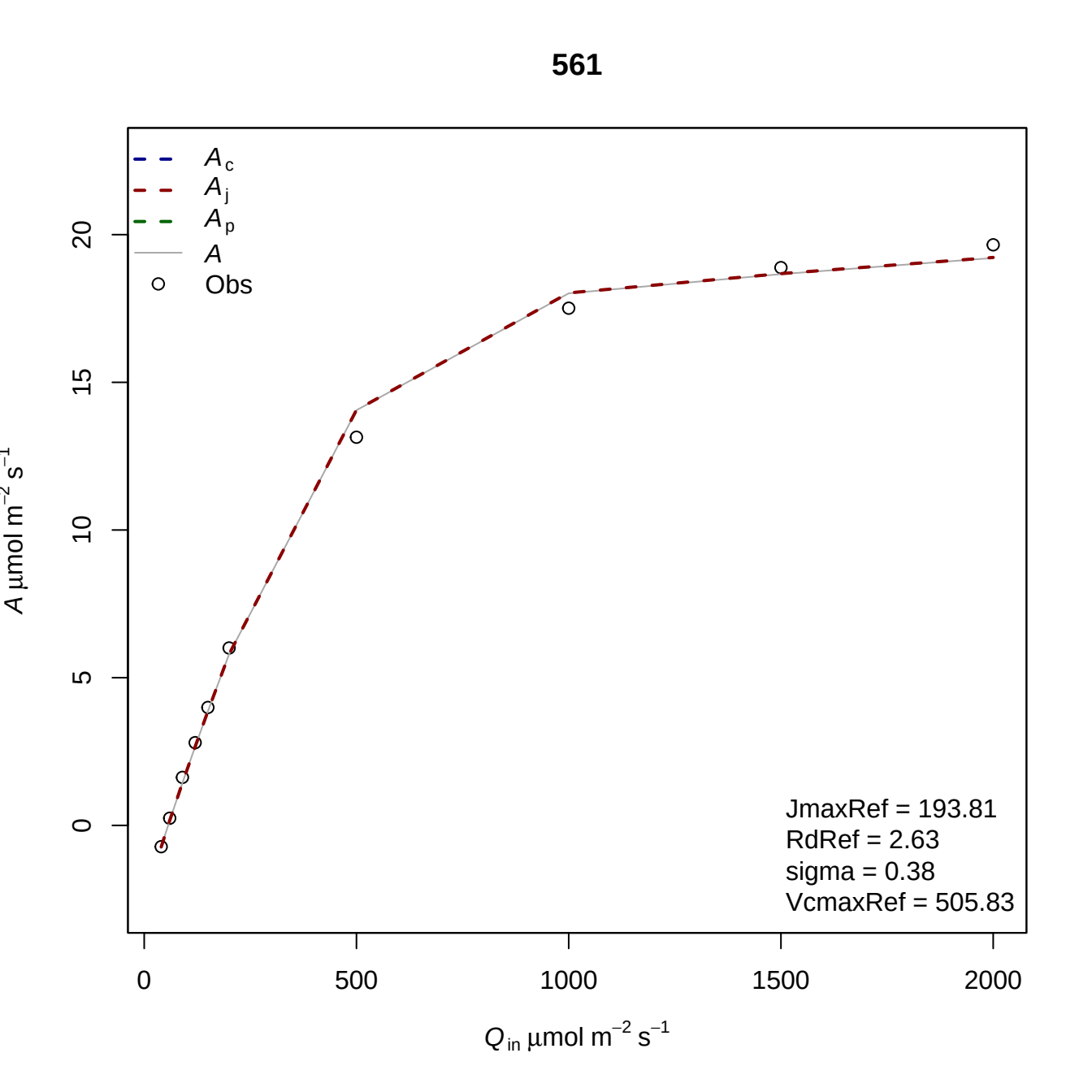

556
A_c
A_j
A_p
A
Obs
JmaxRef = 196.73
RdRef = 3.29
sigma = 0.34
VcmaxRef = 487.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

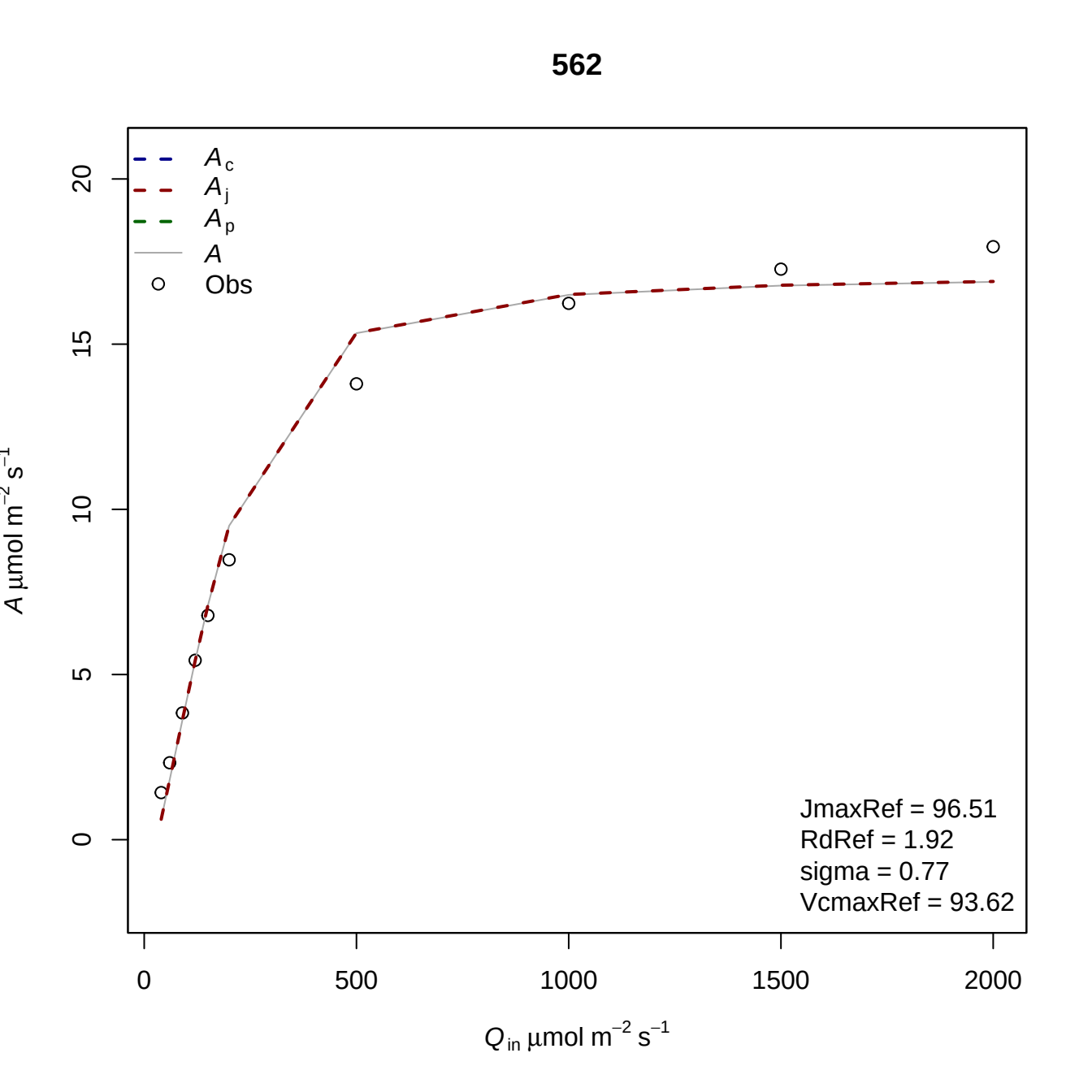

557
A_c
A_j
A_p
A
Obs
JmaxRef = 111.48
RdRef = 1.72
sigma = 1.35
VcmaxRef = 148.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

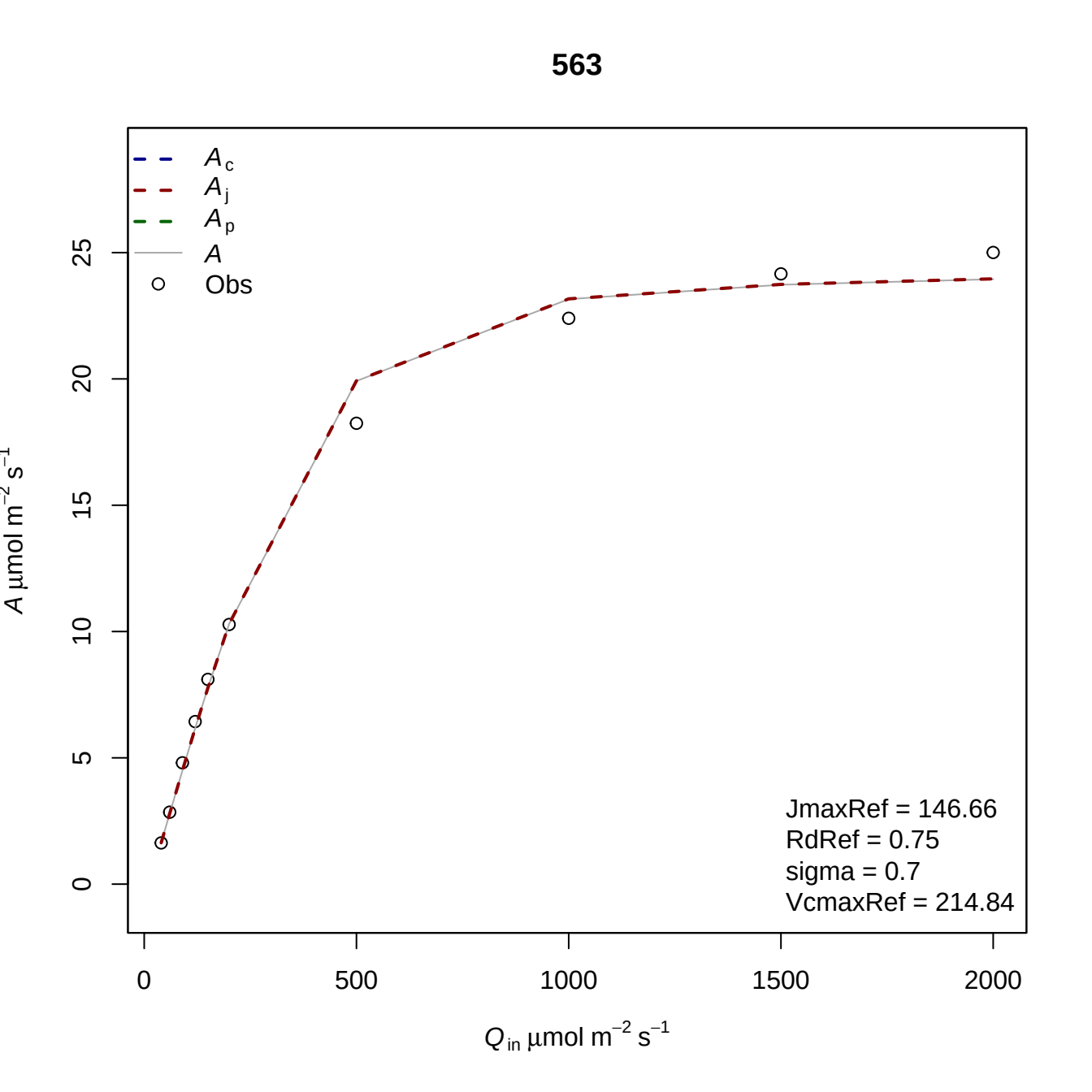

558
A_c
A_j
A_p
A
Obs
JmaxRef = 160.01
RdRef = 1.19
sigma = 0.93
VcmaxRef = 231.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

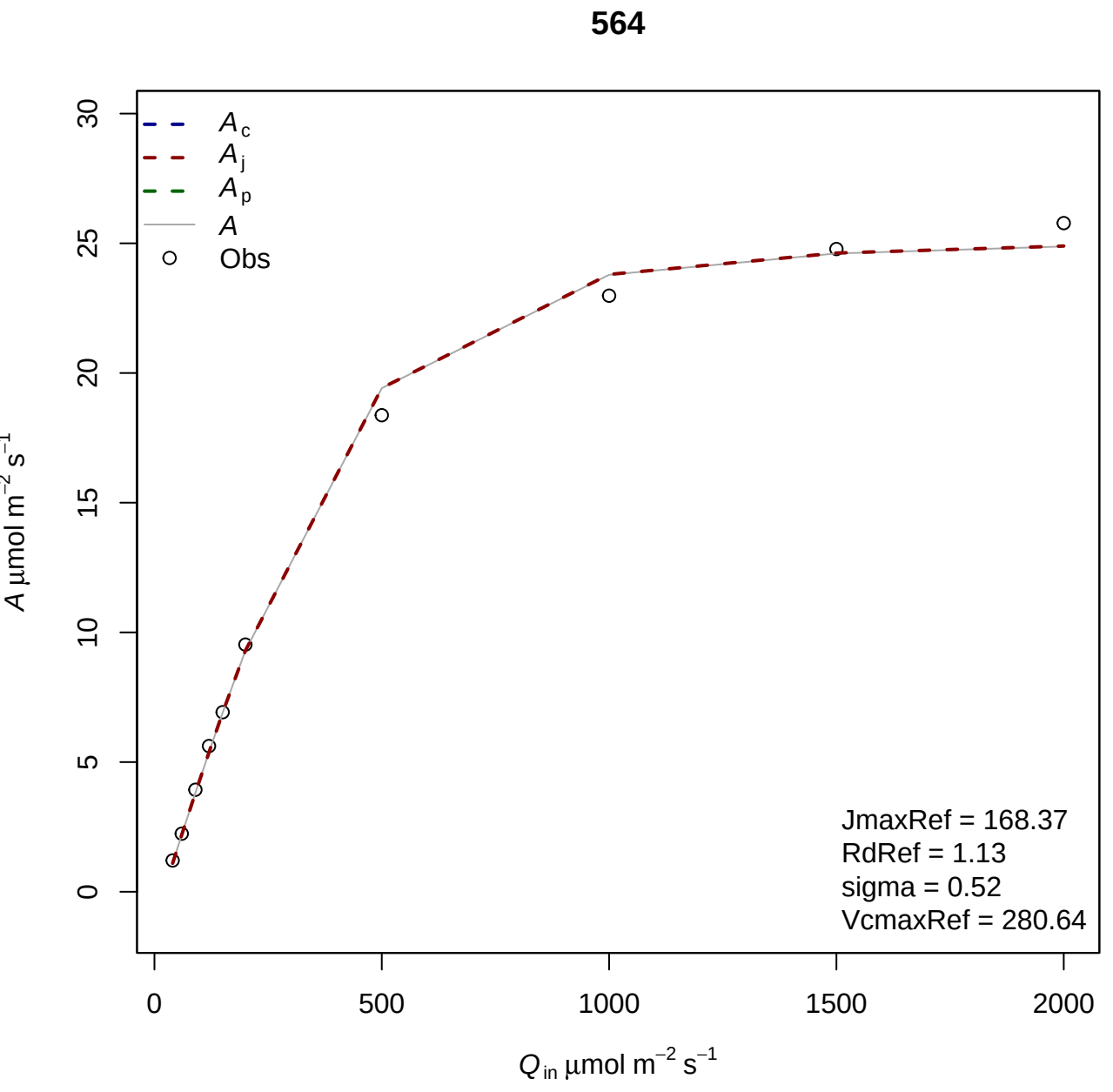

559
A_c
A_j
A_p
A
Obs
JmaxRef = 174.94
RdRef = 2.02
sigma = 0.61
VcmaxRef = 298.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

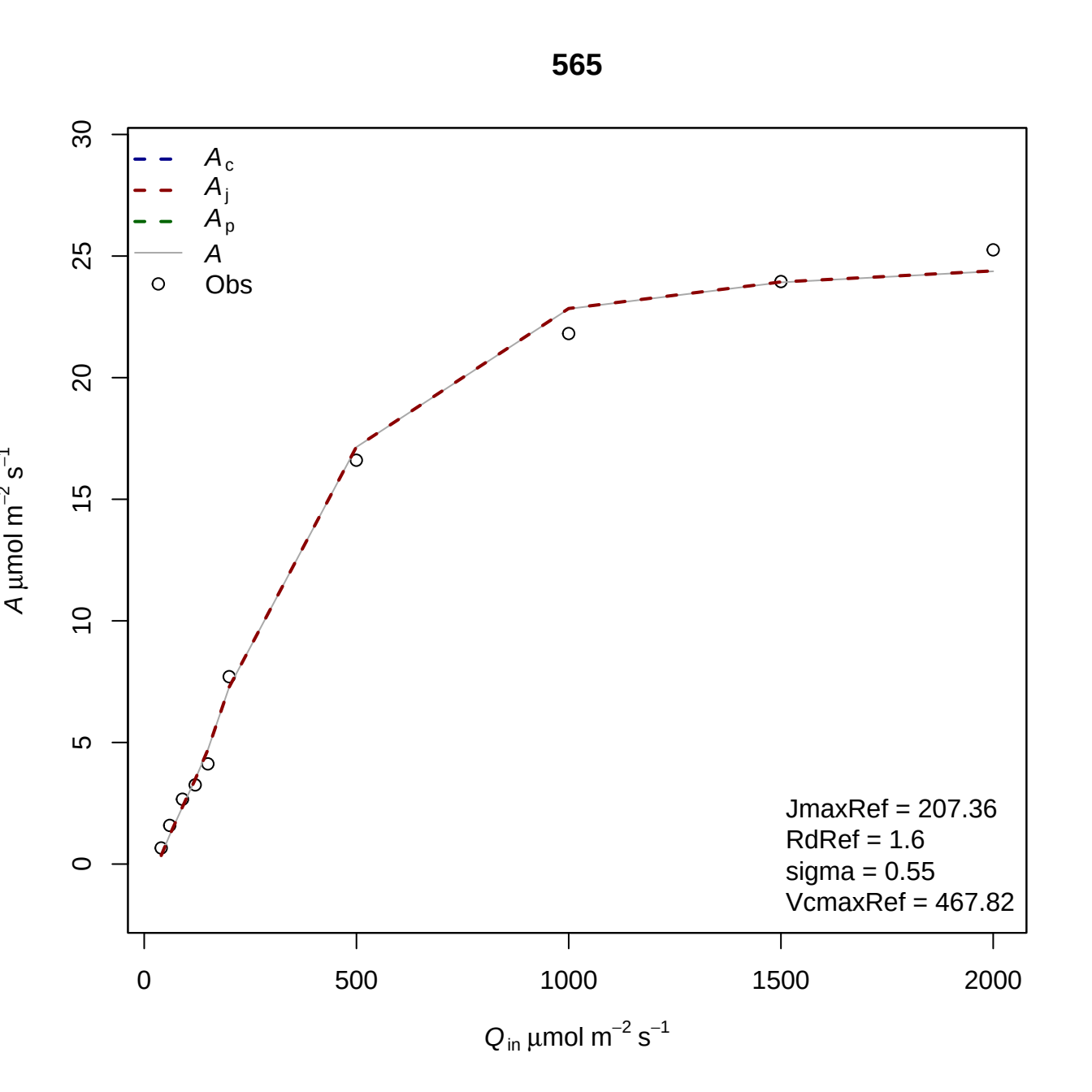

560
A_c
A_j
A_p
A
Obs
JmaxRef = 185.84
RdRef = 1.75
sigma = 0.49
VcmaxRef = 394.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

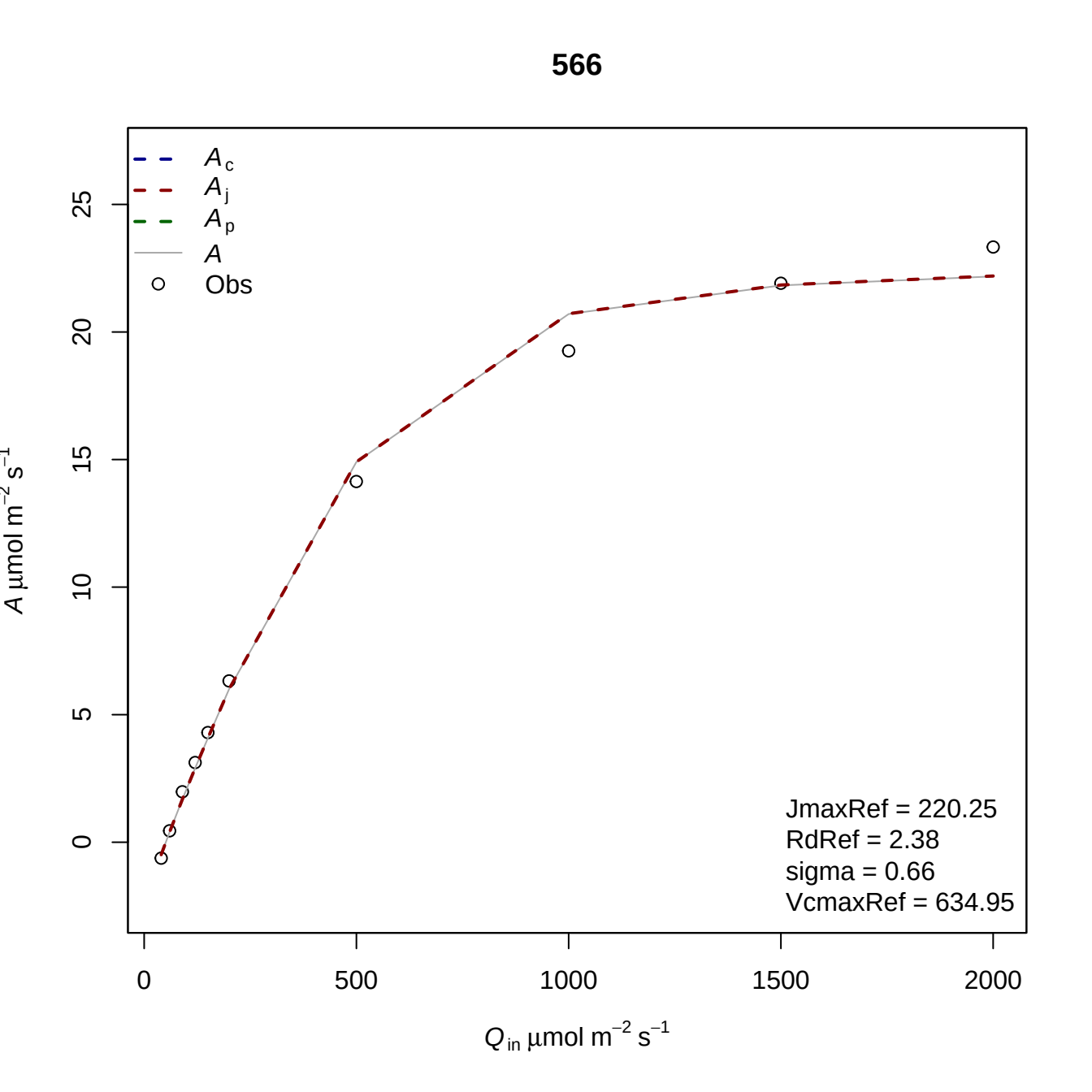

561
A_c
A_j
A_p
A
Obs
JmaxRef = 193.81
RdRef = 2.63
sigma = 0.38
VcmaxRef = 505.83
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

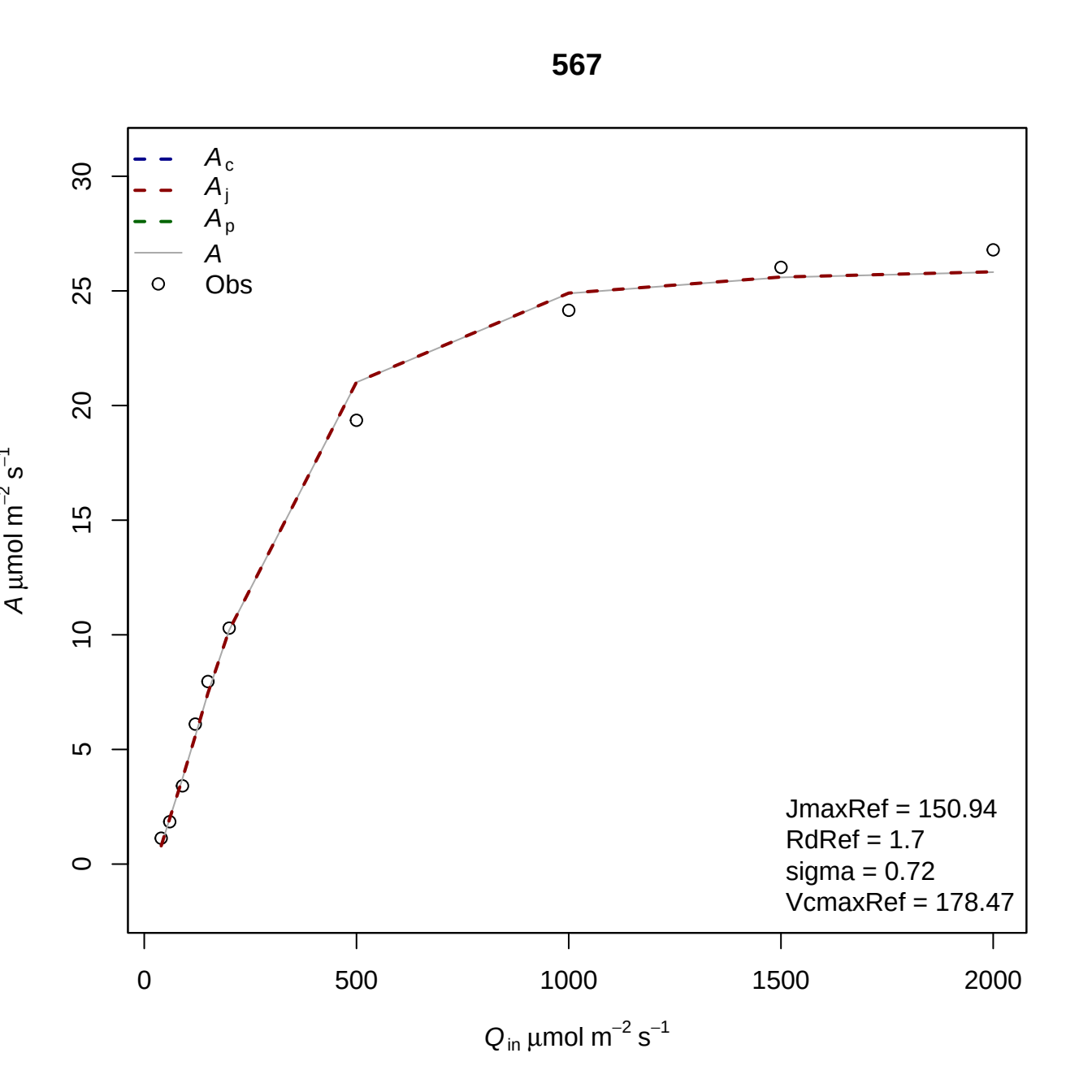

562
A_c
A_j
A_p
A
Obs
JmaxRef = 96.51
RdRef = 1.92
sigma = 0.77
VcmaxRef = 93.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

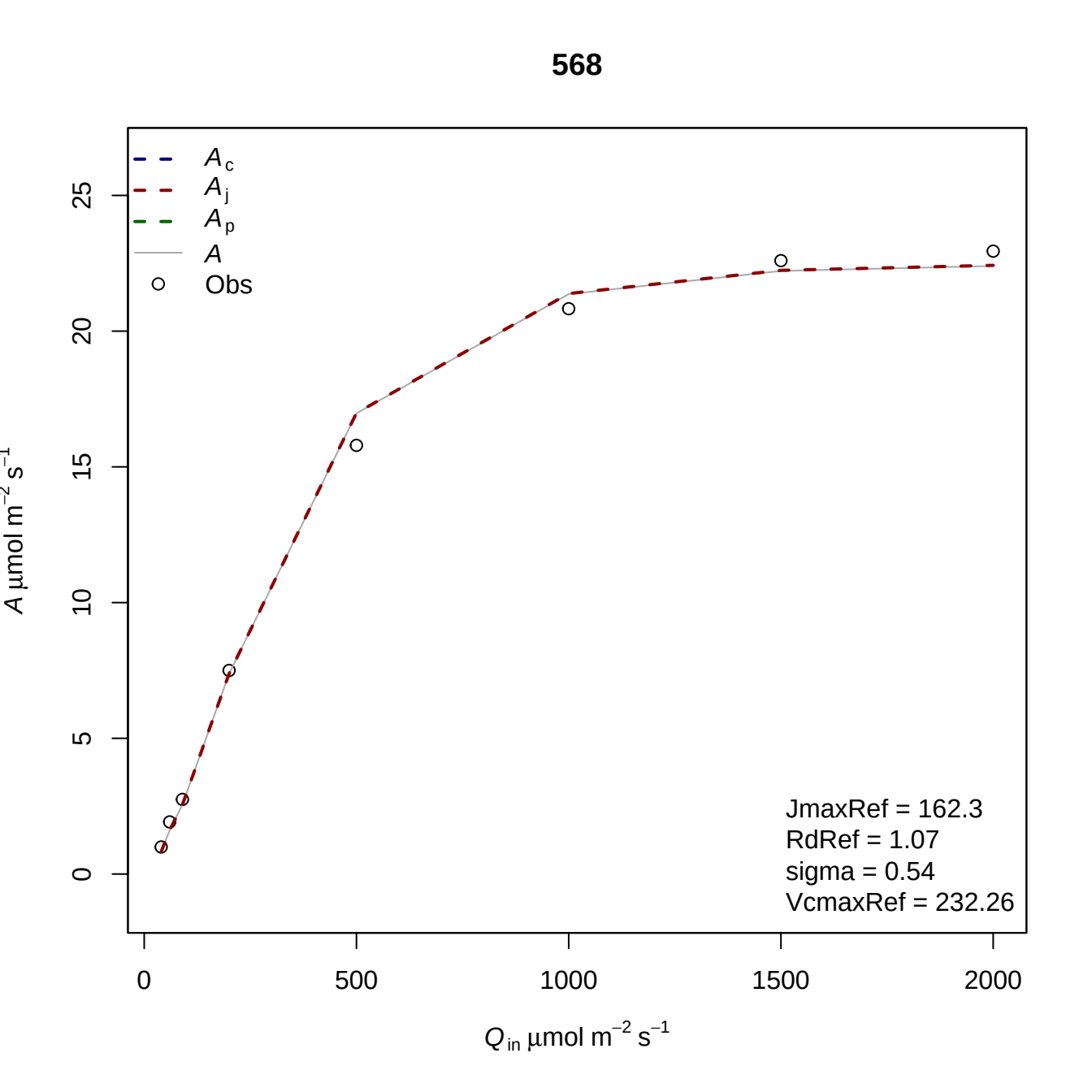

563
A_c
A_j
A_p
A
Obs
JmaxRef = 146.66
RdRef = 0.75
sigma = 0.7
VcmaxRef = 214.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

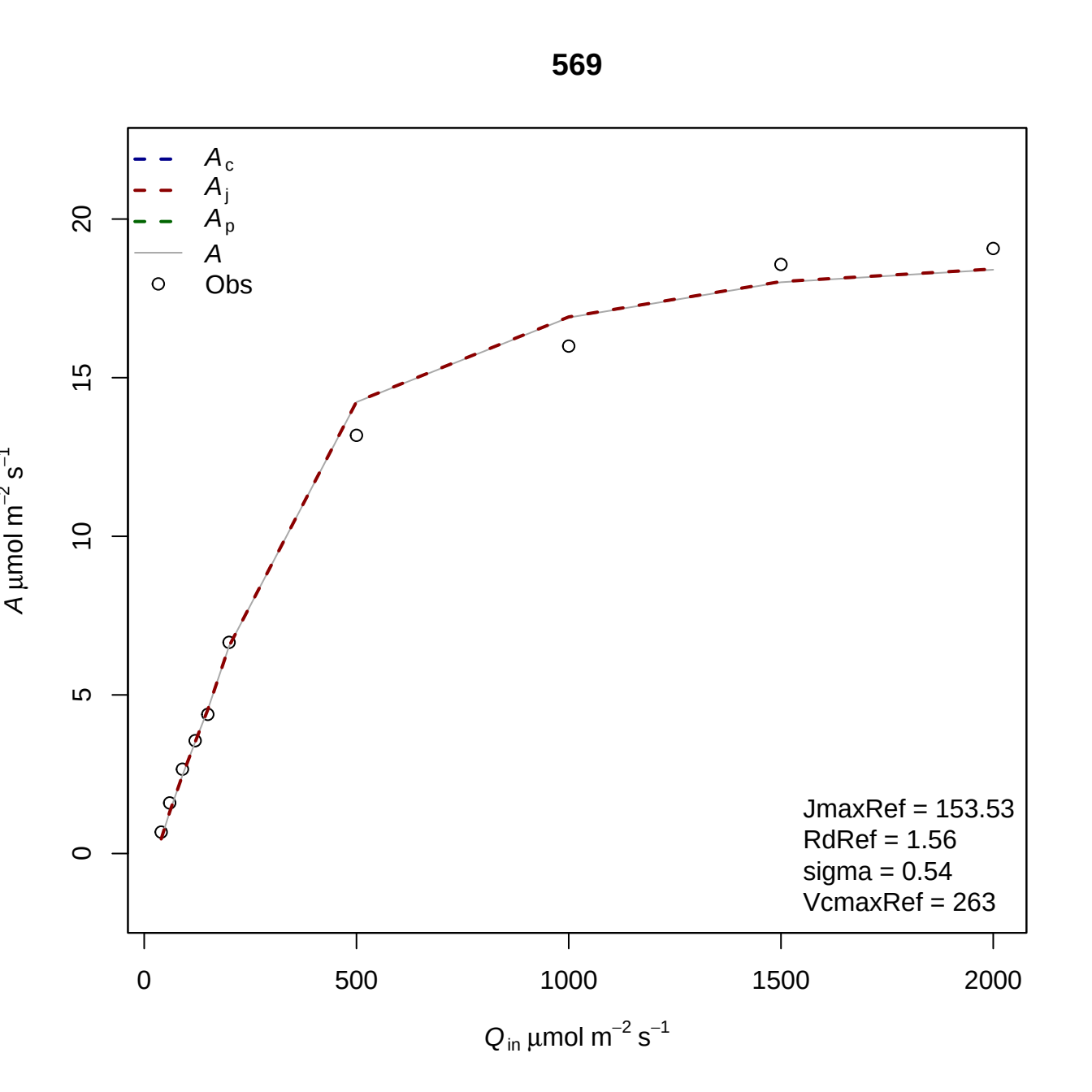

564
A_c
A_j
A_p
A
Obs
JmaxRef = 168.37
RdRef = 1.13
sigma = 0.52
VcmaxRef = 280.64
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

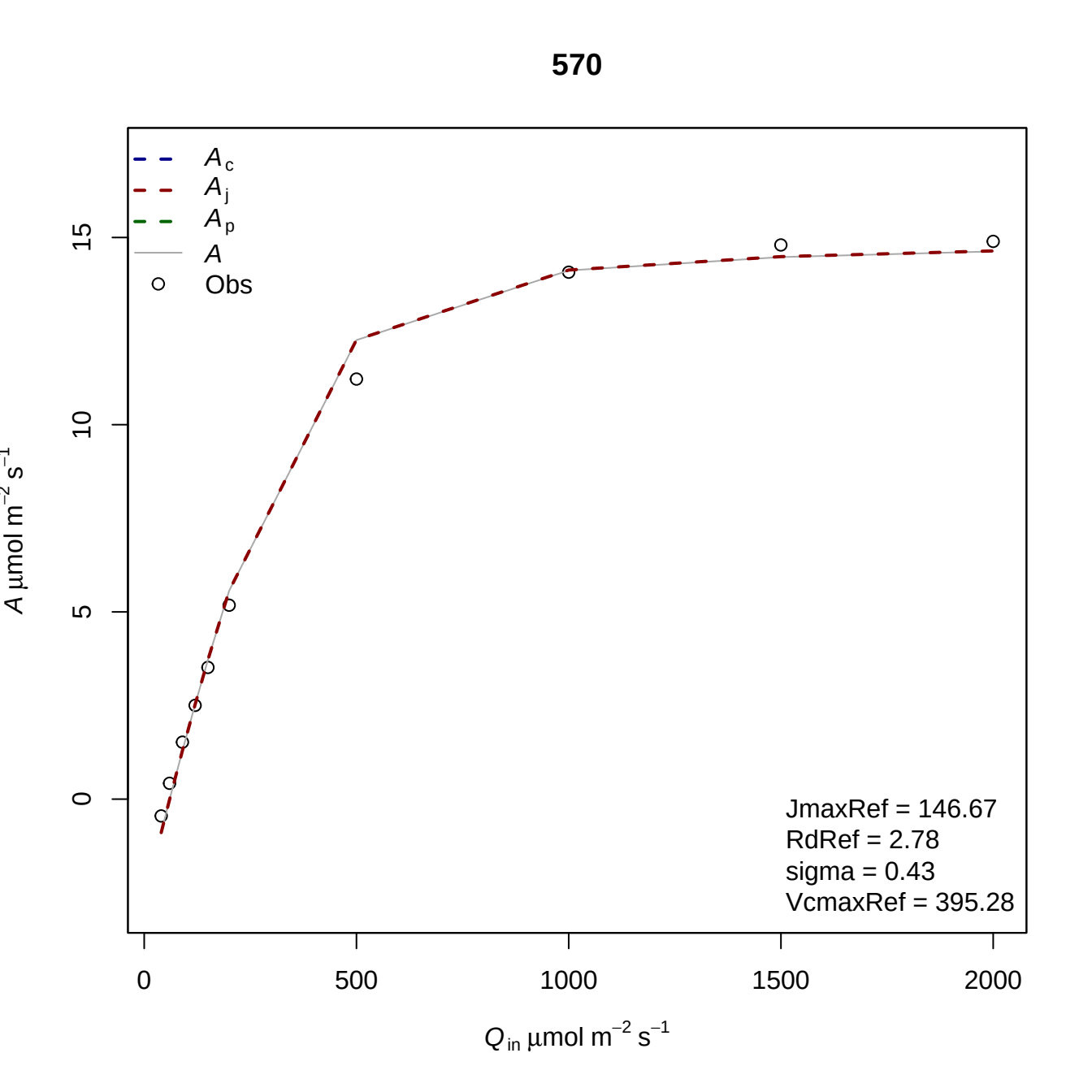

565
A_c
A_j
A_p
A
Obs
JmaxRef = 207.36
RdRef = 1.6
sigma = 0.55
VcmaxRef = 467.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

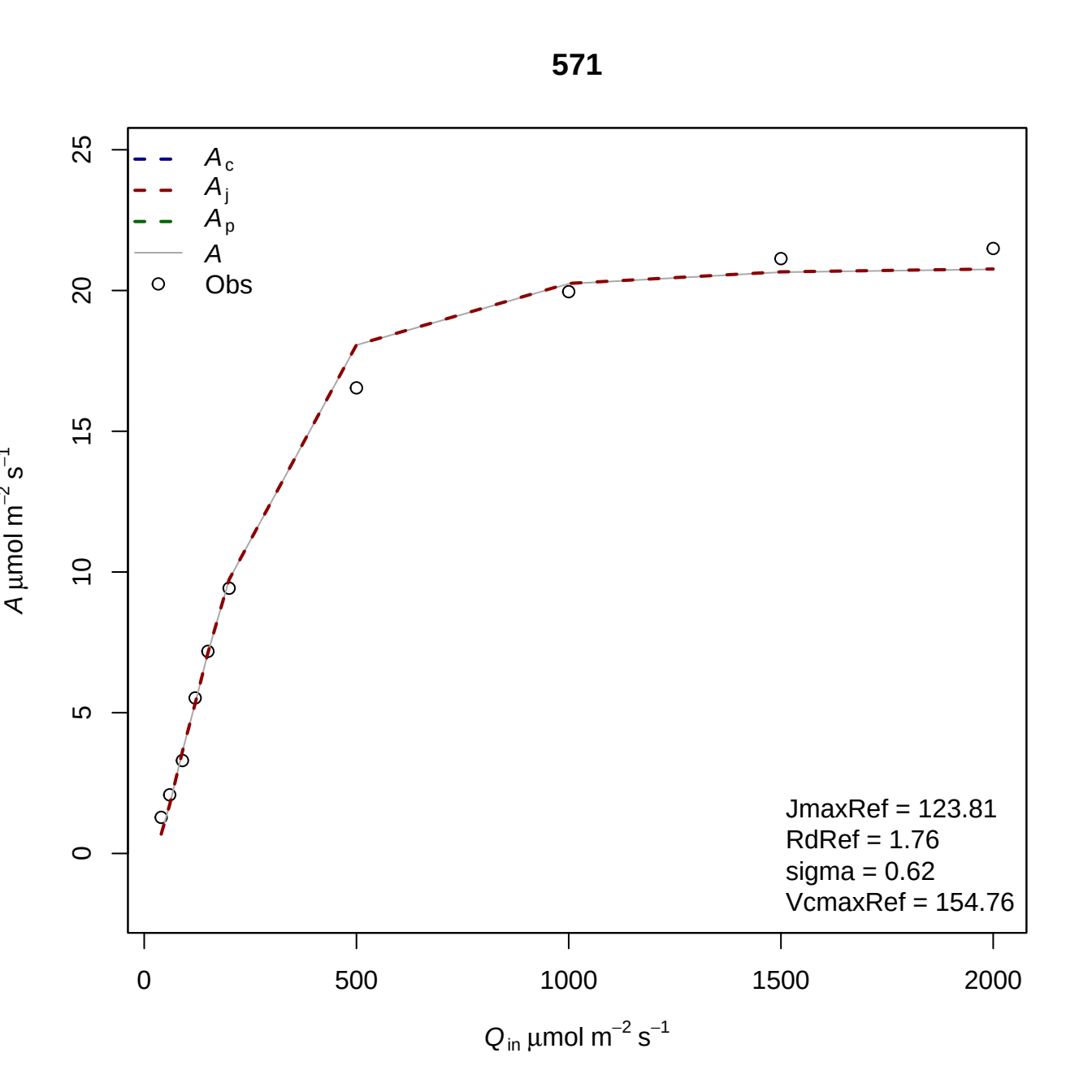

566
A_c
A_j
A_p
A
Obs
JmaxRef = 220.25
RdRef = 2.38
sigma = 0.66
VcmaxRef = 634.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

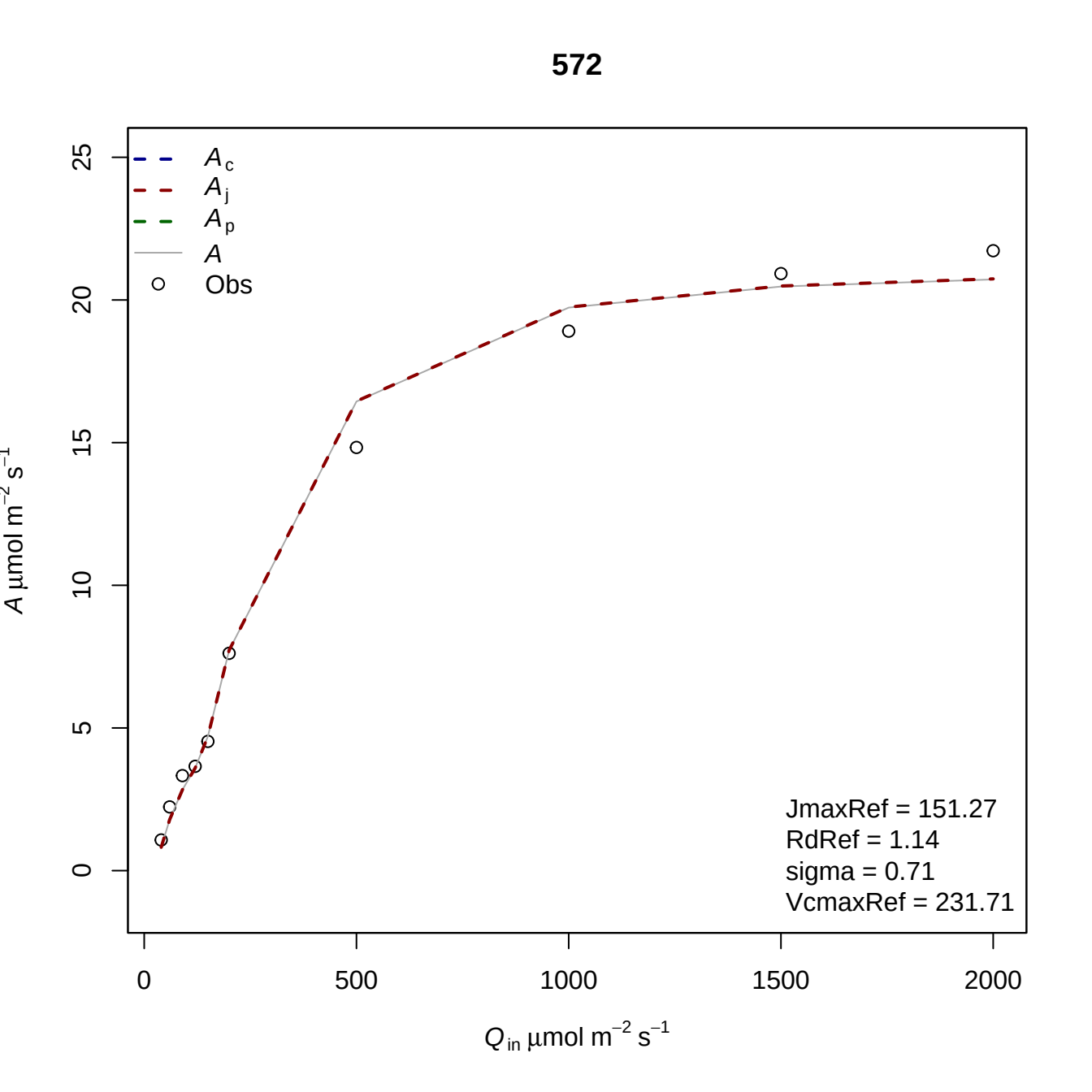

567
A_c
A_j
A_p
A
Obs
JmaxRef = 150.94
RdRef = 1.7
sigma = 0.72
VcmaxRef = 178.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

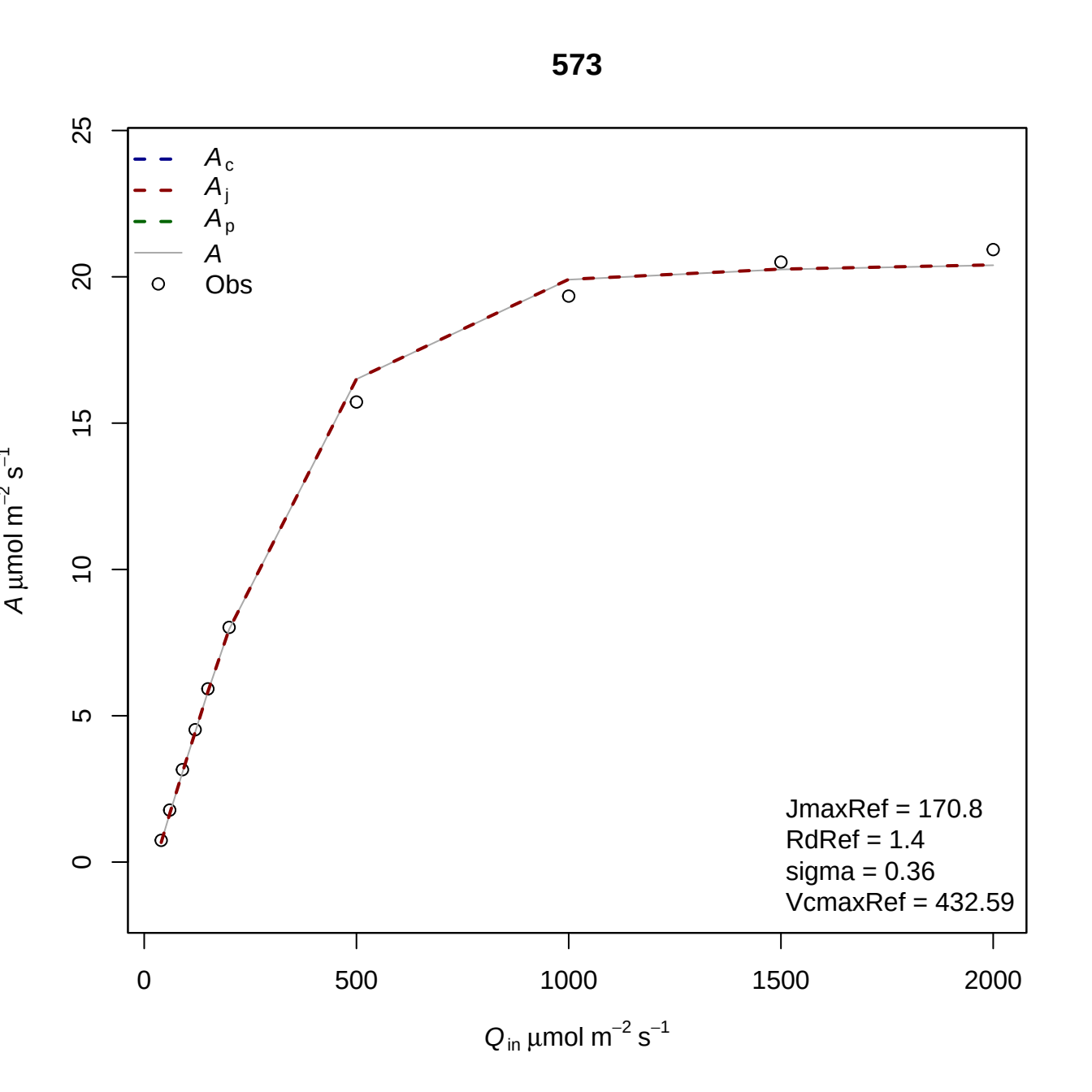

568
A_c
A_j
A_p
A
Obs
JmaxRef = 162.3
RdRef = 1.07
sigma = 0.54
VcmaxRef = 232.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

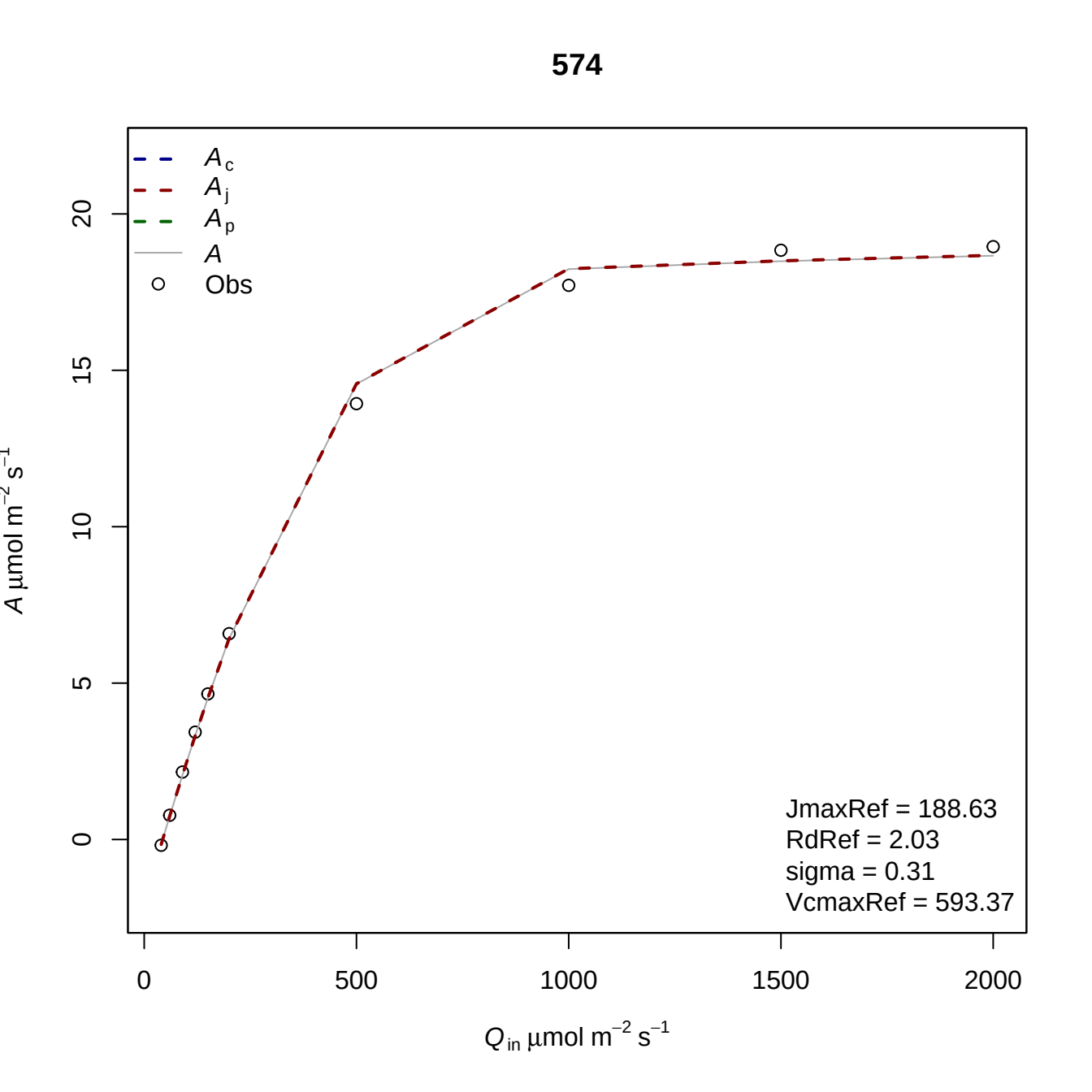

569
A_c
A_j
A_p
A
Obs
JmaxRef = 153.53
RdRef = 1.56
sigma = 0.54
VcmaxRef = 263
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

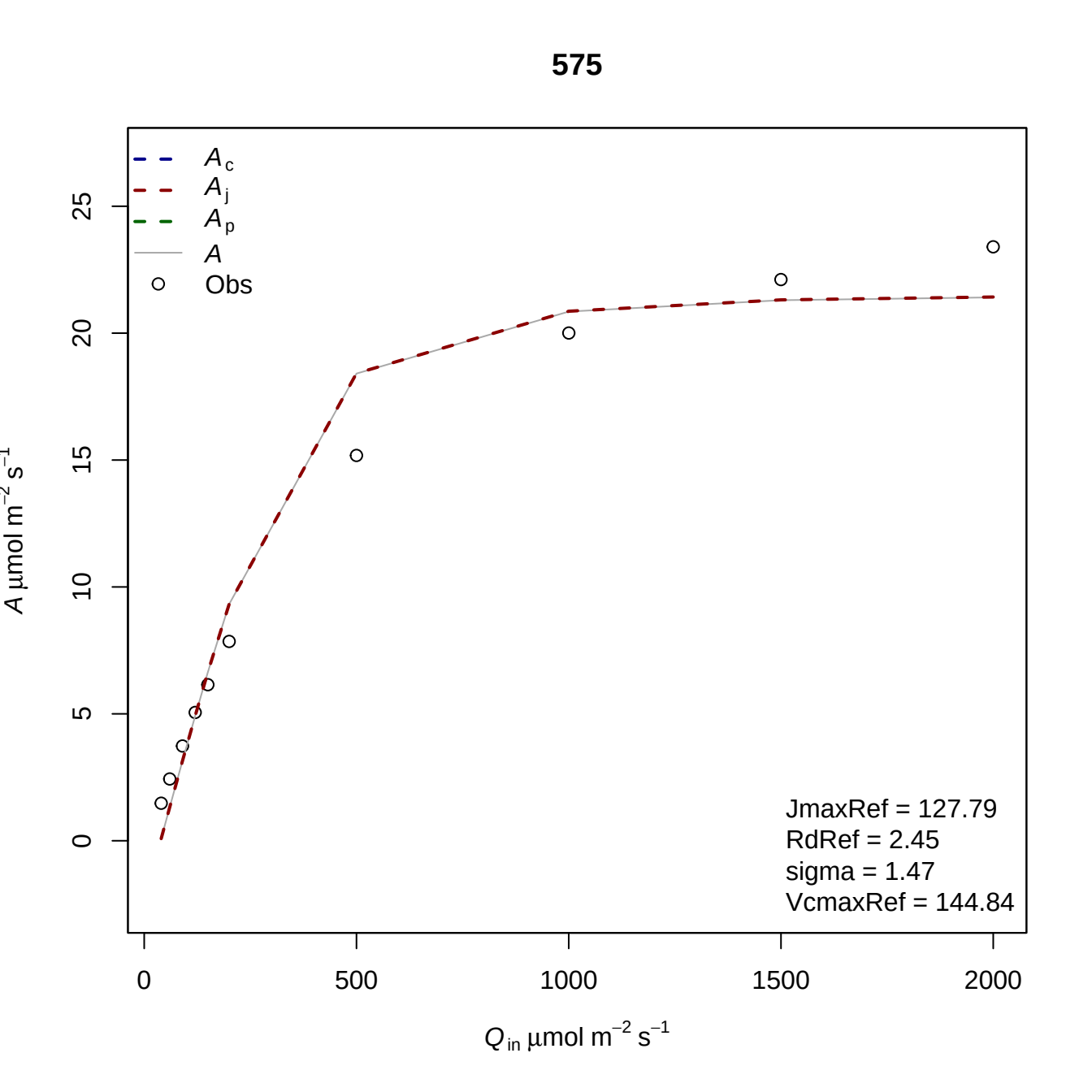

570
A_c
A_j
A_p
A
Obs
JmaxRef = 146.67
RdRef = 2.78
sigma = 0.43
VcmaxRef = 395.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

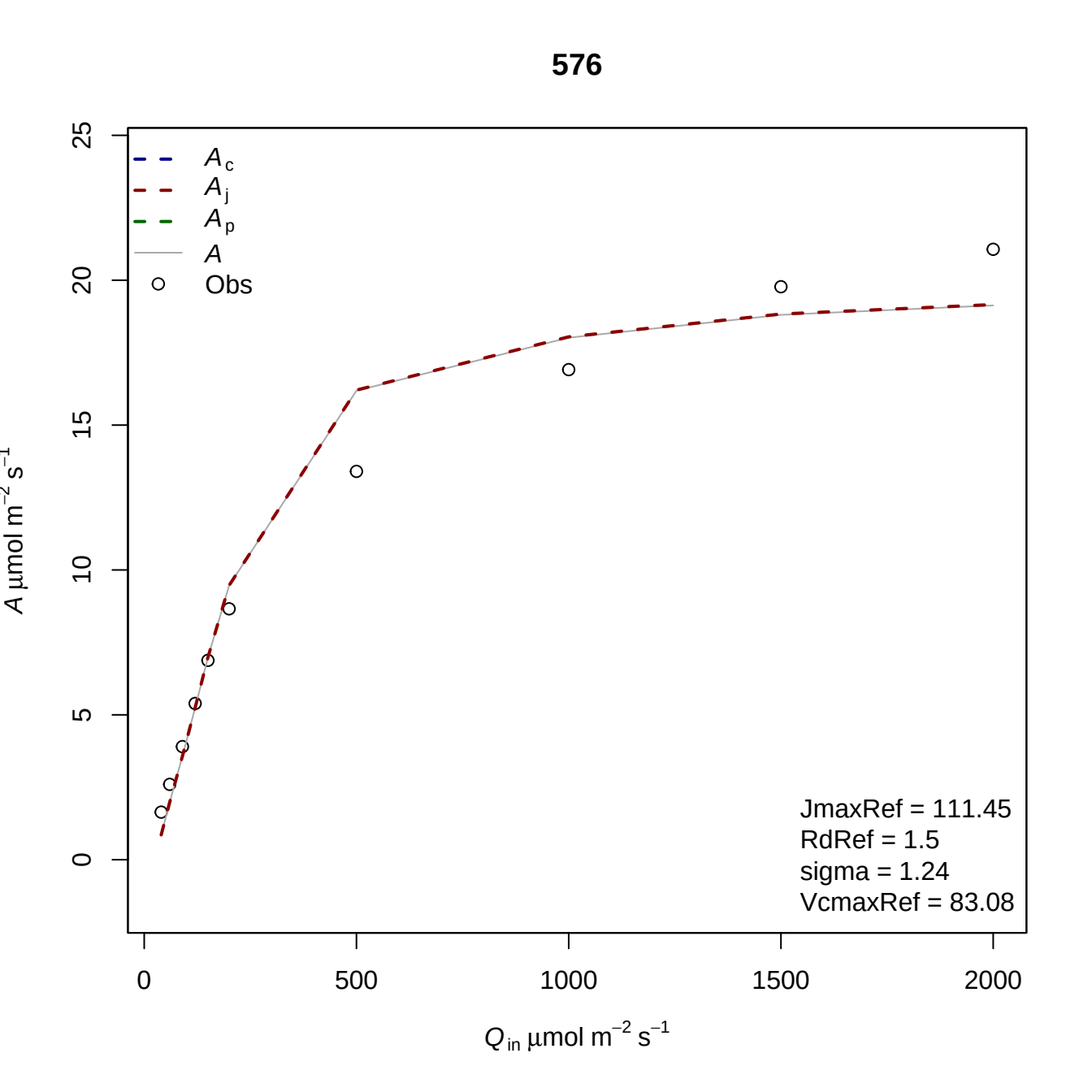

571
A_c
A_j
A_p
A
Obs
JmaxRef = 123.81
RdRef = 1.76
sigma = 0.62
VcmaxRef = 154.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

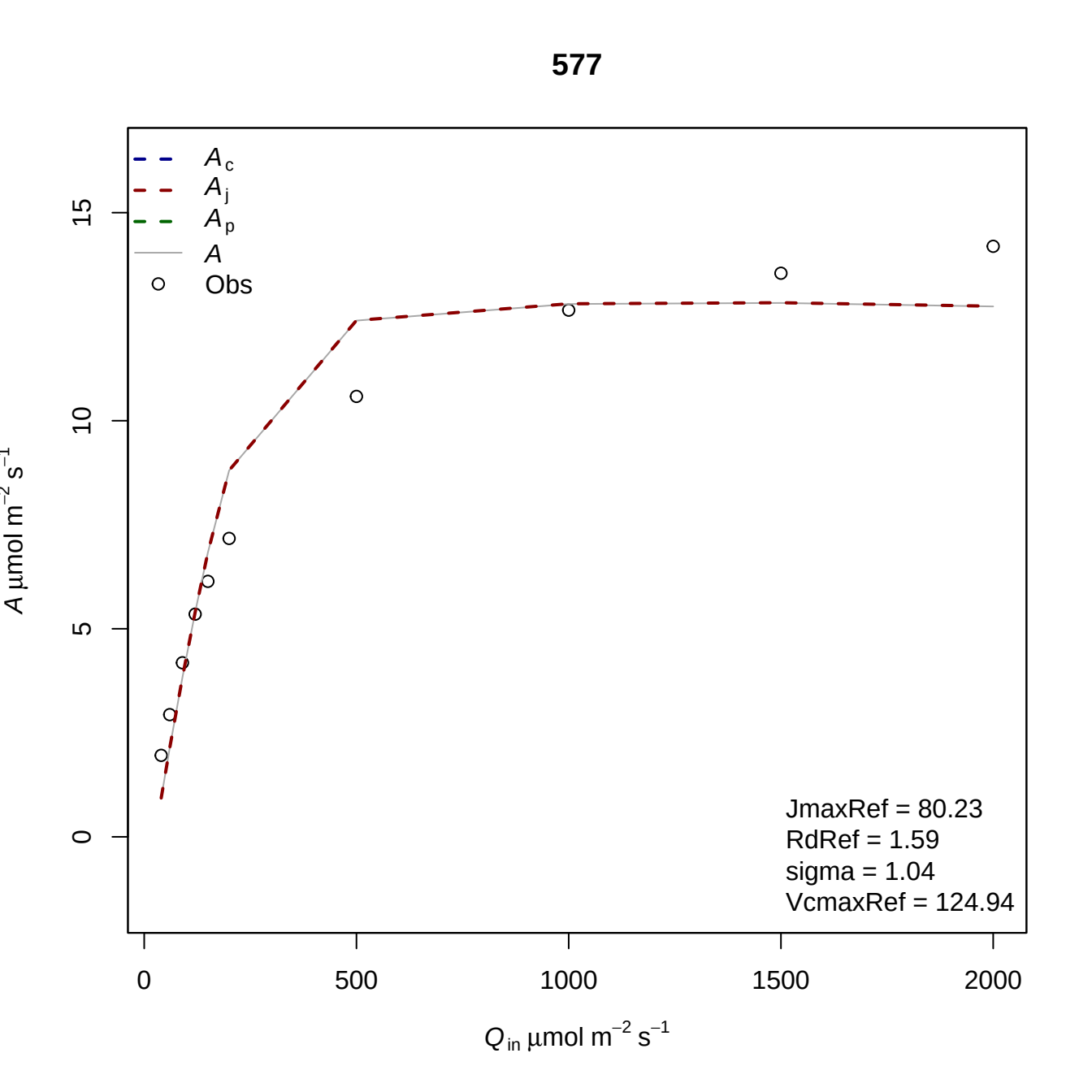

572
A_c
A_j
A_p
A
Obs
JmaxRef = 151.27
RdRef = 1.14
sigma = 0.71
VcmaxRef = 231.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

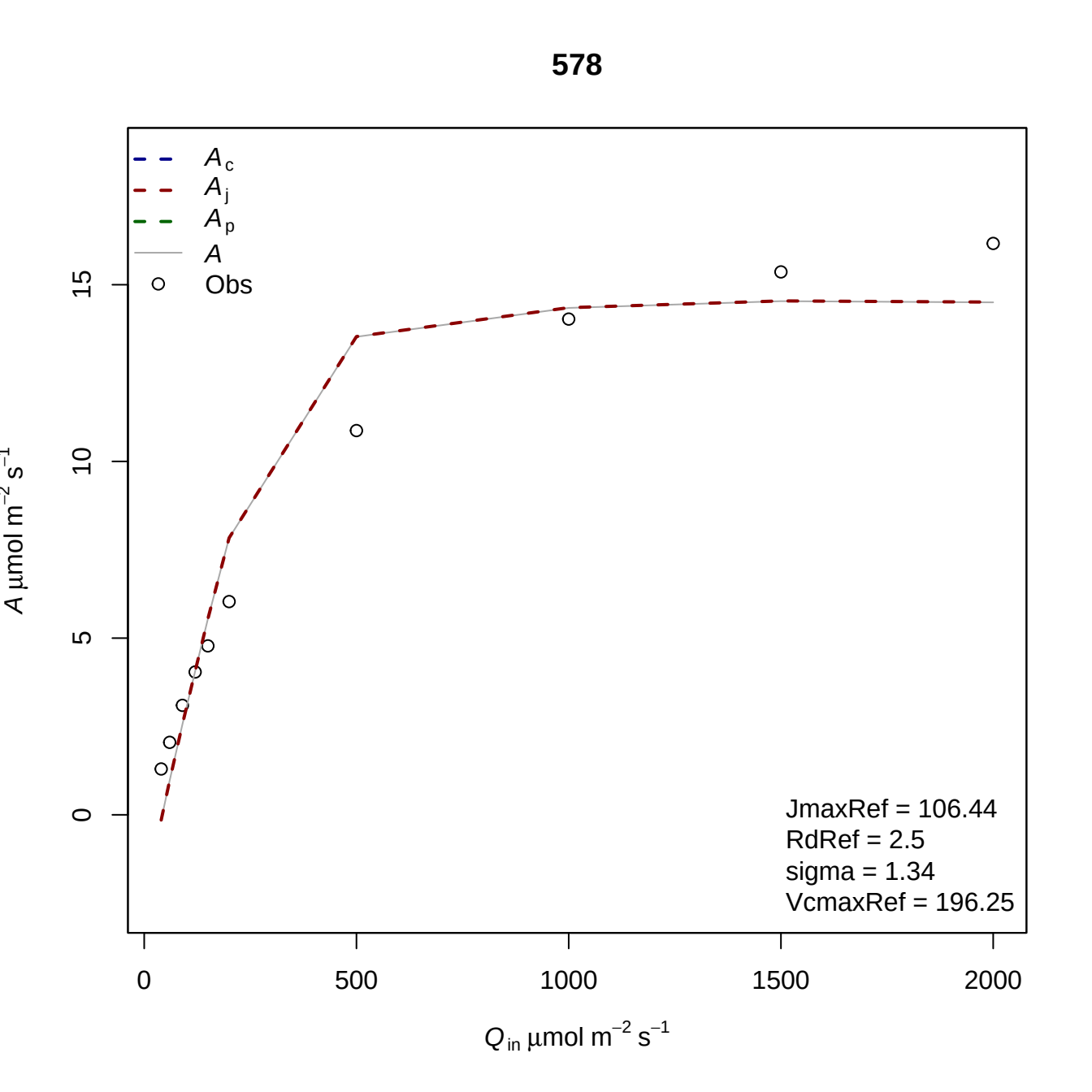

573
A_c
A_j
A_p
A
Obs
JmaxRef = 170.8
RdRef = 1.4
sigma = 0.36
VcmaxRef = 432.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

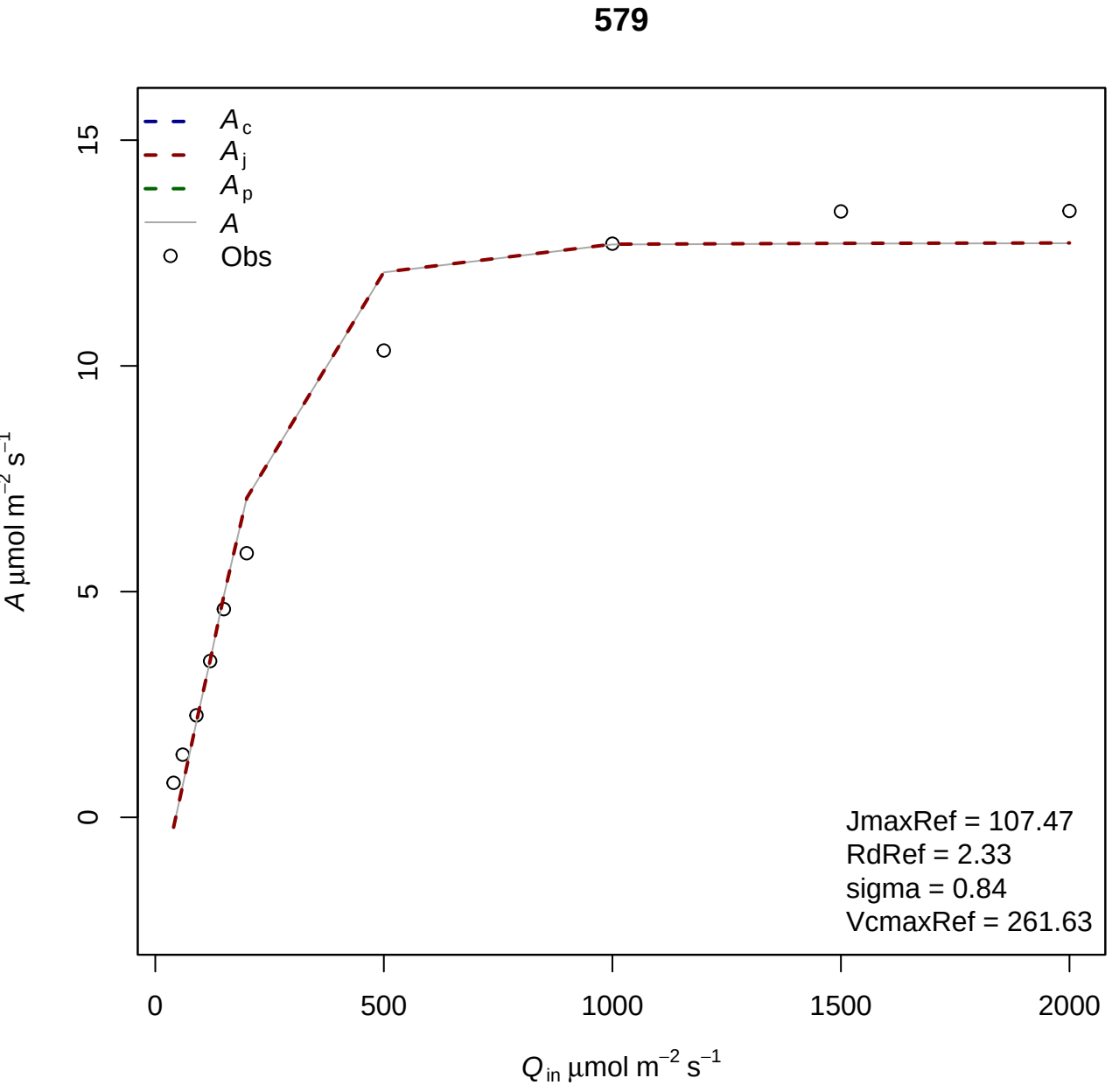

574
A_c
A_j
A_p
A
Obs
JmaxRef = 188.63
RdRef = 2.03
sigma = 0.31
VcmaxRef = 593.37
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

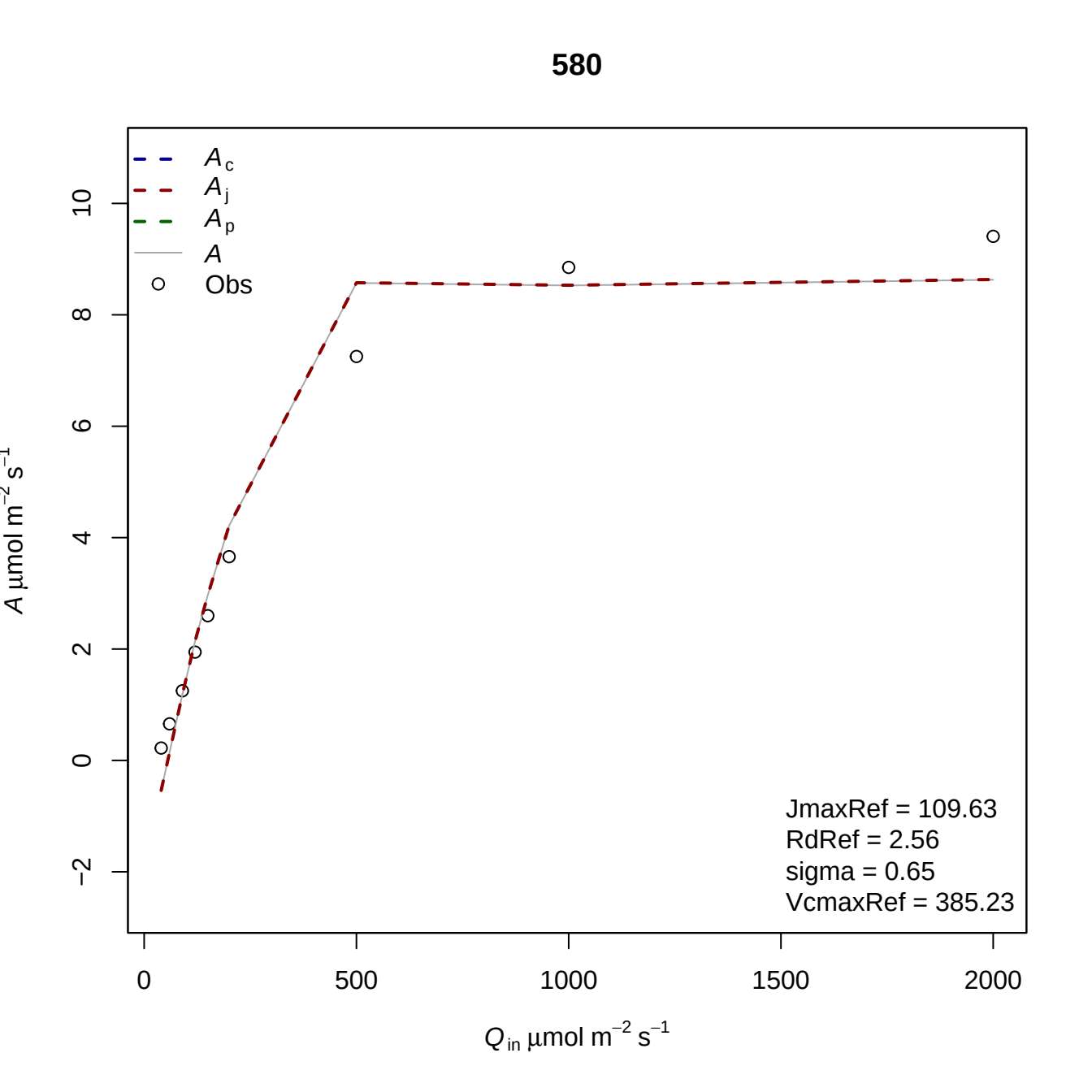

575
A_c
A_j
A_p
A
Obs
JmaxRef = 127.79
RdRef = 2.45
sigma = 1.47
VcmaxRef = 144.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

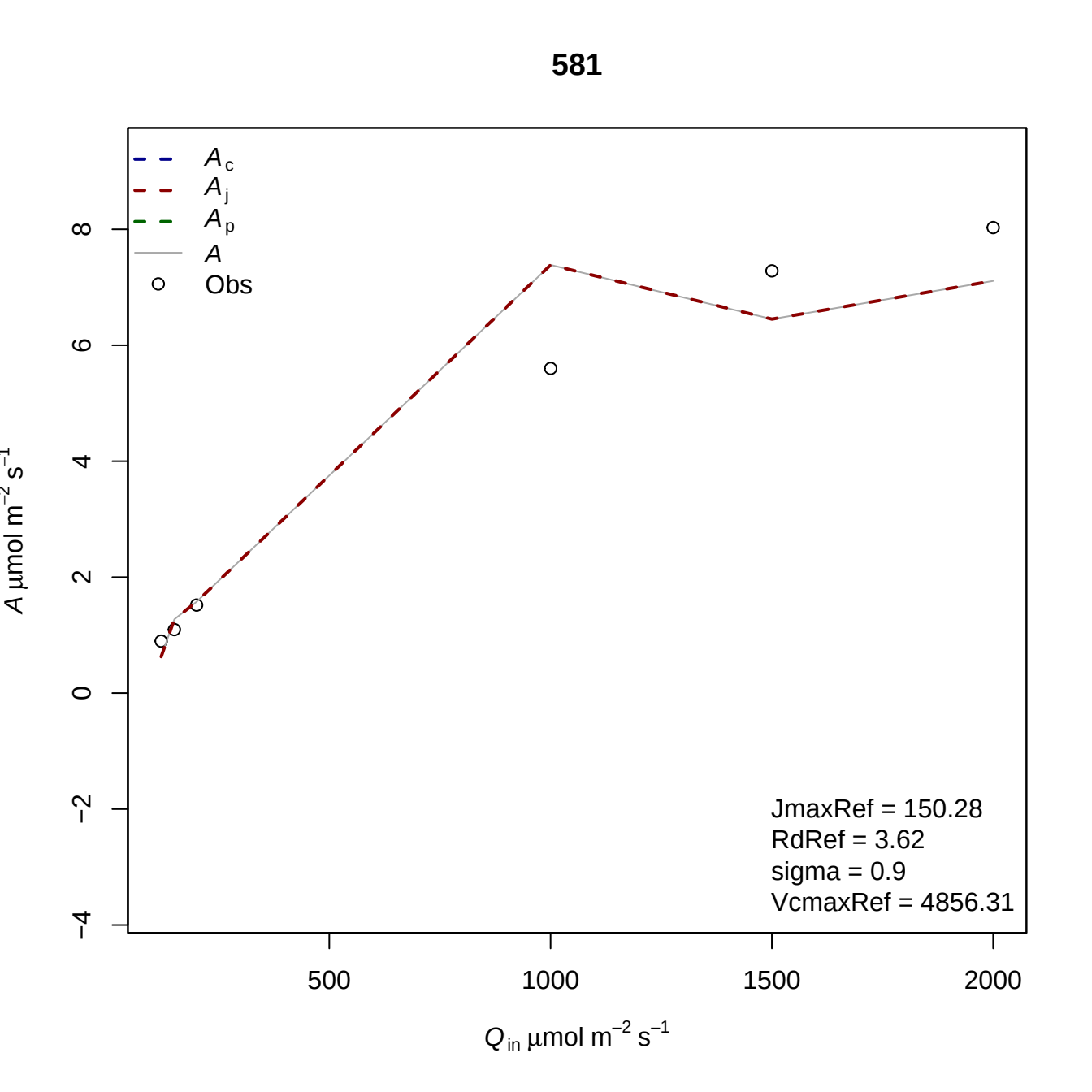

576
A_c
A_j
A_p
A
Obs
JmaxRef = 111.45
RdRef = 1.5
sigma = 1.24
VcmaxRef = 83.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

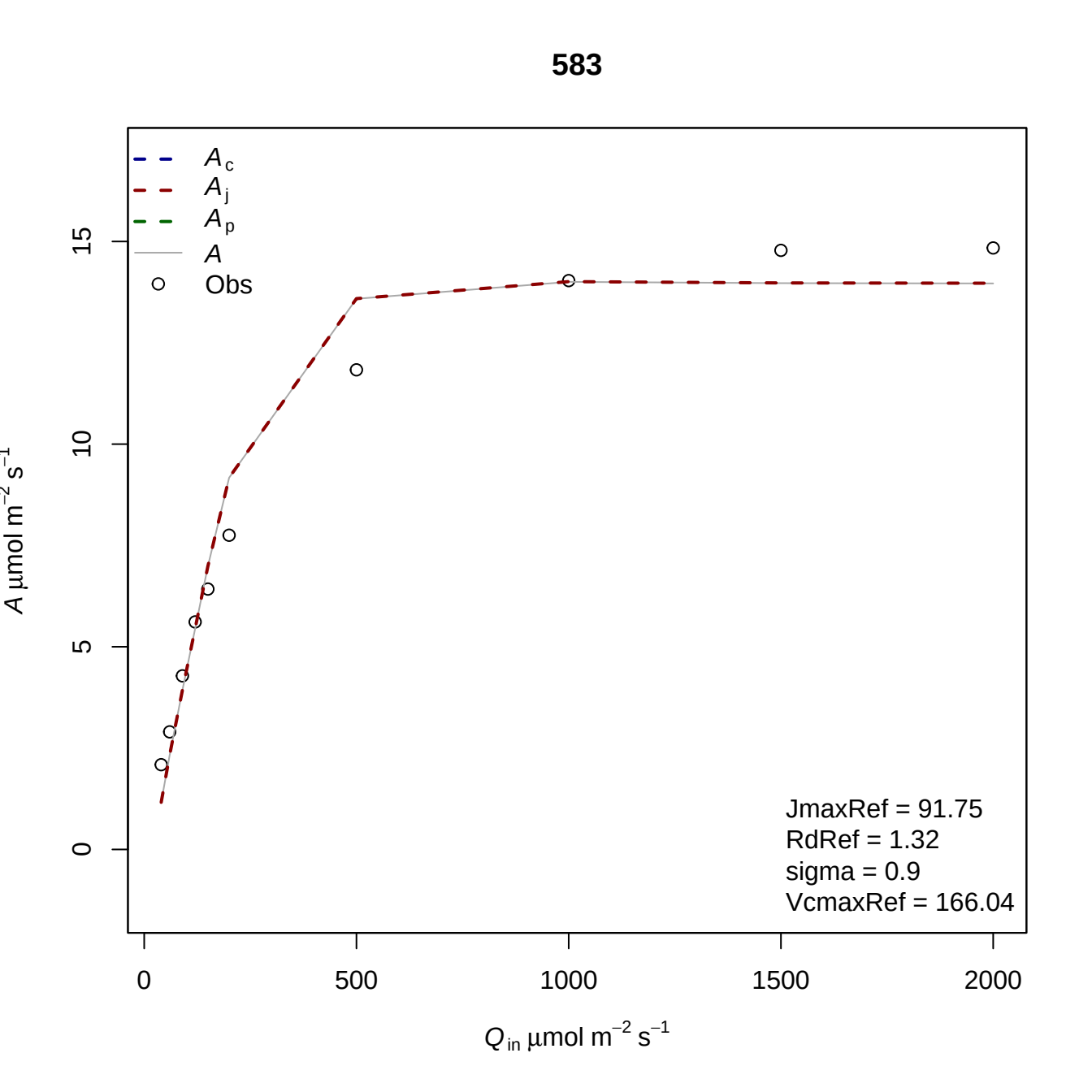

577
A_c
A_j
A_p
A
Obs
JmaxRef = 80.23
RdRef = 1.59
sigma = 1.04
VcmaxRef = 124.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

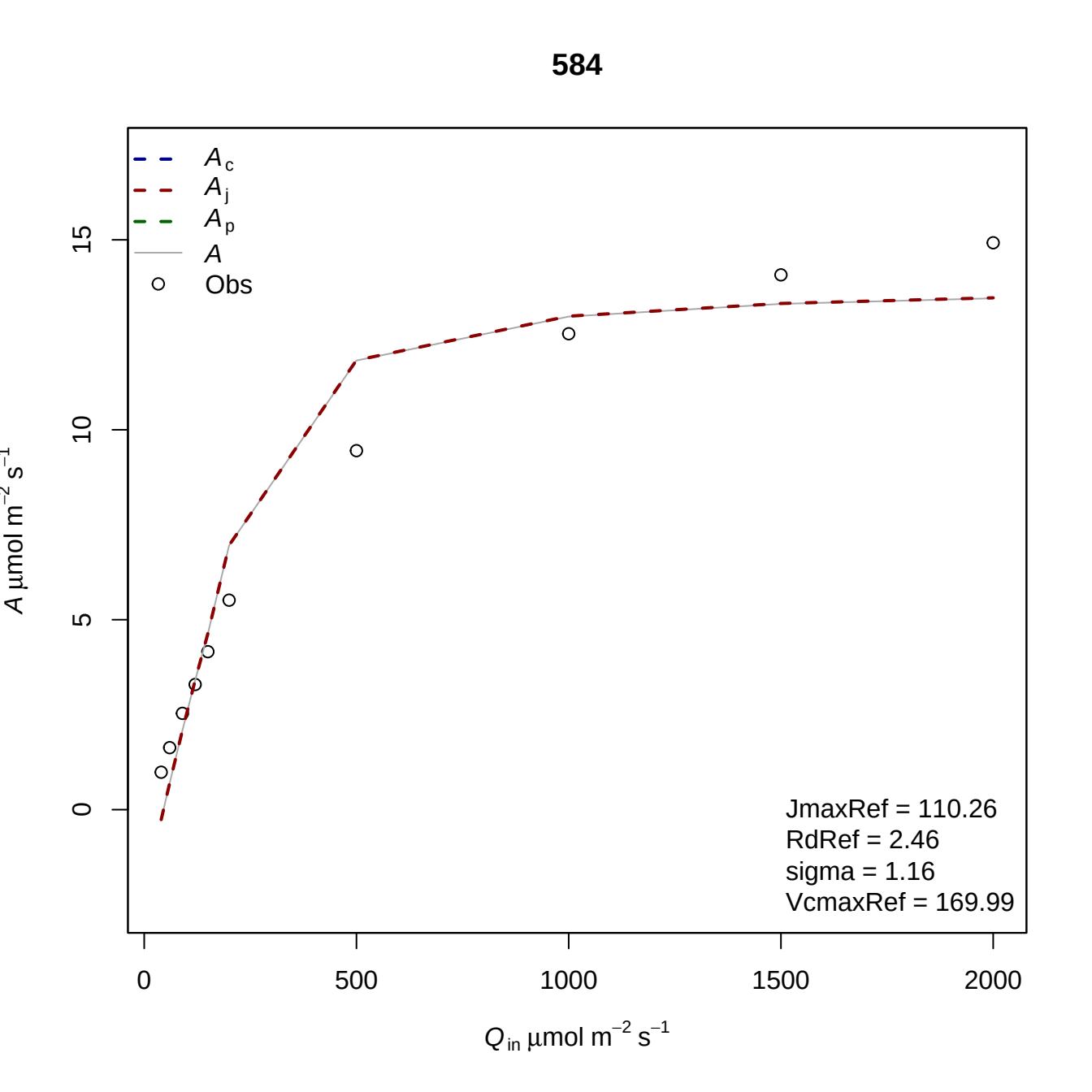

578
A_c
A_j
A_p
A
Obs
JmaxRef = 106.44
RdRef = 2.5
sigma = 1.34
VcmaxRef = 196.25
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

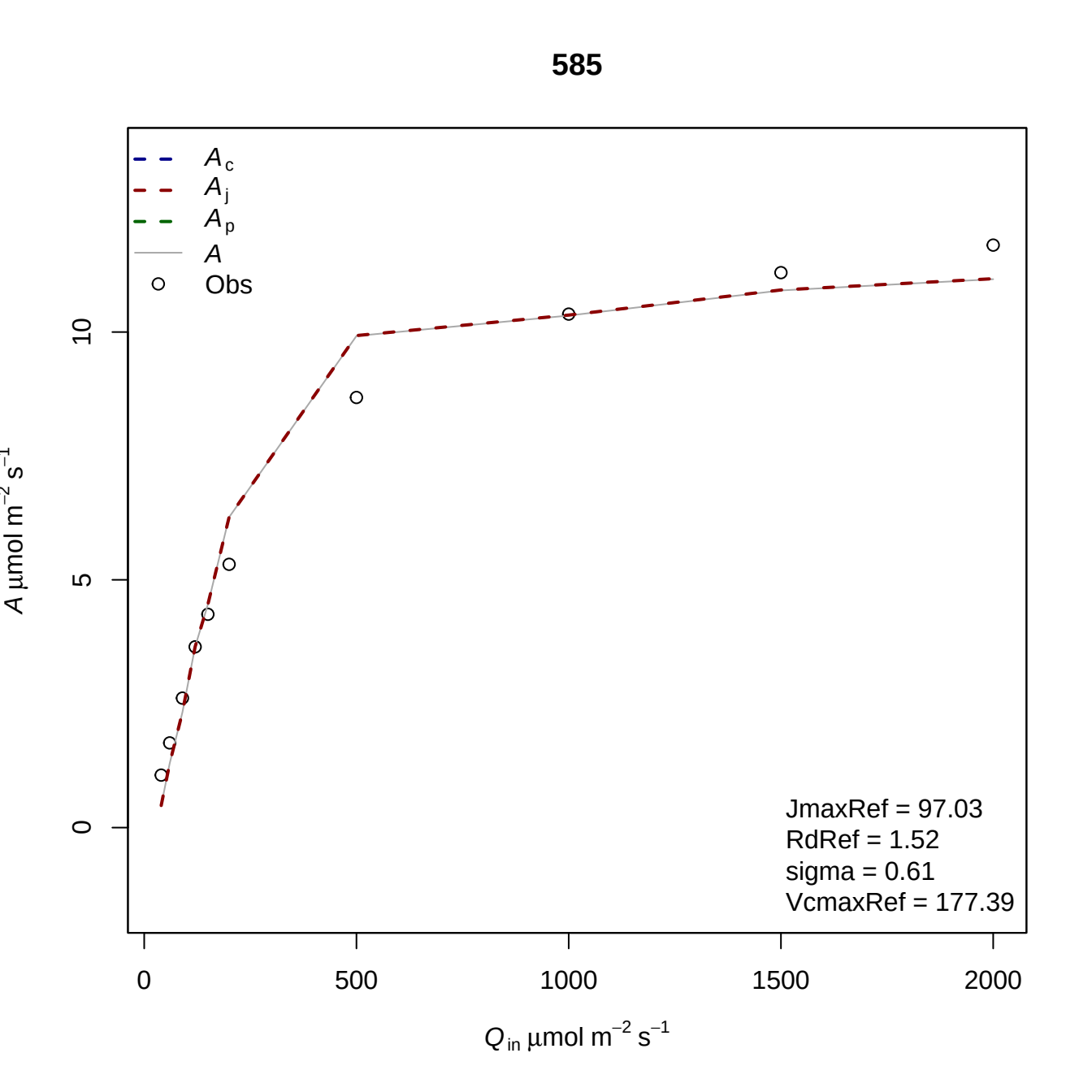

579
A_c
A_j
A_p
A
Obs
JmaxRef = 107.47
RdRef = 2.33
sigma = 0.84
VcmaxRef = 261.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

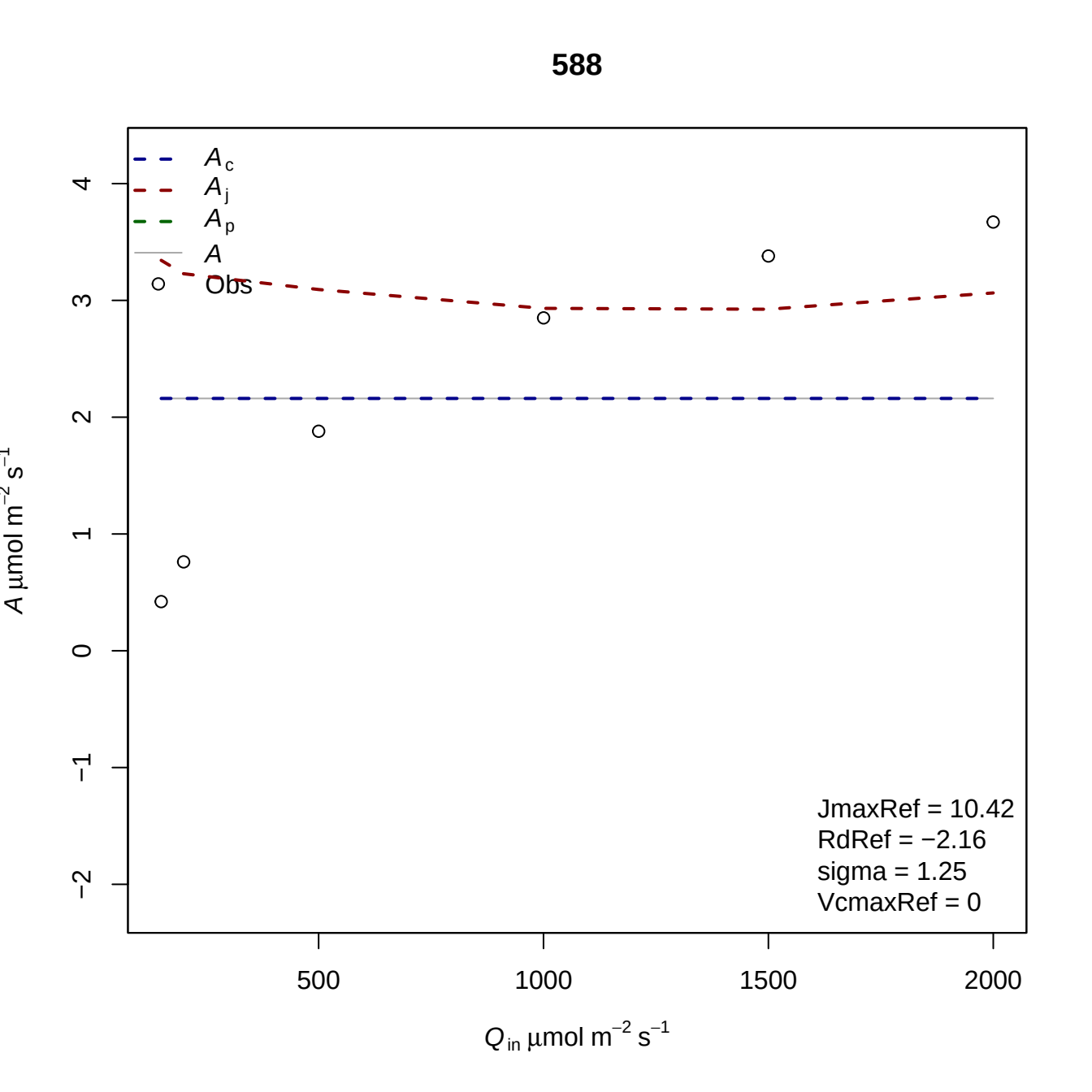

580
A_c
A_j
A_p
A
Obs
JmaxRef = 109.63
RdRef = 2.56
sigma = 0.65
VcmaxRef = 385.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

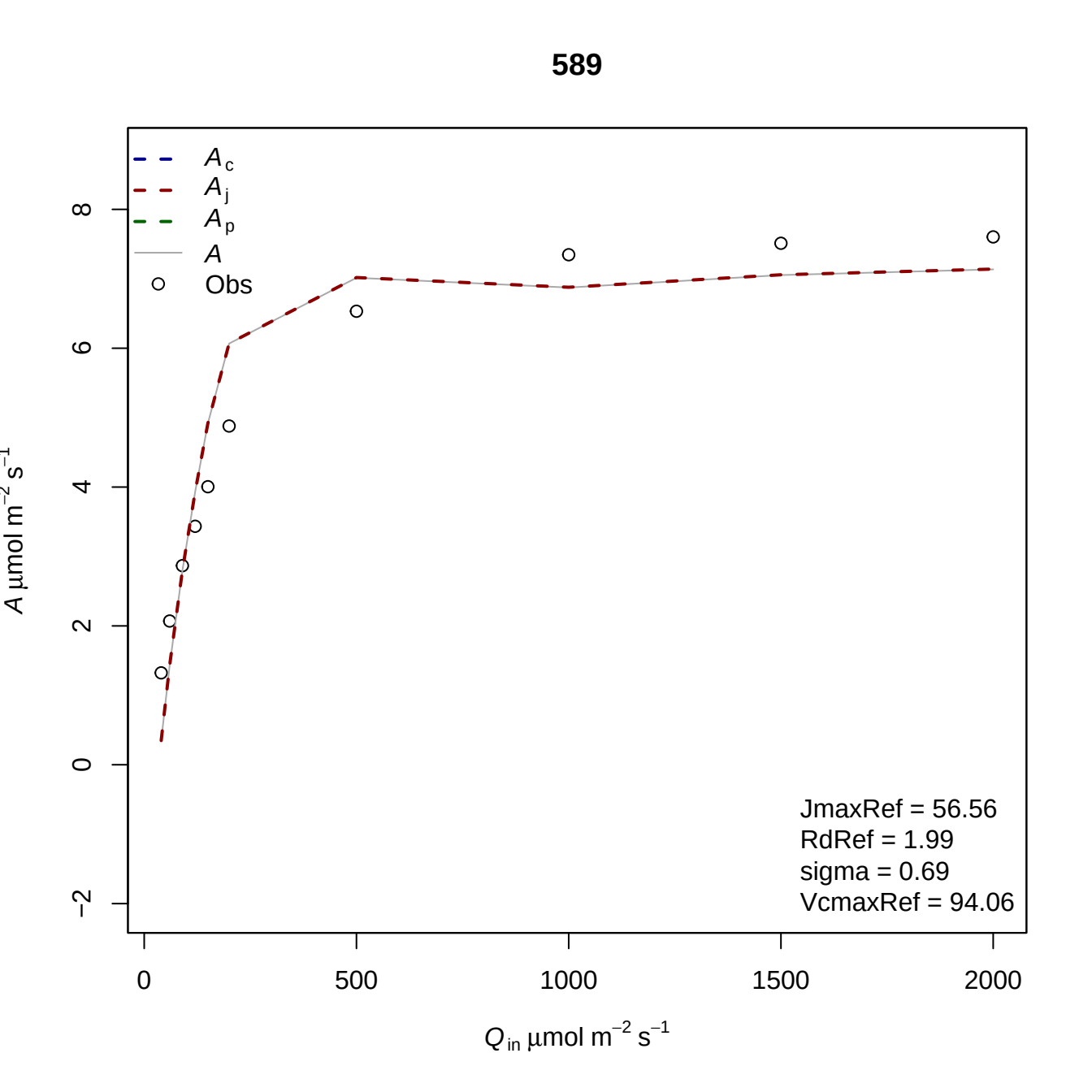

581
A_c
A_j
A_p
A
Obs
JmaxRef = 150.28
RdRef = 3.62
sigma = 0.9
VcmaxRef = 4856.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
500
1000
1500
2000

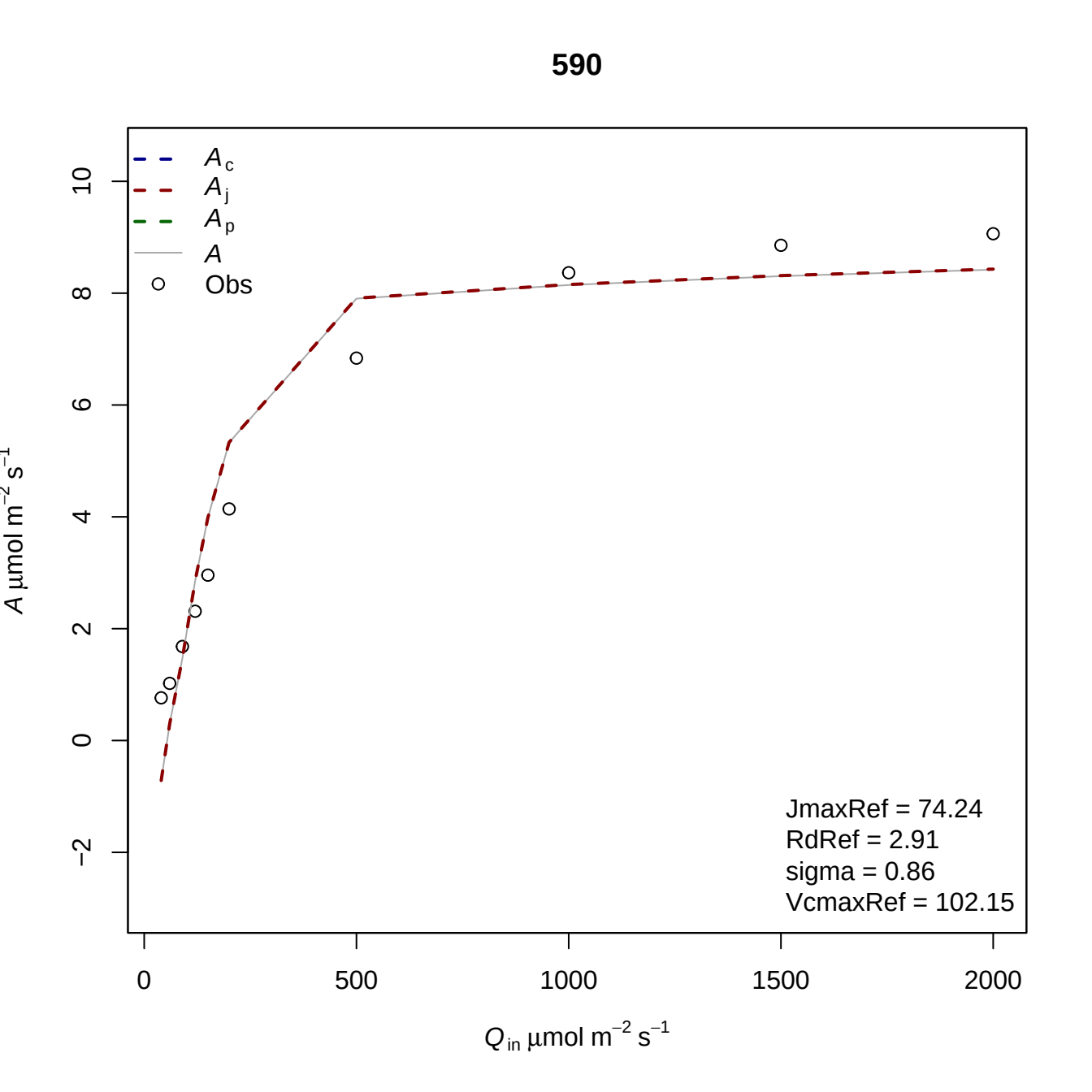

583
A_c
A_j
A_p
A
Obs
JmaxRef = 91.75
RdRef = 1.32
sigma = 0.9
VcmaxRef = 166.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

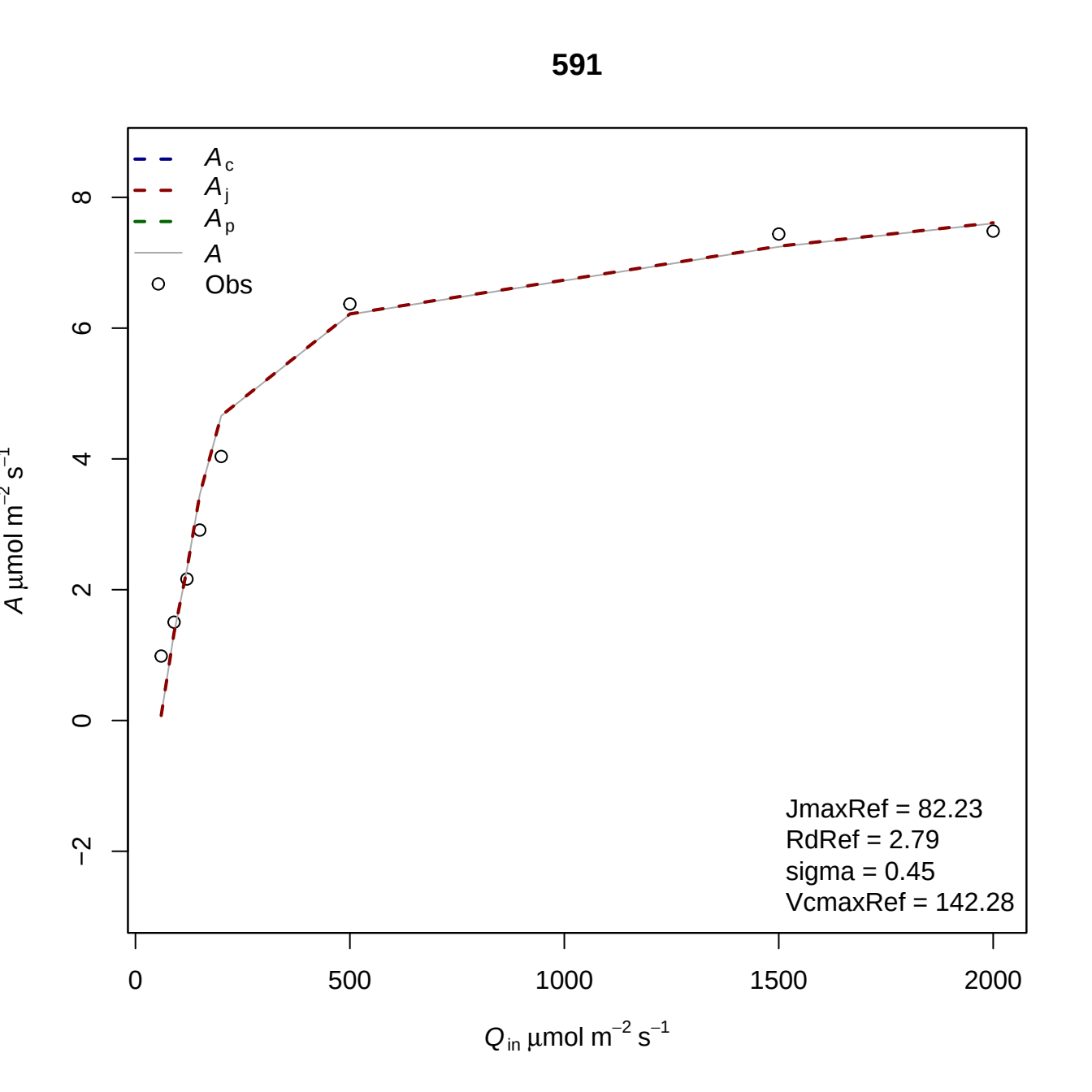

584
A_c
A_j
A_p
A
Obs
JmaxRef = 110.26
RdRef = 2.46
sigma = 1.16
VcmaxRef = 169.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

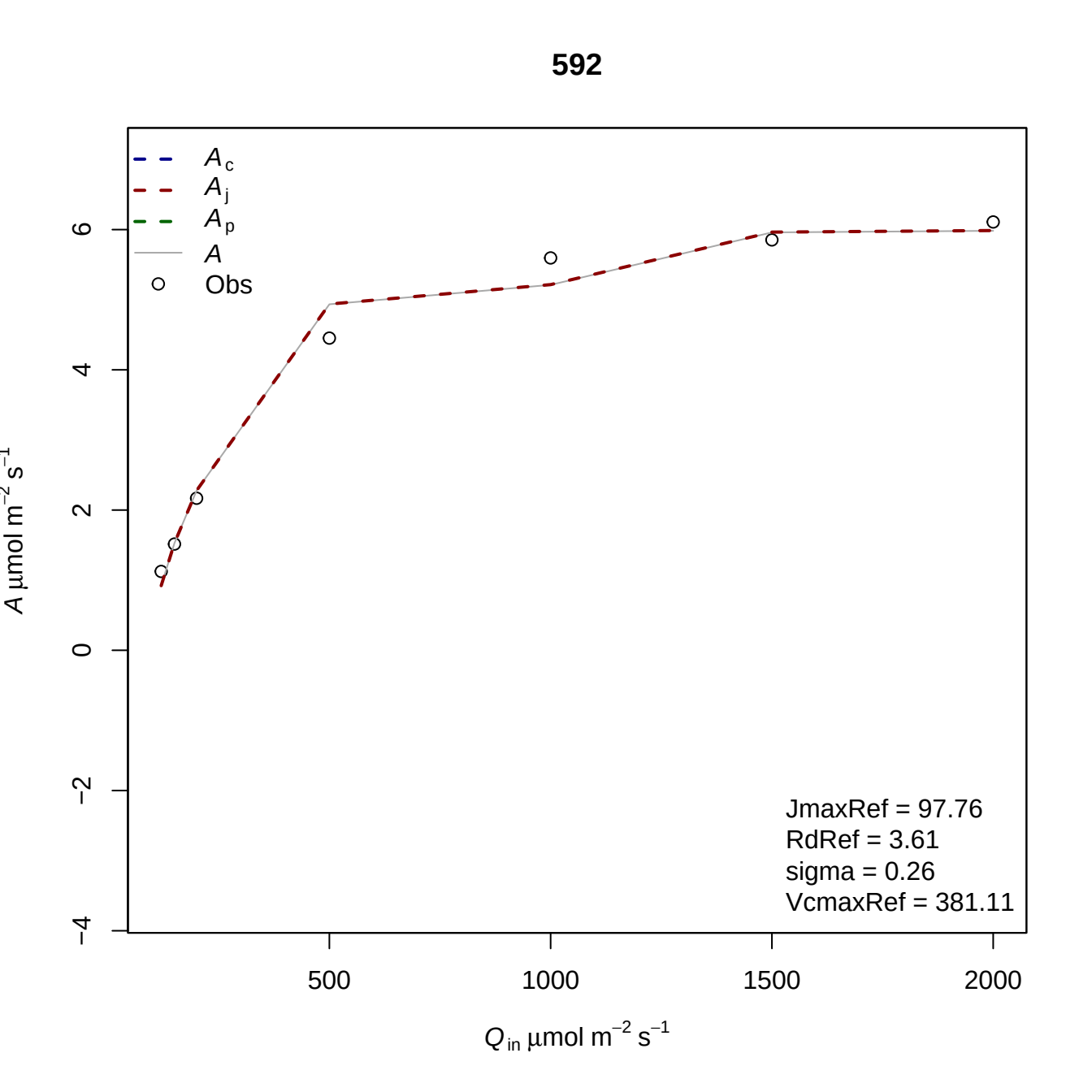

585
A_c
A_j
A_p
A
Obs
JmaxRef = 97.03
RdRef = 1.52
sigma = 0.61
VcmaxRef = 177.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

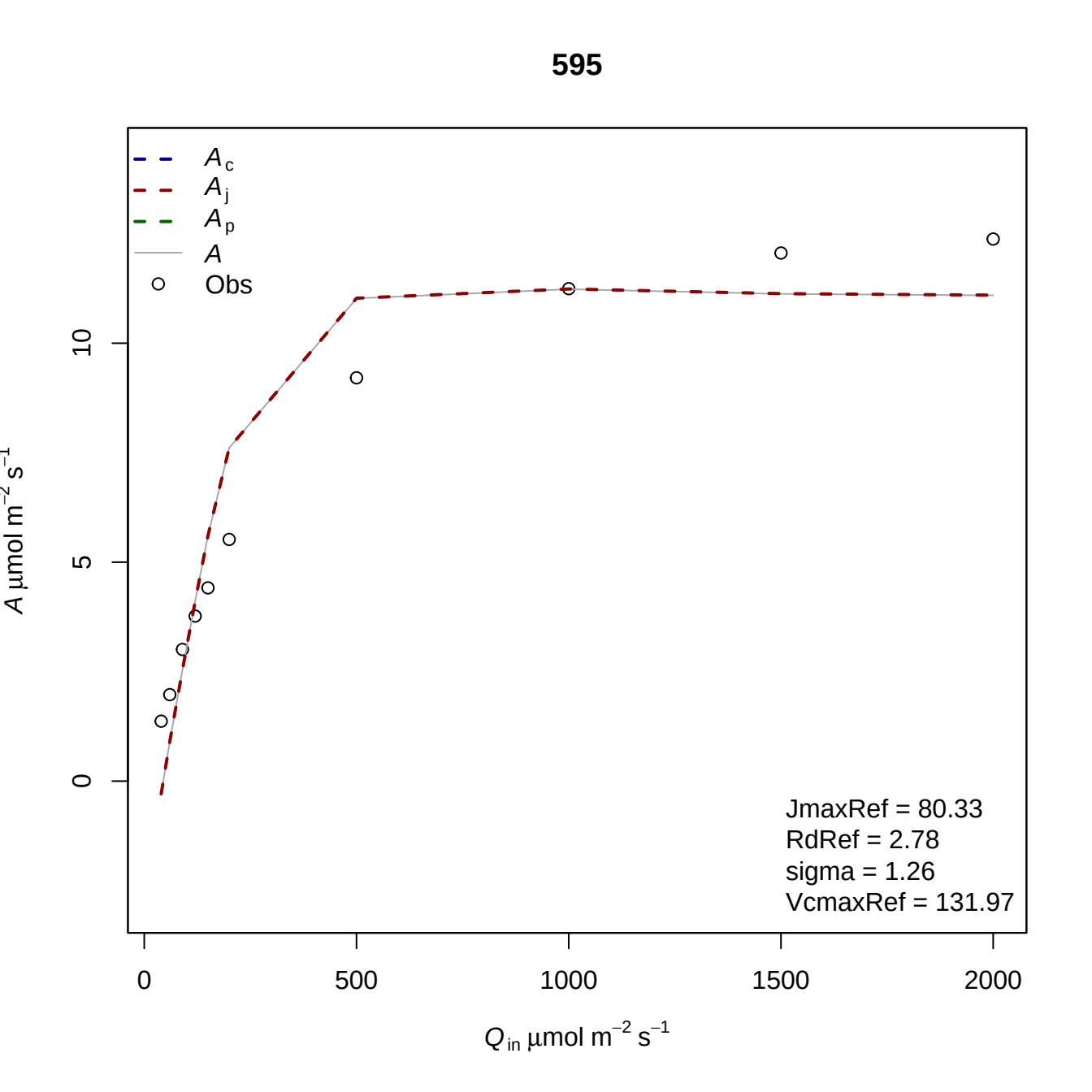

588
A_c
A_j
A_p
A
Obs
JmaxRef = 10.42
RdRef = −2.16
sigma = 1.25
VcmaxRef = 0
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
500
1000
1500
2000

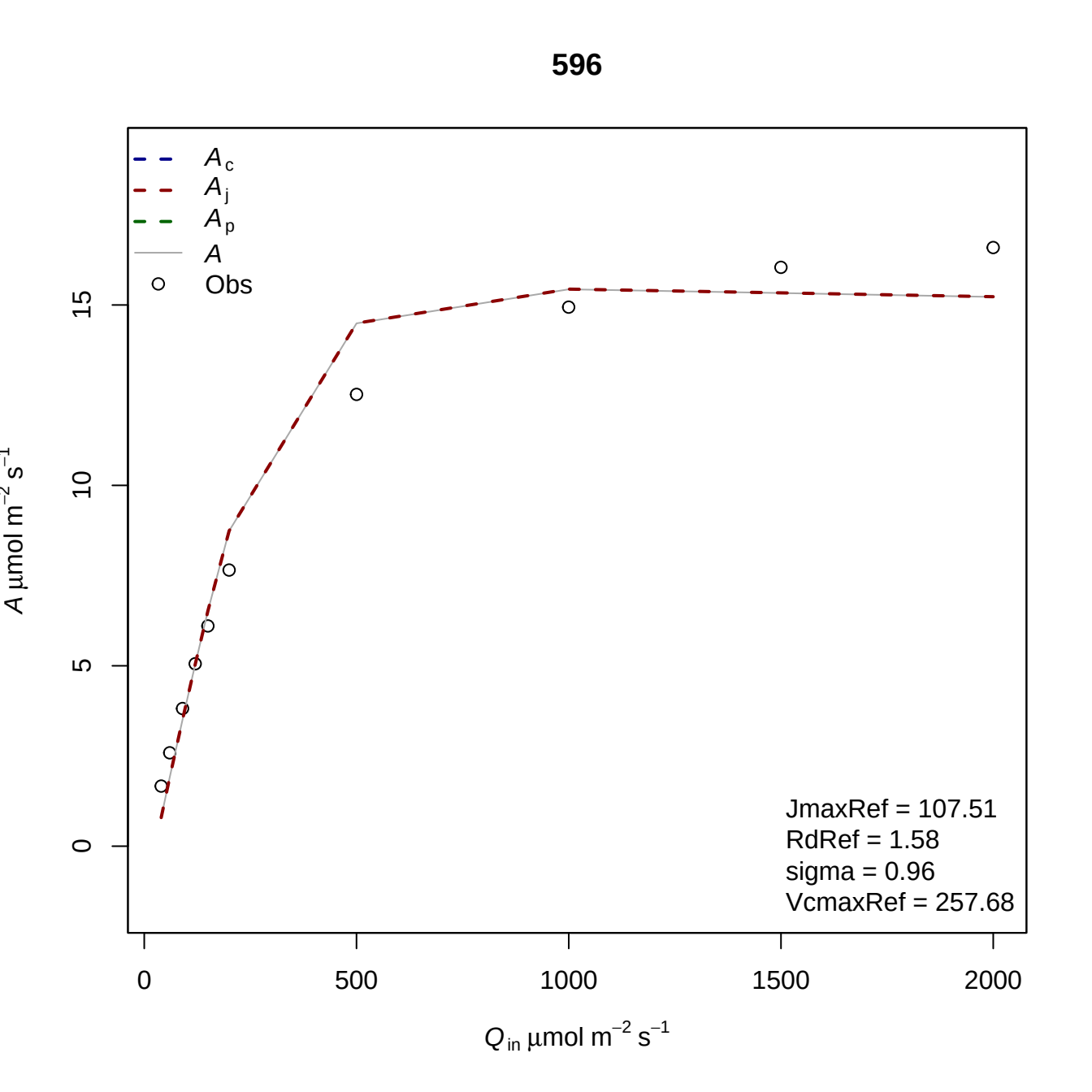

589
A_c
A_j
A_p
A
Obs
JmaxRef = 56.56
RdRef = 1.99
sigma = 0.69
VcmaxRef = 94.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

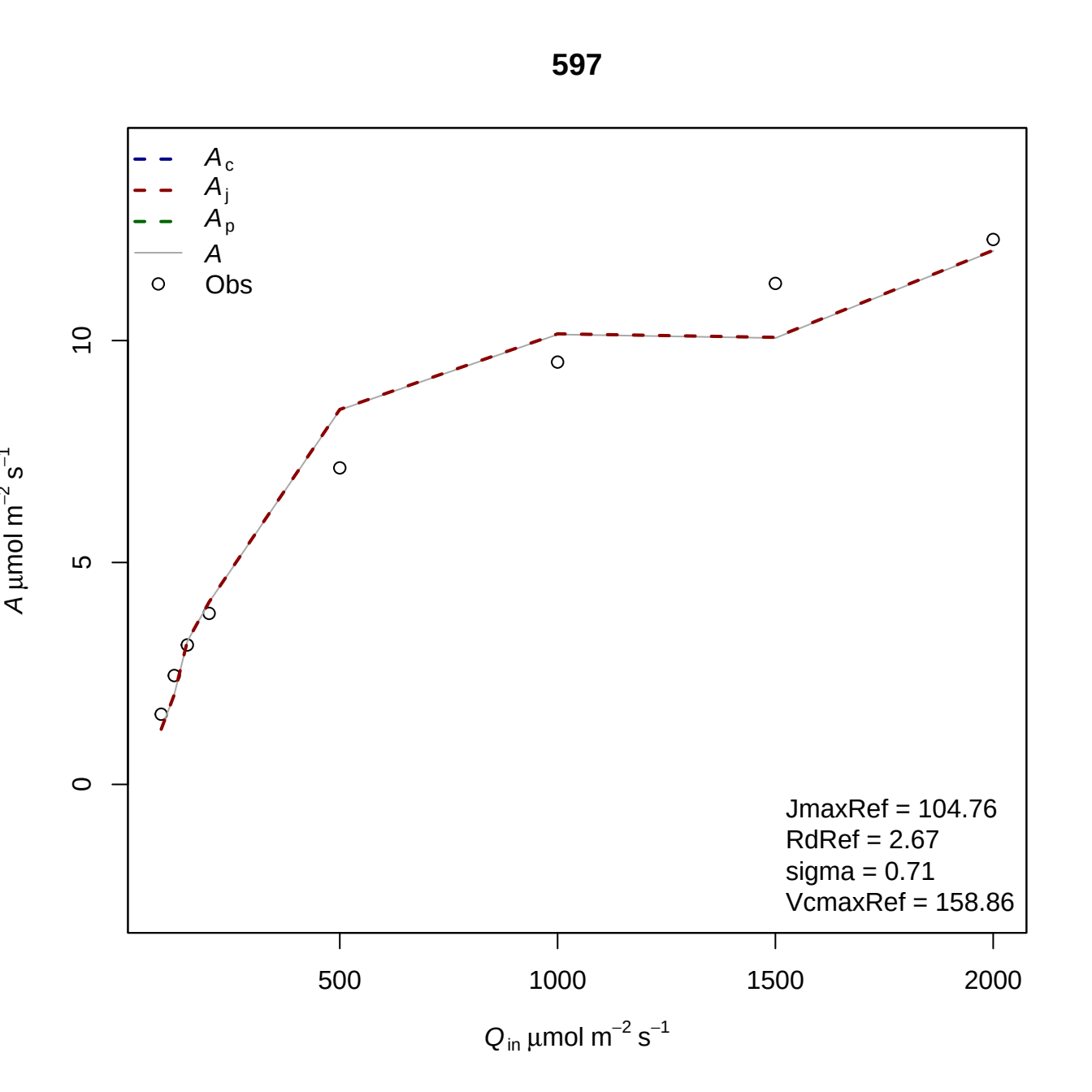

590
A_c
A_j
A_p
A
Obs
JmaxRef = 74.24
RdRef = 2.91
sigma = 0.86
VcmaxRef = 102.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

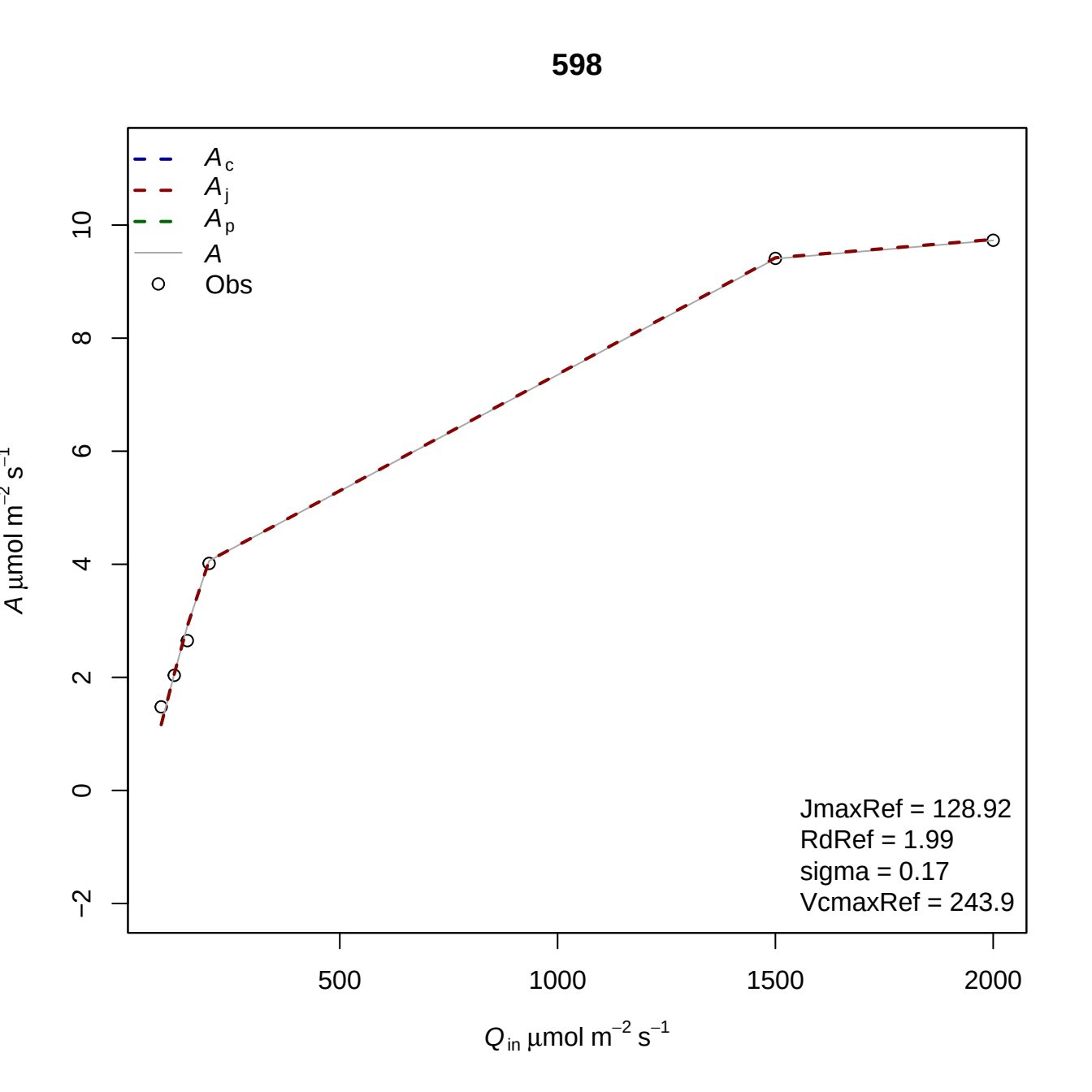

591
A_c
A_j
A_p
A
Obs
JmaxRef = 82.23
RdRef = 2.79
sigma = 0.45
VcmaxRef = 142.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

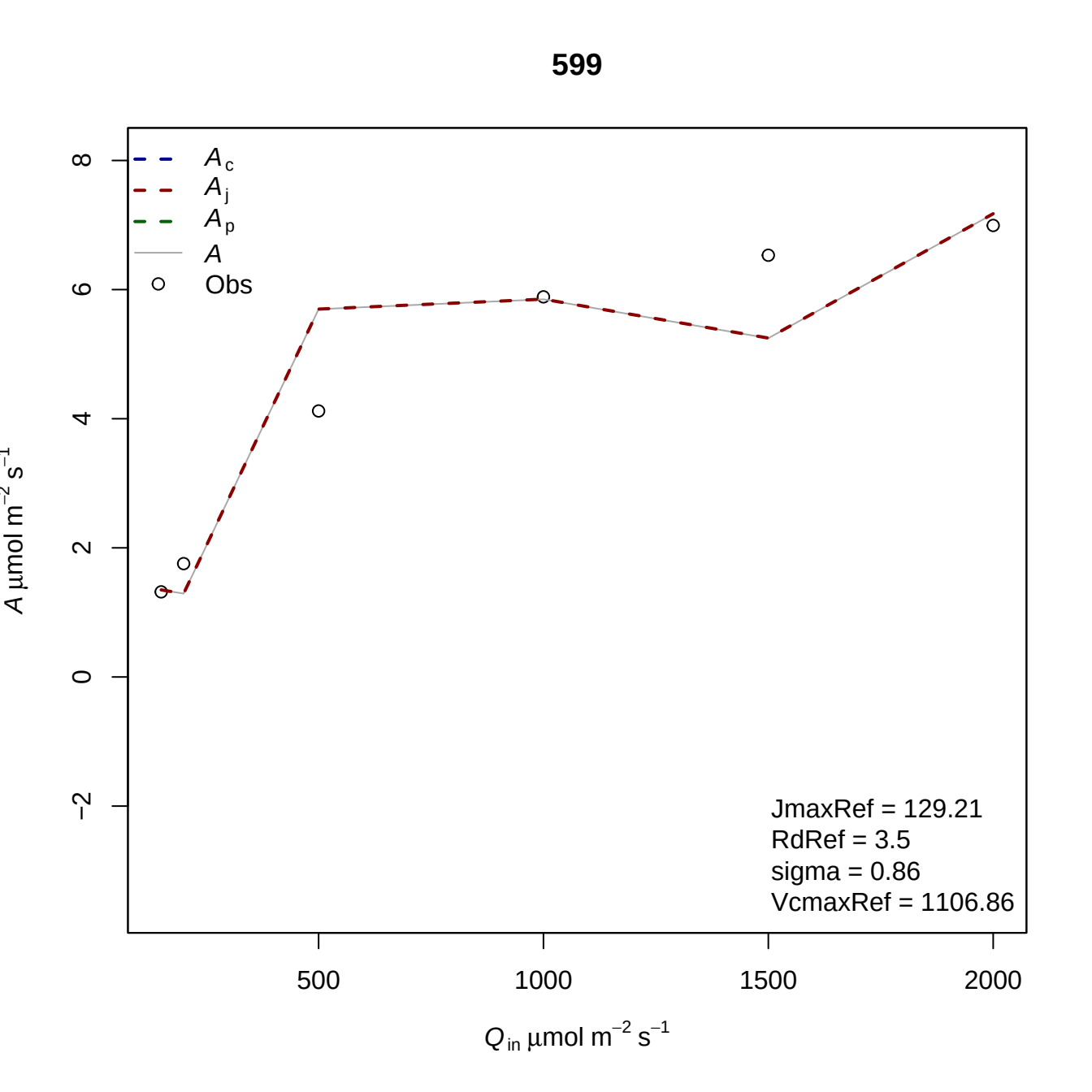

592
A_c
A_j
A_p
A
Obs
JmaxRef = 97.76
RdRef = 3.61
sigma = 0.26
VcmaxRef = 381.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
500
1000
1500
2000

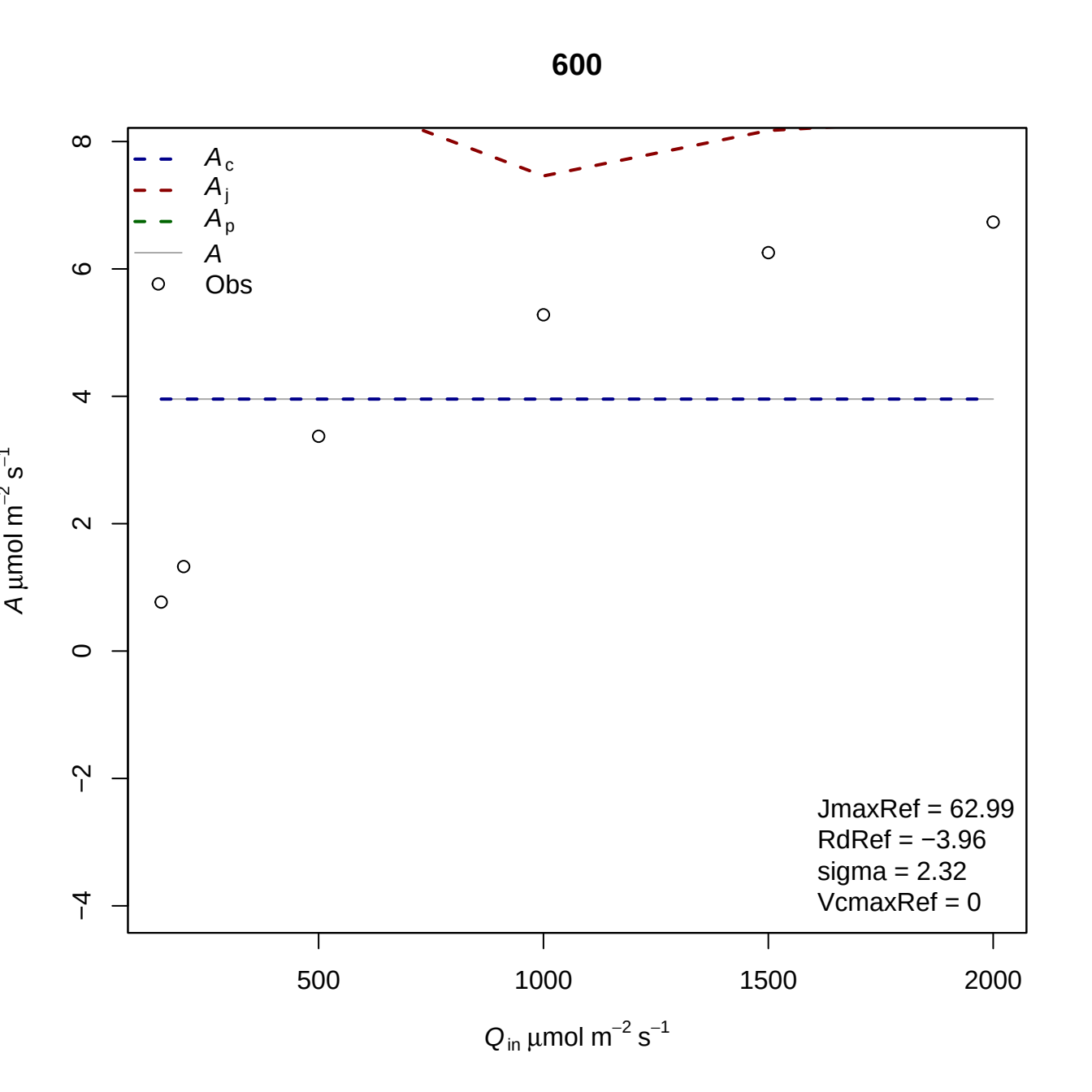

595
A_c
A_j
A_p
A
Obs
JmaxRef = 80.33
RdRef = 2.78
sigma = 1.26
VcmaxRef = 131.97
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

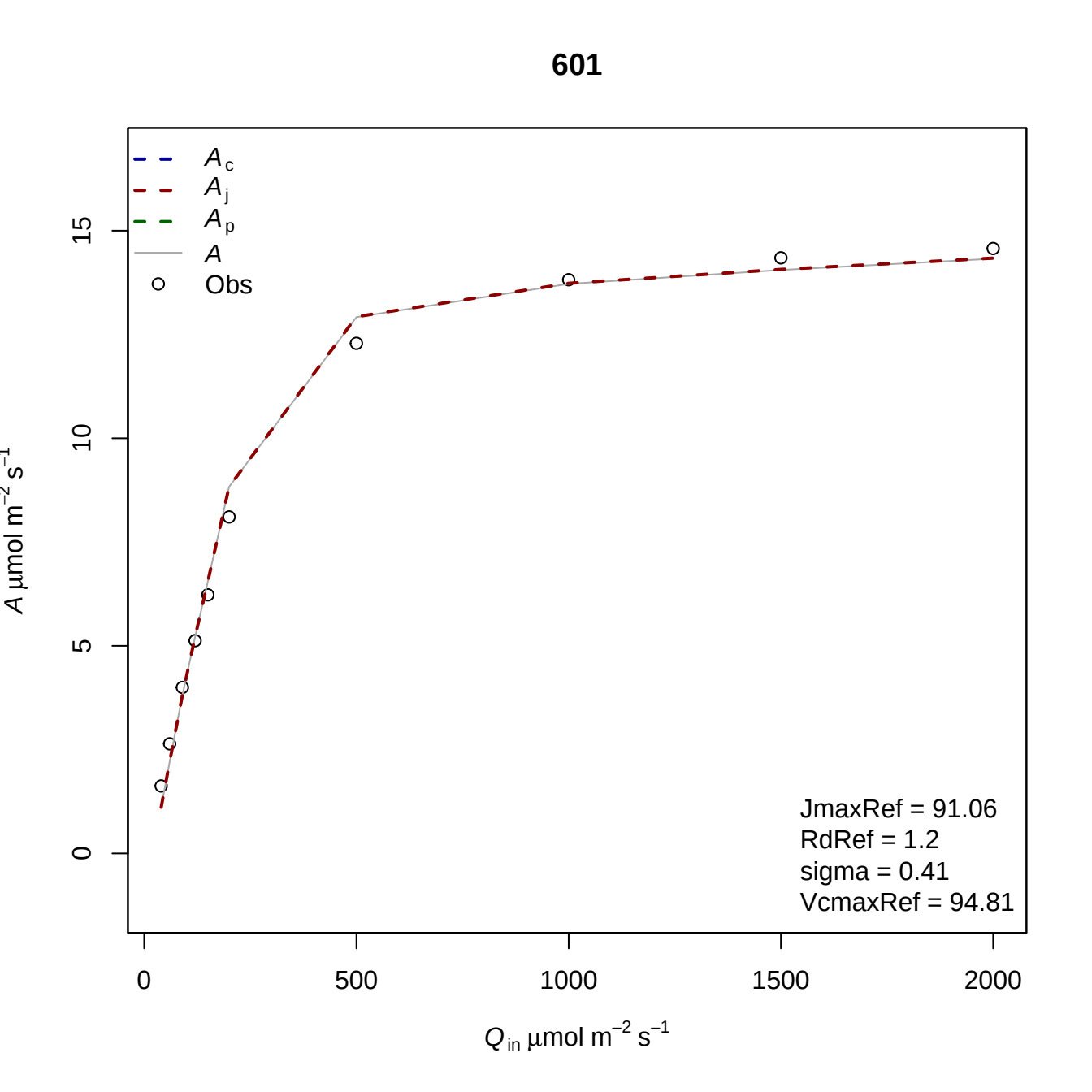

596
A_c
A_j
A_p
A
Obs
JmaxRef = 107.51
RdRef = 1.58
sigma = 0.96
VcmaxRef = 257.68
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

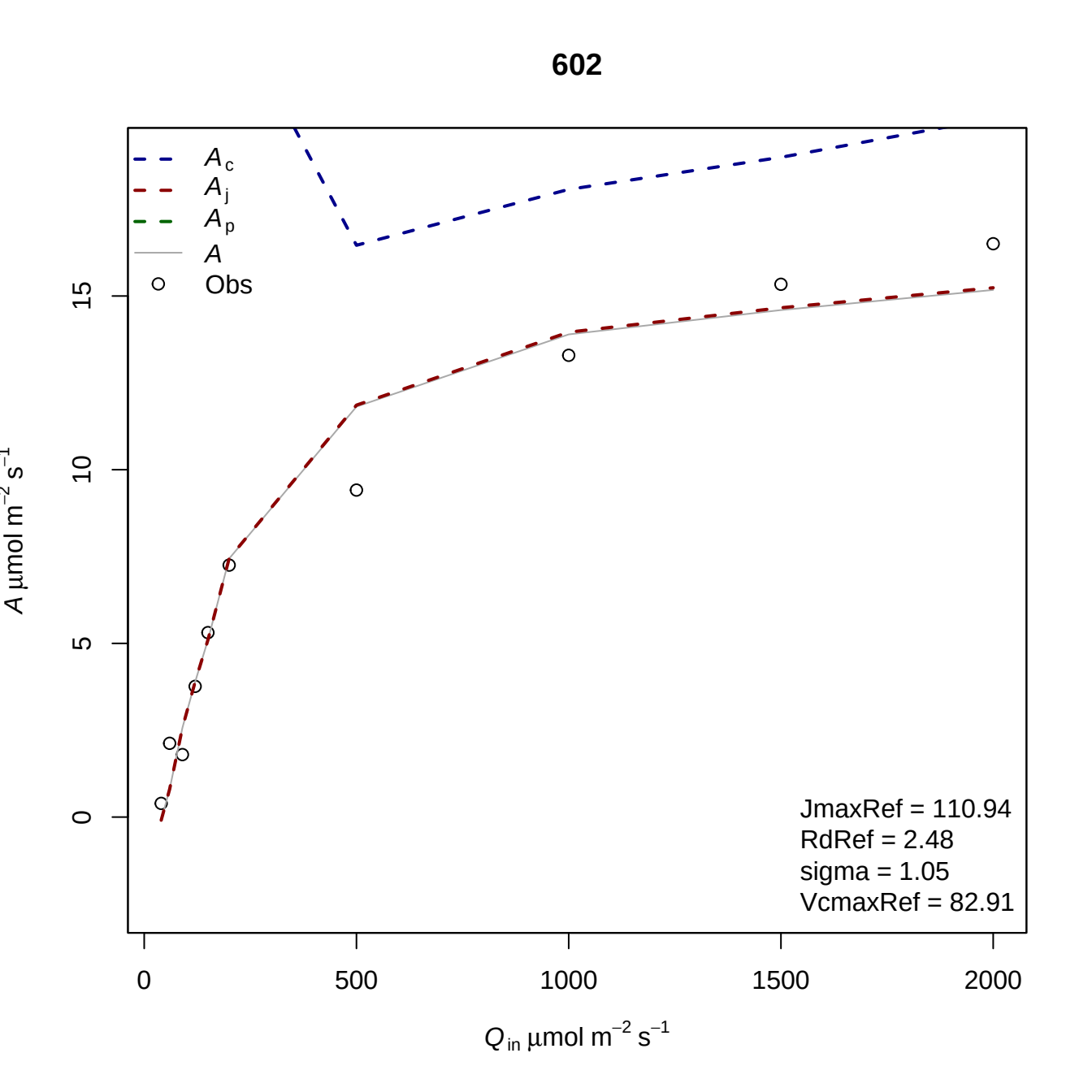

597
A_c
A_j
A_p
A
Obs
JmaxRef = 104.76
RdRef = 2.67
sigma = 0.71
VcmaxRef = 158.86
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

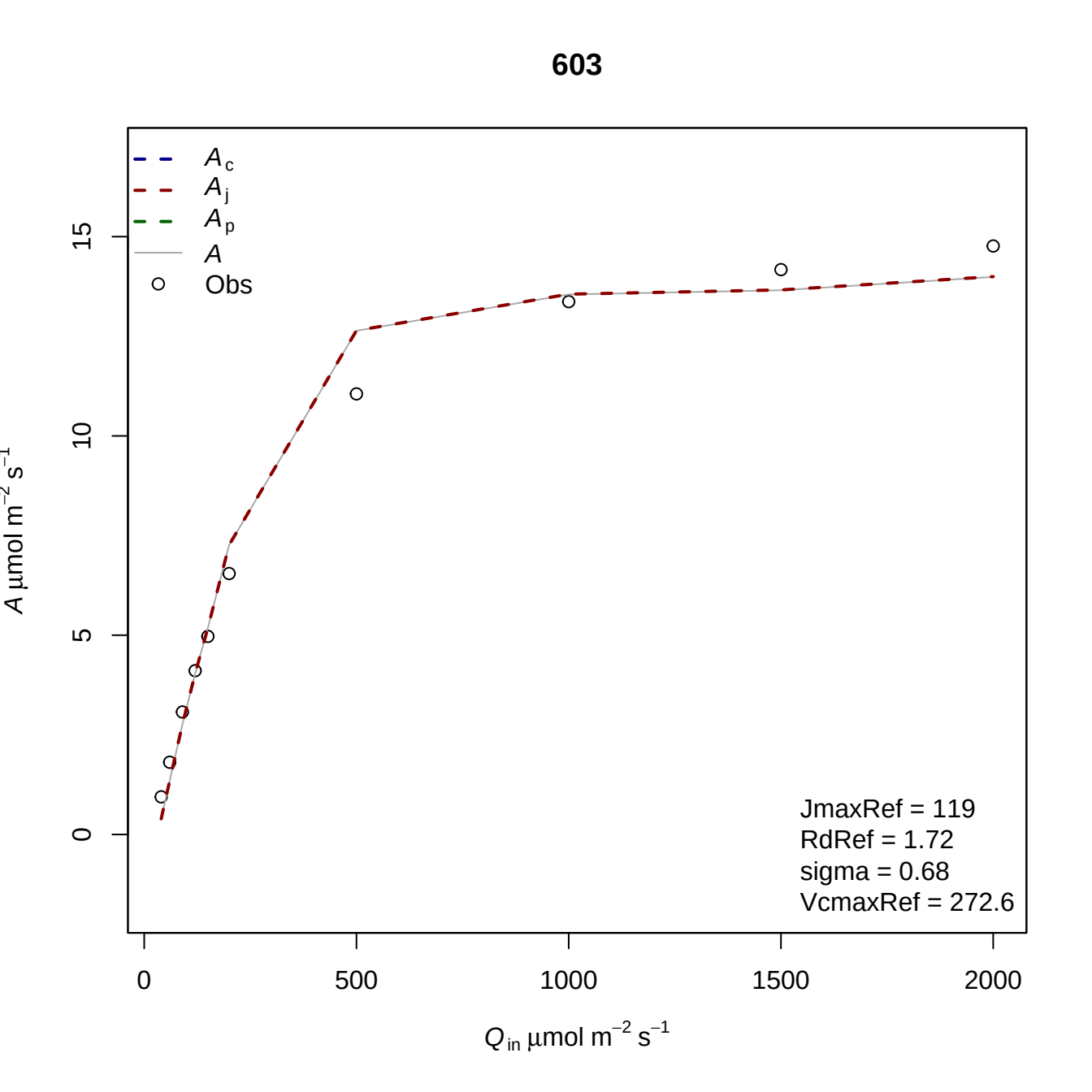

598
A_c
A_j
A_p
A
Obs
JmaxRef = 128.92
RdRef = 1.99
sigma = 0.17
VcmaxRef = 243.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

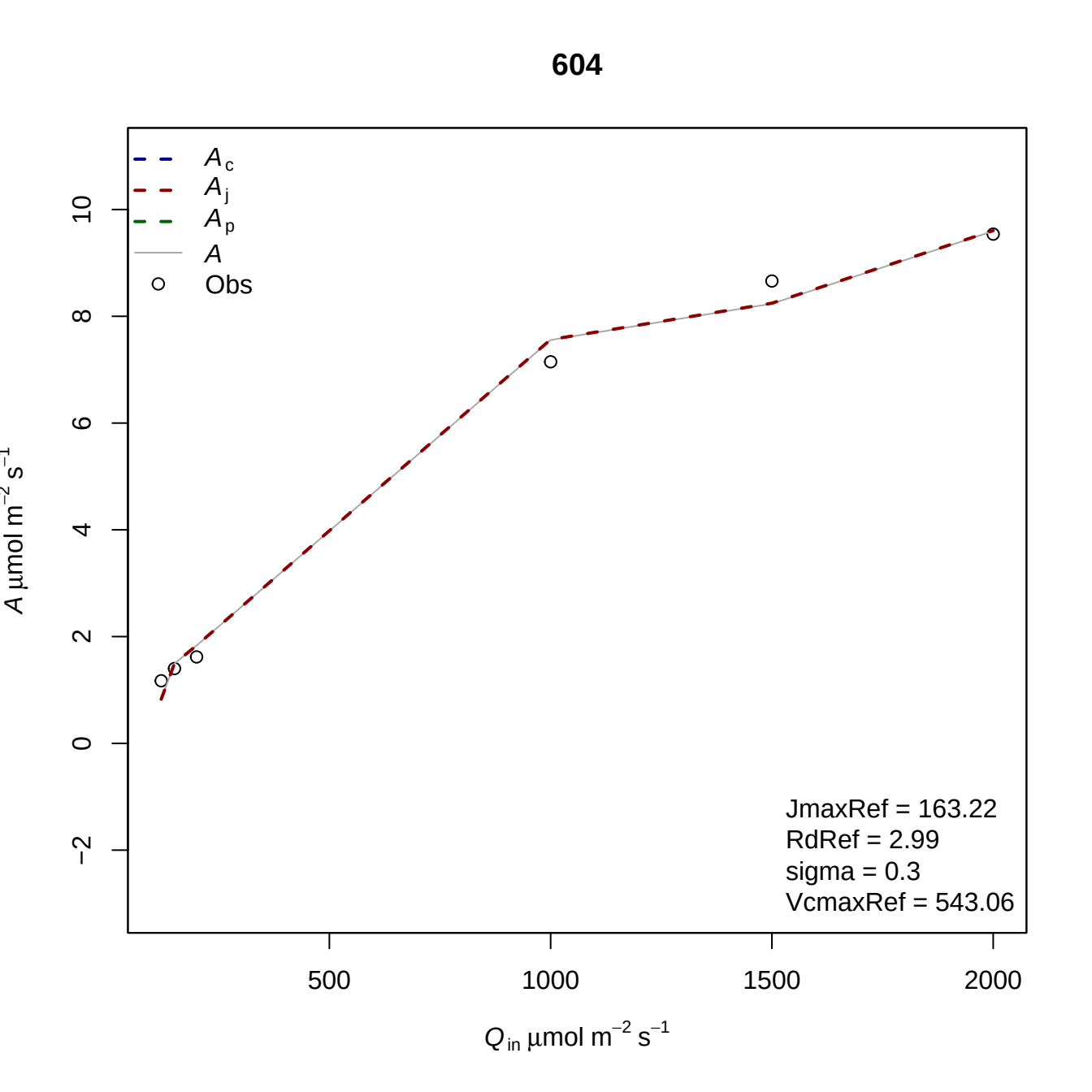

599
A_c
A_j
A_p
A
Obs
8
6
4
2
0
−2
A µmol m^{-2} s^{-1}
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 129.21
RdRef = 3.5
sigma = 0.86
VcmaxRef = 1106.86

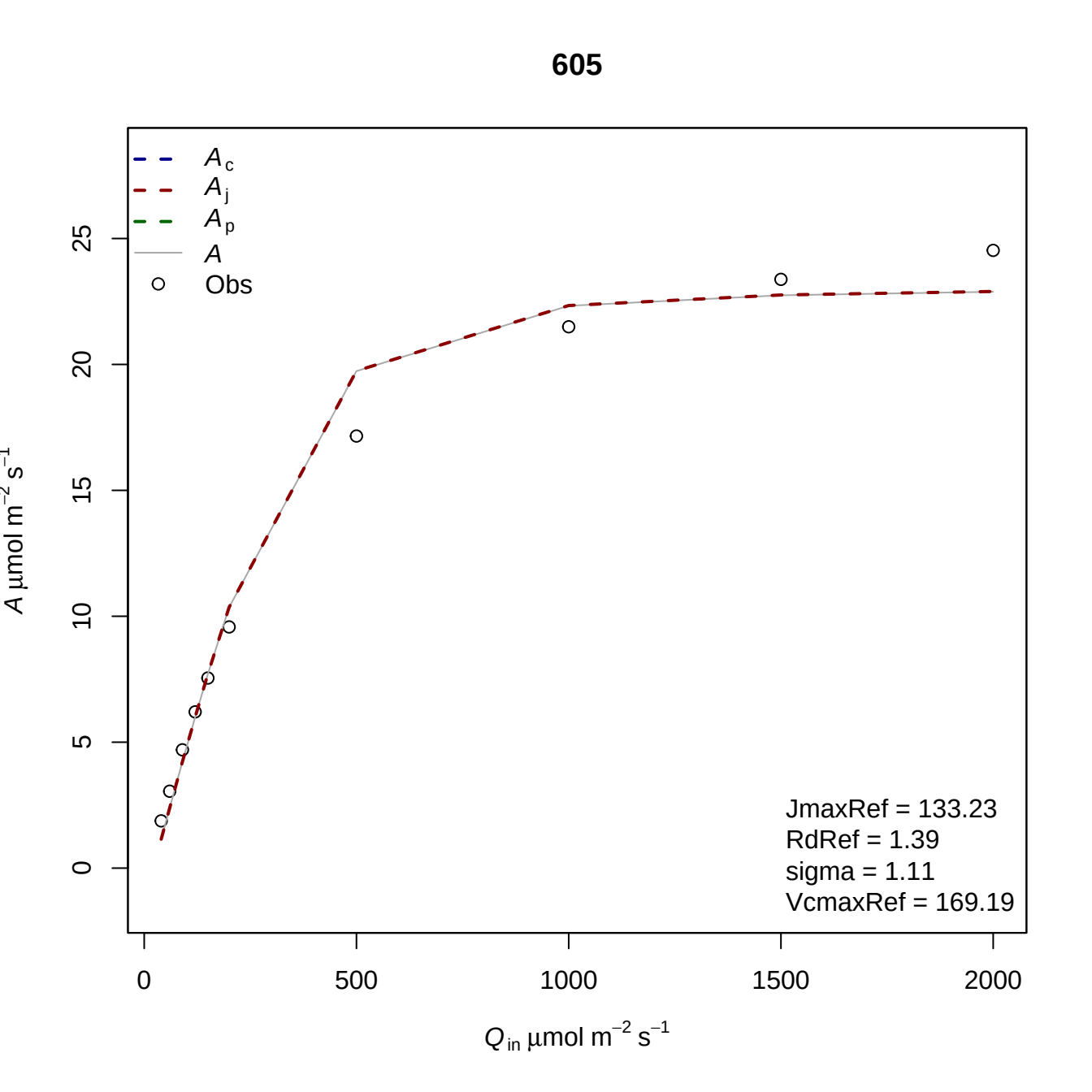

600
A_c
A_j
A_p
A
Obs
JmaxRef = 62.99
RdRef = −3.96
sigma = 2.32
VcmaxRef = 0
8
6
4
2
0
−2
−4
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

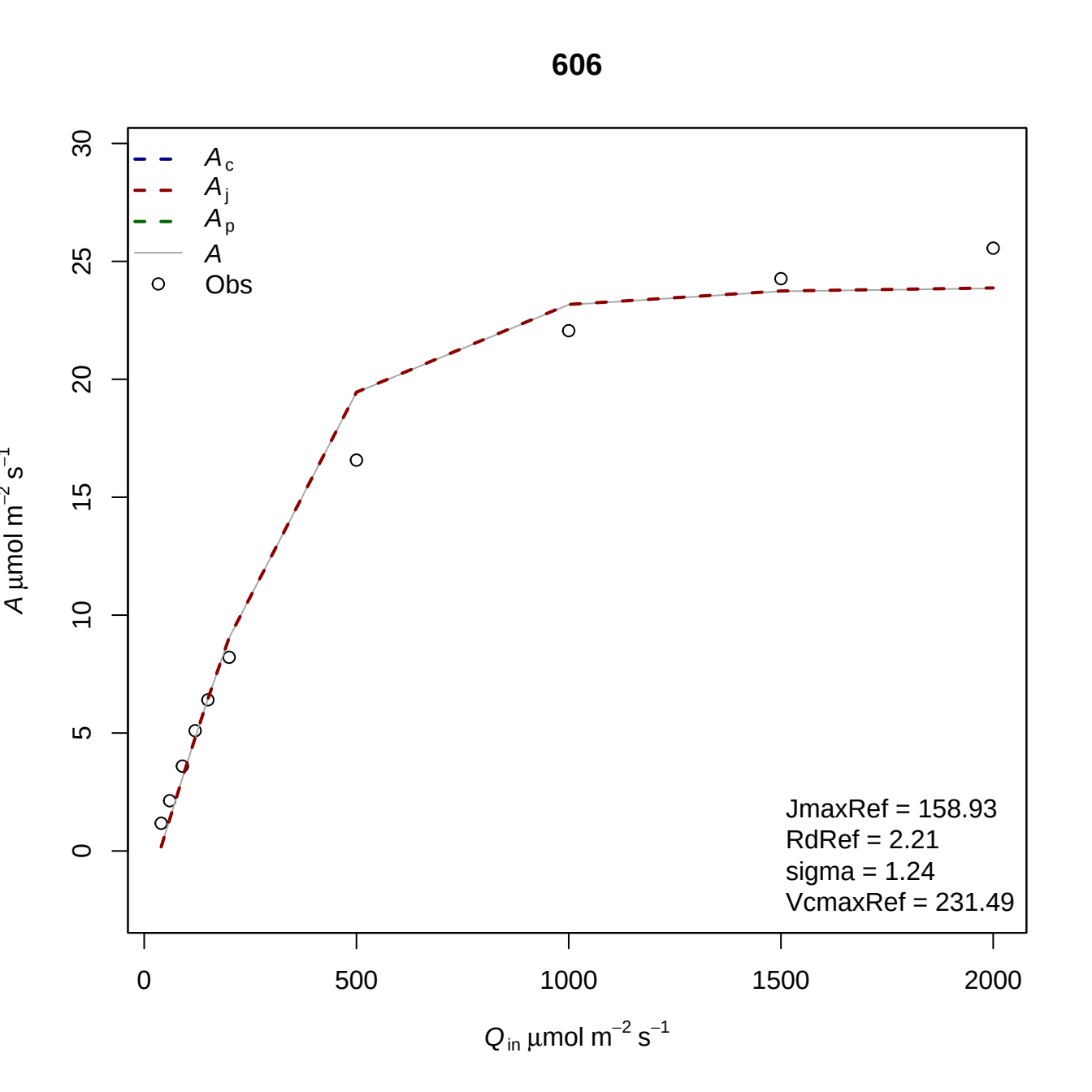

601
A_c
A_j
A_p
A
Obs
JmaxRef = 91.06
RdRef = 1.2
sigma = 0.41
VcmaxRef = 94.81
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

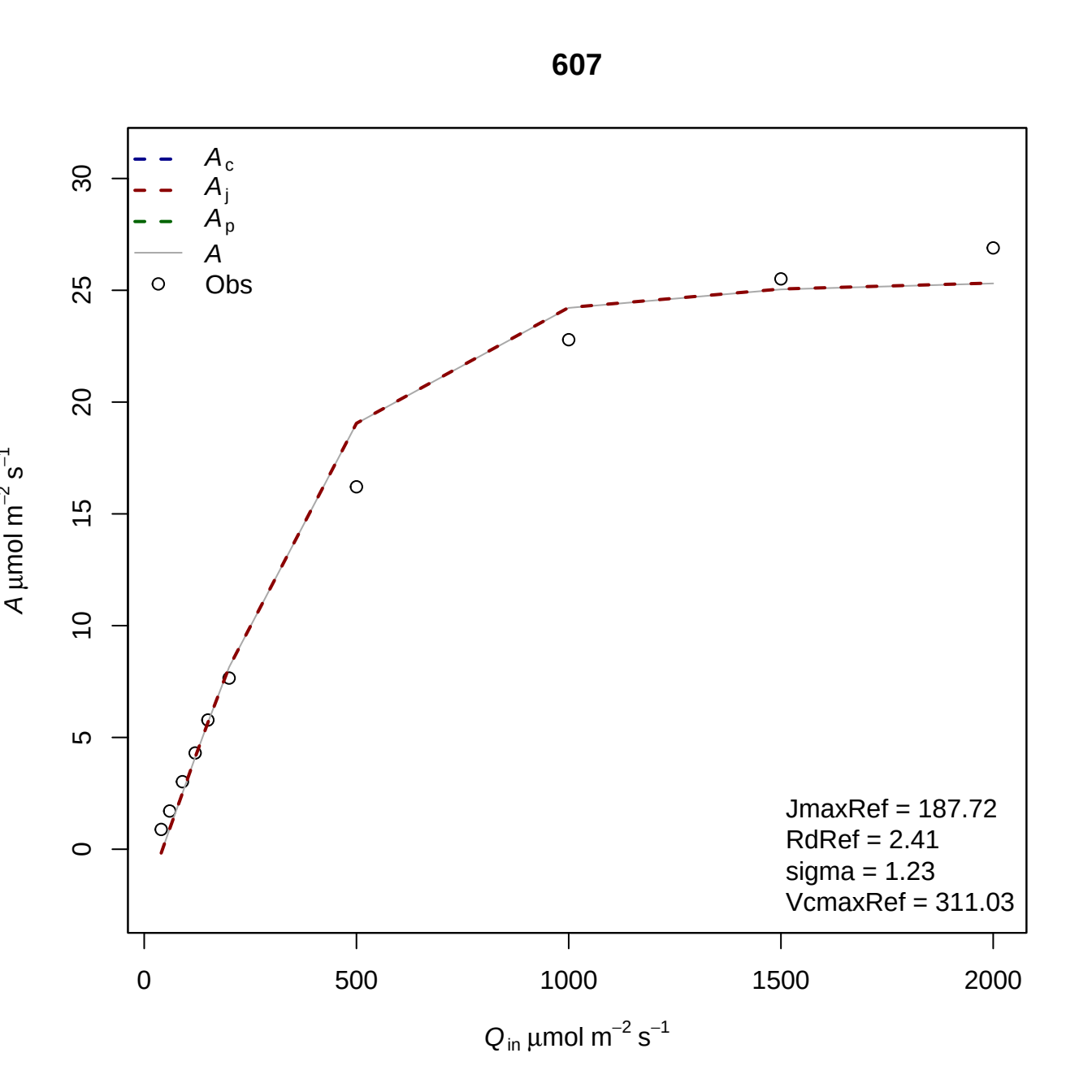

602
A_c
A_j
A_p
A
Obs
JmaxRef = 110.94
RdRef = 2.48
sigma = 1.05
VcmaxRef = 82.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

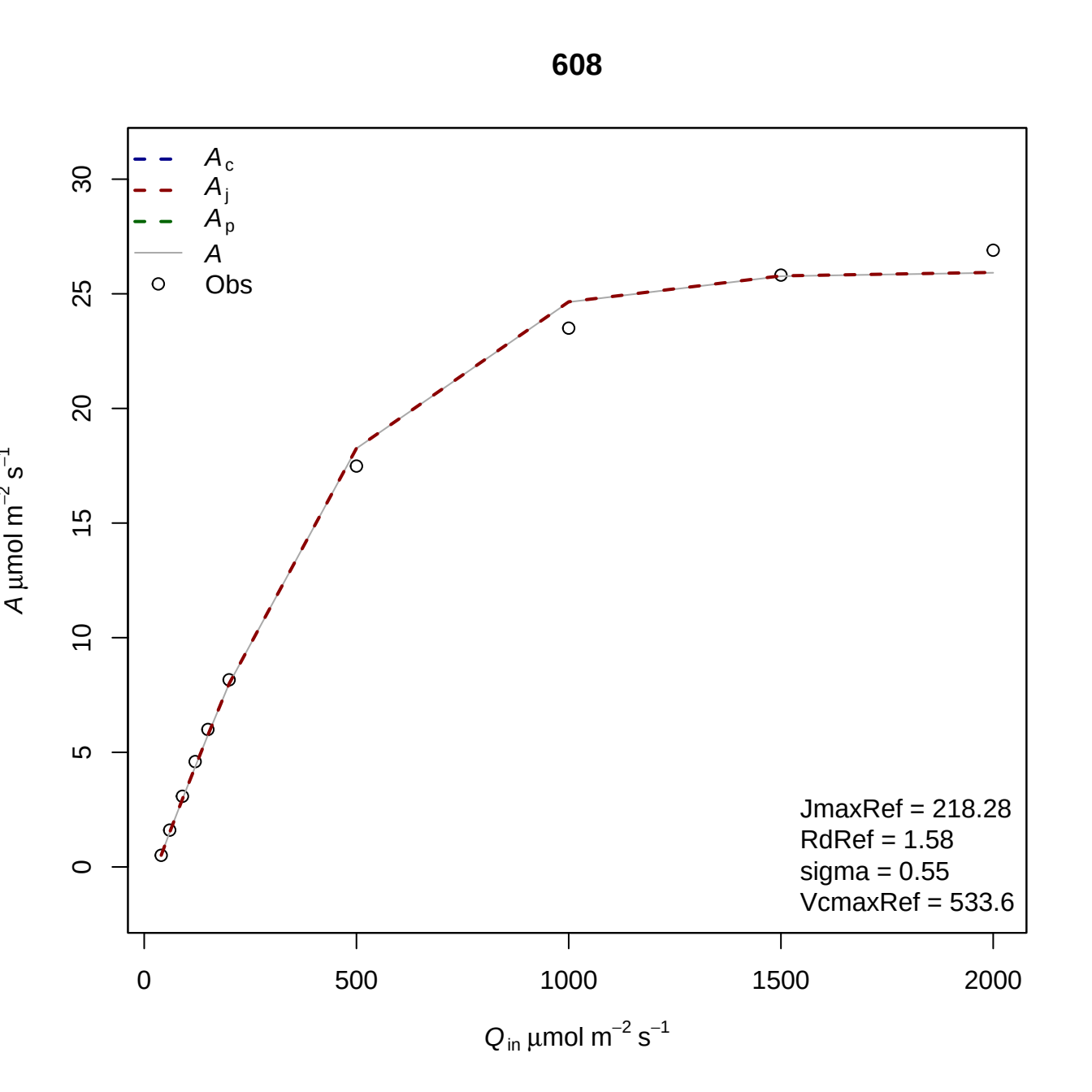

603
A_c
A_j
A_p
A
Obs
JmaxRef = 119
RdRef = 1.72
sigma = 0.68
VcmaxRef = 272.6
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

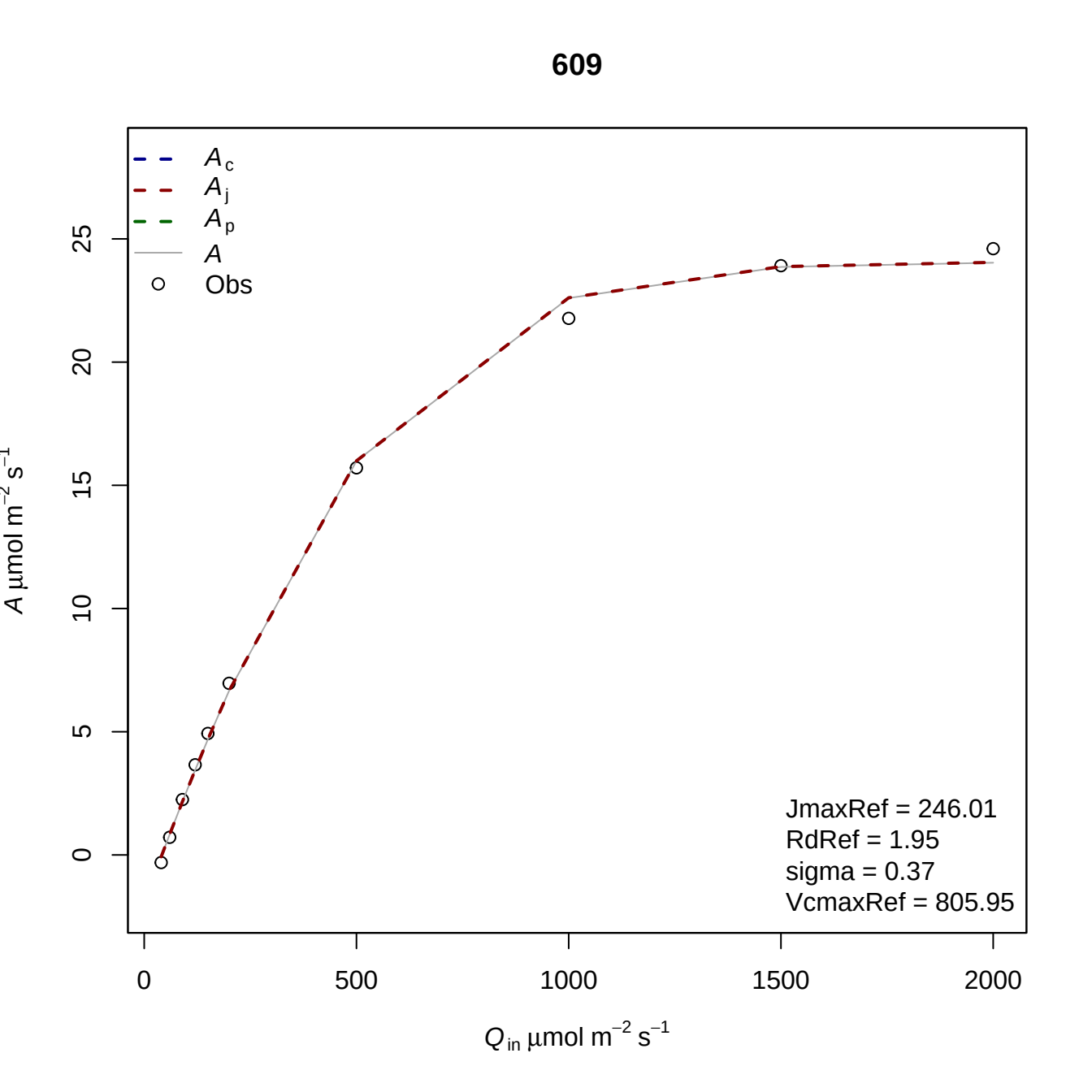

604
A_c
A_j
A_p
A
Obs
JmaxRef = 163.22
RdRef = 2.99
sigma = 0.3
VcmaxRef = 543.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

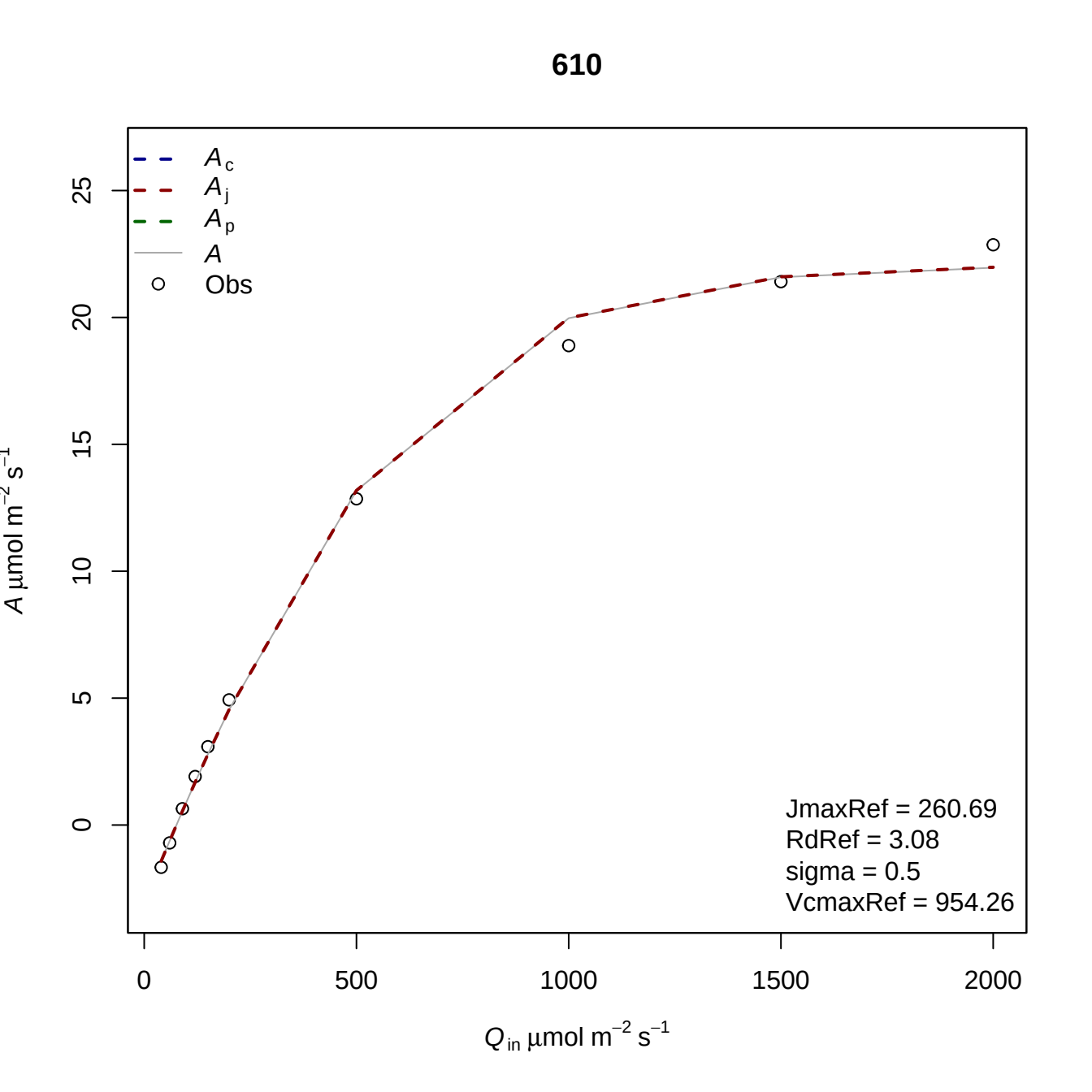

605
A_c
A_j
A_p
A
Obs
JmaxRef = 133.23
RdRef = 1.39
sigma = 1.11
VcmaxRef = 169.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

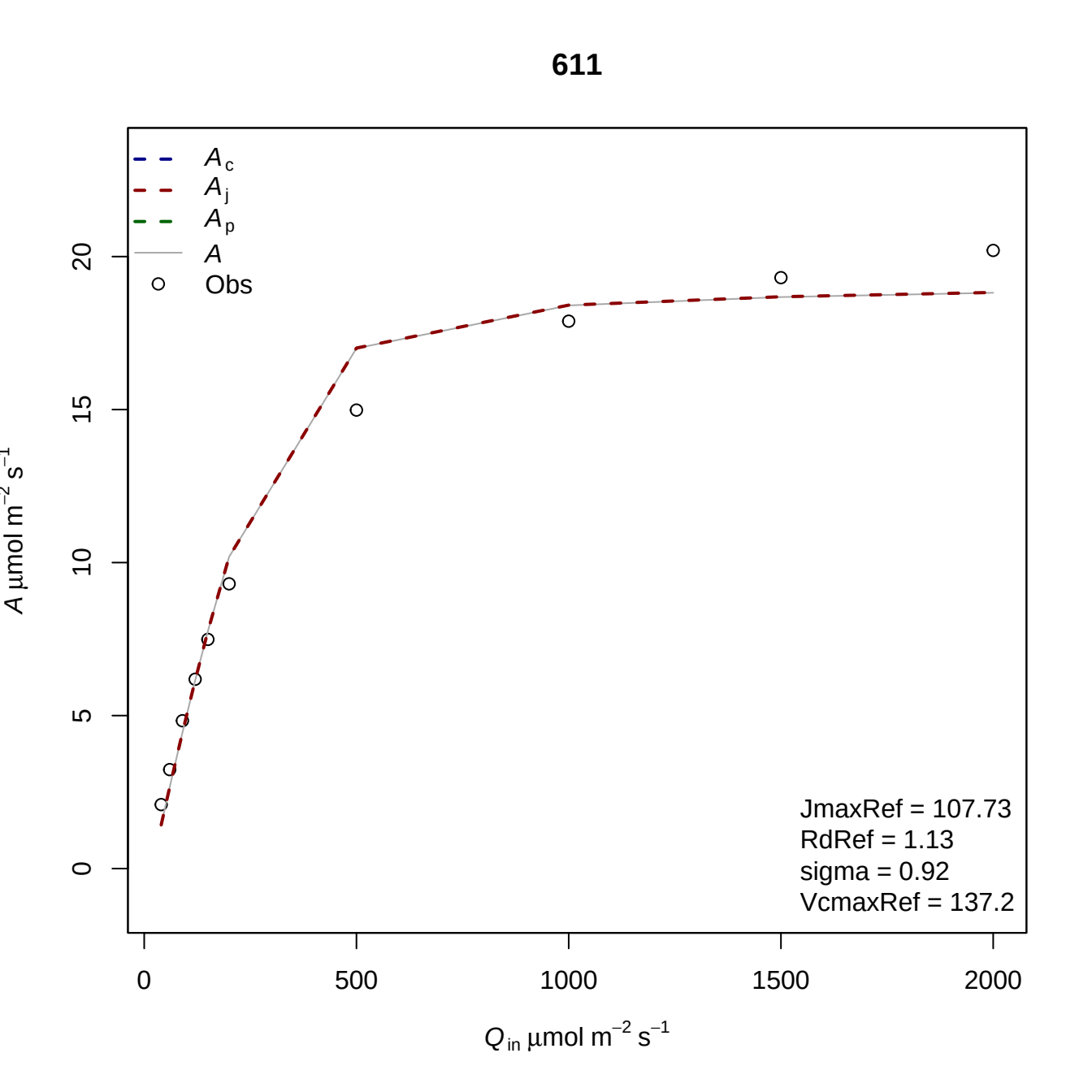

606
A_c
A_j
A_p
A
Obs
JmaxRef = 158.93
RdRef = 2.21
sigma = 1.24
VcmaxRef = 231.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

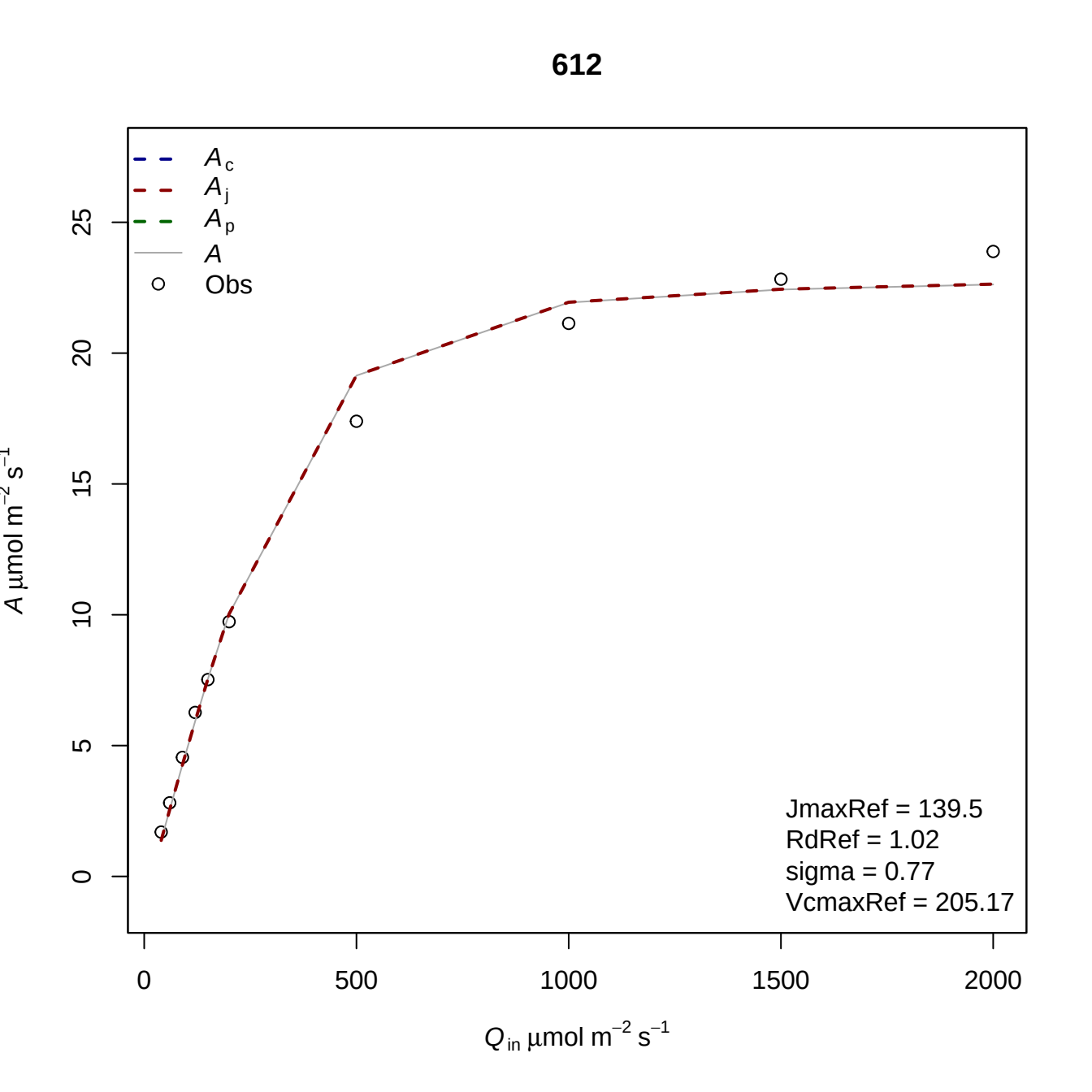

607
A_c
A_j
A_p
A
Obs
JmaxRef = 187.72
RdRef = 2.41
sigma = 1.23
VcmaxRef = 311.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

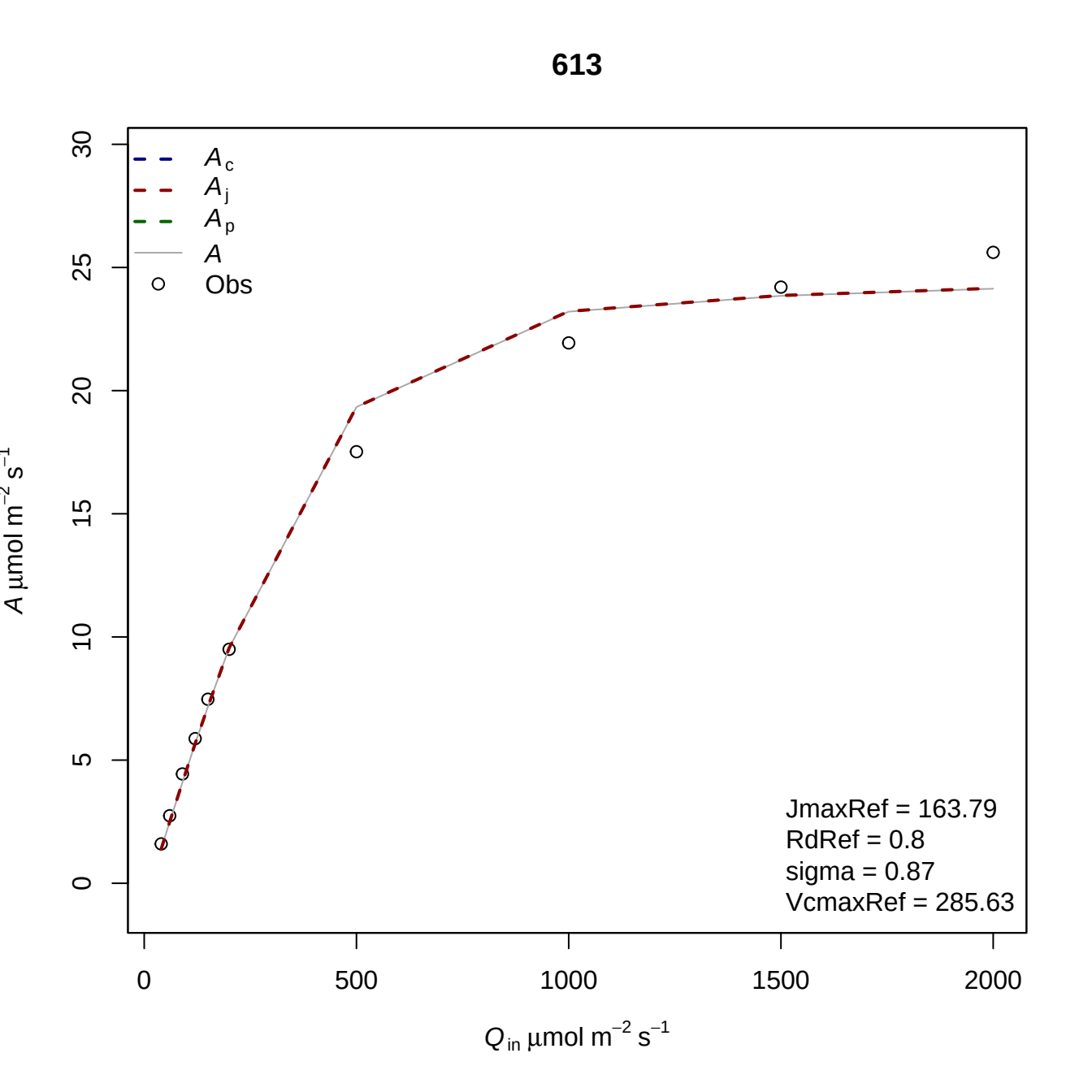

608
A_c
A_j
A_p
A
Obs
JmaxRef = 218.28
RdRef = 1.58
sigma = 0.55
VcmaxRef = 533.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

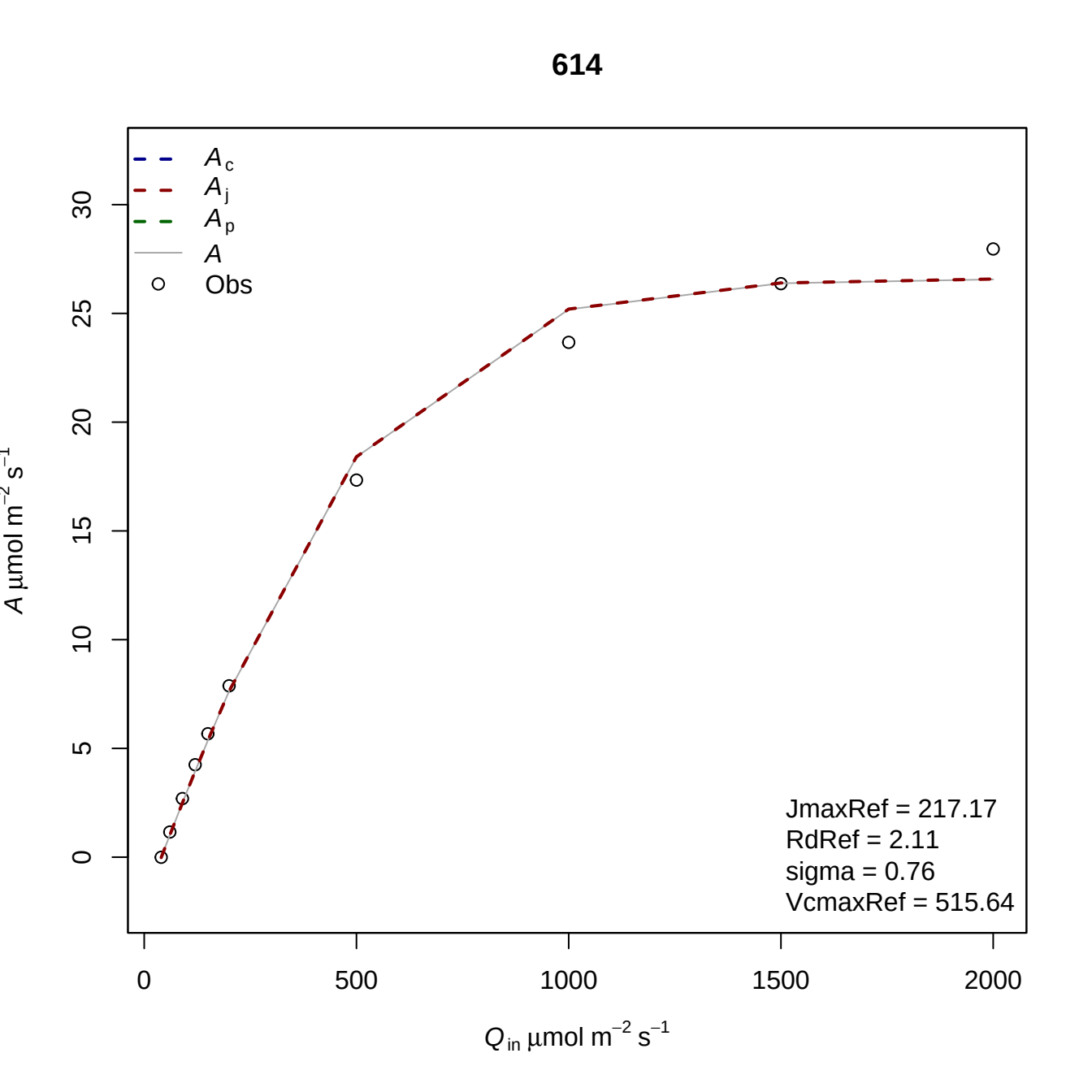

609
A_c
A_j
A_p
A
Obs
JmaxRef = 246.01
RdRef = 1.95
sigma = 0.37
VcmaxRef = 805.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

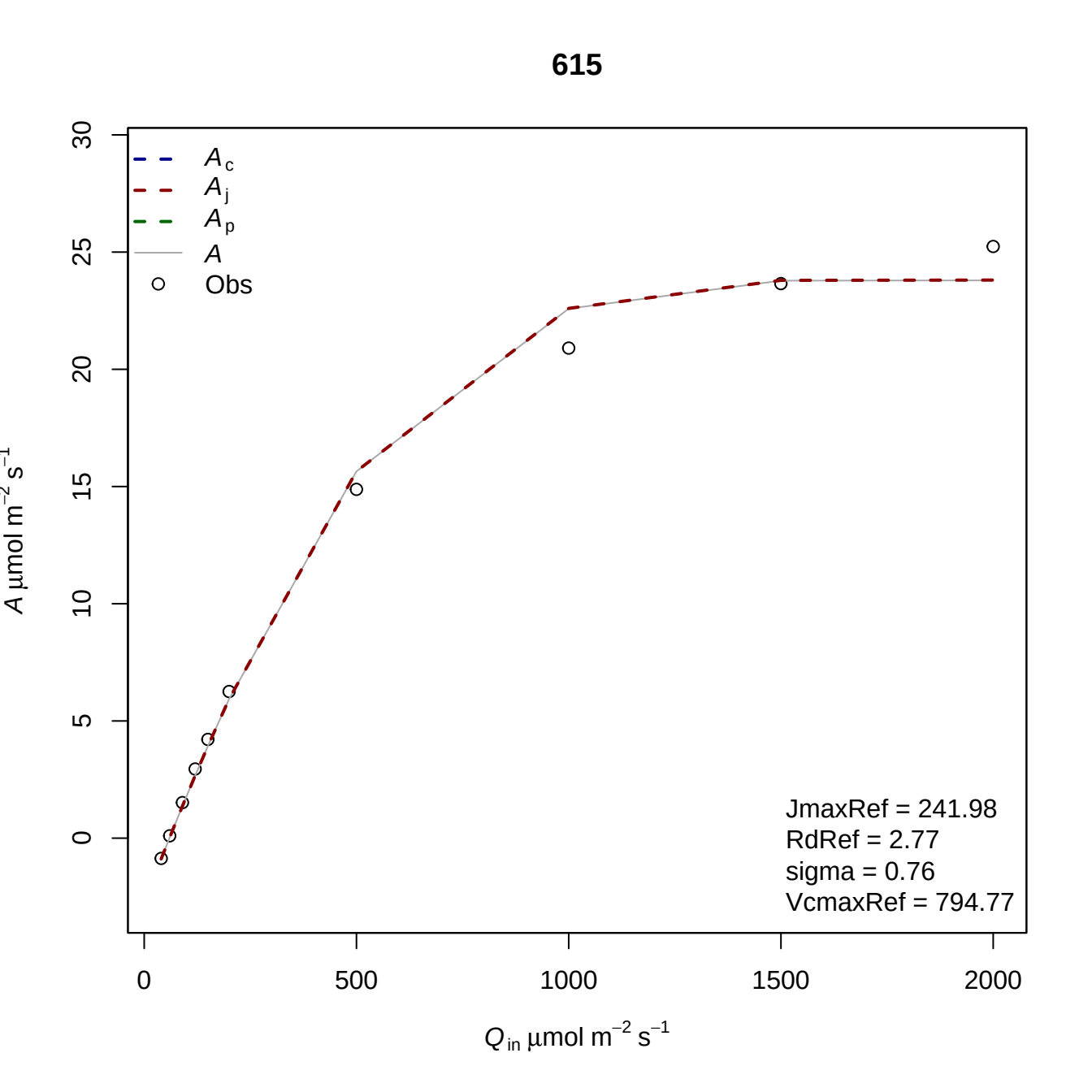

610
A_c
A_j
A_p
A
Obs
JmaxRef = 260.69
RdRef = 3.08
sigma = 0.5
VcmaxRef = 954.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

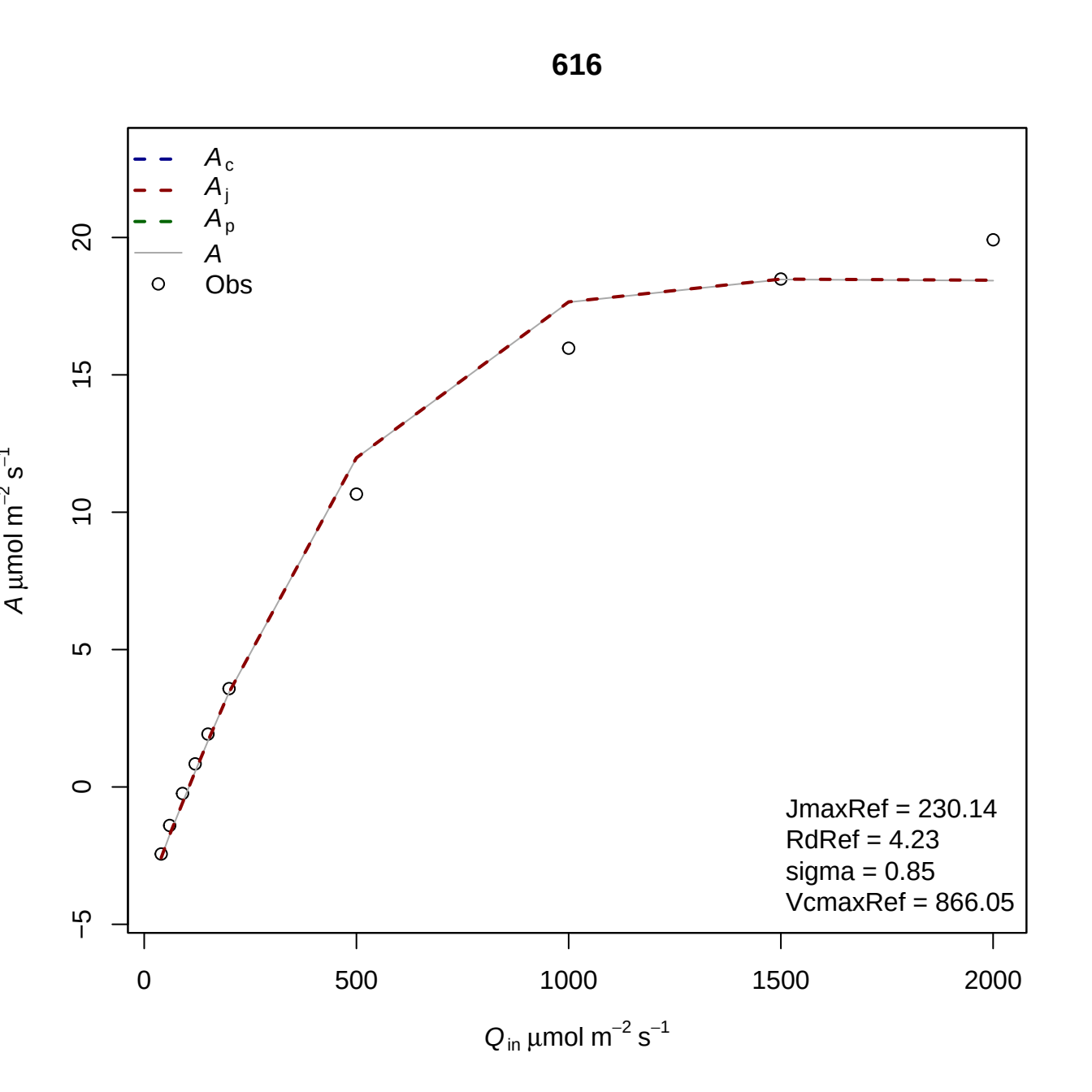

611
A_c
A_j
A_p
A
Obs
JmaxRef = 107.73
RdRef = 1.13
sigma = 0.92
VcmaxRef = 137.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

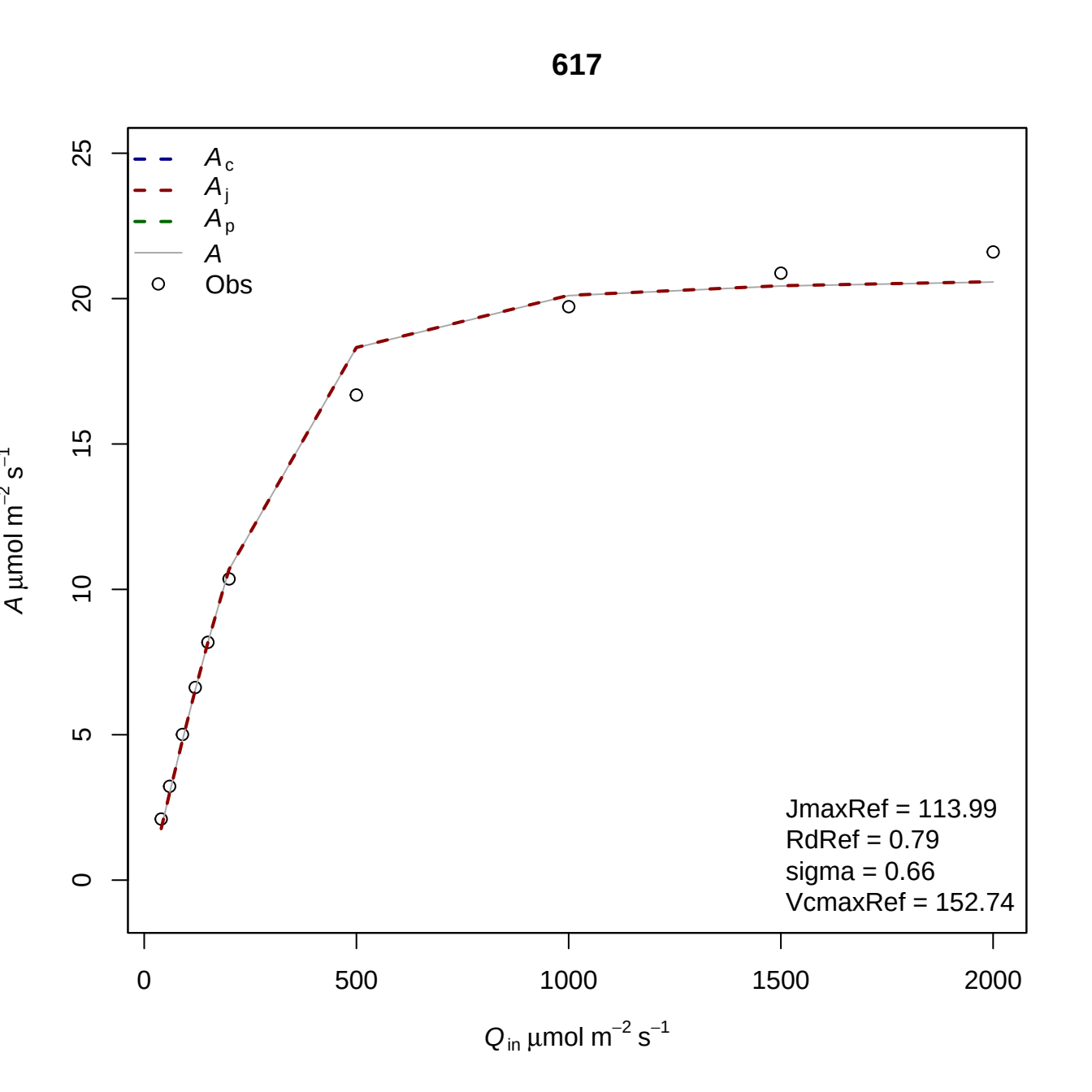

612
A_c
A_j
A_p
A
Obs
JmaxRef = 139.5
RdRef = 1.02
sigma = 0.77
VcmaxRef = 205.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

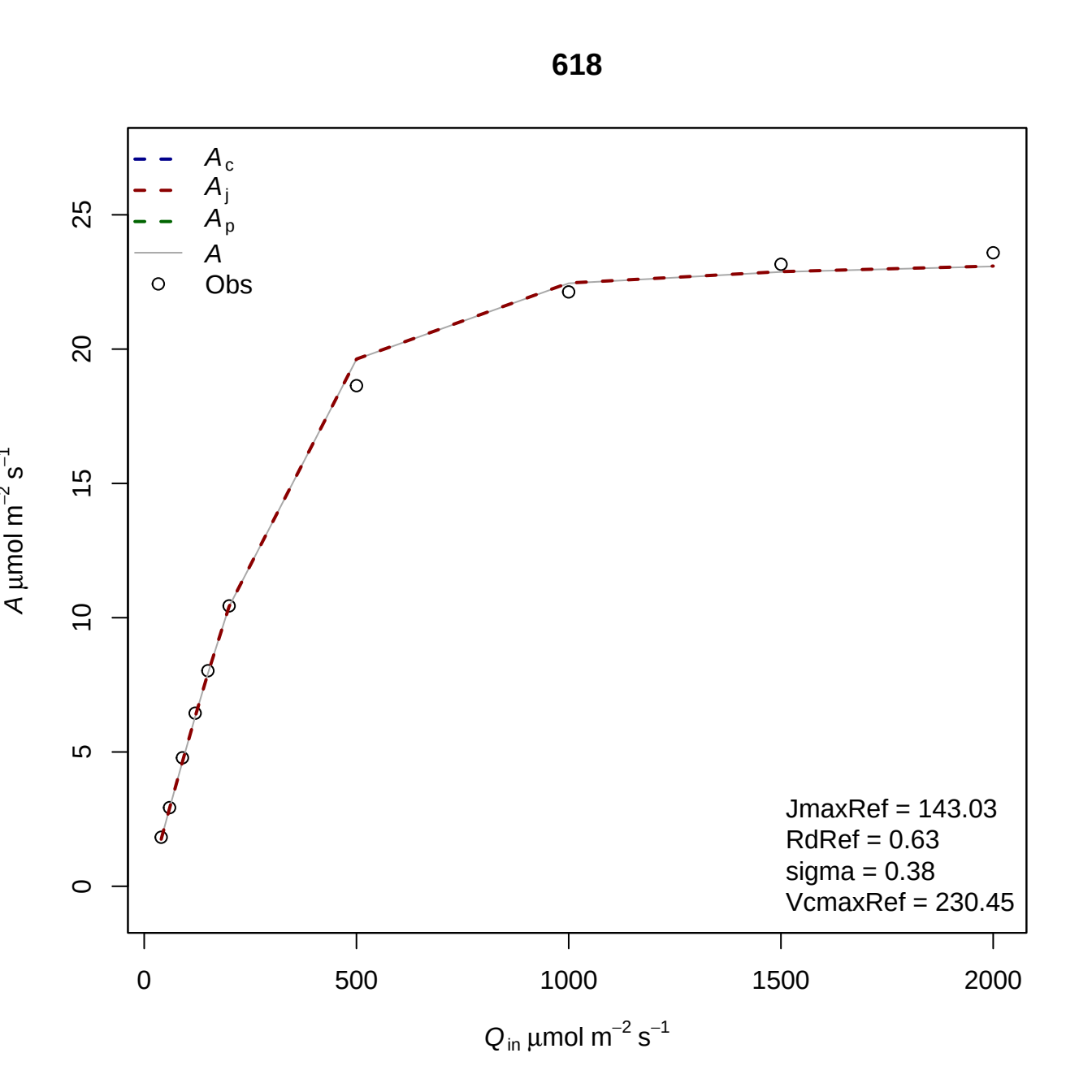

613
A_c
A_j
A_p
A
Obs
JmaxRef = 163.79
RdRef = 0.8
sigma = 0.87
VcmaxRef = 285.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

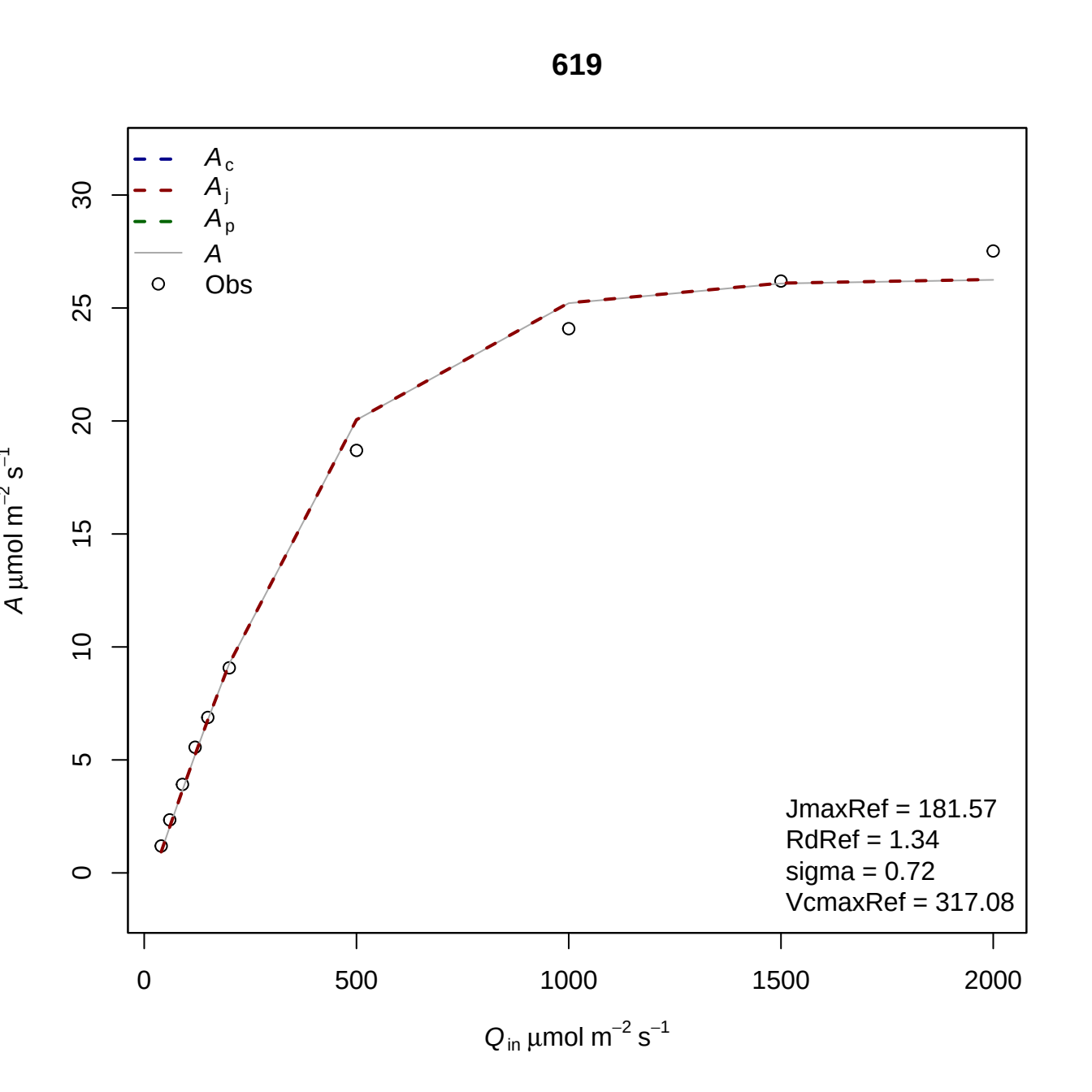

614
A_c
A_j
A_p
A
Obs
JmaxRef = 217.17
RdRef = 2.11
sigma = 0.76
VcmaxRef = 515.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

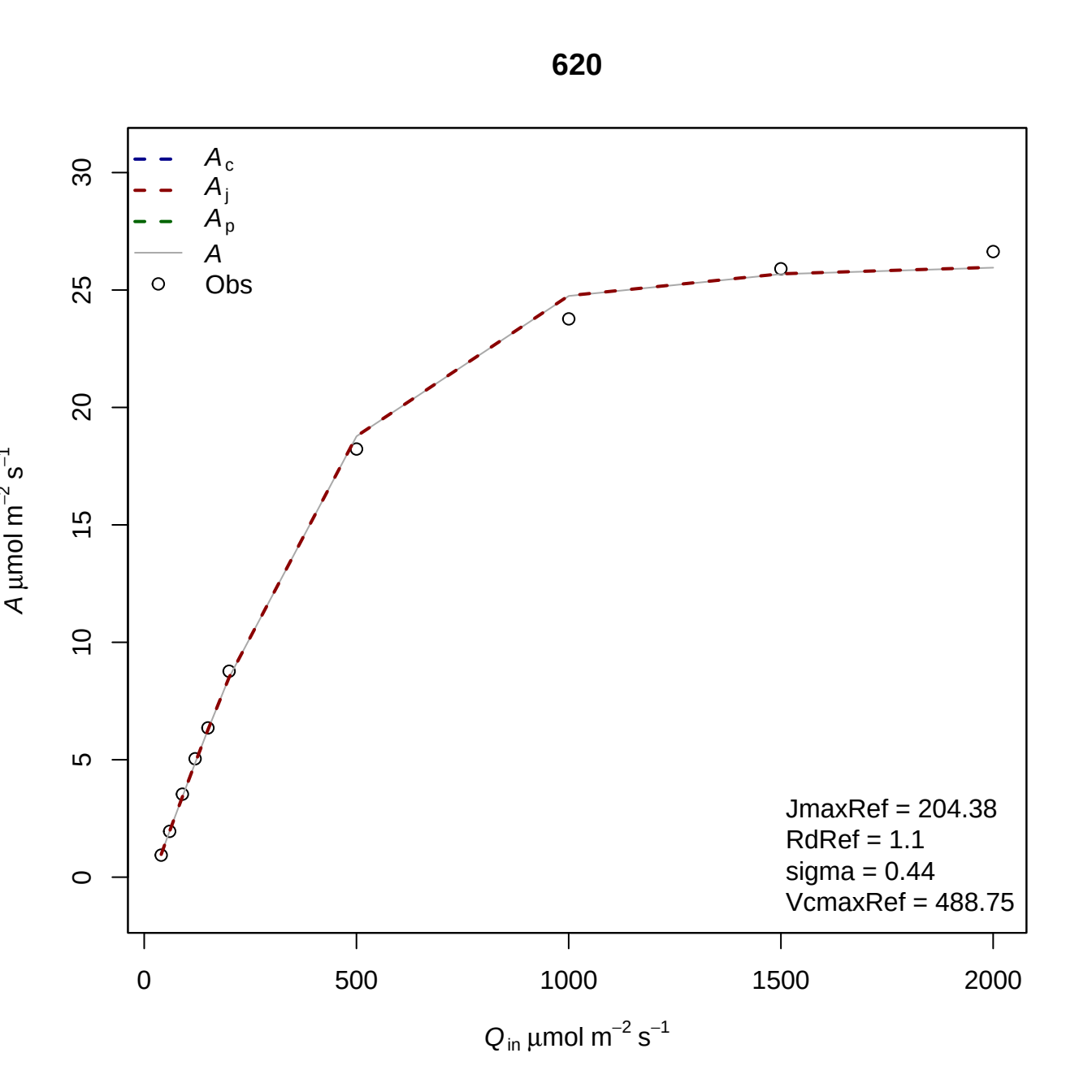

615
A_c
A_j
A_p
A
Obs
JmaxRef = 241.98
RdRef = 2.77
sigma = 0.76
VcmaxRef = 794.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

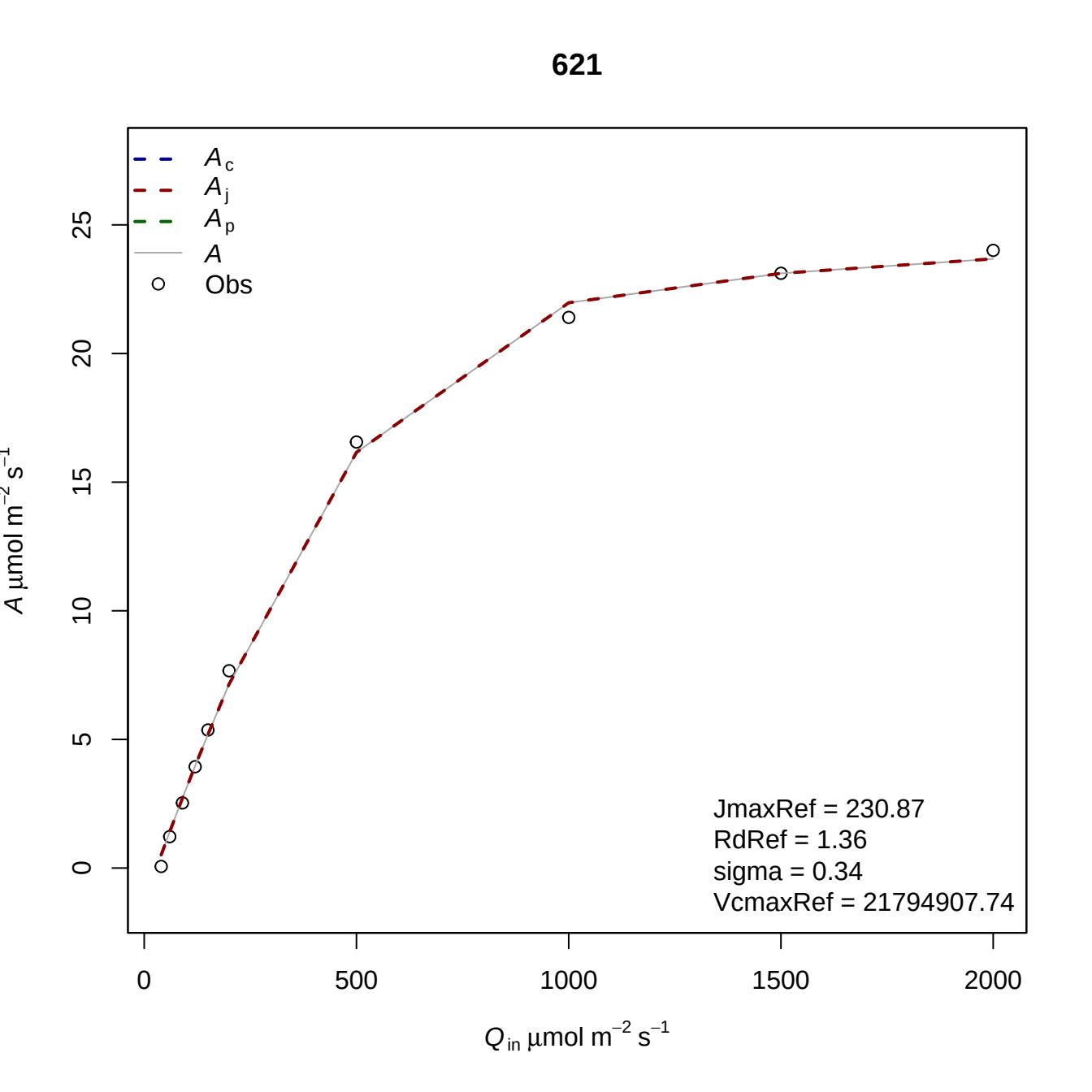

616
A_c
A_j
A_p
A
Obs
JmaxRef = 230.14
RdRef = 4.23
sigma = 0.85
VcmaxRef = 866.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

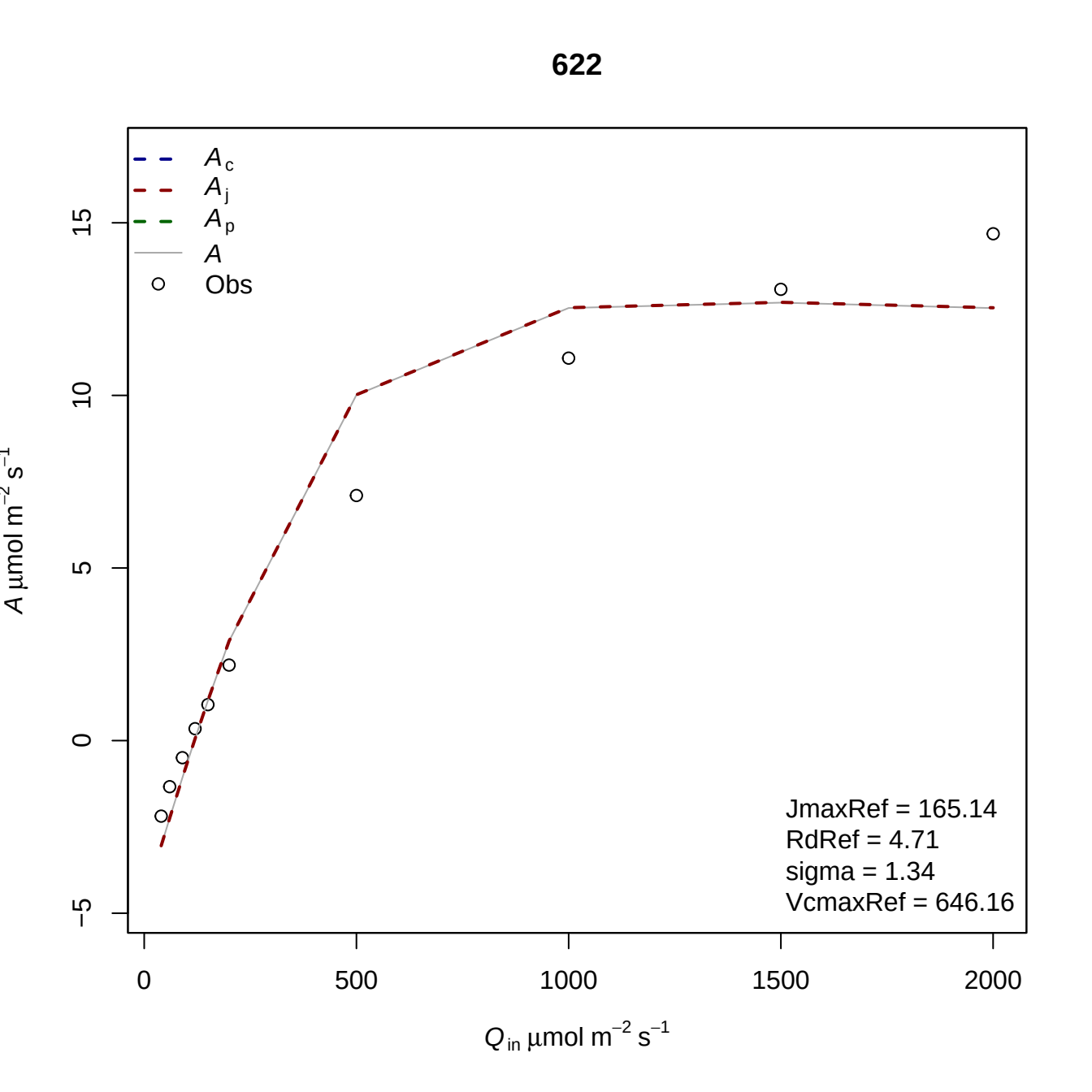

617
A_c
A_j
A_p
A
Obs
JmaxRef = 113.99
RdRef = 0.79
sigma = 0.66
VcmaxRef = 152.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

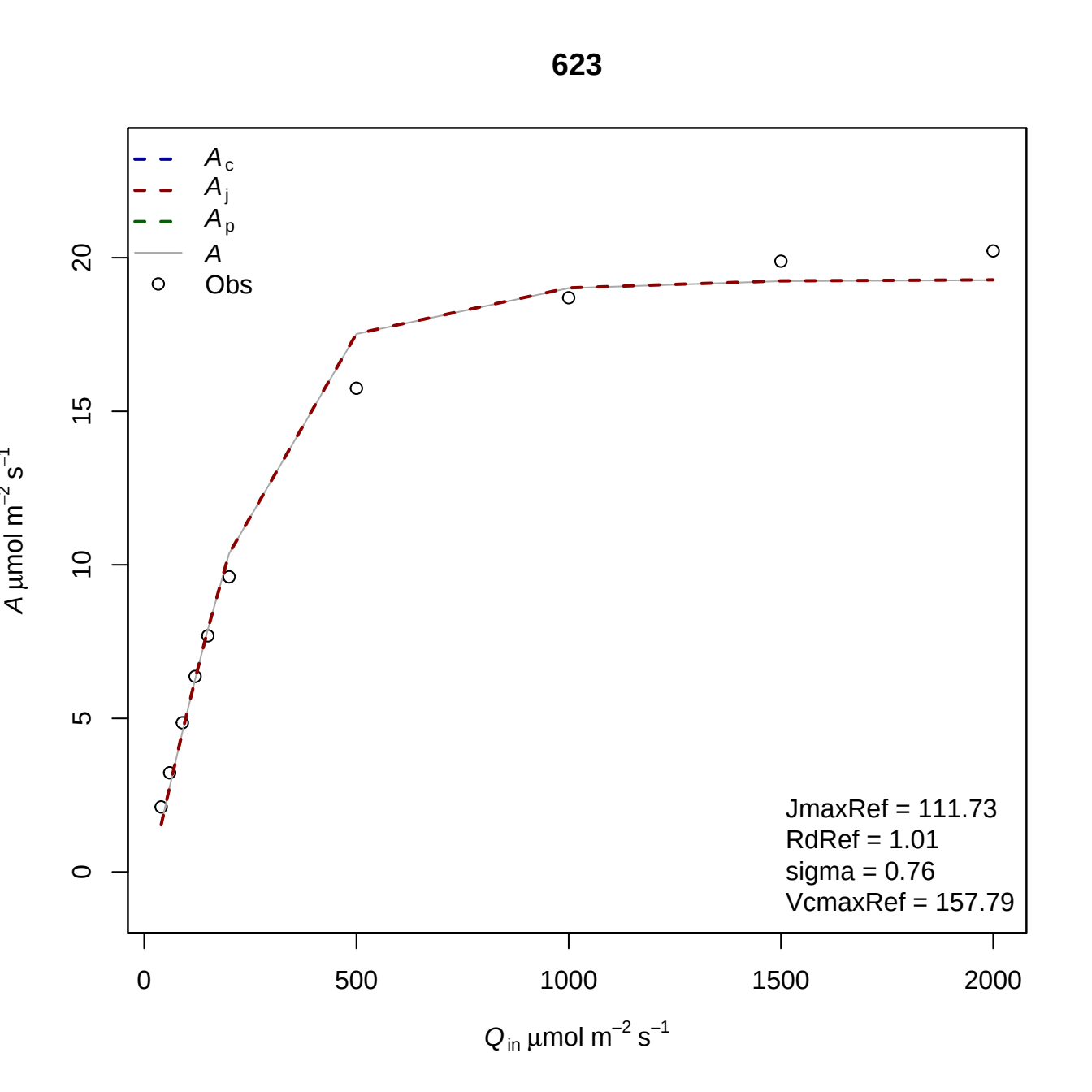

618
A_c
A_j
A_p
A
Obs
JmaxRef = 143.03
RdRef = 0.63
sigma = 0.38
VcmaxRef = 230.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

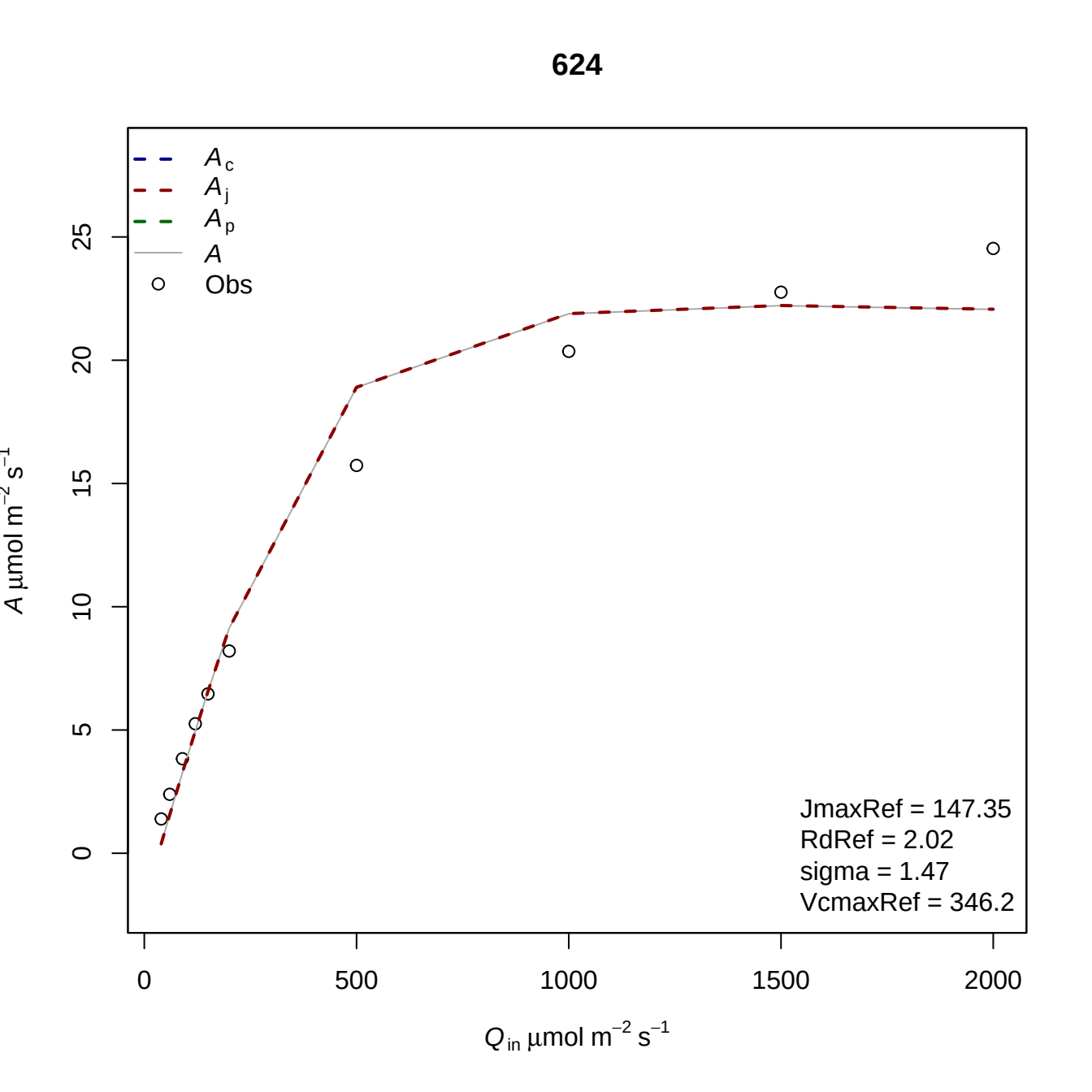

619
A_c
A_j
A_p
A
Obs
JmaxRef = 181.57
RdRef = 1.34
sigma = 0.72
VcmaxRef = 317.08
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

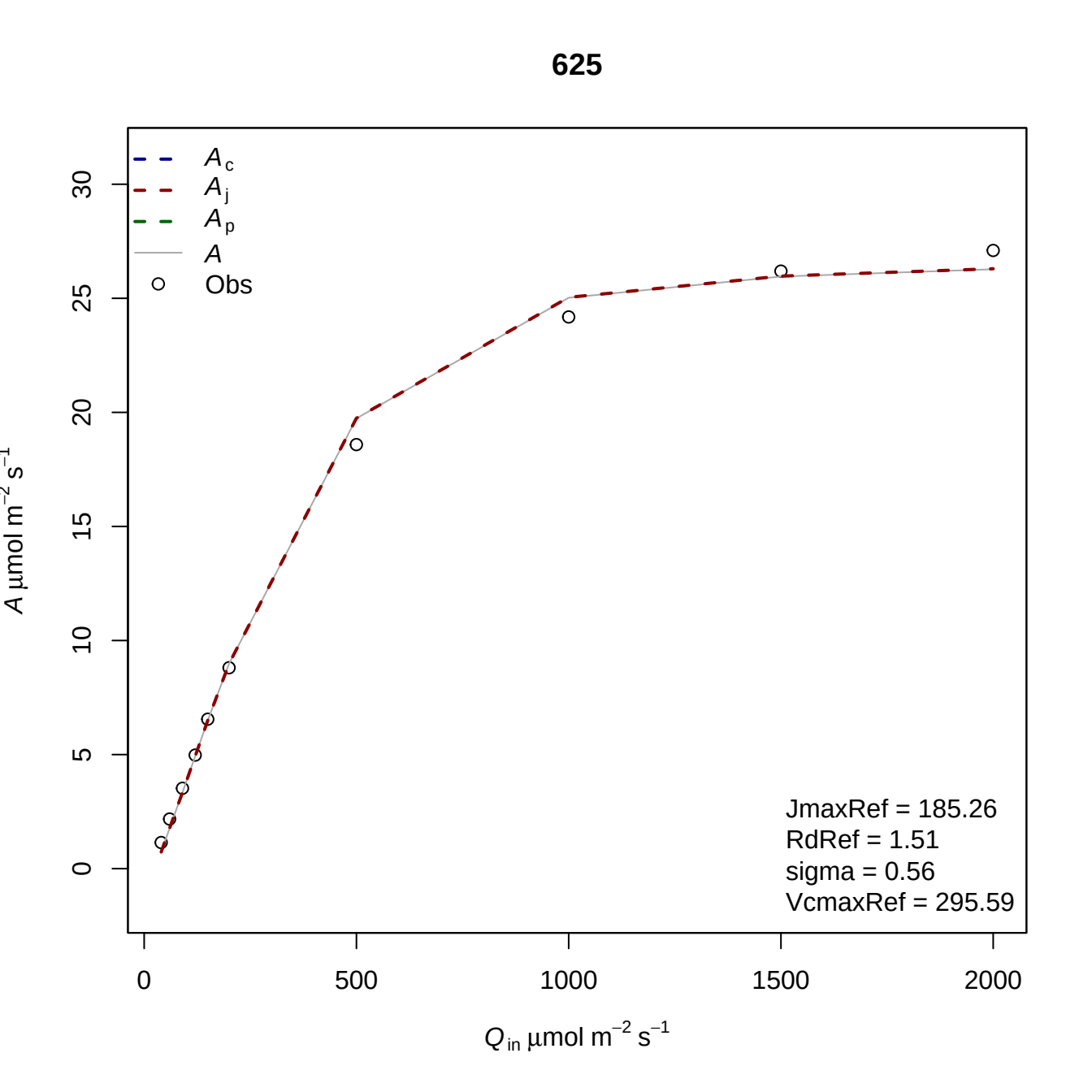

620
A_c
A_j
A_p
A
Obs
JmaxRef = 204.38
RdRef = 1.1
sigma = 0.44
VcmaxRef = 488.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

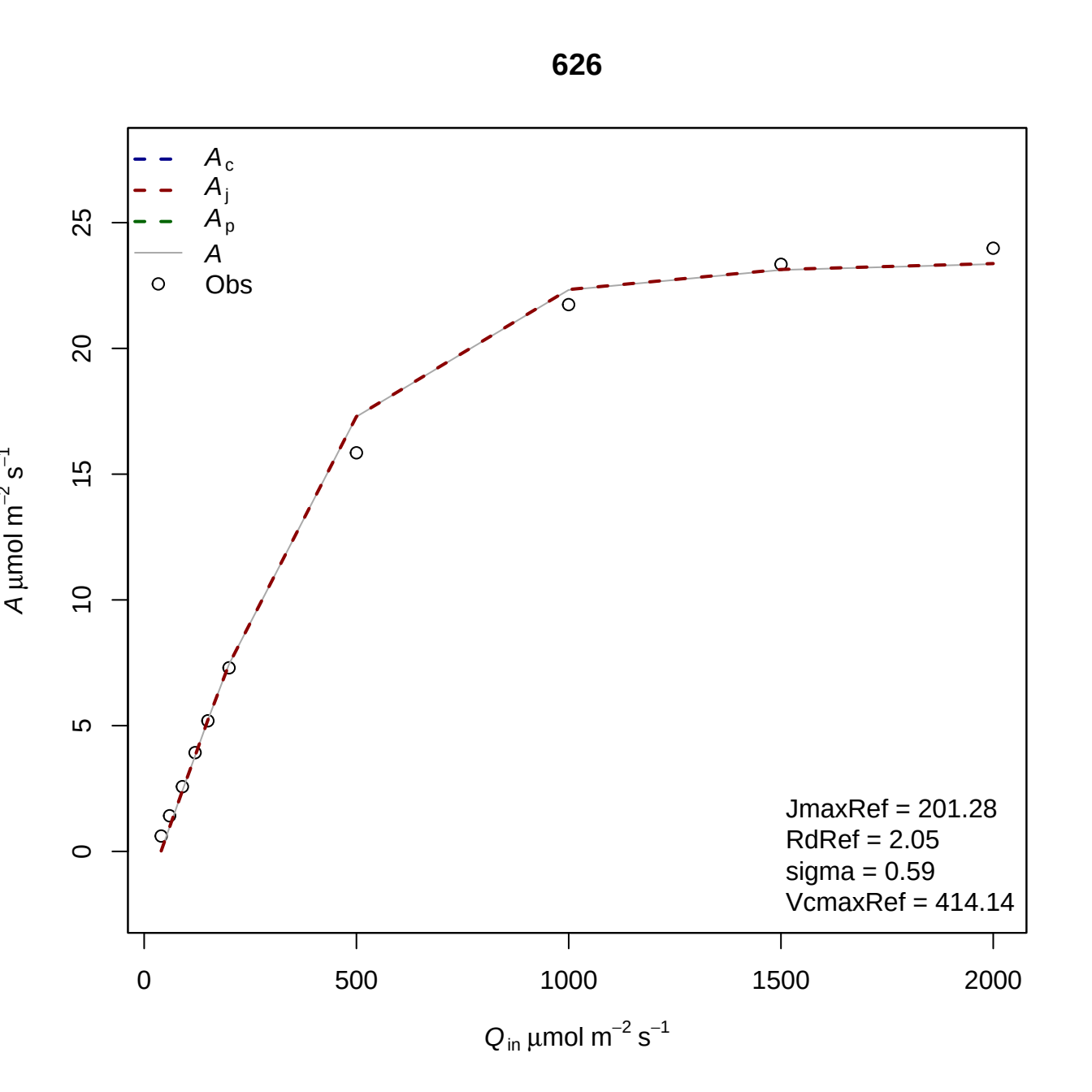

621
A_c
A_j
A_p
A
Obs
JmaxRef = 230.87
RdRef = 1.36
sigma = 0.34
VcmaxRef = 21794907.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

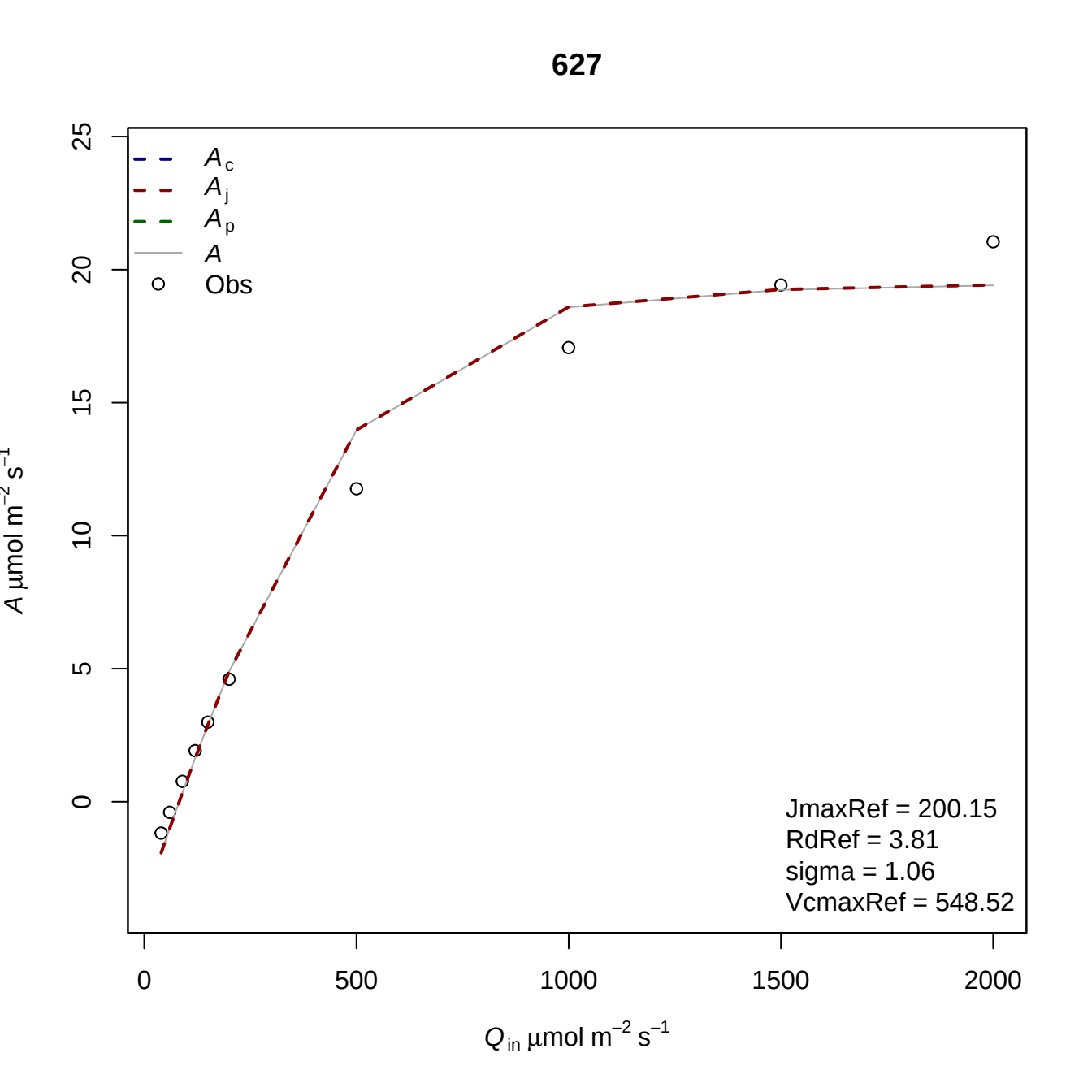

622
A_c
A_j
A_p
A
Obs
JmaxRef = 165.14
RdRef = 4.71
sigma = 1.34
VcmaxRef = 646.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

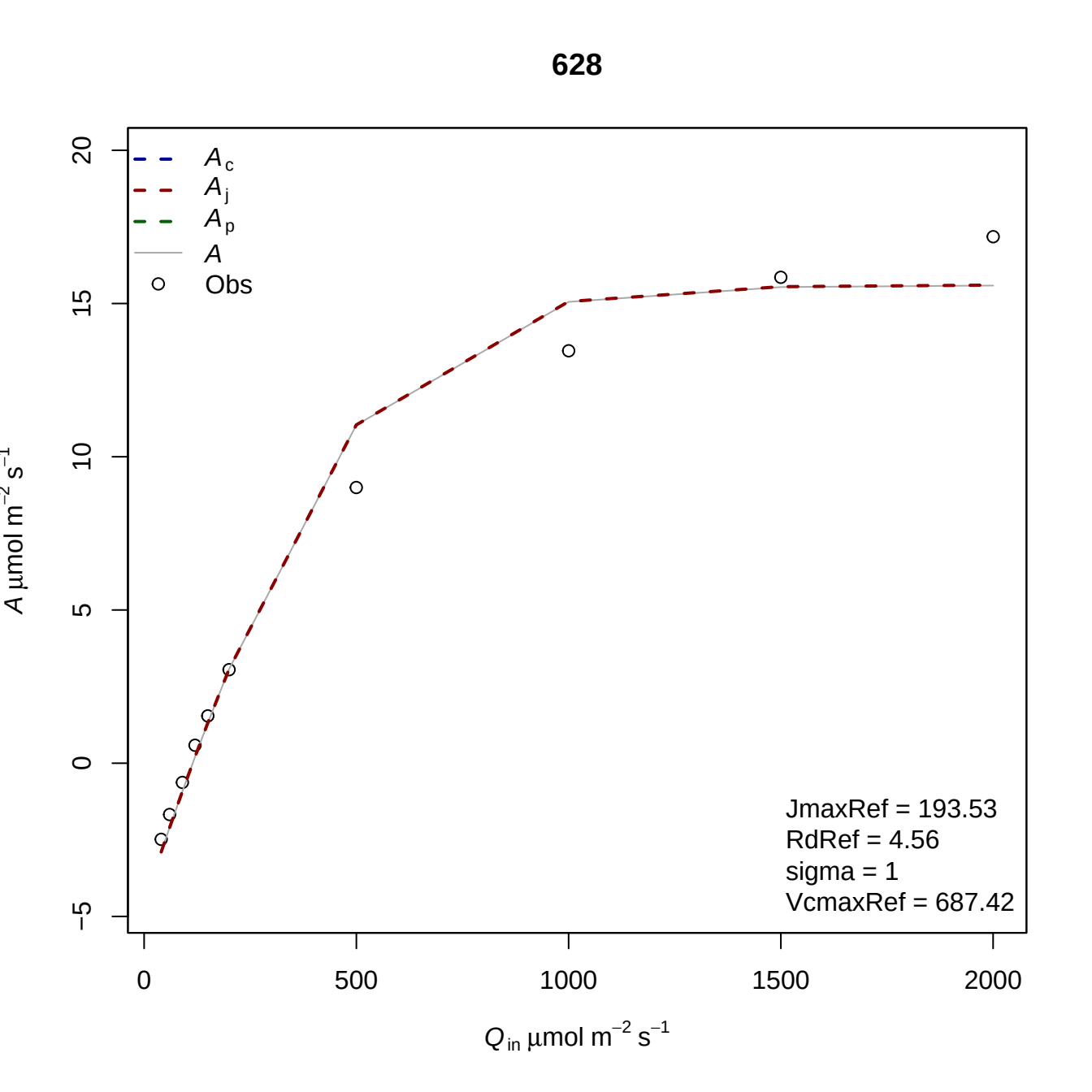

623
A_c
A_j
A_p
A
Obs
JmaxRef = 111.73
RdRef = 1.01
sigma = 0.76
VcmaxRef = 157.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

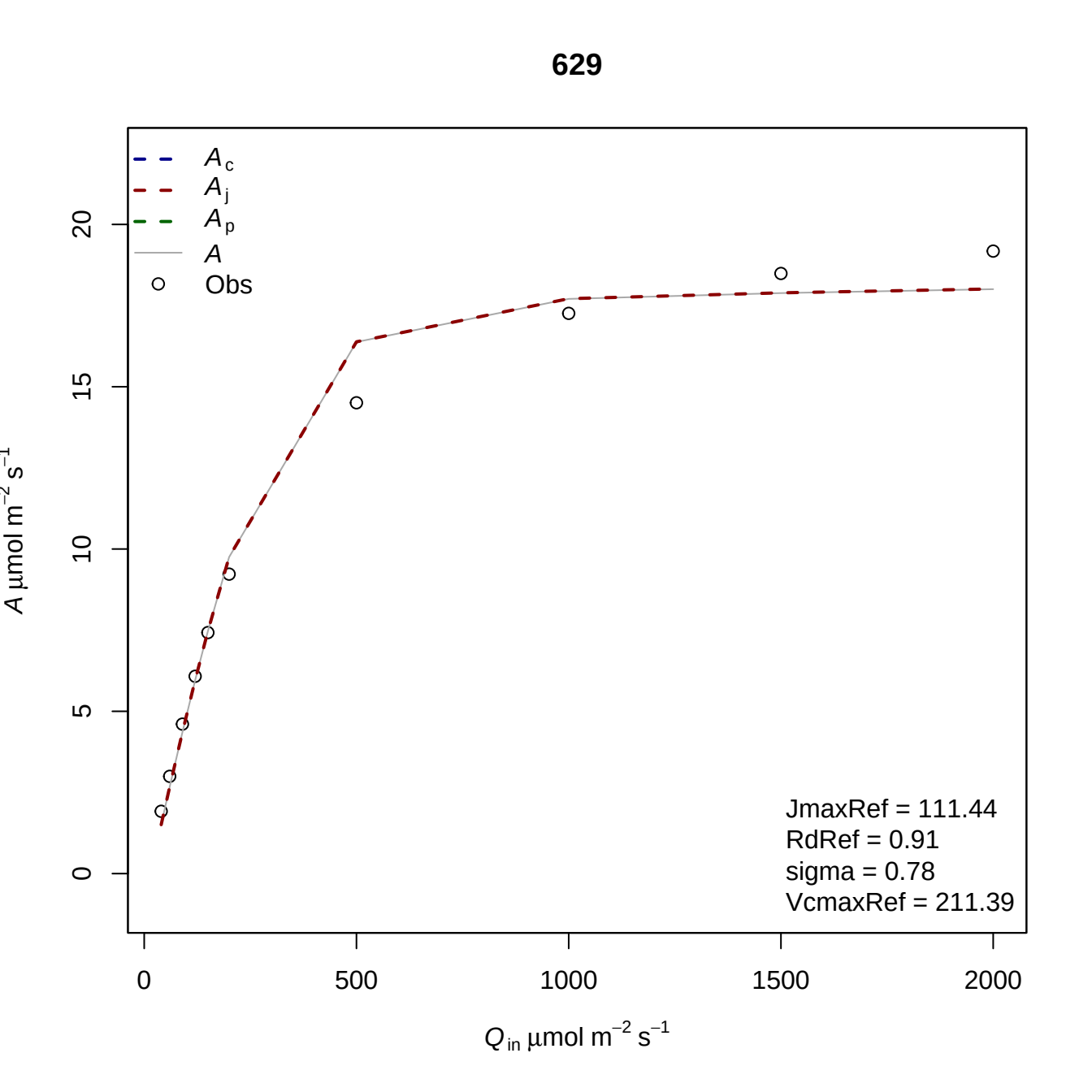

624
A_c
A_j
A_p
A
Obs
JmaxRef = 147.35
RdRef = 2.02
sigma = 1.47
VcmaxRef = 346.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

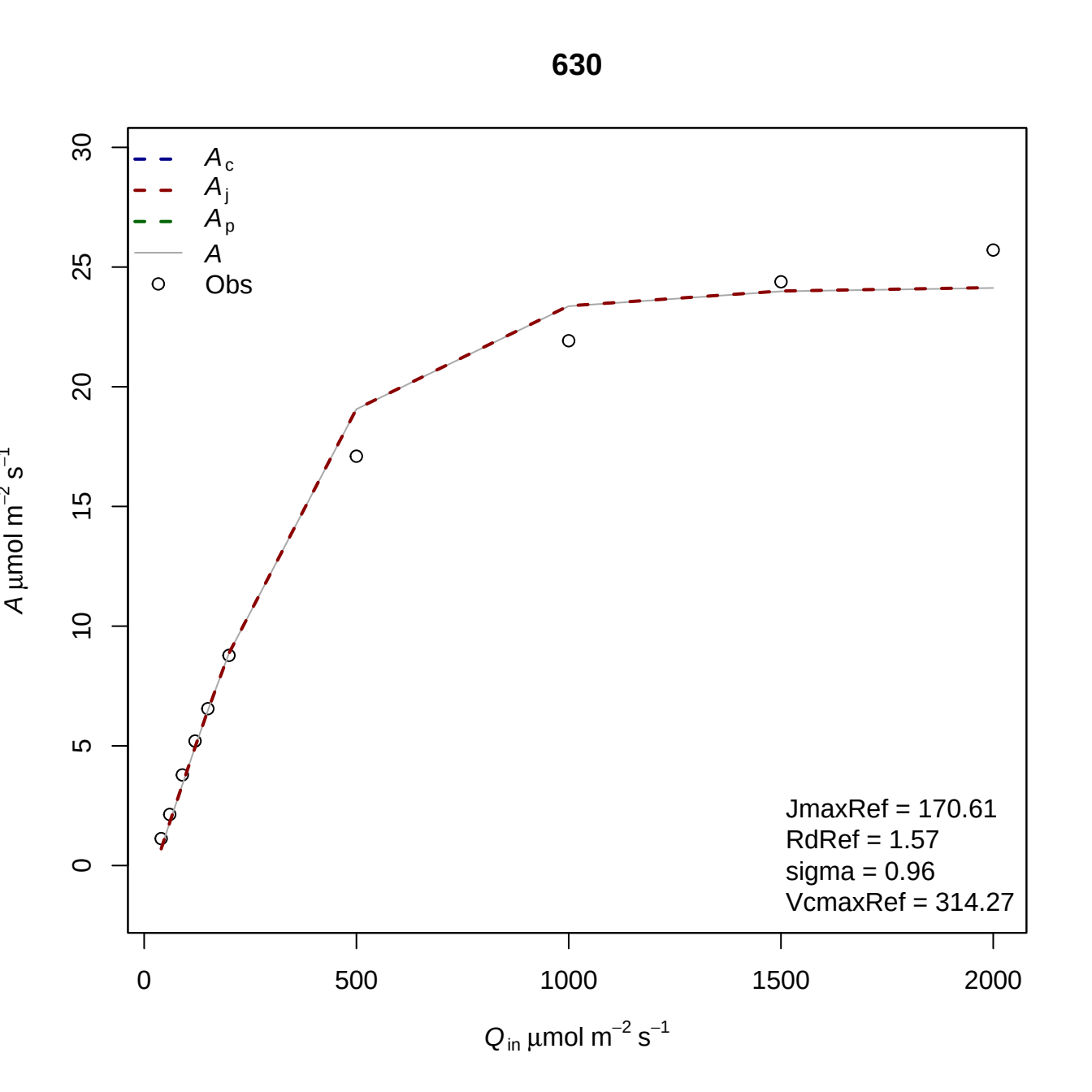

625
A_c
A_j
A_p
A
Obs
JmaxRef = 185.26
RdRef = 1.51
sigma = 0.56
VcmaxRef = 295.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

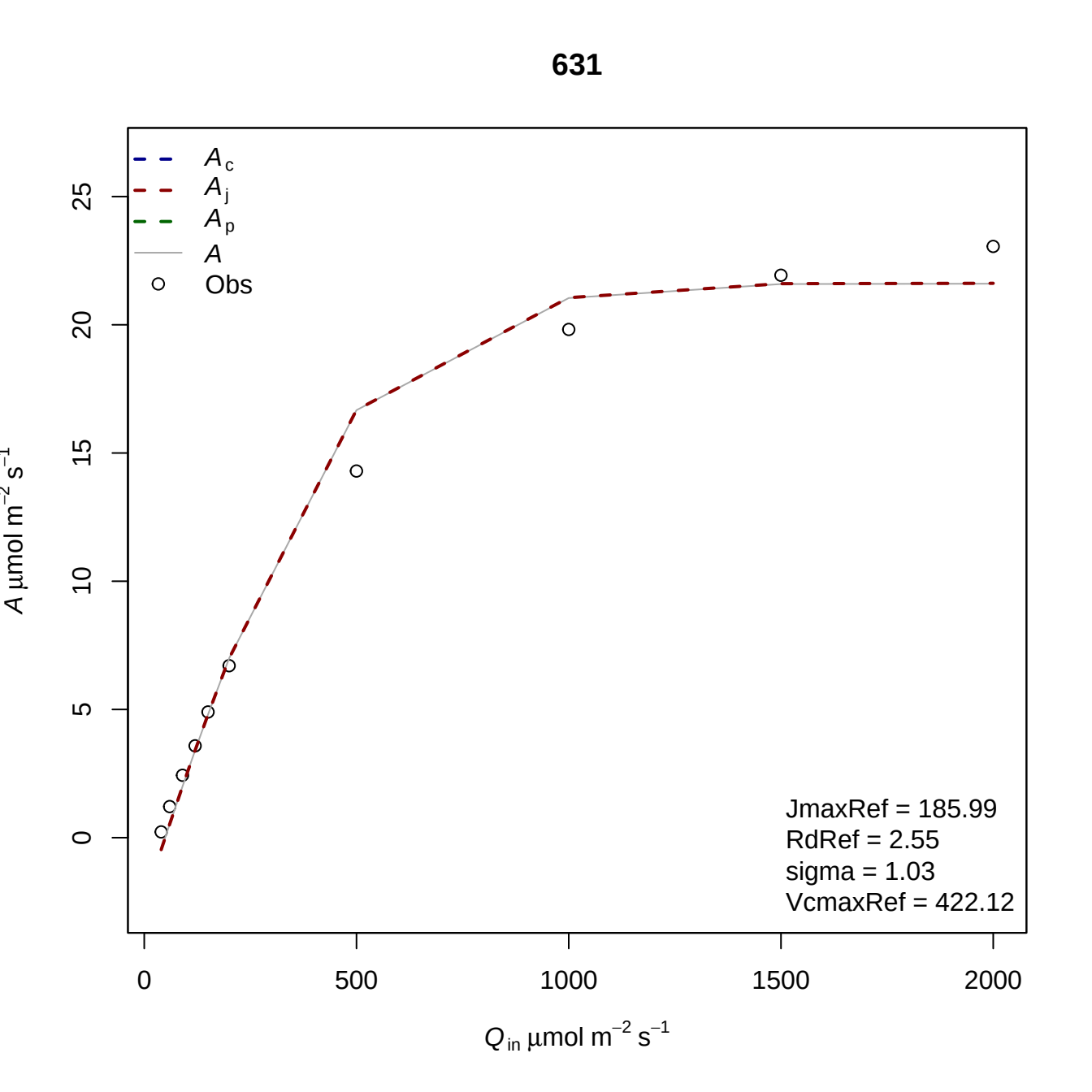

626
A_c
A_j
A_p
A
Obs
JmaxRef = 201.28
RdRef = 2.05
sigma = 0.59
VcmaxRef = 414.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

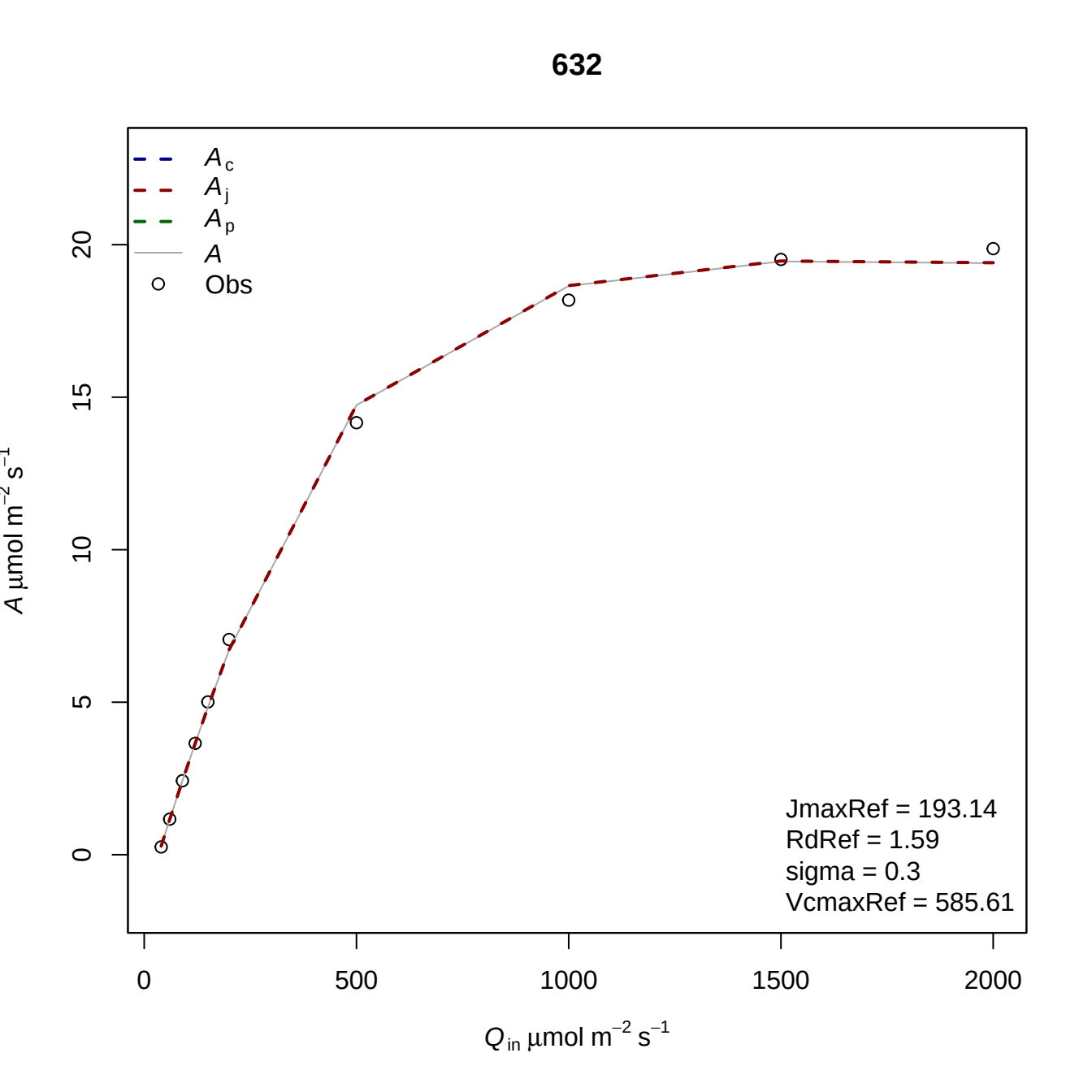

627
A_c
A_j
A_p
A
Obs
JmaxRef = 200.15
RdRef = 3.81
sigma = 1.06
VcmaxRef = 548.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

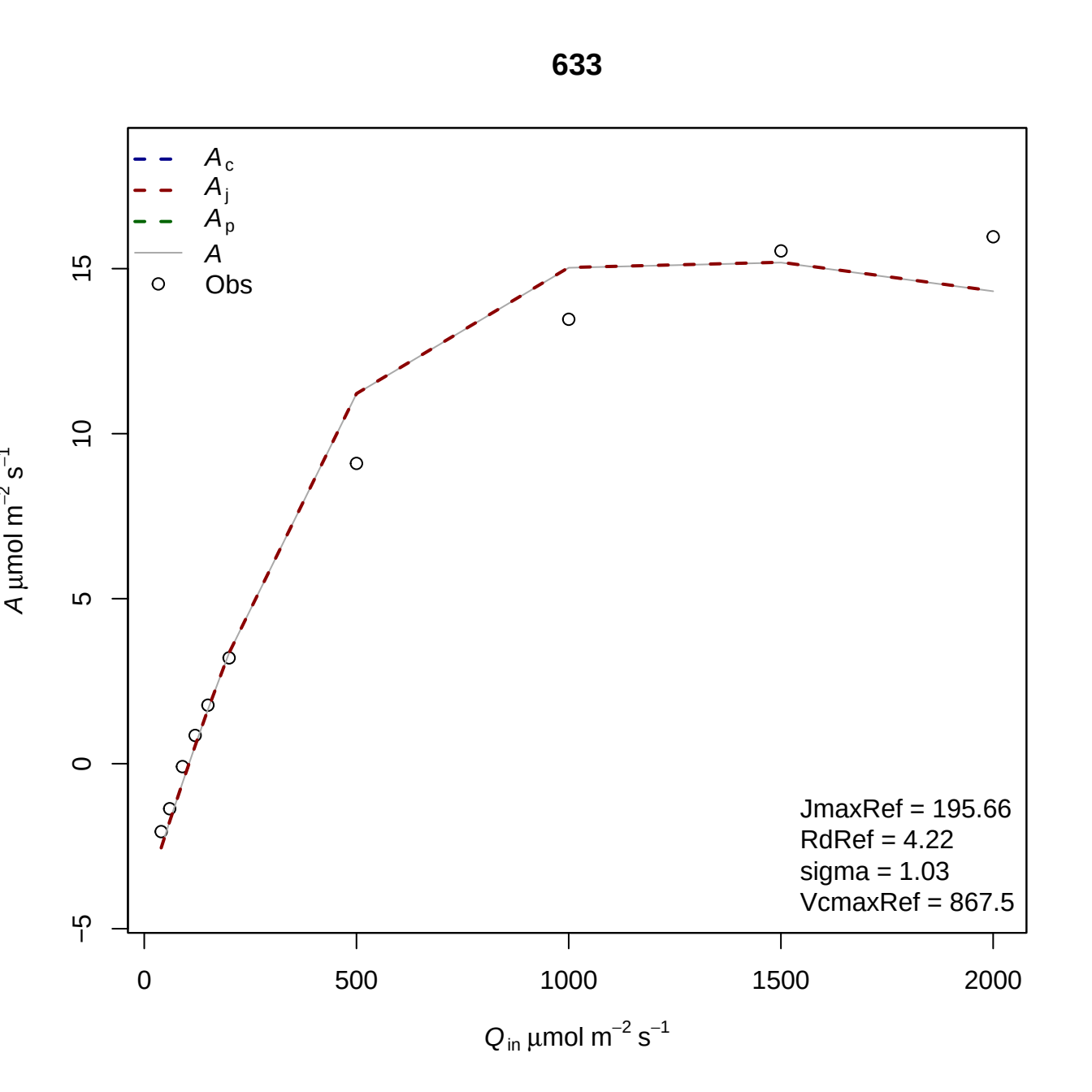

628
A_c
A_j
A_p
A
Obs
JmaxRef = 193.53
RdRef = 4.56
sigma = 1
VcmaxRef = 687.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

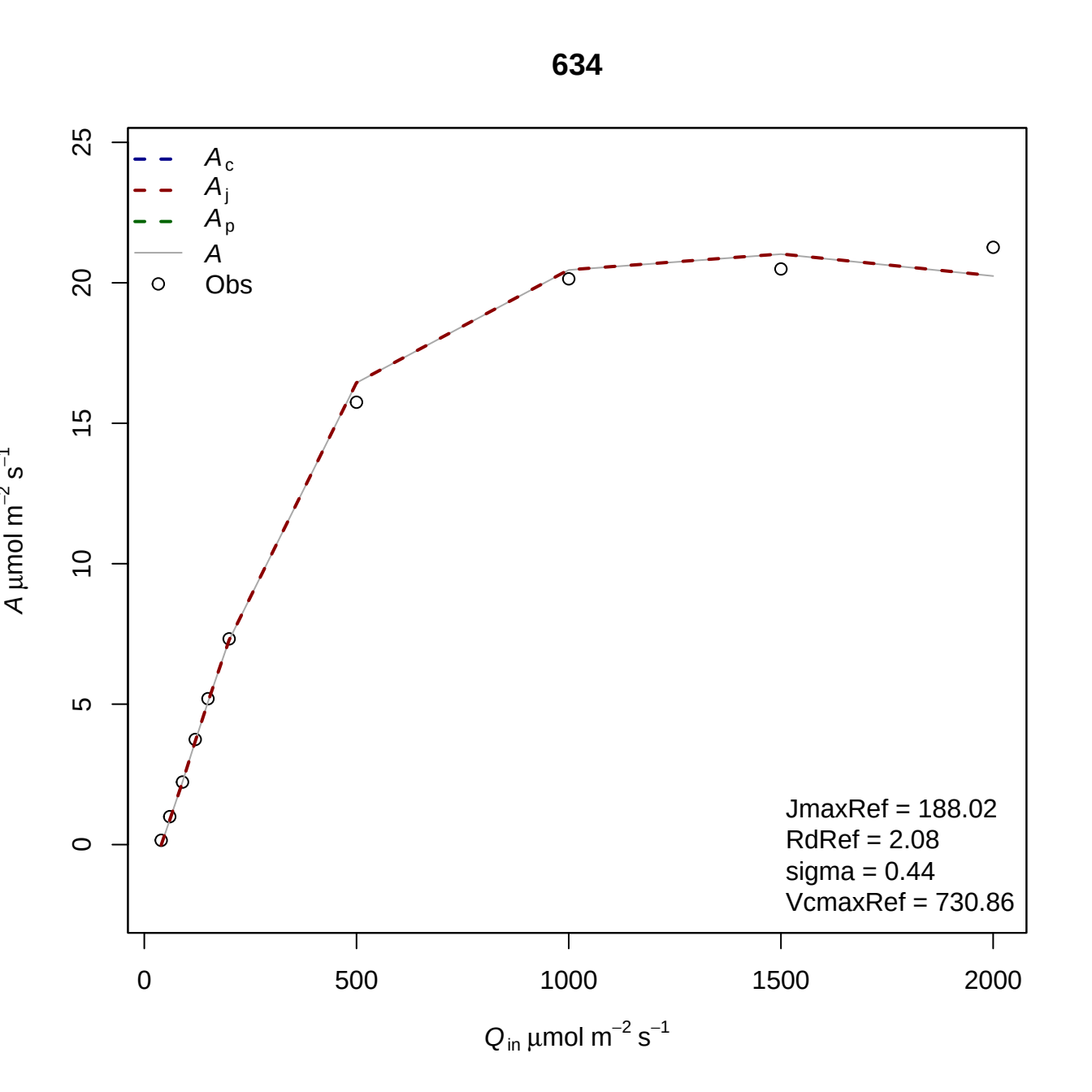

629
A_c
A_j
A_p
A
Obs
JmaxRef = 111.44
RdRef = 0.91
sigma = 0.78
VcmaxRef = 211.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

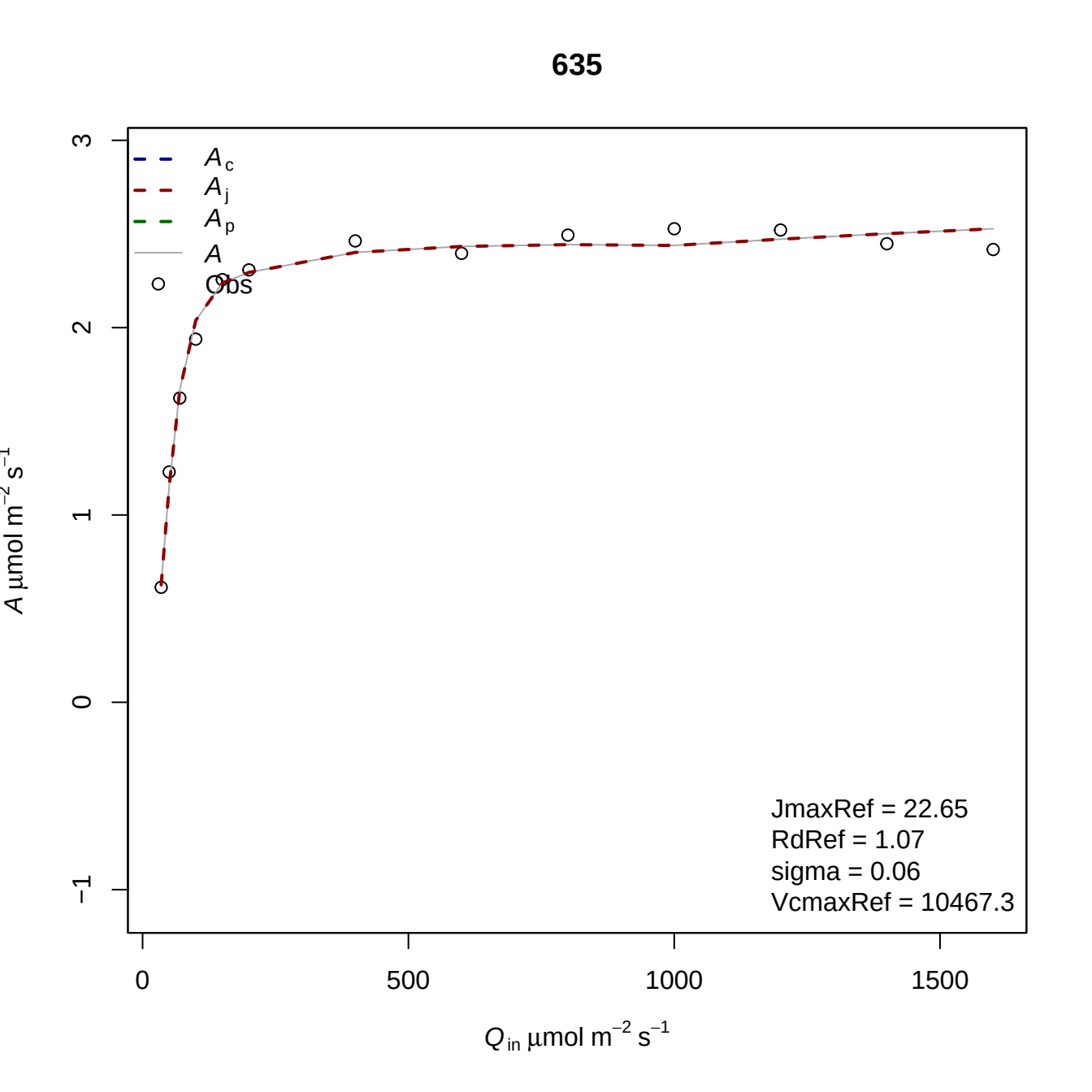

630
A_c
A_j
A_p
A
Obs
JmaxRef = 170.61
RdRef = 1.57
sigma = 0.96
VcmaxRef = 314.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

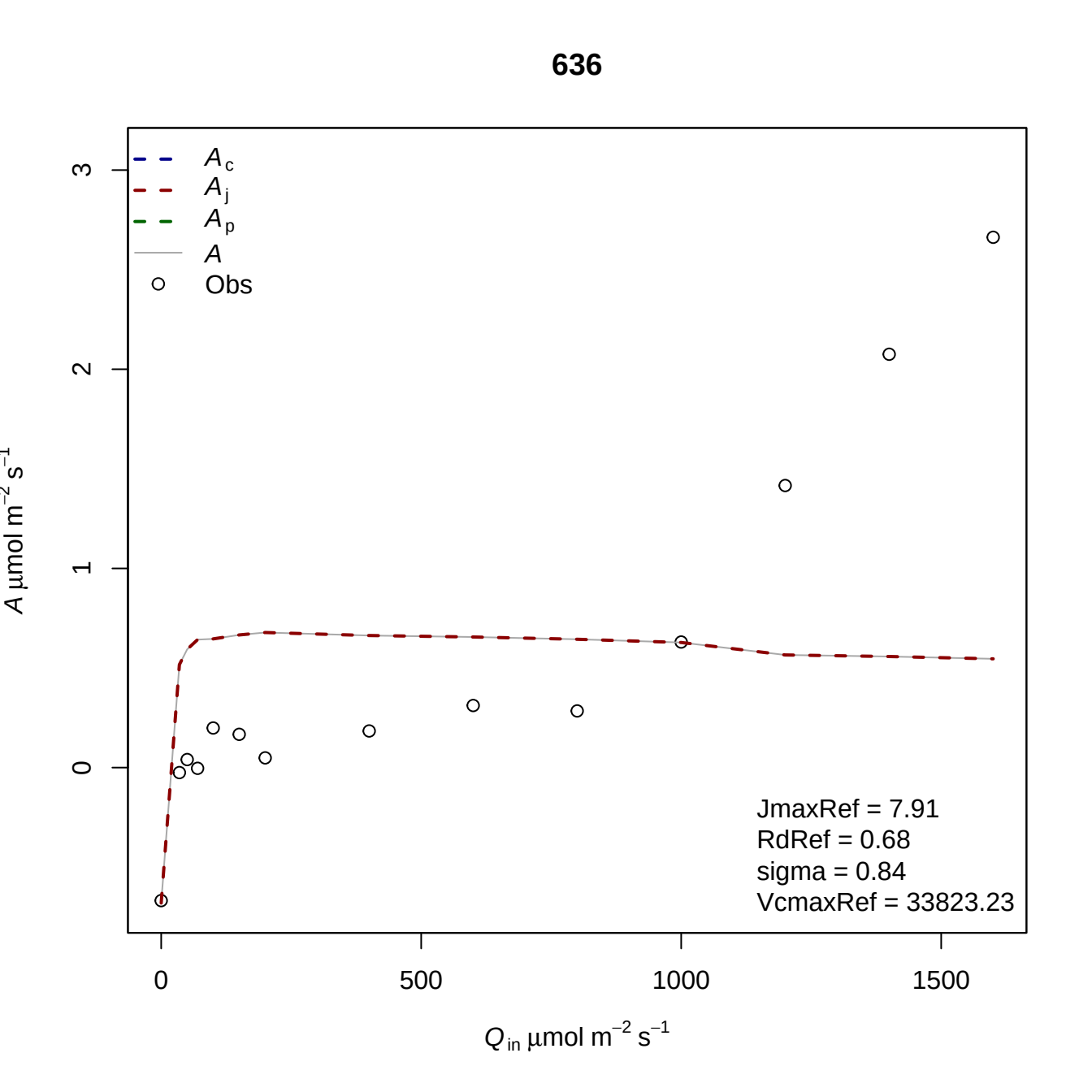

631
A_c
A_j
A_p
A
Obs
JmaxRef = 185.99
RdRef = 2.55
sigma = 1.03
VcmaxRef = 422.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

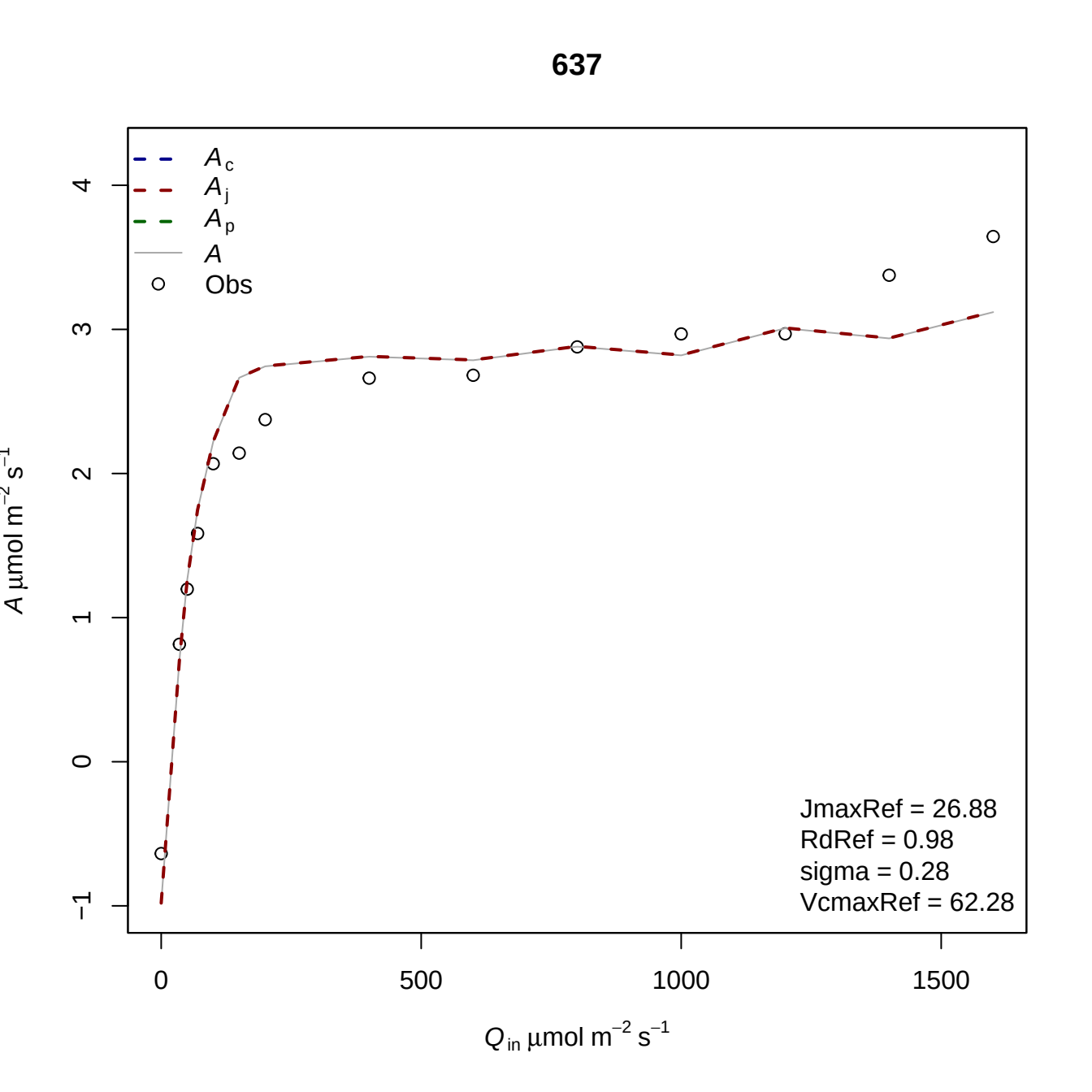

632
A_c
A_j
A_p
A
Obs
JmaxRef = 193.14
RdRef = 1.59
sigma = 0.3
VcmaxRef = 585.61
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

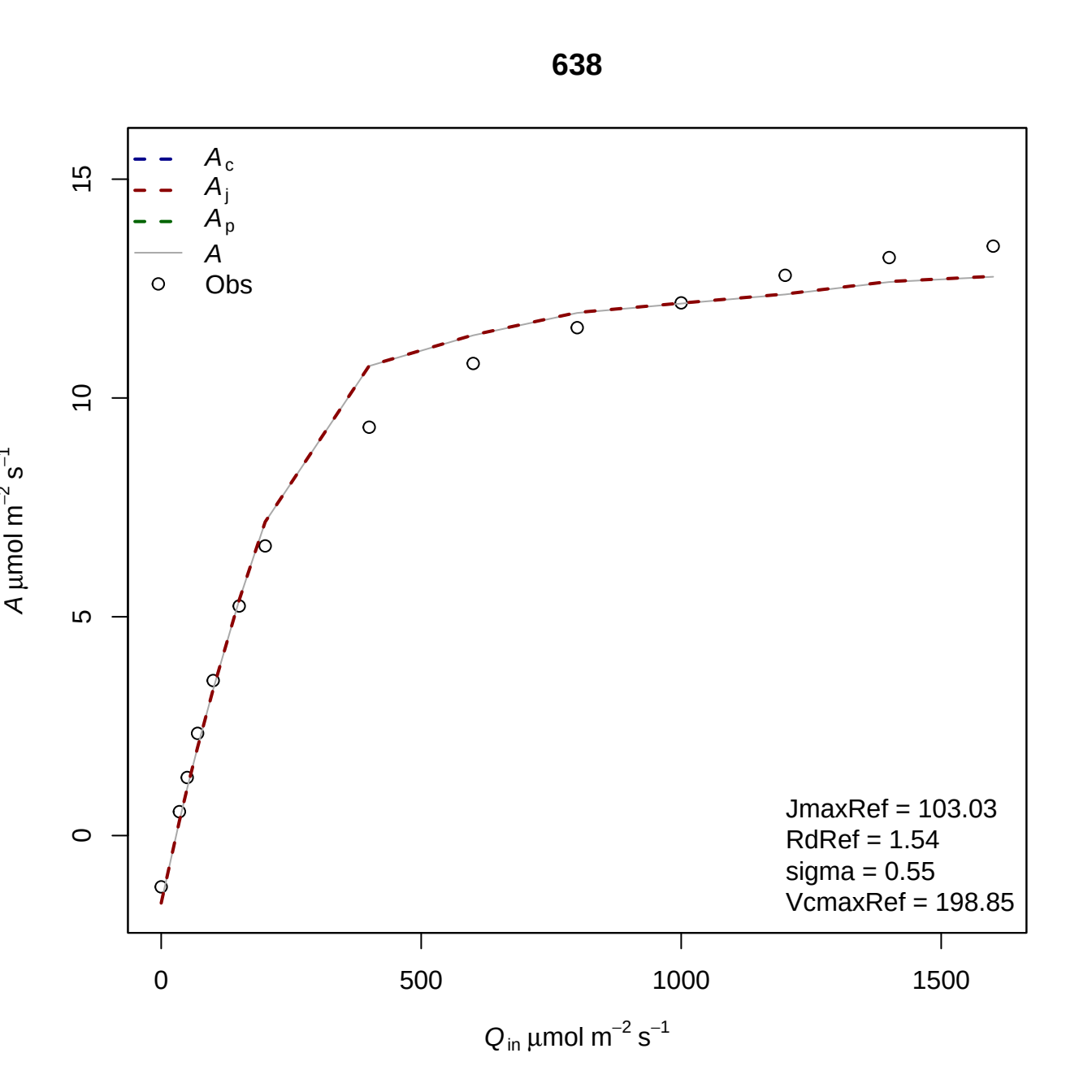

633
A_c
A_j
A_p
A
Obs
JmaxRef = 195.66
RdRef = 4.22
sigma = 1.03
VcmaxRef = 867.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

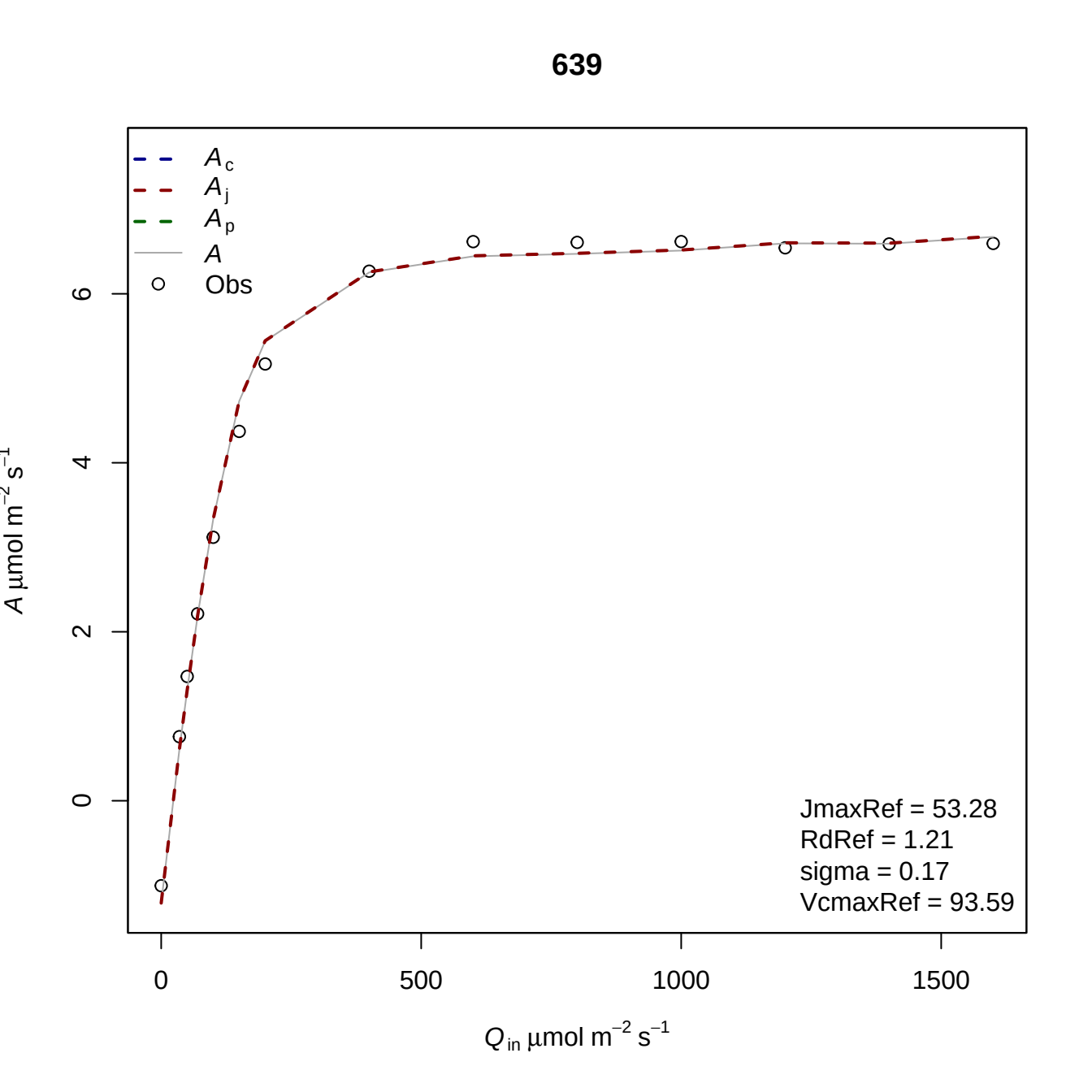

634
A_c
A_j
A_p
A
Obs
JmaxRef = 188.02
RdRef = 2.08
sigma = 0.44
VcmaxRef = 730.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

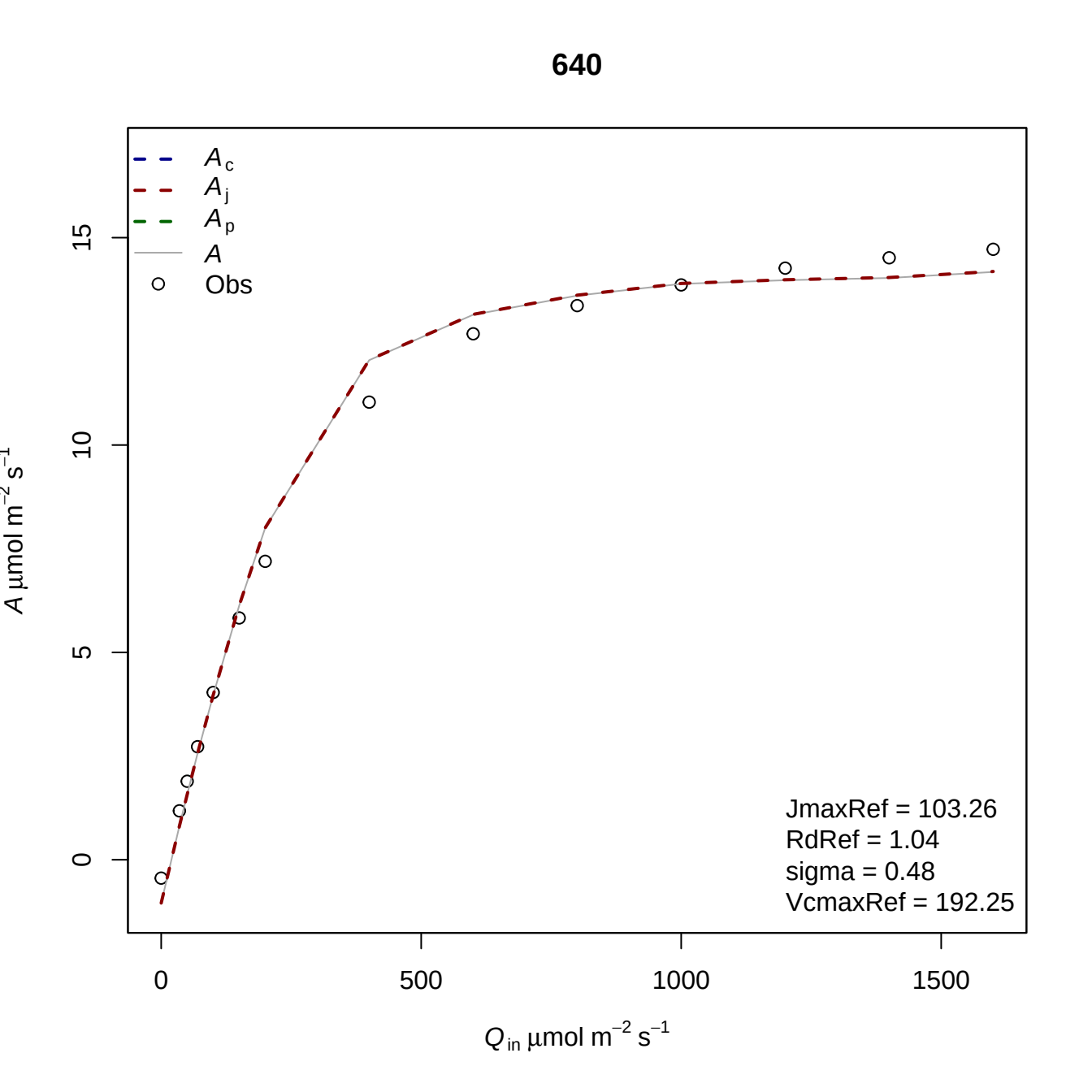

635
A_c
A_j
A_p
A
Obs
JmaxRef = 22.65
RdRef = 1.07
sigma = 0.06
VcmaxRef = 10467.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

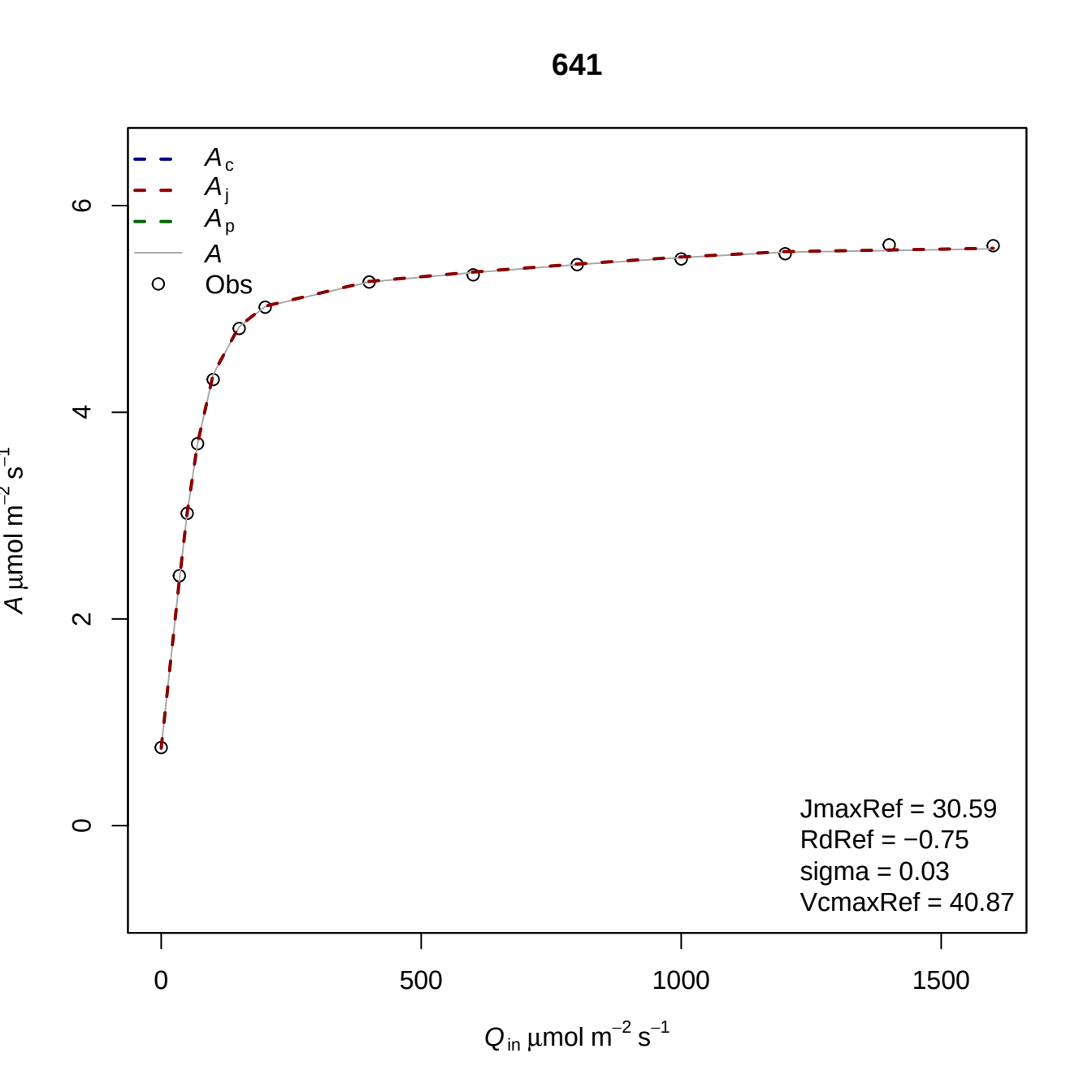

636
A_c
A_j
A_p
A
Obs
JmaxRef = 7.91
RdRef = 0.68
sigma = 0.84
VcmaxRef = 33823.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

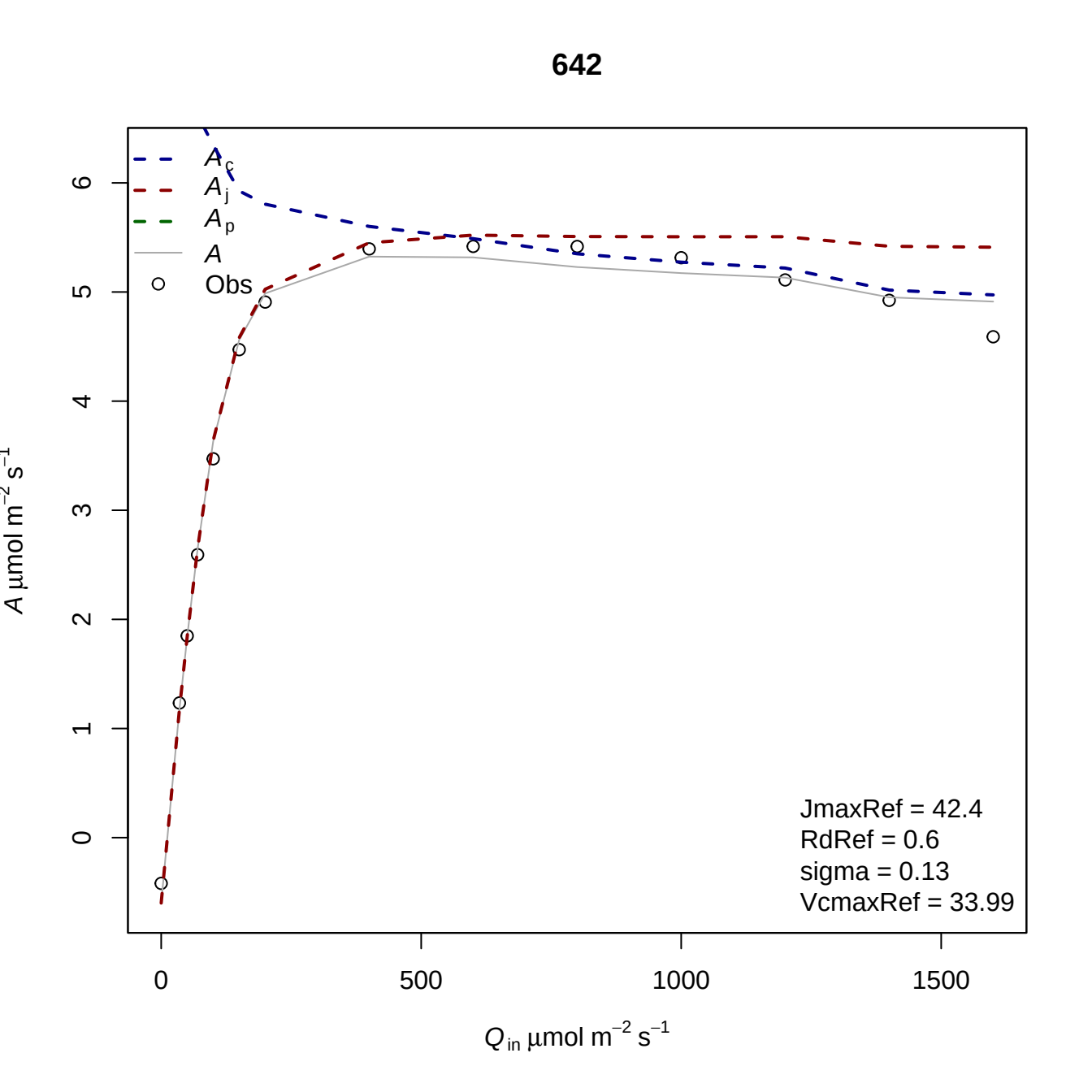

637
A_c
A_j
A_p
A
Obs
JmaxRef = 26.88
RdRef = 0.98
sigma = 0.28
VcmaxRef = 62.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

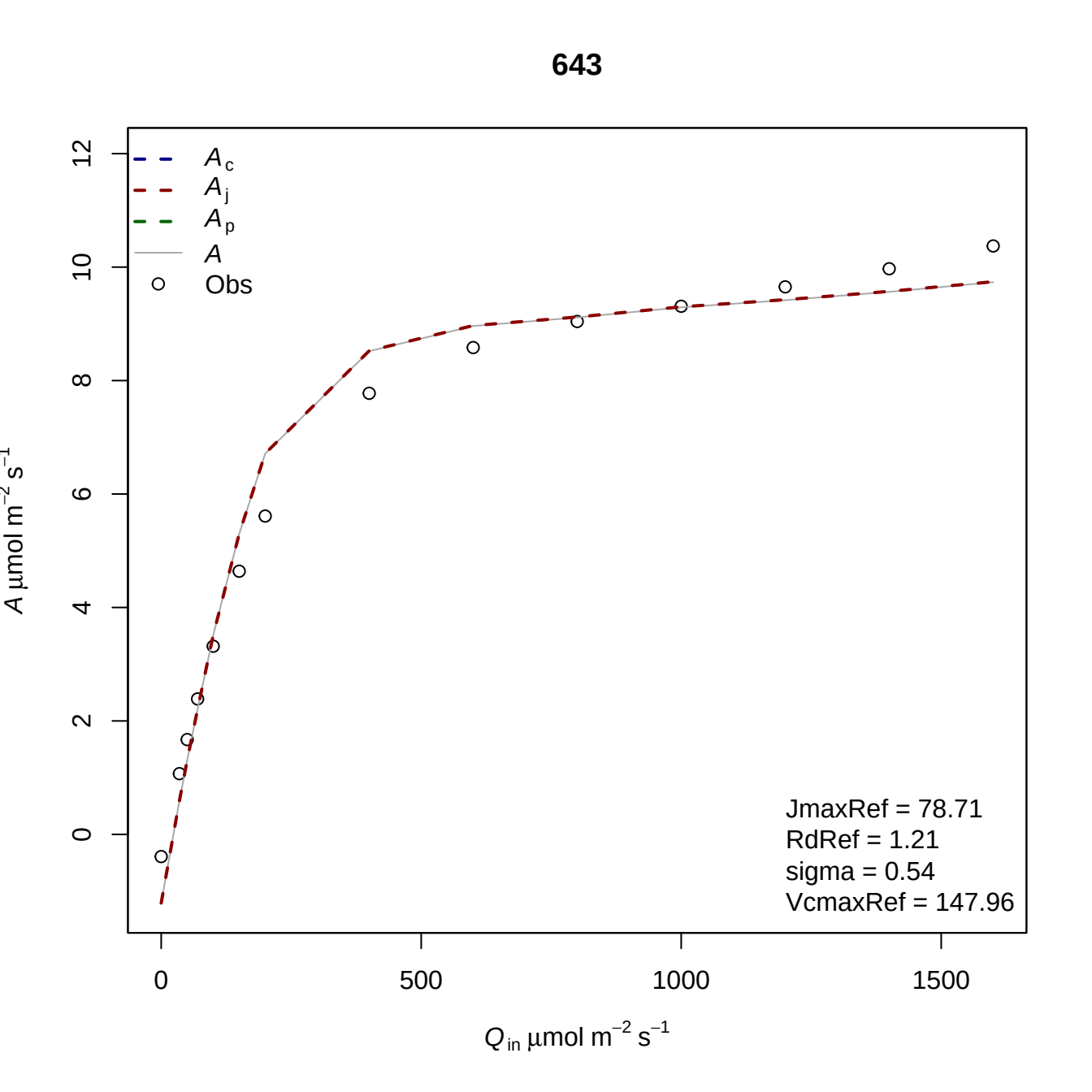

638
A_c
A_j
A_p
A
Obs
JmaxRef = 103.03
RdRef = 1.54
sigma = 0.55
VcmaxRef = 198.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

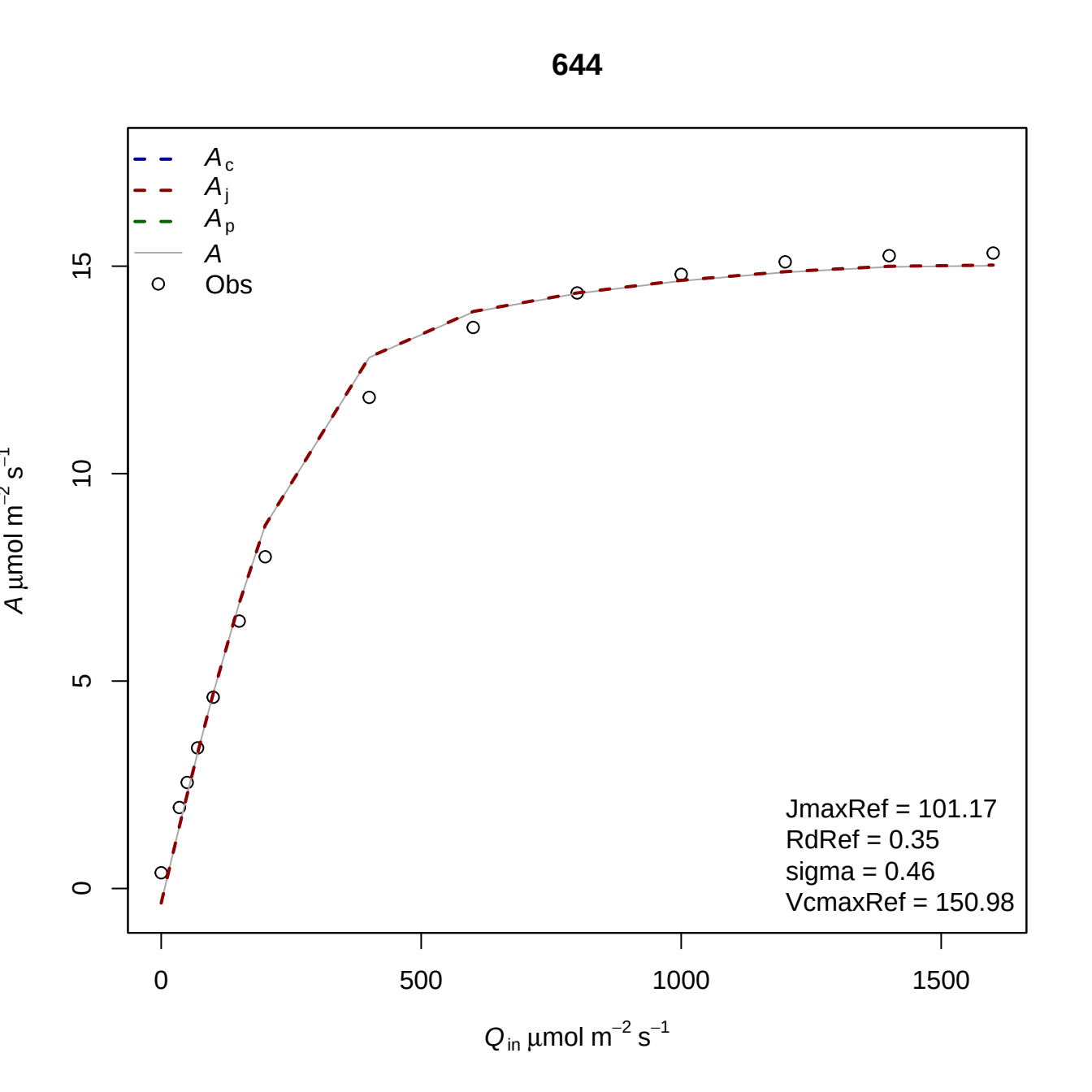

639
A_c
A_j
A_p
A
Obs
JmaxRef = 53.28
RdRef = 1.21
sigma = 0.17
VcmaxRef = 93.59
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

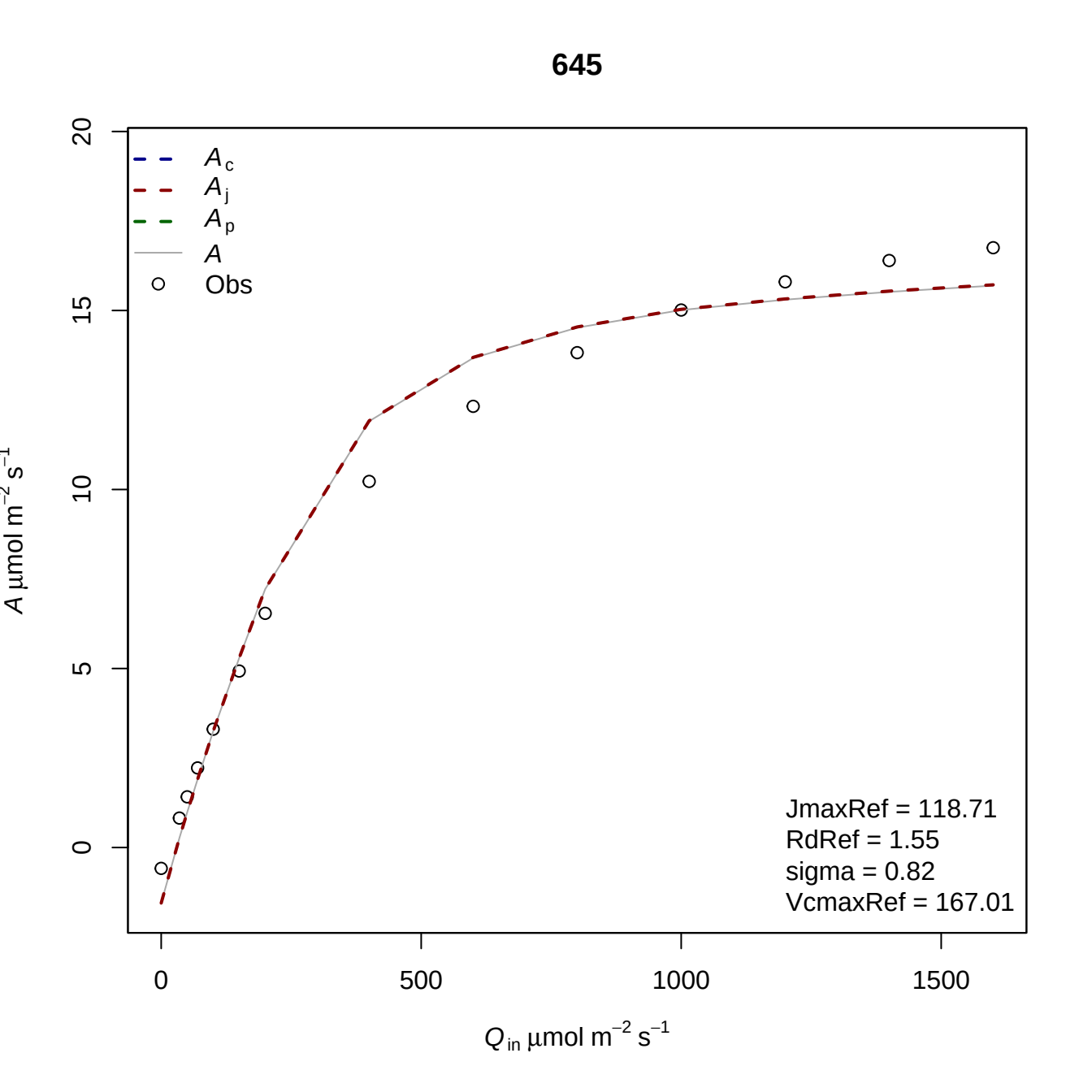

640
A_c
A_j
A_p
A
Obs
JmaxRef = 103.26
RdRef = 1.04
sigma = 0.48
VcmaxRef = 192.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

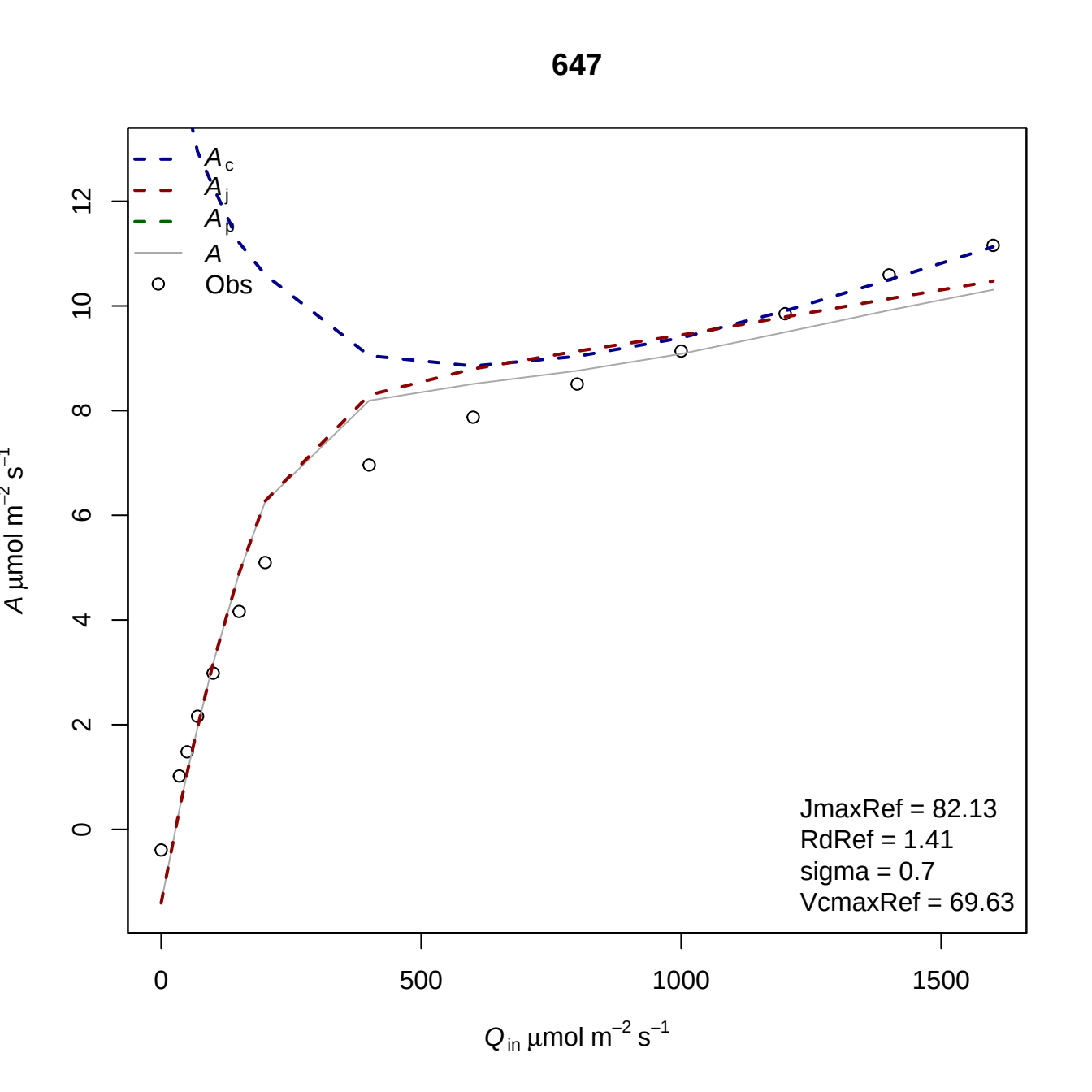

641
A_c
A_j
A_p
A
Obs
JmaxRef = 30.59
RdRef = −0.75
sigma = 0.03
VcmaxRef = 40.87
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

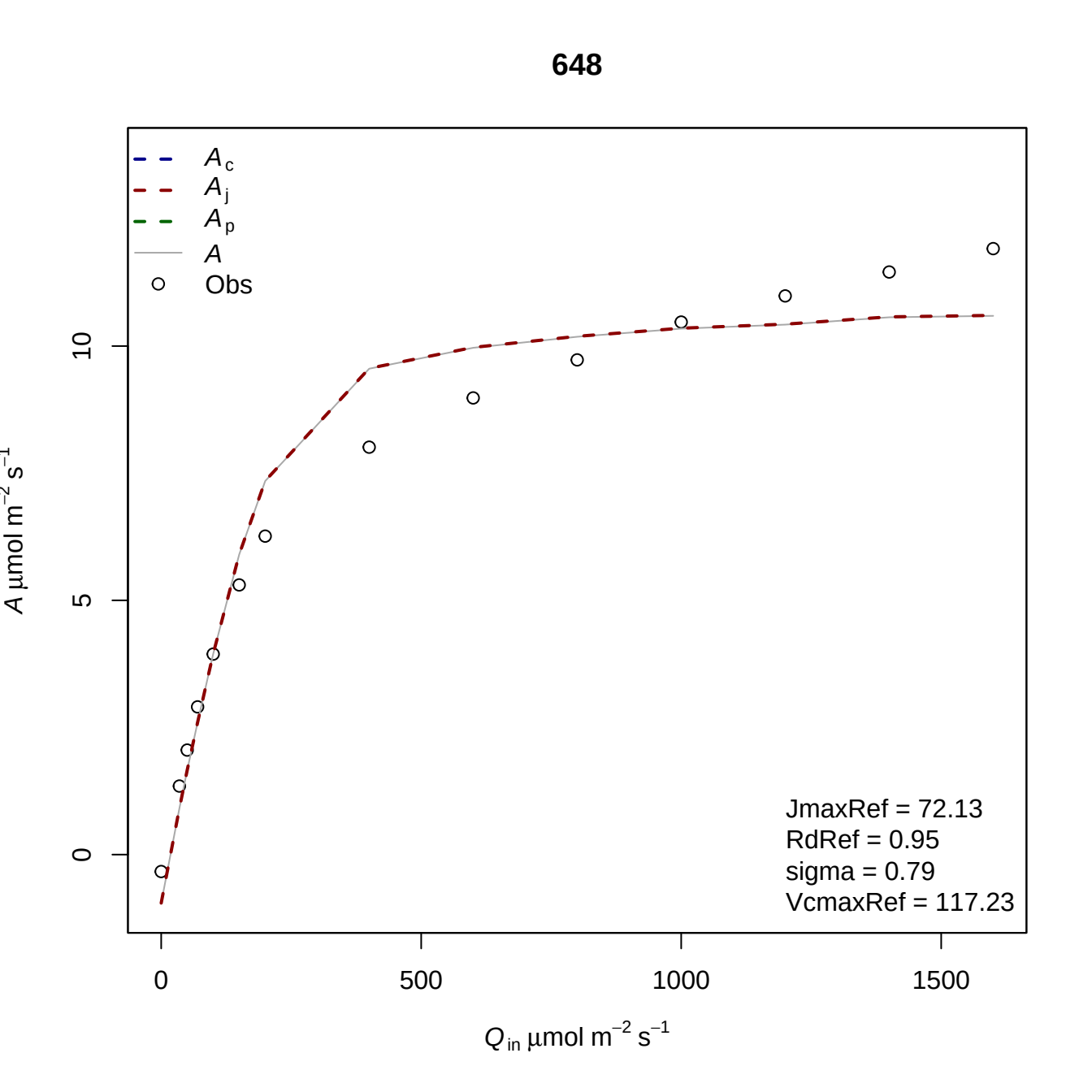

642
A_c
A_j
A_p
A
Obs
JmaxRef = 42.4
RdRef = 0.6
sigma = 0.13
VcmaxRef = 33.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

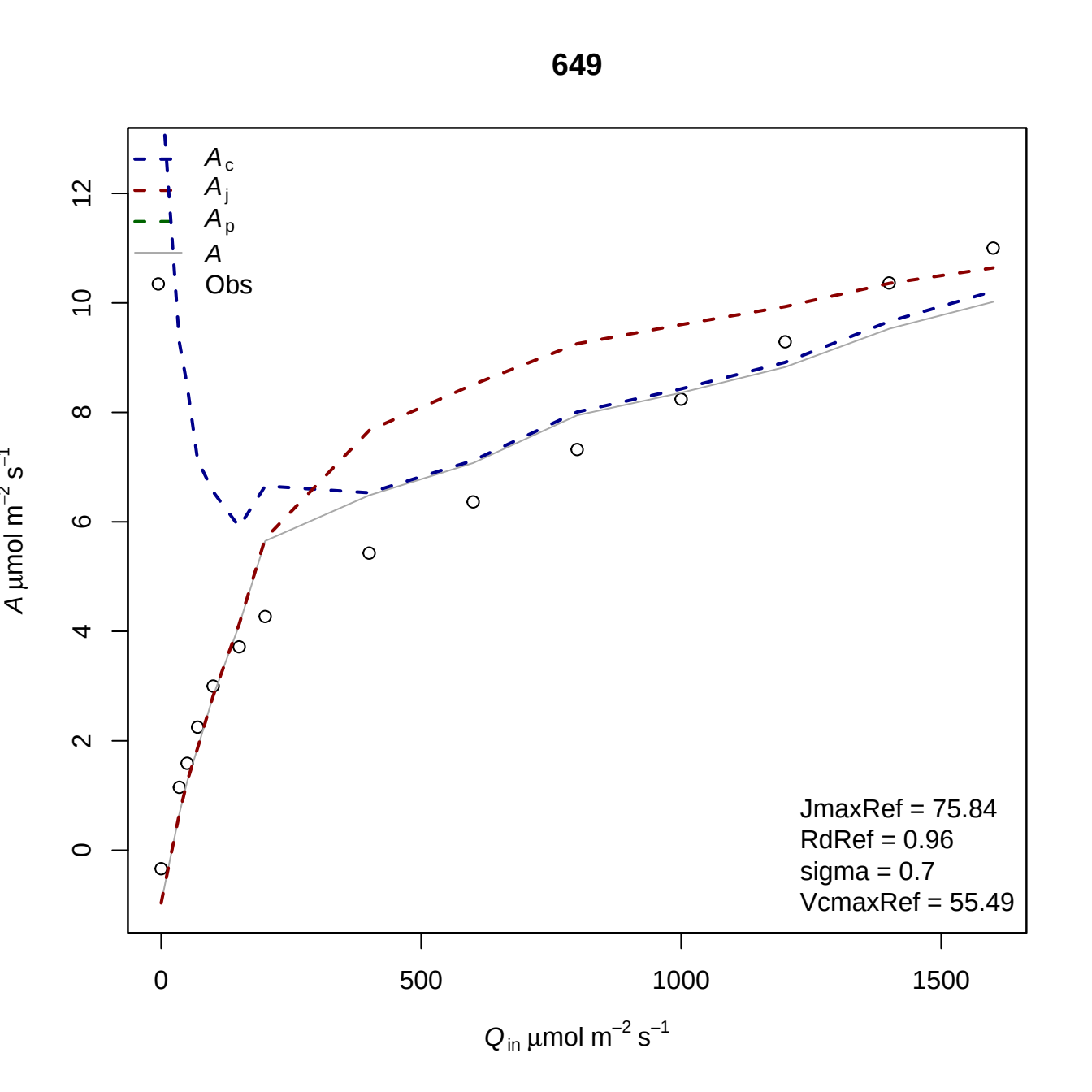

643
A_c
A_j
A_p
A
Obs
JmaxRef = 78.71
RdRef = 1.21
sigma = 0.54
VcmaxRef = 147.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

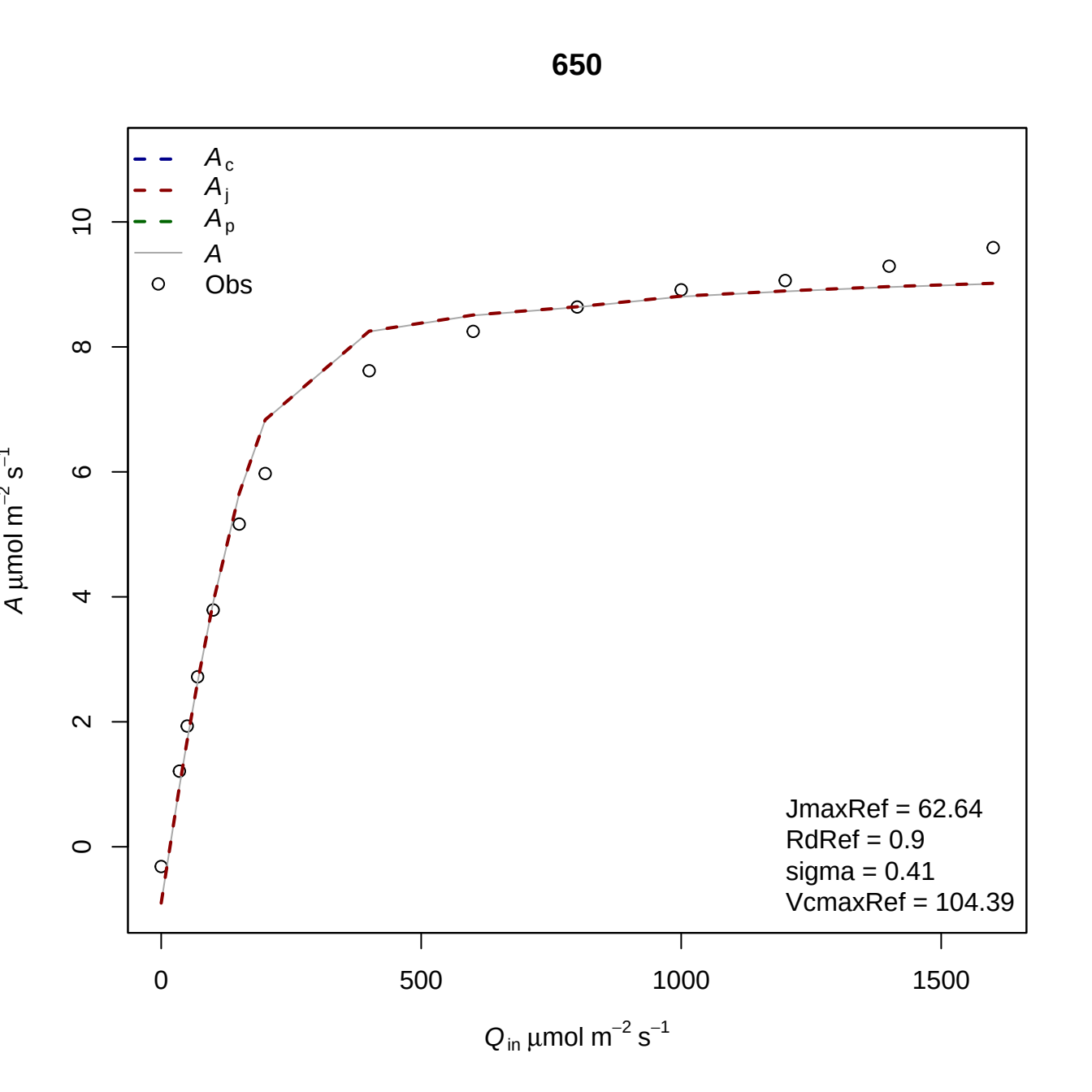

644
A_c
A_j
A_p
A
Obs
JmaxRef = 101.17
RdRef = 0.35
sigma = 0.46
VcmaxRef = 150.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

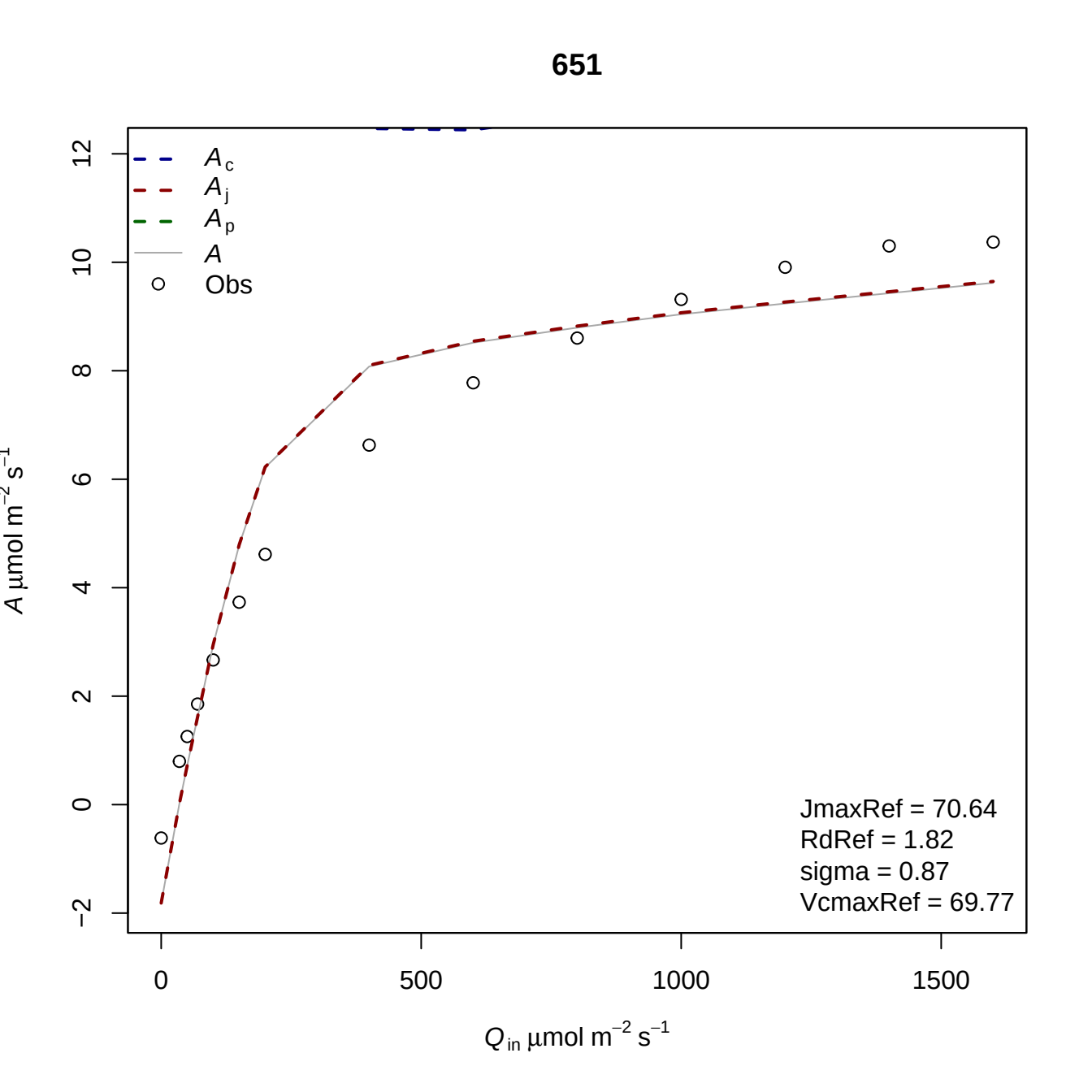

645
A_c
A_j
A_p
A
Obs
JmaxRef = 118.71
RdRef = 1.55
sigma = 0.82
VcmaxRef = 167.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

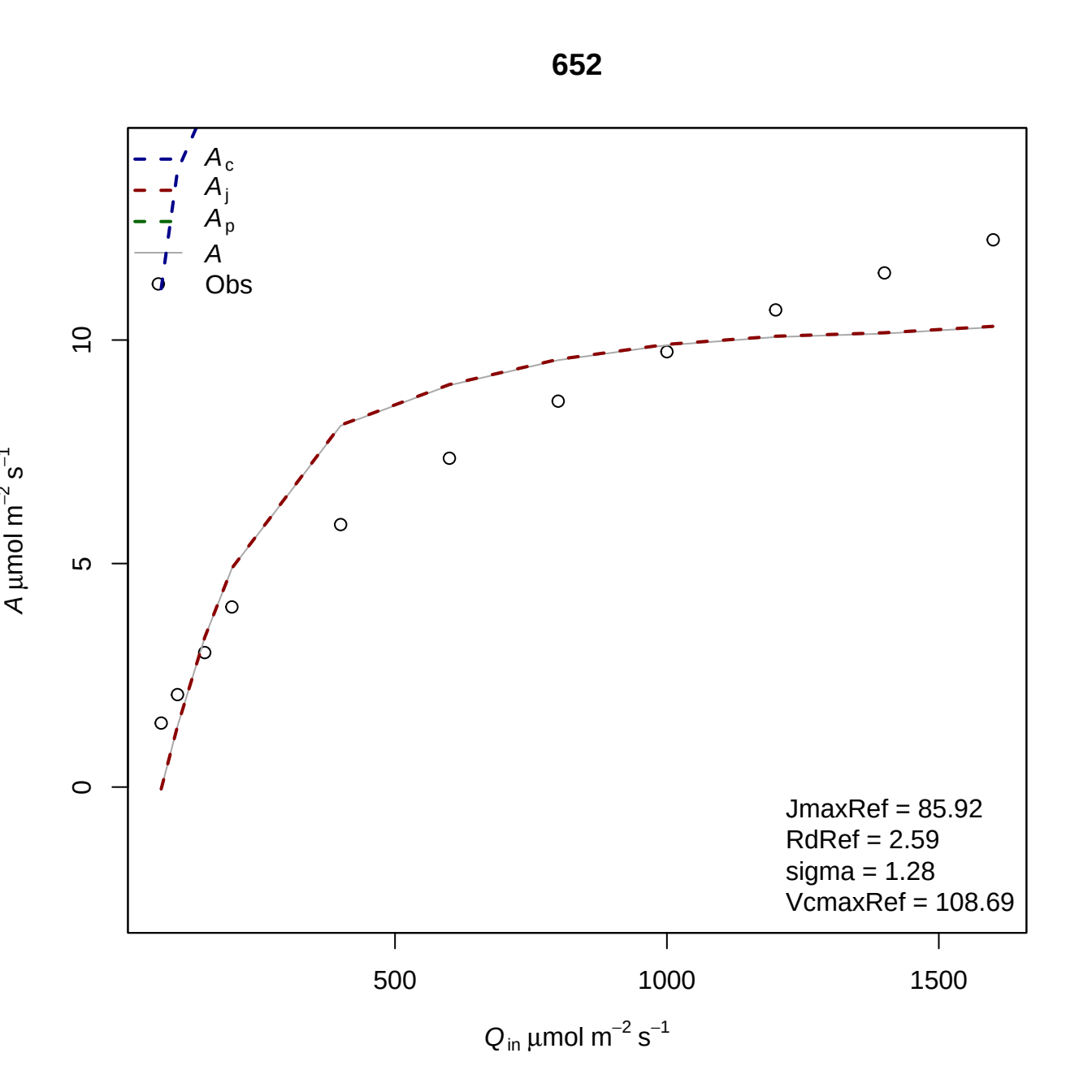

647
A_c
A_j
A_p
A
Obs
JmaxRef = 82.13
RdRef = 1.41
sigma = 0.7
VcmaxRef = 69.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

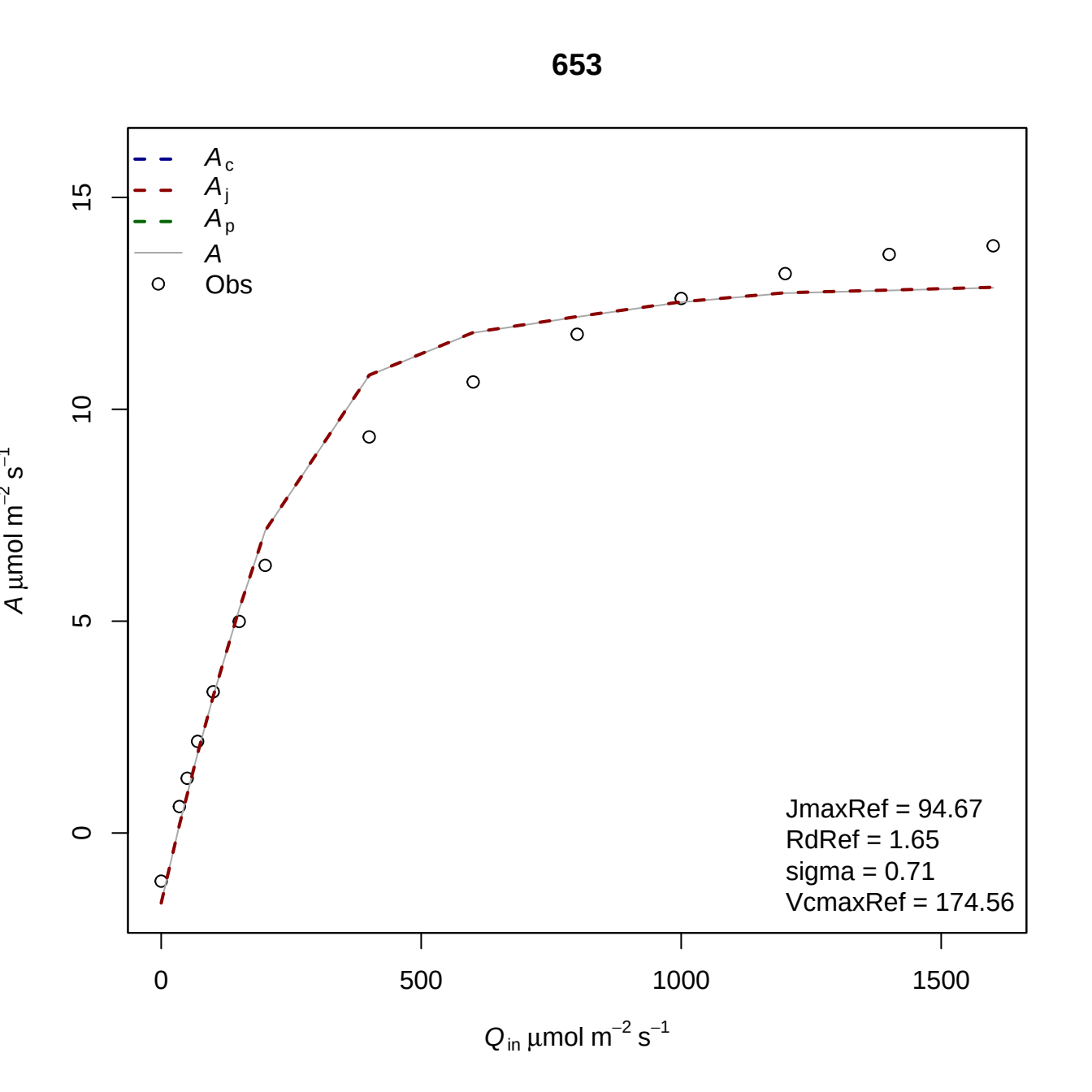

648
A_c
A_j
A_p
A
Obs
JmaxRef = 72.13
RdRef = 0.95
sigma = 0.79
VcmaxRef = 117.23
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

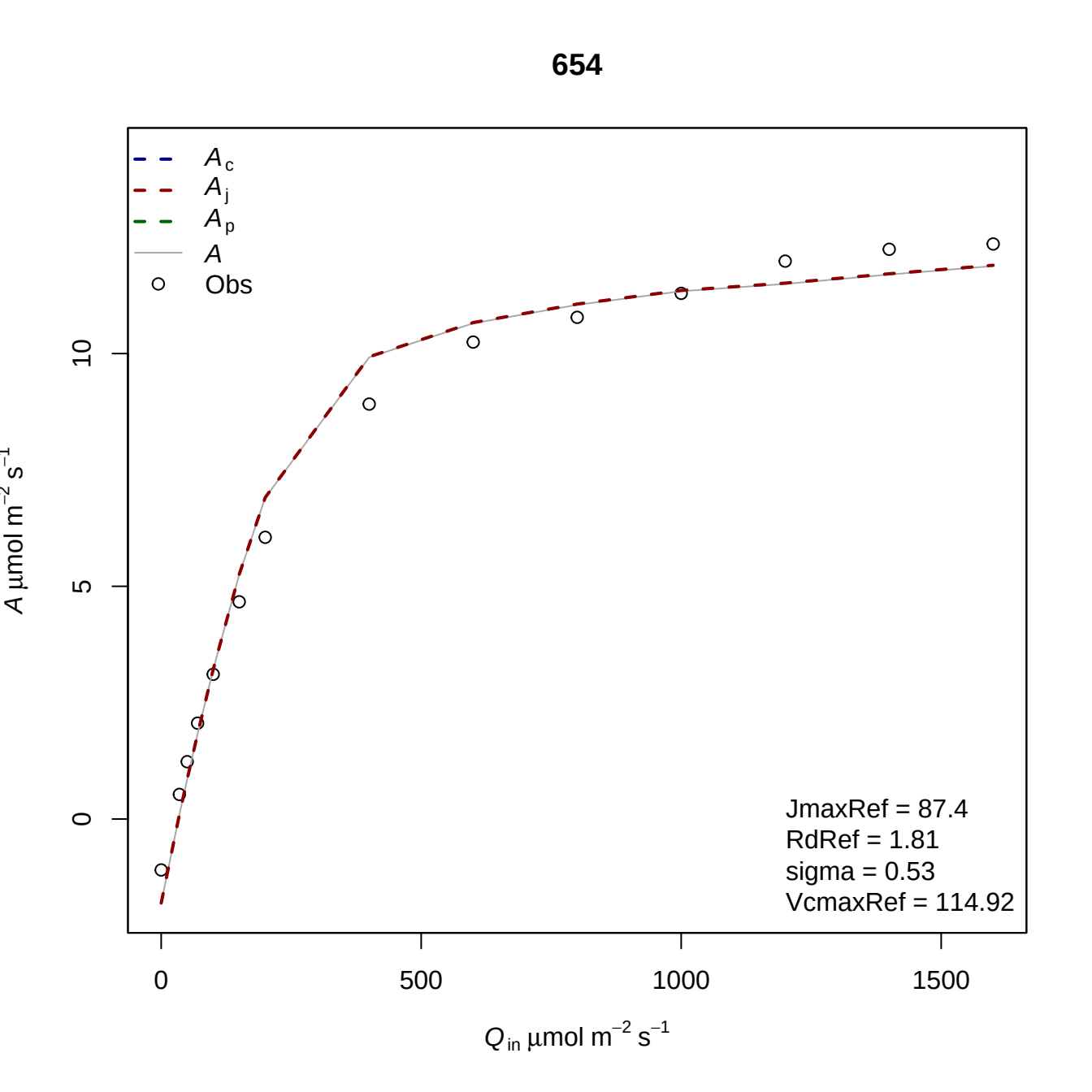

649
A_c
A_j
A_p
A
Obs
JmaxRef = 75.84
RdRef = 0.96
sigma = 0.7
VcmaxRef = 55.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

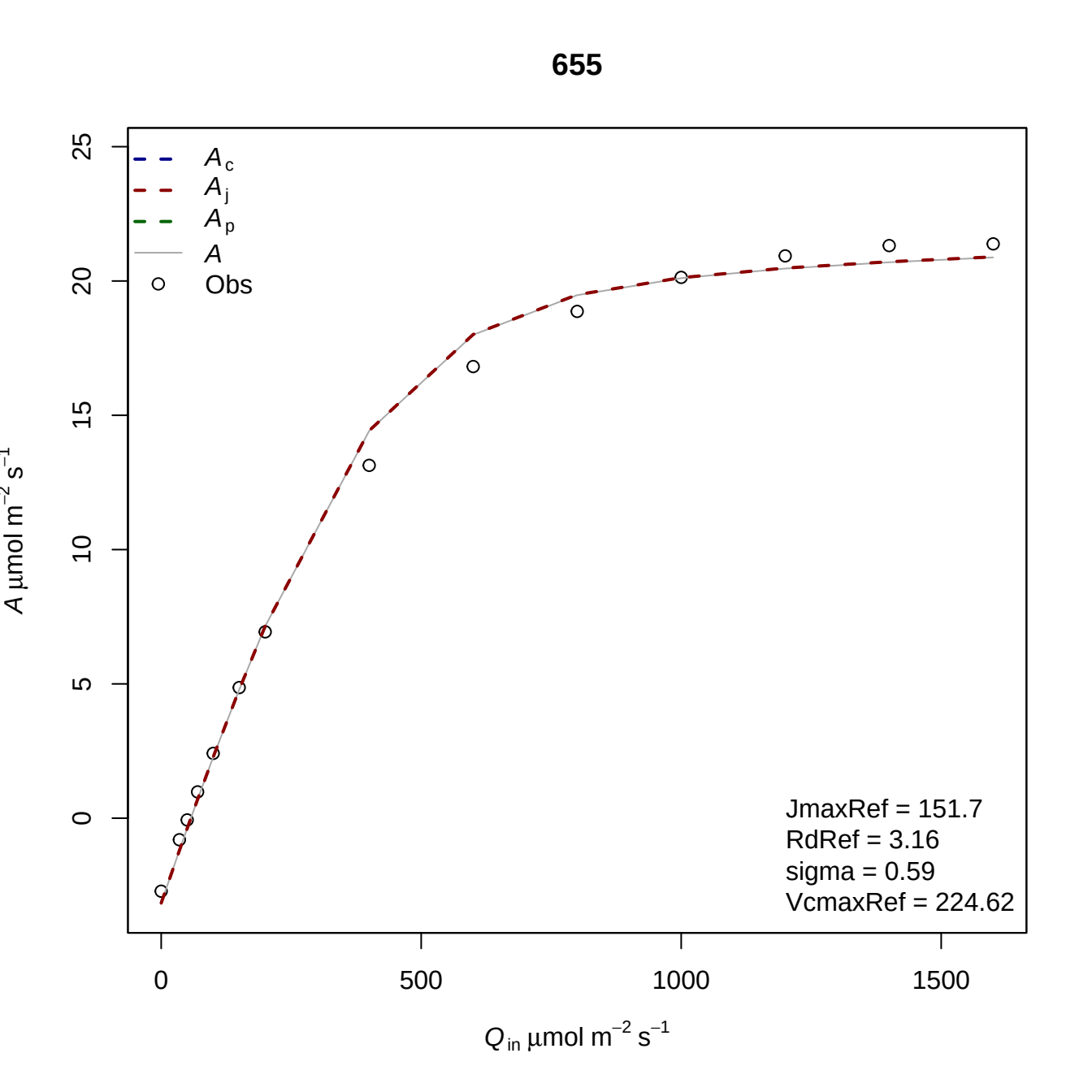

650
A_c
A_j
A_p
A
Obs
JmaxRef = 62.64
RdRef = 0.9
sigma = 0.41
VcmaxRef = 104.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

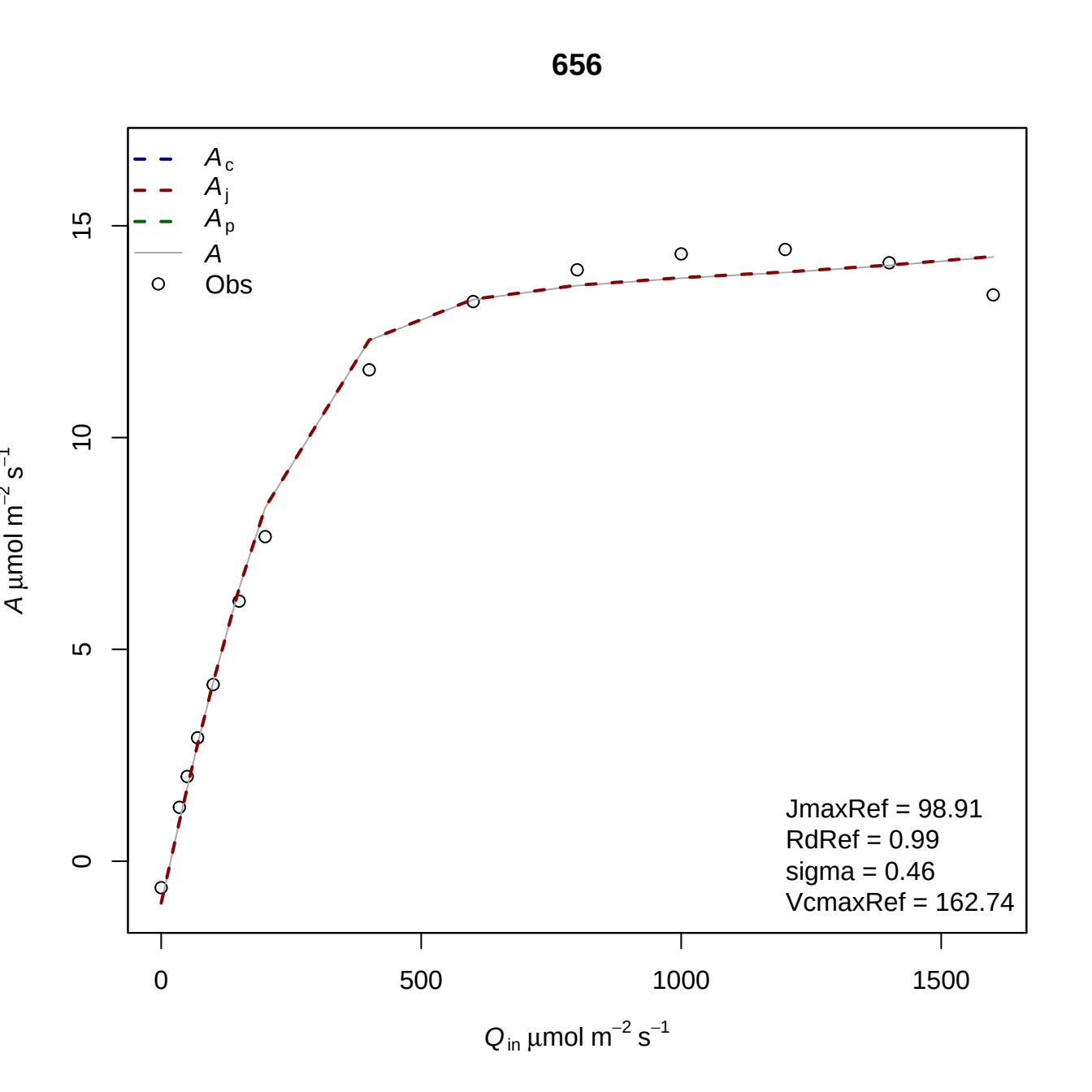

651
A_c
A_j
A_p
A
Obs
JmaxRef = 70.64
RdRef = 1.82
sigma = 0.87
VcmaxRef = 69.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

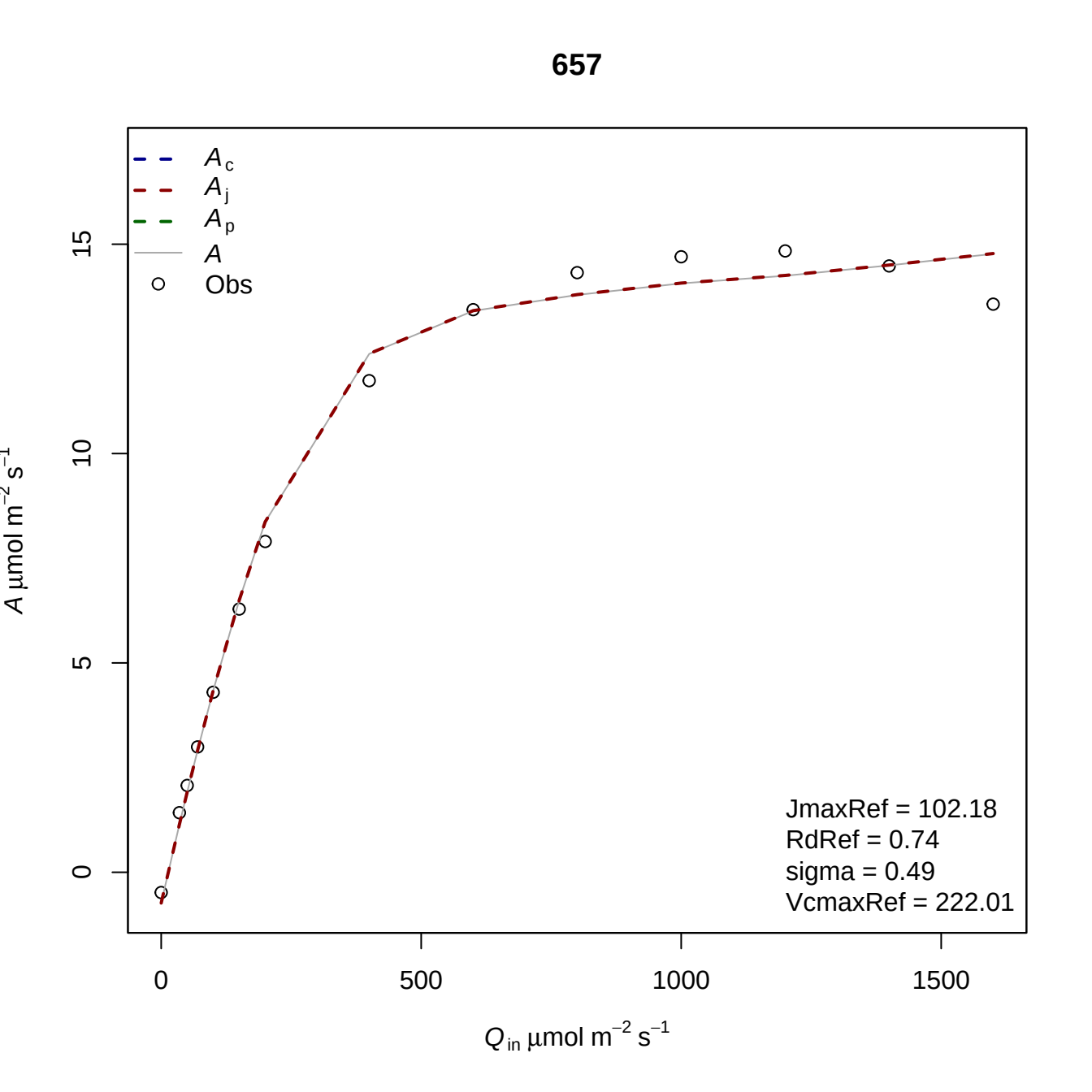

652
A_c
A_j
A_p
A
Obs
JmaxRef = 85.92
RdRef = 2.59
sigma = 1.28
VcmaxRef = 108.69
0
5
10
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

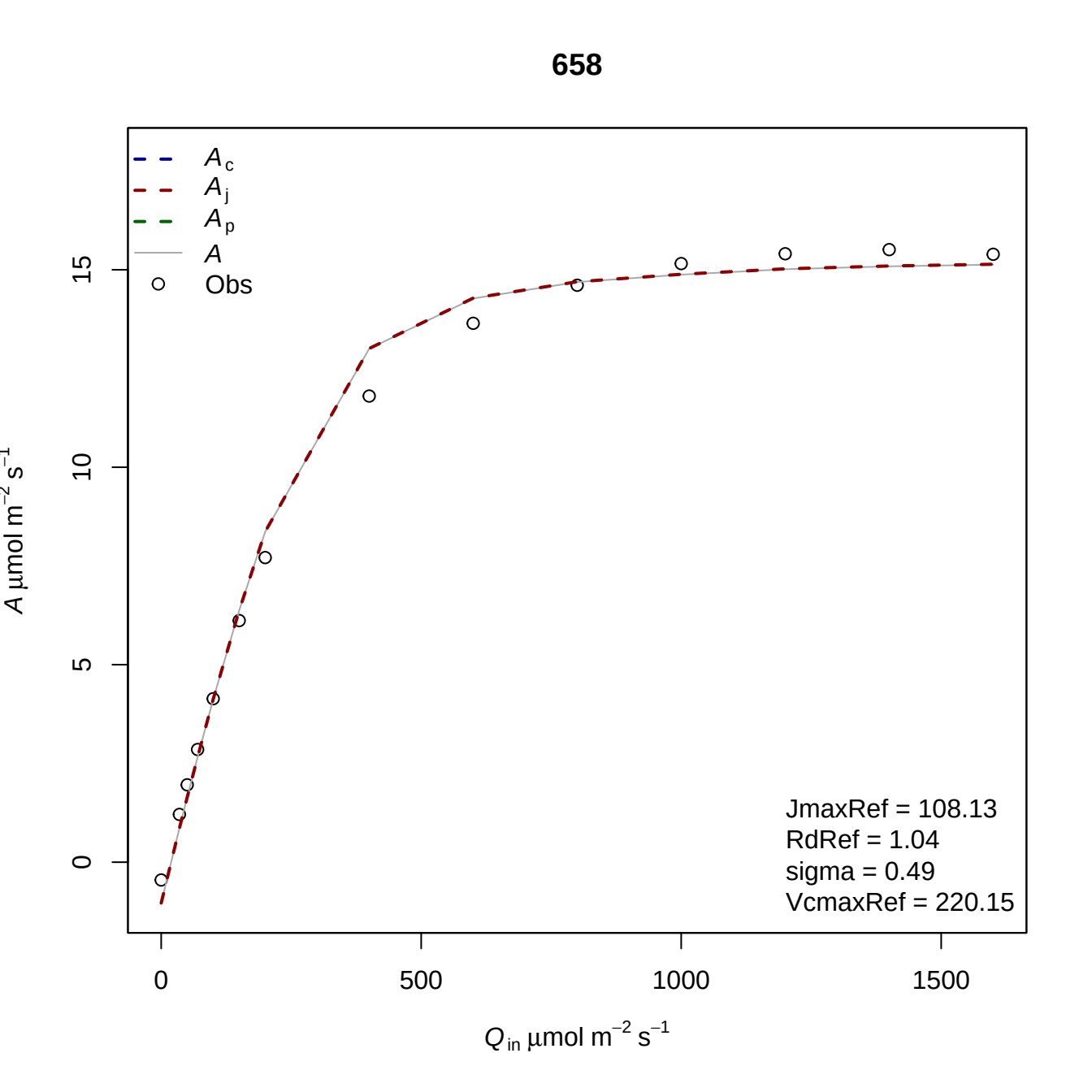

653
A_c
A_j
A_p
A
Obs
JmaxRef = 94.67
RdRef = 1.65
sigma = 0.71
VcmaxRef = 174.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

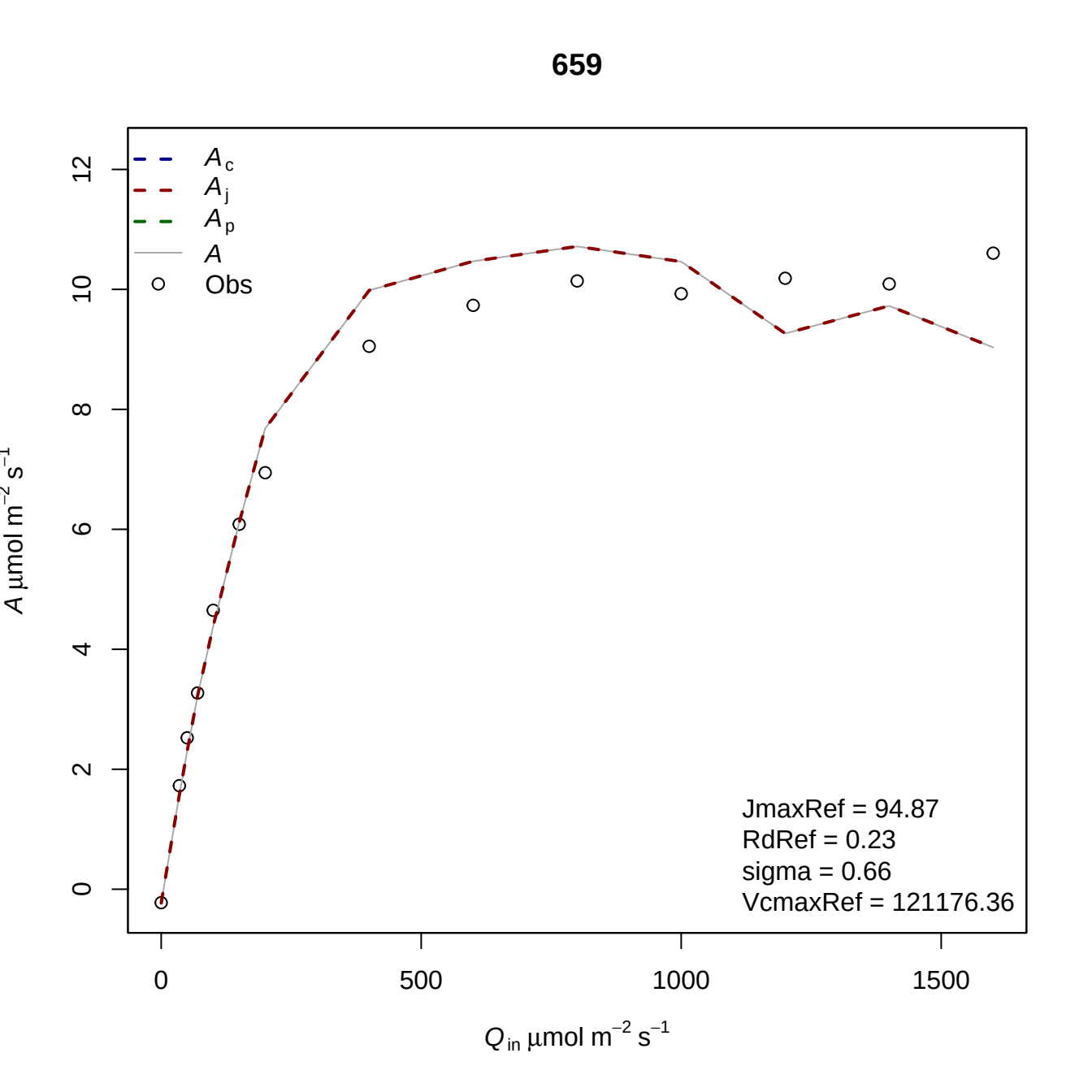

654
A_c
A_j
A_p
A
Obs
JmaxRef = 87.4
RdRef = 1.81
sigma = 0.53
VcmaxRef = 114.92
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

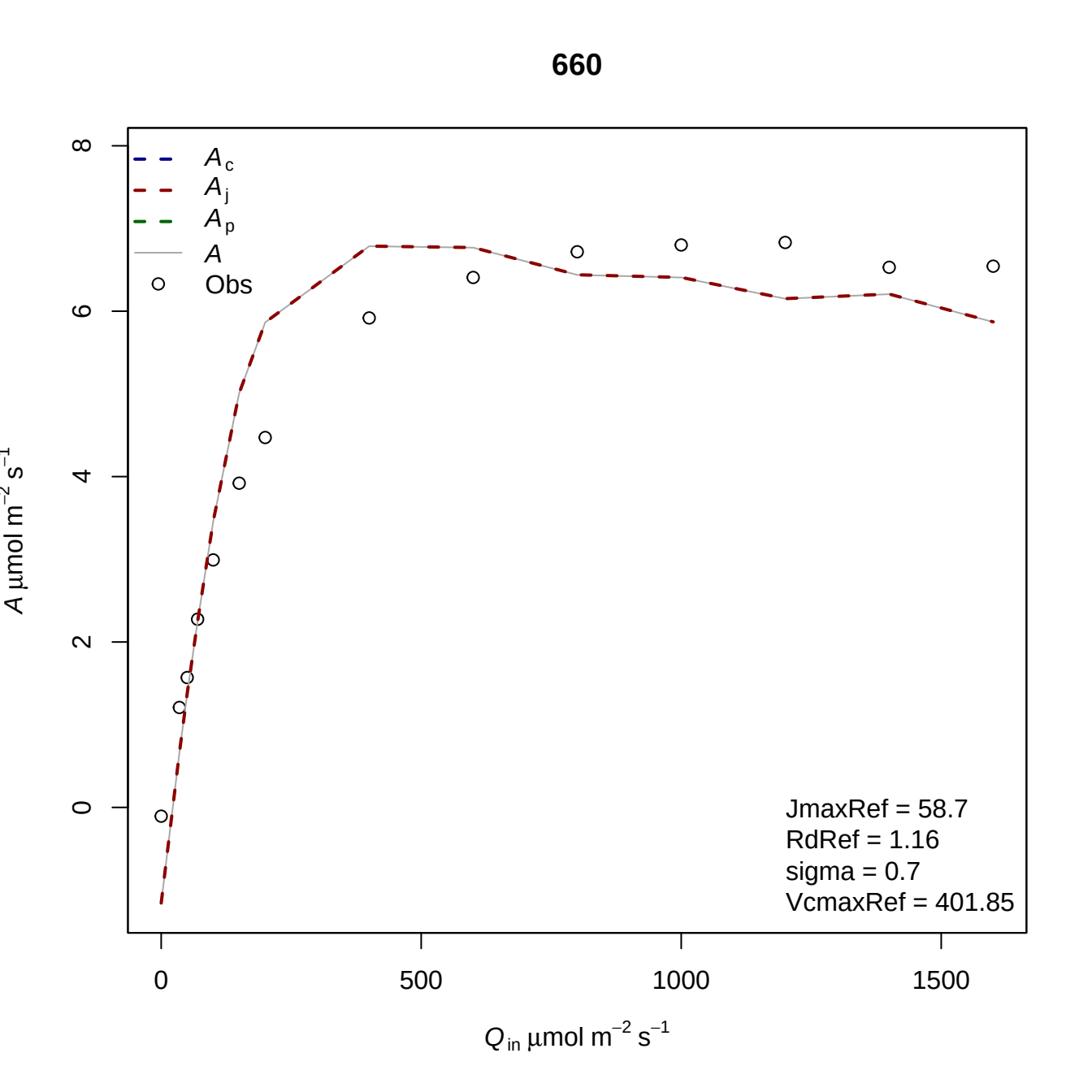

655
A_c
A_j
A_p
A
Obs
JmaxRef = 151.7
RdRef = 3.16
sigma = 0.59
VcmaxRef = 224.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

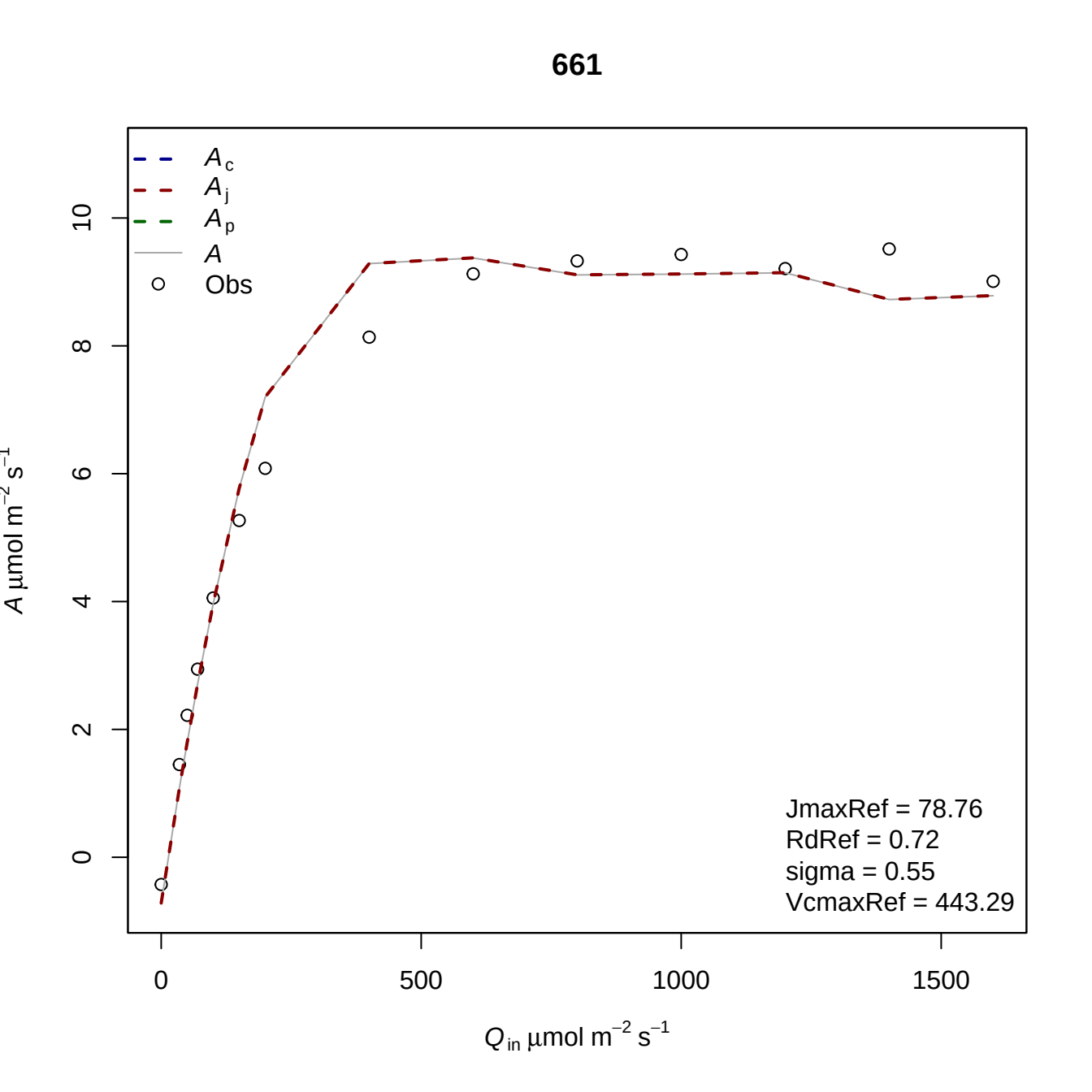

656
A_c
A_j
A_p
A
Obs
JmaxRef = 98.91
RdRef = 0.99
sigma = 0.46
VcmaxRef = 162.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

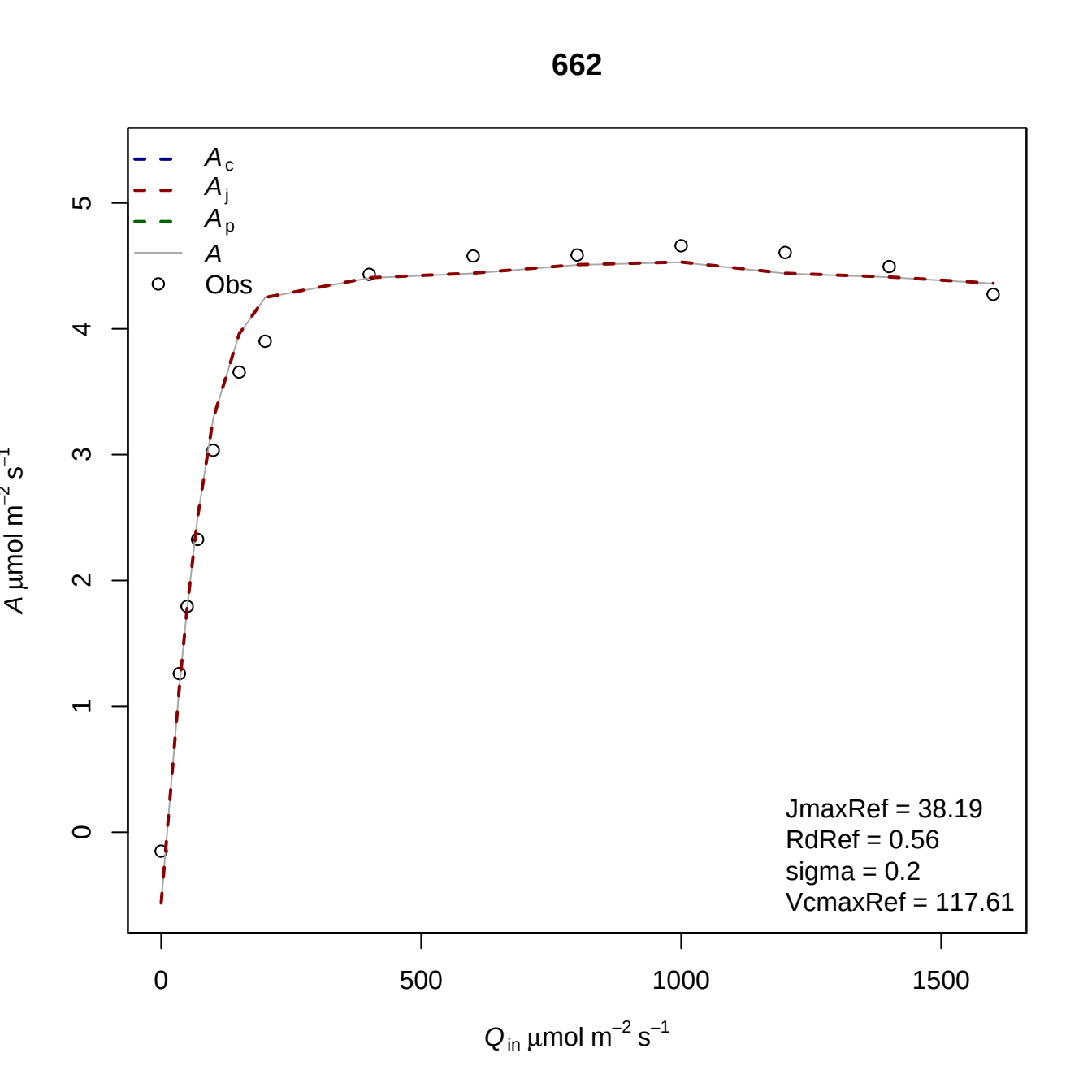

657
A_c
A_j
A_p
A
Obs
JmaxRef = 102.18
RdRef = 0.74
sigma = 0.49
VcmaxRef = 222.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

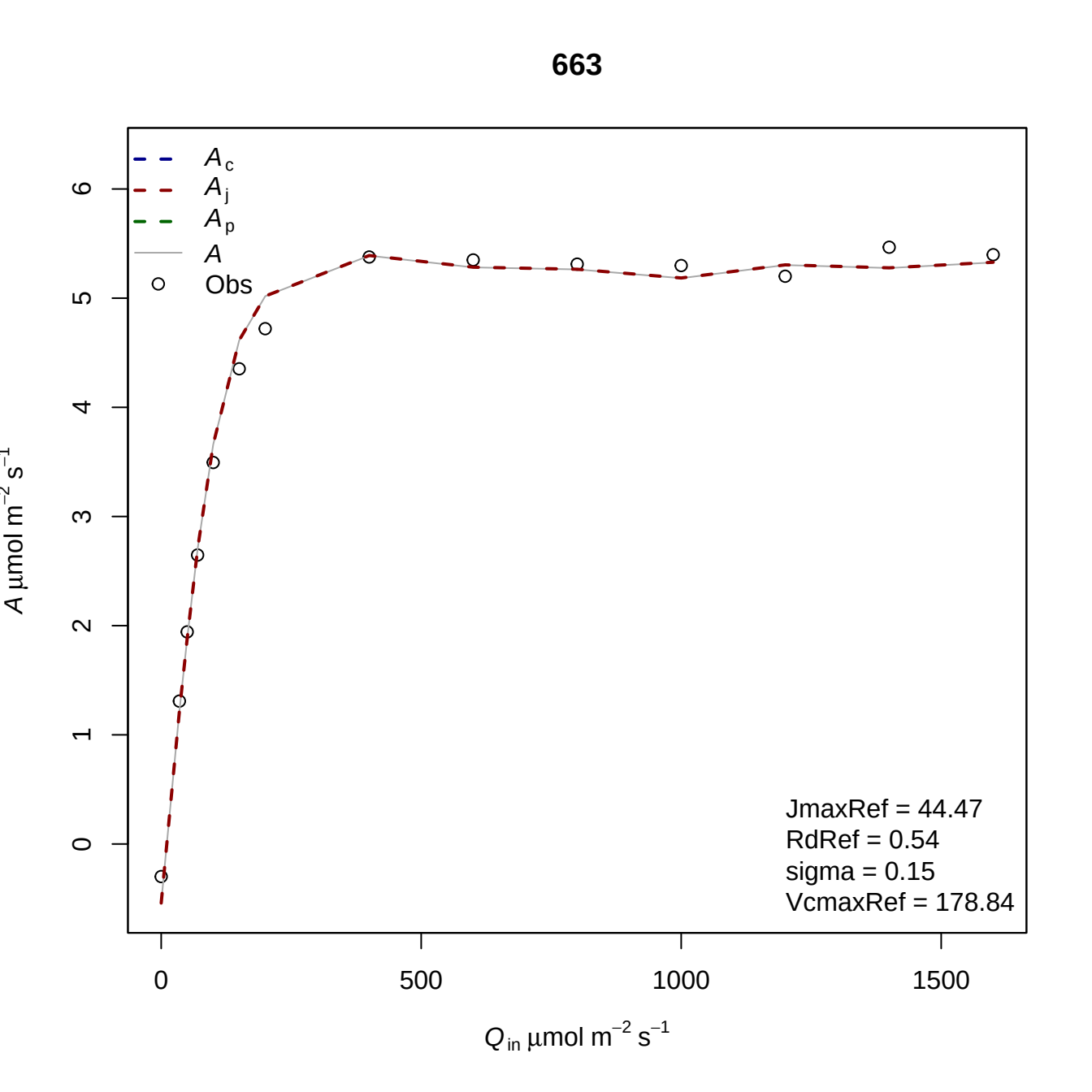

658
A_c
A_j
A_p
A
Obs
JmaxRef = 108.13
RdRef = 1.04
sigma = 0.49
VcmaxRef = 220.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

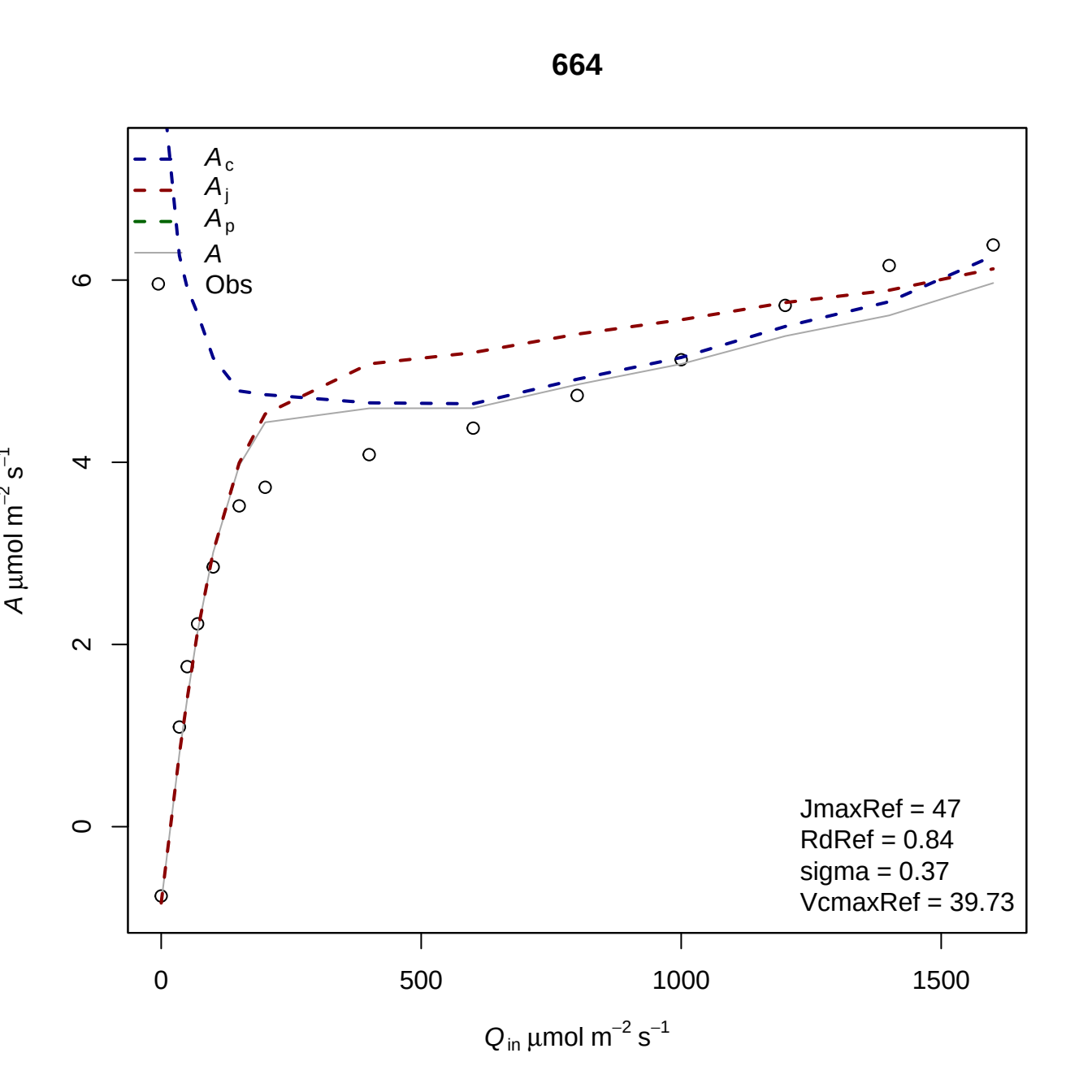

659
A_c
A_j
A_p
A
Obs
JmaxRef = 94.87
RdRef = 0.23
sigma = 0.66
VcmaxRef = 121176.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

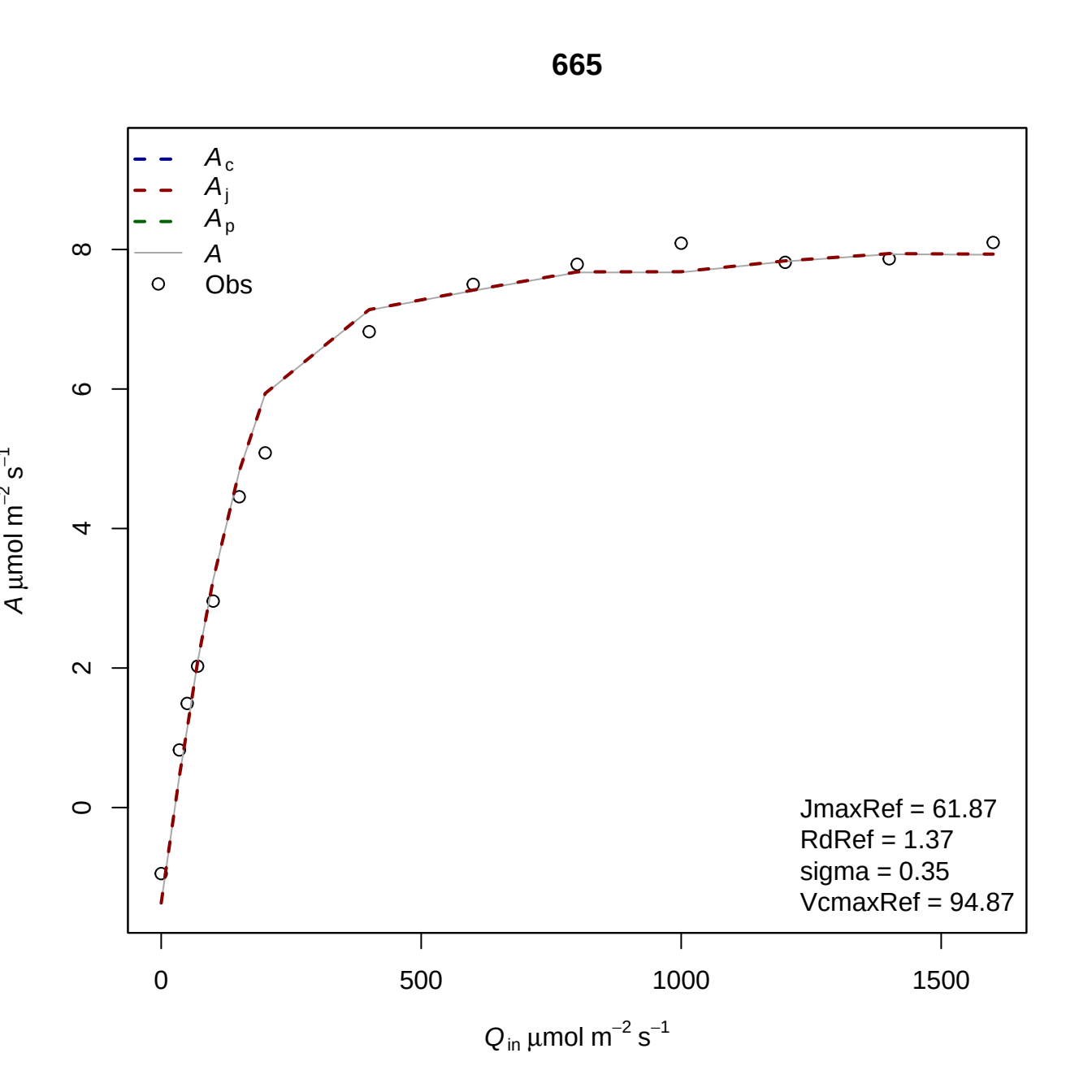

660
A_c
A_j
A_p
A
Obs
JmaxRef = 58.7
RdRef = 1.16
sigma = 0.7
VcmaxRef = 401.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

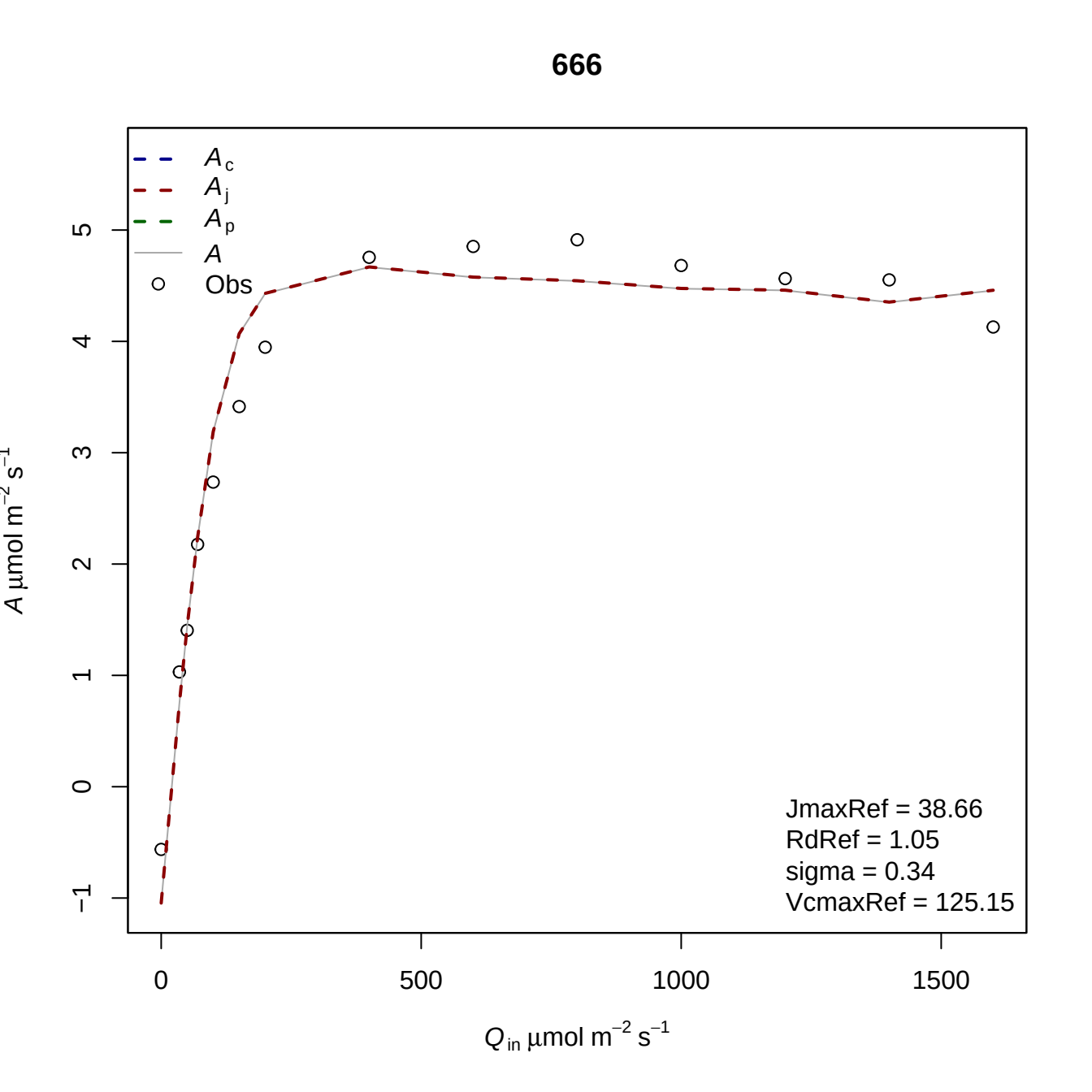

661
A_c
A_j
A_p
A
Obs
JmaxRef = 78.76
RdRef = 0.72
sigma = 0.55
VcmaxRef = 443.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

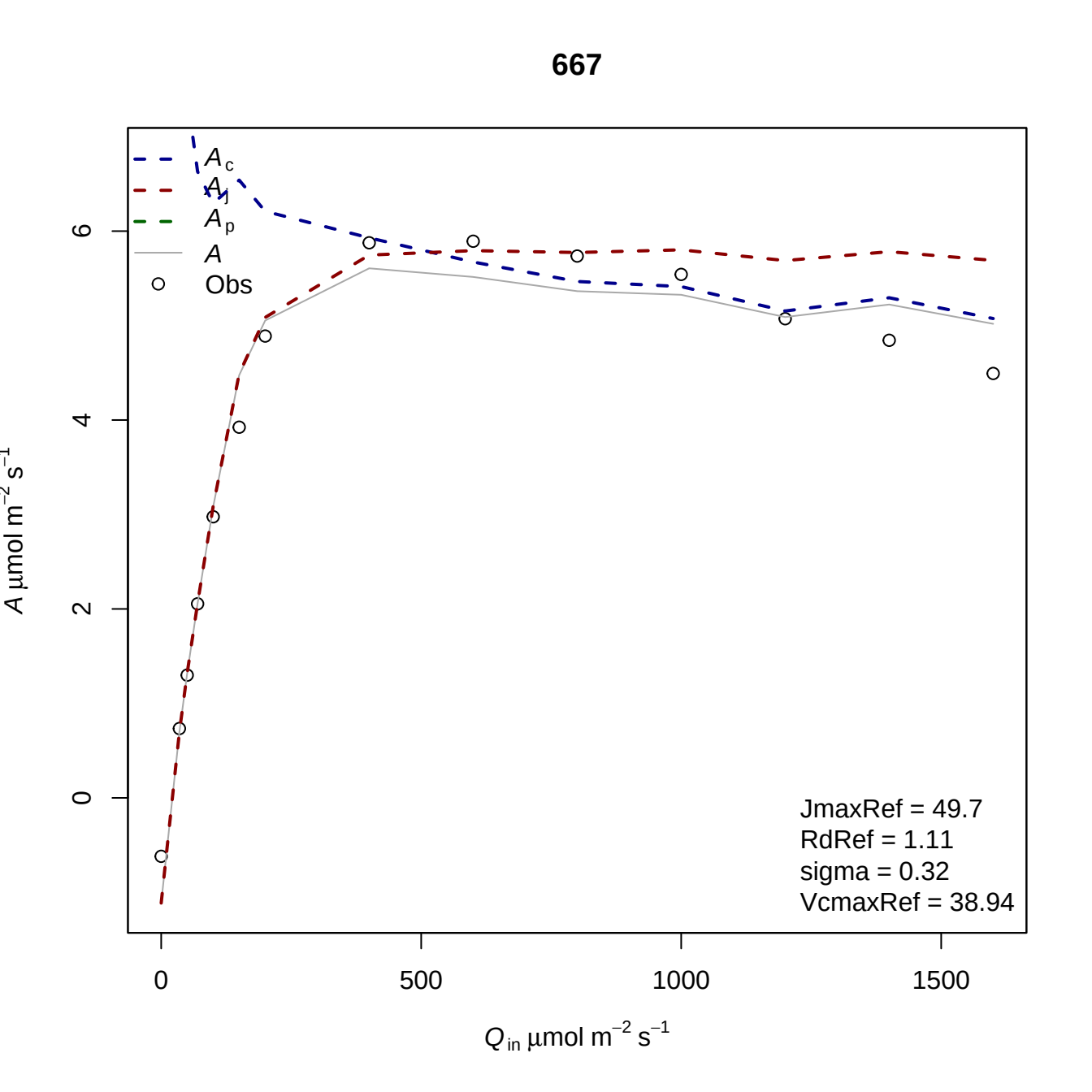

662
A_c
A_j
A_p
A
Obs
JmaxRef = 38.19
RdRef = 0.56
sigma = 0.2
VcmaxRef = 117.61
0
1
2
3
4
5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

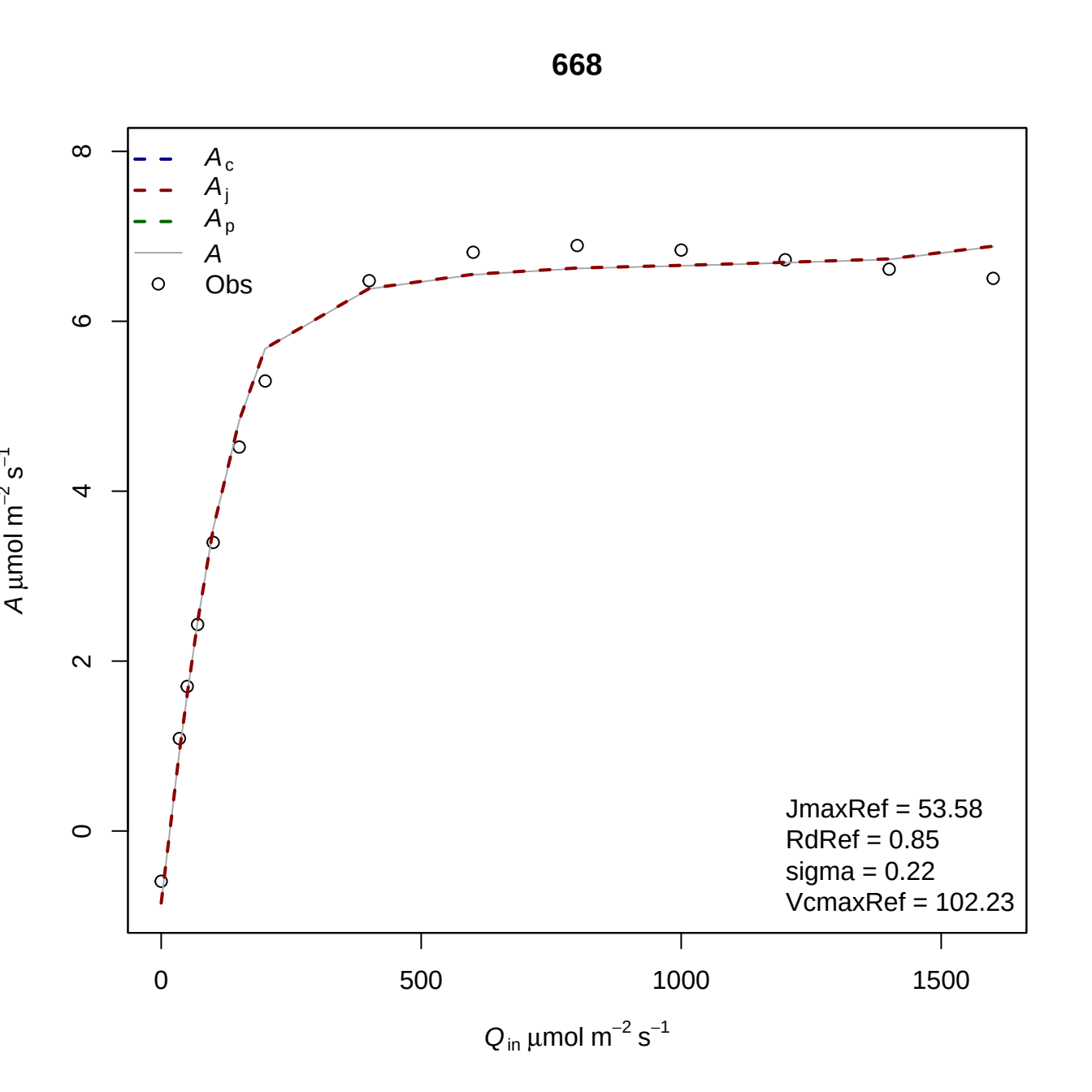

663
A_c
A_j
A_p
A
Obs
JmaxRef = 44.47
RdRef = 0.54
sigma = 0.15
VcmaxRef = 178.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

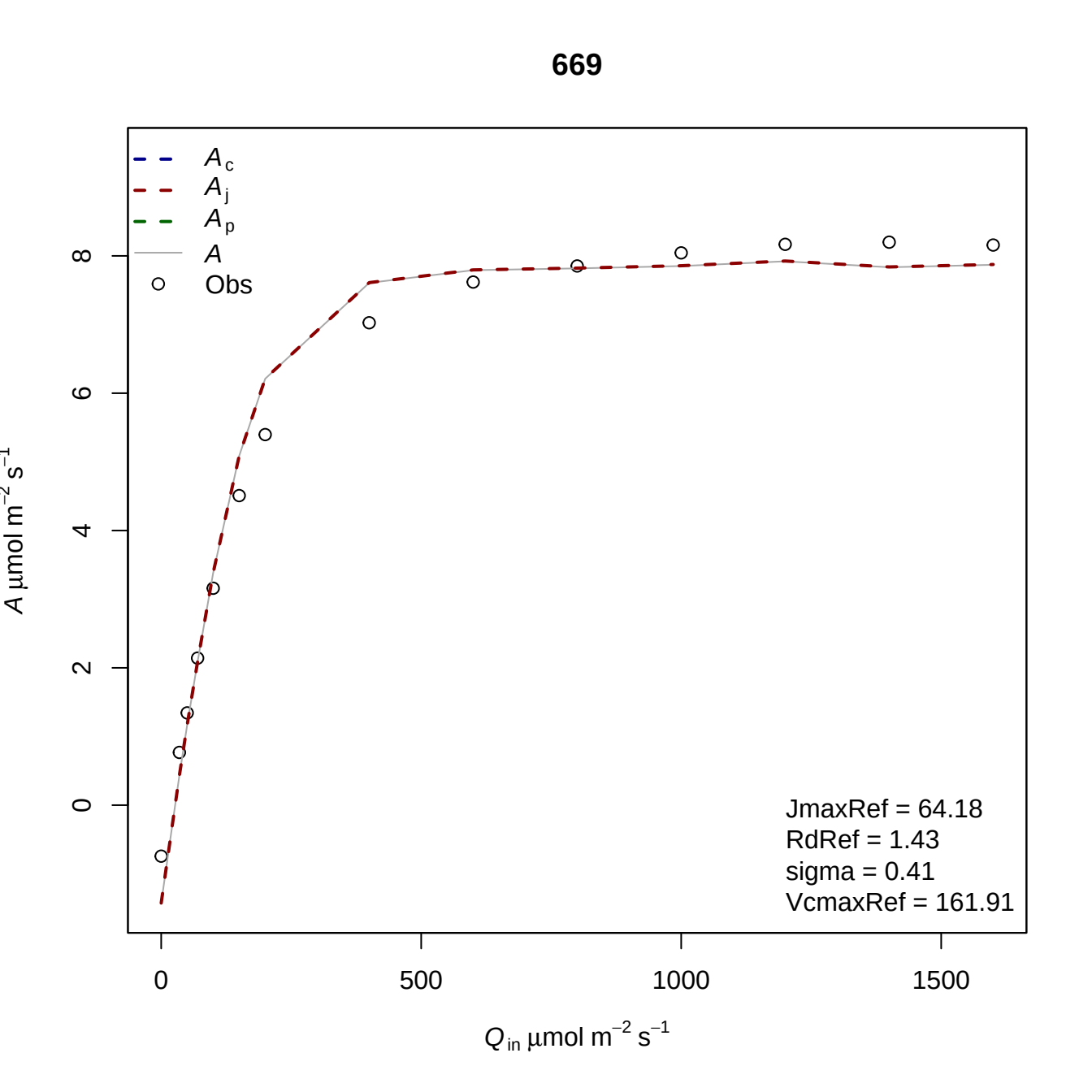

664
A_c
A_j
A_p
A
Obs
JmaxRef = 47
RdRef = 0.84
sigma = 0.37
VcmaxRef = 39.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

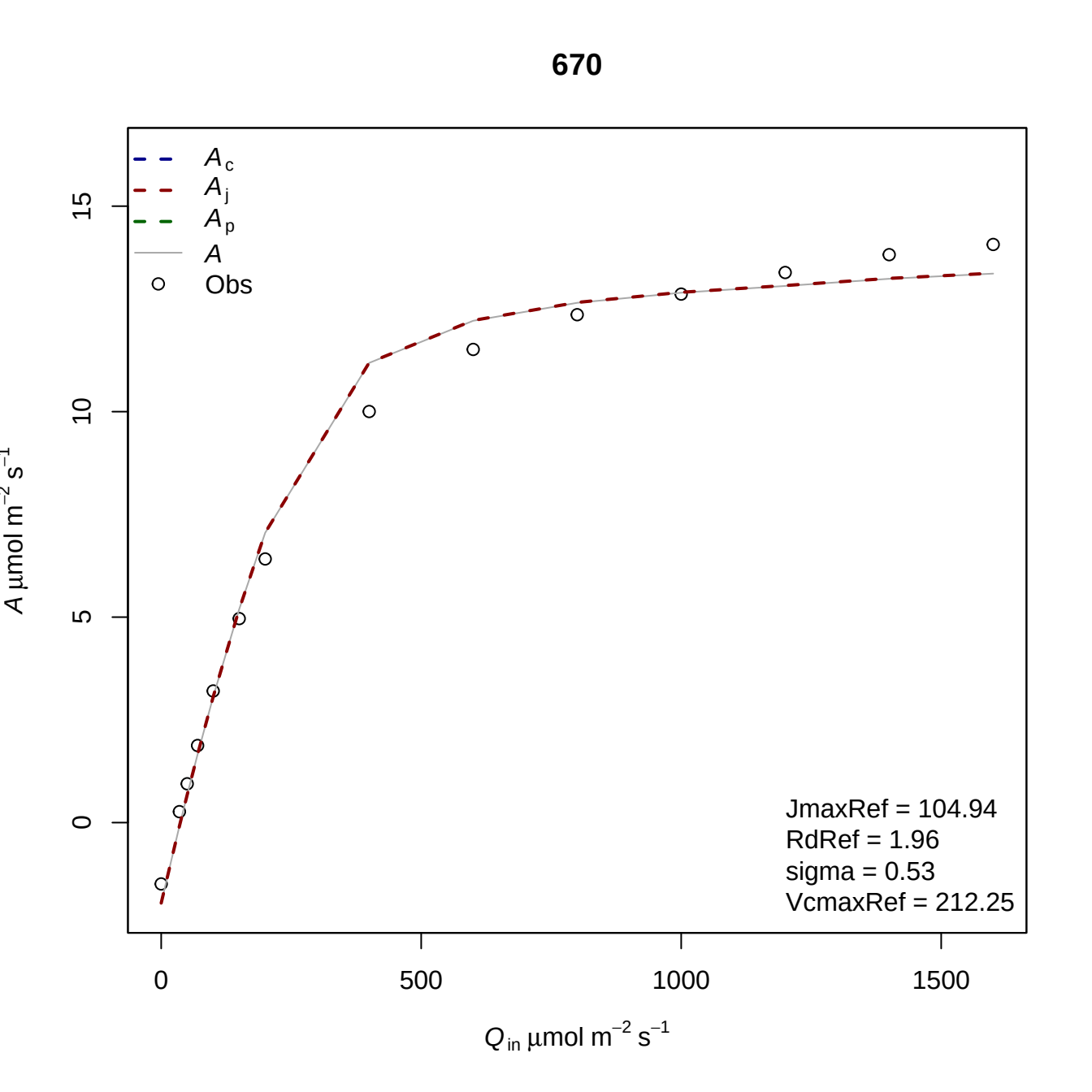

665
A_c
A_j
A_p
A
Obs
JmaxRef = 61.87
RdRef = 1.37
sigma = 0.35
VcmaxRef = 94.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

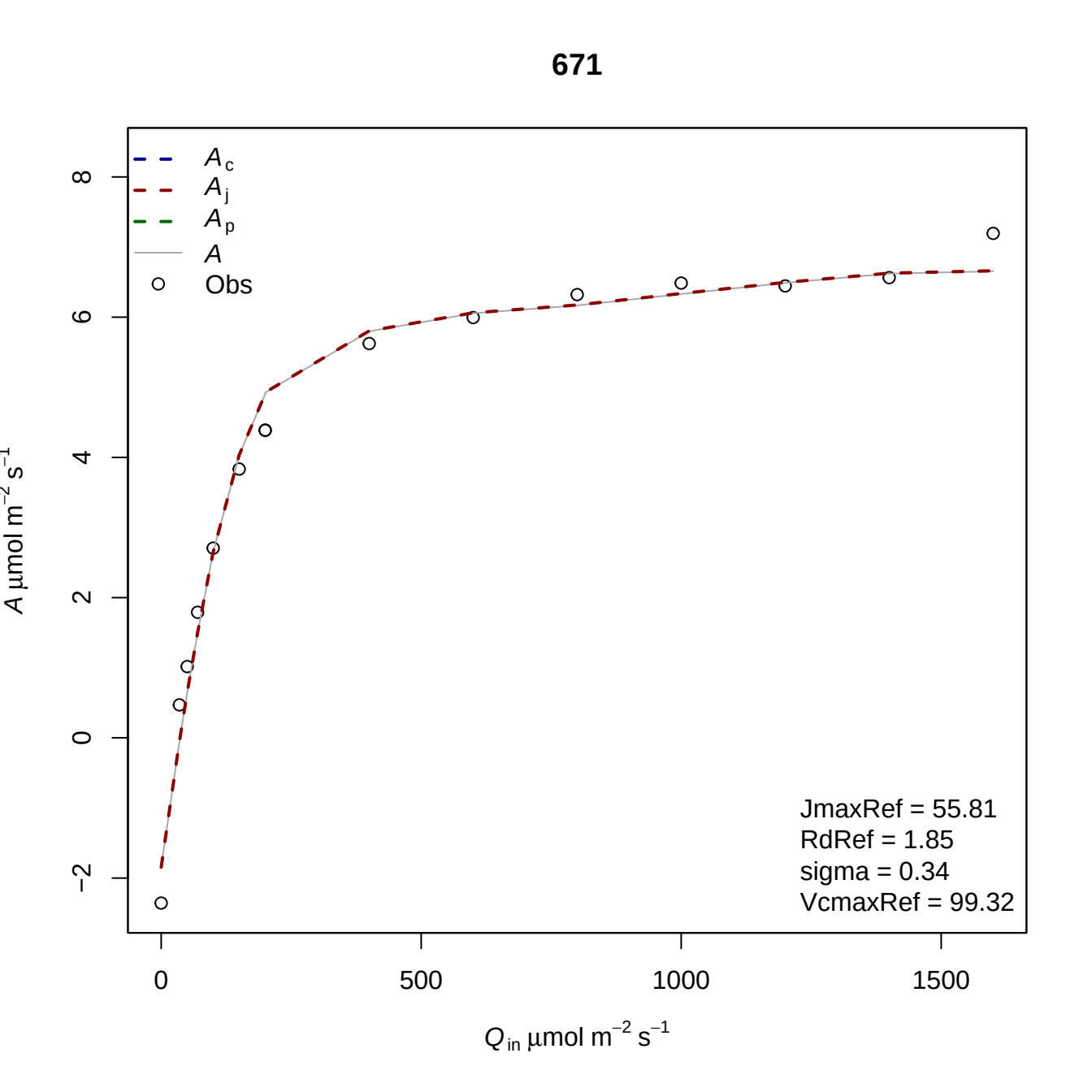

666
A_c
A_j
A_p
A
Obs
JmaxRef = 38.66
RdRef = 1.05
sigma = 0.34
VcmaxRef = 125.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

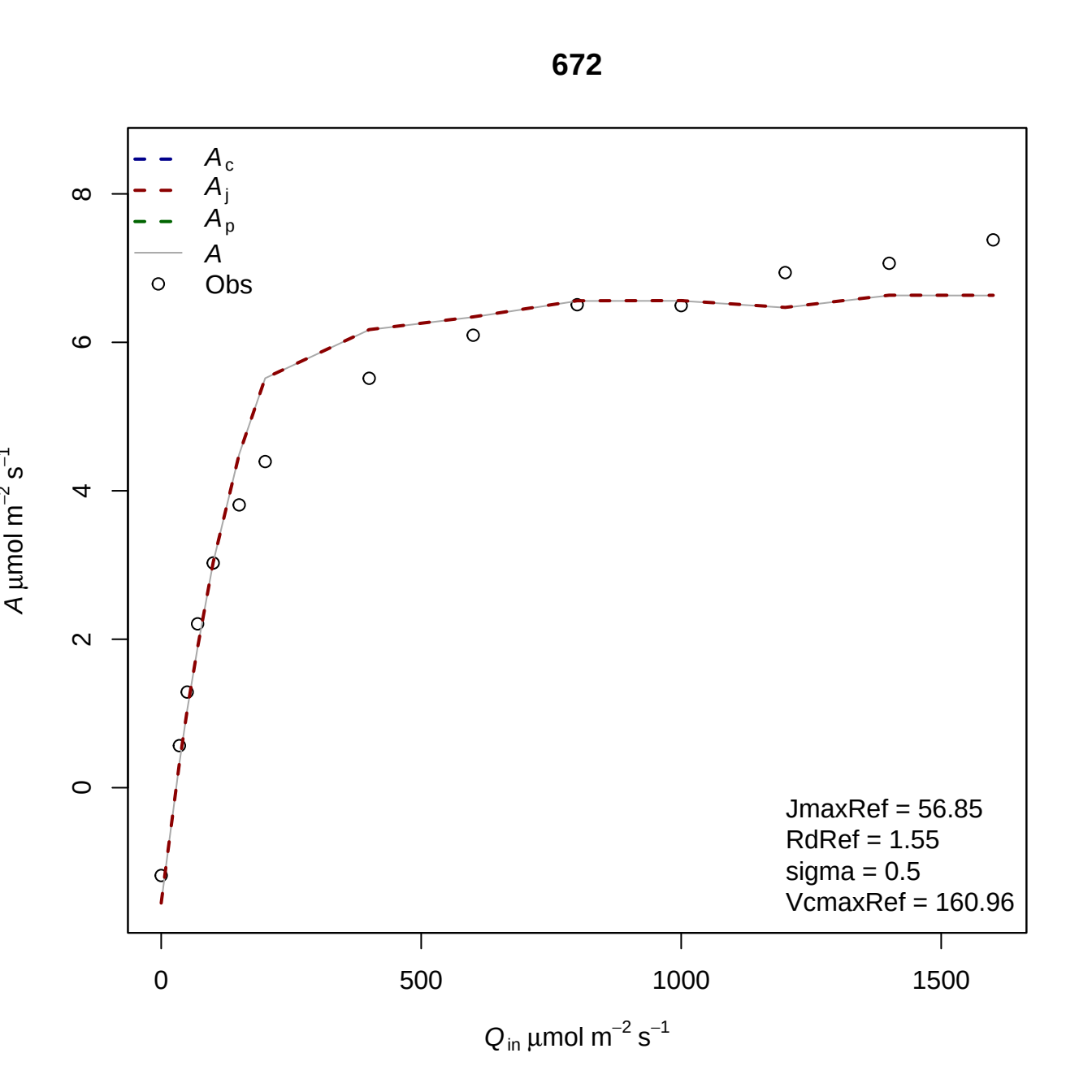

667
A_c
A_j
A_p
A
Obs
JmaxRef = 49.7
RdRef = 1.11
sigma = 0.32
VcmaxRef = 38.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

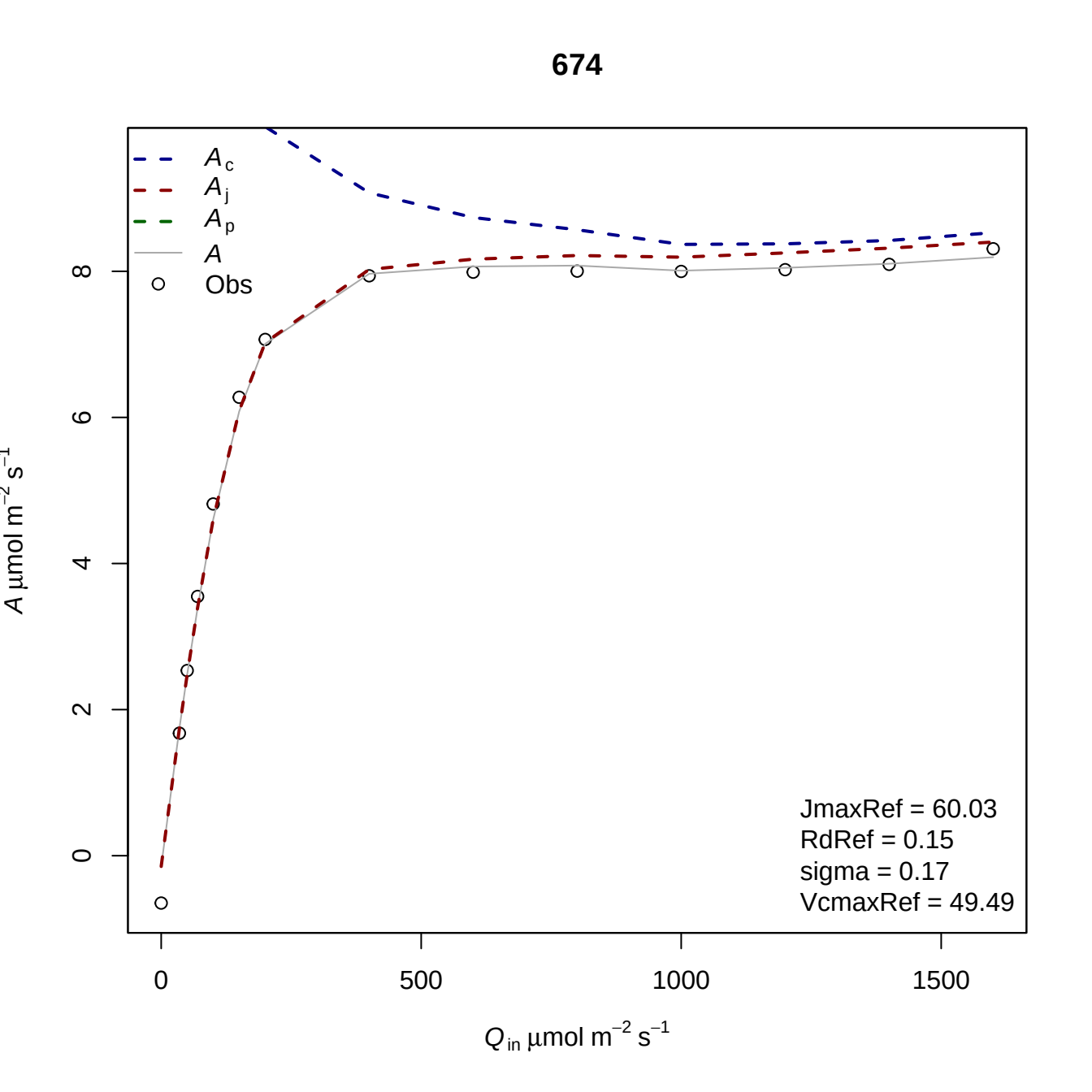

668
A_c
A_j
A_p
A
Obs
JmaxRef = 53.58
RdRef = 0.85
sigma = 0.22
VcmaxRef = 102.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

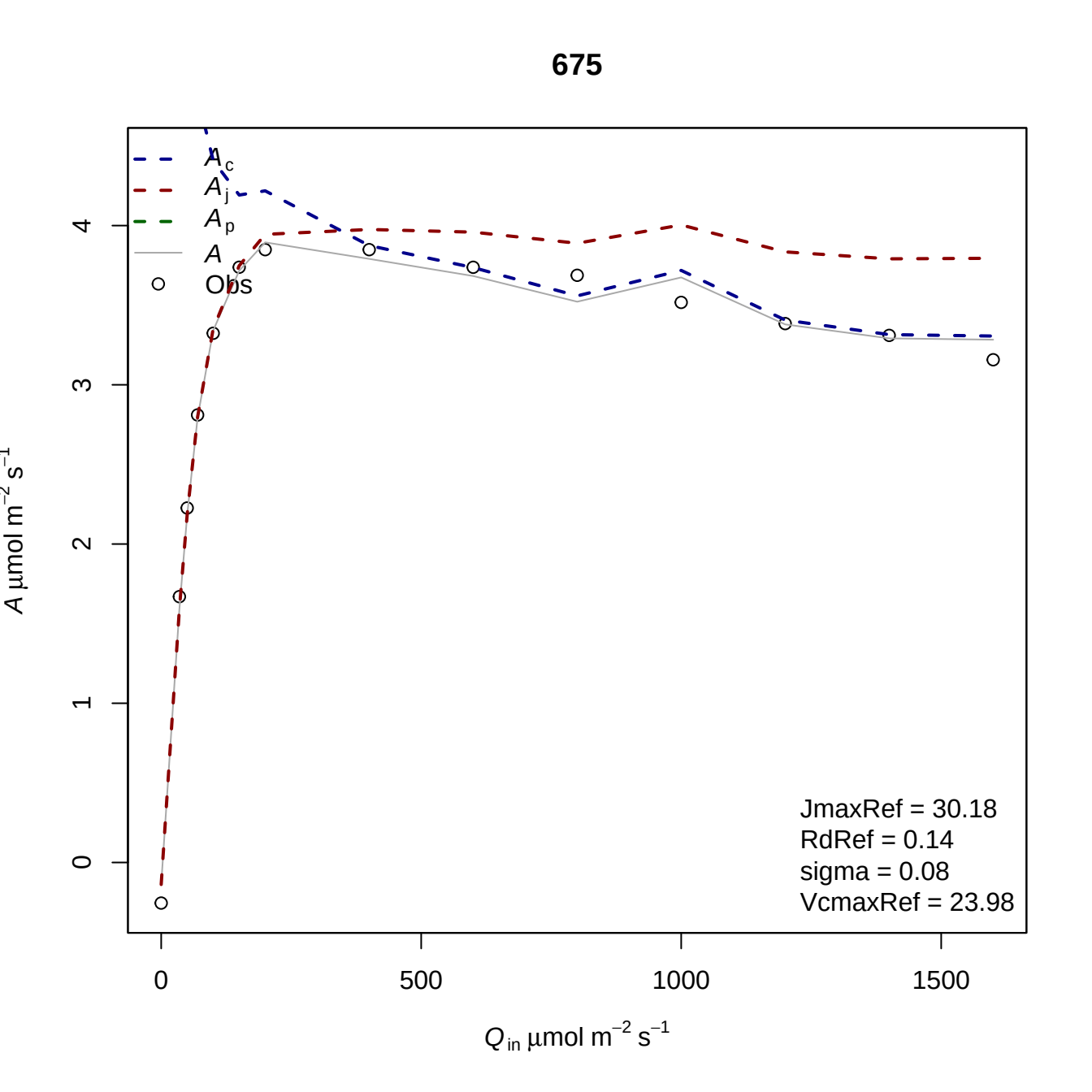

669
A_c
A_j
A_p
A
Obs
JmaxRef = 64.18
RdRef = 1.43
sigma = 0.41
VcmaxRef = 161.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

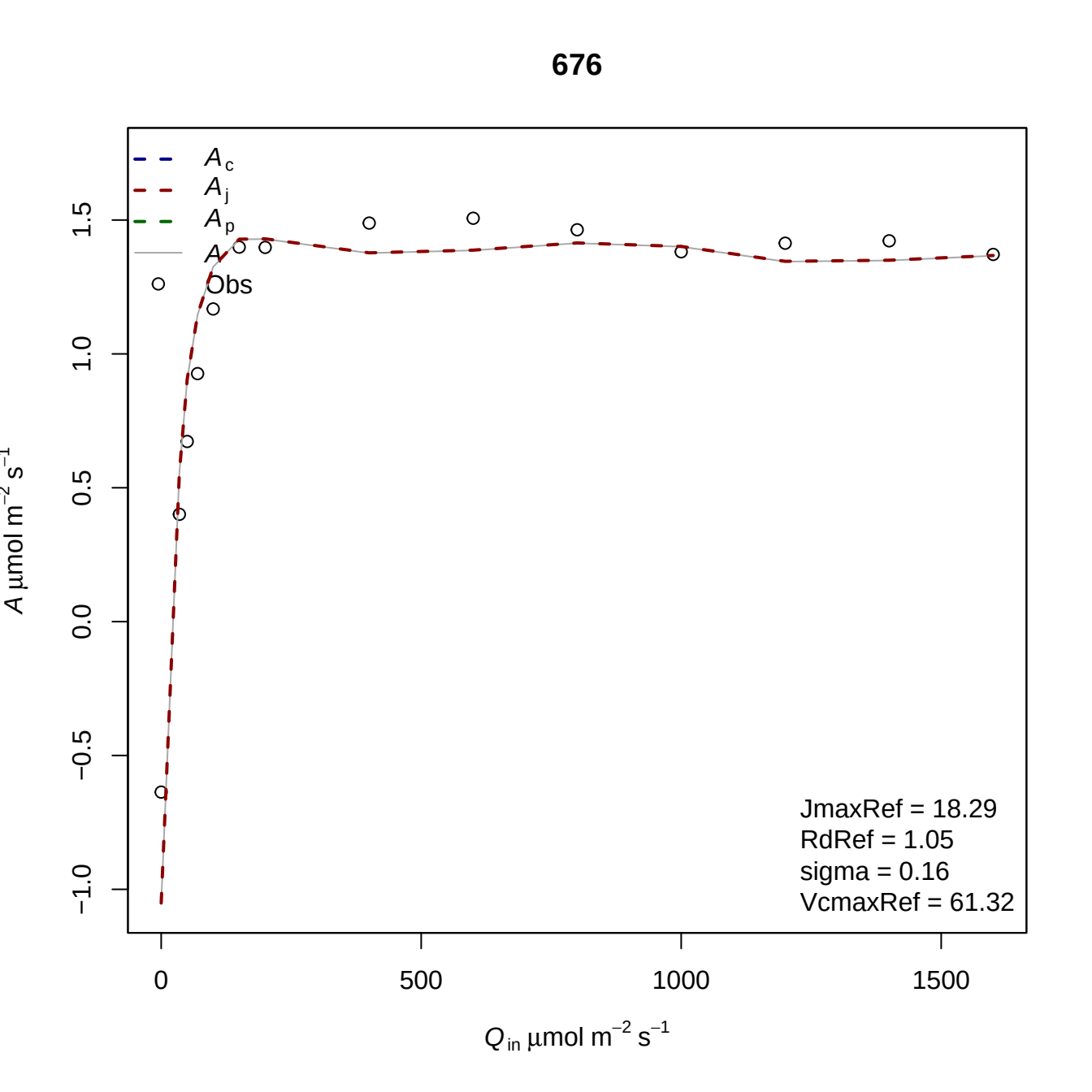

670
A_c
A_j
A_p
A
Obs
JmaxRef = 104.94
RdRef = 1.96
sigma = 0.53
VcmaxRef = 212.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

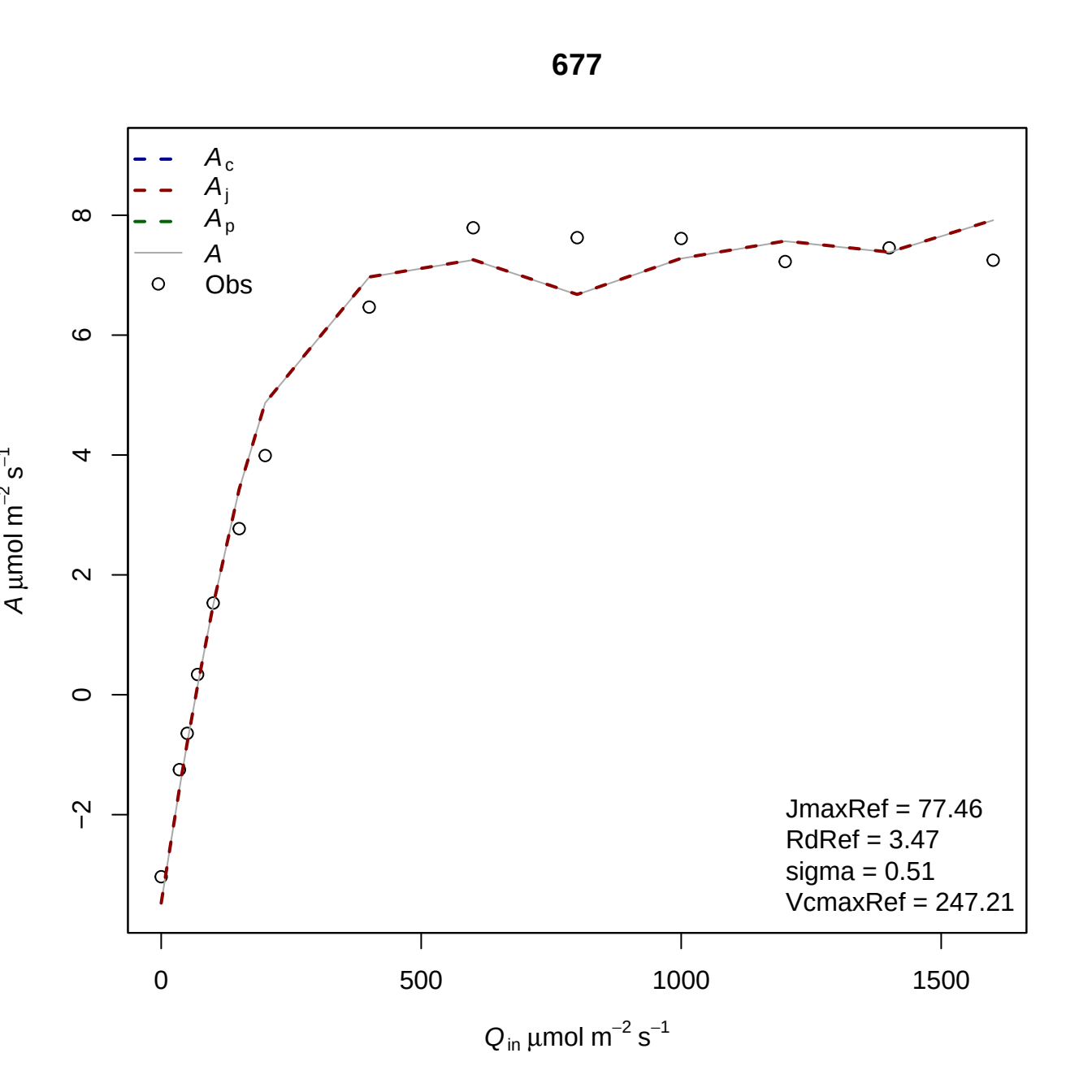

671
A_c
A_j
A_p
A
Obs
JmaxRef = 55.81
RdRef = 1.85
sigma = 0.34
VcmaxRef = 99.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

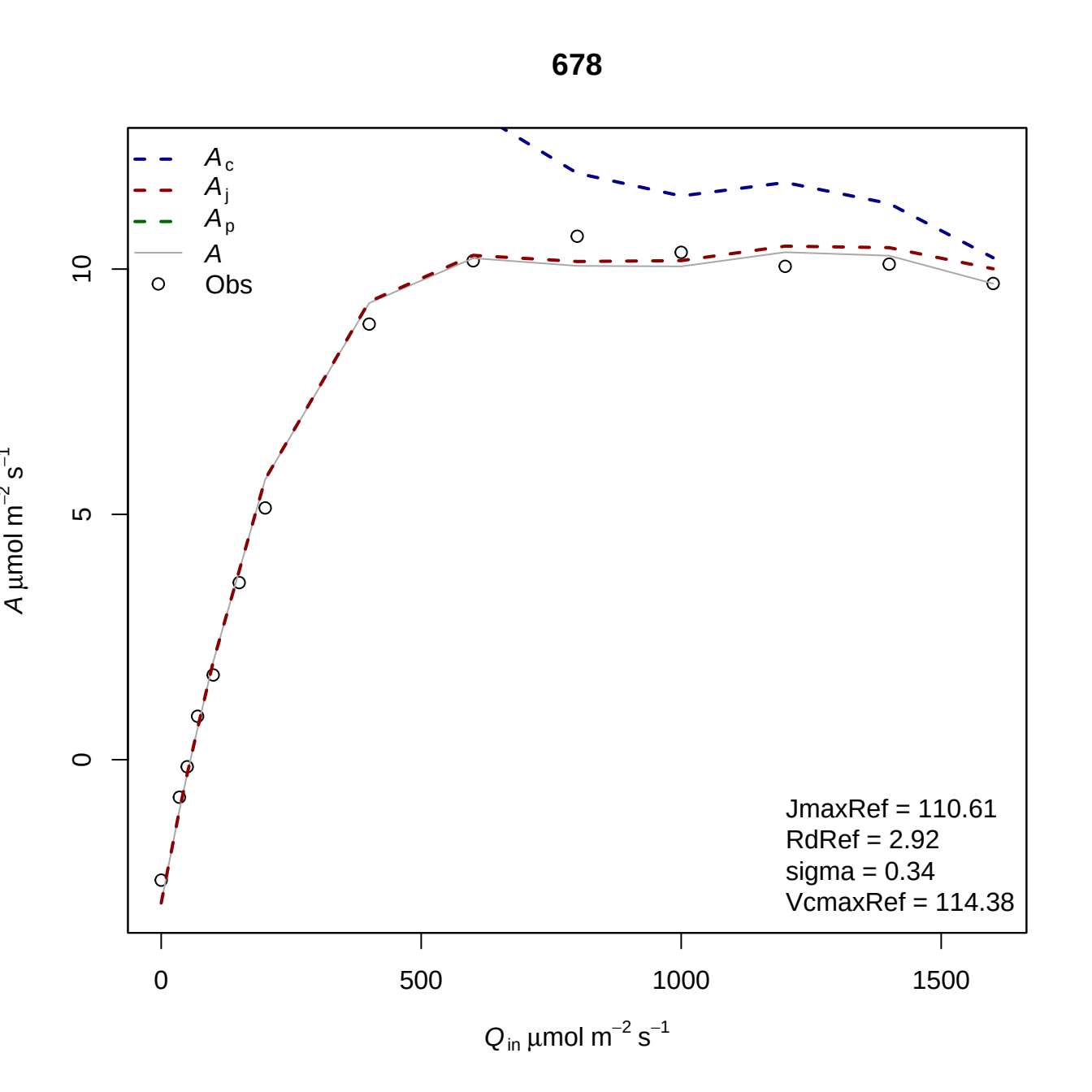

672
A_c
A_j
A_p
A
Obs
JmaxRef = 56.85
RdRef = 1.55
sigma = 0.5
VcmaxRef = 160.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

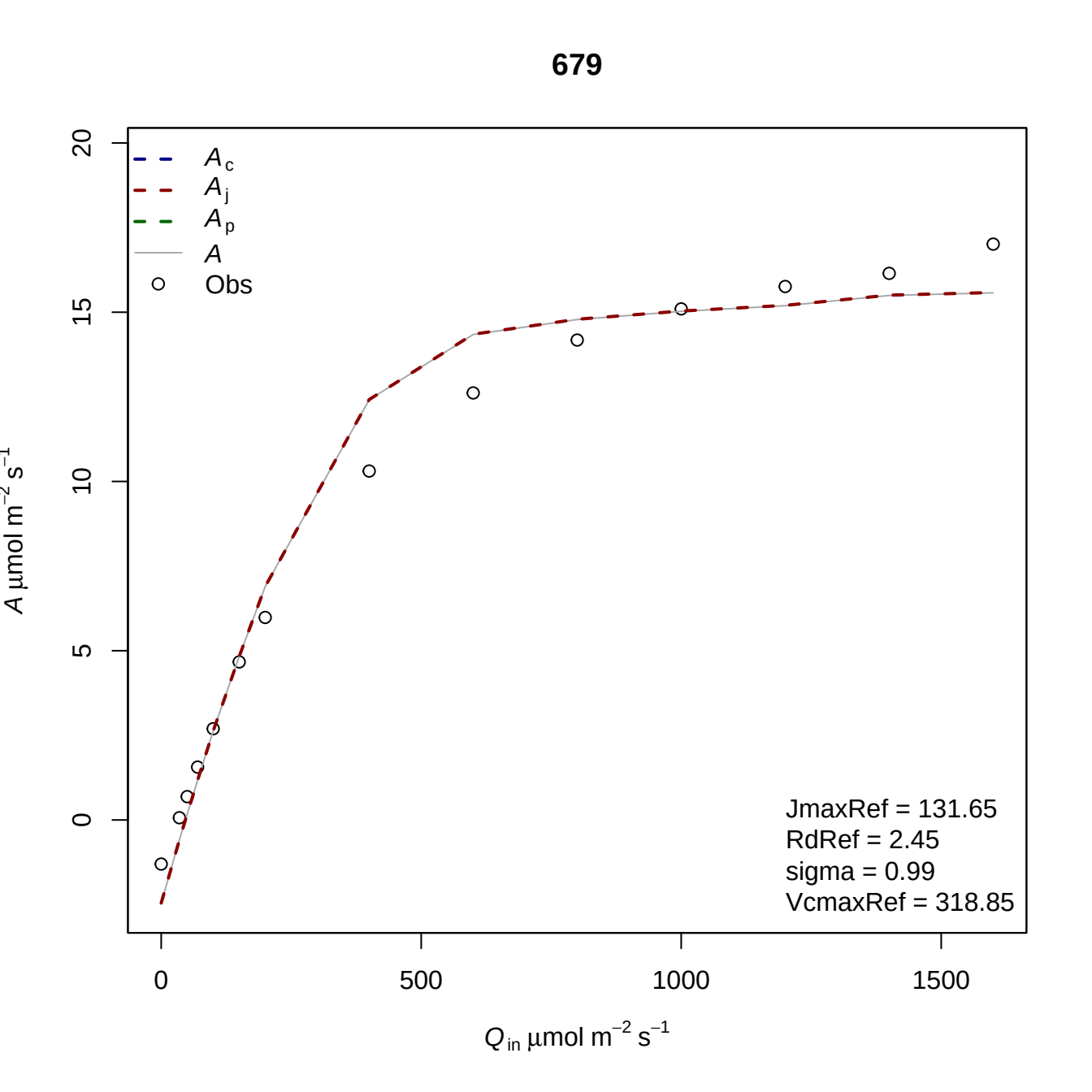

674
A_c
A_j
A_p
A
Obs
JmaxRef = 60.03
RdRef = 0.15
sigma = 0.17
VcmaxRef = 49.49
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

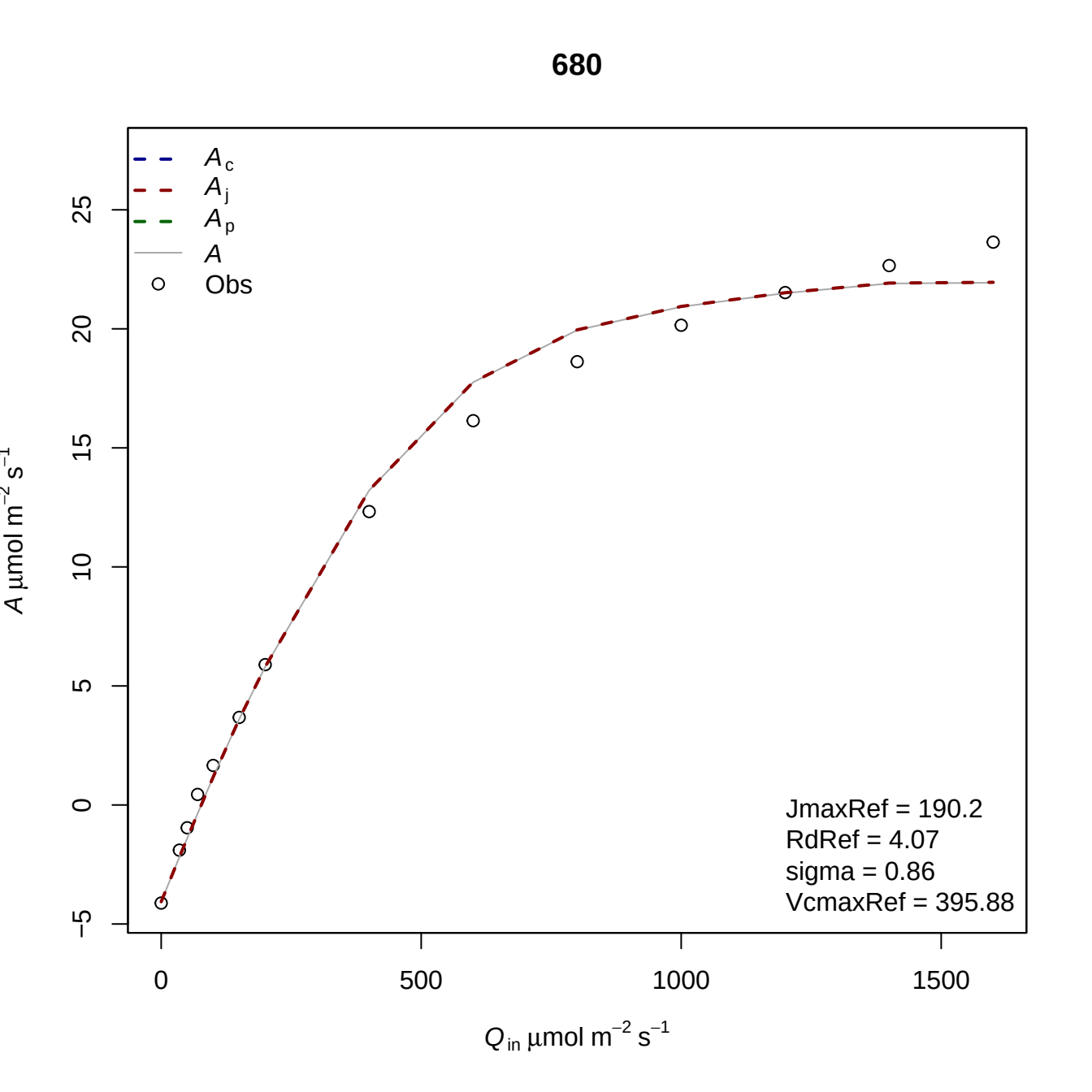

675
A_c
A_j
A_p
A
Obs
JmaxRef = 30.18
RdRef = 0.14
sigma = 0.08
VcmaxRef = 23.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

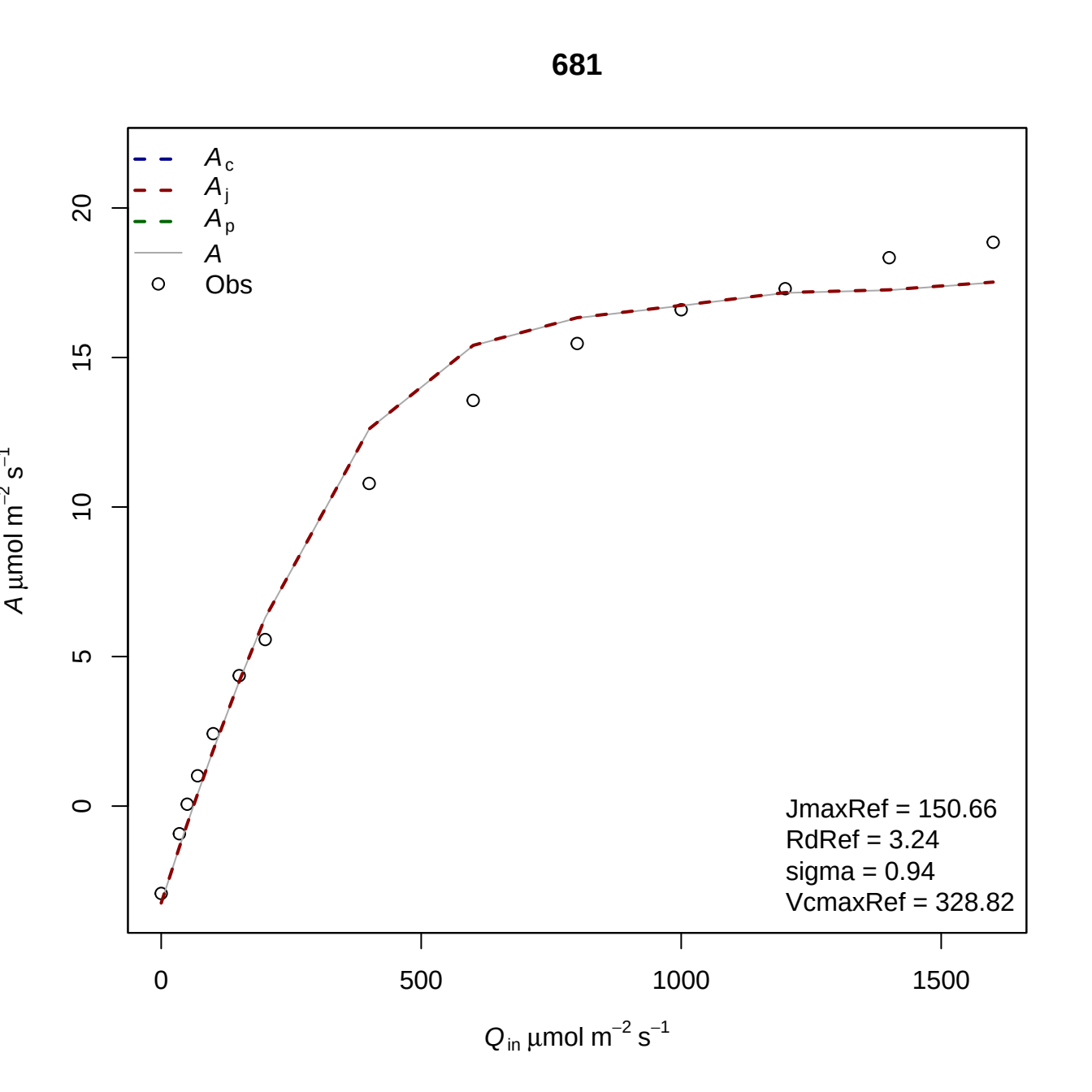

676
A_c
A_j
A_p
A
Obs
JmaxRef = 18.29
RdRef = 1.05
sigma = 0.16
VcmaxRef = 61.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

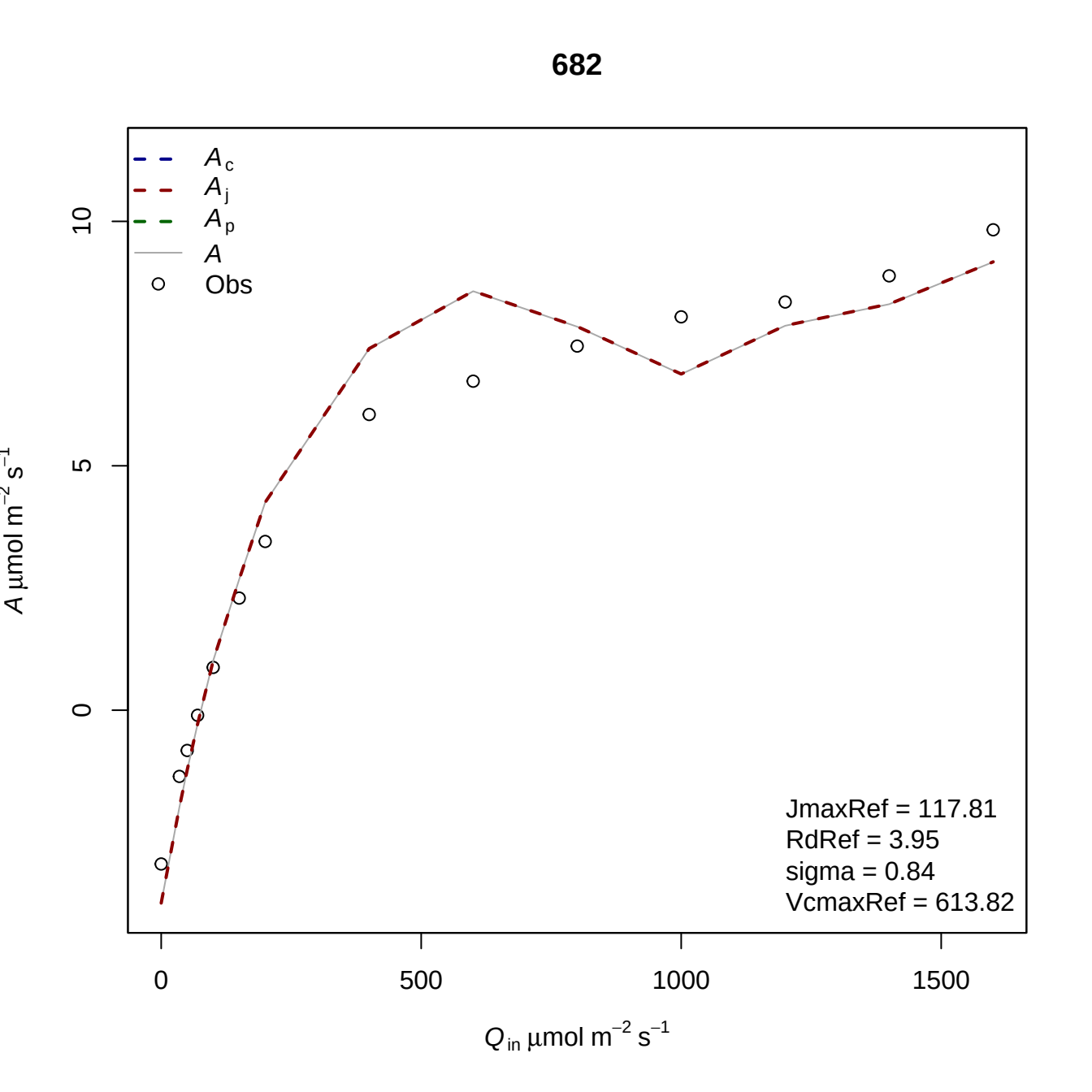

677
A_c
A_j
A_p
A
Obs
JmaxRef = 77.46
RdRef = 3.47
sigma = 0.51
VcmaxRef = 247.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

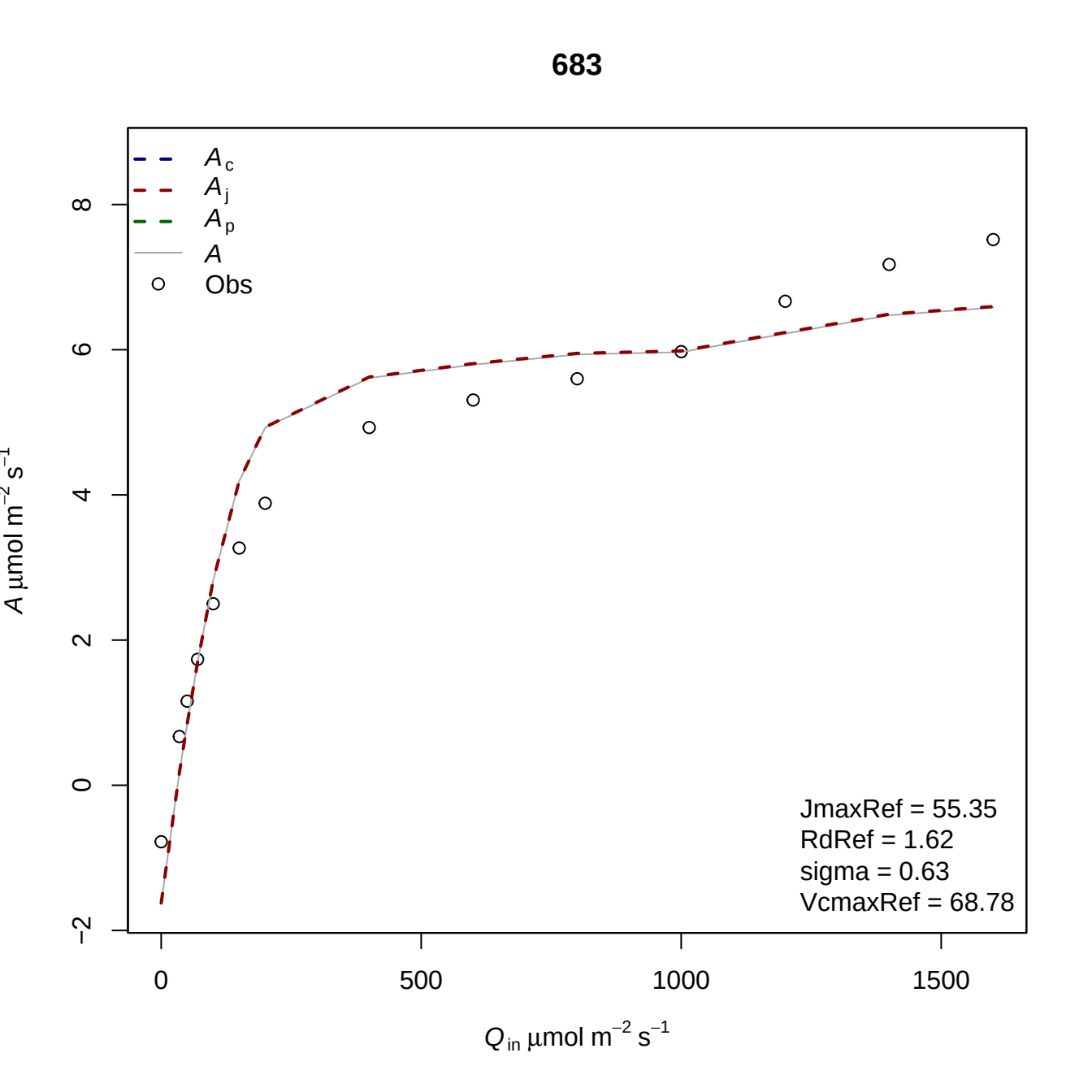

678
A_c
A_j
A_p
A
Obs
JmaxRef = 110.61
RdRef = 2.92
sigma = 0.34
VcmaxRef = 114.38
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

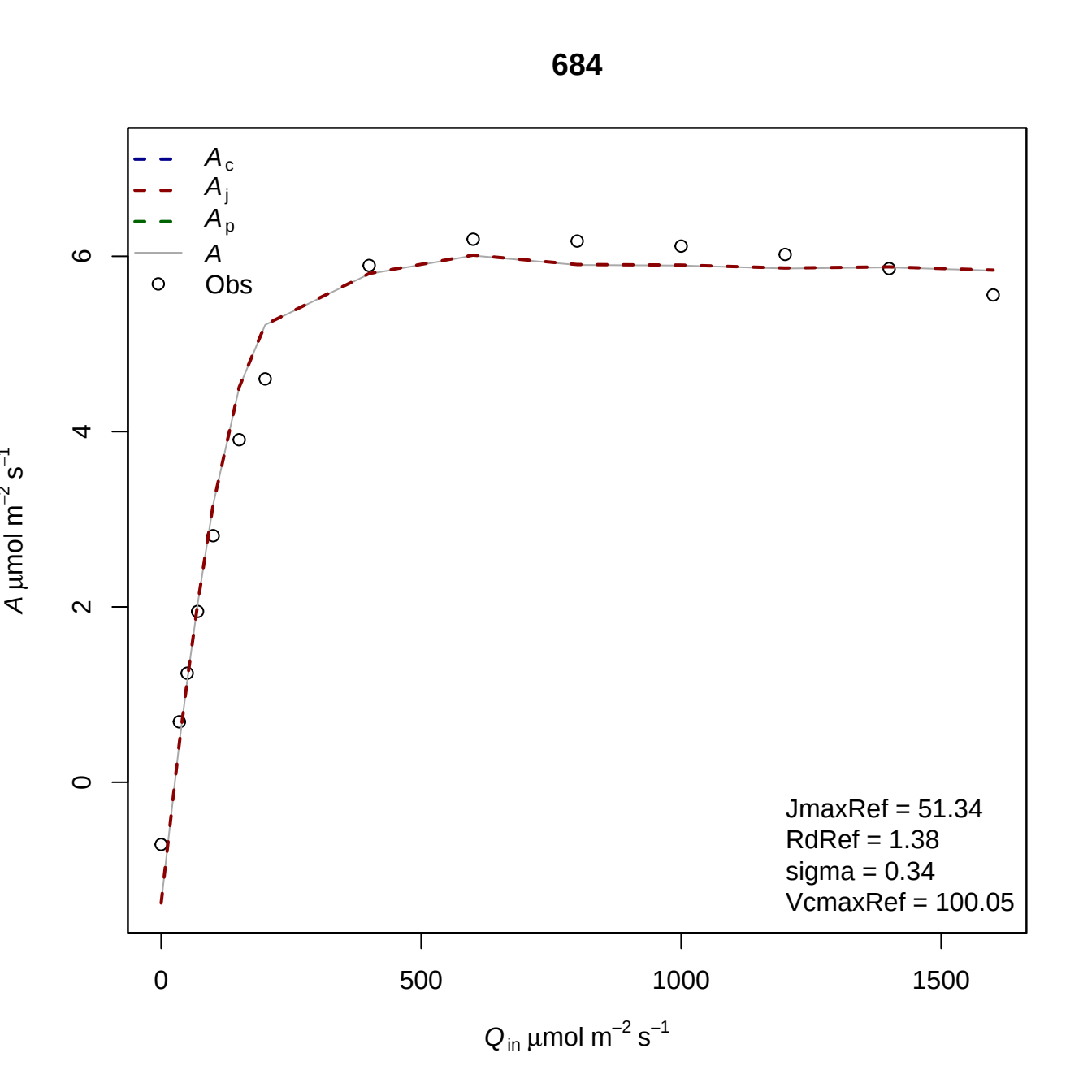

679
A_c
A_j
A_p
A
Obs
JmaxRef = 131.65
RdRef = 2.45
sigma = 0.99
VcmaxRef = 318.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

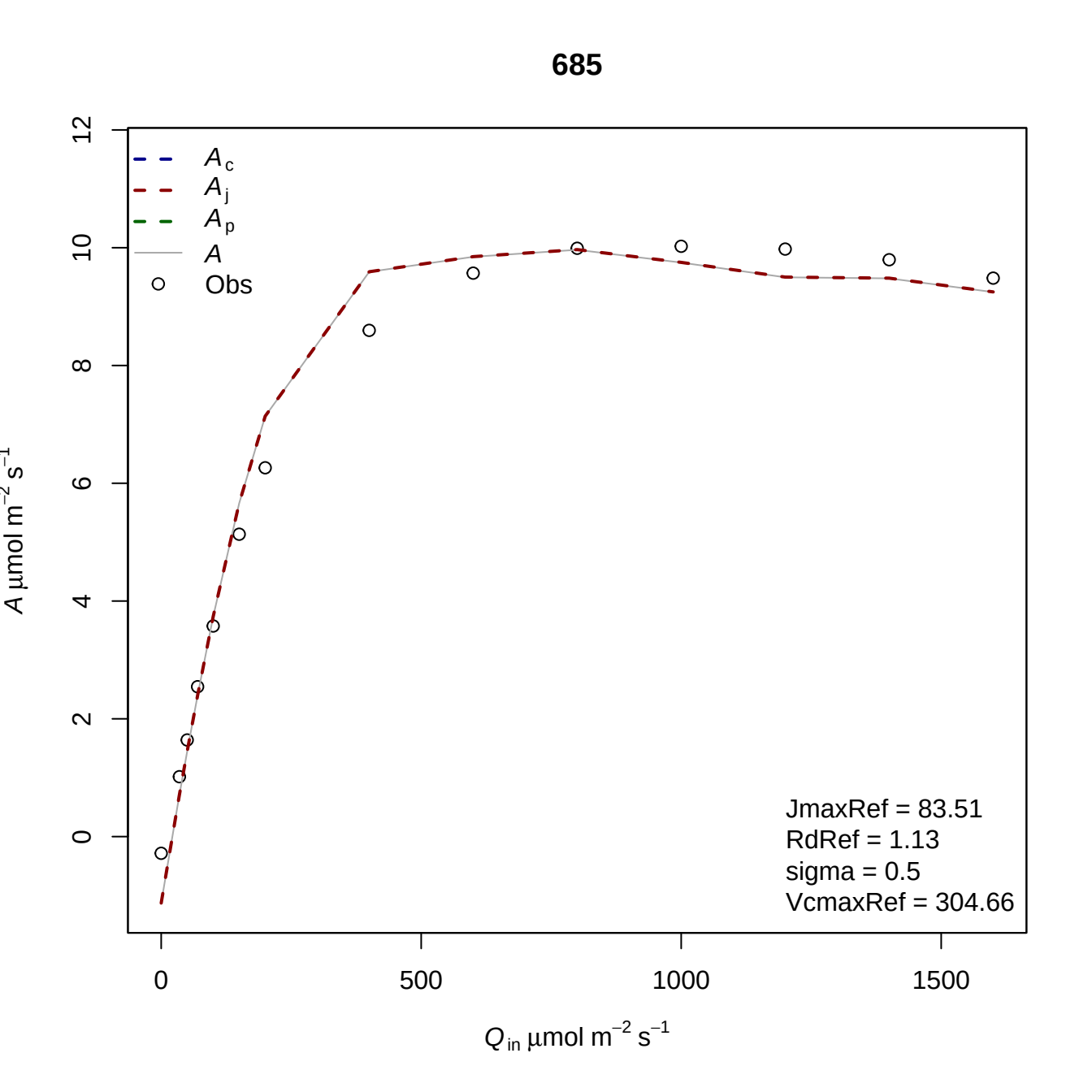

680
A_c
A_j
A_p
A
Obs
JmaxRef = 190.2
RdRef = 4.07
sigma = 0.86
VcmaxRef = 395.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

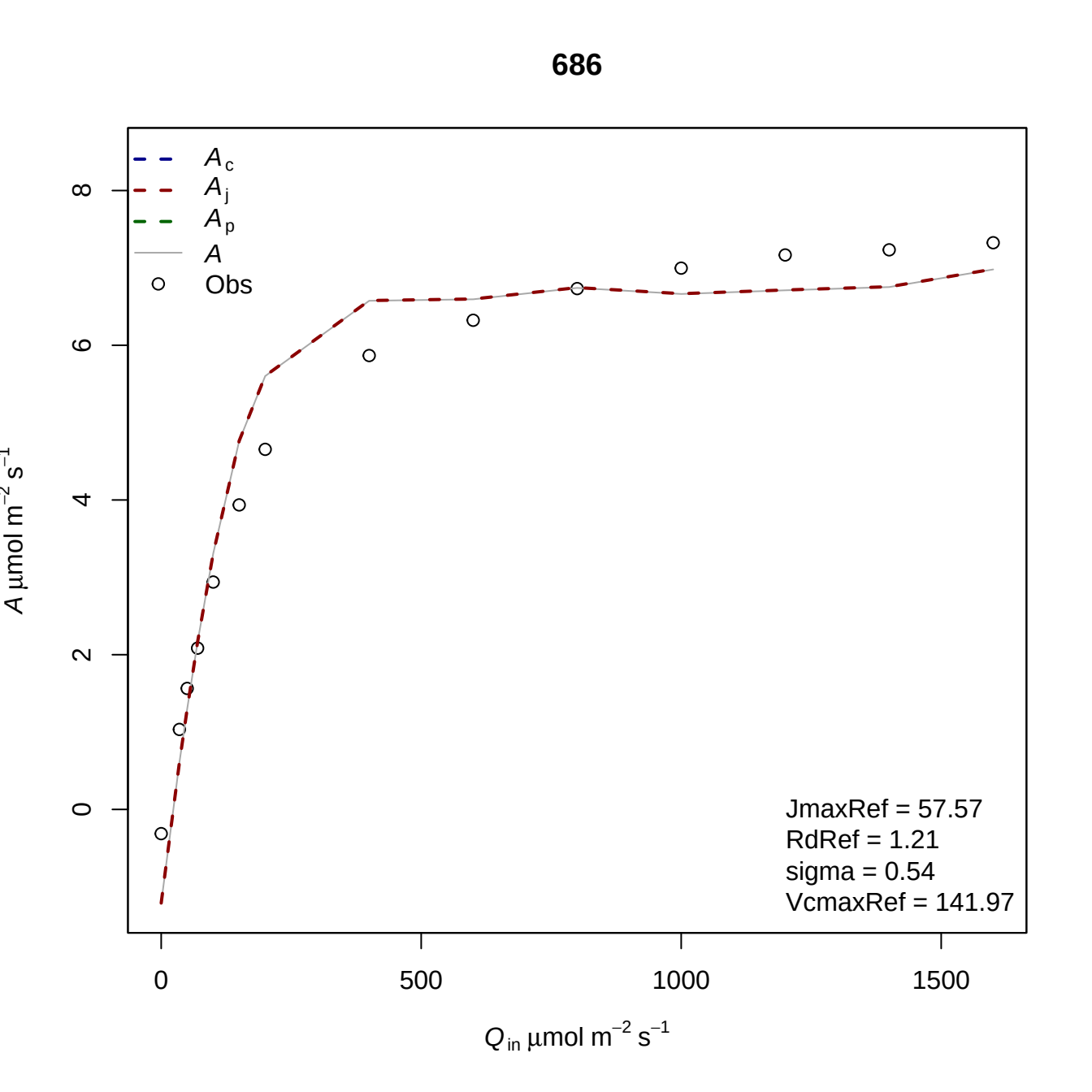

681
A_c
A_j
A_p
A
Obs
JmaxRef = 150.66
RdRef = 3.24
sigma = 0.94
VcmaxRef = 328.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

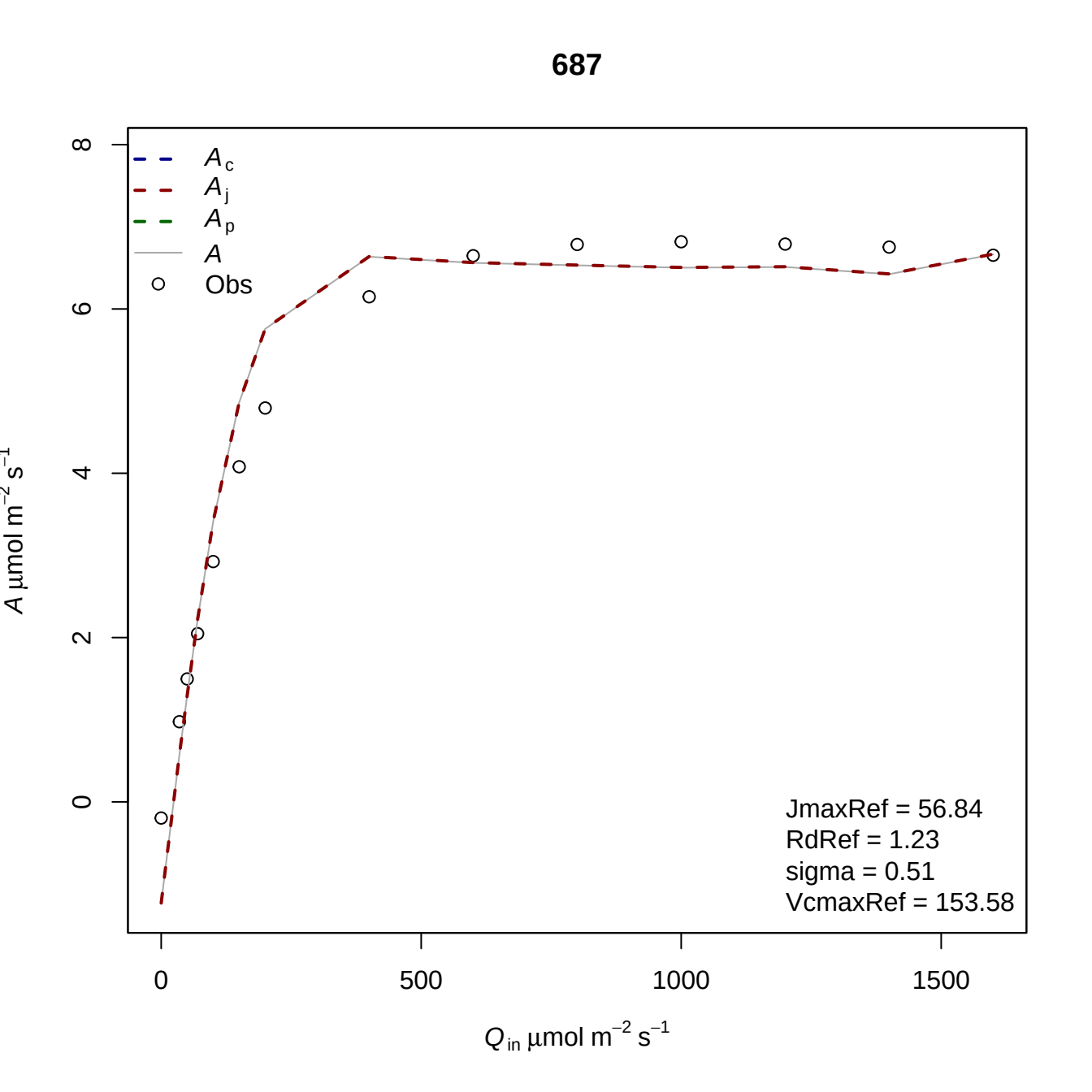

682
A_c
A_j
A_p
A
Obs
JmaxRef = 117.81
RdRef = 3.95
sigma = 0.84
VcmaxRef = 613.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

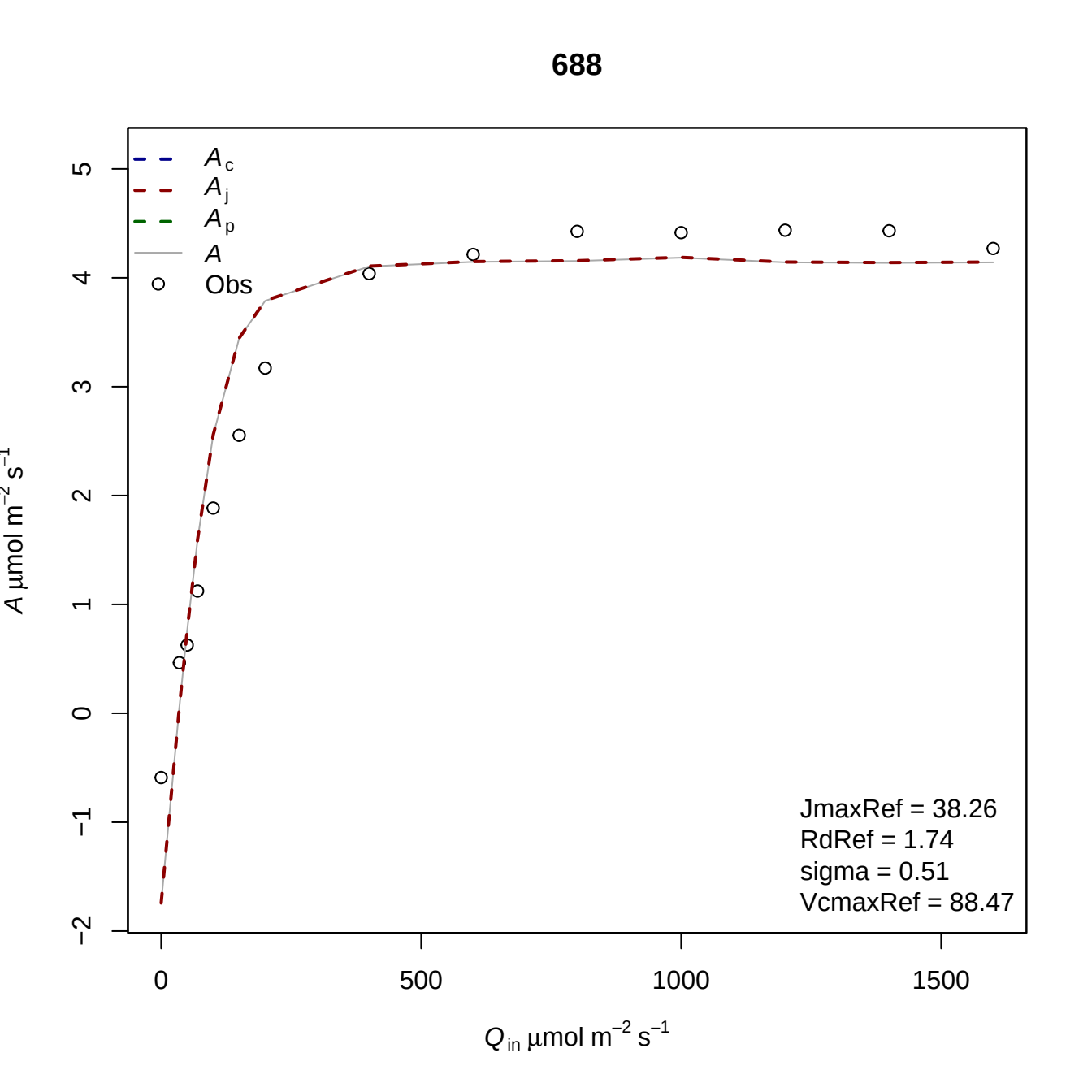

683
A_c
A_j
A_p
A
Obs
JmaxRef = 55.35
RdRef = 1.62
sigma = 0.63
VcmaxRef = 68.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

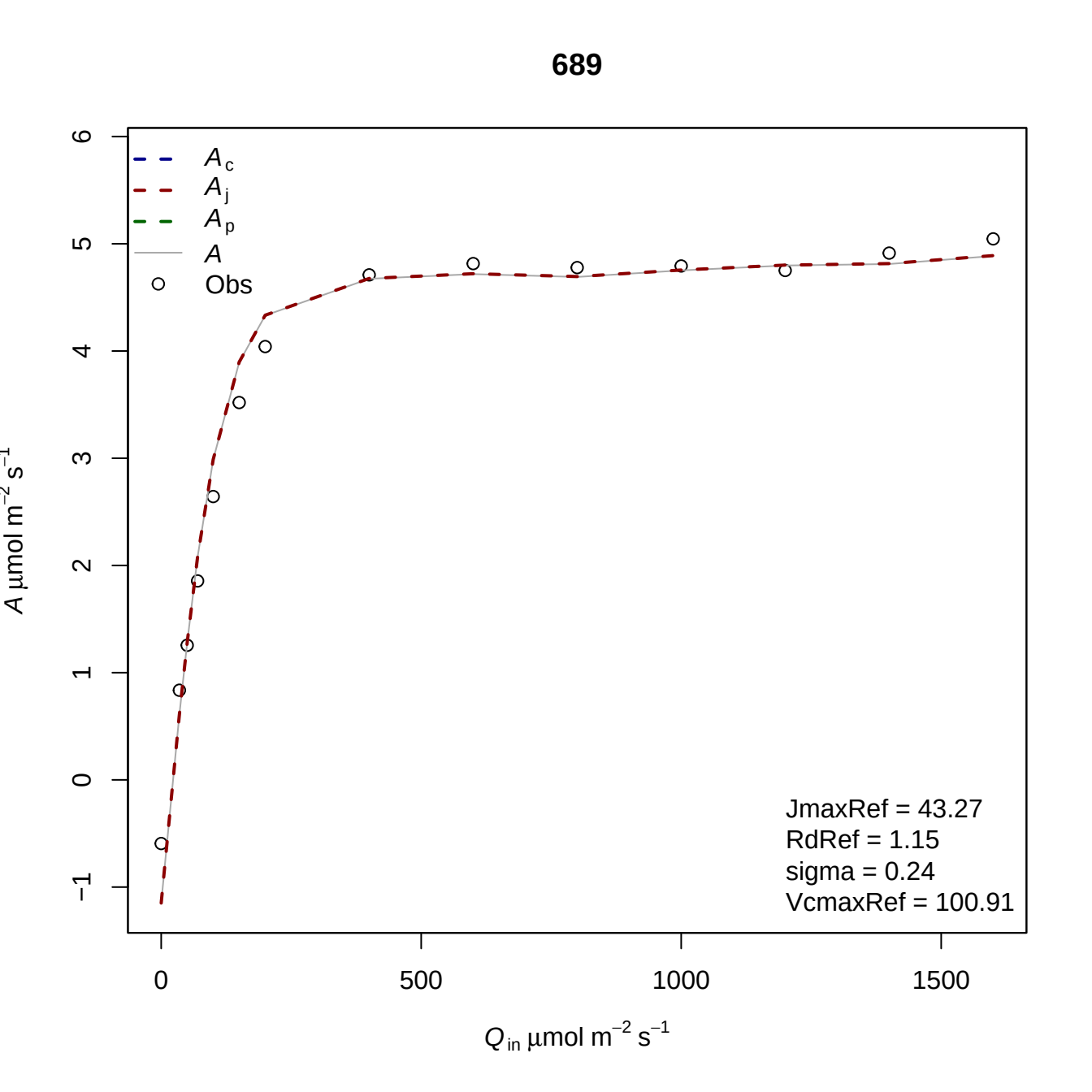

684
A_c
A_j
A_p
A
Obs
JmaxRef = 51.34
RdRef = 1.38
sigma = 0.34
VcmaxRef = 100.05
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

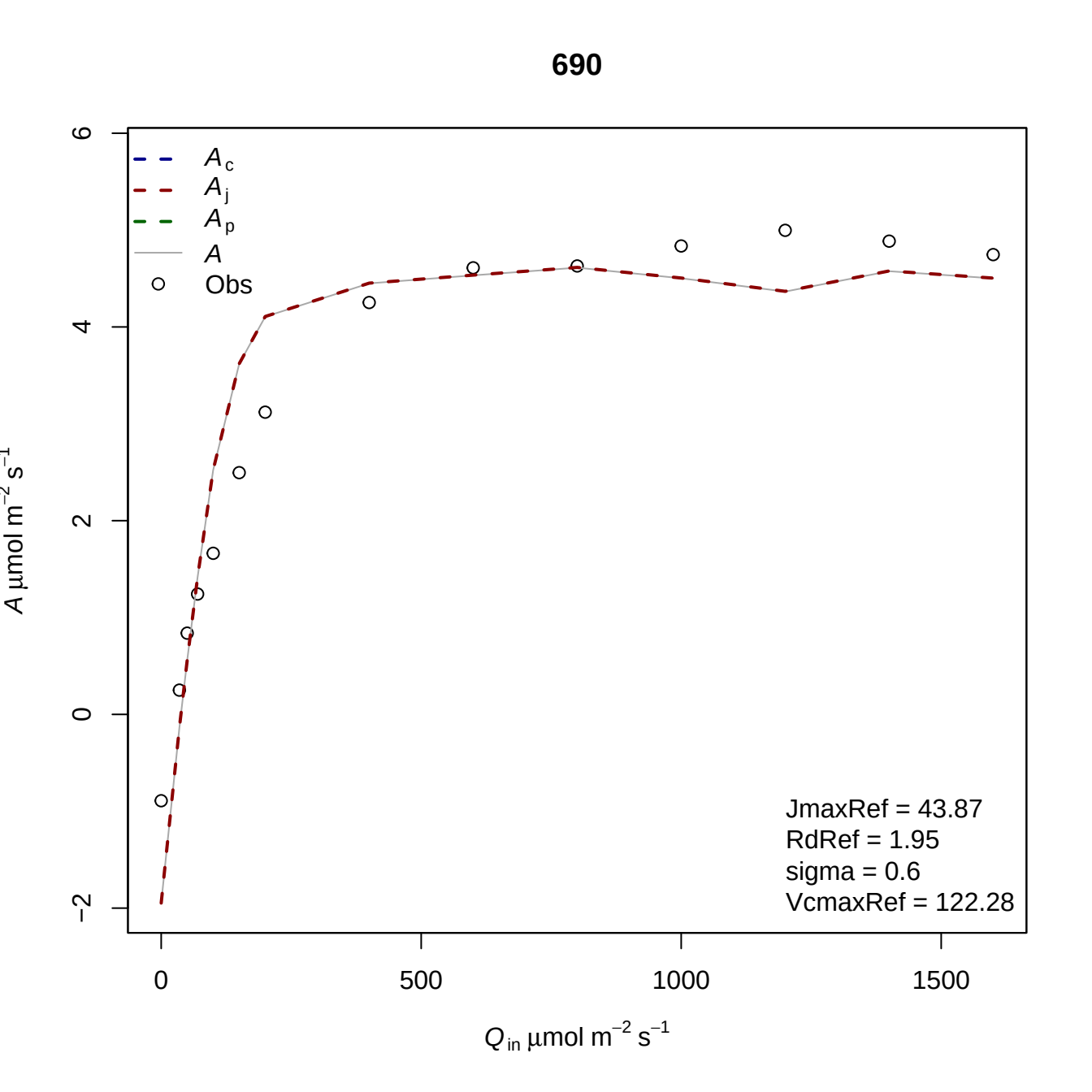

685
A_c
A_j
A_p
A
Obs
JmaxRef = 83.51
RdRef = 1.13
sigma = 0.5
VcmaxRef = 304.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

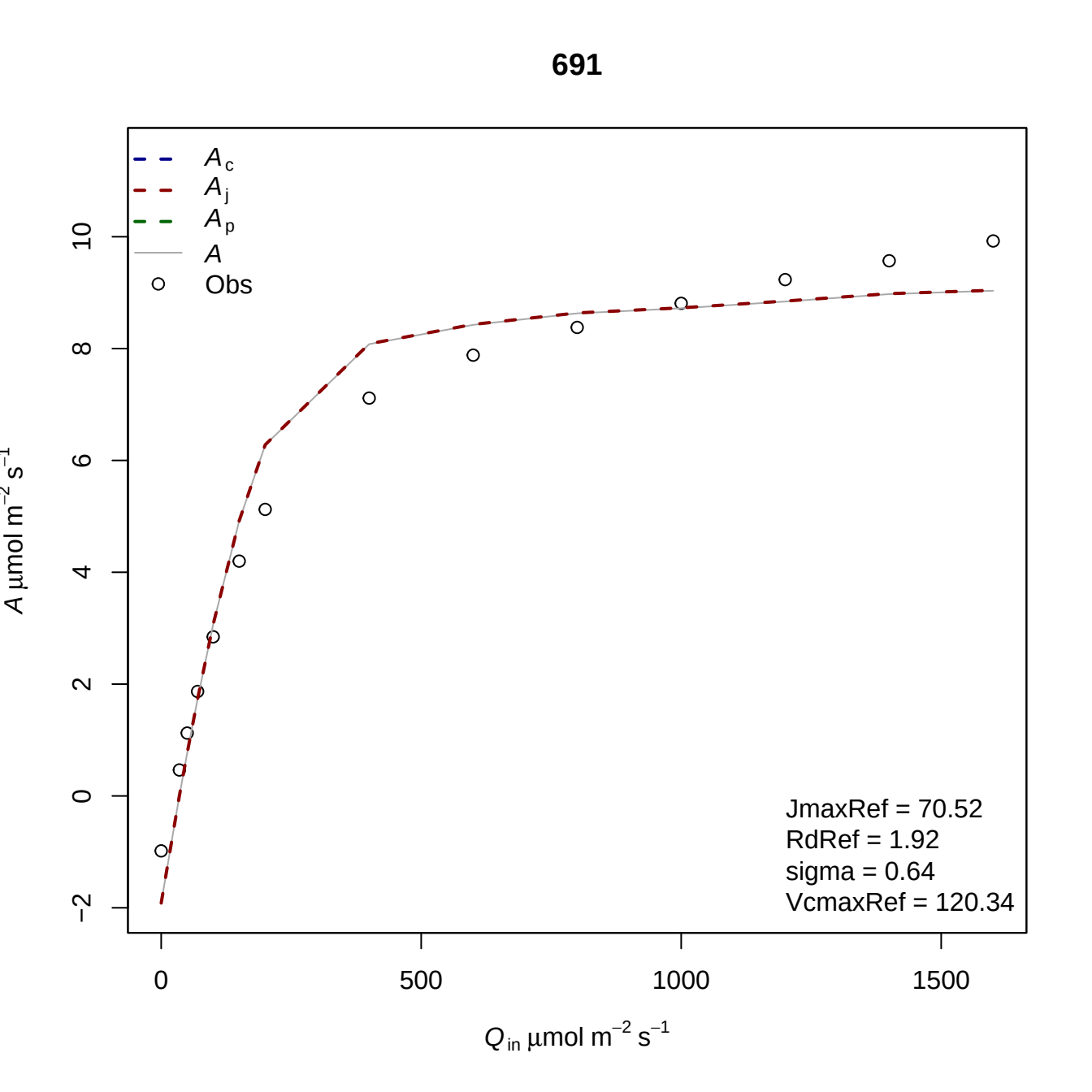

686
A_c
A_j
A_p
A
Obs
JmaxRef = 57.57
RdRef = 1.21
sigma = 0.54
VcmaxRef = 141.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

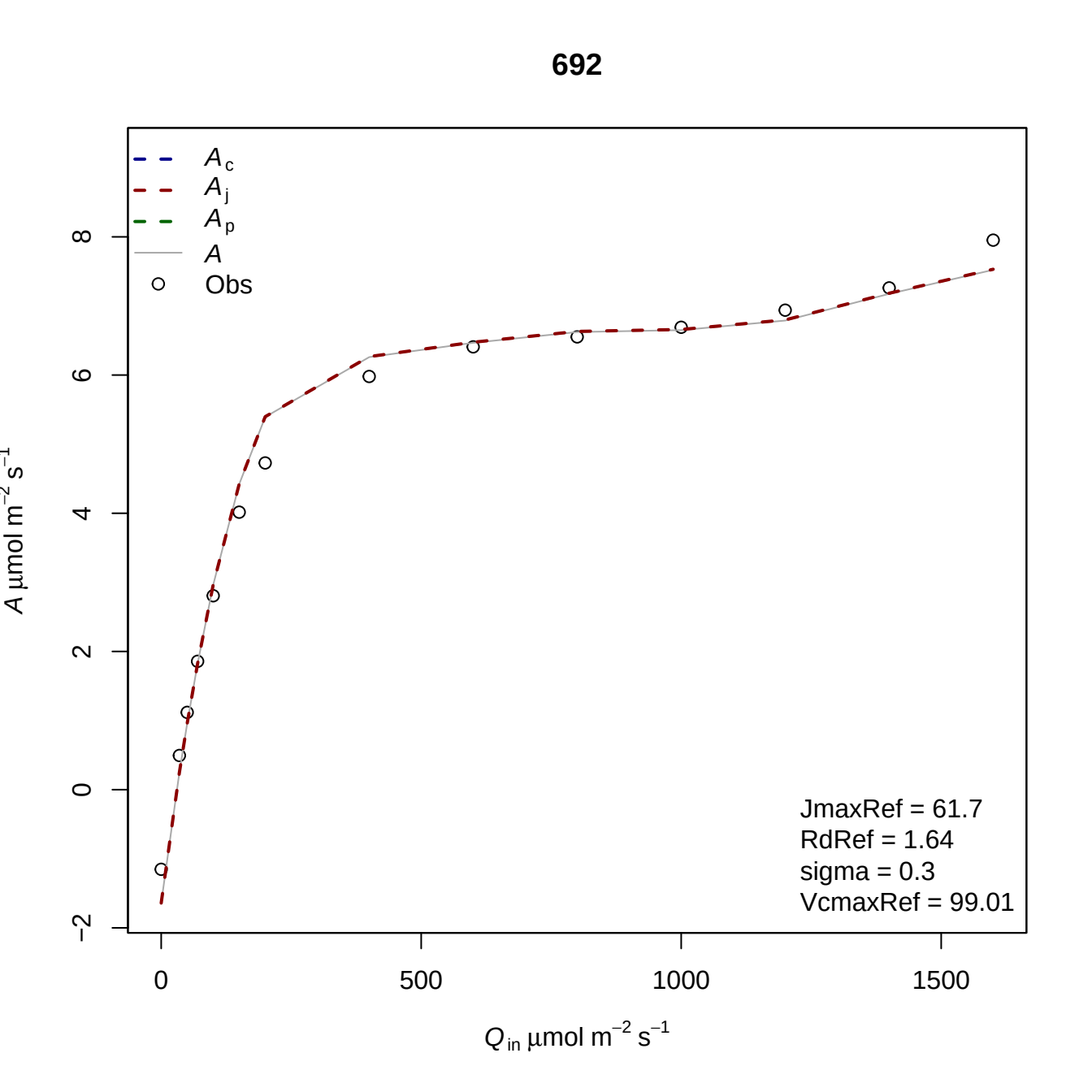

687
A_c
A_j
A_p
A
Obs
JmaxRef = 56.84
RdRef = 1.23
sigma = 0.51
VcmaxRef = 153.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

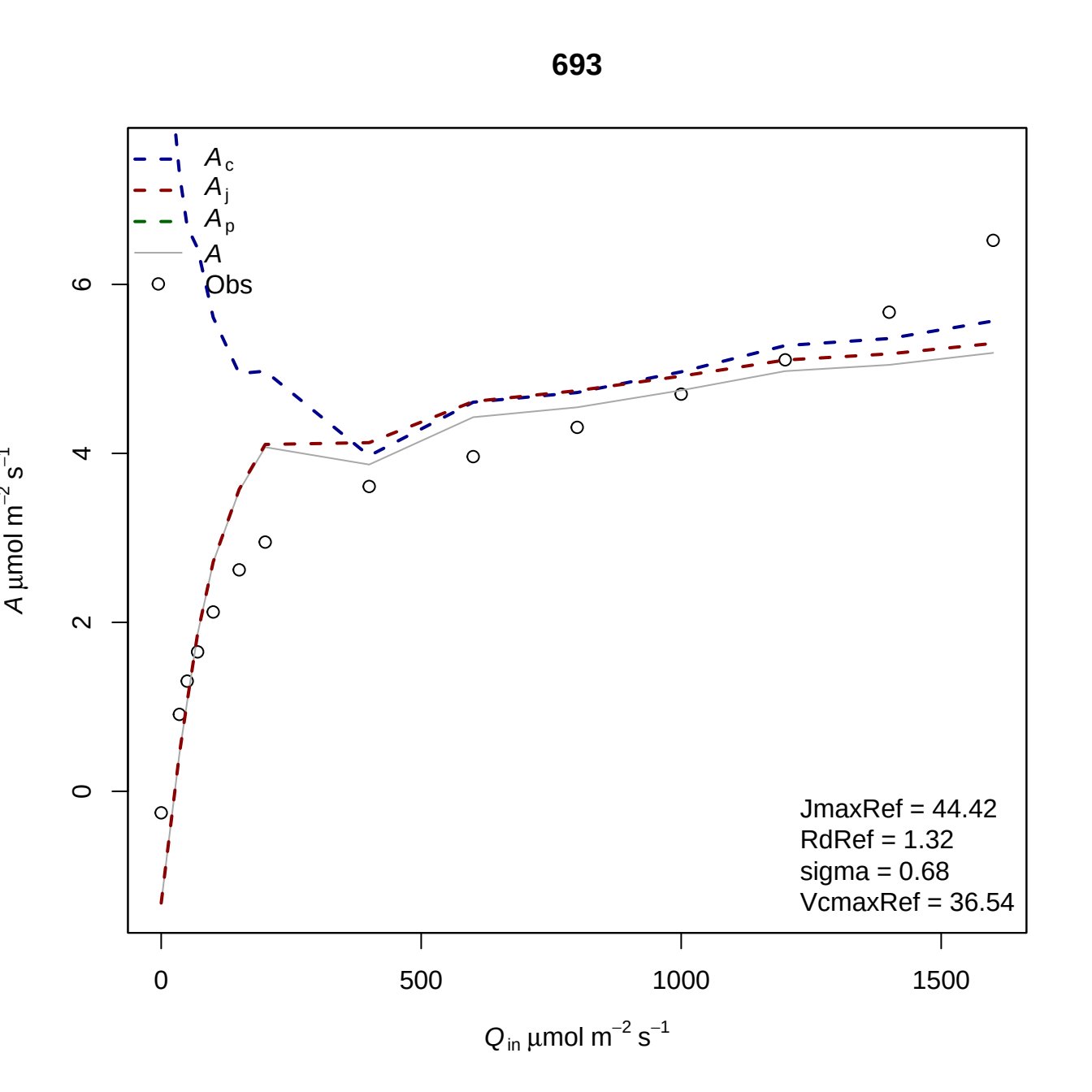

688
A_c
A_j
A_p
A
Obs
JmaxRef = 38.26
RdRef = 1.74
sigma = 0.51
VcmaxRef = 88.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
5
0
500
1000
1500

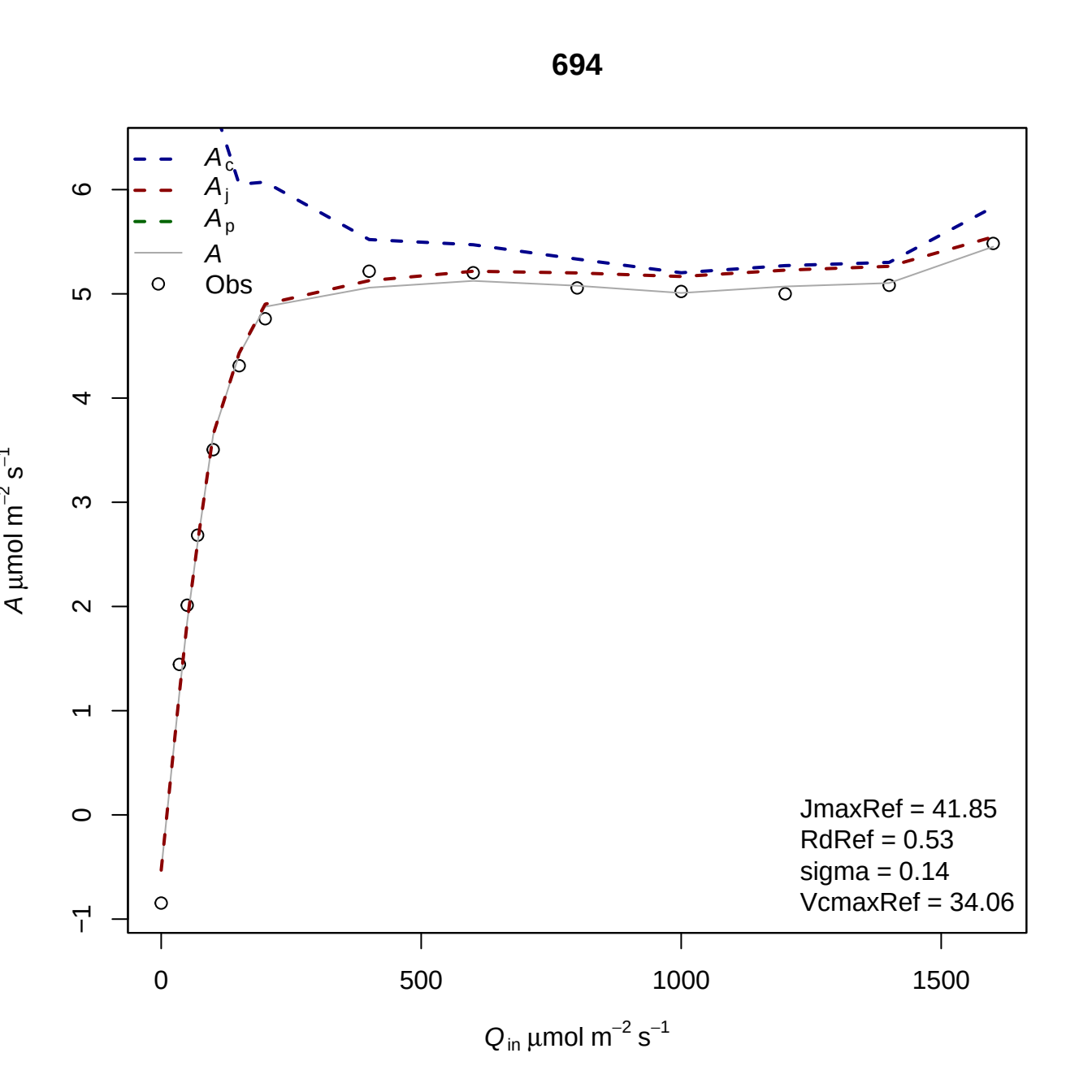

689
A_c
A_j
A_p
A
Obs
JmaxRef = 43.27
RdRef = 1.15
sigma = 0.24
VcmaxRef = 100.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

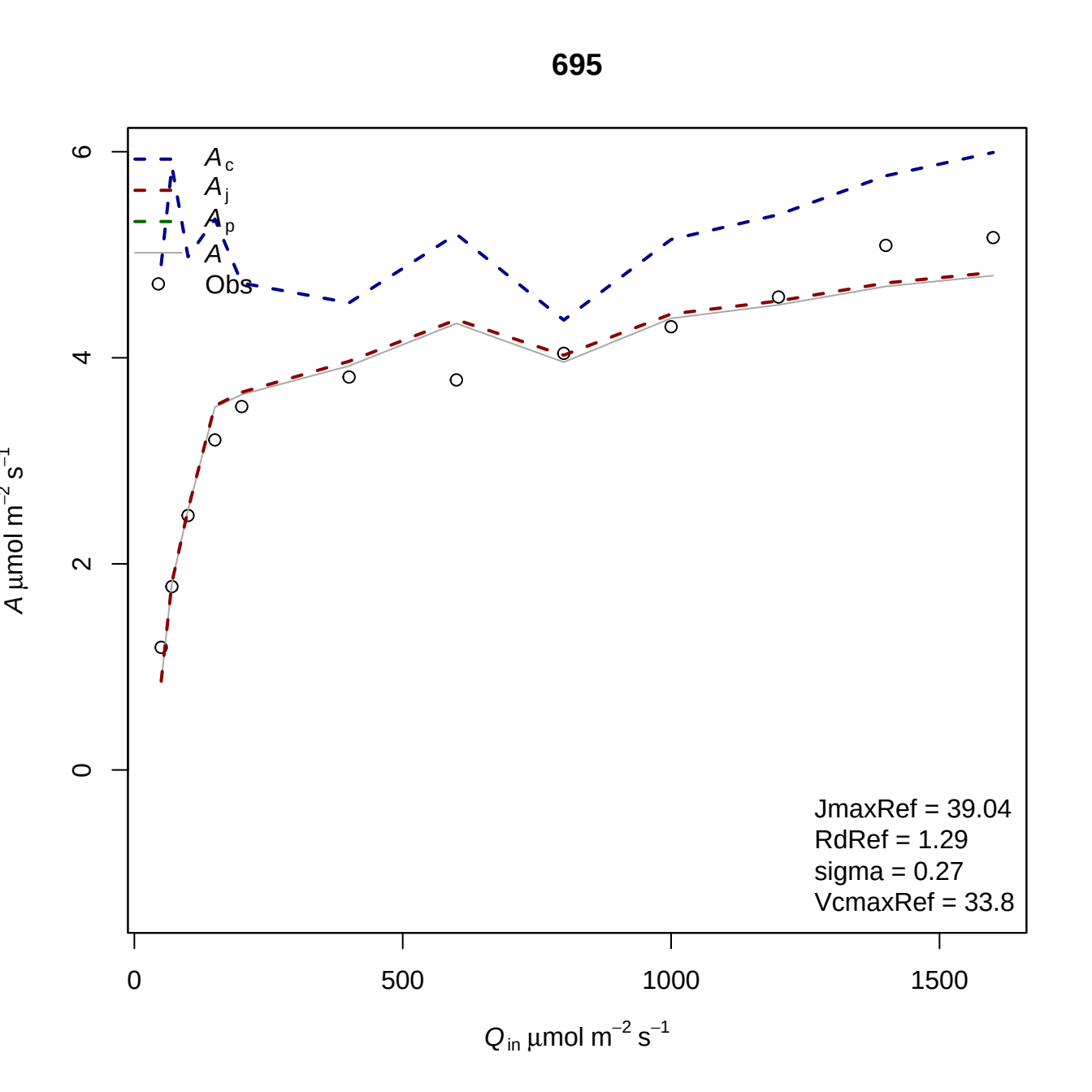

690
A_c
A_j
A_p
A
Obs
JmaxRef = 43.87
RdRef = 1.95
sigma = 0.6
VcmaxRef = 122.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

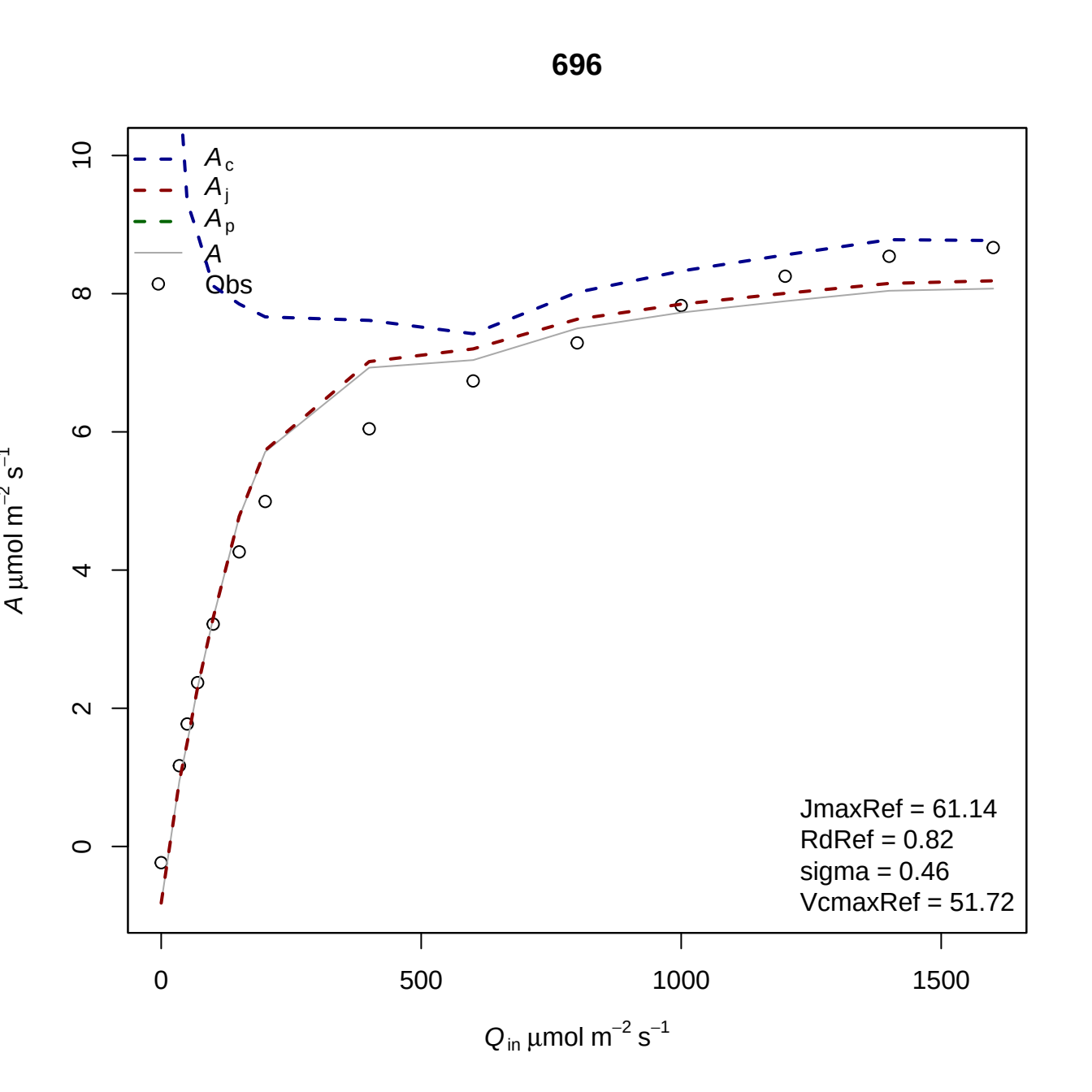

691
A_c
A_j
A_p
A
Obs
JmaxRef = 70.52
RdRef = 1.92
sigma = 0.64
VcmaxRef = 120.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

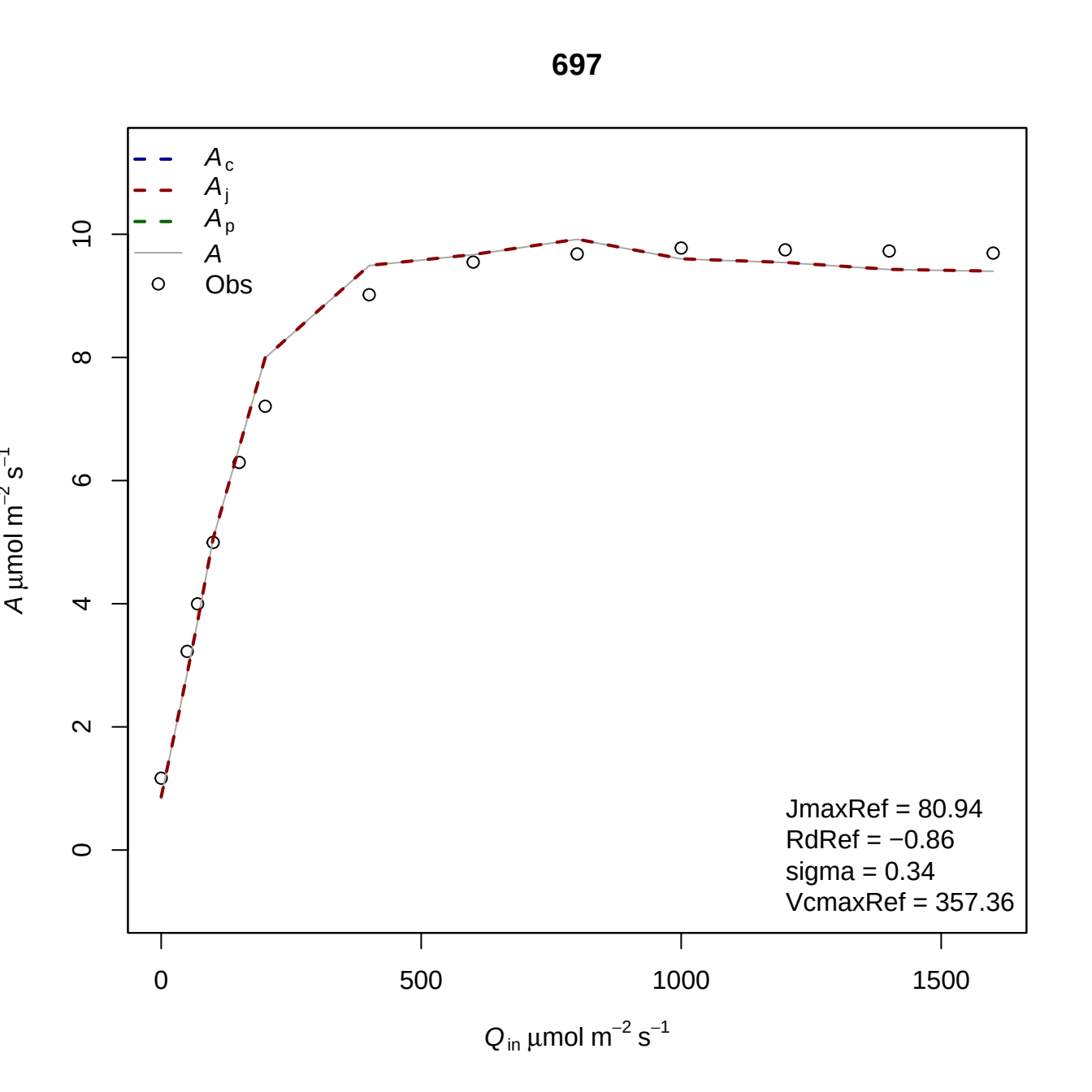

692
A_c
A_j
A_p
A
Obs
JmaxRef = 61.7
RdRef = 1.64
sigma = 0.3
VcmaxRef = 99.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

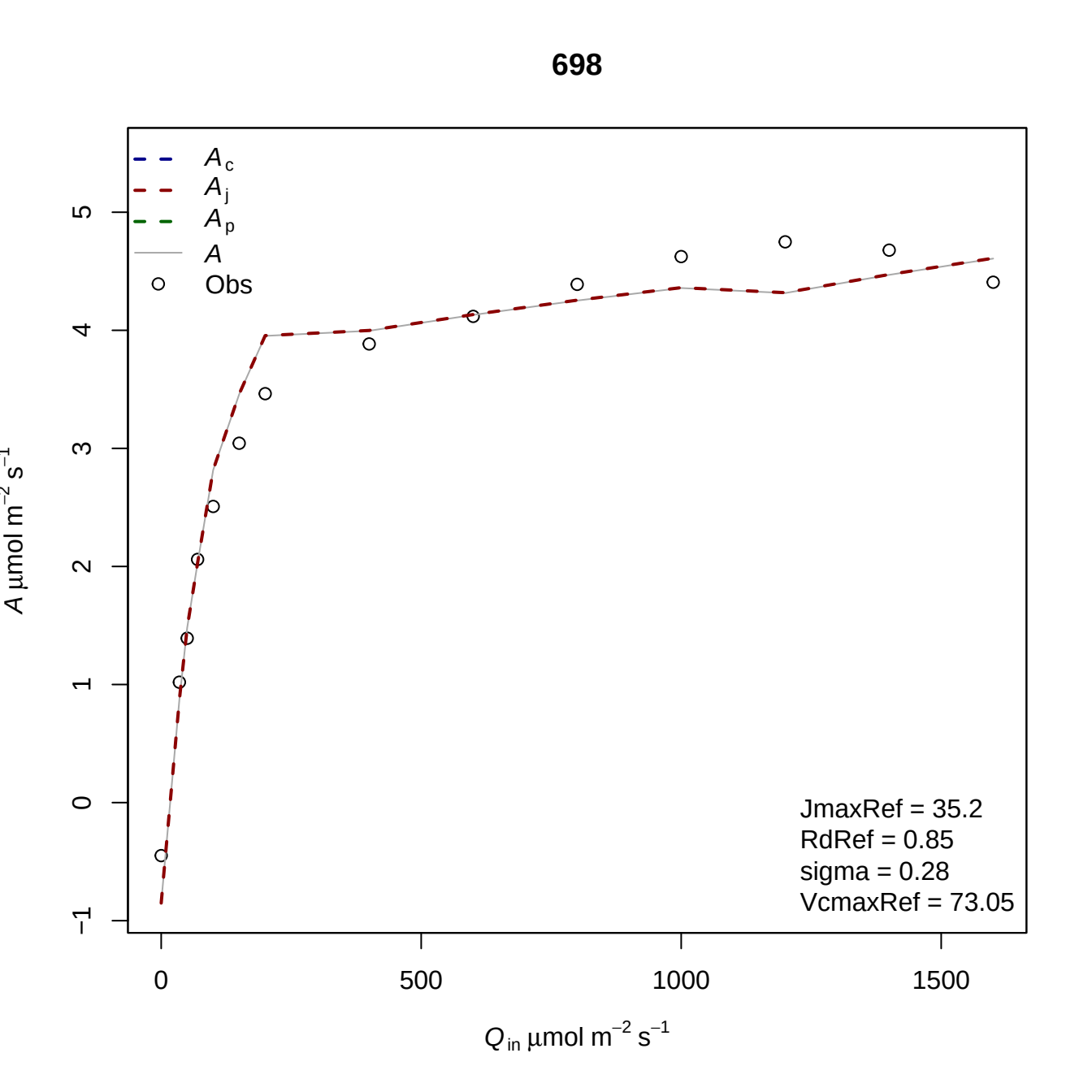

693
A_c
A_j
A_p
A
Obs
JmaxRef = 44.42
RdRef = 1.32
sigma = 0.68
VcmaxRef = 36.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

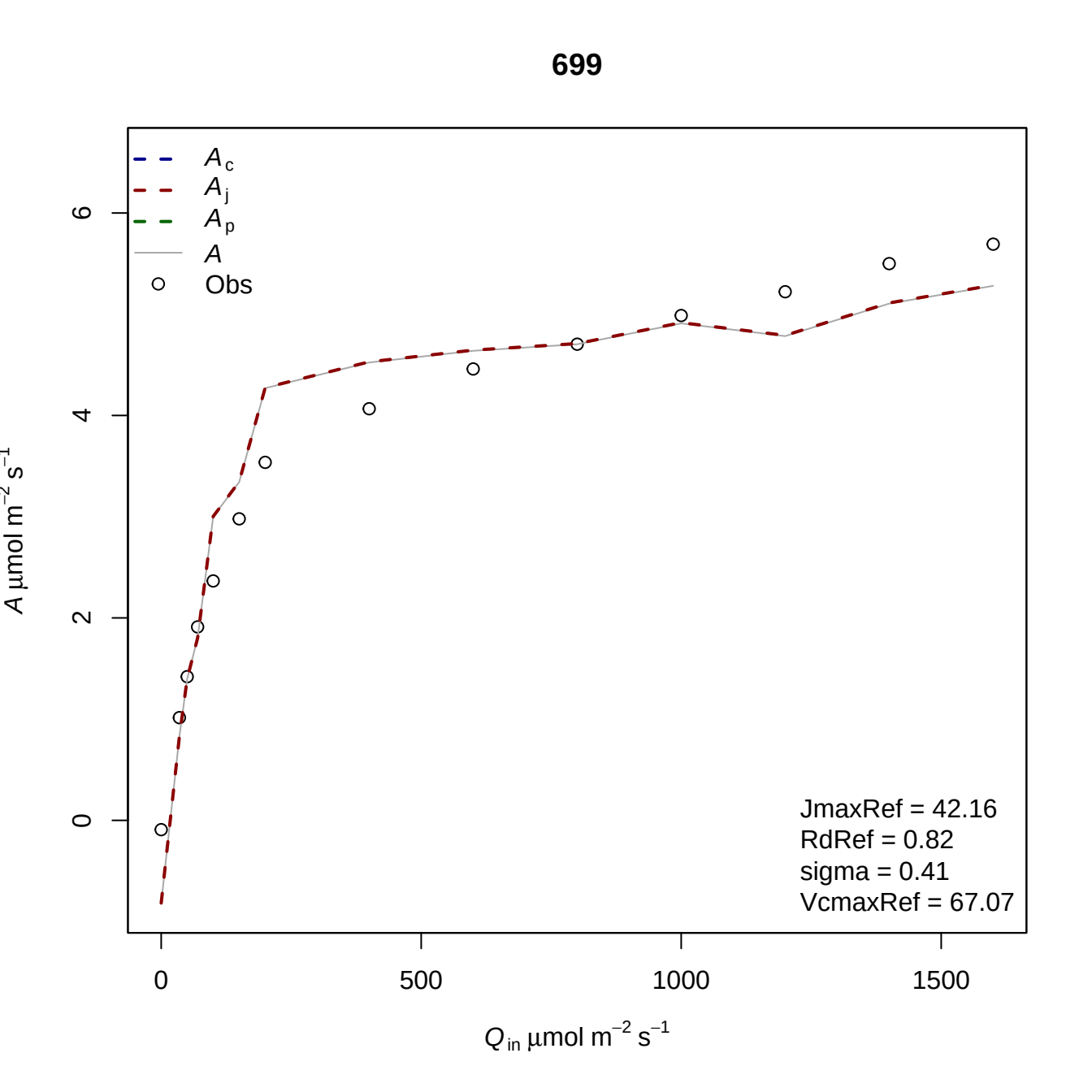

694
A_c
A_j
A_p
A
Obs
JmaxRef = 41.85
RdRef = 0.53
sigma = 0.14
VcmaxRef = 34.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

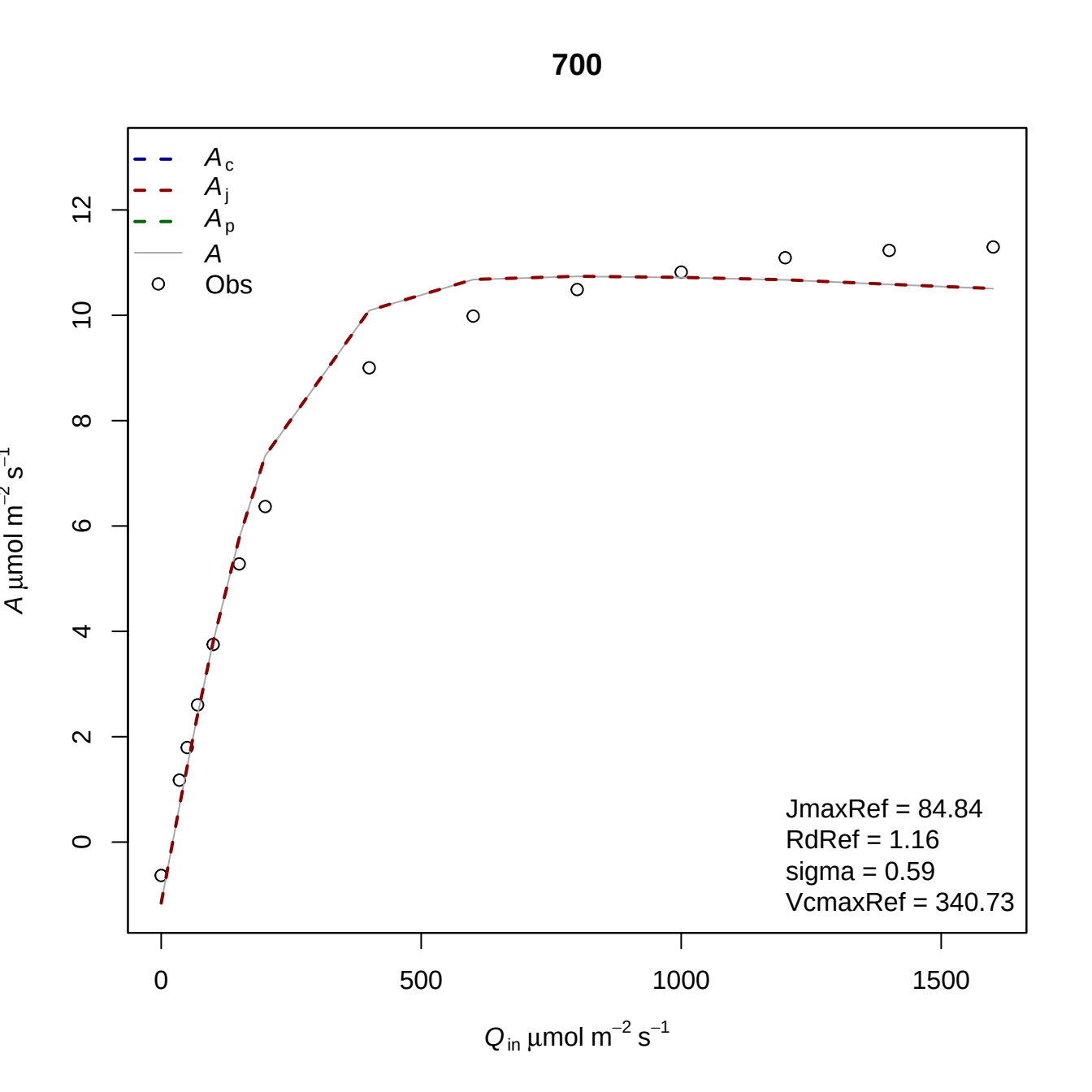

695
A_c
A_j
A_p
A
Obs
JmaxRef = 39.04
RdRef = 1.29
sigma = 0.27
VcmaxRef = 33.8
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

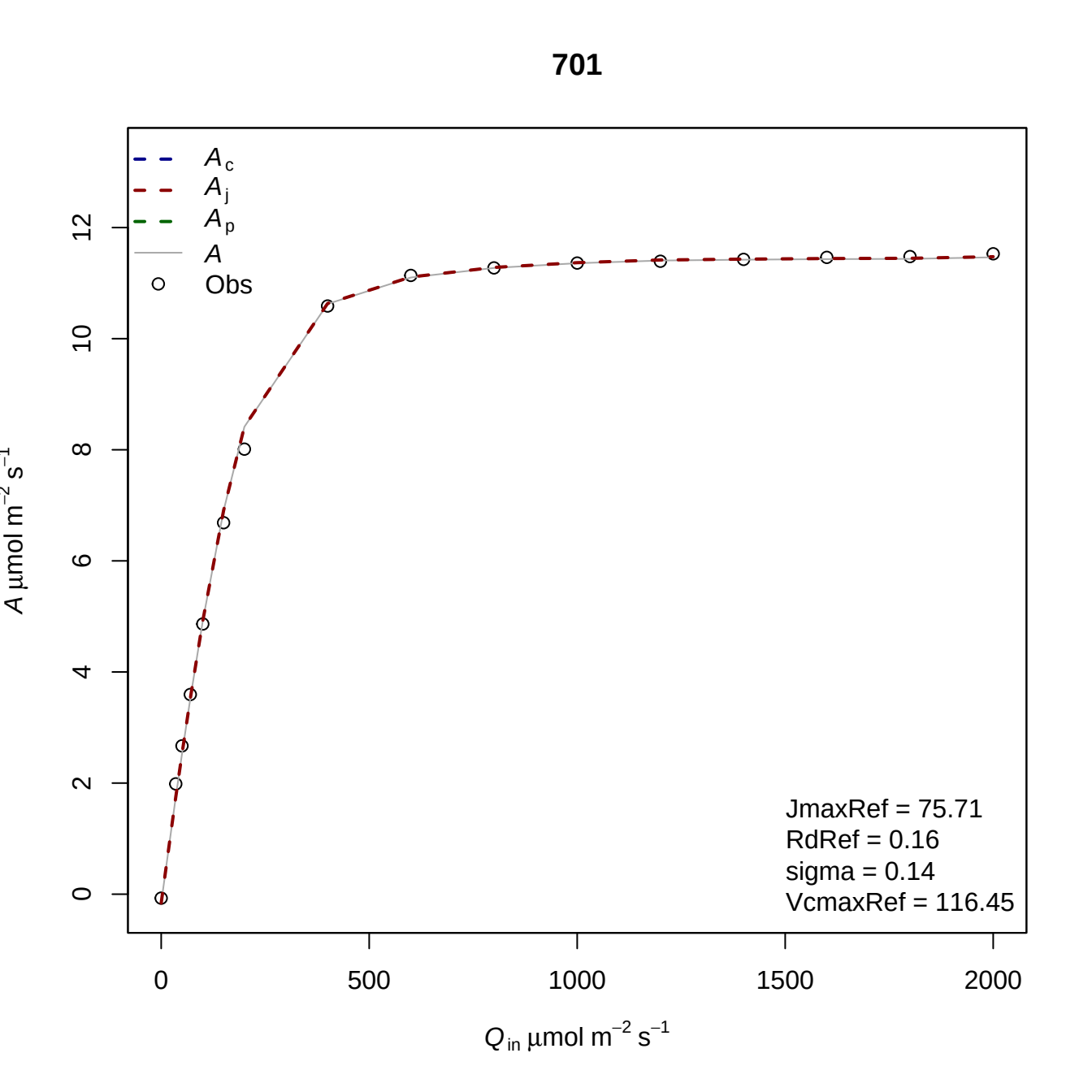

696
A_c
A_j
A_p
A
Qbs
JmaxRef = 61.14
RdRef = 0.82
sigma = 0.46
VcmaxRef = 51.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

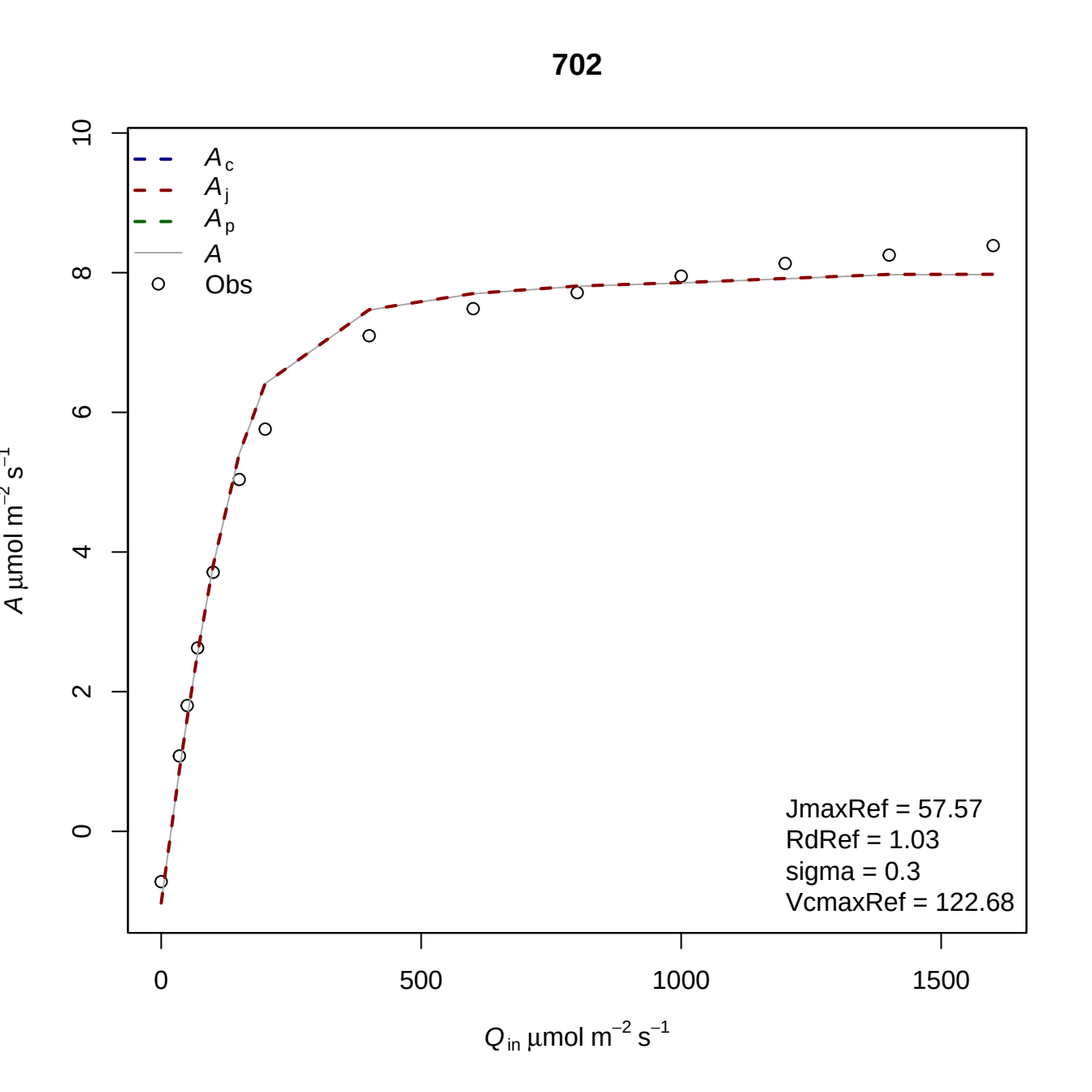

697
A_c
A_j
A_p
A
Obs
JmaxRef = 80.94
RdRef = −0.86
sigma = 0.34
VcmaxRef = 357.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

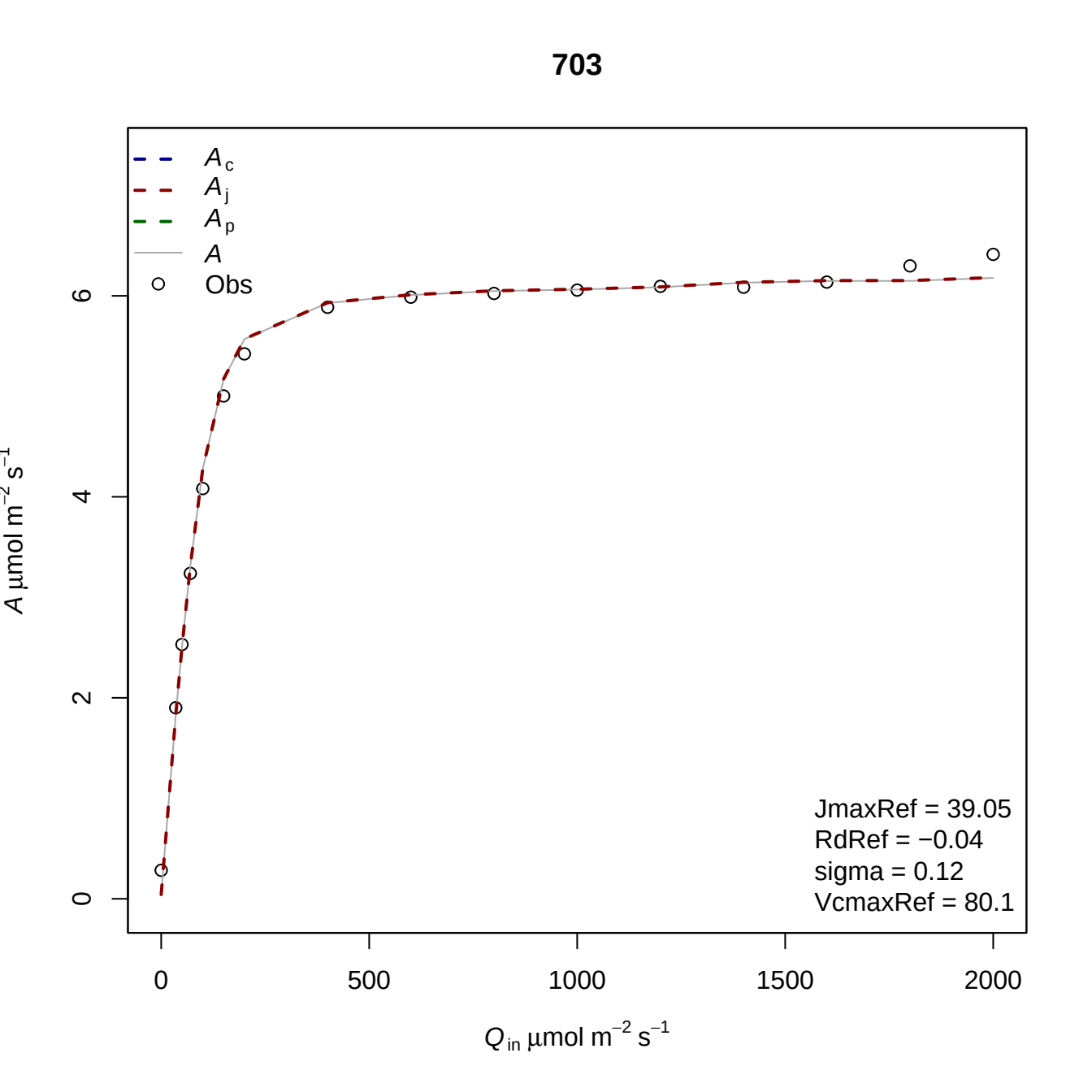

698
A_c
A_j
A_p
A
Obs
JmaxRef = 35.2
RdRef = 0.85
sigma = 0.28
VcmaxRef = 73.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

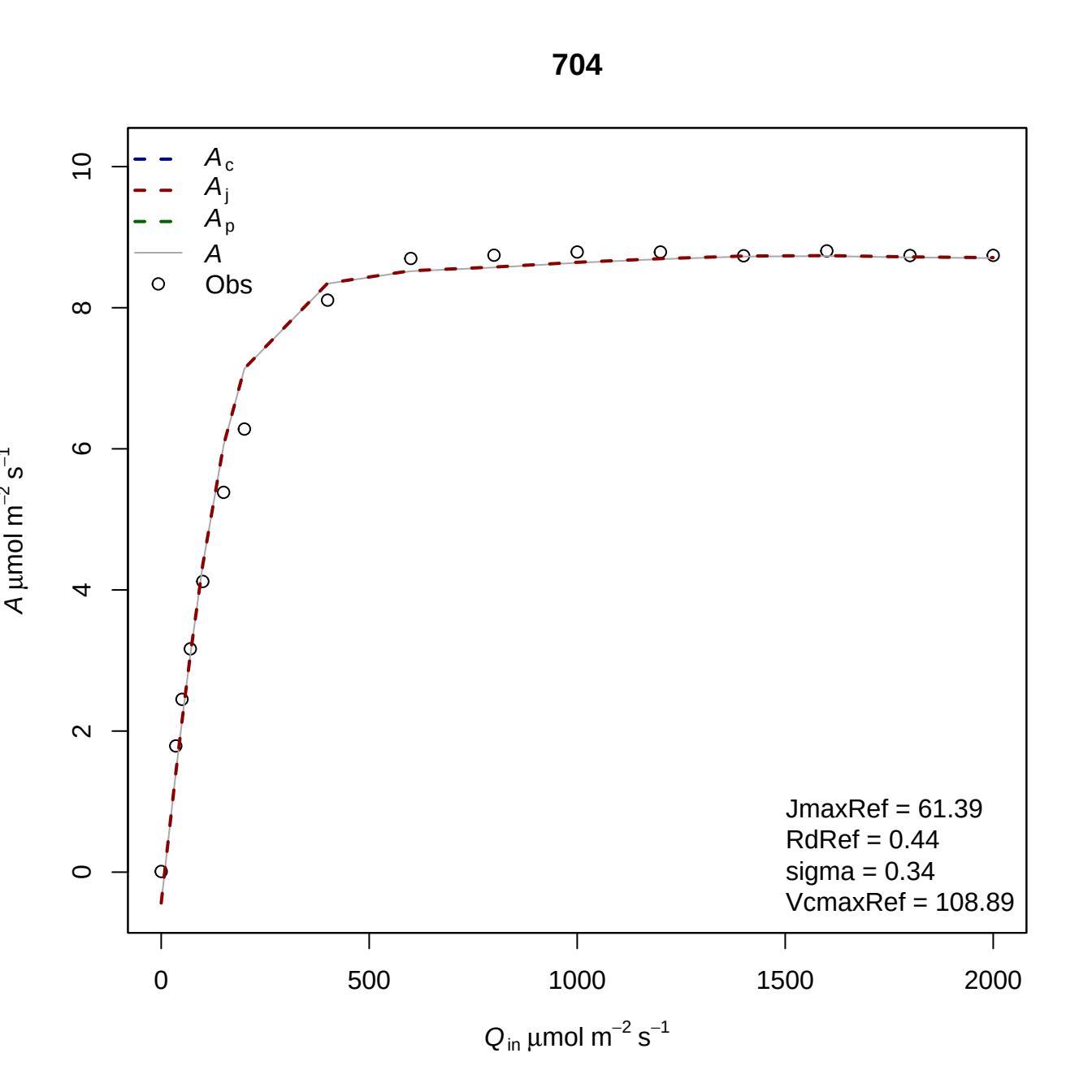

699
A_c
A_j
A_p
A
Obs
JmaxRef = 42.16
RdRef = 0.82
sigma = 0.41
VcmaxRef = 67.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

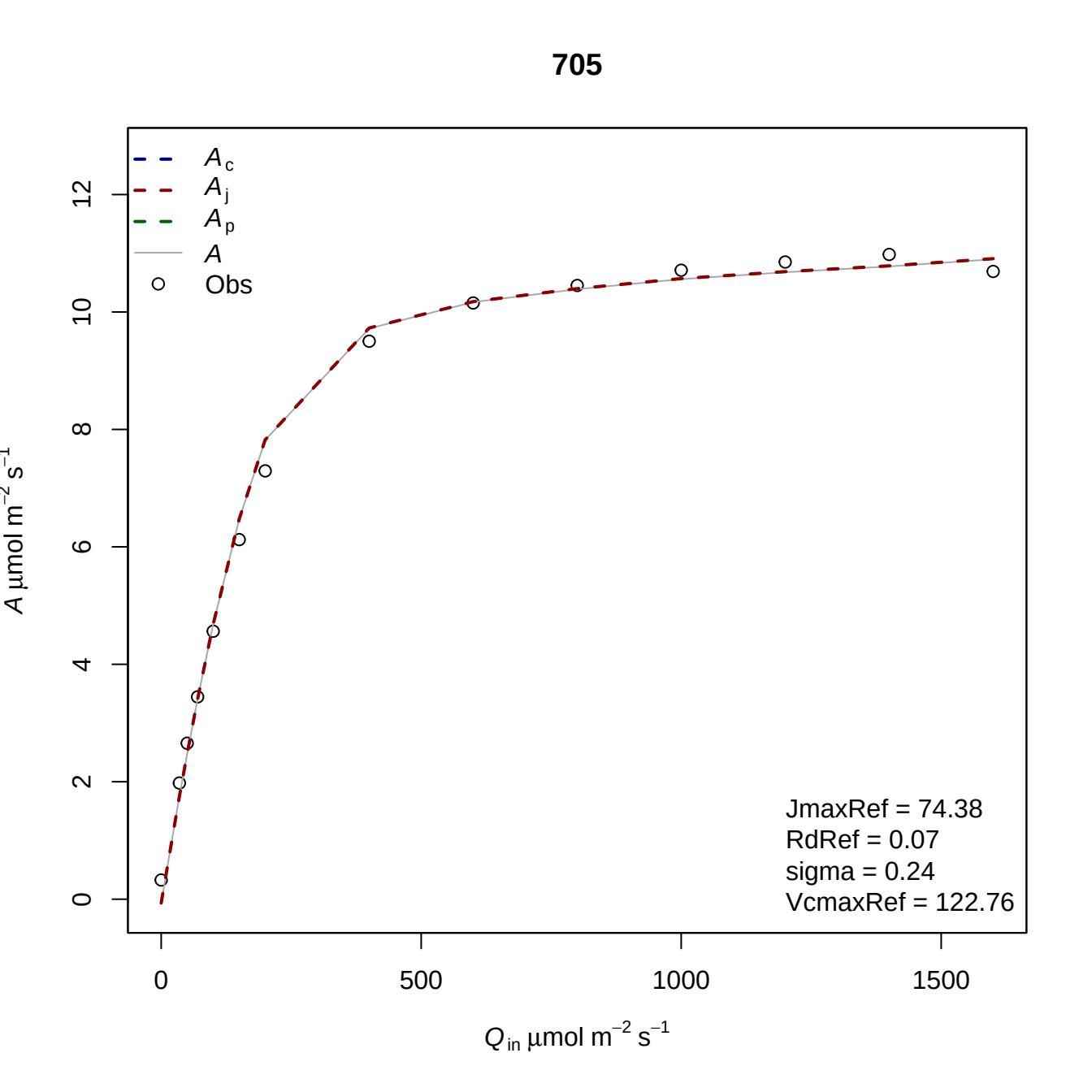

700
A_c
A_j
A_p
A
Obs
JmaxRef = 84.84
RdRef = 1.16
sigma = 0.59
VcmaxRef = 340.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

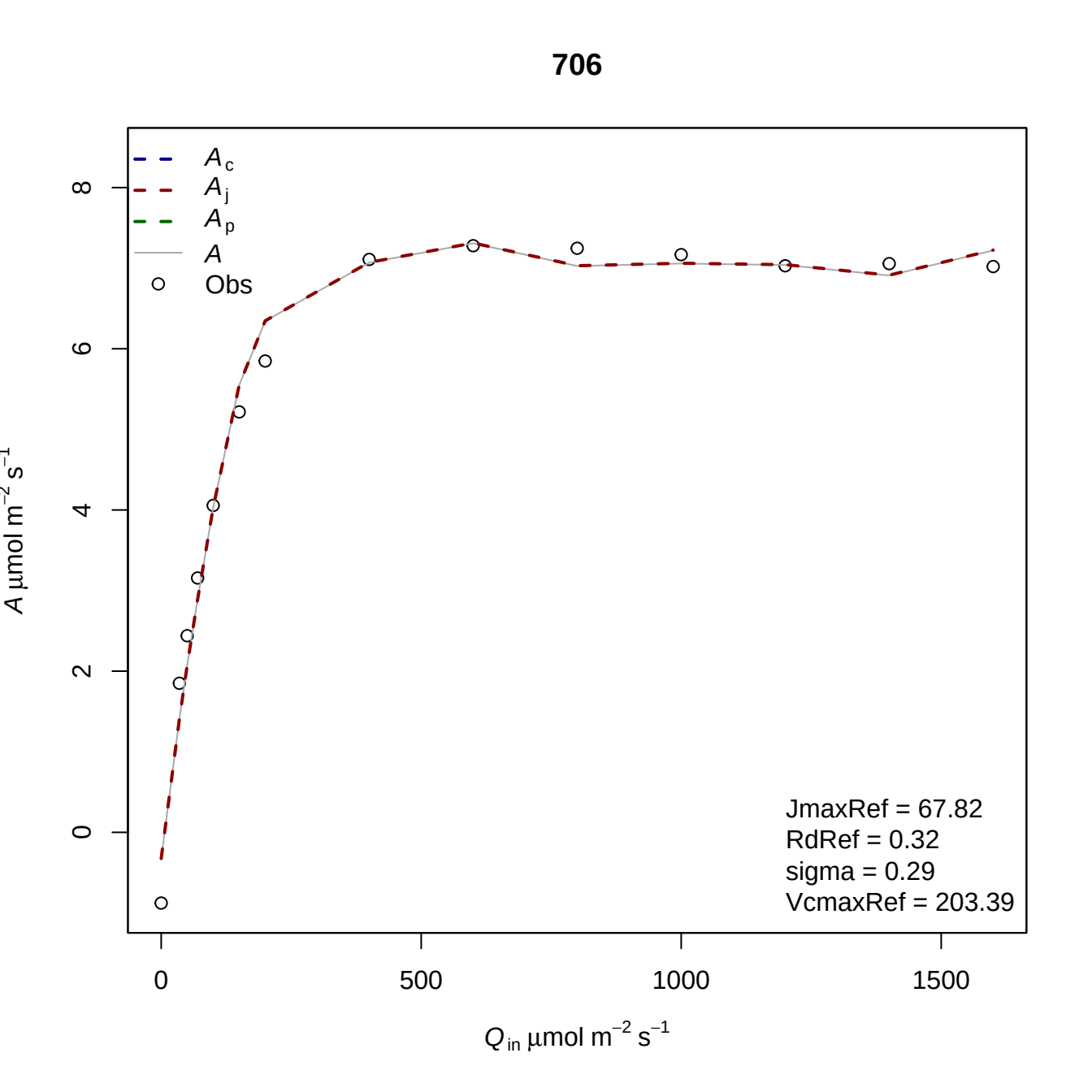

701
A_c
A_j
A_p
A
Obs
JmaxRef = 75.71
RdRef = 0.16
sigma = 0.14
VcmaxRef = 116.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

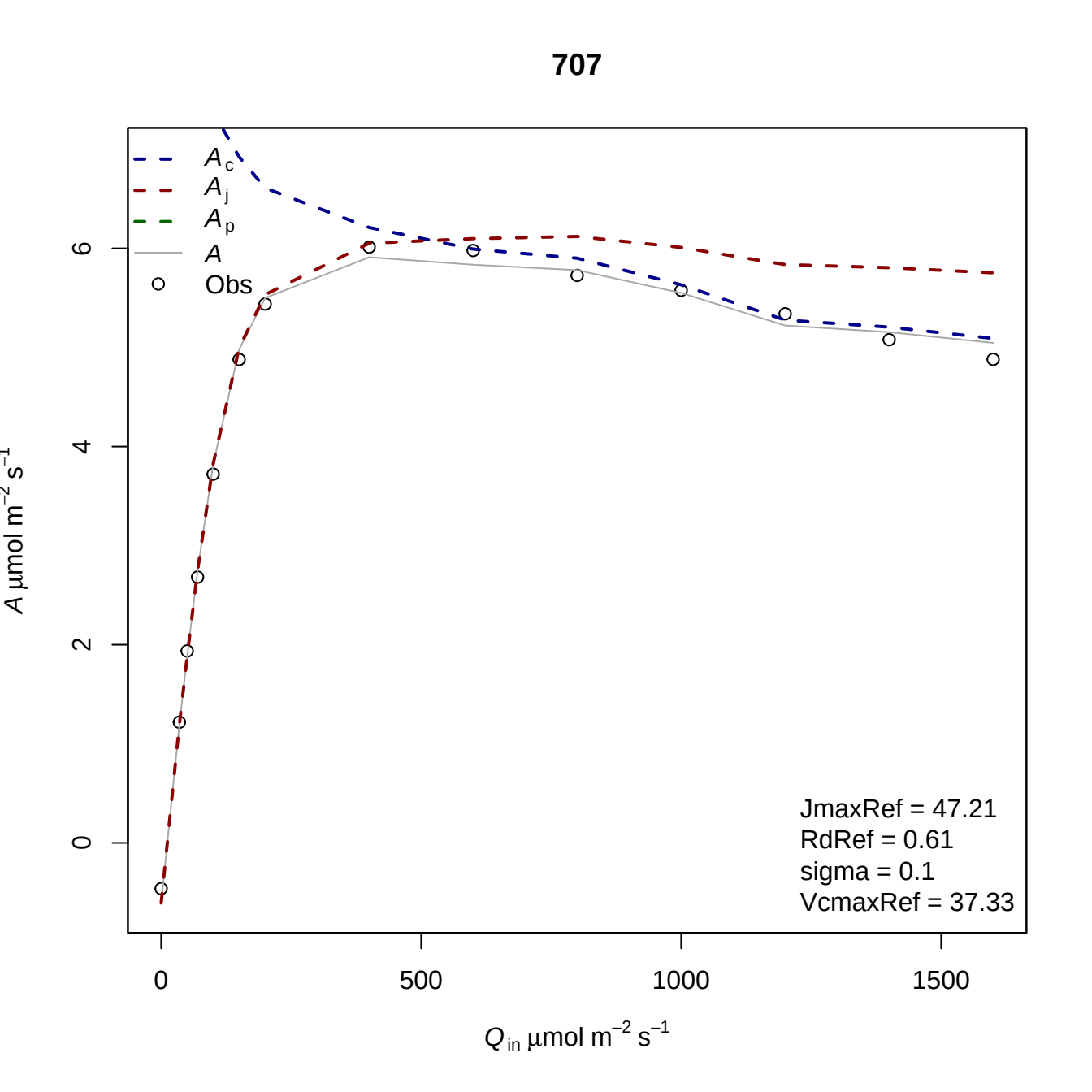

702
A_c
A_j
A_p
A
Obs
JmaxRef = 57.57
RdRef = 1.03
sigma = 0.3
VcmaxRef = 122.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

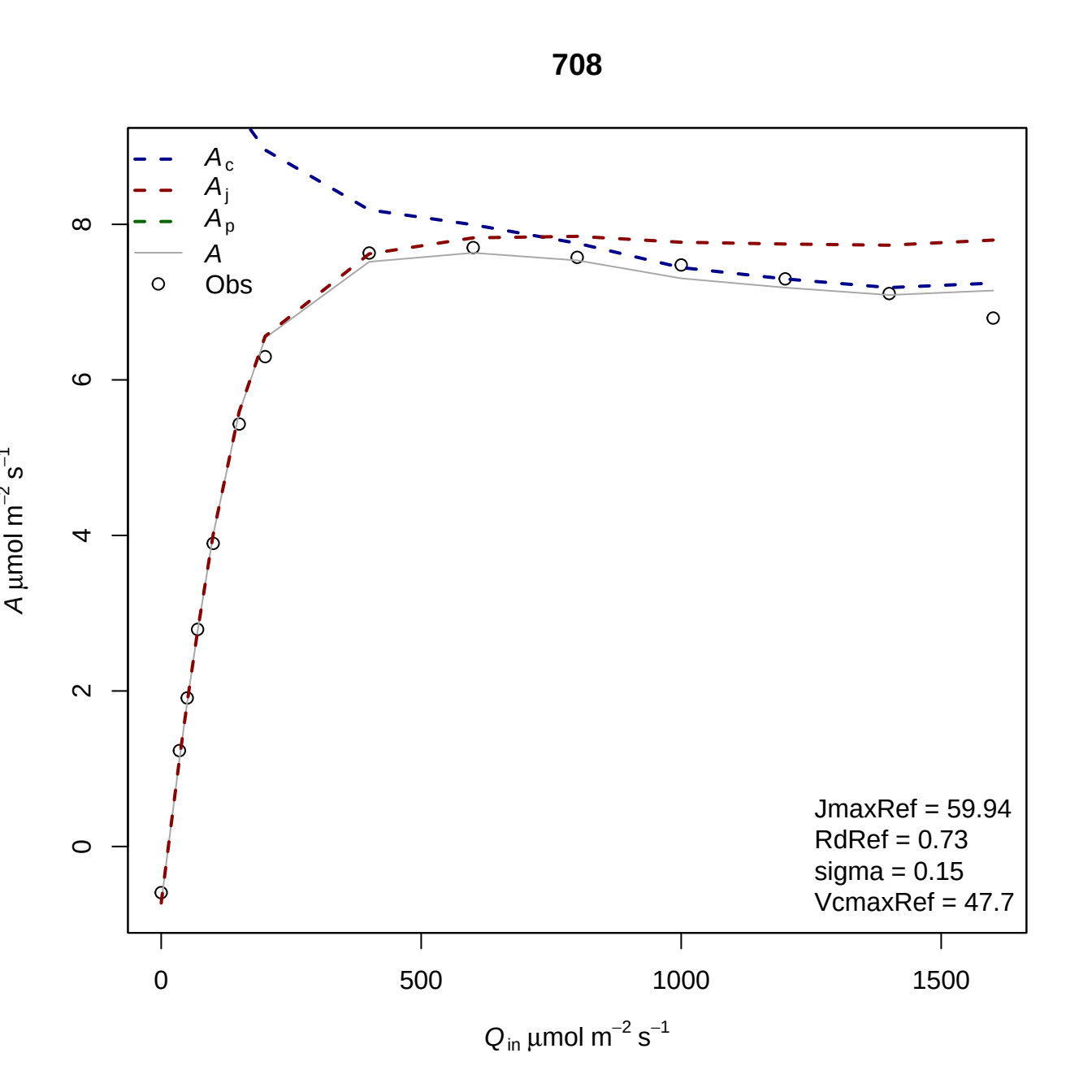

703
A_c
A_j
A_p
A
Obs
JmaxRef = 39.05
RdRef = −0.04
sigma = 0.12
VcmaxRef = 80.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

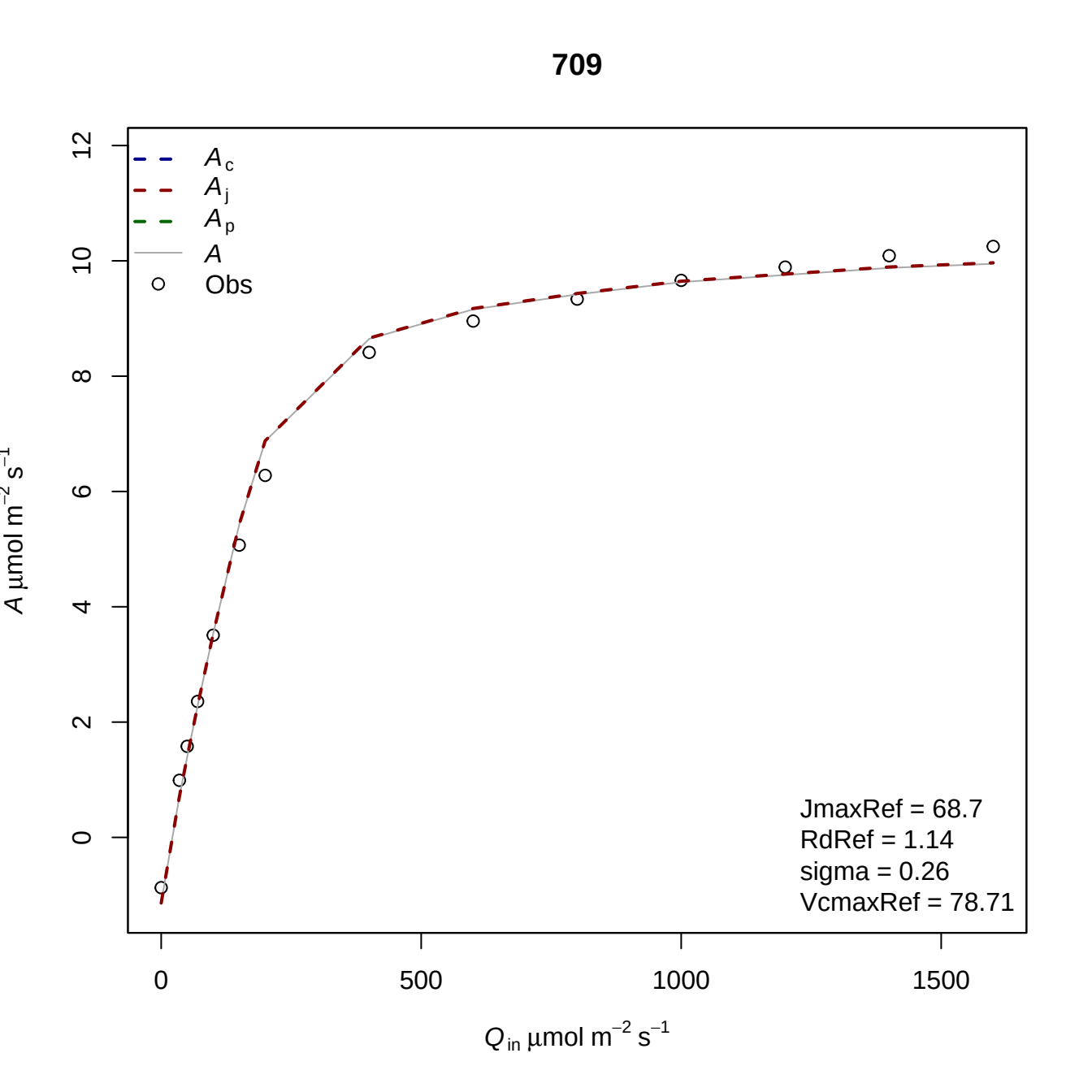

704
A_c
A_j
A_p
A
Obs
JmaxRef = 61.39
RdRef = 0.44
sigma = 0.34
VcmaxRef = 108.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

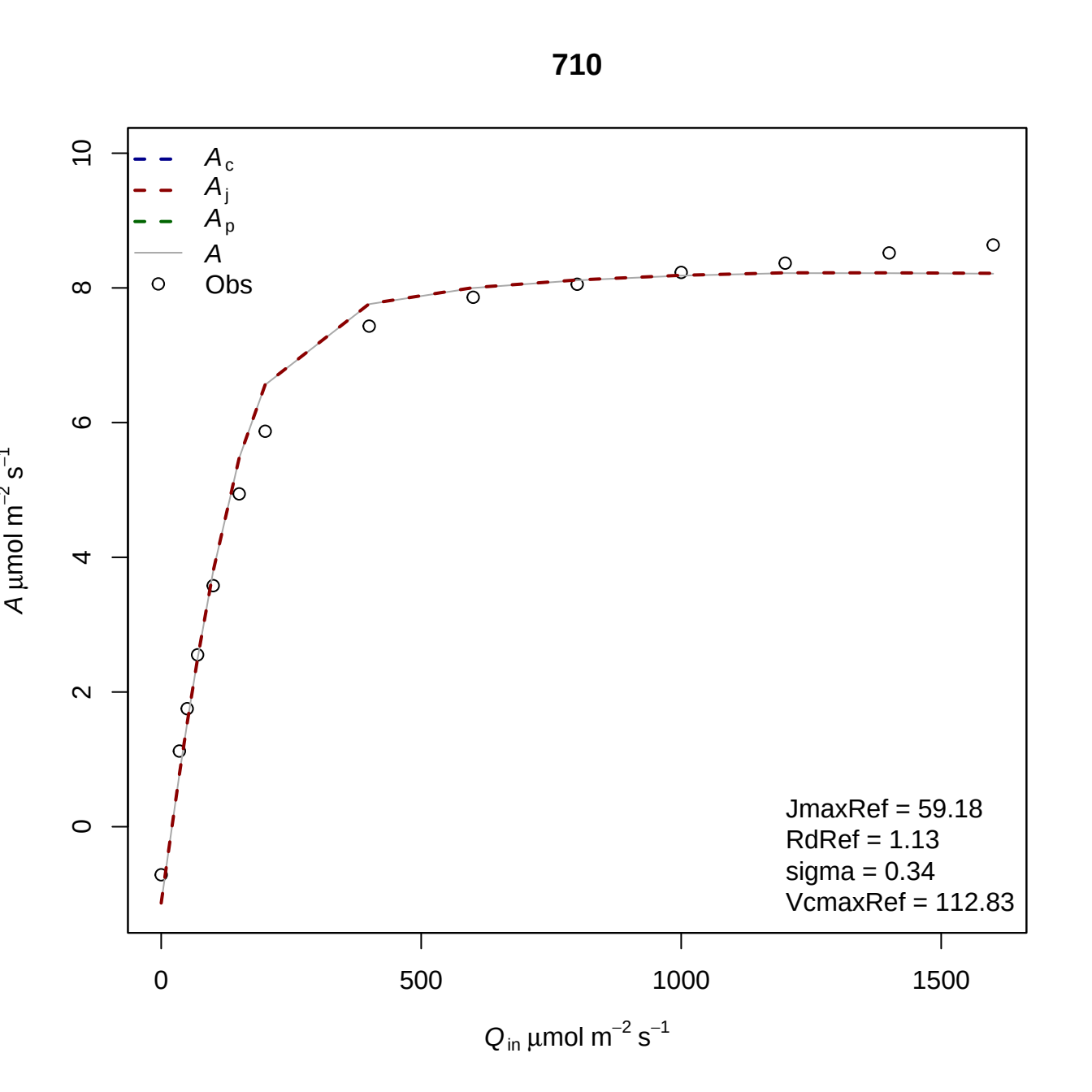

705
A_c
A_j
A_p
A
Obs
JmaxRef = 74.38
RdRef = 0.07
sigma = 0.24
VcmaxRef = 122.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

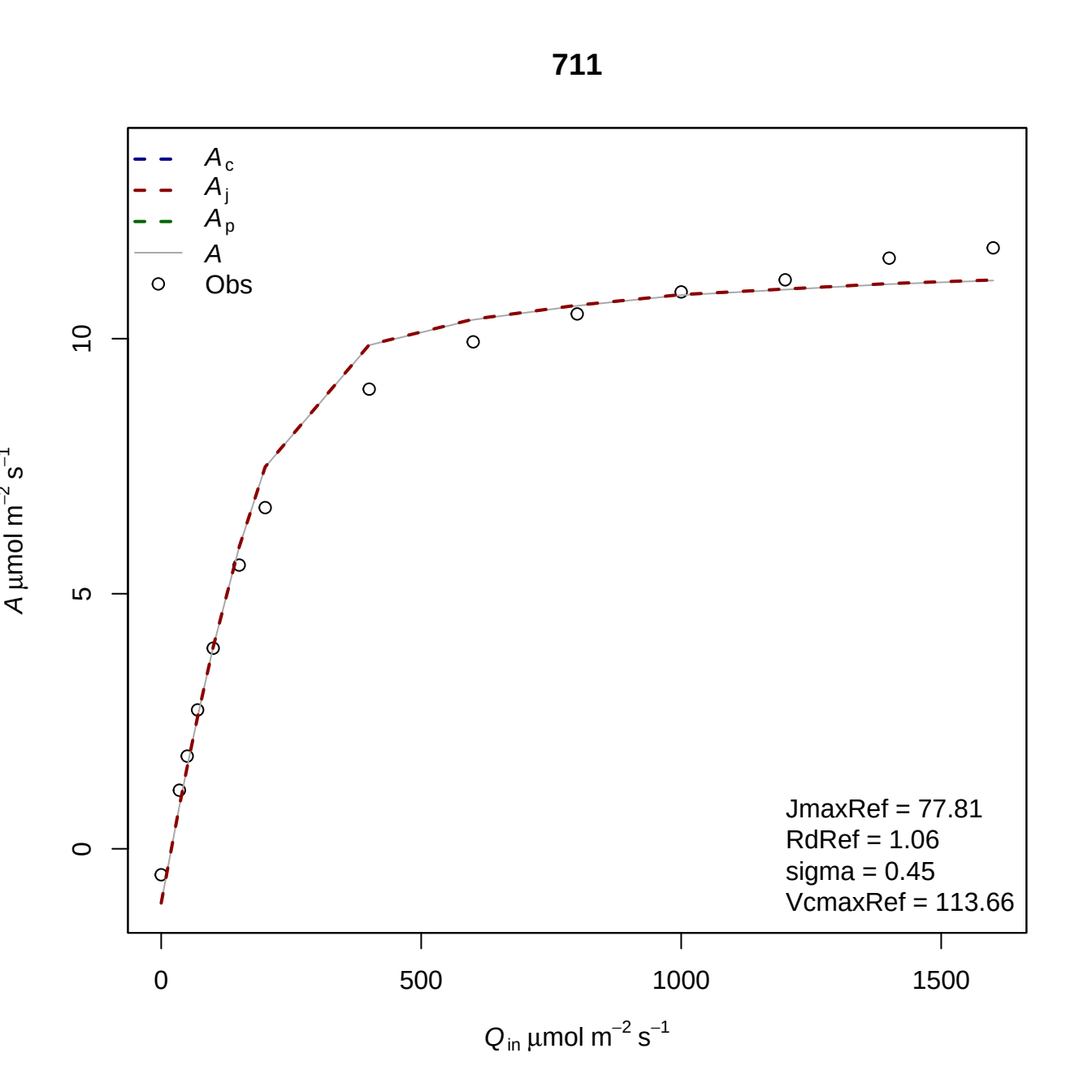

706
A_c
A_j
A_p
A
Obs
JmaxRef = 67.82
RdRef = 0.32
sigma = 0.29
VcmaxRef = 203.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

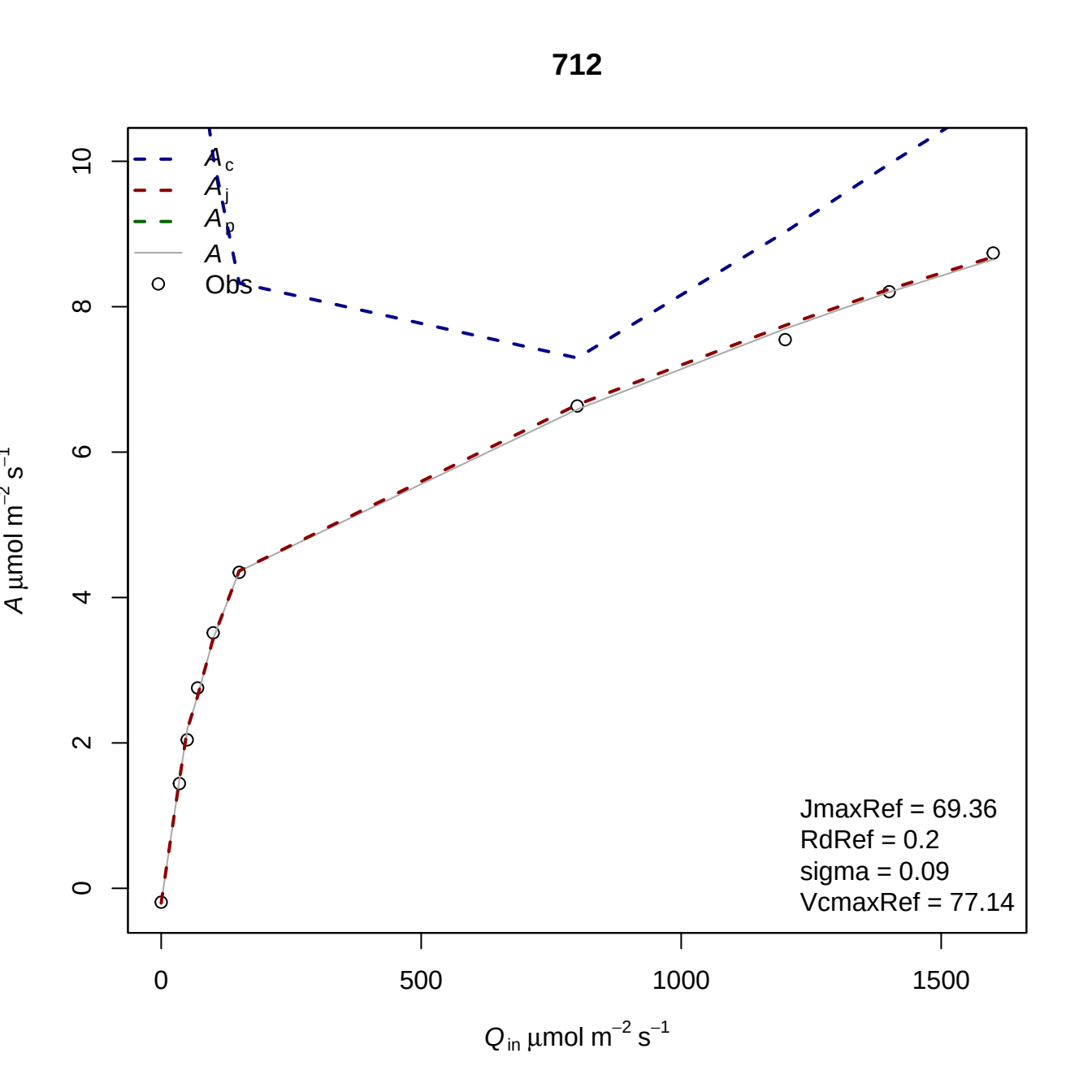

707
A_c
A_j
A_p
A
Obs
JmaxRef = 47.21
RdRef = 0.61
sigma = 0.1
VcmaxRef = 37.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

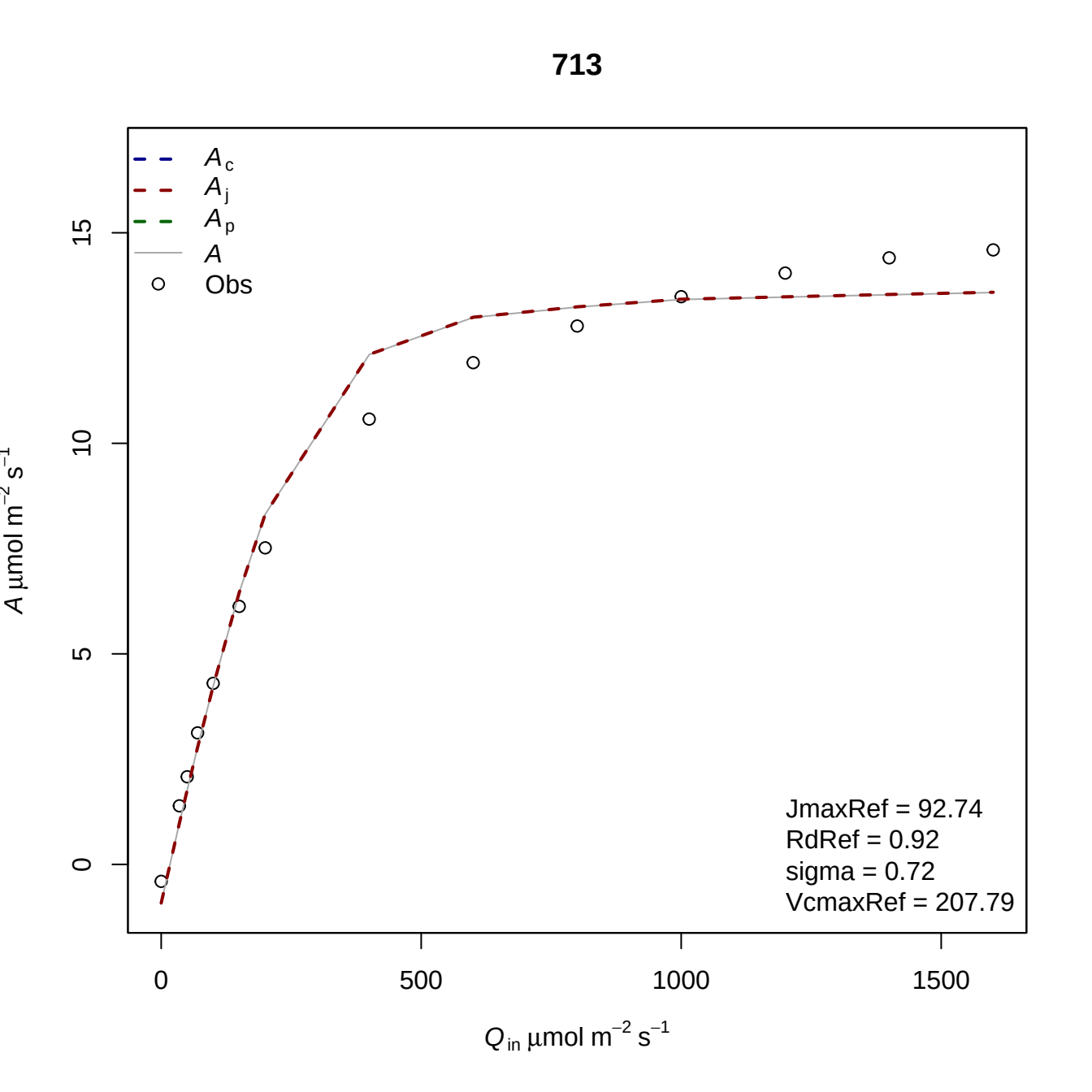

708
A_c
A_j
A_p
A
Obs
JmaxRef = 59.94
RdRef = 0.73
sigma = 0.15
VcmaxRef = 47.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

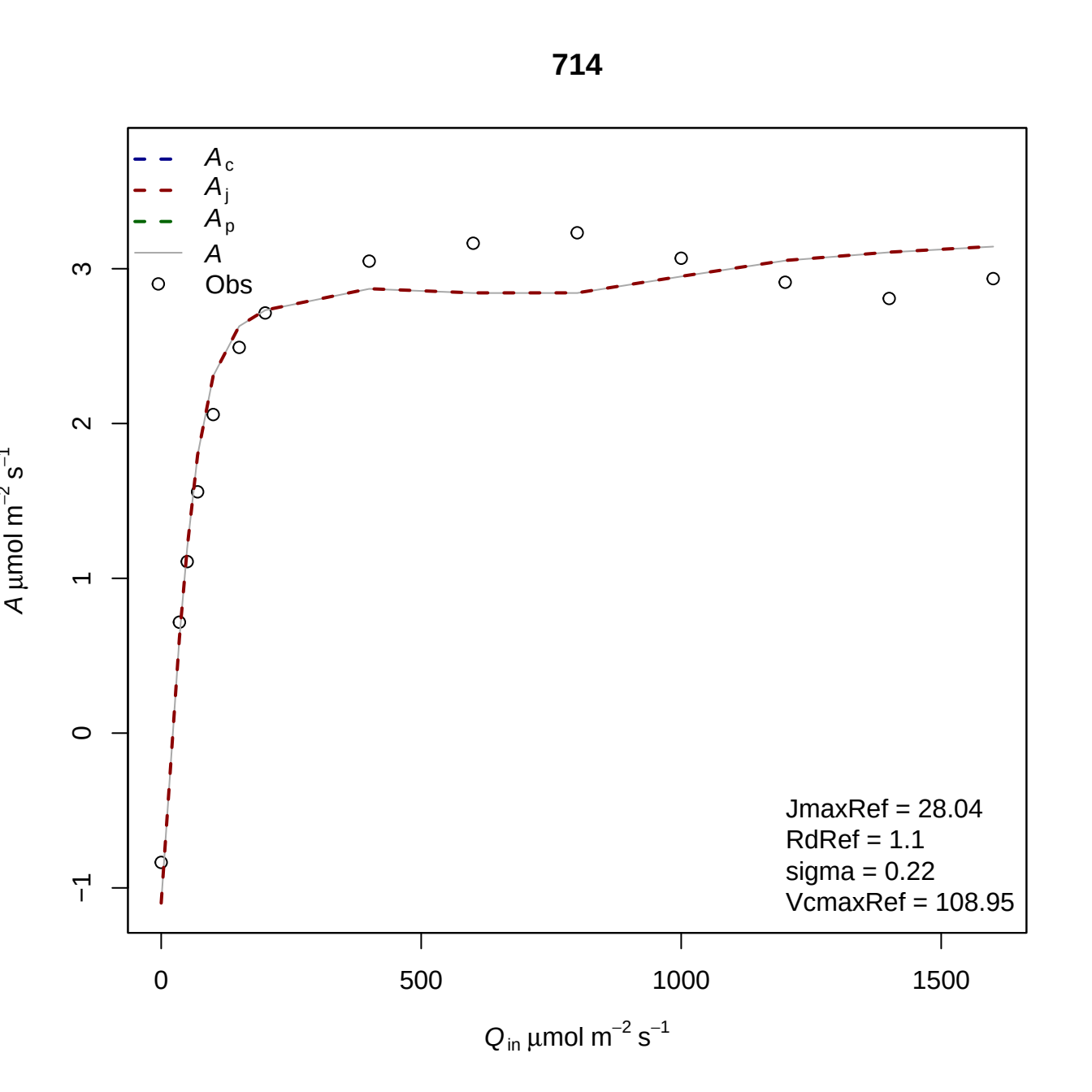

709
A_c
A_j
A_p
A
Obs
JmaxRef = 68.7
RdRef = 1.14
sigma = 0.26
VcmaxRef = 78.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

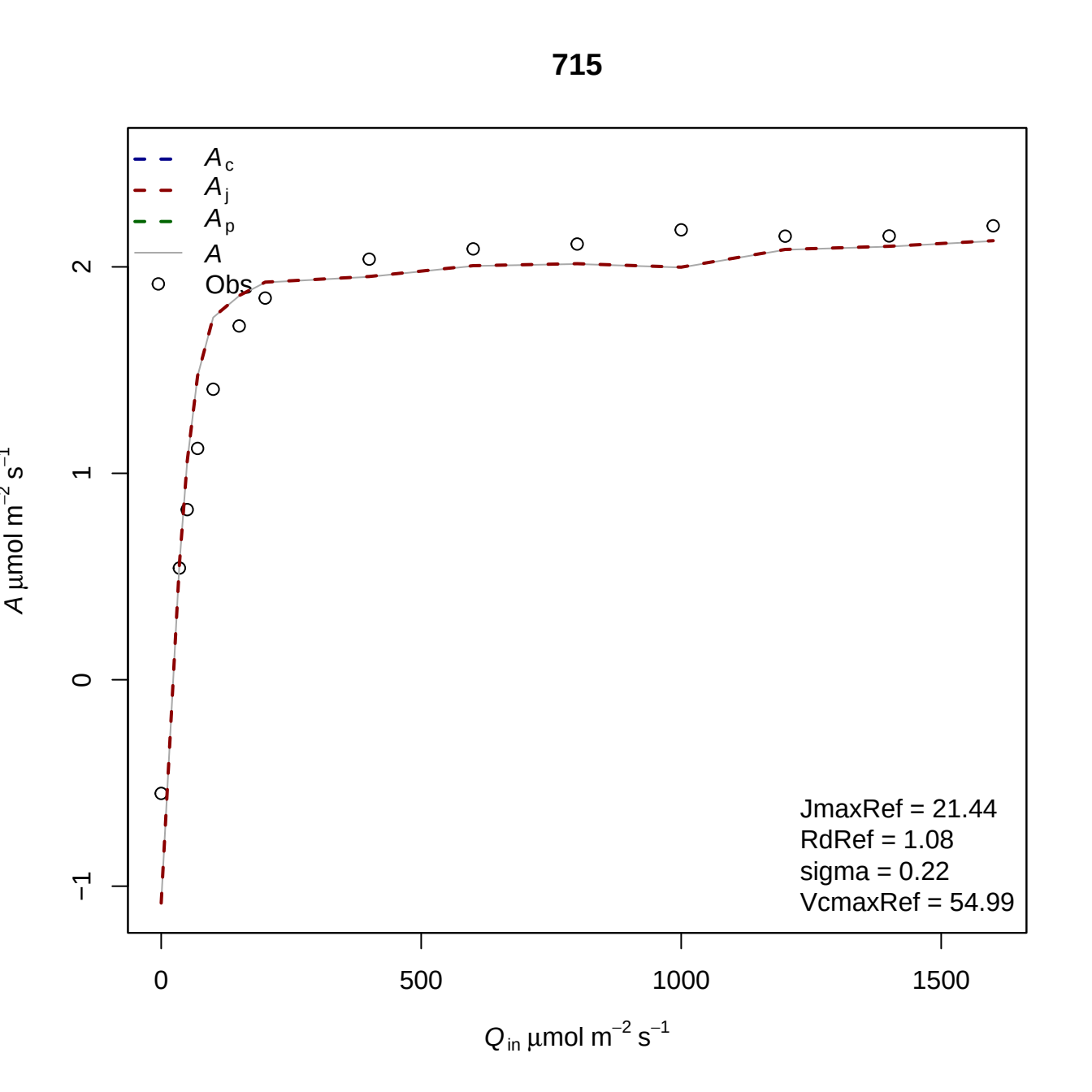

710
A_c
A_j
A_p
A
Obs
JmaxRef = 59.18
RdRef = 1.13
sigma = 0.34
VcmaxRef = 112.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

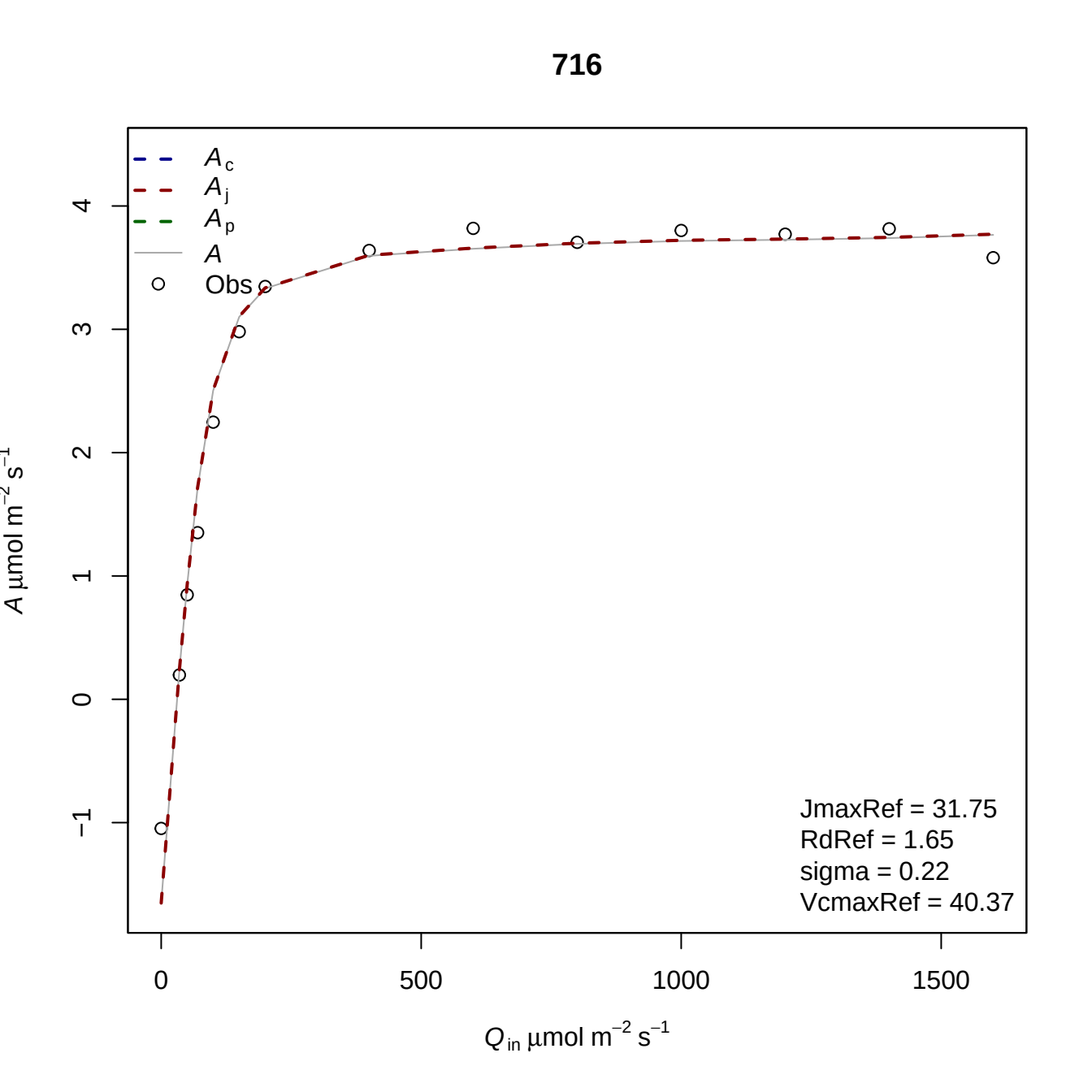

711
A_c
A_j
A_p
A
Obs
JmaxRef = 77.81
RdRef = 1.06
sigma = 0.45
VcmaxRef = 113.66
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

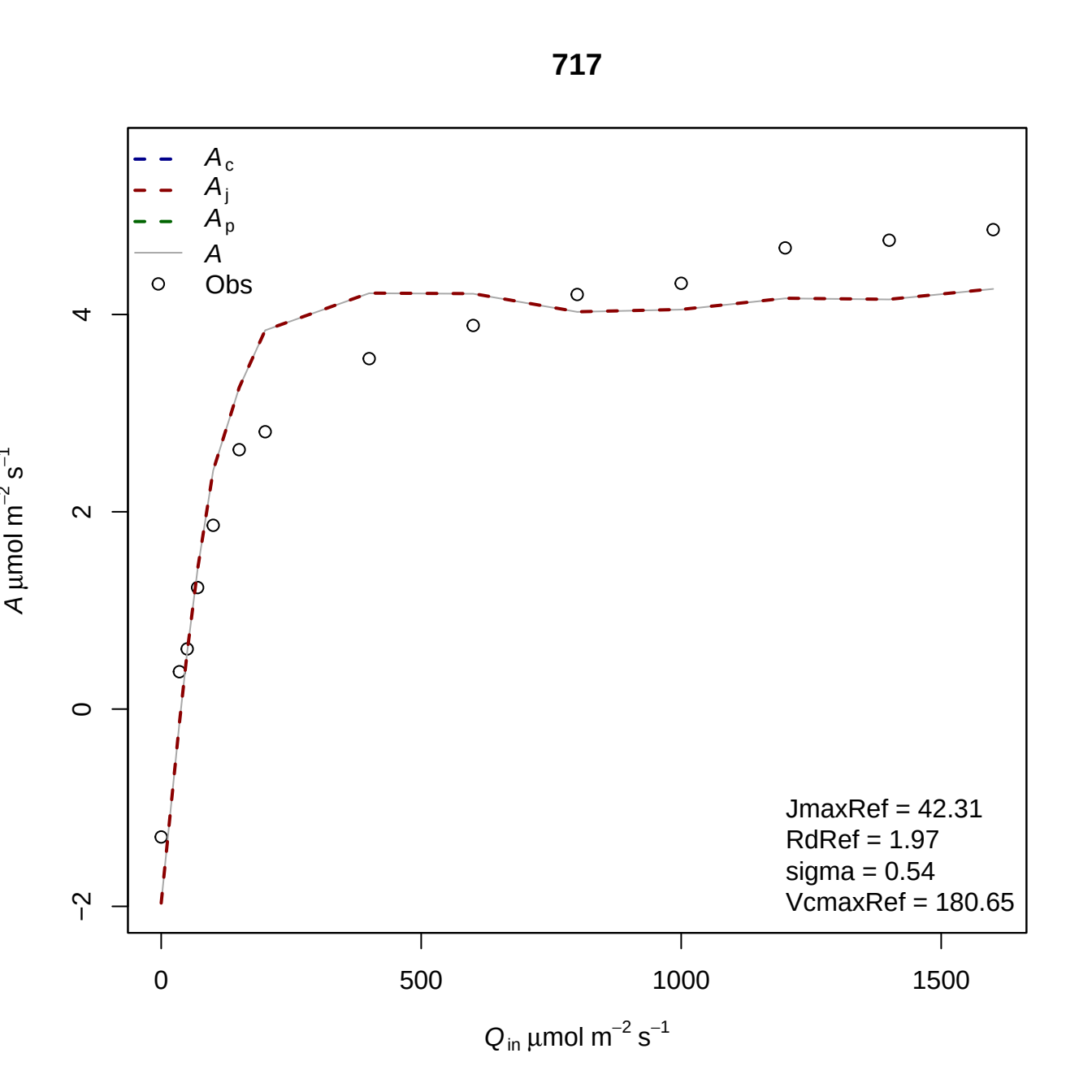

712
A_c
A_j
A_p
A
Obs
JmaxRef = 69.36
RdRef = 0.2
sigma = 0.09
VcmaxRef = 77.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

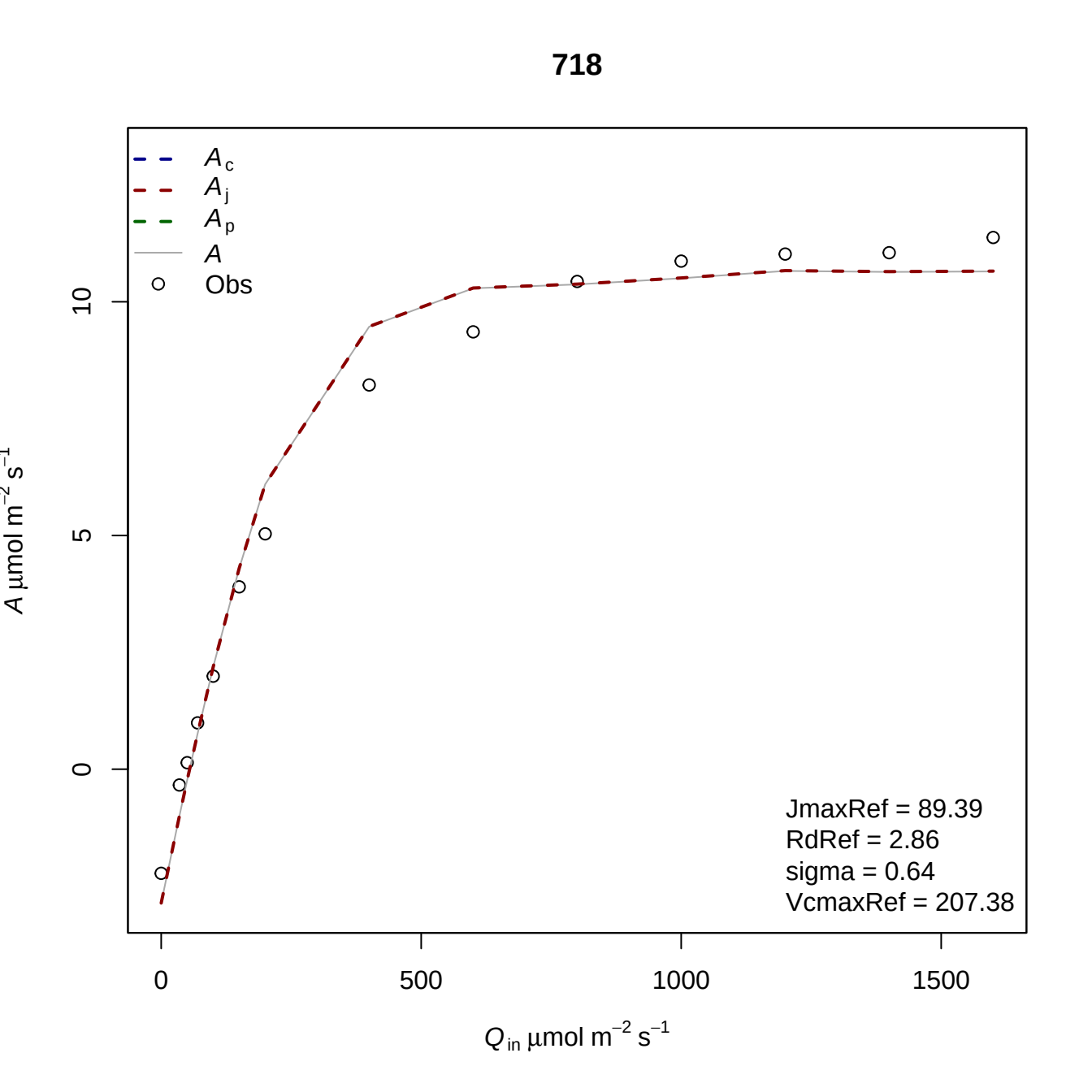

713
A_c
A_j
A_p
A
Obs
JmaxRef = 92.74
RdRef = 0.92
sigma = 0.72
VcmaxRef = 207.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

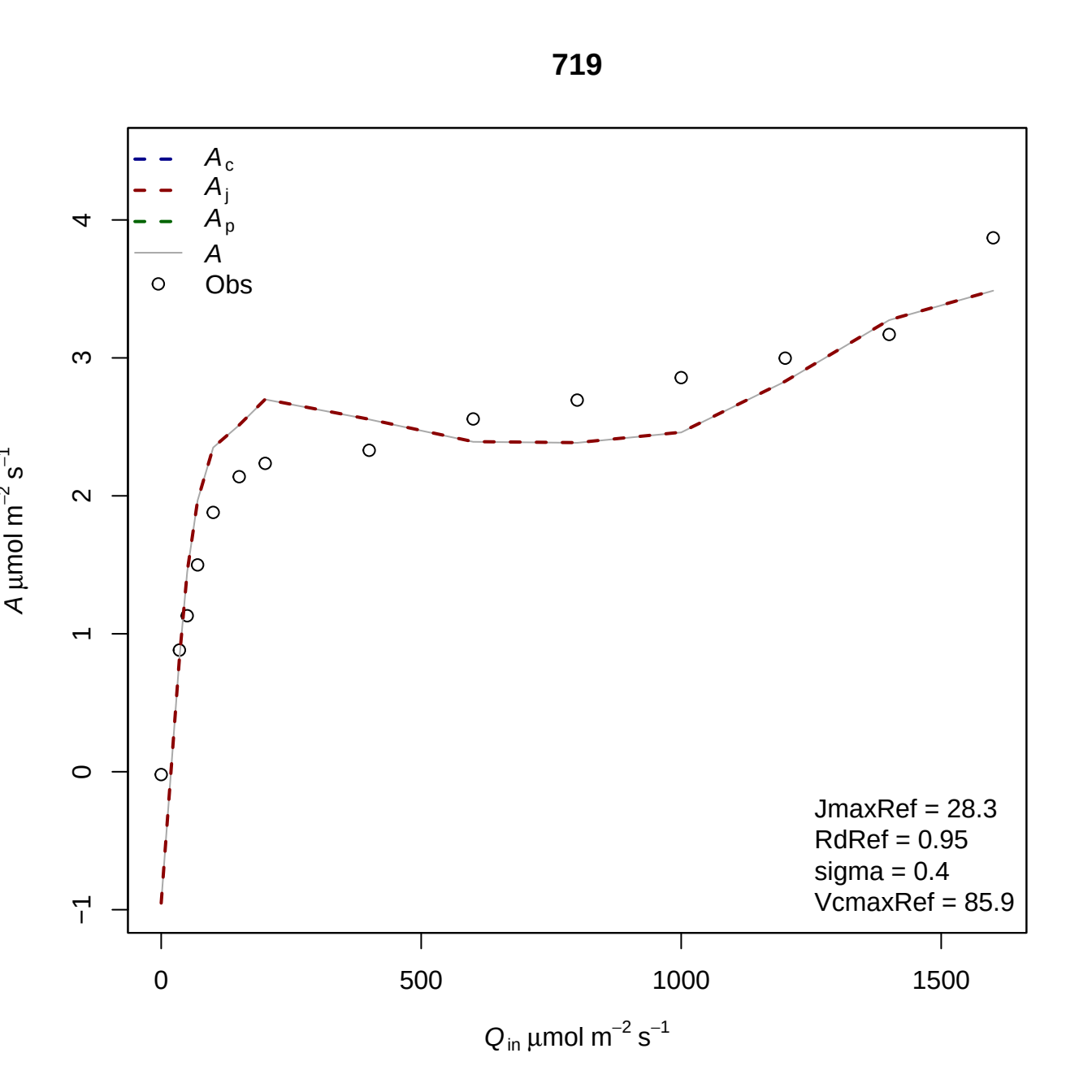

714
A_c
A_j
A_p
A
Obs
JmaxRef = 28.04
RdRef = 1.1
sigma = 0.22
VcmaxRef = 108.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

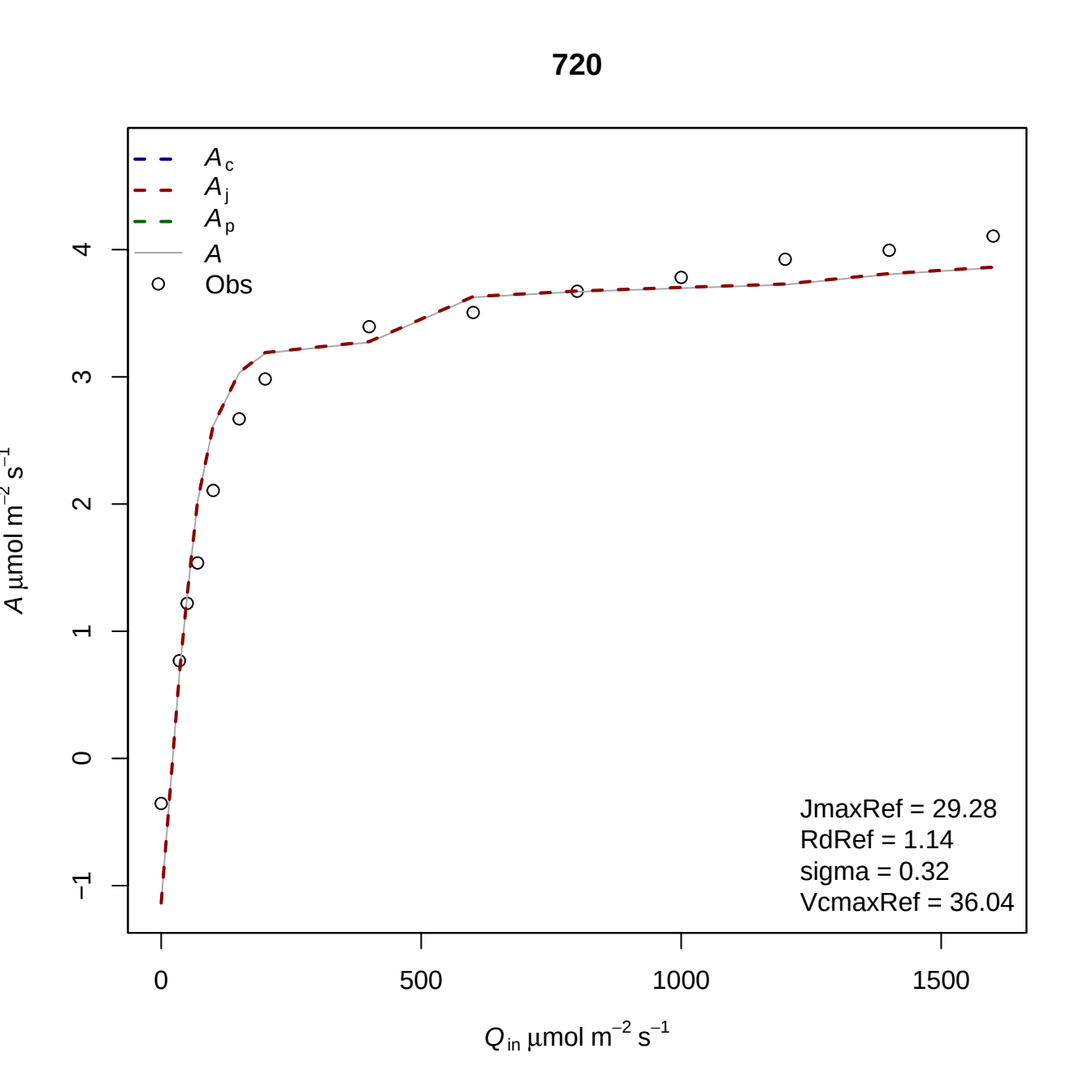

715
A_c
A_j
A_p
A
Obs
JmaxRef = 21.44
RdRef = 1.08
sigma = 0.22
VcmaxRef = 54.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

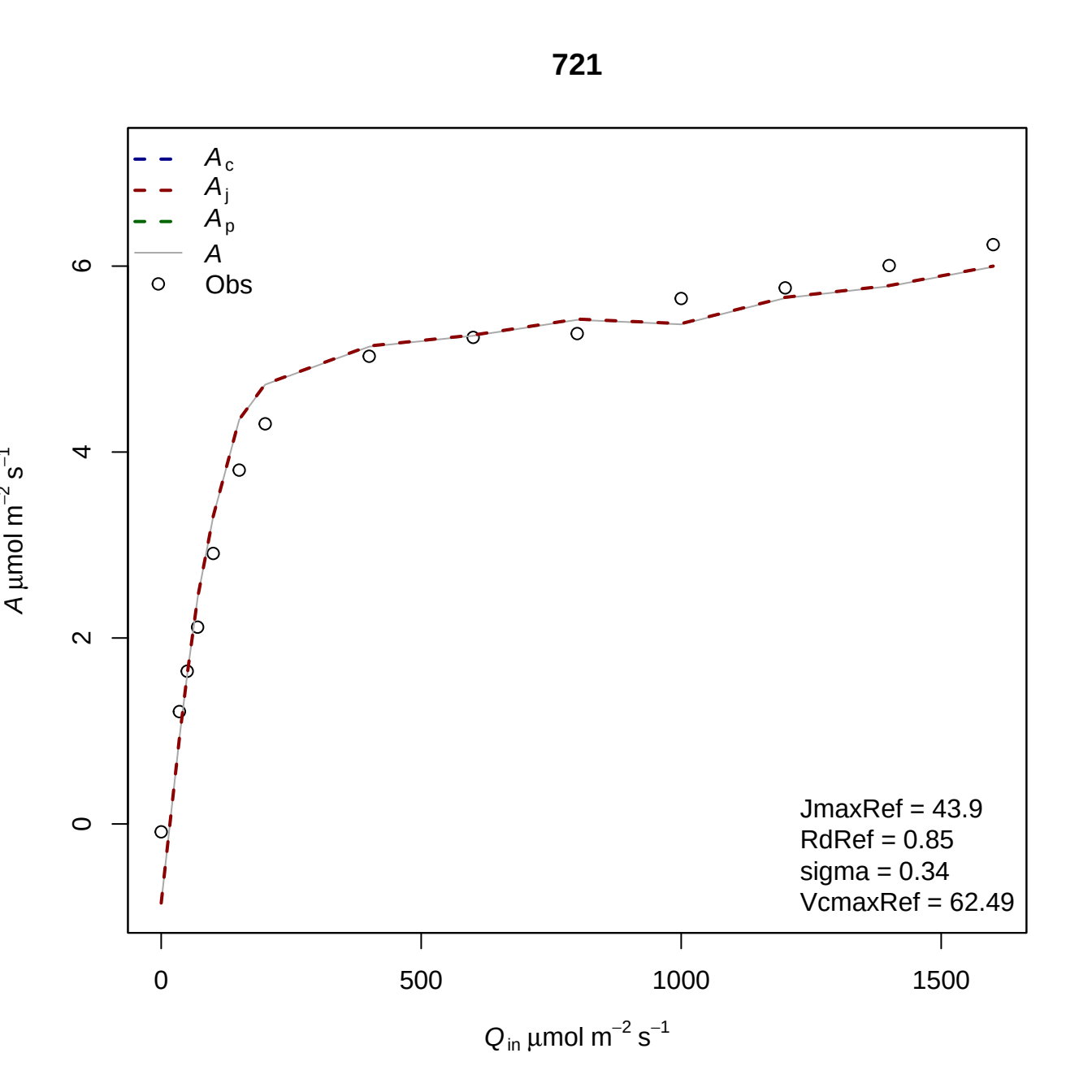

716
A_c
A_j
A_p
A
Obs
JmaxRef = 31.75
RdRef = 1.65
sigma = 0.22
VcmaxRef = 40.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

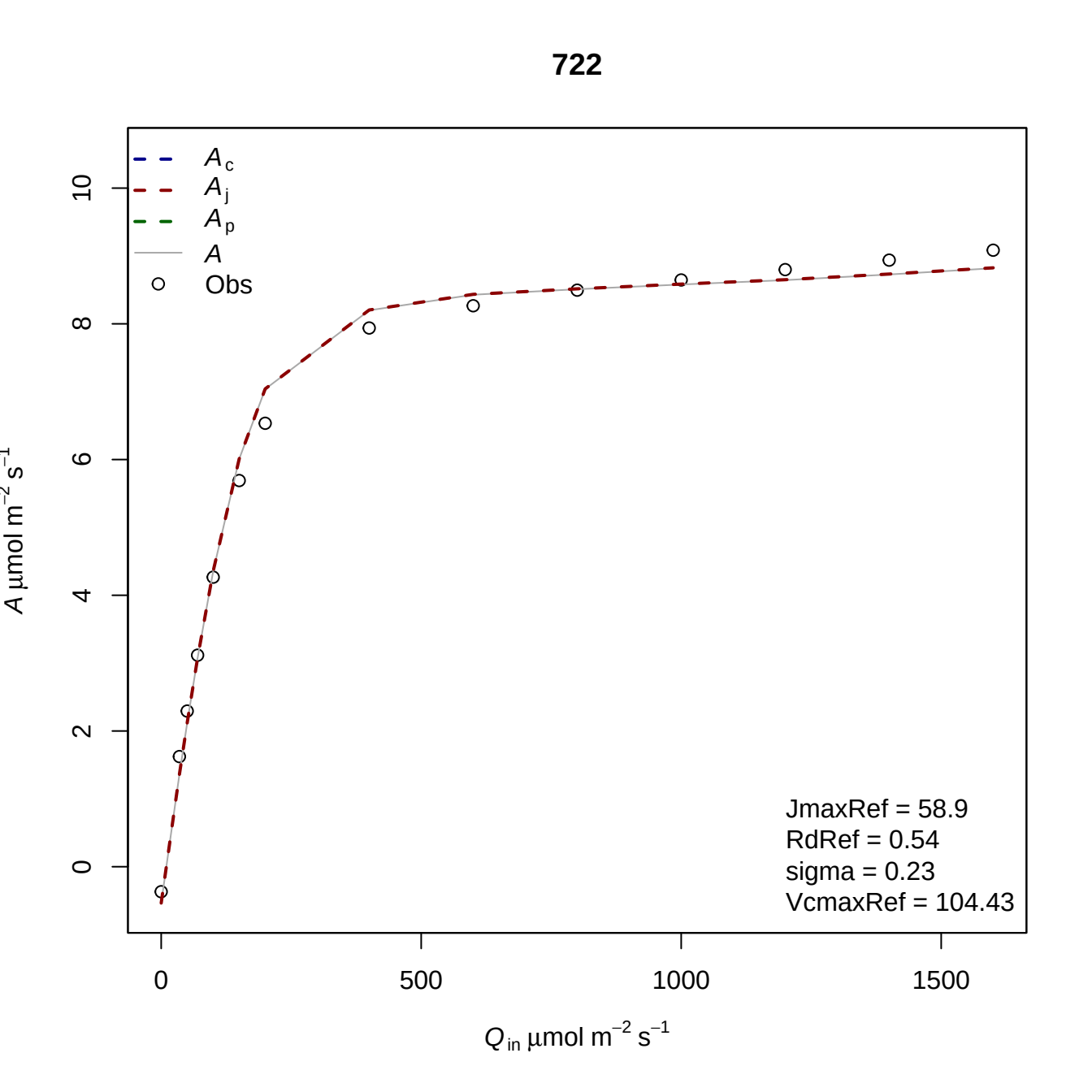

717
A_c
A_j
A_p
A
Obs
JmaxRef = 42.31
RdRef = 1.97
sigma = 0.54
VcmaxRef = 180.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

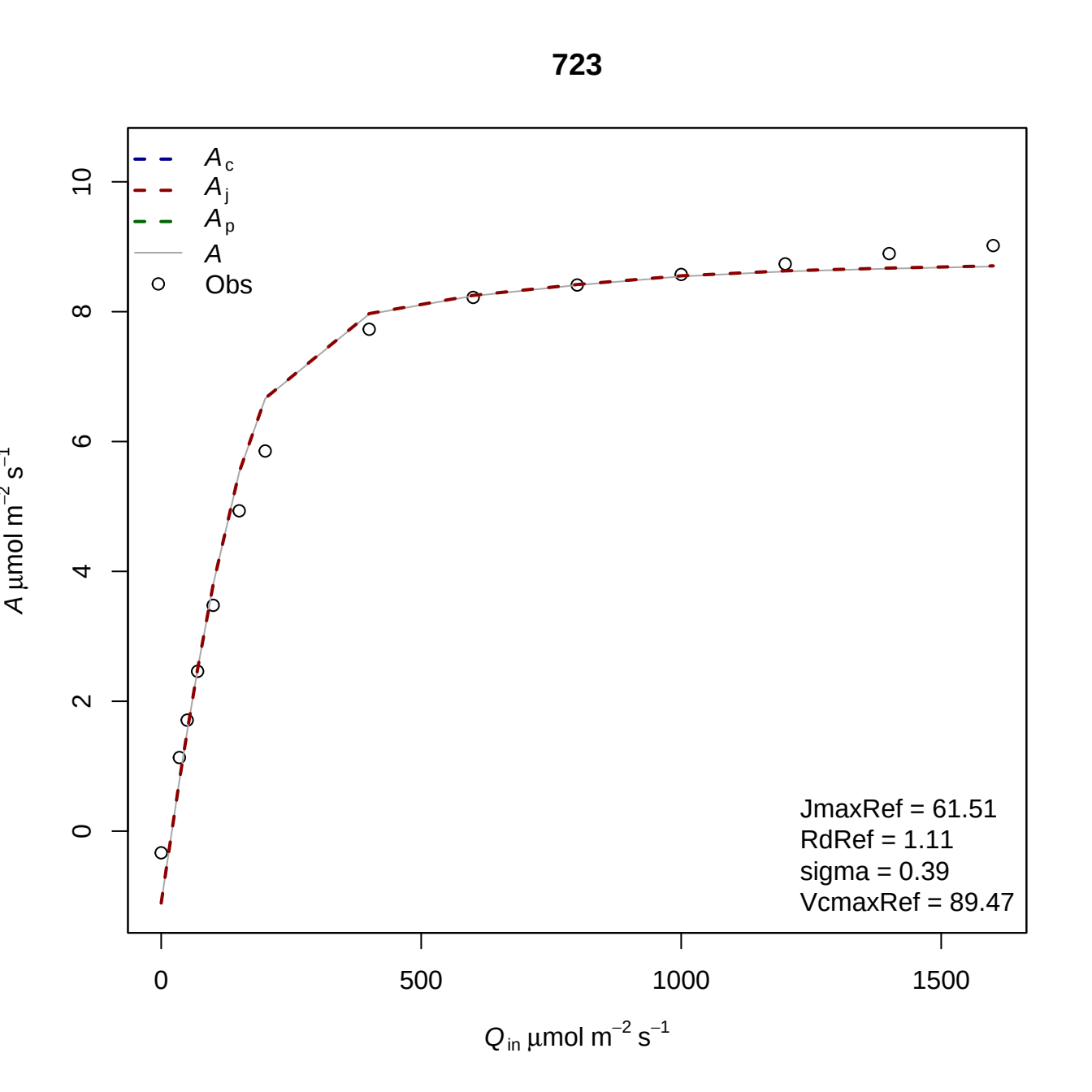

718
A_c
A_j
A_p
A
Obs
JmaxRef = 89.39
RdRef = 2.86
sigma = 0.64
VcmaxRef = 207.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

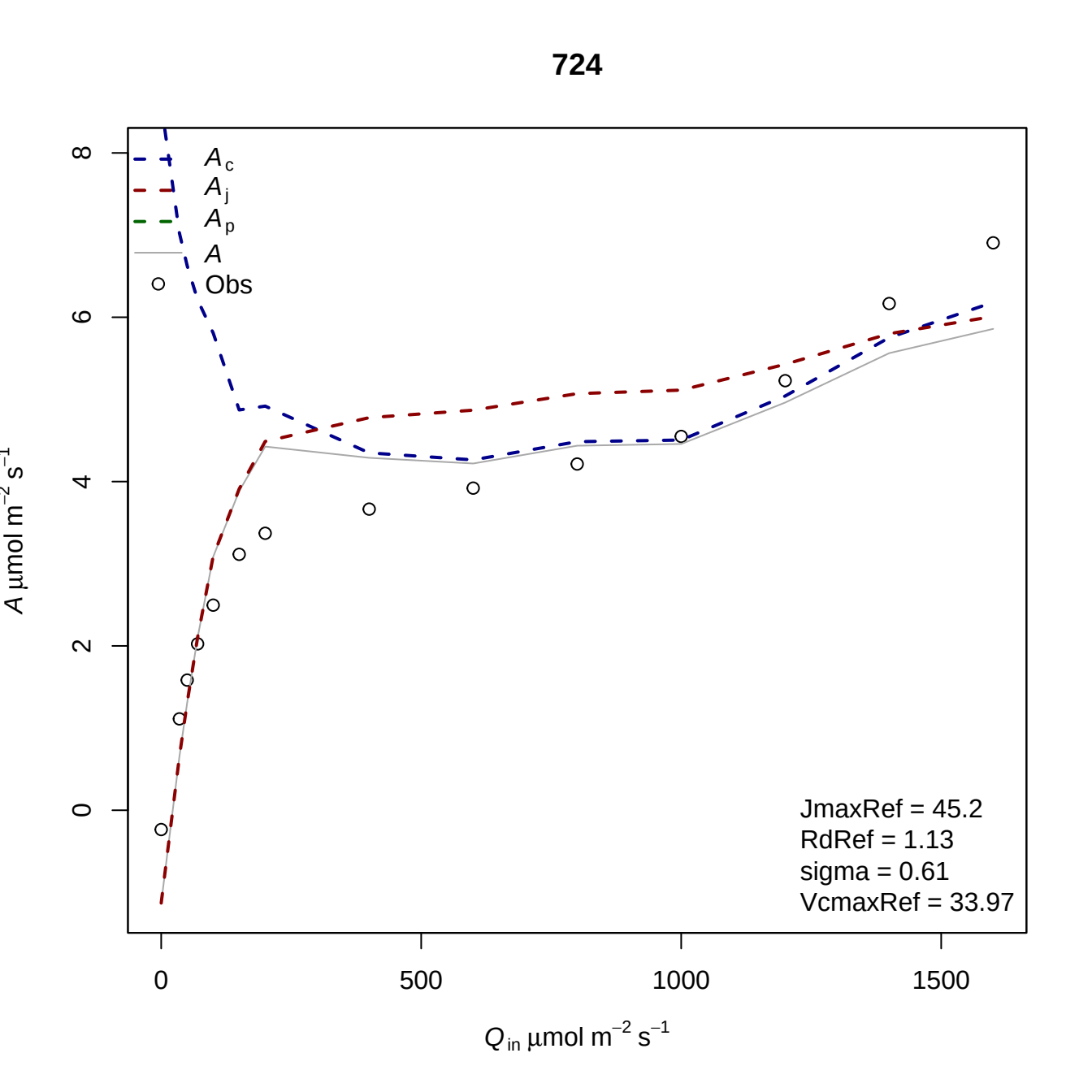

719
A_c
A_j
A_p
A
Obs
JmaxRef = 28.3
RdRef = 0.95
sigma = 0.4
VcmaxRef = 85.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

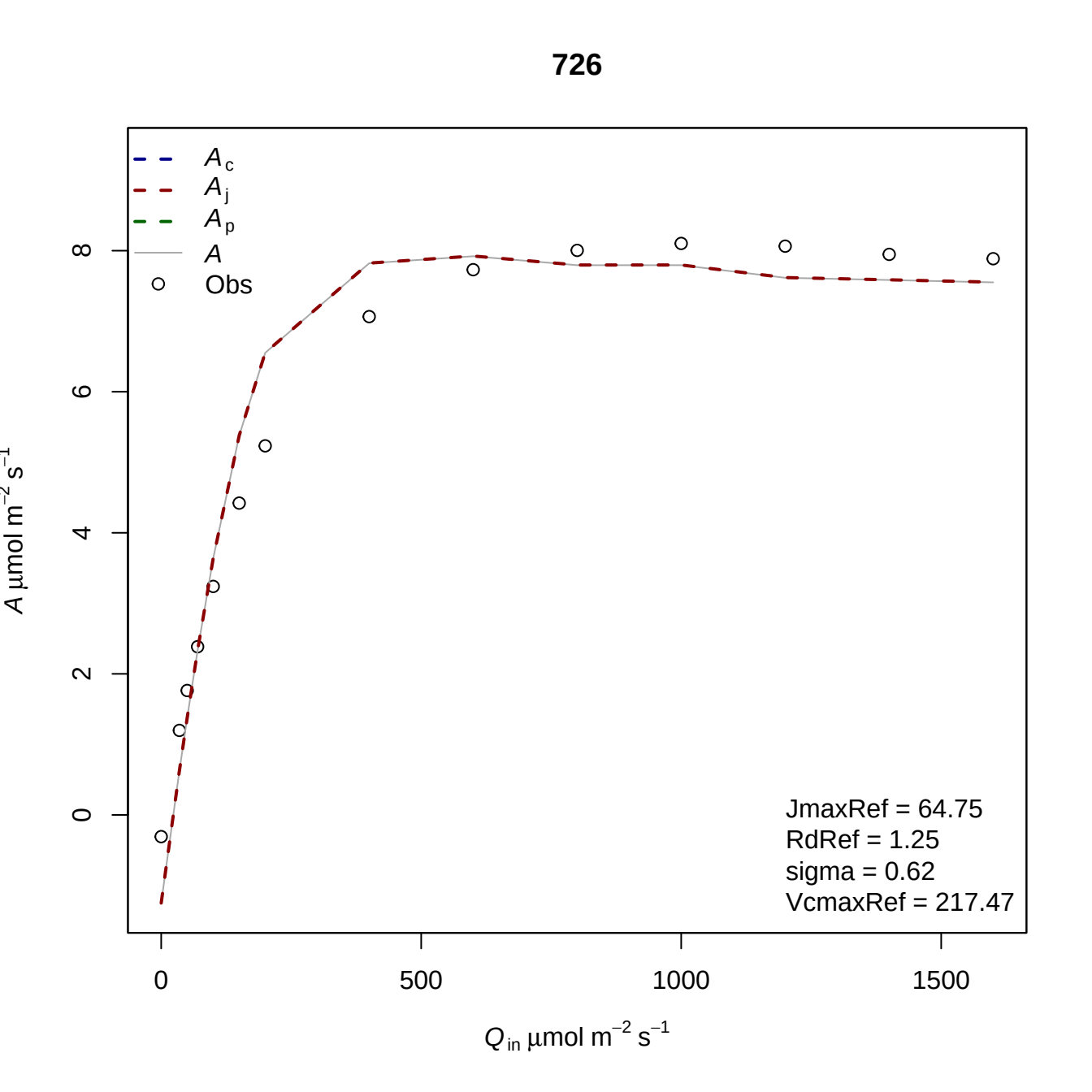

720
A_c
A_j
A_p
A
Obs
JmaxRef = 29.28
RdRef = 1.14
sigma = 0.32
VcmaxRef = 36.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

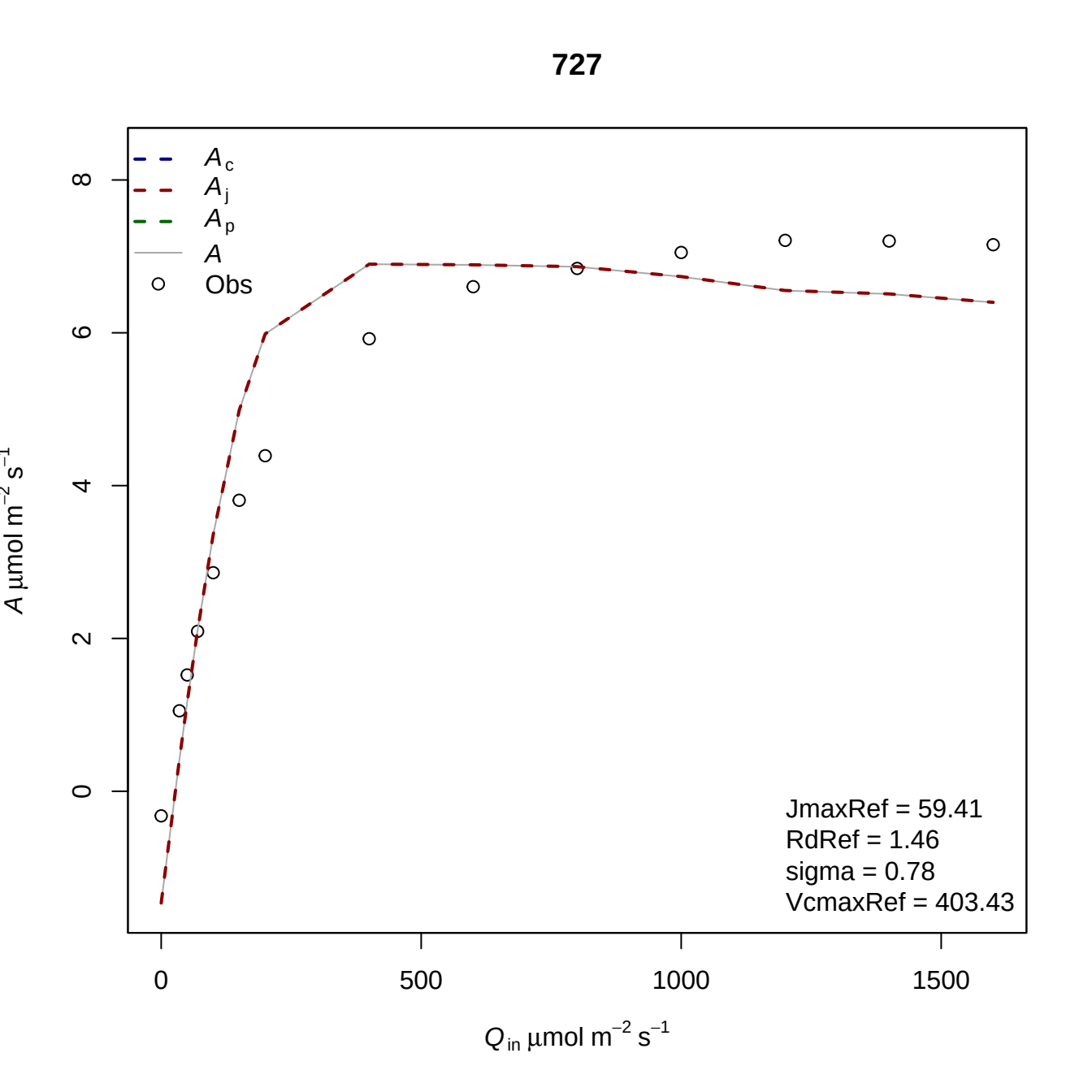

721
A_c
A_j
A_p
A
Obs
JmaxRef = 43.9
RdRef = 0.85
sigma = 0.34
VcmaxRef = 62.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

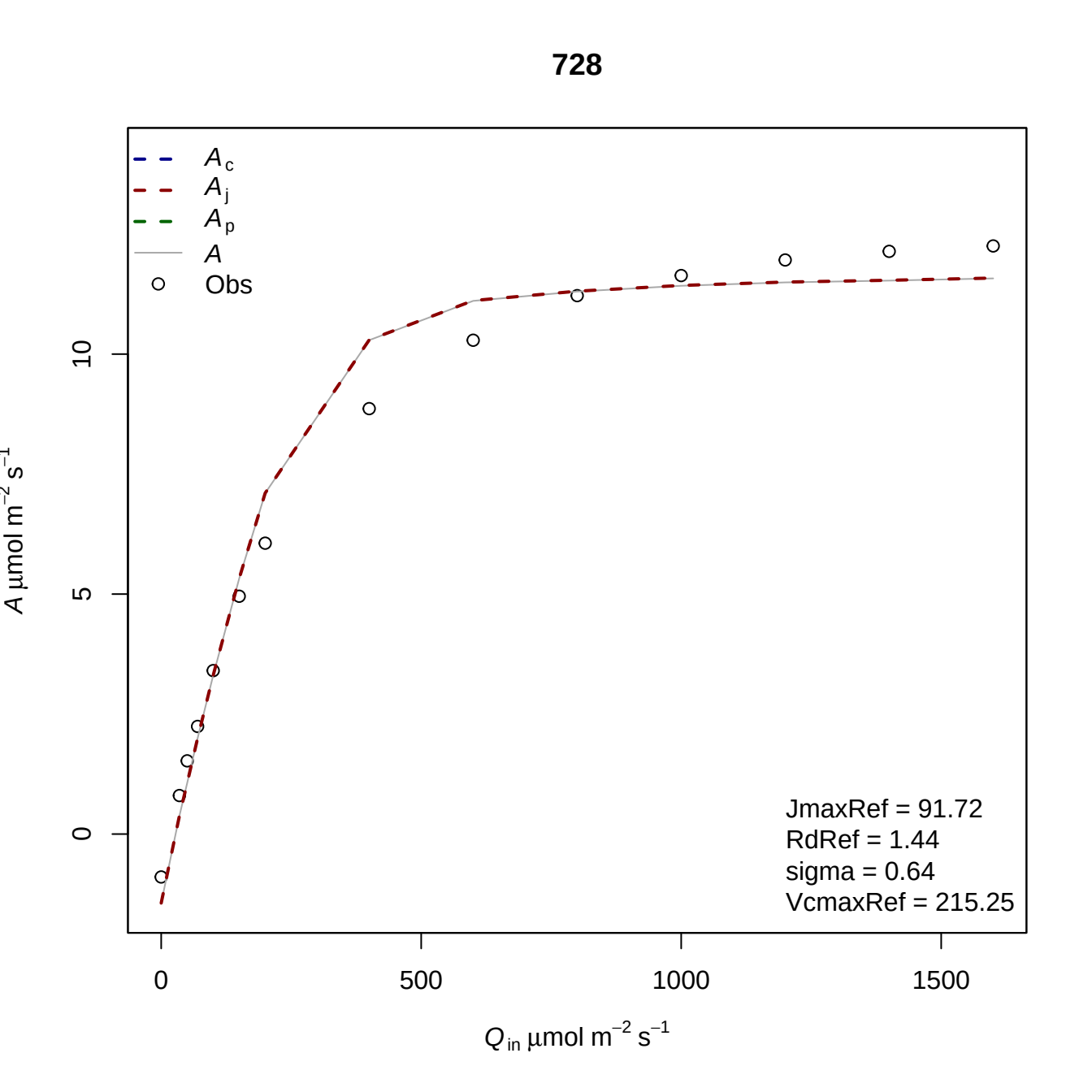

722
A_c
A_j
A_p
A
Obs
JmaxRef = 58.9
RdRef = 0.54
sigma = 0.23
VcmaxRef = 104.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

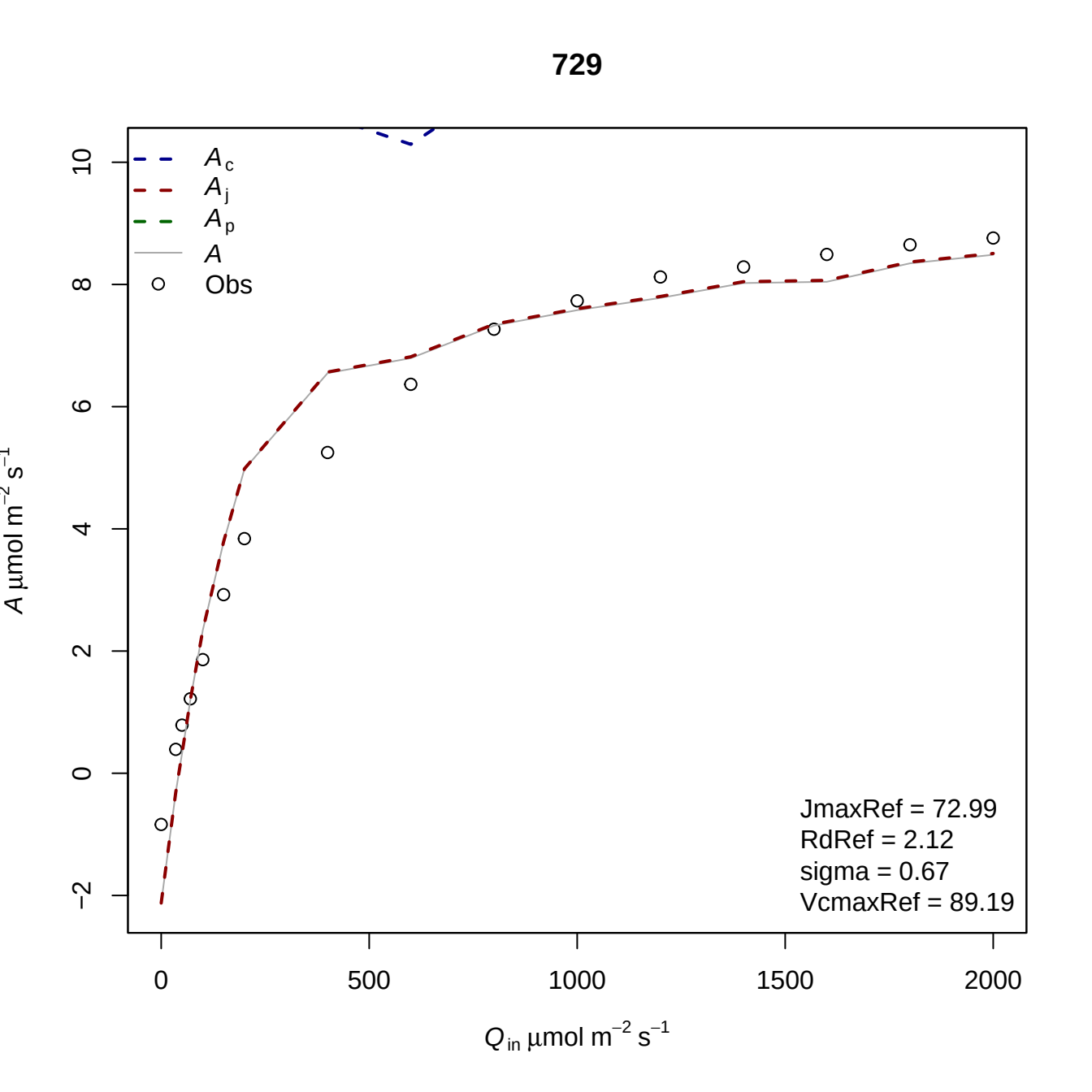

723
A_c
A_j
A_p
A
Obs
JmaxRef = 61.51
RdRef = 1.11
sigma = 0.39
VcmaxRef = 89.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

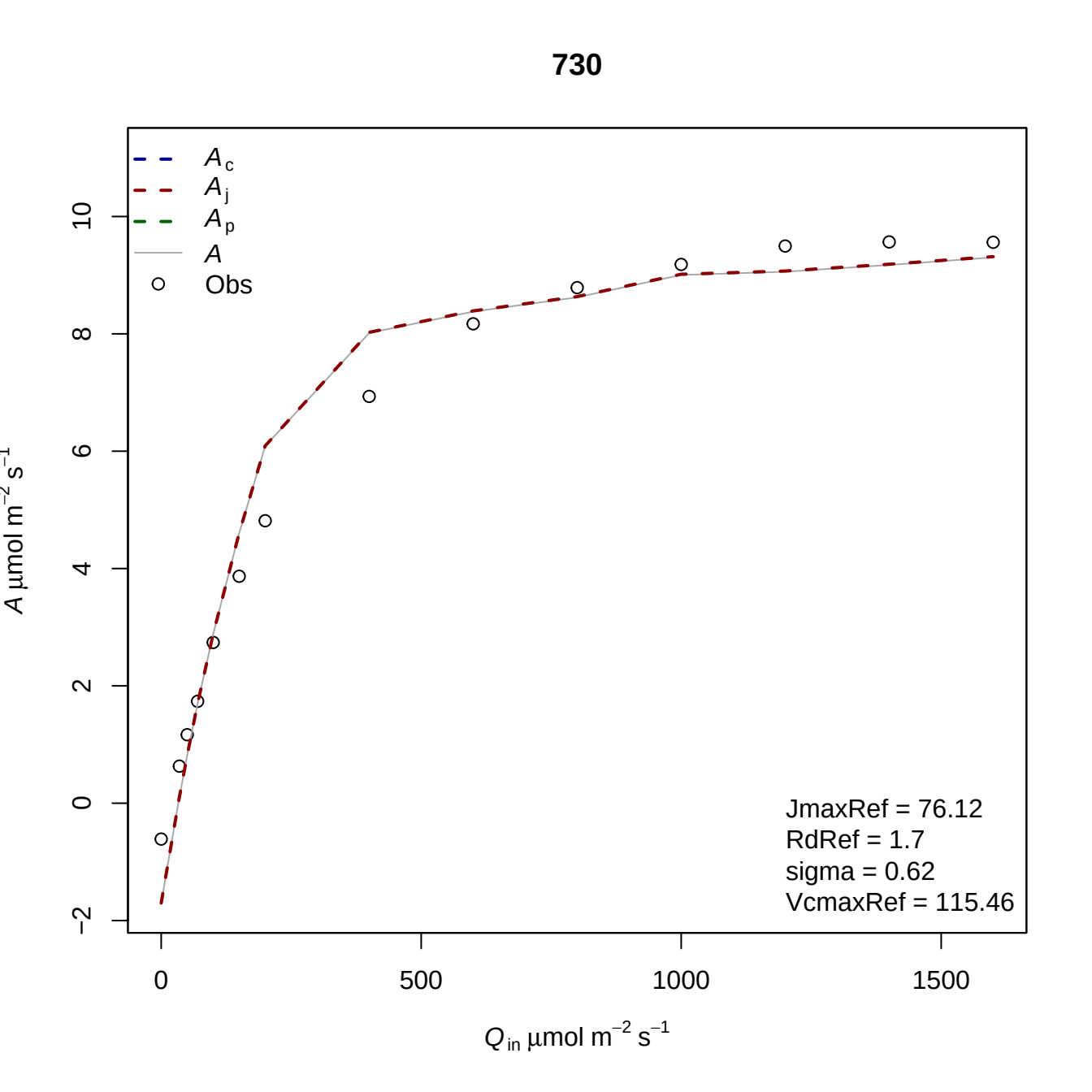

724
A_c
A_j
A_p
A
Obs
JmaxRef = 45.2
RdRef = 1.13
sigma = 0.61
VcmaxRef = 33.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

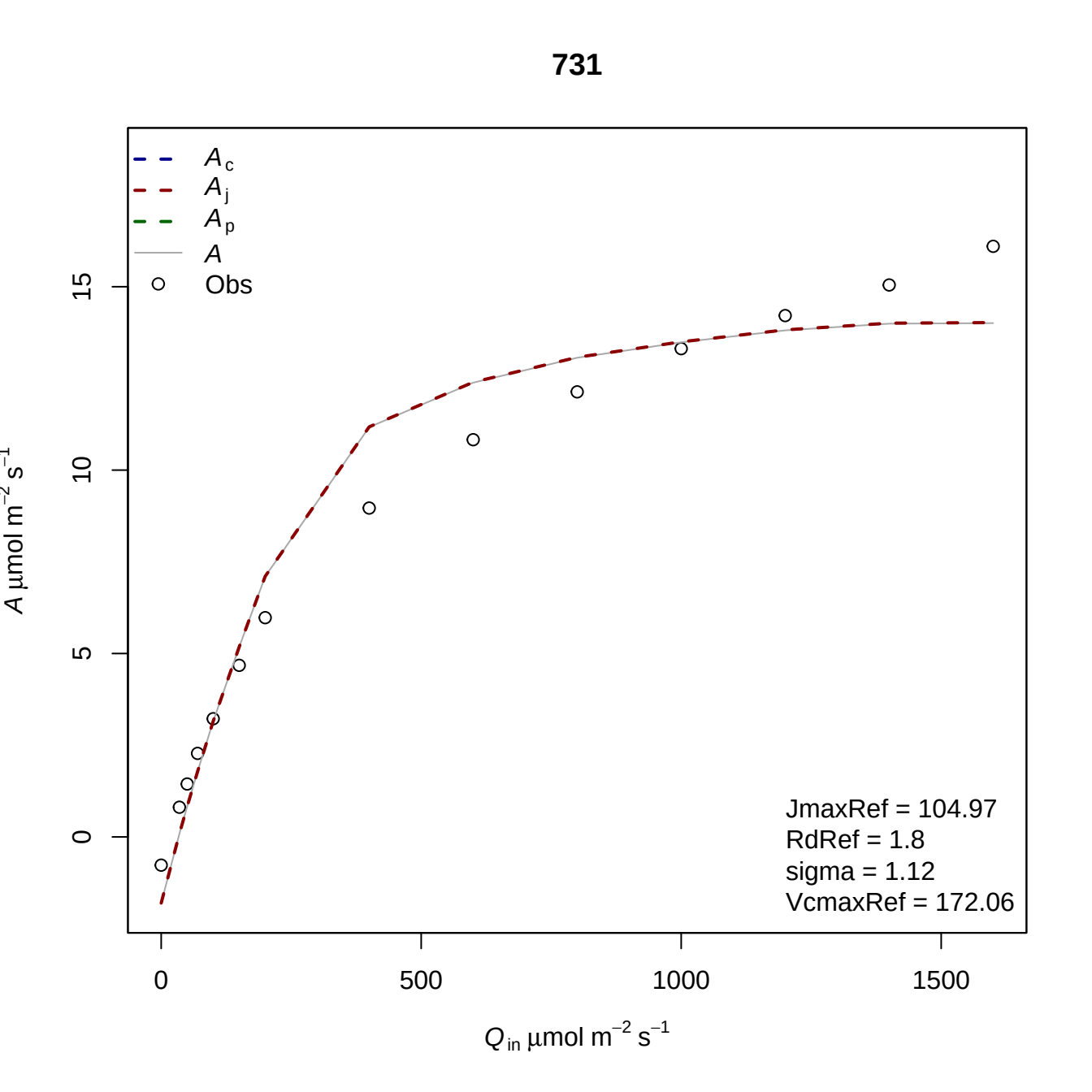

726
A_c
A_j
A_p
A
Obs
JmaxRef = 64.75
RdRef = 1.25
sigma = 0.62
VcmaxRef = 217.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

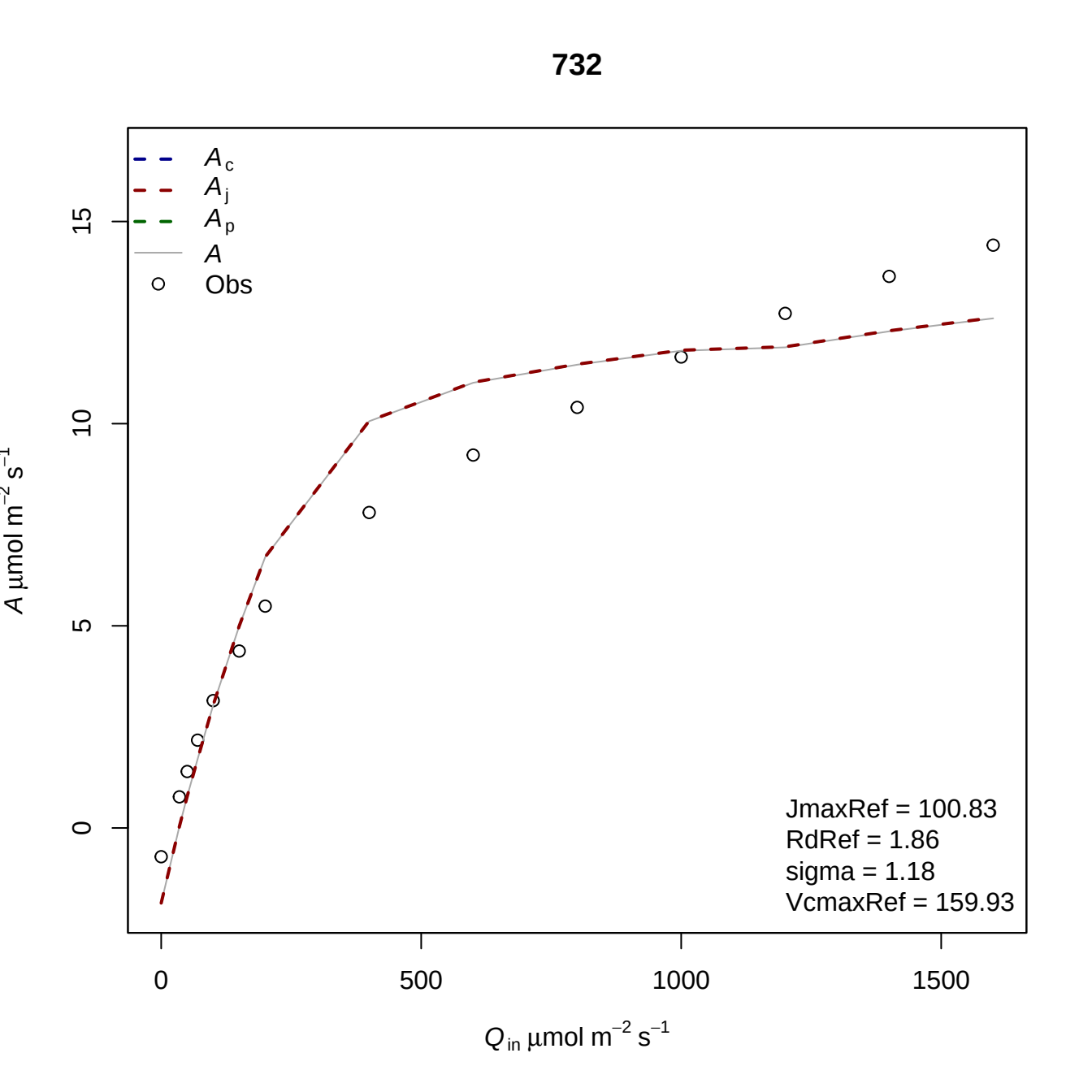

727
A_c
A_j
A_p
A
Obs
JmaxRef = 59.41
RdRef = 1.46
sigma = 0.78
VcmaxRef = 403.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

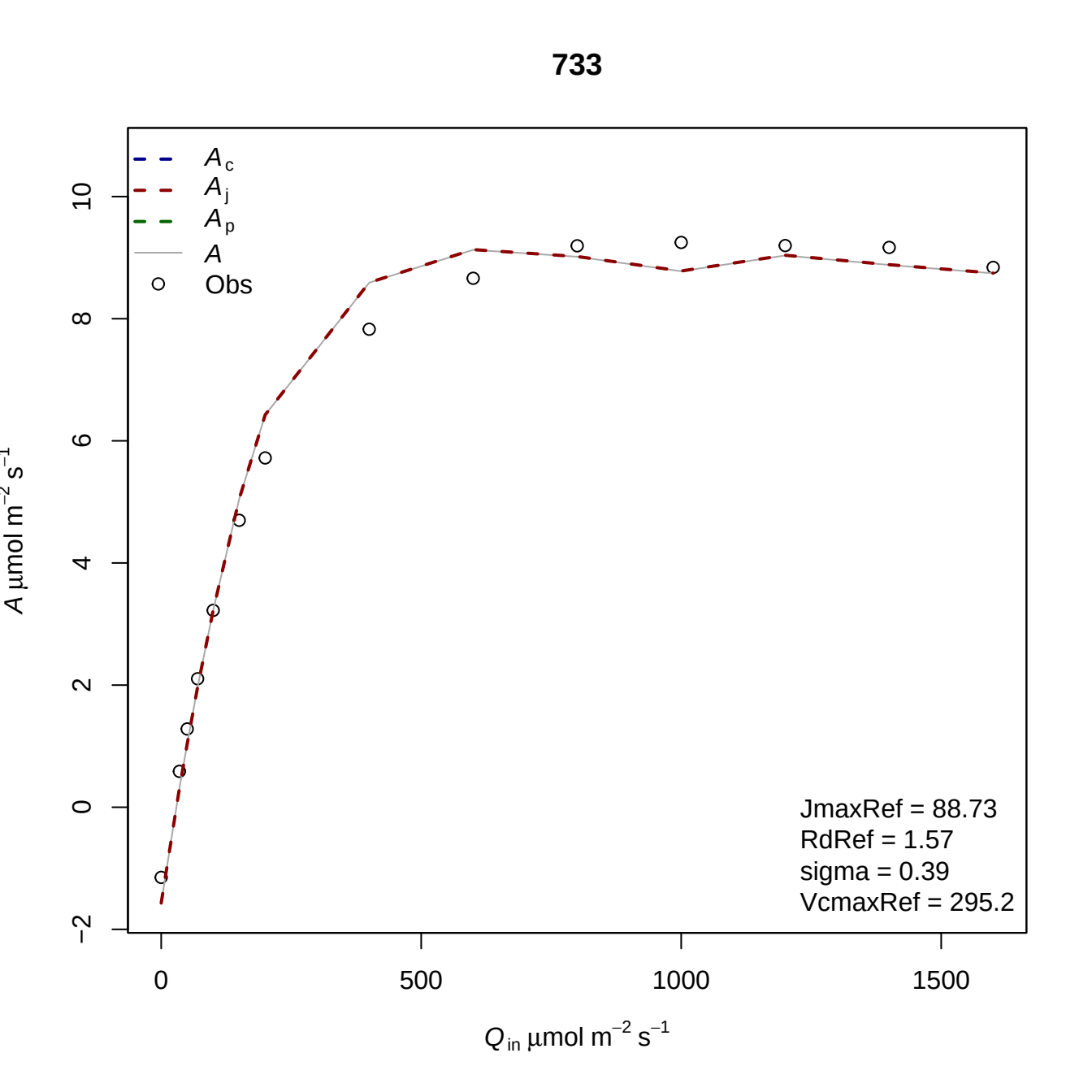

728
A_c
A_j
A_p
A
Obs
JmaxRef = 91.72
RdRef = 1.44
sigma = 0.64
VcmaxRef = 215.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

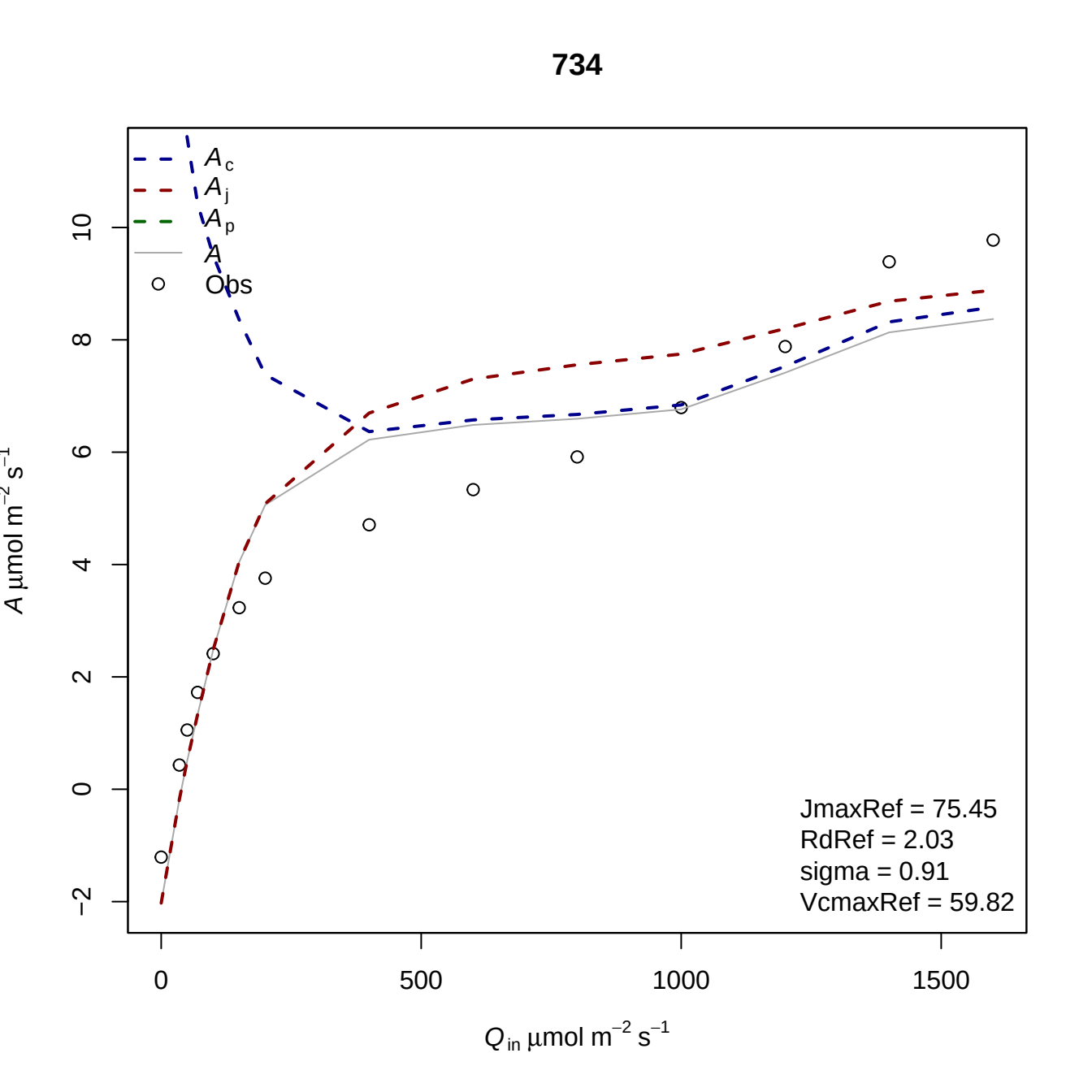

729
A_c
A_j
A_p
A
Obs
JmaxRef = 72.99
RdRef = 2.12
sigma = 0.67
VcmaxRef = 89.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

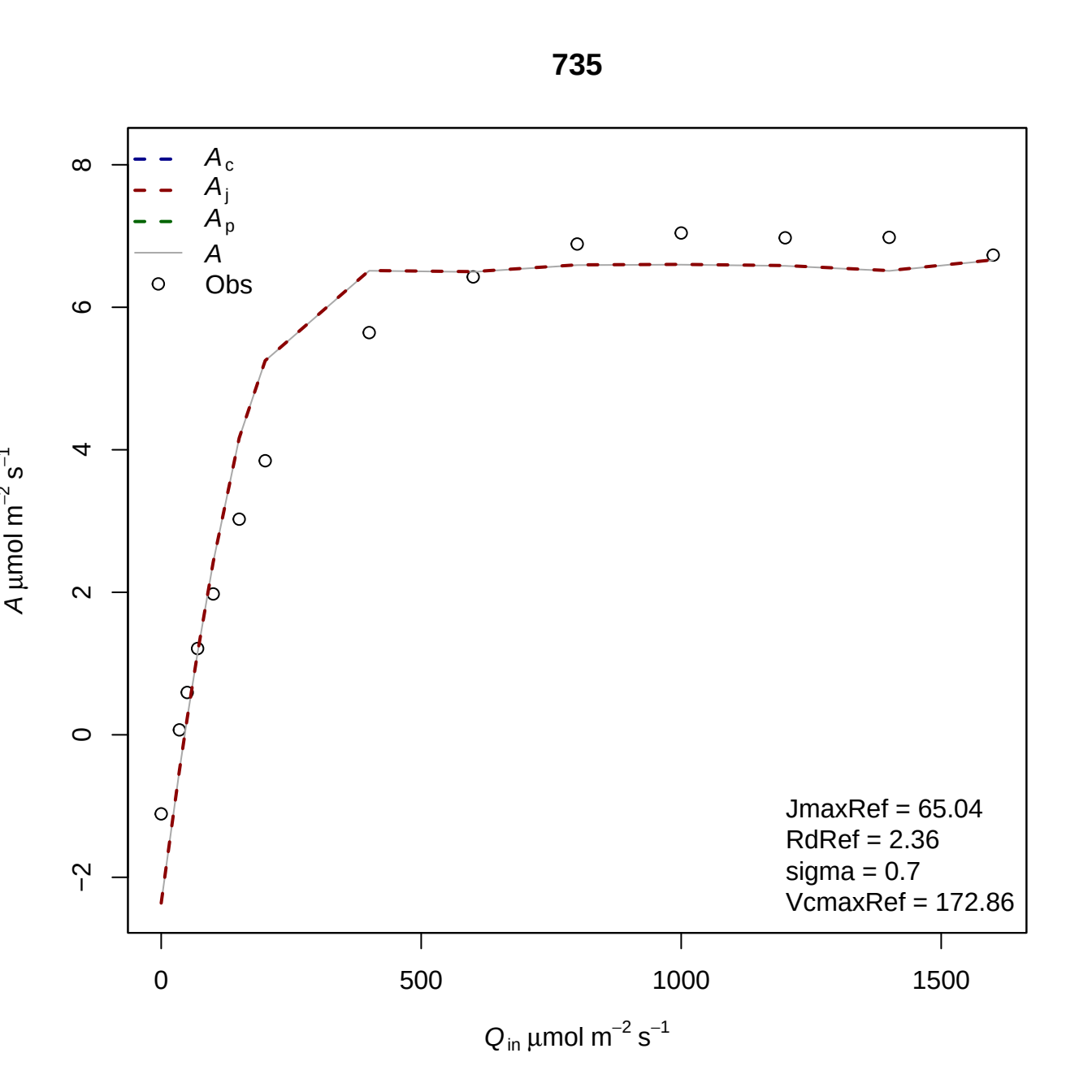

730
A_c
A_j
A_p
A
Obs
JmaxRef = 76.12
RdRef = 1.7
sigma = 0.62
VcmaxRef = 115.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

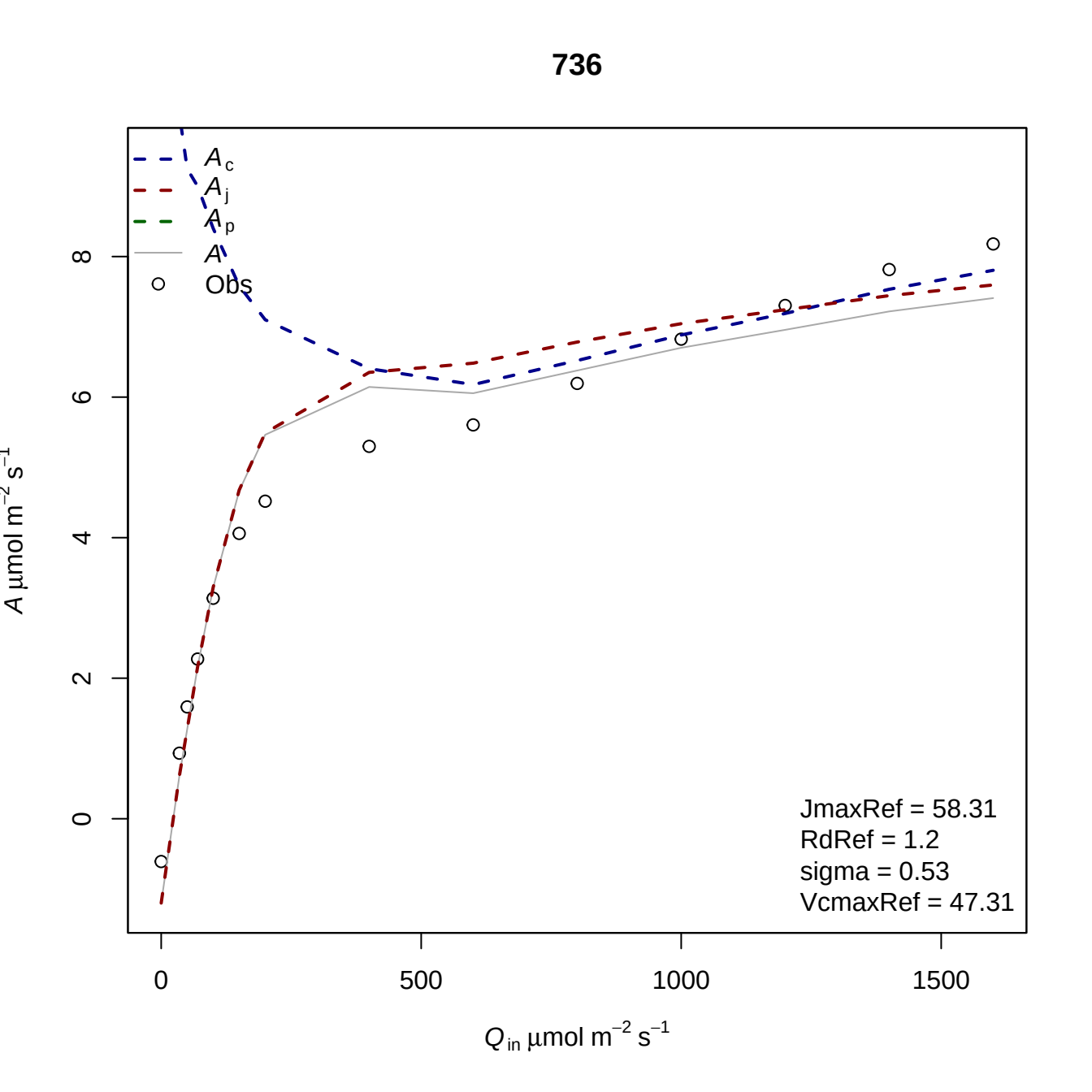

731
A_c
A_j
A_p
A
Obs
JmaxRef = 104.97
RdRef = 1.8
sigma = 1.12
VcmaxRef = 172.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

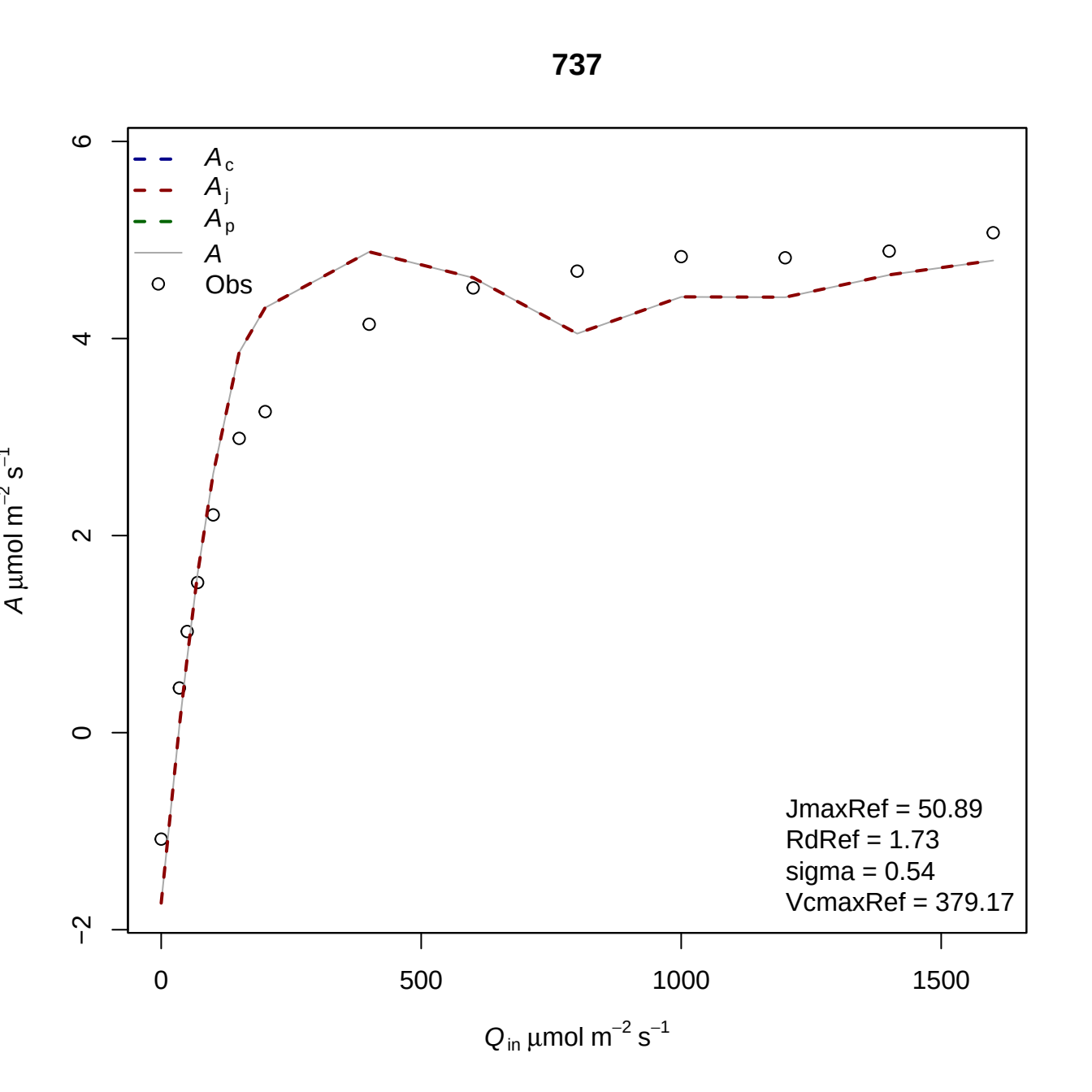

732
A_c
A_j
A_p
A
Obs
JmaxRef = 100.83
RdRef = 1.86
sigma = 1.18
VcmaxRef = 159.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

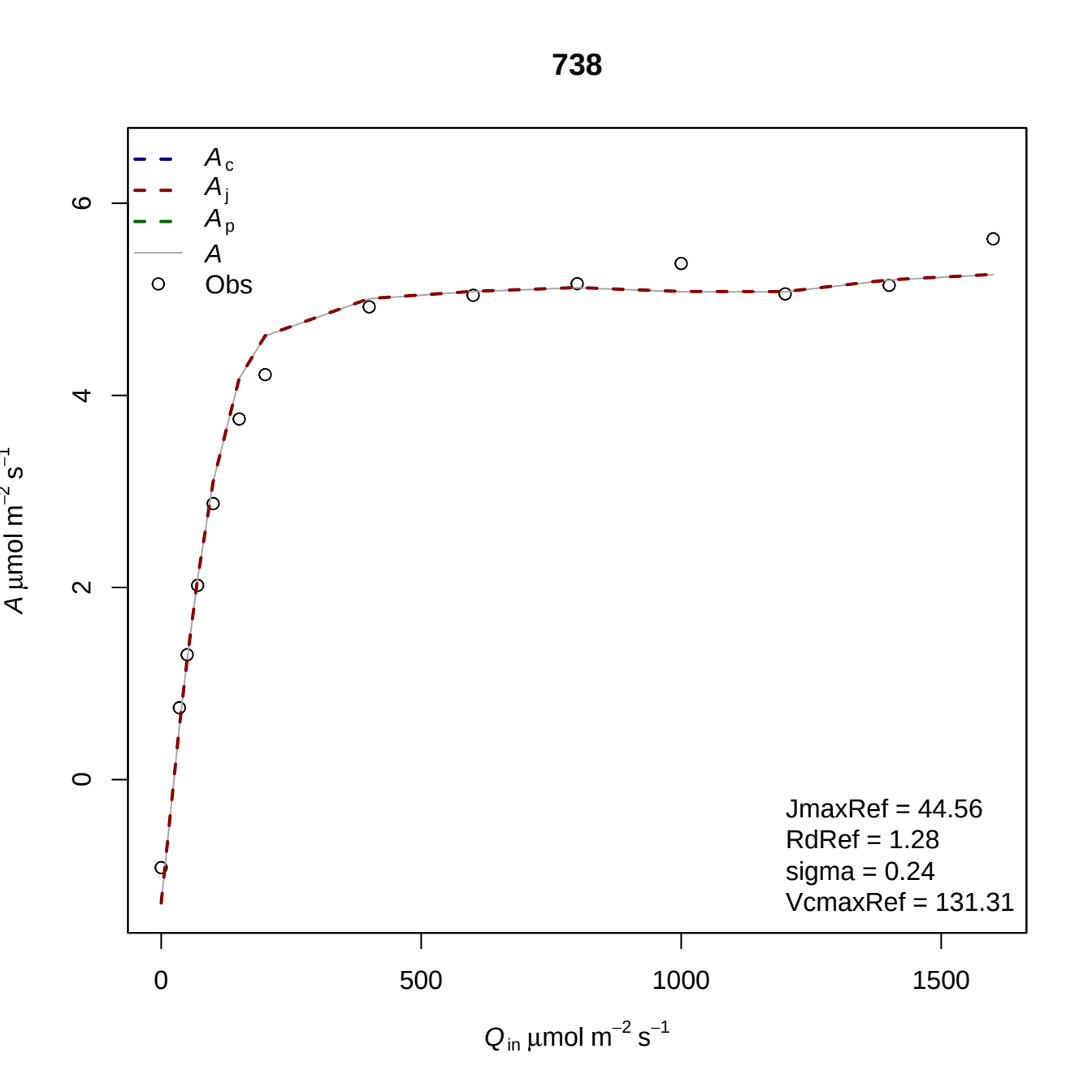

733
A_c
A_j
A_p
A
Obs
JmaxRef = 88.73
RdRef = 1.57
sigma = 0.39
VcmaxRef = 295.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

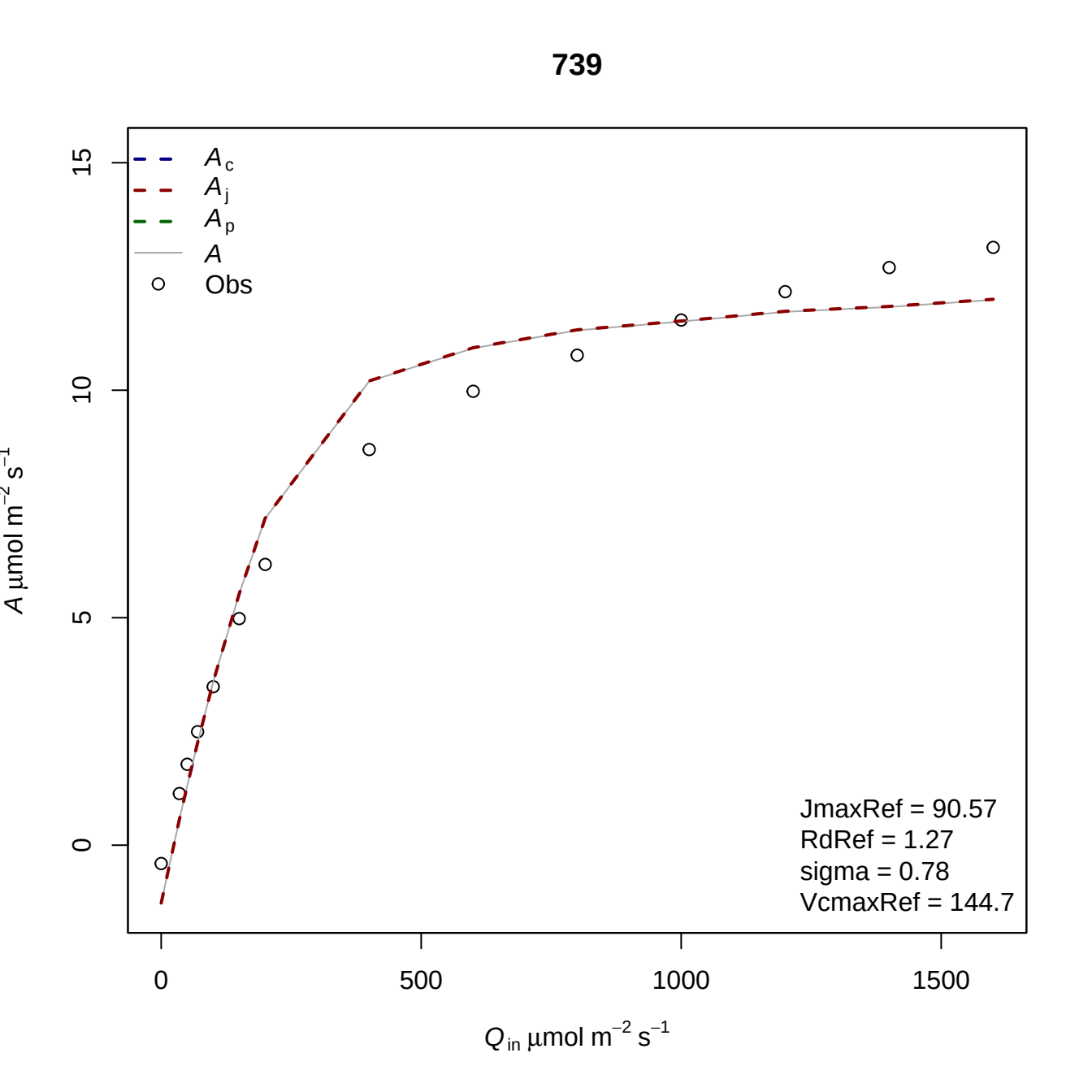

734
A_c
A_j
A_p
A
Obs
JmaxRef = 75.45
RdRef = 2.03
sigma = 0.91
VcmaxRef = 59.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

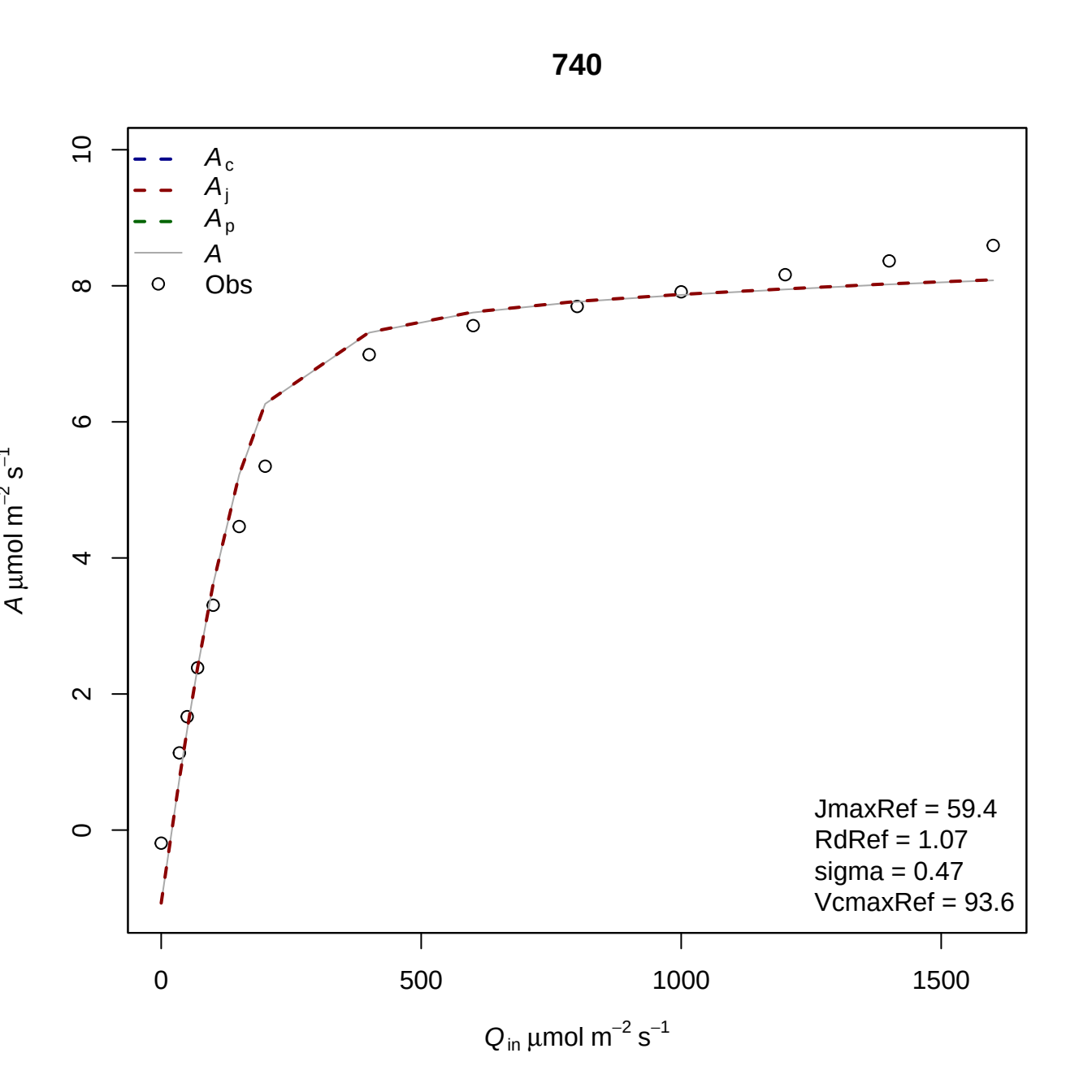

735
A_c
A_j
A_p
A
Obs
JmaxRef = 65.04
RdRef = 2.36
sigma = 0.7
VcmaxRef = 172.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

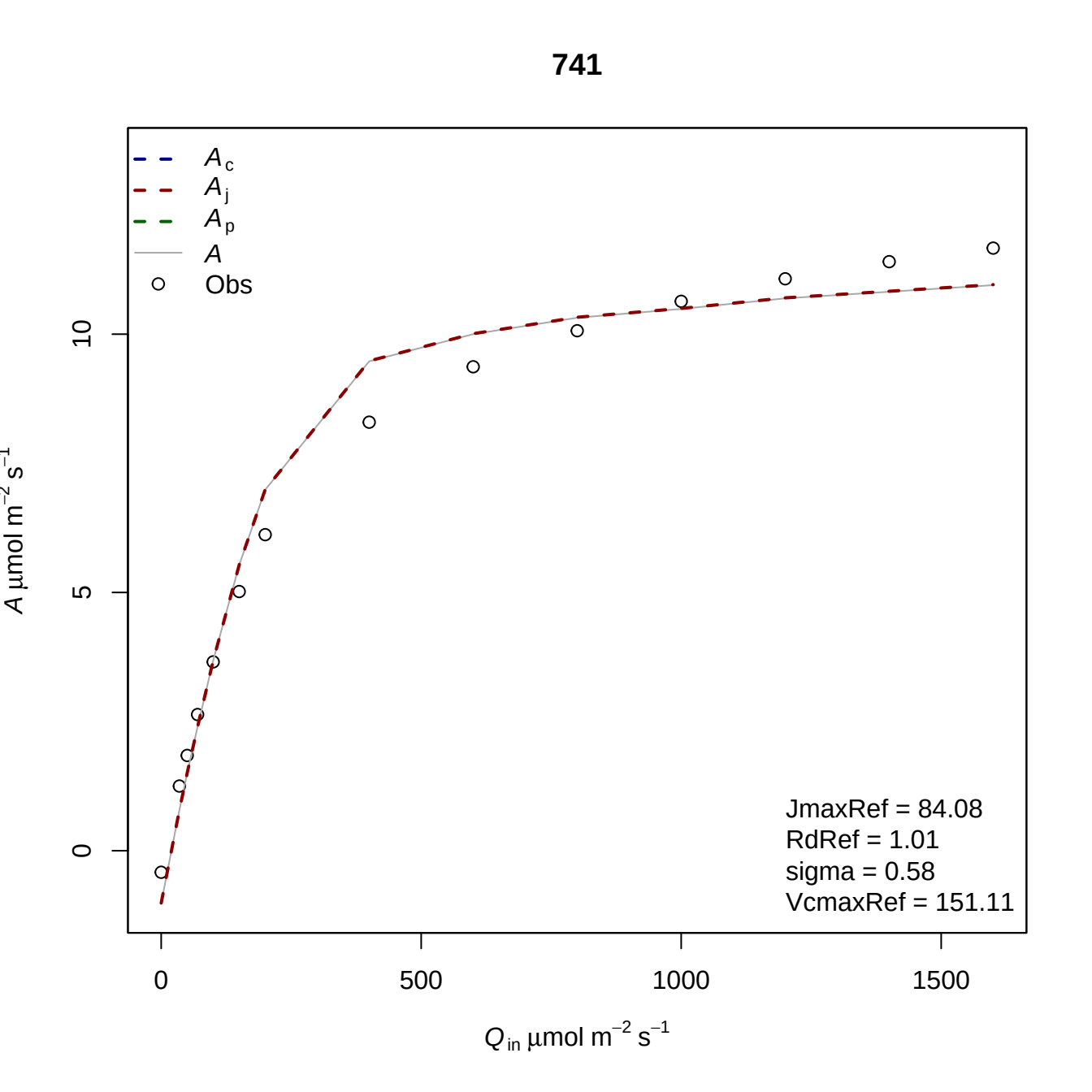

736
A_c
A_j
A_p
A
Obs
JmaxRef = 58.31
RdRef = 1.2
sigma = 0.53
VcmaxRef = 47.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

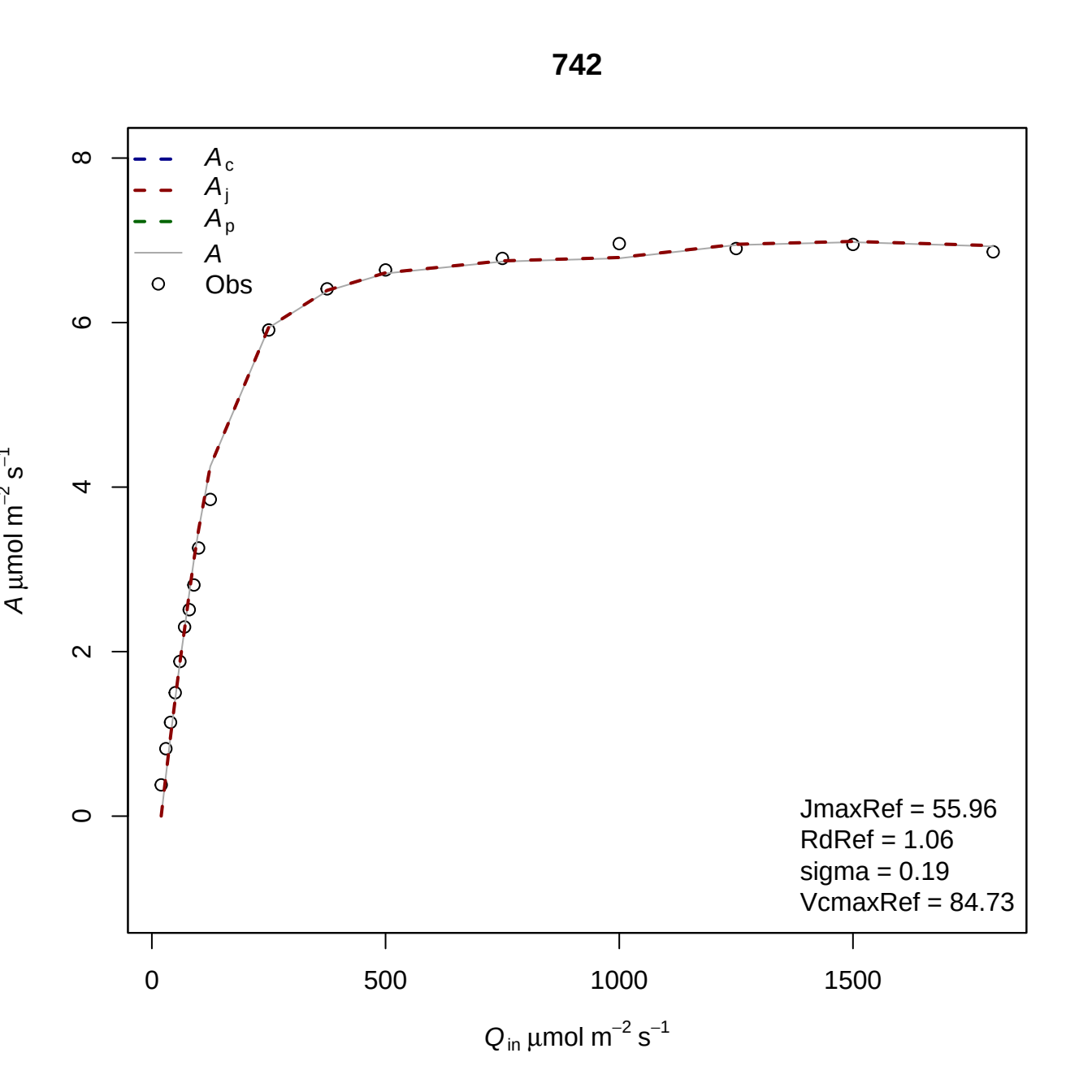

737
A_c
A_j
A_p
A
Obs
JmaxRef = 50.89
RdRef = 1.73
sigma = 0.54
VcmaxRef = 379.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

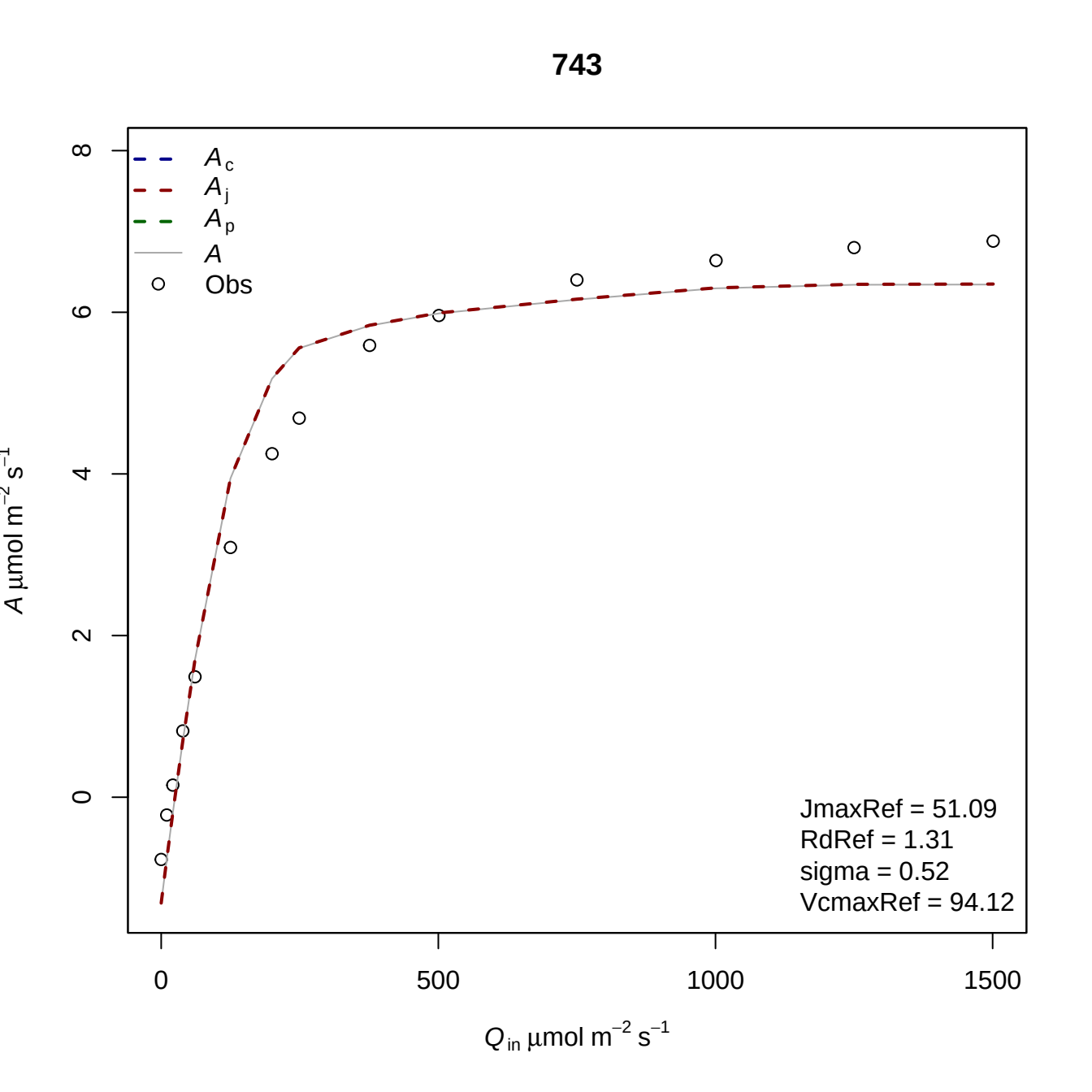

738
A_c
A_j
A_p
A
Obs
JmaxRef = 44.56
RdRef = 1.28
sigma = 0.24
VcmaxRef = 131.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

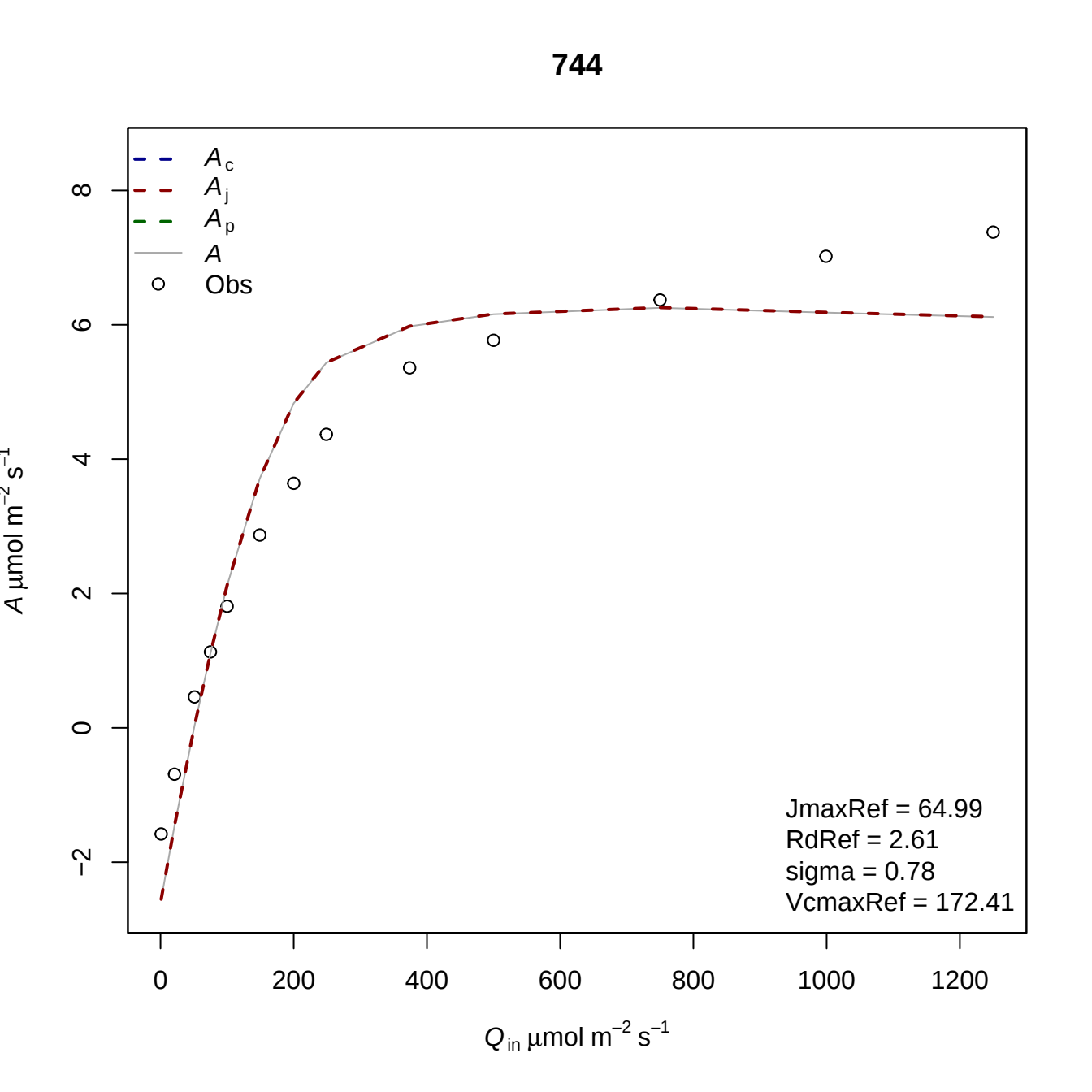

739
A_c
A_j
A_p
A
Obs
JmaxRef = 90.57
RdRef = 1.27
sigma = 0.78
VcmaxRef = 144.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

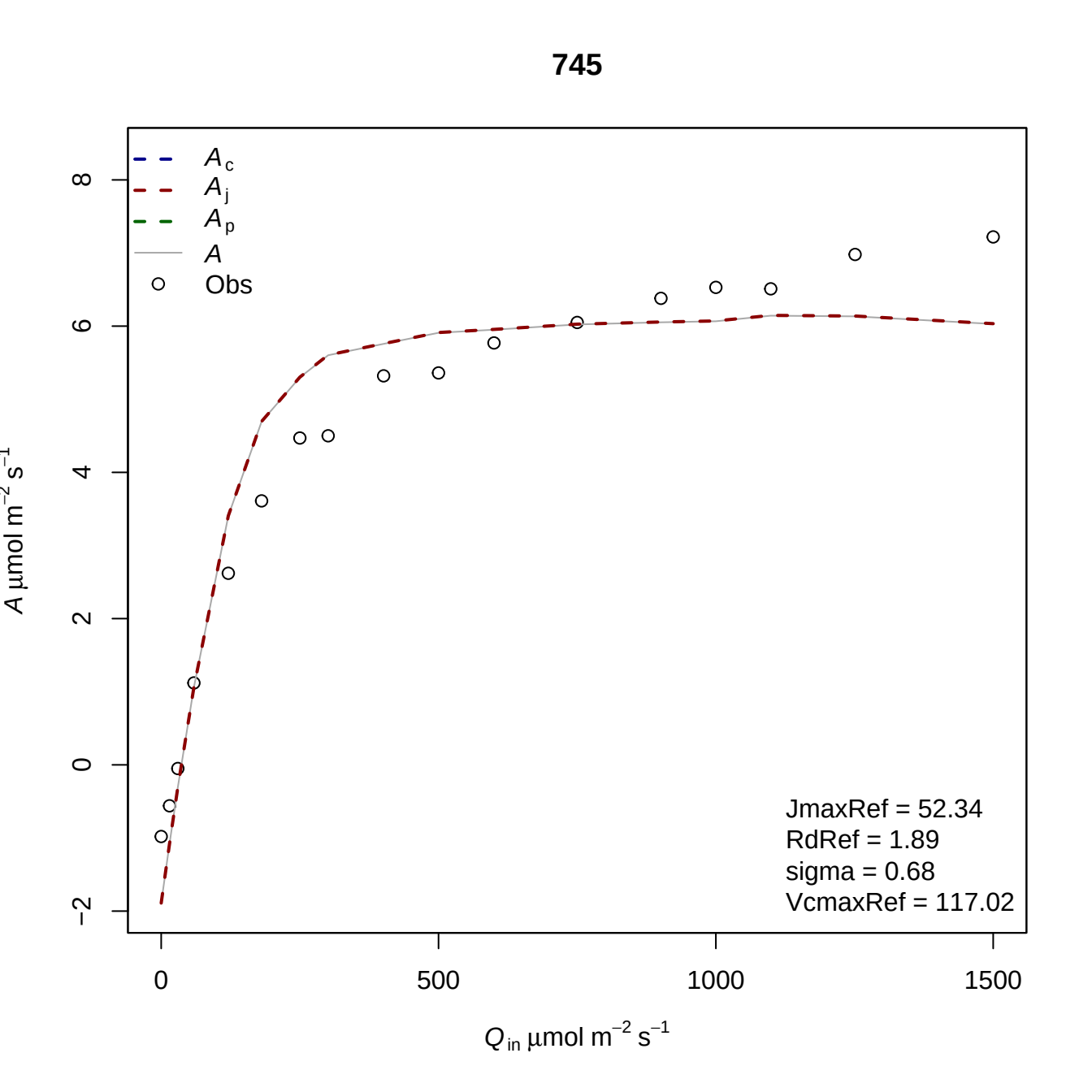

740
A_c
A_j
A_p
A
Obs
JmaxRef = 59.4
RdRef = 1.07
sigma = 0.47
VcmaxRef = 93.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

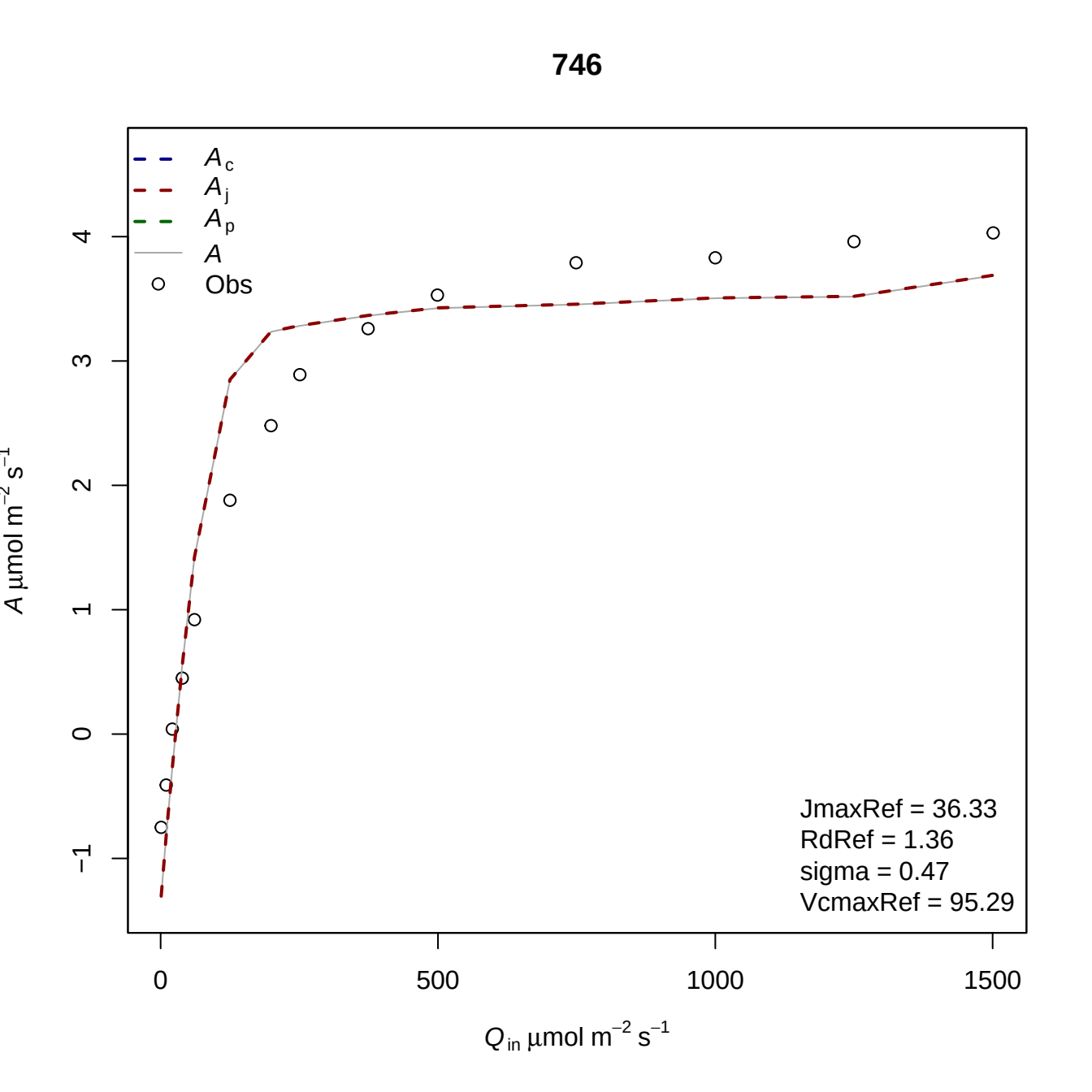

741
A_c
A_j
A_p
A
Obs
JmaxRef = 84.08
RdRef = 1.01
sigma = 0.58
VcmaxRef = 151.11
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

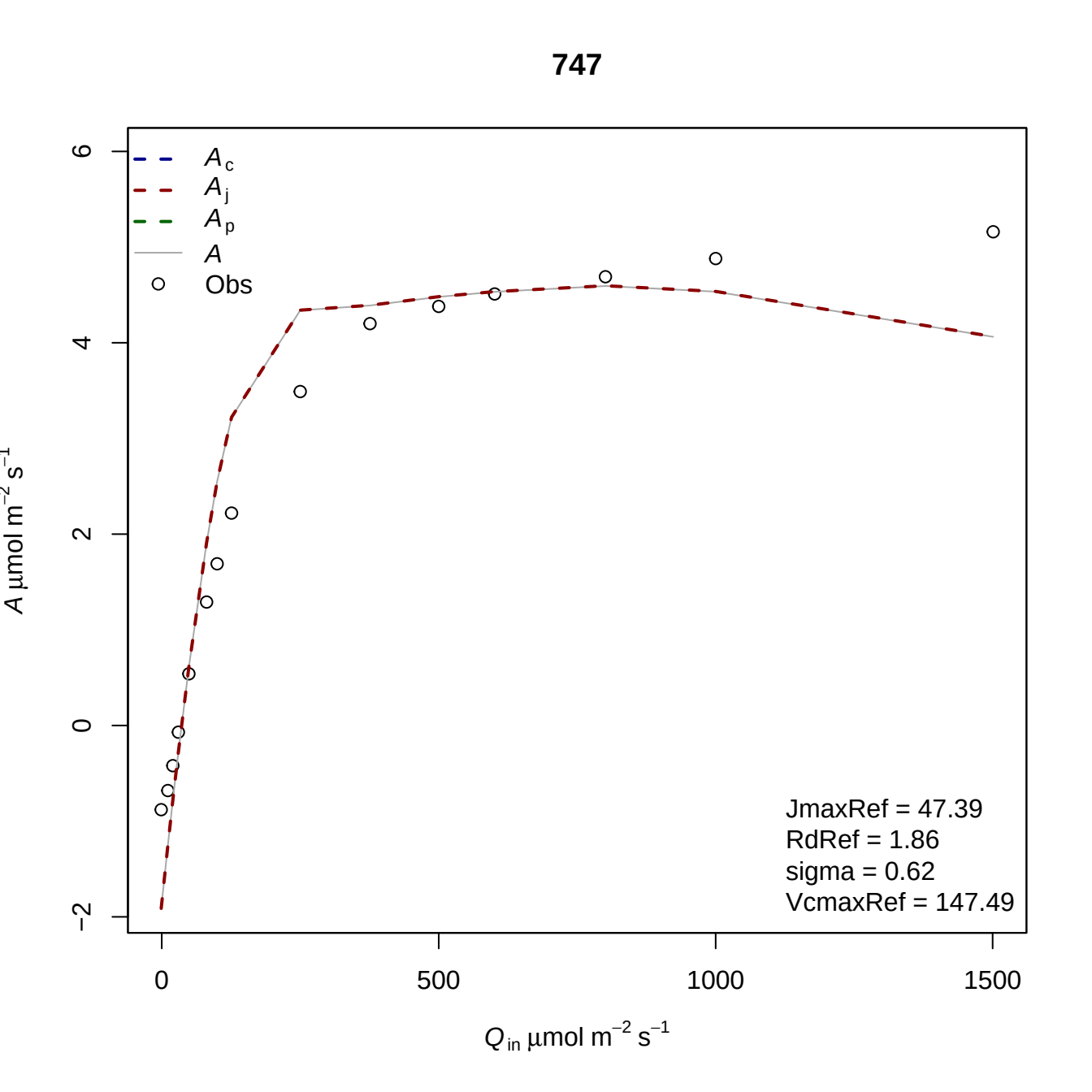

742
A_c
A_j
A_p
A
Obs
JmaxRef = 55.96
RdRef = 1.06
sigma = 0.19
VcmaxRef = 84.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

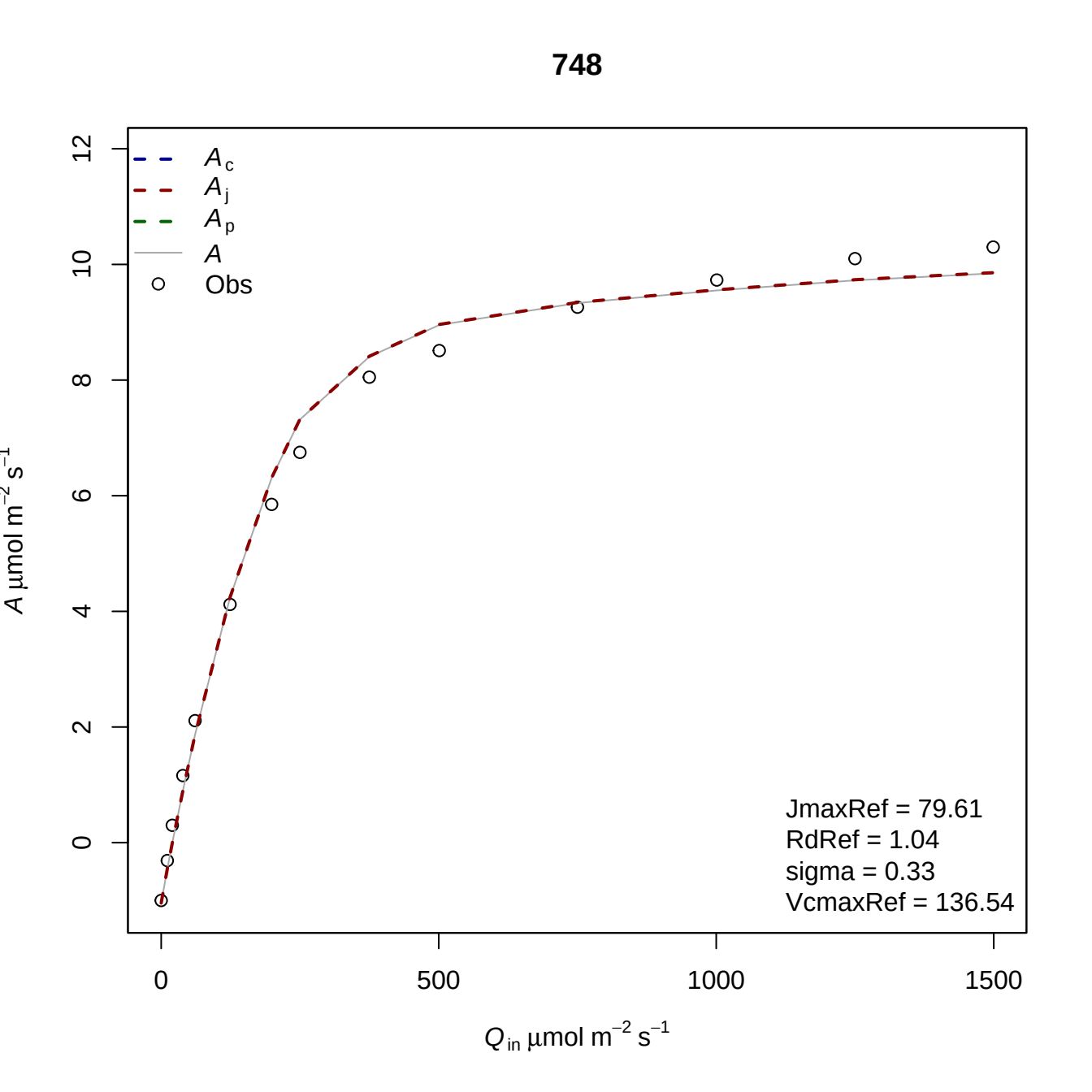

743
A_c
A_j
A_p
A
Obs
JmaxRef = 51.09
RdRef = 1.31
sigma = 0.52
VcmaxRef = 94.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

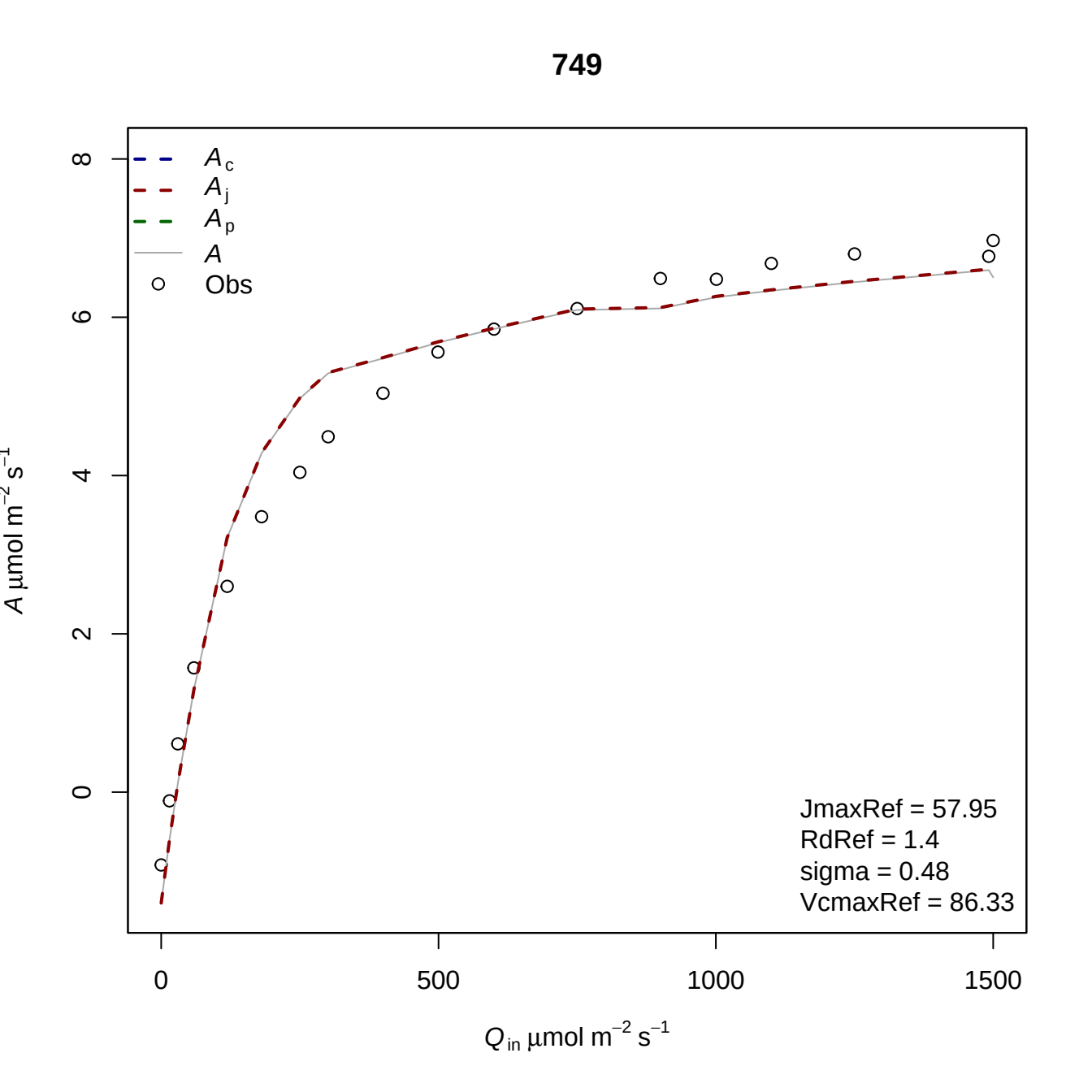

744
A_c
A_j
A_p
A
Obs
JmaxRef = 64.99
RdRef = 2.61
sigma = 0.78
VcmaxRef = 172.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

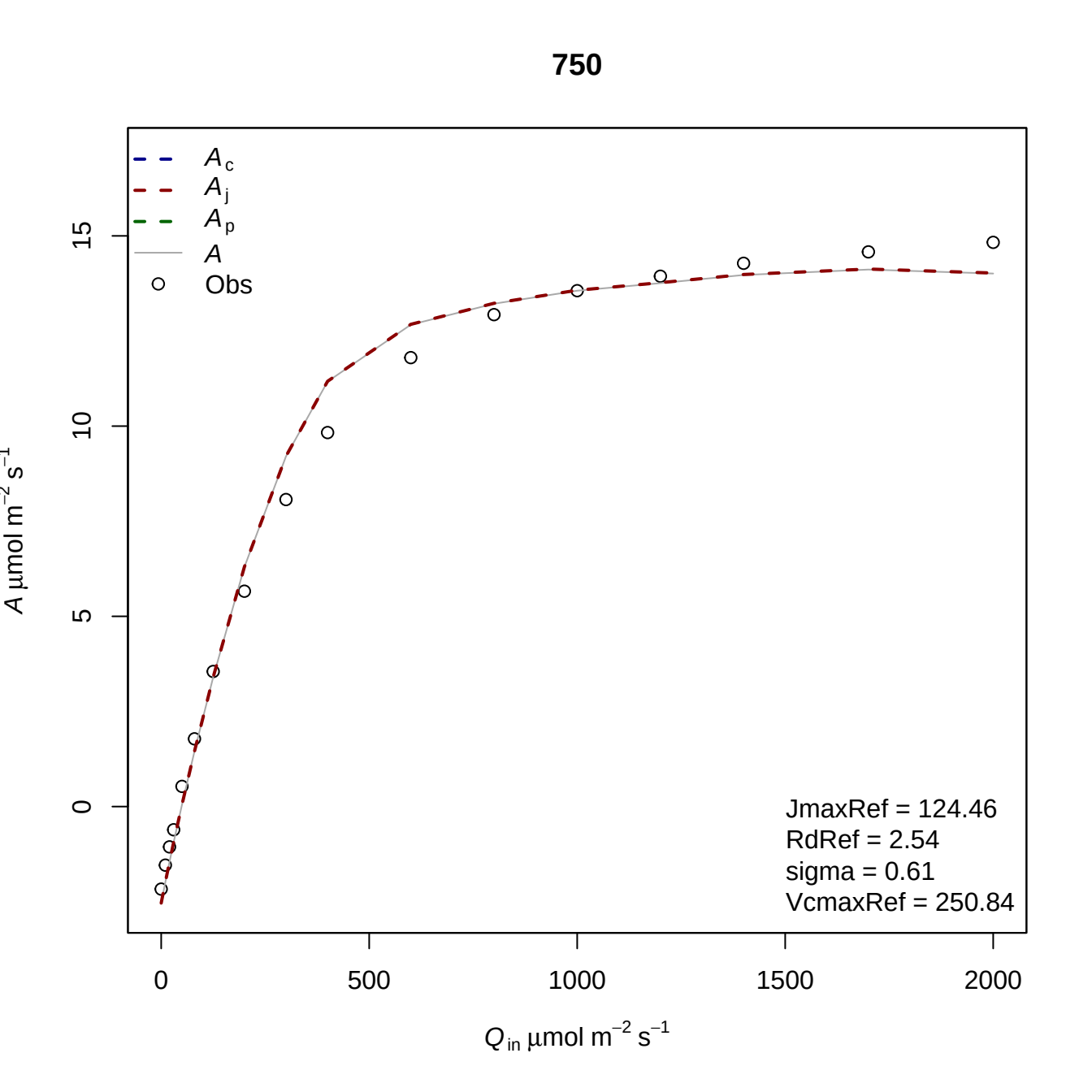

745
A_c
A_j
A_p
A
Obs
JmaxRef = 52.34
RdRef = 1.89
sigma = 0.68
VcmaxRef = 117.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

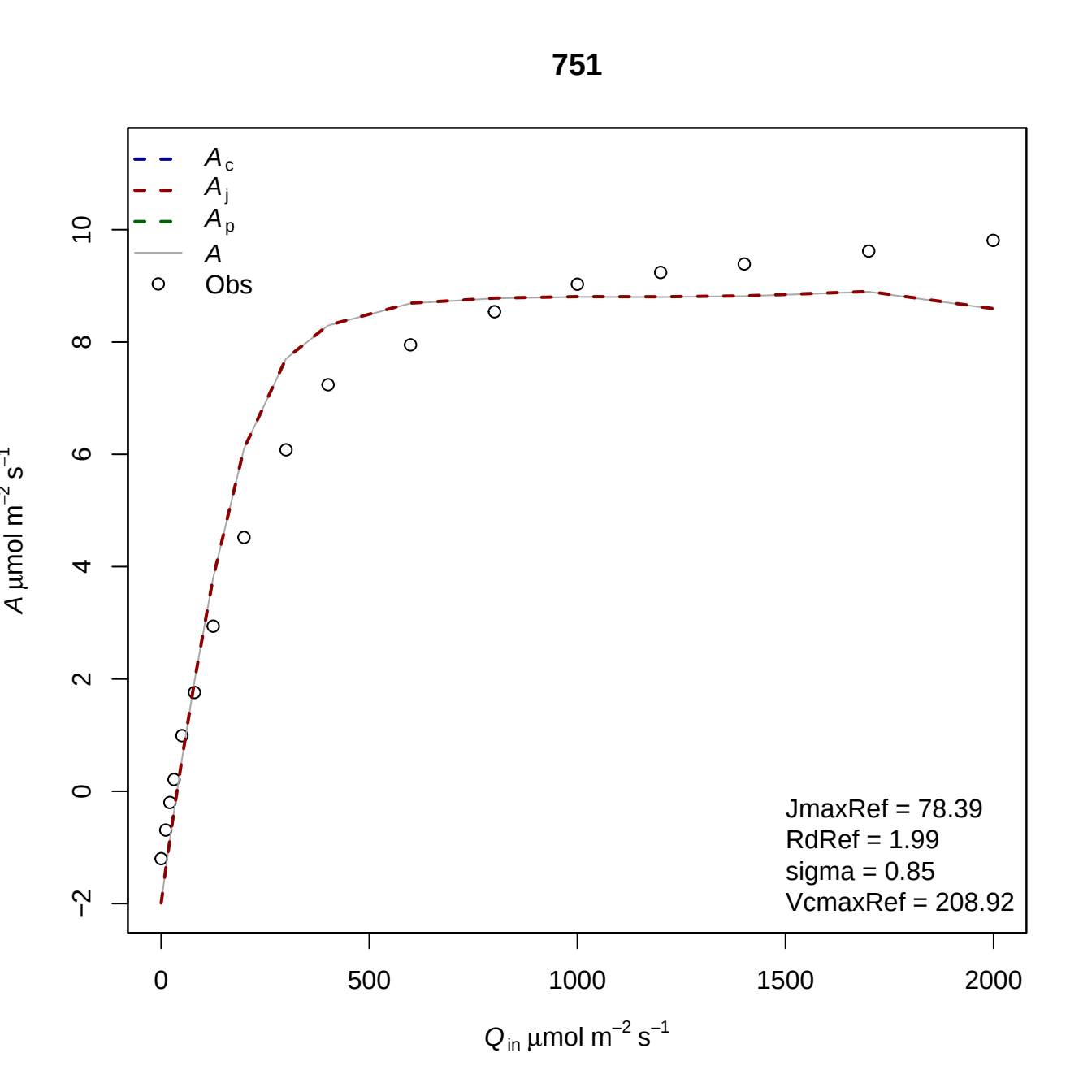

746
A_c
A_j
A_p
A
Obs
JmaxRef = 36.33
RdRef = 1.36
sigma = 0.47
VcmaxRef = 95.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

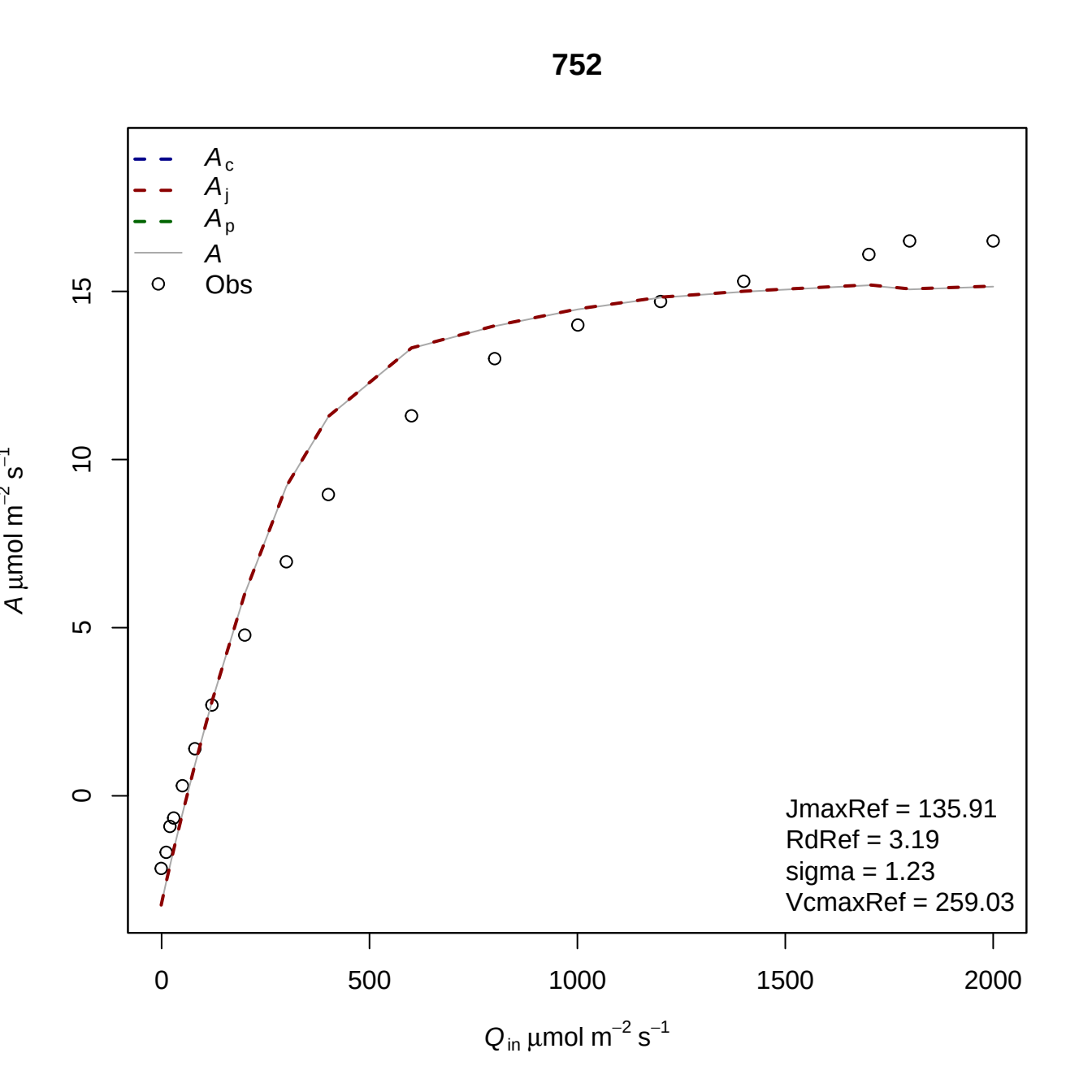

747
A_c
A_j
A_p
A
Obs
JmaxRef = 47.39
RdRef = 1.86
sigma = 0.62
VcmaxRef = 147.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

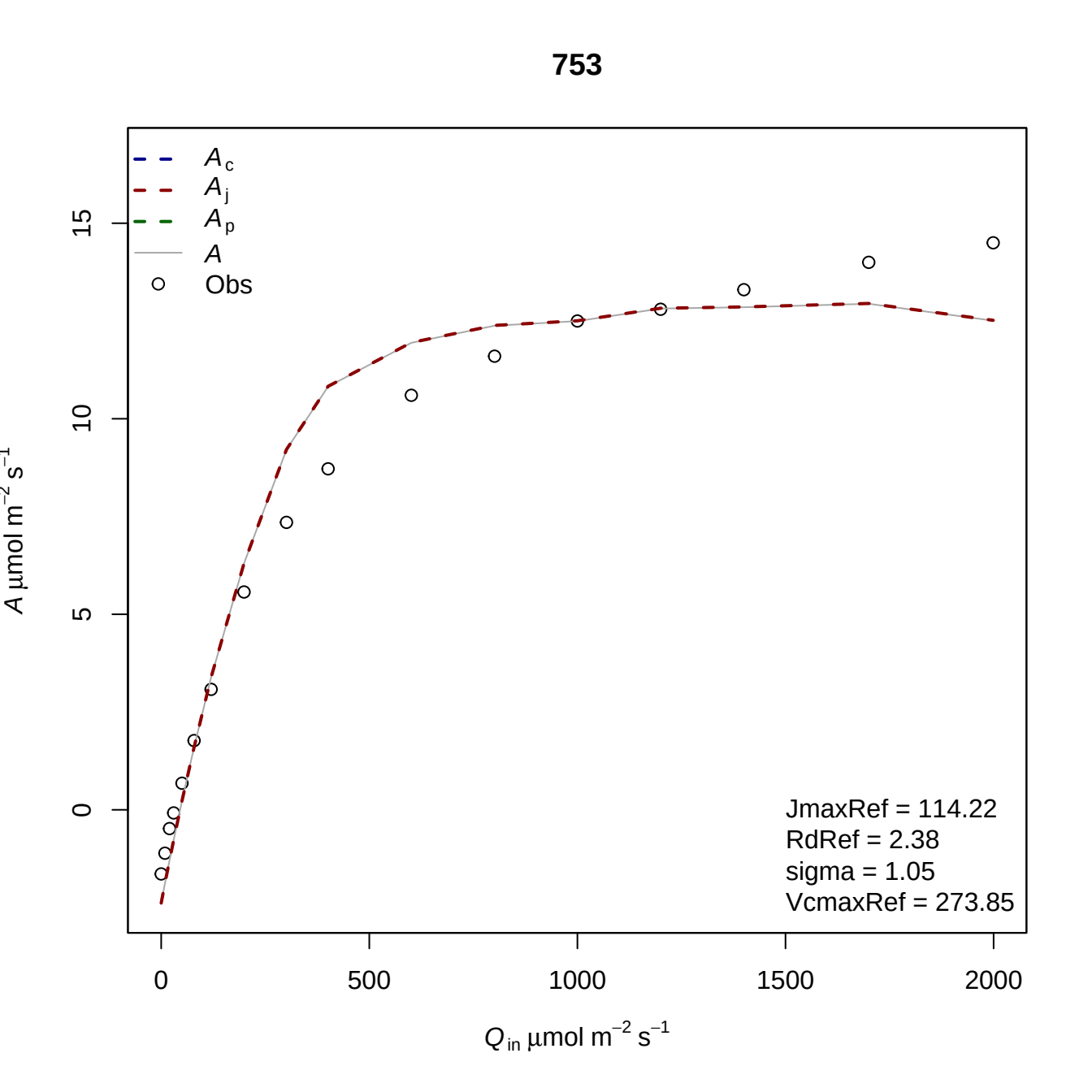

748
A_c
A_j
A_p
A
Obs
JmaxRef = 79.61
RdRef = 1.04
sigma = 0.33
VcmaxRef = 136.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

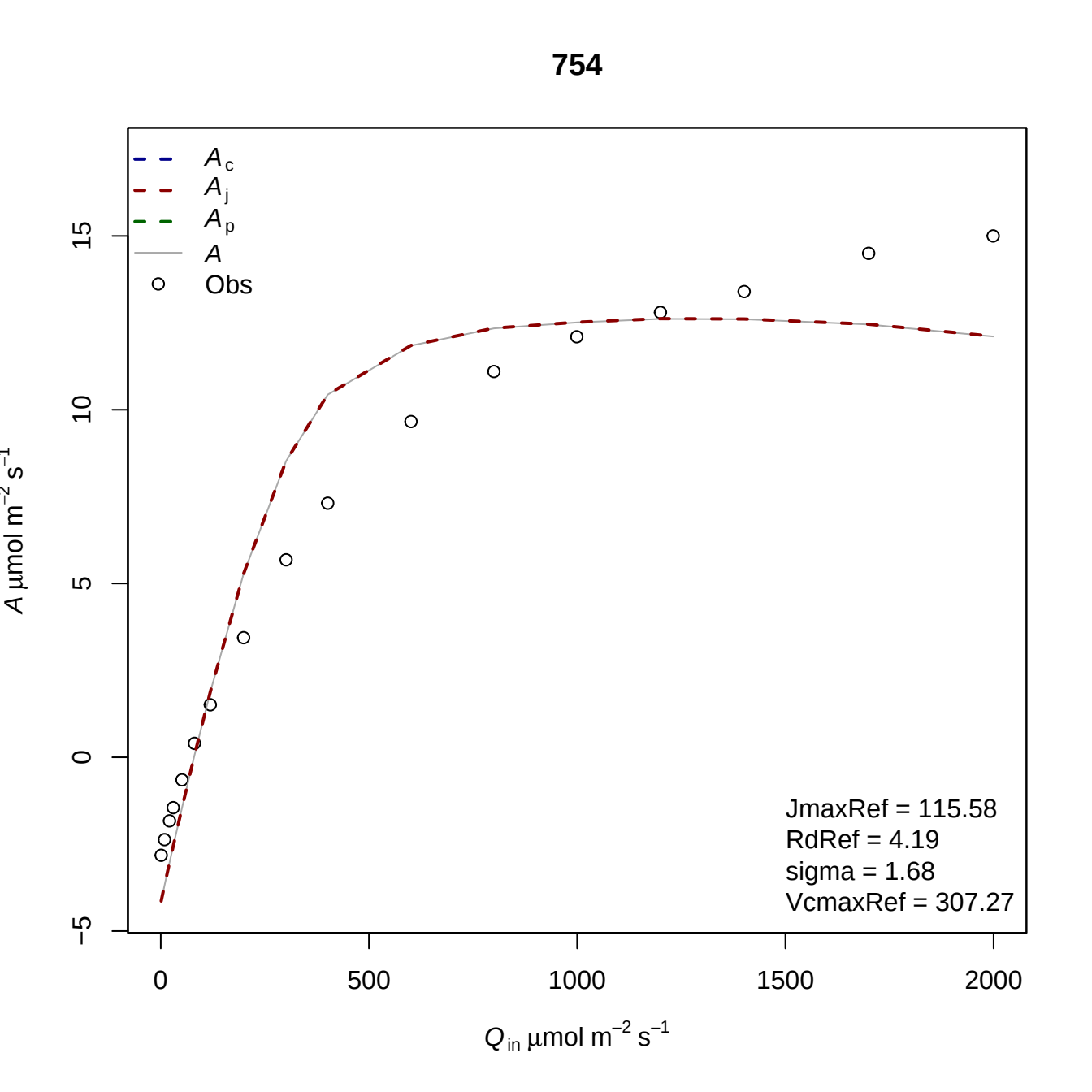

749
A_c
A_j
A_p
A
Obs
JmaxRef = 57.95
RdRef = 1.4
sigma = 0.48
VcmaxRef = 86.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

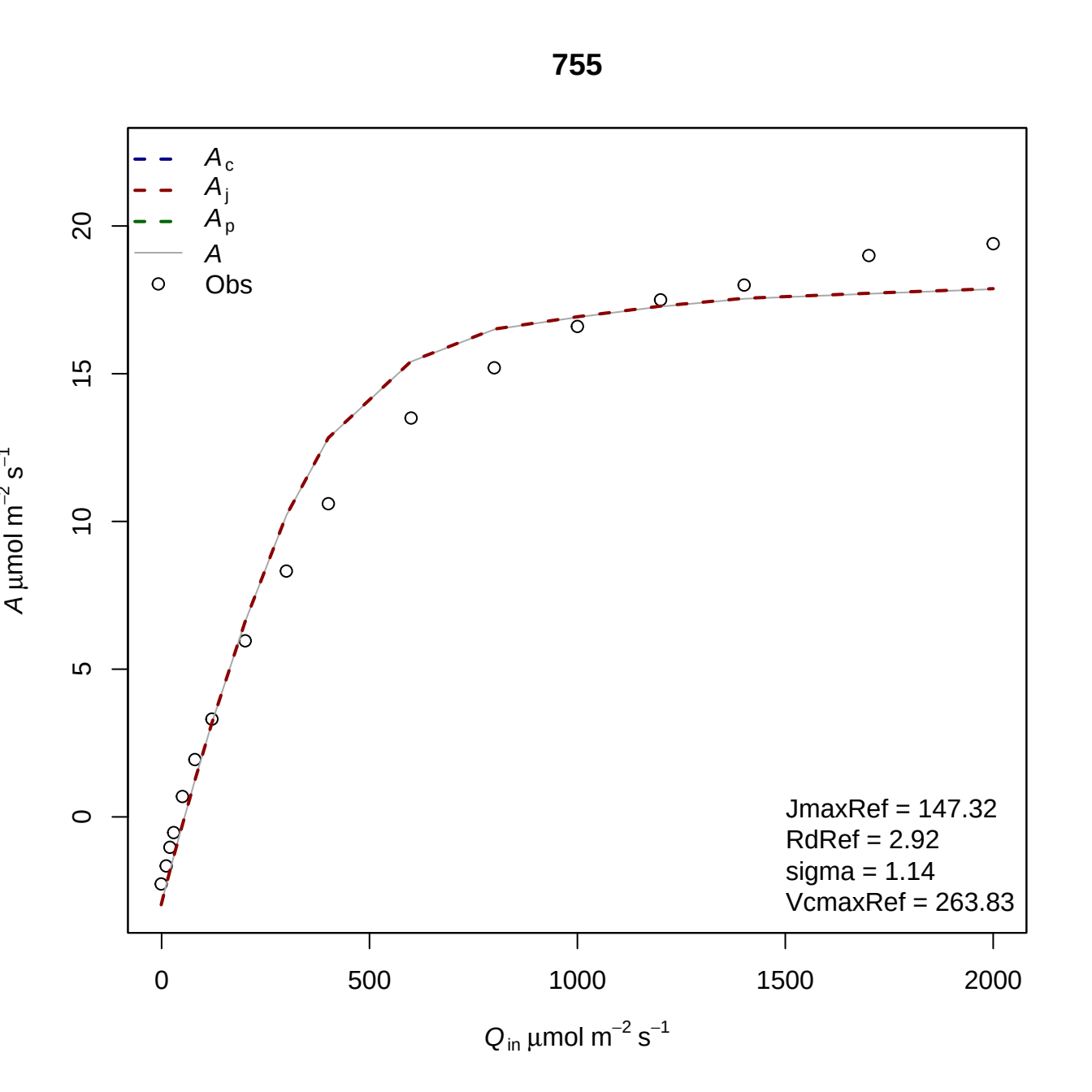

750
A_c
A_j
A_p
A
Obs
JmaxRef = 124.46
RdRef = 2.54
sigma = 0.61
VcmaxRef = 250.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

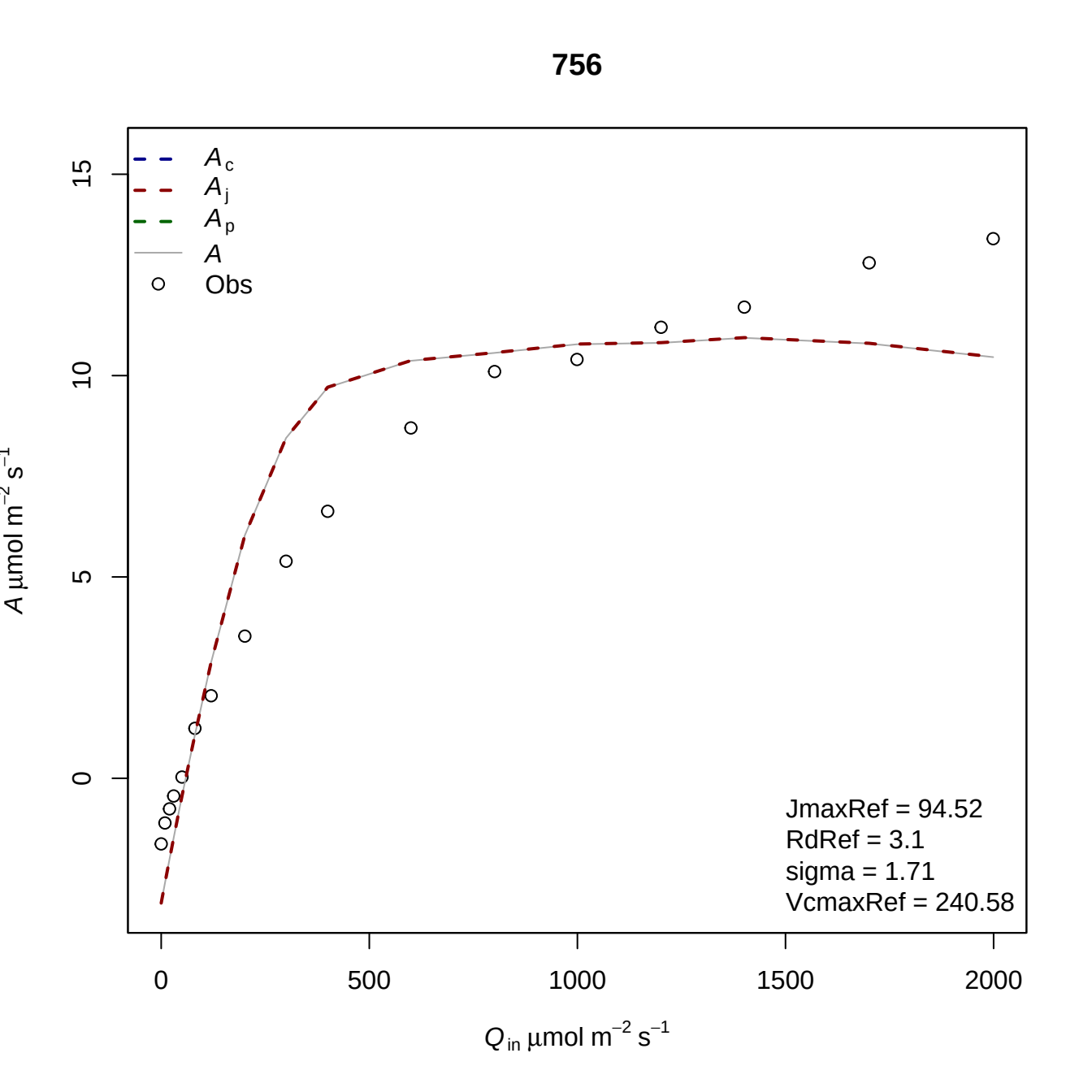

751
A_c
A_j
A_p
A
Obs
JmaxRef = 78.39
RdRef = 1.99
sigma = 0.85
VcmaxRef = 208.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

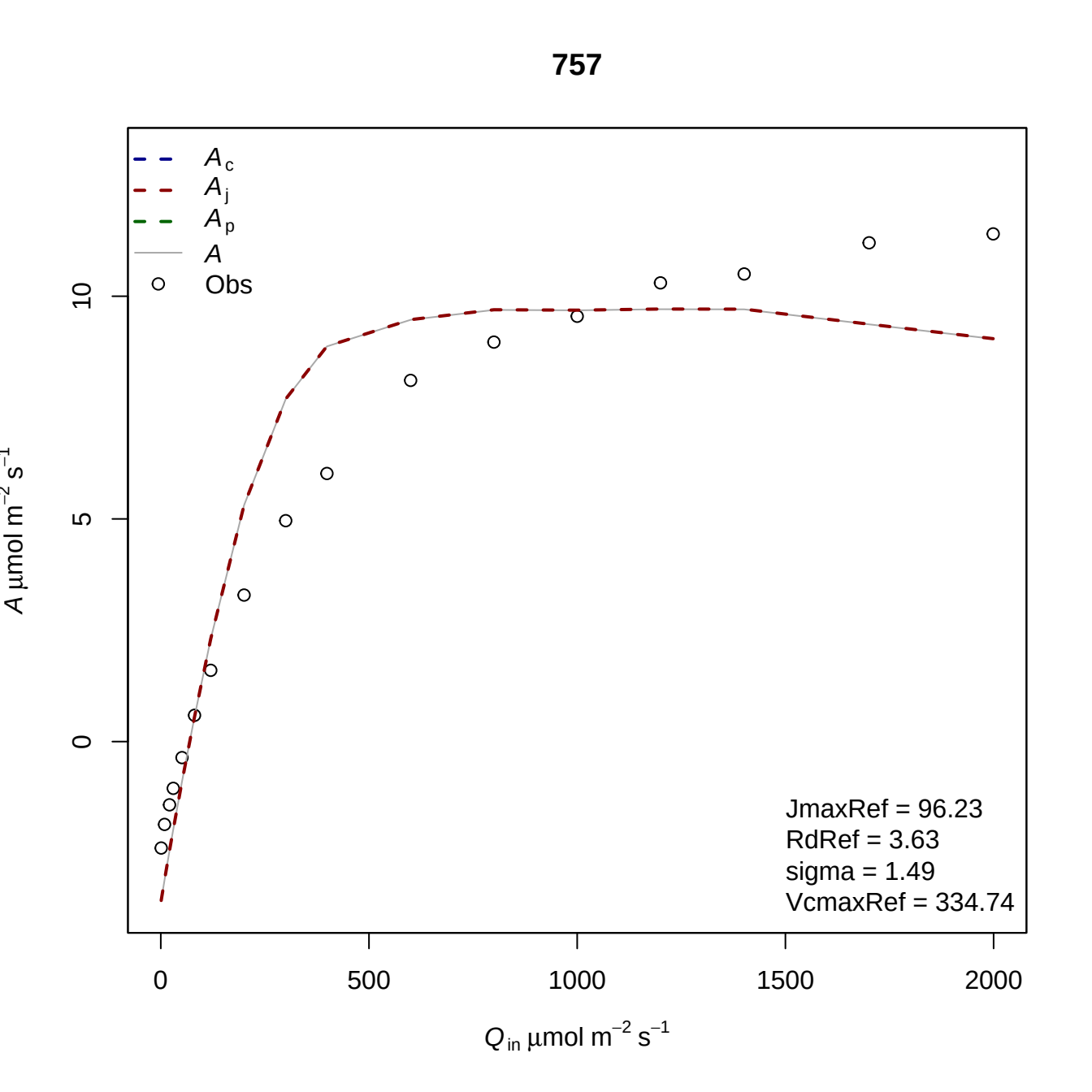

752
A_c
A_j
A_p
A
Obs
JmaxRef = 135.91
RdRef = 3.19
sigma = 1.23
VcmaxRef = 259.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

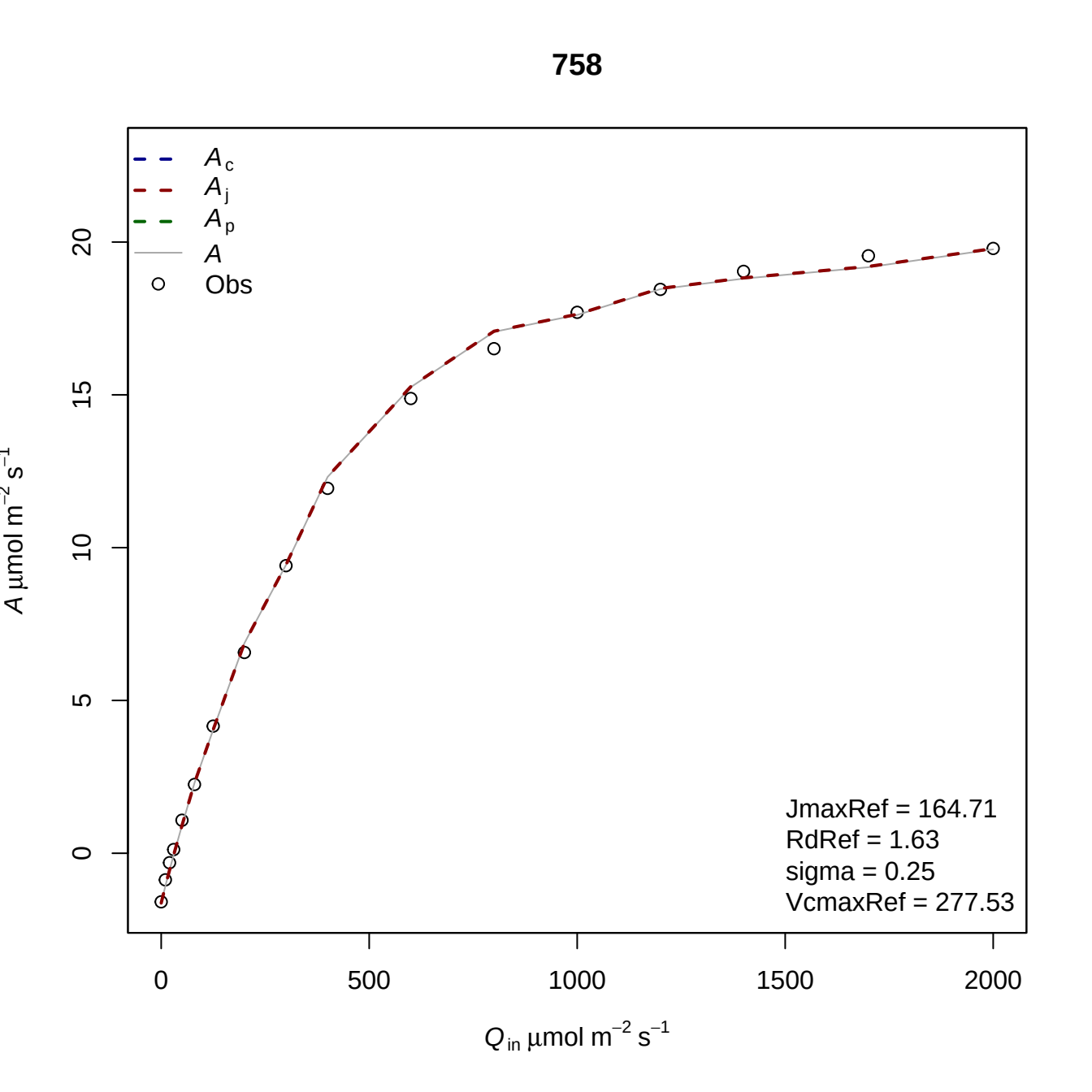

753
A_c
A_j
A_p
A
Obs
JmaxRef = 114.22
RdRef = 2.38
sigma = 1.05
VcmaxRef = 273.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

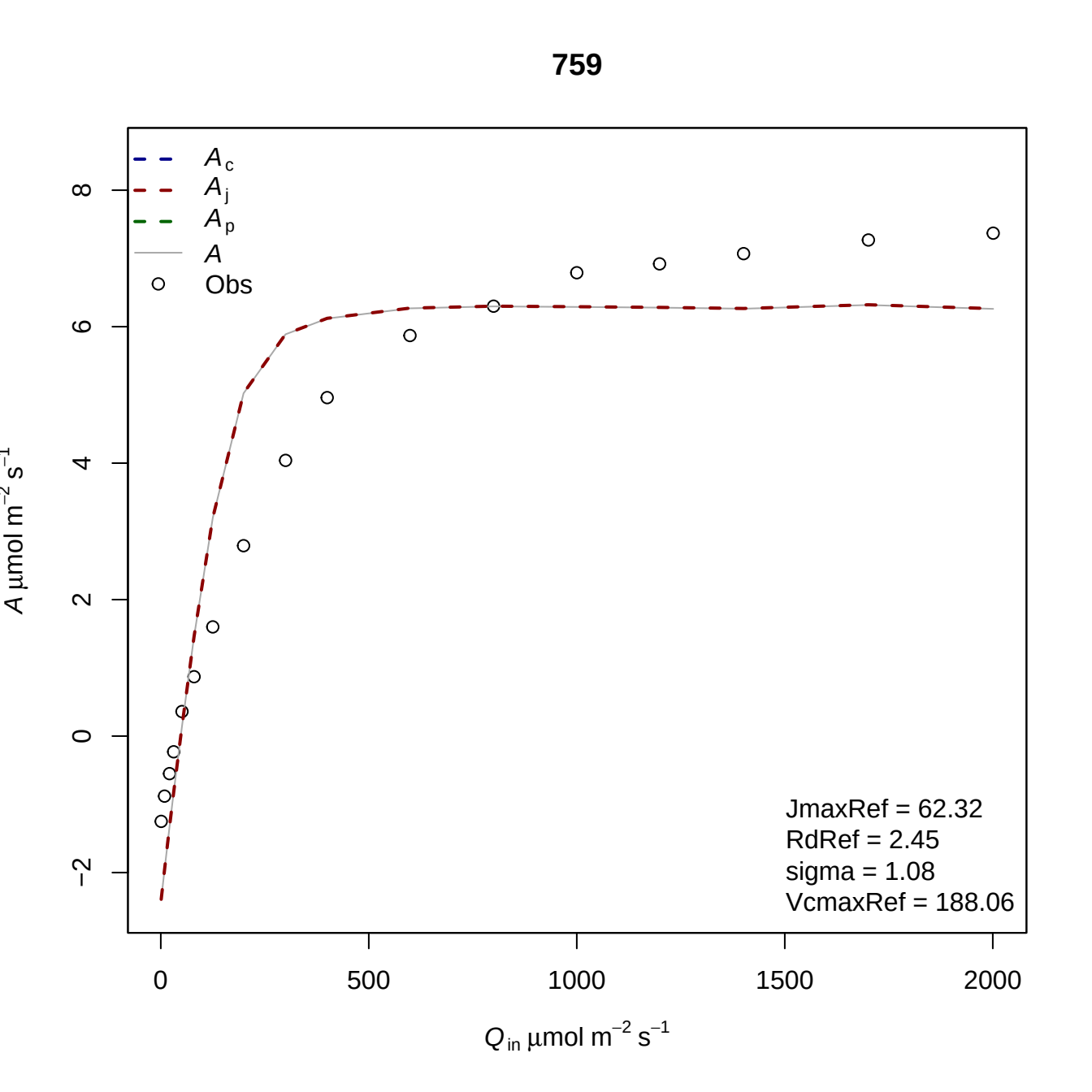

754
A_c
A_j
A_p
A
Obs
JmaxRef = 115.58
RdRef = 4.19
sigma = 1.68
VcmaxRef = 307.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

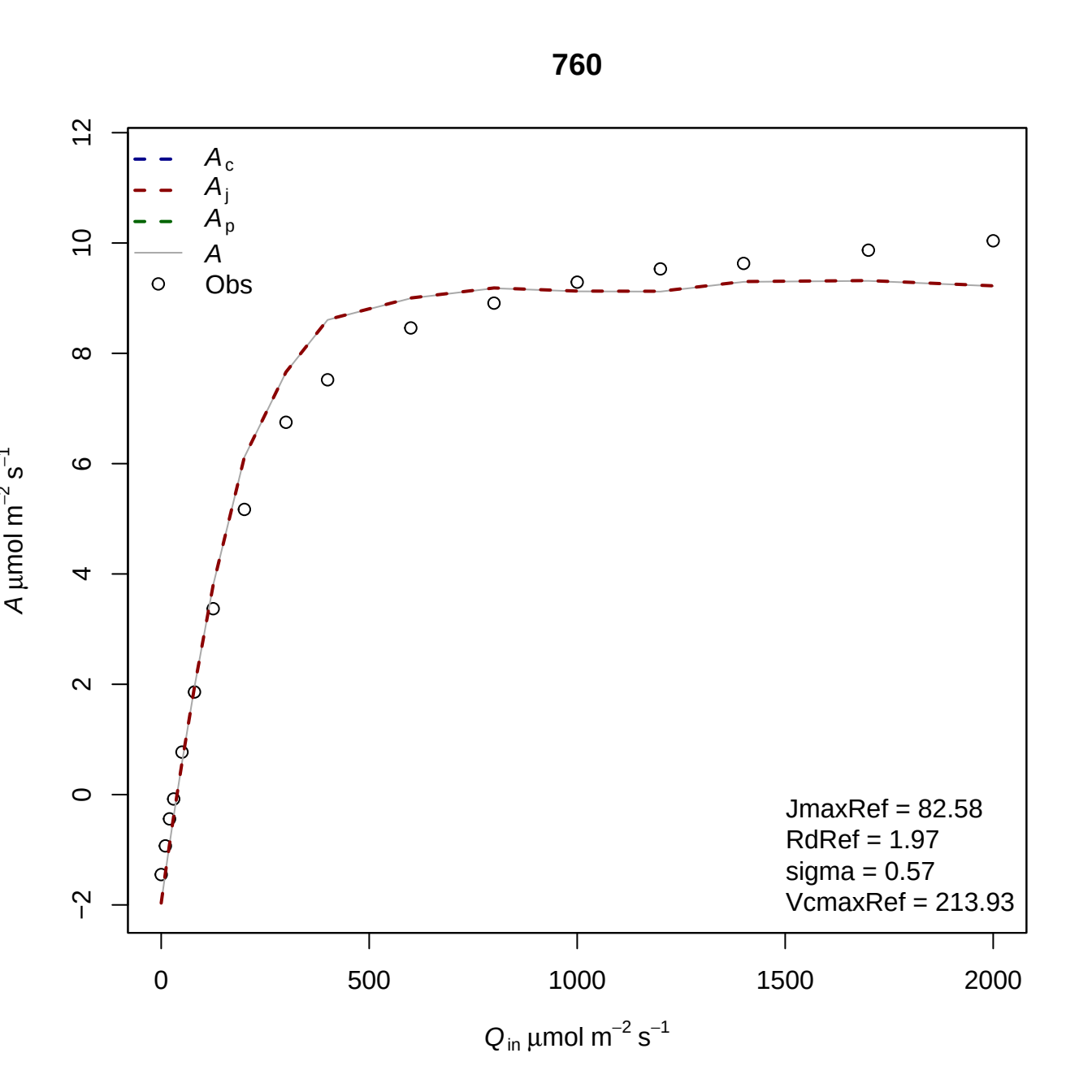

755
A_c
A_j
A_p
A
Obs
JmaxRef = 147.32
RdRef = 2.92
sigma = 1.14
VcmaxRef = 263.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

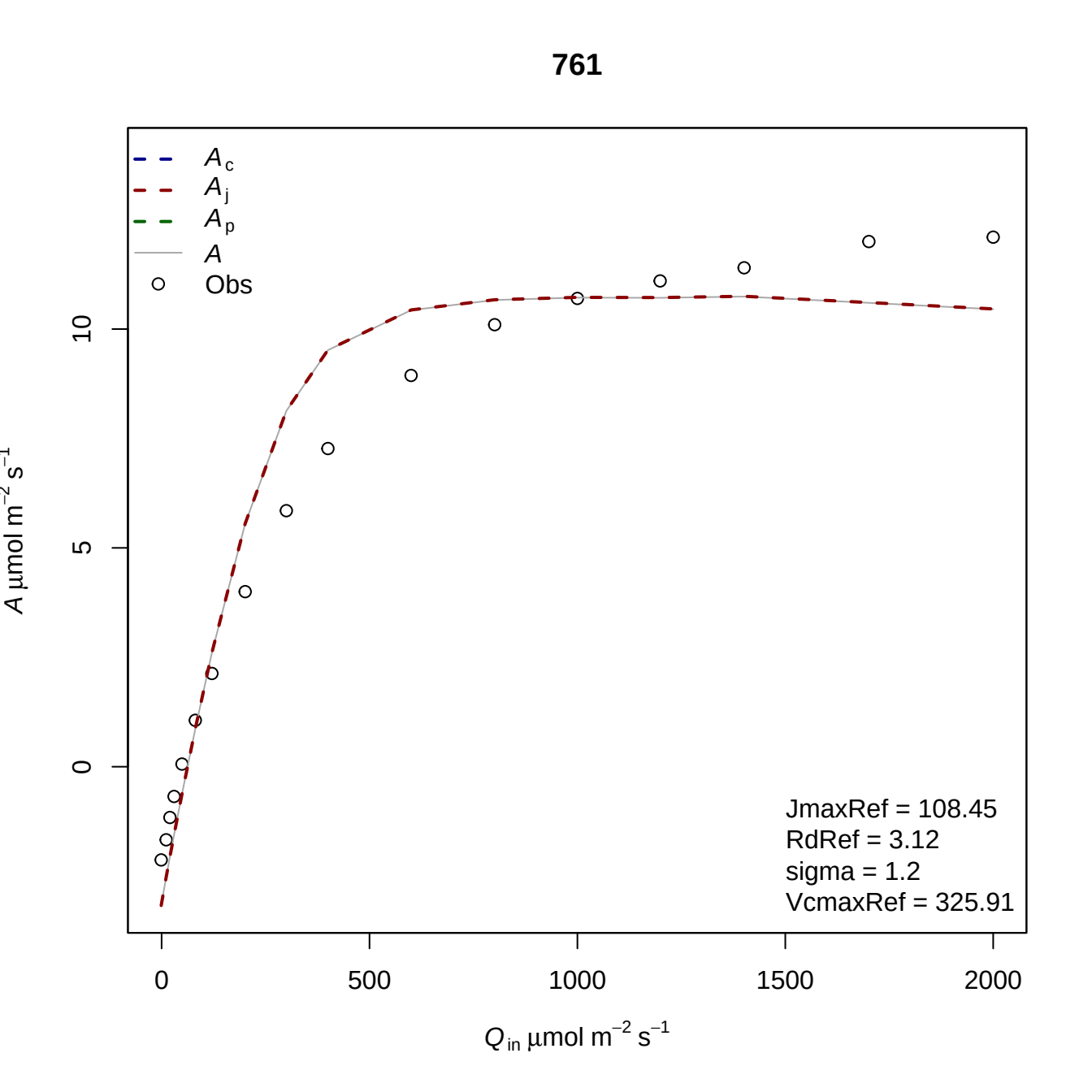

756
A_c
A_j
A_p
A
Obs
JmaxRef = 94.52
RdRef = 3.1
sigma = 1.71
VcmaxRef = 240.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

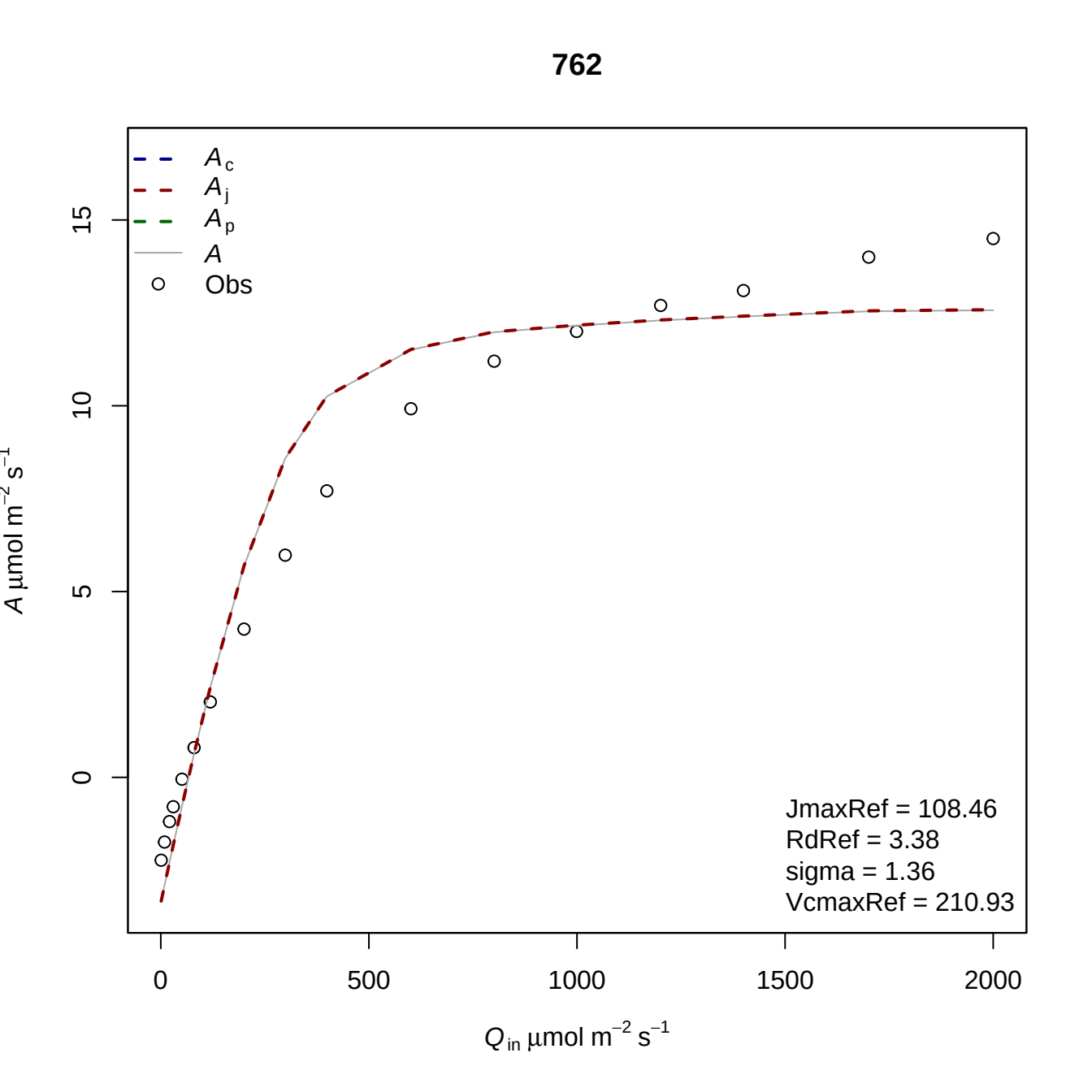

757
A_c
A_j
A_p
A
Obs
JmaxRef = 96.23
RdRef = 3.63
sigma = 1.49
VcmaxRef = 334.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

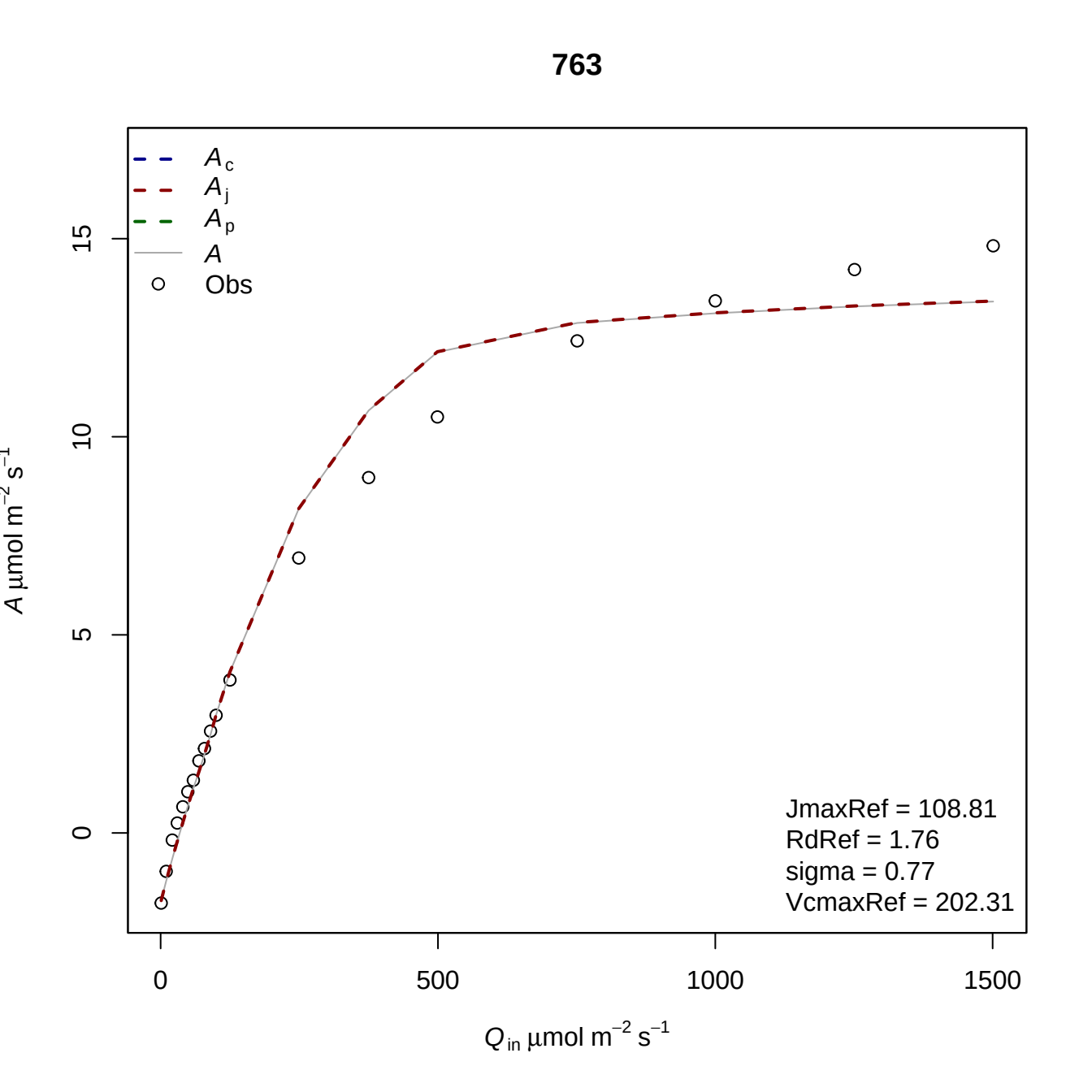

758
A_c
A_j
A_p
A
Obs
JmaxRef = 164.71
RdRef = 1.63
sigma = 0.25
VcmaxRef = 277.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

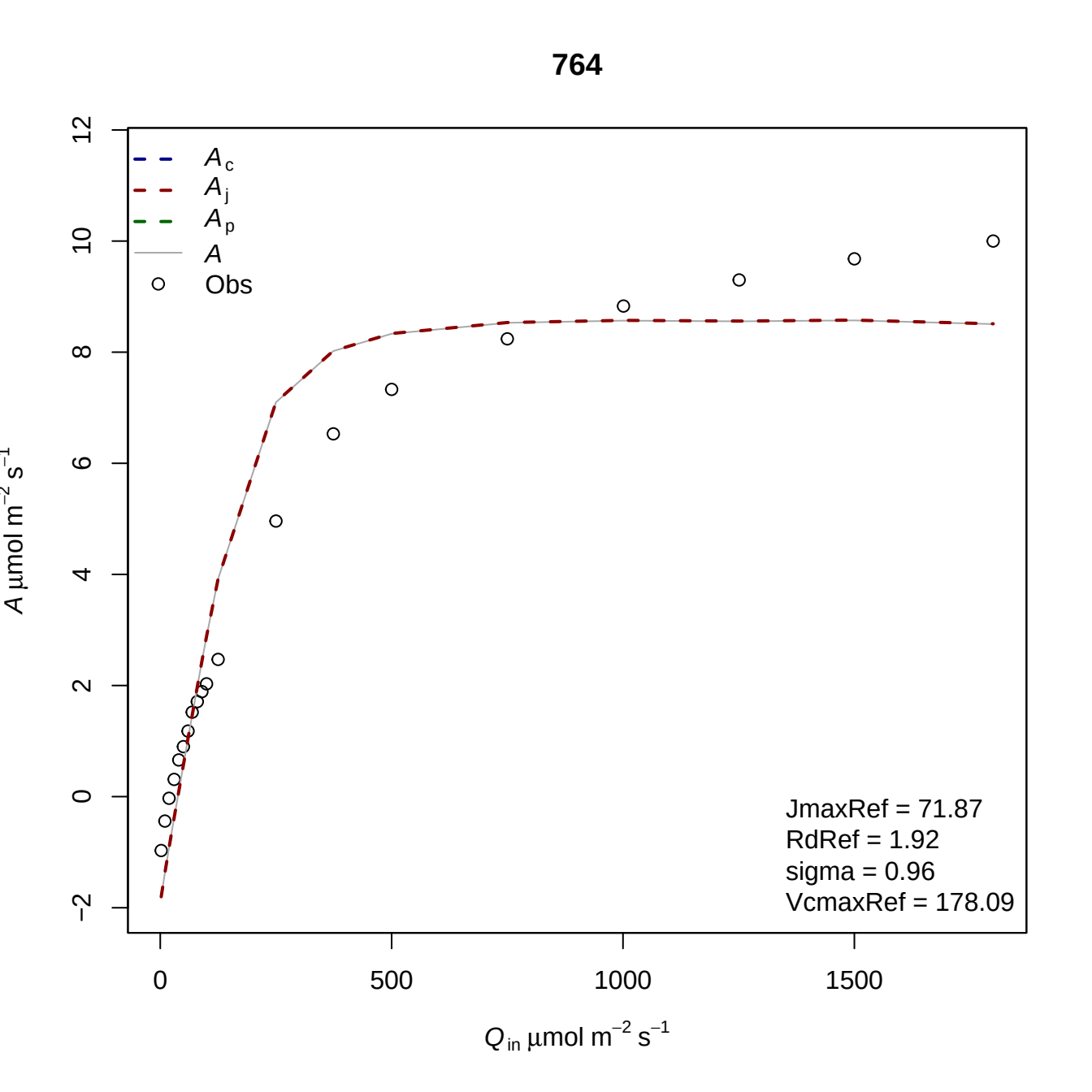

759
A_c
A_j
A_p
A
Obs
JmaxRef = 62.32
RdRef = 2.45
sigma = 1.08
VcmaxRef = 188.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

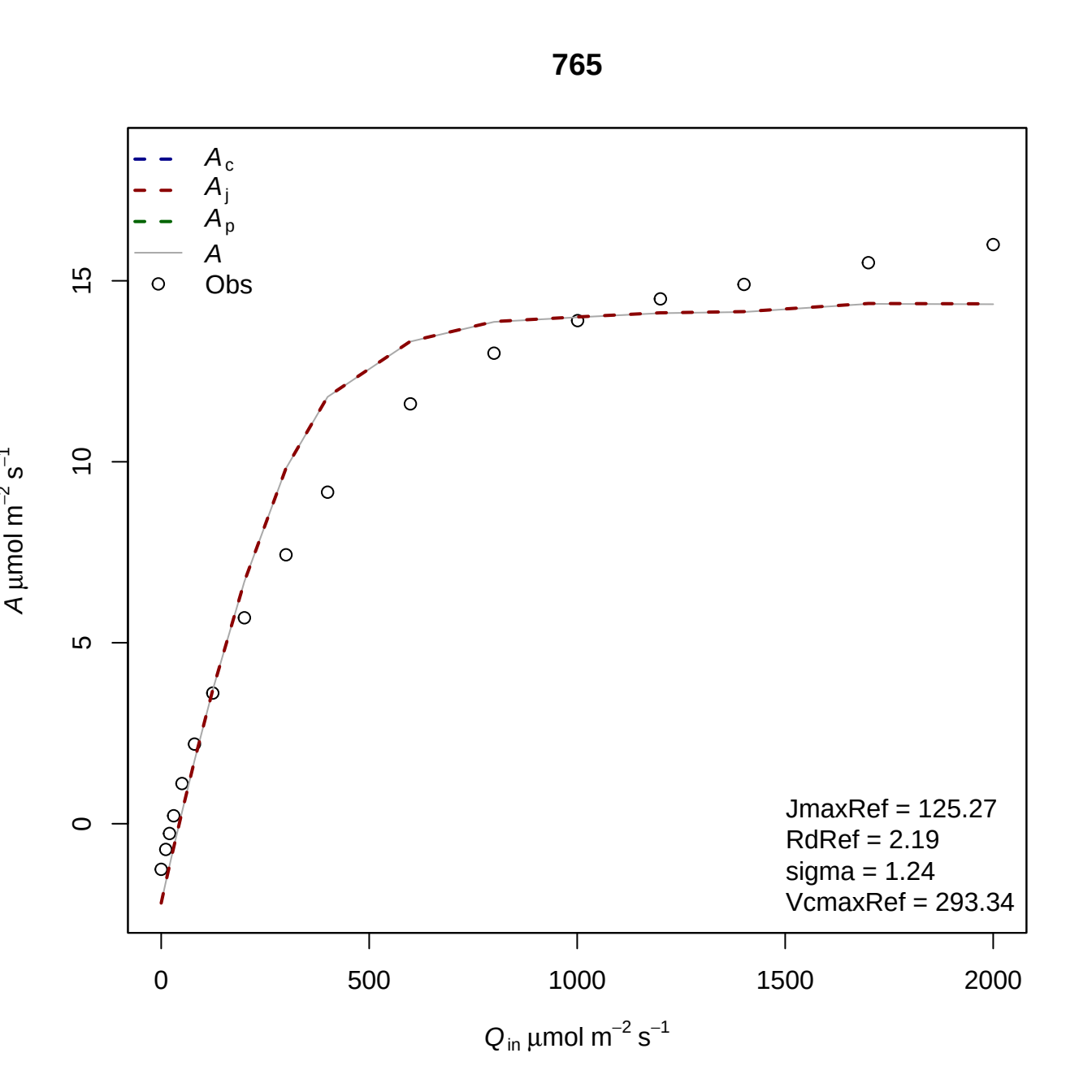

760
A_c
A_j
A_p
A
Obs
JmaxRef = 82.58
RdRef = 1.97
sigma = 0.57
VcmaxRef = 213.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

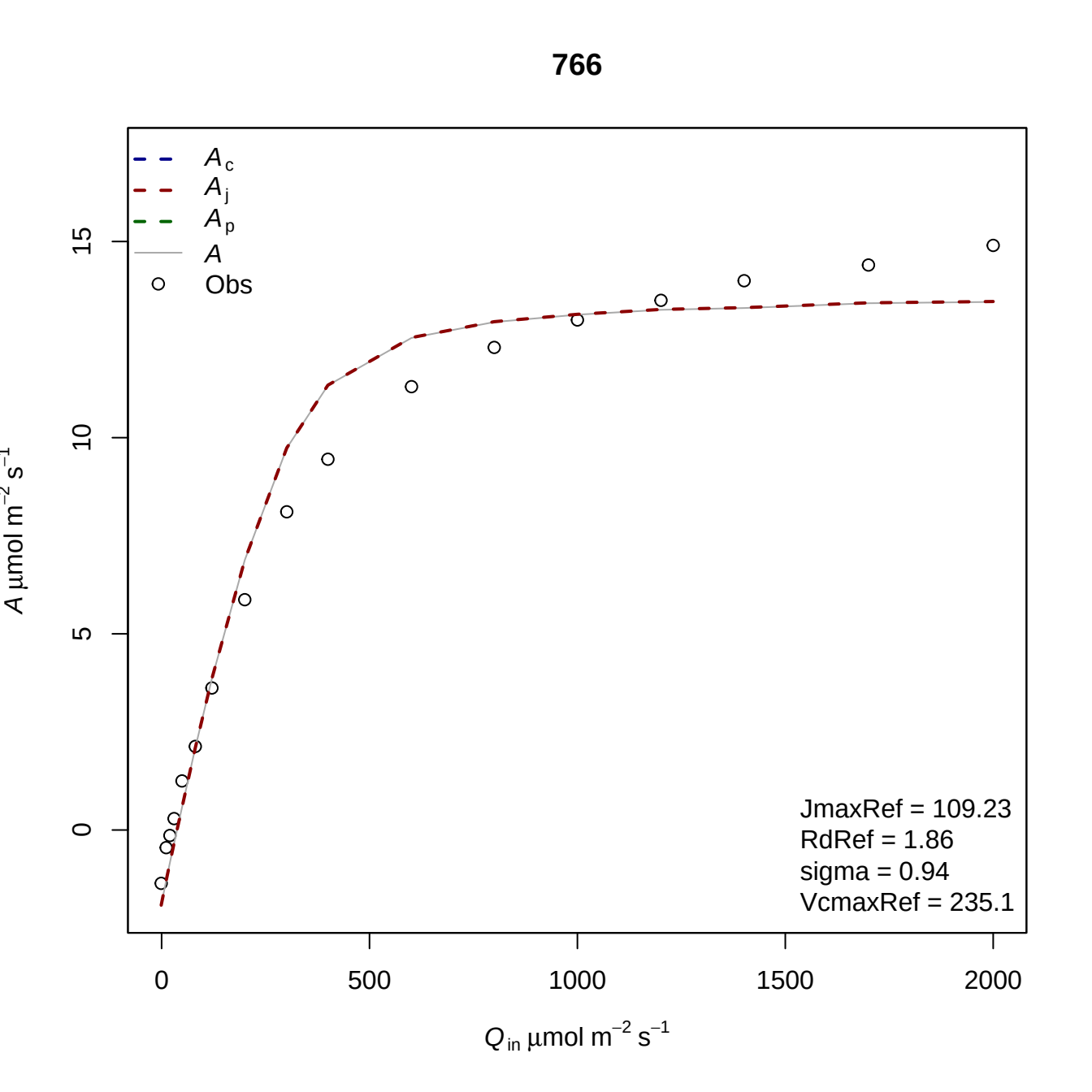

761
A_c
A_j
A_p
A
Obs
JmaxRef = 108.45
RdRef = 3.12
sigma = 1.2
VcmaxRef = 325.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

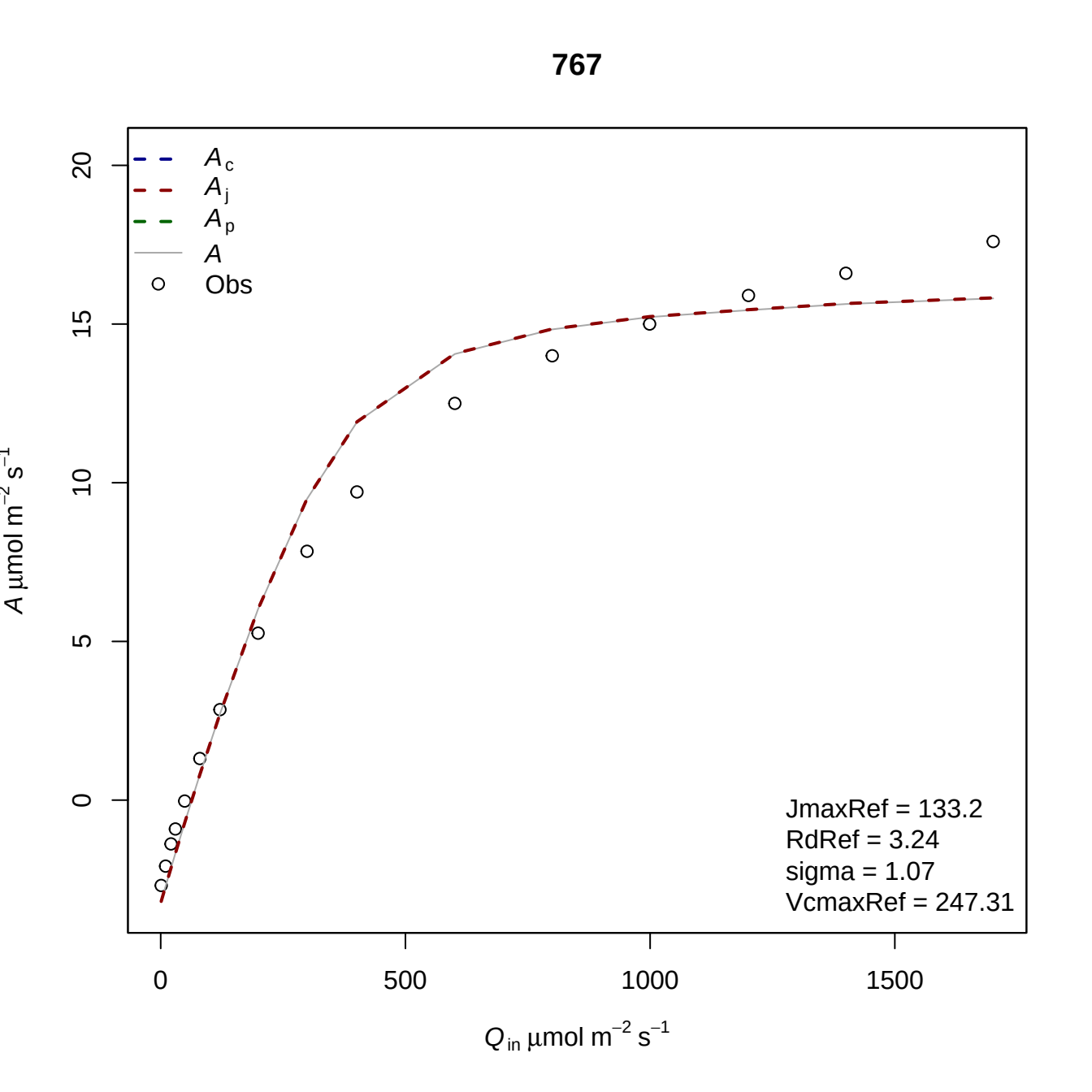

762
A_c
A_j
A_p
A
Obs
JmaxRef = 108.46
RdRef = 3.38
sigma = 1.36
VcmaxRef = 210.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

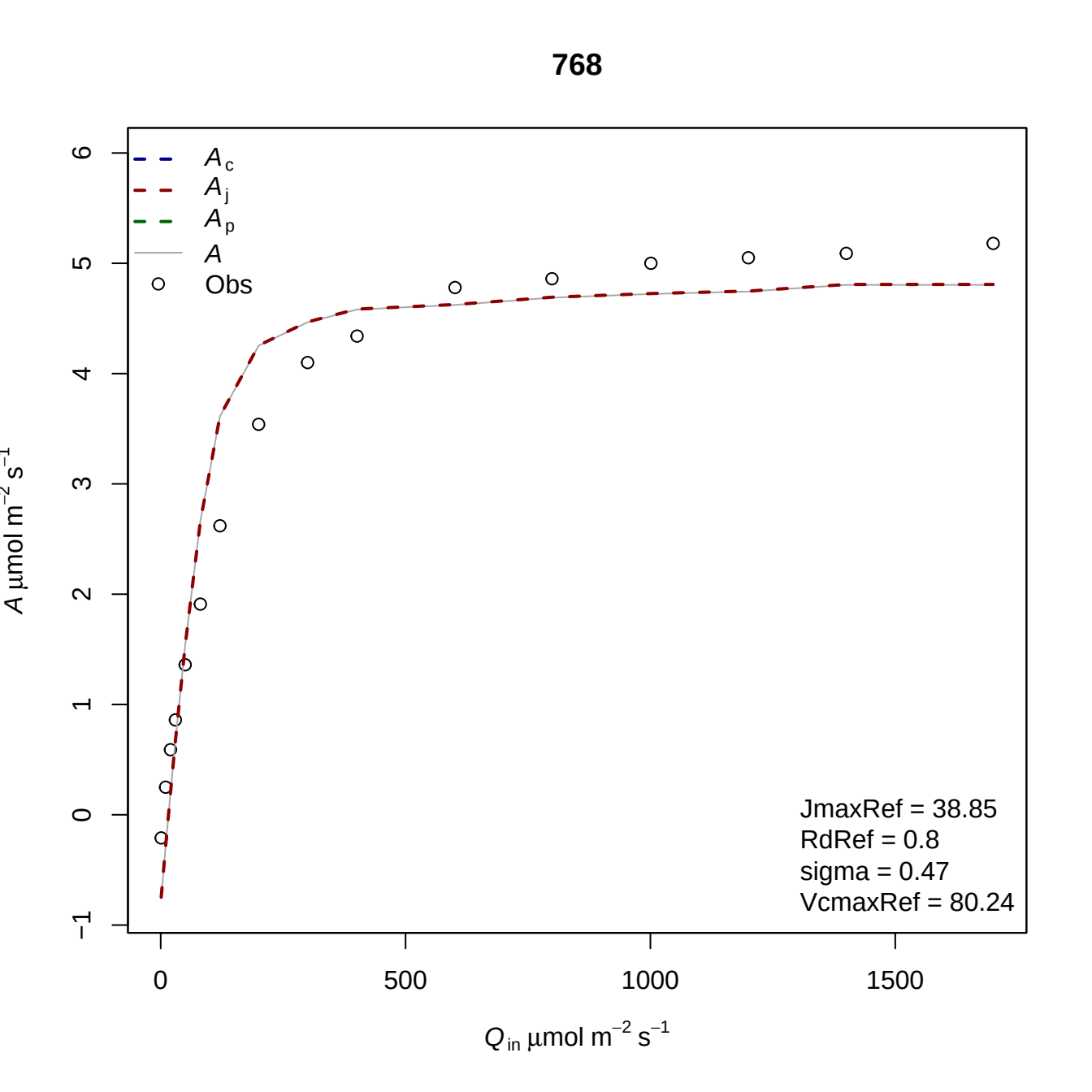

763
A_c
A_j
A_p
A
Obs
JmaxRef = 108.81
RdRef = 1.76
sigma = 0.77
VcmaxRef = 202.31
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

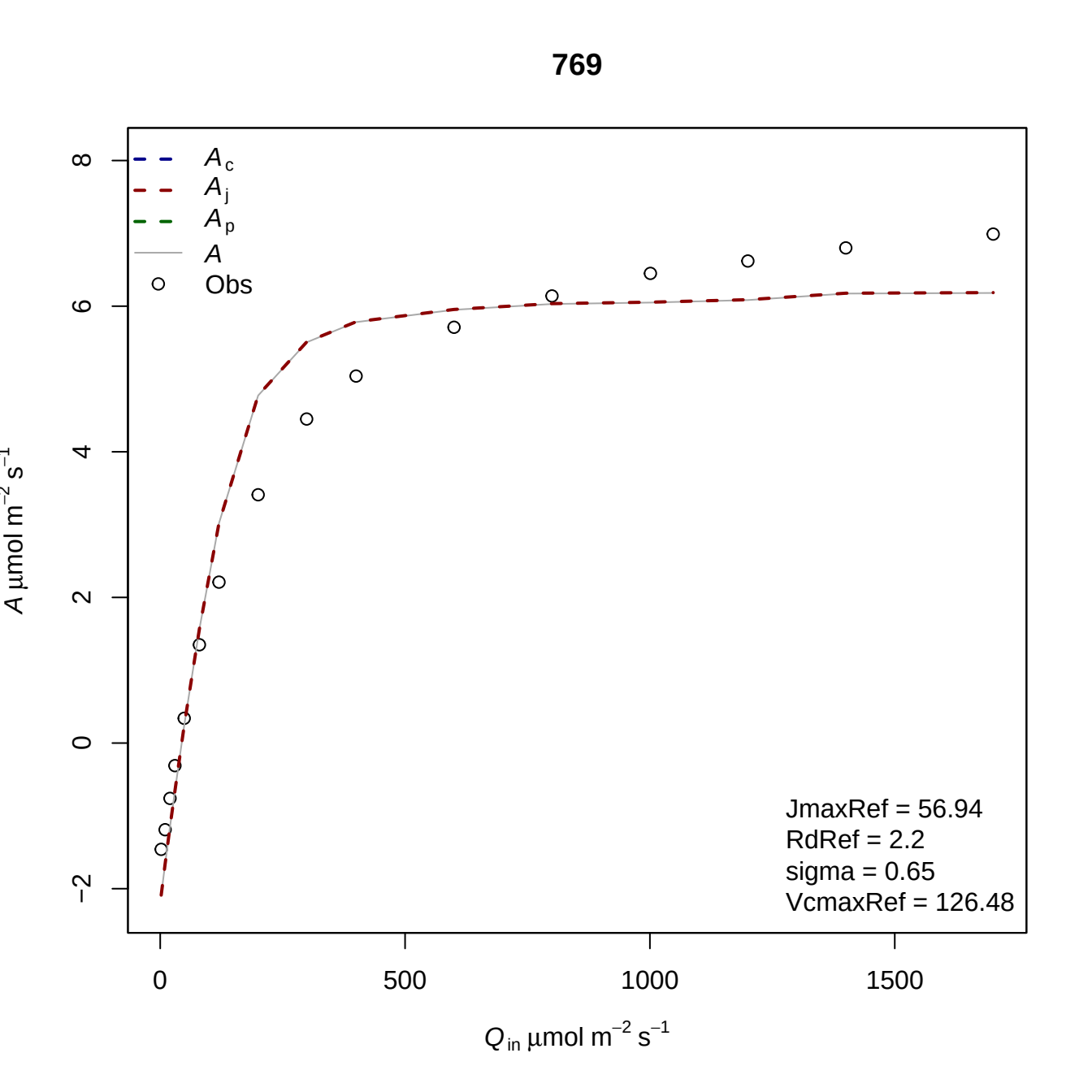

764
A_c
A_j
A_p
A
Obs
JmaxRef = 71.87
RdRef = 1.92
sigma = 0.96
VcmaxRef = 178.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

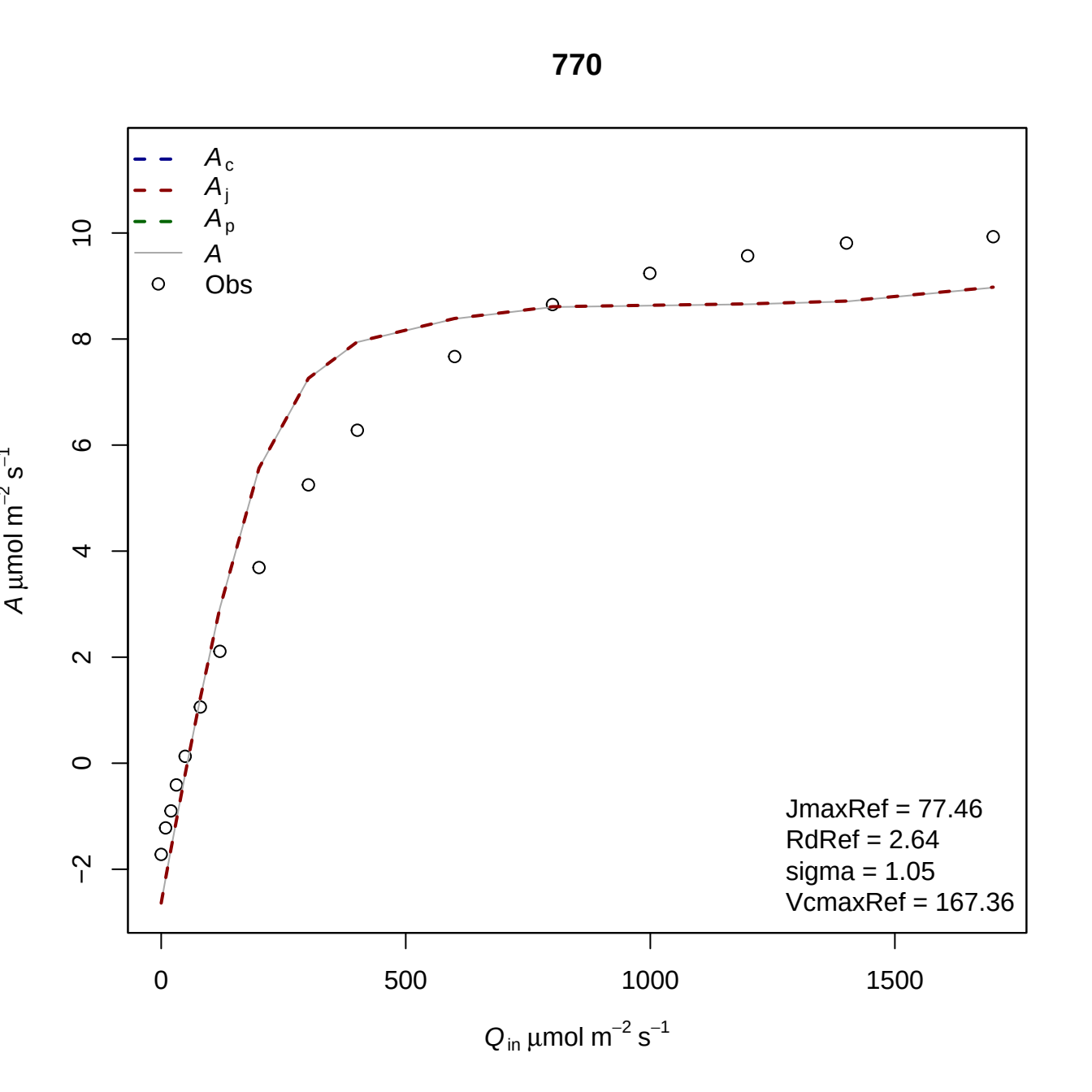

765
A_c
A_j
A_p
A
Obs
JmaxRef = 125.27
RdRef = 2.19
sigma = 1.24
VcmaxRef = 293.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

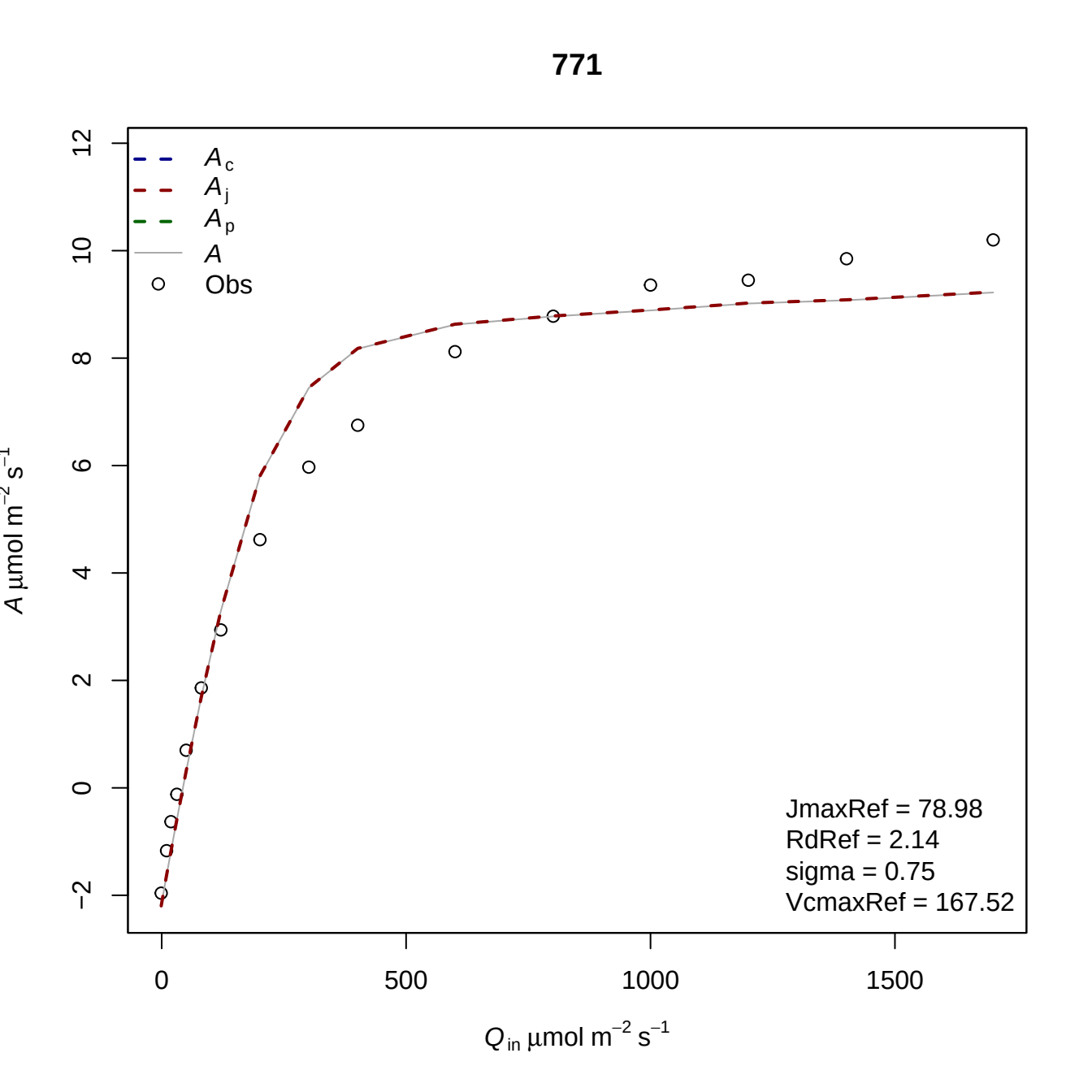

766
A_c
A_j
A_p
A
Obs
JmaxRef = 109.23
RdRef = 1.86
sigma = 0.94
VcmaxRef = 235.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

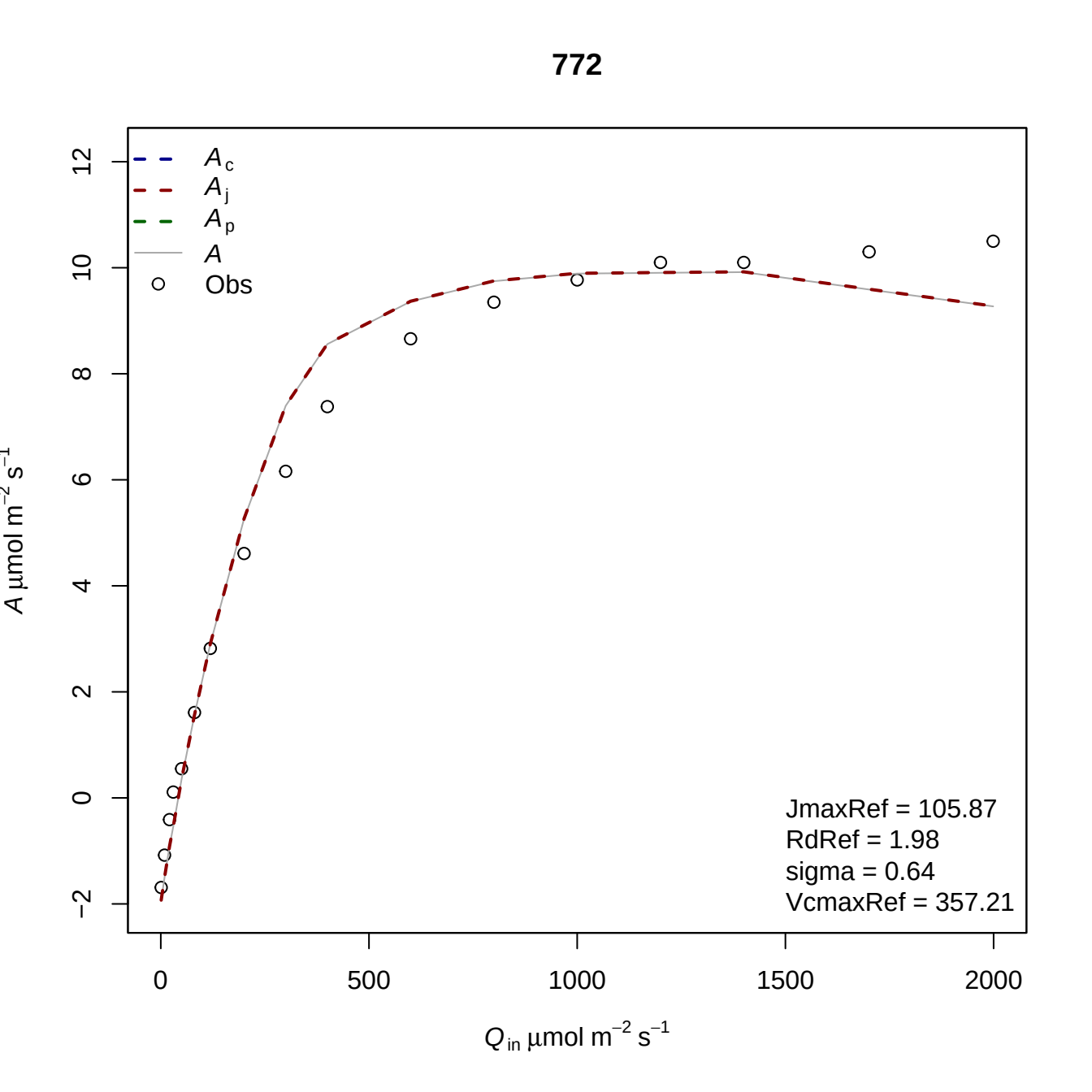

767
A_c
A_j
A_p
A
Obs
JmaxRef = 133.2
RdRef = 3.24
sigma = 1.07
VcmaxRef = 247.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

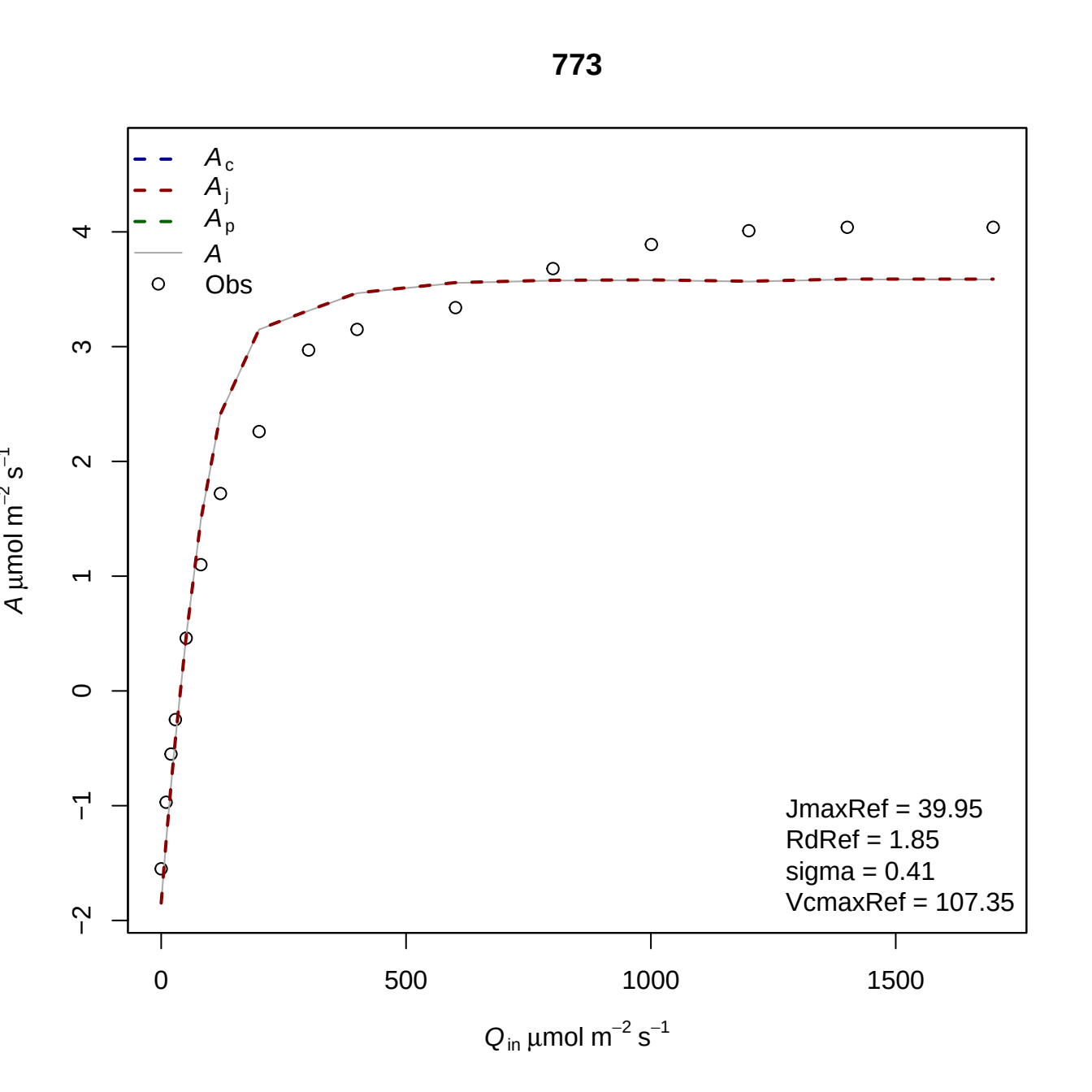

768
A_c
A_j
A_p
A
Obs
JmaxRef = 38.85
RdRef = 0.8
sigma = 0.47
VcmaxRef = 80.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

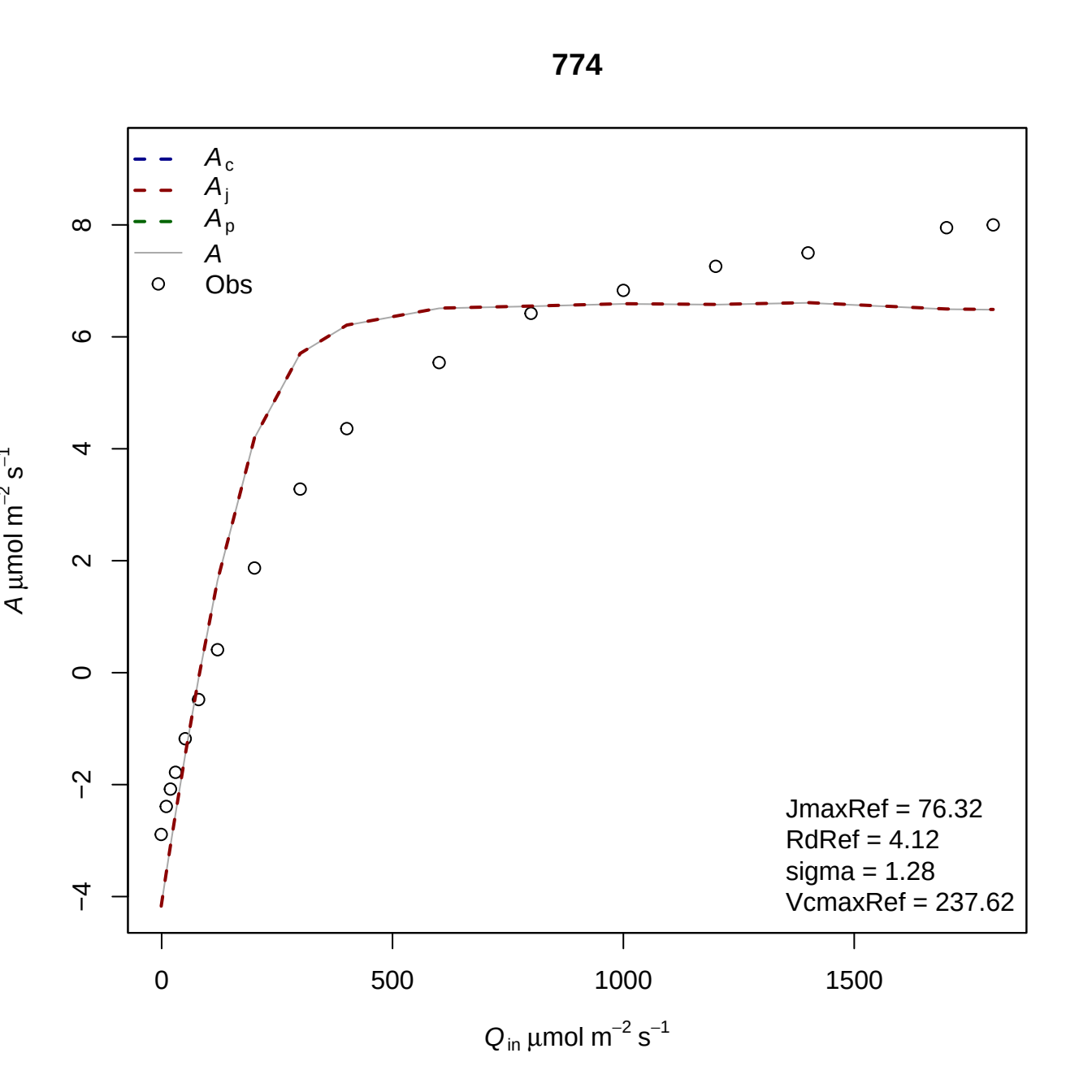

769
A_c
A_j
A_p
A
Obs
JmaxRef = 56.94
RdRef = 2.2
sigma = 0.65
VcmaxRef = 126.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

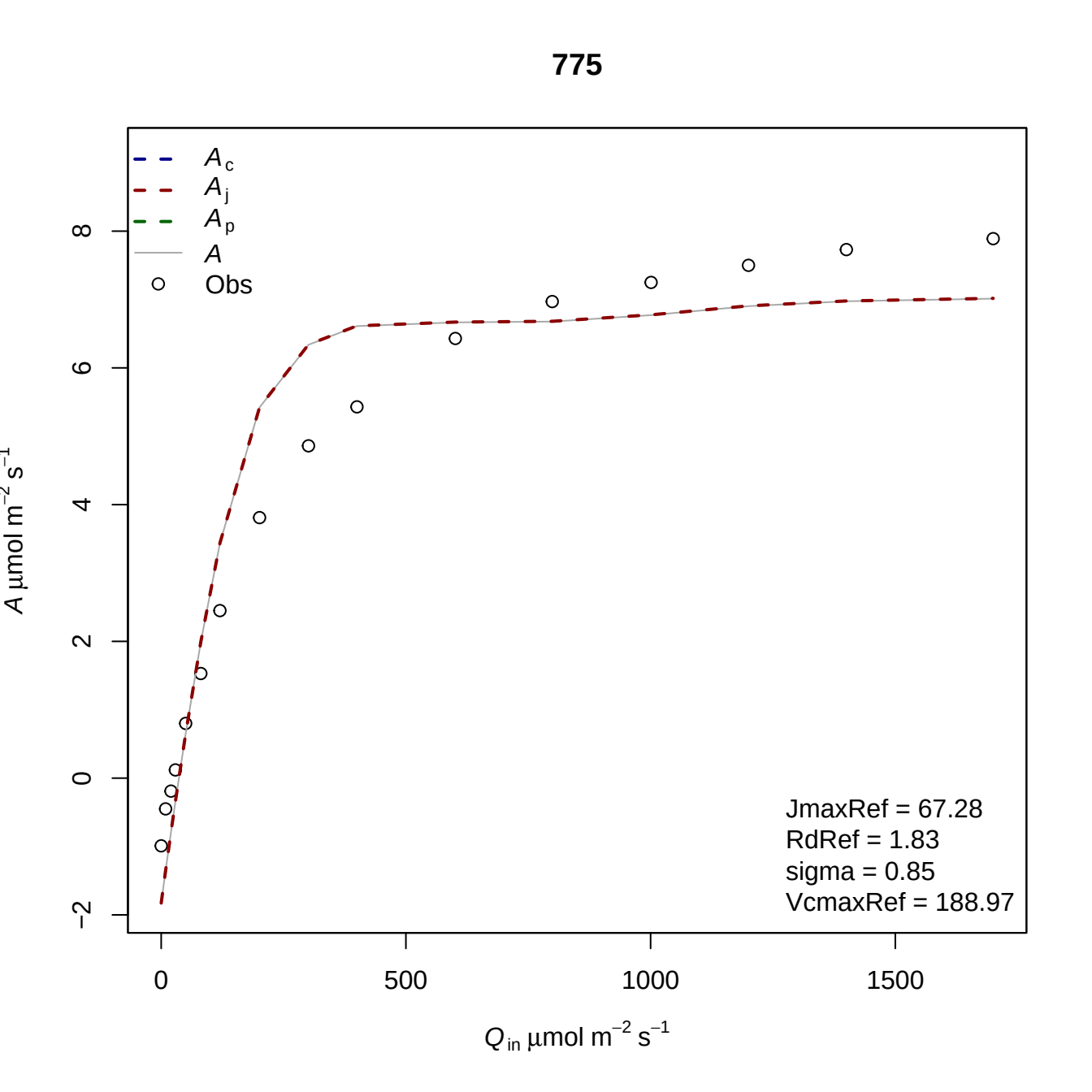

770
A_c
A_j
A_p
A
Obs
JmaxRef = 77.46
RdRef = 2.64
sigma = 1.05
VcmaxRef = 167.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

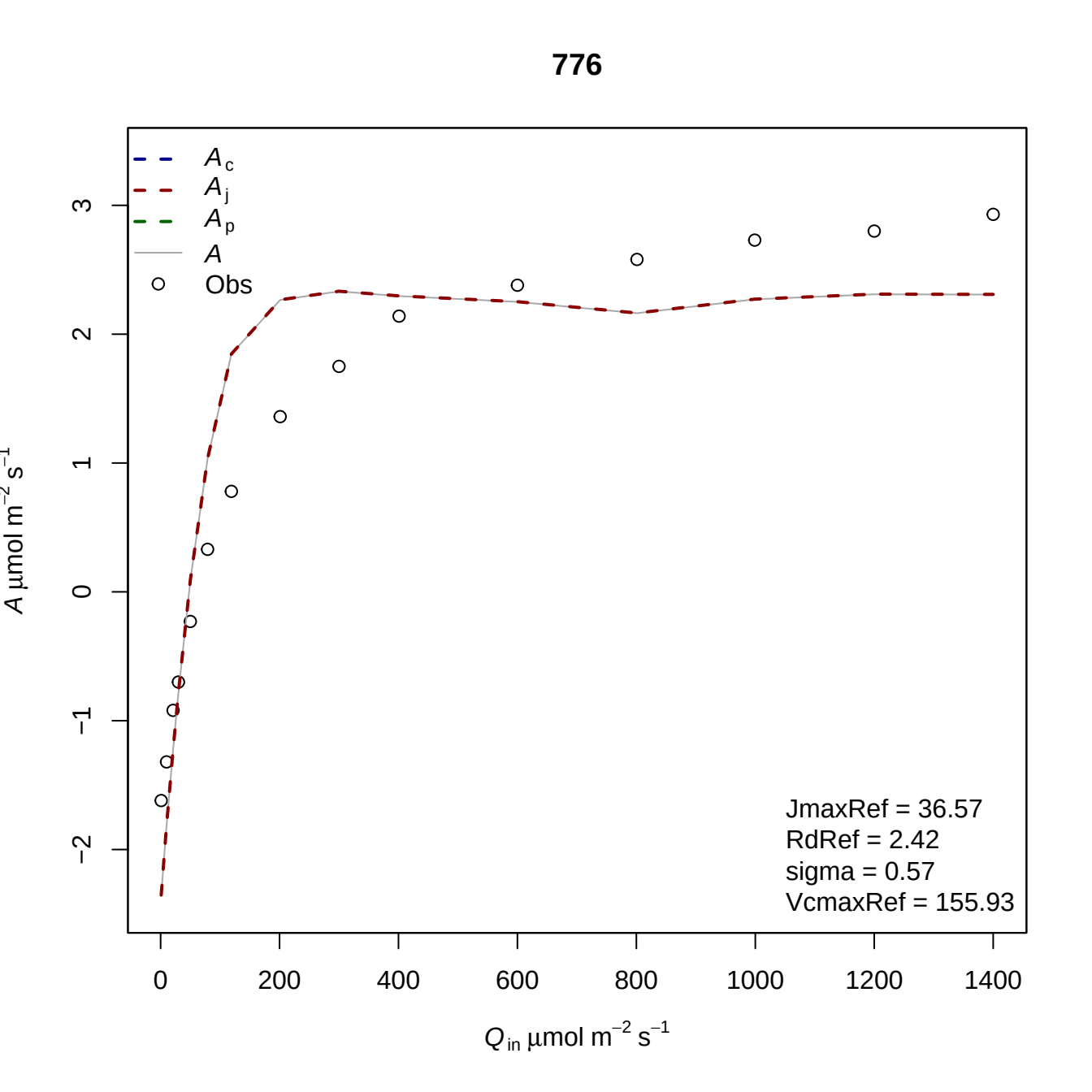

771
A_c
A_j
A_p
A
Obs
JmaxRef = 78.98
RdRef = 2.14
sigma = 0.75
VcmaxRef = 167.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

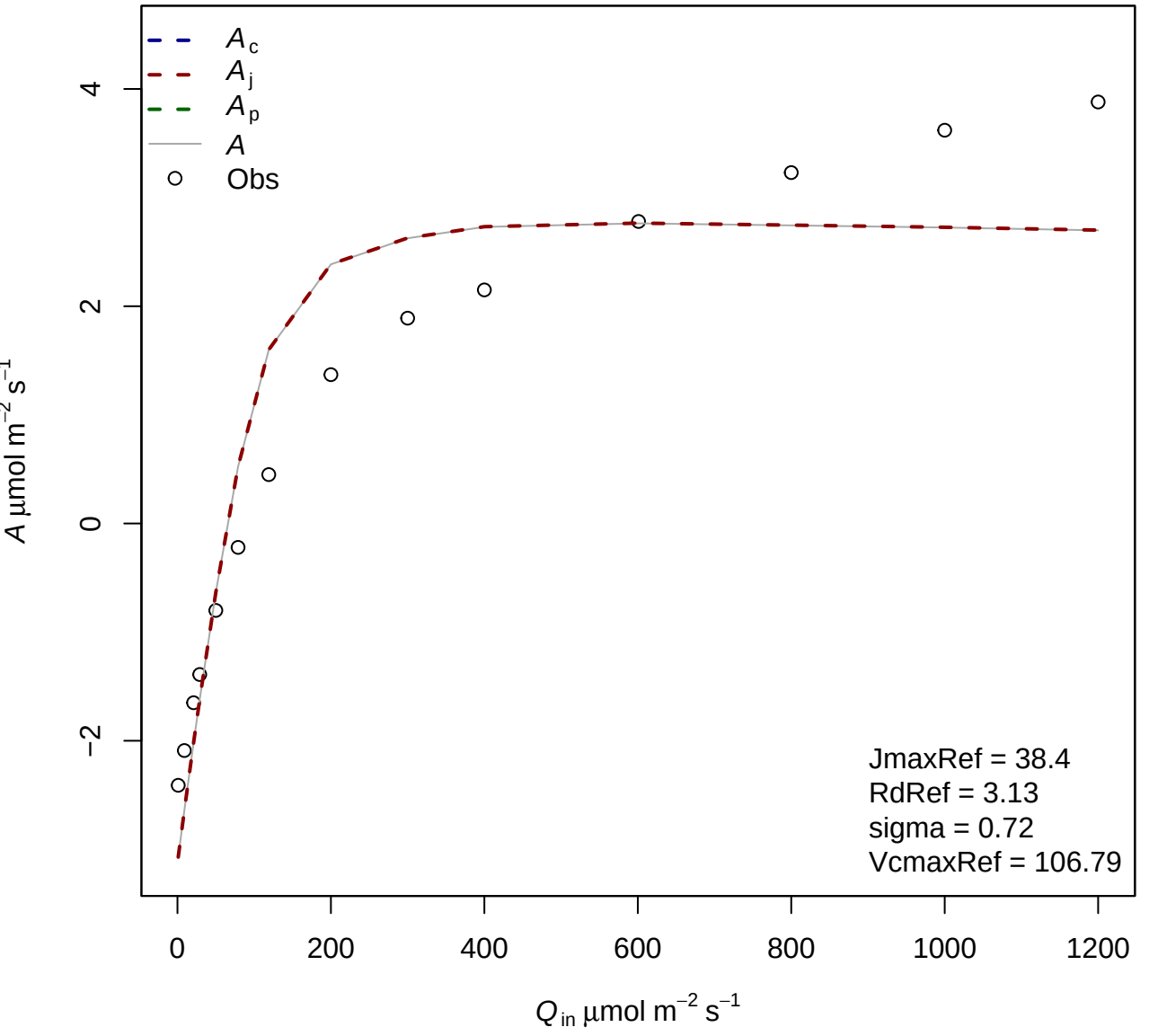

772
A_c
A_j
A_p
A
Obs
JmaxRef = 105.87
RdRef = 1.98
sigma = 0.64
VcmaxRef = 357.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

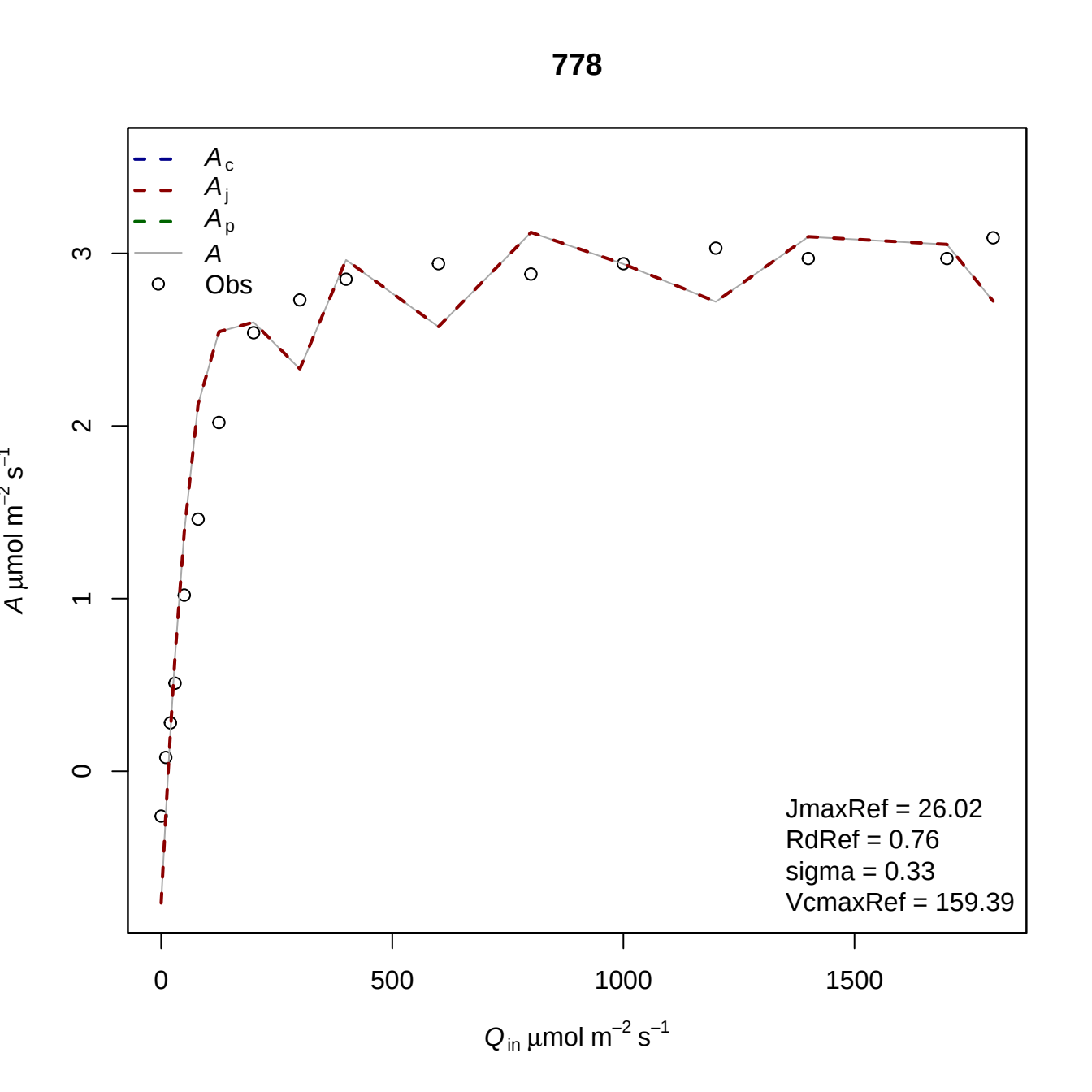

773
A_c
A_j
A_p
A
Obs
JmaxRef = 39.95
RdRef = 1.85
sigma = 0.41
VcmaxRef = 107.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

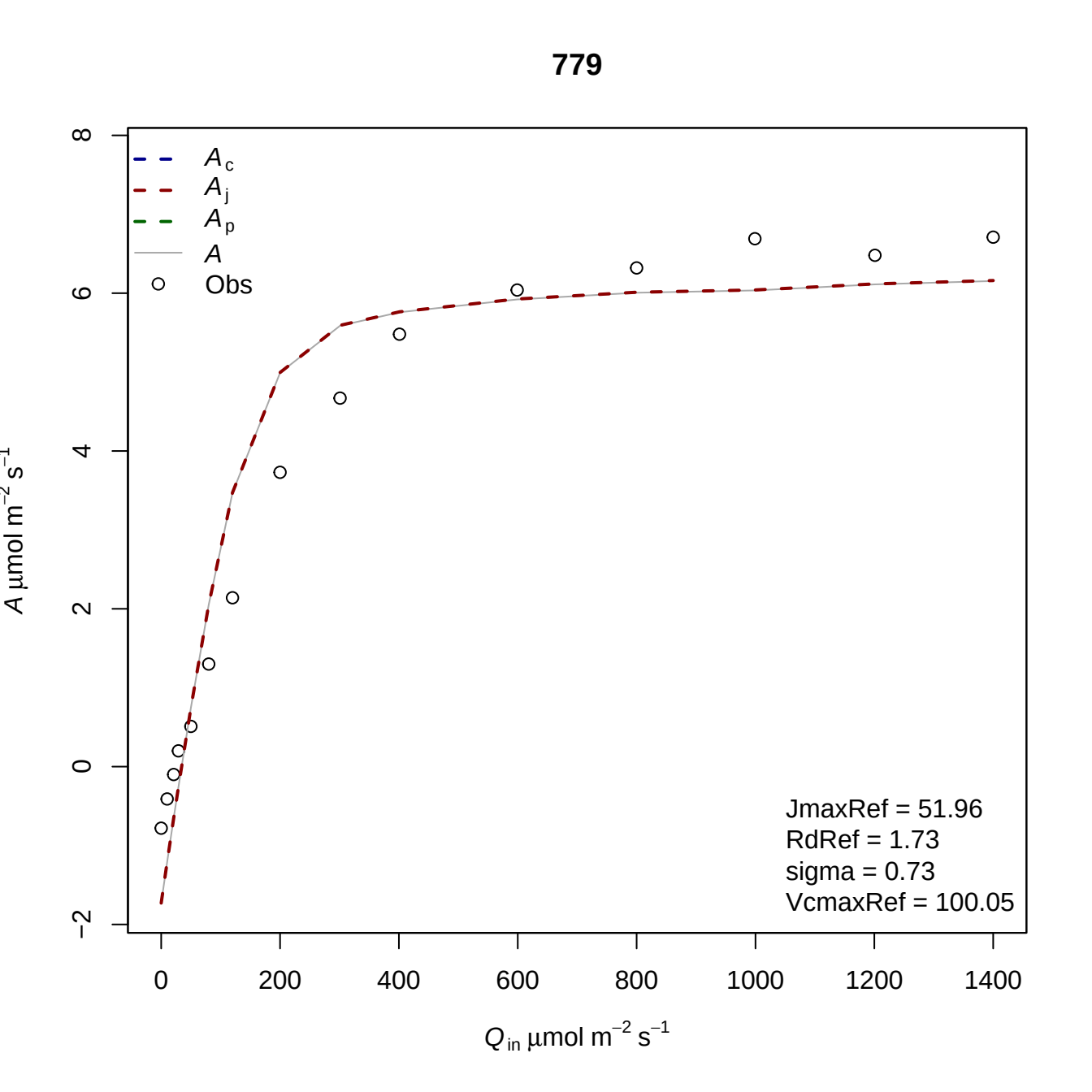

774
A_c
A_j
A_p
A
Obs
JmaxRef = 76.32
RdRef = 4.12
sigma = 1.28
VcmaxRef = 237.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
500
1000
1500

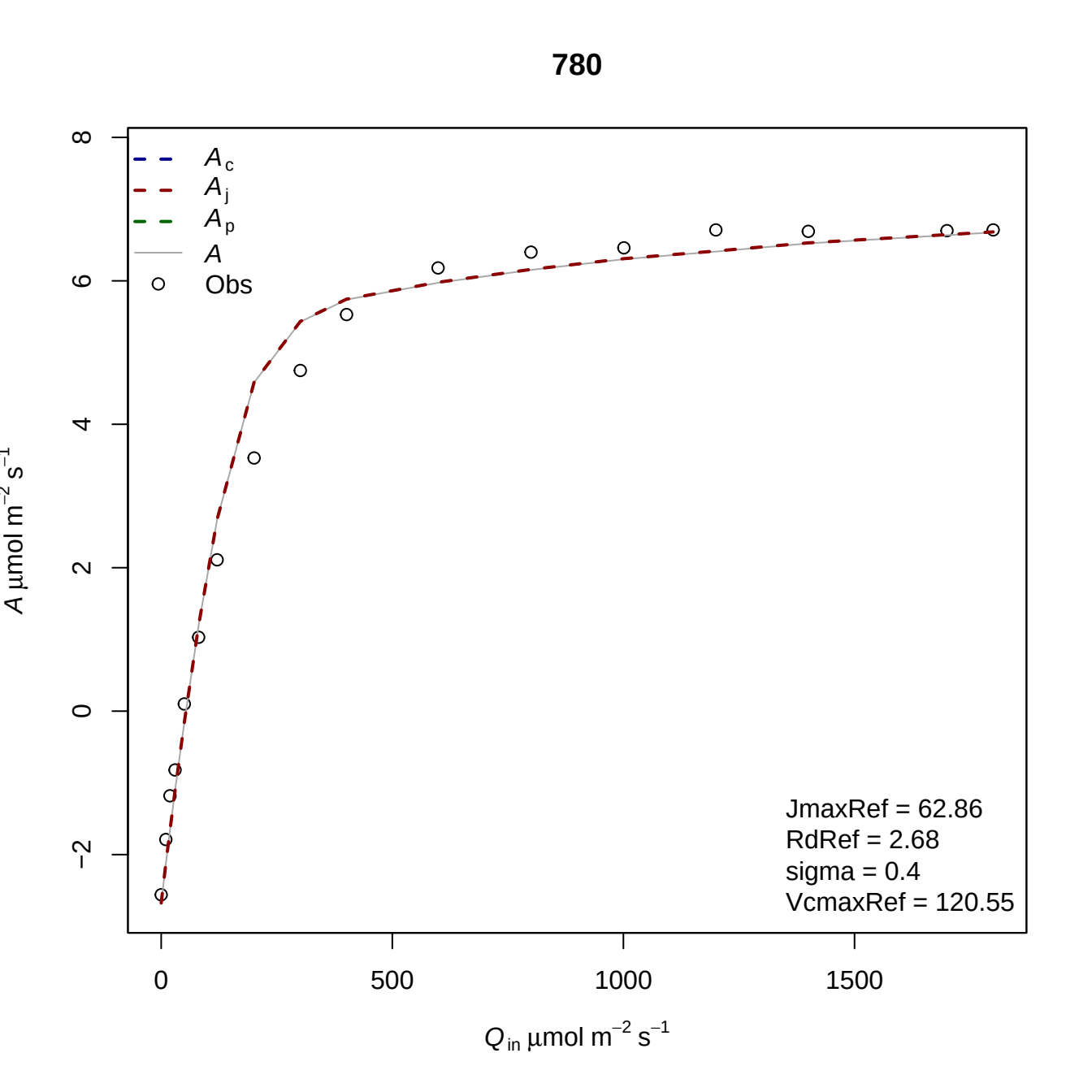

775
A_c
A_j
A_p
A
Obs
JmaxRef = 67.28
RdRef = 1.83
sigma = 0.85
VcmaxRef = 188.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

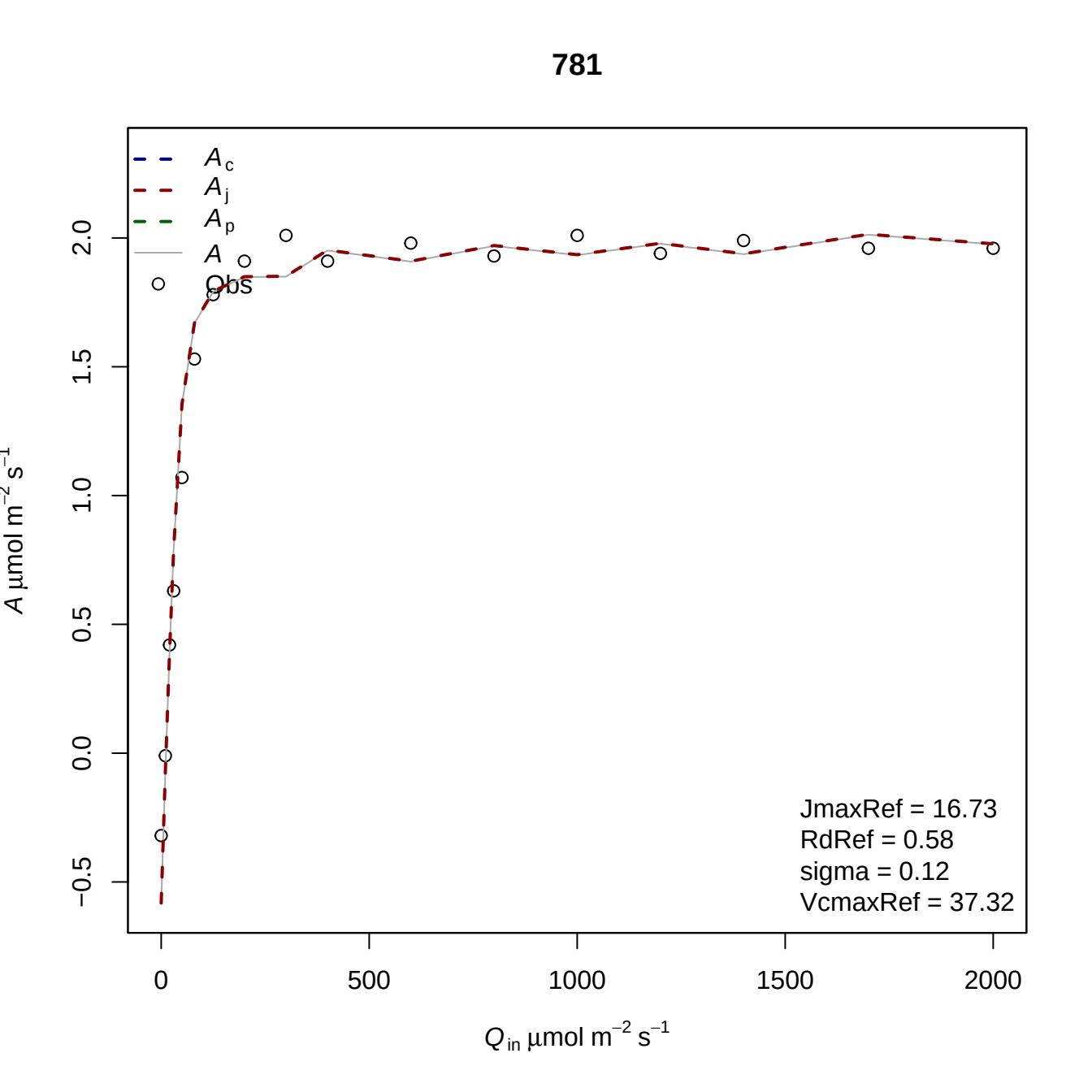

776
A_c
A_j
A_p
A
Obs
JmaxRef = 36.57
RdRef = 2.42
sigma = 0.57
VcmaxRef = 155.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

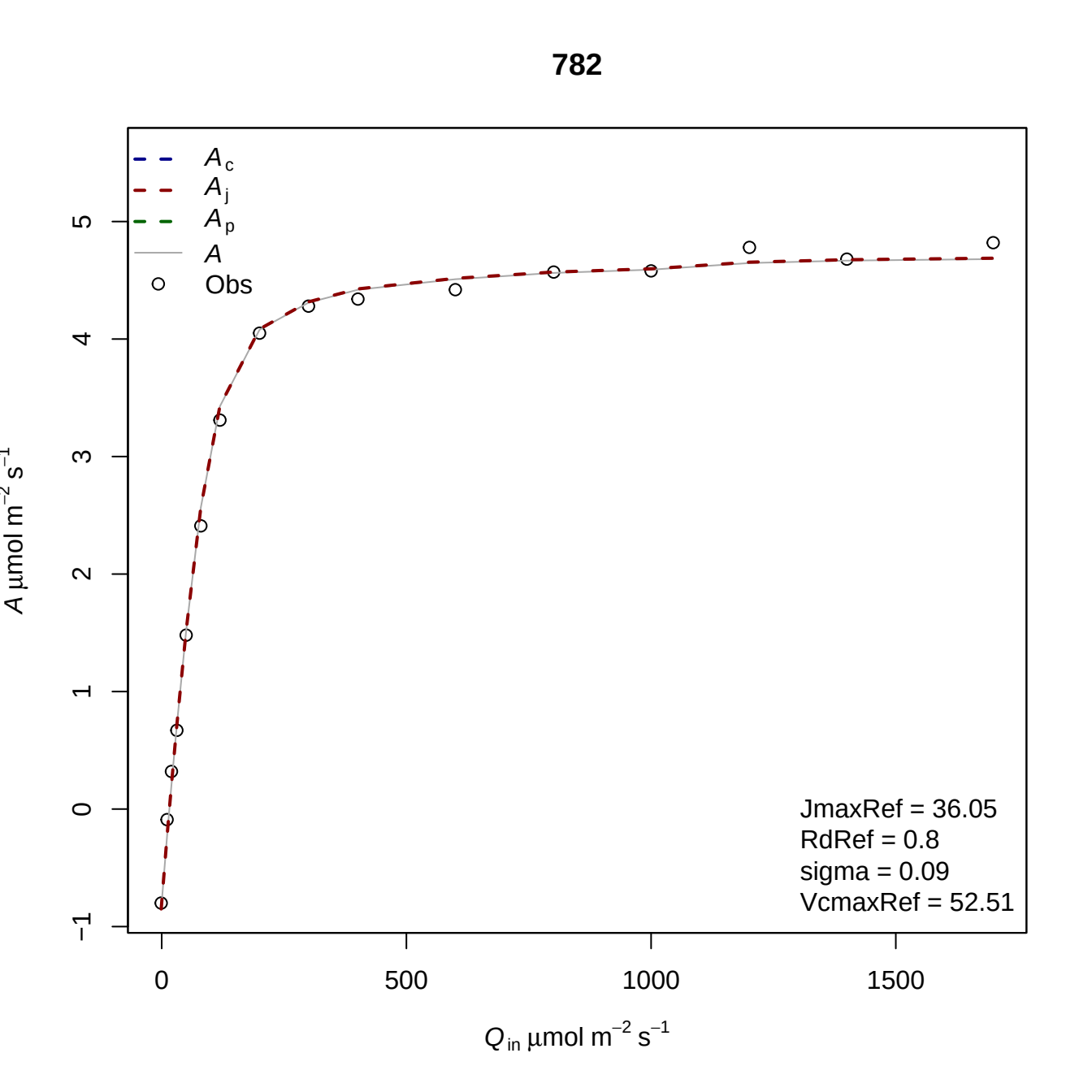

A_c
A_j
A_p
A
Obs
JmaxRef = 38.4
RdRef = 3.13
sigma = 0.72
VcmaxRef = 106.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200

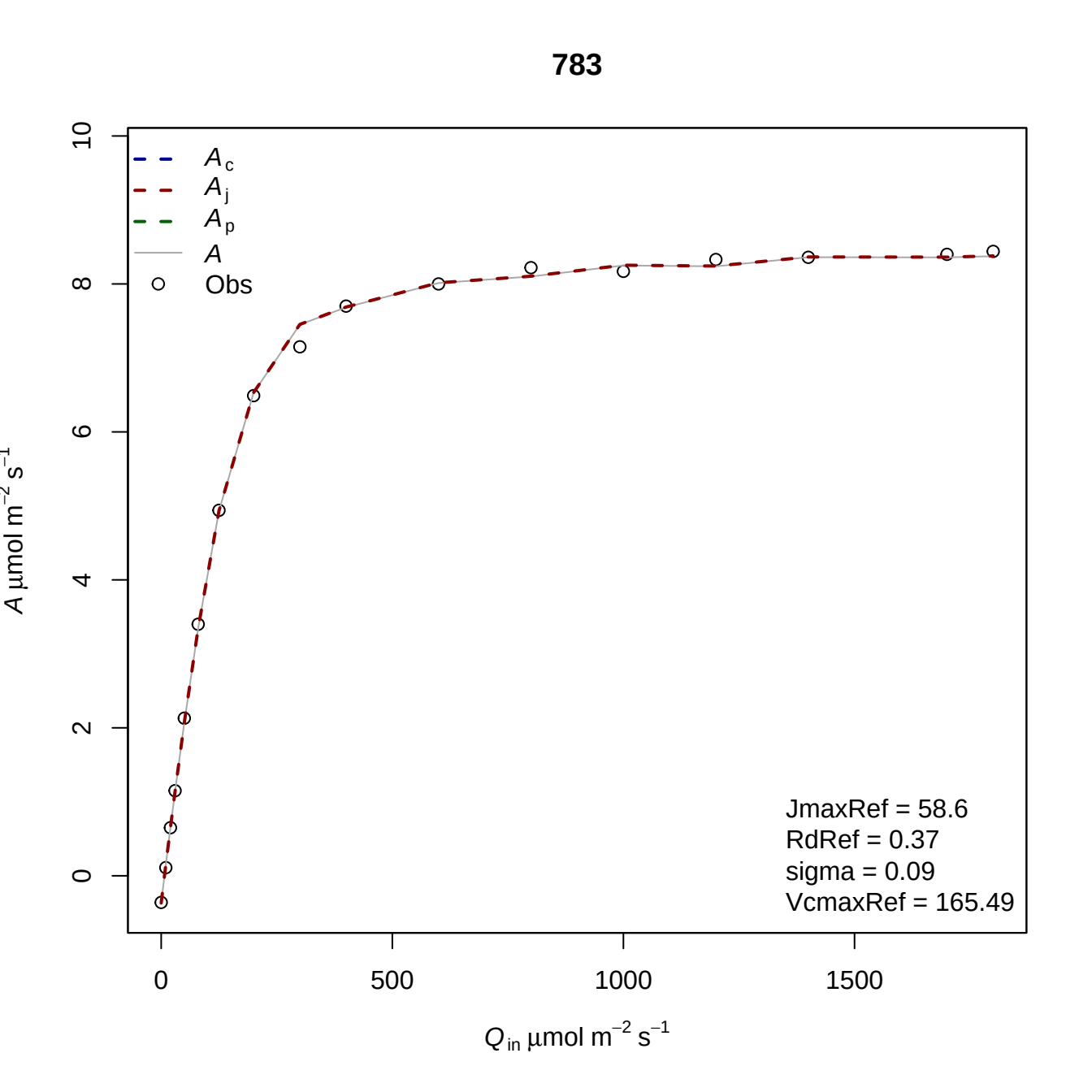

778
A_c
A_j
A_p
A
Obs
JmaxRef = 26.02
RdRef = 0.76
sigma = 0.33
VcmaxRef = 159.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

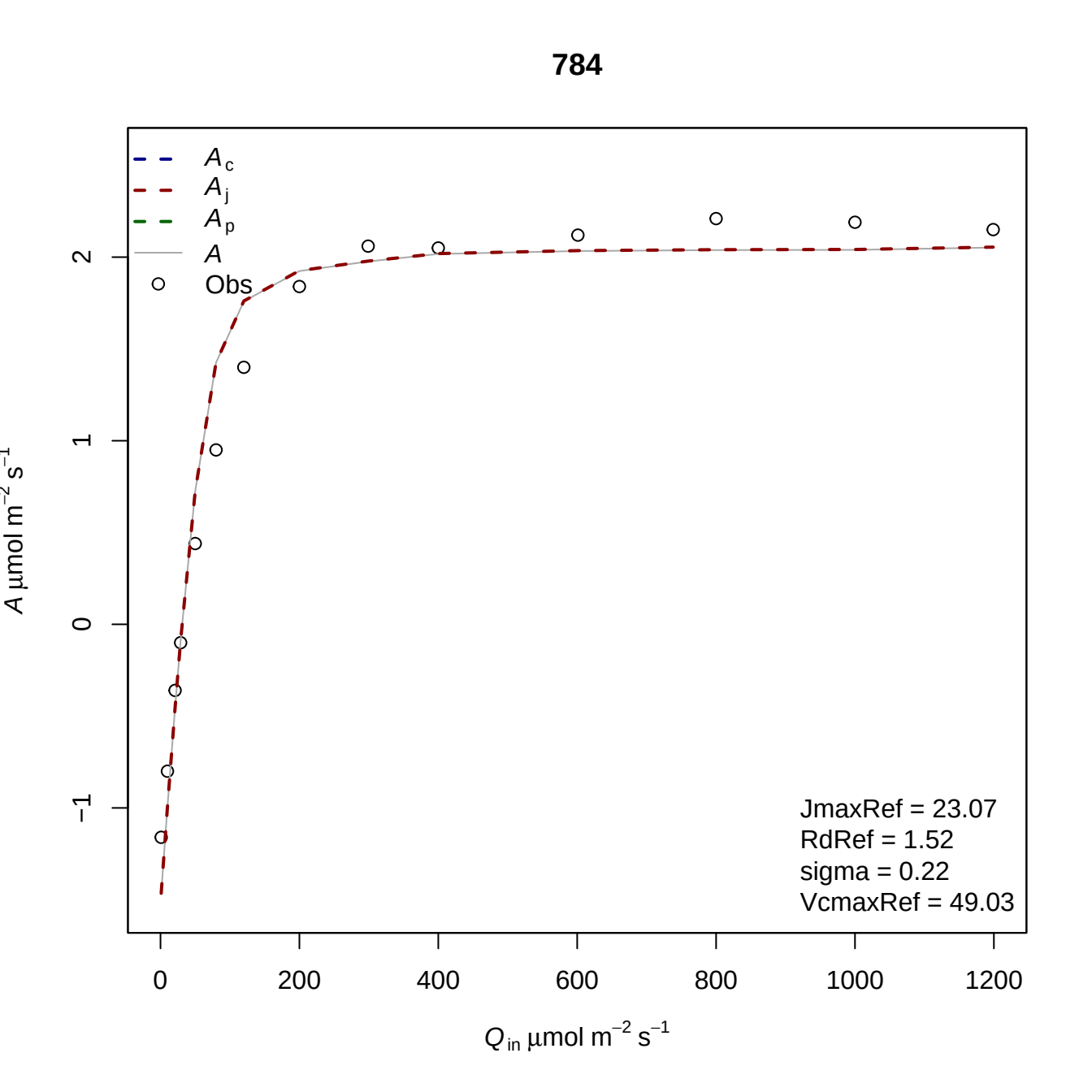

779
A_c
A_j
A_p
A
Obs
JmaxRef = 51.96
RdRef = 1.73
sigma = 0.73
VcmaxRef = 100.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

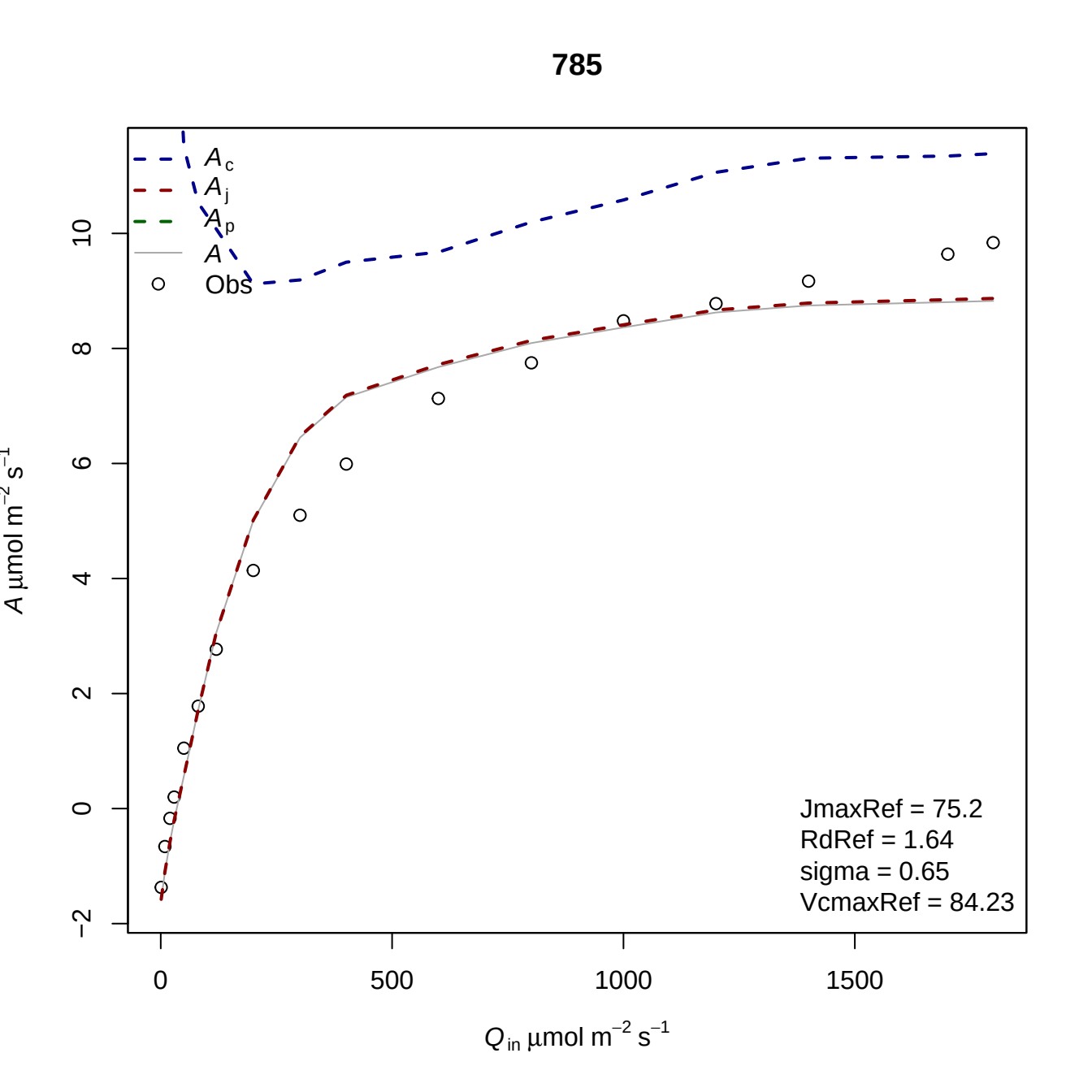

780
A_c
A_j
A_p
A
Obs
JmaxRef = 62.86
RdRef = 2.68
sigma = 0.4
VcmaxRef = 120.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

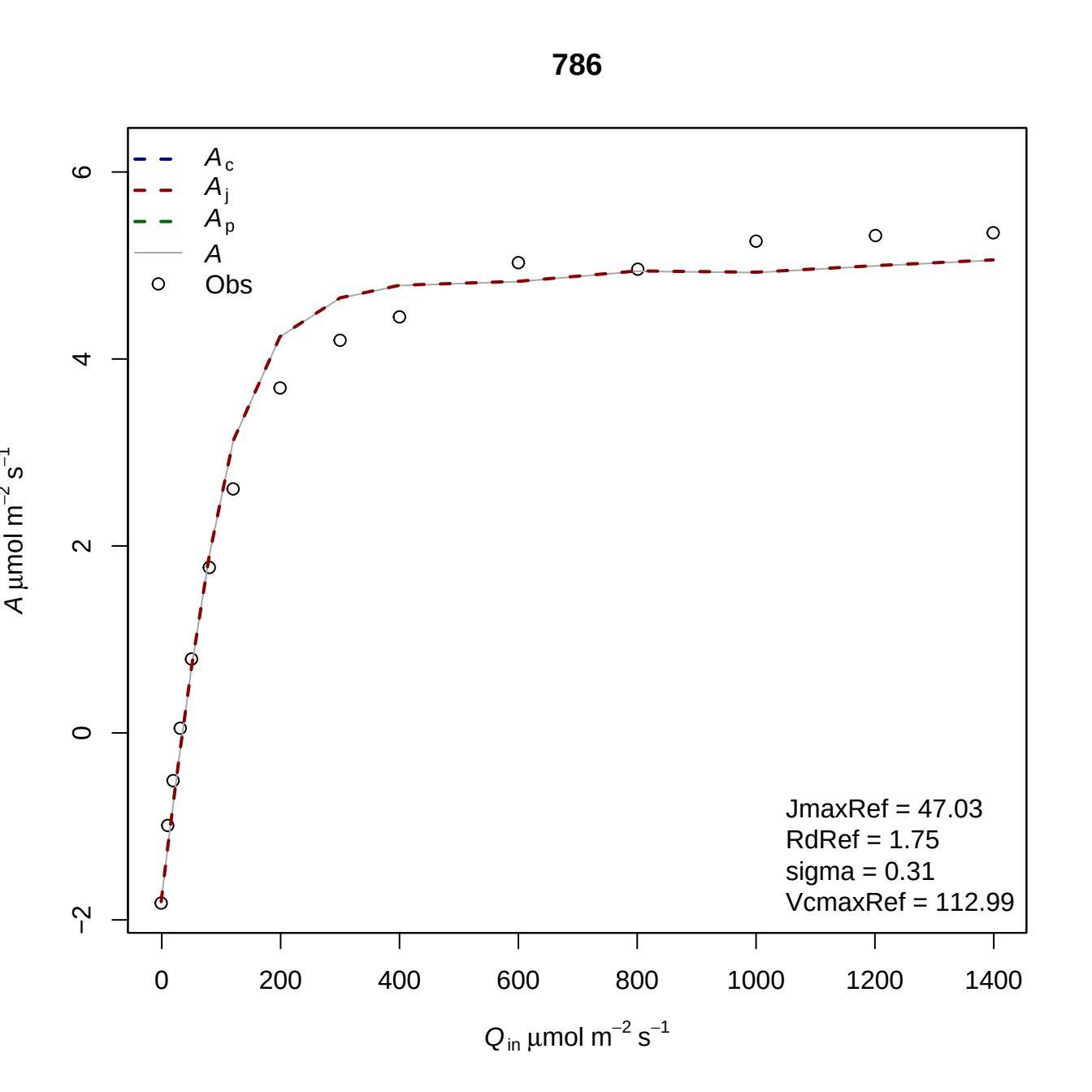

781
A_c
A_j
A_p
A
Obs
JmaxRef = 16.73
RdRef = 0.58
sigma = 0.12
VcmaxRef = 37.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000

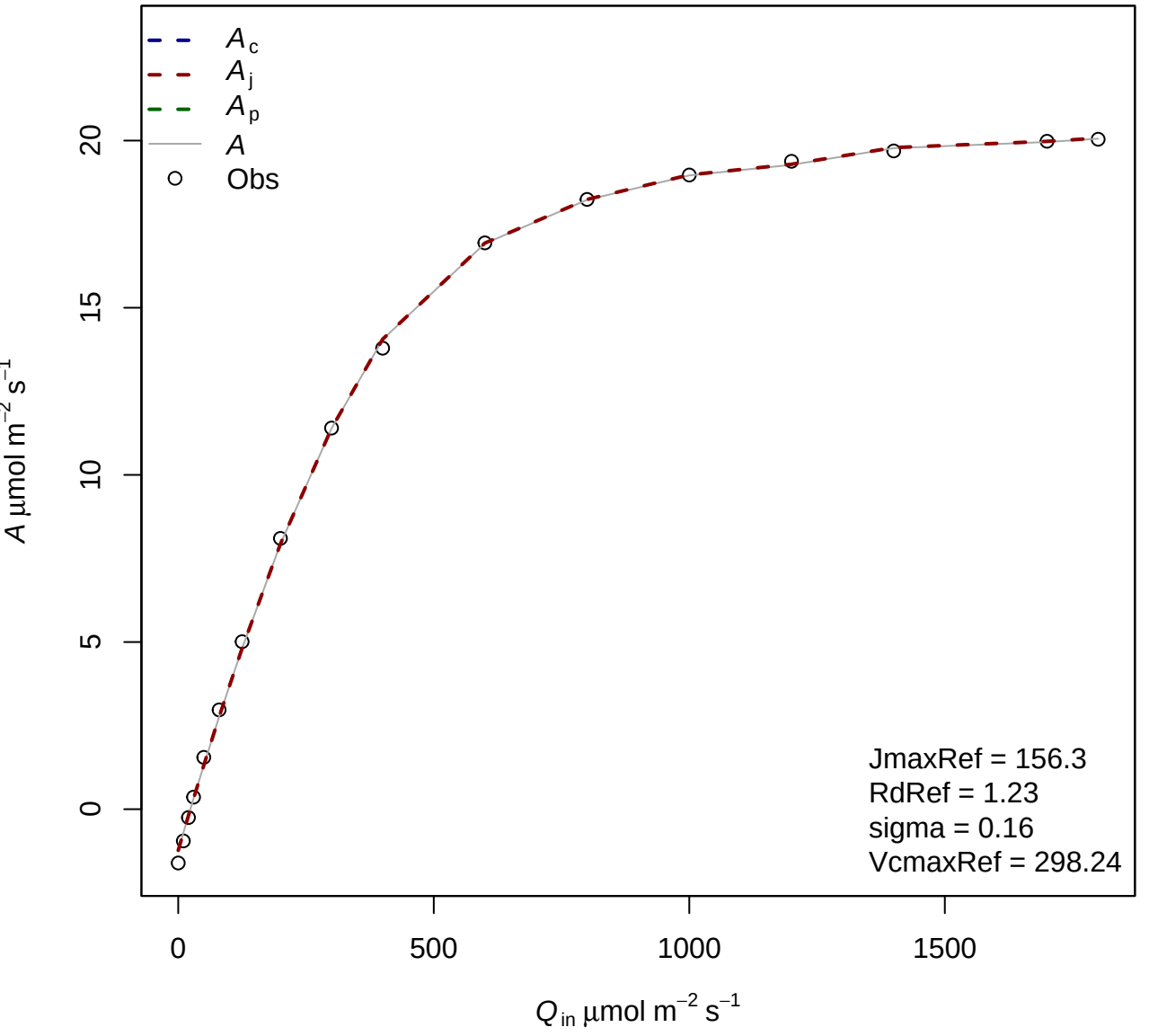

782
A_c
A_j
A_p
A
Obs
JmaxRef = 36.05
RdRef = 0.8
sigma = 0.09
VcmaxRef = 52.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

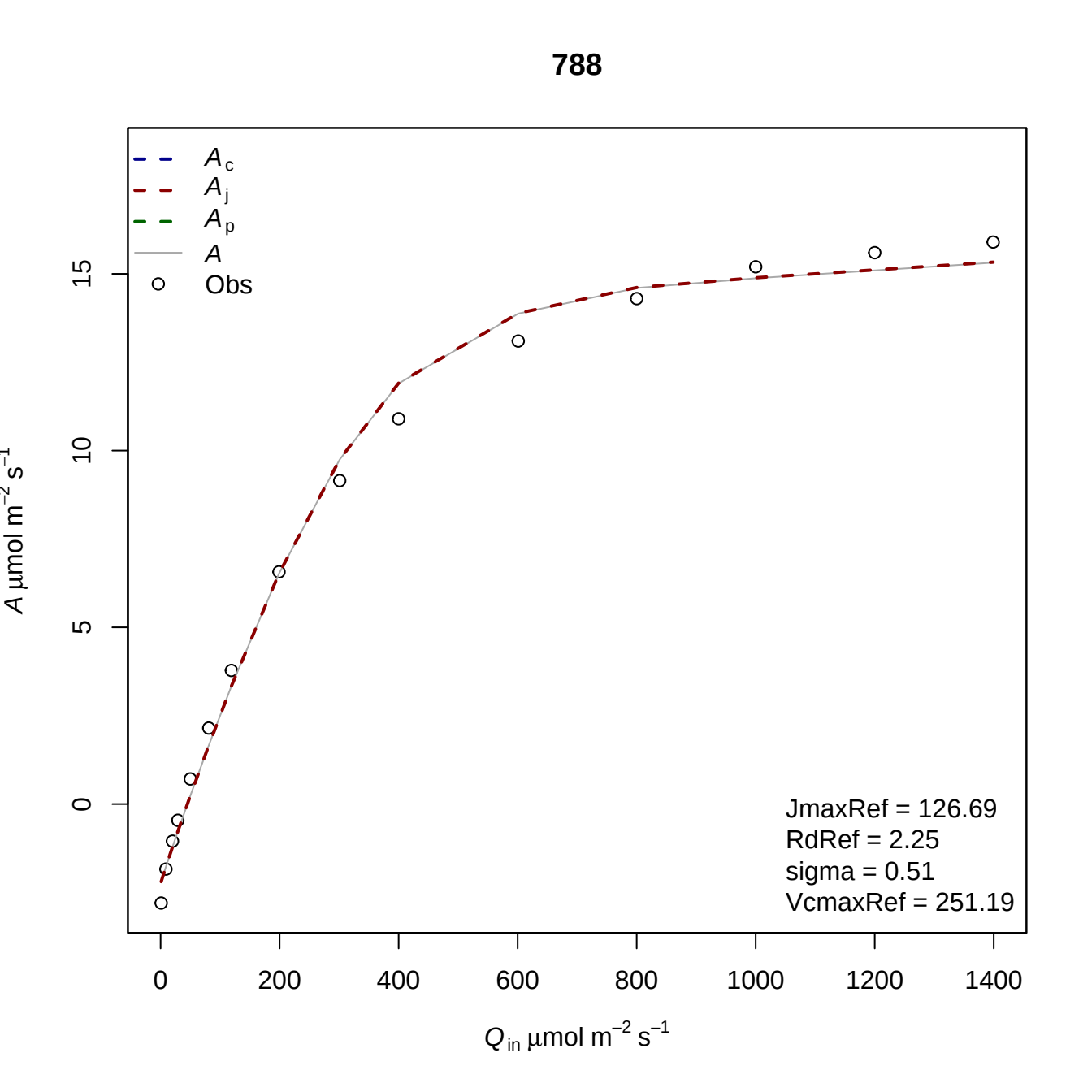

783
A_c
A_j
A_p
A
Obs
JmaxRef = 58.6
RdRef = 0.37
sigma = 0.09
VcmaxRef = 165.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

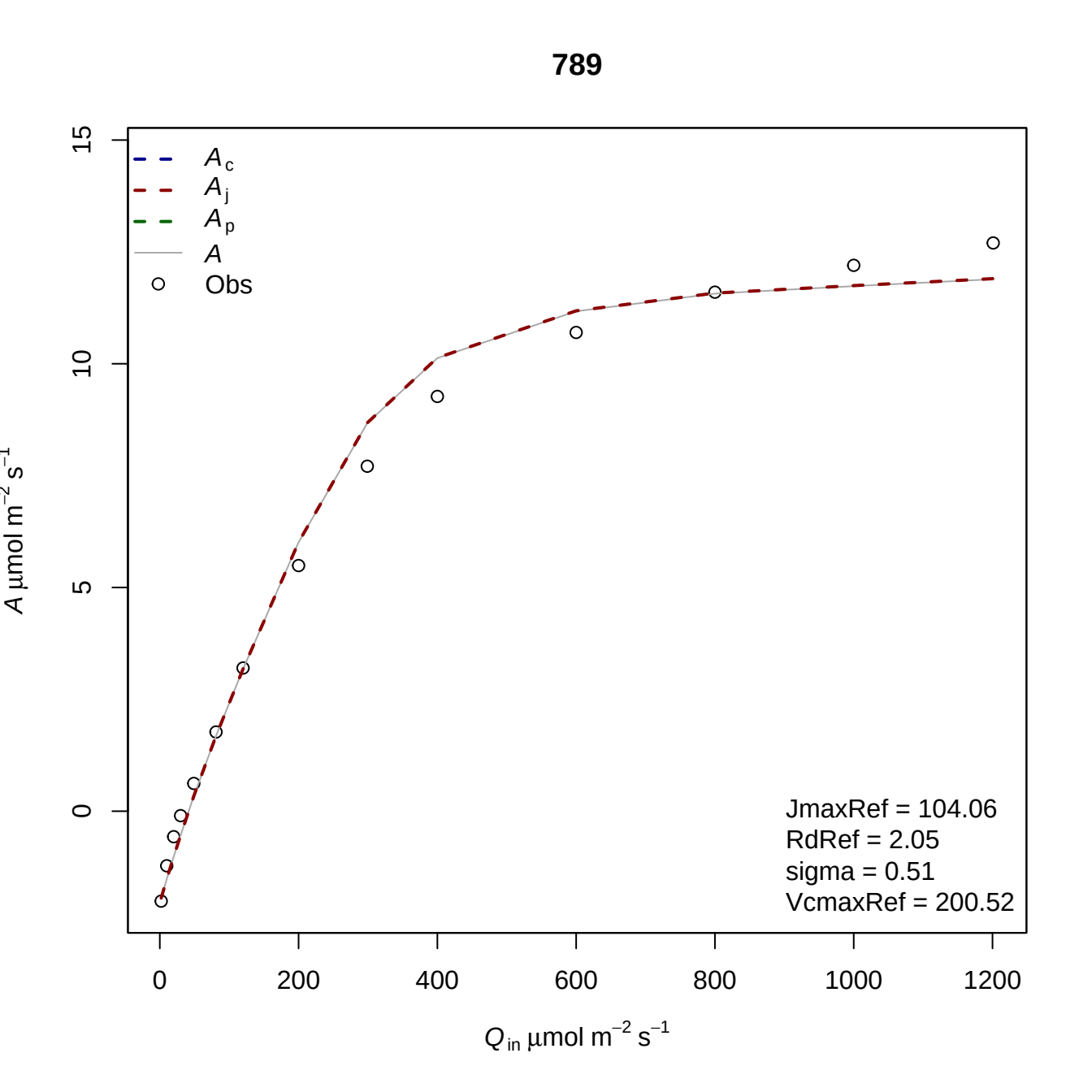

784
A_c
A_j
A_p
A
Obs
JmaxRef = 23.07
RdRef = 1.52
sigma = 0.22
VcmaxRef = 49.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000
1200

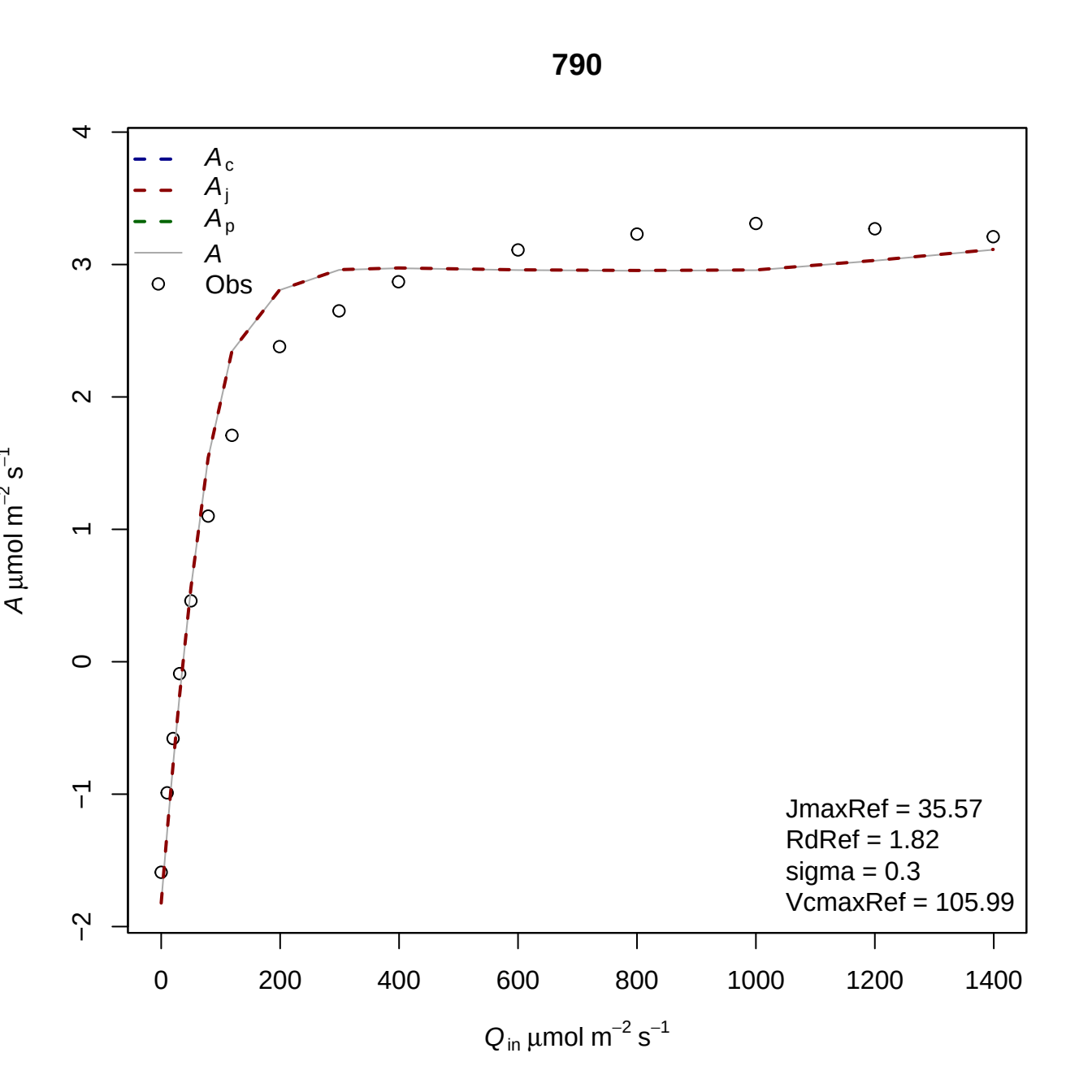

785
A_c
A_j
A_p
A
Obs
JmaxRef = 75.2
RdRef = 1.64
sigma = 0.65
VcmaxRef = 84.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

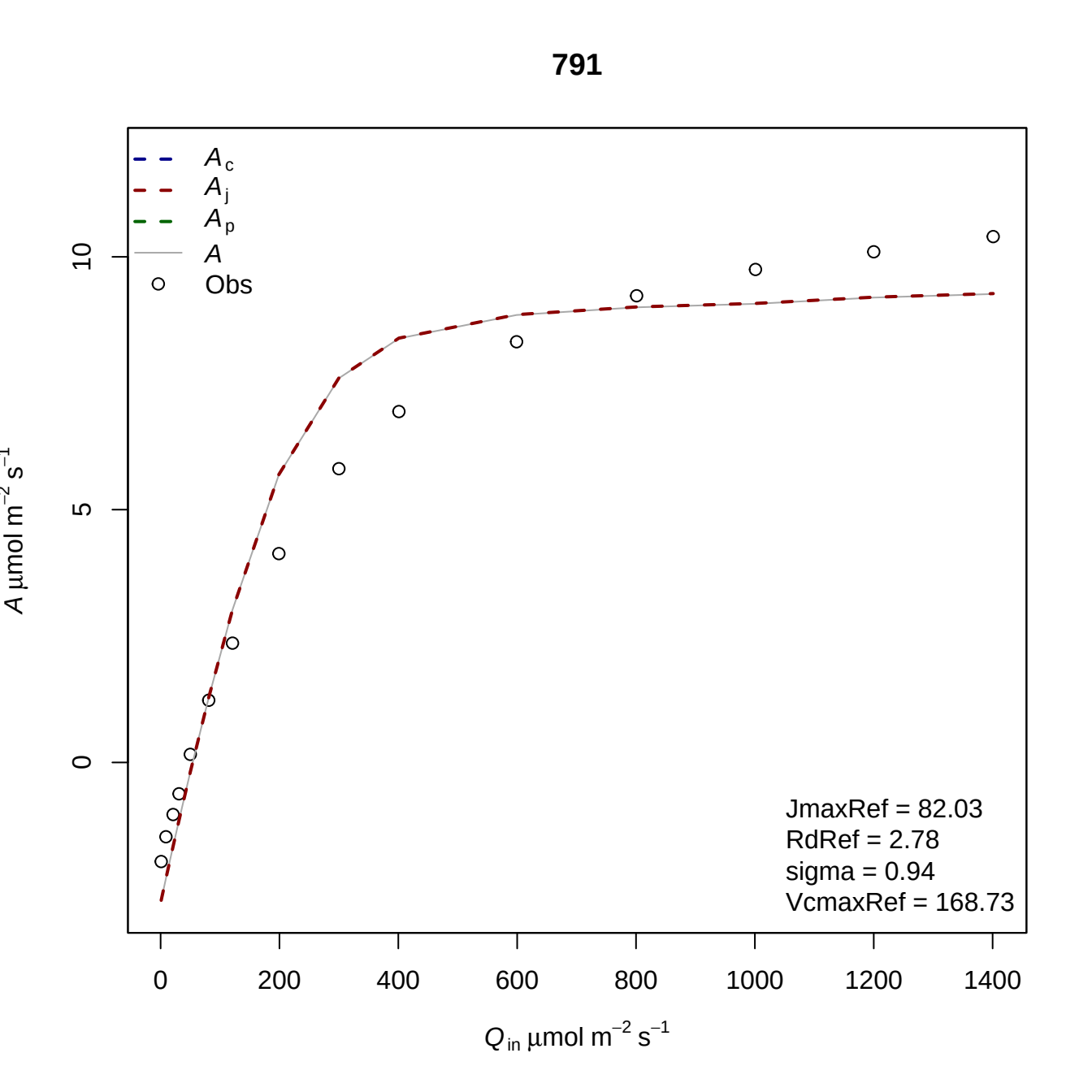

786
A_c
A_j
A_p
A
Obs
JmaxRef = 47.03
RdRef = 1.75
sigma = 0.31
VcmaxRef = 112.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

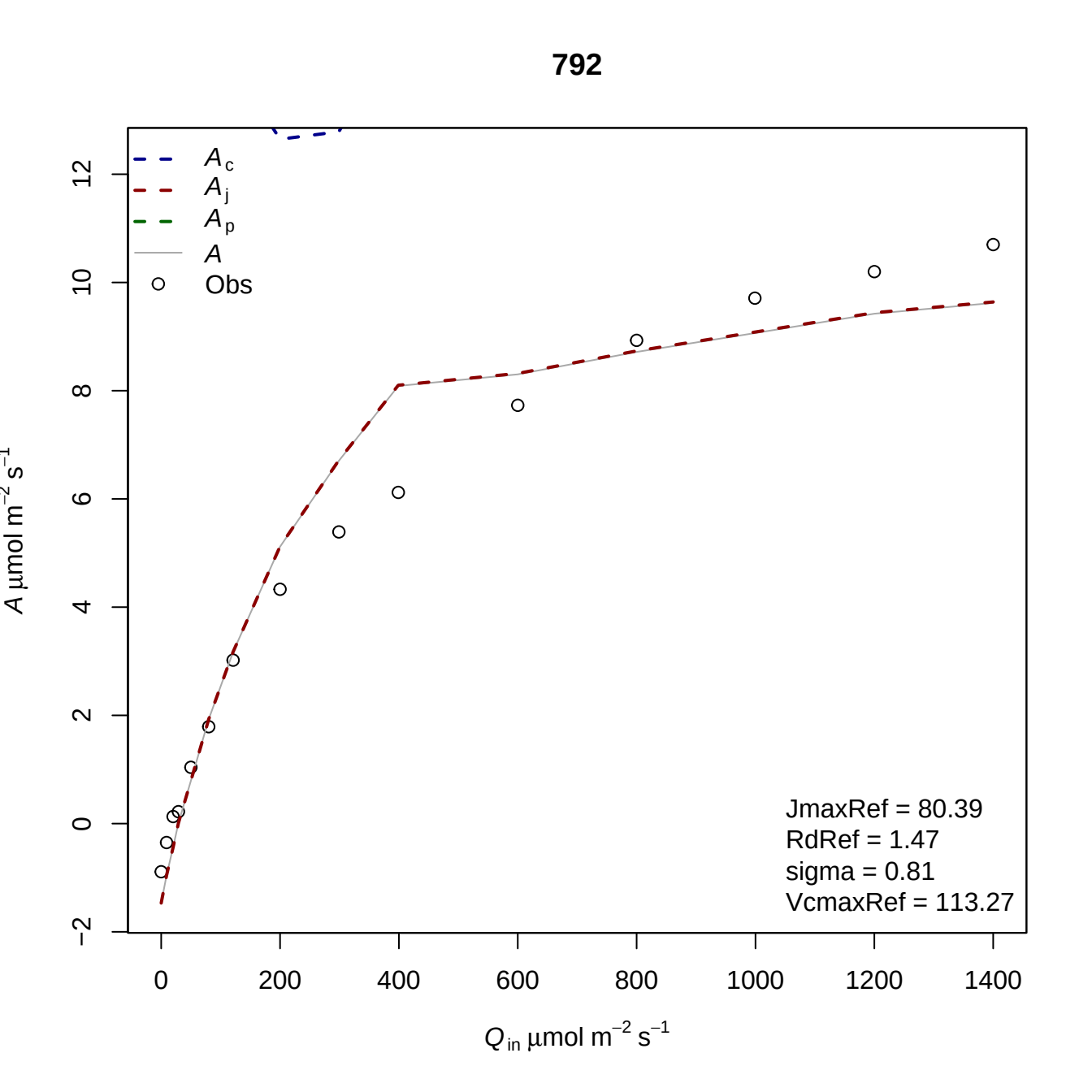

A_c
A_j
A_p
A
Obs
JmaxRef = 156.3
RdRef = 1.23
sigma = 0.16
VcmaxRef = 298.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

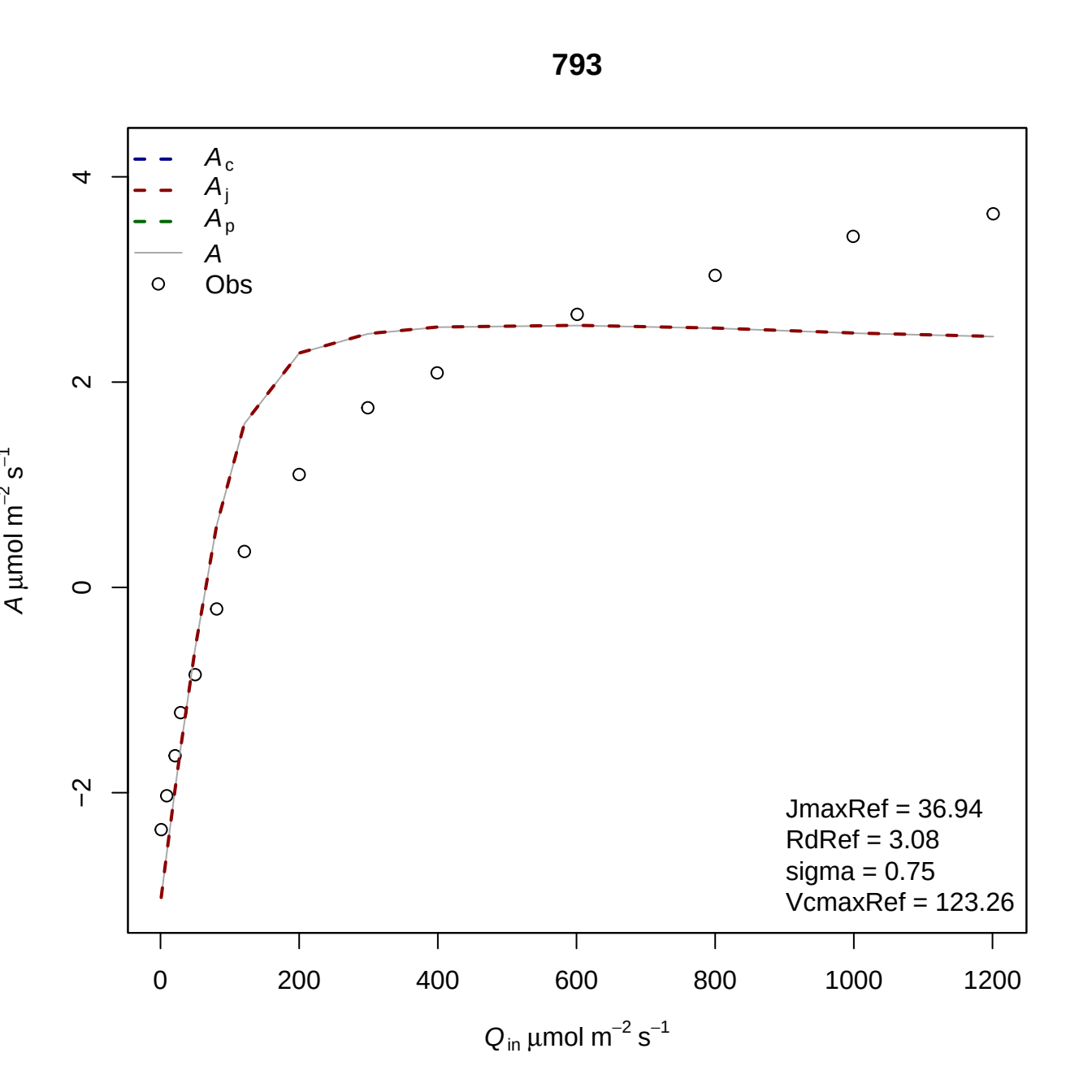

788
A_c
A_j
A_p
A
Obs
JmaxRef = 126.69
RdRef = 2.25
sigma = 0.51
VcmaxRef = 251.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

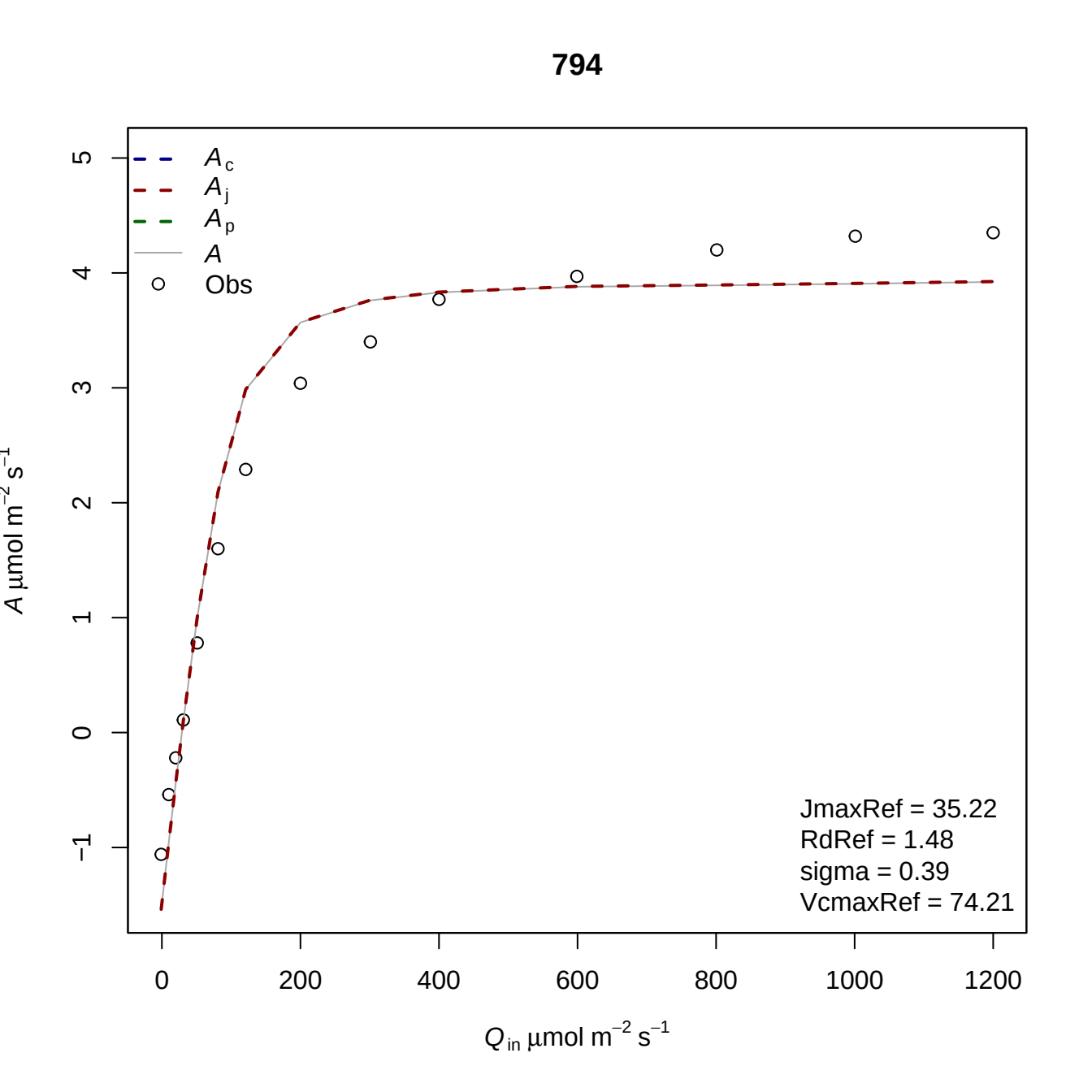

789
A_c
A_j
A_p
A
Obs
JmaxRef = 104.06
RdRef = 2.05
sigma = 0.51
VcmaxRef = 200.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

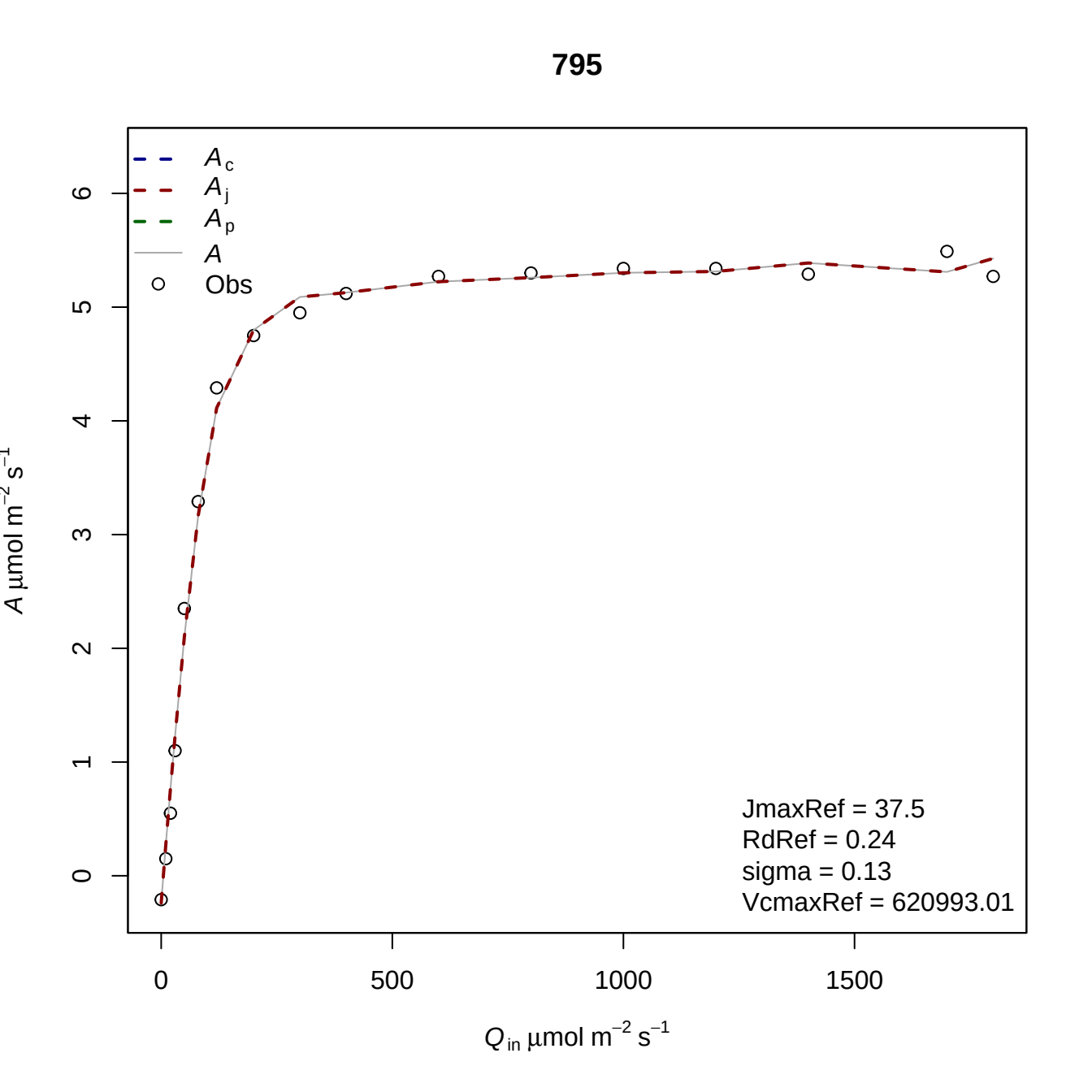

790
A_c
A_j
A_p
A
Obs
JmaxRef = 35.57
RdRef = 1.82
sigma = 0.3
VcmaxRef = 105.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

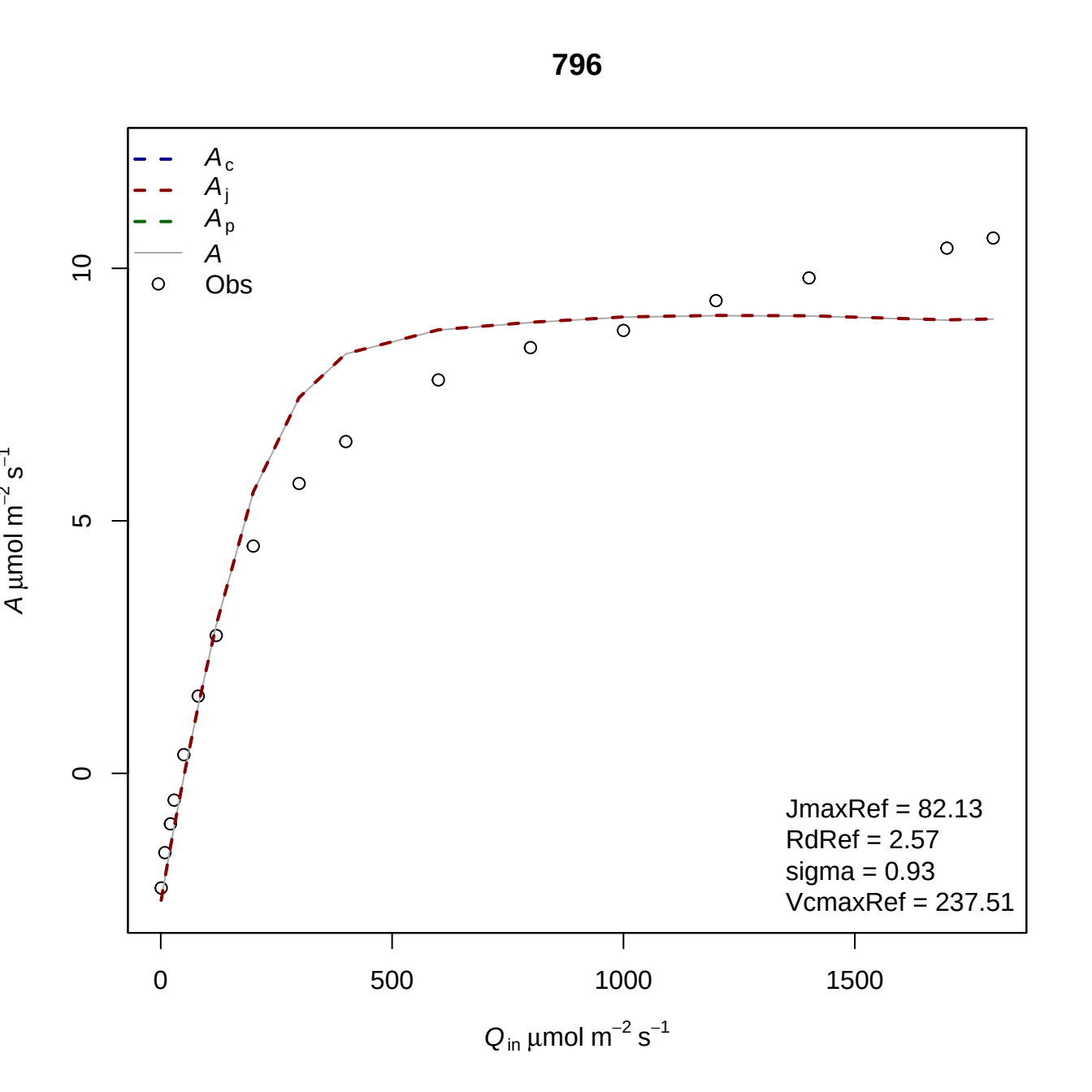

791
A_c
A_j
A_p
A
Obs
JmaxRef = 82.03
RdRef = 2.78
sigma = 0.94
VcmaxRef = 168.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200
1400

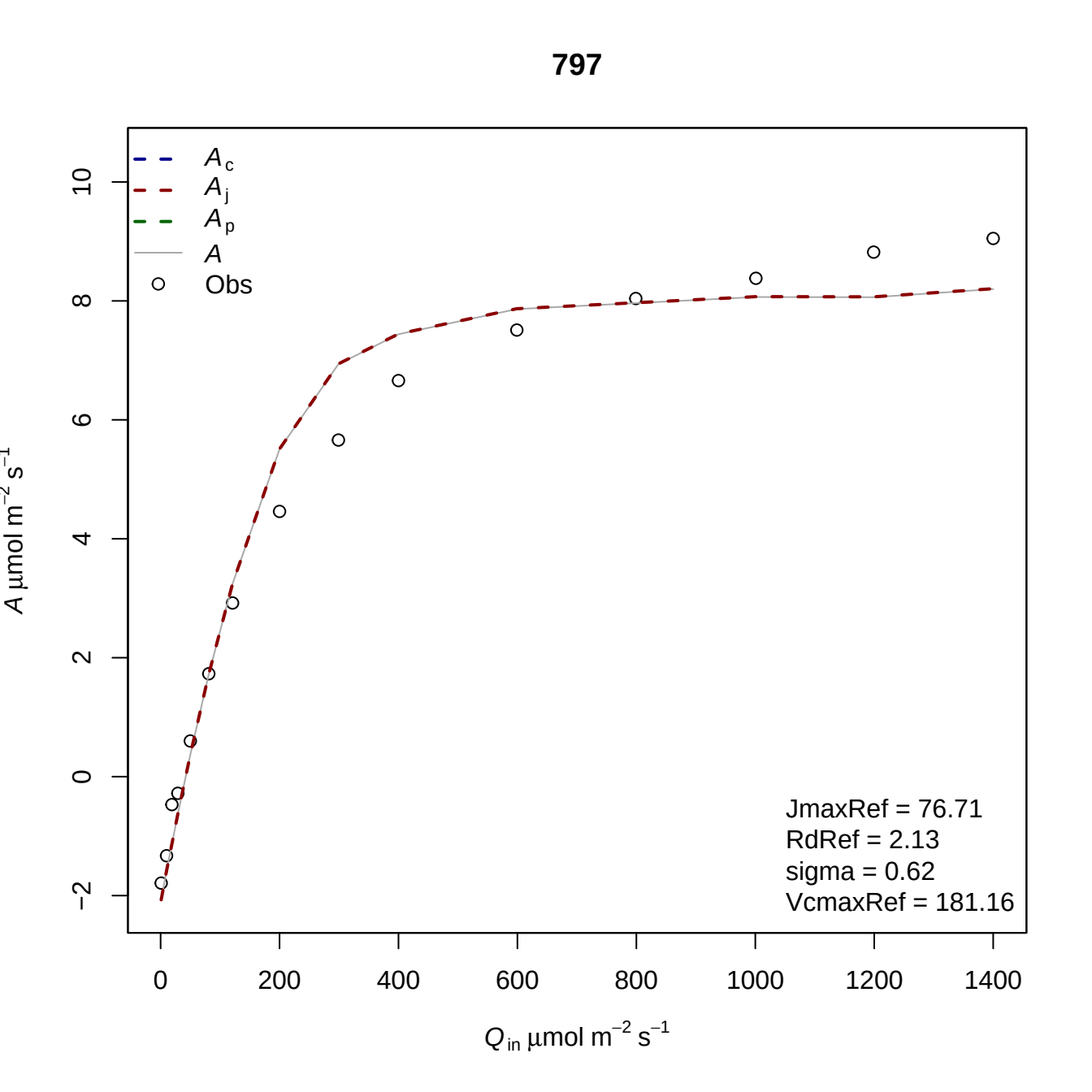

792
A_c
A_j
A_p
A
Obs
JmaxRef = 80.39
RdRef = 1.47
sigma = 0.81
VcmaxRef = 113.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

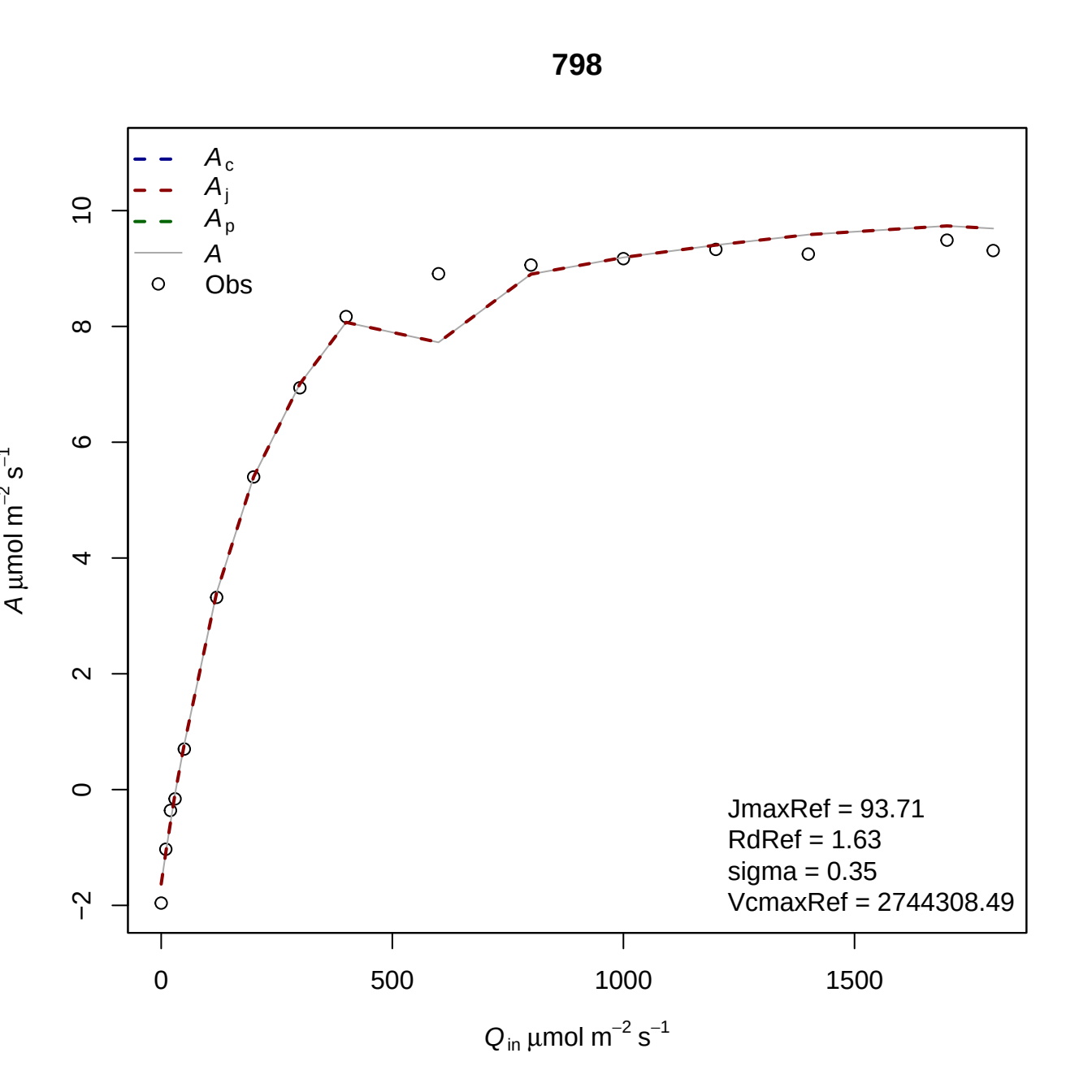

793
A_c
A_j
A_p
A
Obs
JmaxRef = 36.94
RdRef = 3.08
sigma = 0.75
VcmaxRef = 123.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
200
400
600
800
1000
1200

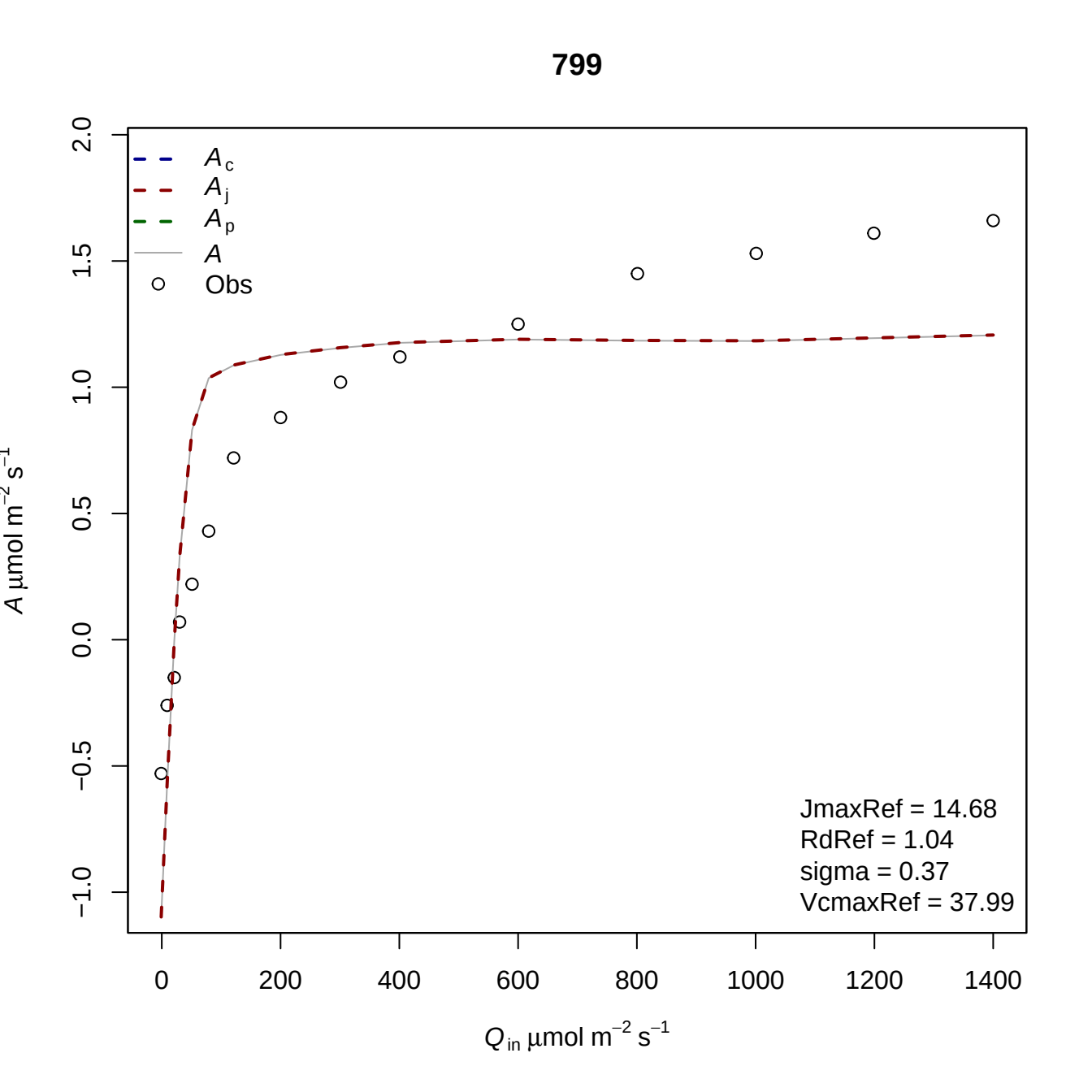

794
A_c
A_j
A_p
A
Obs
JmaxRef = 35.22
RdRef = 1.48
sigma = 0.39
VcmaxRef = 74.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

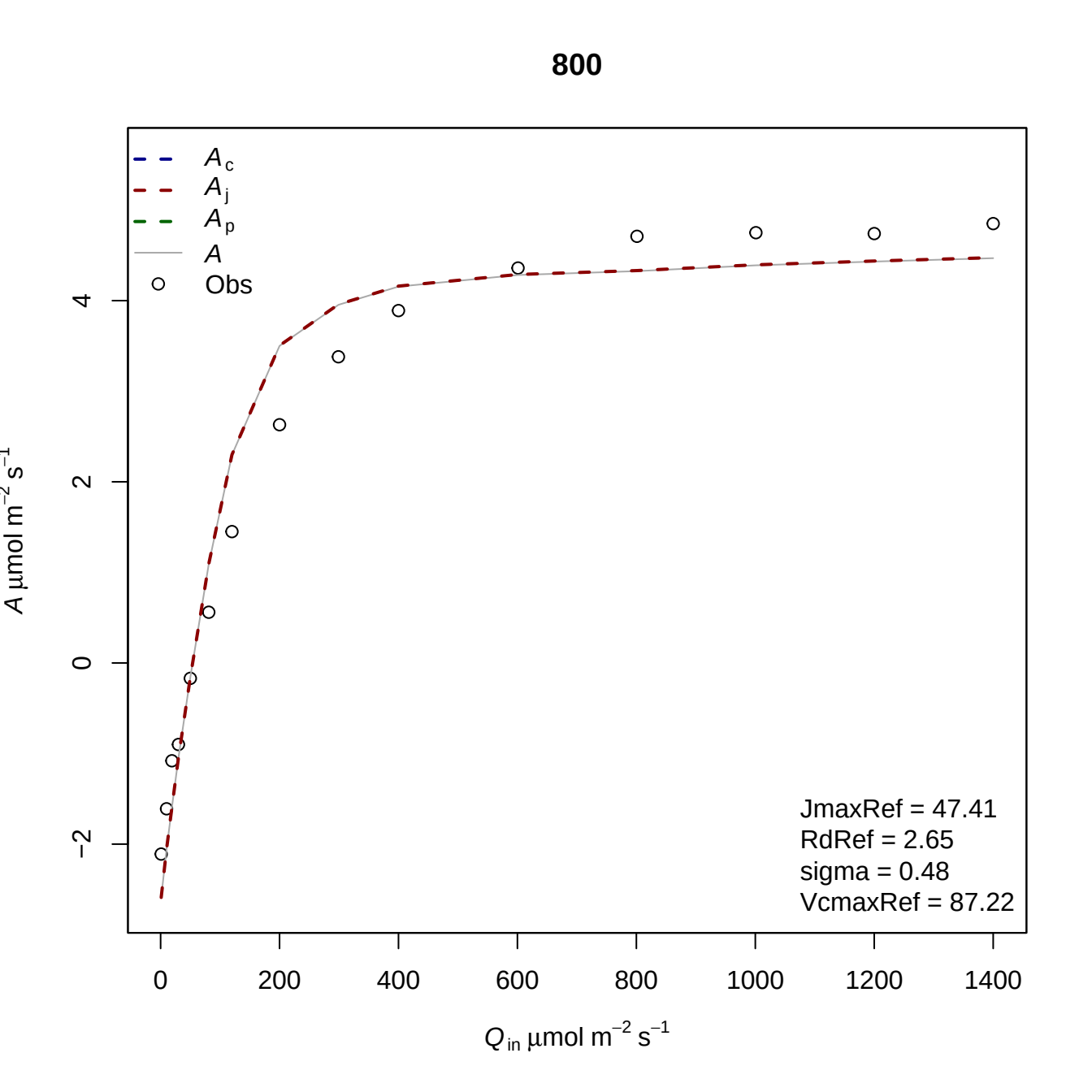

795
A_c
A_j
A_p
A
Obs
JmaxRef = 37.5
RdRef = 0.24
sigma = 0.13
VcmaxRef = 620993.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

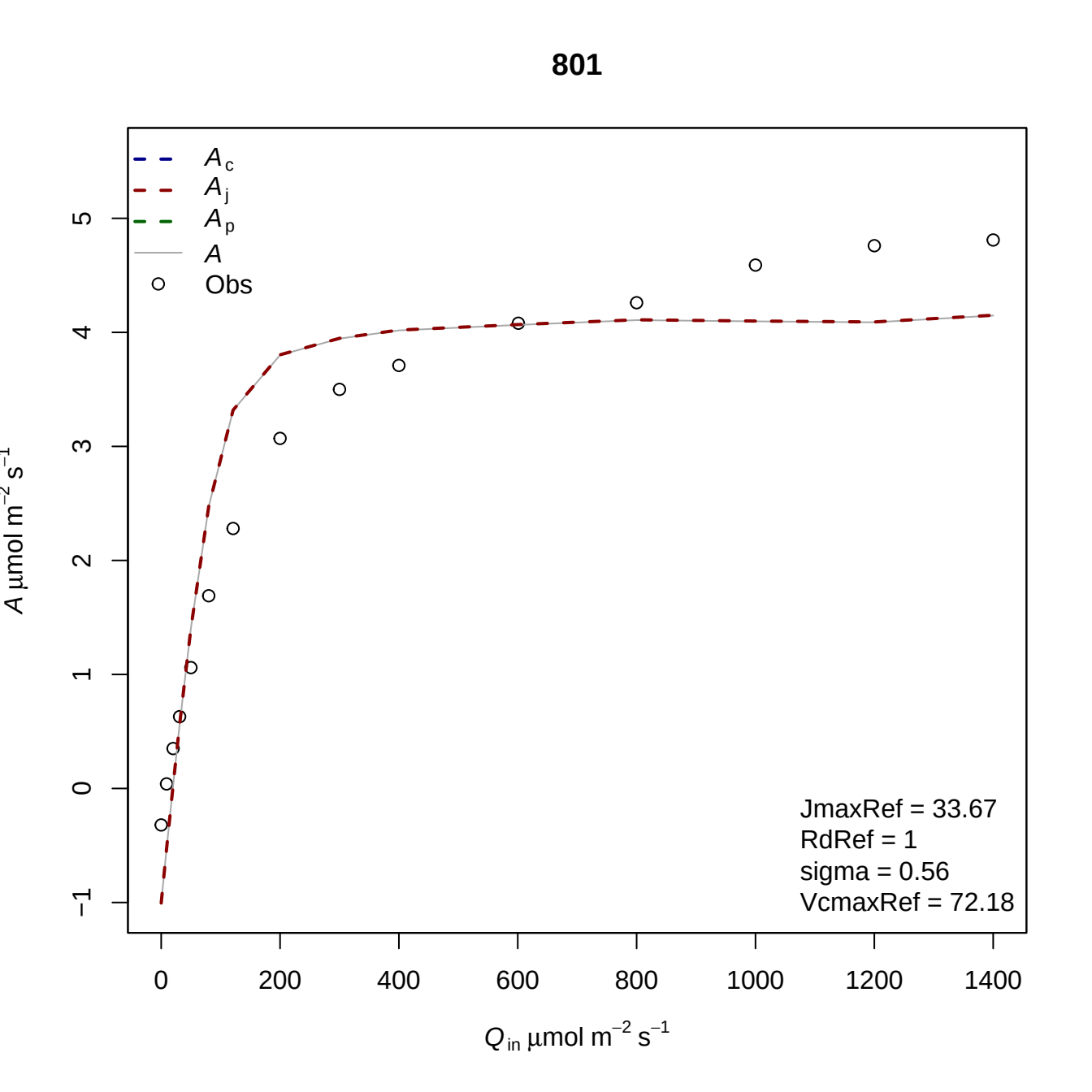

796
A_c
A_j
A_p
A
Obs
JmaxRef = 82.13
RdRef = 2.57
sigma = 0.93
VcmaxRef = 237.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

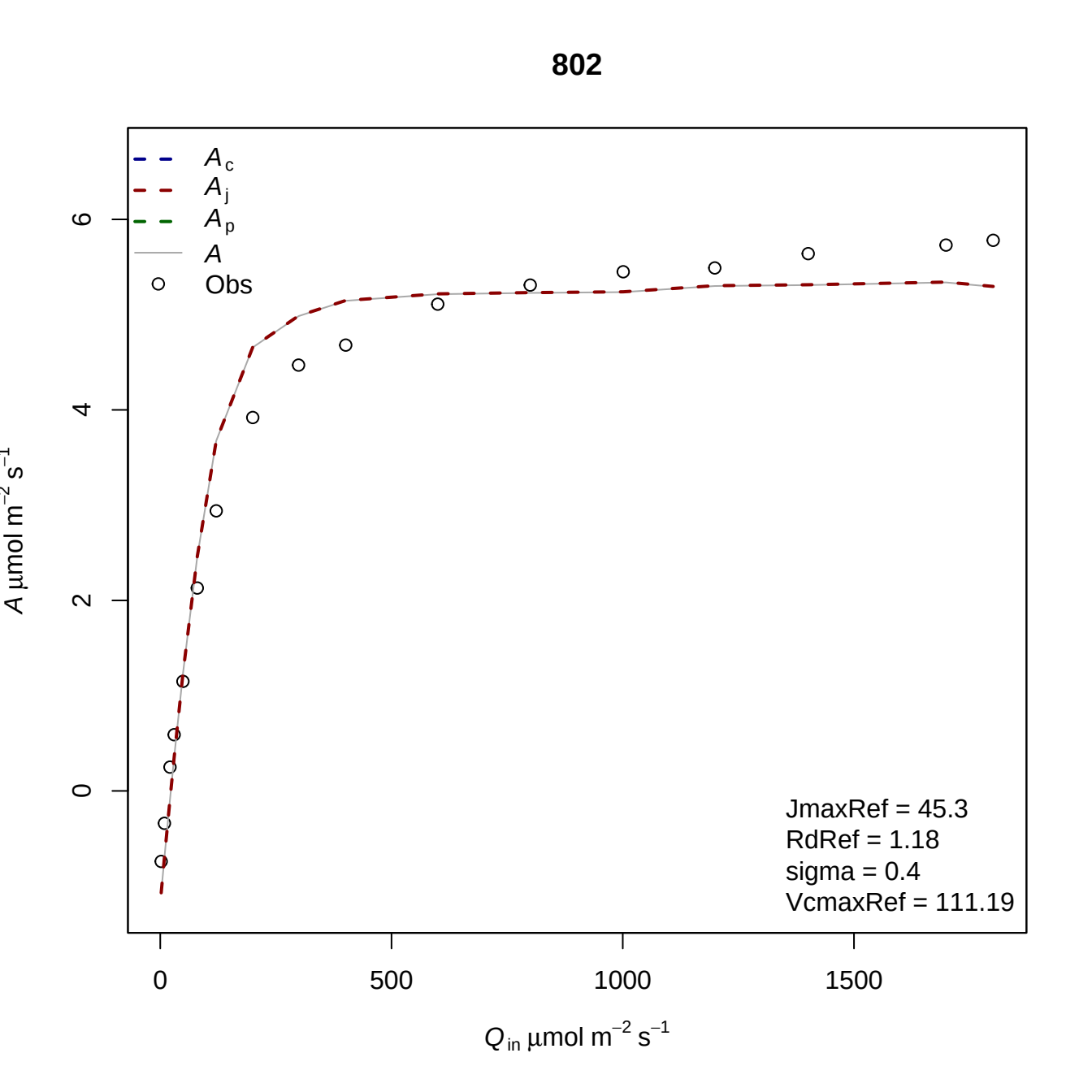

797
A_c
A_j
A_p
A
Obs
JmaxRef = 76.71
RdRef = 2.13
sigma = 0.62
VcmaxRef = 181.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

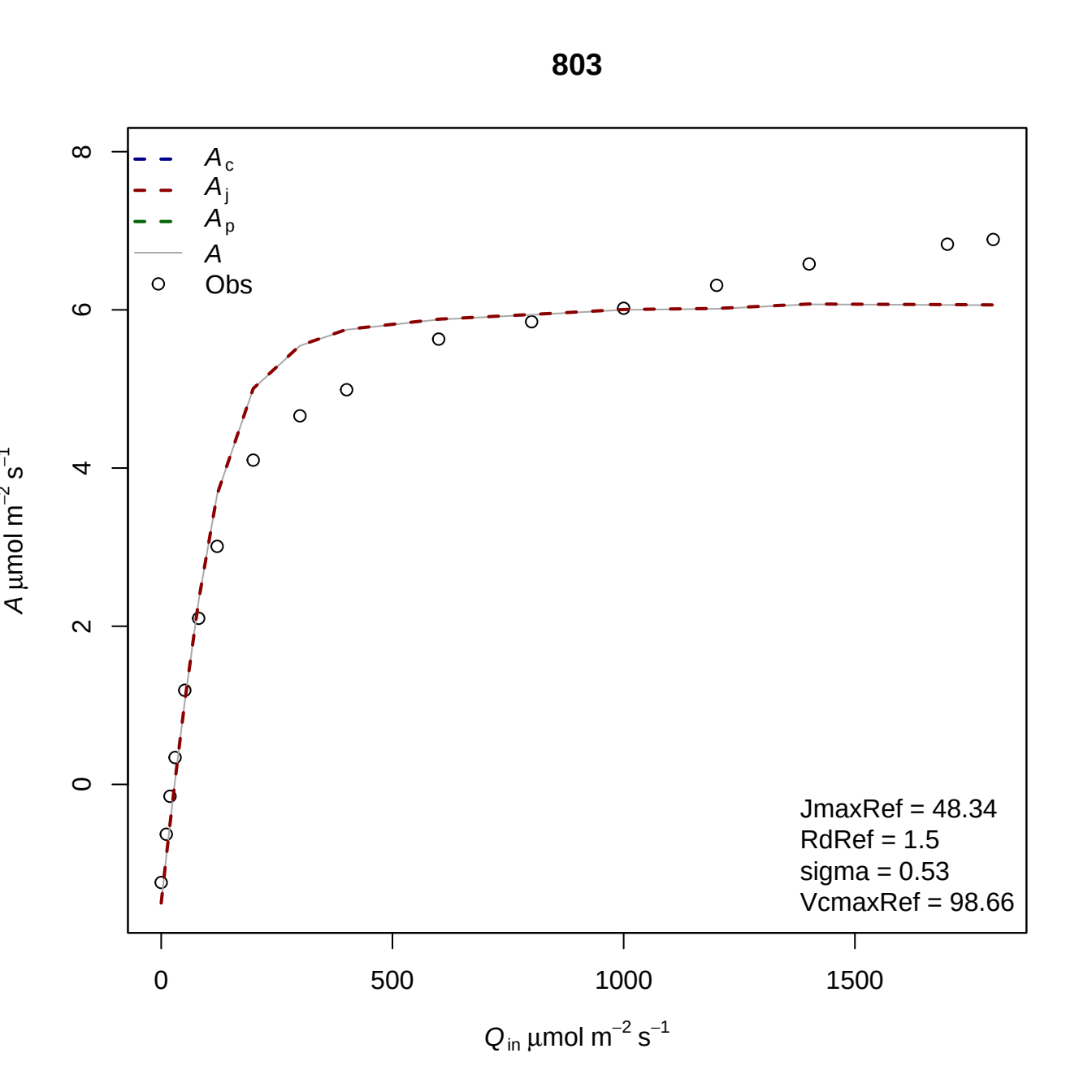

798
A_c
A_j
A_p
A
Obs
JmaxRef = 93.71
RdRef = 1.63
sigma = 0.35
VcmaxRef = 2744308.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

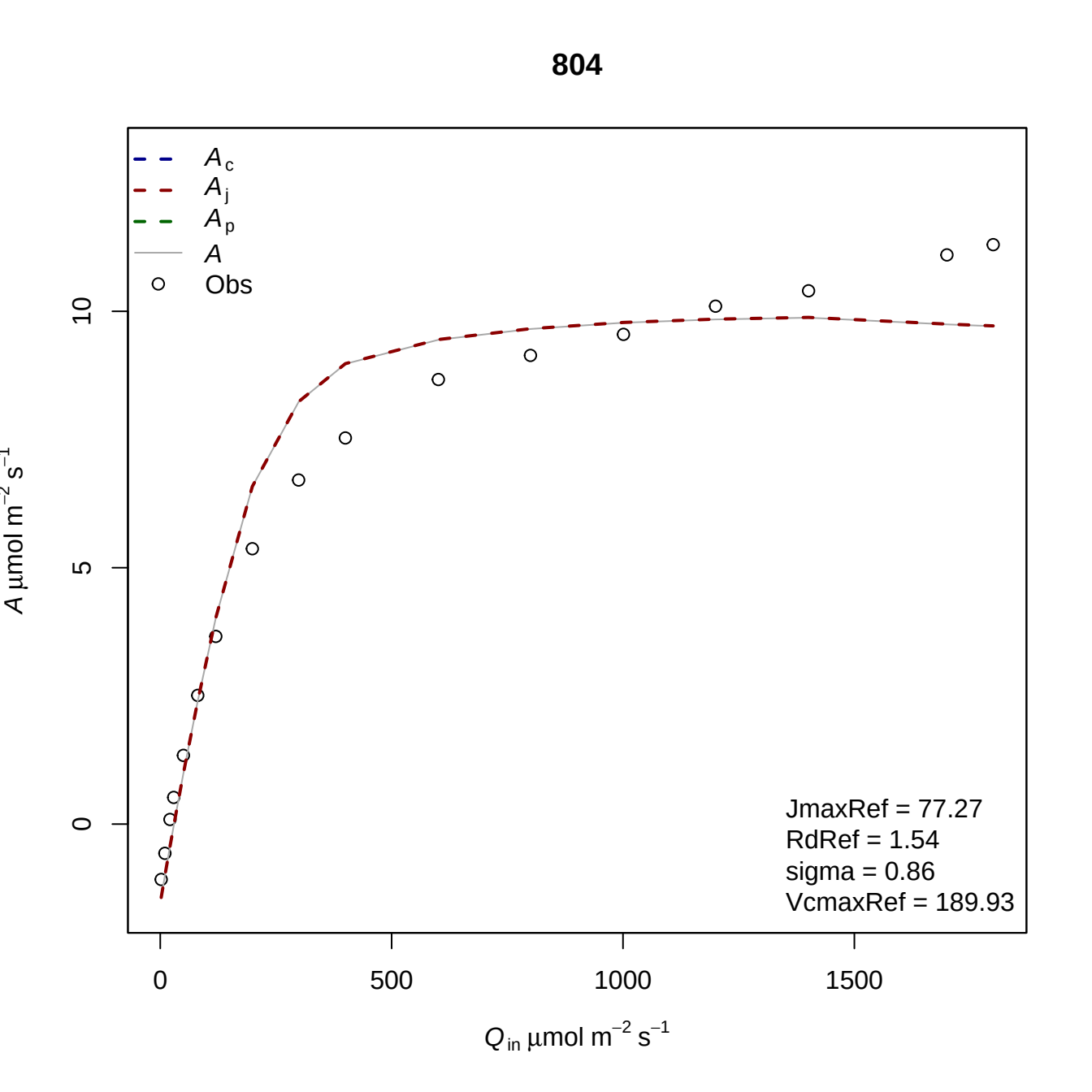

799
A_c
A_j
A_p
A
Obs
JmaxRef = 14.68
RdRef = 1.04
sigma = 0.37
VcmaxRef = 37.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
200
400
600
800
1000
1200
1400

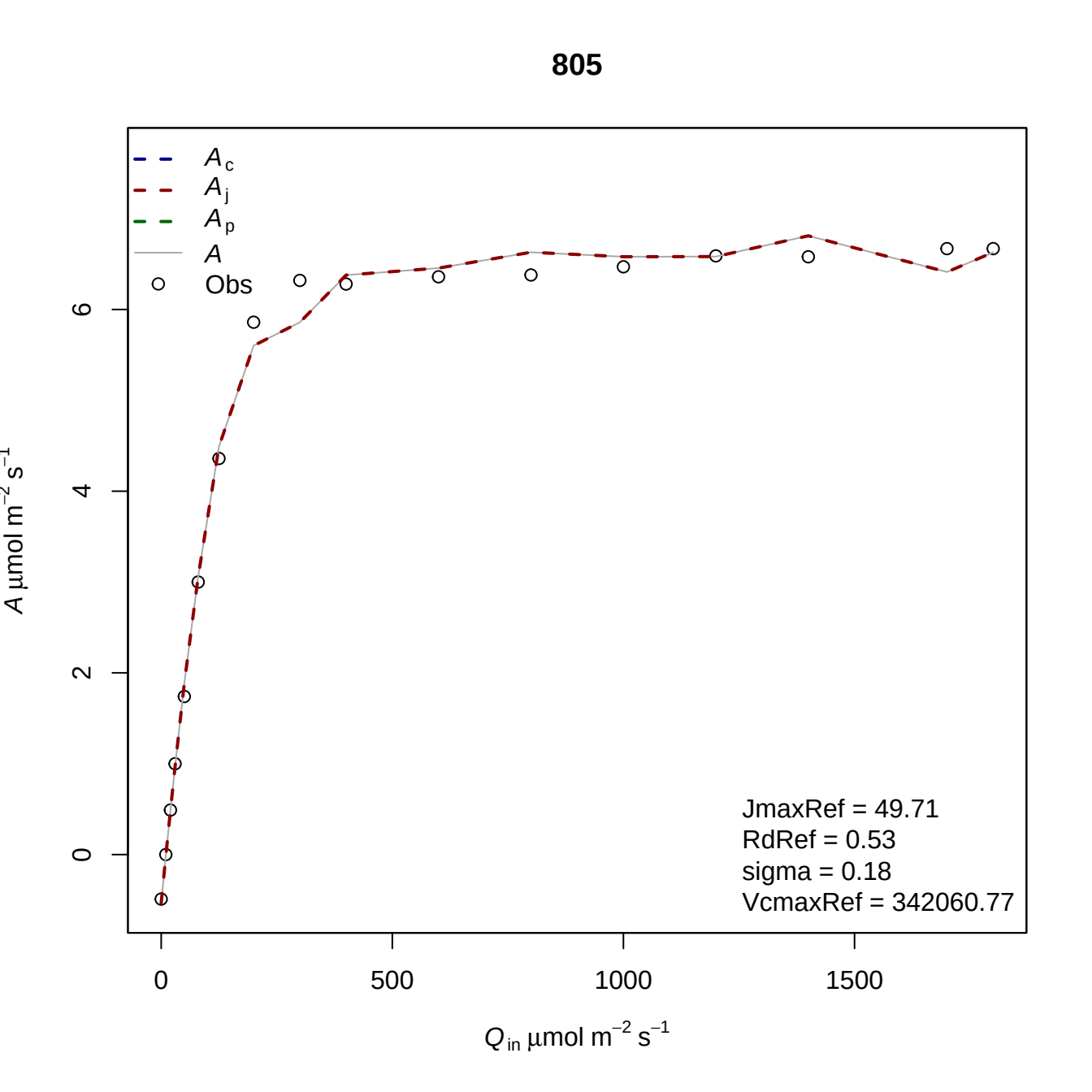

800
A_c
A_j
A_p
A
Obs
JmaxRef = 47.41
RdRef = 2.65
sigma = 0.48
VcmaxRef = 87.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200
1400

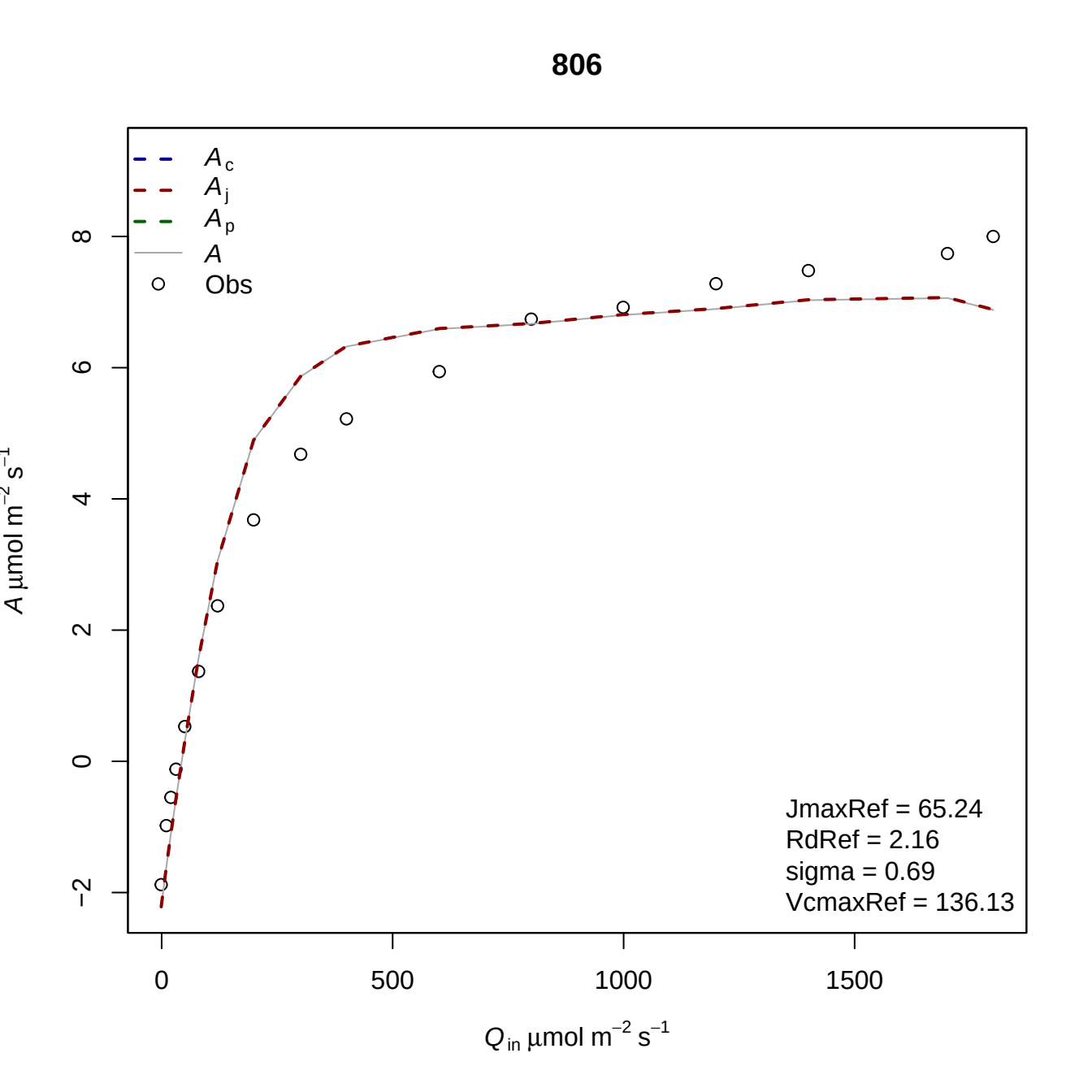

801
A_c
A_j
A_p
A
Obs
JmaxRef = 33.67
RdRef = 1
sigma = 0.56
VcmaxRef = 72.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400

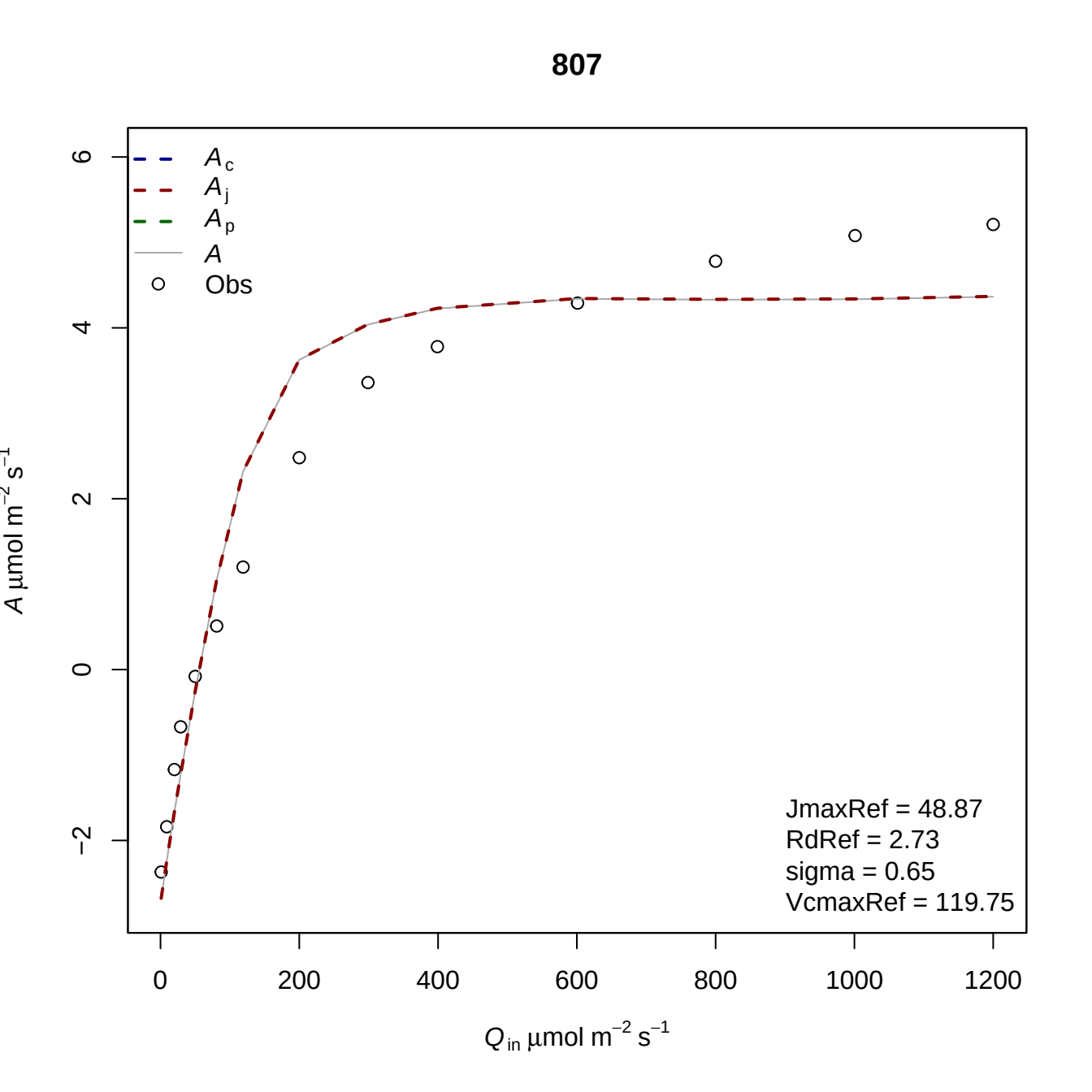

802
A_c
A_j
A_p
A
Obs
JmaxRef = 45.3
RdRef = 1.18
sigma = 0.4
VcmaxRef = 111.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

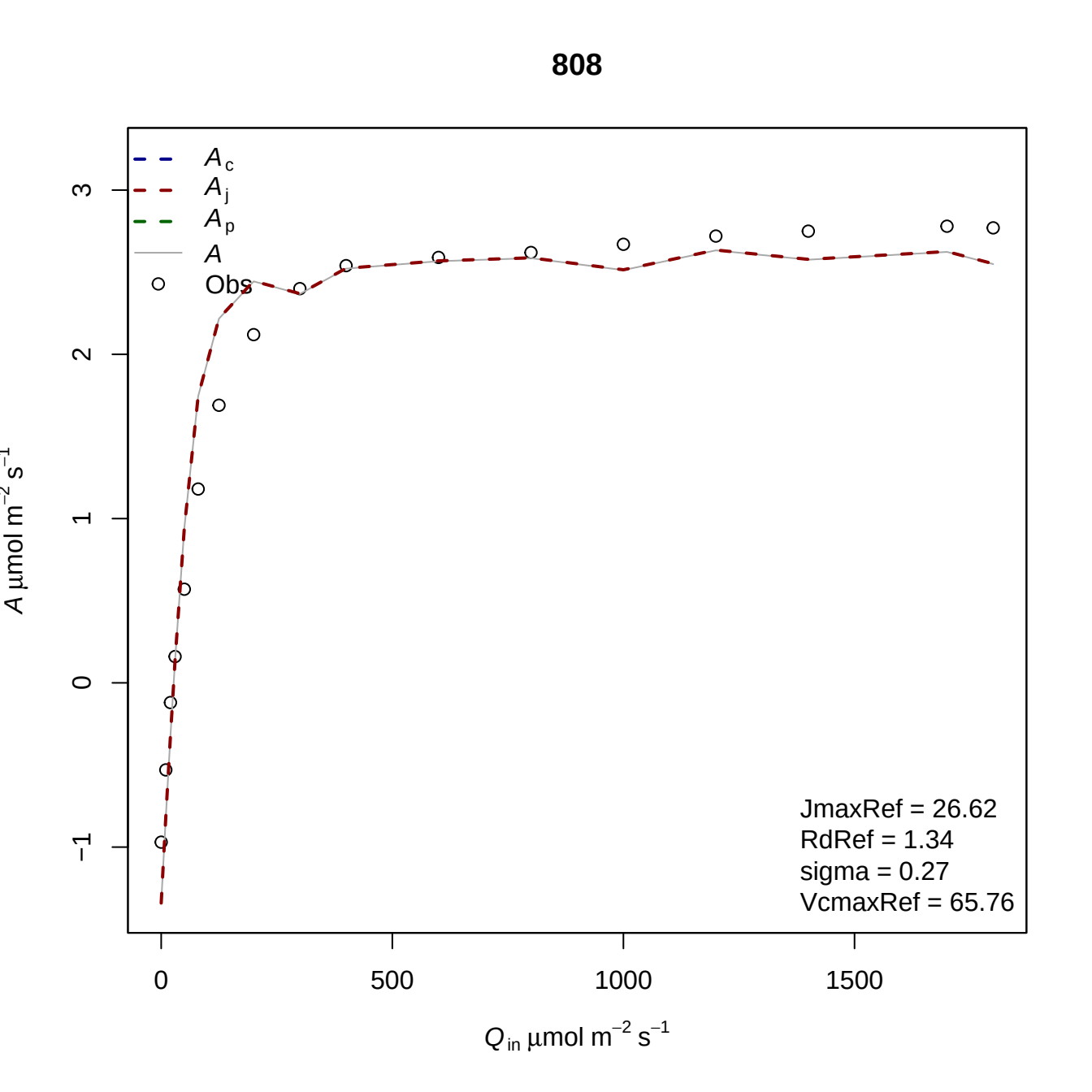

803
A_c
A_j
A_p
A
Obs
JmaxRef = 48.34
RdRef = 1.5
sigma = 0.53
VcmaxRef = 98.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

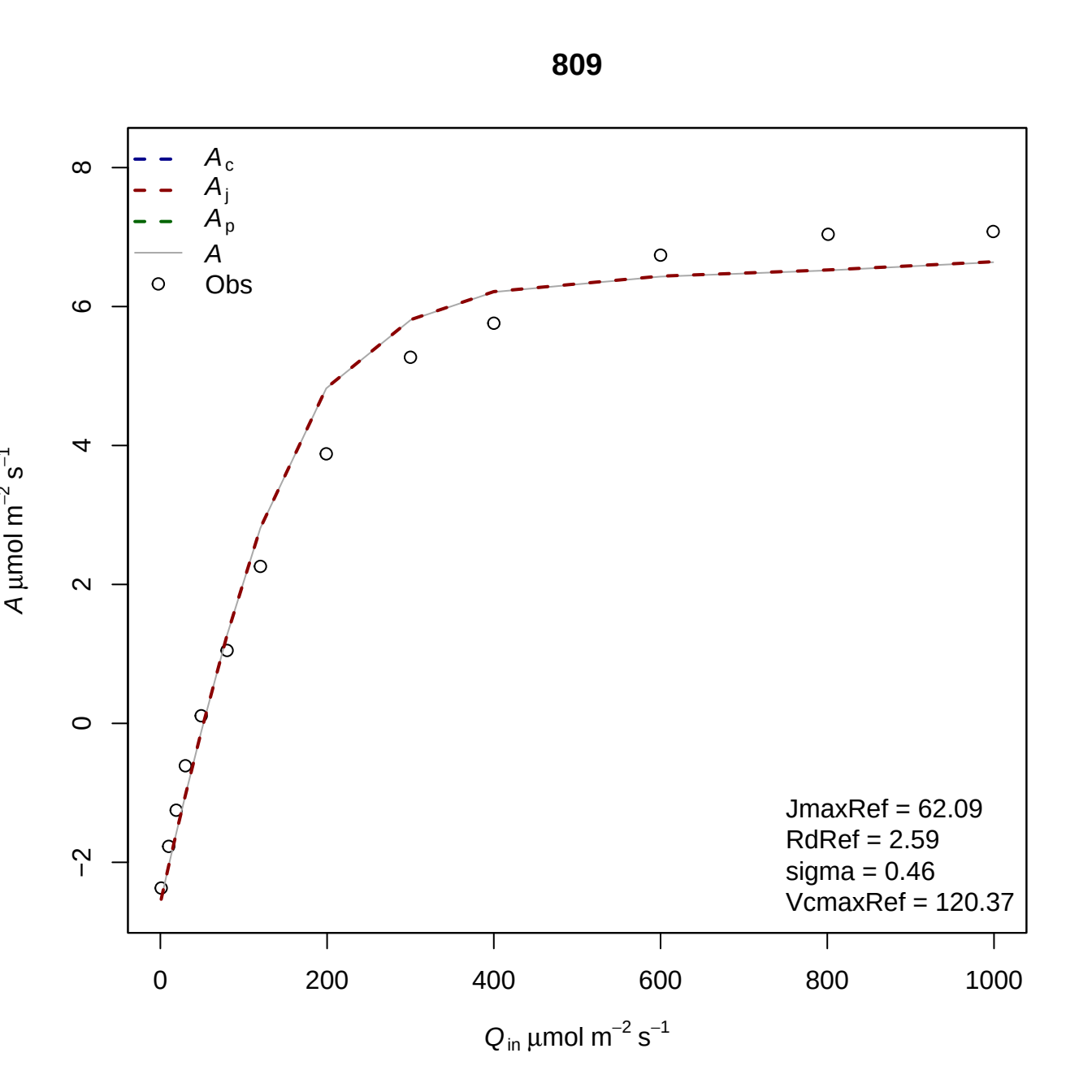

804
A_c
A_j
A_p
A
Obs
JmaxRef = 77.27
RdRef = 1.54
sigma = 0.86
VcmaxRef = 189.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

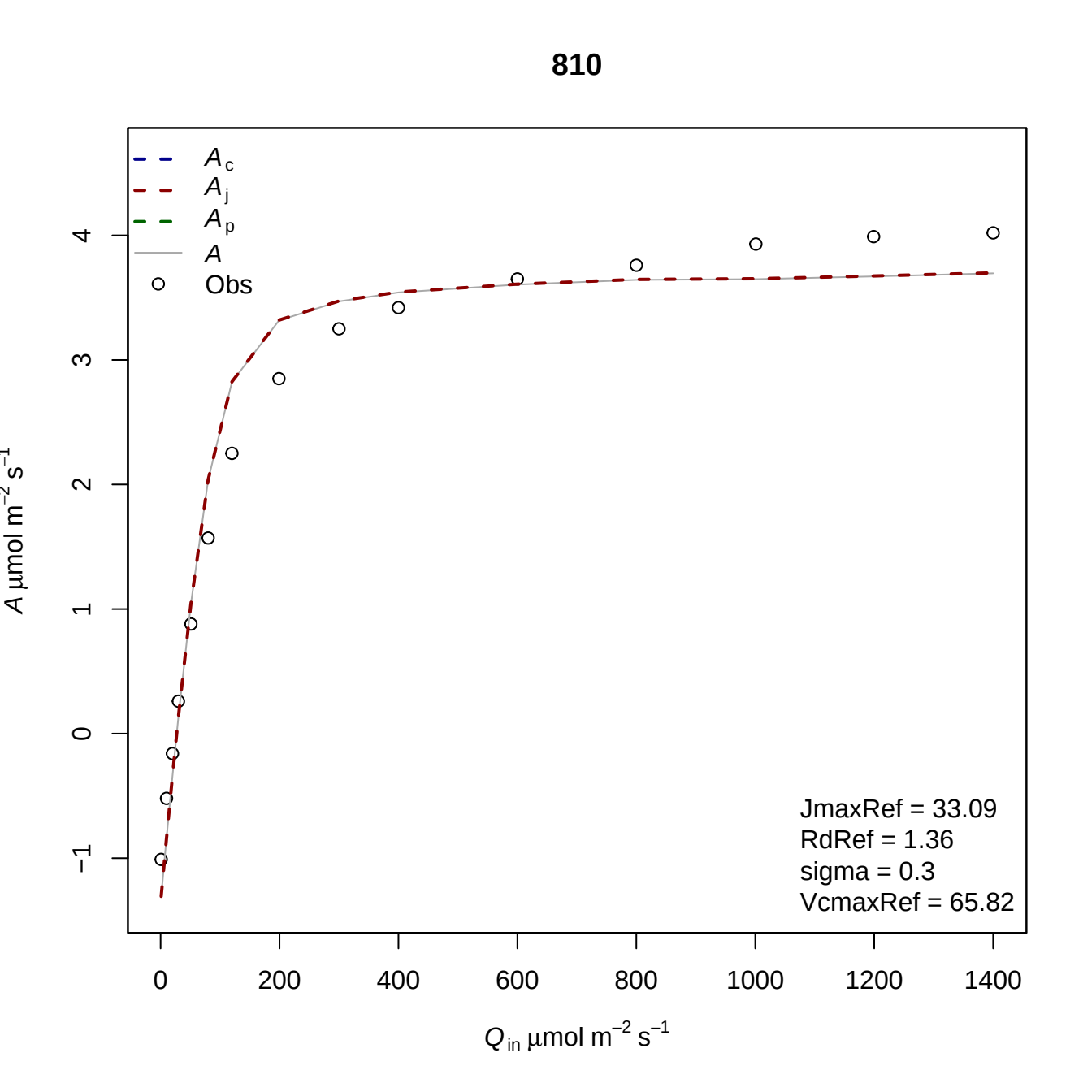

805
A_c
A_j
A_p
A
Obs
JmaxRef = 49.71
RdRef = 0.53
sigma = 0.18
VcmaxRef = 342060.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

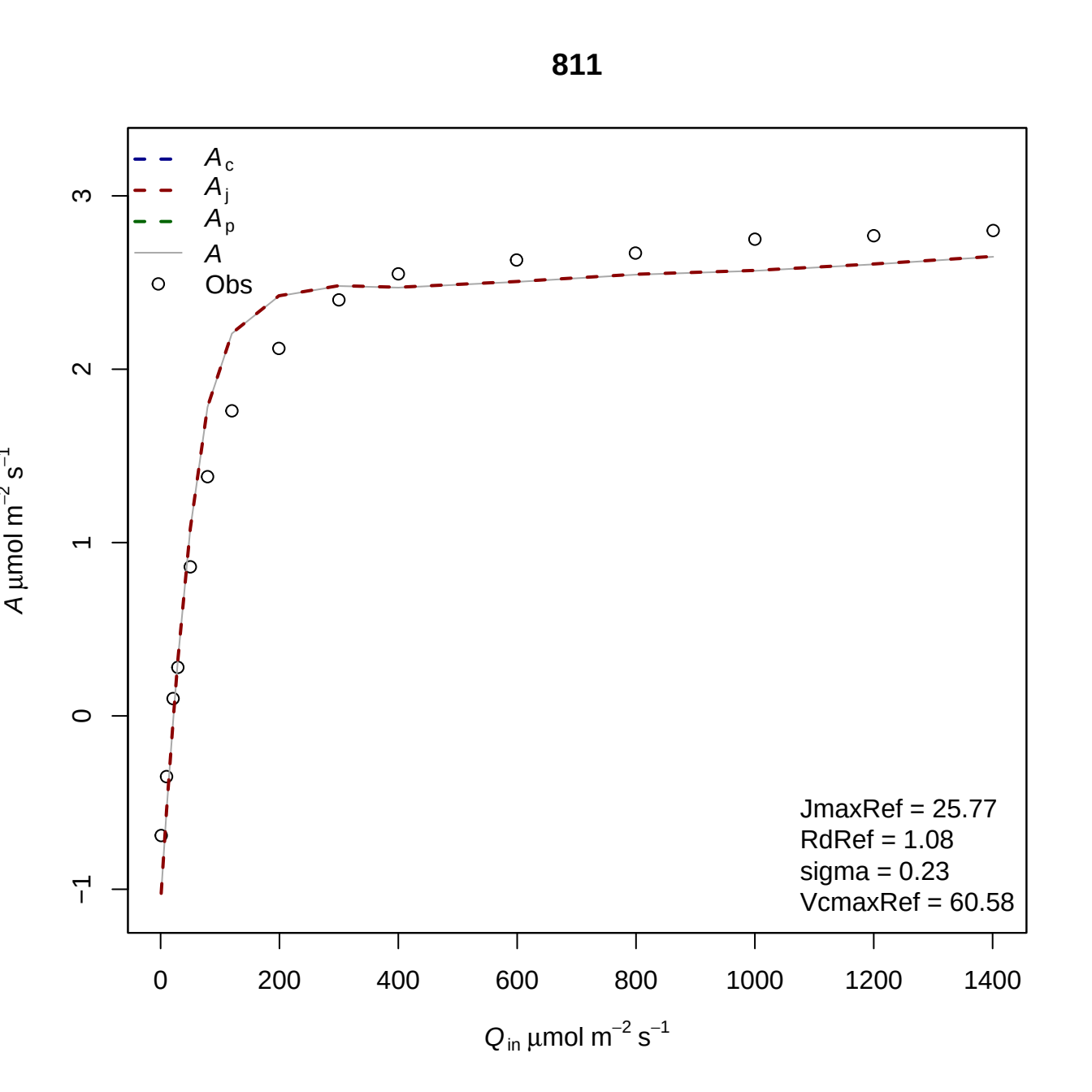

806
A_c
A_j
A_p
A
Obs
JmaxRef = 65.24
RdRef = 2.16
sigma = 0.69
VcmaxRef = 136.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

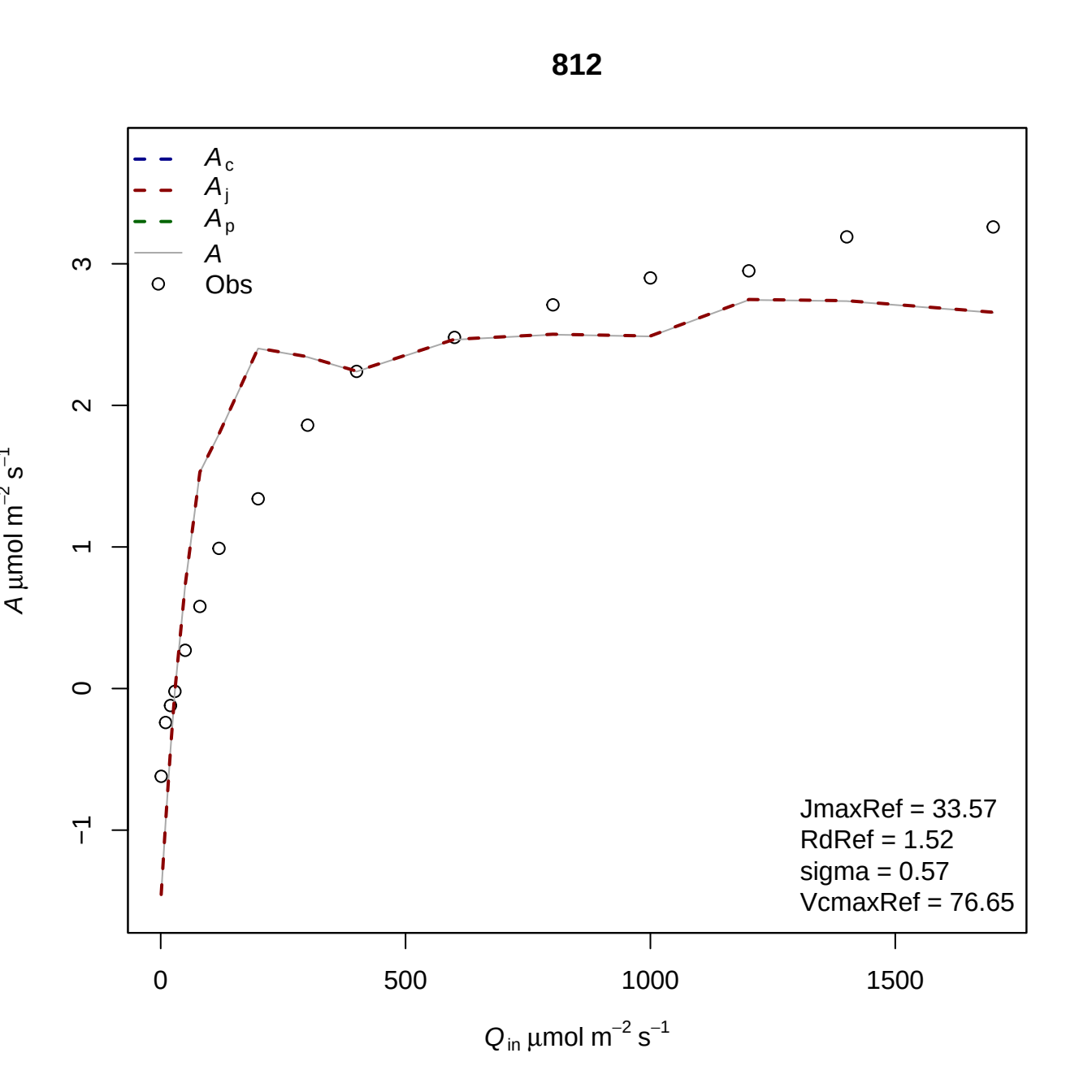

807
A_c
A_j
A_p
A
Obs
JmaxRef = 48.87
RdRef = 2.73
sigma = 0.65
VcmaxRef = 119.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

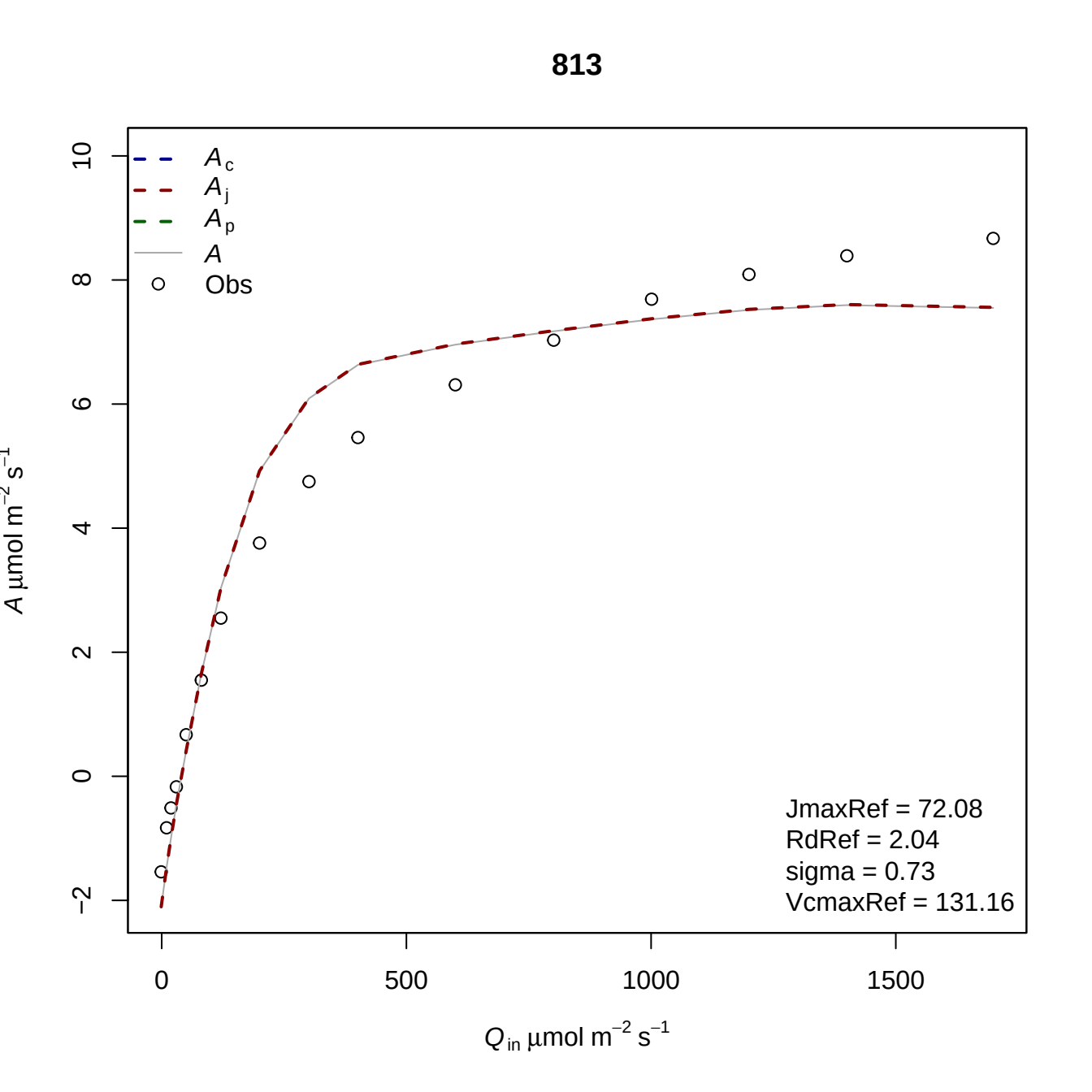

808
A_c
A_j
A_p
A
Obs
JmaxRef = 26.62
RdRef = 1.34
sigma = 0.27
VcmaxRef = 65.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

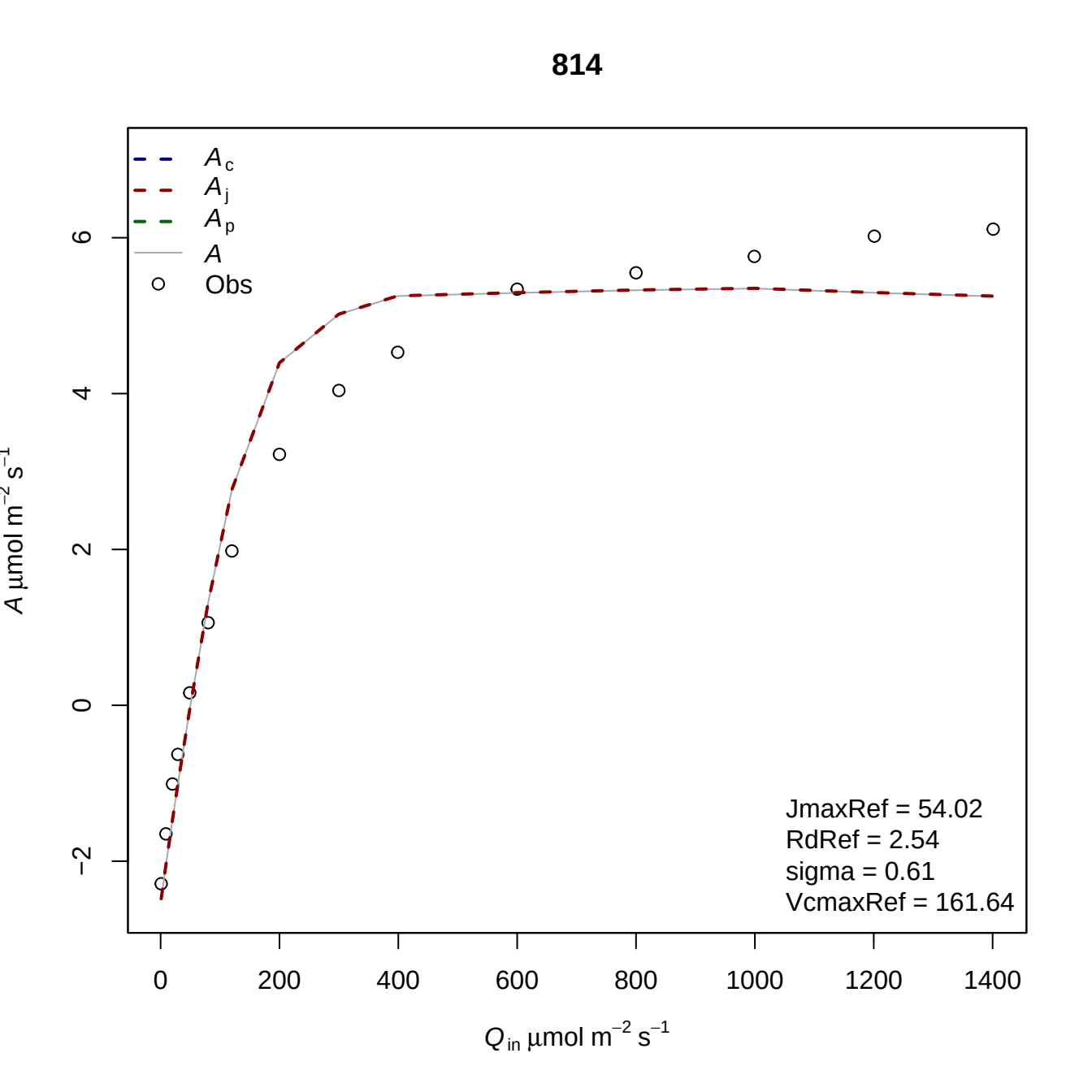

809
A_c
A_j
A_p
A
Obs
JmaxRef = 62.09
RdRef = 2.59
sigma = 0.46
VcmaxRef = 120.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

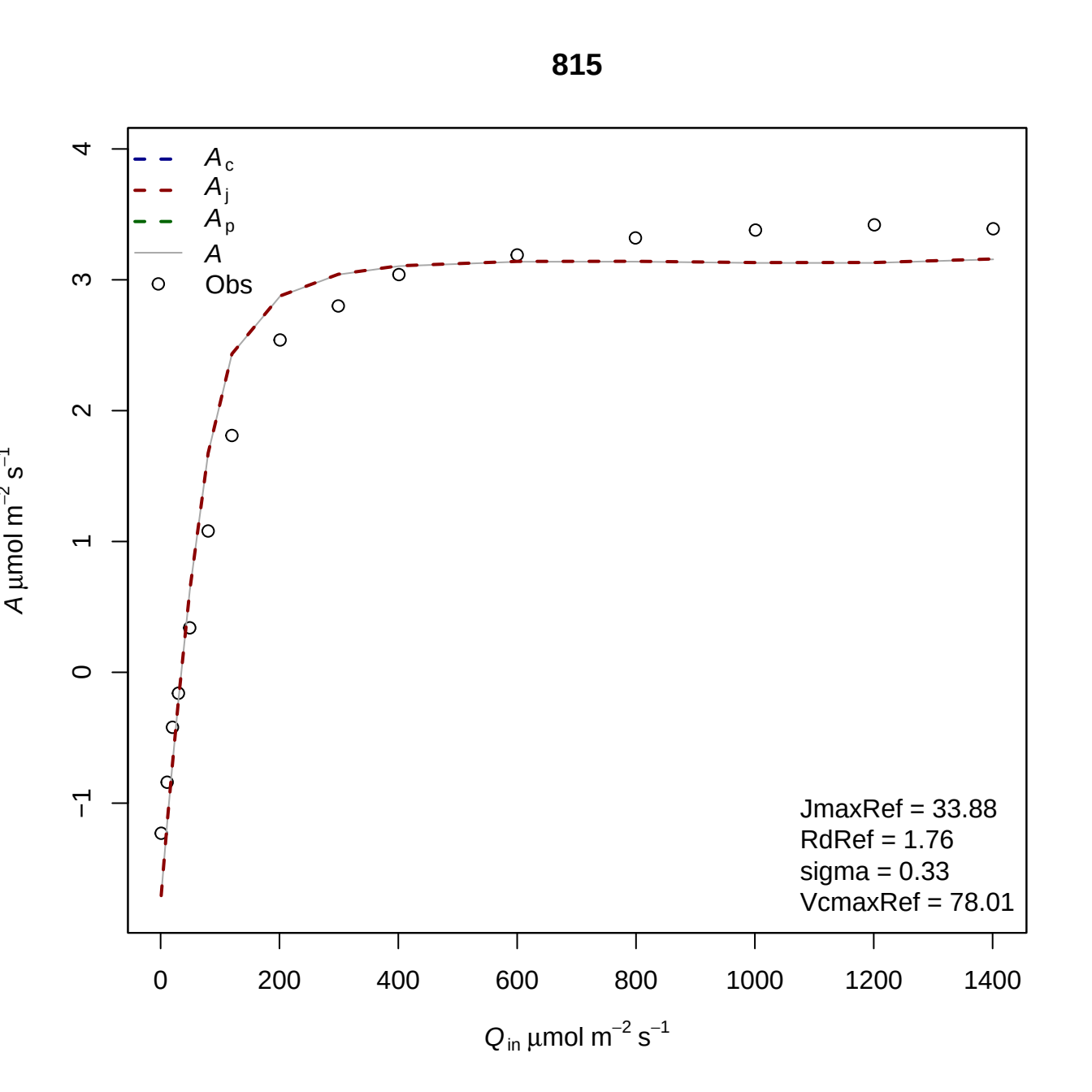

810
A_c
A_j
A_p
A
Obs
JmaxRef = 33.09
RdRef = 1.36
sigma = 0.3
VcmaxRef = 65.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

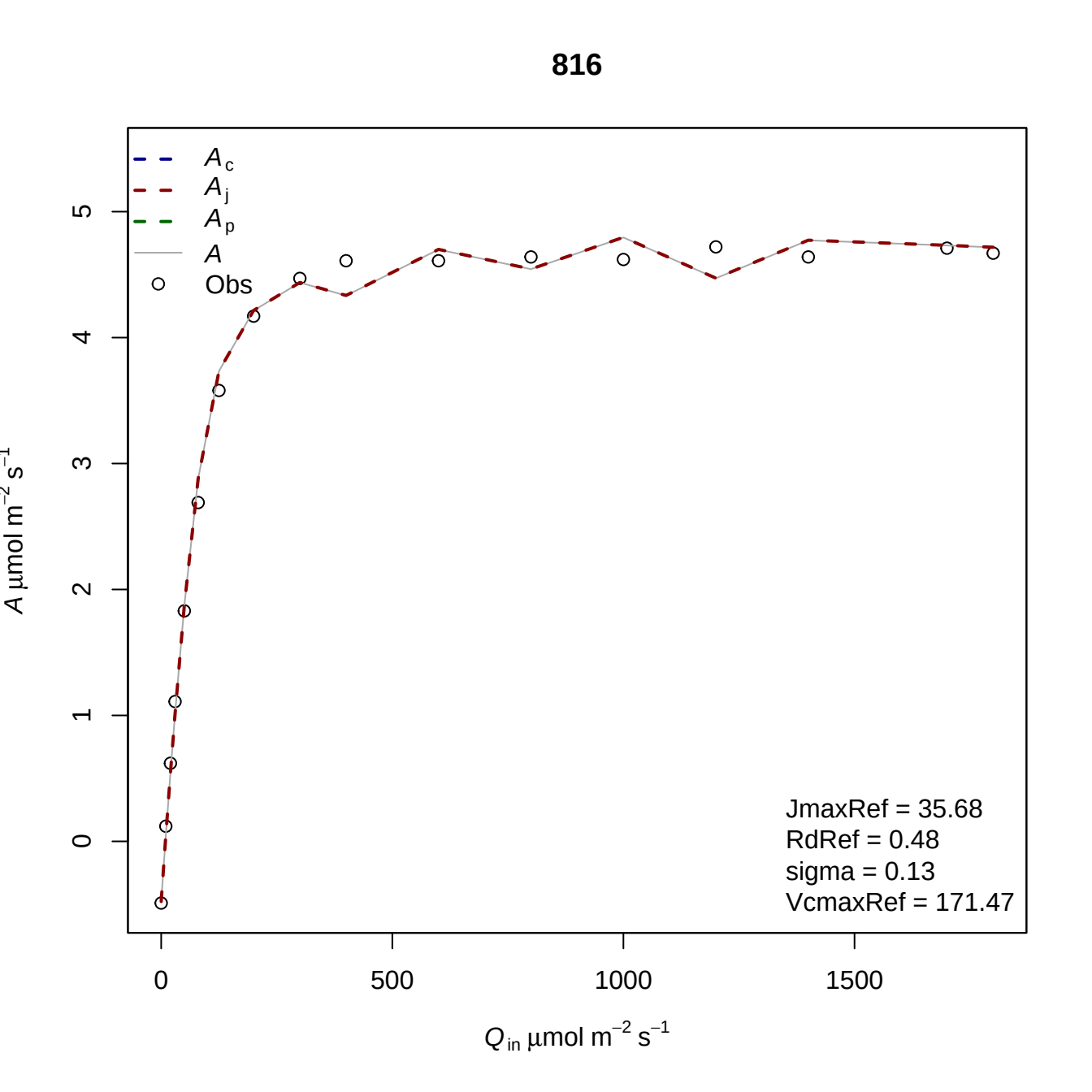

811
A_c
A_j
A_p
A
Obs
JmaxRef = 25.77
RdRef = 1.08
sigma = 0.23
VcmaxRef = 60.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

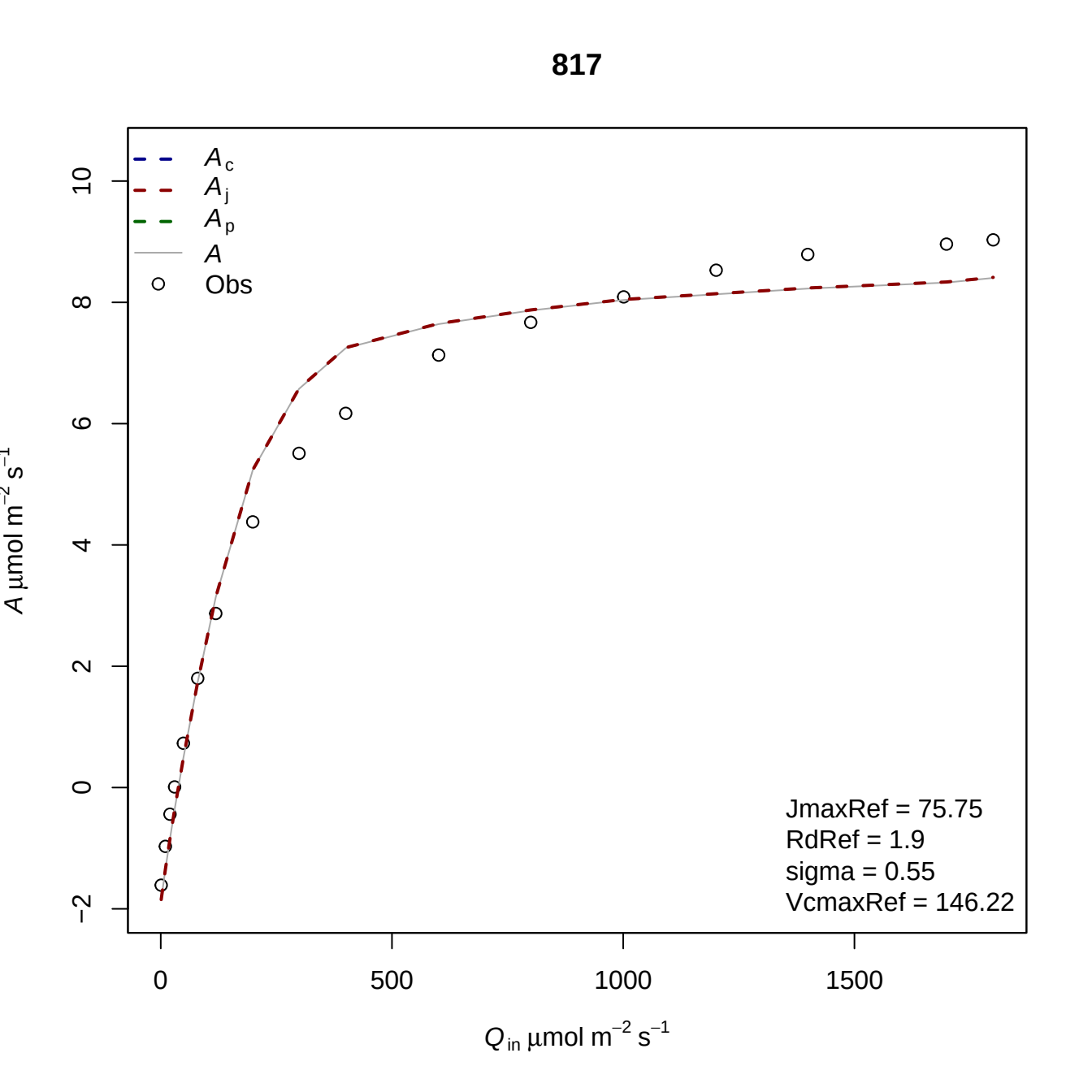

812
A_c
A_j
A_p
A
Obs
JmaxRef = 33.57
RdRef = 1.52
sigma = 0.57
VcmaxRef = 76.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

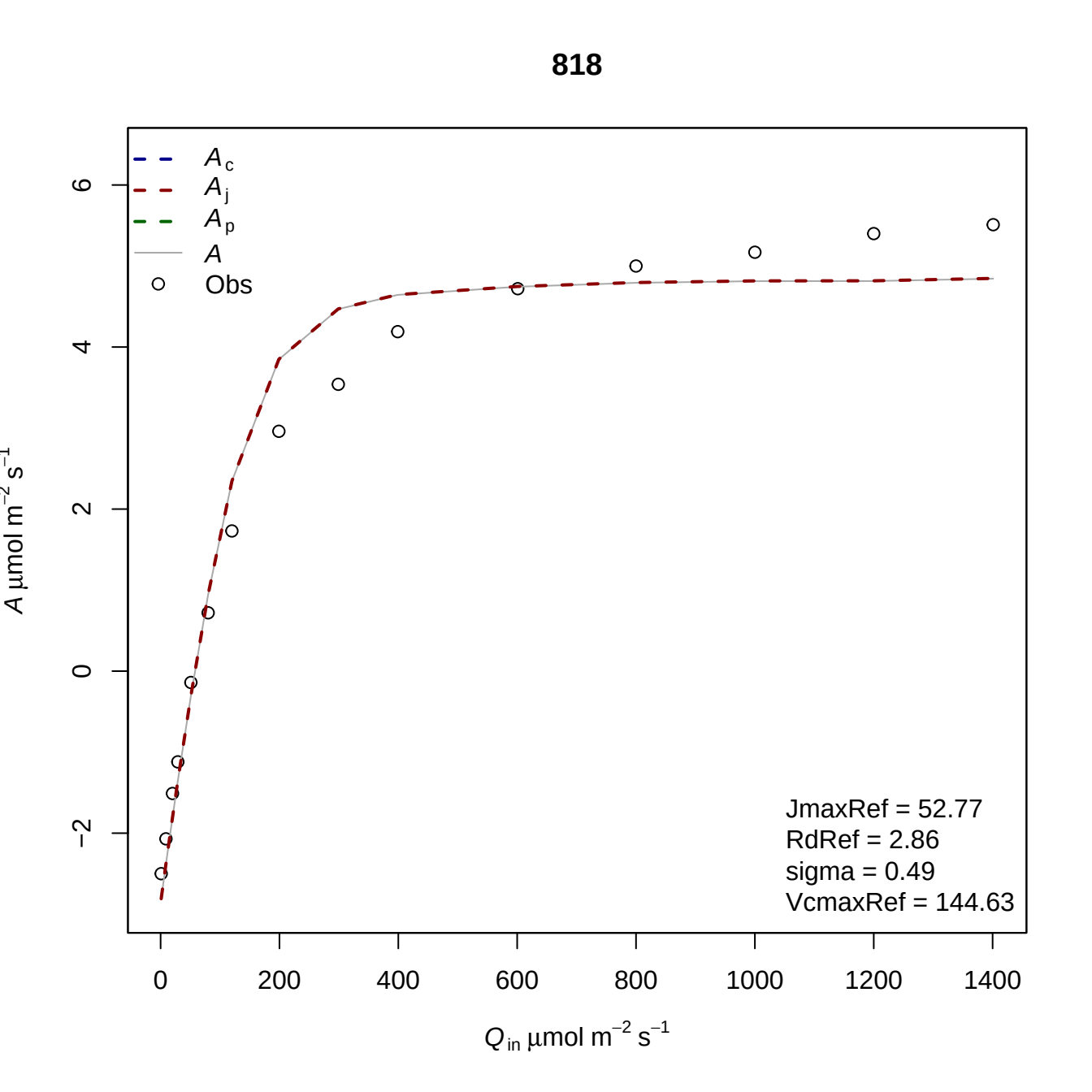

813
A_c
A_j
A_p
A
Obs
JmaxRef = 72.08
RdRef = 2.04
sigma = 0.73
VcmaxRef = 131.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

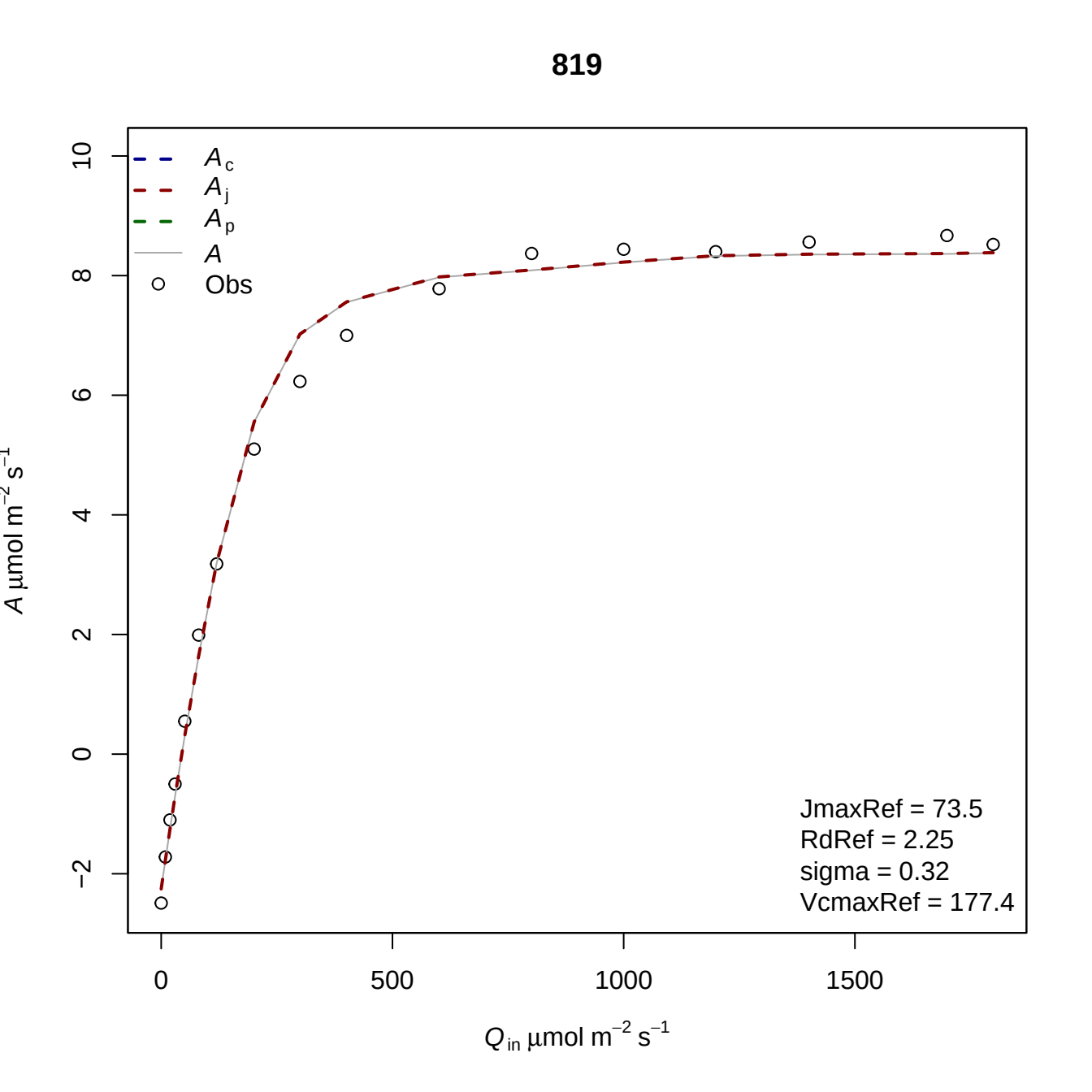

814
A_c
A_j
A_p
A
Obs
JmaxRef = 54.02
RdRef = 2.54
sigma = 0.61
VcmaxRef = 161.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

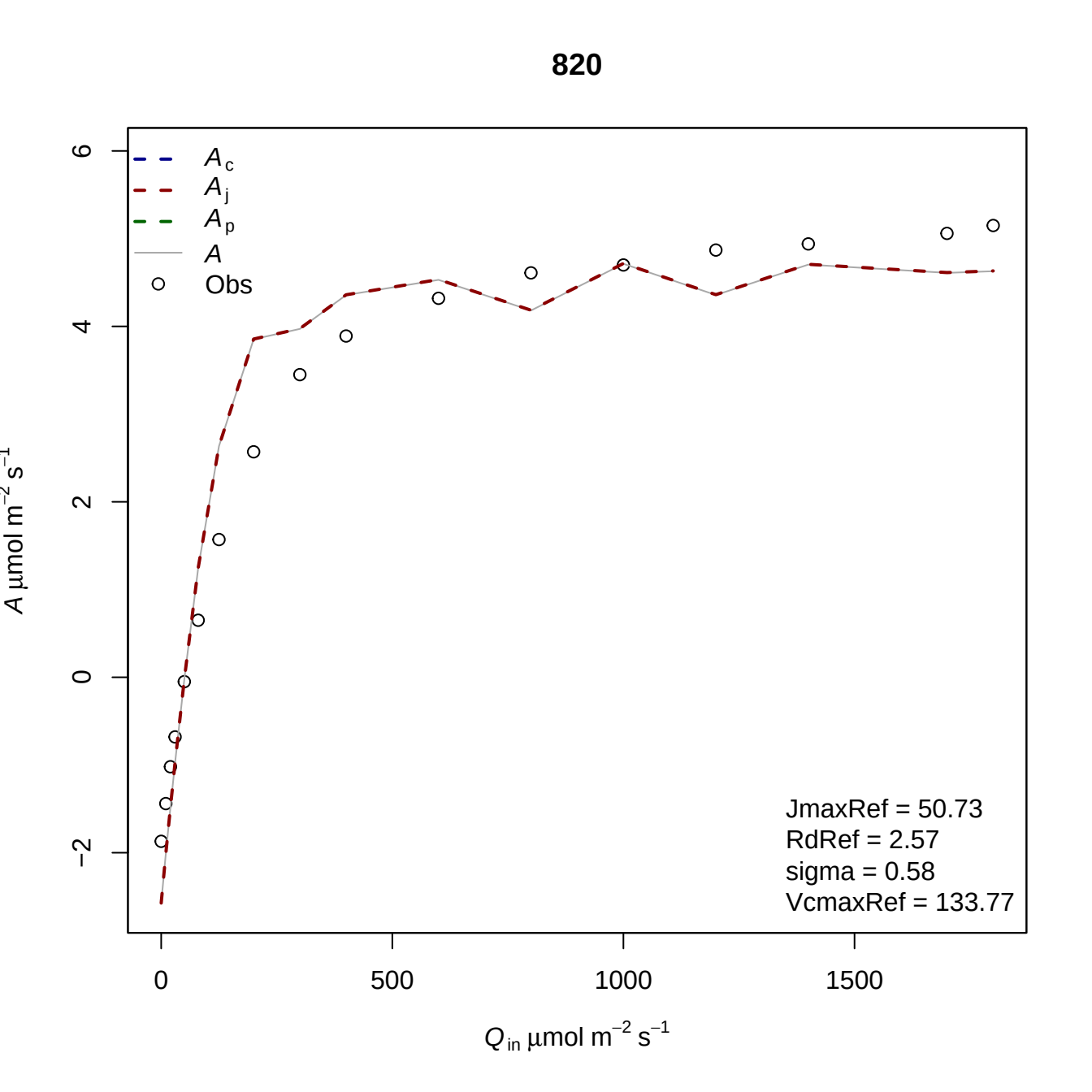

815
A_c
A_j
A_p
A
Obs
JmaxRef = 33.88
RdRef = 1.76
sigma = 0.33
VcmaxRef = 78.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

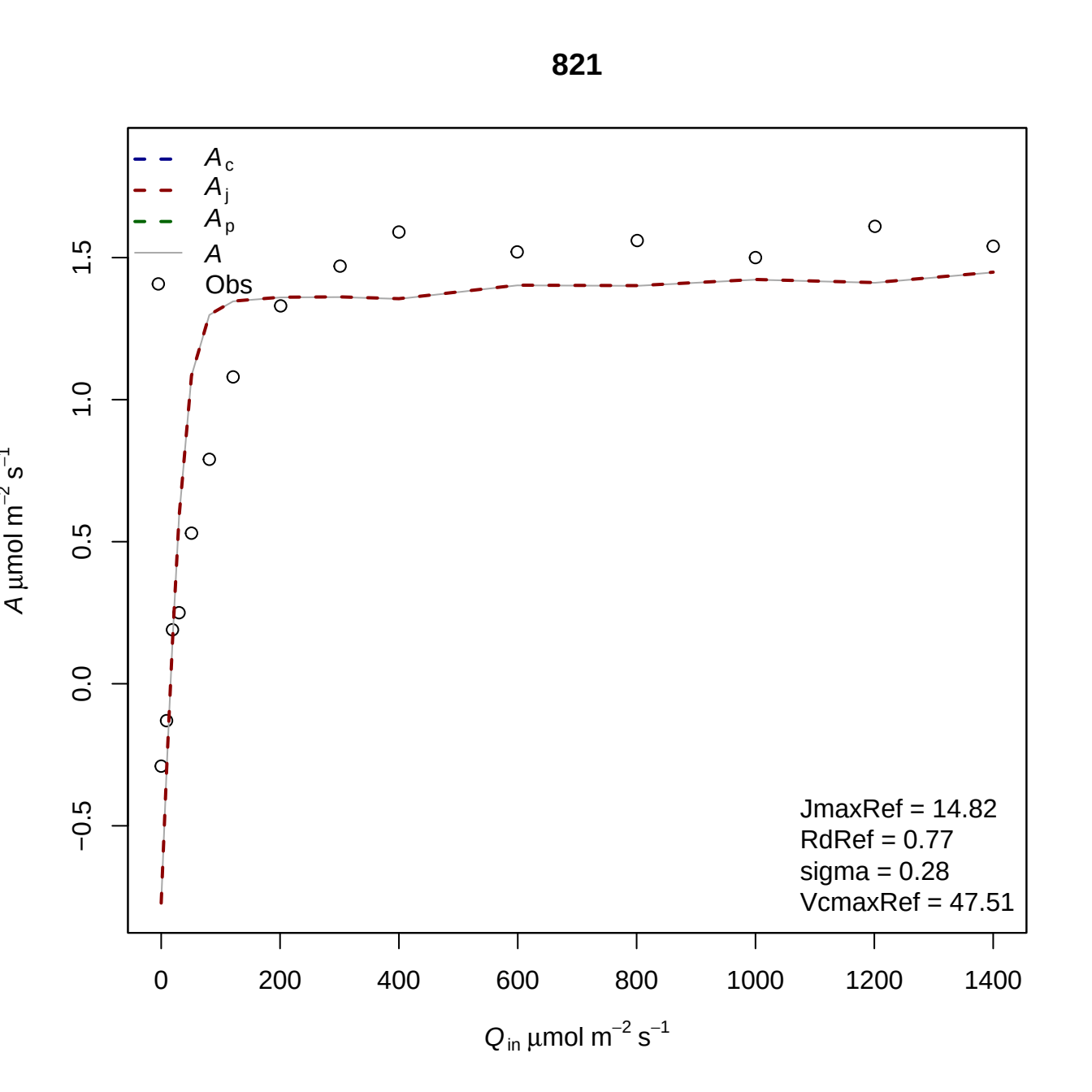

816
A_c
A_j
A_p
A
Obs
JmaxRef = 35.68
RdRef = 0.48
sigma = 0.13
VcmaxRef = 171.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

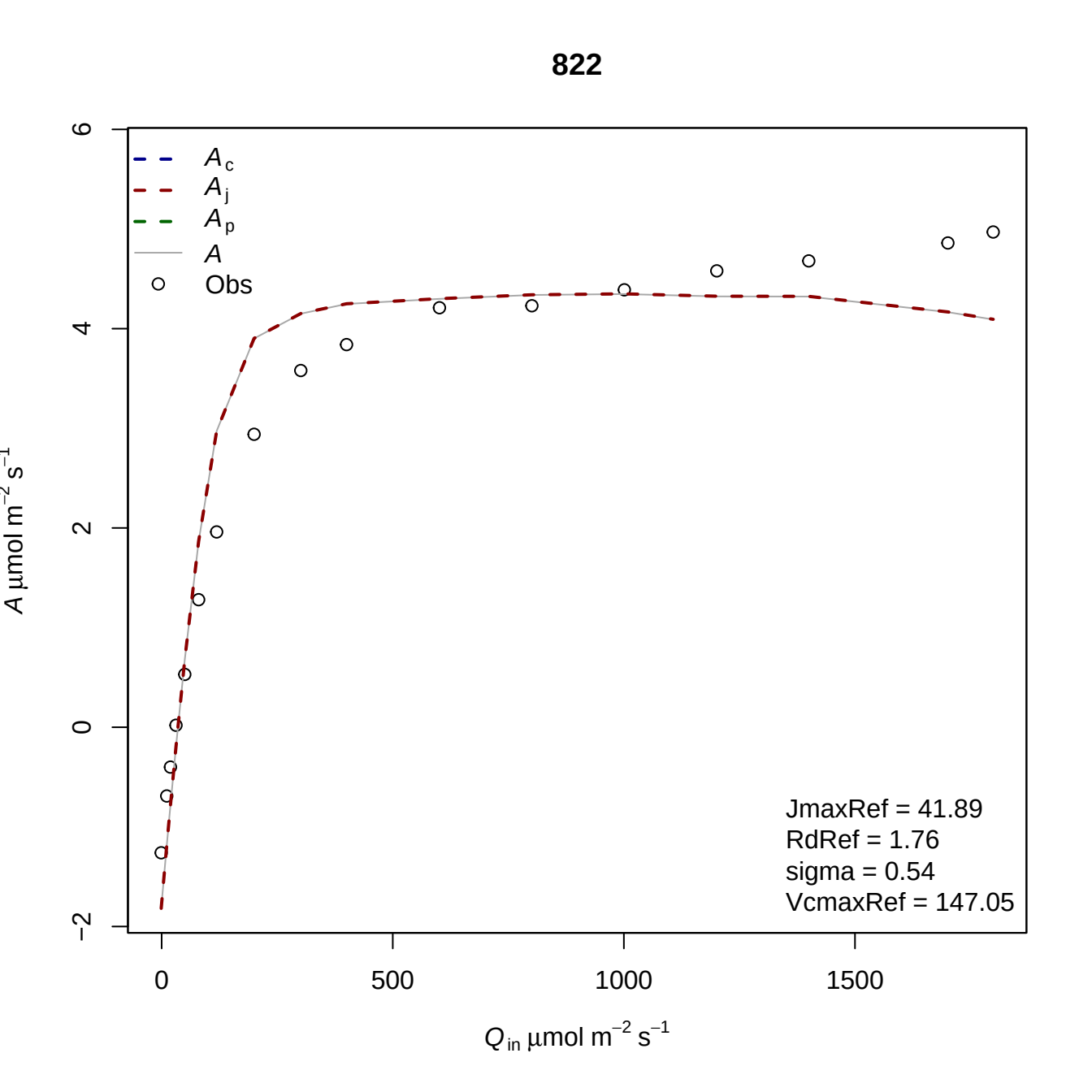

817
A_c
A_j
A_p
A
Obs
JmaxRef = 75.75
RdRef = 1.9
sigma = 0.55
VcmaxRef = 146.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

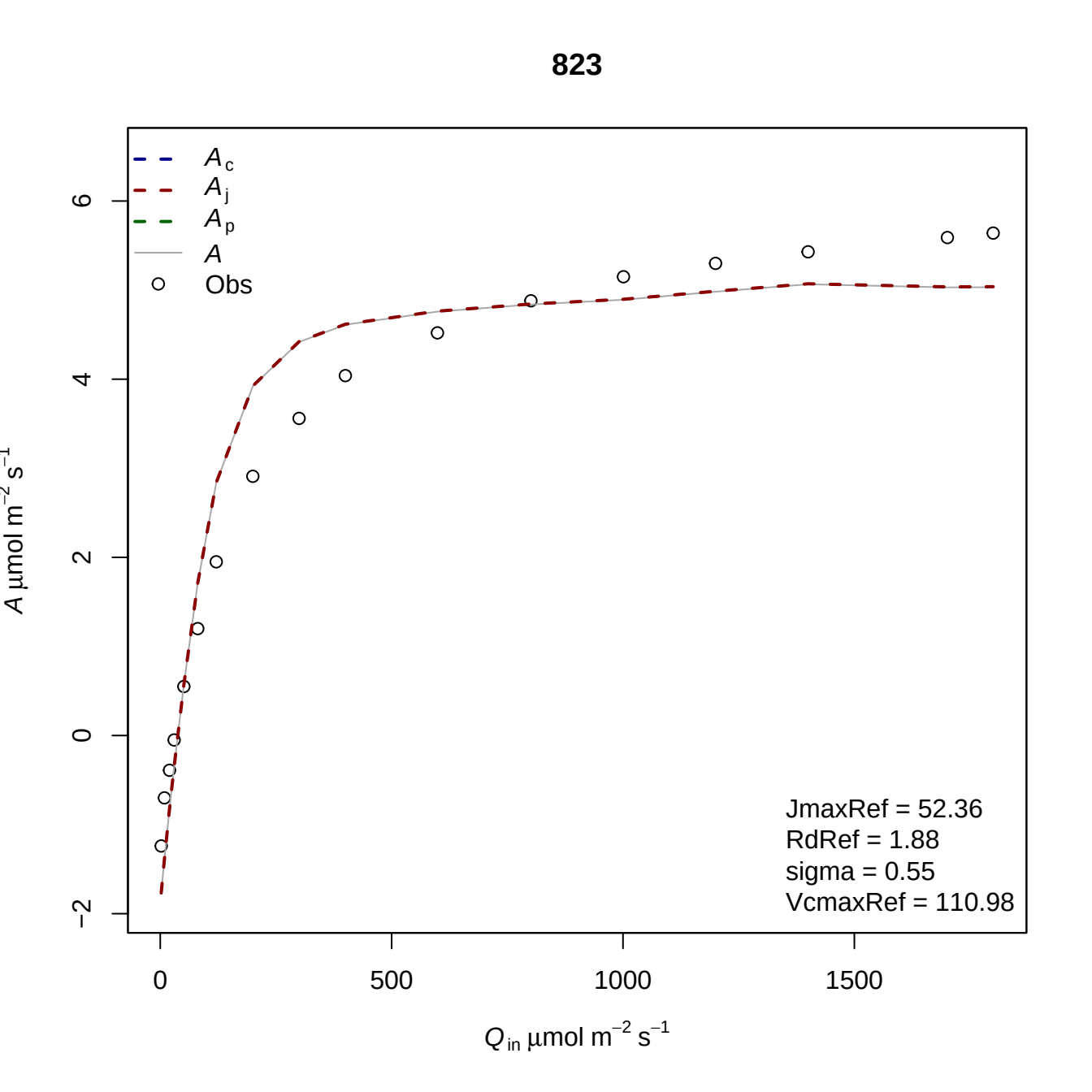

818
A_c
A_j
A_p
A
Obs
JmaxRef = 52.77
RdRef = 2.86
sigma = 0.49
VcmaxRef = 144.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

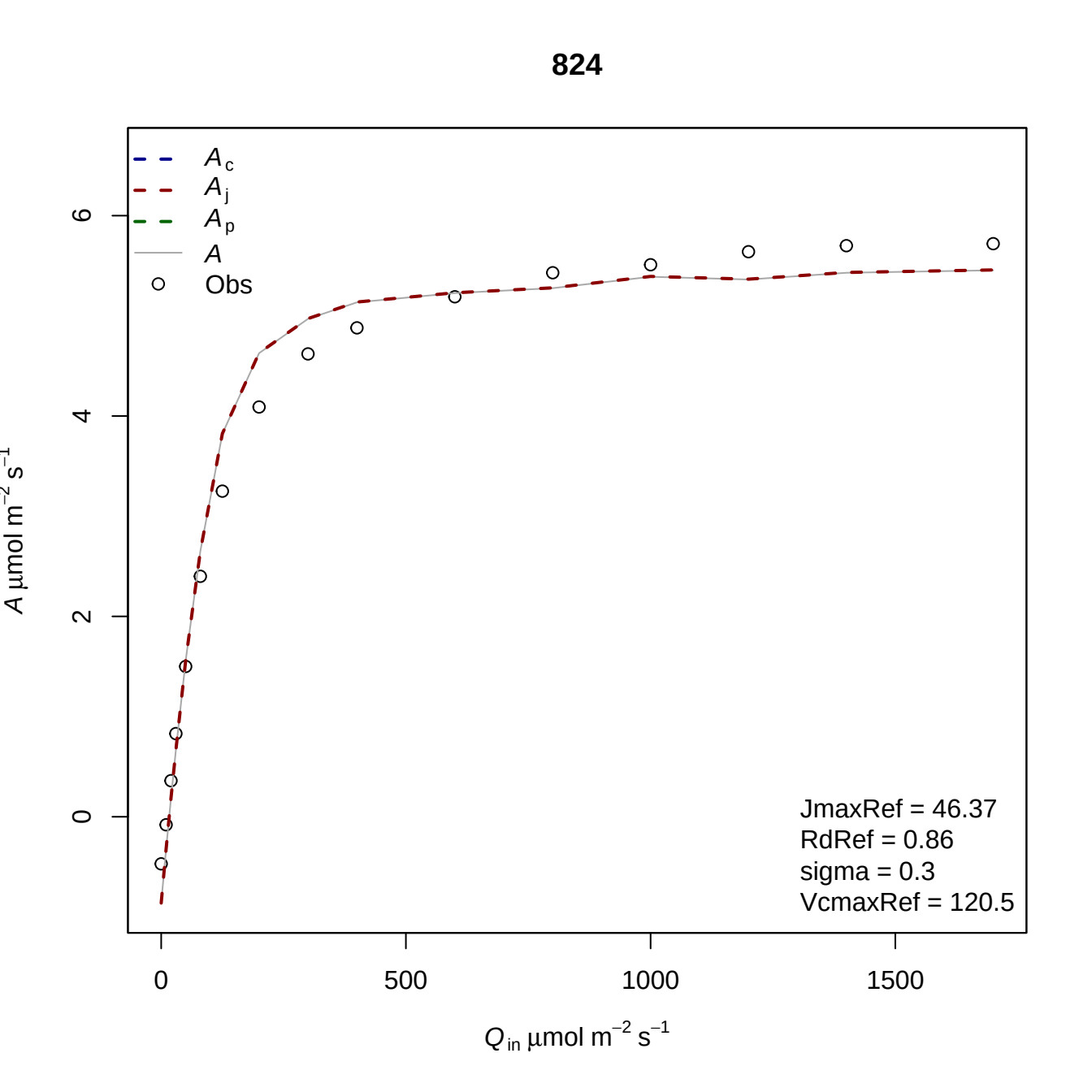

819
A_c
A_j
A_p
A
Obs
JmaxRef = 73.5
RdRef = 2.25
sigma = 0.32
VcmaxRef = 177.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

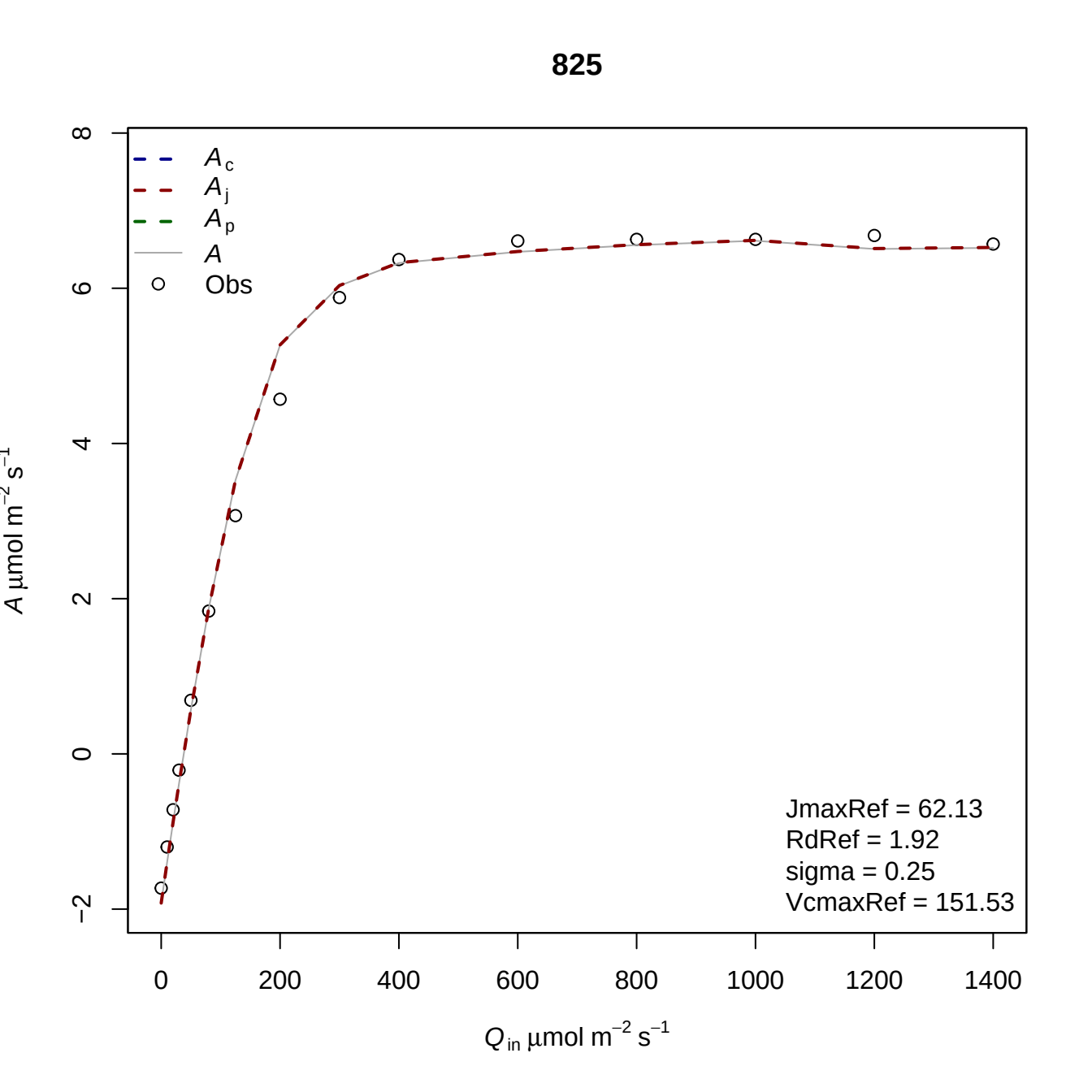

820
A_c
A_j
A_p
A
Obs
JmaxRef = 50.73
RdRef = 2.57
sigma = 0.58
VcmaxRef = 133.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

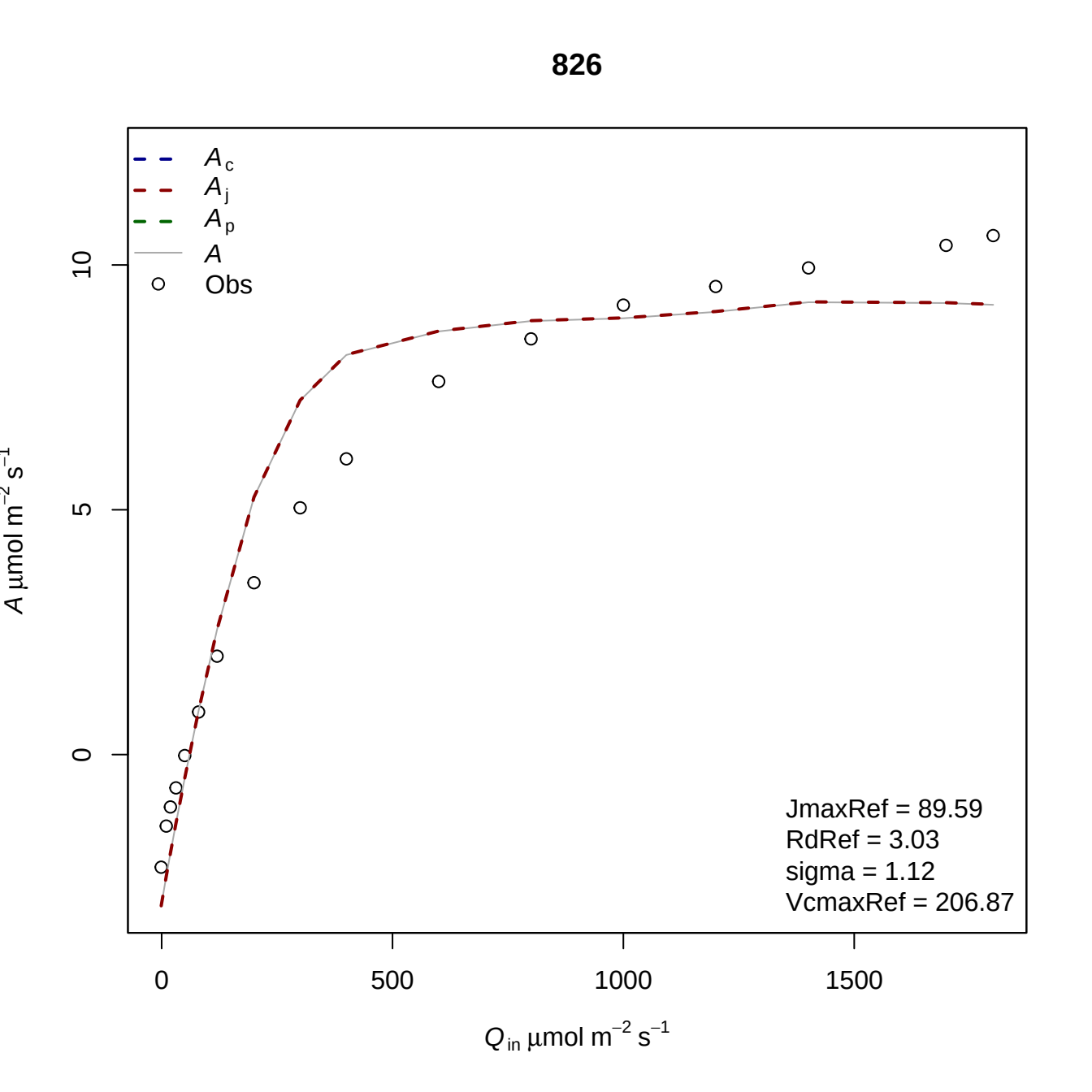

821
A_c
A_j
A_p
A
Obs
JmaxRef = 14.82
RdRef = 0.77
sigma = 0.28
VcmaxRef = 47.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
200
400
600
800
1000
1200
1400

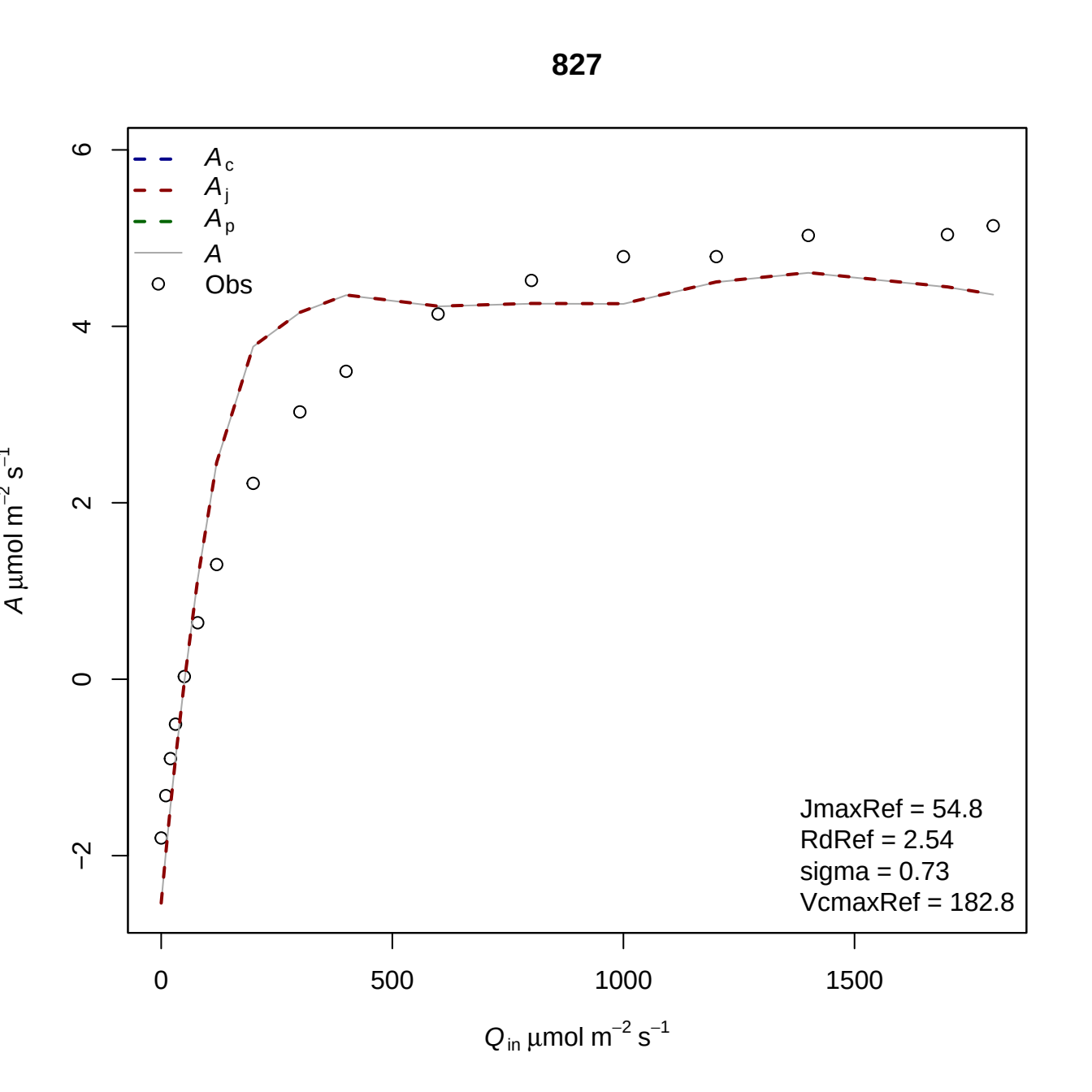

822
A_c
A_j
A_p
A
Obs
JmaxRef = 41.89
RdRef = 1.76
sigma = 0.54
VcmaxRef = 147.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

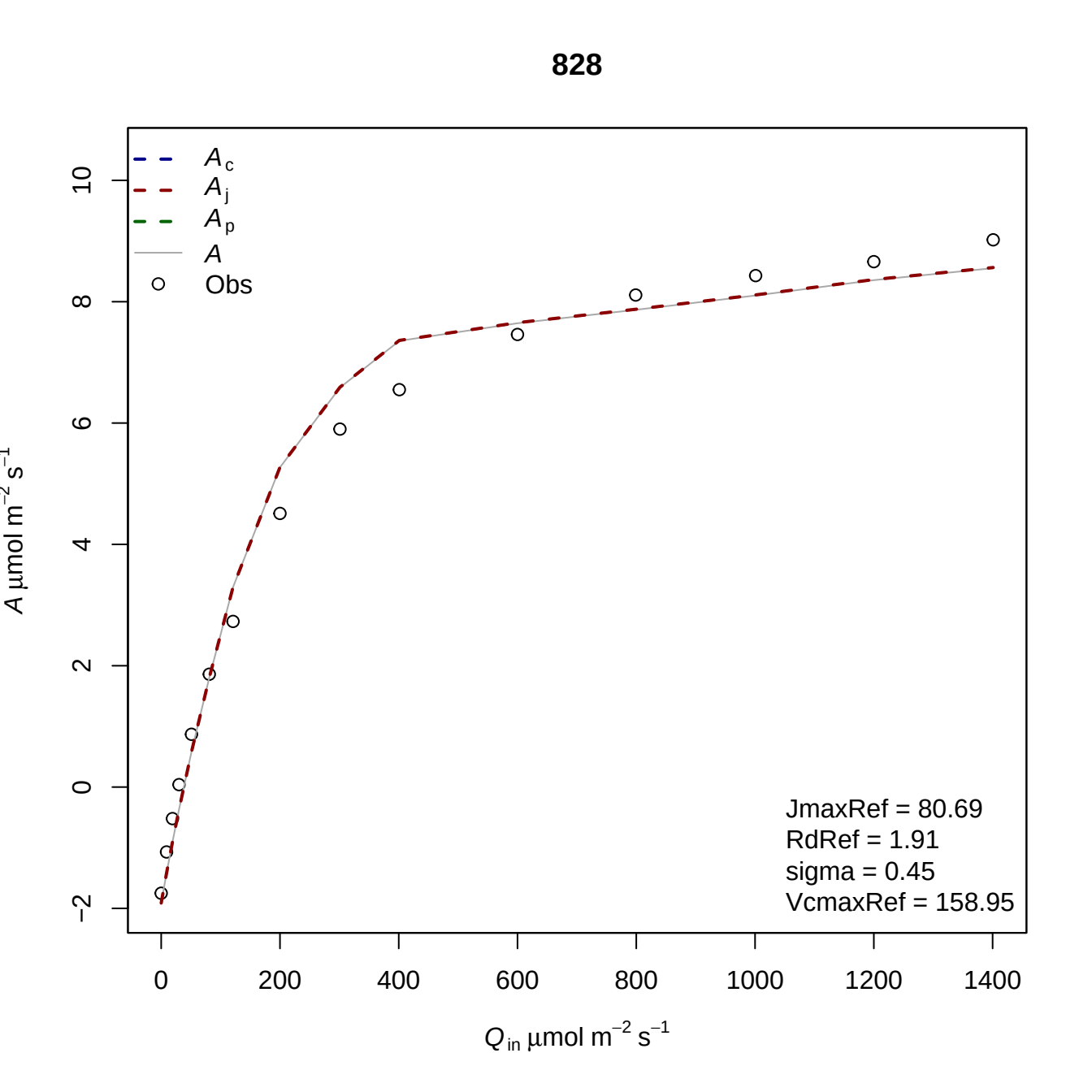

823
A_c
A_j
A_p
A
Obs
JmaxRef = 52.36
RdRef = 1.88
sigma = 0.55
VcmaxRef = 110.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

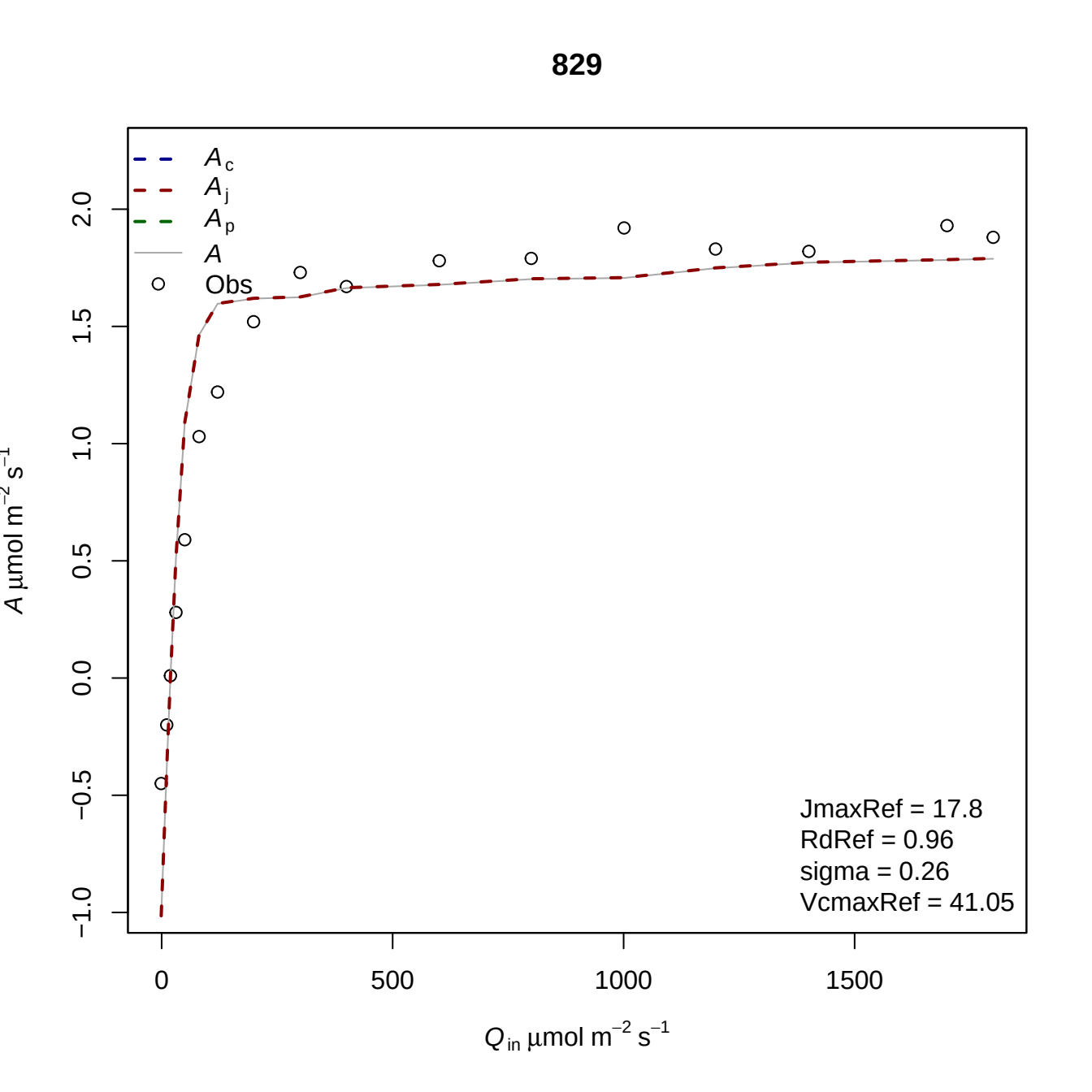

824
A_c
A_j
A_p
A
Obs
JmaxRef = 46.37
RdRef = 0.86
sigma = 0.3
VcmaxRef = 120.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

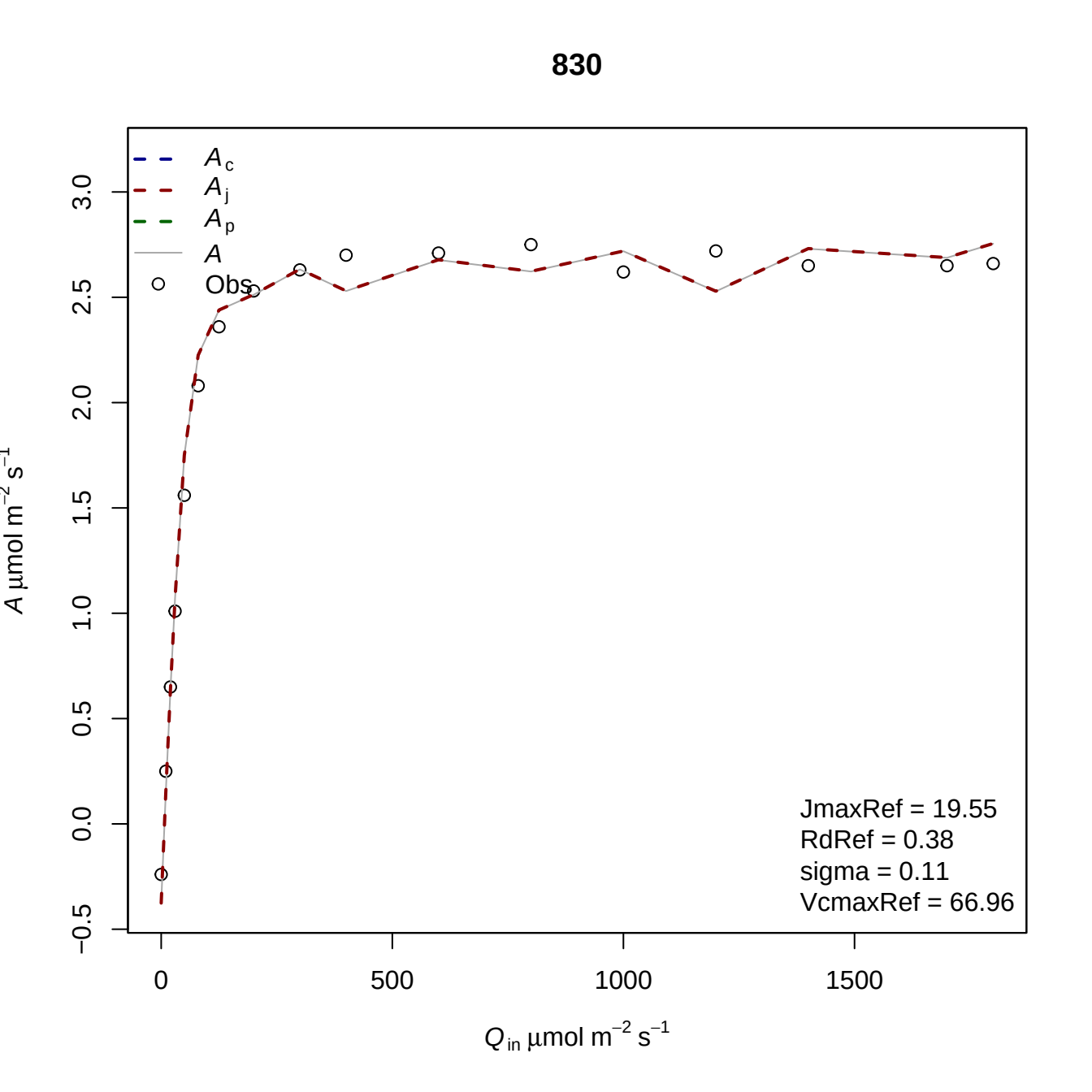

825
A_c
A_j
A_p
A
Obs
JmaxRef = 62.13
RdRef = 1.92
sigma = 0.25
VcmaxRef = 151.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

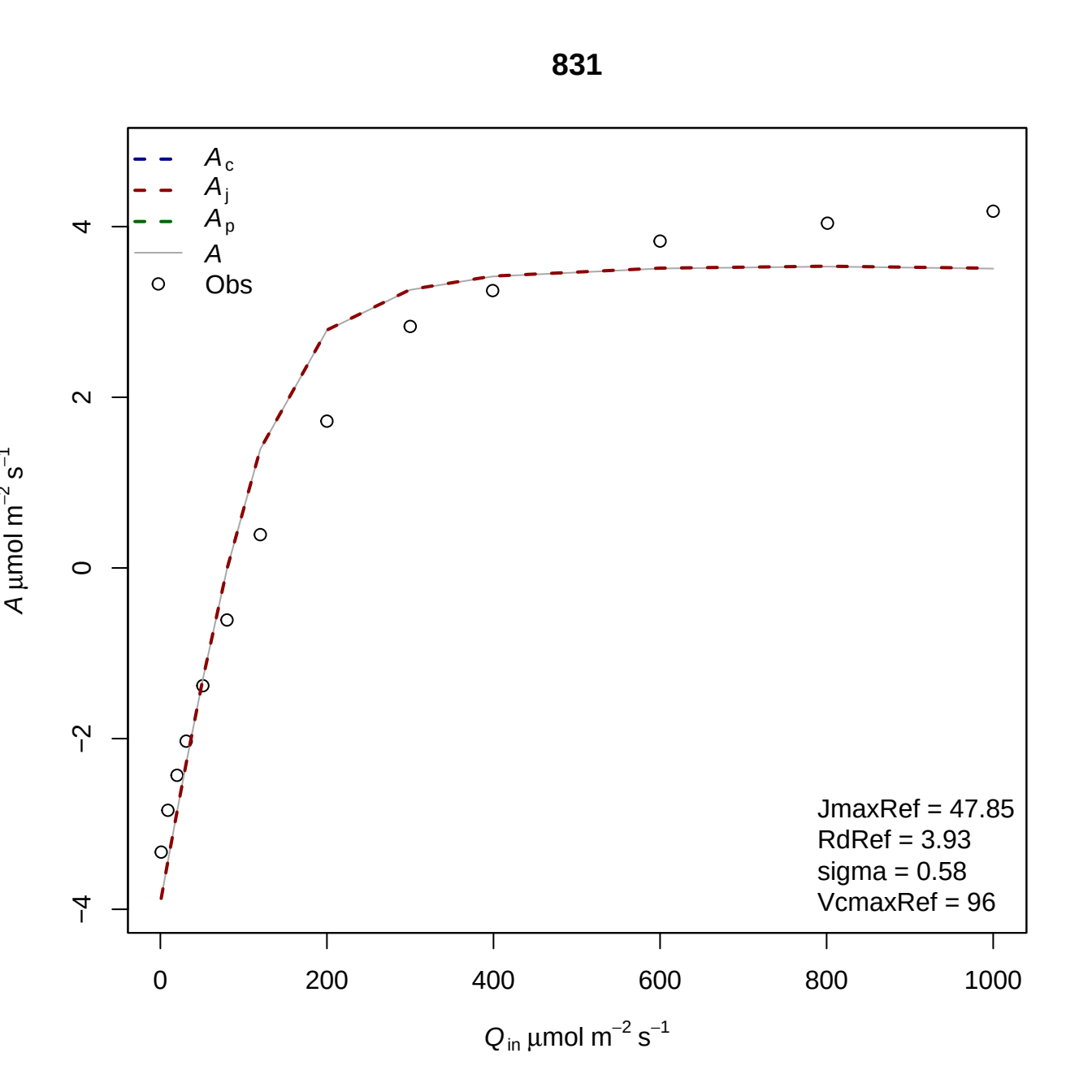

826
A_c
A_j
A_p
A
Obs
JmaxRef = 89.59
RdRef = 3.03
sigma = 1.12
VcmaxRef = 206.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

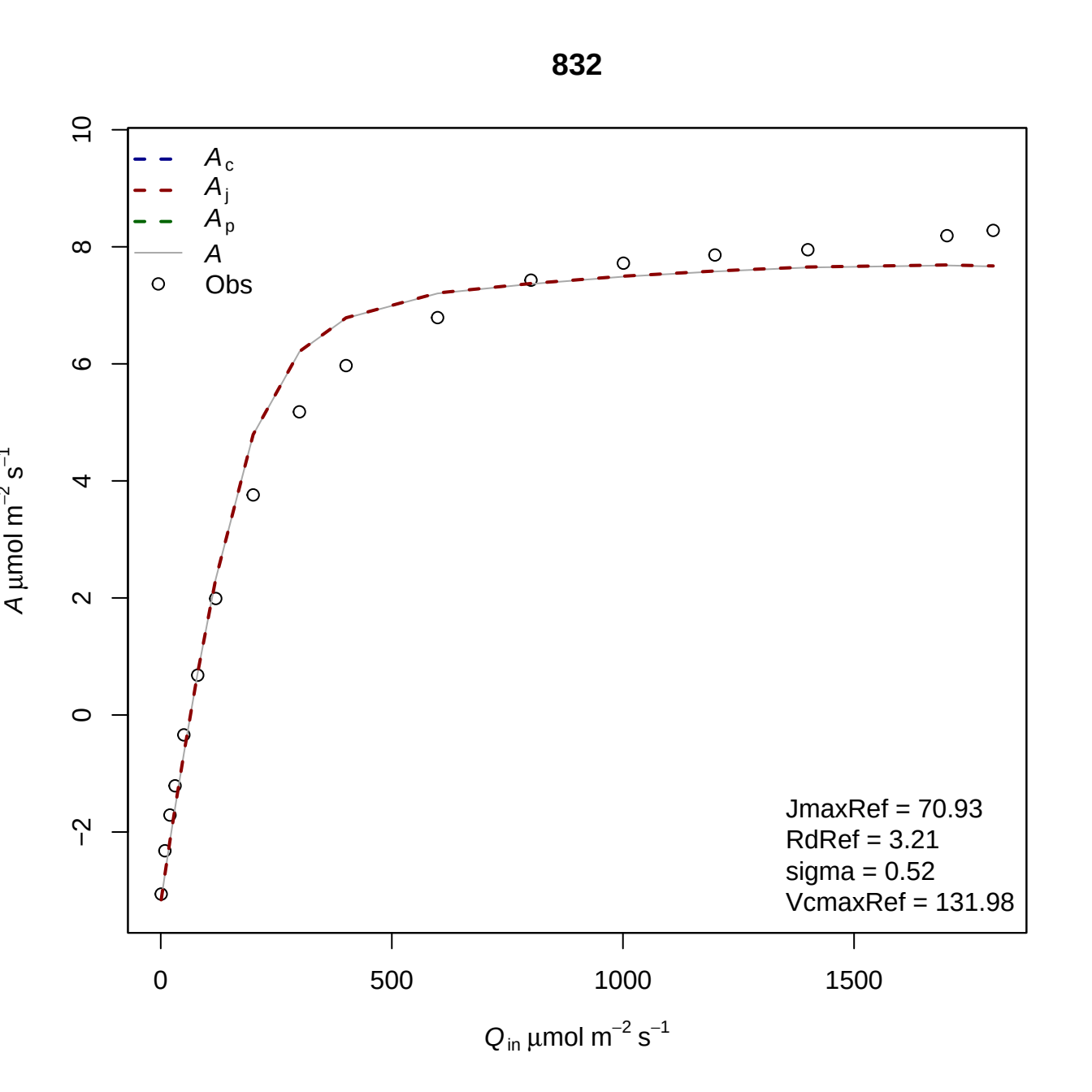

827
A_c
A_j
A_p
A
Obs
JmaxRef = 54.8
RdRef = 2.54
sigma = 0.73
VcmaxRef = 182.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

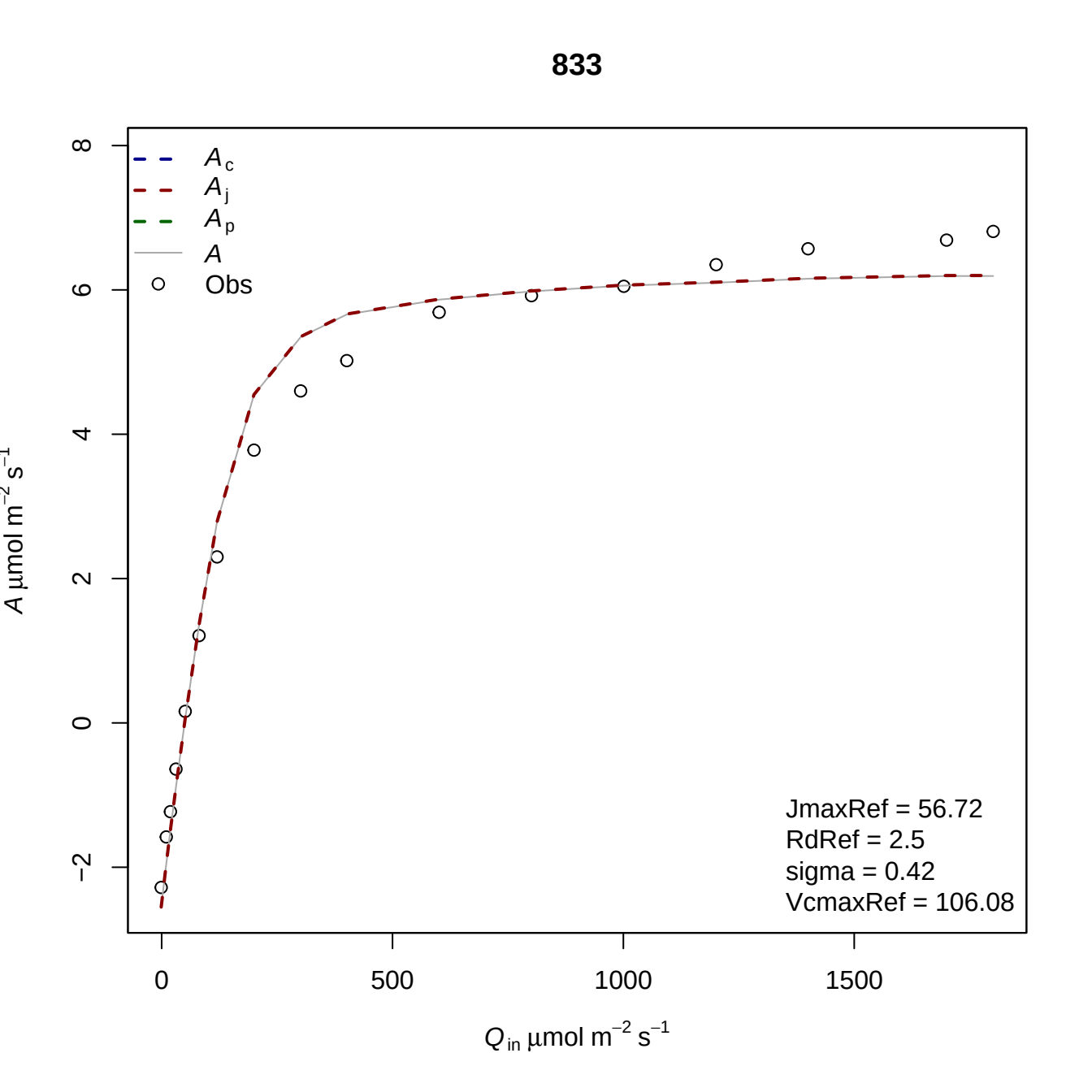

828
A_c
A_j
A_p
A
Obs
JmaxRef = 80.69
RdRef = 1.91
sigma = 0.45
VcmaxRef = 158.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

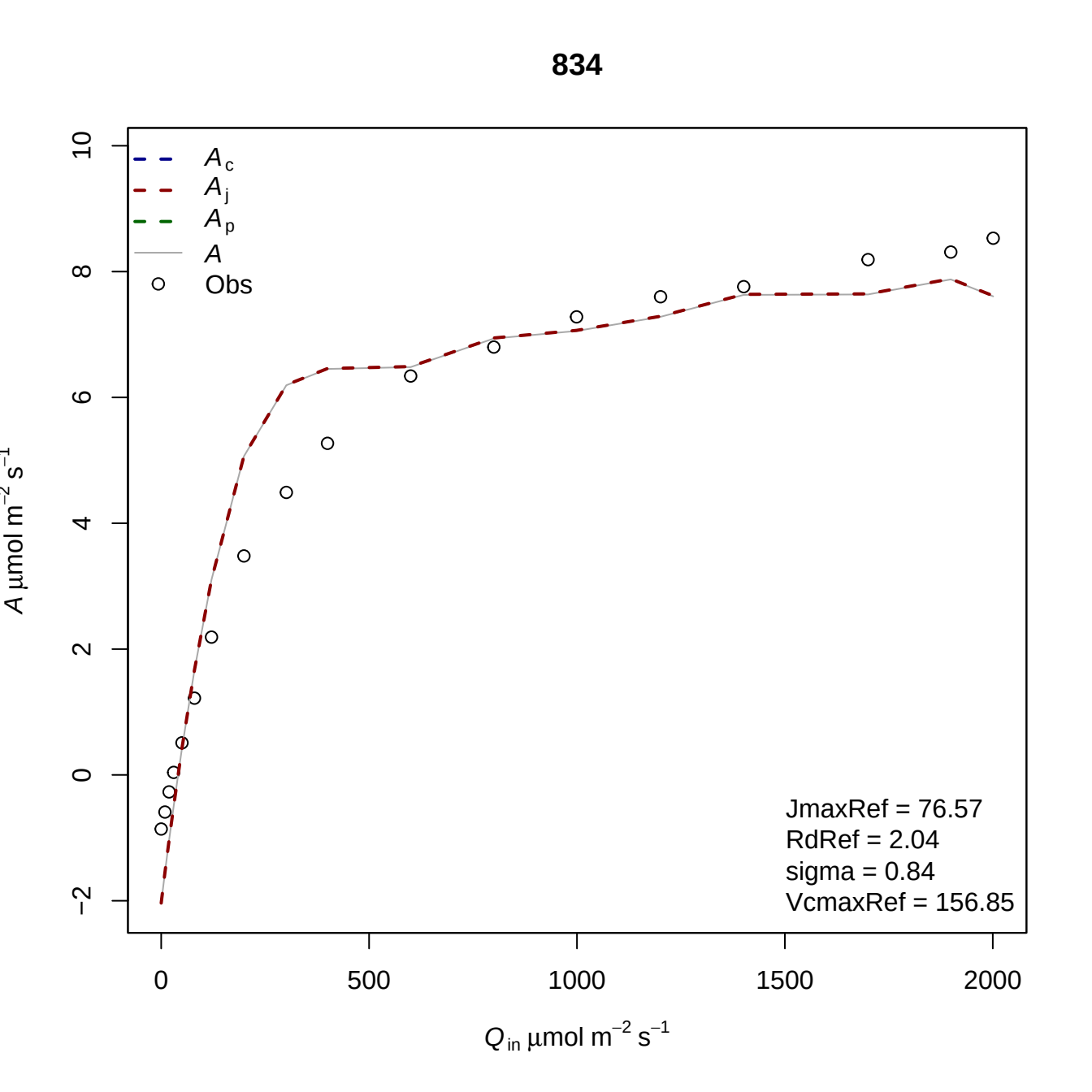

829
A_c
A_j
A_p
A
Obs
JmaxRef = 17.8
RdRef = 0.96
sigma = 0.26
VcmaxRef = 41.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

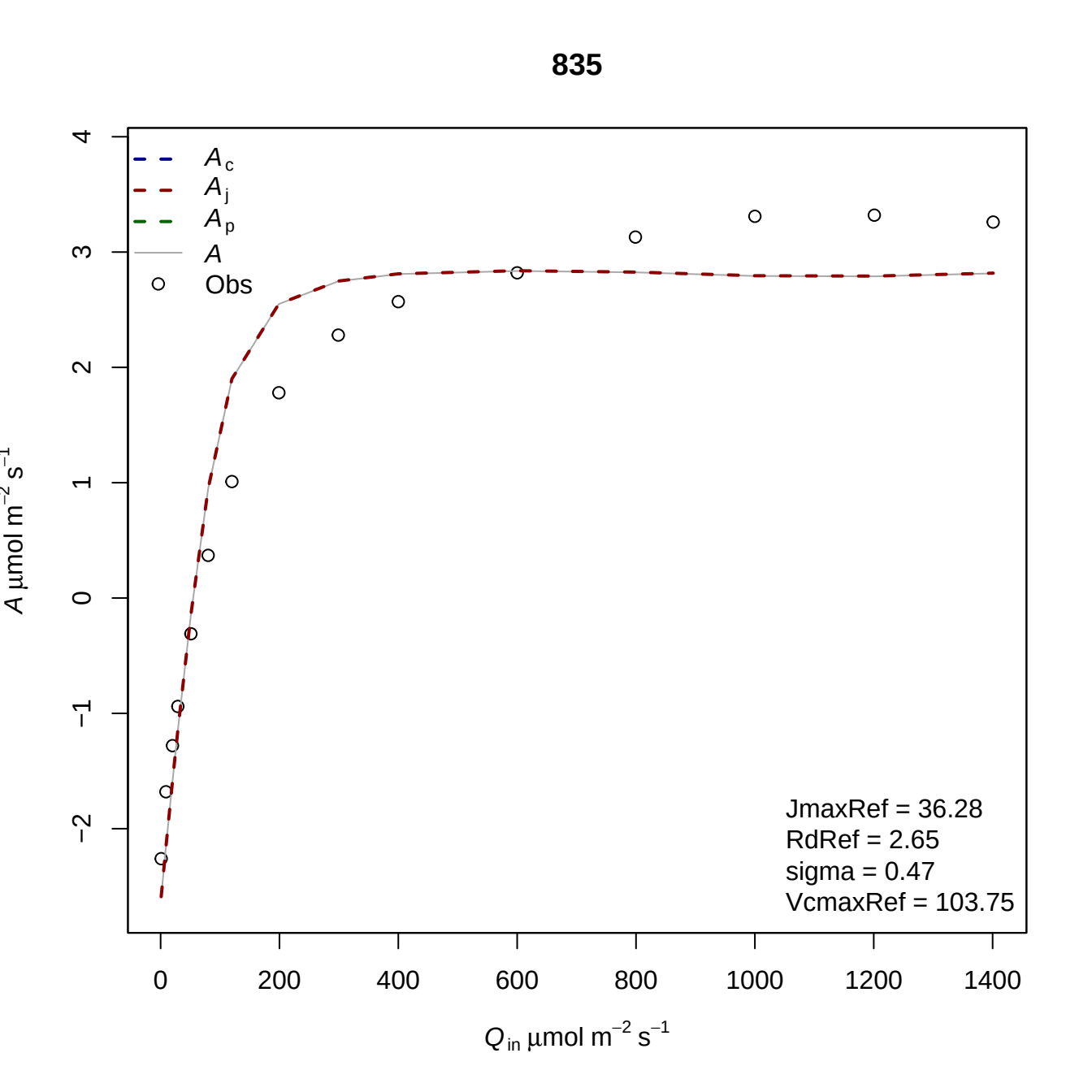

830
A_c
A_j
A_p
A
Obs
JmaxRef = 19.55
RdRef = 0.38
sigma = 0.11
VcmaxRef = 66.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

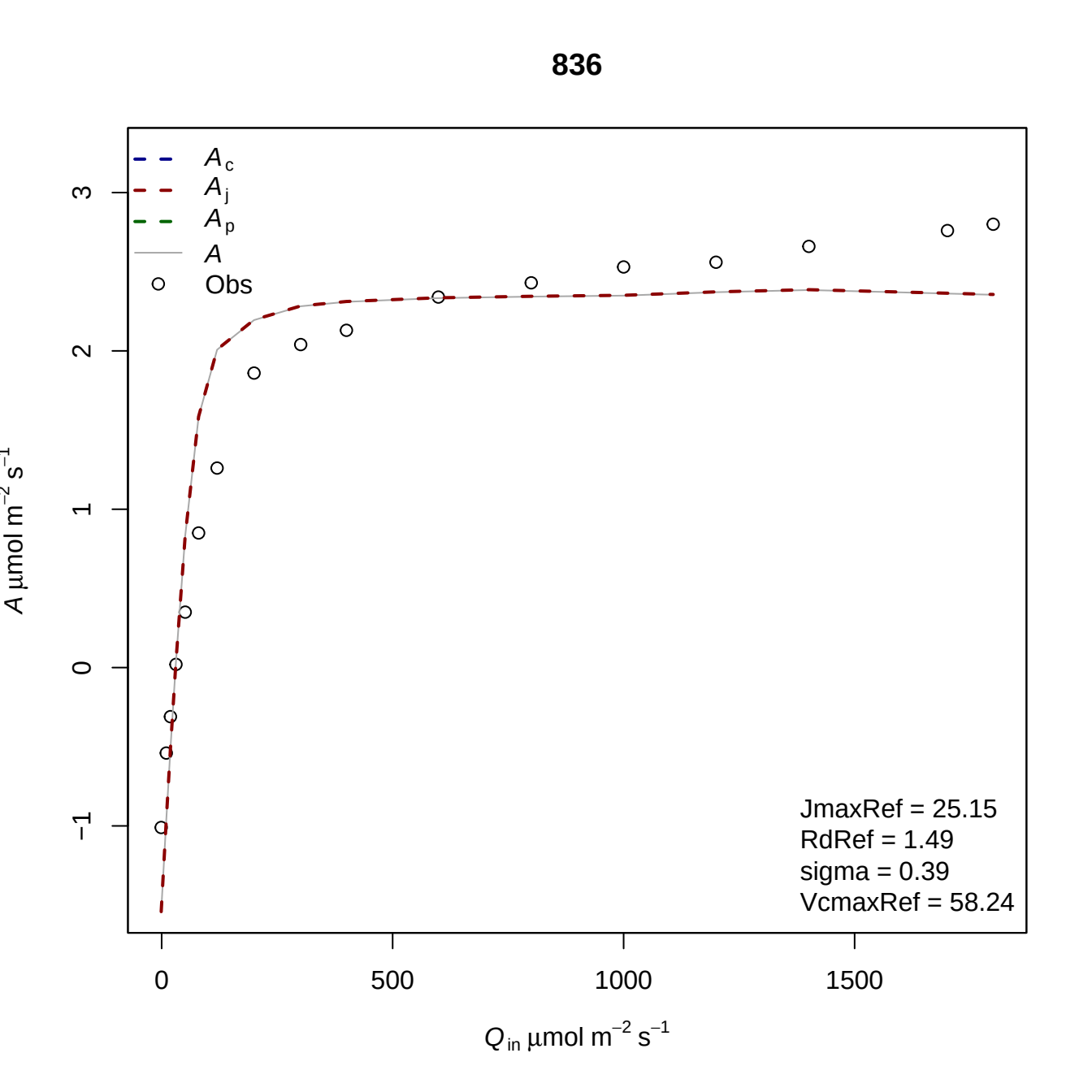

831
A_c
A_j
A_p
A
Obs
JmaxRef = 47.85
RdRef = 3.93
sigma = 0.58
VcmaxRef = 96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
200
400
600
800
1000

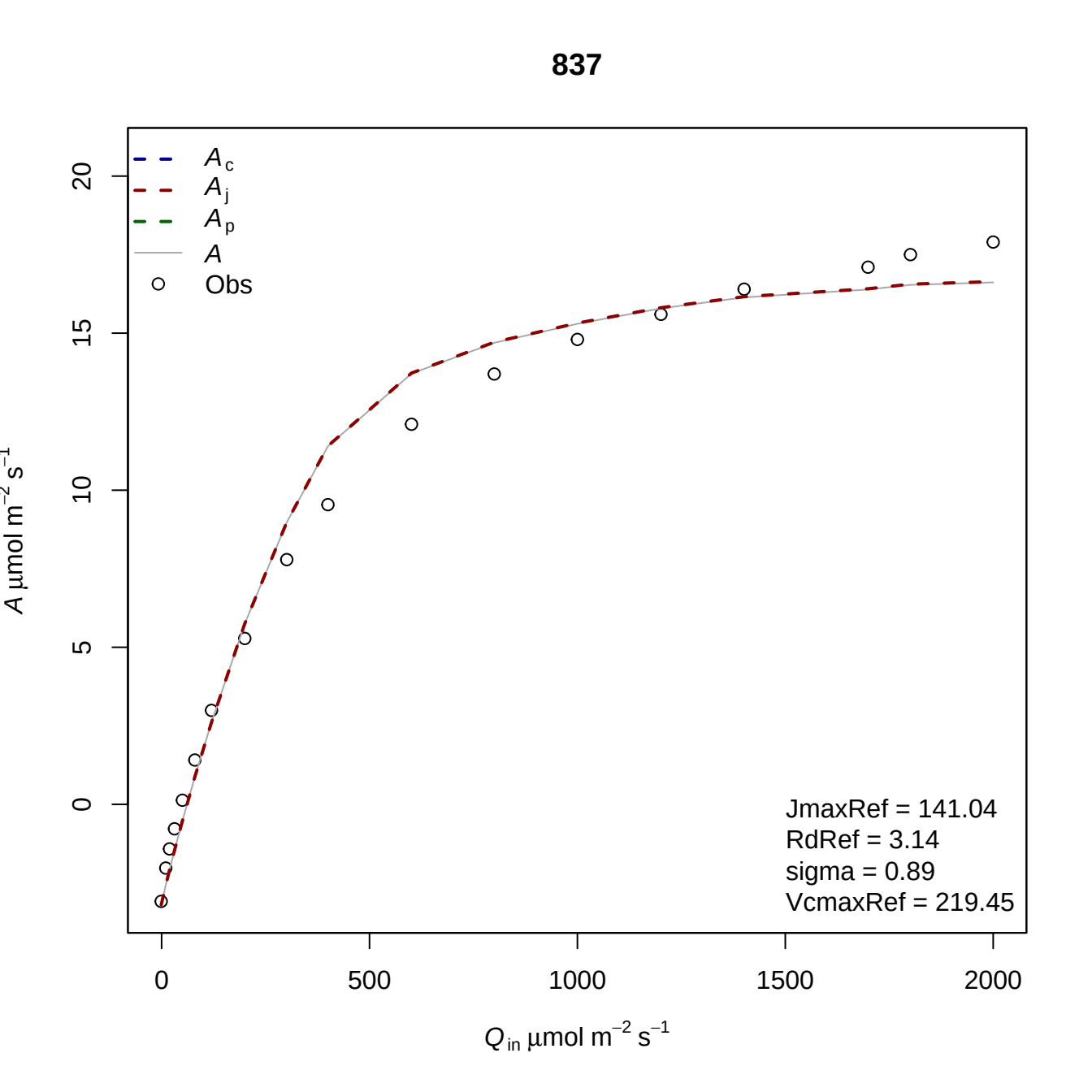

832
A_c
A_j
A_p
A
Obs
JmaxRef = 70.93
RdRef = 3.21
sigma = 0.52
VcmaxRef = 131.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

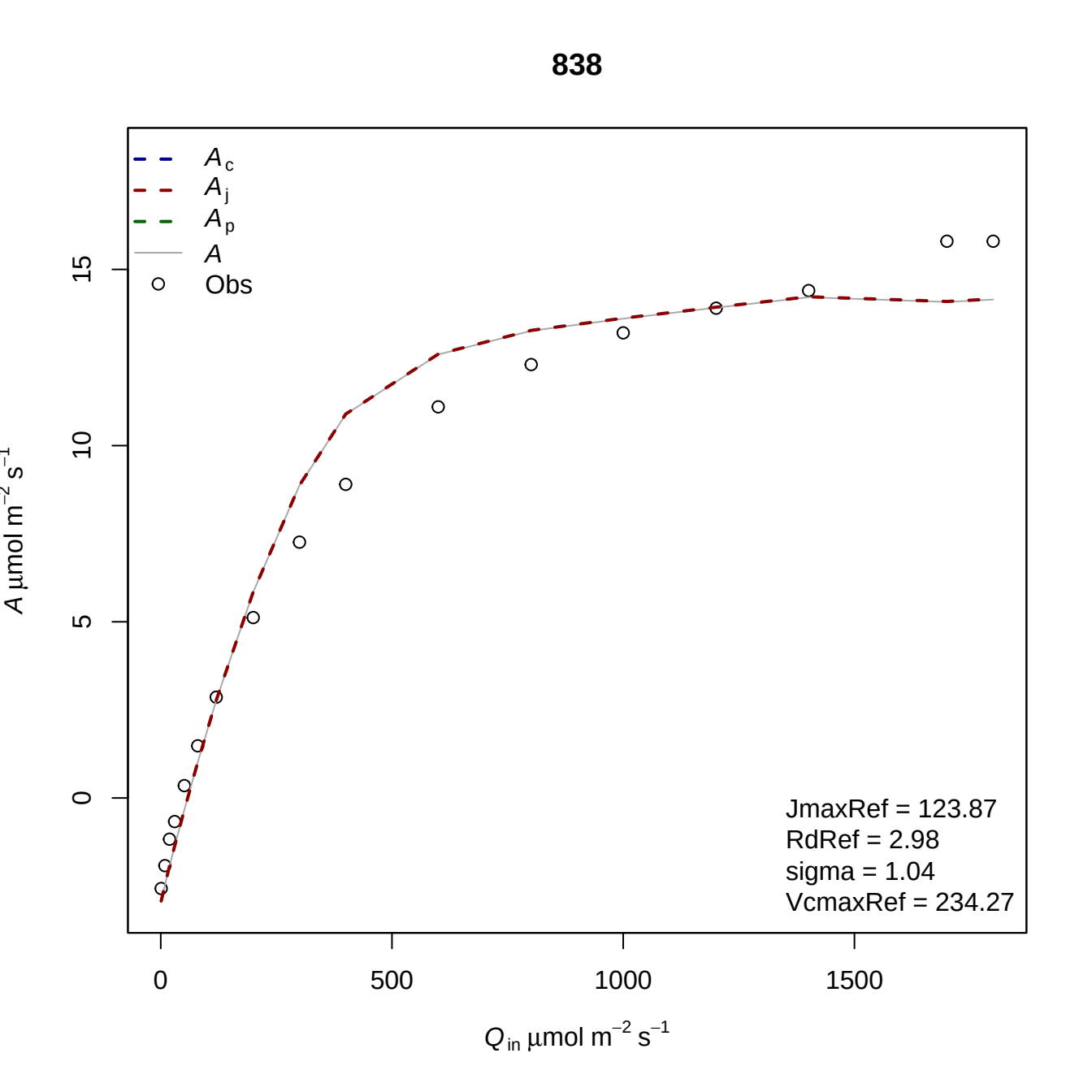

833
A_c
A_j
A_p
A
Obs
JmaxRef = 56.72
RdRef = 2.5
sigma = 0.42
VcmaxRef = 106.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

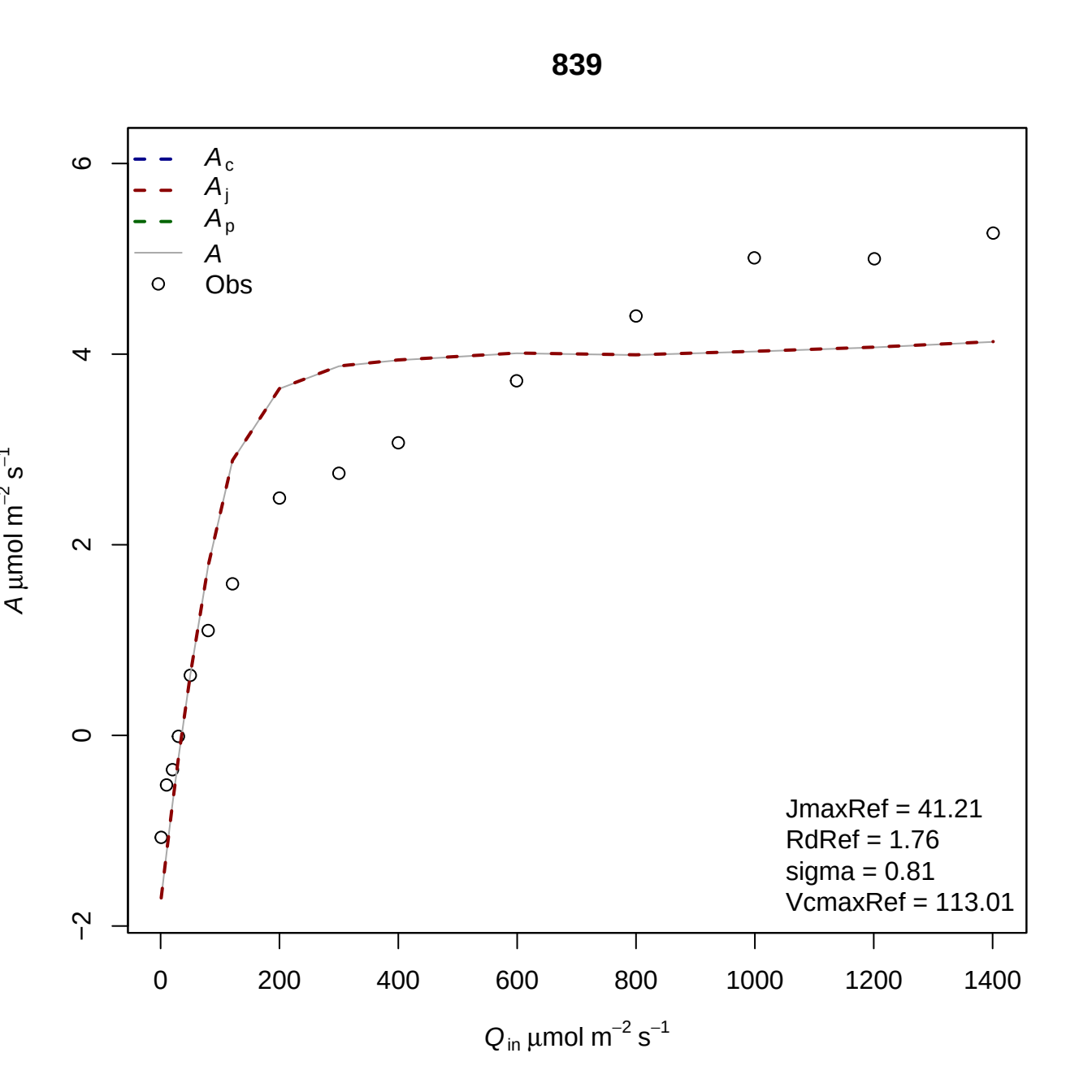

834
A_c
A_j
A_p
A
Obs
JmaxRef = 76.57
RdRef = 2.04
sigma = 0.84
VcmaxRef = 156.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

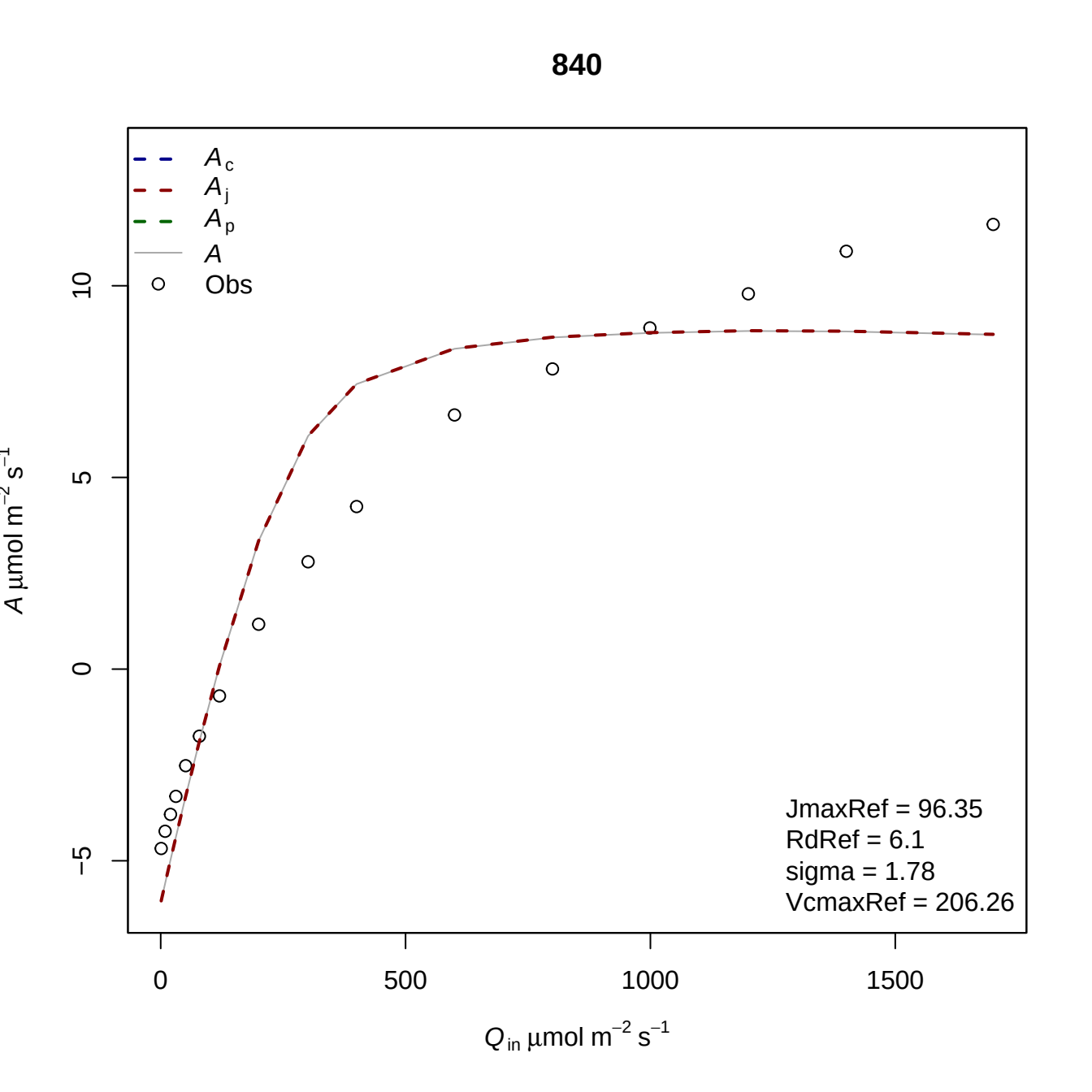

835
A_c
A_j
A_p
A
Obs
JmaxRef = 36.28
RdRef = 2.65
sigma = 0.47
VcmaxRef = 103.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

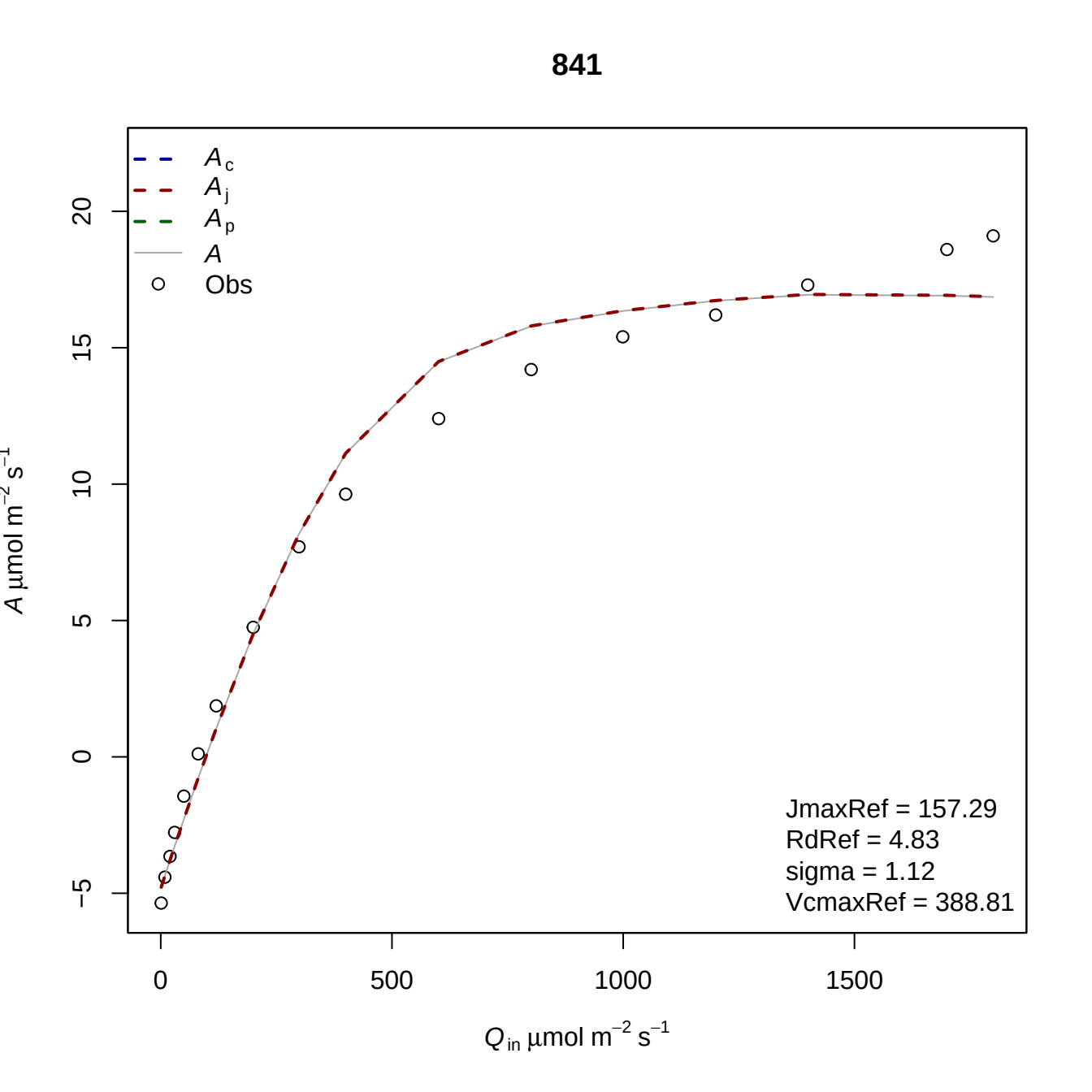

836
A_c
A_j
A_p
A
Obs
JmaxRef = 25.15
RdRef = 1.49
sigma = 0.39
VcmaxRef = 58.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

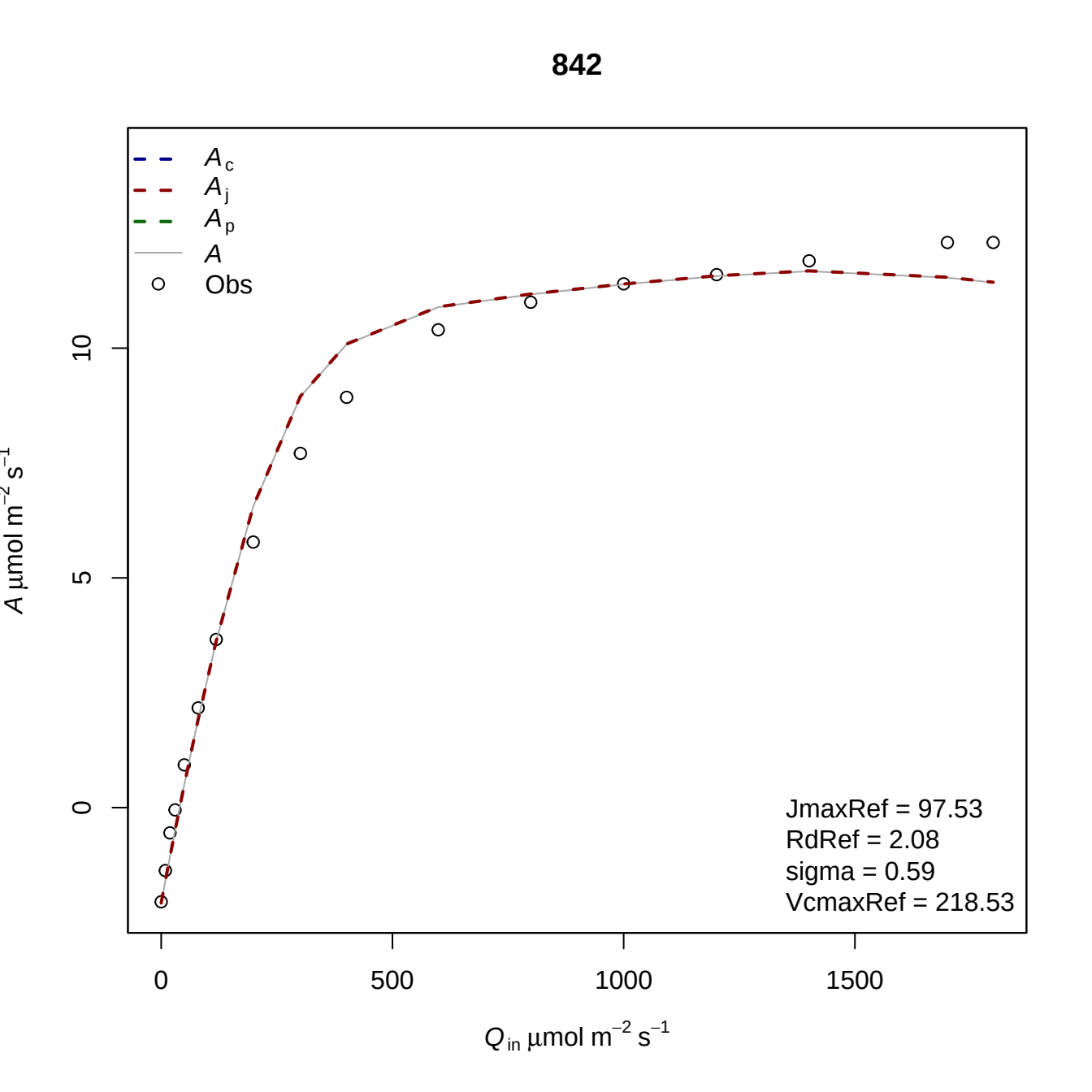

837
A_c
A_j
A_p
A
Obs
JmaxRef = 141.04
RdRef = 3.14
sigma = 0.89
VcmaxRef = 219.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

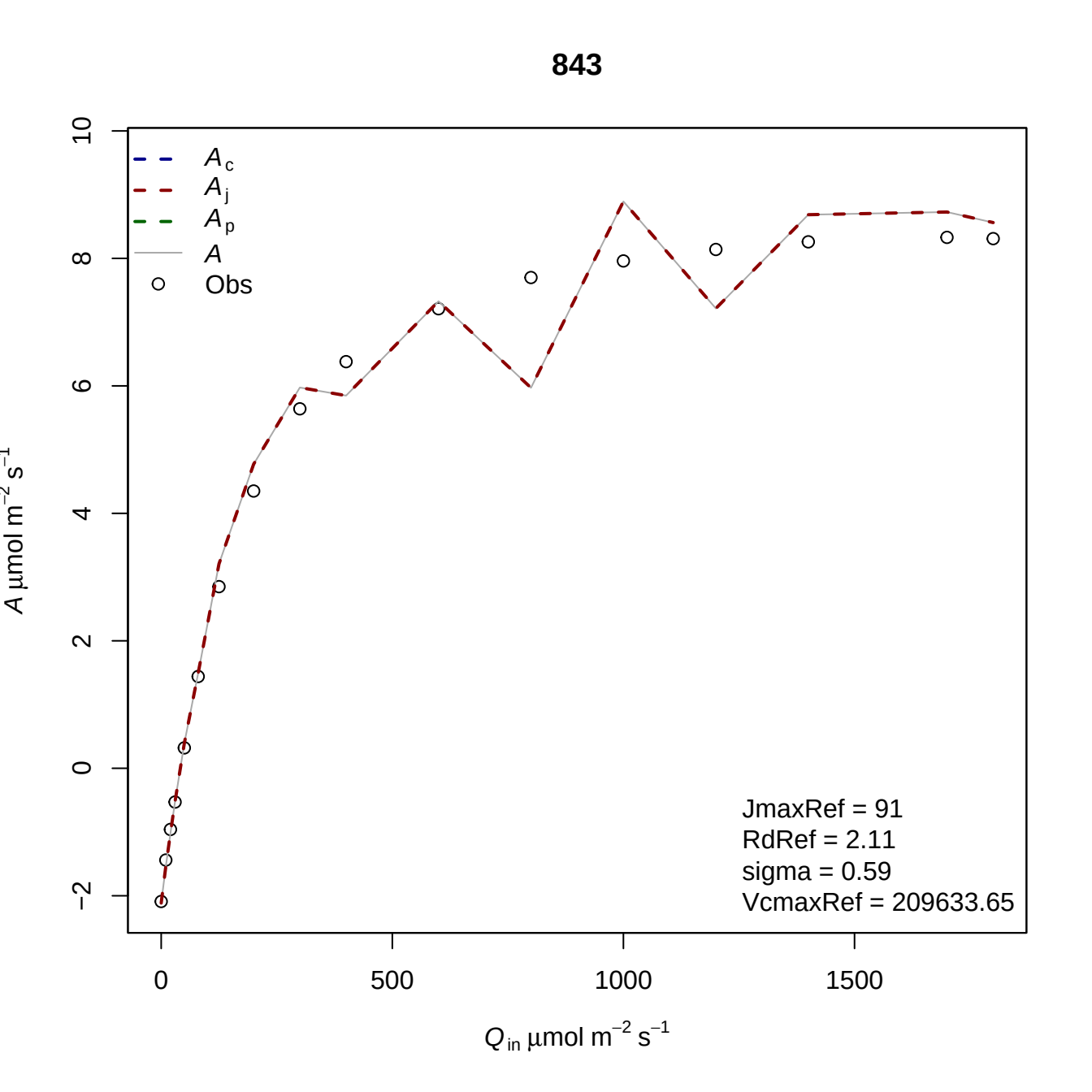

838
A_c
A_j
A_p
A
Obs
JmaxRef = 123.87
RdRef = 2.98
sigma = 1.04
VcmaxRef = 234.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

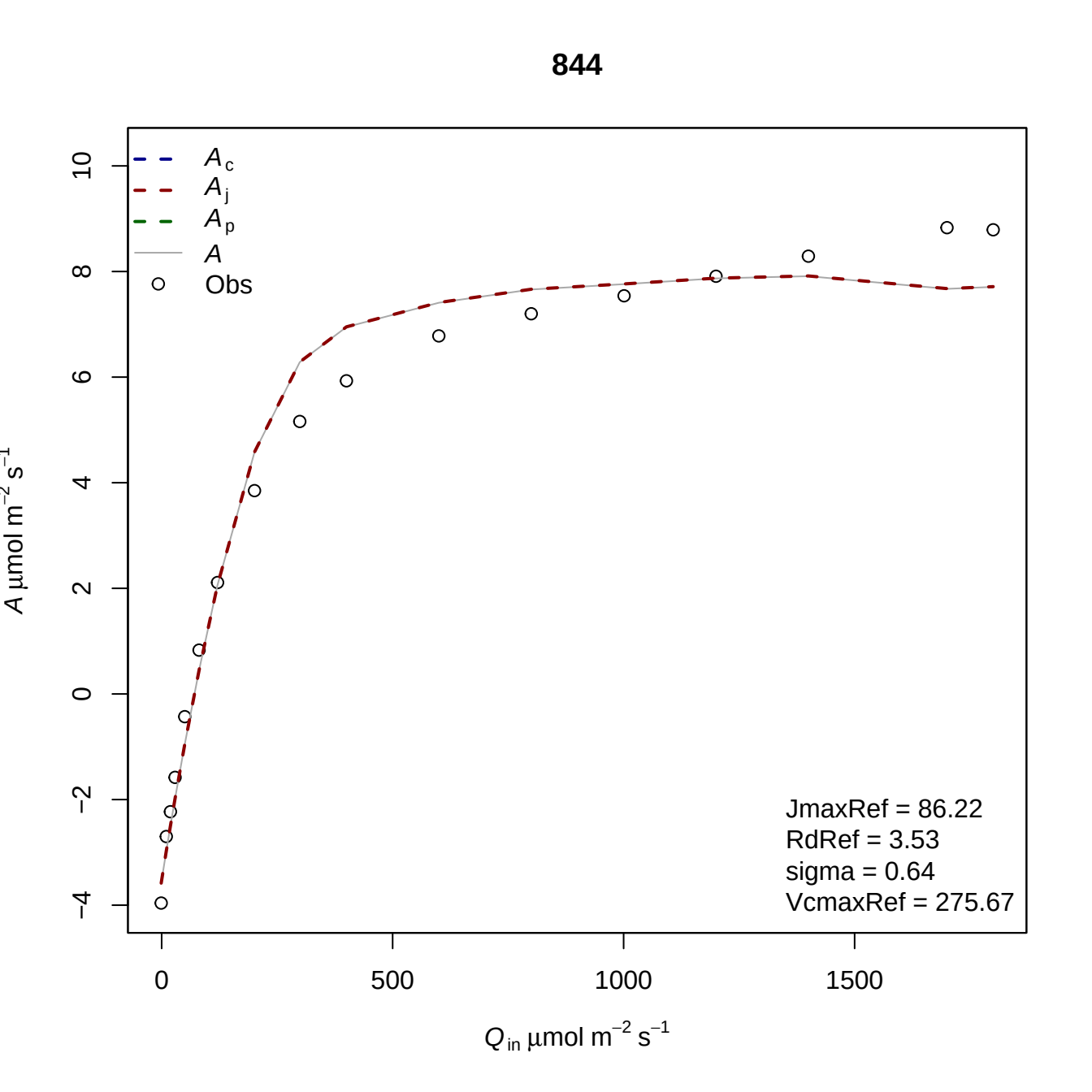

839
A_c
A_j
A_p
A
Obs
JmaxRef = 41.21
RdRef = 1.76
sigma = 0.81
VcmaxRef = 113.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

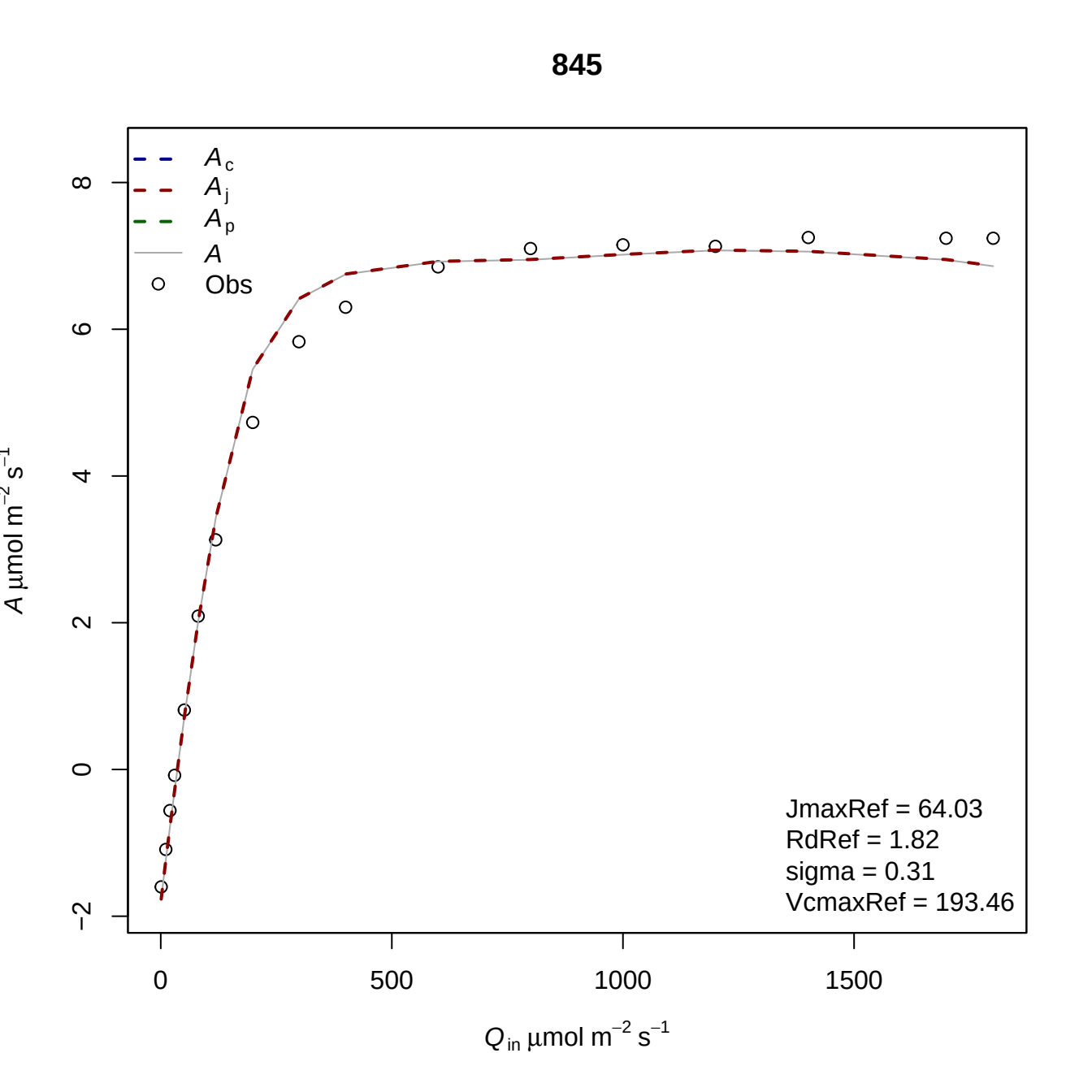

840
A_c
A_j
A_p
A
Obs
JmaxRef = 96.35
RdRef = 6.1
sigma = 1.78
VcmaxRef = 206.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

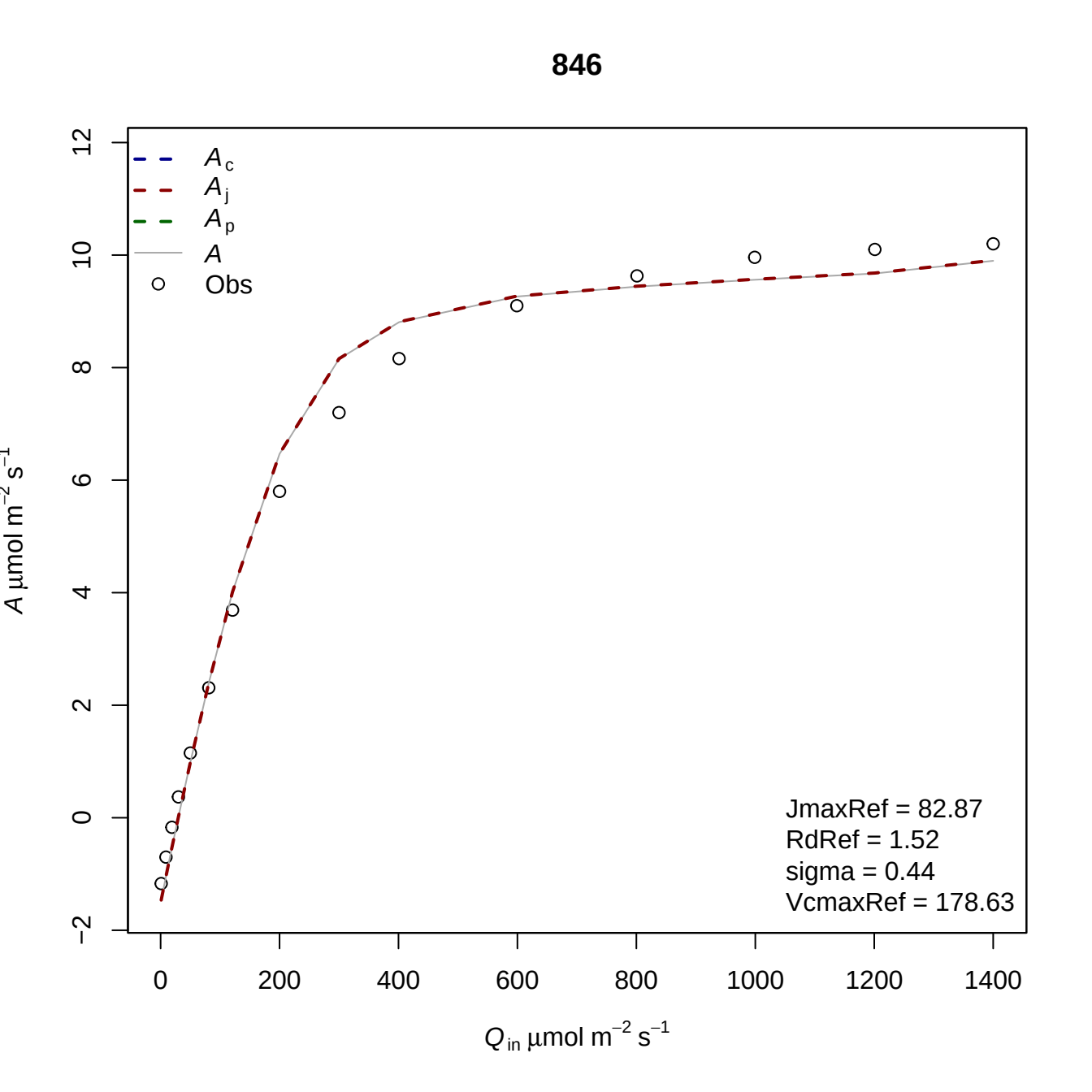

841
A_c
A_j
A_p
A
Obs
JmaxRef = 157.29
RdRef = 4.83
sigma = 1.12
VcmaxRef = 388.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

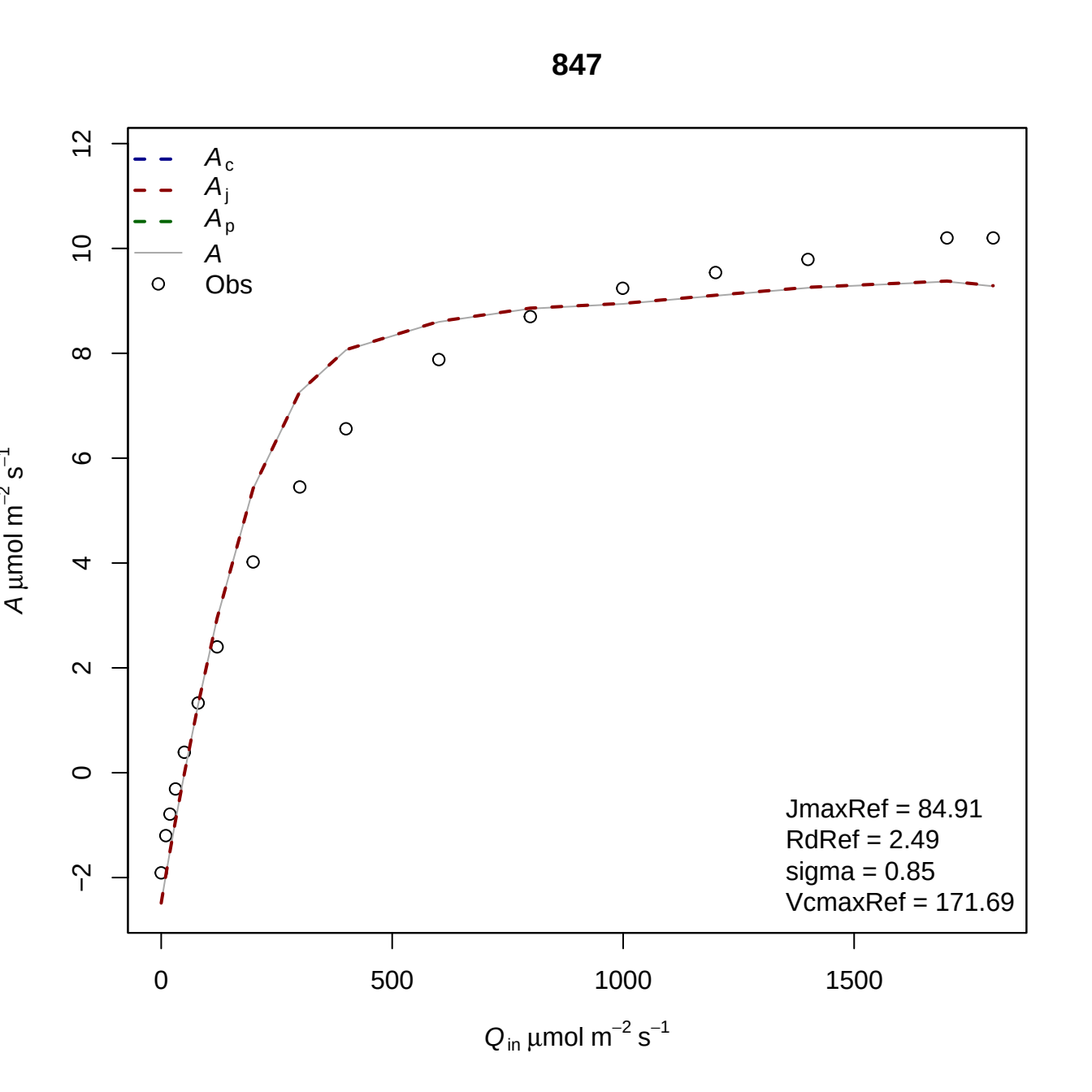

842
A_c
A_j
A_p
A
Obs
JmaxRef = 97.53
RdRef = 2.08
sigma = 0.59
VcmaxRef = 218.53
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

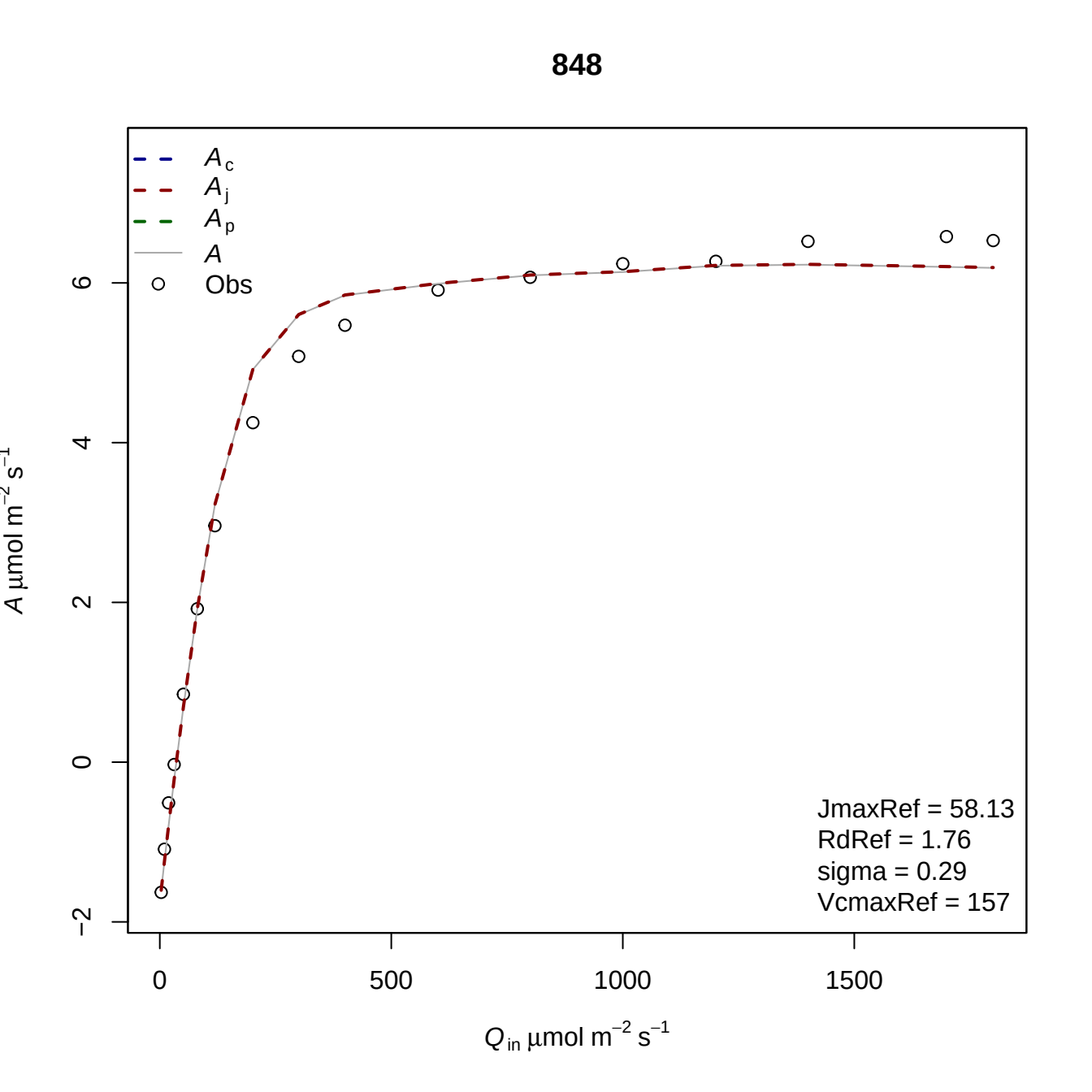

843
A_c
A_j
A_p
A
Obs
JmaxRef = 91
RdRef = 2.11
sigma = 0.59
VcmaxRef = 209633.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

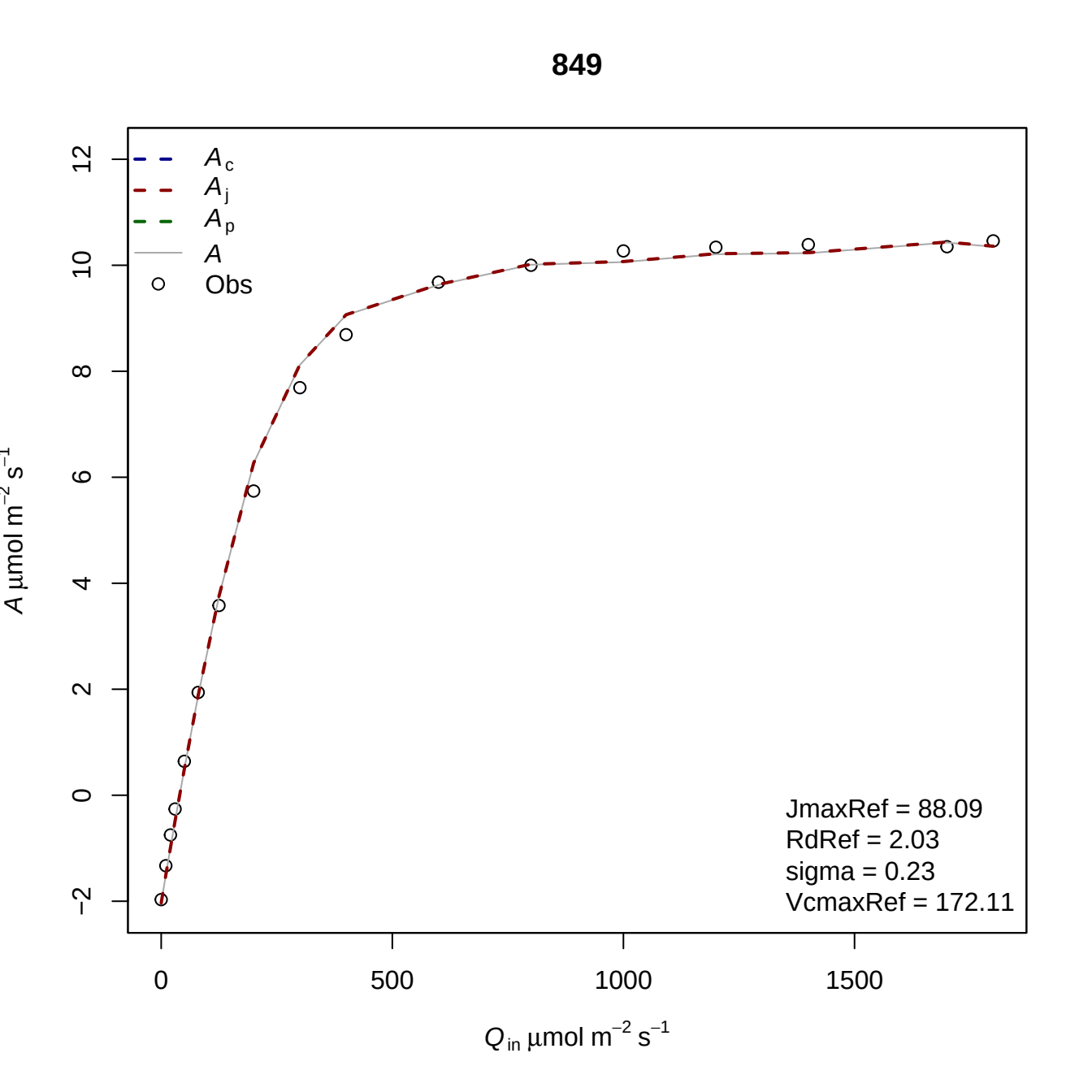

844
A_c
A_j
A_p
A
Obs
JmaxRef = 86.22
RdRef = 3.53
sigma = 0.64
VcmaxRef = 275.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

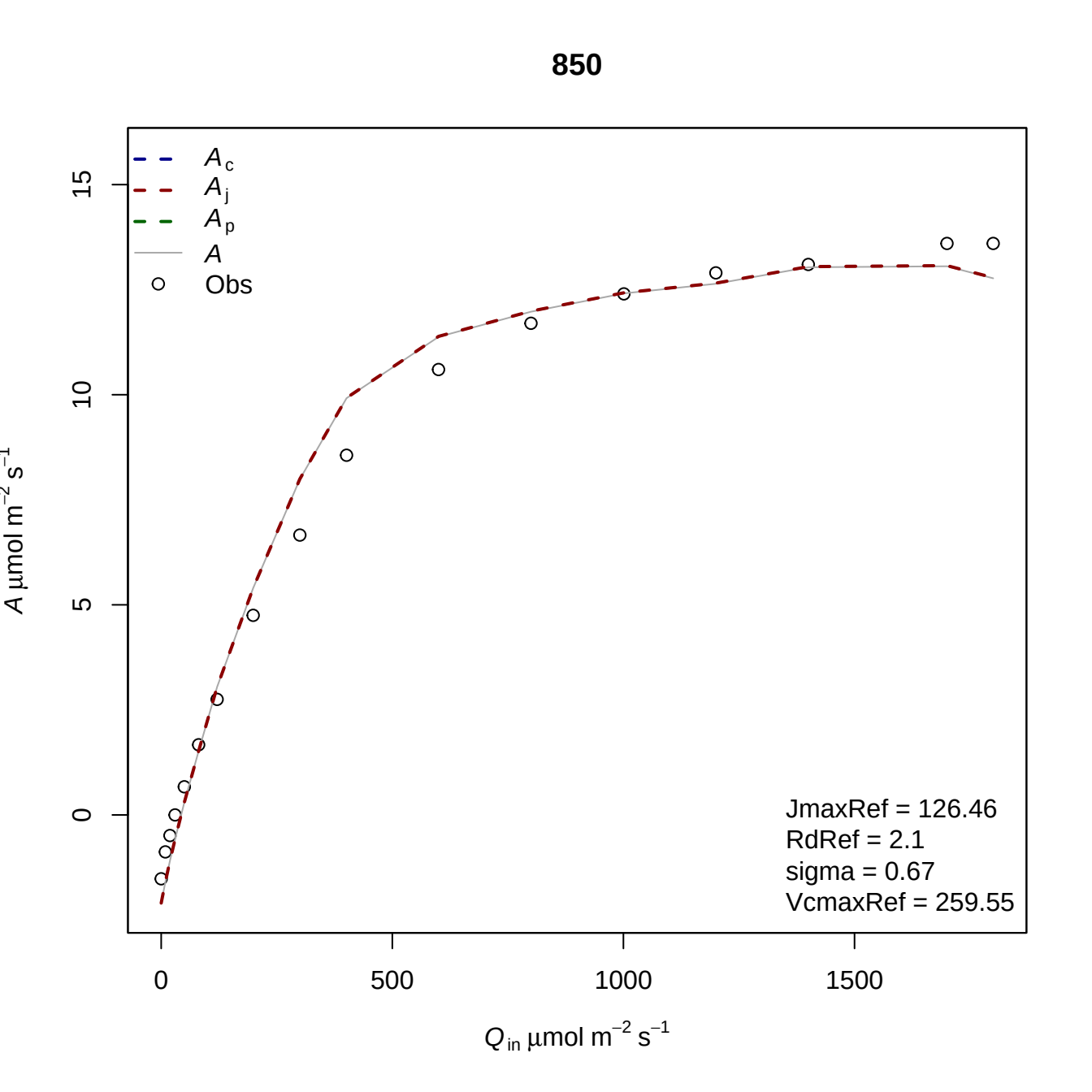

845
A_c
A_j
A_p
A
Obs
JmaxRef = 64.03
RdRef = 1.82
sigma = 0.31
VcmaxRef = 193.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

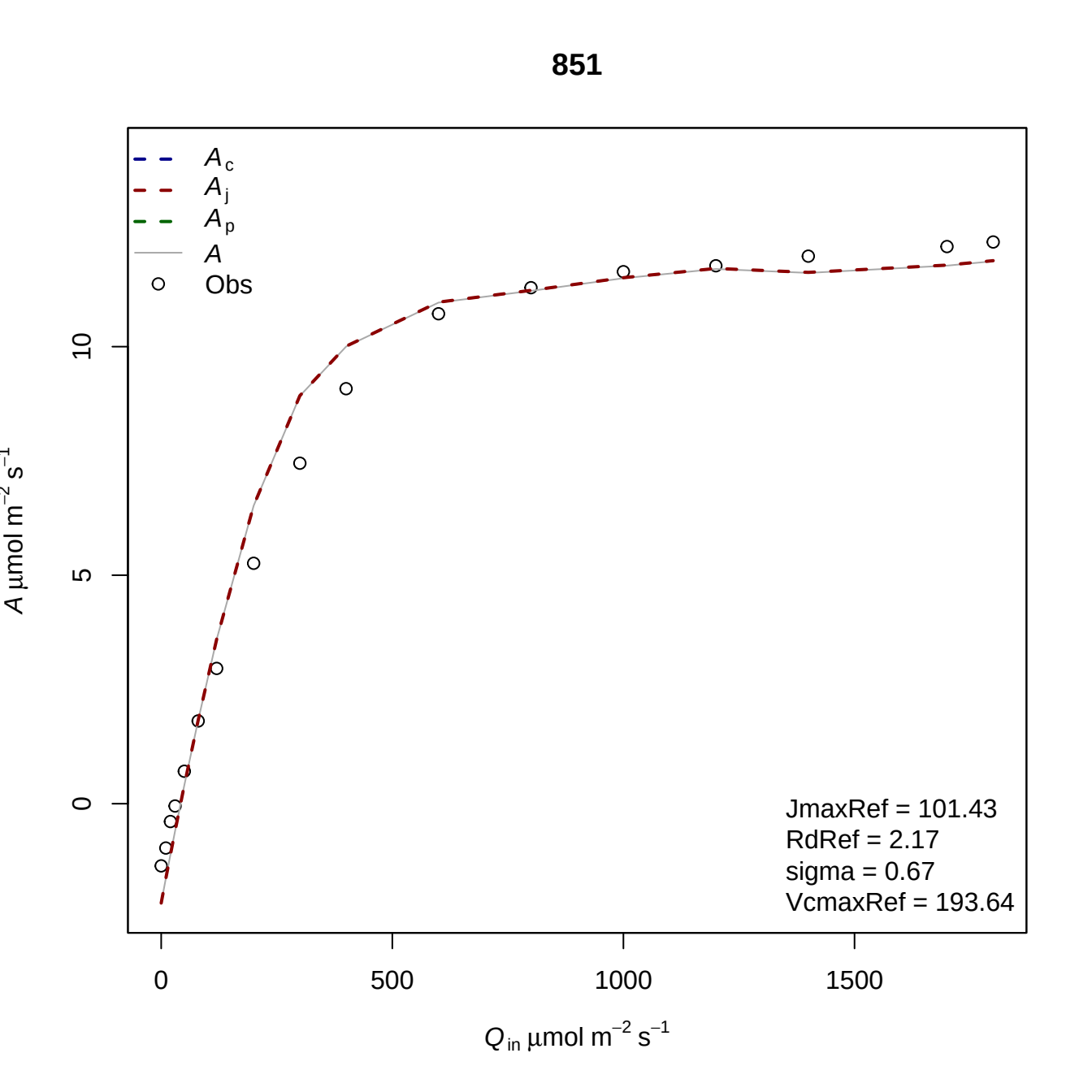

846
A_c
A_j
A_p
A
Obs
JmaxRef = 82.87
RdRef = 1.52
sigma = 0.44
VcmaxRef = 178.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

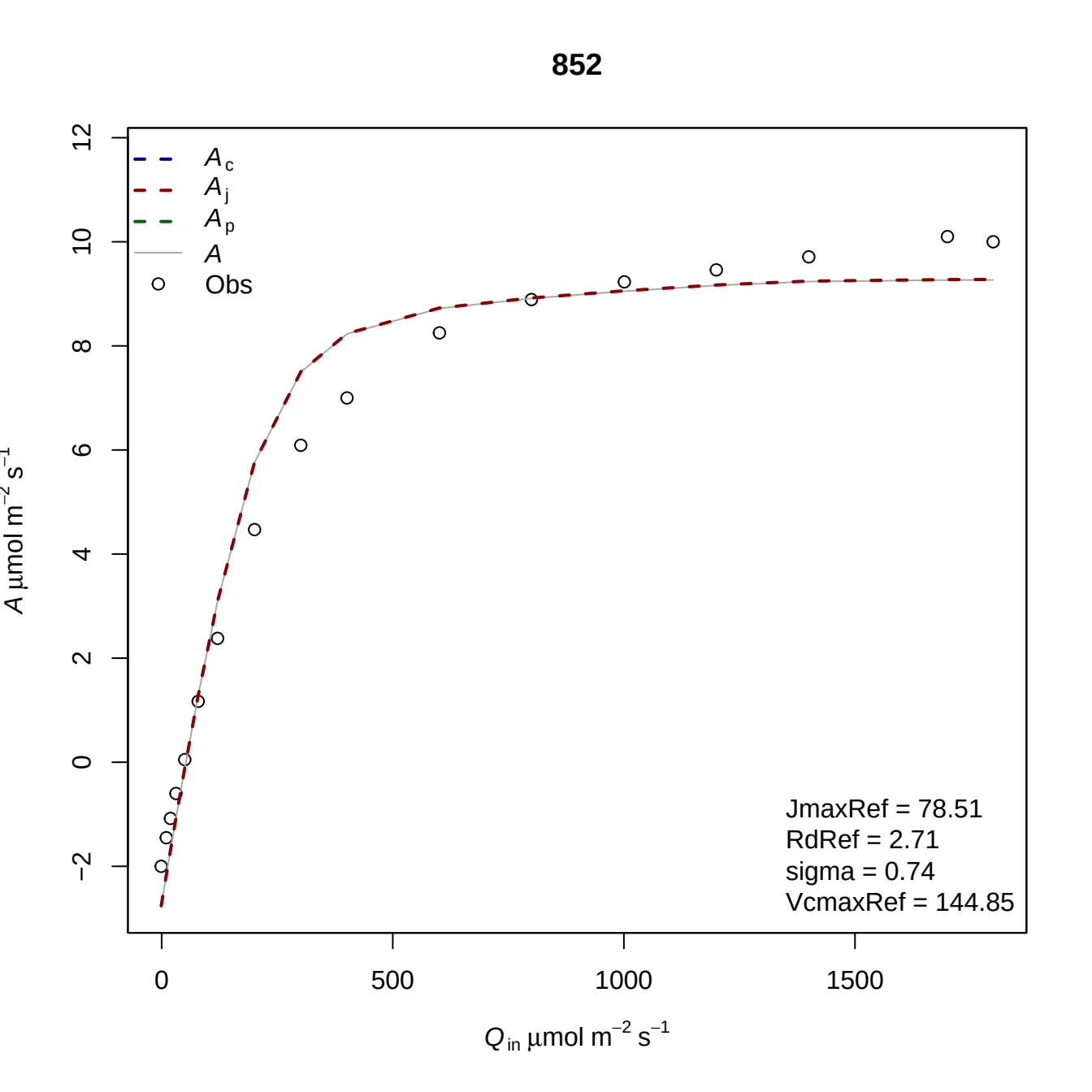

847
A_c
A_j
A_p
A
Obs
JmaxRef = 84.91
RdRef = 2.49
sigma = 0.85
VcmaxRef = 171.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

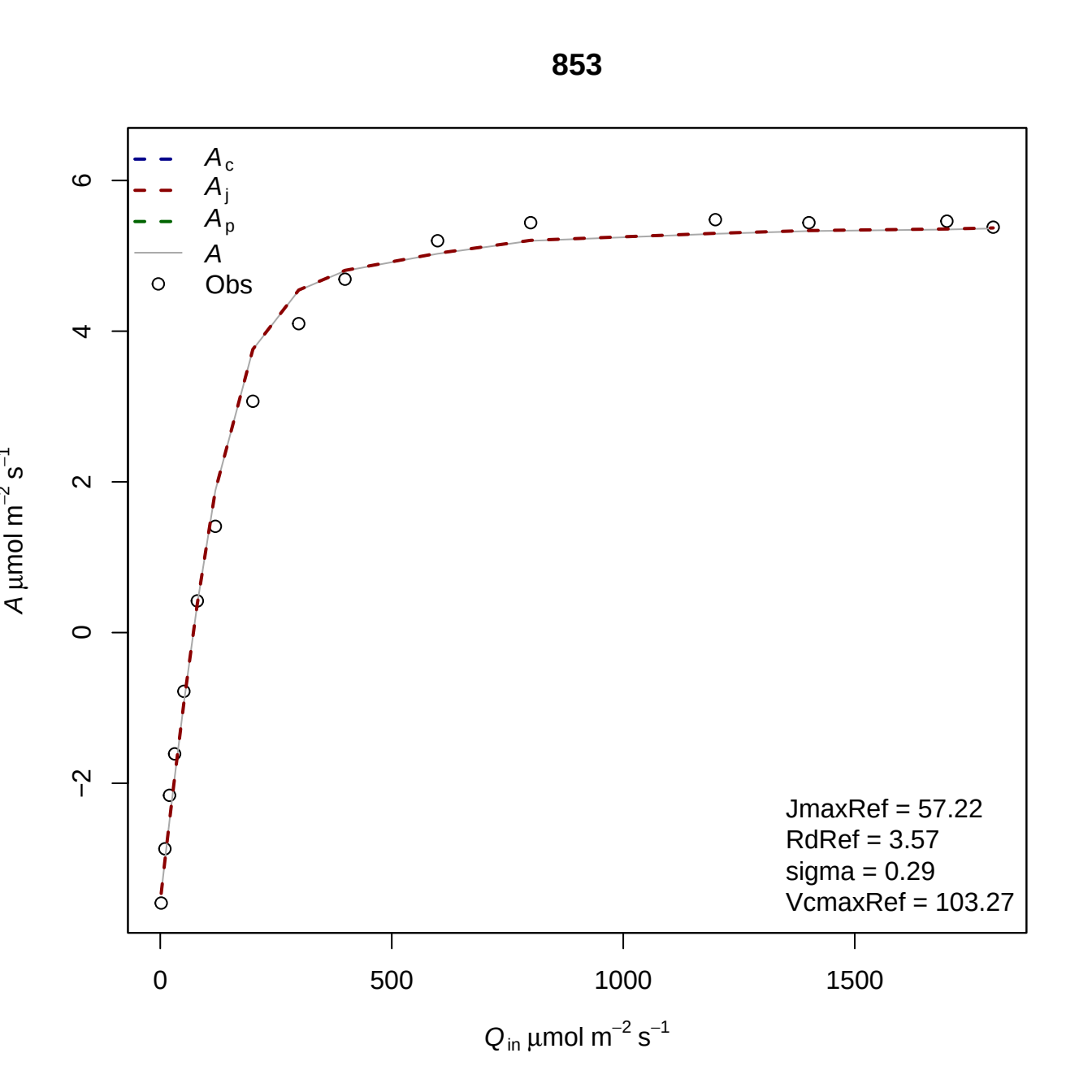

848
A_c
A_j
A_p
A
Obs
JmaxRef = 58.13
RdRef = 1.76
sigma = 0.29
VcmaxRef = 157
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

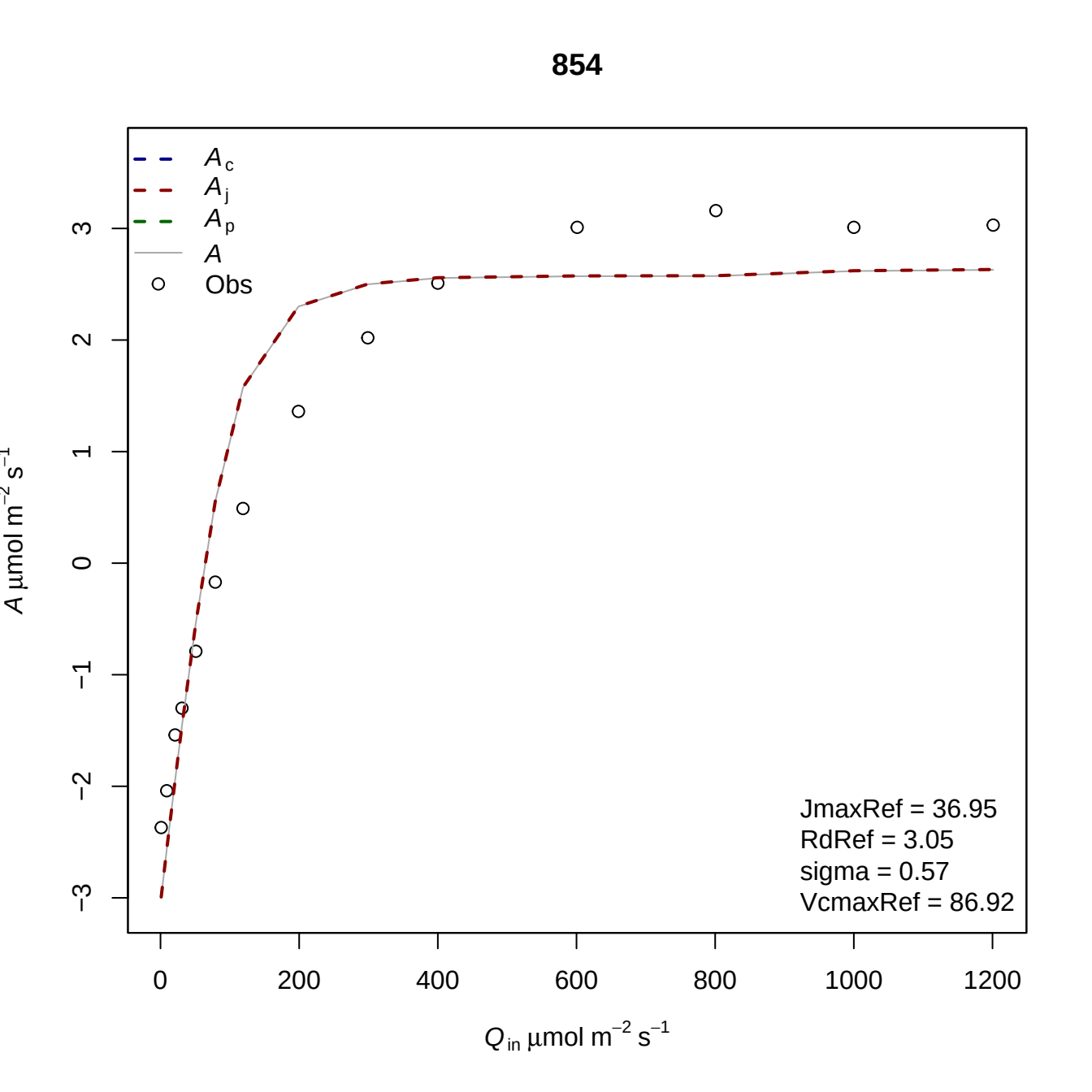

849
A_c
A_j
A_p
A
Obs
JmaxRef = 88.09
RdRef = 2.03
sigma = 0.23
VcmaxRef = 172.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

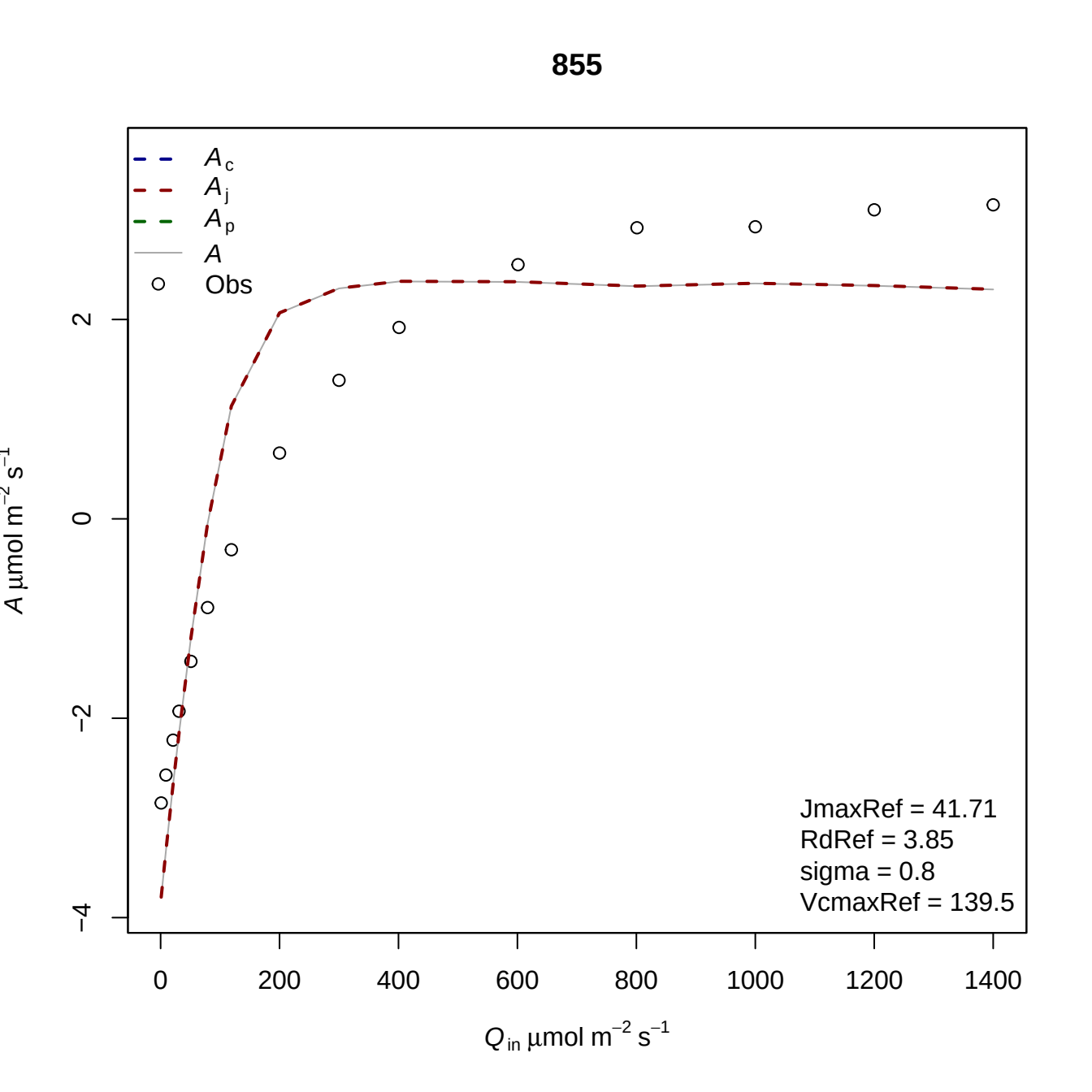

850
A_c
A_j
A_p
A
Obs
JmaxRef = 126.46
RdRef = 2.1
sigma = 0.67
VcmaxRef = 259.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

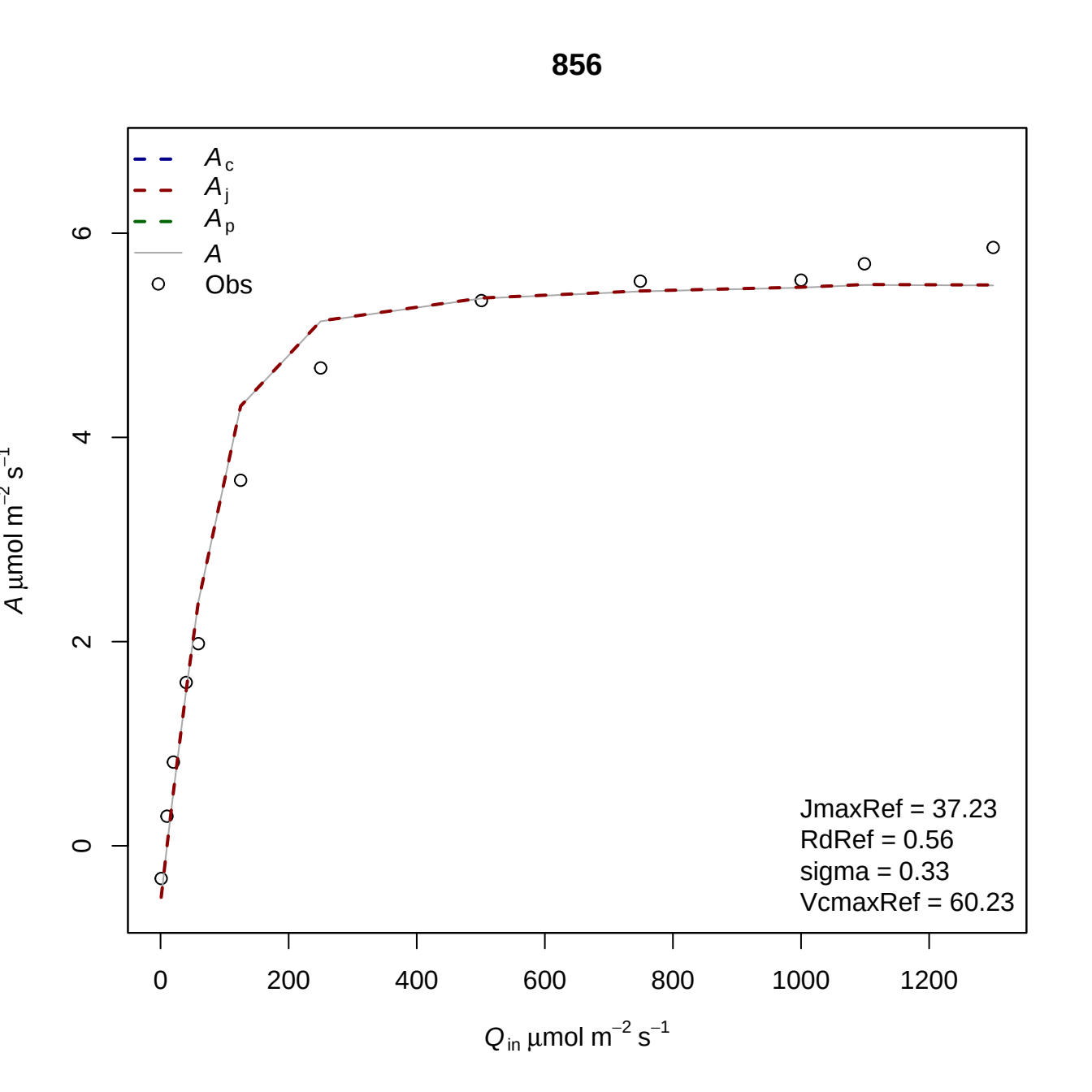

851
A_c
A_j
A_p
A
Obs
JmaxRef = 101.43
RdRef = 2.17
sigma = 0.67
VcmaxRef = 193.64
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

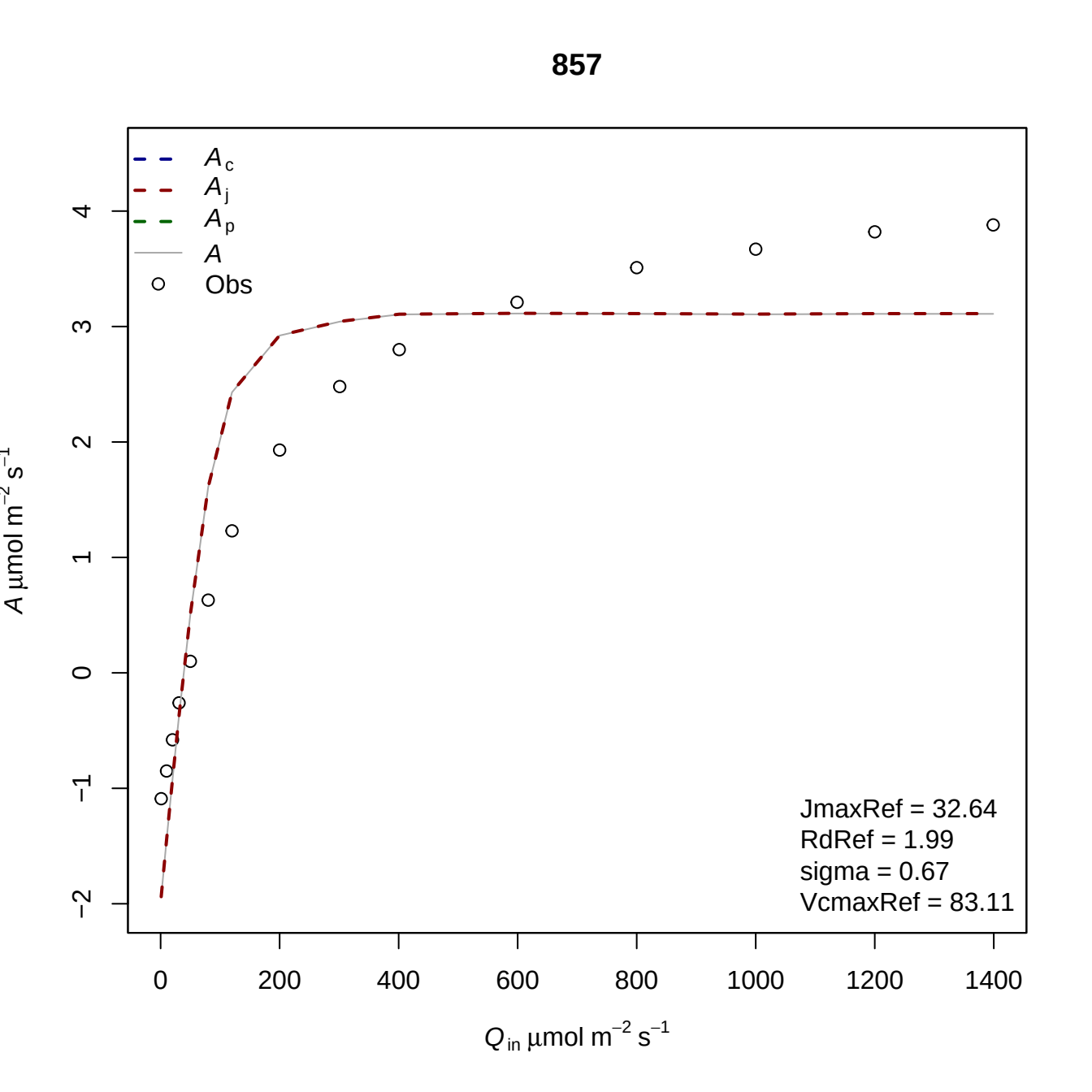

852
A_c
A_j
A_p
A
Obs
JmaxRef = 78.51
RdRef = 2.71
sigma = 0.74
VcmaxRef = 144.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

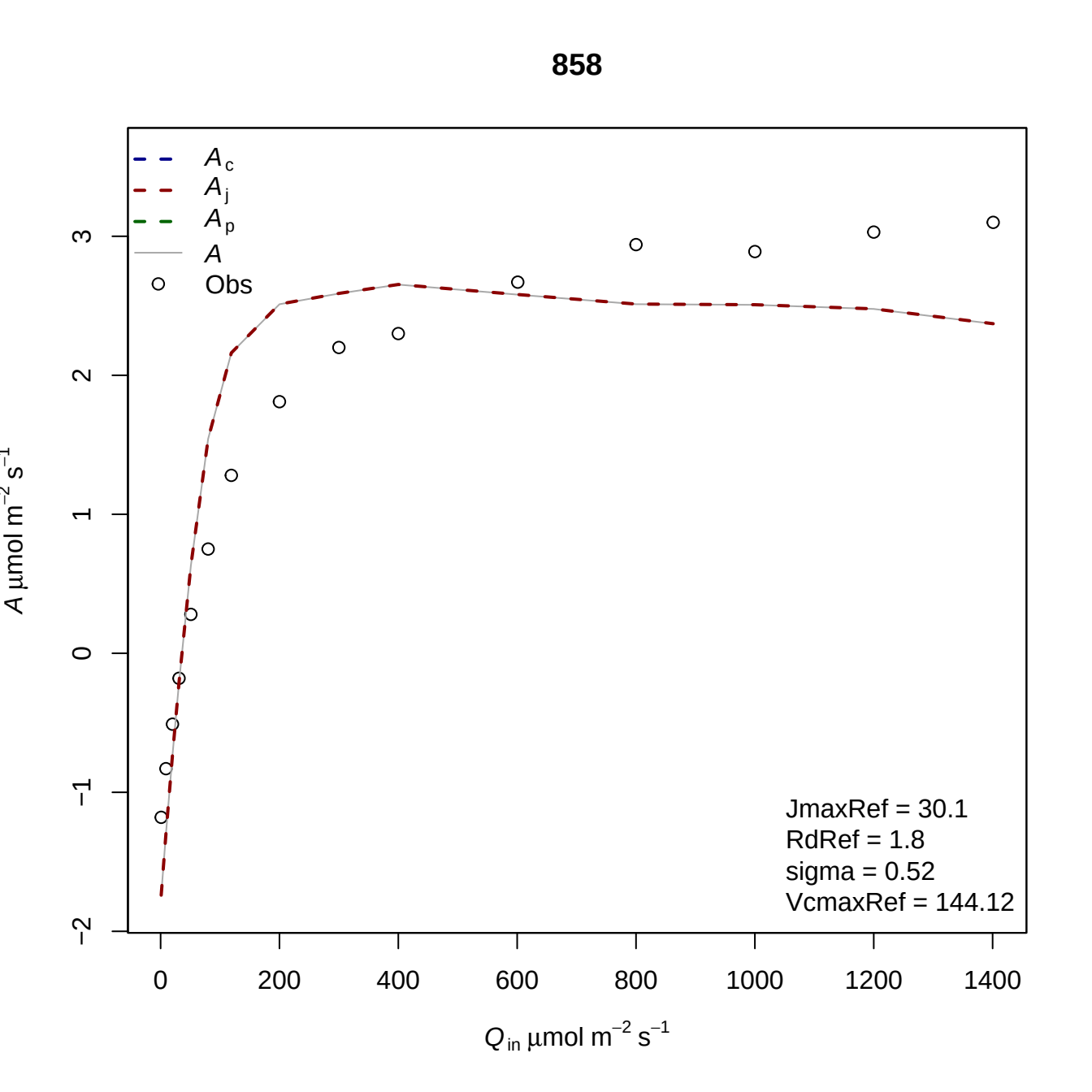

853
A_c
A_j
A_p
A
Obs
JmaxRef = 57.22
RdRef = 3.57
sigma = 0.29
VcmaxRef = 103.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

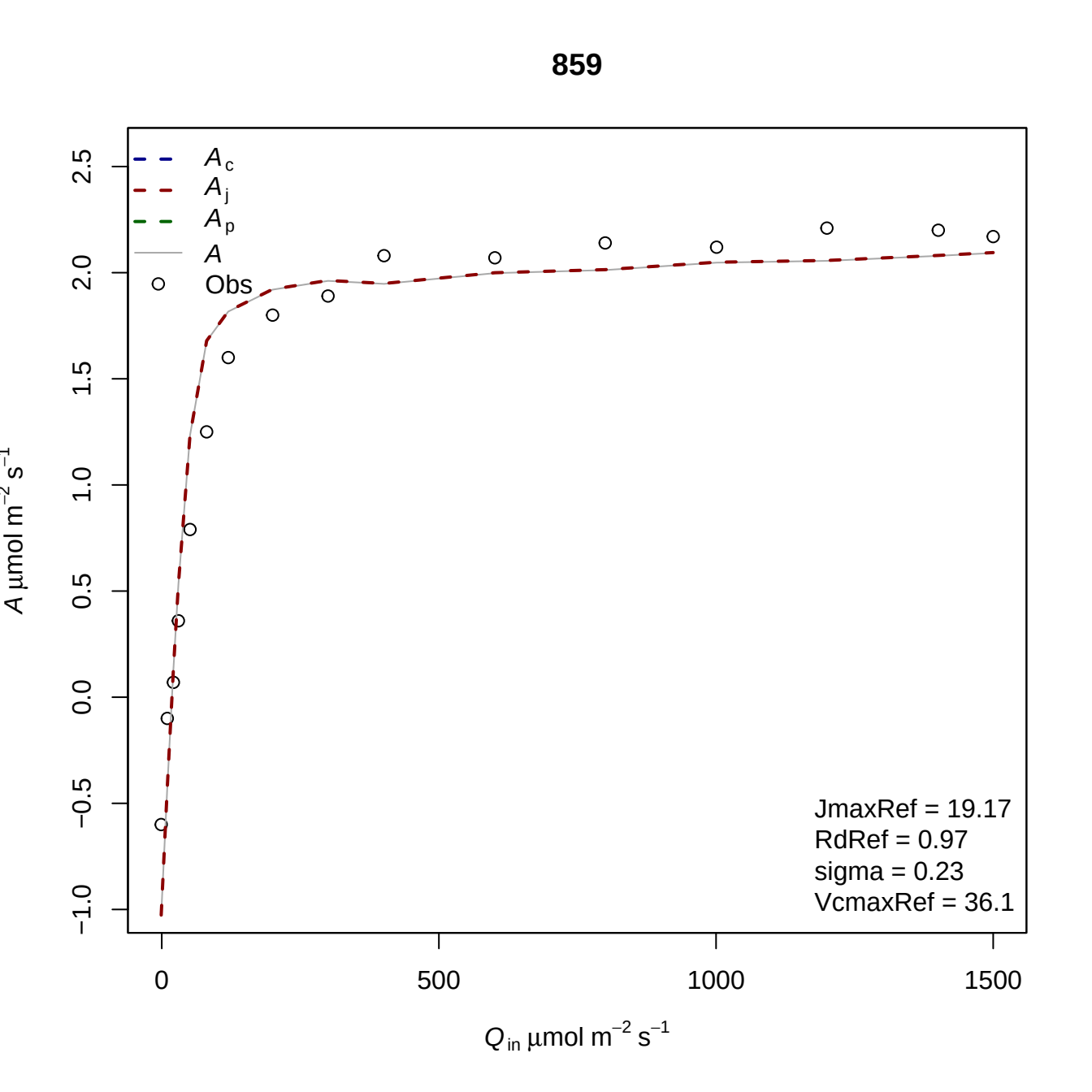

854
A_c
A_j
A_p
A
Obs
JmaxRef = 36.95
RdRef = 3.05
sigma = 0.57
VcmaxRef = 86.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200

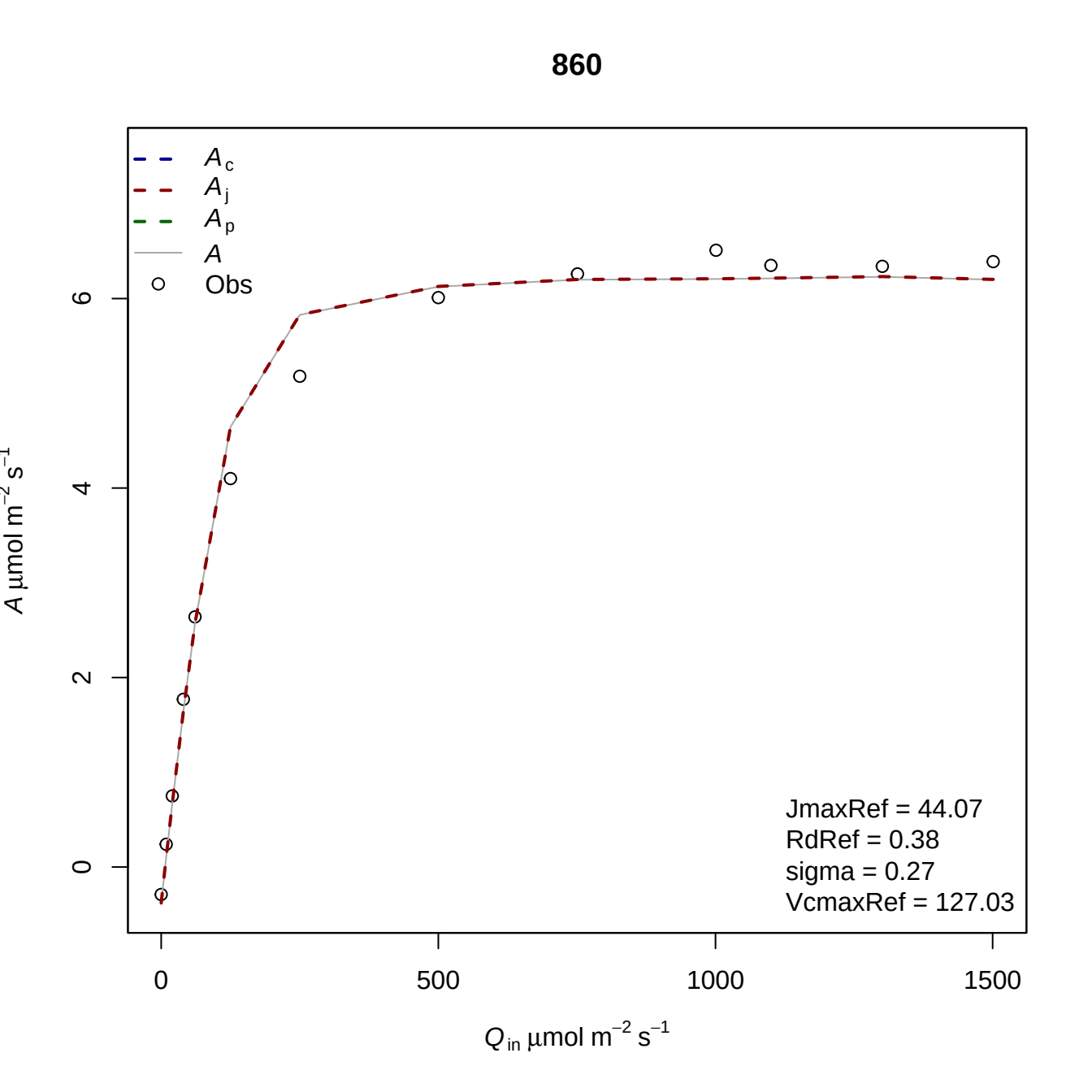

855
A_c
A_j
A_p
A
Obs
JmaxRef = 41.71
RdRef = 3.85
sigma = 0.8
VcmaxRef = 139.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
0
200
400
600
800
1000
1200
1400

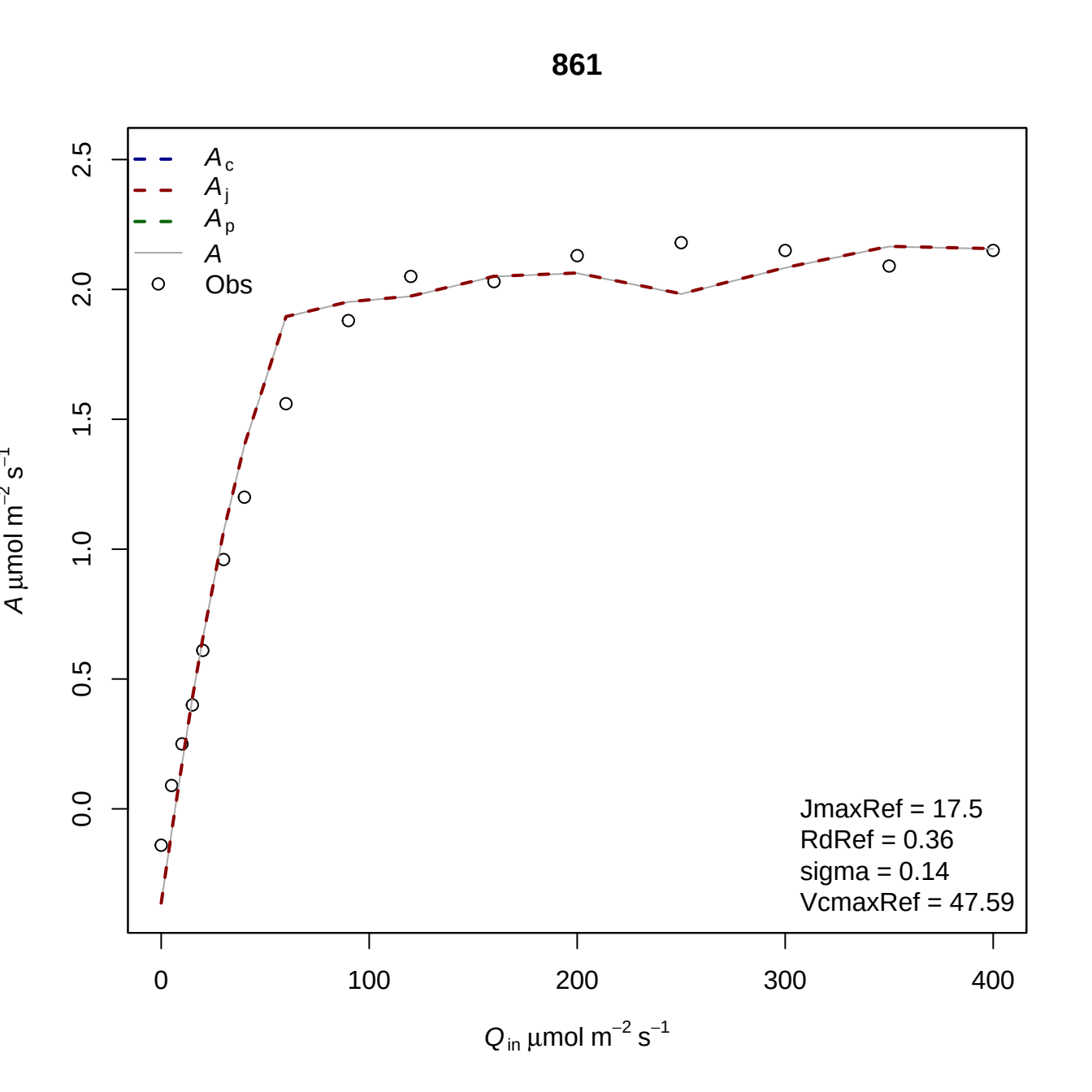

856
A_c
A_j
A_p
A
Obs
JmaxRef = 37.23
RdRef = 0.56
sigma = 0.33
VcmaxRef = 60.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000
1200

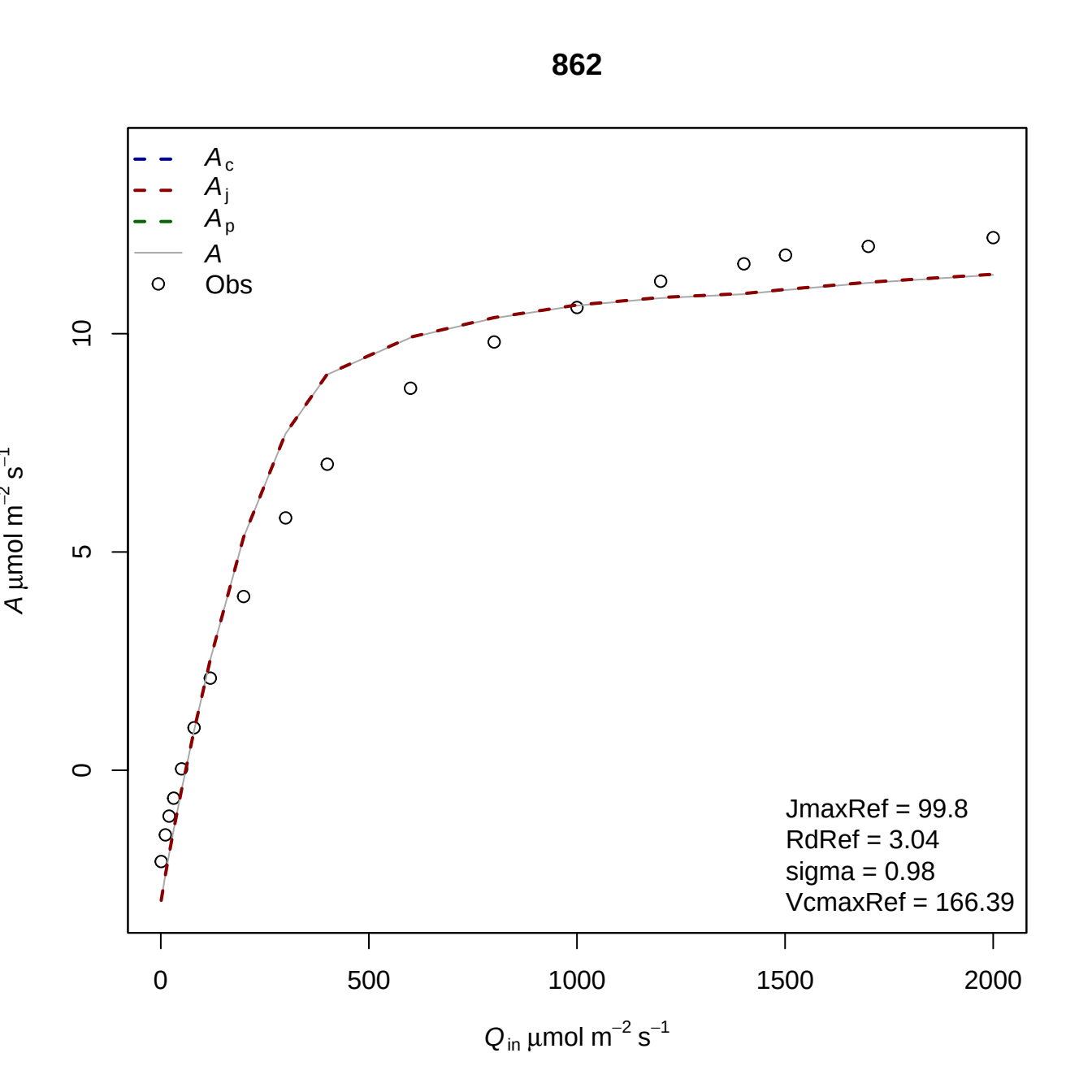

857
A_c
A_j
A_p
A
Obs
JmaxRef = 32.64
RdRef = 1.99
sigma = 0.67
VcmaxRef = 83.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

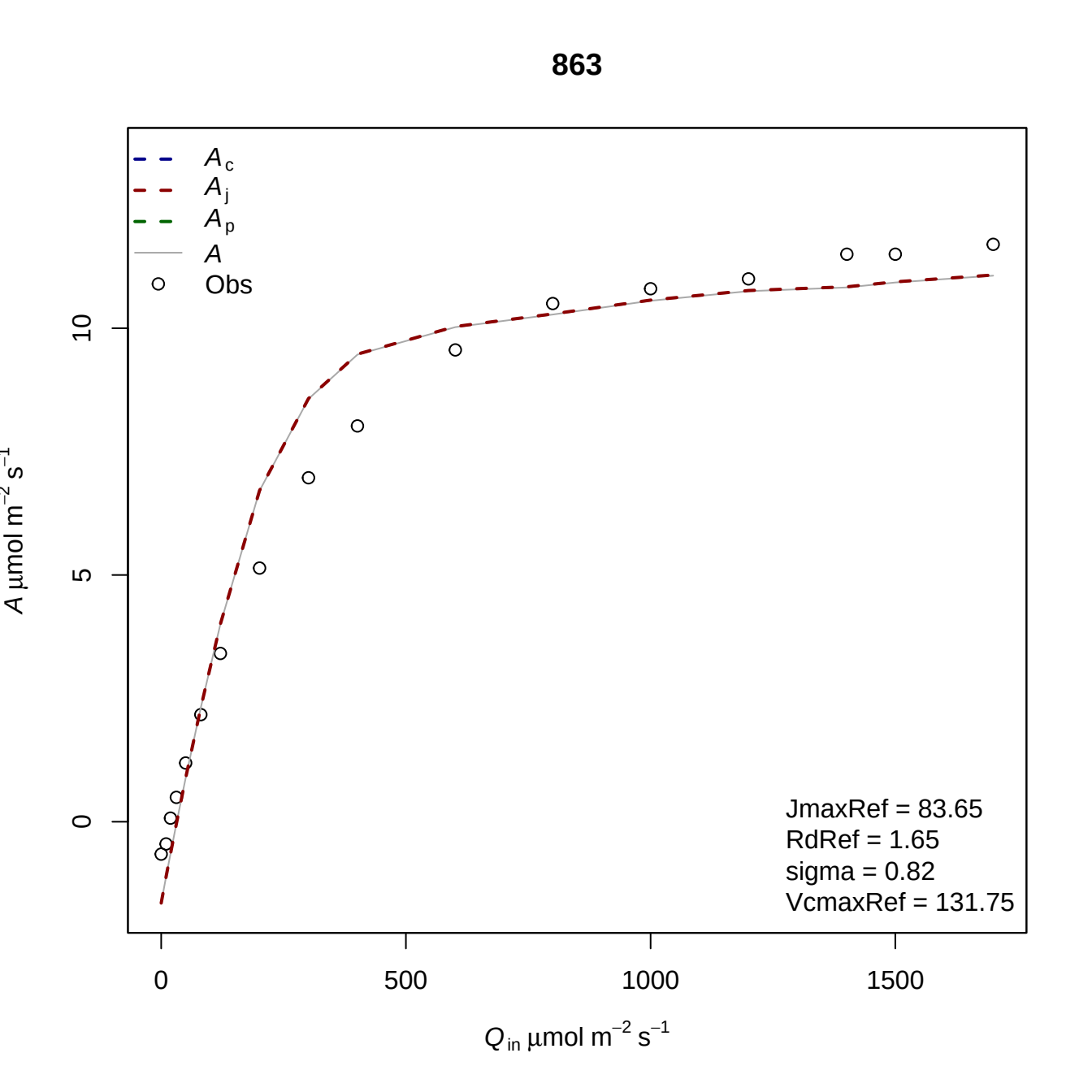

858
A_c
A_j
A_p
A
Obs
JmaxRef = 30.1
RdRef = 1.8
sigma = 0.52
VcmaxRef = 144.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

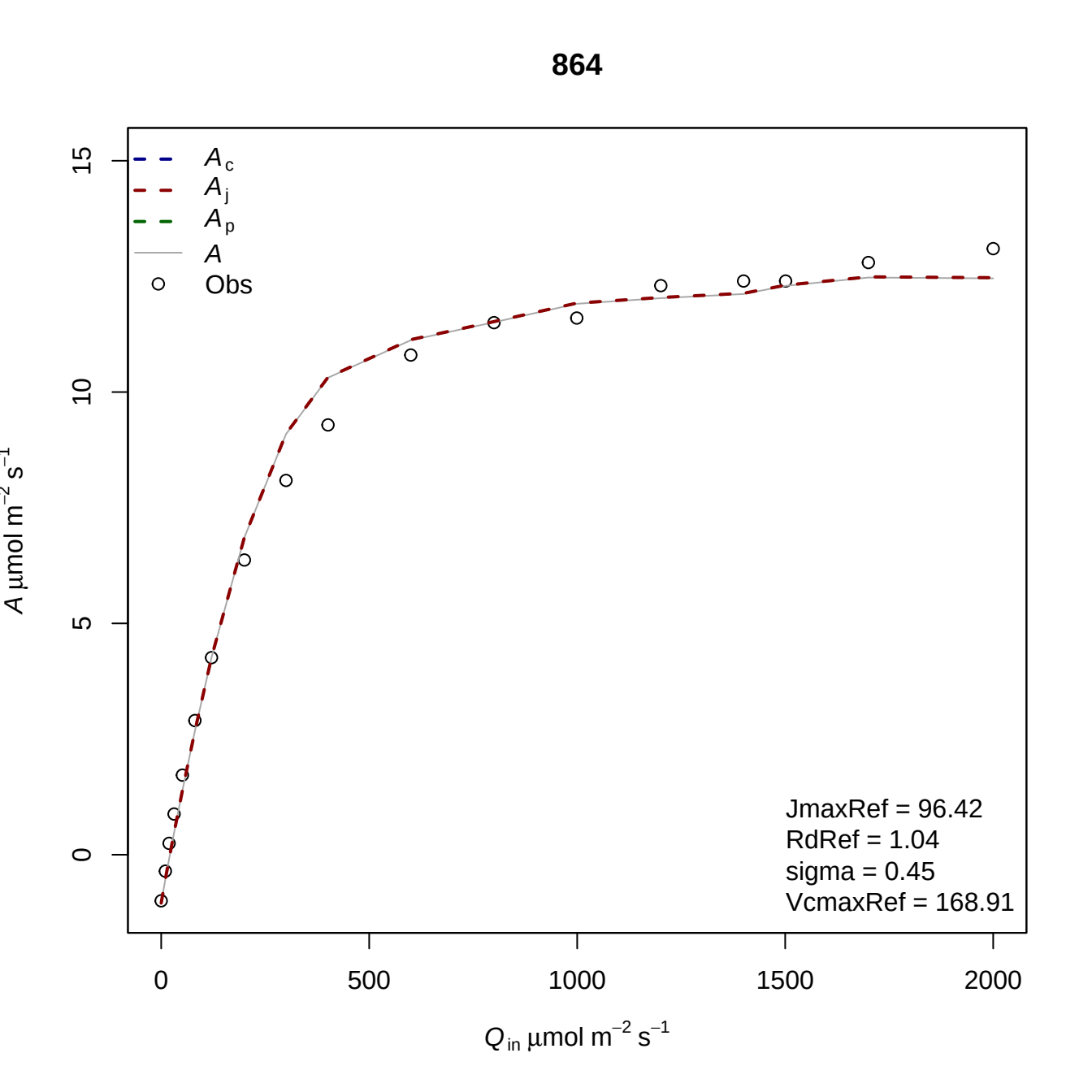

859
A_c
A_j
A_p
A
Obs
JmaxRef = 19.17
RdRef = 0.97
sigma = 0.23
VcmaxRef = 36.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

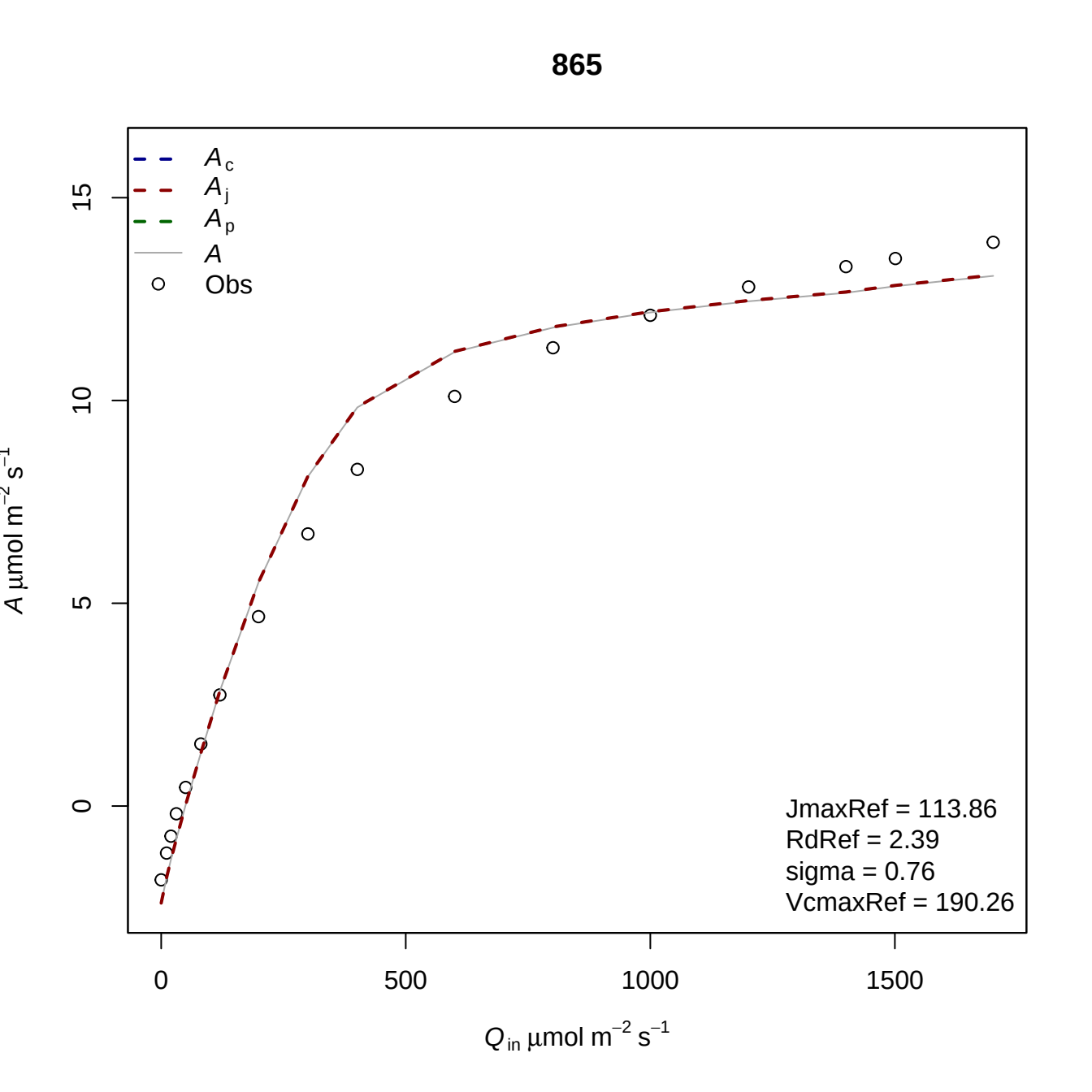

860
A_c
A_j
A_p
A
Obs
JmaxRef = 44.07
RdRef = 0.38
sigma = 0.27
VcmaxRef = 127.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

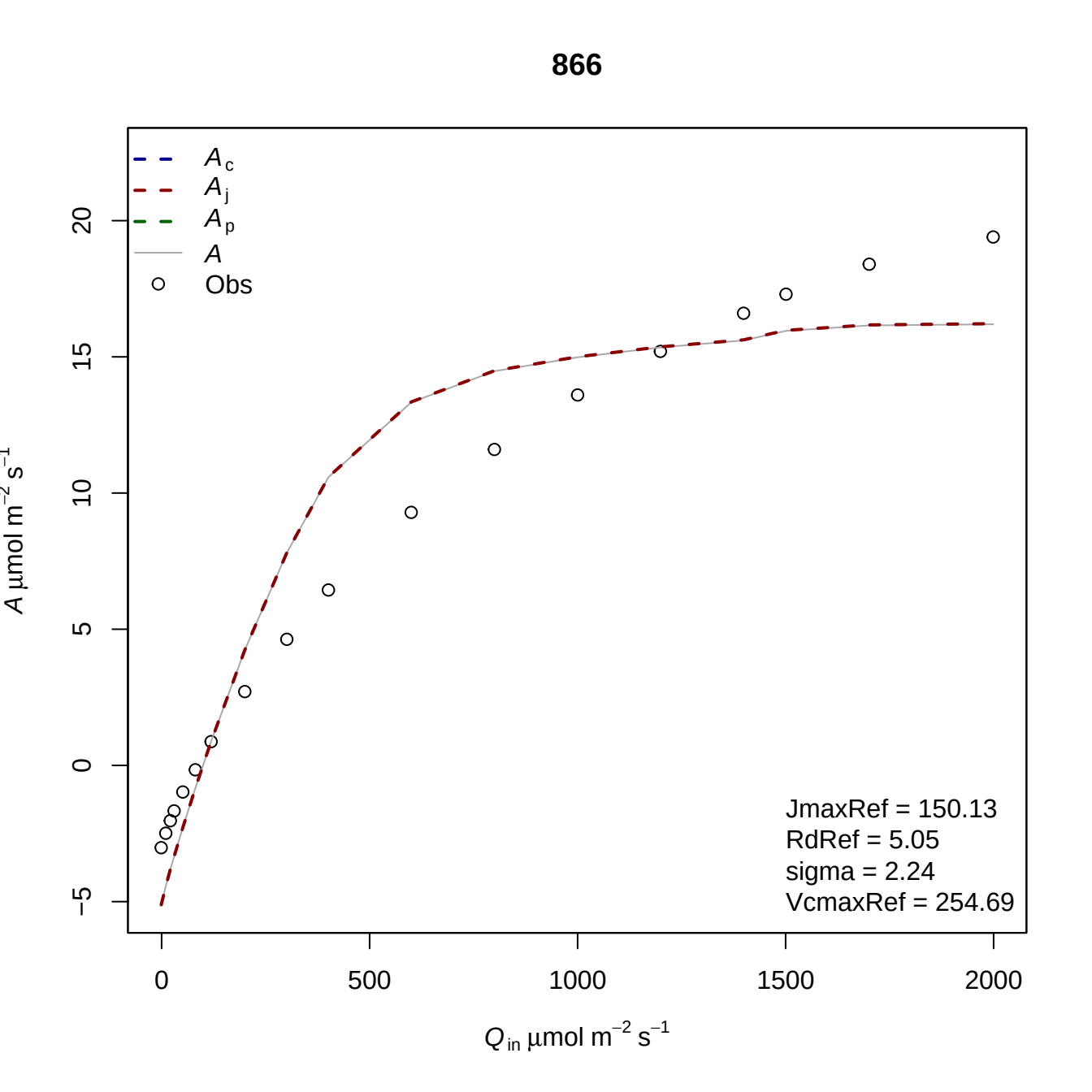

861
A_c
A_j
A_p
A
Obs
JmaxRef = 17.5
RdRef = 0.36
sigma = 0.14
VcmaxRef = 47.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
100
200
300
400

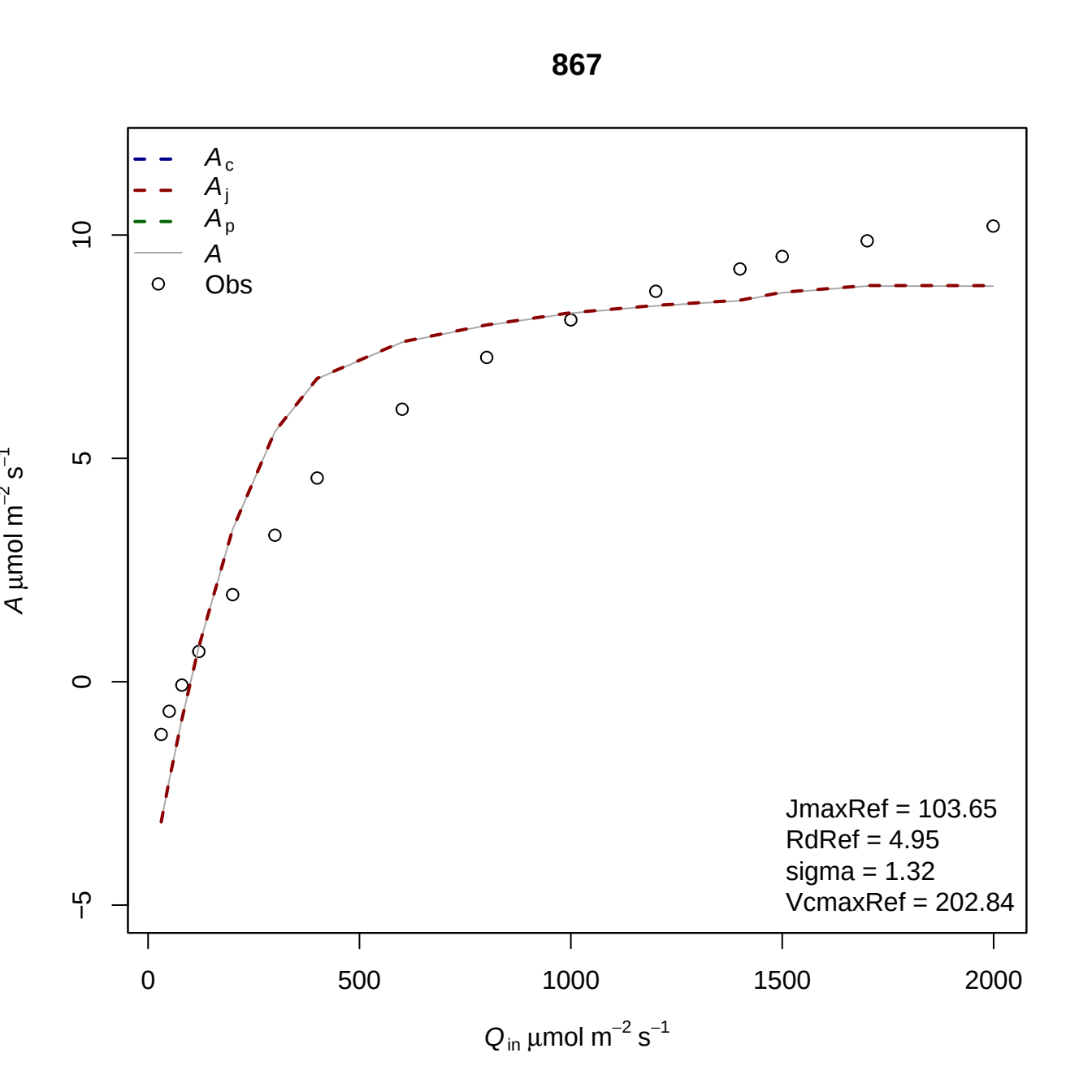

862
A_c
A_j
A_p
A
Obs
JmaxRef = 99.8
RdRef = 3.04
sigma = 0.98
VcmaxRef = 166.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

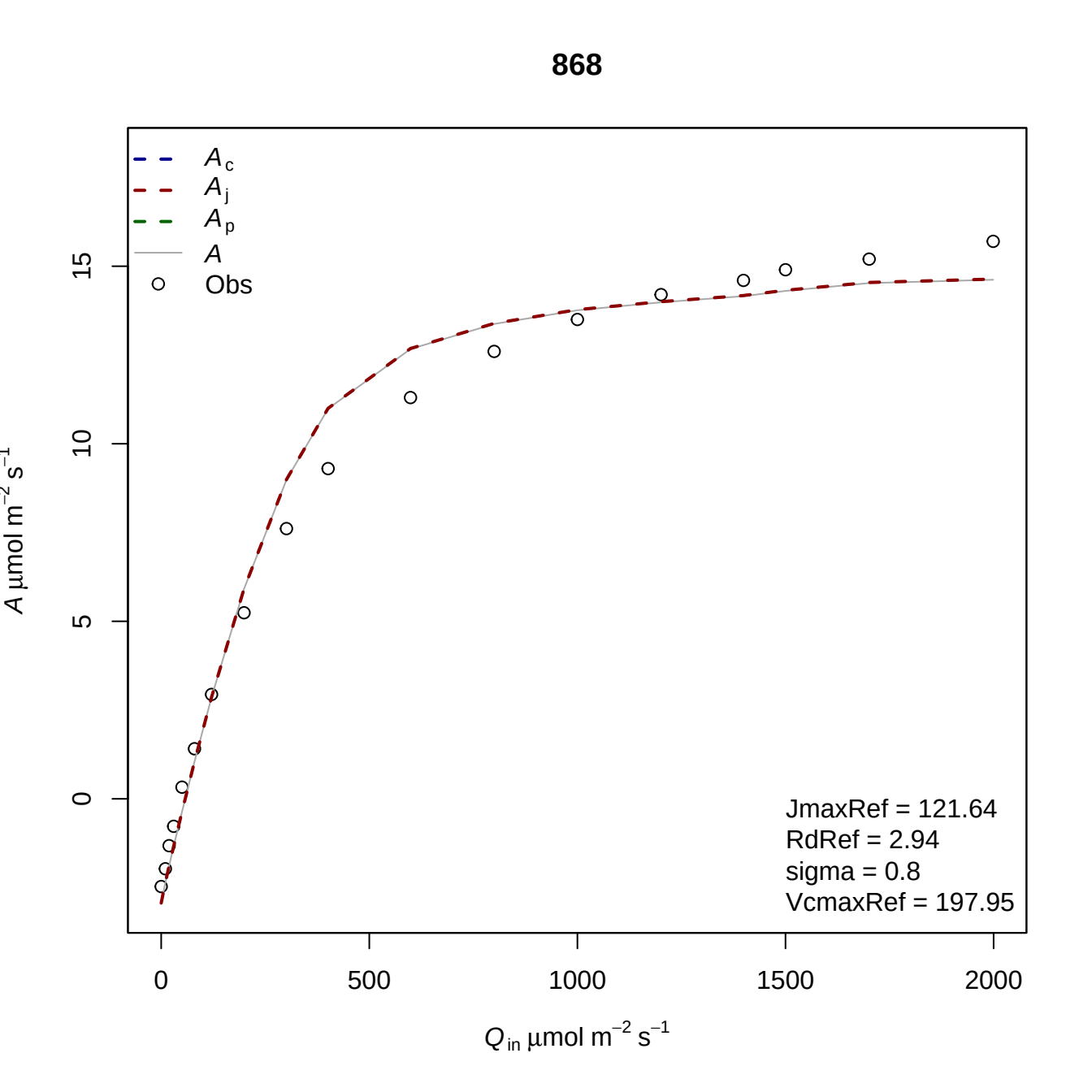

863
A_c
A_j
A_p
A
Obs
JmaxRef = 83.65
RdRef = 1.65
sigma = 0.82
VcmaxRef = 131.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

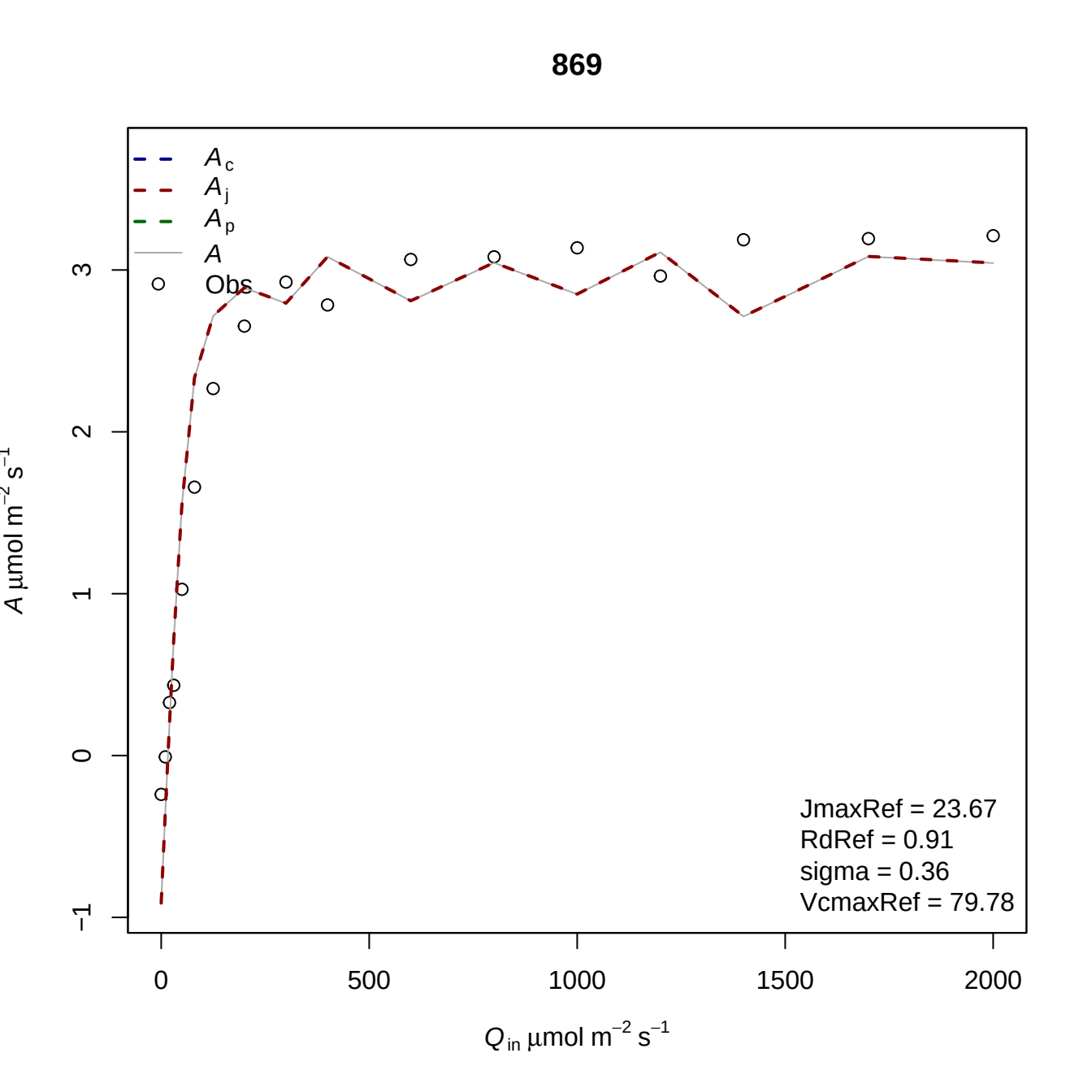

864
A_c
A_j
A_p
A
Obs
JmaxRef = 96.42
RdRef = 1.04
sigma = 0.45
VcmaxRef = 168.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

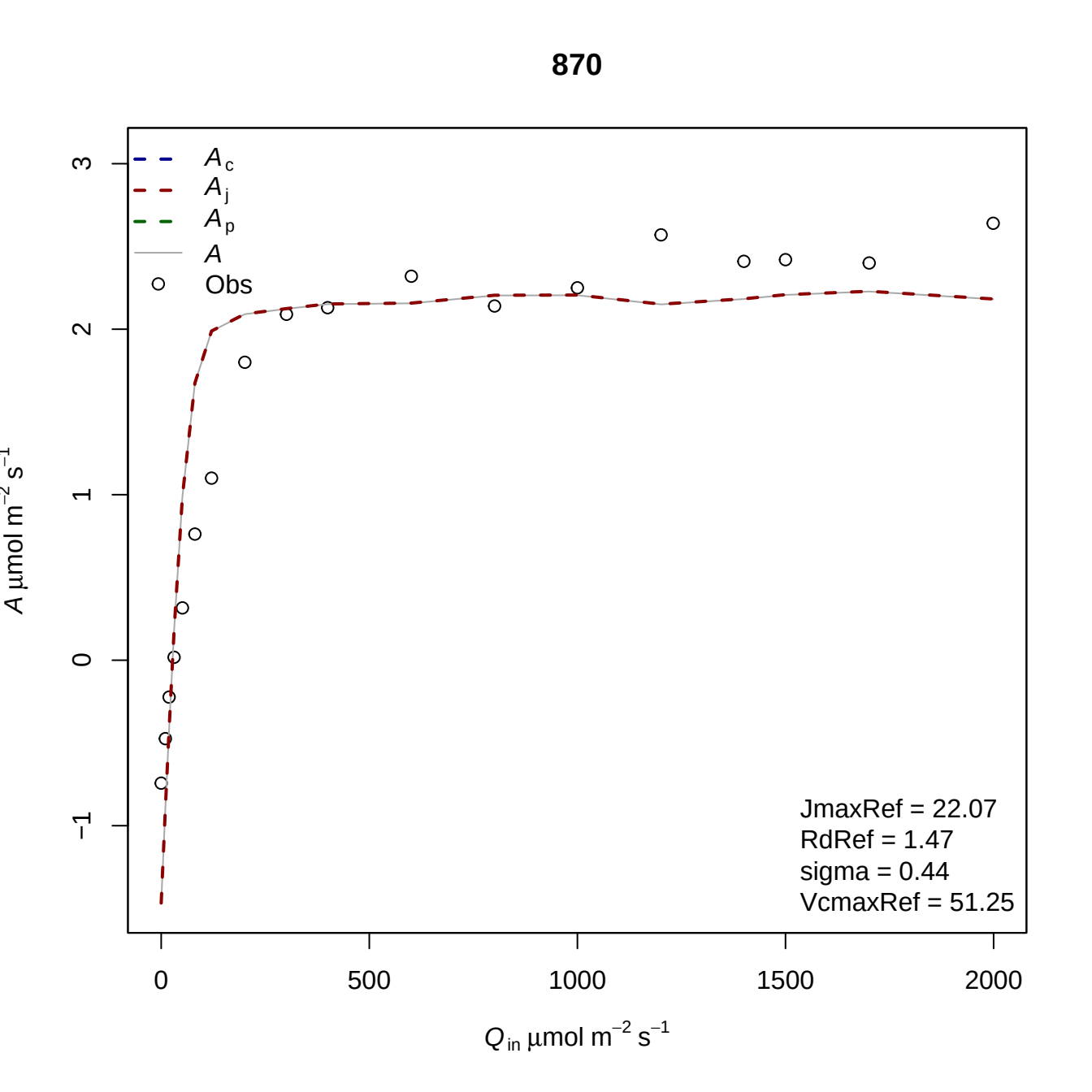

865
A_c
A_j
A_p
A
Obs
JmaxRef = 113.86
RdRef = 2.39
sigma = 0.76
VcmaxRef = 190.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

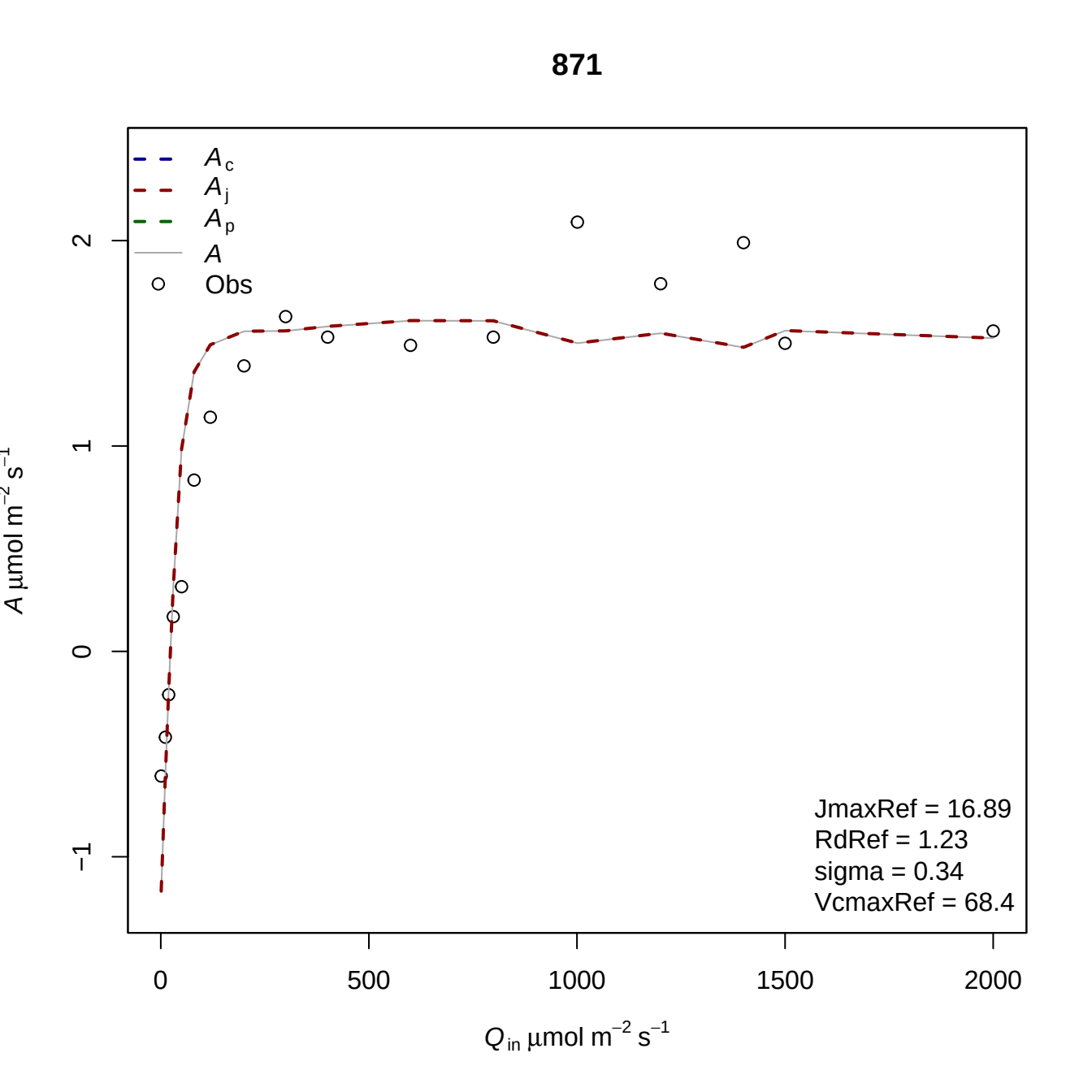

866
A_c
A_j
A_p
A
Obs
JmaxRef = 150.13
RdRef = 5.05
sigma = 2.24
VcmaxRef = 254.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

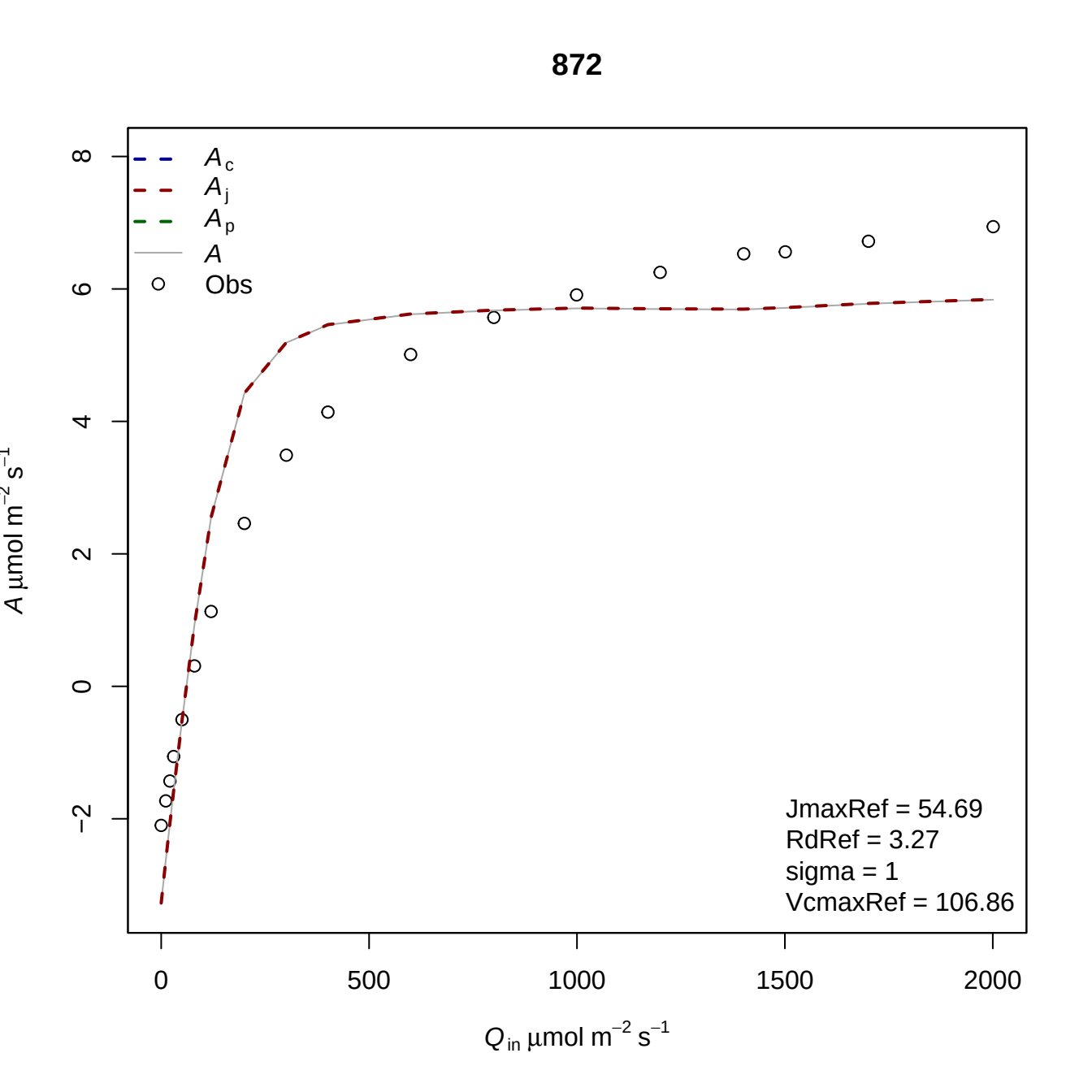

867
A_c
A_j
A_p
A
Obs
JmaxRef = 103.65
RdRef = 4.95
sigma = 1.32
VcmaxRef = 202.84
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

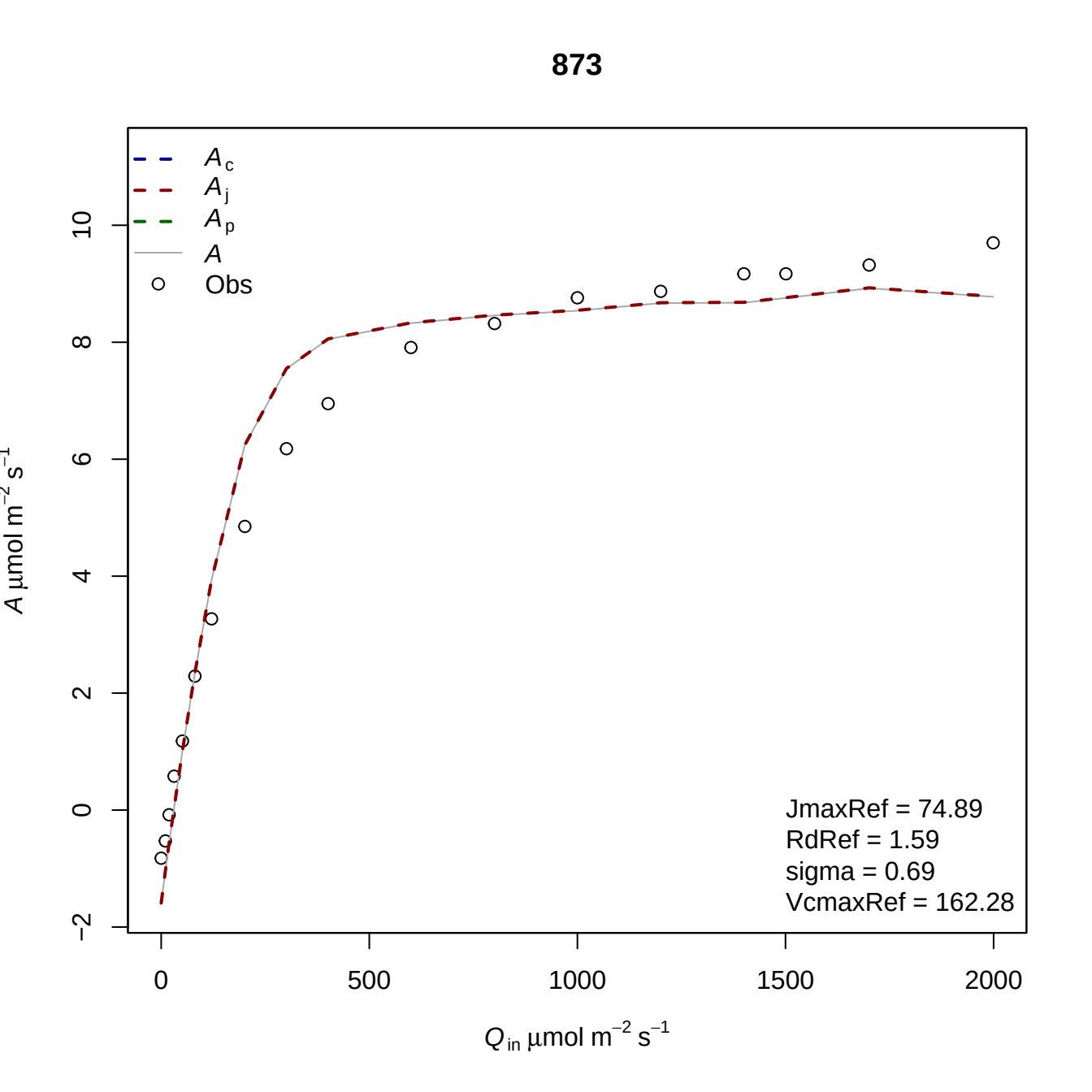

868
A_c
A_j
A_p
A
Obs
JmaxRef = 121.64
RdRef = 2.94
sigma = 0.8
VcmaxRef = 197.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

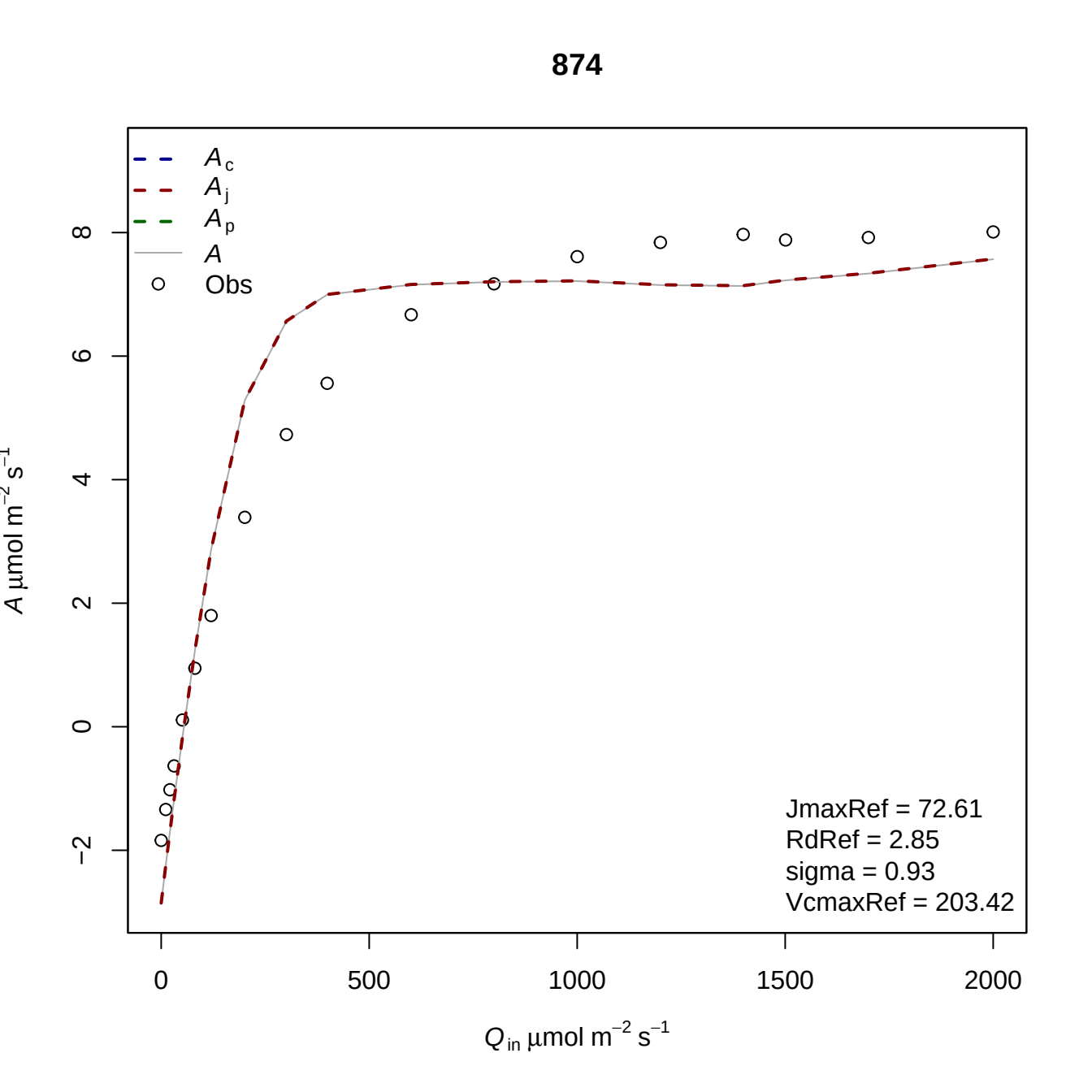

869
A_c
A_j
A_p
A
Obs
JmaxRef = 23.67
RdRef = 0.91
sigma = 0.36
VcmaxRef = 79.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

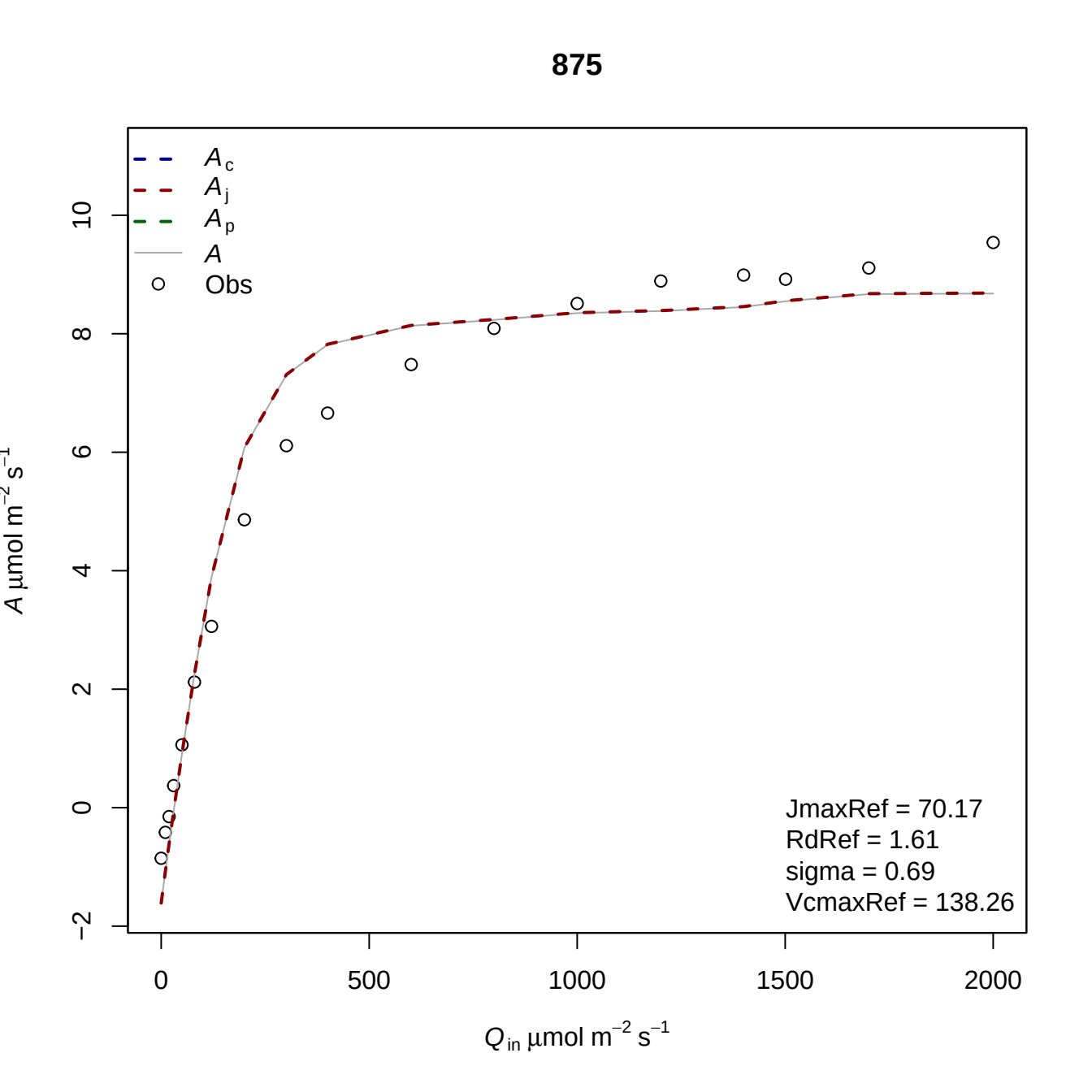

870
A_c
A_j
A_p
A
Obs
JmaxRef = 22.07
RdRef = 1.47
sigma = 0.44
VcmaxRef = 51.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500
2000

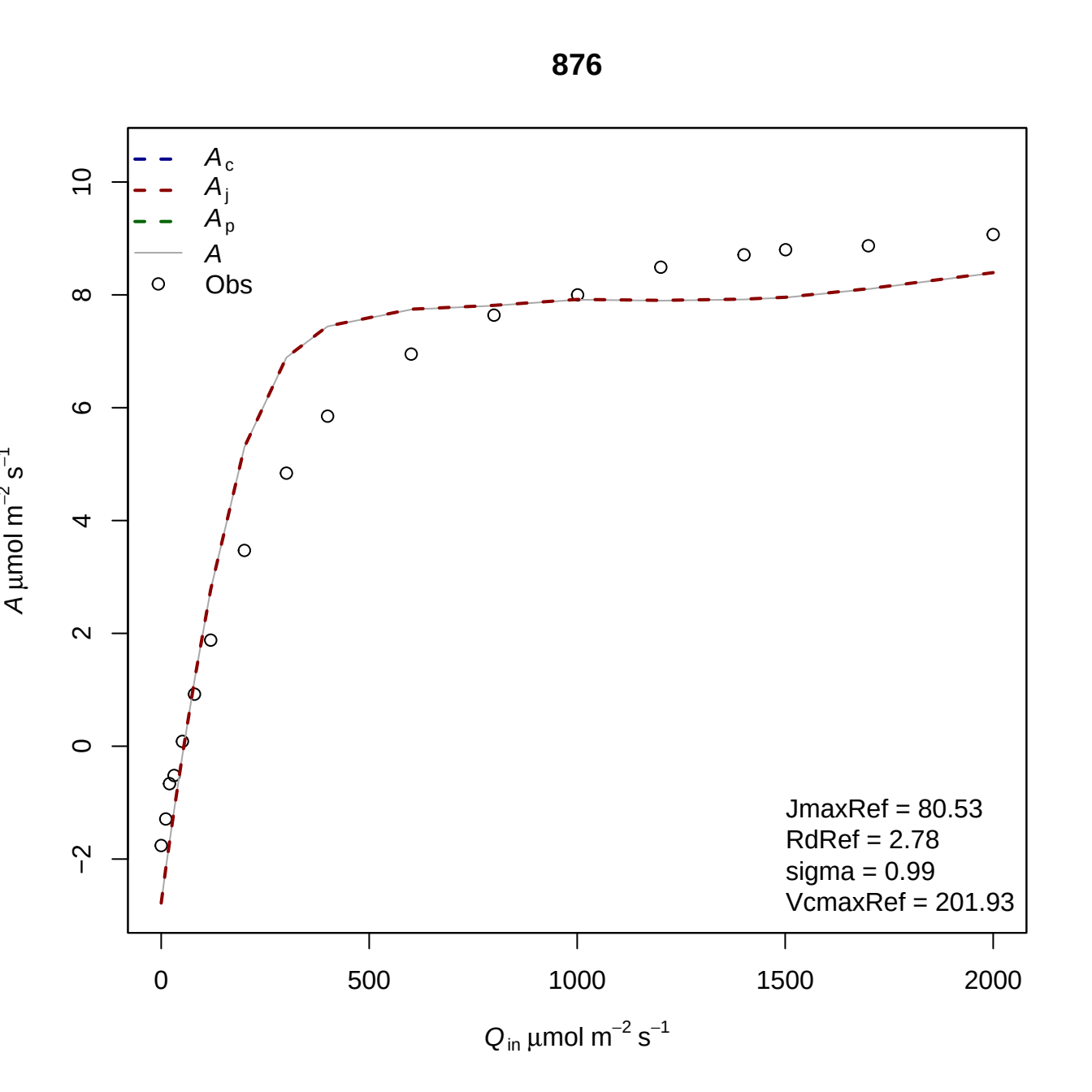

871
A_c
A_j
A_p
A
Obs
JmaxRef = 16.89
RdRef = 1.23
sigma = 0.34
VcmaxRef = 68.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

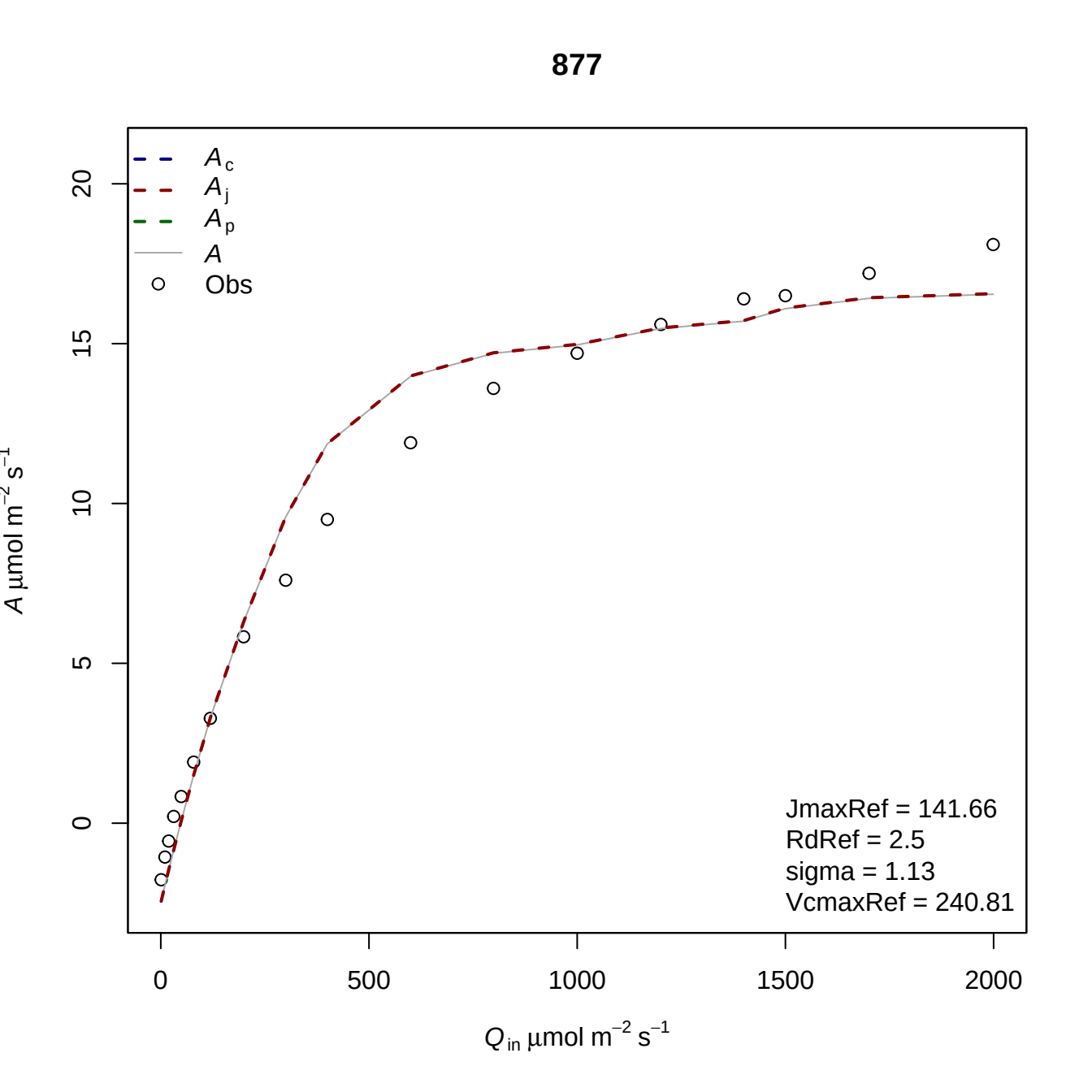

872
A_c
A_j
A_p
A
Obs
JmaxRef = 54.69
RdRef = 3.27
sigma = 1
VcmaxRef = 106.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

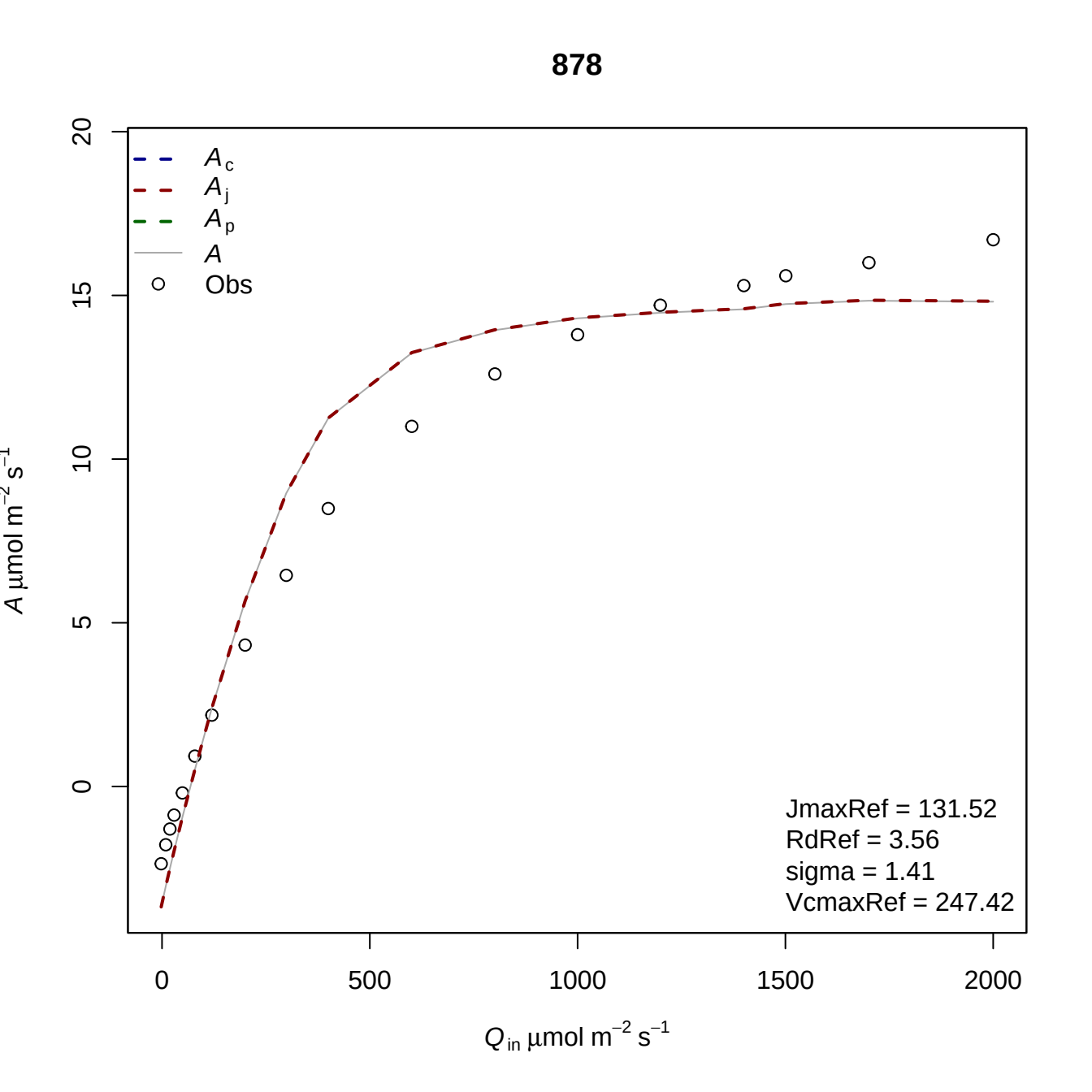

873
A_c
A_j
A_p
A
Obs
JmaxRef = 74.89
RdRef = 1.59
sigma = 0.69
VcmaxRef = 162.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

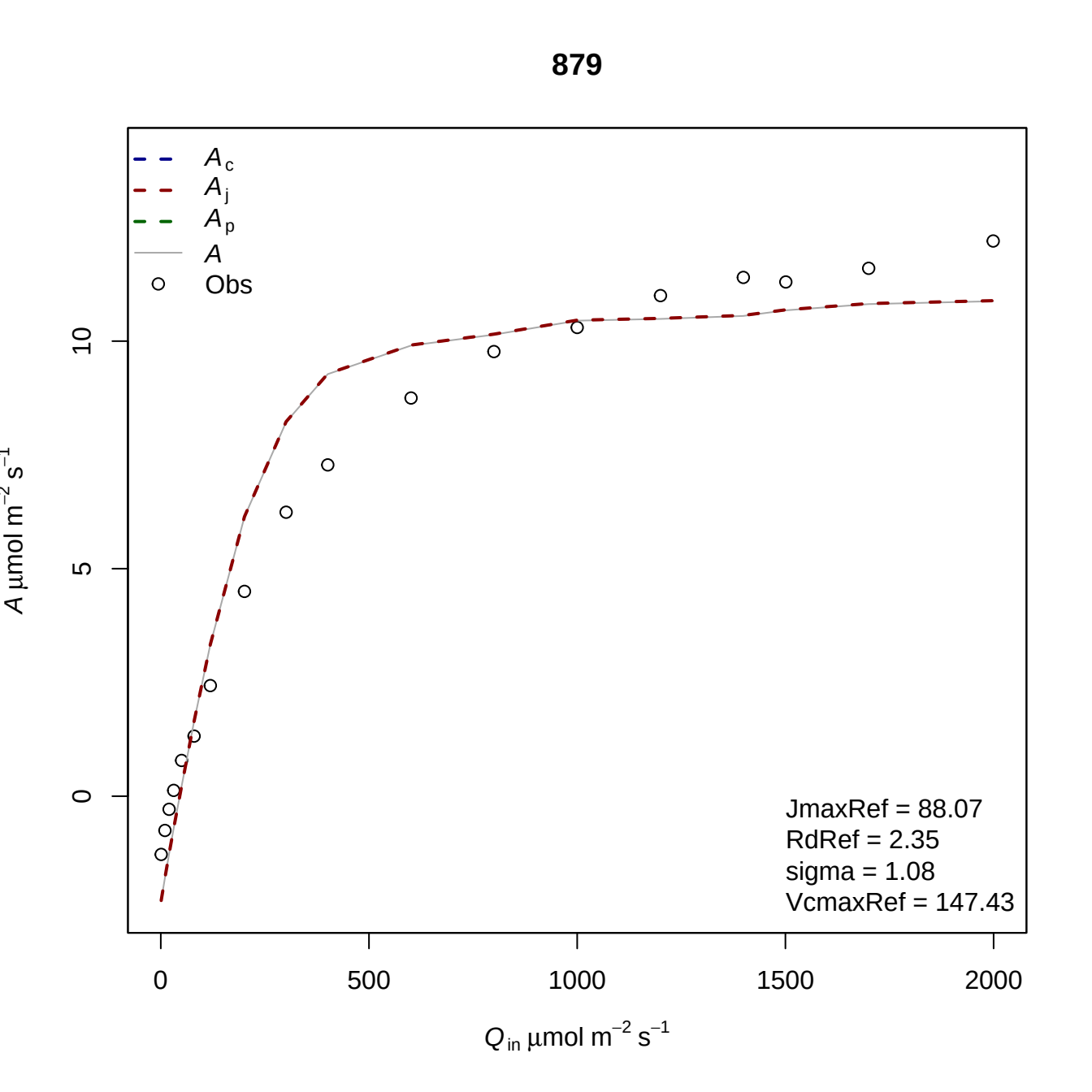

874
A_c
A_j
A_p
A
Obs
JmaxRef = 72.61
RdRef = 2.85
sigma = 0.93
VcmaxRef = 203.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

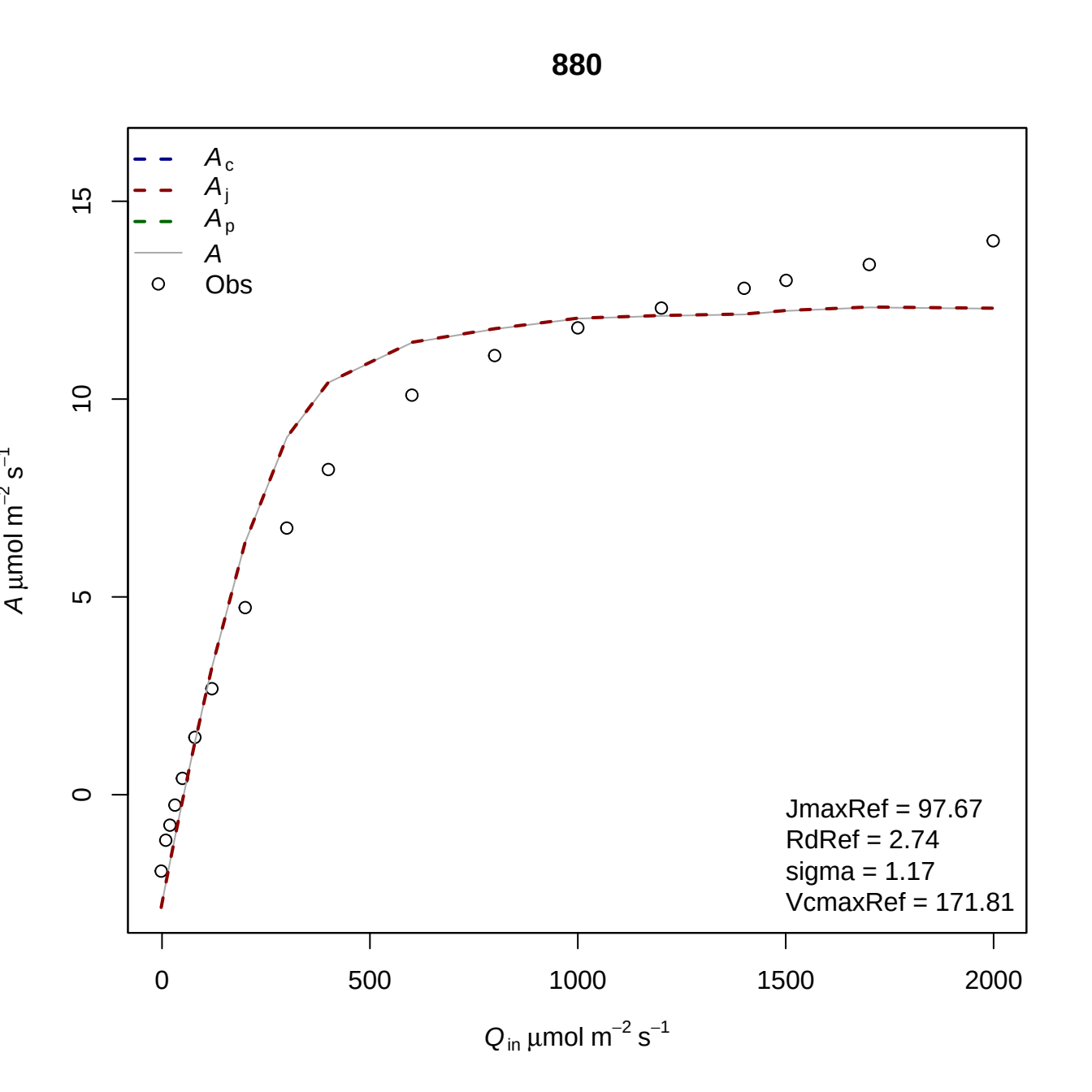

875
A_c
A_j
A_p
A
Obs
JmaxRef = 70.17
RdRef = 1.61
sigma = 0.69
VcmaxRef = 138.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

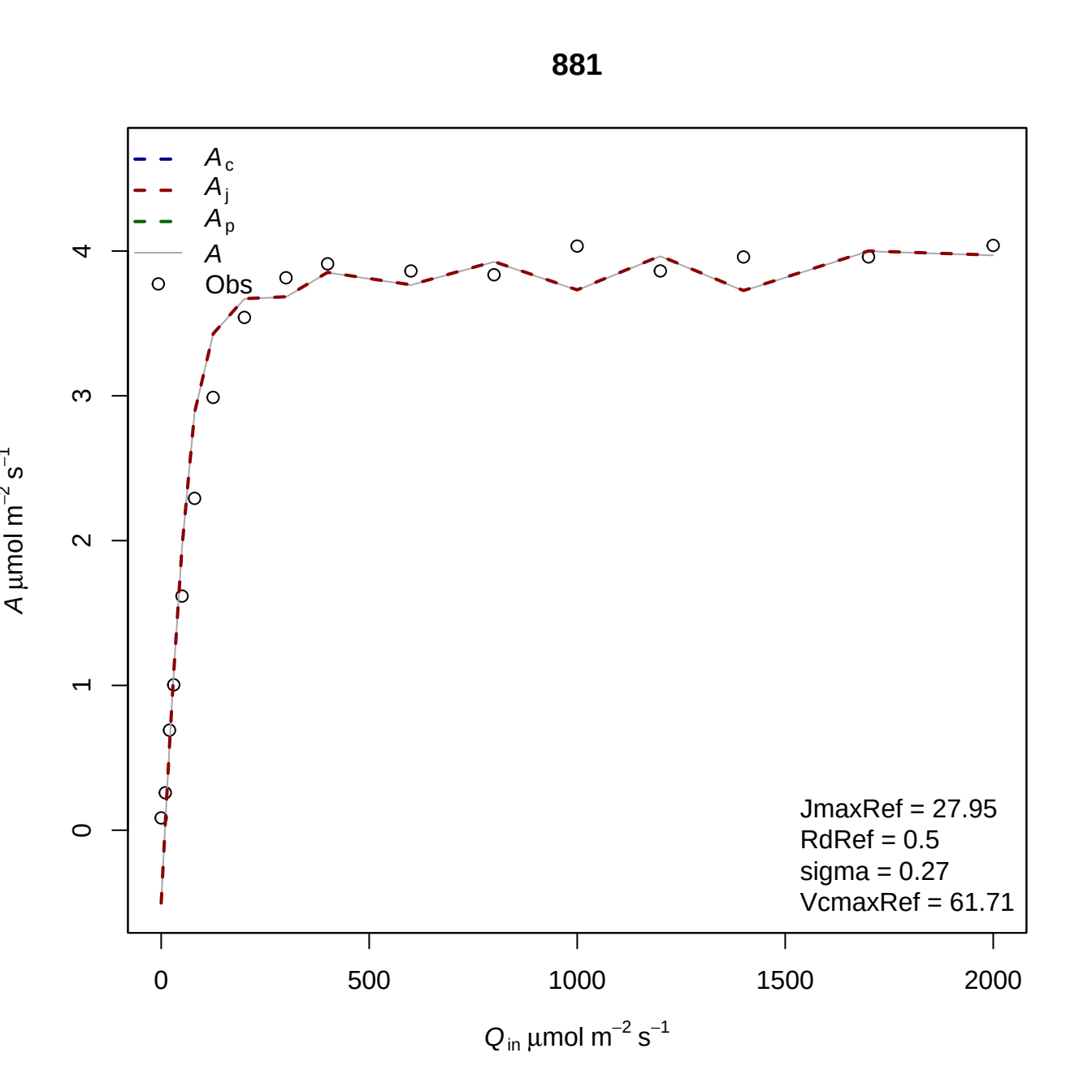

876
A_c
A_j
A_p
A
Obs
JmaxRef = 80.53
RdRef = 2.78
sigma = 0.99
VcmaxRef = 201.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

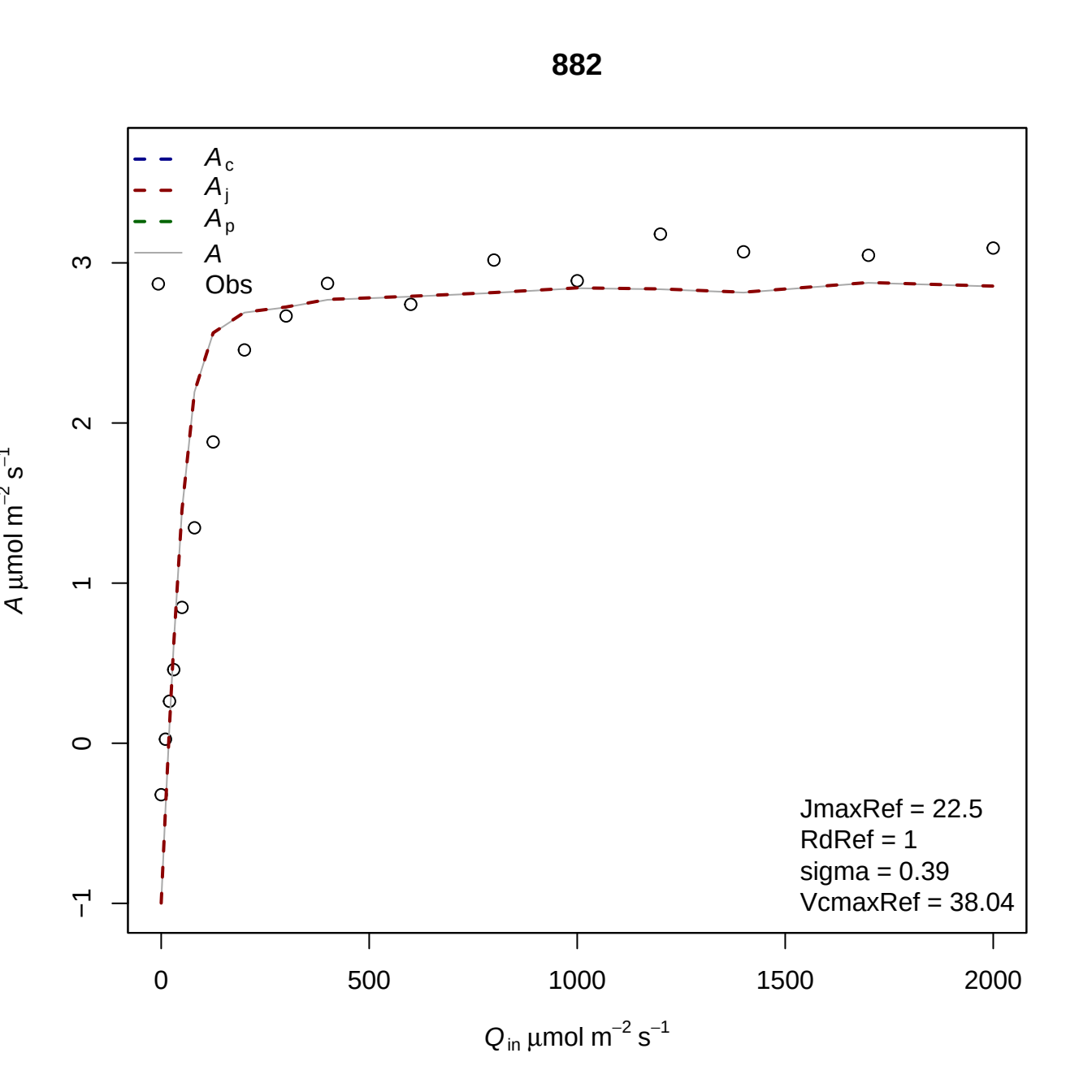

877
A_c
A_j
A_p
A
Obs
JmaxRef = 141.66
RdRef = 2.5
sigma = 1.13
VcmaxRef = 240.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

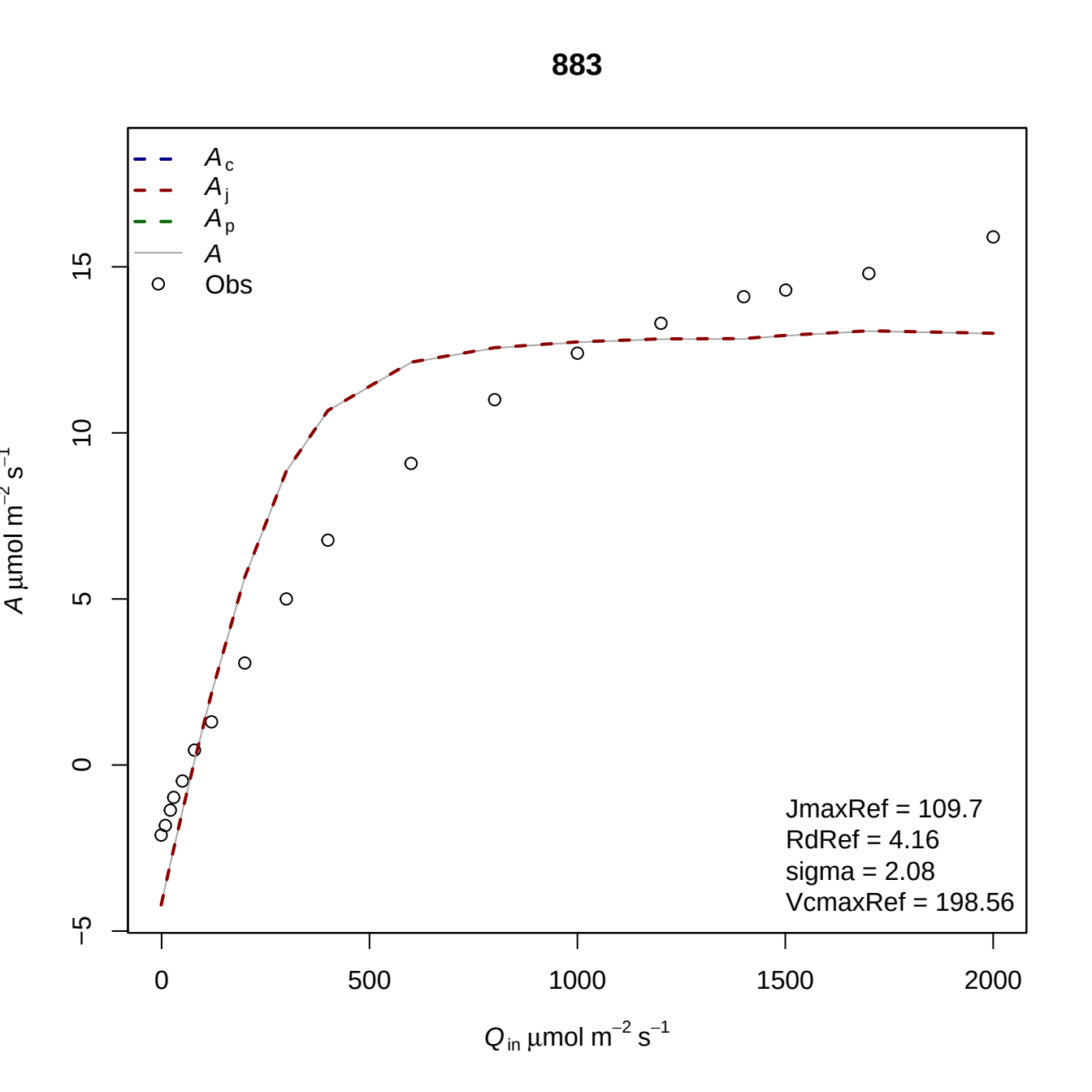

878
A_c
A_j
A_p
A
Obs
JmaxRef = 131.52
RdRef = 3.56
sigma = 1.41
VcmaxRef = 247.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

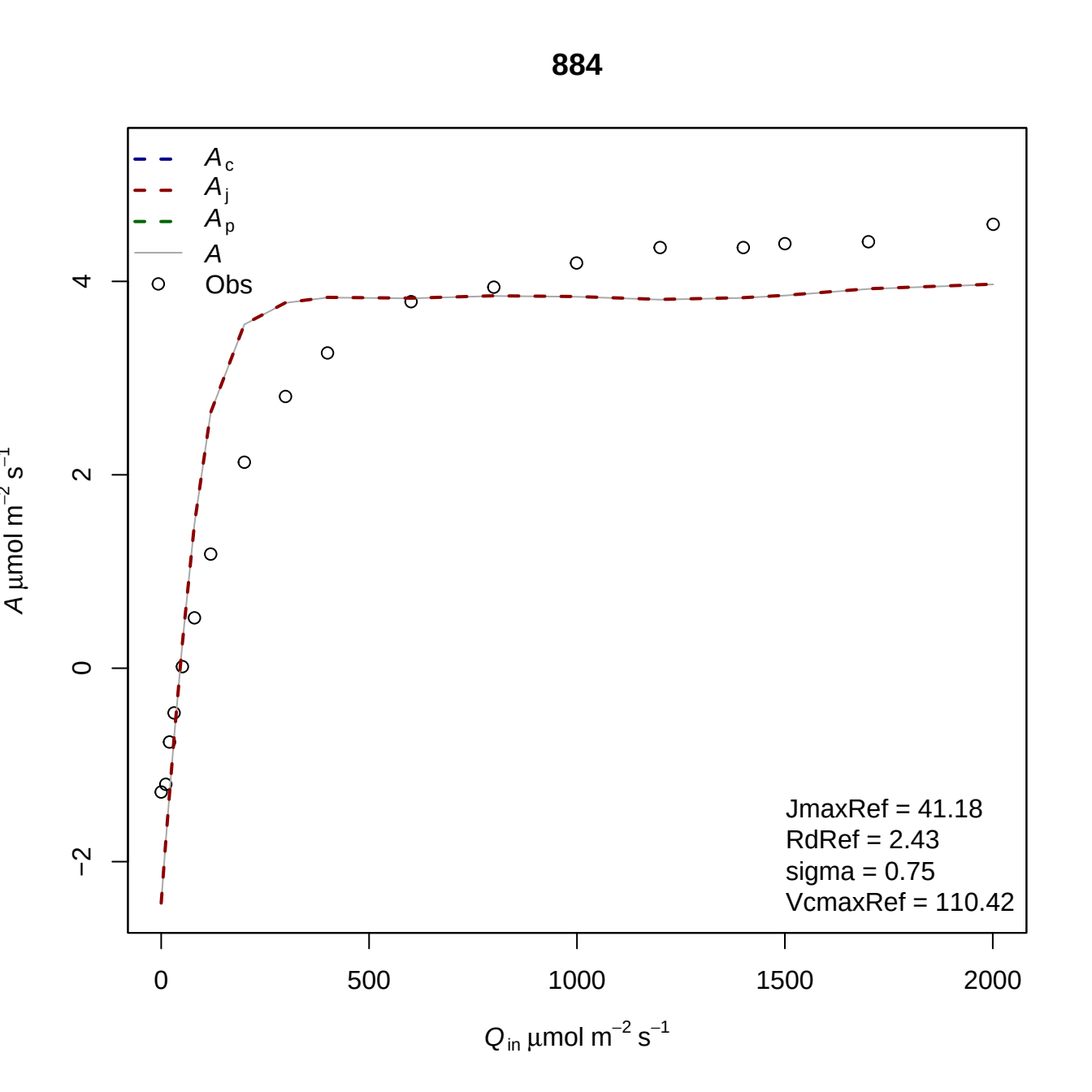

879
A_c
A_j
A_p
A
Obs
JmaxRef = 88.07
RdRef = 2.35
sigma = 1.08
VcmaxRef = 147.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

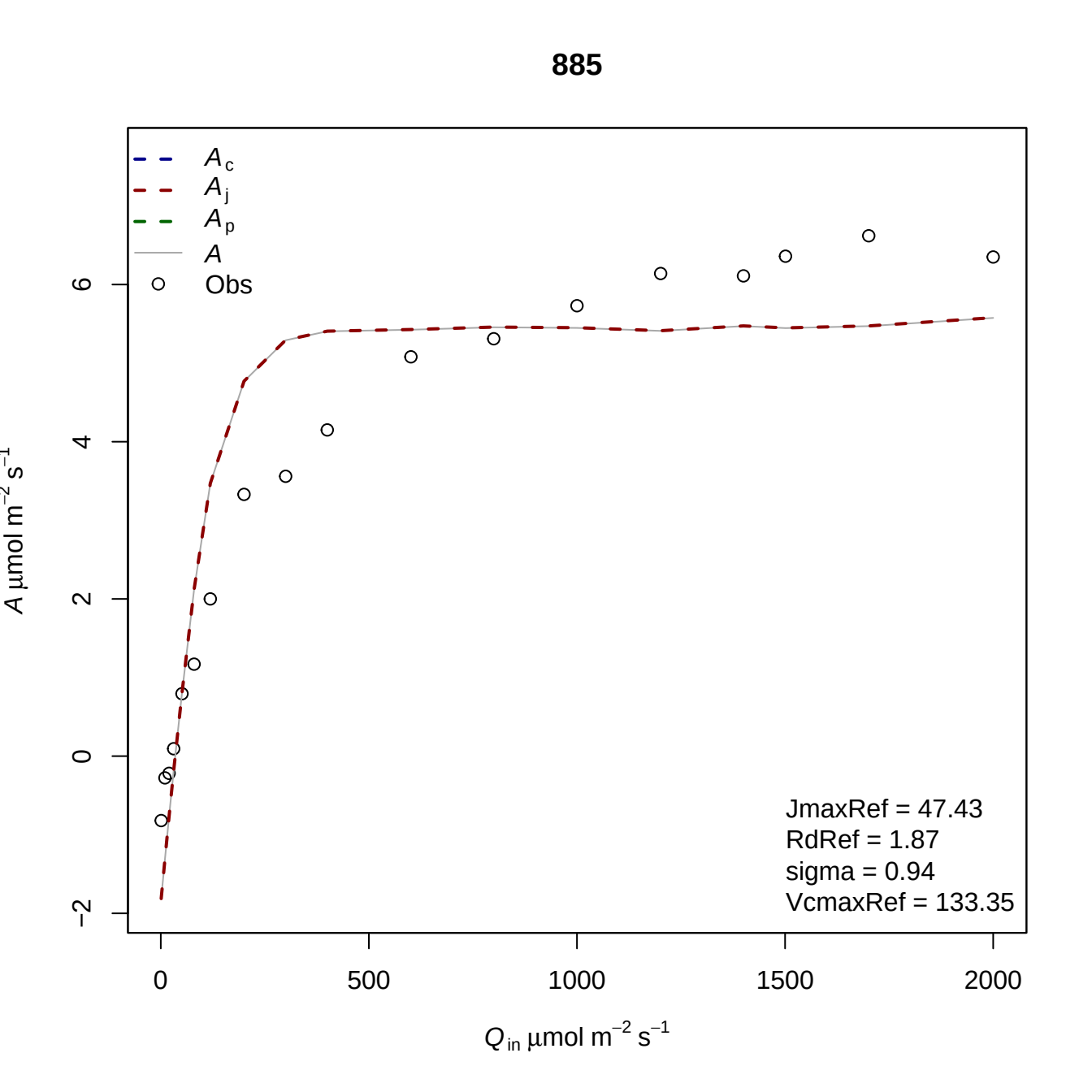

880
A_c
A_j
A_p
A
Obs
JmaxRef = 97.67
RdRef = 2.74
sigma = 1.17
VcmaxRef = 171.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

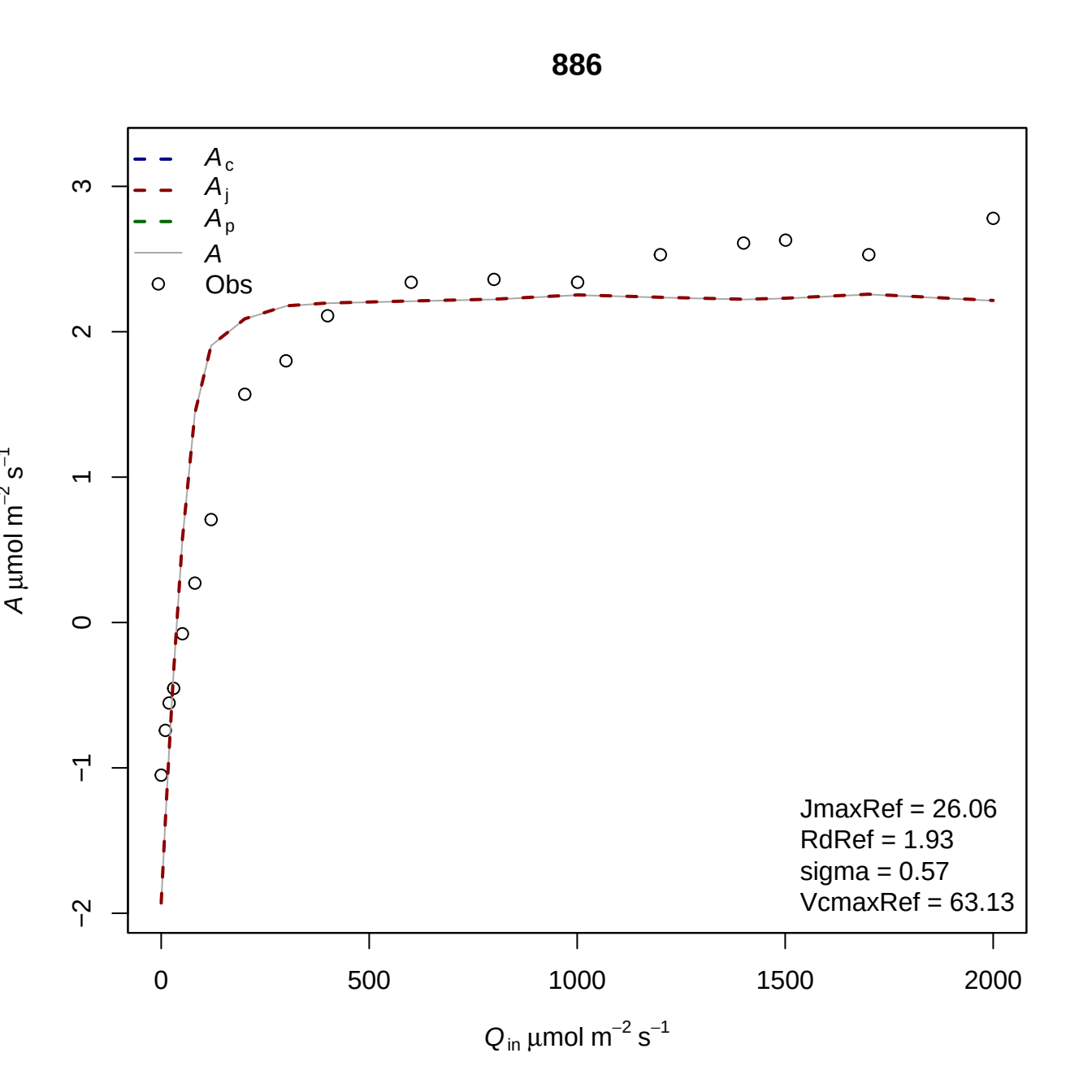

881
A_c
A_j
A_p
A
Obs
JmaxRef = 27.95
RdRef = 0.5
sigma = 0.27
VcmaxRef = 61.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

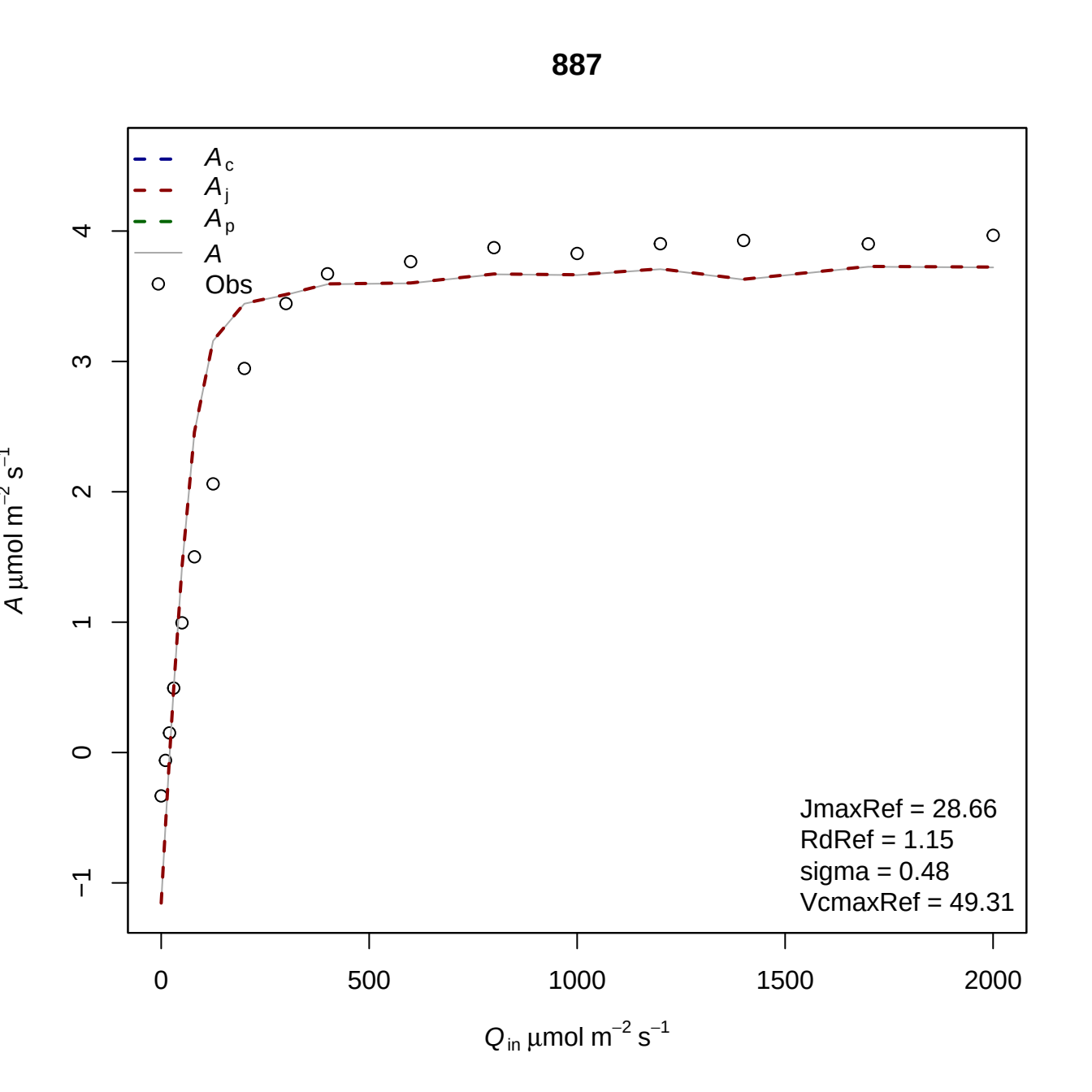

882
A_c
A_j
A_p
A
Obs
JmaxRef = 22.5
RdRef = 1
sigma = 0.39
VcmaxRef = 38.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

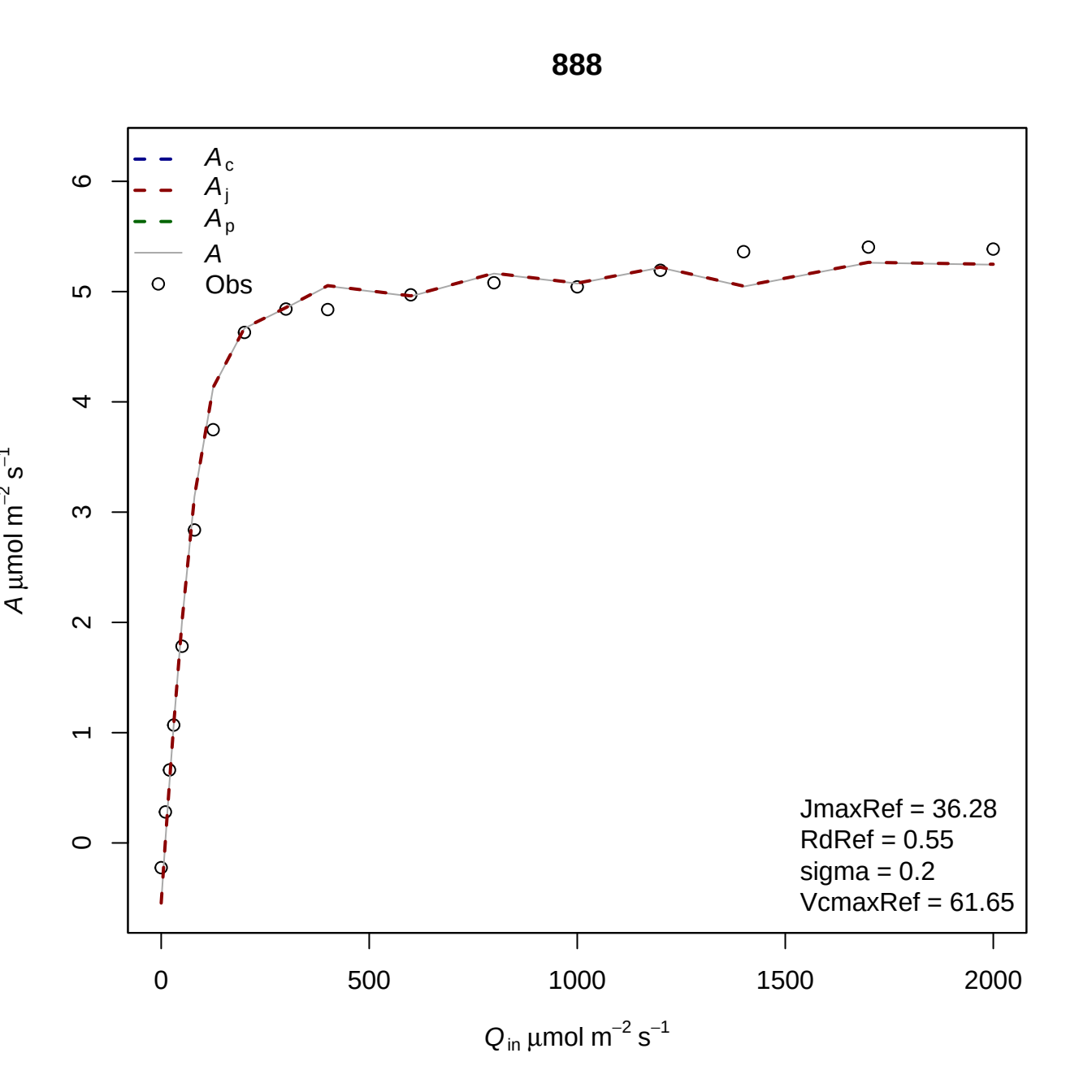

883
A_c
A_j
A_p
A
Obs
JmaxRef = 109.7
RdRef = 4.16
sigma = 2.08
VcmaxRef = 198.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

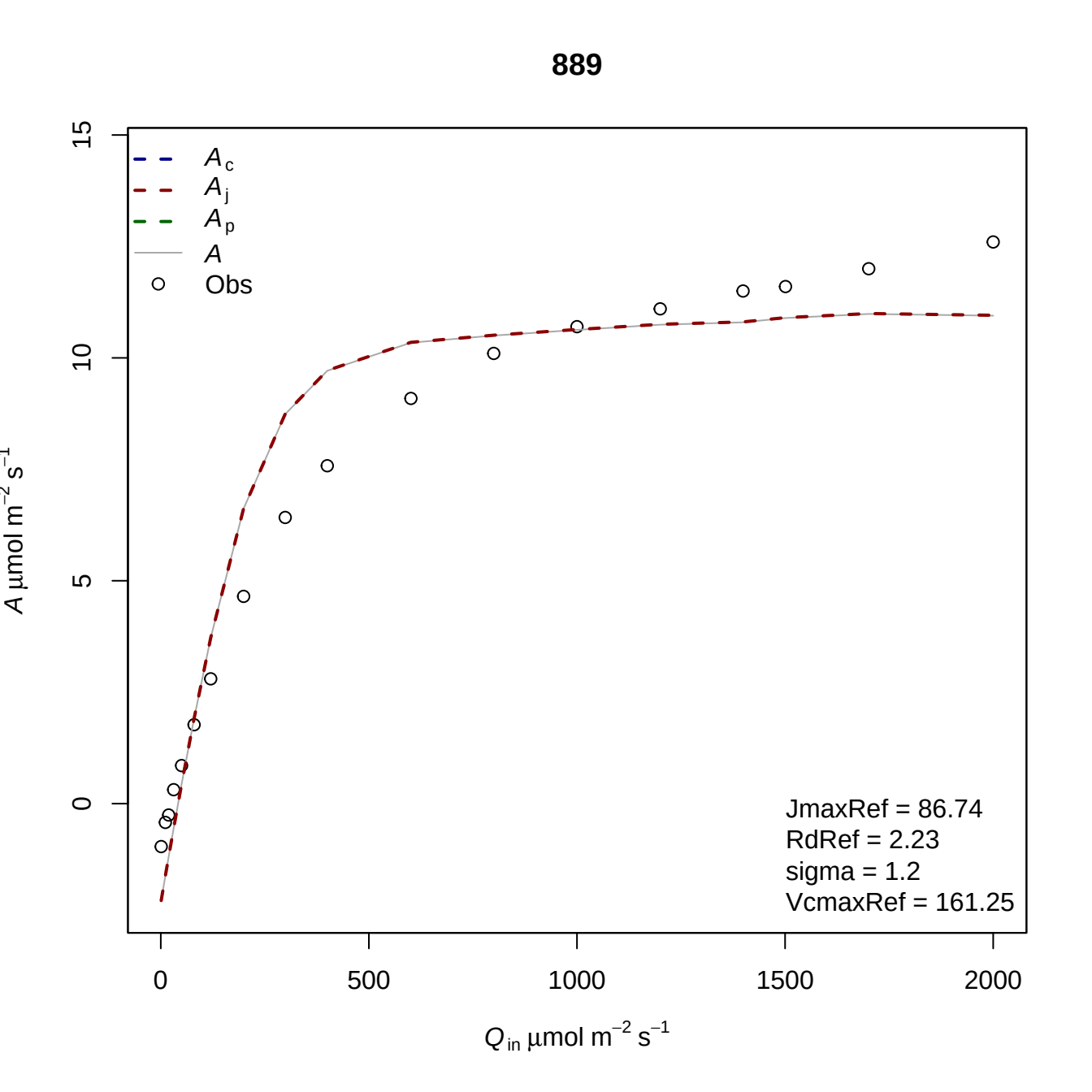

884
A_c
A_j
A_p
A
Obs
JmaxRef = 41.18
RdRef = 2.43
sigma = 0.75
VcmaxRef = 110.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

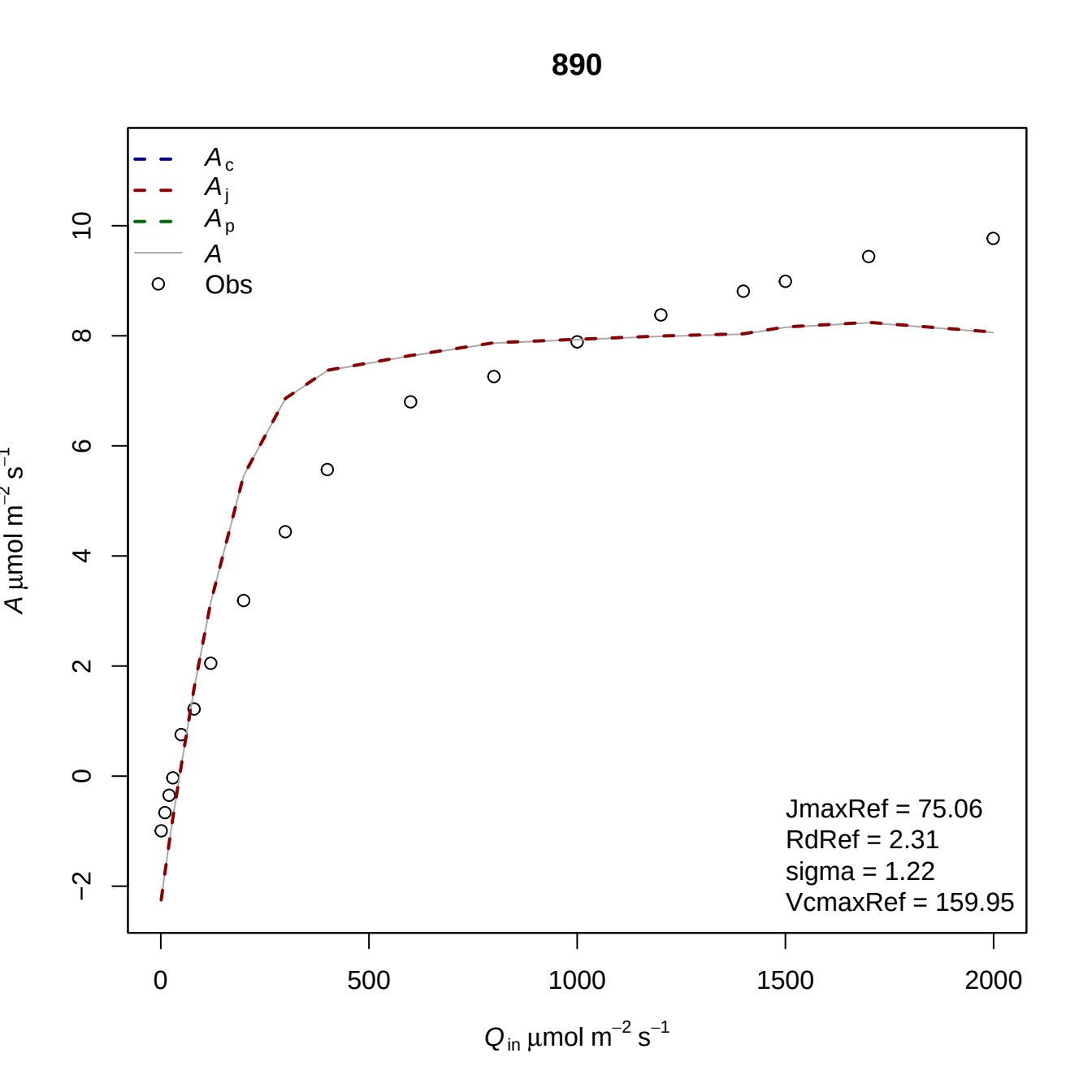

885
A_c
A_j
A_p
A
Obs
JmaxRef = 47.43
RdRef = 1.87
sigma = 0.94
VcmaxRef = 133.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

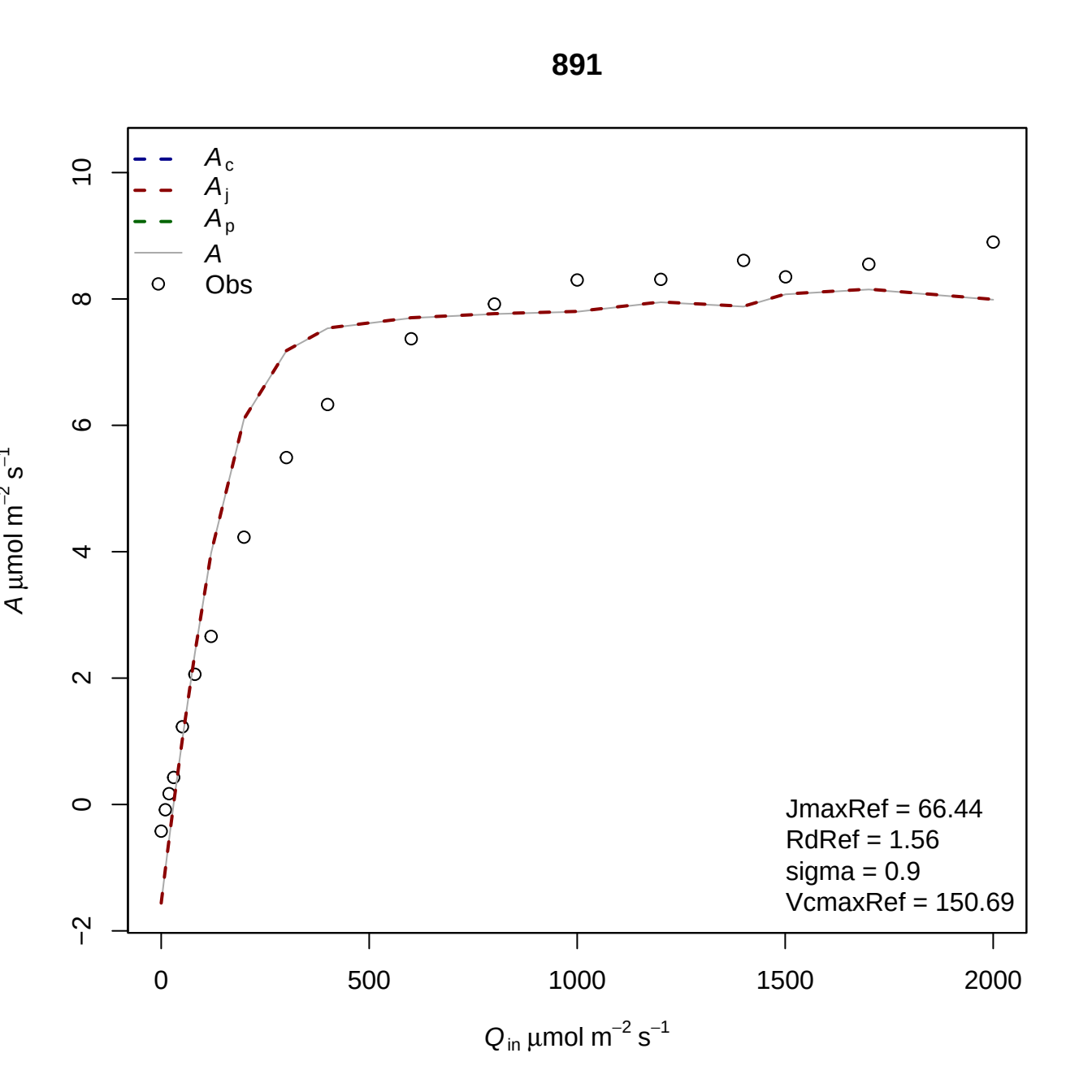

886
A_c
A_j
A_p
A
Obs
JmaxRef = 26.06
RdRef = 1.93
sigma = 0.57
VcmaxRef = 63.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500
2000

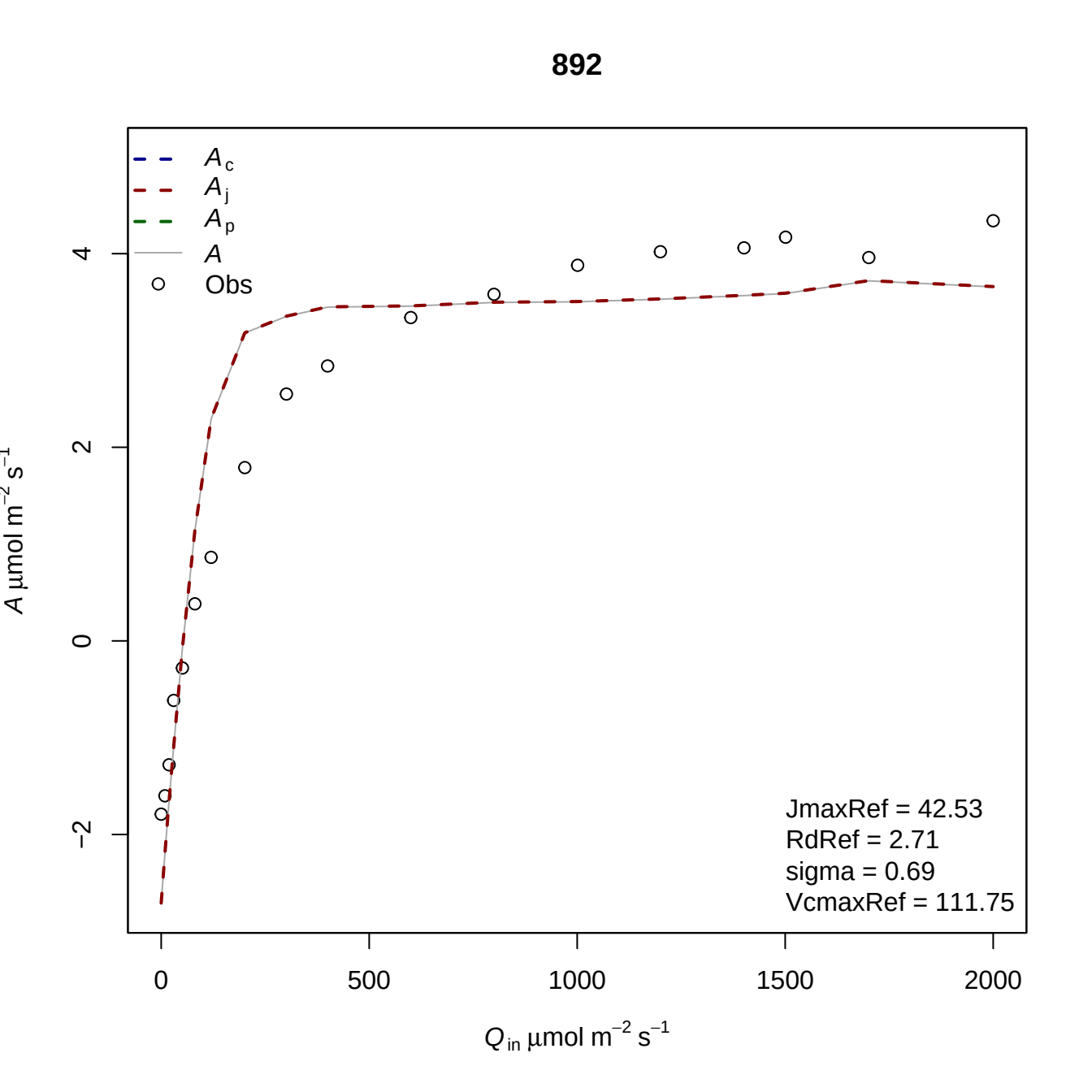

887
A_c
A_j
A_p
A
Obs
JmaxRef = 28.66
RdRef = 1.15
sigma = 0.48
VcmaxRef = 49.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

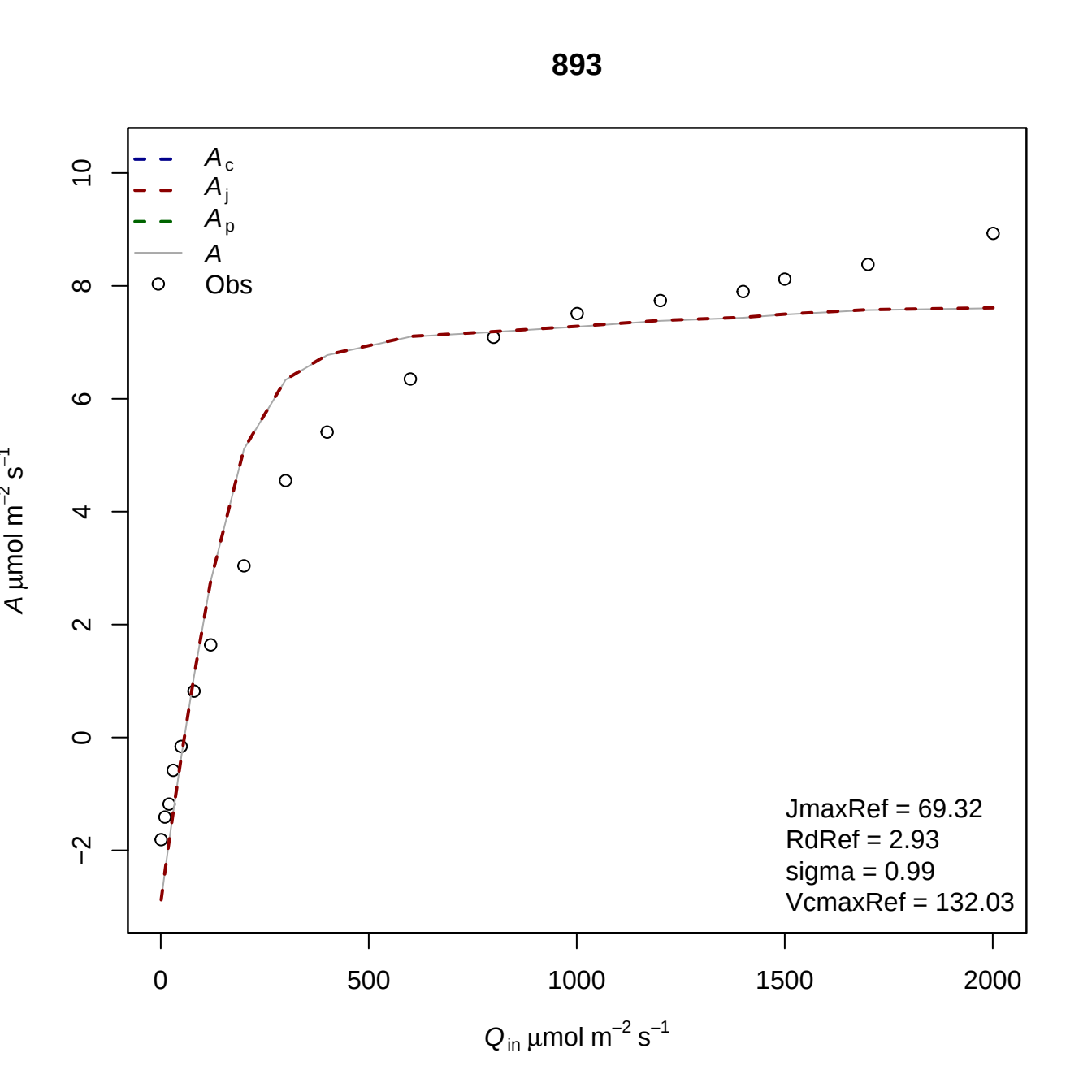

888
A_c
A_j
A_p
A
Obs
JmaxRef = 36.28
RdRef = 0.55
sigma = 0.2
VcmaxRef = 61.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

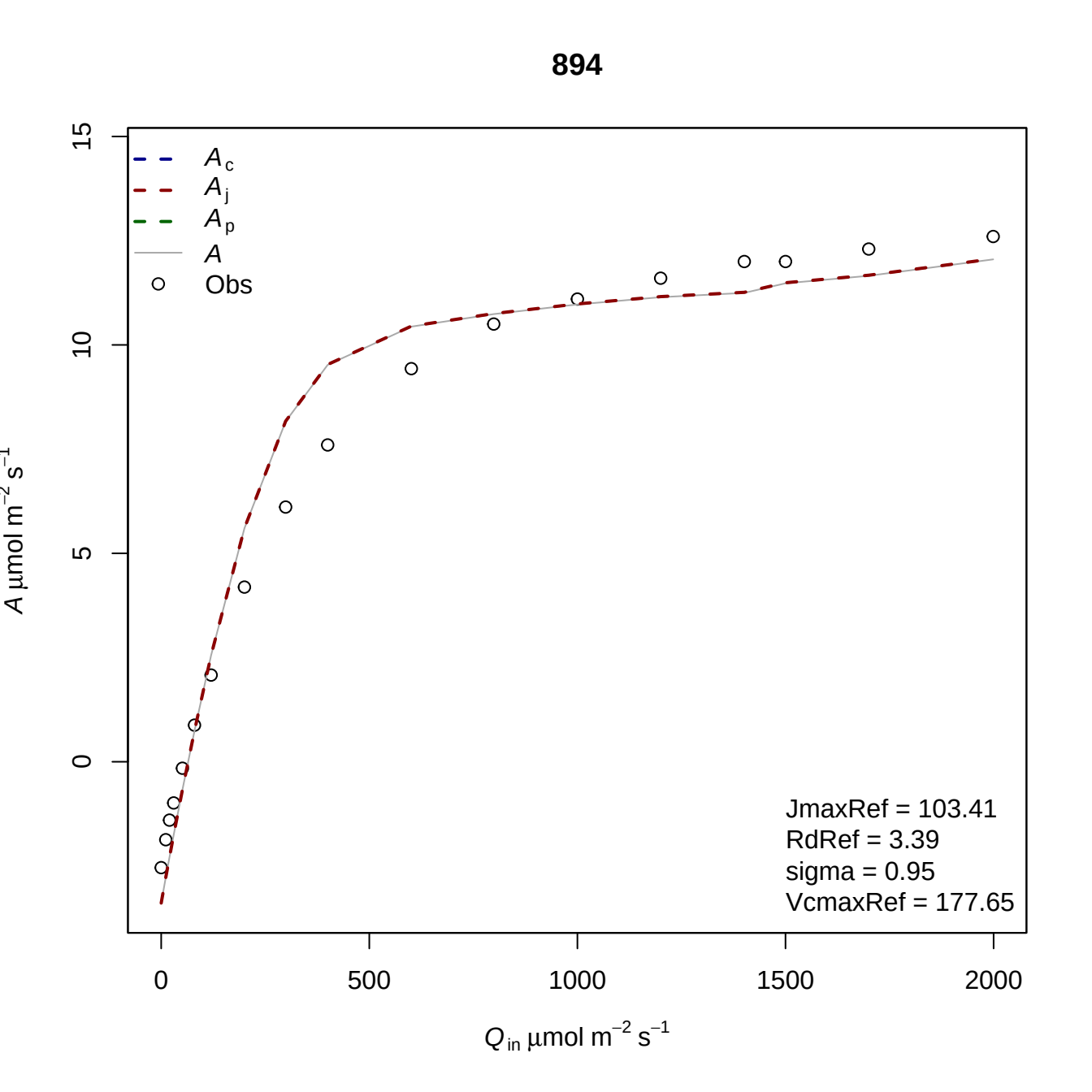

889
A_c
A_j
A_p
A
Obs
JmaxRef = 86.74
RdRef = 2.23
sigma = 1.2
VcmaxRef = 161.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

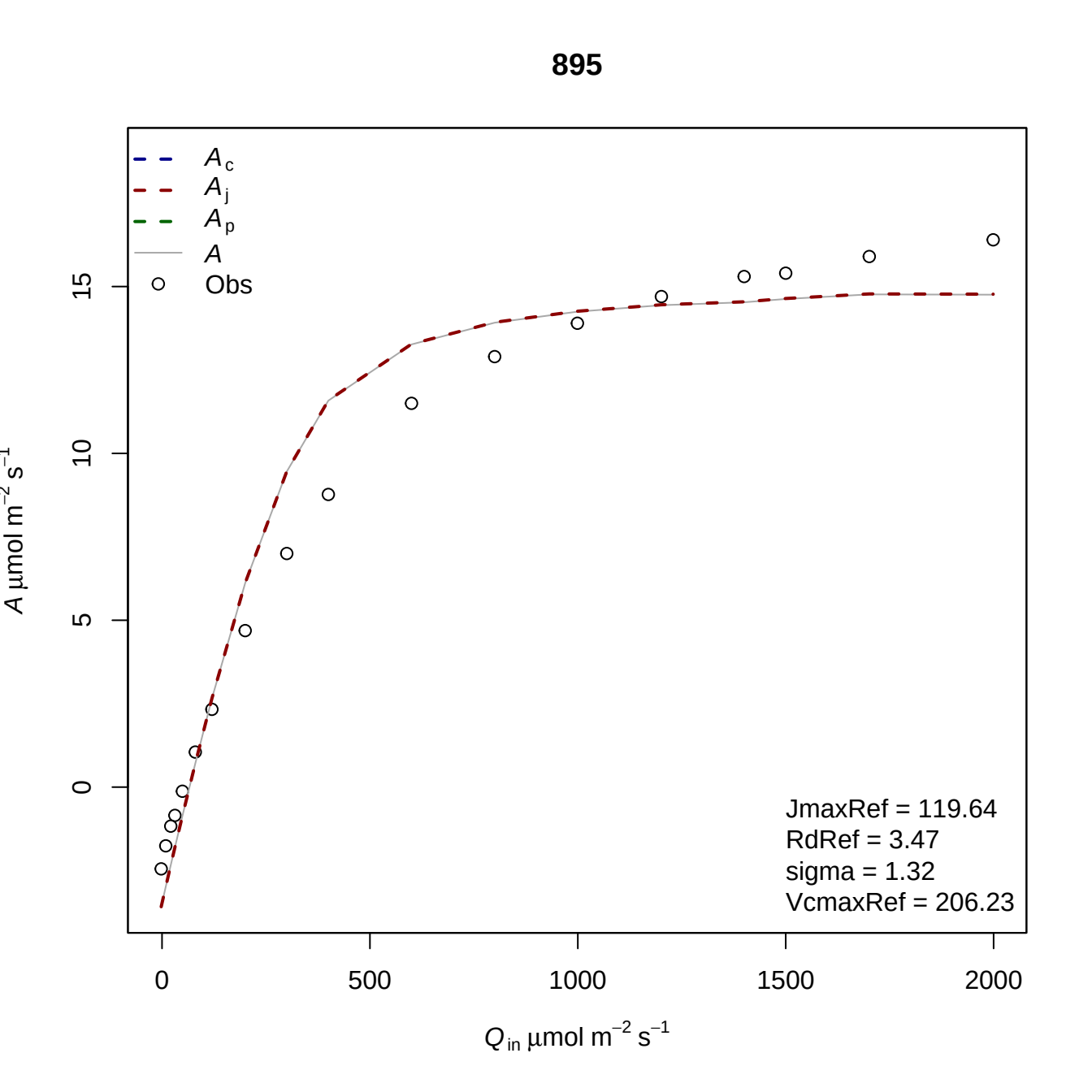

890
A_c
A_j
A_p
A
Obs
JmaxRef = 75.06
RdRef = 2.31
sigma = 1.22
VcmaxRef = 159.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

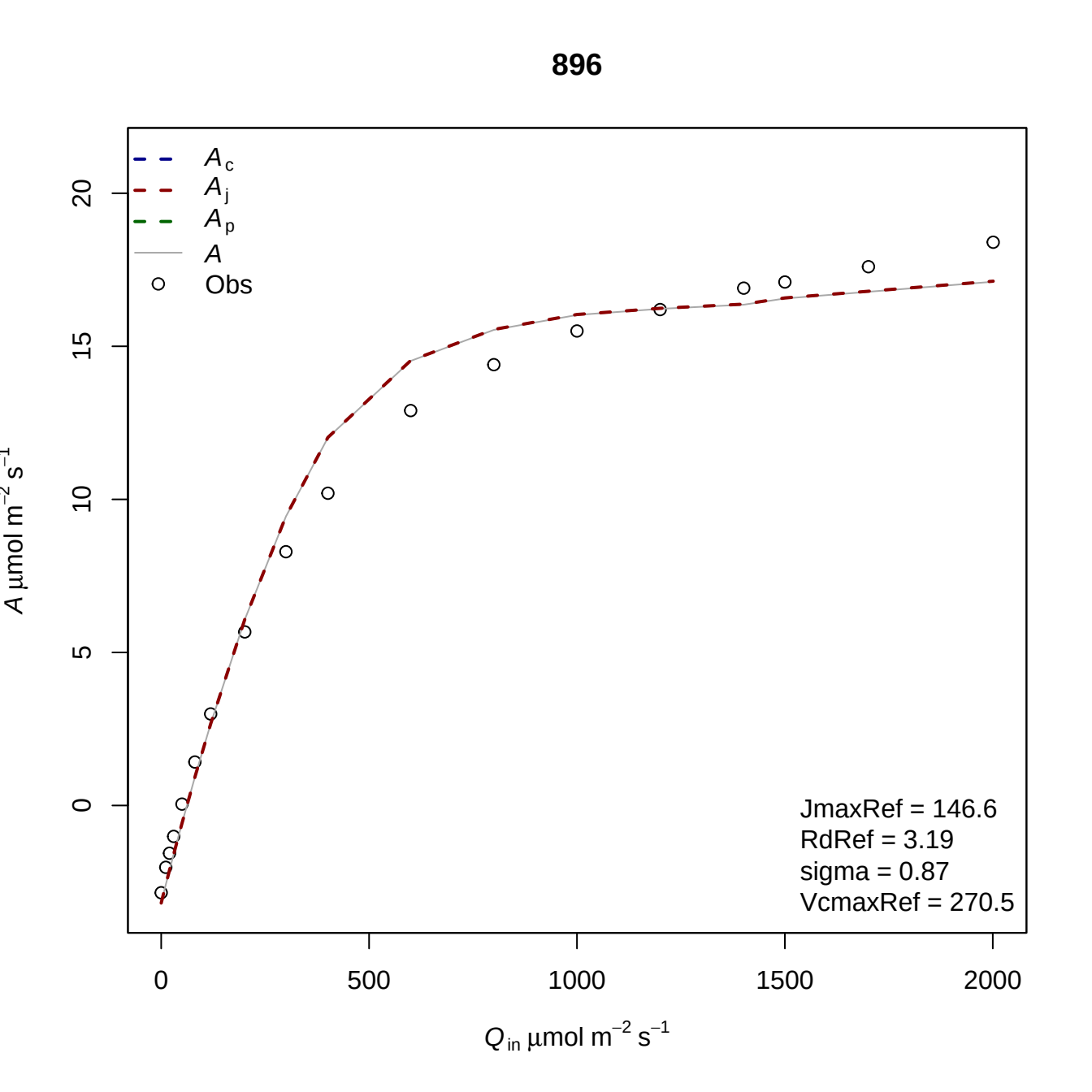

891
A_c
A_j
A_p
A
Obs
JmaxRef = 66.44
RdRef = 1.56
sigma = 0.9
VcmaxRef = 150.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

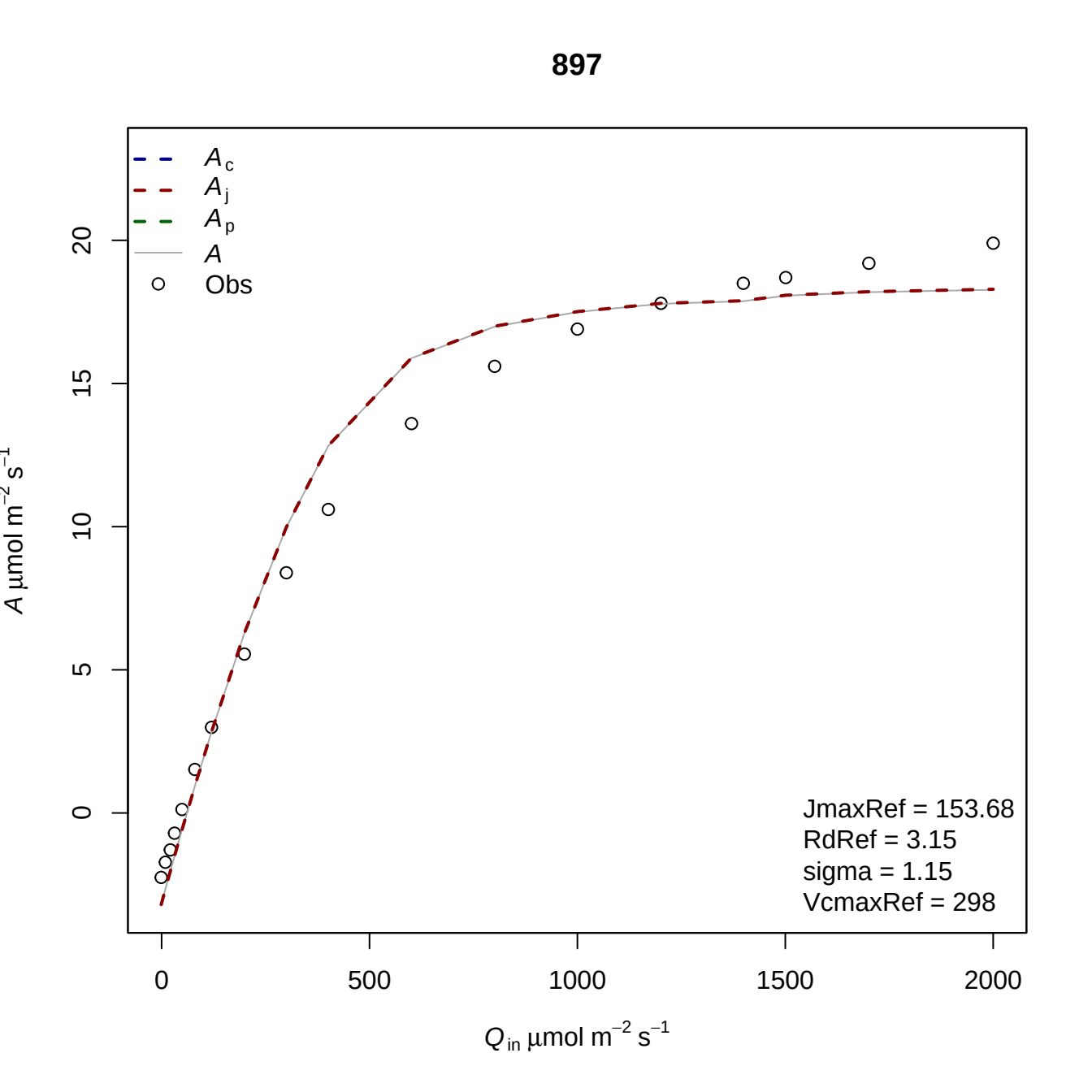

892
A_c
A_j
A_p
A
Obs
JmaxRef = 42.53
RdRef = 2.71
sigma = 0.69
VcmaxRef = 111.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

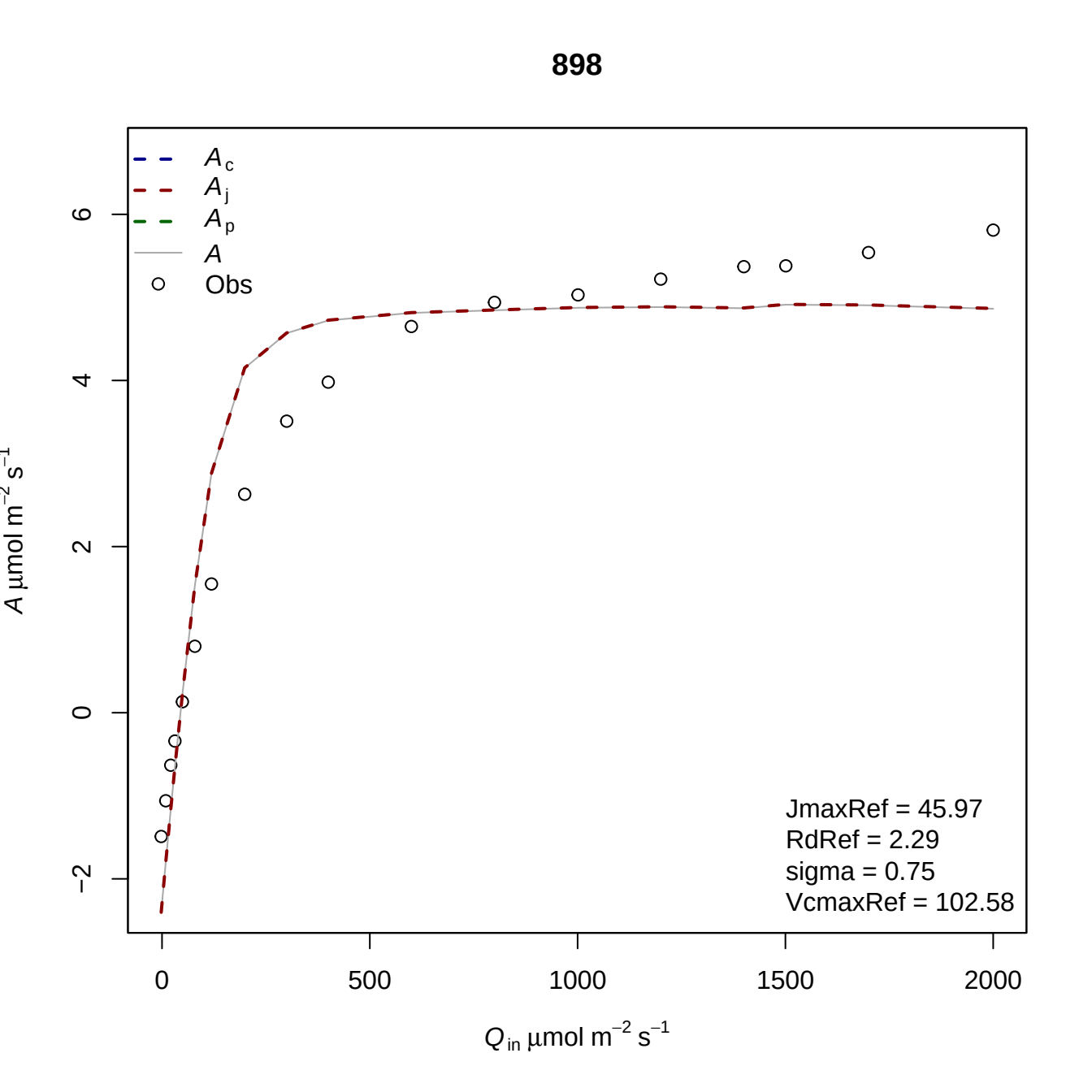

893
A_c
A_j
A_p
A
Obs
JmaxRef = 69.32
RdRef = 2.93
sigma = 0.99
VcmaxRef = 132.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

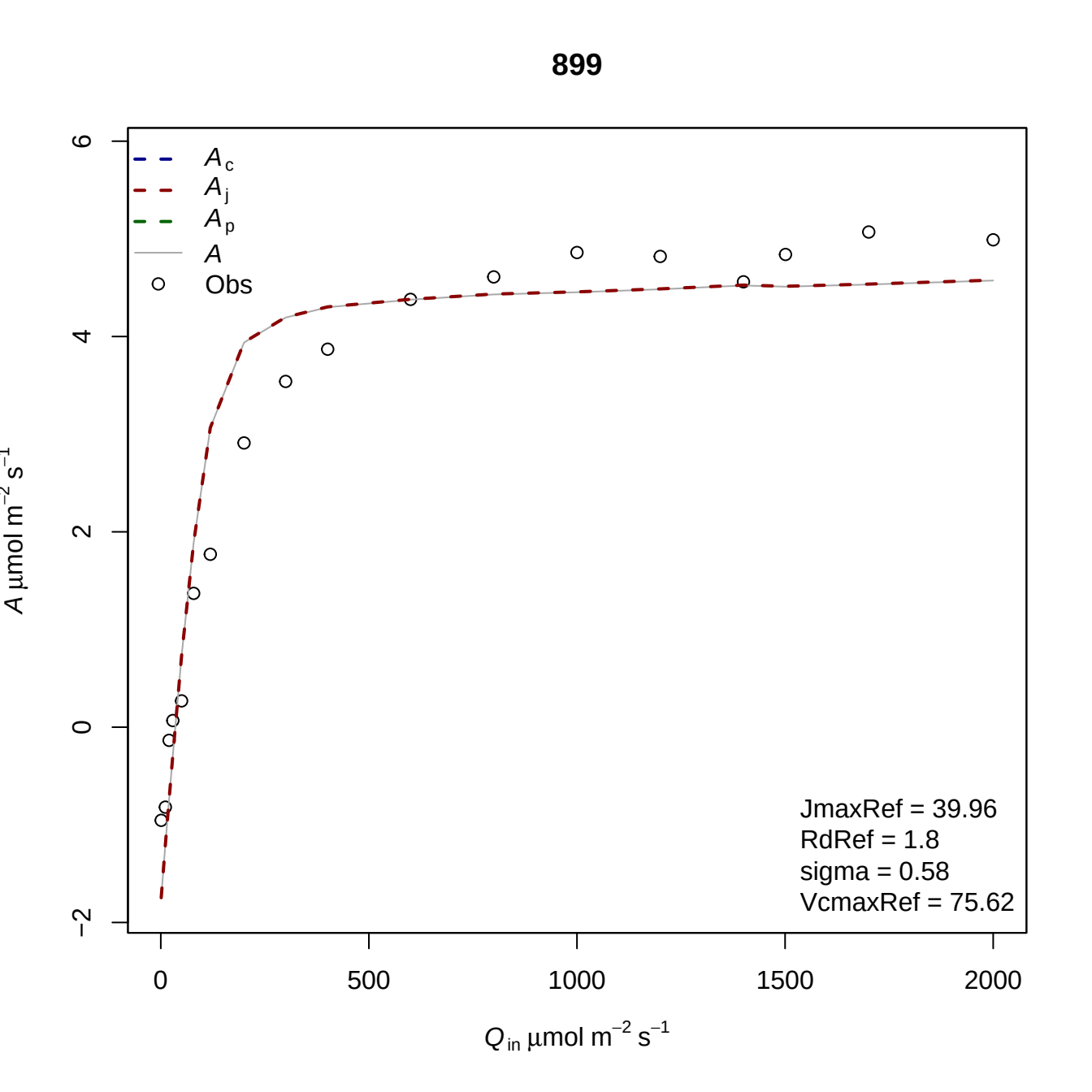

894
A_c
A_j
A_p
A
Obs
JmaxRef = 103.41
RdRef = 3.39
sigma = 0.95
VcmaxRef = 177.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

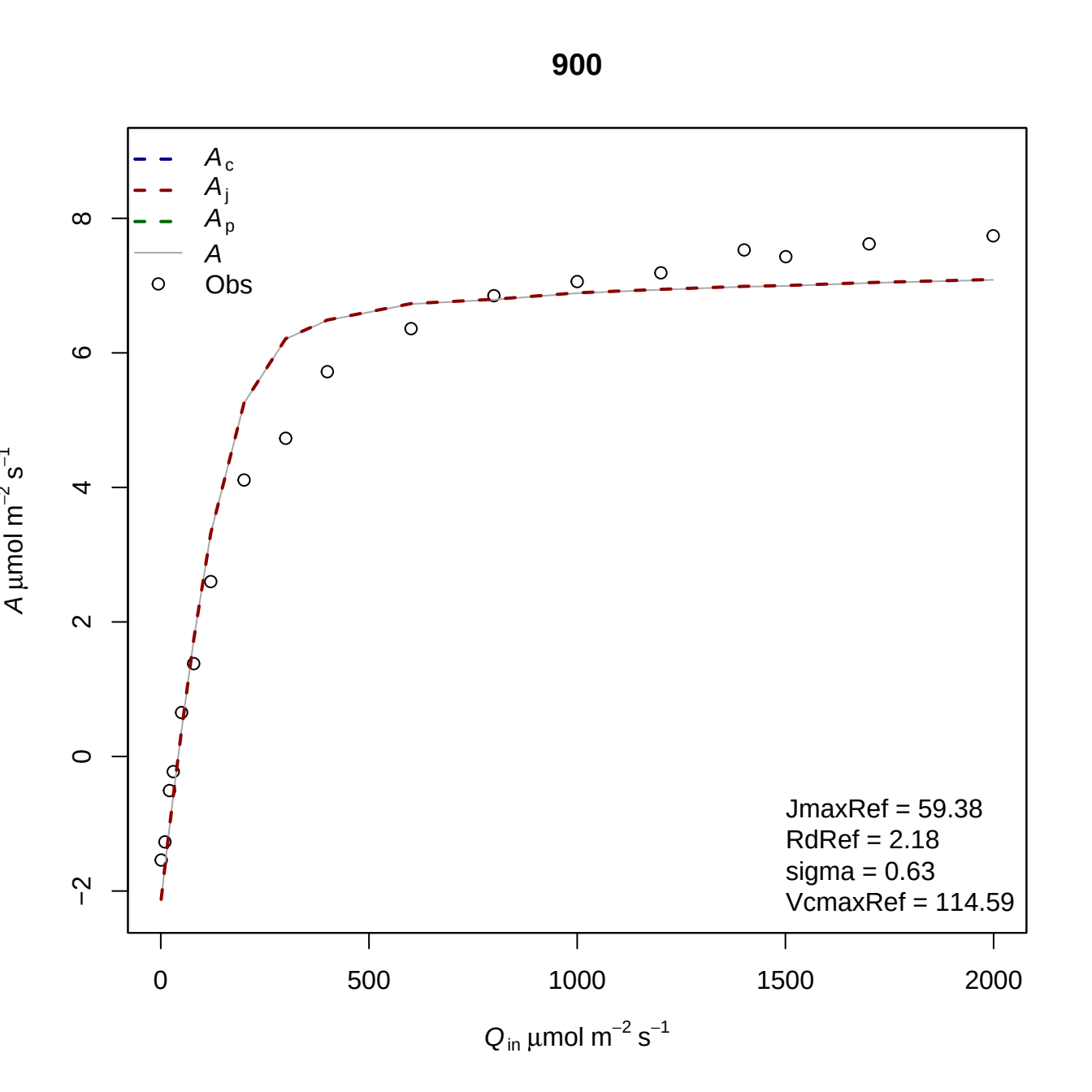

895
A_c
A_j
A_p
A
Obs
JmaxRef = 119.64
RdRef = 3.47
sigma = 1.32
VcmaxRef = 206.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

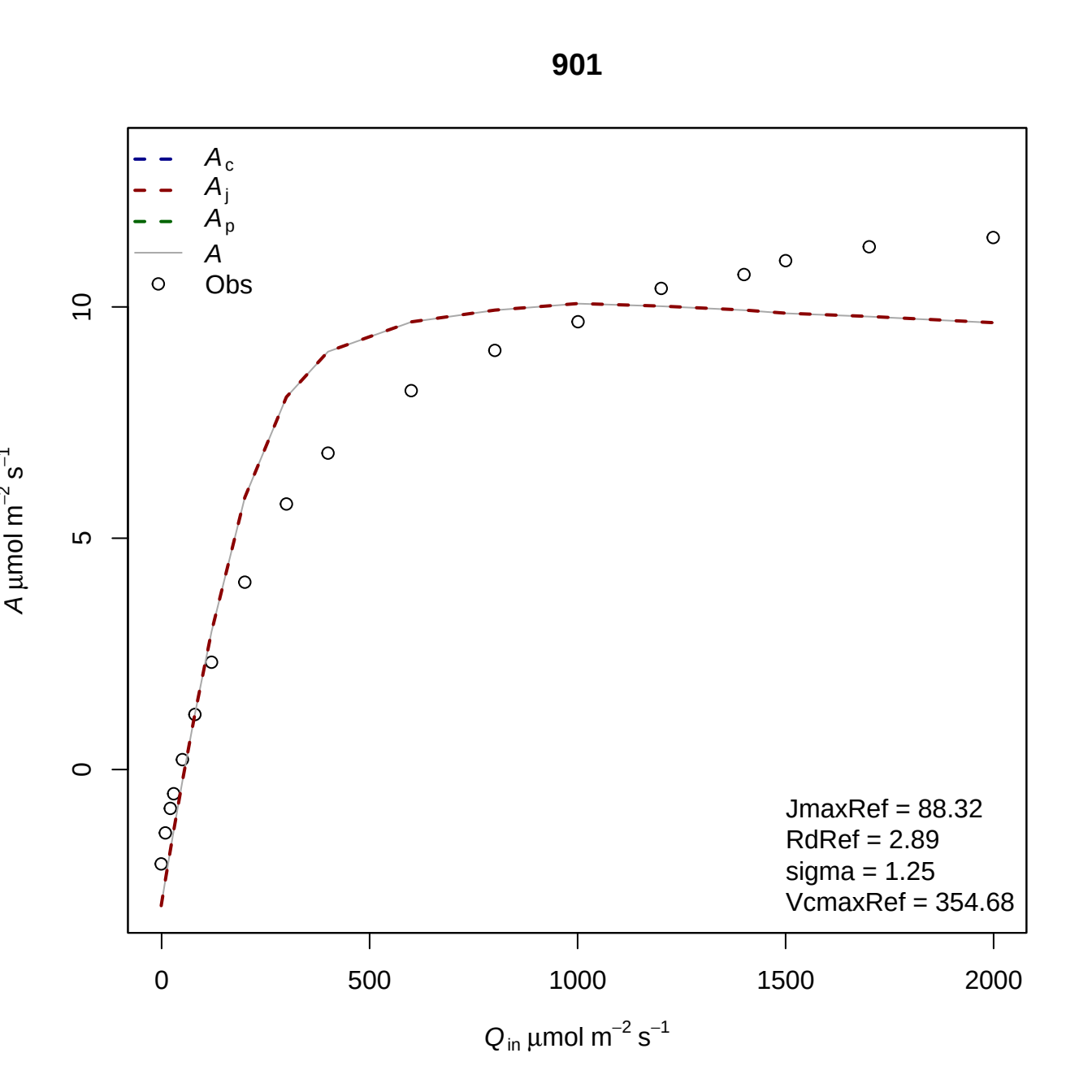

896
A_c
A_j
A_p
A
Obs
JmaxRef = 146.6
RdRef = 3.19
sigma = 0.87
VcmaxRef = 270.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

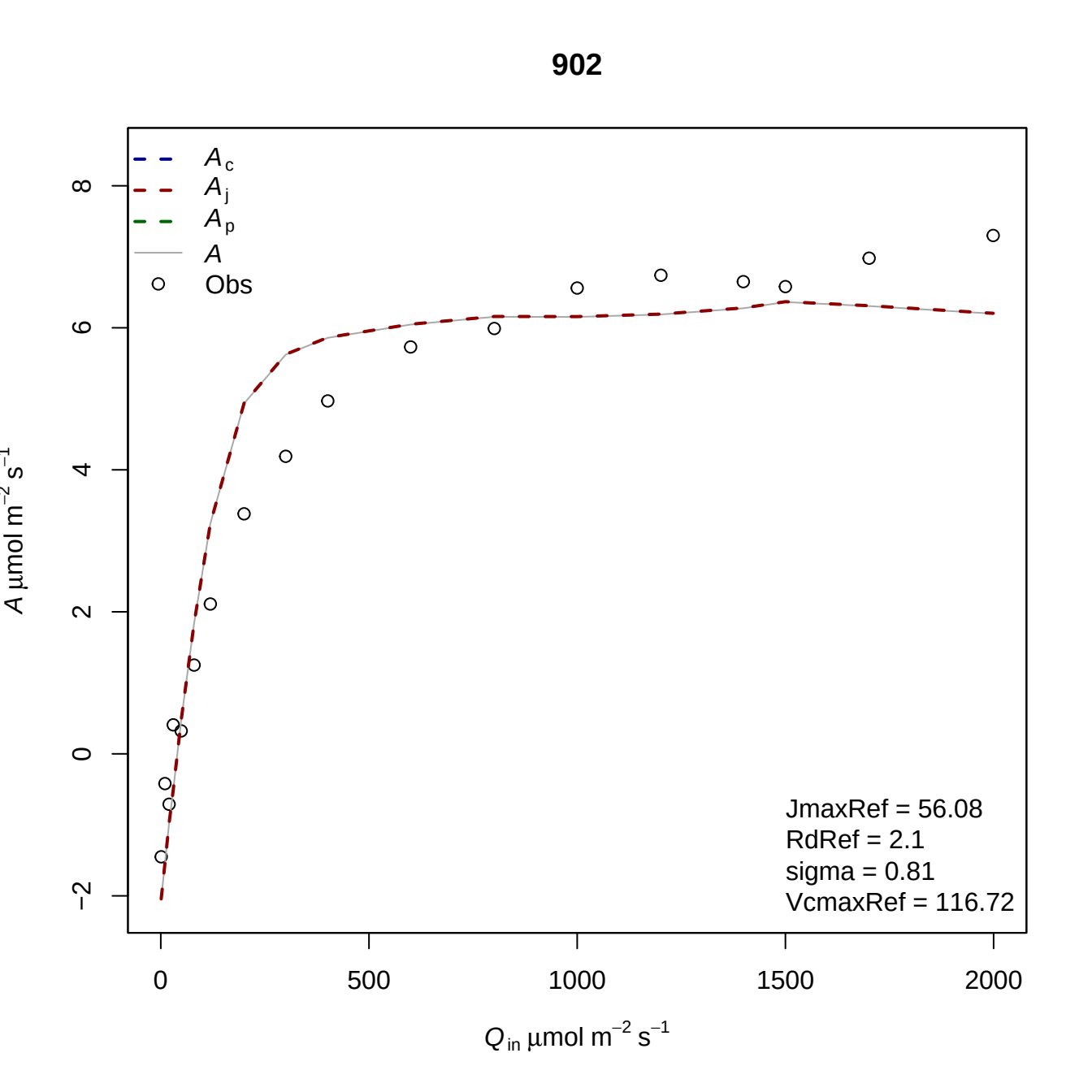

897
A_c
A_j
A_p
A
Obs
JmaxRef = 153.68
RdRef = 3.15
sigma = 1.15
VcmaxRef = 298
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

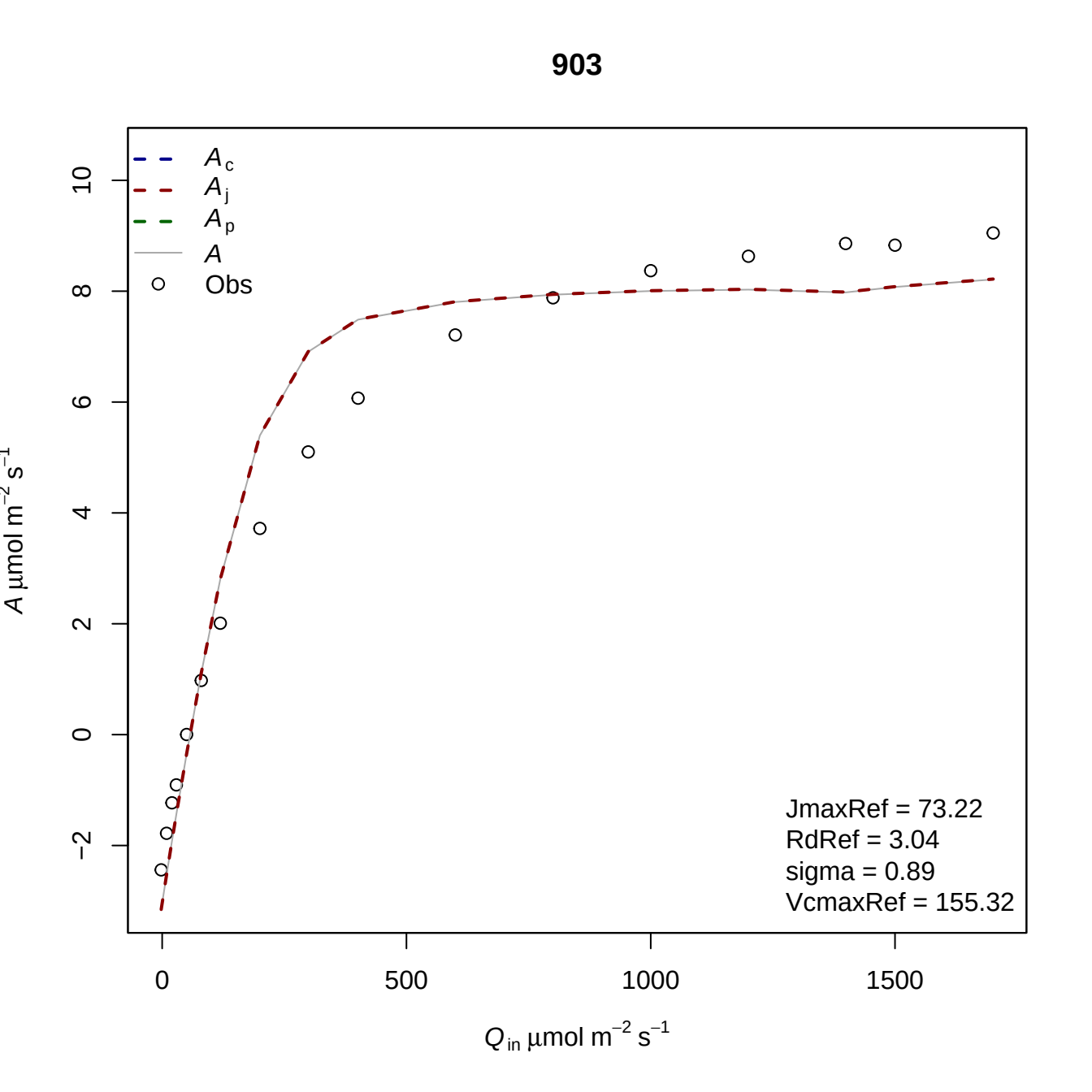

898
A_c
A_j
A_p
A
Obs
JmaxRef = 45.97
RdRef = 2.29
sigma = 0.75
VcmaxRef = 102.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

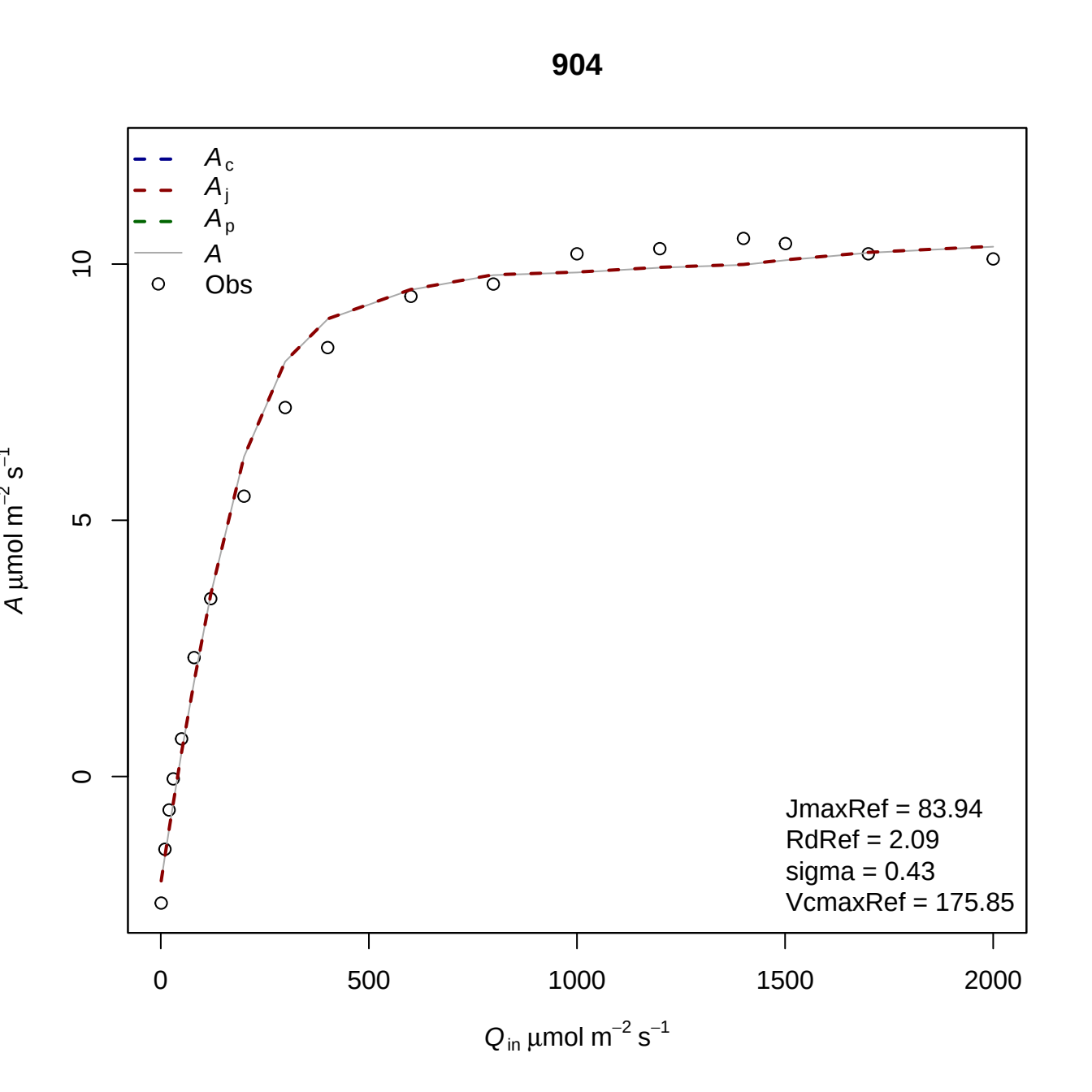

899
A_c
A_j
A_p
A
Obs
JmaxRef = 39.96
RdRef = 1.8
sigma = 0.58
VcmaxRef = 75.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

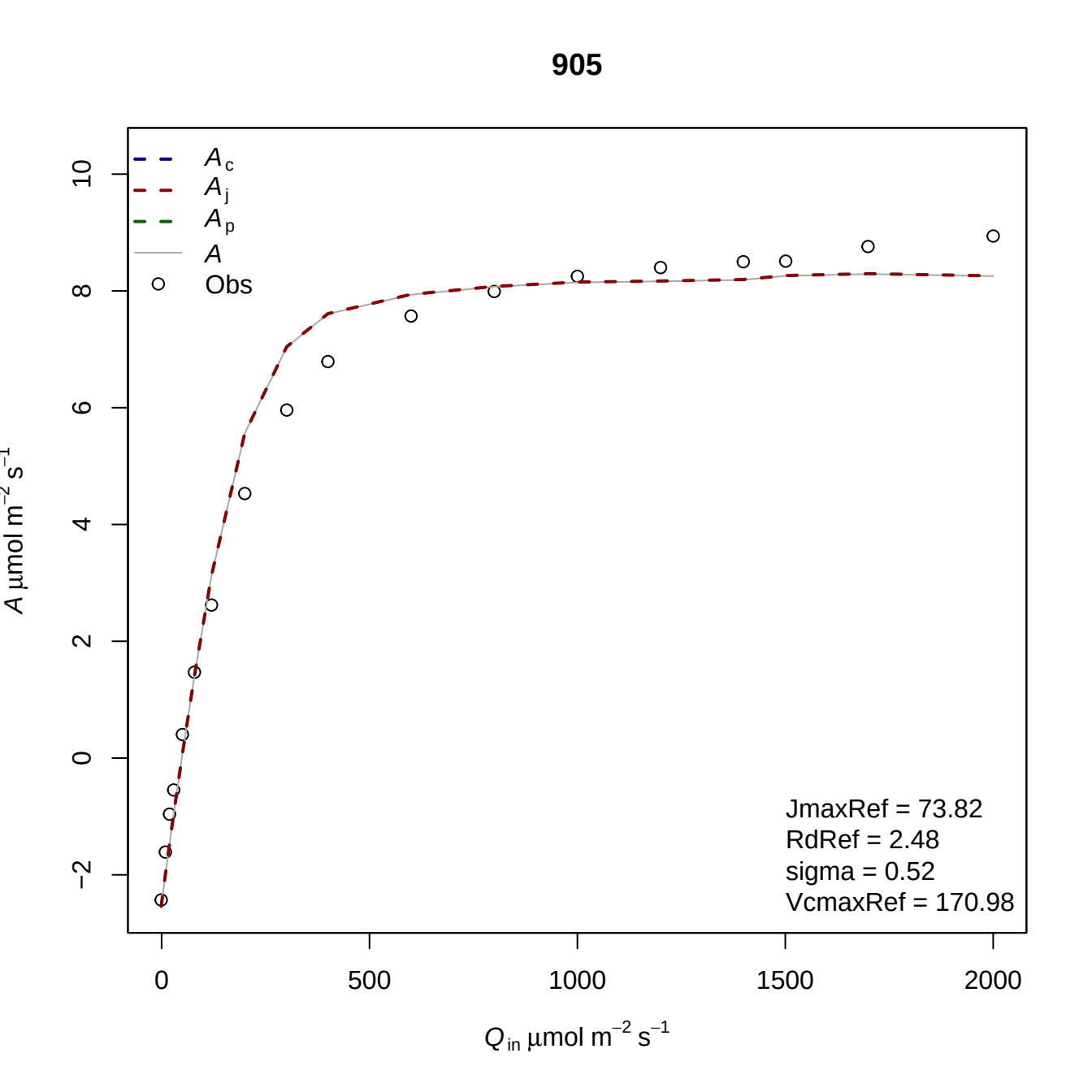

900
A_c
A_j
A_p
A
Obs
JmaxRef = 59.38
RdRef = 2.18
sigma = 0.63
VcmaxRef = 114.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

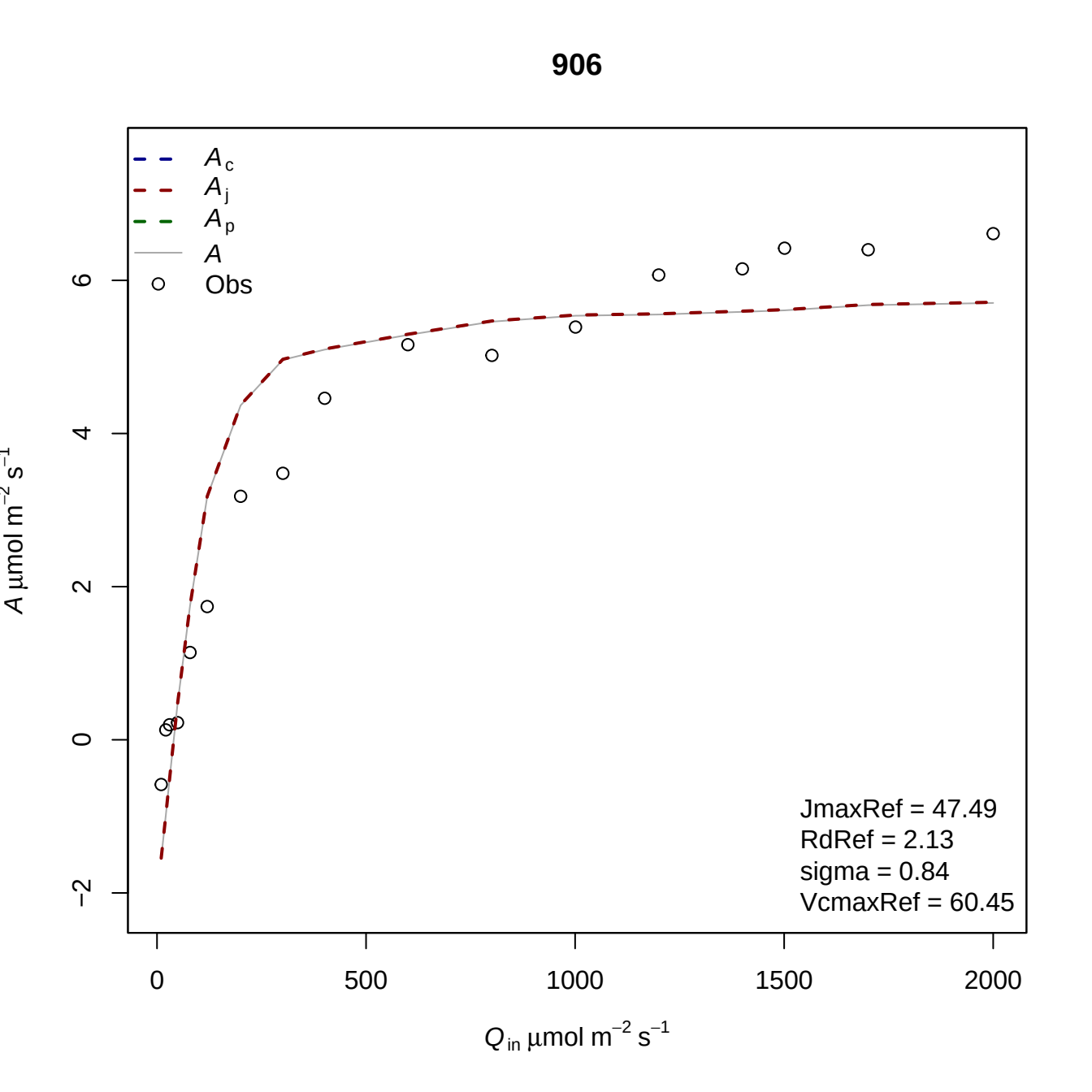

901
A_c
A_j
A_p
A
Obs
JmaxRef = 88.32
RdRef = 2.89
sigma = 1.25
VcmaxRef = 354.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

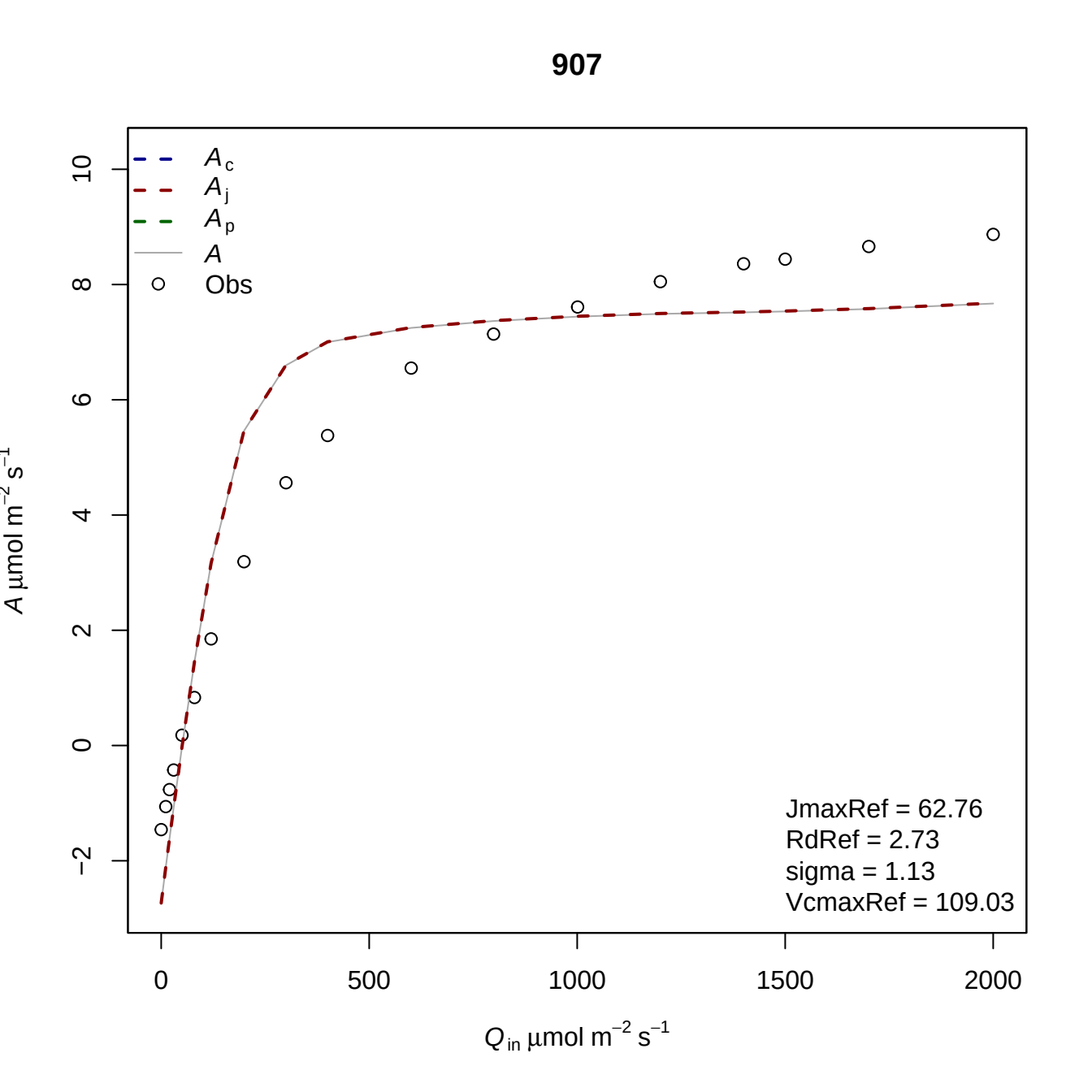

902
A_c
A_j
A_p
A
Obs
JmaxRef = 56.08
RdRef = 2.1
sigma = 0.81
VcmaxRef = 116.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

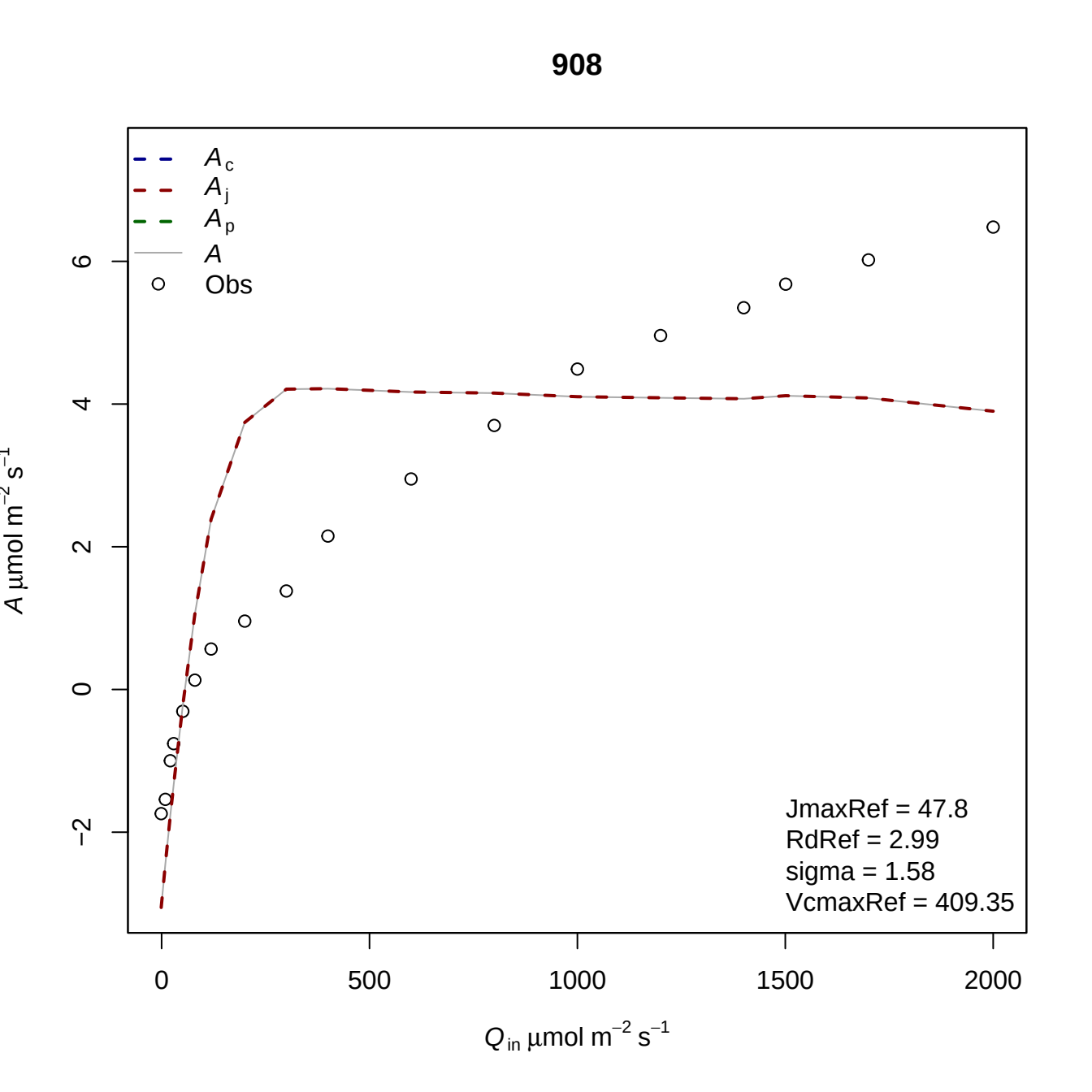

903
A_c
A_j
A_p
A
Obs
JmaxRef = 73.22
RdRef = 3.04
sigma = 0.89
VcmaxRef = 155.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

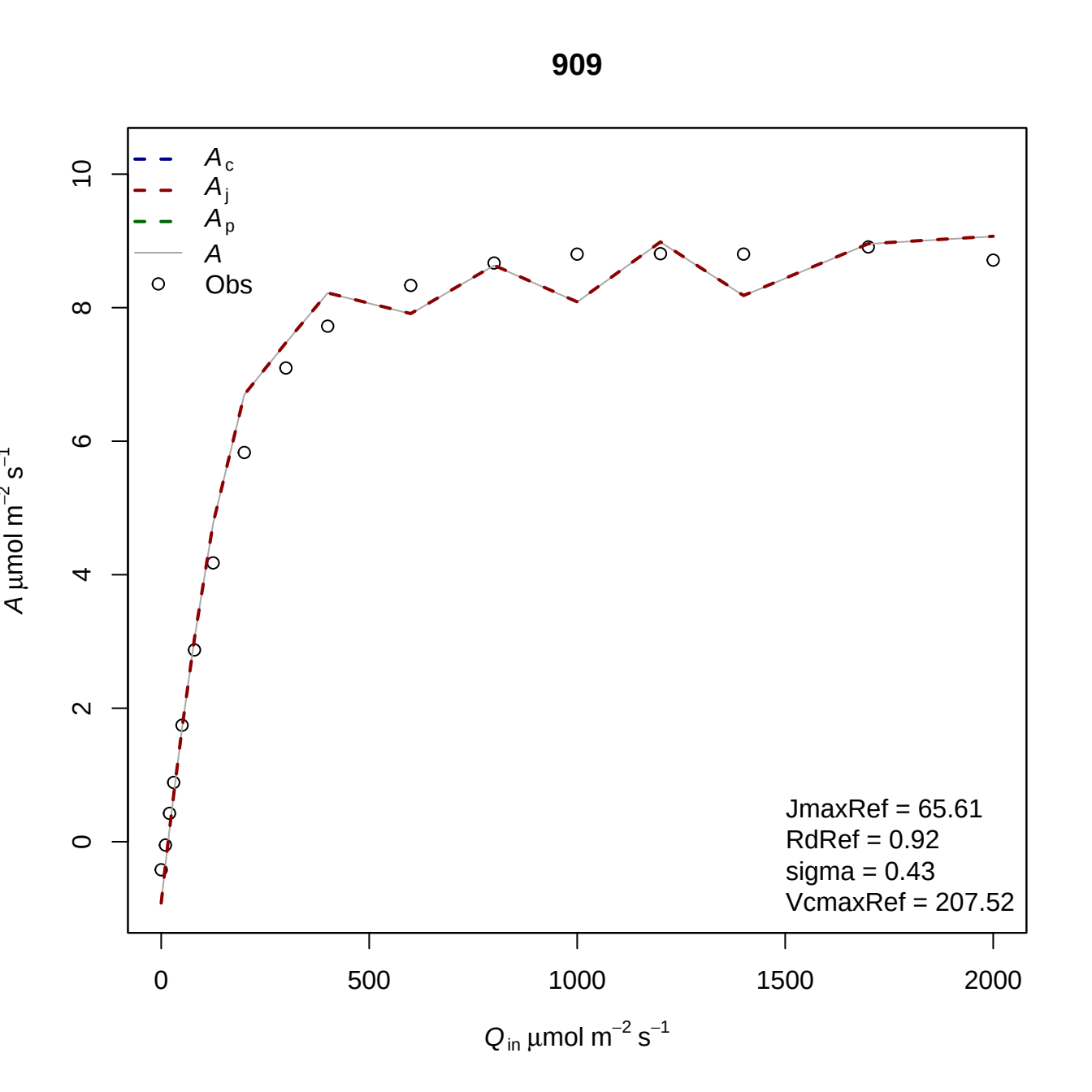

904
A_c
A_j
A_p
A
Obs
JmaxRef = 83.94
RdRef = 2.09
sigma = 0.43
VcmaxRef = 175.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

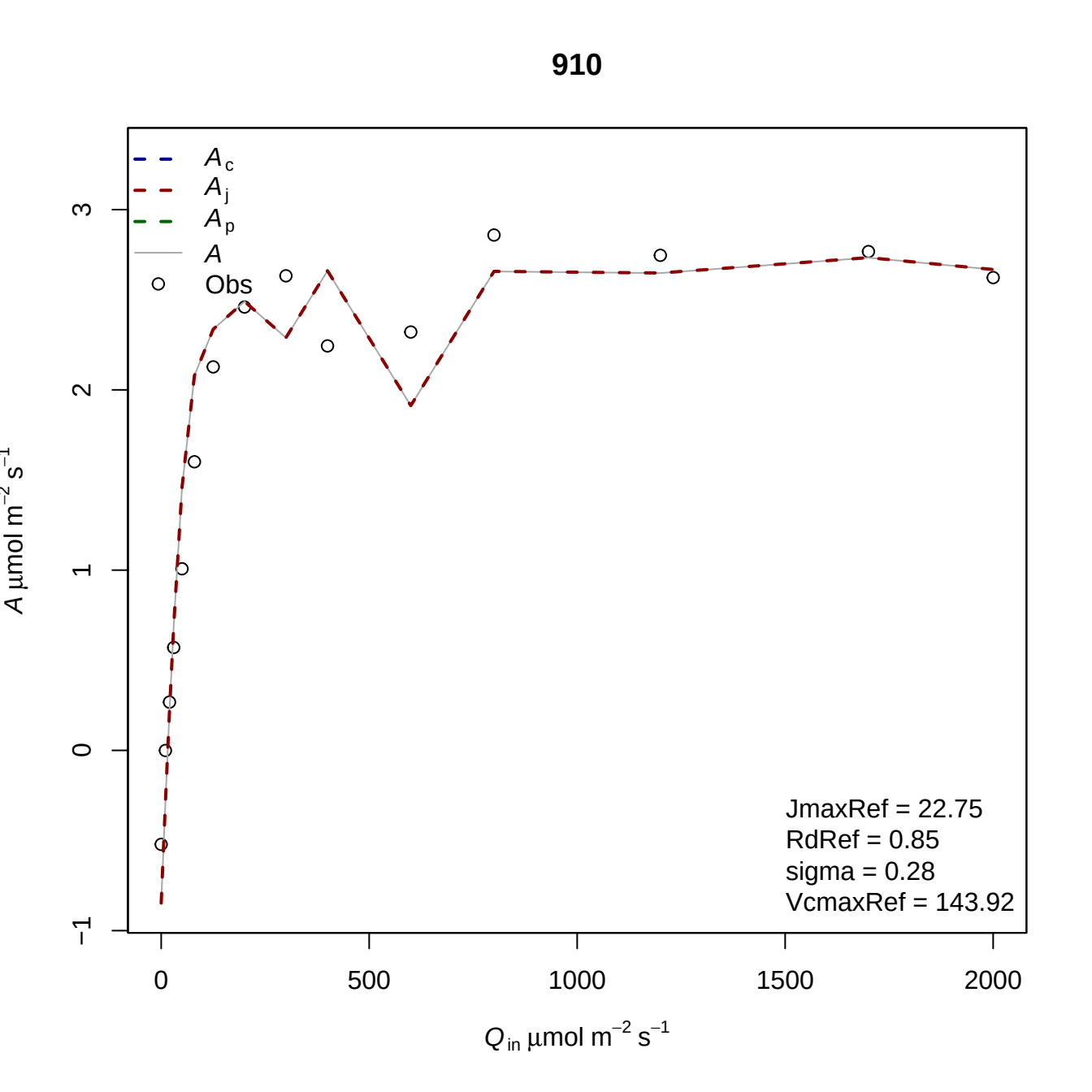

905
A_c
A_j
A_p
A
Obs
JmaxRef = 73.82
RdRef = 2.48
sigma = 0.52
VcmaxRef = 170.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

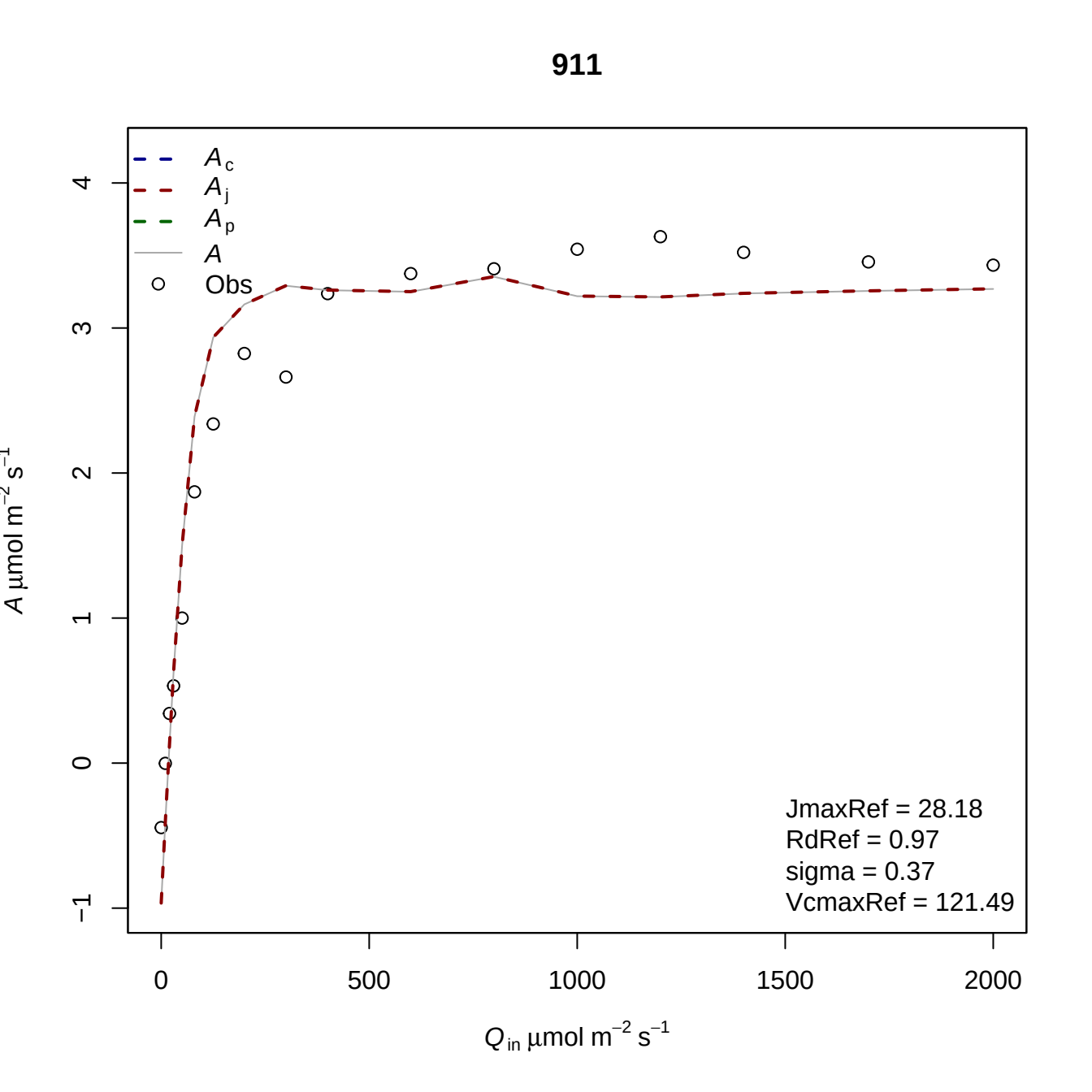

906
A_c
A_j
A_p
A
Obs
JmaxRef = 47.49
RdRef = 2.13
sigma = 0.84
VcmaxRef = 60.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

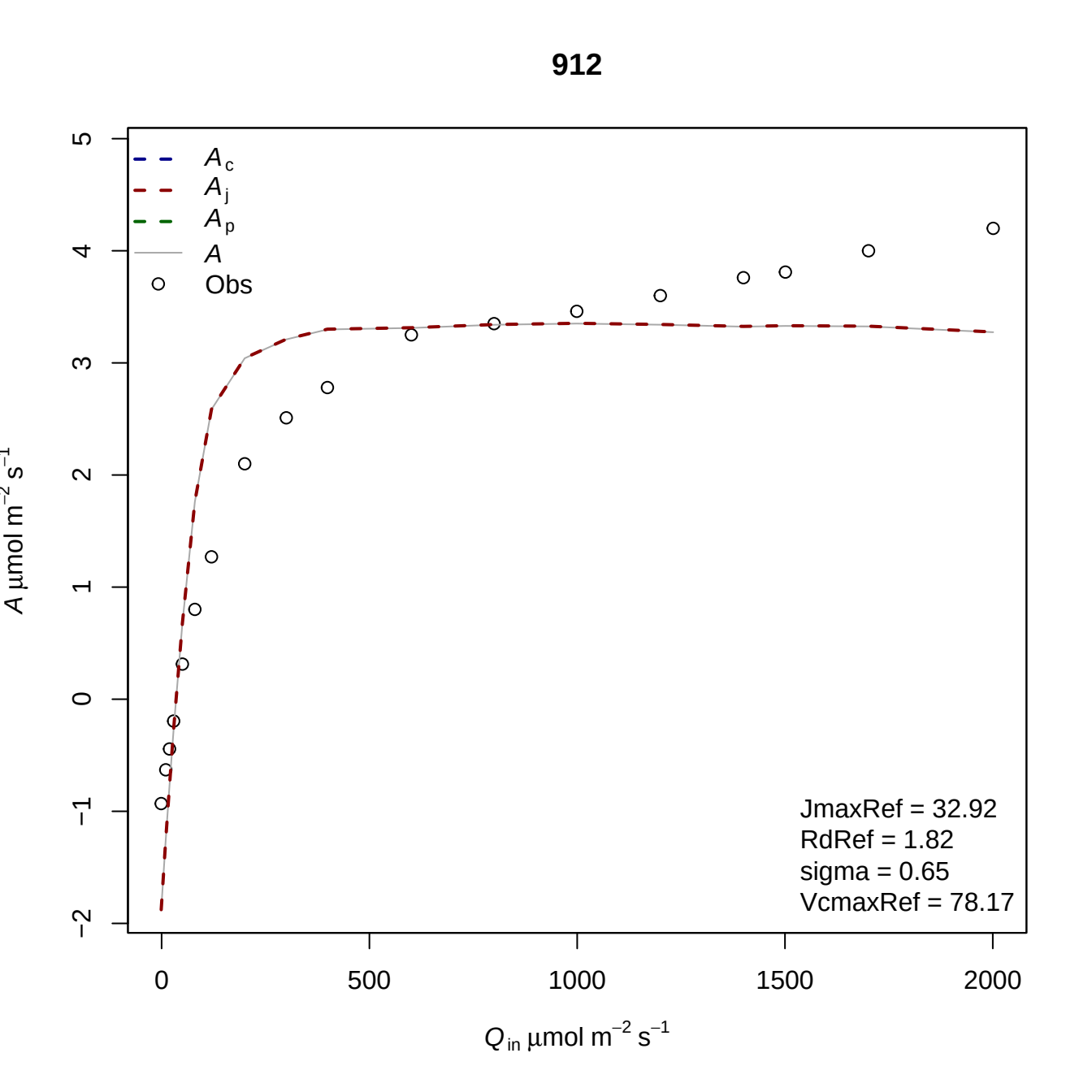

907
A_c
A_j
A_p
A
Obs
JmaxRef = 62.76
RdRef = 2.73
sigma = 1.13
VcmaxRef = 109.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

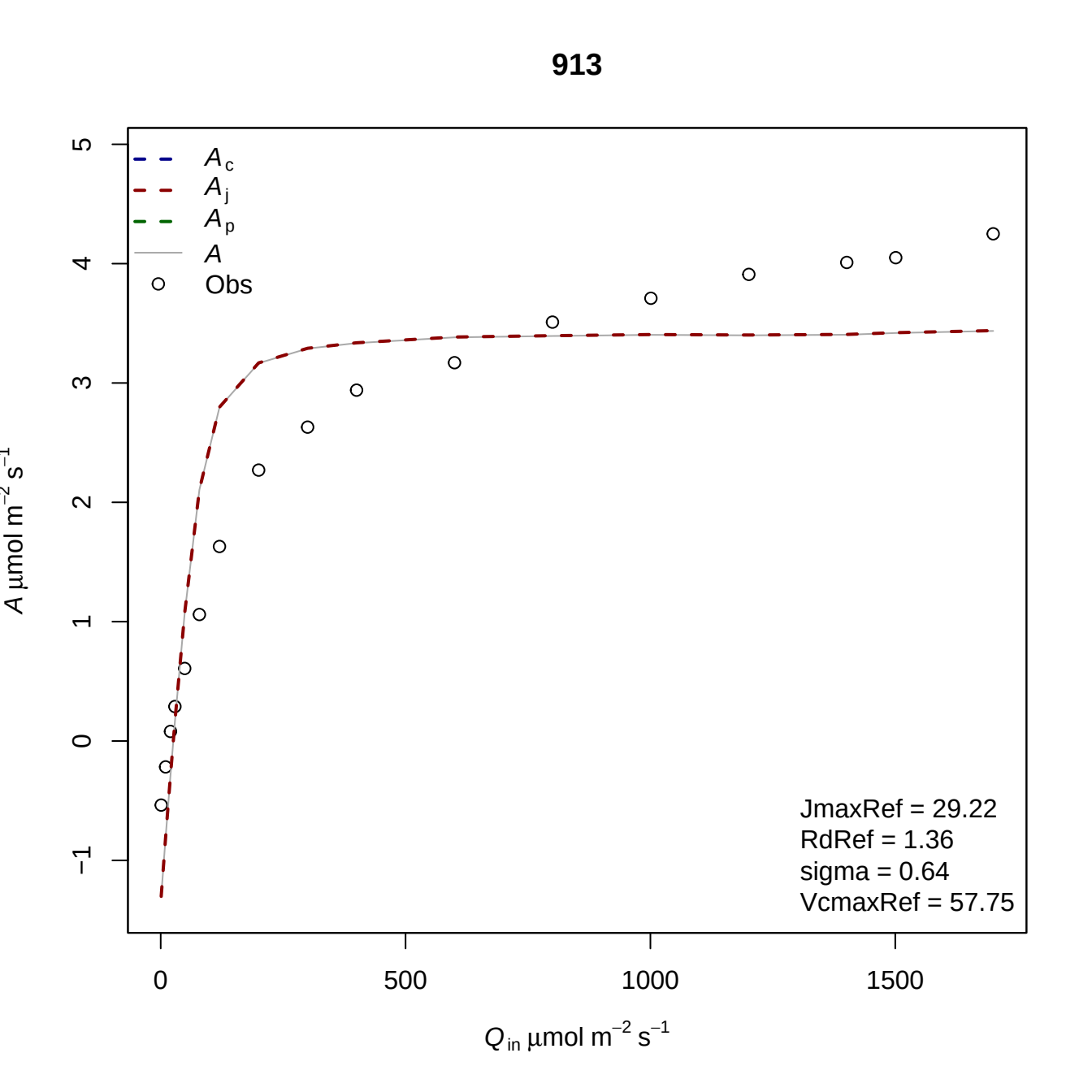

908
A_c
A_j
A_p
A
Obs
JmaxRef = 47.8
RdRef = 2.99
sigma = 1.58
VcmaxRef = 409.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

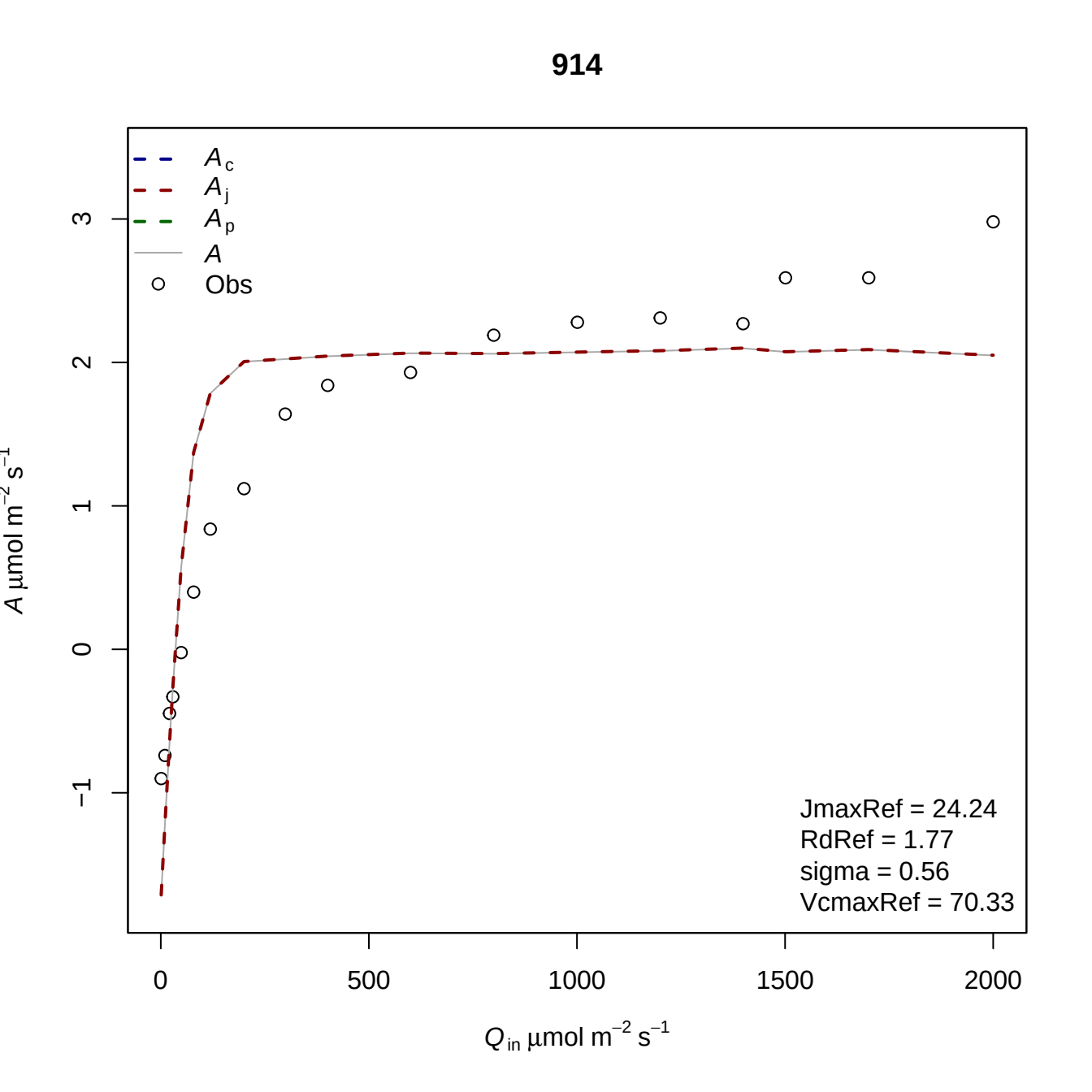

909
A_c
A_j
A_p
A
Obs
JmaxRef = 65.61
RdRef = 0.92
sigma = 0.43
VcmaxRef = 207.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

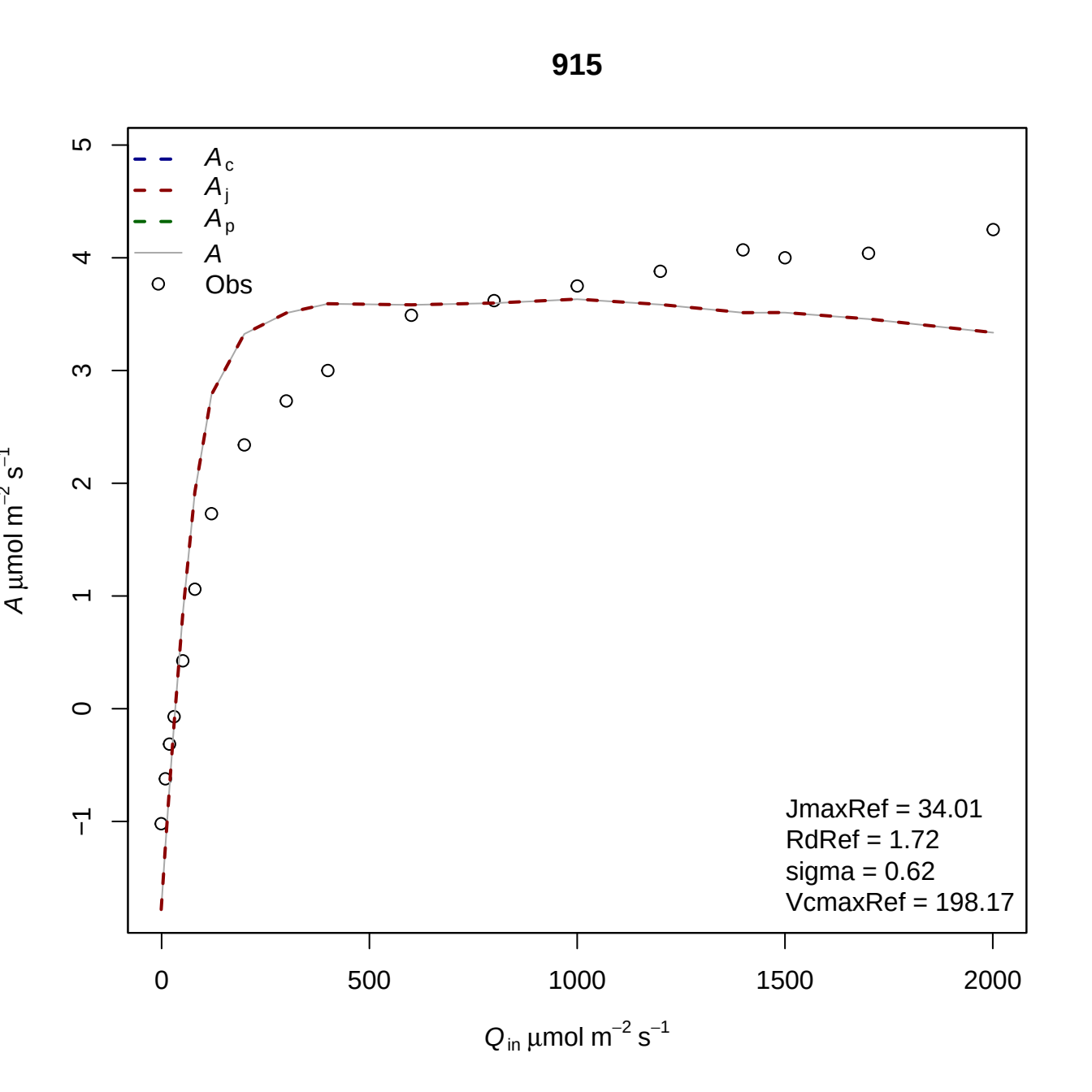

910
A_c
A_j
A_p
A
Obs
JmaxRef = 22.75
RdRef = 0.85
sigma = 0.28
VcmaxRef = 143.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

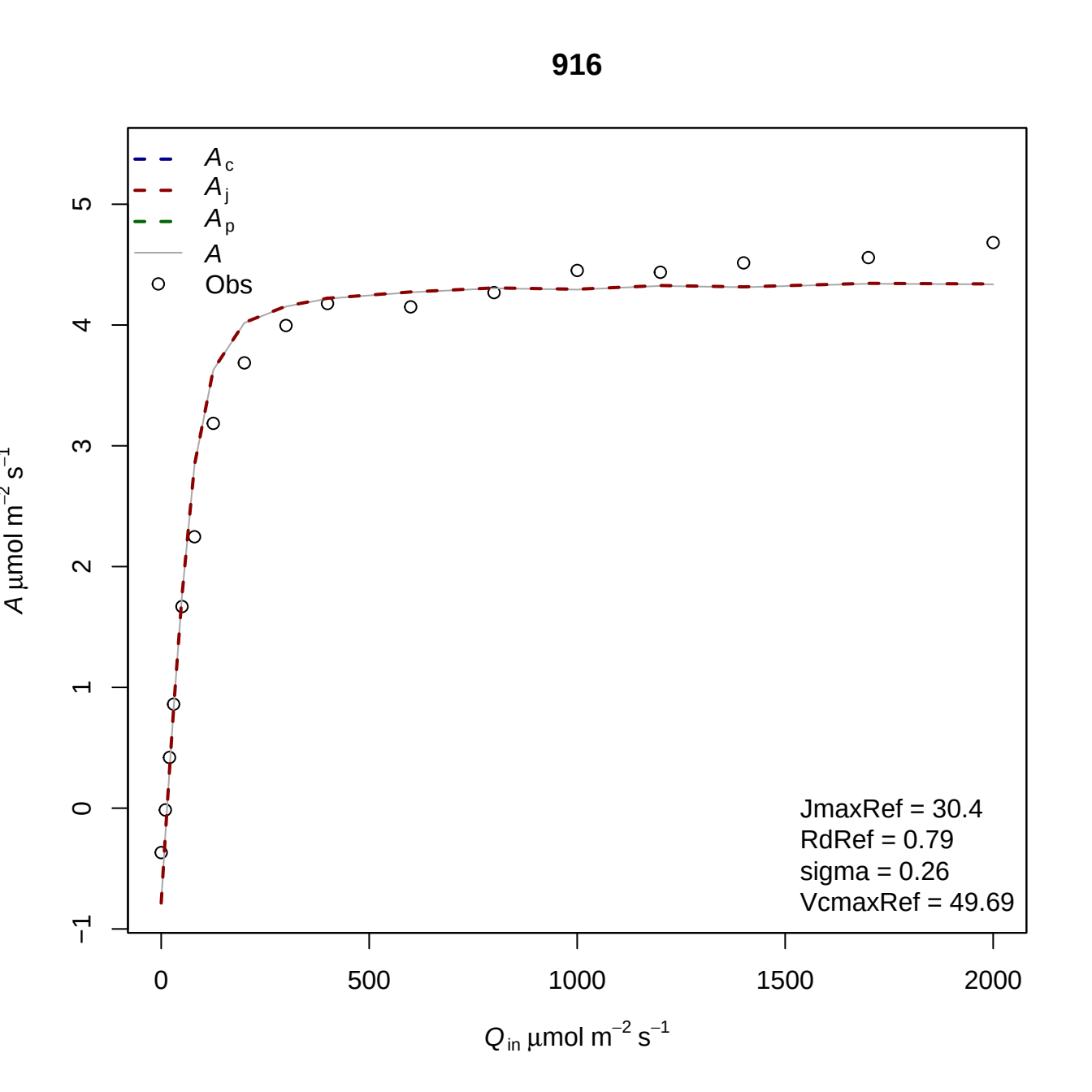

911
A_c
A_j
A_p
A
Obs
JmaxRef = 28.18
RdRef = 0.97
sigma = 0.37
VcmaxRef = 121.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

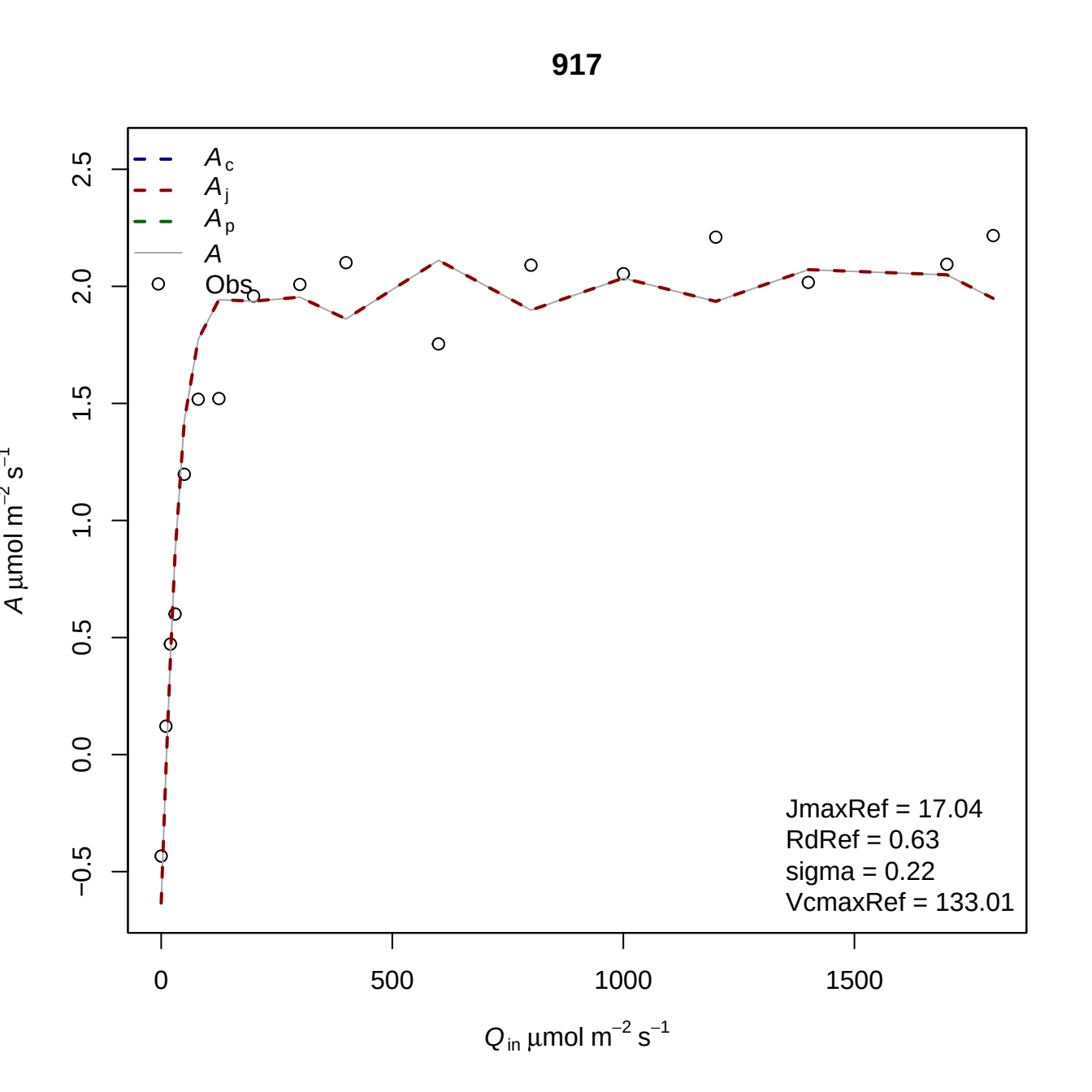

912
A_c
A_j
A_p
A
Obs
JmaxRef = 32.92
RdRef = 1.82
sigma = 0.65
VcmaxRef = 78.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

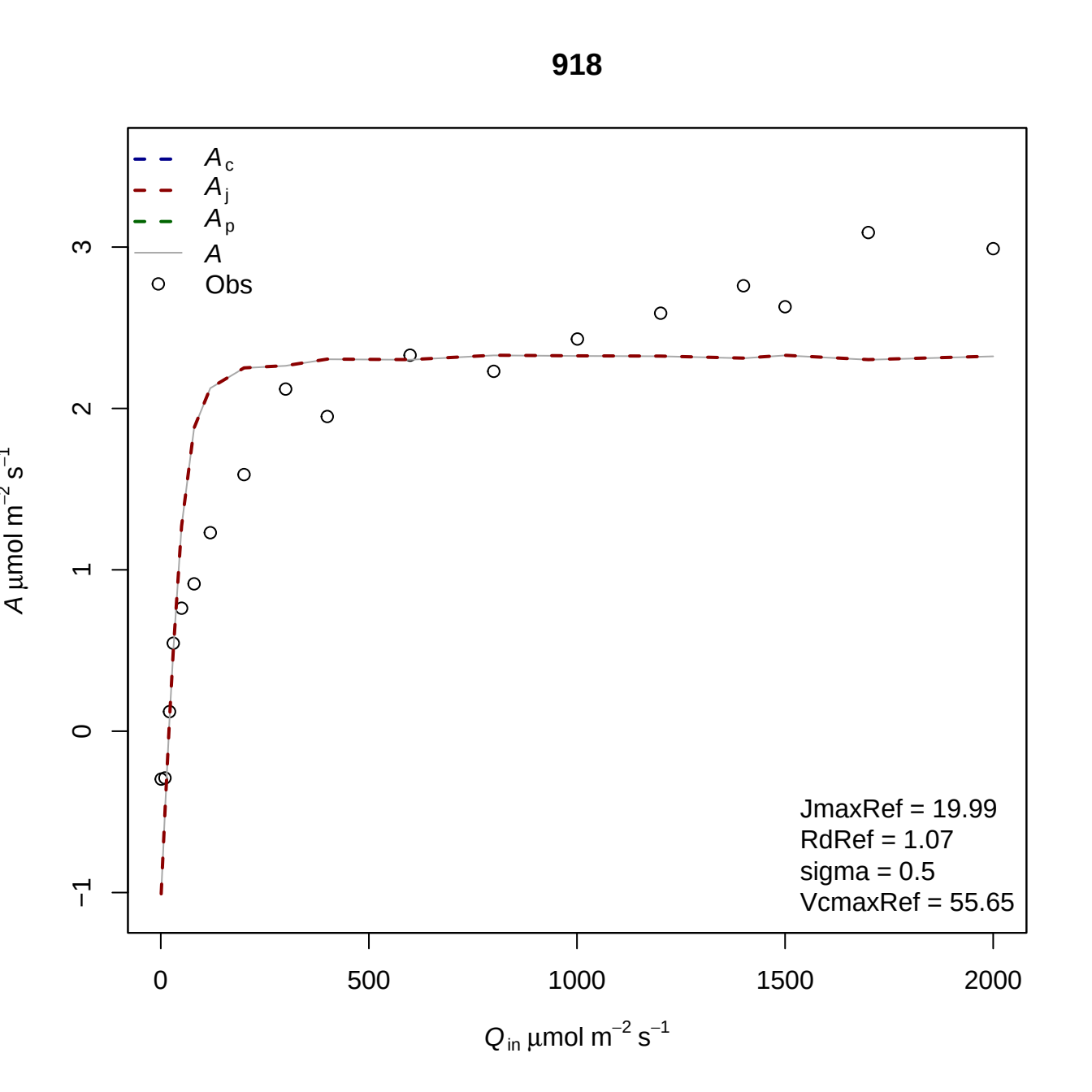

913
A_c
A_j
A_p
A
Obs
JmaxRef = 29.22
RdRef = 1.36
sigma = 0.64
VcmaxRef = 57.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

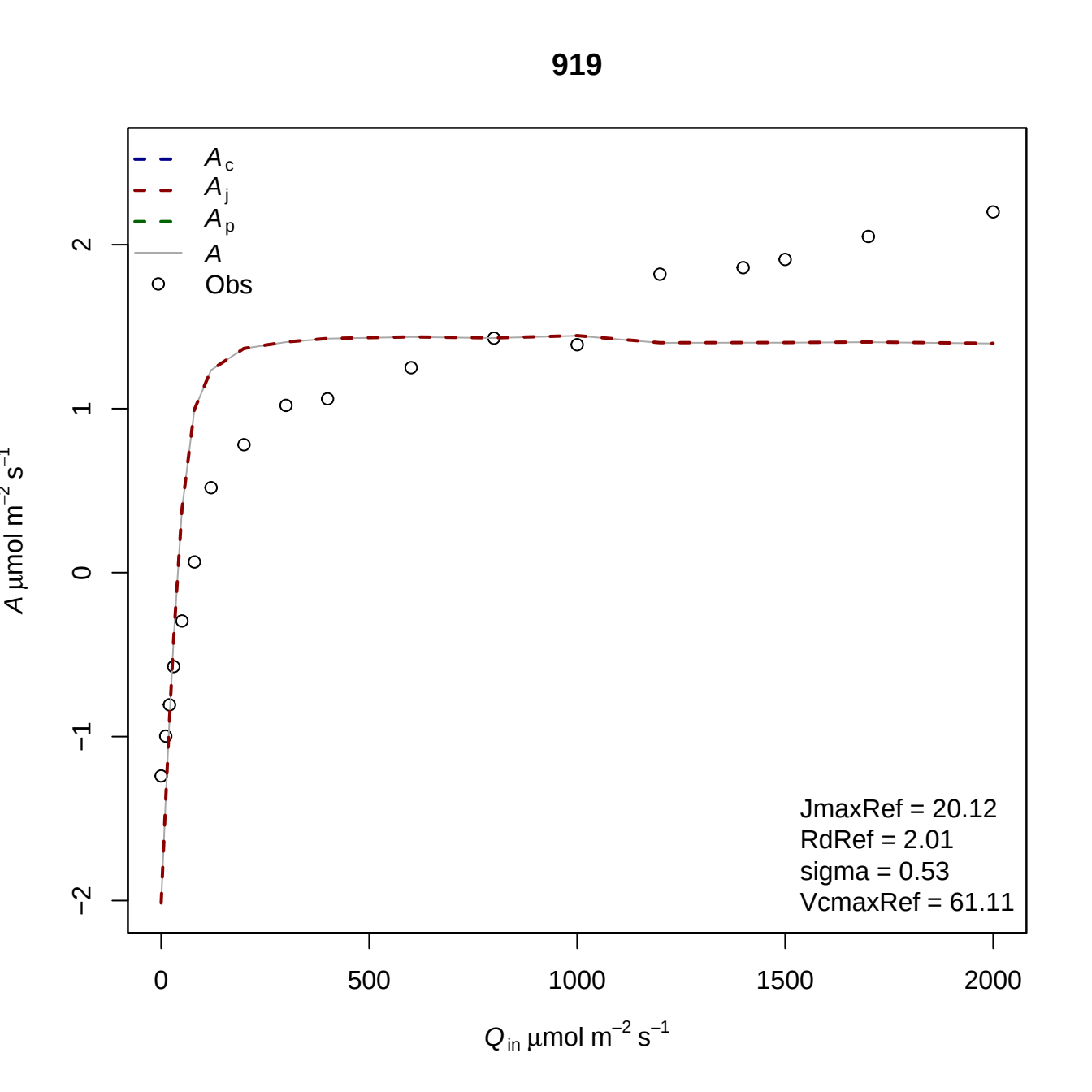

914
A_c
A_j
A_p
A
Obs
JmaxRef = 24.24
RdRef = 1.77
sigma = 0.56
VcmaxRef = 70.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

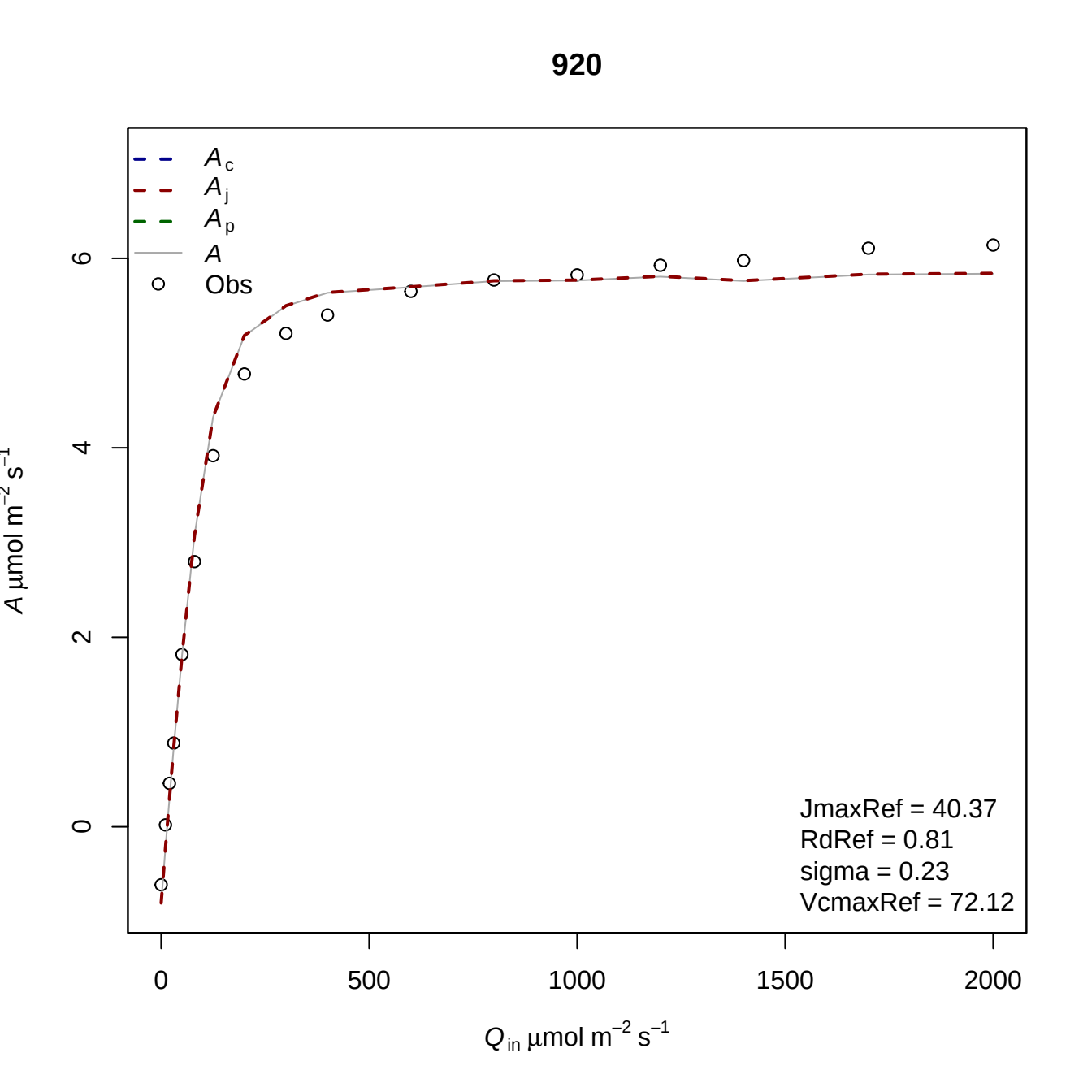

915
A_c
A_j
A_p
A
Obs
JmaxRef = 34.01
RdRef = 1.72
sigma = 0.62
VcmaxRef = 198.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

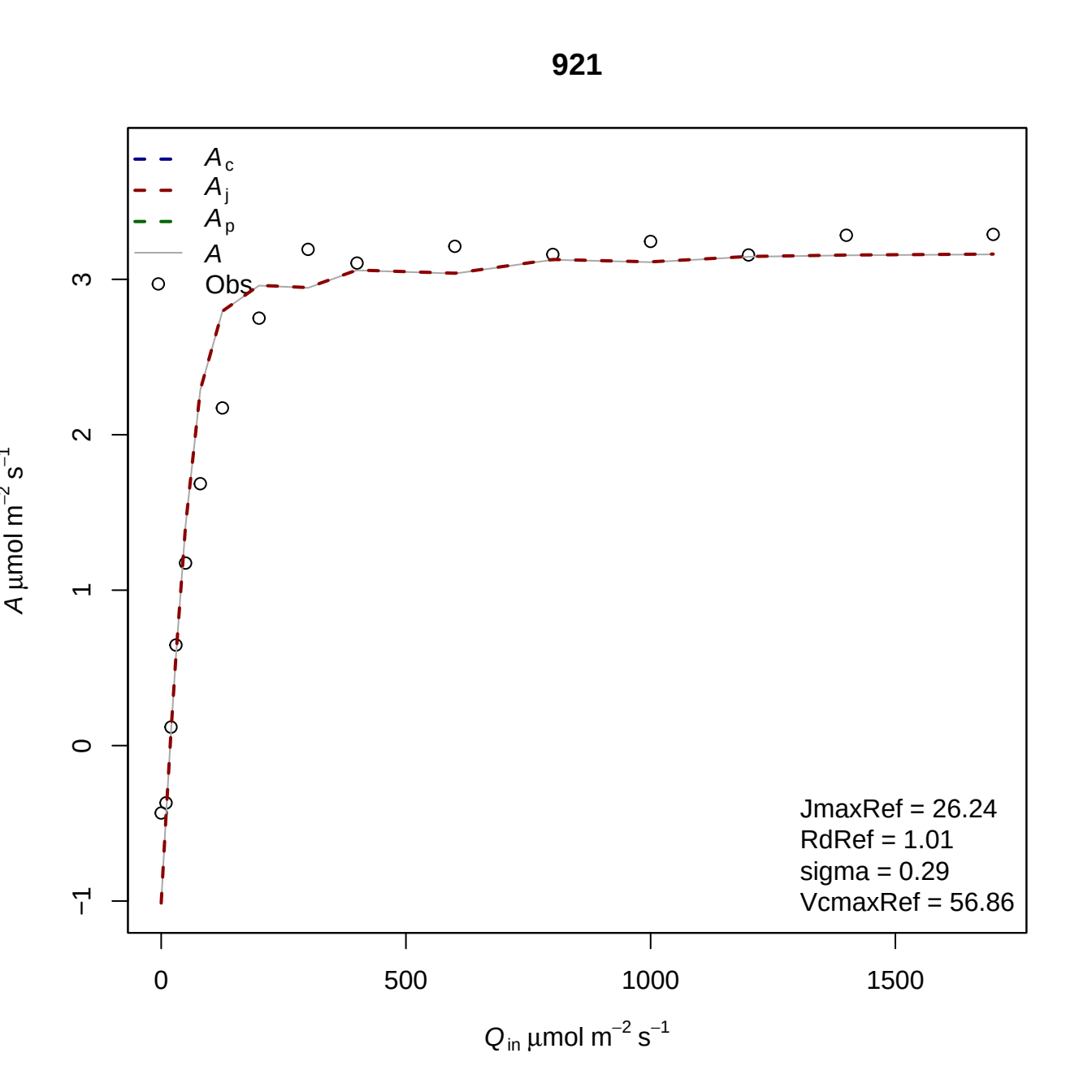

916
A_c
A_j
A_p
A
Obs
JmaxRef = 30.4
RdRef = 0.79
sigma = 0.26
VcmaxRef = 49.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

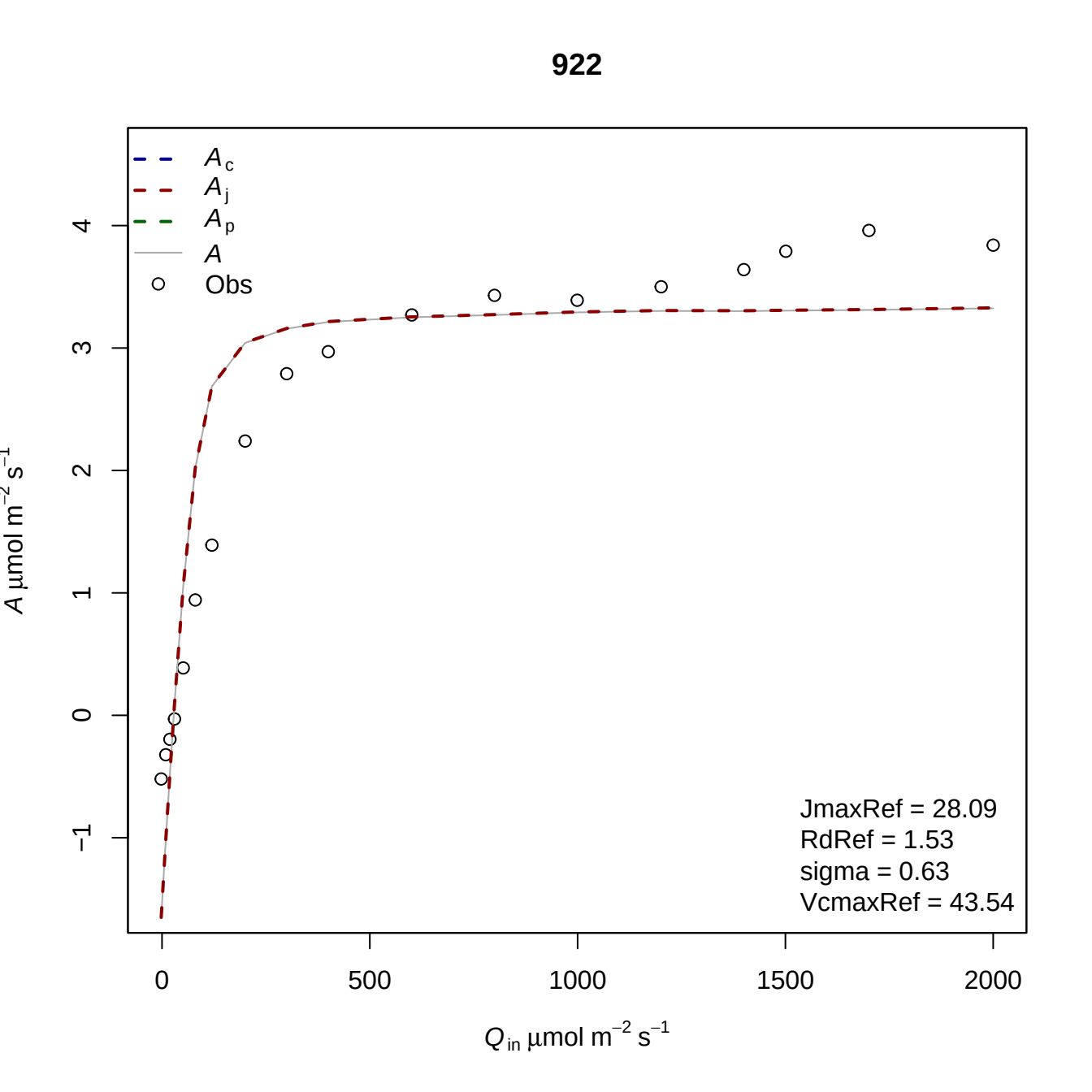

917
A_c
A_j
A_p
A
Obs
JmaxRef = 17.04
RdRef = 0.63
sigma = 0.22
VcmaxRef = 133.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

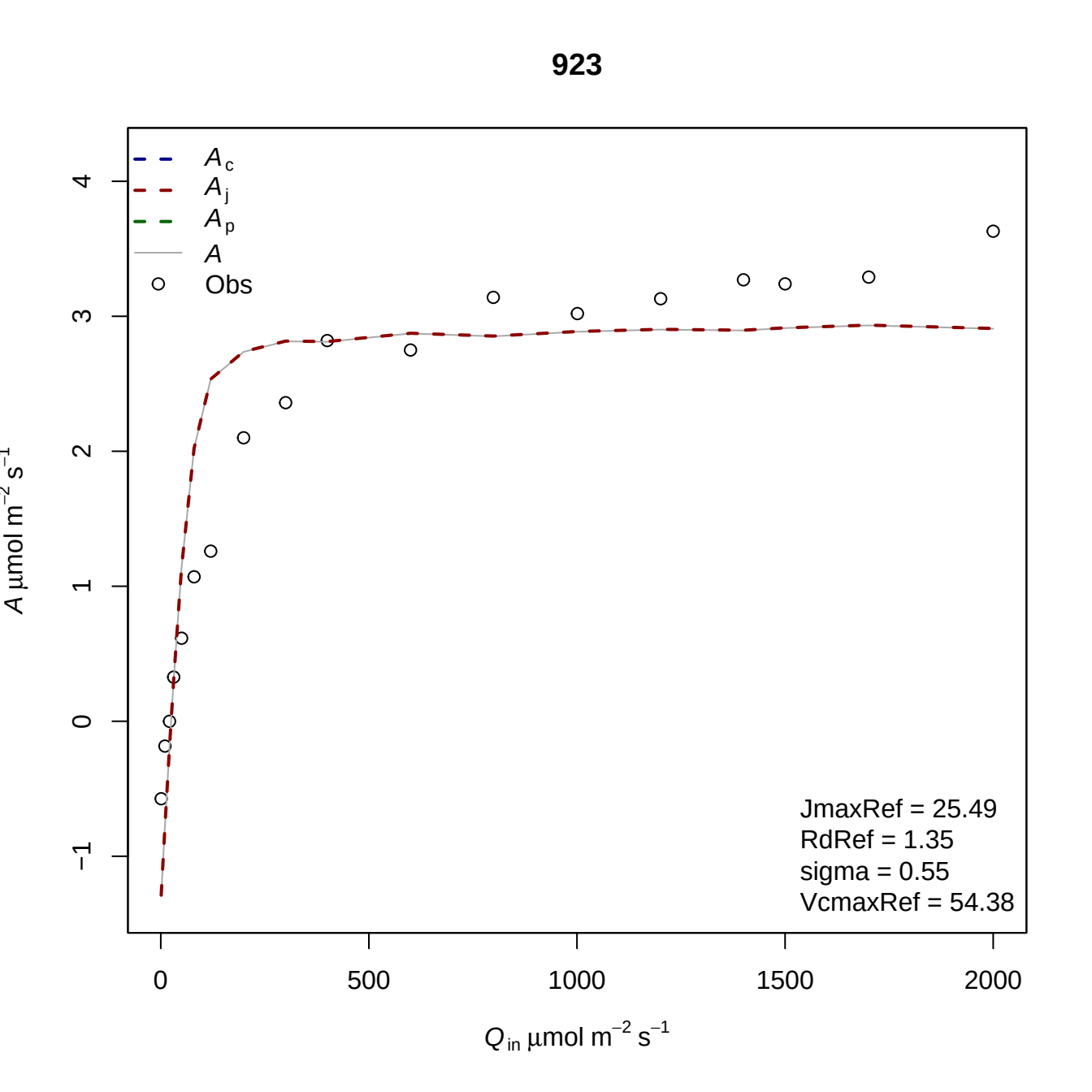

918
A_c
A_j
A_p
A
Obs
JmaxRef = 19.99
RdRef = 1.07
sigma = 0.5
VcmaxRef = 55.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

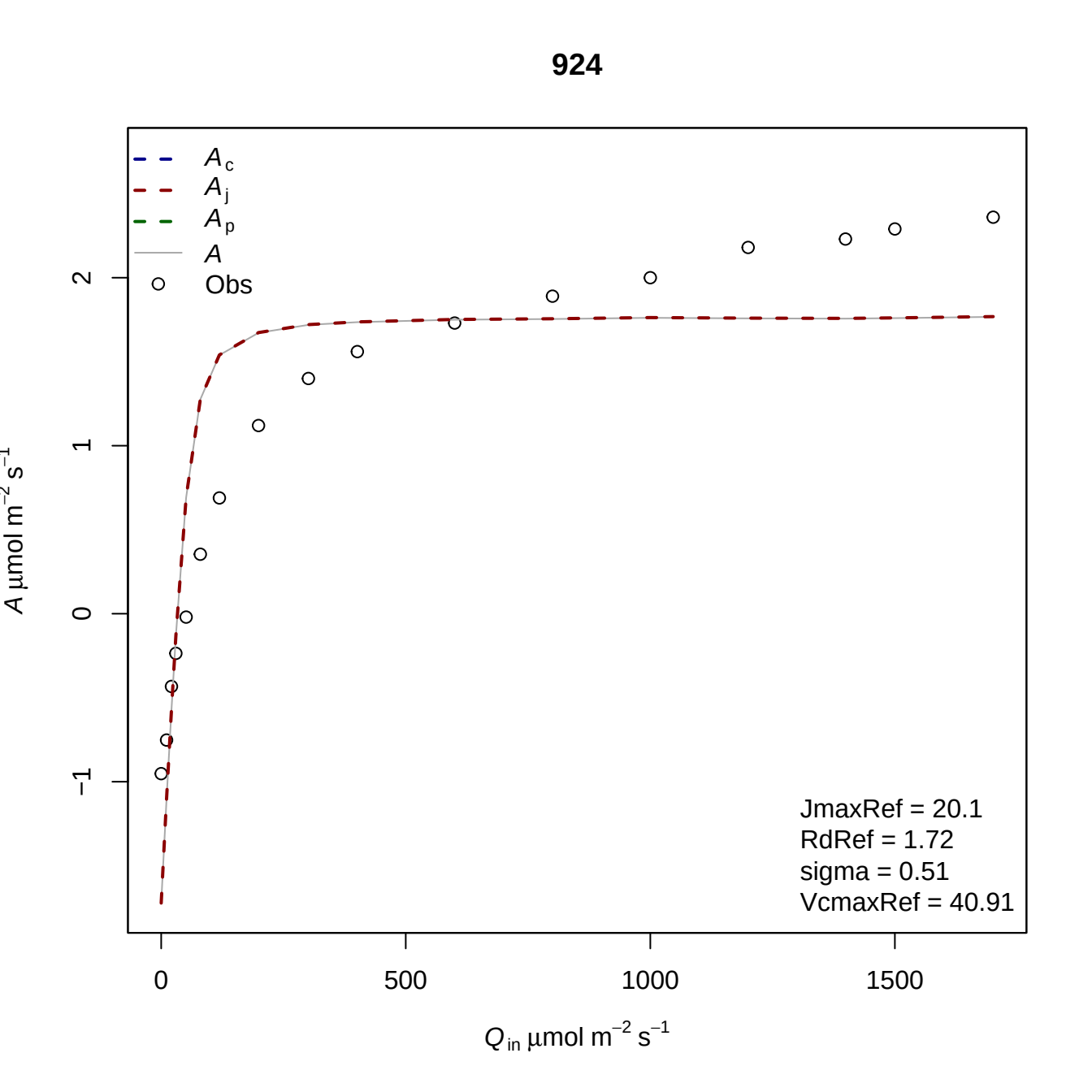

919
A_c
A_j
A_p
A
Obs
JmaxRef = 20.12
RdRef = 2.01
sigma = 0.53
VcmaxRef = 61.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
0
500
1000
1500
2000

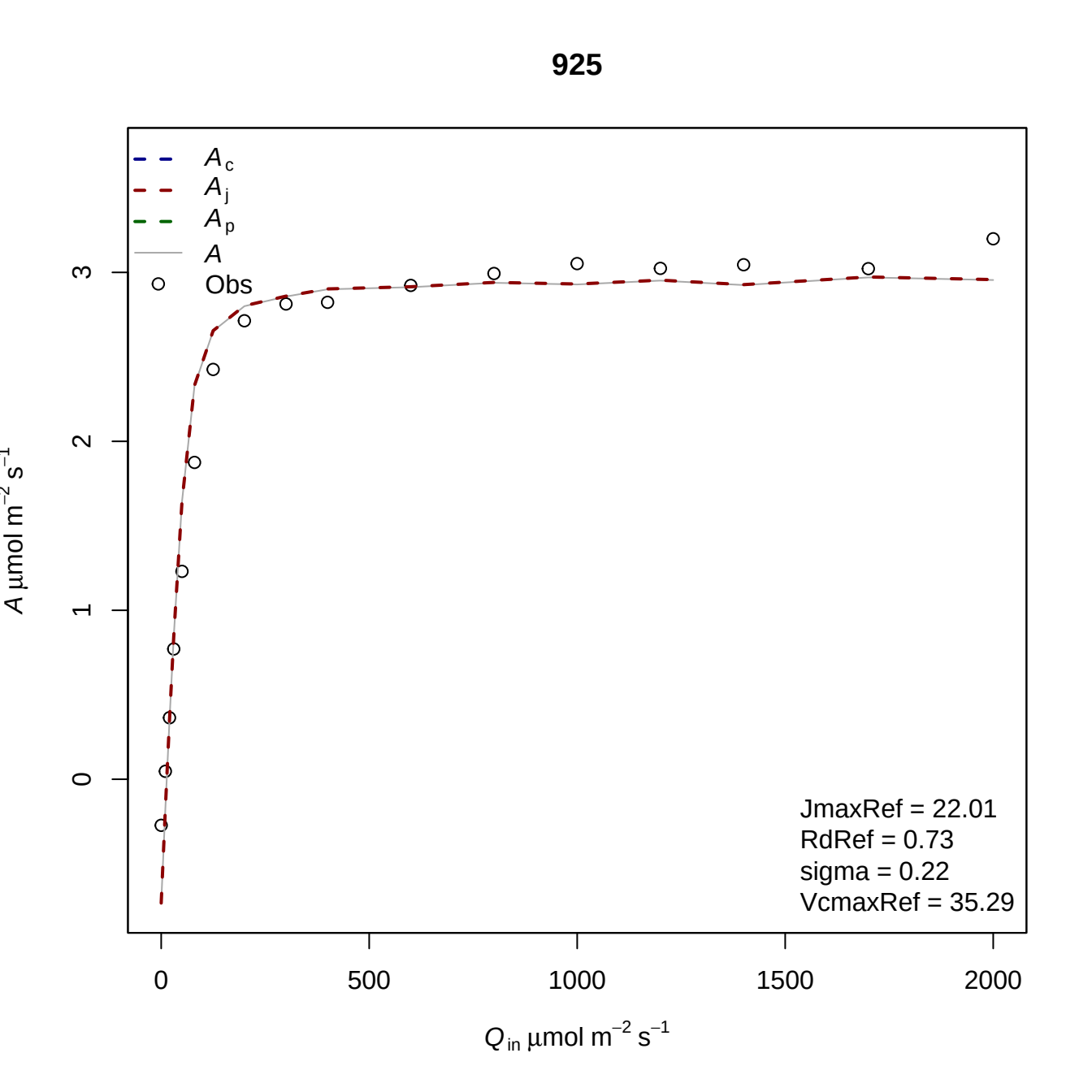

920
A_c
A_j
A_p
A
Obs
JmaxRef = 40.37
RdRef = 0.81
sigma = 0.23
VcmaxRef = 72.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

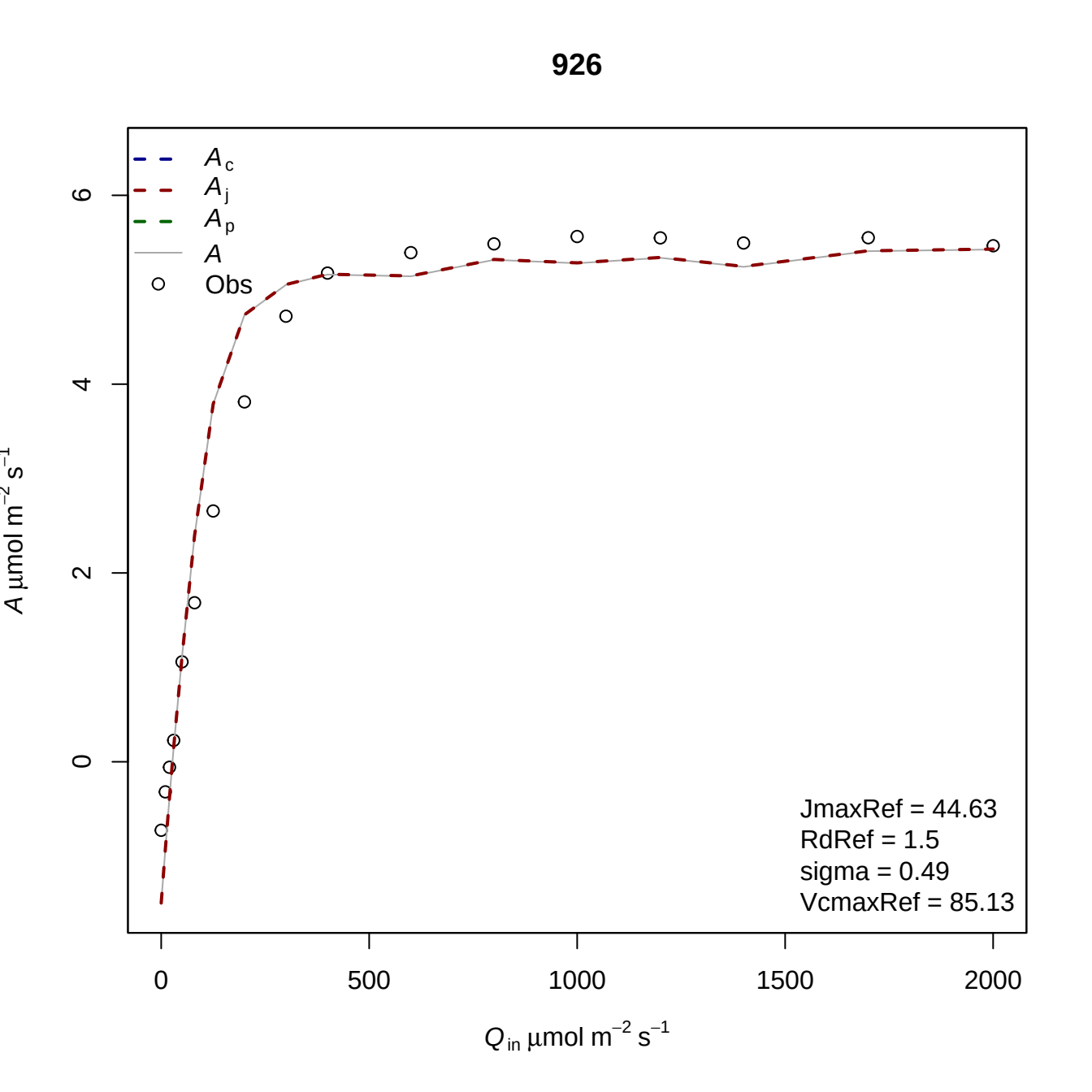

921
A_c
A_j
A_p
A
Obs
JmaxRef = 26.24
RdRef = 1.01
sigma = 0.29
VcmaxRef = 56.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

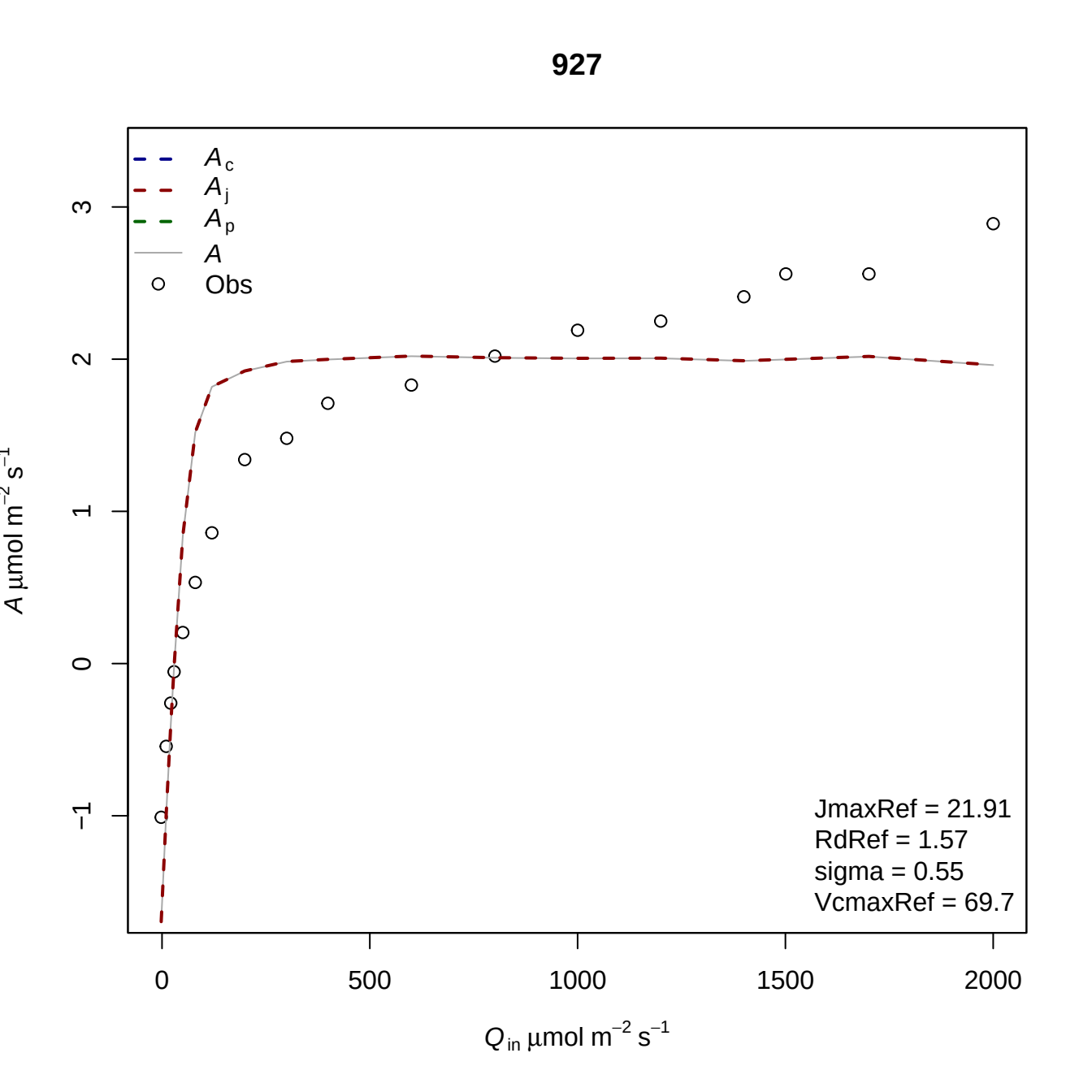

922
A_c
A_j
A_p
A
Obs
JmaxRef = 28.09
RdRef = 1.53
sigma = 0.63
VcmaxRef = 43.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

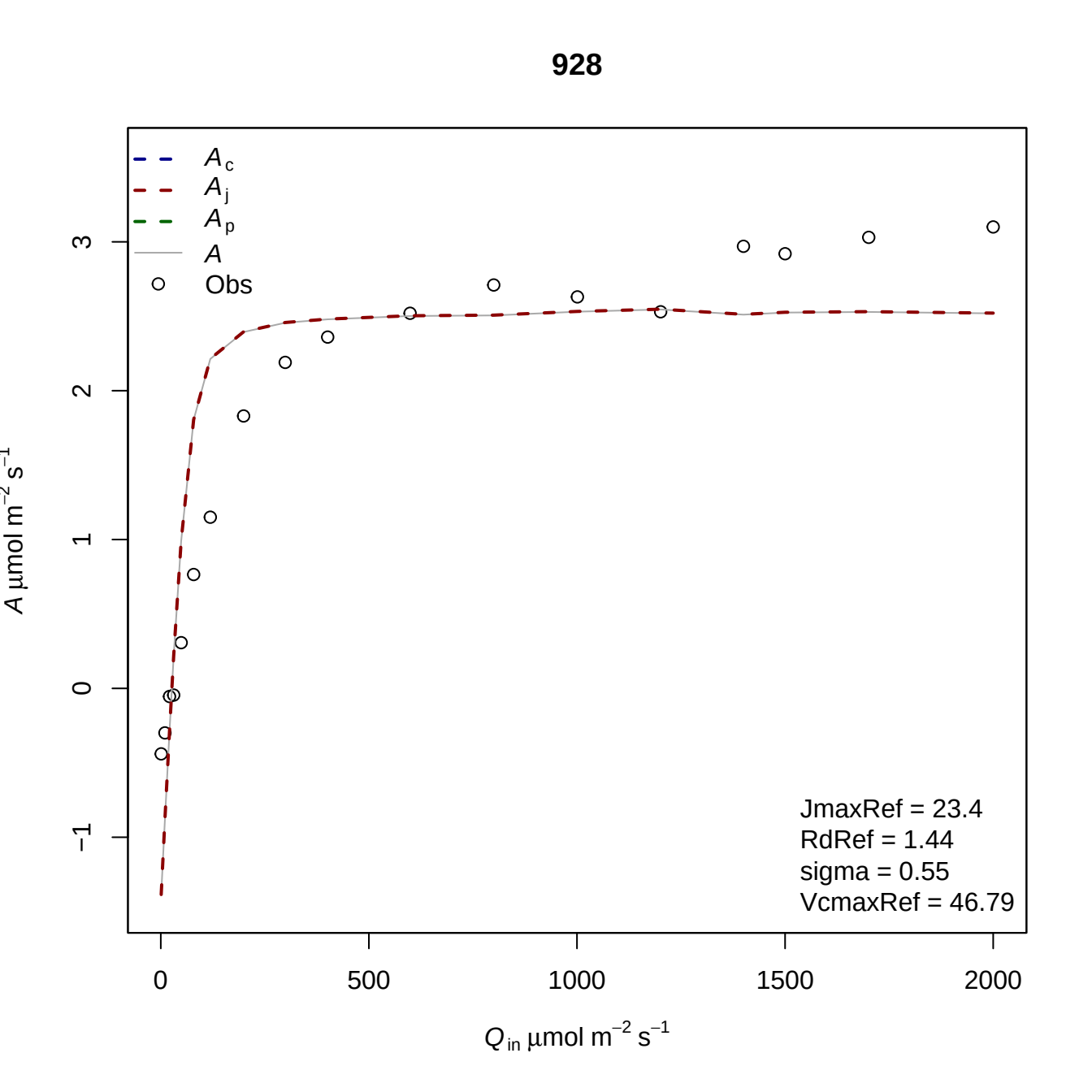

923
A_c
A_j
A_p
A
Obs
JmaxRef = 25.49
RdRef = 1.35
sigma = 0.55
VcmaxRef = 54.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

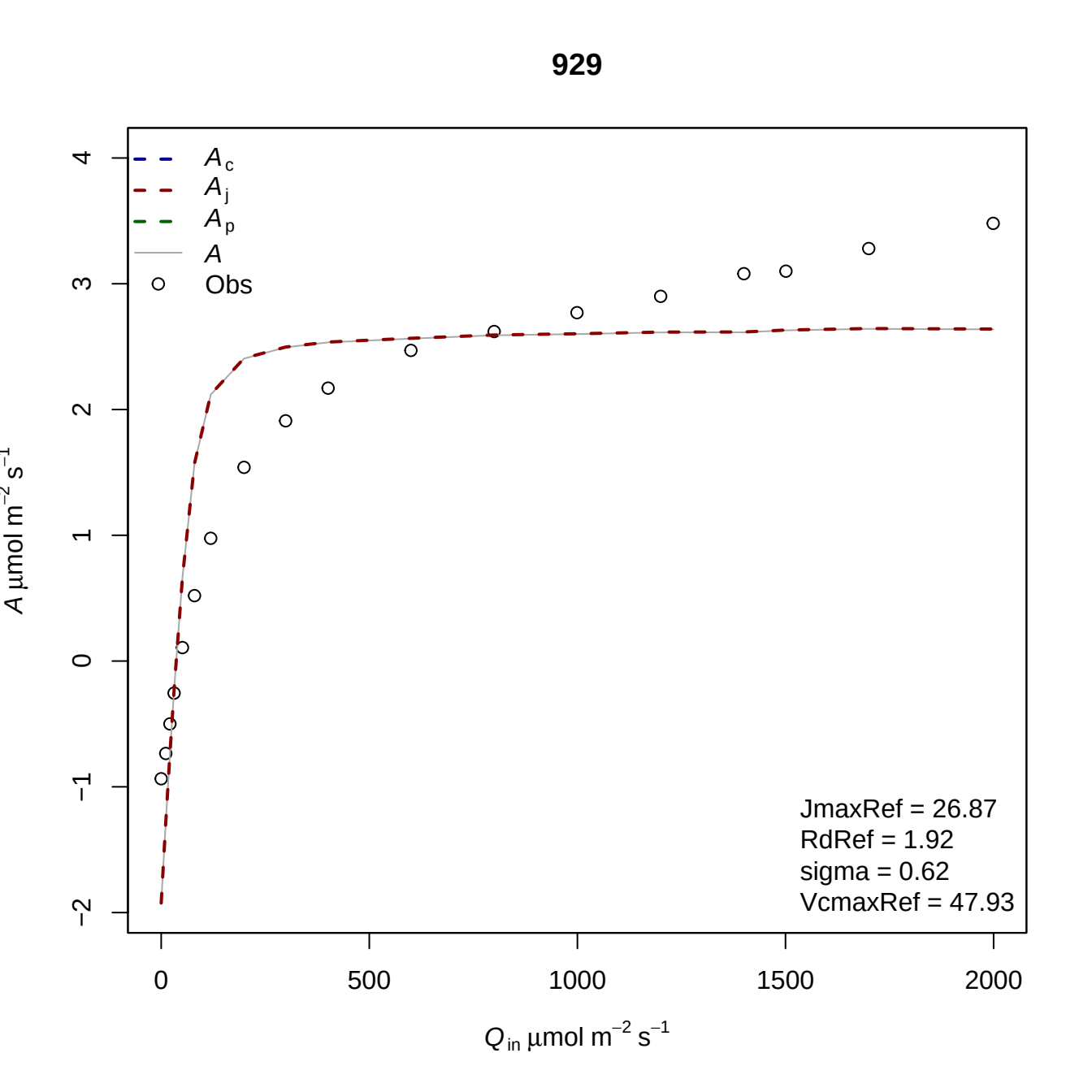

924
A_c
A_j
A_p
A
Obs
JmaxRef = 20.1
RdRef = 1.72
sigma = 0.51
VcmaxRef = 40.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

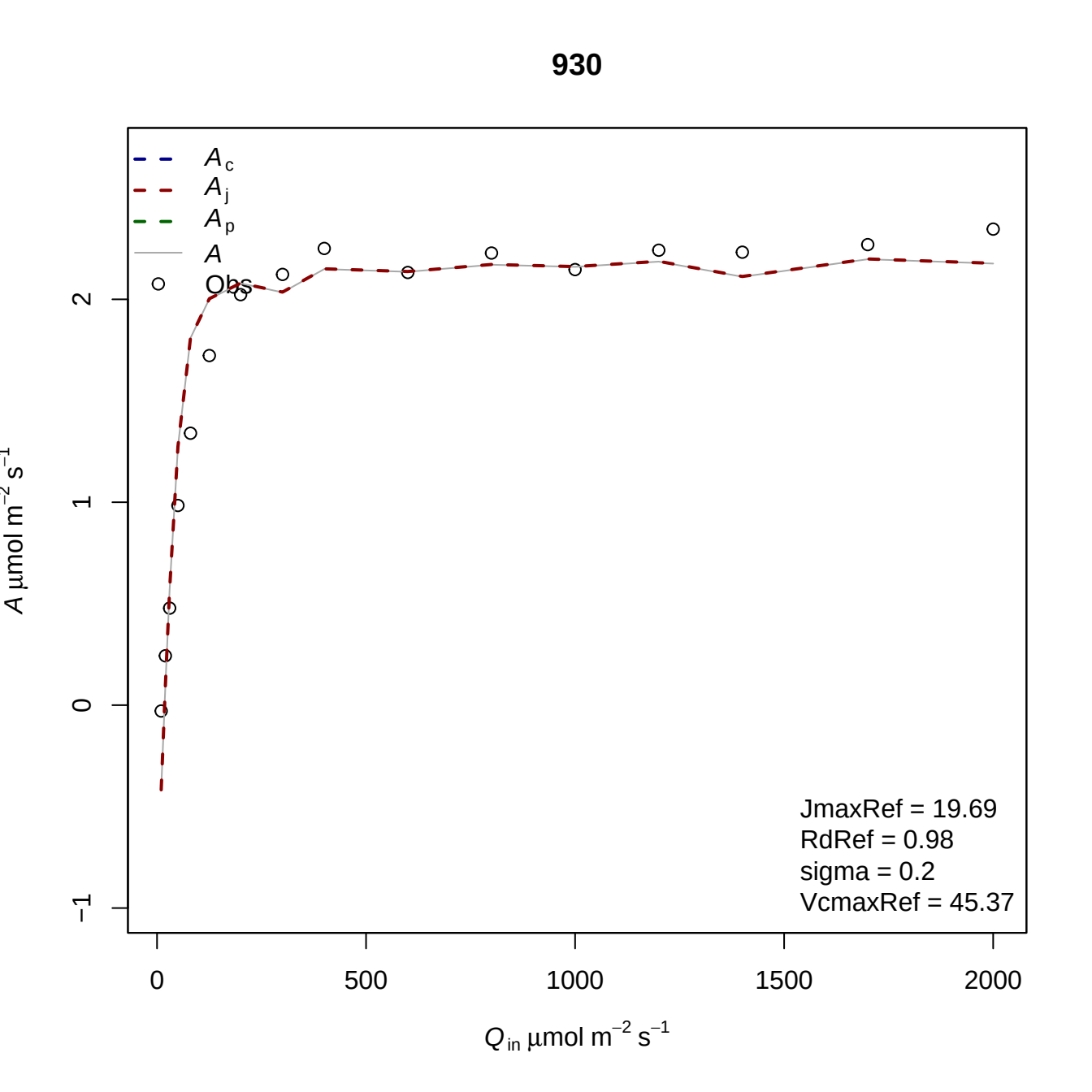

925
A_c
A_j
A_p
A
Obs
JmaxRef = 22.01
RdRef = 0.73
sigma = 0.22
VcmaxRef = 35.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

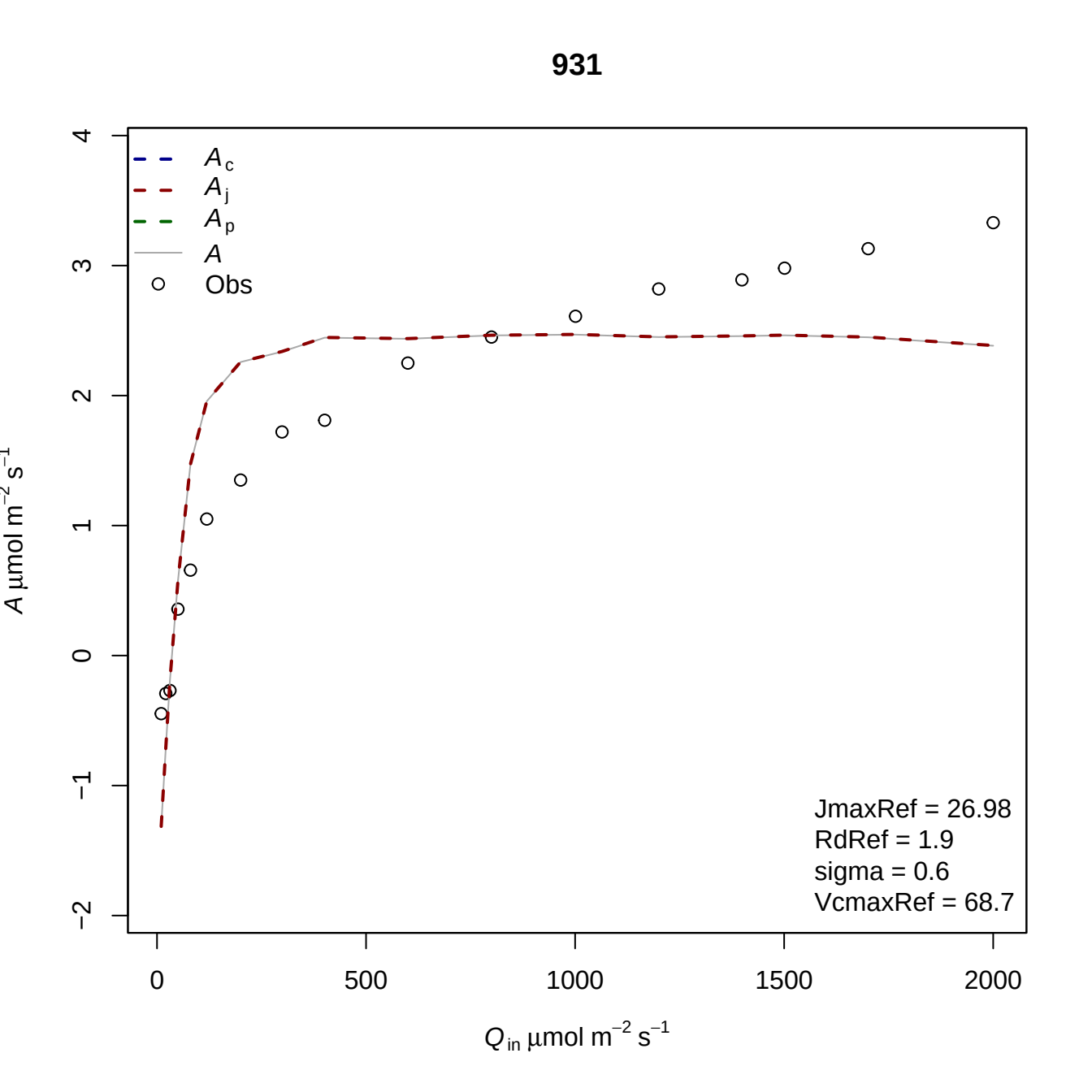

926
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 44.63
RdRef = 1.5
sigma = 0.49
VcmaxRef = 85.13

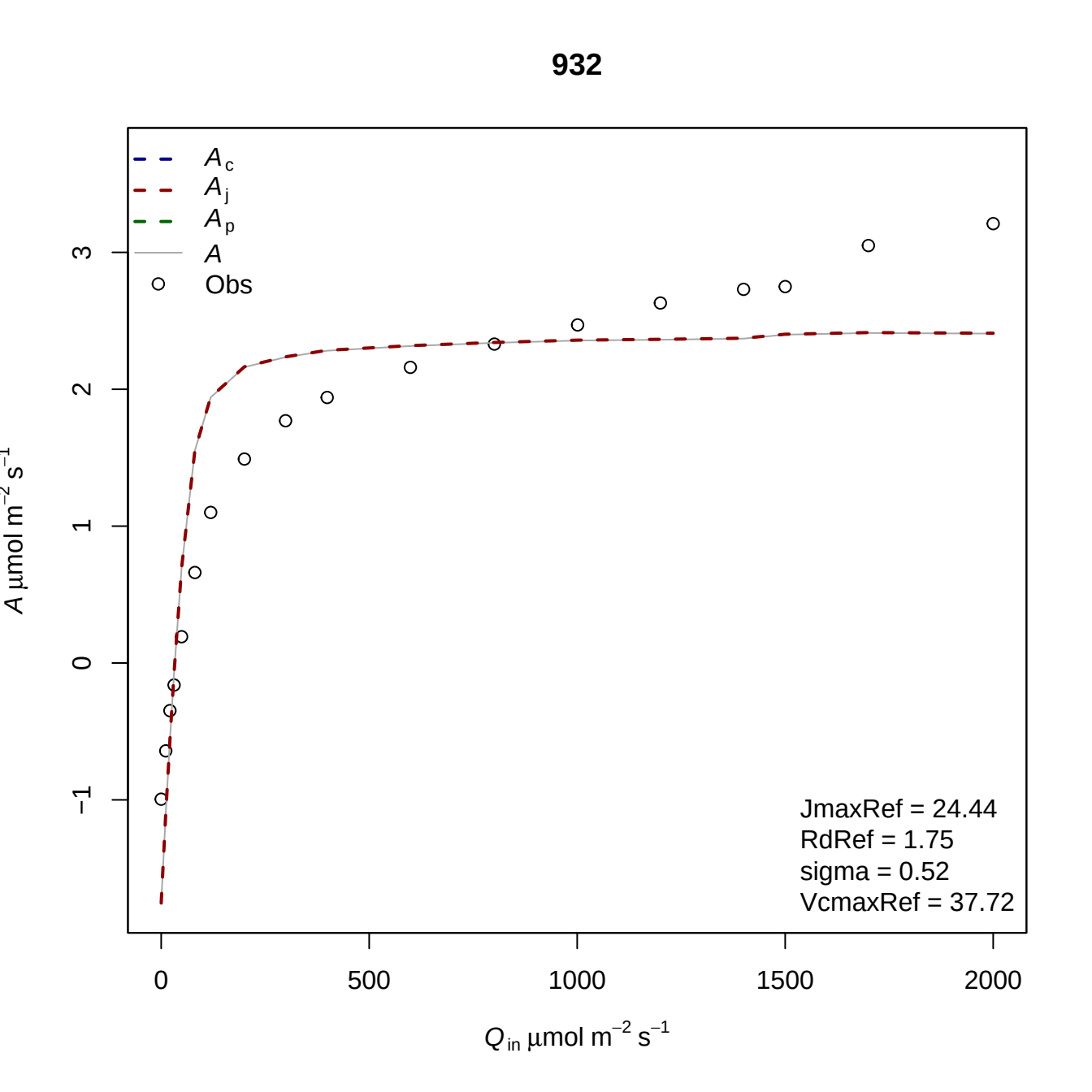

927
A_c
A_j
A_p
A
Obs
JmaxRef = 21.91
RdRef = 1.57
sigma = 0.55
VcmaxRef = 69.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

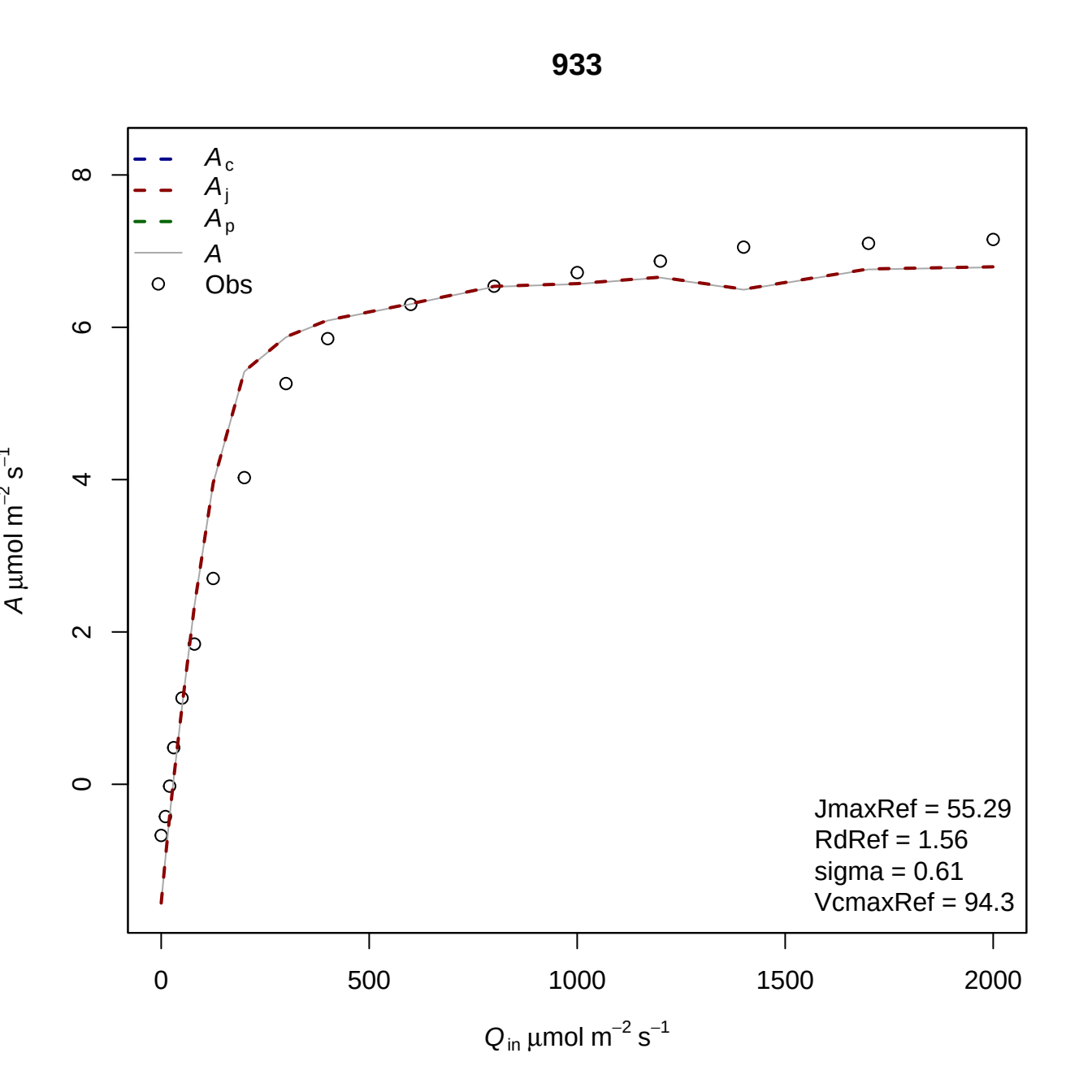

928
A_c
A_j
A_p
A
Obs
JmaxRef = 23.4
RdRef = 1.44
sigma = 0.55
VcmaxRef = 46.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

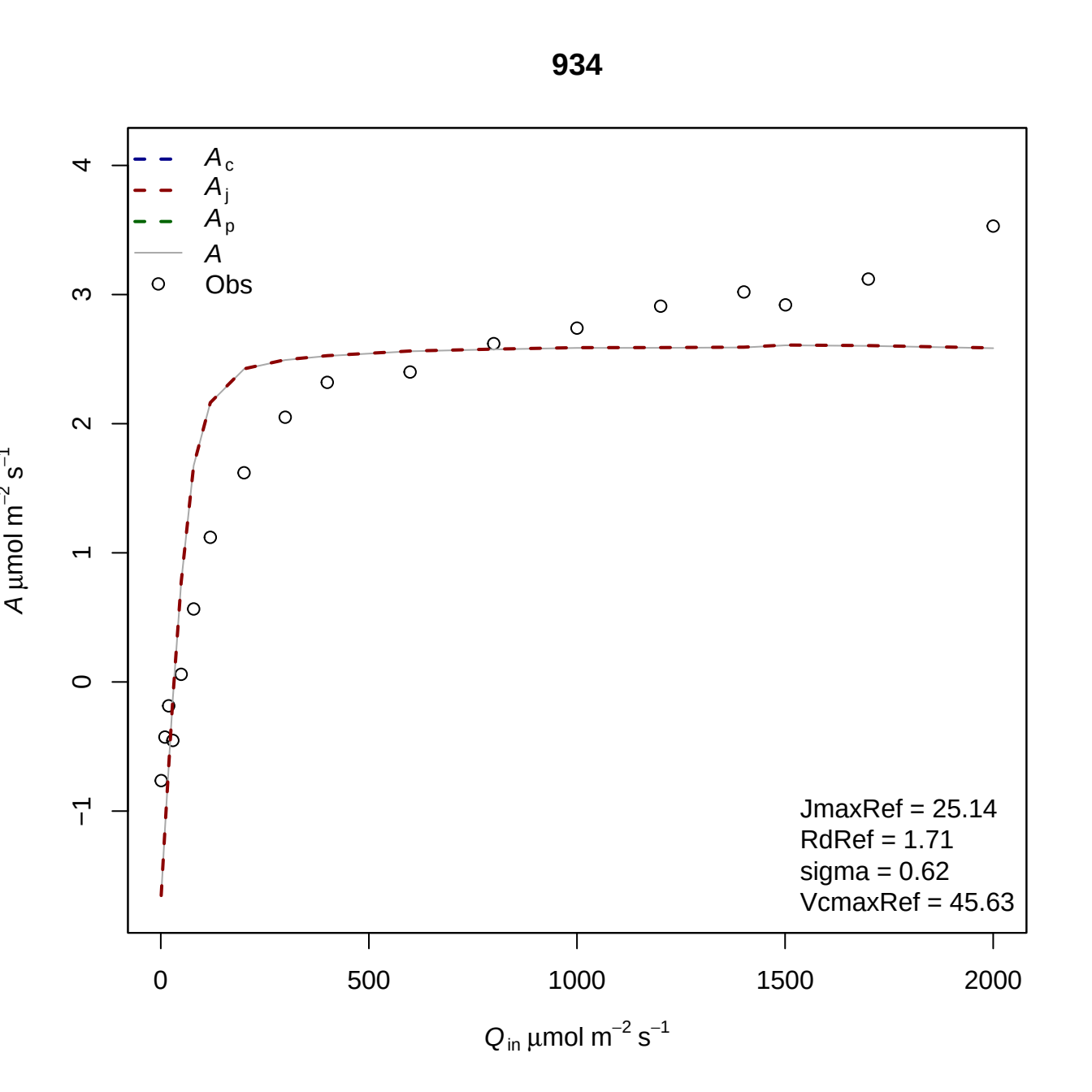

929
A_c
A_j
A_p
A
Obs
JmaxRef = 26.87
RdRef = 1.92
sigma = 0.62
VcmaxRef = 47.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

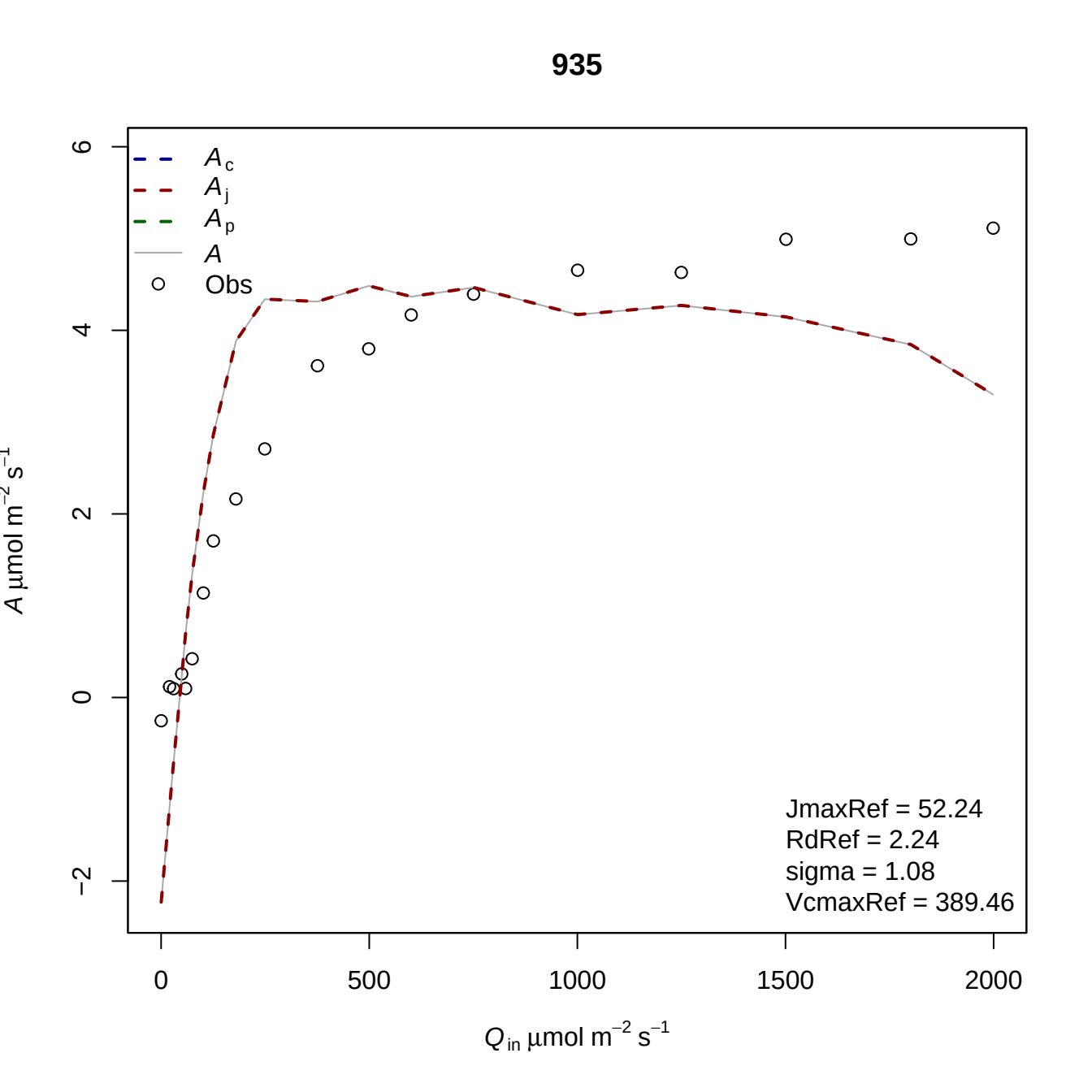

930
A_c
A_j
A_p
A
Obs
JmaxRef = 19.69
RdRef = 0.98
sigma = 0.2
VcmaxRef = 45.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

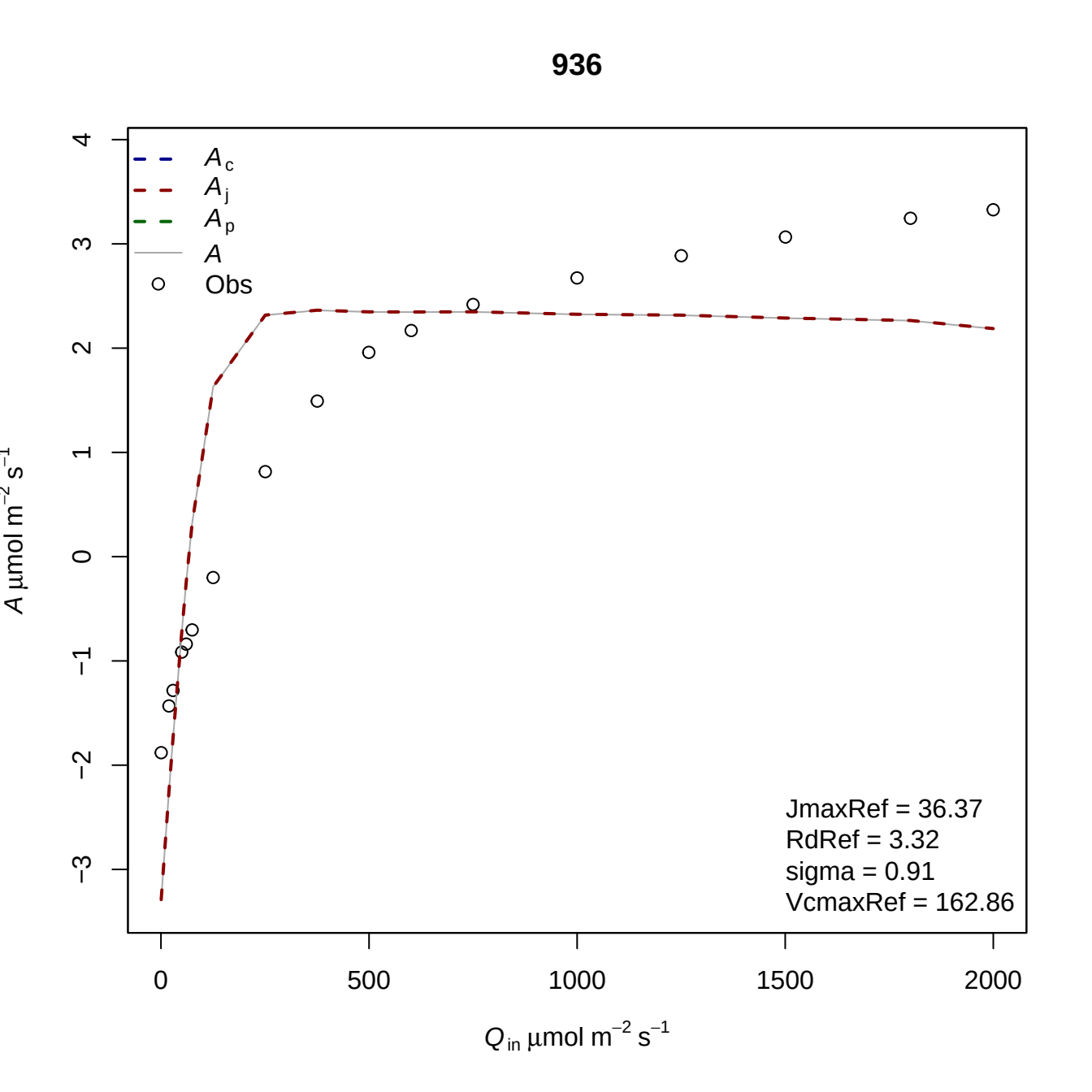

931
A_c
A_j
A_p
A
Obs
JmaxRef = 26.98
RdRef = 1.9
sigma = 0.6
VcmaxRef = 68.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

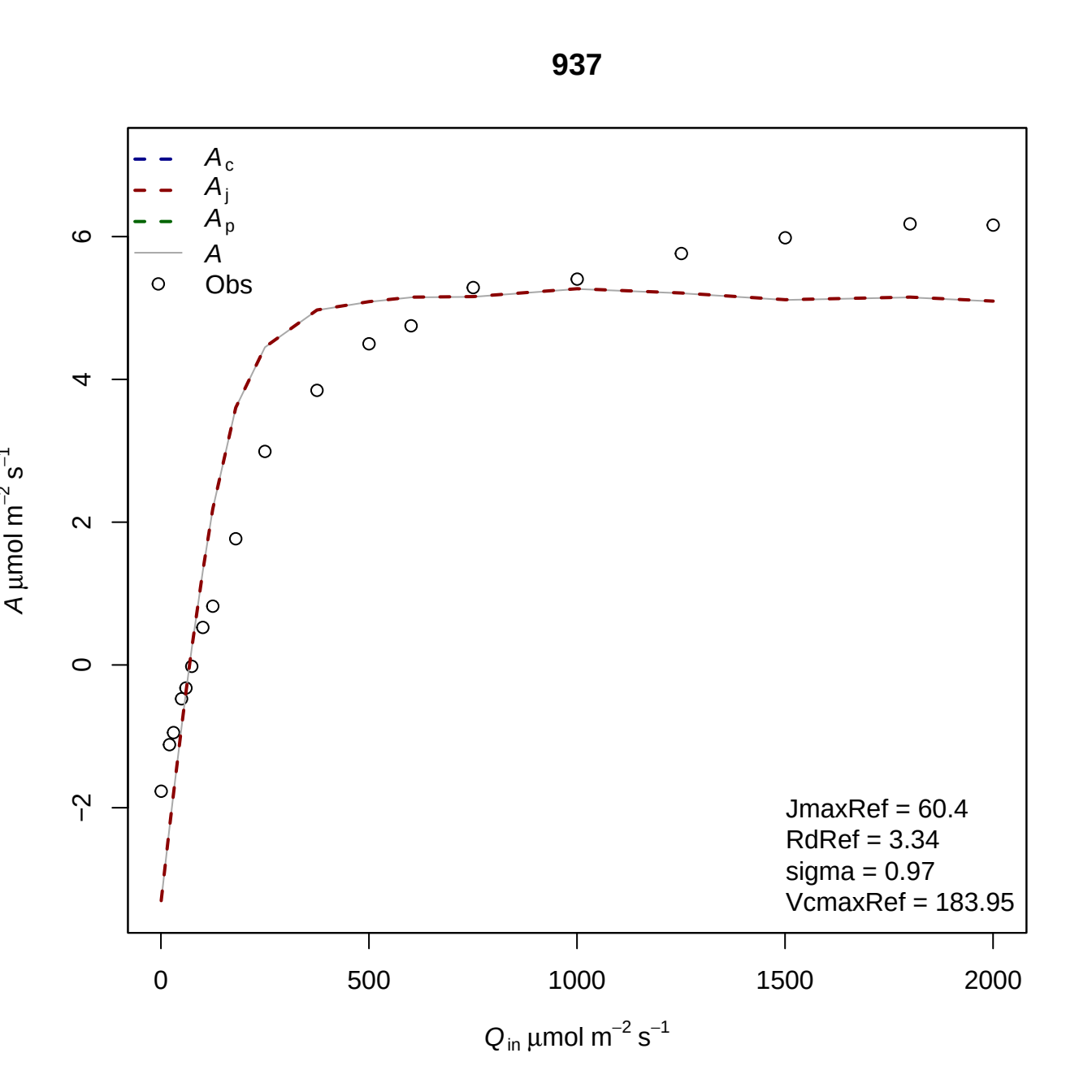

932
A_c
A_j
A_p
A
Obs
JmaxRef = 24.44
RdRef = 1.75
sigma = 0.52
VcmaxRef = 37.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

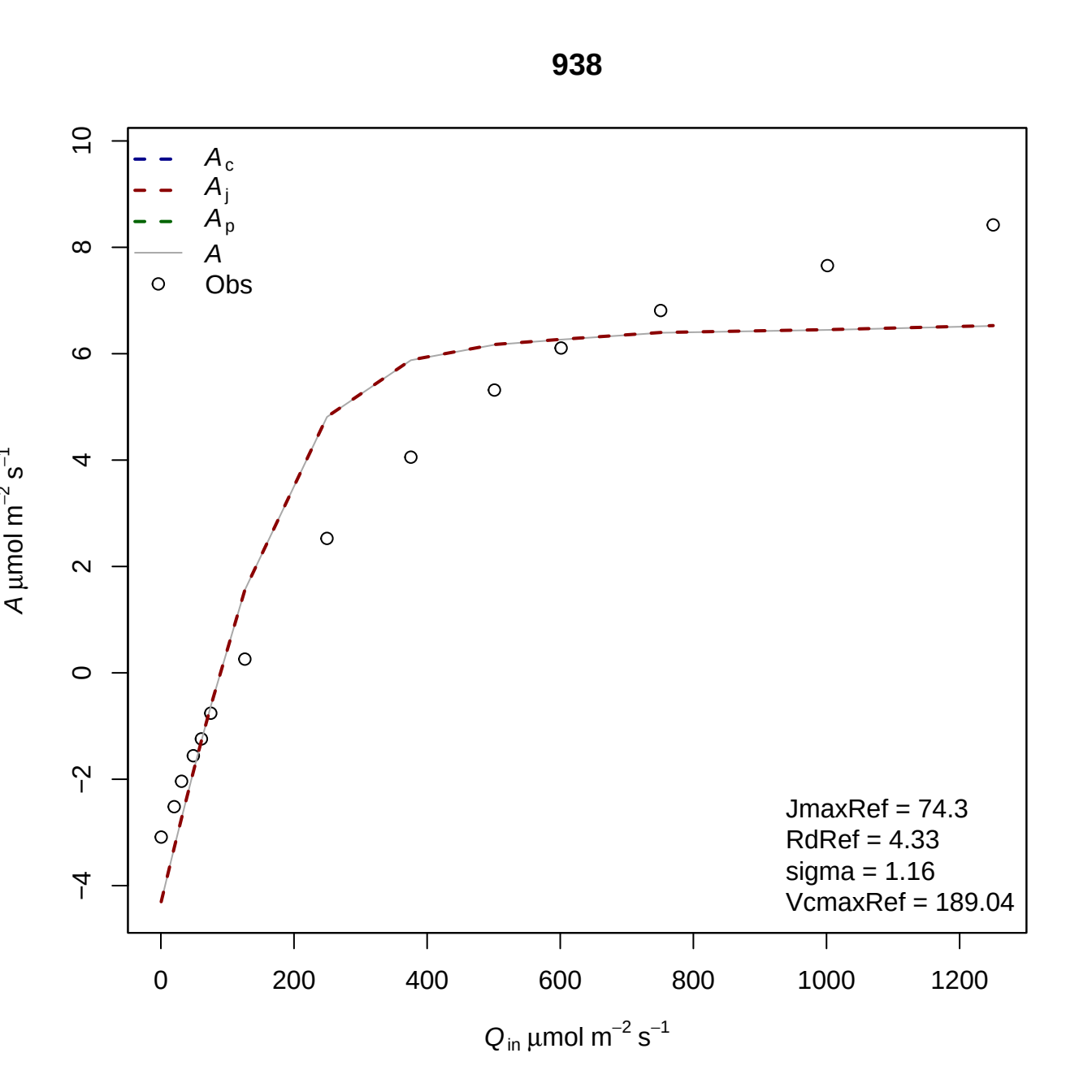

933
A_c
A_j
A_p
A
Obs
JmaxRef = 55.29
RdRef = 1.56
sigma = 0.61
VcmaxRef = 94.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

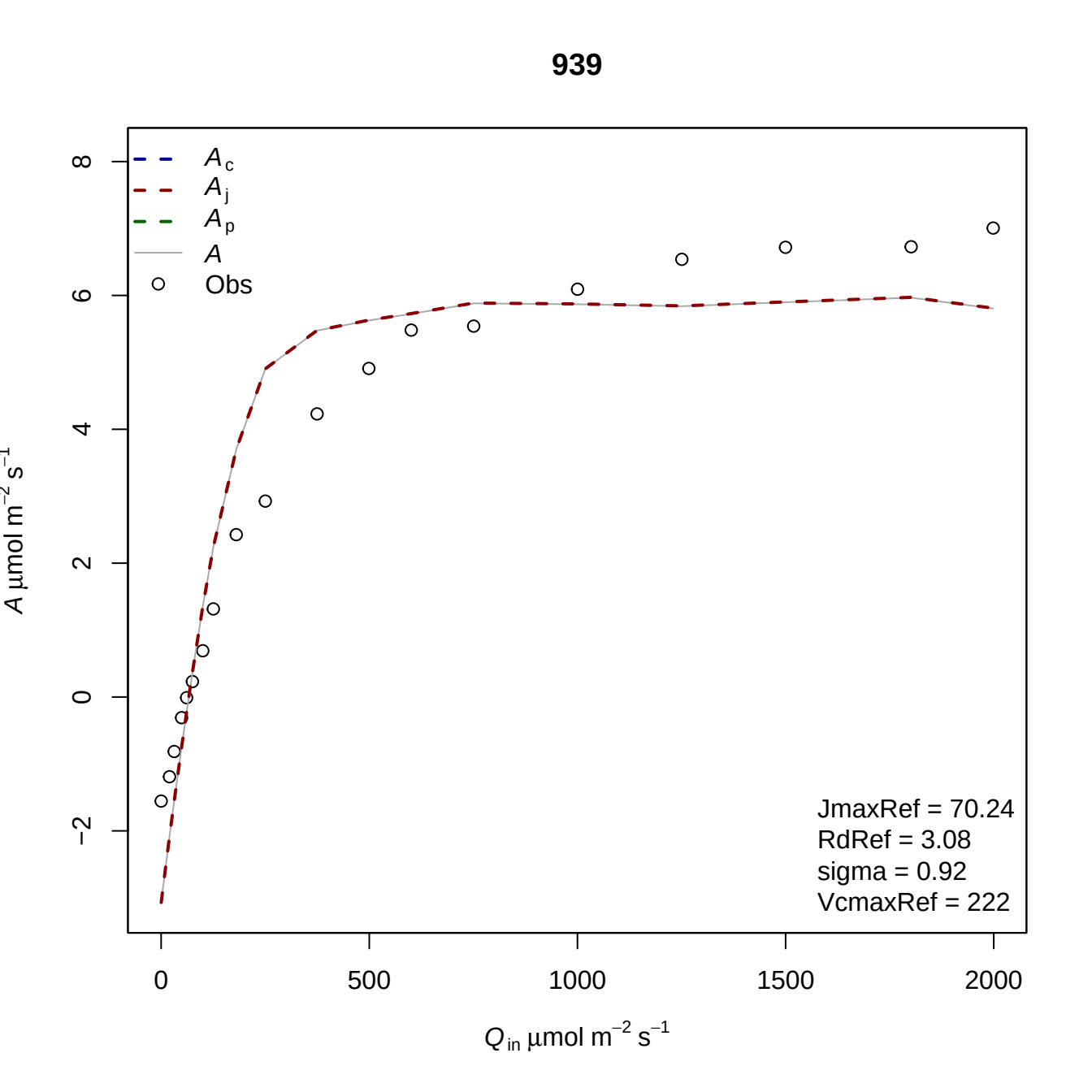

934
A_c
A_j
A_p
A
Obs
JmaxRef = 25.14
RdRef = 1.71
sigma = 0.62
VcmaxRef = 45.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

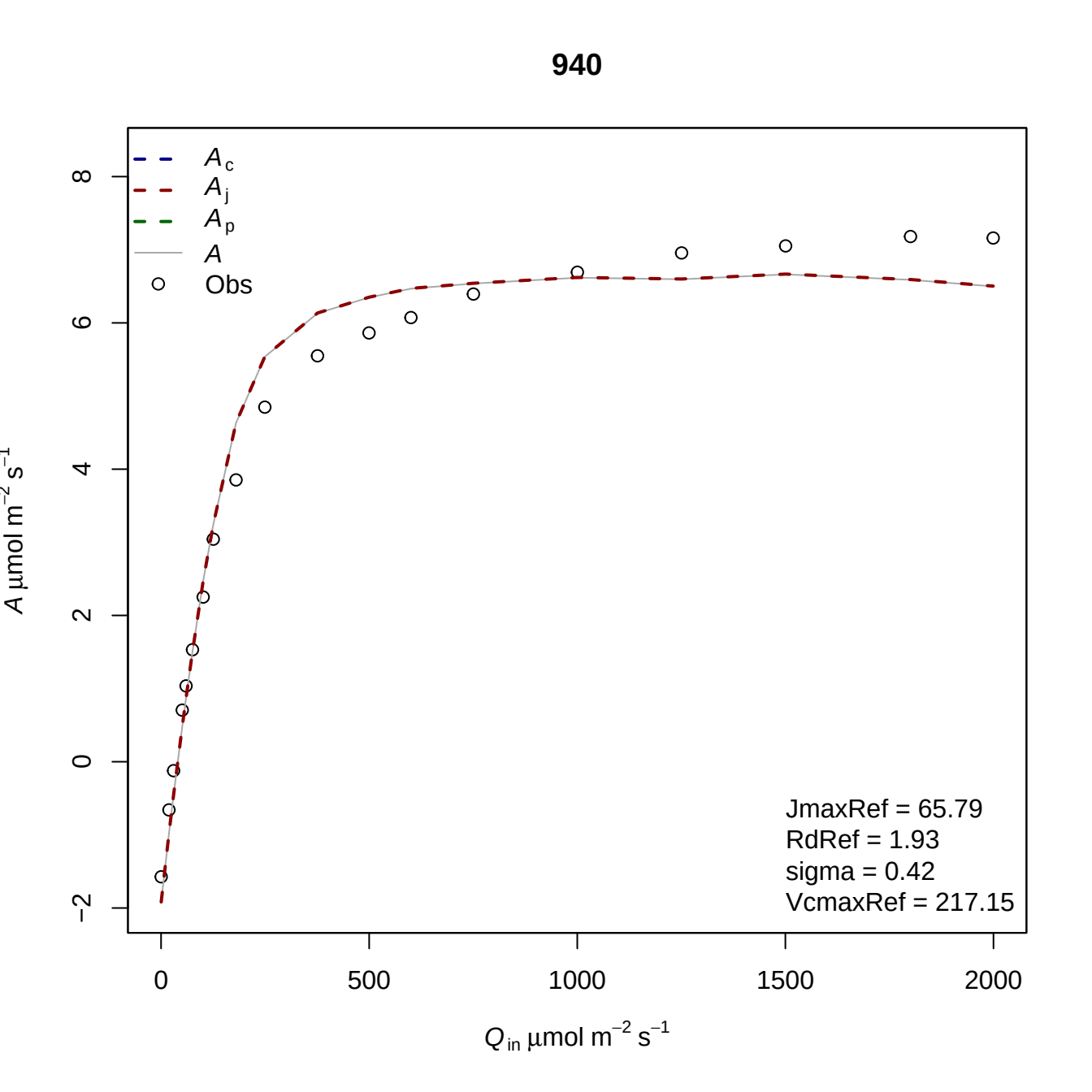

935
A_c
A_j
A_p
A
Obs
JmaxRef = 52.24
RdRef = 2.24
sigma = 1.08
VcmaxRef = 389.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

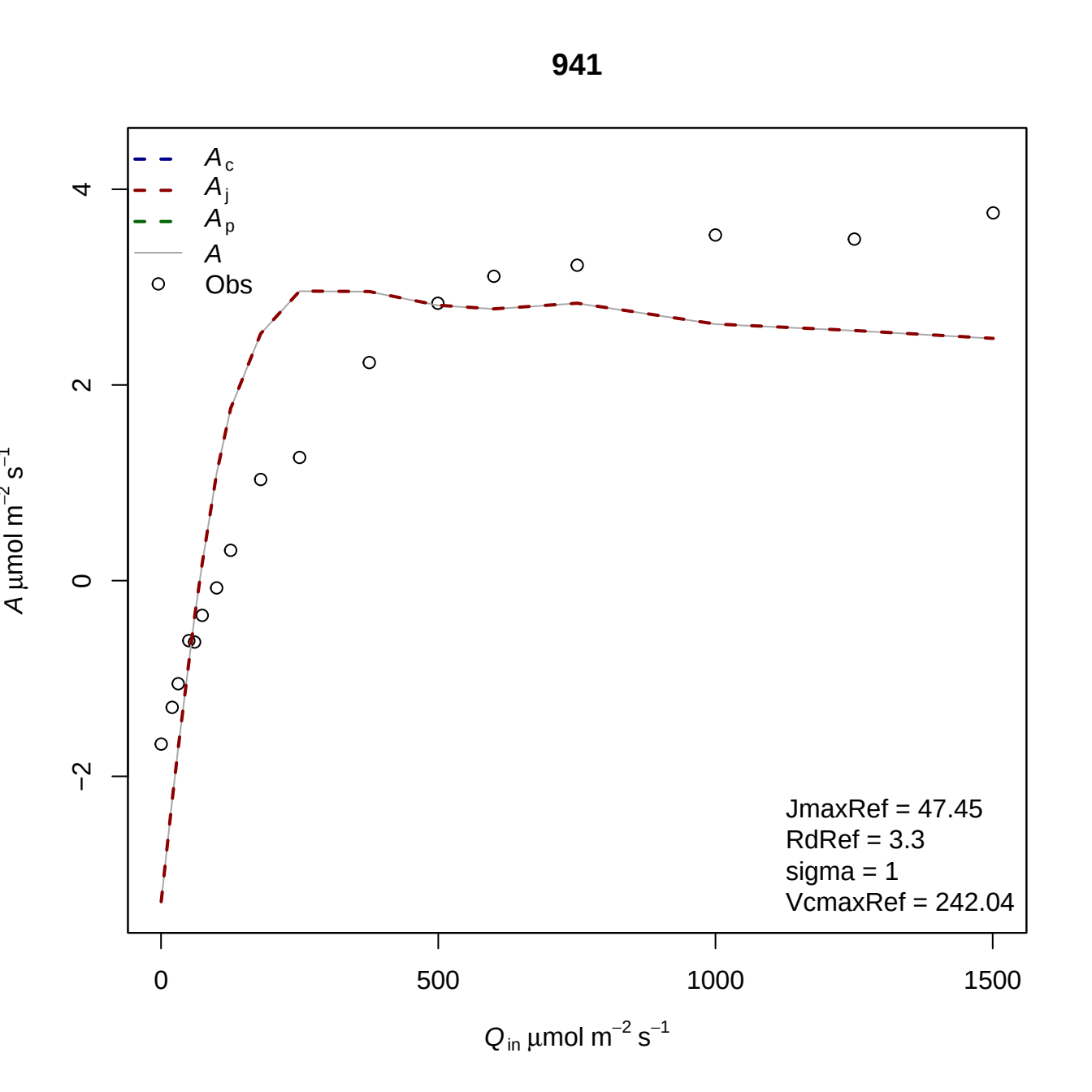

936
A_c
A_j
A_p
A
Obs
JmaxRef = 36.37
RdRef = 3.32
sigma = 0.91
VcmaxRef = 162.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500
2000

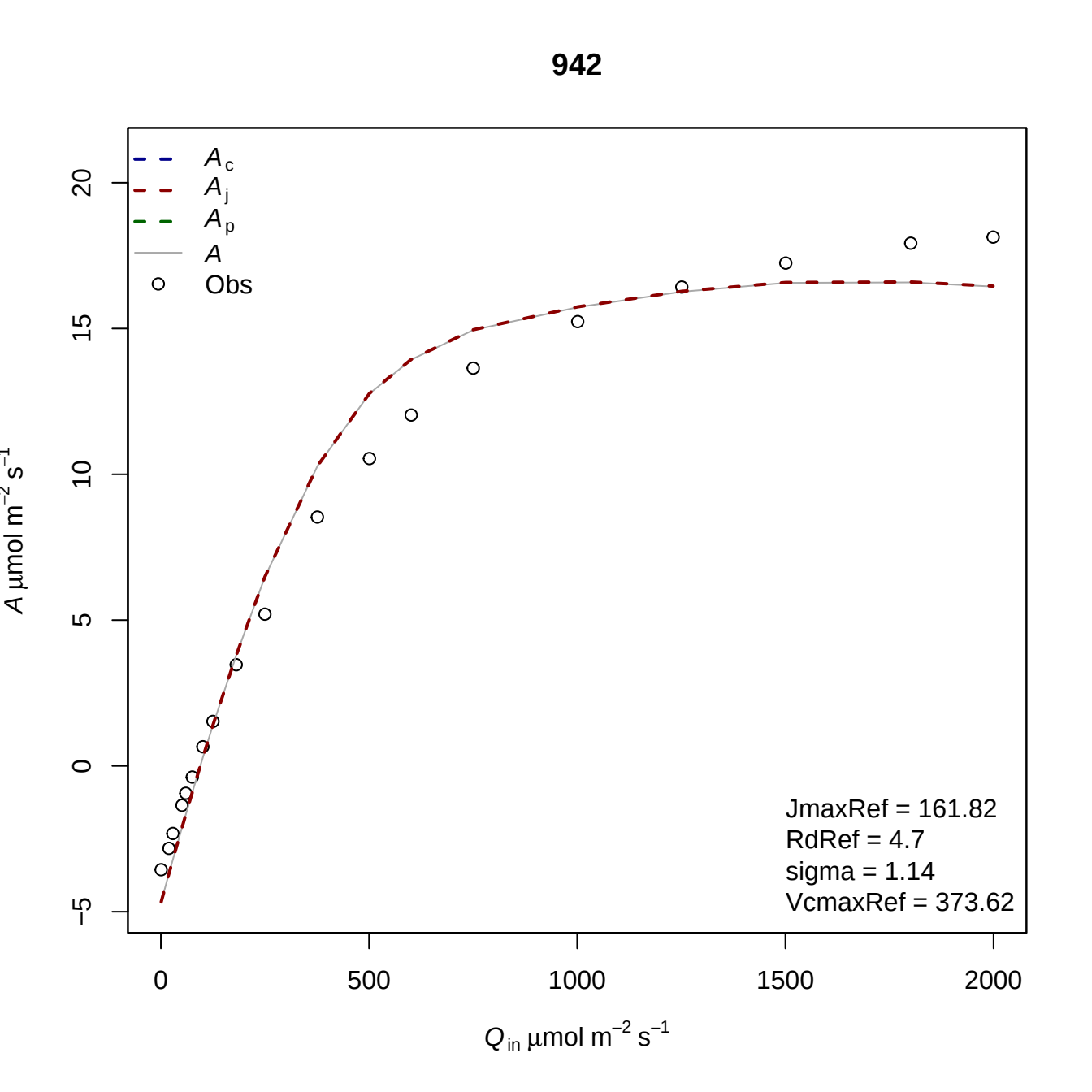

937
A_c
A_j
A_p
A
Obs
JmaxRef = 60.4
RdRef = 3.34
sigma = 0.97
VcmaxRef = 183.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

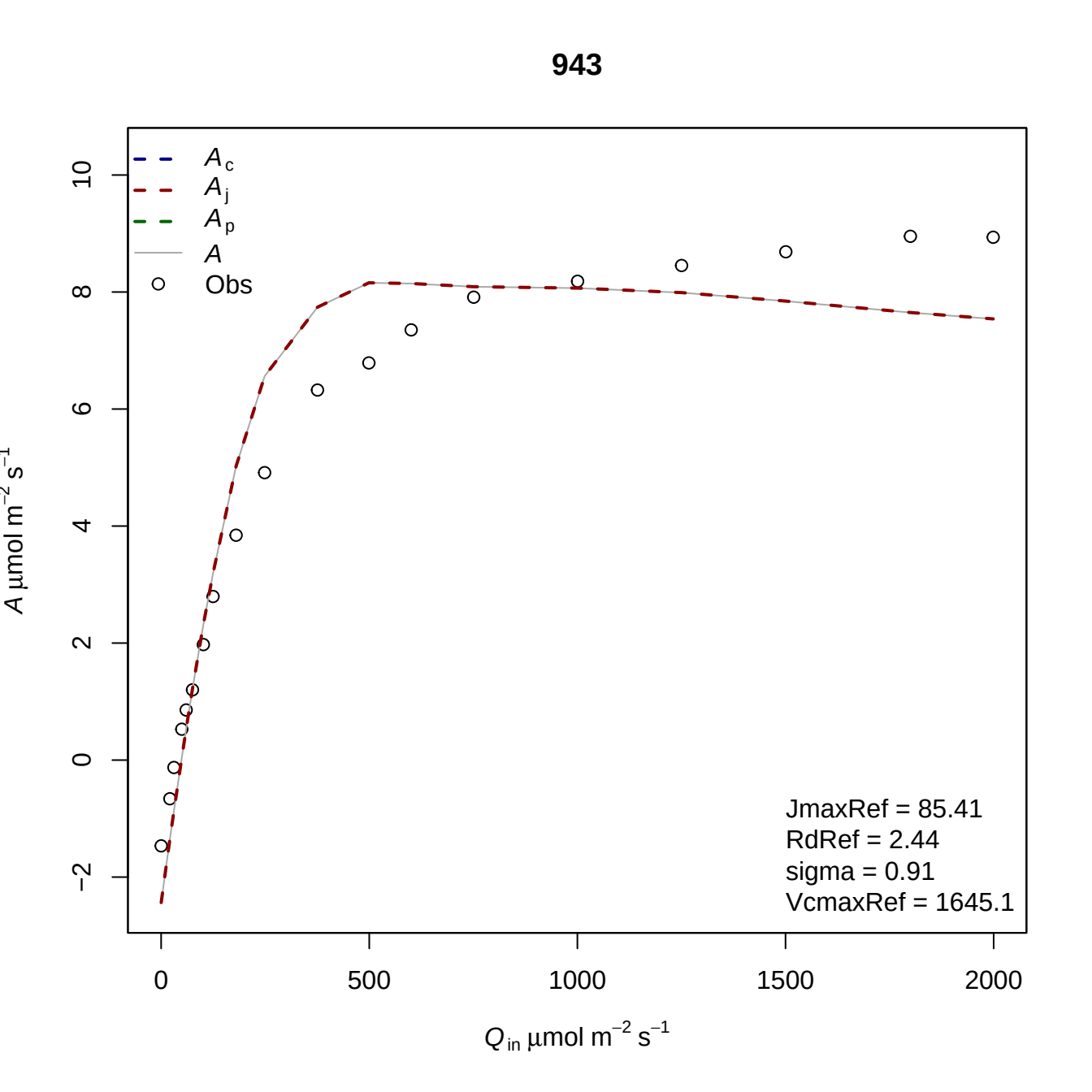

938
A_c
A_j
A_p
A
Obs
JmaxRef = 74.3
RdRef = 4.33
sigma = 1.16
VcmaxRef = 189.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

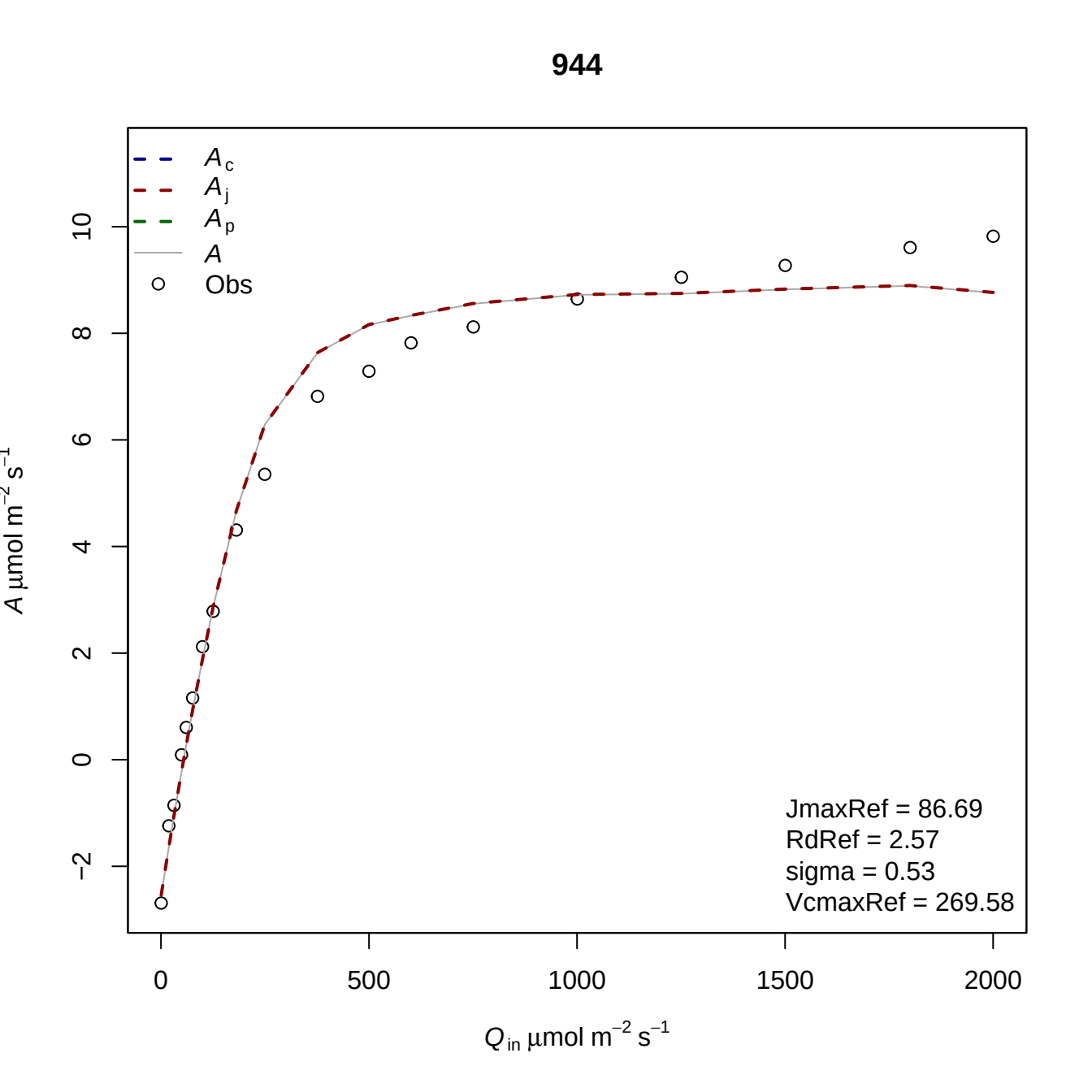

939
A_c
A_j
A_p
A
Obs
JmaxRef = 70.24
RdRef = 3.08
sigma = 0.92
VcmaxRef = 222
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

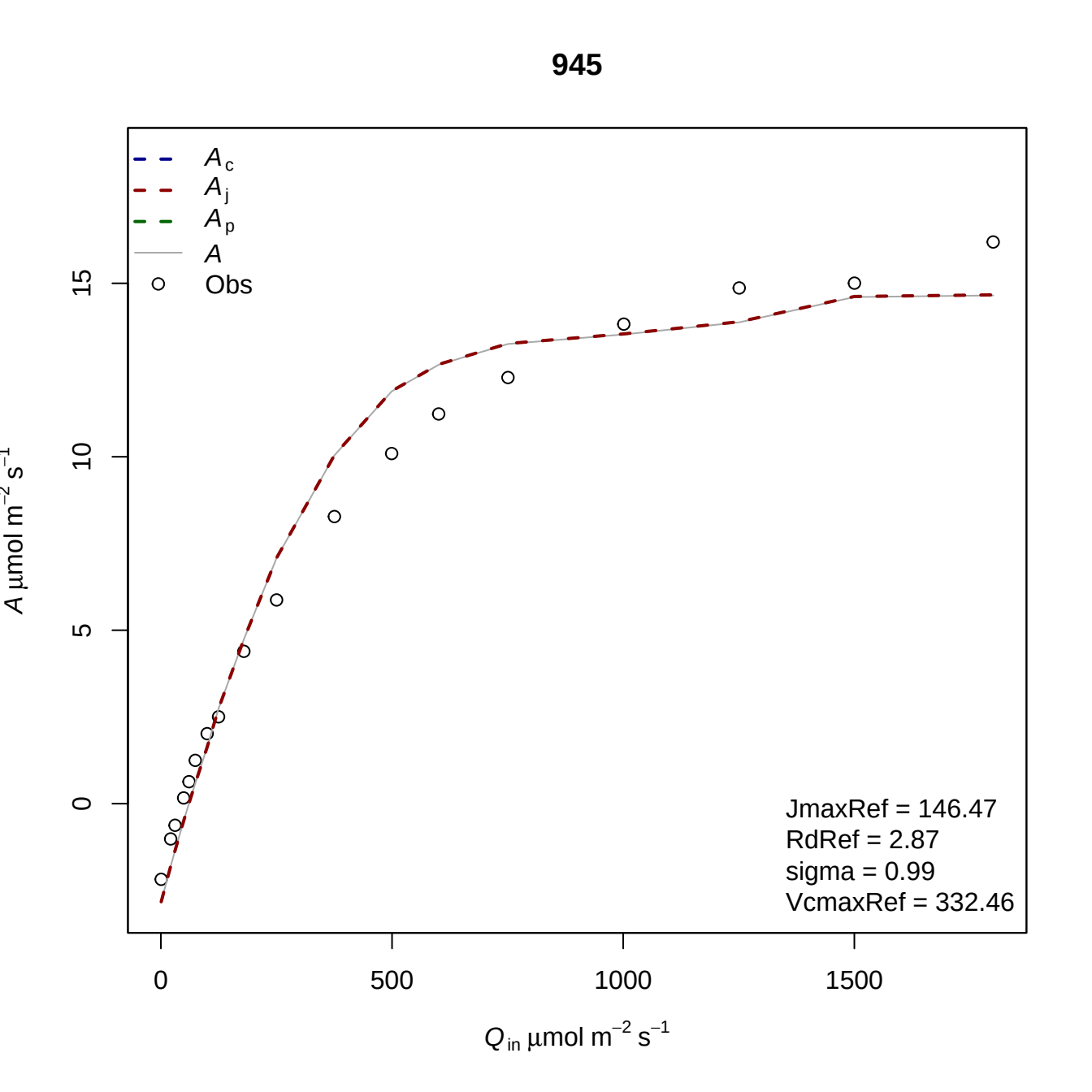

940
A_c
A_j
A_p
A
Obs
JmaxRef = 65.79
RdRef = 1.93
sigma = 0.42
VcmaxRef = 217.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

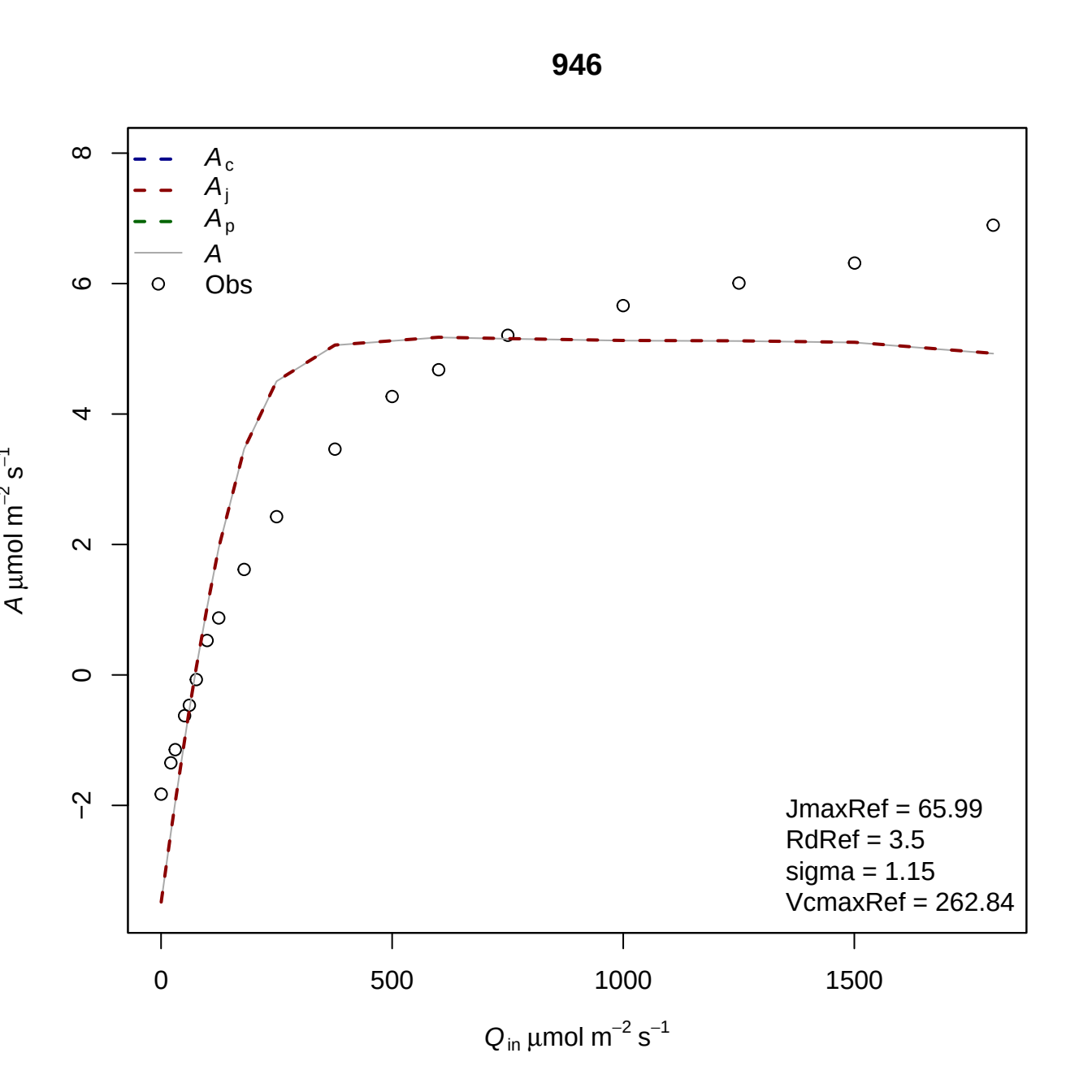

941
A_c
A_j
A_p
A
Obs
JmaxRef = 47.45
RdRef = 3.3
sigma = 1
VcmaxRef = 242.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
500
1000
1500

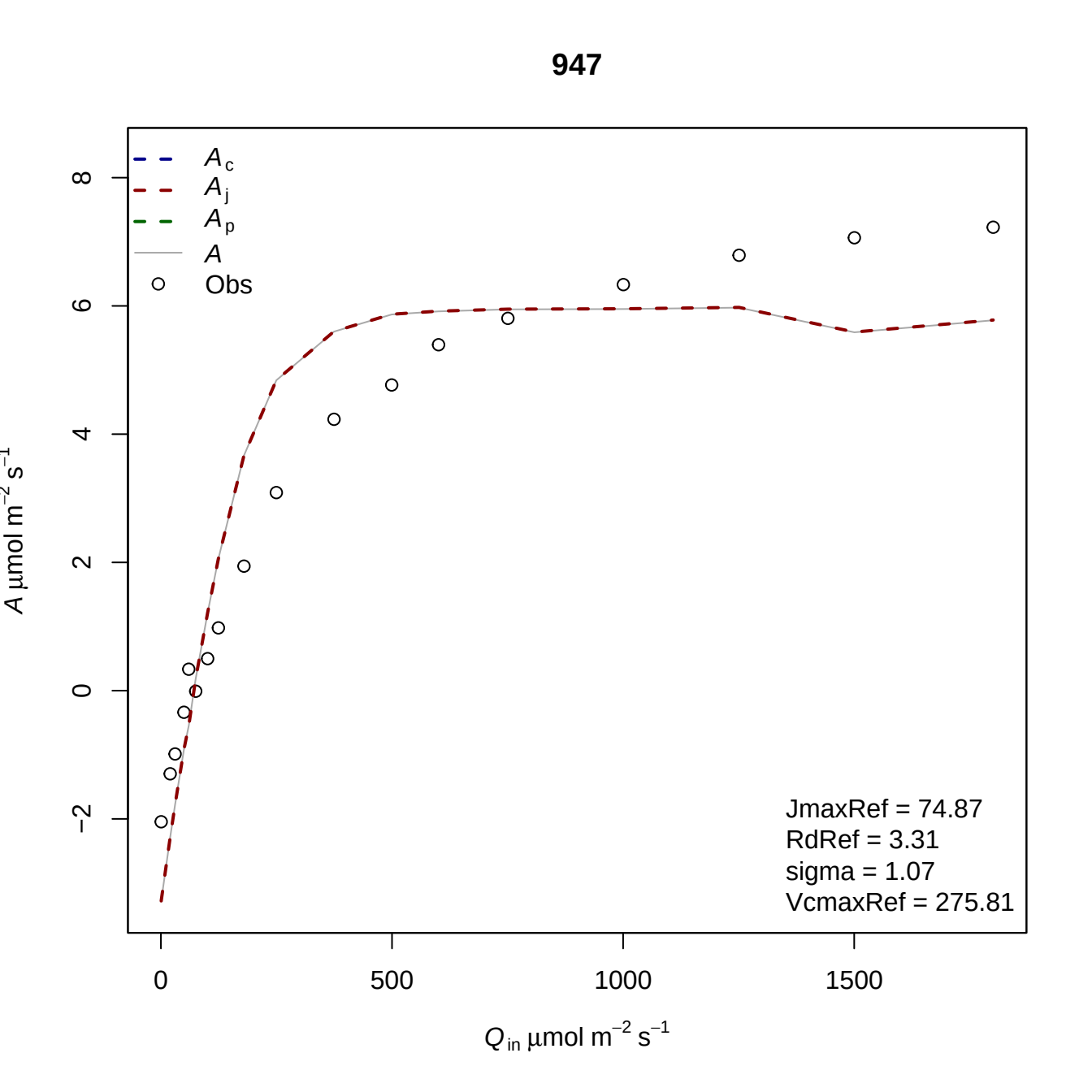

942
A_c
A_j
A_p
A
Obs
JmaxRef = 161.82
RdRef = 4.7
sigma = 1.14
VcmaxRef = 373.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

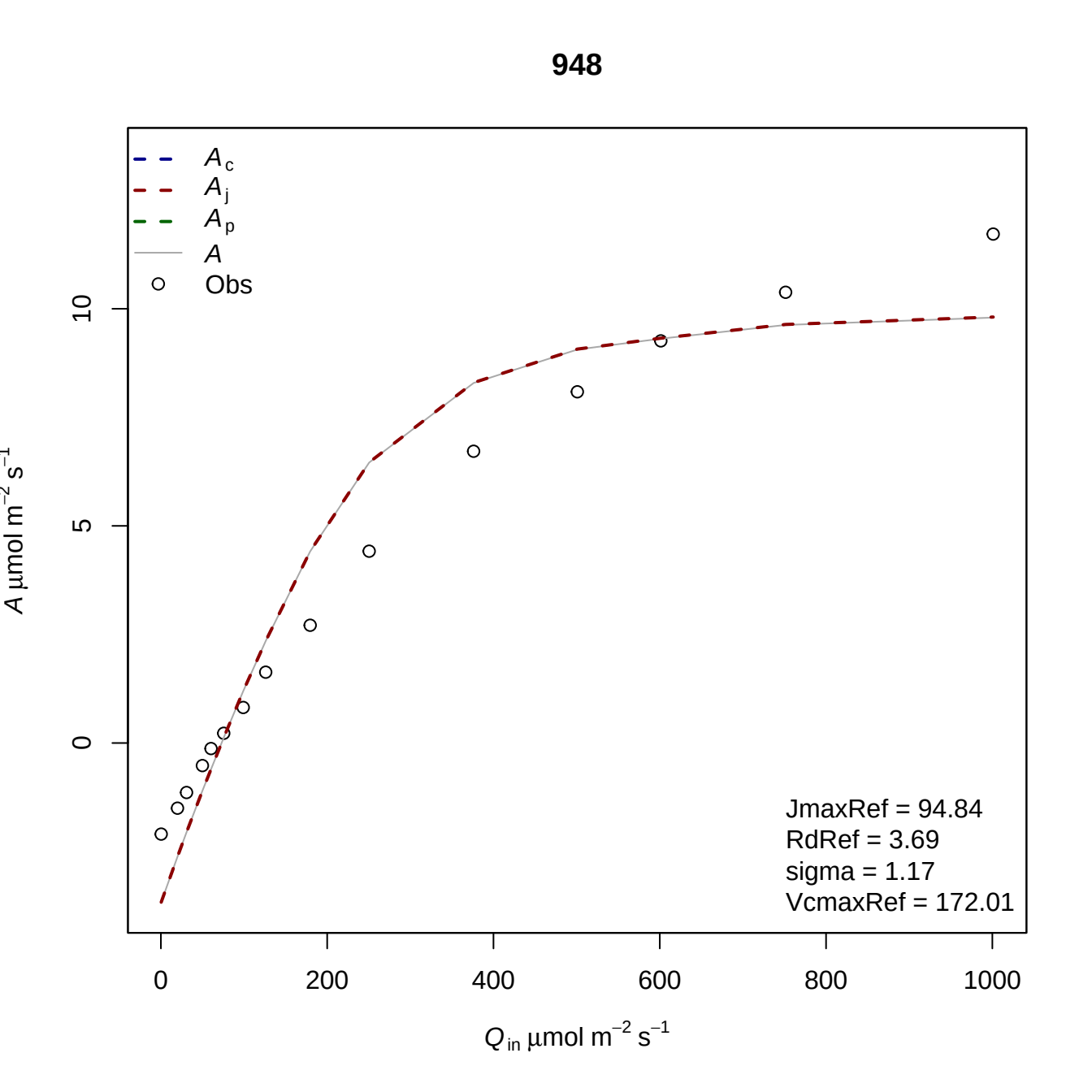

943
A_c
A_j
A_p
A
Obs
JmaxRef = 85.41
RdRef = 2.44
sigma = 0.91
VcmaxRef = 1645.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

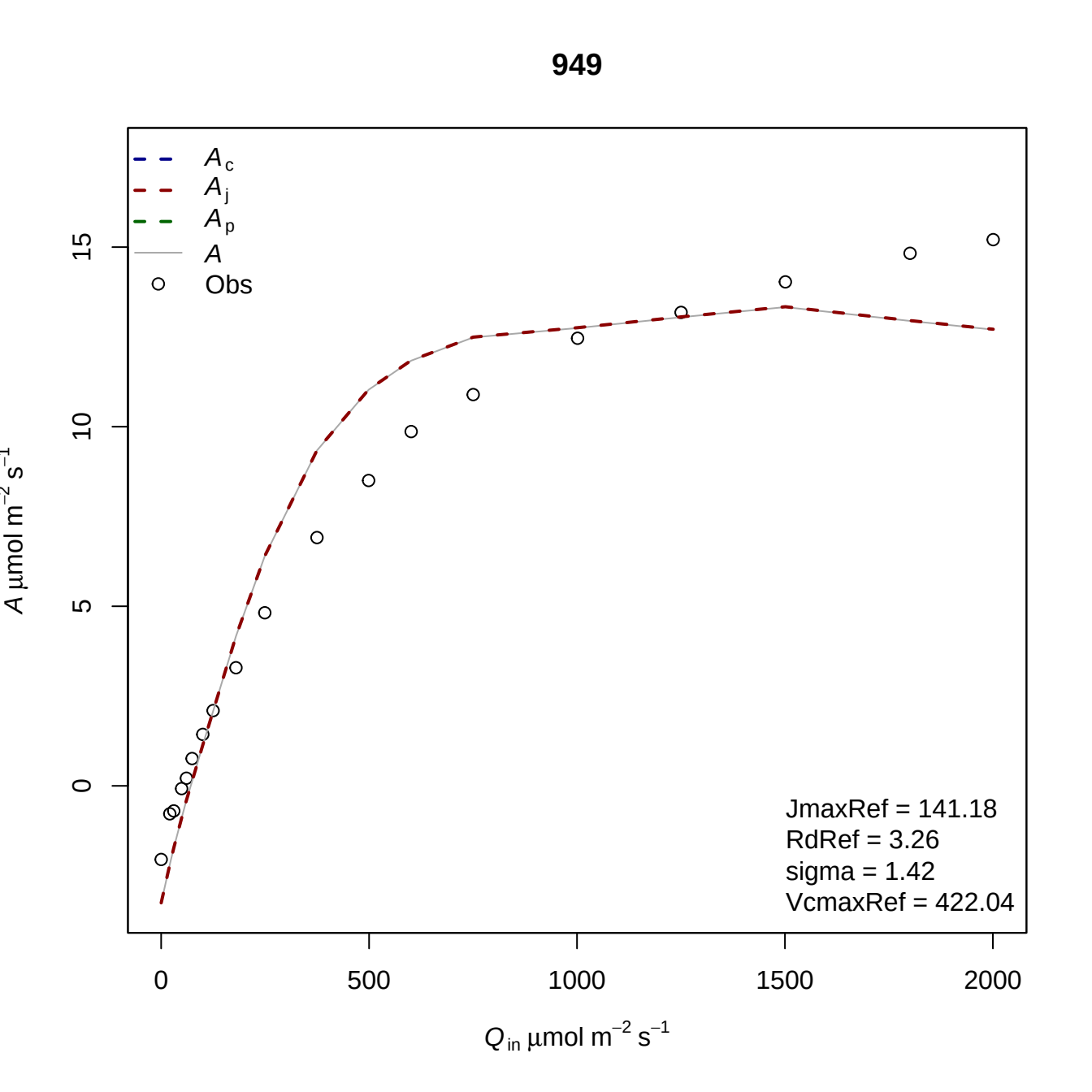

944
A_c
A_j
A_p
A
Obs
JmaxRef = 86.69
RdRef = 2.57
sigma = 0.53
VcmaxRef = 269.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

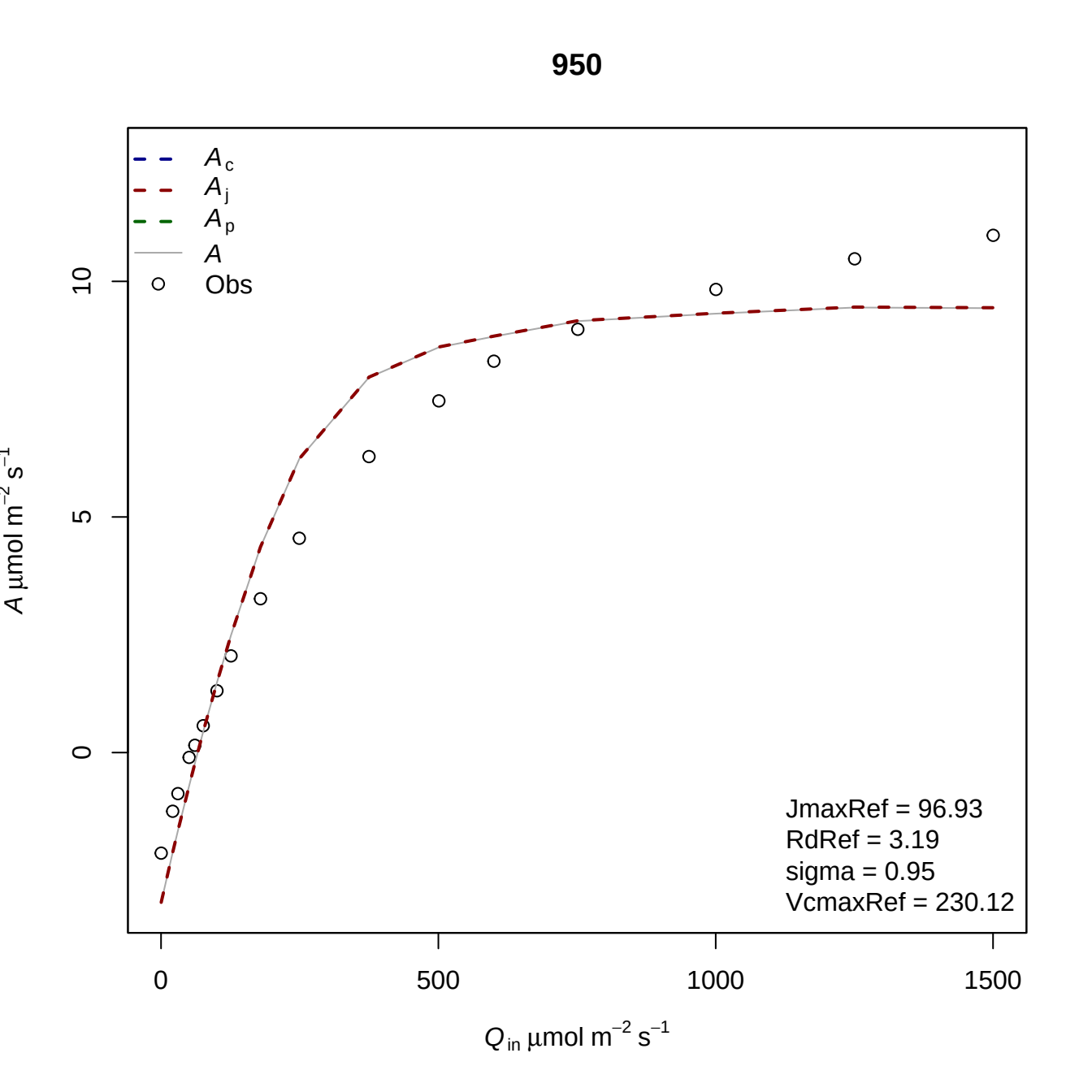

945
A_c
A_j
A_p
A
Obs
JmaxRef = 146.47
RdRef = 2.87
sigma = 0.99
VcmaxRef = 332.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

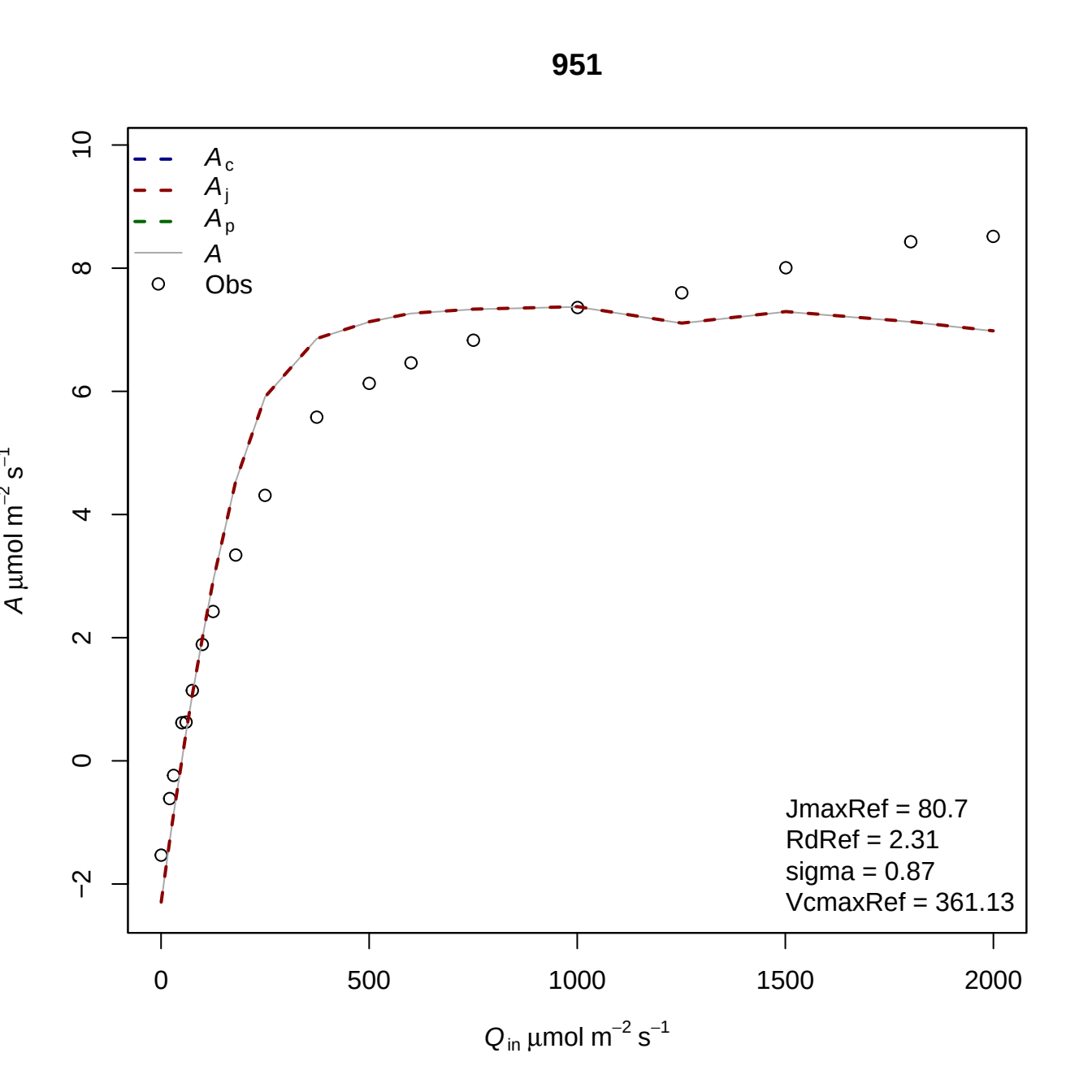

946
A_c
A_j
A_p
A
Obs
JmaxRef = 65.99
RdRef = 3.5
sigma = 1.15
VcmaxRef = 262.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

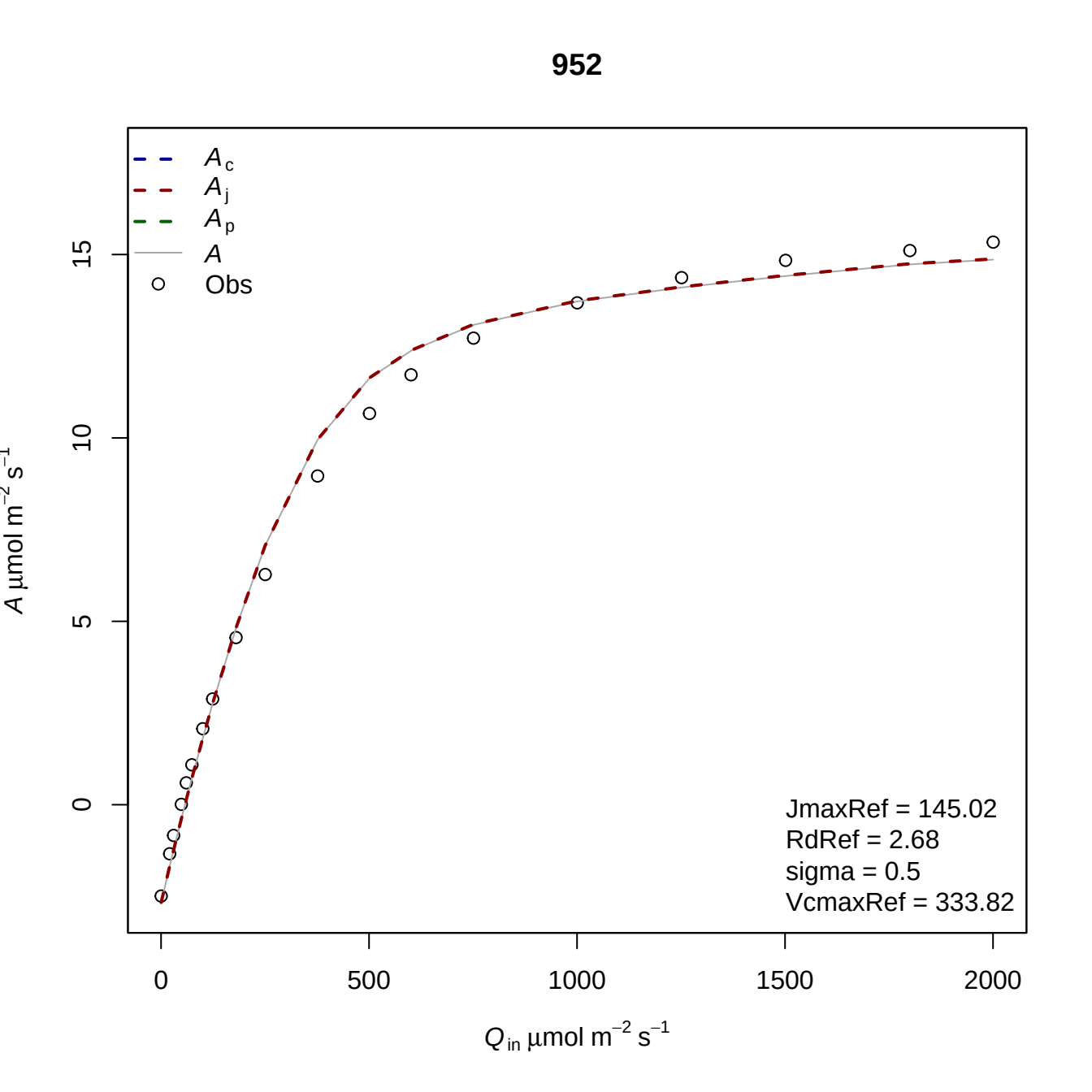

947
A_c
A_j
A_p
A
Obs
JmaxRef = 74.87
RdRef = 3.31
sigma = 1.07
VcmaxRef = 275.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

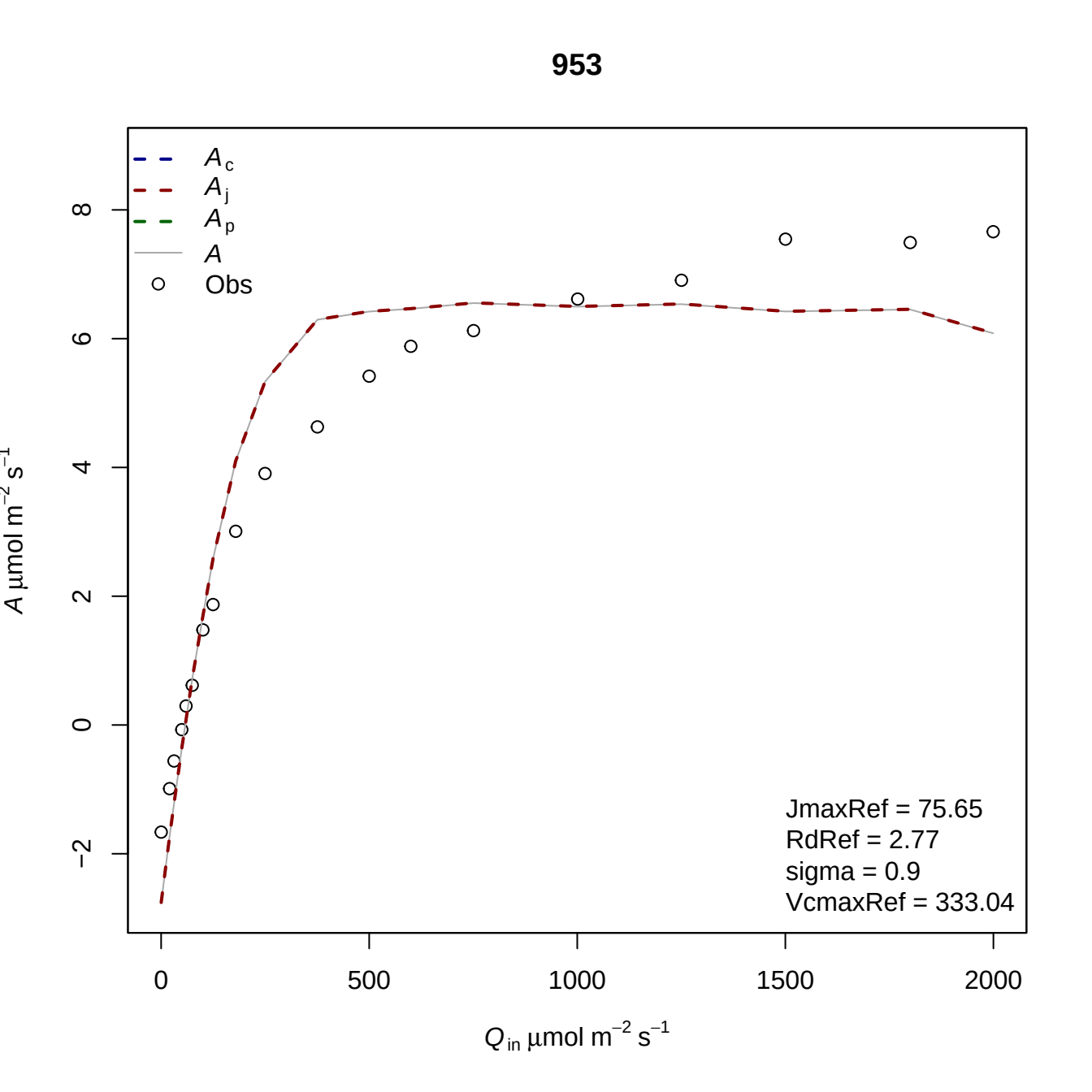

948
A_c
A_j
A_p
A
Obs
JmaxRef = 94.84
RdRef = 3.69
sigma = 1.17
VcmaxRef = 172.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

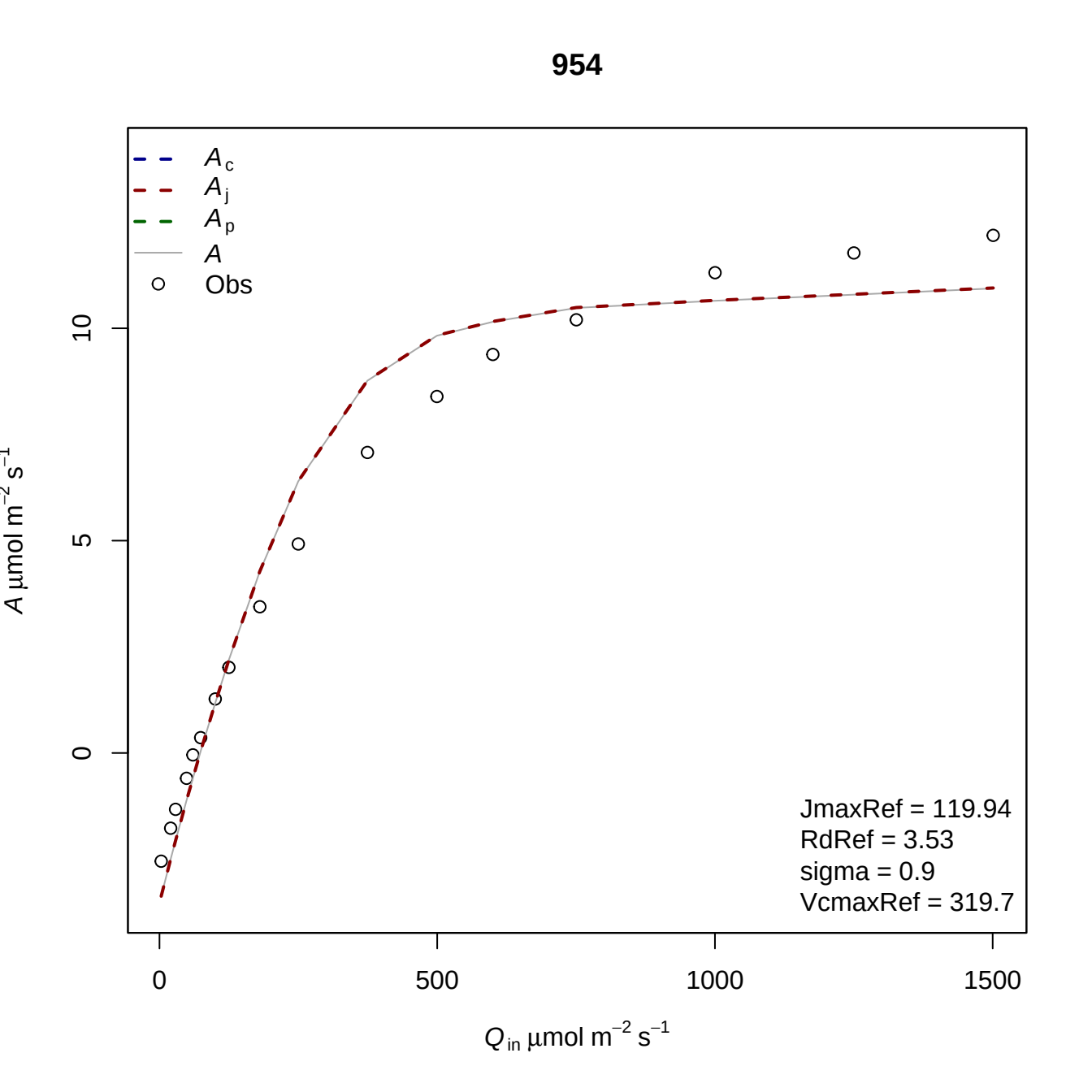

949
A_c
A_j
A_p
A
Obs
JmaxRef = 141.18
RdRef = 3.26
sigma = 1.42
VcmaxRef = 422.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

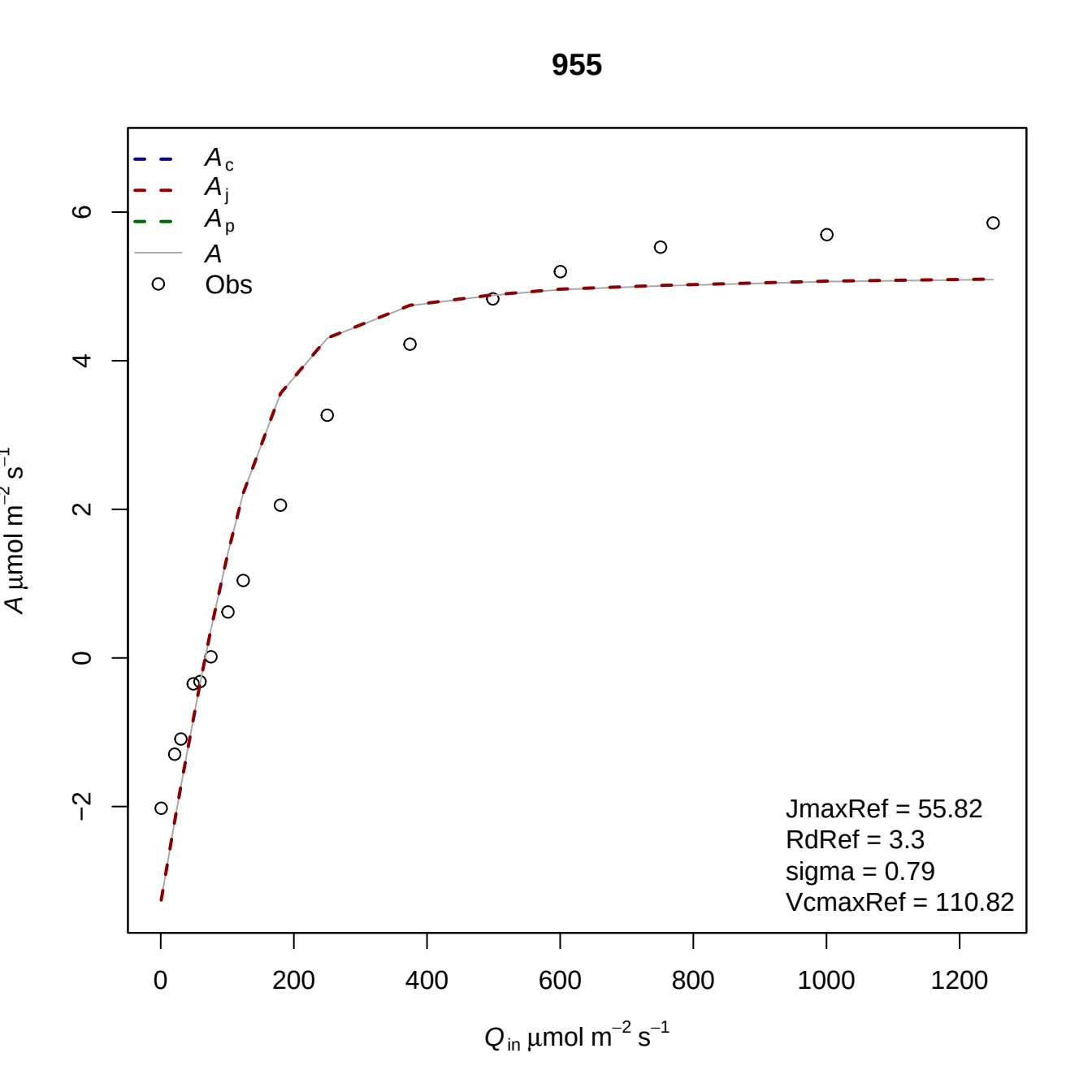

950
A_c
A_j
A_p
A
Obs
JmaxRef = 96.93
RdRef = 3.19
sigma = 0.95
VcmaxRef = 230.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

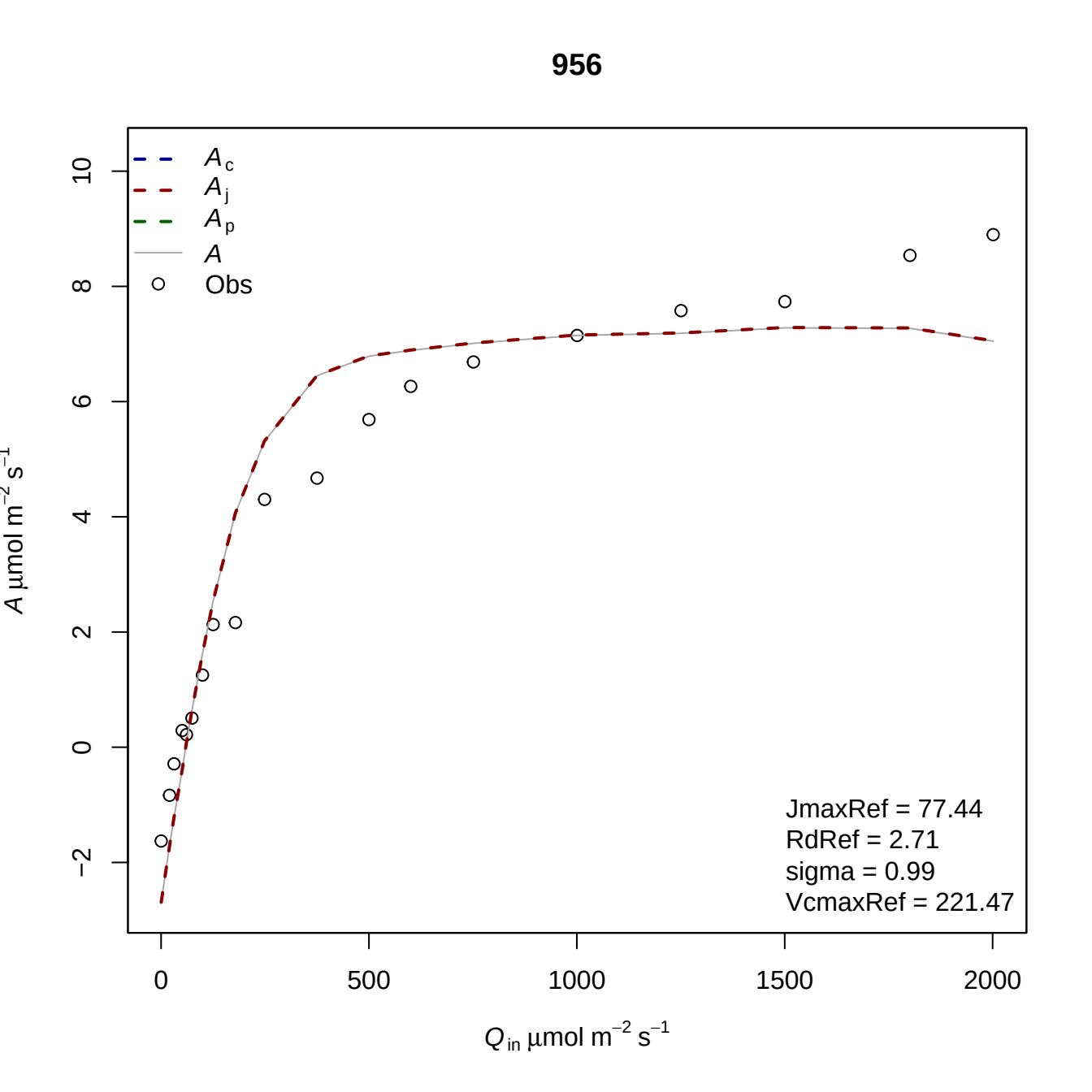

951
A_c
A_j
A_p
A
Obs
JmaxRef = 80.7
RdRef = 2.31
sigma = 0.87
VcmaxRef = 361.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

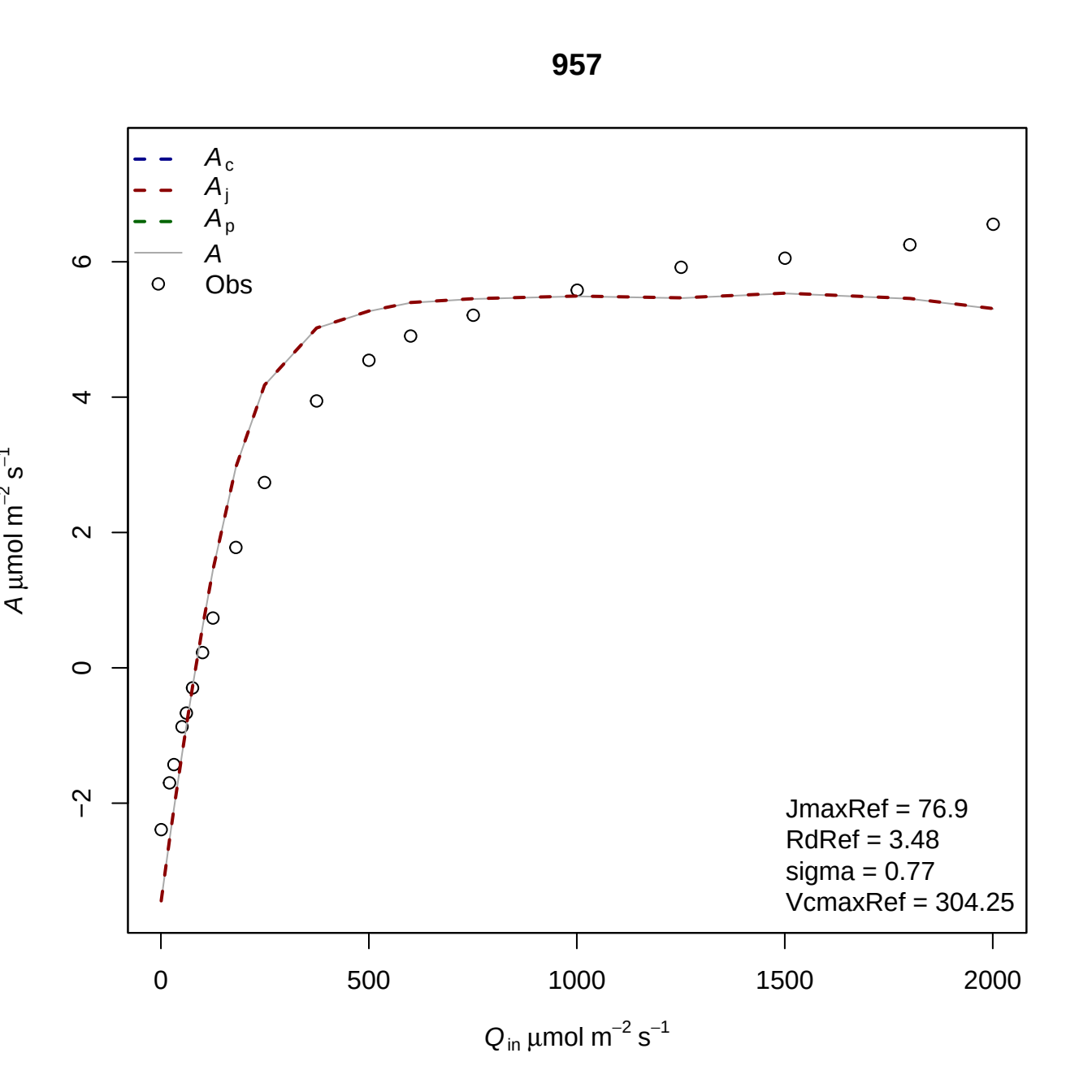

952
A_c
A_j
A_p
A
Obs
JmaxRef = 145.02
RdRef = 2.68
sigma = 0.5
VcmaxRef = 333.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

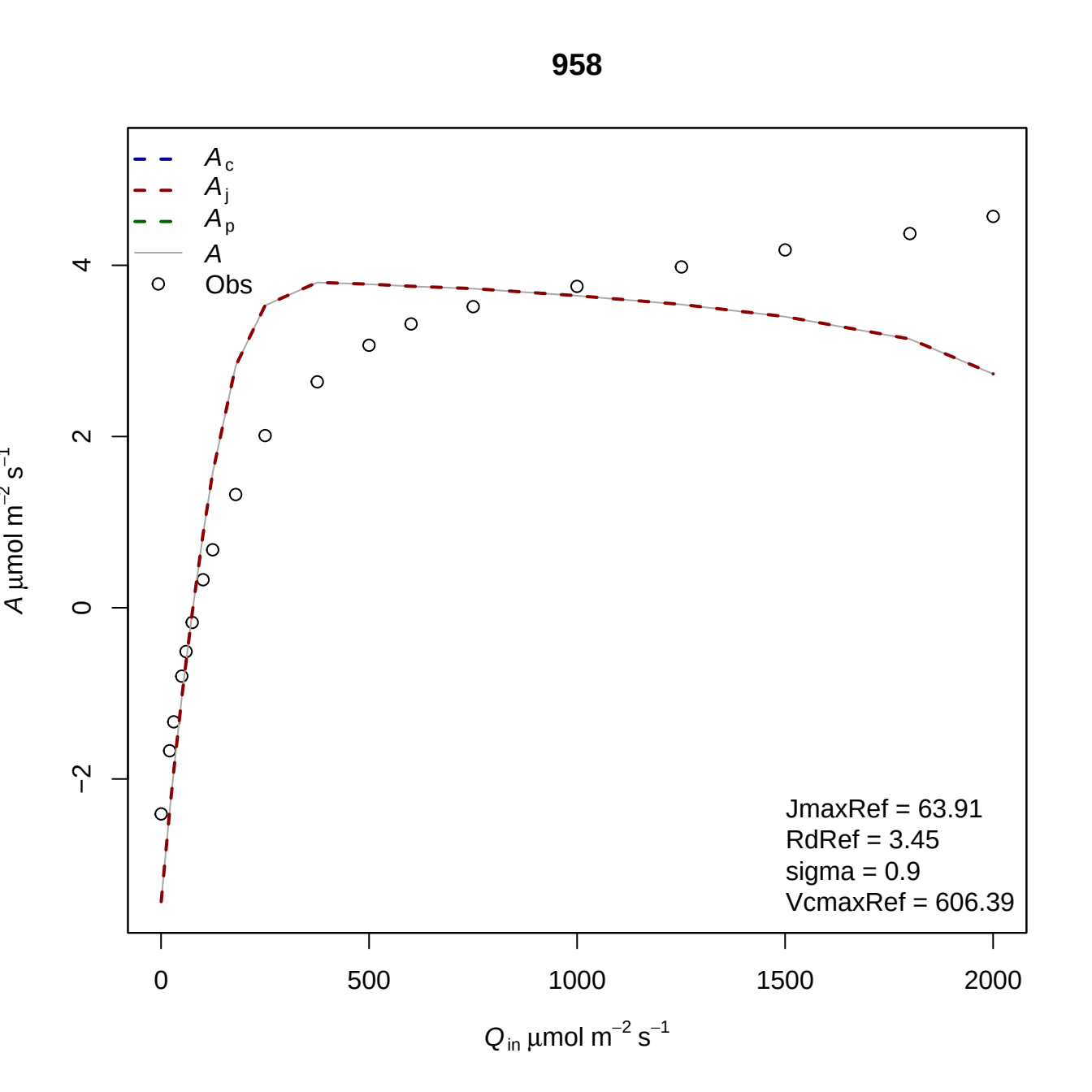

953
A_c
A_j
A_p
A
Obs
JmaxRef = 75.65
RdRef = 2.77
sigma = 0.9
VcmaxRef = 333.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

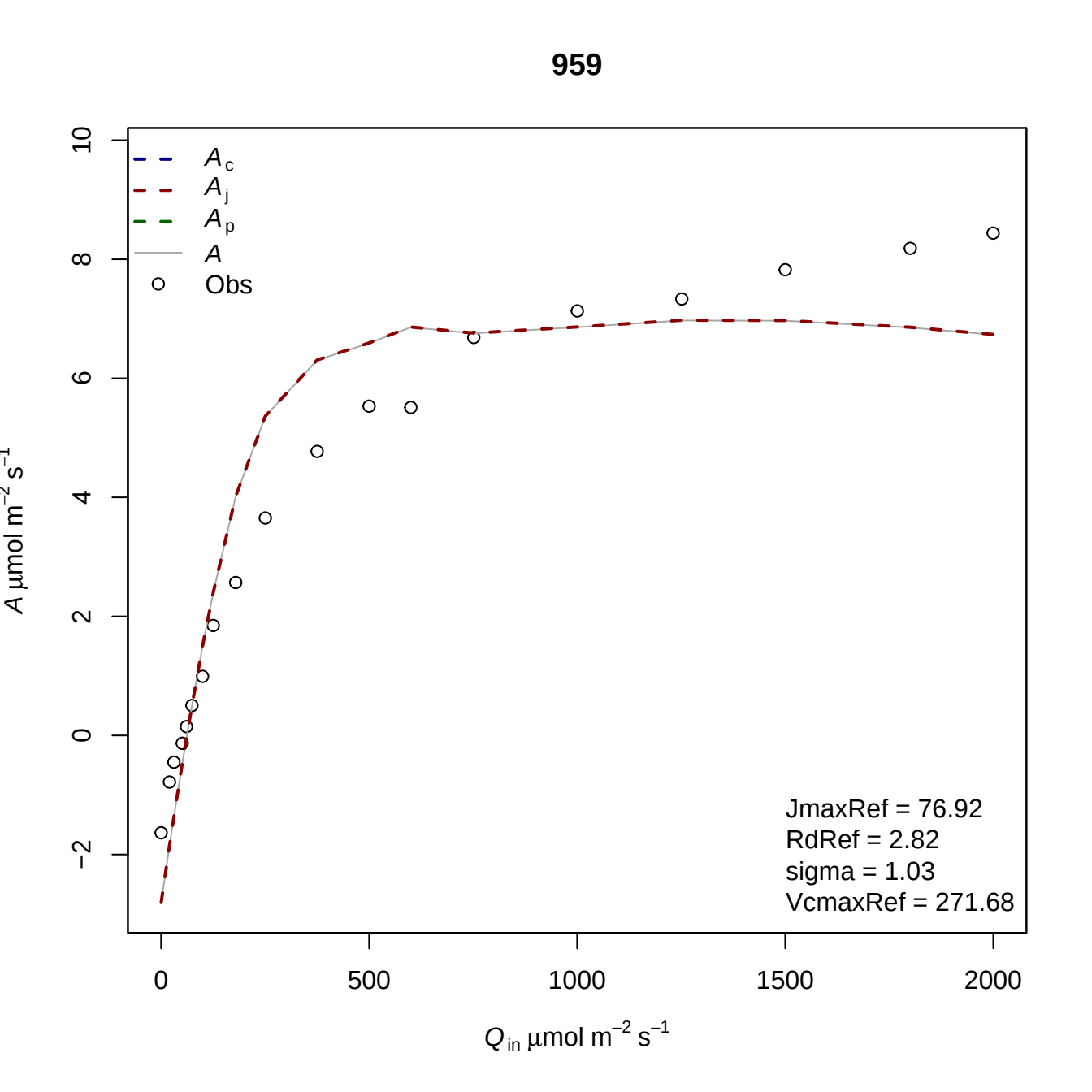

954
A_c
A_j
A_p
A
Obs
JmaxRef = 119.94
RdRef = 3.53
sigma = 0.9
VcmaxRef = 319.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

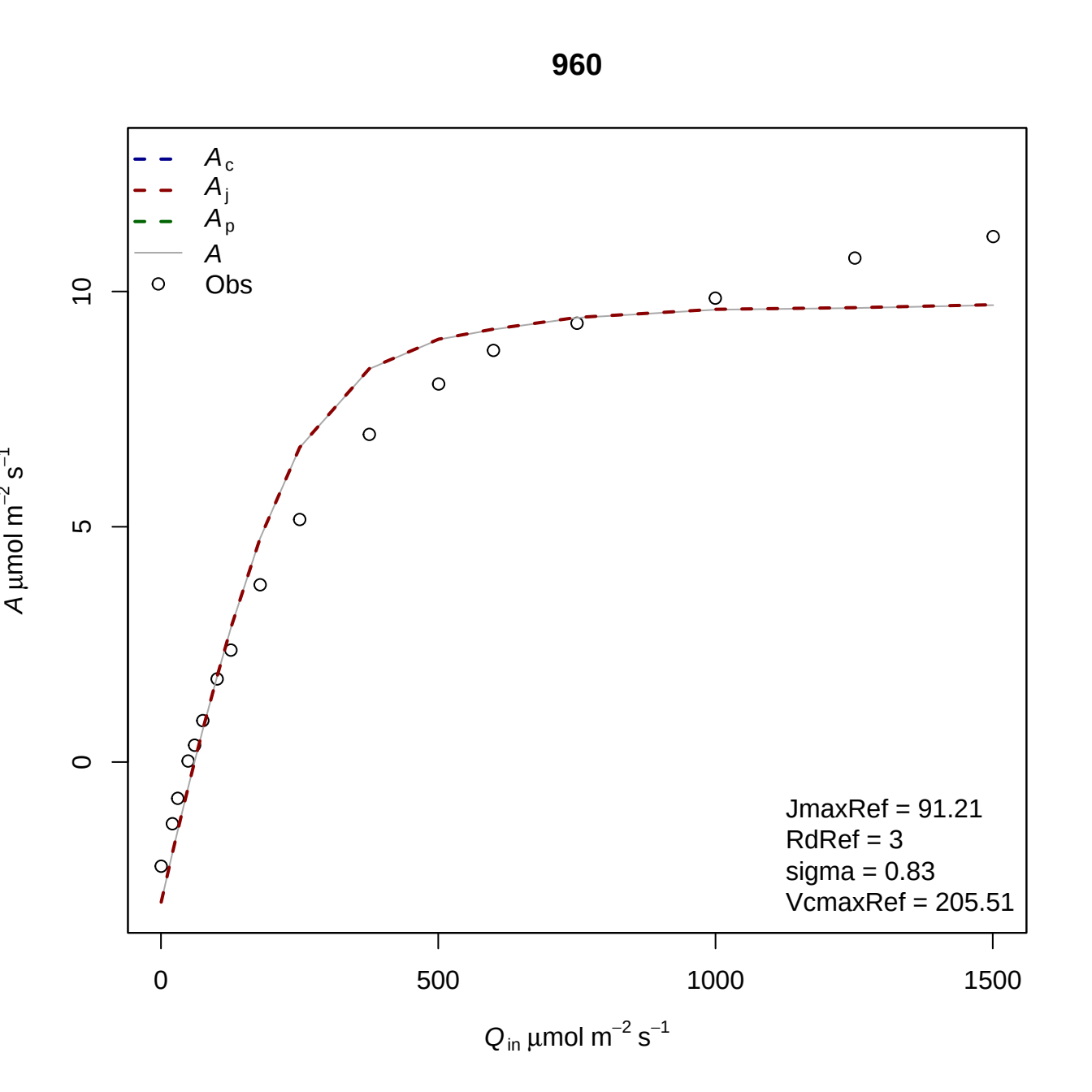

955
A_c
A_j
A_p
A
Obs
JmaxRef = 55.82
RdRef = 3.3
sigma = 0.79
VcmaxRef = 110.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

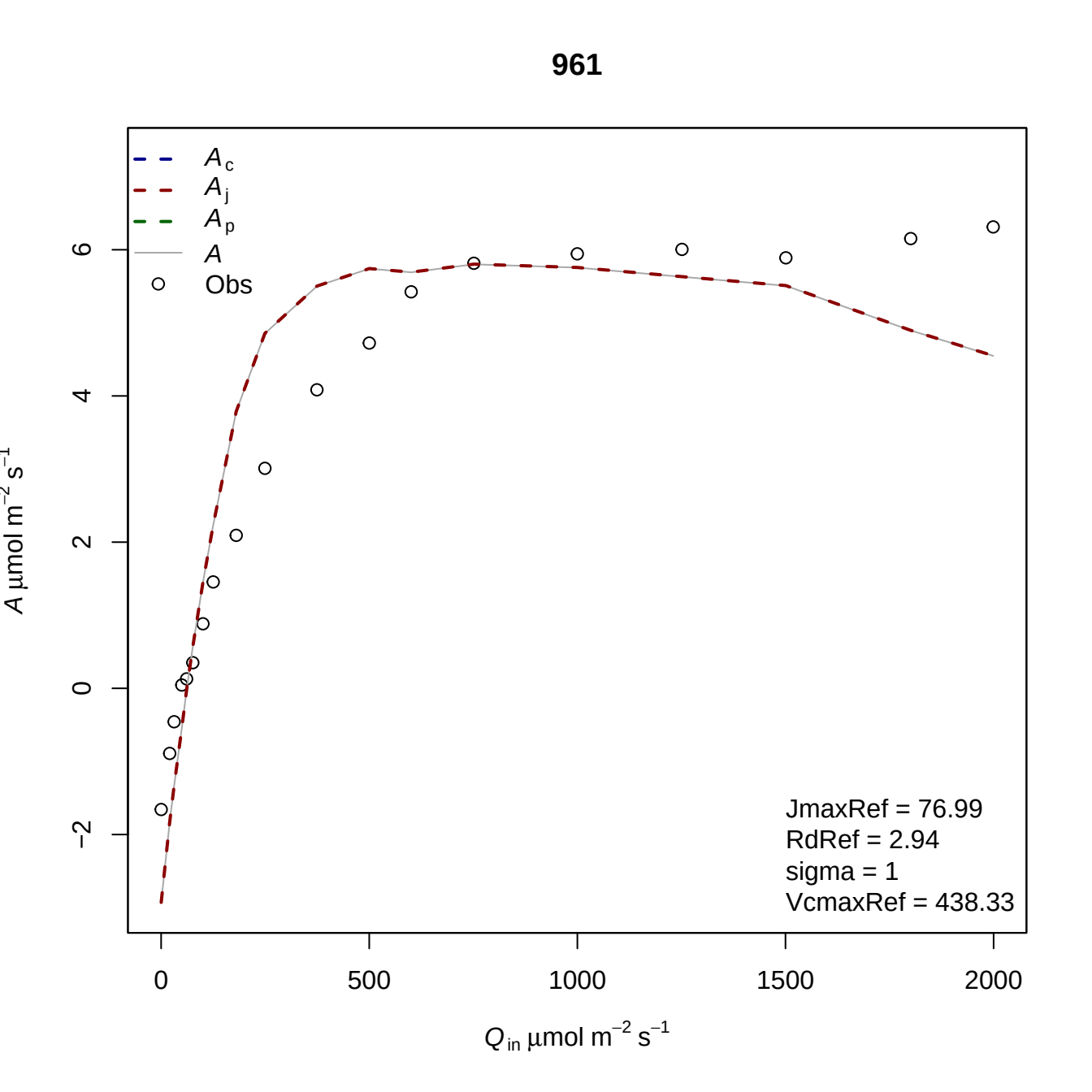

956
A_c
A_j
A_p
A
Obs
JmaxRef = 77.44
RdRef = 2.71
sigma = 0.99
VcmaxRef = 221.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

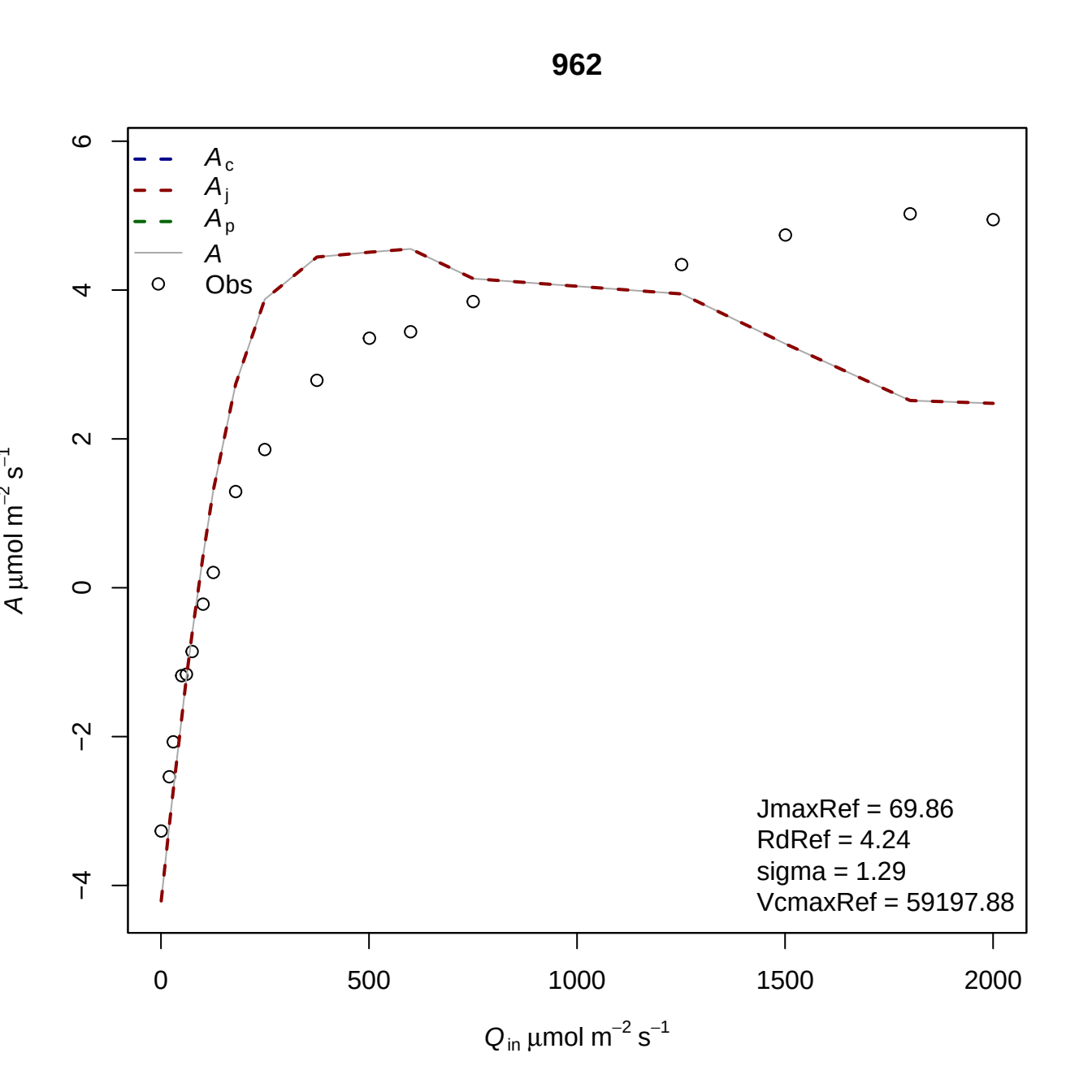

957
A_c
A_j
A_p
A
Obs
JmaxRef = 76.9
RdRef = 3.48
sigma = 0.77
VcmaxRef = 304.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

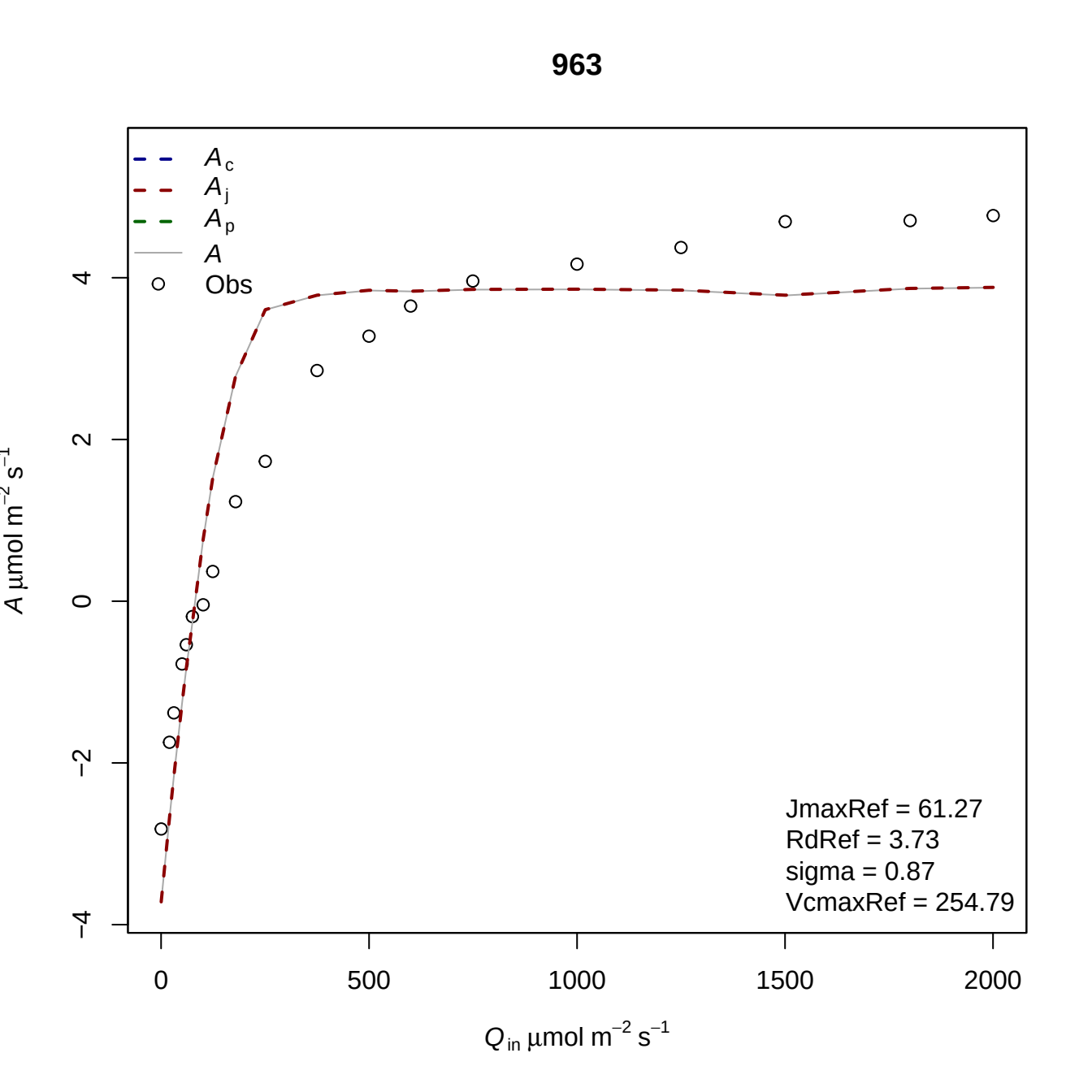

958
A_c
A_j
A_p
A
Obs
JmaxRef = 63.91
RdRef = 3.45
sigma = 0.9
VcmaxRef = 606.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

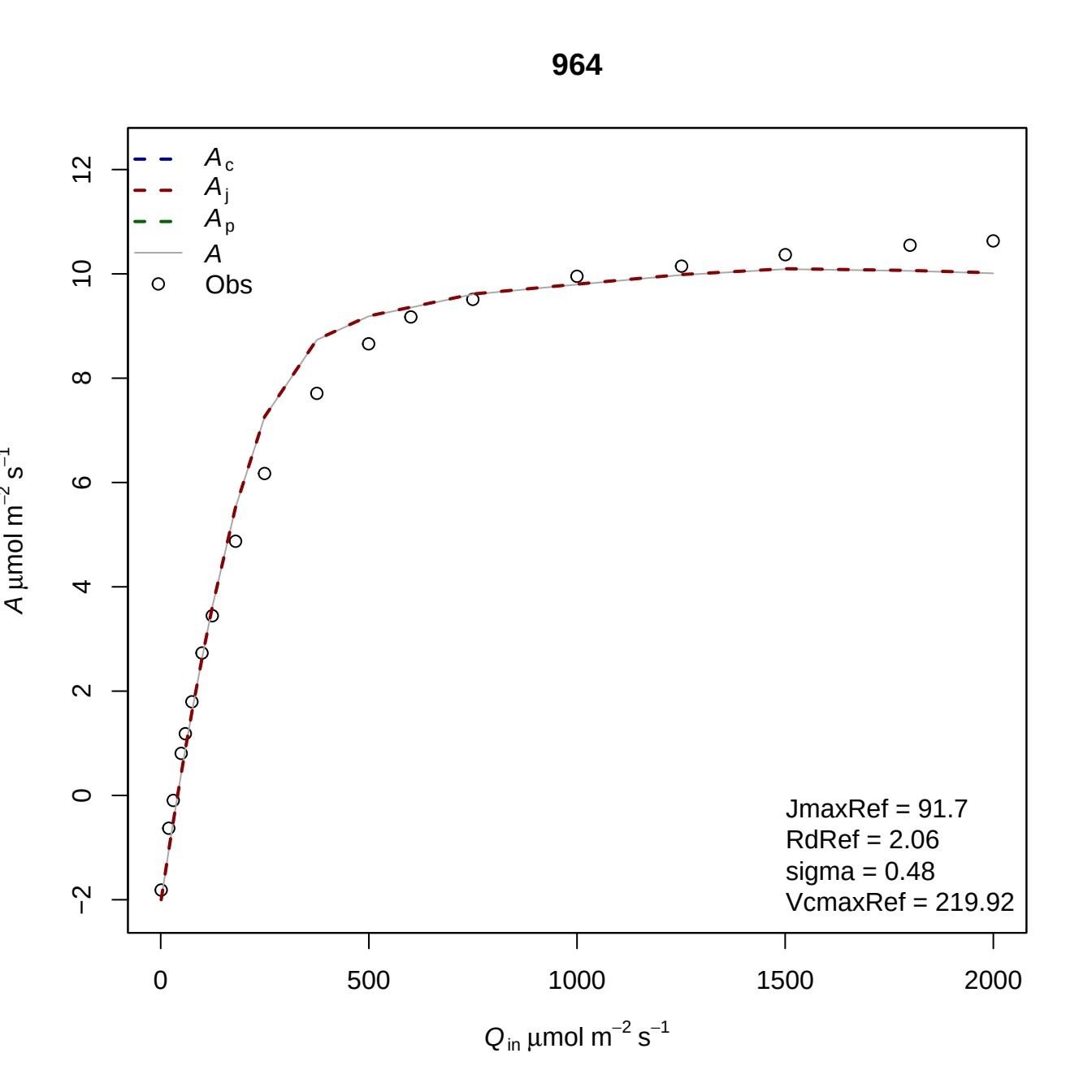

959
A_c
A_j
A_p
A
Obs
JmaxRef = 76.92
RdRef = 2.82
sigma = 1.03
VcmaxRef = 271.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

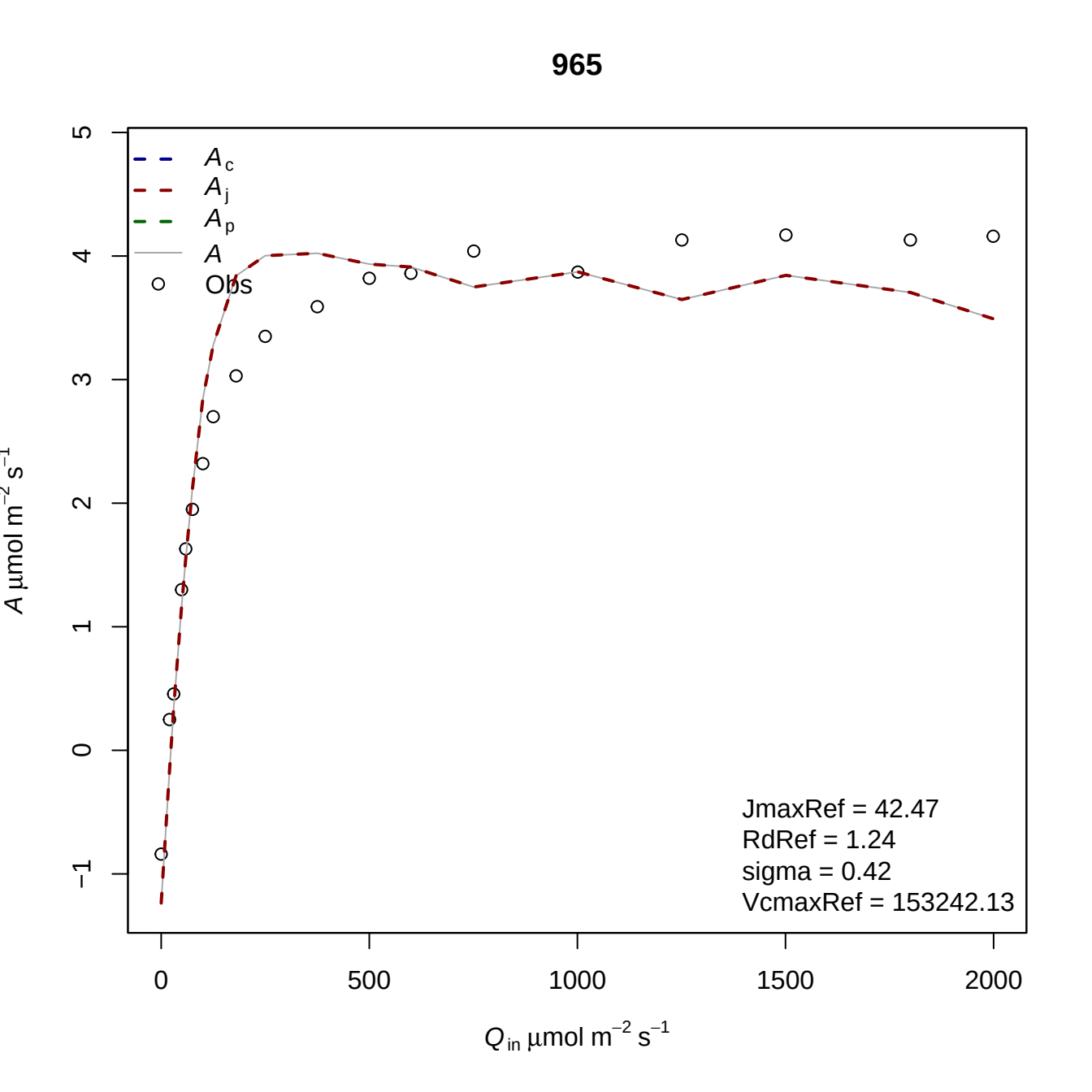

960
A_c
A_j
A_p
A
Obs
JmaxRef = 91.21
RdRef = 3
sigma = 0.83
VcmaxRef = 205.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

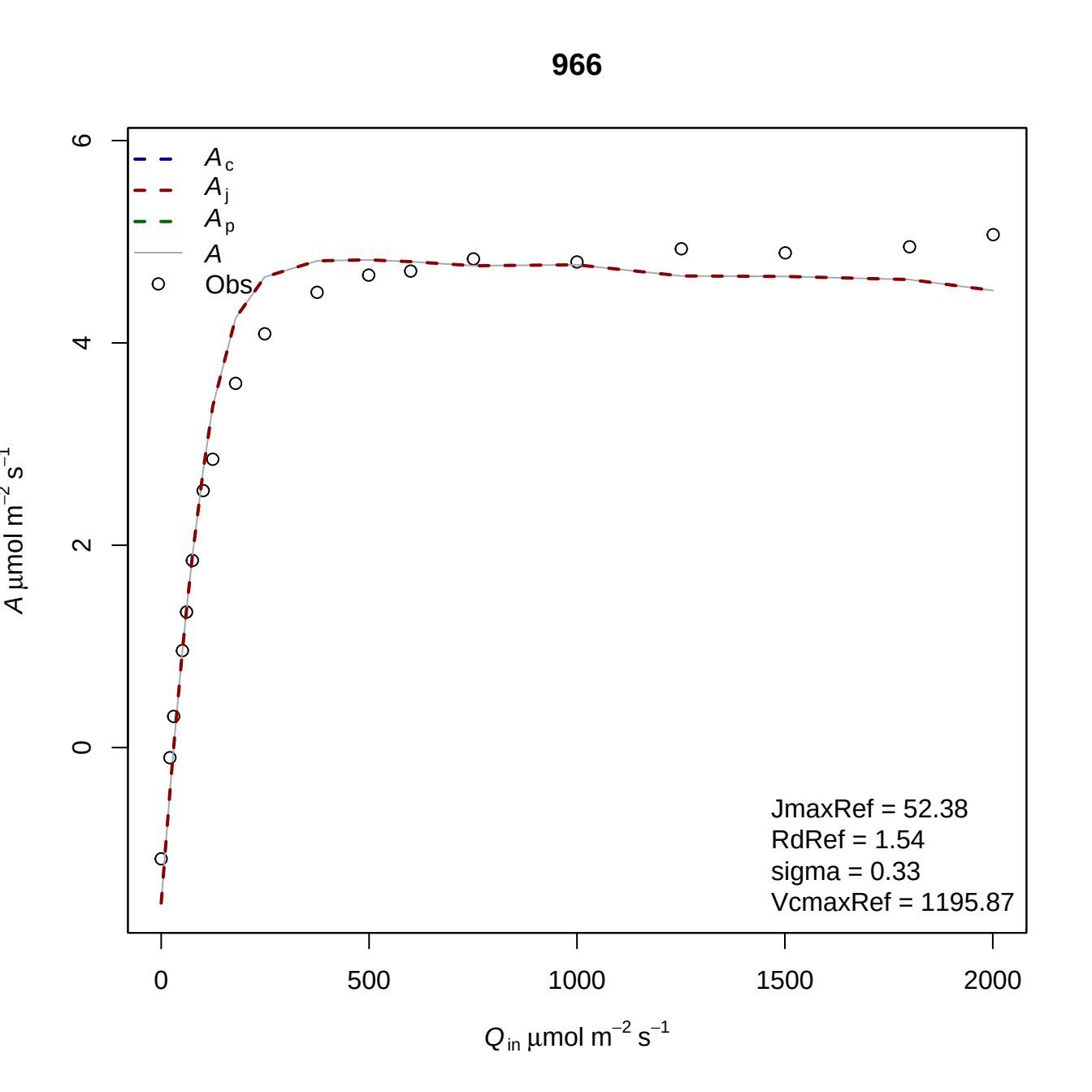

961
A_c
A_j
A_p
A
Obs
JmaxRef = 76.99
RdRef = 2.94
sigma = 1
VcmaxRef = 438.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

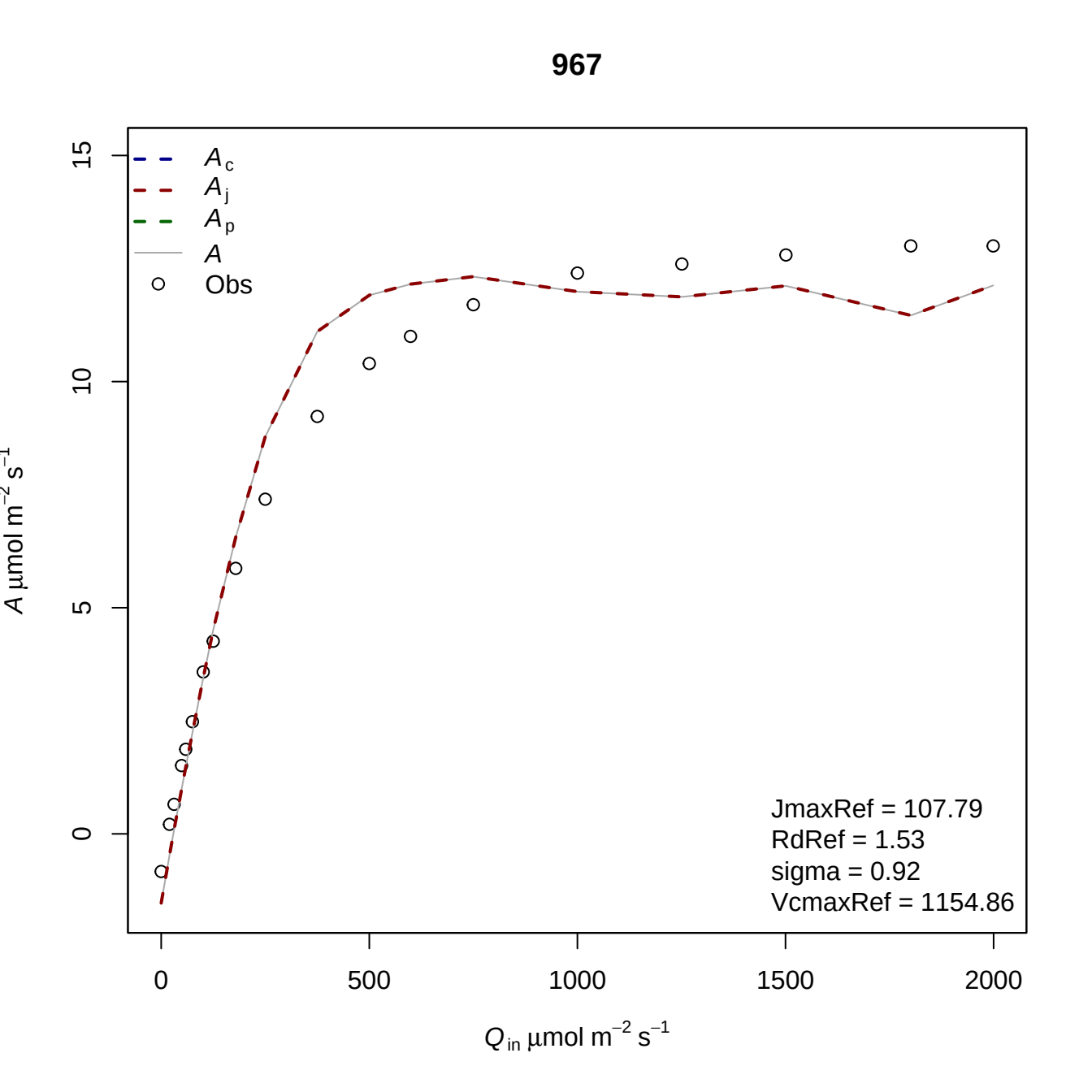

962
A_c
A_j
A_p
A
Obs
JmaxRef = 69.86
RdRef = 4.24
sigma = 1.29
VcmaxRef = 59197.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

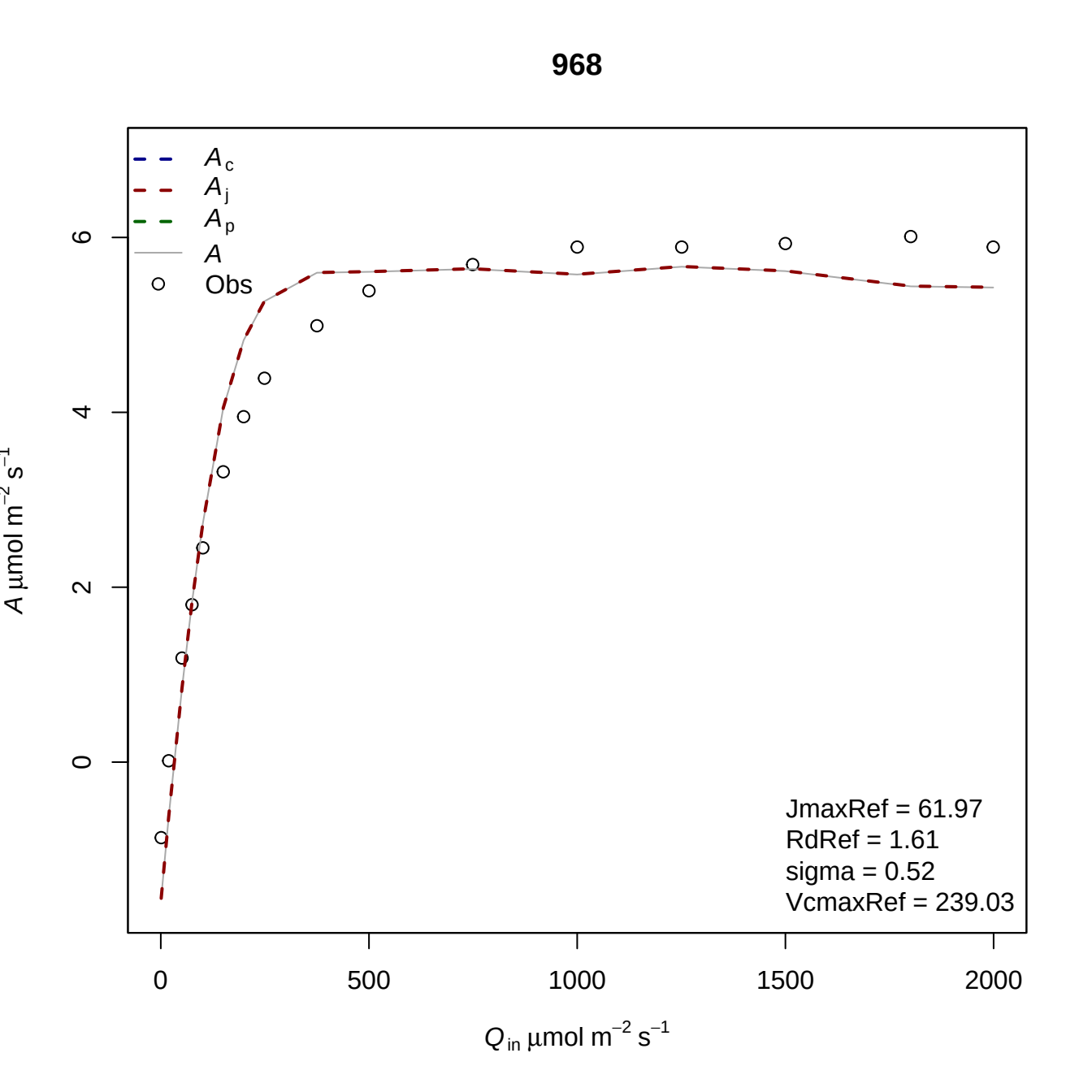

963
A_c
A_j
A_p
A
Obs
JmaxRef = 61.27
RdRef = 3.73
sigma = 0.87
VcmaxRef = 254.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
0
500
1000
1500
2000

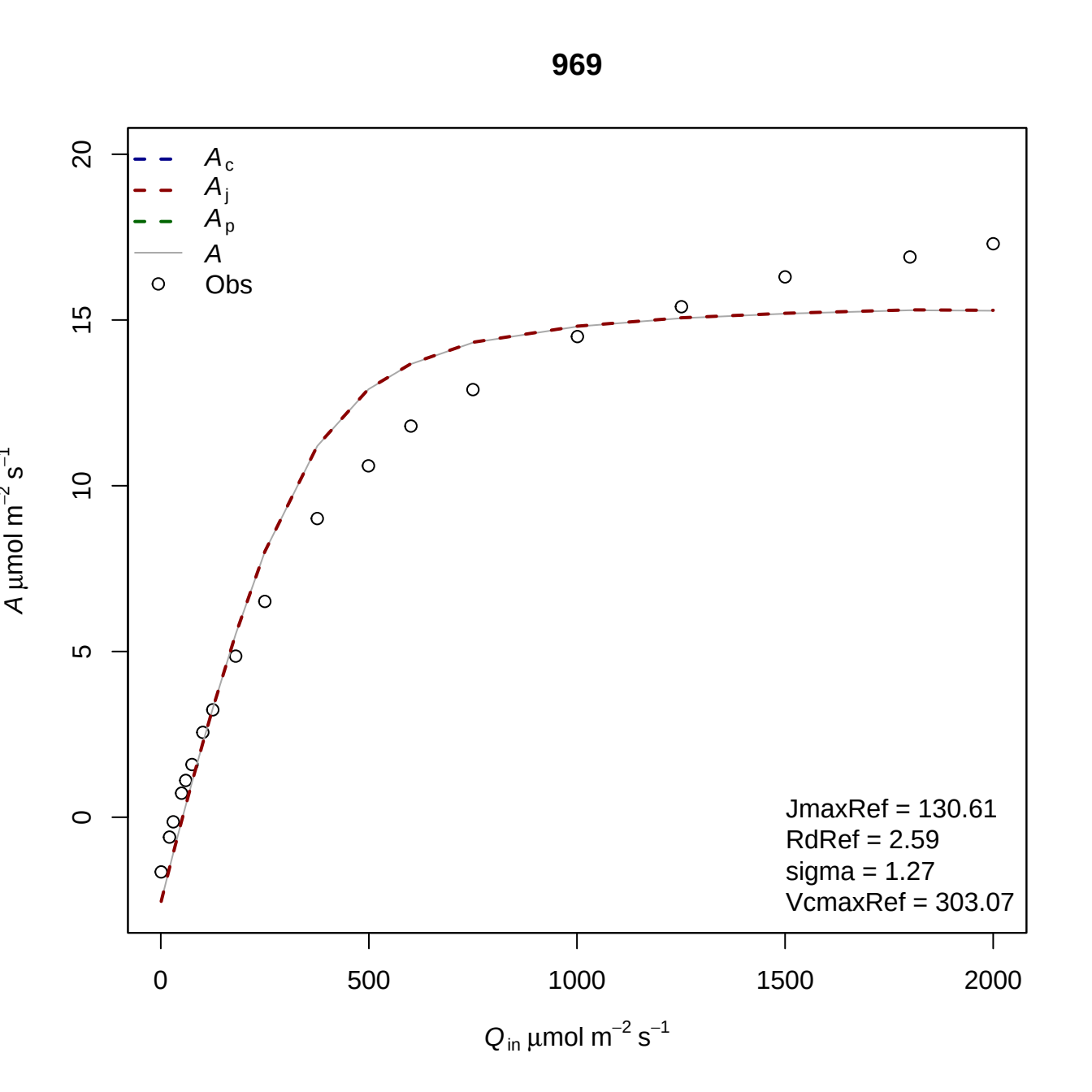

964
A_c
A_j
A_p
A
Obs
JmaxRef = 91.7
RdRef = 2.06
sigma = 0.48
VcmaxRef = 219.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

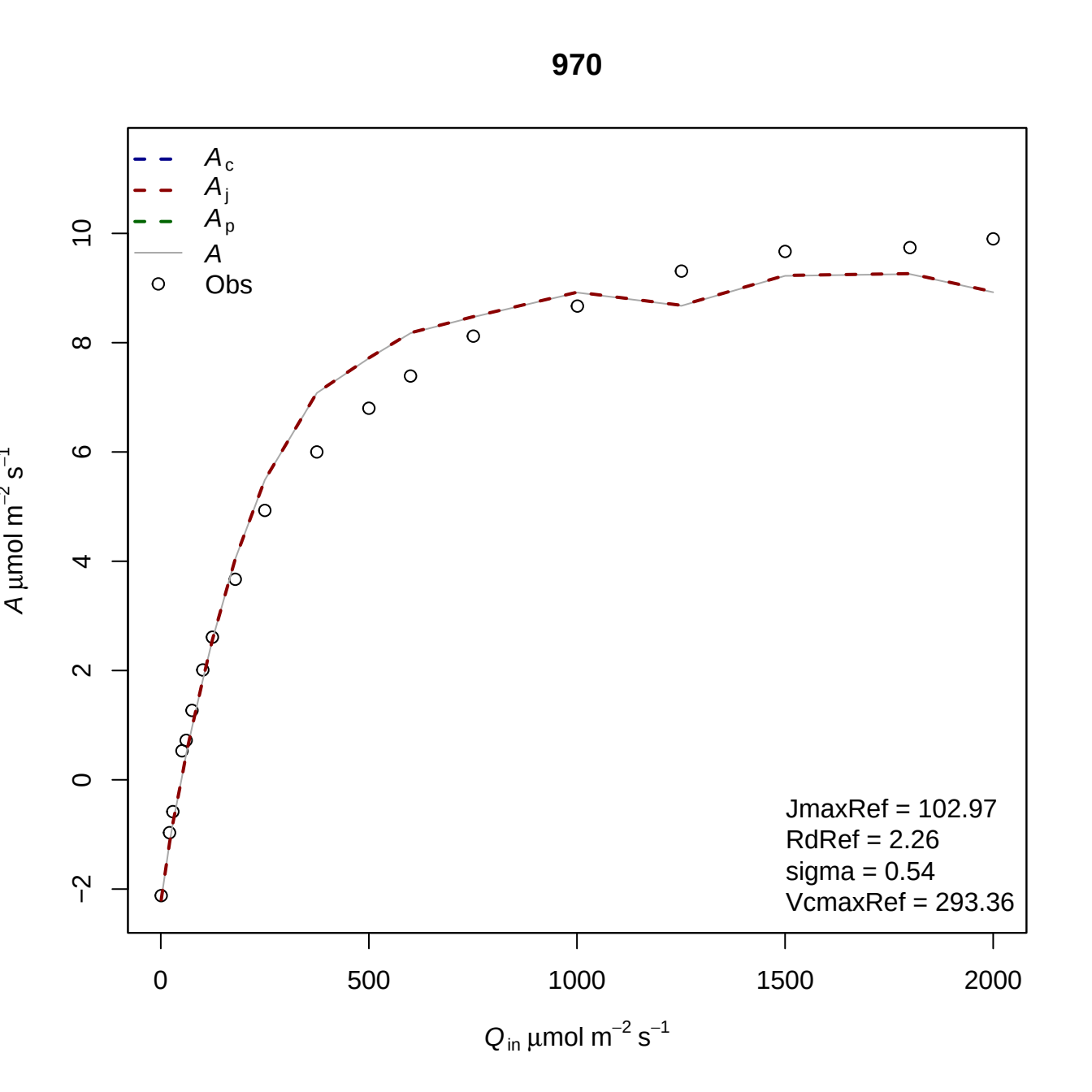

965
A_c
A_j
A_p
A
Obs
JmaxRef = 42.47
RdRef = 1.24
sigma = 0.42
VcmaxRef = 153242.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

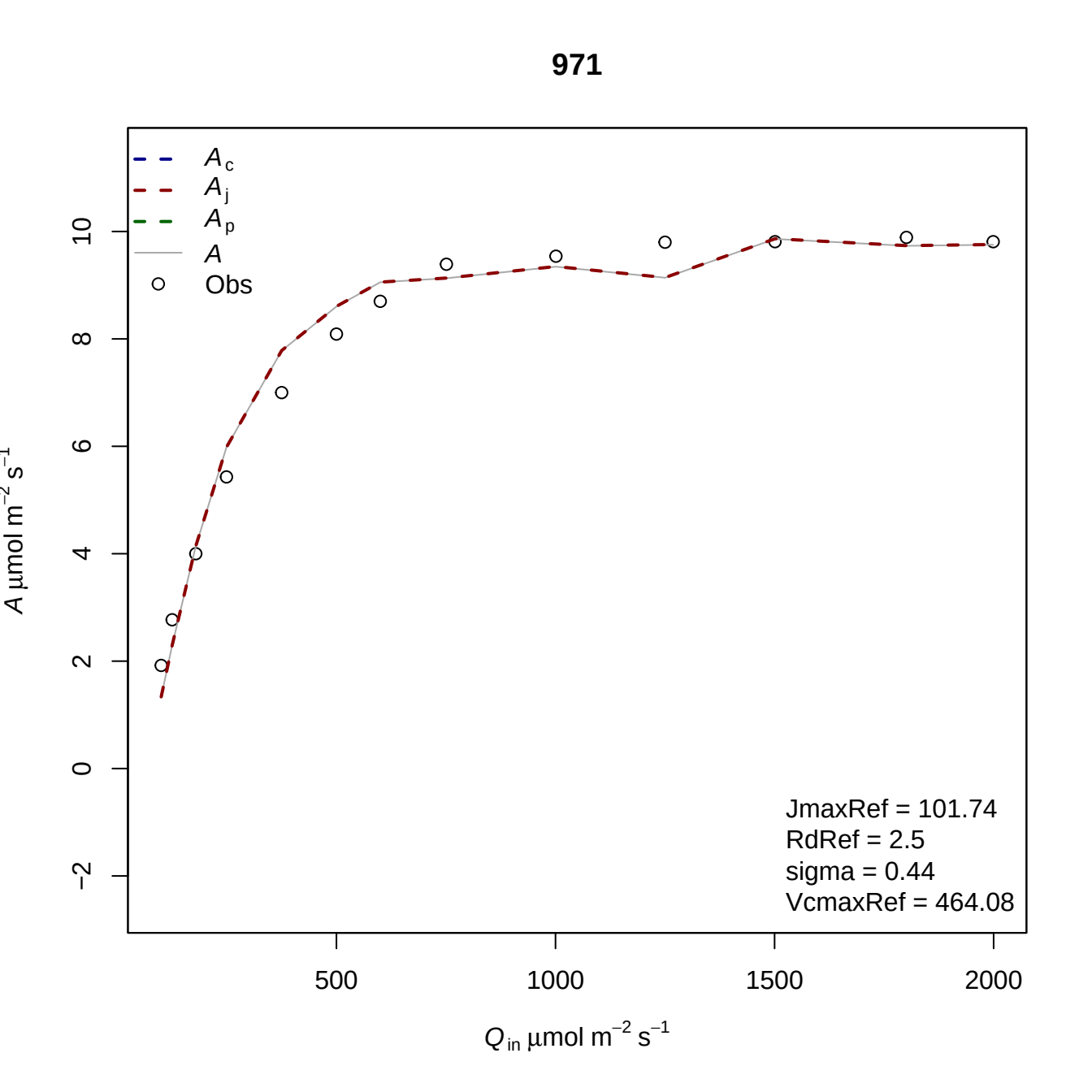

966
A_c
A_j
A_p
A
Obs
JmaxRef = 52.38
RdRef = 1.54
sigma = 0.33
VcmaxRef = 1195.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

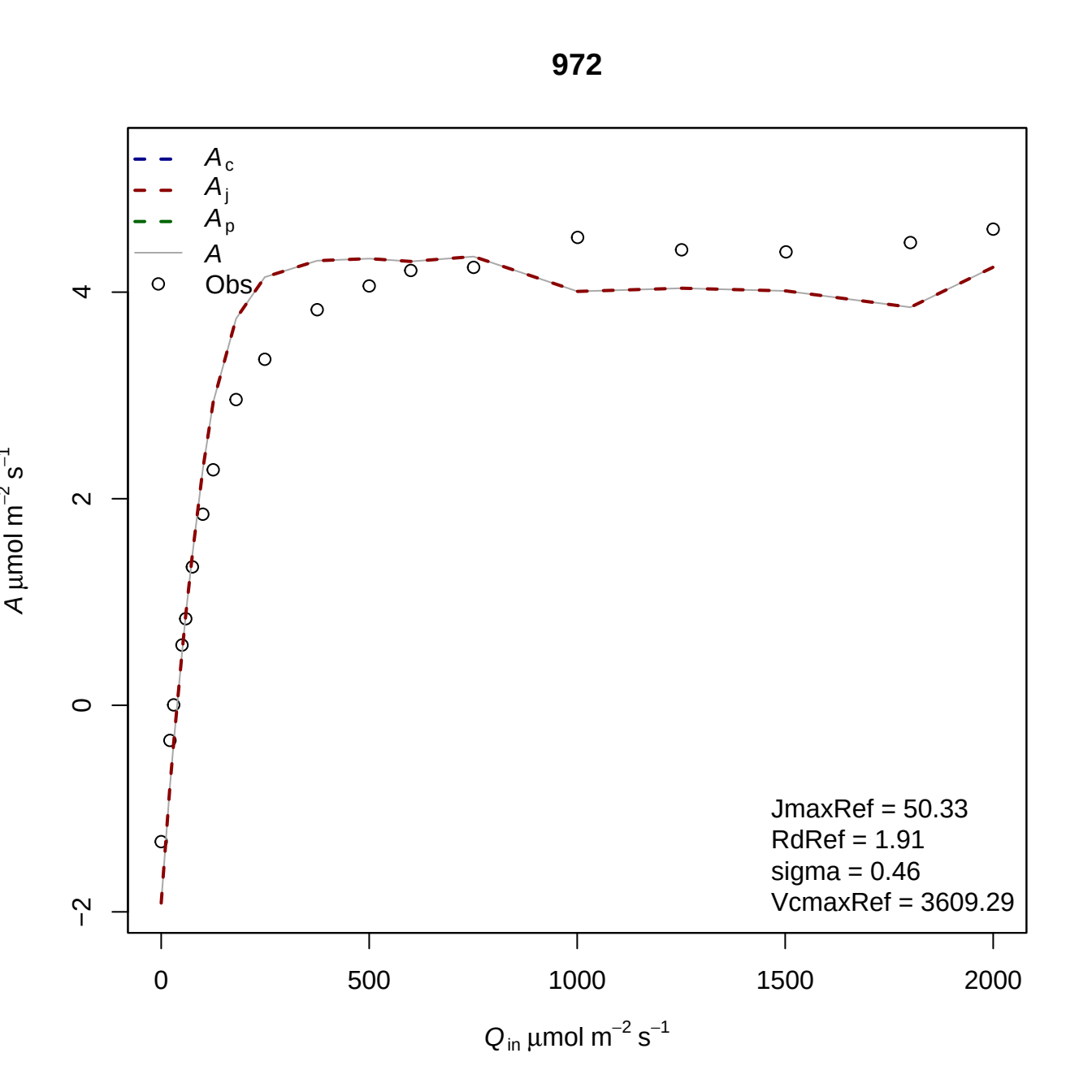

967
A_c
A_j
A_p
A
Obs
JmaxRef = 107.79
RdRef = 1.53
sigma = 0.92
VcmaxRef = 1154.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

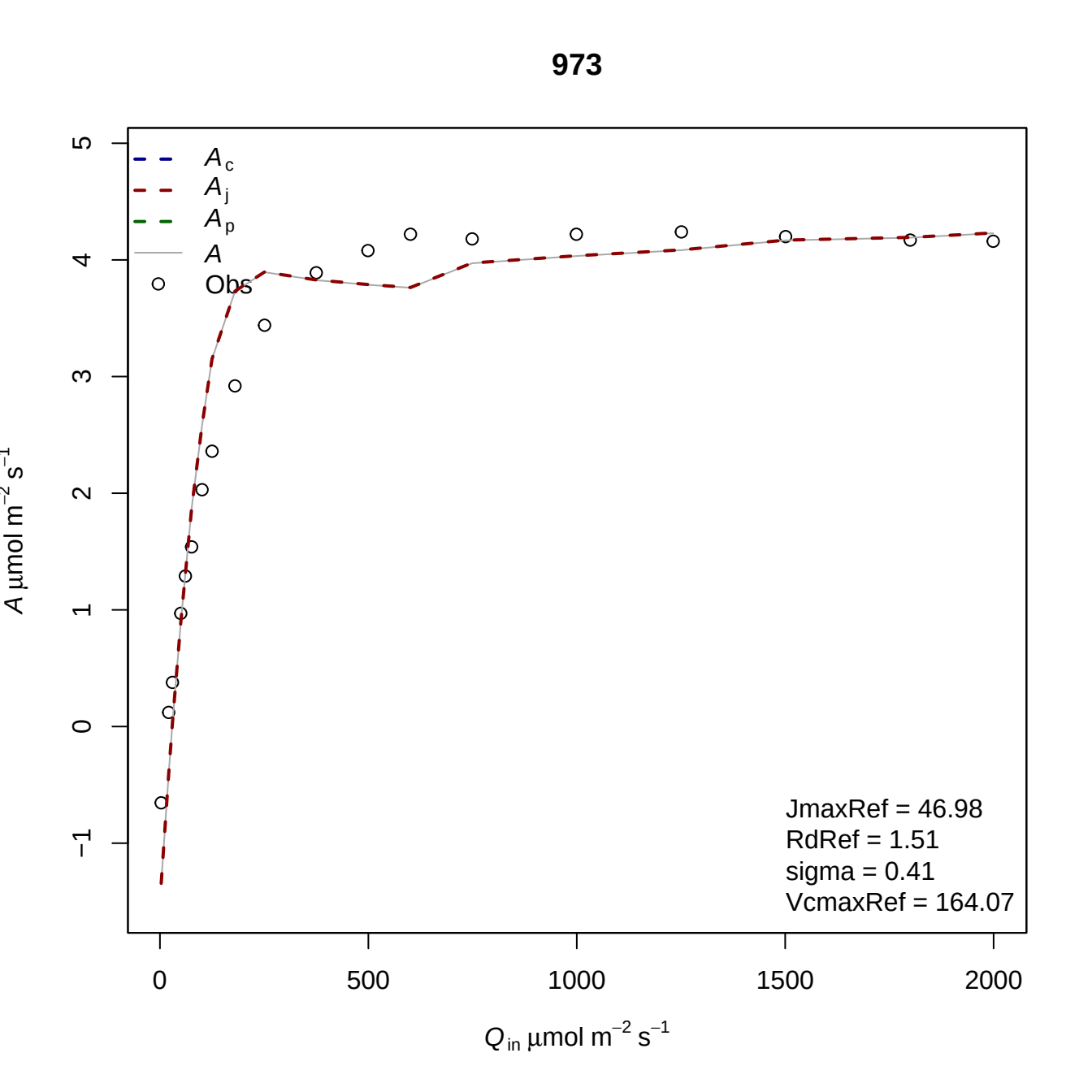

968
A_c
A_j
A_p
A
Obs
JmaxRef = 61.97
RdRef = 1.61
sigma = 0.52
VcmaxRef = 239.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

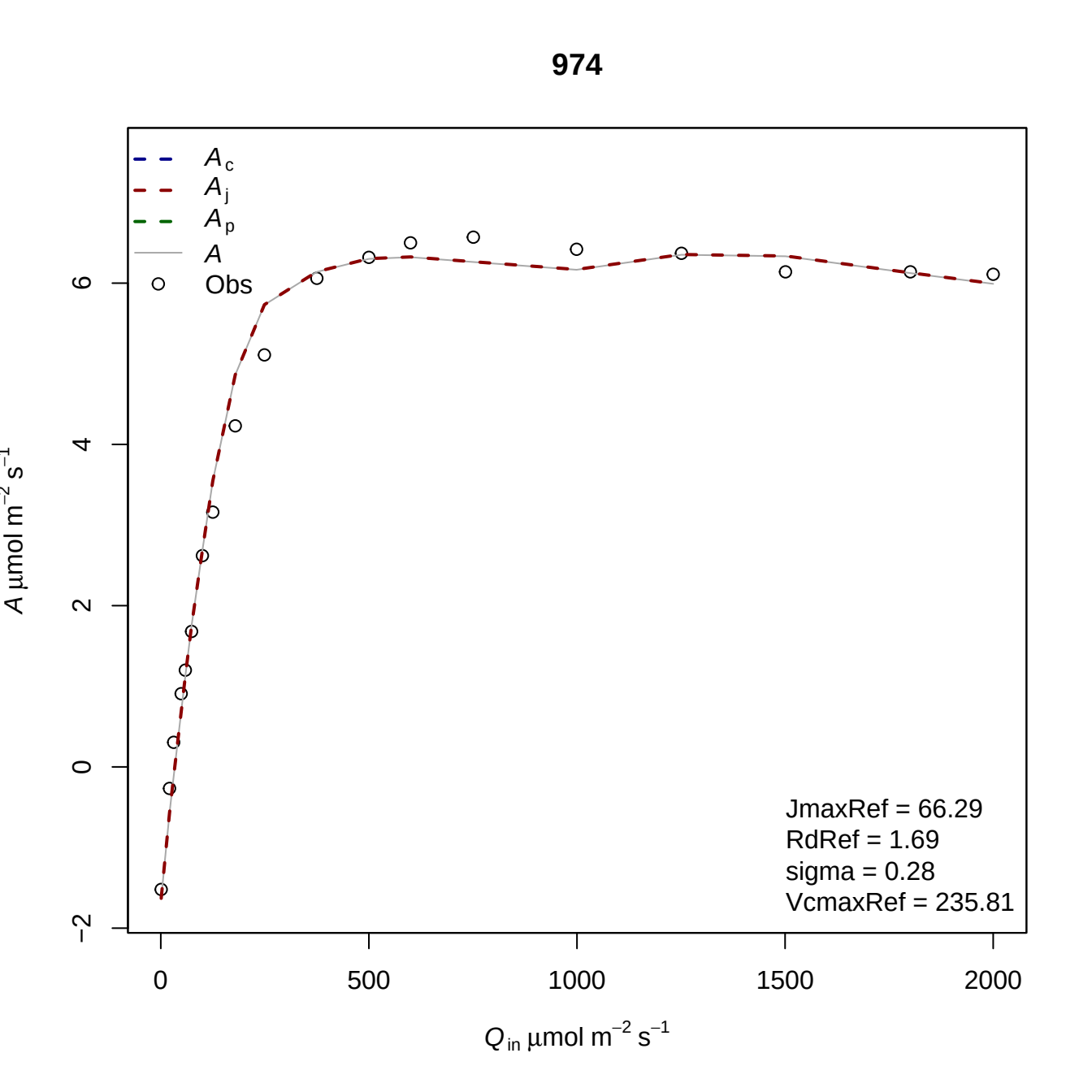

969
A_c
A_j
A_p
A
Obs
JmaxRef = 130.61
RdRef = 2.59
sigma = 1.27
VcmaxRef = 303.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

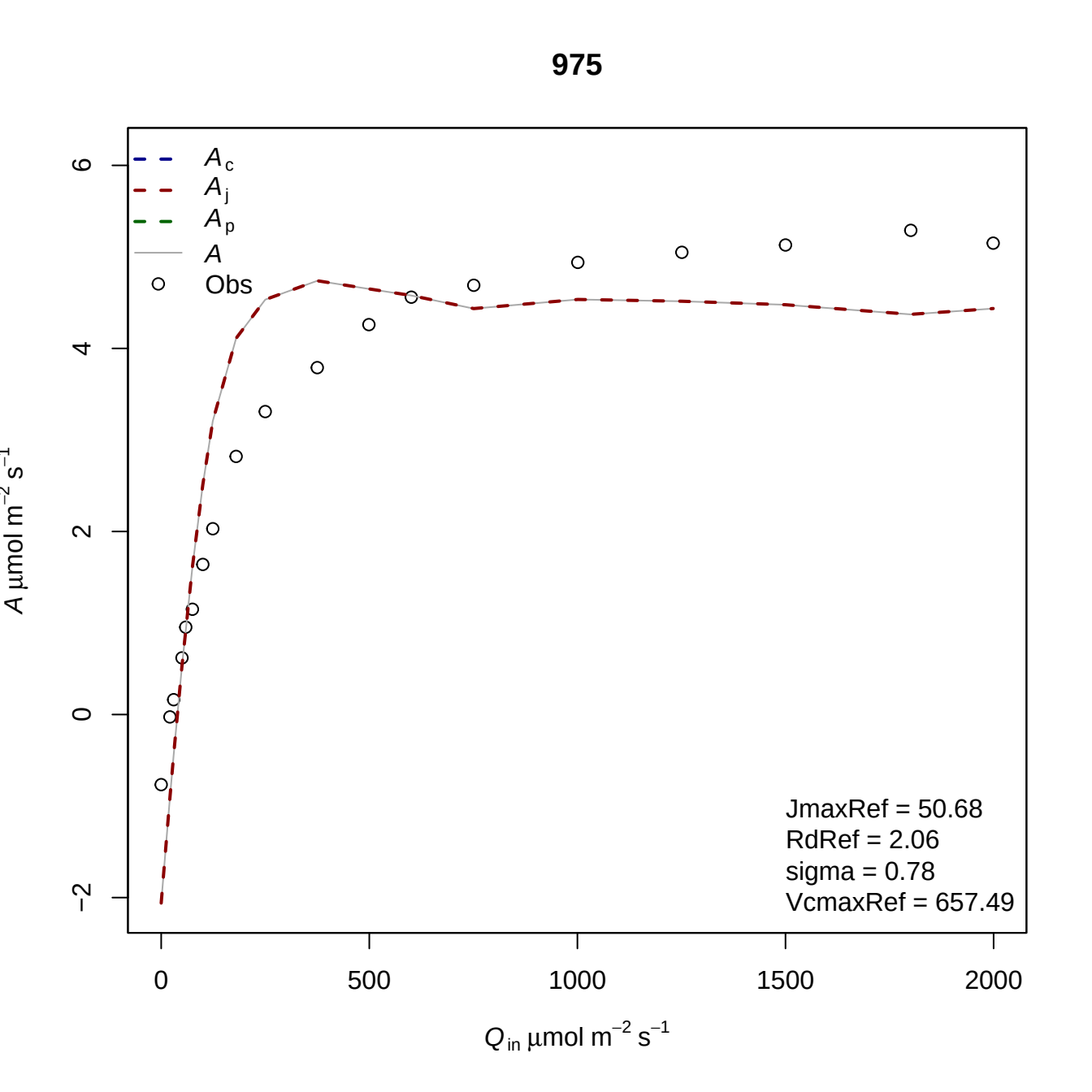

970
A_c
A_j
A_p
A
Obs
JmaxRef = 102.97
RdRef = 2.26
sigma = 0.54
VcmaxRef = 293.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

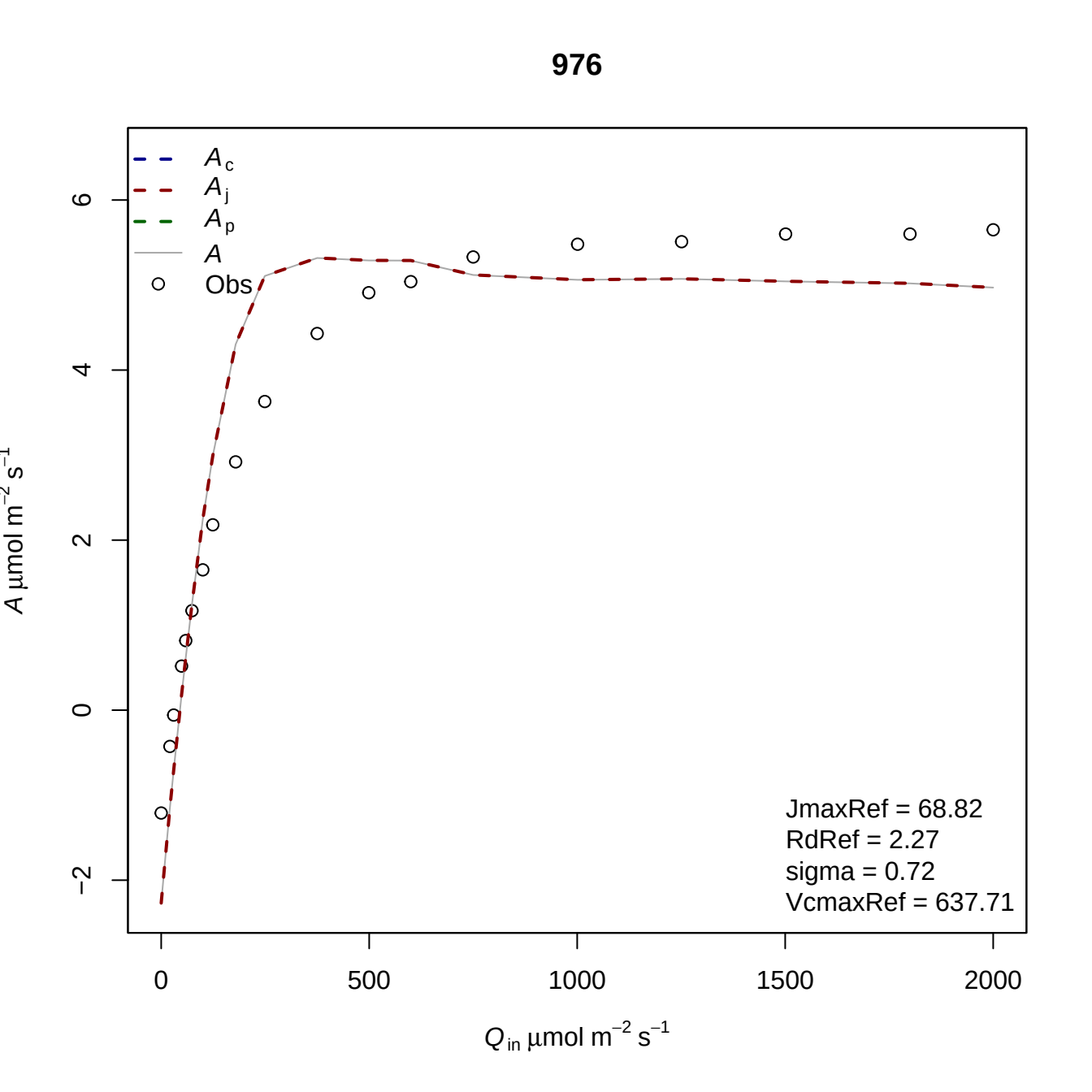

971
A_c
A_j
A_p
A
Obs
JmaxRef = 101.74
RdRef = 2.5
sigma = 0.44
VcmaxRef = 464.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

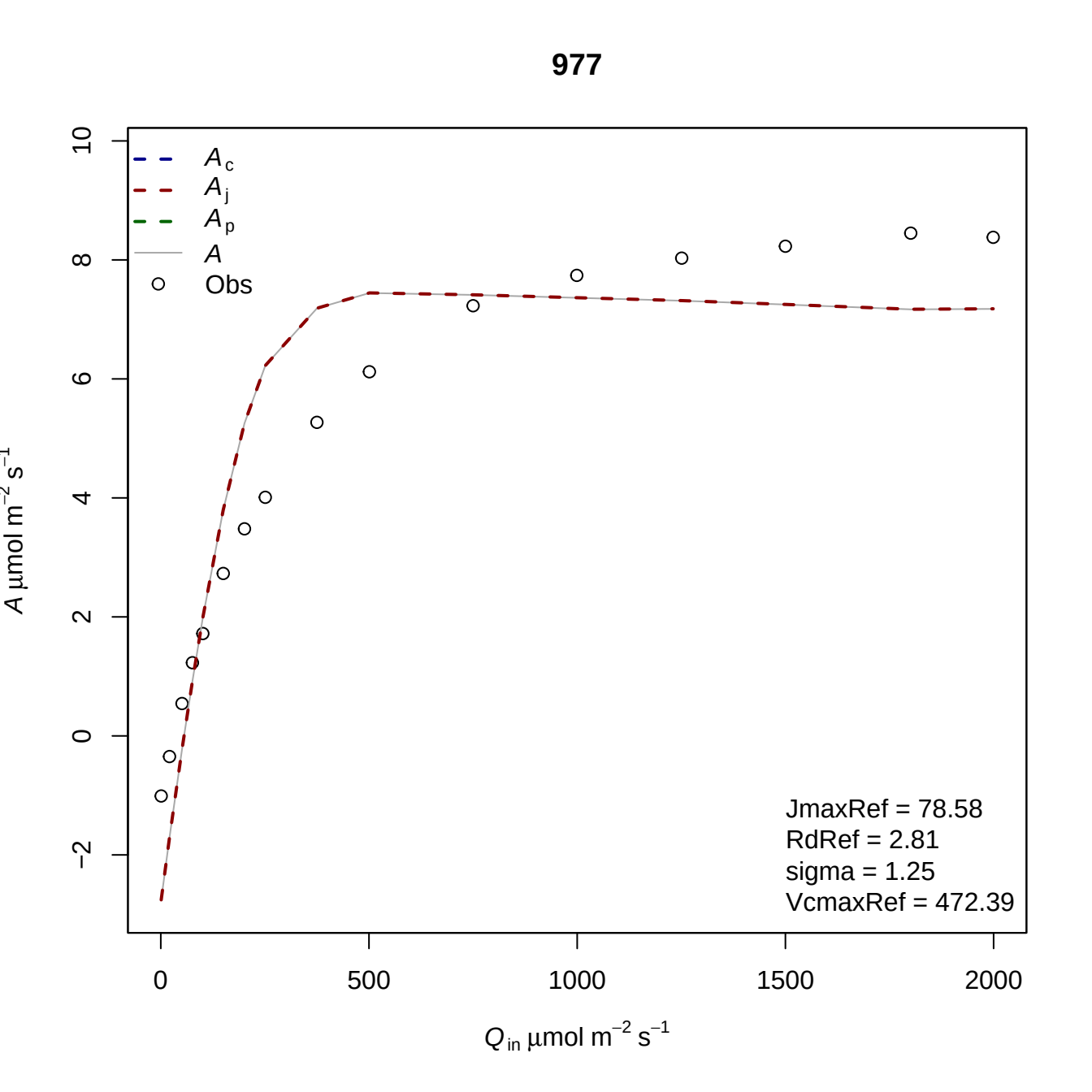

972
A_c
A_j
A_p
A
Obs
JmaxRef = 50.33
RdRef = 1.91
sigma = 0.46
VcmaxRef = 3609.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

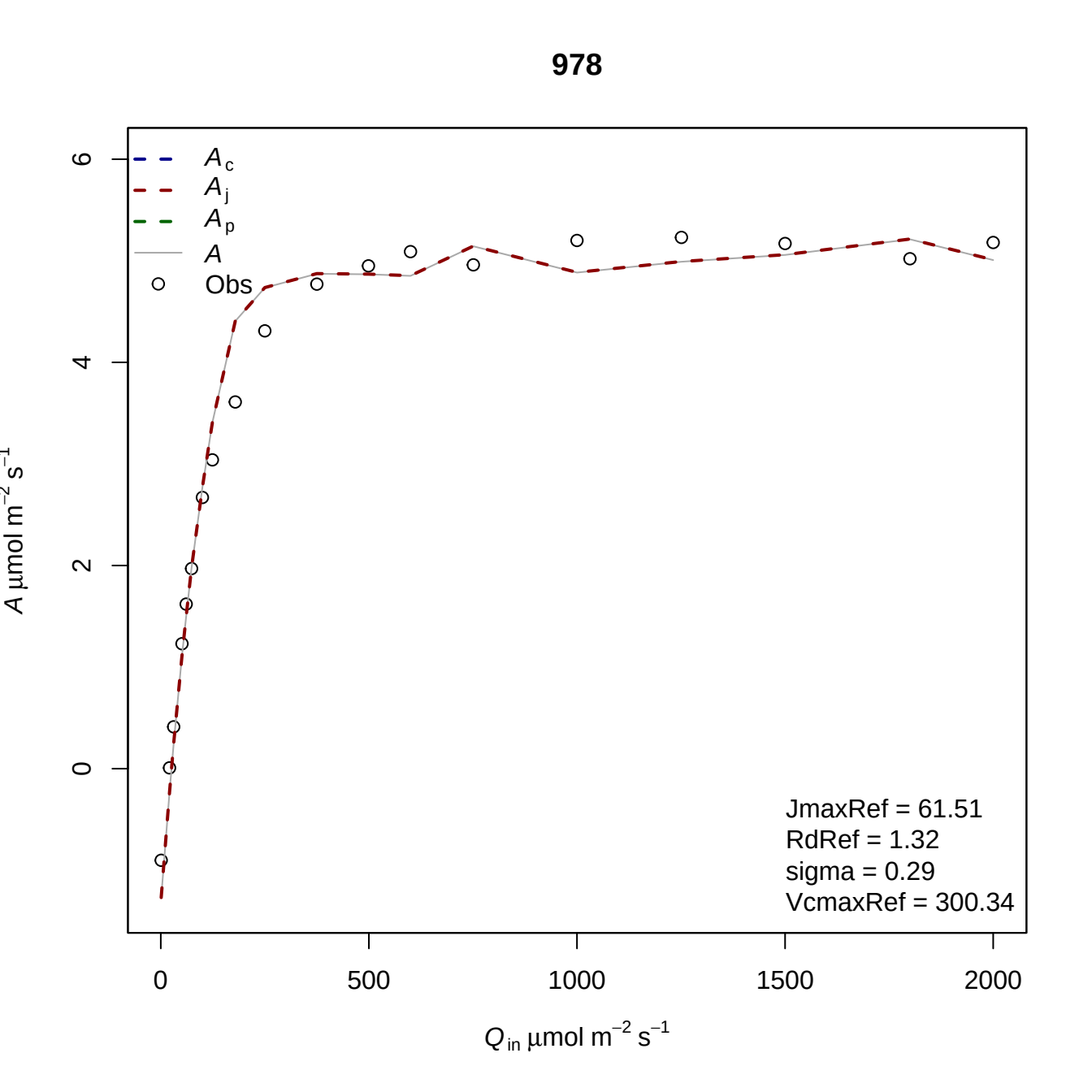

973
A_c
A_j
A_p
A
Obs
JmaxRef = 46.98
RdRef = 1.51
sigma = 0.41
VcmaxRef = 164.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

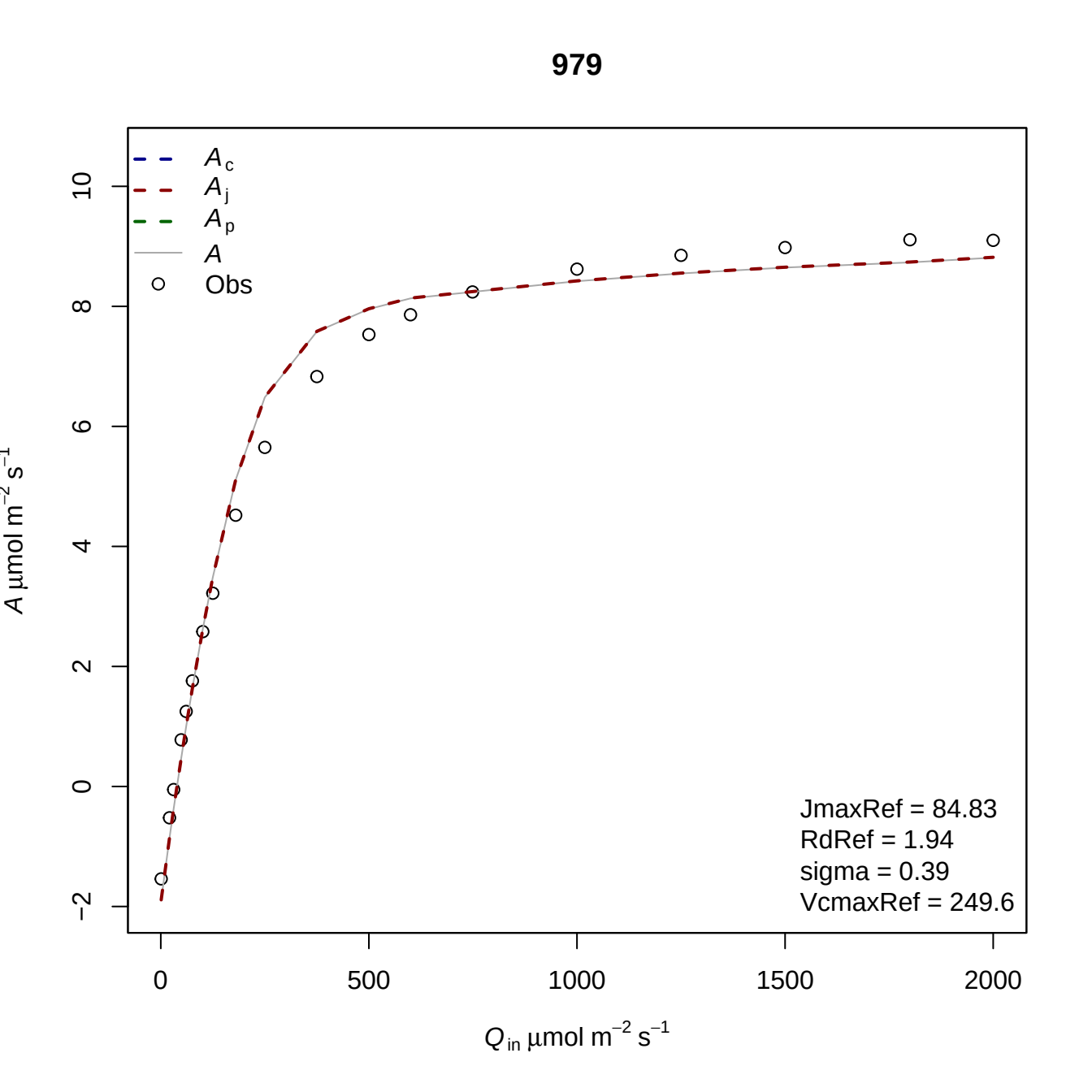

974
A_c
A_j
A_p
A
Obs
JmaxRef = 66.29
RdRef = 1.69
sigma = 0.28
VcmaxRef = 235.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

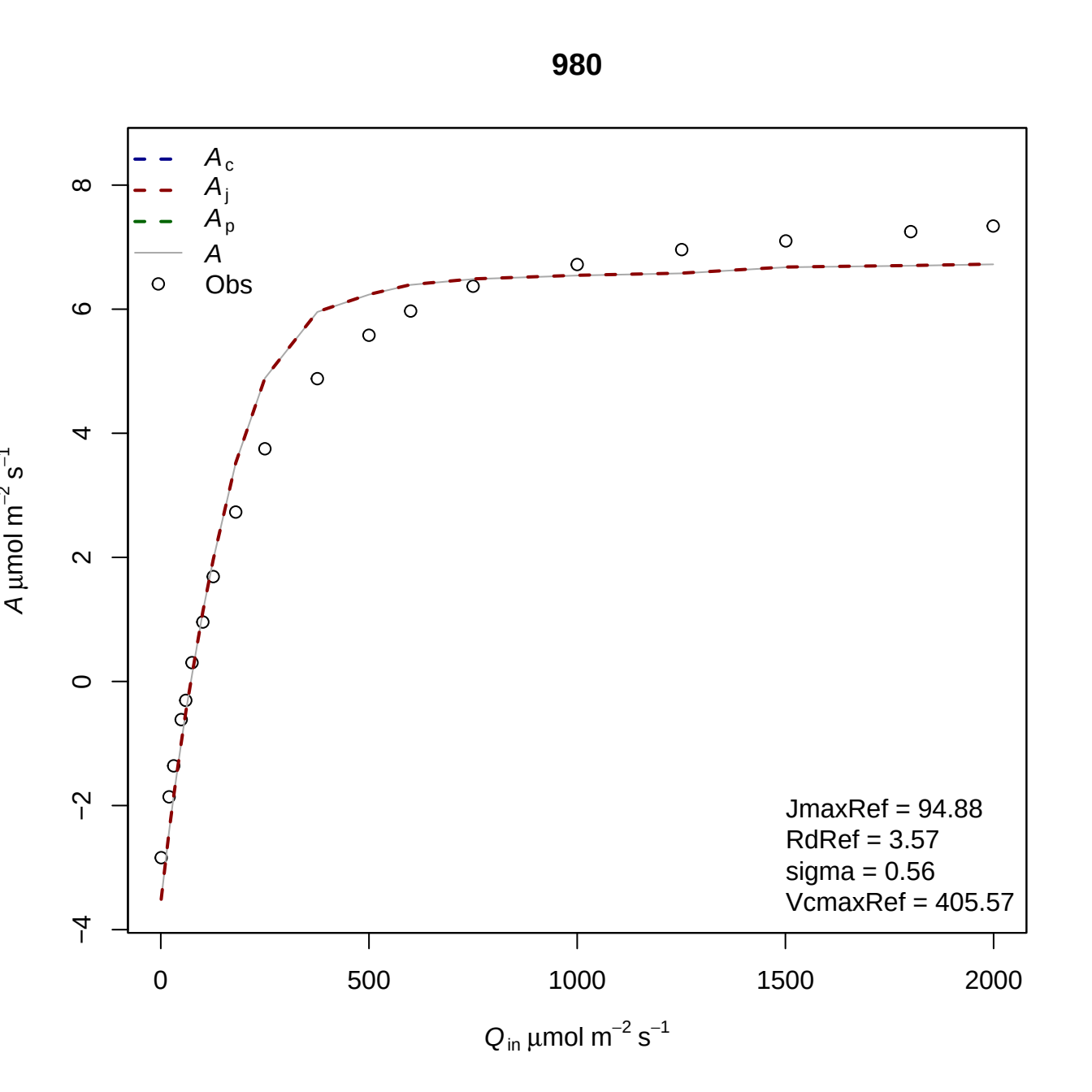

975
A_c
A_j
A_p
A
Obs
JmaxRef = 50.68
RdRef = 2.06
sigma = 0.78
VcmaxRef = 657.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

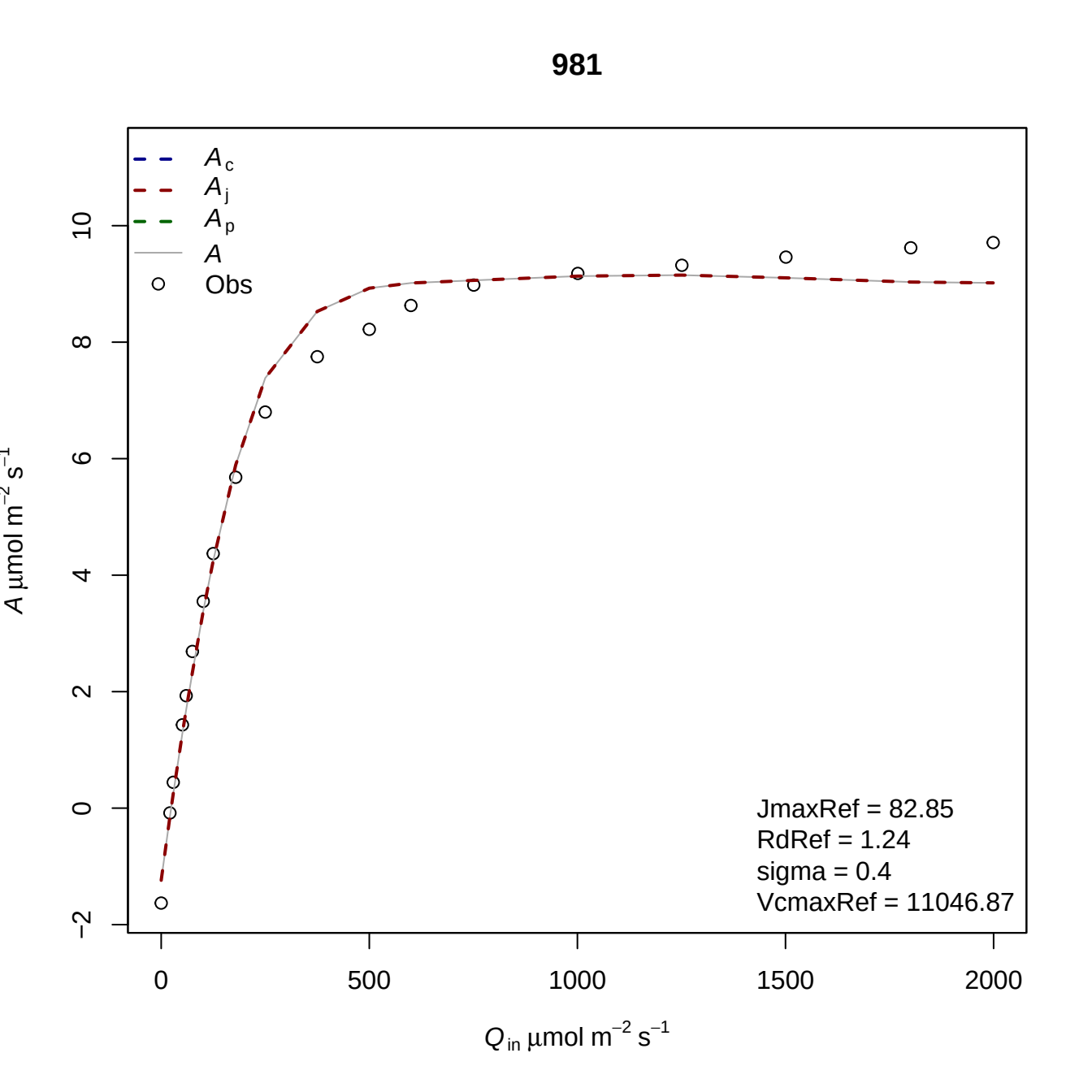

976
A_c
A_j
A_p
A
Obs
JmaxRef = 68.82
RdRef = 2.27
sigma = 0.72
VcmaxRef = 637.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

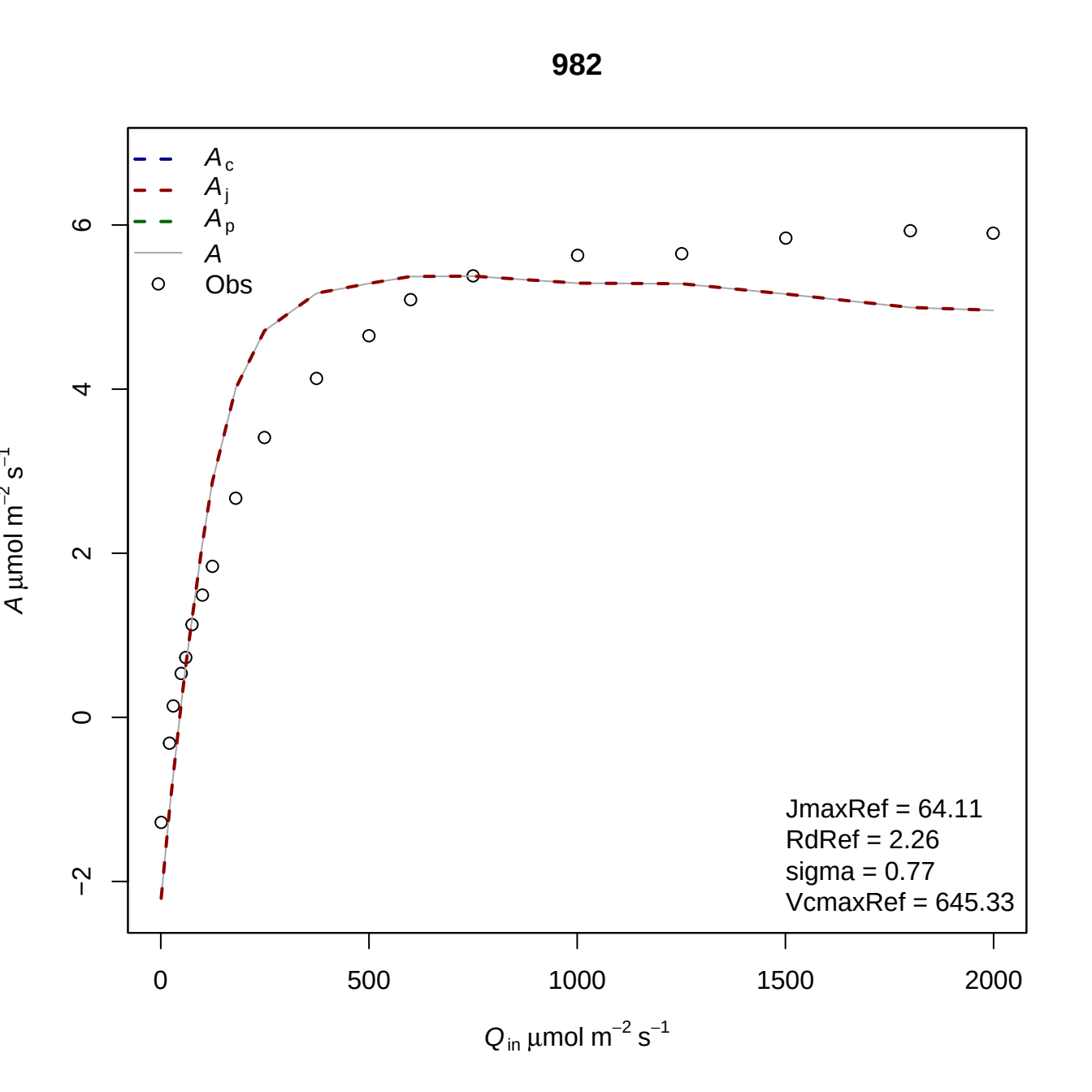

977
A_c
A_j
A_p
A
Obs
JmaxRef = 78.58
RdRef = 2.81
sigma = 1.25
VcmaxRef = 472.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

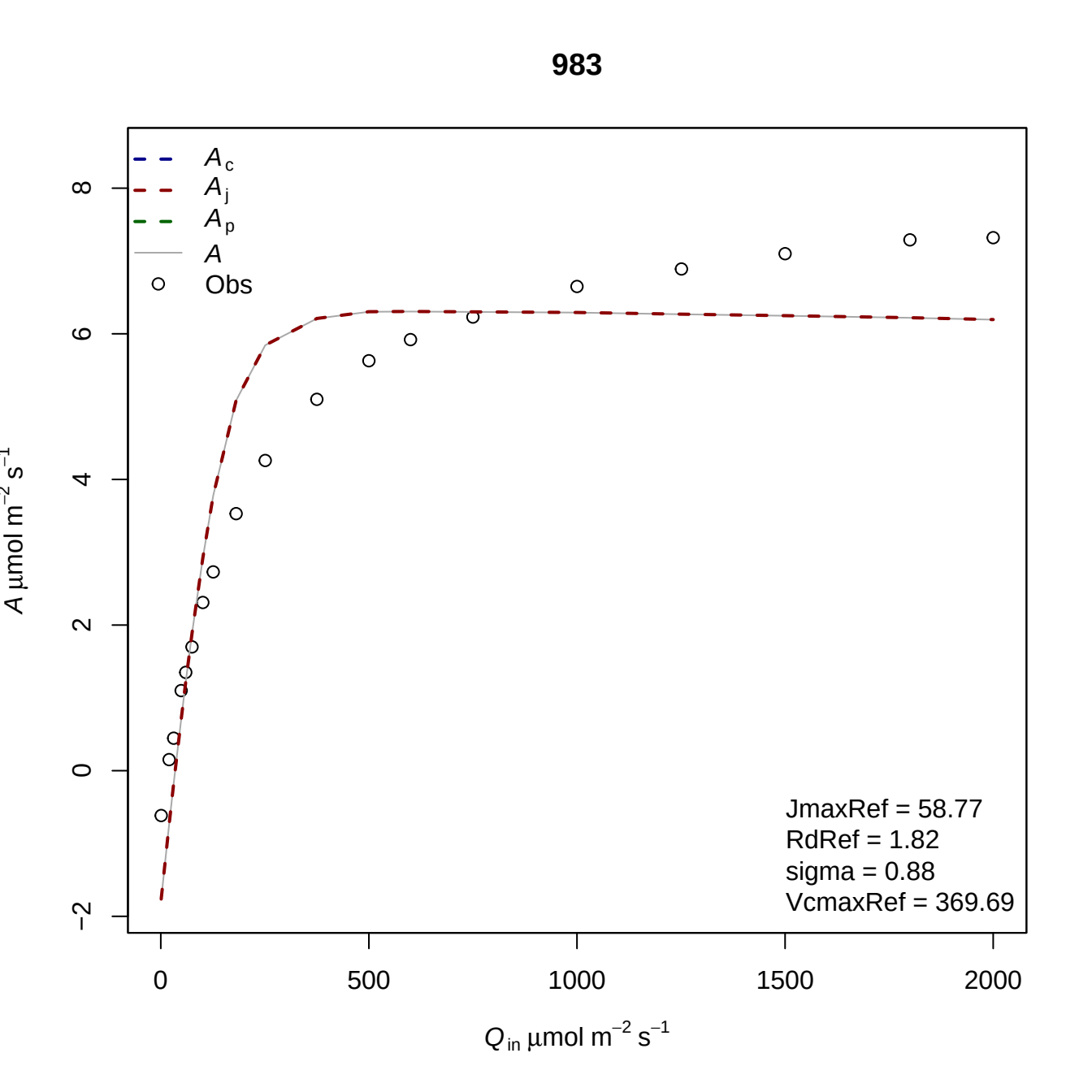

978
A_c
A_j
A_p
A
Obs
JmaxRef = 61.51
RdRef = 1.32
sigma = 0.29
VcmaxRef = 300.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

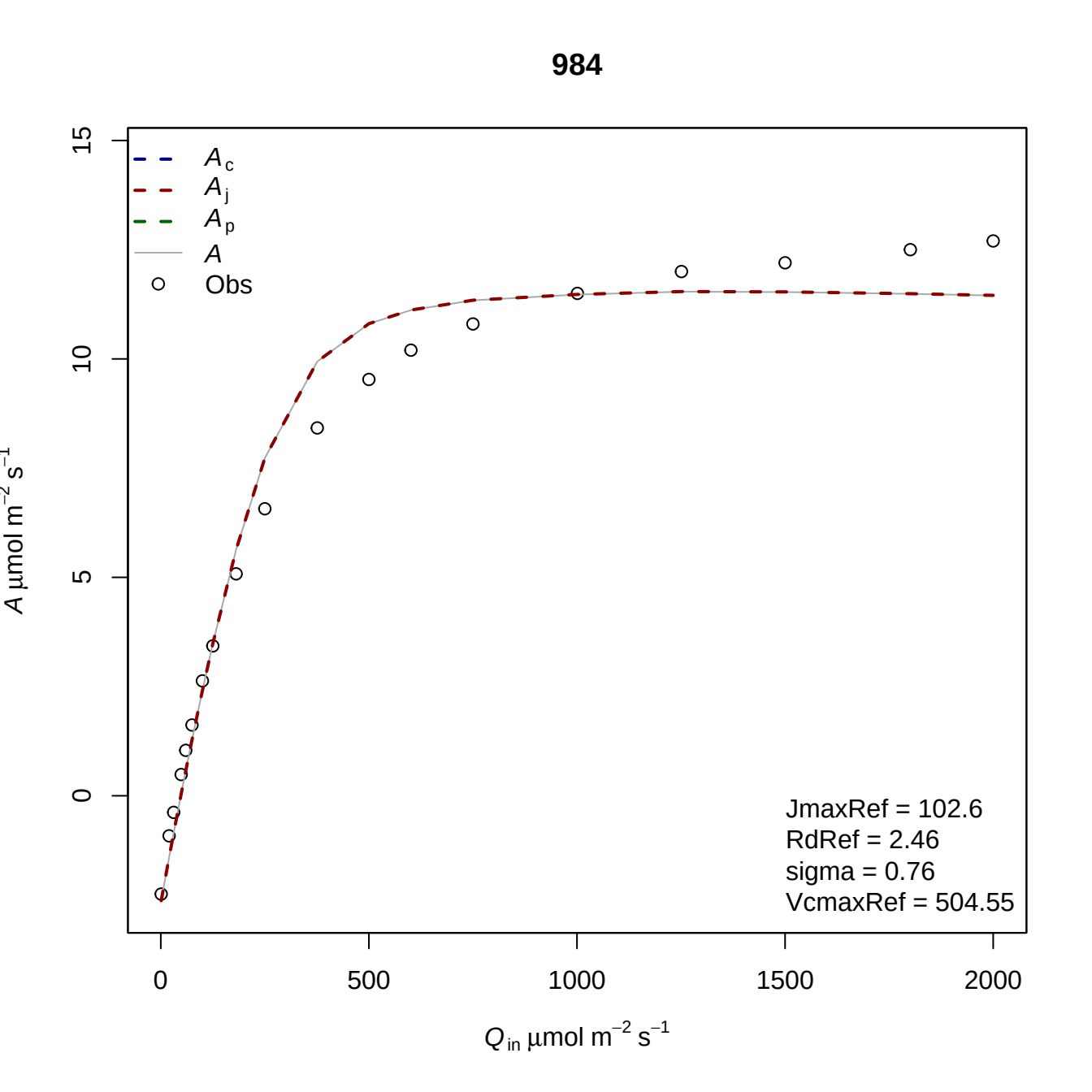

979
A_c
A_j
A_p
A
Obs
JmaxRef = 84.83
RdRef = 1.94
sigma = 0.39
VcmaxRef = 249.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

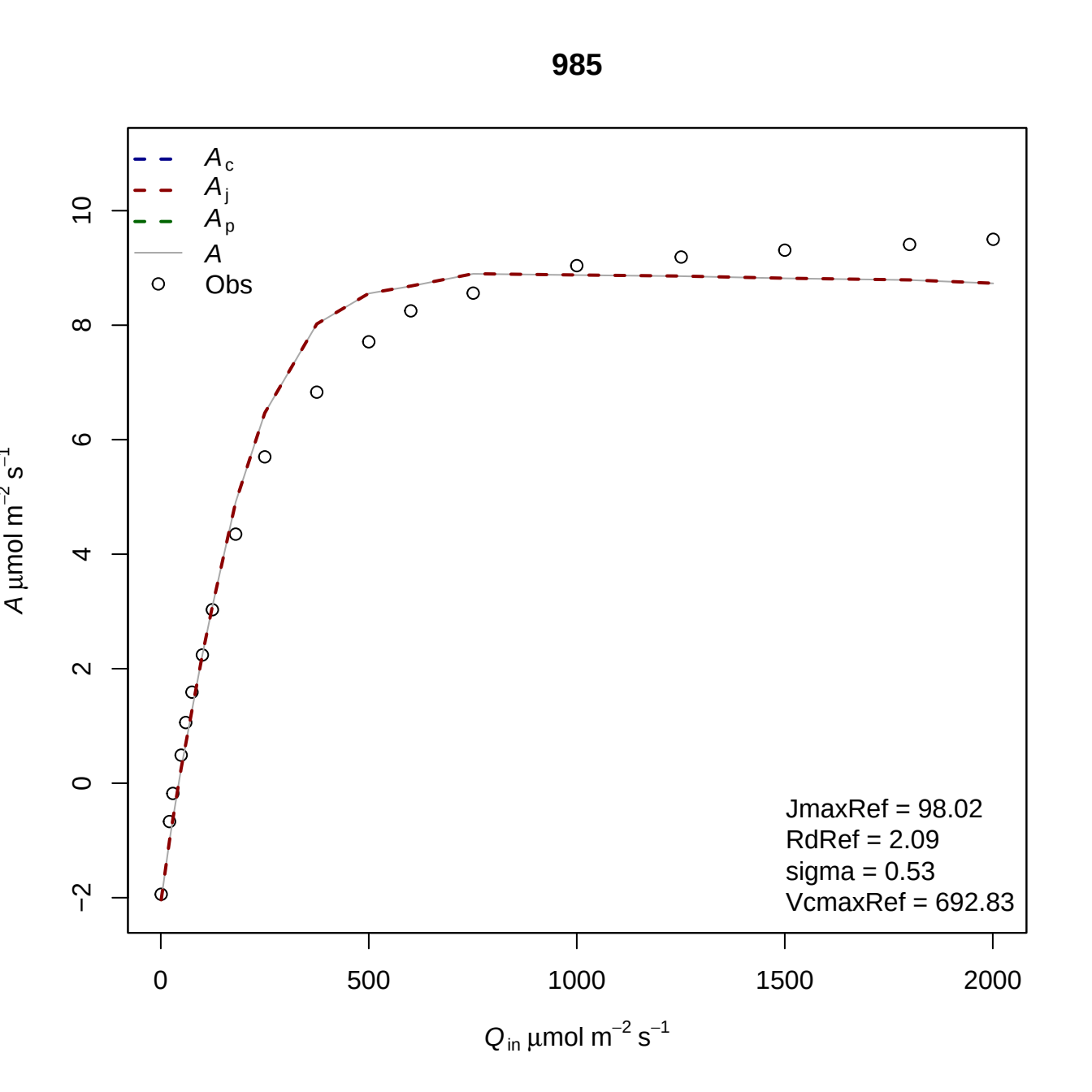

980
A_c
A_j
A_p
A
Obs
JmaxRef = 94.88
RdRef = 3.57
sigma = 0.56
VcmaxRef = 405.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

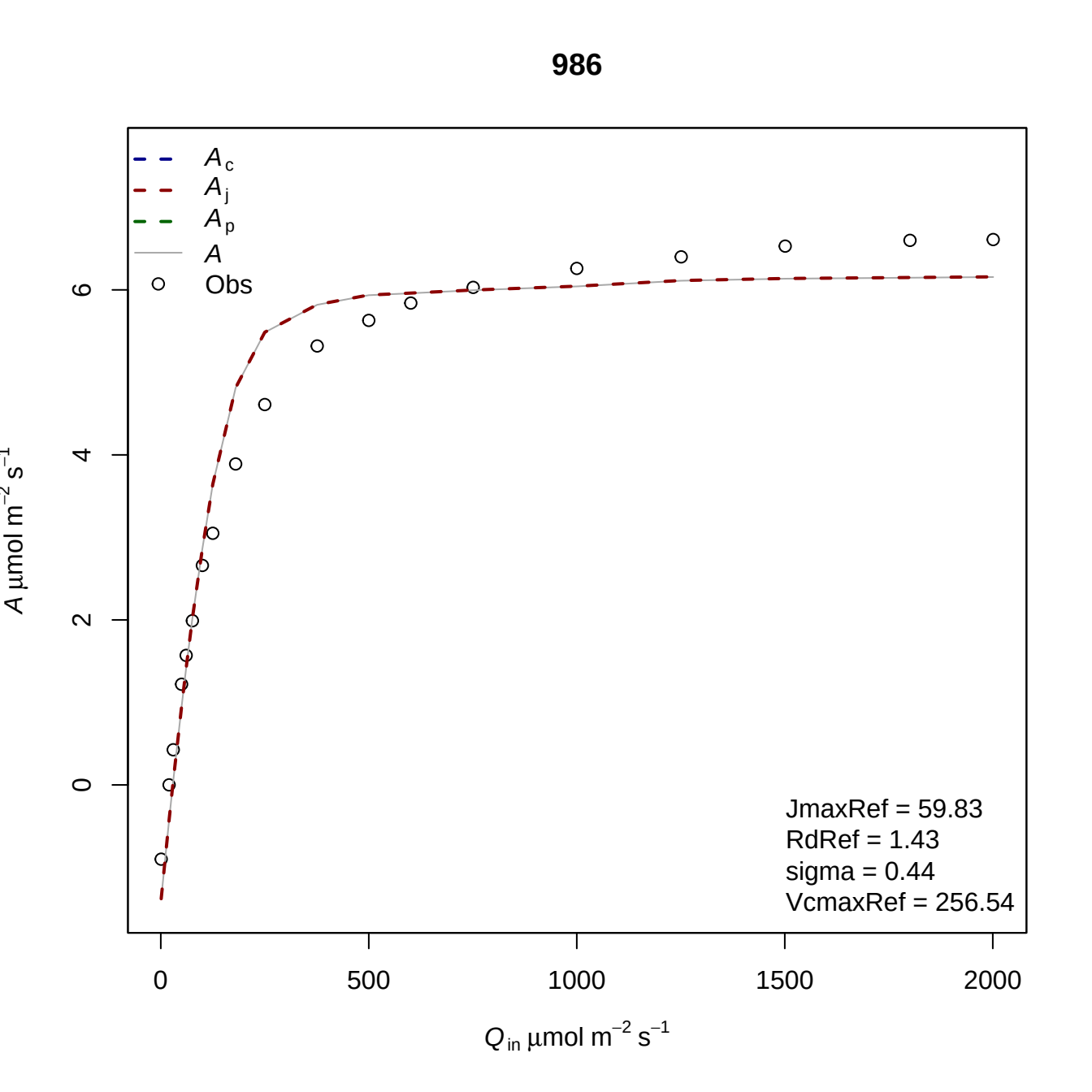

981
A_c
A_j
A_p
A
Obs
JmaxRef = 82.85
RdRef = 1.24
sigma = 0.4
VcmaxRef = 11046.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

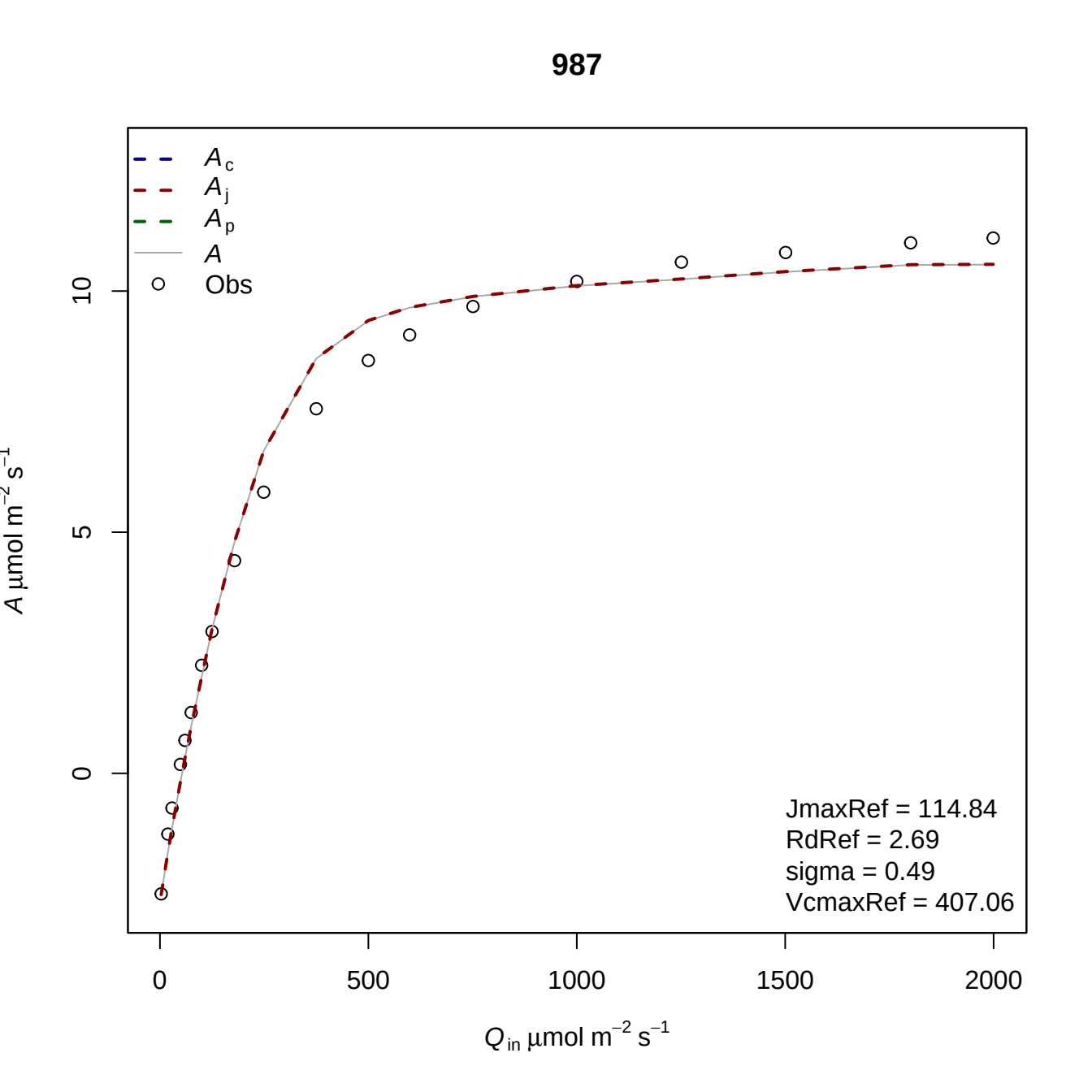

982
A_c
A_j
A_p
A
Obs
JmaxRef = 64.11
RdRef = 2.26
sigma = 0.77
VcmaxRef = 645.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

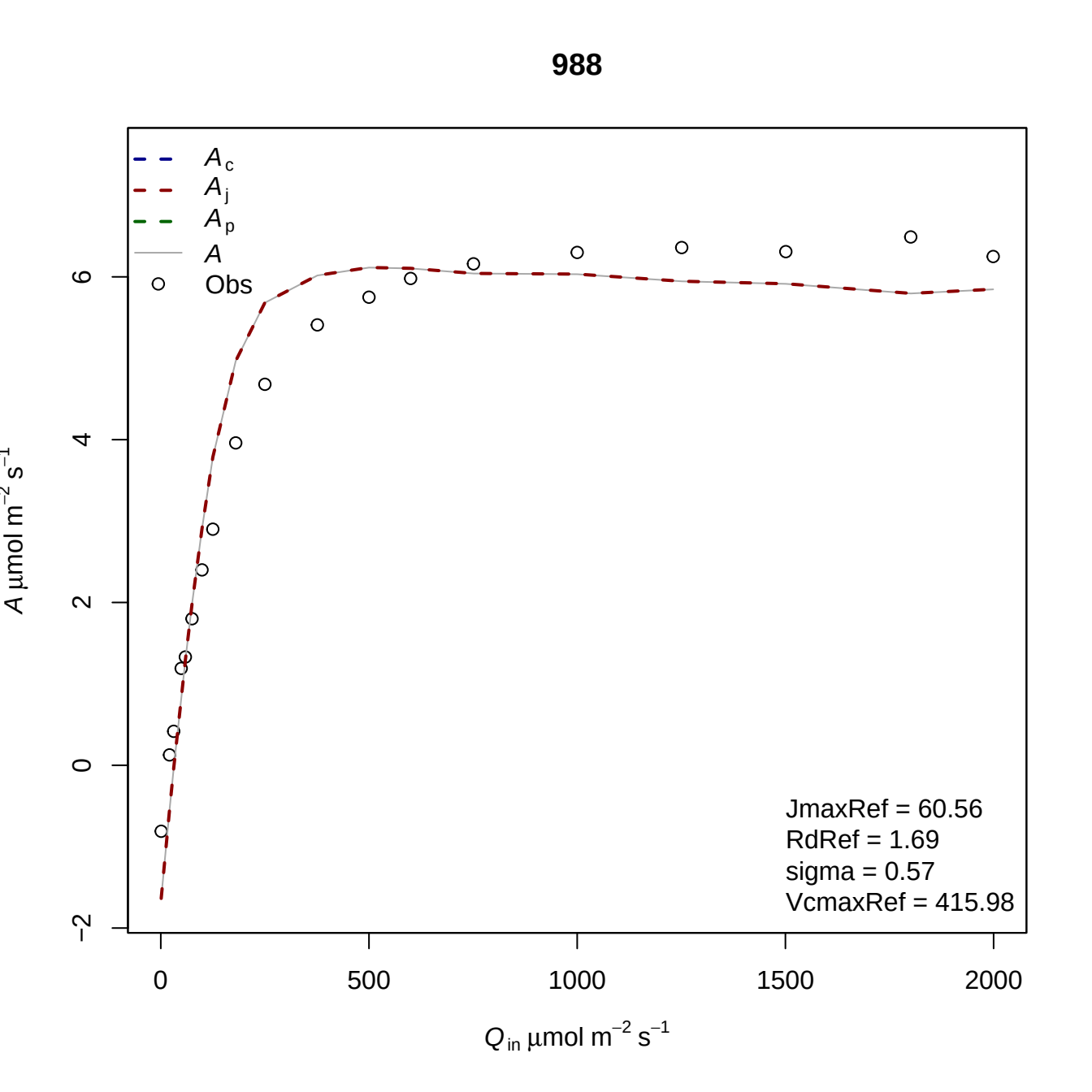

983
A_c
A_j
A_p
A
Obs
JmaxRef = 58.77
RdRef = 1.82
sigma = 0.88
VcmaxRef = 369.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

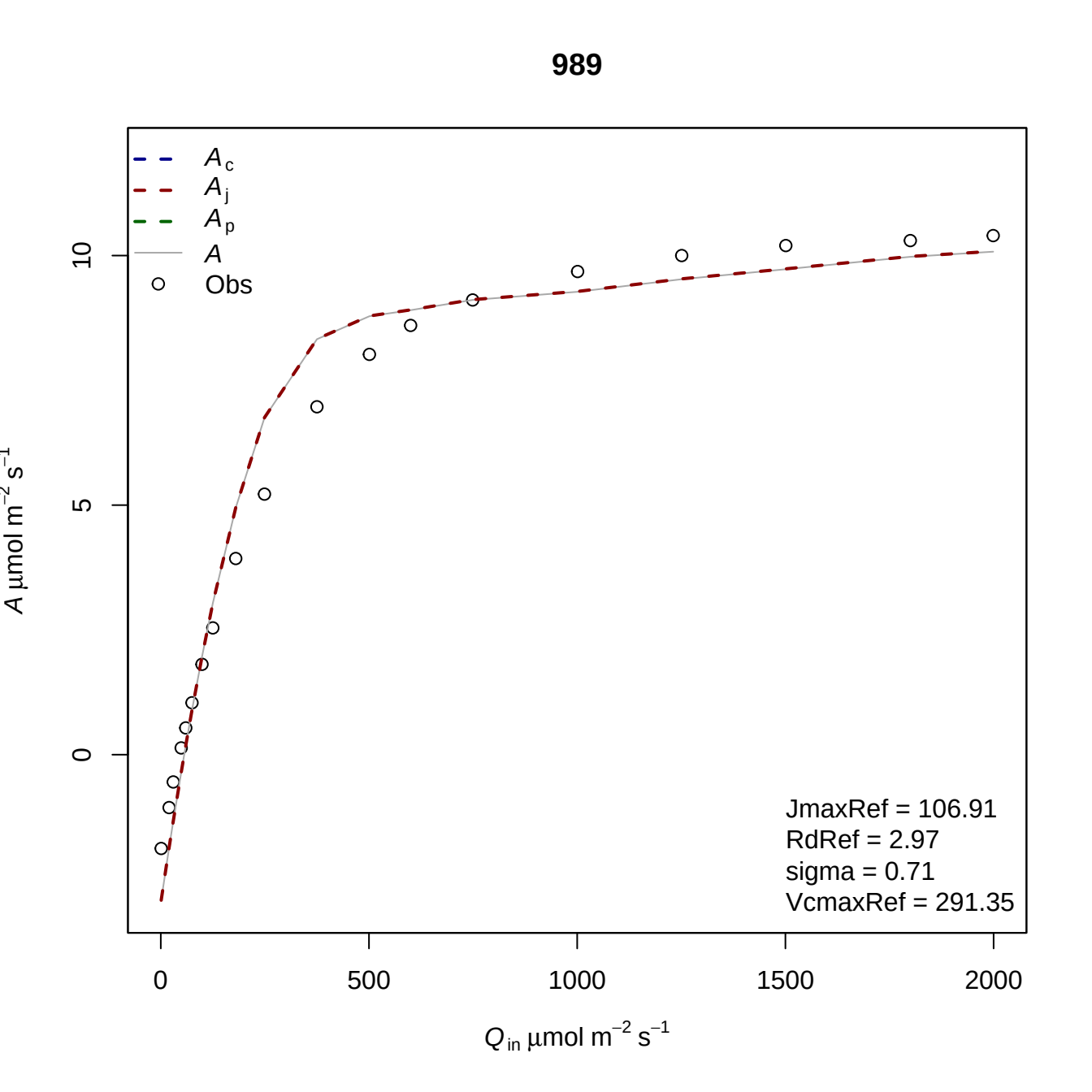

984
A_c
A_j
A_p
A
Obs
JmaxRef = 102.6
RdRef = 2.46
sigma = 0.76
VcmaxRef = 504.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

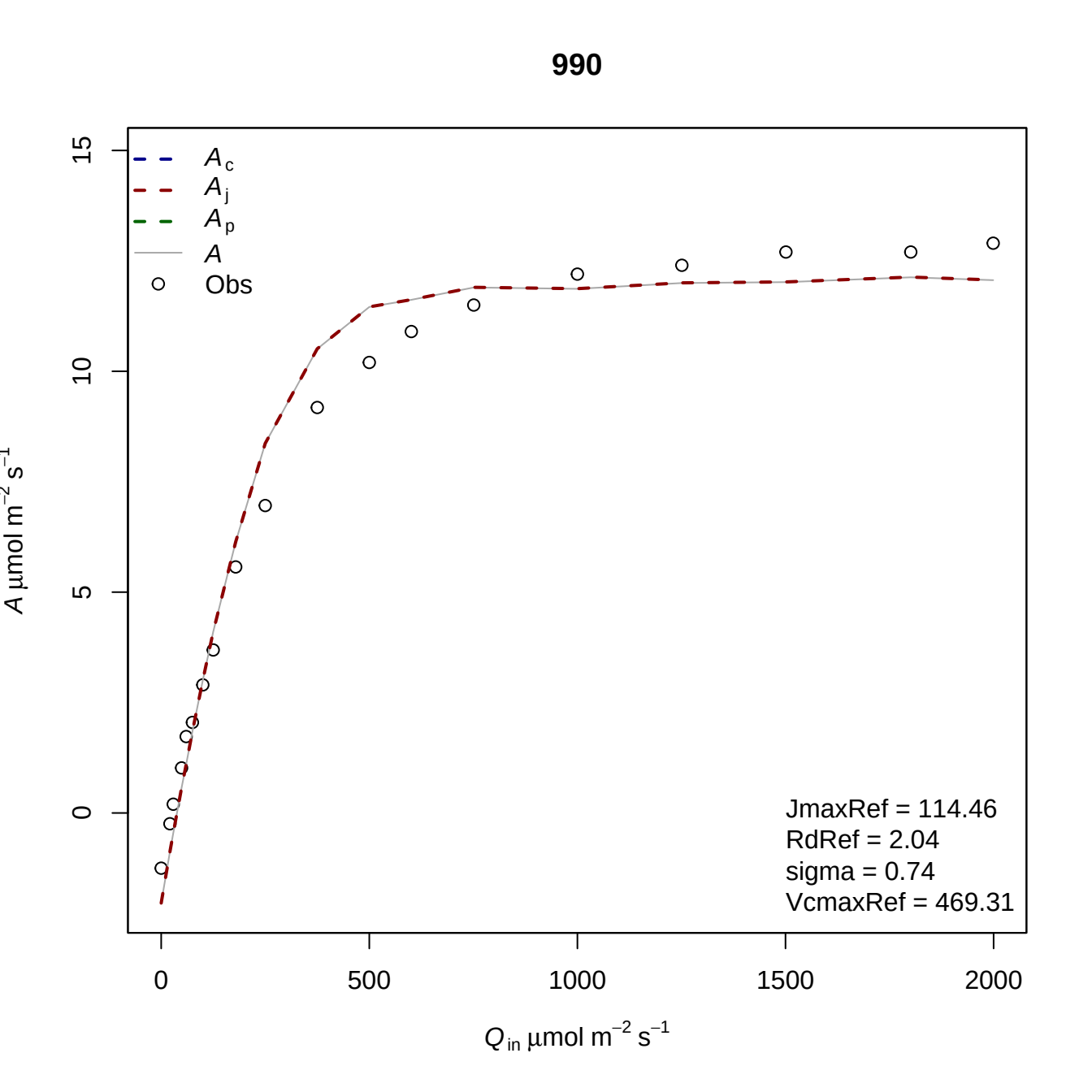

985
A_c
A_j
A_p
A
Obs
JmaxRef = 98.02
RdRef = 2.09
sigma = 0.53
VcmaxRef = 692.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

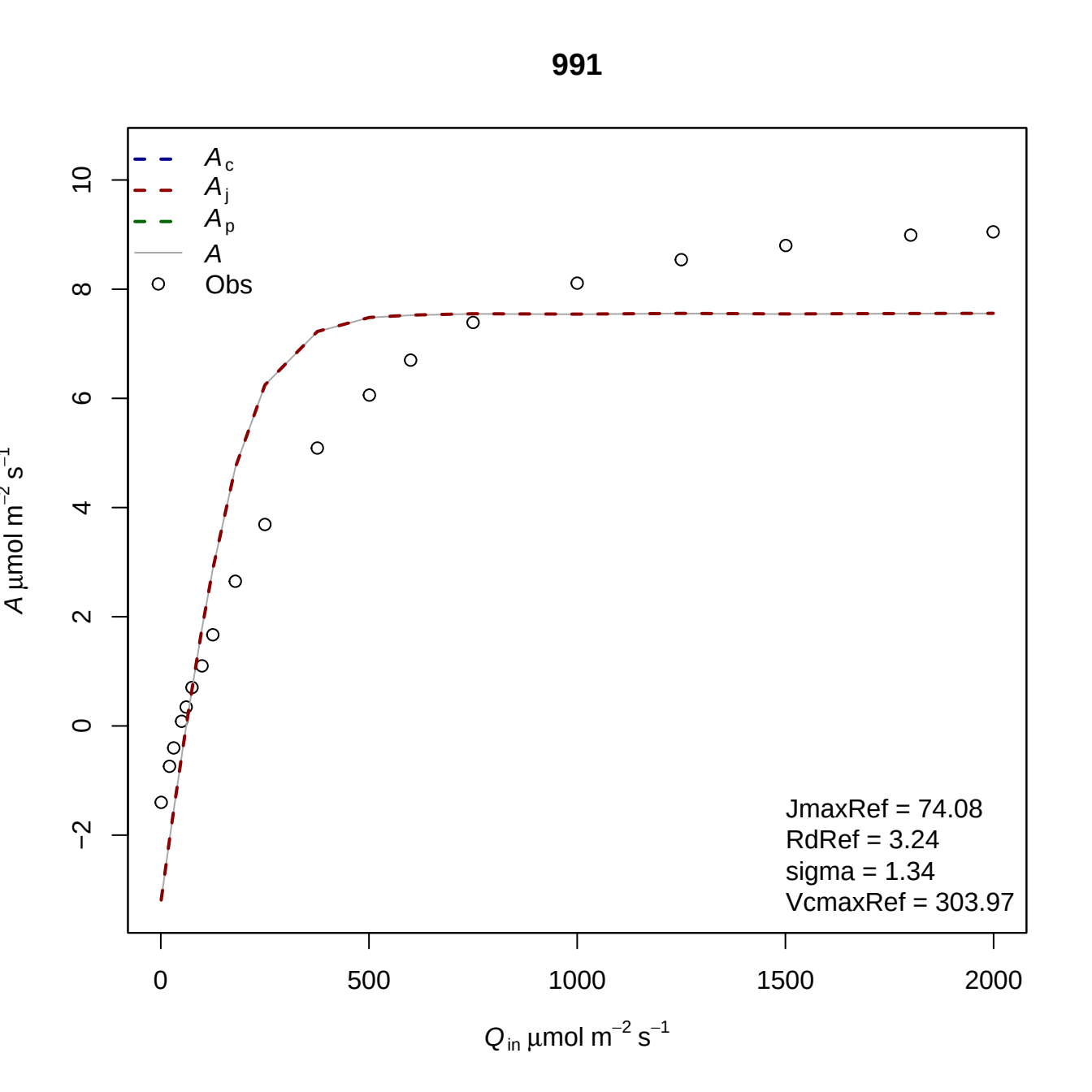

986
A_c
A_j
A_p
A
Obs
JmaxRef = 59.83
RdRef = 1.43
sigma = 0.44
VcmaxRef = 256.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

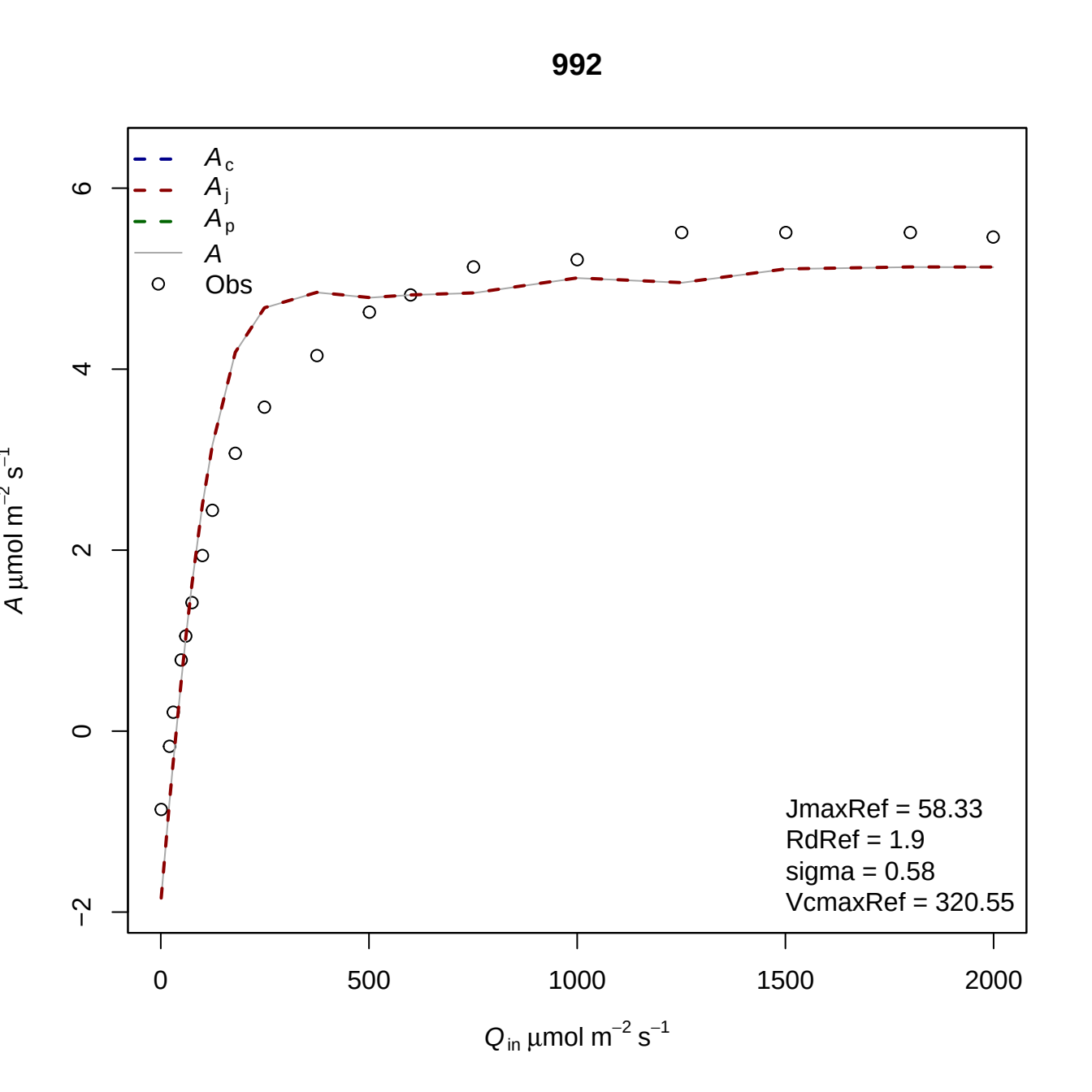

987
A_c
A_j
A_p
A
Obs
JmaxRef = 114.84
RdRef = 2.69
sigma = 0.49
VcmaxRef = 407.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

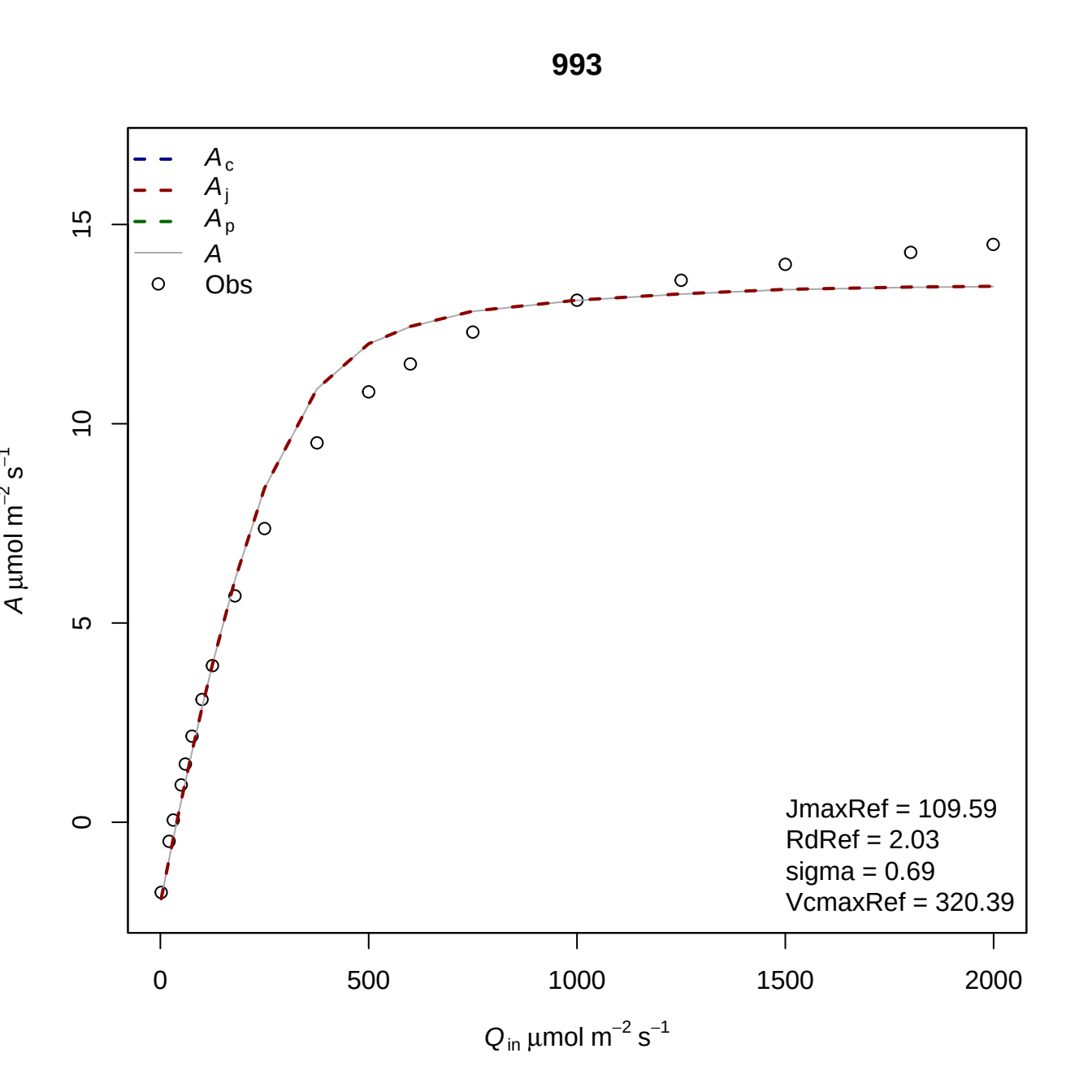

988
A_c
A_j
A_p
A
Obs
JmaxRef = 60.56
RdRef = 1.69
sigma = 0.57
VcmaxRef = 415.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

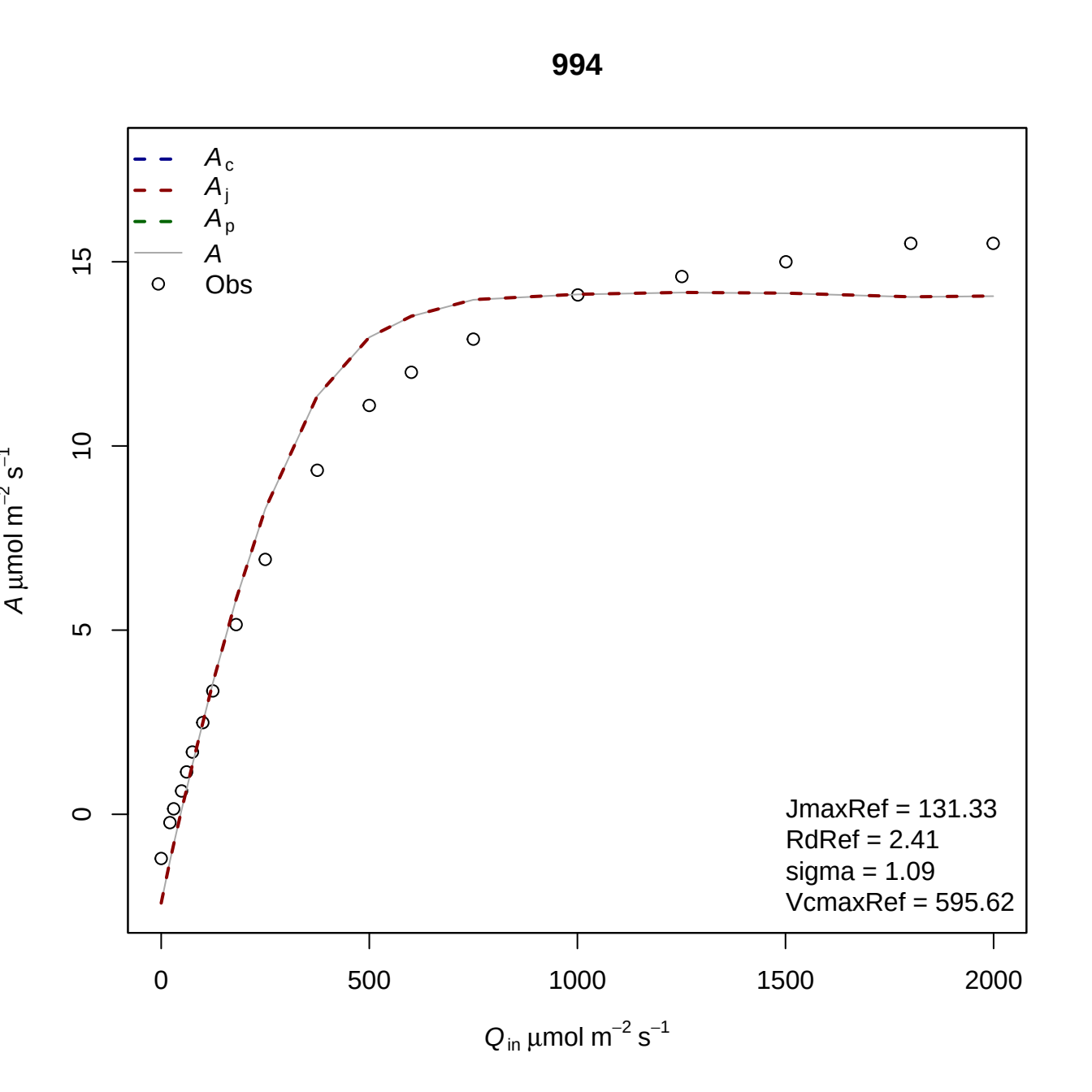

989
A_c
A_j
A_p
A
Obs
JmaxRef = 106.91
RdRef = 2.97
sigma = 0.71
VcmaxRef = 291.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

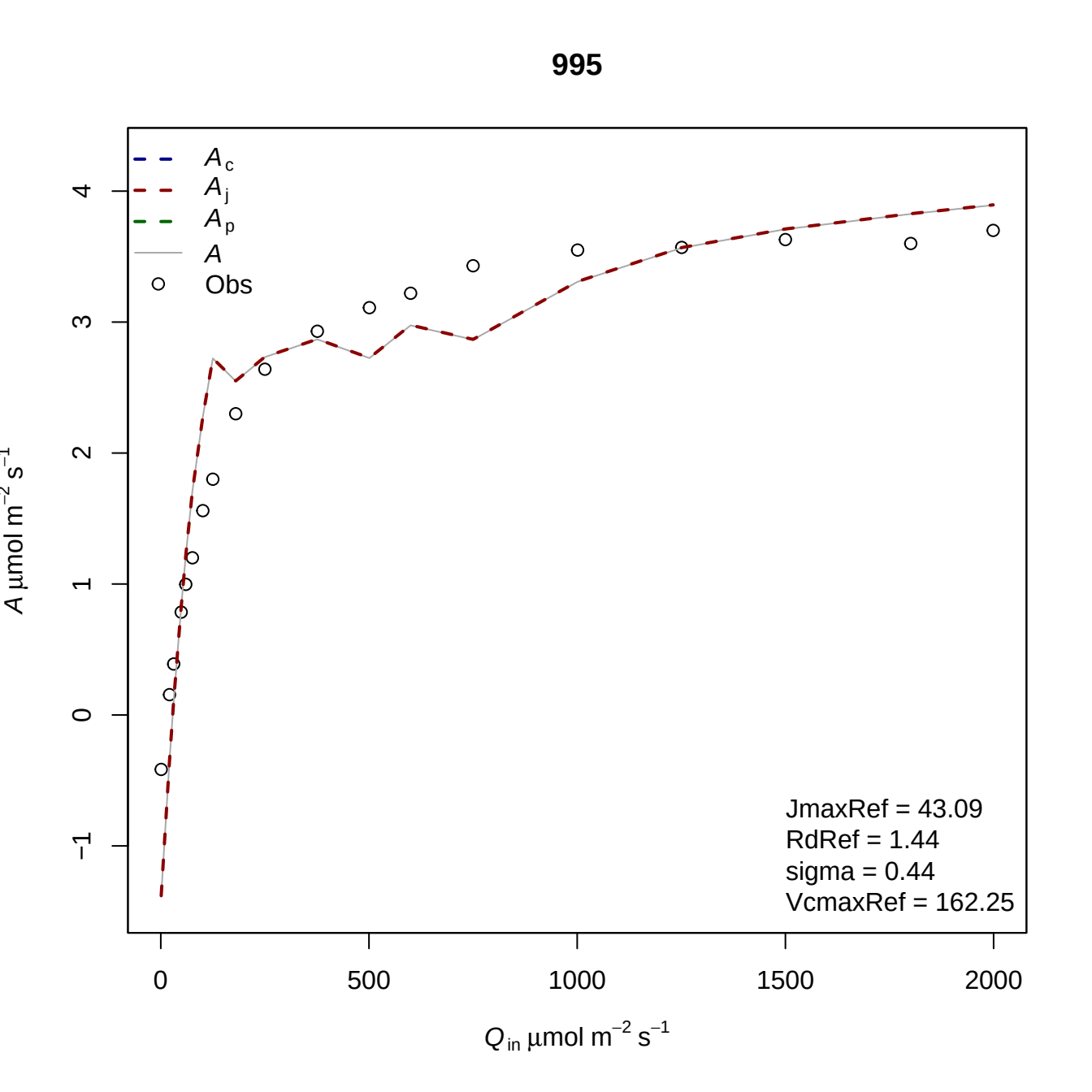

990
A_c
A_j
A_p
A
Obs
JmaxRef = 114.46
RdRef = 2.04
sigma = 0.74
VcmaxRef = 469.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

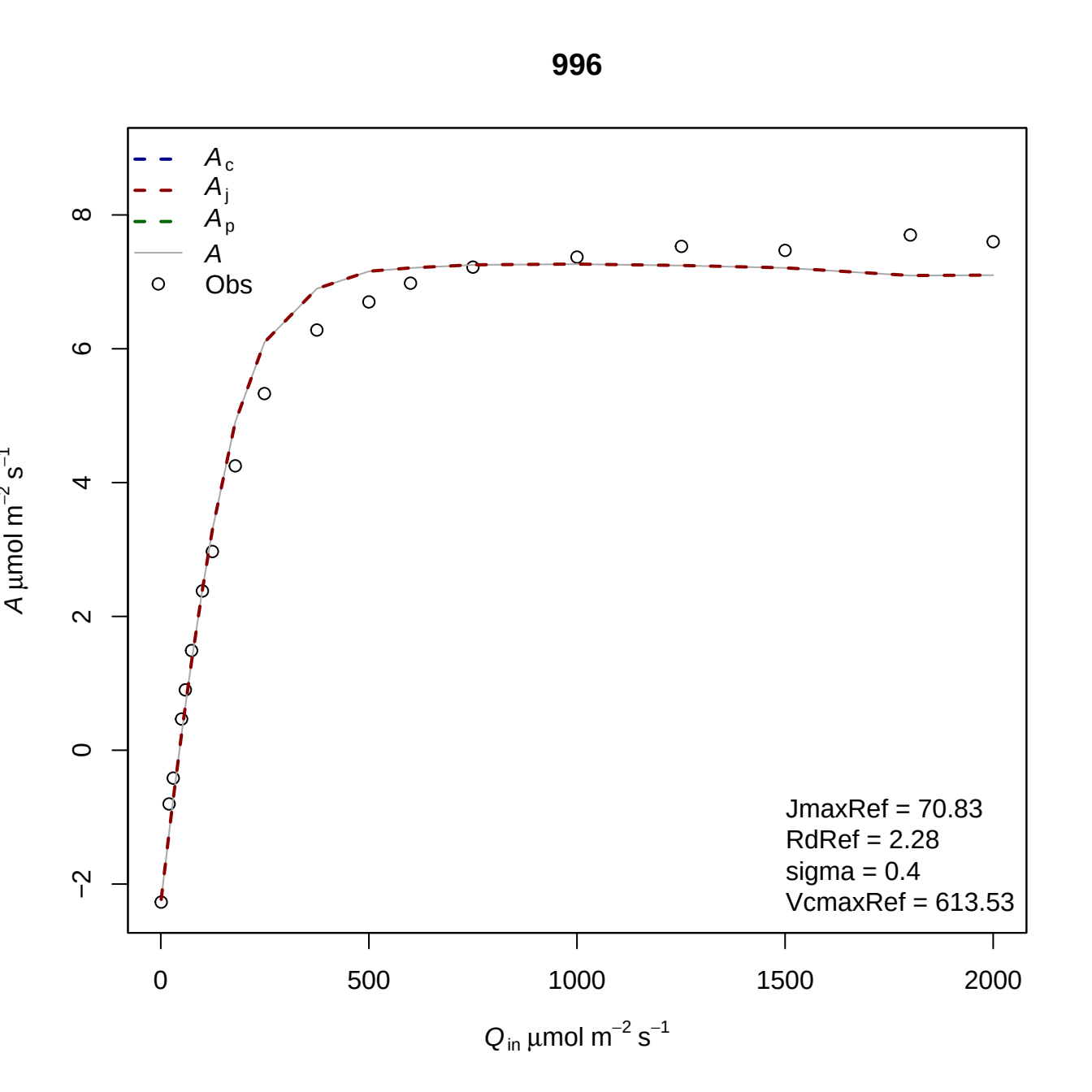

991
A_c
A_j
A_p
A
Obs
JmaxRef = 74.08
RdRef = 3.24
sigma = 1.34
VcmaxRef = 303.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

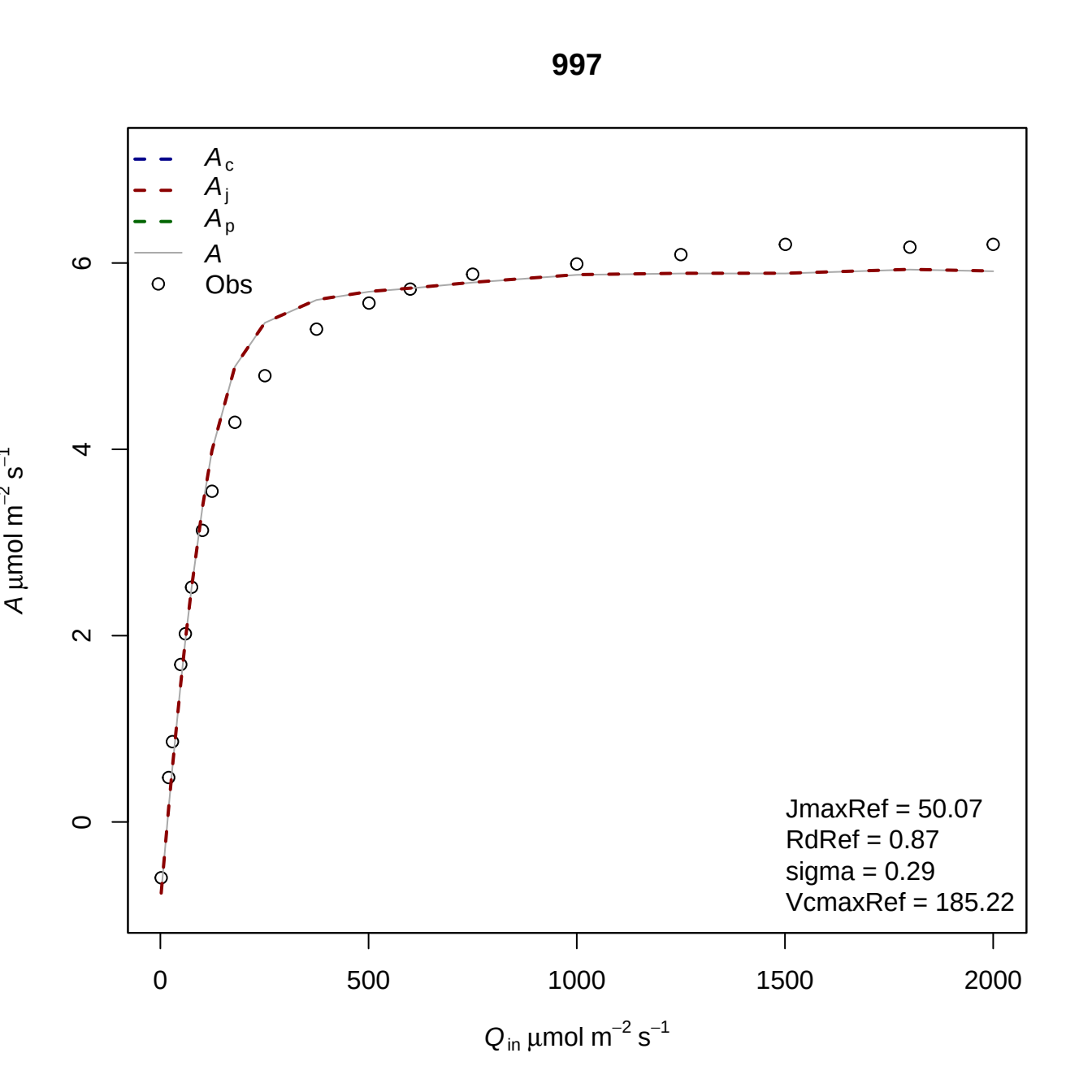

992
A_c
A_j
A_p
A
Obs
JmaxRef = 58.33
RdRef = 1.9
sigma = 0.58
VcmaxRef = 320.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

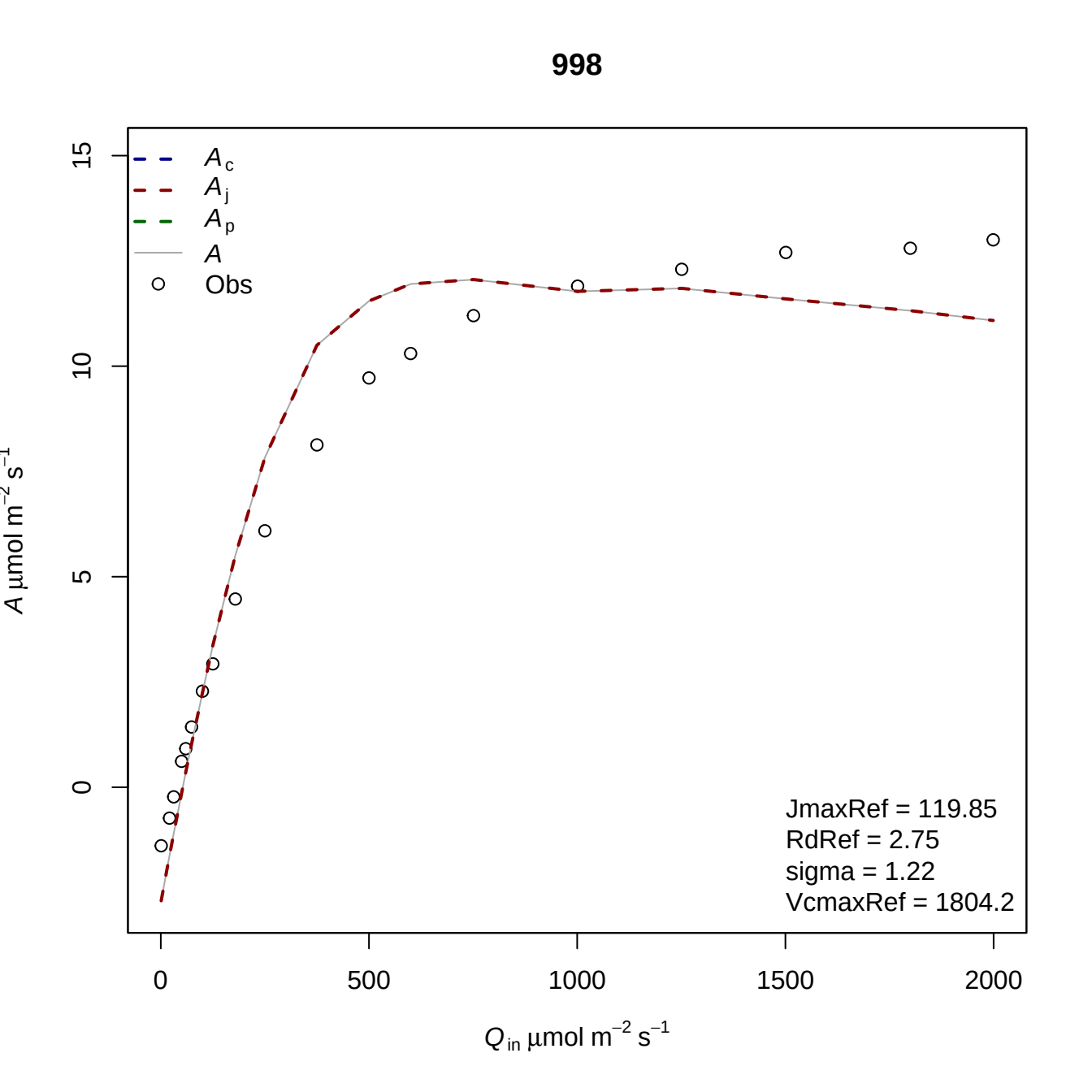

993
A_c
A_j
A_p
A
Obs
JmaxRef = 109.59
RdRef = 2.03
sigma = 0.69
VcmaxRef = 320.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

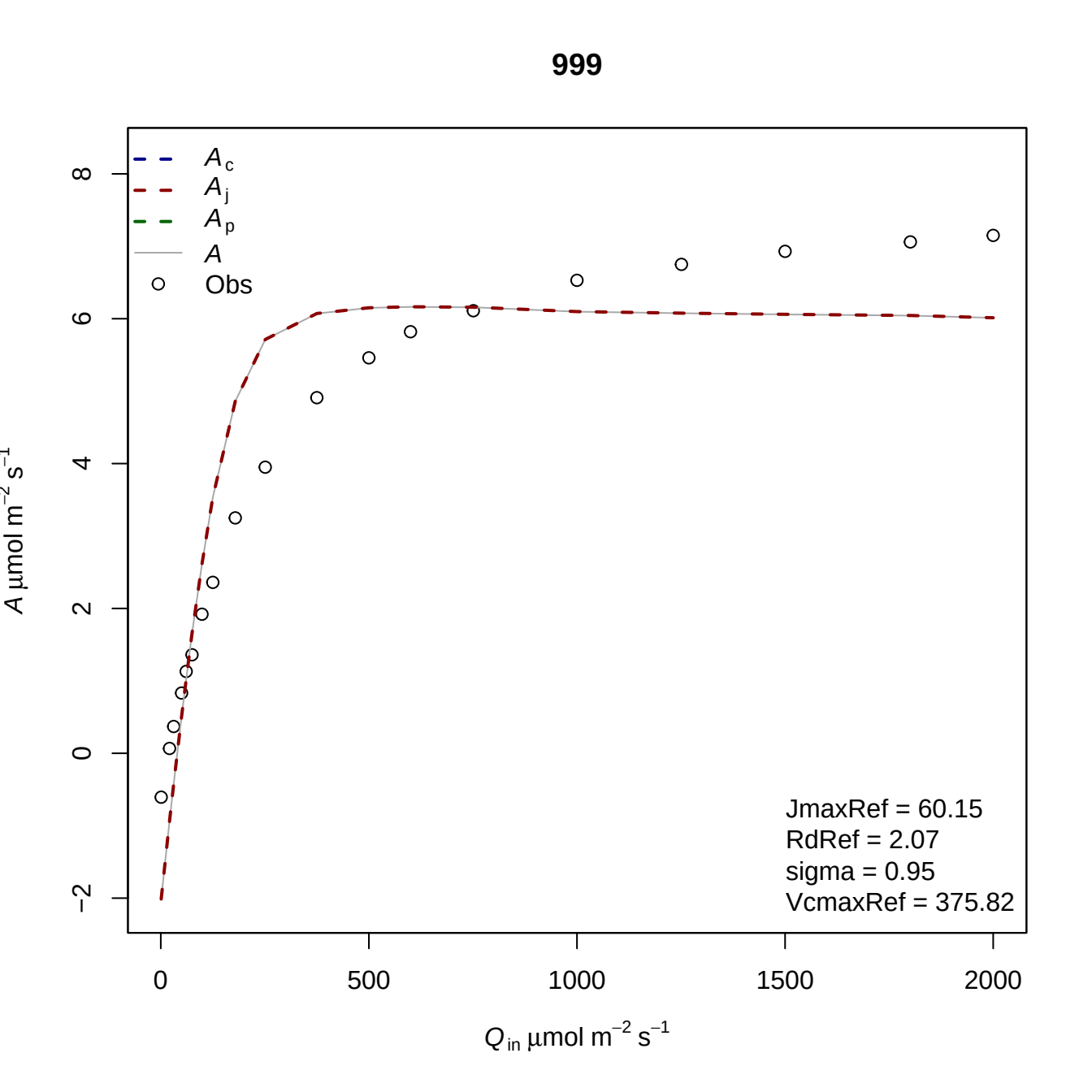

994
A_c
A_j
A_p
A
Obs
JmaxRef = 131.33
RdRef = 2.41
sigma = 1.09
VcmaxRef = 595.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

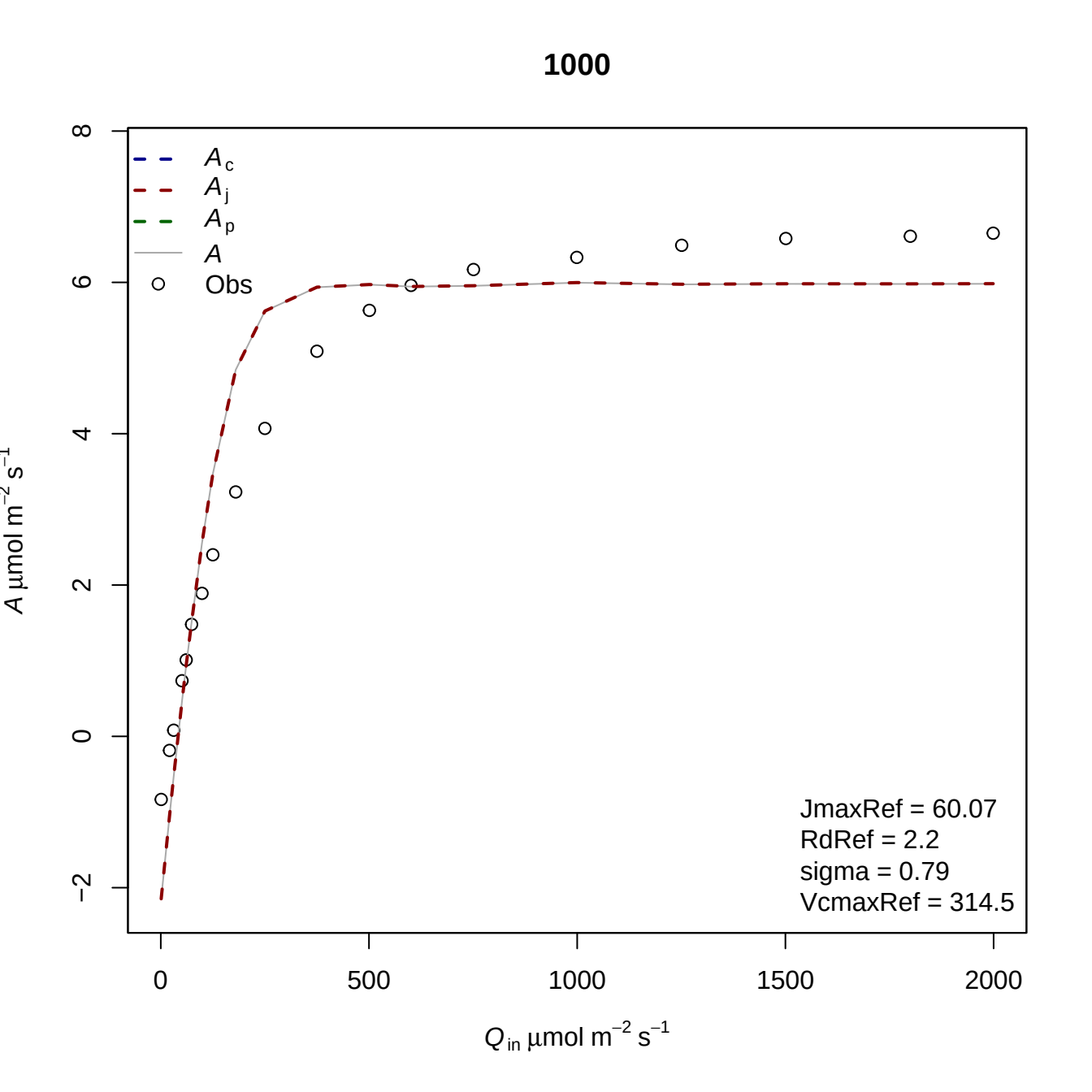

995
A_c
A_j
A_p
A
Obs
JmaxRef = 43.09
RdRef = 1.44
sigma = 0.44
VcmaxRef = 162.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

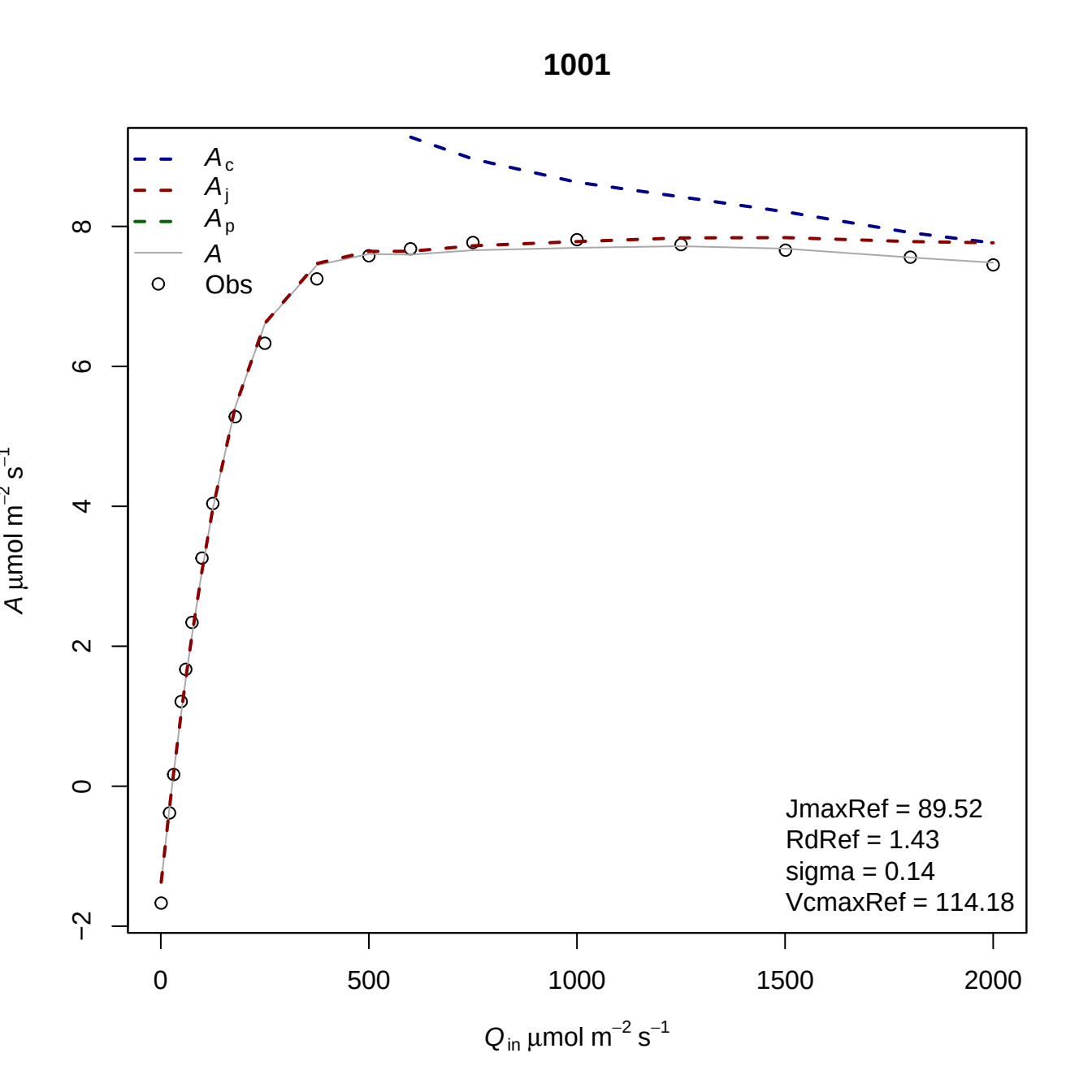

996
A_c
A_j
A_p
A
Obs
JmaxRef = 70.83
RdRef = 2.28
sigma = 0.4
VcmaxRef = 613.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

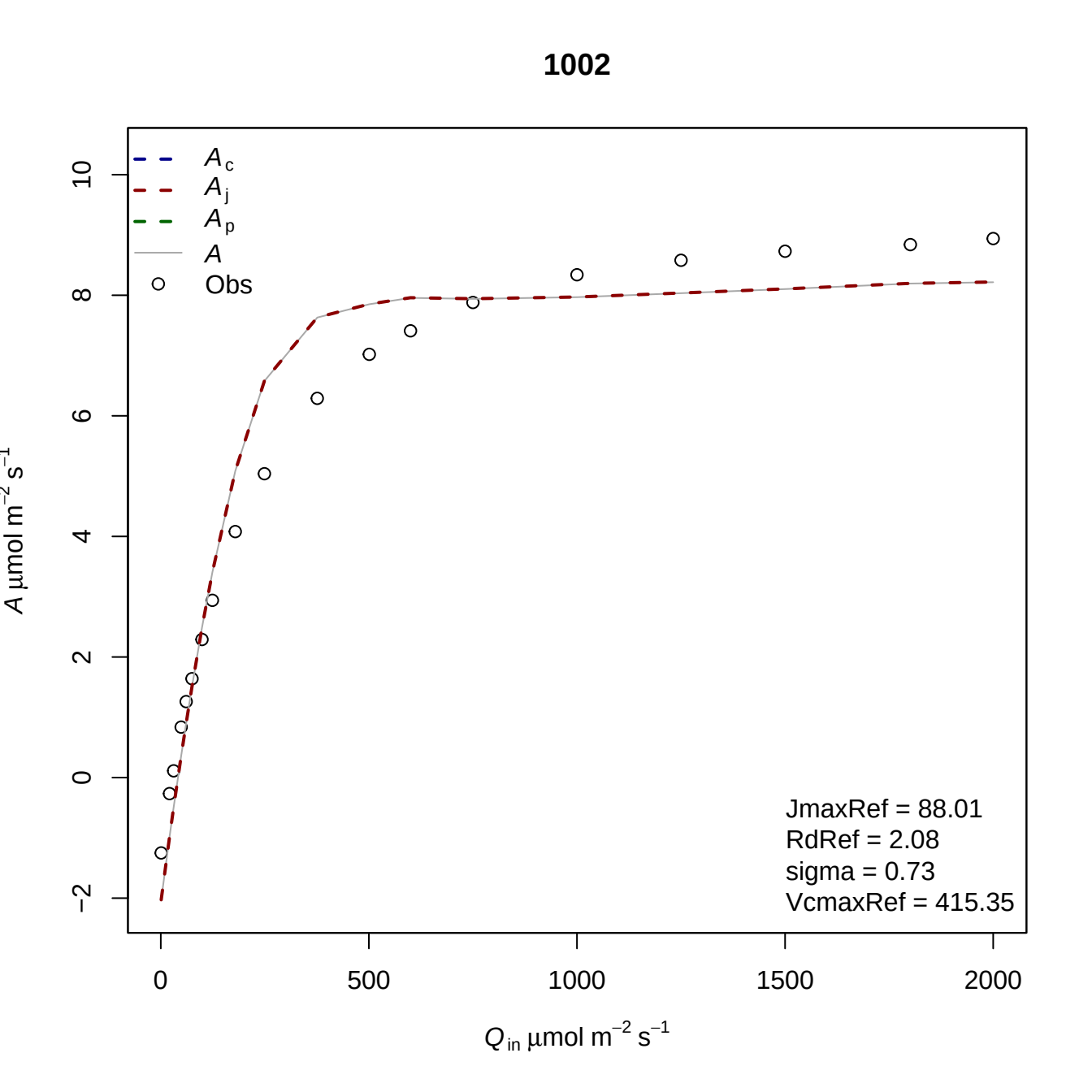

997
A_c
A_j
A_p
A
Obs
JmaxRef = 50.07
RdRef = 0.87
sigma = 0.29
VcmaxRef = 185.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

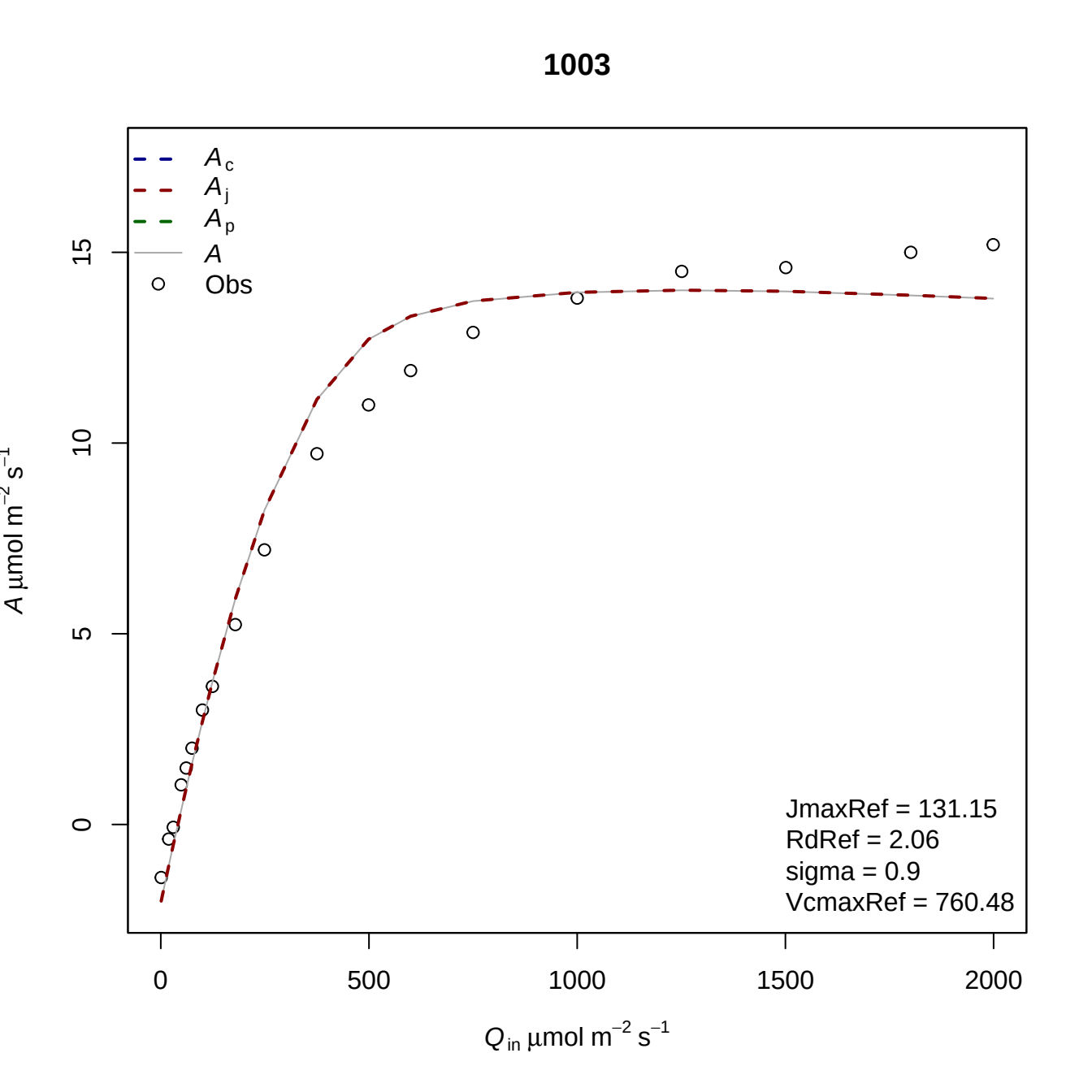

998
A_c
A_j
A_p
A
Obs
JmaxRef = 119.85
RdRef = 2.75
sigma = 1.22
VcmaxRef = 1804.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

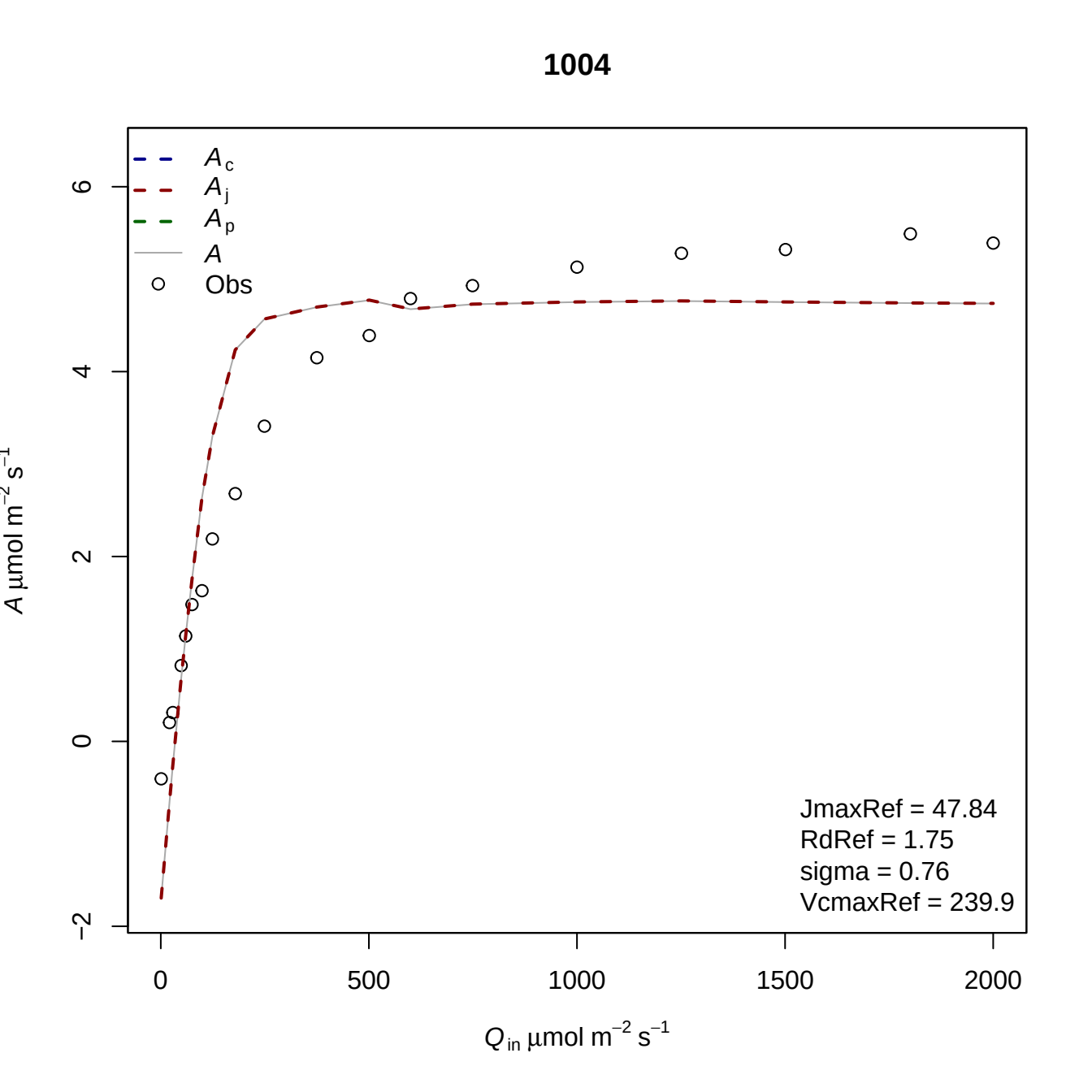

999
A_c
A_j
A_p
A
Obs
JmaxRef = 60.15
RdRef = 2.07
sigma = 0.95
VcmaxRef = 375.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

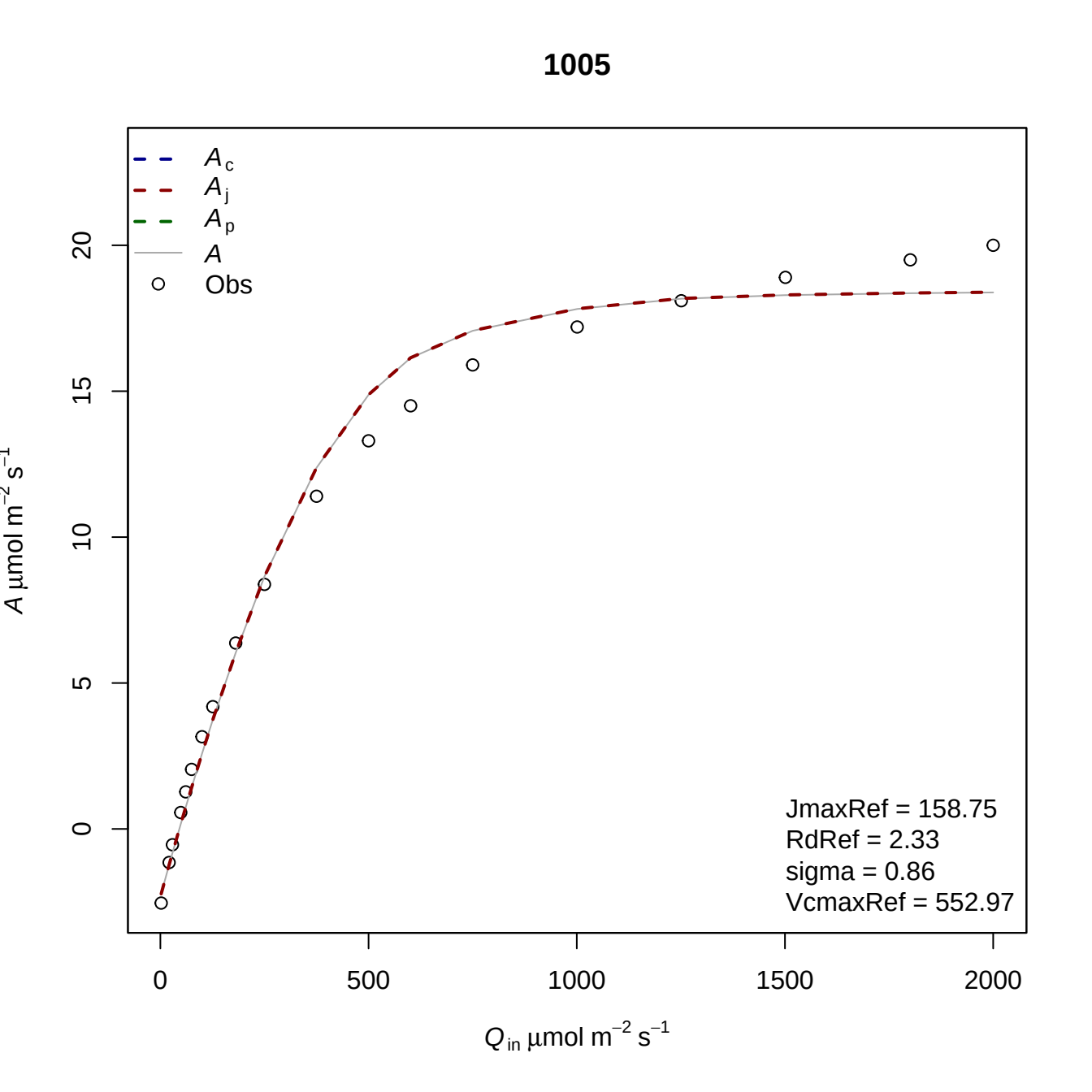

1000
A_c
A_j
A_p
A
Obs
JmaxRef = 60.07
RdRef = 2.2
sigma = 0.79
VcmaxRef = 314.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

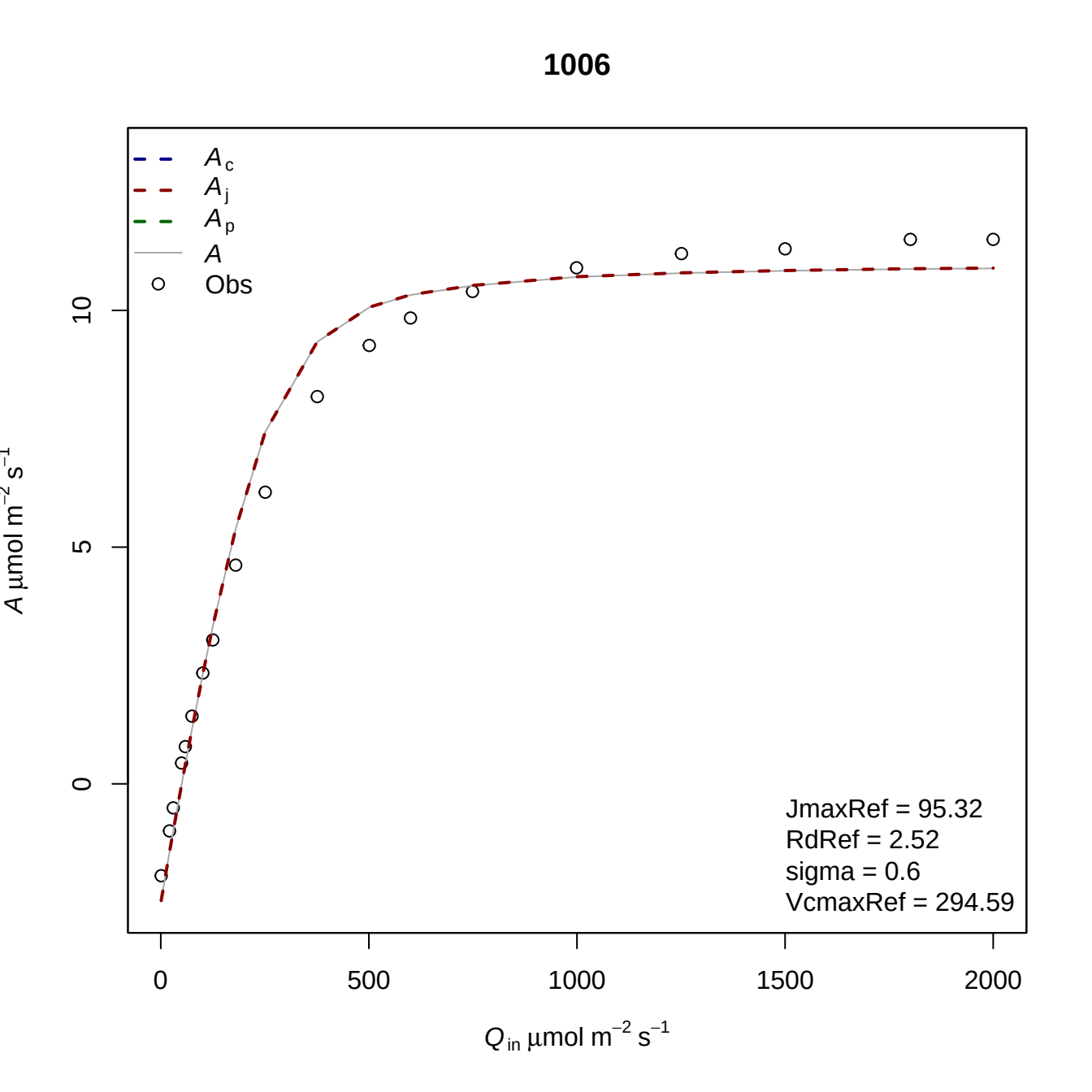

1001
A_c
A_j
A_p
A
Obs
JmaxRef = 89.52
RdRef = 1.43
sigma = 0.14
VcmaxRef = 114.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

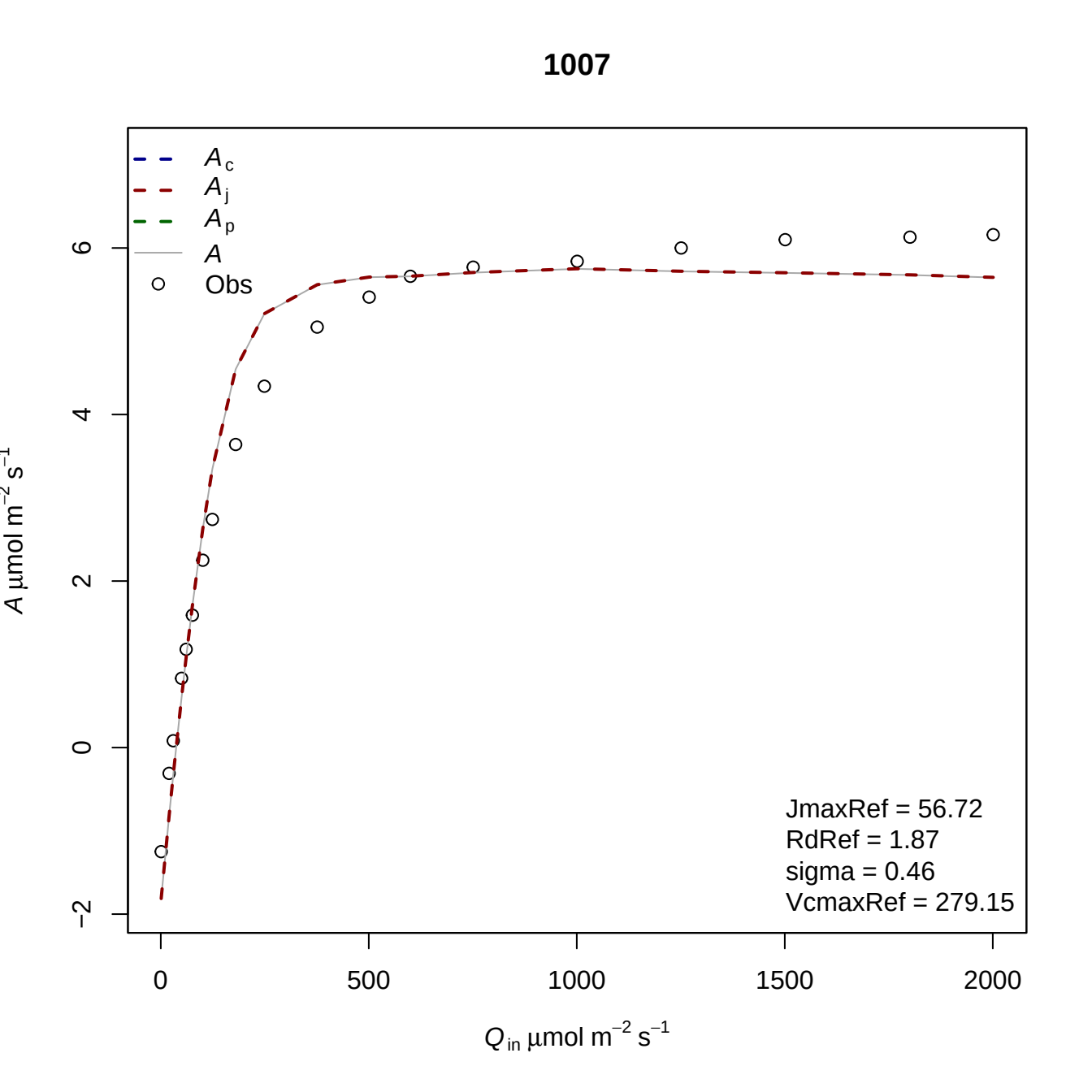

1002
A_c
A_j
A_p
A
Obs
JmaxRef = 88.01
RdRef = 2.08
sigma = 0.73
VcmaxRef = 415.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

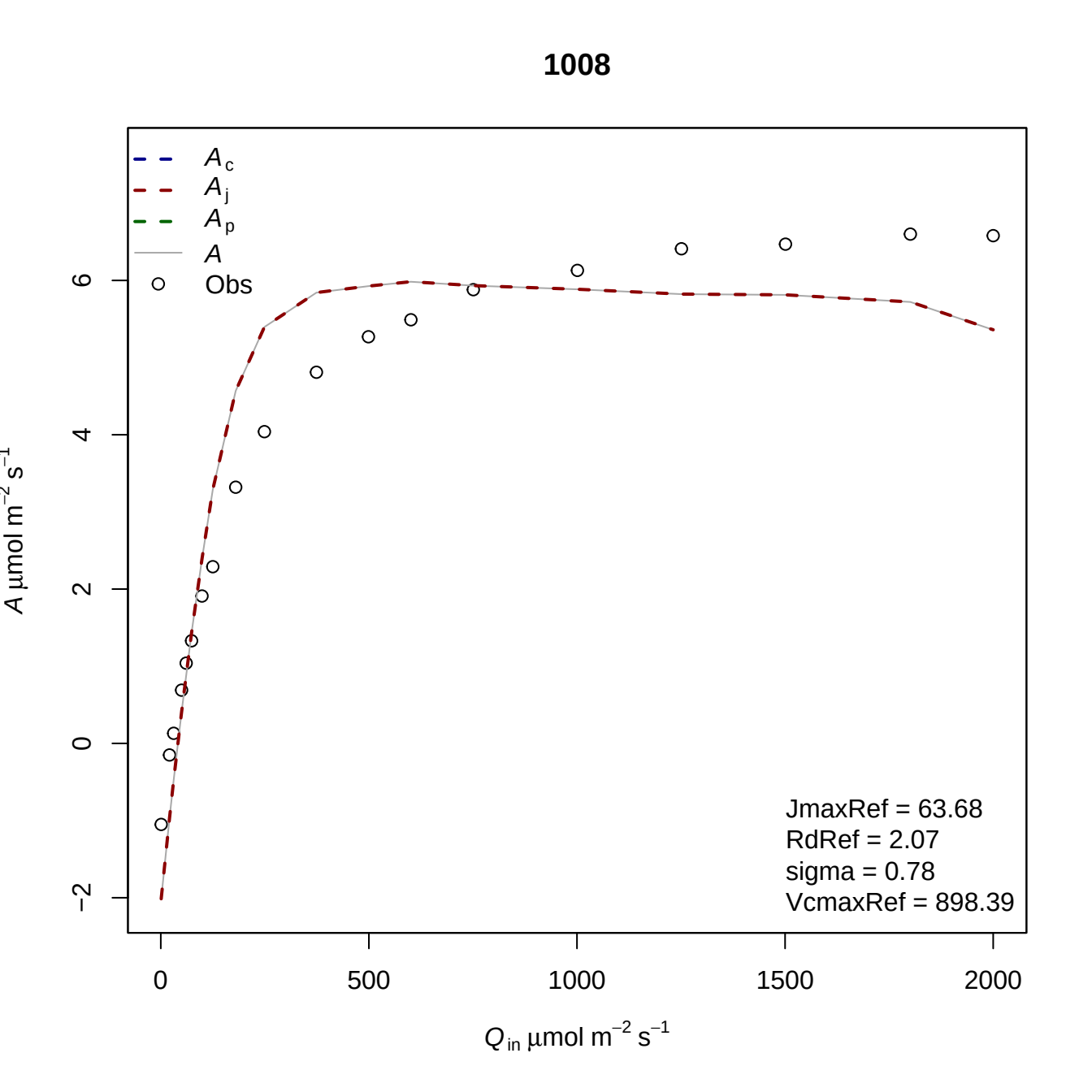

1003
A_c
A_j
A_p
A
Obs
JmaxRef = 131.15
RdRef = 2.06
sigma = 0.9
VcmaxRef = 760.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

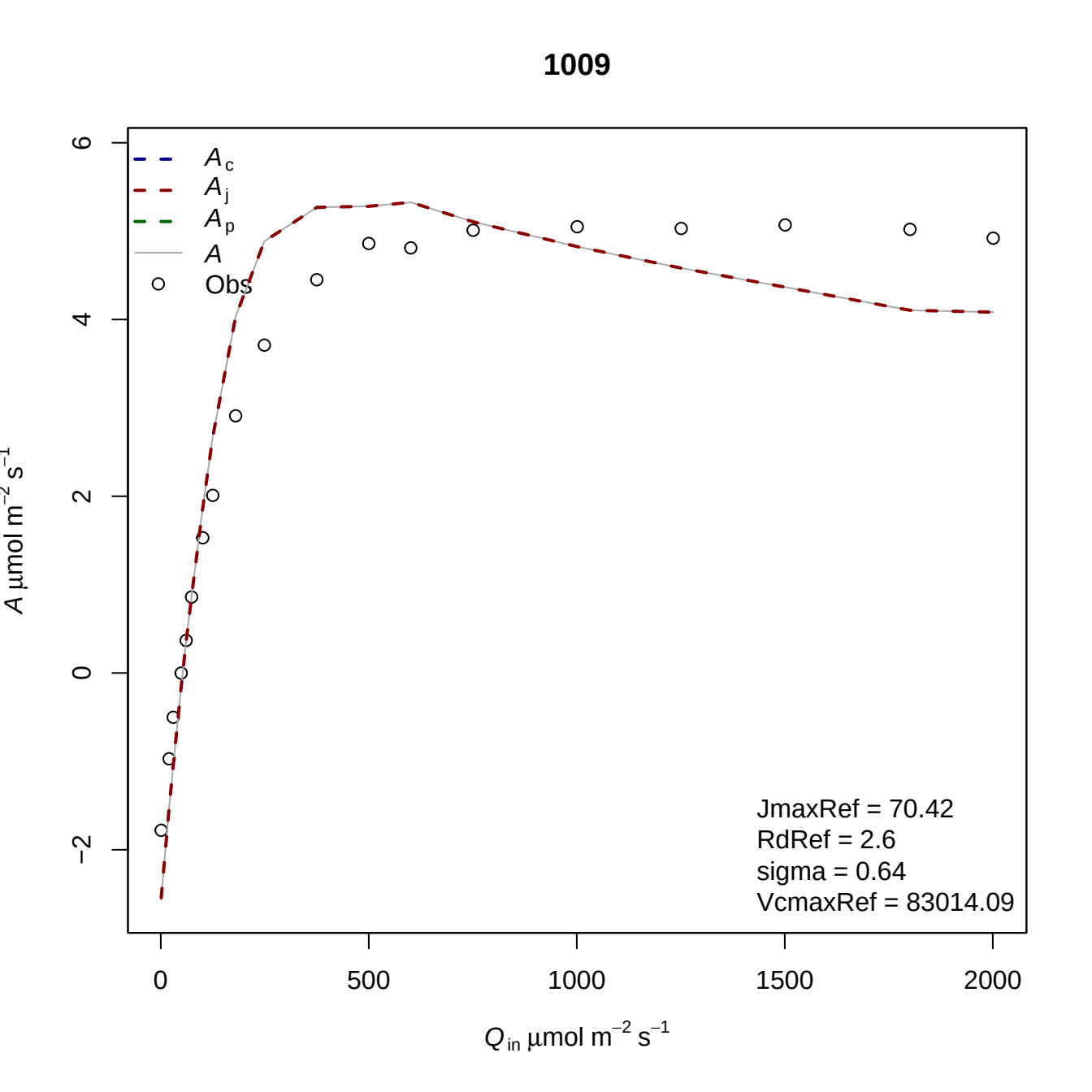

1004
A_c
A_j
A_p
A
Obs
JmaxRef = 47.84
RdRef = 1.75
sigma = 0.76
VcmaxRef = 239.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

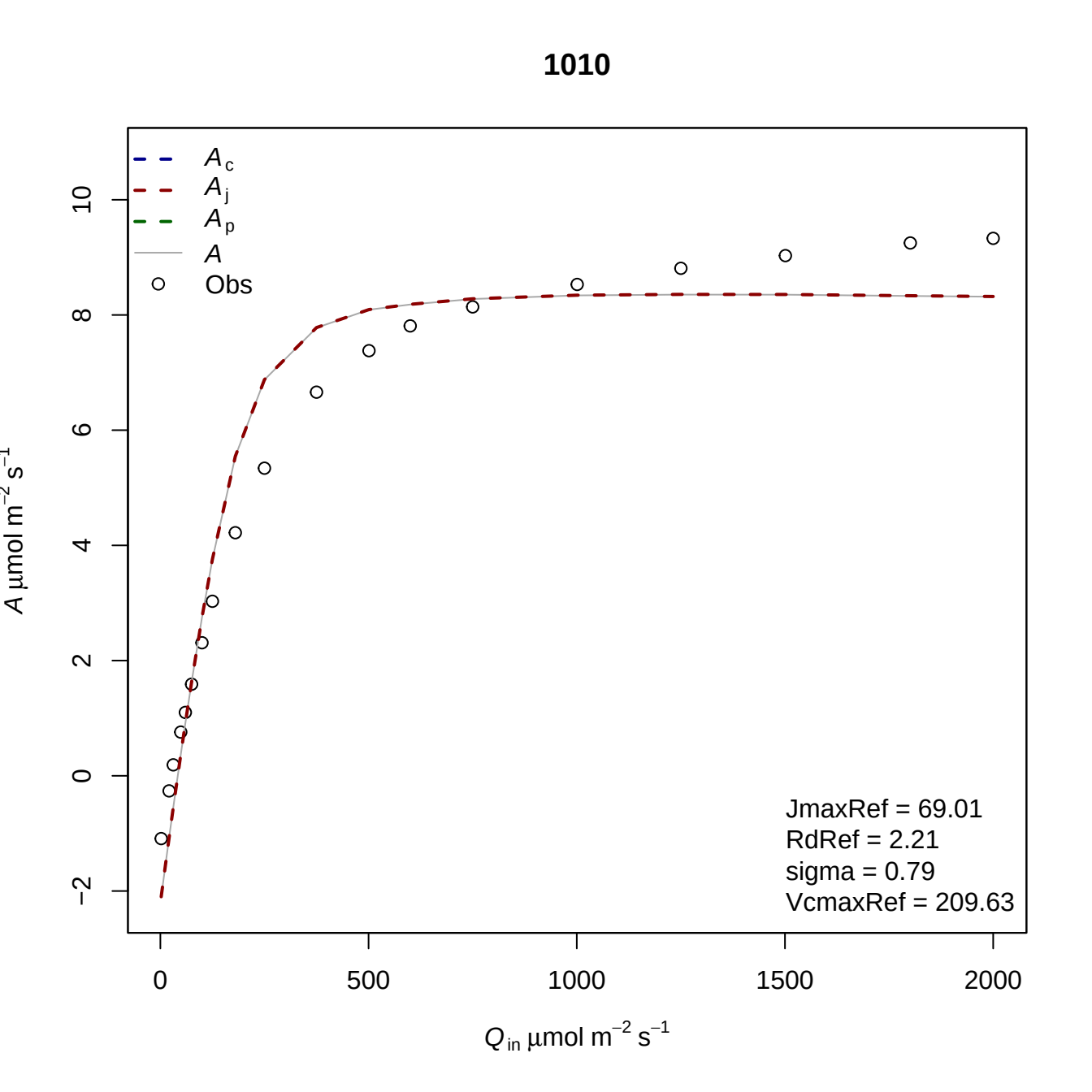

1005
A_c
A_j
A_p
A
Obs
JmaxRef = 158.75
RdRef = 2.33
sigma = 0.86
VcmaxRef = 552.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

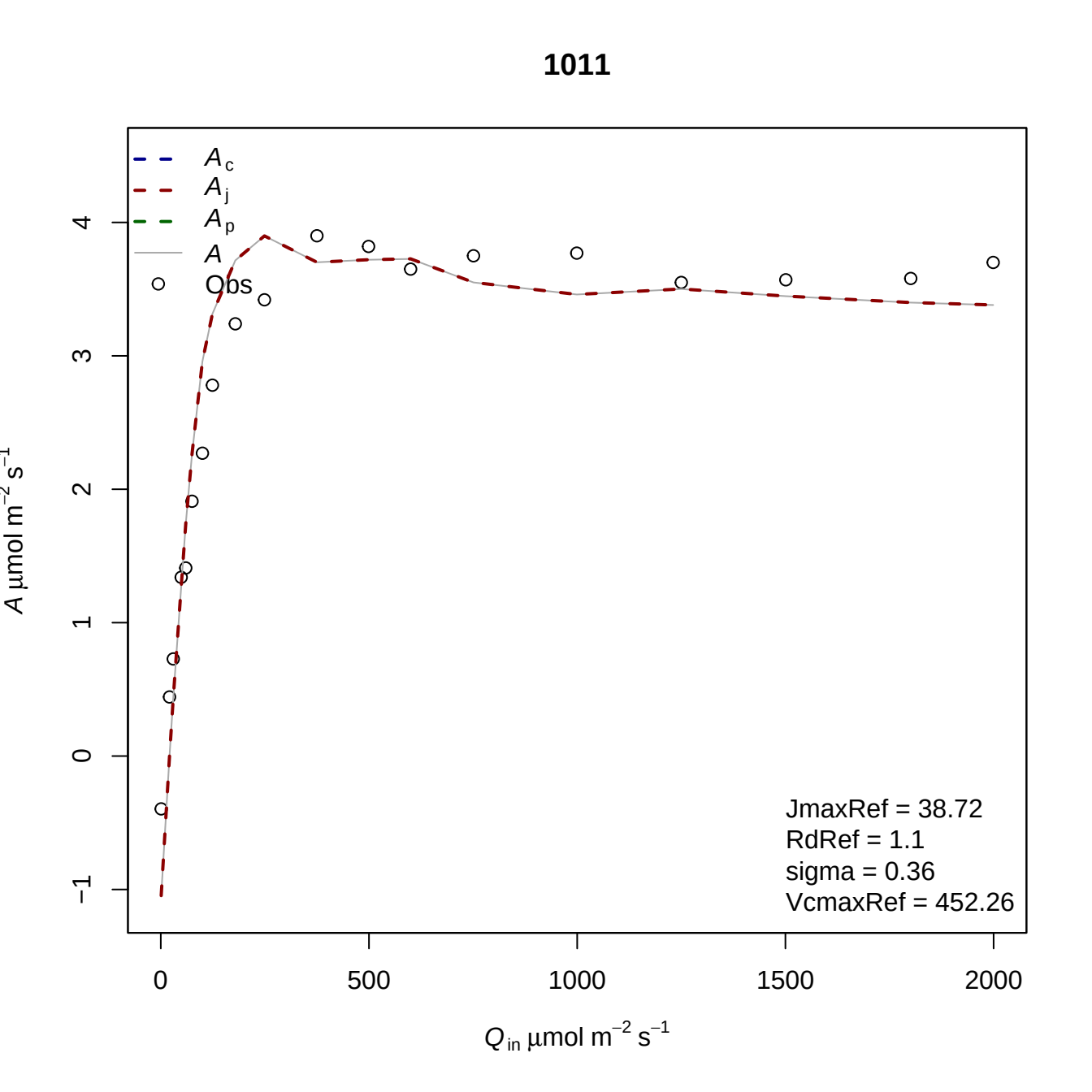

1006
A_c
A_j
A_p
A
Obs
JmaxRef = 95.32
RdRef = 2.52
sigma = 0.6
VcmaxRef = 294.59
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

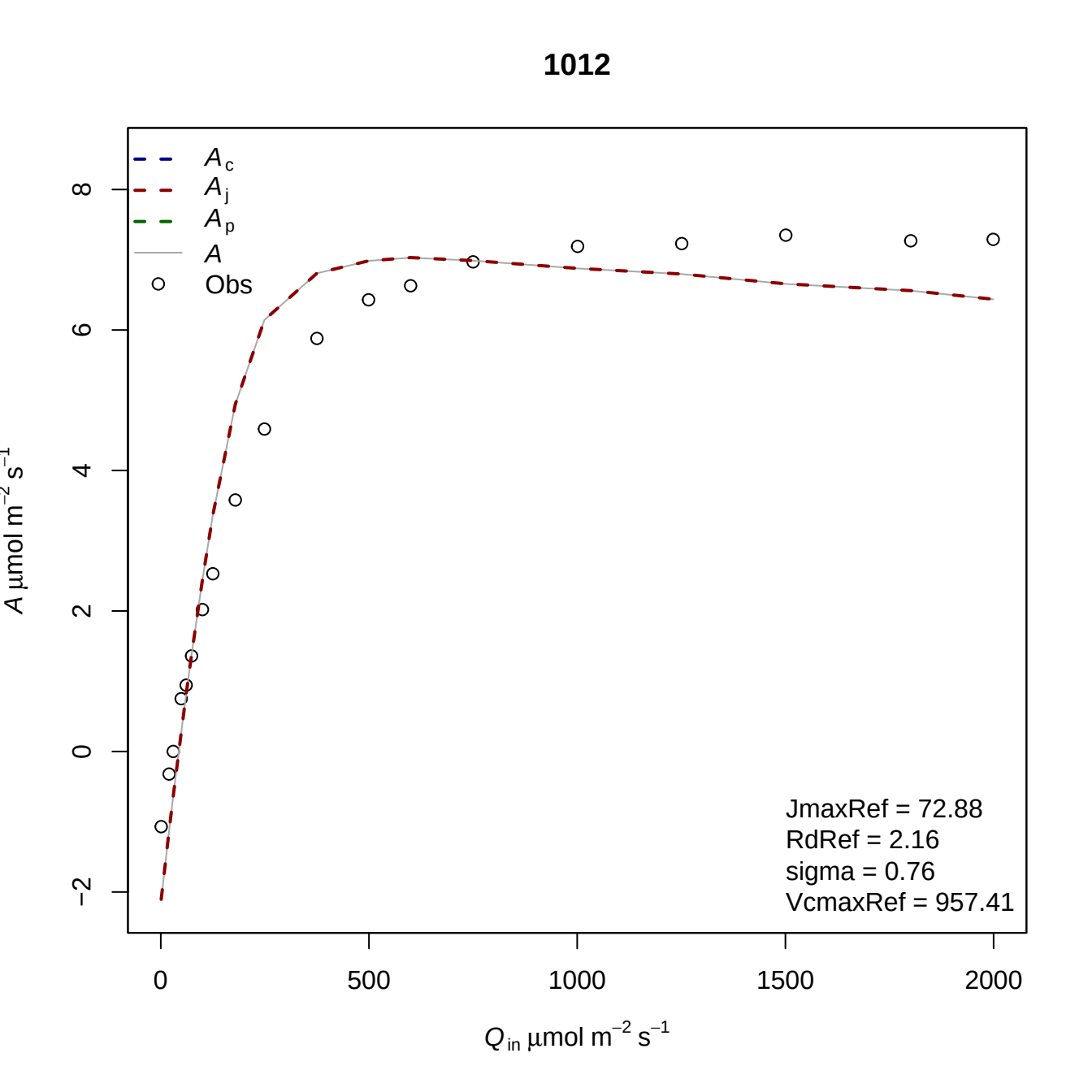

1007
A_c
A_j
A_p
A
Obs
JmaxRef = 56.72
RdRef = 1.87
sigma = 0.46
VcmaxRef = 279.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

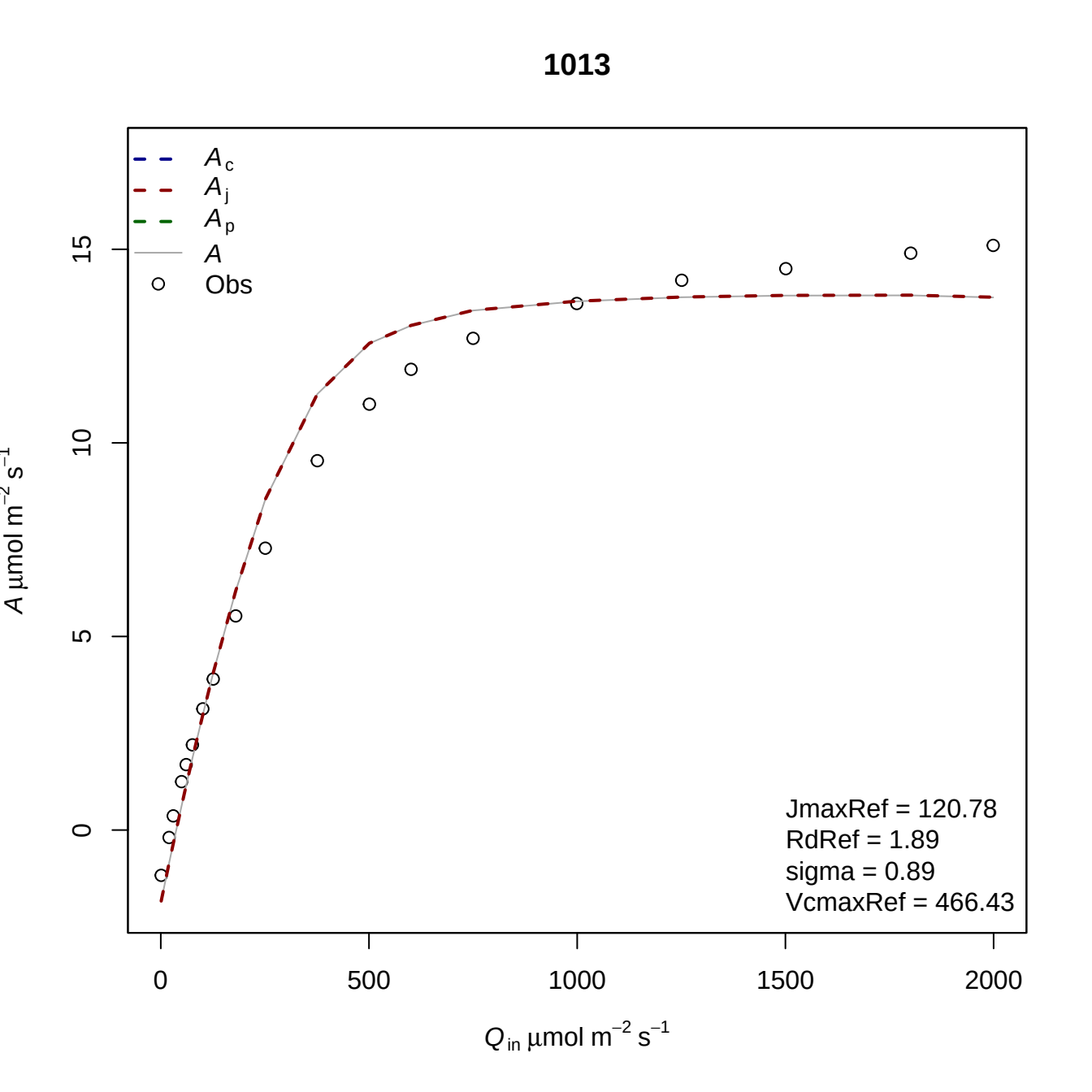

1008
A_c
A_j
A_p
A
Obs
JmaxRef = 63.68
RdRef = 2.07
sigma = 0.78
VcmaxRef = 898.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

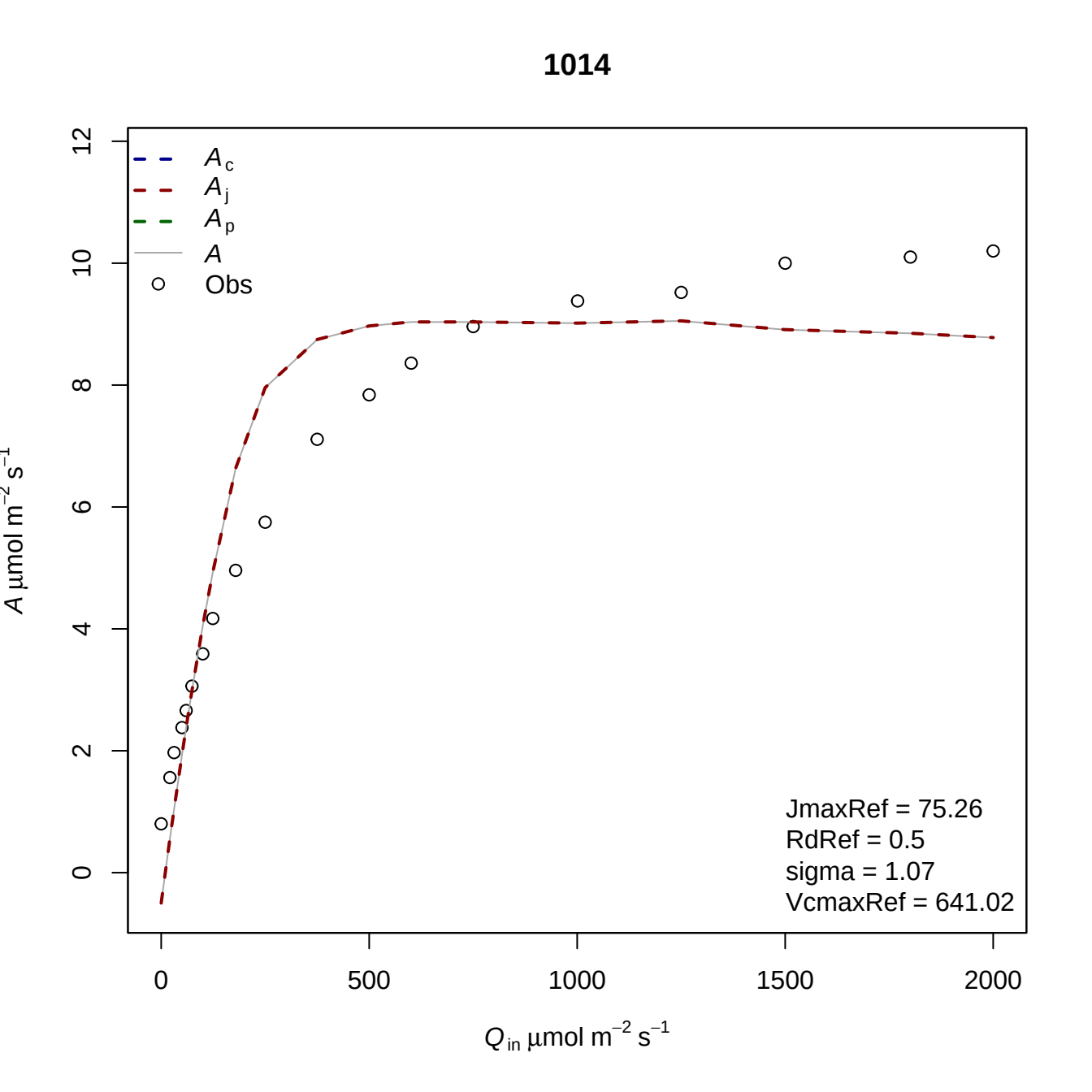

1009
A_c
A_j
A_p
A
Obs
JmaxRef = 70.42
RdRef = 2.6
sigma = 0.64
VcmaxRef = 83014.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

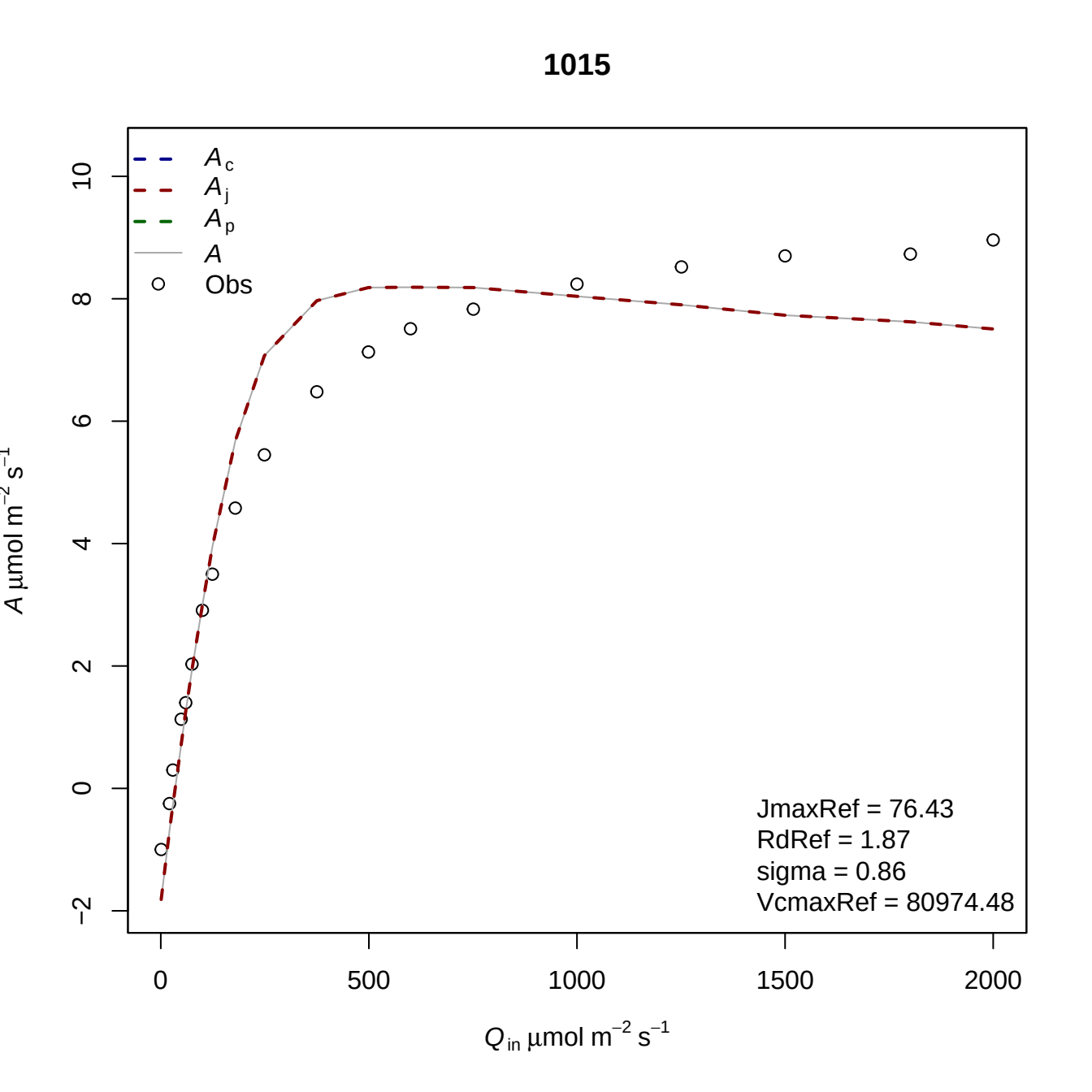

1010
A_c
A_j
A_p
A
Obs
JmaxRef = 69.01
RdRef = 2.21
sigma = 0.79
VcmaxRef = 209.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

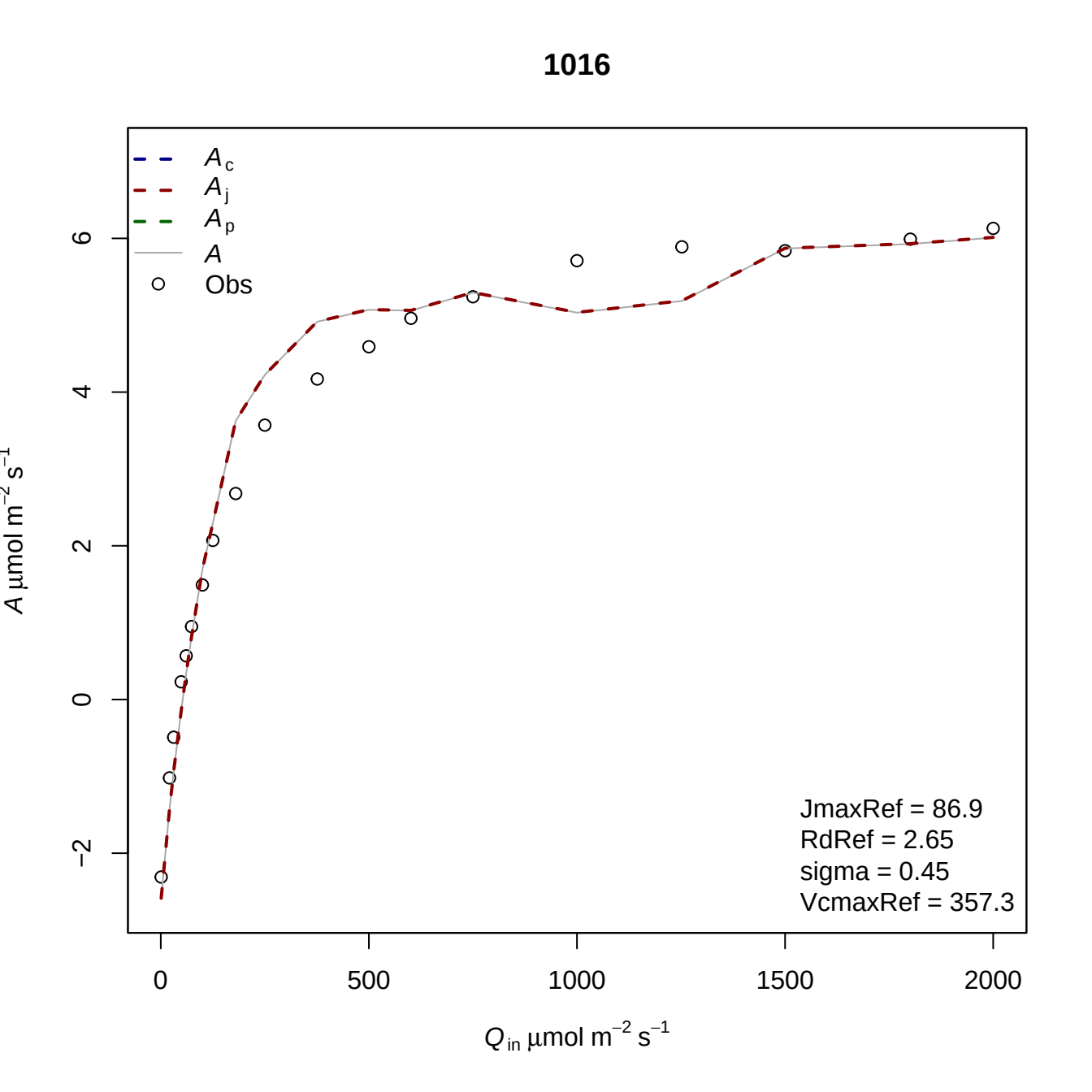

1011
A_c
A_j
A_p
A
Obs
JmaxRef = 38.72
RdRef = 1.1
sigma = 0.36
VcmaxRef = 452.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

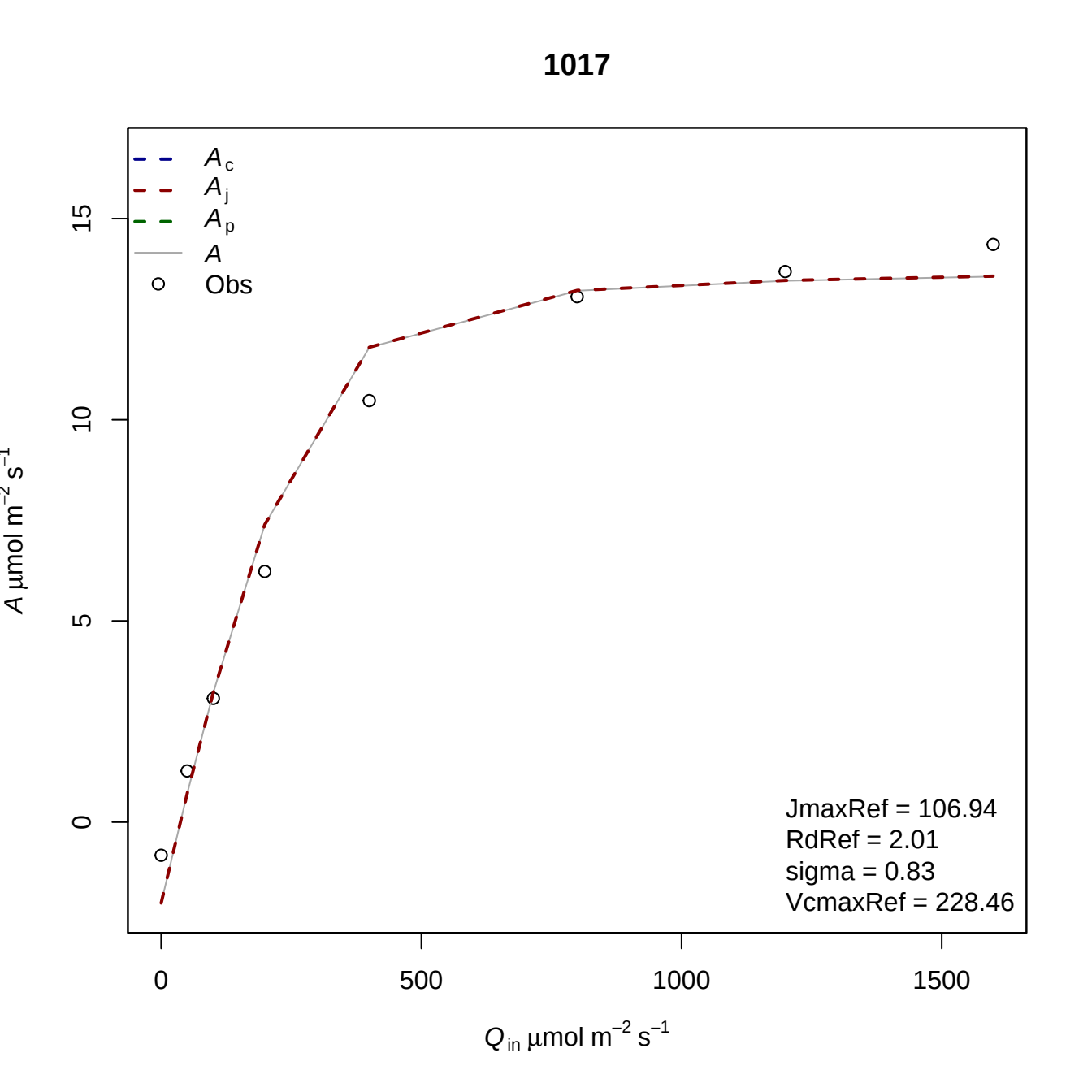

1012
A_c
A_j
A_p
A
Obs
JmaxRef = 72.88
RdRef = 2.16
sigma = 0.76
VcmaxRef = 957.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

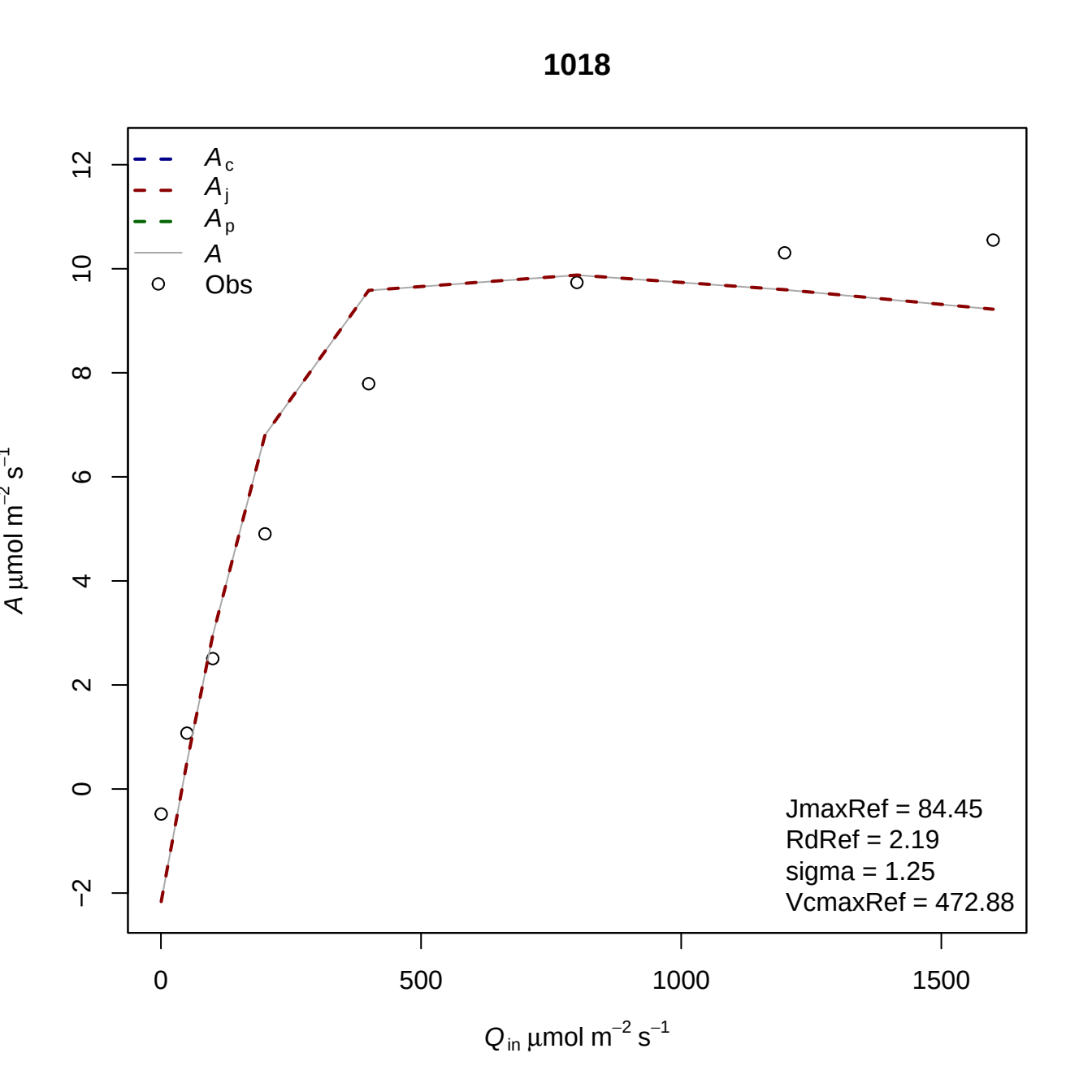

1013
A_c
A_j
A_p
A
Obs
JmaxRef = 120.78
RdRef = 1.89
sigma = 0.89
VcmaxRef = 466.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

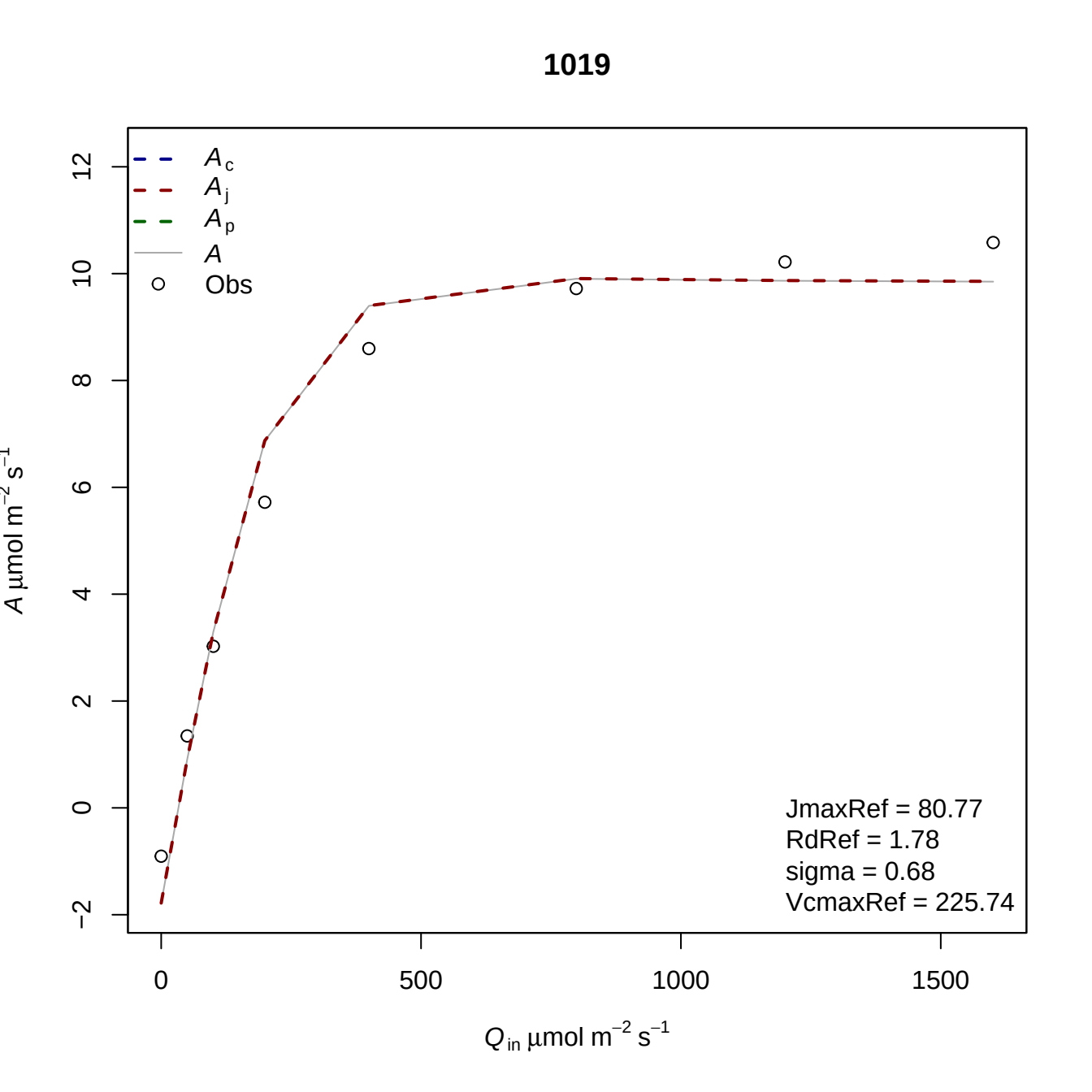

1014
A_c
A_j
A_p
A
Obs
JmaxRef = 75.26
RdRef = 0.5
sigma = 1.07
VcmaxRef = 641.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

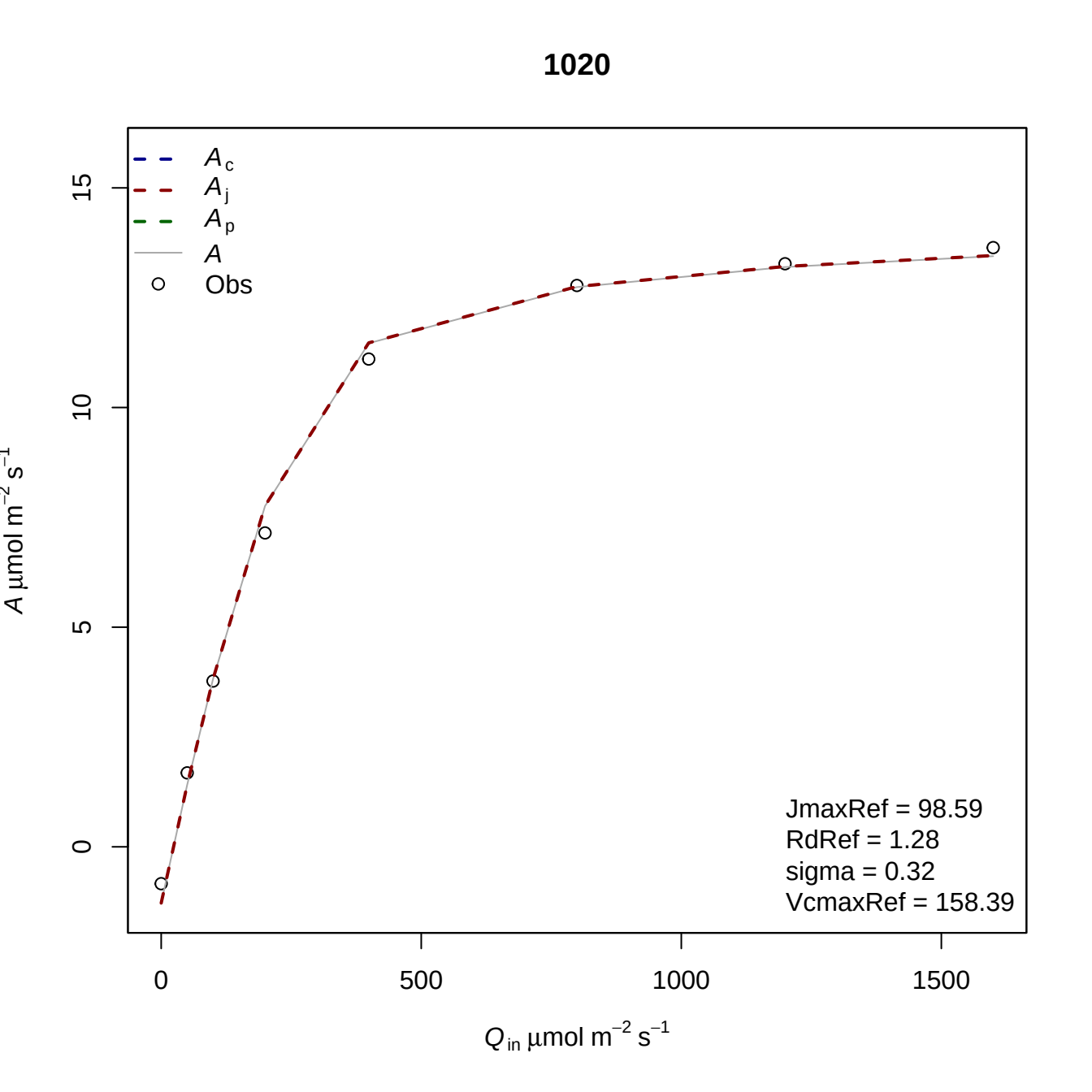

1015
A_c
A_j
A_p
A
Obs
JmaxRef = 76.43
RdRef = 1.87
sigma = 0.86
VcmaxRef = 80974.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

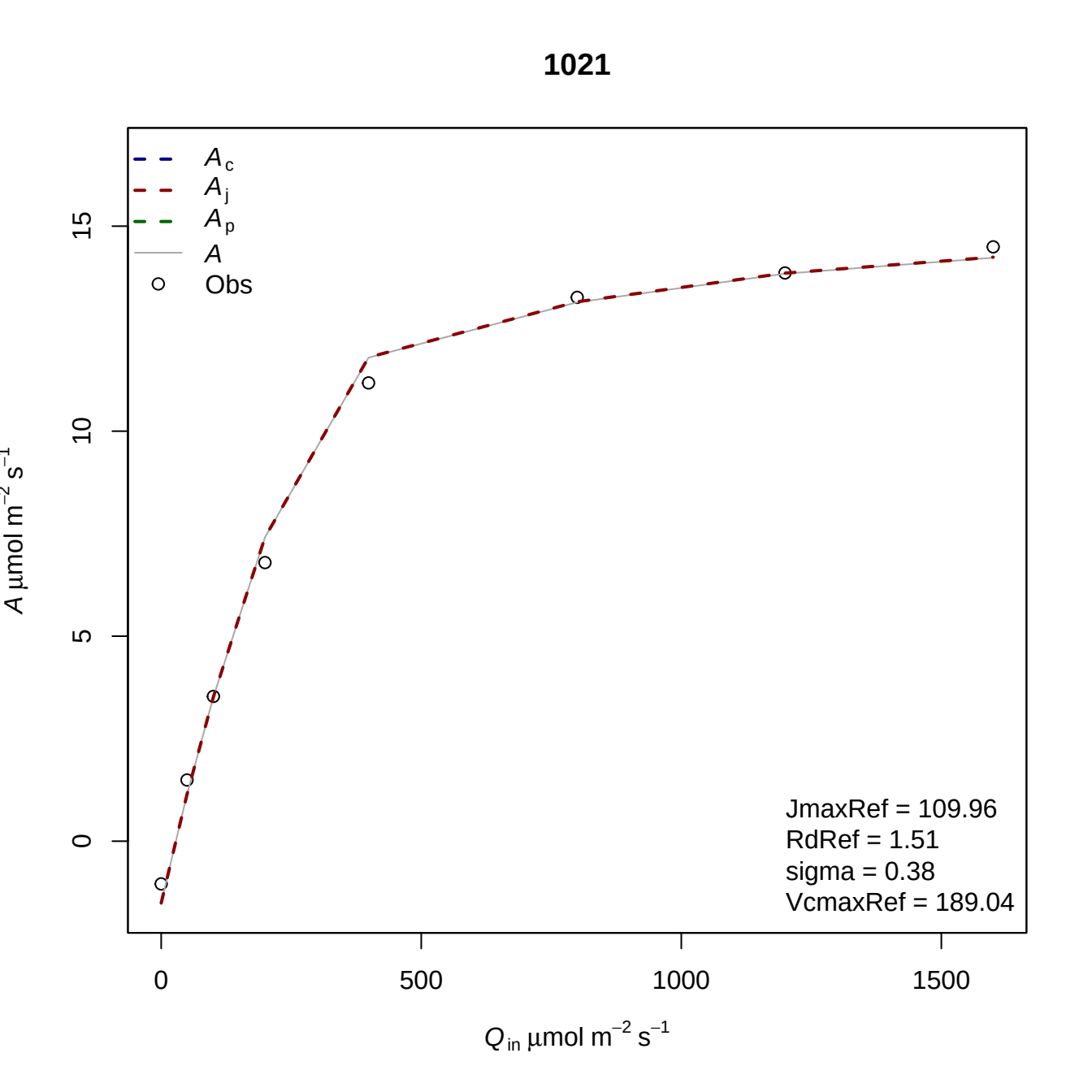

1016
A_c
A_j
A_p
A
Obs
JmaxRef = 86.9
RdRef = 2.65
sigma = 0.45
VcmaxRef = 357.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

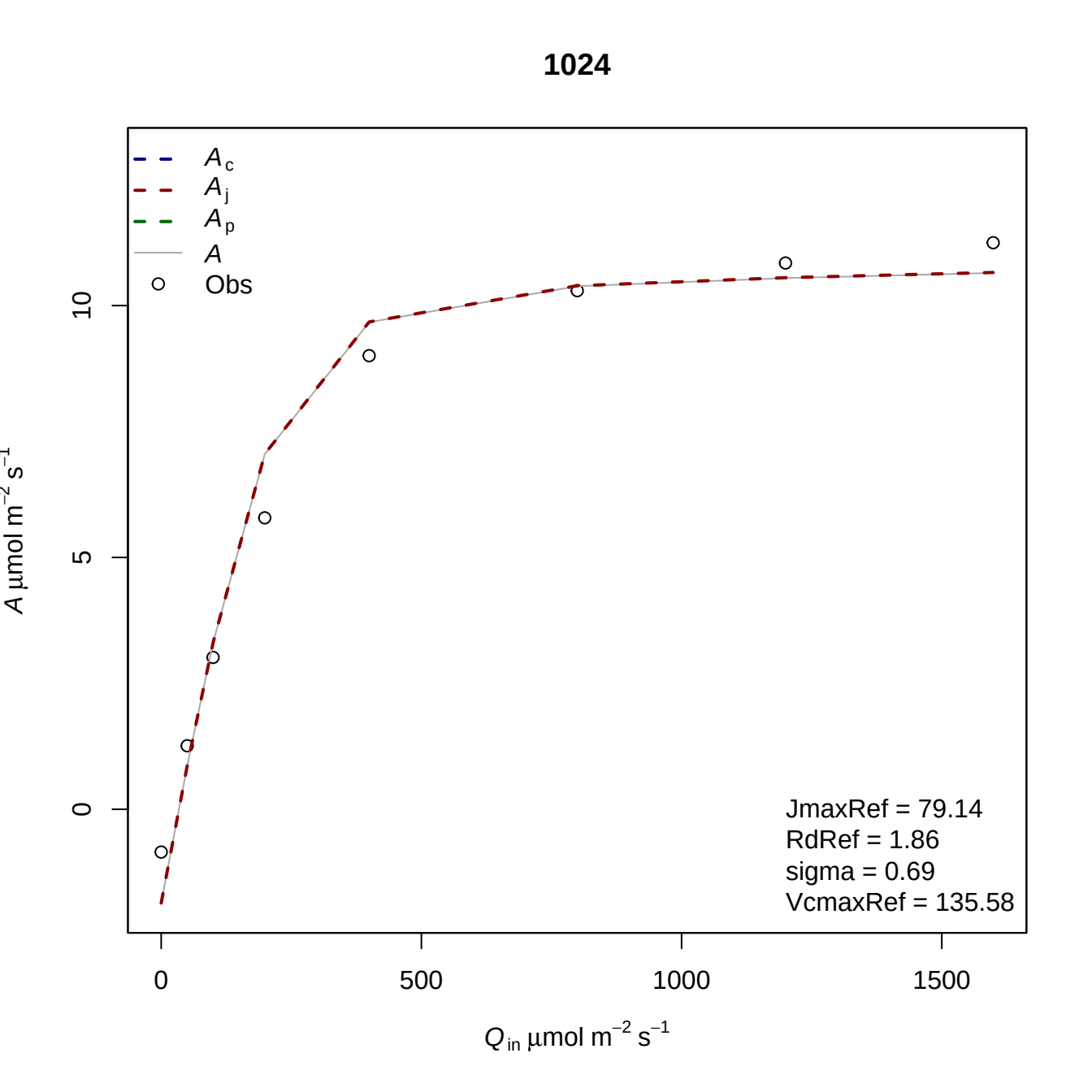

1017
A_c
A_j
A_p
A
Obs
JmaxRef = 106.94
RdRef = 2.01
sigma = 0.83
VcmaxRef = 228.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

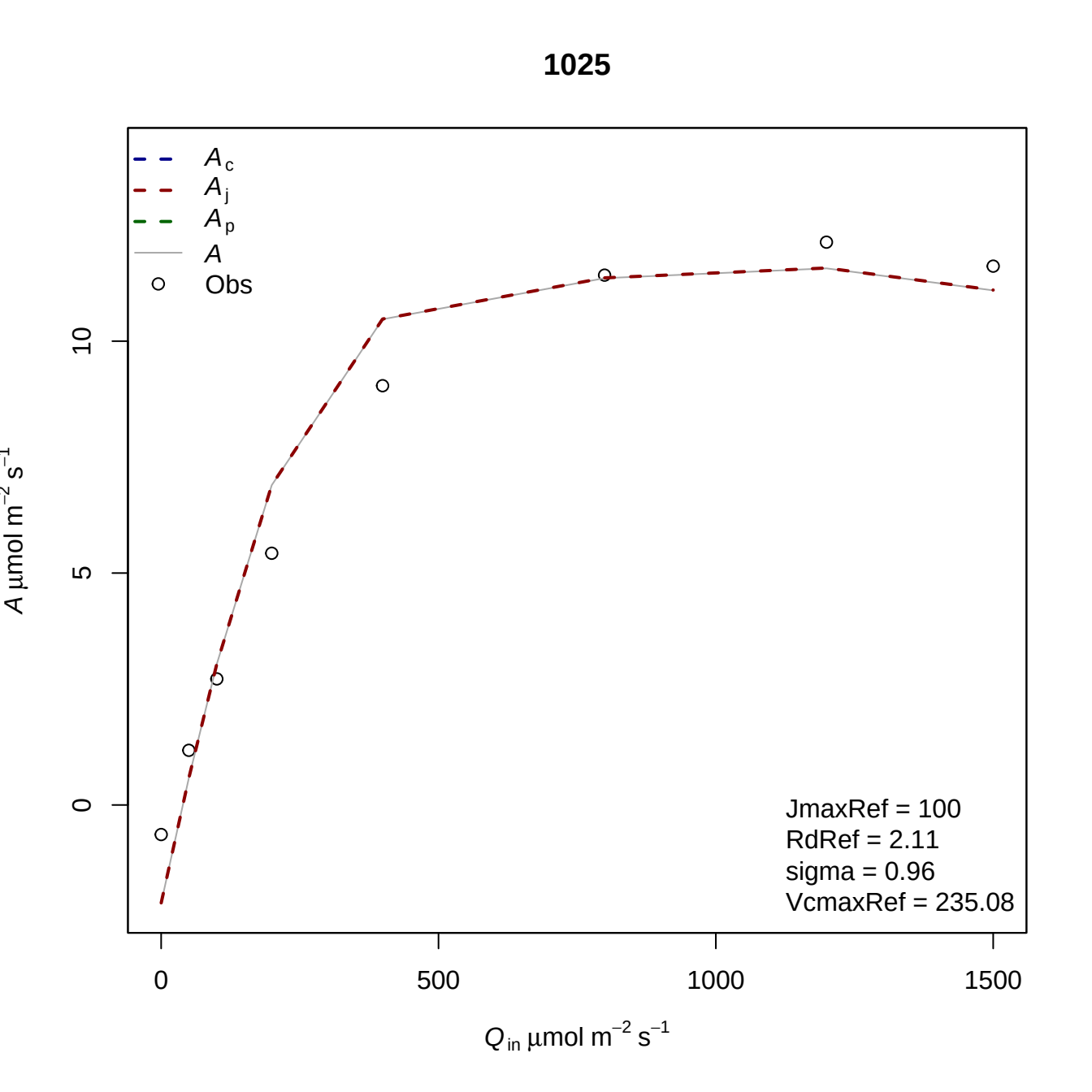

1018
A_c
A_j
A_p
A
Obs
JmaxRef = 84.45
RdRef = 2.19
sigma = 1.25
VcmaxRef = 472.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

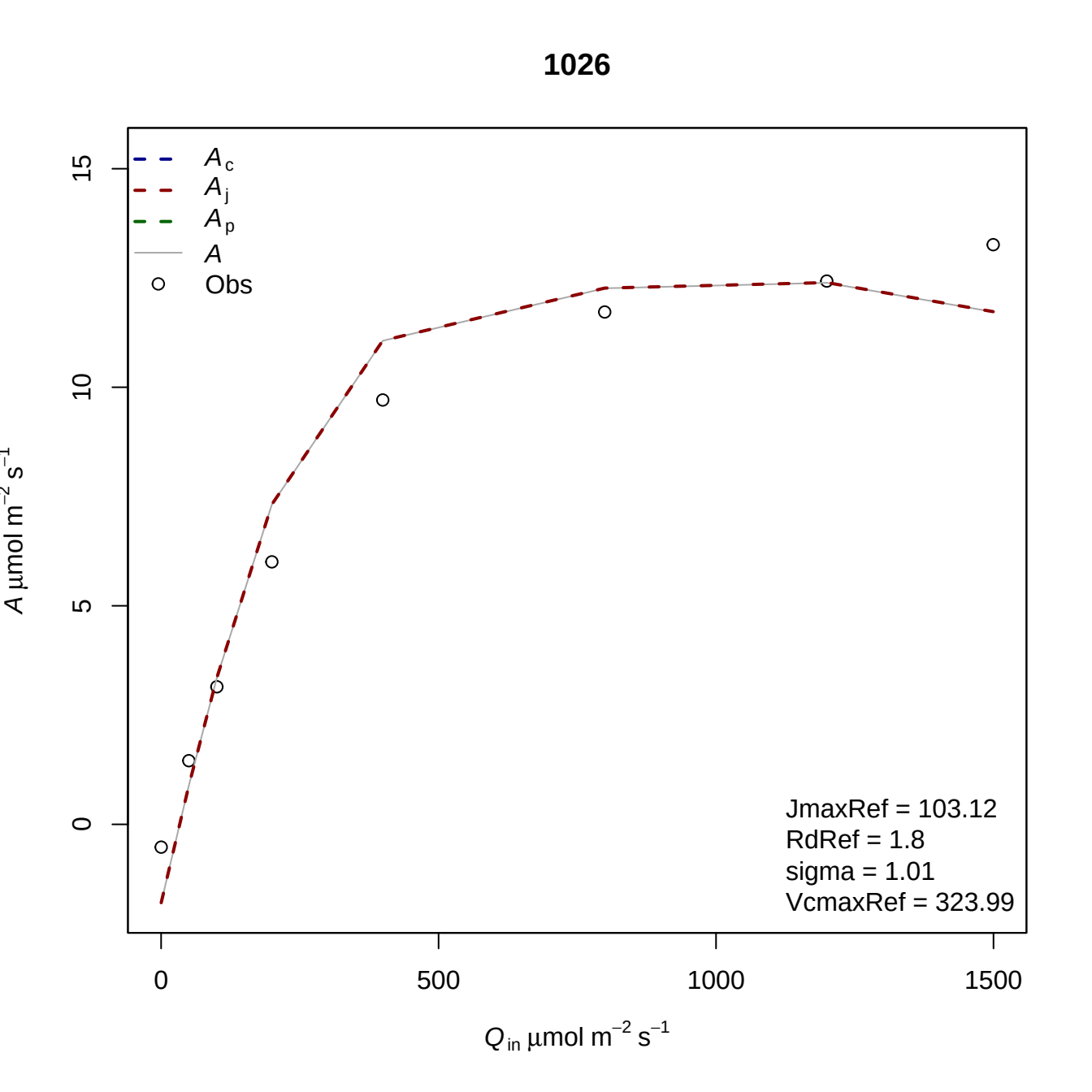

1019
A_c
A_j
A_p
A
Obs
JmaxRef = 80.77
RdRef = 1.78
sigma = 0.68
VcmaxRef = 225.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

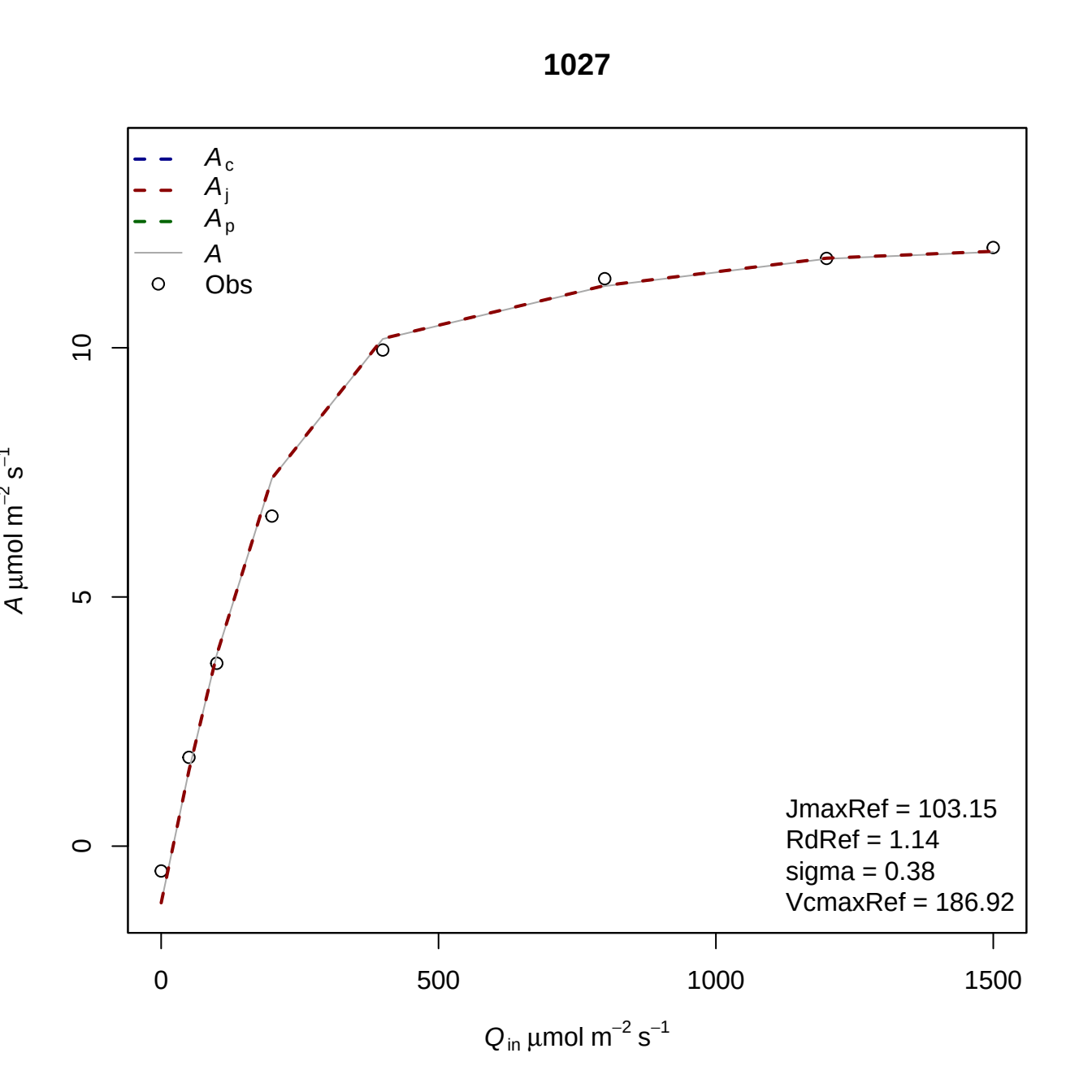

1020
A_c
A_j
A_p
A
Obs
JmaxRef = 98.59
RdRef = 1.28
sigma = 0.32
VcmaxRef = 158.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

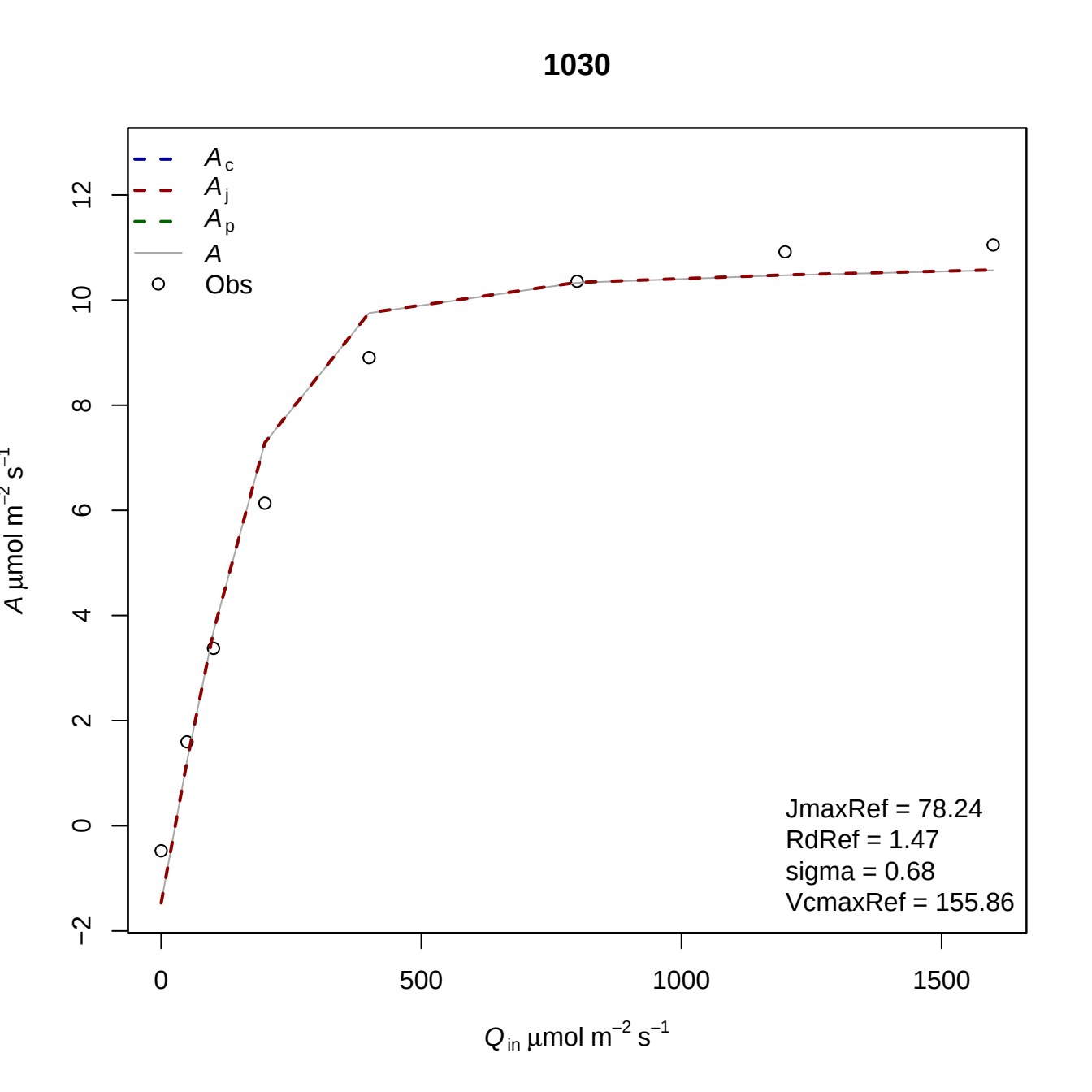

1021
A_c
A_j
A_p
A
Obs
JmaxRef = 109.96
RdRef = 1.51
sigma = 0.38
VcmaxRef = 189.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

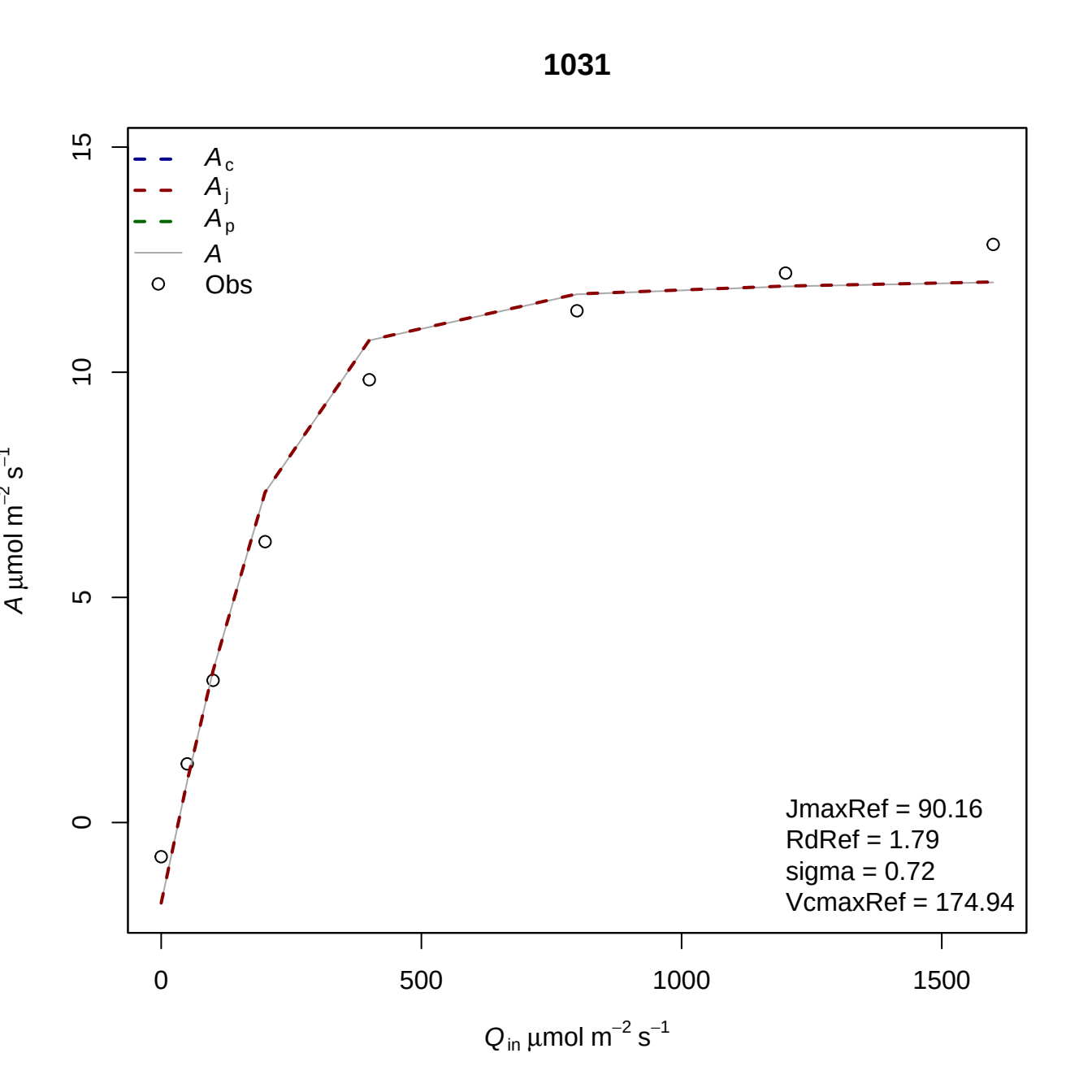

1024
A_c
A_j
A_p
A
Obs
JmaxRef = 79.14
RdRef = 1.86
sigma = 0.69
VcmaxRef = 135.58
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

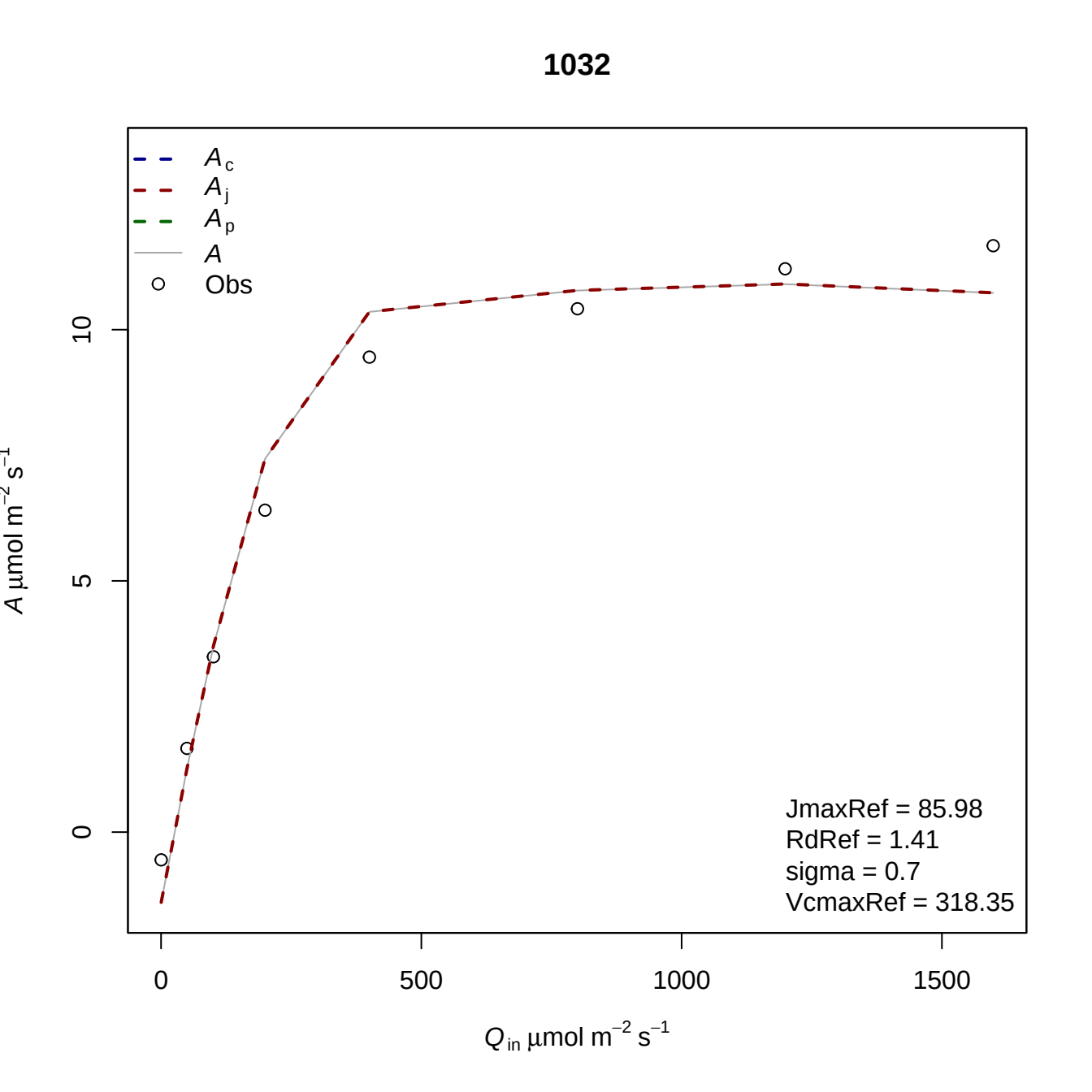

1025
A_c
A_j
A_p
A
Obs
JmaxRef = 100
RdRef = 2.11
sigma = 0.96
VcmaxRef = 235.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

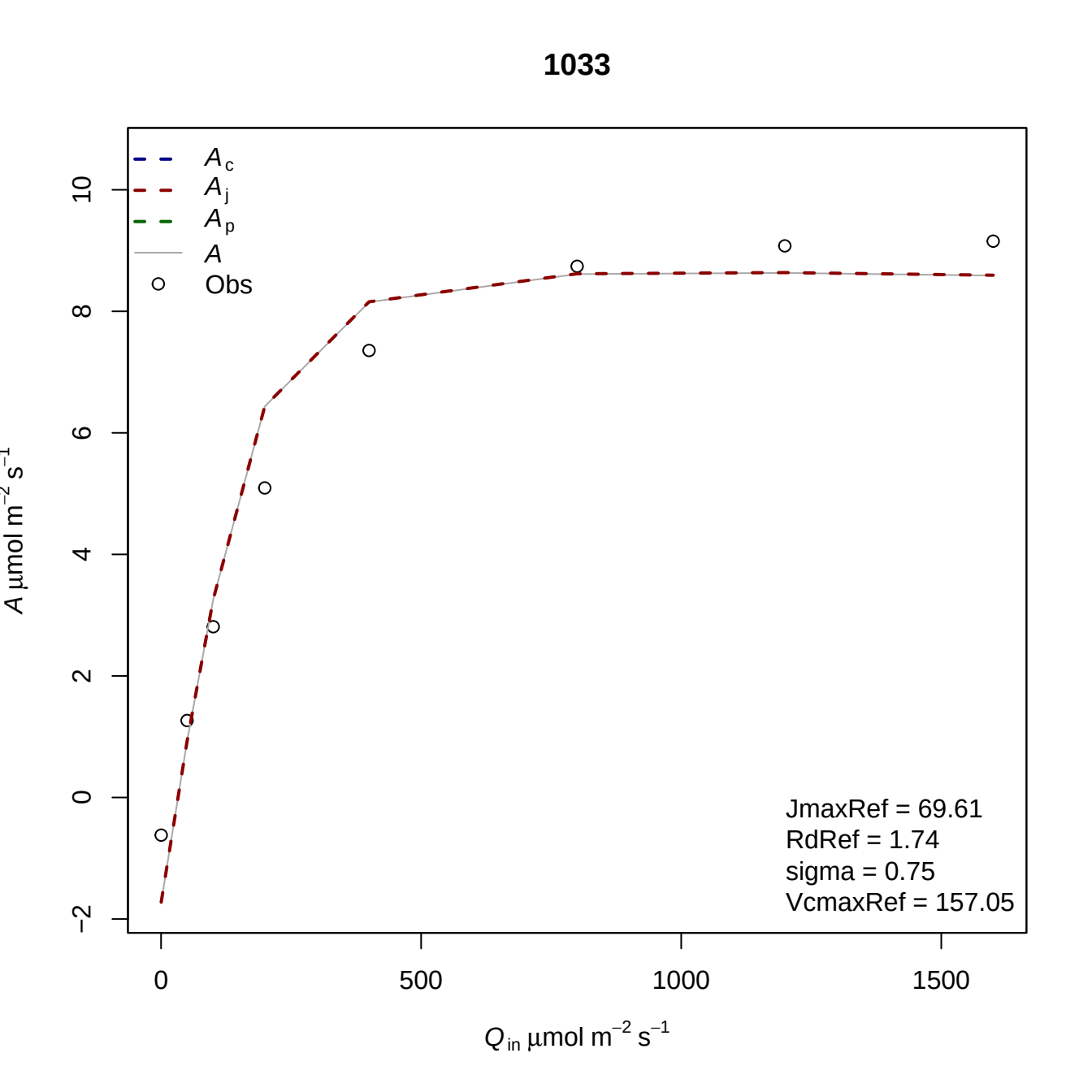

1026
A_c
A_j
A_p
A
Obs
JmaxRef = 103.12
RdRef = 1.8
sigma = 1.01
VcmaxRef = 323.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

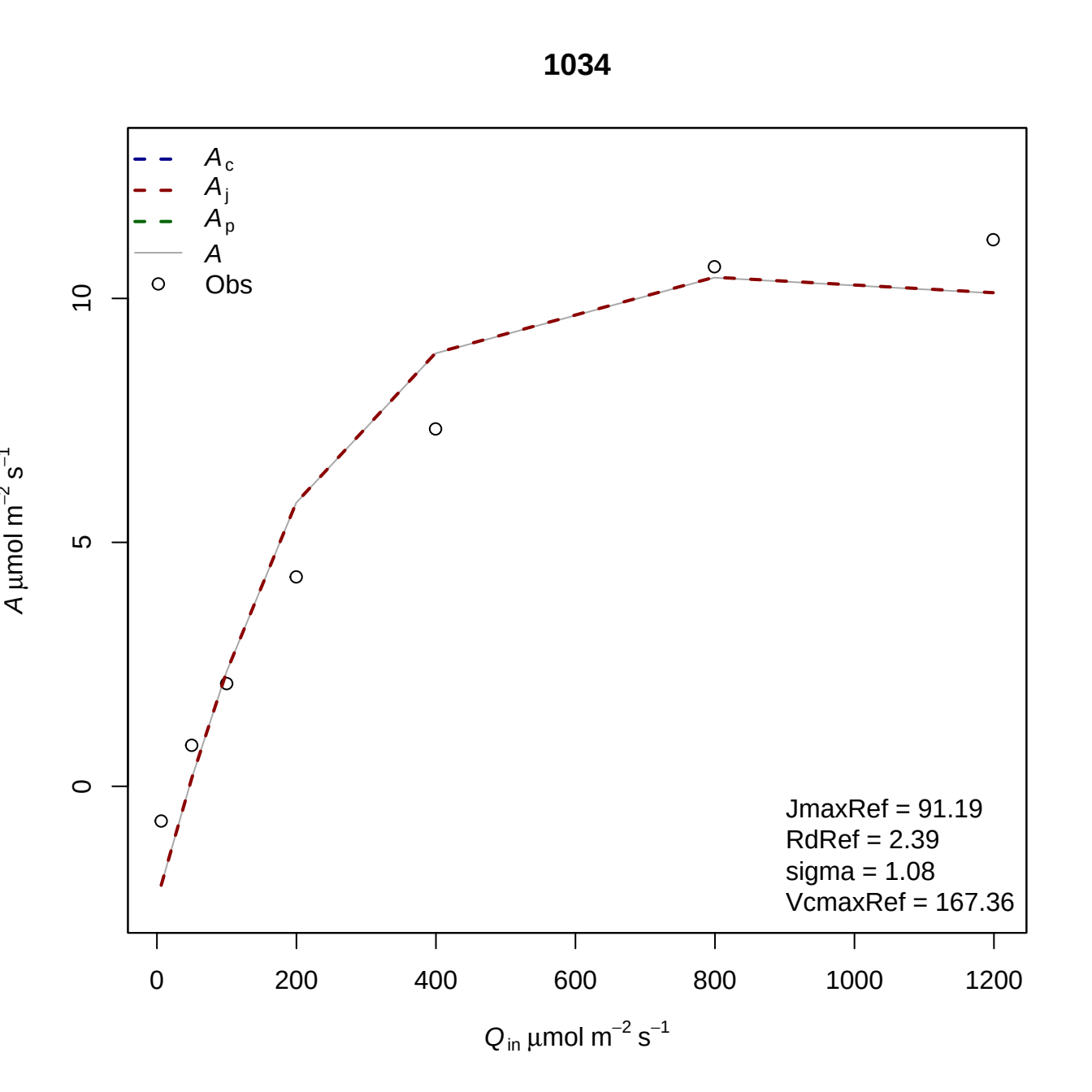

1027
A_c
A_j
A_p
A
Obs
JmaxRef = 103.15
RdRef = 1.14
sigma = 0.38
VcmaxRef = 186.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

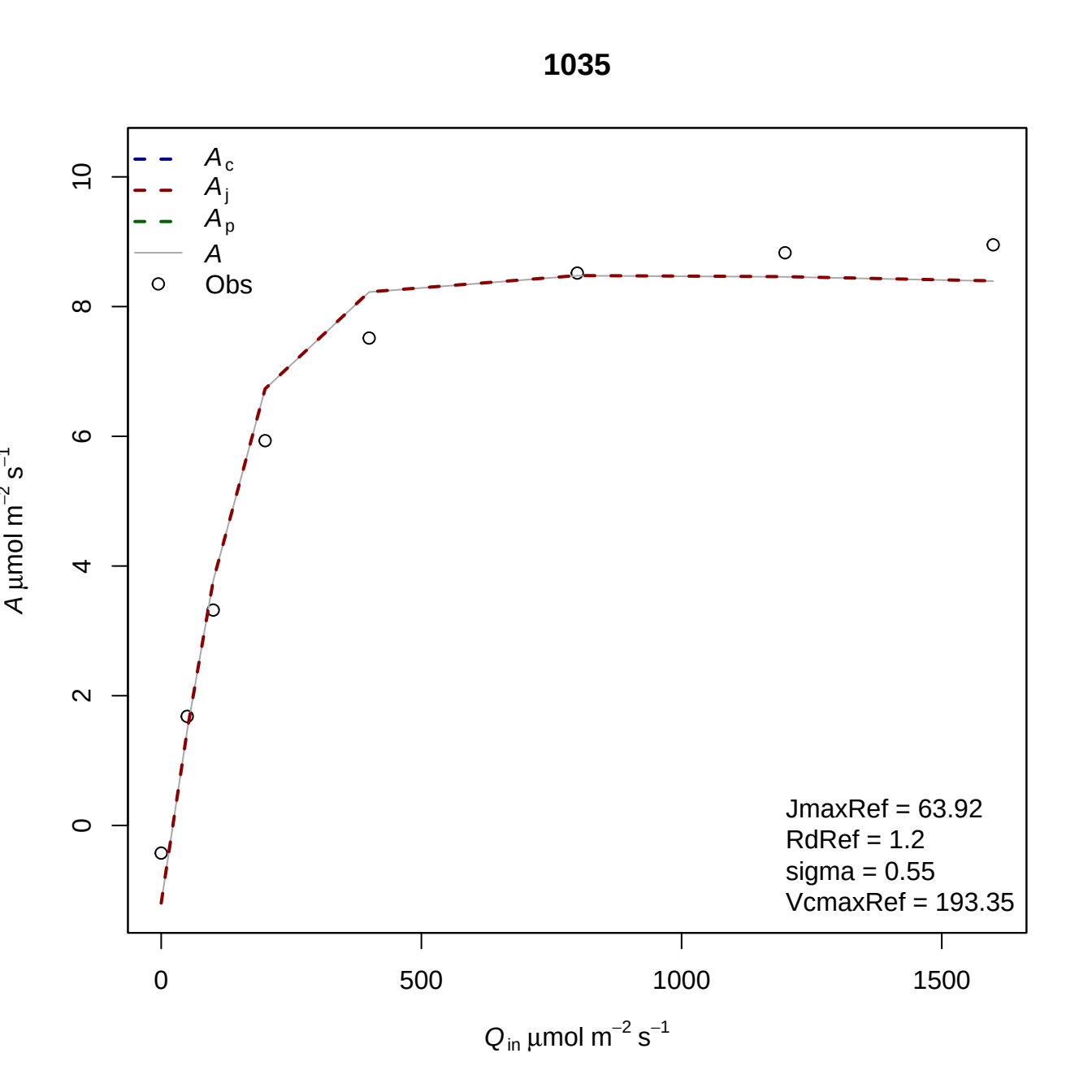

1030
A_c
A_j
A_p
A
Obs
JmaxRef = 78.24
RdRef = 1.47
sigma = 0.68
VcmaxRef = 155.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

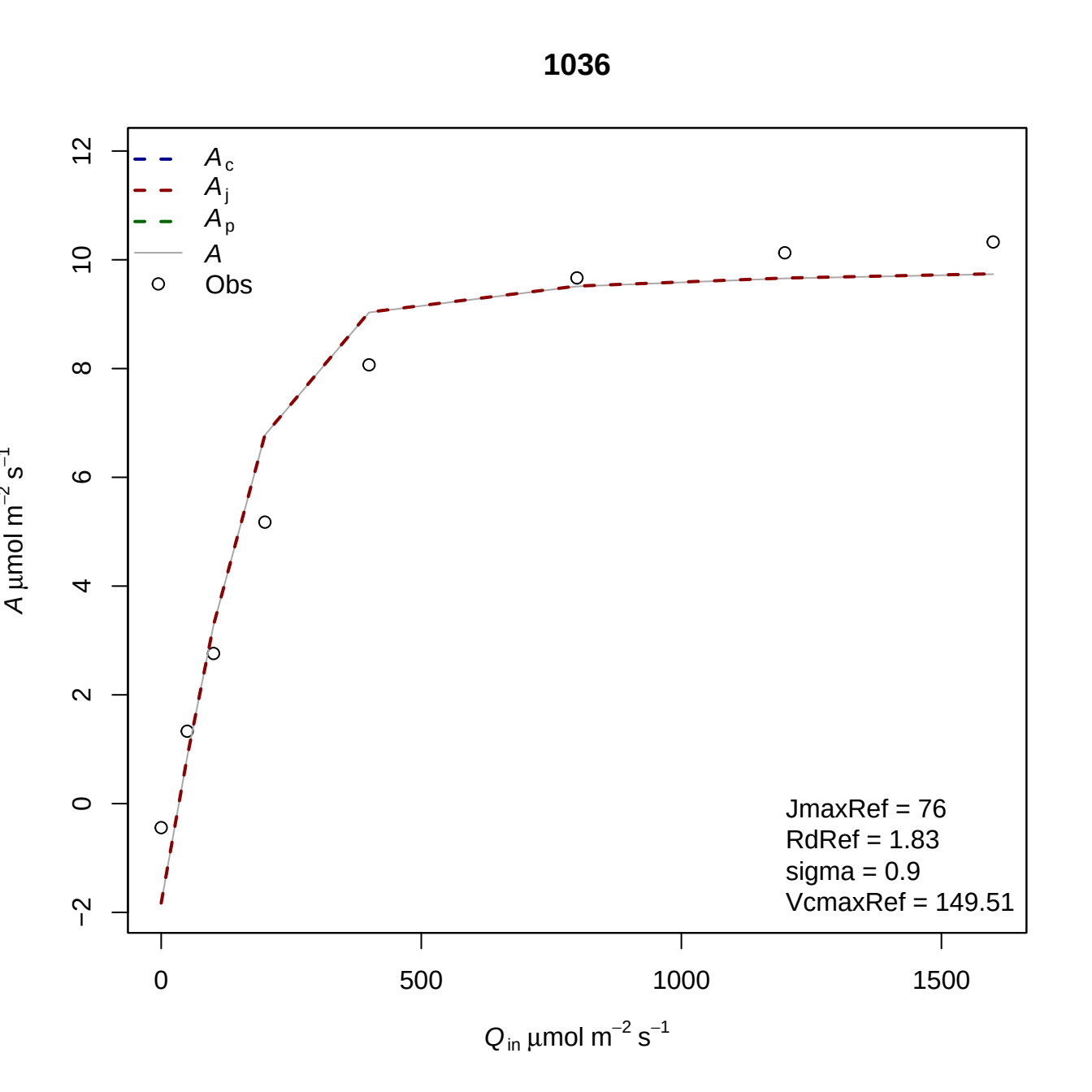

1031
A_c
A_j
A_p
A
Obs
JmaxRef = 90.16
RdRef = 1.79
sigma = 0.72
VcmaxRef = 174.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

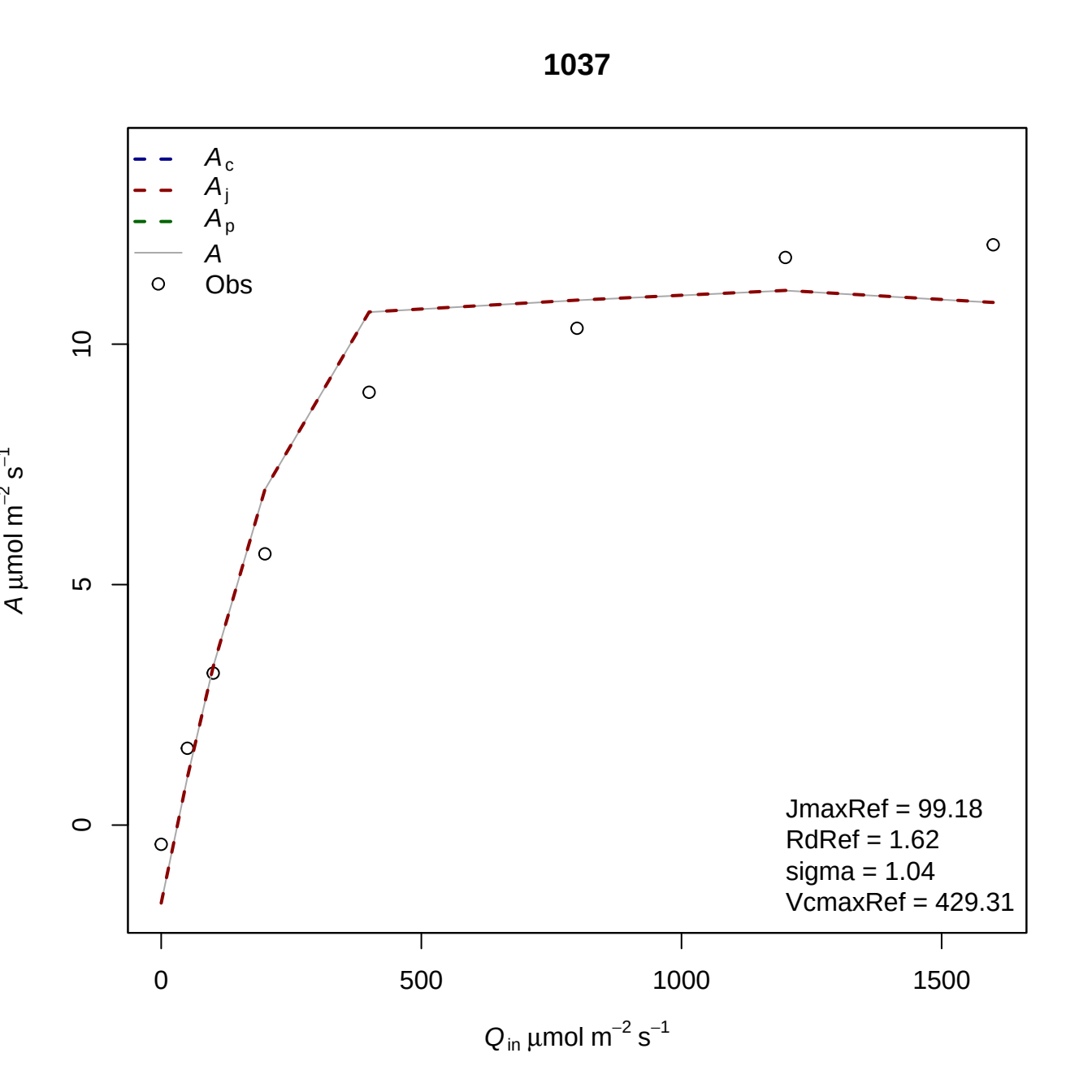

1032
A_c
A_j
A_p
A
Obs
JmaxRef = 85.98
RdRef = 1.41
sigma = 0.7
VcmaxRef = 318.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

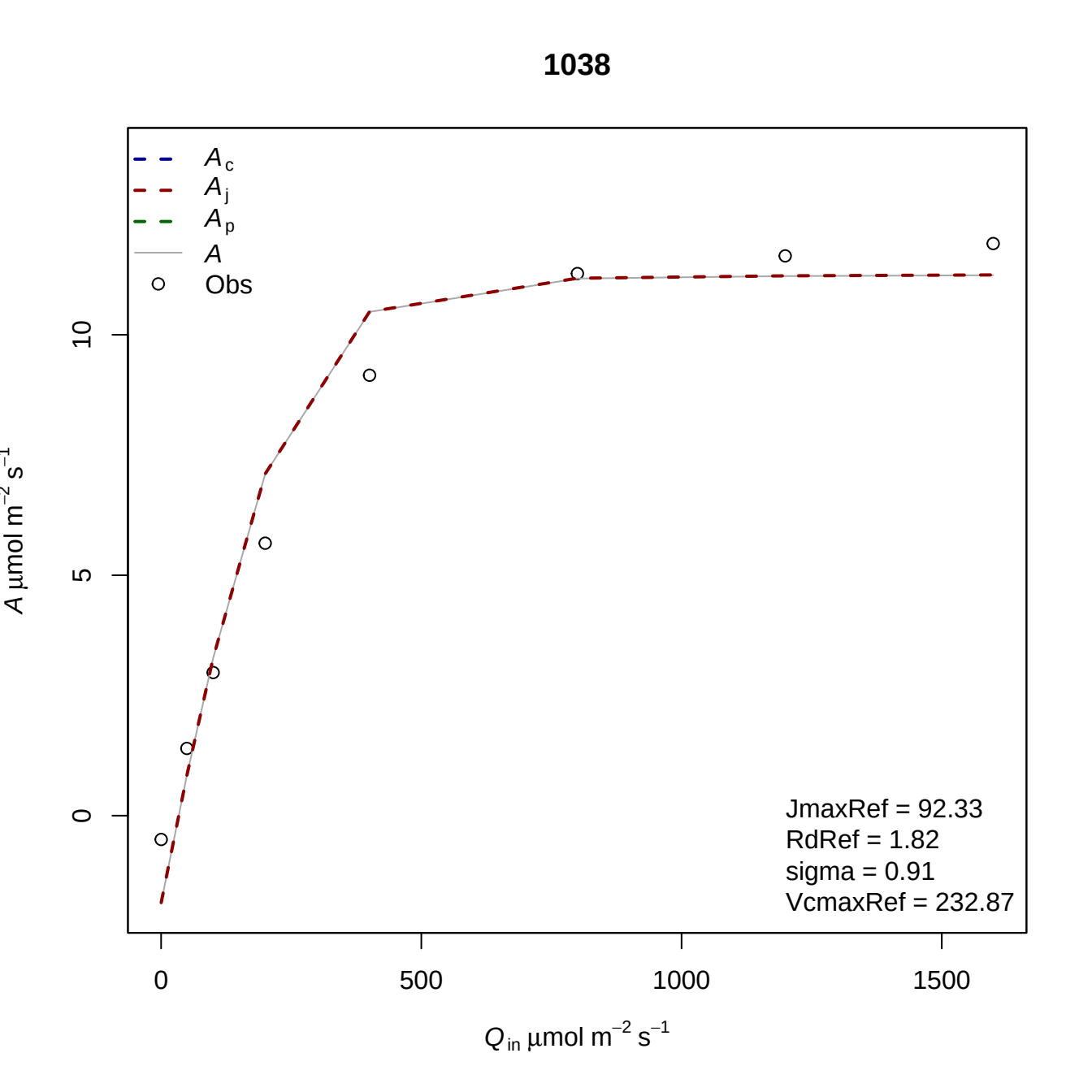

1033
A_c
A_j
A_p
A
Obs
JmaxRef = 69.61
RdRef = 1.74
sigma = 0.75
VcmaxRef = 157.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

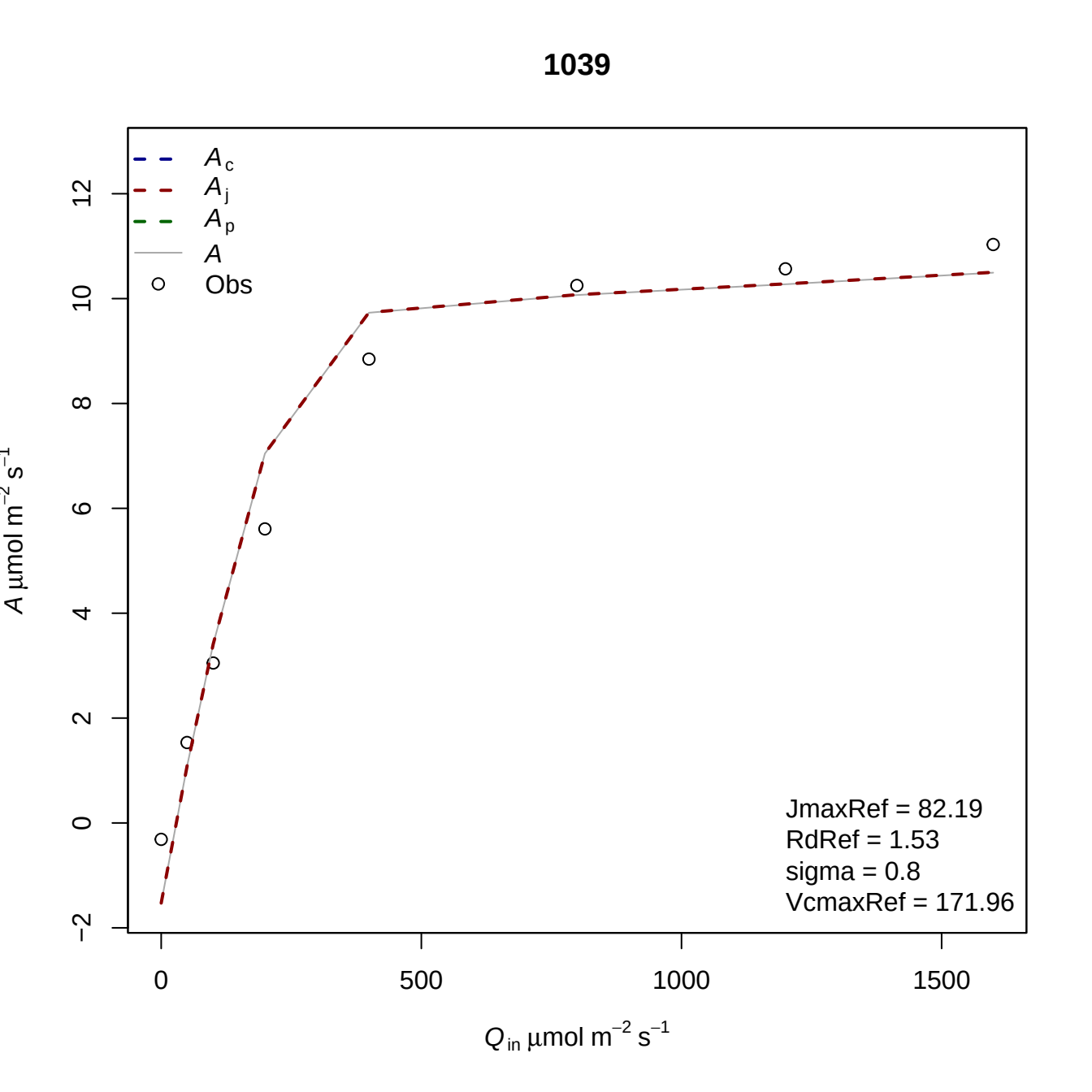

1034
A_c
A_j
A_p
A
Obs
JmaxRef = 91.19
RdRef = 2.39
sigma = 1.08
VcmaxRef = 167.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

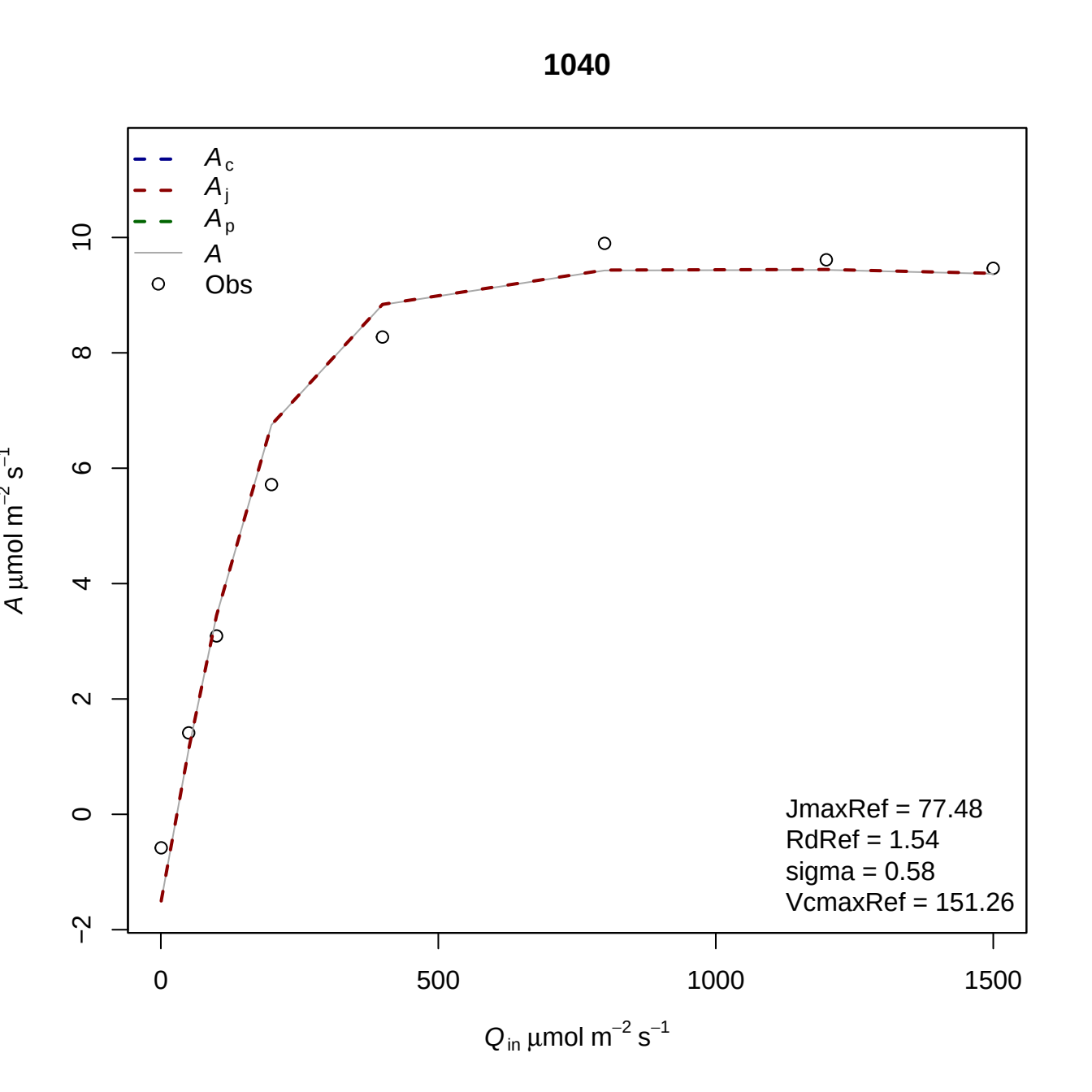

1035
A_c
A_j
A_p
A
Obs
JmaxRef = 63.92
RdRef = 1.2
sigma = 0.55
VcmaxRef = 193.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

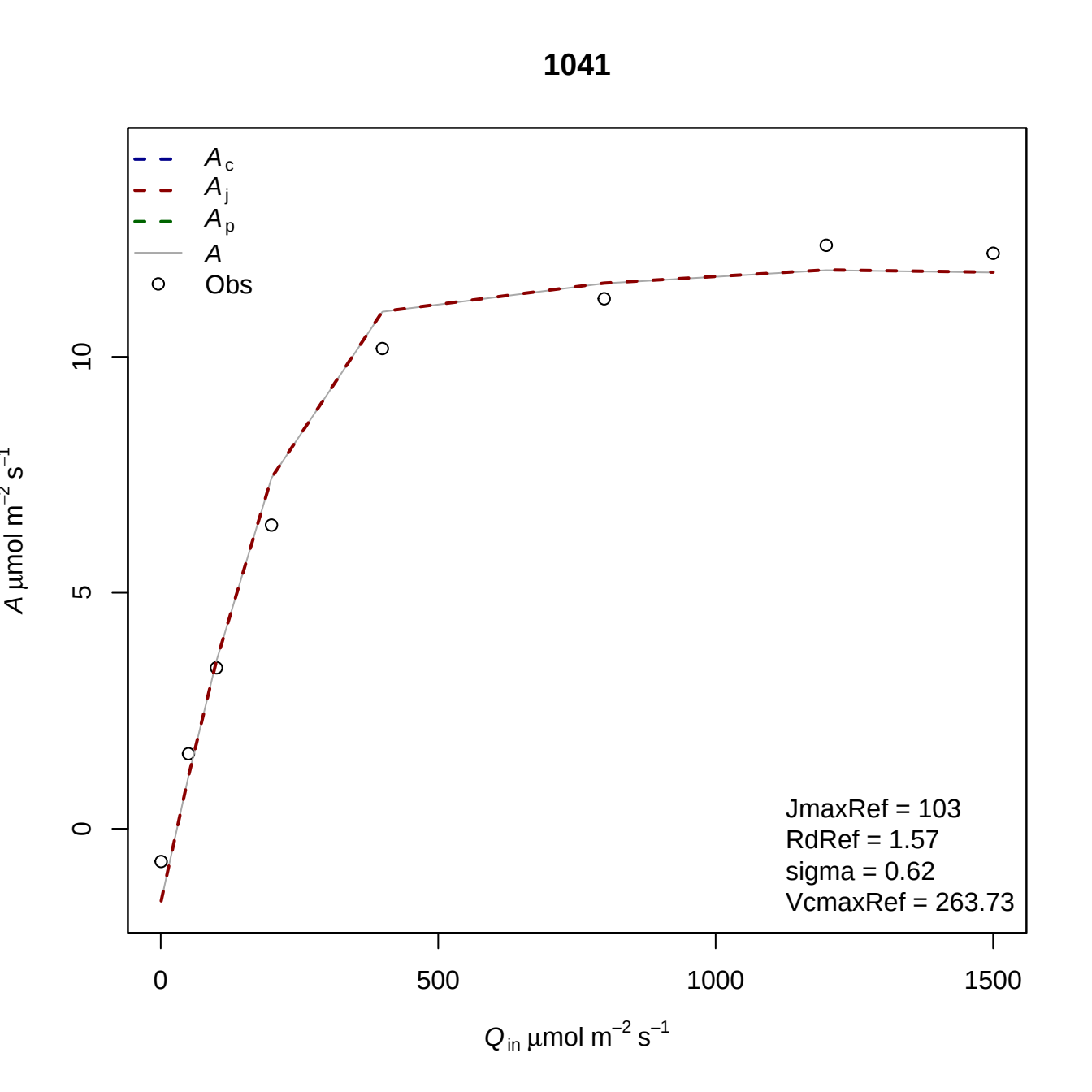

1036
A_c
A_j
A_p
A
Obs
JmaxRef = 76
RdRef = 1.83
sigma = 0.9
VcmaxRef = 149.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

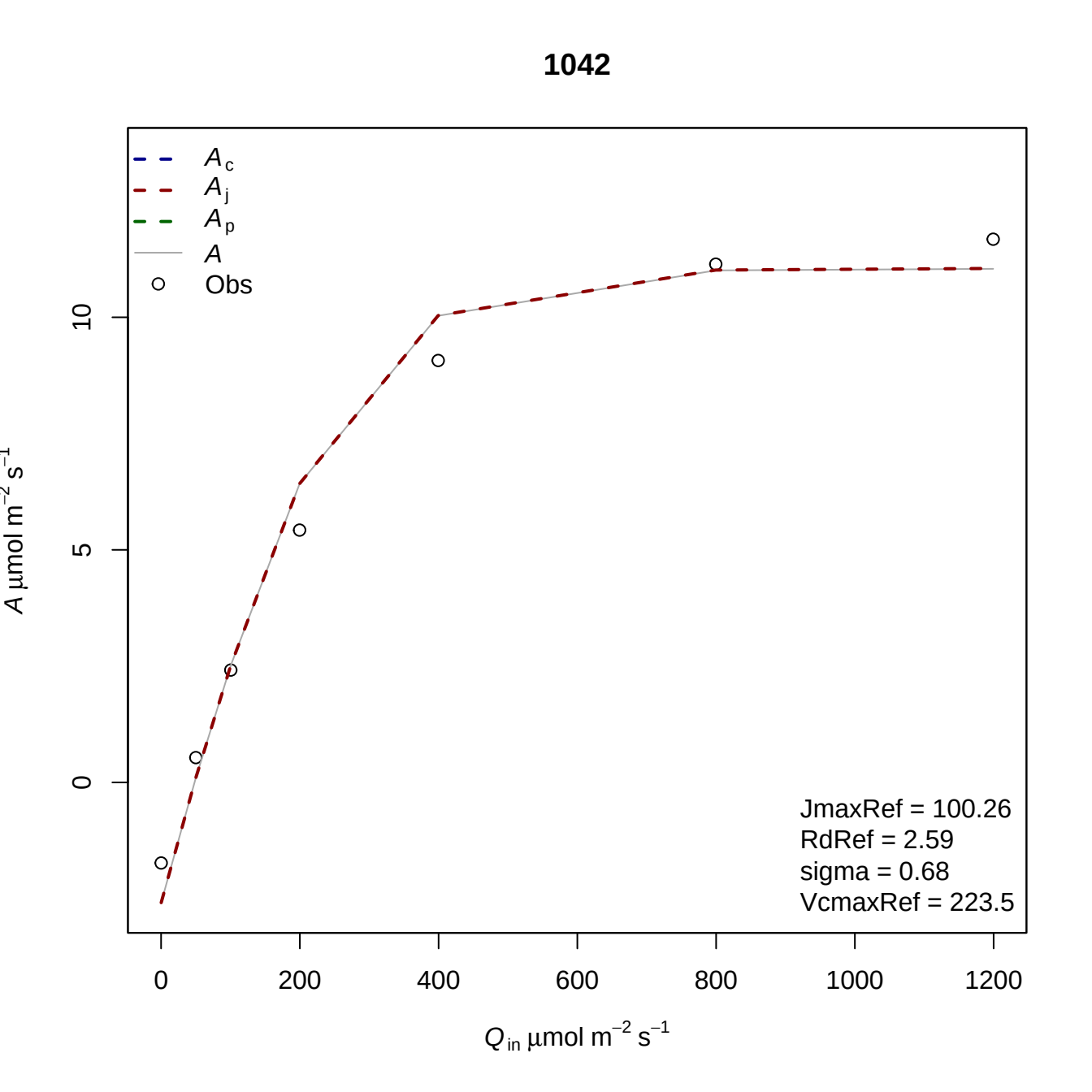

1037
A_c
A_j
A_p
A
Obs
JmaxRef = 99.18
RdRef = 1.62
sigma = 1.04
VcmaxRef = 429.31
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

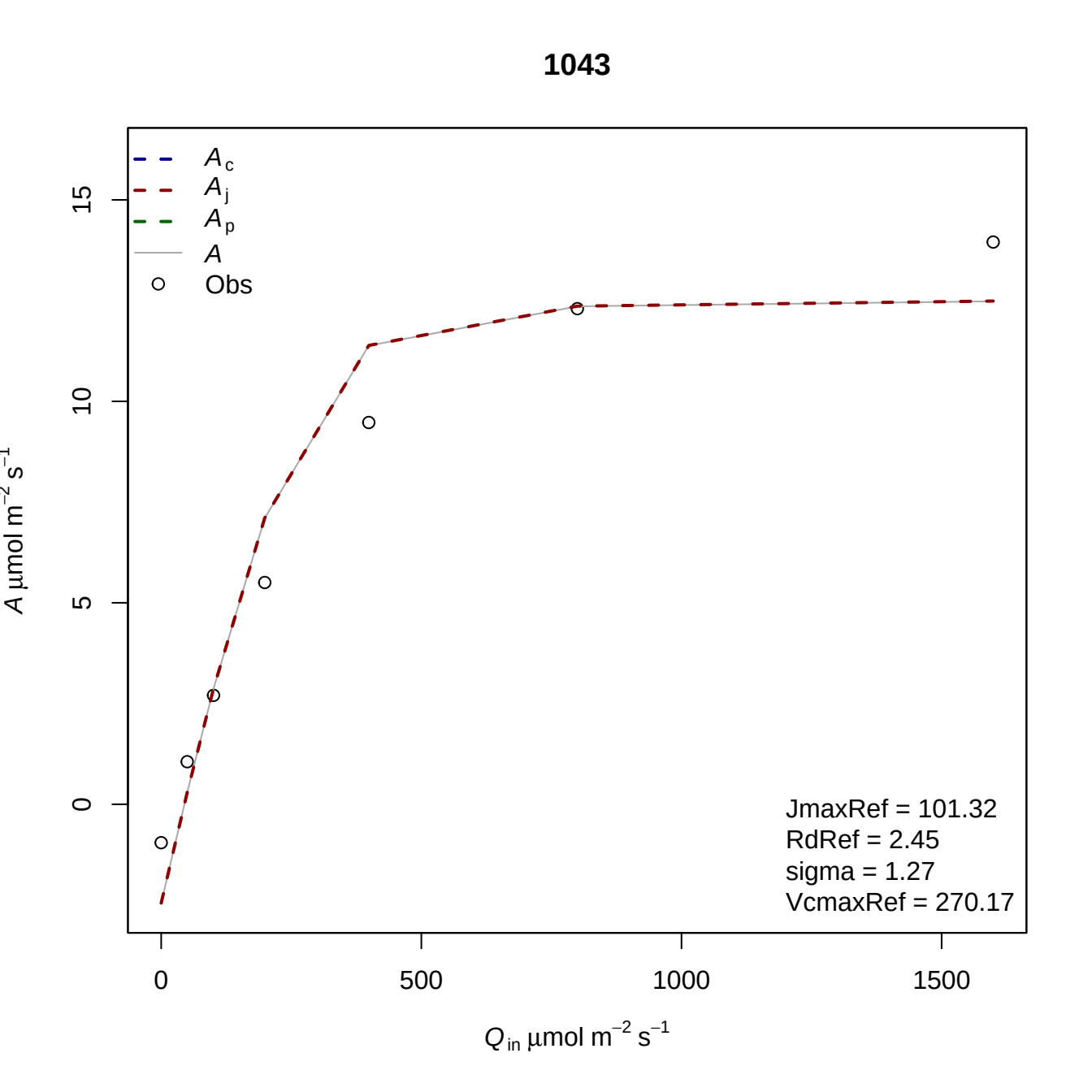

1038
A_c
A_j
A_p
A
Obs
JmaxRef = 92.33
RdRef = 1.82
sigma = 0.91
VcmaxRef = 232.87
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

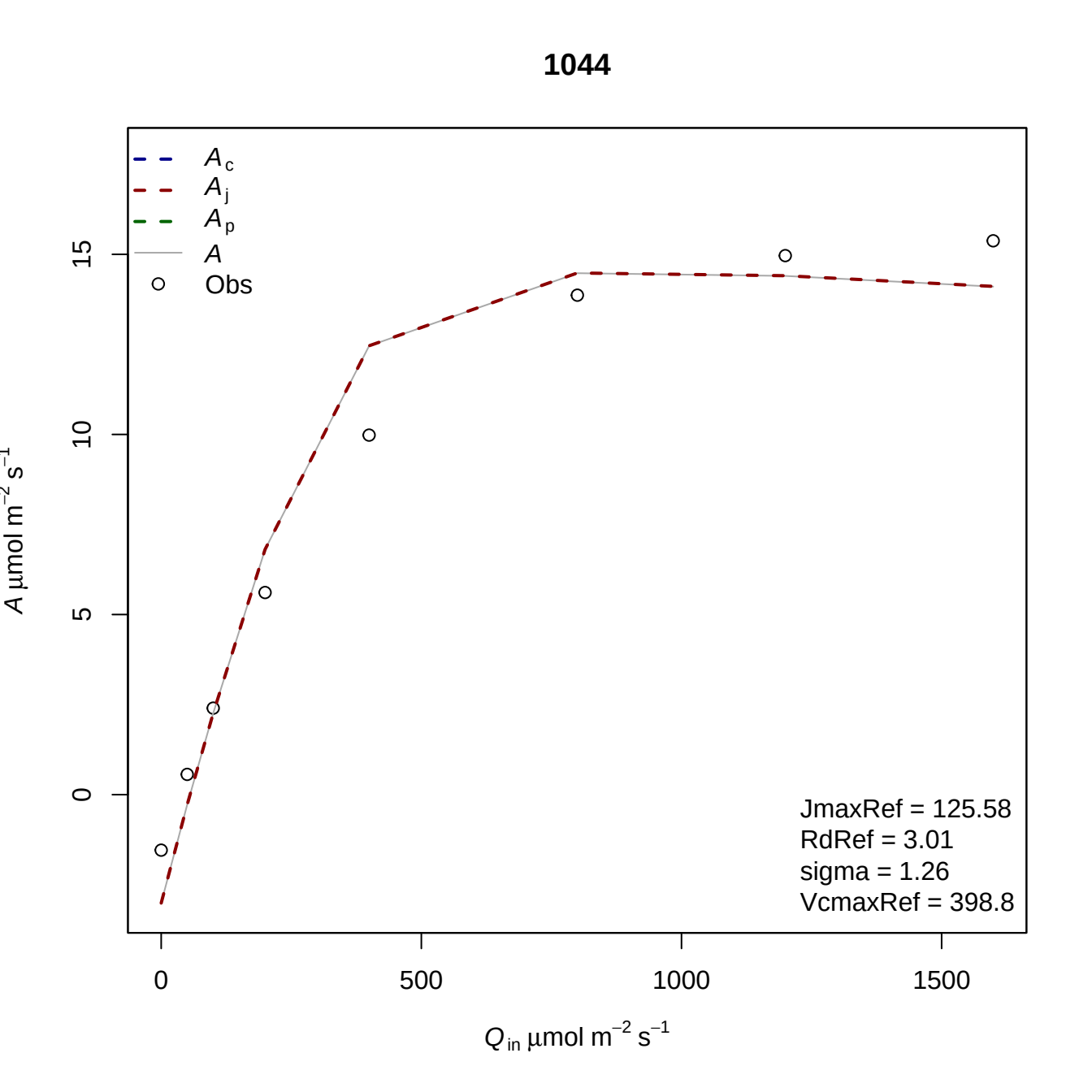

1039
A_c
A_j
A_p
A
Obs
JmaxRef = 82.19
RdRef = 1.53
sigma = 0.8
VcmaxRef = 171.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

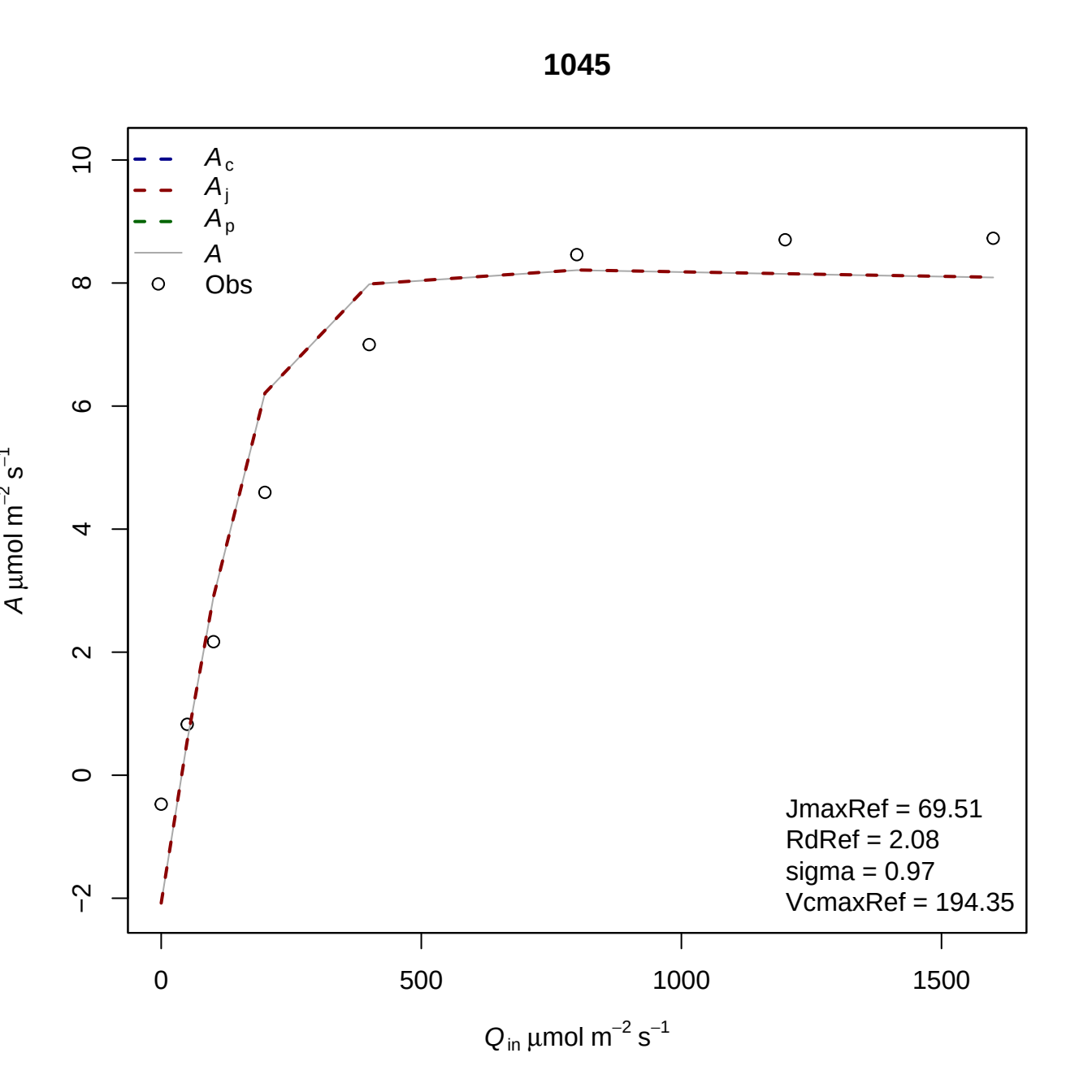

1040
A_c
A_j
A_p
A
Obs
JmaxRef = 77.48
RdRef = 1.54
sigma = 0.58
VcmaxRef = 151.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

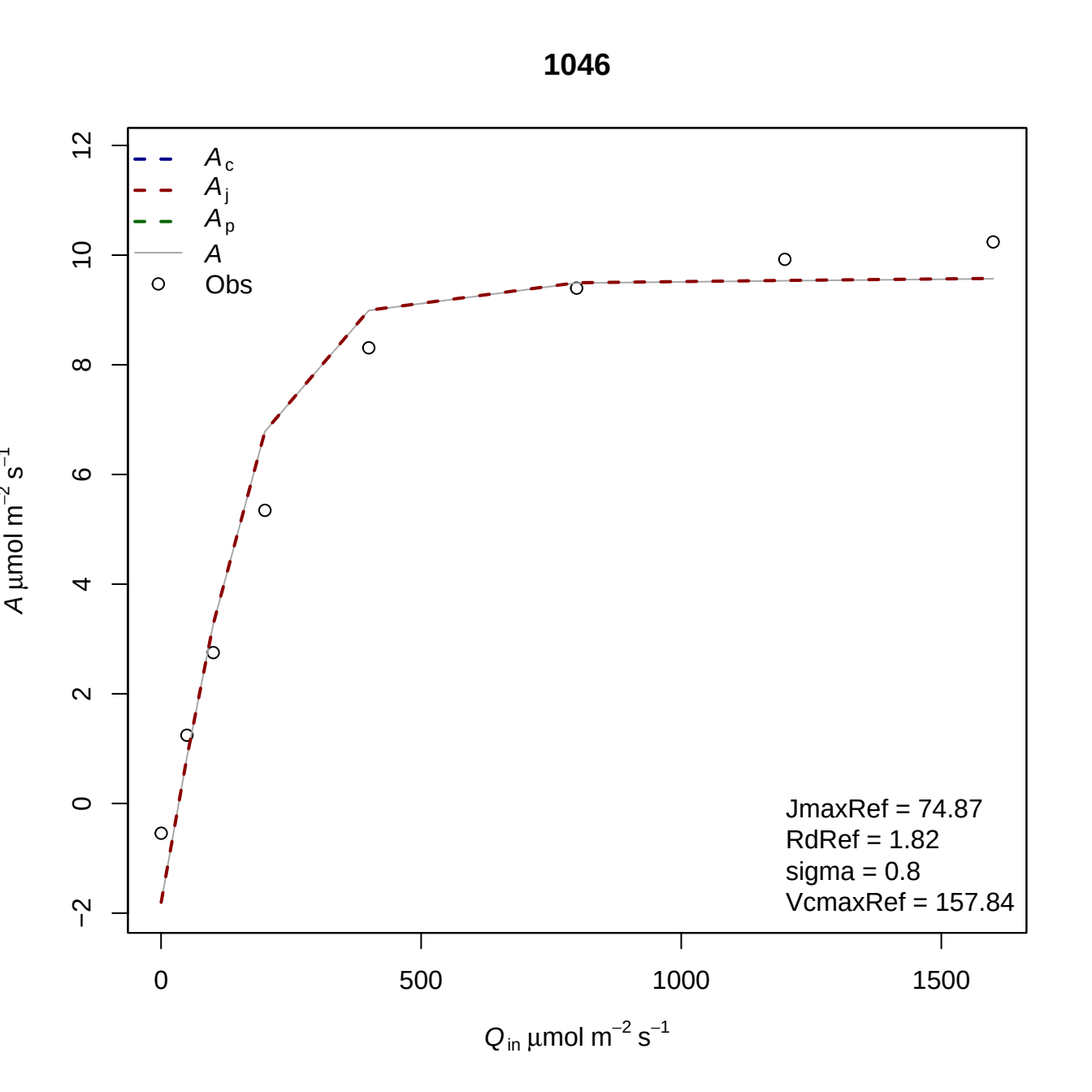

1041
A_c
A_j
A_p
A
Obs
JmaxRef = 103
RdRef = 1.57
sigma = 0.62
VcmaxRef = 263.73
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

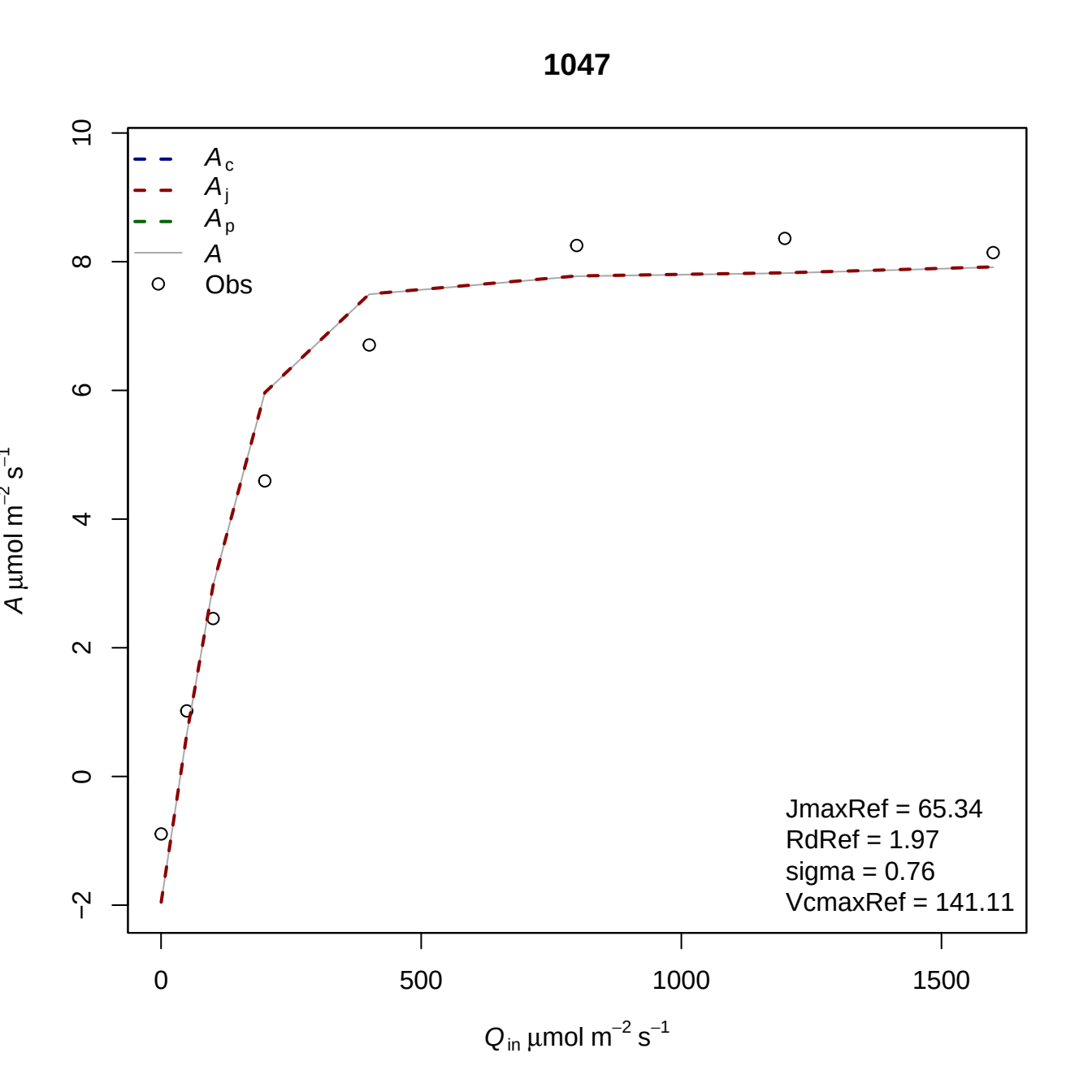

1042
A_c
A_j
A_p
A
Obs
JmaxRef = 100.26
RdRef = 2.59
sigma = 0.68
VcmaxRef = 223.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

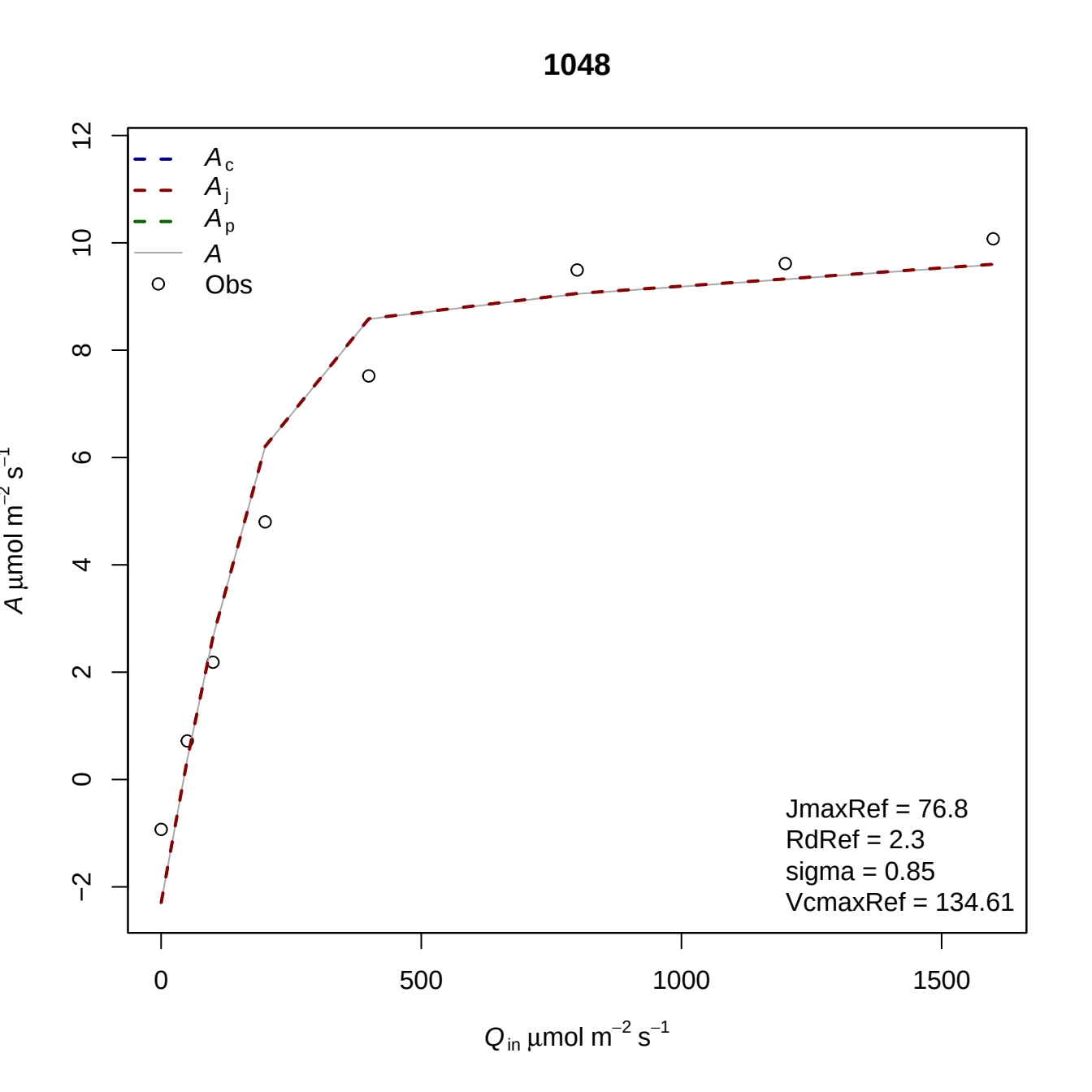

1043
A_c
A_j
A_p
A
Obs
JmaxRef = 101.32
RdRef = 2.45
sigma = 1.27
VcmaxRef = 270.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

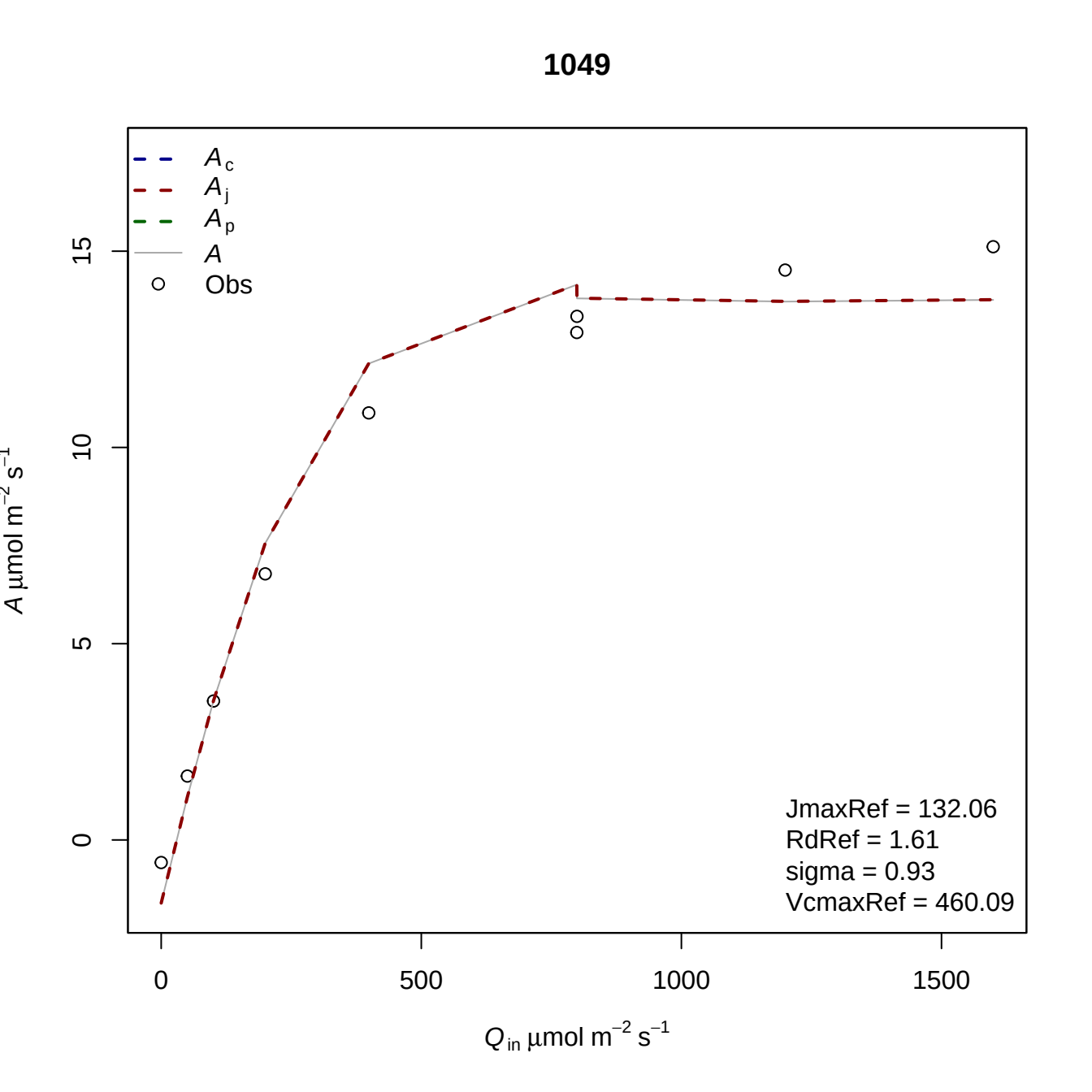

1044
A_c
A_j
A_p
A
Obs
JmaxRef = 125.58
RdRef = 3.01
sigma = 1.26
VcmaxRef = 398.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

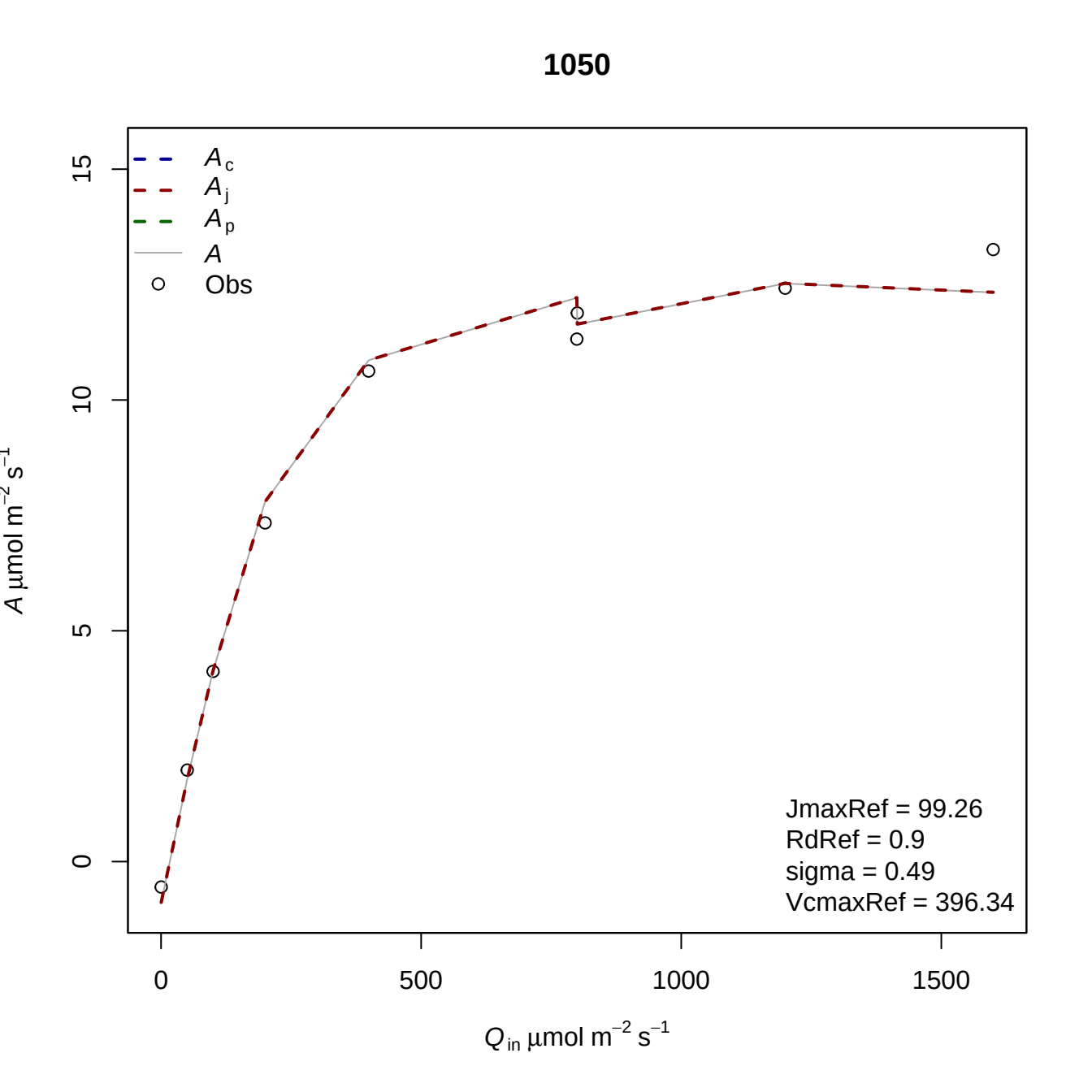

1045
A_c
A_j
A_p
A
Obs
JmaxRef = 69.51
RdRef = 2.08
sigma = 0.97
VcmaxRef = 194.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

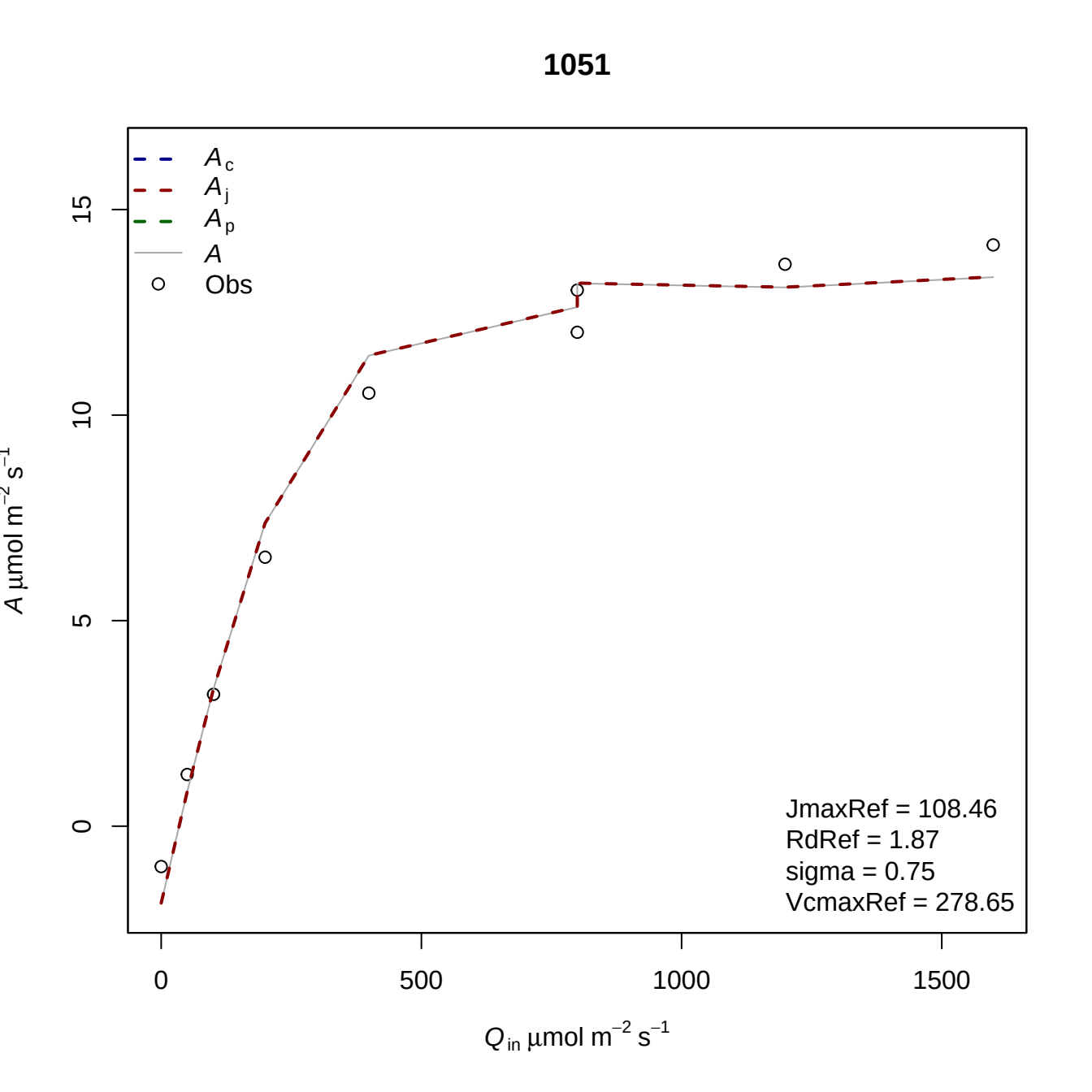

1046
A_c
A_j
A_p
A
Obs
JmaxRef = 74.87
RdRef = 1.82
sigma = 0.8
VcmaxRef = 157.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

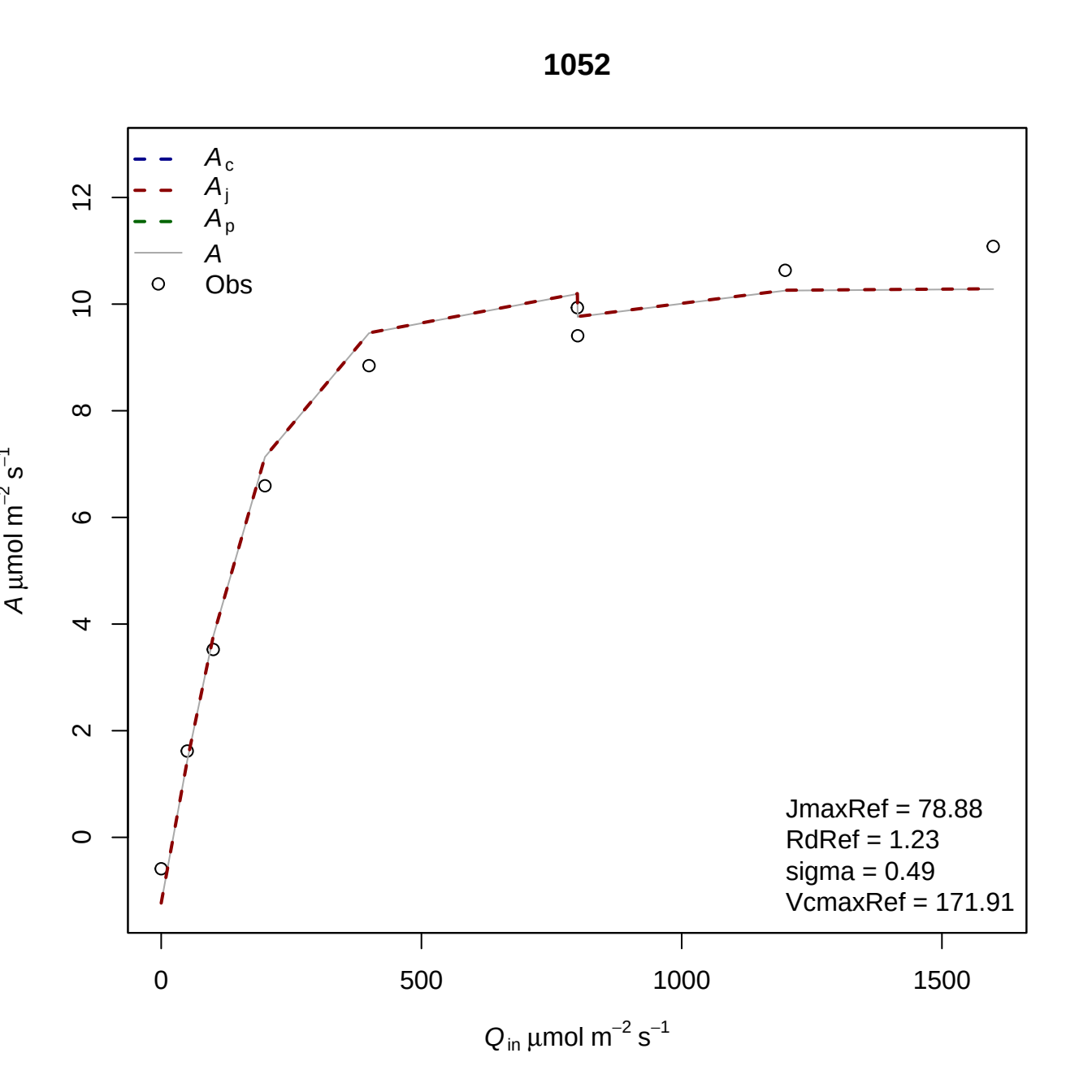

1047
A_c
A_j
A_p
A
Obs
JmaxRef = 65.34
RdRef = 1.97
sigma = 0.76
VcmaxRef = 141.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

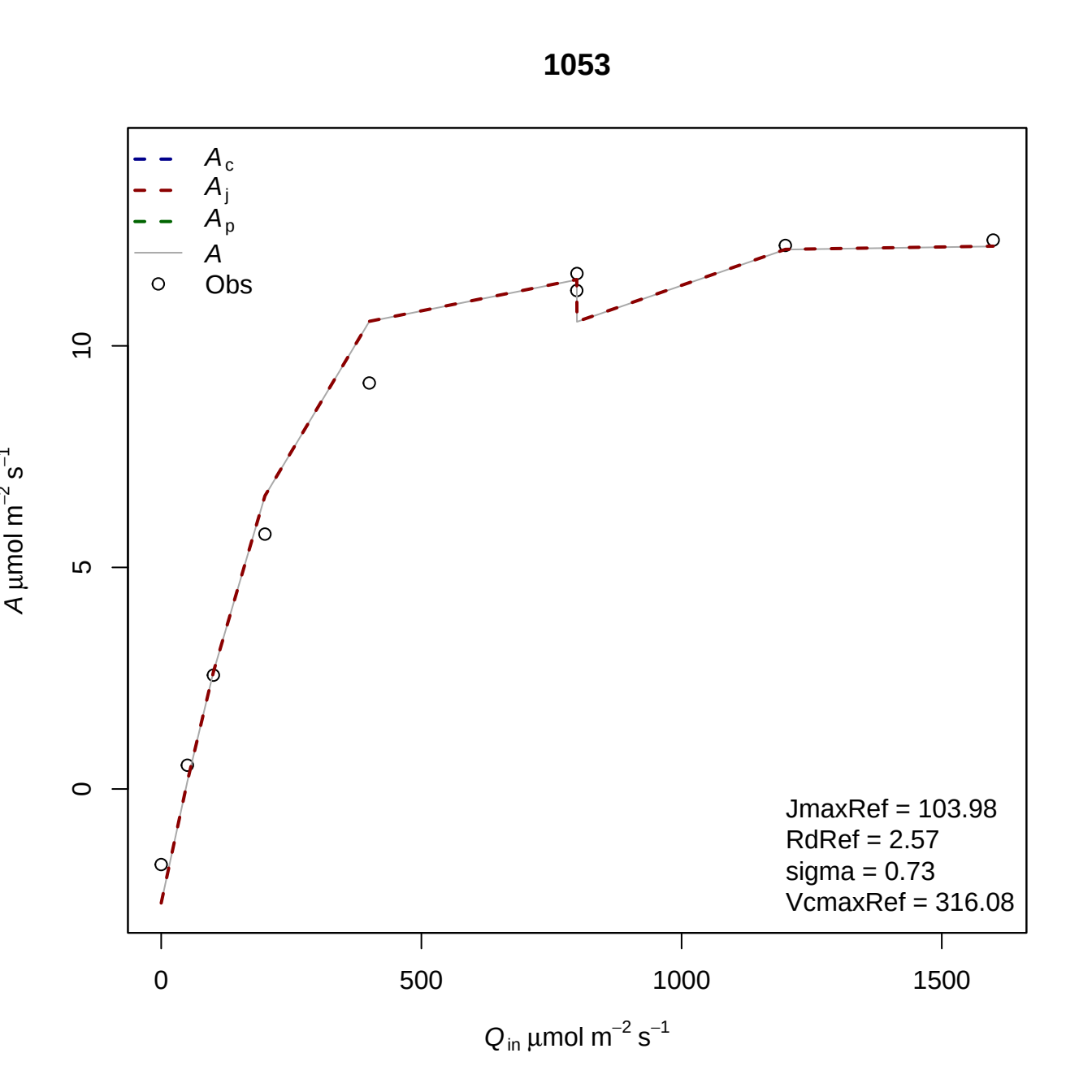

1048
A_c
A_j
A_p
A
Obs
JmaxRef = 76.8
RdRef = 2.3
sigma = 0.85
VcmaxRef = 134.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

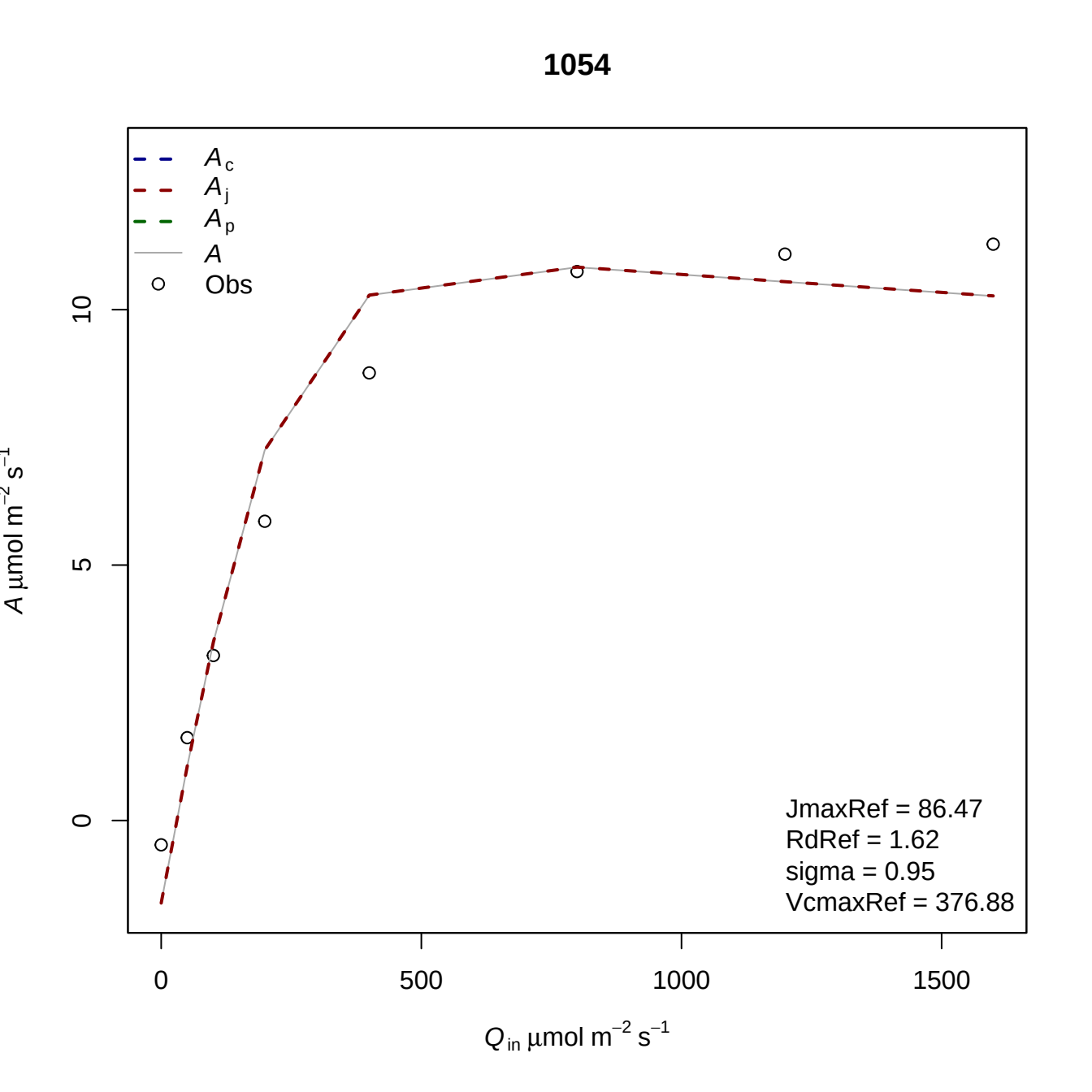

1049
A_c
A_j
A_p
A
Obs
JmaxRef = 132.06
RdRef = 1.61
sigma = 0.93
VcmaxRef = 460.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

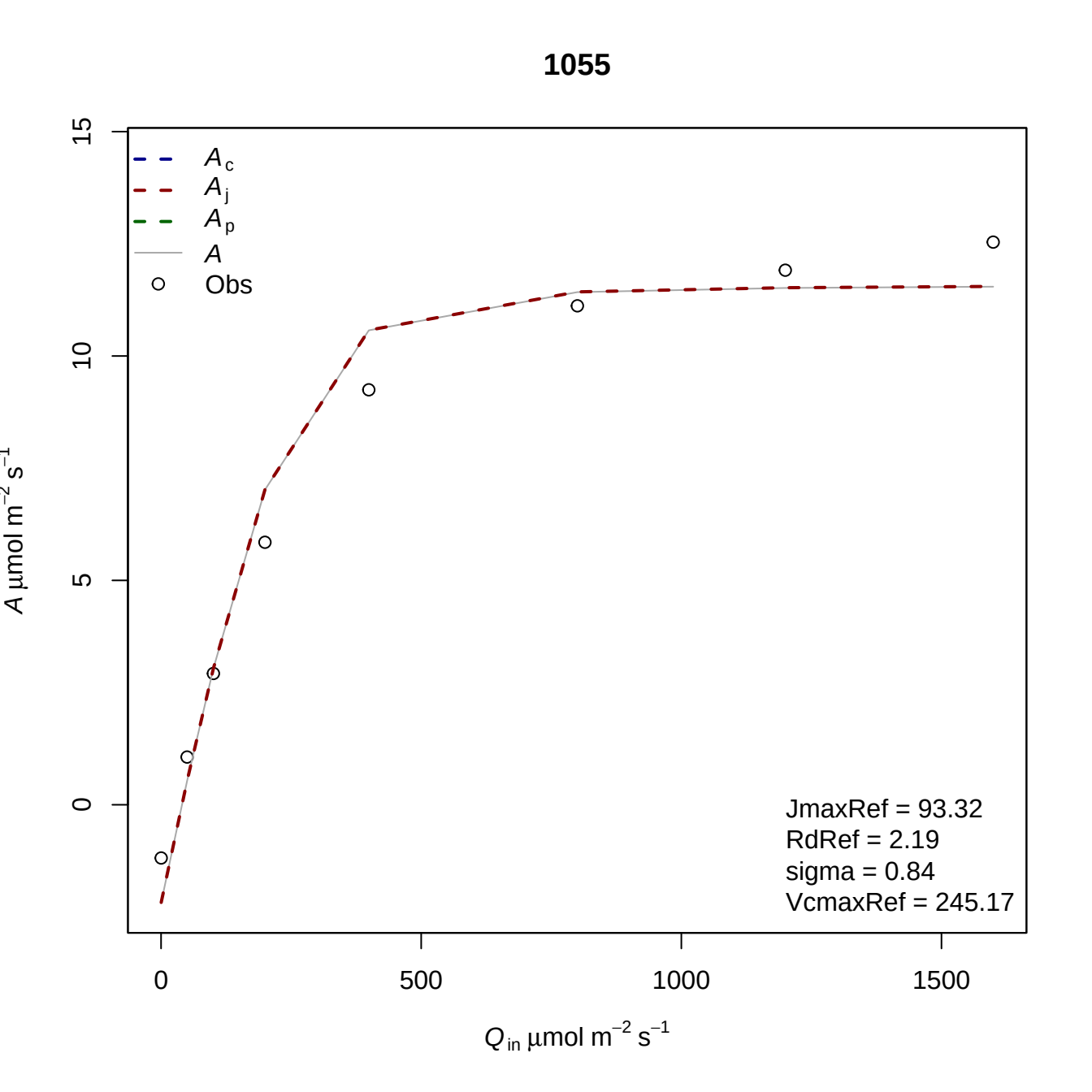

1050
A_c
A_j
A_p
A
Obs
JmaxRef = 99.26
RdRef = 0.9
sigma = 0.49
VcmaxRef = 396.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

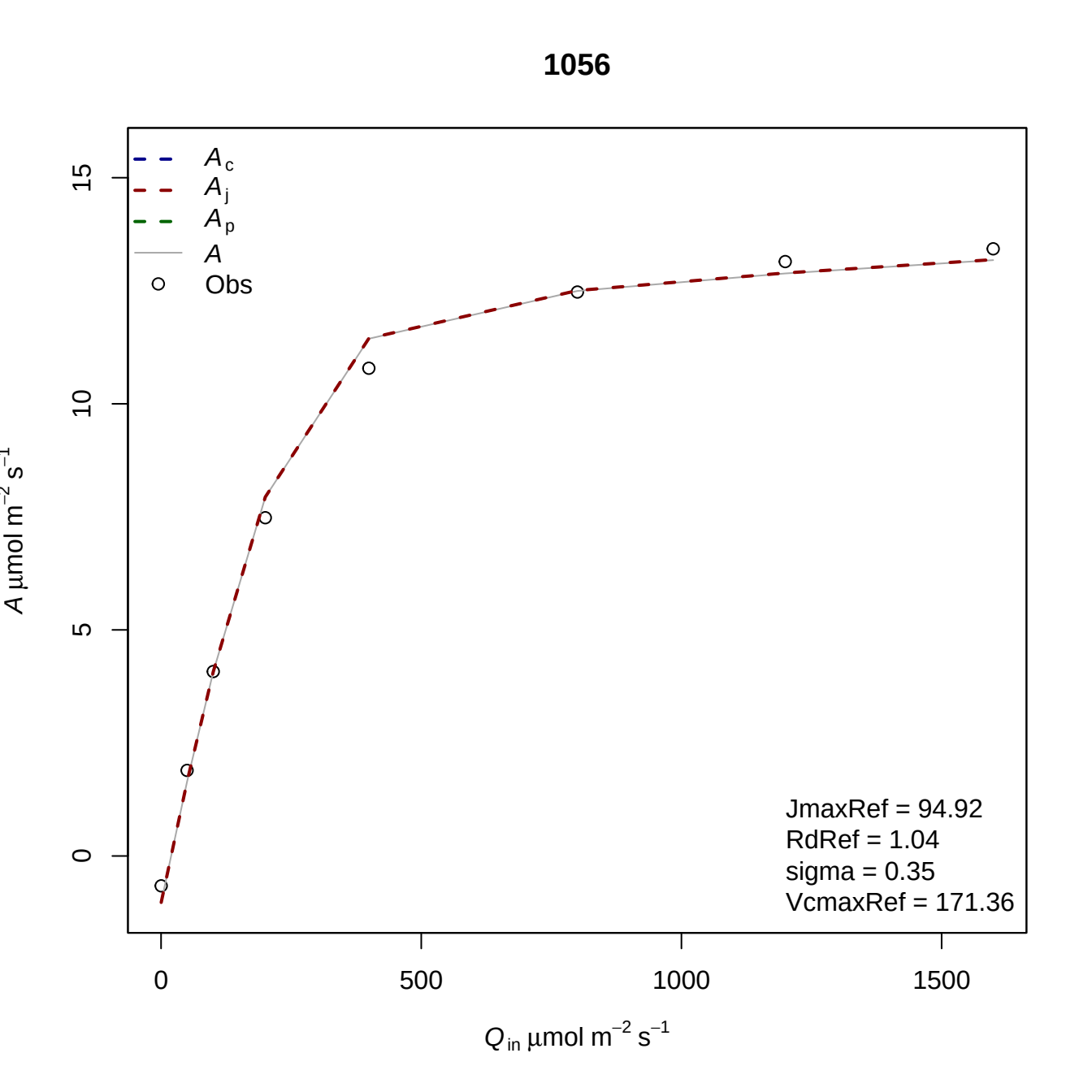

1051
A_c
A_j
A_p
A
Obs
JmaxRef = 108.46
RdRef = 1.87
sigma = 0.75
VcmaxRef = 278.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

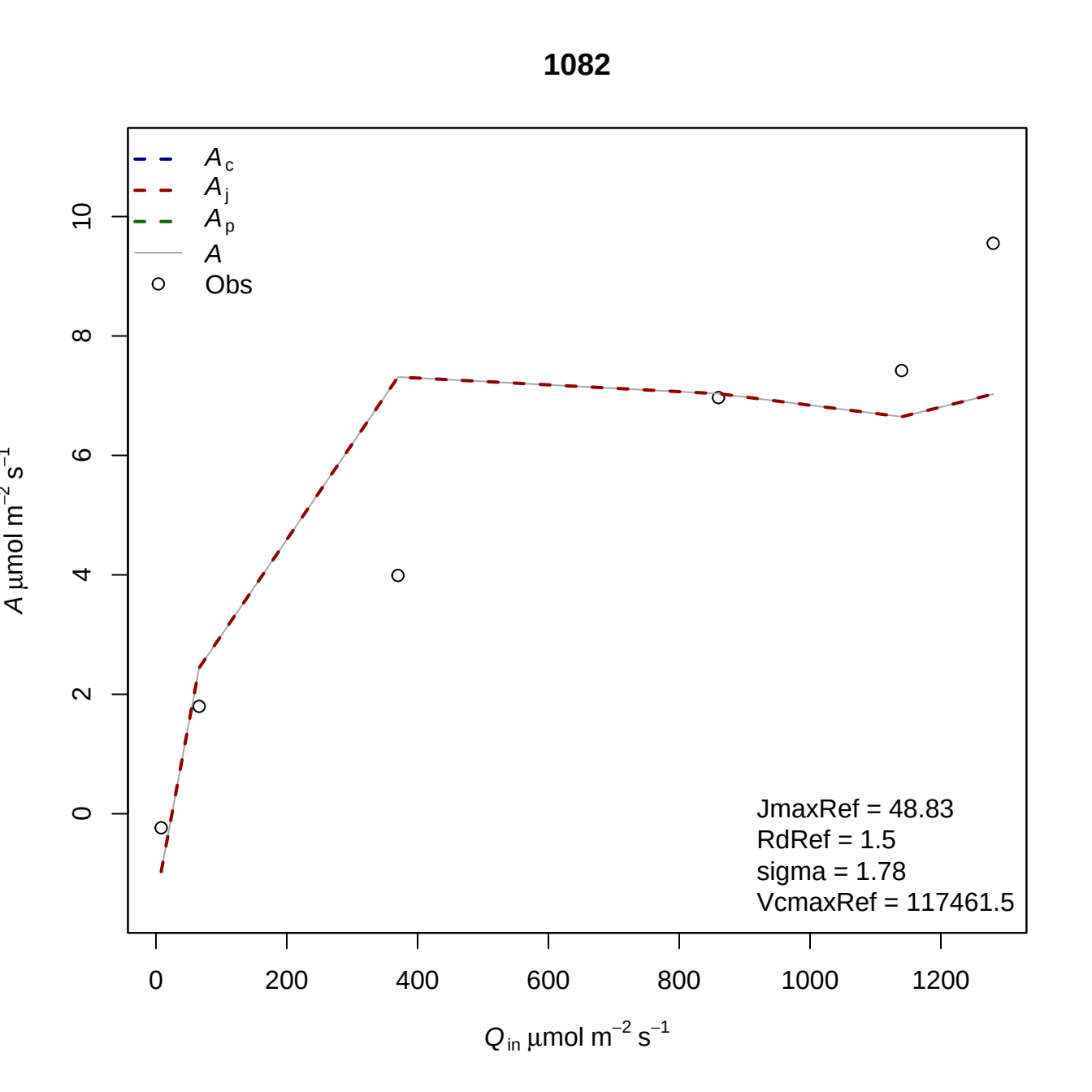

1052
A_c
A_j
A_p
A
Obs
JmaxRef = 78.88
RdRef = 1.23
sigma = 0.49
VcmaxRef = 171.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

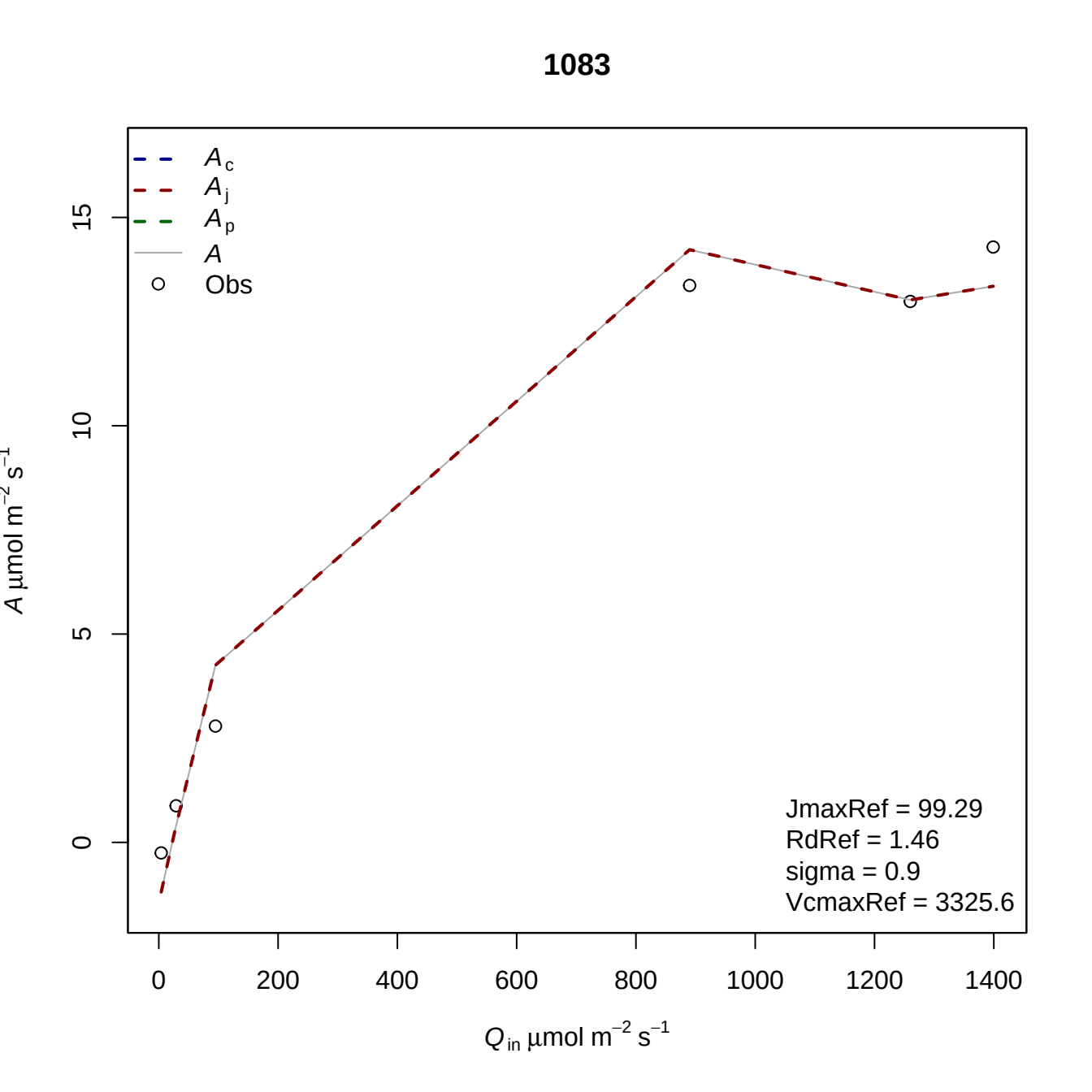

1053
A_c
A_j
A_p
A
Obs
JmaxRef = 103.98
RdRef = 2.57
sigma = 0.73
VcmaxRef = 316.08
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

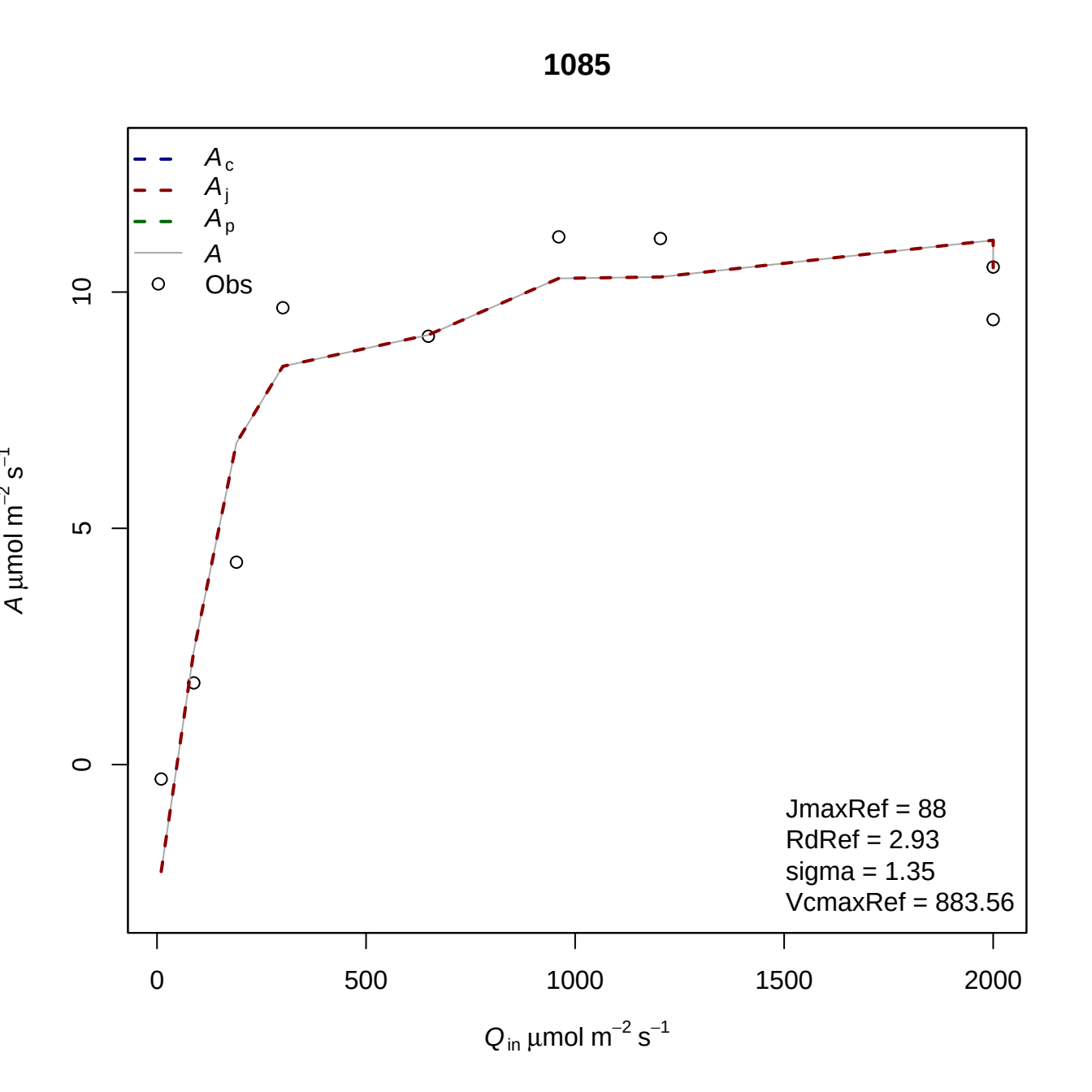

1054
A_c
A_j
A_p
A
Obs
JmaxRef = 86.47
RdRef = 1.62
sigma = 0.95
VcmaxRef = 376.88
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

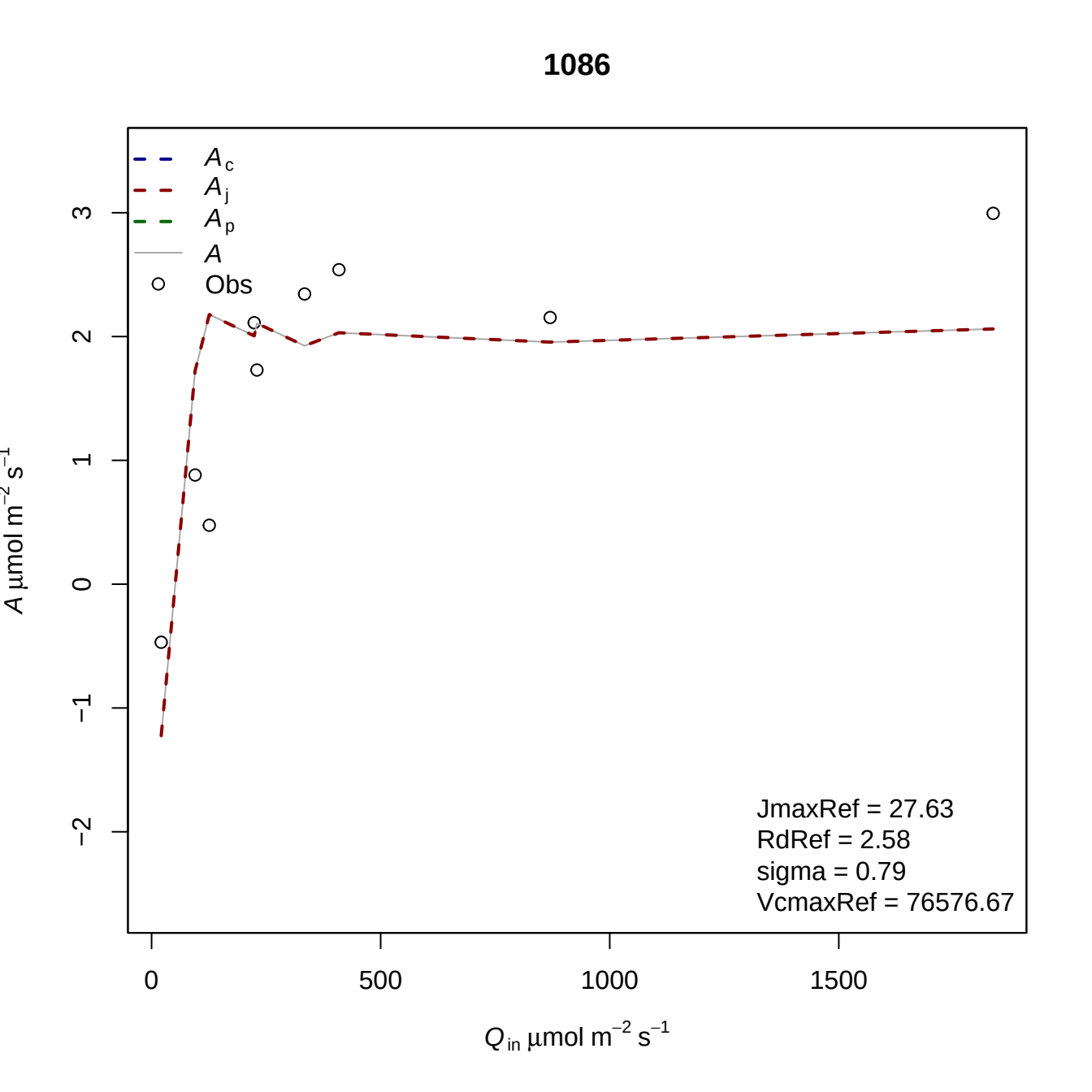

1055
A_c
A_j
A_p
A
Obs
JmaxRef = 93.32
RdRef = 2.19
sigma = 0.84
VcmaxRef = 245.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

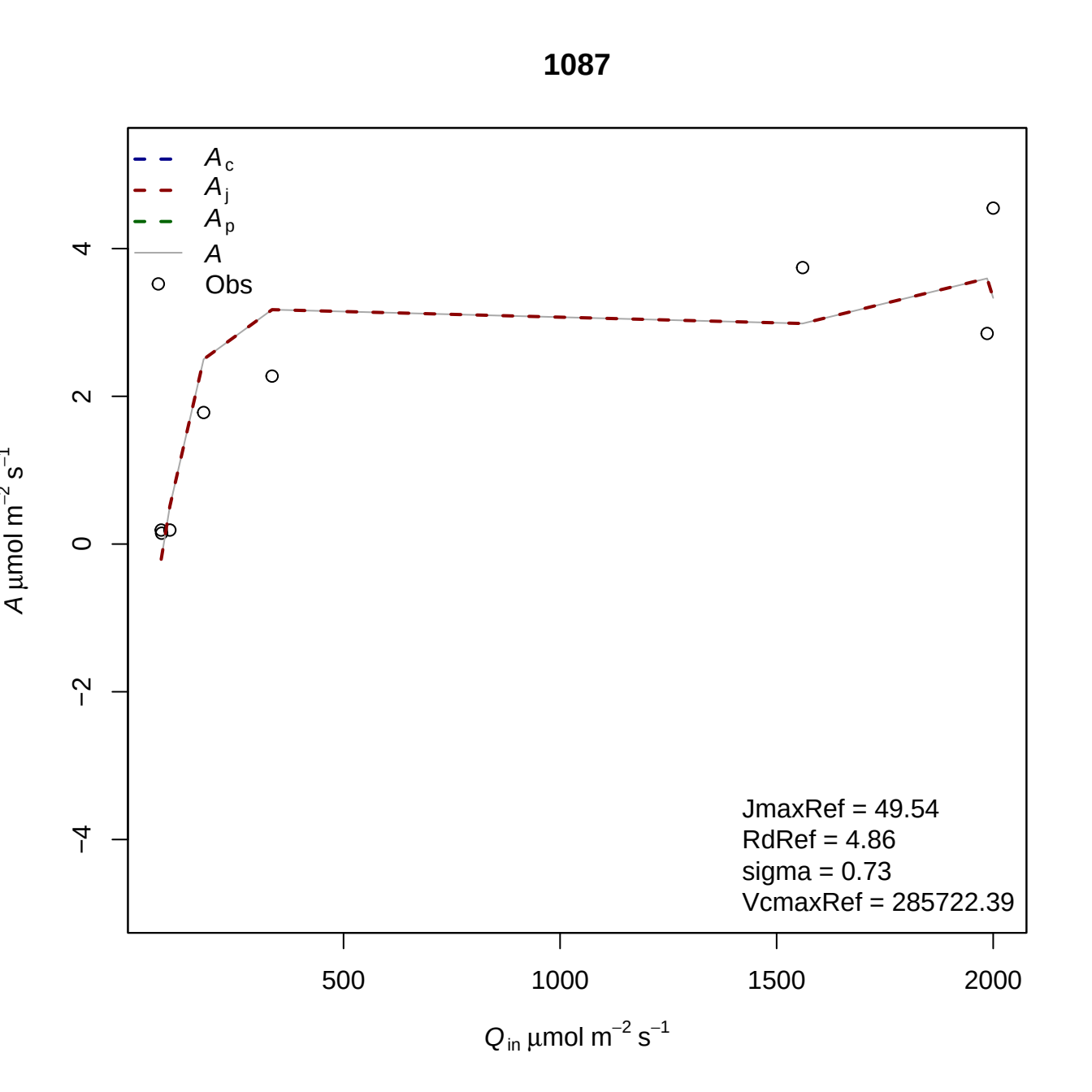

1056
A_c
A_j
A_p
A
Obs
JmaxRef = 94.92
RdRef = 1.04
sigma = 0.35
VcmaxRef = 171.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

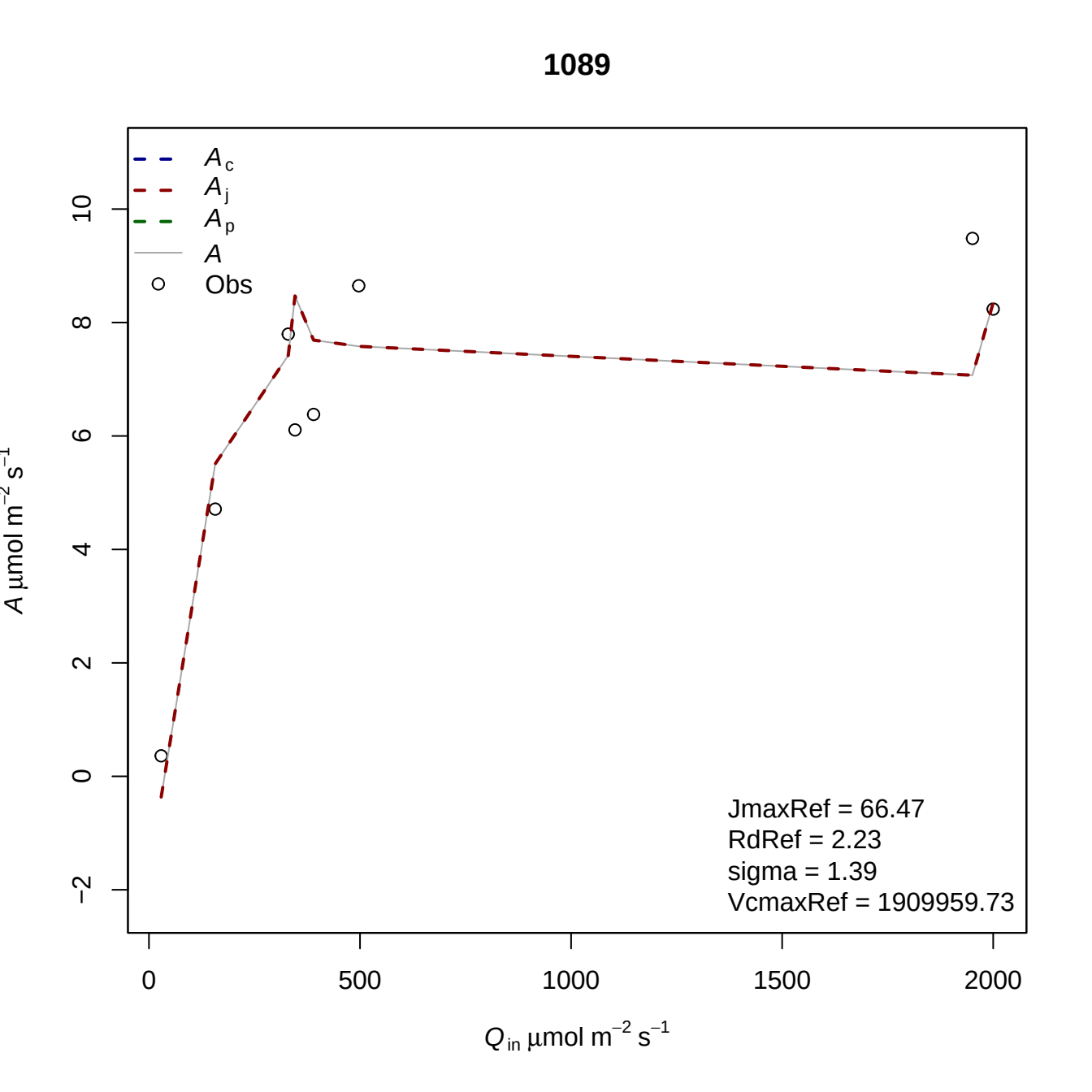

1082
A_c
A_j
A_p
A
Obs
JmaxRef = 48.83
RdRef = 1.5
sigma = 1.78
VcmaxRef = 117461.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

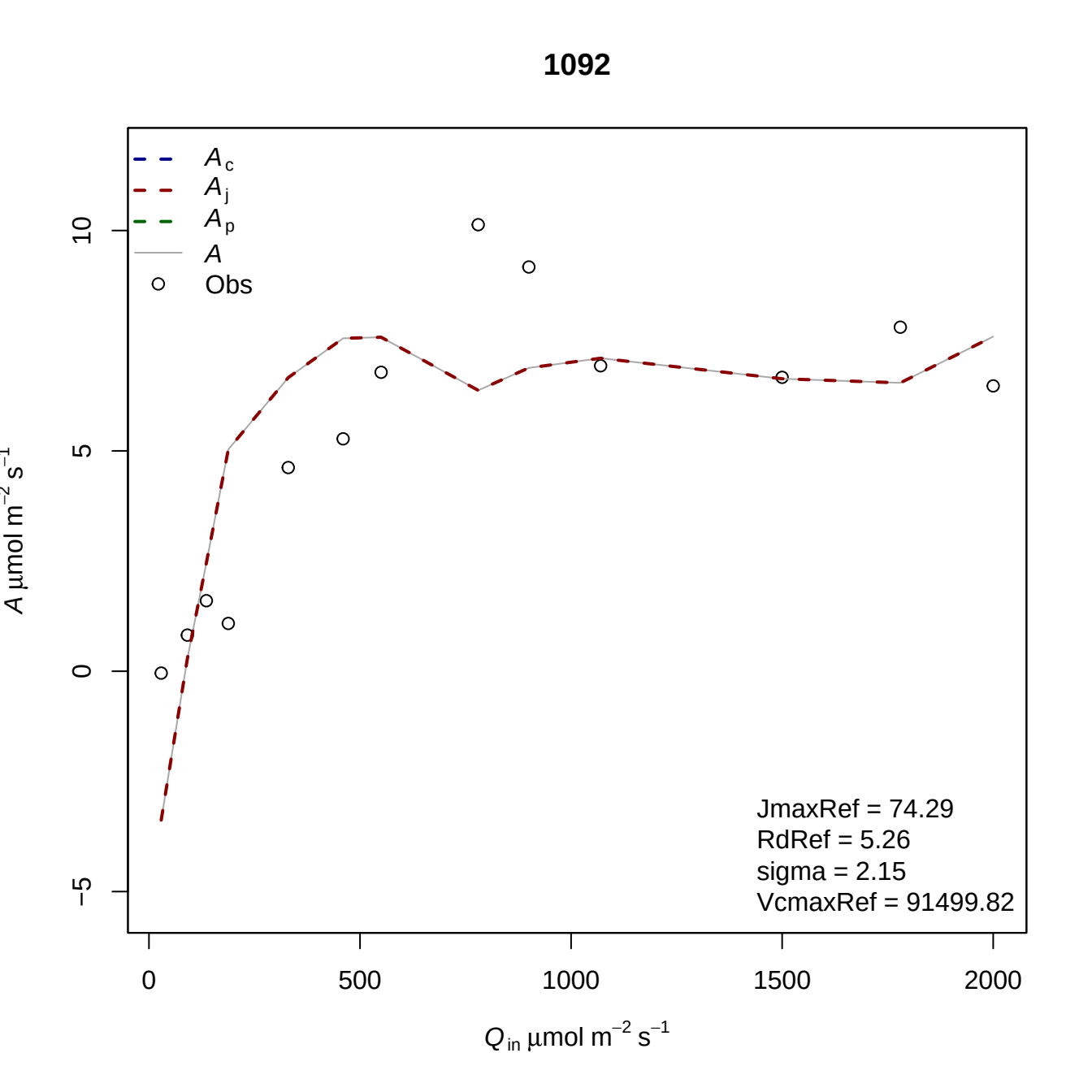

1083
A_c
A_j
A_p
A
Obs
JmaxRef = 99.29
RdRef = 1.46
sigma = 0.9
VcmaxRef = 3325.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

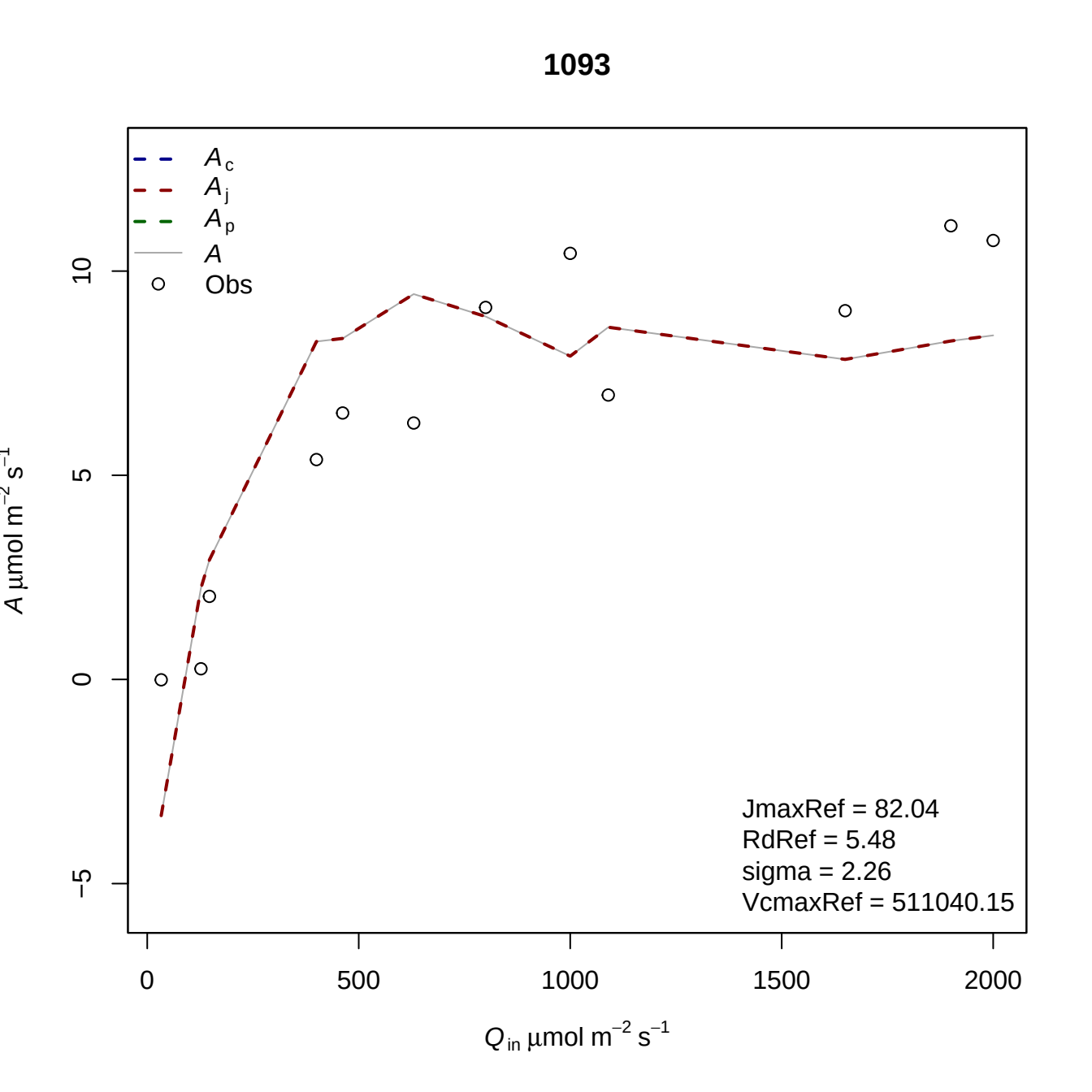

1085
A_c
A_j
A_p
A
Obs
JmaxRef = 88
RdRef = 2.93
sigma = 1.35
VcmaxRef = 883.56
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

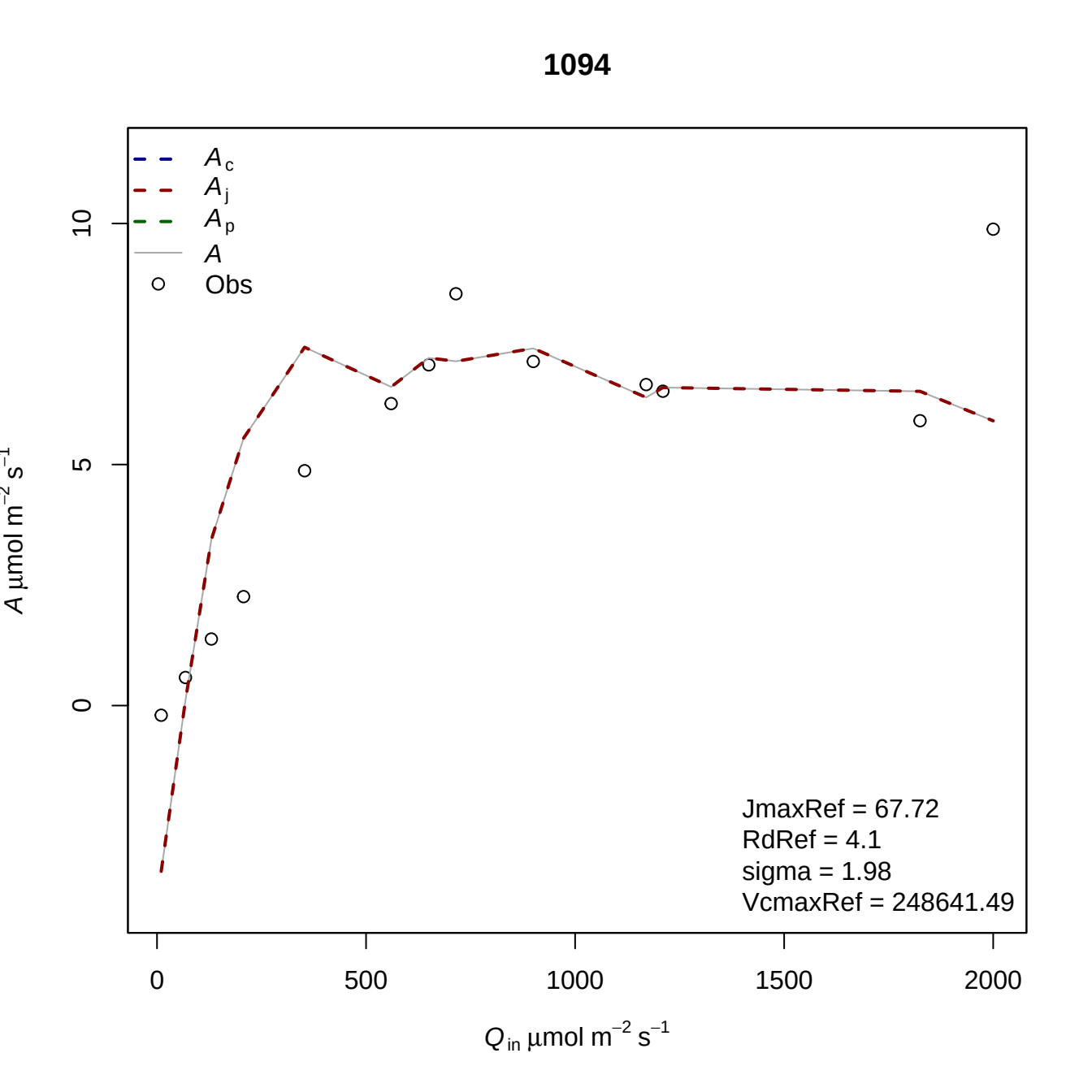

1086
A_c
A_j
A_p
A
Obs
JmaxRef = 27.63
RdRef = 2.58
sigma = 0.79
VcmaxRef = 76576.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500

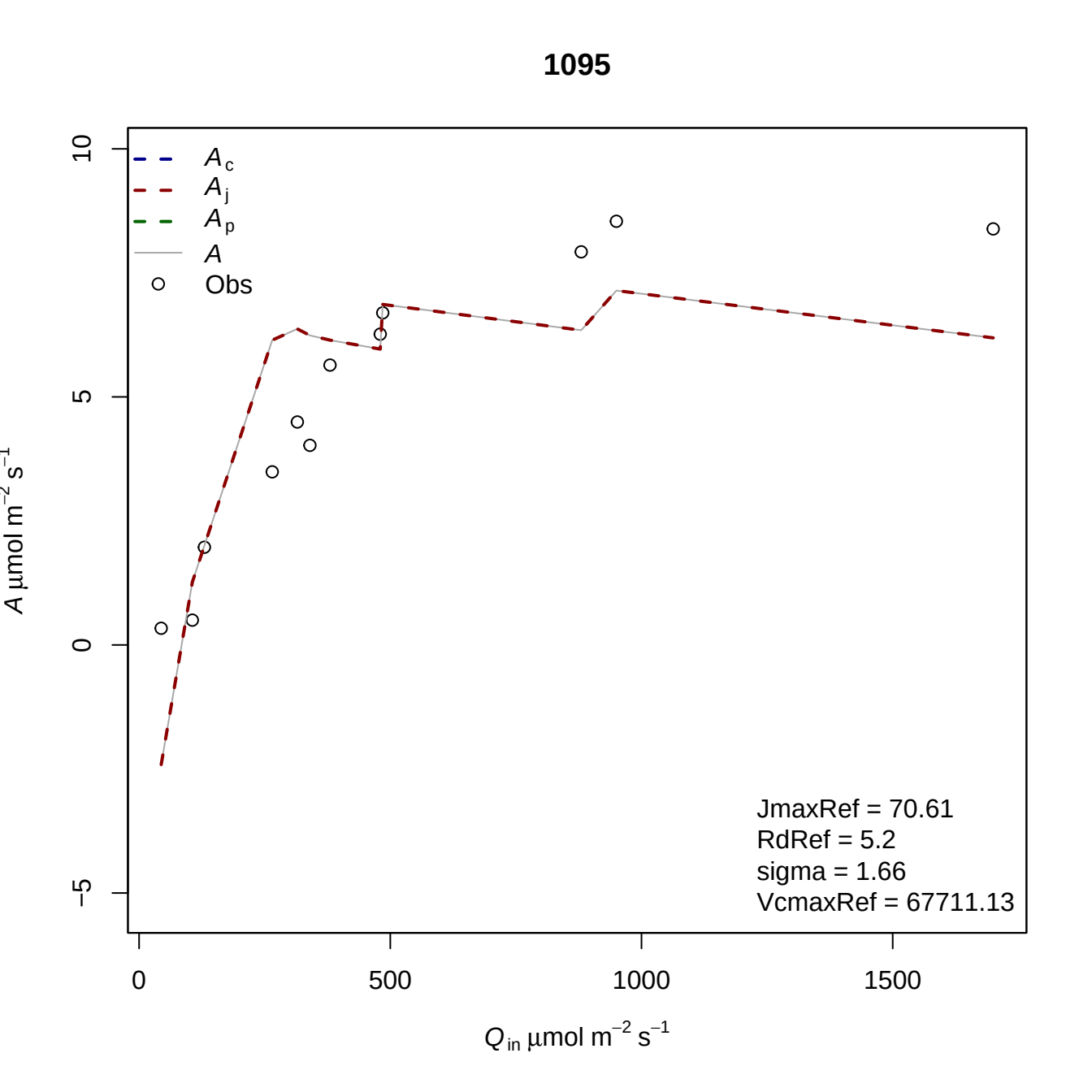

1087
A_c
A_j
A_p
A
Obs
JmaxRef = 49.54
RdRef = 4.86
sigma = 0.73
VcmaxRef = 285722.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
500
1000
1500
2000

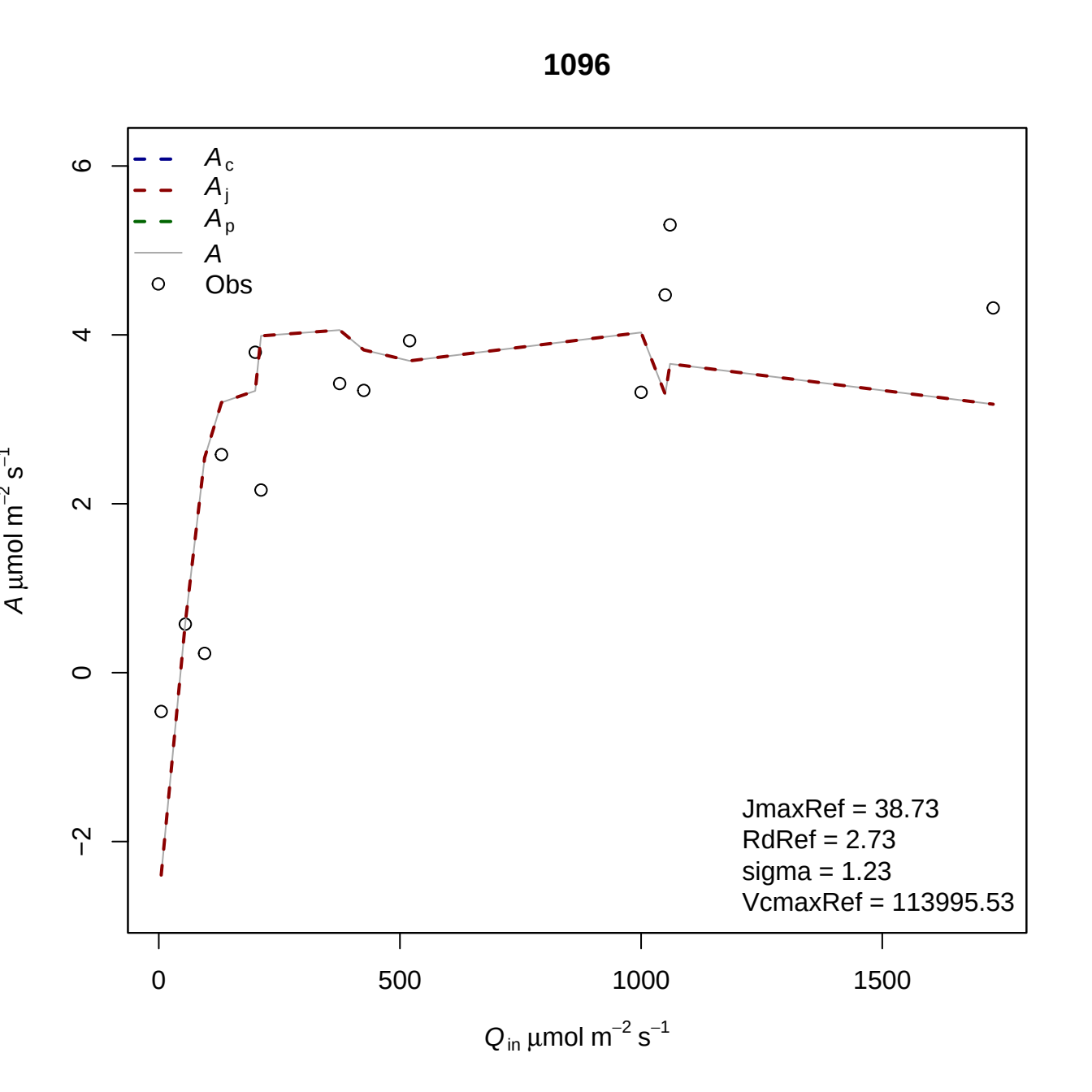

1089
A_c
A_j
A_p
A
Obs
JmaxRef = 66.47
RdRef = 2.23
sigma = 1.39
VcmaxRef = 1909959.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

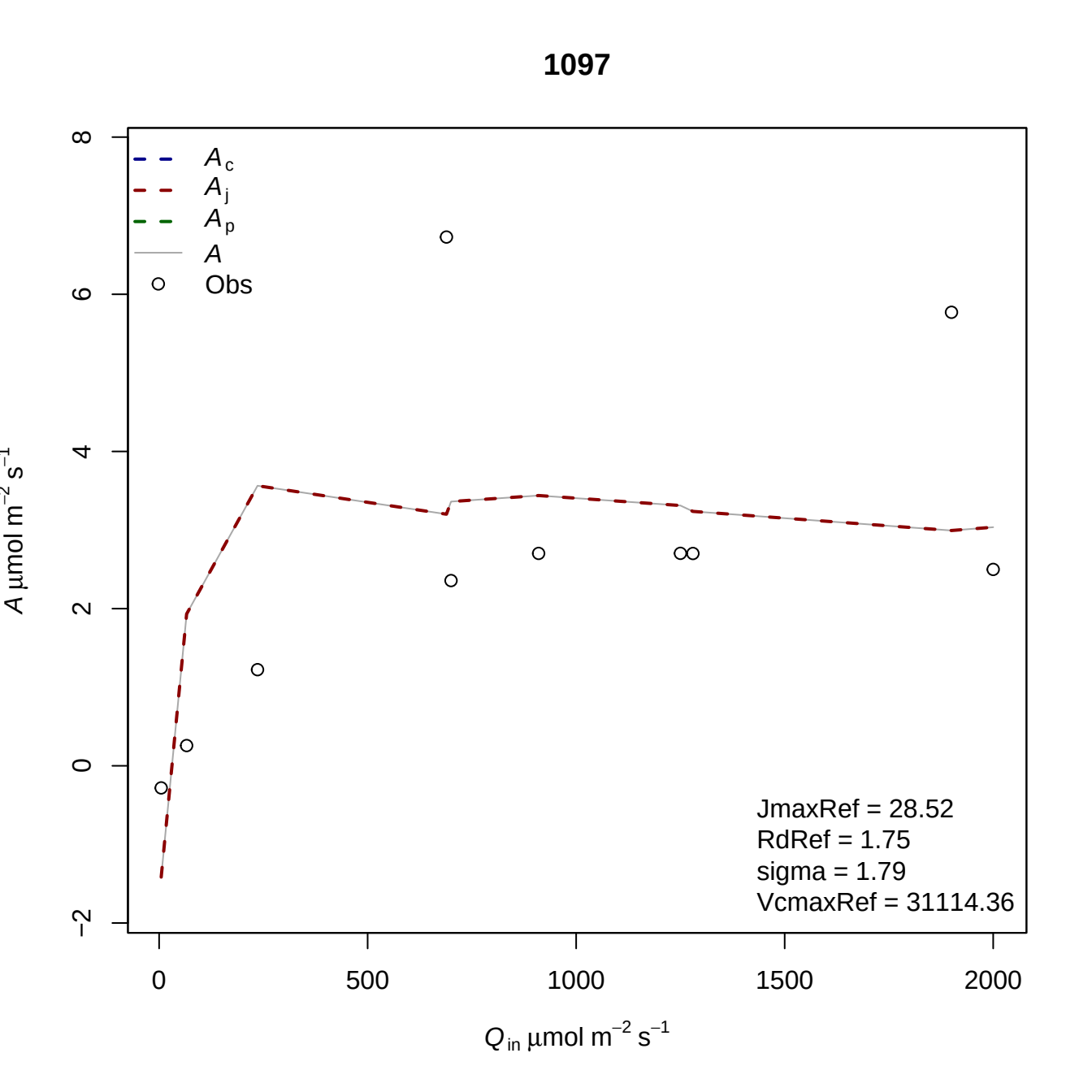

1092
A_c
A_j
A_p
A
Obs
JmaxRef = 74.29
RdRef = 5.26
sigma = 2.15
VcmaxRef = 91499.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

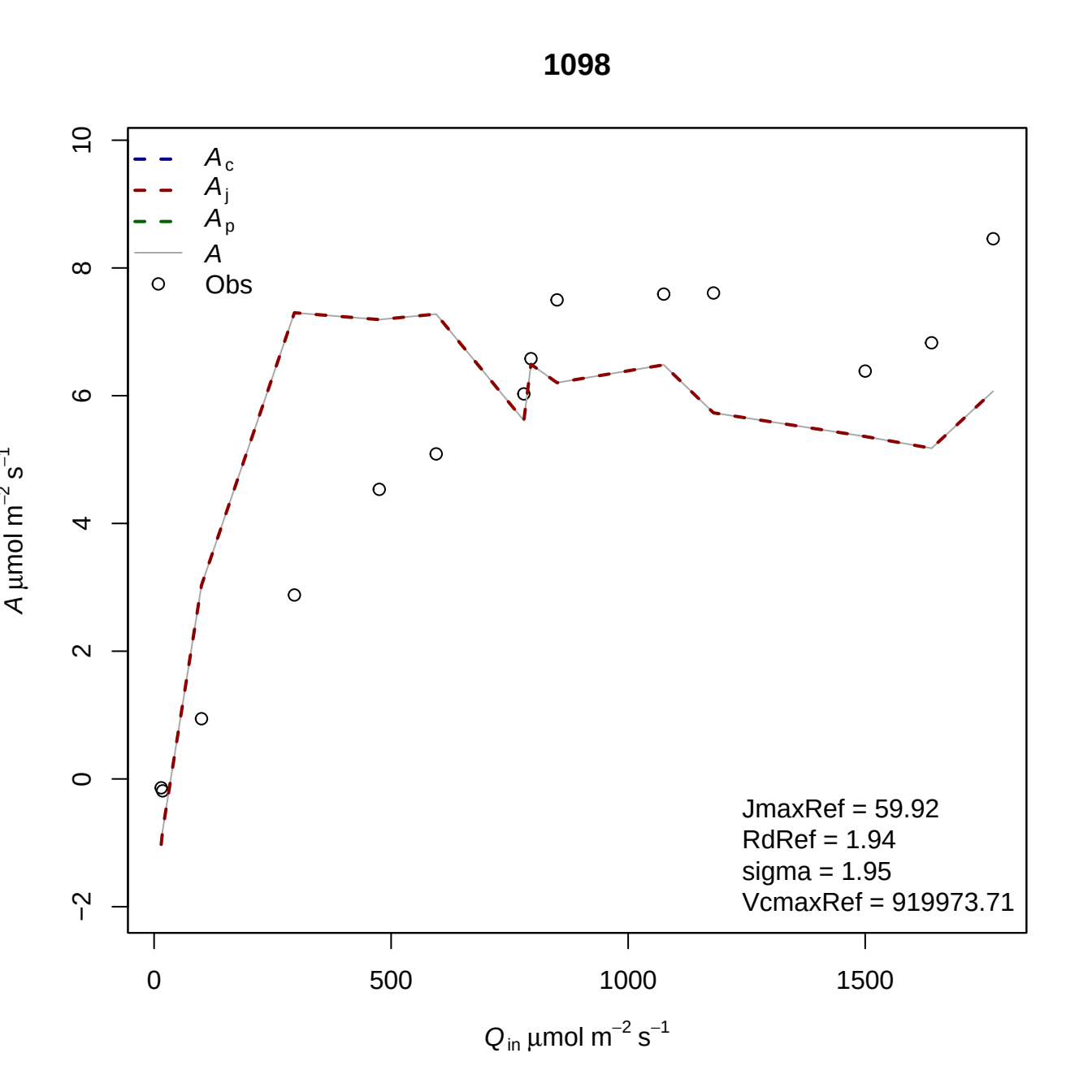

1093
A_c
A_j
A_p
A
Obs
JmaxRef = 82.04
RdRef = 5.48
sigma = 2.26
VcmaxRef = 511040.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

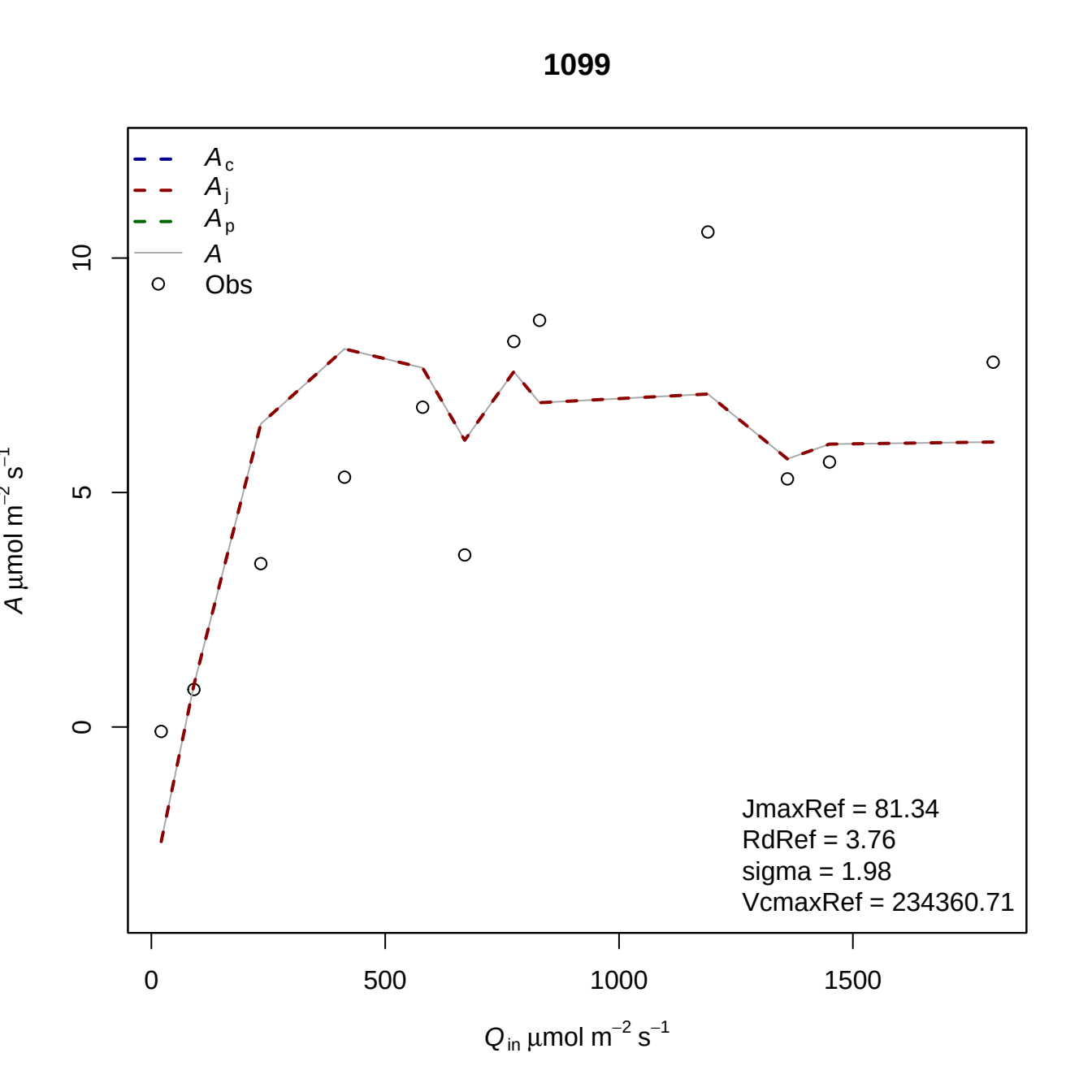

1094
A_c
A_j
A_p
A
Obs
JmaxRef = 67.72
RdRef = 4.1
sigma = 1.98
VcmaxRef = 248641.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

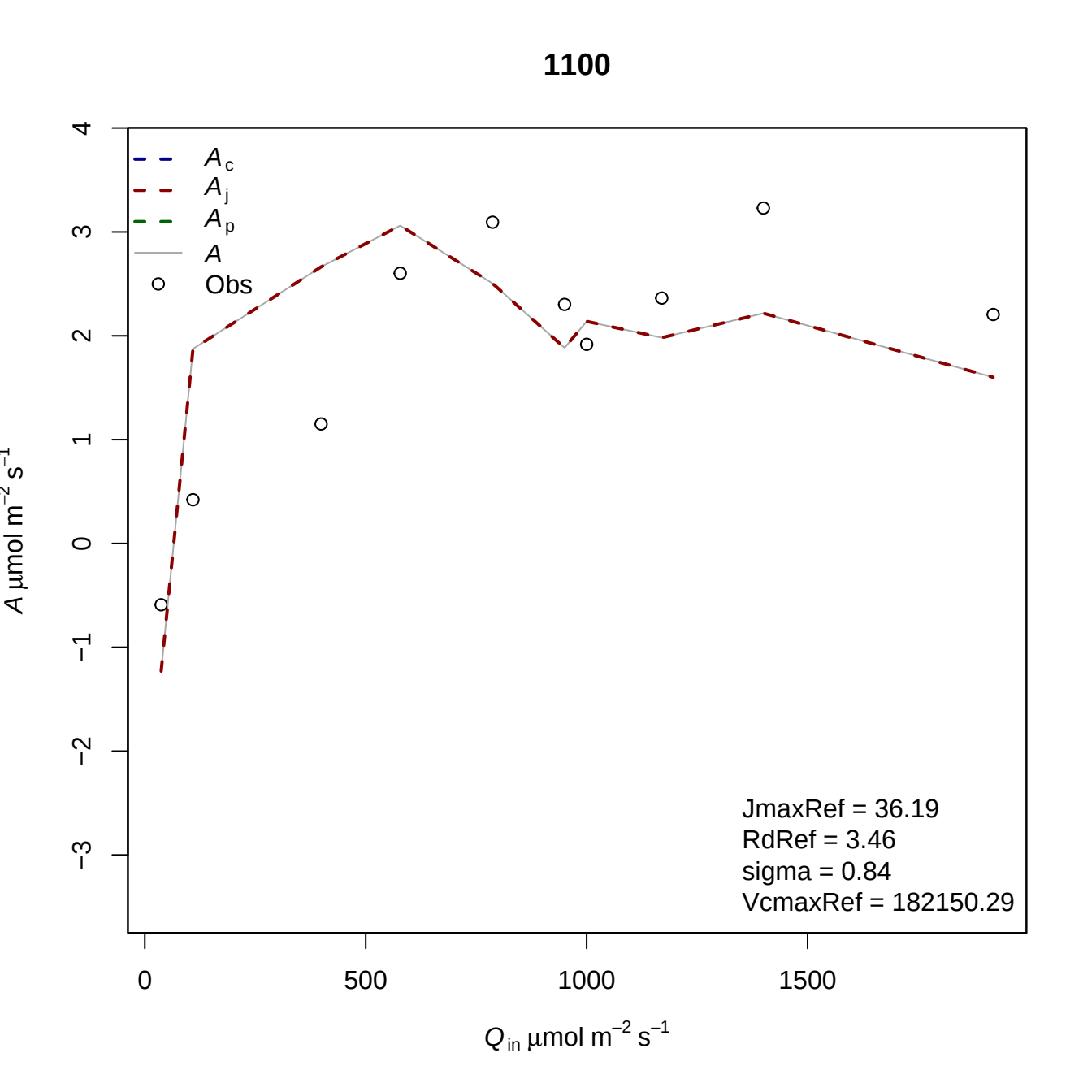

1095
A_c
A_j
A_p
A
Obs
JmaxRef = 70.61
RdRef = 5.2
sigma = 1.66
VcmaxRef = 67711.13
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

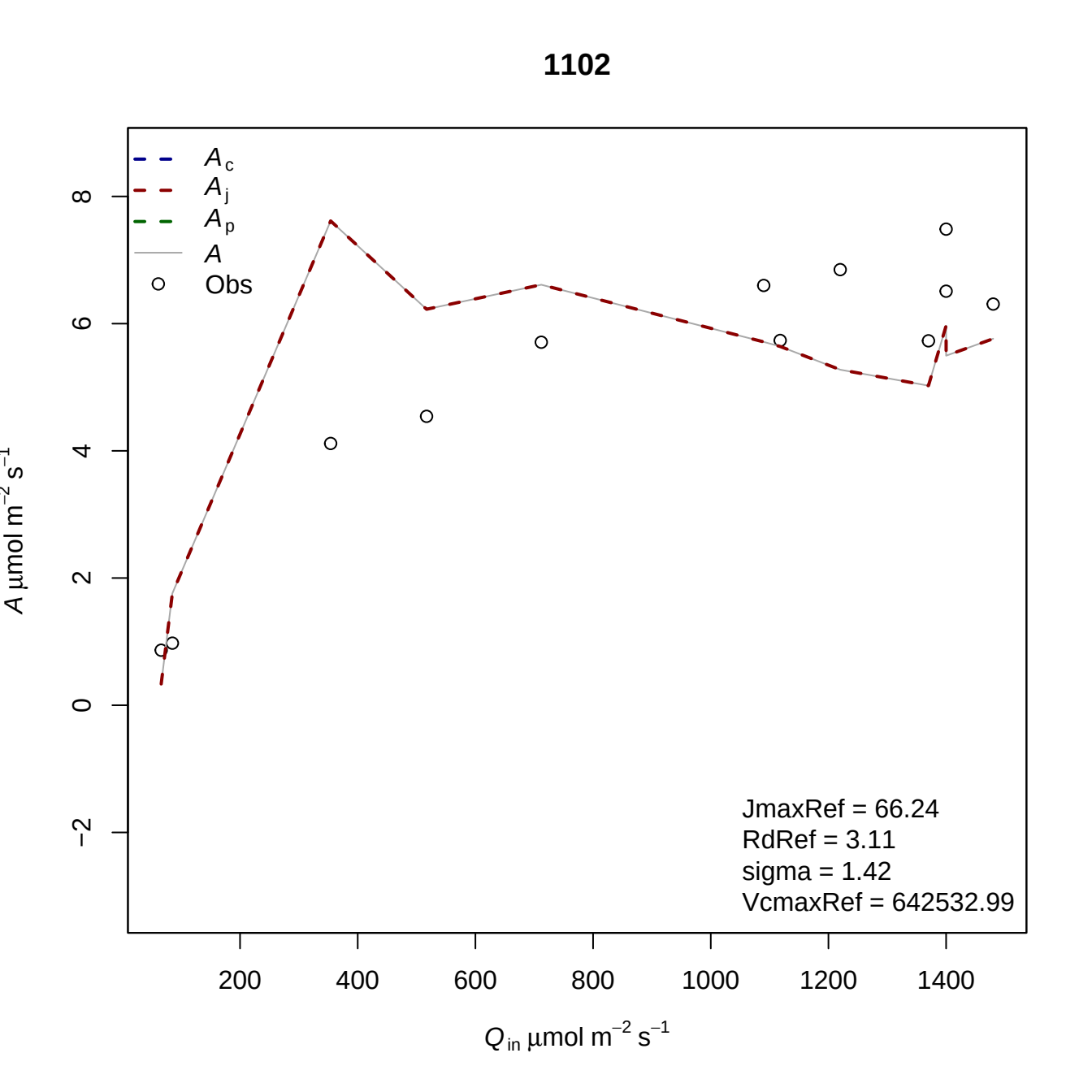

1096
A_c
A_j
A_p
A
Obs
JmaxRef = 38.73
RdRef = 2.73
sigma = 1.23
VcmaxRef = 113995.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

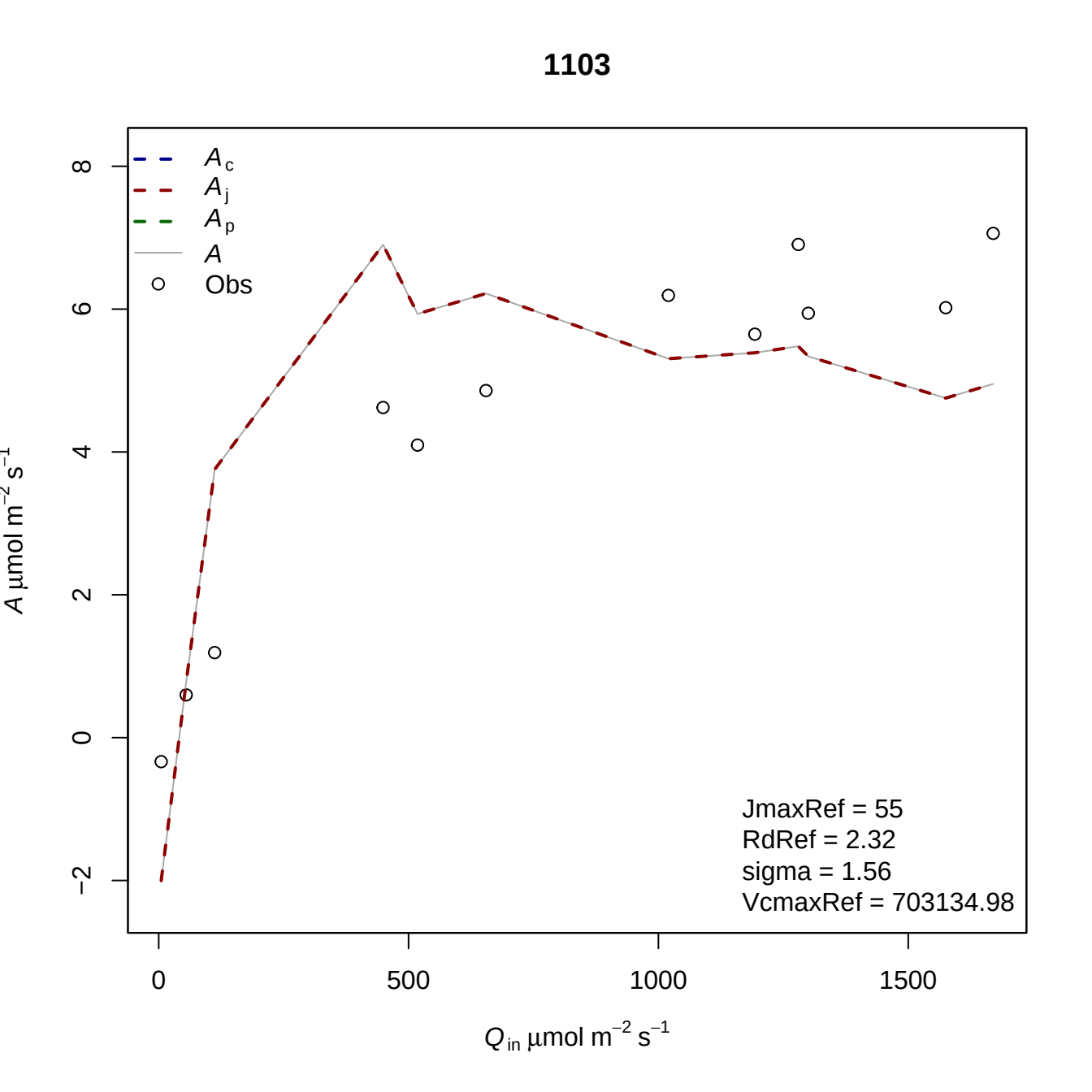

1097
A_c
A_j
A_p
A
Obs
JmaxRef = 28.52
RdRef = 1.75
sigma = 1.79
VcmaxRef = 31114.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

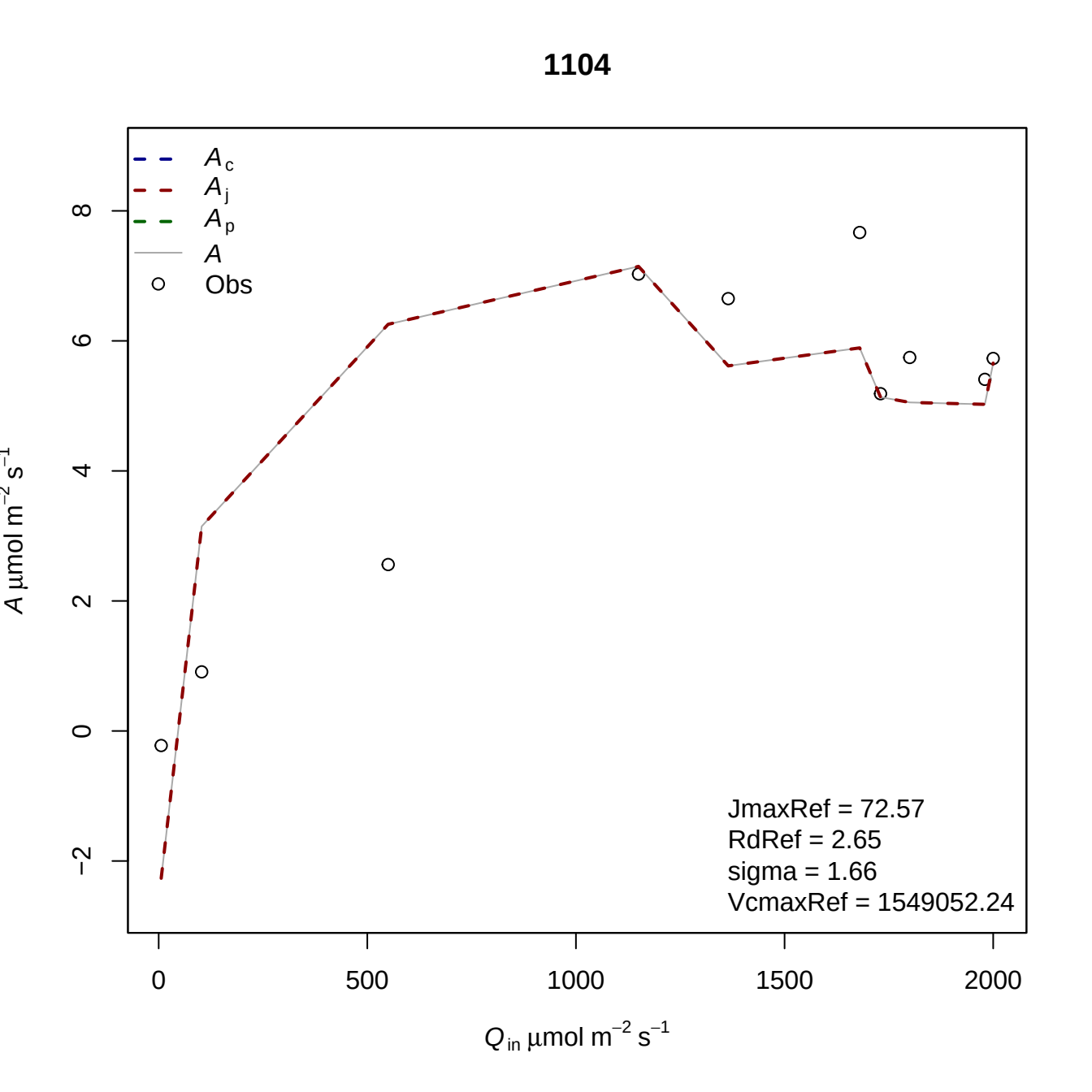

1098
A_c
A_j
A_p
A
Obs
JmaxRef = 59.92
RdRef = 1.94
sigma = 1.95
VcmaxRef = 919973.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

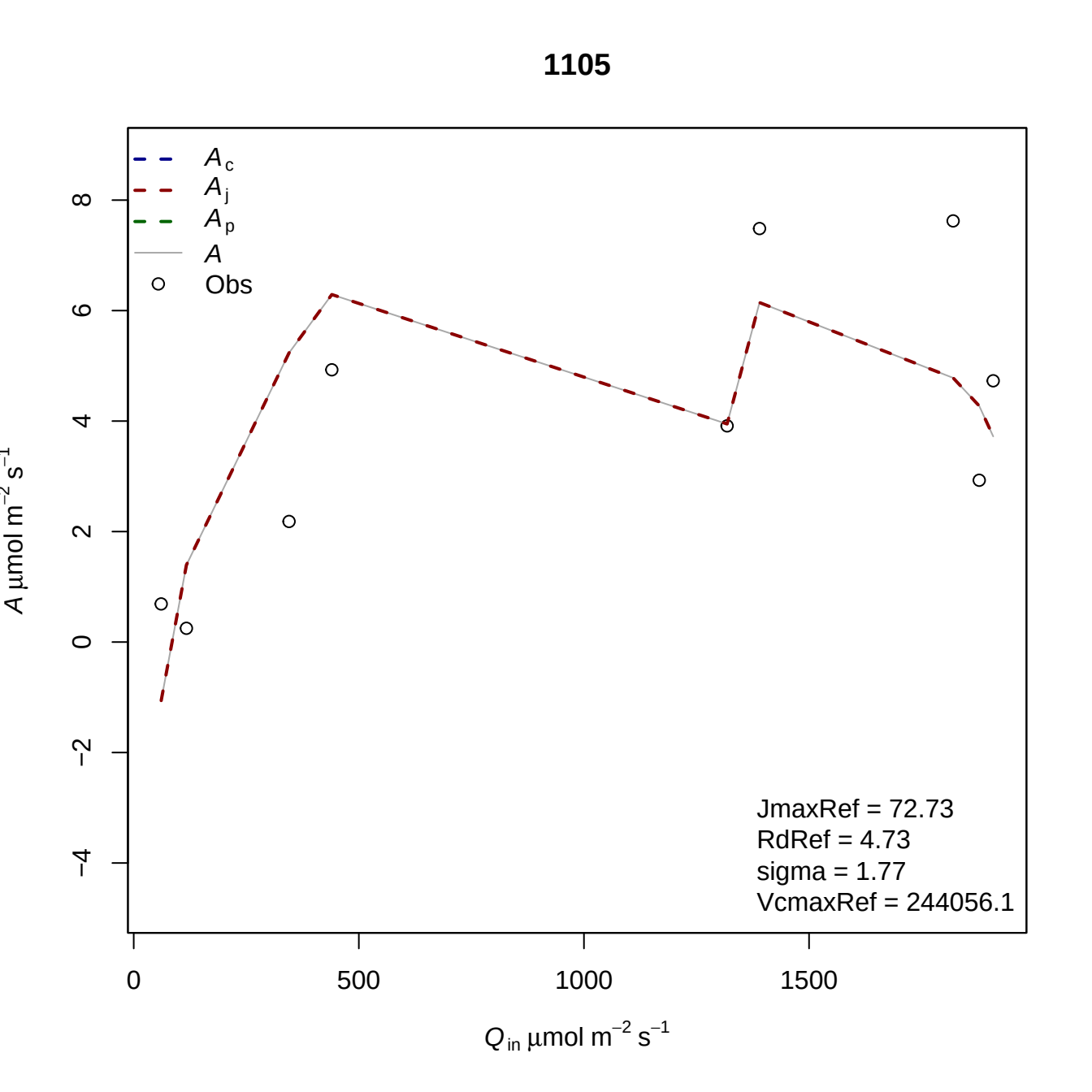

1099
A_c
A_j
A_p
A
Obs
JmaxRef = 81.34
RdRef = 3.76
sigma = 1.98
VcmaxRef = 234360.71
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

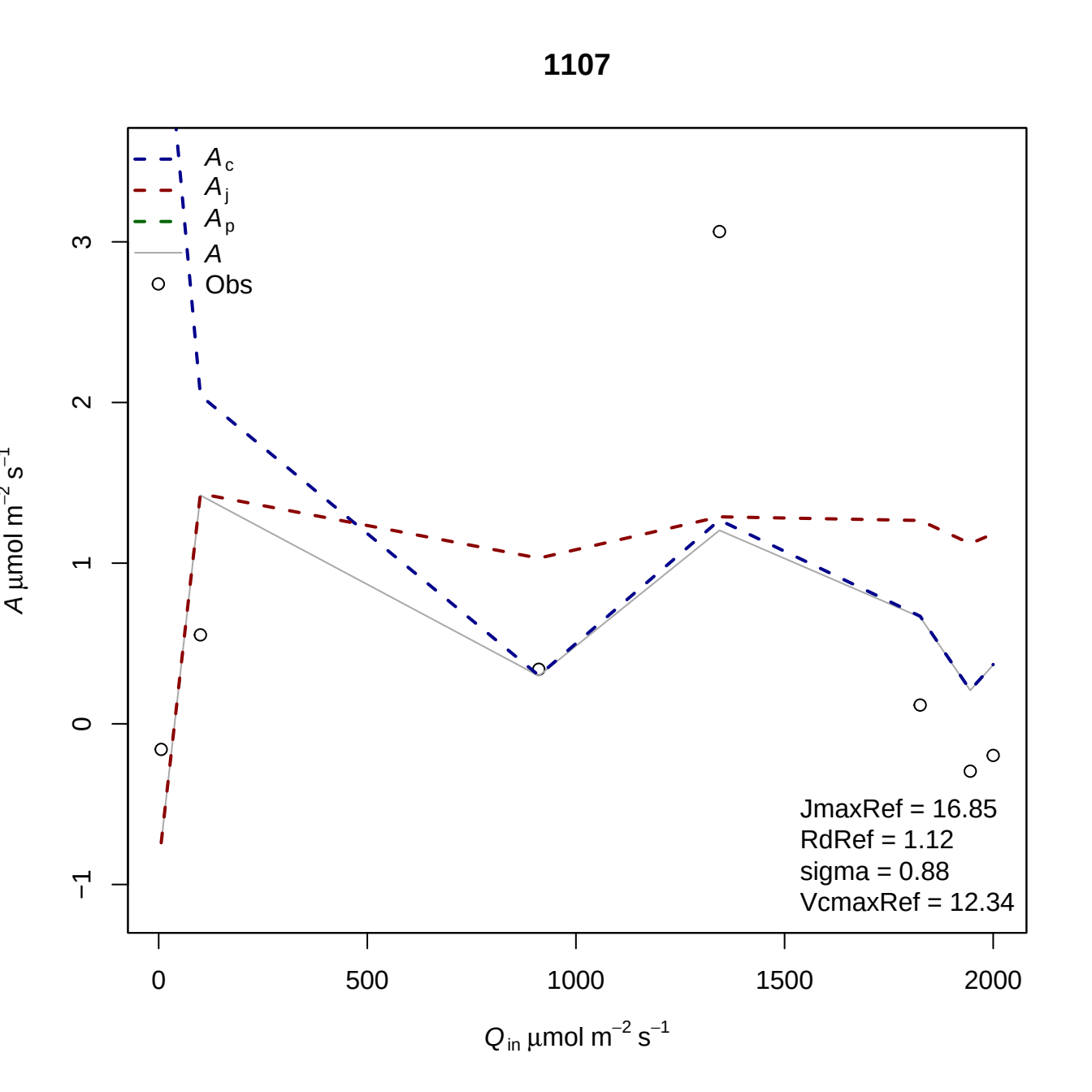

1100
A_c
A_j
A_p
A
Obs
JmaxRef = 36.19
RdRef = 3.46
sigma = 0.84
VcmaxRef = 182150.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

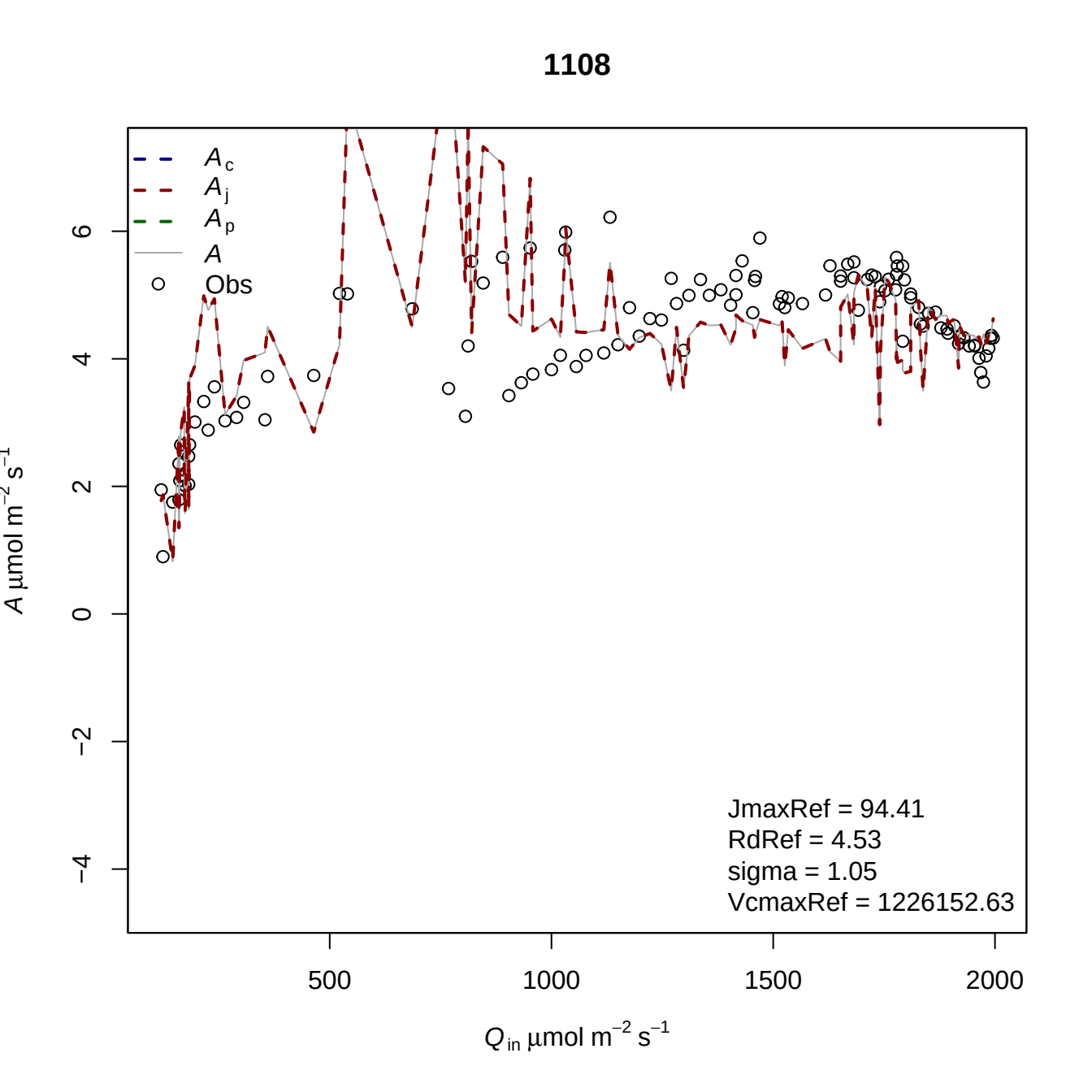

1102
A_c
A_j
A_p
A
Obs
JmaxRef = 66.24
RdRef = 3.11
sigma = 1.42
VcmaxRef = 642532.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

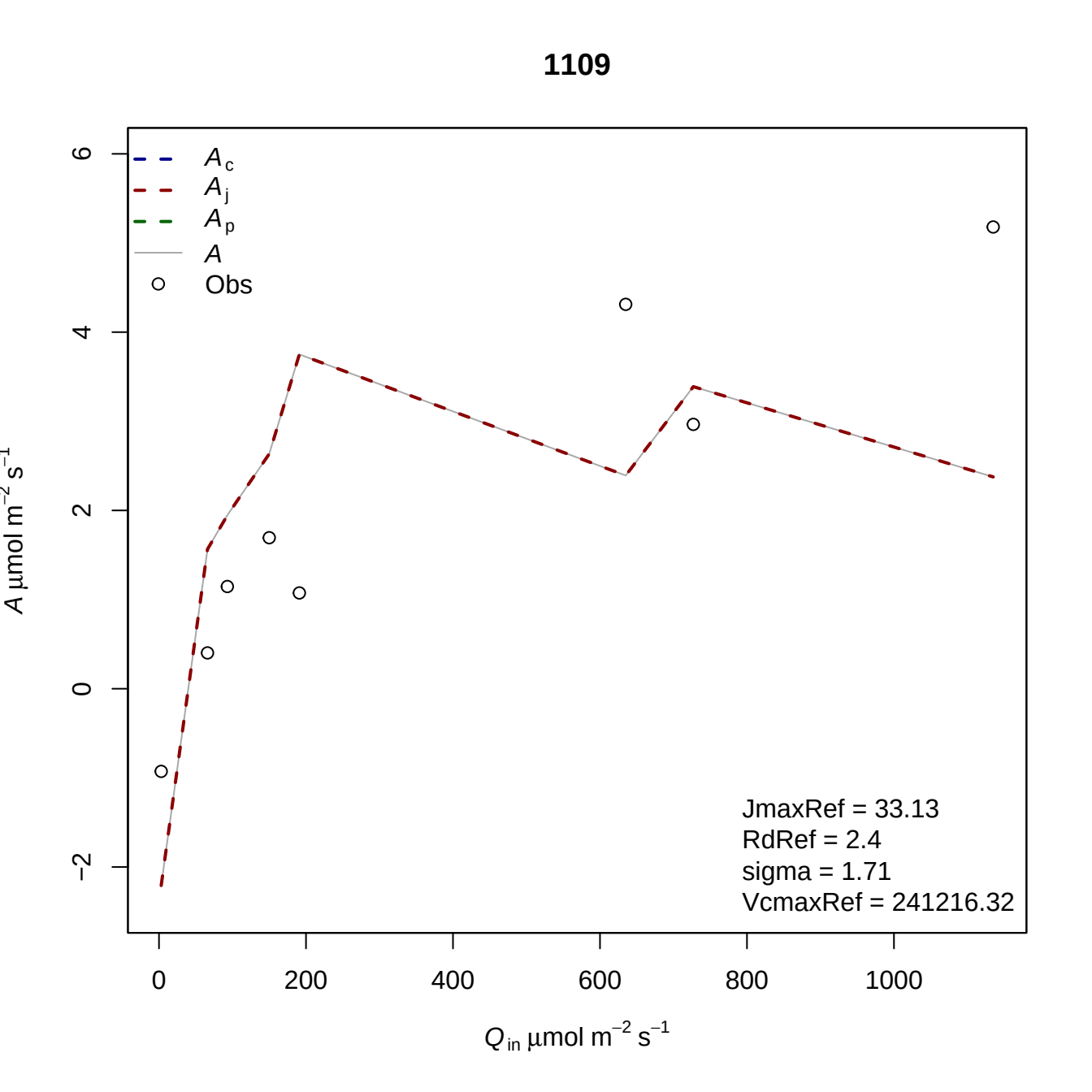

1103
A_c
A_j
A_p
A
Obs
JmaxRef = 55
RdRef = 2.32
sigma = 1.56
VcmaxRef = 703134.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

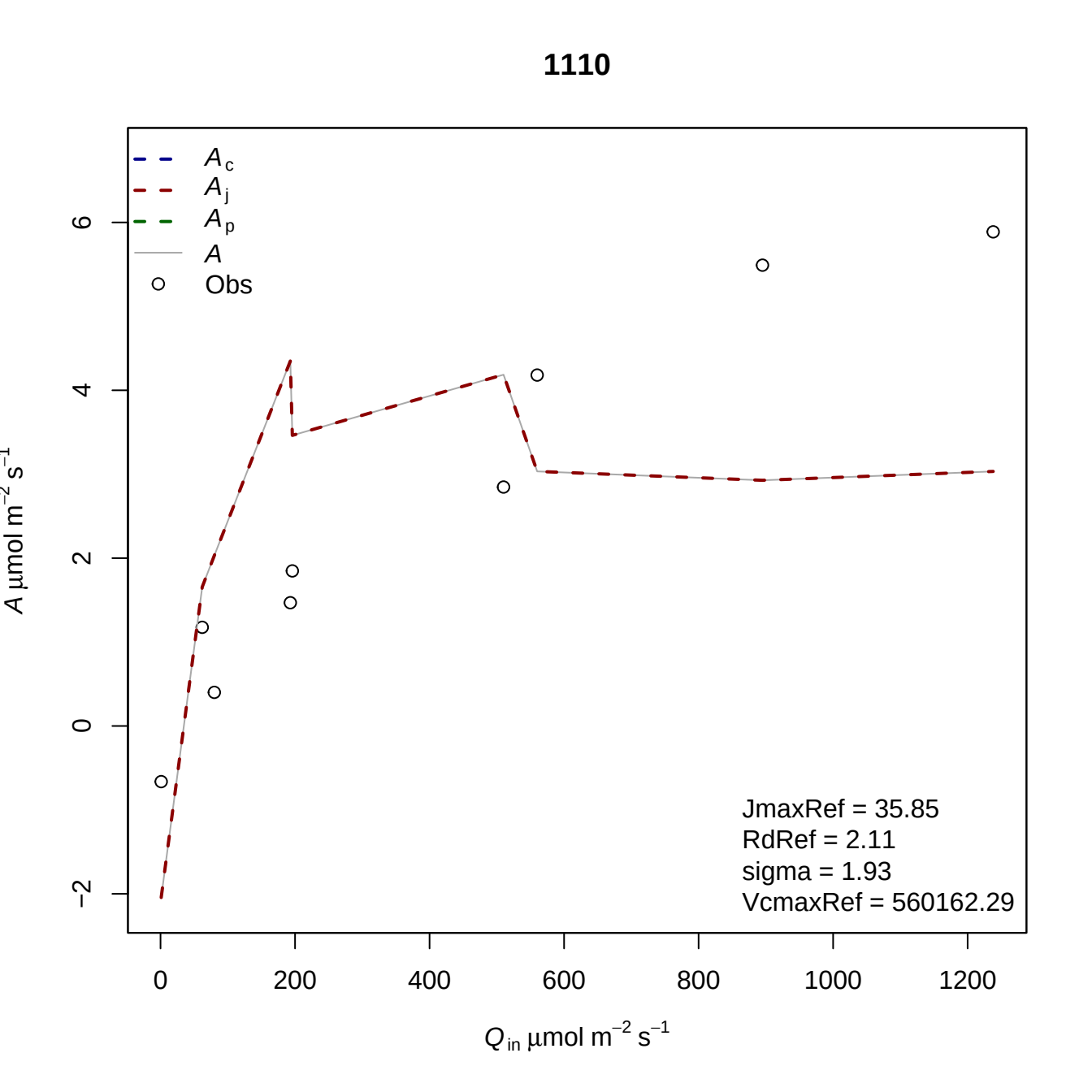

1104
A_c
A_j
A_p
A
Obs
JmaxRef = 72.57
RdRef = 2.65
sigma = 1.66
VcmaxRef = 1549052.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

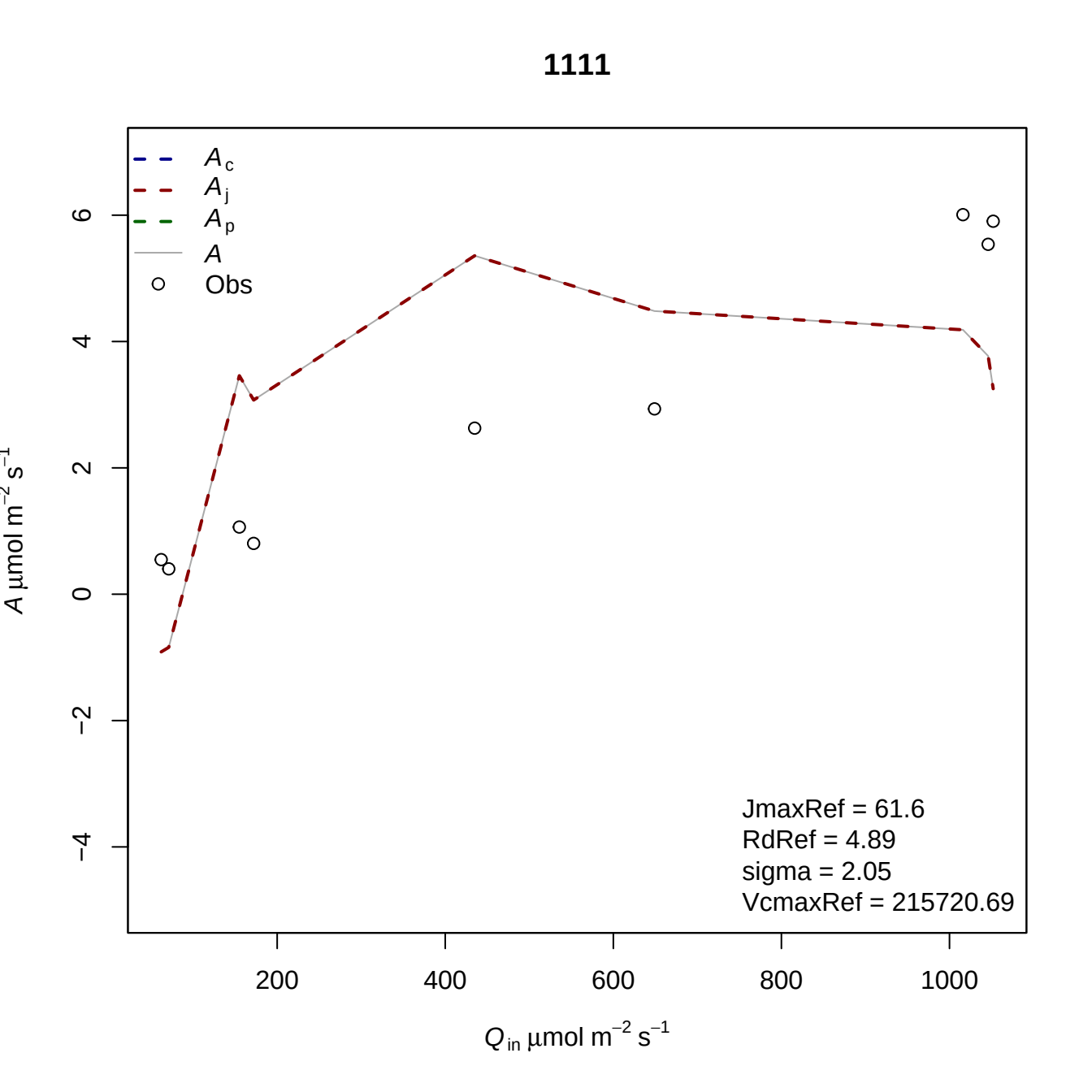

1105
A_c
A_j
A_p
A
Obs
JmaxRef = 72.73
RdRef = 4.73
sigma = 1.77
VcmaxRef = 244056.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

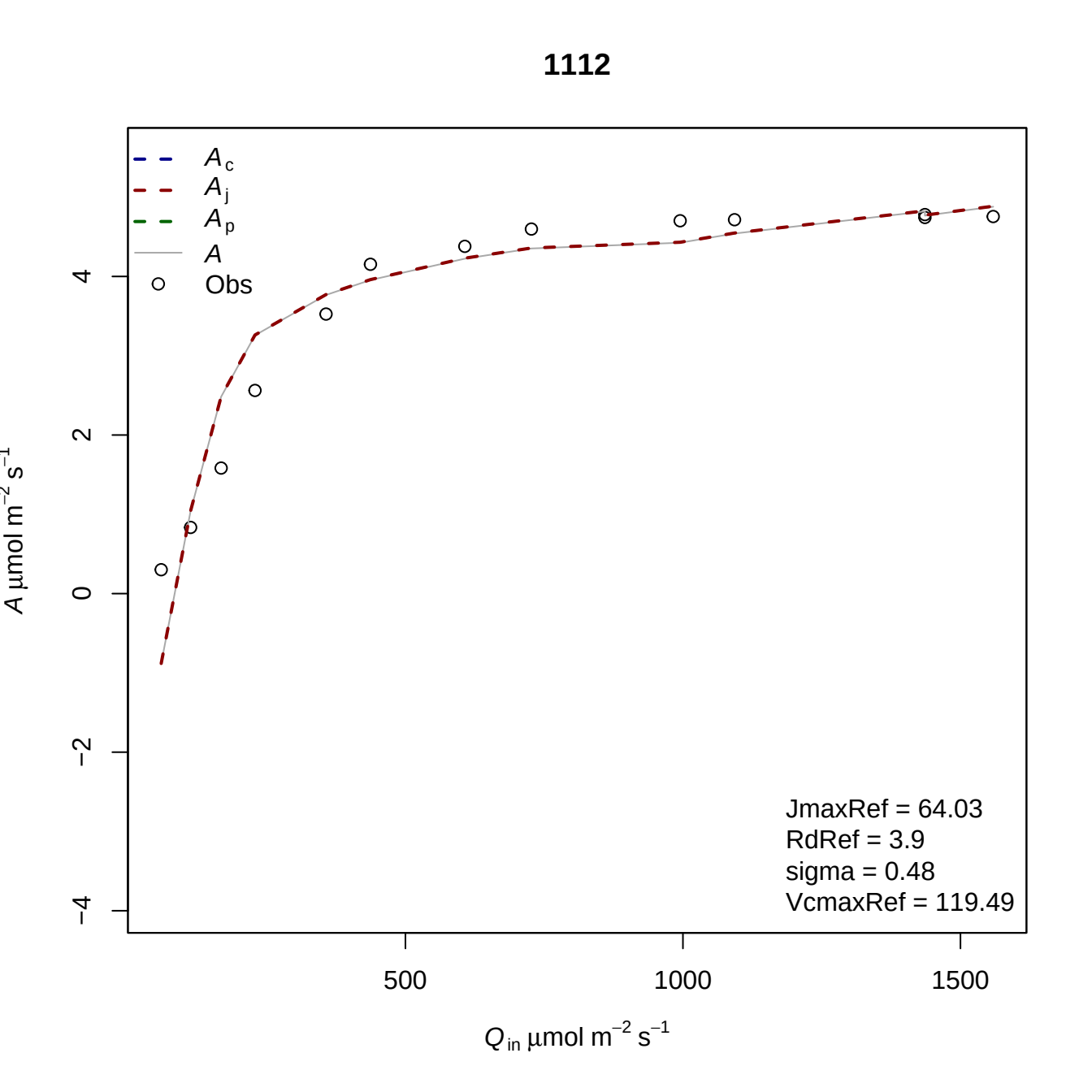

1107
A_c
A_j
A_p
A
Obs
JmaxRef = 16.85
RdRef = 1.12
sigma = 0.88
VcmaxRef = 12.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

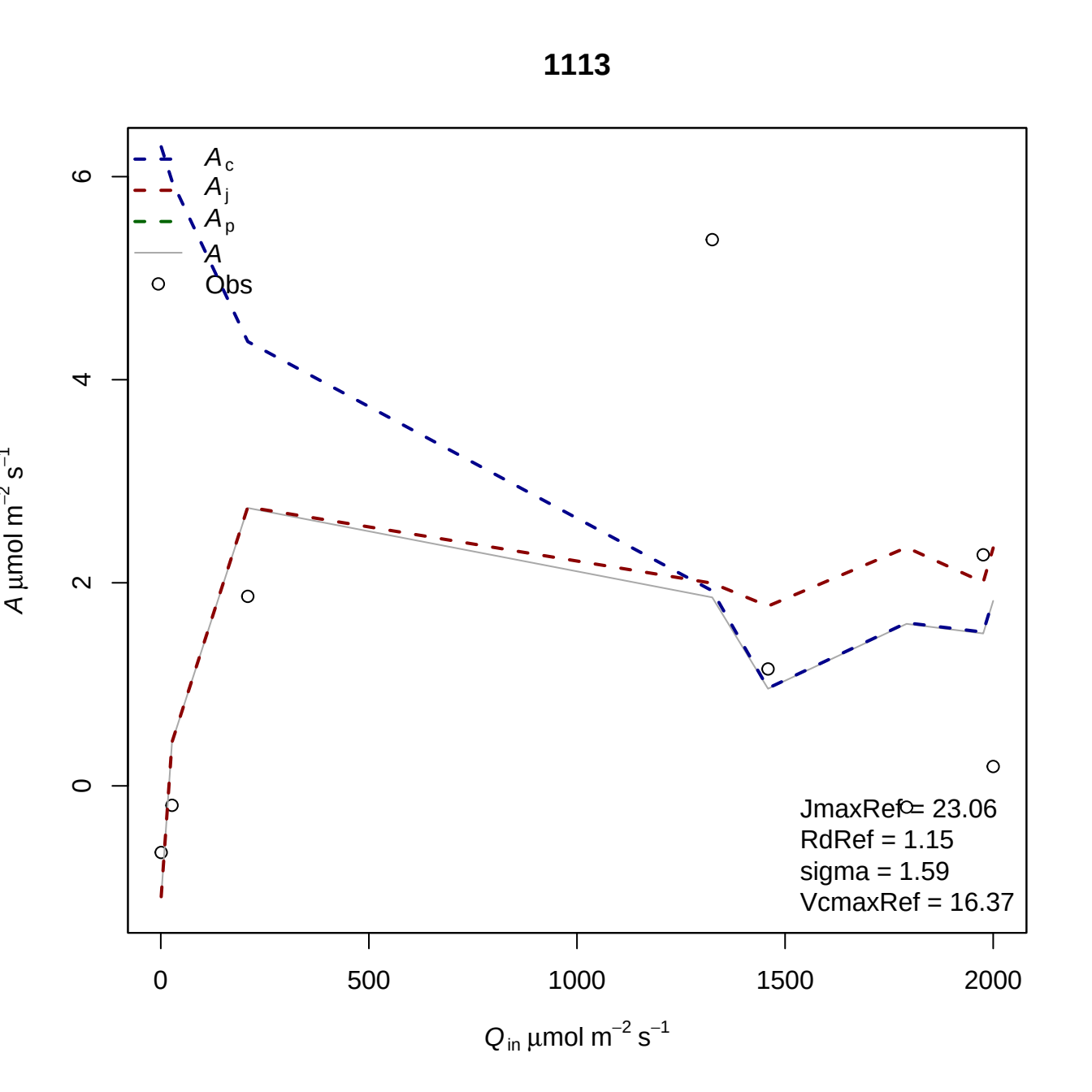

1108
A_c
A_j
A_p
A
Obs
JmaxRef = 94.41
RdRef = 4.53
sigma = 1.05
VcmaxRef = 1226152.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
500
1000
1500
2000

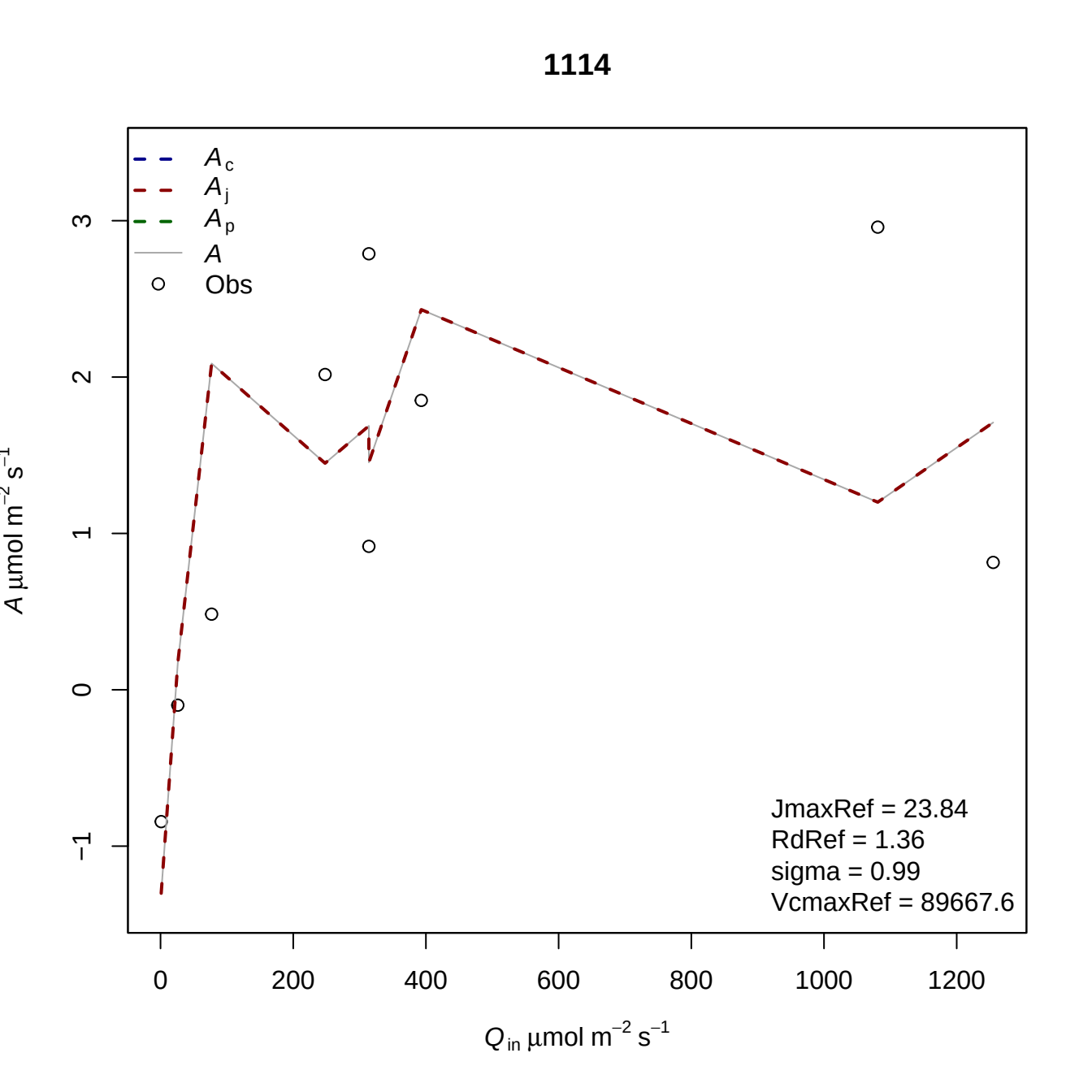

1109
A_c
A_j
A_p
A
Obs
JmaxRef = 33.13
RdRef = 2.4
sigma = 1.71
VcmaxRef = 241216.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

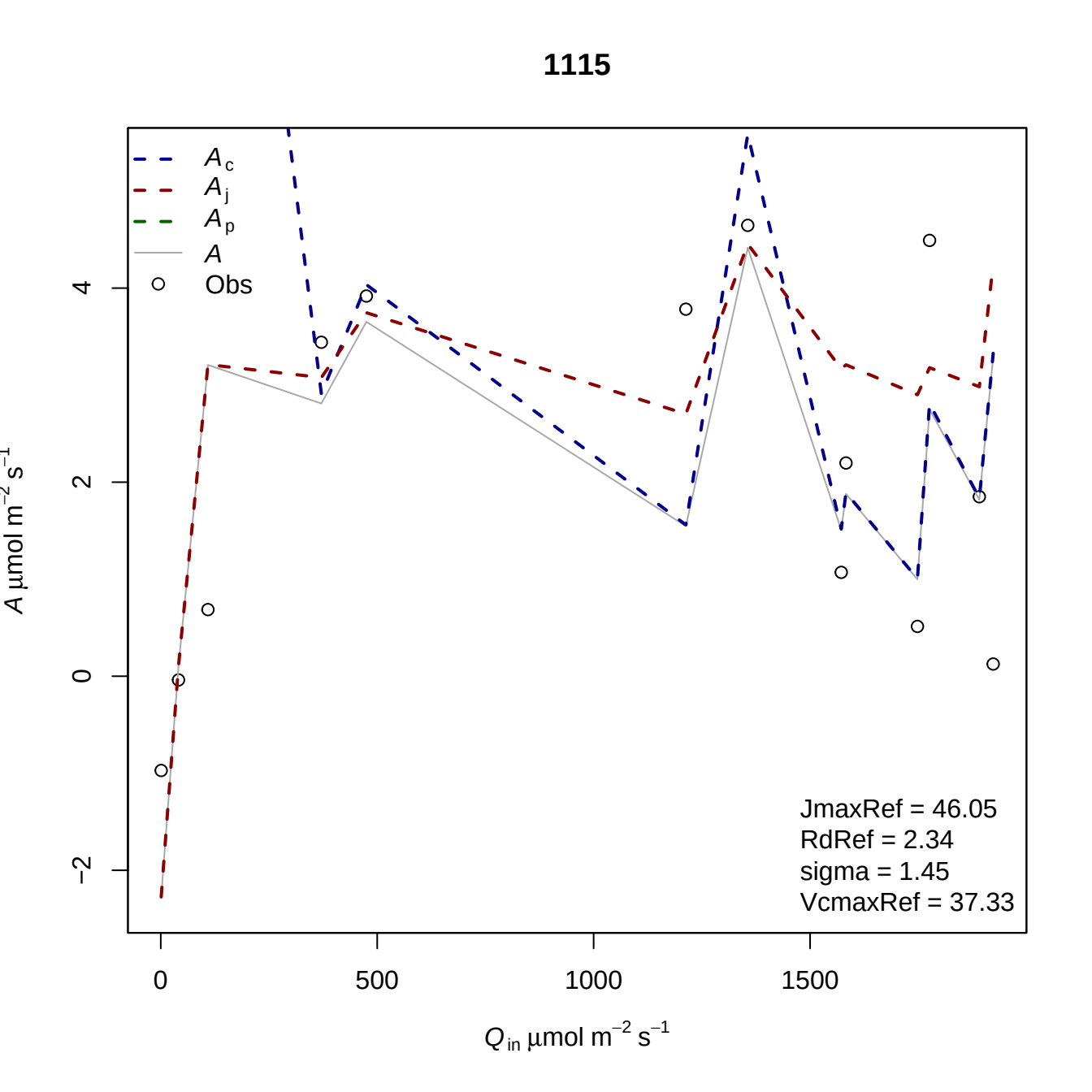

1110
A_c
A_j
A_p
A
Obs
JmaxRef = 35.85
RdRef = 2.11
sigma = 1.93
VcmaxRef = 560162.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

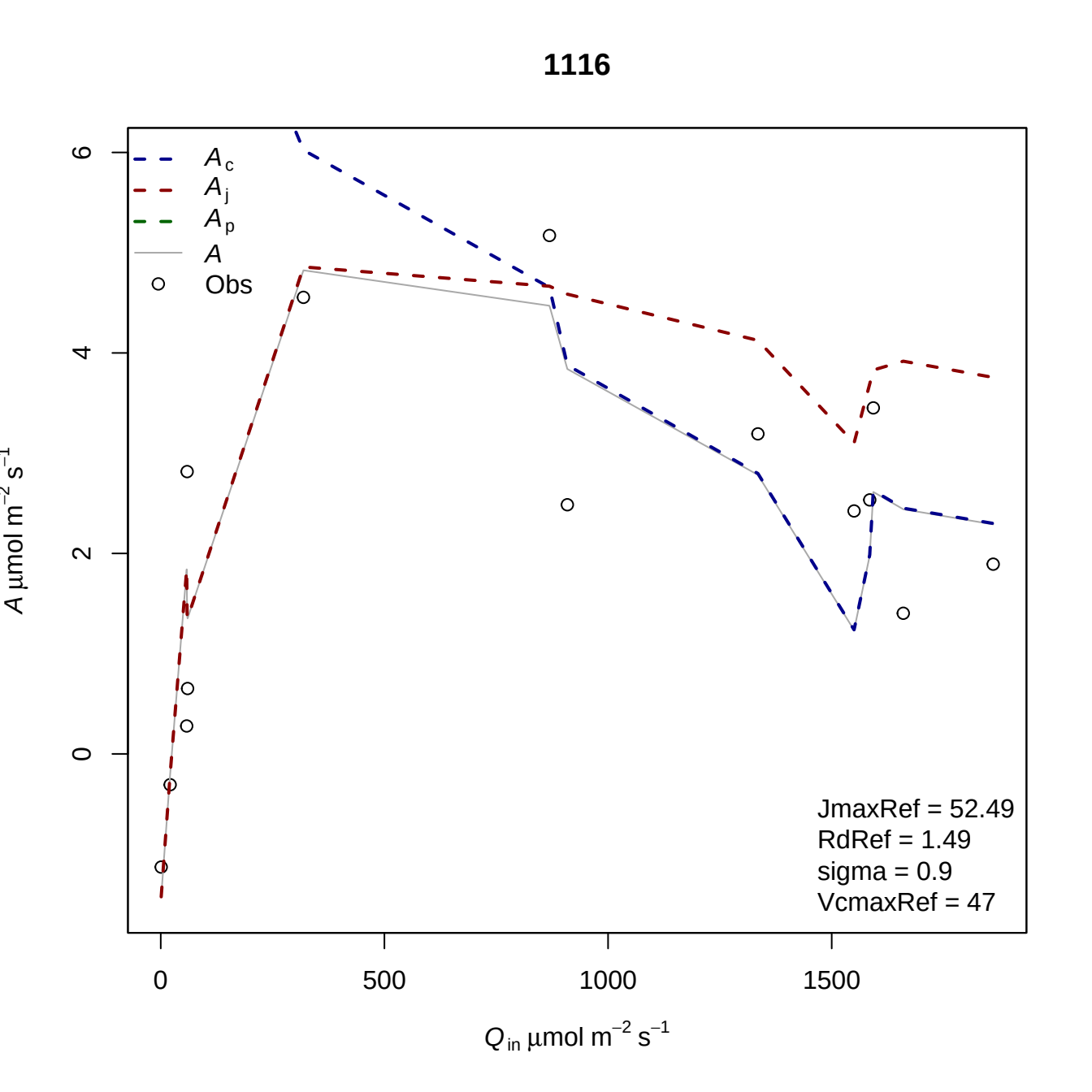

1111
A_c
A_j
A_p
A
Obs
JmaxRef = 61.6
RdRef = 4.89
sigma = 2.05
VcmaxRef = 215720.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
200
400
600
800
1000

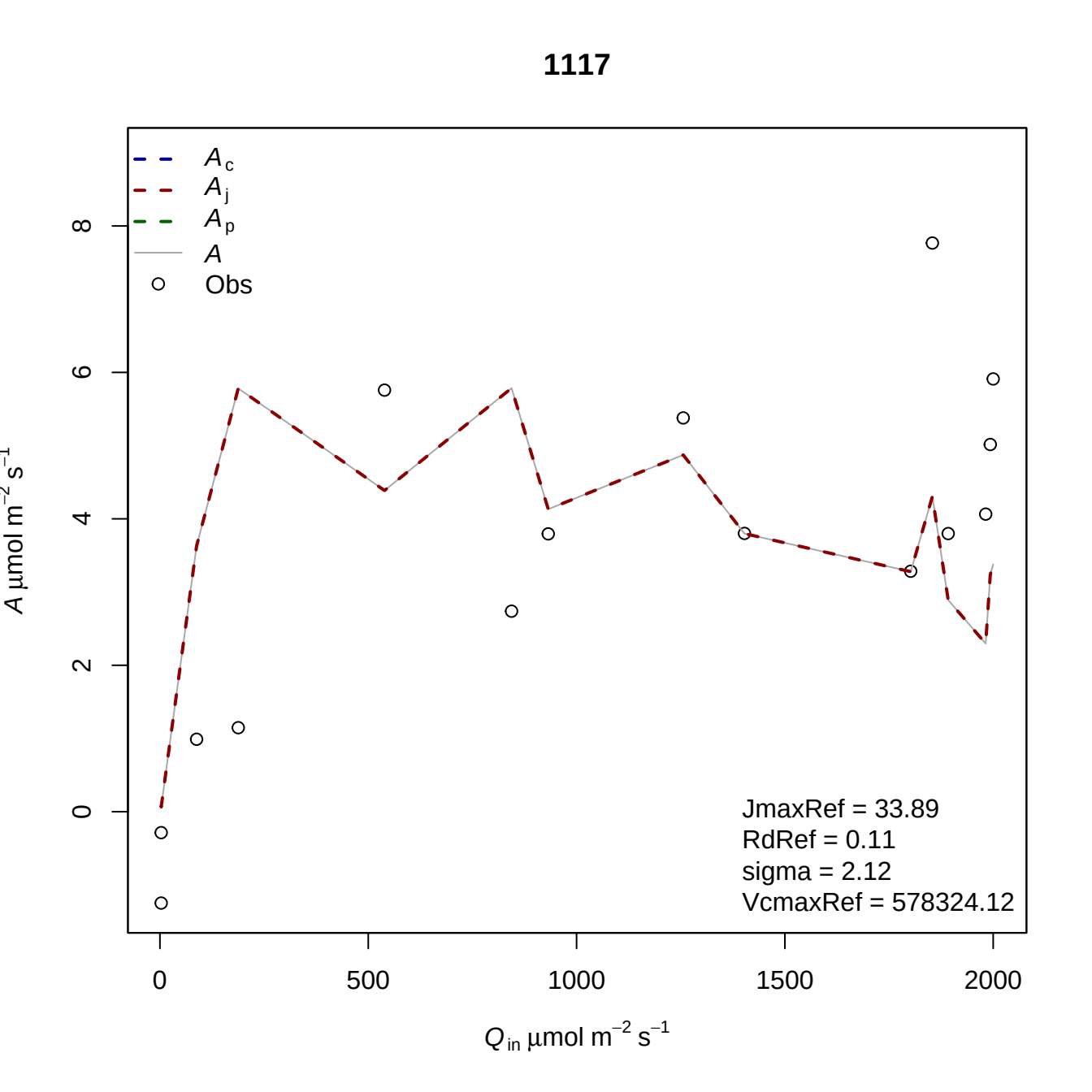

1112
A_c
A_j
A_p
A
Obs
JmaxRef = 64.03
RdRef = 3.9
sigma = 0.48
VcmaxRef = 119.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
500
1000
1500

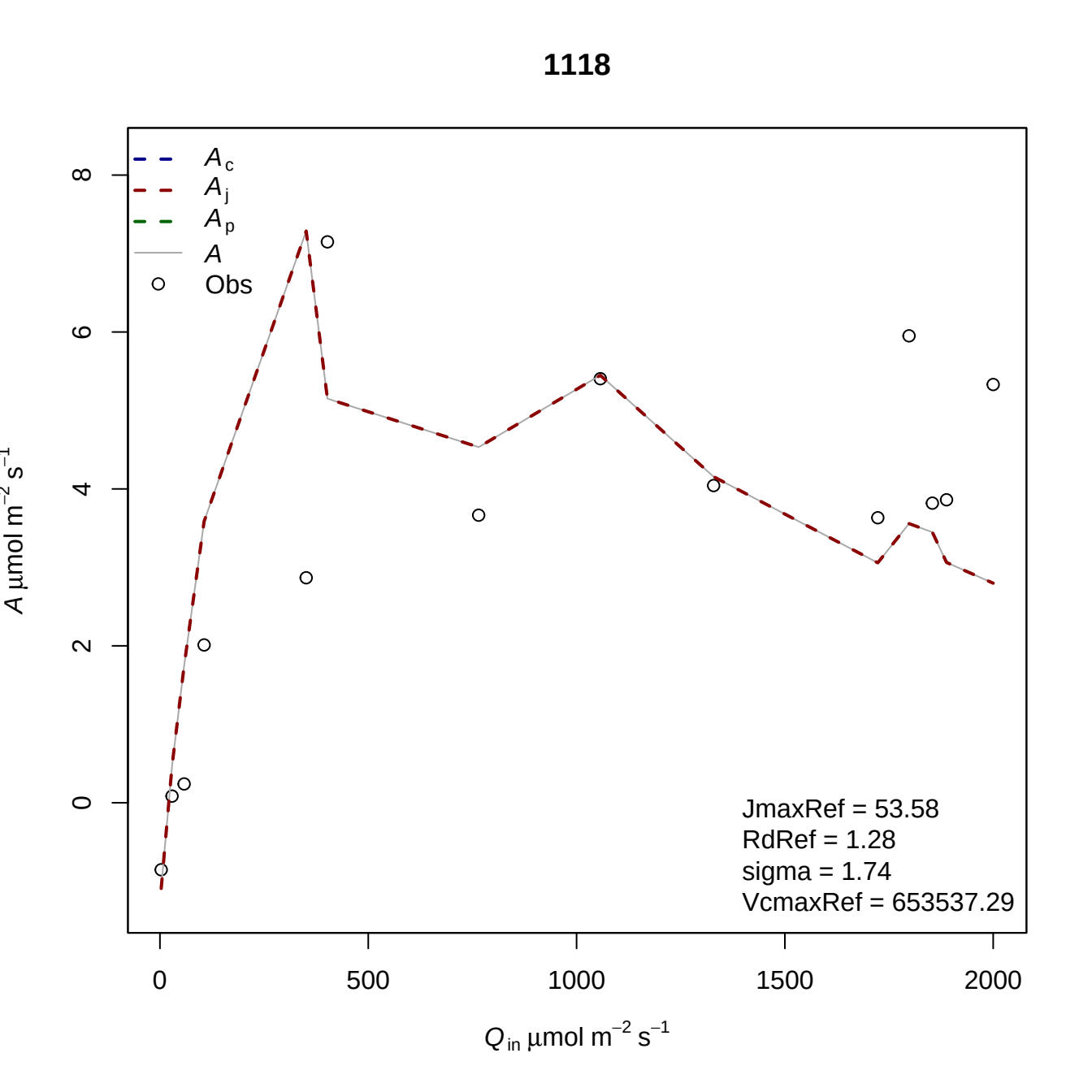

1113
A_c
A_j
A_p
A
Obs
JmaxRef = 23.06
RdRef = 1.15
sigma = 1.59
VcmaxRef = 16.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

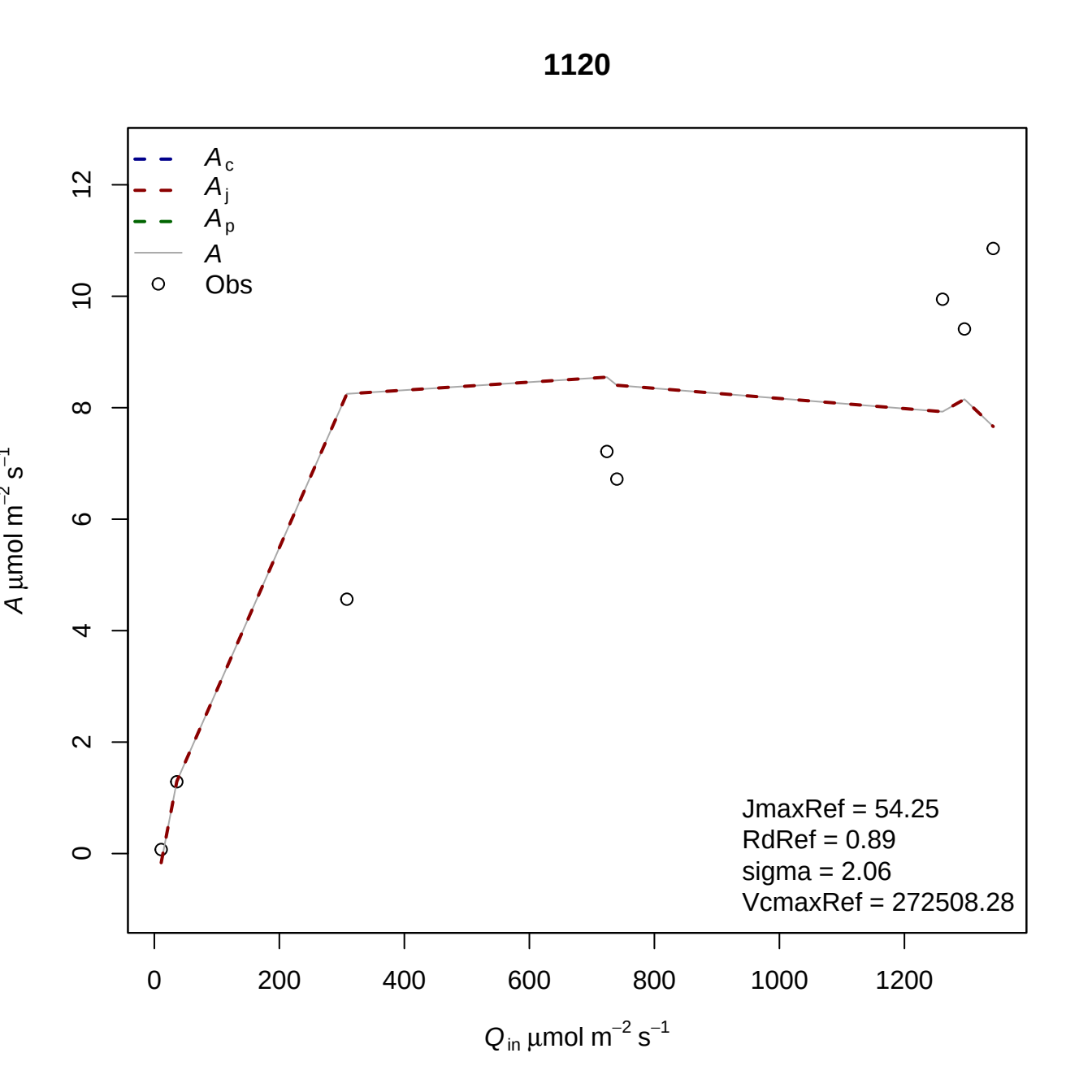

1114
A_c
A_j
A_p
A
Obs
JmaxRef = 23.84
RdRef = 1.36
sigma = 0.99
VcmaxRef = 89667.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200

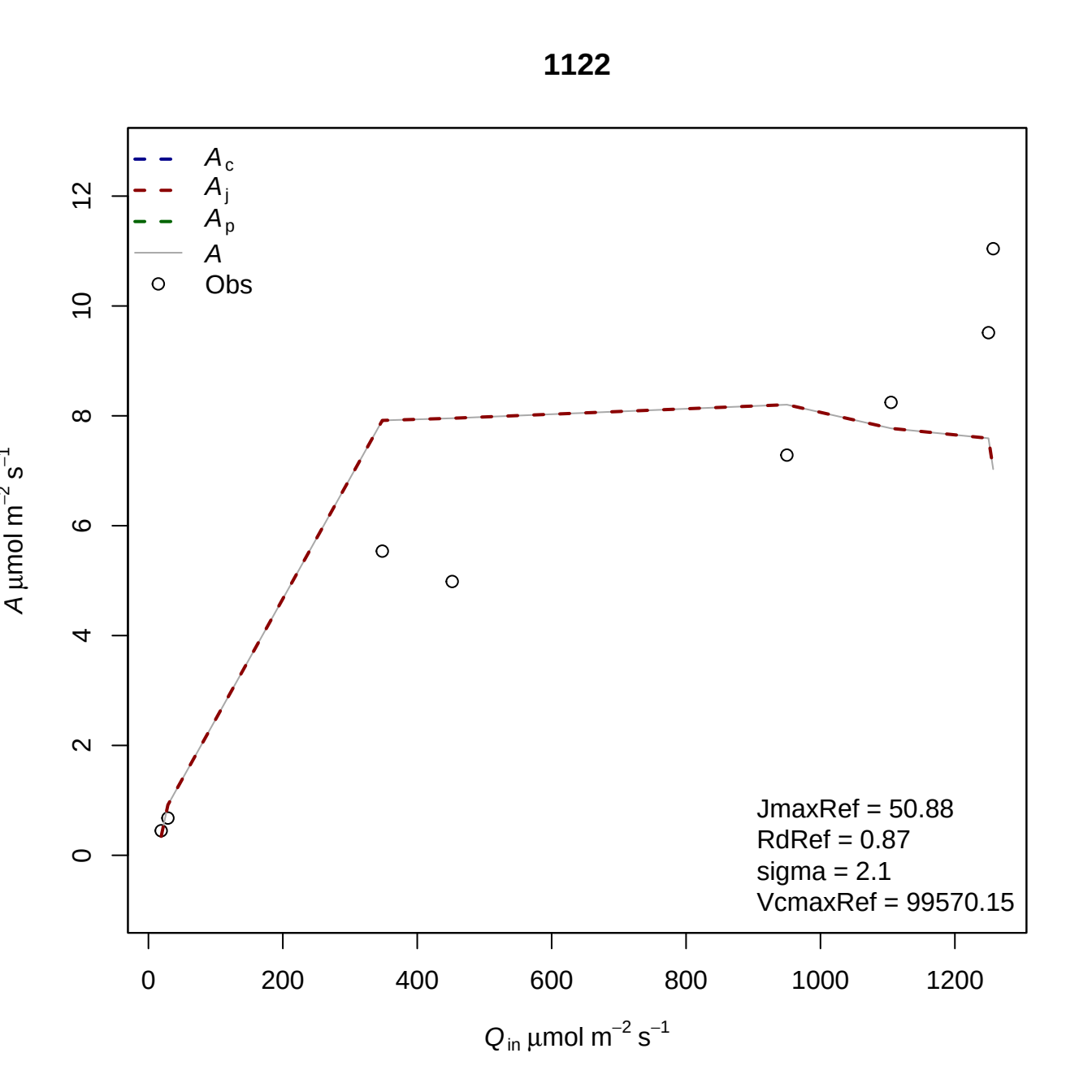

1115
A_c
A_j
A_p
A
Obs
JmaxRef = 46.05
RdRef = 2.34
sigma = 1.45
VcmaxRef = 37.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

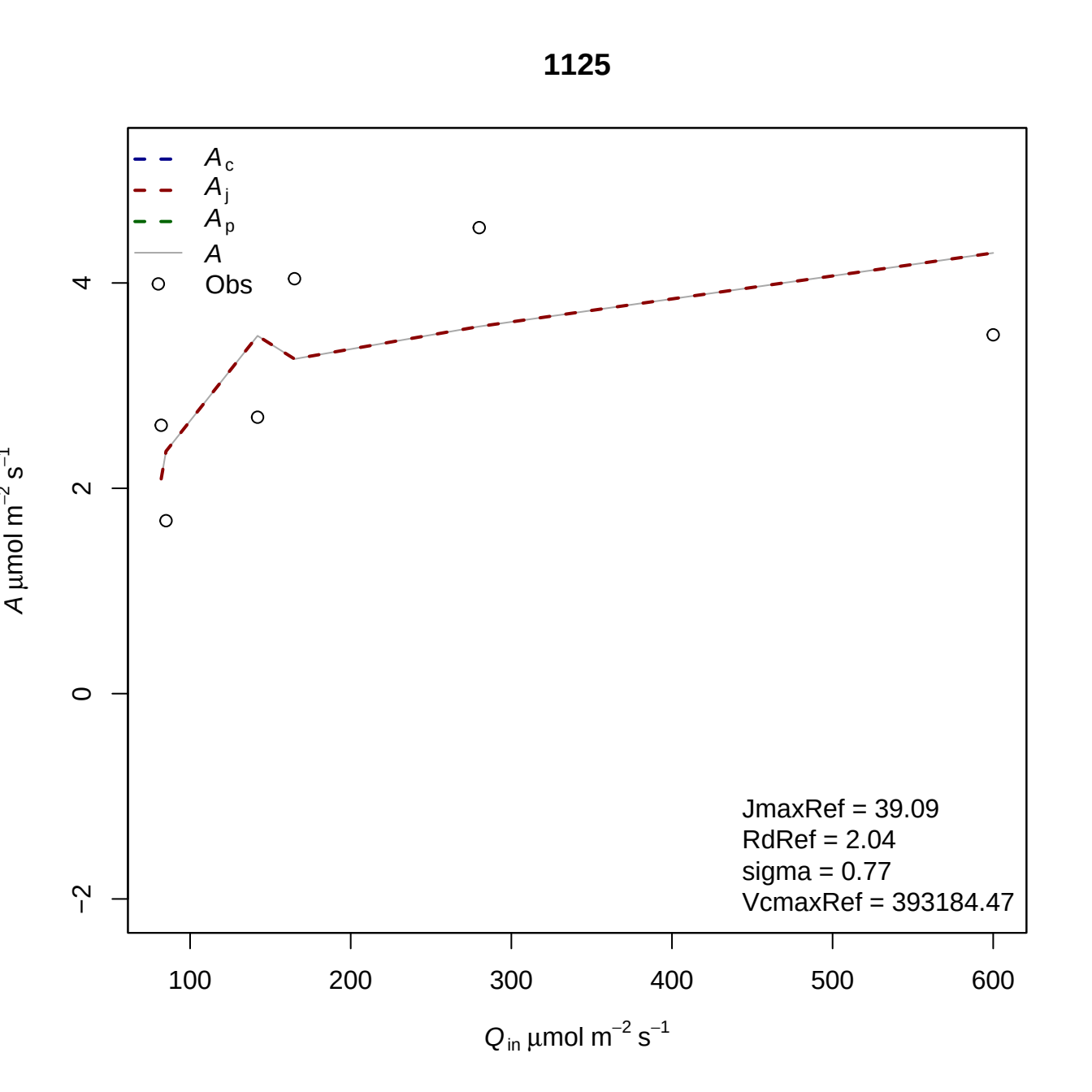

1116
A_c
A_j
A_p
A
Obs
JmaxRef = 52.49
RdRef = 1.49
sigma = 0.9
VcmaxRef = 47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

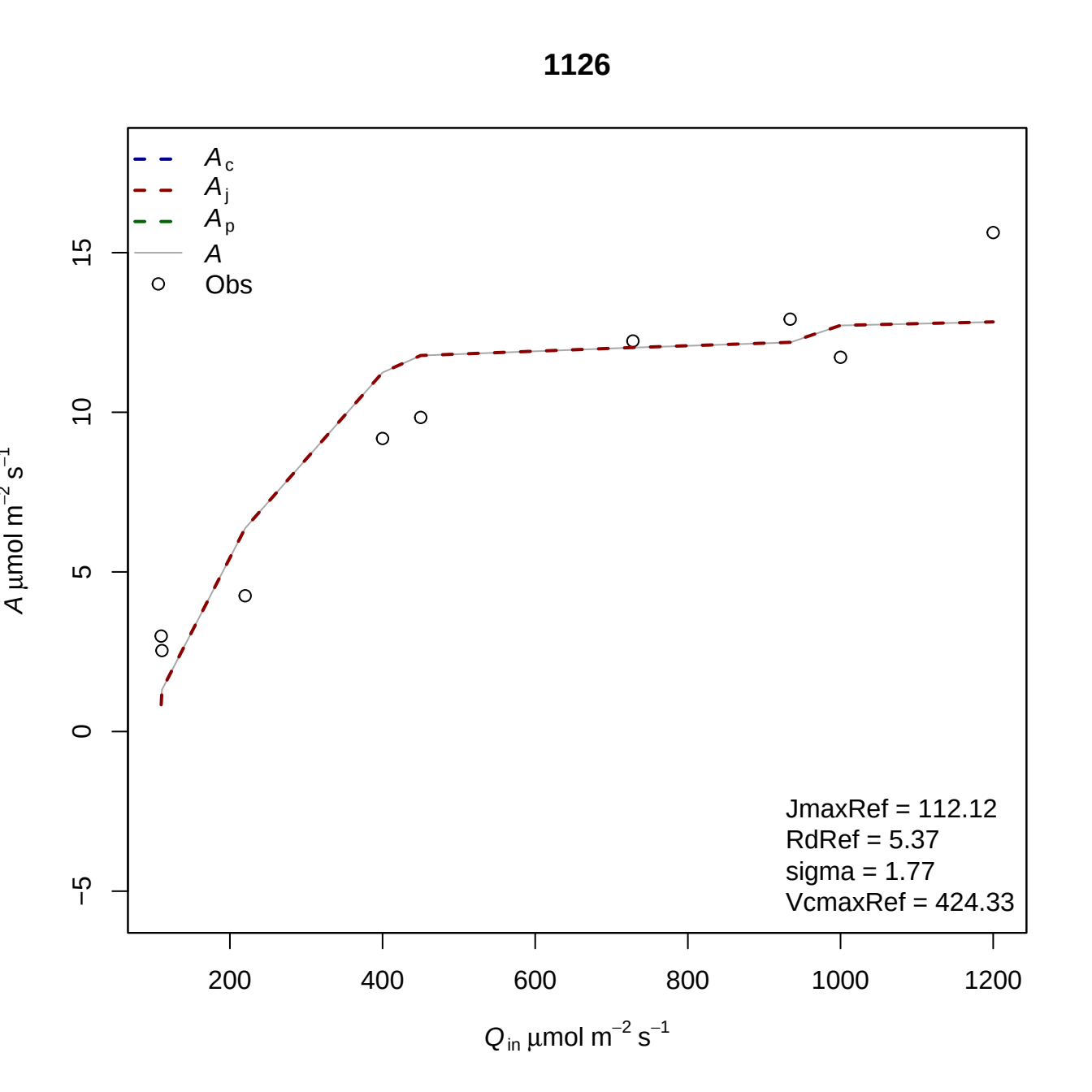

1117
A_c
A_j
A_p
A
Obs
JmaxRef = 33.89
RdRef = 0.11
sigma = 2.12
VcmaxRef = 578324.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

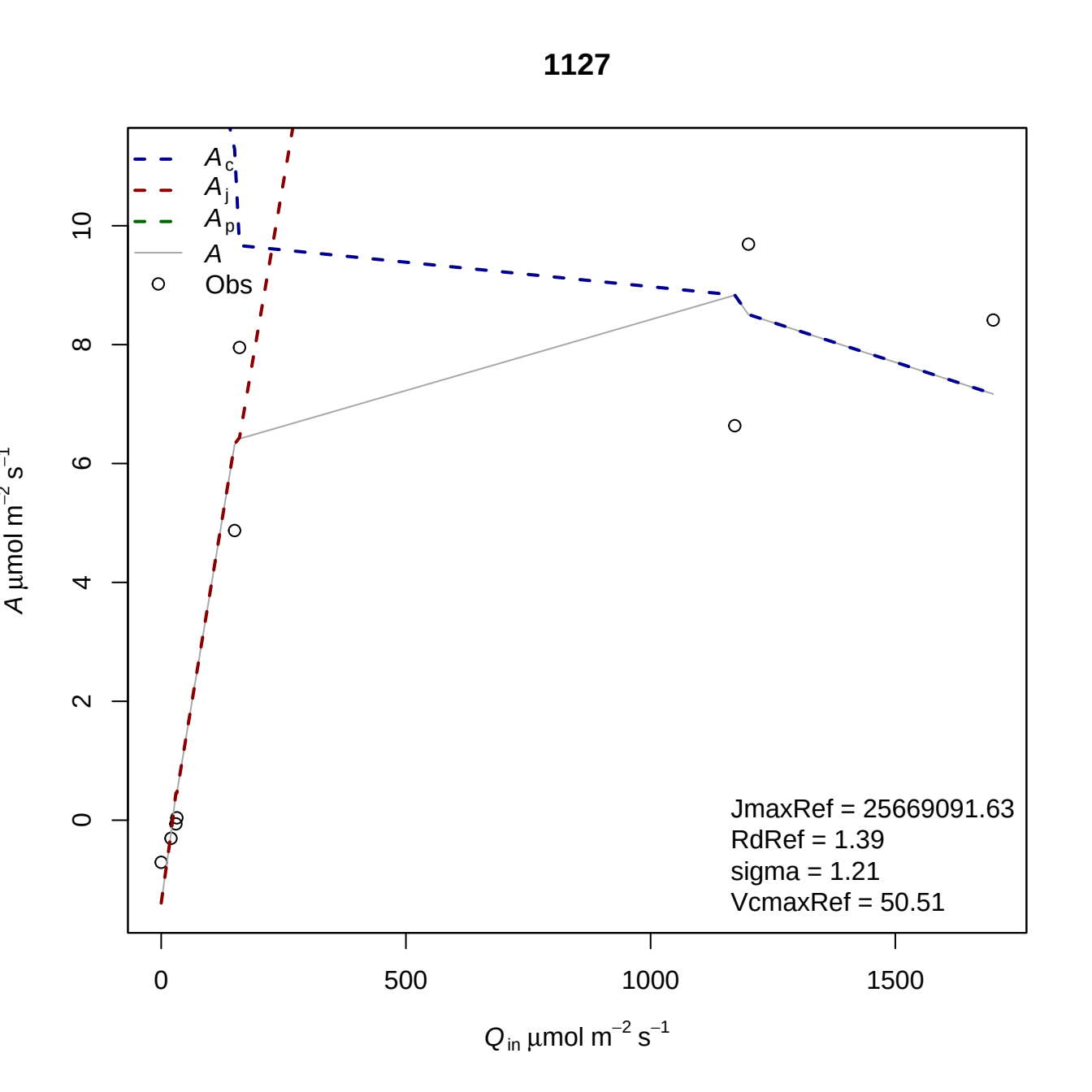

1118
A_c
A_j
A_p
A
Obs
JmaxRef = 53.58
RdRef = 1.28
sigma = 1.74
VcmaxRef = 653537.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

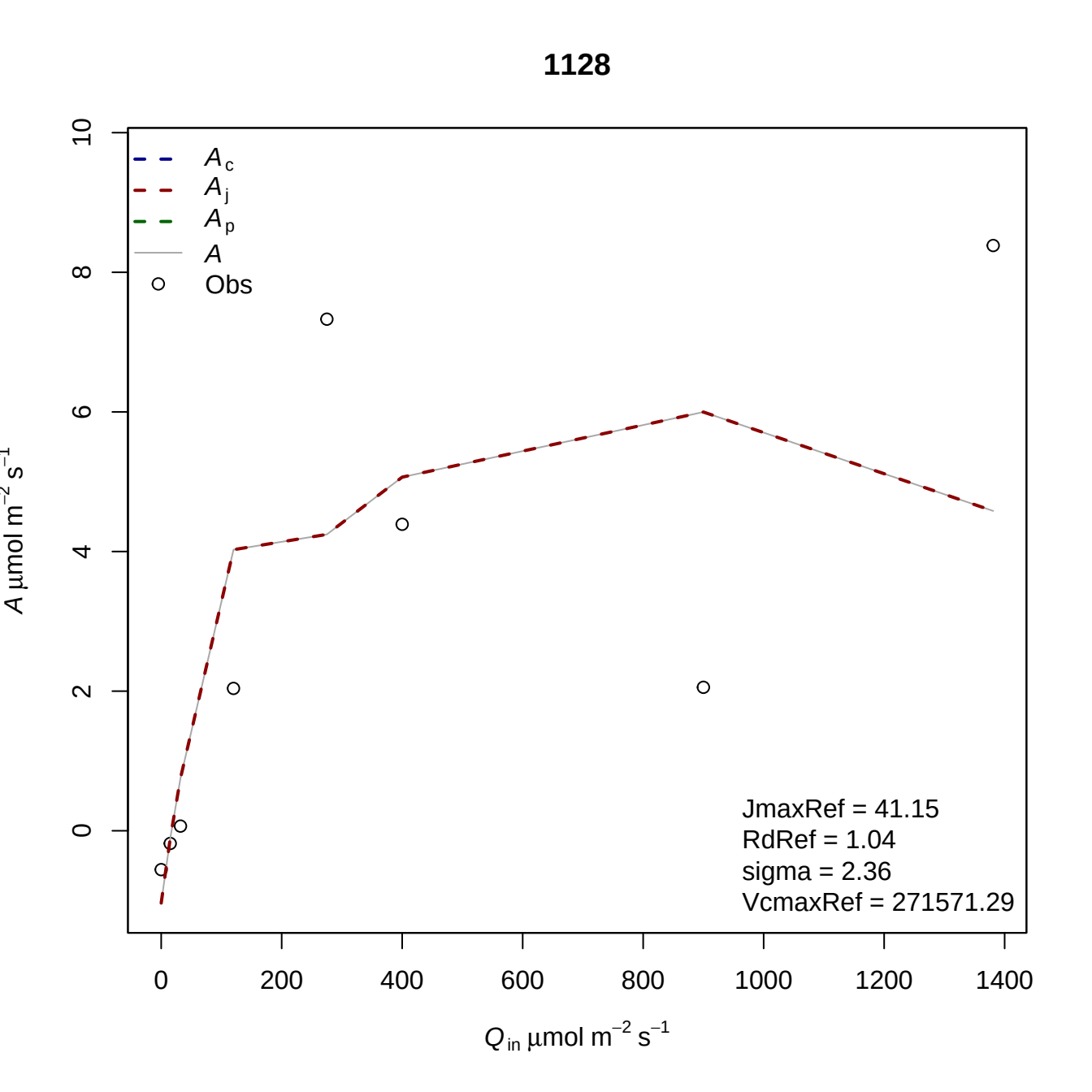

1120
A_c
A_j
A_p
A
Obs
JmaxRef = 54.25
RdRef = 0.89
sigma = 2.06
VcmaxRef = 272508.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

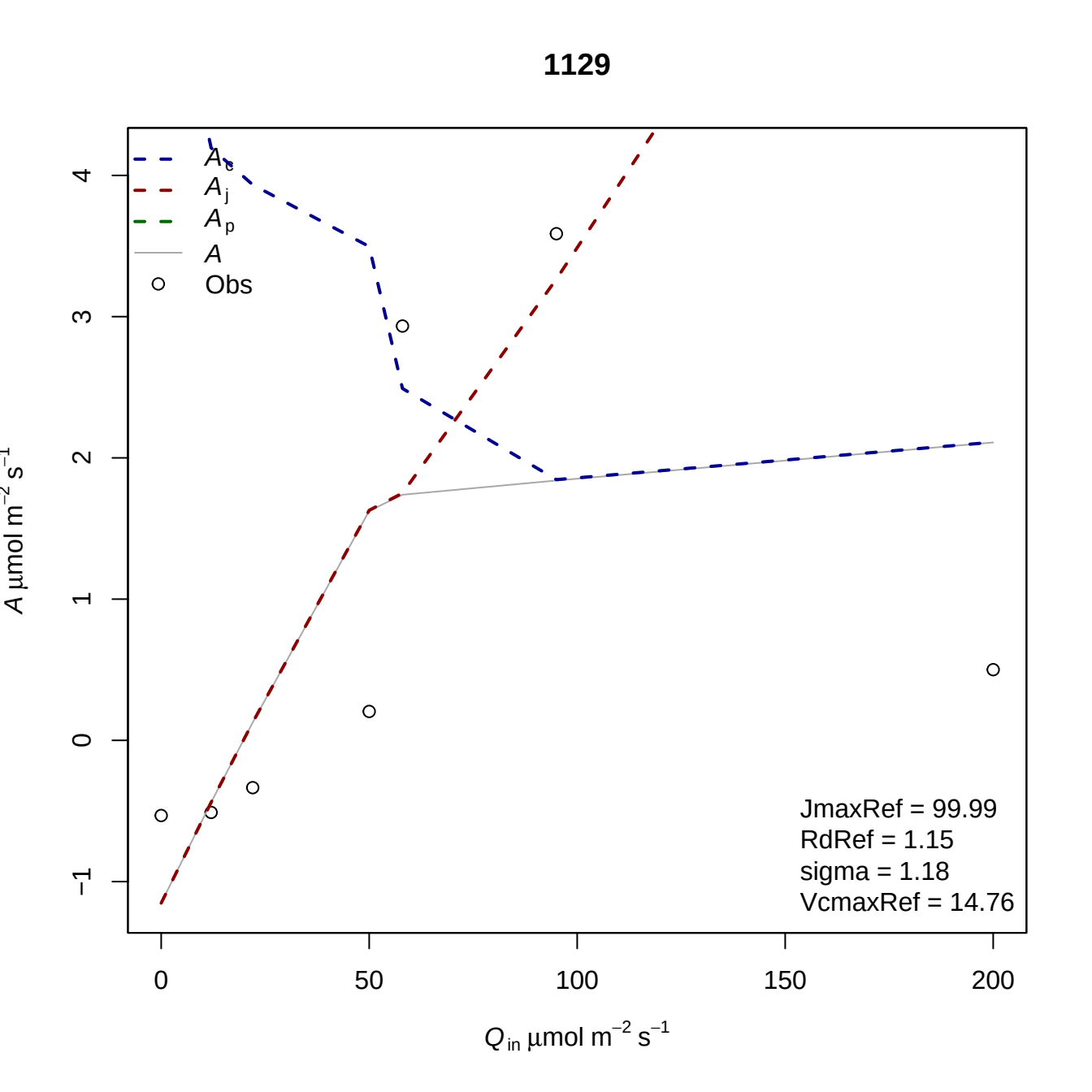

1122
A_c
A_j
A_p
A
Obs
JmaxRef = 50.88
RdRef = 0.87
sigma = 2.1
VcmaxRef = 99570.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

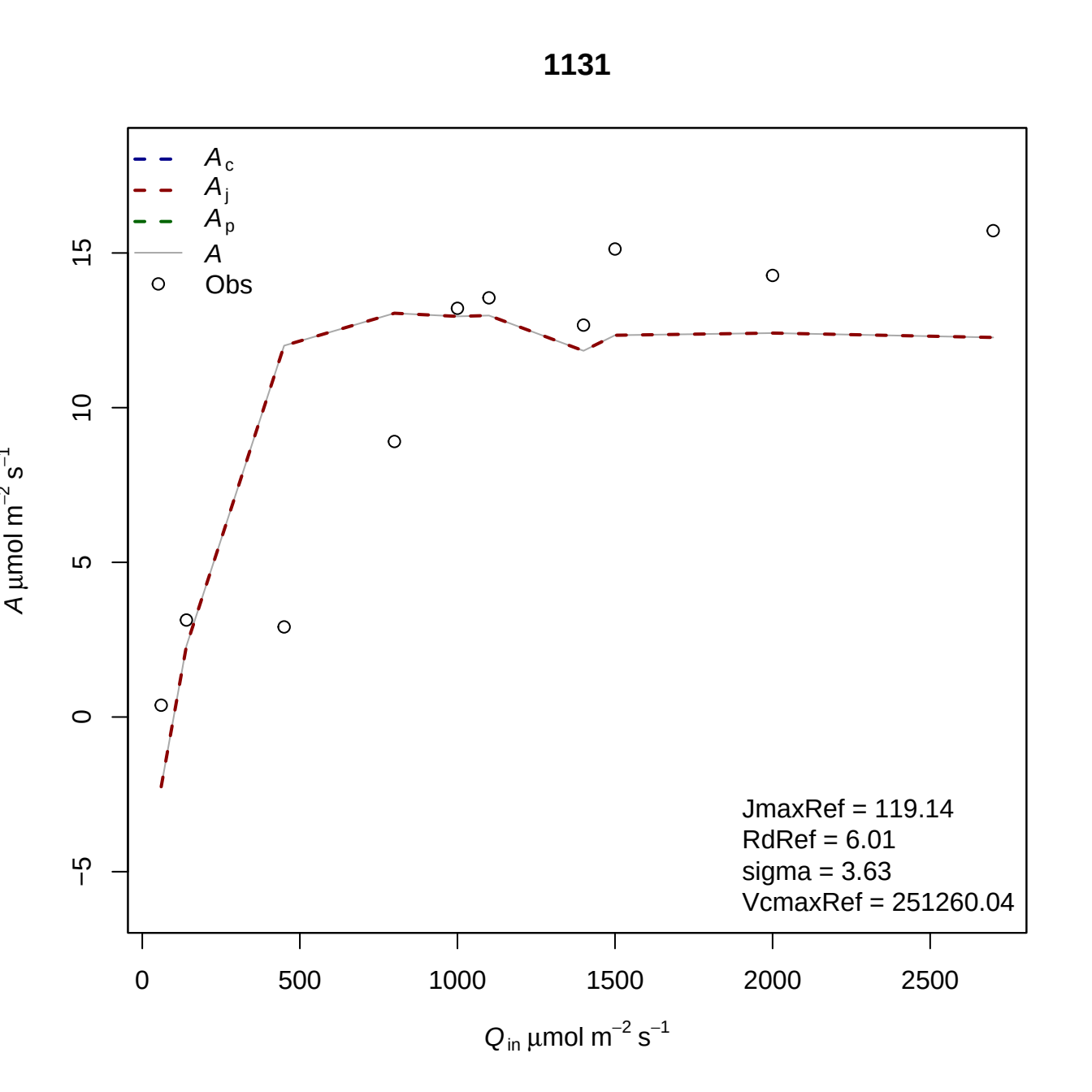

1125
A_c
A_j
A_p
A
Obs
JmaxRef = 39.09
RdRef = 2.04
sigma = 0.77
VcmaxRef = 393184.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
100
200
300
400
500
600

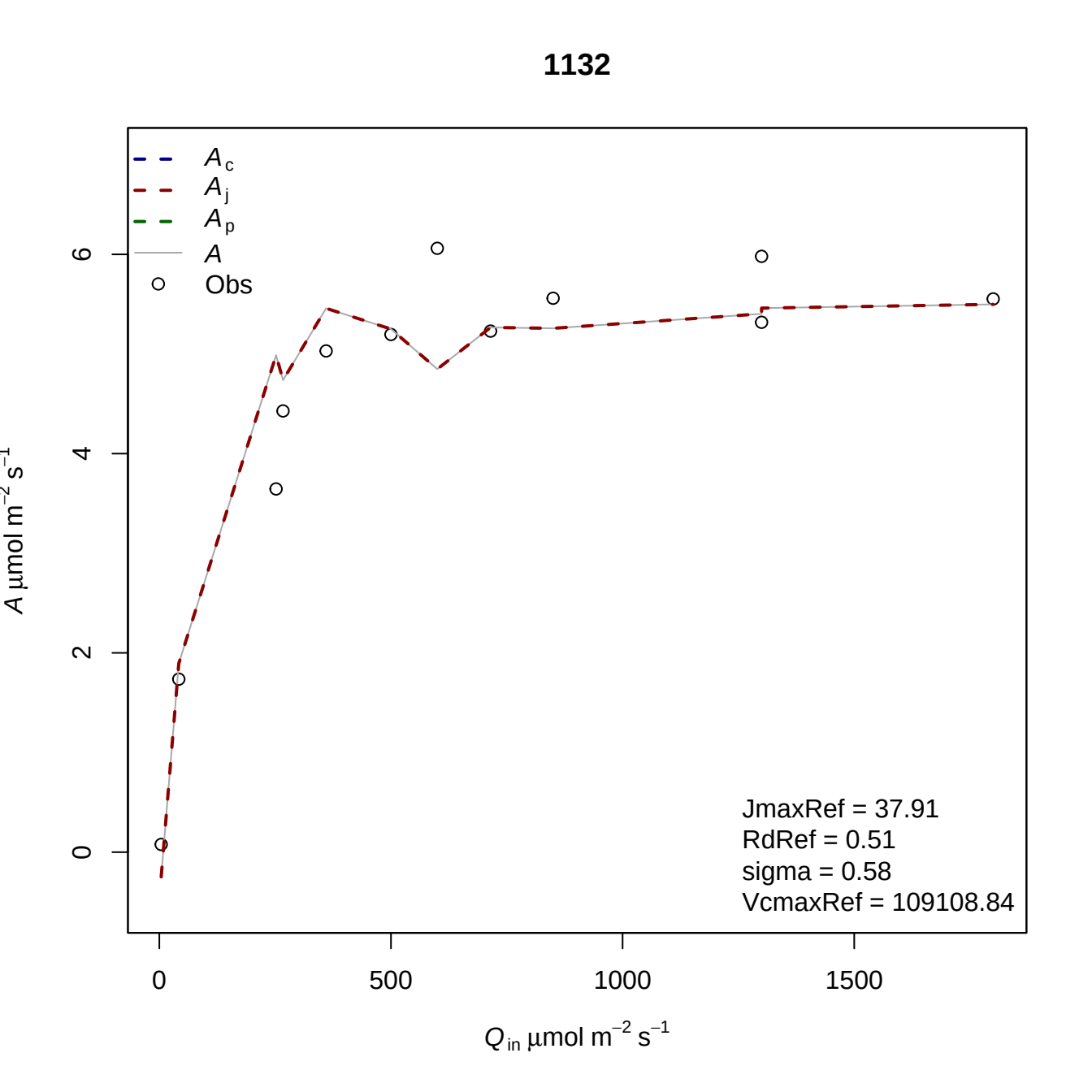

1126
A_c
A_j
A_p
A
Obs
JmaxRef = 112.12
RdRef = 5.37
sigma = 1.77
VcmaxRef = 424.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
200
400
600
800
1000
1200

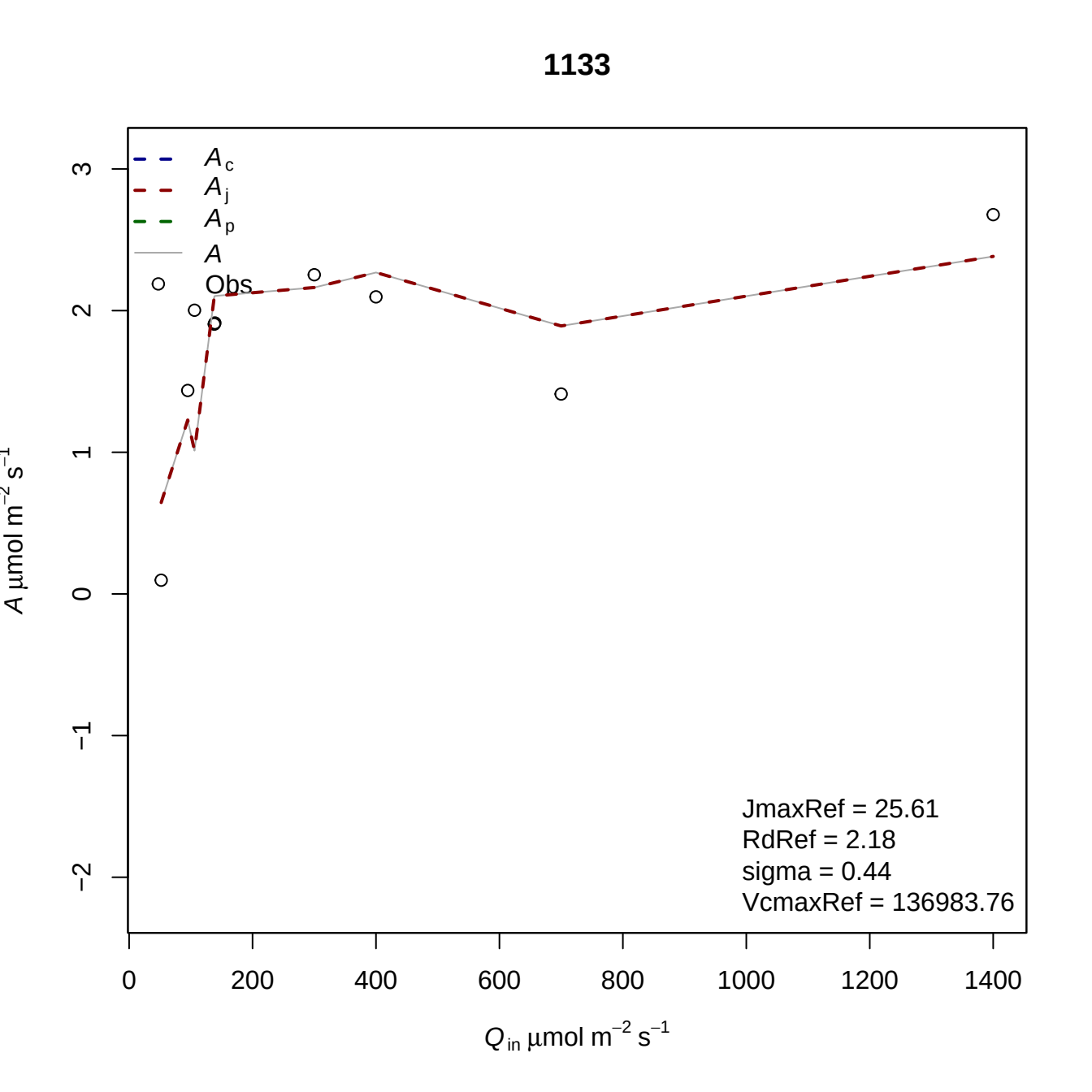

1127
A_c
A_j
A_p
A
Obs
JmaxRef = 25669091.63
RdRef = 1.39
sigma = 1.21
VcmaxRef = 50.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

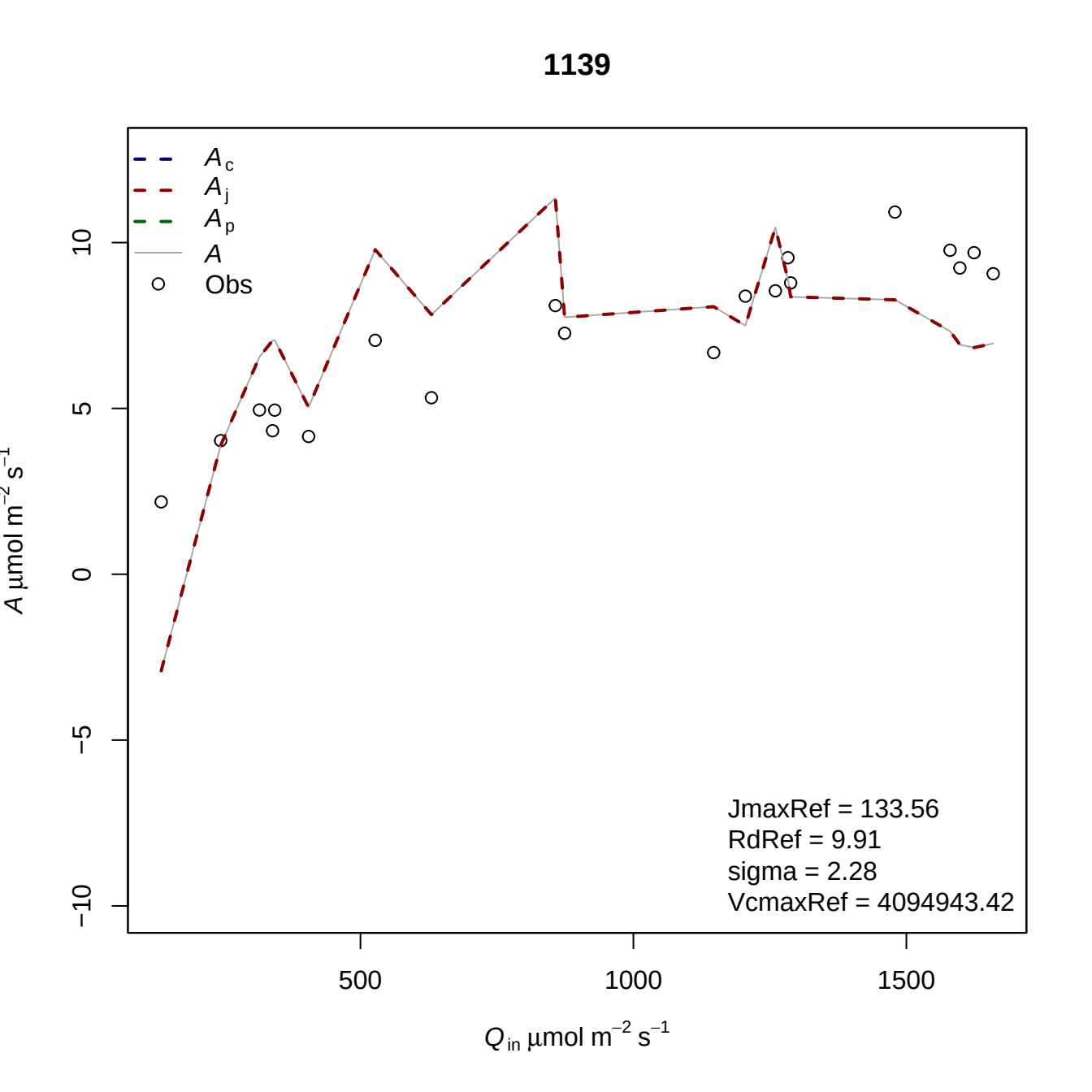

1128
A_c
A_j
A_p
A
Obs
JmaxRef = 41.15
RdRef = 1.04
sigma = 2.36
VcmaxRef = 271571.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

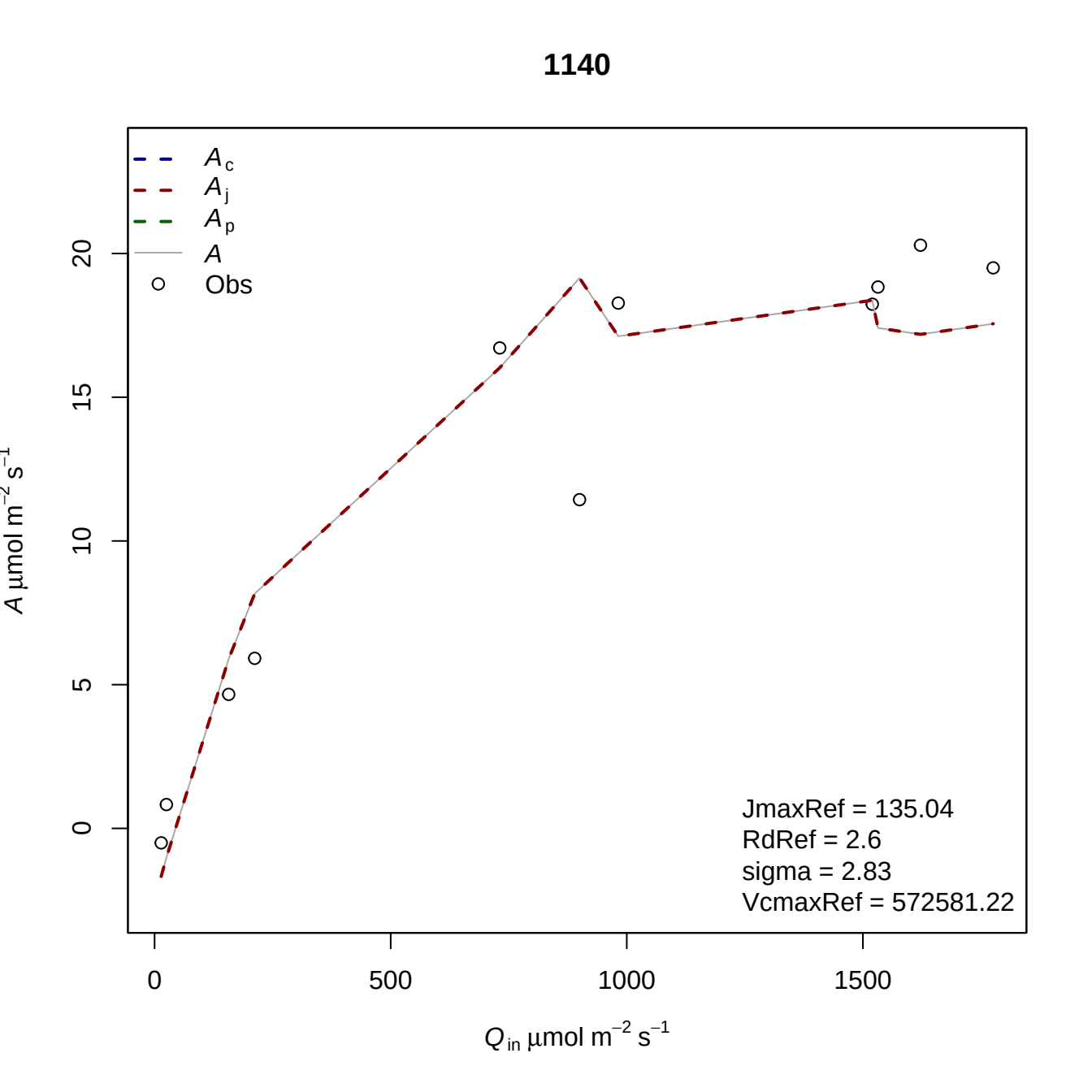

1129
A_c
A_j
A_p
A
Obs
JmaxRef = 99.99
RdRef = 1.15
sigma = 1.18
VcmaxRef = 14.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
50
100
150
200

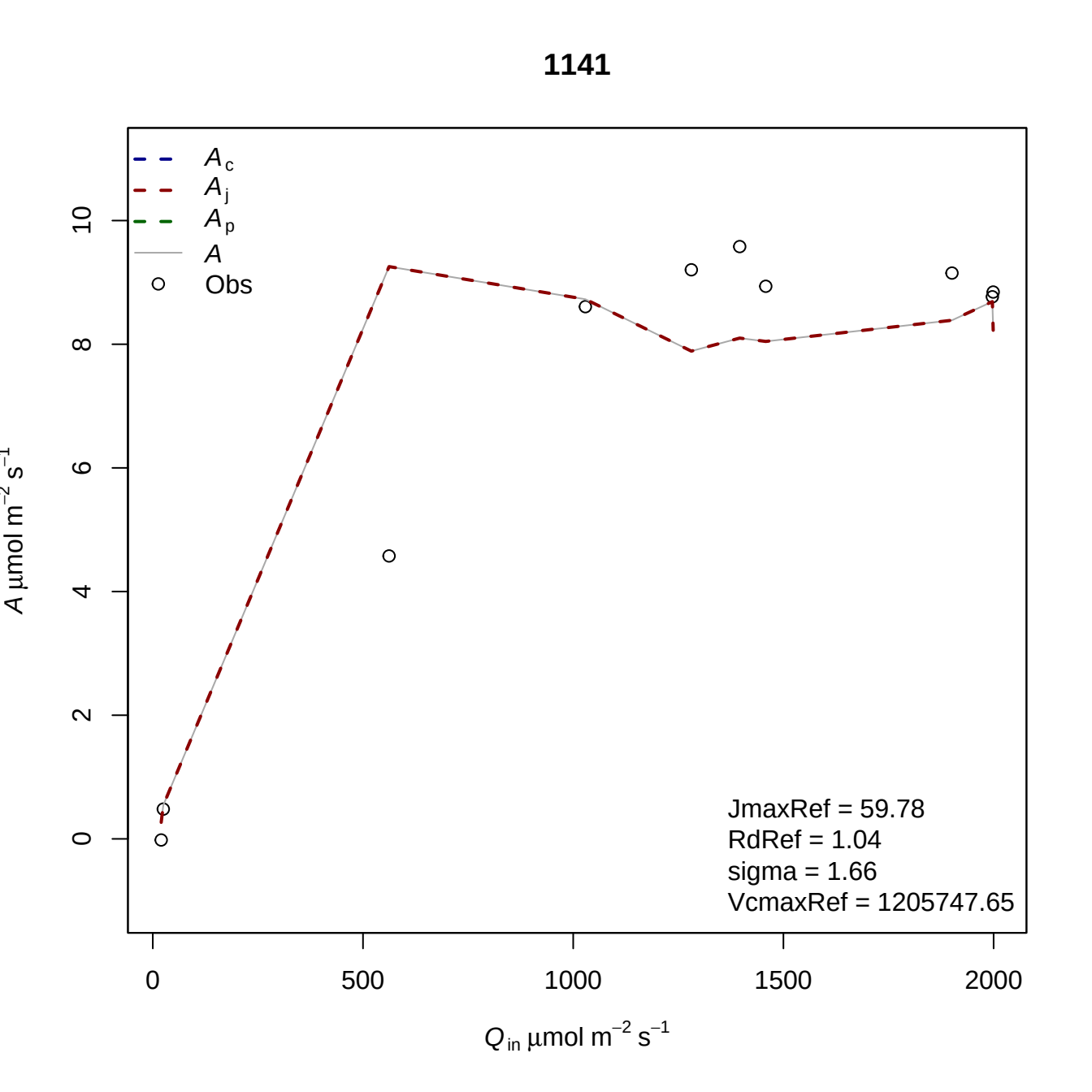

1131
A_c
A_j
A_p
A
Obs
JmaxRef = 119.14
RdRef = 6.01
sigma = 3.63
VcmaxRef = 251260.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000
2500

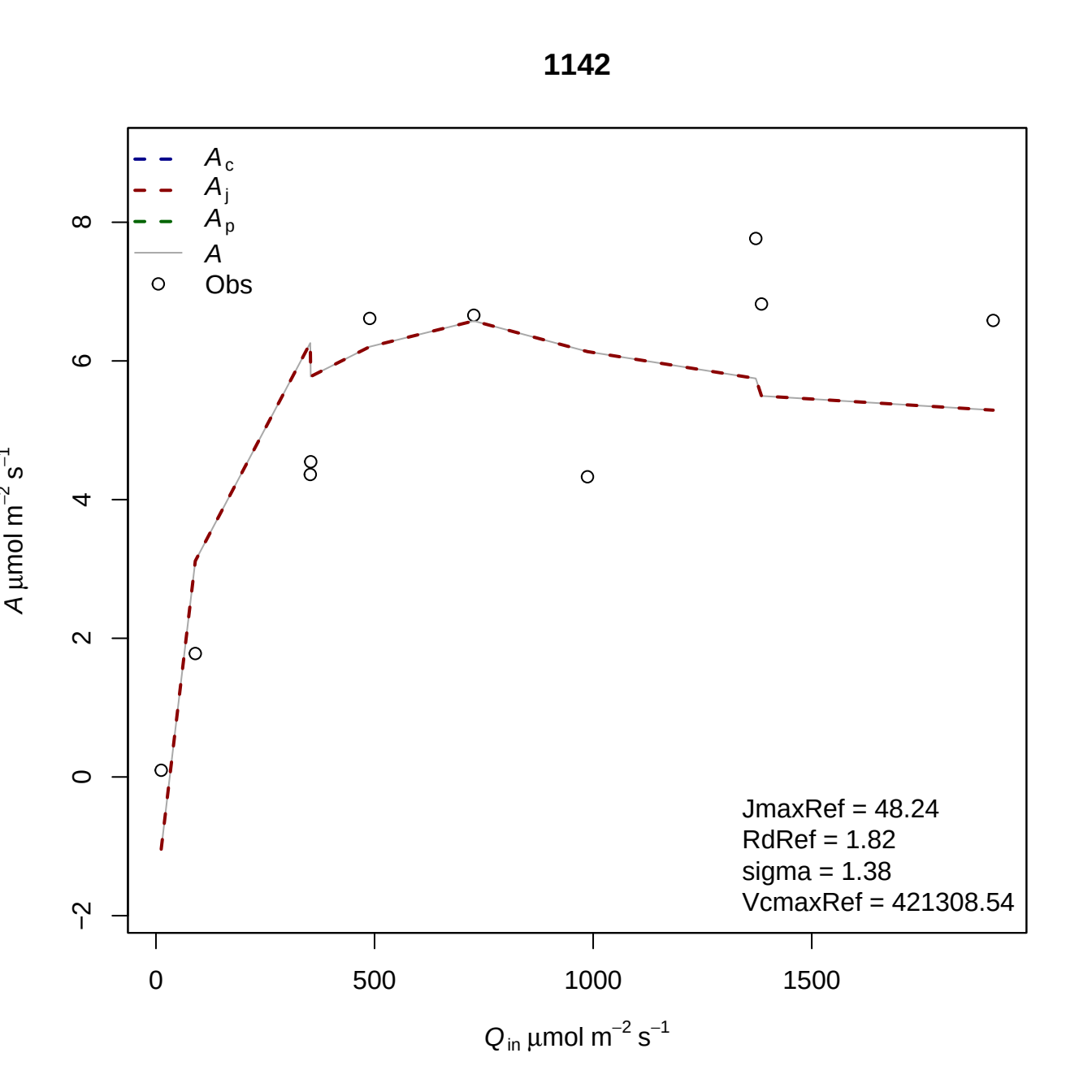

1132
A_c
A_j
A_p
A
Obs
JmaxRef = 37.91
RdRef = 0.51
sigma = 0.58
VcmaxRef = 109108.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

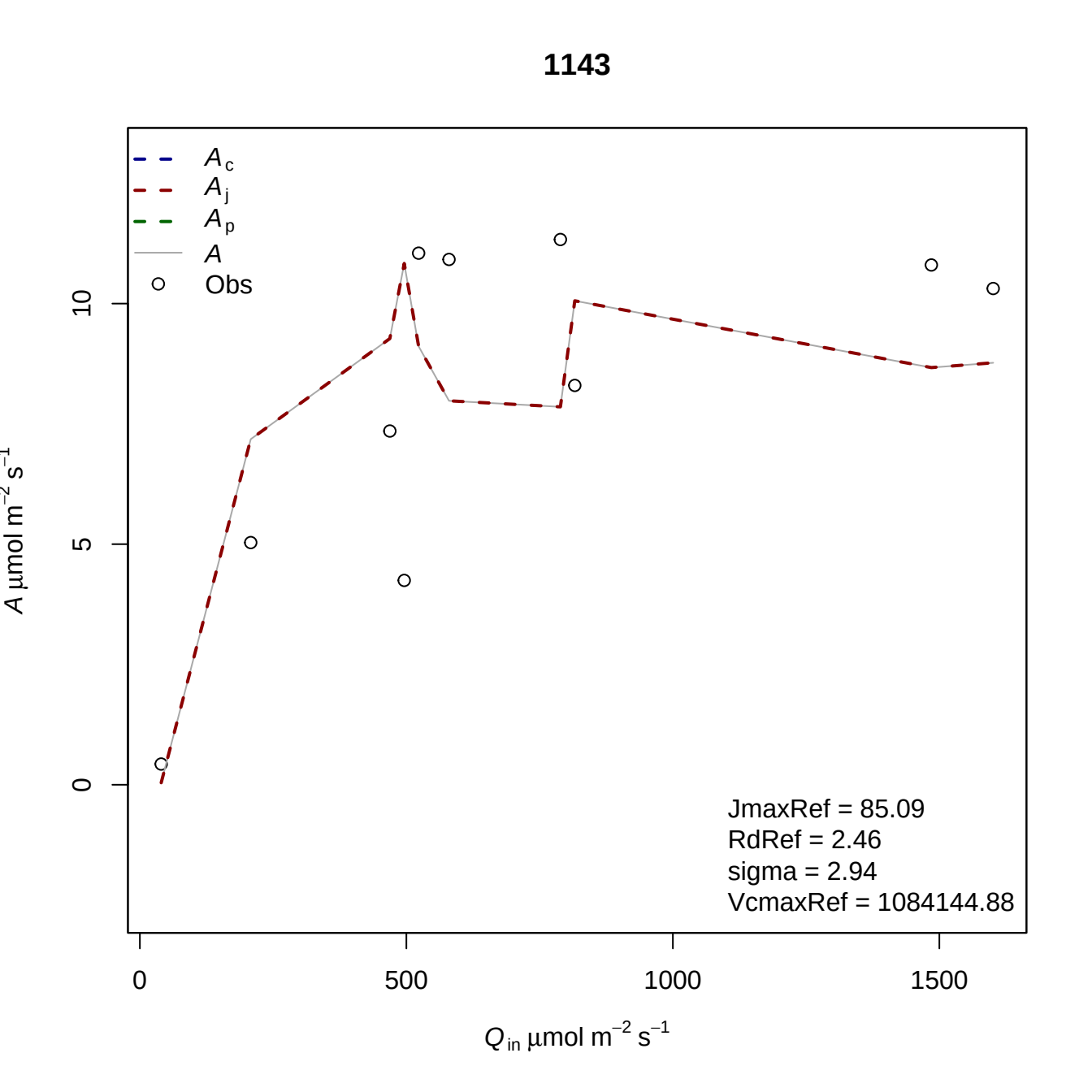

1133
A_c
A_j
A_p
A
Obs
JmaxRef = 25.61
RdRef = 2.18
sigma = 0.44
VcmaxRef = 136983.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

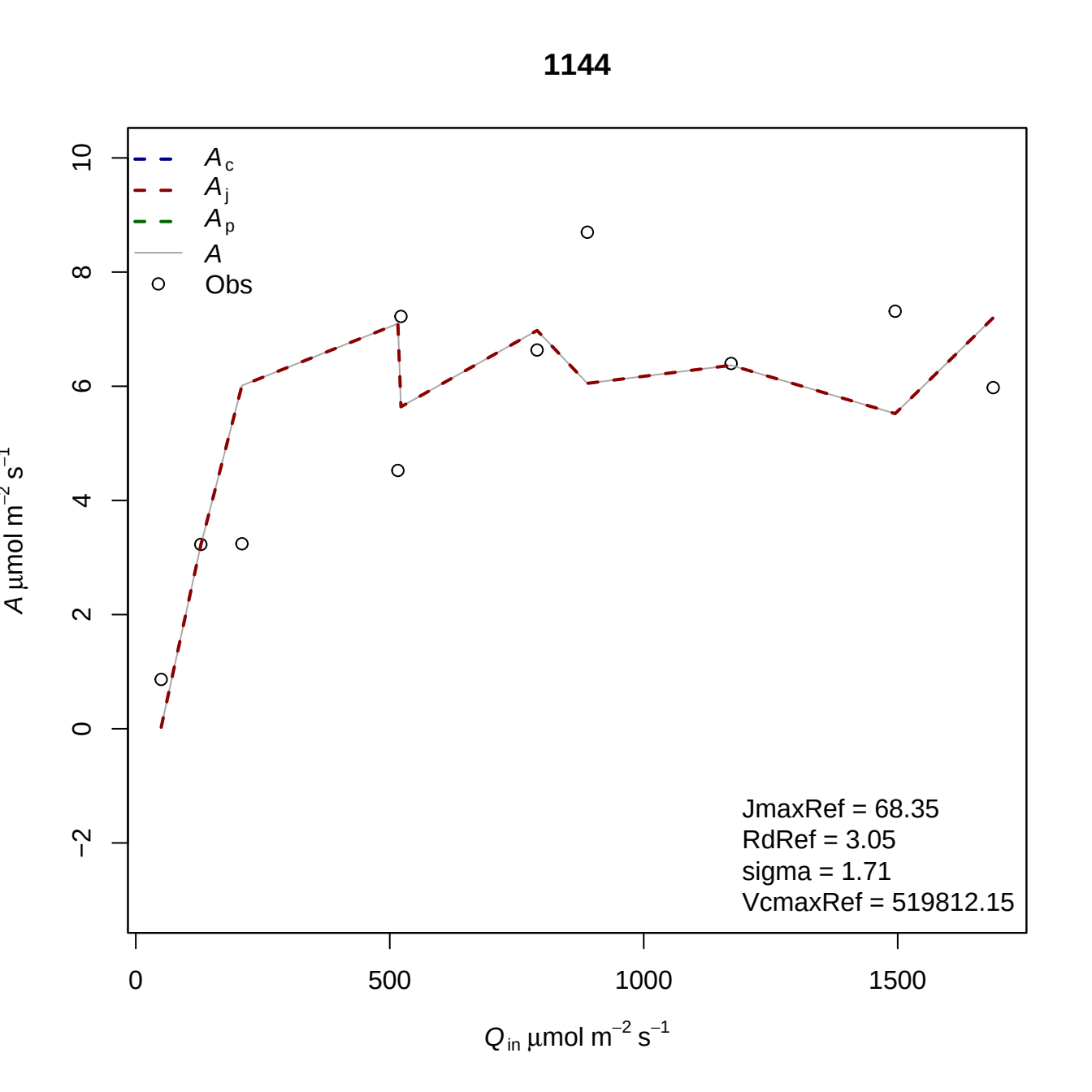

1139
A_c
A_j
A_p
A
Obs
JmaxRef = 133.56
RdRef = 9.91
sigma = 2.28
VcmaxRef = 4094943.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
−10
500
1000
1500

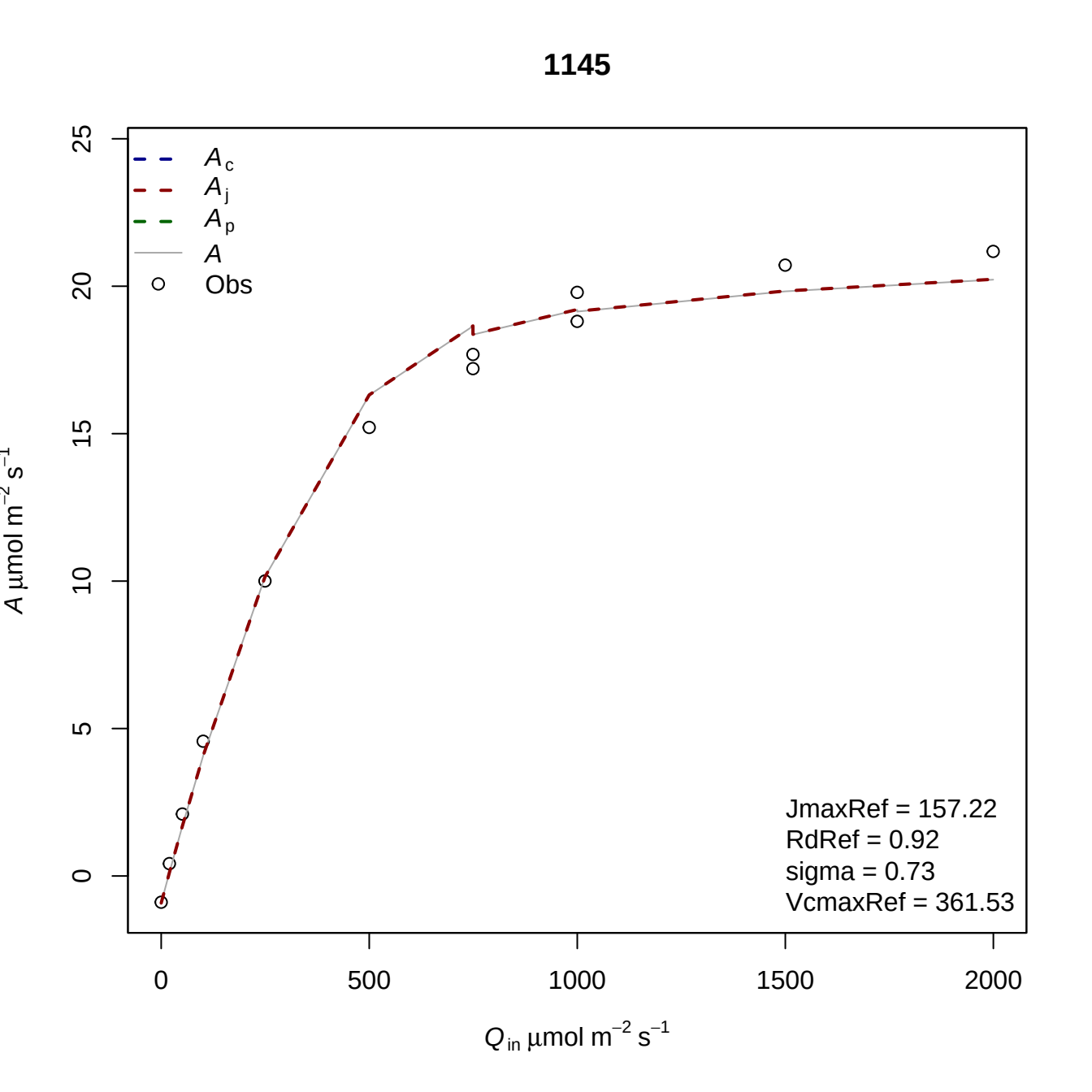

1140
A_c
A_j
A_p
A
Obs
JmaxRef = 135.04
RdRef = 2.6
sigma = 2.83
VcmaxRef = 572581.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

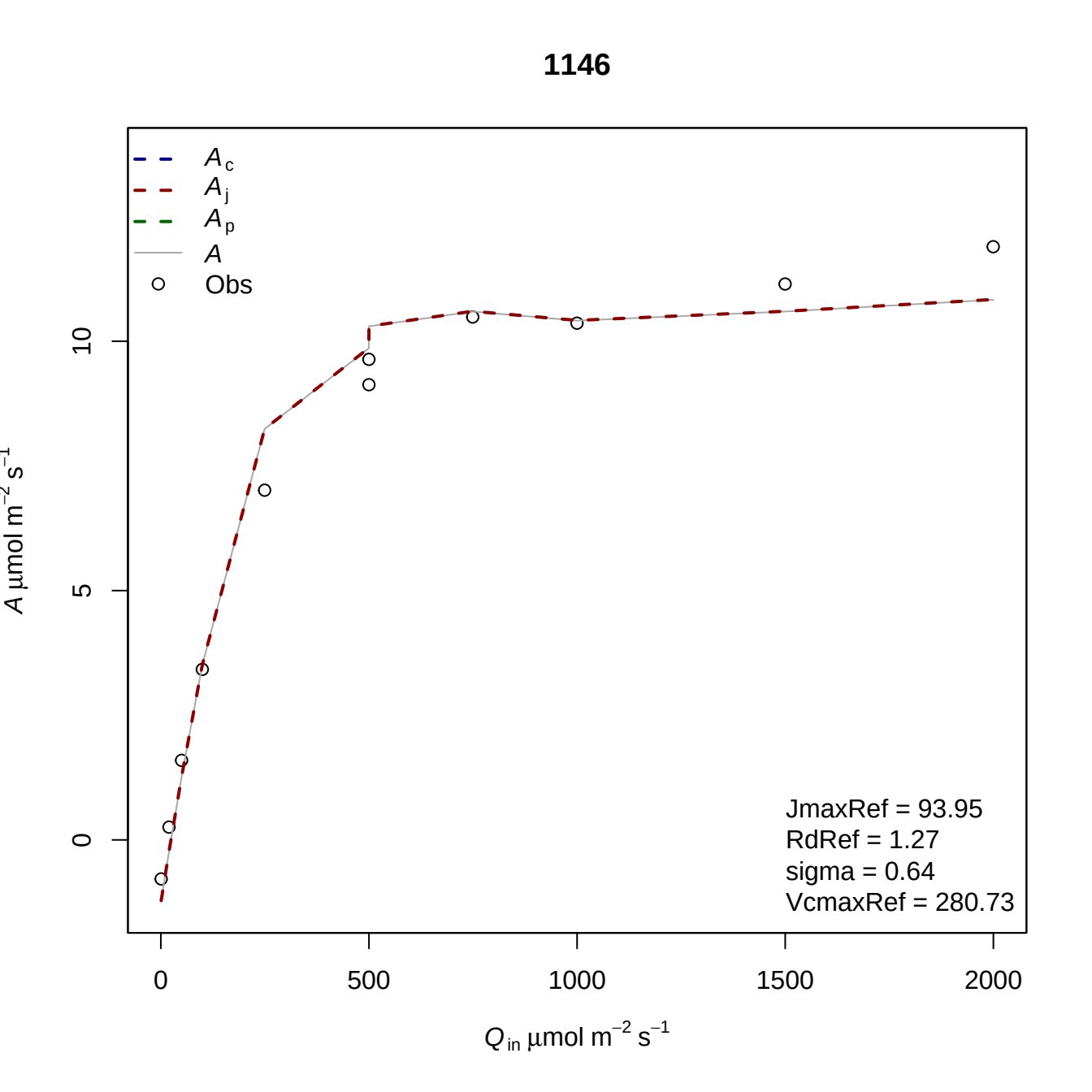

1141
A_c
A_j
A_p
A
Obs
JmaxRef = 59.78
RdRef = 1.04
sigma = 1.66
VcmaxRef = 1205747.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

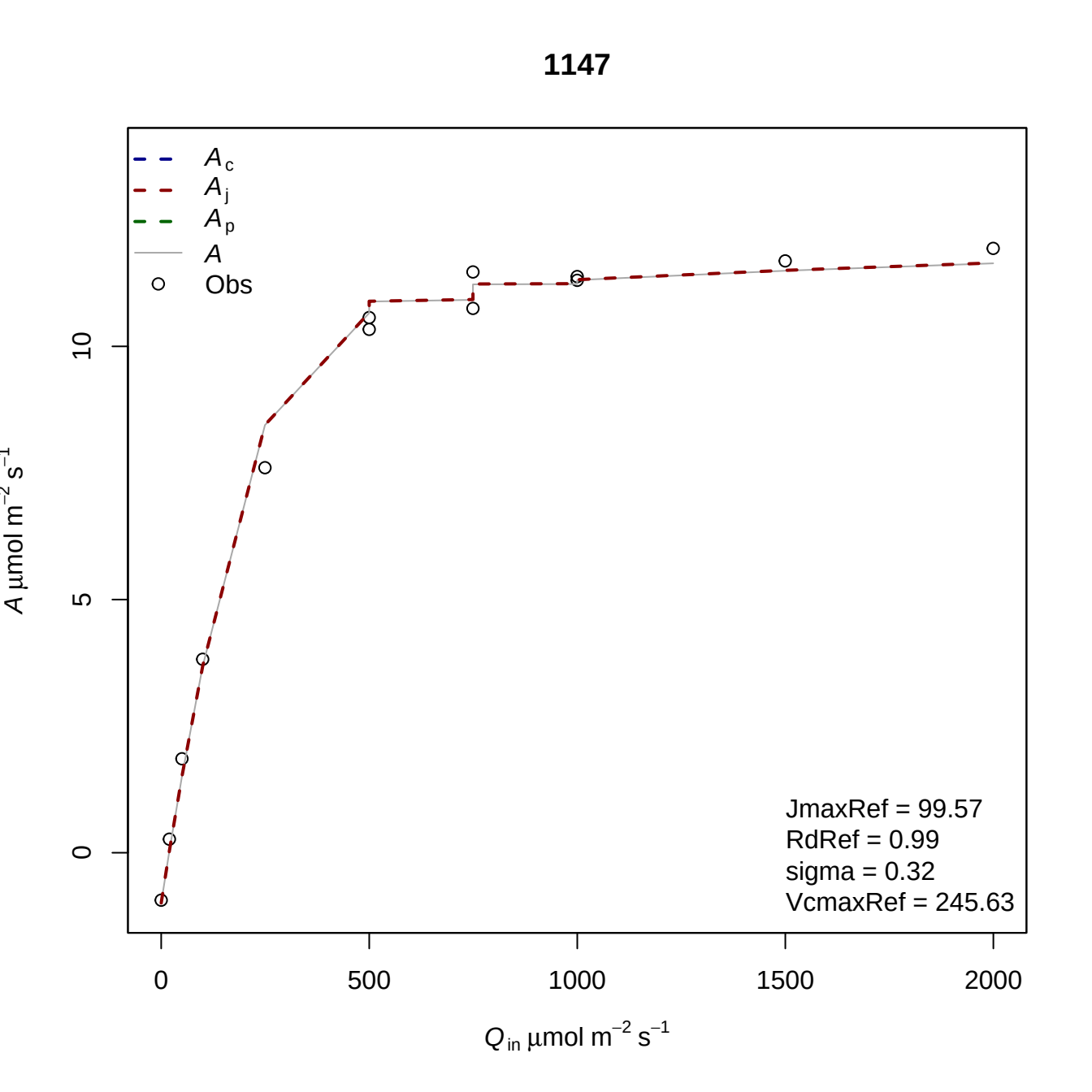

1142
A_c
A_j
A_p
A
Obs
JmaxRef = 48.24
RdRef = 1.82
sigma = 1.38
VcmaxRef = 421308.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

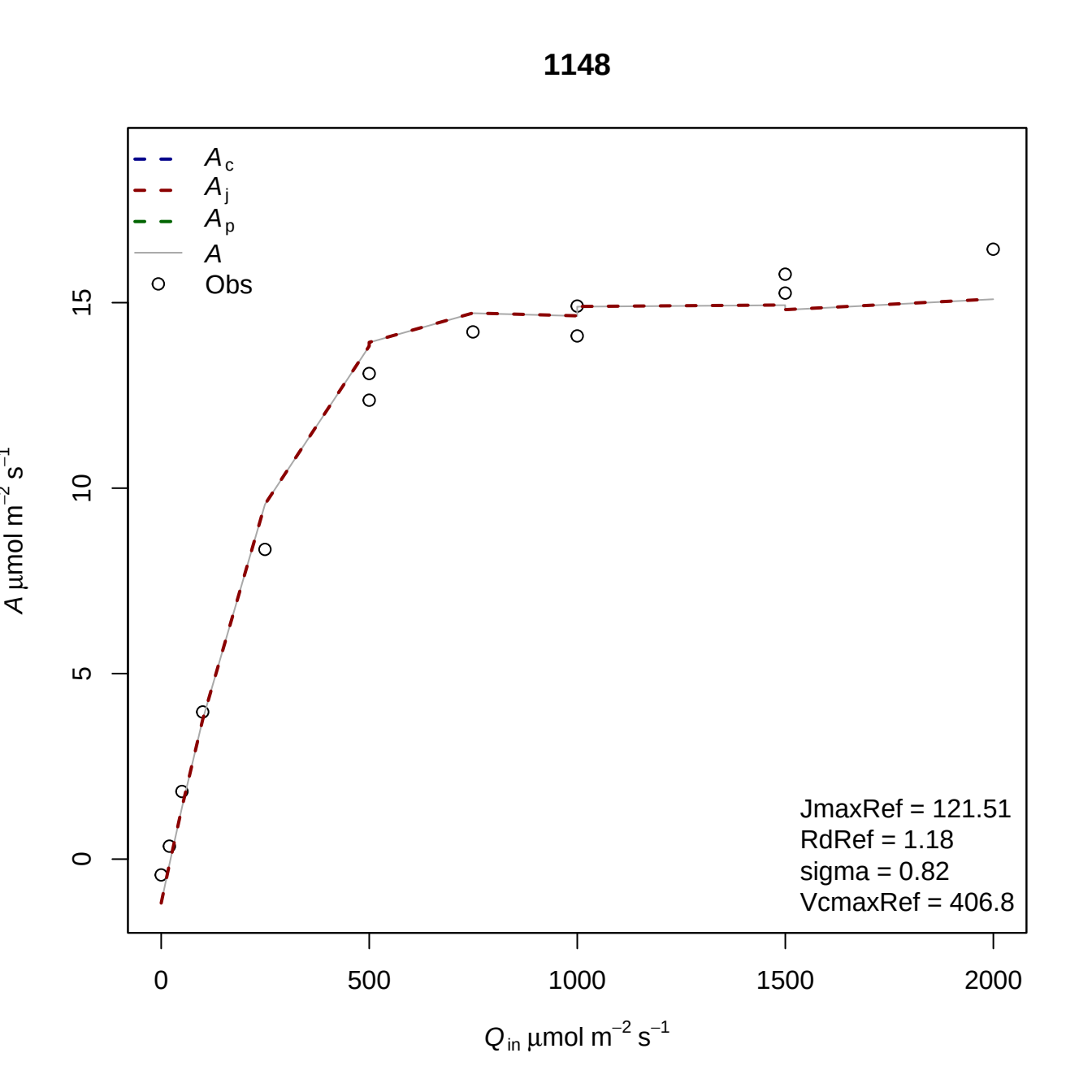

1143
A_c
A_j
A_p
A
Obs
JmaxRef = 85.09
RdRef = 2.46
sigma = 2.94
VcmaxRef = 1084144.88
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

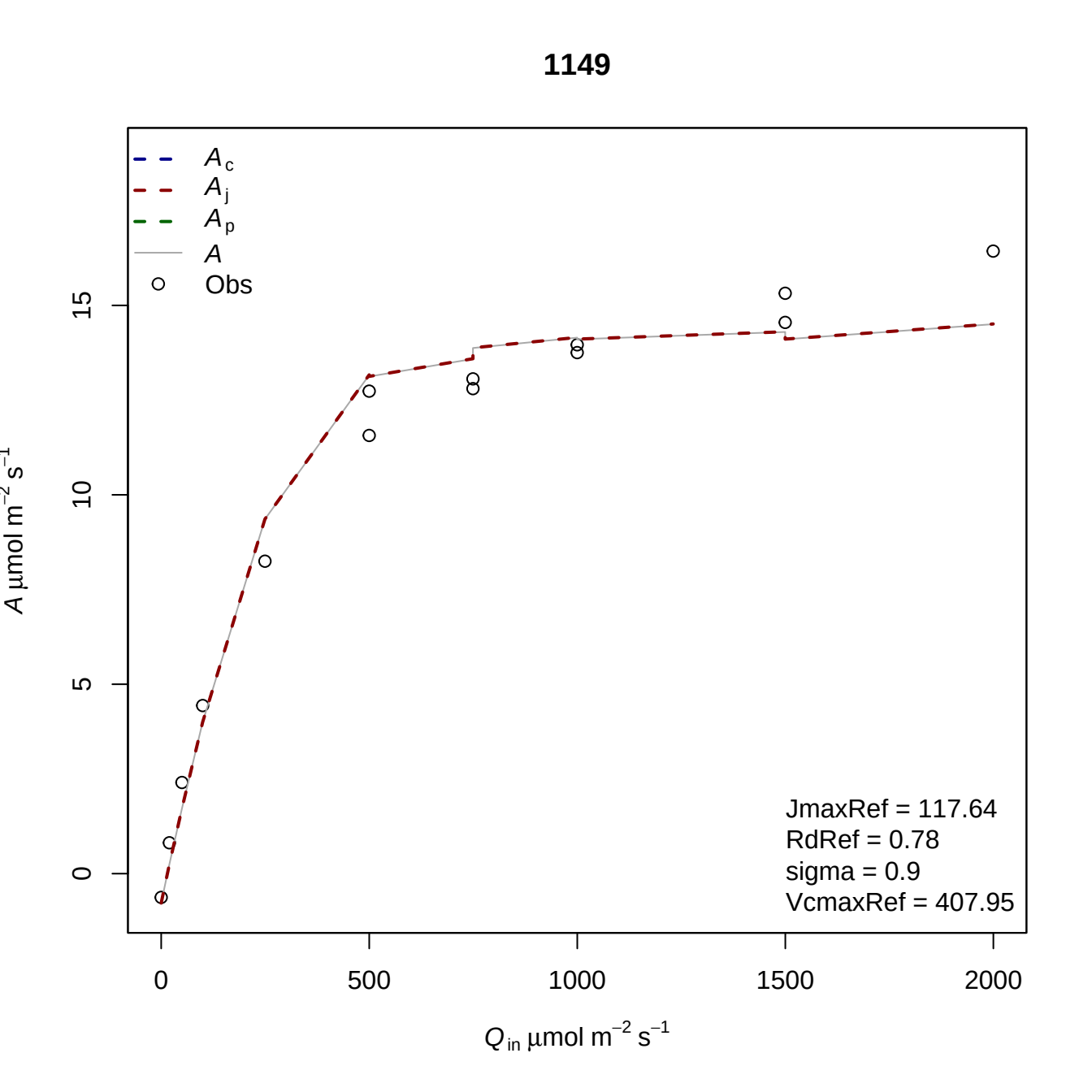

1144
A_c
A_j
A_p
A
Obs
JmaxRef = 68.35
RdRef = 3.05
sigma = 1.71
VcmaxRef = 519812.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

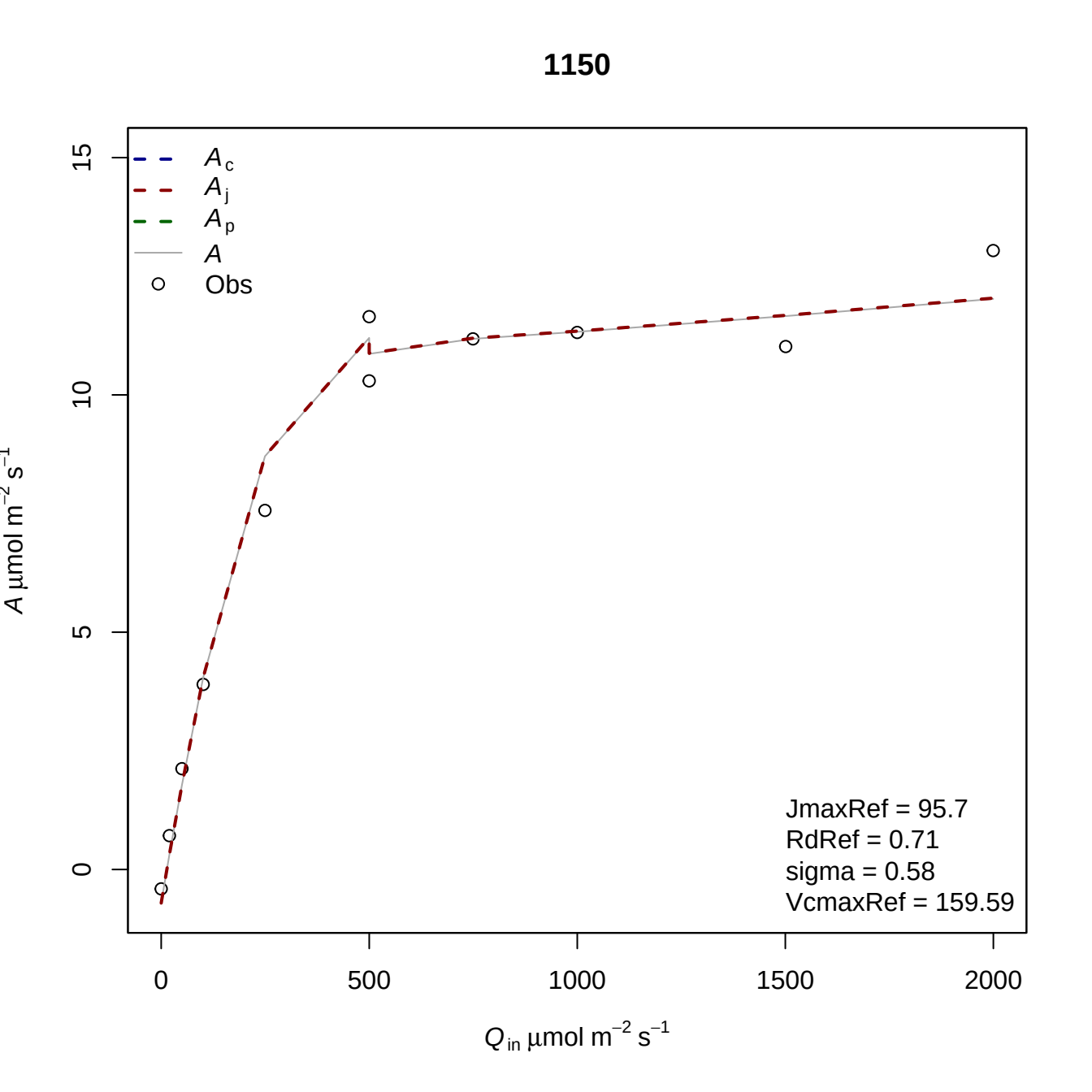

1145
A_c
A_j
A_p
A
Obs
JmaxRef = 157.22
RdRef = 0.92
sigma = 0.73
VcmaxRef = 361.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

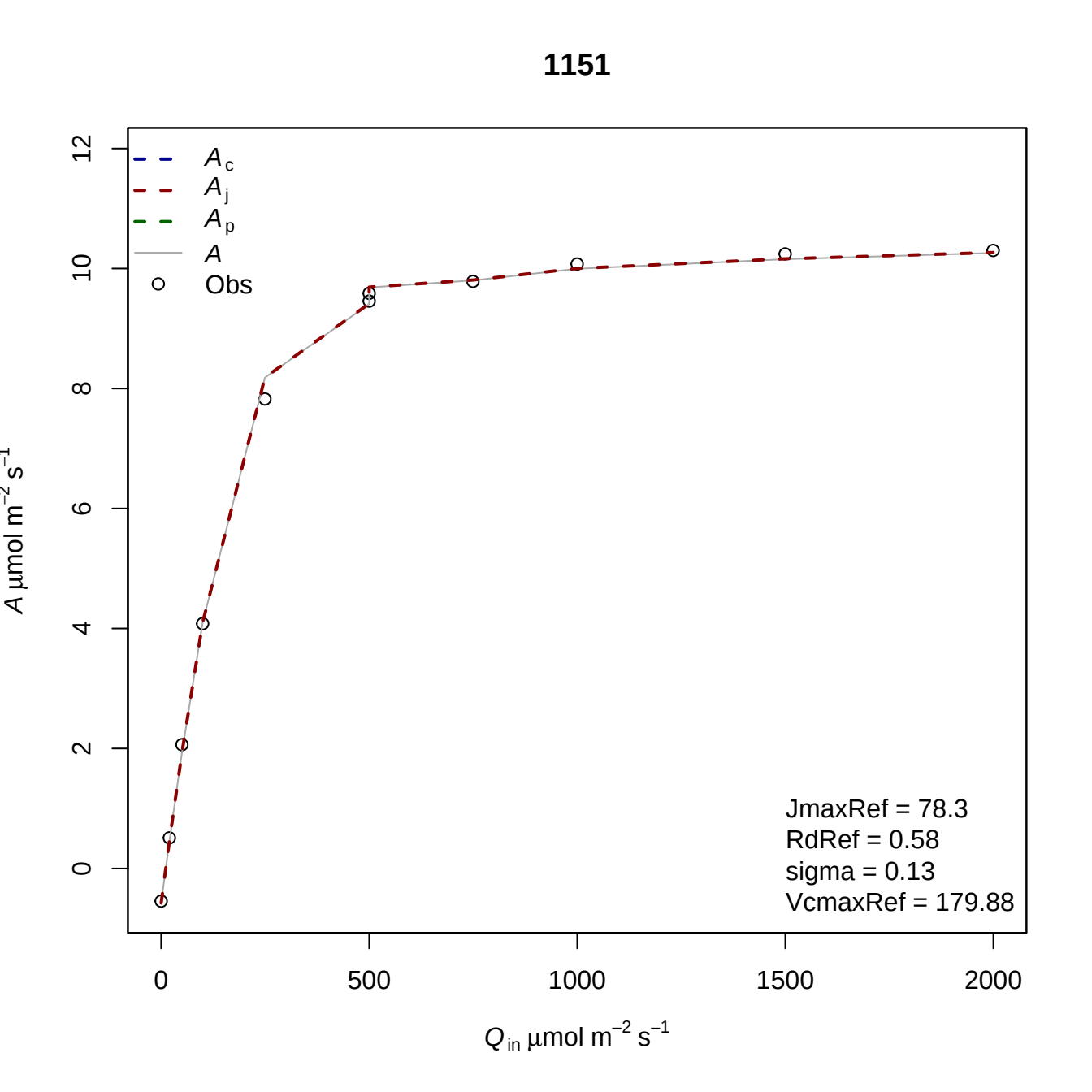

1146
A_c
A_j
A_p
A
Obs
JmaxRef = 93.95
RdRef = 1.27
sigma = 0.64
VcmaxRef = 280.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

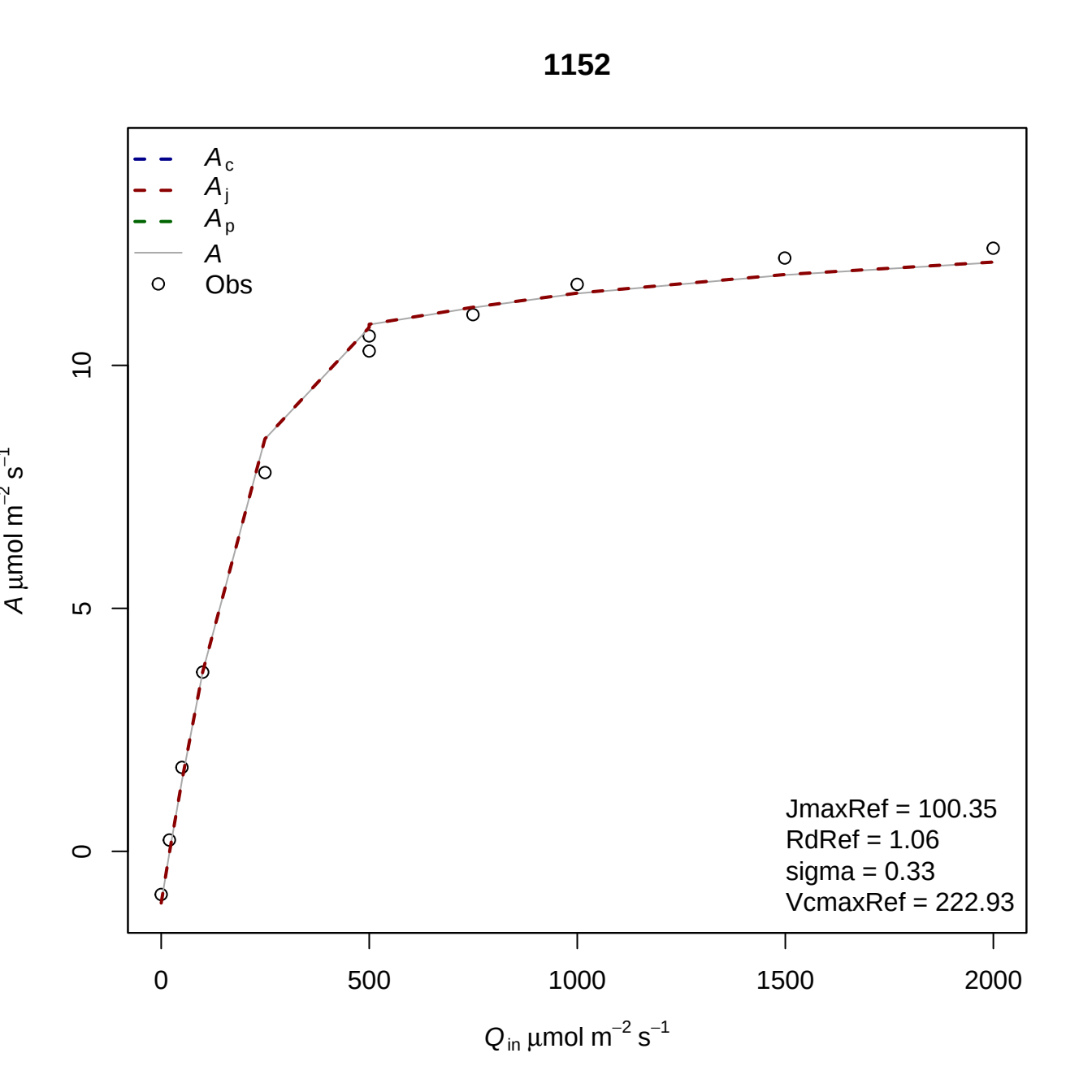

1147
A_c
A_j
A_p
A
Obs
JmaxRef = 99.57
RdRef = 0.99
sigma = 0.32
VcmaxRef = 245.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

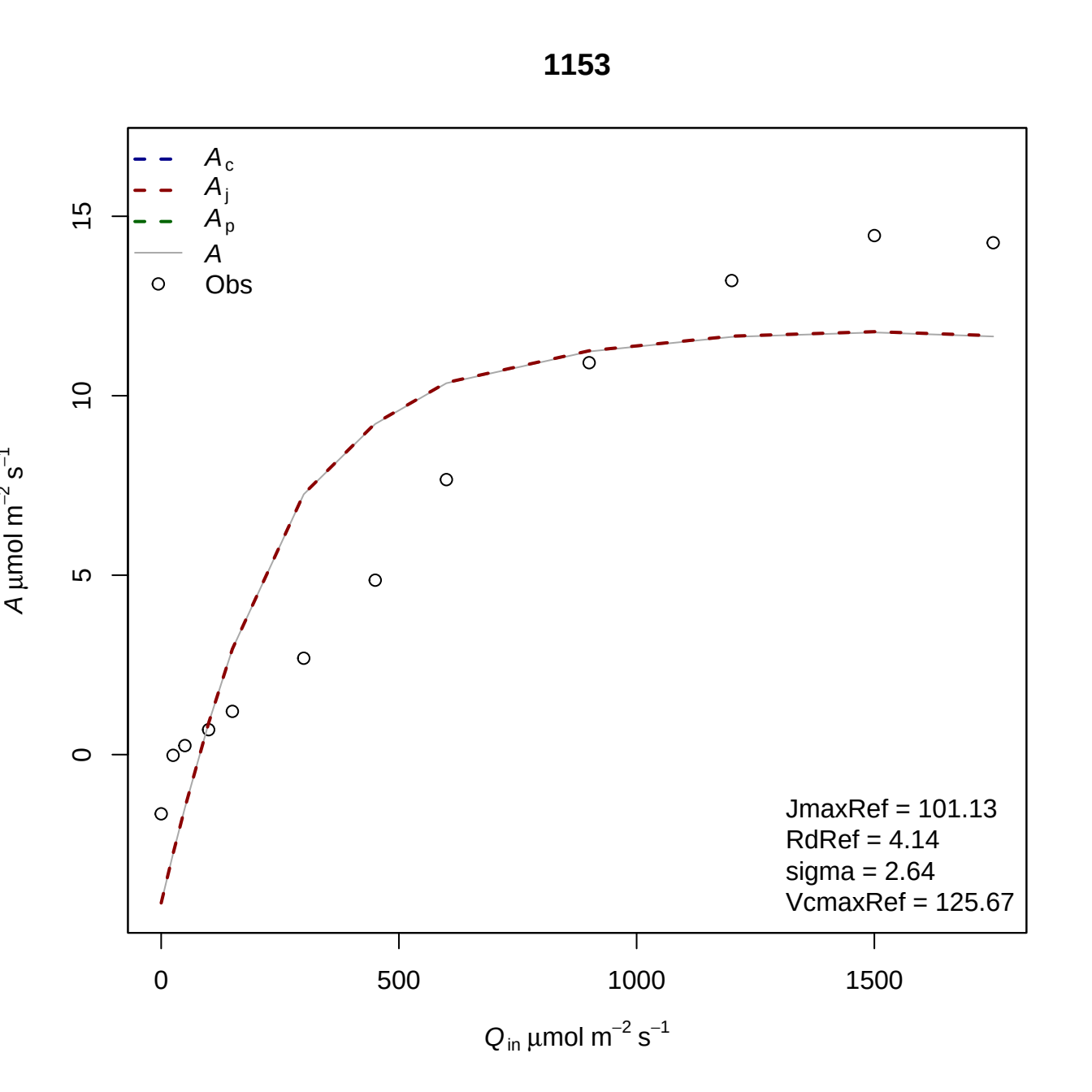

1148
A_c
A_j
A_p
A
Obs
JmaxRef = 121.51
RdRef = 1.18
sigma = 0.82
VcmaxRef = 406.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

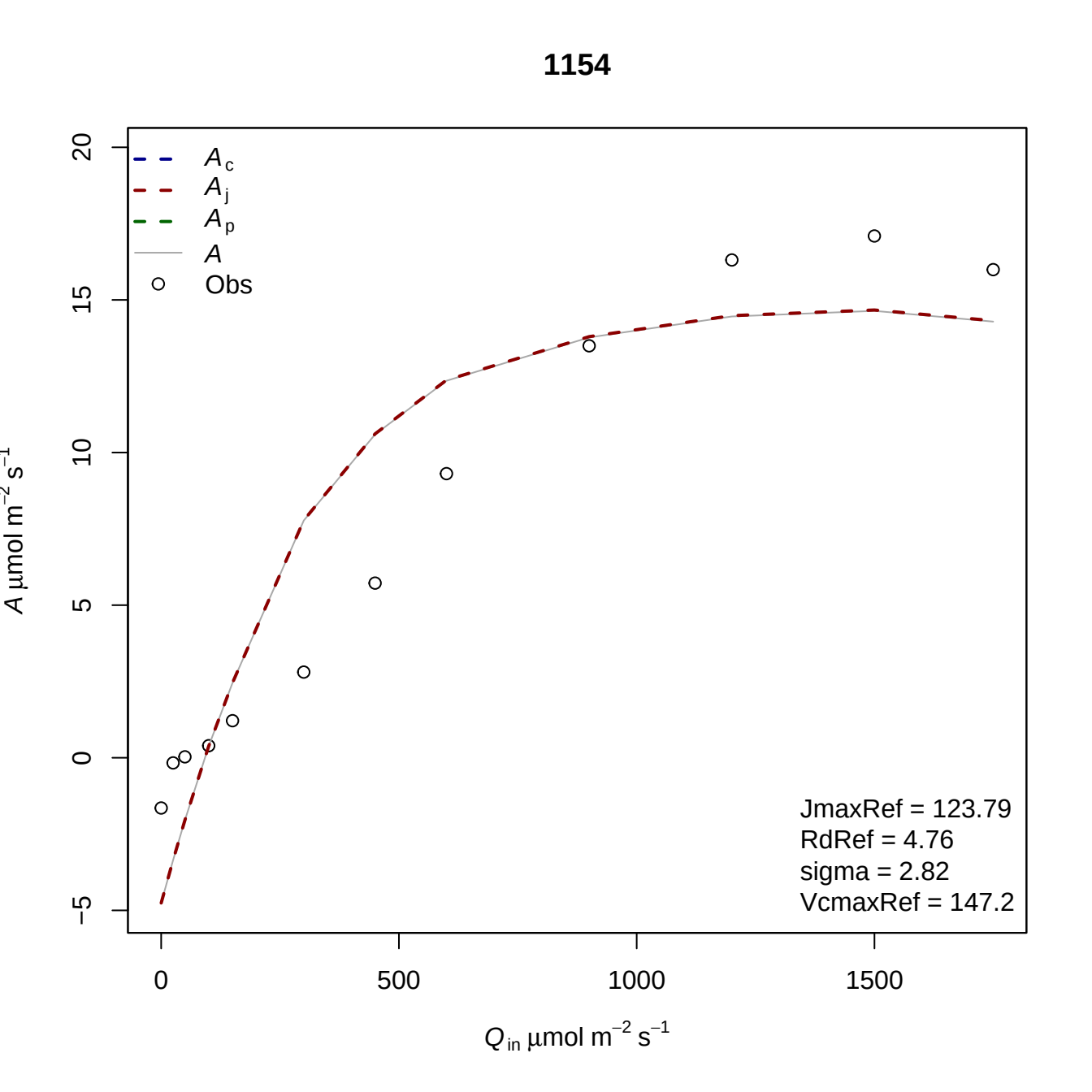

1149
A_c
A_j
A_p
A
Obs
JmaxRef = 117.64
RdRef = 0.78
sigma = 0.9
VcmaxRef = 407.95
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

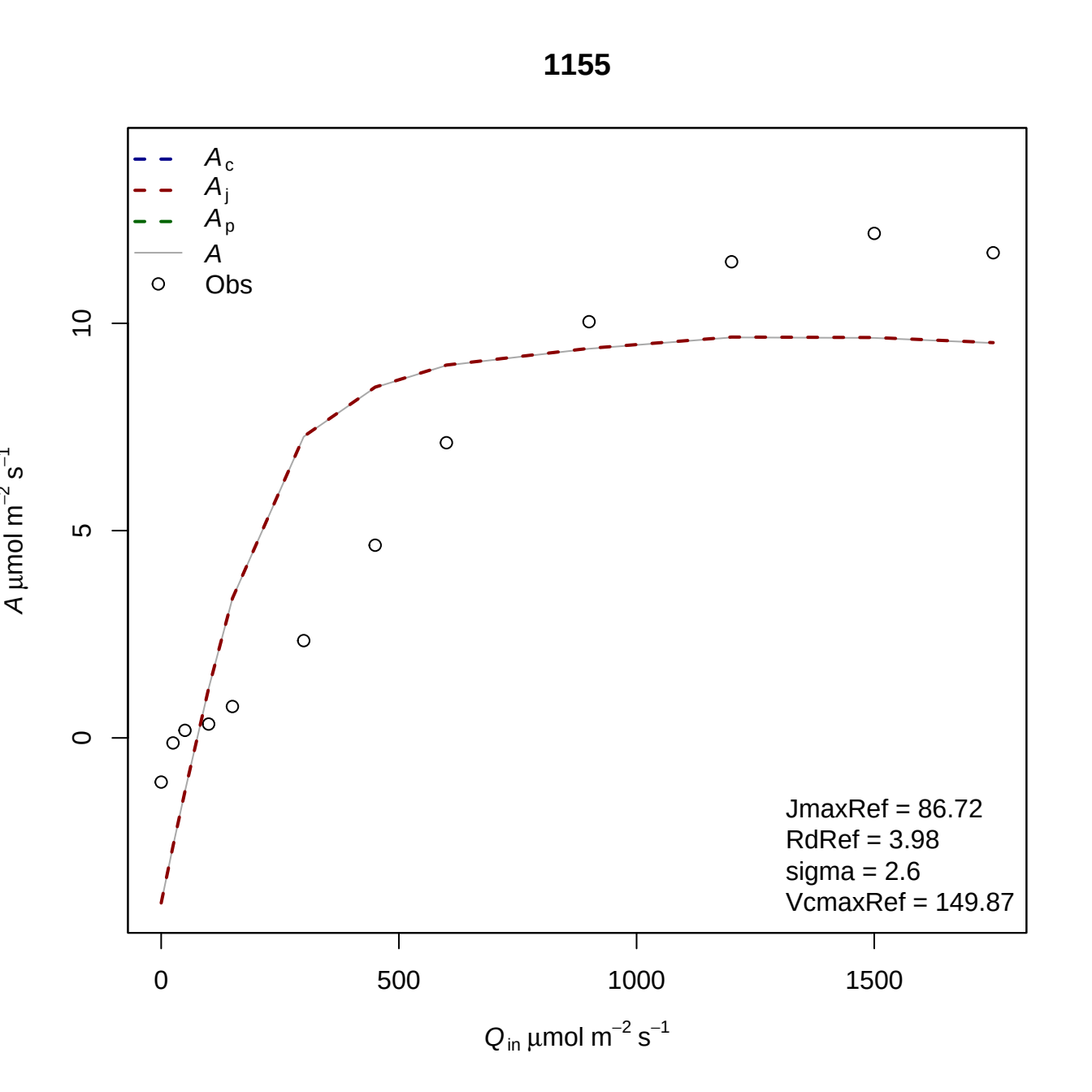

1150
A_c
A_j
A_p
A
Obs
JmaxRef = 95.7
RdRef = 0.71
sigma = 0.58
VcmaxRef = 159.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

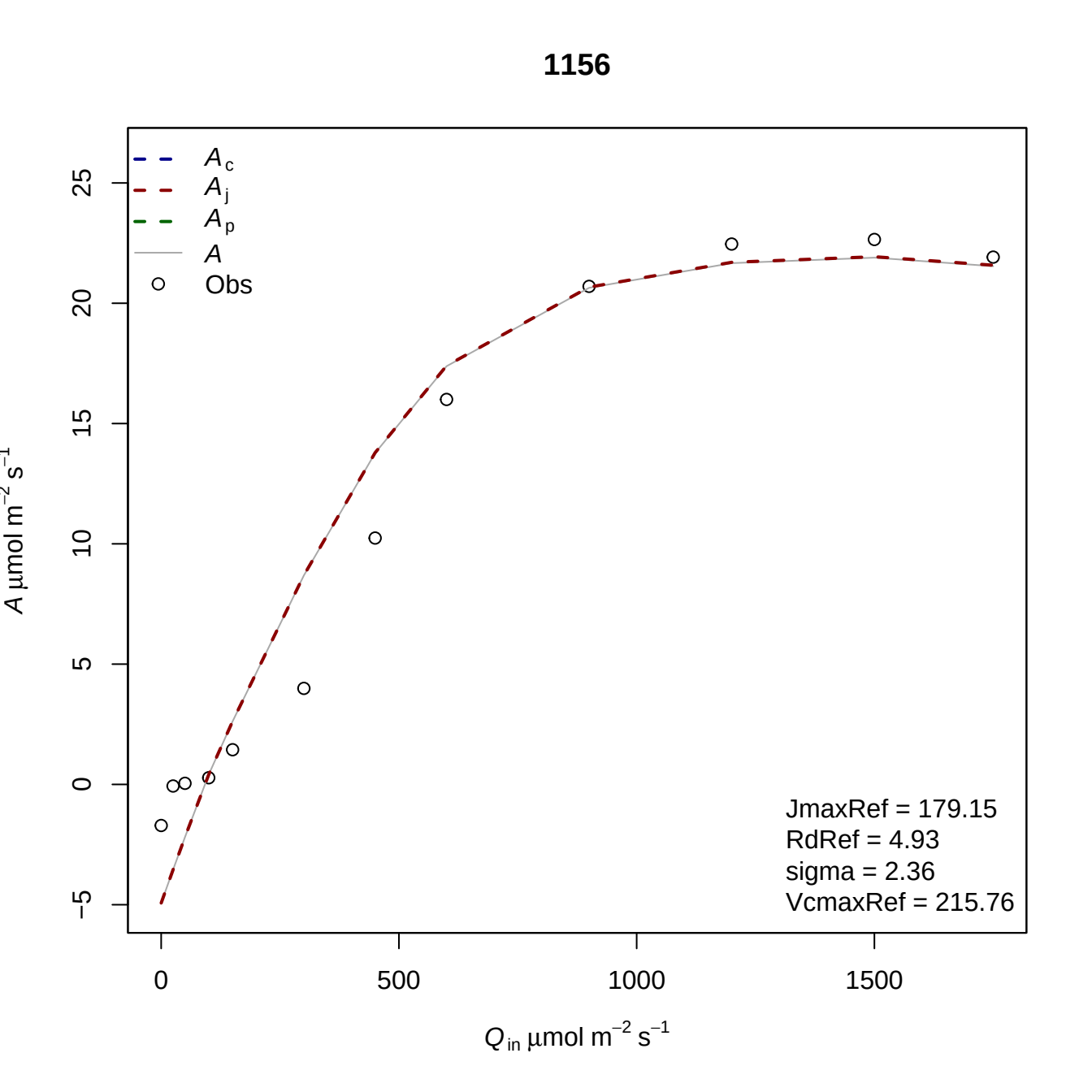

1151
A_c
A_j
A_p
A
Obs
JmaxRef = 78.3
RdRef = 0.58
sigma = 0.13
VcmaxRef = 179.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

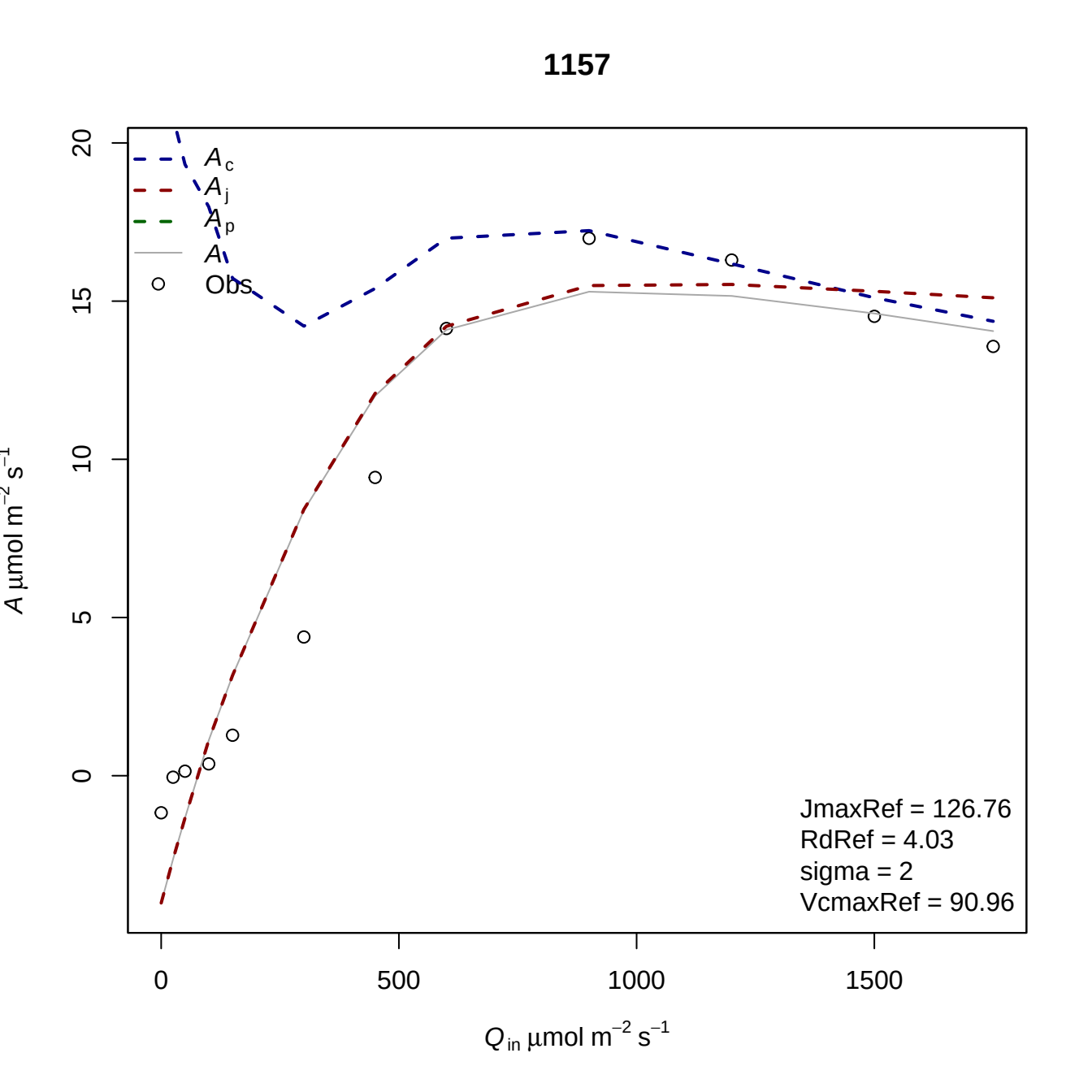

1152
A_c
A_j
A_p
A
Obs
JmaxRef = 100.35
RdRef = 1.06
sigma = 0.33
VcmaxRef = 222.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

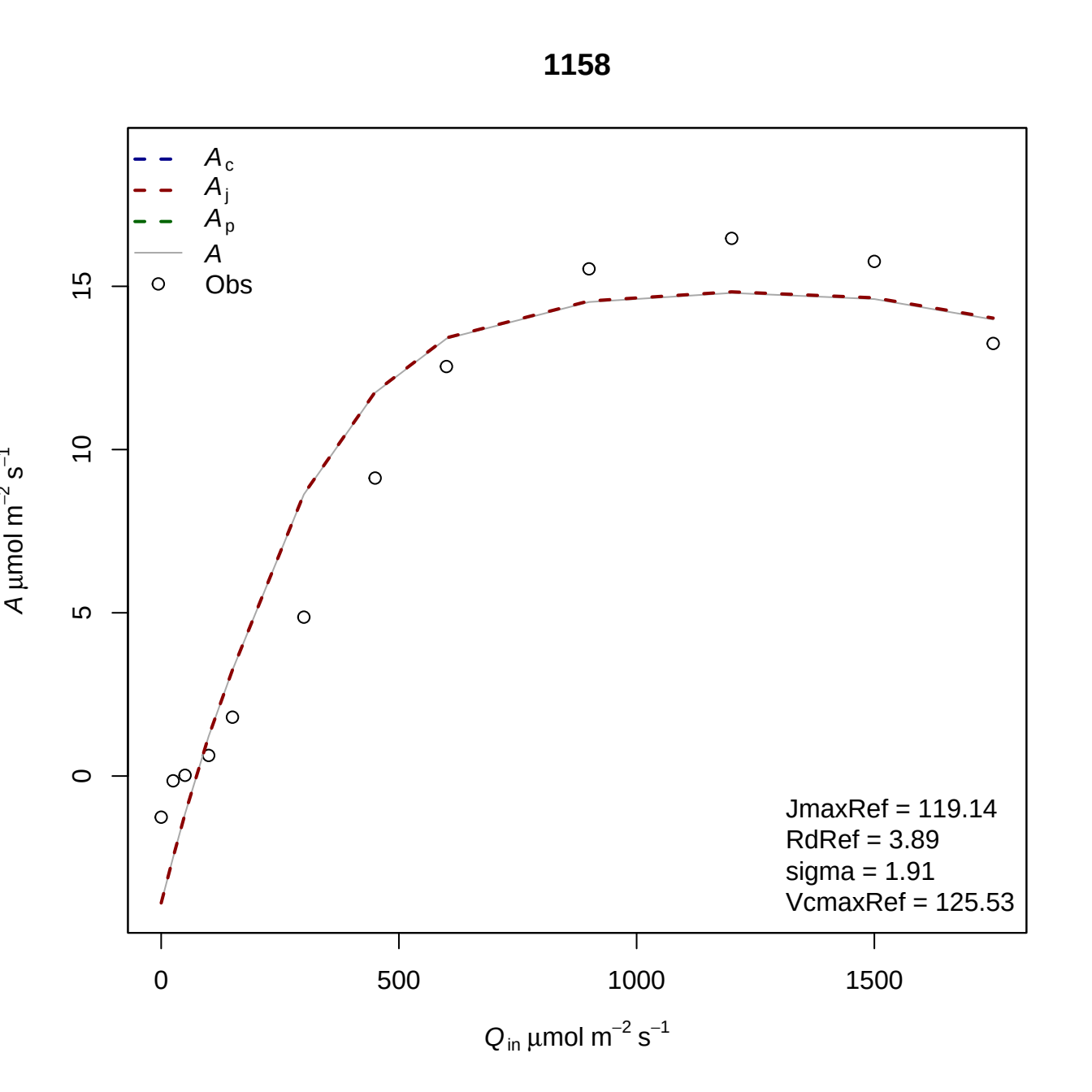

1153
A_c
A_j
A_p
A
Obs
JmaxRef = 101.13
RdRef = 4.14
sigma = 2.64
VcmaxRef = 125.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

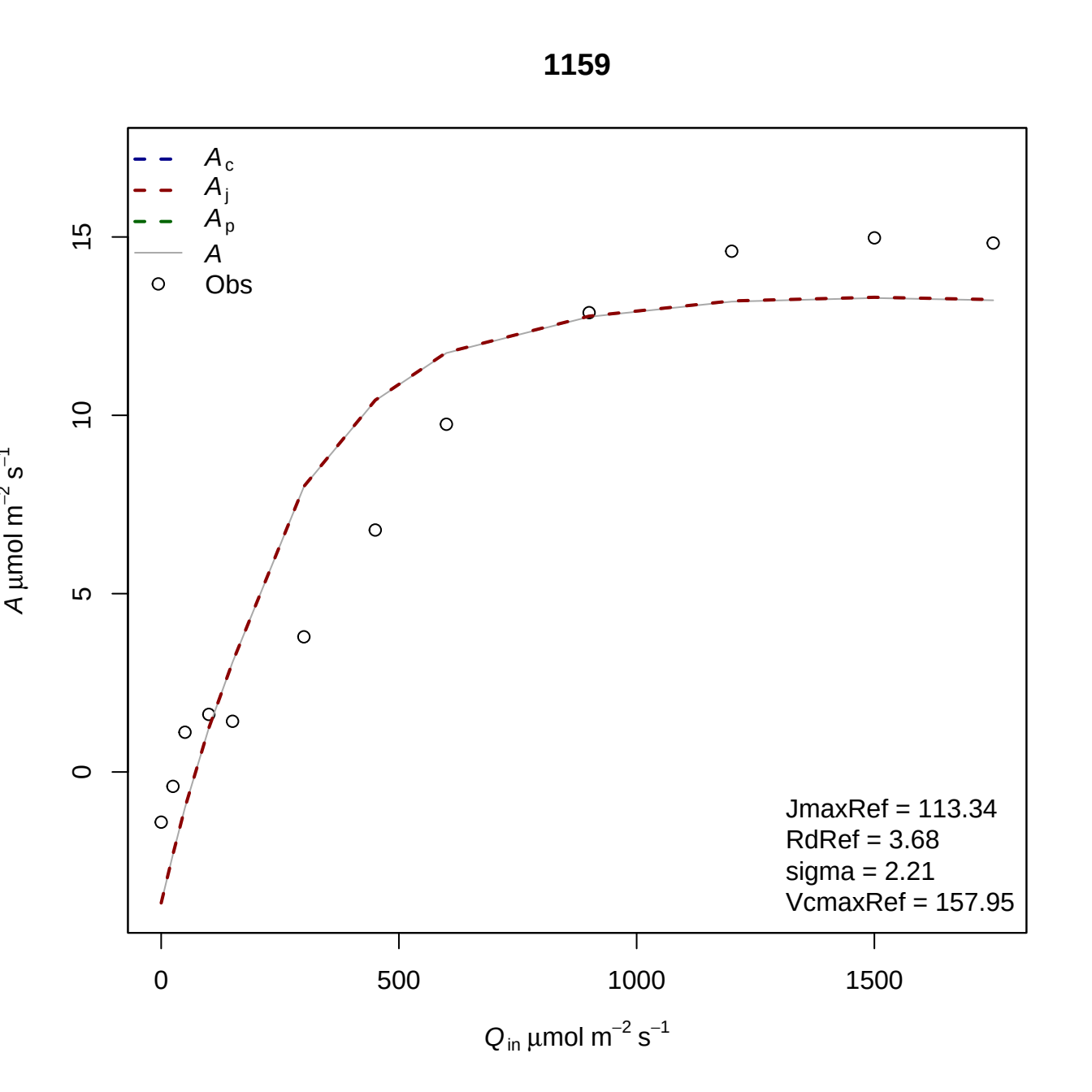

1154
A_c
A_j
A_p
A
Obs
JmaxRef = 123.79
RdRef = 4.76
sigma = 2.82
VcmaxRef = 147.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

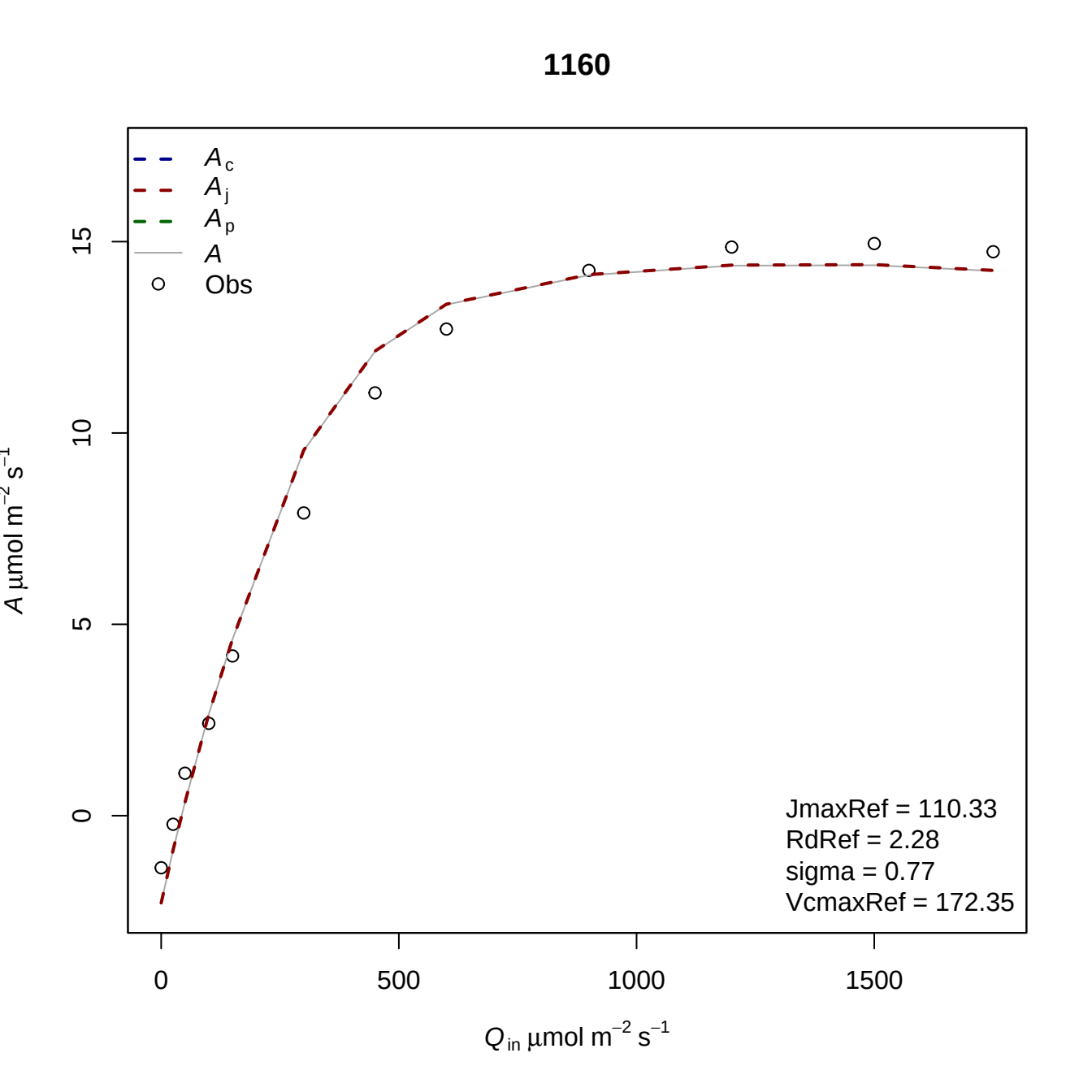

1155
A_c
A_j
A_p
A
Obs
JmaxRef = 86.72
RdRef = 3.98
sigma = 2.6
VcmaxRef = 149.87
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

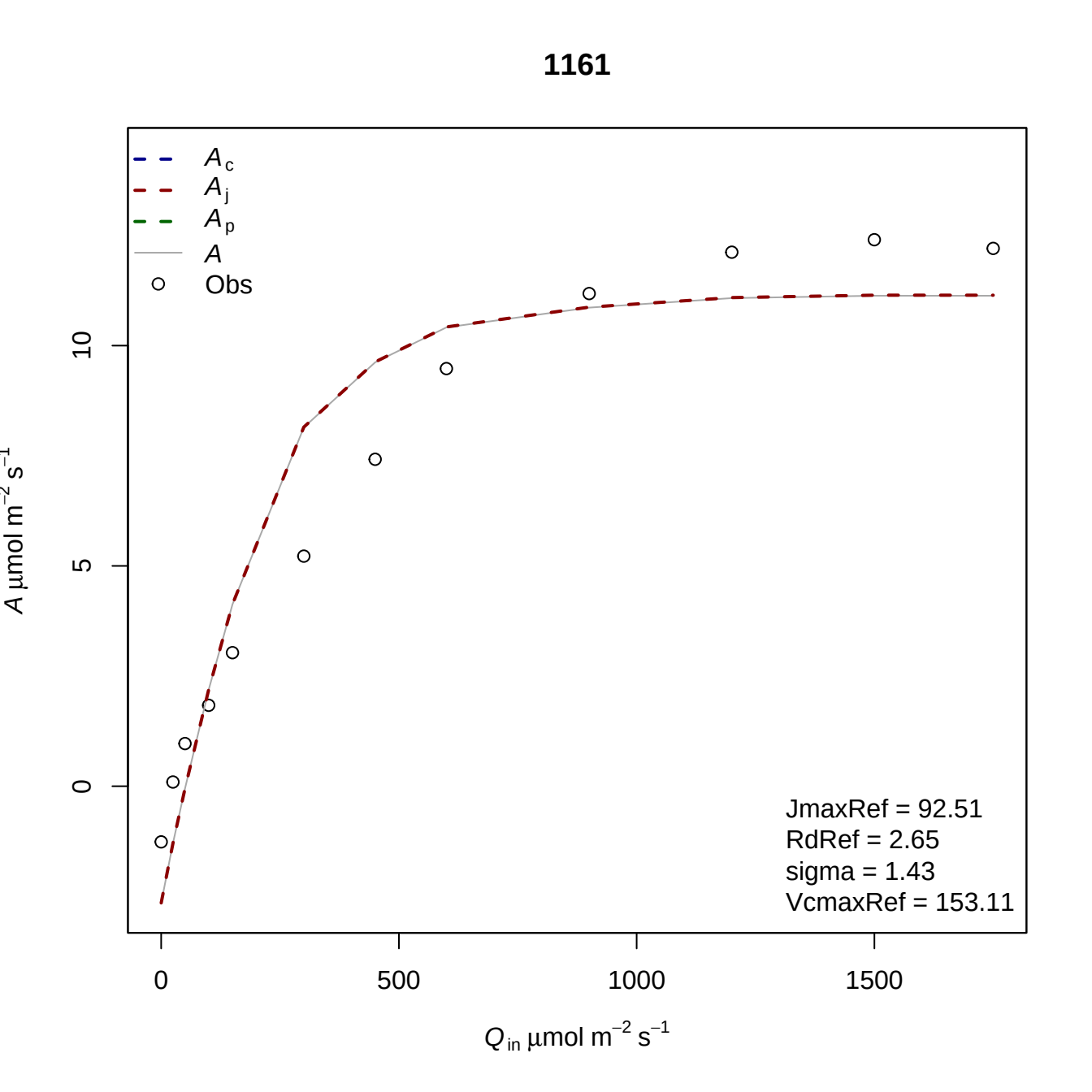

1156
A_c
A_j
A_p
A
Obs
JmaxRef = 179.15
RdRef = 4.93
sigma = 2.36
VcmaxRef = 215.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500

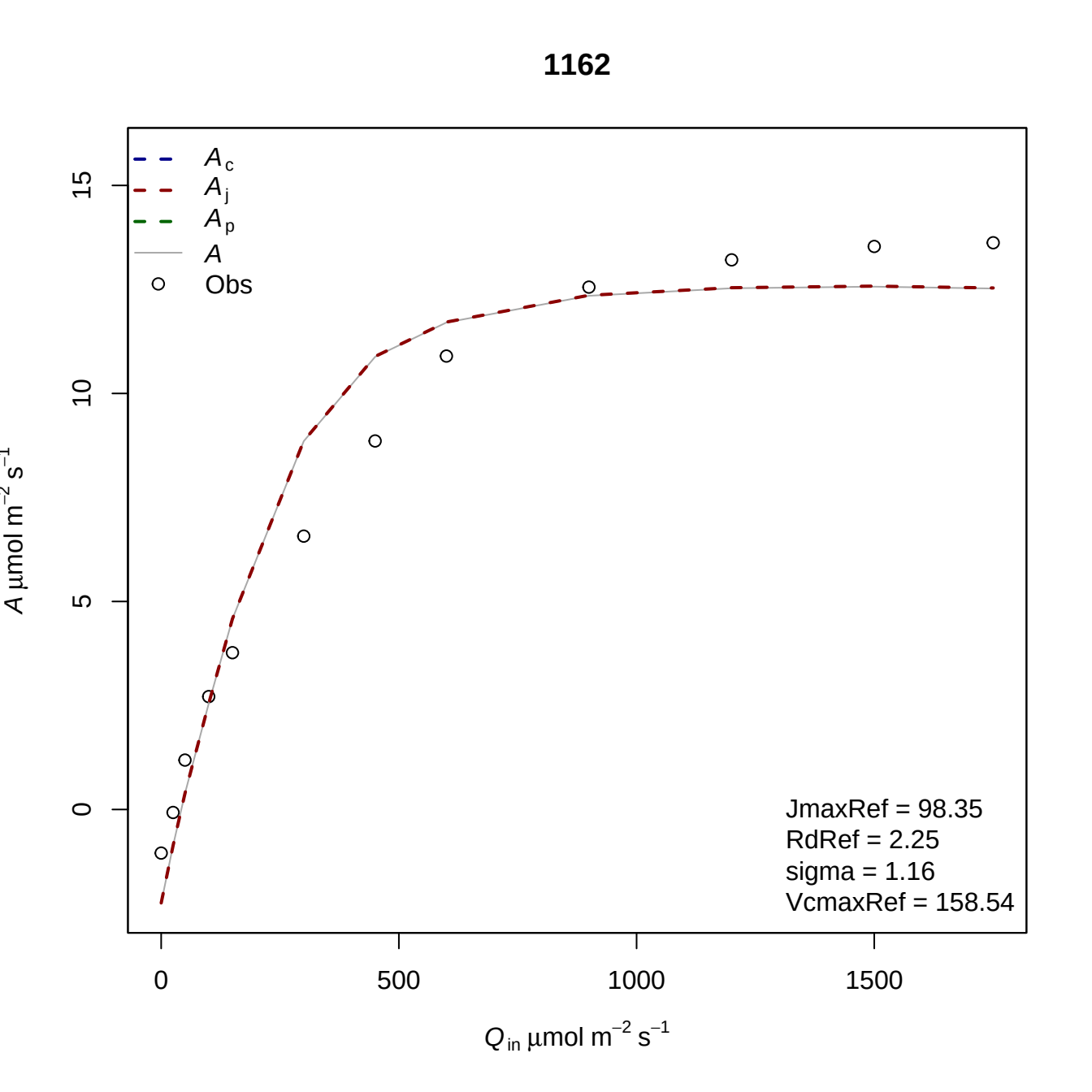

1157
A_c
A_j
A_p
A
Obs
JmaxRef = 126.76
RdRef = 4.03
sigma = 2
VcmaxRef = 90.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

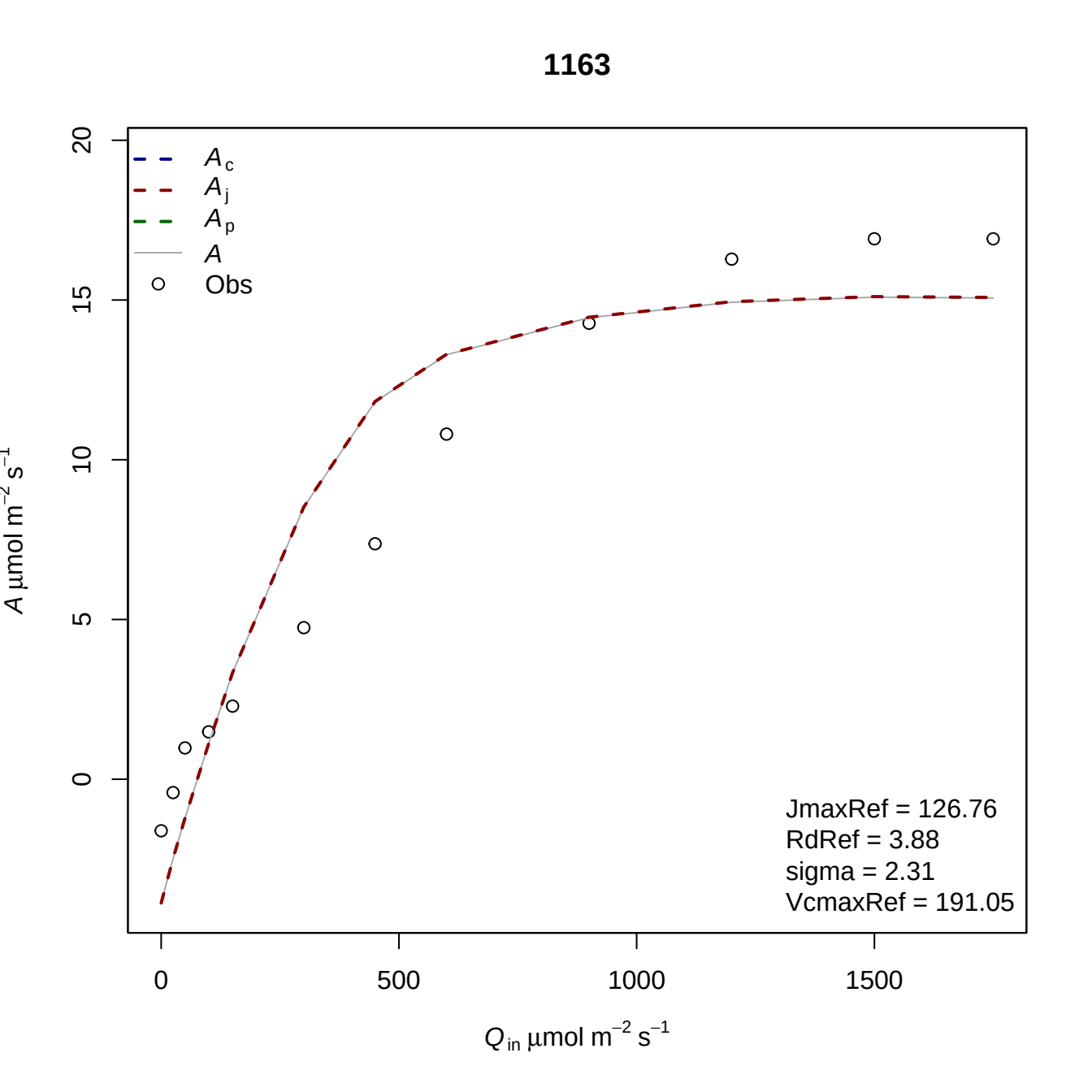

1158
A_c
A_j
A_p
A
Obs
JmaxRef = 119.14
RdRef = 3.89
sigma = 1.91
VcmaxRef = 125.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

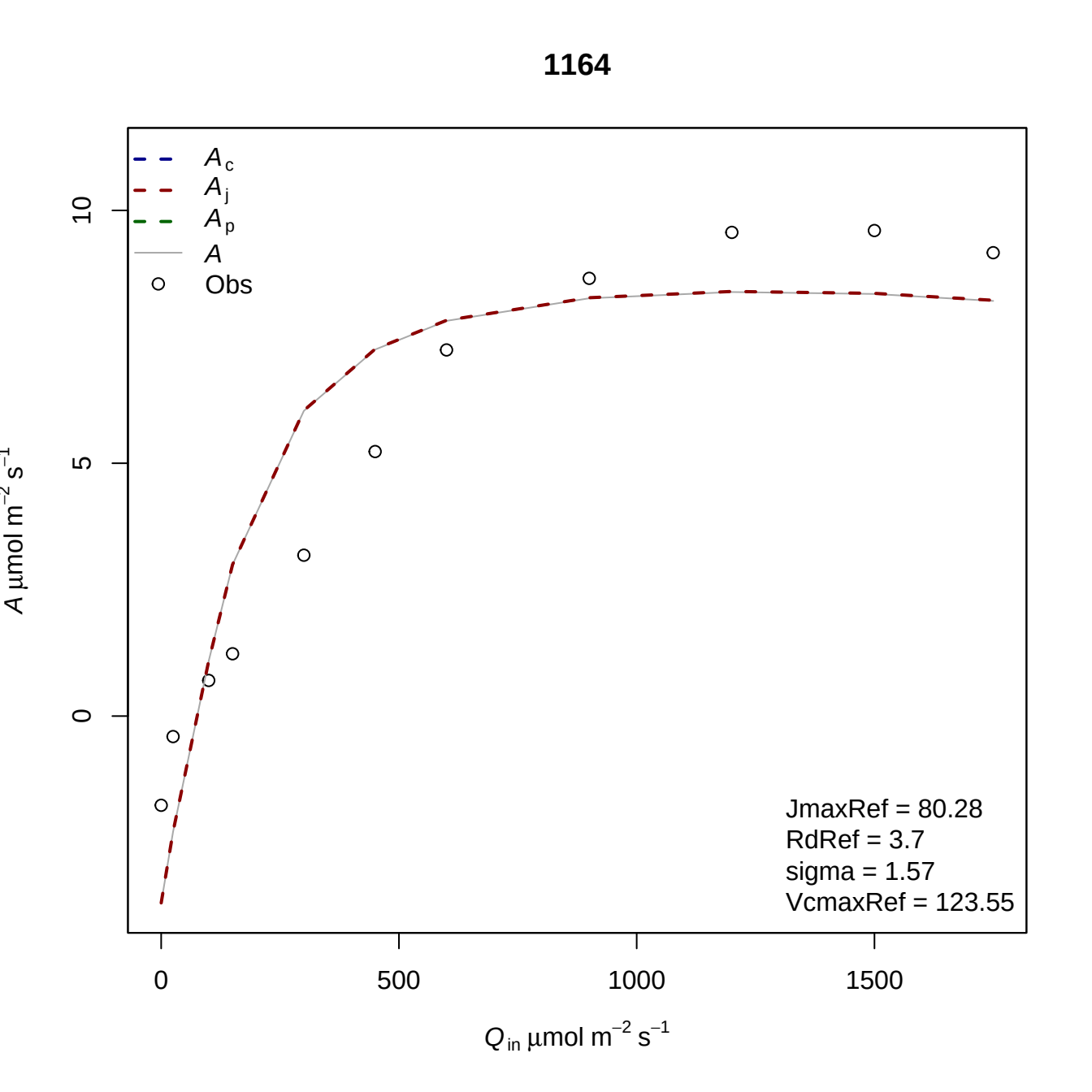

1159
A_c
A_j
A_p
A
Obs
JmaxRef = 113.34
RdRef = 3.68
sigma = 2.21
VcmaxRef = 157.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

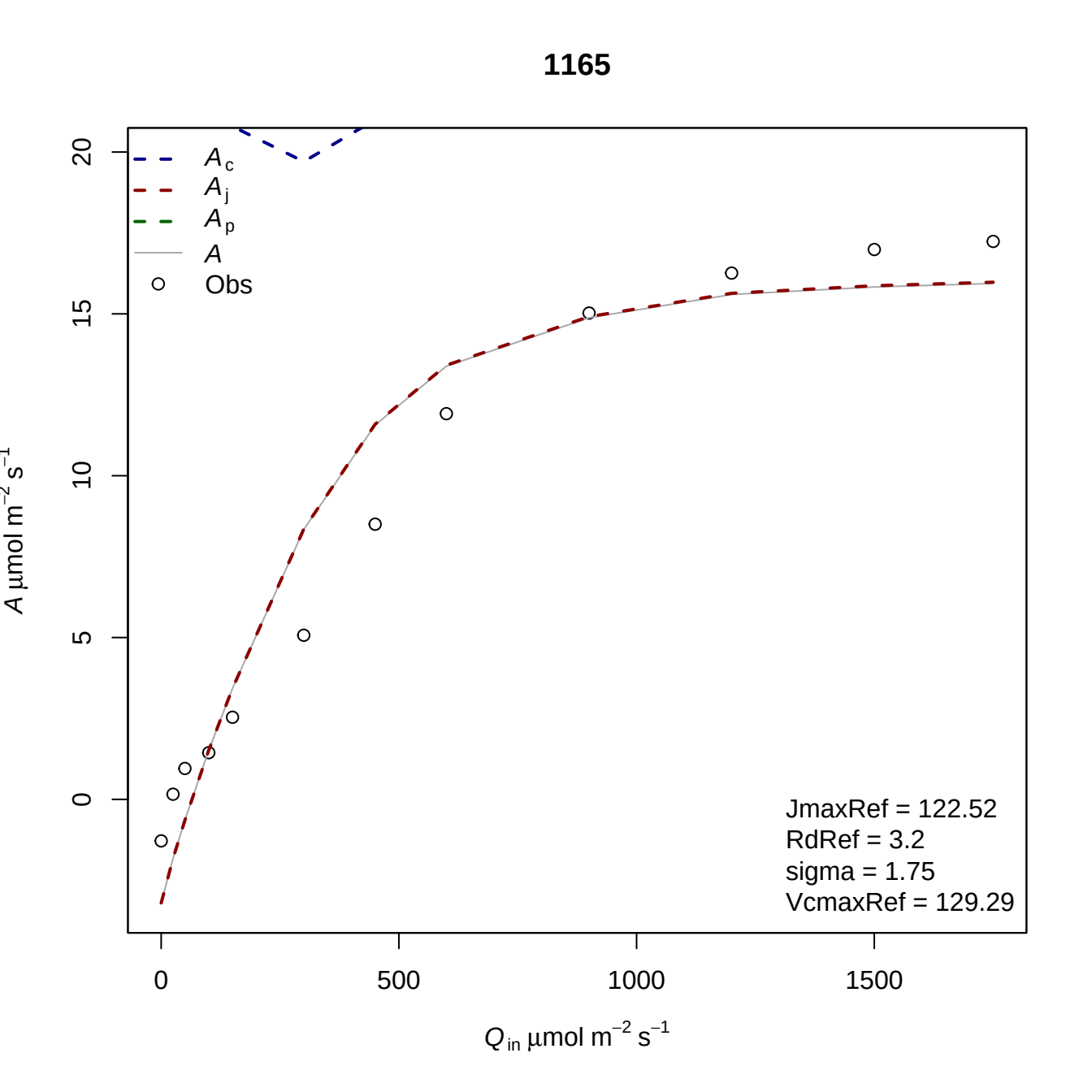

1160
A_c
A_j
A_p
A
Obs
JmaxRef = 110.33
RdRef = 2.28
sigma = 0.77
VcmaxRef = 172.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

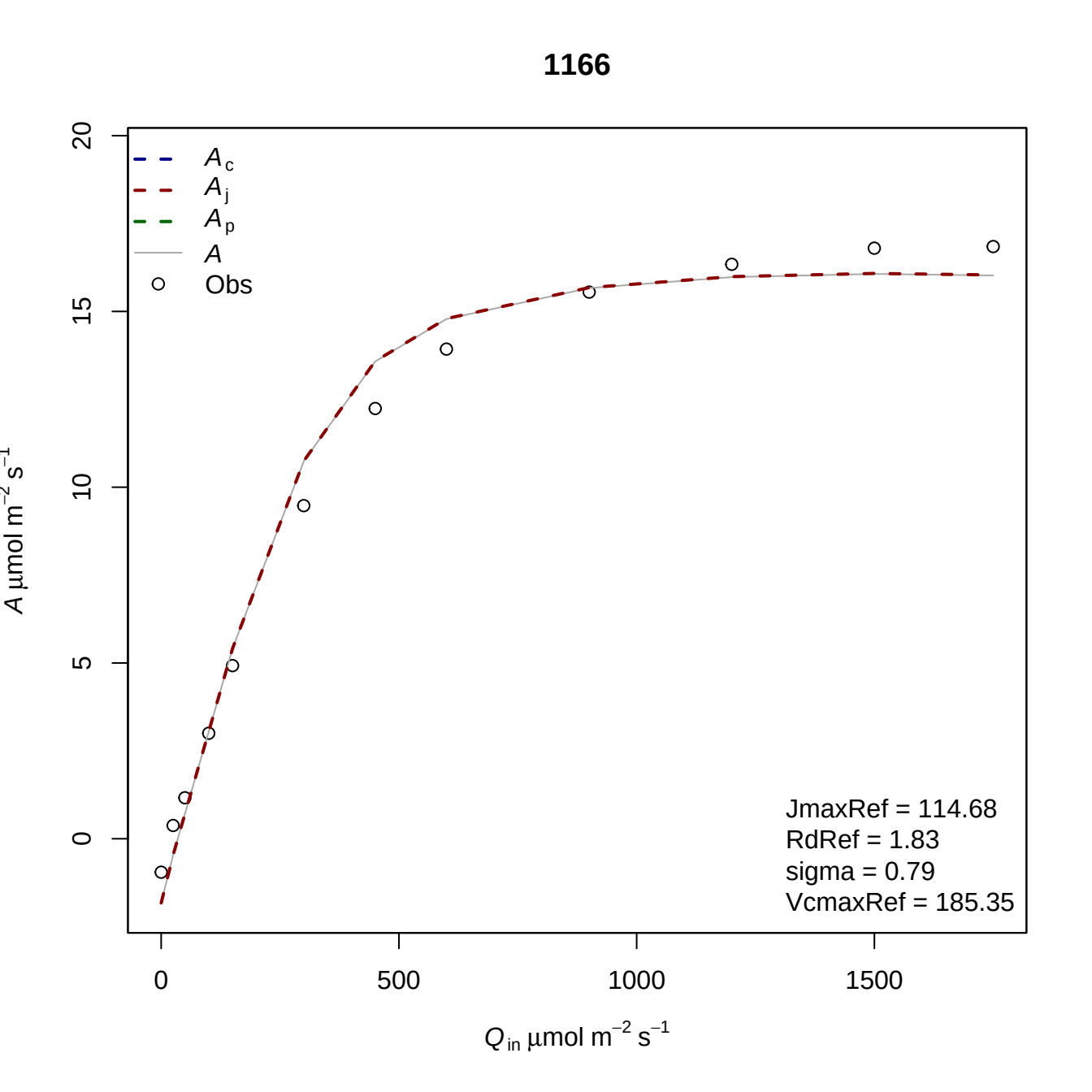

1161
A_c
A_j
A_p
A
Obs
JmaxRef = 92.51
RdRef = 2.65
sigma = 1.43
VcmaxRef = 153.11
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

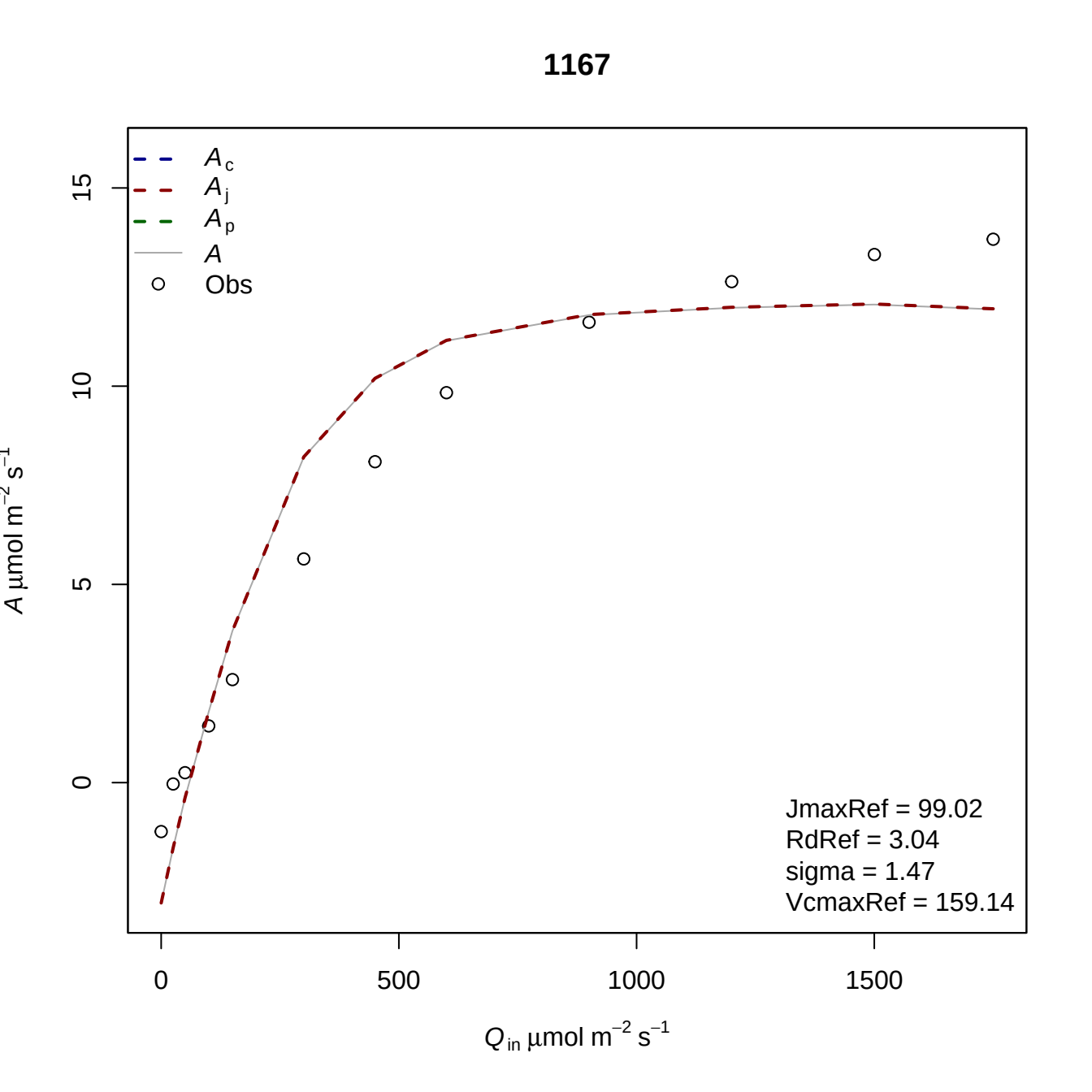

1162
A_c
A_j
A_p
A
Obs
JmaxRef = 98.35
RdRef = 2.25
sigma = 1.16
VcmaxRef = 158.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

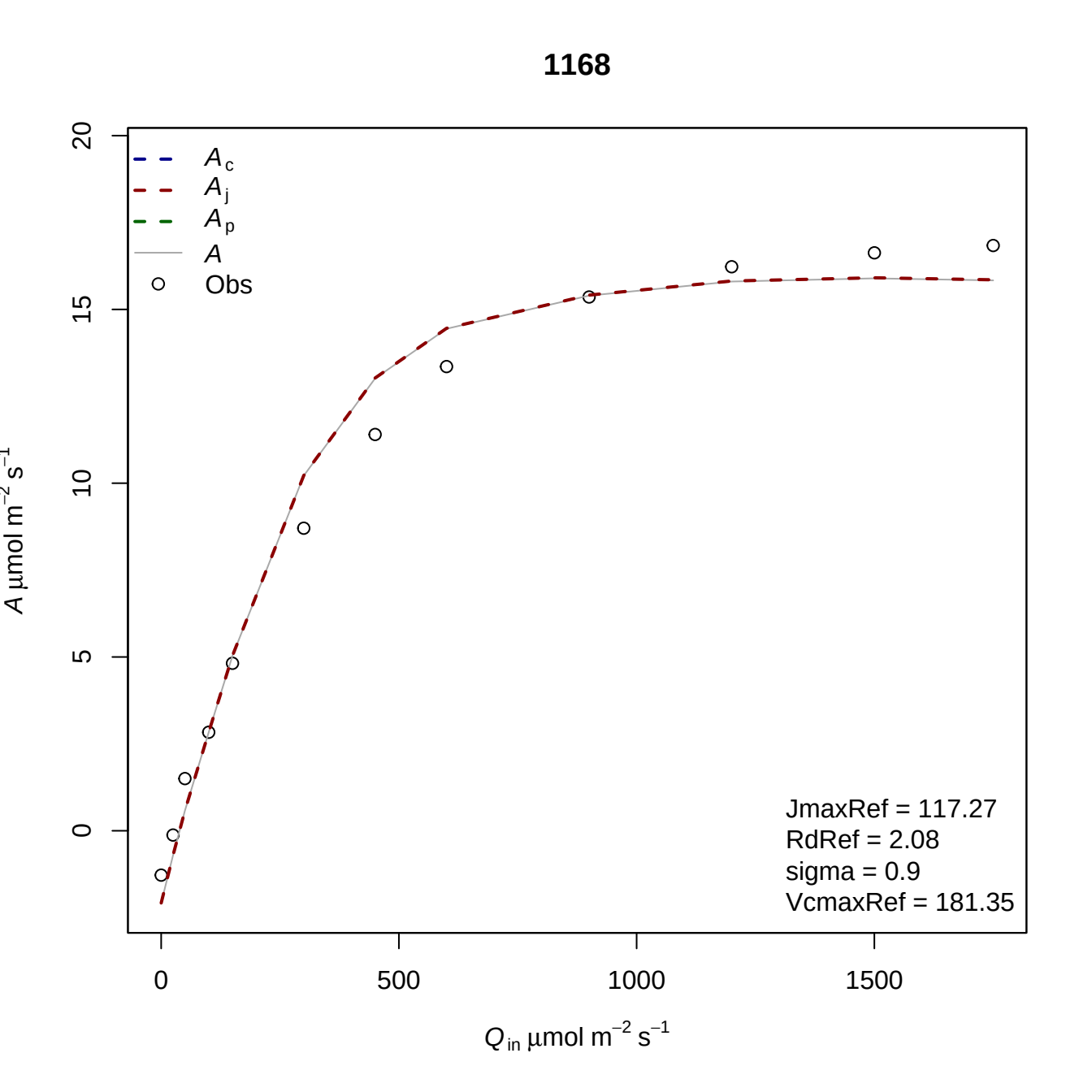

1163
A_c
A_j
A_p
A
Obs
JmaxRef = 126.76
RdRef = 3.88
sigma = 2.31
VcmaxRef = 191.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

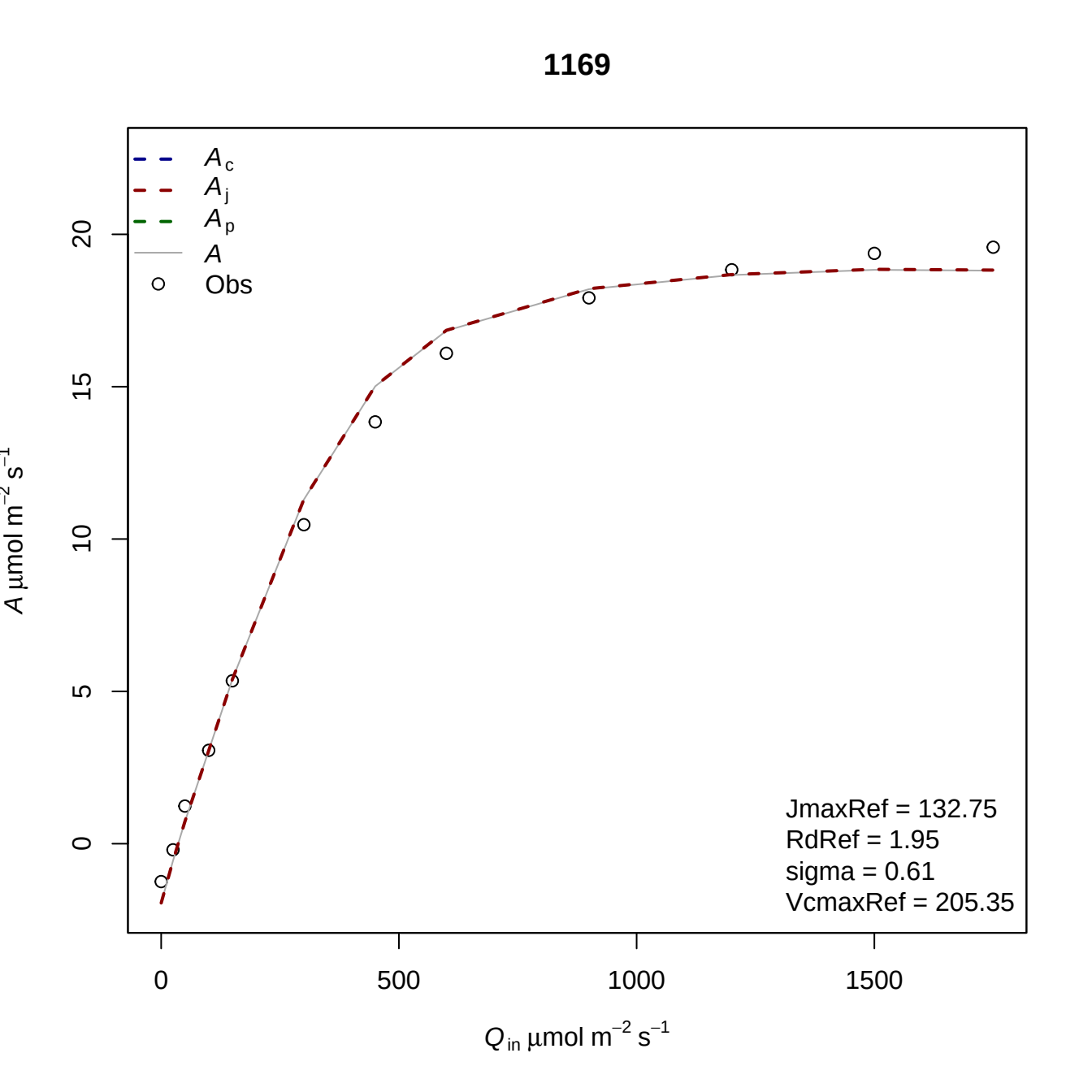

1164
A_c
A_j
A_p
A
Obs
JmaxRef = 80.28
RdRef = 3.7
sigma = 1.57
VcmaxRef = 123.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

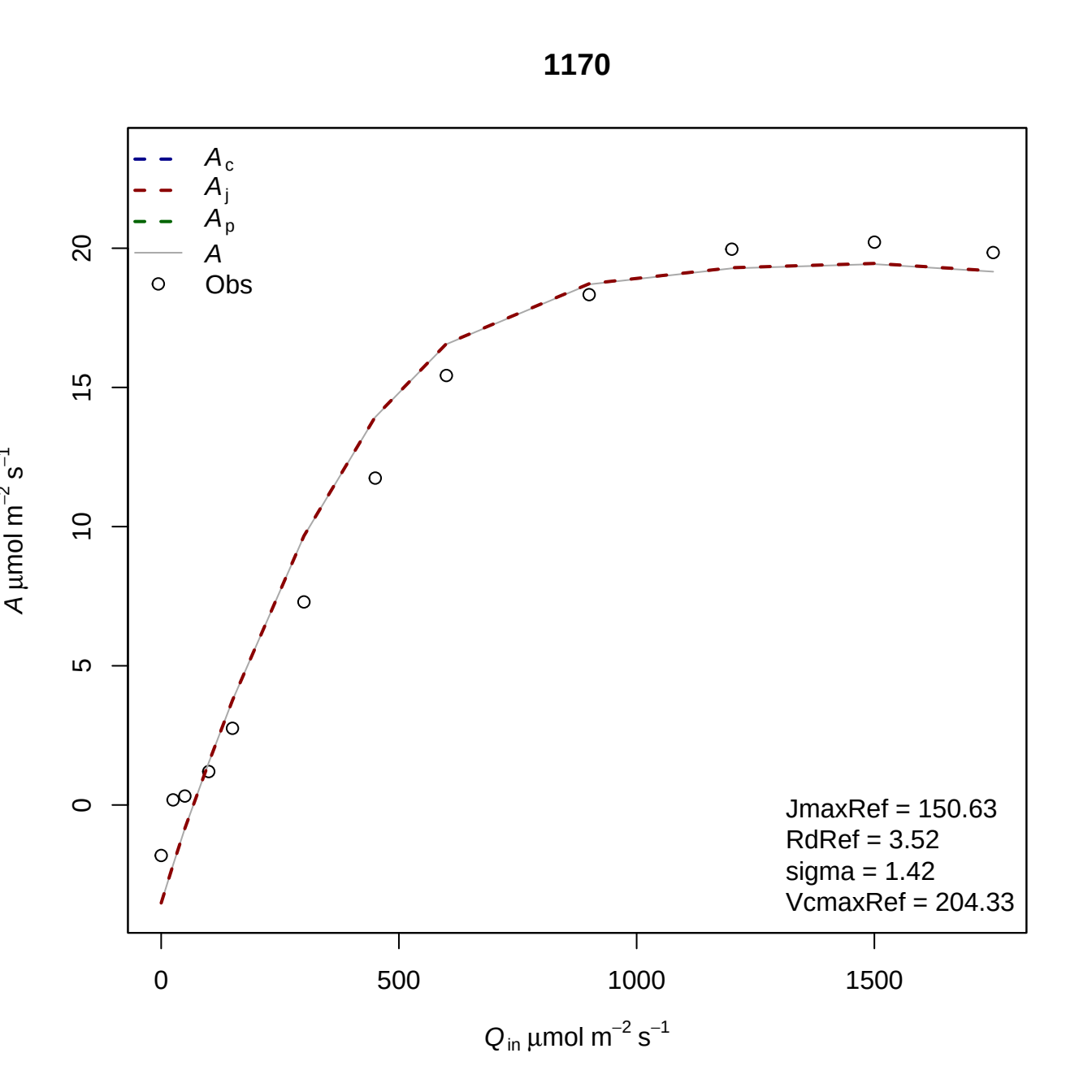

1165
A_c
A_j
A_p
A
Obs
JmaxRef = 122.52
RdRef = 3.2
sigma = 1.75
VcmaxRef = 129.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

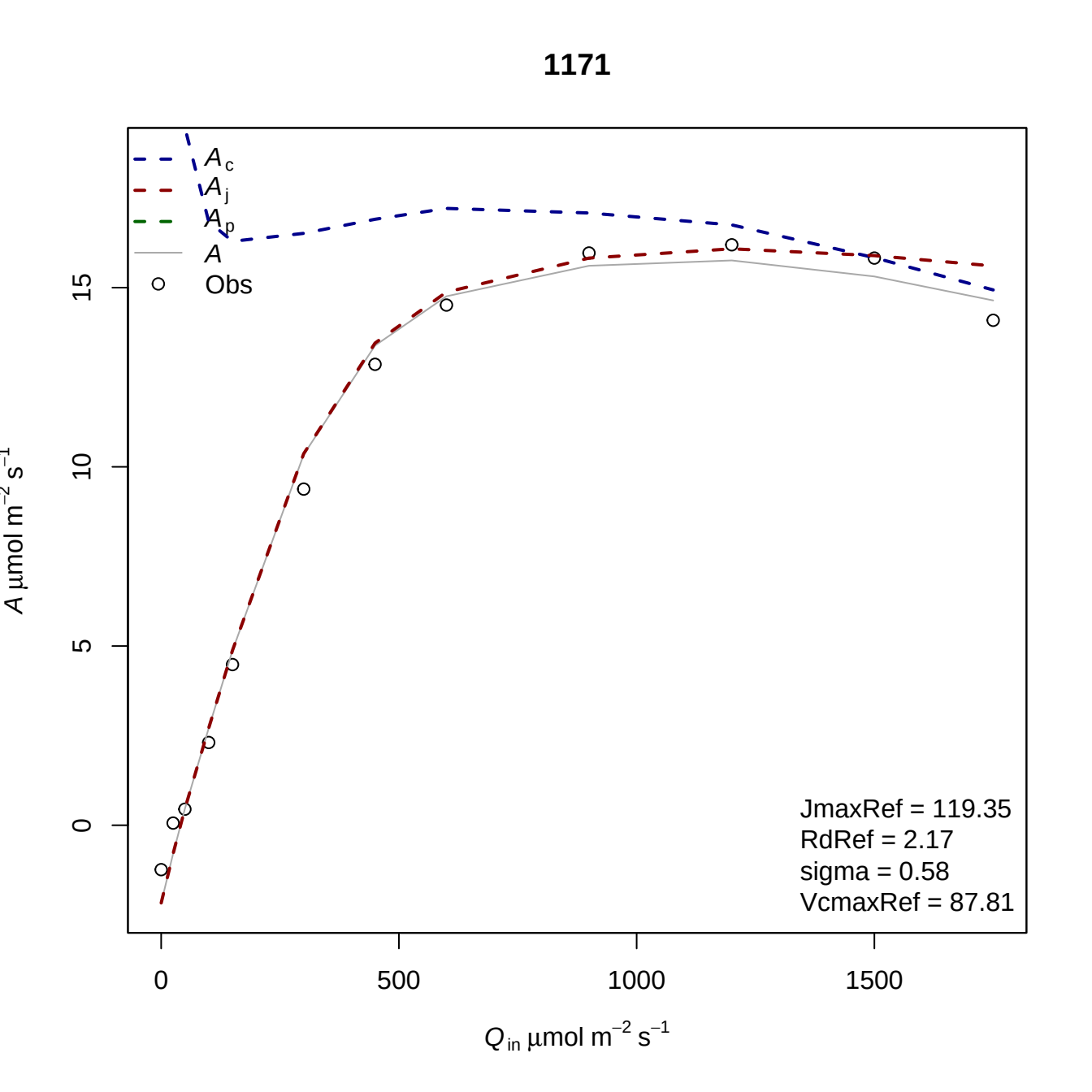

1166
A_c
A_j
A_p
A
Obs
JmaxRef = 114.68
RdRef = 1.83
sigma = 0.79
VcmaxRef = 185.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

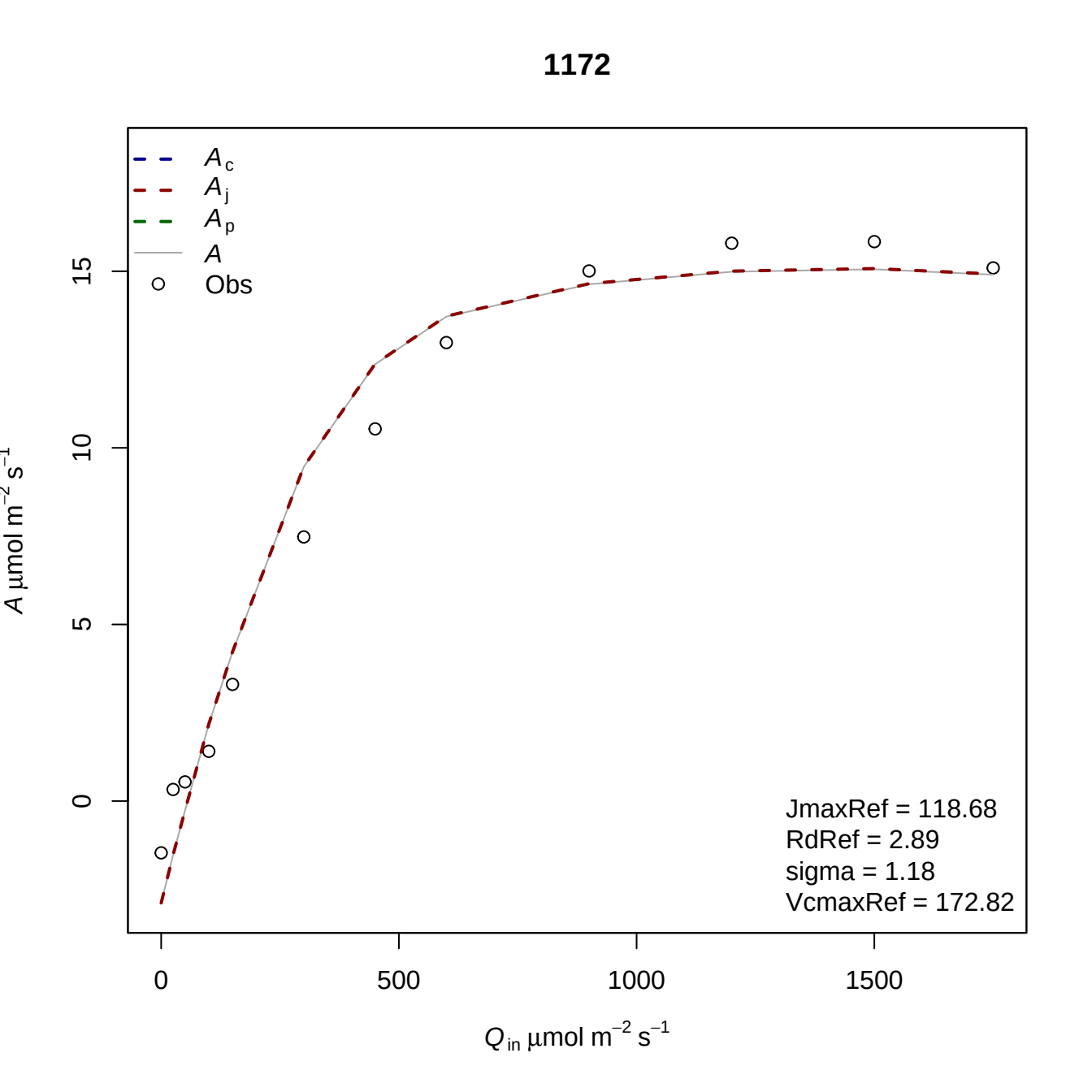

1167
A_c
A_j
A_p
A
Obs
JmaxRef = 99.02
RdRef = 3.04
sigma = 1.47
VcmaxRef = 159.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

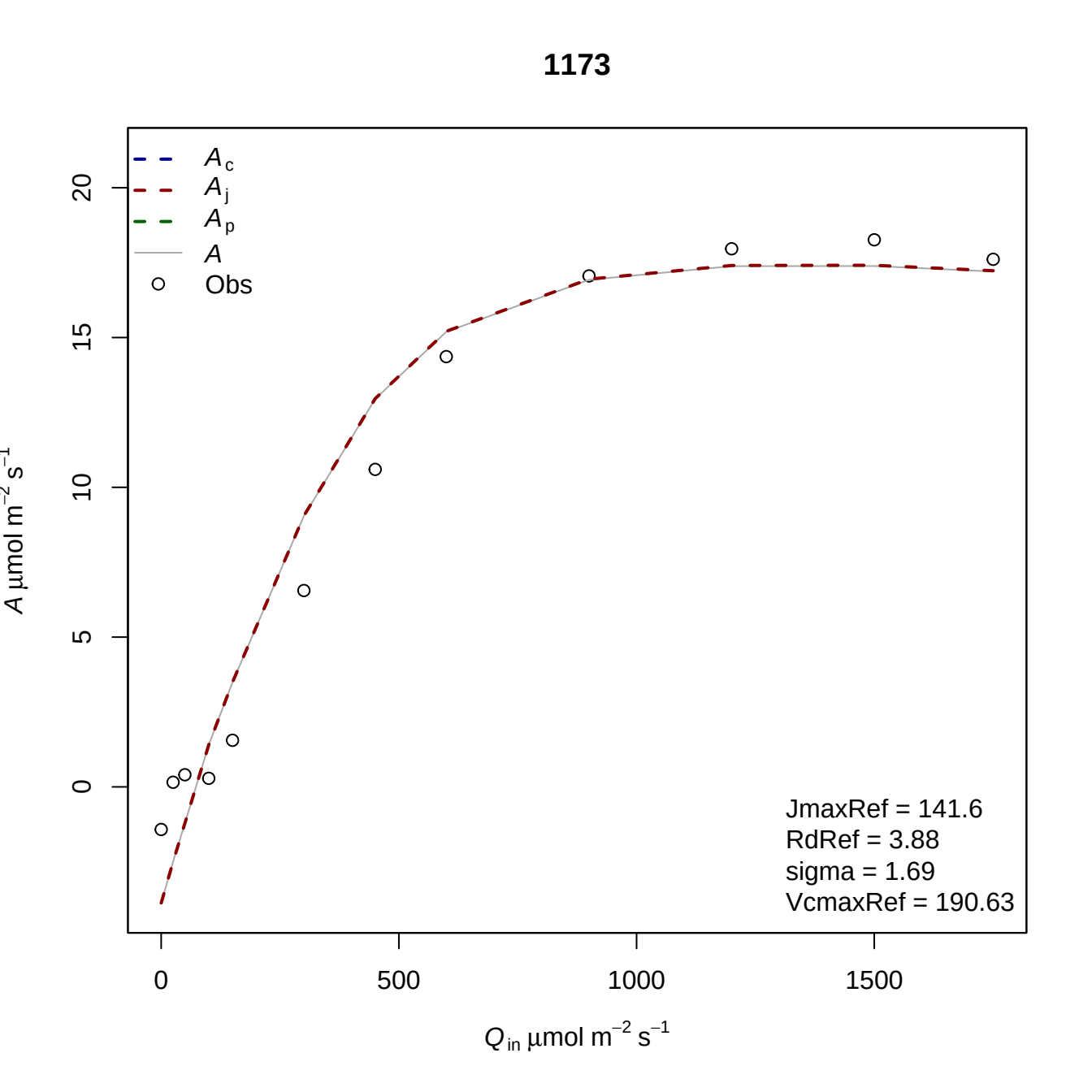

1168
A_c
A_j
A_p
A
Obs
JmaxRef = 117.27
RdRef = 2.08
sigma = 0.9
VcmaxRef = 181.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

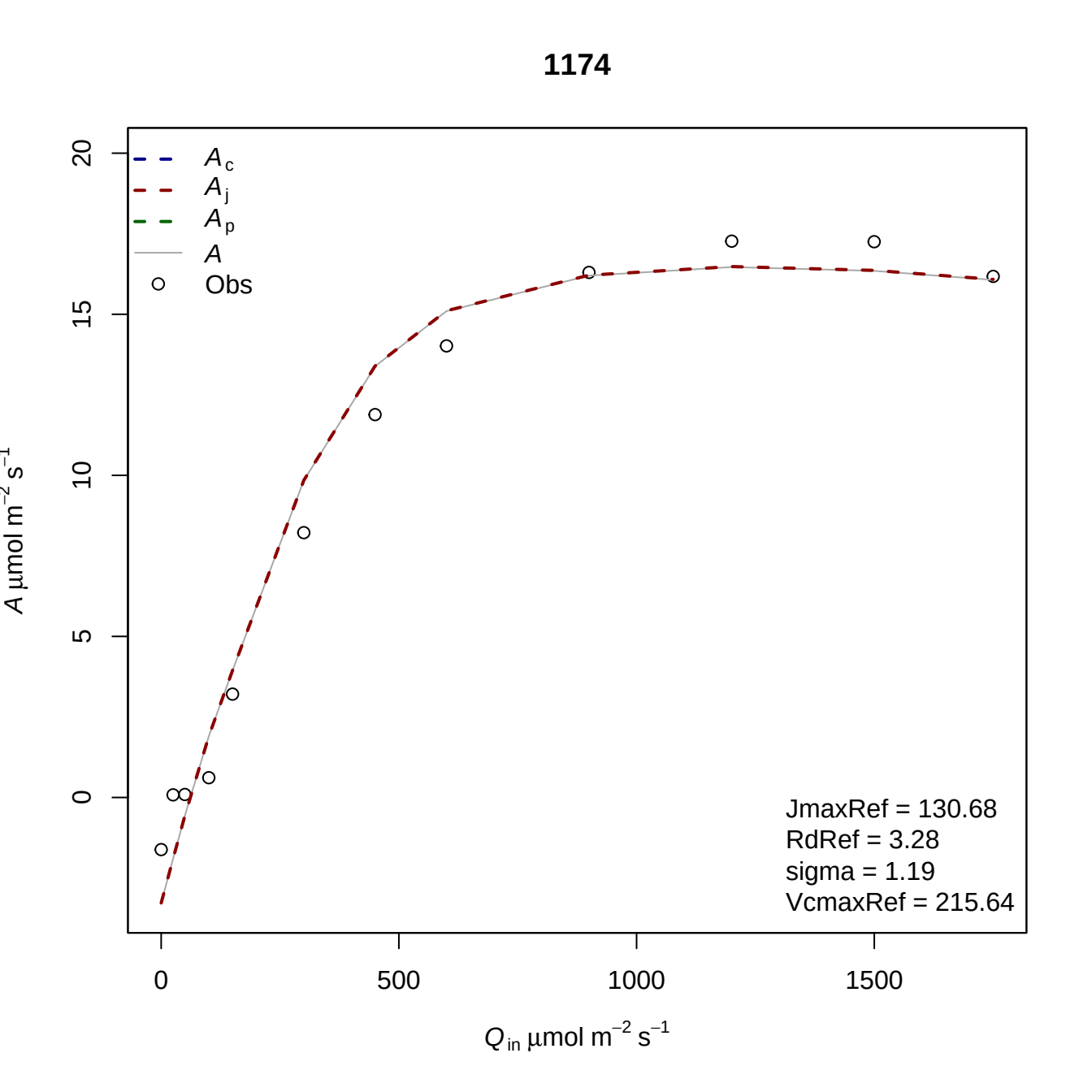

1169
A_c
A_j
A_p
A
Obs
JmaxRef = 132.75
RdRef = 1.95
sigma = 0.61
VcmaxRef = 205.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

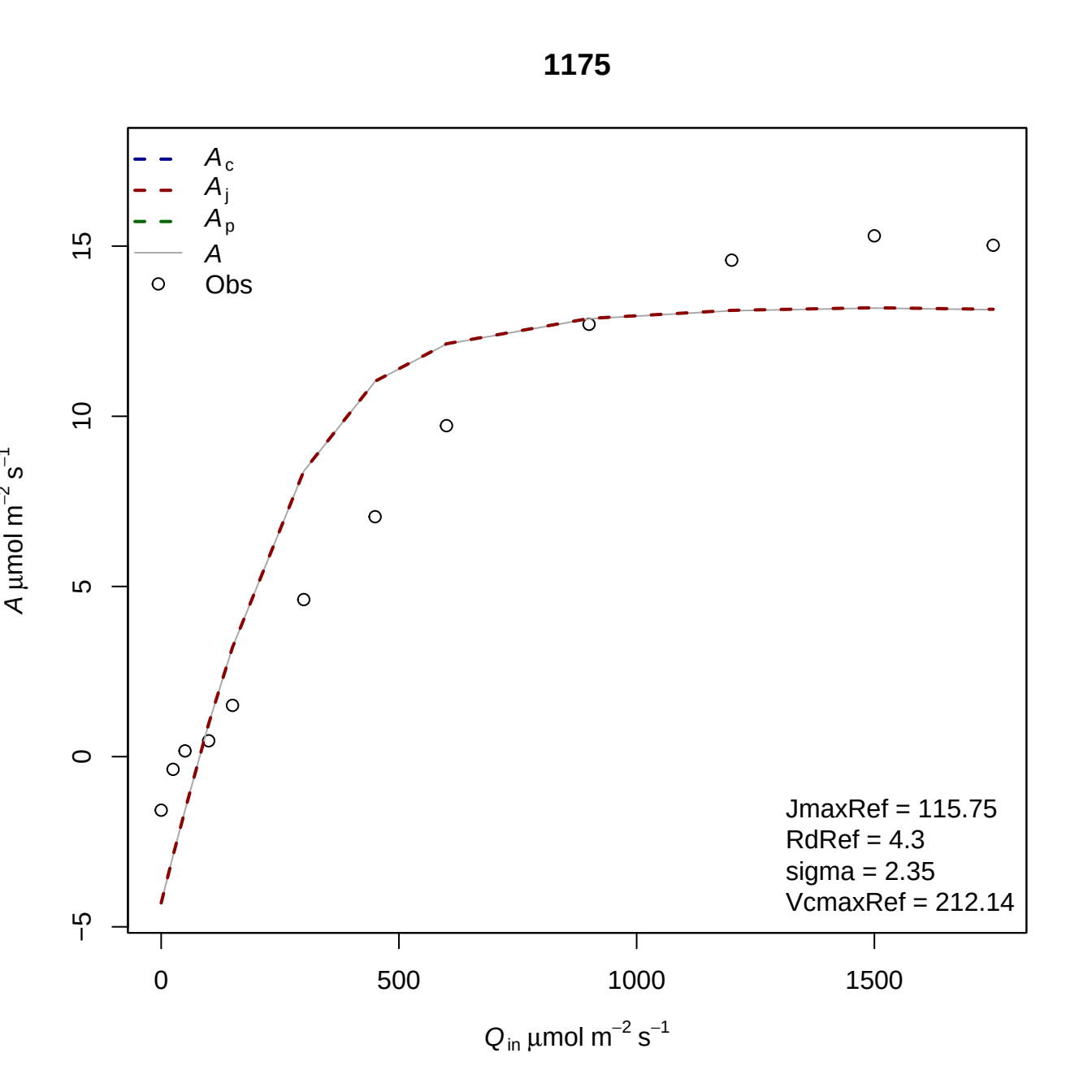

1170
A_c
A_j
A_p
A
Obs
JmaxRef = 150.63
RdRef = 3.52
sigma = 1.42
VcmaxRef = 204.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

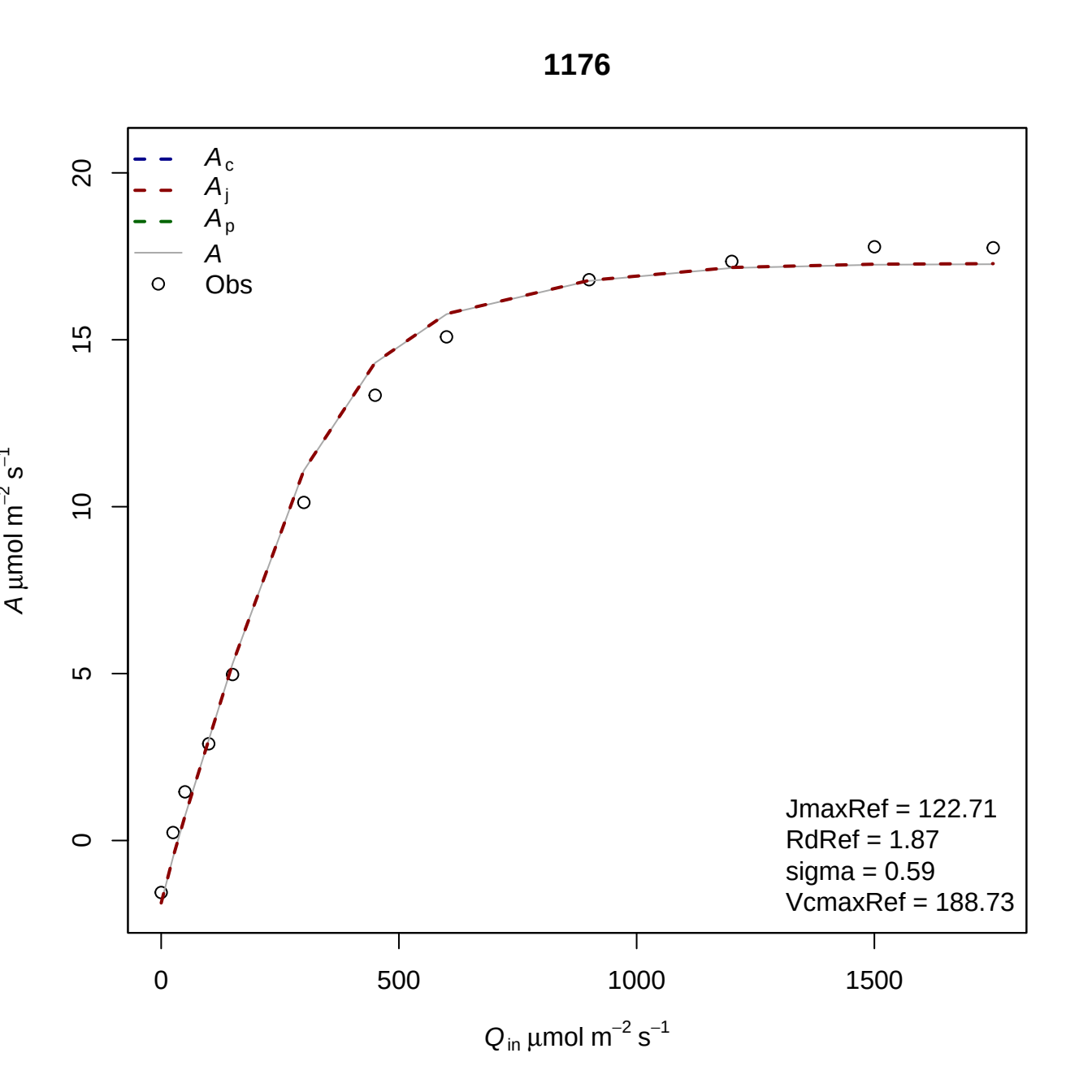

1171
A_c
A_j
A_p
A
Obs
JmaxRef = 119.35
RdRef = 2.17
sigma = 0.58
VcmaxRef = 87.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

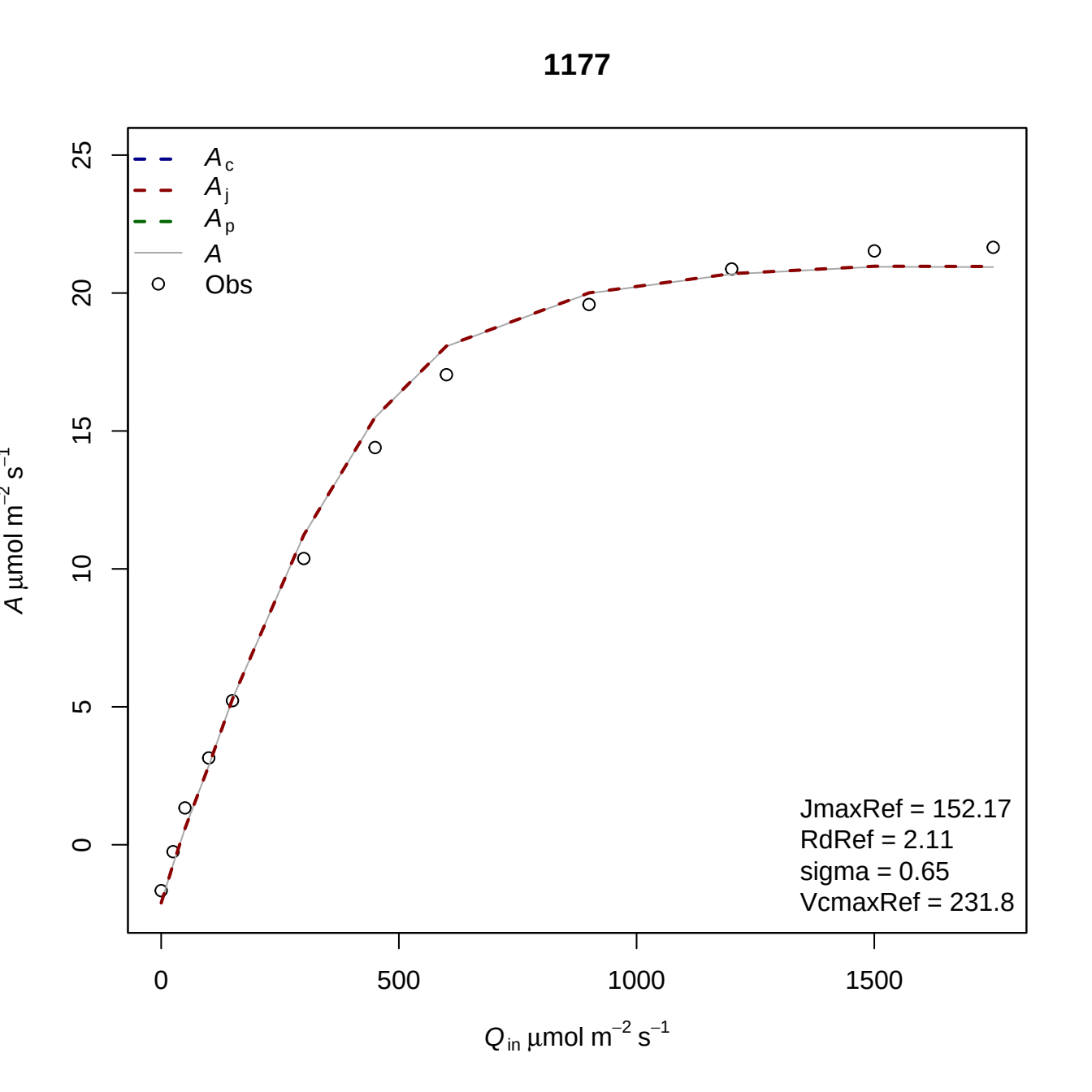

1172
A_c
A_j
A_p
A
Obs
JmaxRef = 118.68
RdRef = 2.89
sigma = 1.18
VcmaxRef = 172.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

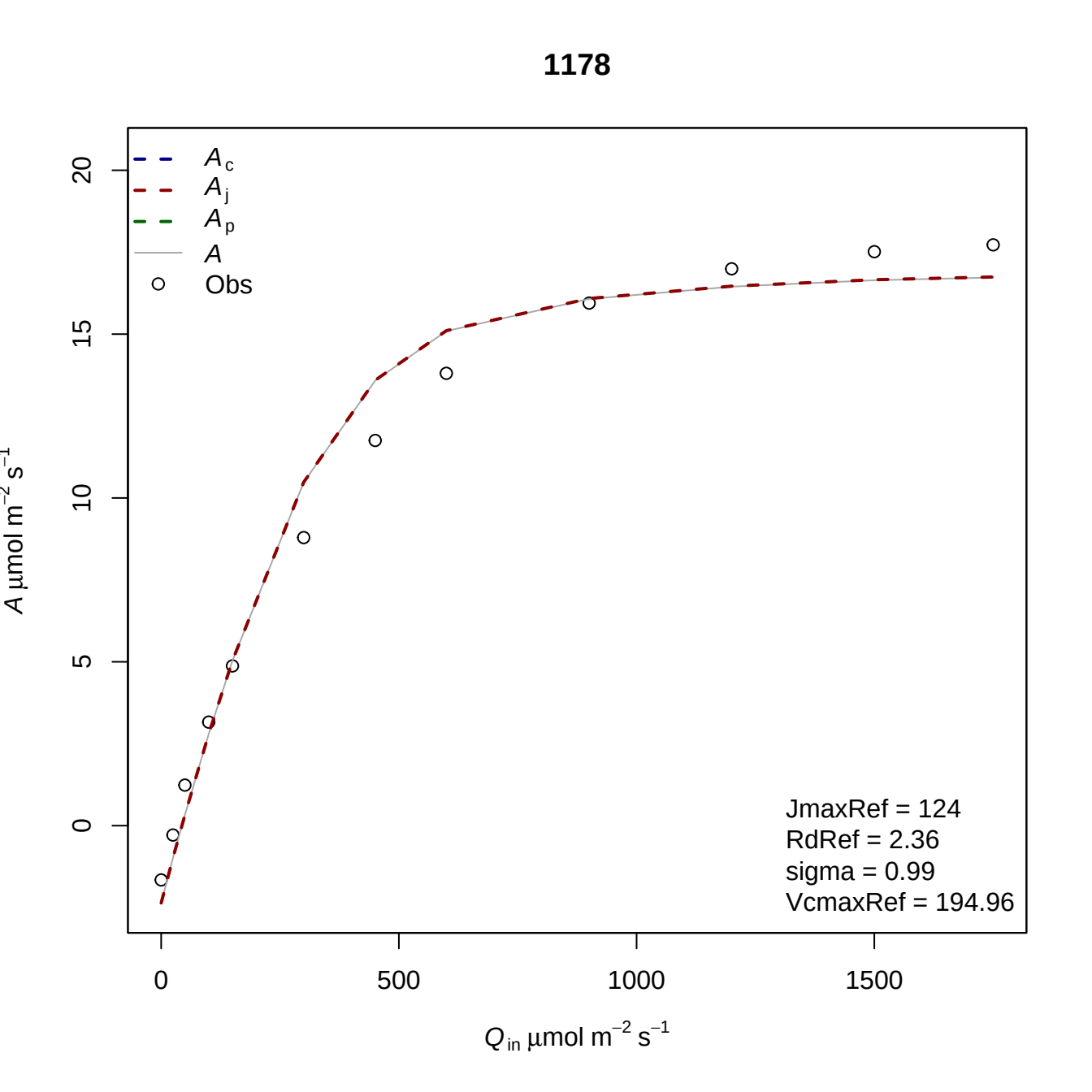

1173
A_c
A_j
A_p
A
Obs
JmaxRef = 141.6
RdRef = 3.88
sigma = 1.69
VcmaxRef = 190.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

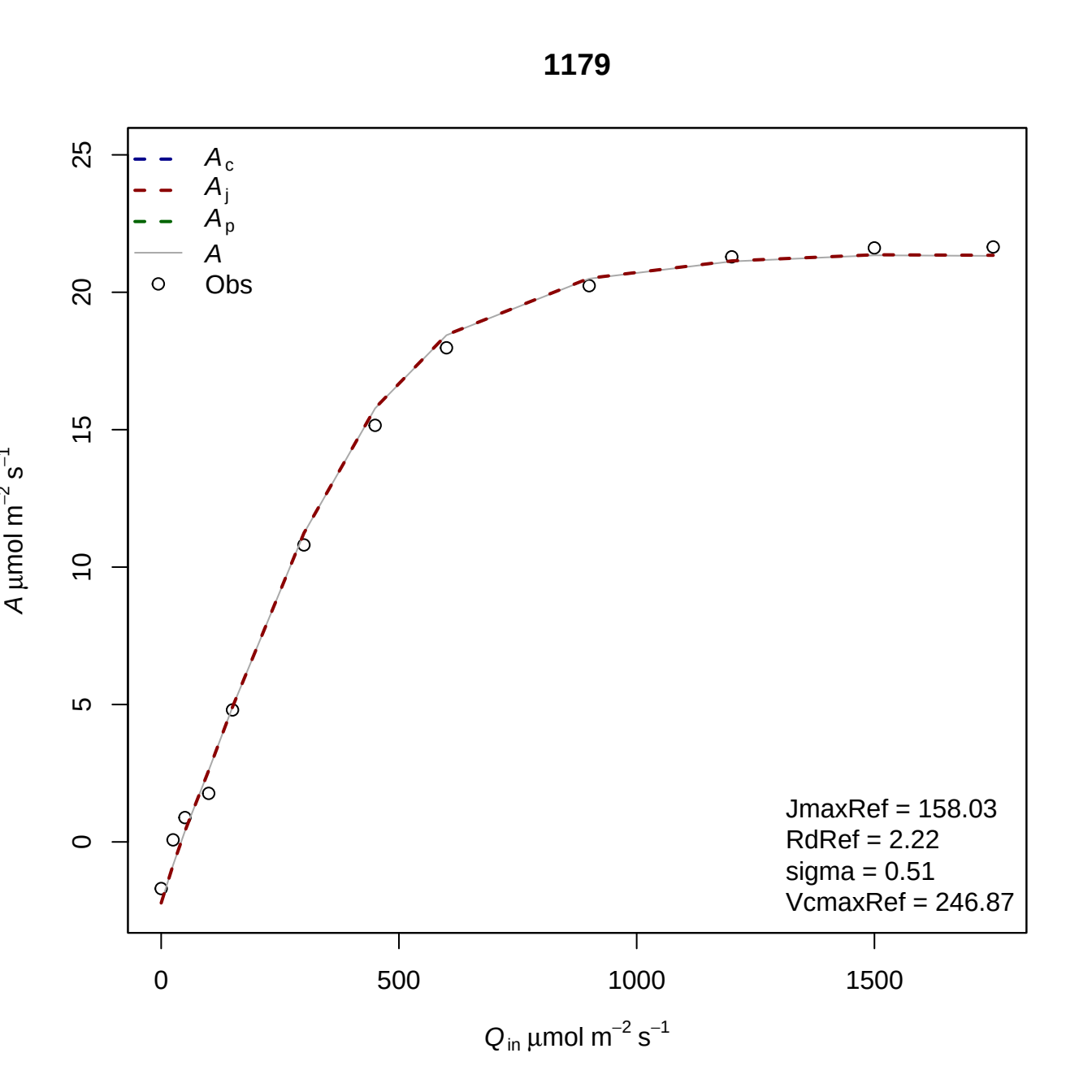

1174
A_c
A_j
A_p
A
Obs
JmaxRef = 130.68
RdRef = 3.28
sigma = 1.19
VcmaxRef = 215.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

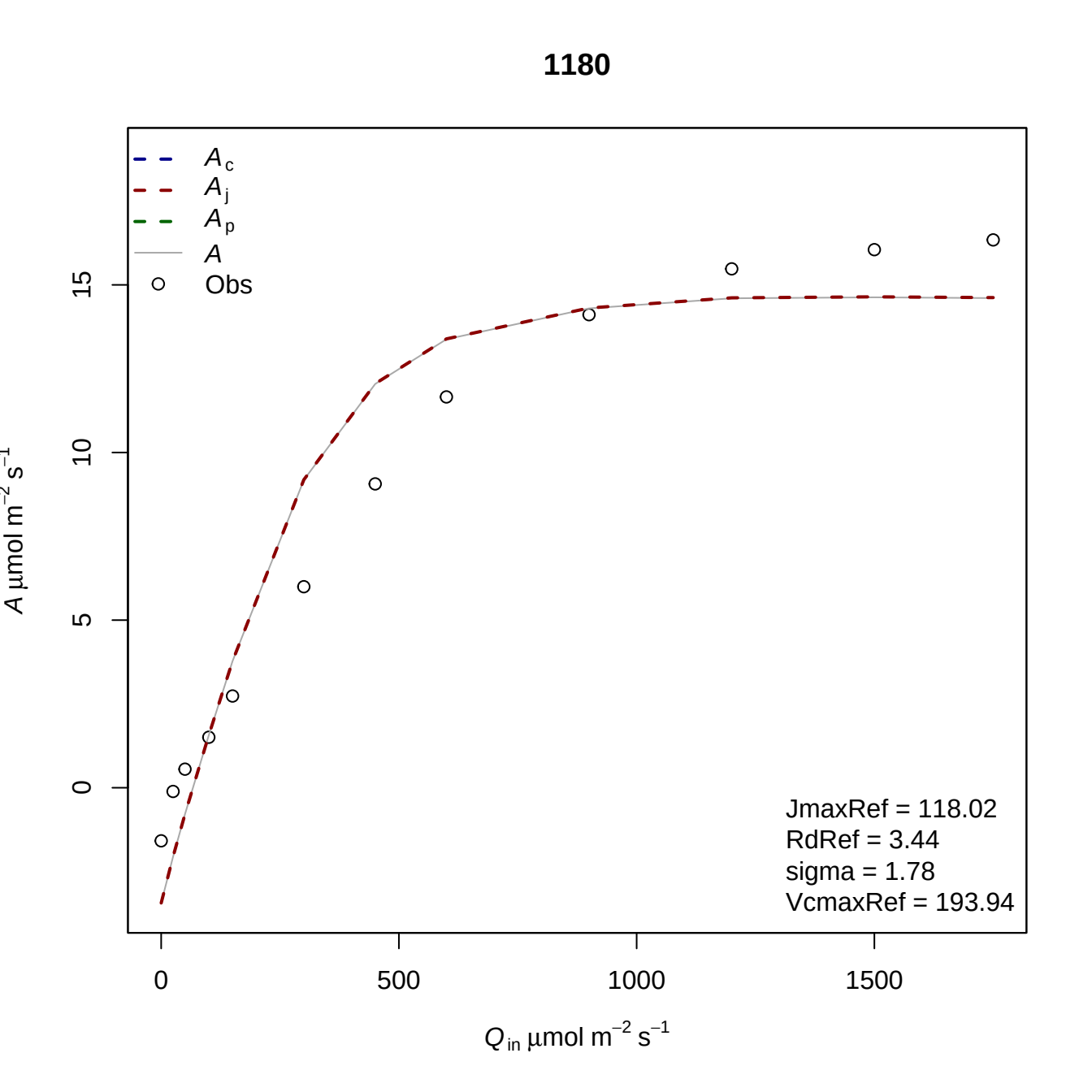

1175
A_c
A_j
A_p
A
Obs
JmaxRef = 115.75
RdRef = 4.3
sigma = 2.35
VcmaxRef = 212.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500

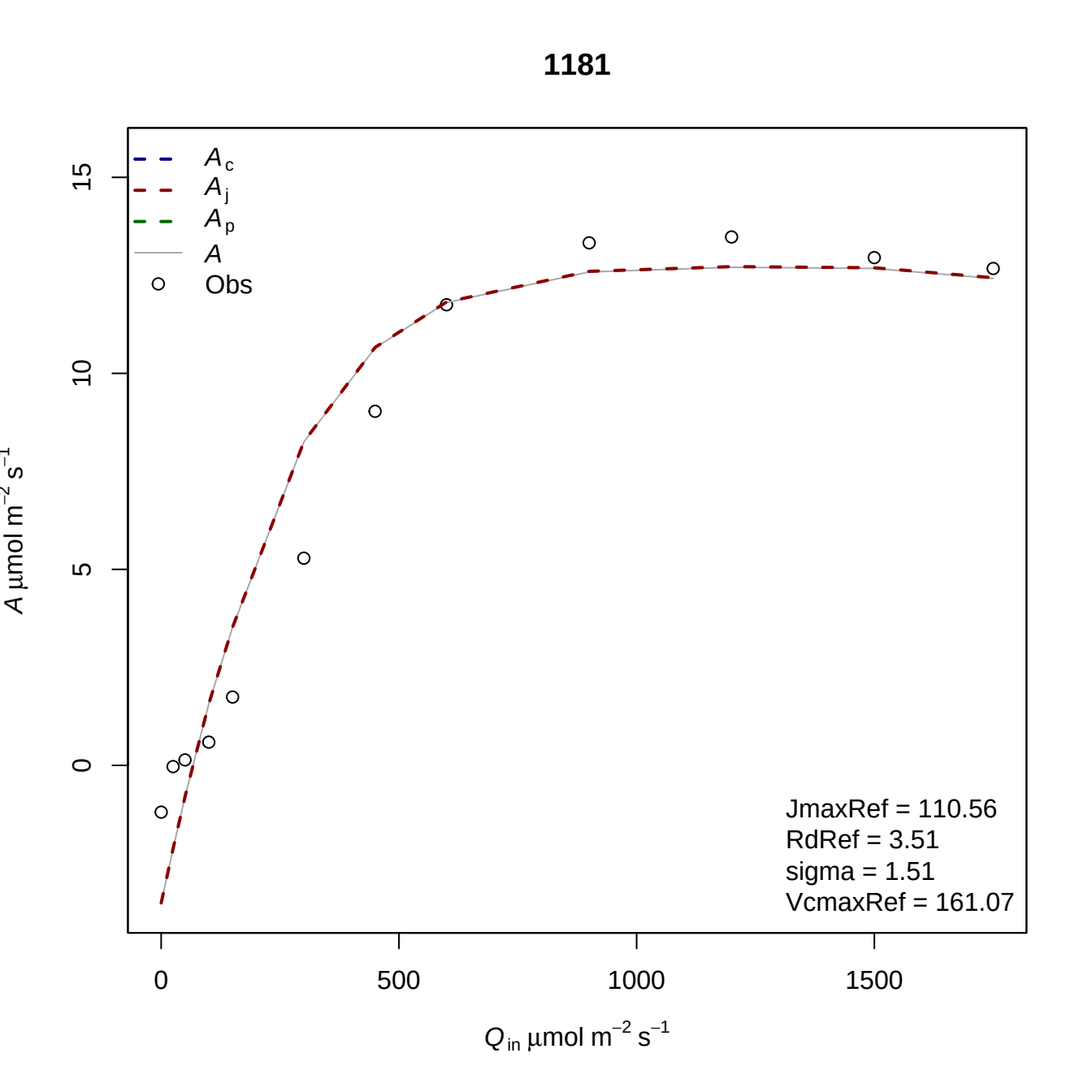

1176
A_c
A_j
A_p
A
Obs
JmaxRef = 122.71
RdRef = 1.87
sigma = 0.59
VcmaxRef = 188.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

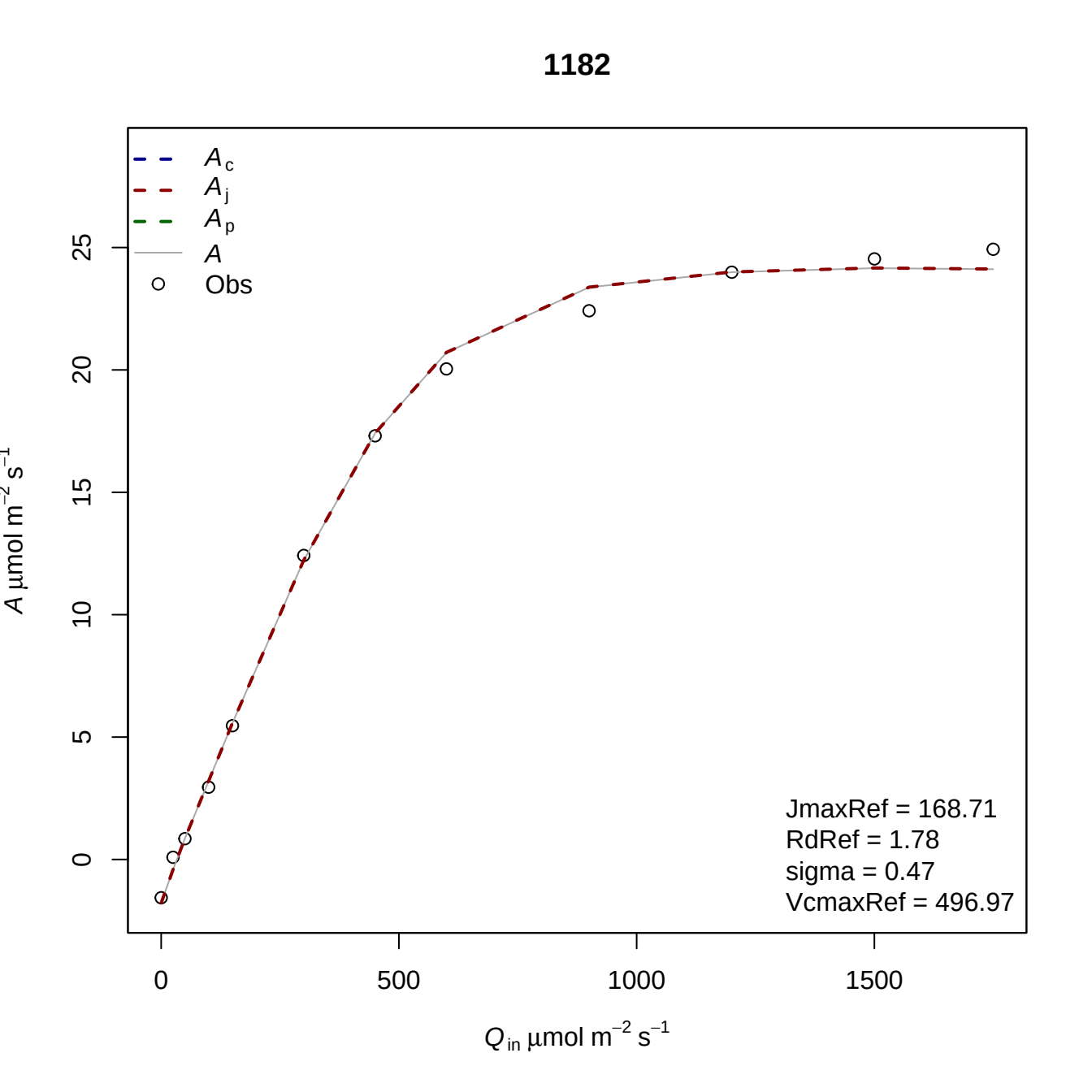

1177
A_c
A_j
A_p
A
Obs
JmaxRef = 152.17
RdRef = 2.11
sigma = 0.65
VcmaxRef = 231.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

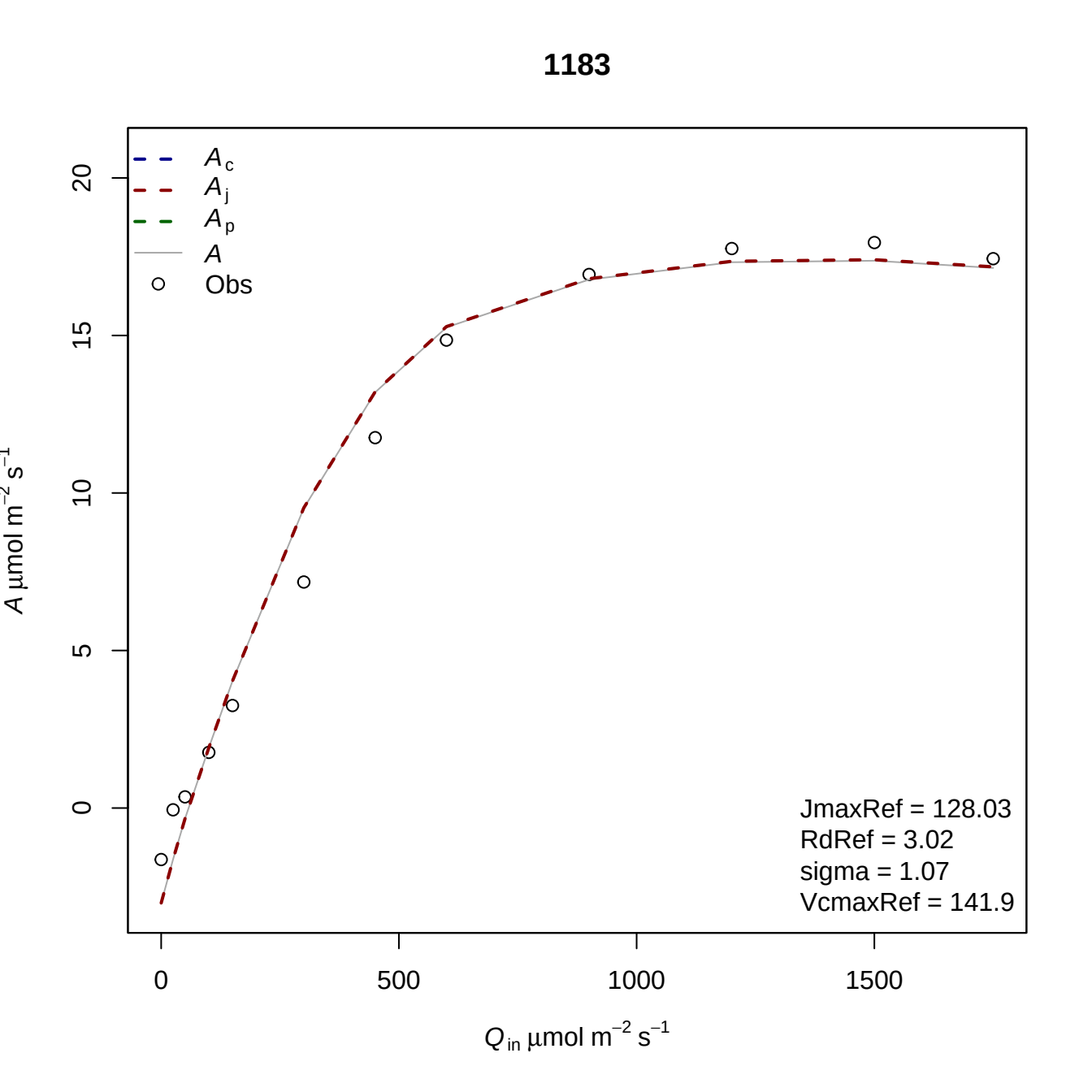

1178
A_c
A_j
A_p
A
Obs
JmaxRef = 124
RdRef = 2.36
sigma = 0.99
VcmaxRef = 194.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

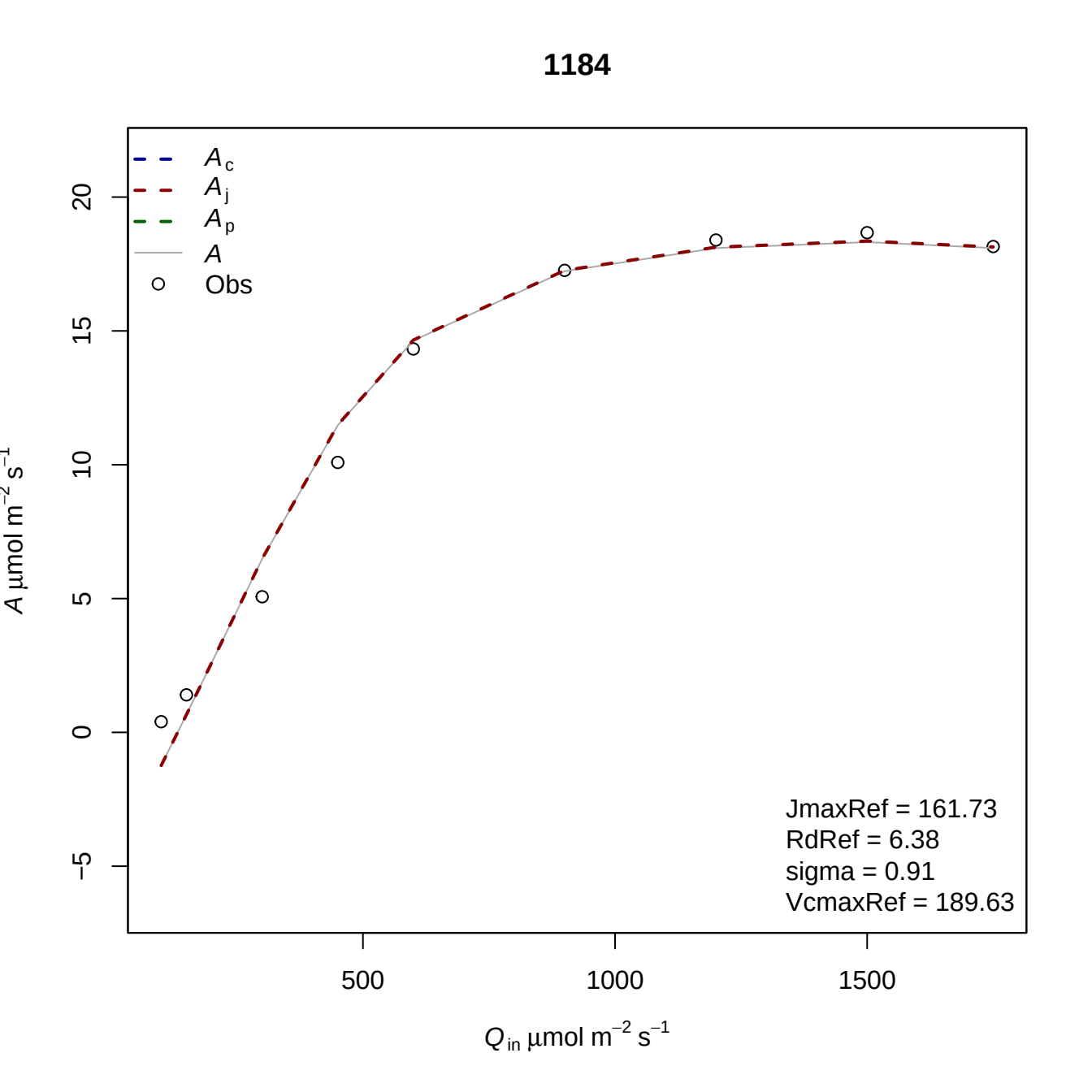

1179
A_c
A_j
A_p
A
Obs
JmaxRef = 158.03
RdRef = 2.22
sigma = 0.51
VcmaxRef = 246.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

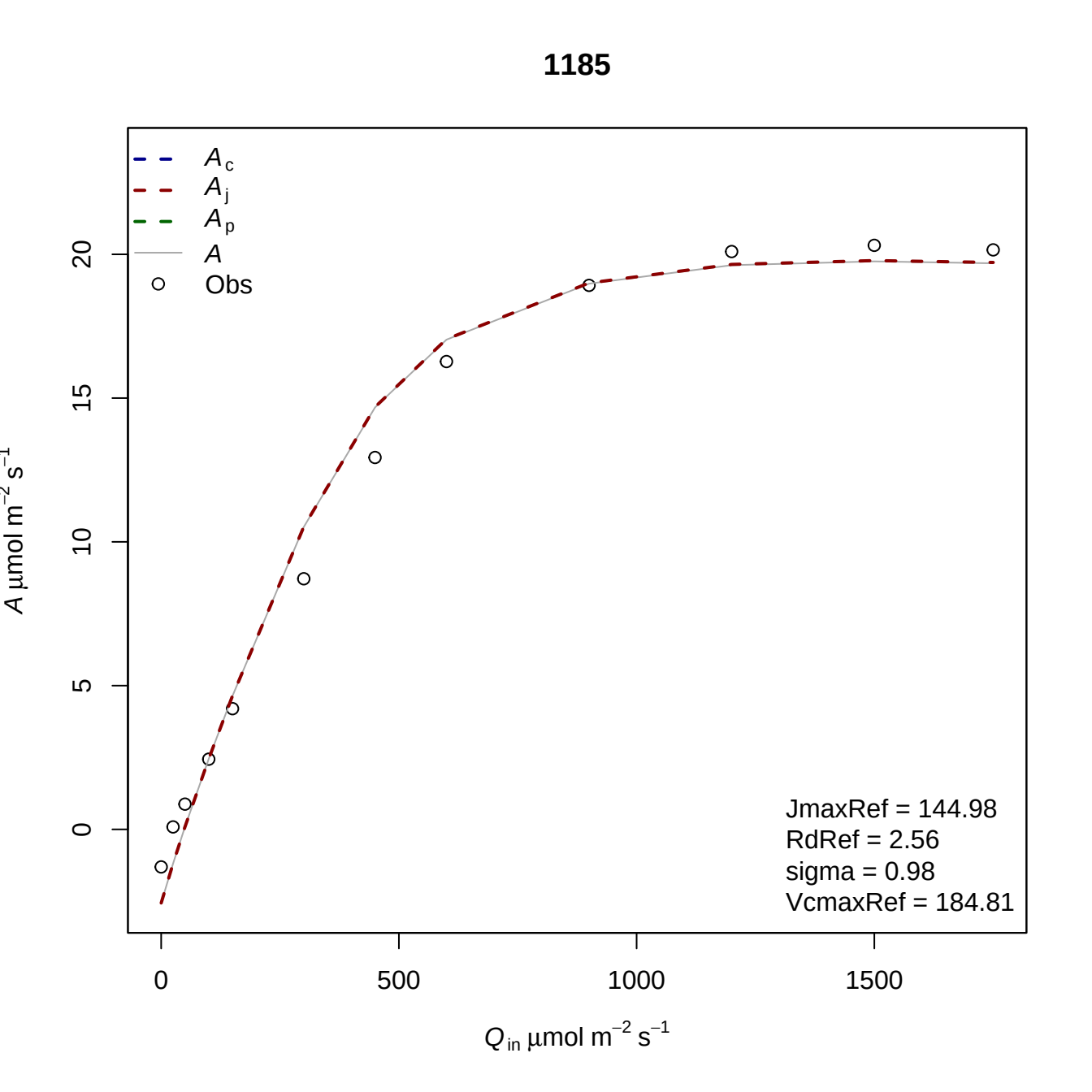

1180
A_c
A_j
A_p
A
Obs
JmaxRef = 118.02
RdRef = 3.44
sigma = 1.78
VcmaxRef = 193.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

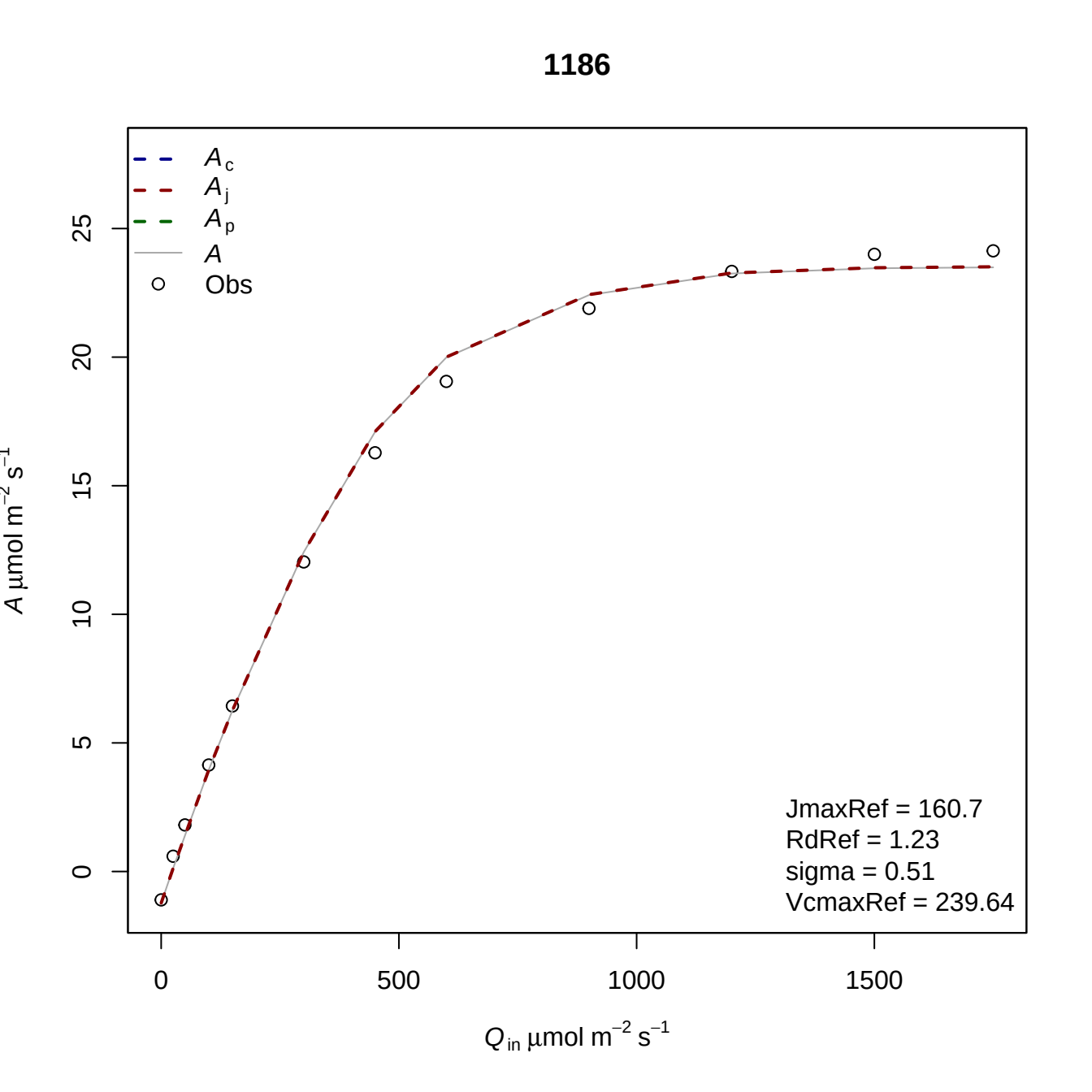

1181
A_c
A_j
A_p
A
Obs
JmaxRef = 110.56
RdRef = 3.51
sigma = 1.51
VcmaxRef = 161.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

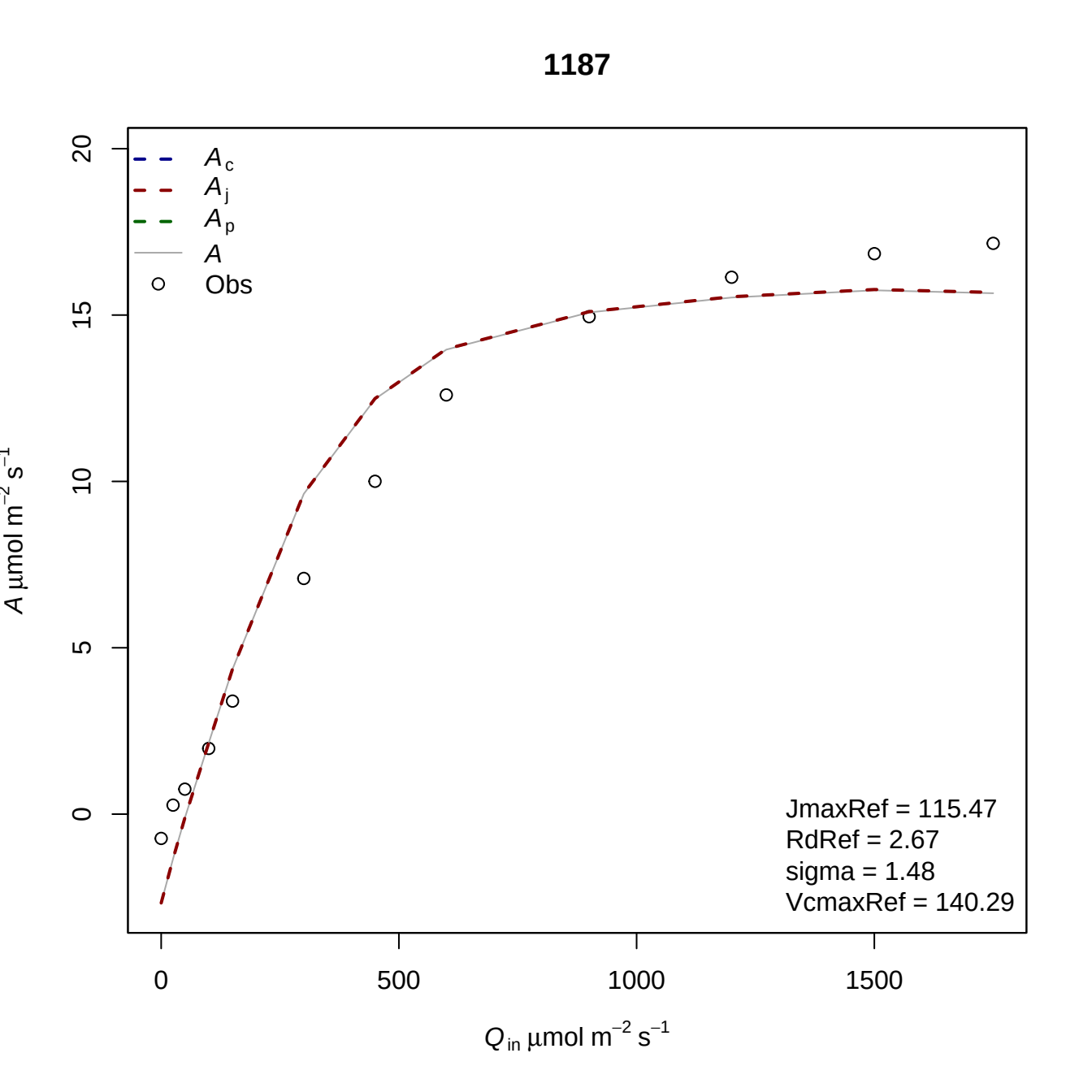

1182
A_c
A_j
A_p
A
Obs
JmaxRef = 168.71
RdRef = 1.78
sigma = 0.47
VcmaxRef = 496.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

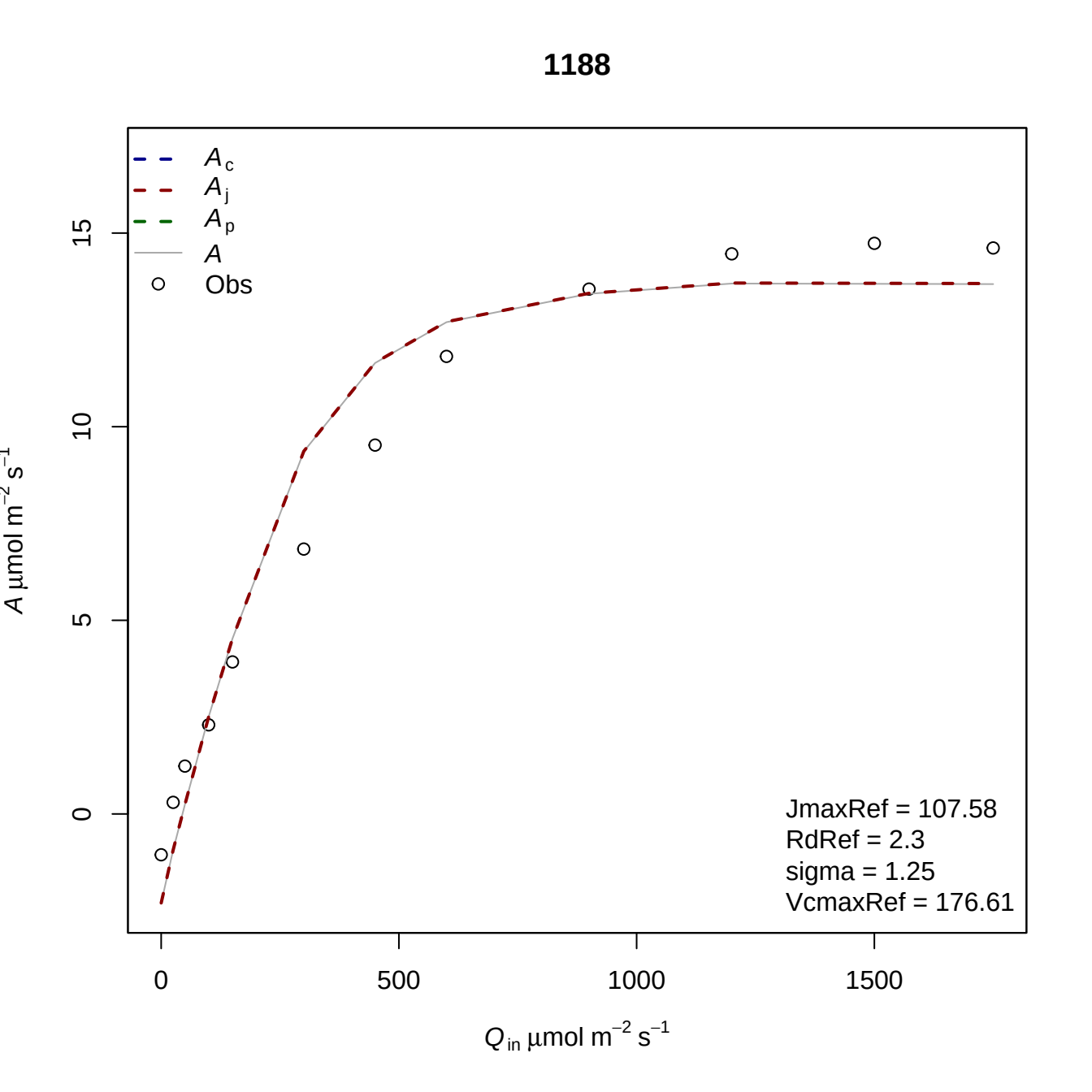

1183
A_c
A_j
A_p
A
Obs
JmaxRef = 128.03
RdRef = 3.02
sigma = 1.07
VcmaxRef = 141.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

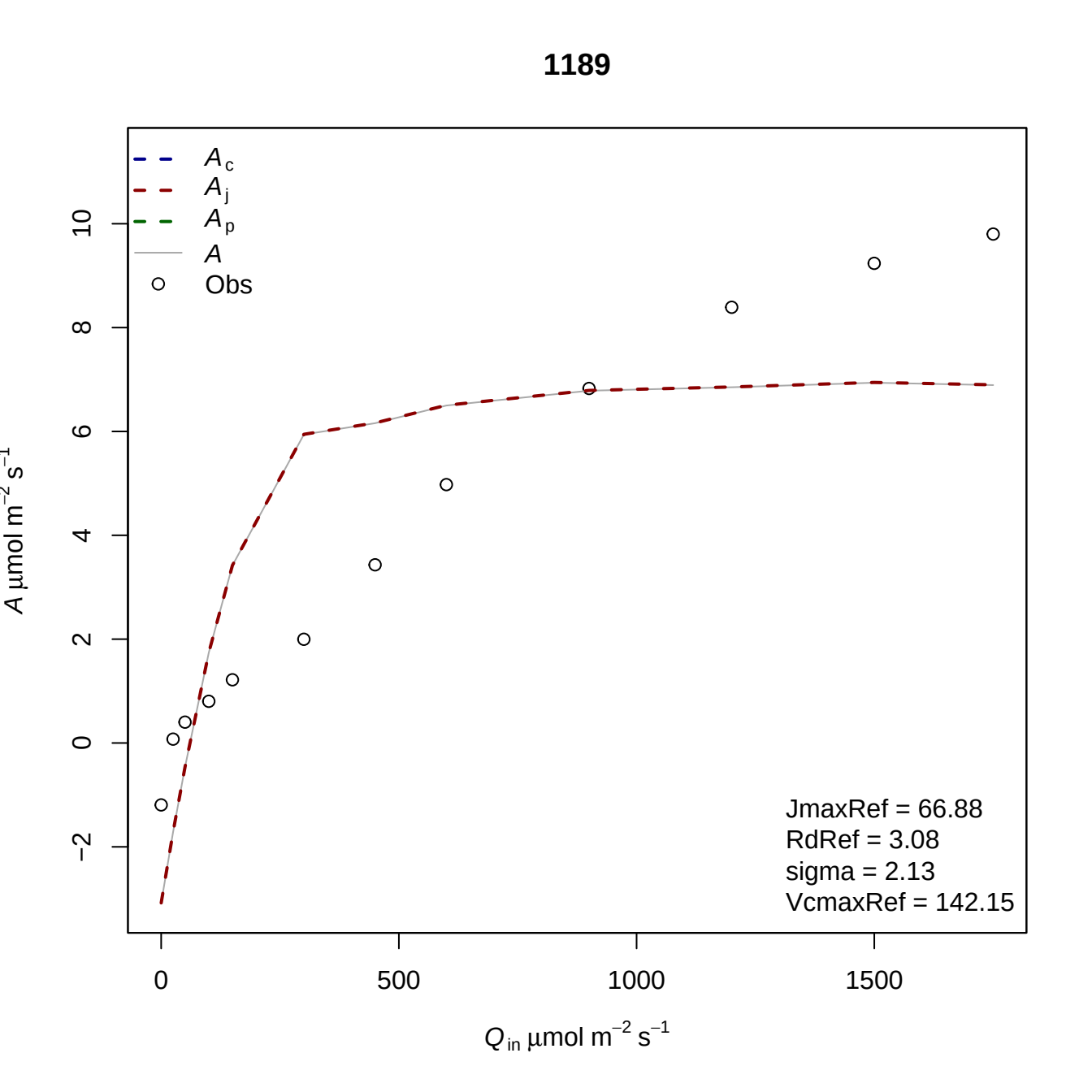

1184
A_c
A_j
A_p
A
Obs
JmaxRef = 161.73
RdRef = 6.38
sigma = 0.91
VcmaxRef = 189.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
500
1000
1500

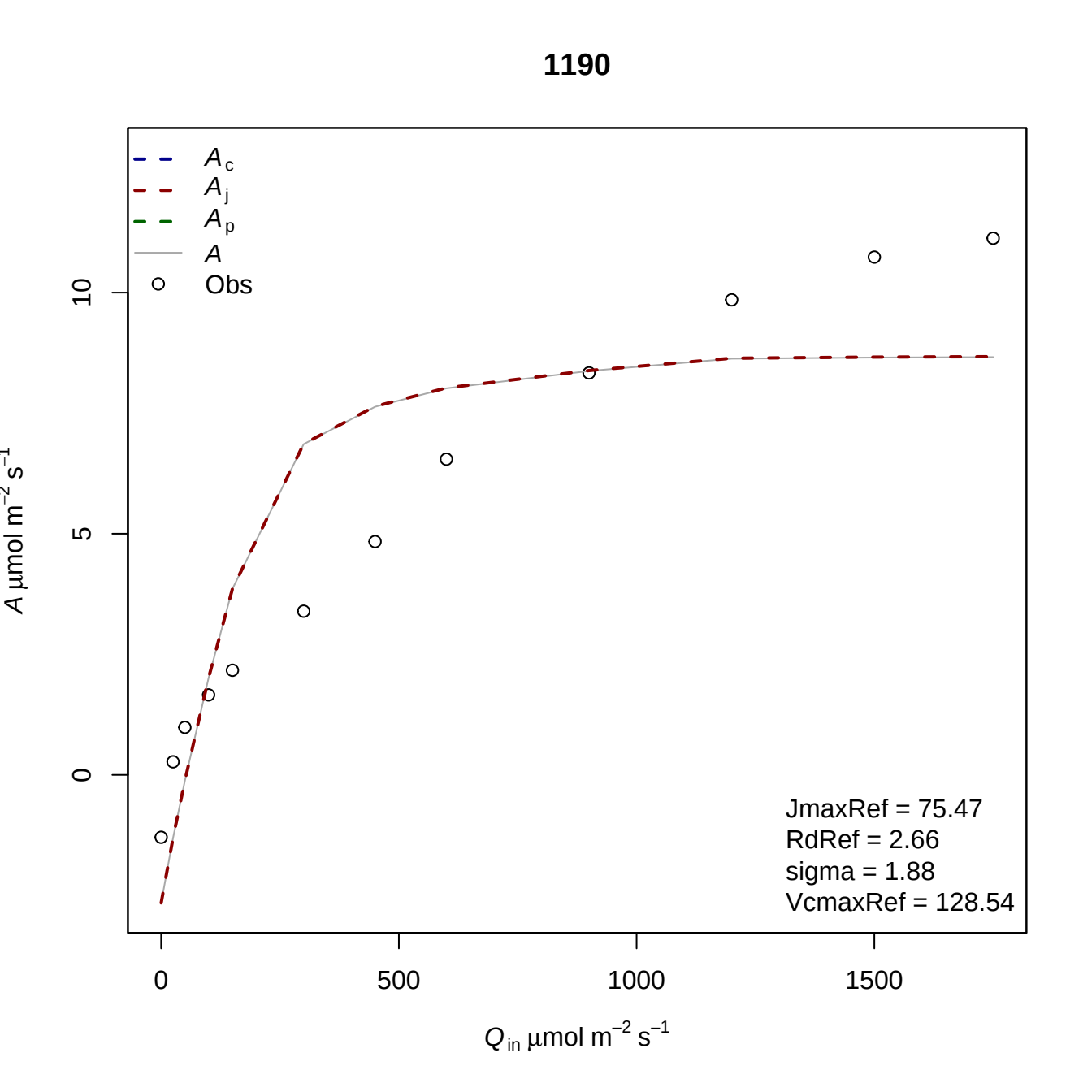

1185
A_c
A_j
A_p
A
Obs
JmaxRef = 144.98
RdRef = 2.56
sigma = 0.98
VcmaxRef = 184.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

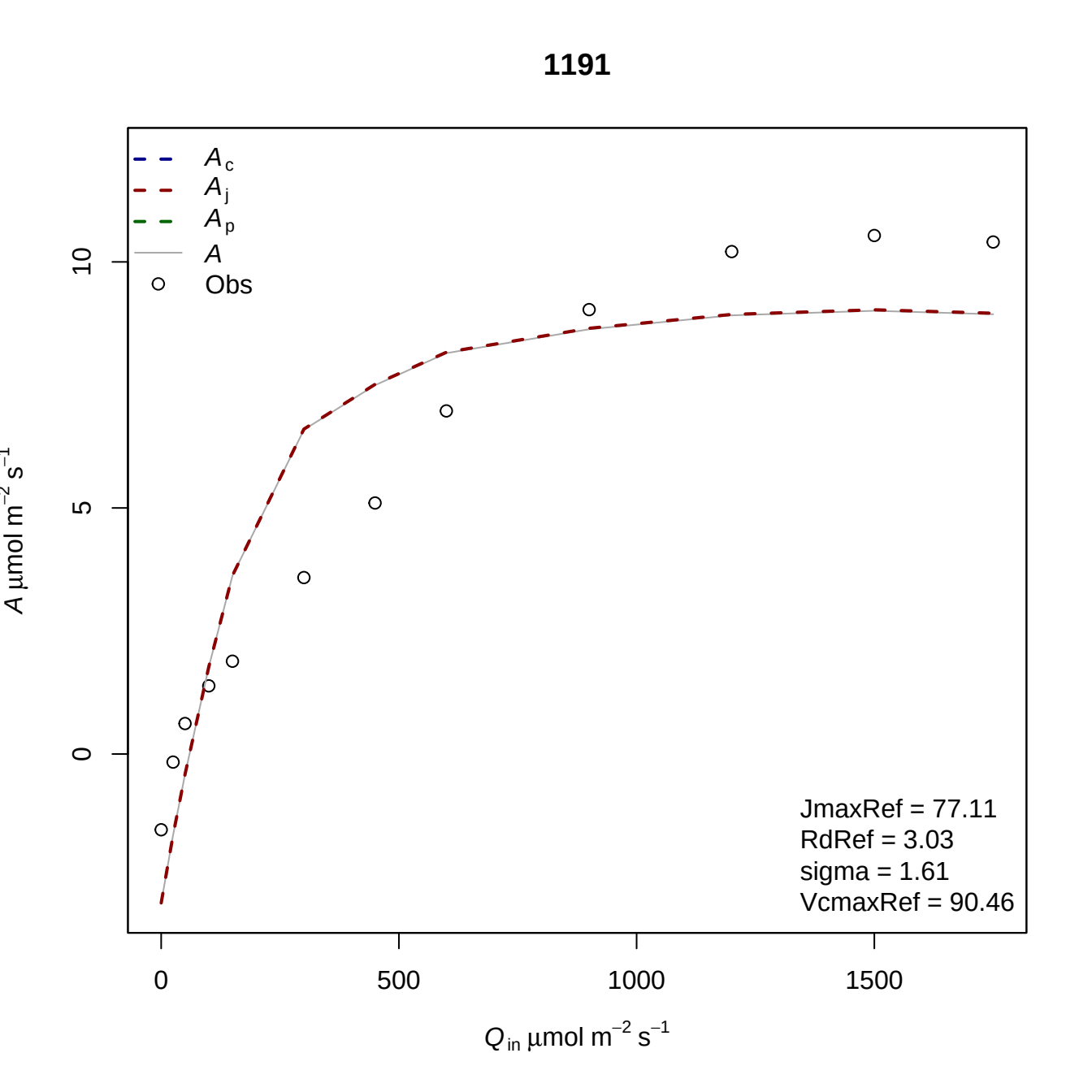

1186
A_c
A_j
A_p
A
Obs
JmaxRef = 160.7
RdRef = 1.23
sigma = 0.51
VcmaxRef = 239.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

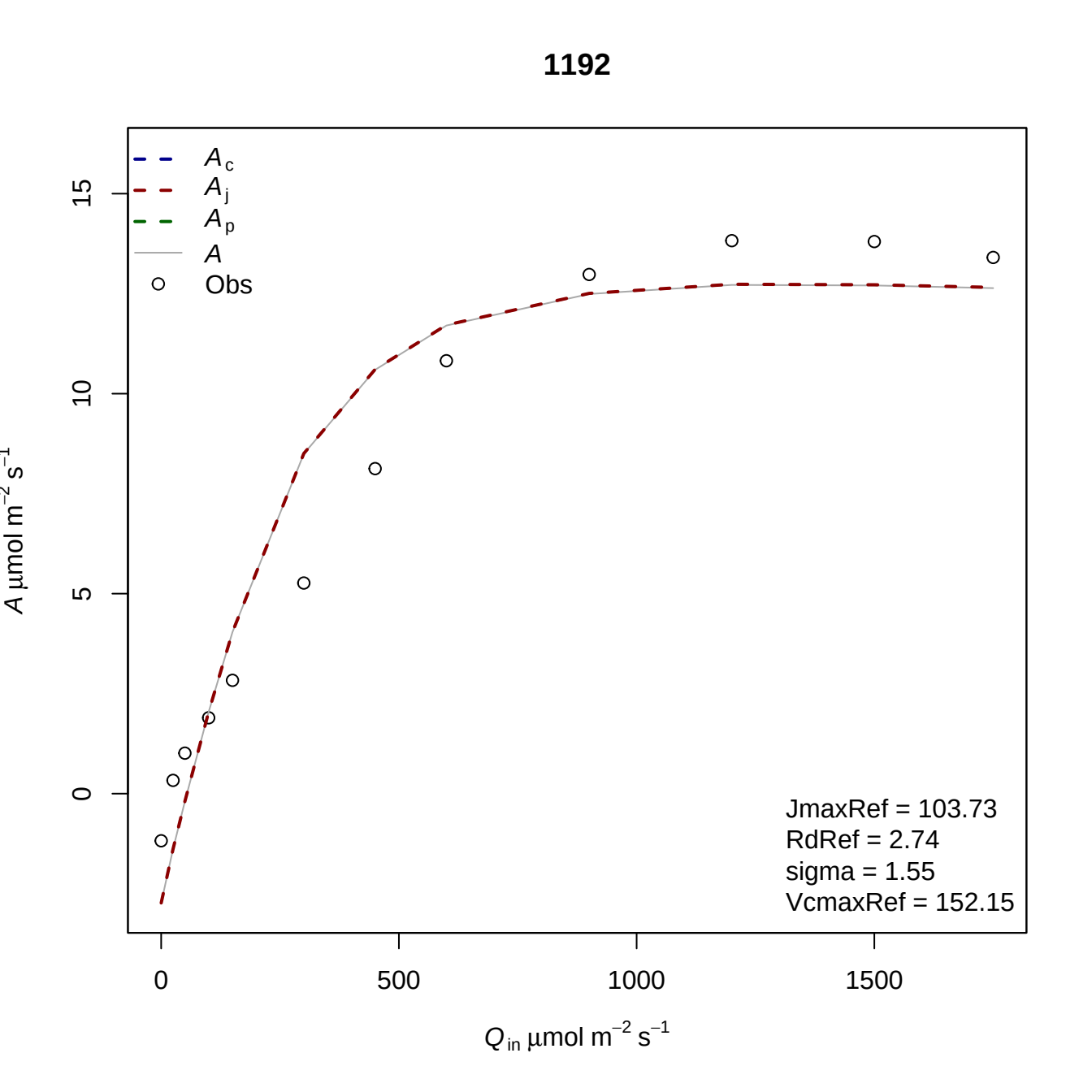

1187
A_c
A_j
A_p
A
Obs
JmaxRef = 115.47
RdRef = 2.67
sigma = 1.48
VcmaxRef = 140.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

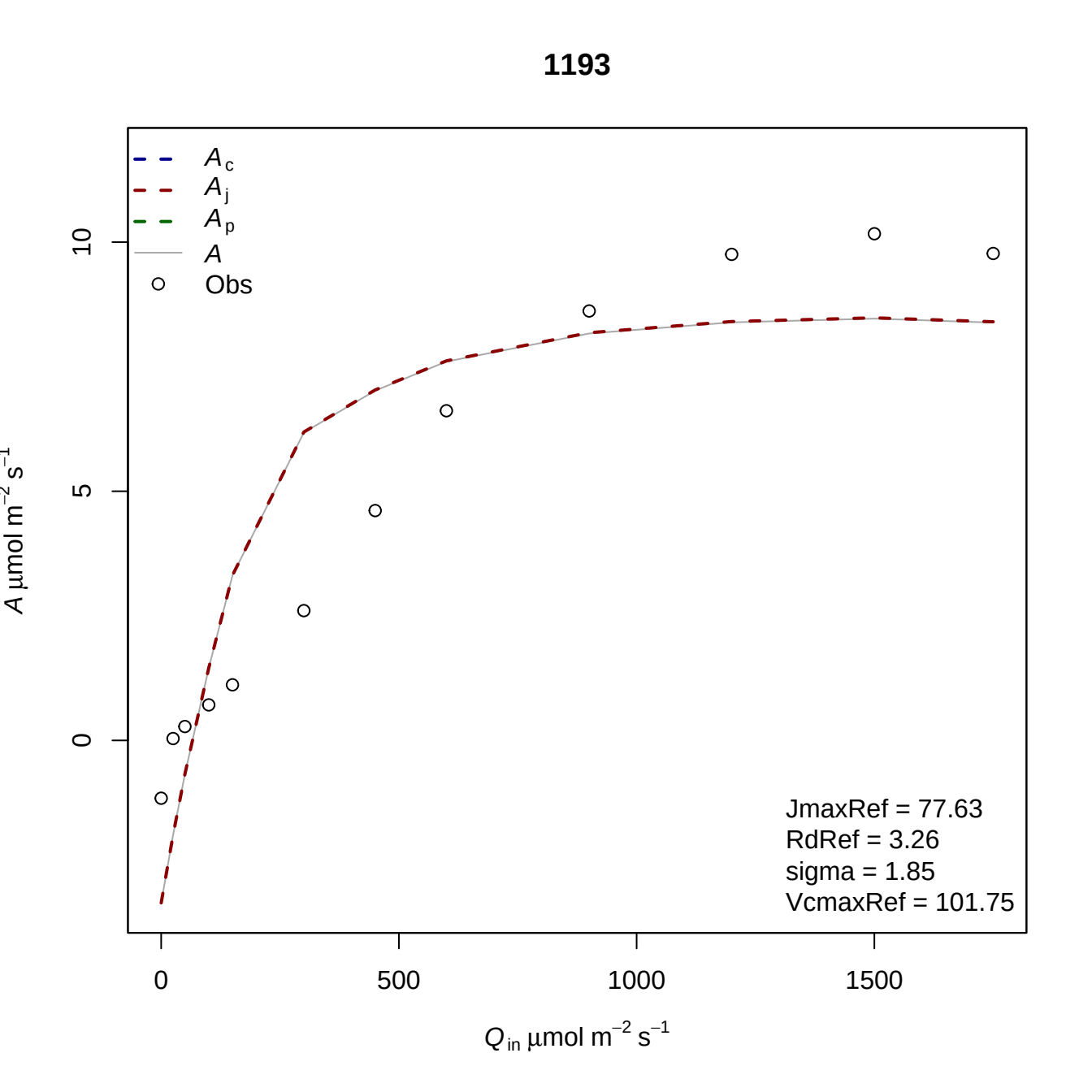

1188
A_c
A_j
A_p
A
Obs
JmaxRef = 107.58
RdRef = 2.3
sigma = 1.25
VcmaxRef = 176.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

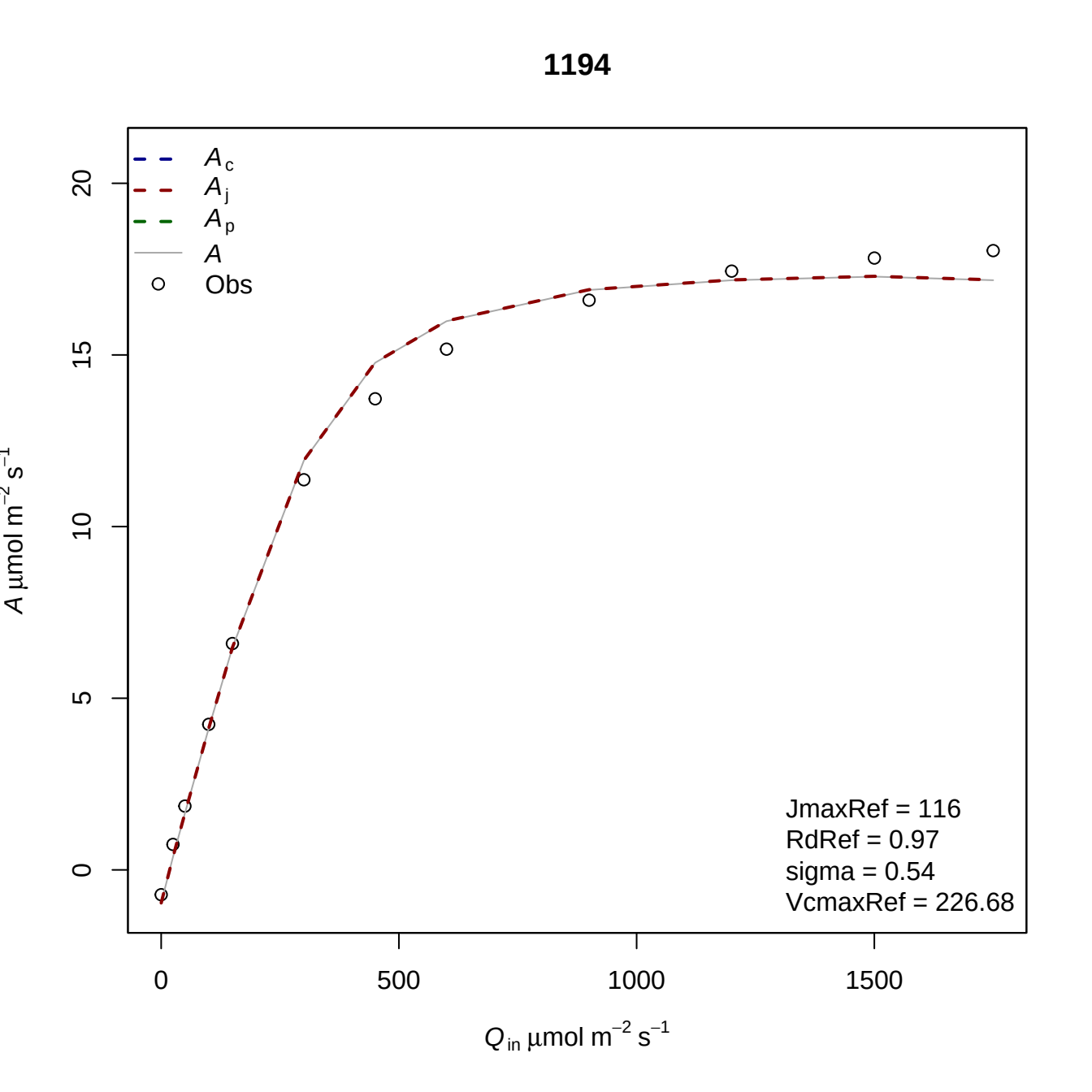

1189
A_c
A_j
A_p
A
Obs
JmaxRef = 66.88
RdRef = 3.08
sigma = 2.13
VcmaxRef = 142.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

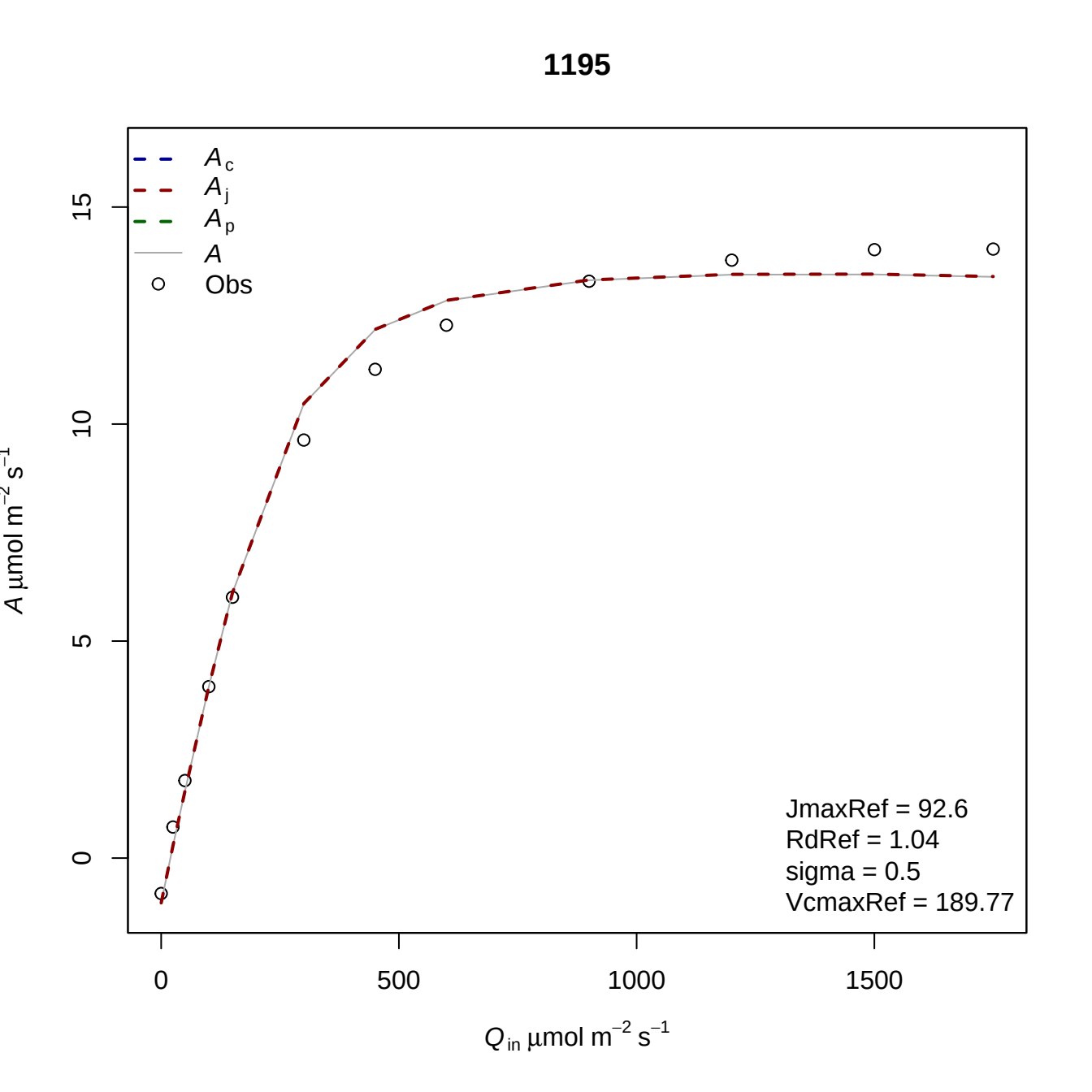

1190
A_c
A_j
A_p
A
Obs
JmaxRef = 75.47
RdRef = 2.66
sigma = 1.88
VcmaxRef = 128.54
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

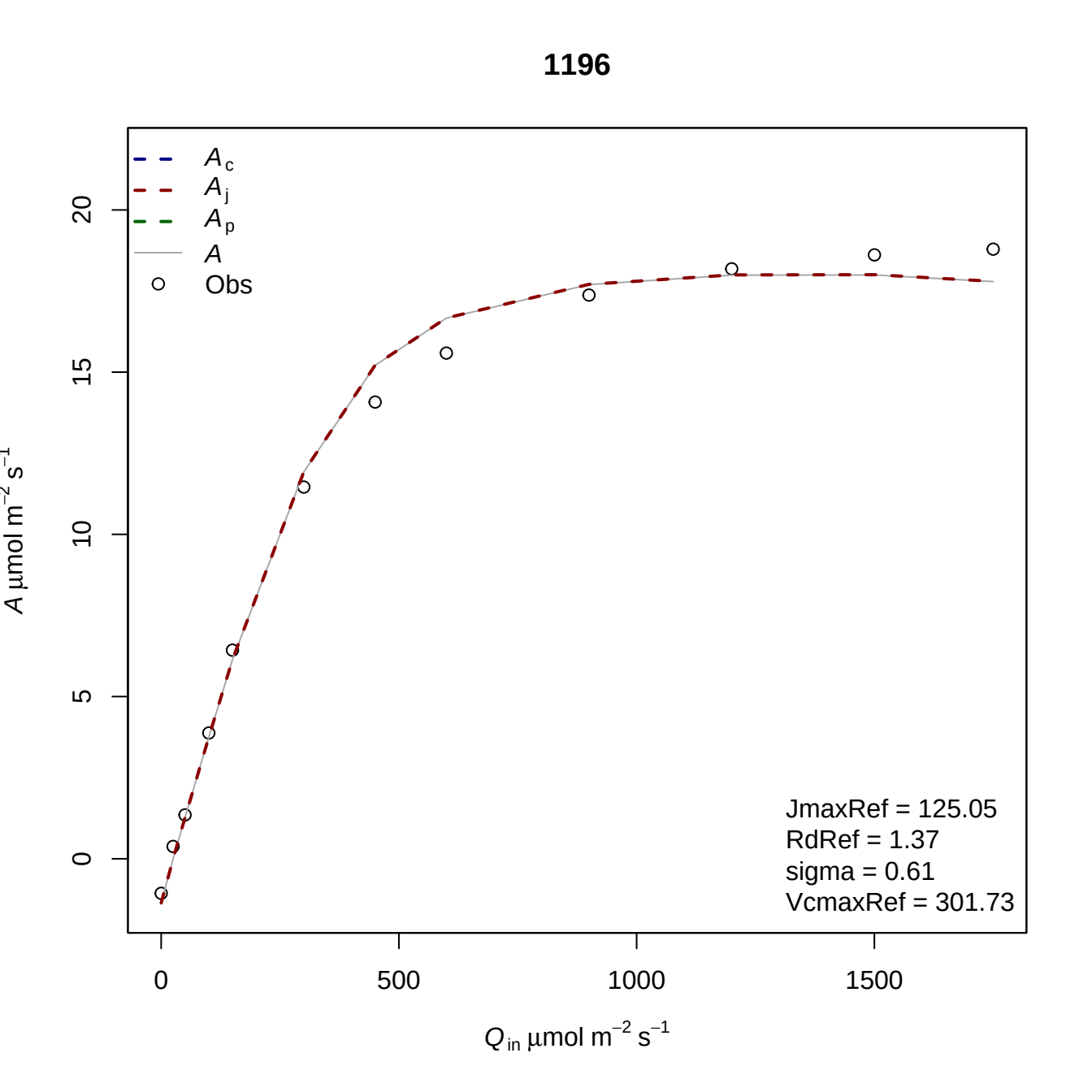

1191
A_c
A_j
A_p
A
Obs
JmaxRef = 77.11
RdRef = 3.03
sigma = 1.61
VcmaxRef = 90.46
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

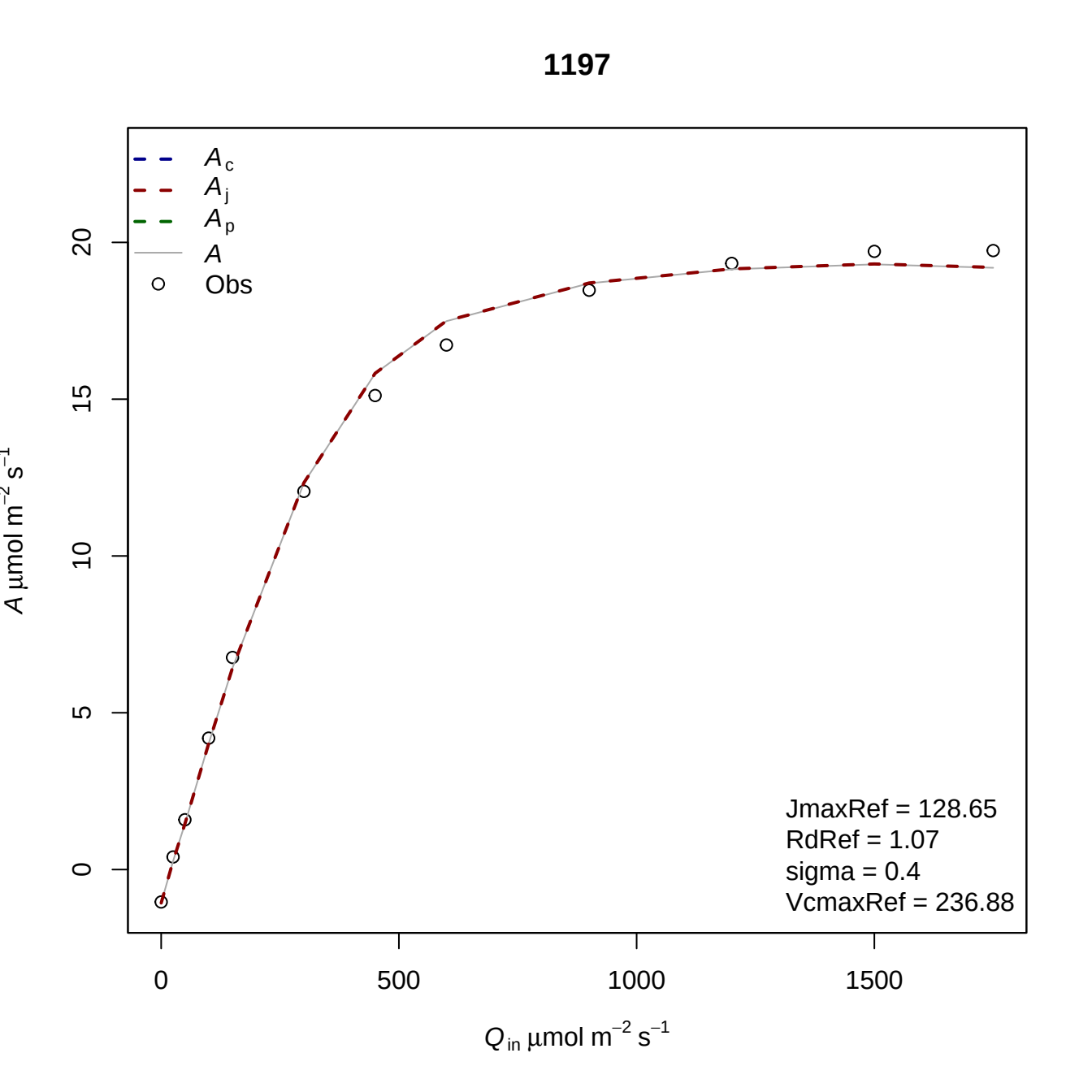

1192
A_c
A_j
A_p
A
Obs
JmaxRef = 103.73
RdRef = 2.74
sigma = 1.55
VcmaxRef = 152.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

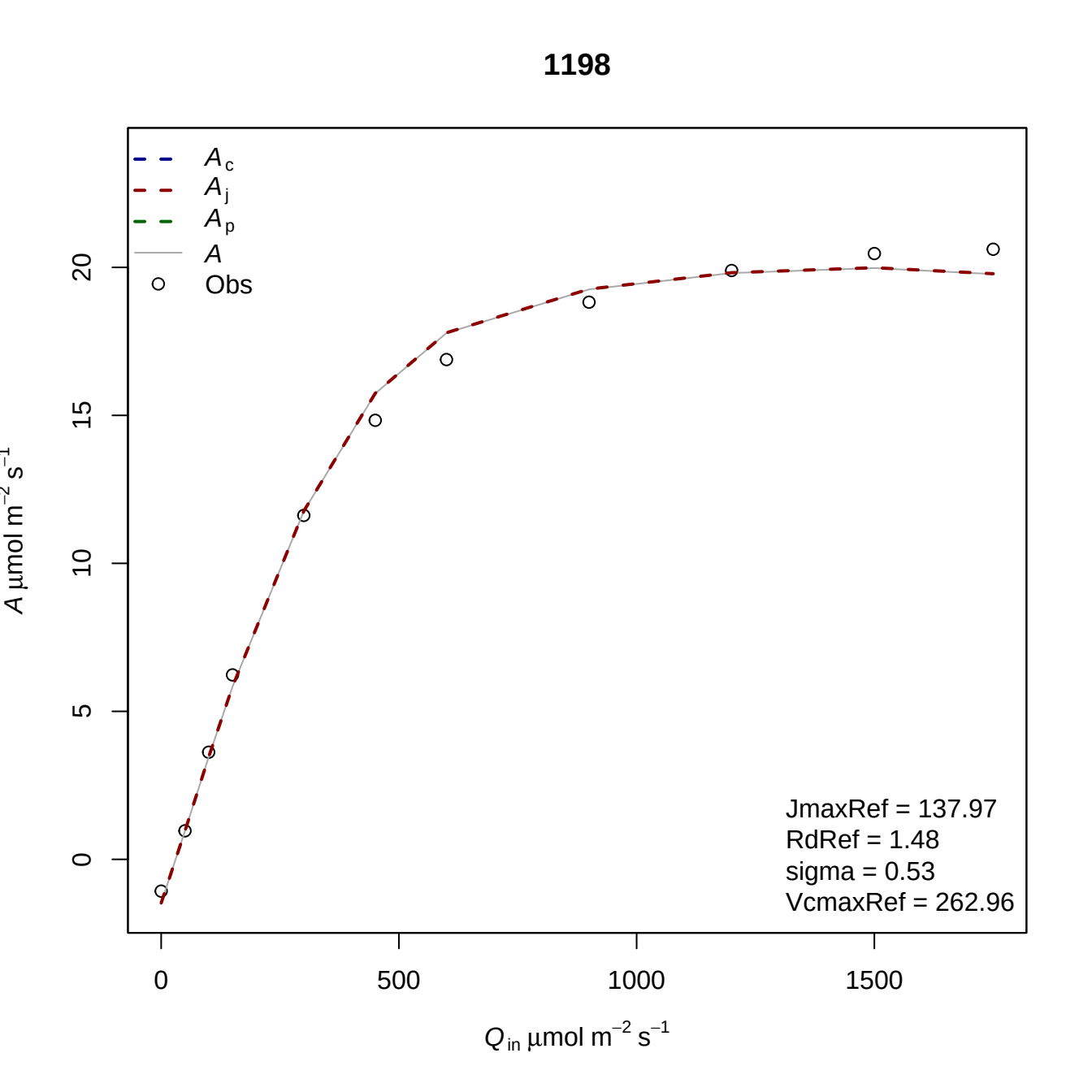

1193
A_c
A_j
A_p
A
Obs
JmaxRef = 77.63
RdRef = 3.26
sigma = 1.85
VcmaxRef = 101.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

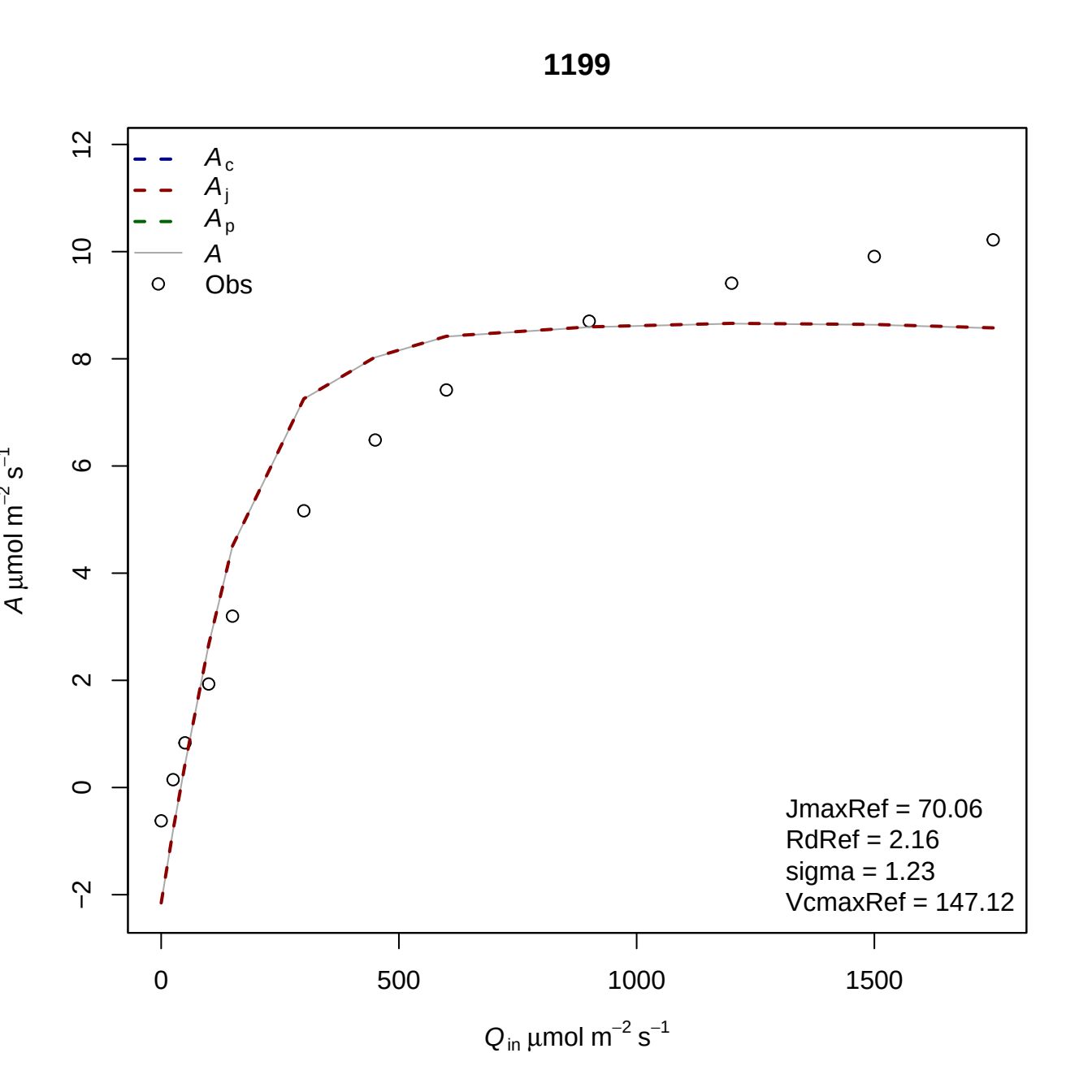

1194
A_c
A_j
A_p
A
Obs
JmaxRef = 116
RdRef = 0.97
sigma = 0.54
VcmaxRef = 226.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

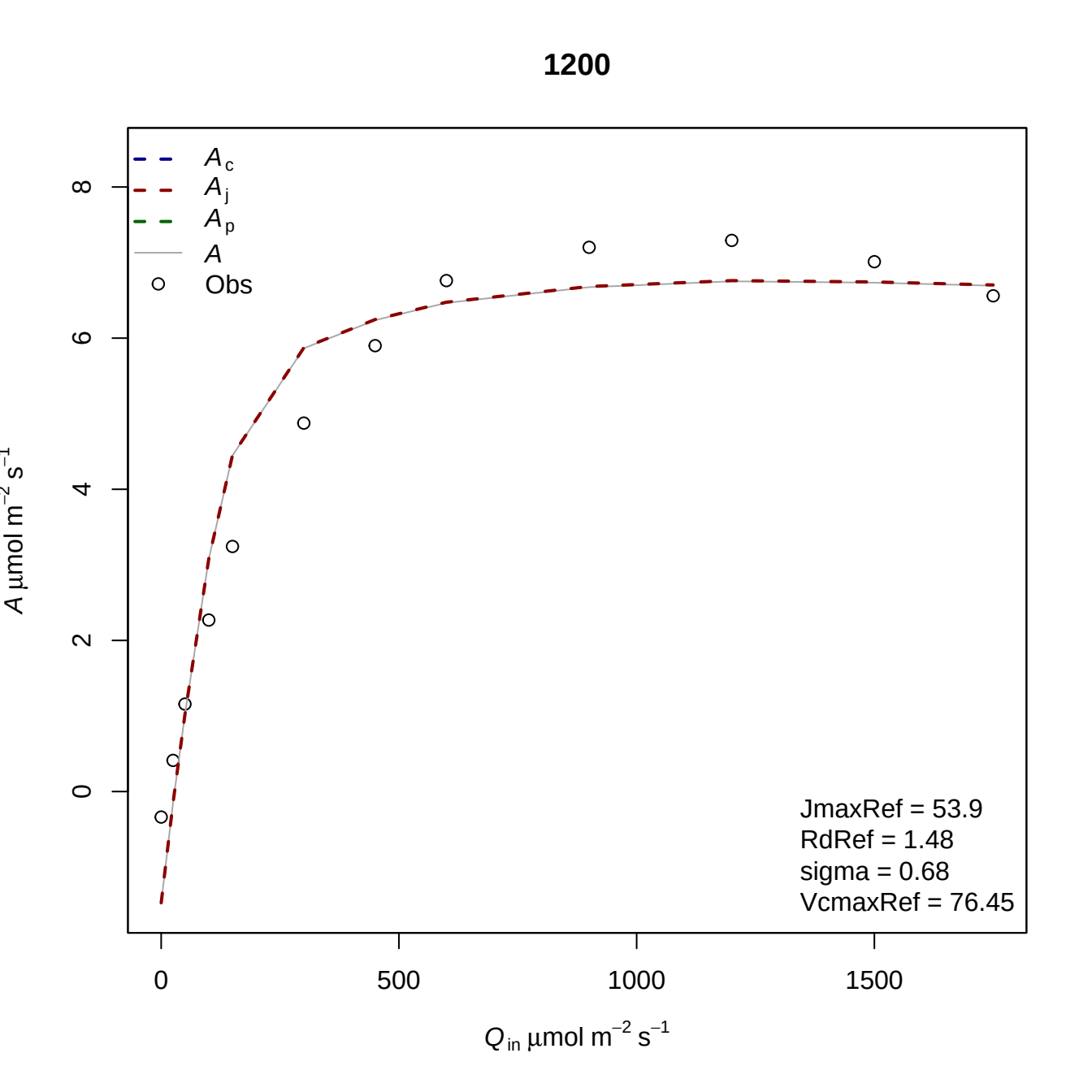

1195
A_c
A_j
A_p
A
Obs
JmaxRef = 92.6
RdRef = 1.04
sigma = 0.5
VcmaxRef = 189.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

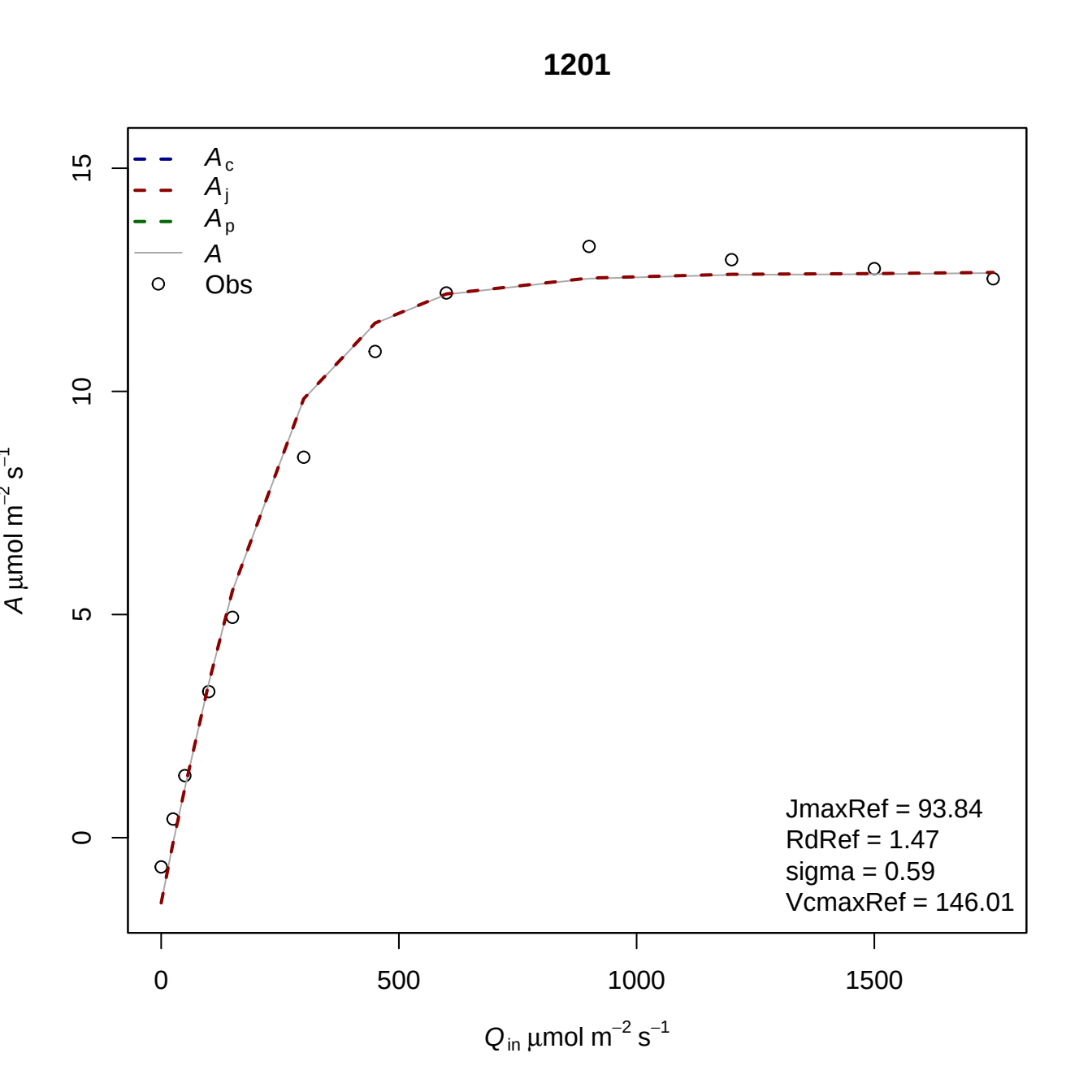

1196
A_c
A_j
A_p
A
Obs
JmaxRef = 125.05
RdRef = 1.37
sigma = 0.61
VcmaxRef = 301.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

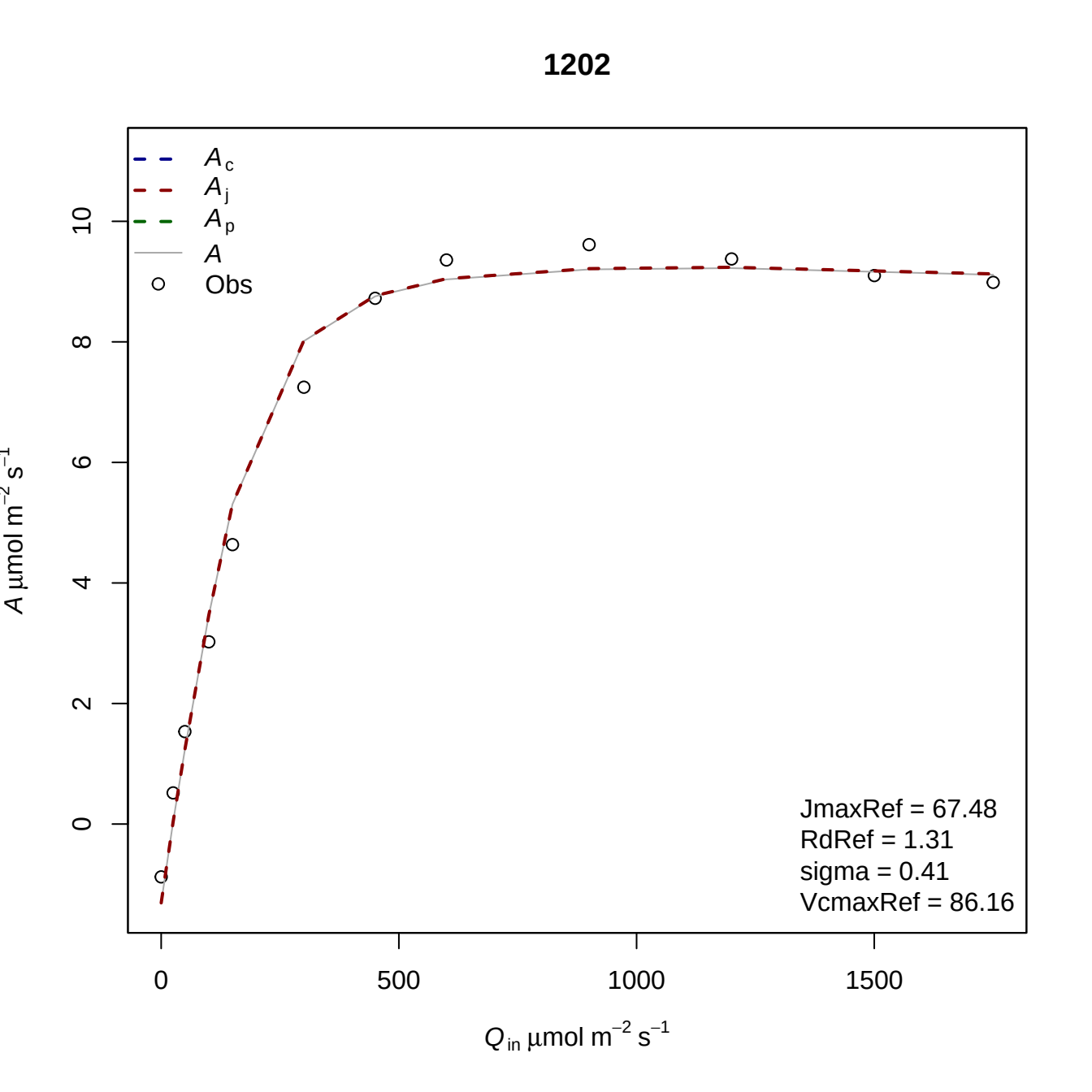

1197
A_c
A_j
A_p
A
Obs
JmaxRef = 128.65
RdRef = 1.07
sigma = 0.4
VcmaxRef = 236.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

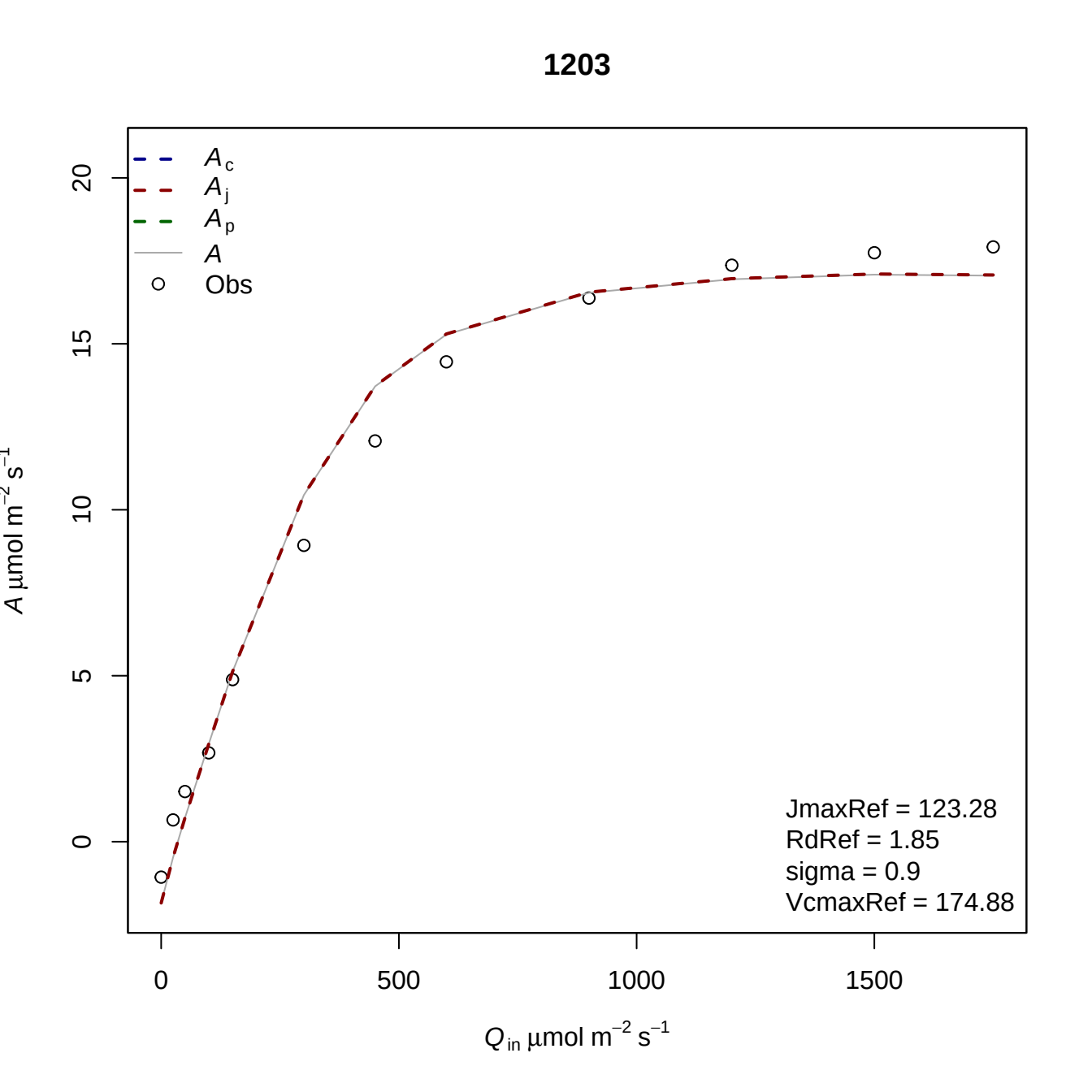

1198
A_c
A_j
A_p
A
Obs
JmaxRef = 137.97
RdRef = 1.48
sigma = 0.53
VcmaxRef = 262.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

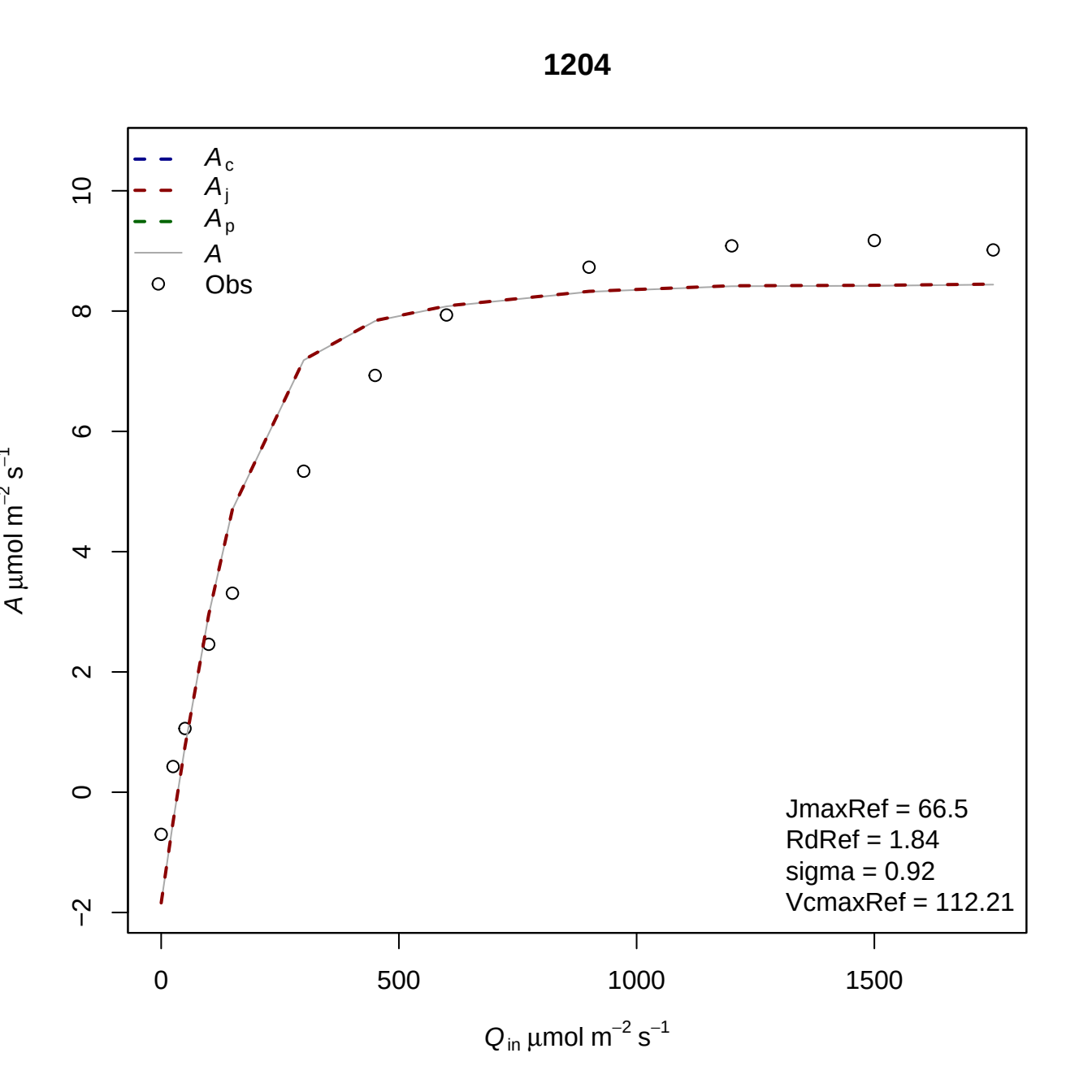

1199
A_c
A_j
A_p
A
Obs
JmaxRef = 70.06
RdRef = 2.16
sigma = 1.23
VcmaxRef = 147.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

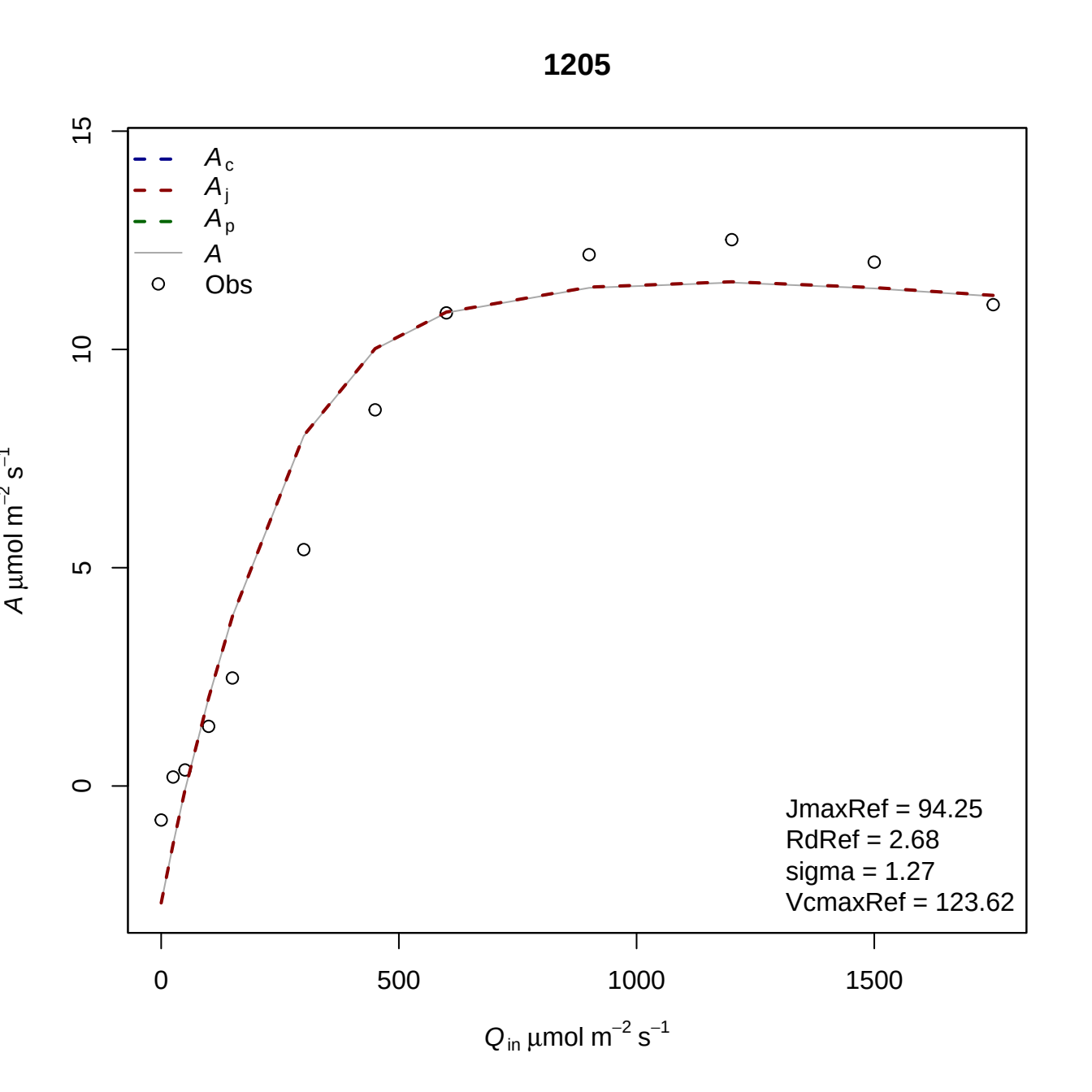

1200
A_c
A_j
A_p
A
Obs
JmaxRef = 53.9
RdRef = 1.48
sigma = 0.68
VcmaxRef = 76.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

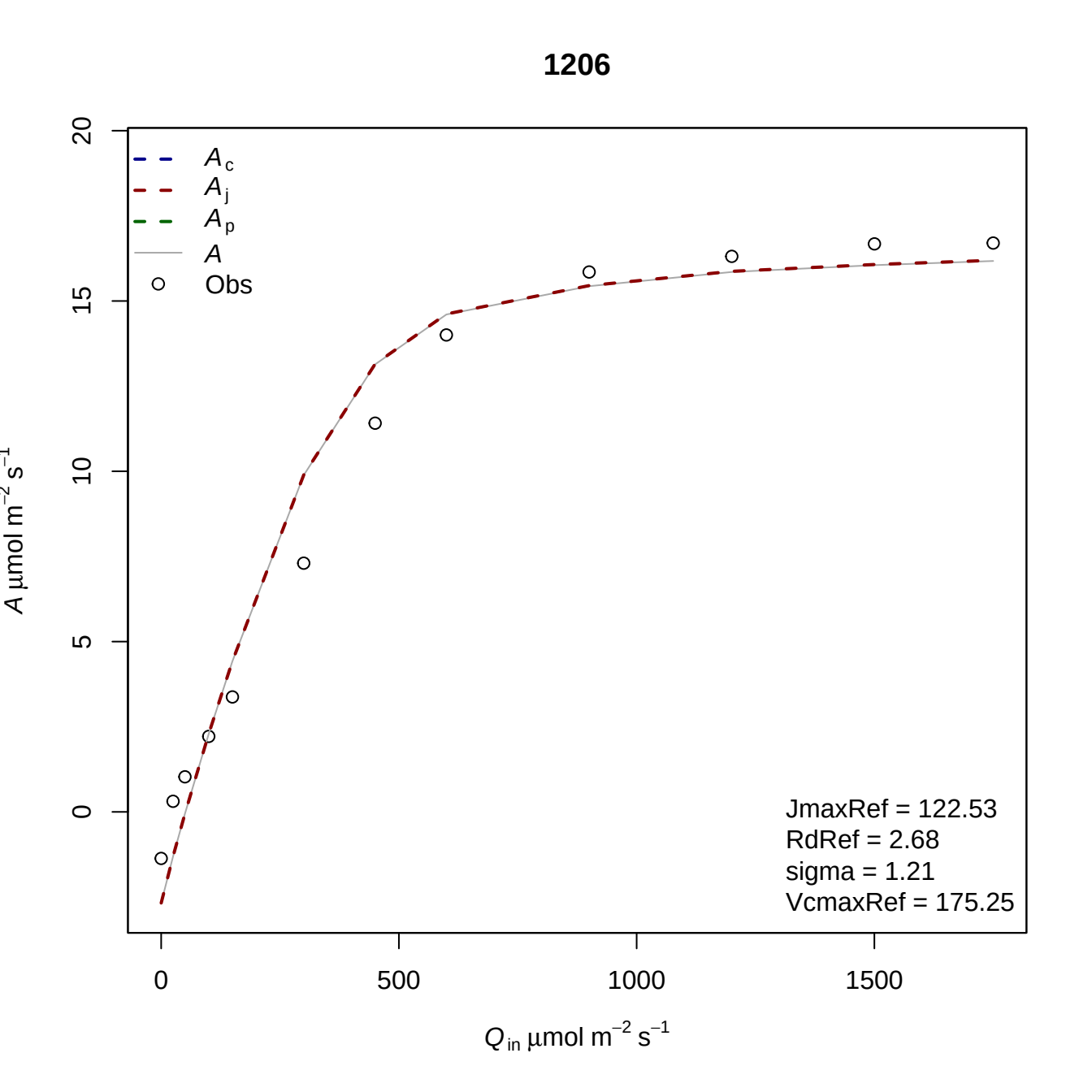

1201
A_c
A_j
A_p
A
Obs
JmaxRef = 93.84
RdRef = 1.47
sigma = 0.59
VcmaxRef = 146.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

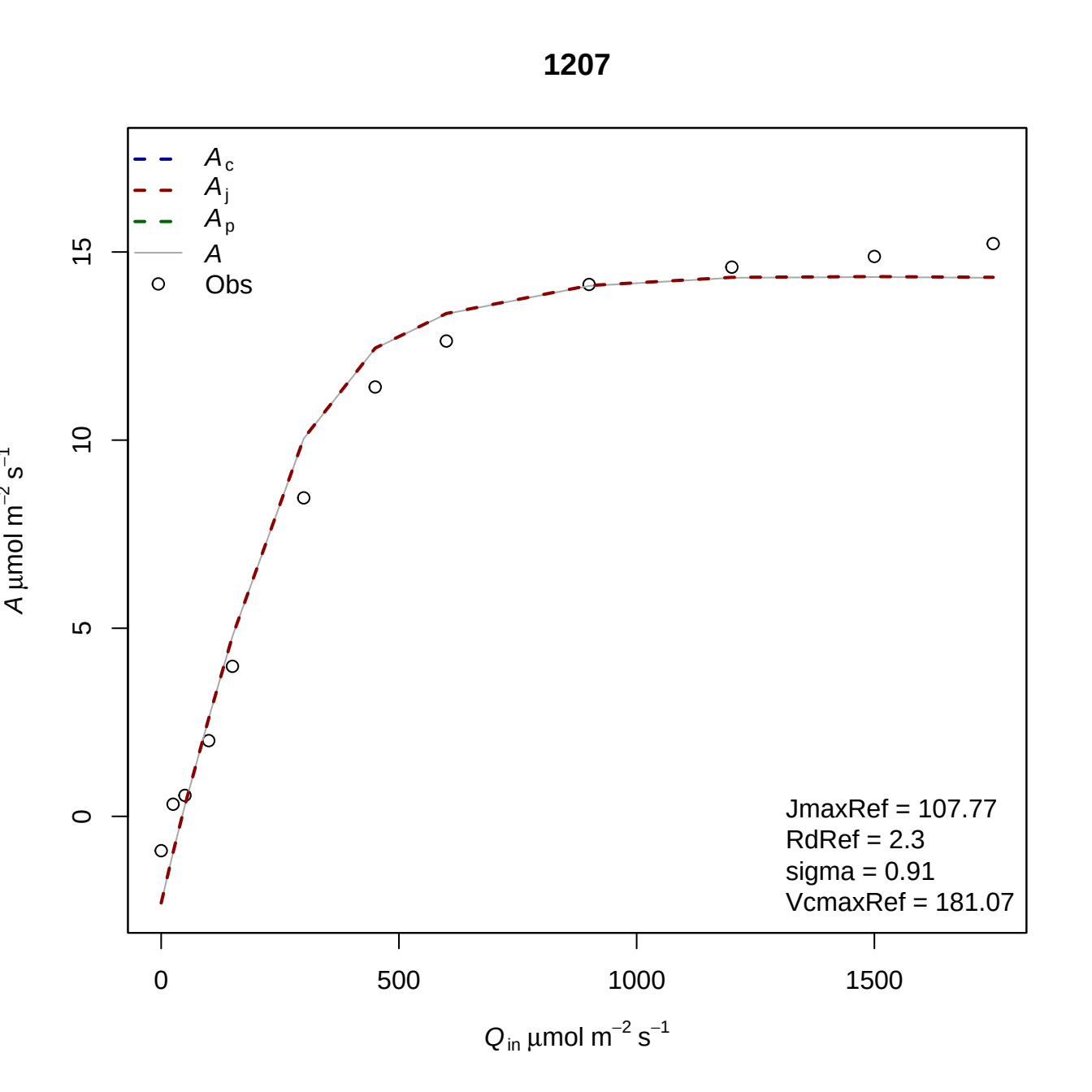

1202
A_c
A_j
A_p
A
Obs
JmaxRef = 67.48
RdRef = 1.31
sigma = 0.41
VcmaxRef = 86.16
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

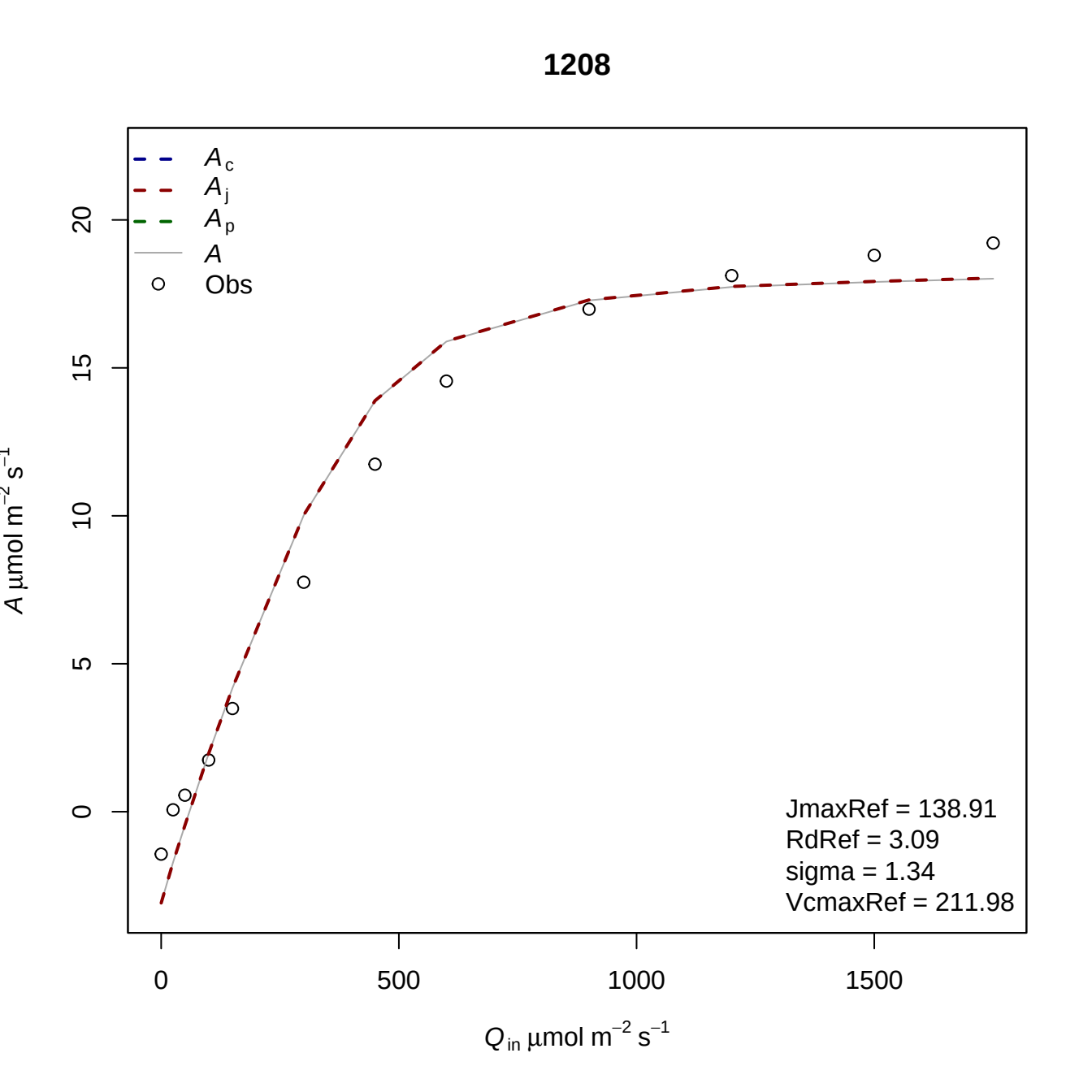

1203
A_c
A_j
A_p
A
Obs
JmaxRef = 123.28
RdRef = 1.85
sigma = 0.9
VcmaxRef = 174.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

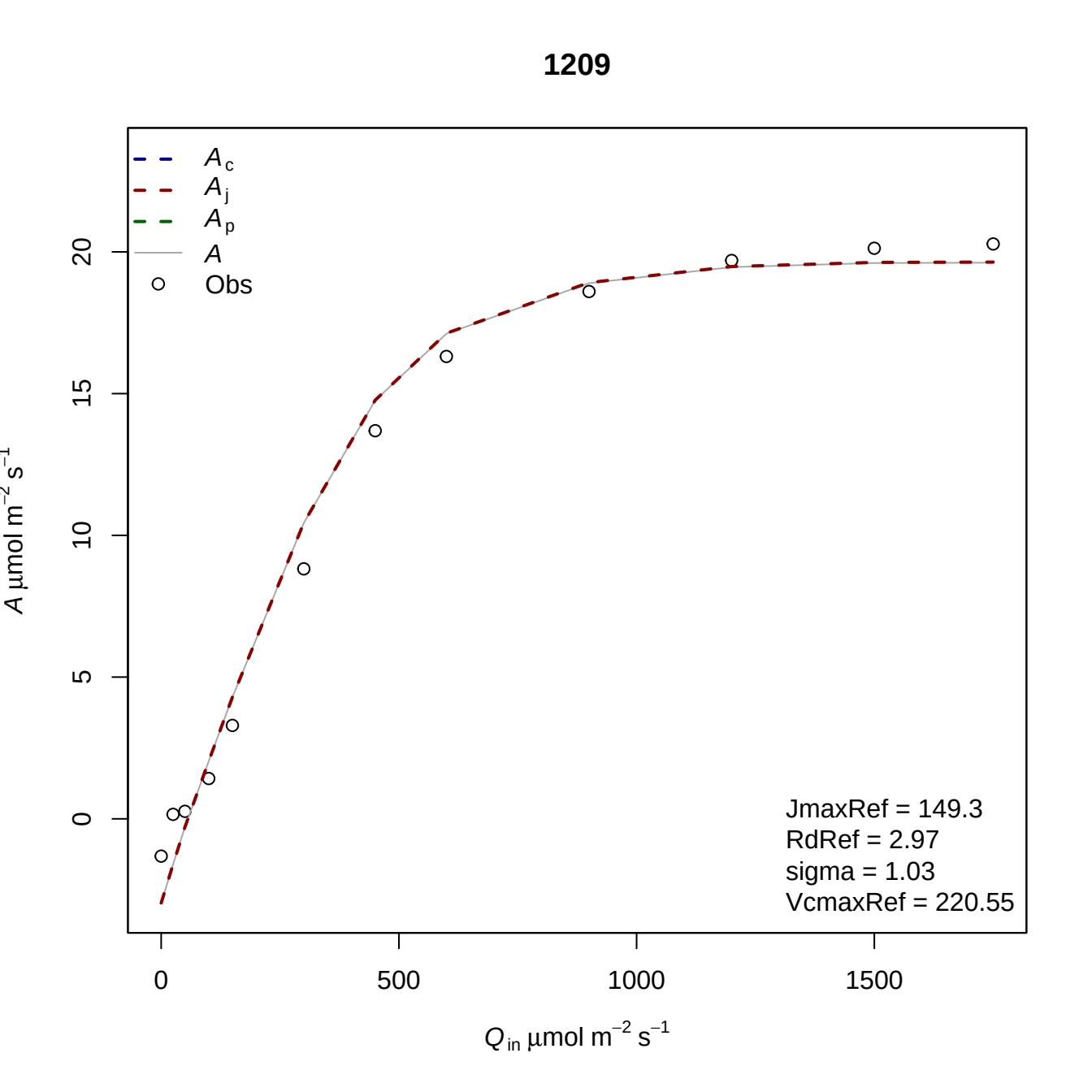

1204
A_c
A_j
A_p
A
Obs
JmaxRef = 66.5
RdRef = 1.84
sigma = 0.92
VcmaxRef = 112.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

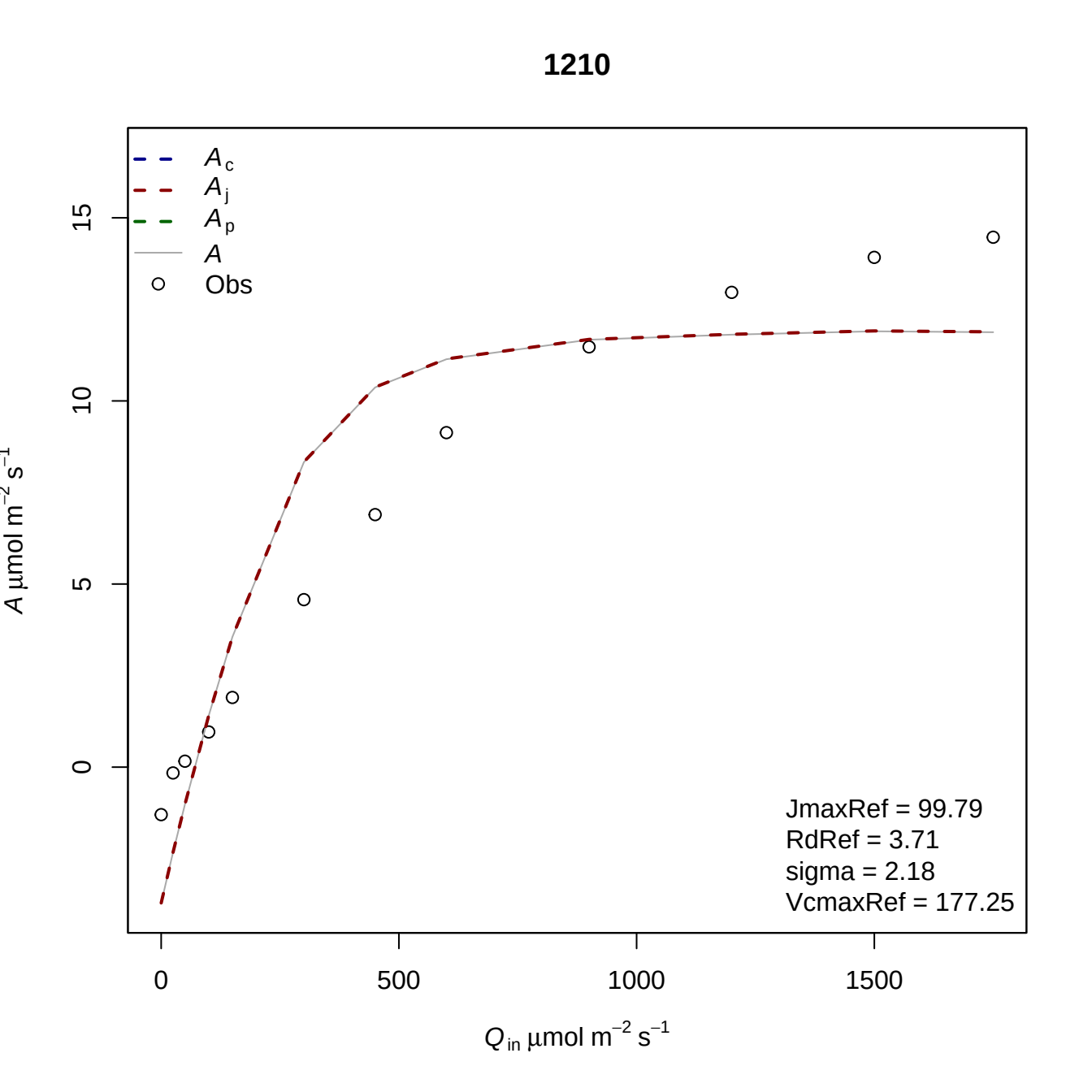

1205
A_c
A_j
A_p
A
Obs
JmaxRef = 94.25
RdRef = 2.68
sigma = 1.27
VcmaxRef = 123.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

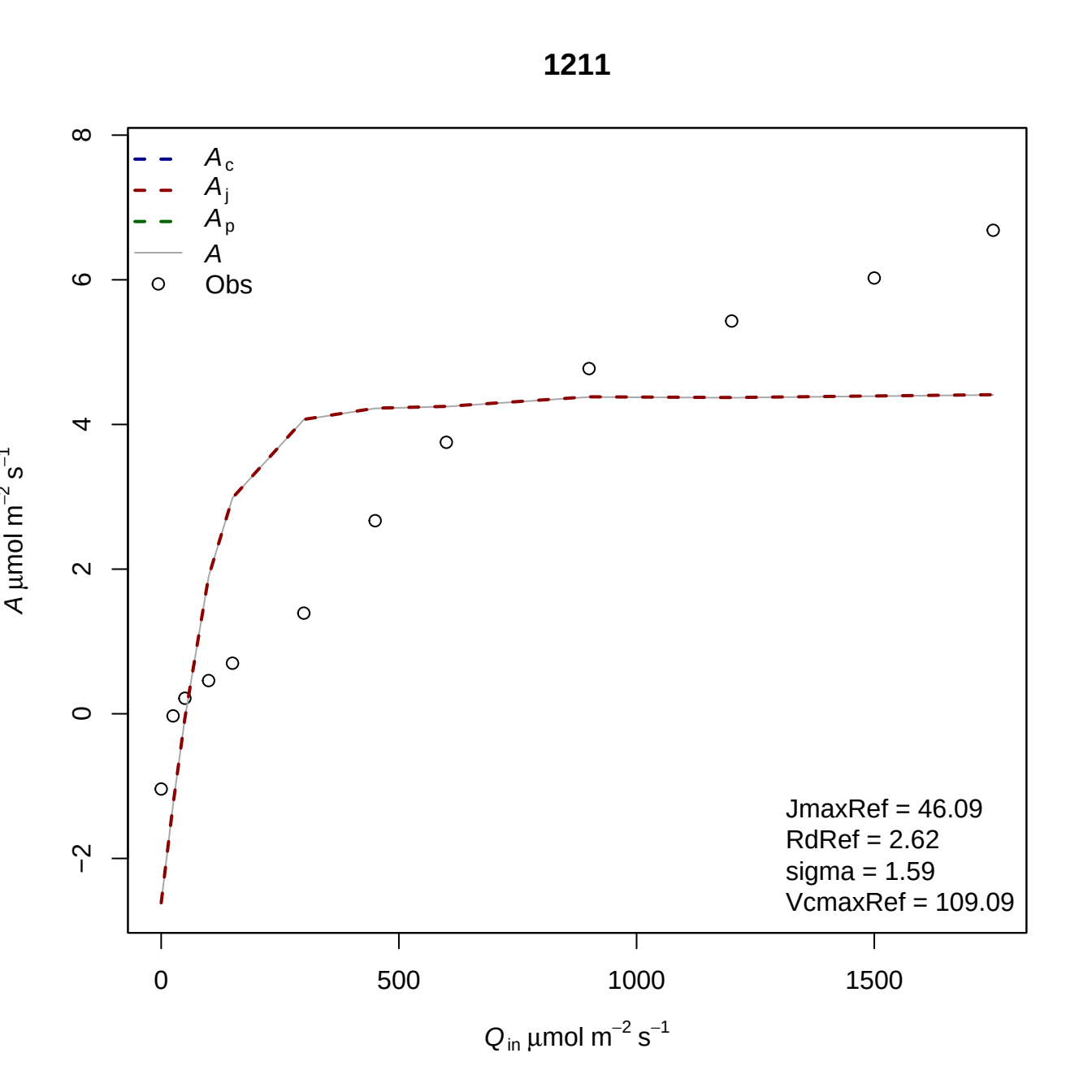

1206
A_c
A_j
A_p
A
Obs
JmaxRef = 122.53
RdRef = 2.68
sigma = 1.21
VcmaxRef = 175.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

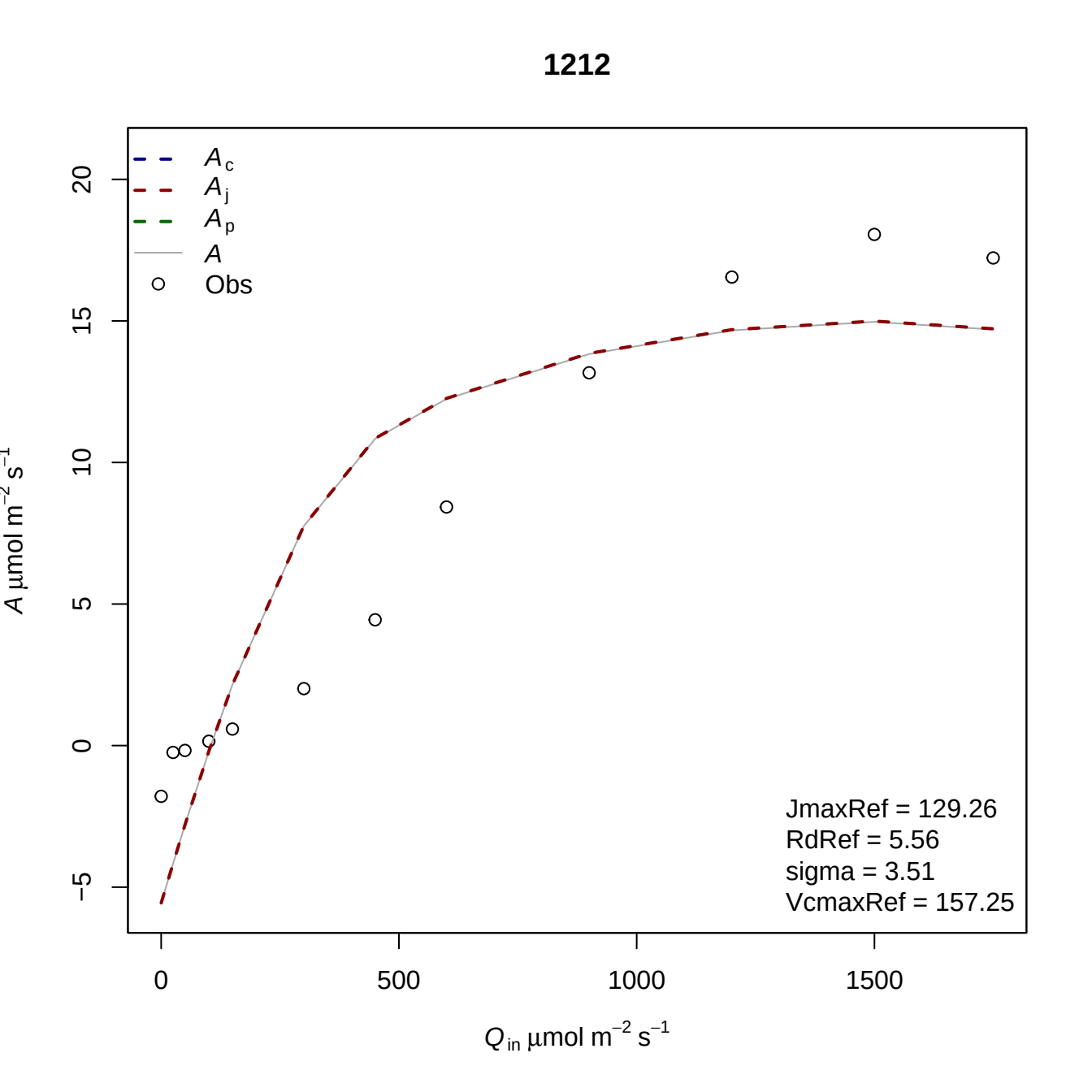

1207
A_c
A_j
A_p
A
Obs
JmaxRef = 107.77
RdRef = 2.3
sigma = 0.91
VcmaxRef = 181.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

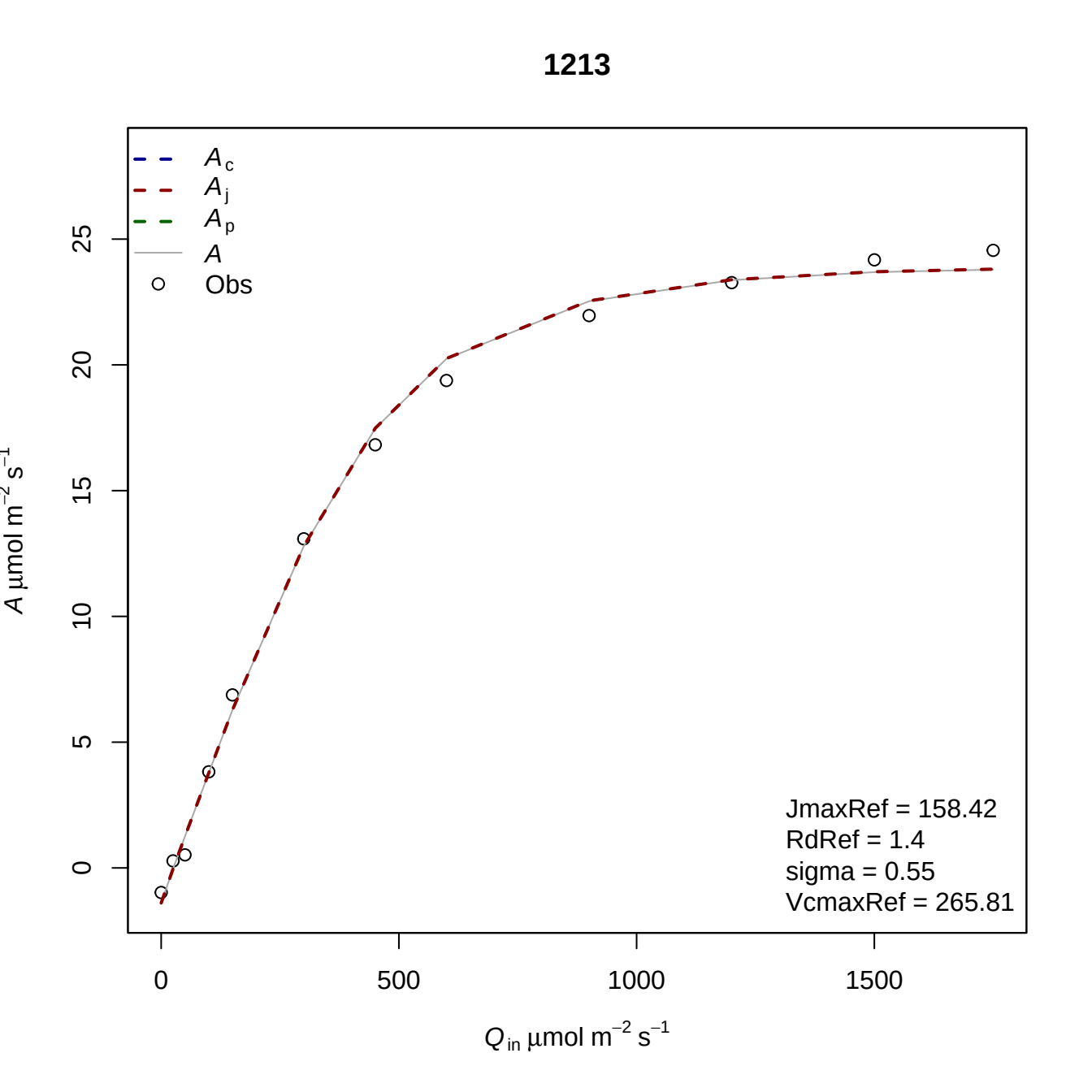

1208
A_c
A_j
A_p
A
Obs
JmaxRef = 138.91
RdRef = 3.09
sigma = 1.34
VcmaxRef = 211.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

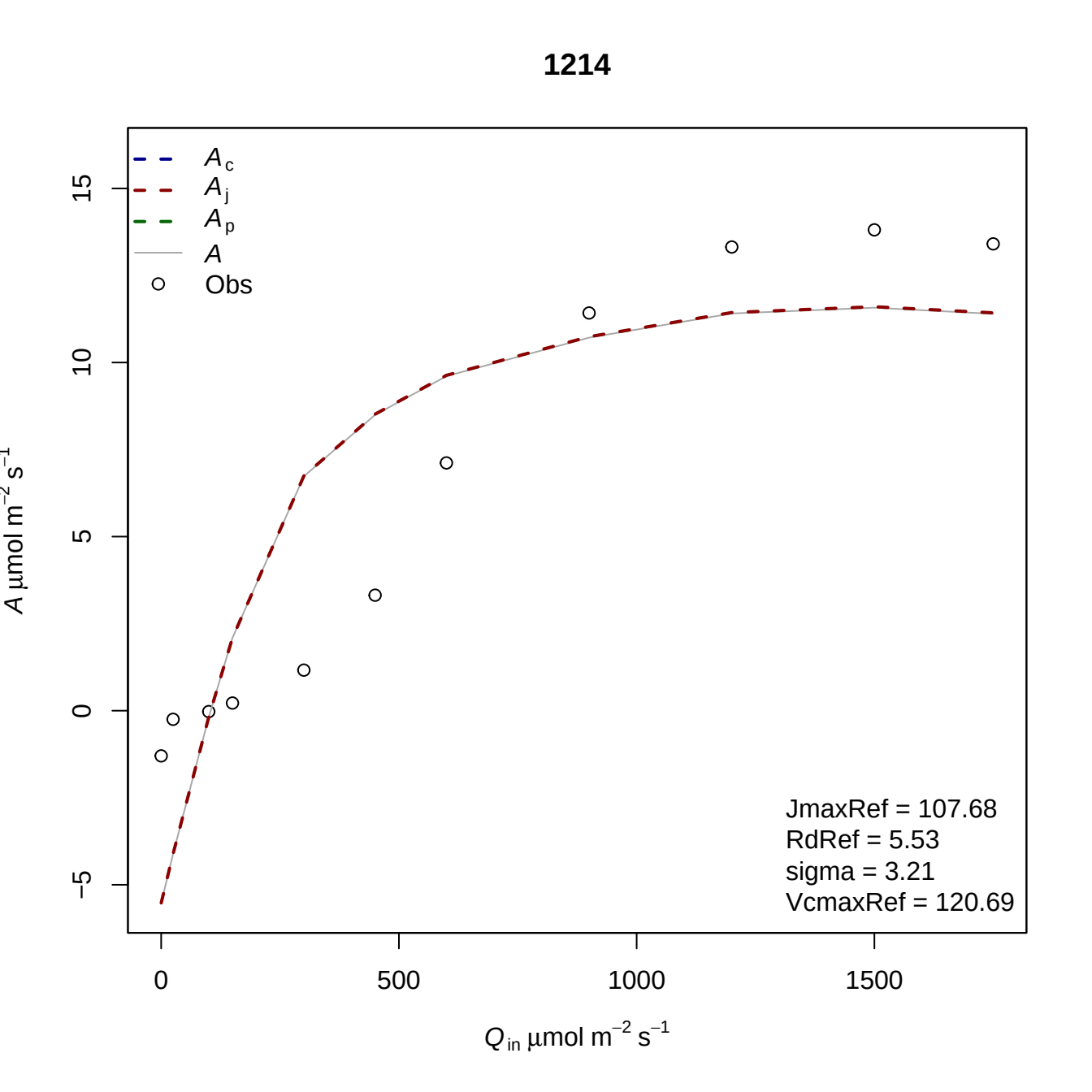

1209
A_c
A_j
A_p
A
Obs
JmaxRef = 149.3
RdRef = 2.97
sigma = 1.03
VcmaxRef = 220.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

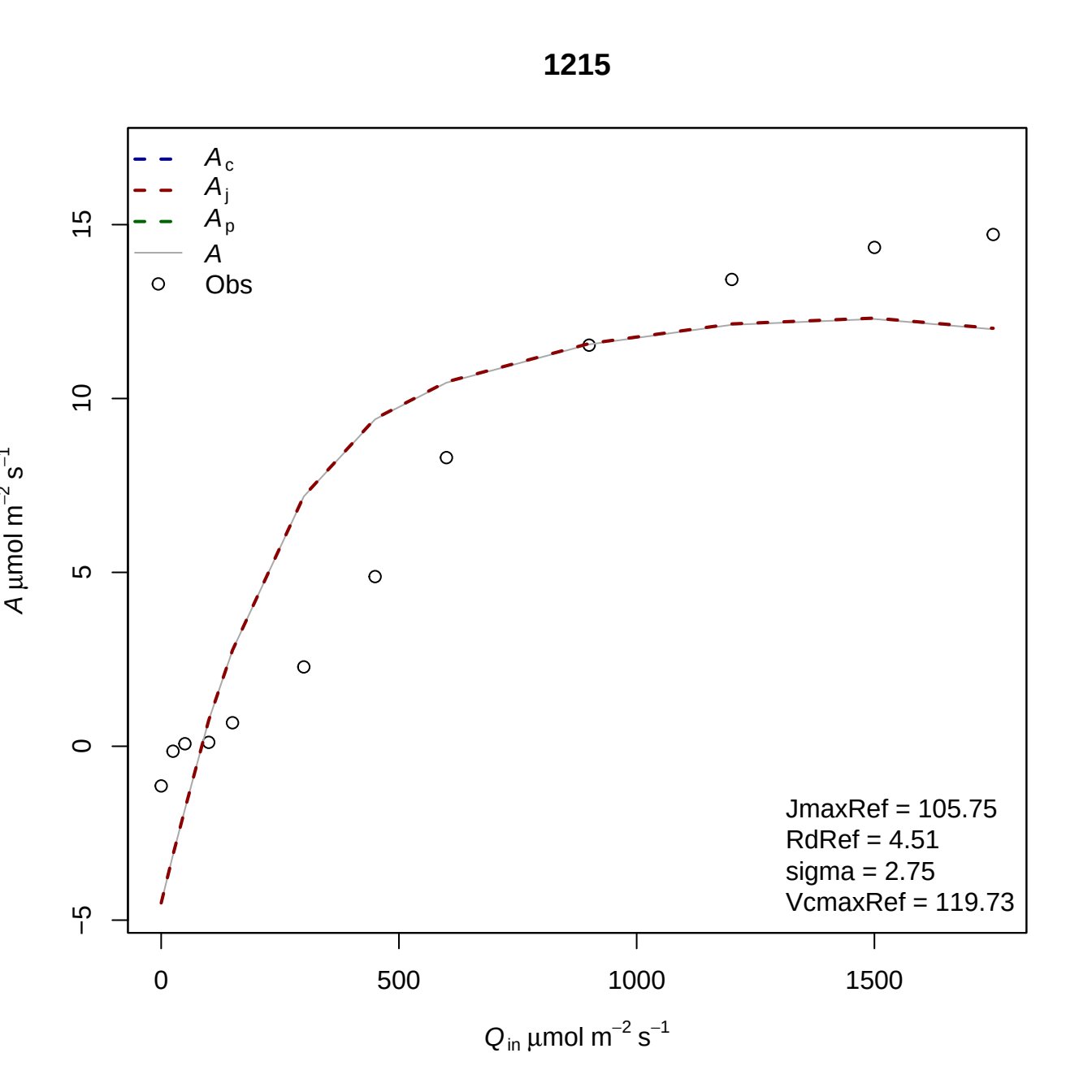

1210
A_c
A_j
A_p
A
Obs
JmaxRef = 99.79
RdRef = 3.71
sigma = 2.18
VcmaxRef = 177.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

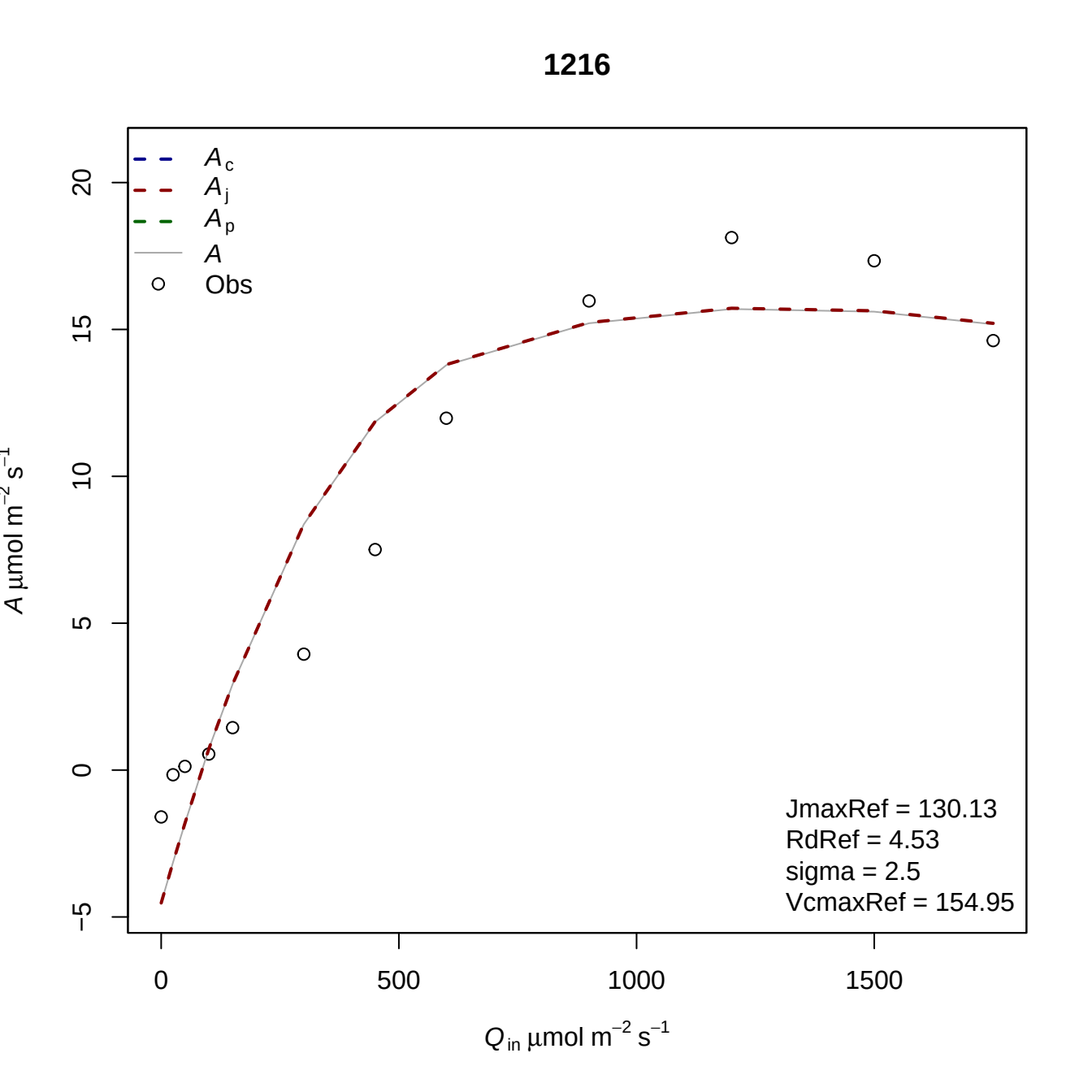

1211
A_c
A_j
A_p
A
Obs
JmaxRef = 46.09
RdRef = 2.62
sigma = 1.59
VcmaxRef = 109.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

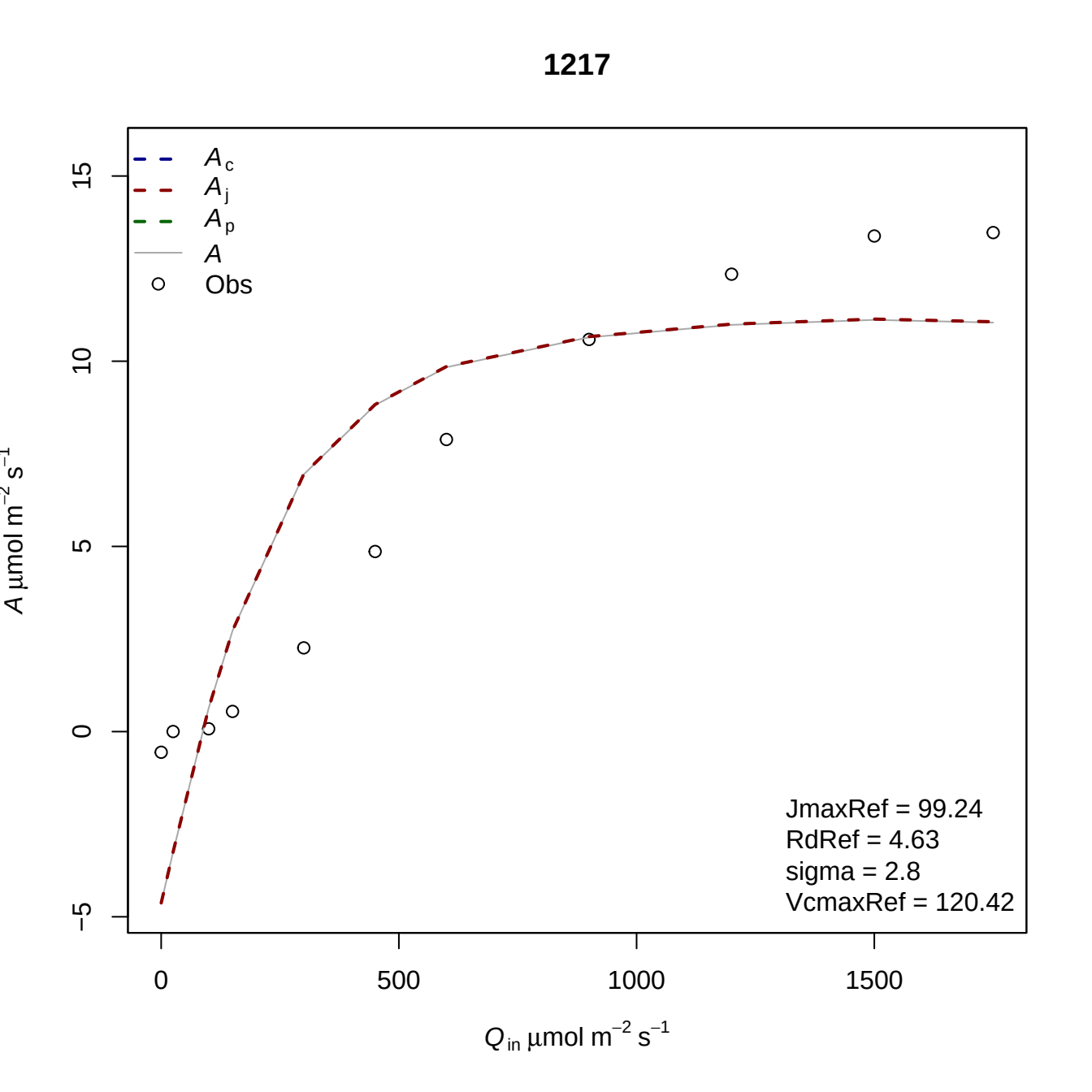

1212
A_c
A_j
A_p
A
Obs
JmaxRef = 129.26
RdRef = 5.56
sigma = 3.51
VcmaxRef = 157.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

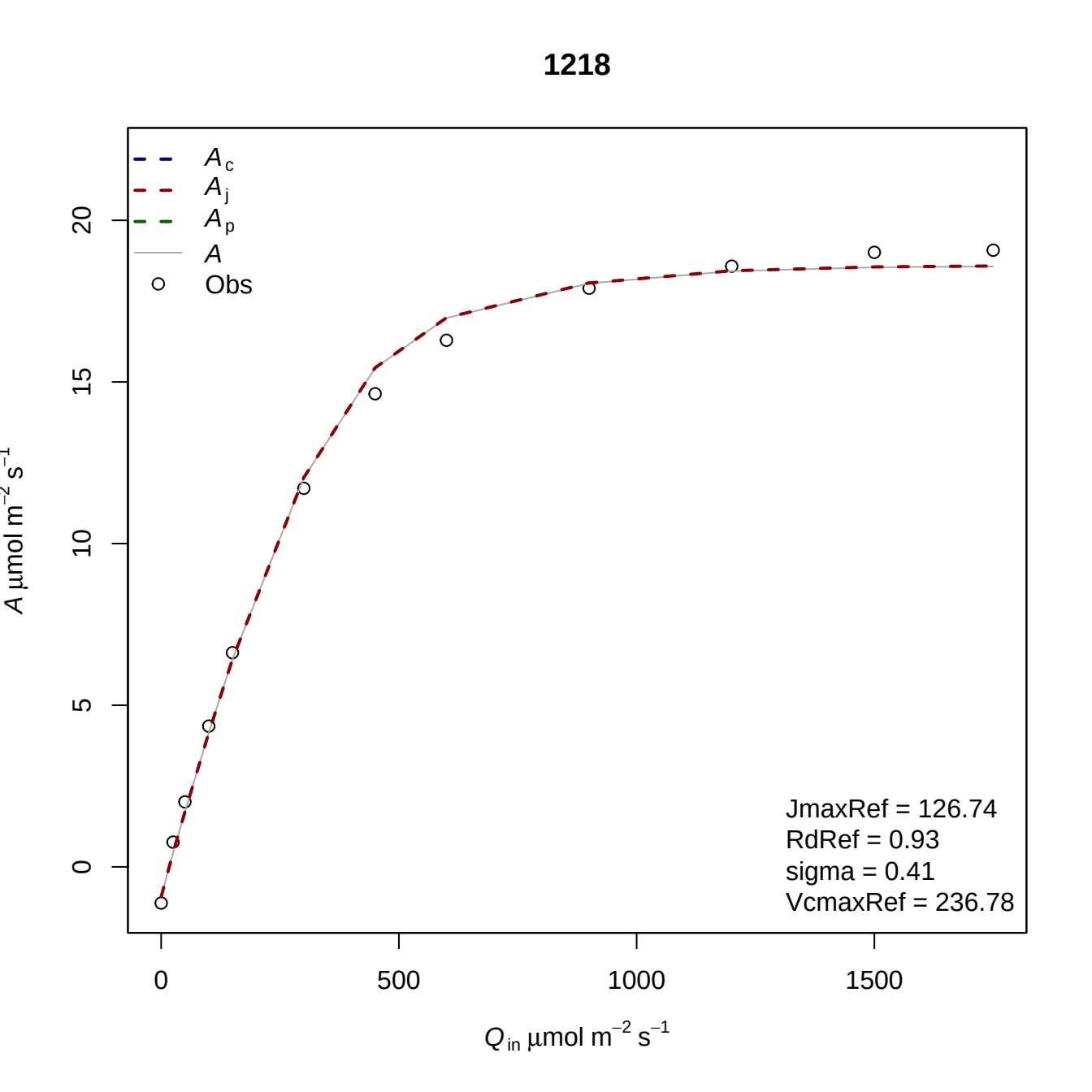

1213
A_c
A_j
A_p
A
Obs
JmaxRef = 158.42
RdRef = 1.4
sigma = 0.55
VcmaxRef = 265.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

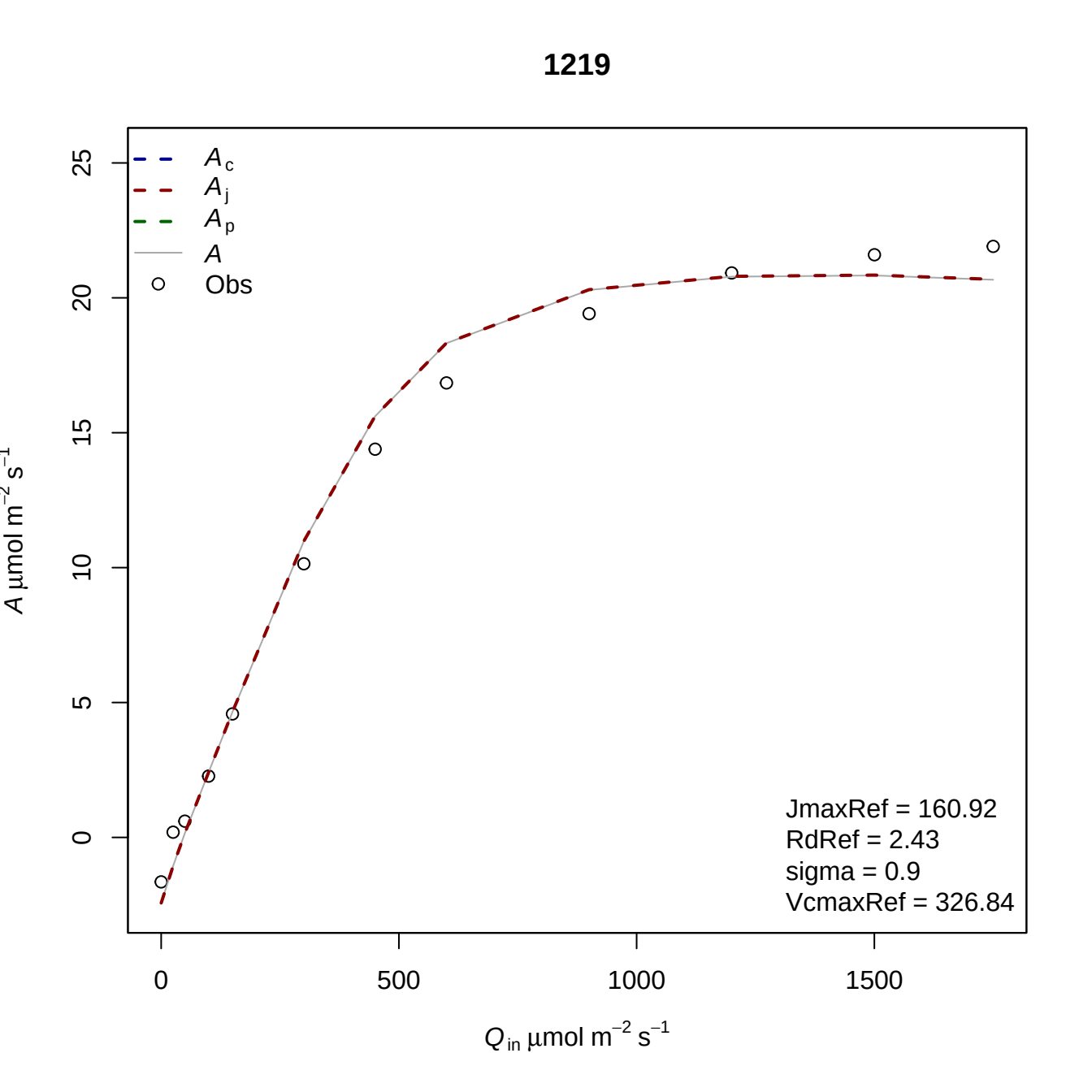

1214
A_c
A_j
A_p
A
Obs
JmaxRef = 107.68
RdRef = 5.53
sigma = 3.21
VcmaxRef = 120.69
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

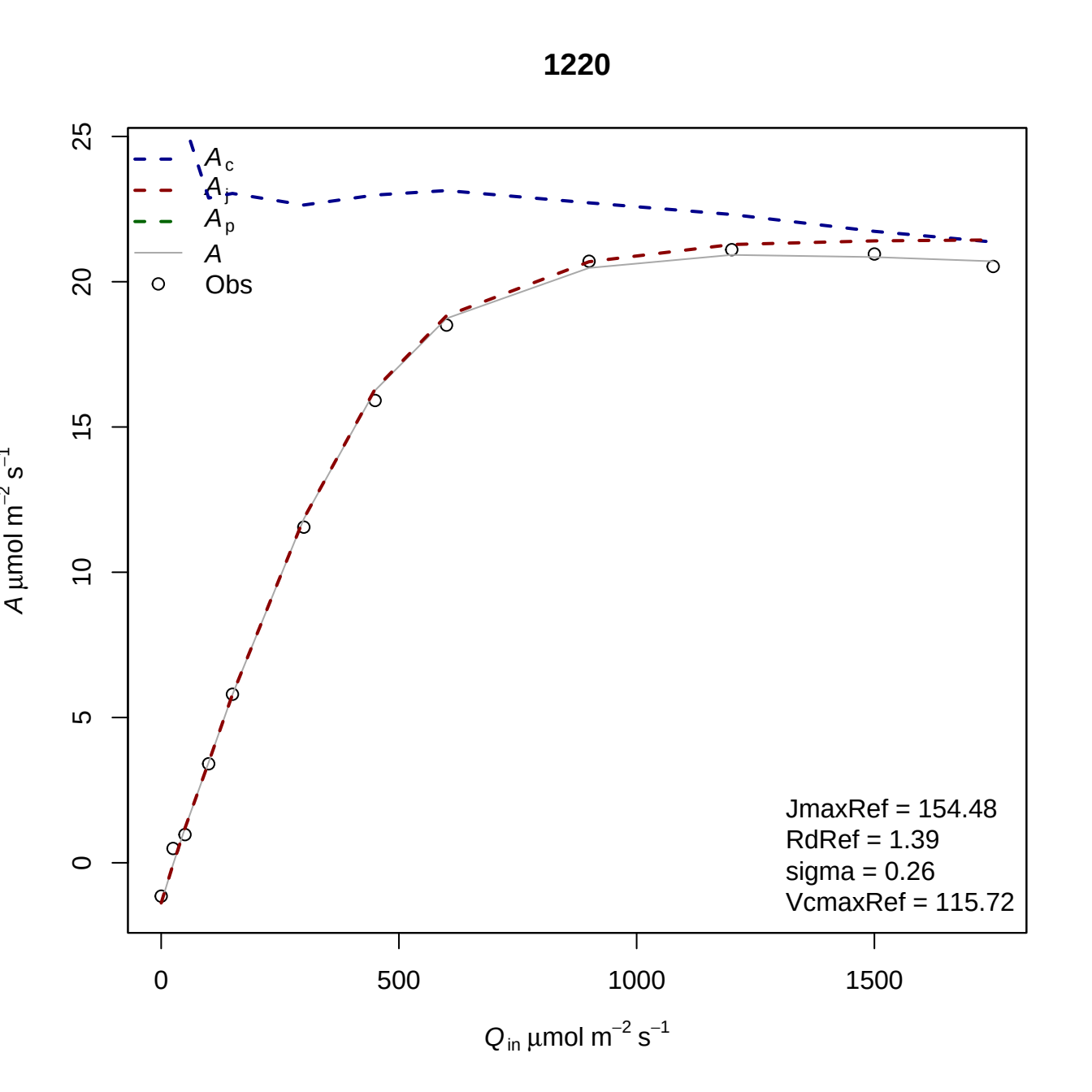

1215
A_c
A_j
A_p
A
Obs
JmaxRef = 105.75
RdRef = 4.51
sigma = 2.75
VcmaxRef = 119.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500

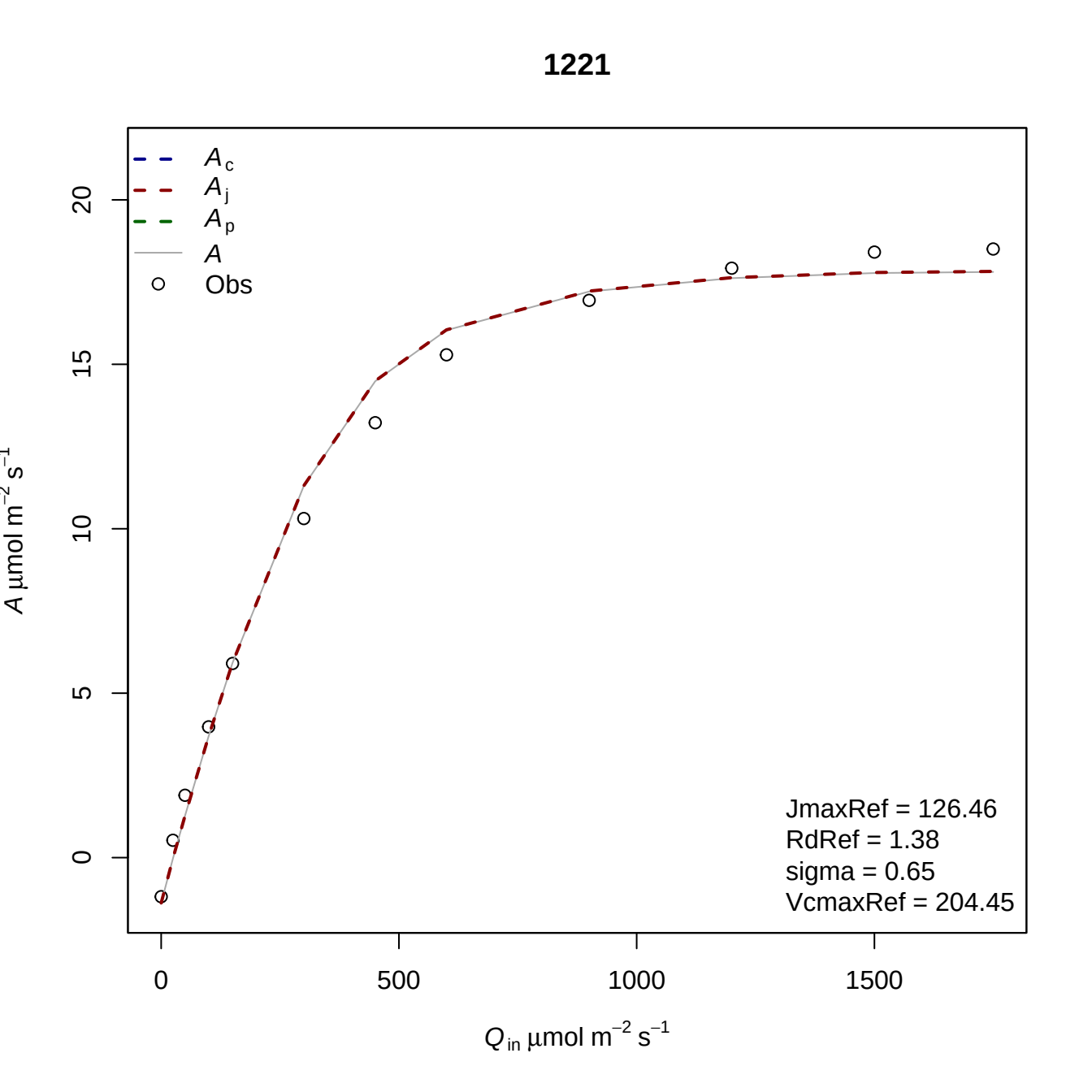

1216
A_c
A_j
A_p
A
Obs
JmaxRef = 130.13
RdRef = 4.53
sigma = 2.5
VcmaxRef = 154.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

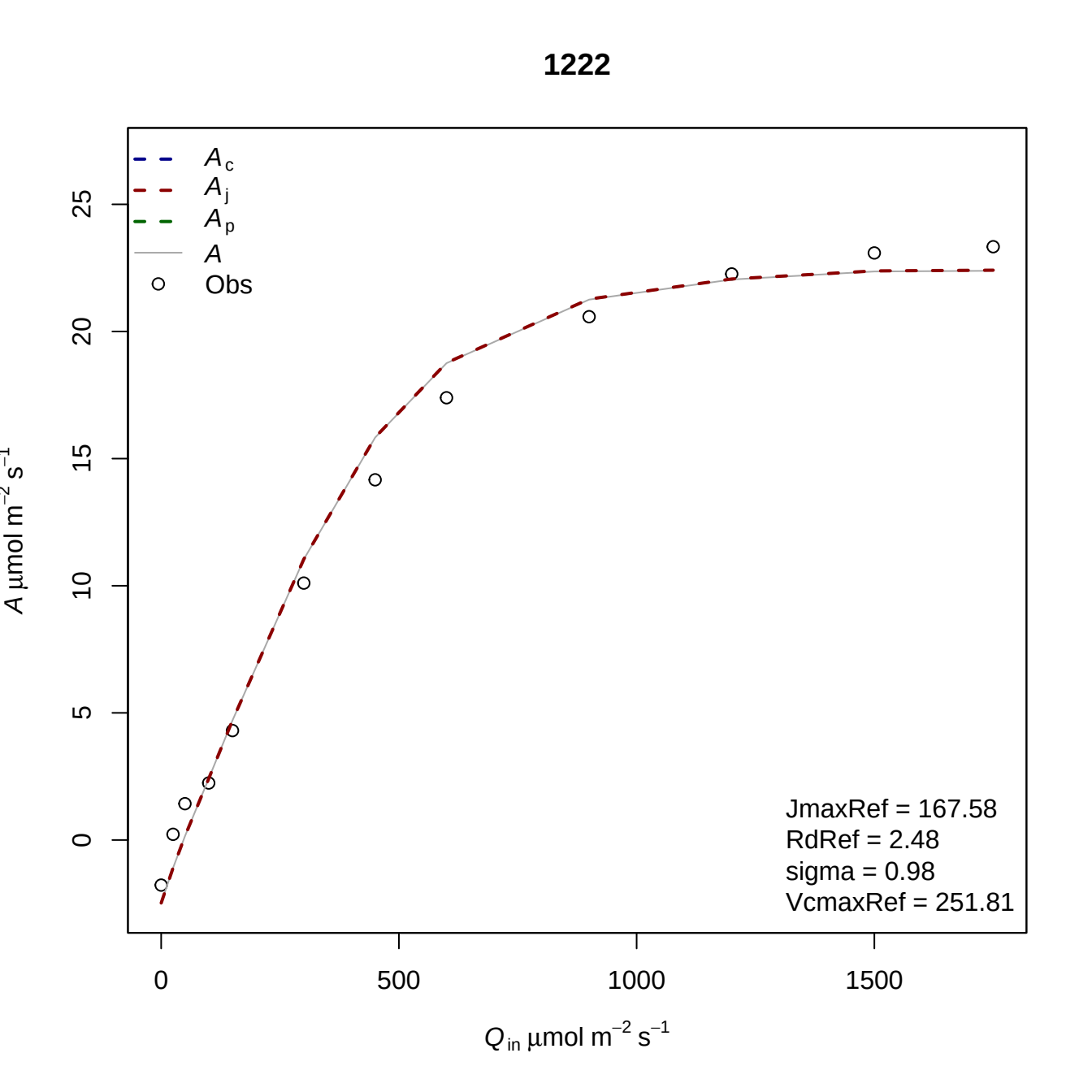

1217
A_c
A_j
A_p
A
Obs
JmaxRef = 99.24
RdRef = 4.63
sigma = 2.8
VcmaxRef = 120.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500

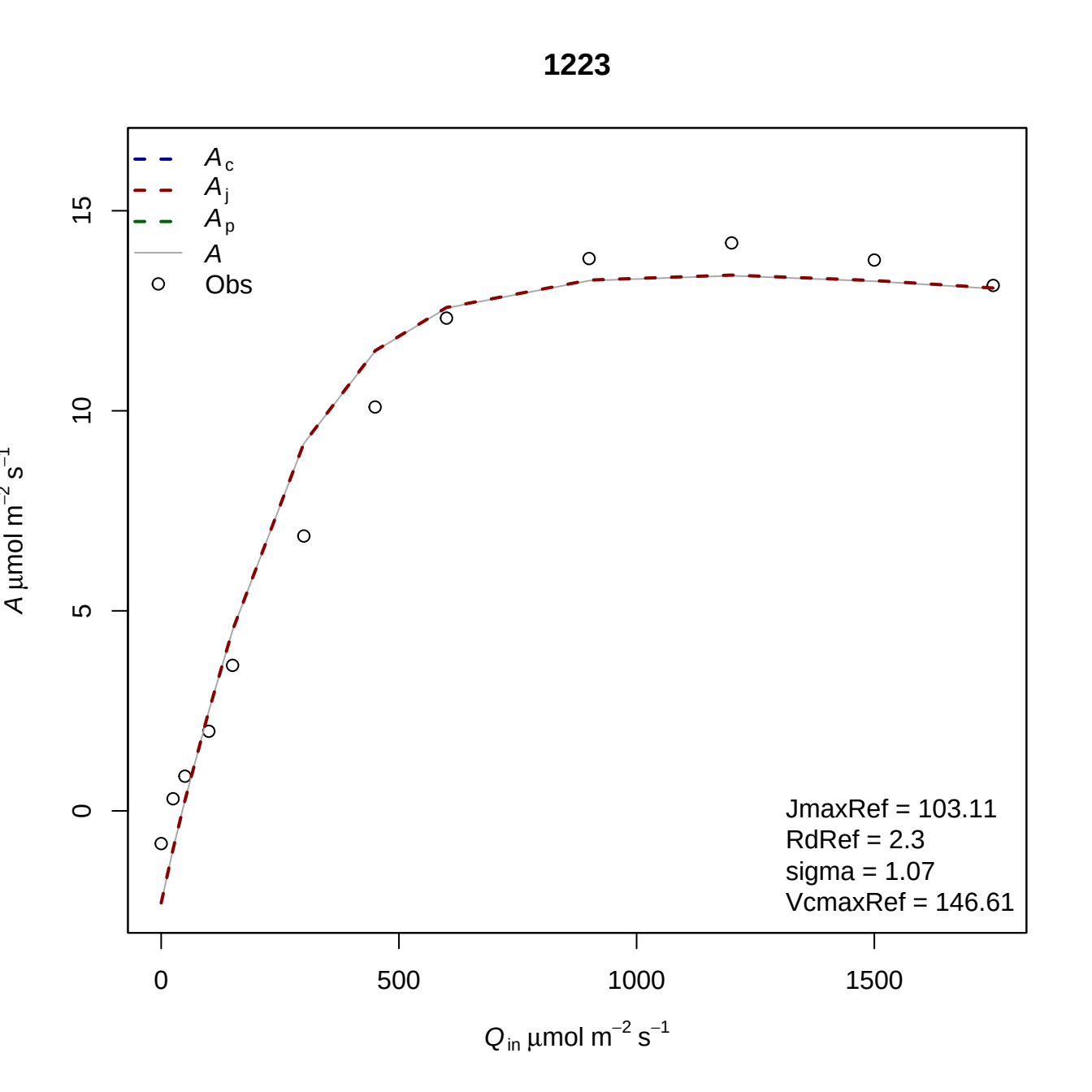

1218
A_c
A_j
A_p
A
Obs
JmaxRef = 126.74
RdRef = 0.93
sigma = 0.41
VcmaxRef = 236.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

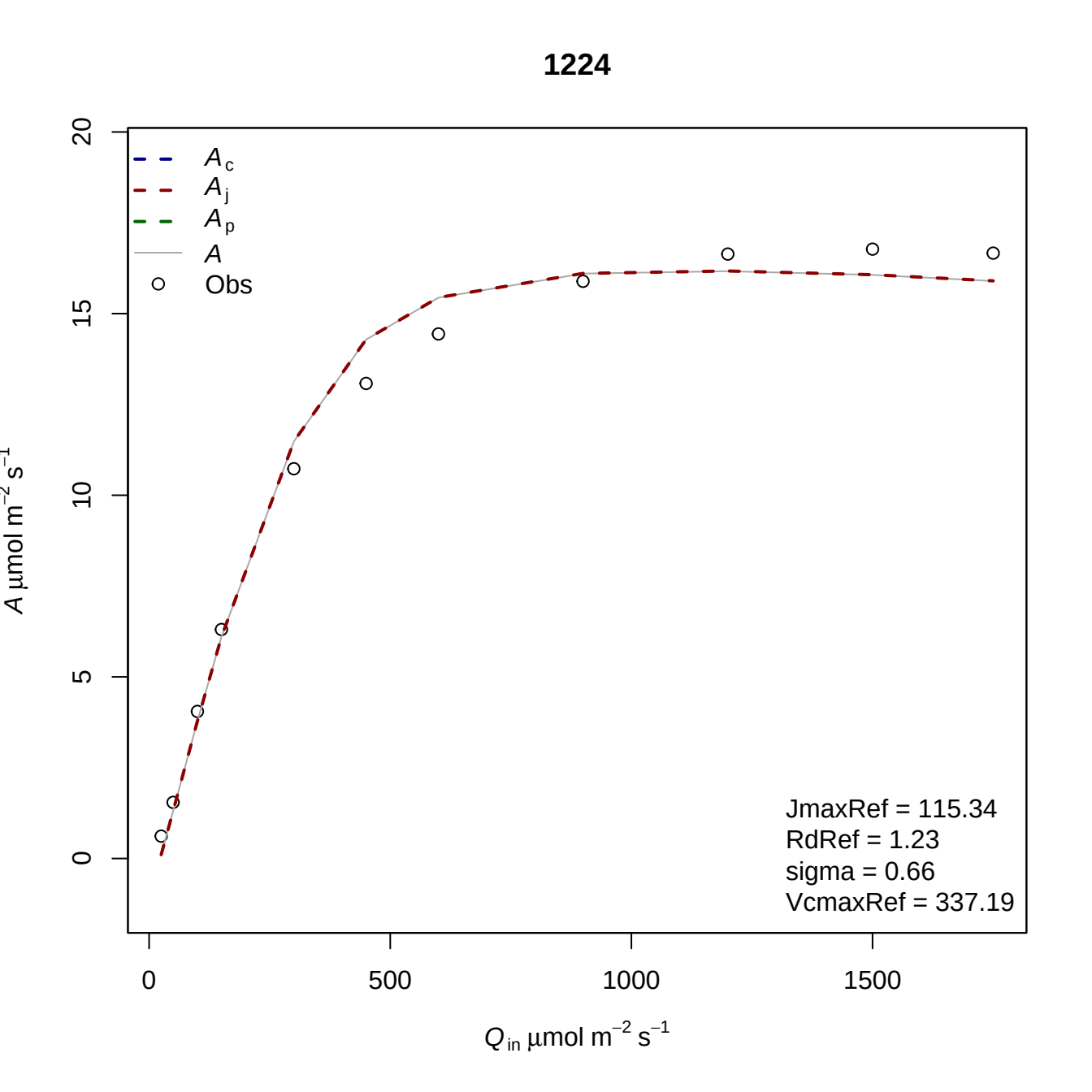

1219
A_c
A_j
A_p
A
Obs
JmaxRef = 160.92
RdRef = 2.43
sigma = 0.9
VcmaxRef = 326.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

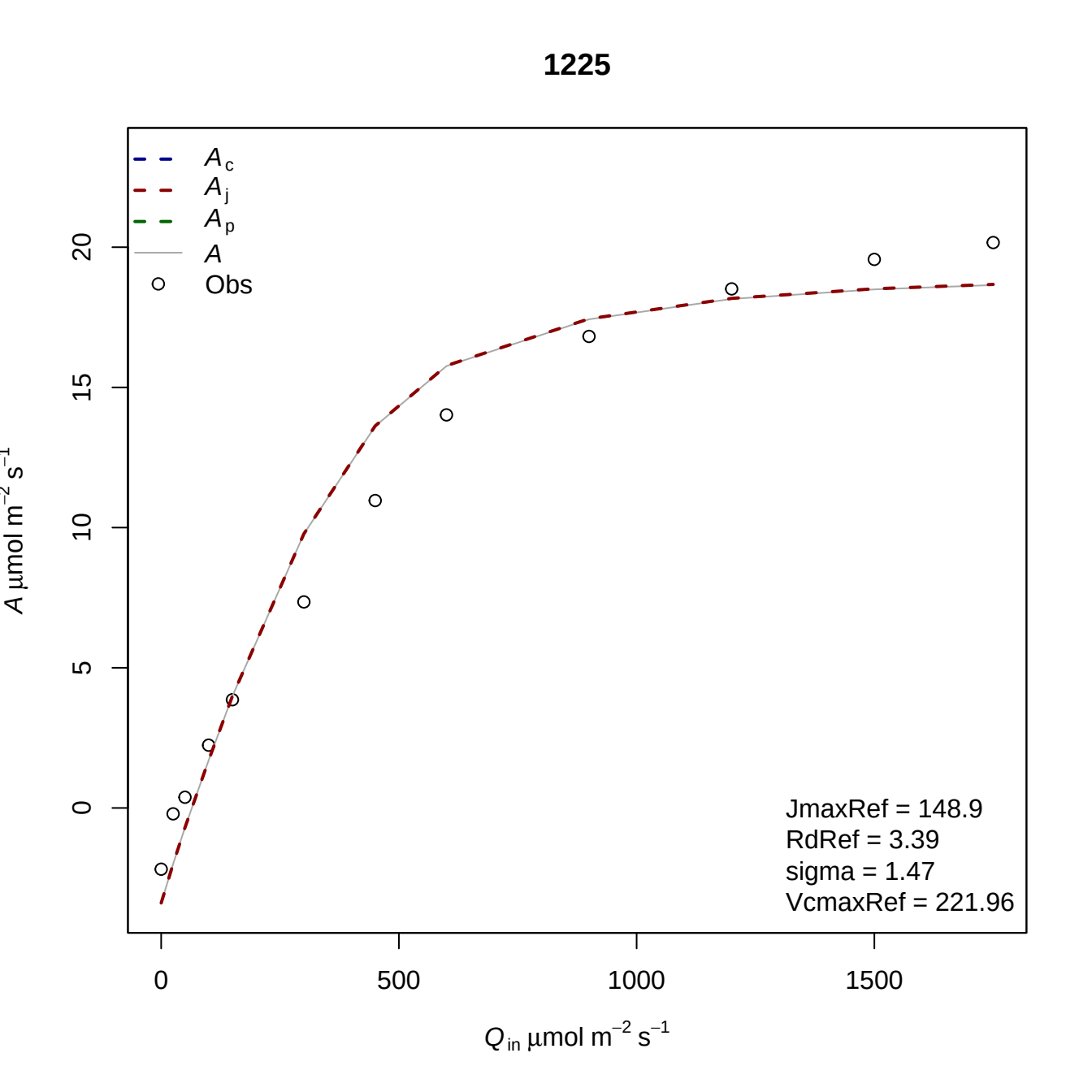

1220
A_c
A_j
A_p
A
Obs
JmaxRef = 154.48
RdRef = 1.39
sigma = 0.26
VcmaxRef = 115.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

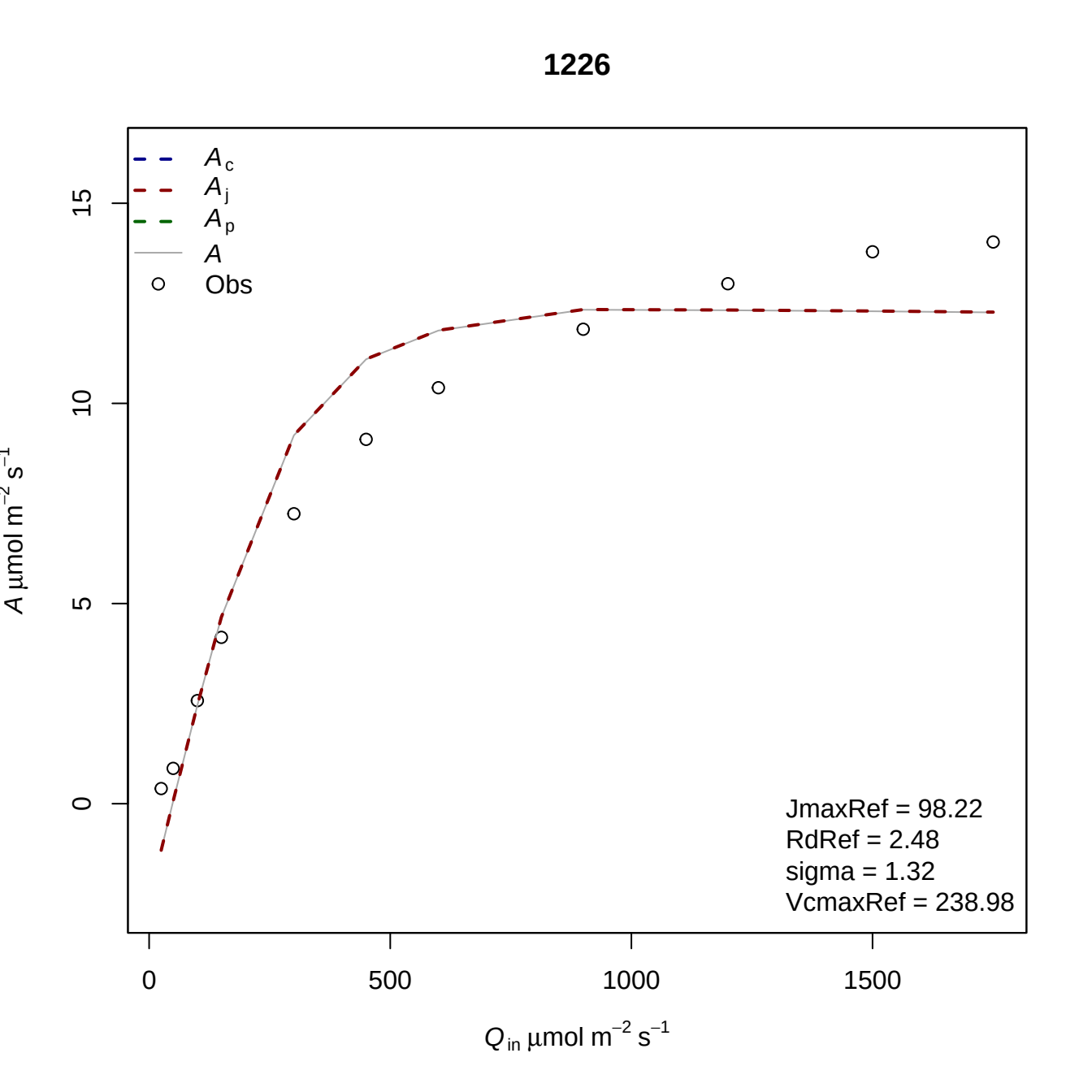

1221
A_c
A_j
A_p
A
Obs
JmaxRef = 126.46
RdRef = 1.38
sigma = 0.65
VcmaxRef = 204.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

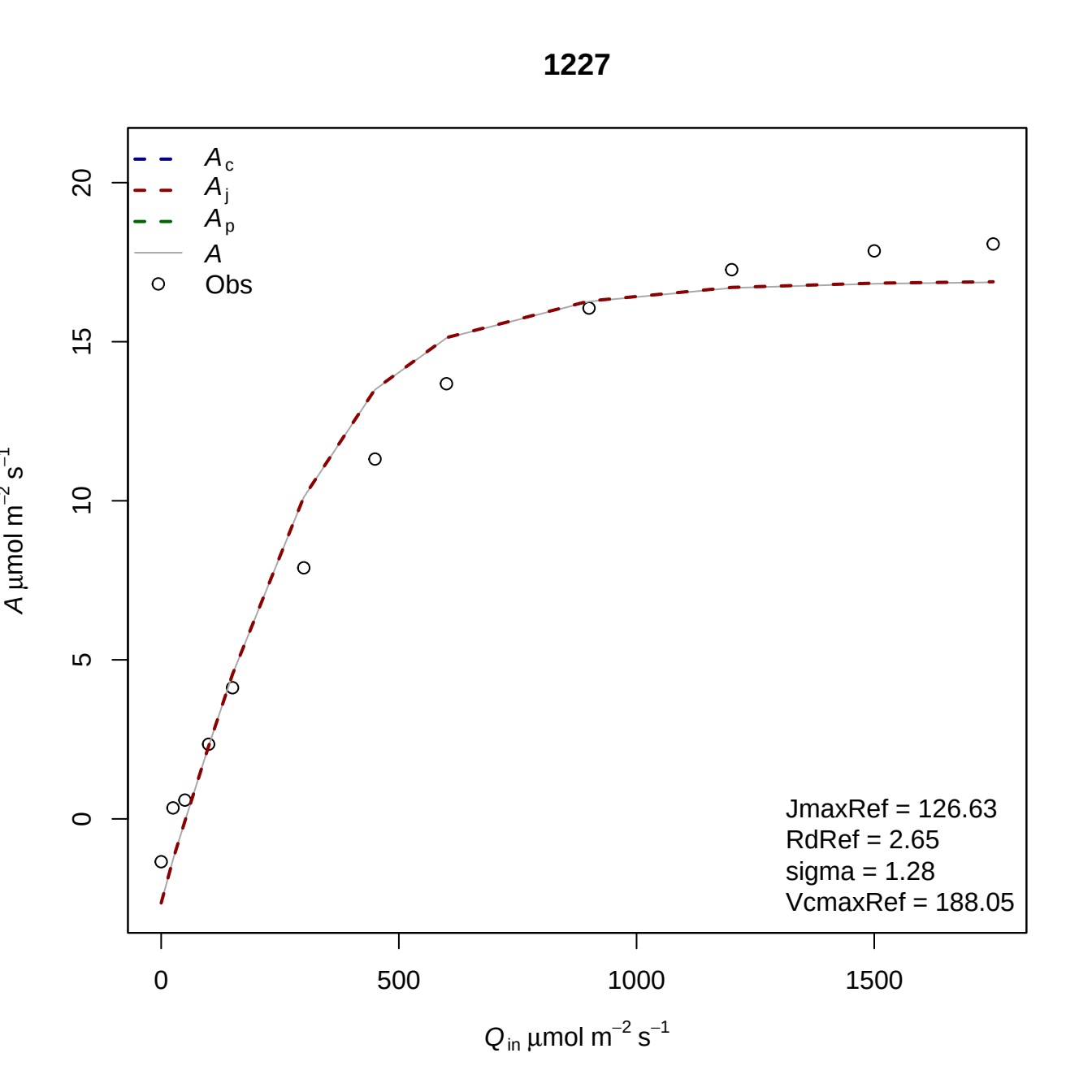

1222
A_c
A_j
A_p
A
Obs
JmaxRef = 167.58
RdRef = 2.48
sigma = 0.98
VcmaxRef = 251.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

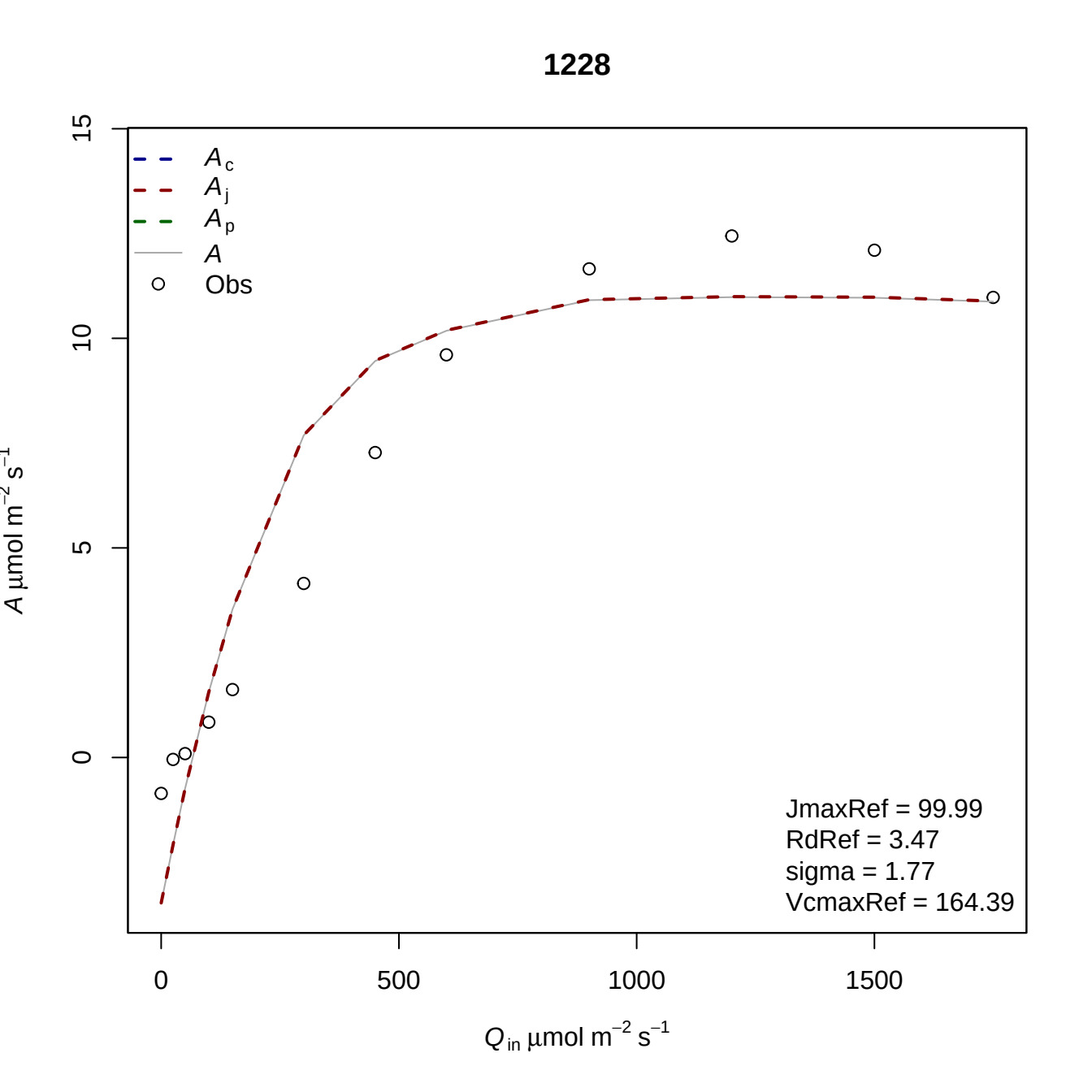

1223
A_c
A_j
A_p
A
Obs
JmaxRef = 103.11
RdRef = 2.3
sigma = 1.07
VcmaxRef = 146.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

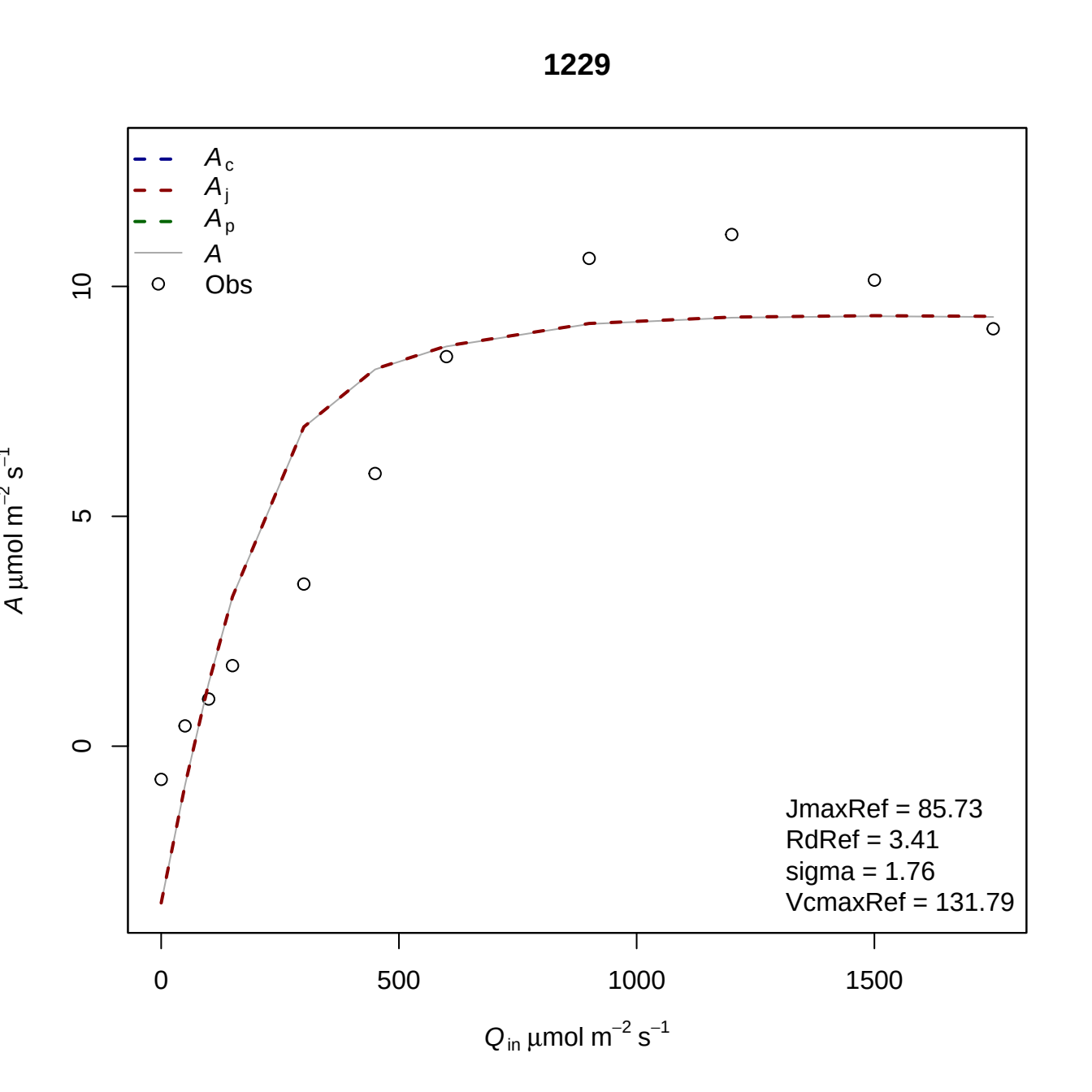

1224
A_c
A_j
A_p
A
Obs
JmaxRef = 115.34
RdRef = 1.23
sigma = 0.66
VcmaxRef = 337.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

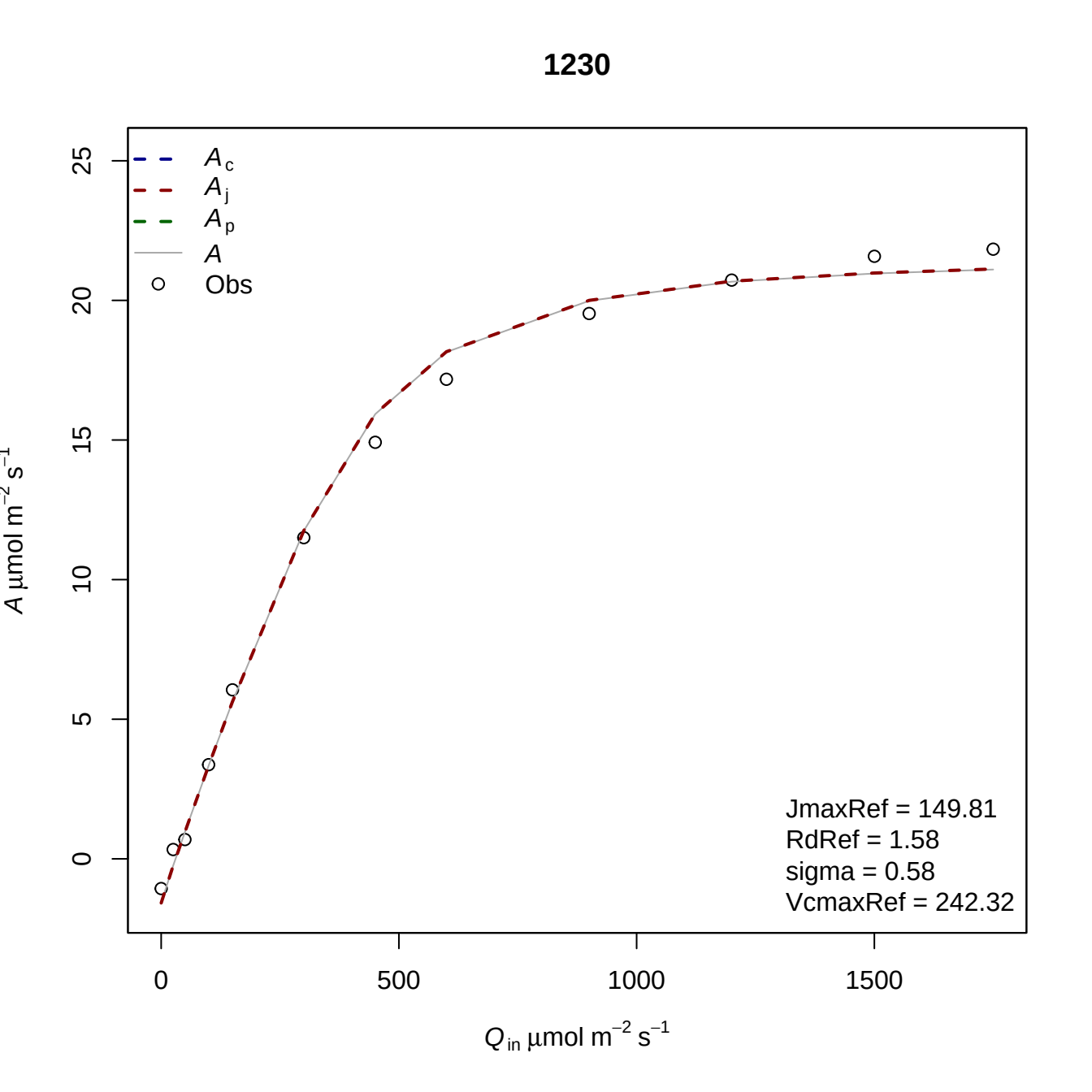

1225
A_c
A_j
A_p
A
Obs
JmaxRef = 148.9
RdRef = 3.39
sigma = 1.47
VcmaxRef = 221.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

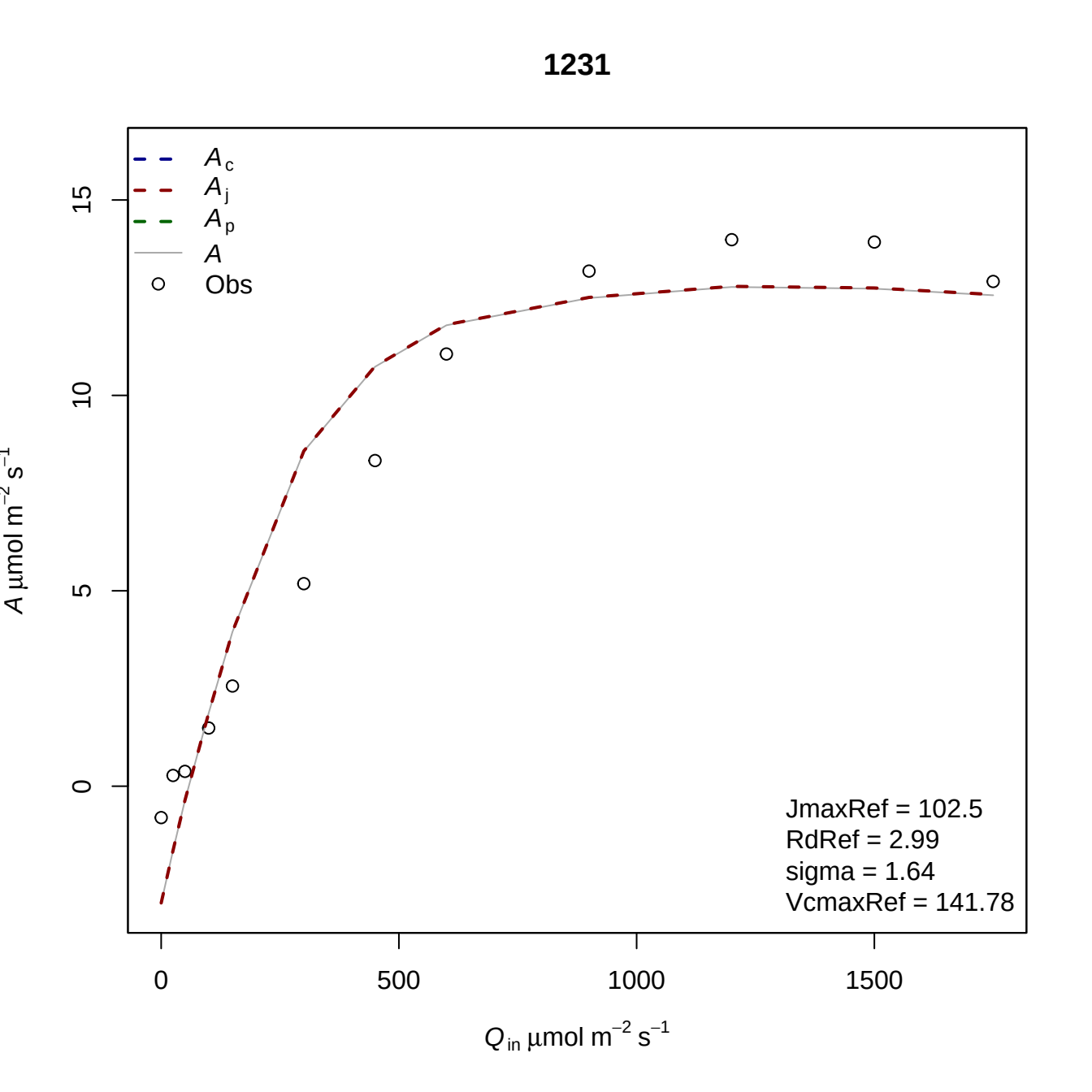

1226
A_c
A_j
A_p
A
Obs
JmaxRef = 98.22
RdRef = 2.48
sigma = 1.32
VcmaxRef = 238.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

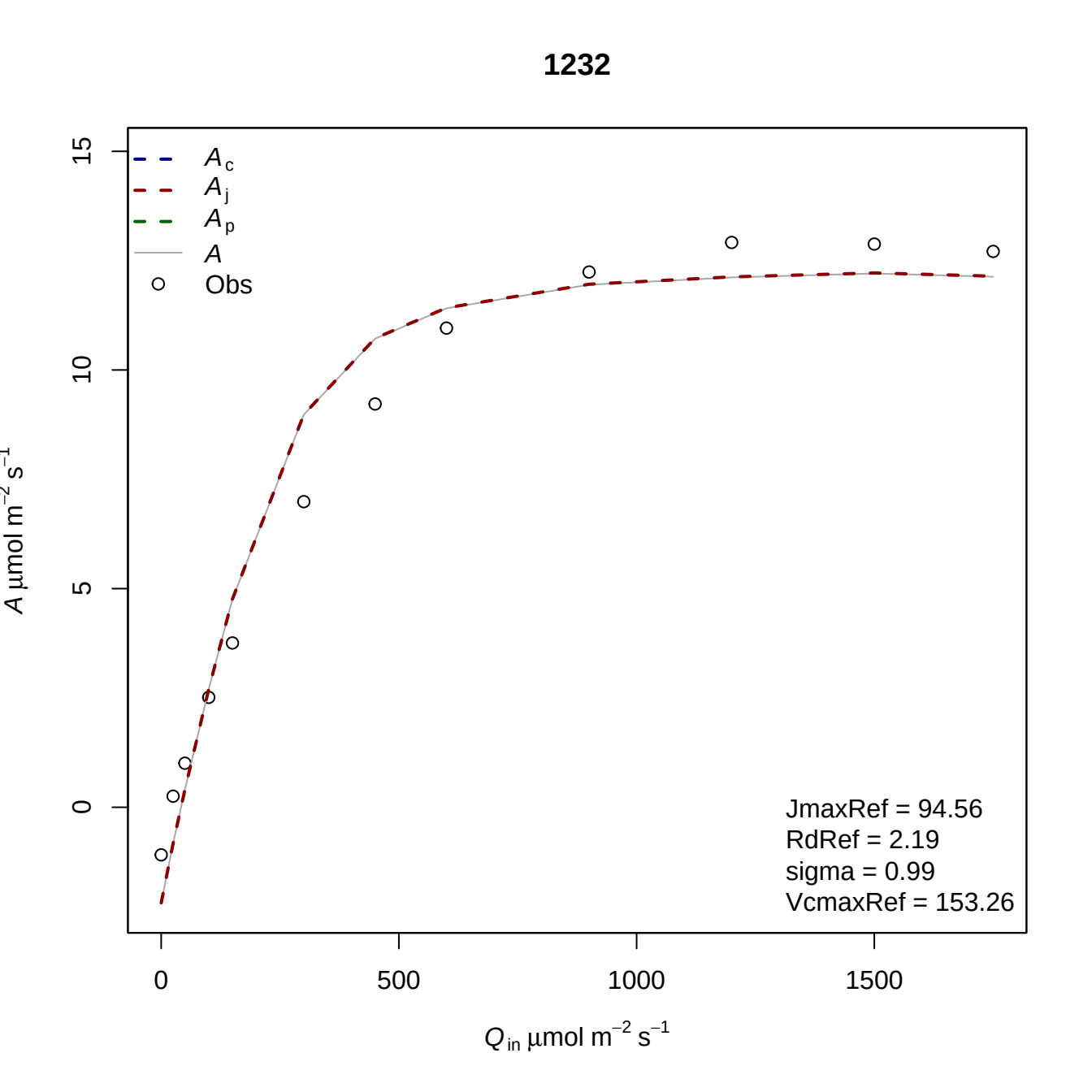

1227
A_c
A_j
A_p
A
Obs
JmaxRef = 126.63
RdRef = 2.65
sigma = 1.28
VcmaxRef = 188.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

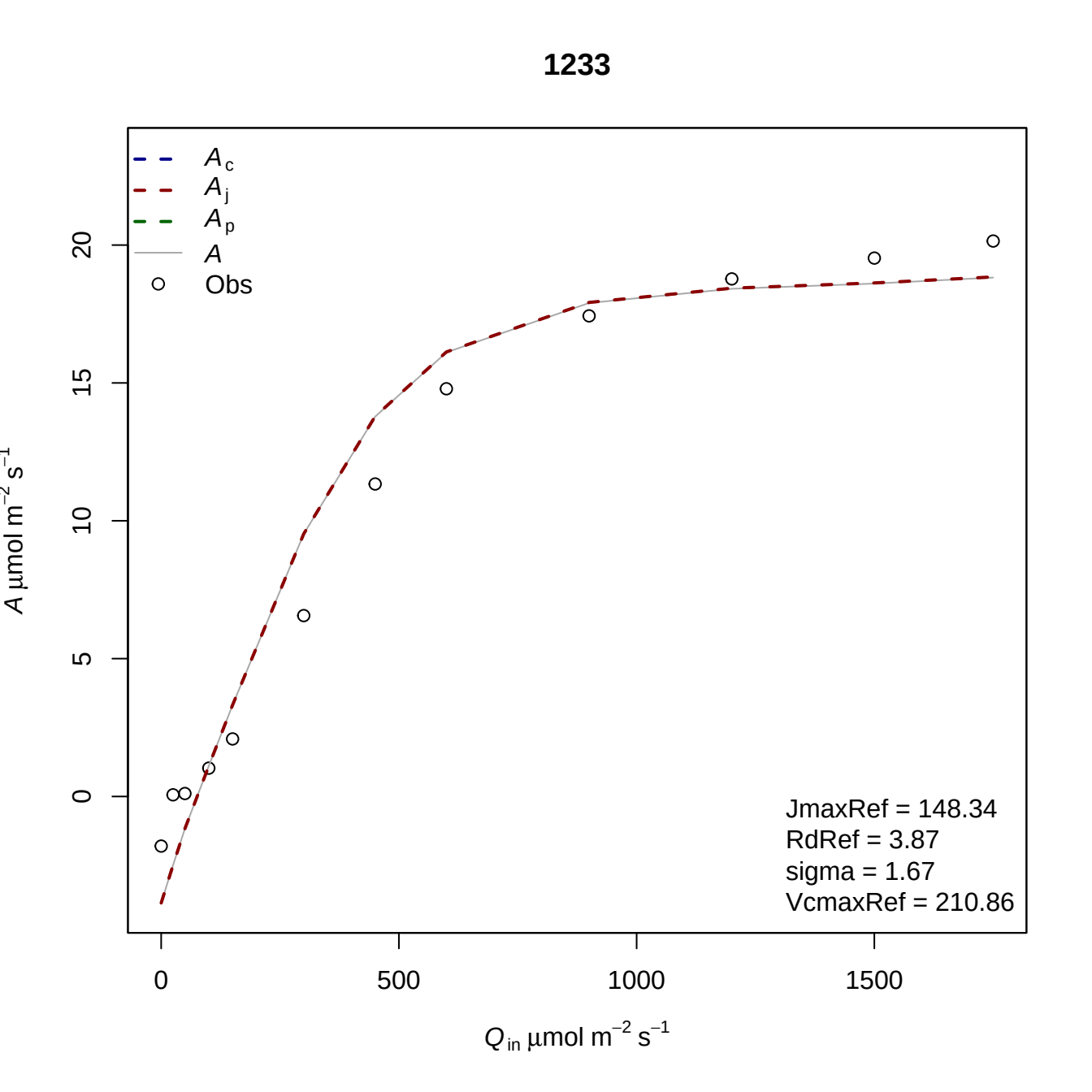

1228
A_c
A_j
A_p
A
Obs
JmaxRef = 99.99
RdRef = 3.47
sigma = 1.77
VcmaxRef = 164.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

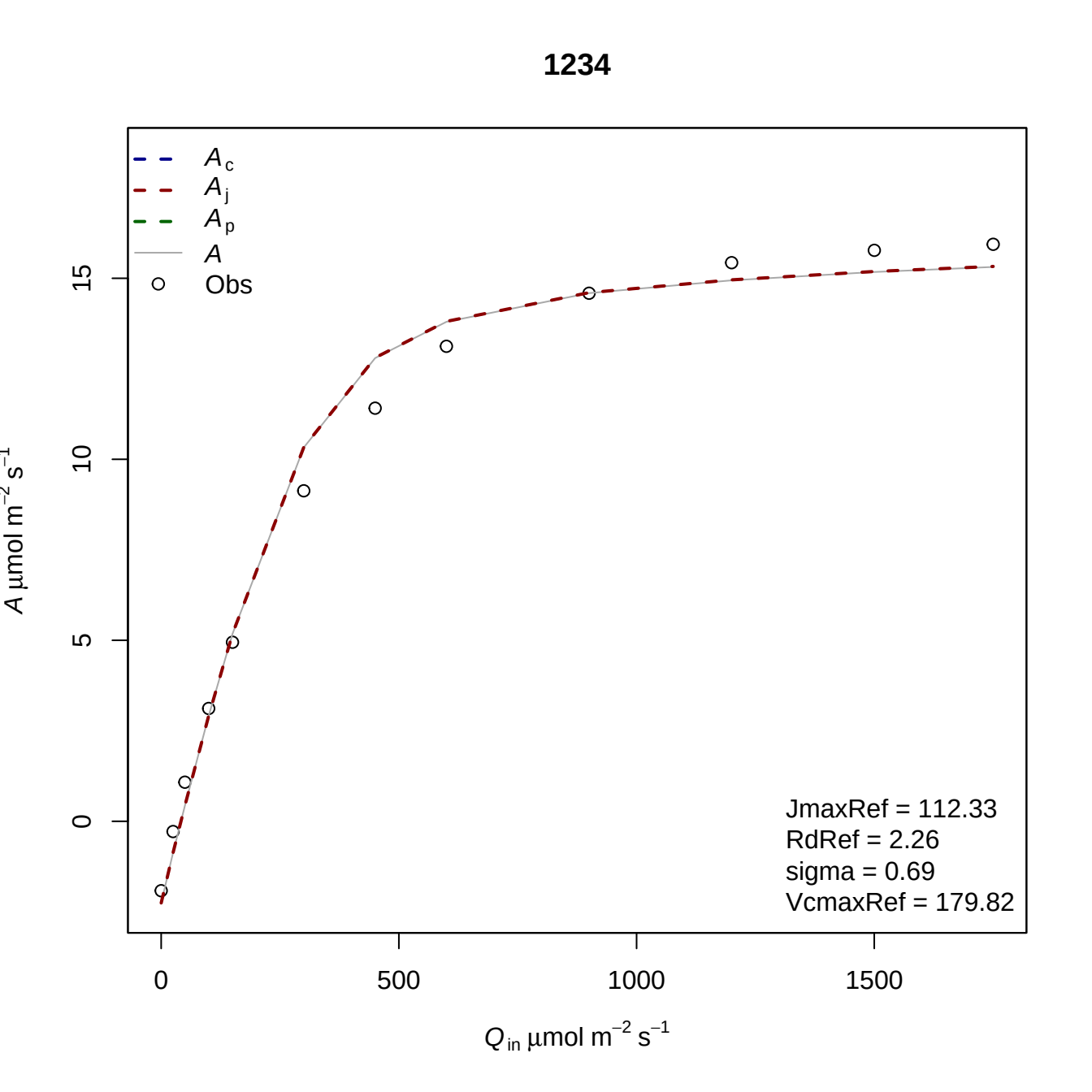

1229
A_c
A_j
A_p
A
Obs
JmaxRef = 85.73
RdRef = 3.41
sigma = 1.76
VcmaxRef = 131.79
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

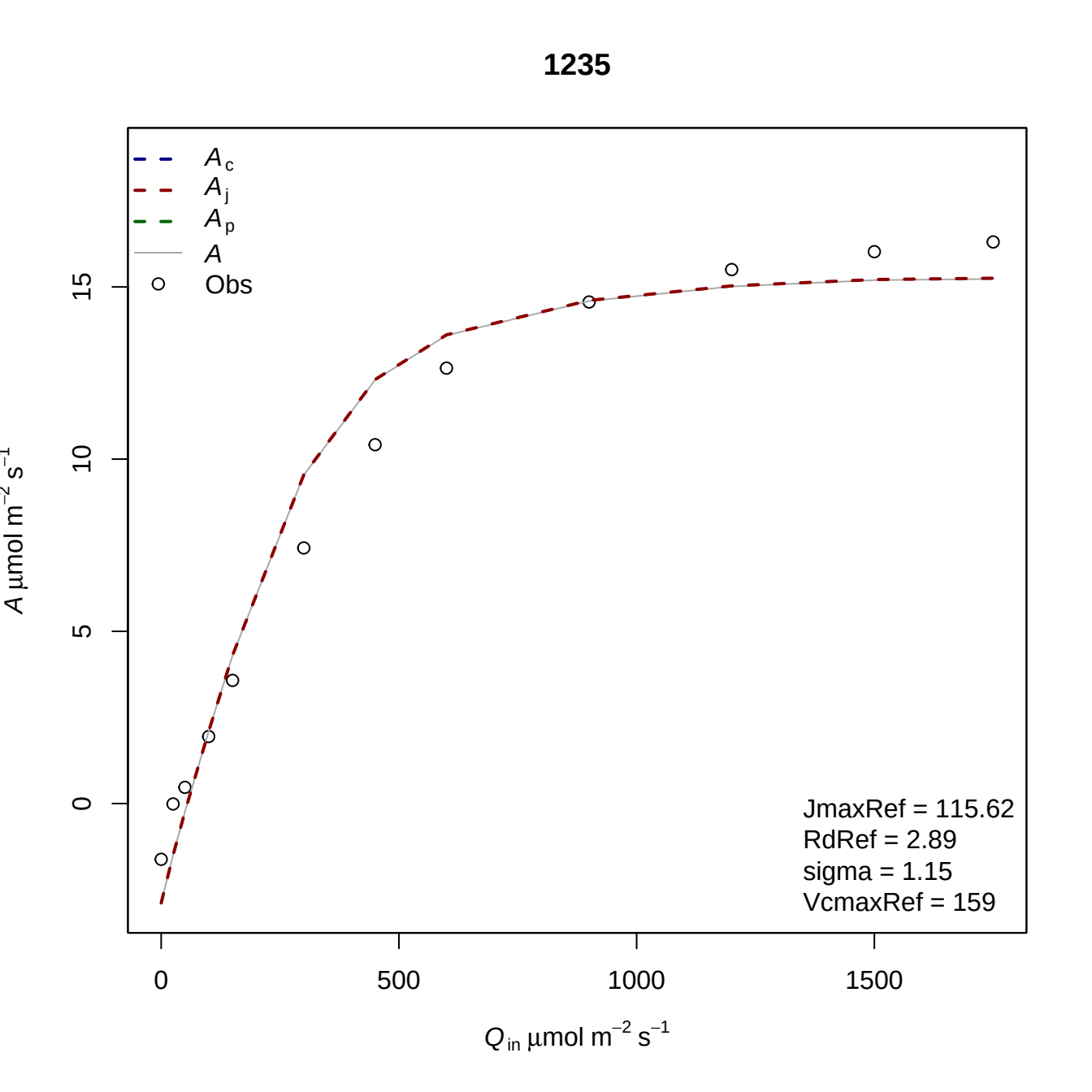

1230
A_c
A_j
A_p
A
Obs
JmaxRef = 149.81
RdRef = 1.58
sigma = 0.58
VcmaxRef = 242.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

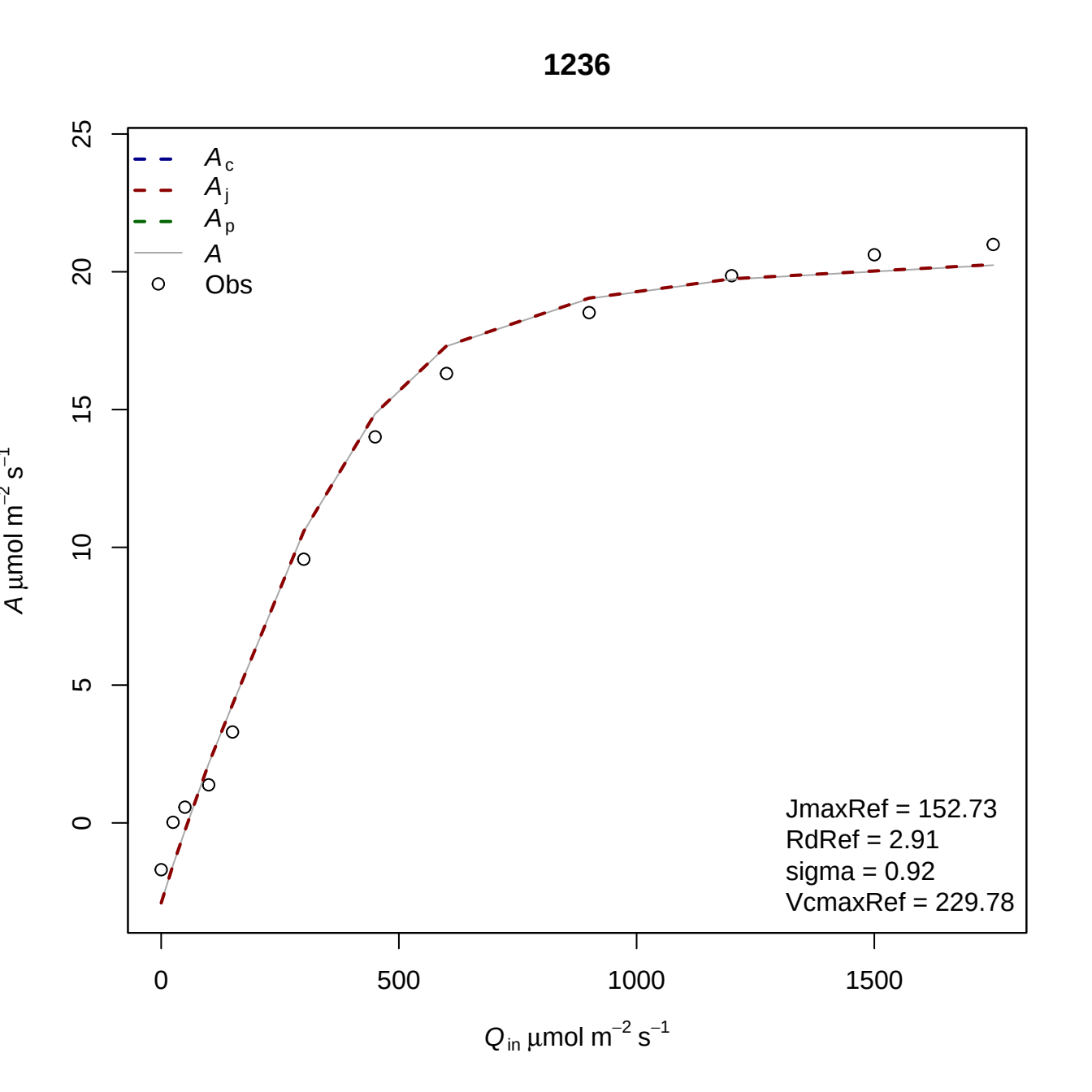

1231
A_c
A_j
A_p
A
Obs
JmaxRef = 102.5
RdRef = 2.99
sigma = 1.64
VcmaxRef = 141.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

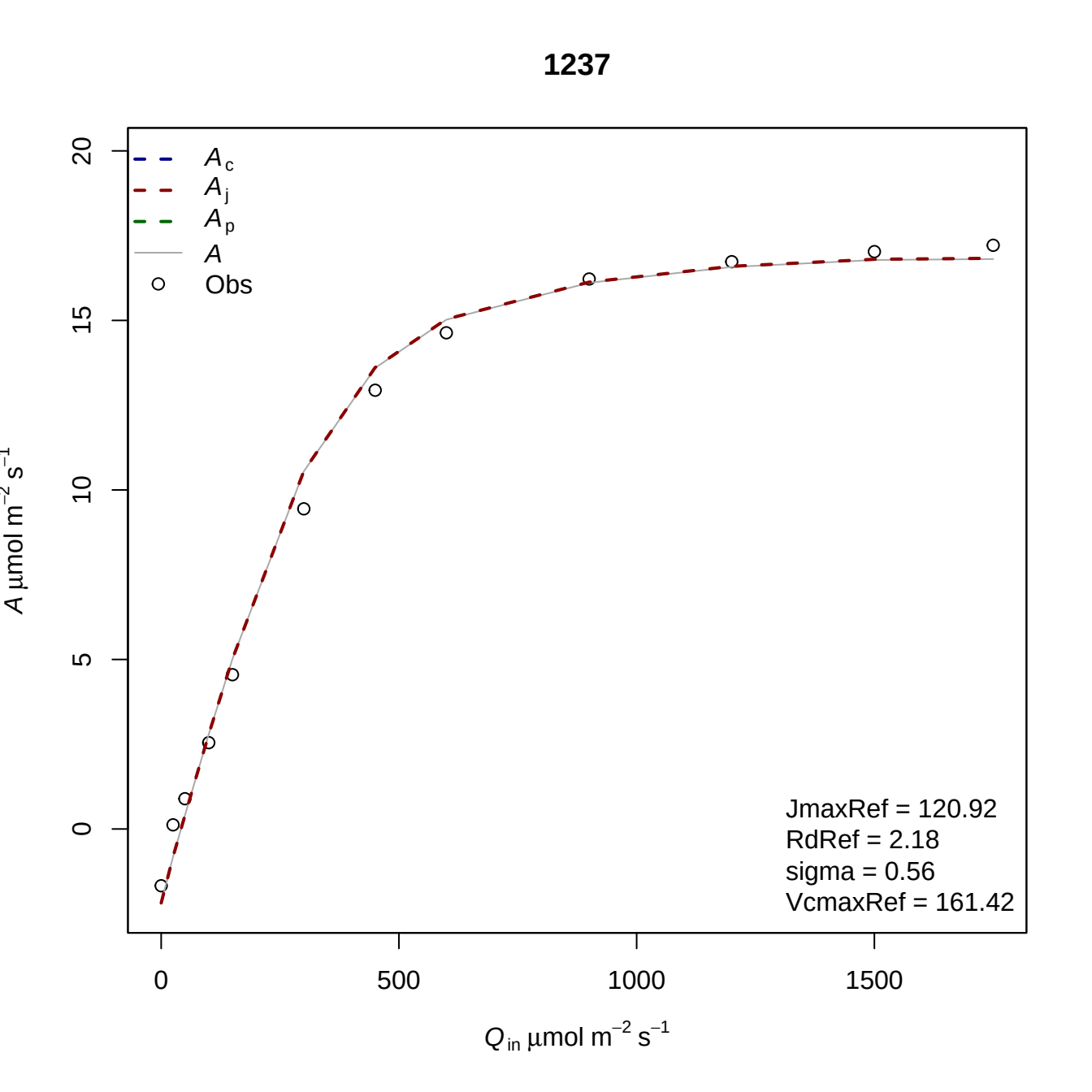

1232
A_c
A_j
A_p
A
Obs
JmaxRef = 94.56
RdRef = 2.19
sigma = 0.99
VcmaxRef = 153.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

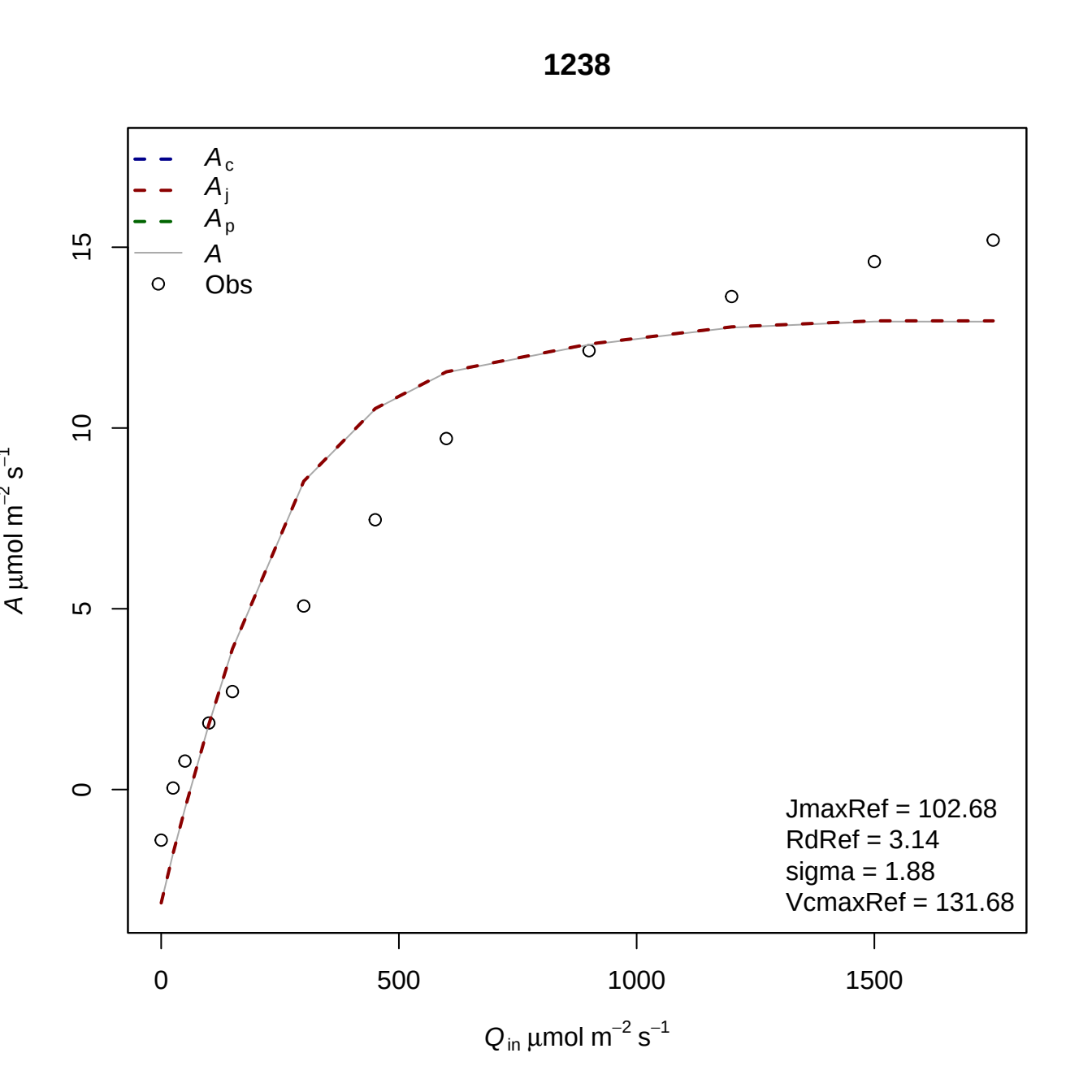

1233
A_c
A_j
A_p
A
Obs
JmaxRef = 148.34
RdRef = 3.87
sigma = 1.67
VcmaxRef = 210.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

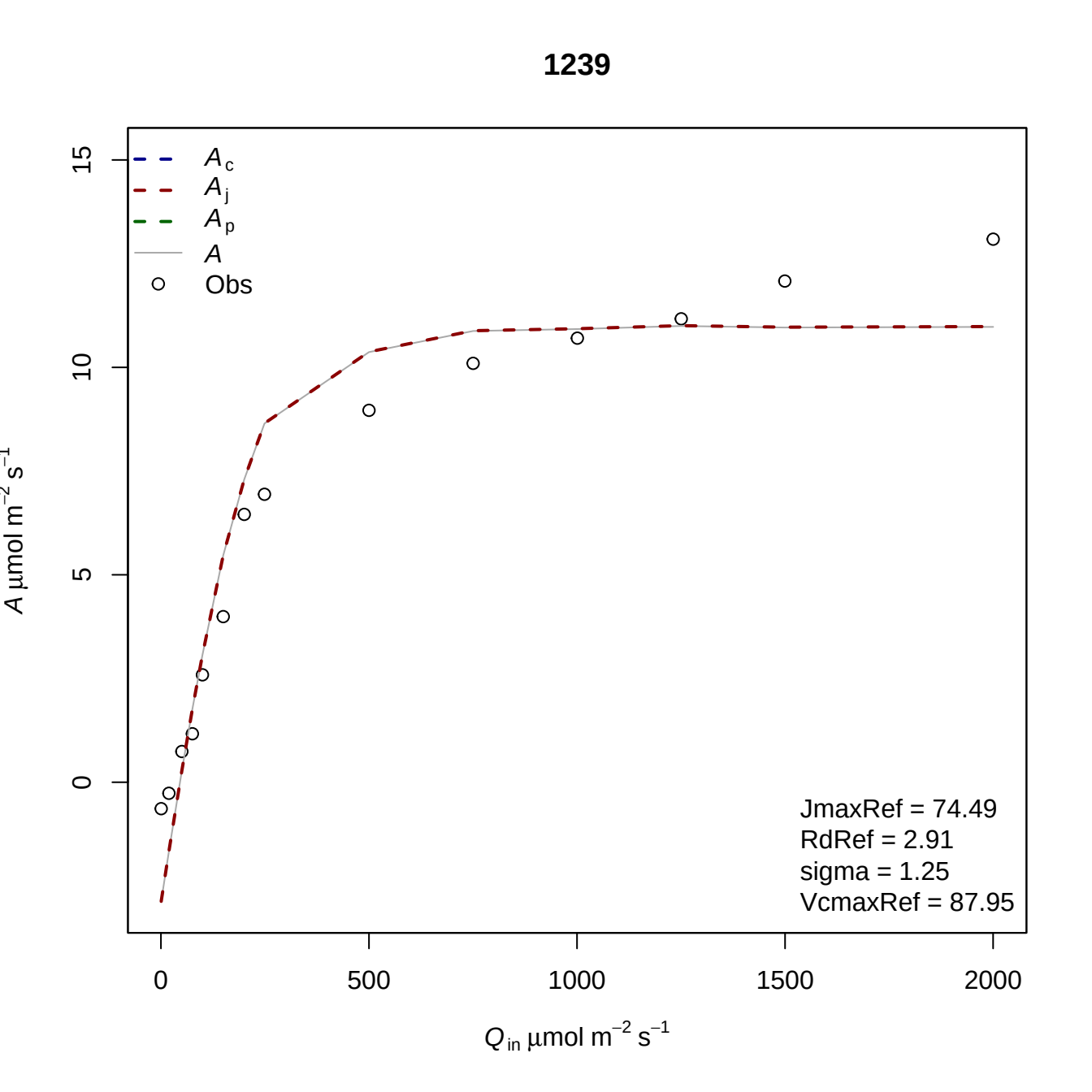

1234
A_c
A_j
A_p
A
Obs
JmaxRef = 112.33
RdRef = 2.26
sigma = 0.69
VcmaxRef = 179.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

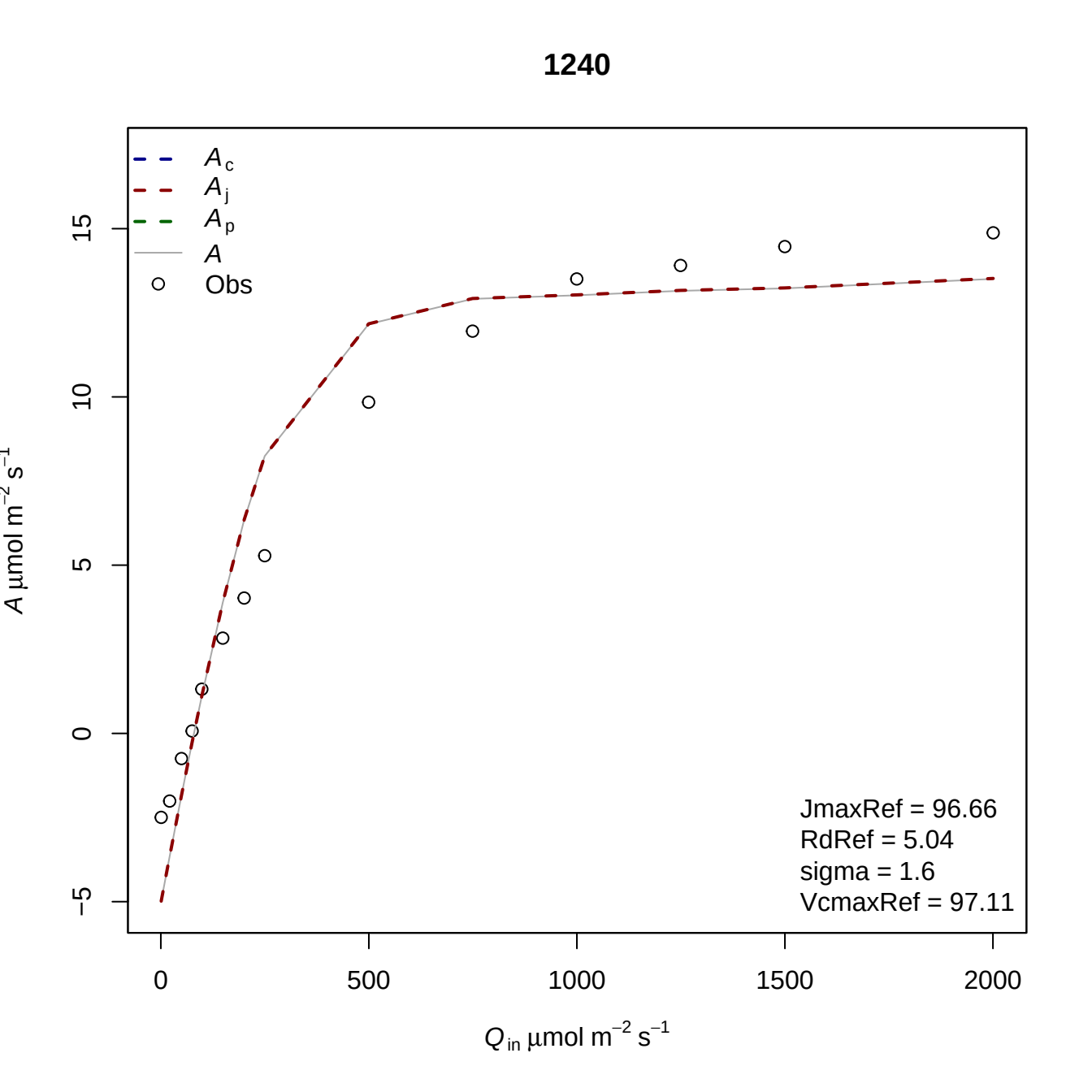

1235
A_c
A_j
A_p
A
Obs
JmaxRef = 115.62
RdRef = 2.89
sigma = 1.15
VcmaxRef = 159
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

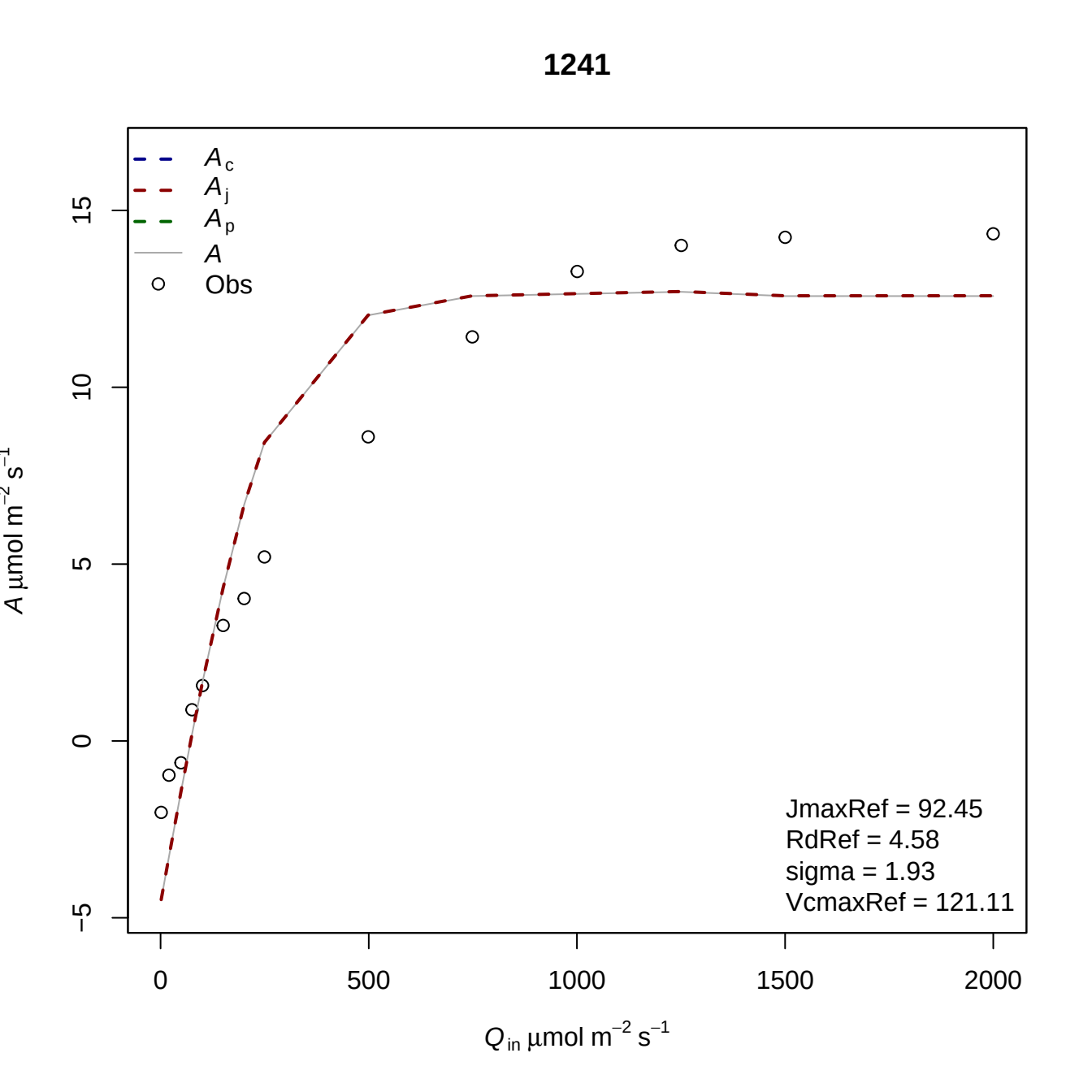

1236
A_c
A_j
A_p
A
Obs
JmaxRef = 152.73
RdRef = 2.91
sigma = 0.92
VcmaxRef = 229.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

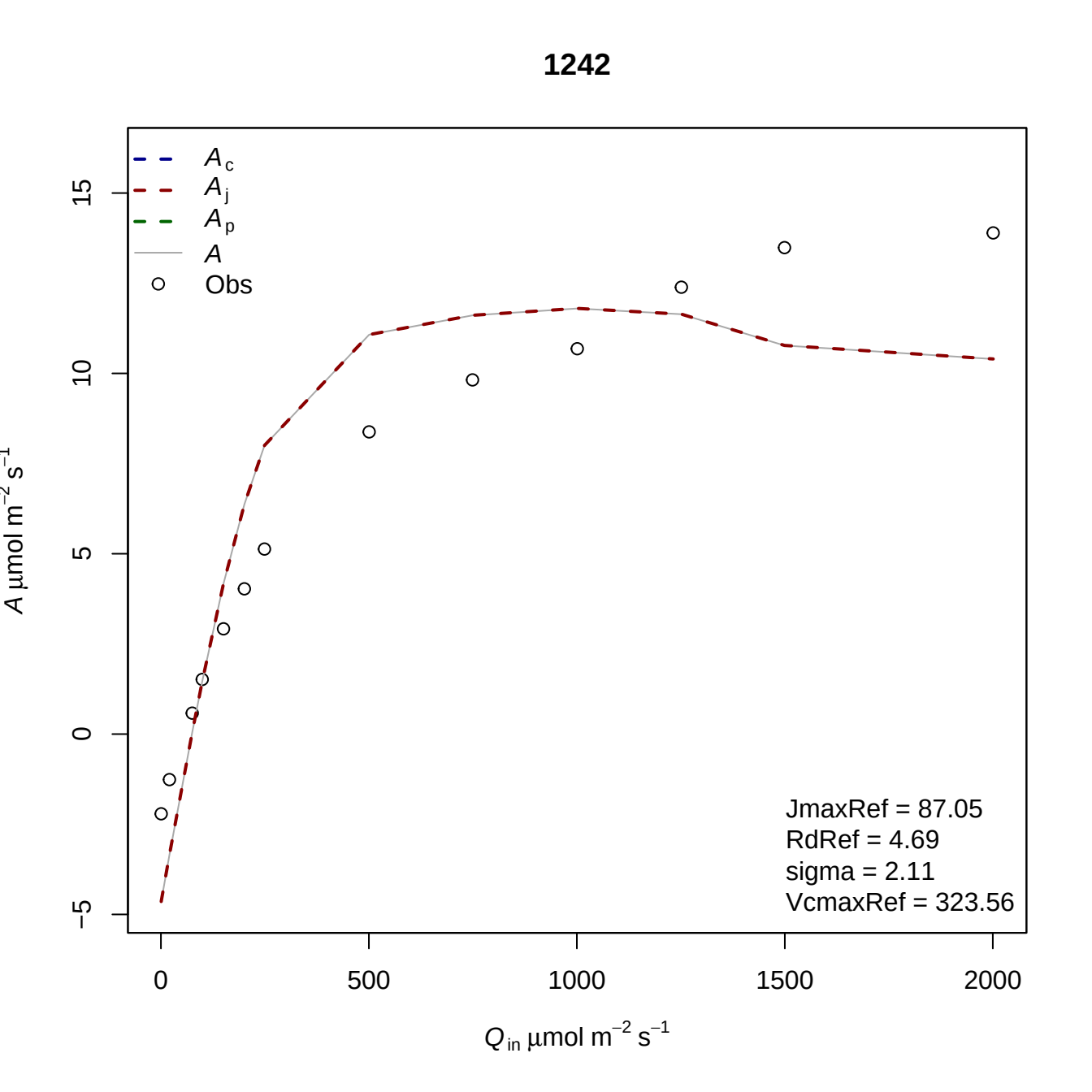

1237
A_c
A_j
A_p
A
Obs
JmaxRef = 120.92
RdRef = 2.18
sigma = 0.56
VcmaxRef = 161.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

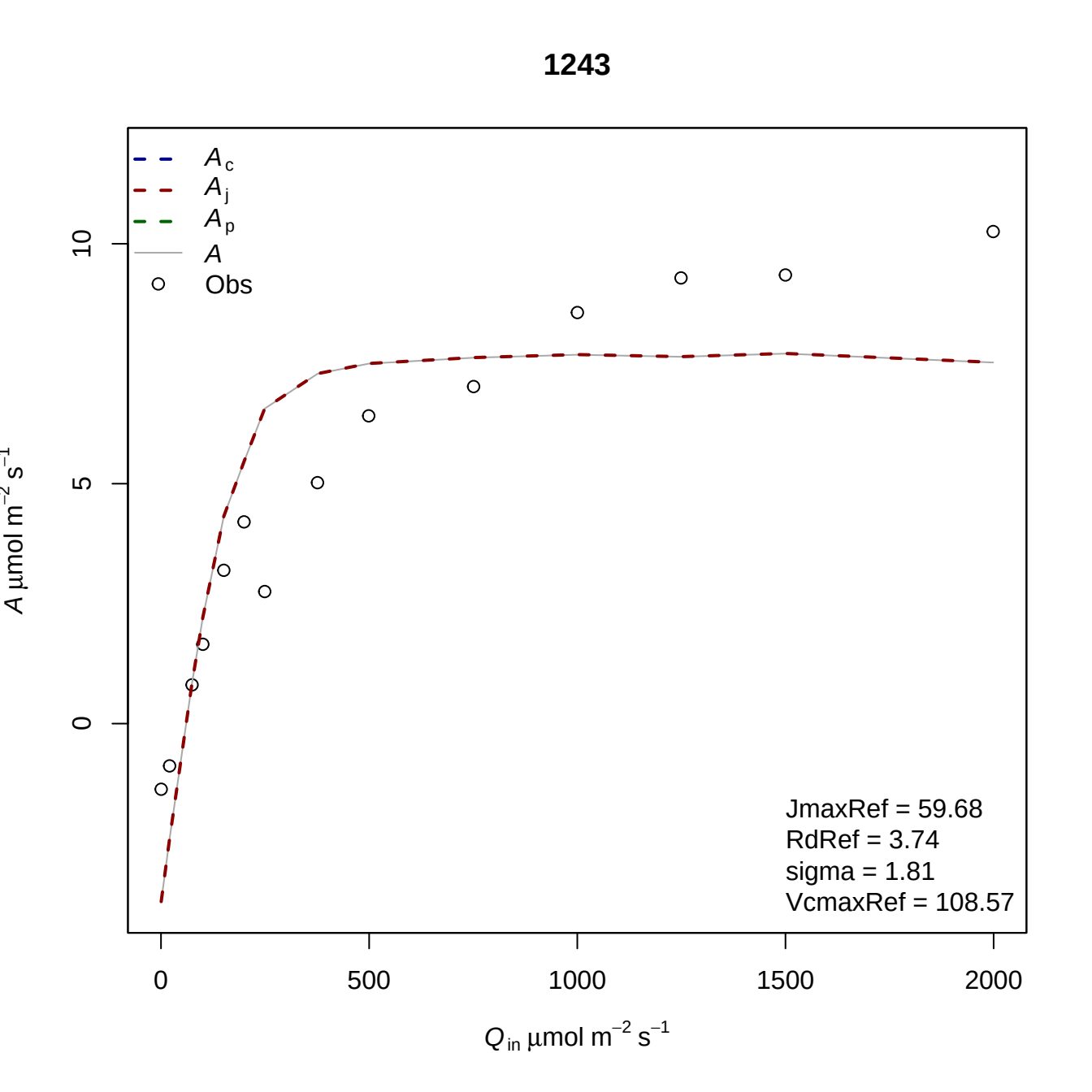

1238
A_c
A_j
A_p
A
Obs
JmaxRef = 102.68
RdRef = 3.14
sigma = 1.88
VcmaxRef = 131.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

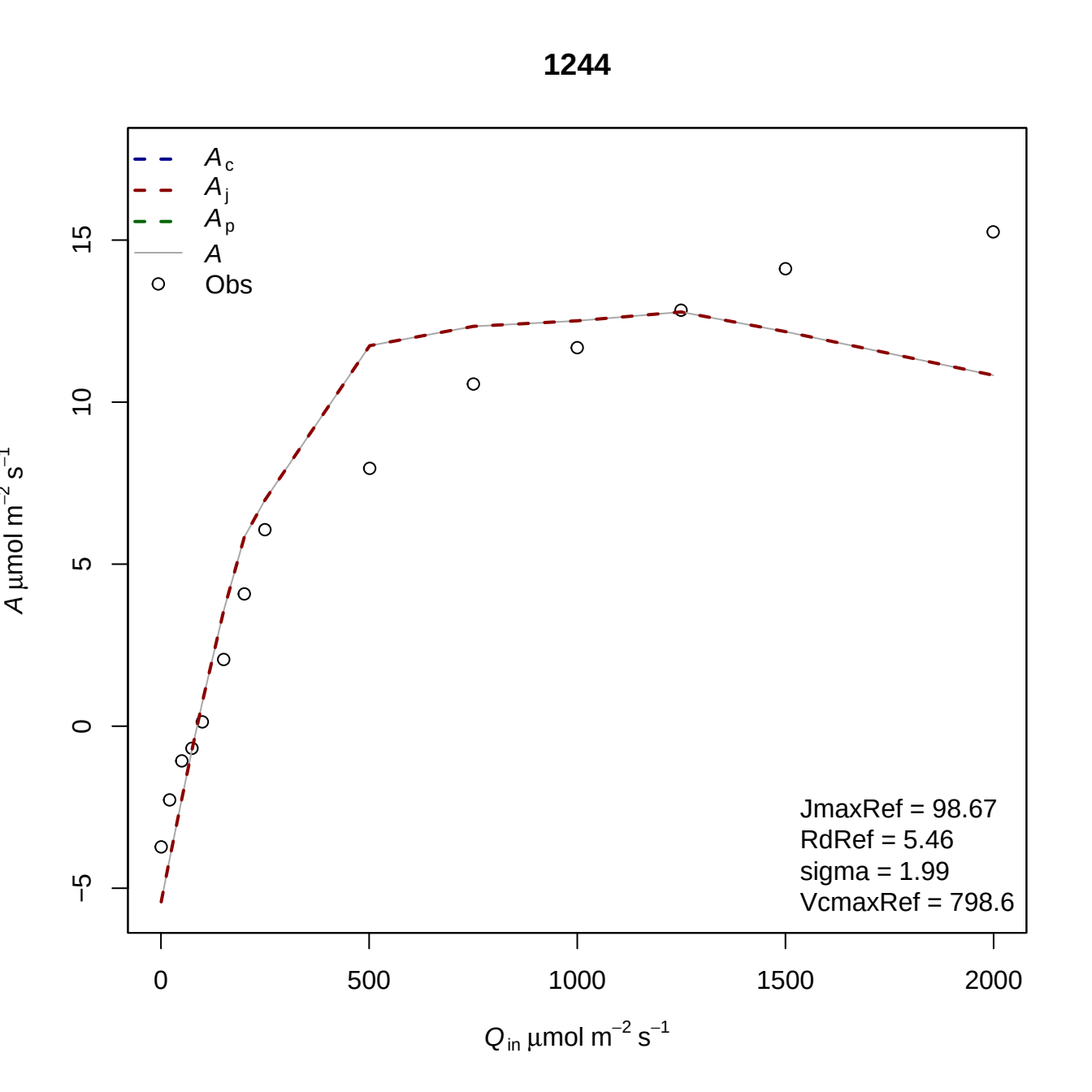

1239
A_c
A_j
A_p
A
Obs
JmaxRef = 74.49
RdRef = 2.91
sigma = 1.25
VcmaxRef = 87.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

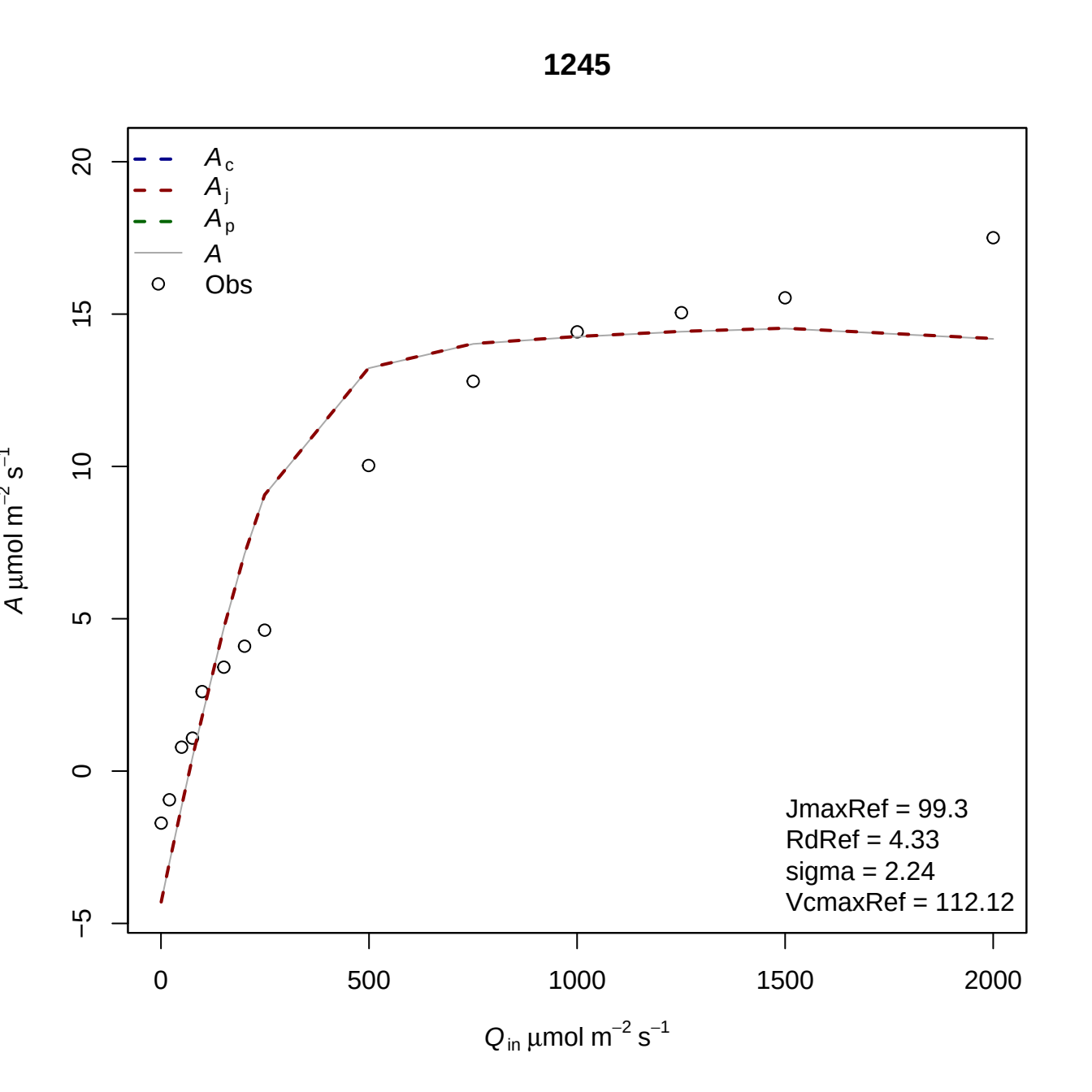

1240
A_c
A_j
A_p
A
Obs
JmaxRef = 96.66
RdRef = 5.04
sigma = 1.6
VcmaxRef = 97.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

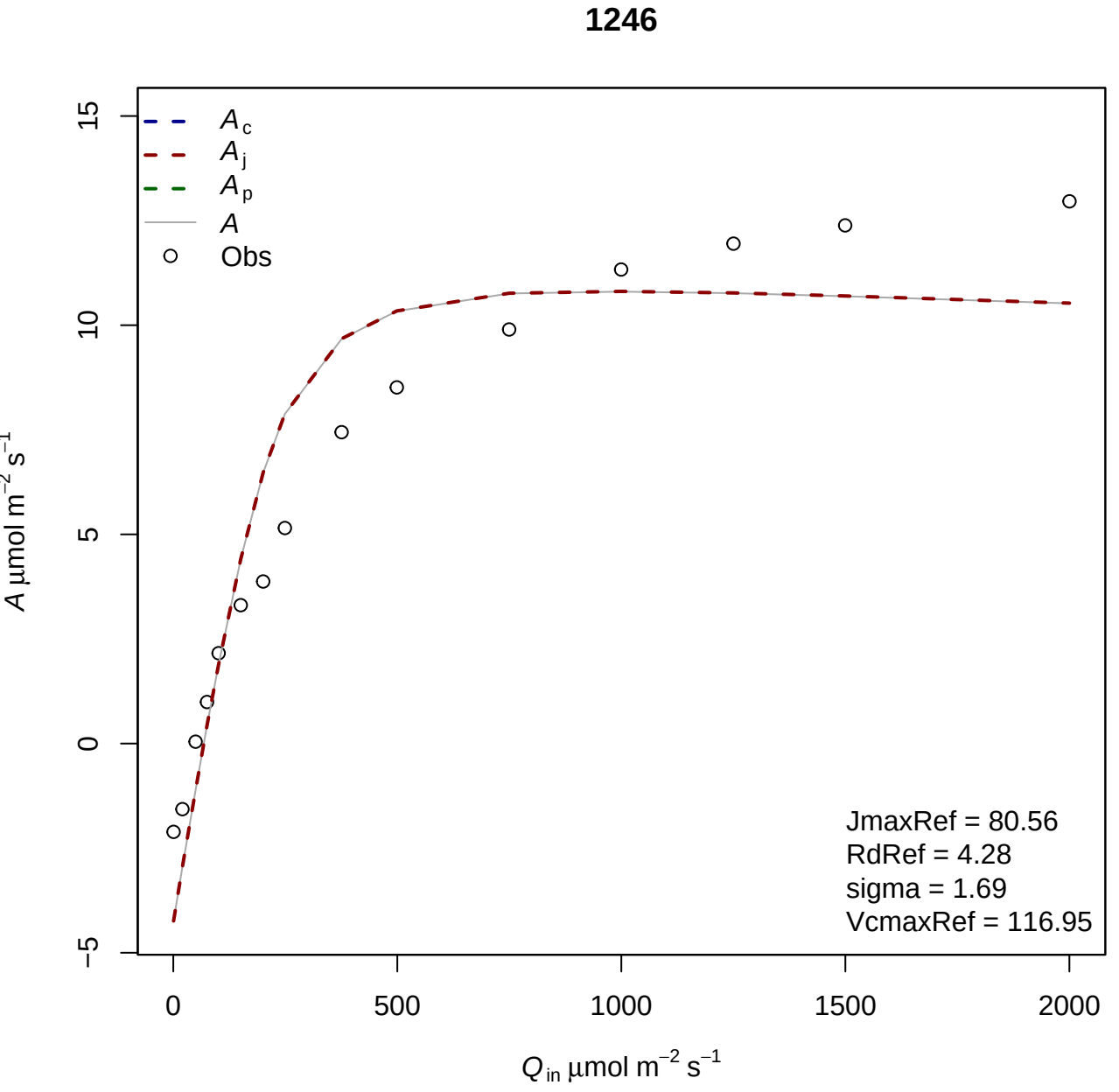

1241
A_c
A_j
A_p
A
Obs
JmaxRef = 92.45
RdRef = 4.58
sigma = 1.93
VcmaxRef = 121.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

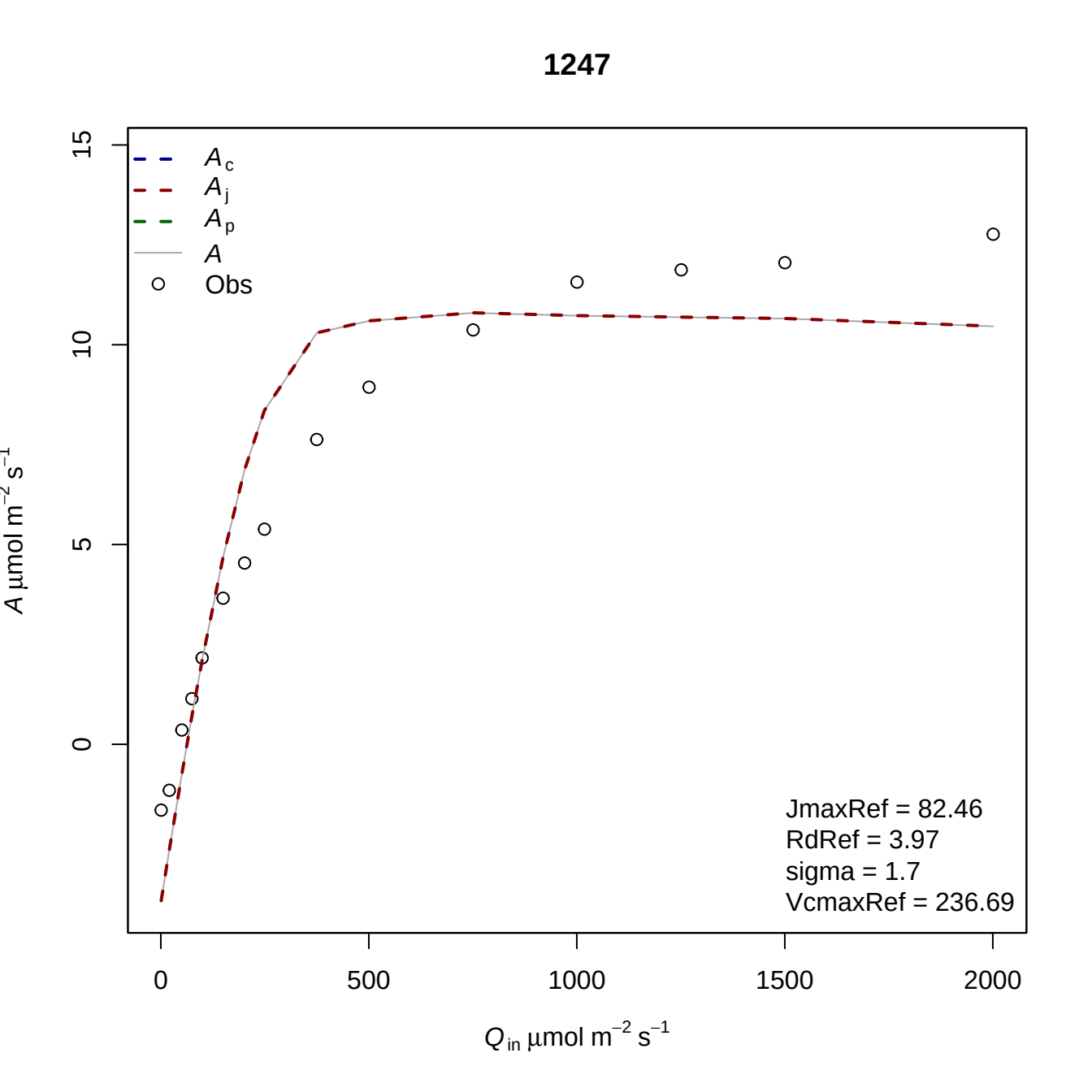

1242
A_c
A_j
A_p
A
Obs
JmaxRef = 87.05
RdRef = 4.69
sigma = 2.11
VcmaxRef = 323.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

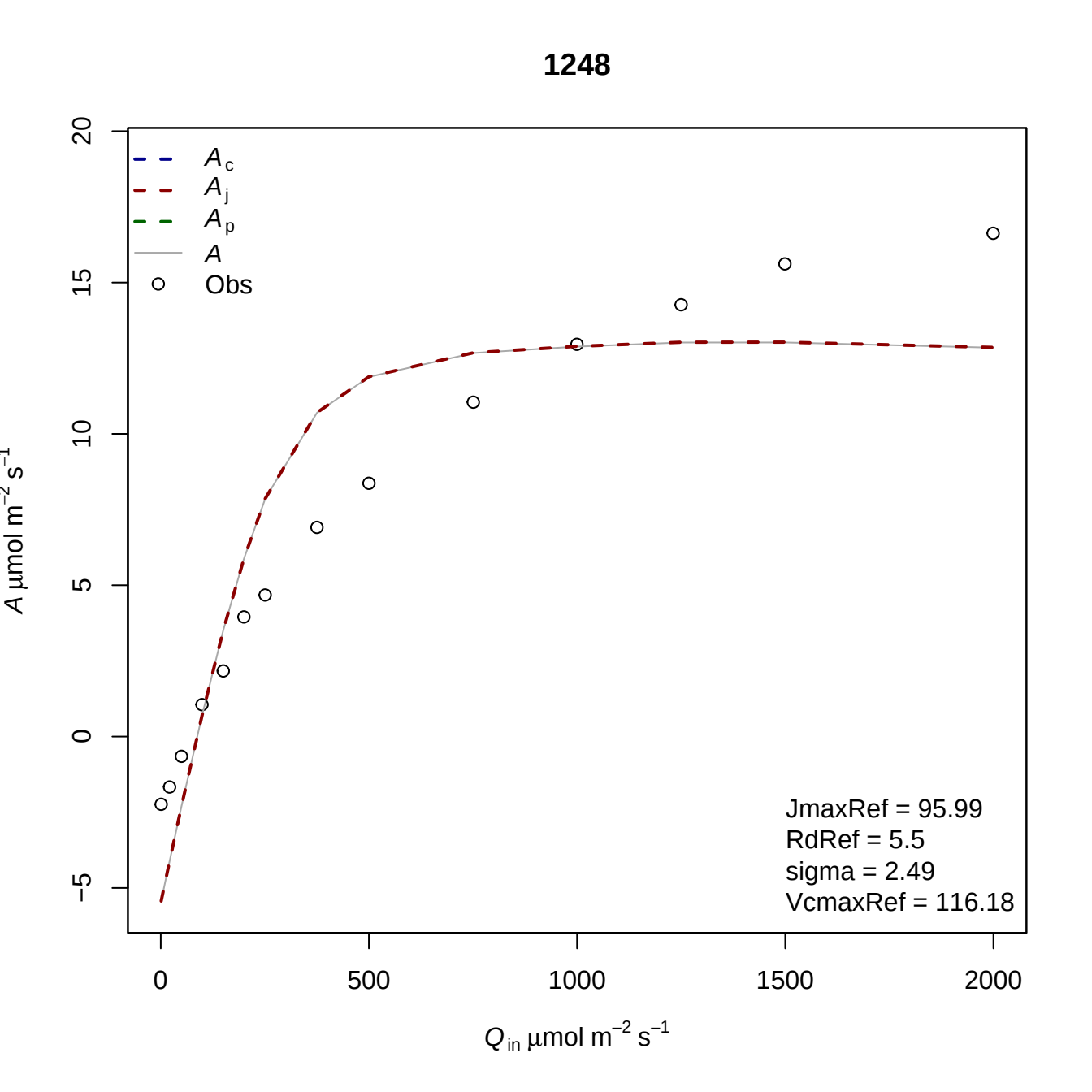

1243
A_c
A_j
A_p
A
Obs
JmaxRef = 59.68
RdRef = 3.74
sigma = 1.81
VcmaxRef = 108.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

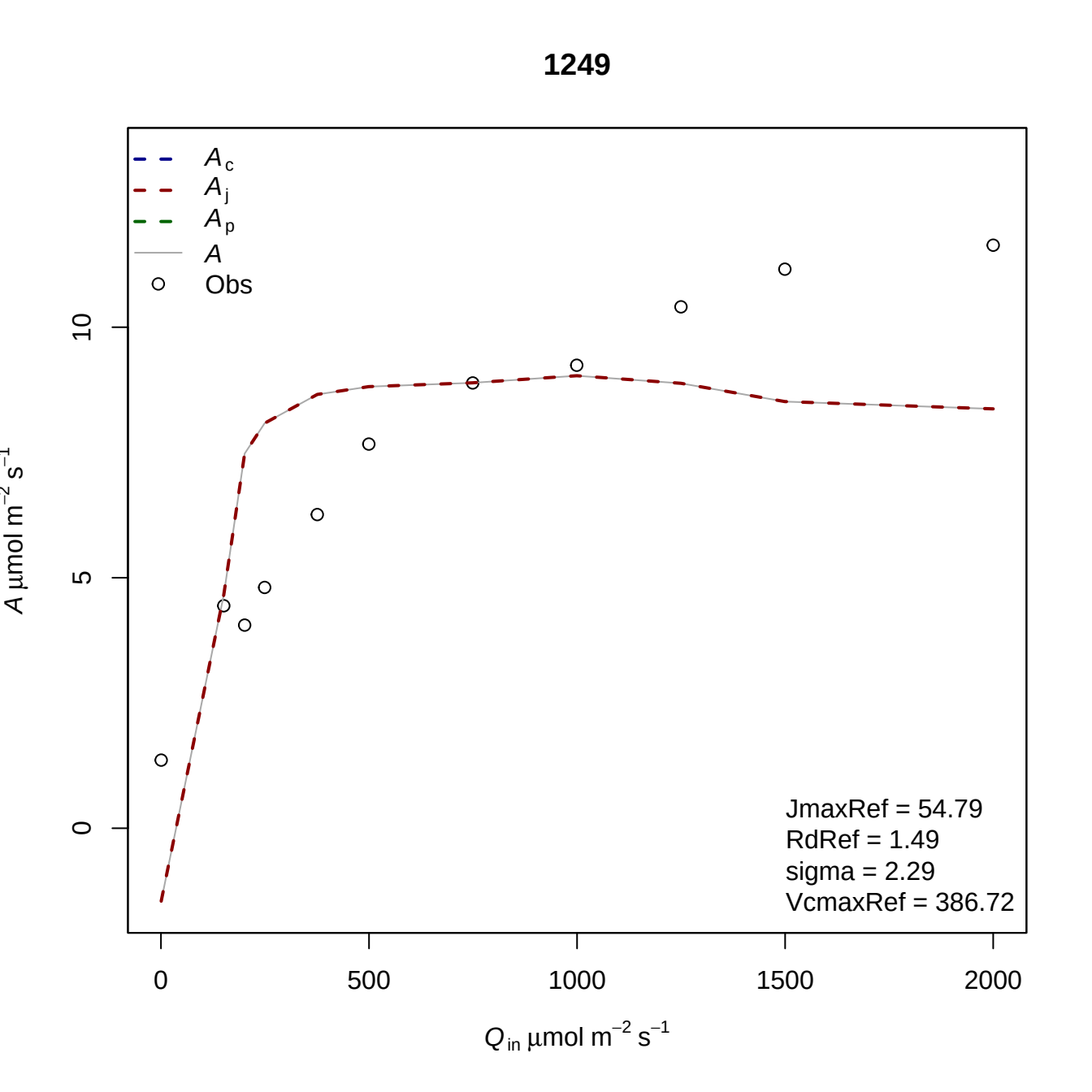

1244
A_c
A_j
A_p
A
Obs
JmaxRef = 98.67
RdRef = 5.46
sigma = 1.99
VcmaxRef = 798.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

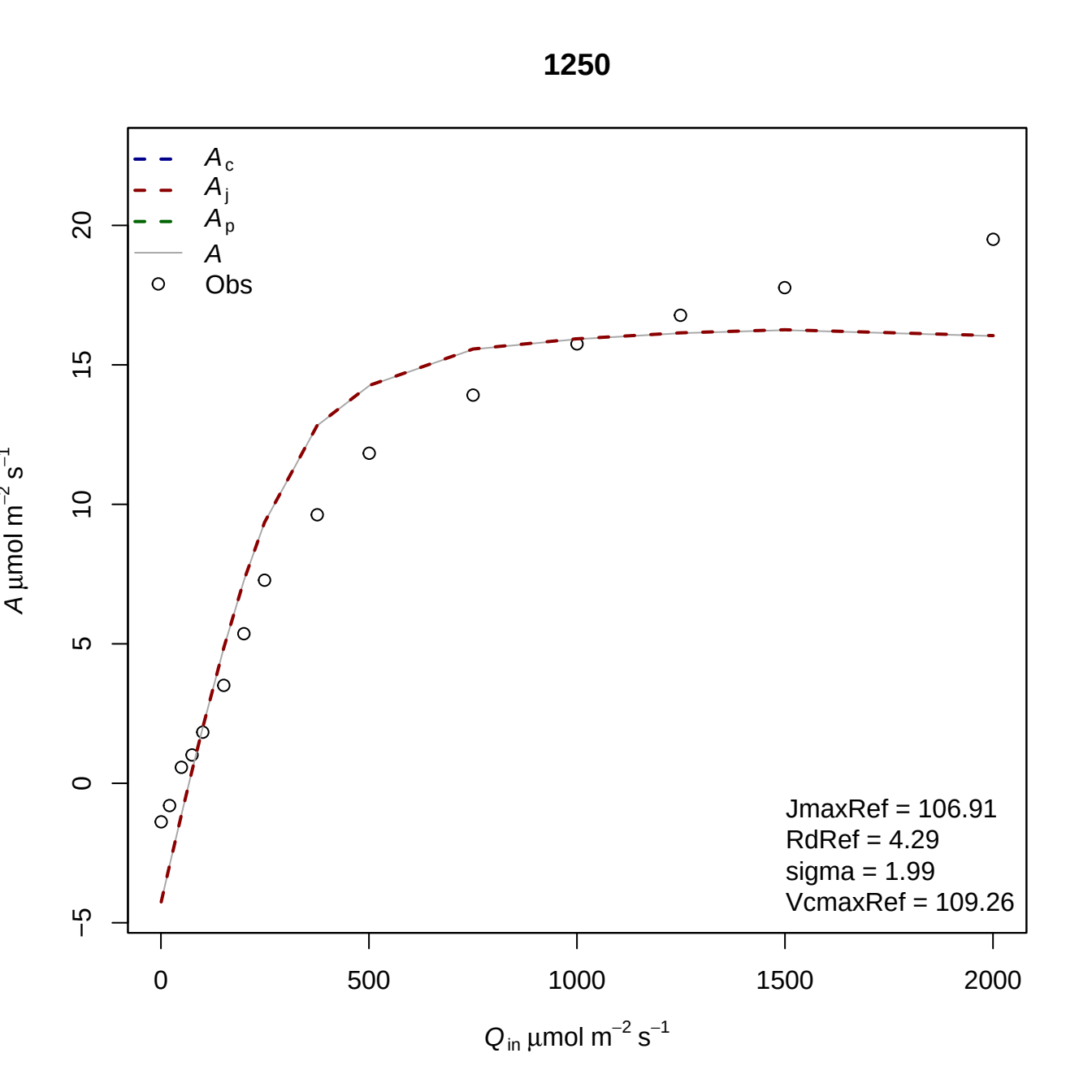

1245
A_c
A_j
A_p
A
Obs
JmaxRef = 99.3
RdRef = 4.33
sigma = 2.24
VcmaxRef = 112.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

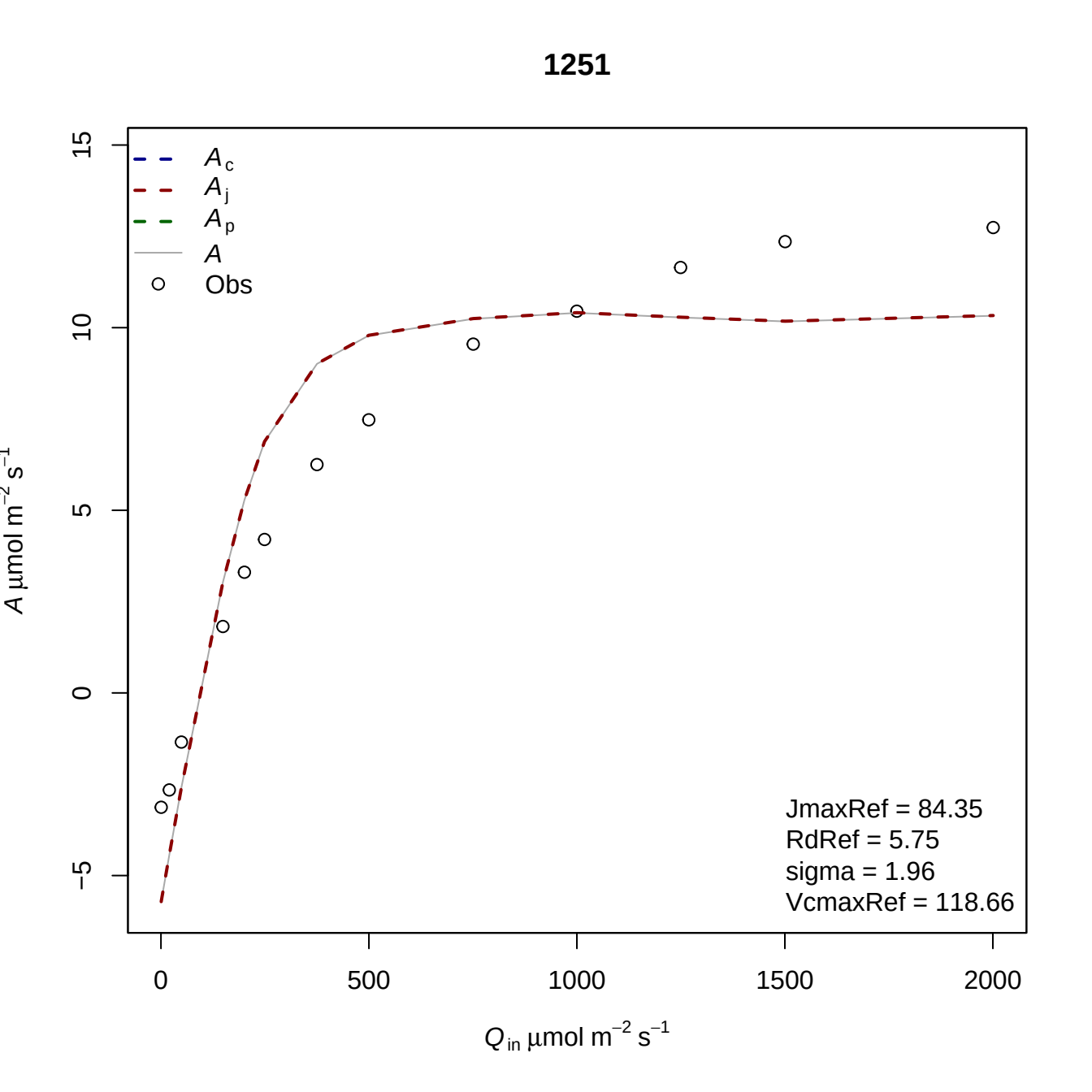

1246
A_c
A_j
A_p
A
Obs
JmaxRef = 80.56
RdRef = 4.28
sigma = 1.69
VcmaxRef = 116.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

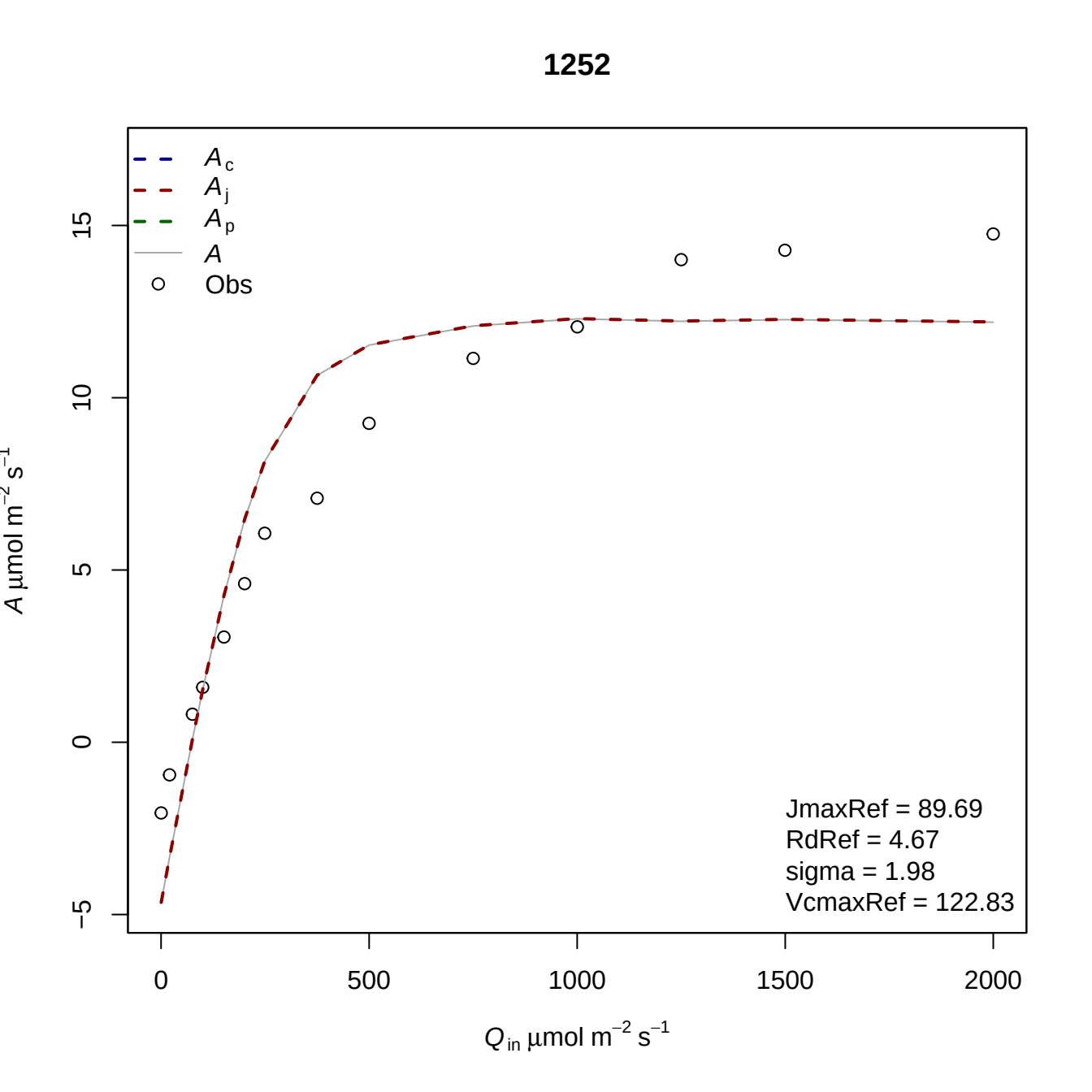

1247
A_c
A_j
A_p
A
Obs
JmaxRef = 82.46
RdRef = 3.97
sigma = 1.7
VcmaxRef = 236.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

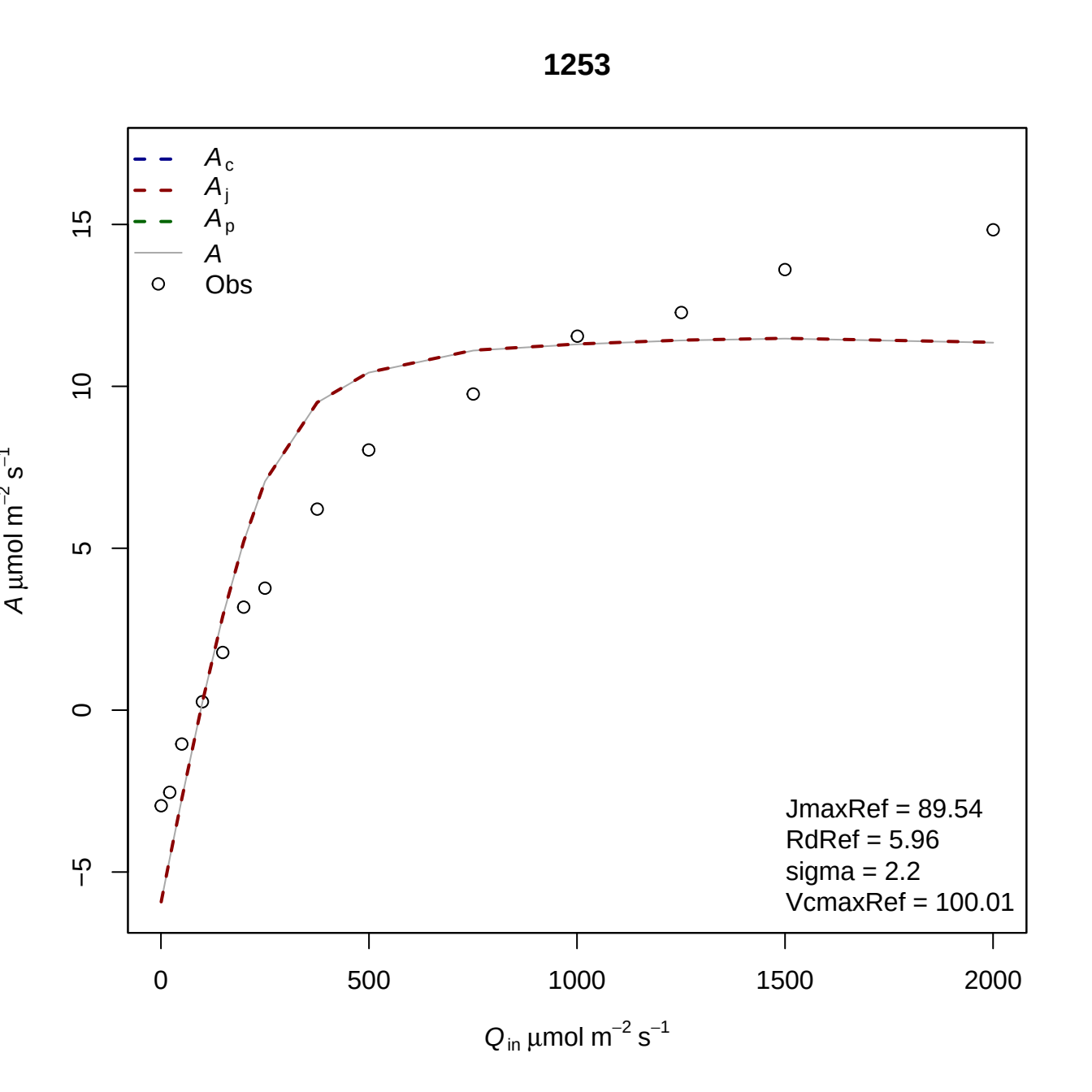

1248
A_c
A_j
A_p
A
Obs
JmaxRef = 95.99
RdRef = 5.5
sigma = 2.49
VcmaxRef = 116.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

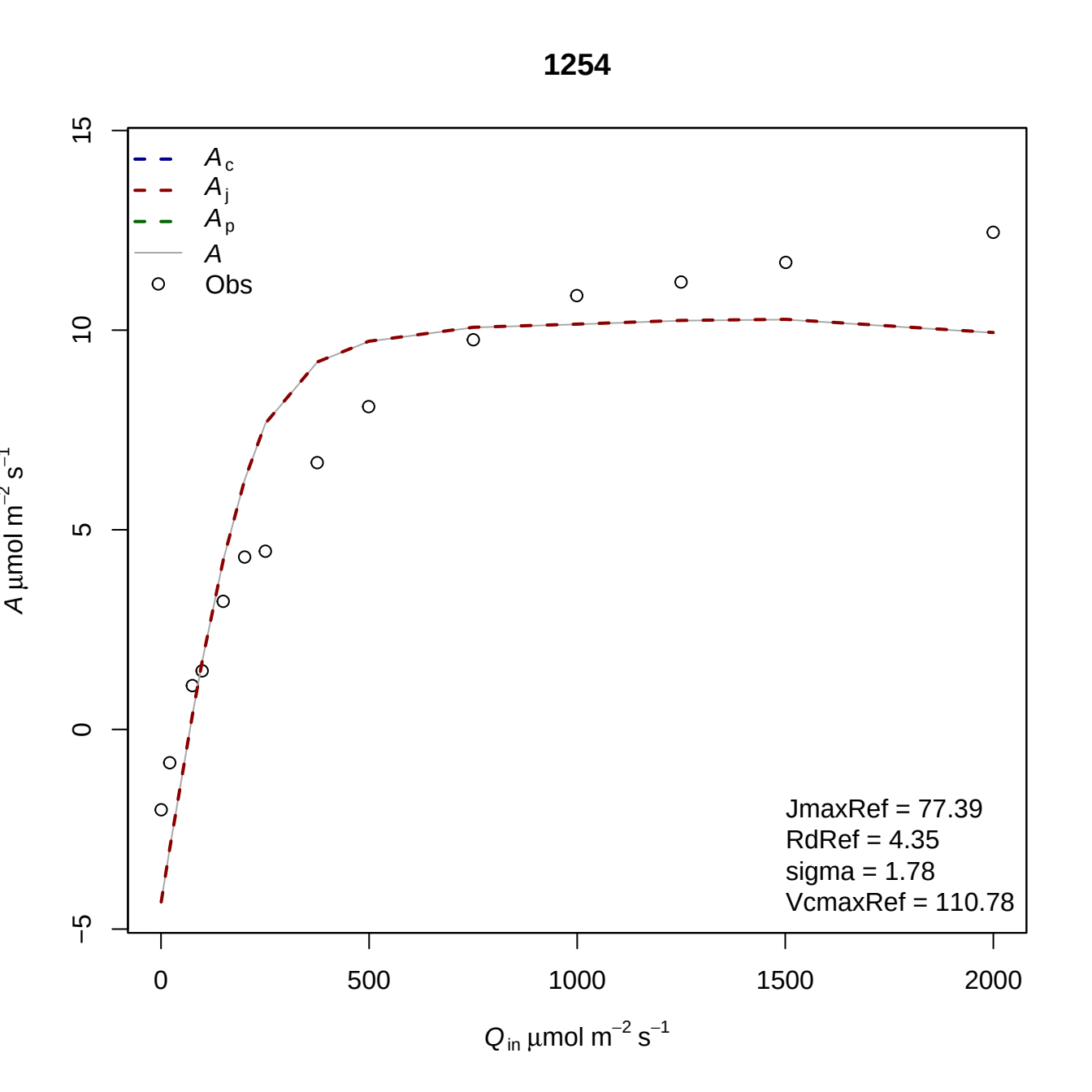

1249
A_c
A_j
A_p
A
Obs
JmaxRef = 54.79
RdRef = 1.49
sigma = 2.29
VcmaxRef = 386.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

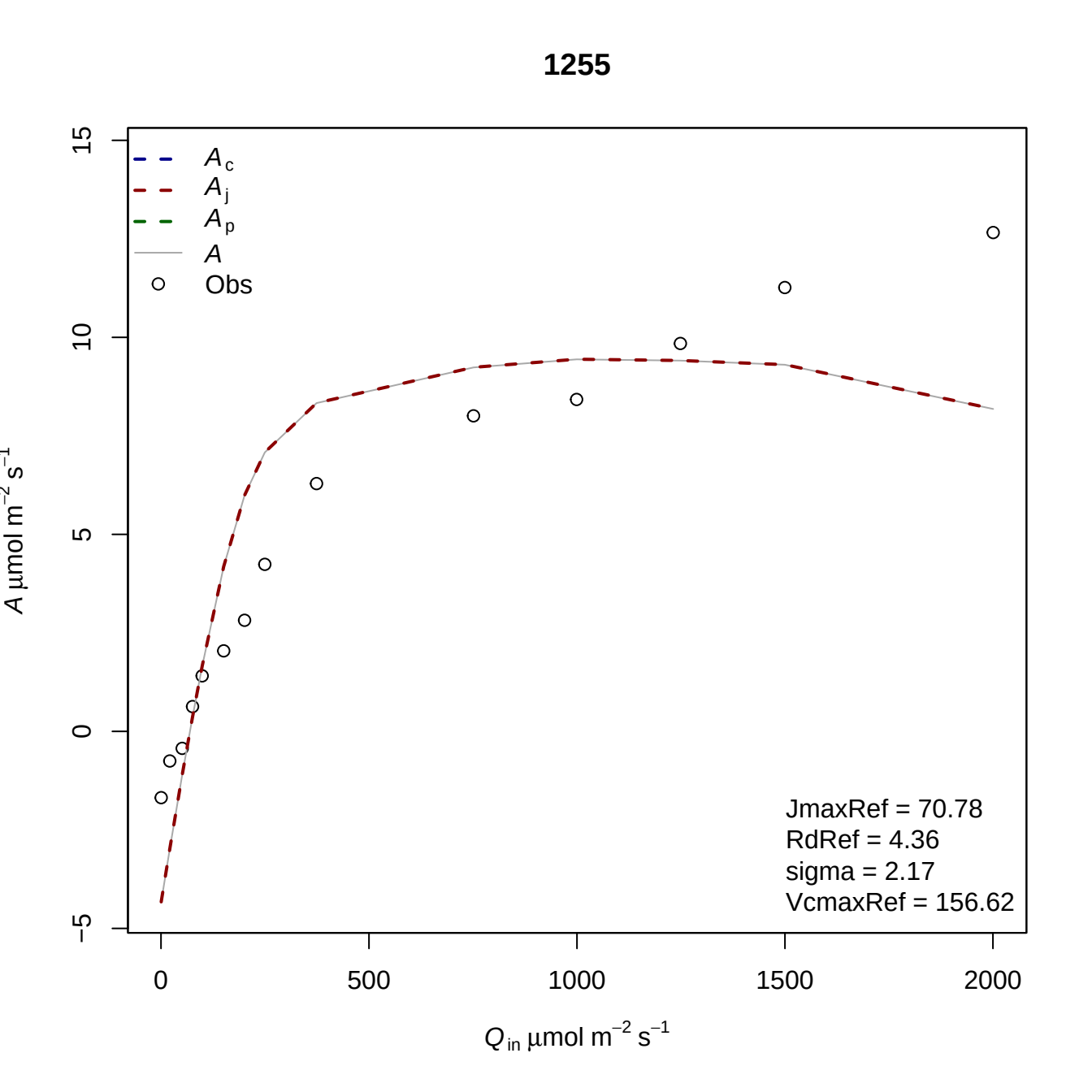

1250
A_c
A_j
A_p
A
Obs
JmaxRef = 106.91
RdRef = 4.29
sigma = 1.99
VcmaxRef = 109.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

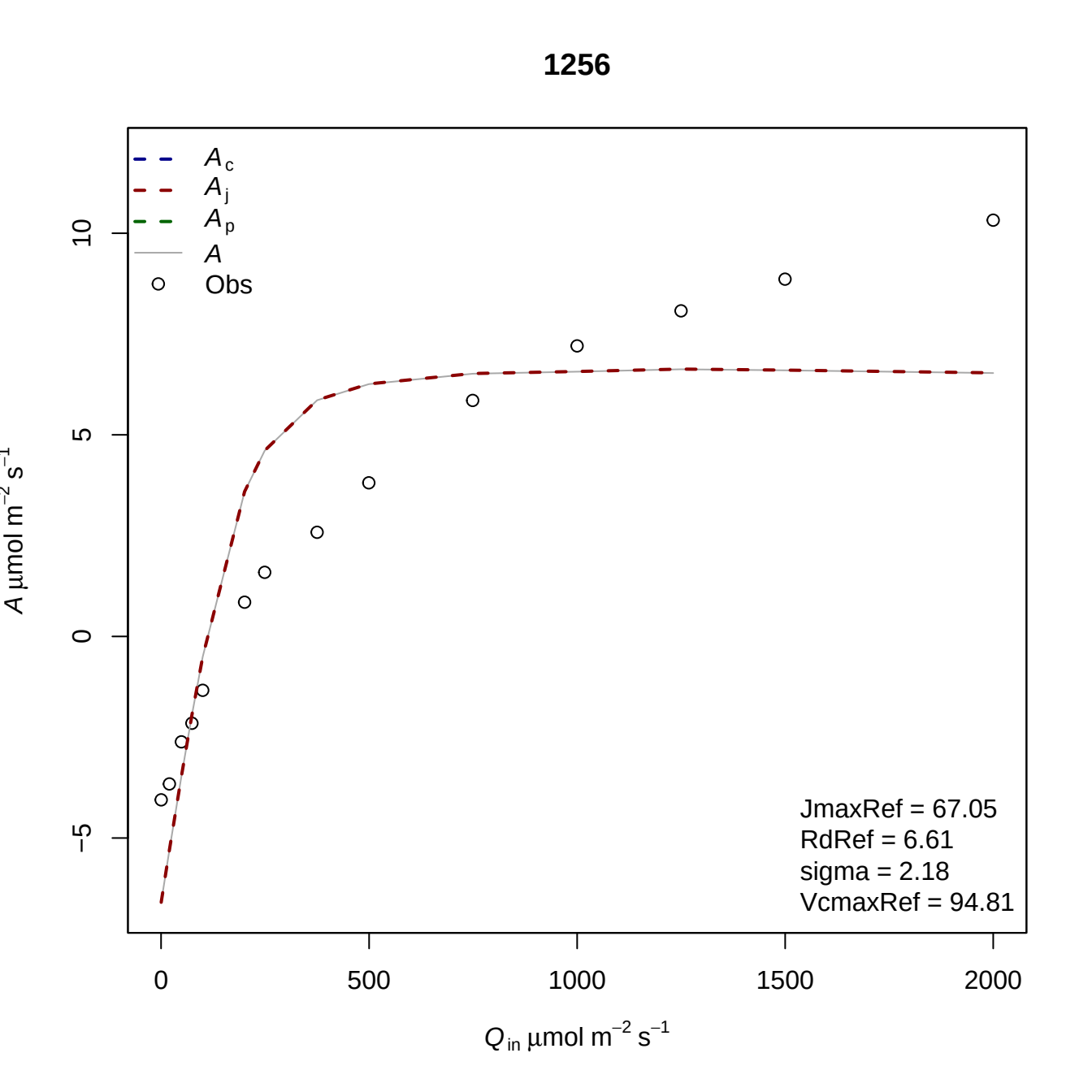

1251
A_c
A_j
A_p
A
Obs
JmaxRef = 84.35
RdRef = 5.75
sigma = 1.96
VcmaxRef = 118.66
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

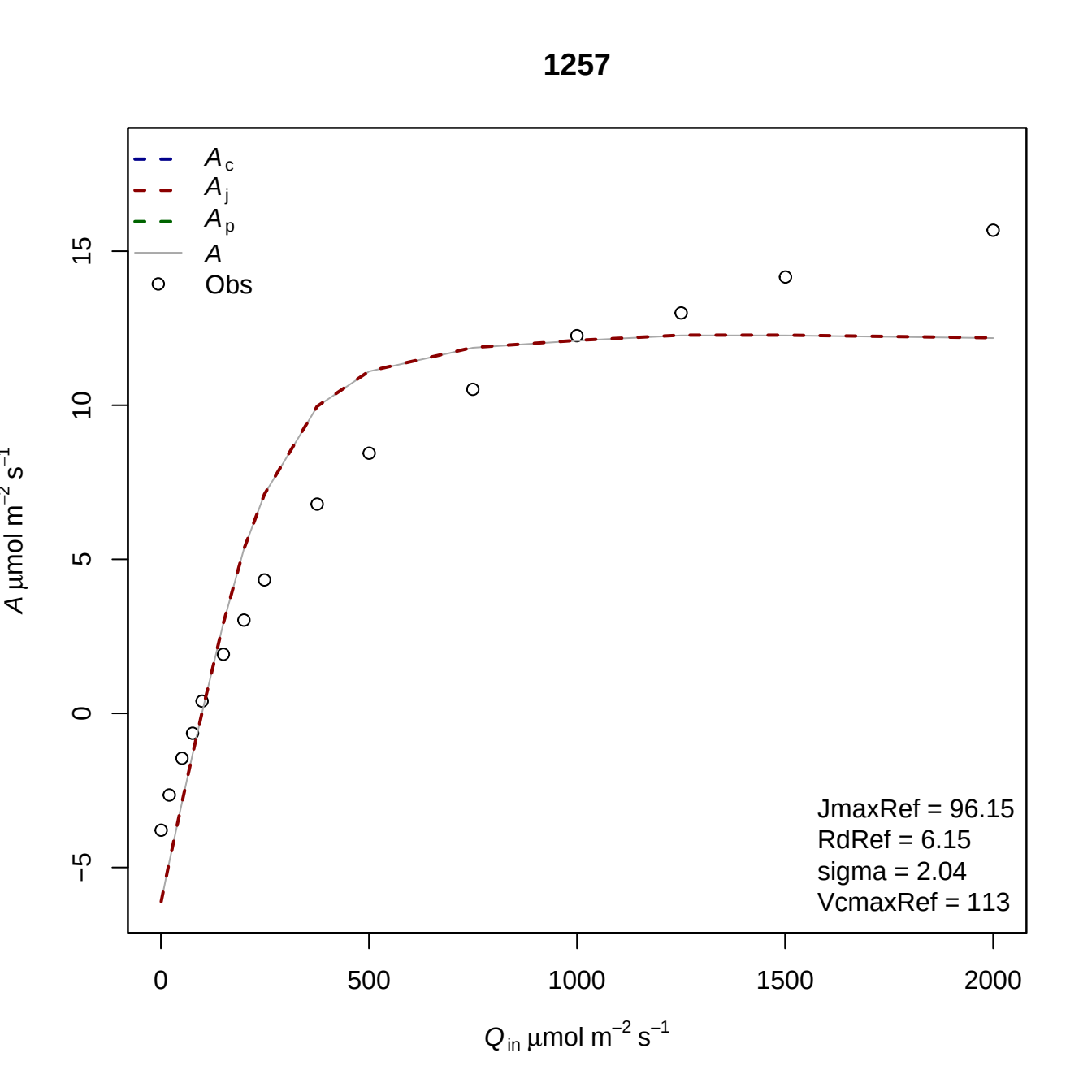

1252
A_c
A_j
A_p
A
Obs
JmaxRef = 89.69
RdRef = 4.67
sigma = 1.98
VcmaxRef = 122.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

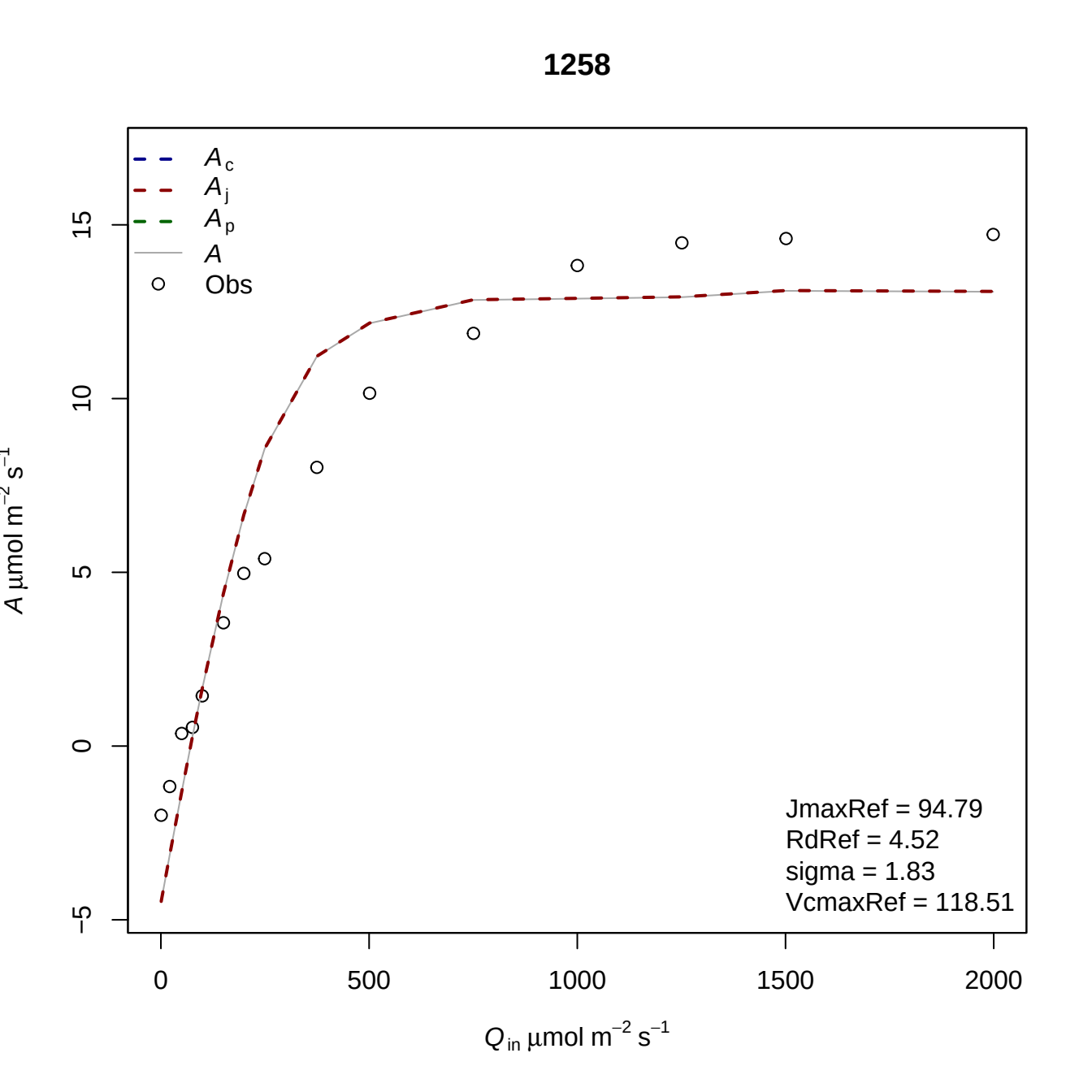

1253
A_c
A_j
A_p
A
Obs
JmaxRef = 89.54
RdRef = 5.96
sigma = 2.2
VcmaxRef = 100.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

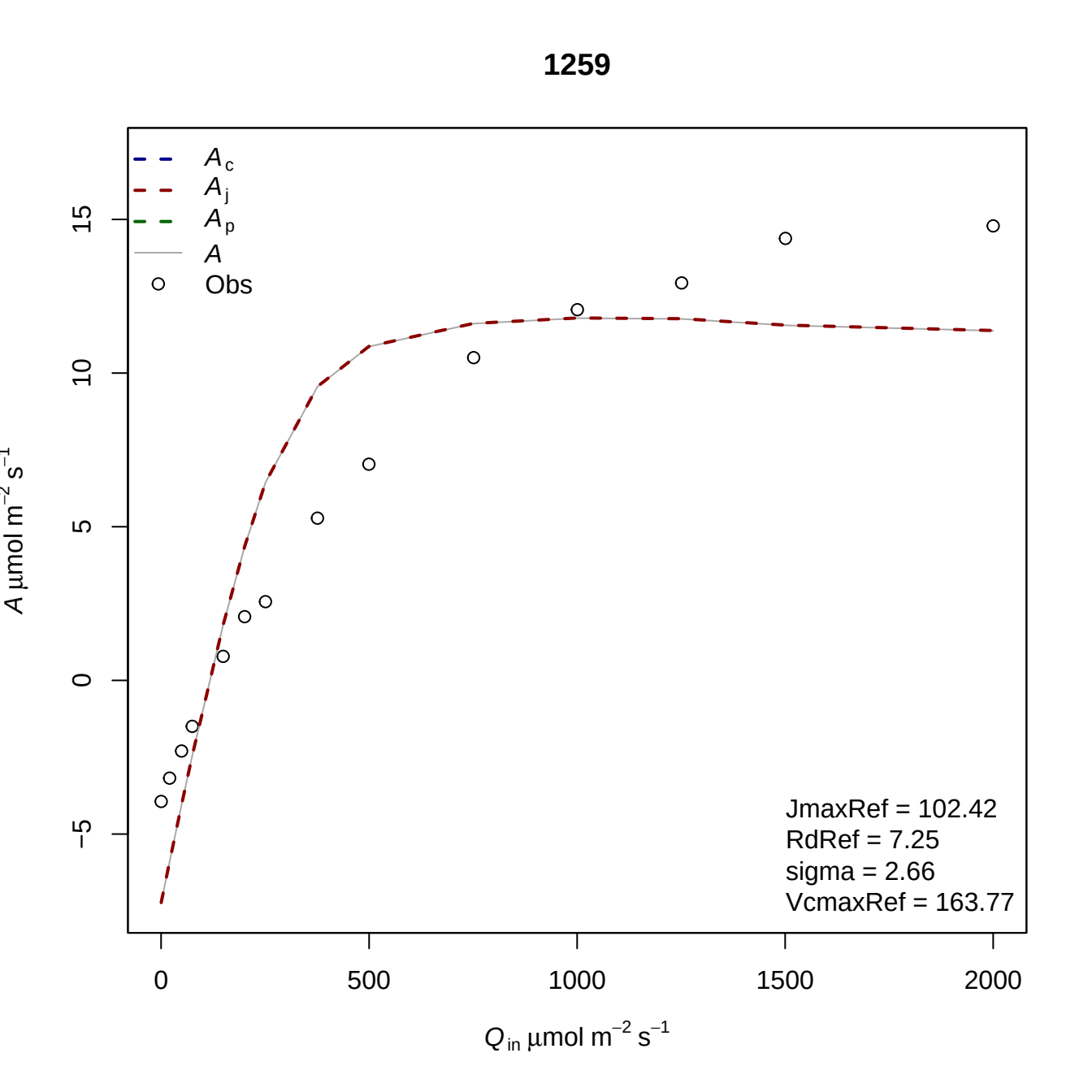

1254
A_c
A_j
A_p
A
Obs
JmaxRef = 77.39
RdRef = 4.35
sigma = 1.78
VcmaxRef = 110.78
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

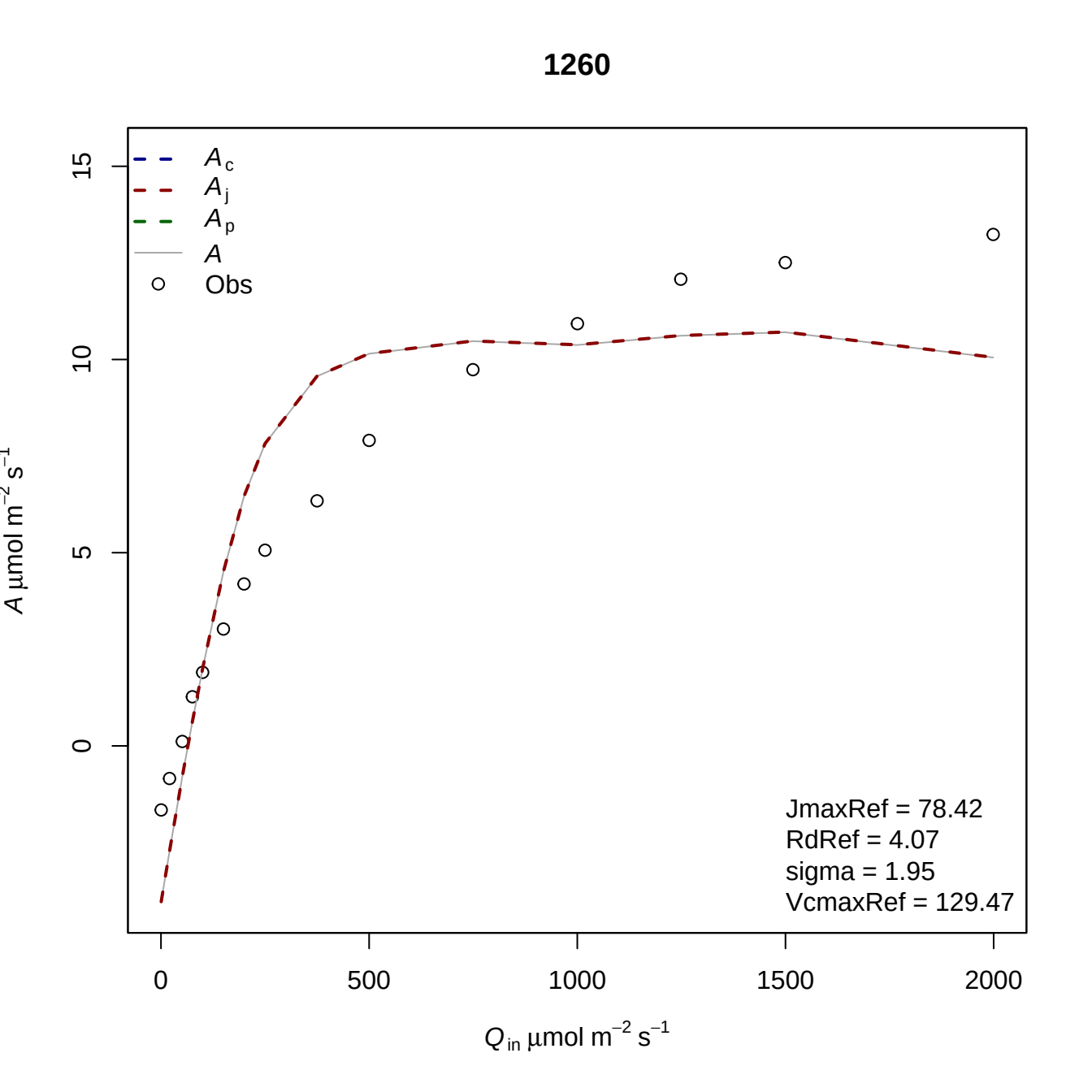

1255
A_c
A_j
A_p
A
Obs
JmaxRef = 70.78
RdRef = 4.36
sigma = 2.17
VcmaxRef = 156.62
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

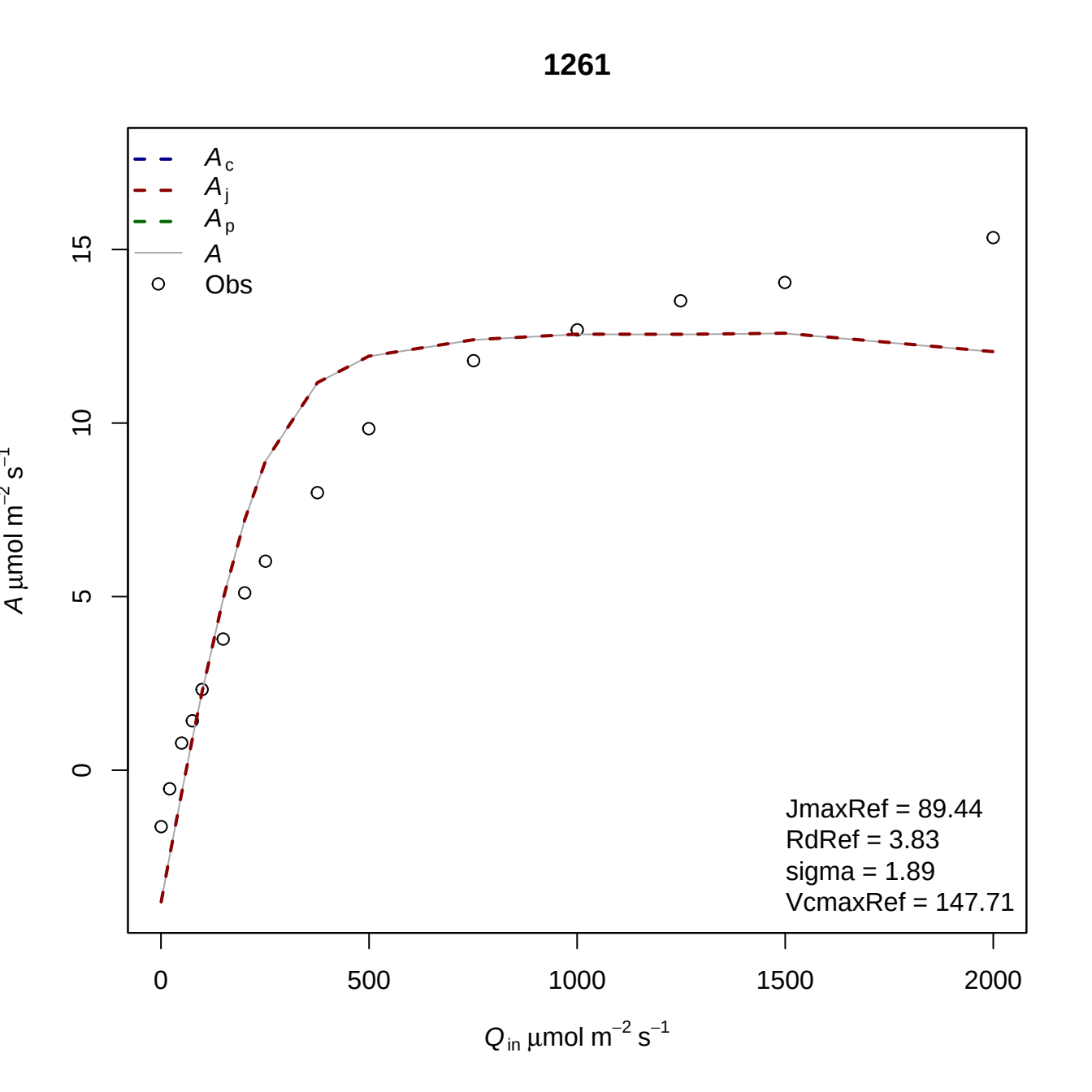

1256
A_c
A_j
A_p
A
Obs
JmaxRef = 67.05
RdRef = 6.61
sigma = 2.18
VcmaxRef = 94.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

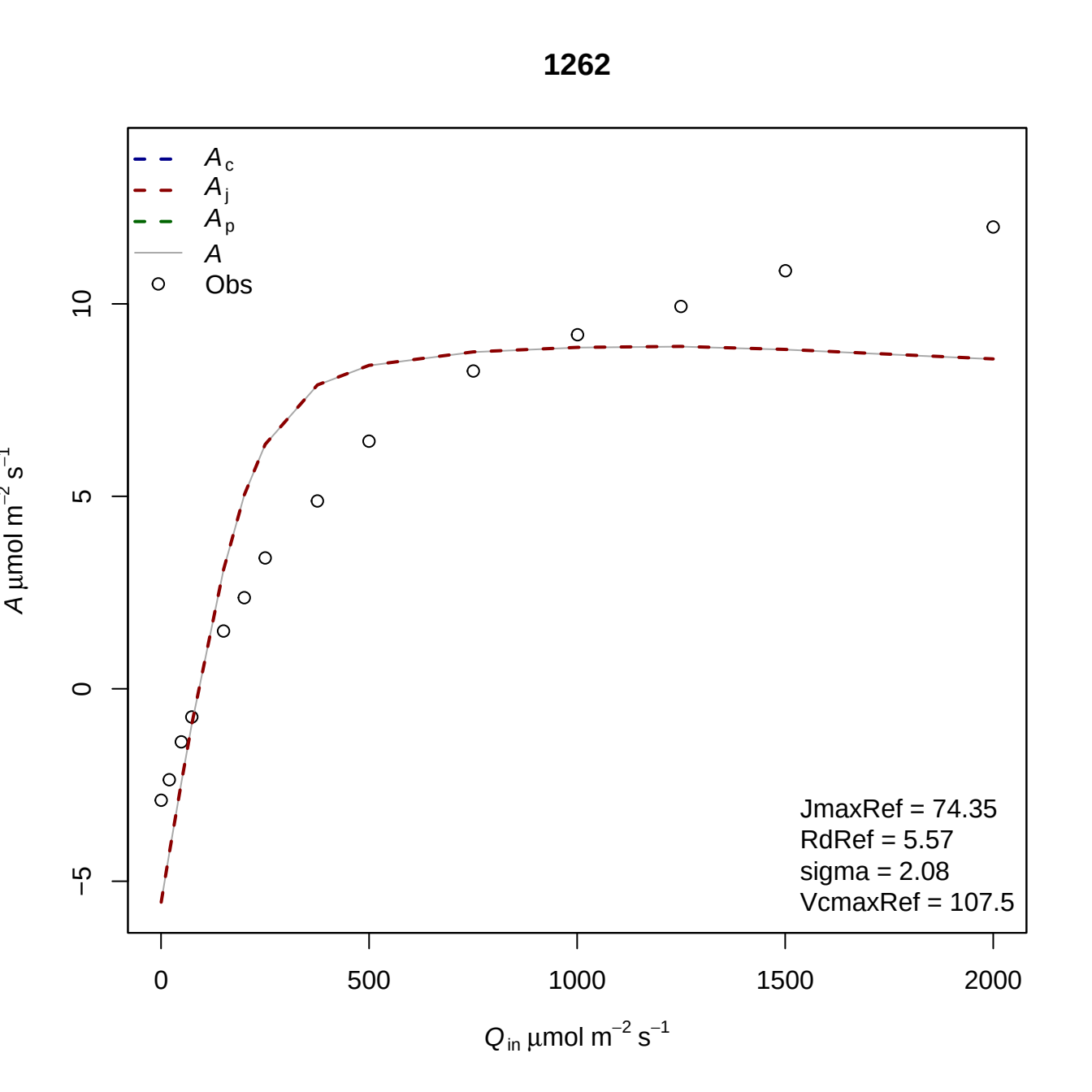

1257
A_c
A_j
A_p
A
Obs
JmaxRef = 96.15
RdRef = 6.15
sigma = 2.04
VcmaxRef = 113
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

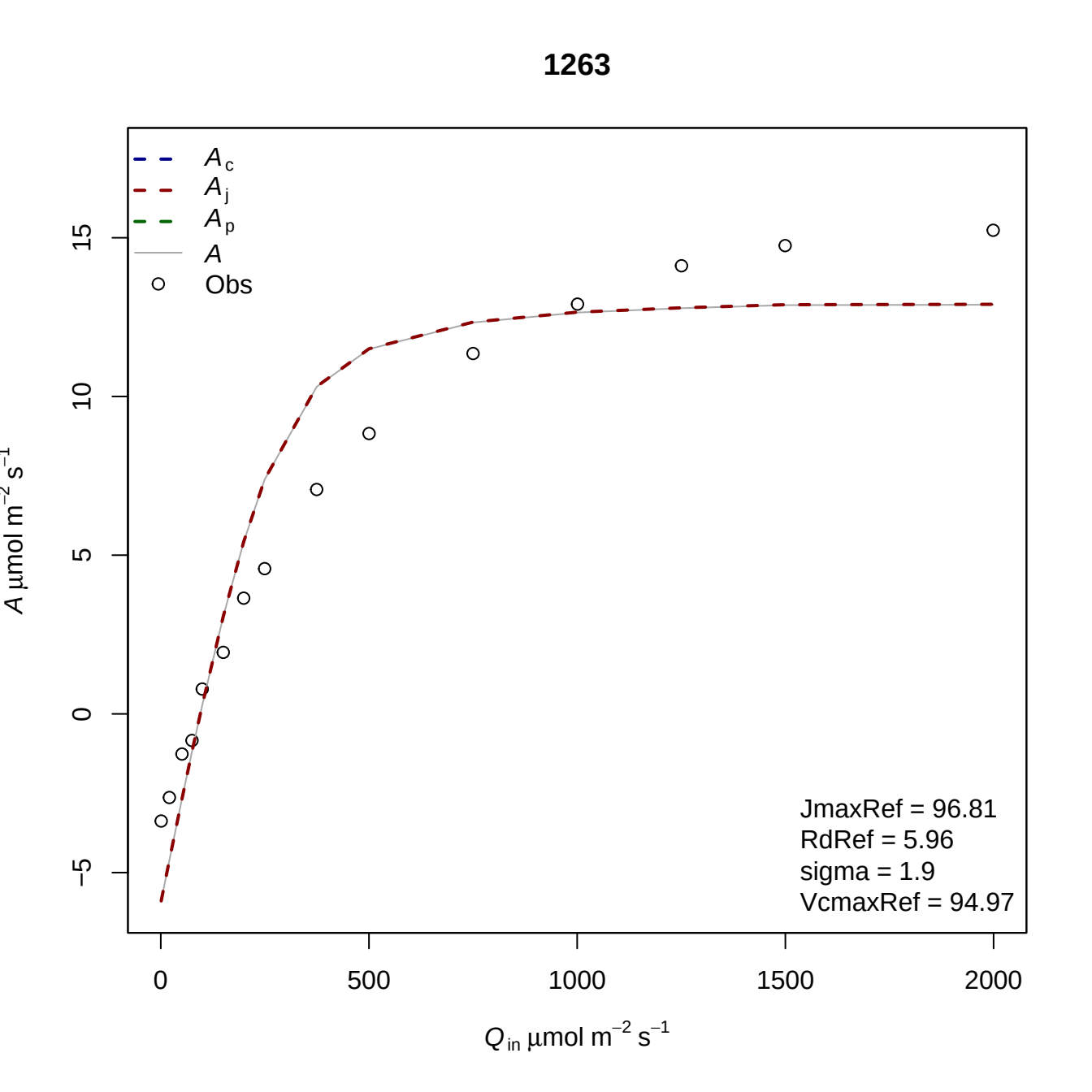

1258
A_c
A_j
A_p
A
Obs
JmaxRef = 94.79
RdRef = 4.52
sigma = 1.83
VcmaxRef = 118.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

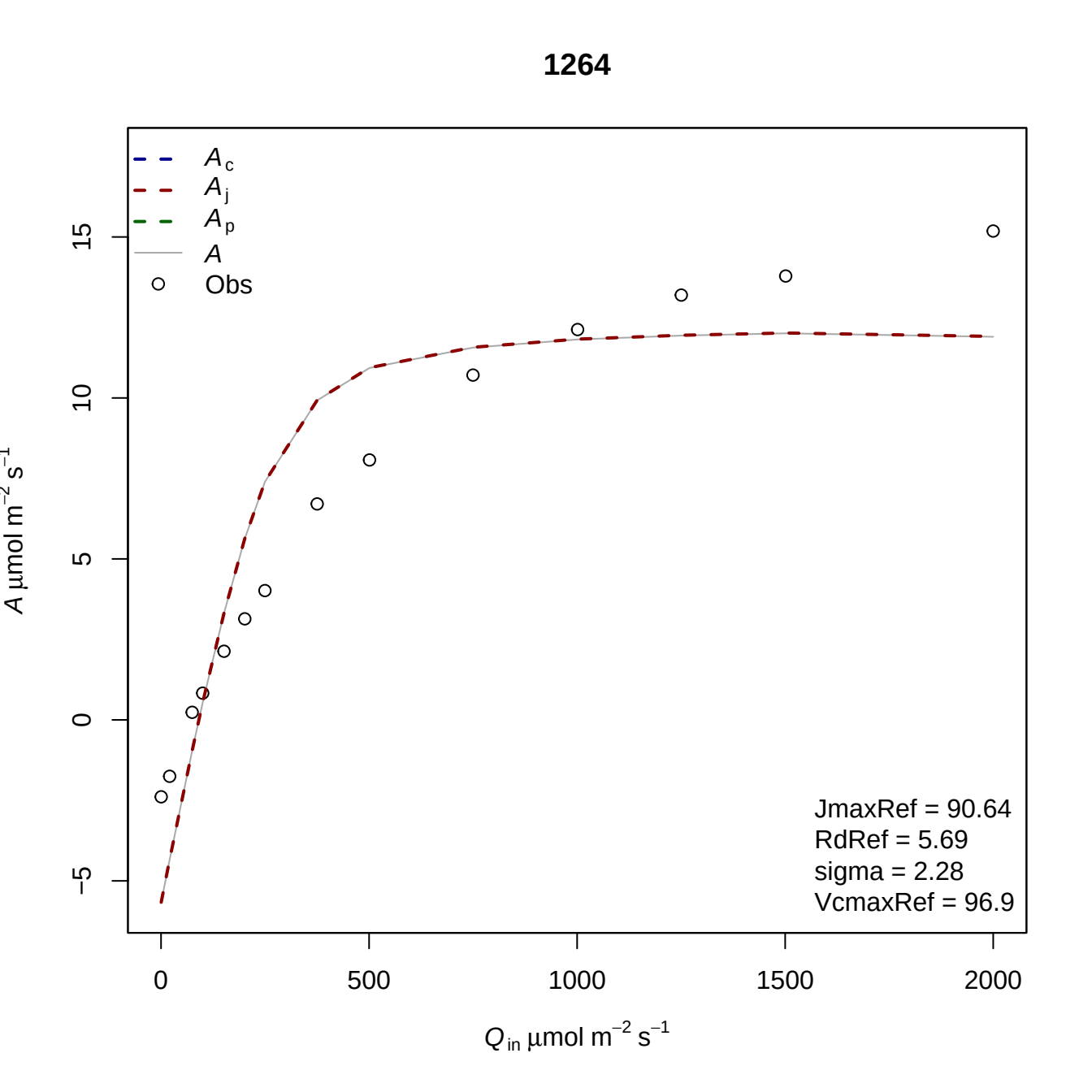

1259
A_c
A_j
A_p
A
Obs
JmaxRef = 102.42
RdRef = 7.25
sigma = 2.66
VcmaxRef = 163.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

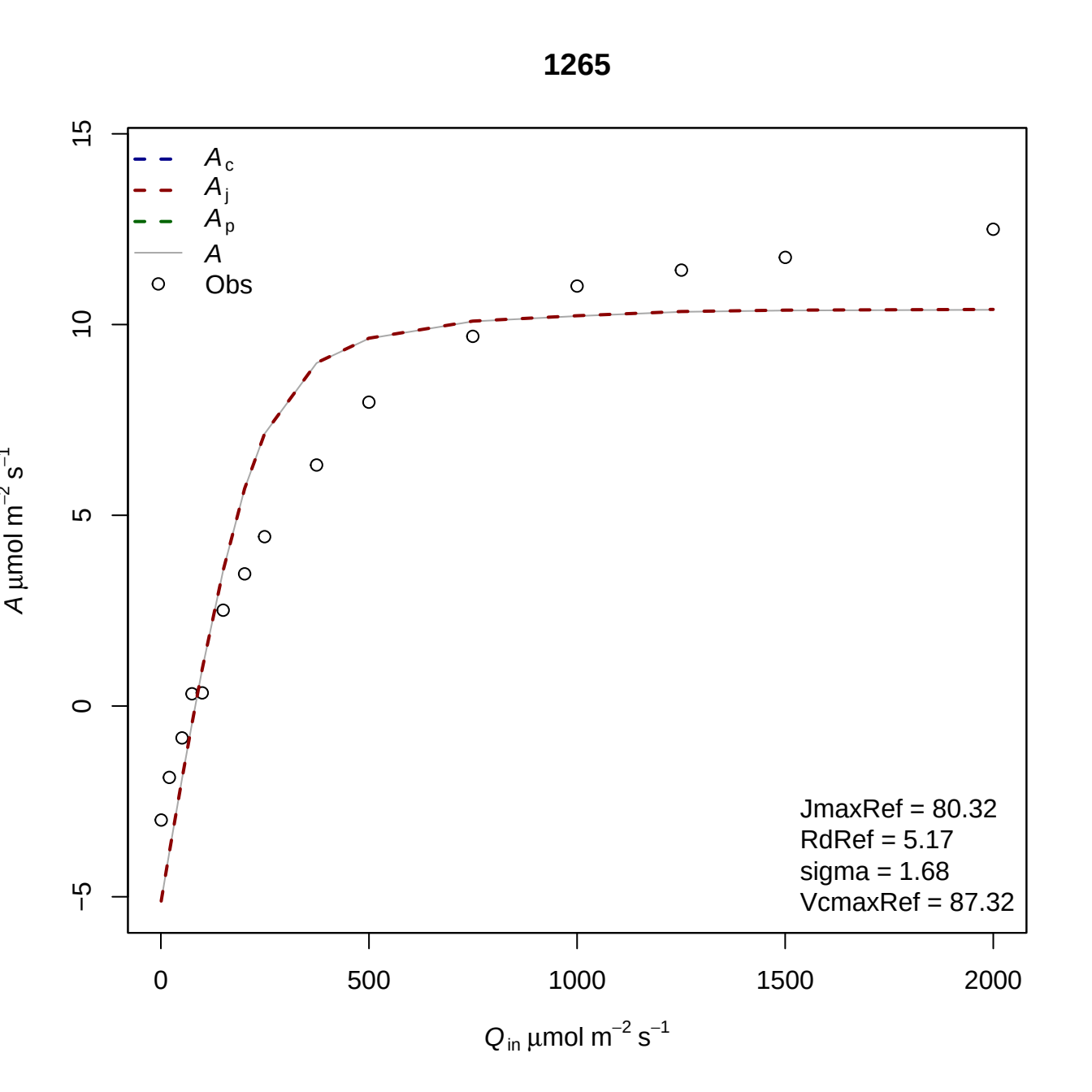

1260
A_c
A_j
A_p
A
Obs
JmaxRef = 78.42
RdRef = 4.07
sigma = 1.95
VcmaxRef = 129.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

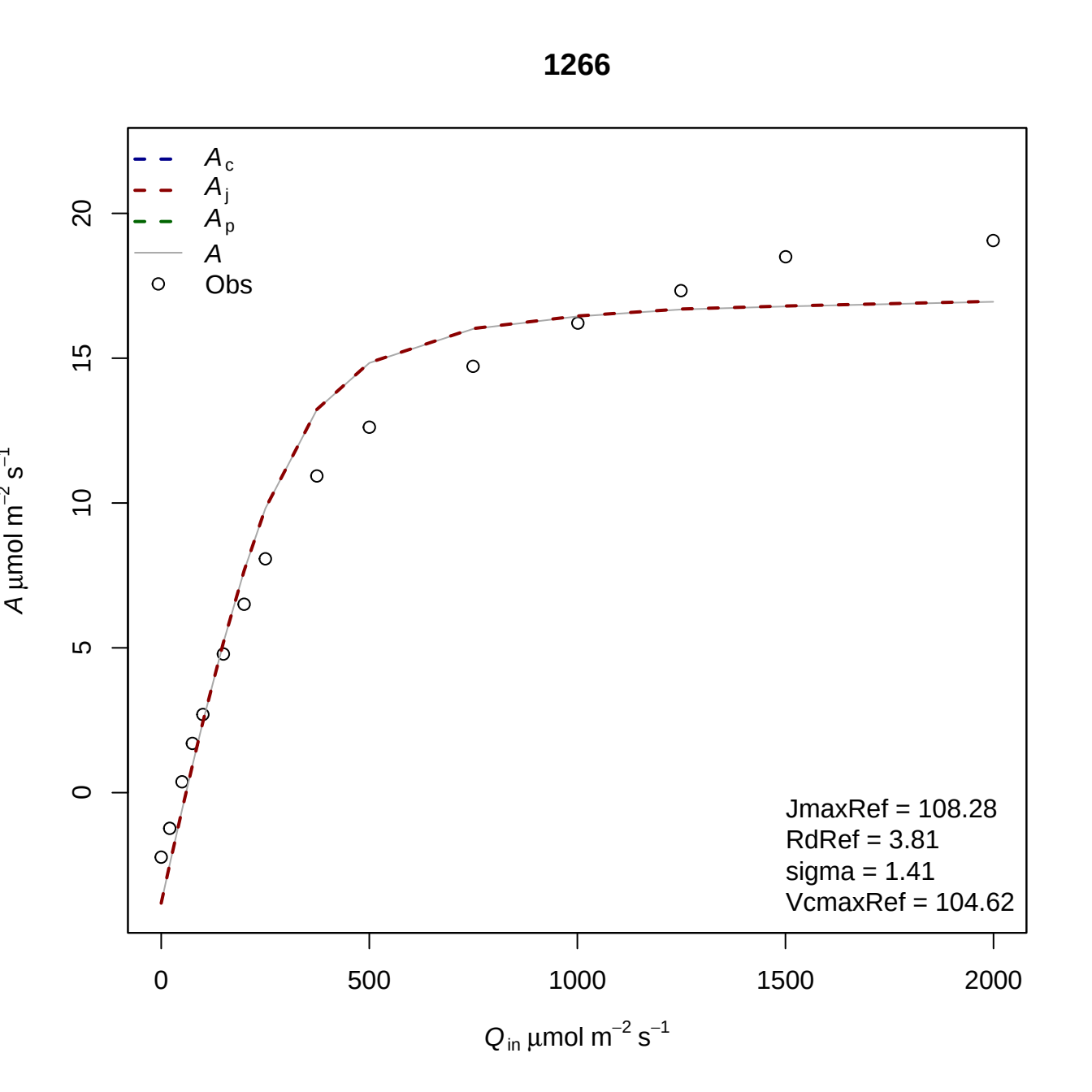

1261
A_c
A_j
A_p
A
Obs
JmaxRef = 89.44
RdRef = 3.83
sigma = 1.89
VcmaxRef = 147.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

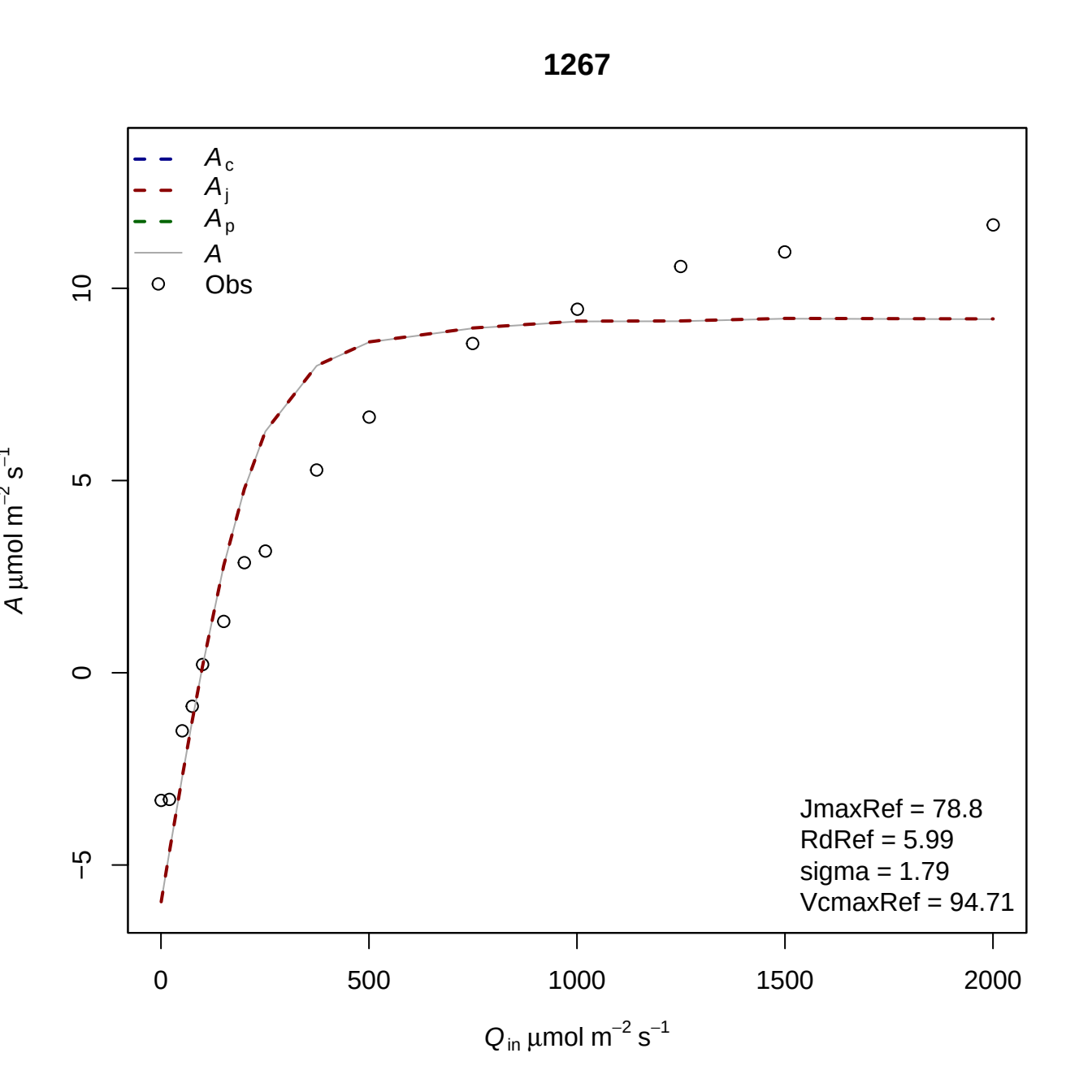

1262
A_c
A_j
A_p
A
Obs
JmaxRef = 74.35
RdRef = 5.57
sigma = 2.08
VcmaxRef = 107.5
A µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

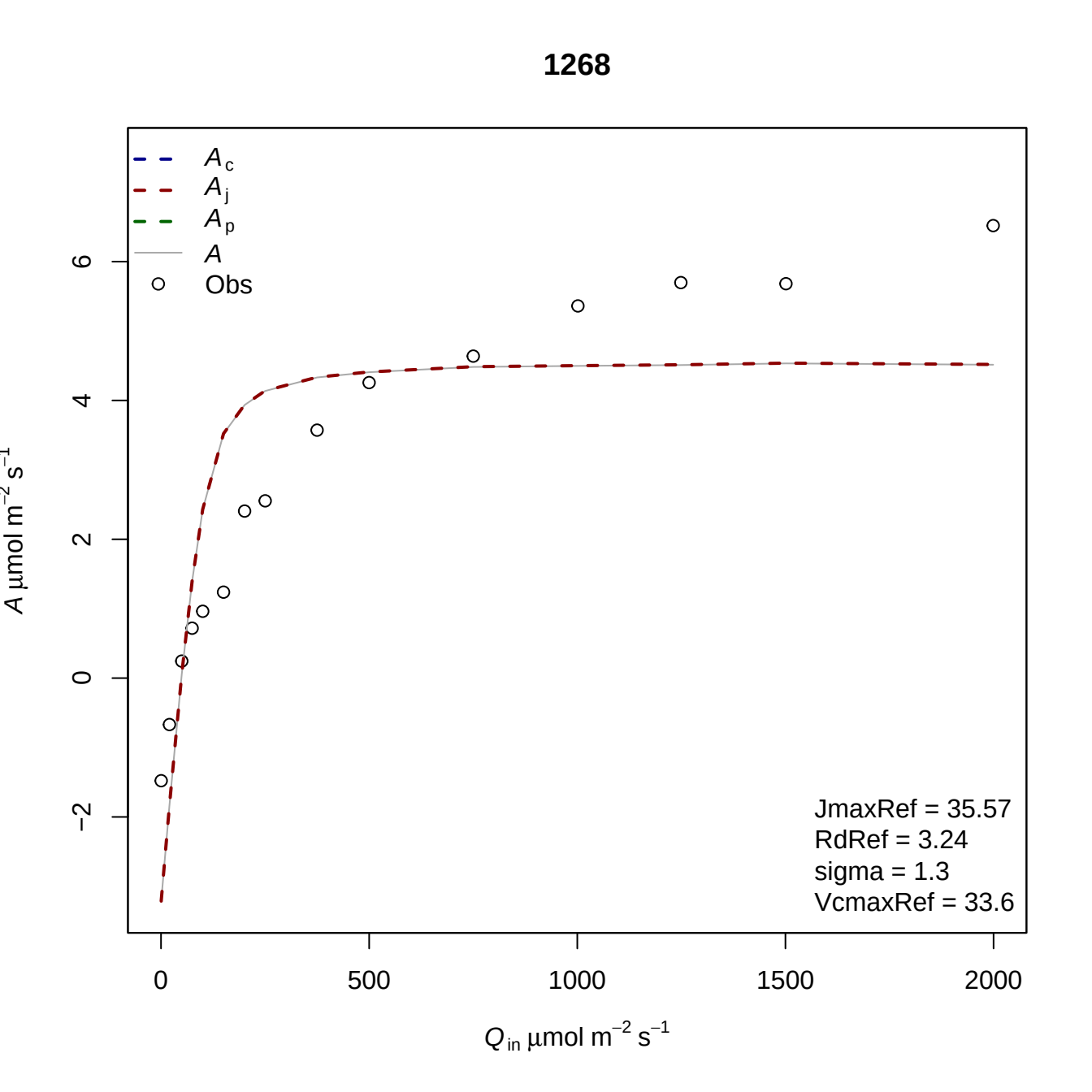

1263
A_c
A_j
A_p
A
Obs
JmaxRef = 96.81
RdRef = 5.96
sigma = 1.9
VcmaxRef = 94.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

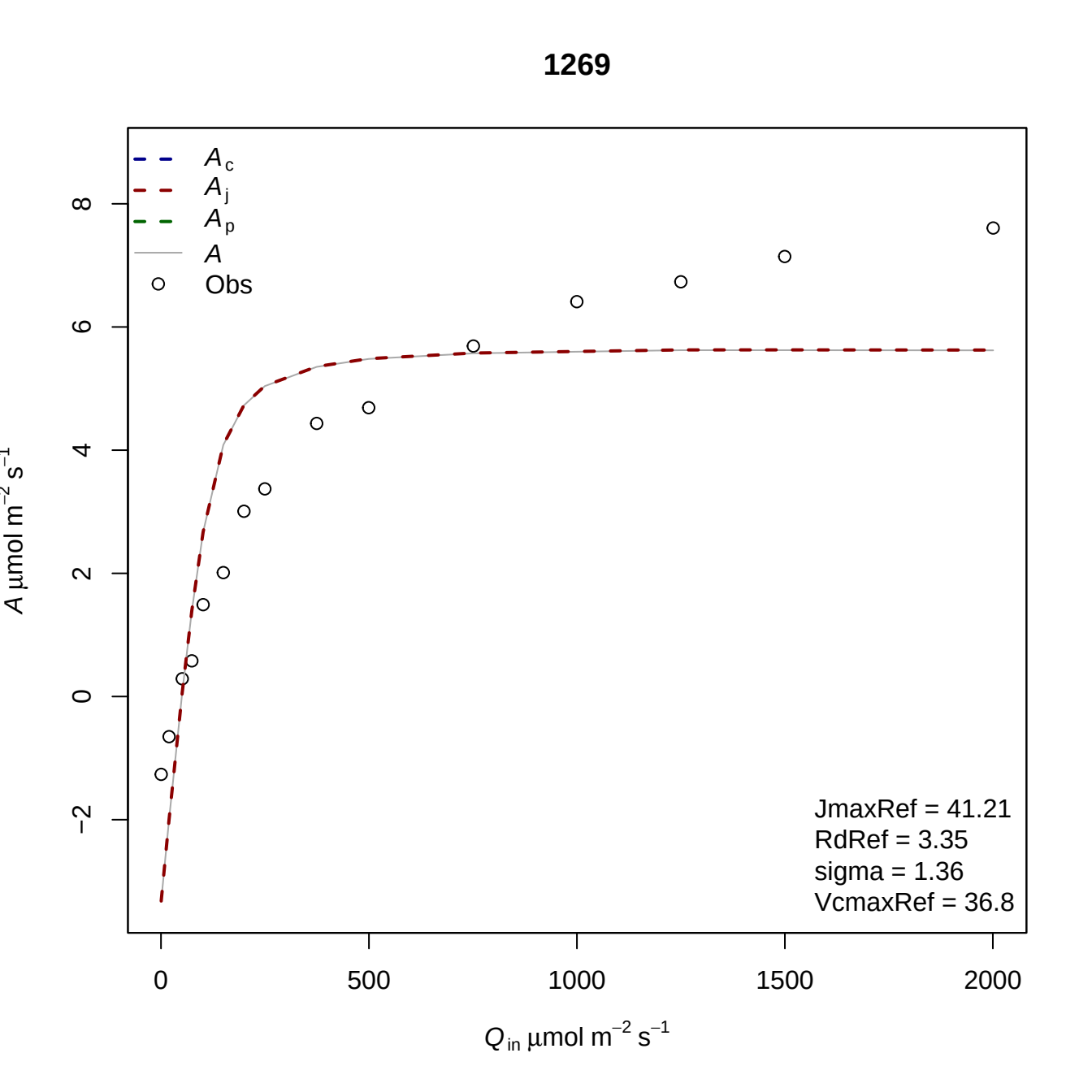

1264
A_c
A_j
A_p
A
Obs
JmaxRef = 90.64
RdRef = 5.69
sigma = 2.28
VcmaxRef = 96.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

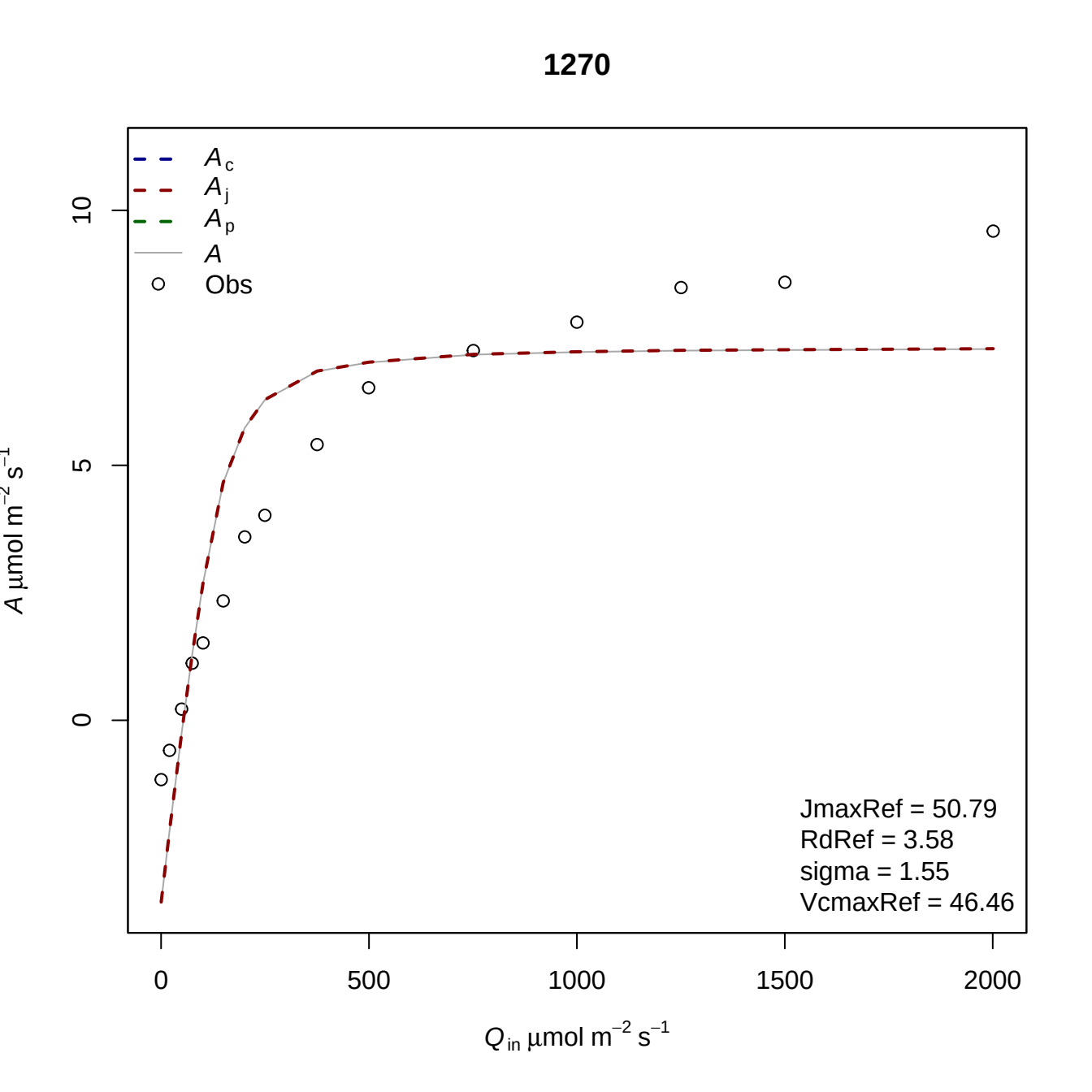

1265
A_c
A_j
A_p
A
Obs
JmaxRef = 80.32
RdRef = 5.17
sigma = 1.68
VcmaxRef = 87.32
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

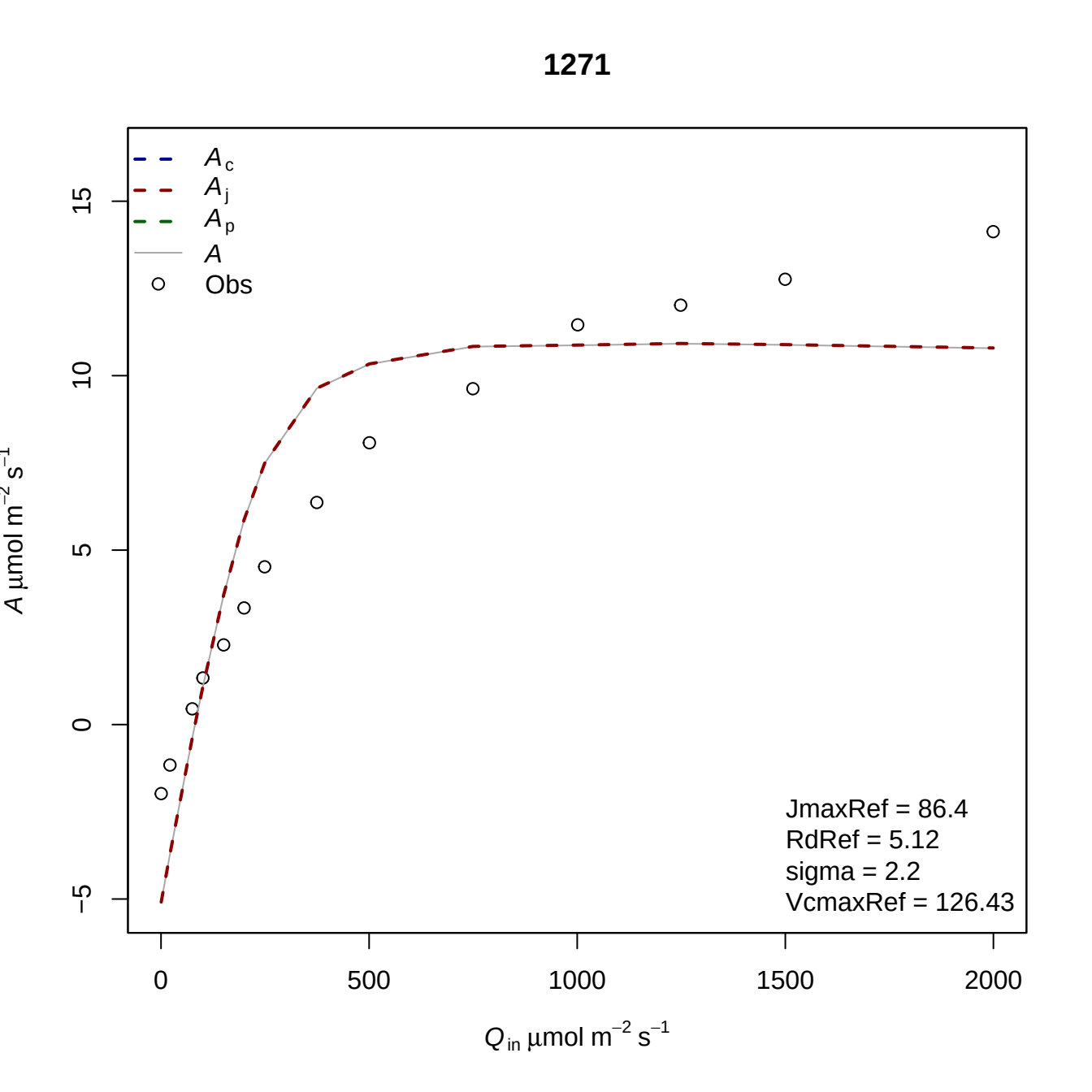

1266
A_c
A_j
A_p
A
Obs
JmaxRef = 108.28
RdRef = 3.81
sigma = 1.41
VcmaxRef = 104.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

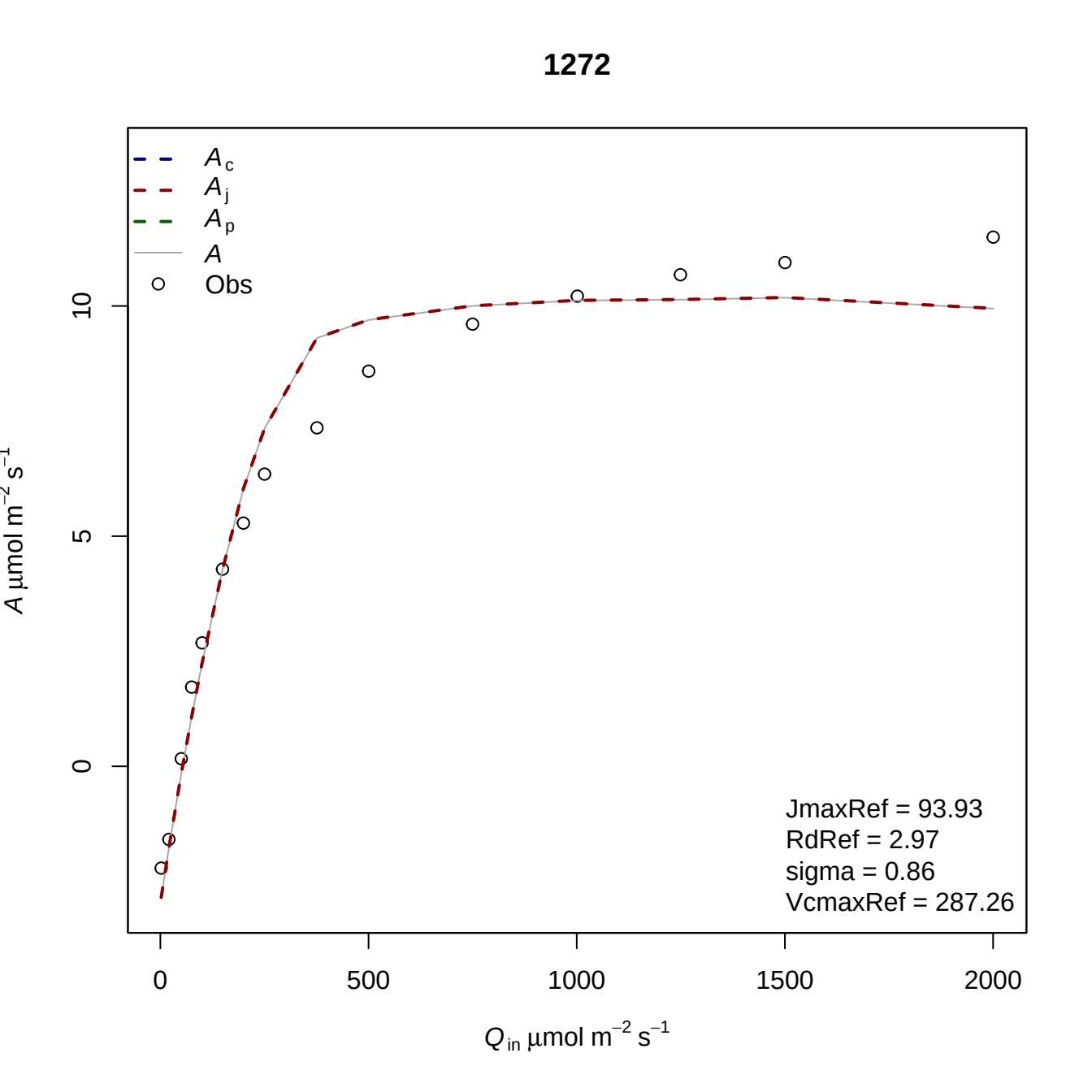

1267
A_c
A_j
A_p
A
Obs
JmaxRef = 78.8
RdRef = 5.99
sigma = 1.79
VcmaxRef = 94.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

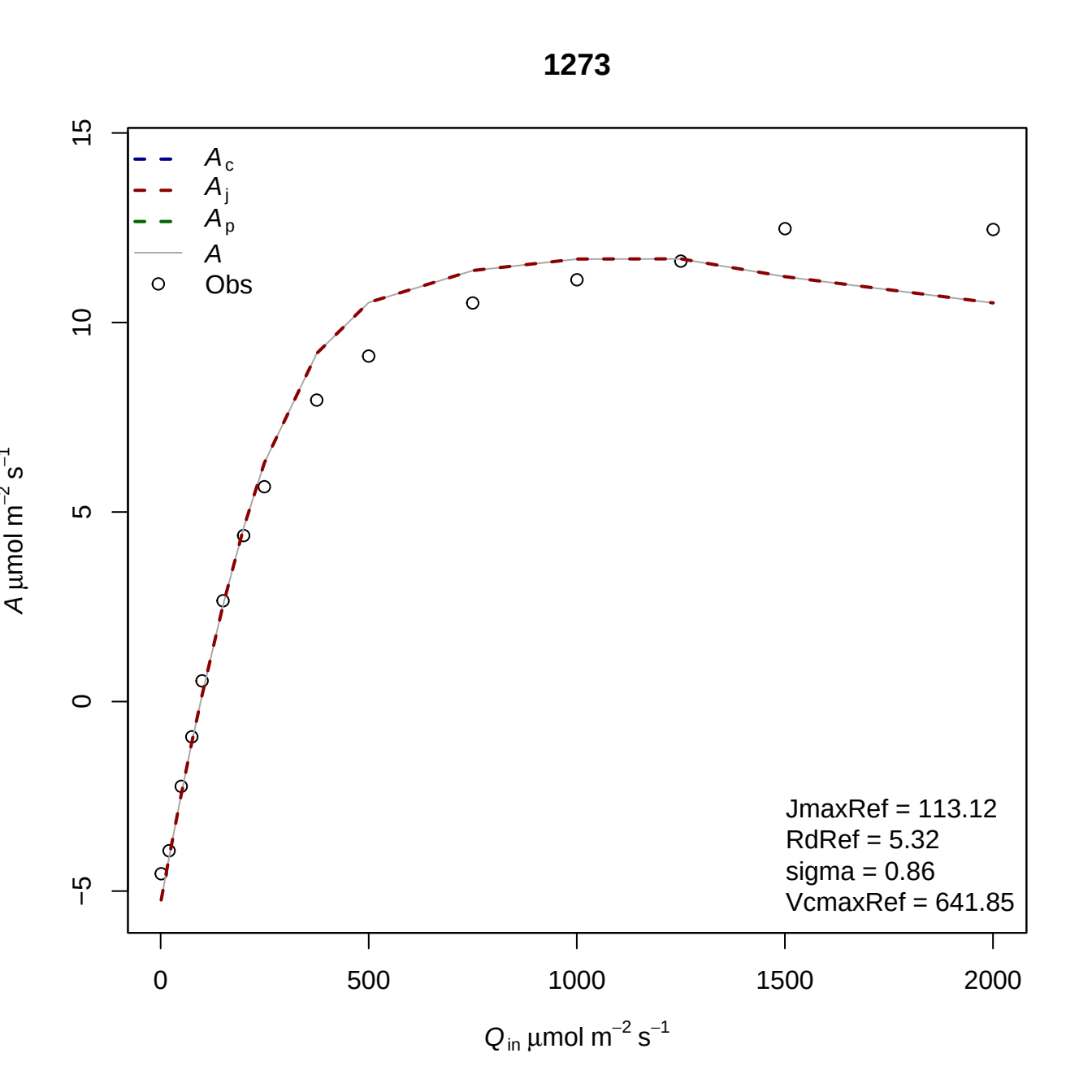

1268
A_c
A_j
A_p
A
Obs
JmaxRef = 35.57
RdRef = 3.24
sigma = 1.3
VcmaxRef = 33.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

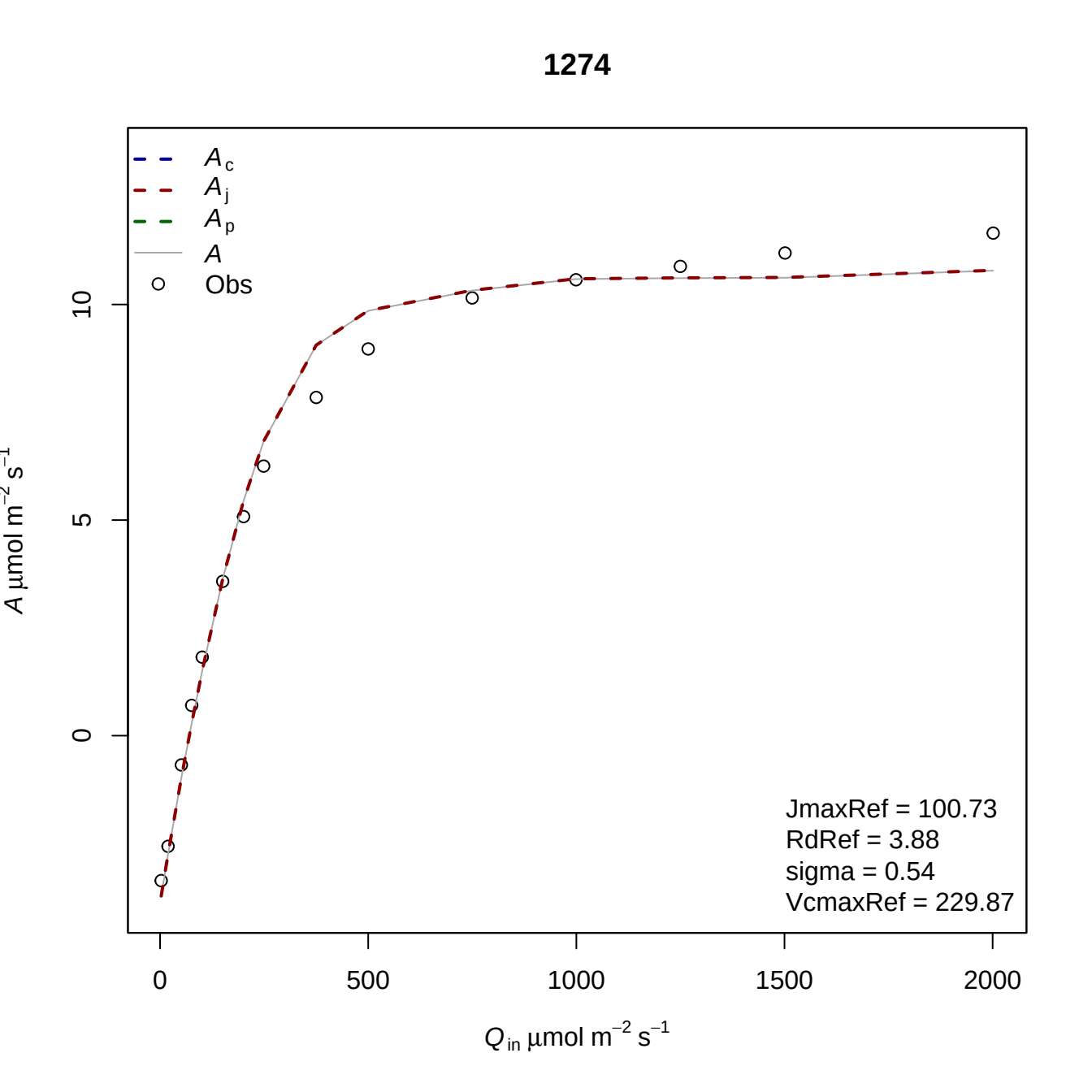

1269
A_c
A_j
A_p
A
Obs
JmaxRef = 41.21
RdRef = 3.35
sigma = 1.36
VcmaxRef = 36.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

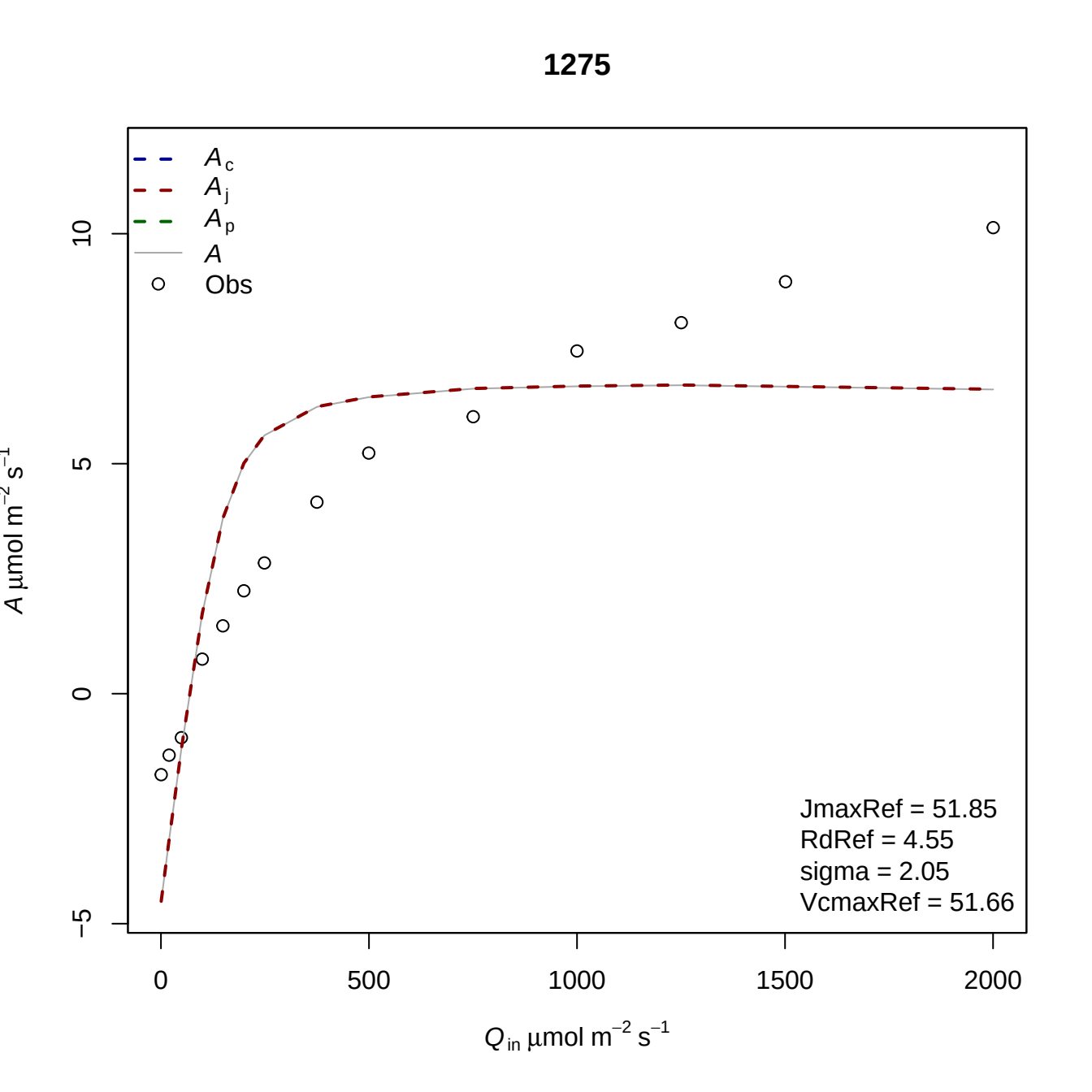

1270
A_c
A_j
A_p
A
Obs
JmaxRef = 50.79
RdRef = 3.58
sigma = 1.55
VcmaxRef = 46.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

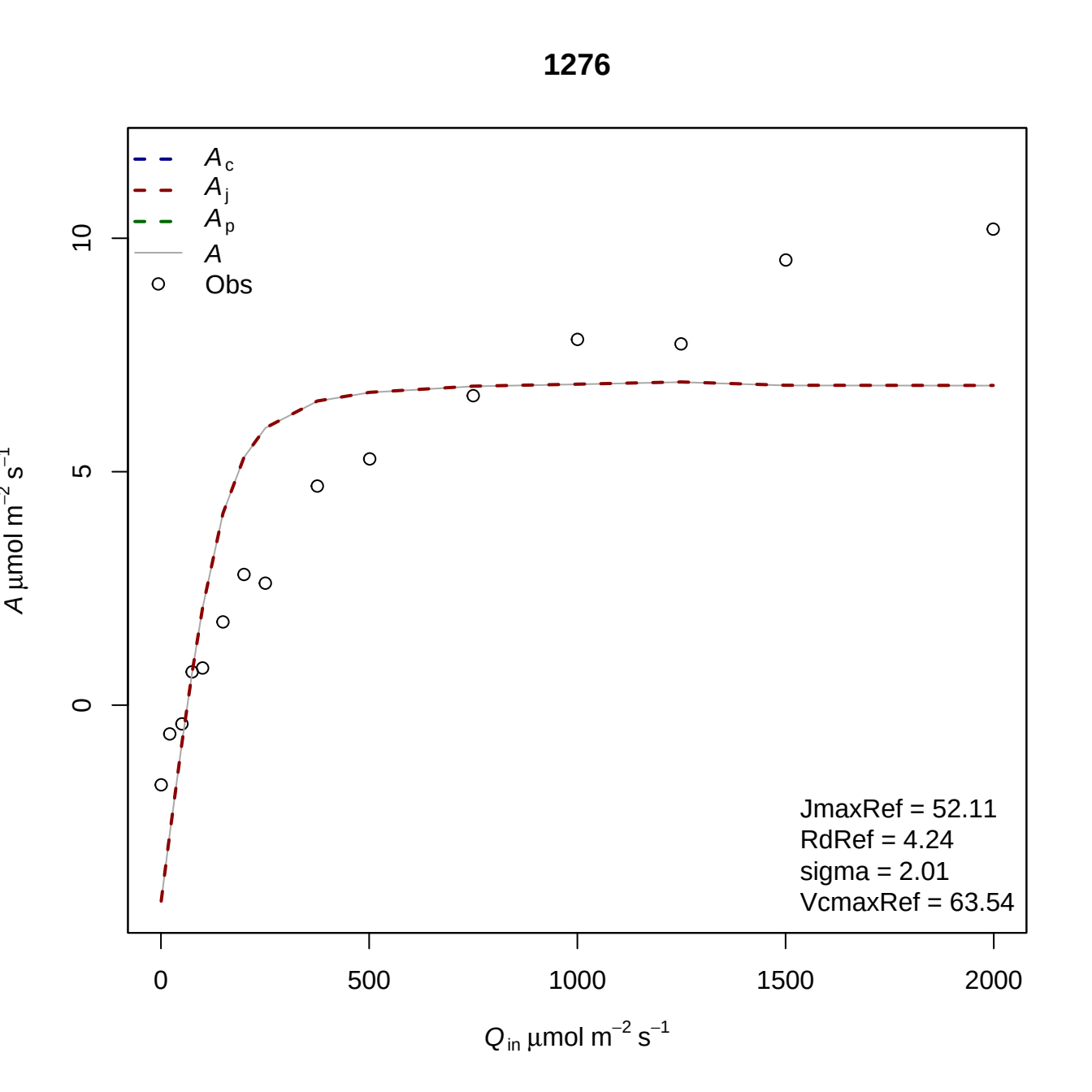

1271
A_c
A_j
A_p
A
Obs
JmaxRef = 86.4
RdRef = 5.12
sigma = 2.2
VcmaxRef = 126.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

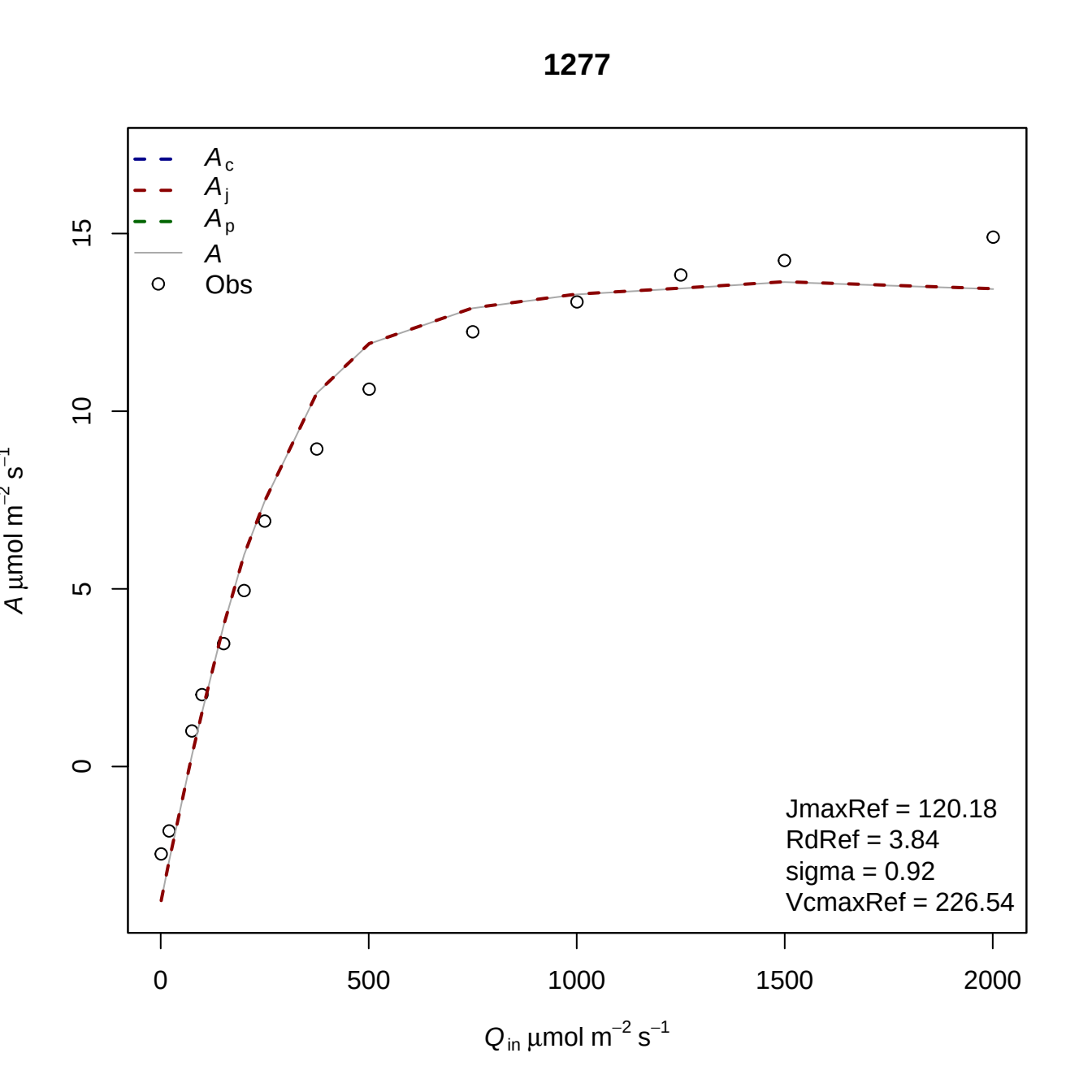

1272
A_c
A_j
A_p
A
Obs
JmaxRef = 93.93
RdRef = 2.97
sigma = 0.86
VcmaxRef = 287.26
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

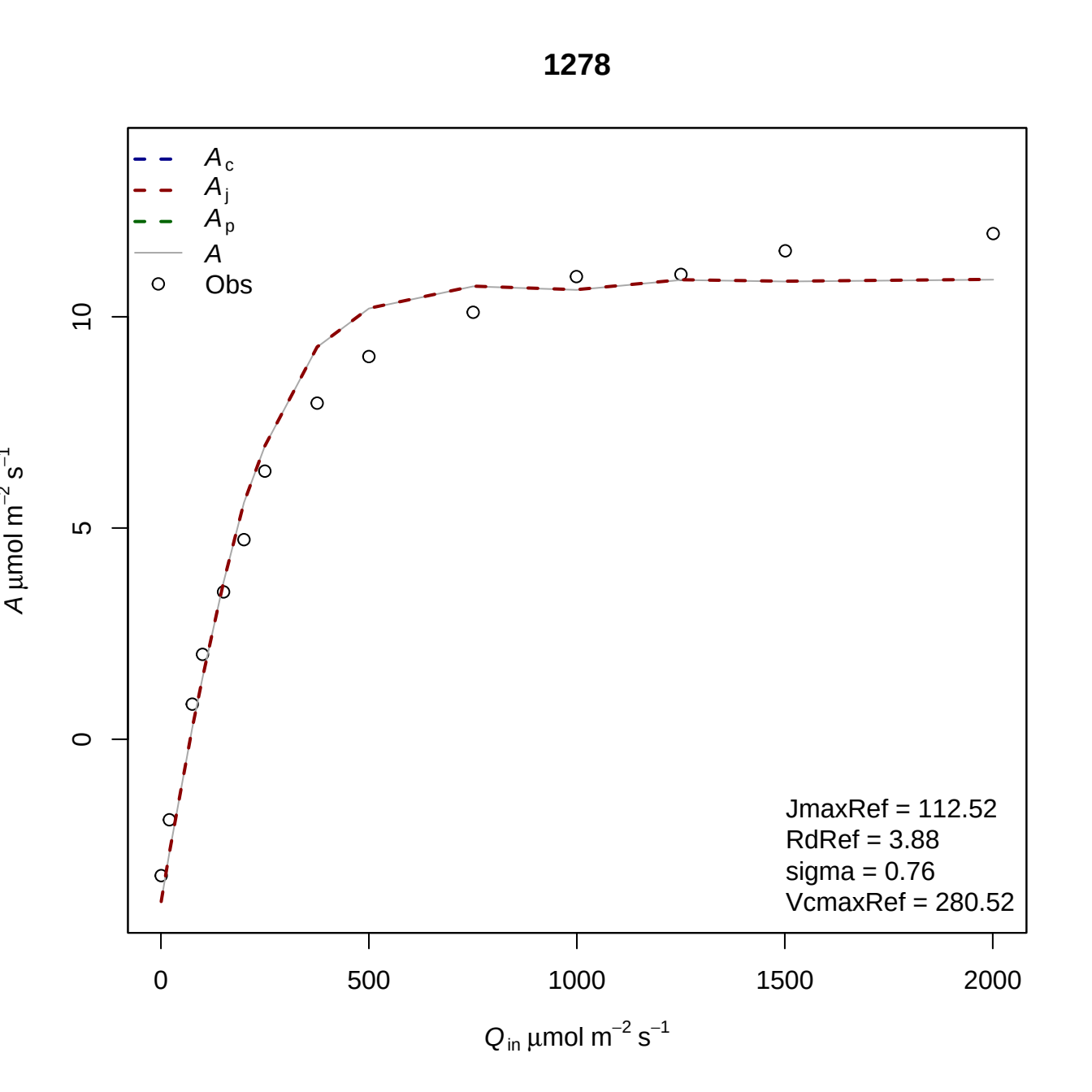

1273
A_c
A_j
A_p
A
Obs
JmaxRef = 113.12
RdRef = 5.32
sigma = 0.86
VcmaxRef = 641.85
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

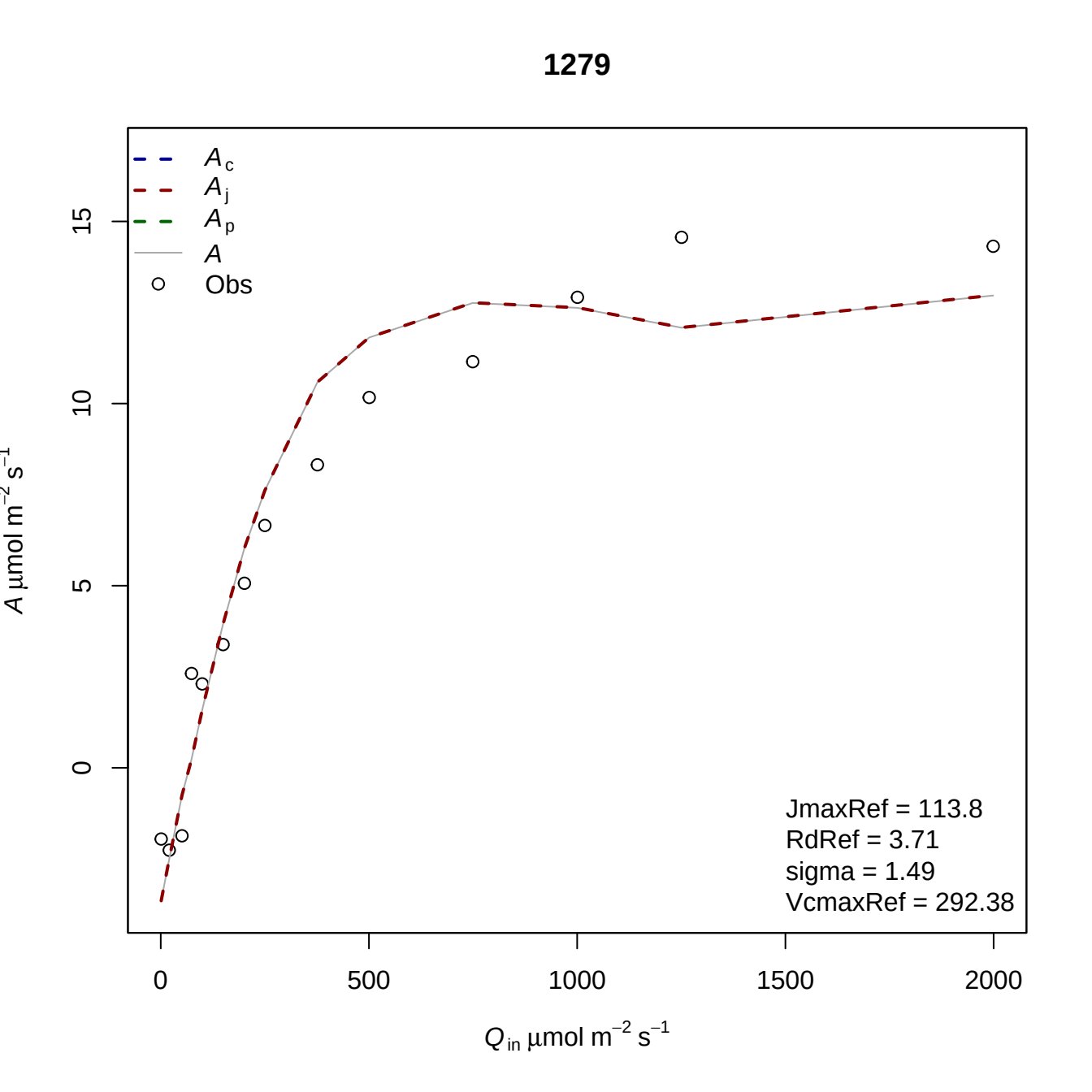

1274
A_c
A_j
A_p
A
Obs
JmaxRef = 100.73
RdRef = 3.88
sigma = 0.54
VcmaxRef = 229.87
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

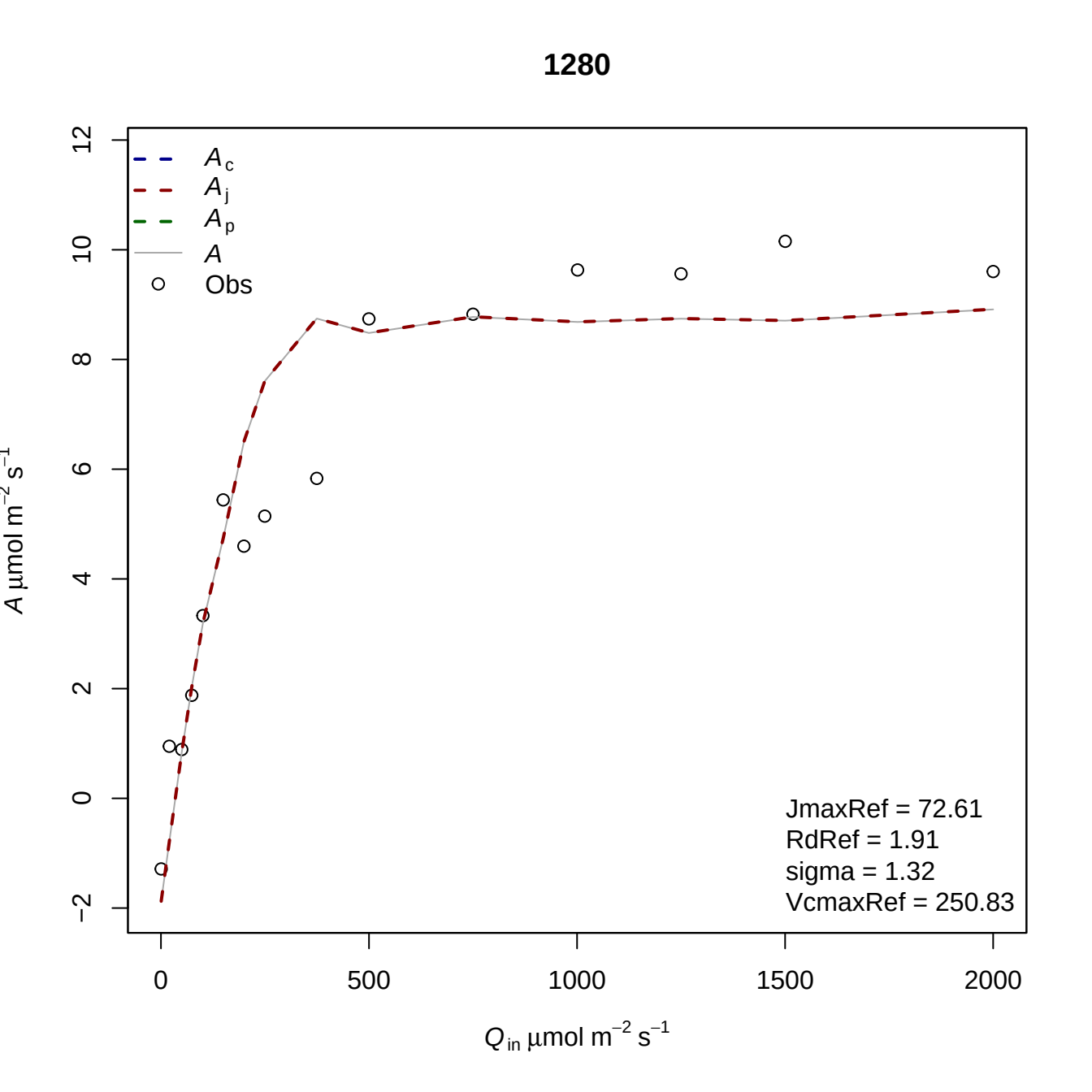

1275
A_c
A_j
A_p
A
Obs
JmaxRef = 51.85
RdRef = 4.55
sigma = 2.05
VcmaxRef = 51.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

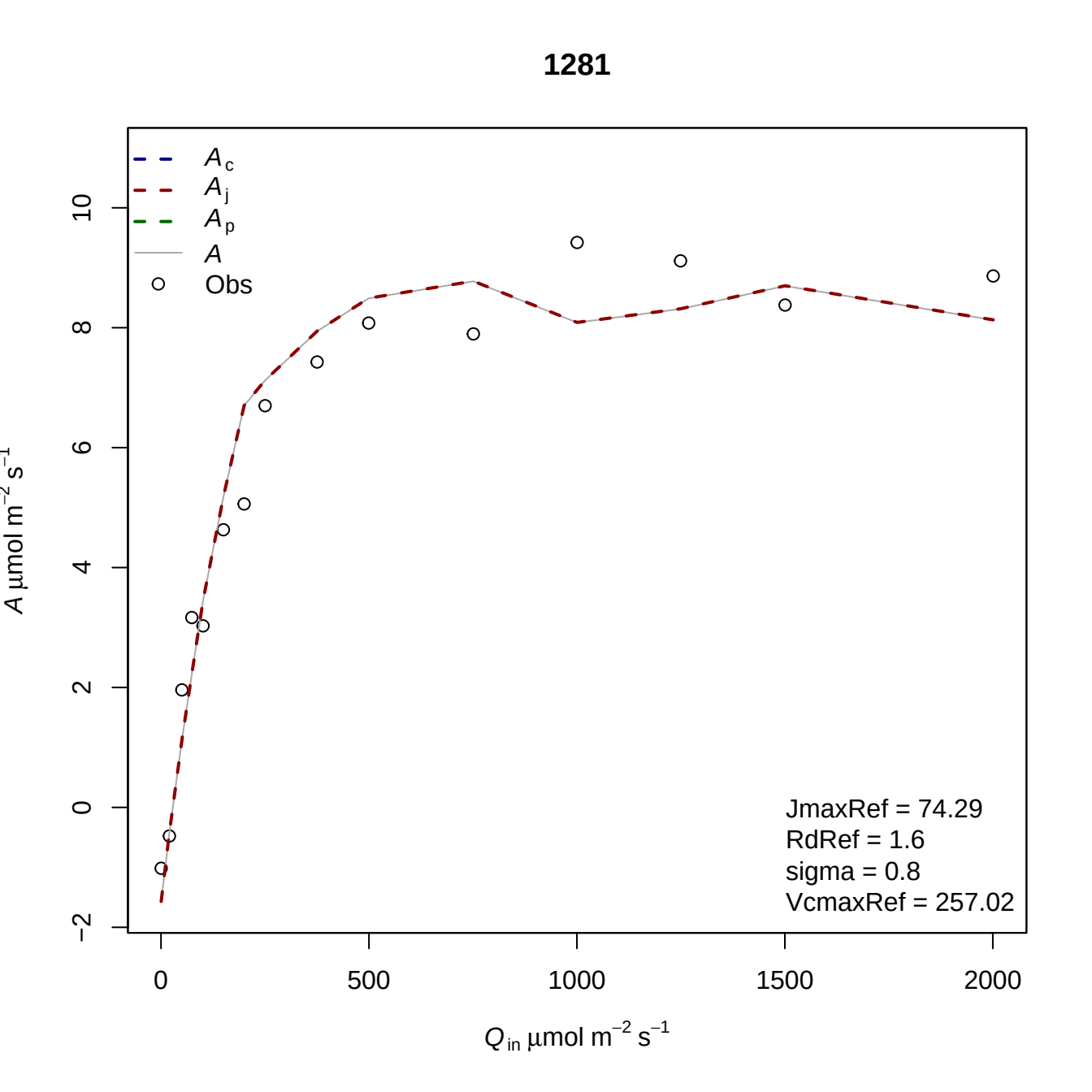

1276
A_c
A_j
A_p
A
Obs
JmaxRef = 52.11
RdRef = 4.24
sigma = 2.01
VcmaxRef = 63.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

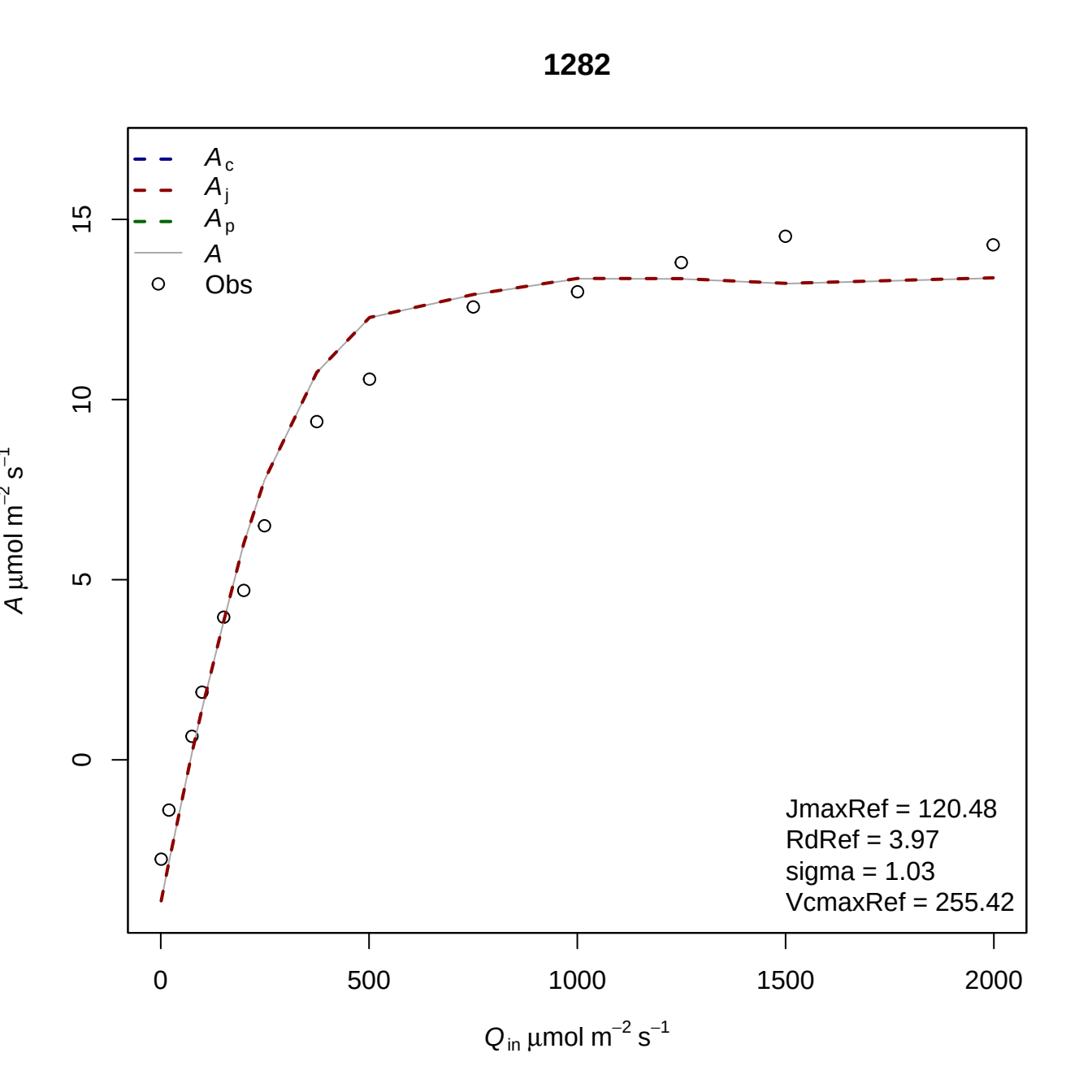

1277
A_c
A_j
A_p
A
Obs
JmaxRef = 120.18
RdRef = 3.84
sigma = 0.92
VcmaxRef = 226.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

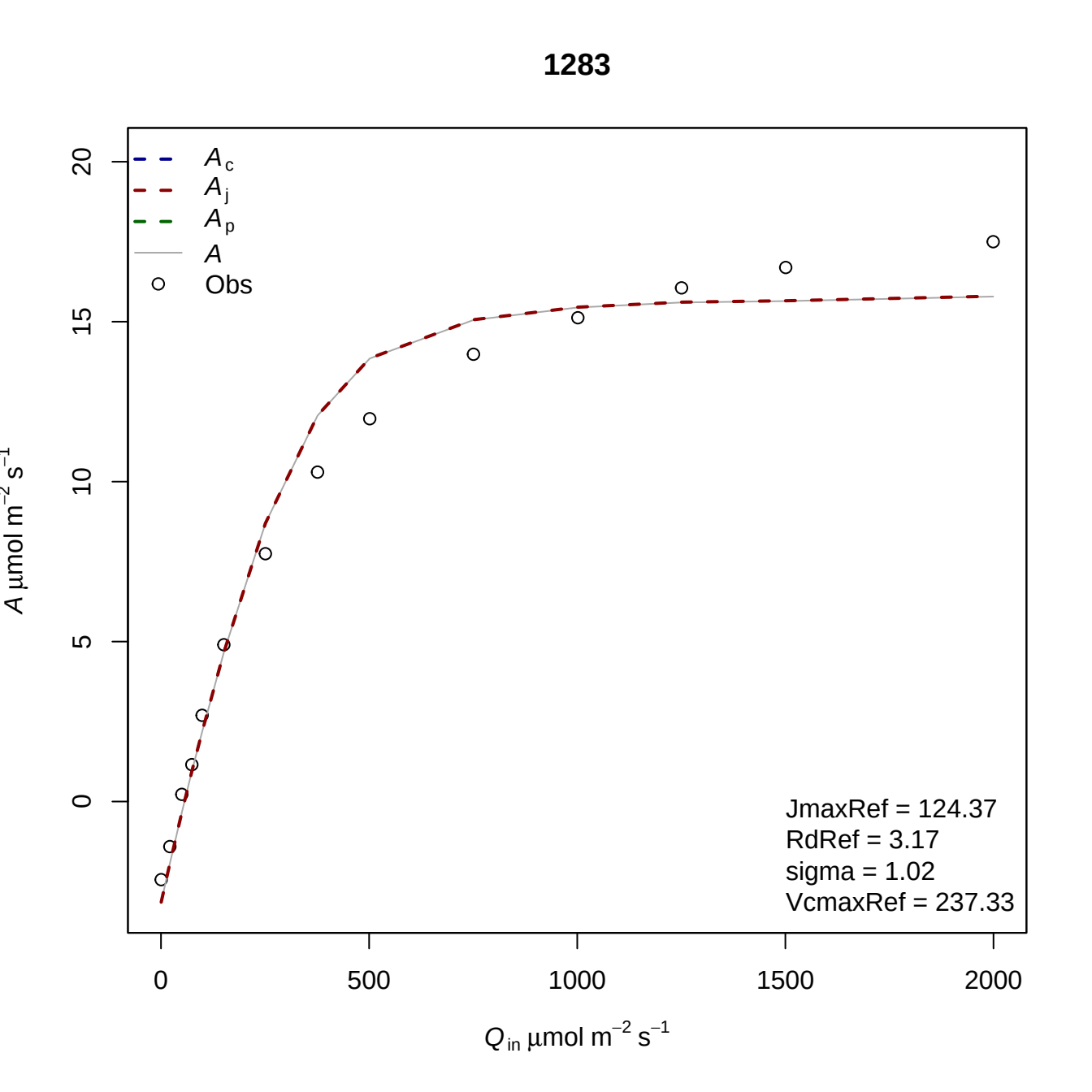

1278
A_c
A_j
A_p
A
Obs
JmaxRef = 112.52
RdRef = 3.88
sigma = 0.76
VcmaxRef = 280.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

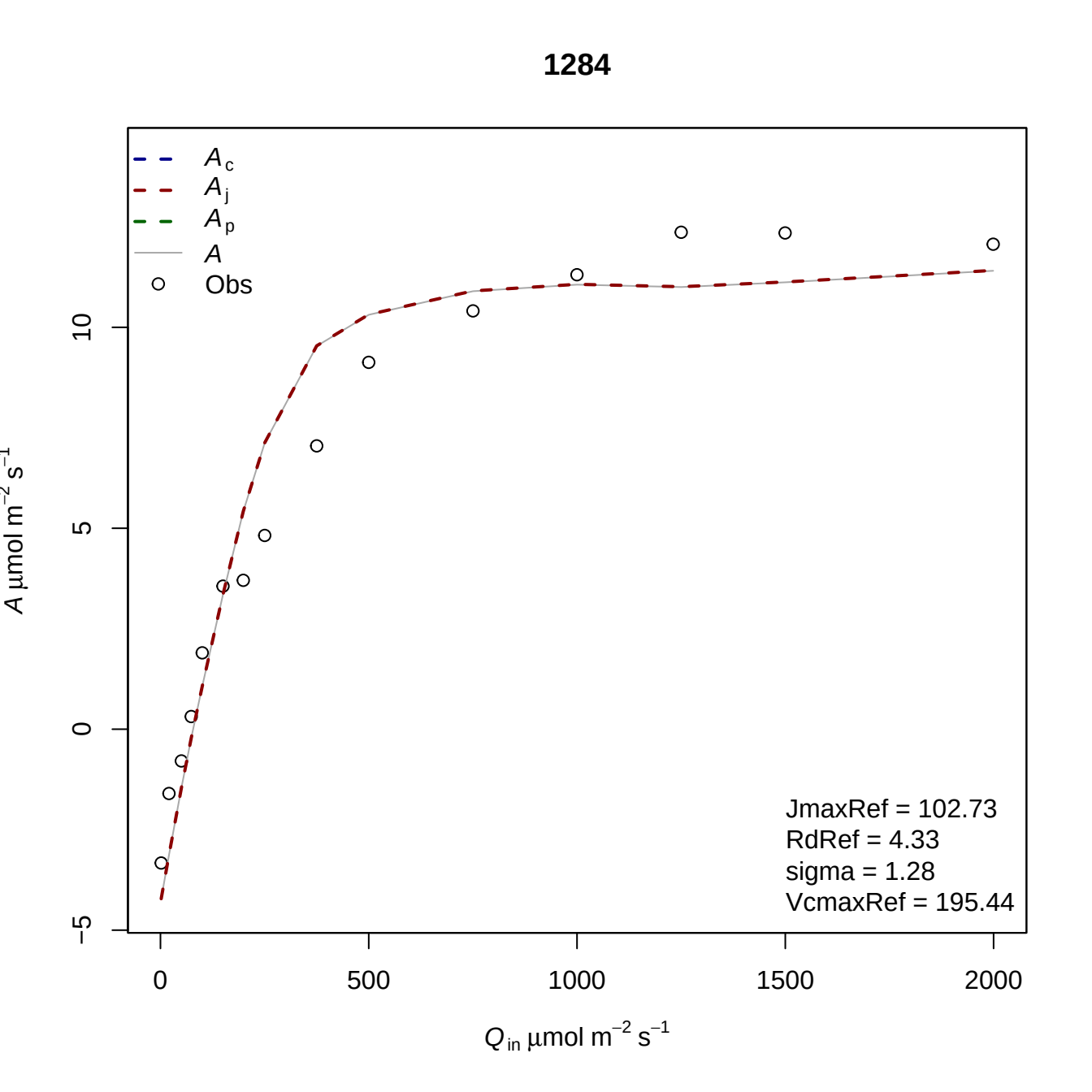

1279
A_c
A_j
A_p
A
Obs
JmaxRef = 113.8
RdRef = 3.71
sigma = 1.49
VcmaxRef = 292.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

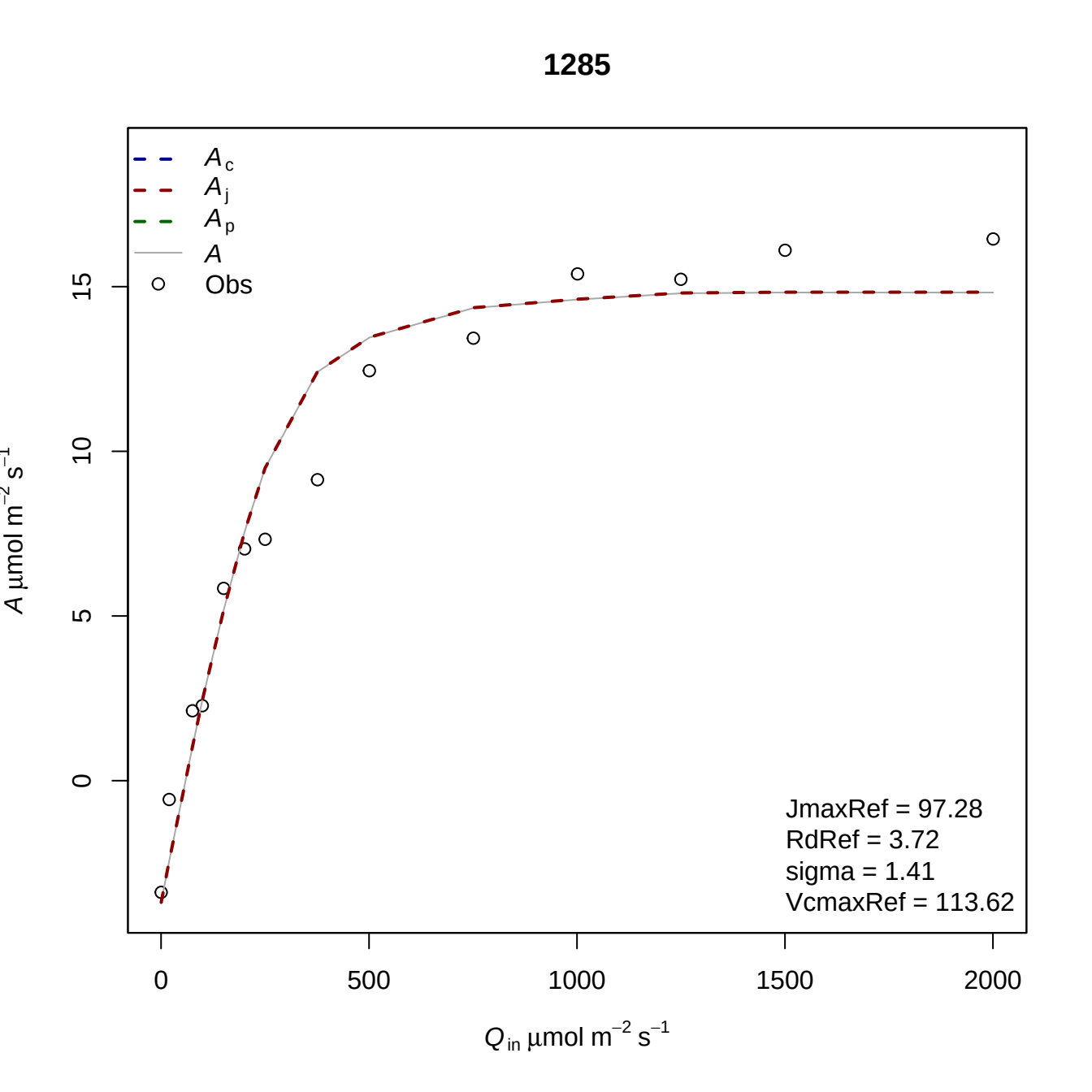

1280
A_c
A_j
A_p
A
Obs
JmaxRef = 72.61
RdRef = 1.91
sigma = 1.32
VcmaxRef = 250.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

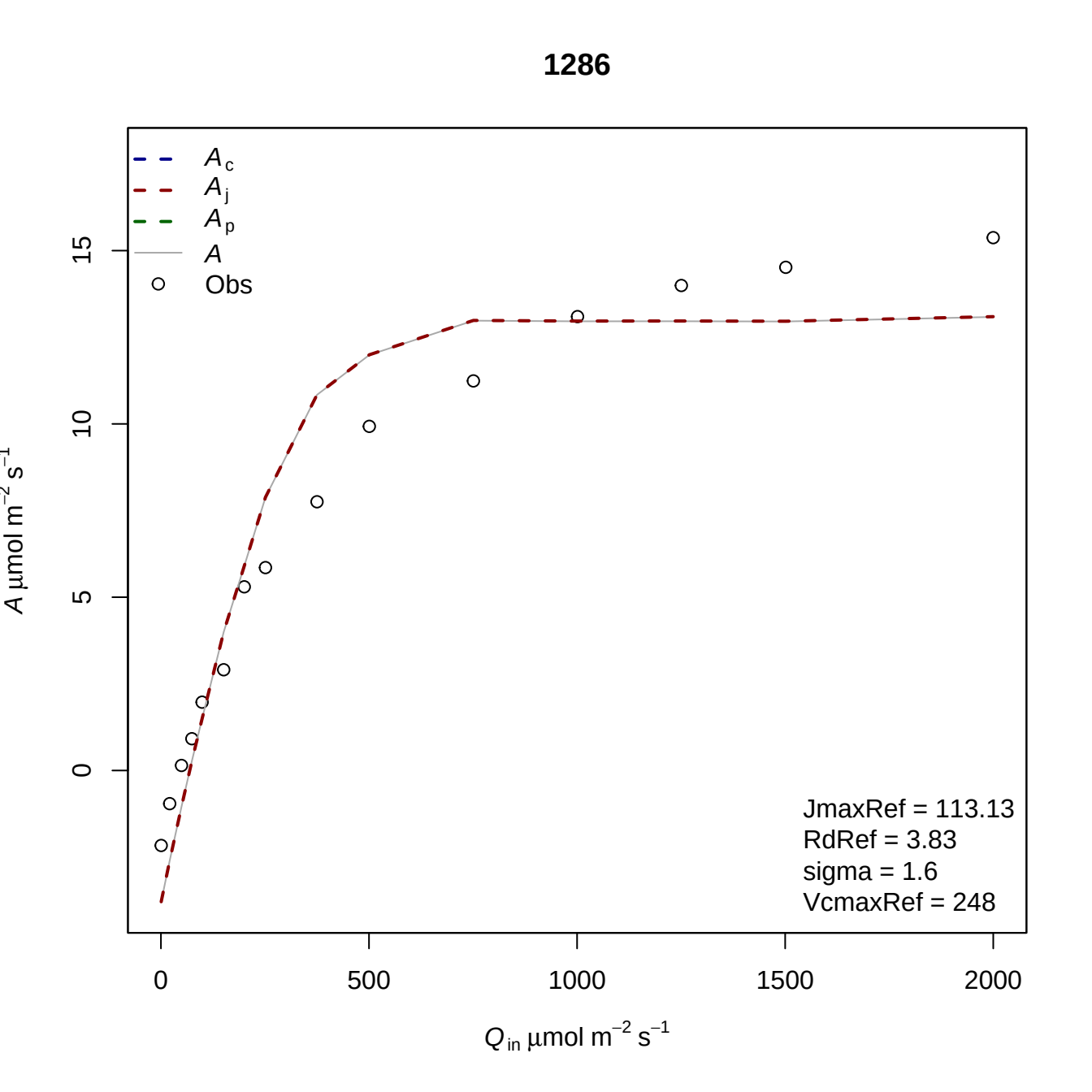

1281
A_c
A_j
A_p
A
Obs
JmaxRef = 74.29
RdRef = 1.6
sigma = 0.8
VcmaxRef = 257.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

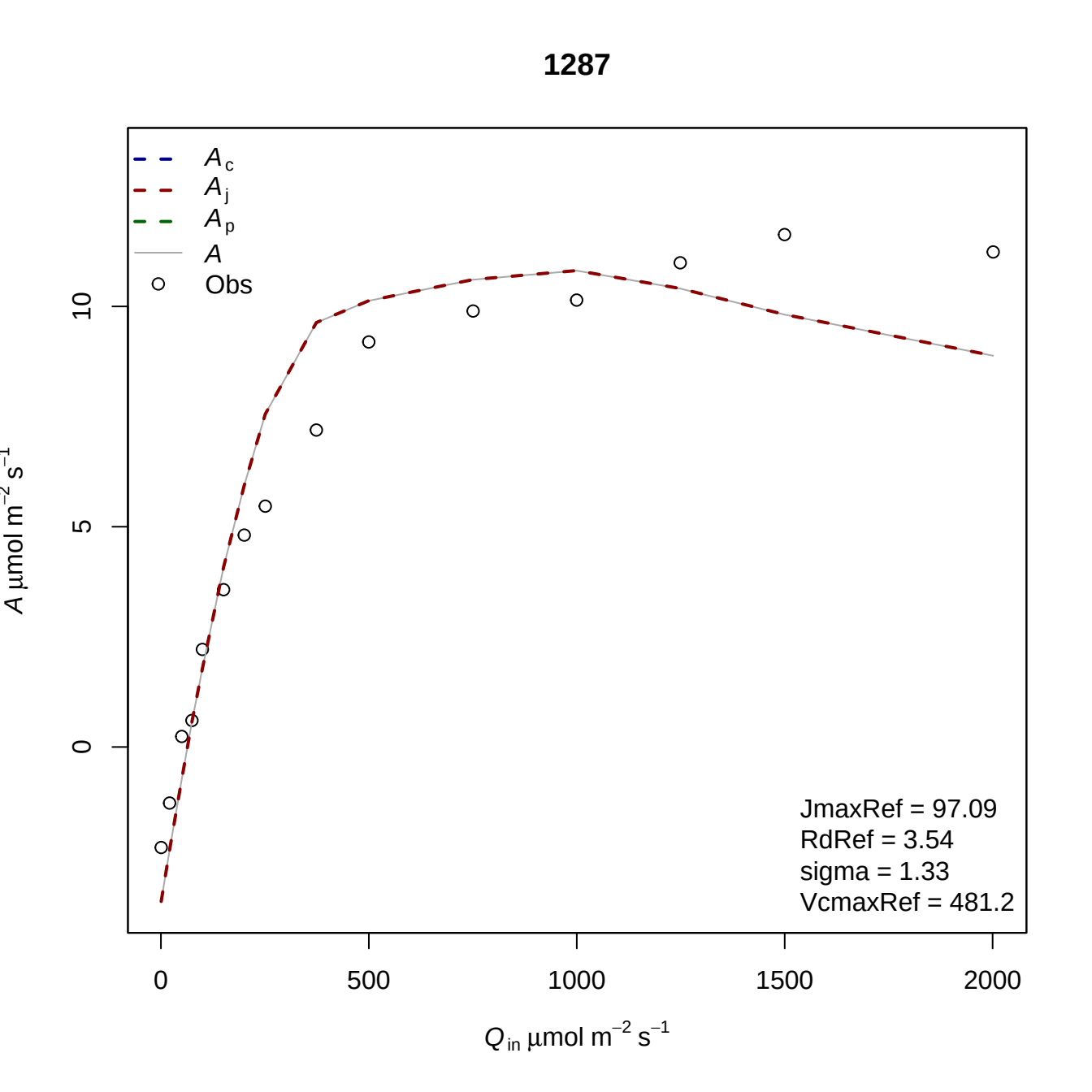

1282
A_c
A_j
A_p
A
Obs
JmaxRef = 120.48
RdRef = 3.97
sigma = 1.03
VcmaxRef = 255.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

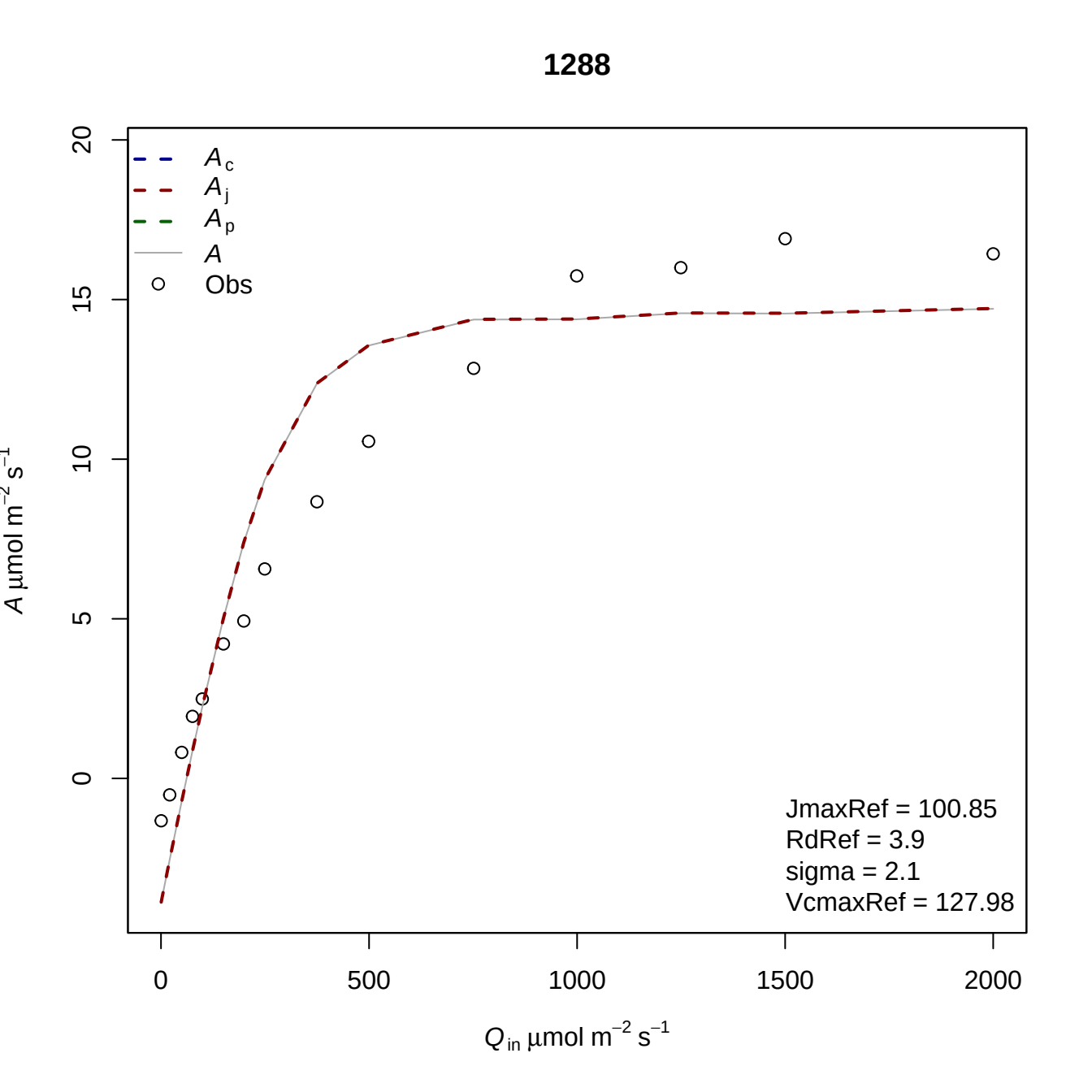

1283
A_c
A_j
A_p
A
Obs
JmaxRef = 124.37
RdRef = 3.17
sigma = 1.02
VcmaxRef = 237.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

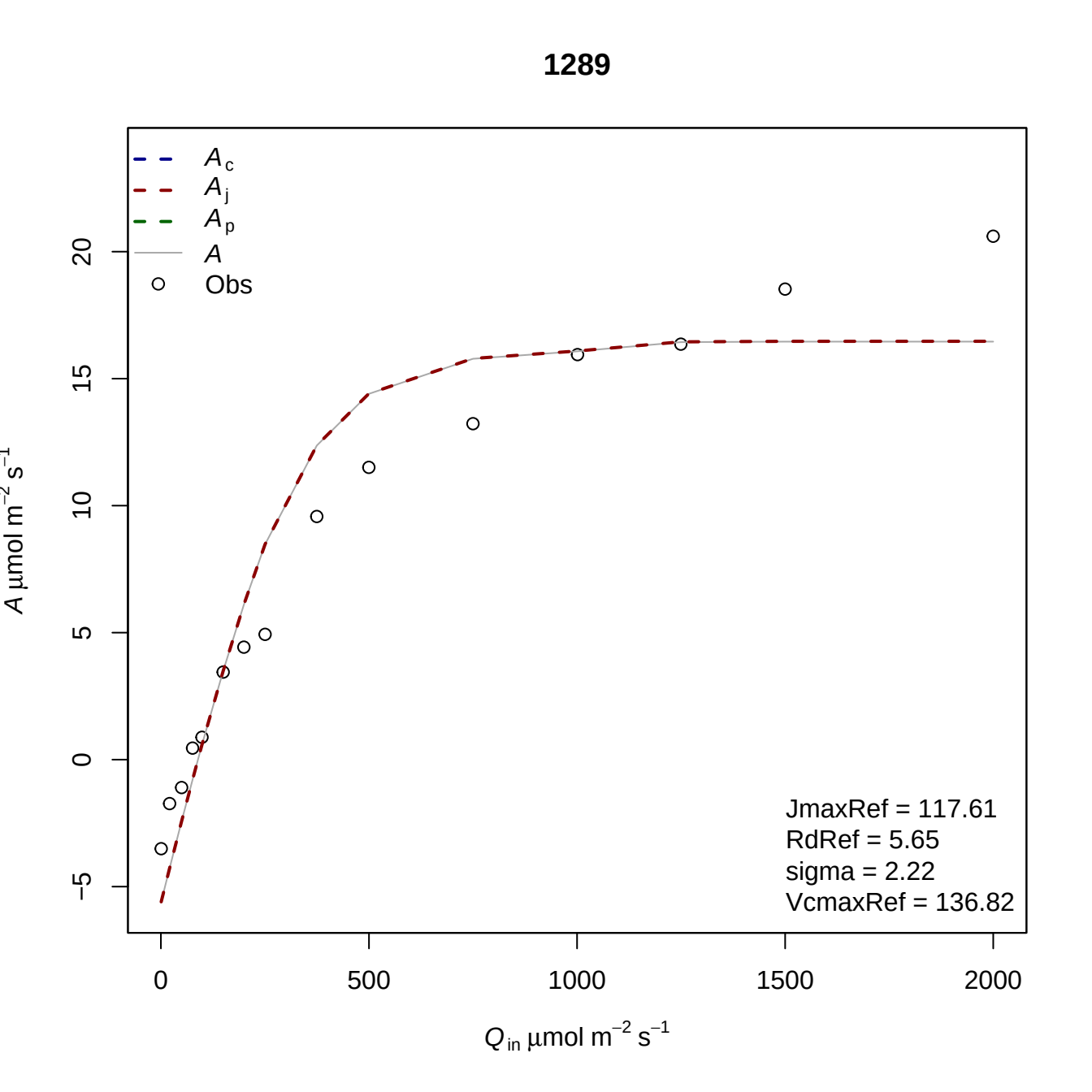

1284
A_c
A_j
A_p
A
Obs
JmaxRef = 102.73
RdRef = 4.33
sigma = 1.28
VcmaxRef = 195.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

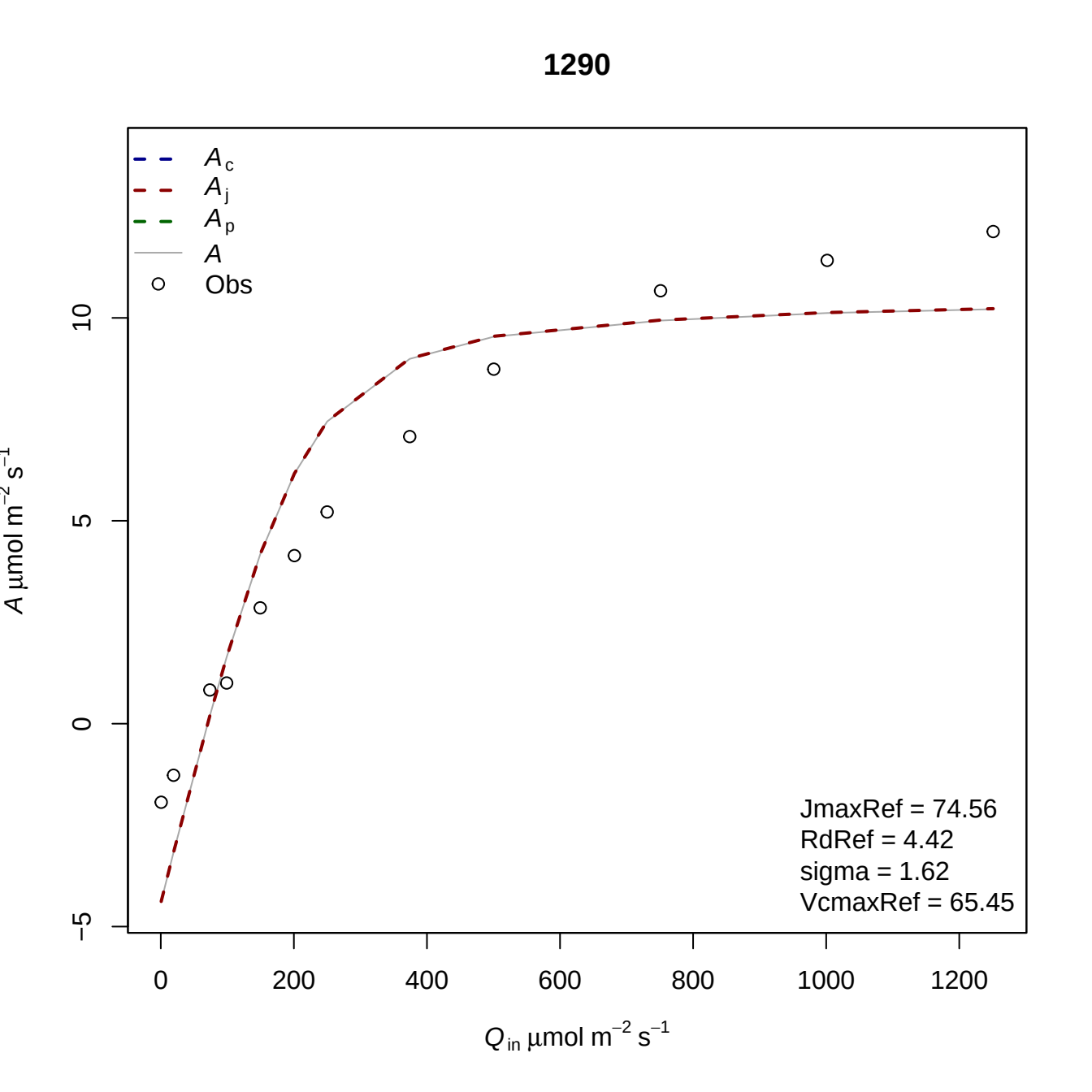

1285
A_c
A_j
A_p
A
Obs
JmaxRef = 97.28
RdRef = 3.72
sigma = 1.41
VcmaxRef = 113.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

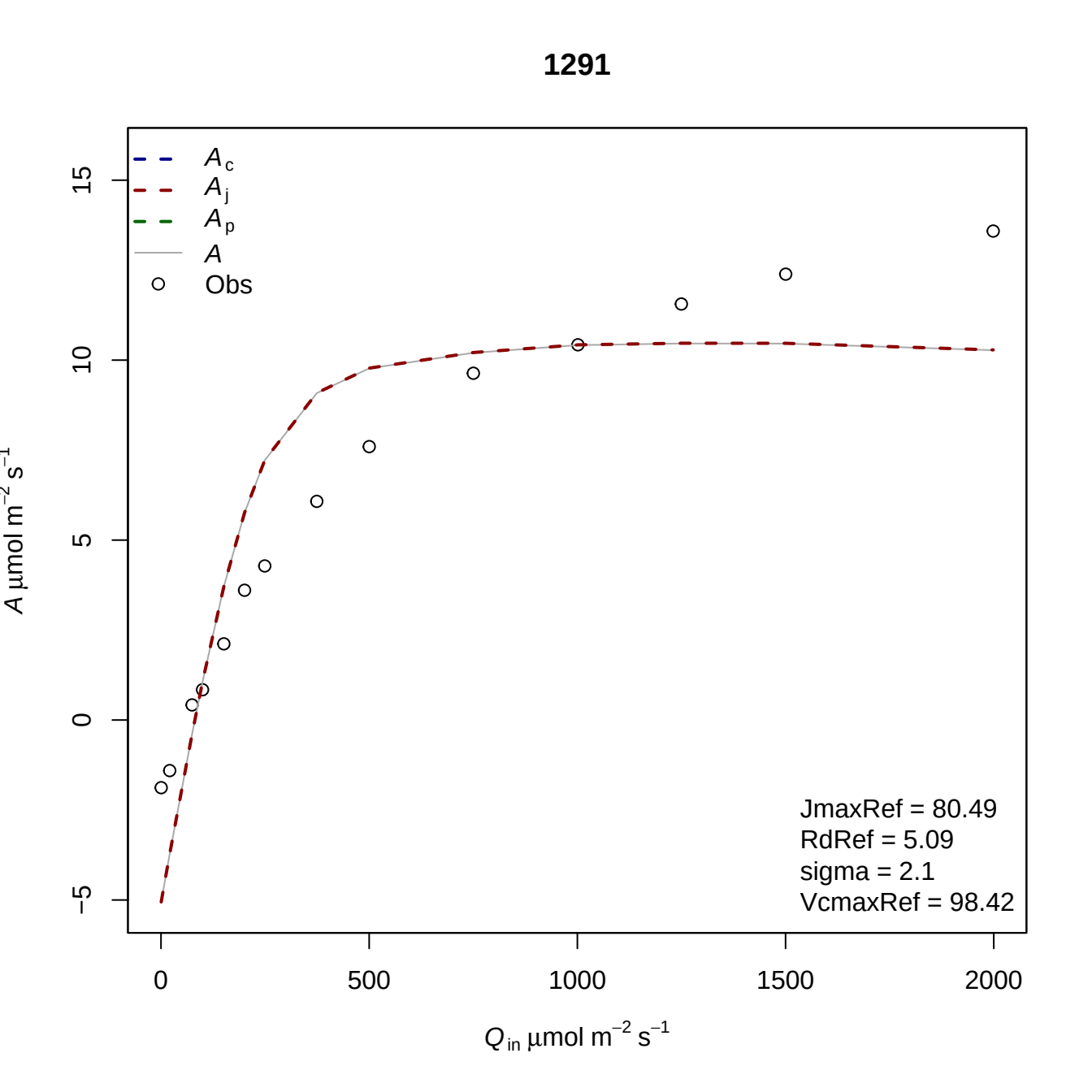

1286
A_c
A_j
A_p
A
Obs
JmaxRef = 113.13
RdRef = 3.83
sigma = 1.6
VcmaxRef = 248
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

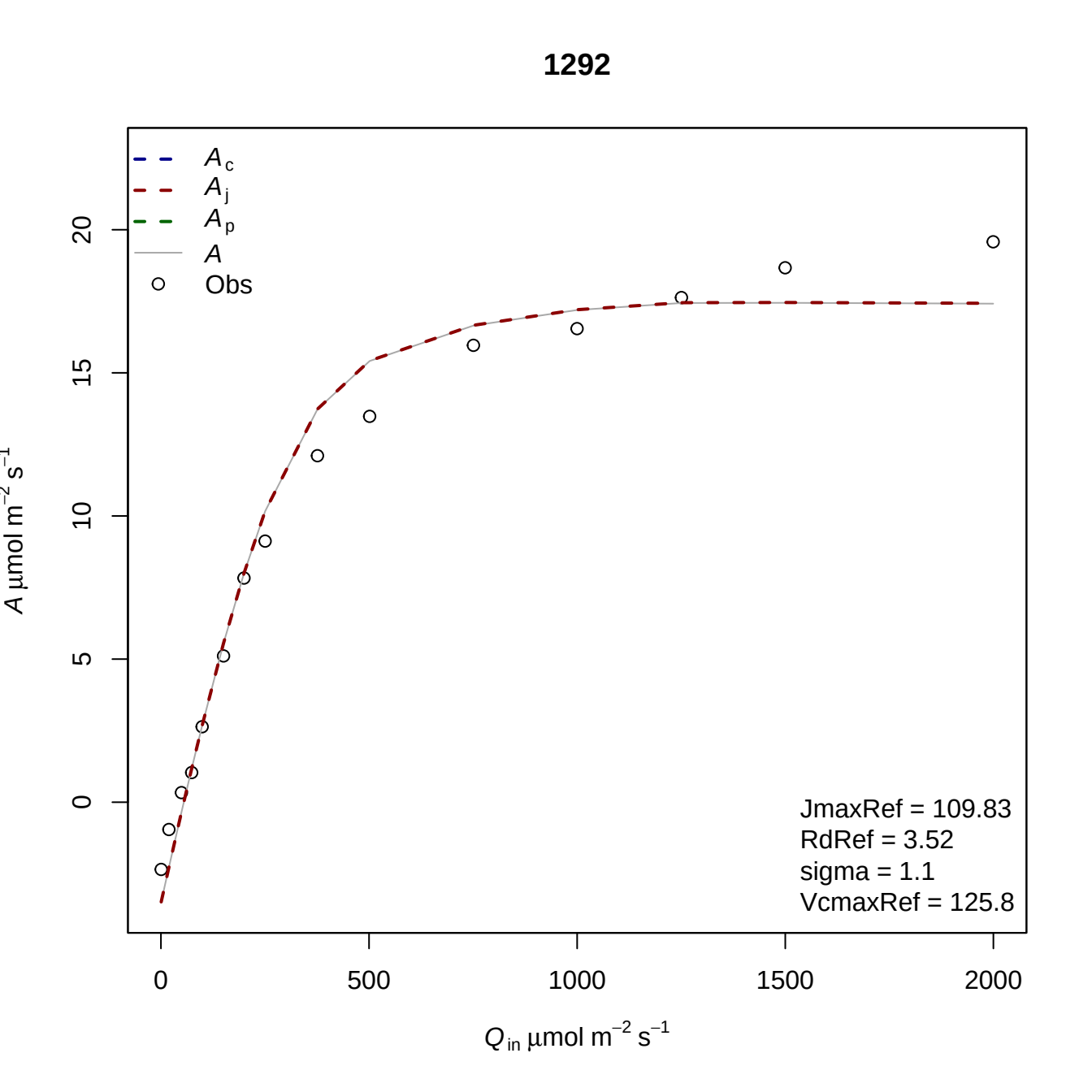

1287
A_c
A_j
A_p
A
Obs
JmaxRef = 97.09
RdRef = 3.54
sigma = 1.33
VcmaxRef = 481.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

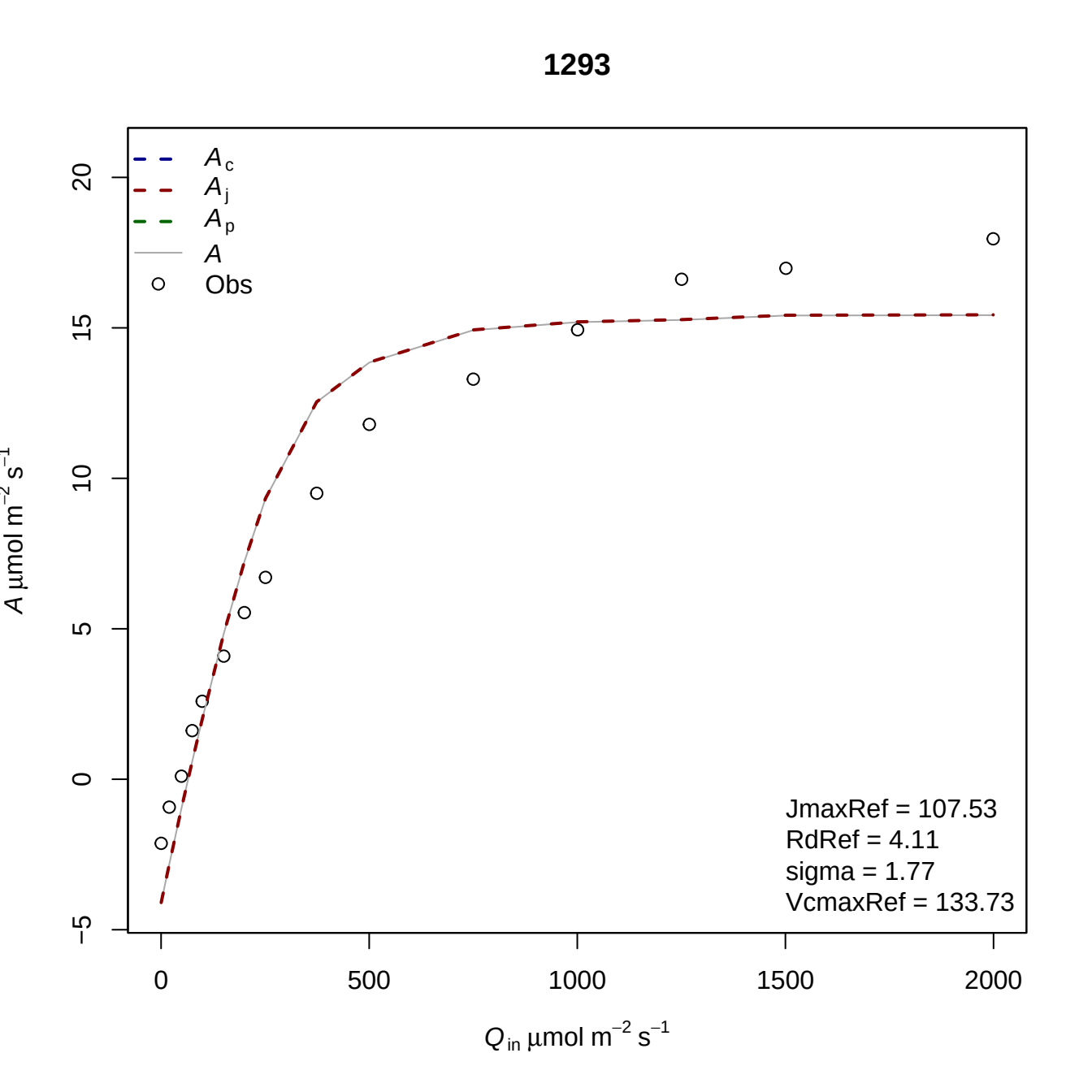

1288
A_c
A_j
A_p
A
Obs
JmaxRef = 100.85
RdRef = 3.9
sigma = 2.1
VcmaxRef = 127.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

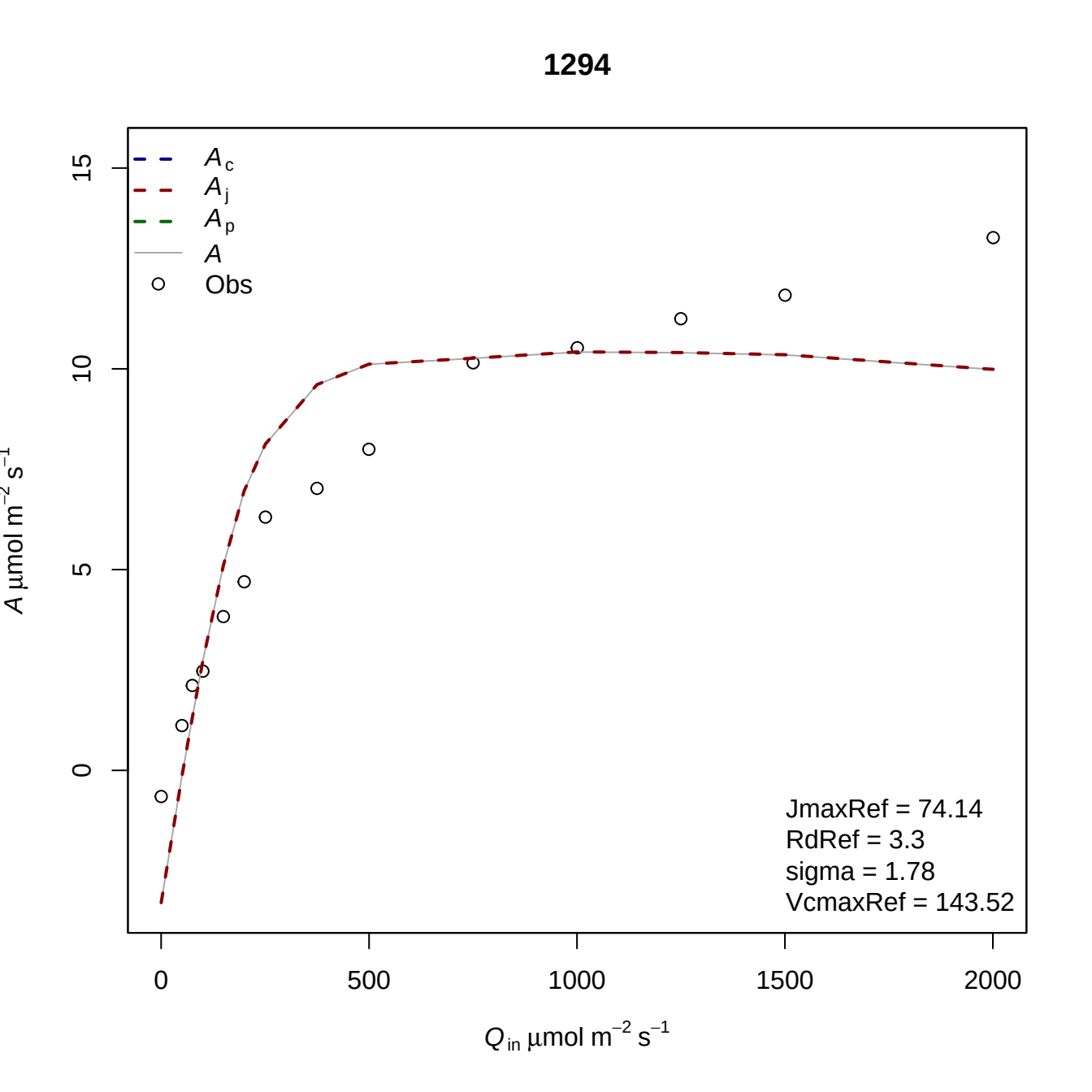

1289
A_c
A_j
A_p
A
Obs
JmaxRef = 117.61
RdRef = 5.65
sigma = 2.22
VcmaxRef = 136.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

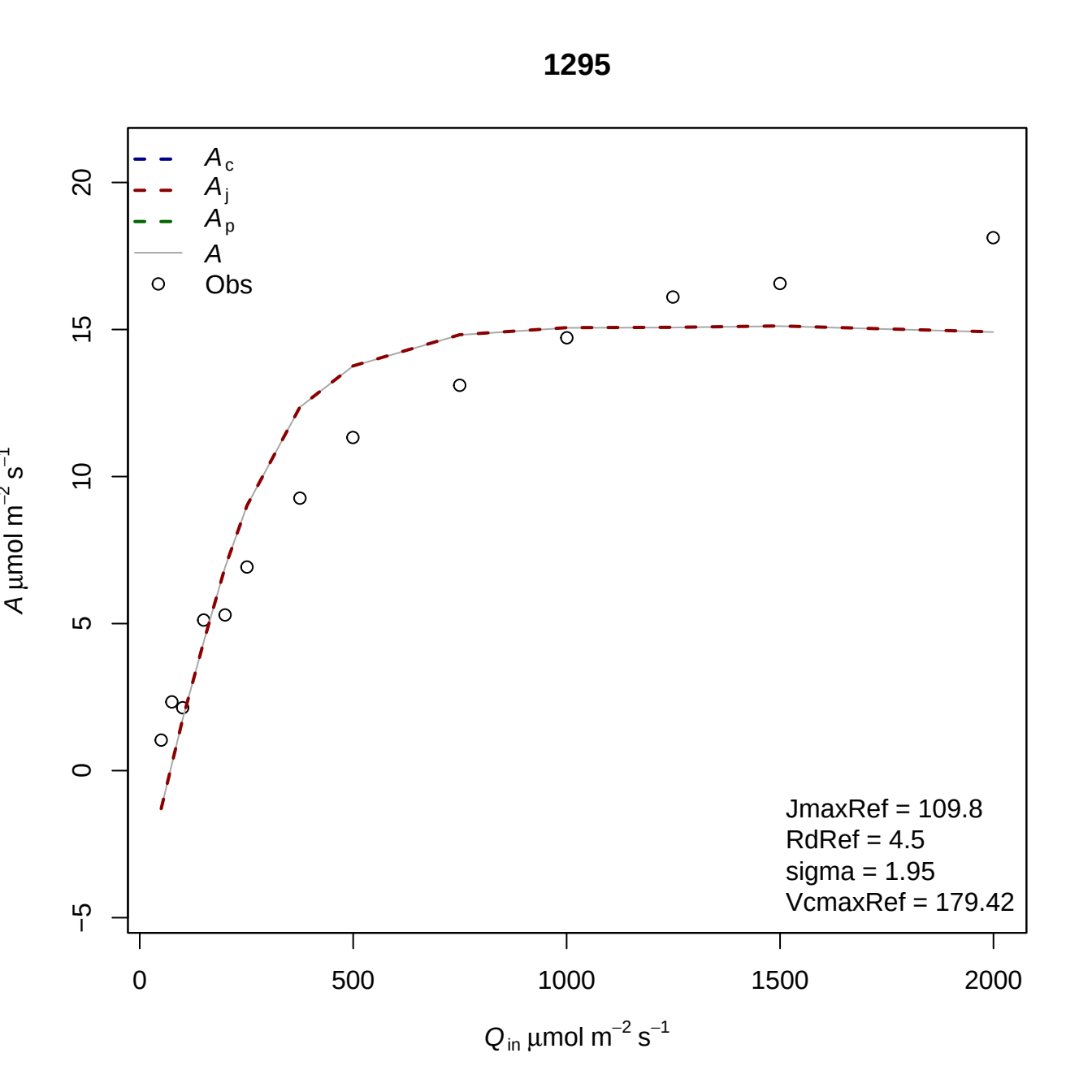

1290
A_c
A_j
A_p
A
Obs
JmaxRef = 74.56
RdRef = 4.42
sigma = 1.62
VcmaxRef = 65.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
200
400
600
800
1000
1200

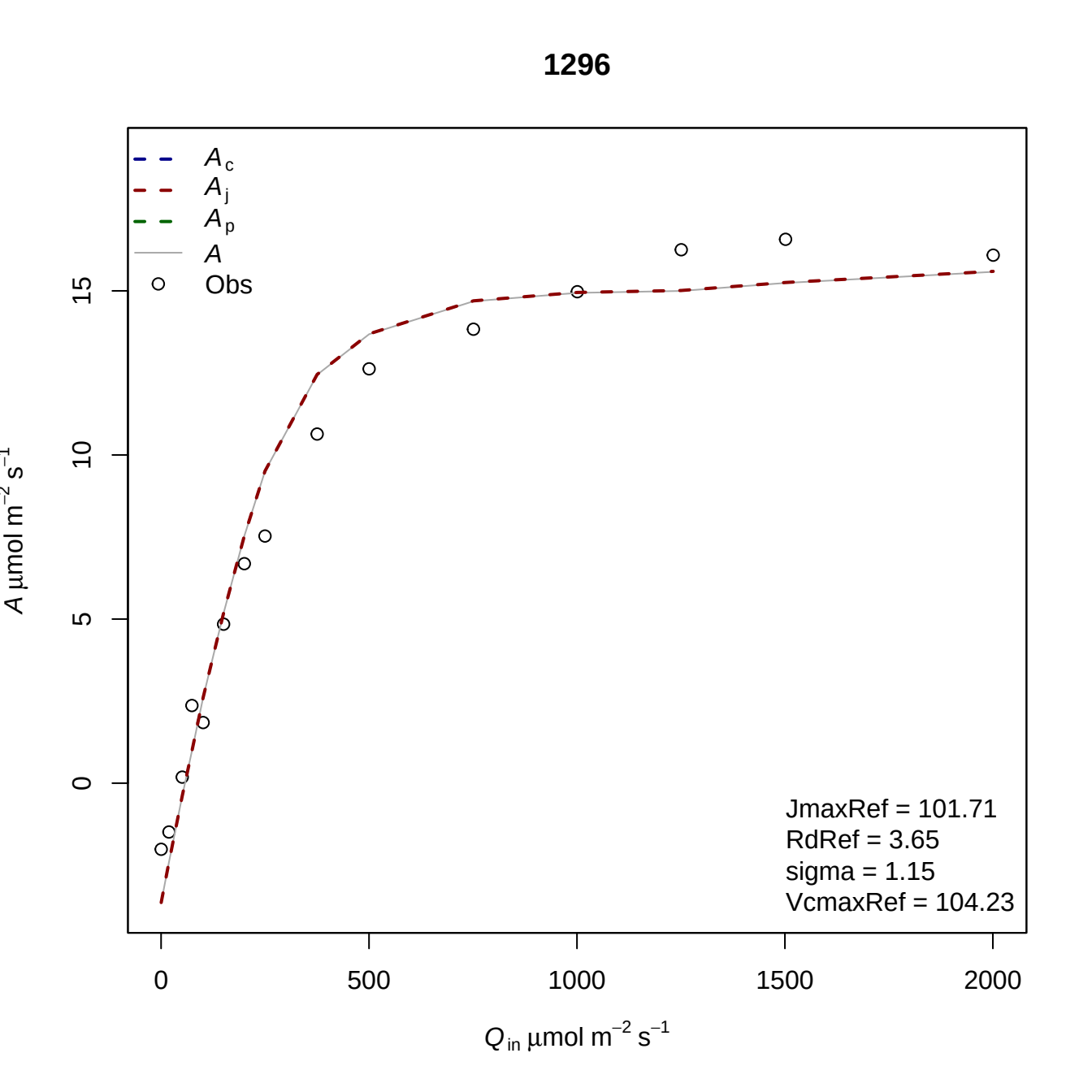

1291
A_c
A_j
A_p
A
Obs
JmaxRef = 80.49
RdRef = 5.09
sigma = 2.1
VcmaxRef = 98.42
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

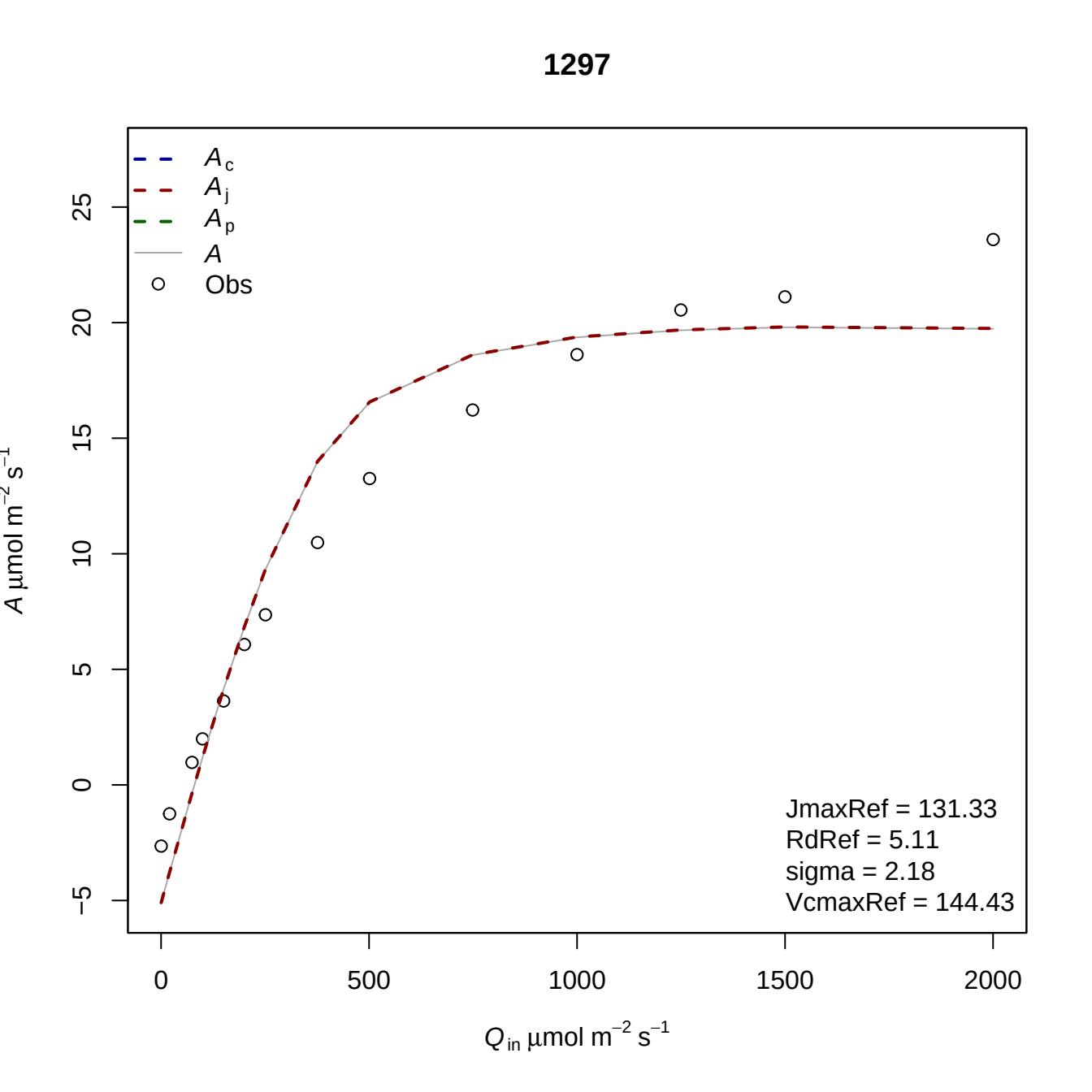

1292
A_c
A_j
A_p
A
Obs
JmaxRef = 109.83
RdRef = 3.52
sigma = 1.1
VcmaxRef = 125.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

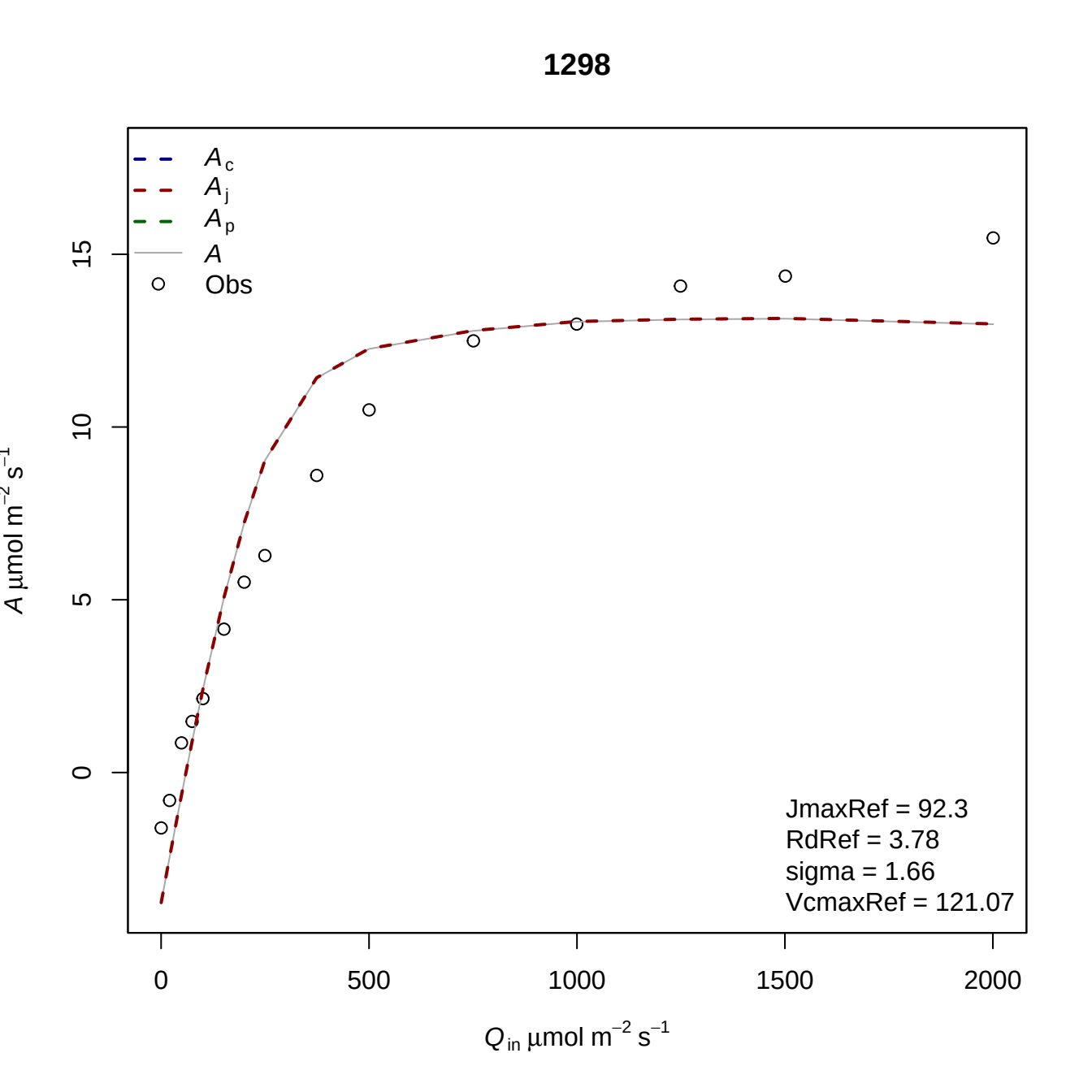

1293
A_c
A_j
A_p
A
Obs
JmaxRef = 107.53
RdRef = 4.11
sigma = 1.77
VcmaxRef = 133.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

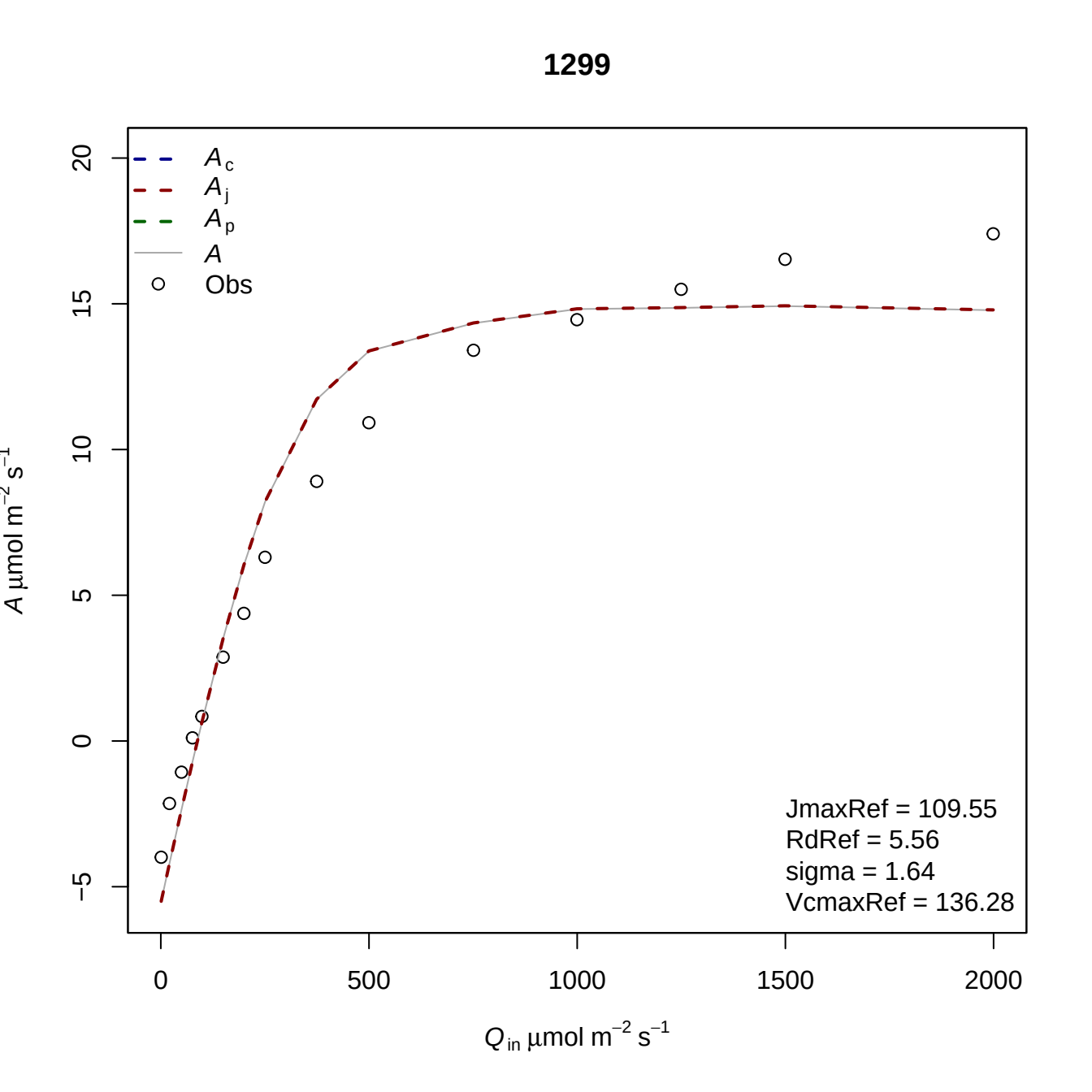

1294
A_c
A_j
A_p
A
Obs
JmaxRef = 74.14
RdRef = 3.3
sigma = 1.78
VcmaxRef = 143.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

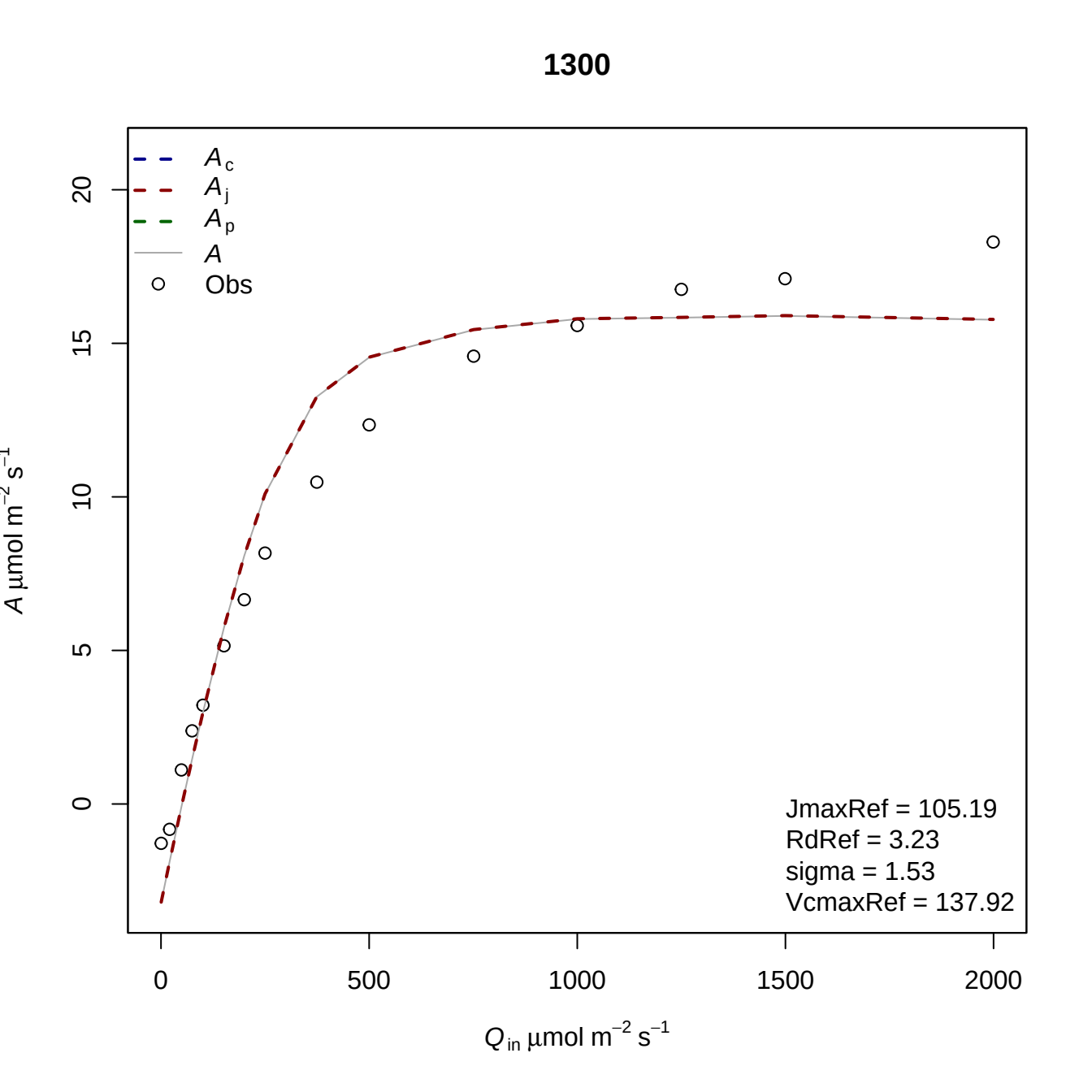

1295
A_c
A_j
A_p
A
Obs
JmaxRef = 109.8
RdRef = 4.5
sigma = 1.95
VcmaxRef = 179.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

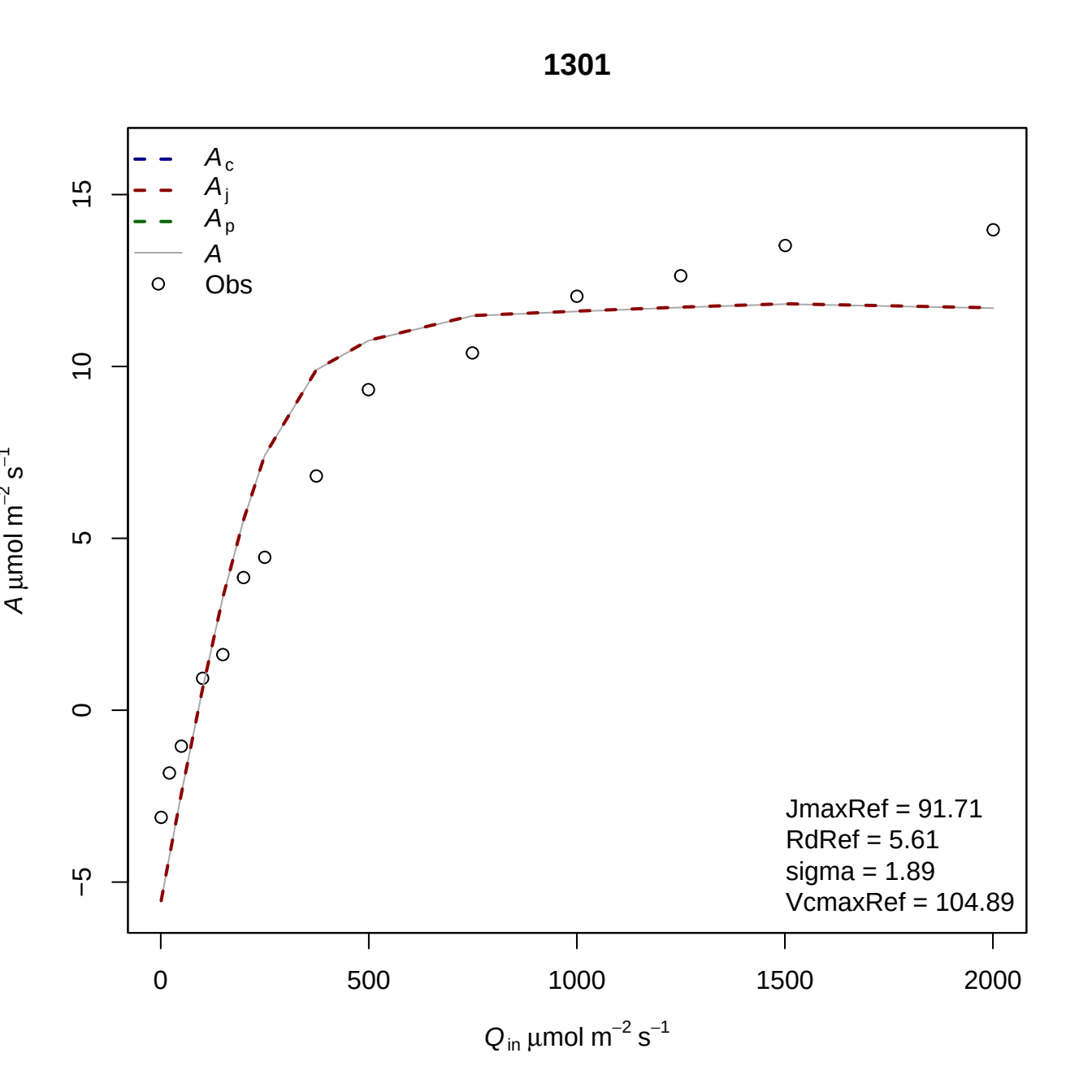

1296
A_c
A_j
A_p
A
Obs
JmaxRef = 101.71
RdRef = 3.65
sigma = 1.15
VcmaxRef = 104.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

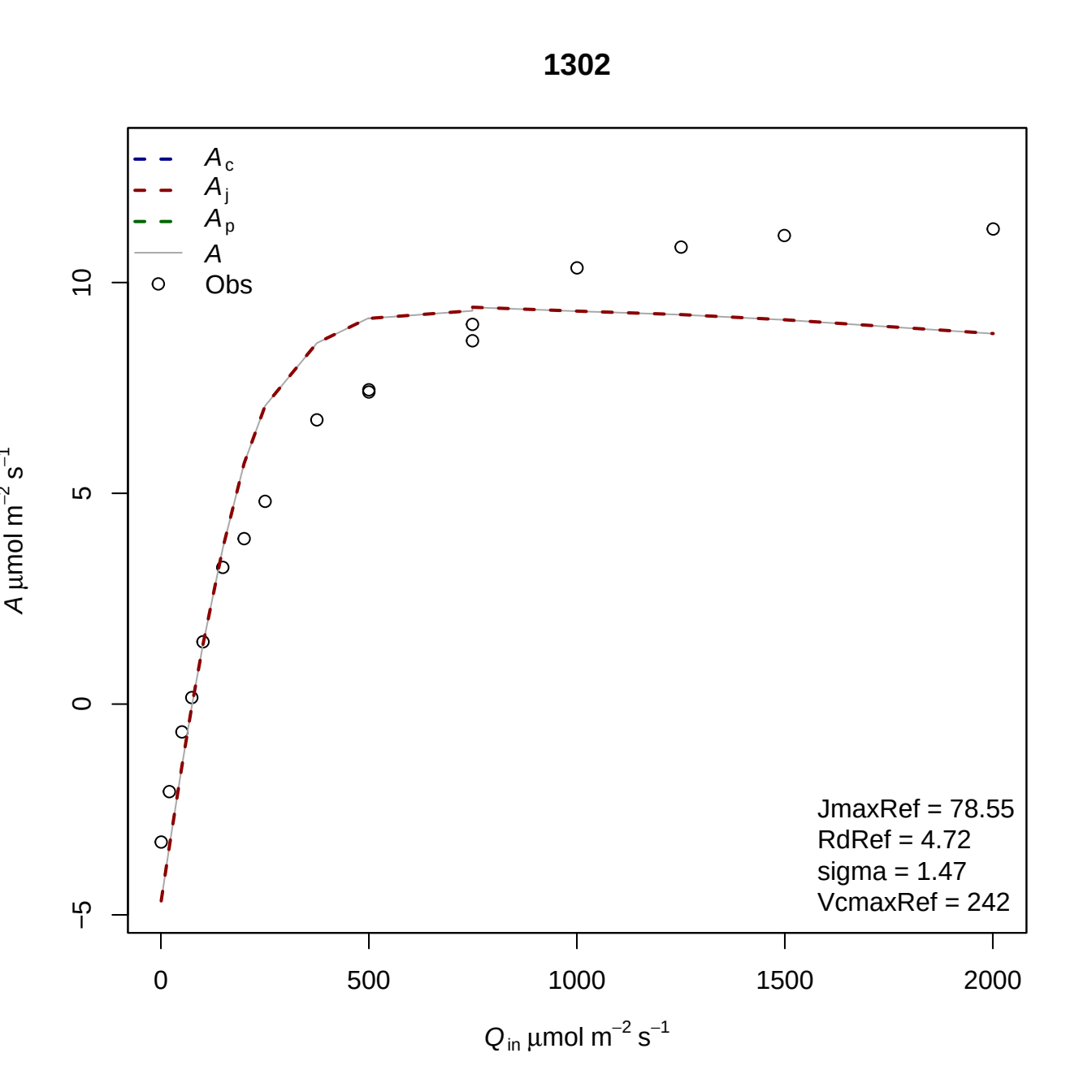

1297
A_c
A_j
A_p
A
Obs
JmaxRef = 131.33
RdRef = 5.11
sigma = 2.18
VcmaxRef = 144.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

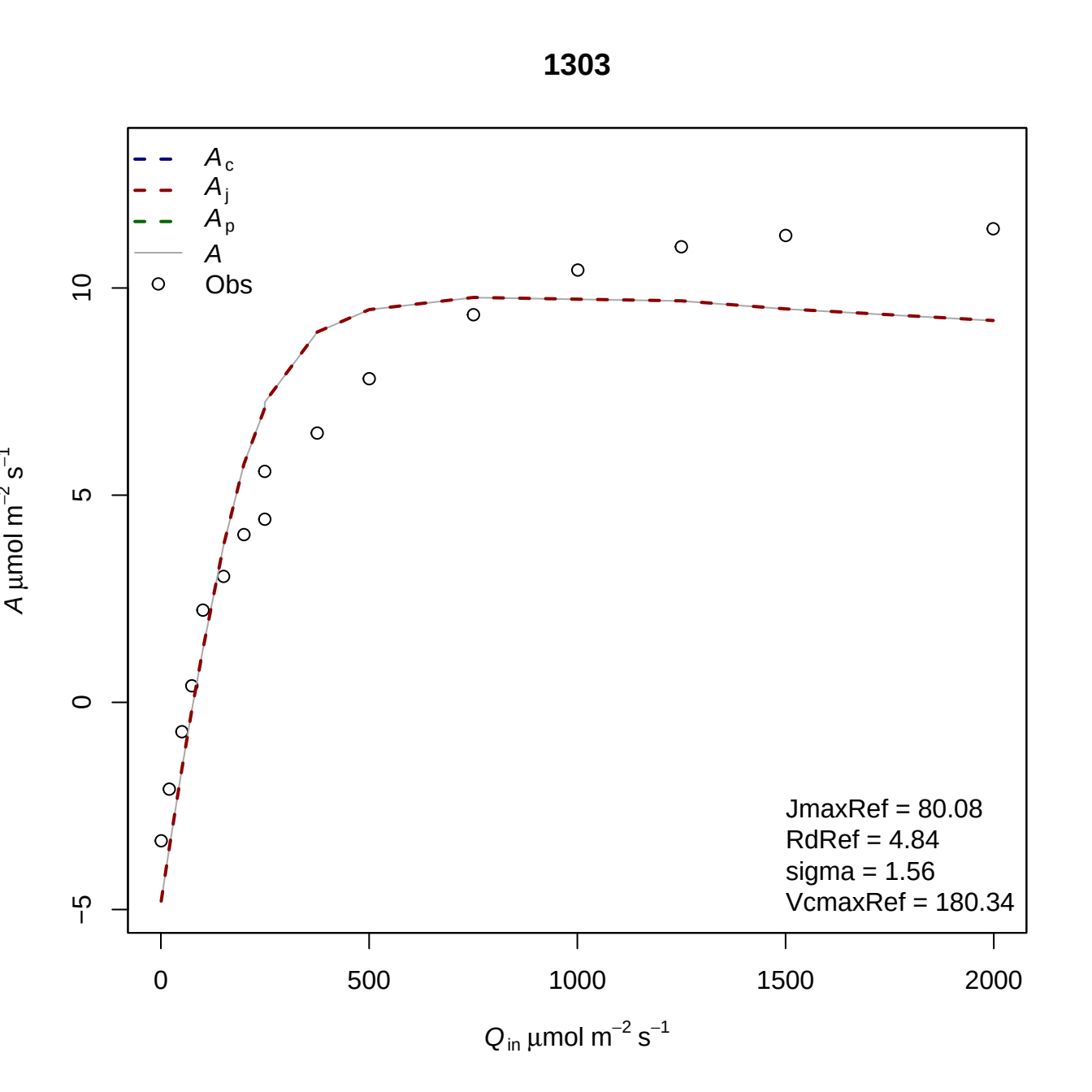

1298
A_c
A_j
A_p
A
Obs
JmaxRef = 92.3
RdRef = 3.78
sigma = 1.66
VcmaxRef = 121.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

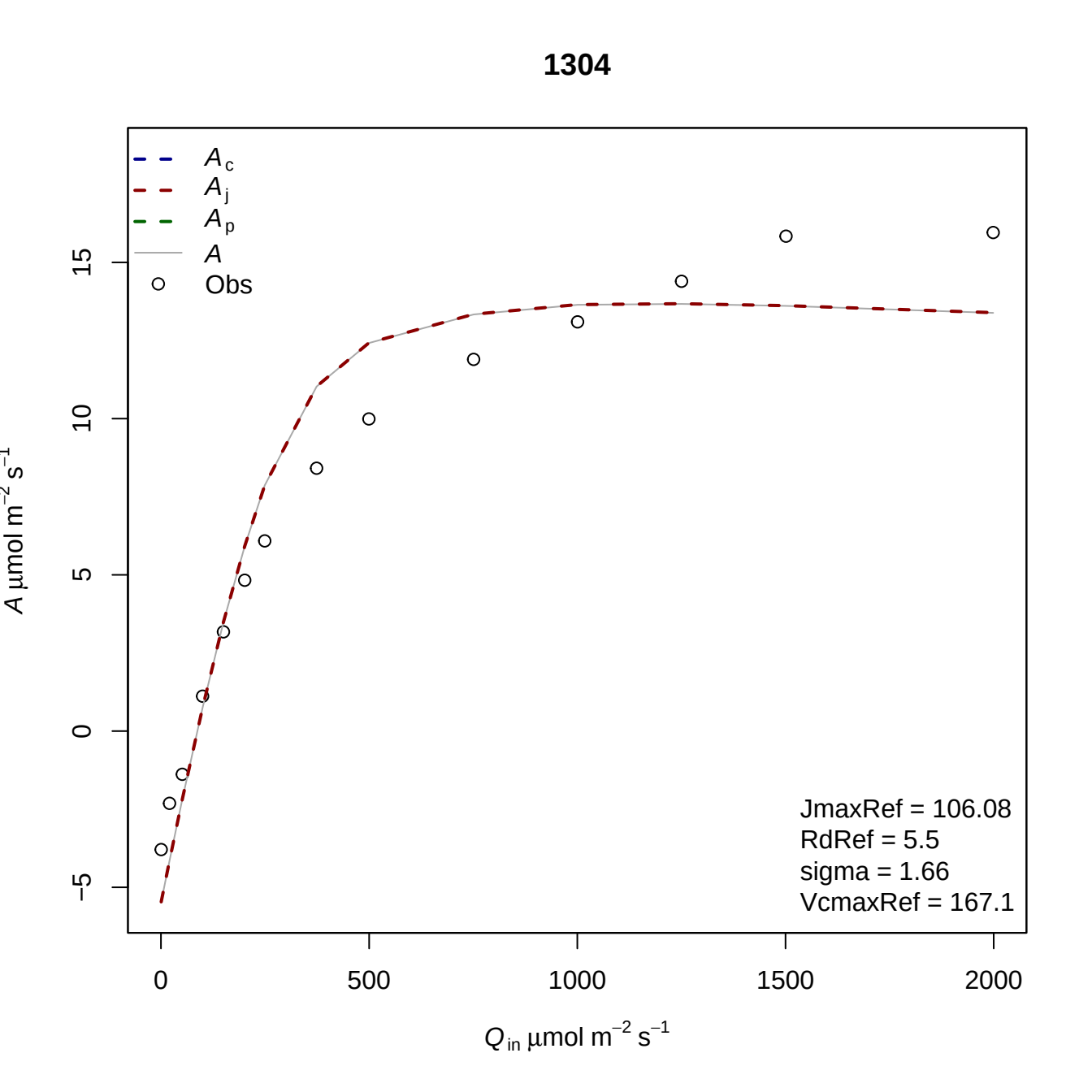

1299
A_c
A_j
A_p
A
Obs
JmaxRef = 109.55
RdRef = 5.56
sigma = 1.64
VcmaxRef = 136.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

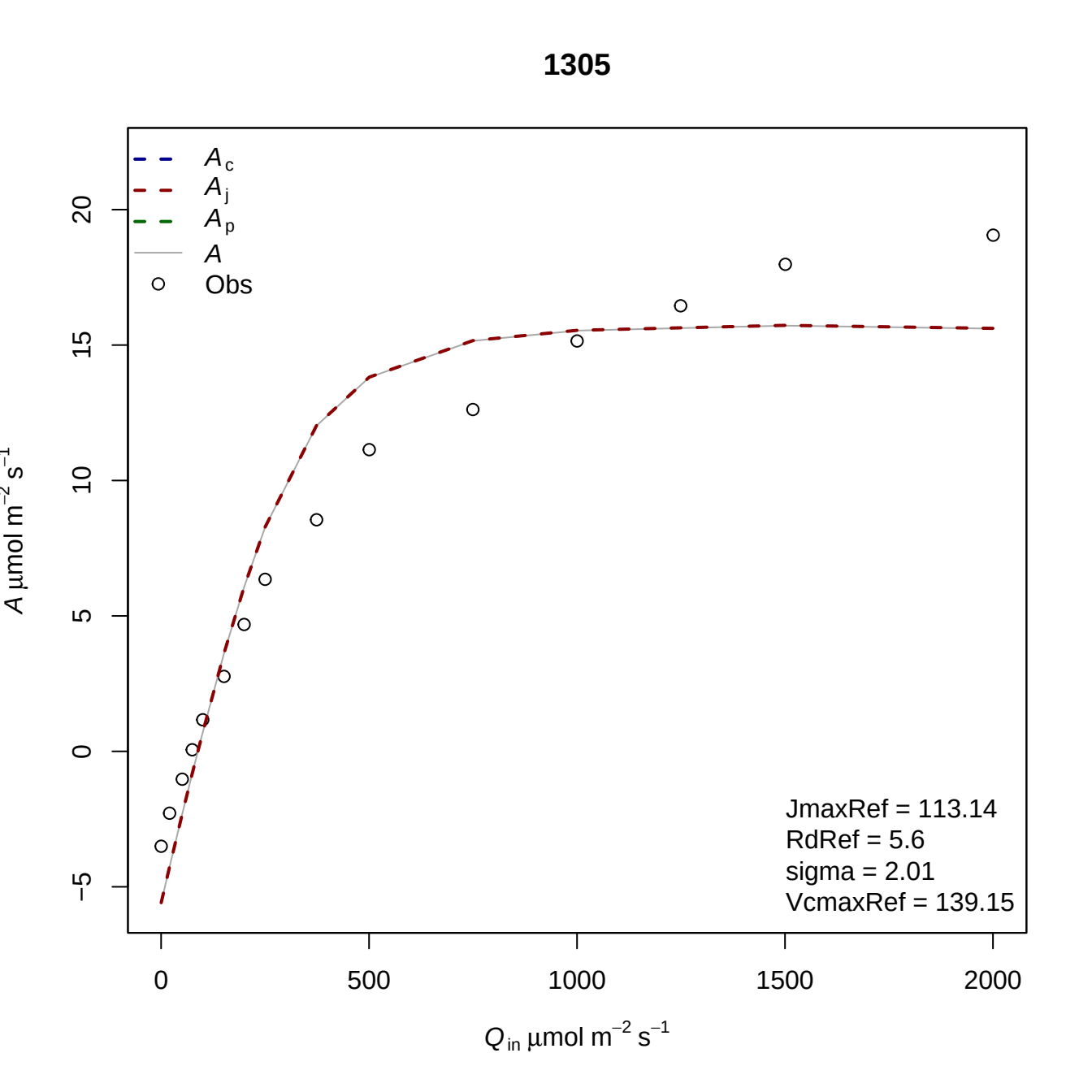

1300
A_c
A_j
A_p
A
Obs
JmaxRef = 105.19
RdRef = 3.23
sigma = 1.53
VcmaxRef = 137.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

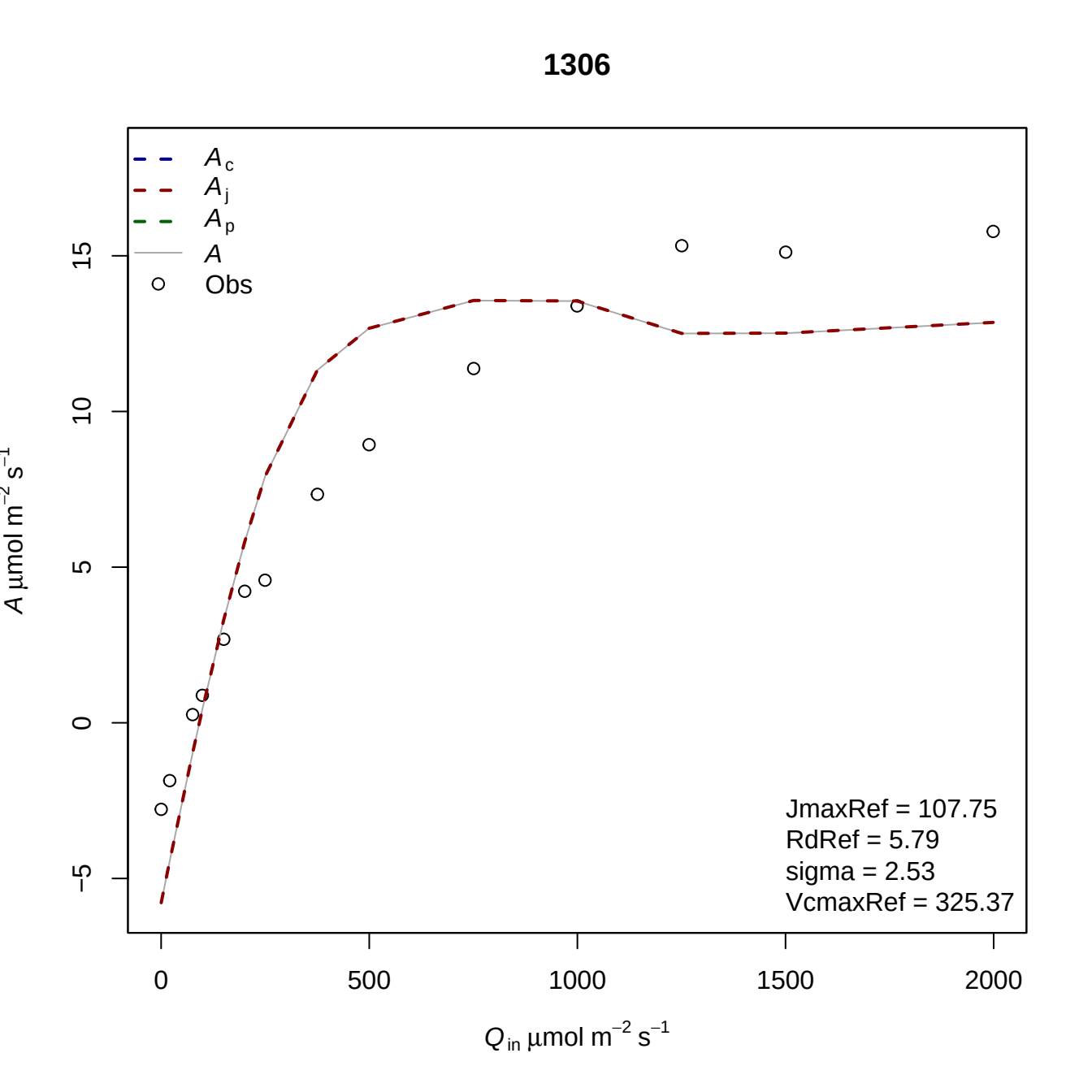

1301
A_c
A_j
A_p
A
Obs
JmaxRef = 91.71
RdRef = 5.61
sigma = 1.89
VcmaxRef = 104.89
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

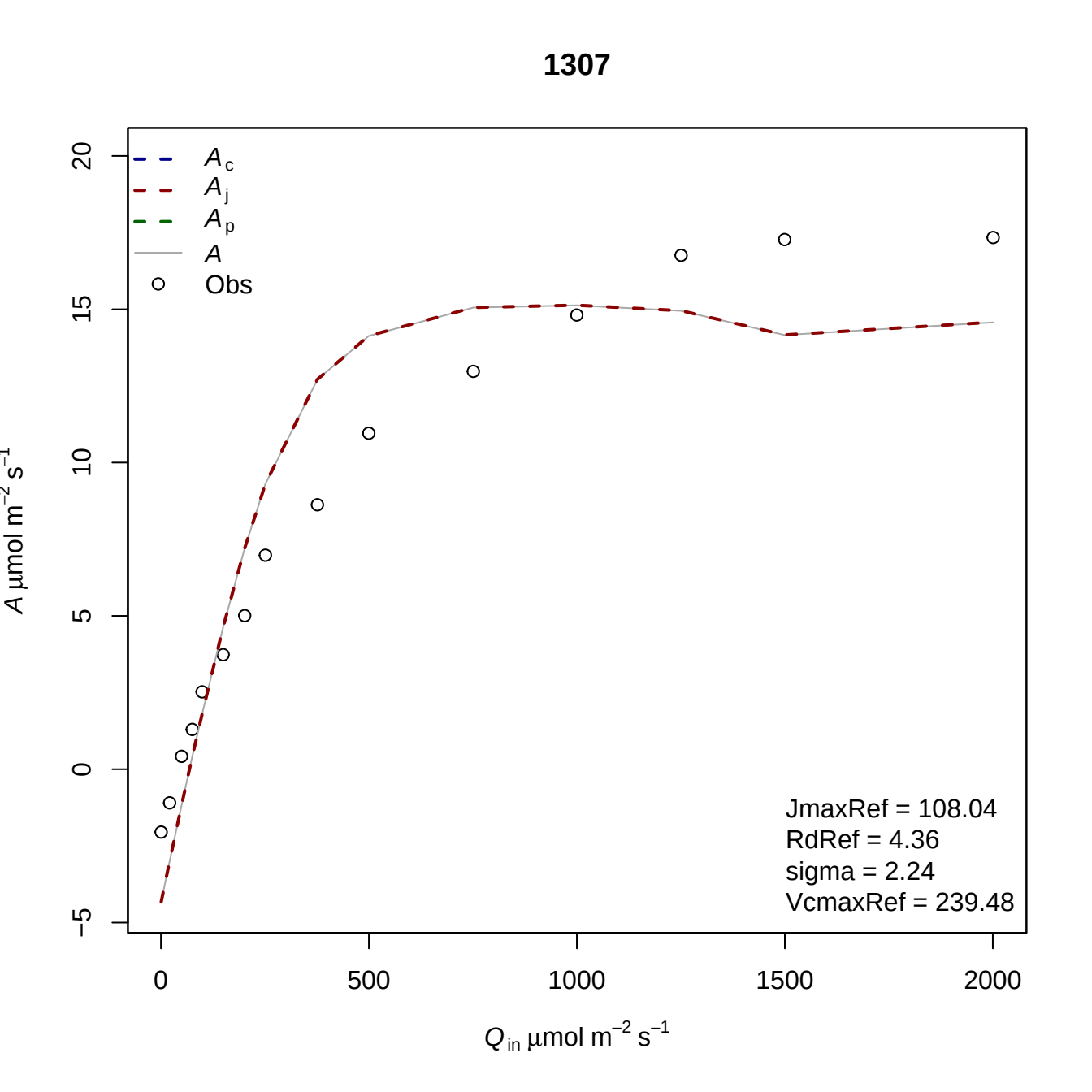

1302
A_c
A_j
A_p
A
Obs
JmaxRef = 78.55
RdRef = 4.72
sigma = 1.47
VcmaxRef = 242
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

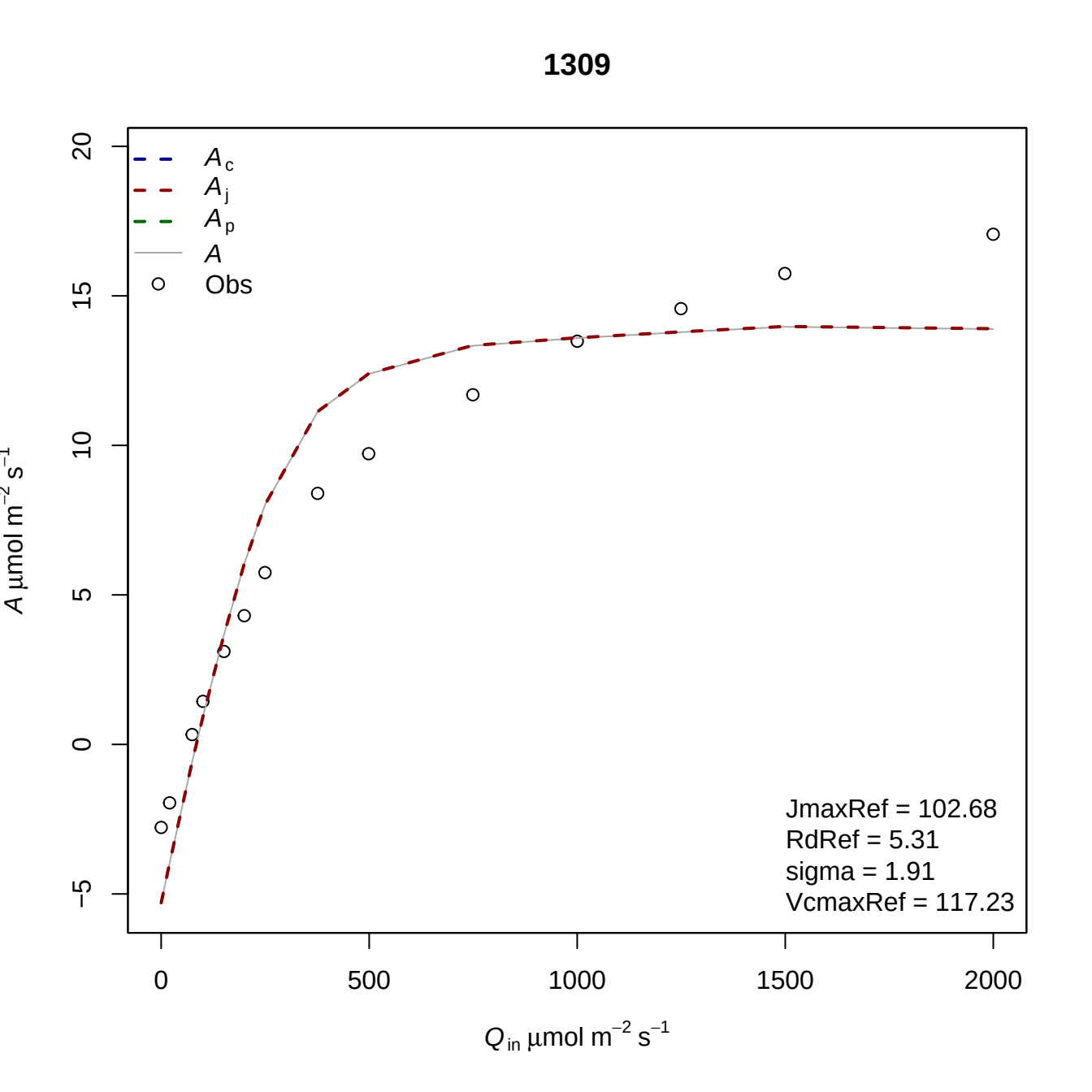

1303
A_c
A_j
A_p
A
Obs
JmaxRef = 80.08
RdRef = 4.84
sigma = 1.56
VcmaxRef = 180.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

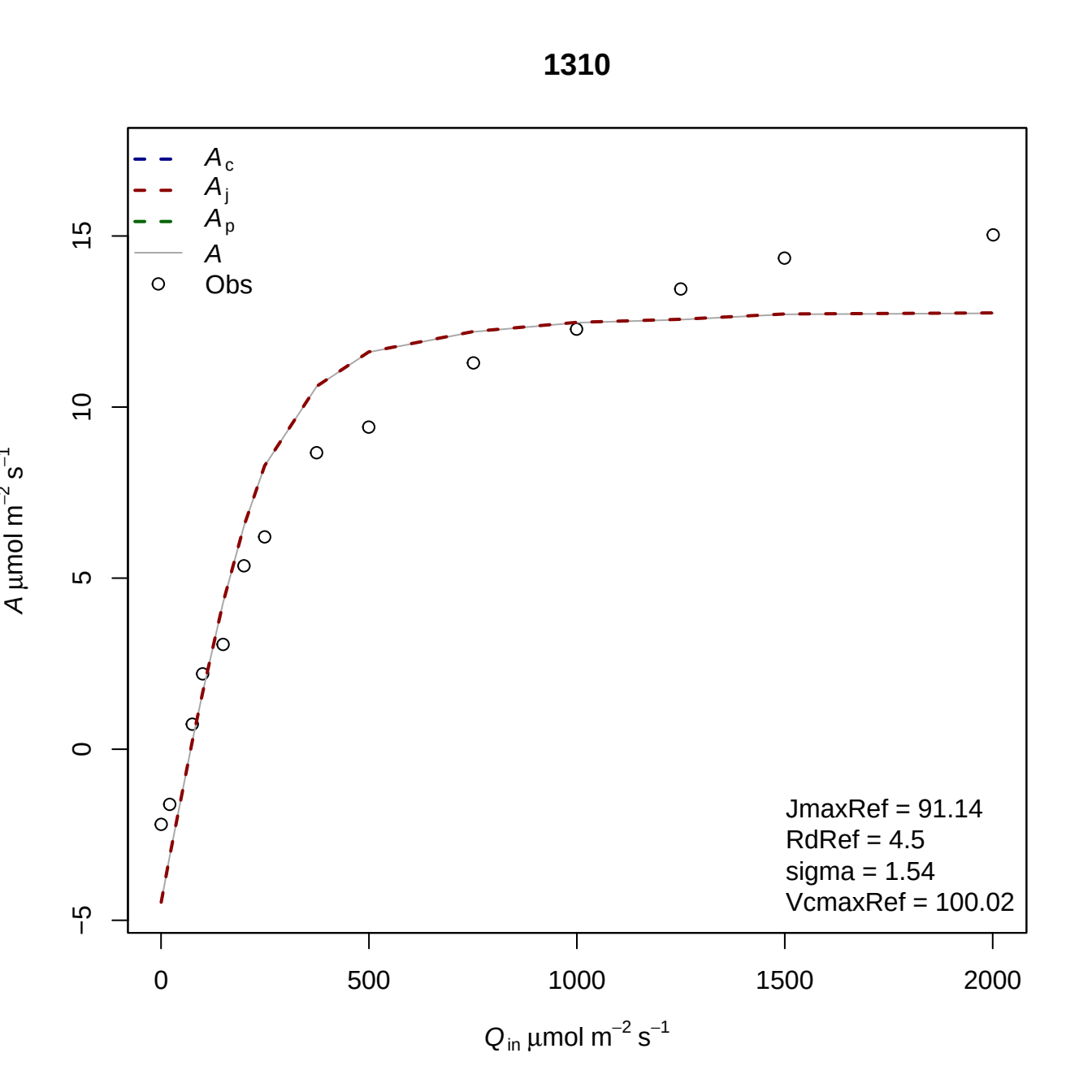

1304
A_c
A_j
A_p
A
Obs
JmaxRef = 106.08
RdRef = 5.5
sigma = 1.66
VcmaxRef = 167.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

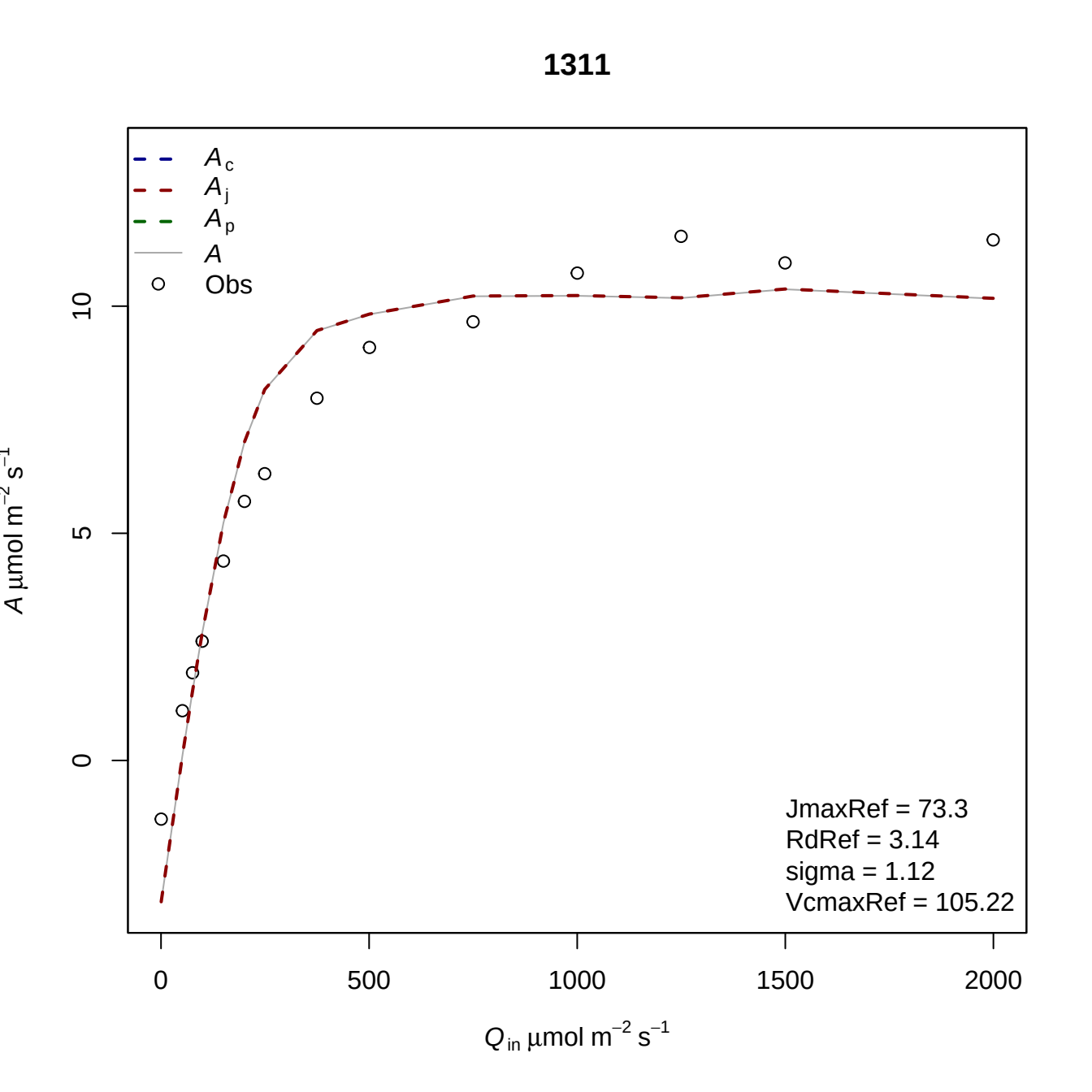

1305
A_c
A_j
A_p
A
Obs
JmaxRef = 113.14
RdRef = 5.6
sigma = 2.01
VcmaxRef = 139.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

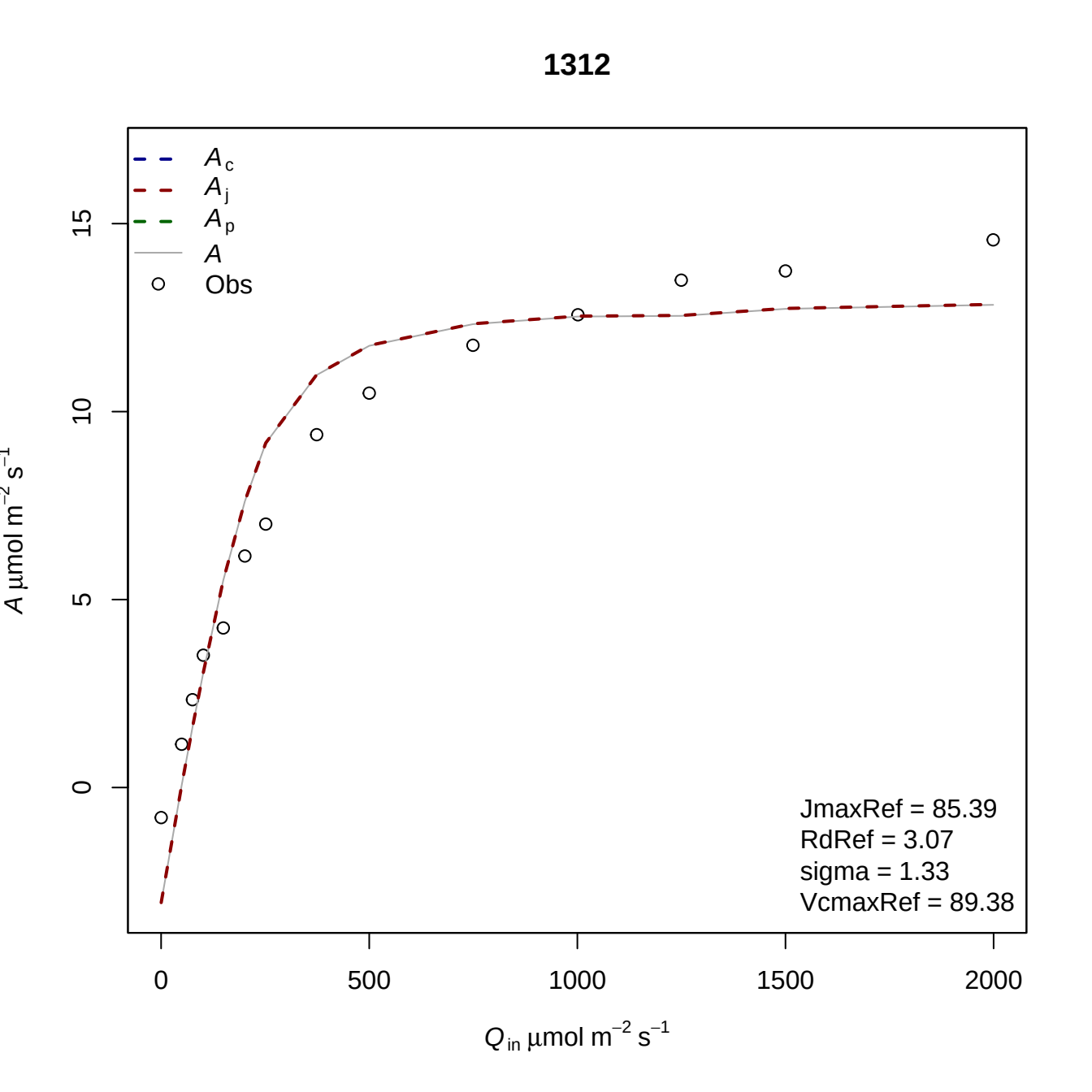

1306
A_c
A_j
A_p
A
Obs
JmaxRef = 107.75
RdRef = 5.79
sigma = 2.53
VcmaxRef = 325.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

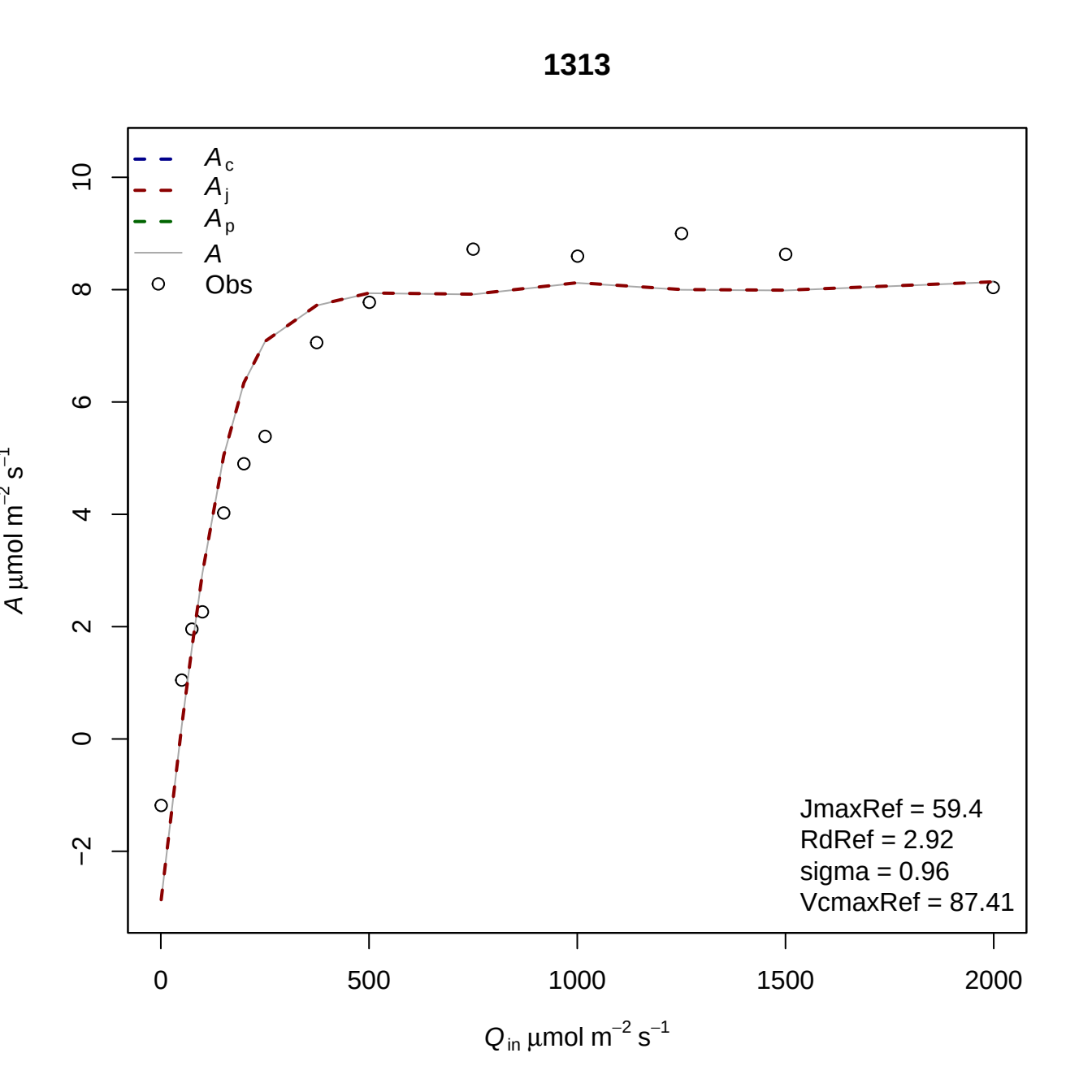

1307
A_c
A_j
A_p
A
Obs
JmaxRef = 108.04
RdRef = 4.36
sigma = 2.24
VcmaxRef = 239.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

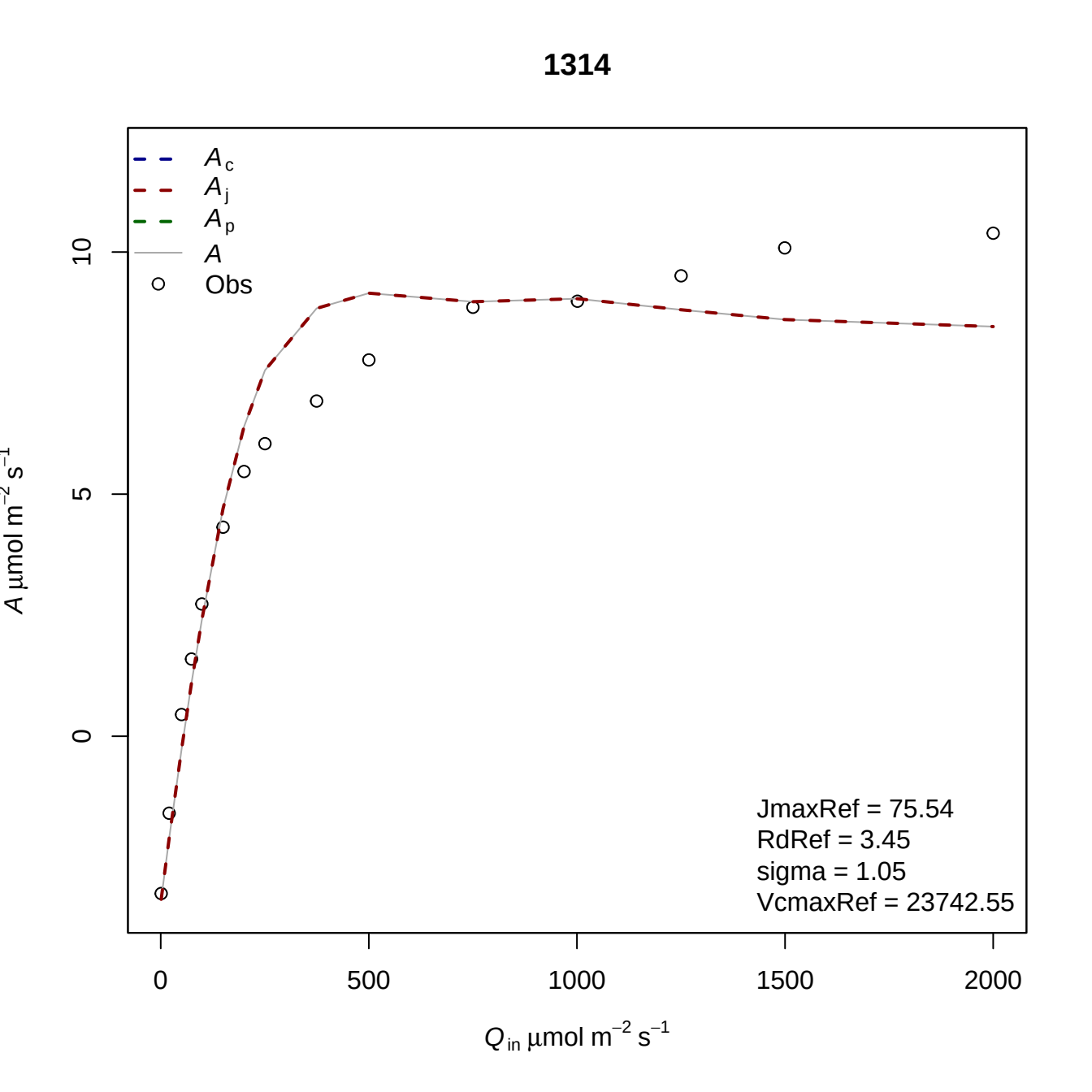

1309
A_c
A_j
A_p
A
Obs
JmaxRef = 102.68
RdRef = 5.31
sigma = 1.91
VcmaxRef = 117.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

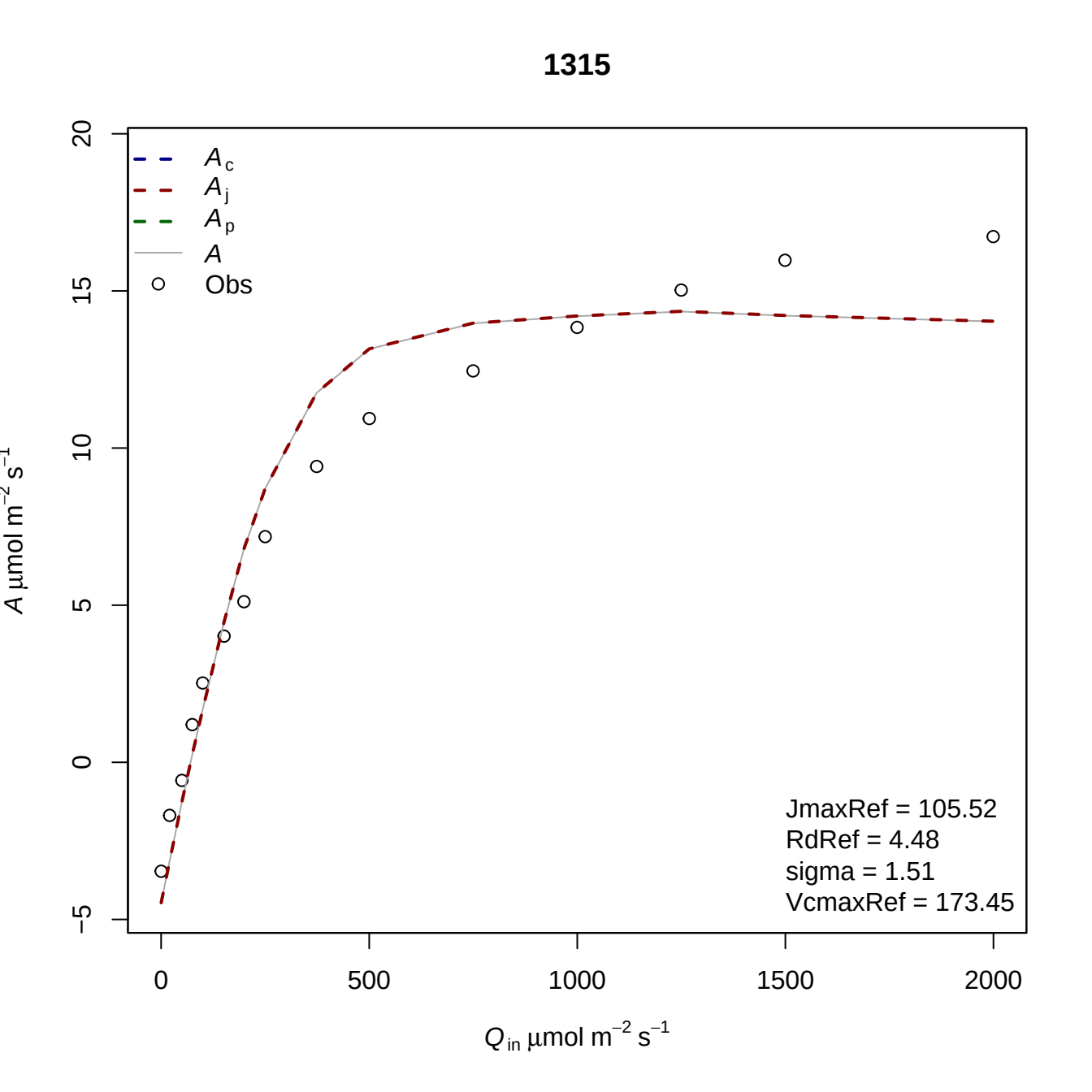

1310
A_c
A_j
A_p
A
Obs
JmaxRef = 91.14
RdRef = 4.5
sigma = 1.54
VcmaxRef = 100.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

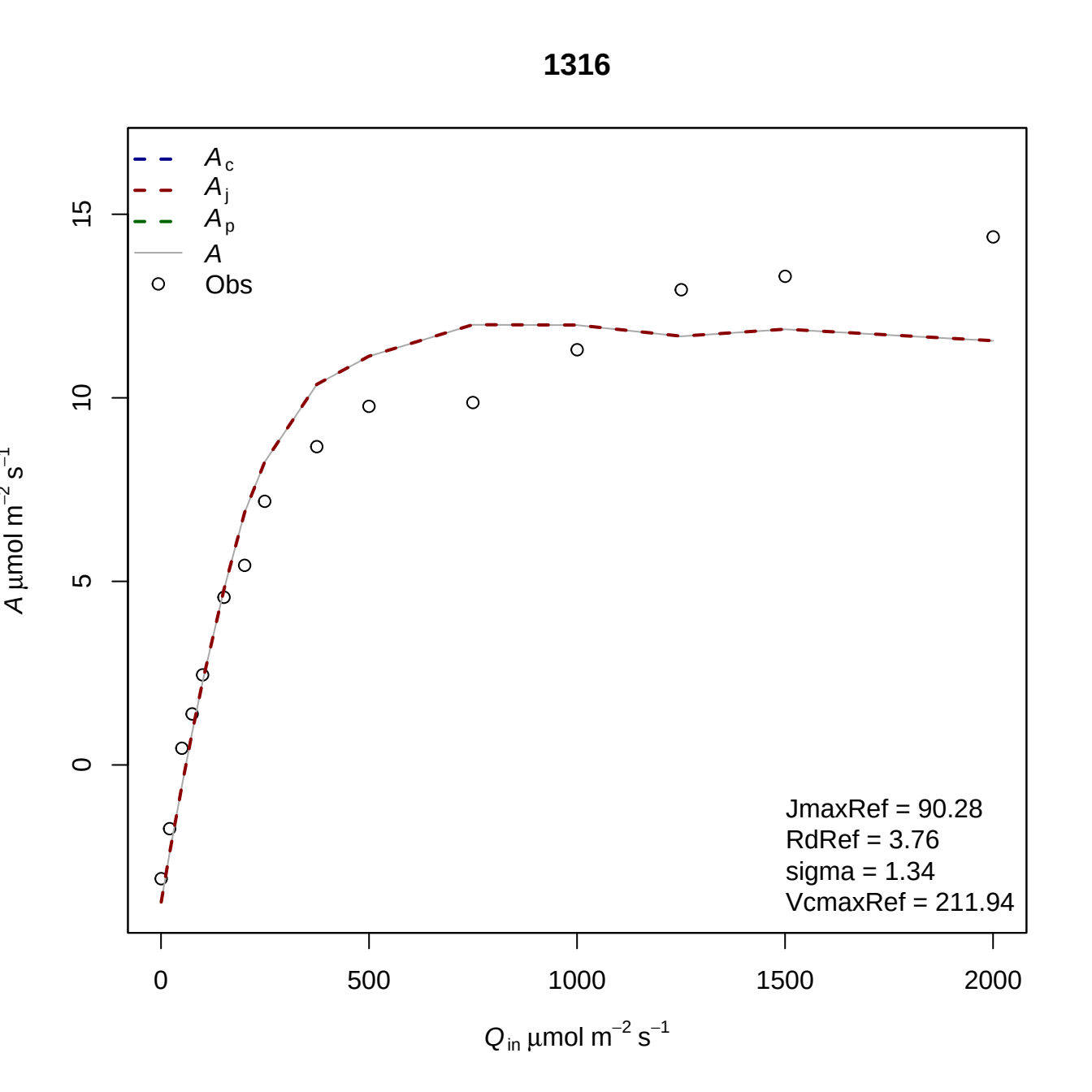

1311
A_c
A_j
A_p
A
Obs
JmaxRef = 73.3
RdRef = 3.14
sigma = 1.12
VcmaxRef = 105.22
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

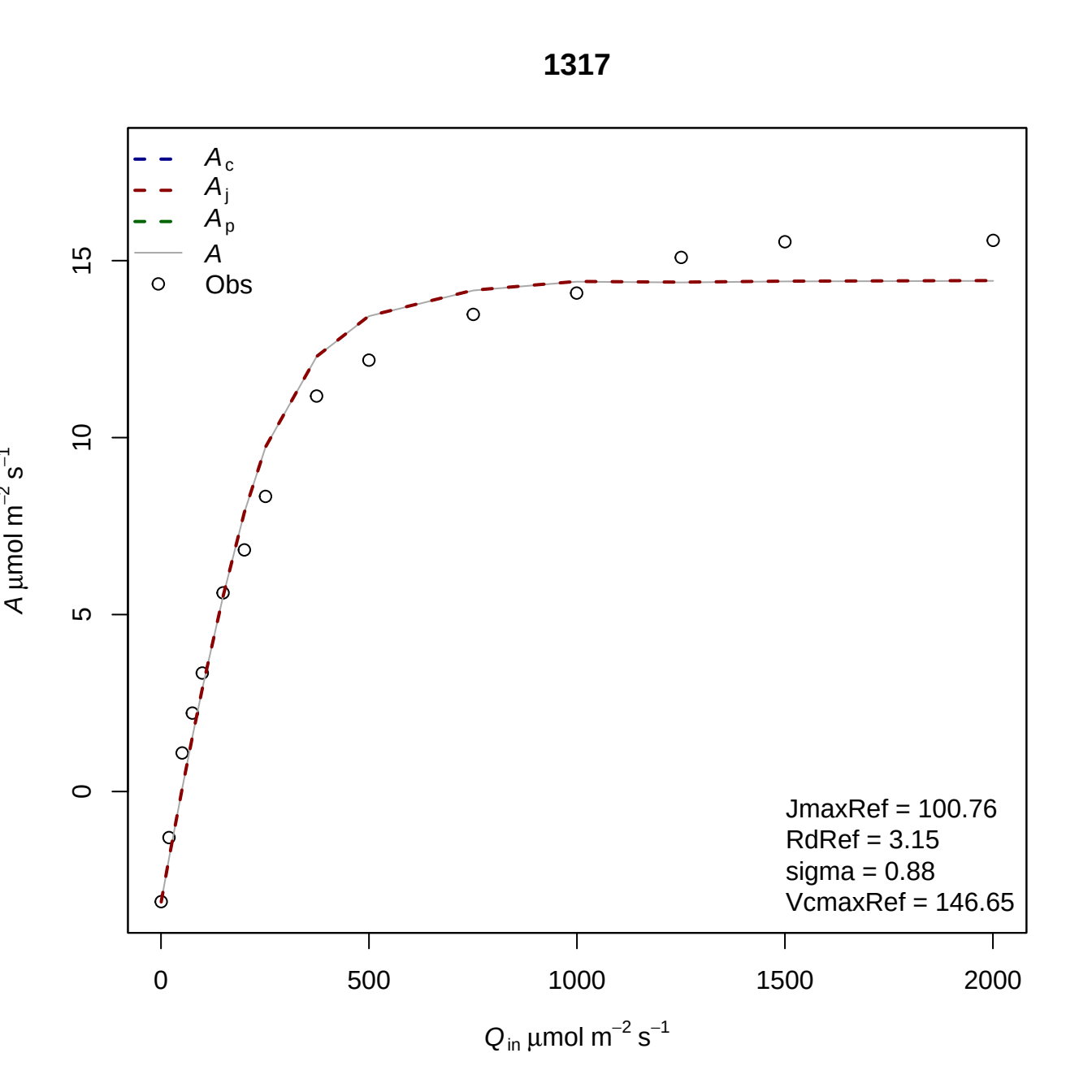

1312
A_c
A_j
A_p
A
Obs
JmaxRef = 85.39
RdRef = 3.07
sigma = 1.33
VcmaxRef = 89.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

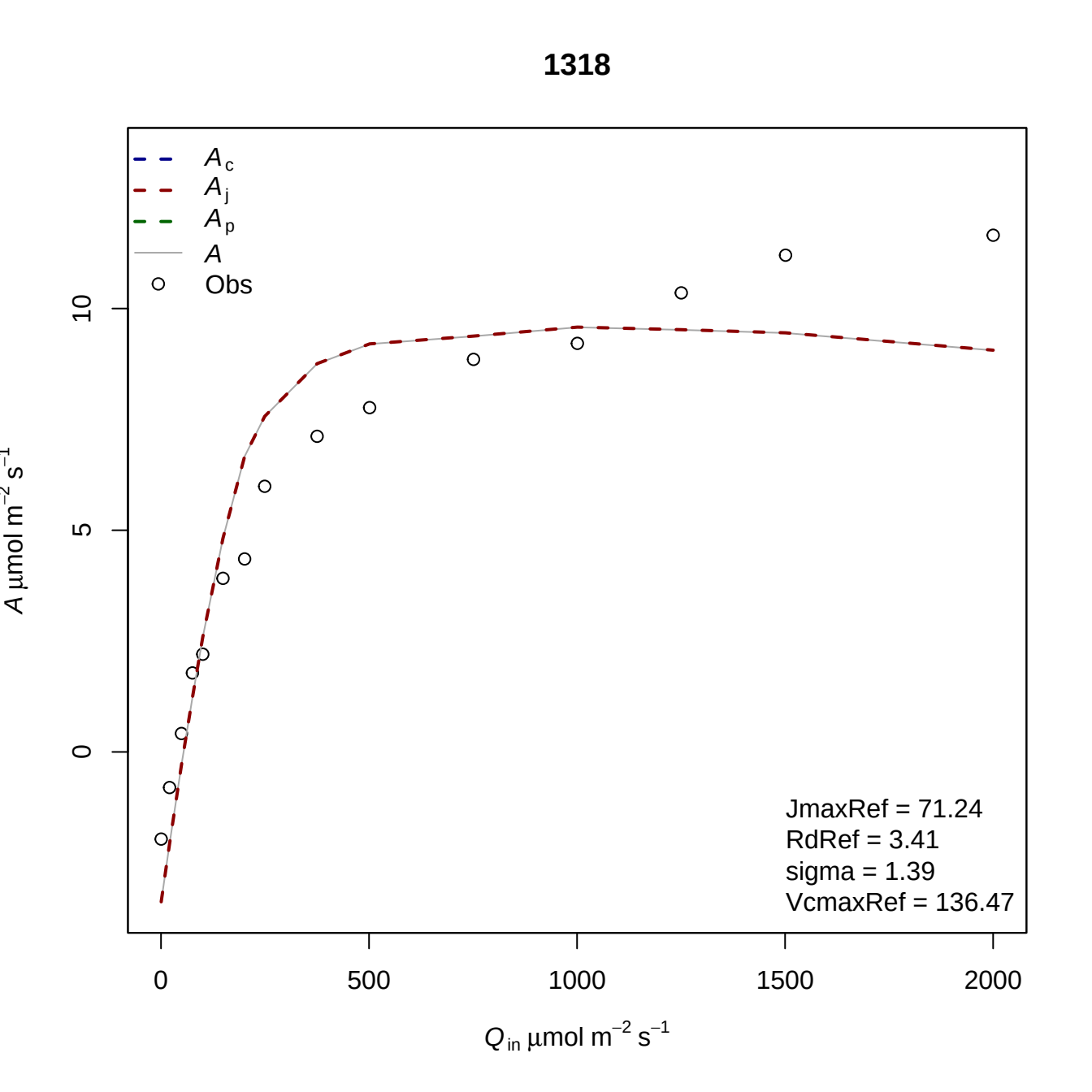

1313
A_c
A_j
A_p
A
Obs
JmaxRef = 59.4
RdRef = 2.92
sigma = 0.96
VcmaxRef = 87.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

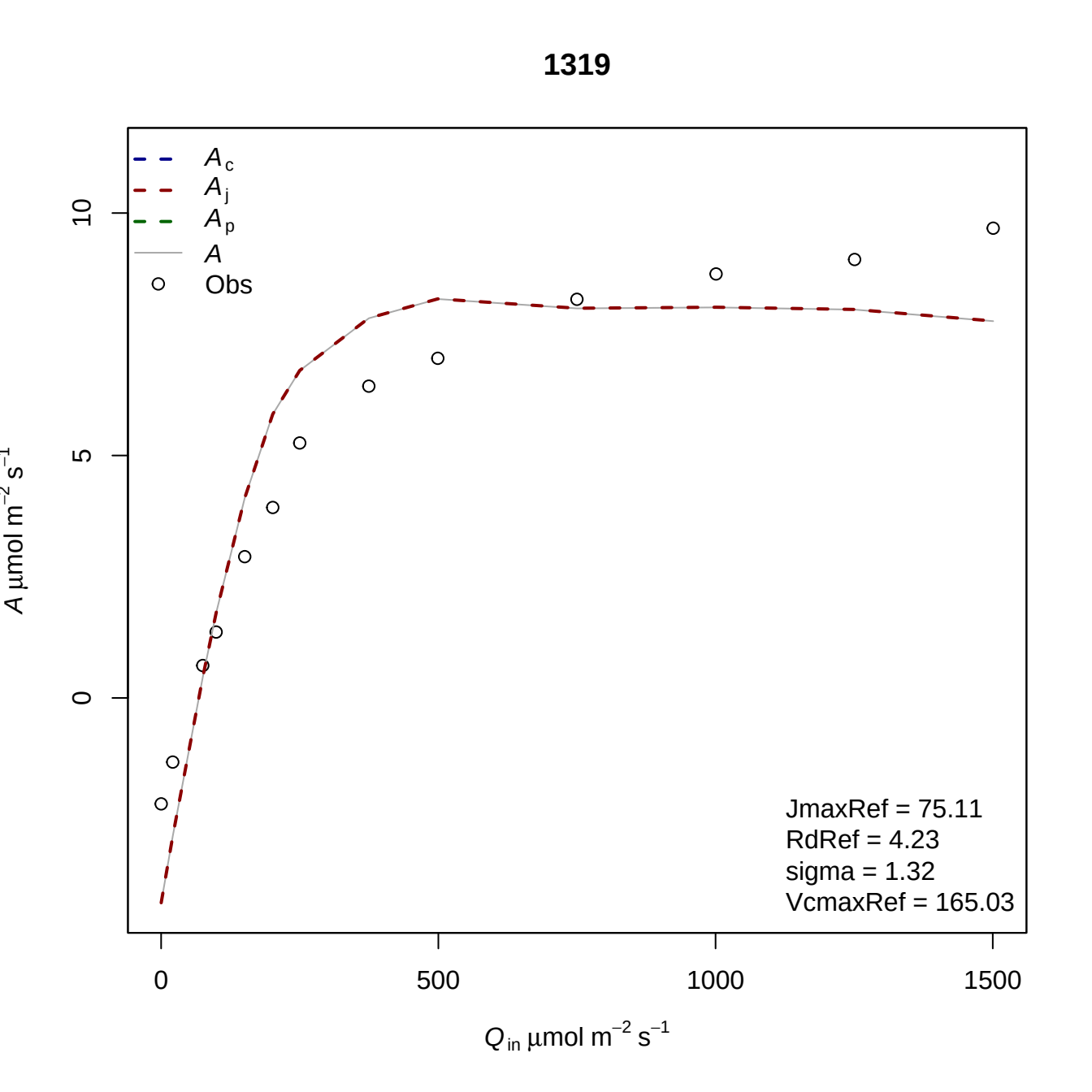

1314
A_c
A_j
A_p
A
Obs
JmaxRef = 75.54
RdRef = 3.45
sigma = 1.05
VcmaxRef = 23742.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

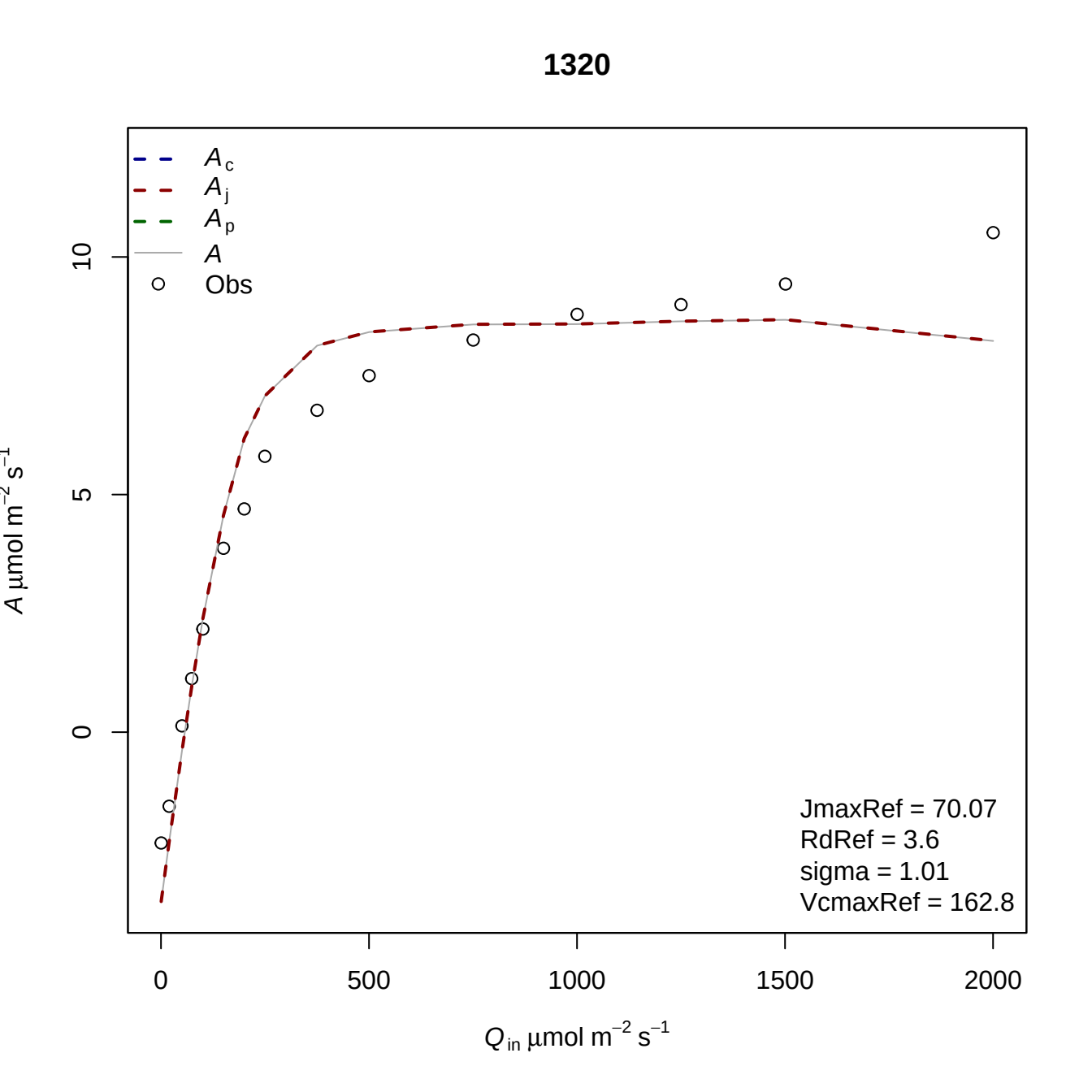

1315
A_c
A_j
A_p
A
Obs
JmaxRef = 105.52
RdRef = 4.48
sigma = 1.51
VcmaxRef = 173.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

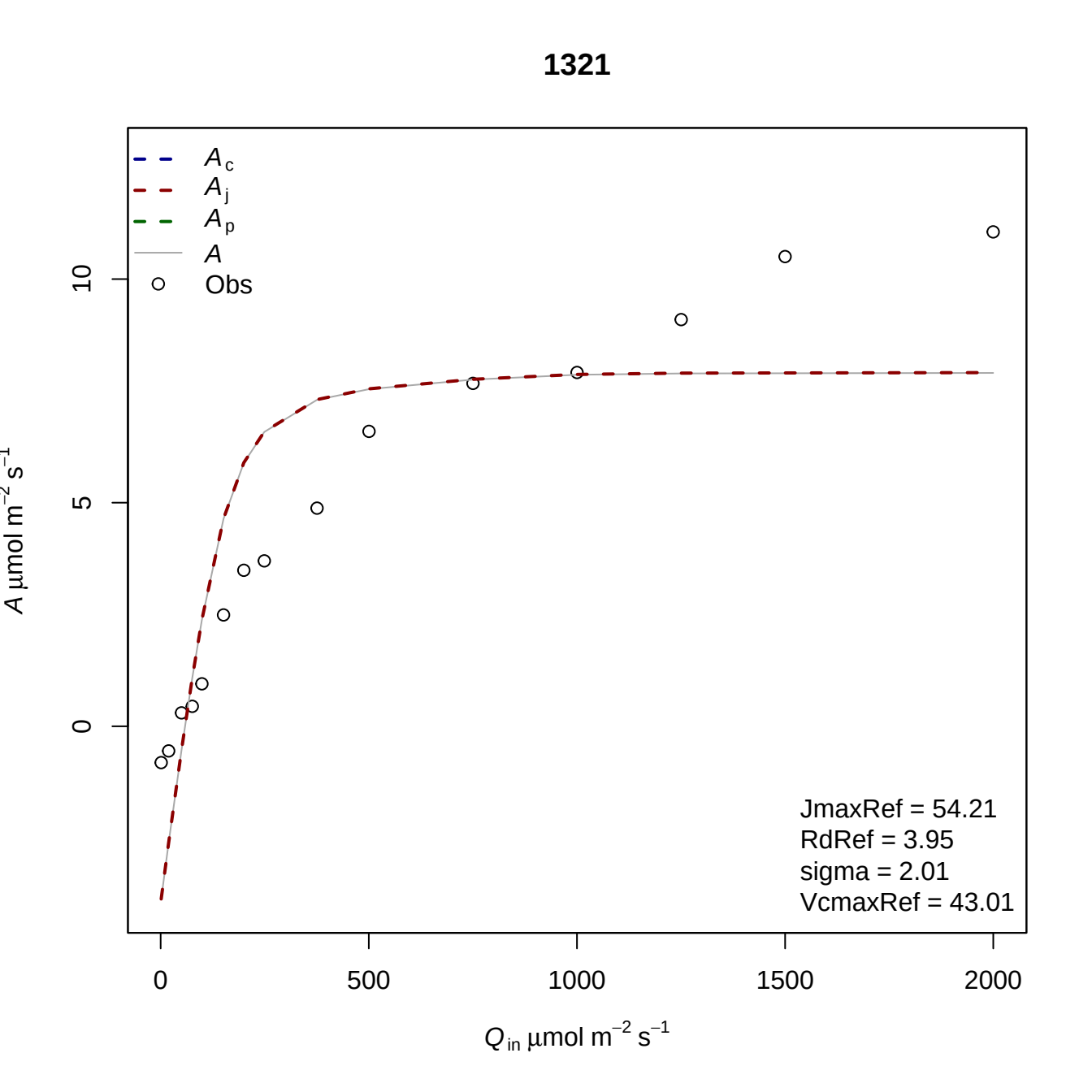

1316
A_c
A_j
A_p
A
Obs
JmaxRef = 90.28
RdRef = 3.76
sigma = 1.34
VcmaxRef = 211.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

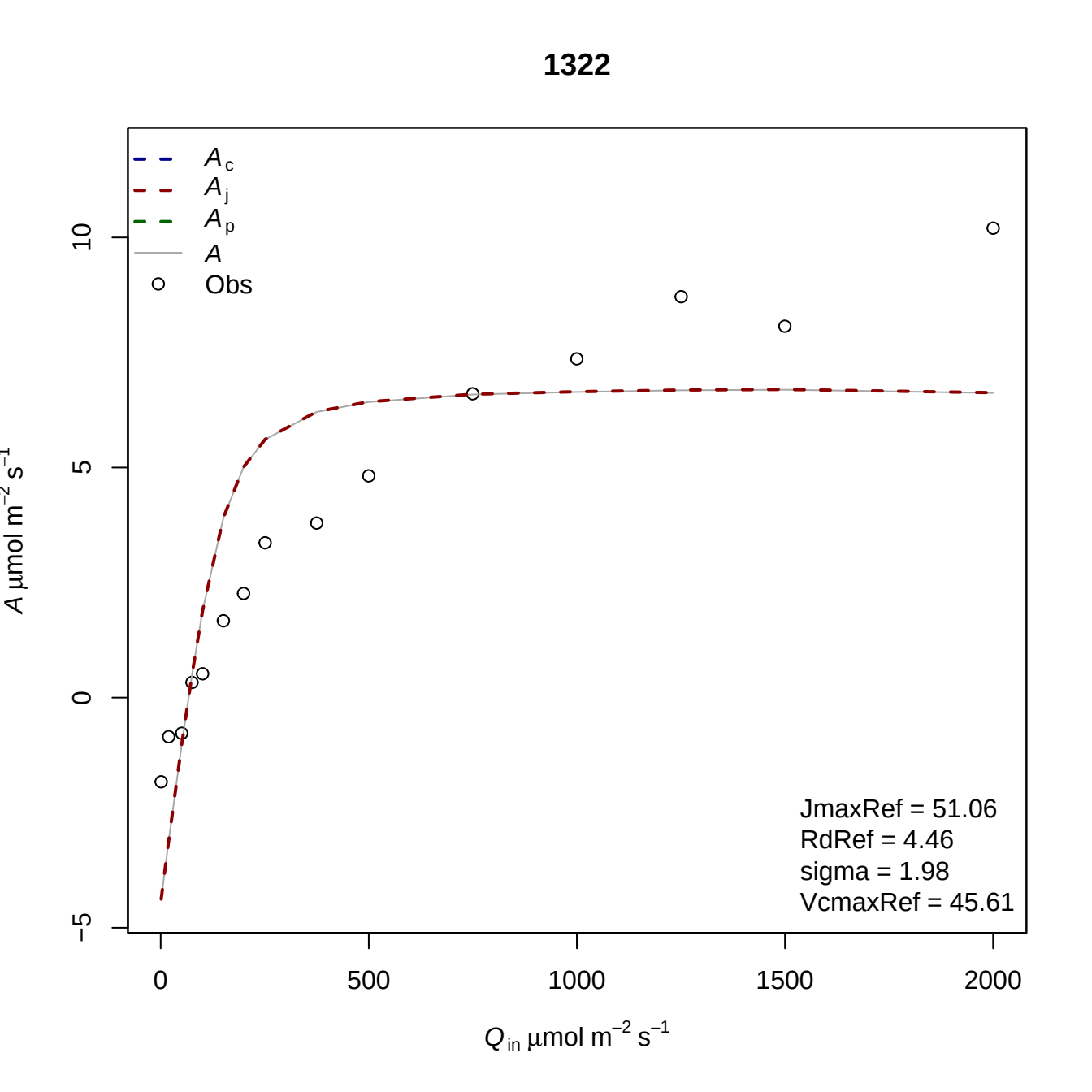

1317
A_c
A_j
A_p
A
Obs
JmaxRef = 100.76
RdRef = 3.15
sigma = 0.88
VcmaxRef = 146.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

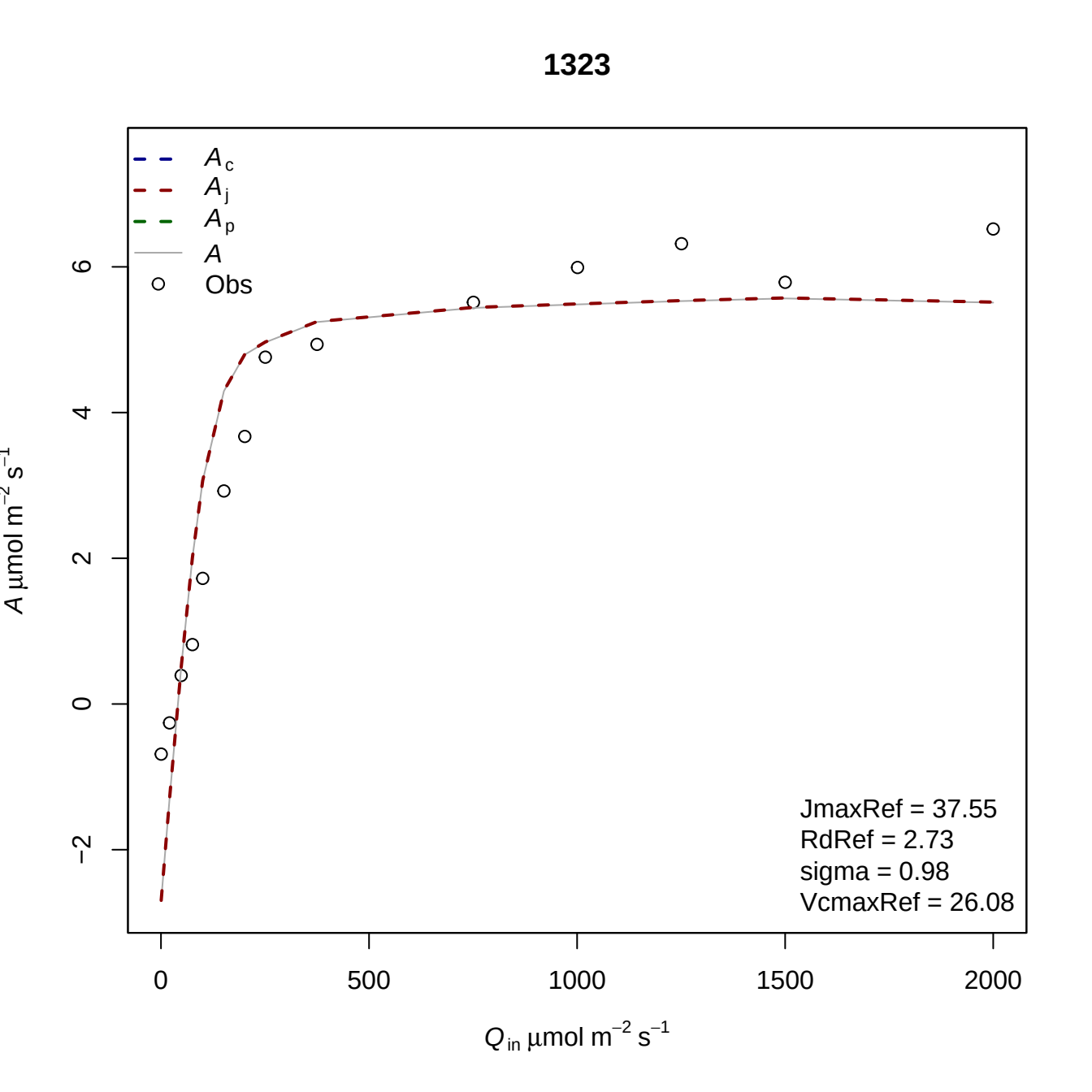

1318
A_c
A_j
A_p
A
Obs
JmaxRef = 71.24
RdRef = 3.41
sigma = 1.39
VcmaxRef = 136.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

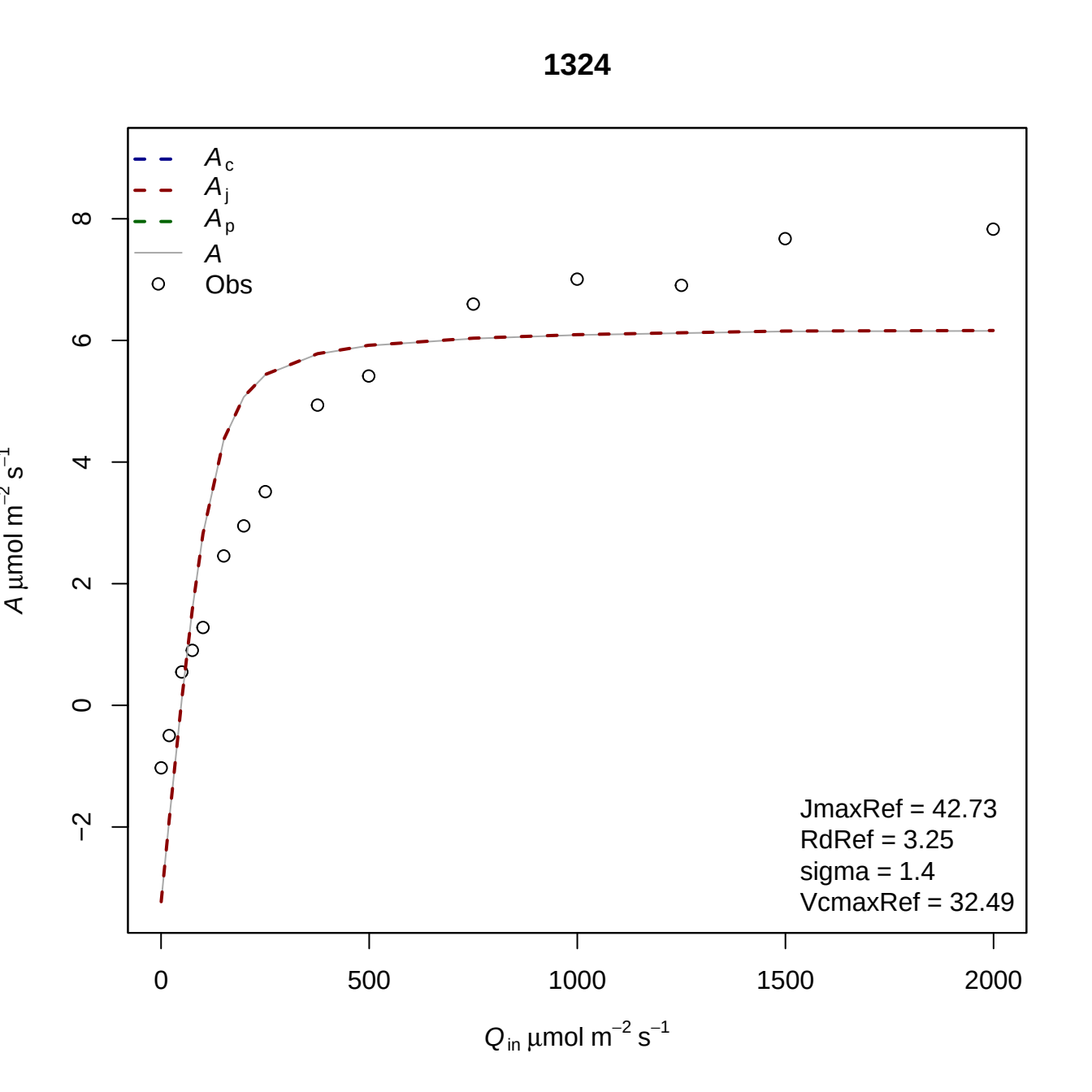

1319
A_c
A_j
A_p
A
Obs
JmaxRef = 75.11
RdRef = 4.23
sigma = 1.32
VcmaxRef = 165.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

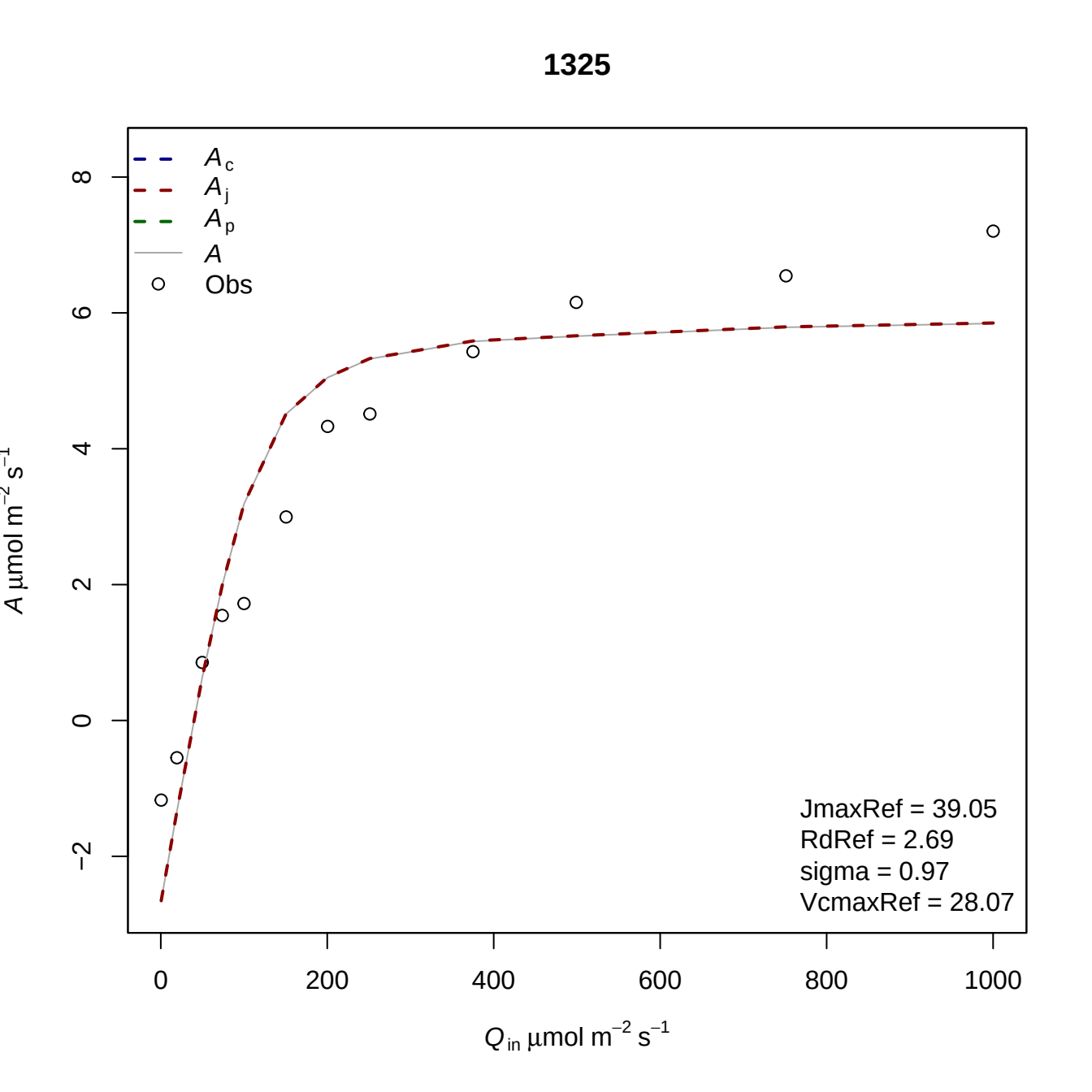

1320
A_c
A_j
A_p
A
Obs
JmaxRef = 70.07
RdRef = 3.6
sigma = 1.01
VcmaxRef = 162.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

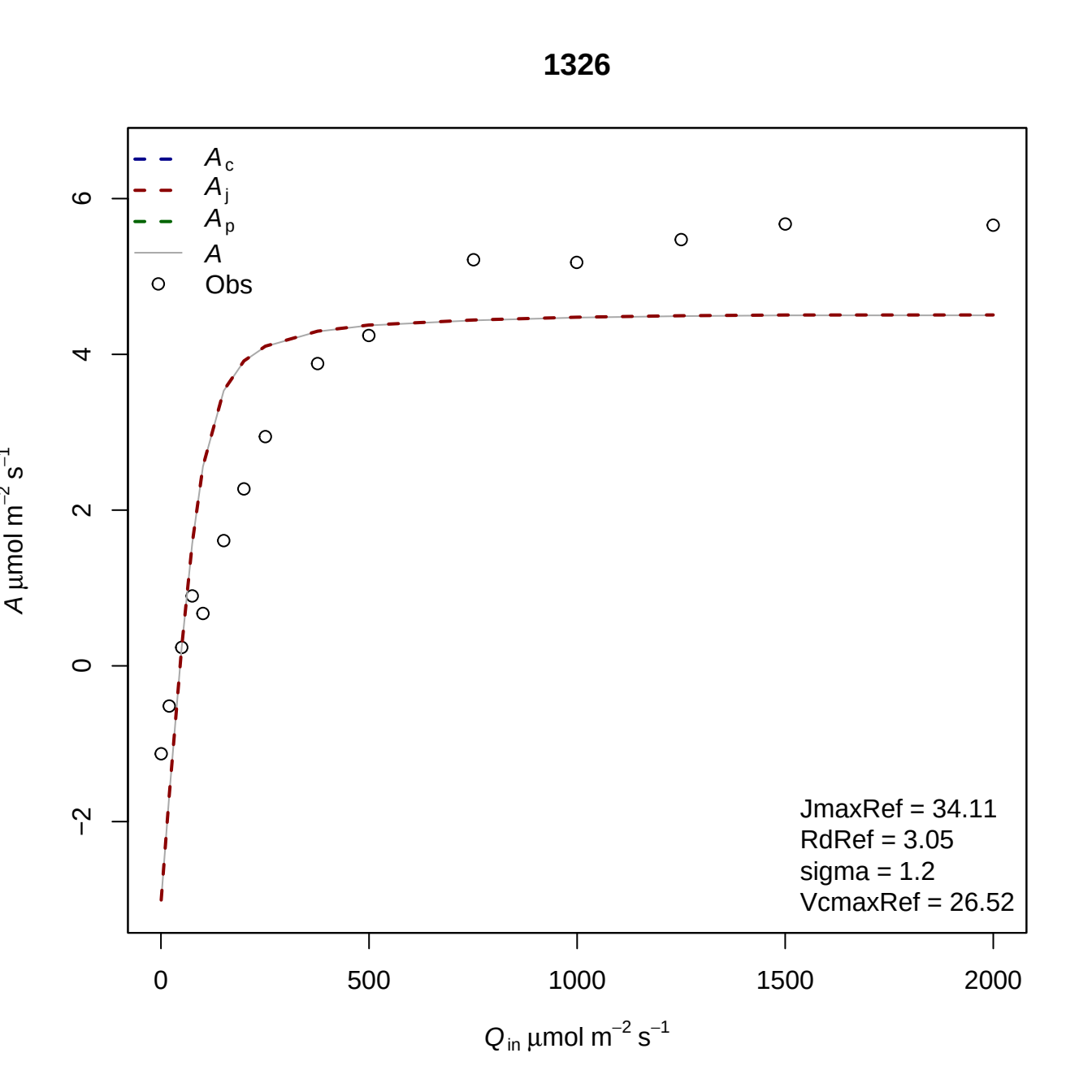

1321
A_c
A_j
A_p
A
Obs
JmaxRef = 54.21
RdRef = 3.95
sigma = 2.01
VcmaxRef = 43.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

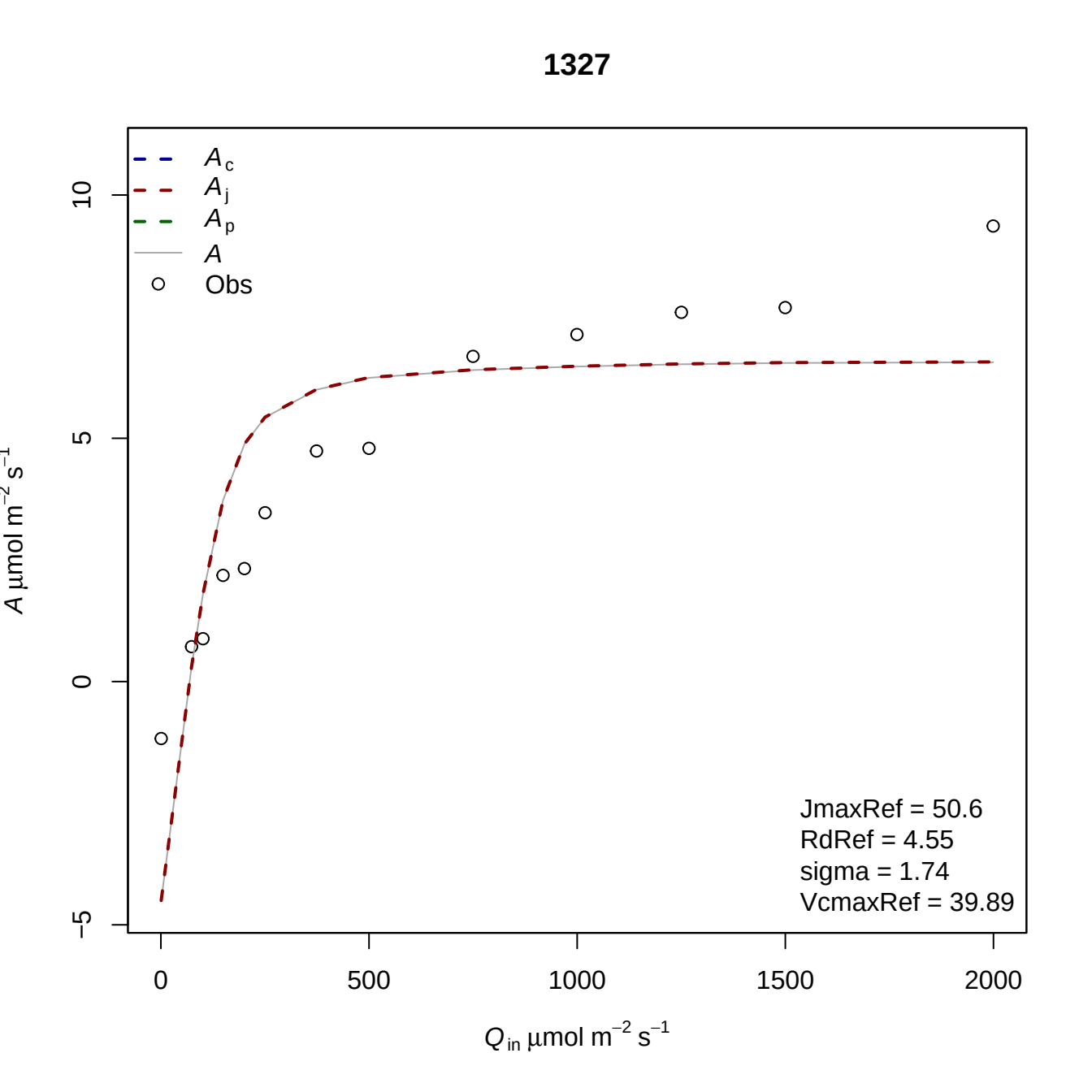

1322
A_c
A_j
A_p
A
Obs
JmaxRef = 51.06
RdRef = 4.46
sigma = 1.98
VcmaxRef = 45.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

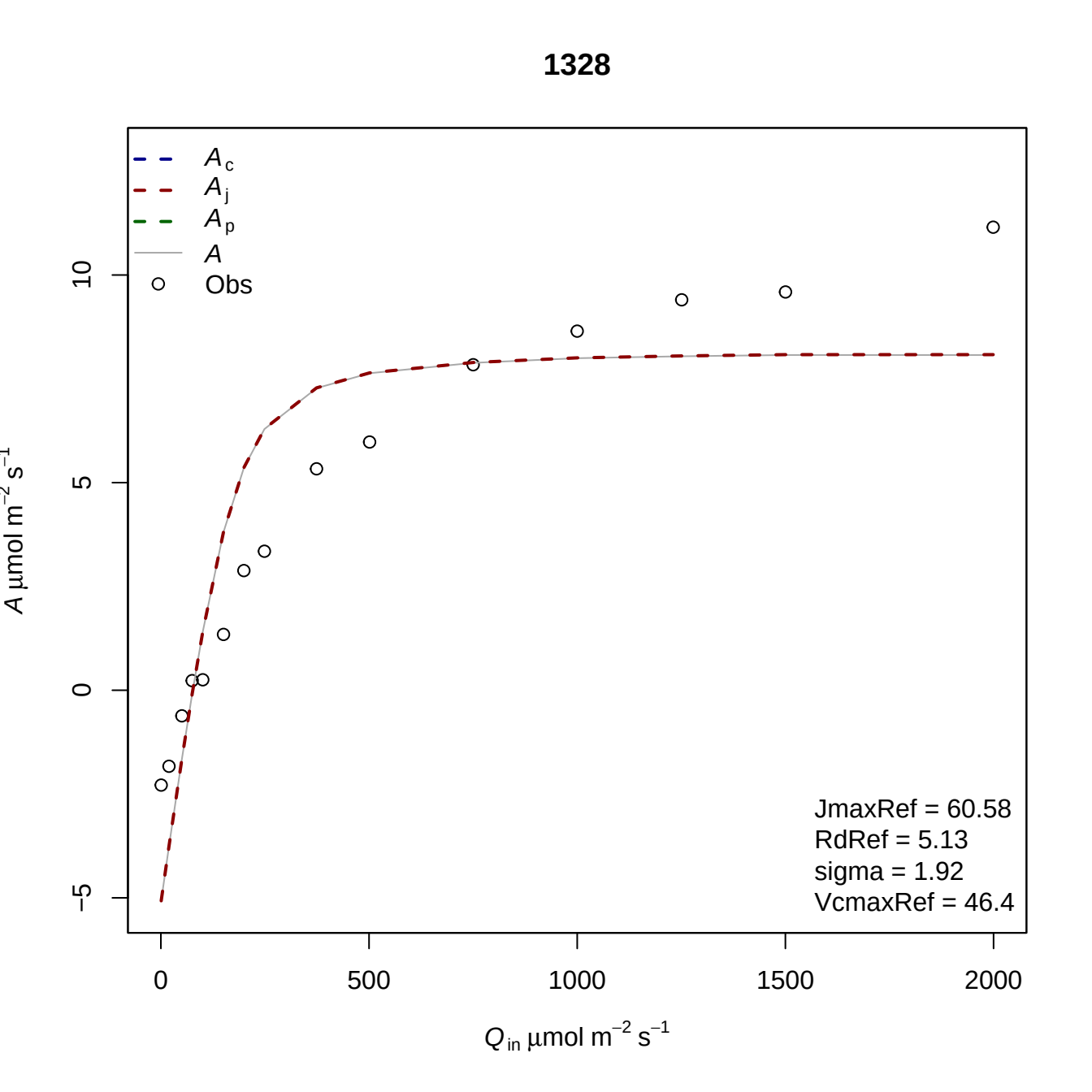

1323
A_c
A_j
A_p
A
Obs
JmaxRef = 37.55
RdRef = 2.73
sigma = 0.98
VcmaxRef = 26.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

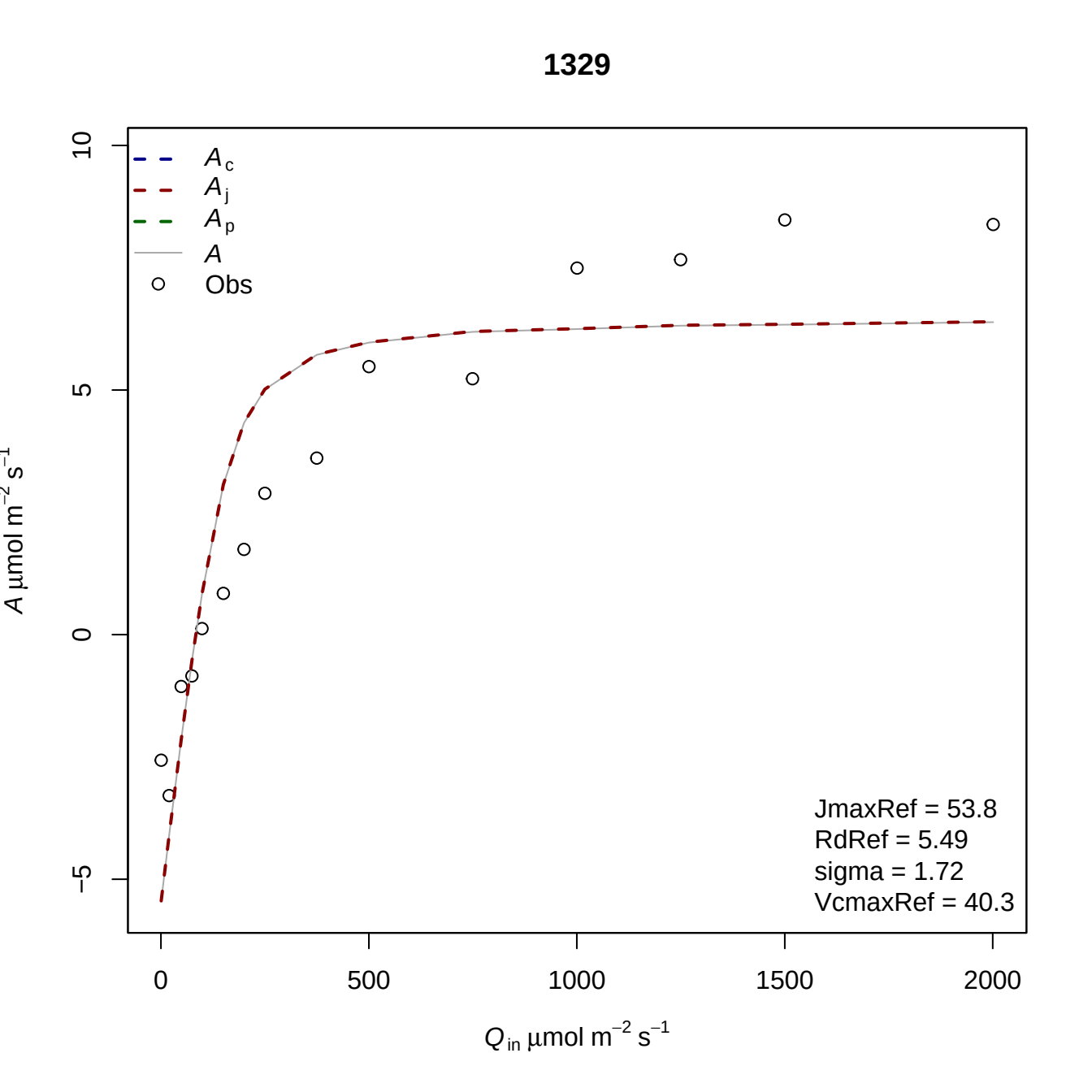

1324
A_c
A_j
A_p
A
Obs
JmaxRef = 42.73
RdRef = 3.25
sigma = 1.4
VcmaxRef = 32.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

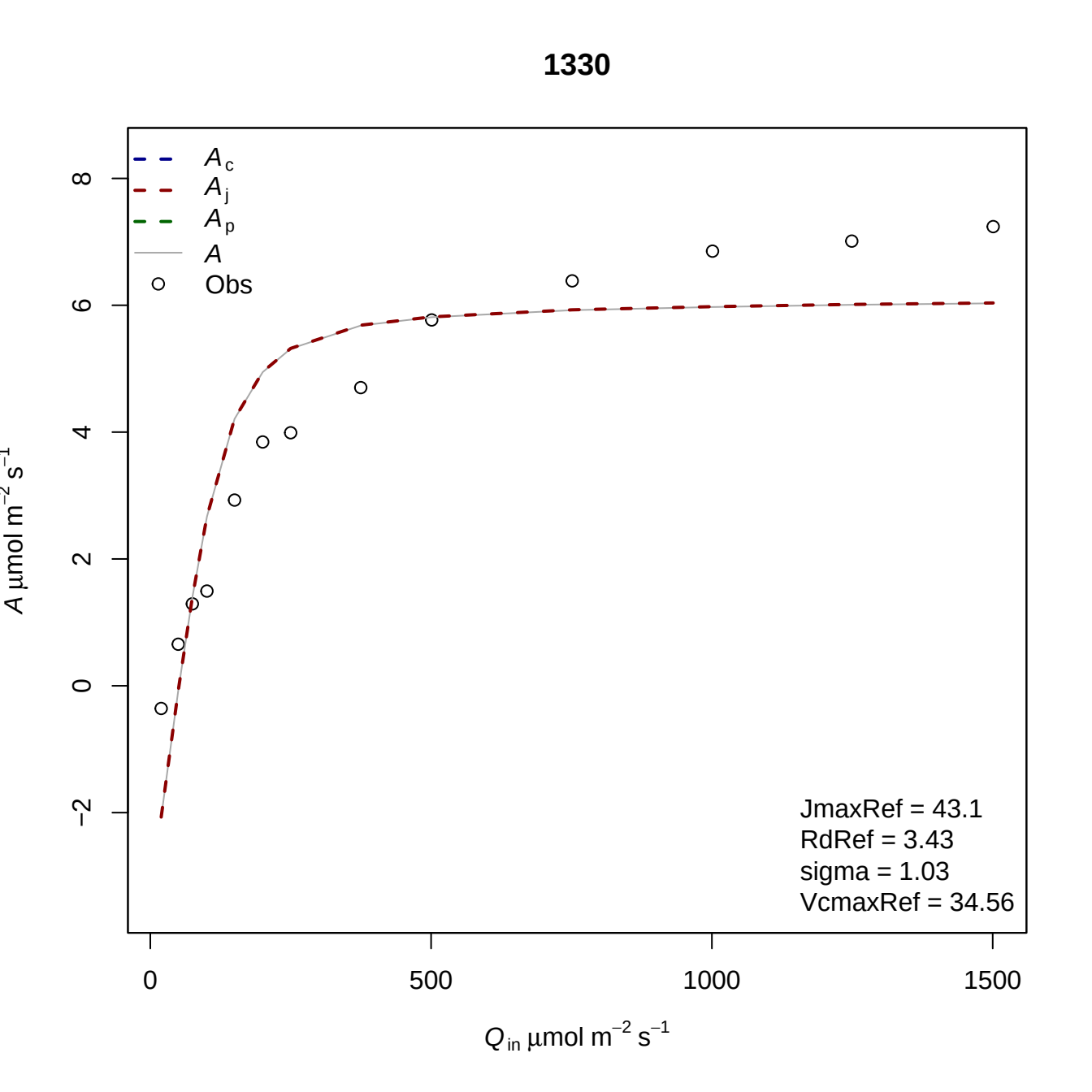

1325
A_c
A_j
A_p
A
Obs
JmaxRef = 39.05
RdRef = 2.69
sigma = 0.97
VcmaxRef = 28.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

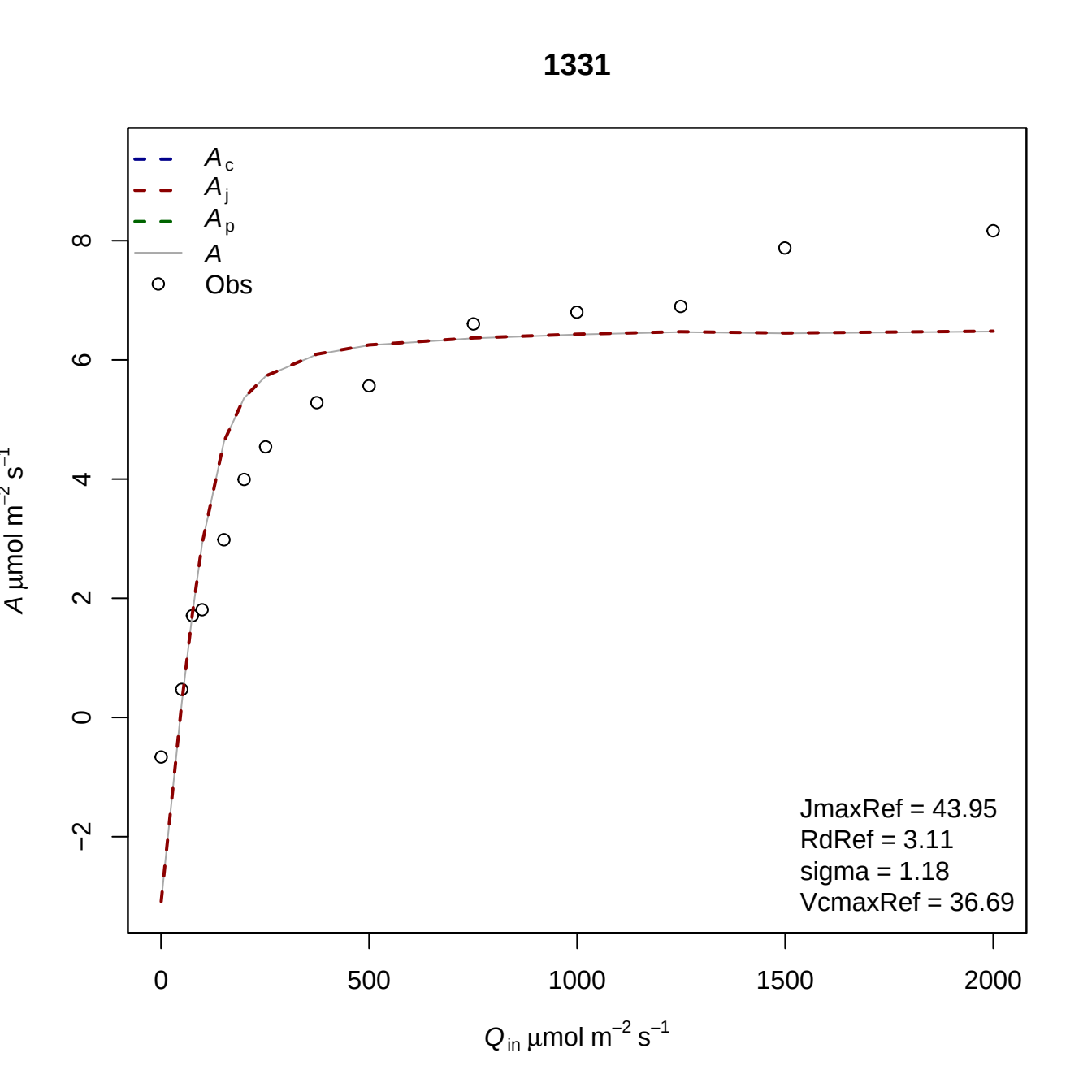

1326
A_c
A_j
A_p
A
Obs
JmaxRef = 34.11
RdRef = 3.05
sigma = 1.2
VcmaxRef = 26.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

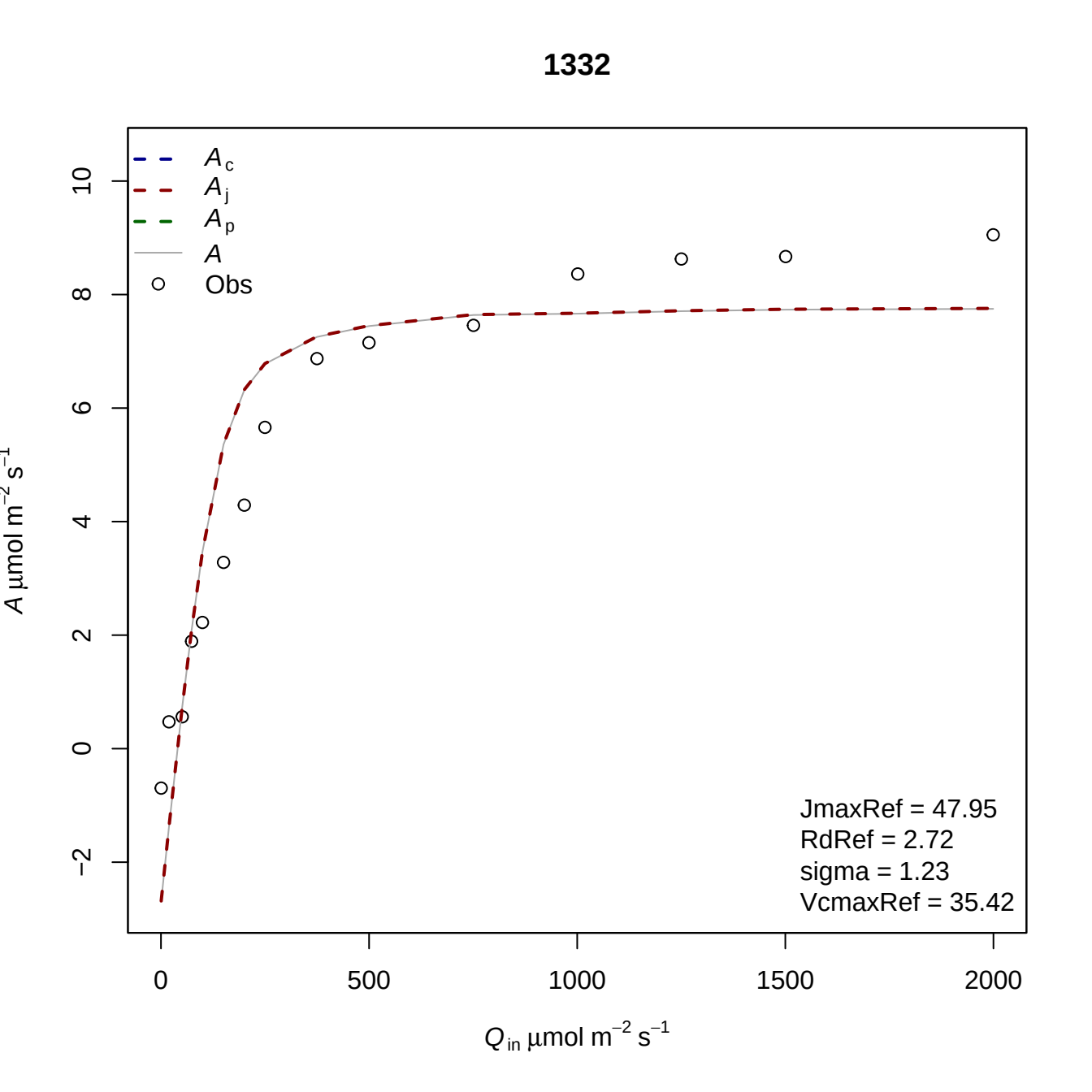

1327
A_c
A_j
A_p
A
Obs
JmaxRef = 50.6
RdRef = 4.55
sigma = 1.74
VcmaxRef = 39.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

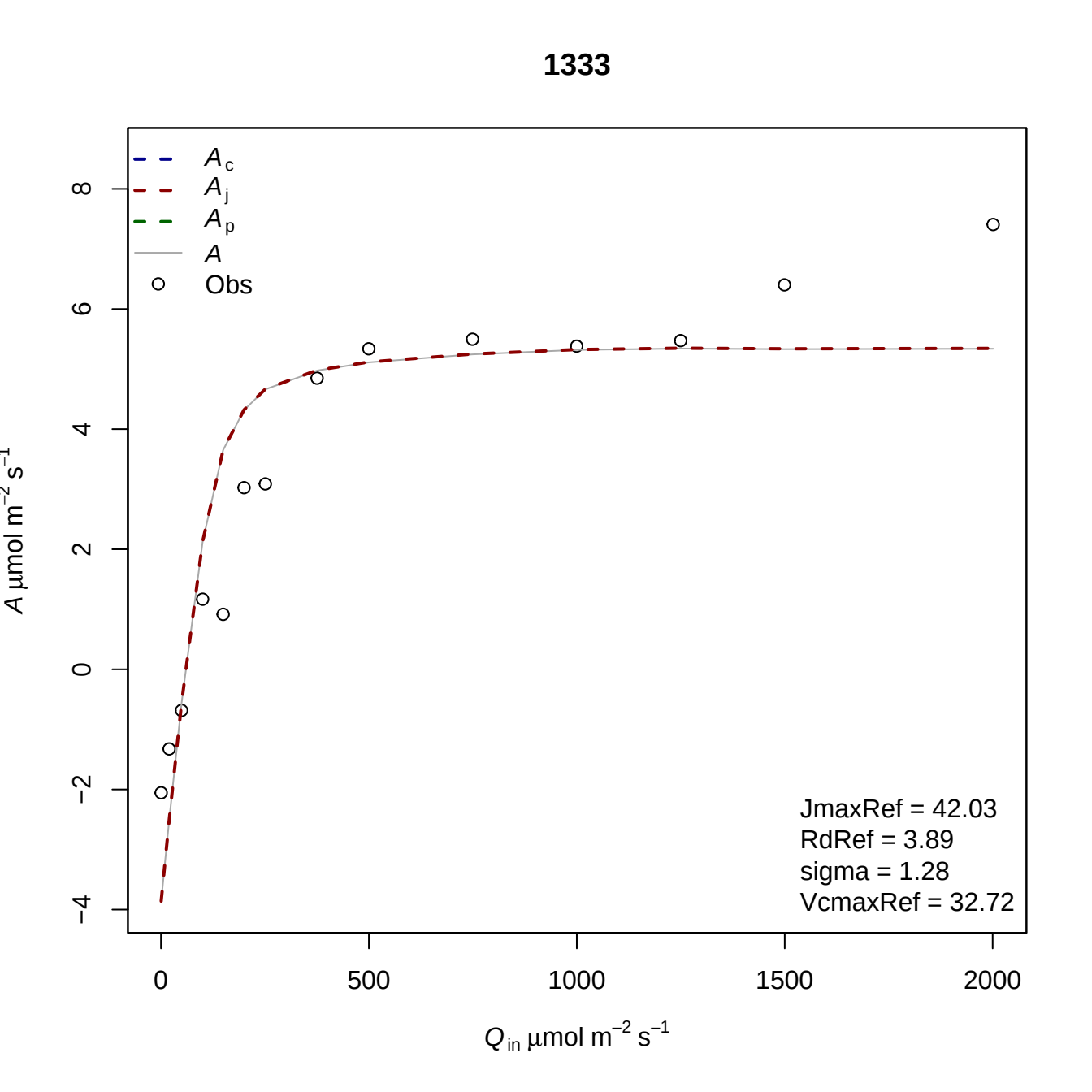

1328
A_c
A_j
A_p
A
Obs
JmaxRef = 60.58
RdRef = 5.13
sigma = 1.92
VcmaxRef = 46.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

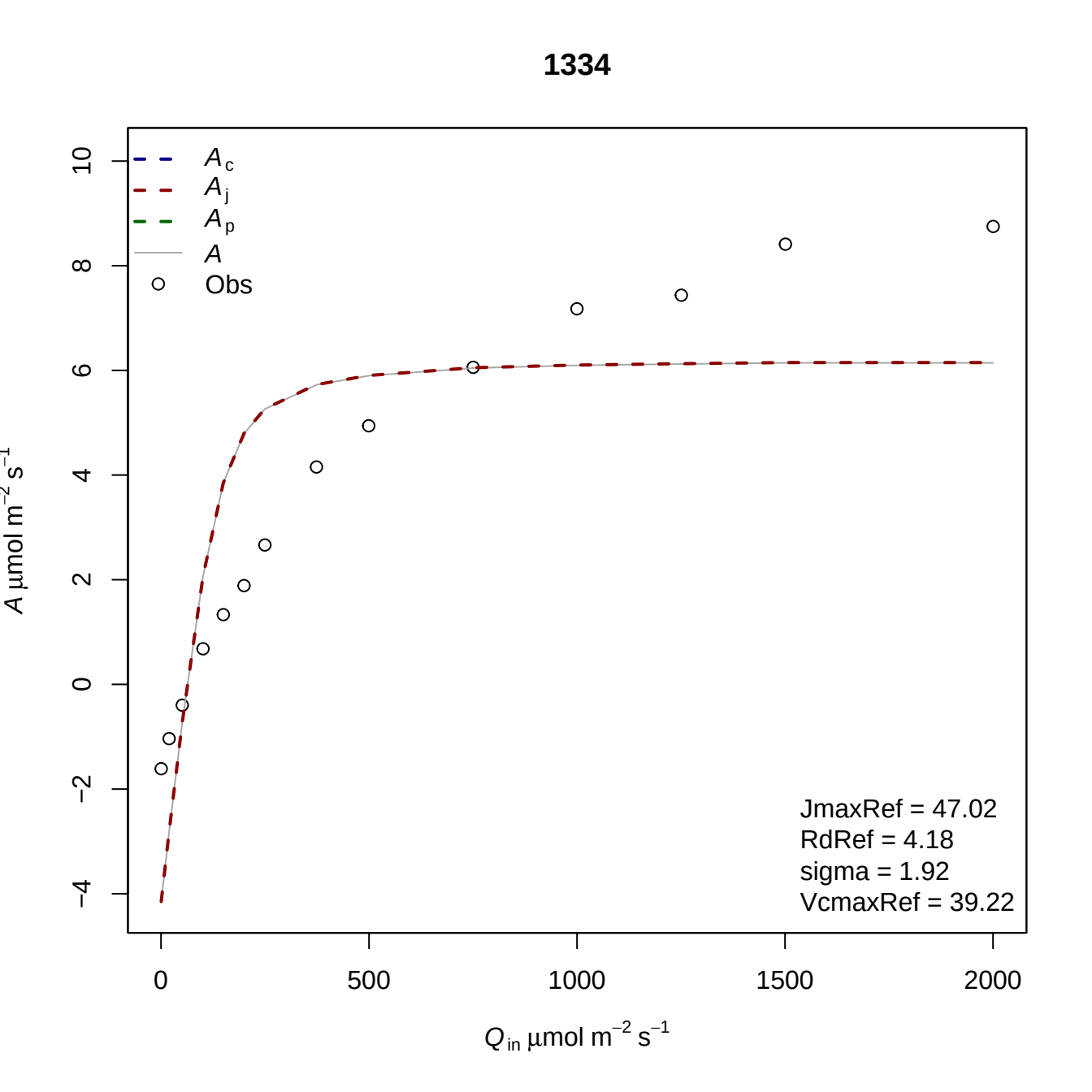

1329
A_c
A_j
A_p
A
Obs
JmaxRef = 53.8
RdRef = 5.49
sigma = 1.72
VcmaxRef = 40.3
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

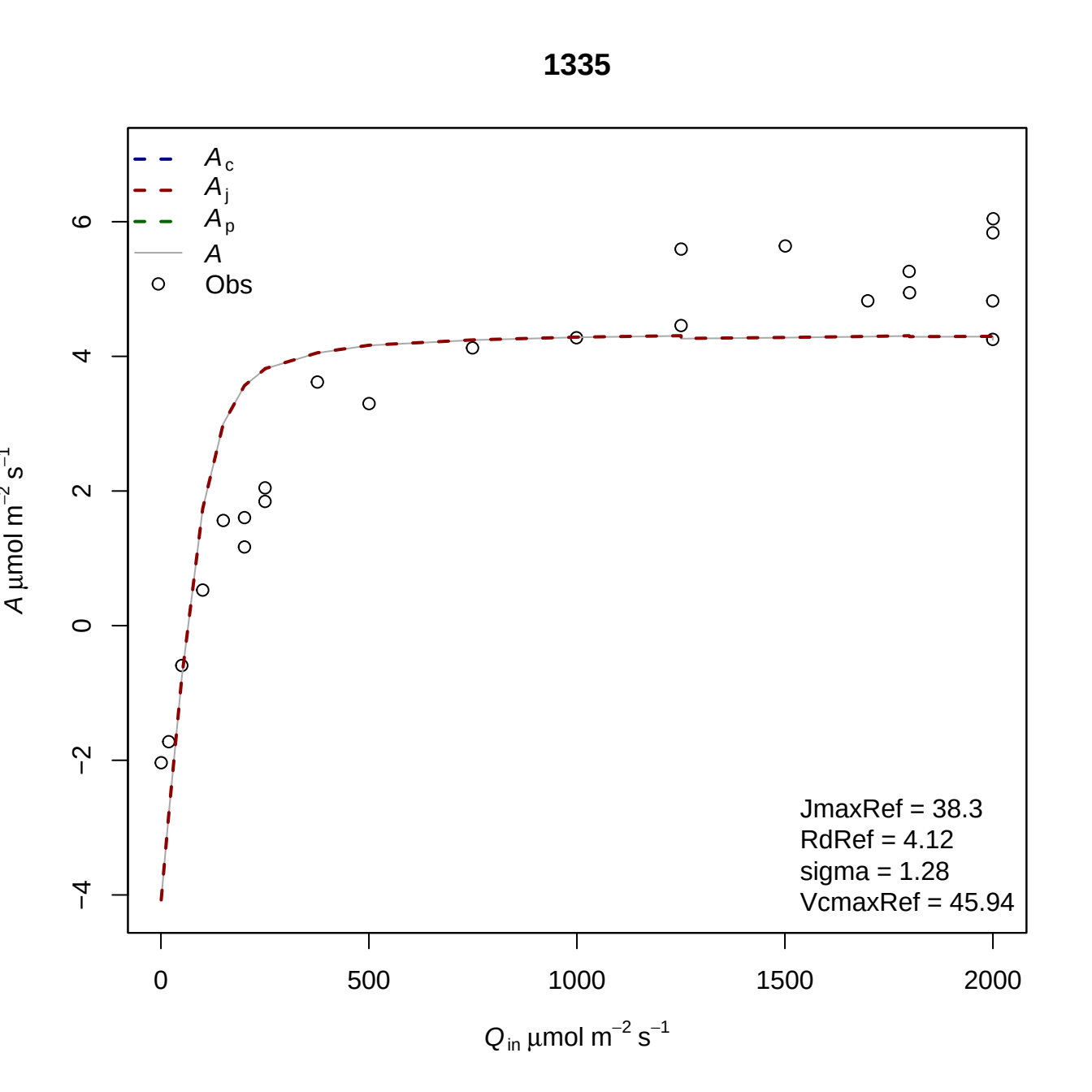

1330
A_c
A_j
A_p
A
Obs
JmaxRef = 43.1
RdRef = 3.43
sigma = 1.03
VcmaxRef = 34.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

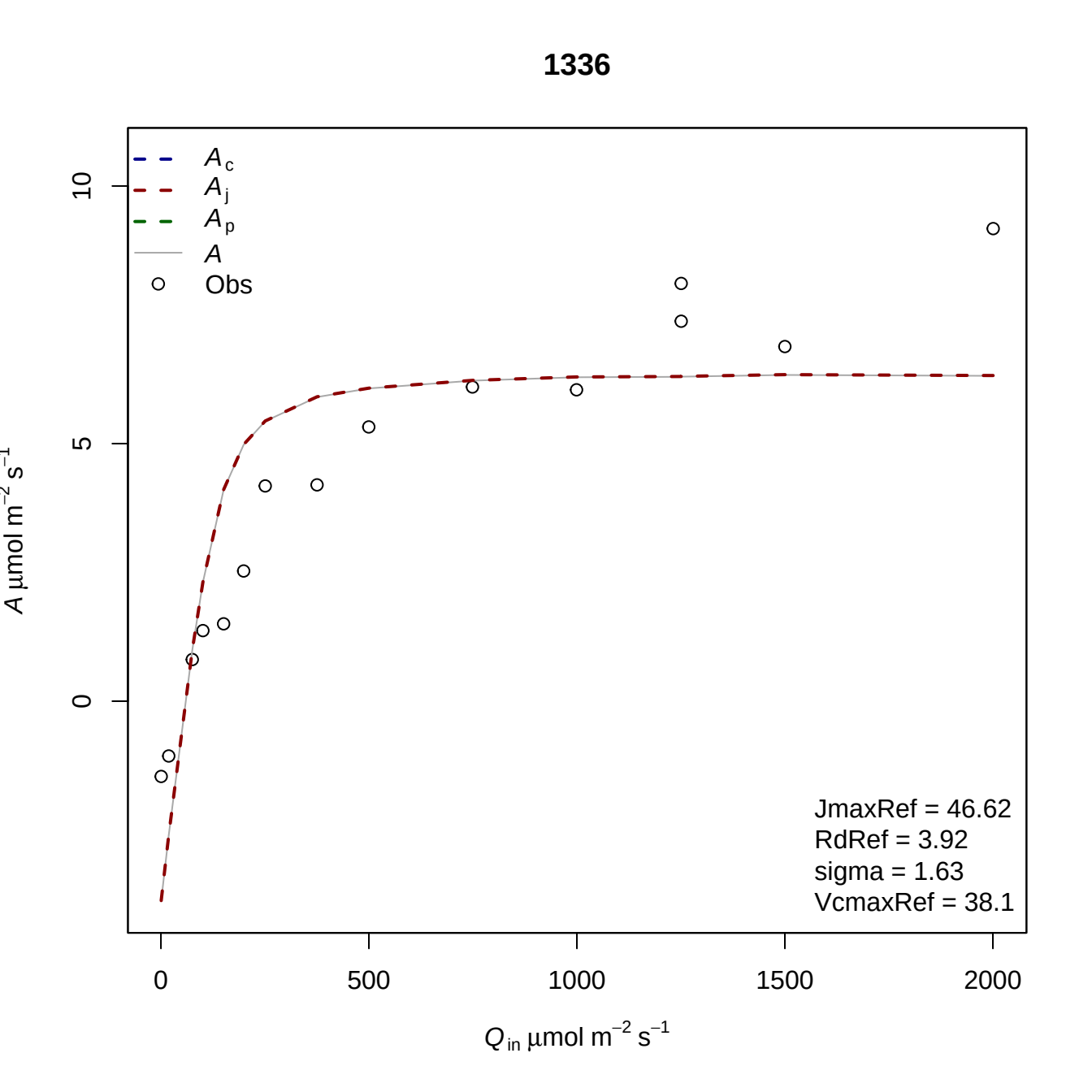

1331
A_c
A_j
A_p
A
Obs
JmaxRef = 43.95
RdRef = 3.11
sigma = 1.18
VcmaxRef = 36.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

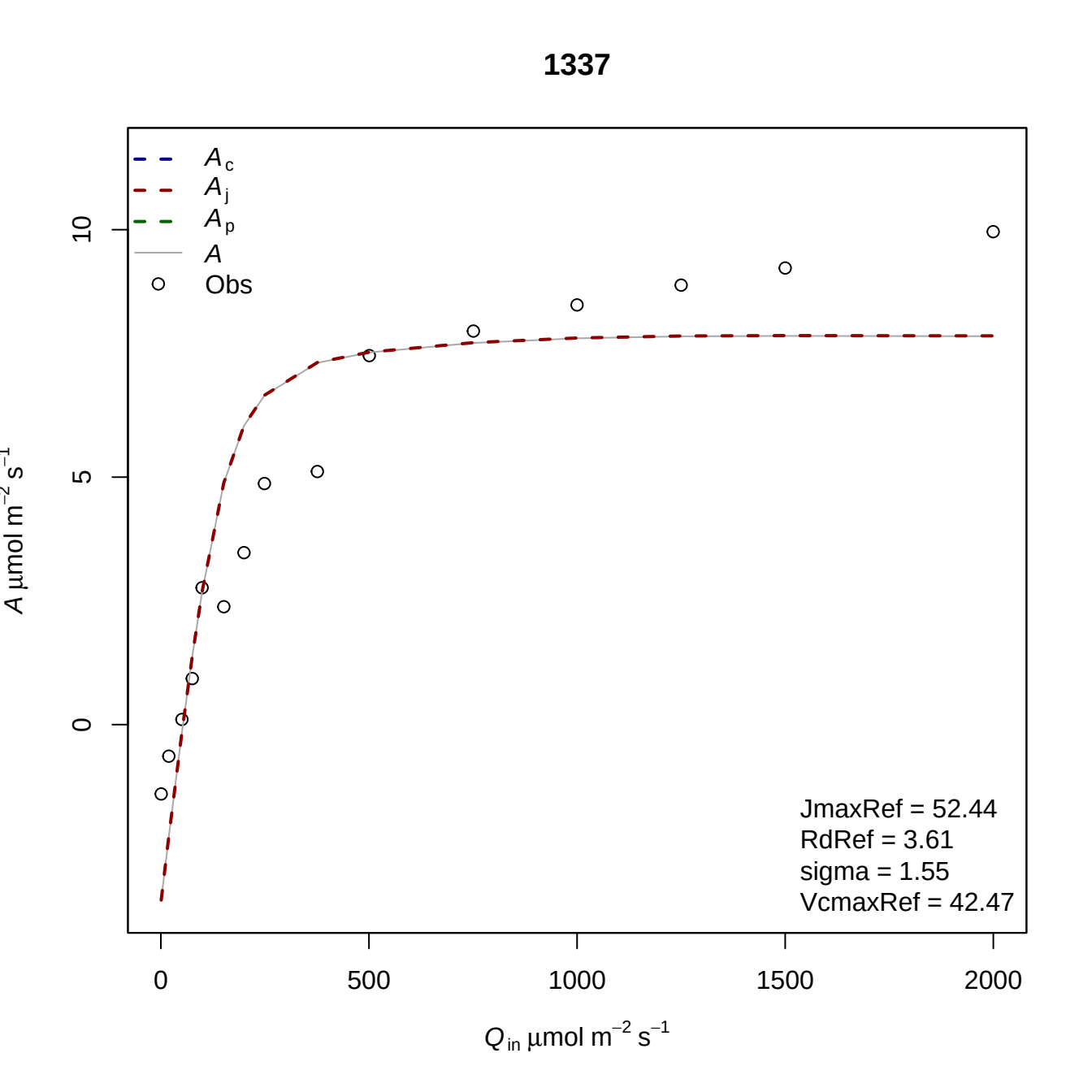

1332
A_c
A_j
A_p
A
Obs
JmaxRef = 47.95
RdRef = 2.72
sigma = 1.23
VcmaxRef = 35.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

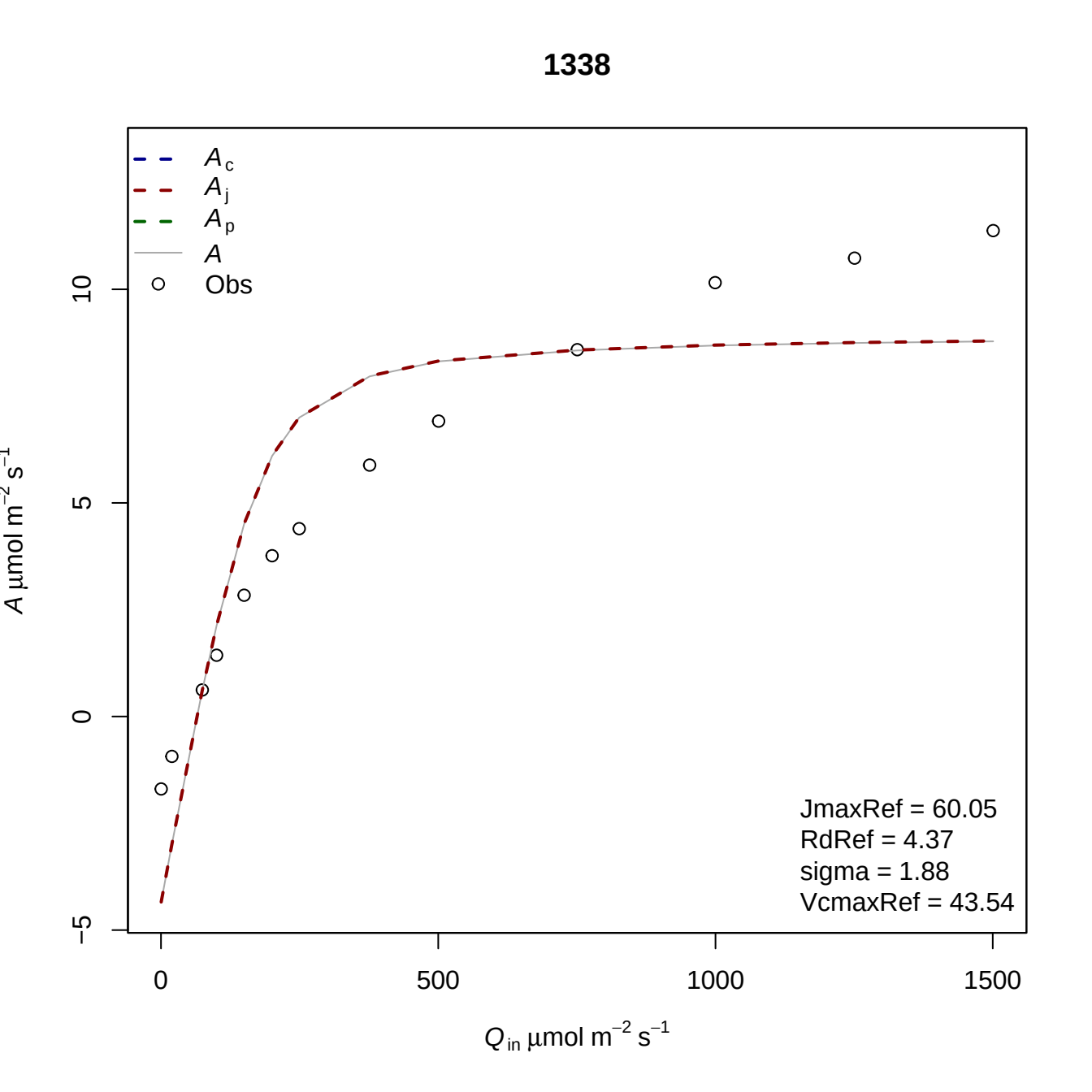

1333
A_c
A_j
A_p
A
Obs
JmaxRef = 42.03
RdRef = 3.89
sigma = 1.28
VcmaxRef = 32.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

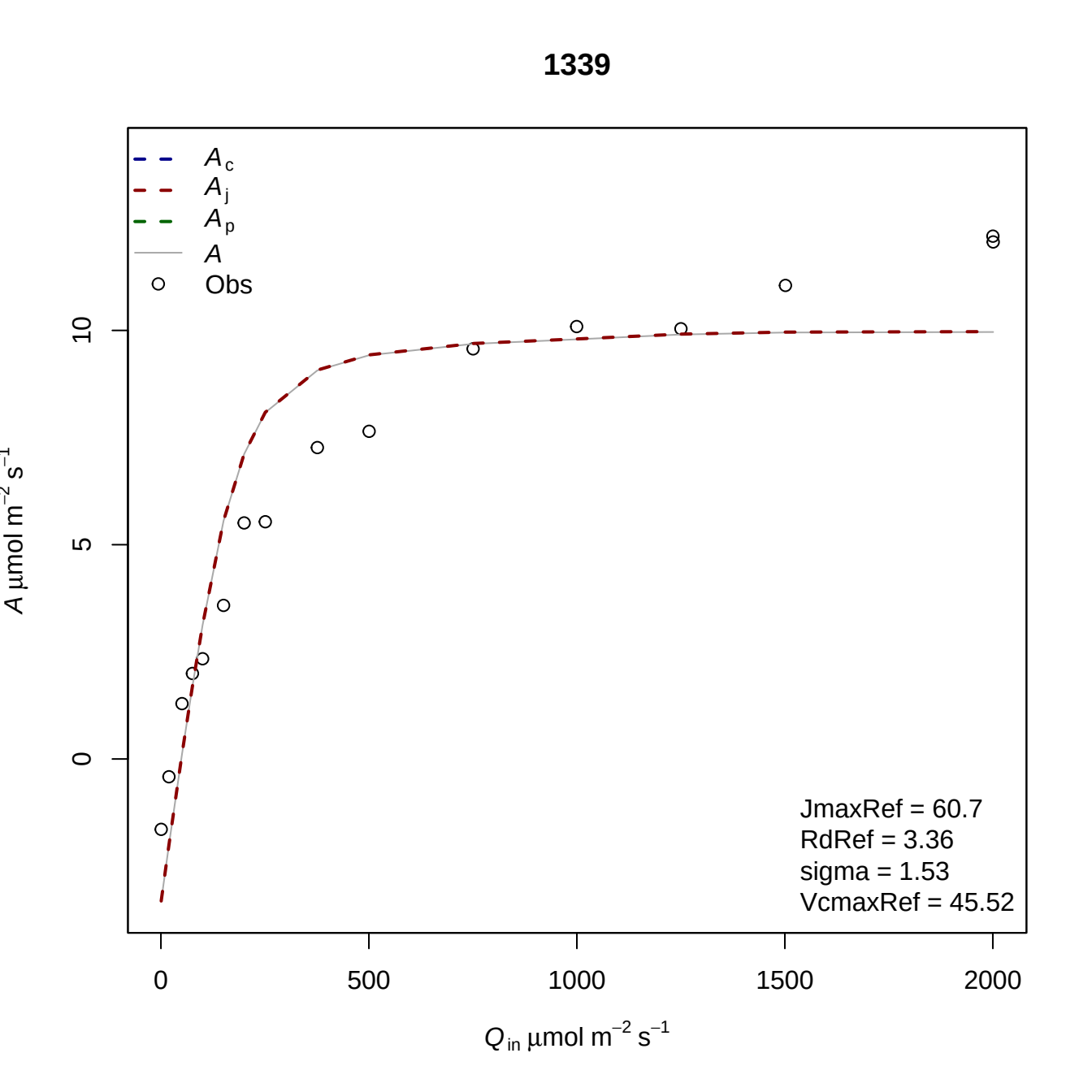

1334
A_c
A_j
A_p
A
Obs
JmaxRef = 47.02
RdRef = 4.18
sigma = 1.92
VcmaxRef = 39.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

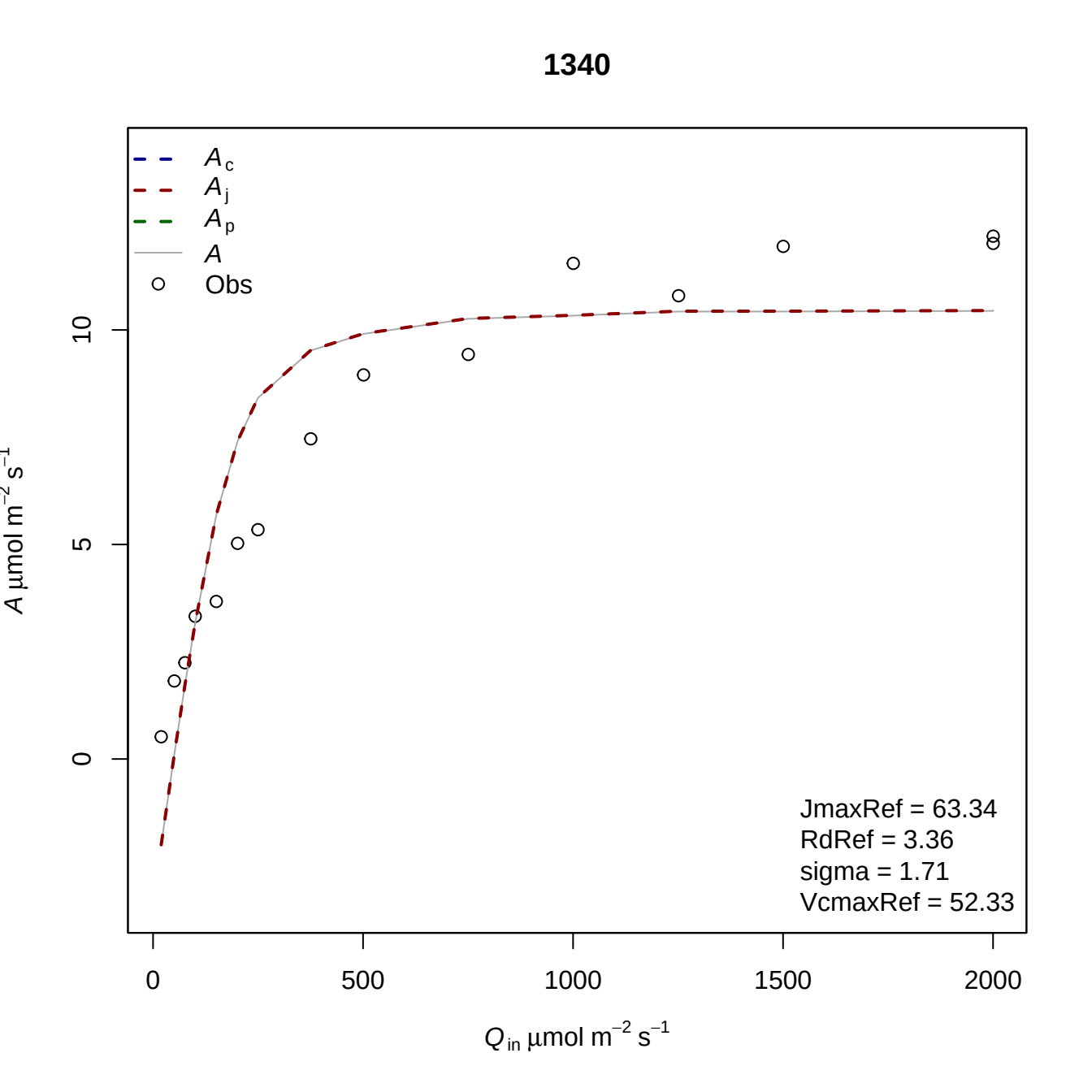

1335
A_c
A_j
A_p
A
Obs
JmaxRef = 38.3
RdRef = 4.12
sigma = 1.28
VcmaxRef = 45.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

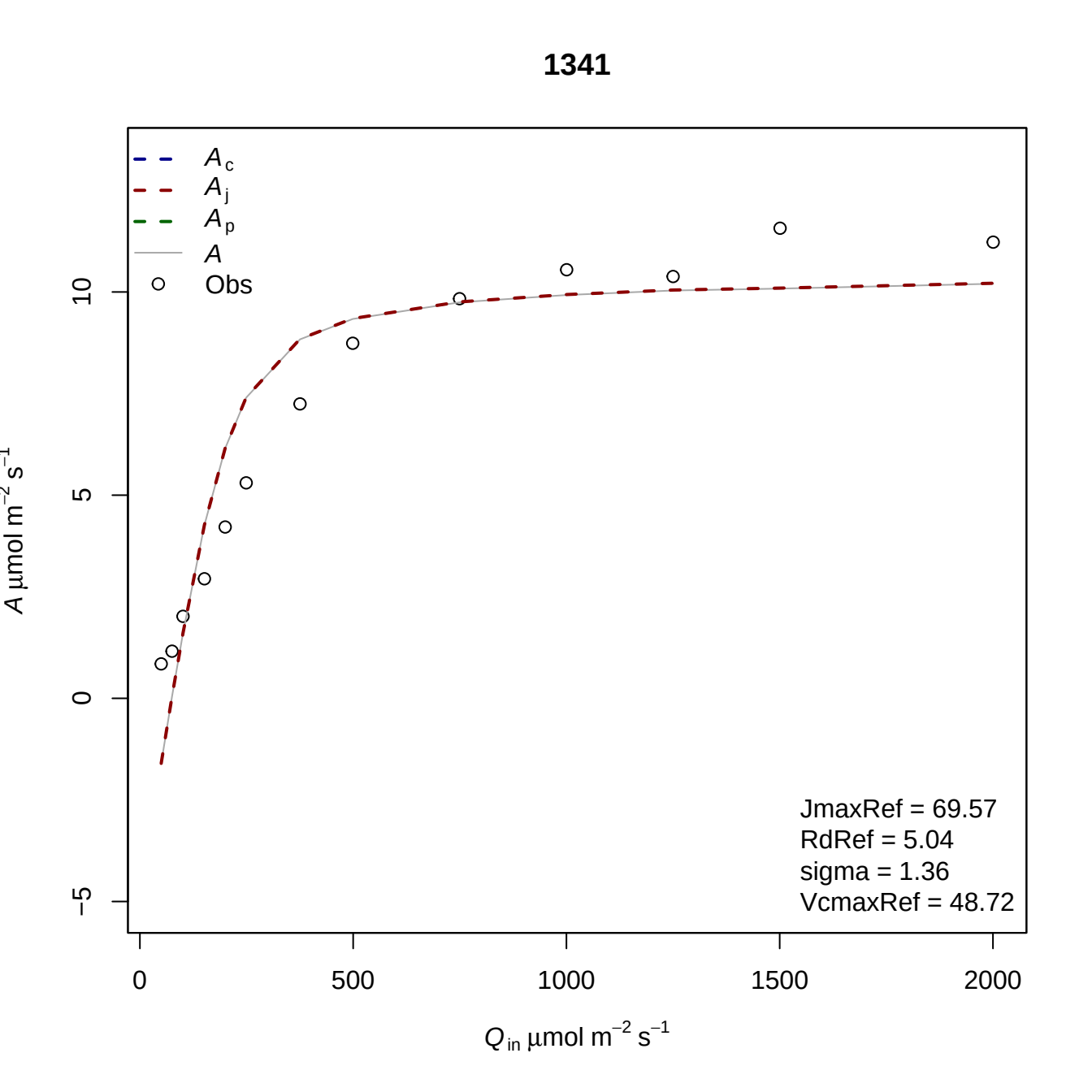

1336
A_c
A_j
A_p
A
Obs
JmaxRef = 46.62
RdRef = 3.92
sigma = 1.63
VcmaxRef = 38.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

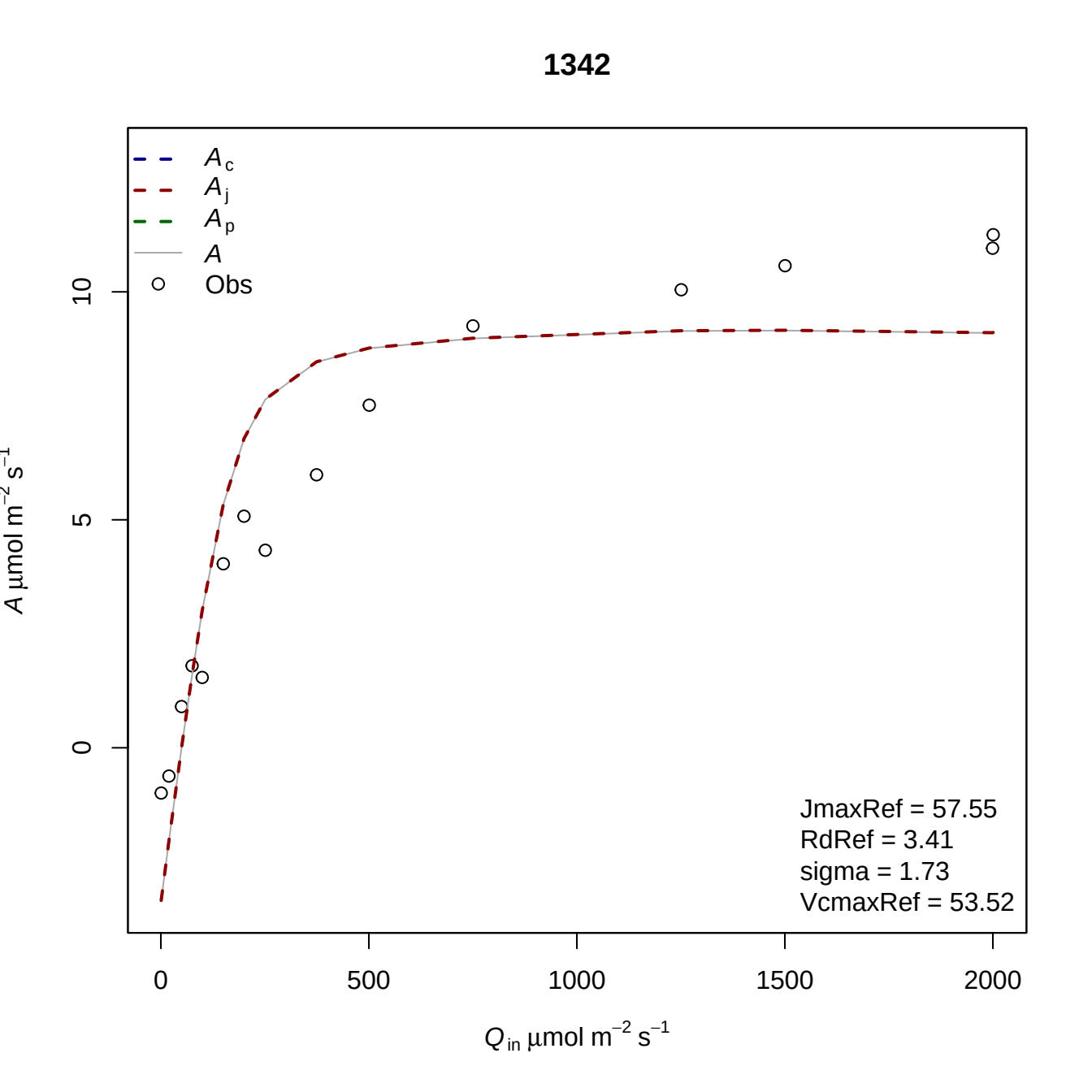

1337
A_c
A_j
A_p
A
Obs
JmaxRef = 52.44
RdRef = 3.61
sigma = 1.55
VcmaxRef = 42.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

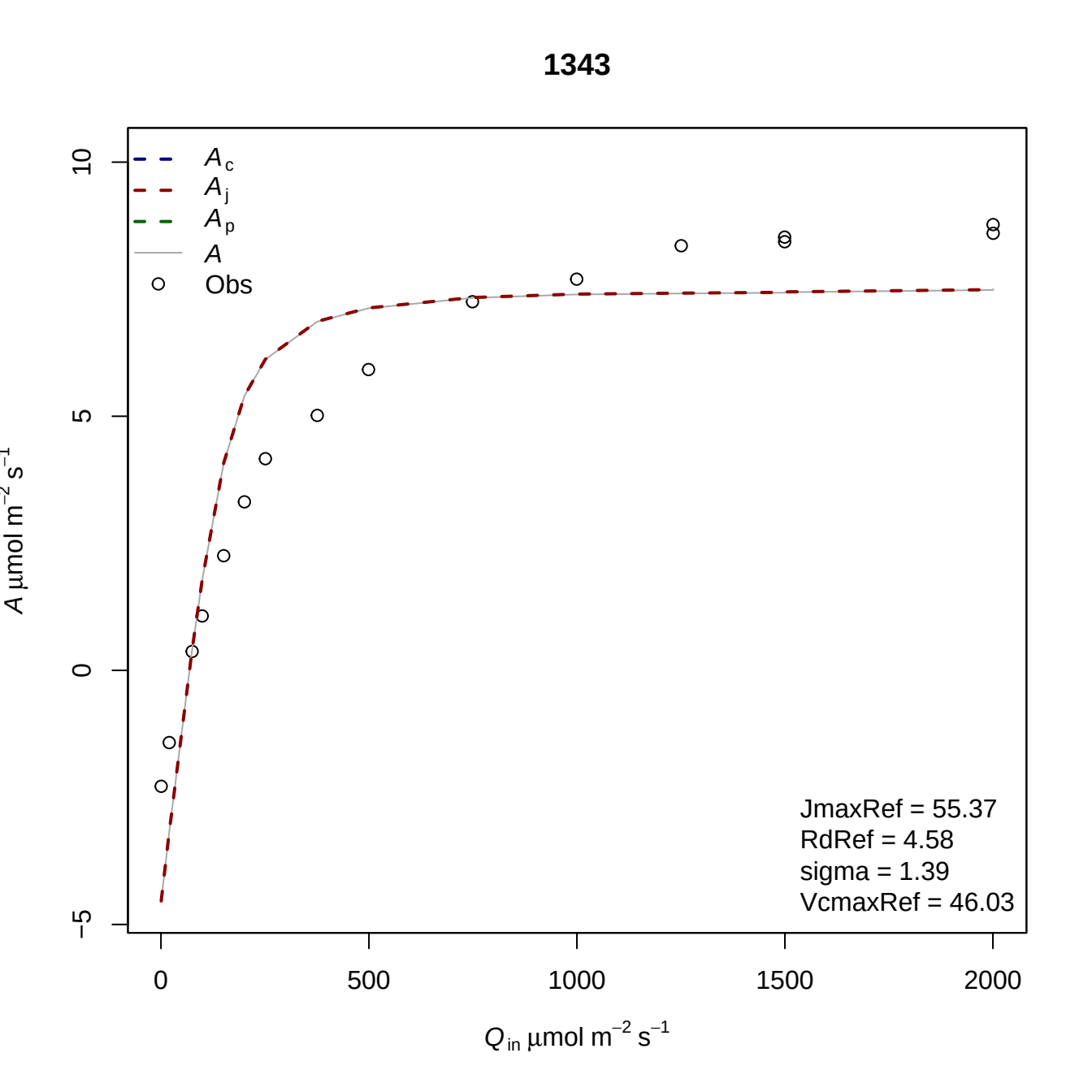

1338
A_c
A_j
A_p
A
Obs
JmaxRef = 60.05
RdRef = 4.37
sigma = 1.88
VcmaxRef = 43.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

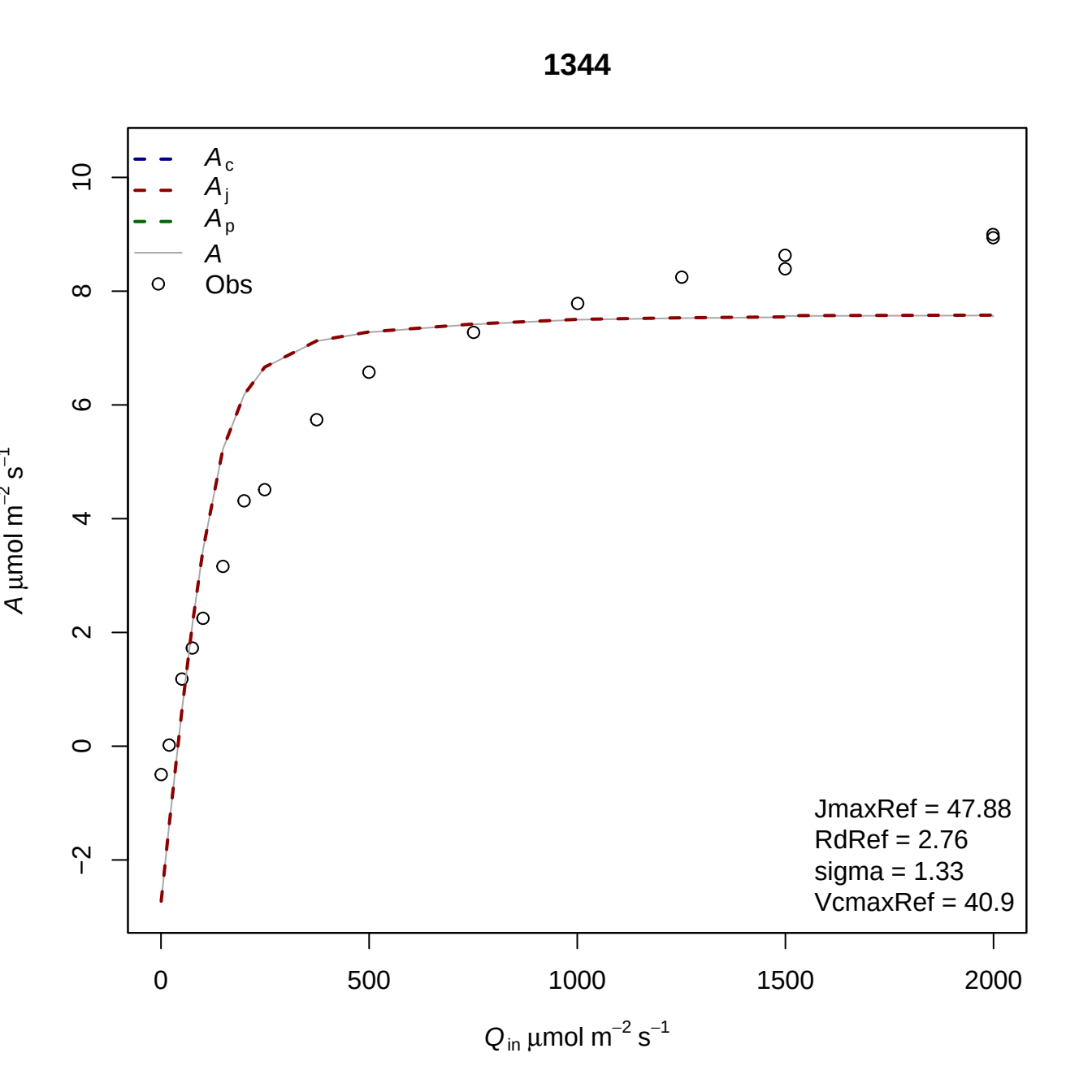

1339
A_c
A_j
A_p
A
Obs
JmaxRef = 60.7
RdRef = 3.36
sigma = 1.53
VcmaxRef = 45.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

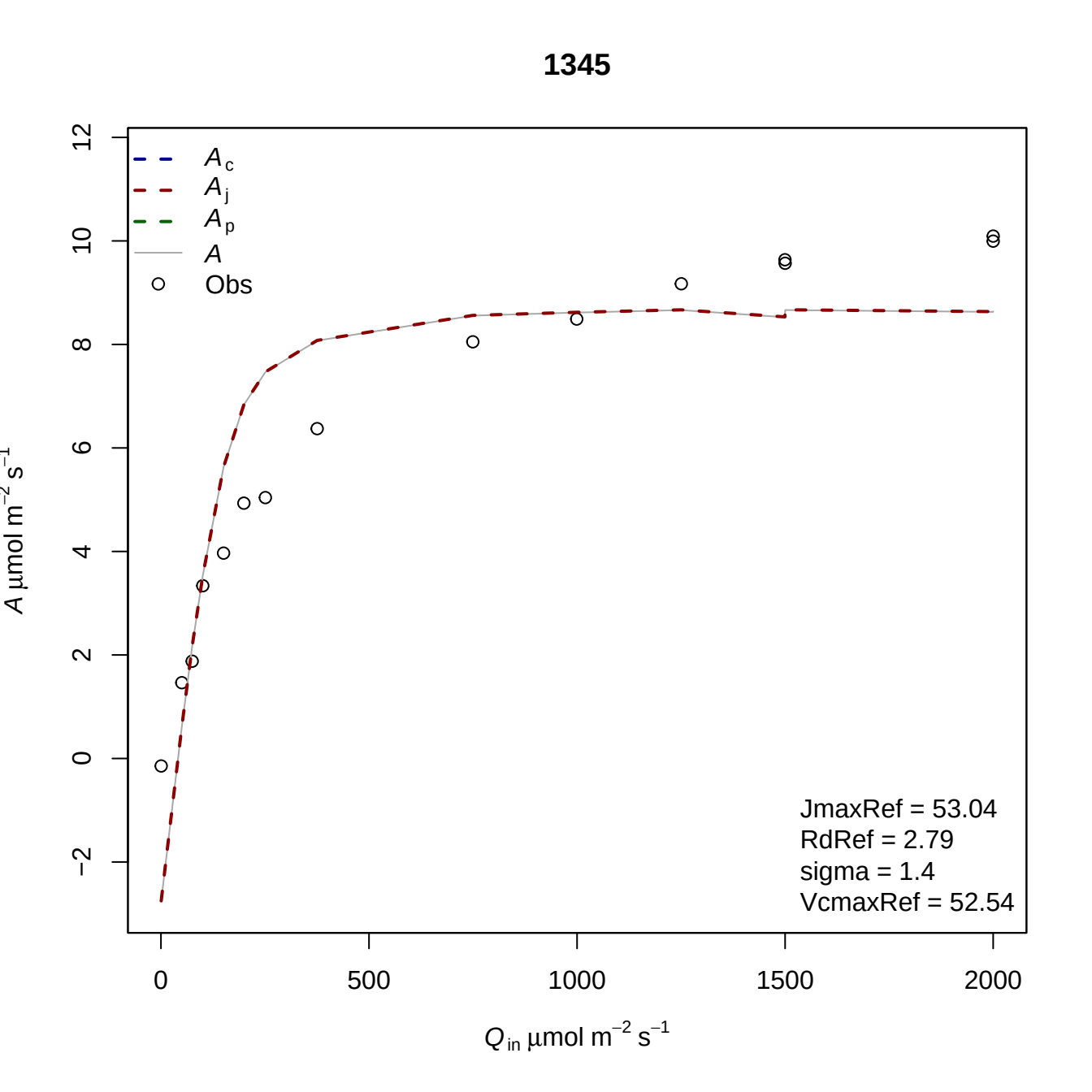

1340
A_c
A_j
A_p
A
Obs
JmaxRef = 63.34
RdRef = 3.36
sigma = 1.71
VcmaxRef = 52.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

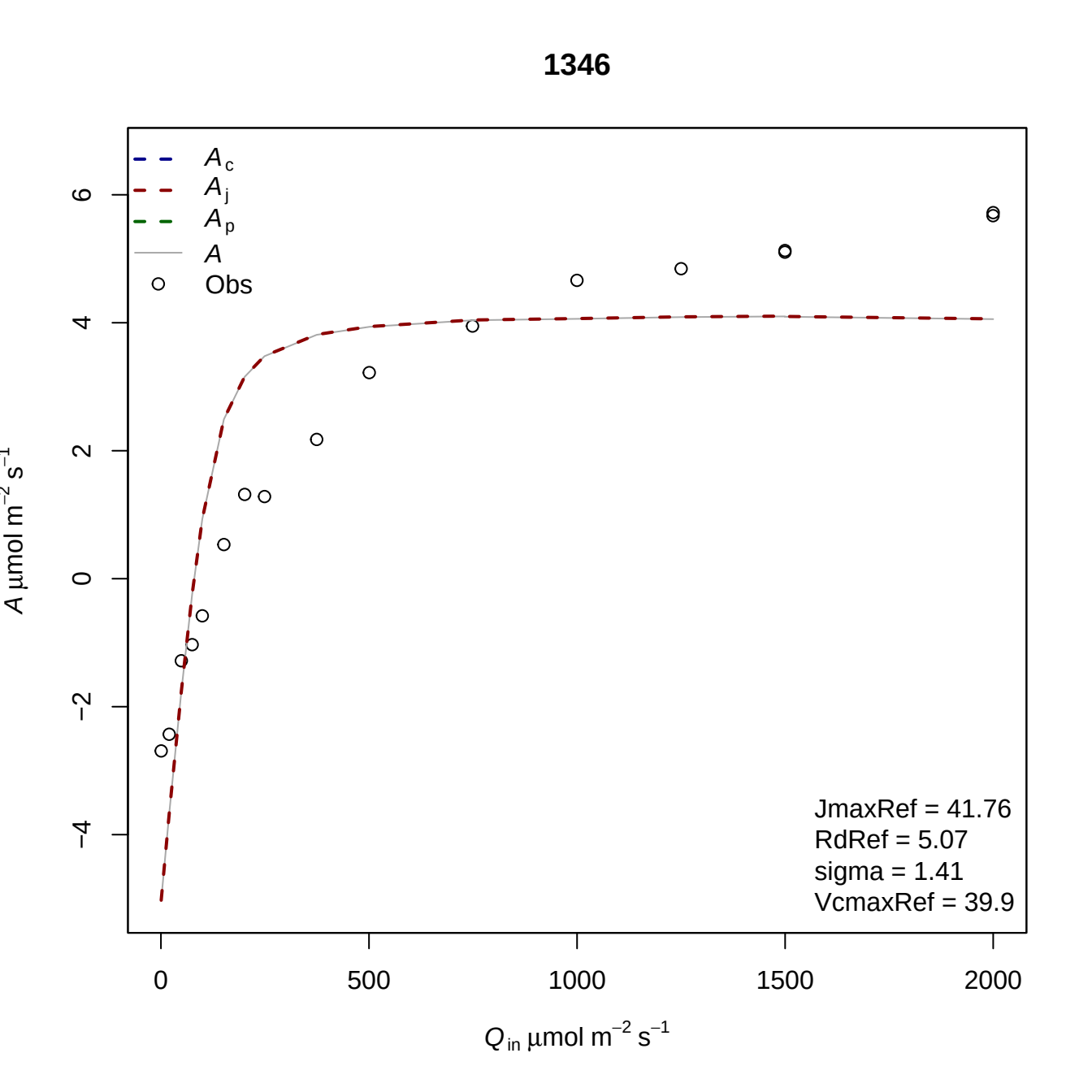

1341
A_c
A_j
A_p
A
Obs
JmaxRef = 69.57
RdRef = 5.04
sigma = 1.36
VcmaxRef = 48.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

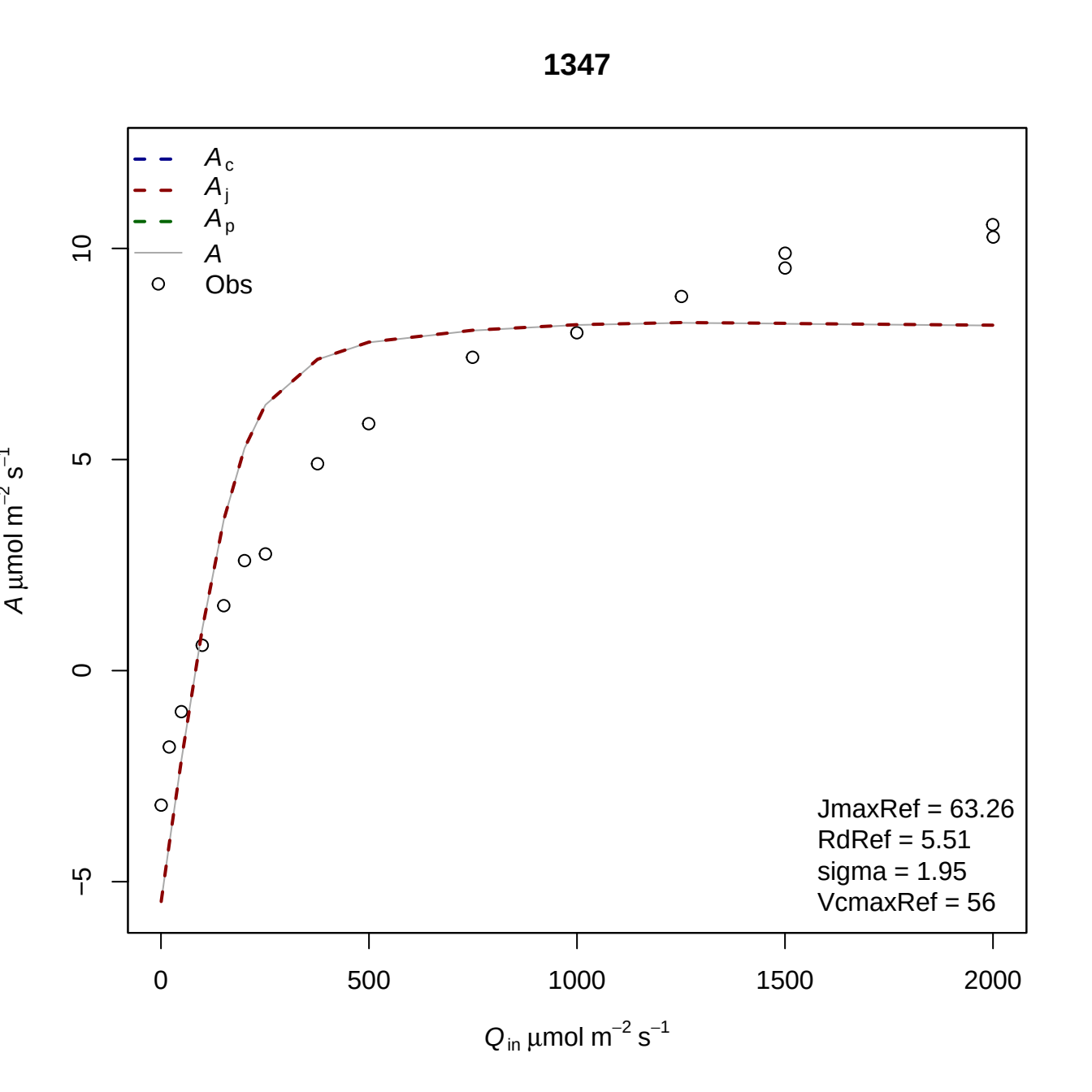

1342
A_c
A_j
A_p
A
Obs
JmaxRef = 57.55
RdRef = 3.41
sigma = 1.73
VcmaxRef = 53.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

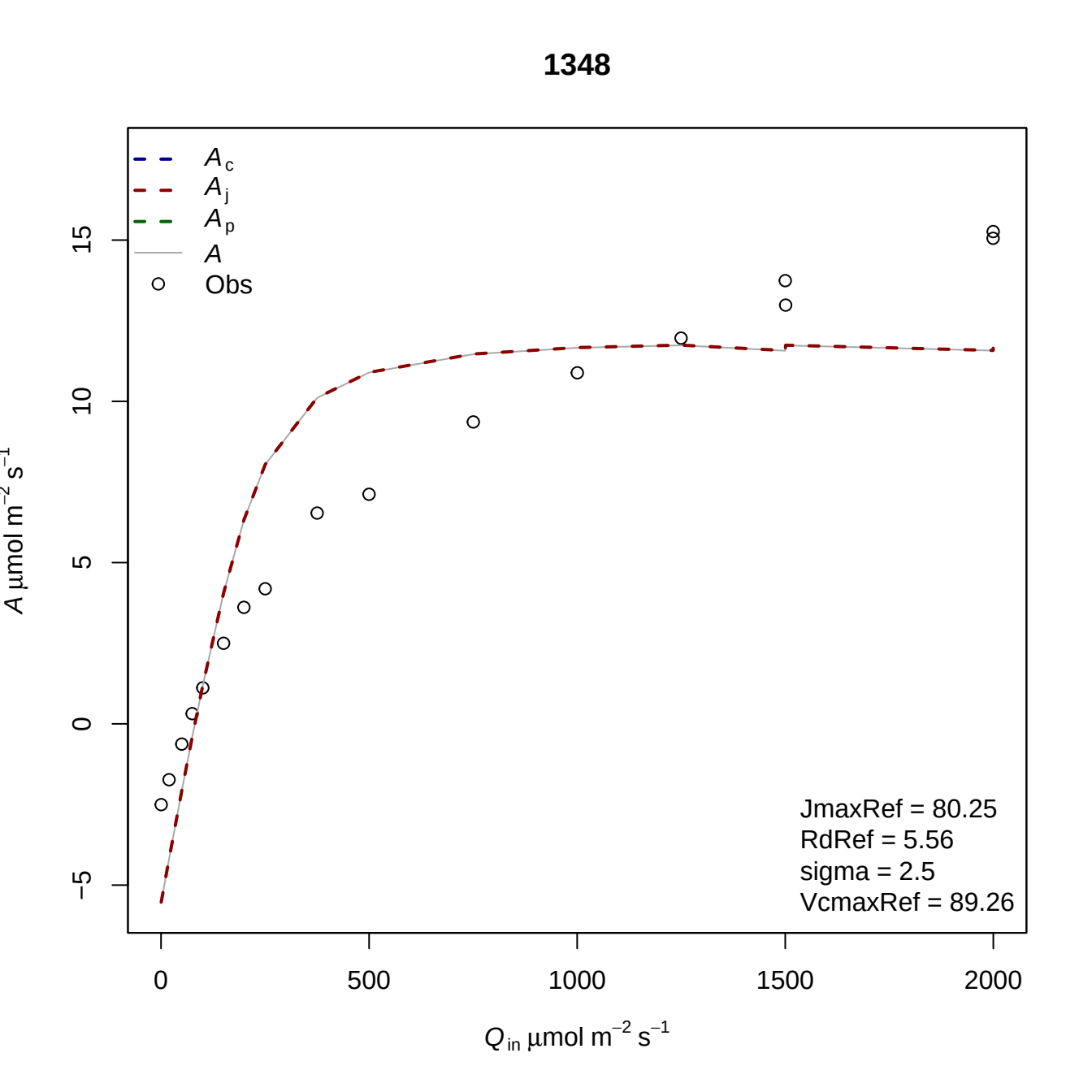

1343
A_c
A_j
A_p
A
Obs
JmaxRef = 55.37
RdRef = 4.58
sigma = 1.39
VcmaxRef = 46.03
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

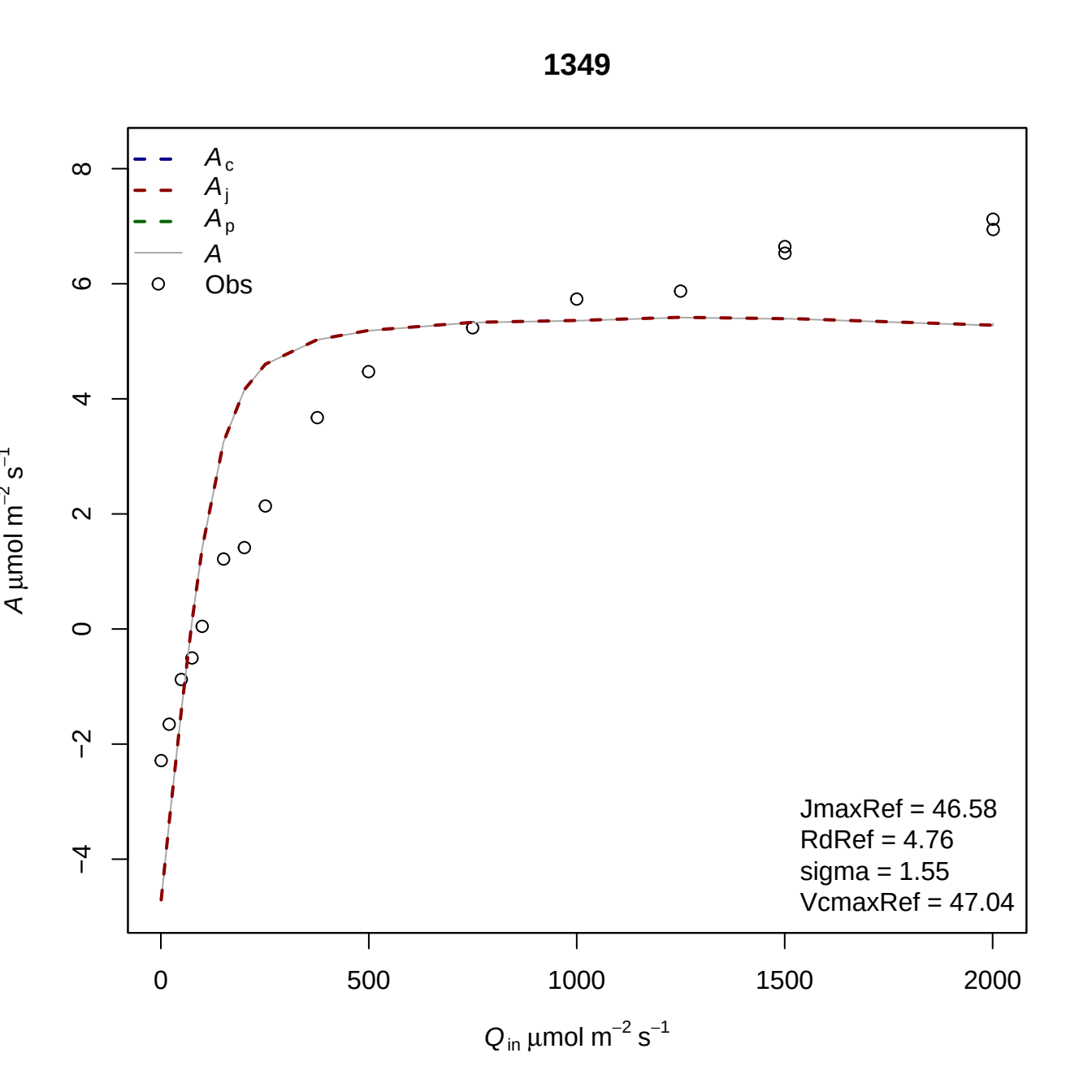

1344
A_c
A_j
A_p
A
Obs
JmaxRef = 47.88
RdRef = 2.76
sigma = 1.33
VcmaxRef = 40.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

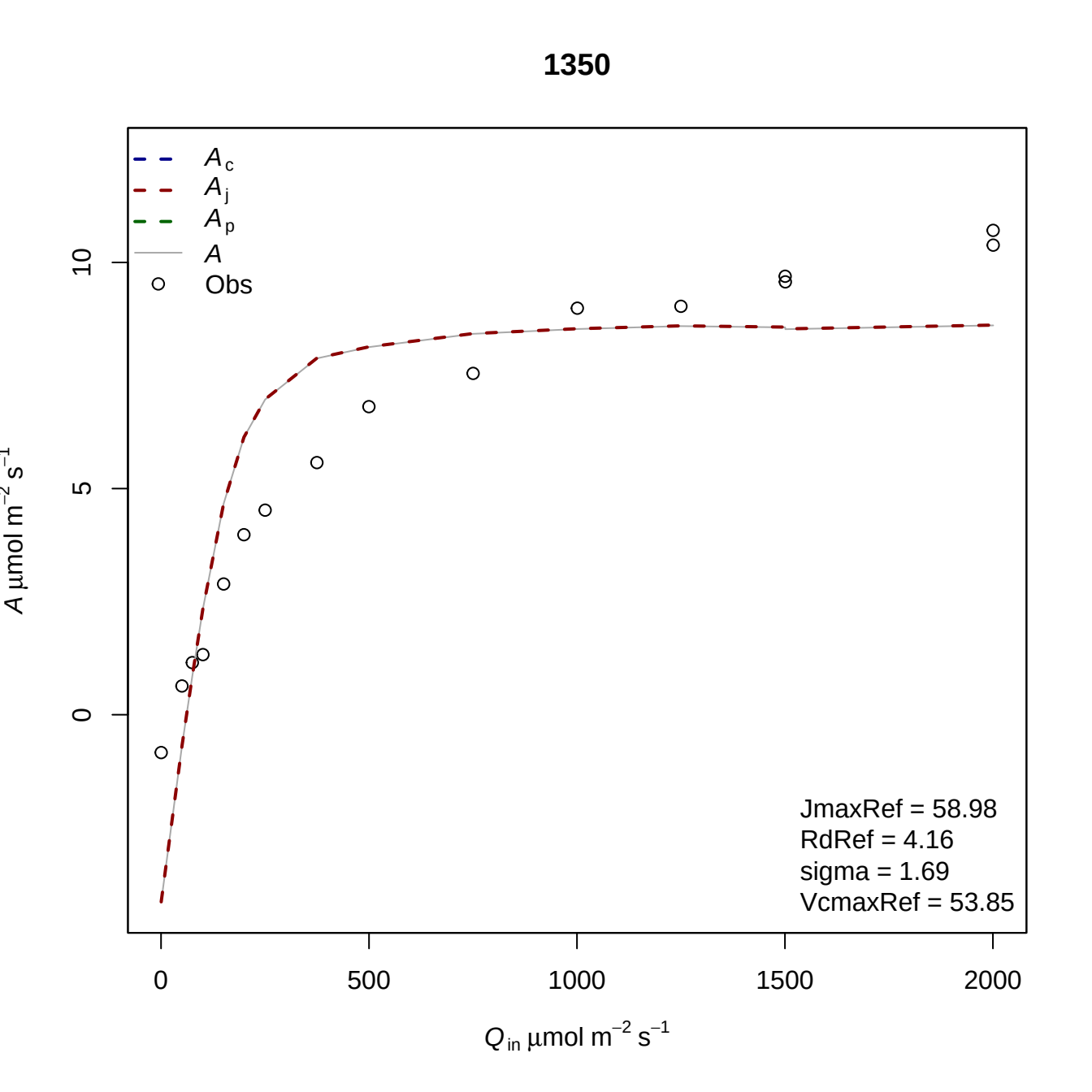

1345
A_c
A_j
A_p
A
Obs
JmaxRef = 53.04
RdRef = 2.79
sigma = 1.4
VcmaxRef = 52.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

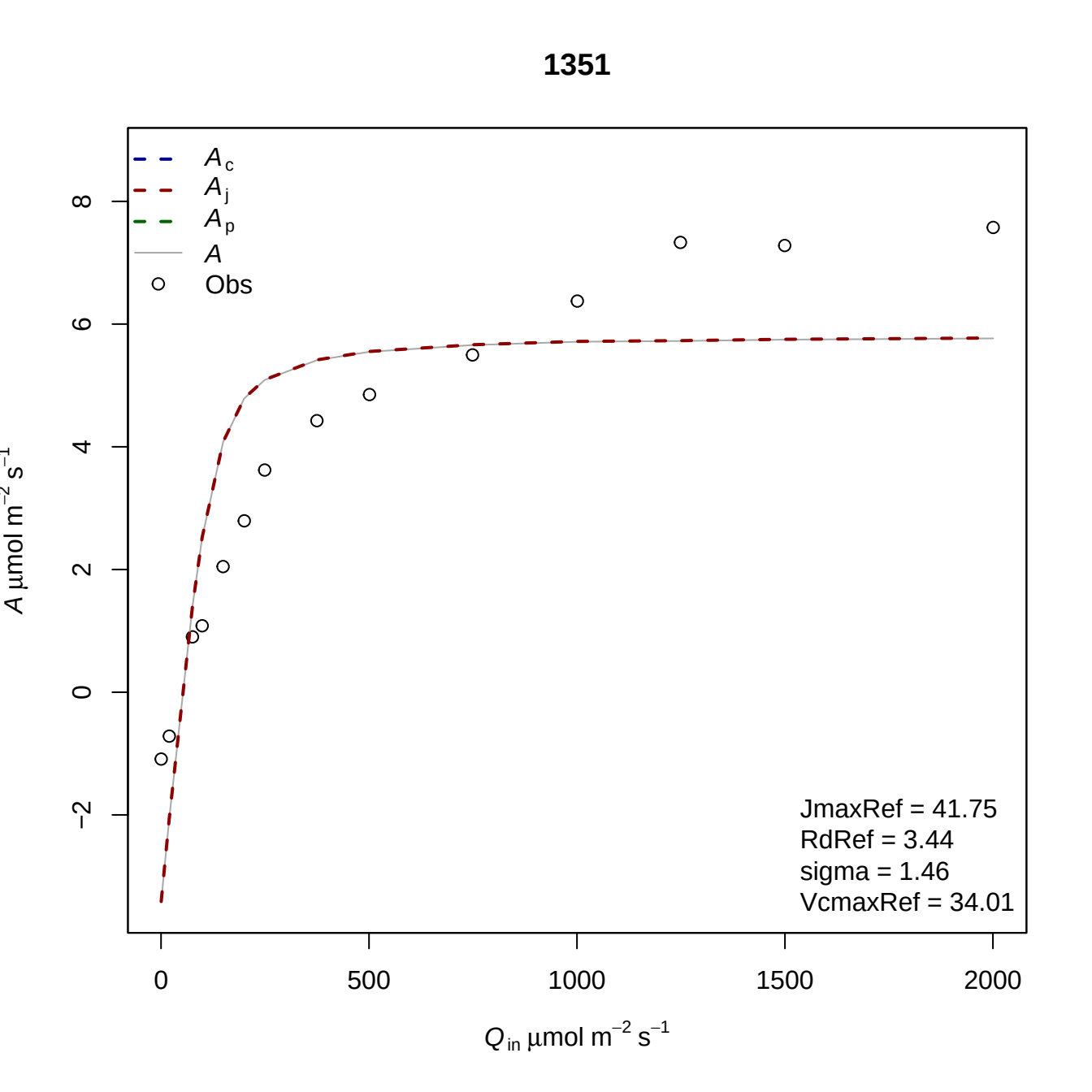

1346
A_c
A_j
A_p
A
Obs
JmaxRef = 41.76
RdRef = 5.07
sigma = 1.41
VcmaxRef = 39.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

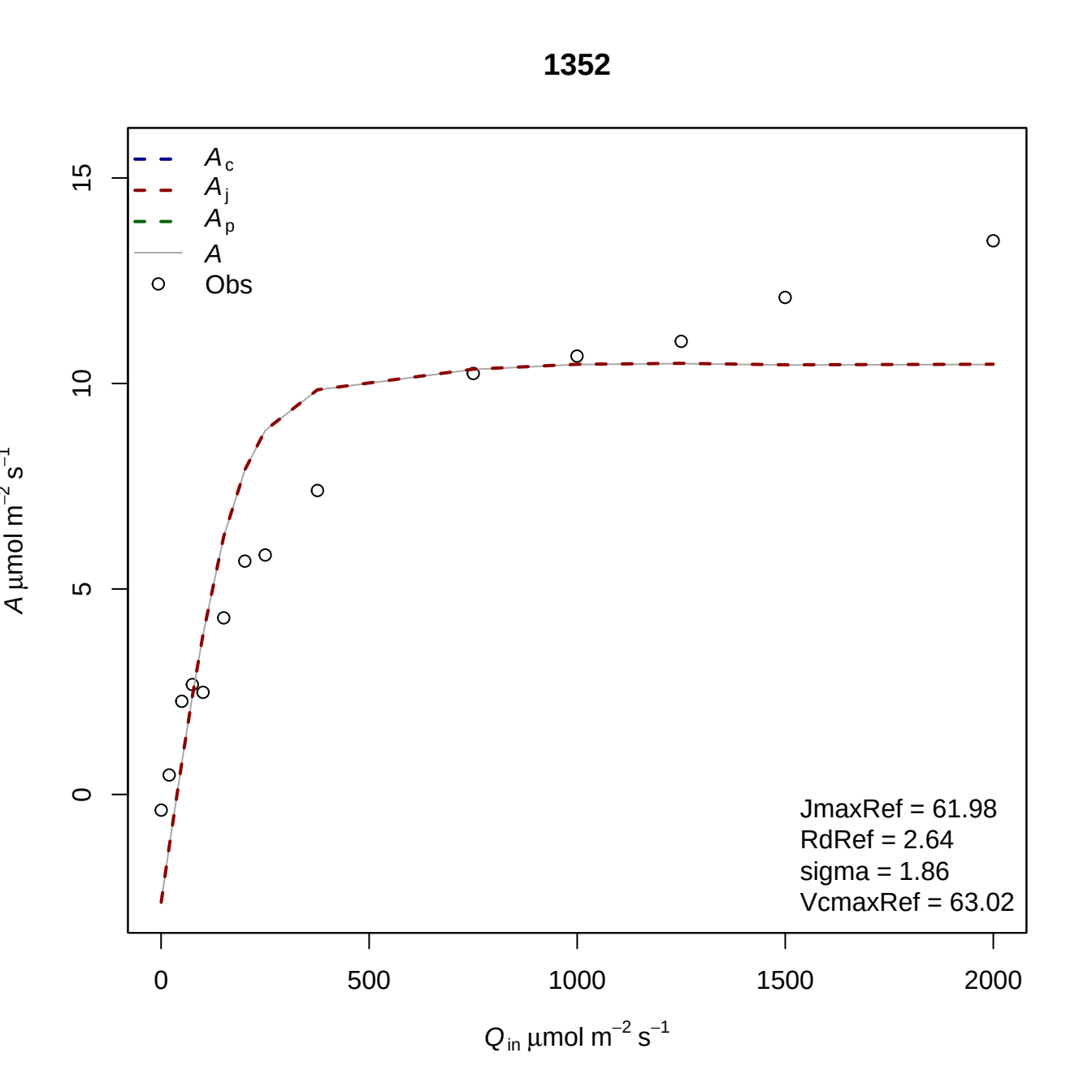

1347
A_c
A_j
A_p
A
Obs
JmaxRef = 63.26
RdRef = 5.51
sigma = 1.95
VcmaxRef = 56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

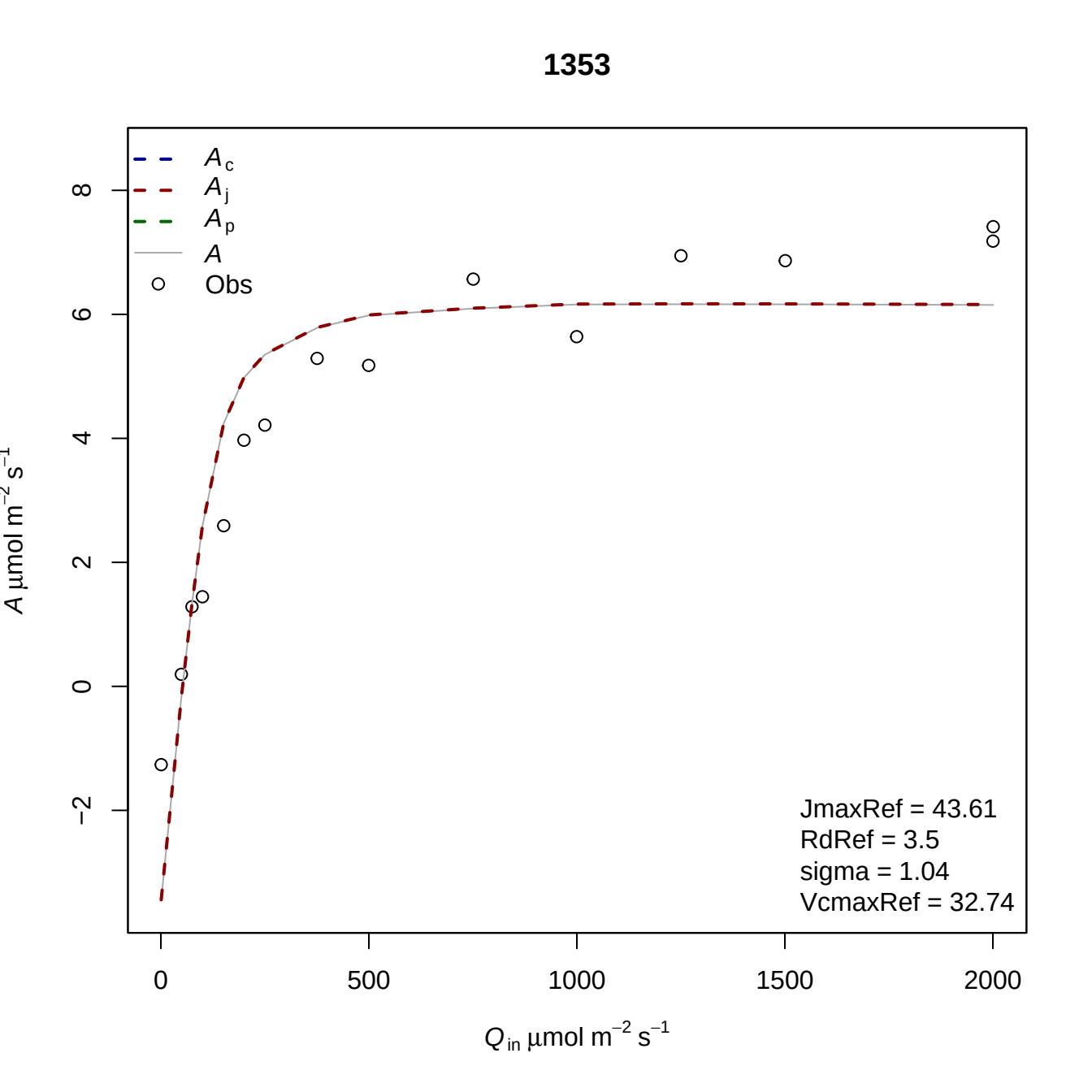

1348
A_c
A_j
A_p
A
Obs
JmaxRef = 80.25
RdRef = 5.56
sigma = 2.5
VcmaxRef = 89.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

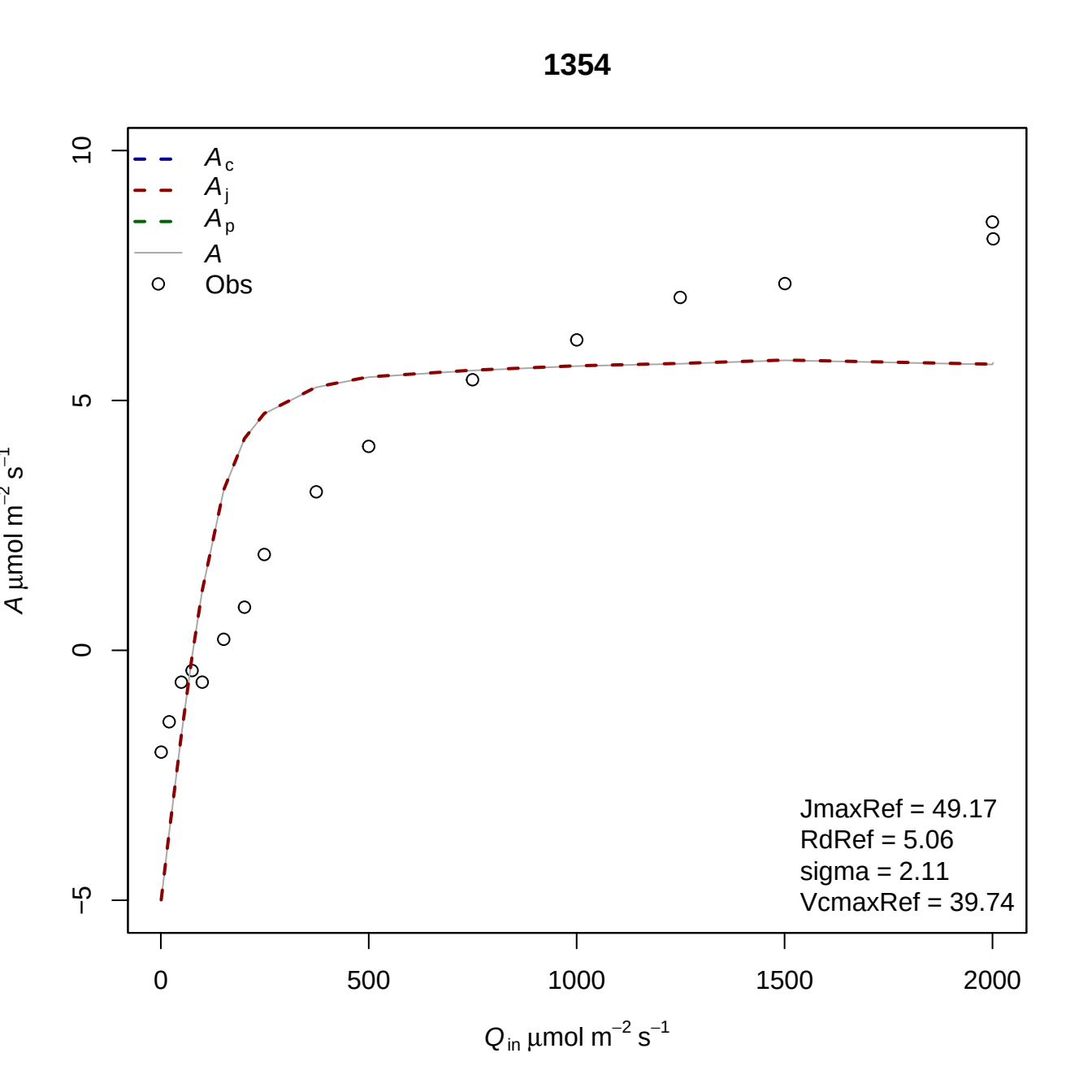

1349
A_c
A_j
A_p
A
Obs
JmaxRef = 46.58
RdRef = 4.76
sigma = 1.55
VcmaxRef = 47.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

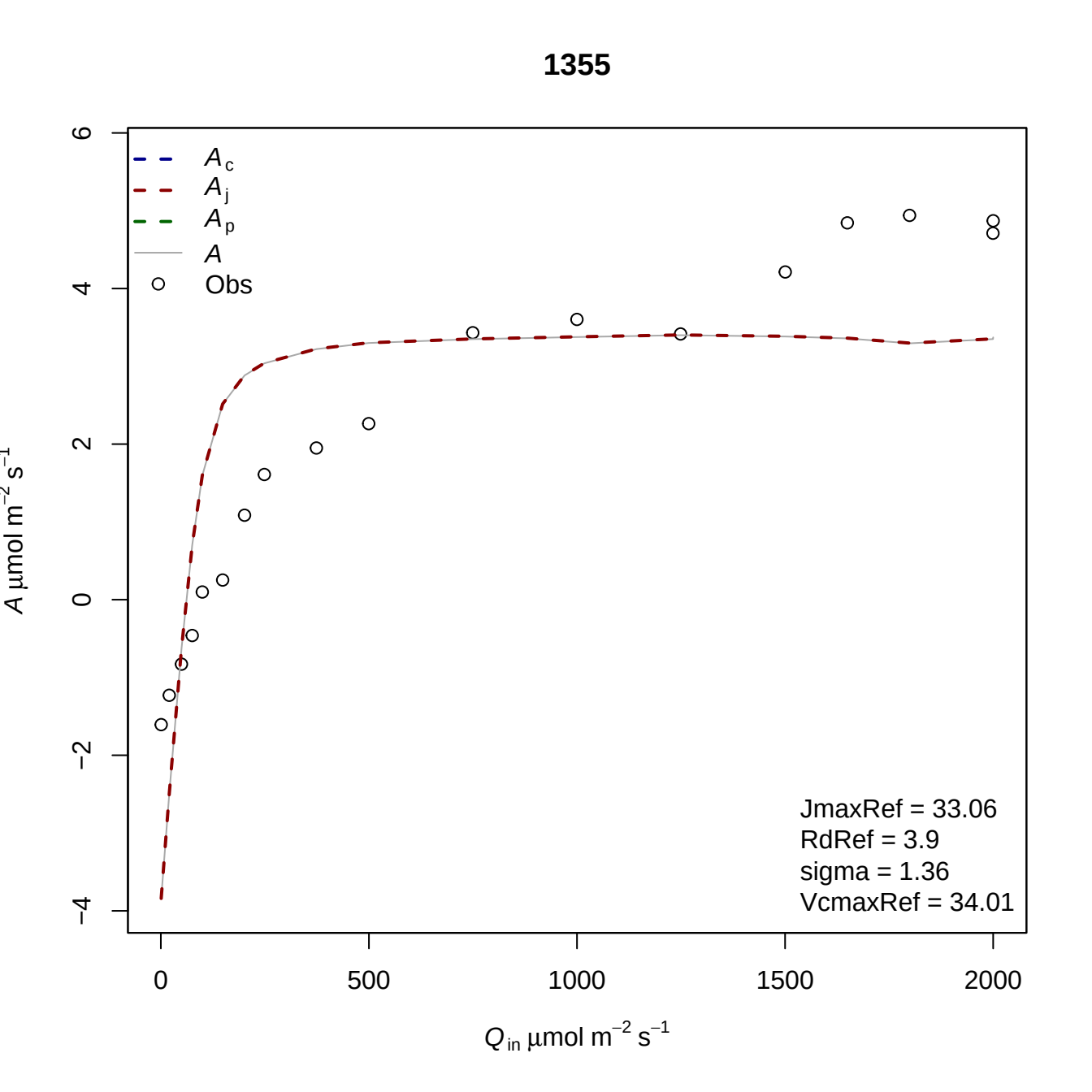

1350
A_c
A_j
A_p
A
Obs
JmaxRef = 58.98
RdRef = 4.16
sigma = 1.69
VcmaxRef = 53.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

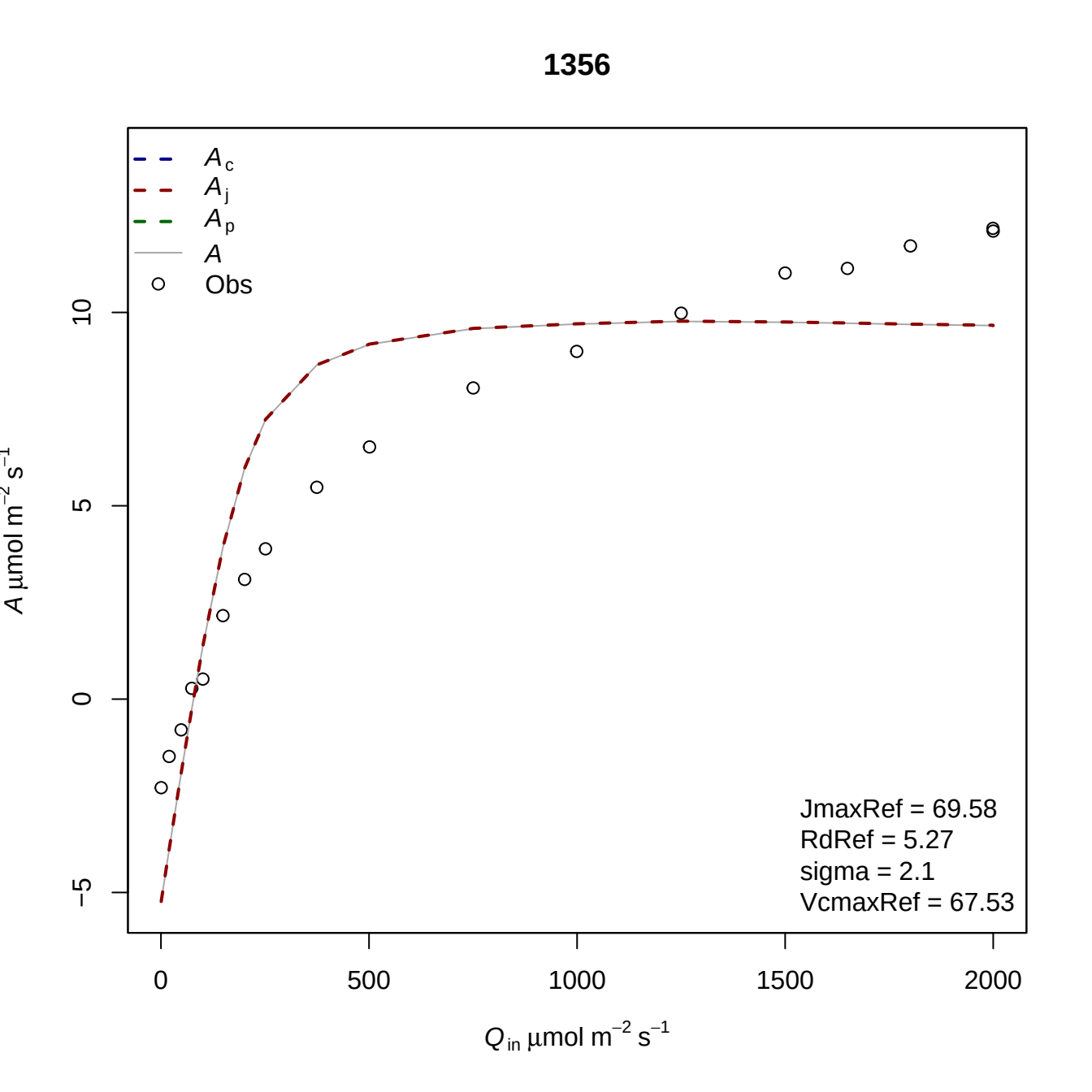

1351
A_c
A_j
A_p
A
Obs
JmaxRef = 41.75
RdRef = 3.44
sigma = 1.46
VcmaxRef = 34.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

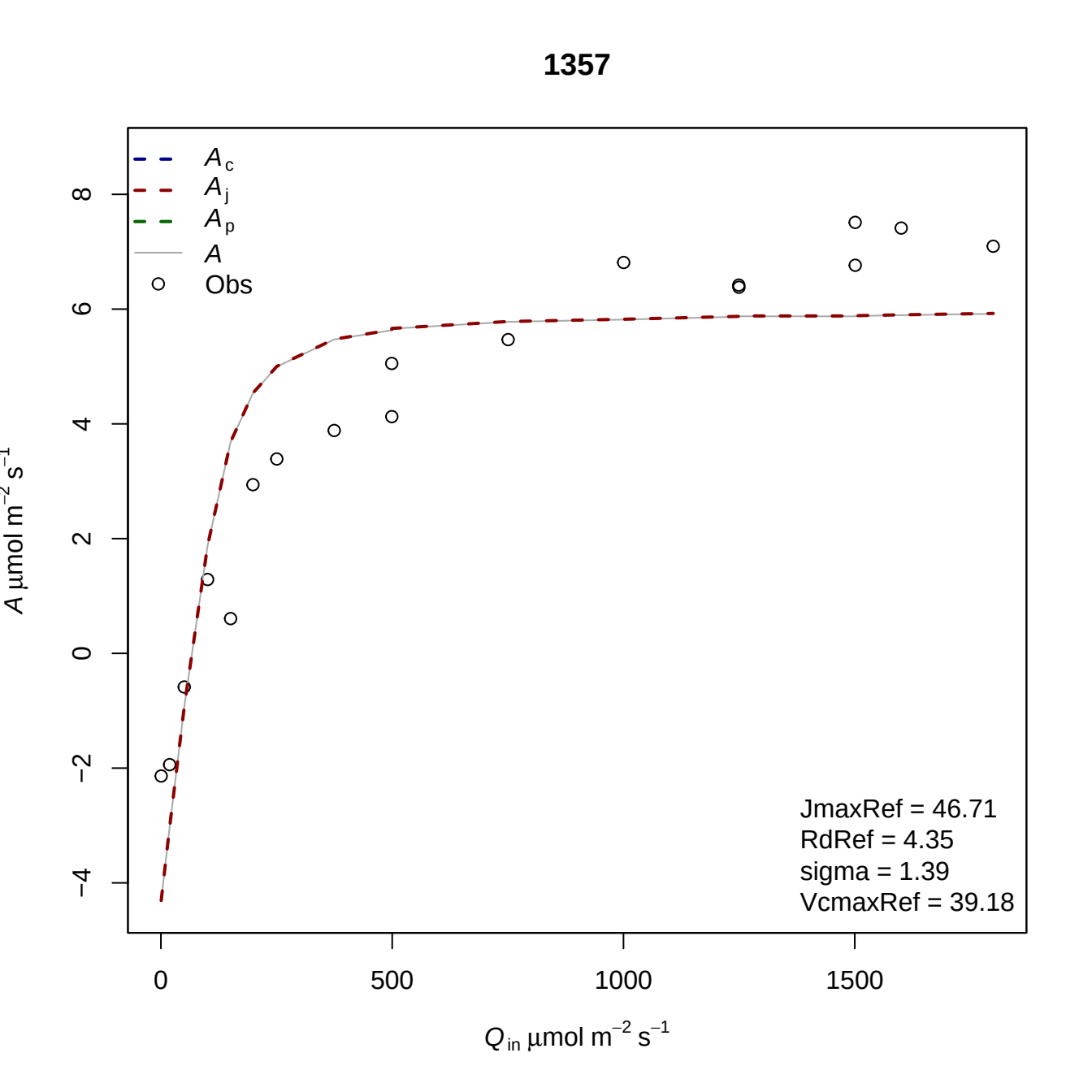

1352
A_c
A_j
A_p
A
Obs
JmaxRef = 61.98
RdRef = 2.64
sigma = 1.86
VcmaxRef = 63.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

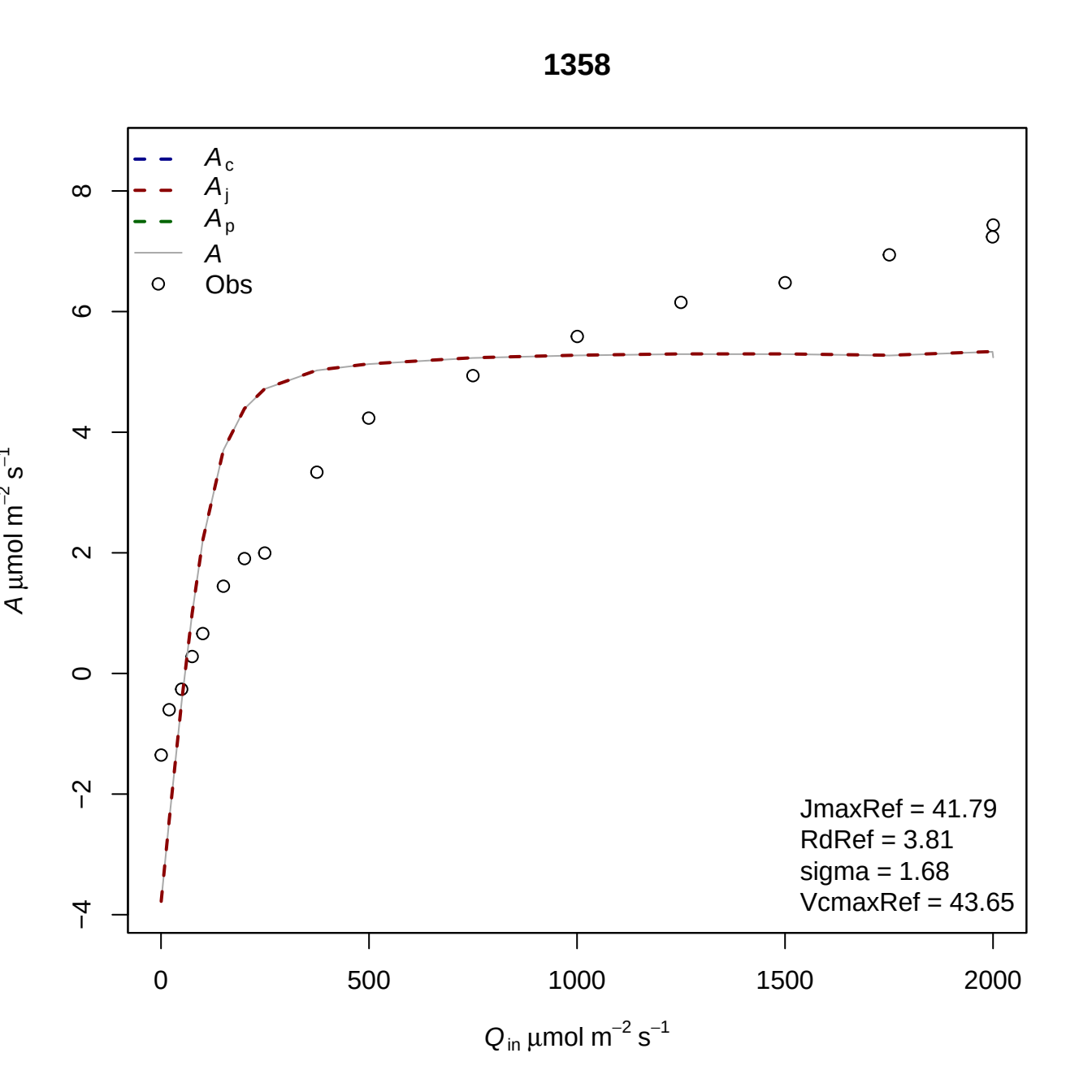

1353
A_c
A_j
A_p
A
Obs
JmaxRef = 43.61
RdRef = 3.5
sigma = 1.04
VcmaxRef = 32.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

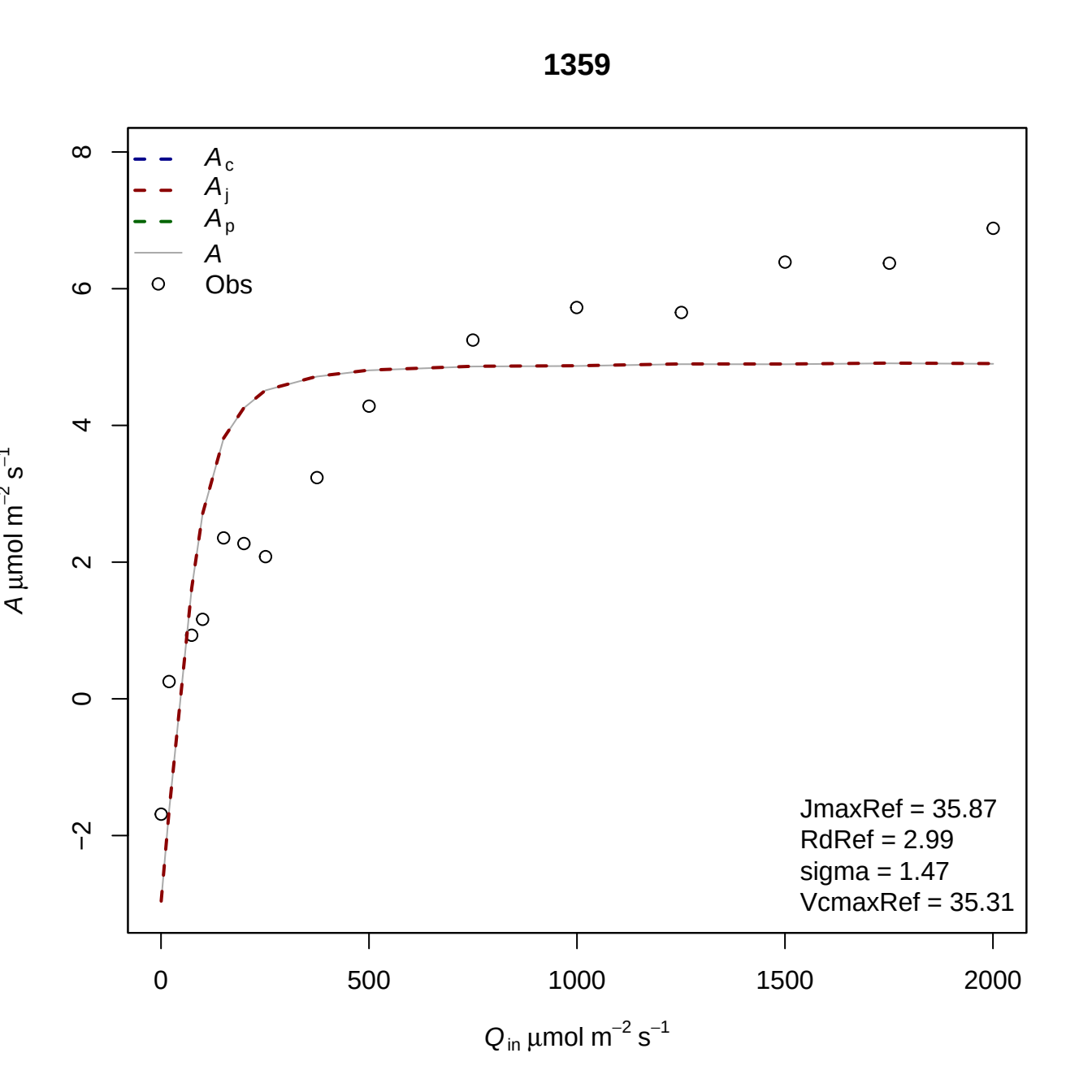

1354
A_c
A_j
A_p
A
Obs
JmaxRef = 49.17
RdRef = 5.06
sigma = 2.11
VcmaxRef = 39.74
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

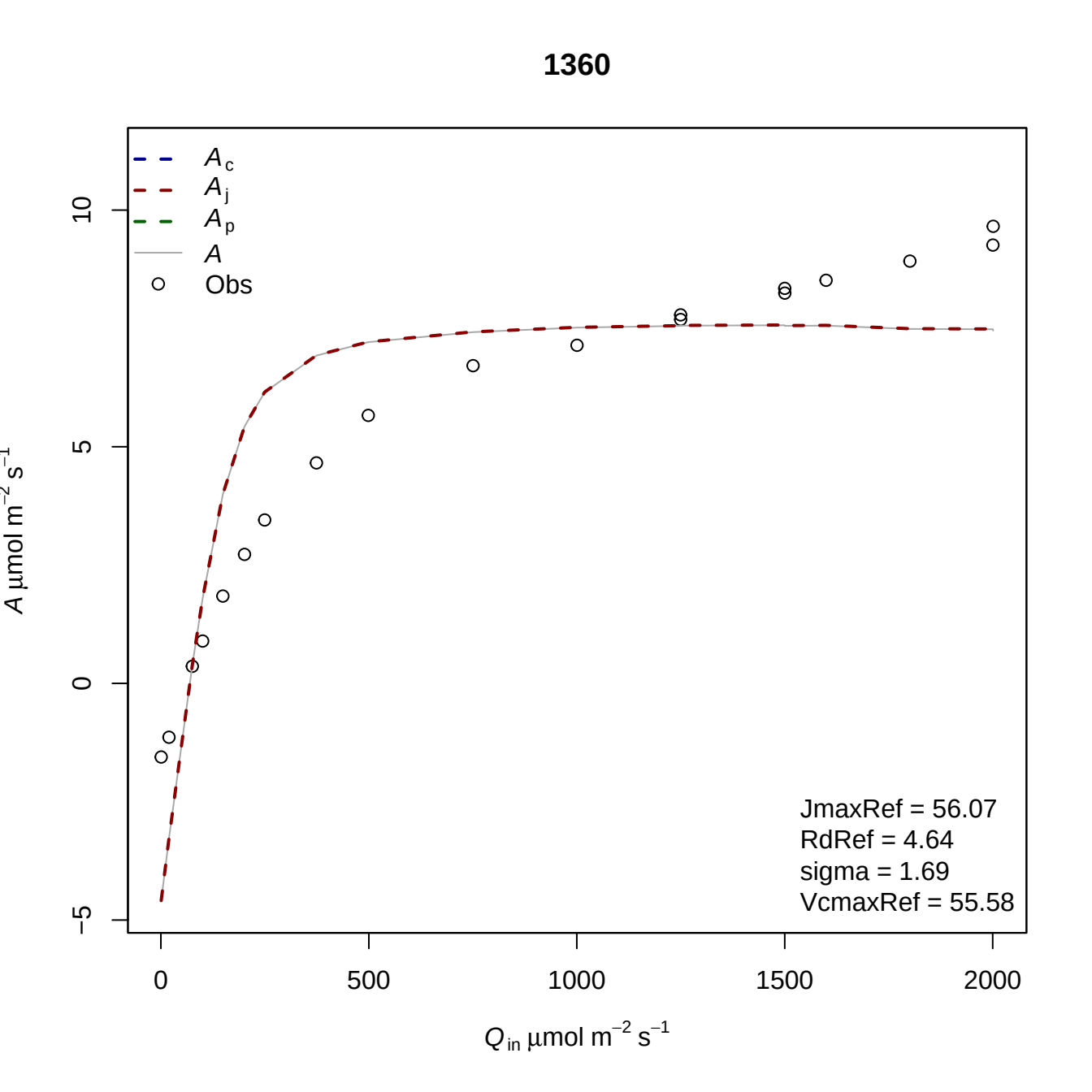

1355
A_c
A_j
A_p
A
Obs
JmaxRef = 33.06
RdRef = 3.9
sigma = 1.36
VcmaxRef = 34.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

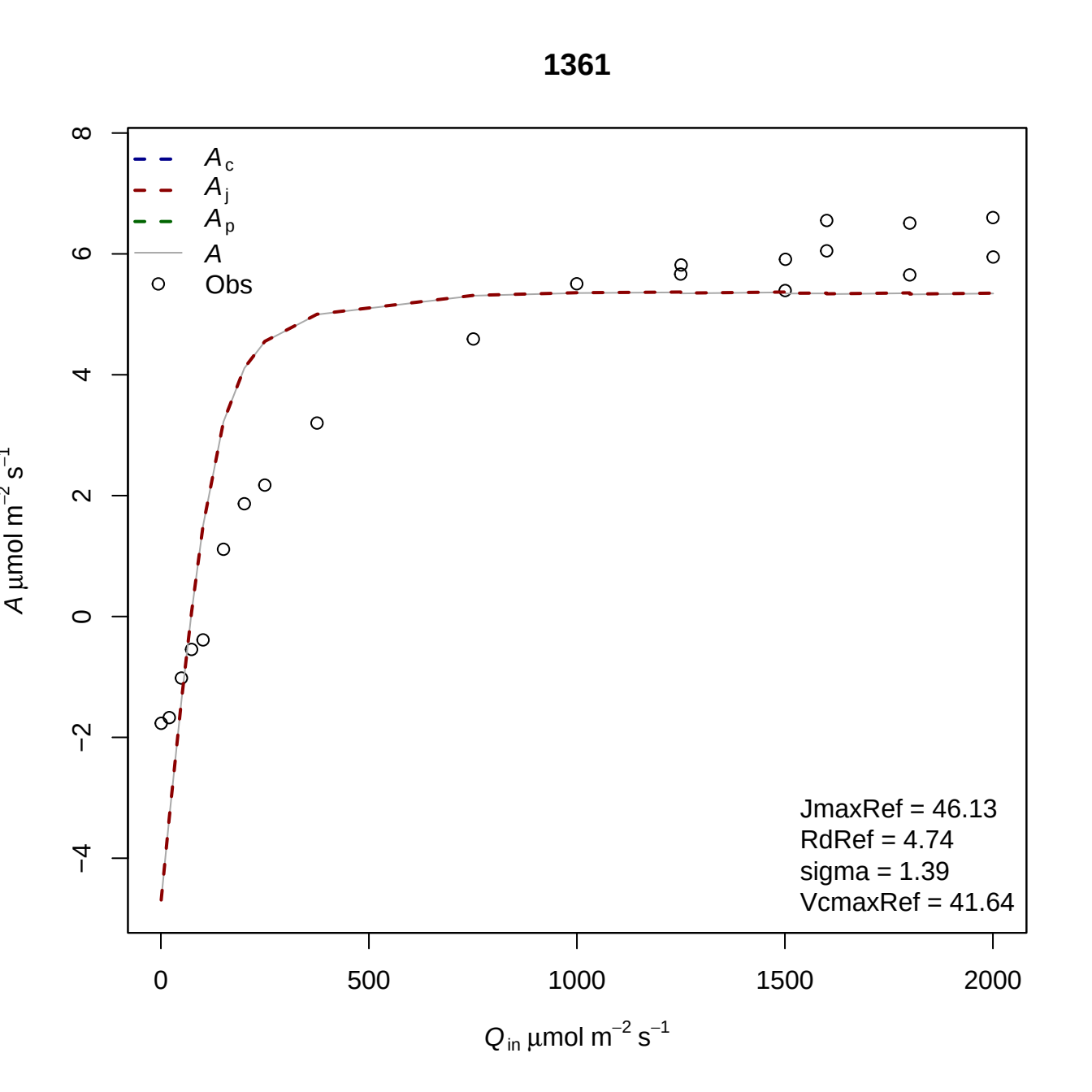

1356
A_c
A_j
A_p
A
Obs
JmaxRef = 69.58
RdRef = 5.27
sigma = 2.1
VcmaxRef = 67.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

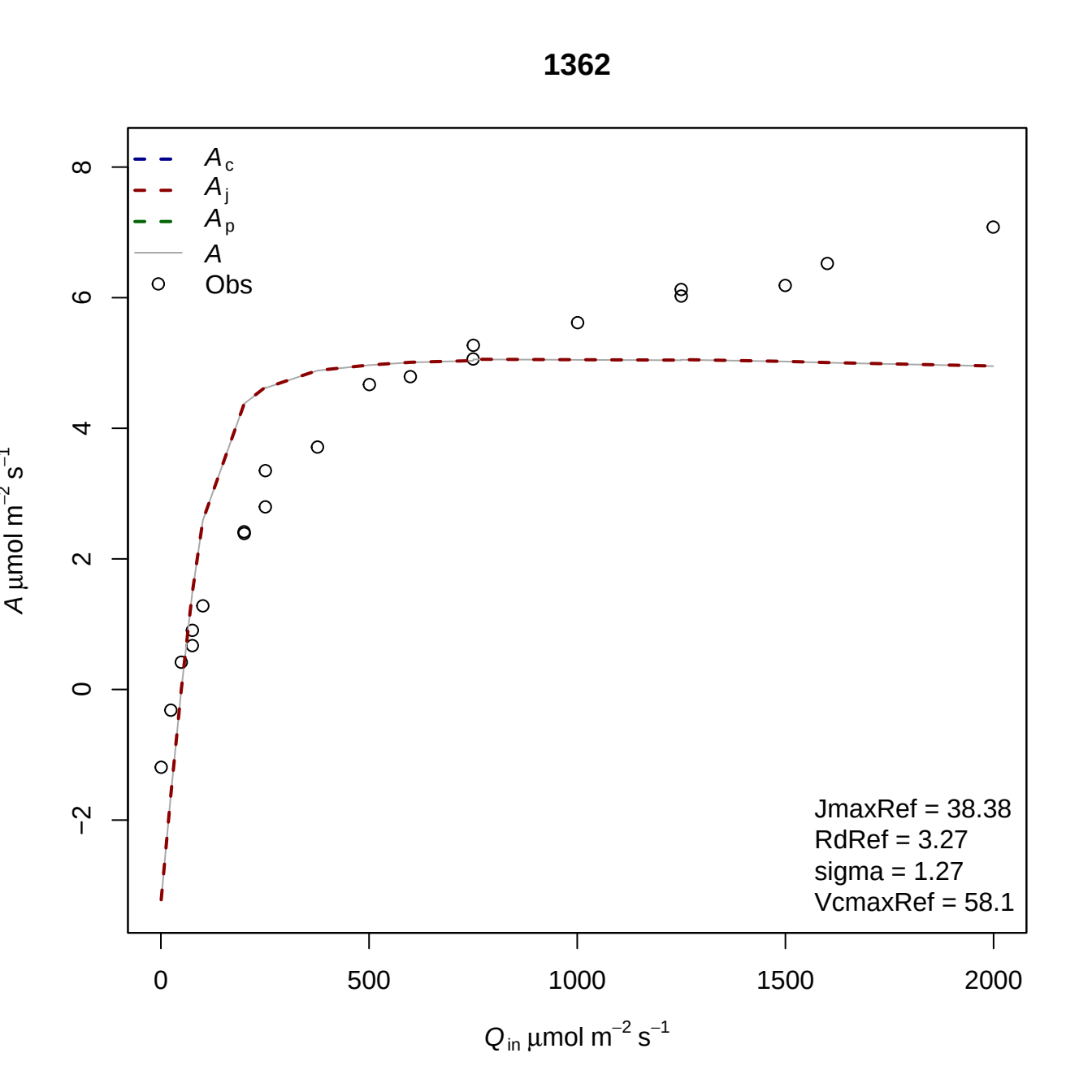

1357
A_c
A_j
A_p
A
Obs
JmaxRef = 46.71
RdRef = 4.35
sigma = 1.39
VcmaxRef = 39.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

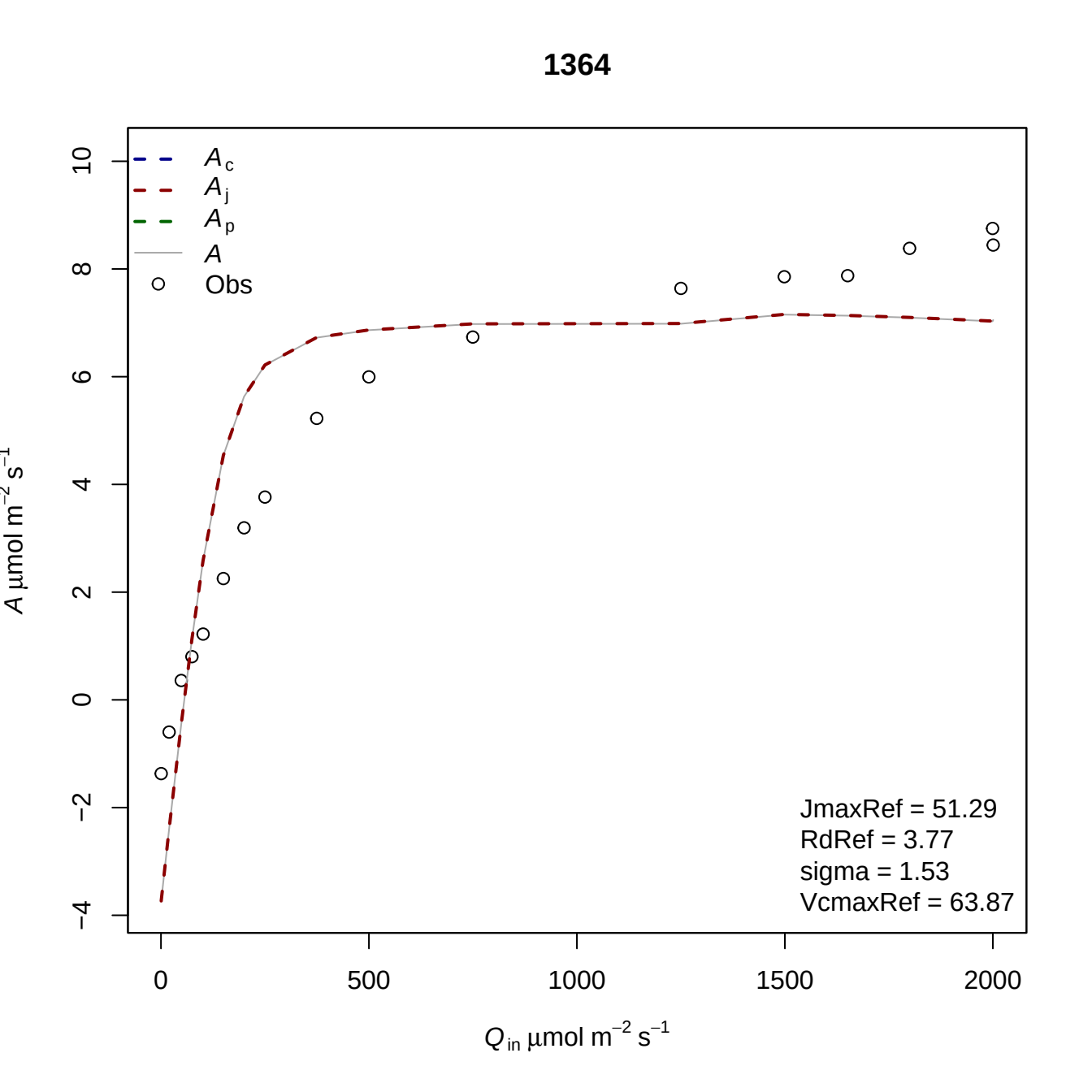

1358
A_c
A_j
A_p
A
Obs
JmaxRef = 41.79
RdRef = 3.81
sigma = 1.68
VcmaxRef = 43.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

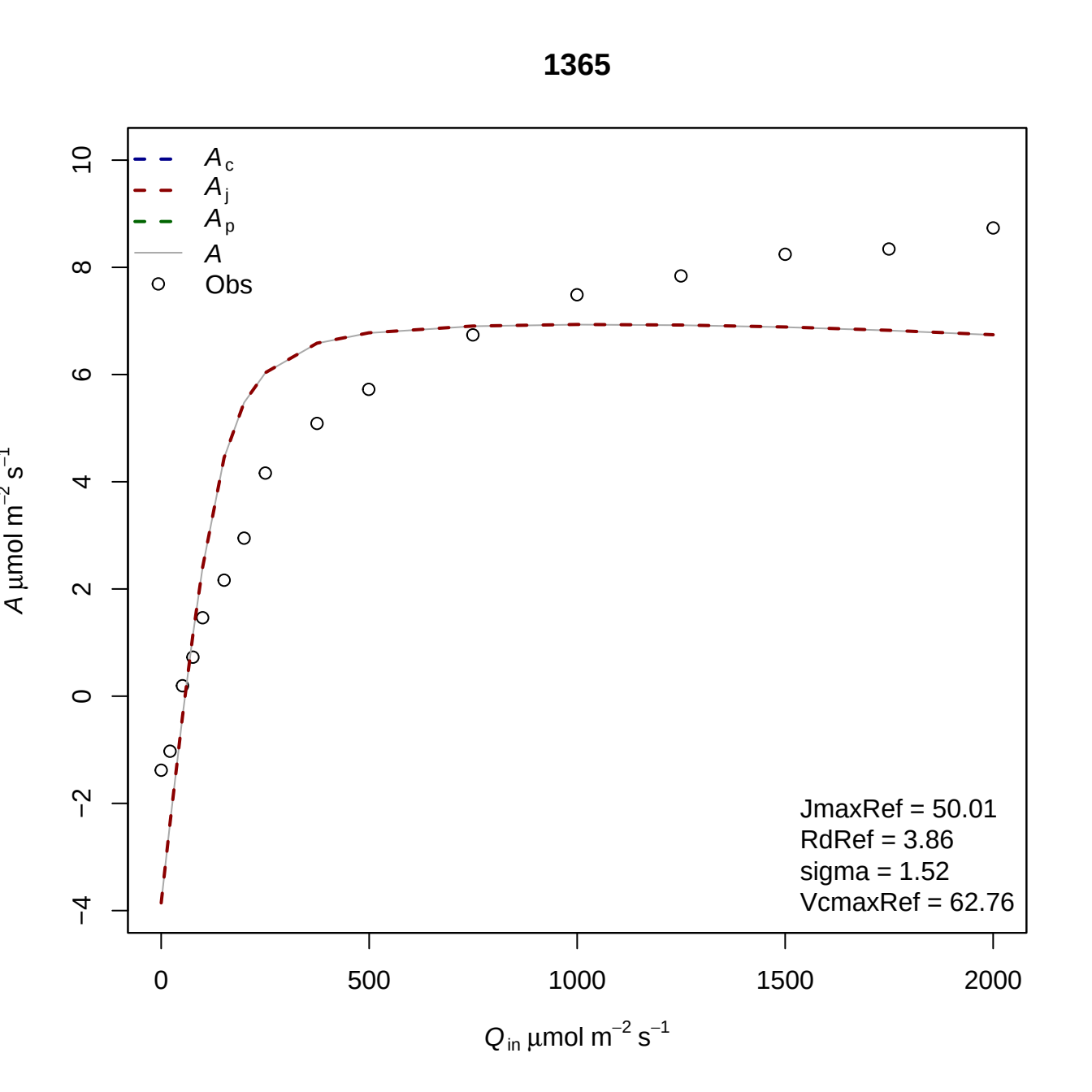

1359
A_c
A_j
A_p
A
Obs
JmaxRef = 35.87
RdRef = 2.99
sigma = 1.47
VcmaxRef = 35.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

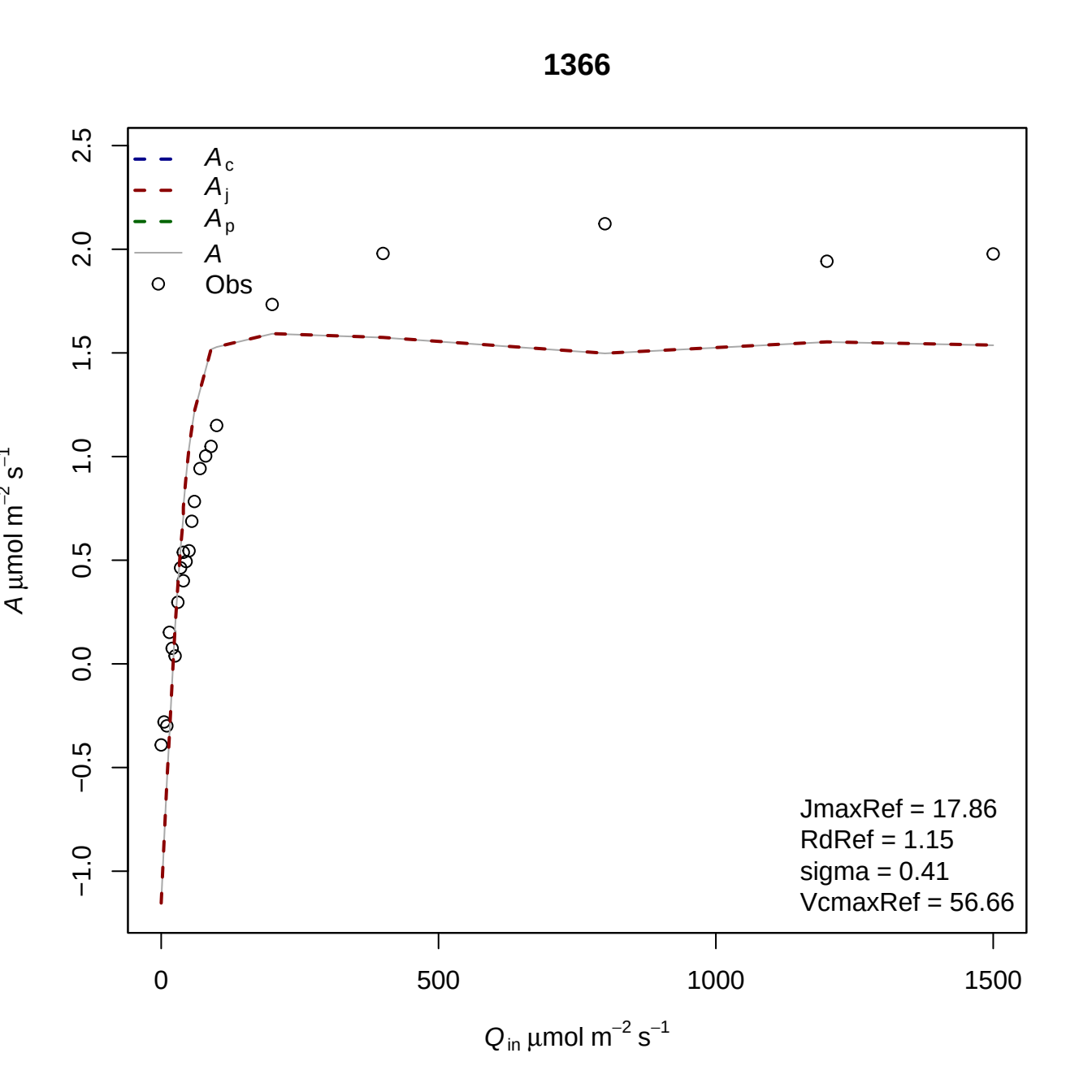

1360
A_c
A_j
A_p
A
Obs
JmaxRef = 56.07
RdRef = 4.64
sigma = 1.69
VcmaxRef = 55.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

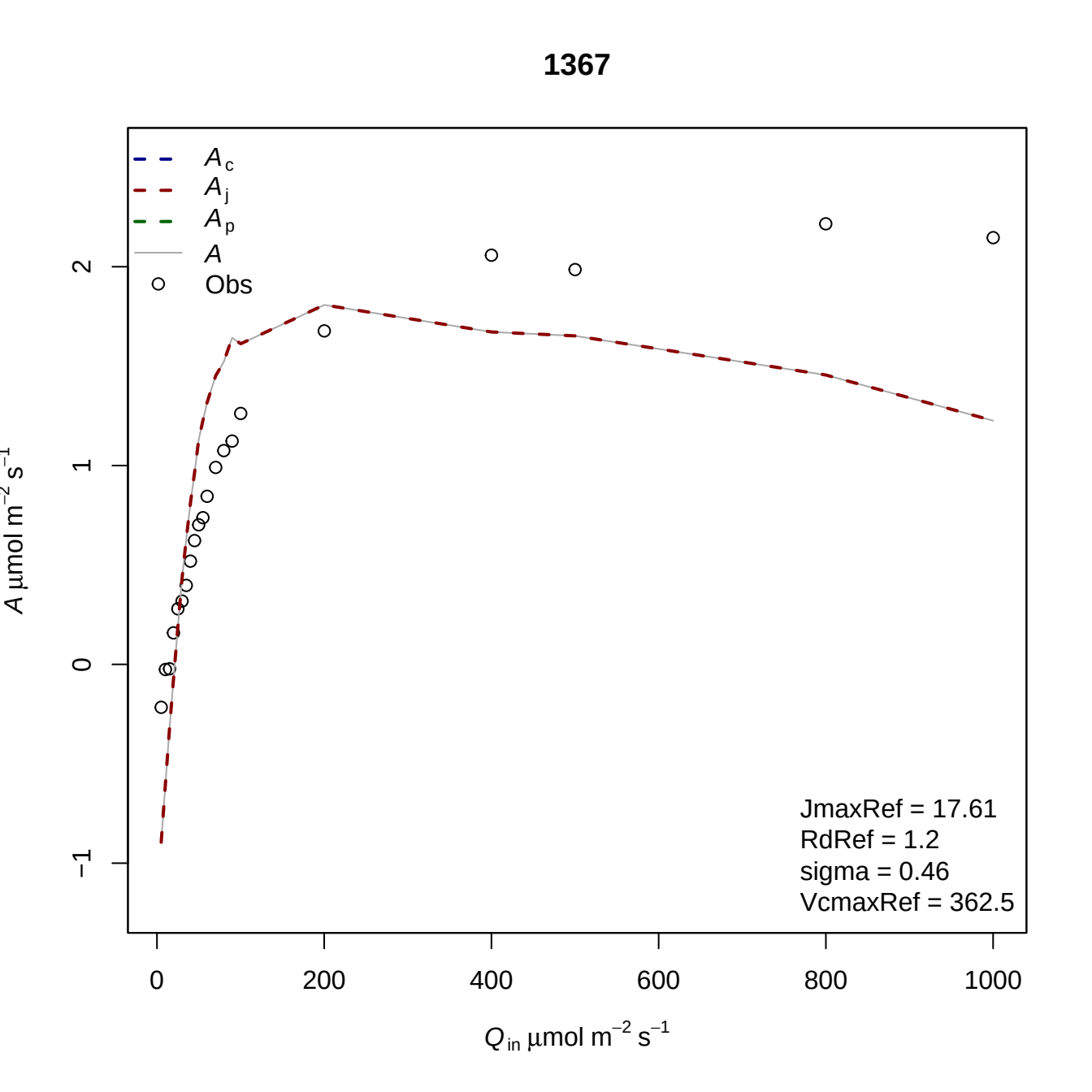

1361
A_c
A_j
A_p
A
Obs
JmaxRef = 46.13
RdRef = 4.74
sigma = 1.39
VcmaxRef = 41.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

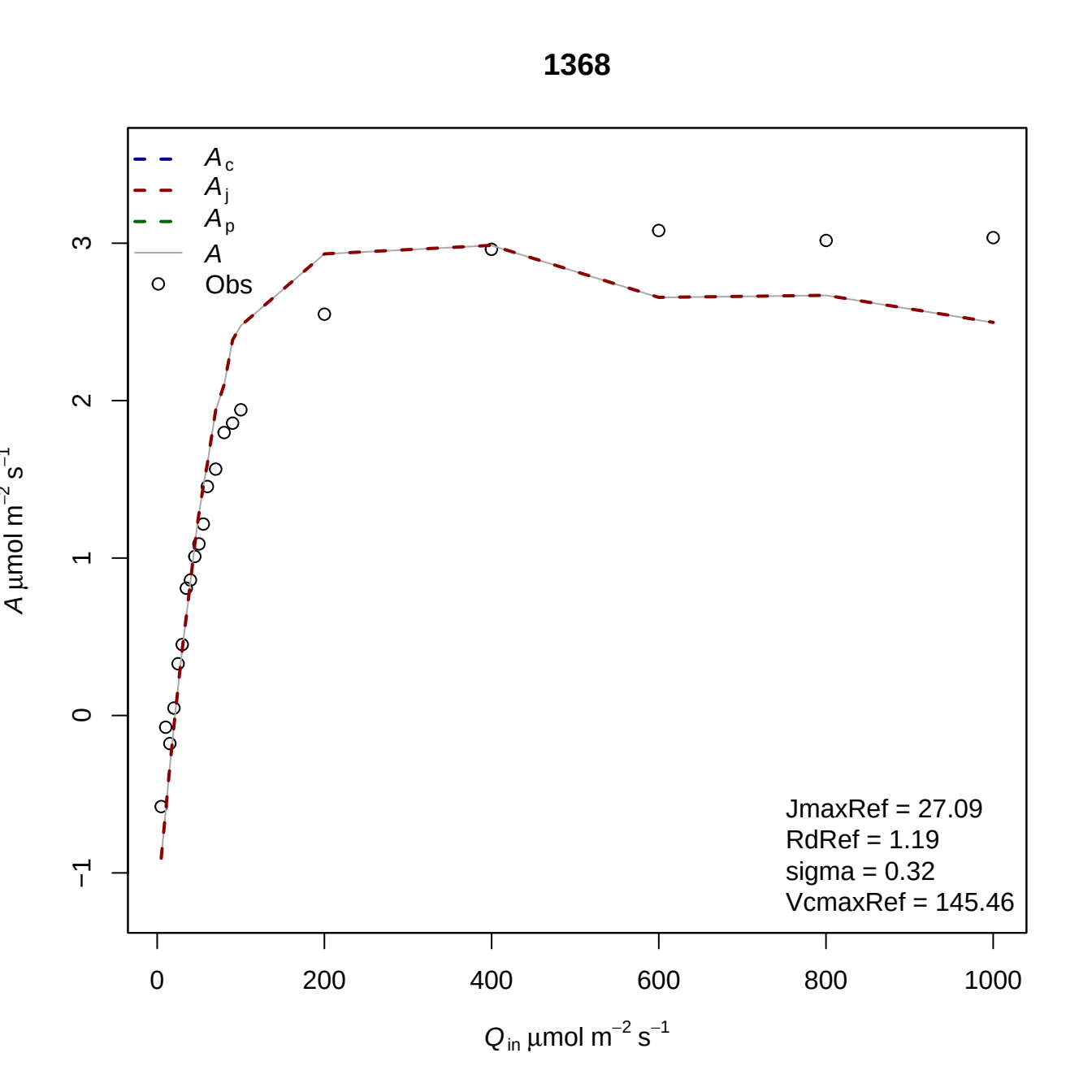

1362
A_c
A_j
A_p
A
Obs
JmaxRef = 38.38
RdRef = 3.27
sigma = 1.27
VcmaxRef = 58.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

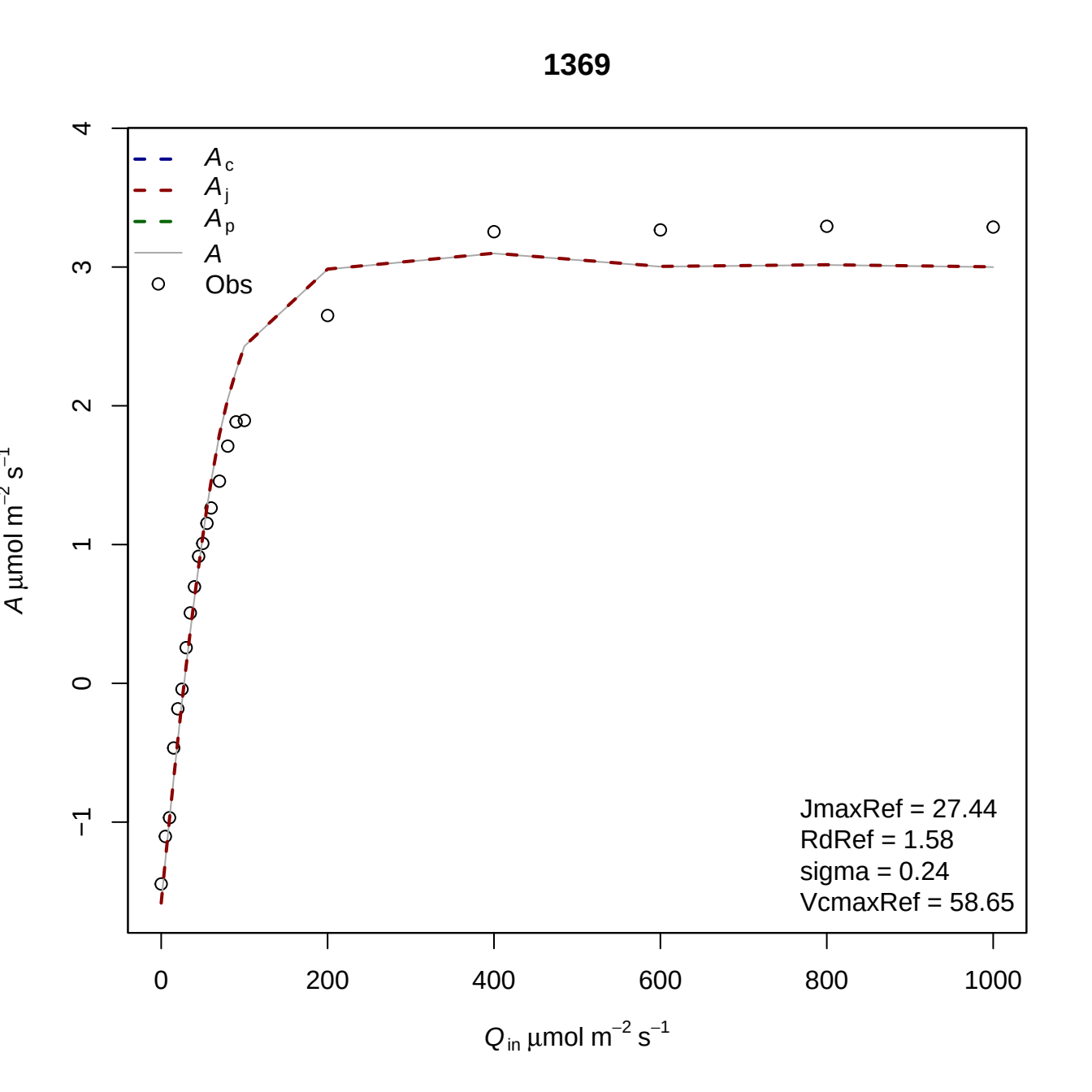

1364
A_c
A_j
A_p
A
Obs
JmaxRef = 51.29
RdRef = 3.77
sigma = 1.53
VcmaxRef = 63.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

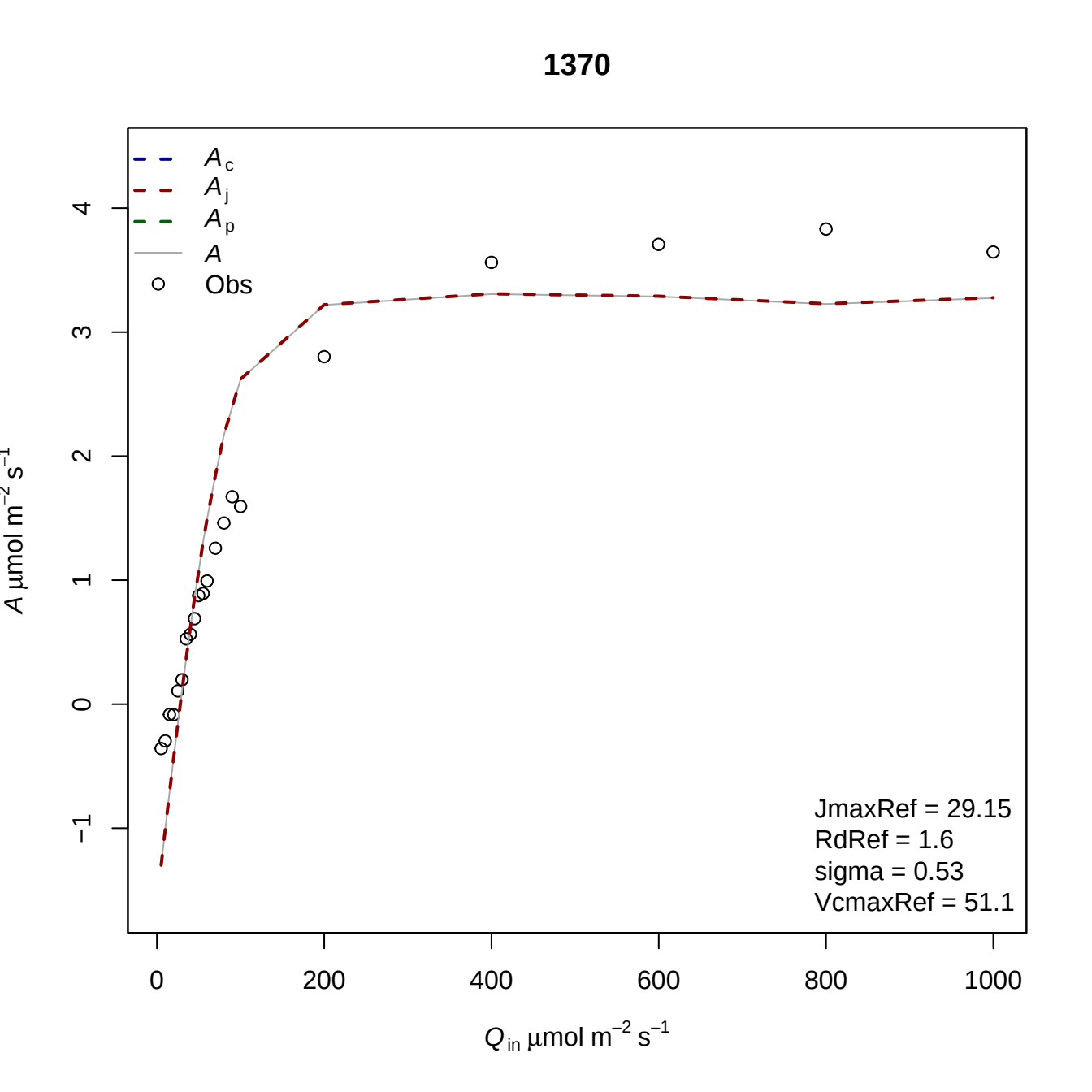

1365
A_c
A_j
A_p
A
Obs
JmaxRef = 50.01
RdRef = 3.86
sigma = 1.52
VcmaxRef = 62.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

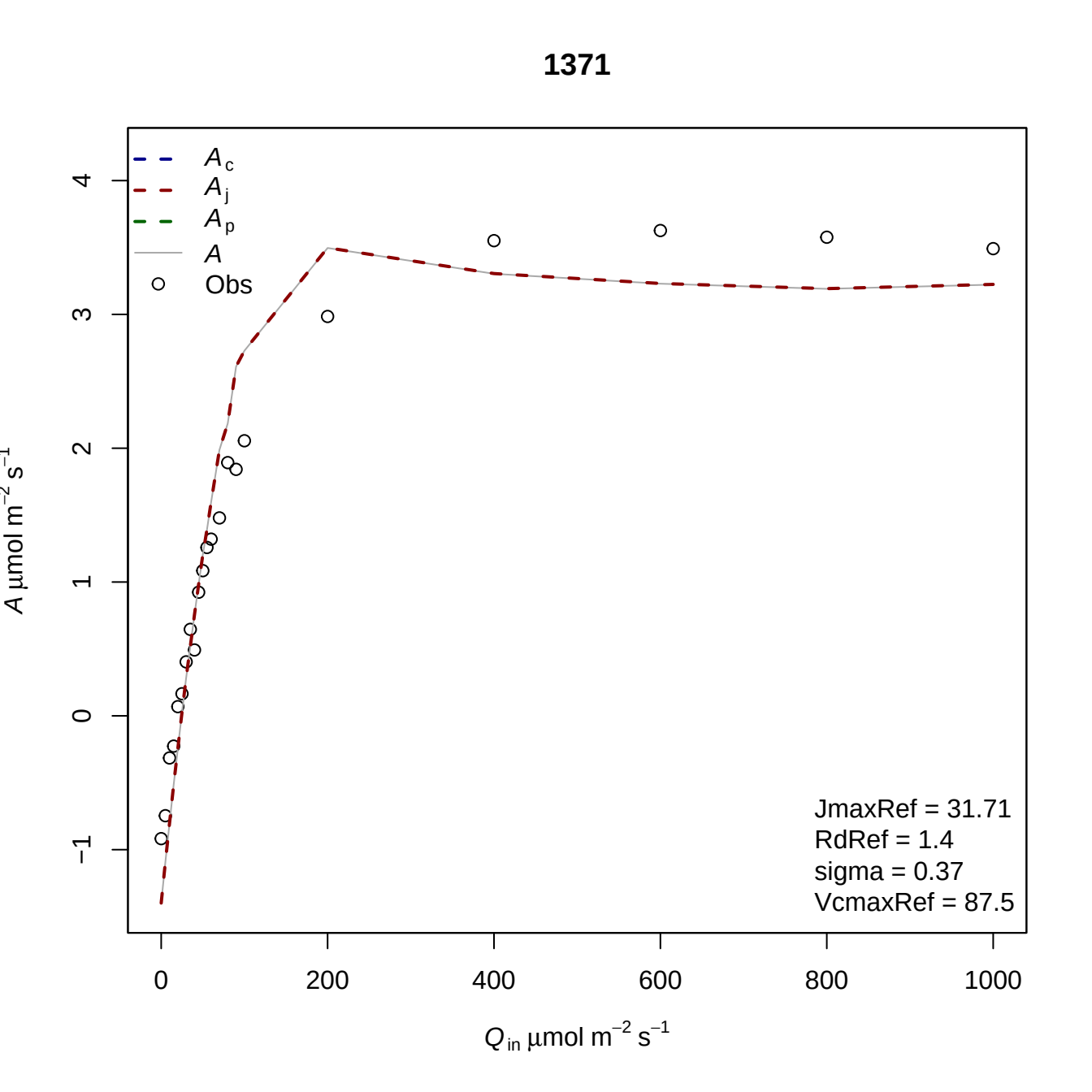

1366
A_c
A_j
A_p
A
Obs
JmaxRef = 17.86
RdRef = 1.15
sigma = 0.41
VcmaxRef = 56.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

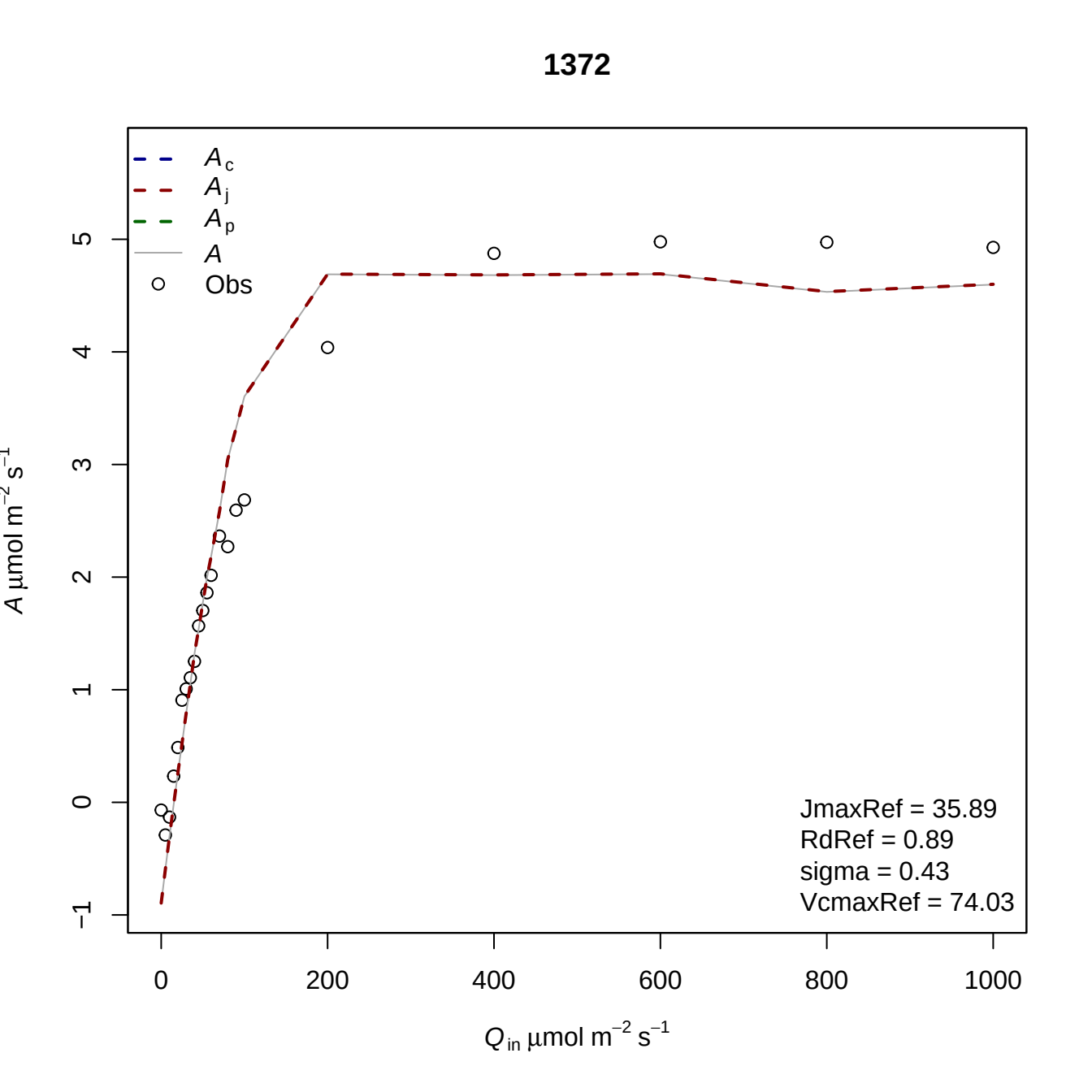

1367
A_c
A_j
A_p
A
Obs
JmaxRef = 17.61
RdRef = 1.2
sigma = 0.46
VcmaxRef = 362.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000

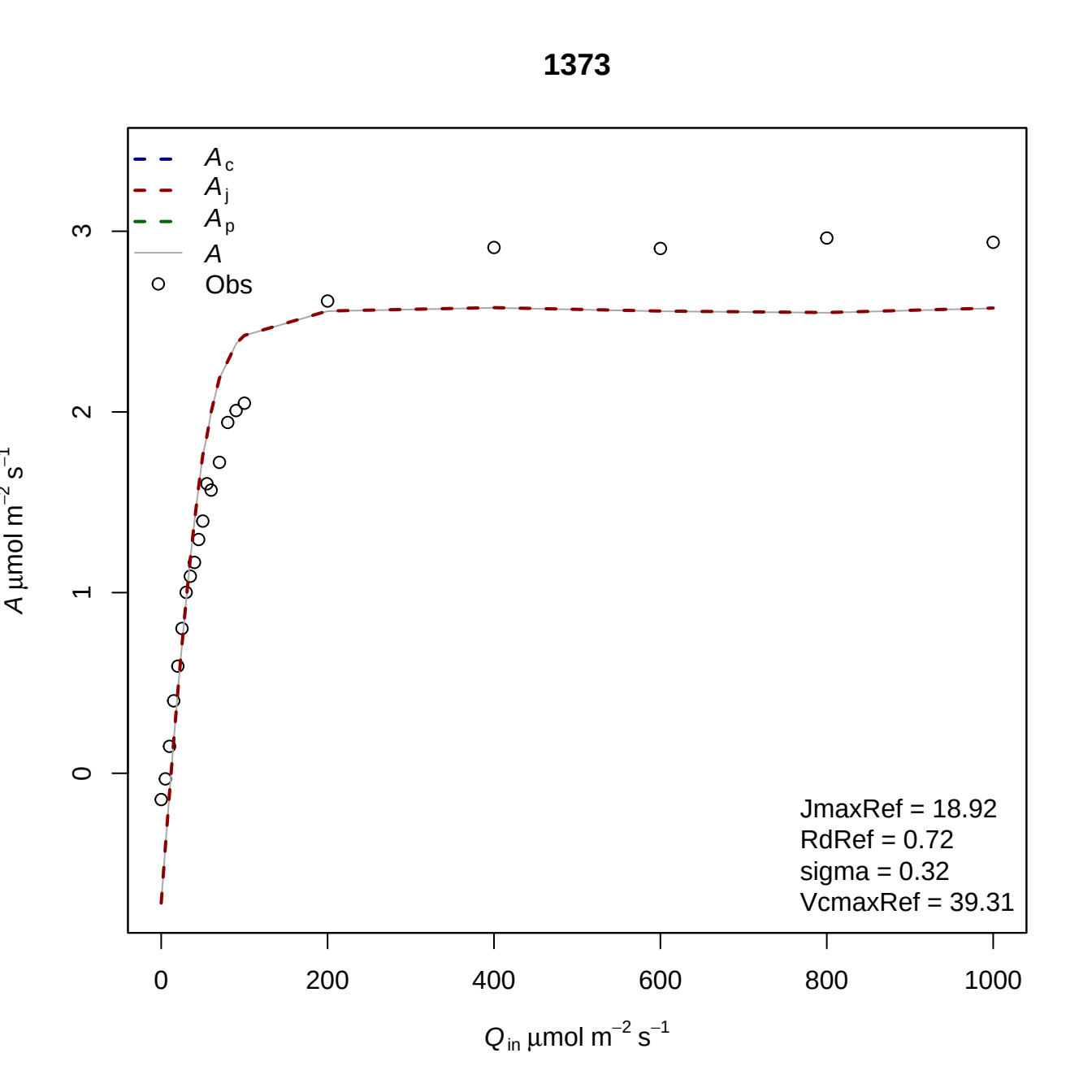

1368
A_c
A_j
A_p
A
Obs
JmaxRef = 27.09
RdRef = 1.19
sigma = 0.32
VcmaxRef = 145.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

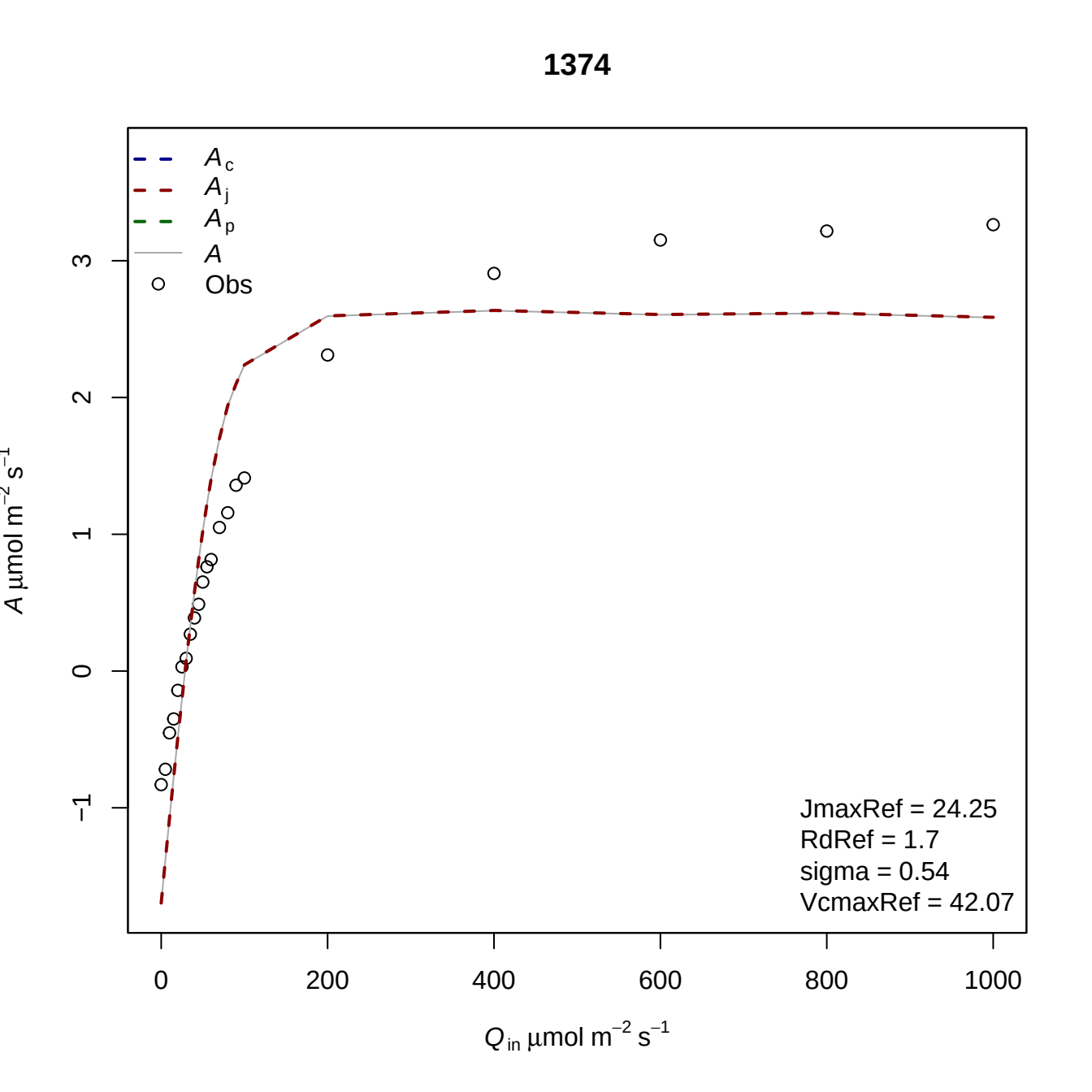

1369
A_c
A_j
A_p
A
Obs
JmaxRef = 27.44
RdRef = 1.58
sigma = 0.24
VcmaxRef = 58.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000

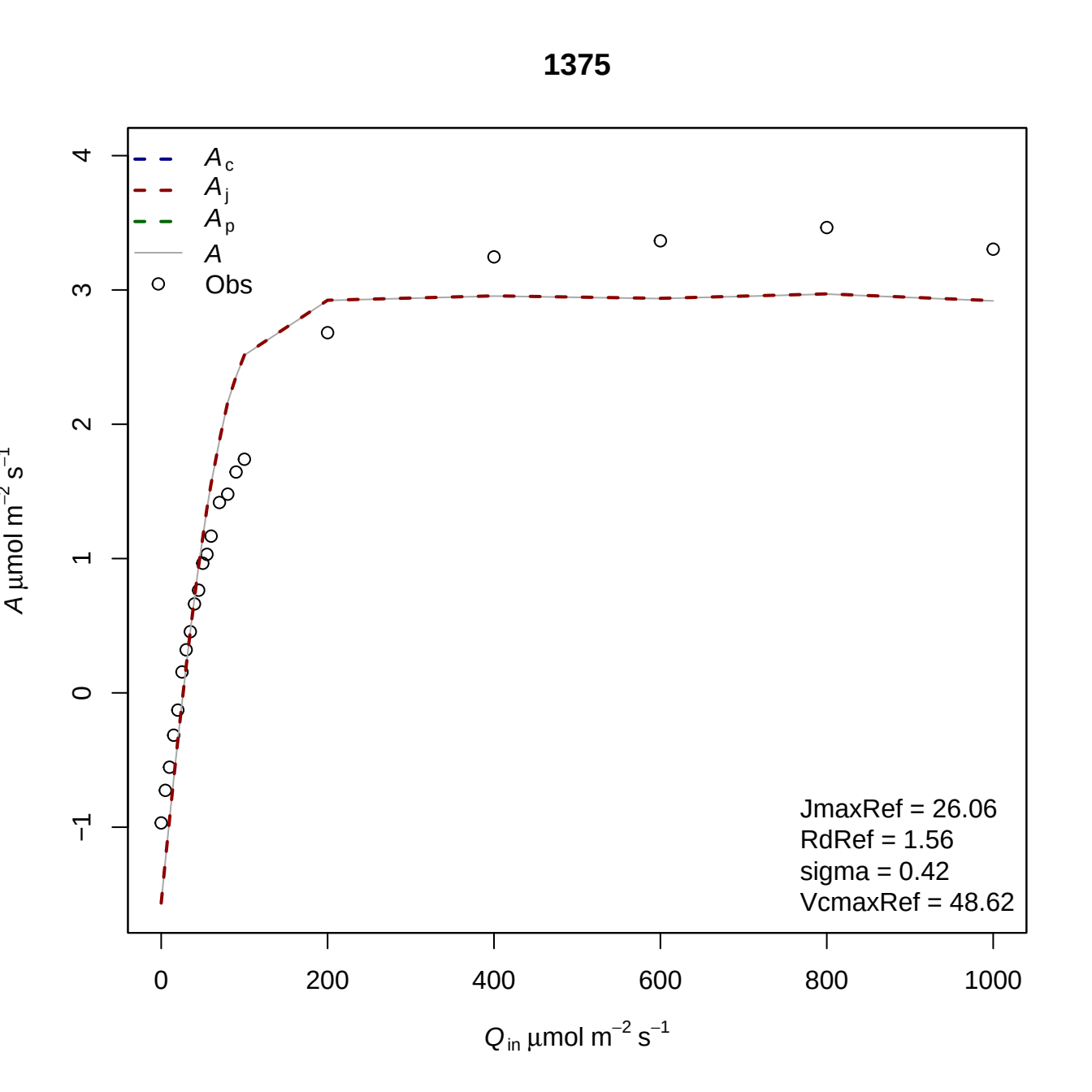

1370
A_c
A_j
A_p
A
Obs
JmaxRef = 29.15
RdRef = 1.6
sigma = 0.53
VcmaxRef = 51.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

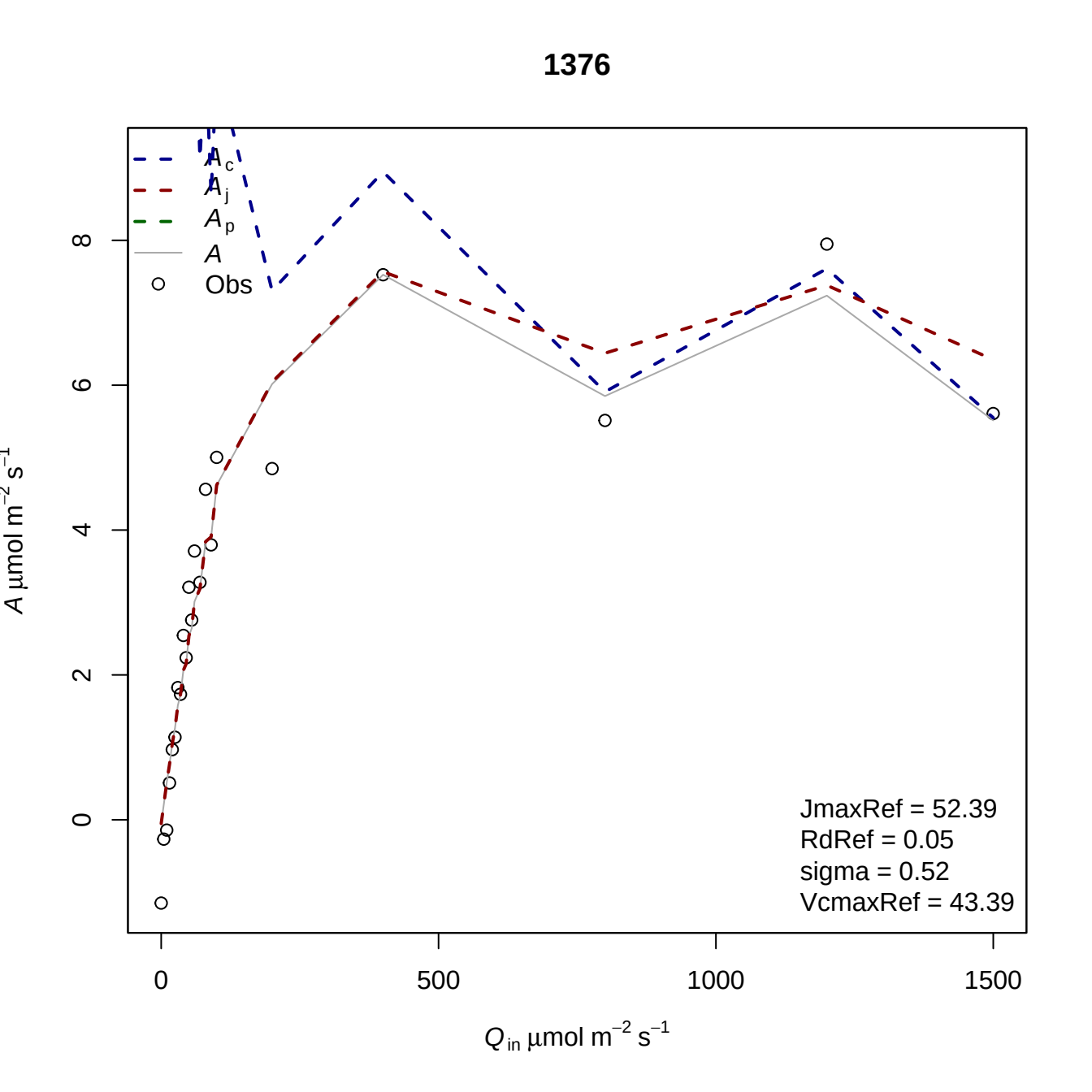

1371
A_c
A_j
A_p
A
Obs
JmaxRef = 31.71
RdRef = 1.4
sigma = 0.37
VcmaxRef = 87.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

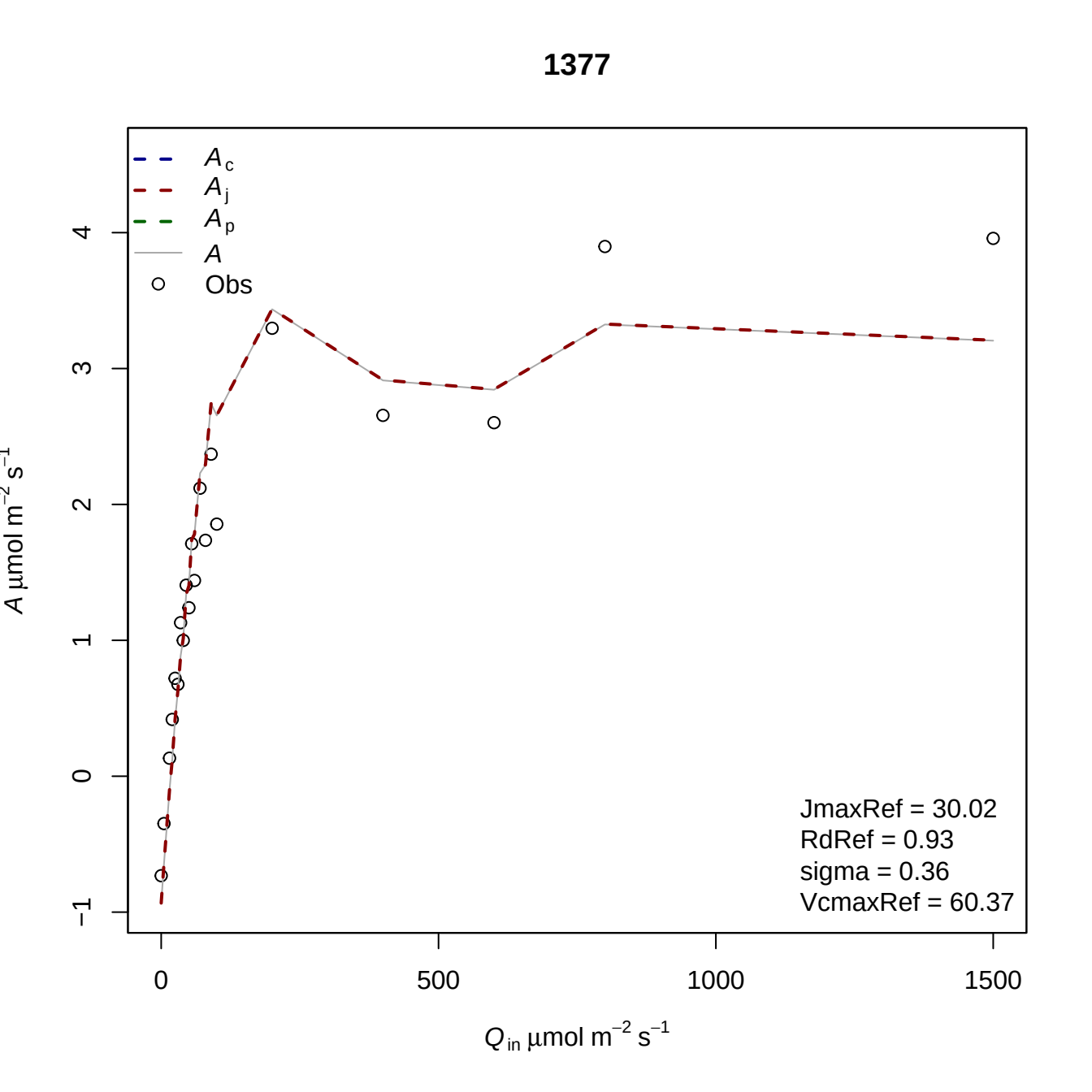

1372
A_c
A_j
A_p
A
Obs
JmaxRef = 35.89
RdRef = 0.89
sigma = 0.43
VcmaxRef = 74.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000

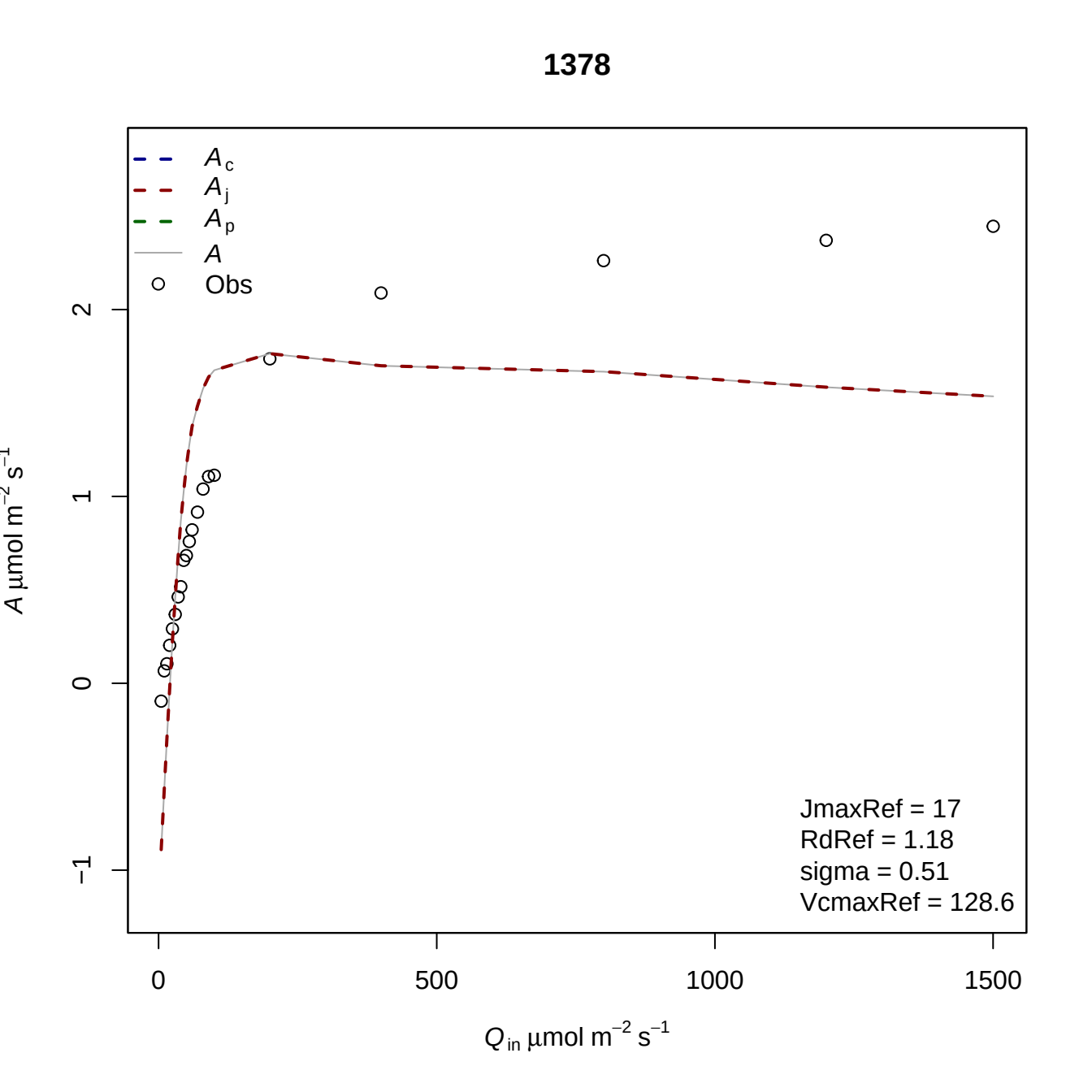

1373
A_c
A_j
A_p
A
Obs
JmaxRef = 18.92
RdRef = 0.72
sigma = 0.32
VcmaxRef = 39.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

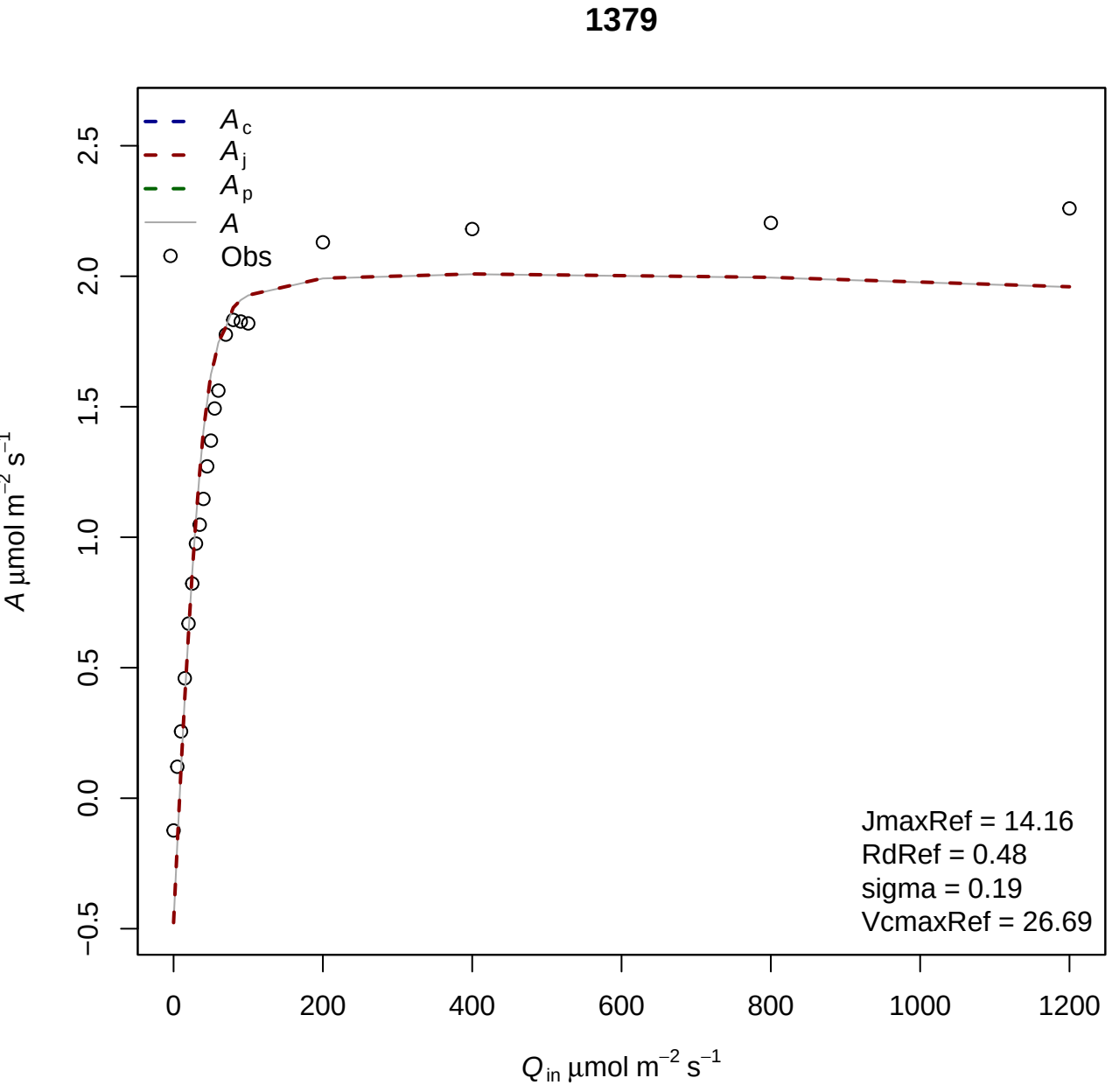

1374
A_c
A_j
A_p
A
Obs
JmaxRef = 24.25
RdRef = 1.7
sigma = 0.54
VcmaxRef = 42.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

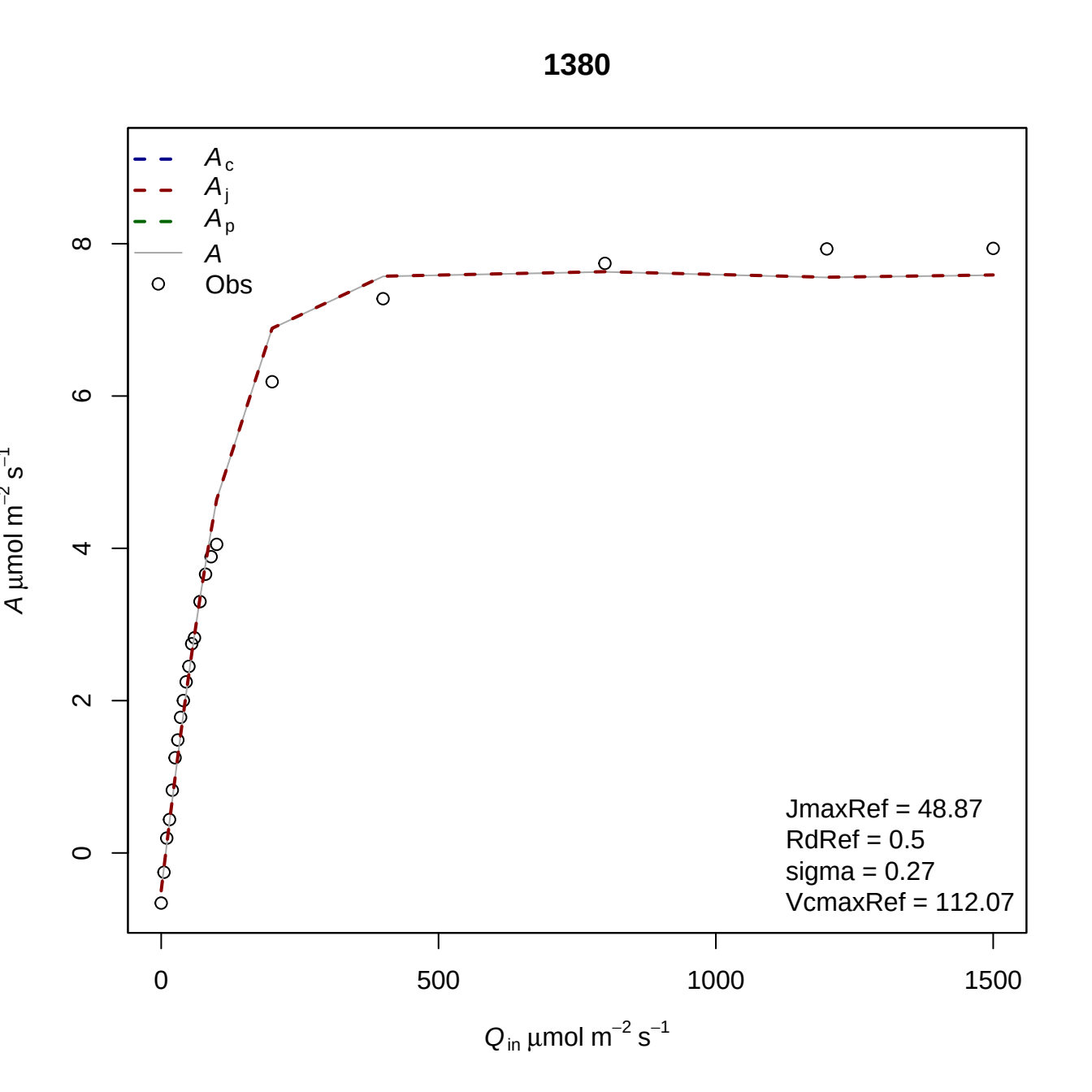

1375
A_c
A_j
A_p
A
Obs
JmaxRef = 26.06
RdRef = 1.56
sigma = 0.42
VcmaxRef = 48.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

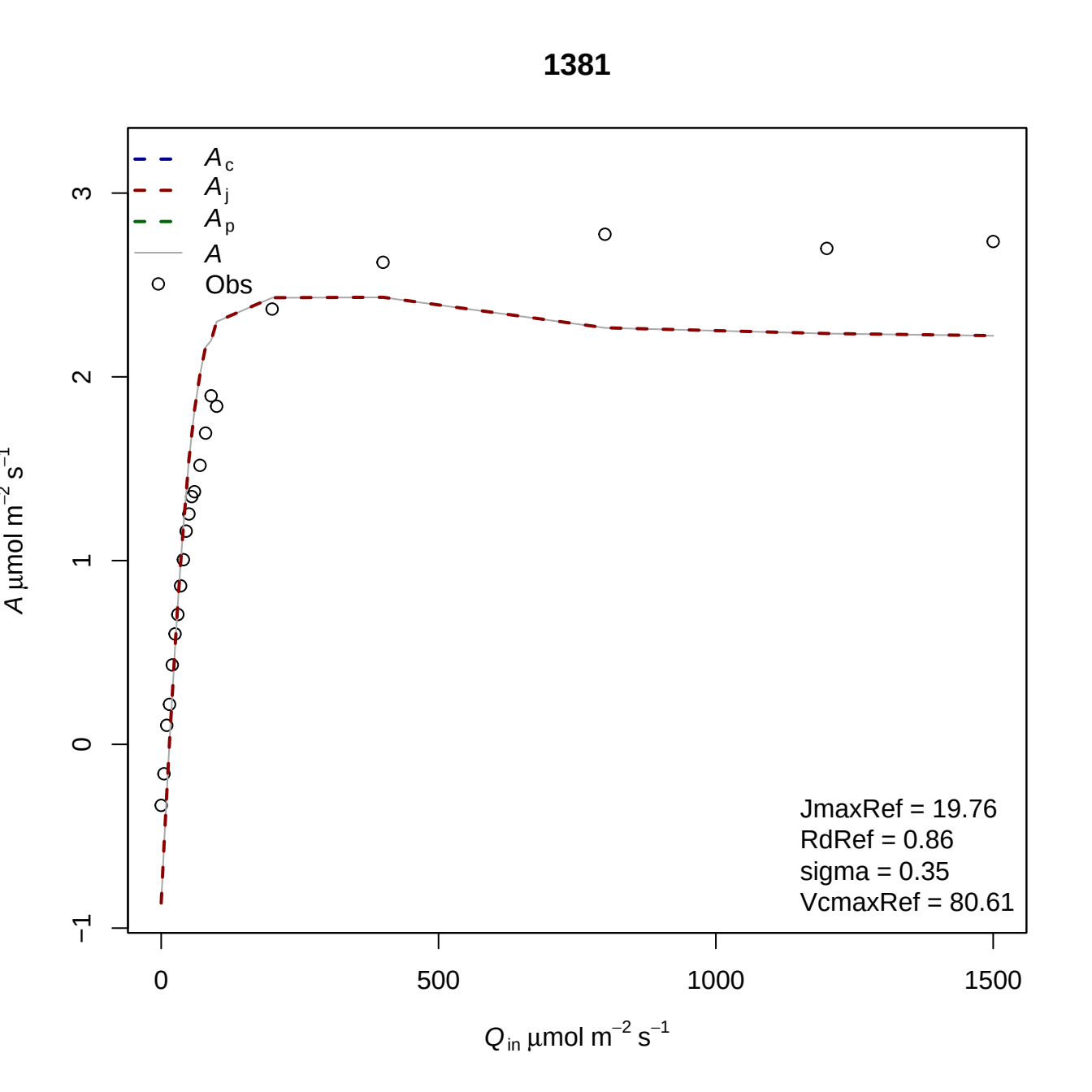

1376
A_c
A_j
A_p
A
Obs
JmaxRef = 52.39
RdRef = 0.05
sigma = 0.52
VcmaxRef = 43.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

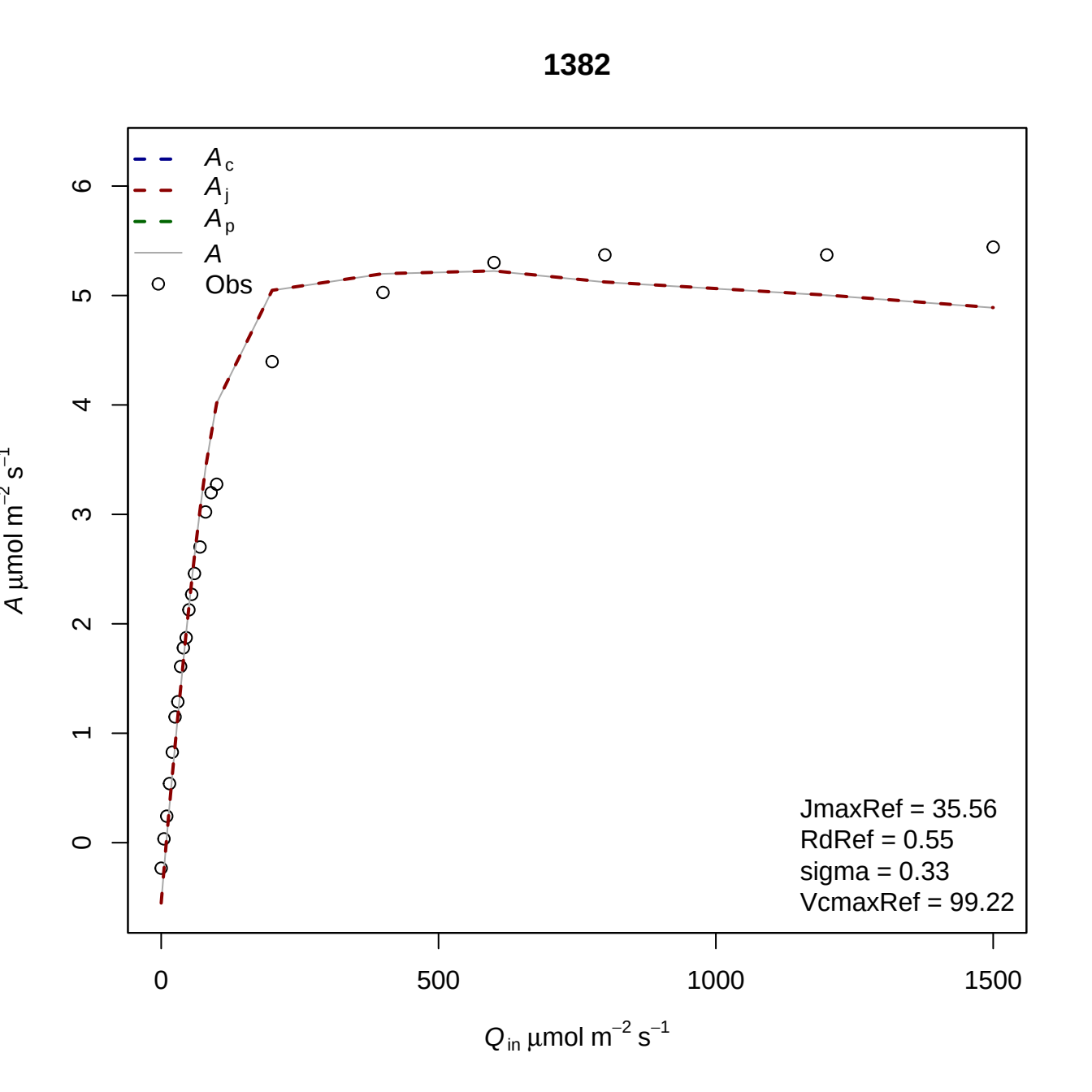

1377
A_c
A_j
A_p
A
Obs
JmaxRef = 30.02
RdRef = 0.93
sigma = 0.36
VcmaxRef = 60.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

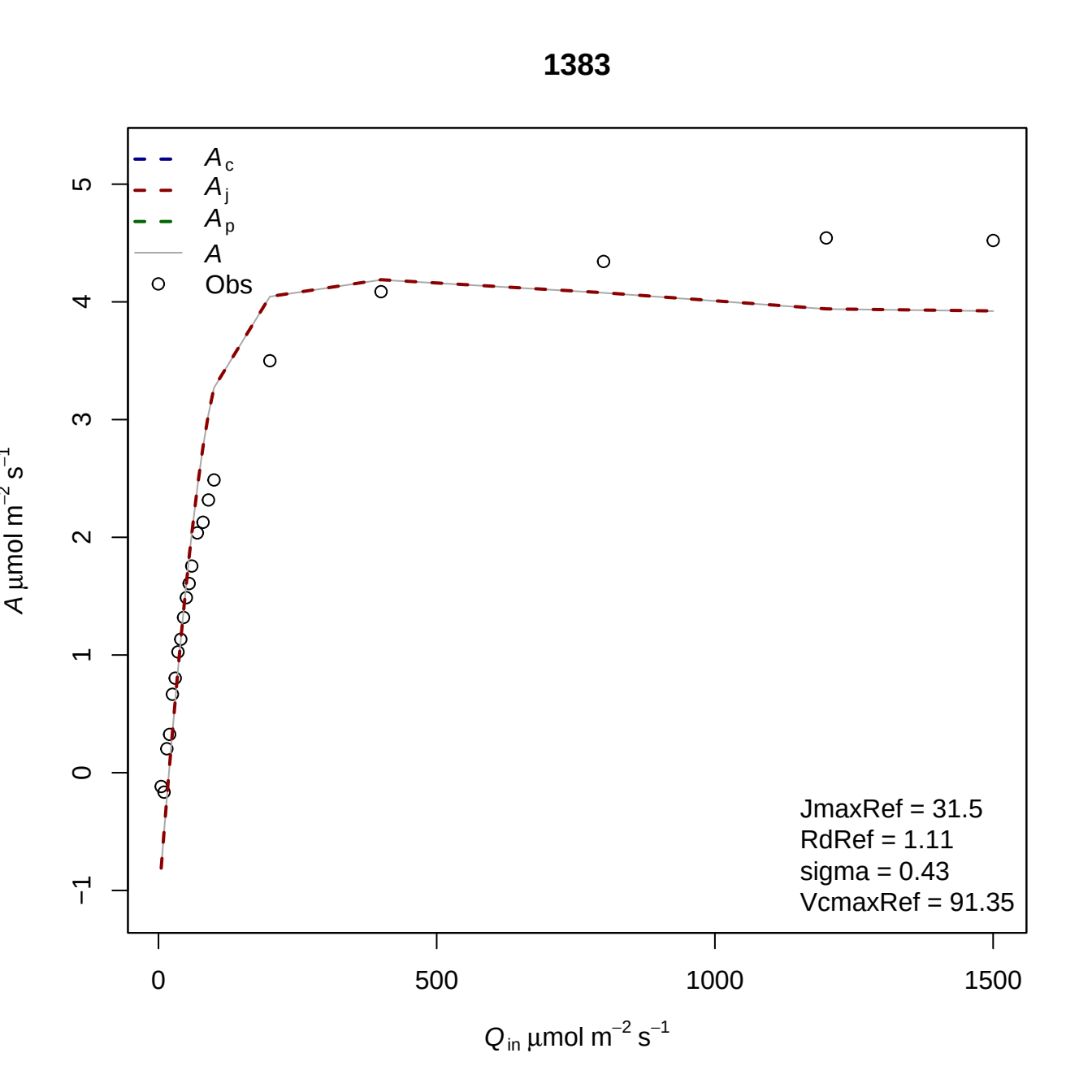

1378
A_c
A_j
A_p
A
Obs
JmaxRef = 17
RdRef = 1.18
sigma = 0.51
VcmaxRef = 128.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

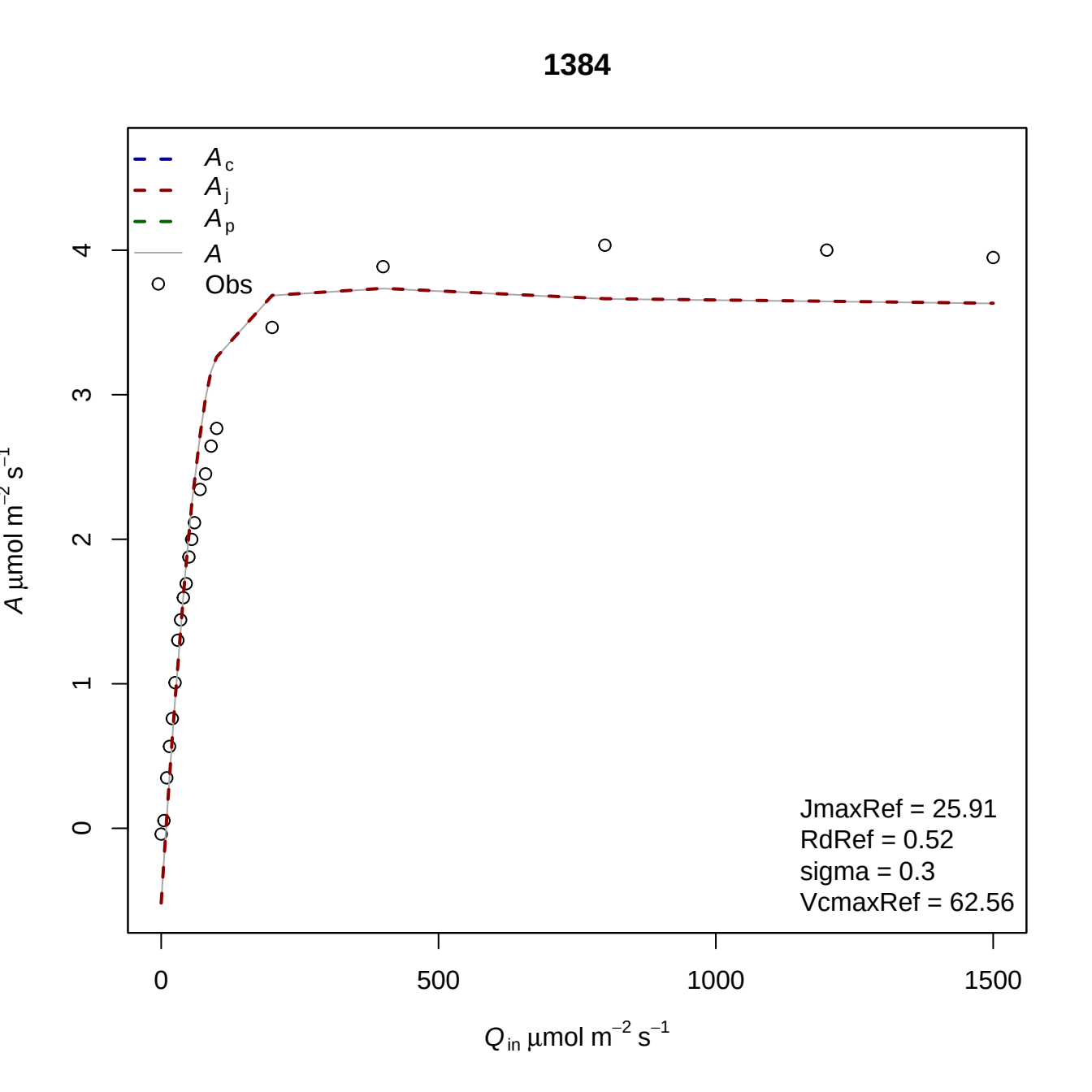

1379
A_c
A_j
A_p
A
Obs
JmaxRef = 14.16
RdRef = 0.48
sigma = 0.19
VcmaxRef = 26.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000
1200

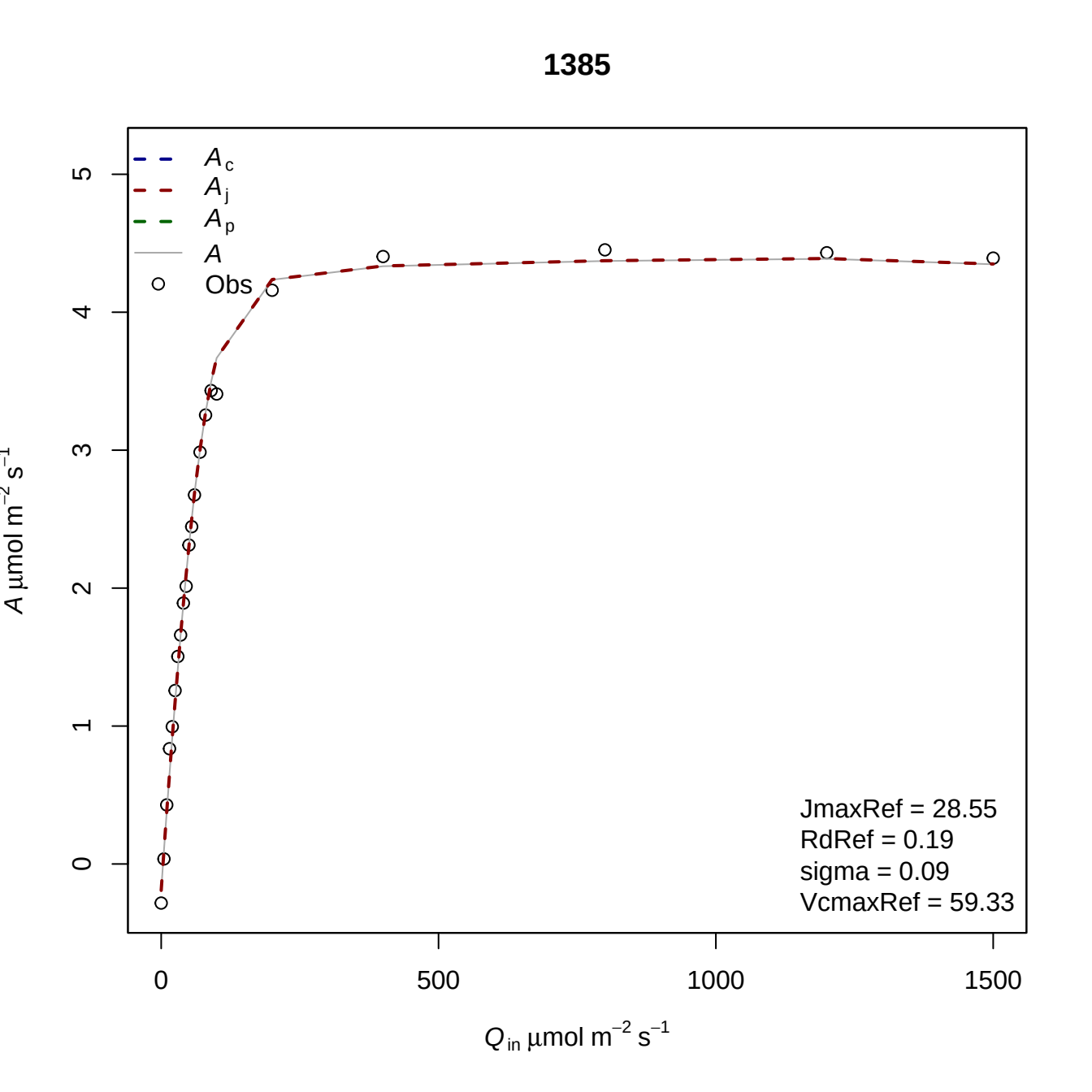

1380
A_c
A_j
A_p
A
Obs
JmaxRef = 48.87
RdRef = 0.5
sigma = 0.27
VcmaxRef = 112.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

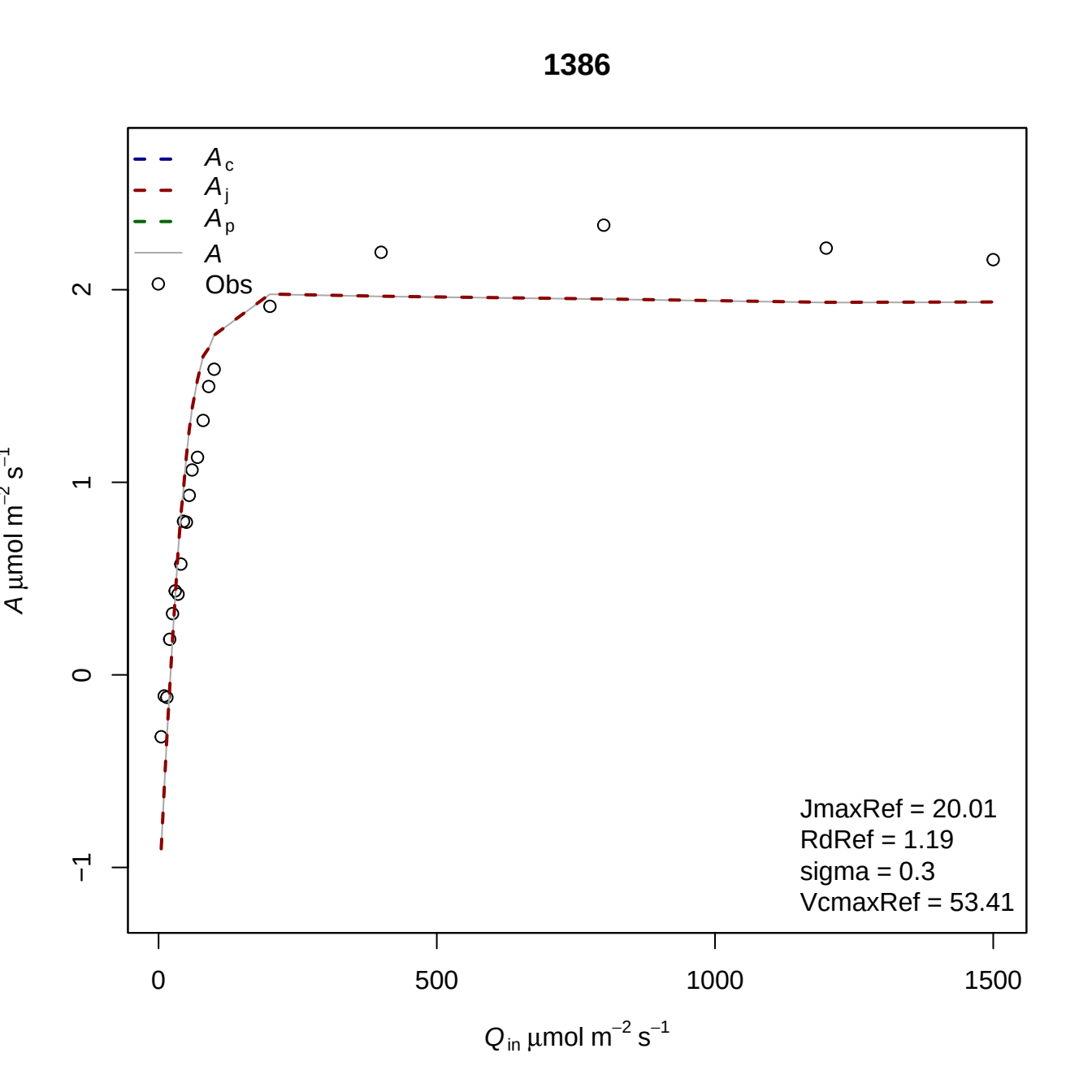

1381
A_c
A_j
A_p
A
Obs
JmaxRef = 19.76
RdRef = 0.86
sigma = 0.35
VcmaxRef = 80.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

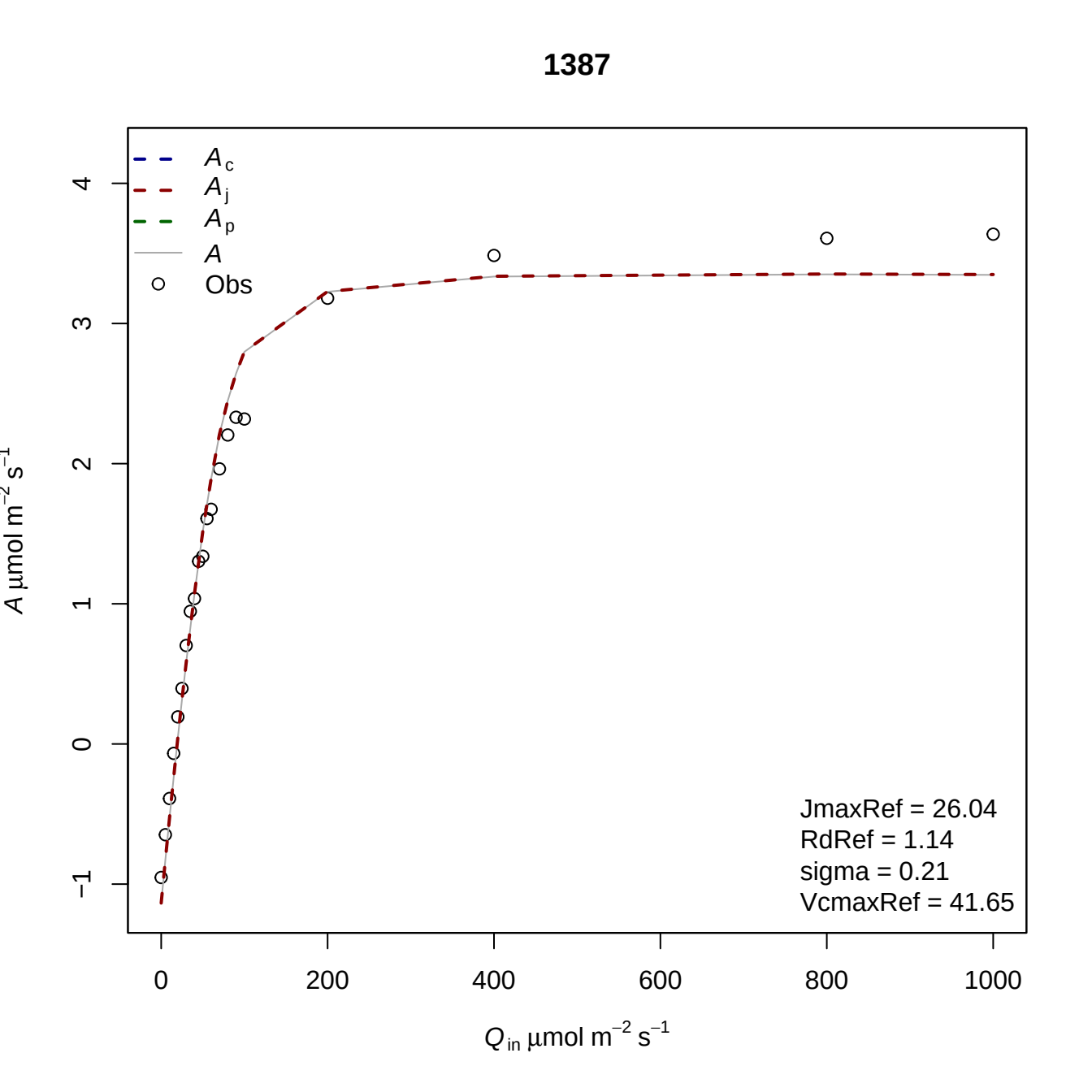

1382
A_c
A_j
A_p
A
Obs
JmaxRef = 35.56
RdRef = 0.55
sigma = 0.33
VcmaxRef = 99.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

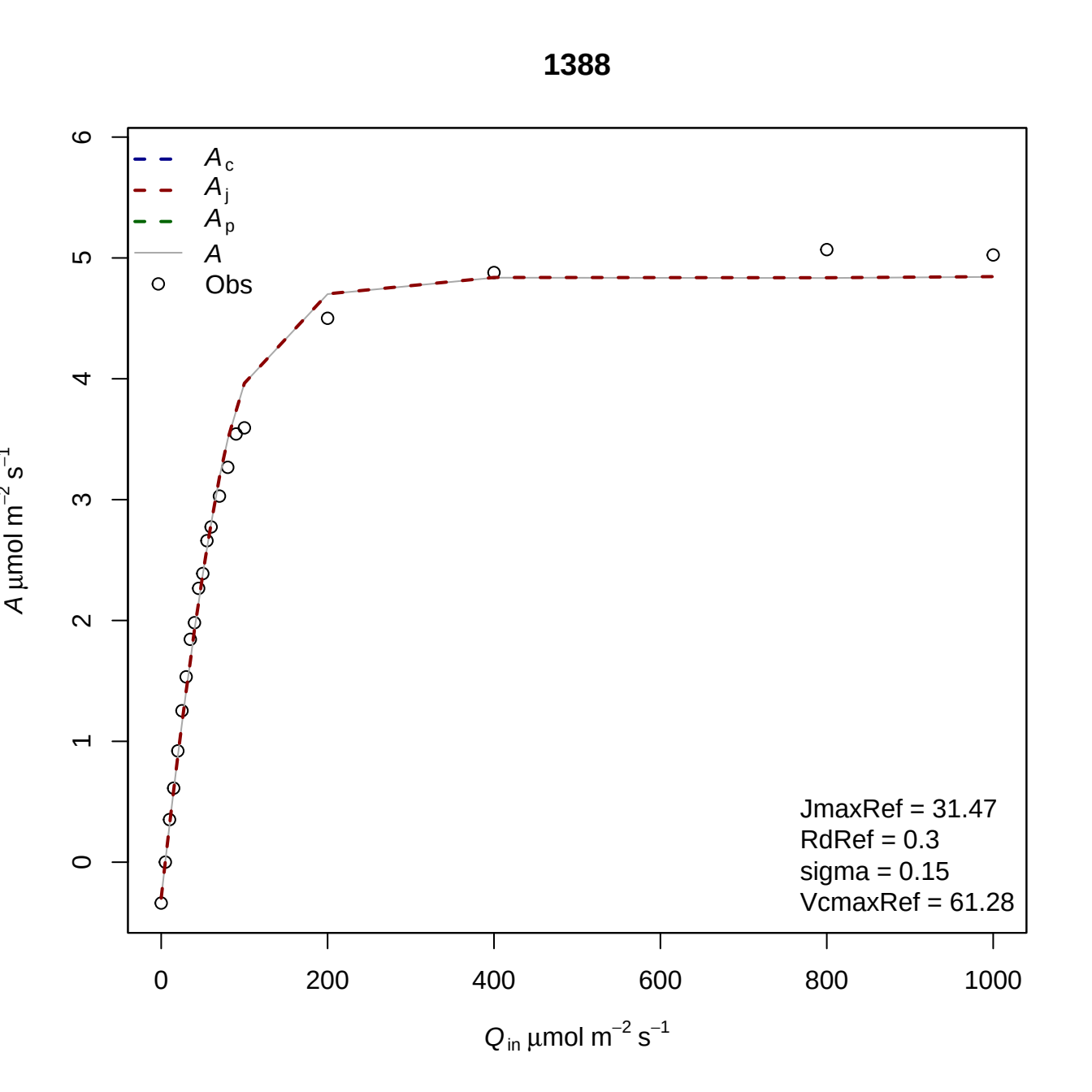

1383
A_c
A_j
A_p
A
Obs
JmaxRef = 31.5
RdRef = 1.11
sigma = 0.43
VcmaxRef = 91.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

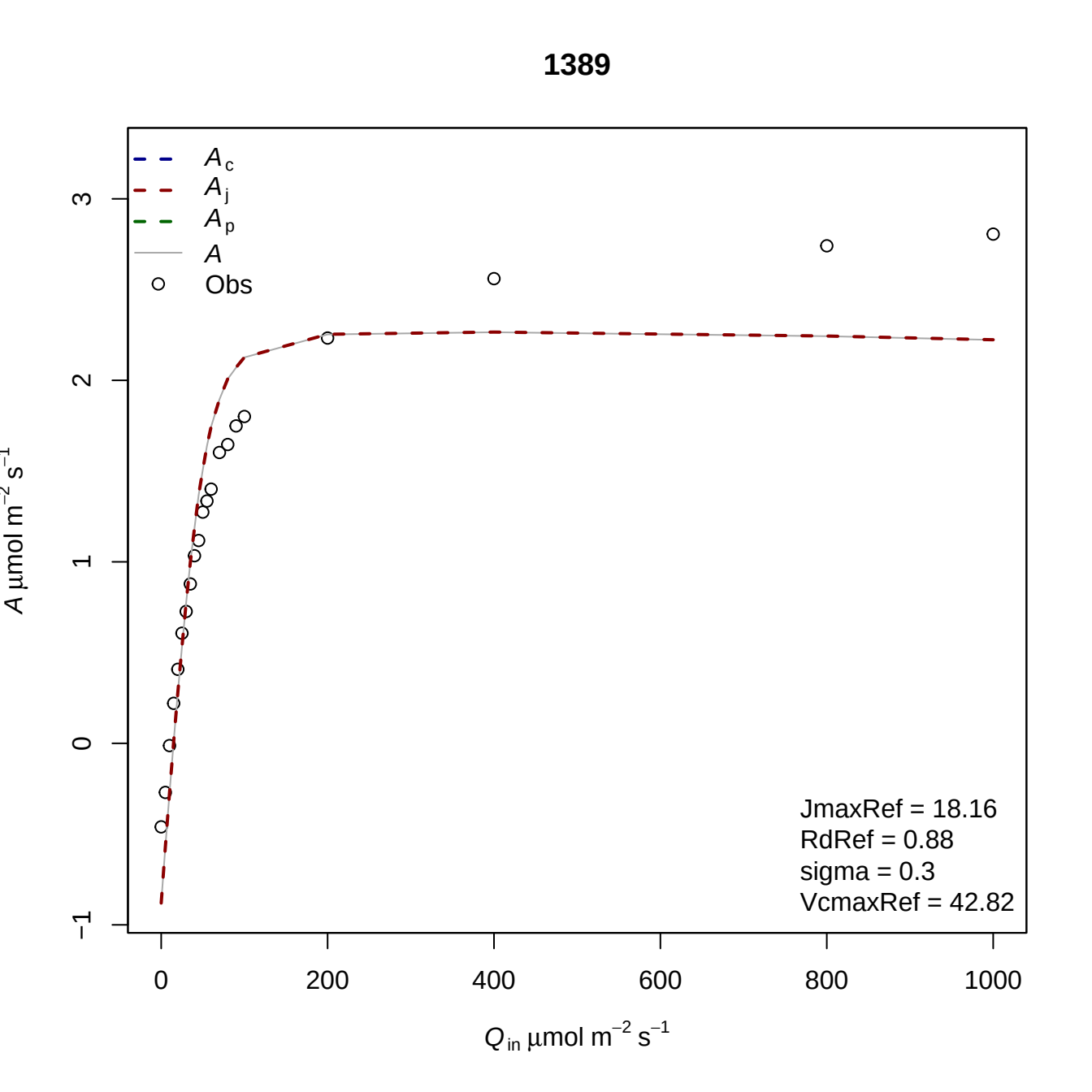

1384
A_c
A_j
A_p
A
Obs
JmaxRef = 25.91
RdRef = 0.52
sigma = 0.3
VcmaxRef = 62.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

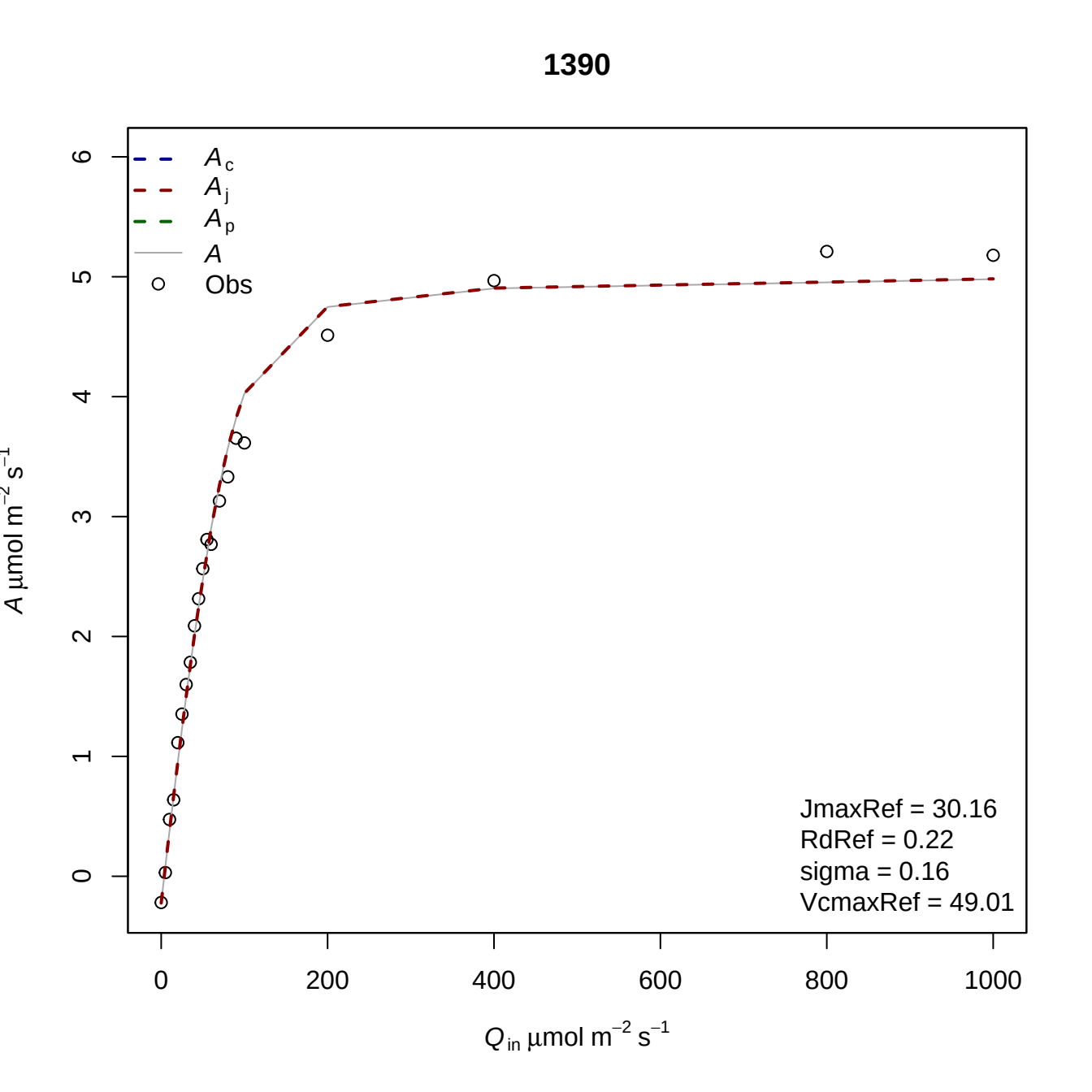

1385
A_c
A_j
A_p
A
Obs
JmaxRef = 28.55
RdRef = 0.19
sigma = 0.09
VcmaxRef = 59.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

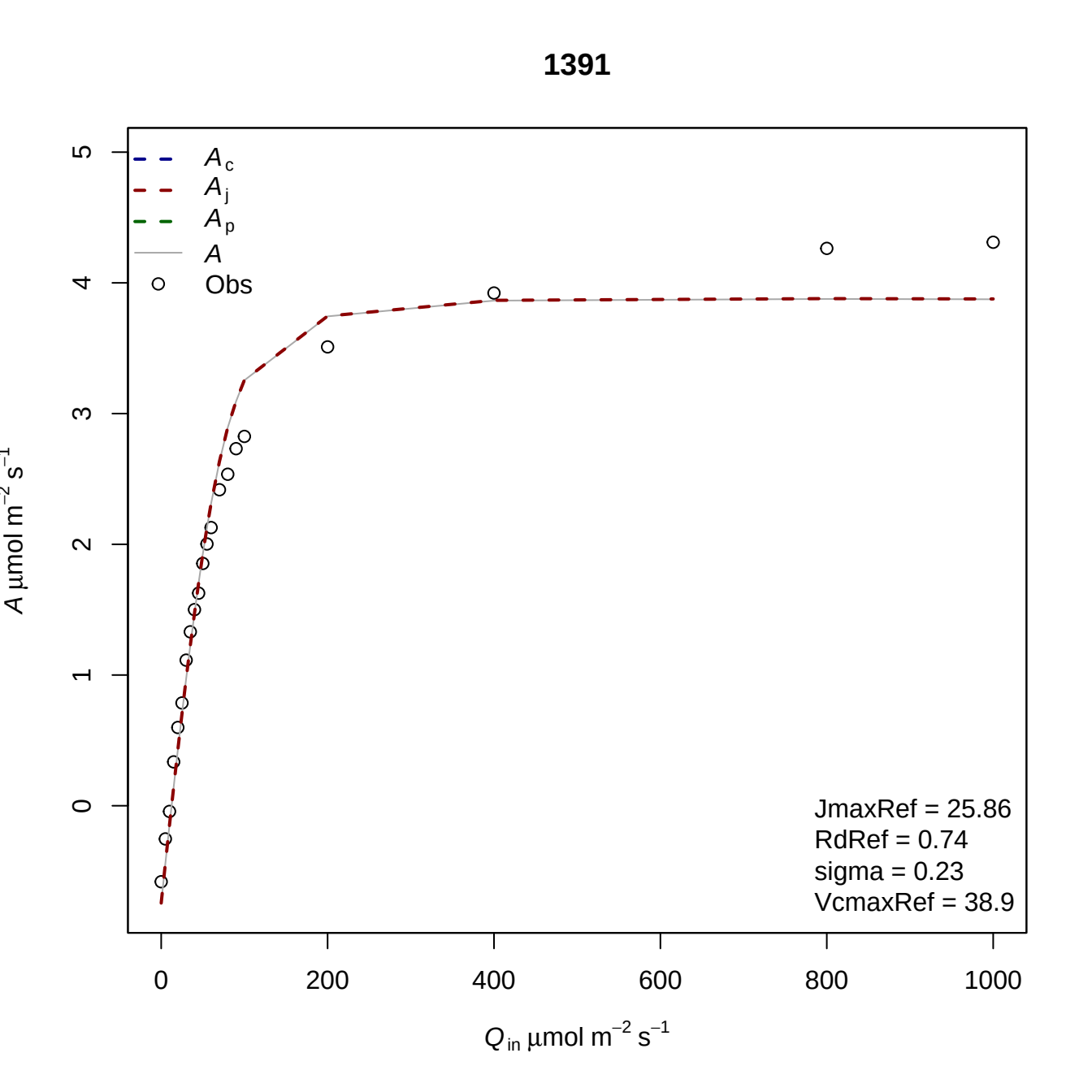

1386
A_c
A_j
A_p
A
Obs
JmaxRef = 20.01
RdRef = 1.19
sigma = 0.3
VcmaxRef = 53.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

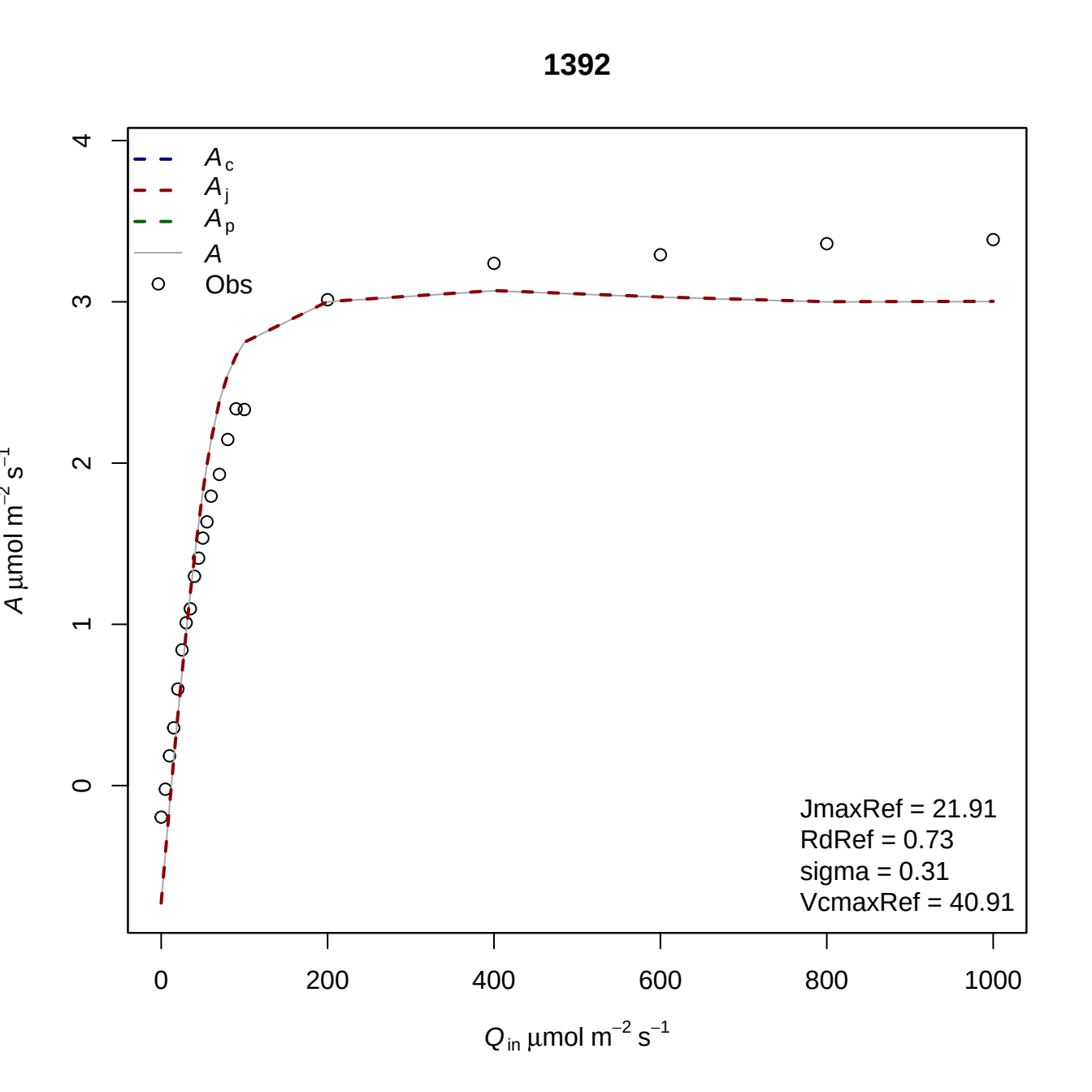

1387
A_c
A_j
A_p
A
Obs
JmaxRef = 26.04
RdRef = 1.14
sigma = 0.21
VcmaxRef = 41.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

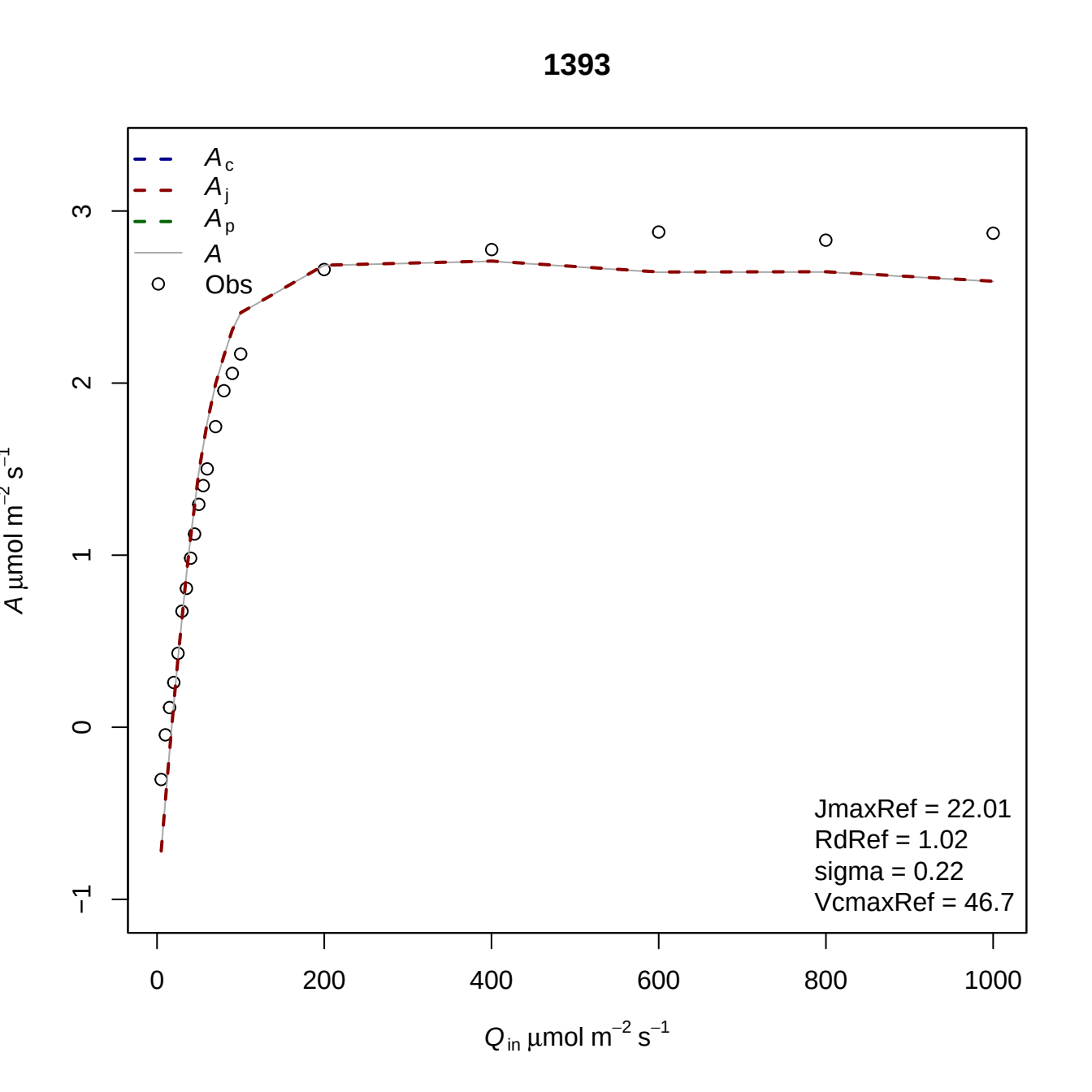

1388
A_c
A_j
A_p
A
Obs
JmaxRef = 31.47
RdRef = 0.3
sigma = 0.15
VcmaxRef = 61.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

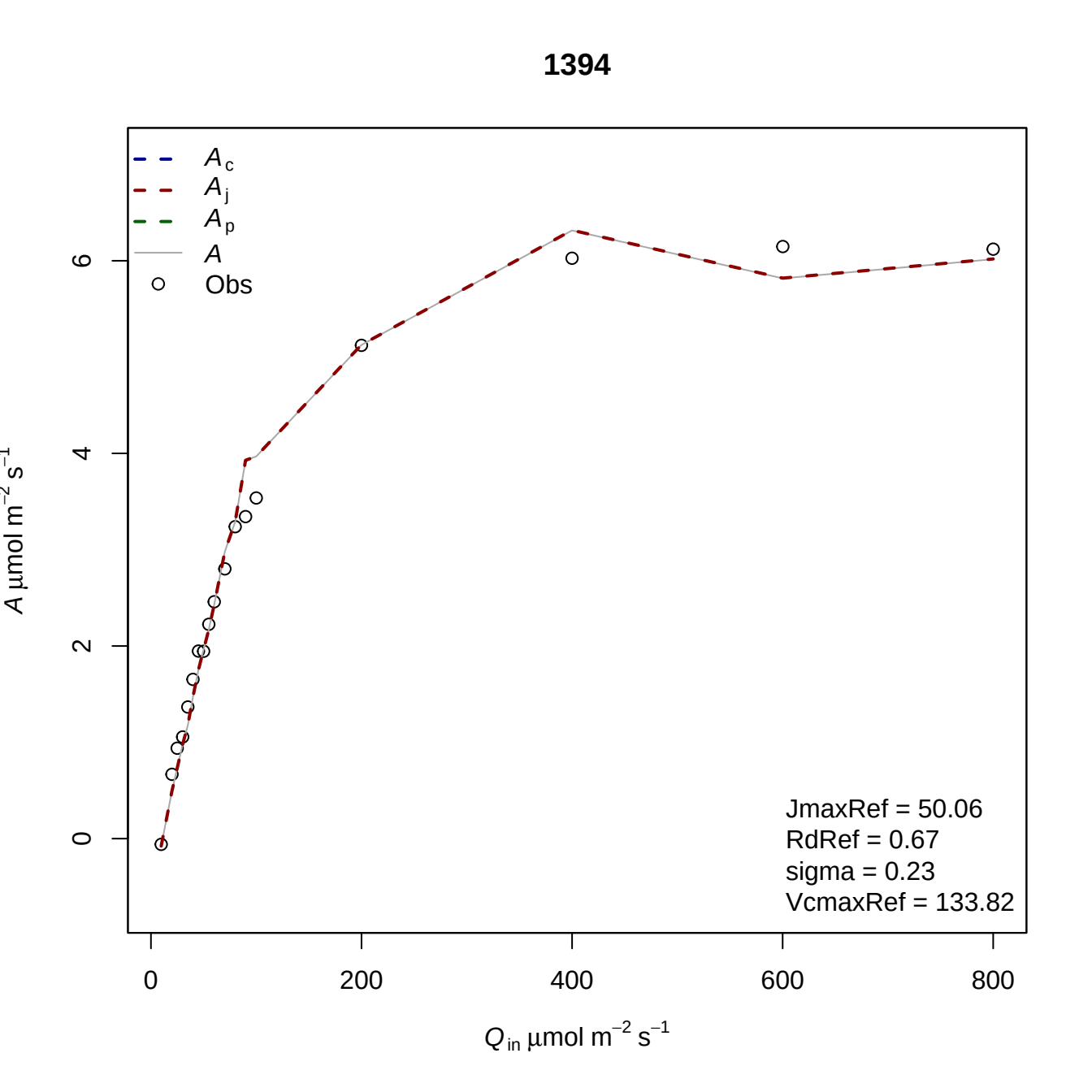

1389
A_c
A_j
A_p
A
Obs
JmaxRef = 18.16
RdRef = 0.88
sigma = 0.3
VcmaxRef = 42.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

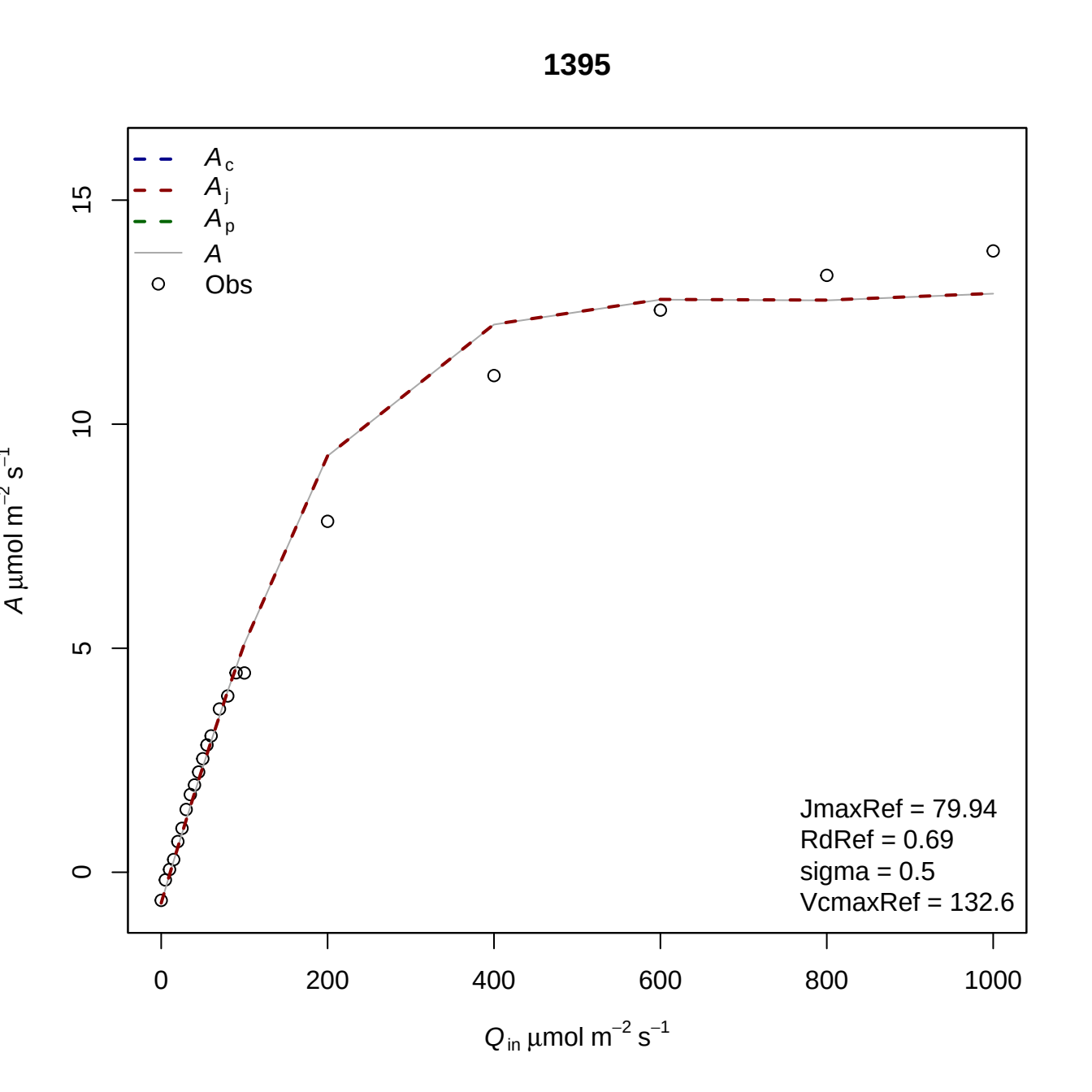

1390
A_c
A_j
A_p
A
Obs
JmaxRef = 30.16
RdRef = 0.22
sigma = 0.16
VcmaxRef = 49.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

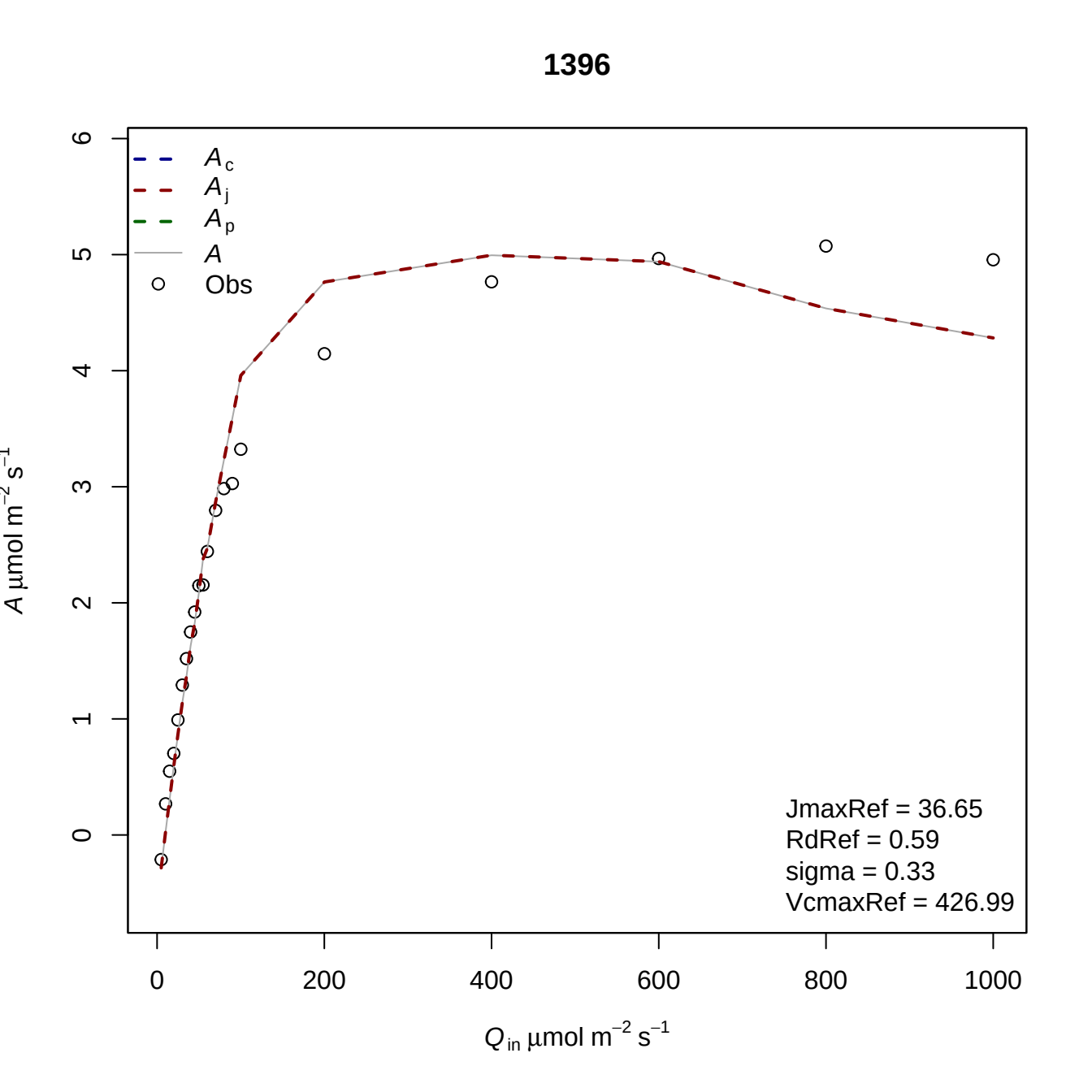

1391
A_c
A_j
A_p
A
Obs
JmaxRef = 25.86
RdRef = 0.74
sigma = 0.23
VcmaxRef = 38.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

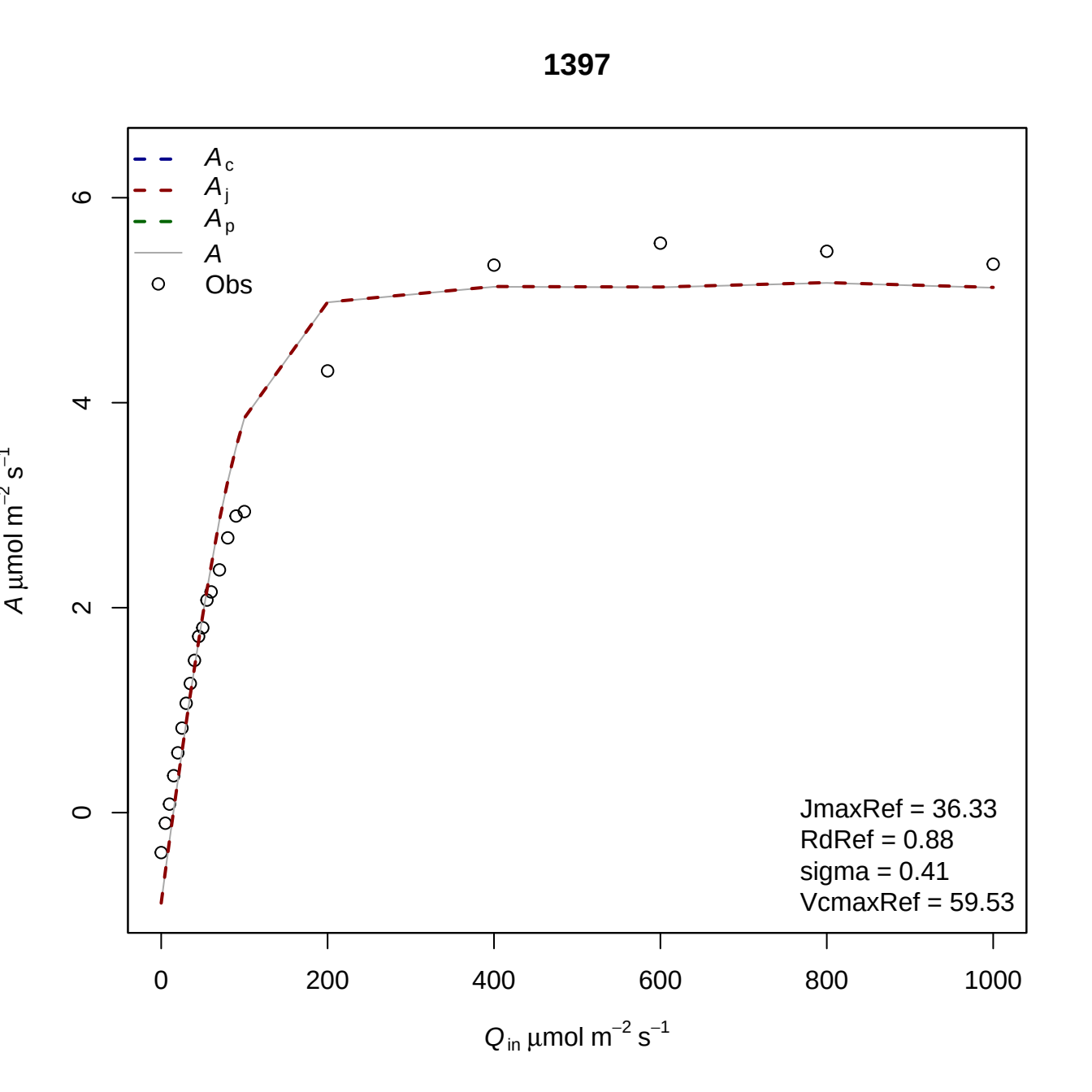

1392
A_c
A_j
A_p
A
Obs
JmaxRef = 21.91
RdRef = 0.73
sigma = 0.31
VcmaxRef = 40.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
200
400
600
800
1000

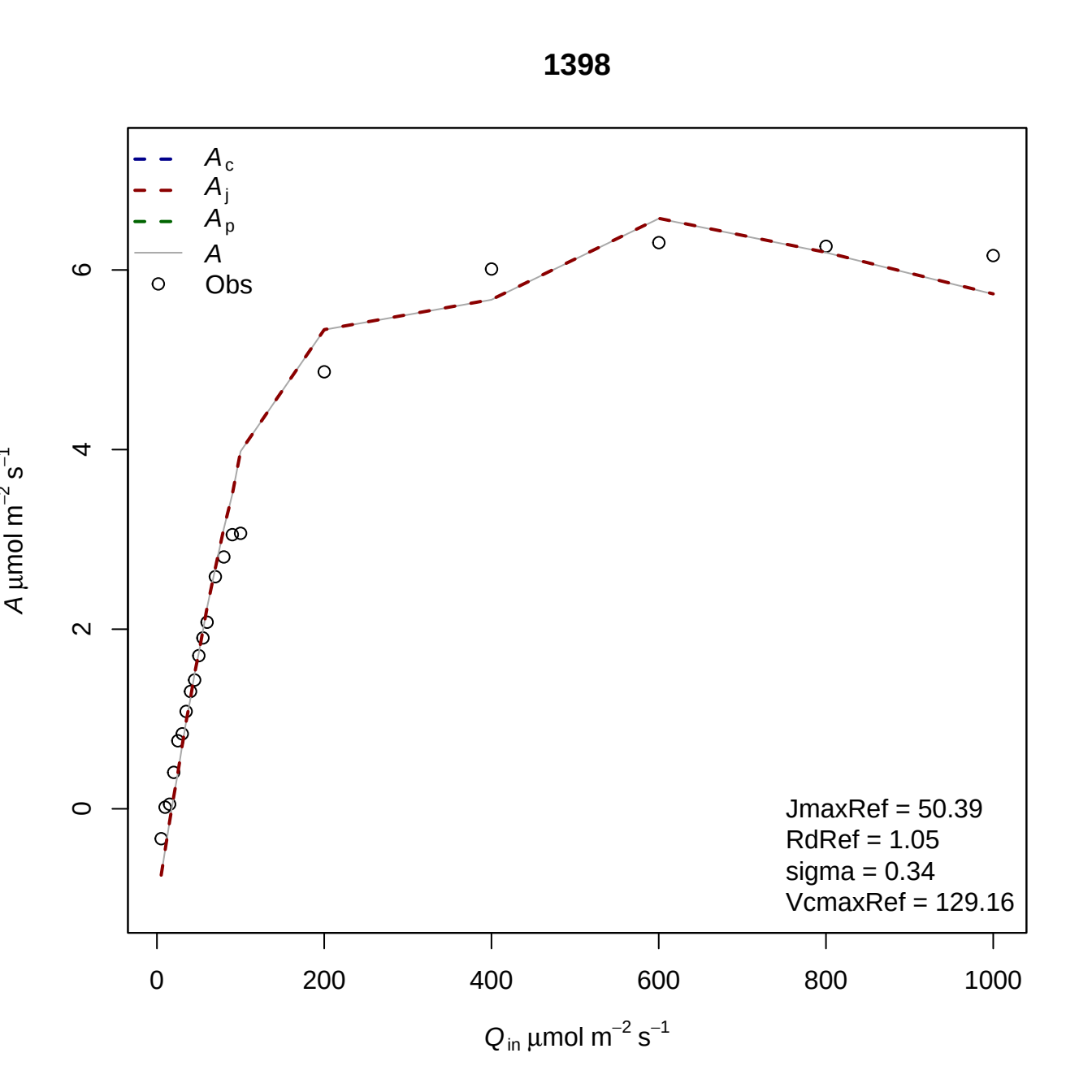

1393
A_c
A_j
A_p
A
Obs
JmaxRef = 22.01
RdRef = 1.02
sigma = 0.22
VcmaxRef = 46.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

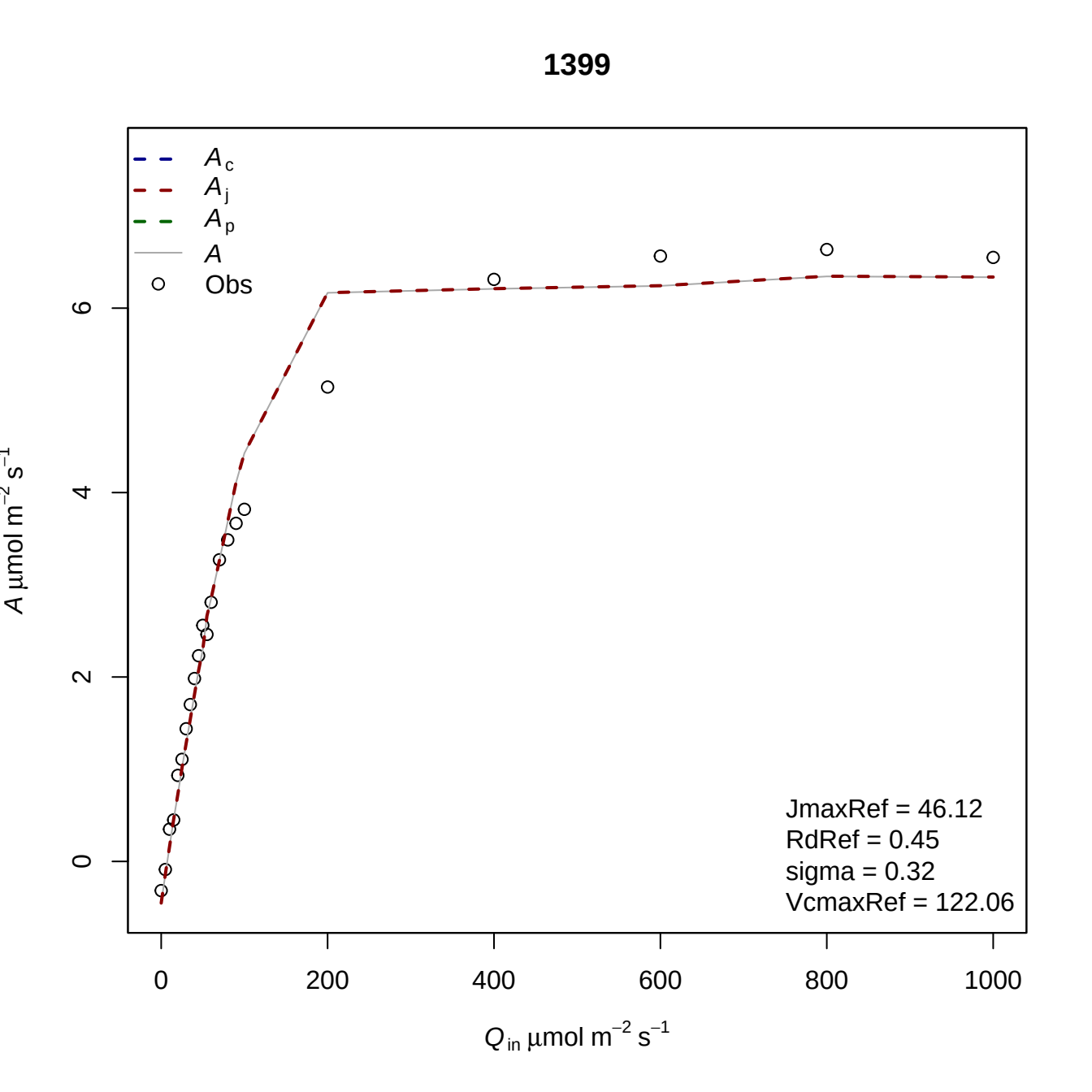

1394
A_c
A_j
A_p
A
Obs
JmaxRef = 50.06
RdRef = 0.67
sigma = 0.23
VcmaxRef = 133.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800

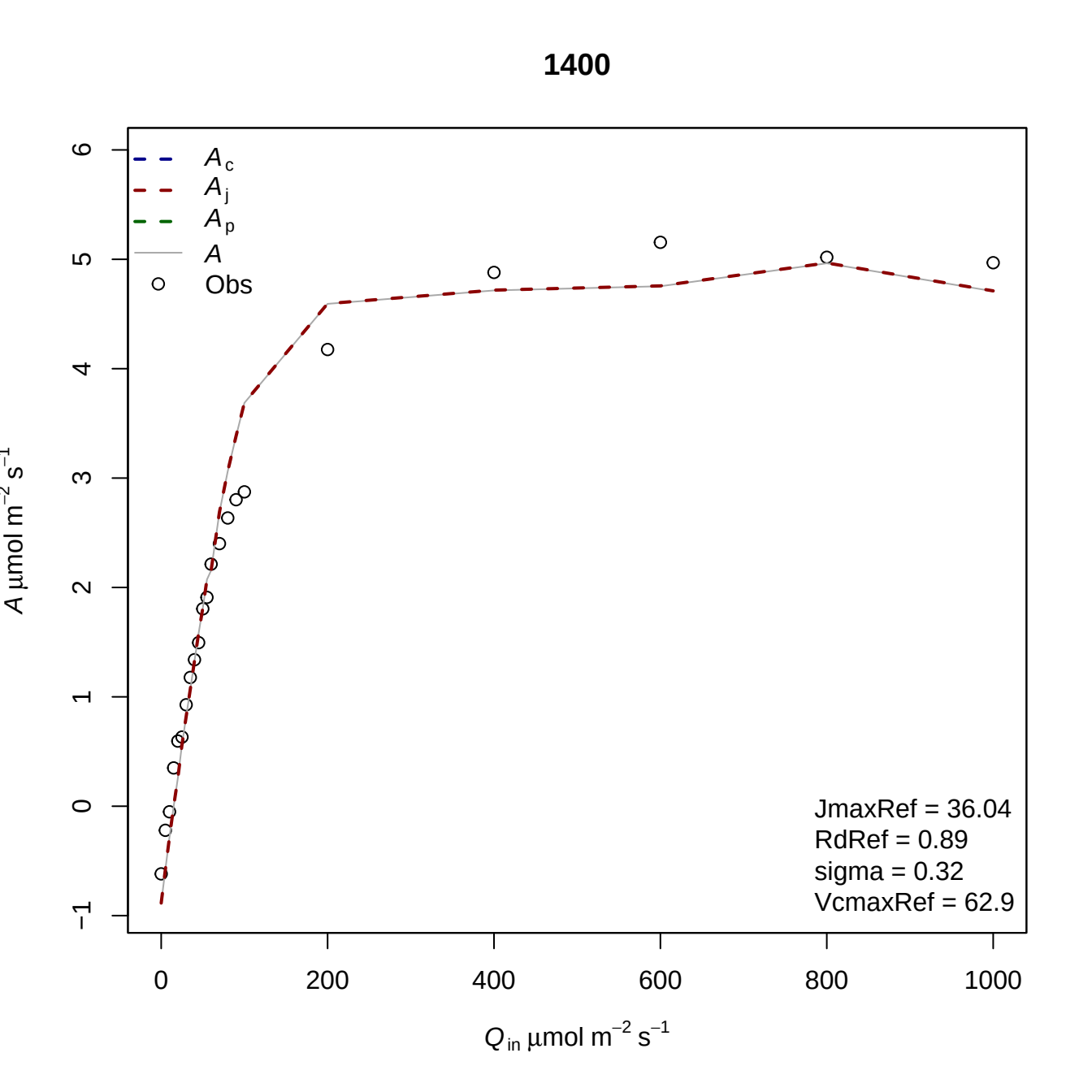

1395
A_c
A_j
A_p
A
Obs
JmaxRef = 79.94
RdRef = 0.69
sigma = 0.5
VcmaxRef = 132.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

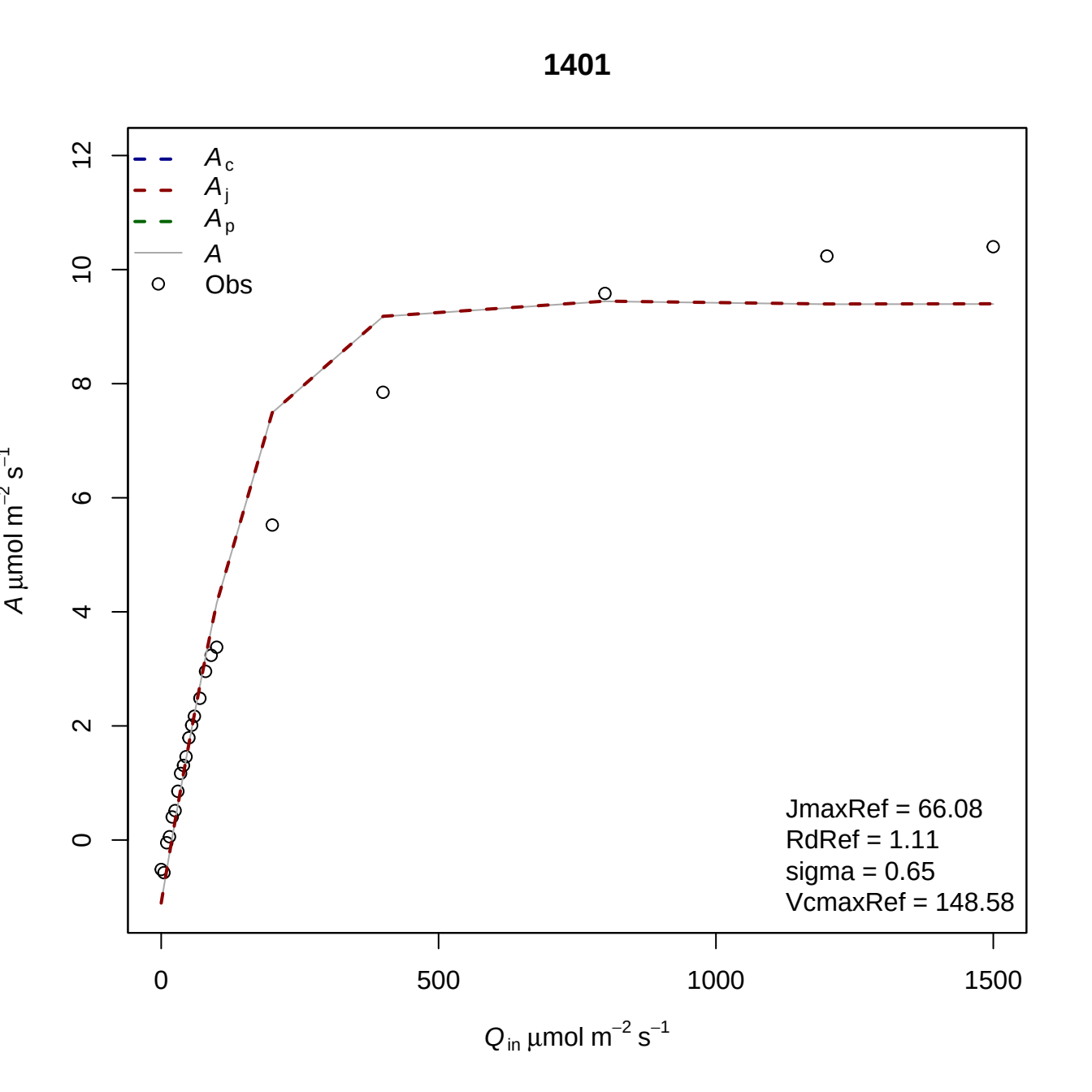

1396
A_c
A_j
A_p
A
Obs
JmaxRef = 36.65
RdRef = 0.59
sigma = 0.33
VcmaxRef = 426.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

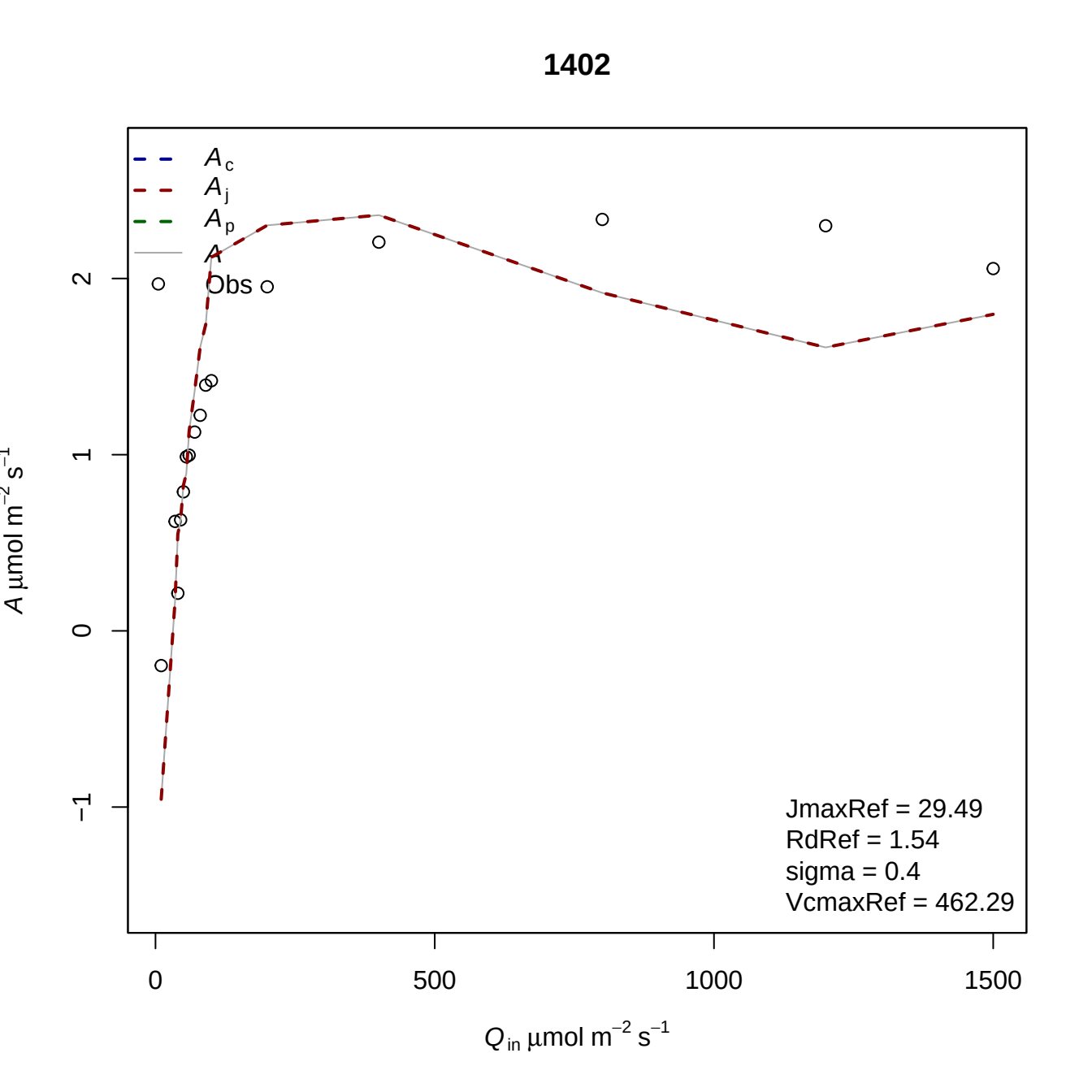

1397
A_c
A_j
A_p
A
Obs
JmaxRef = 36.33
RdRef = 0.88
sigma = 0.41
VcmaxRef = 59.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

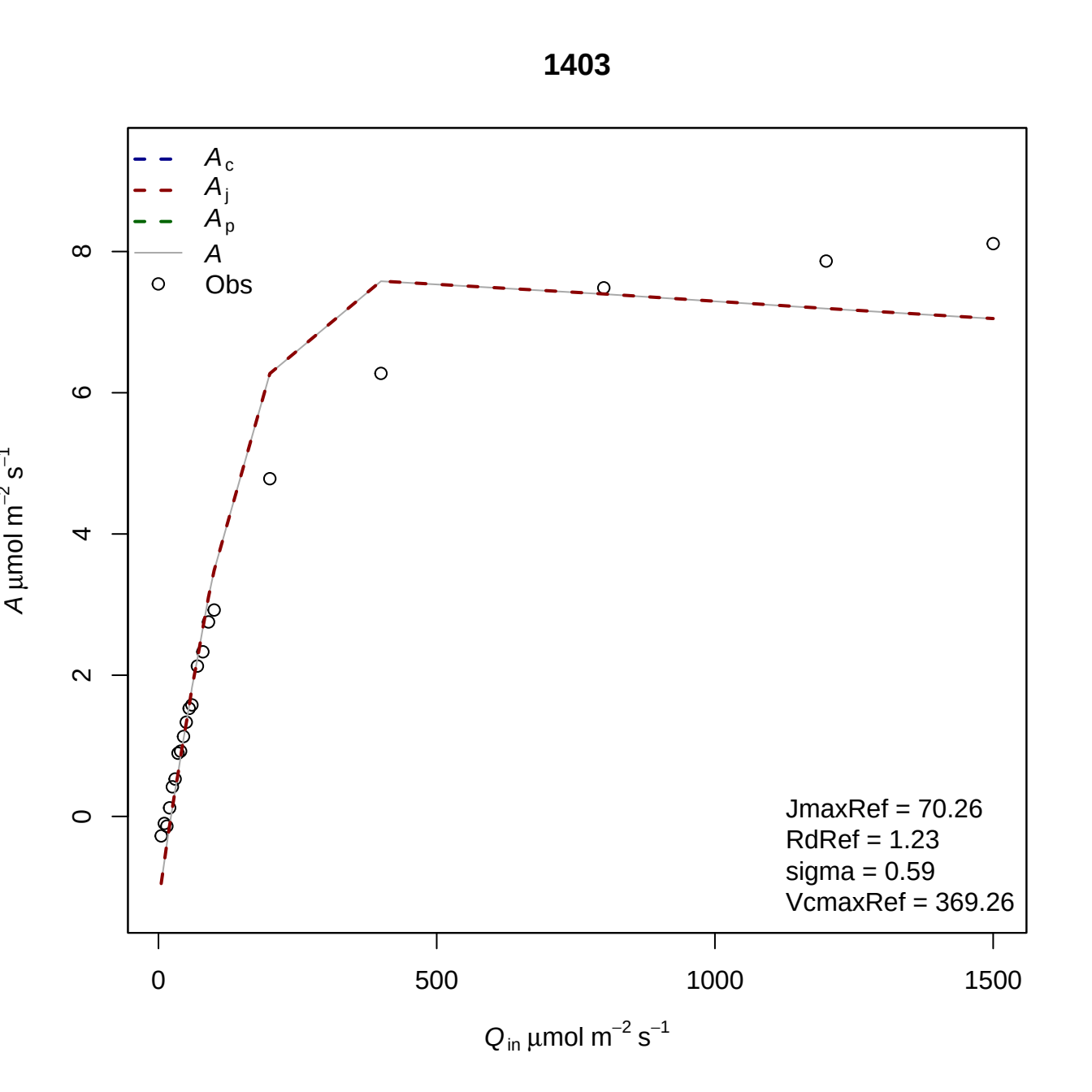

1398
A_c
A_j
A_p
A
Obs
JmaxRef = 50.39
RdRef = 1.05
sigma = 0.34
VcmaxRef = 129.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

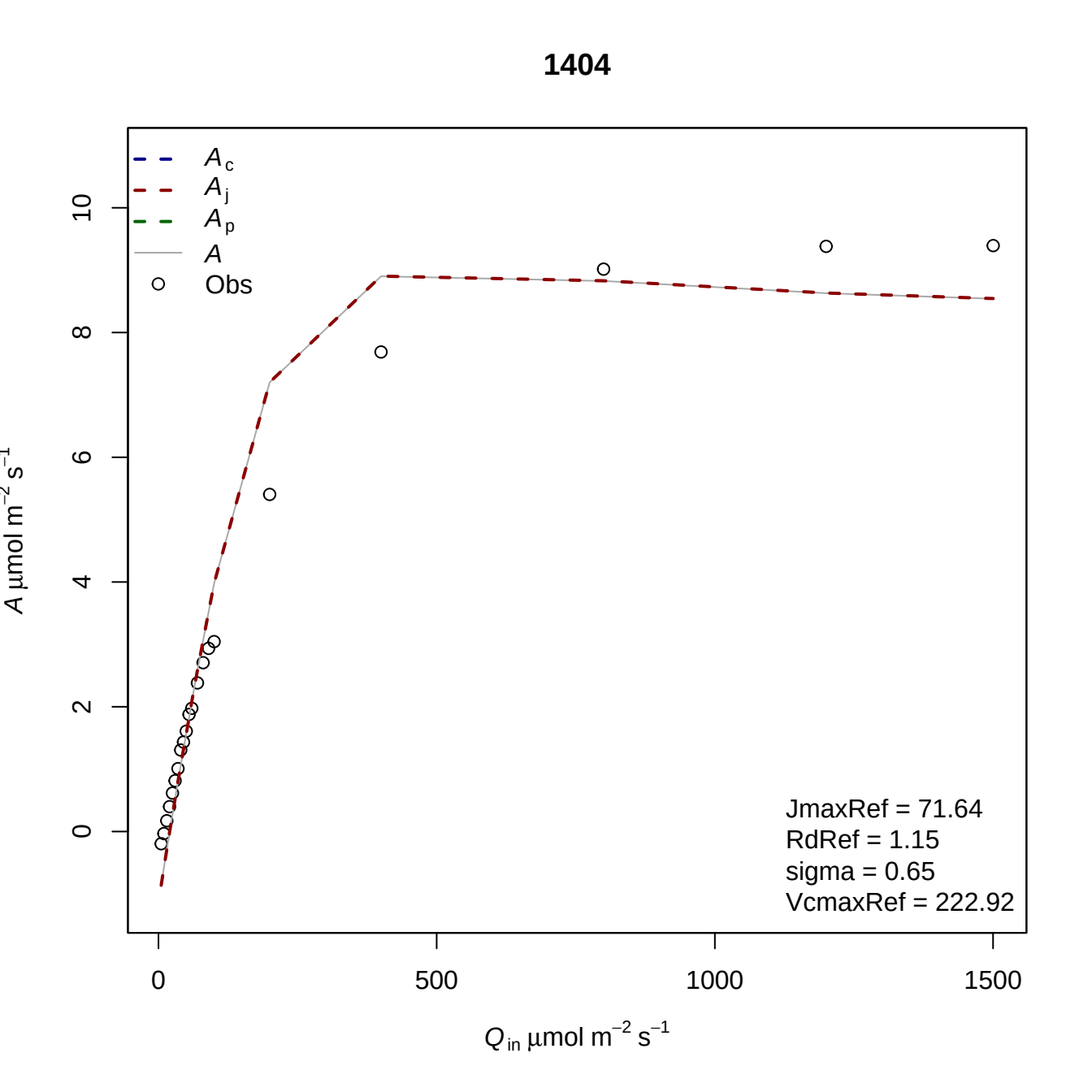

1399
A_c
A_j
A_p
A
Obs
JmaxRef = 46.12
RdRef = 0.45
sigma = 0.32
VcmaxRef = 122.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

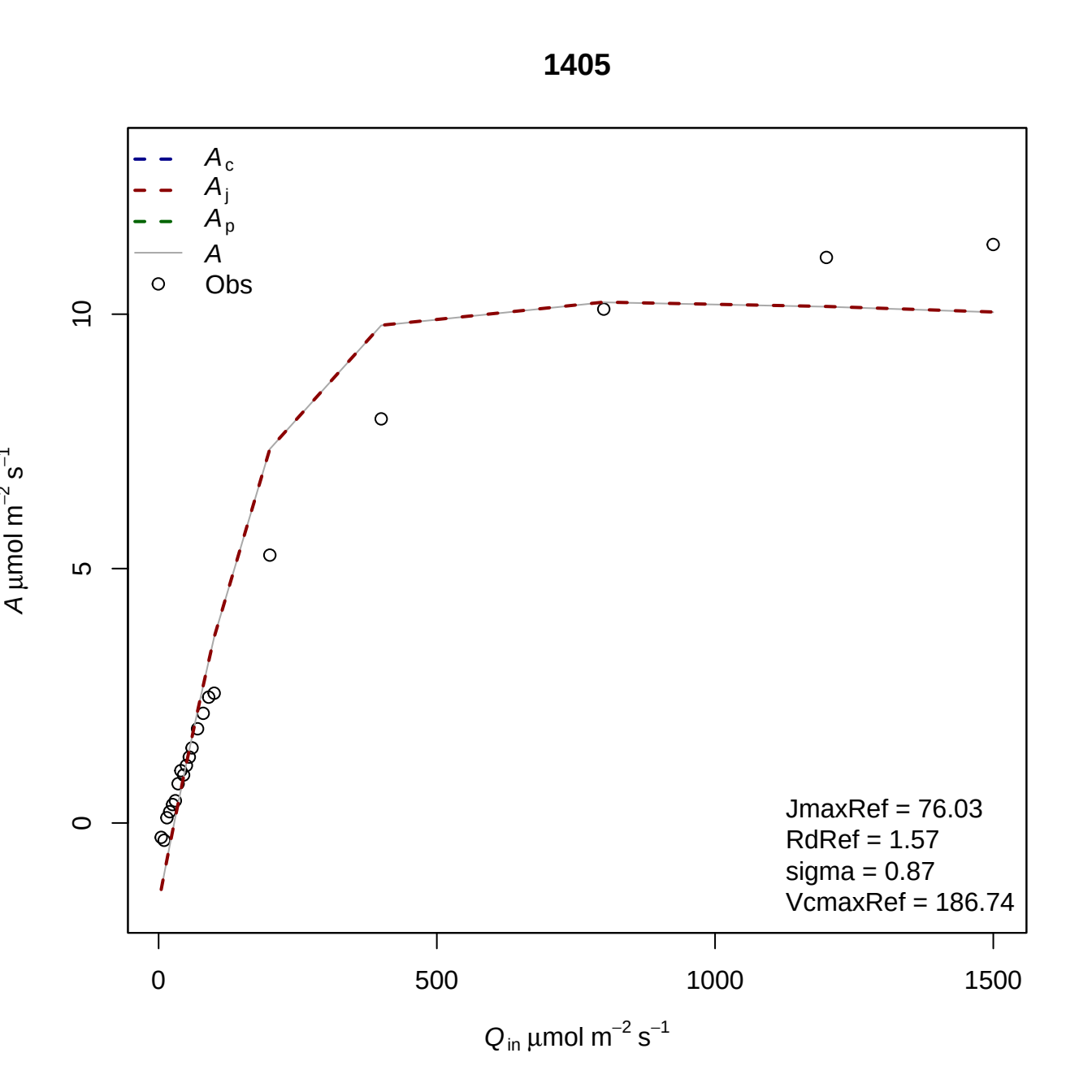

1400
A_c
A_j
A_p
A
Obs
JmaxRef = 36.04
RdRef = 0.89
sigma = 0.32
VcmaxRef = 62.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
200
400
600
800
1000

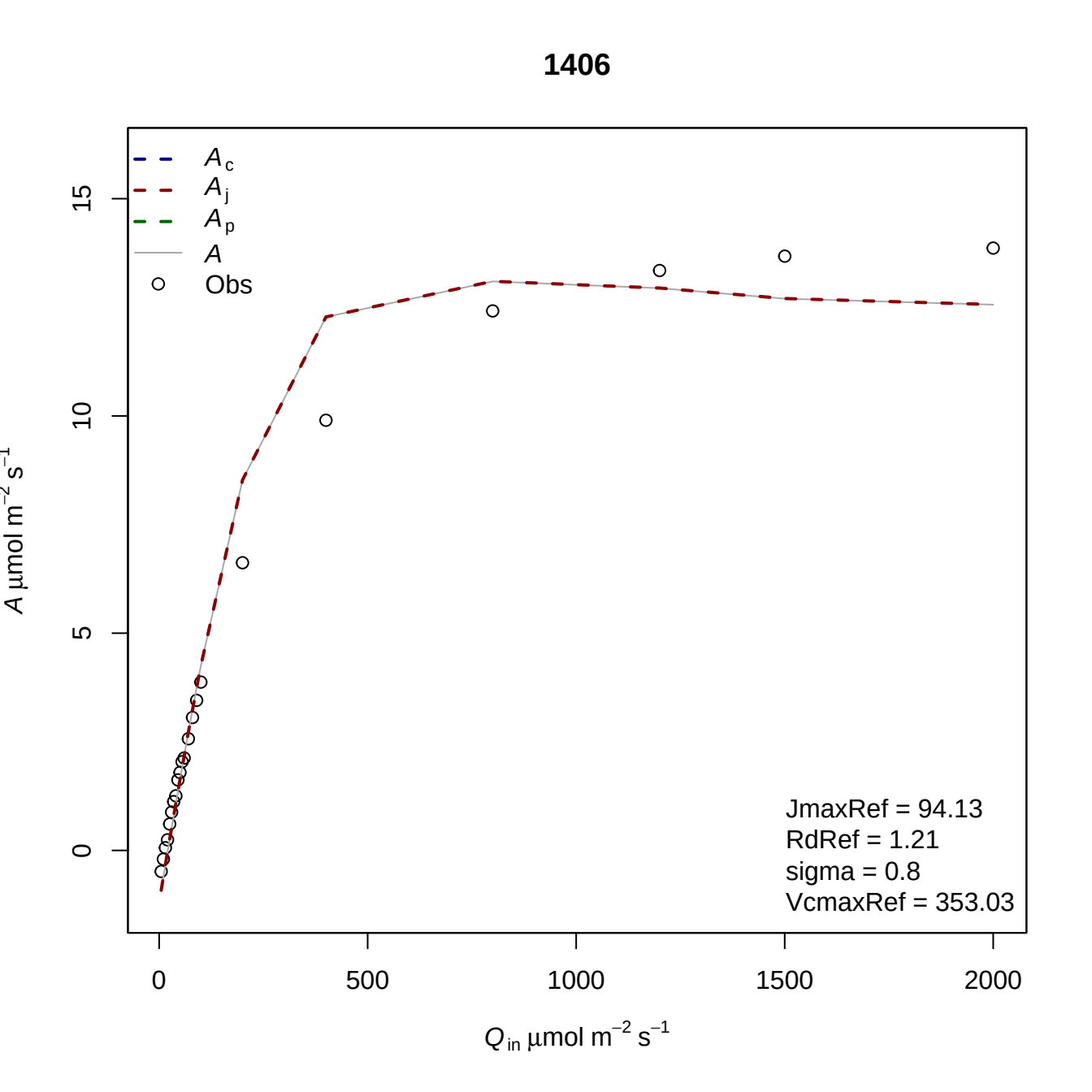

1401
A_c
A_j
A_p
A
Obs
JmaxRef = 66.08
RdRef = 1.11
sigma = 0.65
VcmaxRef = 148.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

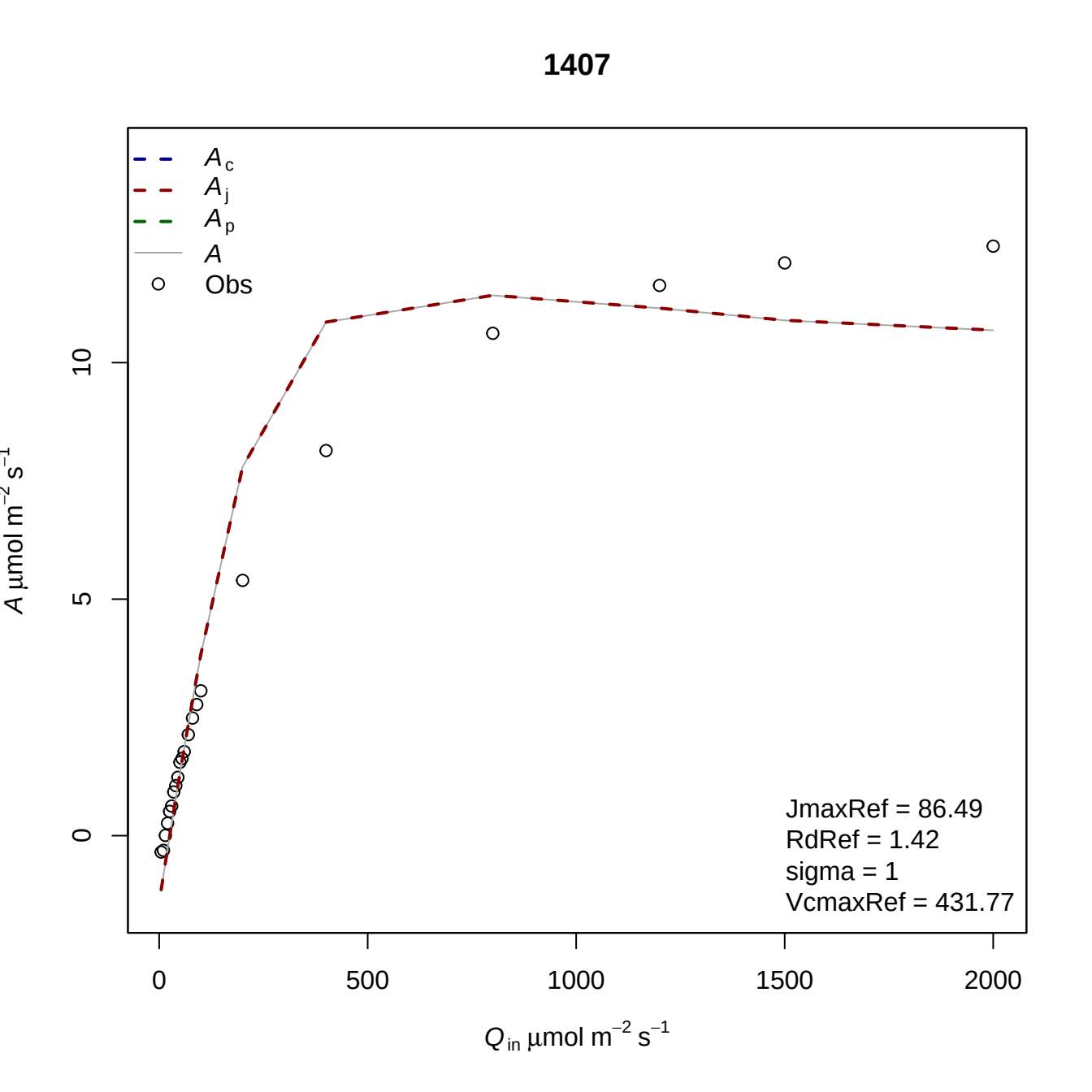

1402
A_c
A_j
A_p
A
Obs
JmaxRef = 29.49
RdRef = 1.54
sigma = 0.4
VcmaxRef = 462.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

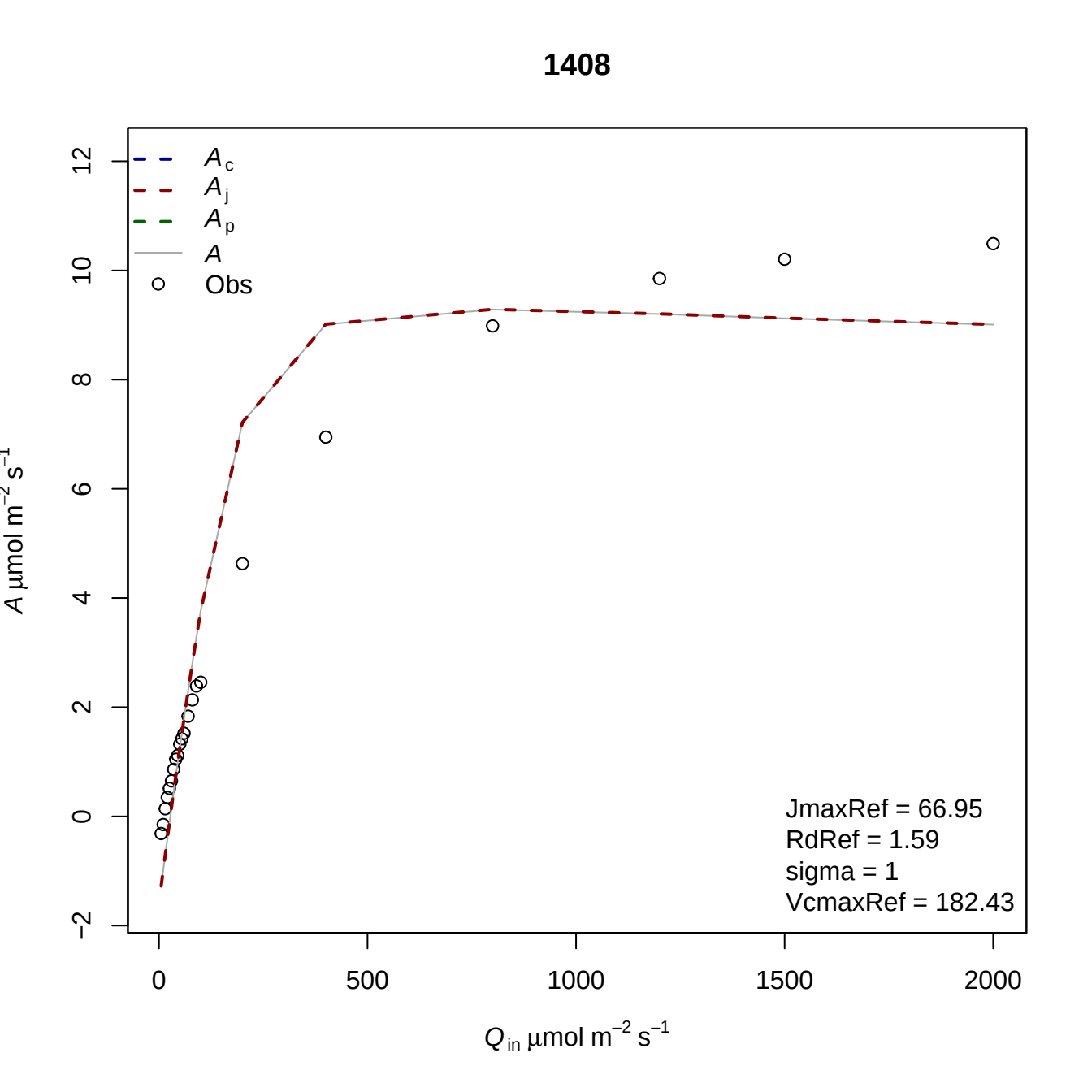

1403
A_c
A_j
A_p
A
Obs
JmaxRef = 70.26
RdRef = 1.23
sigma = 0.59
VcmaxRef = 369.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

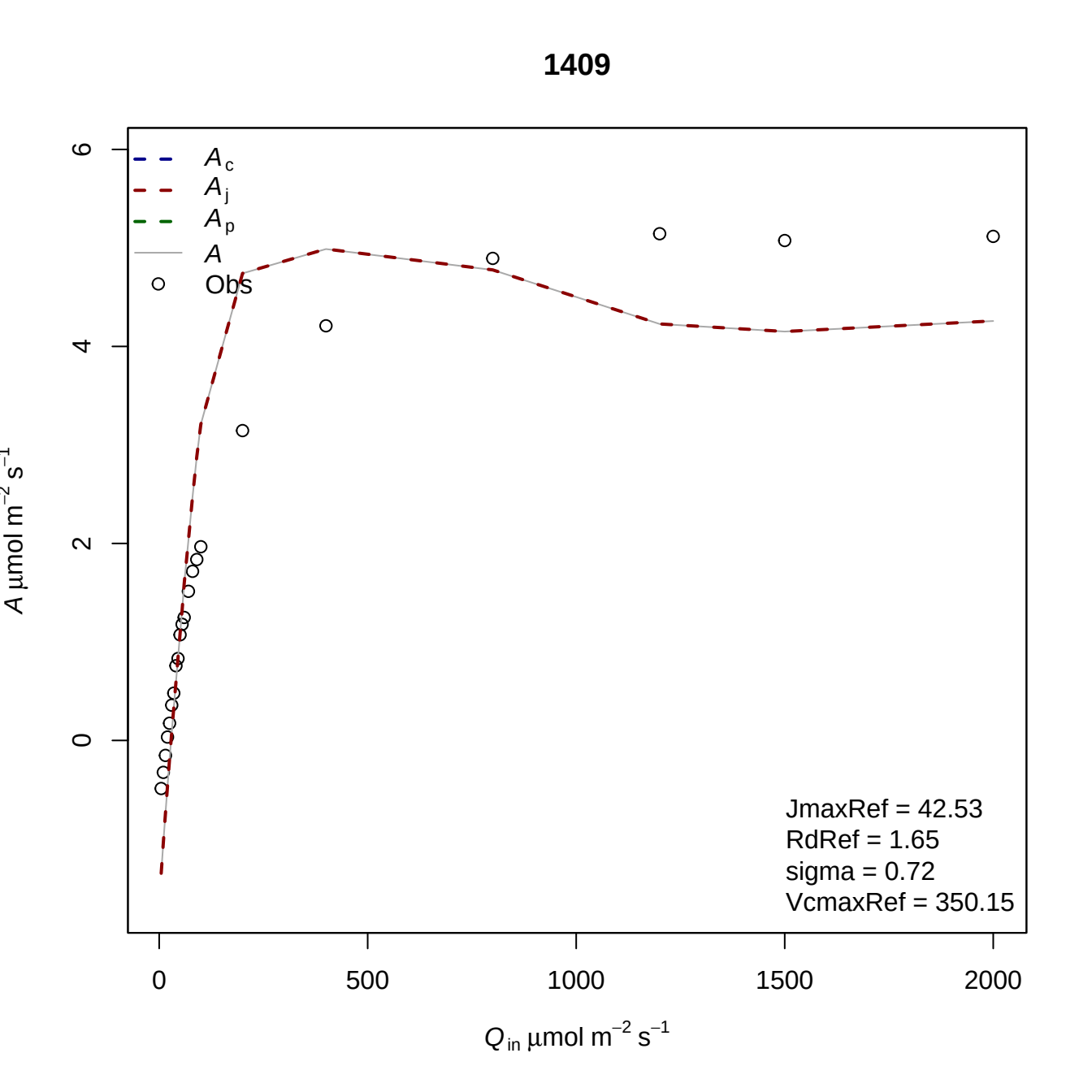

1404
A_c
A_j
A_p
A
Obs
JmaxRef = 71.64
RdRef = 1.15
sigma = 0.65
VcmaxRef = 222.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

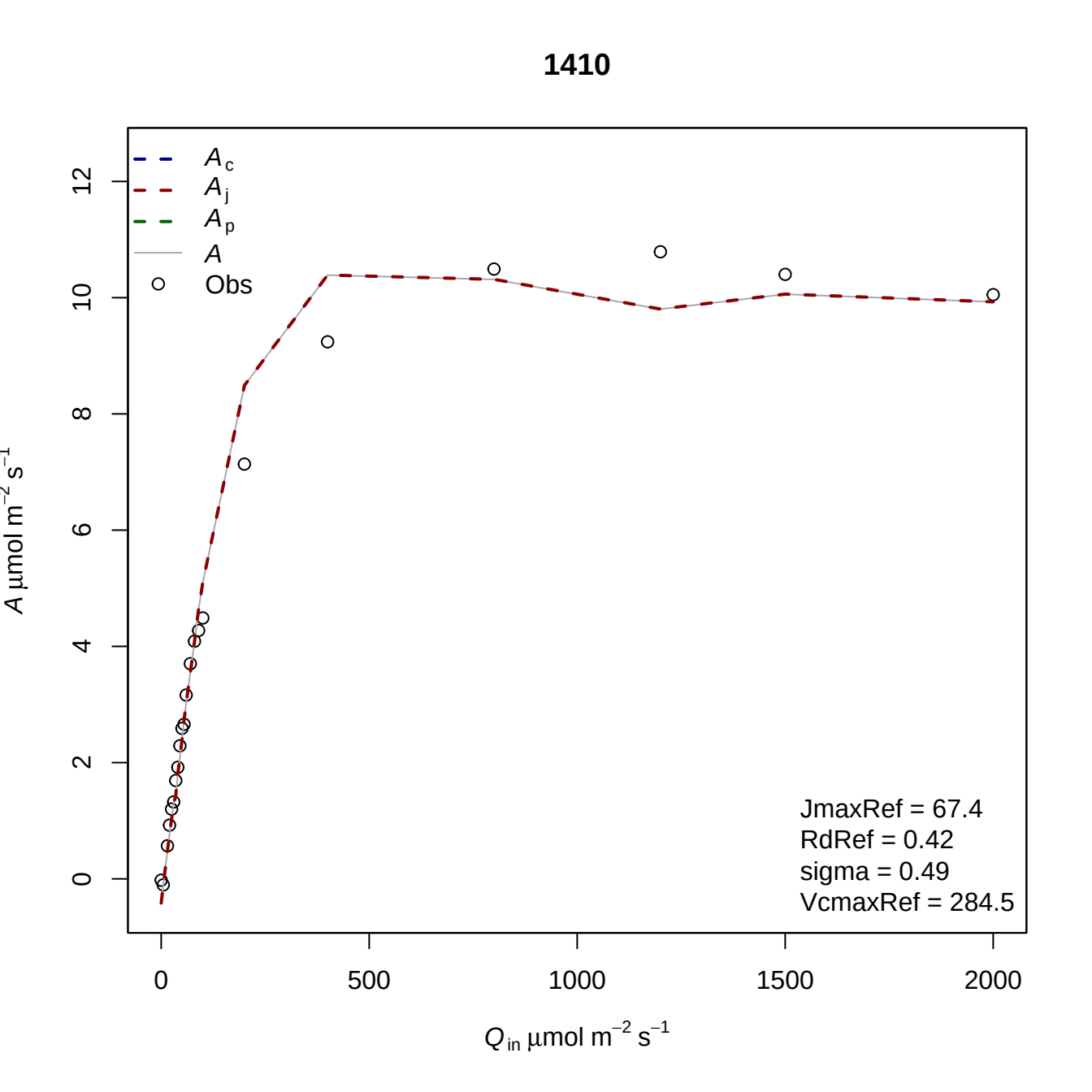

1405
A_c
A_j
A_p
A
Obs
JmaxRef = 76.03
RdRef = 1.57
sigma = 0.87
VcmaxRef = 186.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

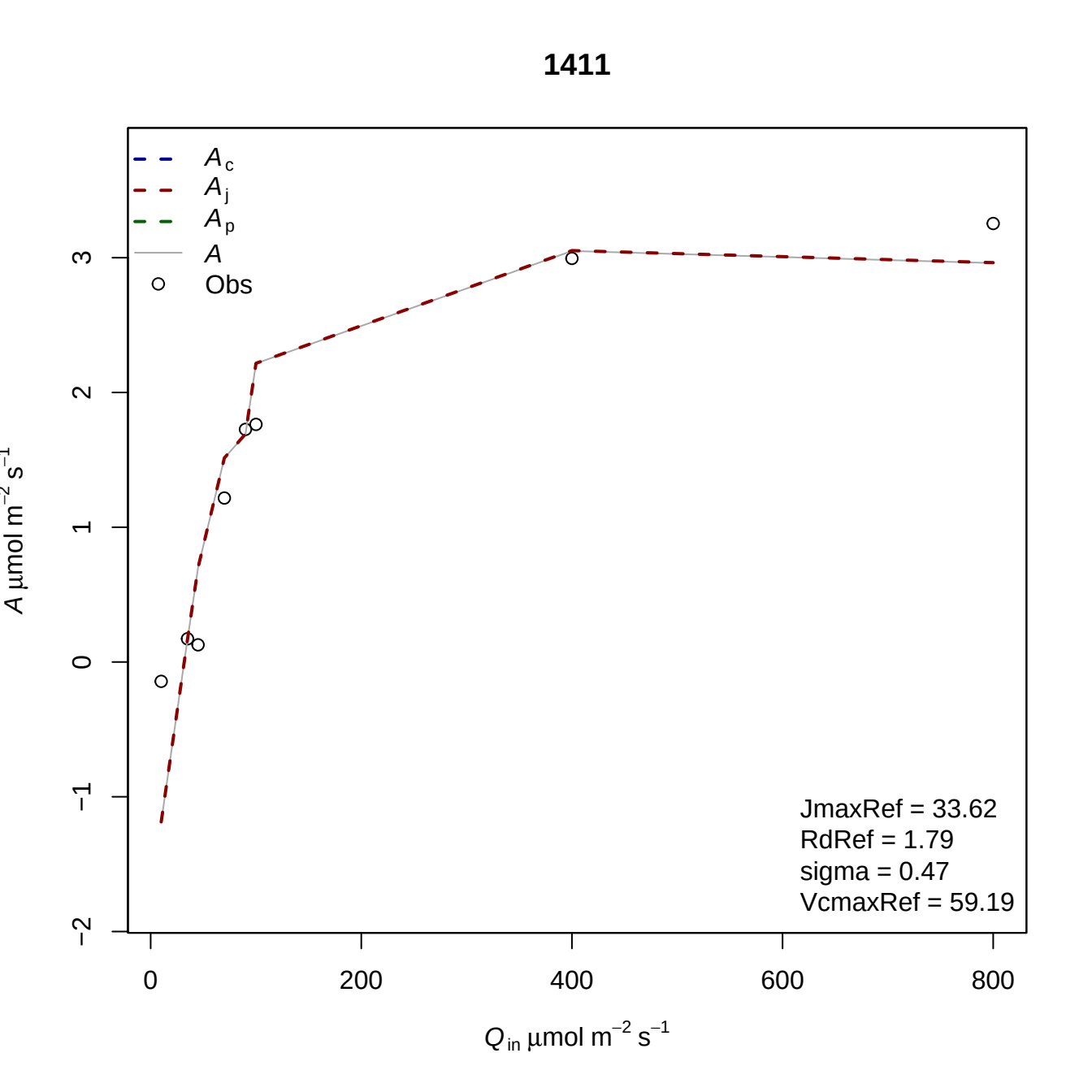

1406
A_c
A_j
A_p
A
Obs
JmaxRef = 94.13
RdRef = 1.21
sigma = 0.8
VcmaxRef = 353.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

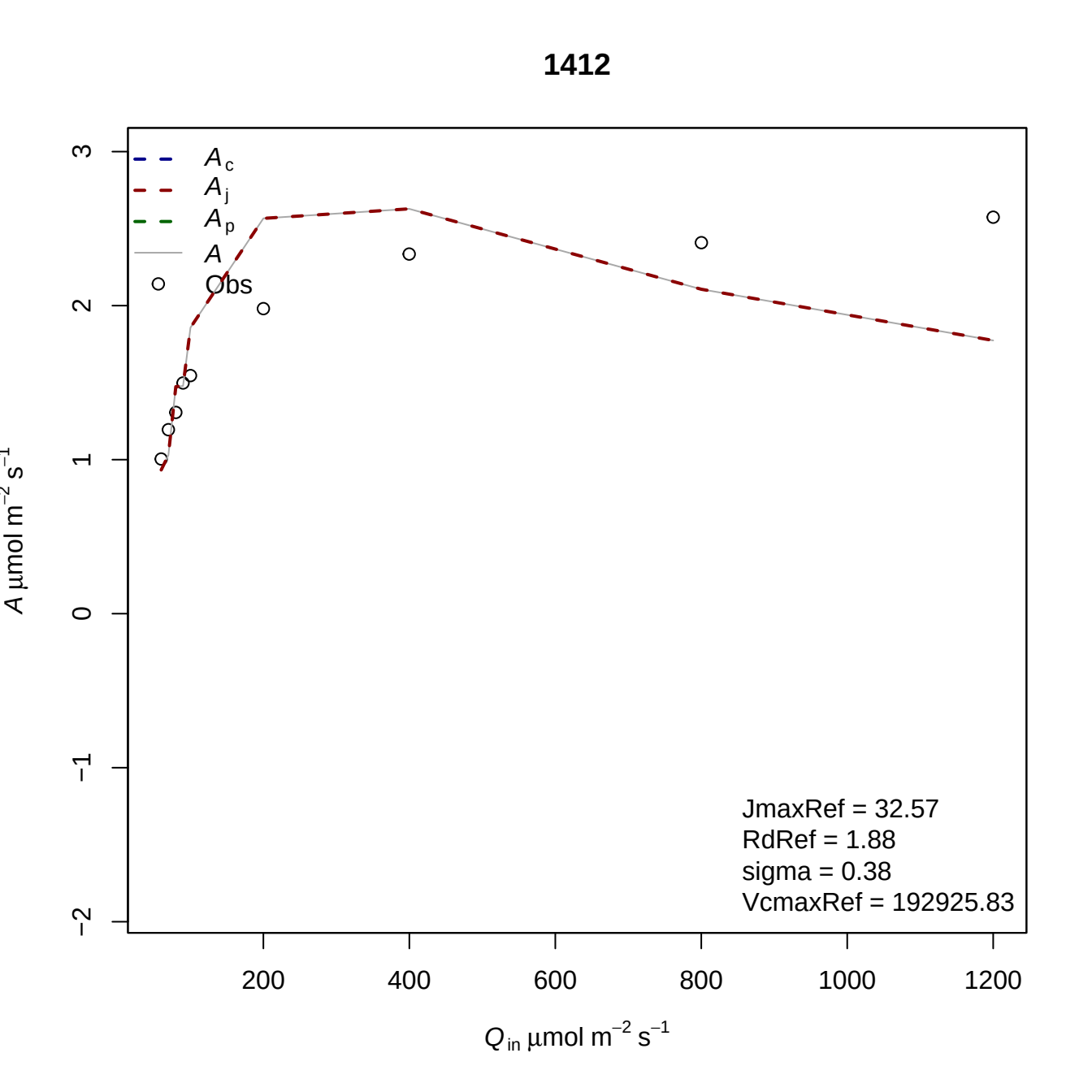

1407
A_c
A_j
A_p
A
Obs
JmaxRef = 86.49
RdRef = 1.42
sigma = 1
VcmaxRef = 431.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

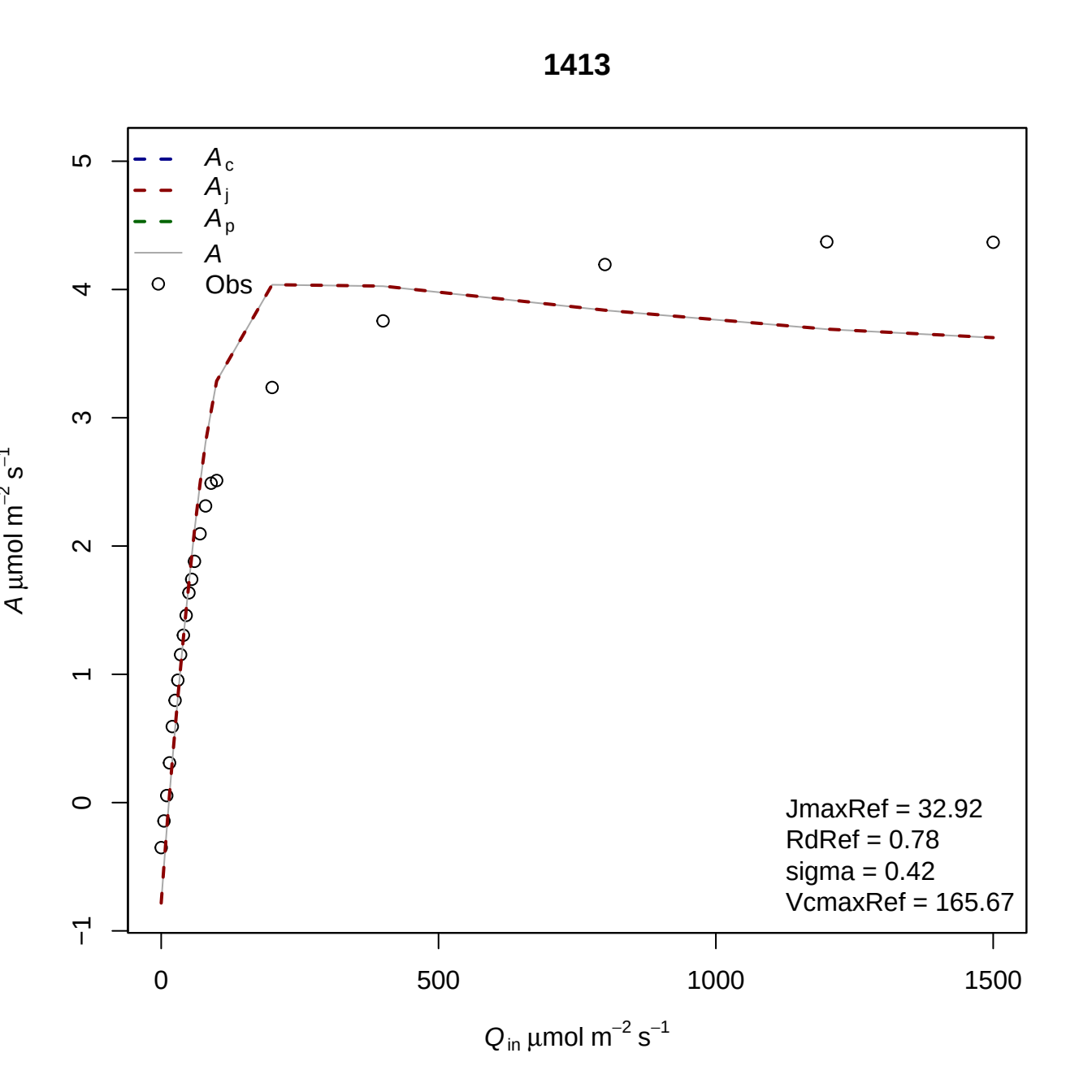

1408
A_c
A_j
A_p
A
Obs
JmaxRef = 66.95
RdRef = 1.59
sigma = 1
VcmaxRef = 182.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

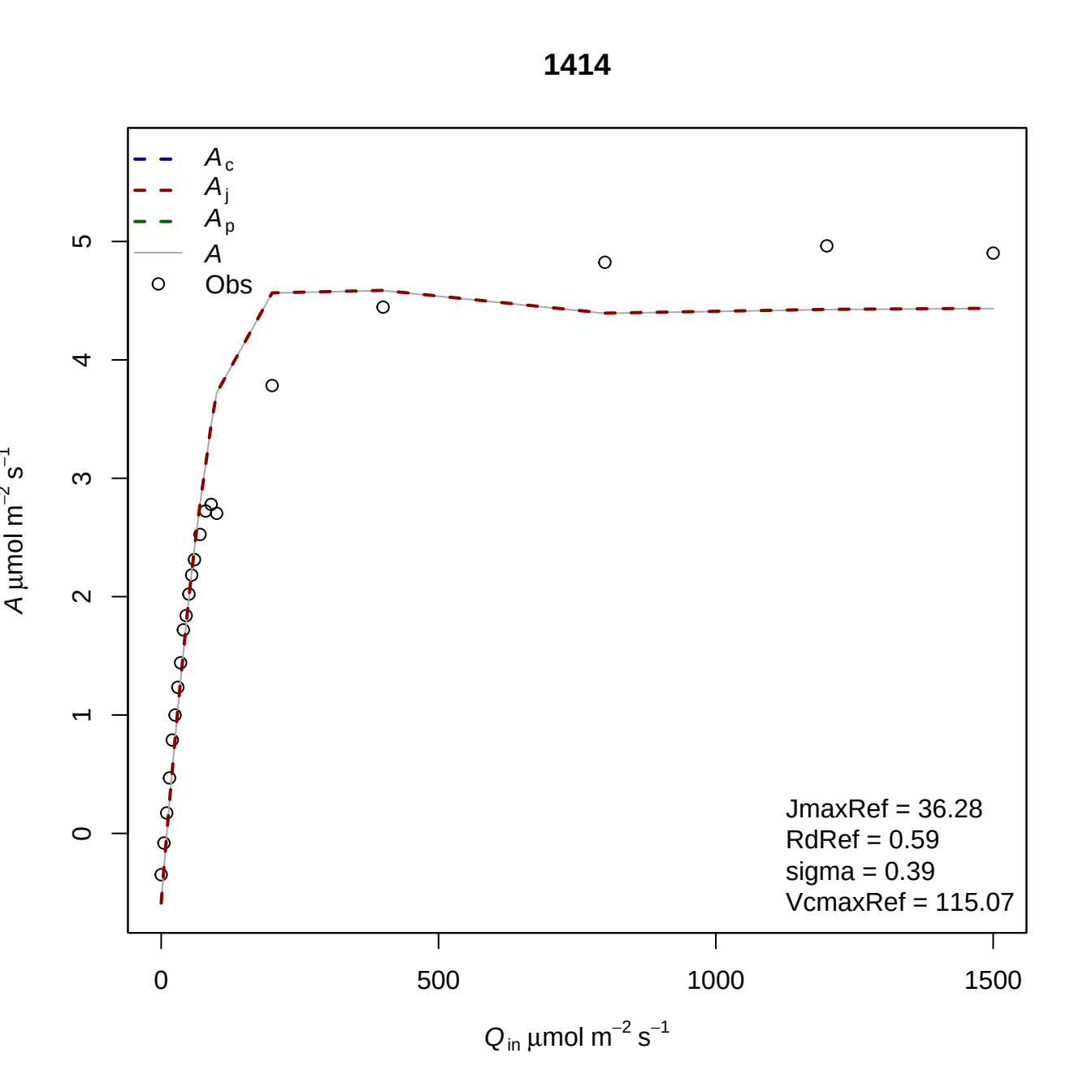

1409
A_c
A_j
A_p
A
Obs
JmaxRef = 42.53
RdRef = 1.65
sigma = 0.72
VcmaxRef = 350.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

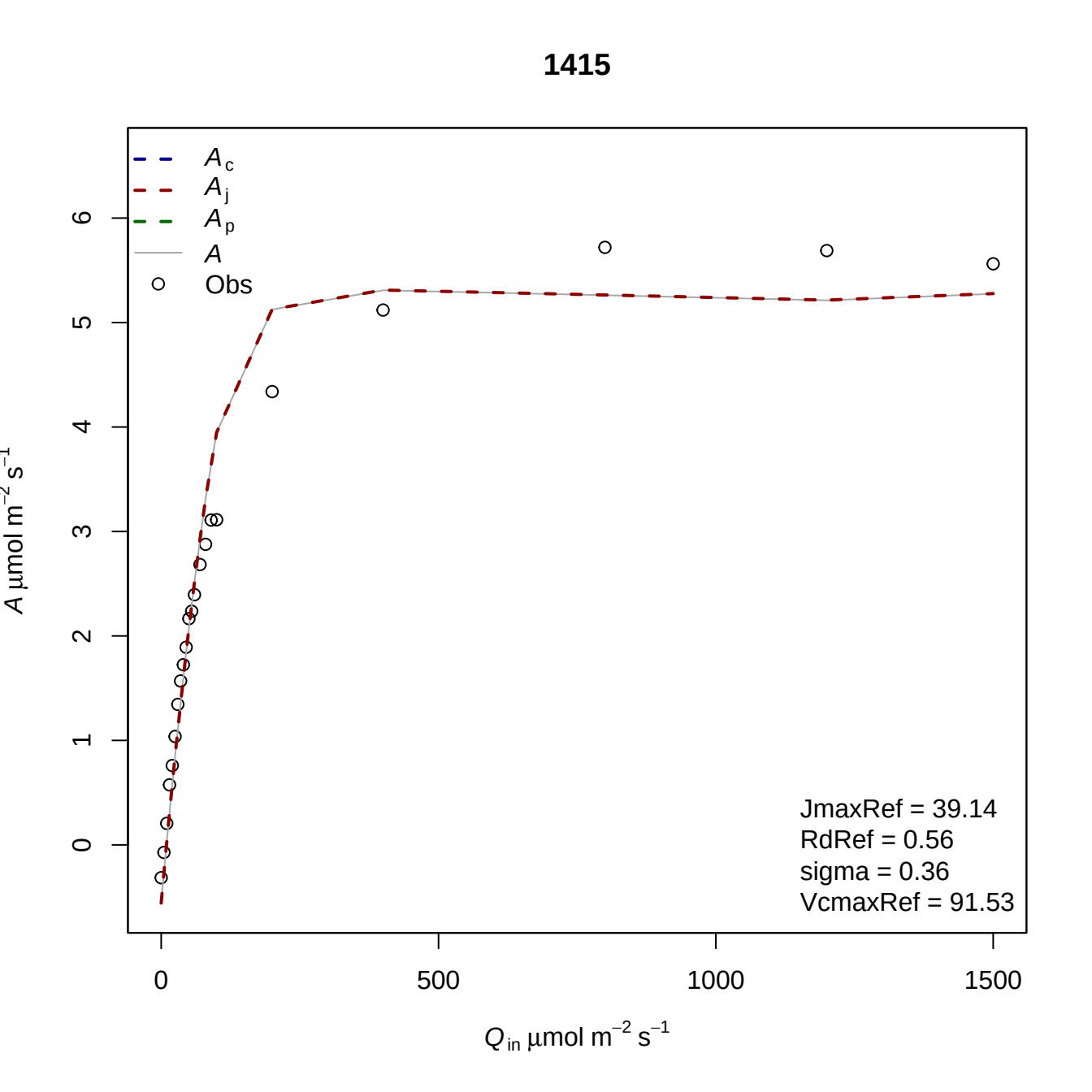

1410
A_c
A_j
A_p
A
Obs
JmaxRef = 67.4
RdRef = 0.42
sigma = 0.49
VcmaxRef = 284.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

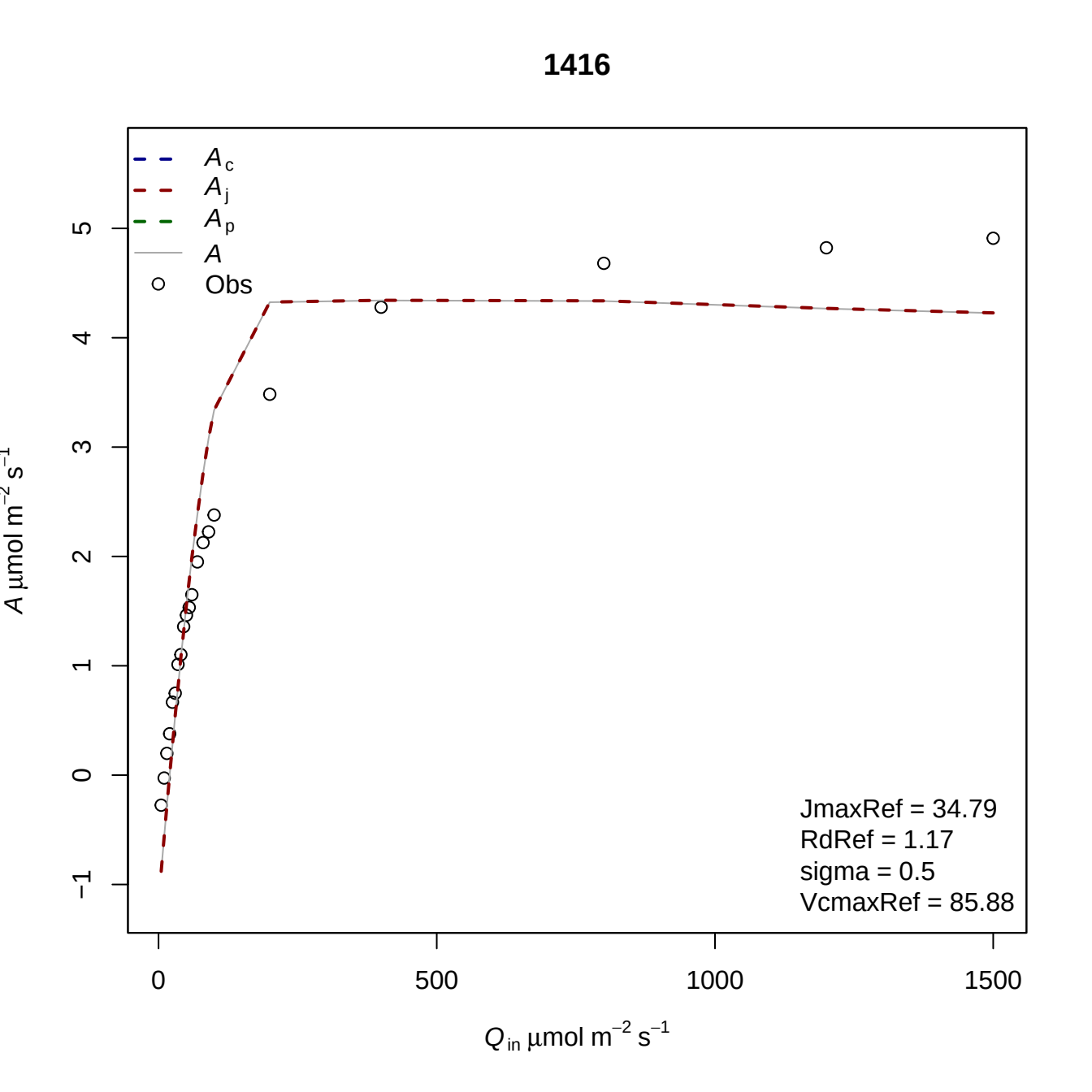

1411
A_c
A_j
A_p
A
Obs
JmaxRef = 33.62
RdRef = 1.79
sigma = 0.47
VcmaxRef = 59.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800

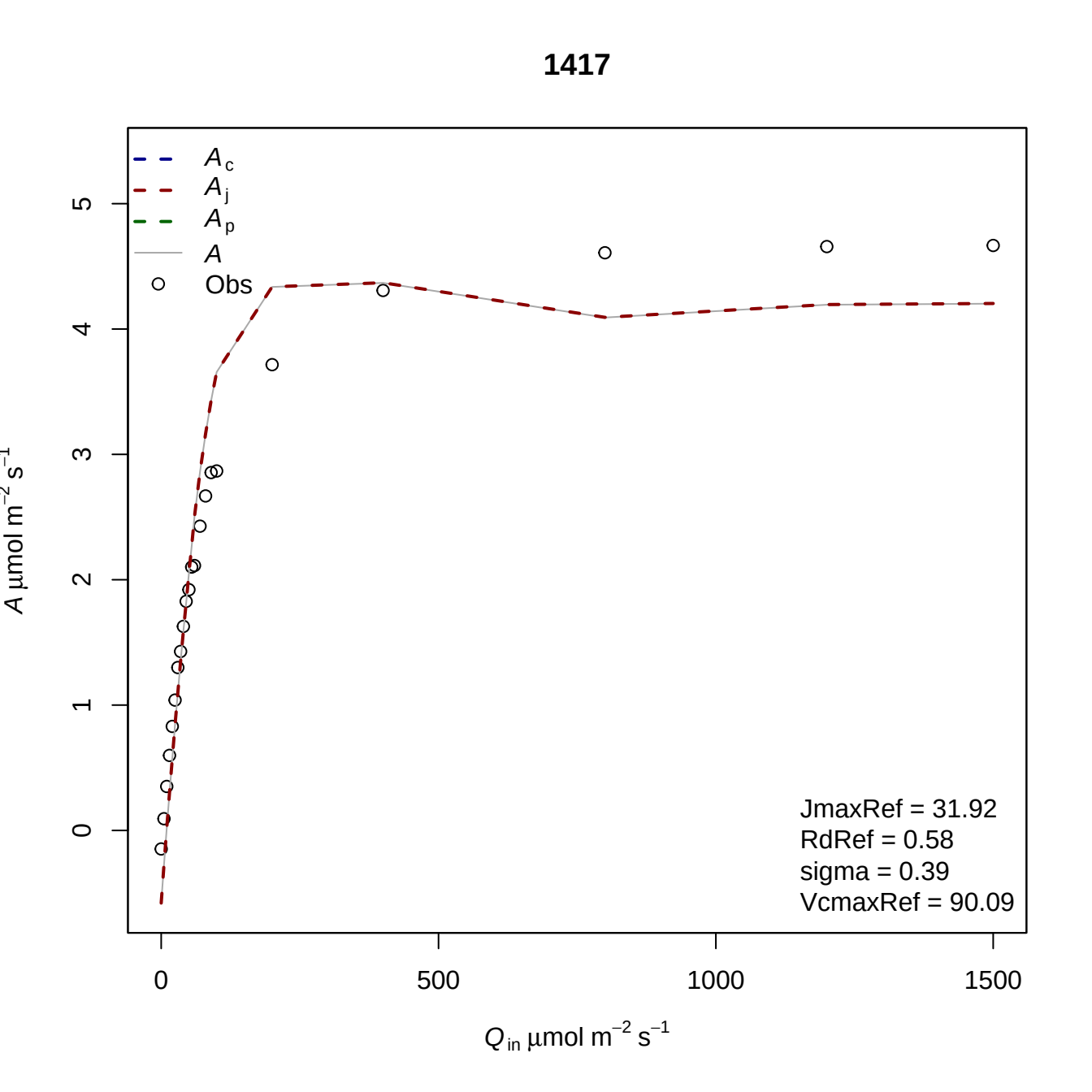

1412
A_c
A_j
A_p
A
Obs
JmaxRef = 32.57
RdRef = 1.88
sigma = 0.38
VcmaxRef = 192925.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
200
400
600
800
1000
1200

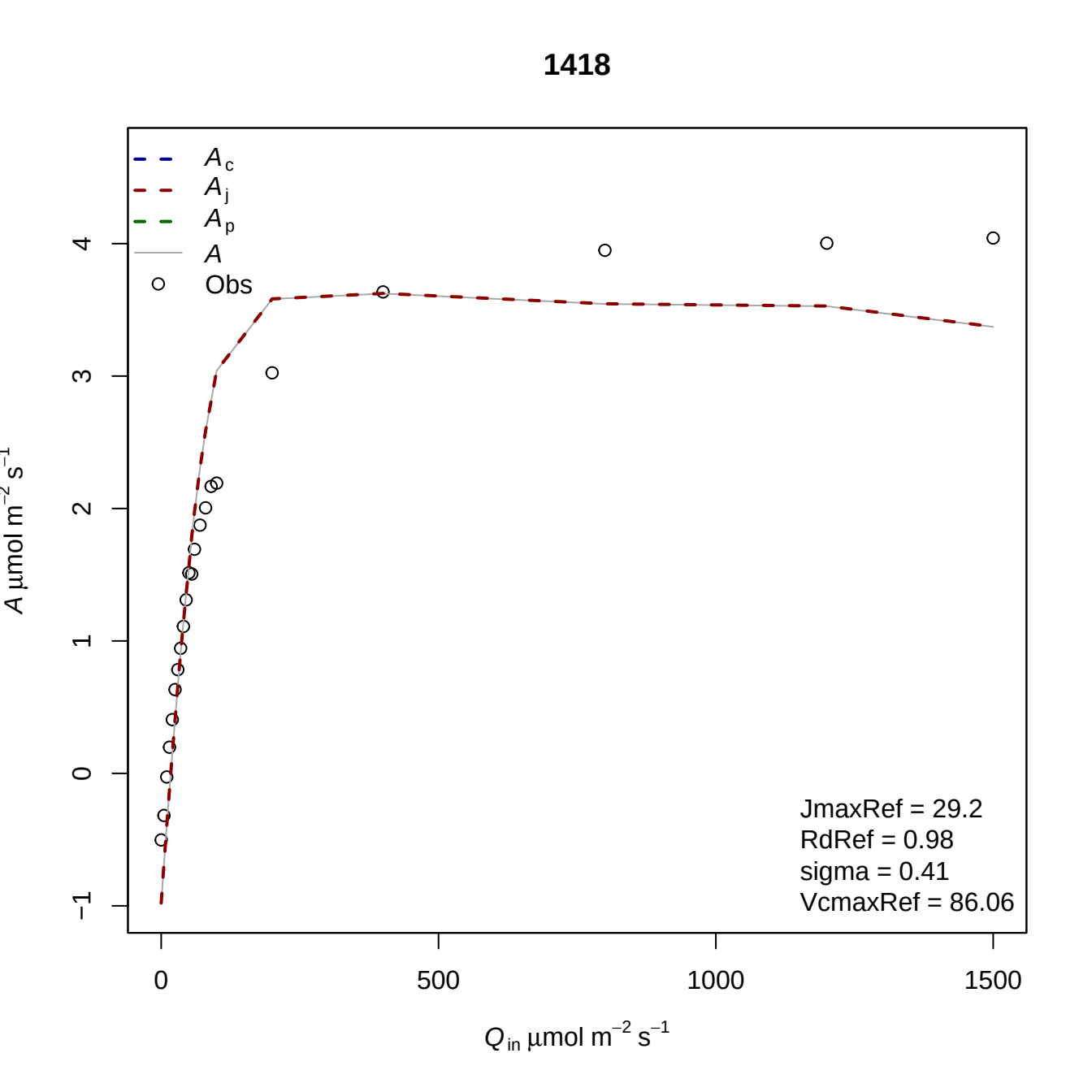

1413
A_c
A_j
A_p
A
Obs
JmaxRef = 32.92
RdRef = 0.78
sigma = 0.42
VcmaxRef = 165.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

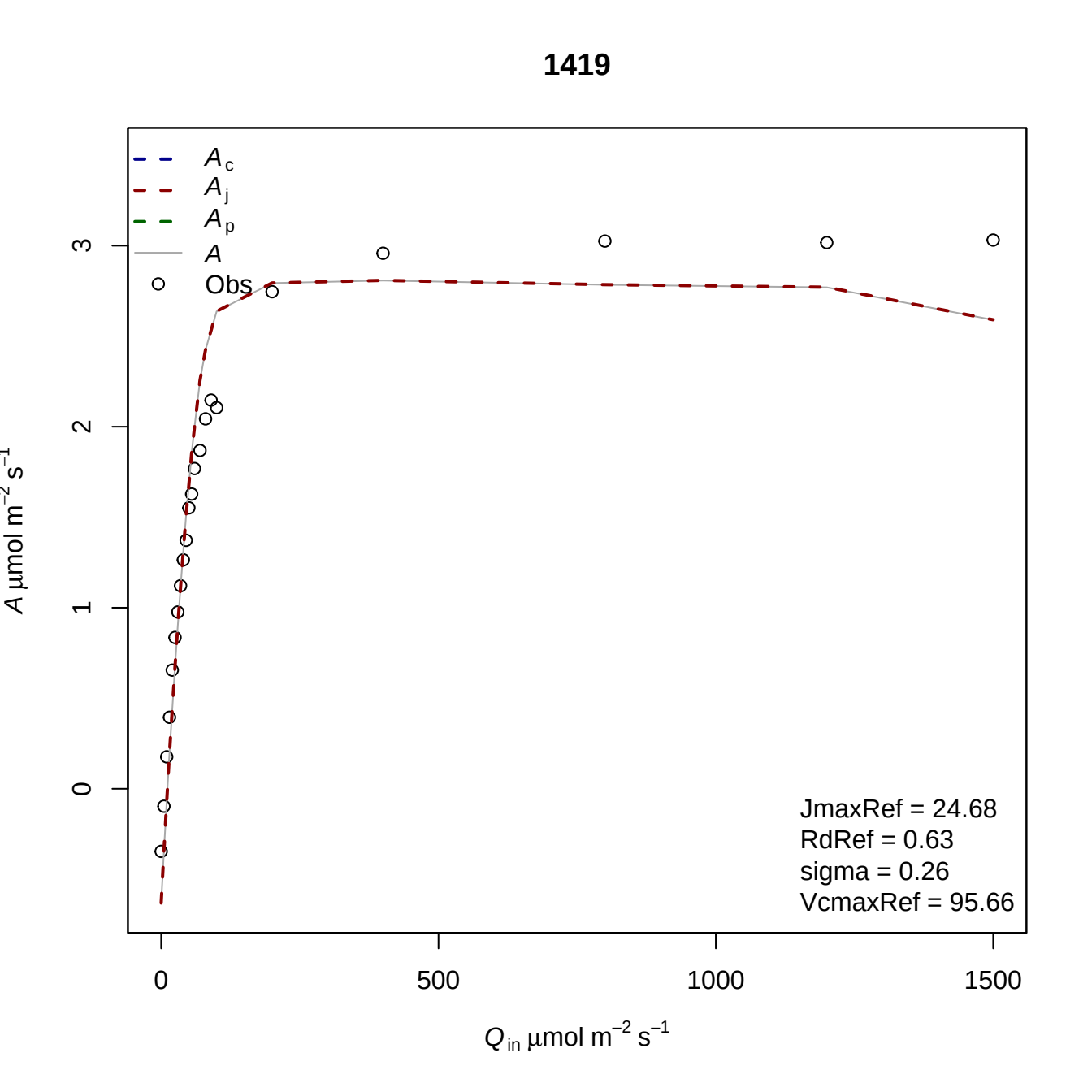

1414
A_c
A_j
A_p
A
Obs
JmaxRef = 36.28
RdRef = 0.59
sigma = 0.39
VcmaxRef = 115.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

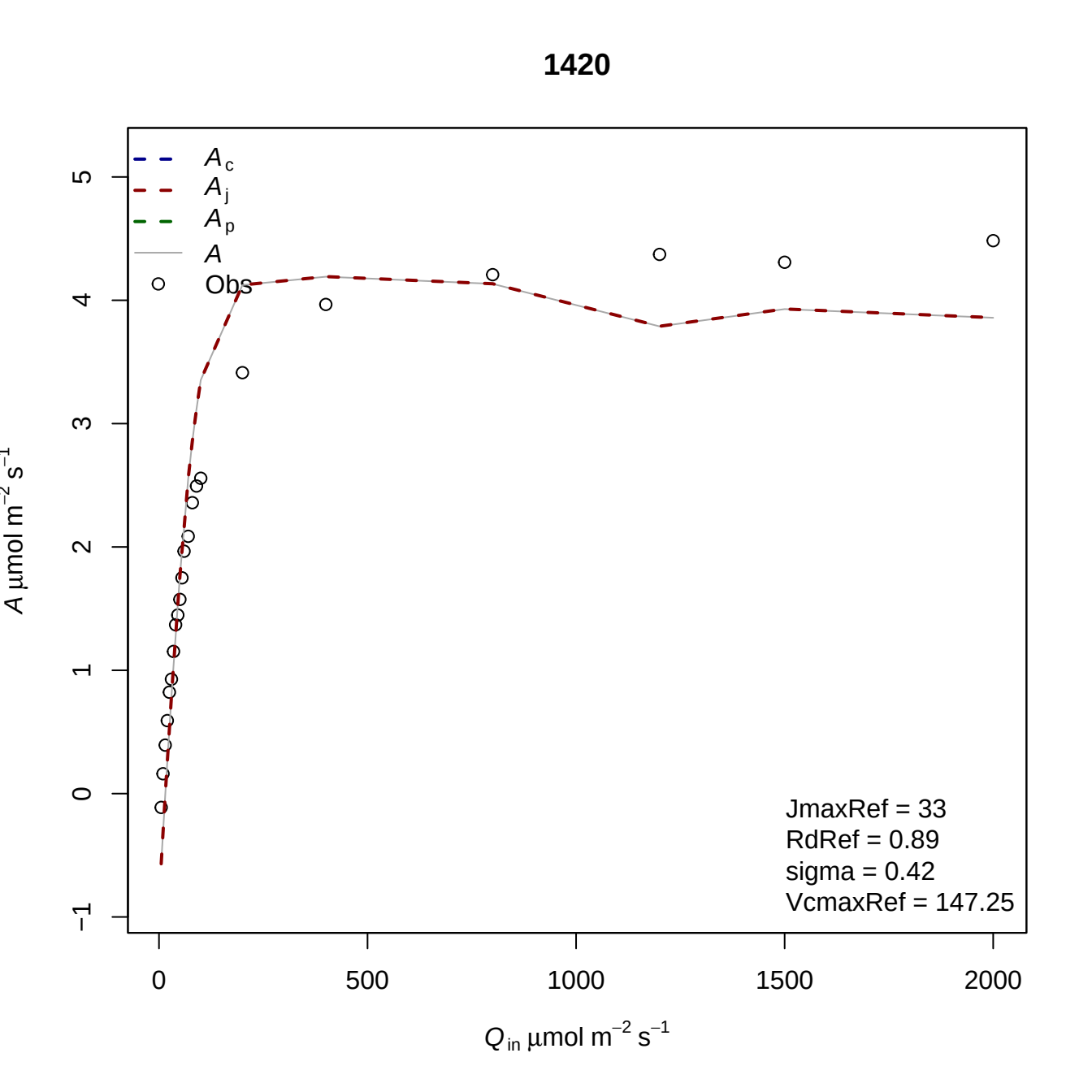

1415
A_c
A_j
A_p
A
Obs
JmaxRef = 39.14
RdRef = 0.56
sigma = 0.36
VcmaxRef = 91.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

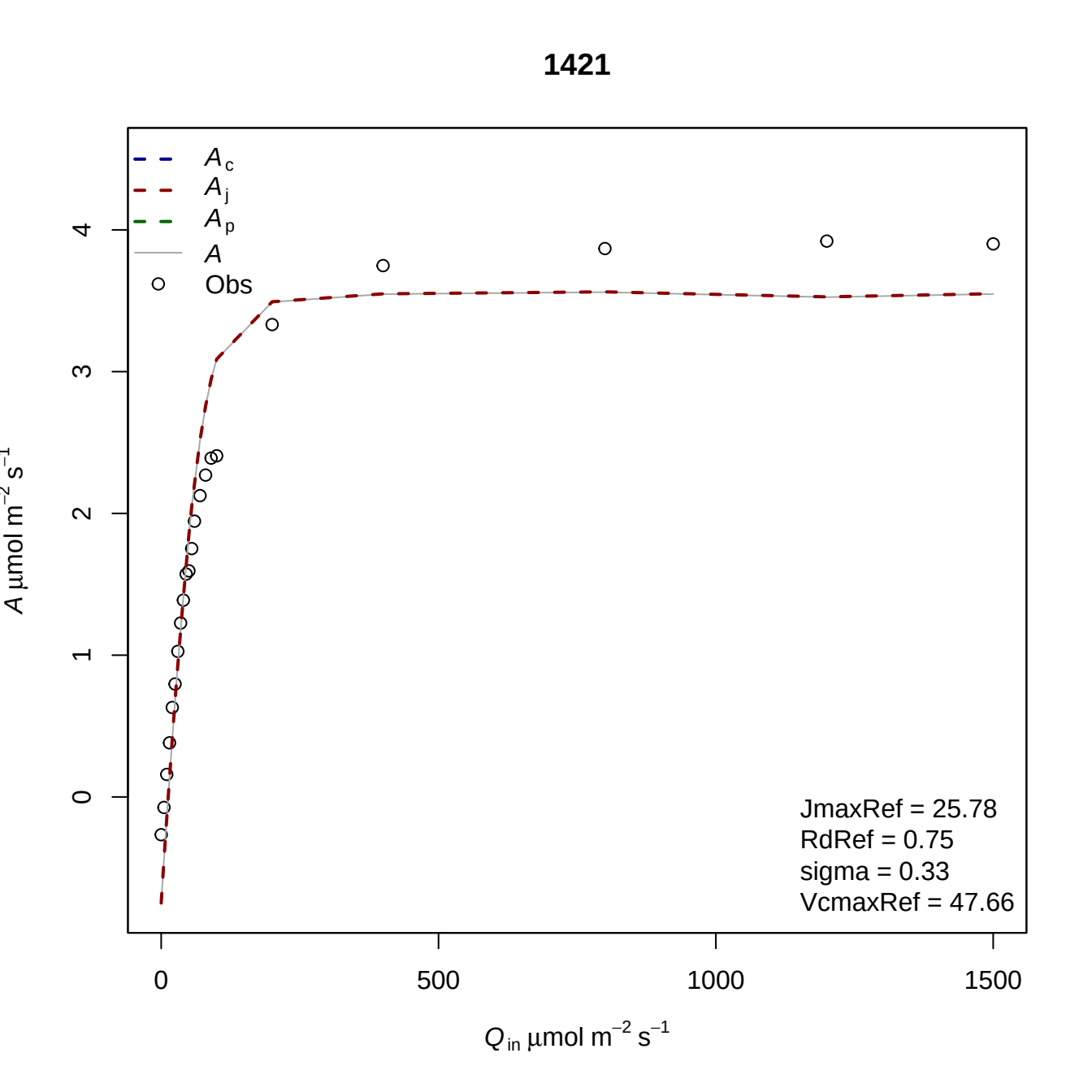

1416
A_c
A_j
A_p
A
Obs
JmaxRef = 34.79
RdRef = 1.17
sigma = 0.5
VcmaxRef = 85.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

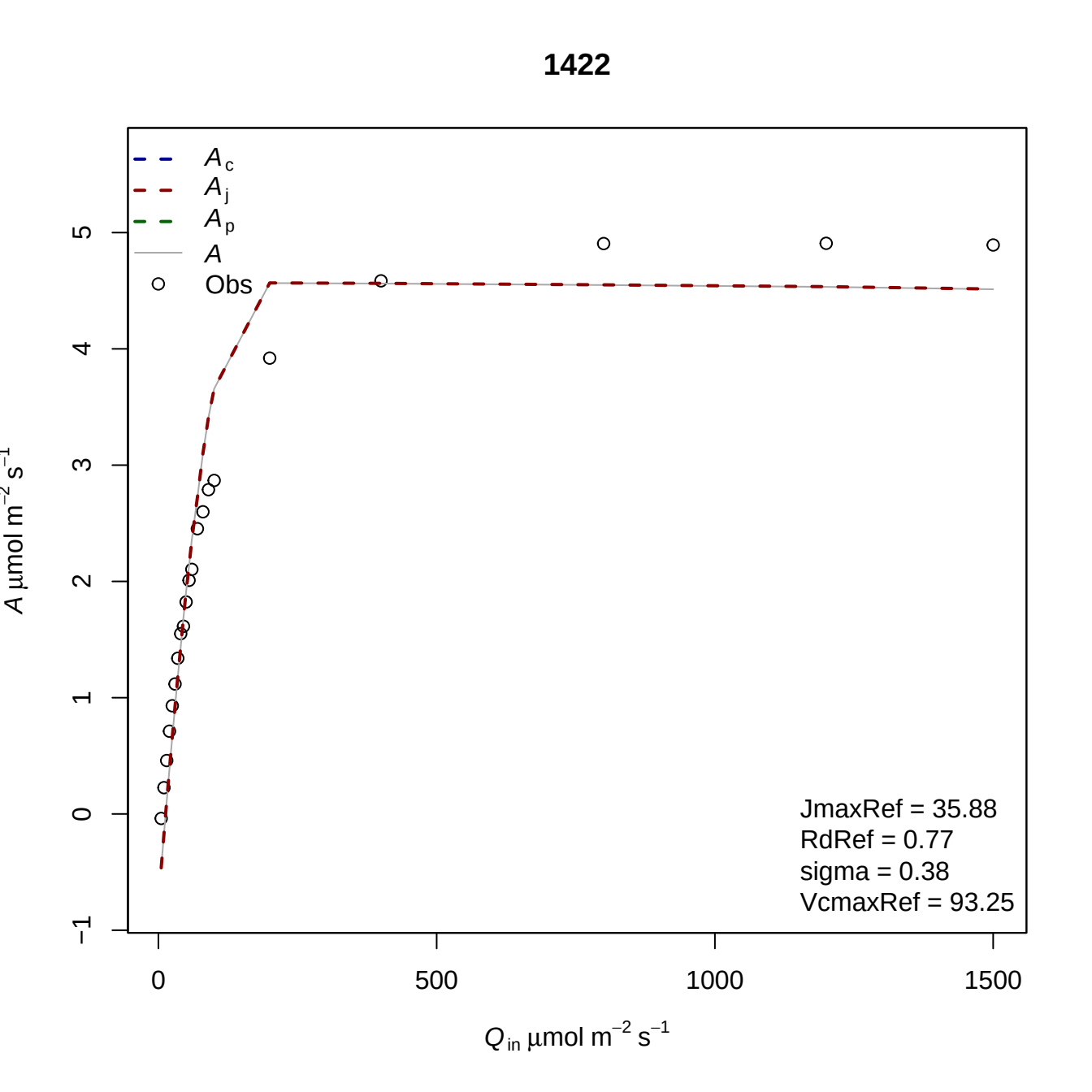

1417
A_c
A_j
A_p
A
Obs
JmaxRef = 31.92
RdRef = 0.58
sigma = 0.39
VcmaxRef = 90.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

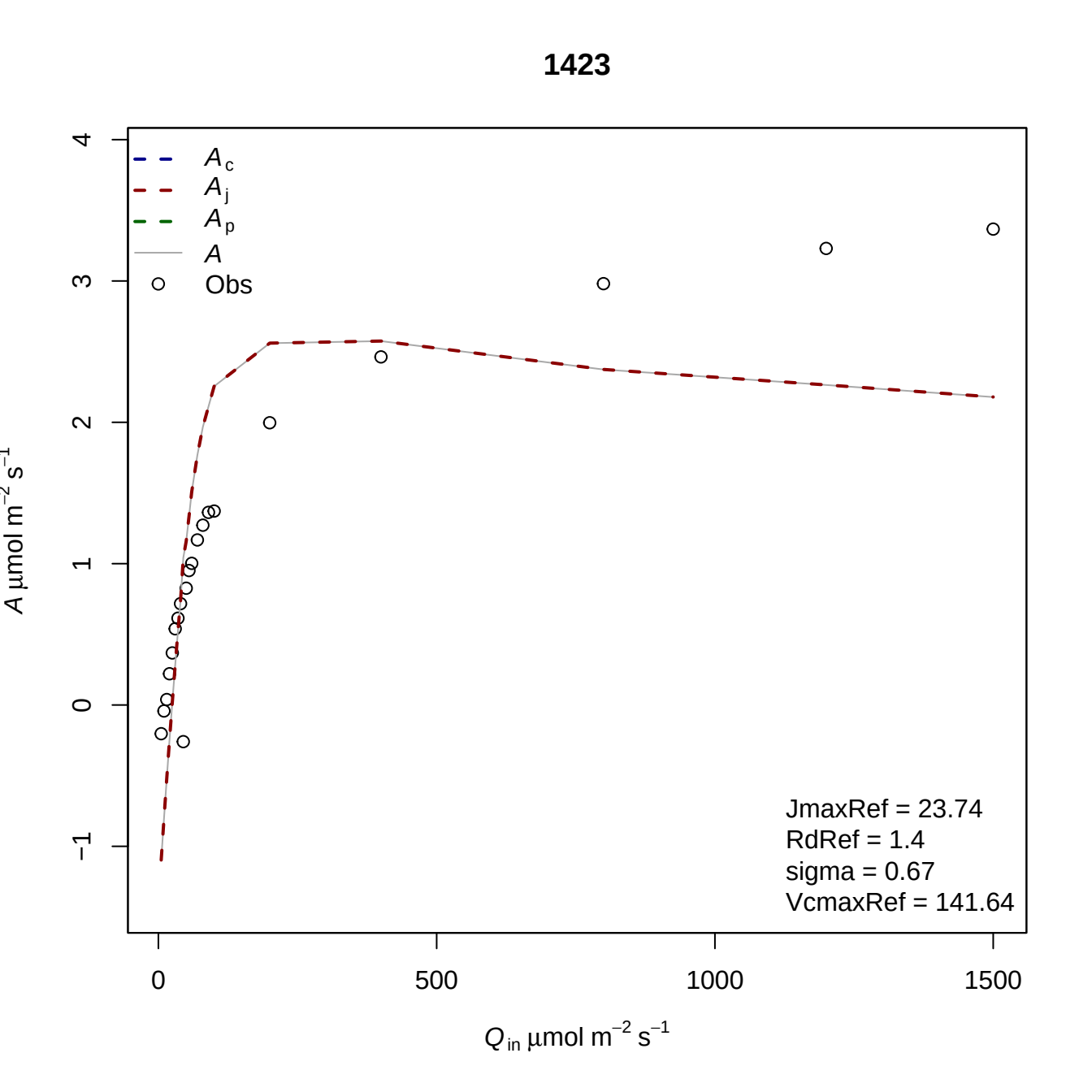

1418
A_c
A_j
A_p
A
Obs
JmaxRef = 29.2
RdRef = 0.98
sigma = 0.41
VcmaxRef = 86.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

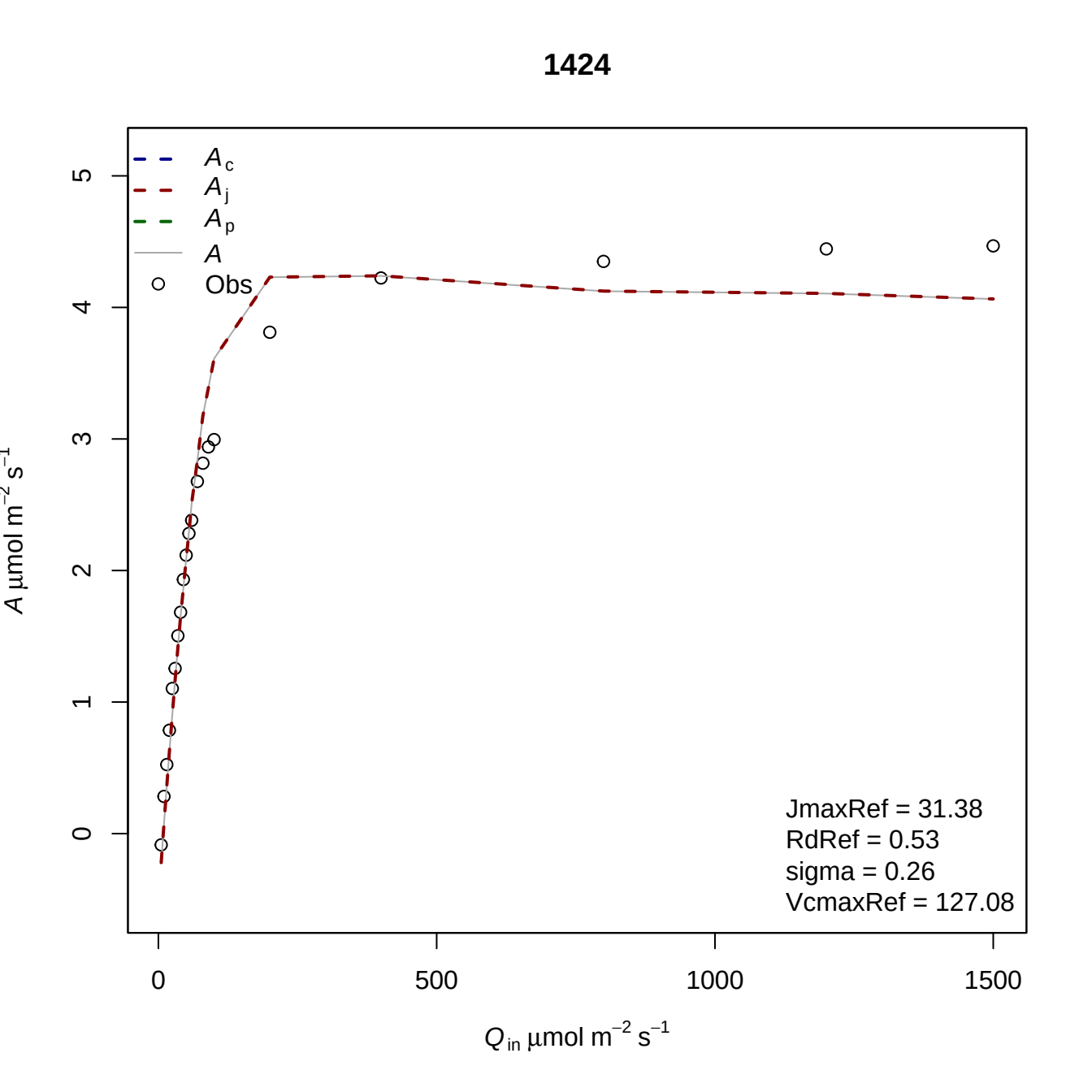

1419
A_c
A_j
A_p
A
Obs
JmaxRef = 24.68
RdRef = 0.63
sigma = 0.26
VcmaxRef = 95.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

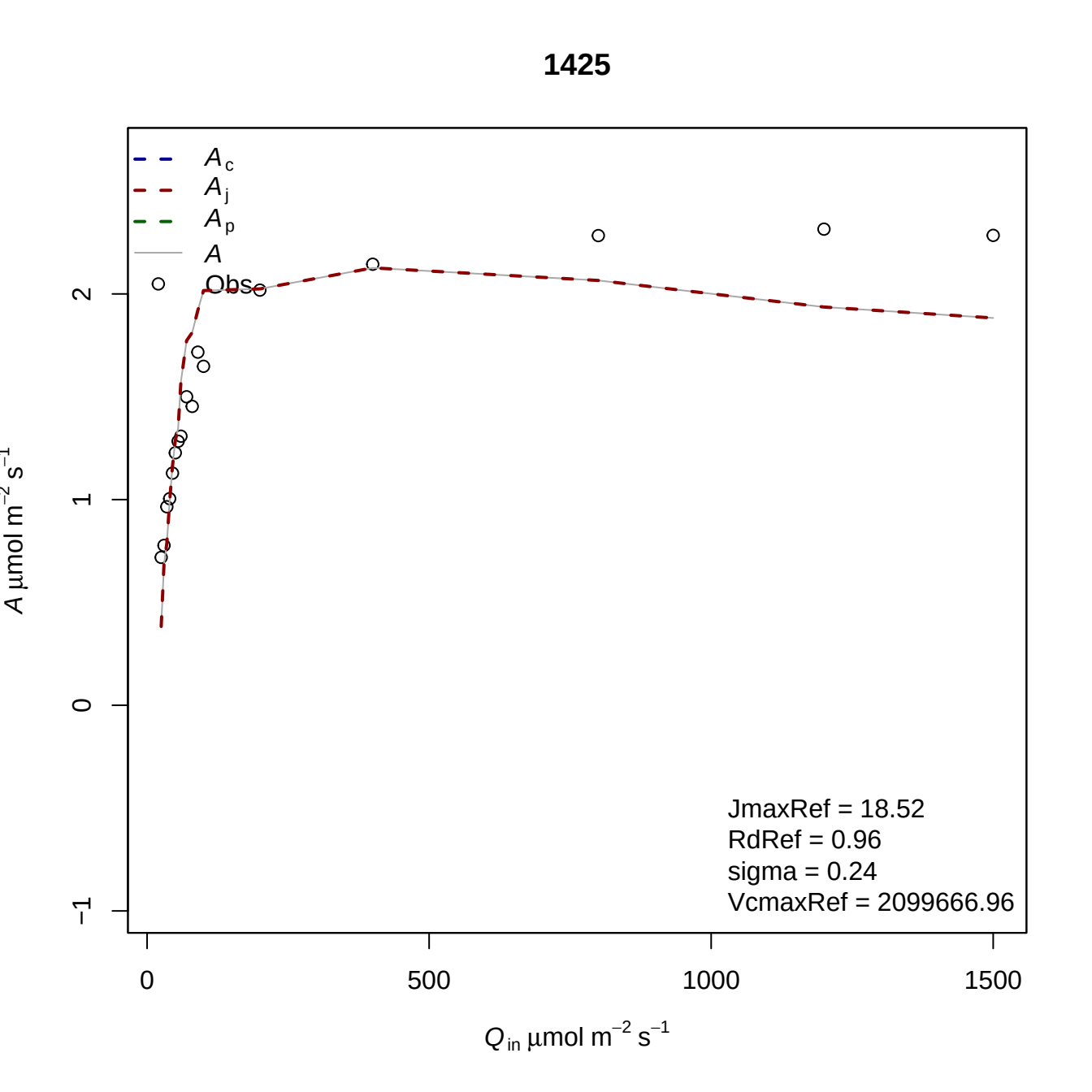

1420
A_c
A_j
A_p
A
Obs
JmaxRef = 33
RdRef = 0.89
sigma = 0.42
VcmaxRef = 147.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

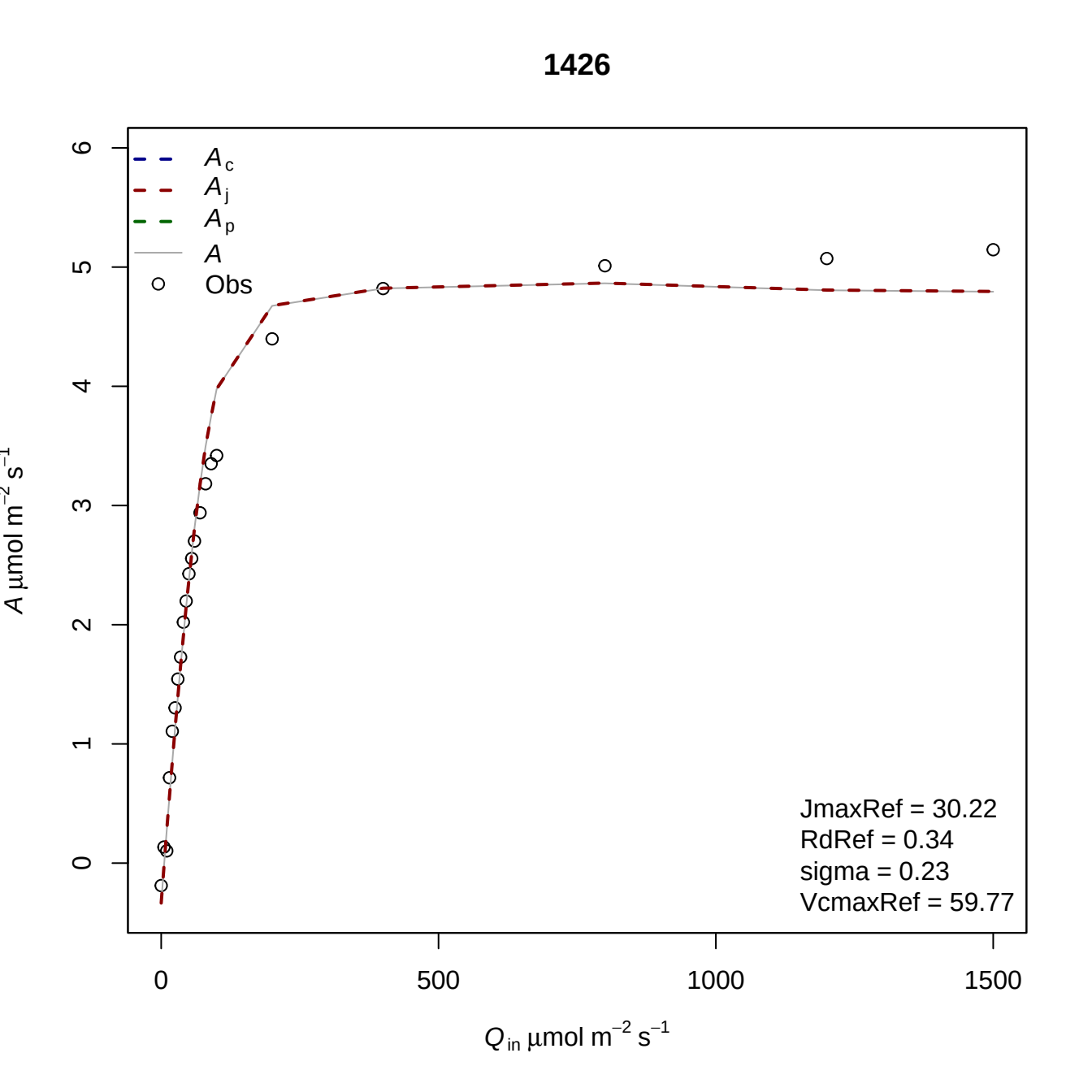

1421
A_c
A_j
A_p
A
Obs
JmaxRef = 25.78
RdRef = 0.75
sigma = 0.33
VcmaxRef = 47.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

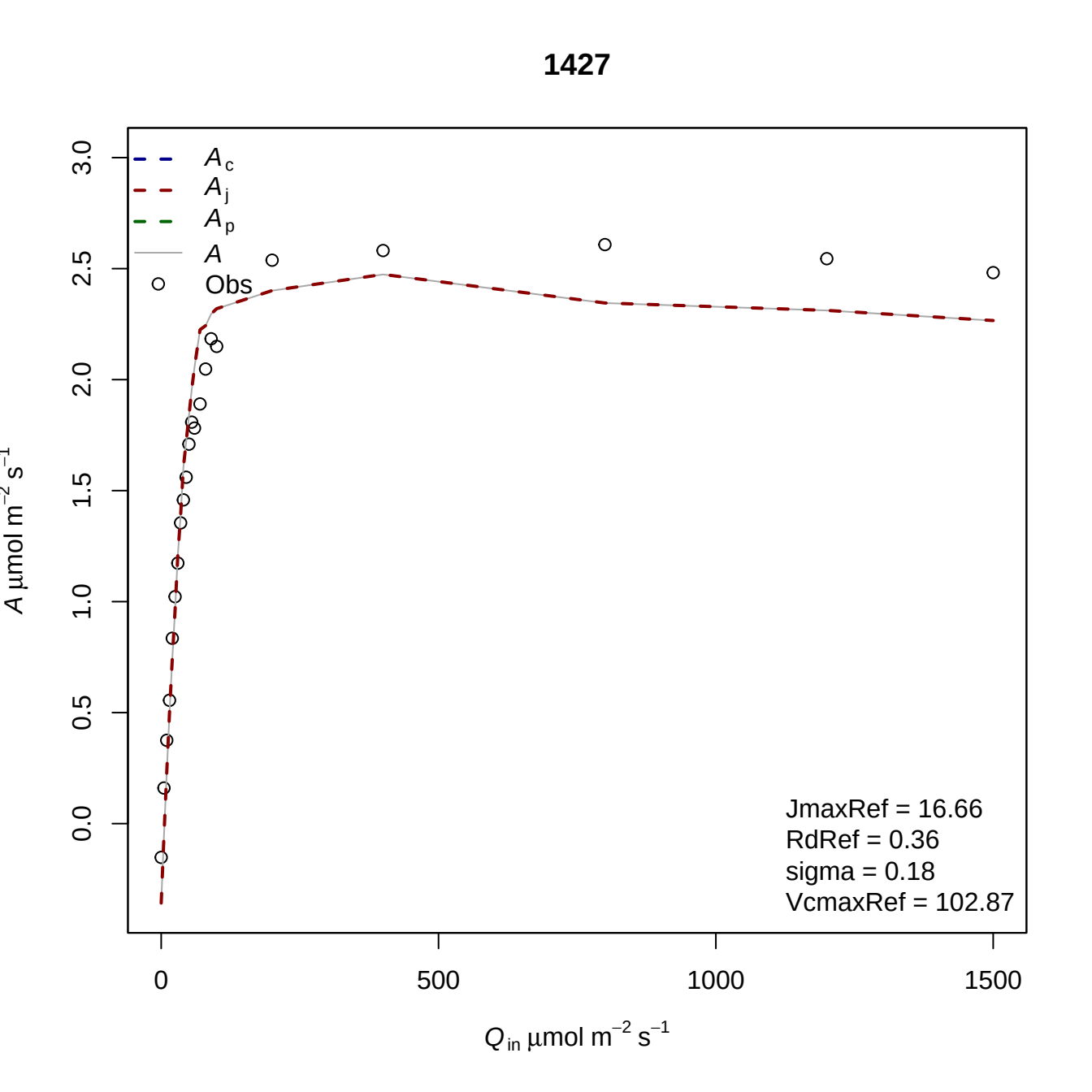

1422
A_c
A_j
A_p
A
Obs
JmaxRef = 35.88
RdRef = 0.77
sigma = 0.38
VcmaxRef = 93.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

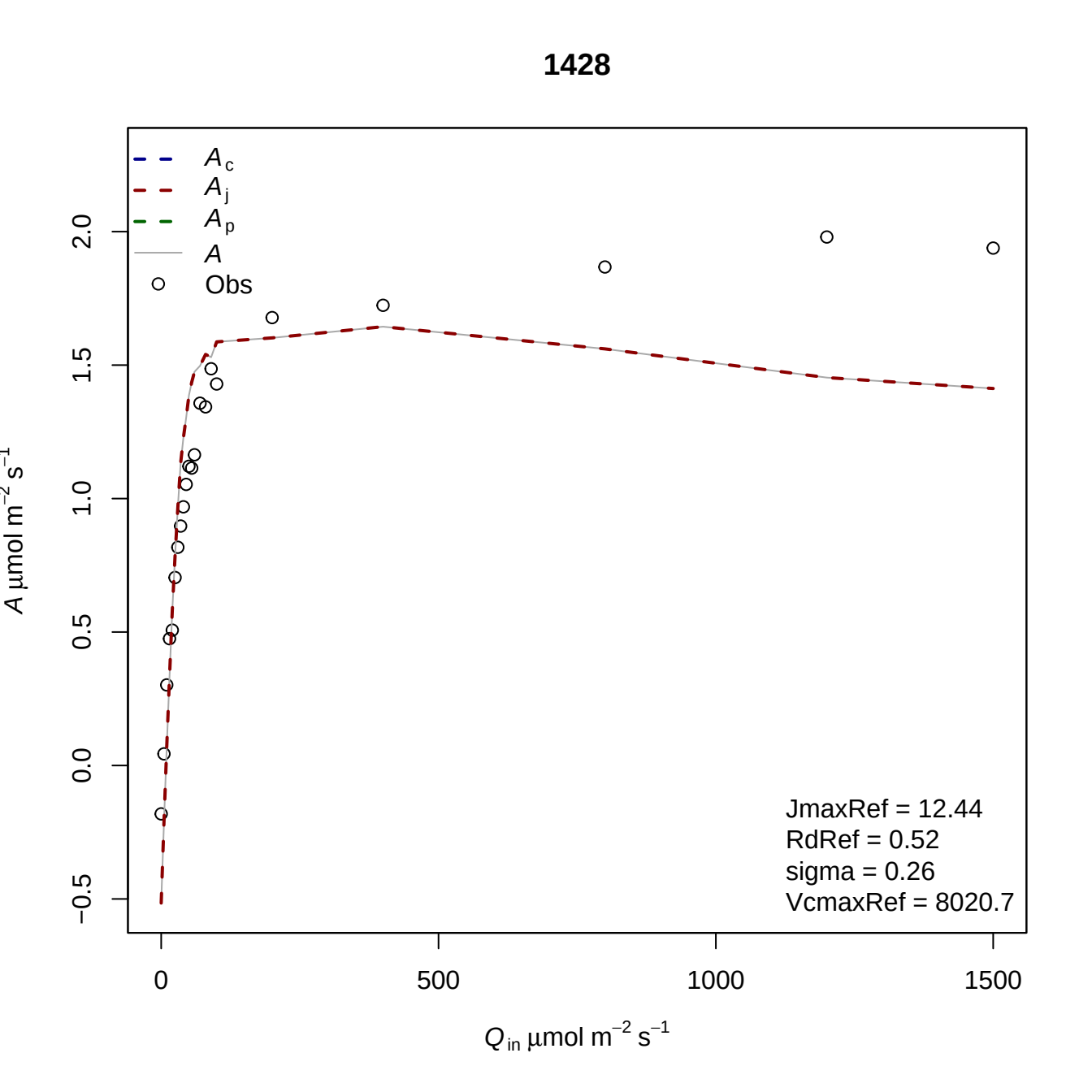

1423
A_c
A_j
A_p
A
Obs
JmaxRef = 23.74
RdRef = 1.4
sigma = 0.67
VcmaxRef = 141.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

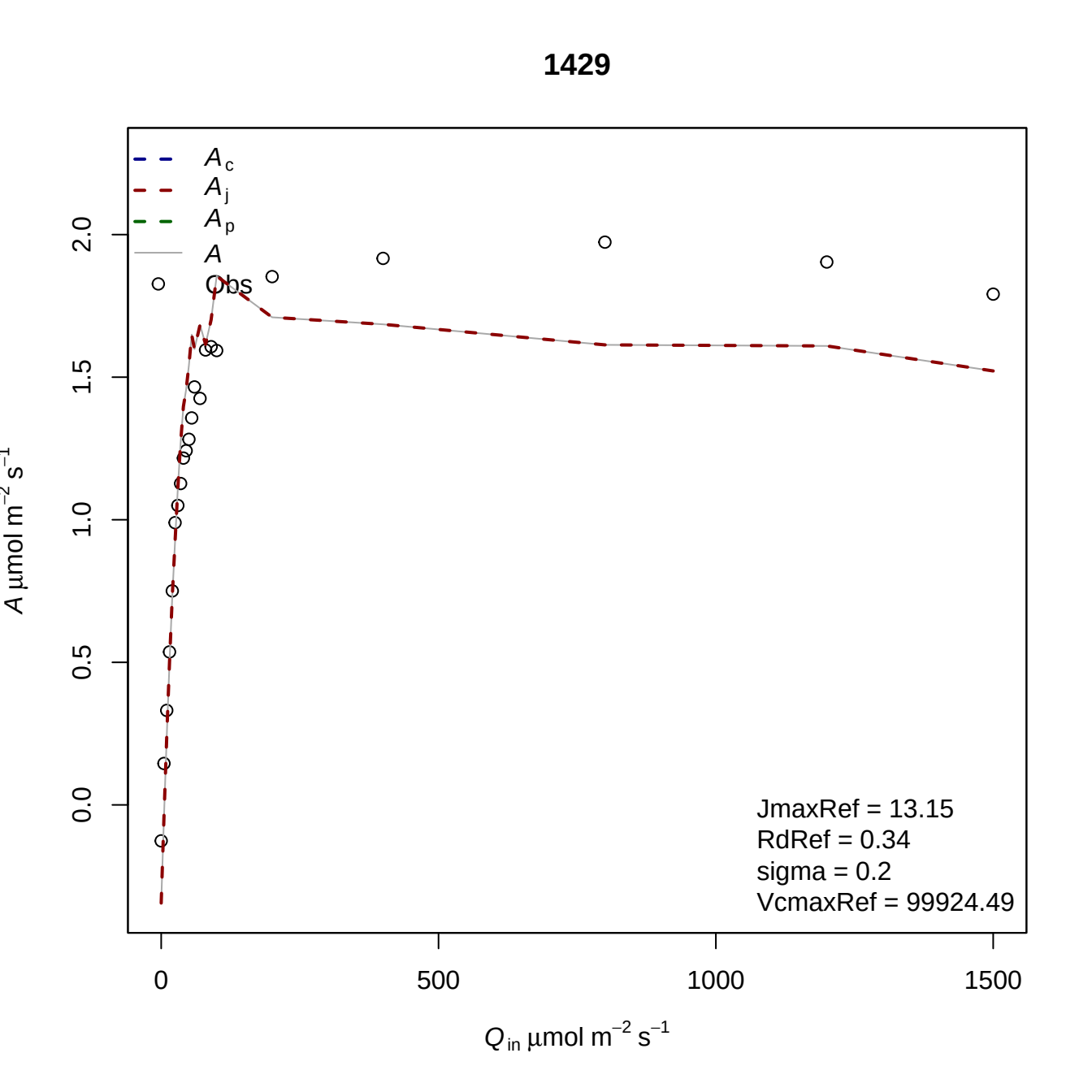

1424
A_c
A_j
A_p
A
Obs
JmaxRef = 31.38
RdRef = 0.53
sigma = 0.26
VcmaxRef = 127.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

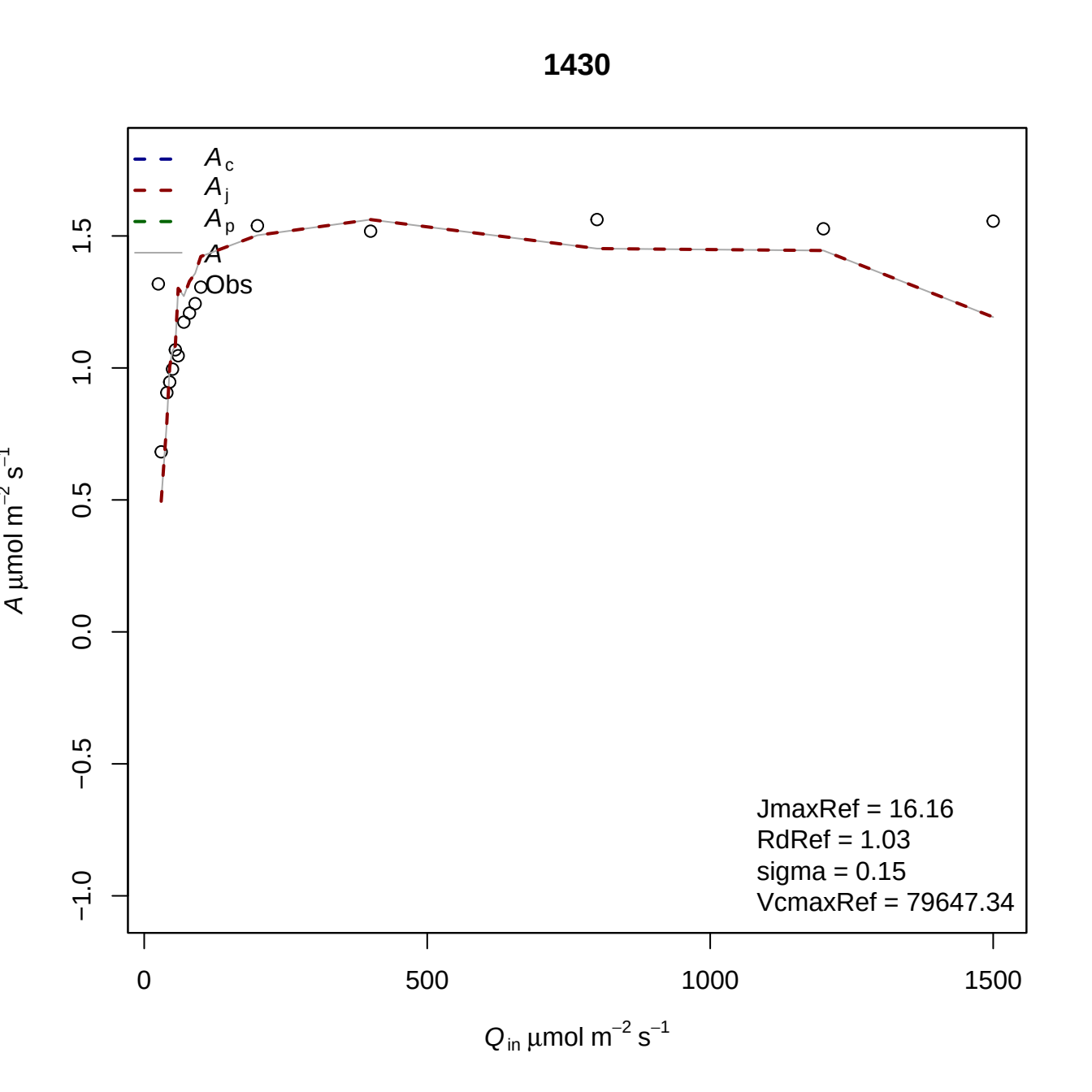

1425
A_c
A_j
A_p
A
Obs
JmaxRef = 18.52
RdRef = 0.96
sigma = 0.24
VcmaxRef = 2099666.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

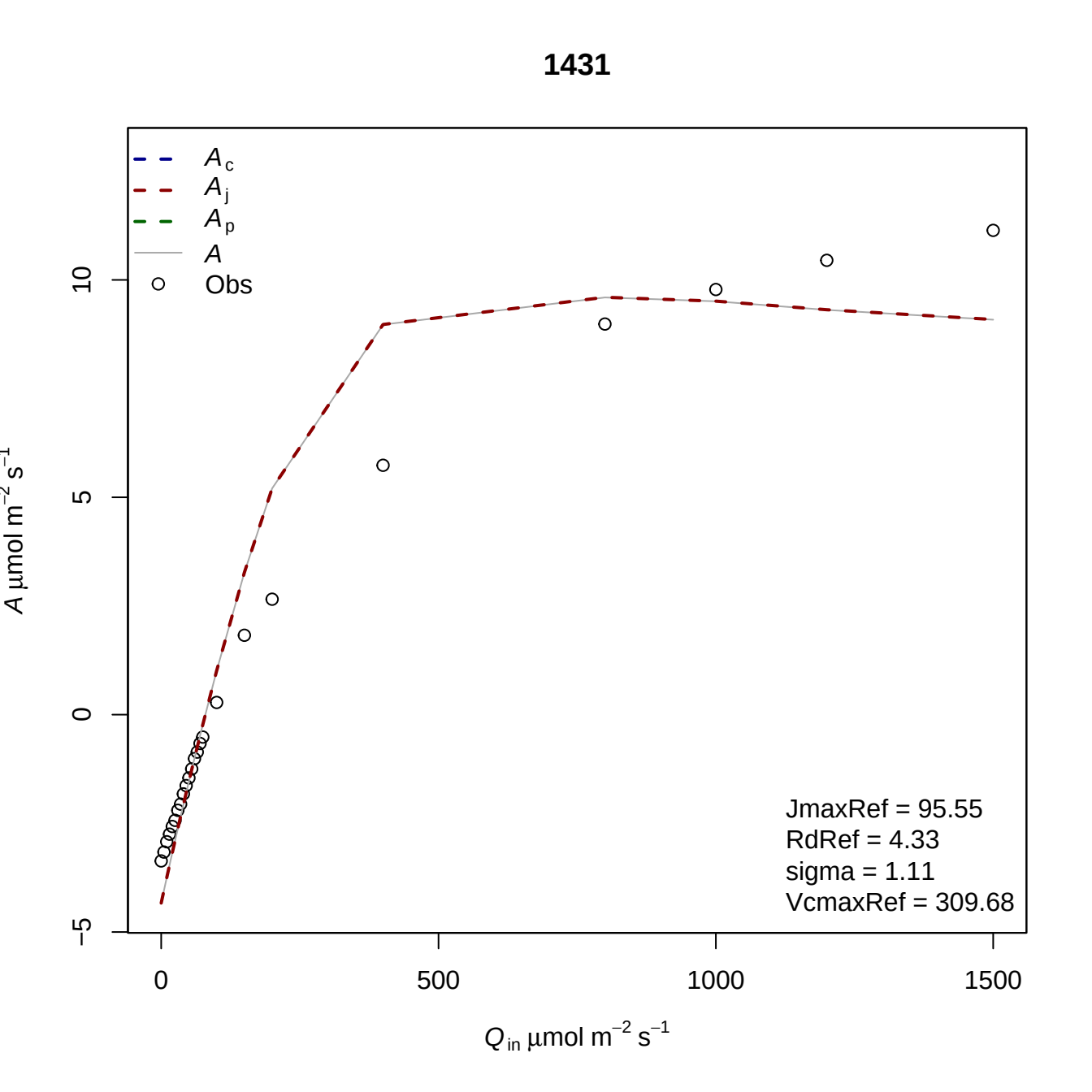

1426
A_c
A_j
A_p
A
Obs
JmaxRef = 30.22
RdRef = 0.34
sigma = 0.23
VcmaxRef = 59.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

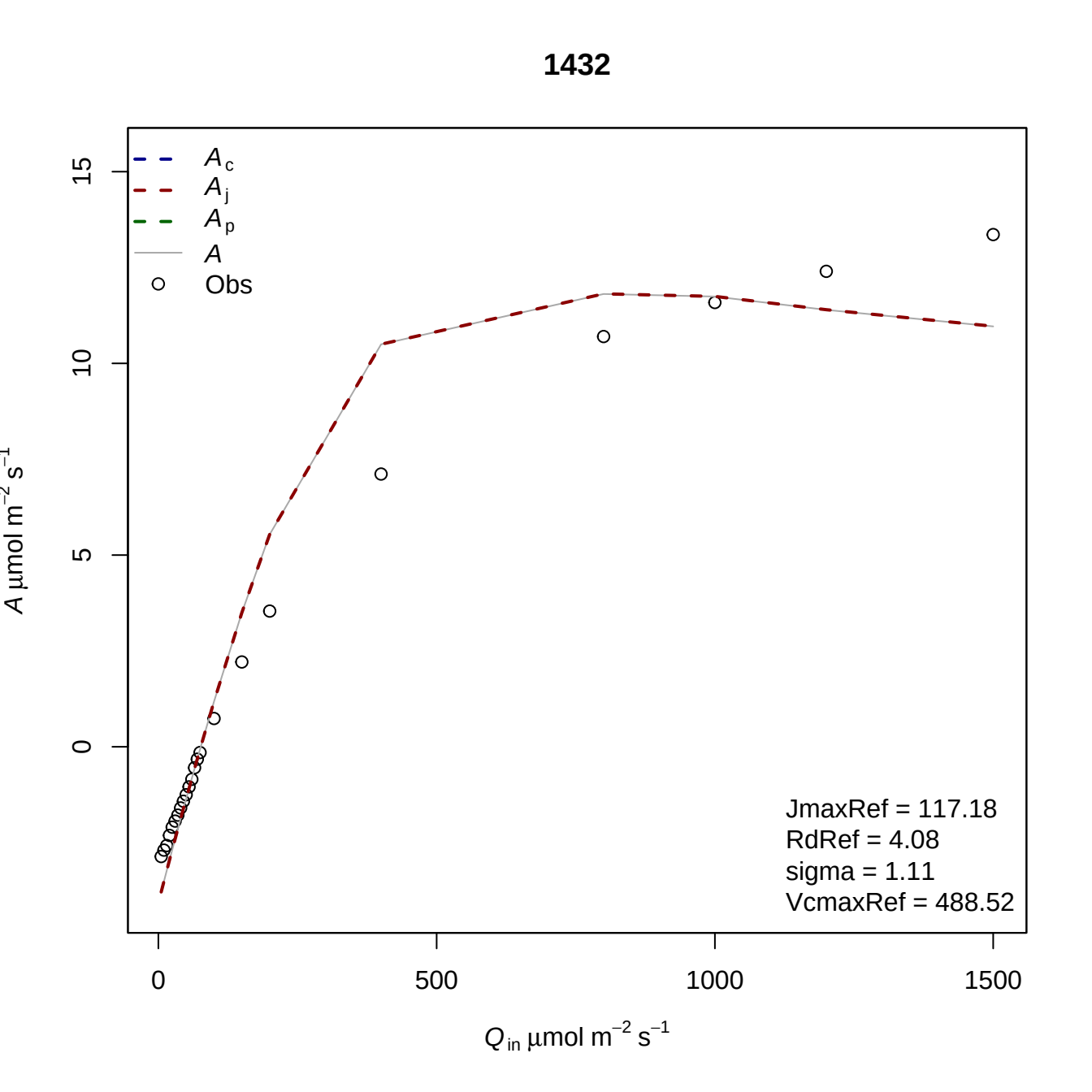

1427
A_c
A_j
A_p
A
Obs
JmaxRef = 16.66
RdRef = 0.36
sigma = 0.18
VcmaxRef = 102.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

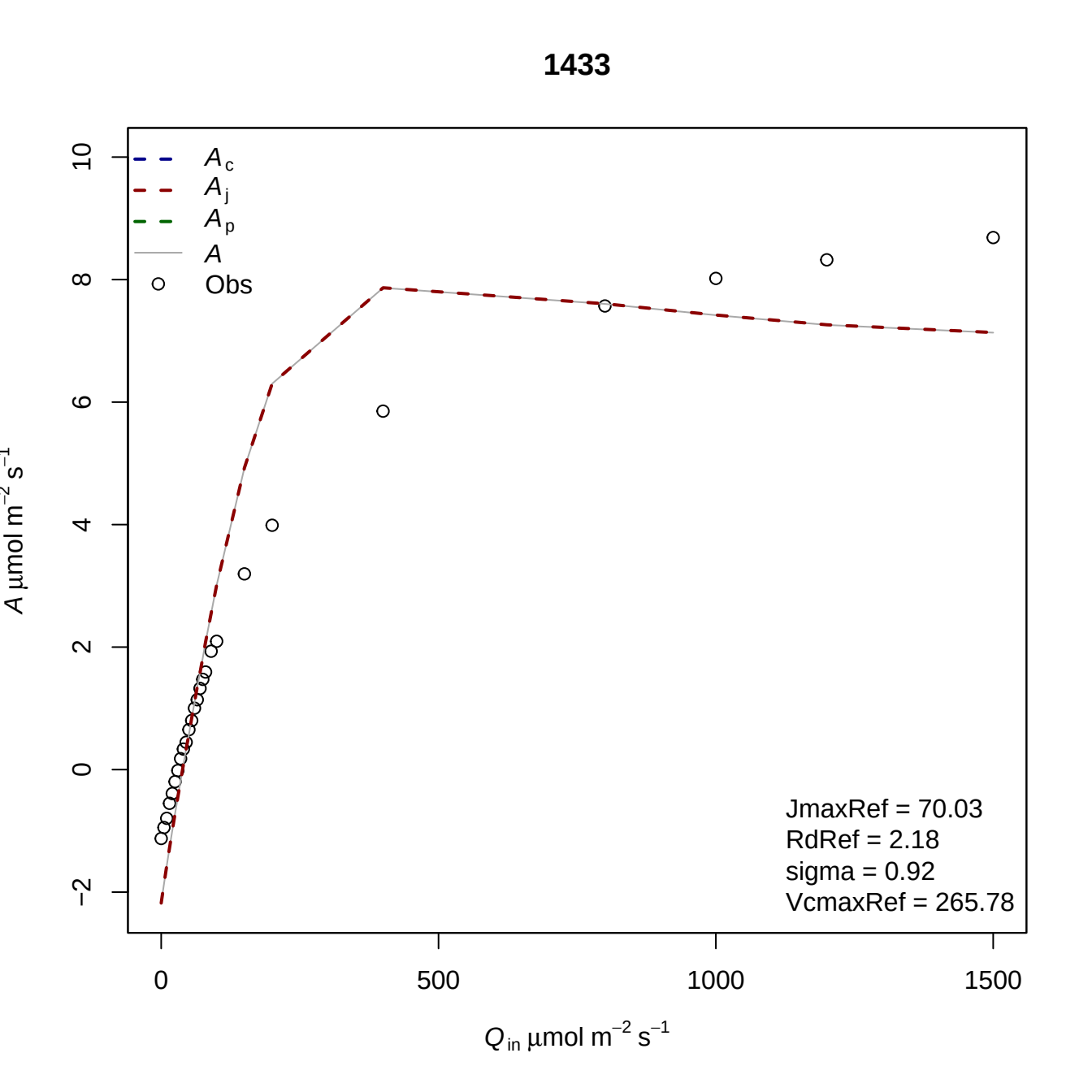

1428
A_c
A_j
A_p
A
Obs
JmaxRef = 12.44
RdRef = 0.52
sigma = 0.26
VcmaxRef = 8020.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

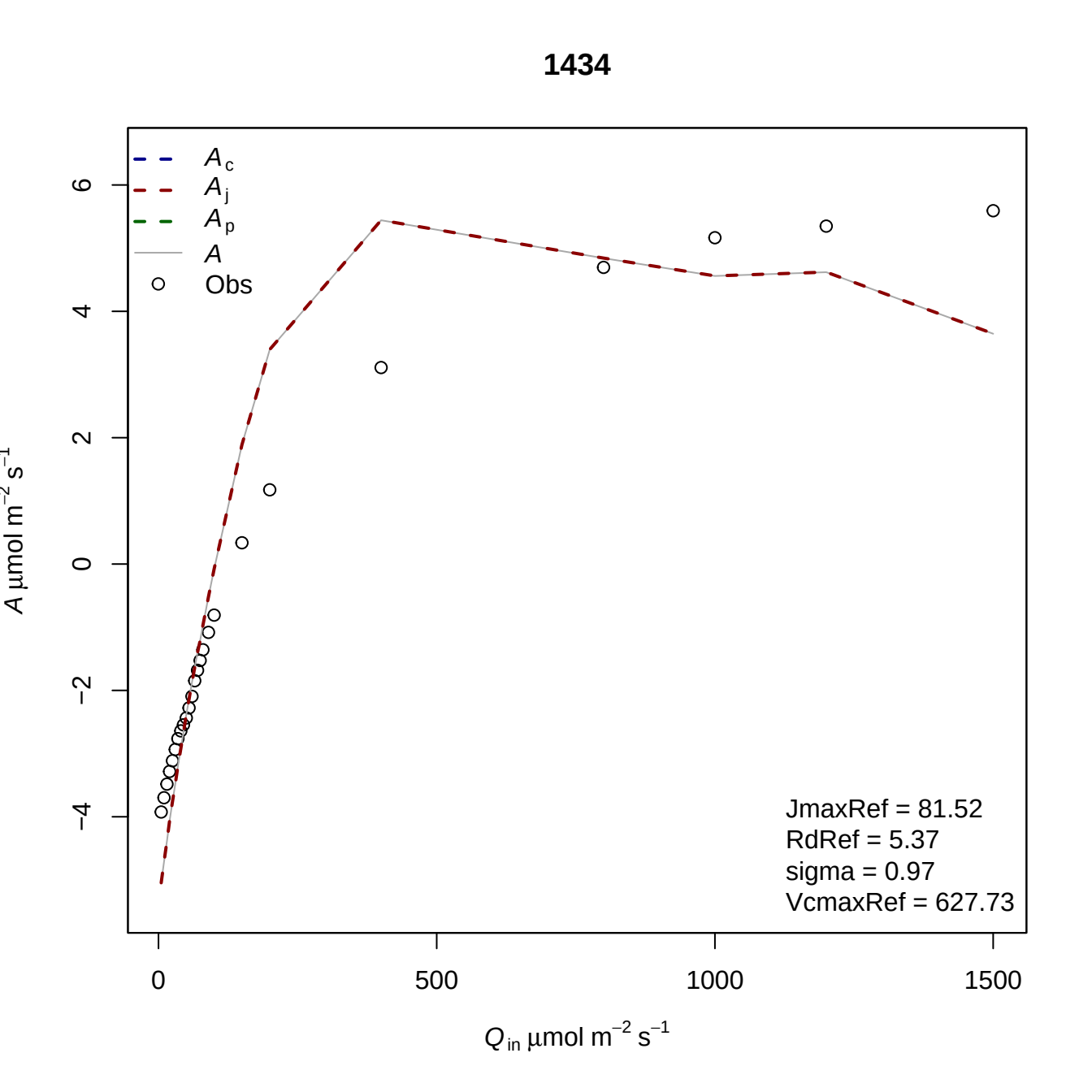

1429
A_c
A_j
A_p
A
Obs
JmaxRef = 13.15
RdRef = 0.34
sigma = 0.2
VcmaxRef = 99924.49
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

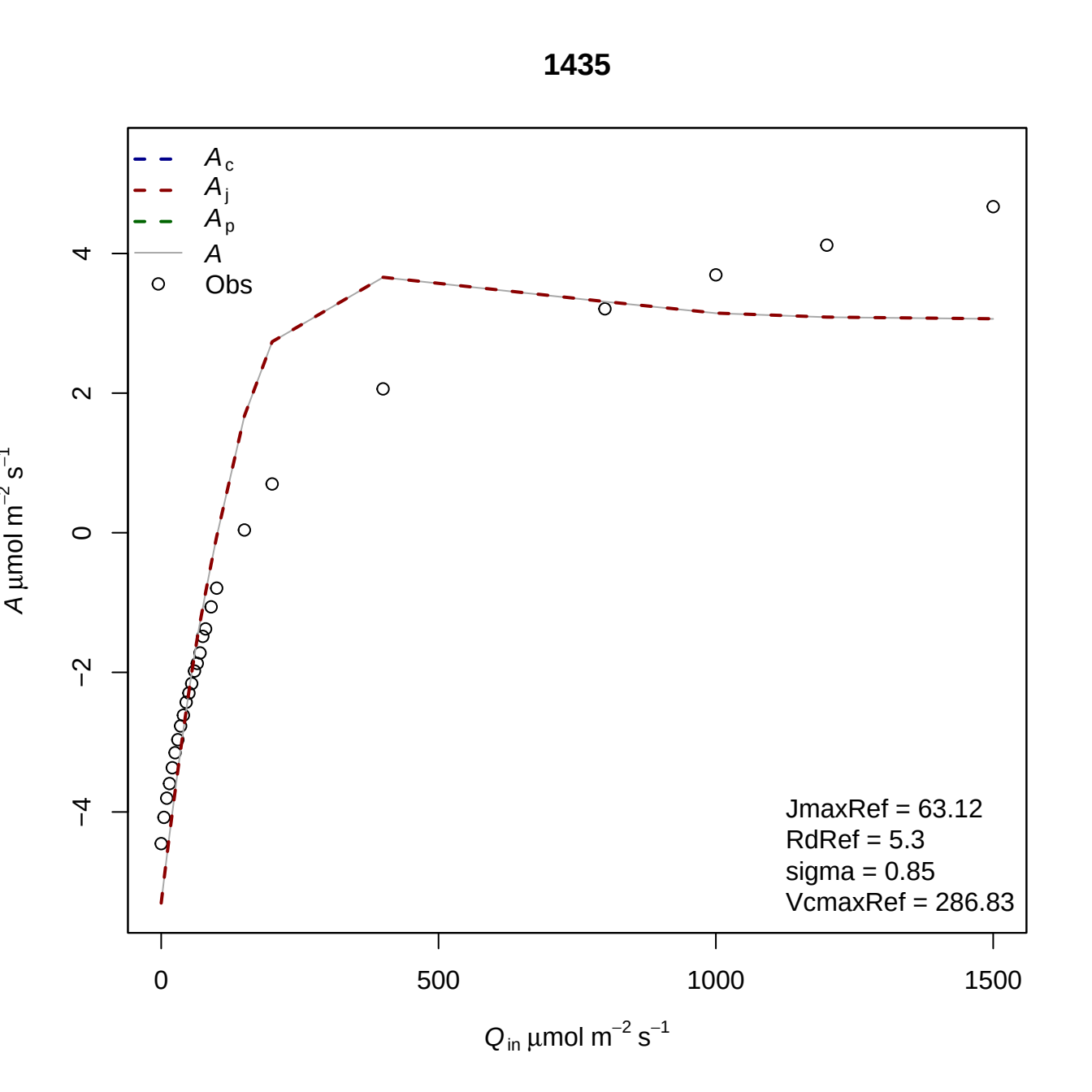

1430
A_c
A_j
A_p
A
Obs
JmaxRef = 16.16
RdRef = 1.03
sigma = 0.15
VcmaxRef = 79647.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1.0
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500

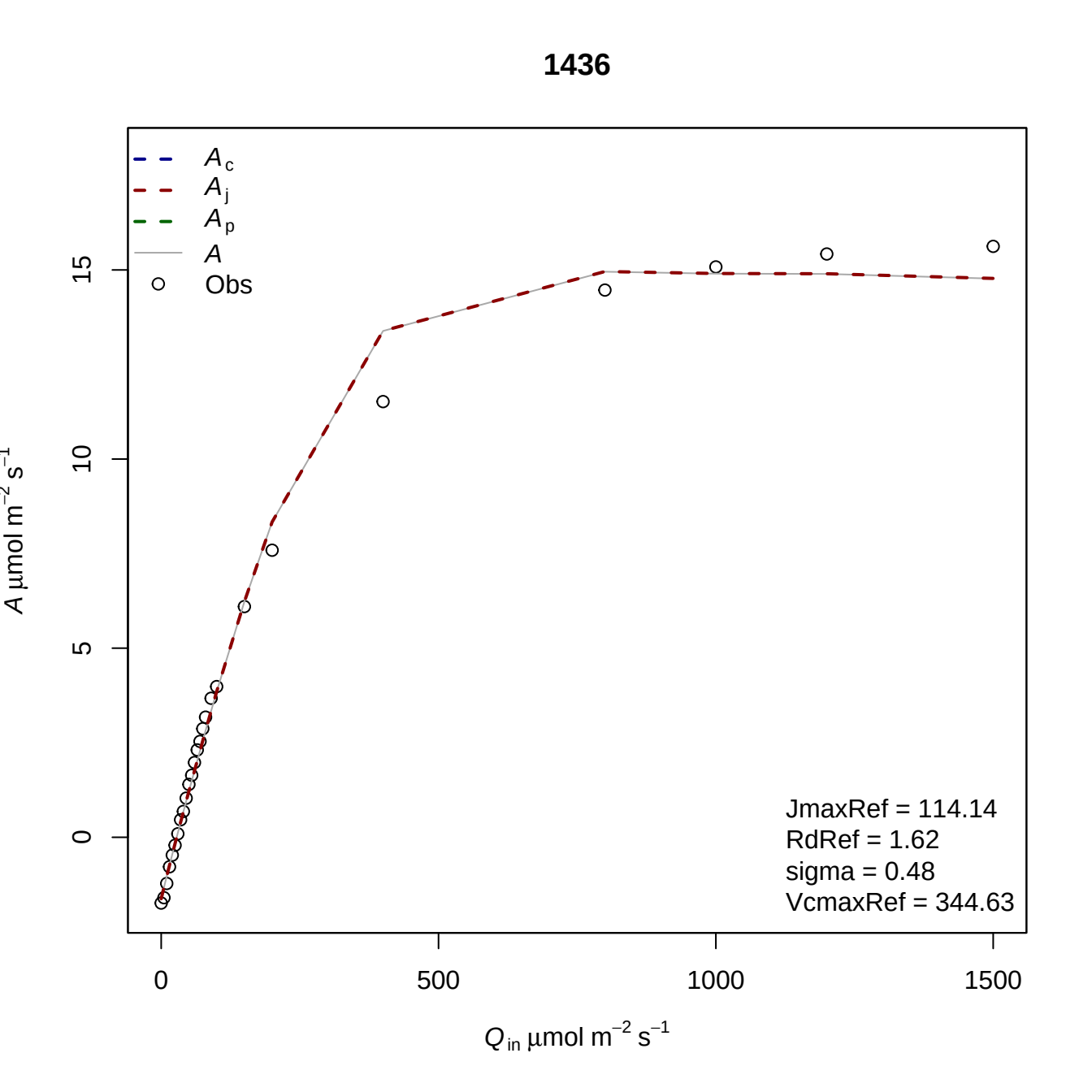

1431
A_c
A_j
A_p
A
Obs
JmaxRef = 95.55
RdRef = 4.33
sigma = 1.11
VcmaxRef = 309.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

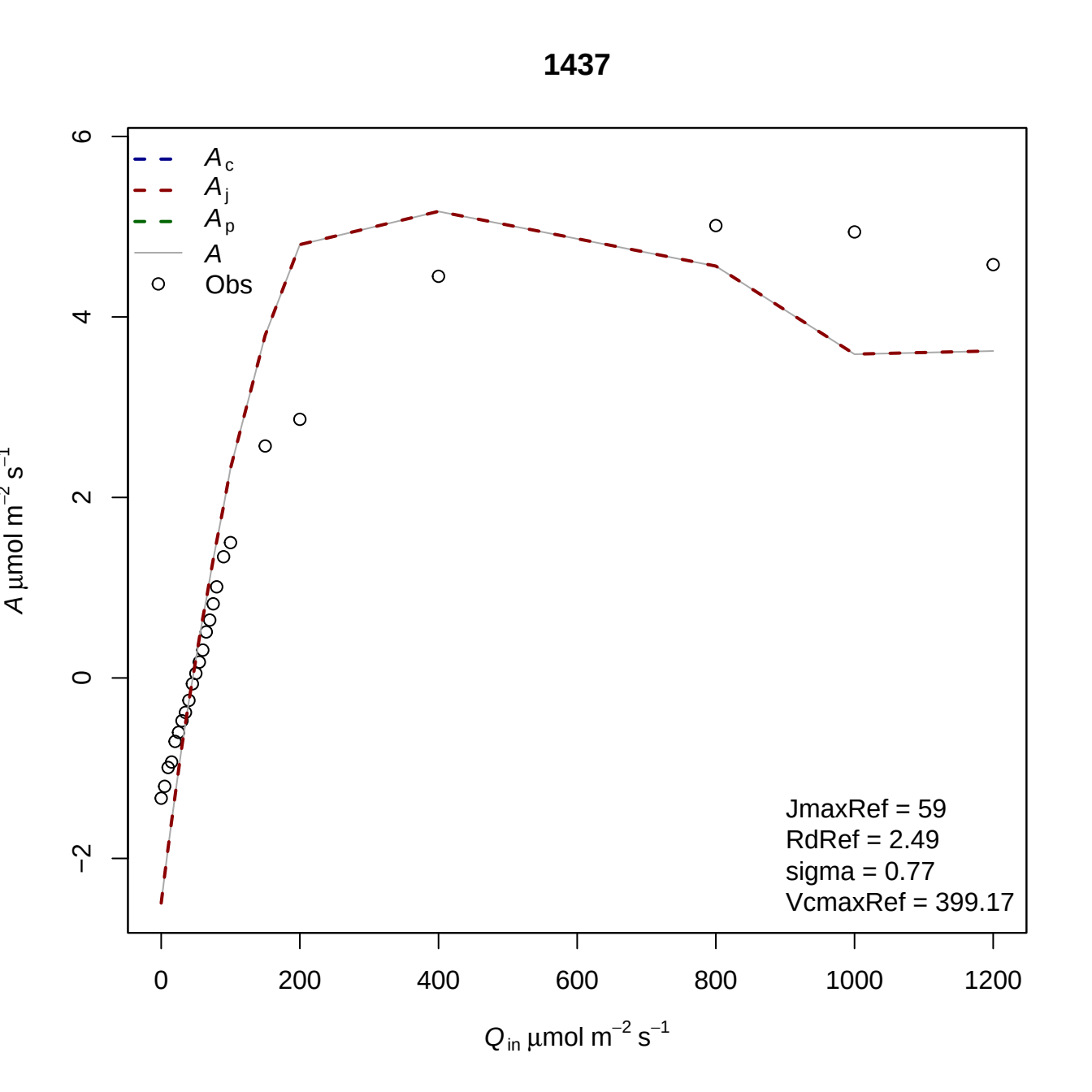

1432
A_c
A_j
A_p
A
Obs
JmaxRef = 117.18
RdRef = 4.08
sigma = 1.11
VcmaxRef = 488.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

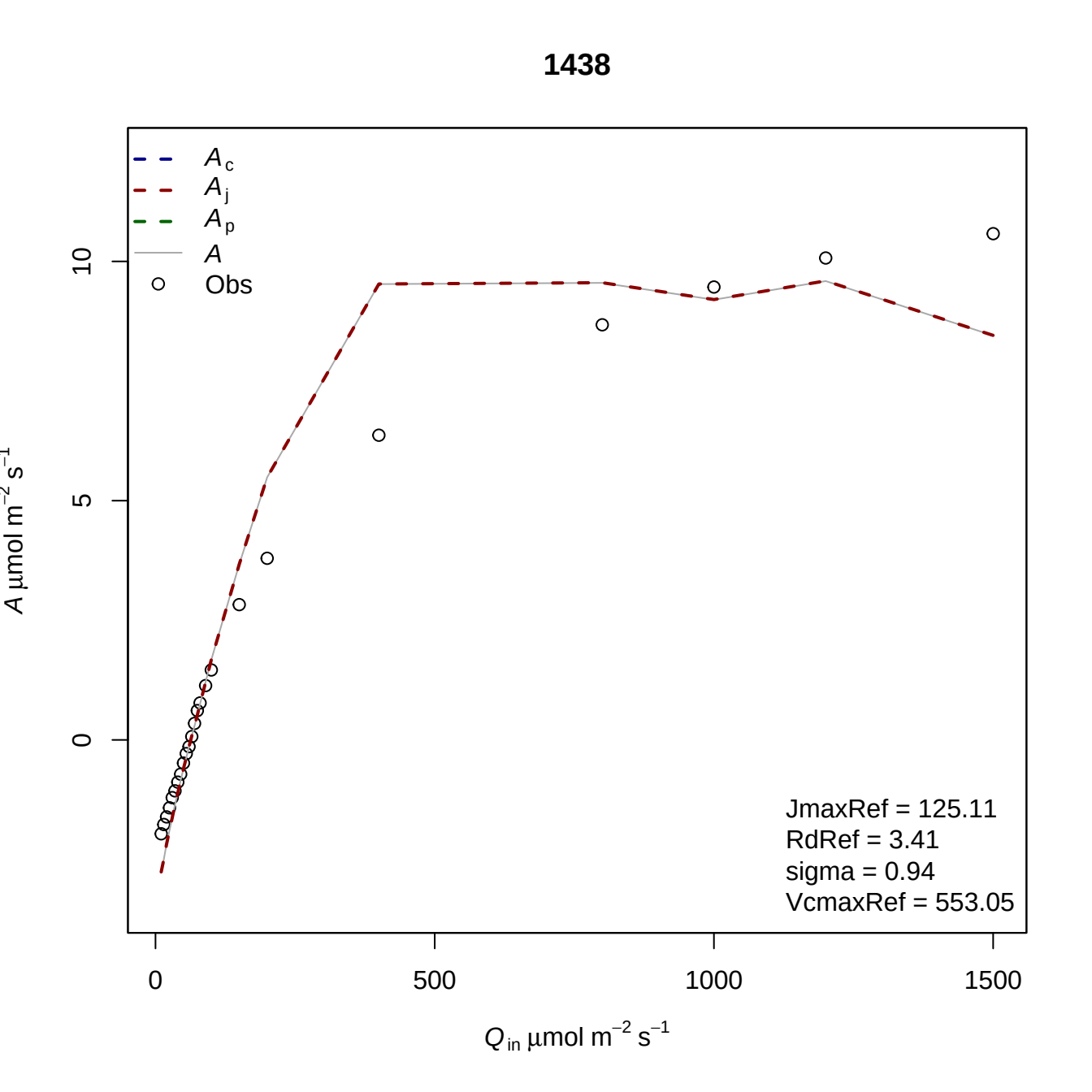

1433
A_c
A_j
A_p
A
Obs
JmaxRef = 70.03
RdRef = 2.18
sigma = 0.92
VcmaxRef = 265.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

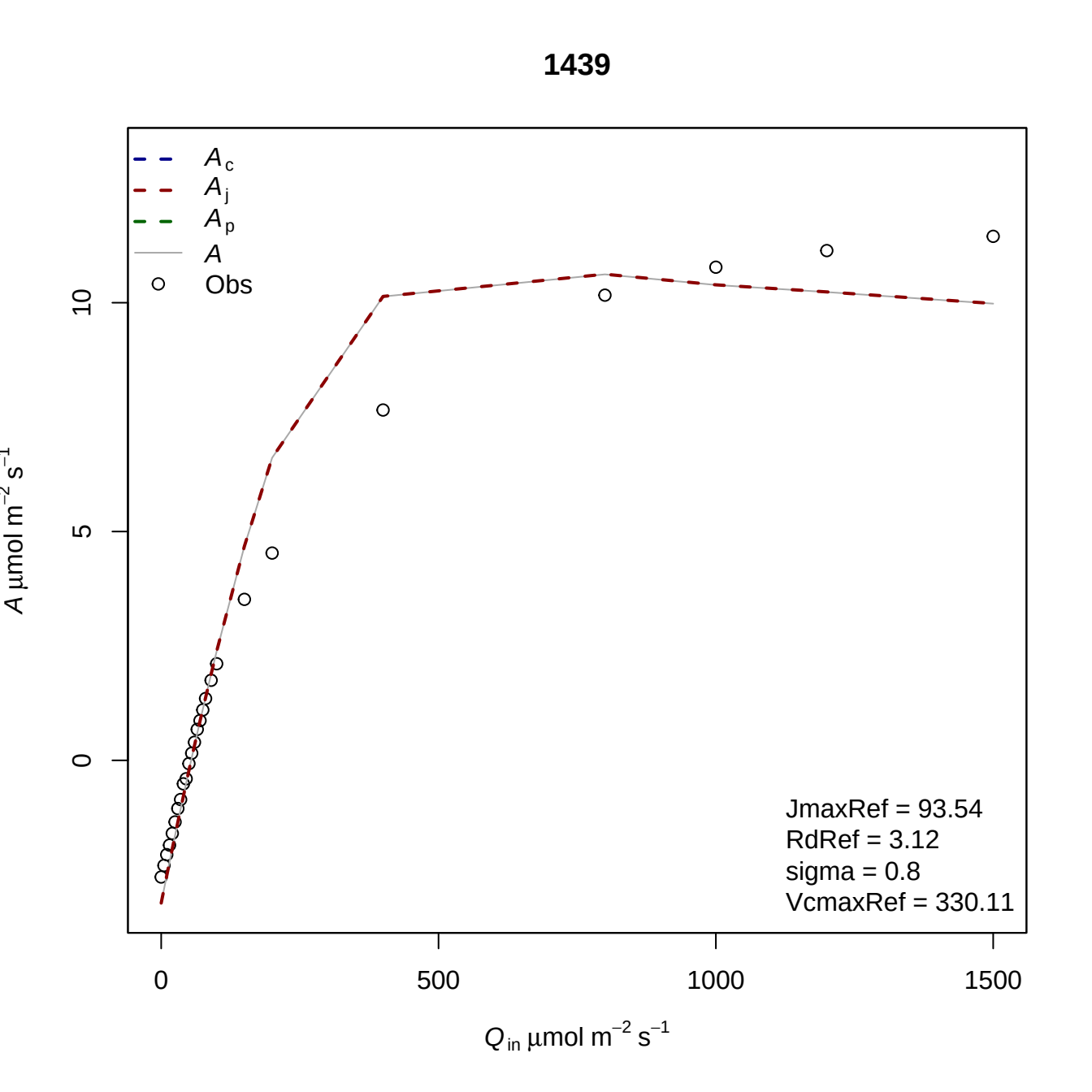

1434
A_c
A_j
A_p
A
Obs
JmaxRef = 81.52
RdRef = 5.37
sigma = 0.97
VcmaxRef = 627.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

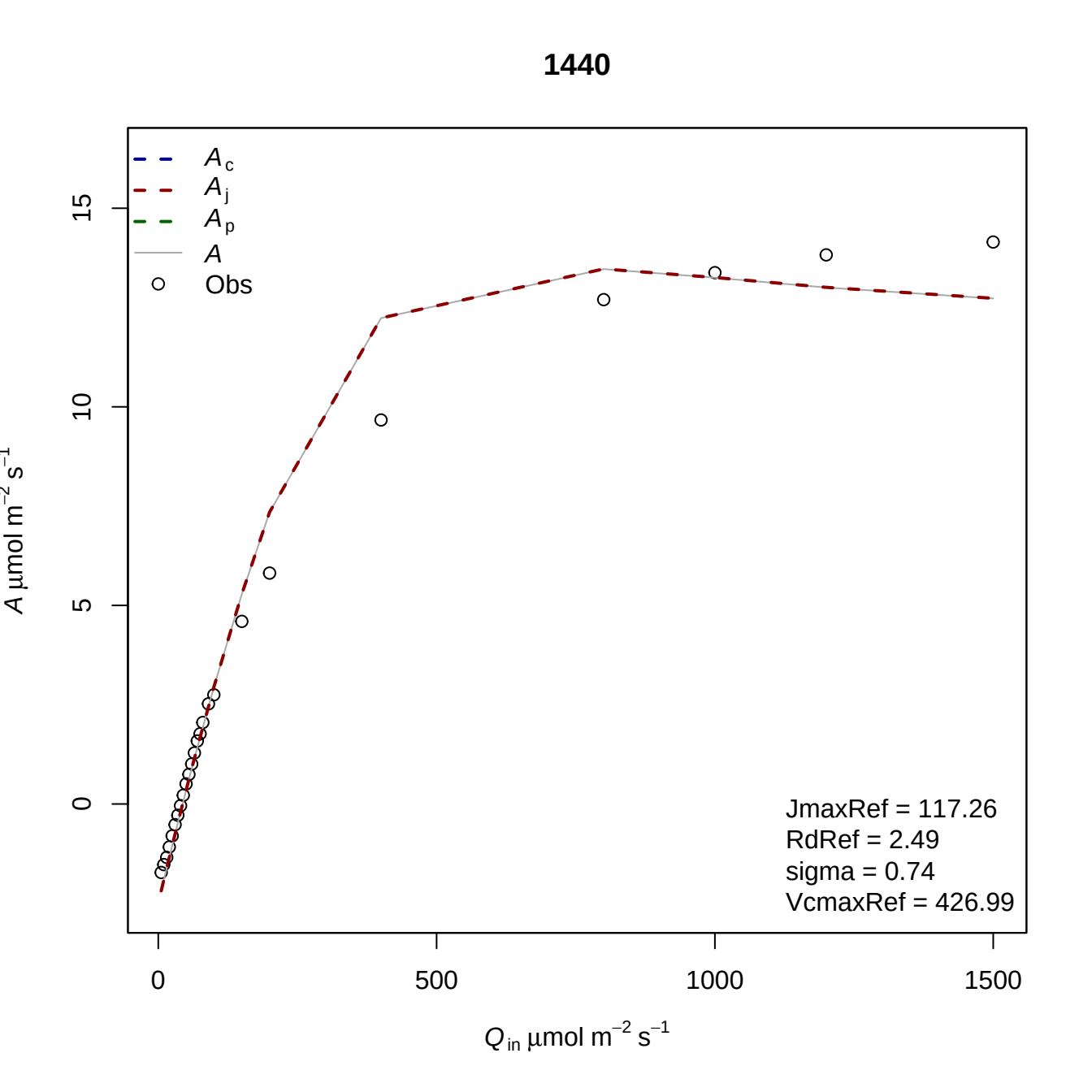

1435
A_c
A_j
A_p
A
Obs
JmaxRef = 63.12
RdRef = 5.3
sigma = 0.85
VcmaxRef = 286.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

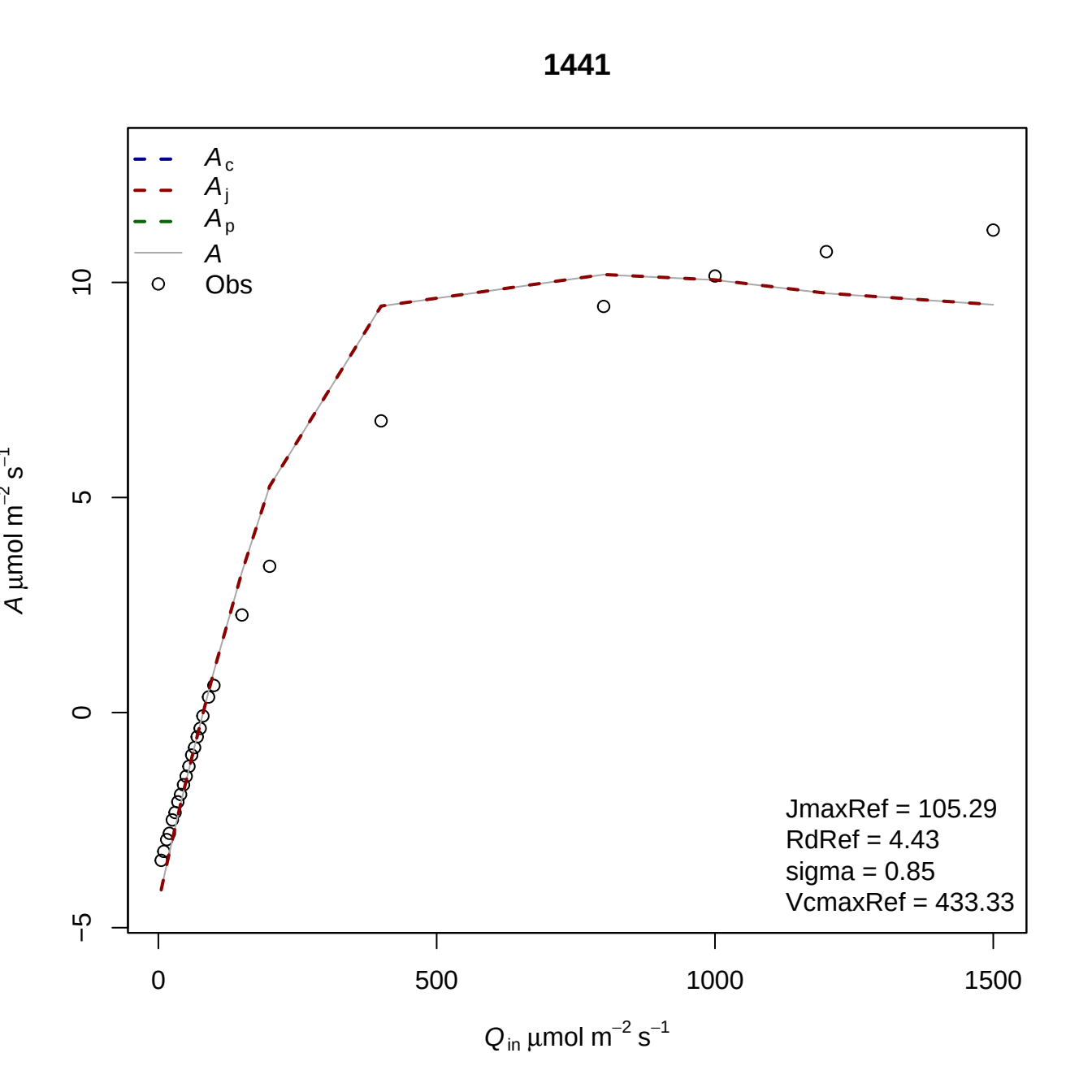

1436
A_c
A_j
A_p
A
Obs
JmaxRef = 114.14
RdRef = 1.62
sigma = 0.48
VcmaxRef = 344.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

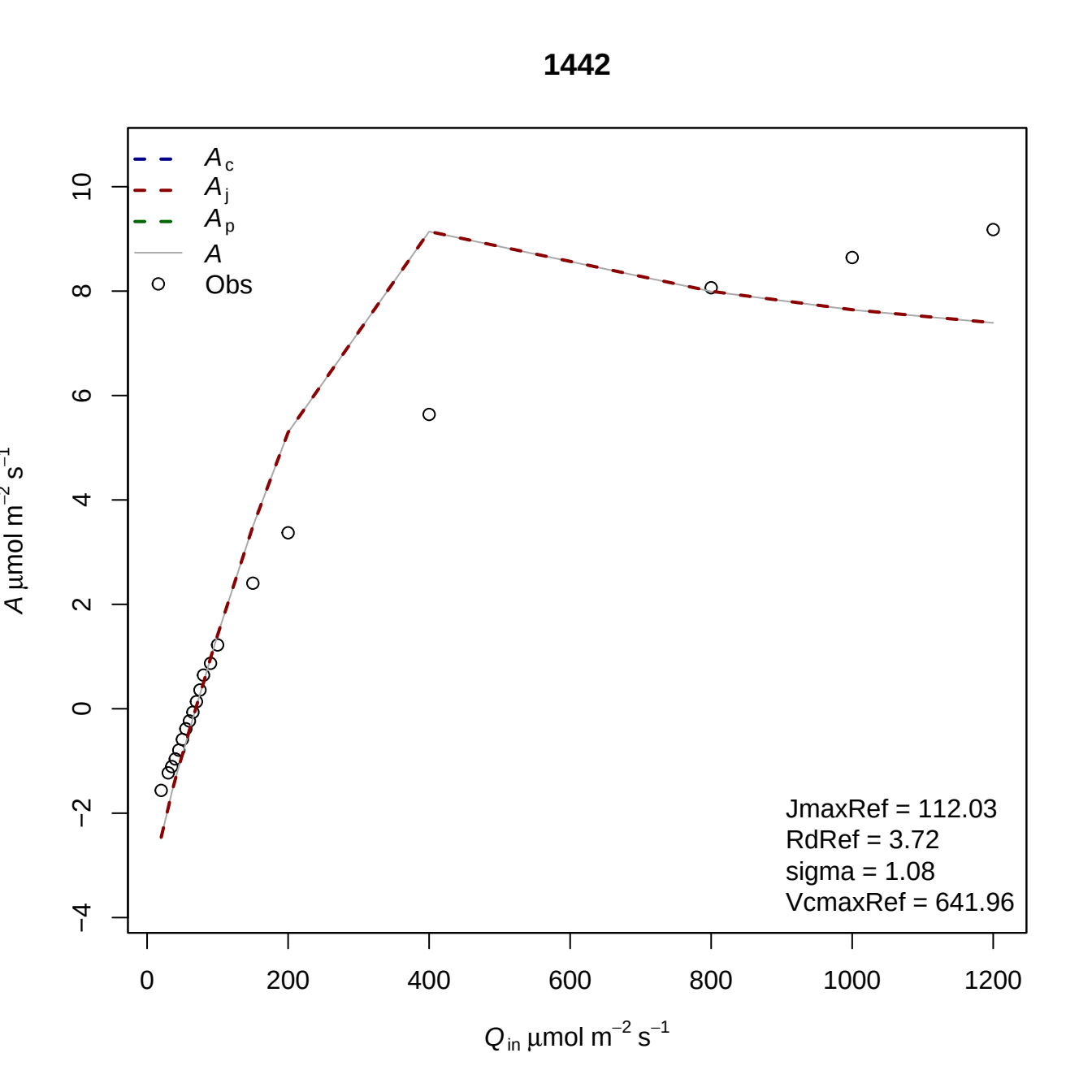

1437
A_c
A_j
A_p
A
Obs
JmaxRef = 59
RdRef = 2.49
sigma = 0.77
VcmaxRef = 399.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

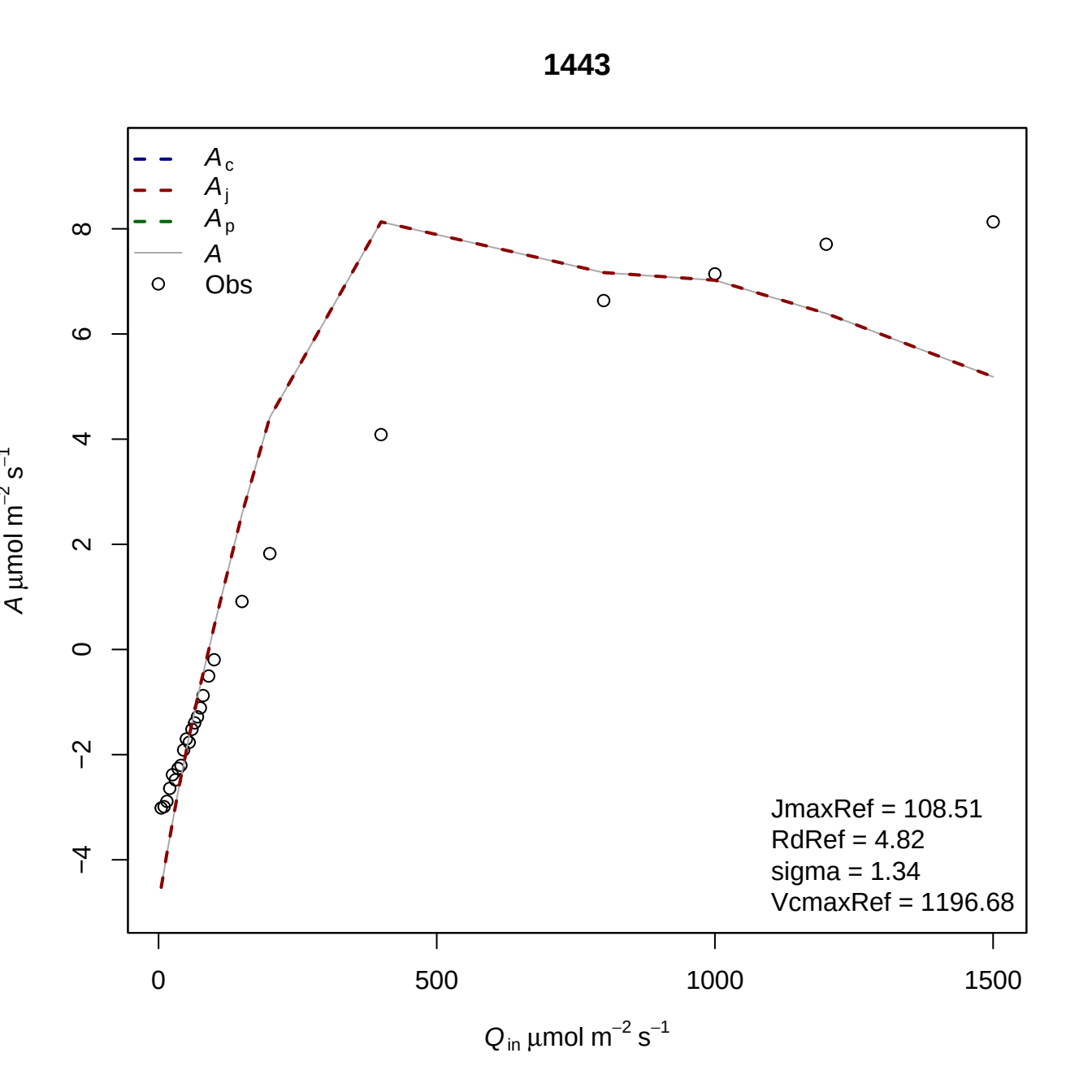

1438
A_c
A_j
A_p
A
Obs
JmaxRef = 125.11
RdRef = 3.41
sigma = 0.94
VcmaxRef = 553.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

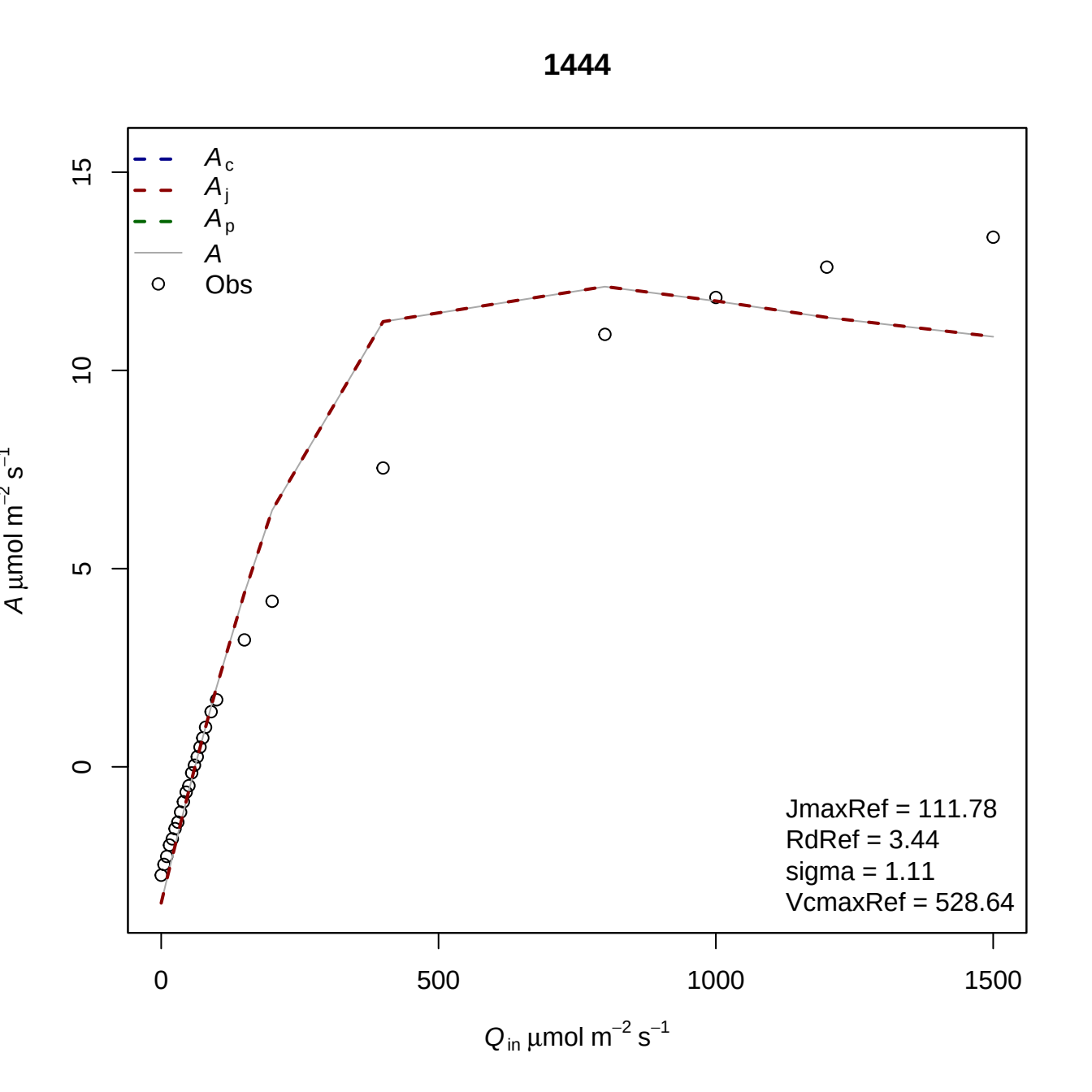

1439
A_c
A_j
A_p
A
Obs
JmaxRef = 93.54
RdRef = 3.12
sigma = 0.8
VcmaxRef = 330.11
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

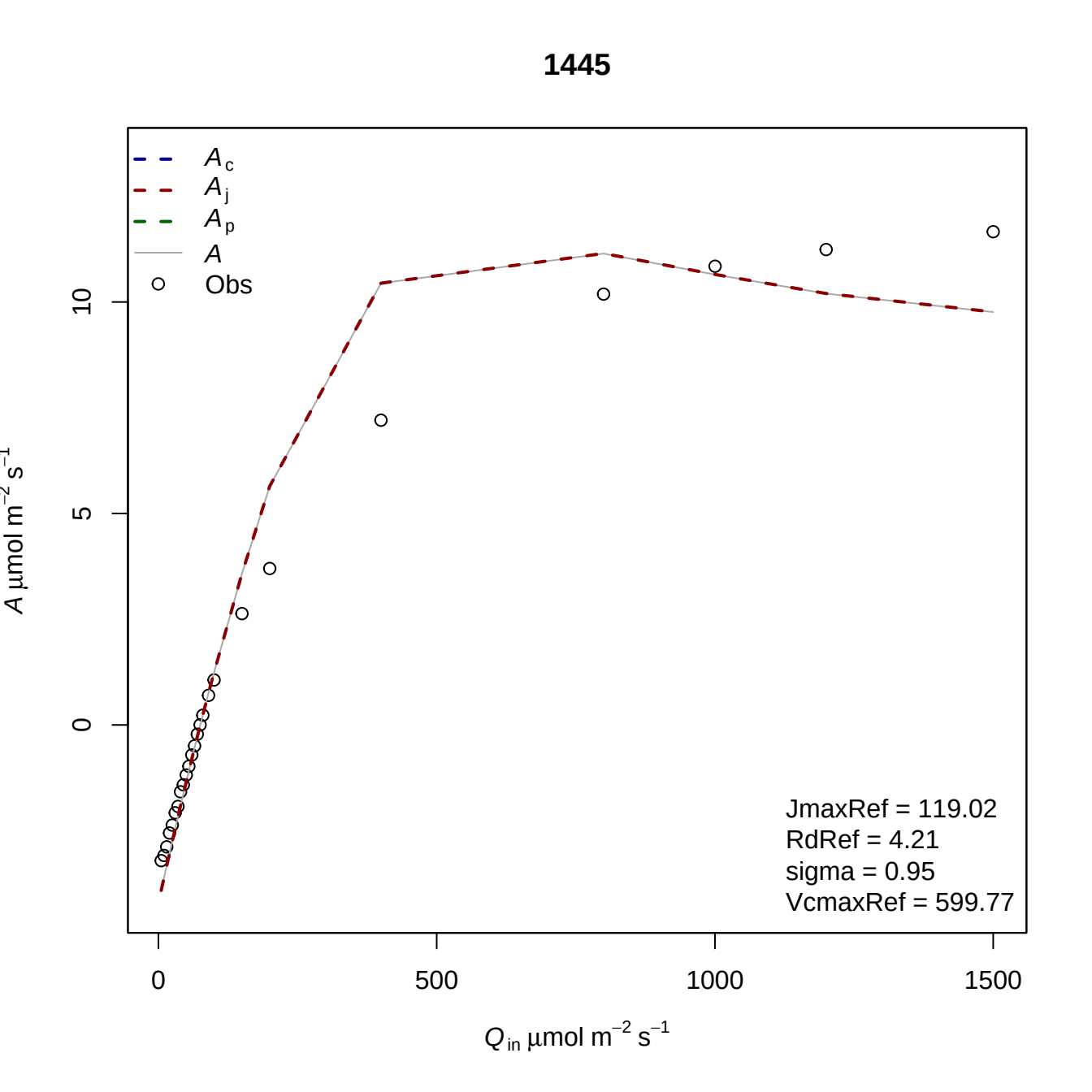

1440
A_c
A_j
A_p
A
Obs
JmaxRef = 117.26
RdRef = 2.49
sigma = 0.74
VcmaxRef = 426.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

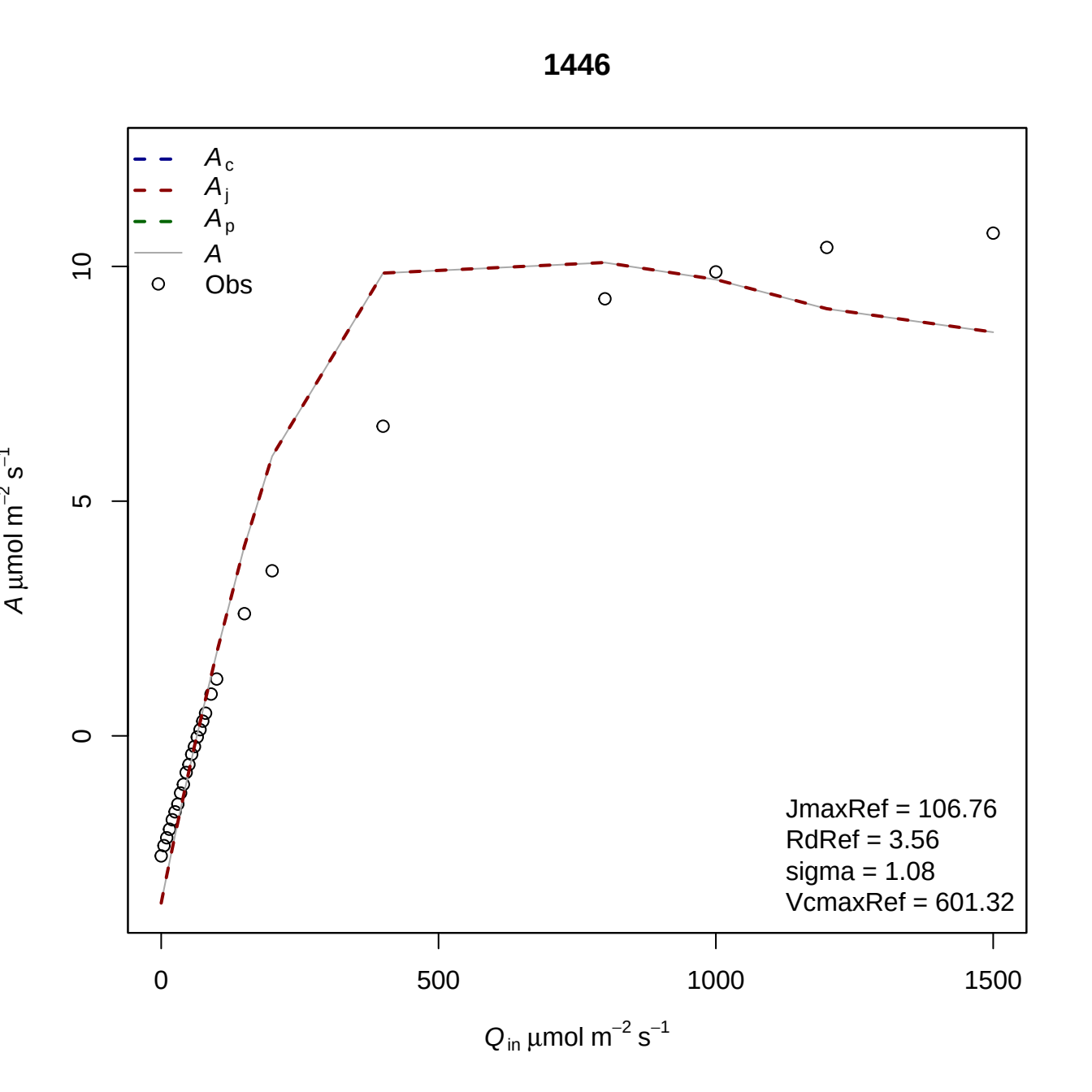

1441
A_c
A_j
A_p
A
Obs
JmaxRef = 105.29
RdRef = 4.43
sigma = 0.85
VcmaxRef = 433.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

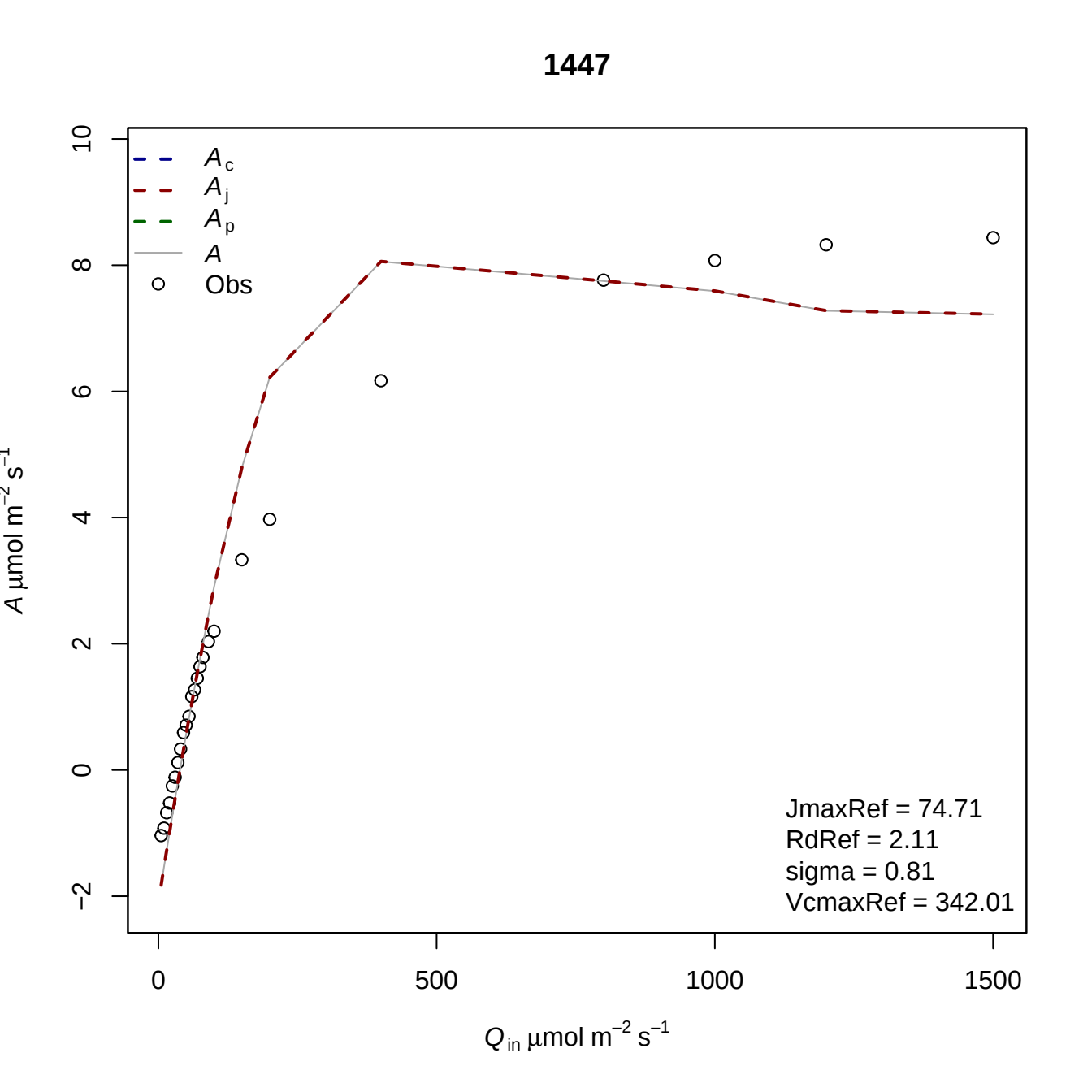

1442
A_c
A_j
A_p
A
Obs
JmaxRef = 112.03
RdRef = 3.72
sigma = 1.08
VcmaxRef = 641.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

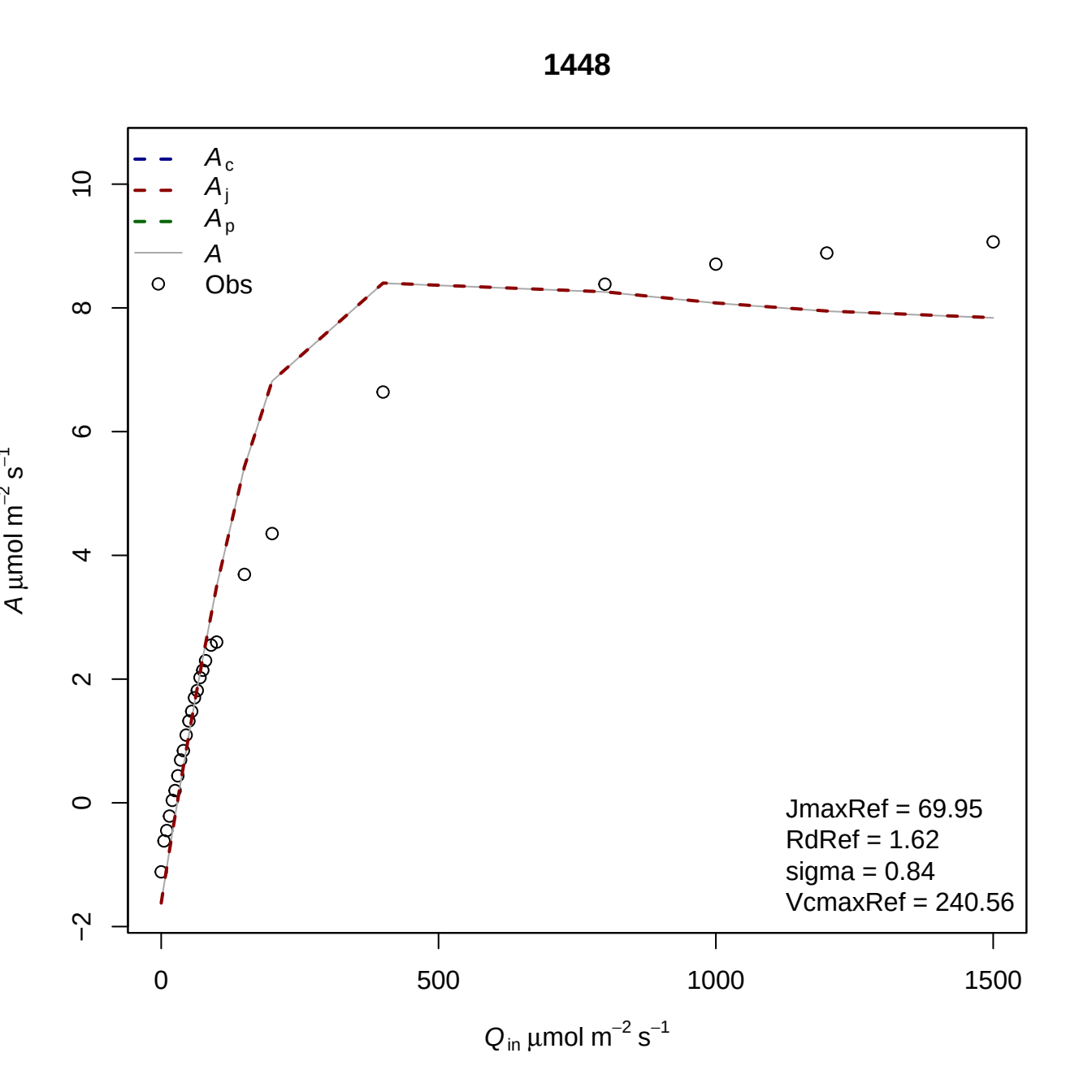

1443
A_c
A_j
A_p
A
Obs
JmaxRef = 108.51
RdRef = 4.82
sigma = 1.34
VcmaxRef = 1196.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

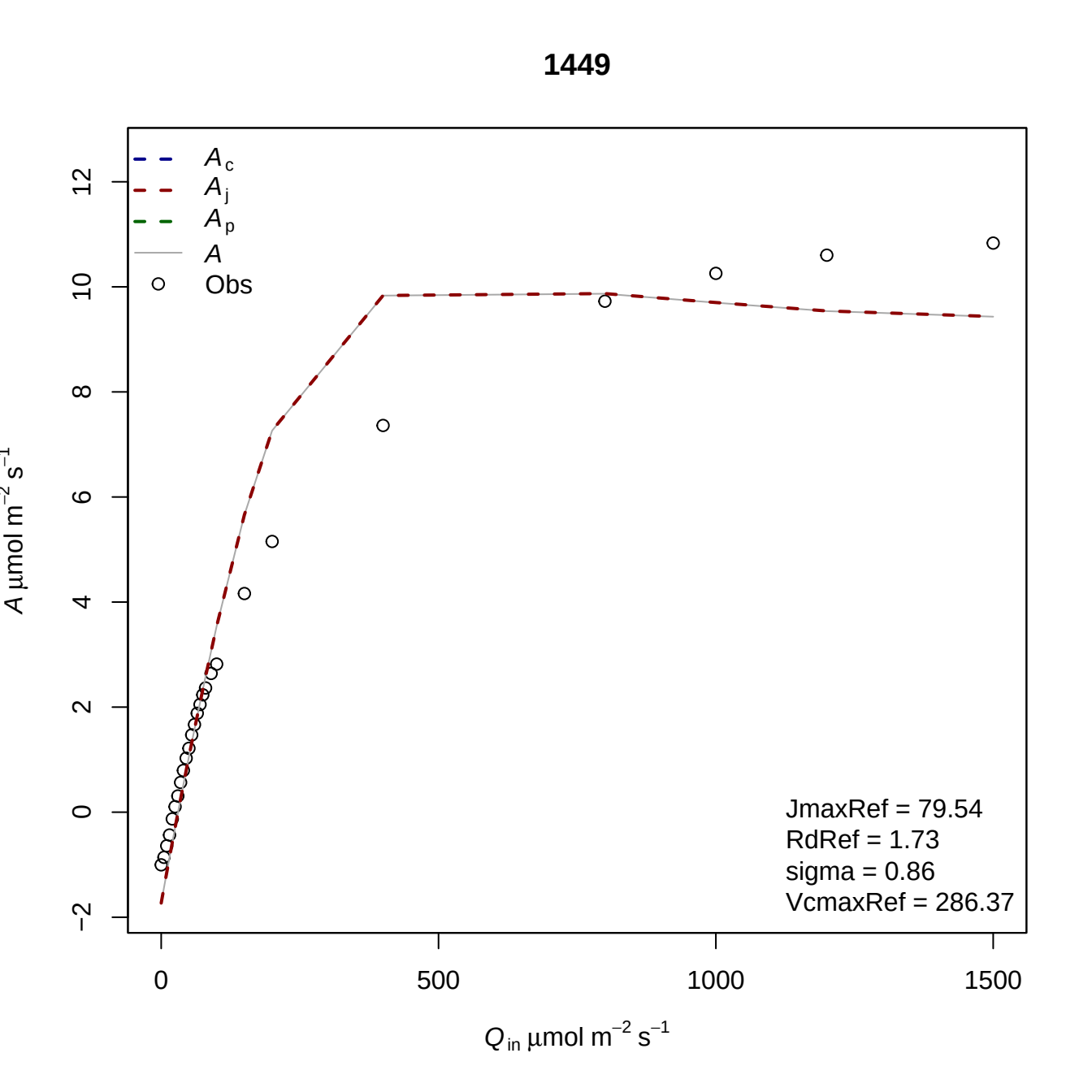

1444
A_c
A_j
A_p
A
Obs
JmaxRef = 111.78
RdRef = 3.44
sigma = 1.11
VcmaxRef = 528.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

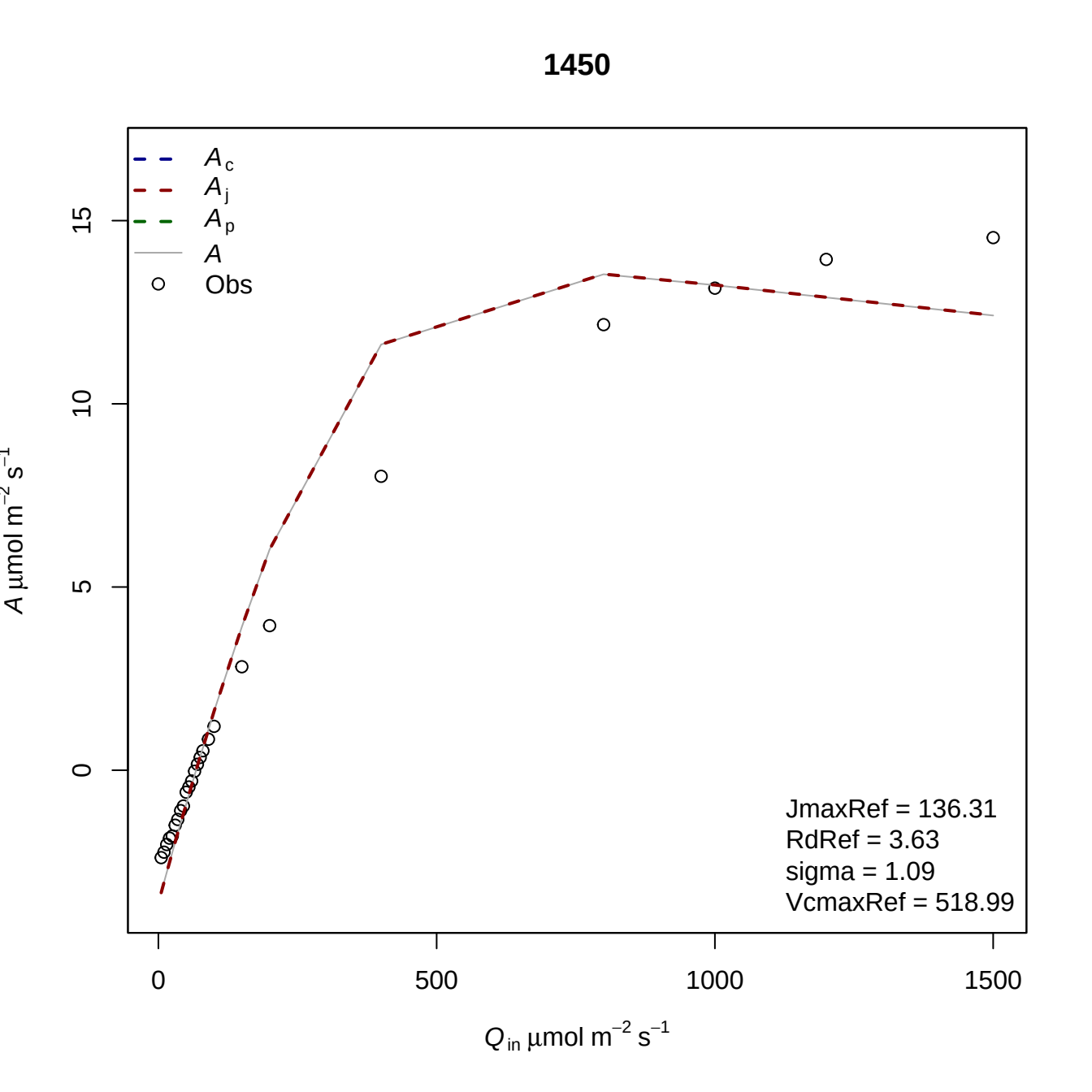

1445
A_c
A_j
A_p
A
Obs
JmaxRef = 119.02
RdRef = 4.21
sigma = 0.95
VcmaxRef = 599.77
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

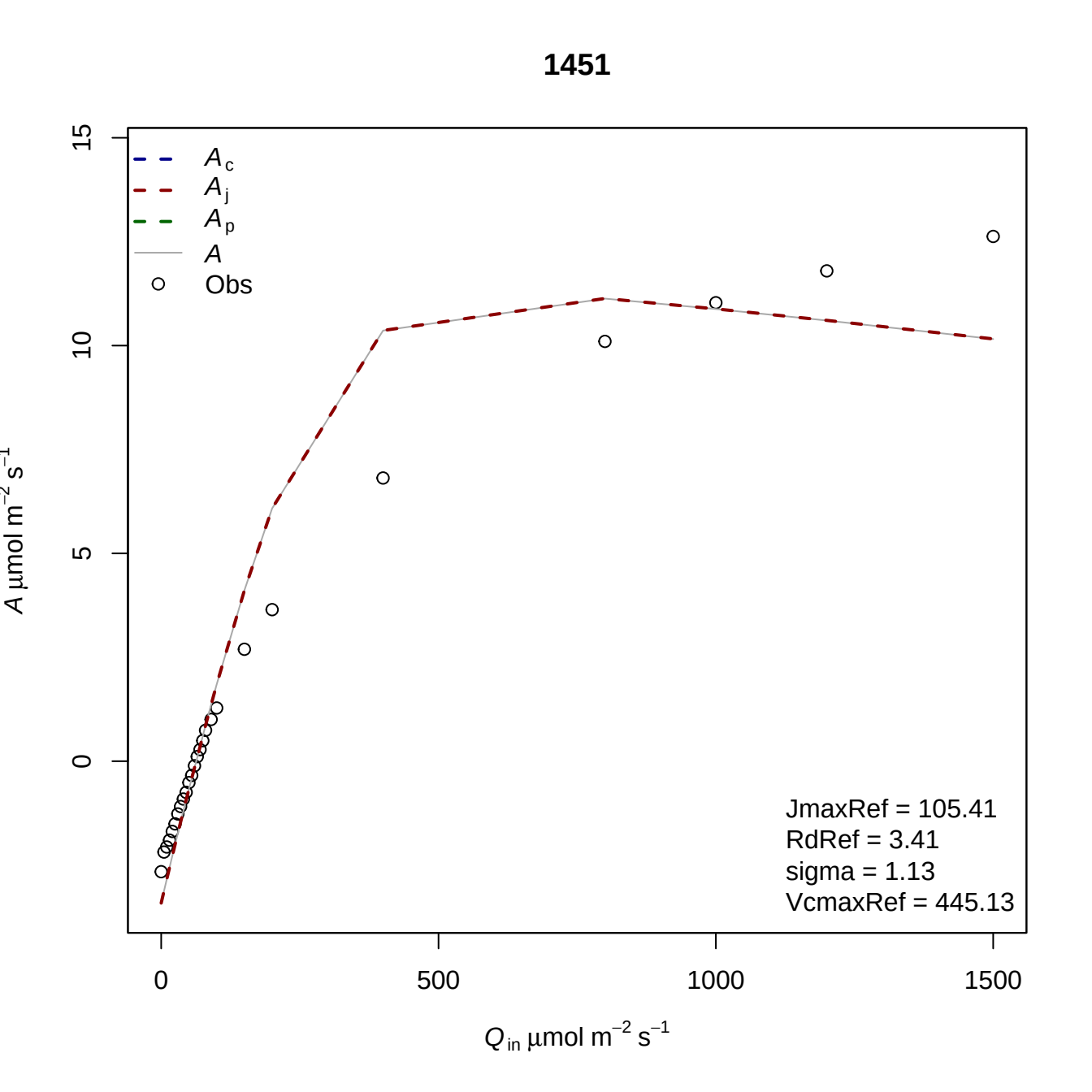

1446
A_c
A_j
A_p
A
Obs
JmaxRef = 106.76
RdRef = 3.56
sigma = 1.08
VcmaxRef = 601.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

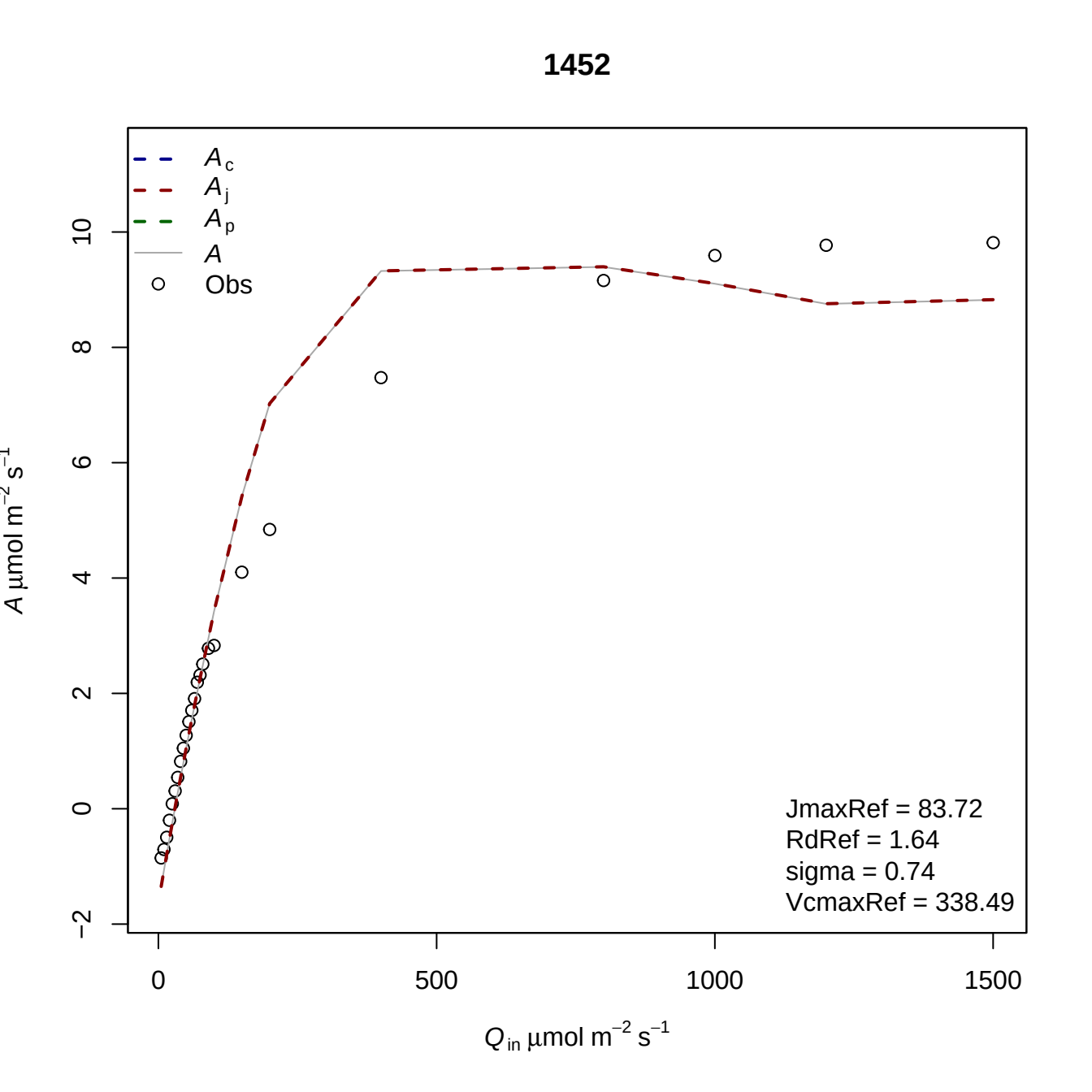

1447
A_c
A_j
A_p
A
Obs
JmaxRef = 74.71
RdRef = 2.11
sigma = 0.81
VcmaxRef = 342.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

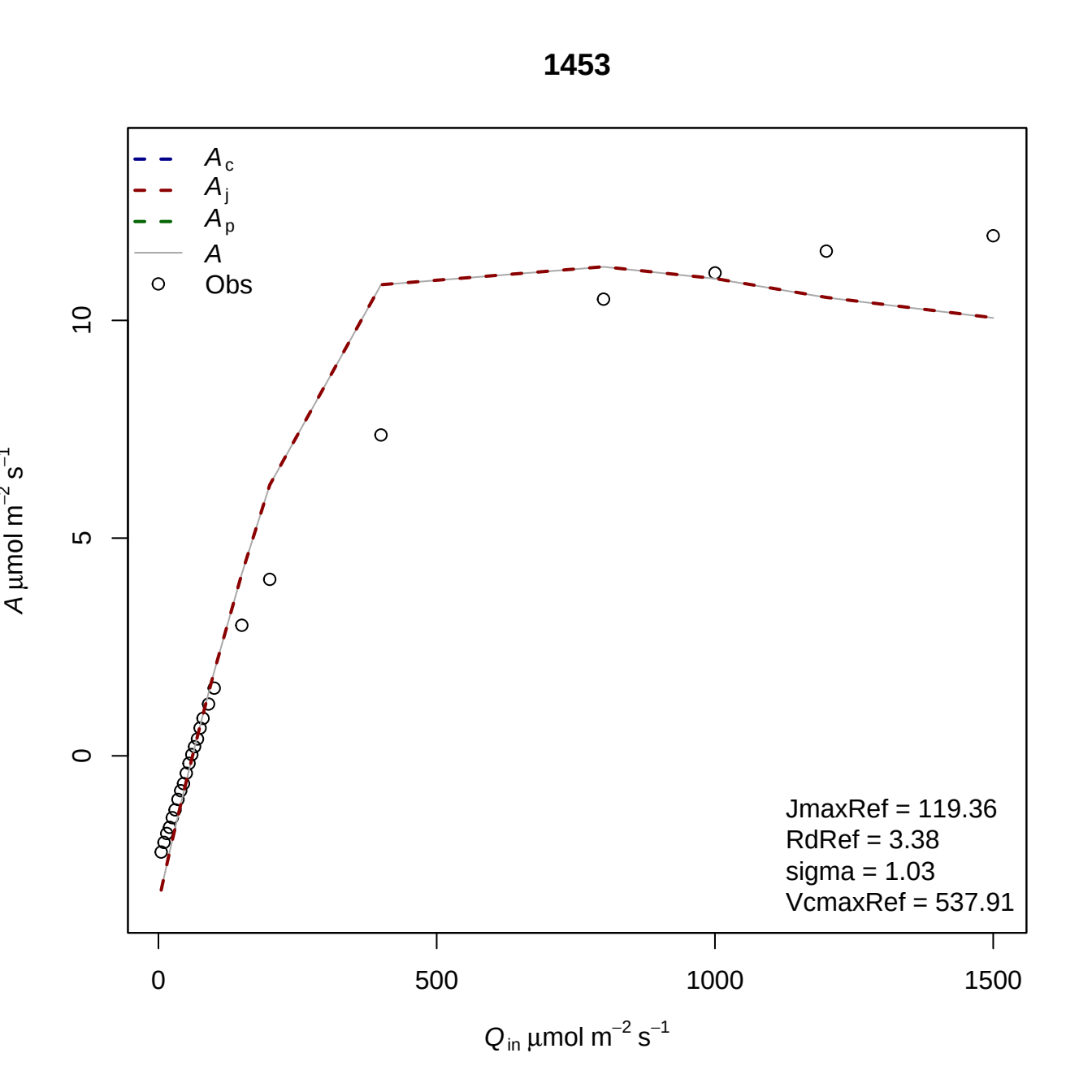

1448
A_c
A_j
A_p
A
Obs
JmaxRef = 69.95
RdRef = 1.62
sigma = 0.84
VcmaxRef = 240.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

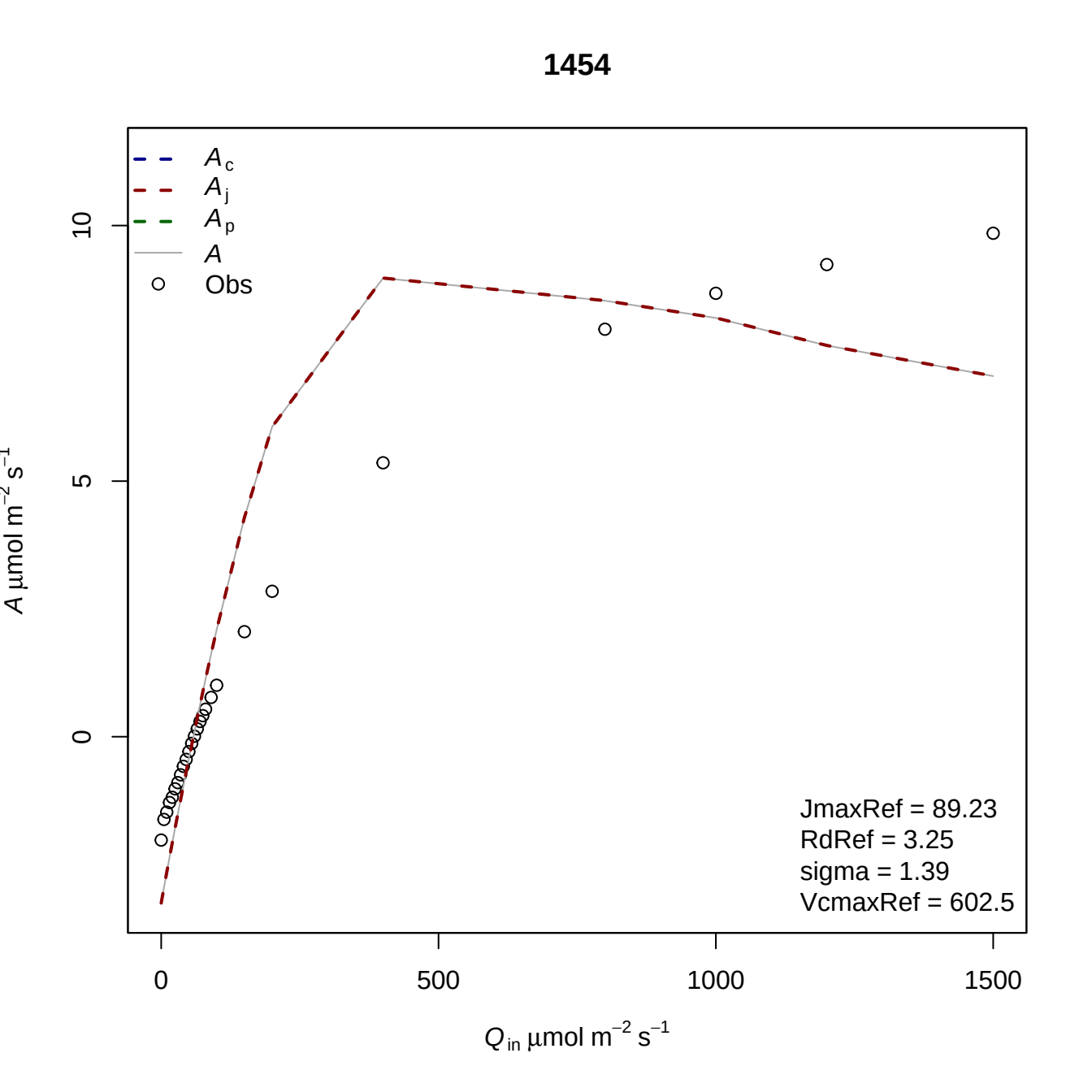

1449
A_c
A_j
A_p
A
Obs
JmaxRef = 79.54
RdRef = 1.73
sigma = 0.86
VcmaxRef = 286.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

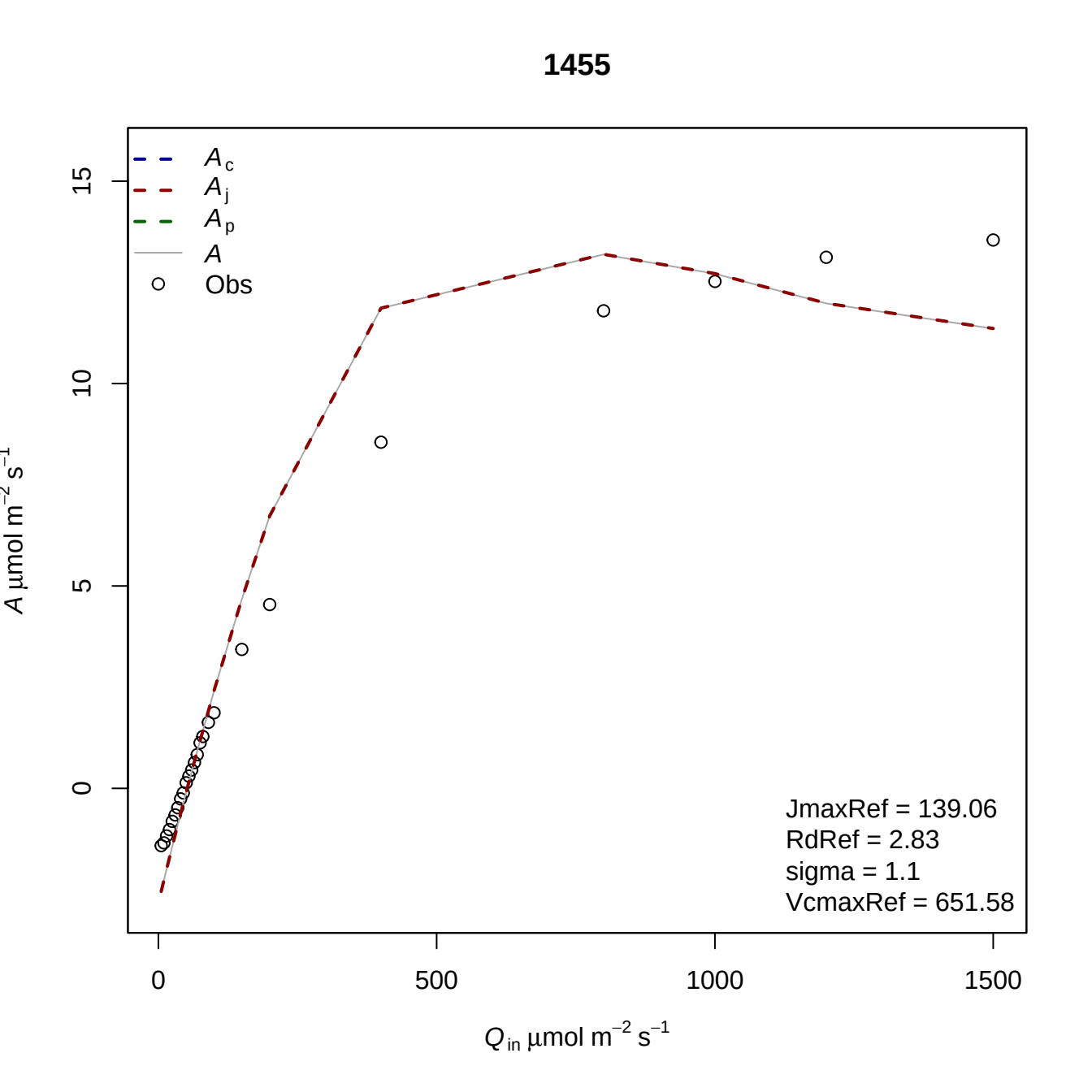

1450
A_c
A_j
A_p
A
Obs
JmaxRef = 136.31
RdRef = 3.63
sigma = 1.09
VcmaxRef = 518.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

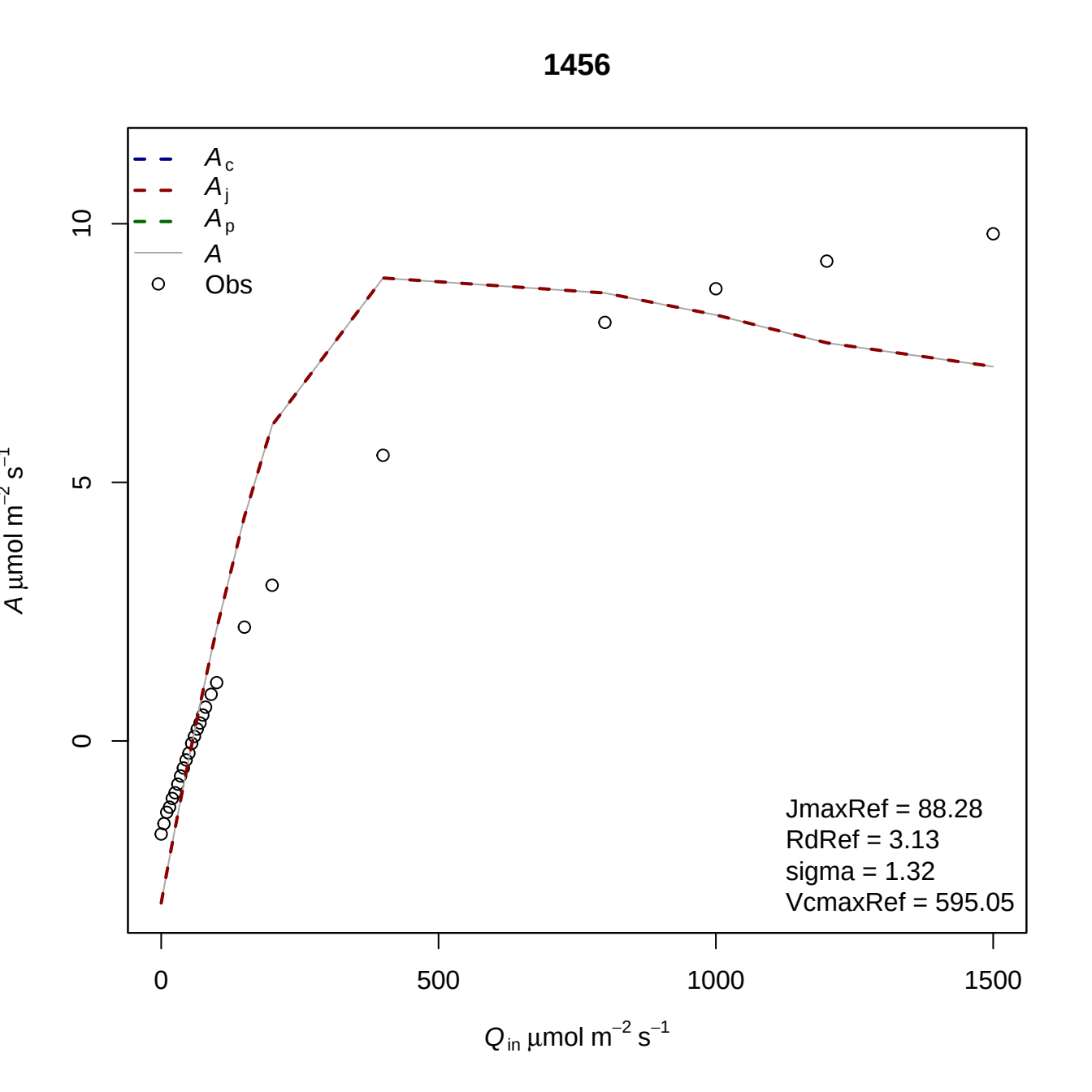

1451
A_c
A_j
A_p
A
Obs
JmaxRef = 105.41
RdRef = 3.41
sigma = 1.13
VcmaxRef = 445.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

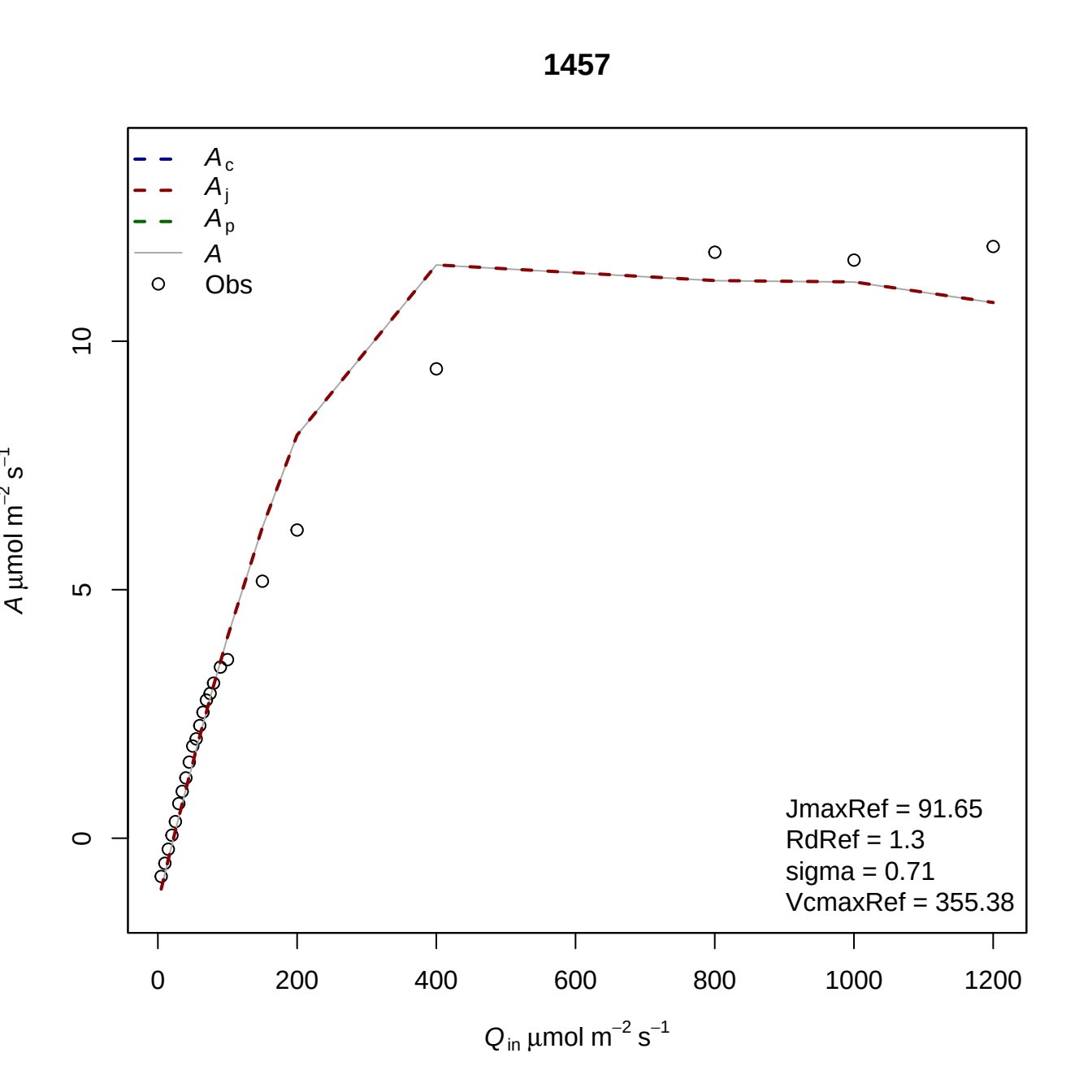

1452
A_c
A_j
A_p
A
Obs
JmaxRef = 83.72
RdRef = 1.64
sigma = 0.74
VcmaxRef = 338.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

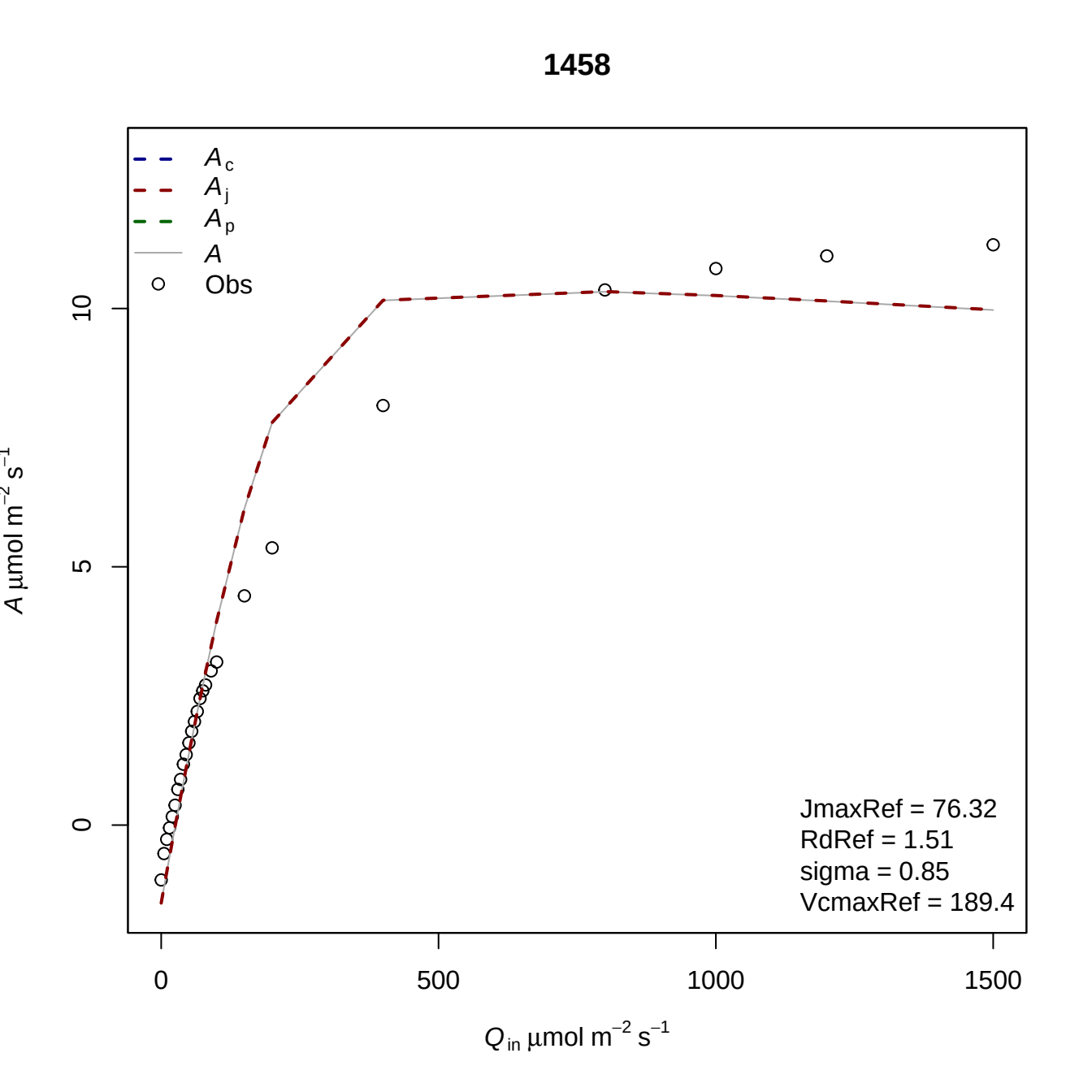

1453
A_c
A_j
A_p
A
Obs
JmaxRef = 119.36
RdRef = 3.38
sigma = 1.03
VcmaxRef = 537.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

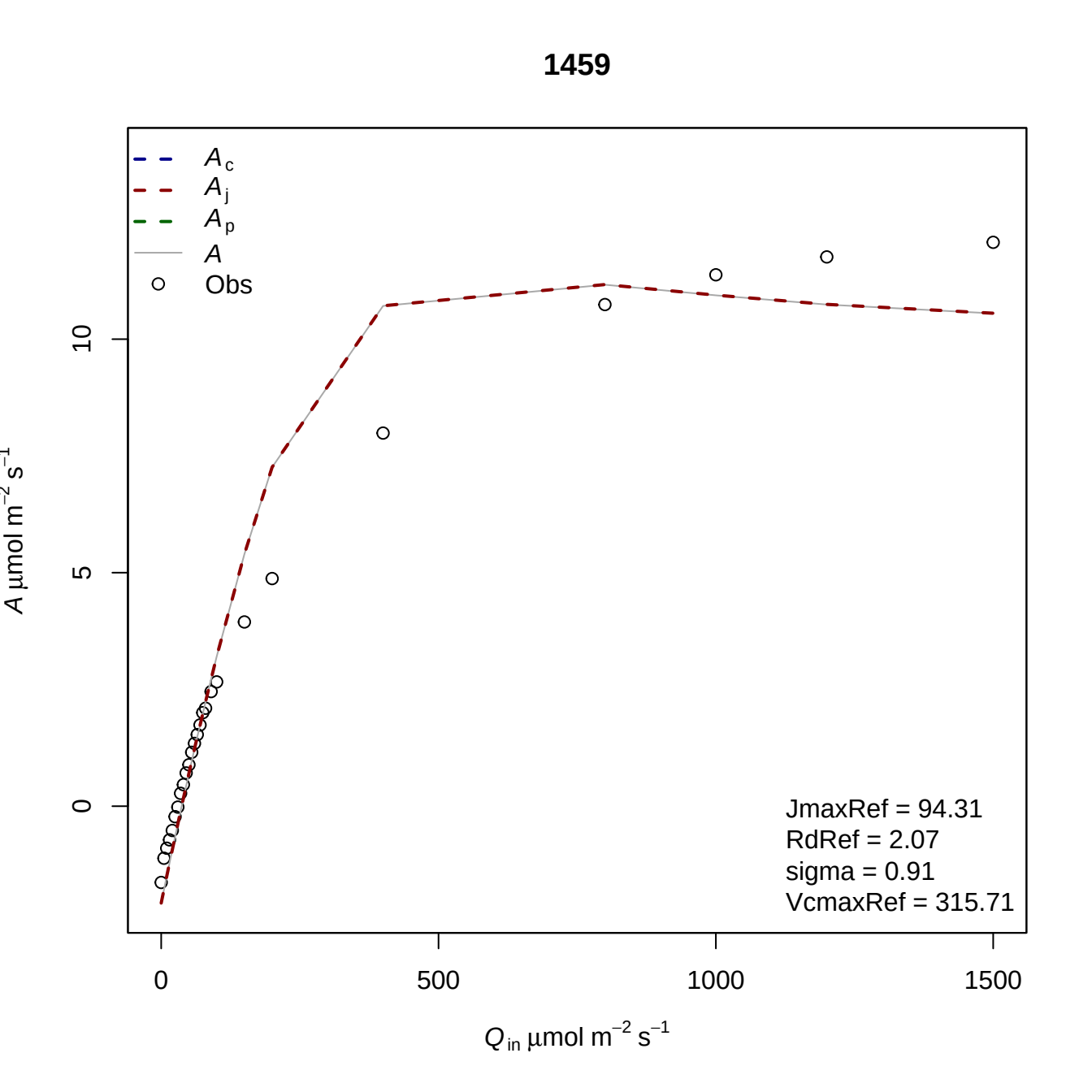

1454
A_c
A_j
A_p
A
Obs
JmaxRef = 89.23
RdRef = 3.25
sigma = 1.39
VcmaxRef = 602.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

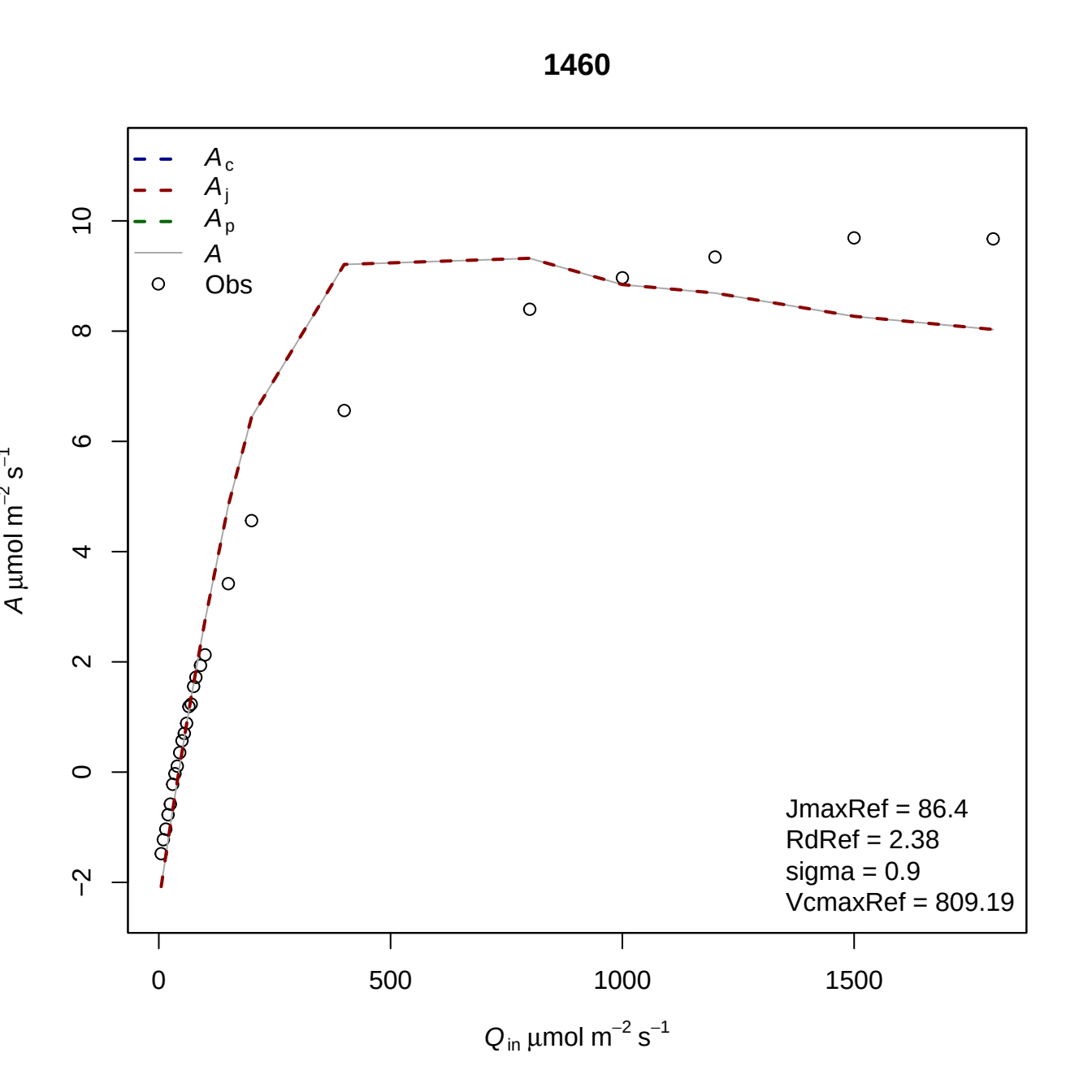

1455
A_c
A_j
A_p
A
Obs
JmaxRef = 139.06
RdRef = 2.83
sigma = 1.1
VcmaxRef = 651.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

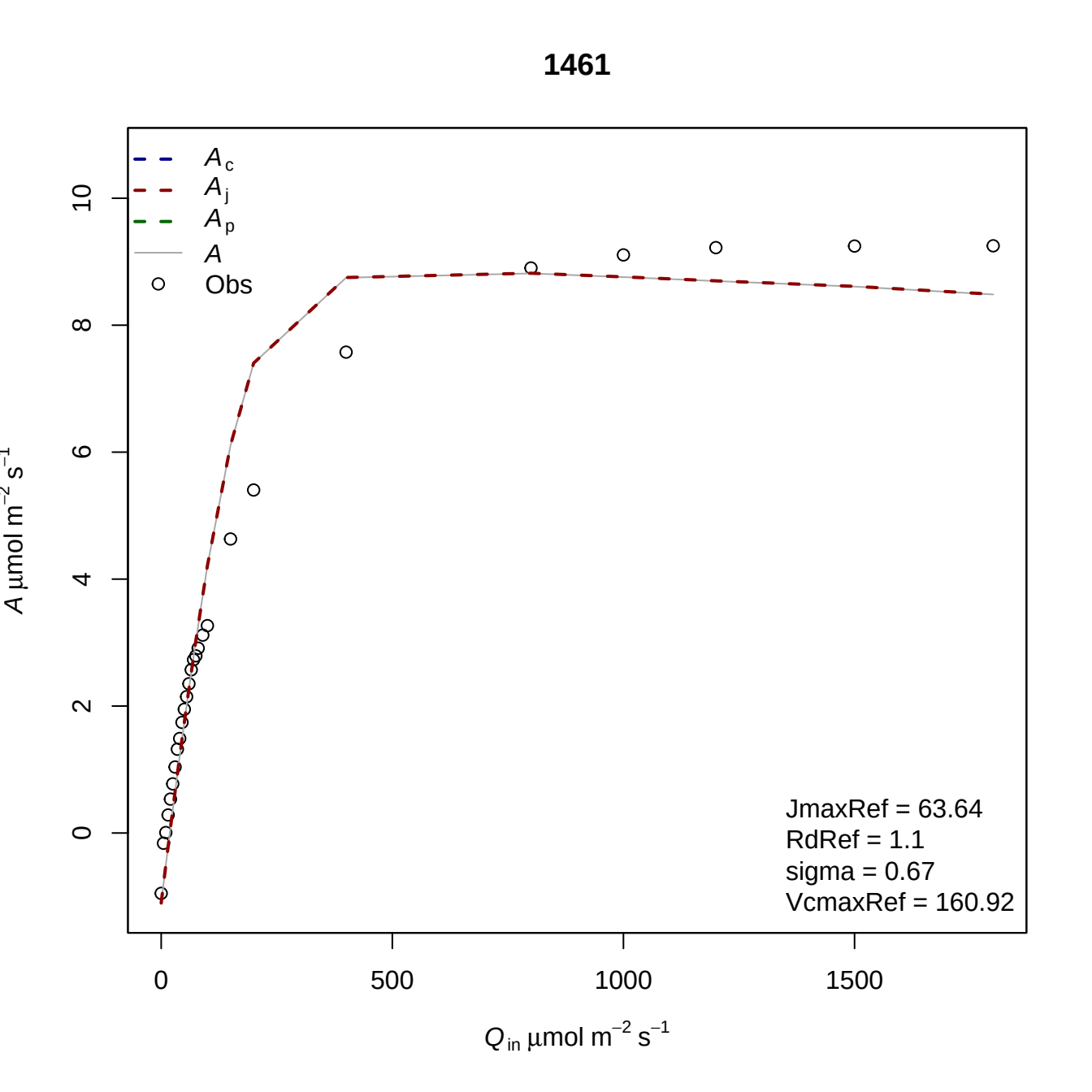

1456
A_c
A_j
A_p
A
Obs
JmaxRef = 88.28
RdRef = 3.13
sigma = 1.32
VcmaxRef = 595.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

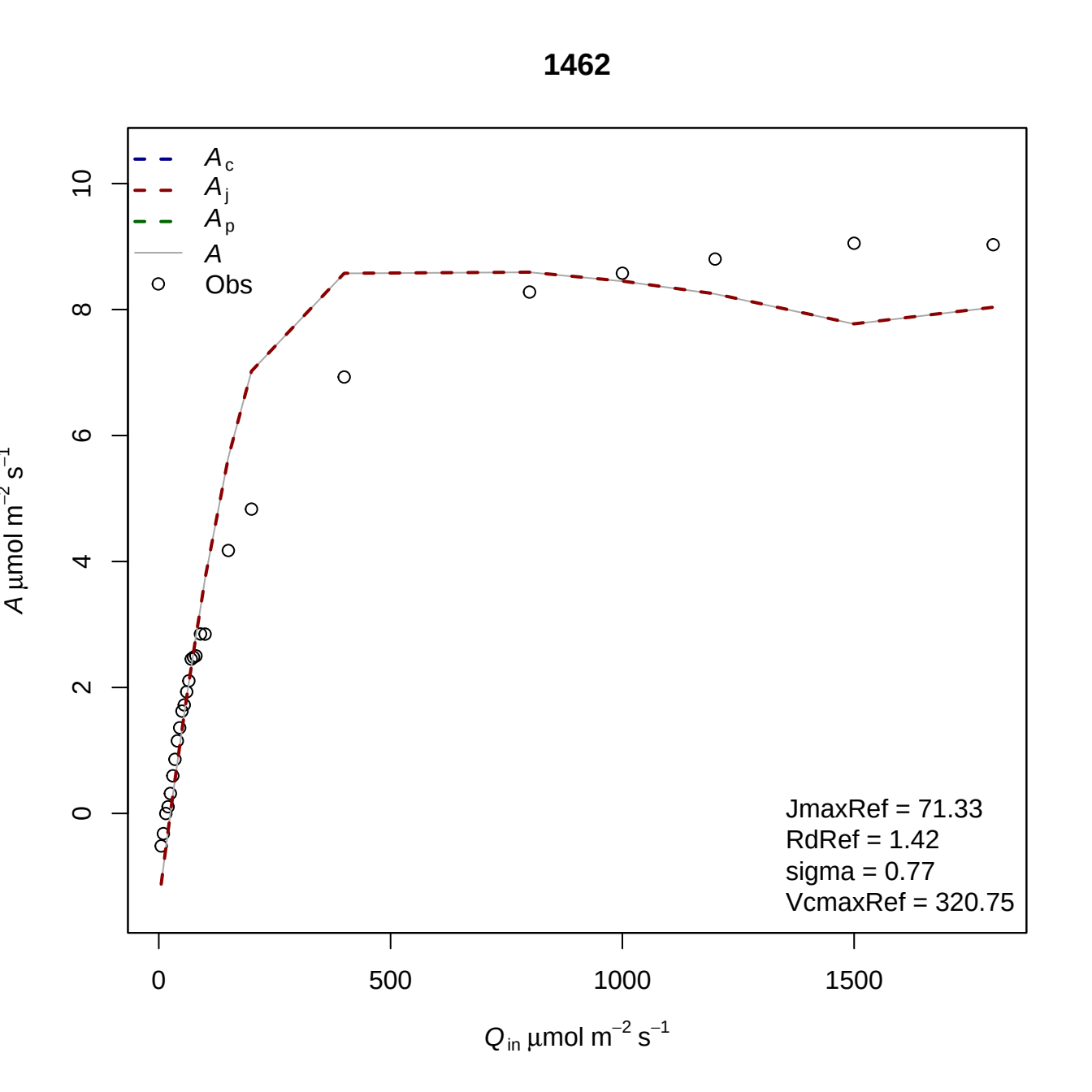

1457
A_c
A_j
A_p
A
Obs
JmaxRef = 91.65
RdRef = 1.3
sigma = 0.71
VcmaxRef = 355.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

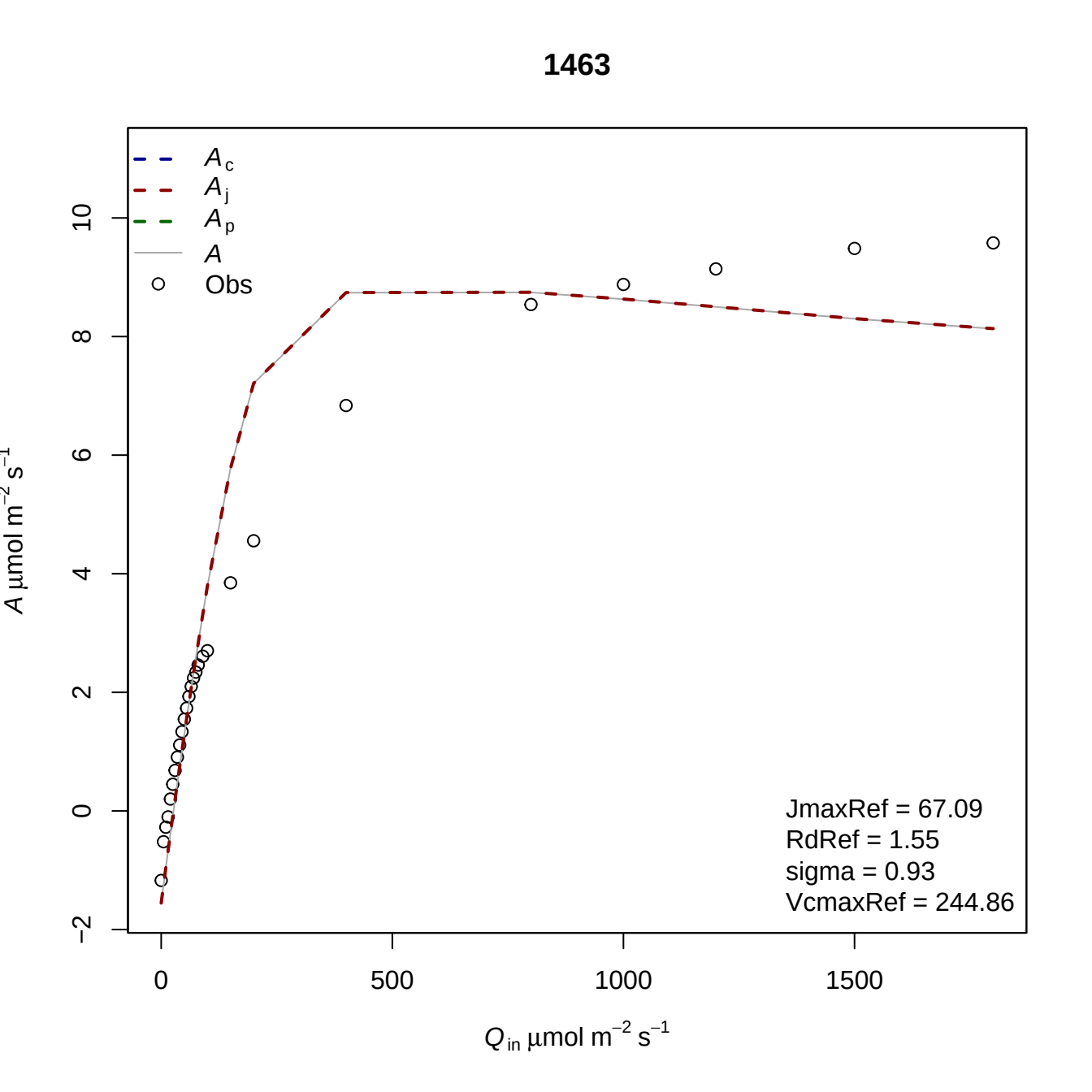

1458
A_c
A_j
A_p
A
Obs
JmaxRef = 76.32
RdRef = 1.51
sigma = 0.85
VcmaxRef = 189.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

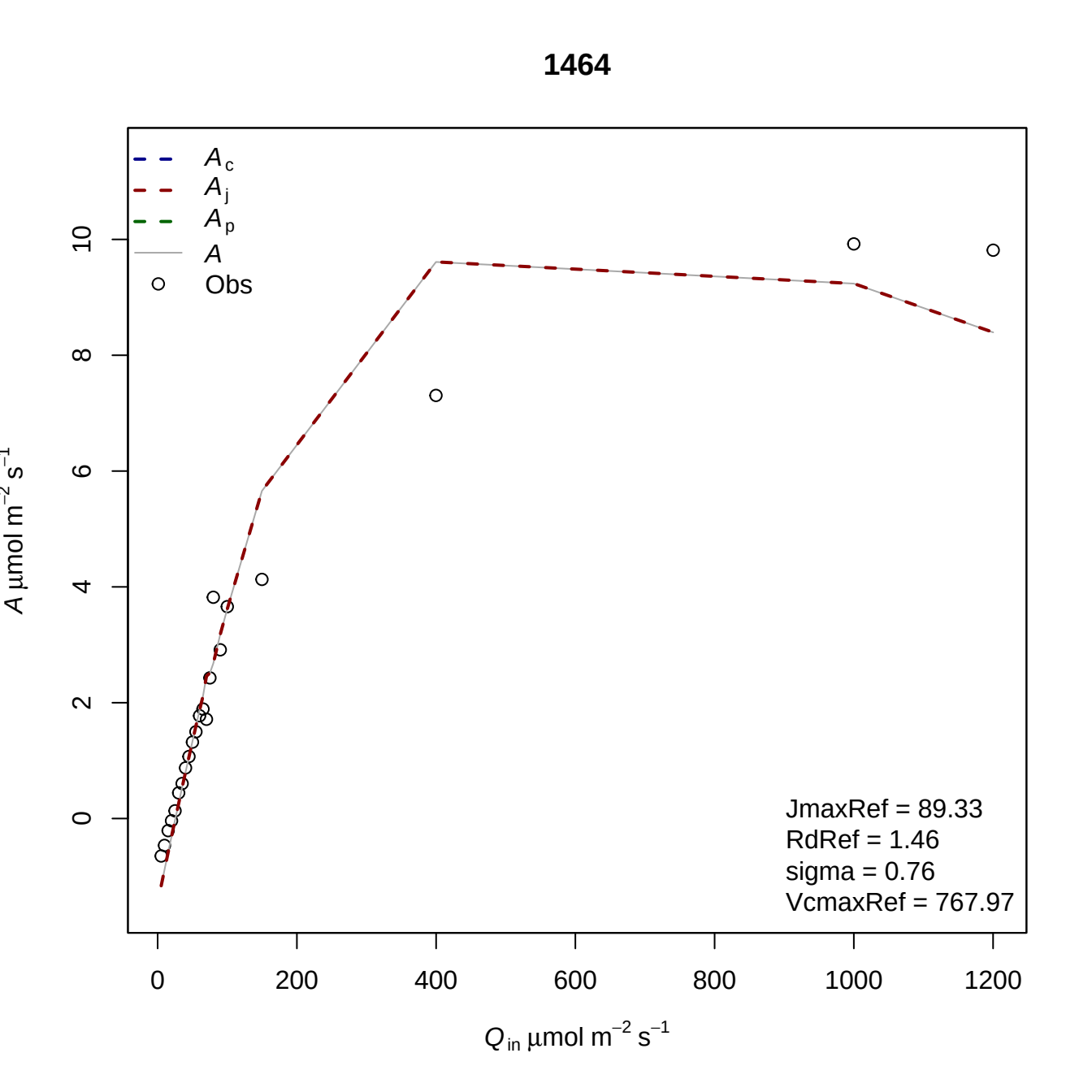

1459
A_c
A_j
A_p
A
Obs
JmaxRef = 94.31
RdRef = 2.07
sigma = 0.91
VcmaxRef = 315.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

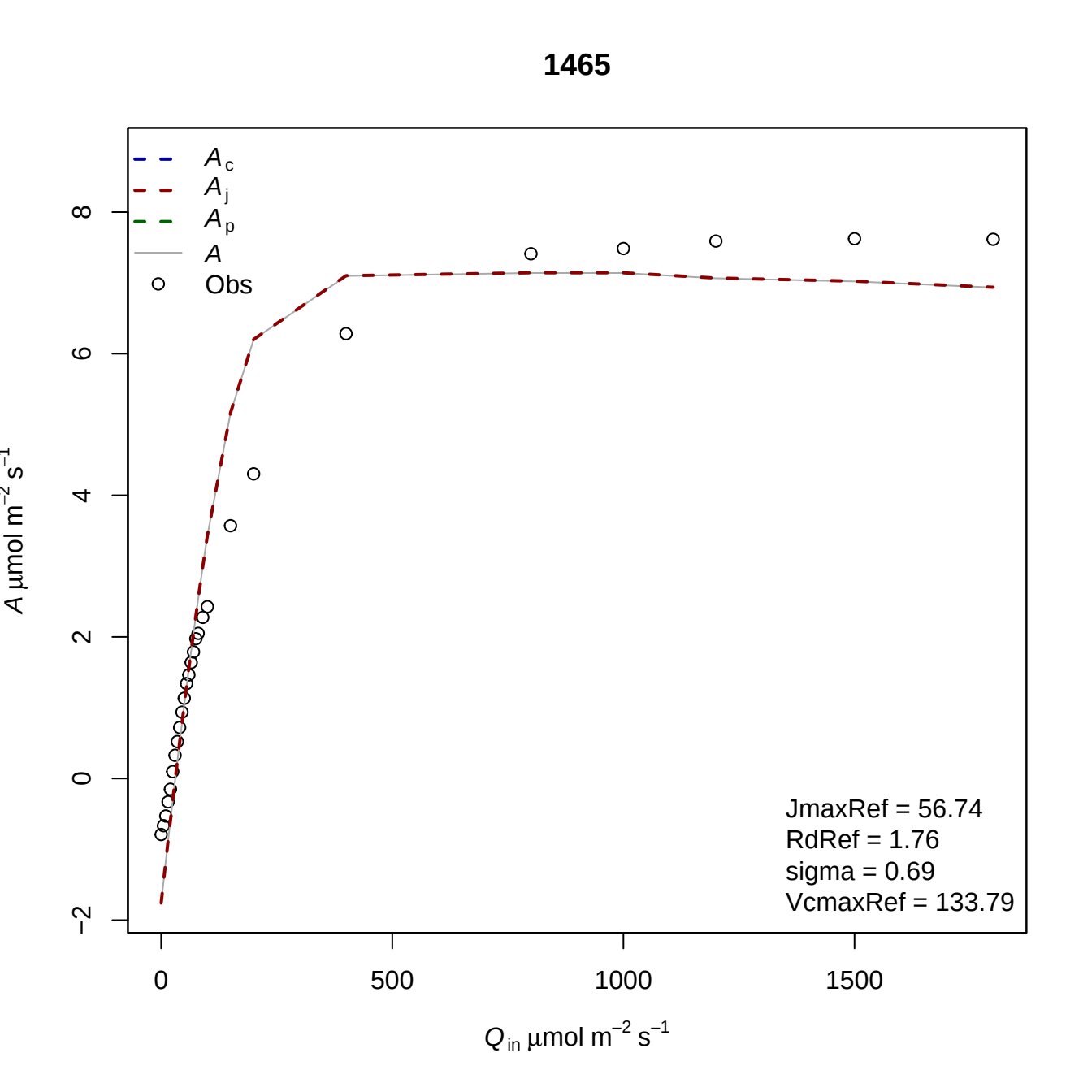

1460
A_c
A_j
A_p
A
Obs
JmaxRef = 86.4
RdRef = 2.38
sigma = 0.9
VcmaxRef = 809.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

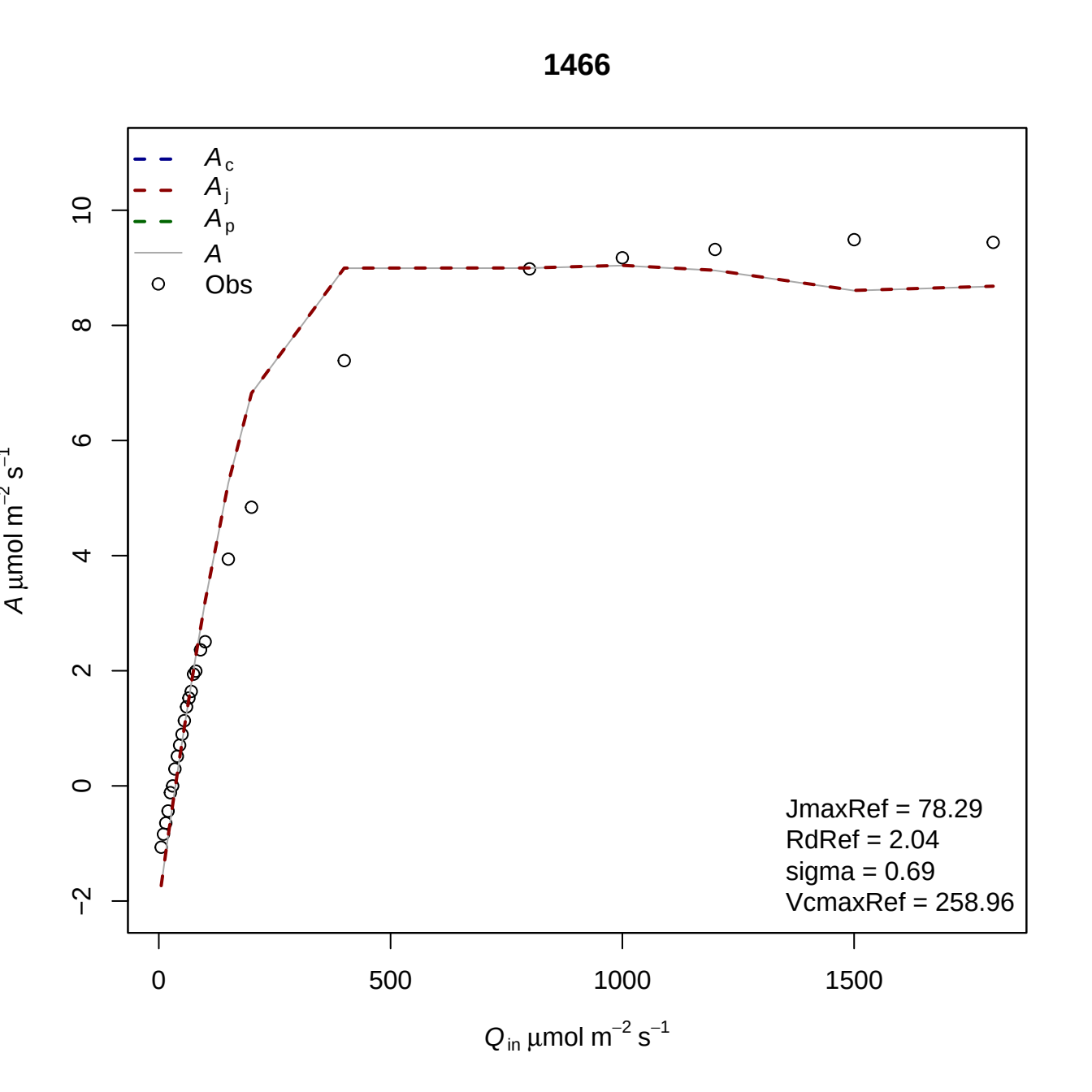

1461
A_c
A_j
A_p
A
Obs
JmaxRef = 63.64
RdRef = 1.1
sigma = 0.67
VcmaxRef = 160.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

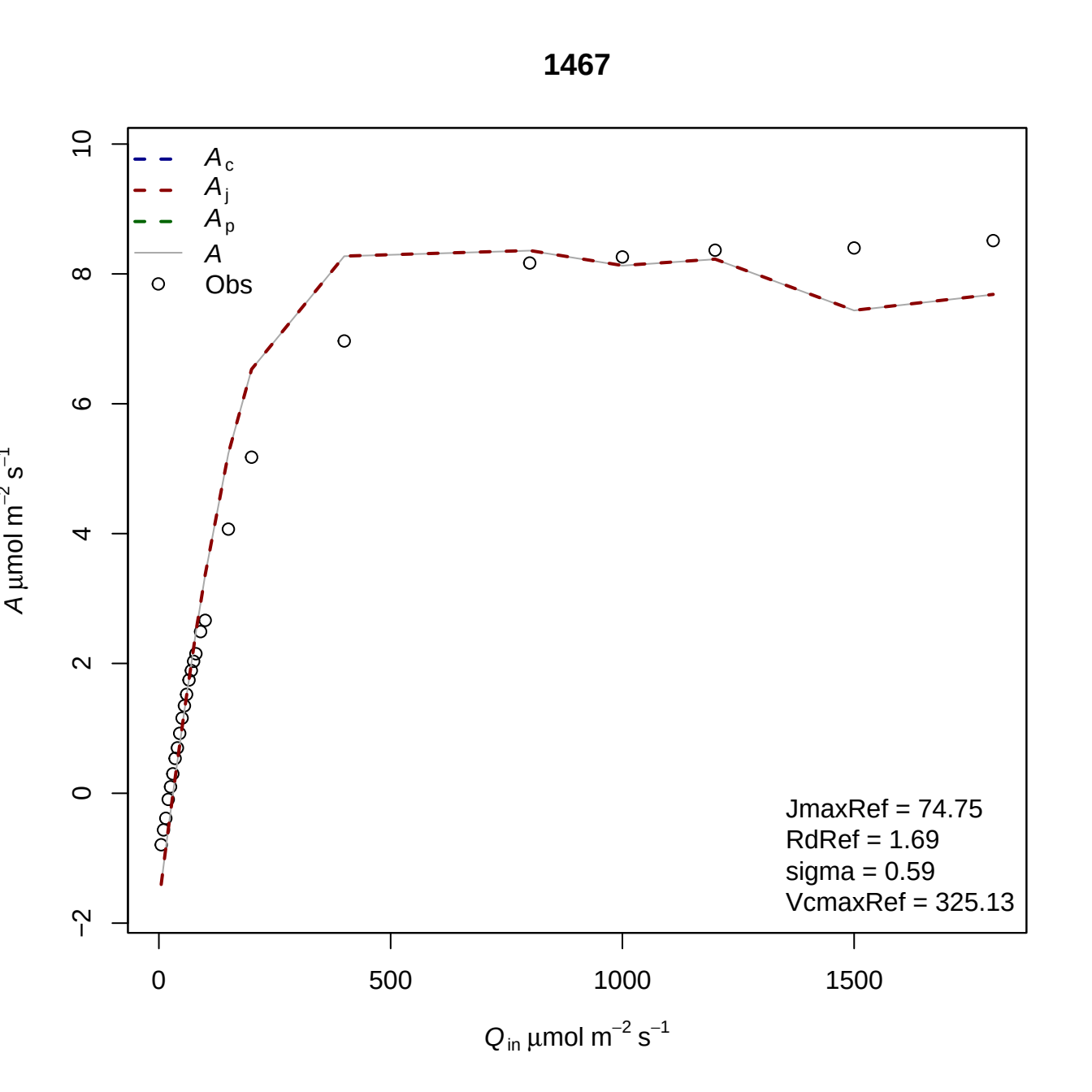

1462
A_c
A_j
A_p
A
Obs
JmaxRef = 71.33
RdRef = 1.42
sigma = 0.77
VcmaxRef = 320.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

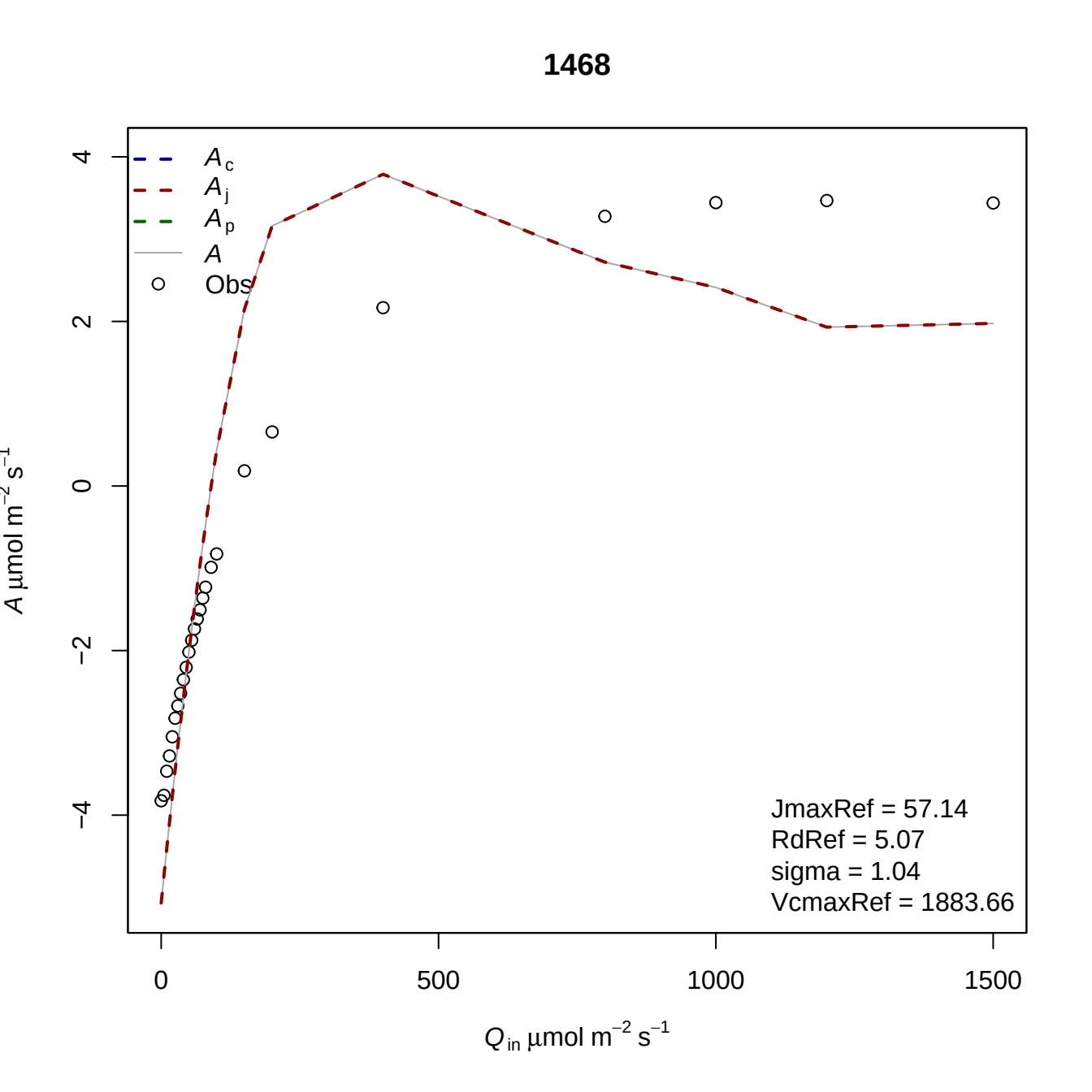

1463
A_c
A_j
A_p
A
Obs
JmaxRef = 67.09
RdRef = 1.55
sigma = 0.93
VcmaxRef = 244.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

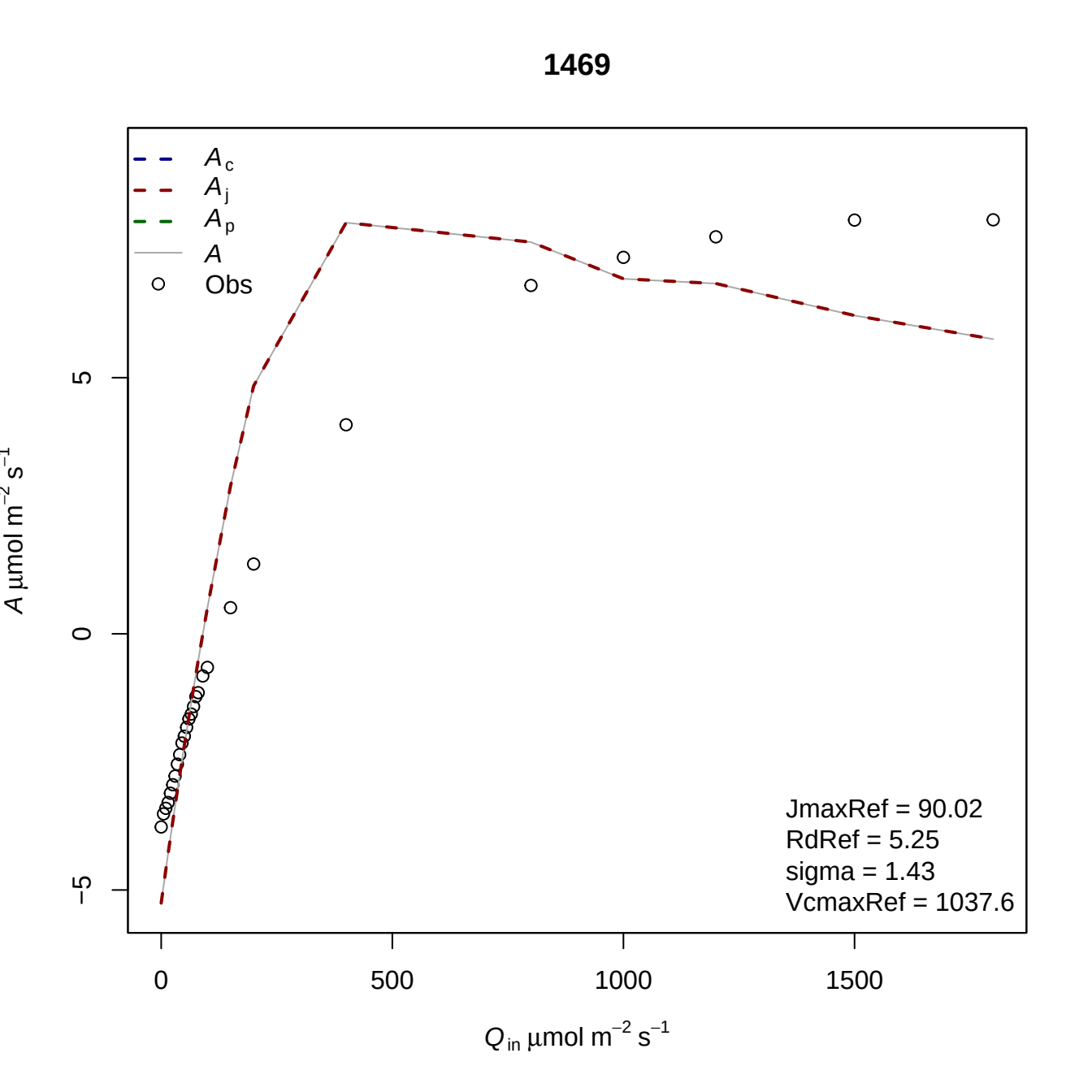

1464
A_c
A_j
A_p
A
Obs
JmaxRef = 89.33
RdRef = 1.46
sigma = 0.76
VcmaxRef = 767.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

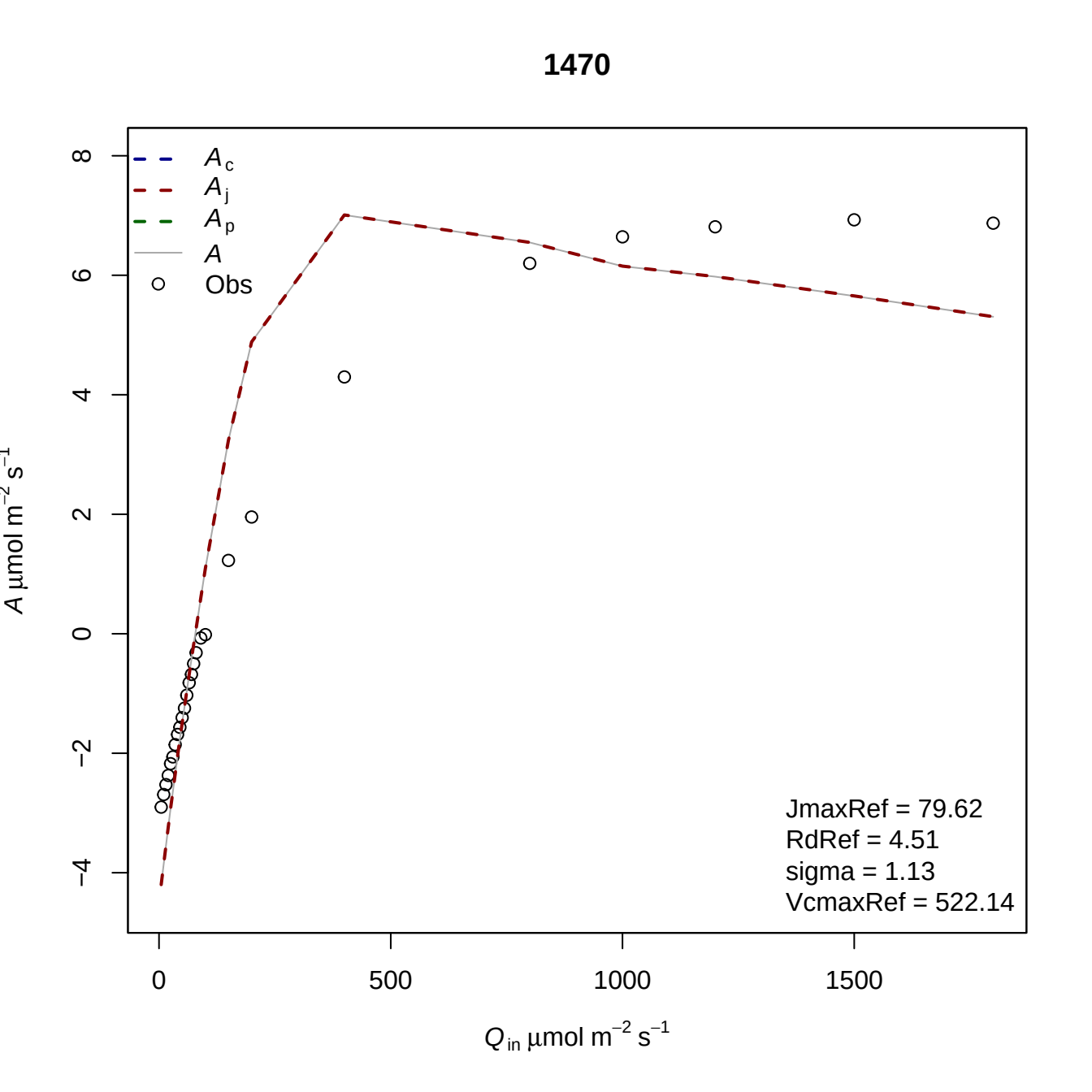

1465
A_c
A_j
A_p
A
Obs
JmaxRef = 56.74
RdRef = 1.76
sigma = 0.69
VcmaxRef = 133.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

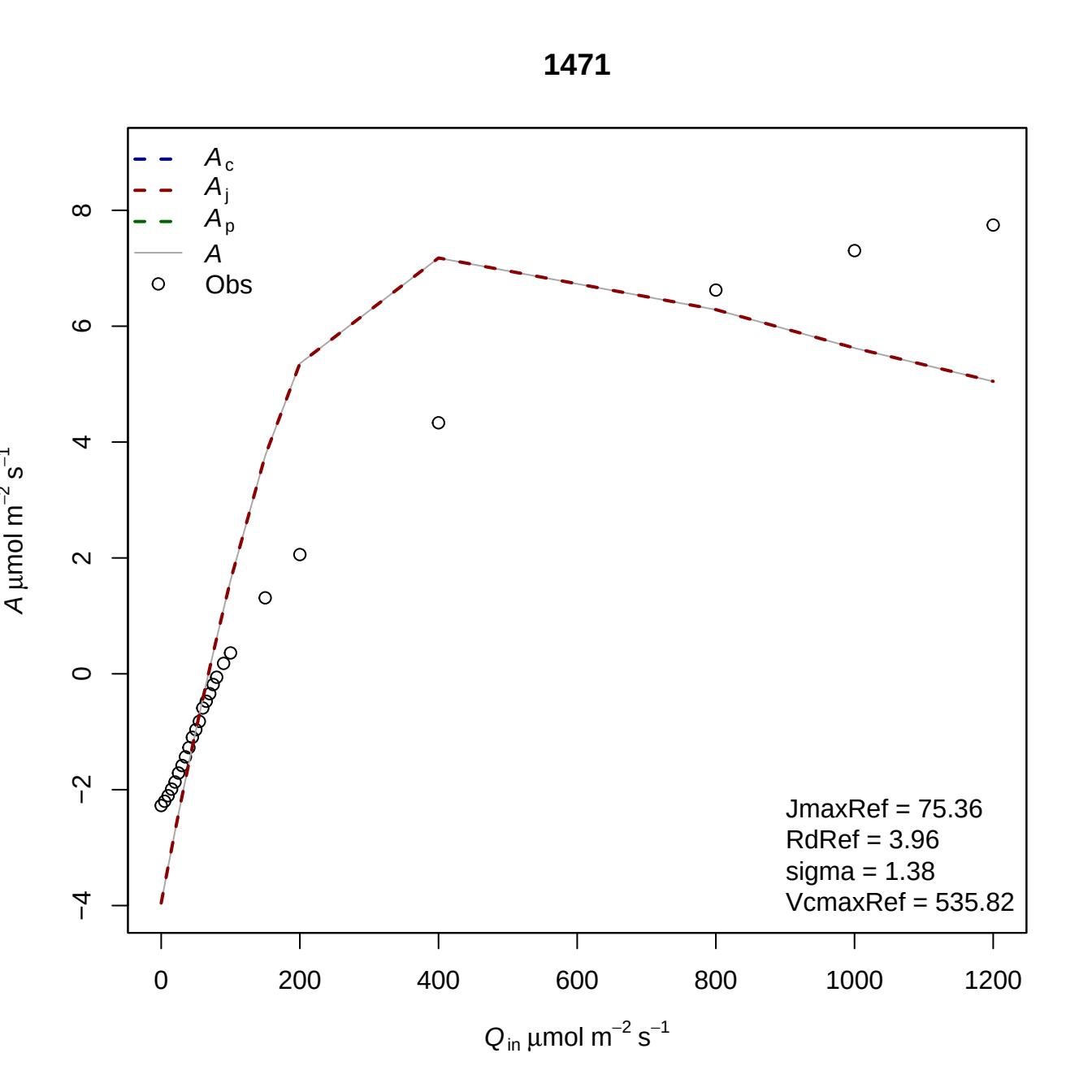

1466
A_c
A_j
A_p
A
Obs
JmaxRef = 78.29
RdRef = 2.04
sigma = 0.69
VcmaxRef = 258.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

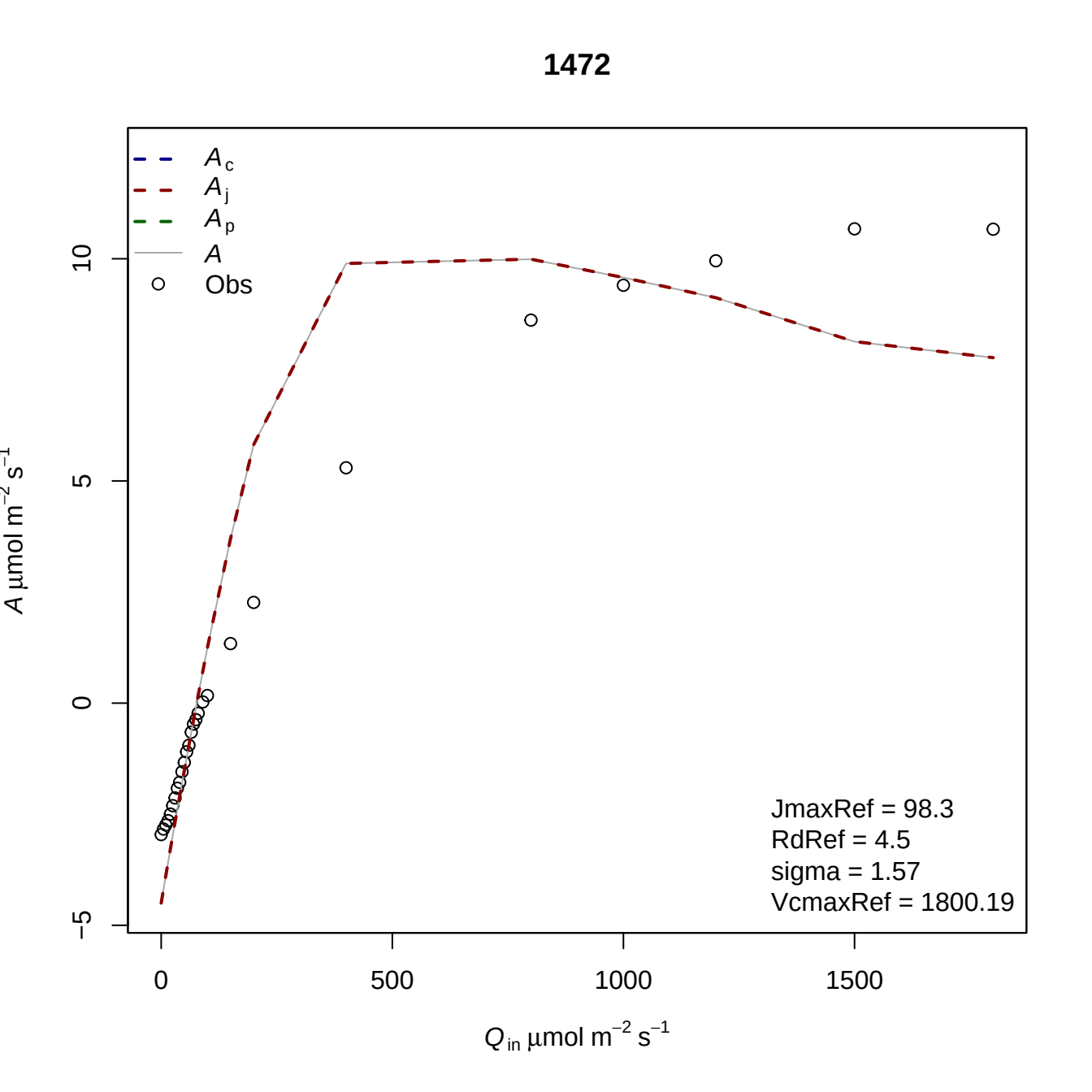

1467
A_c
A_j
A_p
A
Obs
JmaxRef = 74.75
RdRef = 1.69
sigma = 0.59
VcmaxRef = 325.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

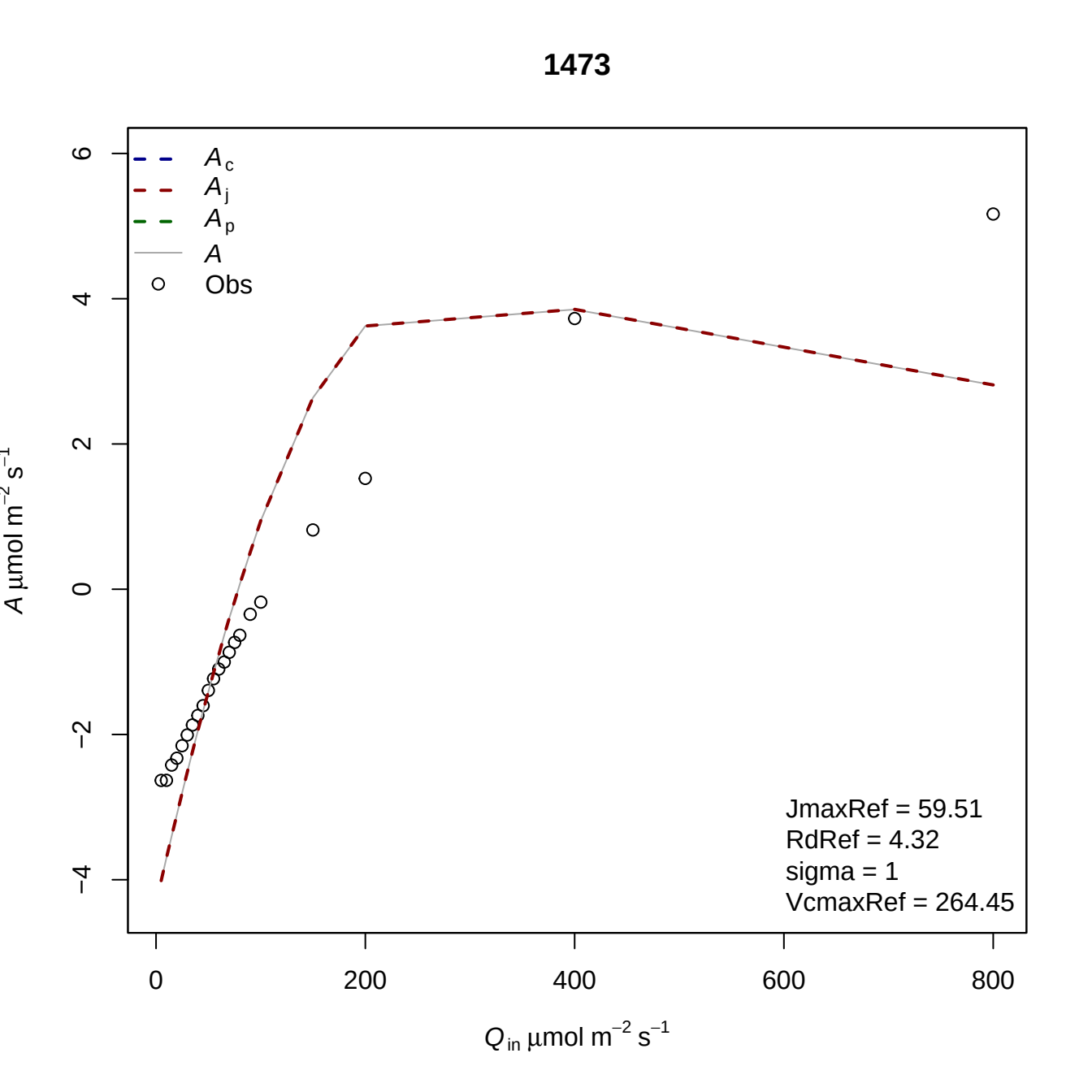

1468
A_c
A_j
A_p
A
Obs
JmaxRef = 57.14
RdRef = 5.07
sigma = 1.04
VcmaxRef = 1883.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

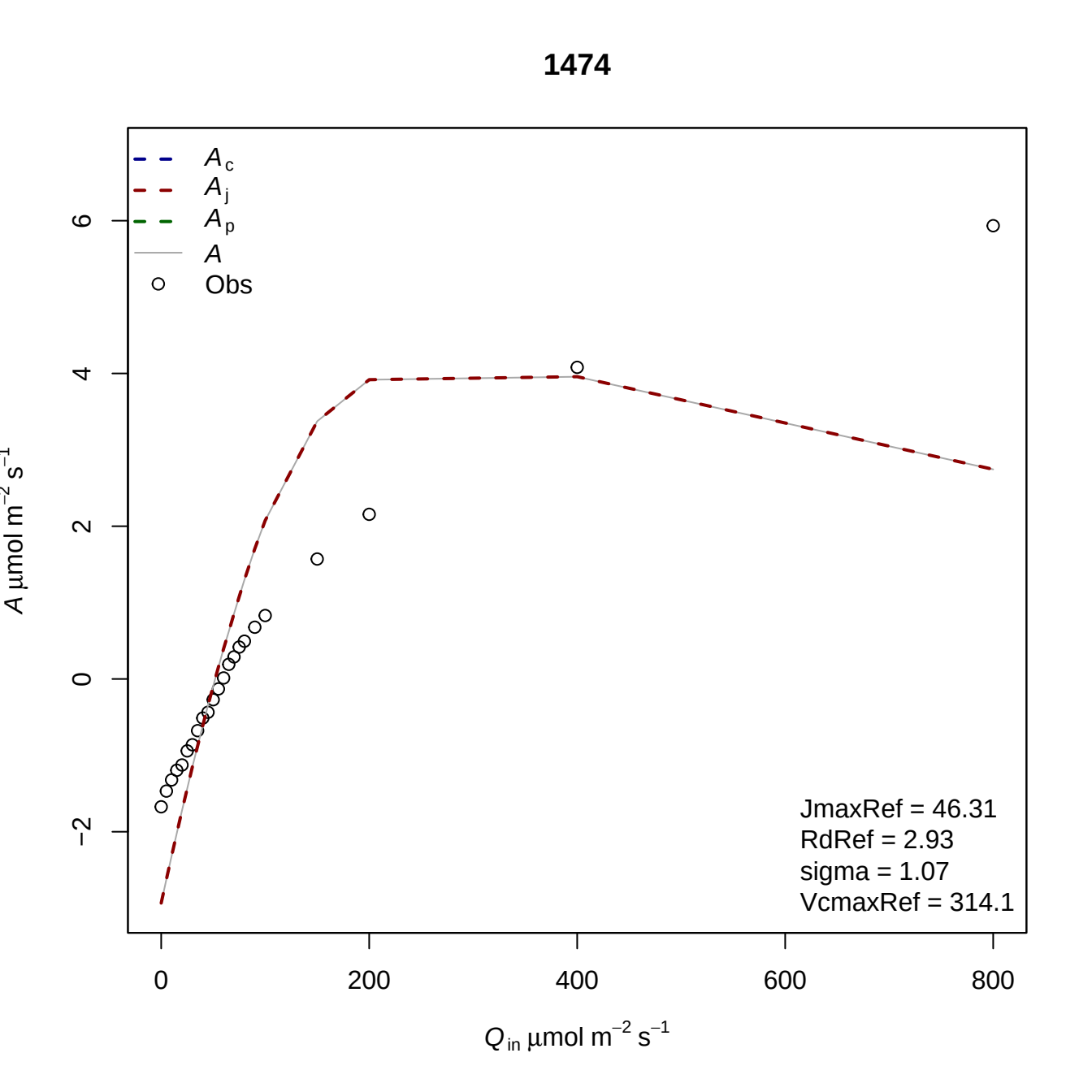

1469
A_c
A_j
A_p
A
Obs
JmaxRef = 90.02
RdRef = 5.25
sigma = 1.43
VcmaxRef = 1037.6
A µmol m^{-2} s^{-1}
−5
0
5
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

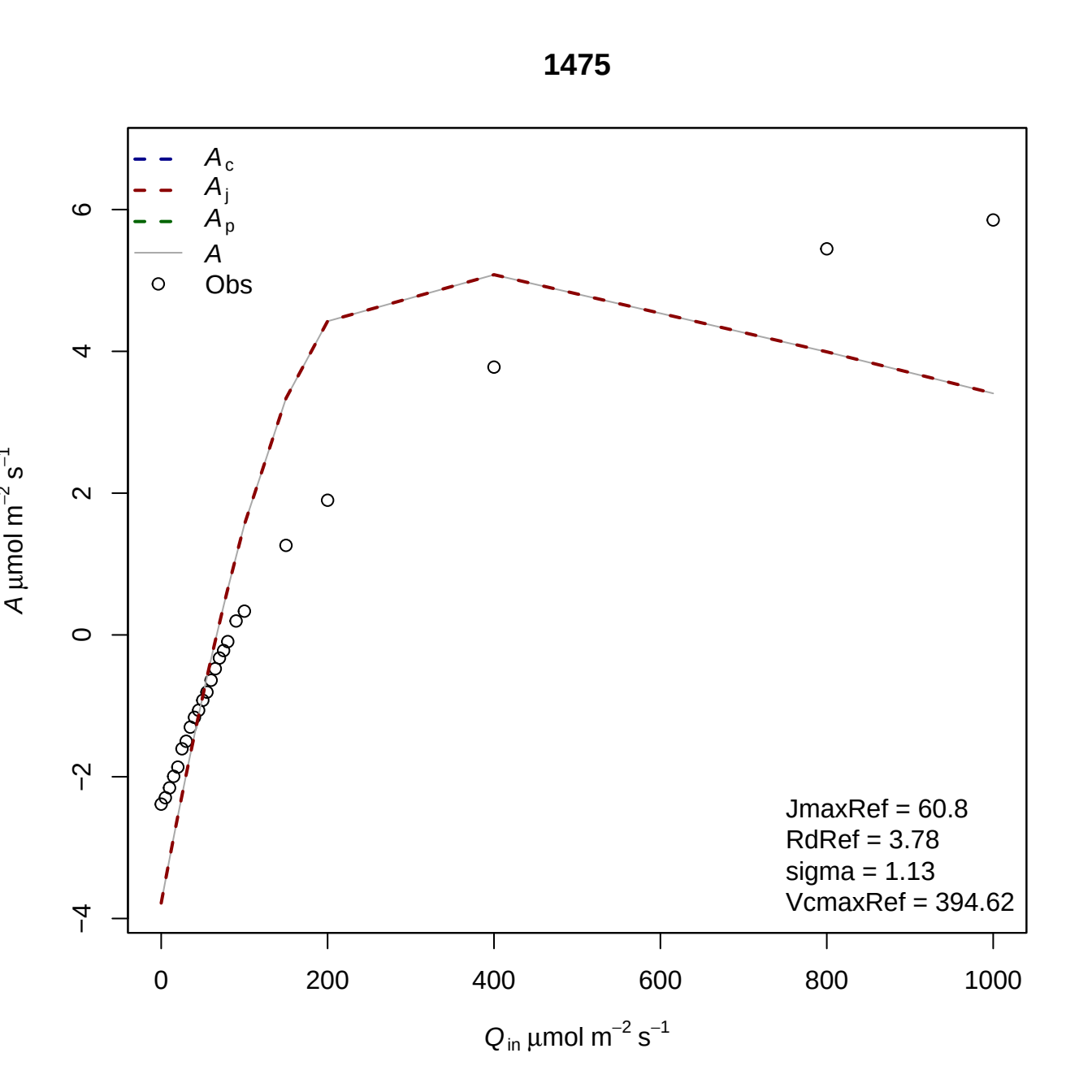

1470
A_c
A_j
A_p
A
Obs
JmaxRef = 79.62
RdRef = 4.51
sigma = 1.13
VcmaxRef = 522.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

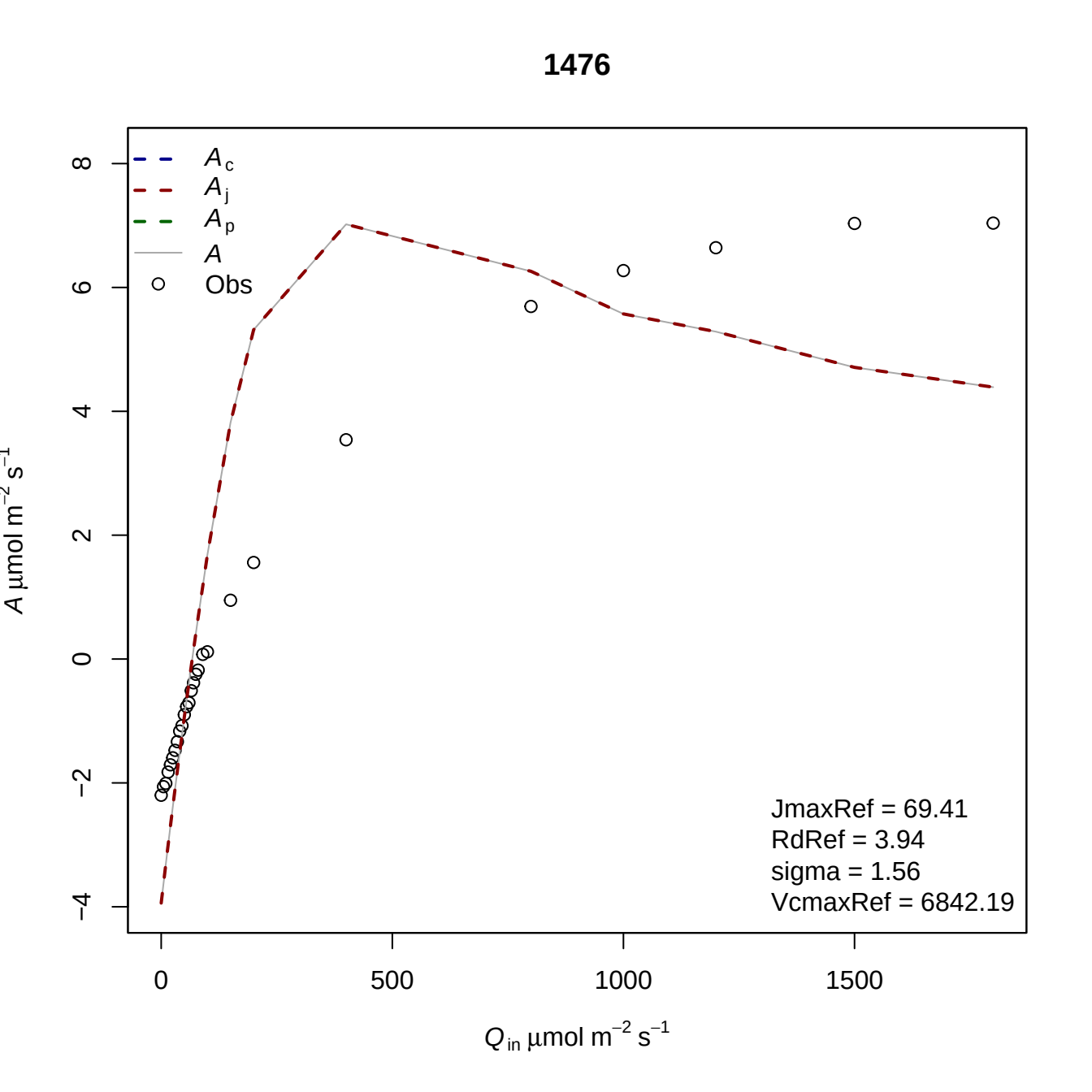

1471
A_c
A_j
A_p
A
Obs
JmaxRef = 75.36
RdRef = 3.96
sigma = 1.38
VcmaxRef = 535.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

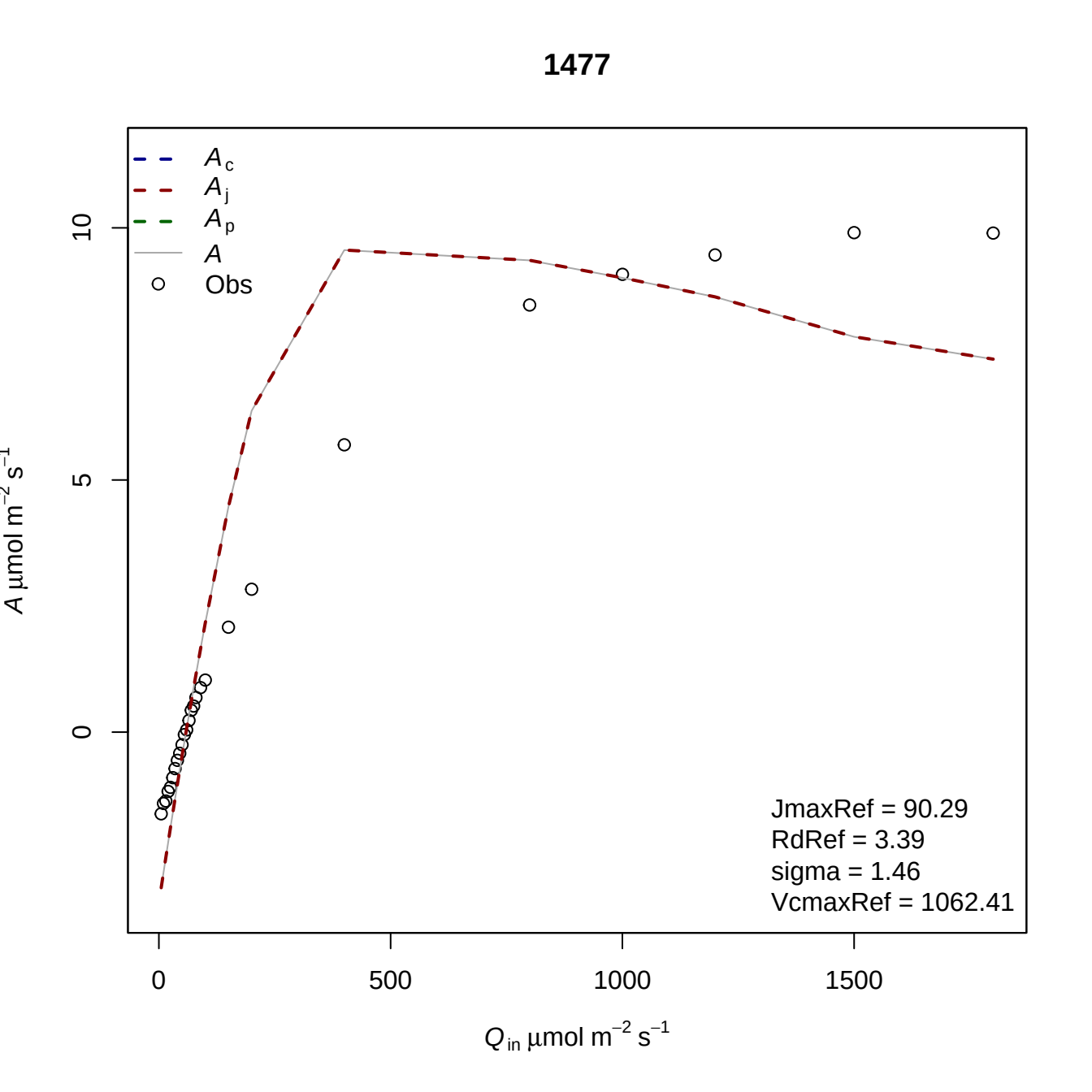

1472
A_c
A_j
A_p
A
Obs
JmaxRef = 98.3
RdRef = 4.5
sigma = 1.57
VcmaxRef = 1800.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

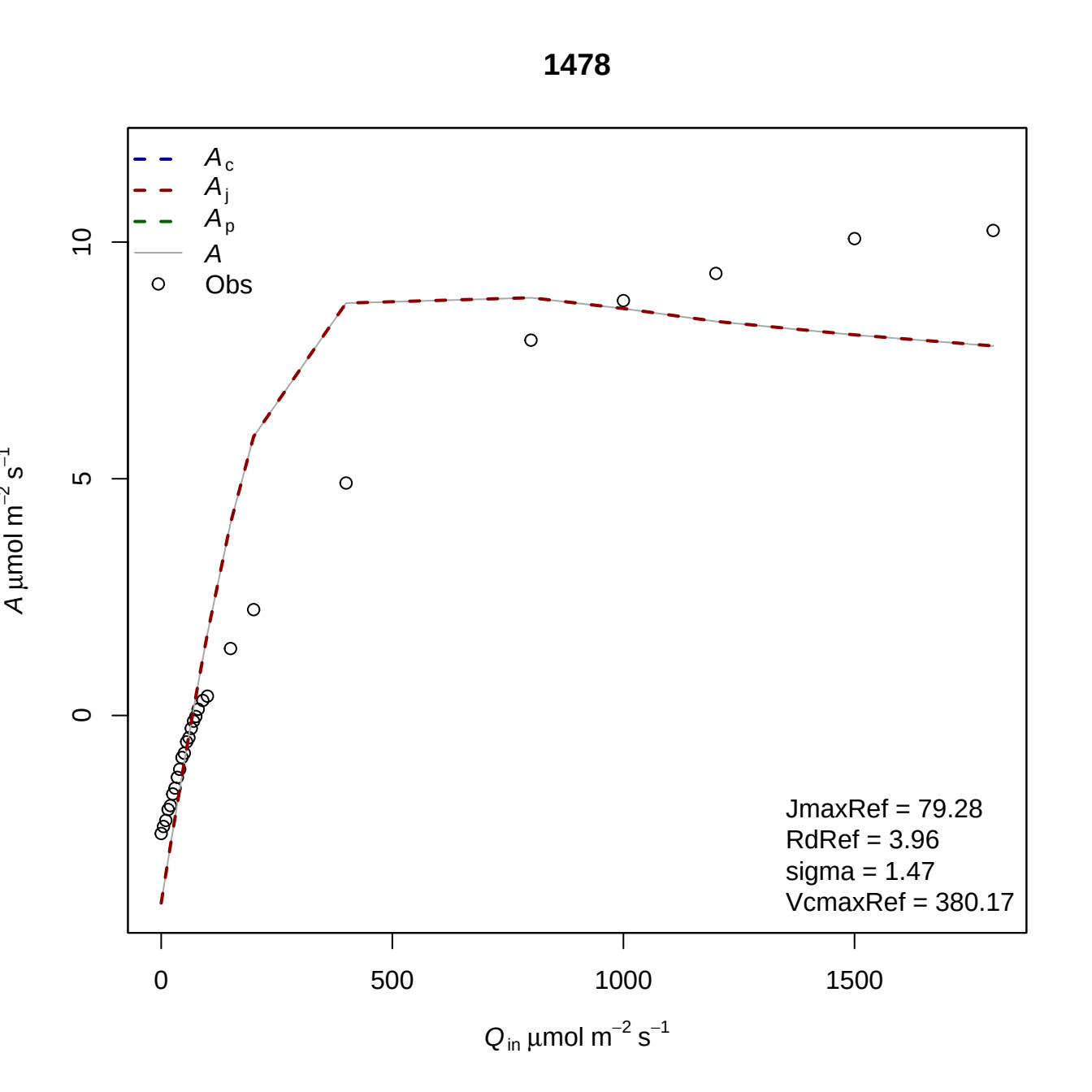

1473
A_c
A_j
A_p
A
Obs
JmaxRef = 59.51
RdRef = 4.32
sigma = 1
VcmaxRef = 264.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800

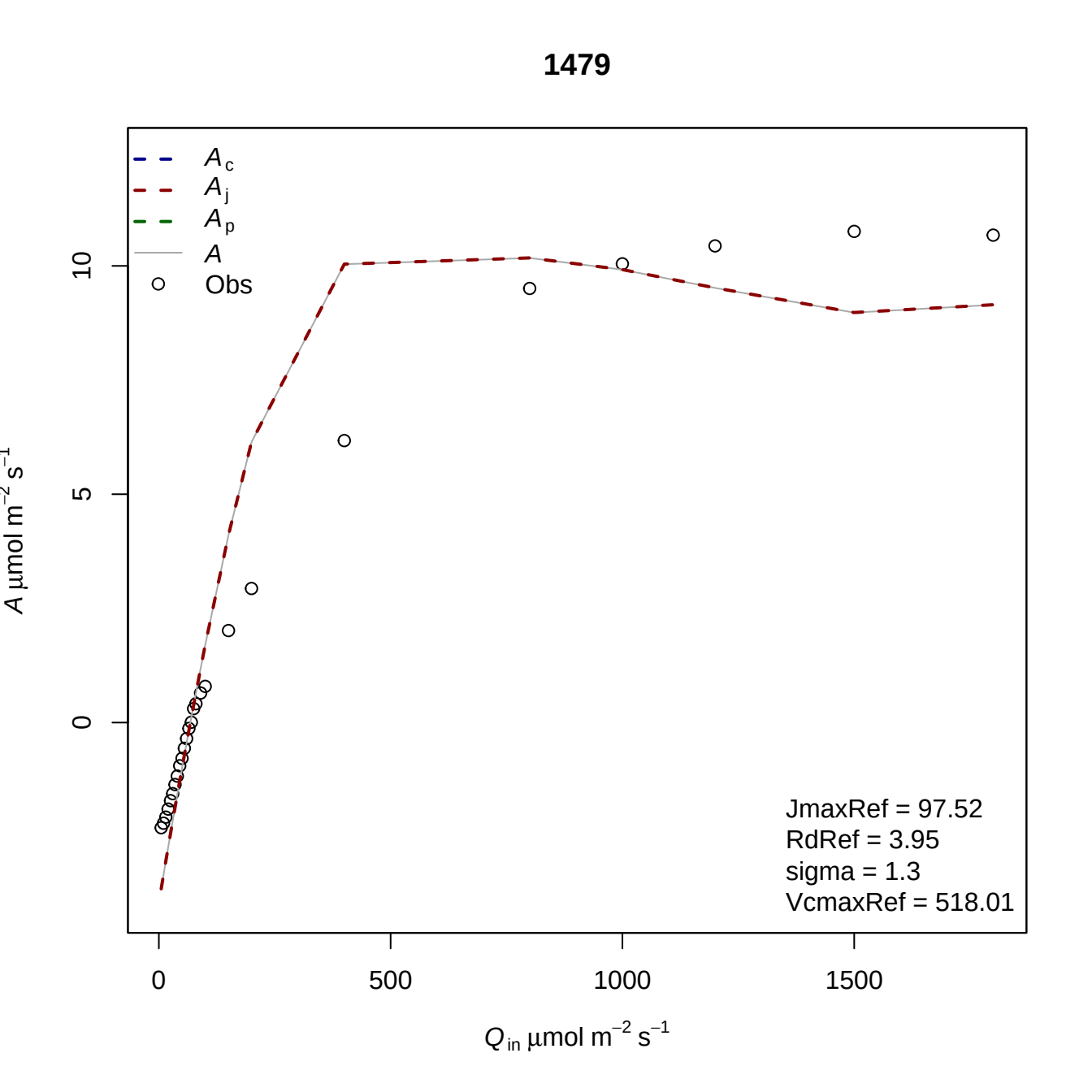

1474
A_c
A_j
A_p
A
Obs
JmaxRef = 46.31
RdRef = 2.93
sigma = 1.07
VcmaxRef = 314.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

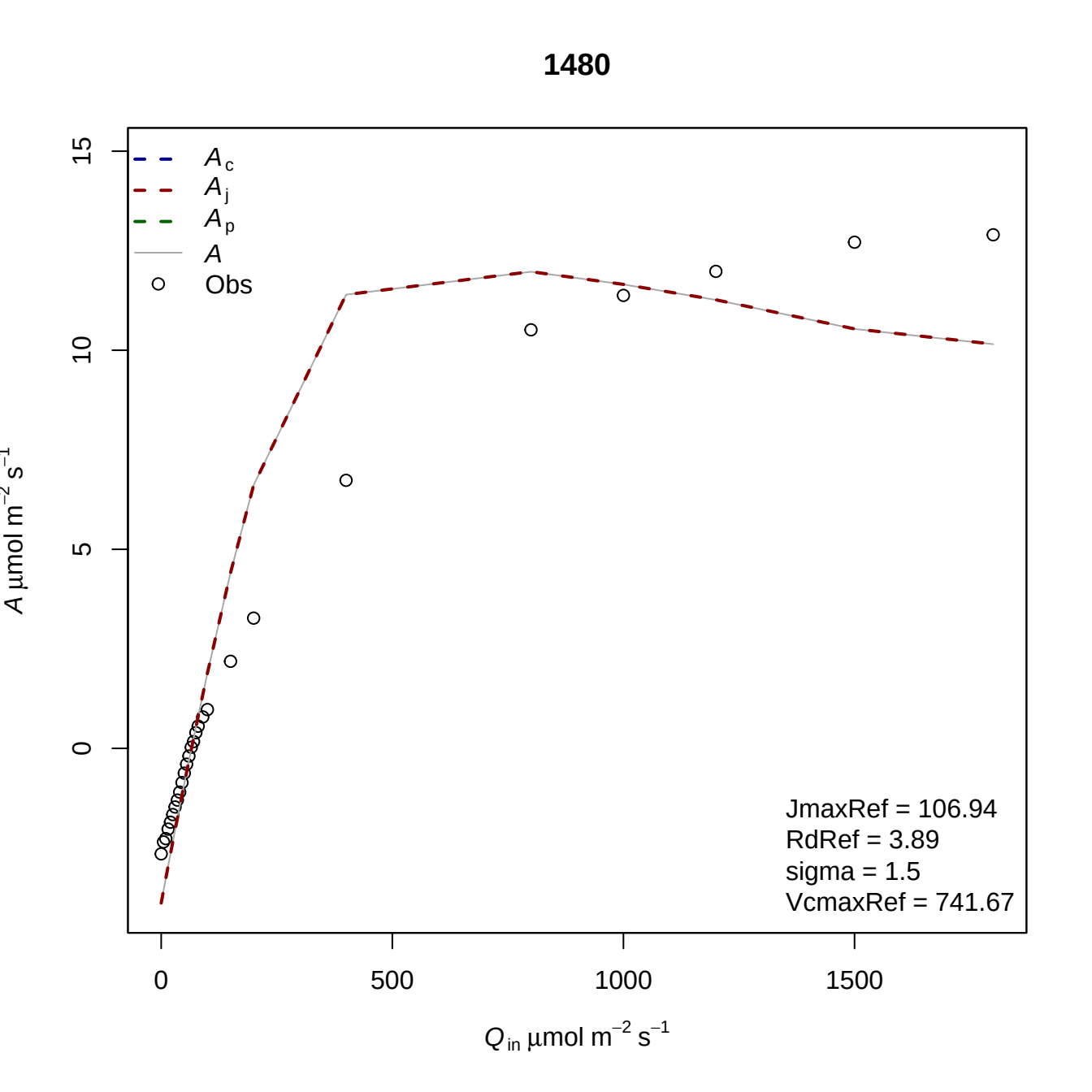

1475
A_c
A_j
A_p
A
Obs
JmaxRef = 60.8
RdRef = 3.78
sigma = 1.13
VcmaxRef = 394.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000

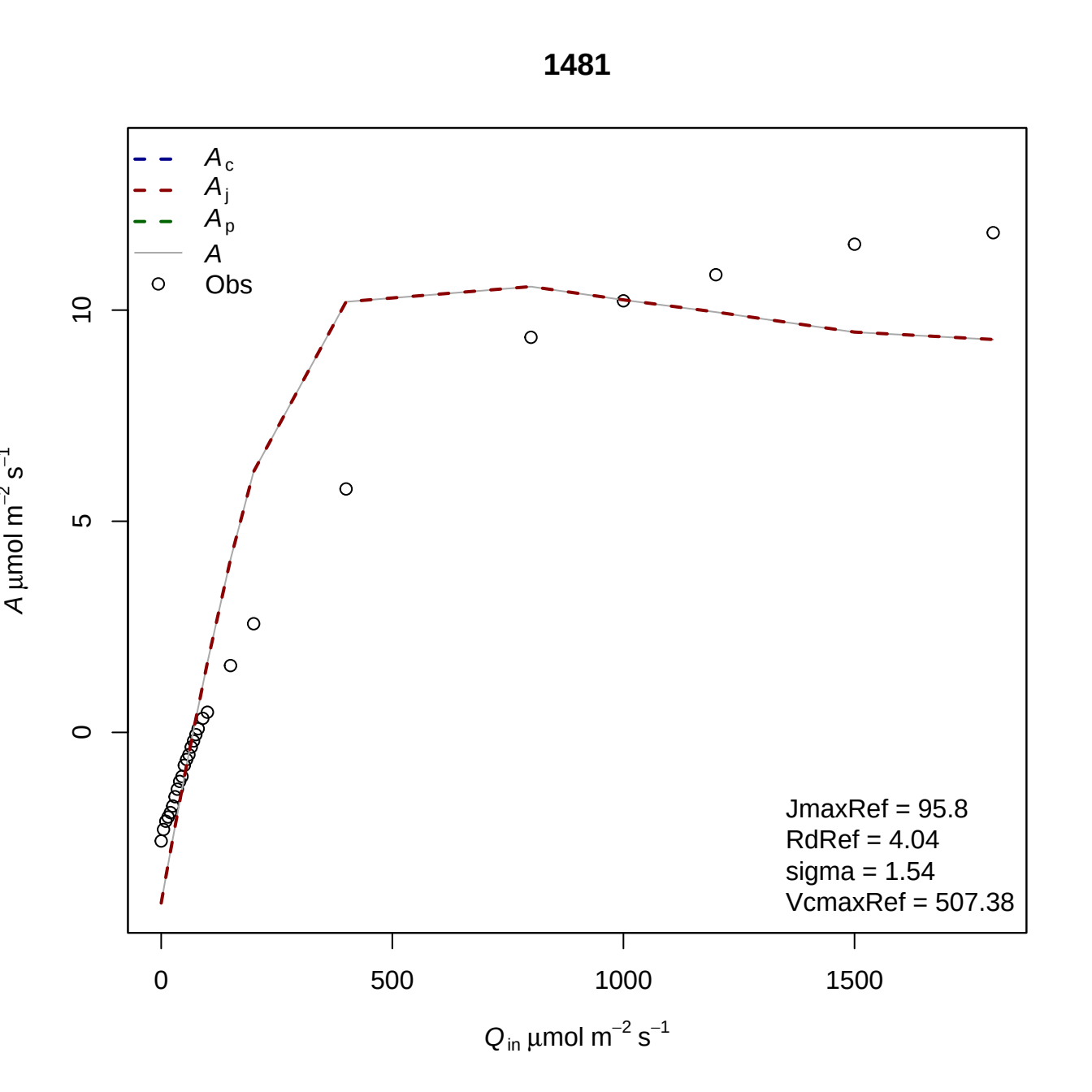

1476
A_c
A_j
A_p
A
Obs
JmaxRef = 69.41
RdRef = 3.94
sigma = 1.56
VcmaxRef = 6842.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

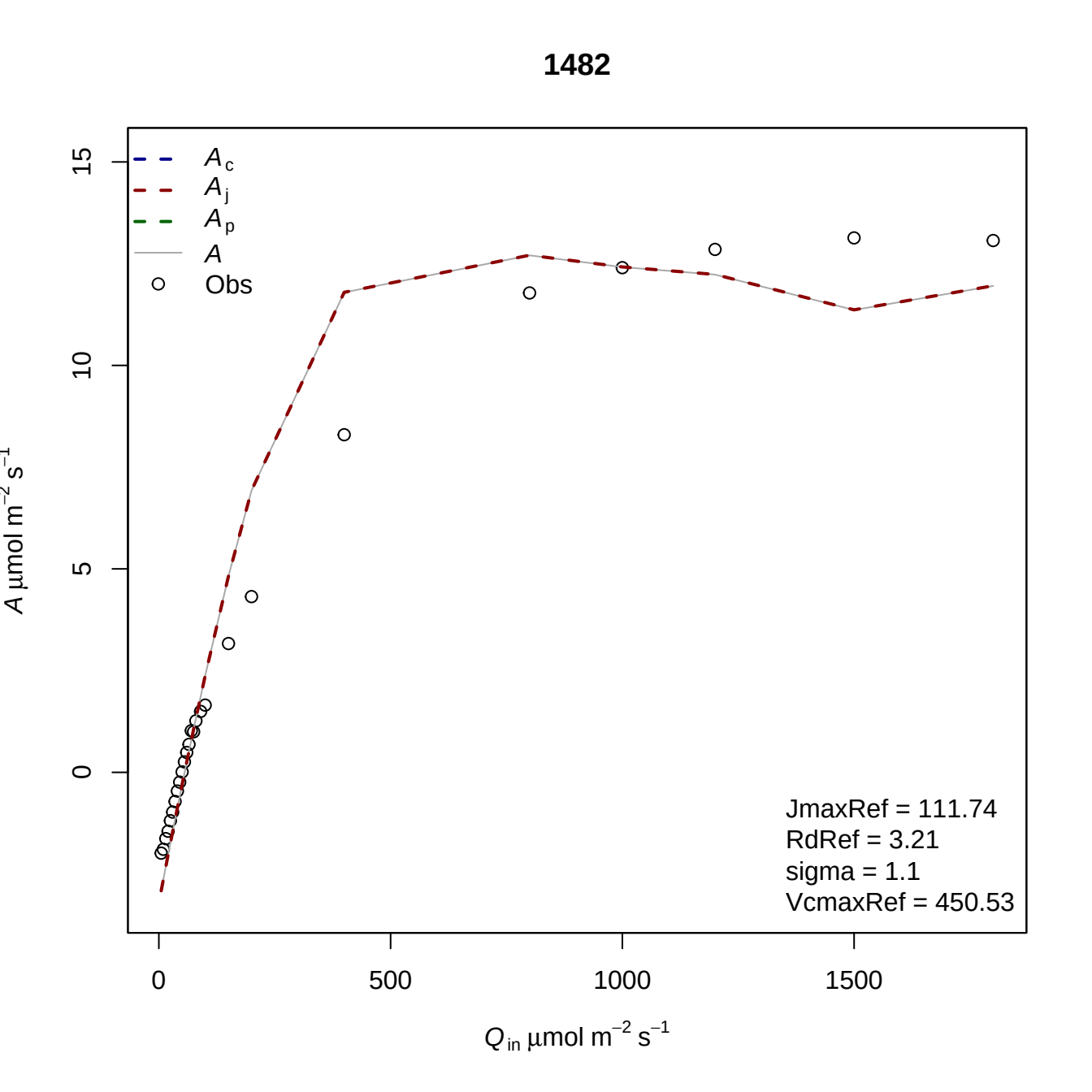

1477
A_c
A_j
A_p
A
Obs
JmaxRef = 90.29
RdRef = 3.39
sigma = 1.46
VcmaxRef = 1062.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

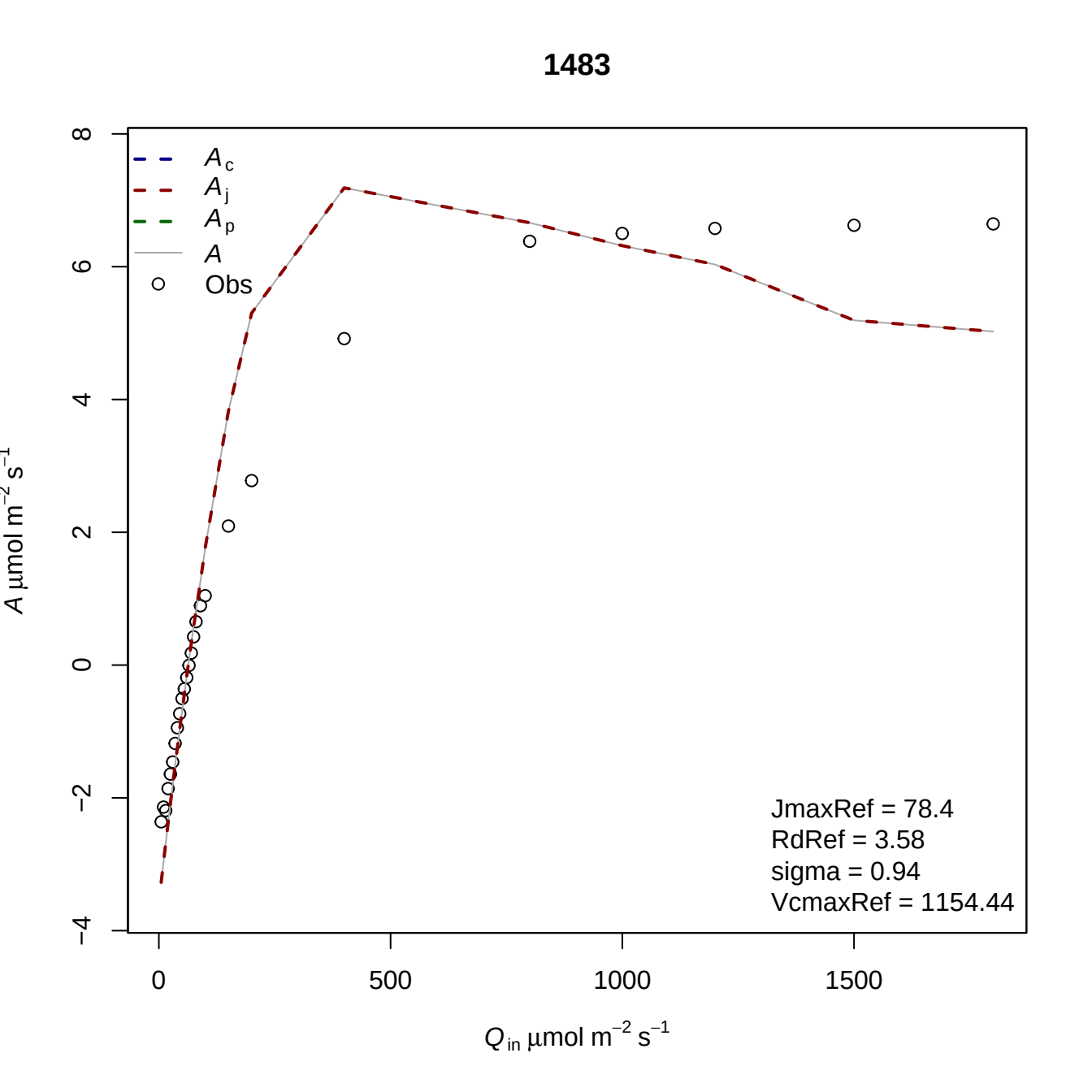

1478
A_c
A_j
A_p
A
Obs
JmaxRef = 79.28
RdRef = 3.96
sigma = 1.47
VcmaxRef = 380.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

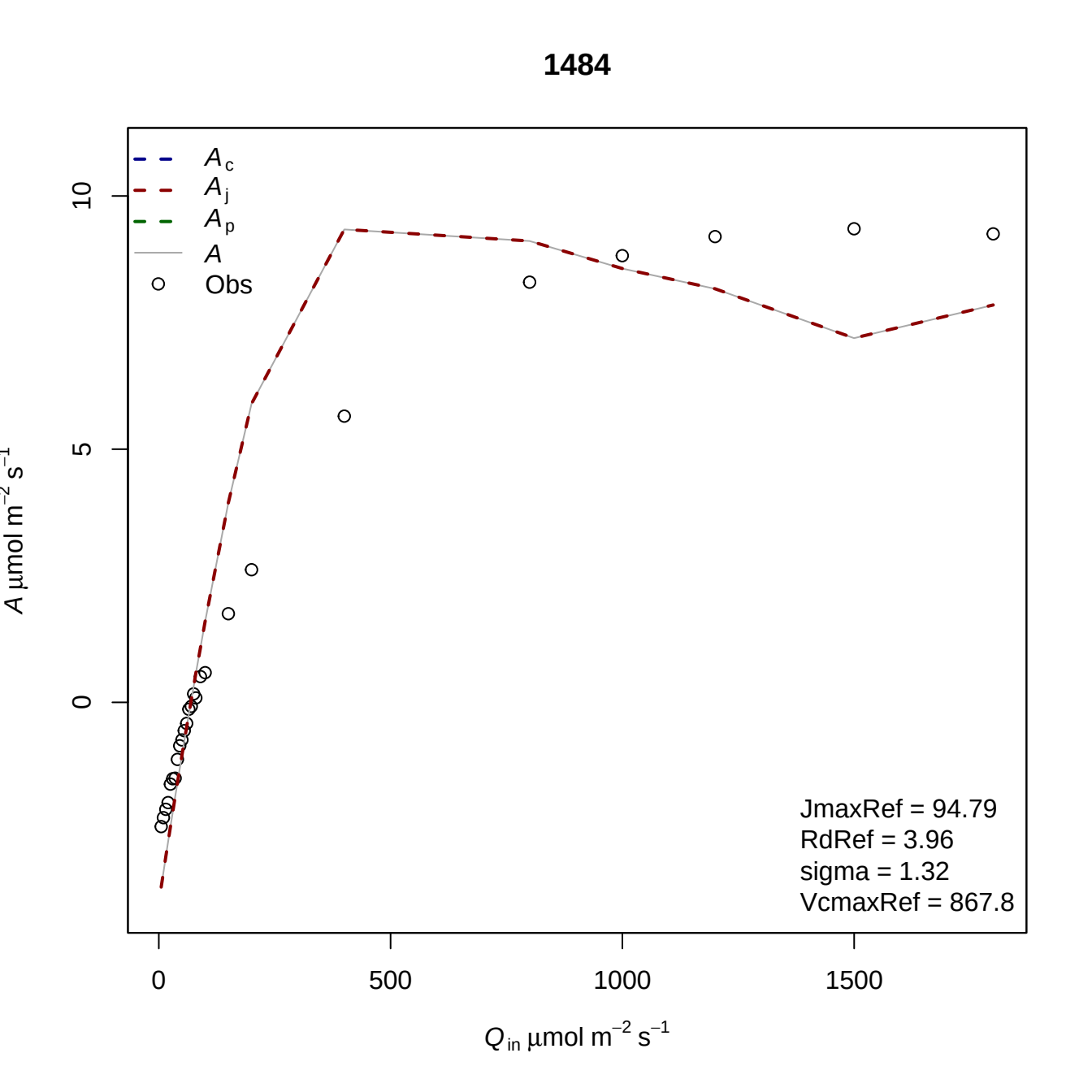

1479
A_c
A_j
A_p
A
Obs
JmaxRef = 97.52
RdRef = 3.95
sigma = 1.3
VcmaxRef = 518.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

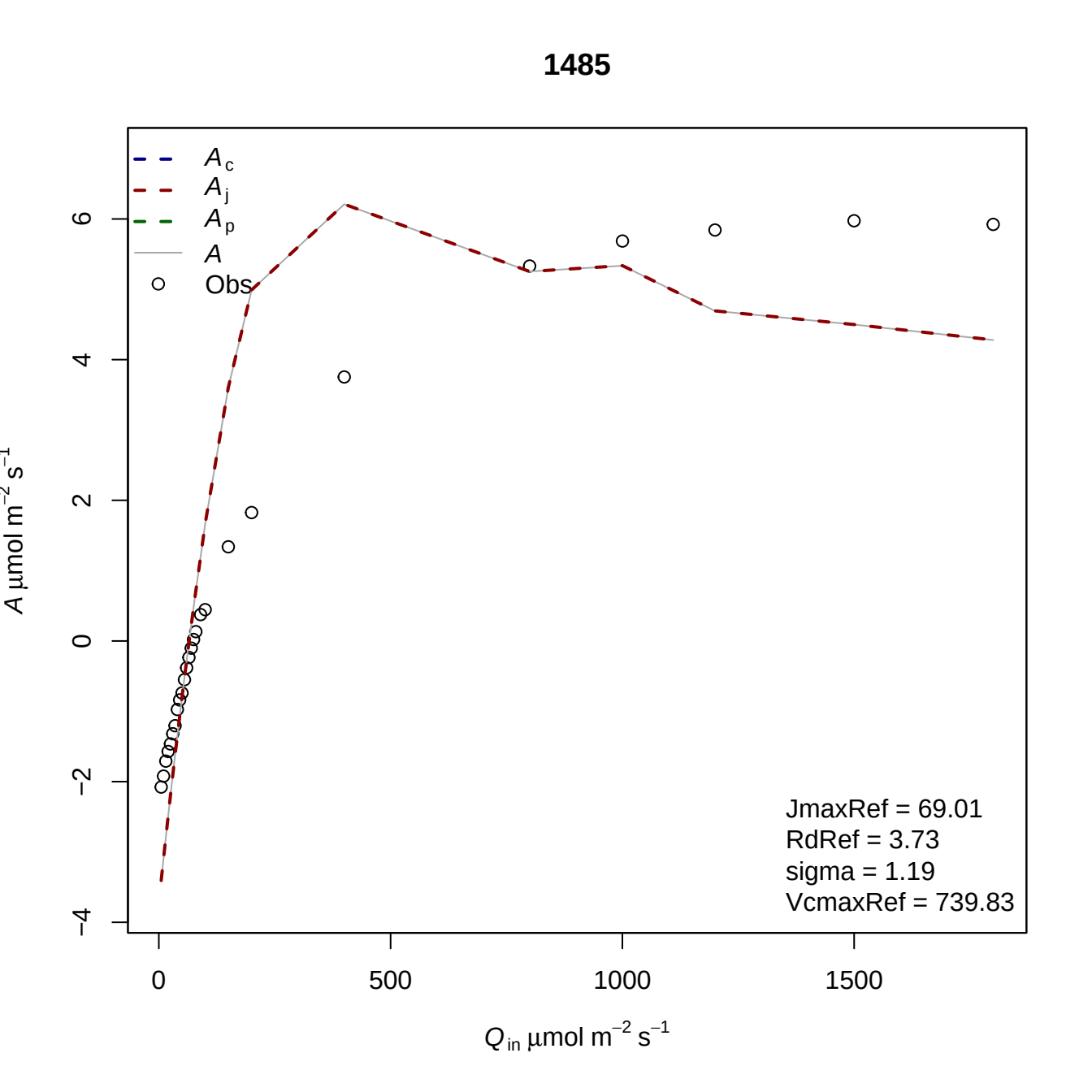

1480
A_c
A_j
A_p
A
Obs
JmaxRef = 106.94
RdRef = 3.89
sigma = 1.5
VcmaxRef = 741.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

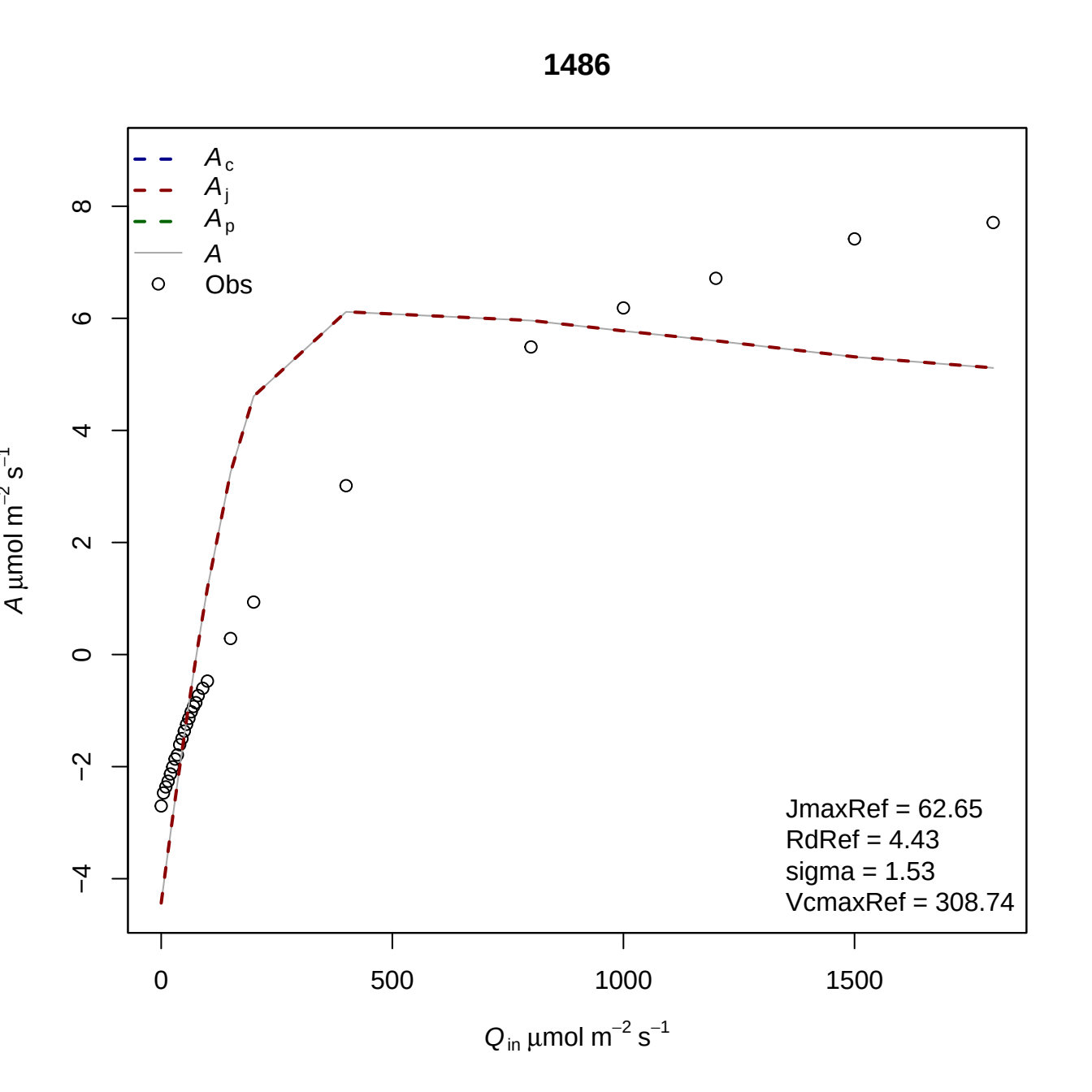

1481
A_c
A_j
A_p
A
Obs
JmaxRef = 95.8
RdRef = 4.04
sigma = 1.54
VcmaxRef = 507.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

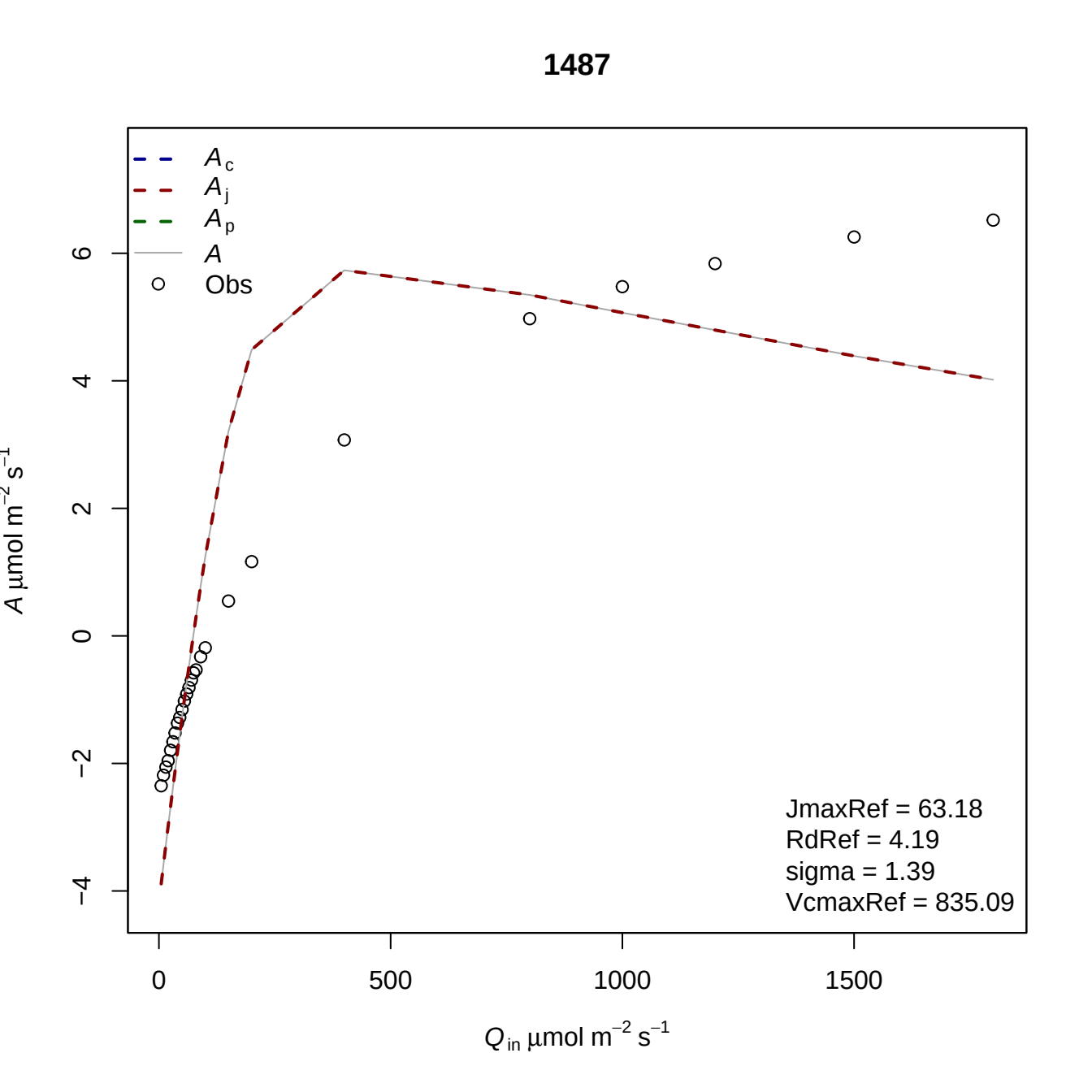

1482
A_c
A_j
A_p
A
Obs
JmaxRef = 111.74
RdRef = 3.21
sigma = 1.1
VcmaxRef = 450.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

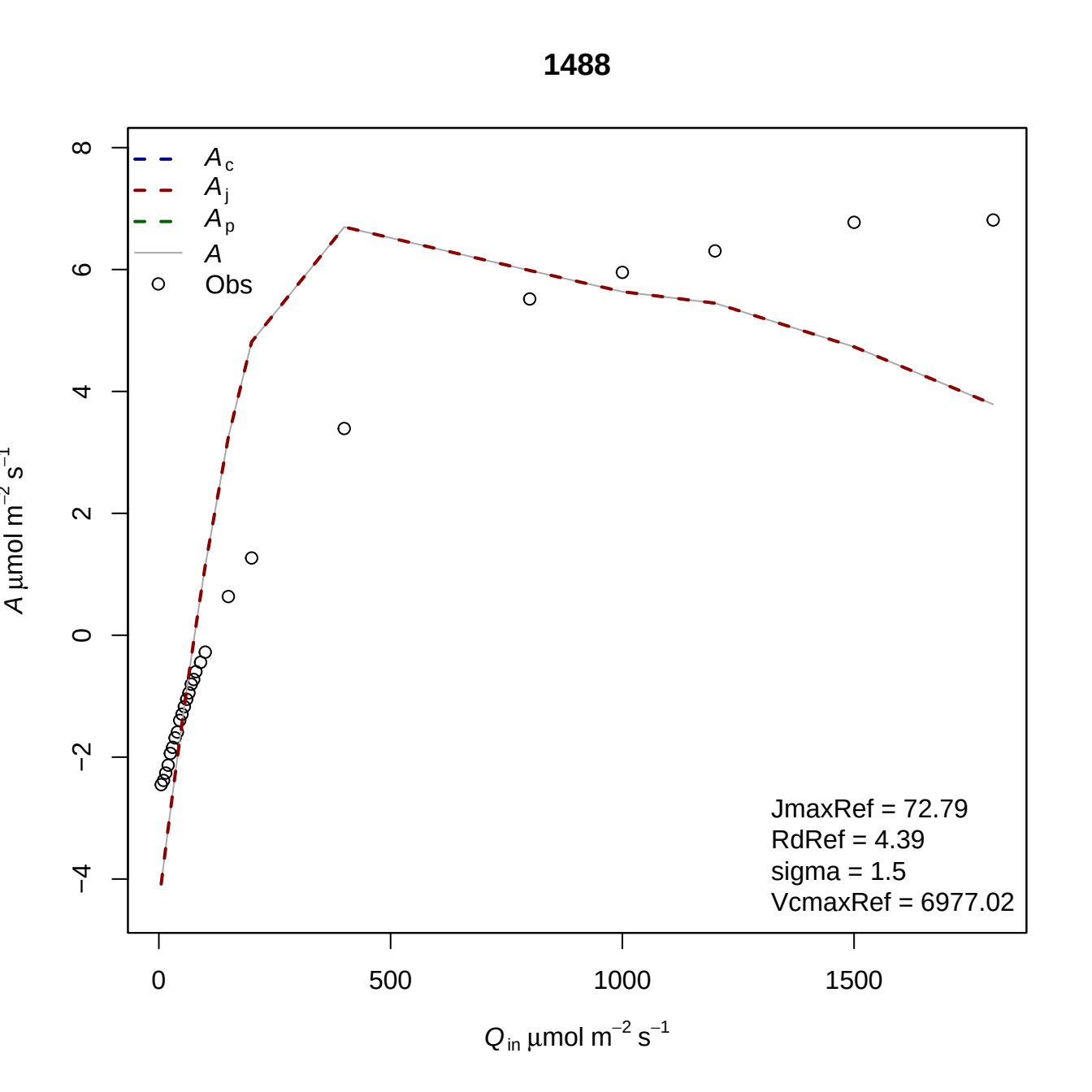

1483
A_c
A_j
A_p
A
Obs
JmaxRef = 78.4
RdRef = 3.58
sigma = 0.94
VcmaxRef = 1154.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

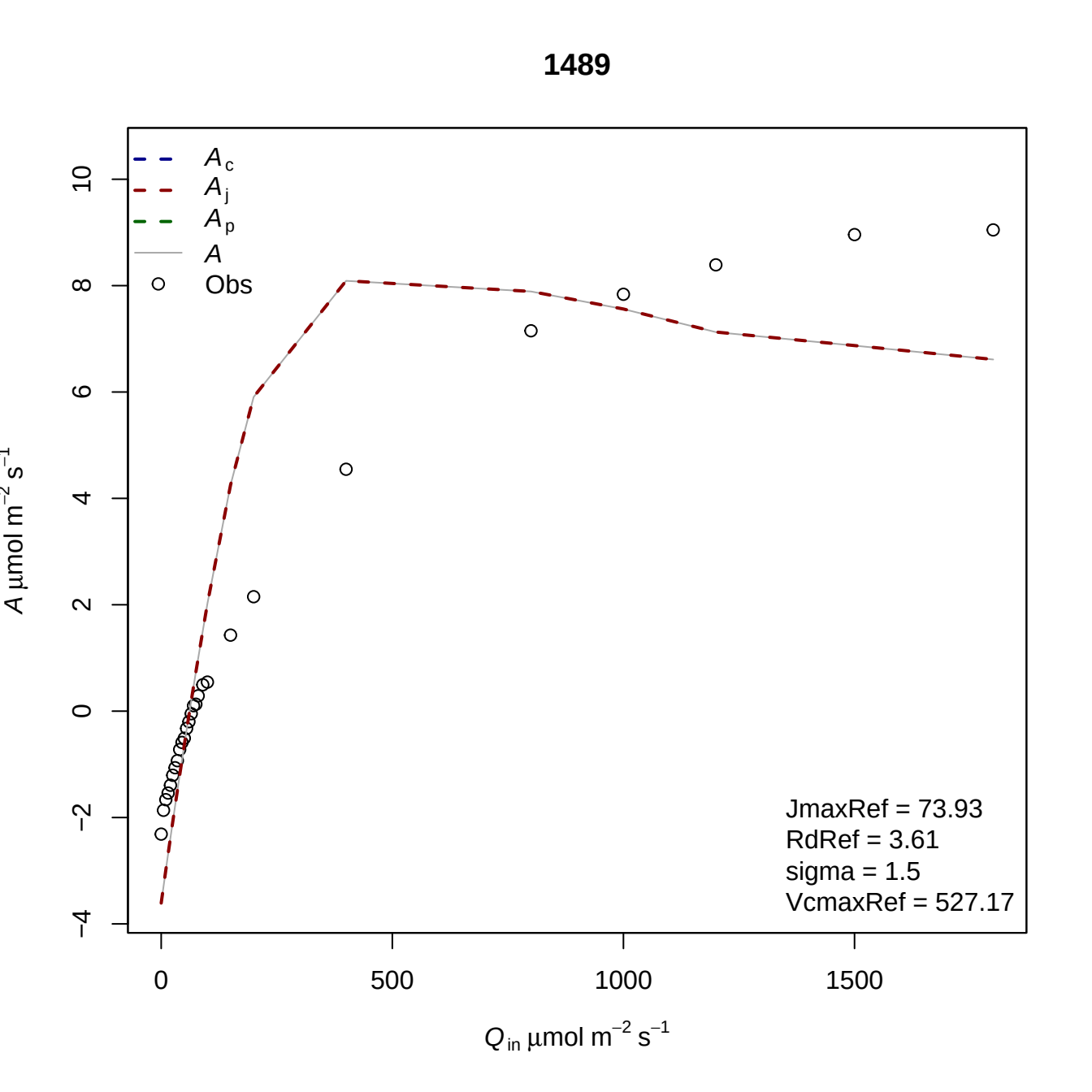

1484
A_c
A_j
A_p
A
Obs
JmaxRef = 94.79
RdRef = 3.96
sigma = 1.32
VcmaxRef = 867.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

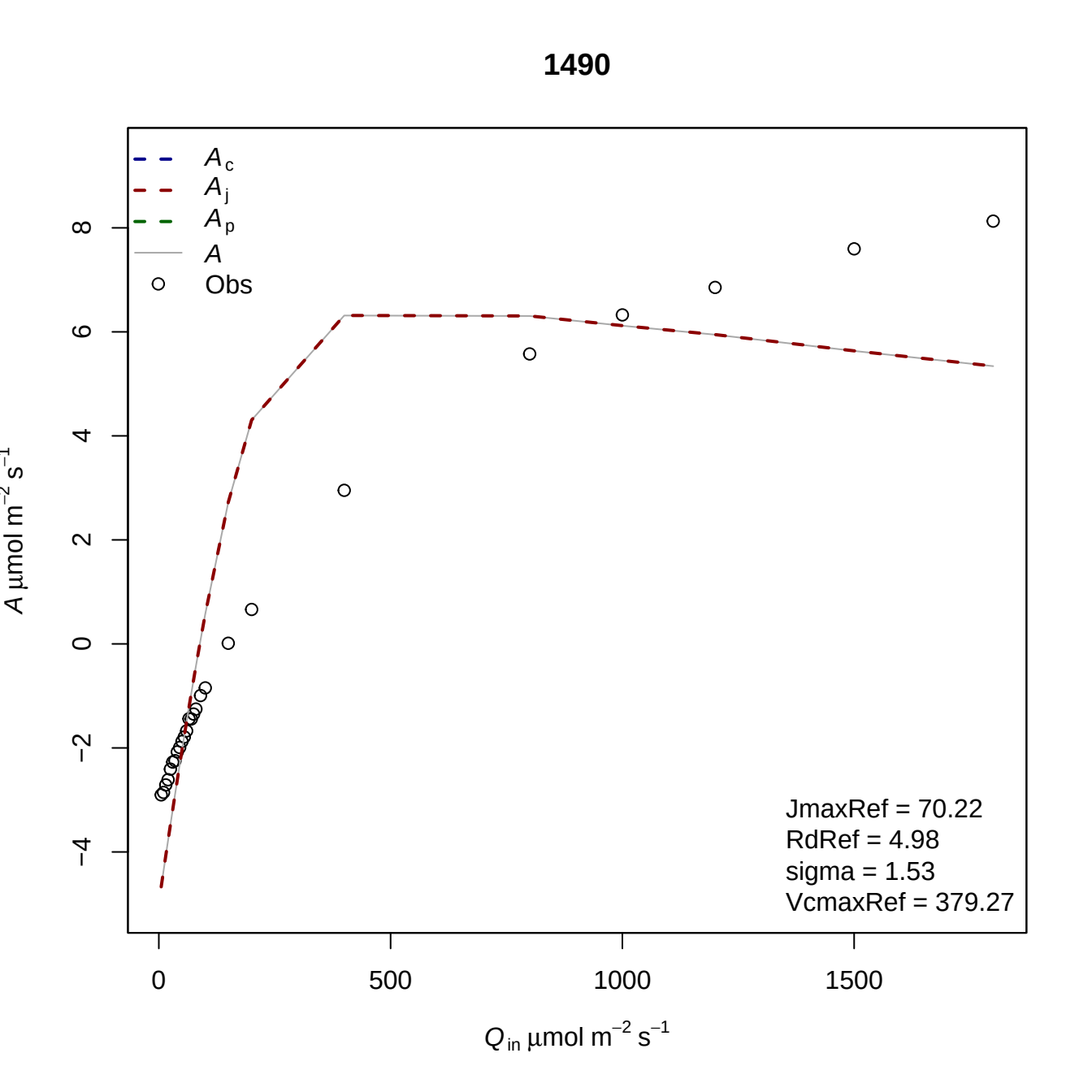

1485
A_c
A_j
A_p
A
Obs
JmaxRef = 69.01
RdRef = 3.73
sigma = 1.19
VcmaxRef = 739.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

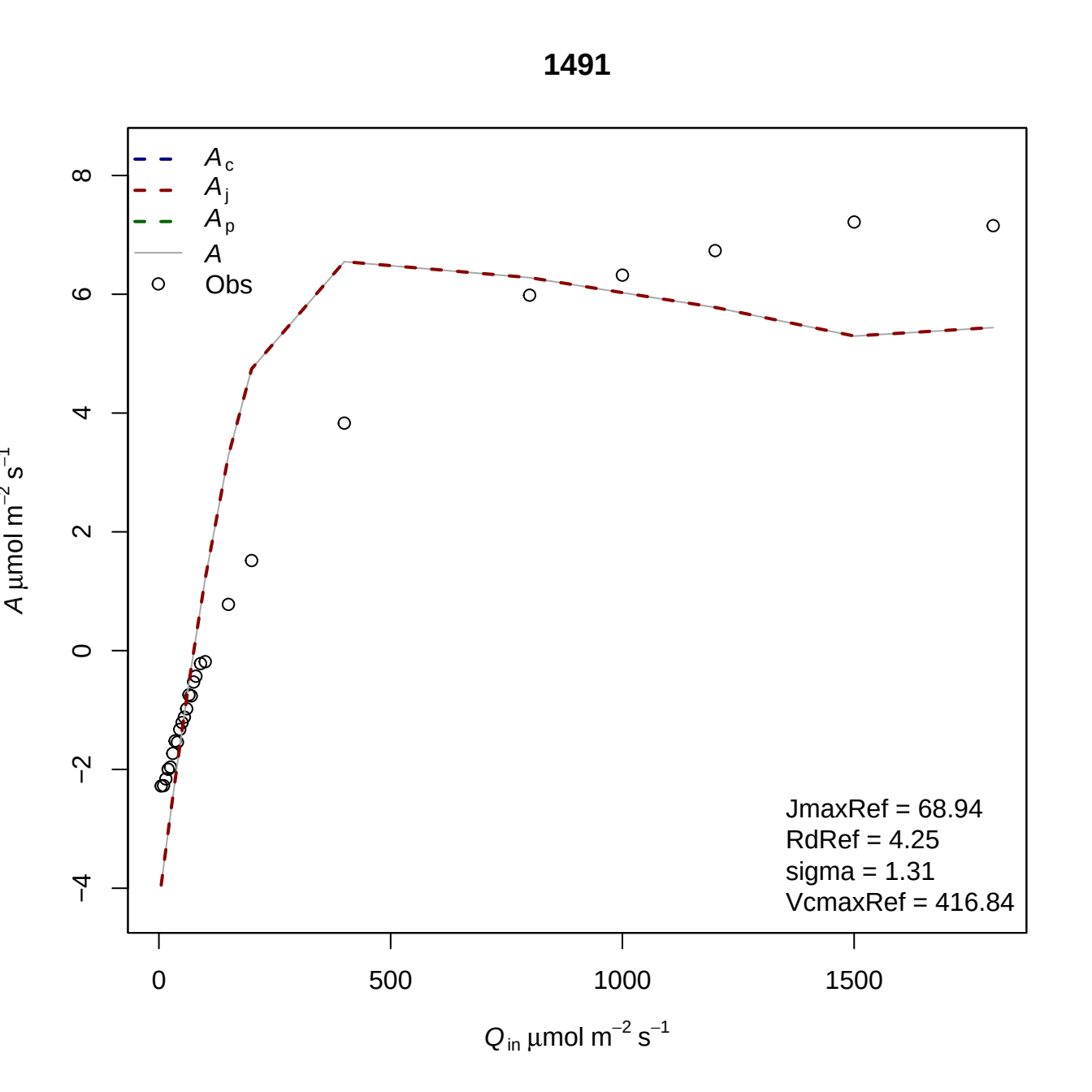

1486
A_c
A_j
A_p
A
Obs
JmaxRef = 62.65
RdRef = 4.43
sigma = 1.53
VcmaxRef = 308.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

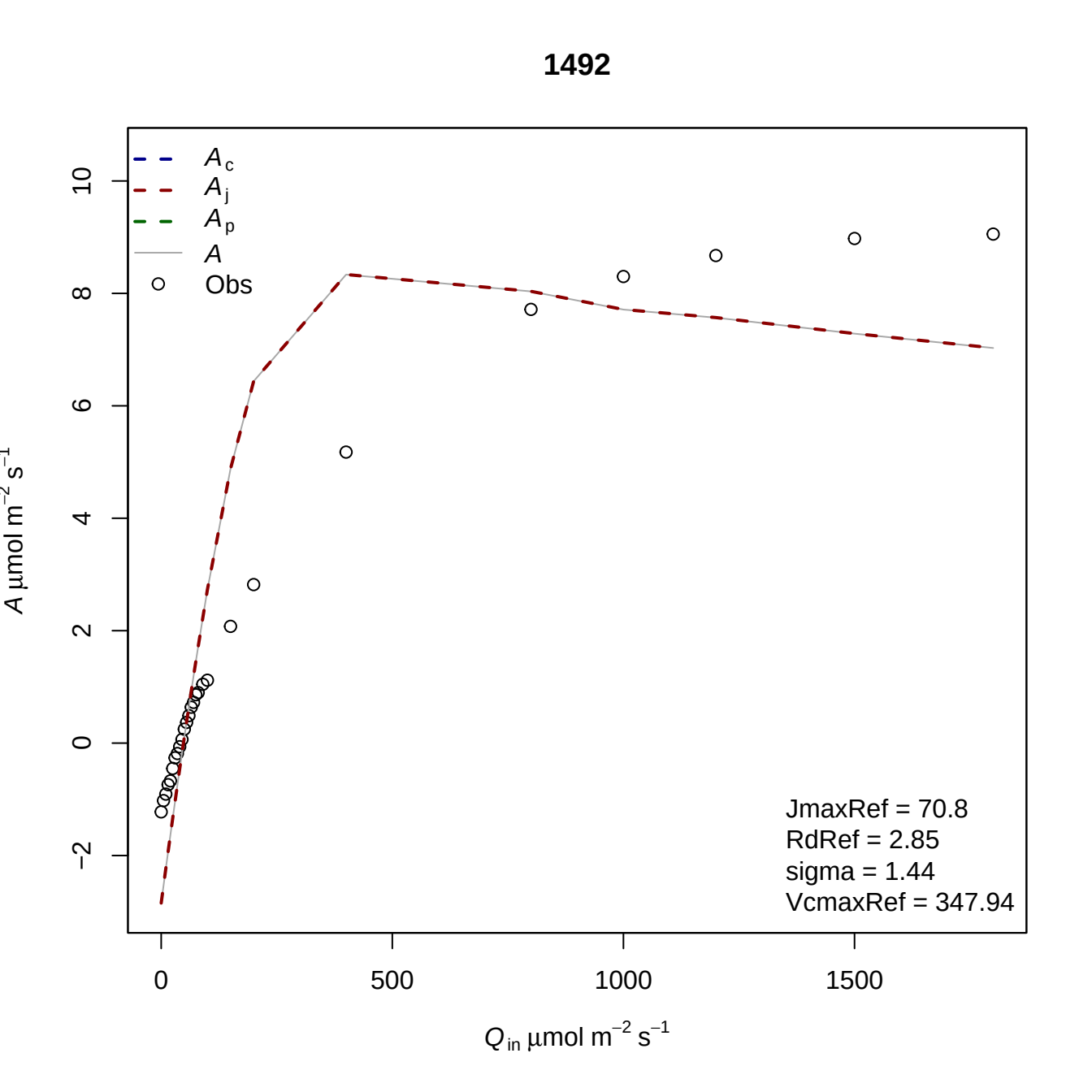

1487
A_c
A_j
A_p
A
Obs
JmaxRef = 63.18
RdRef = 4.19
sigma = 1.39
VcmaxRef = 835.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

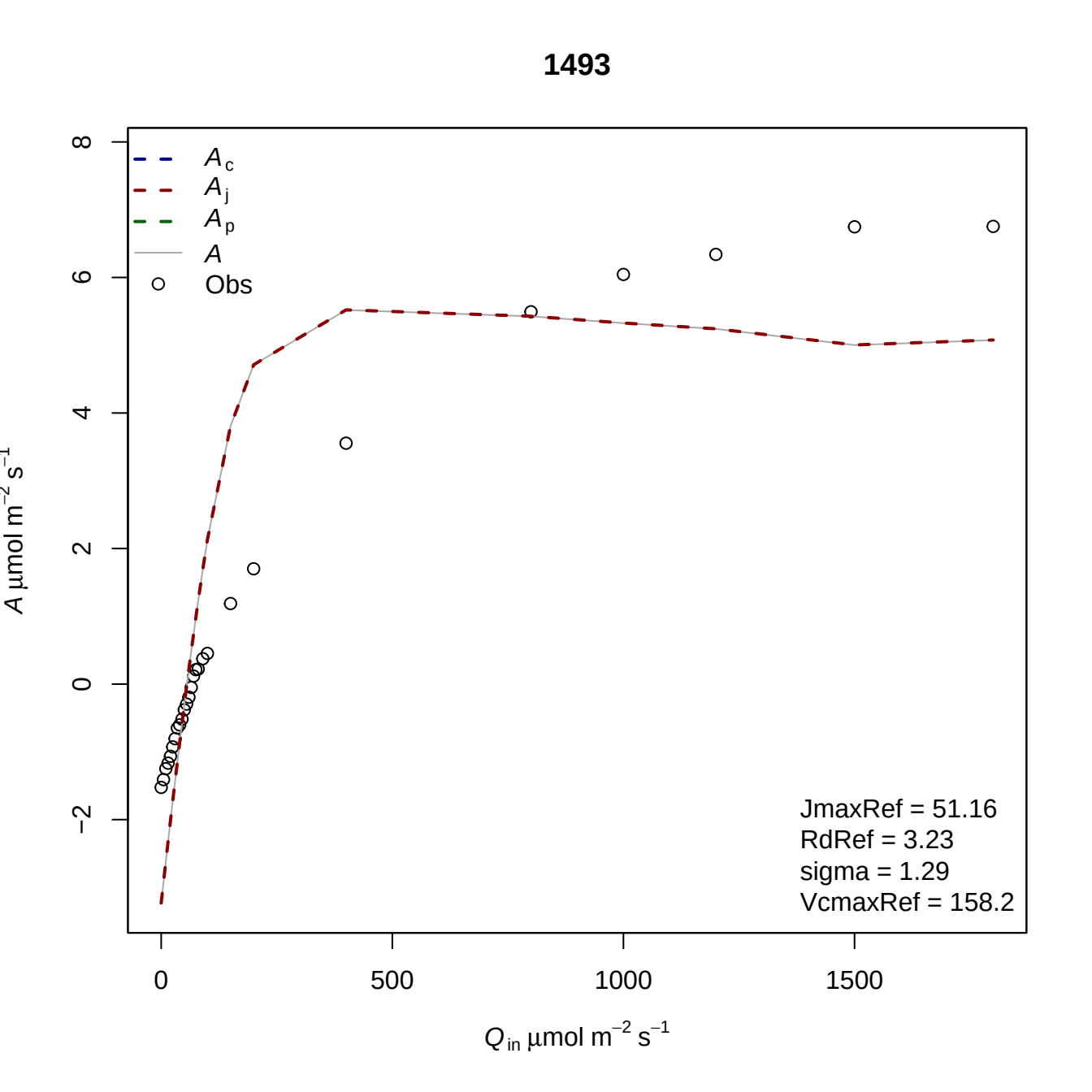

1488
A_c
A_j
A_p
A
Obs
JmaxRef = 72.79
RdRef = 4.39
sigma = 1.5
VcmaxRef = 6977.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

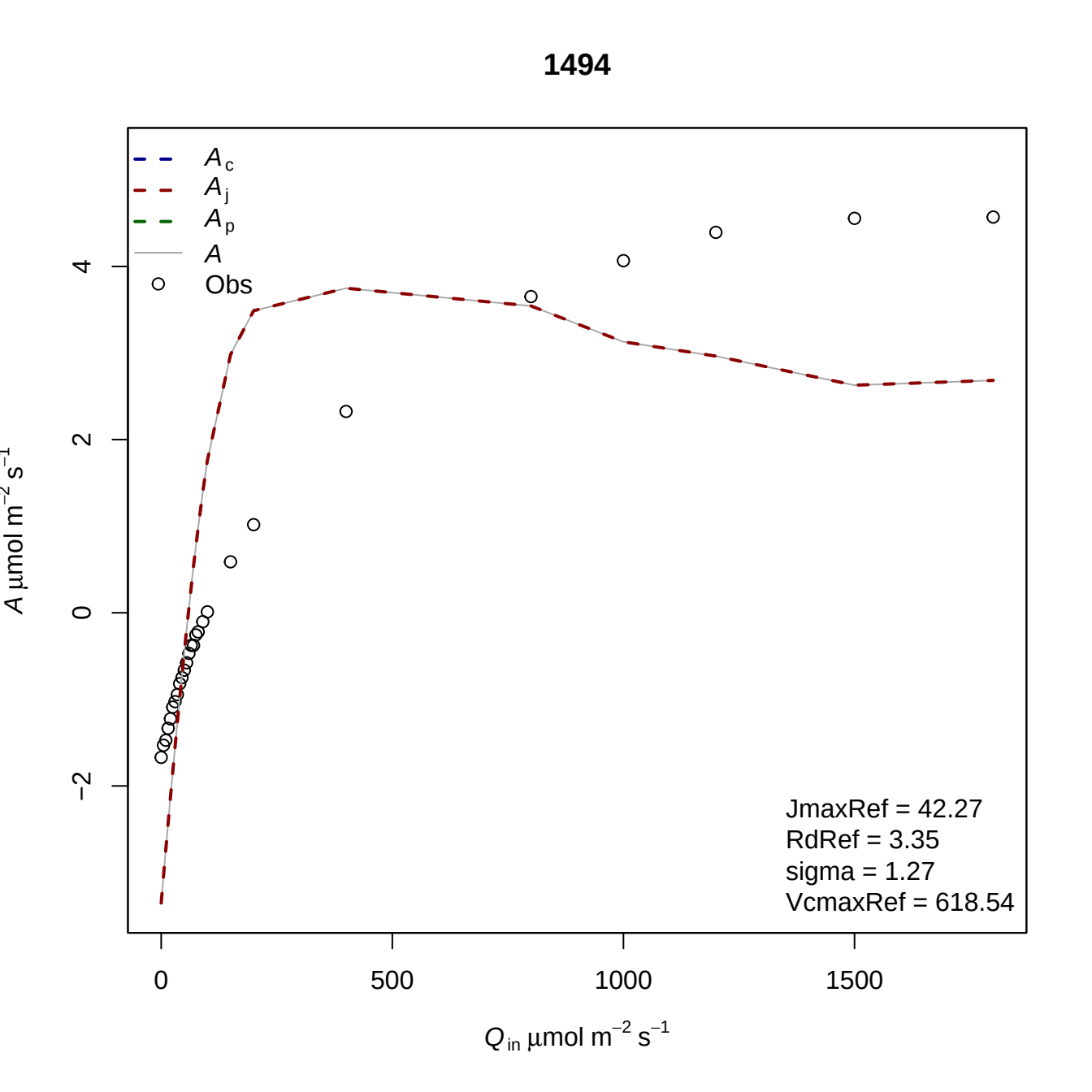

1489
A_c
A_j
A_p
A
Obs
JmaxRef = 73.93
RdRef = 3.61
sigma = 1.5
VcmaxRef = 527.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

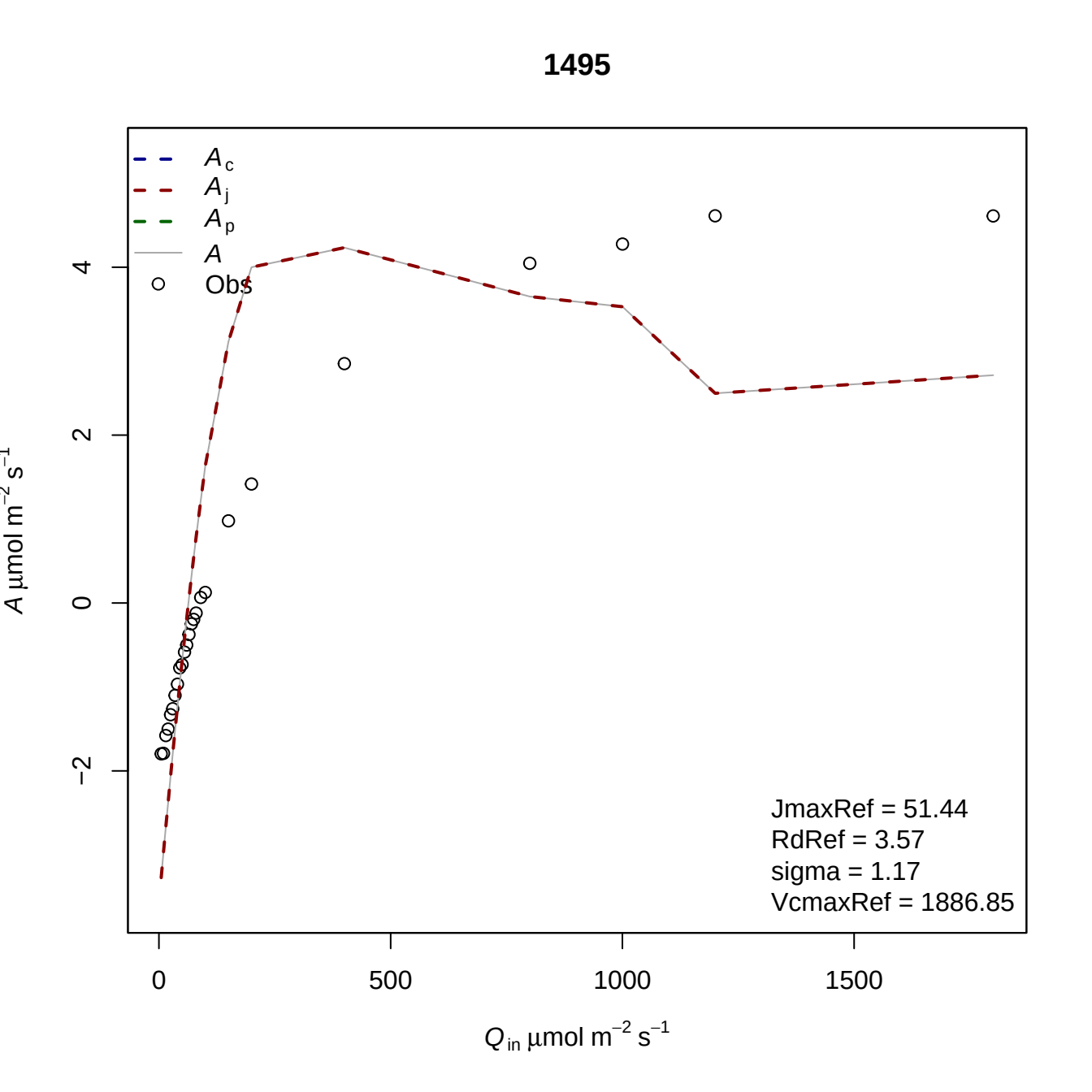

1490
A_c
A_j
A_p
A
Obs
JmaxRef = 70.22
RdRef = 4.98
sigma = 1.53
VcmaxRef = 379.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

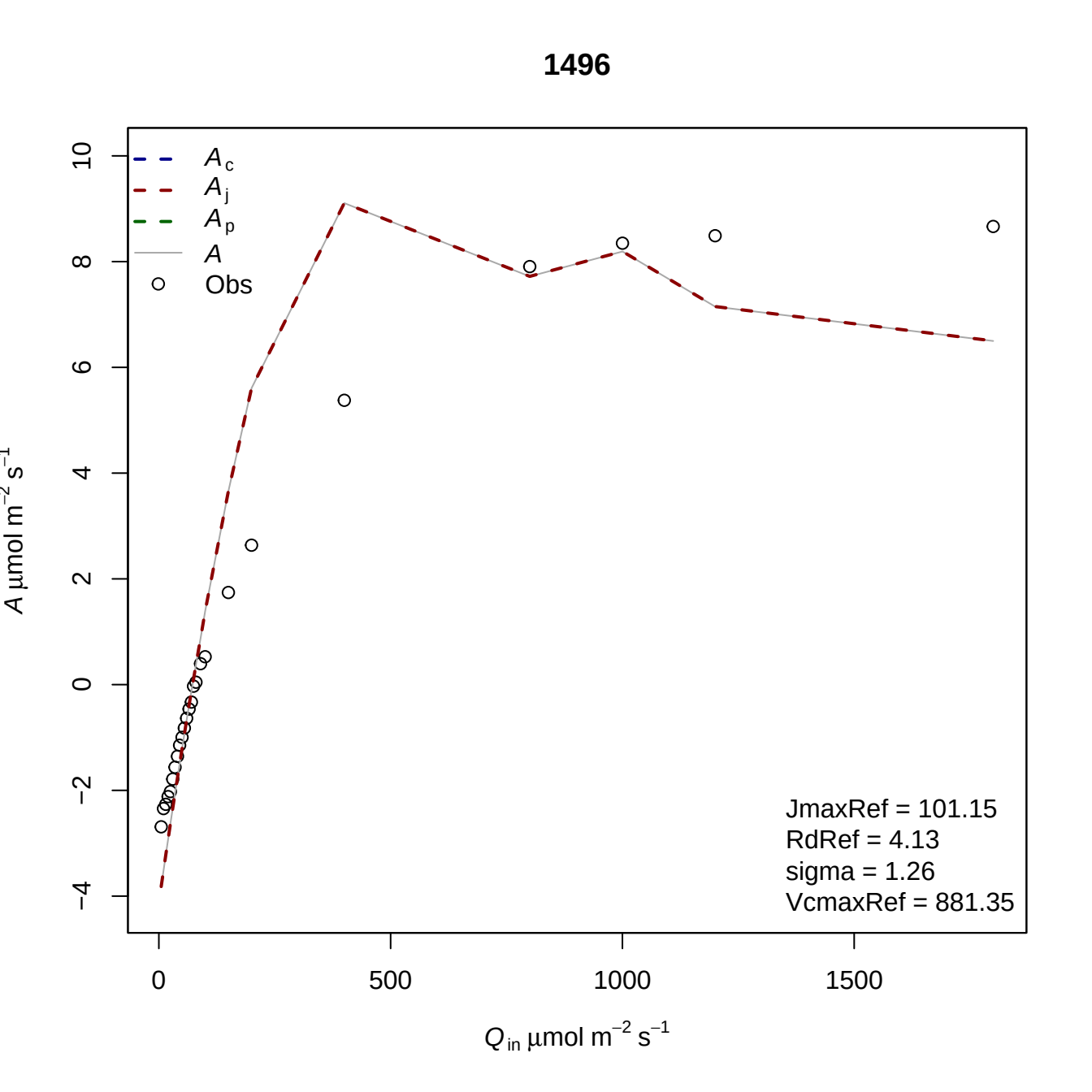

1491
A_c
A_j
A_p
A
Obs
JmaxRef = 68.94
RdRef = 4.25
sigma = 1.31
VcmaxRef = 416.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

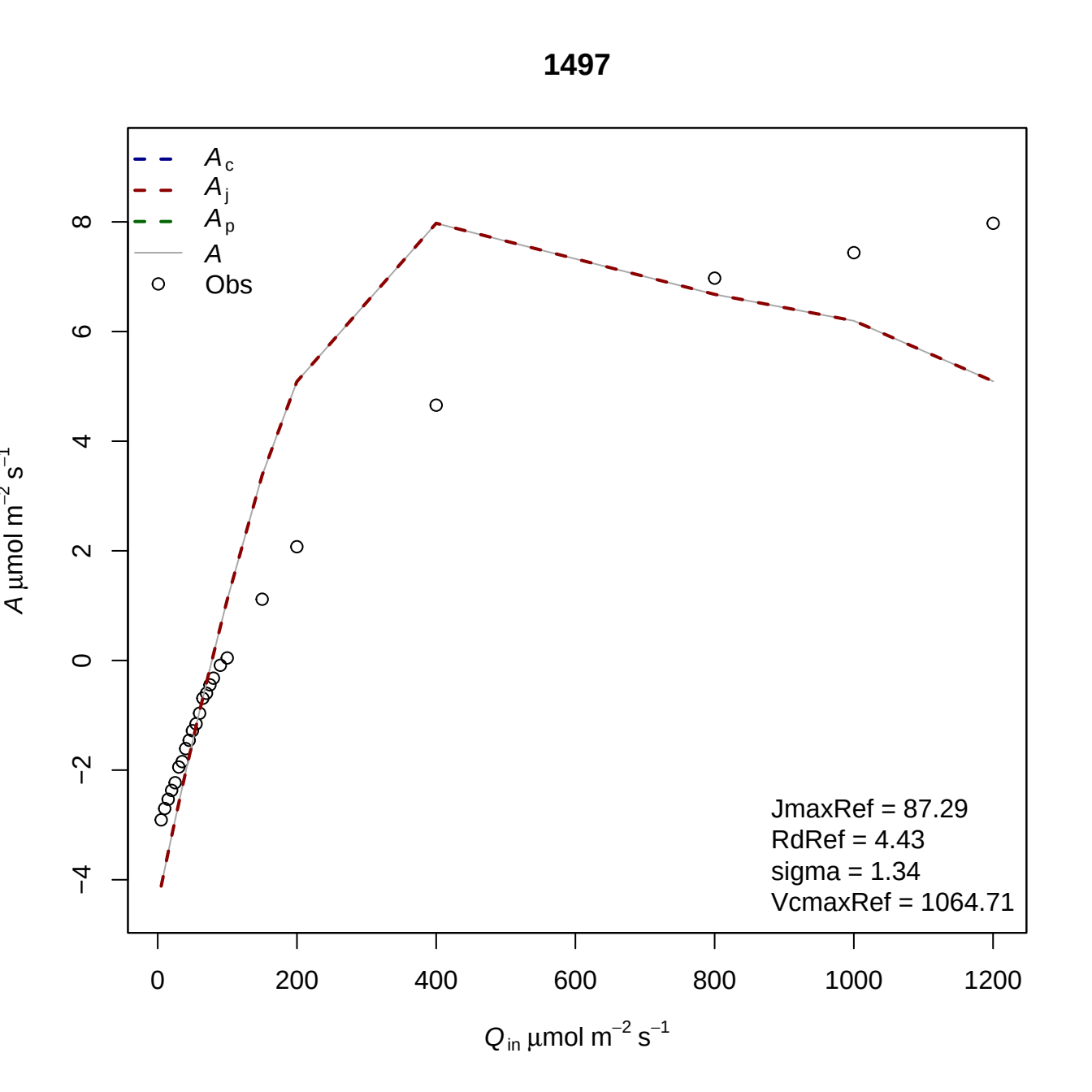

1492
A_c
A_j
A_p
A
Obs
JmaxRef = 70.8
RdRef = 2.85
sigma = 1.44
VcmaxRef = 347.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

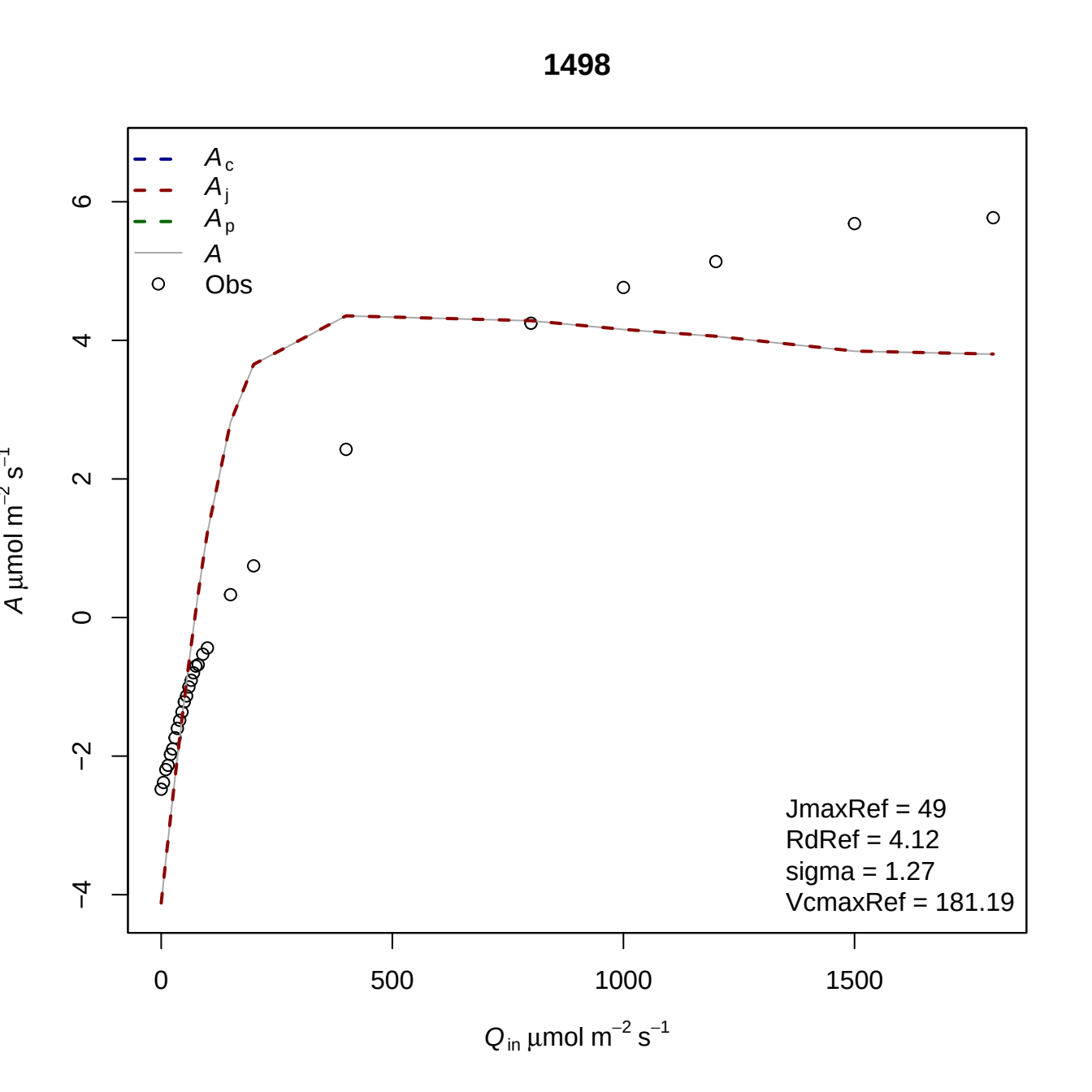

1493
A_c
A_j
A_p
A
Obs
JmaxRef = 51.16
RdRef = 3.23
sigma = 1.29
VcmaxRef = 158.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

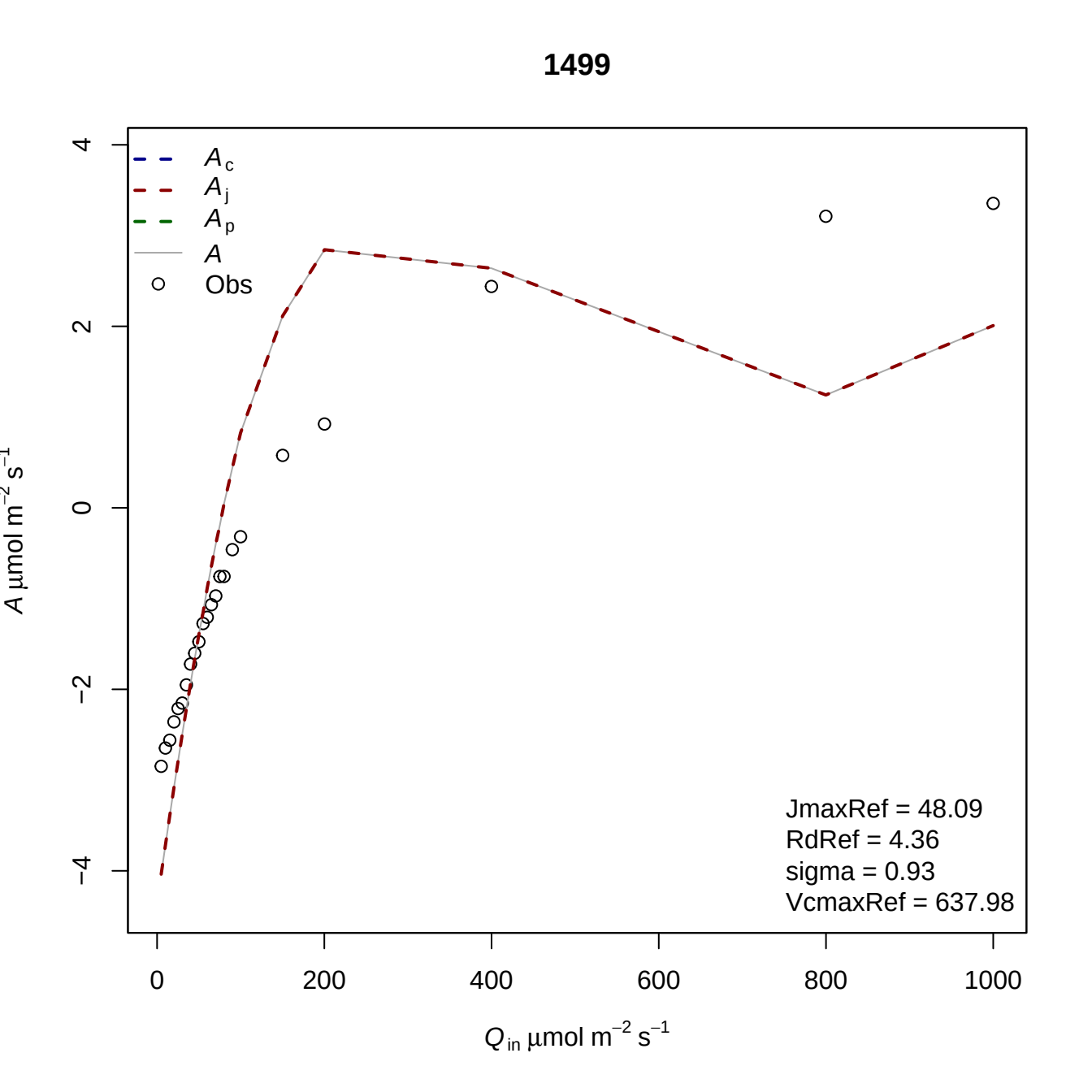

1494
A_c
A_j
A_p
A
Obs
JmaxRef = 42.27
RdRef = 3.35
sigma = 1.27
VcmaxRef = 618.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

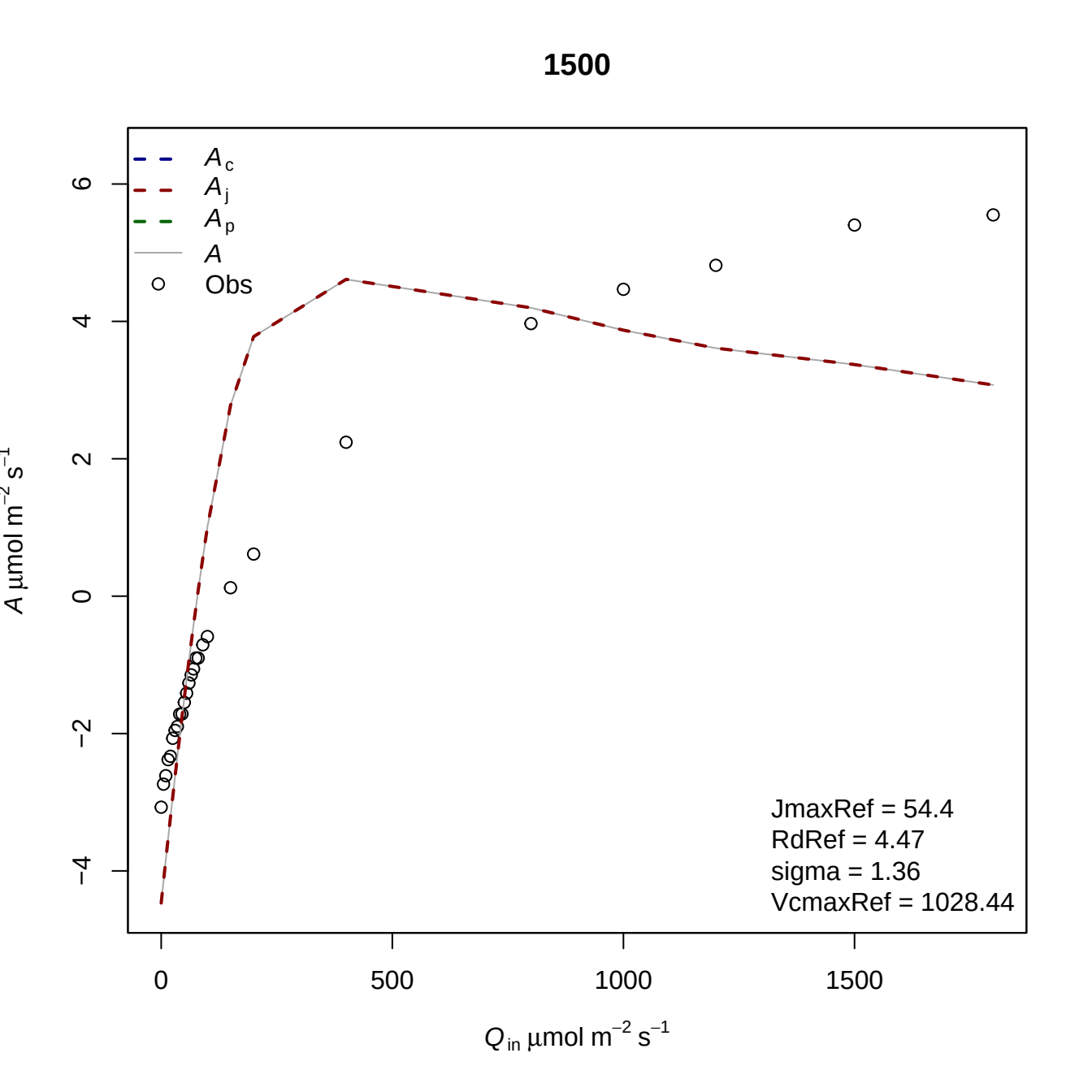

1495
A_c
A_j
A_p
A
Obs
JmaxRef = 51.44
RdRef = 3.57
sigma = 1.17
VcmaxRef = 1886.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

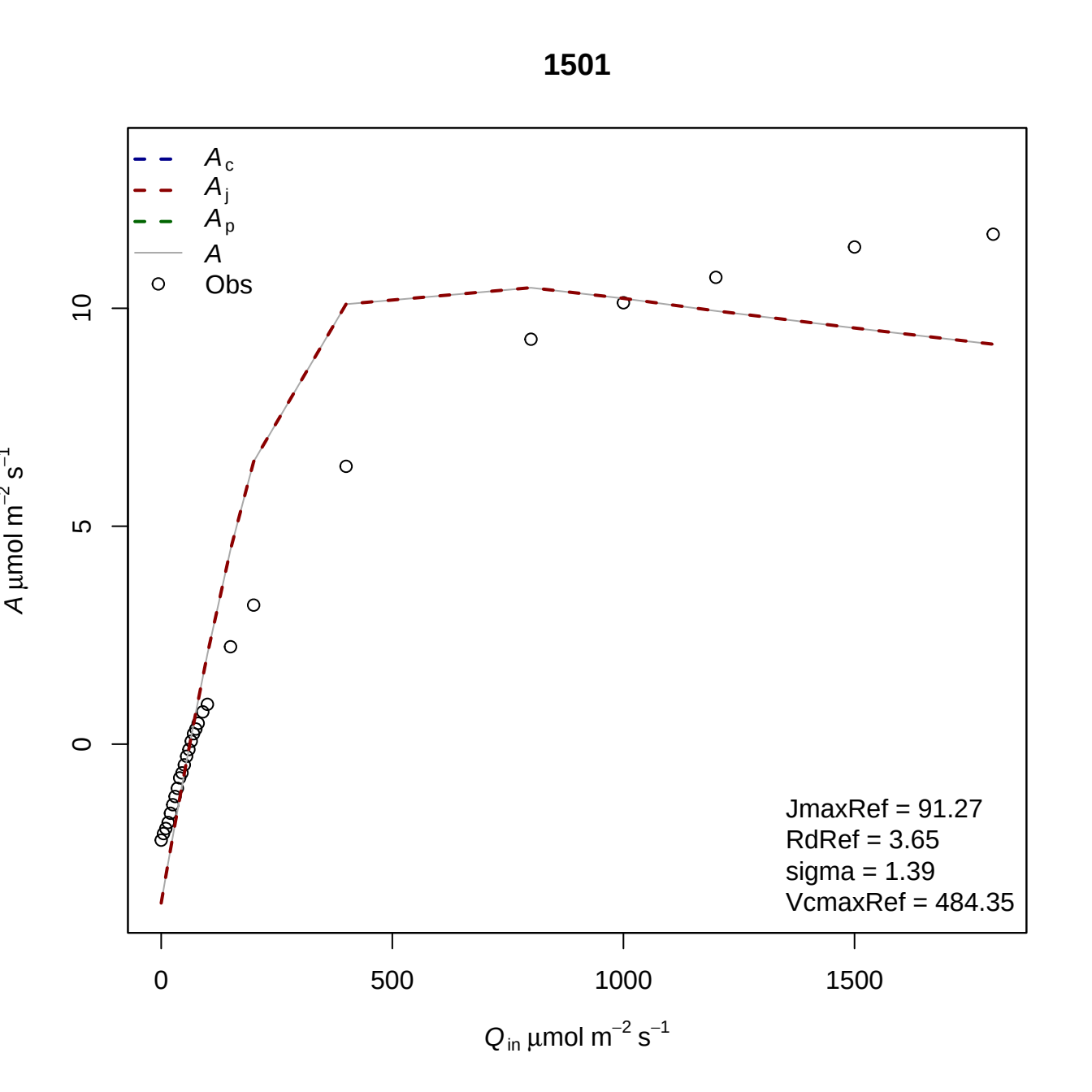

1496
A_c
A_j
A_p
A
Obs
JmaxRef = 101.15
RdRef = 4.13
sigma = 1.26
VcmaxRef = 881.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

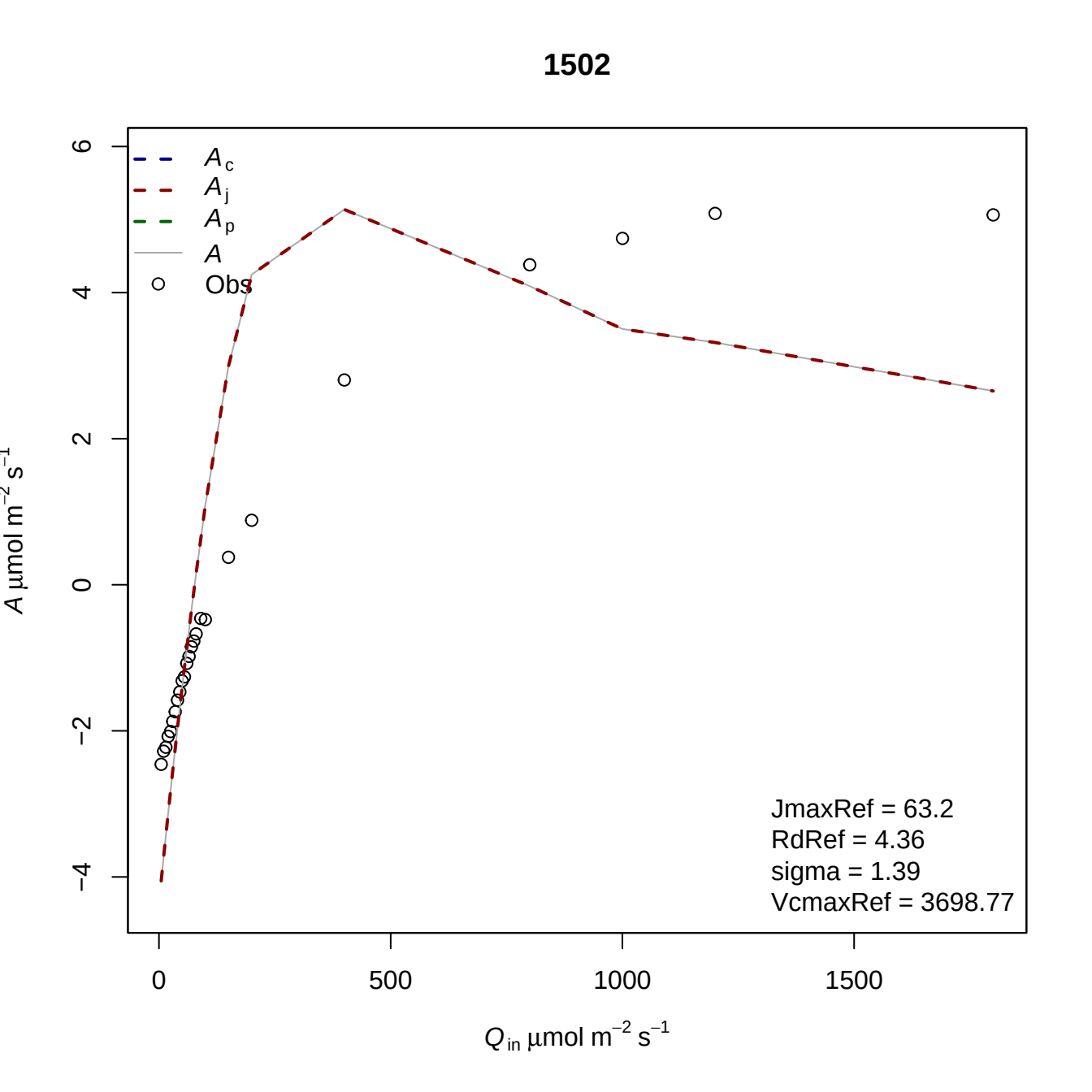

1497
A_c
A_j
A_p
A
Obs
JmaxRef = 87.29
RdRef = 4.43
sigma = 1.34
VcmaxRef = 1064.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

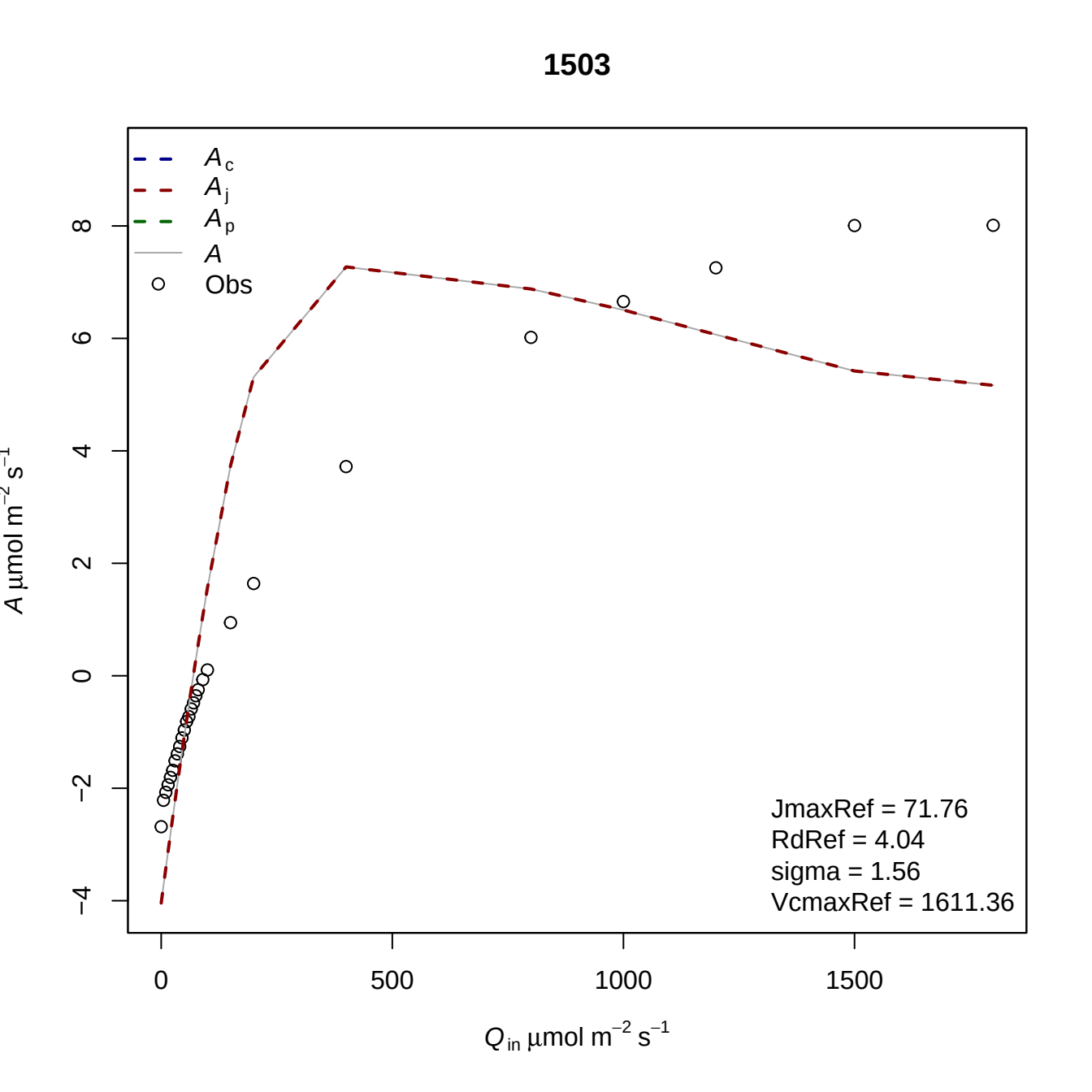

1498
A_c
A_j
A_p
A
Obs
JmaxRef = 49
RdRef = 4.12
sigma = 1.27
VcmaxRef = 181.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

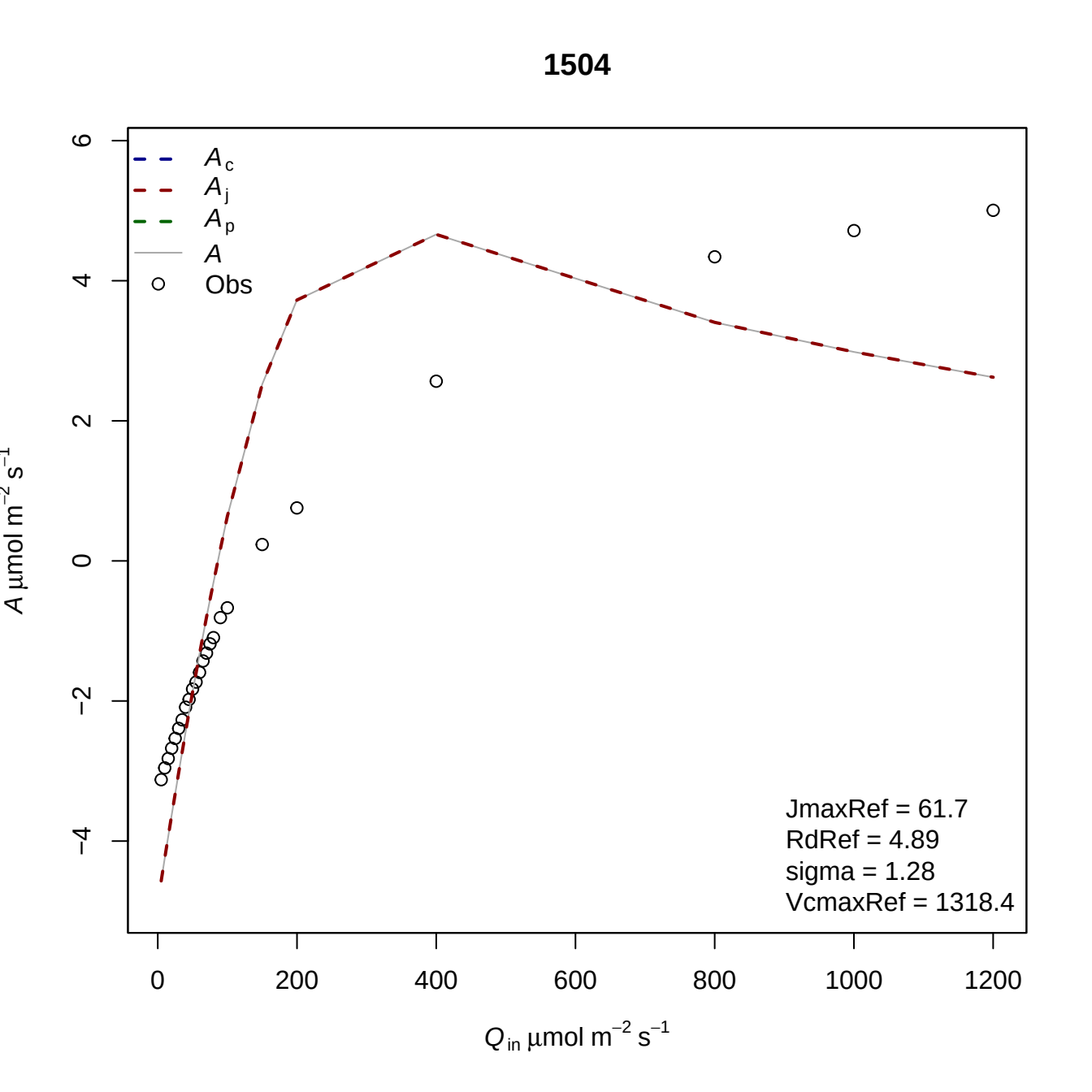

1499
A_c
A_j
A_p
A
Obs
JmaxRef = 48.09
RdRef = 4.36
sigma = 0.93
VcmaxRef = 637.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
200
400
600
800
1000

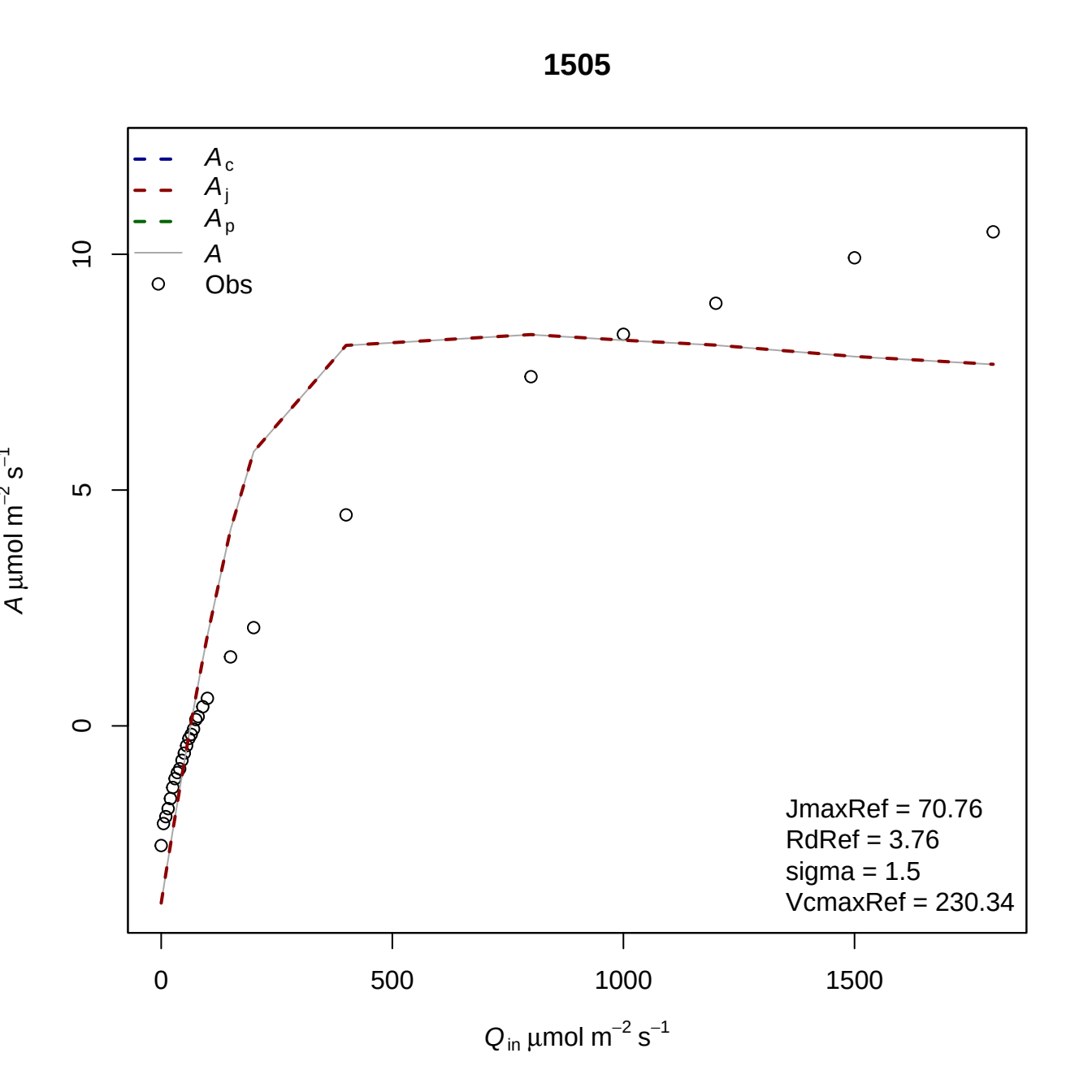

1500
A_c
A_j
A_p
A
Obs
JmaxRef = 54.4
RdRef = 4.47
sigma = 1.36
VcmaxRef = 1028.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

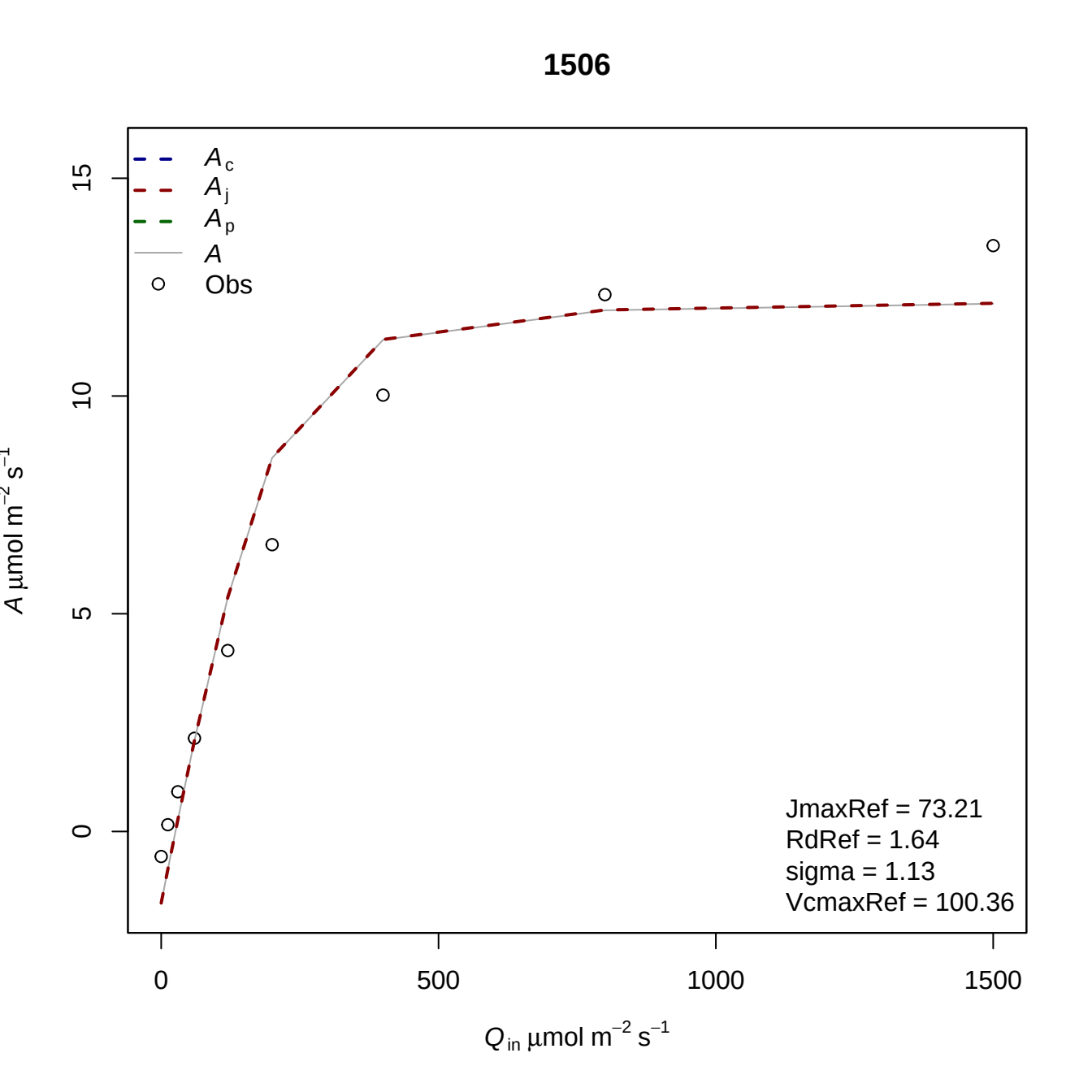

1501
A_c
A_j
A_p
A
Obs
JmaxRef = 91.27
RdRef = 3.65
sigma = 1.39
VcmaxRef = 484.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

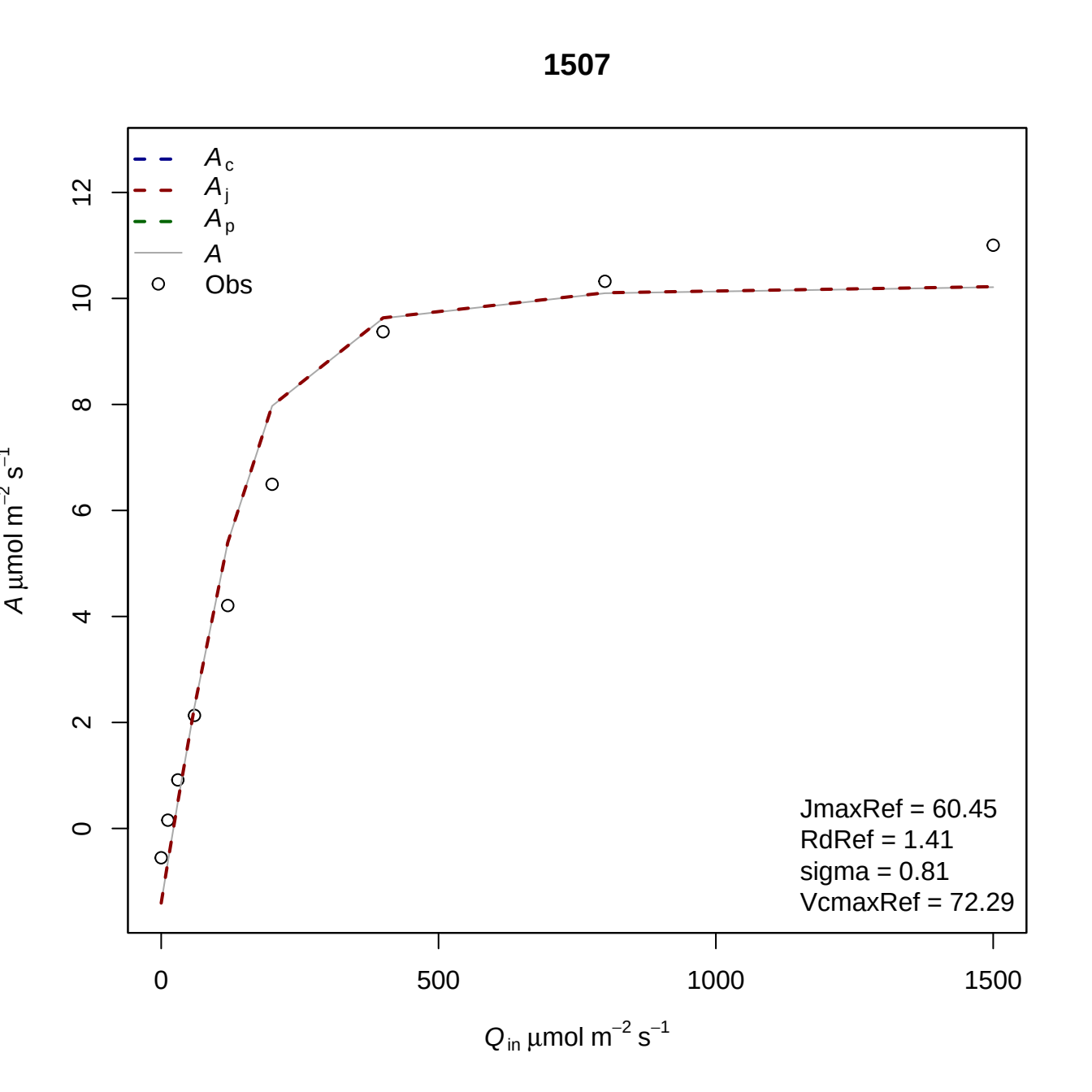

1502
A_c
A_j
A_p
A
Obs
JmaxRef = 63.2
RdRef = 4.36
sigma = 1.39
VcmaxRef = 3698.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

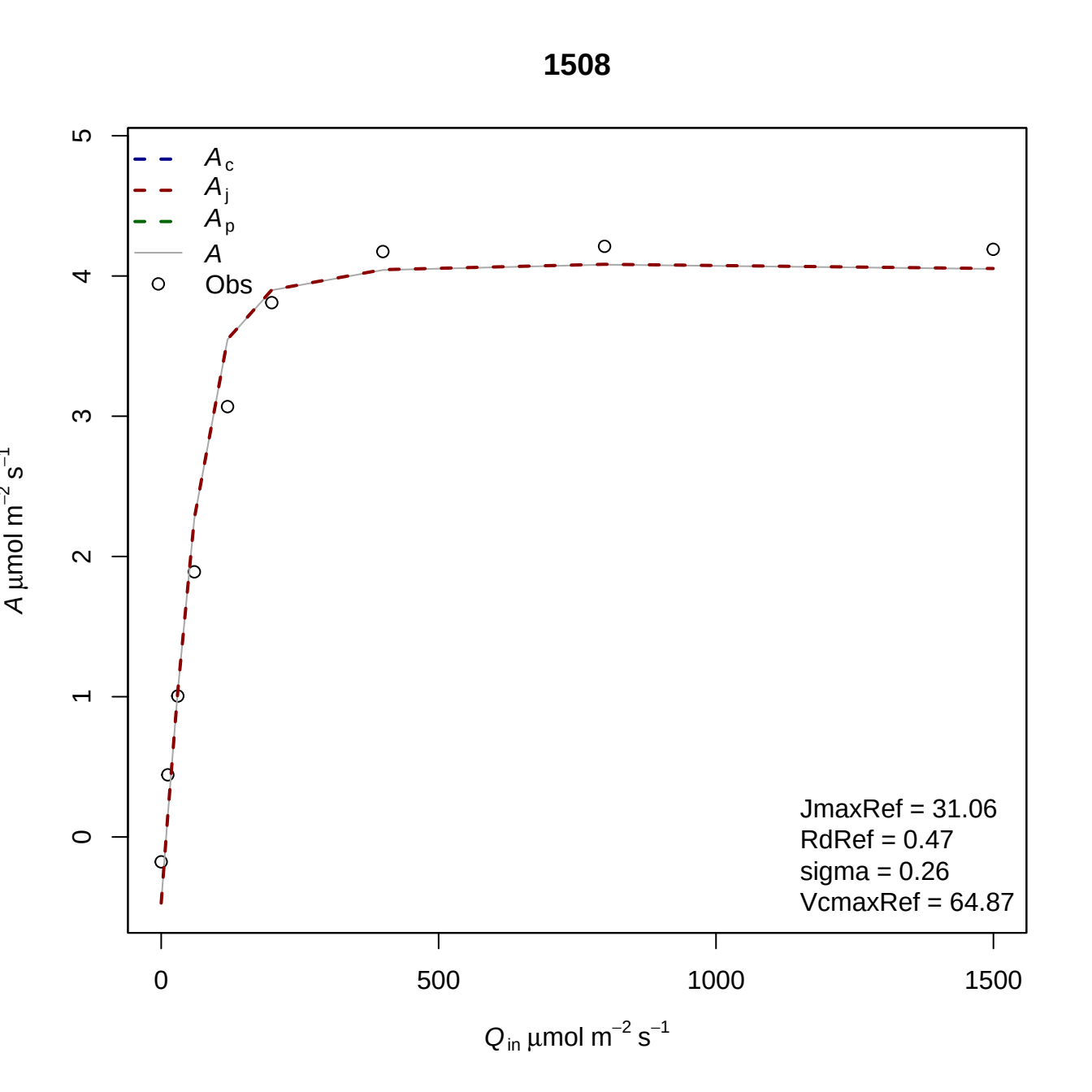

1503
A_c
A_j
A_p
A
Obs
JmaxRef = 71.76
RdRef = 4.04
sigma = 1.56
VcmaxRef = 1611.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

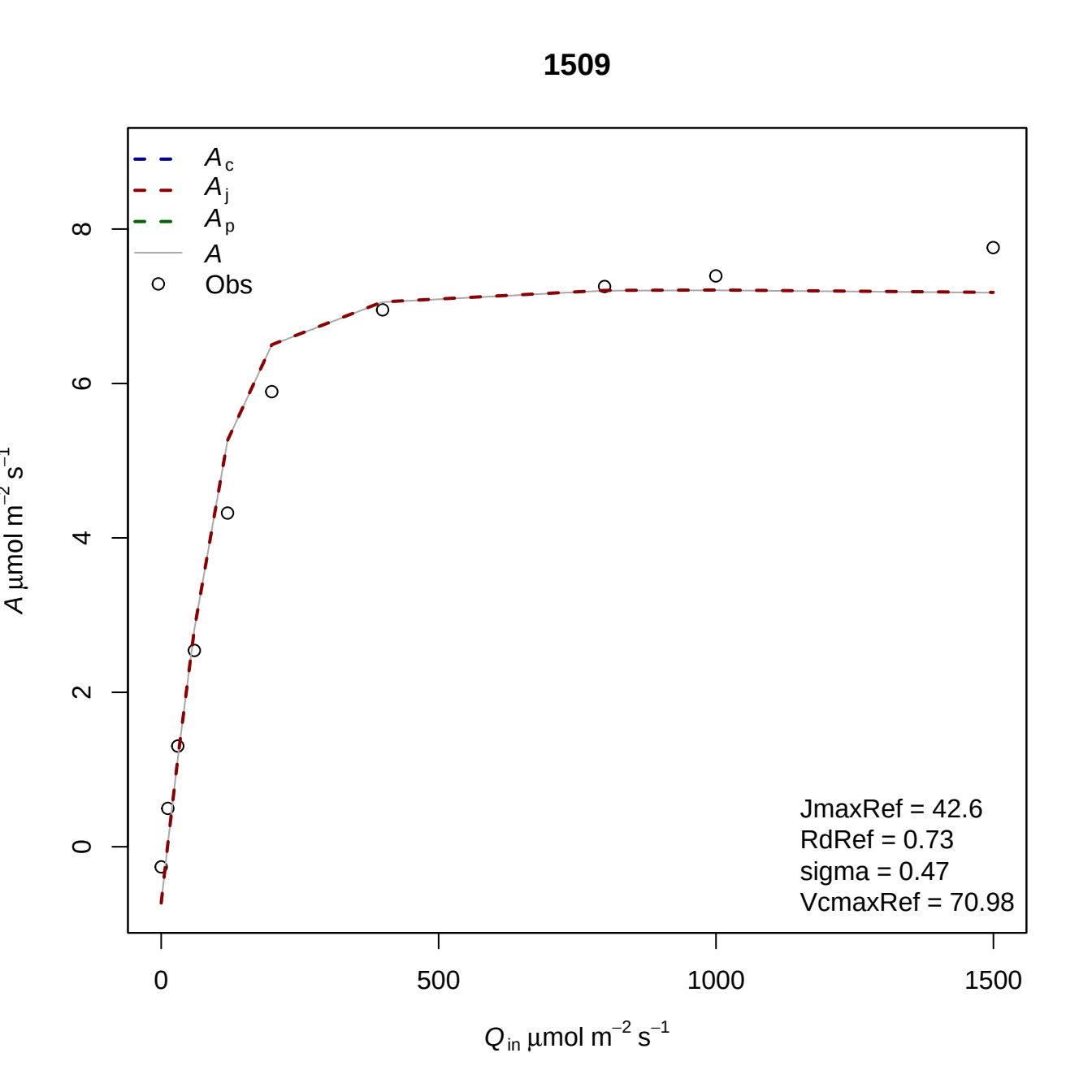

1504
A_c
A_j
A_p
A
Obs
JmaxRef = 61.7
RdRef = 4.89
sigma = 1.28
VcmaxRef = 1318.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

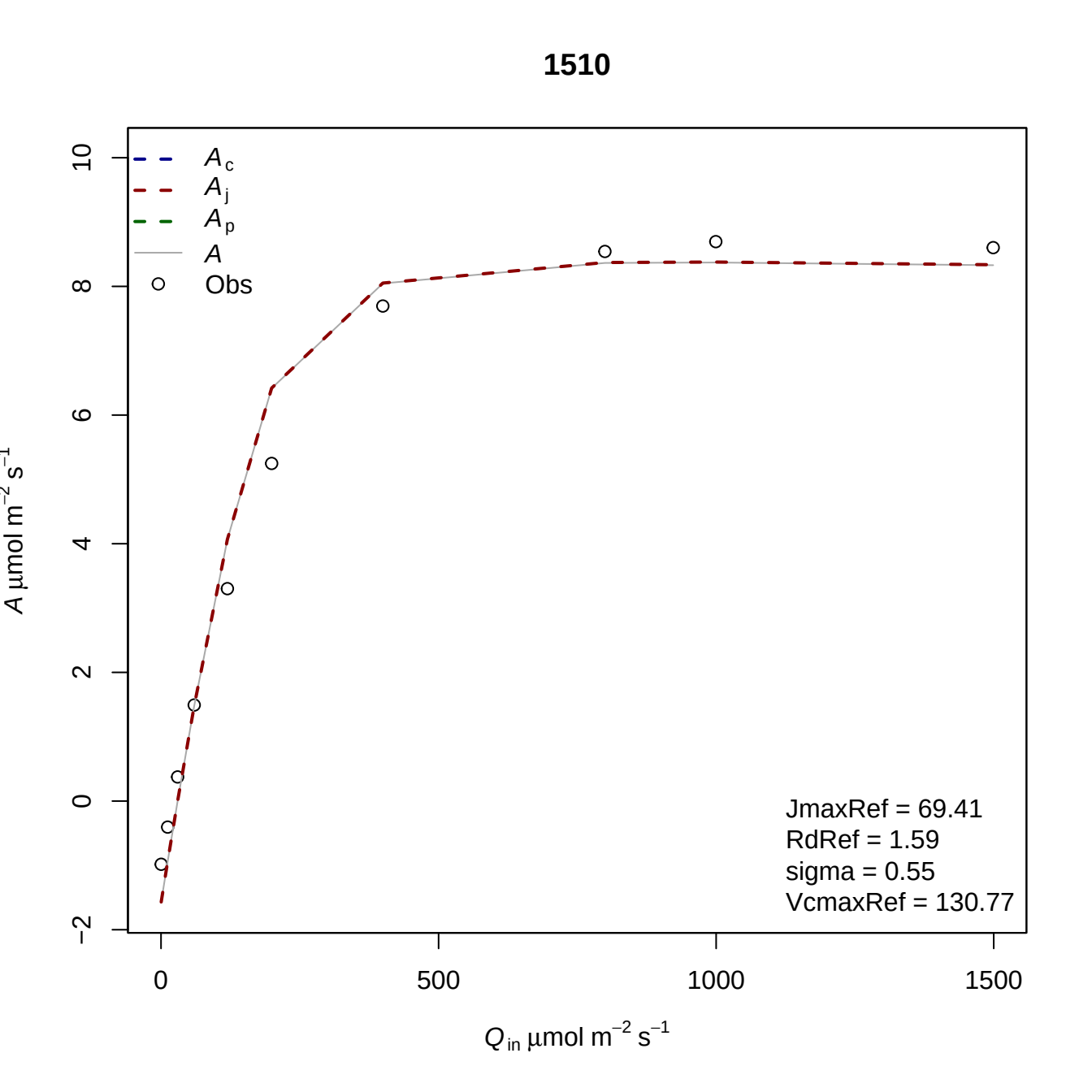

1505
A_c
A_j
A_p
A
Obs
JmaxRef = 70.76
RdRef = 3.76
sigma = 1.5
VcmaxRef = 230.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

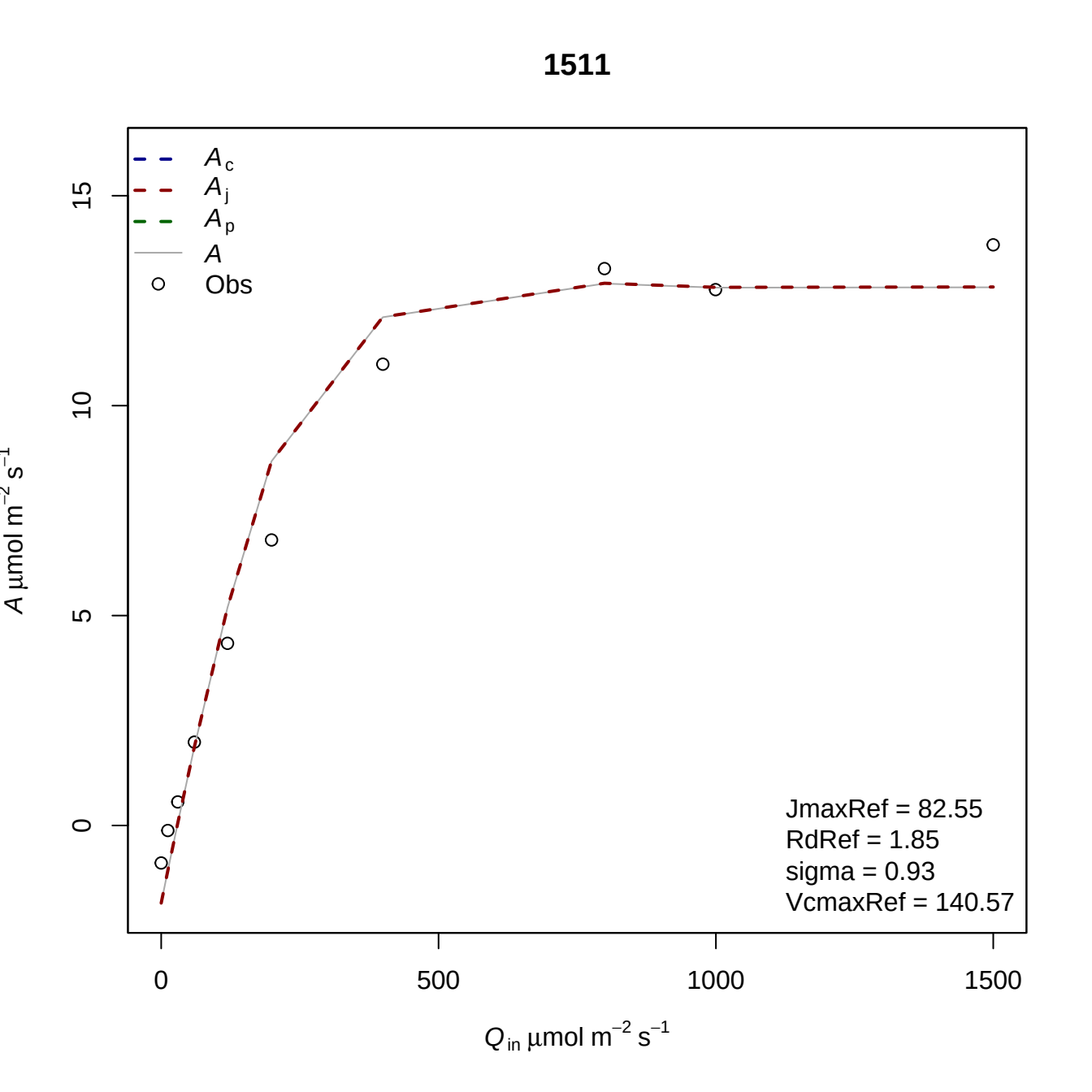

1506
A_c
A_j
A_p
A
Obs
JmaxRef = 73.21
RdRef = 1.64
sigma = 1.13
VcmaxRef = 100.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

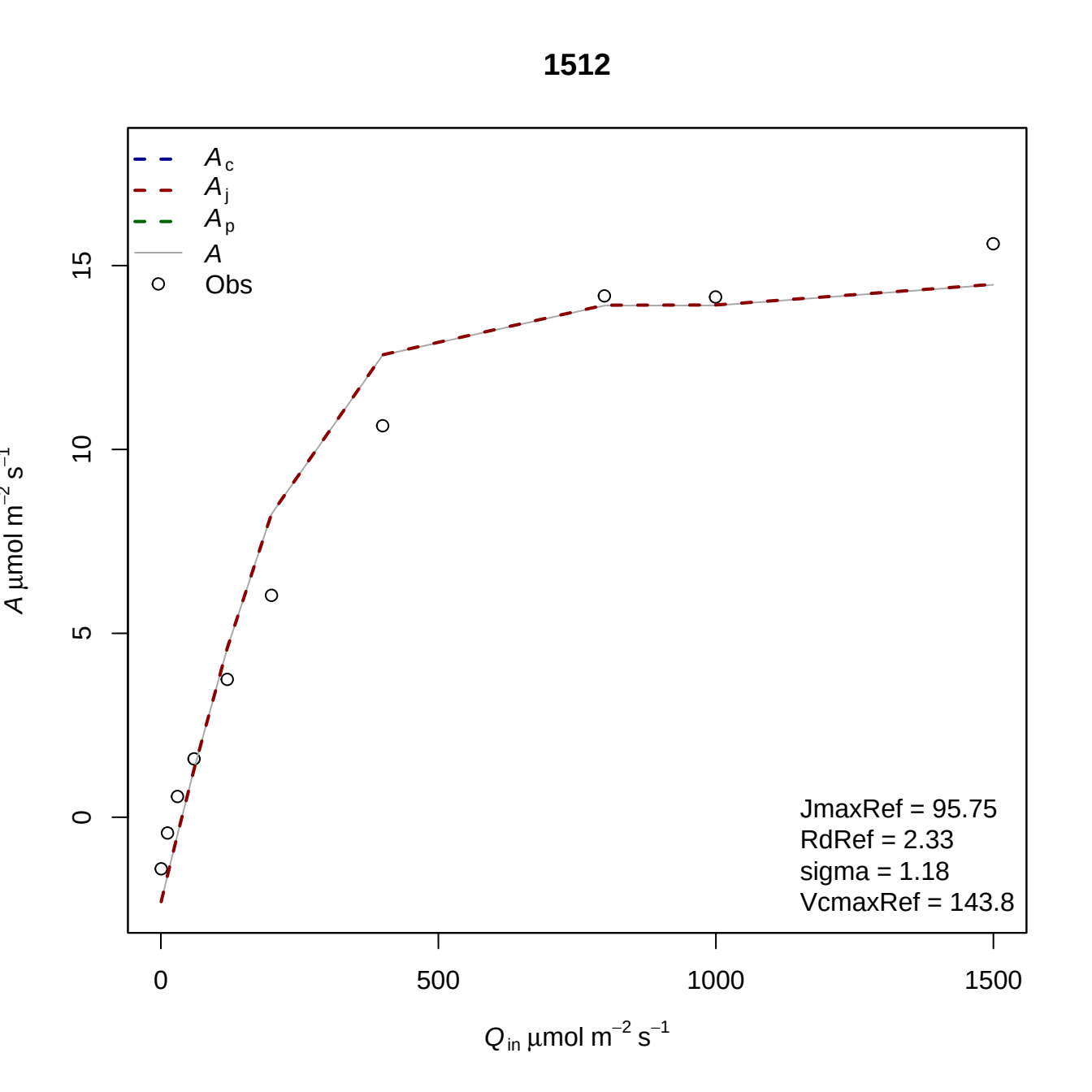

1507
A_c
A_j
A_p
A
Obs
JmaxRef = 60.45
RdRef = 1.41
sigma = 0.81
VcmaxRef = 72.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

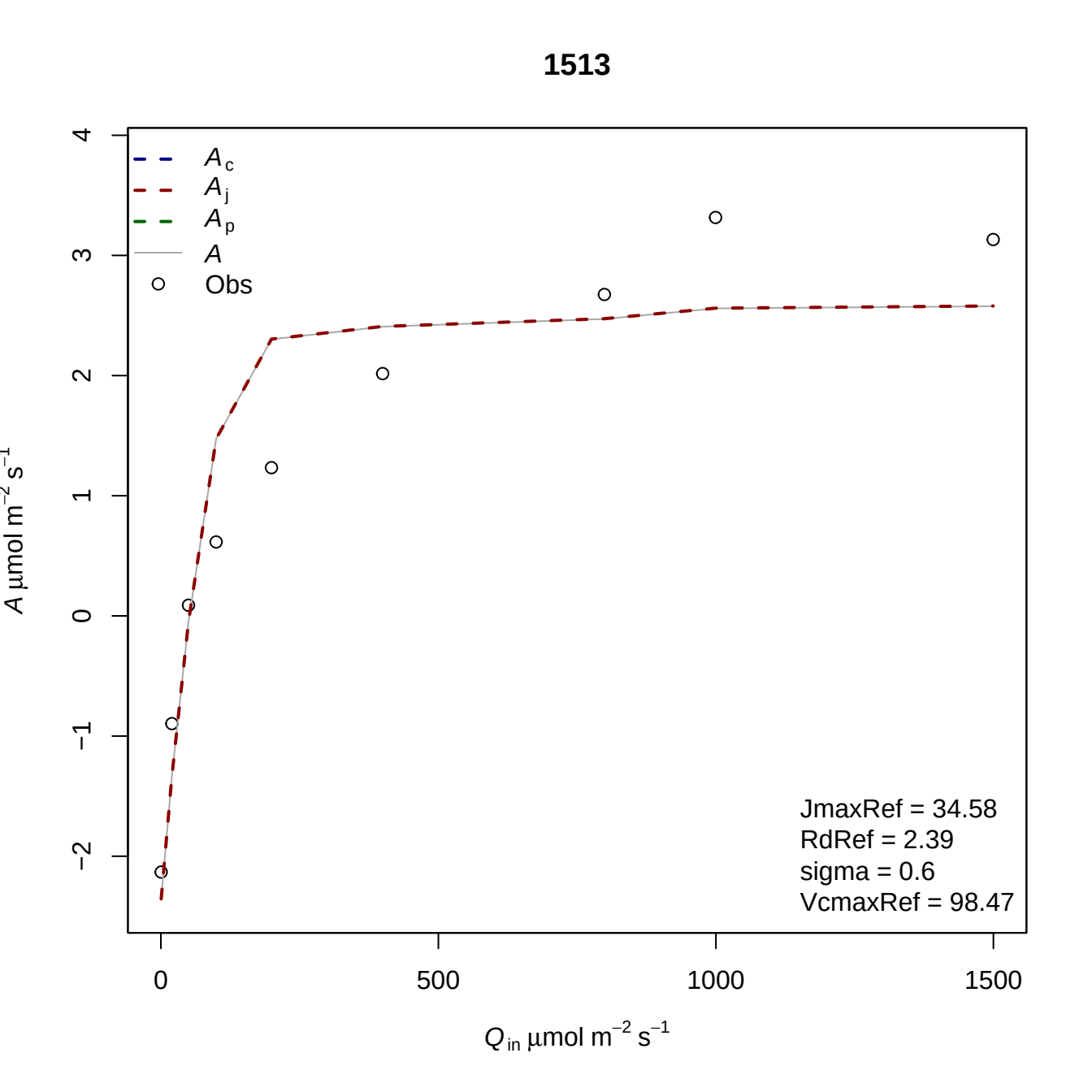

1508
A_c
A_j
A_p
A
Obs
JmaxRef = 31.06
RdRef = 0.47
sigma = 0.26
VcmaxRef = 64.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

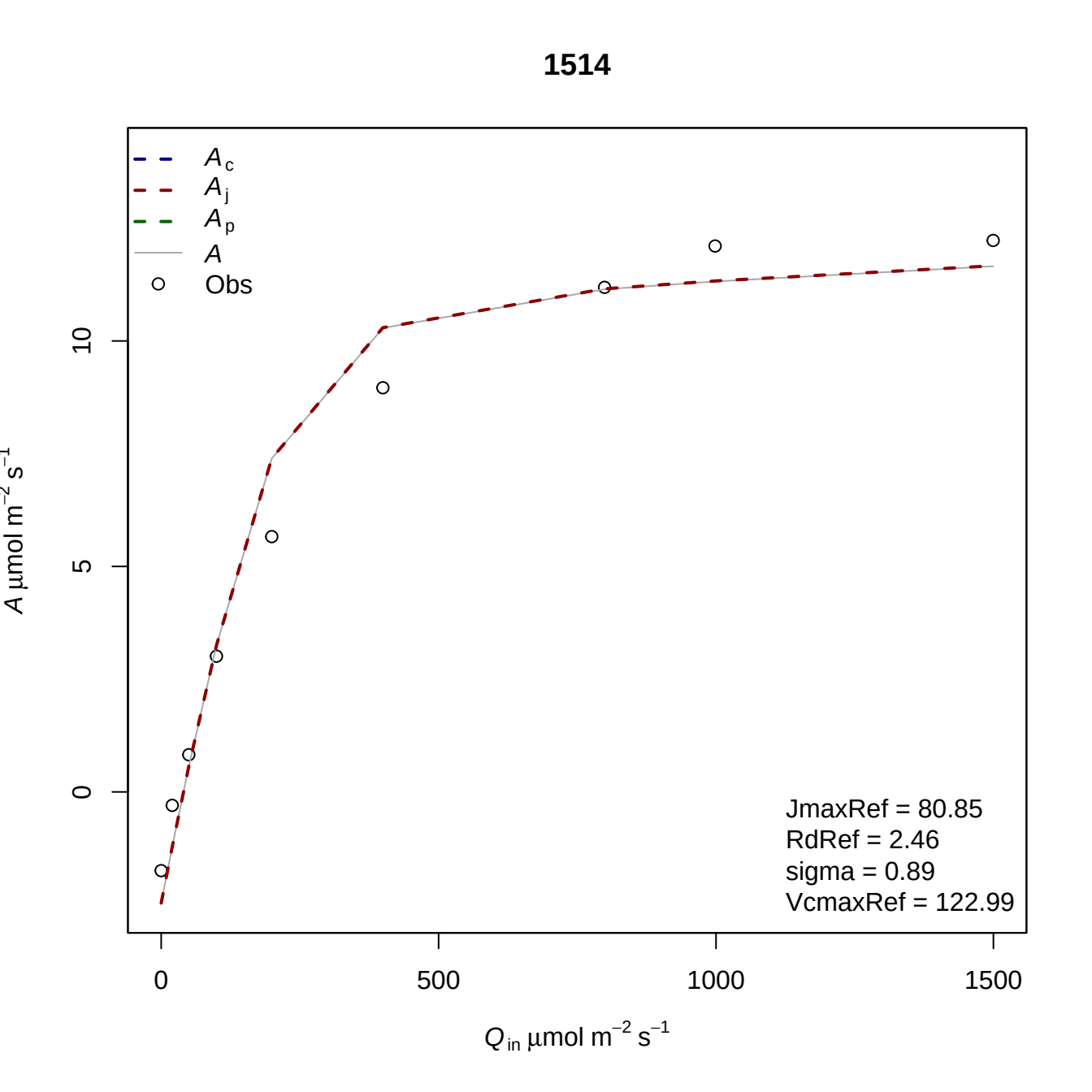

1509
A_c
A_j
A_p
A
Obs
JmaxRef = 42.6
RdRef = 0.73
sigma = 0.47
VcmaxRef = 70.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

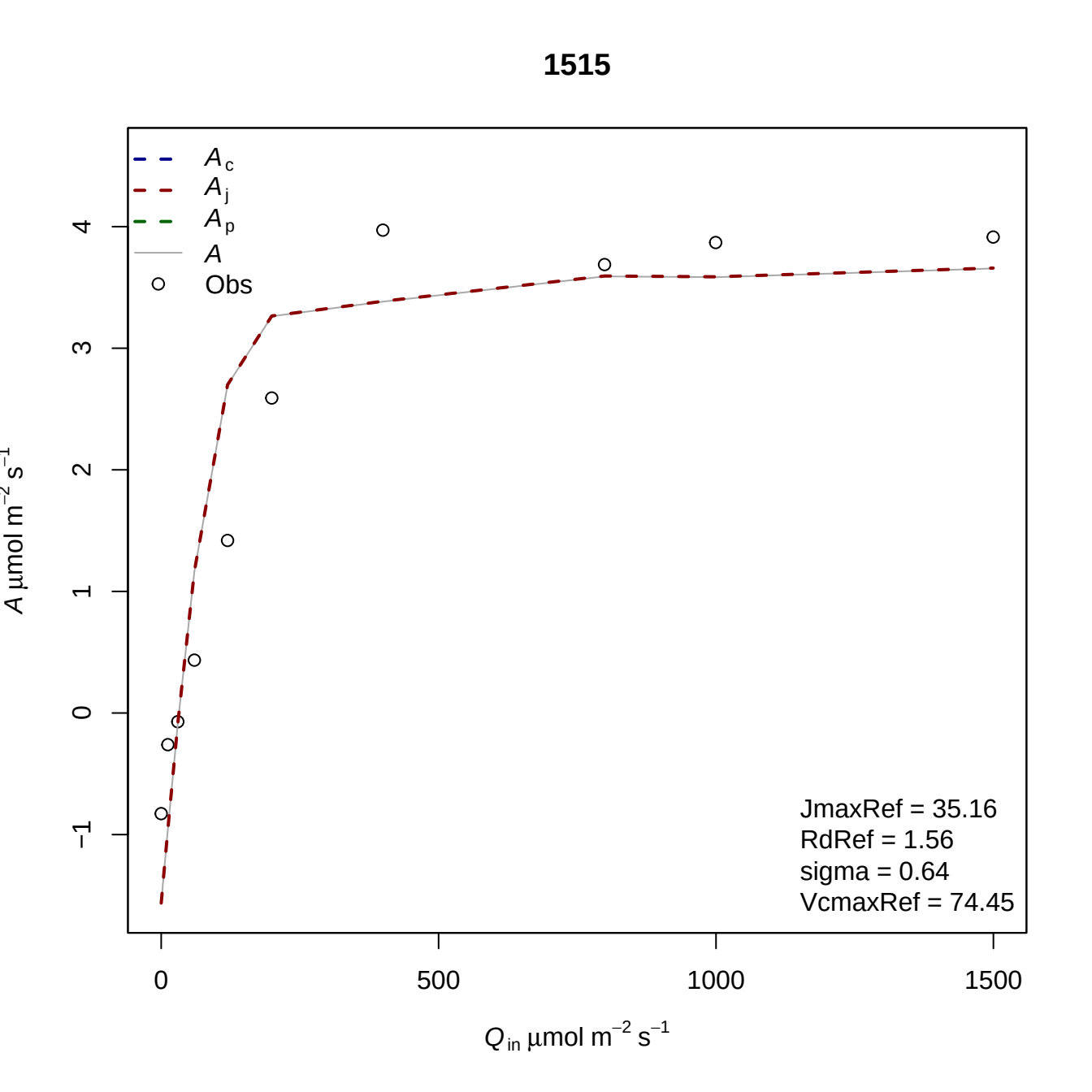

1510
A_c
A_j
A_p
A
Obs
JmaxRef = 69.41
RdRef = 1.59
sigma = 0.55
VcmaxRef = 130.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

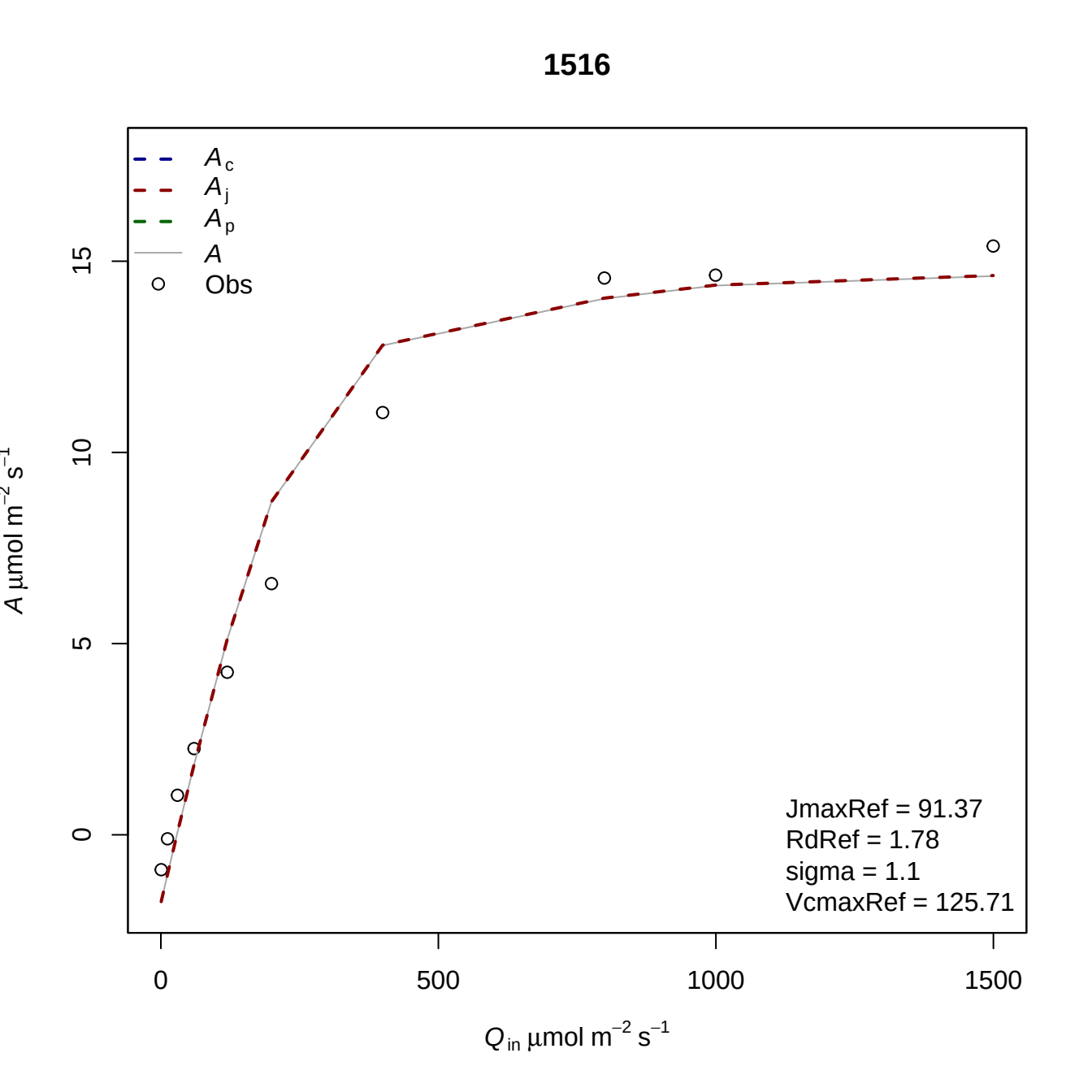

1511
A_c
A_j
A_p
A
Obs
JmaxRef = 82.55
RdRef = 1.85
sigma = 0.93
VcmaxRef = 140.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

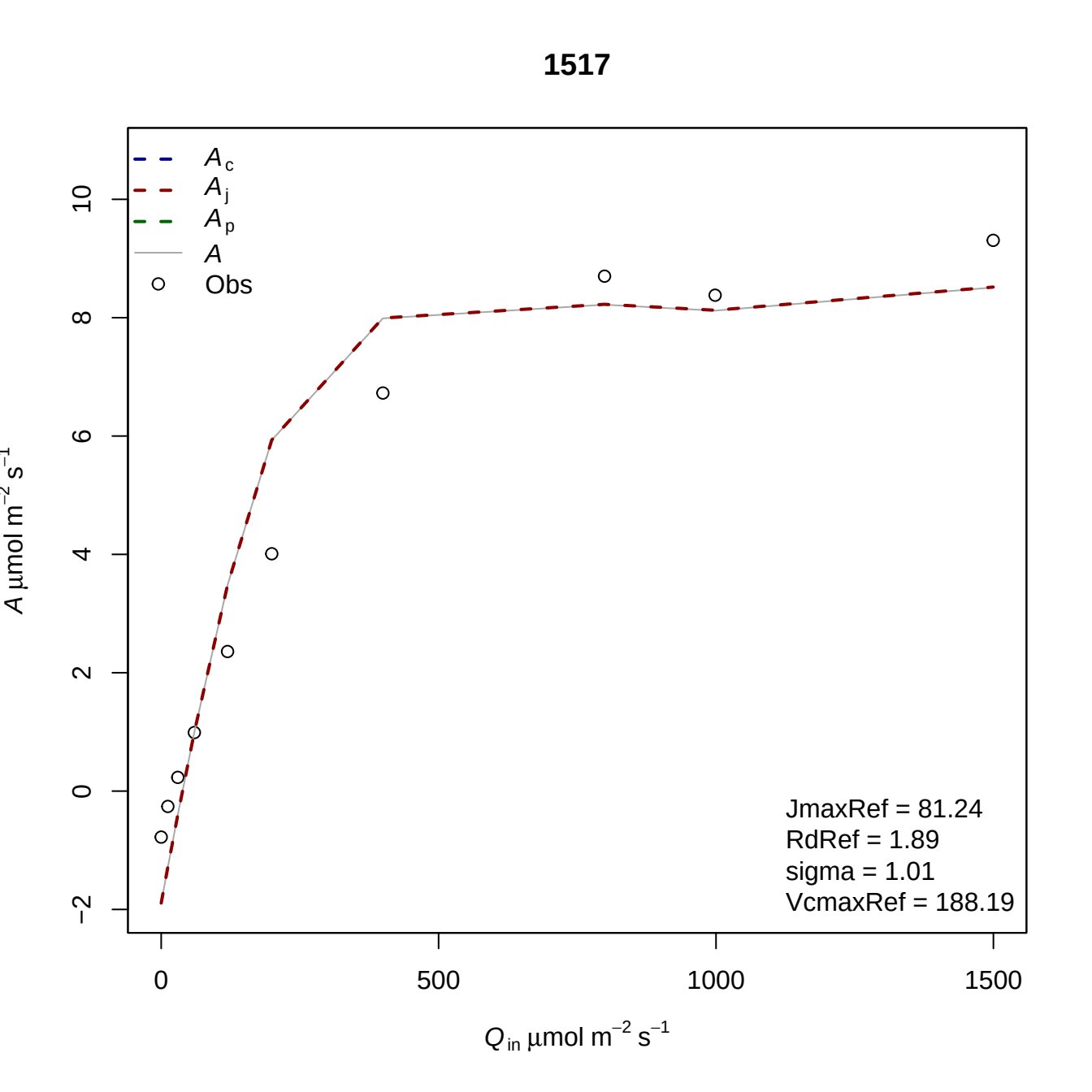

1512
A_c
A_j
A_p
A
Obs
JmaxRef = 95.75
RdRef = 2.33
sigma = 1.18
VcmaxRef = 143.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

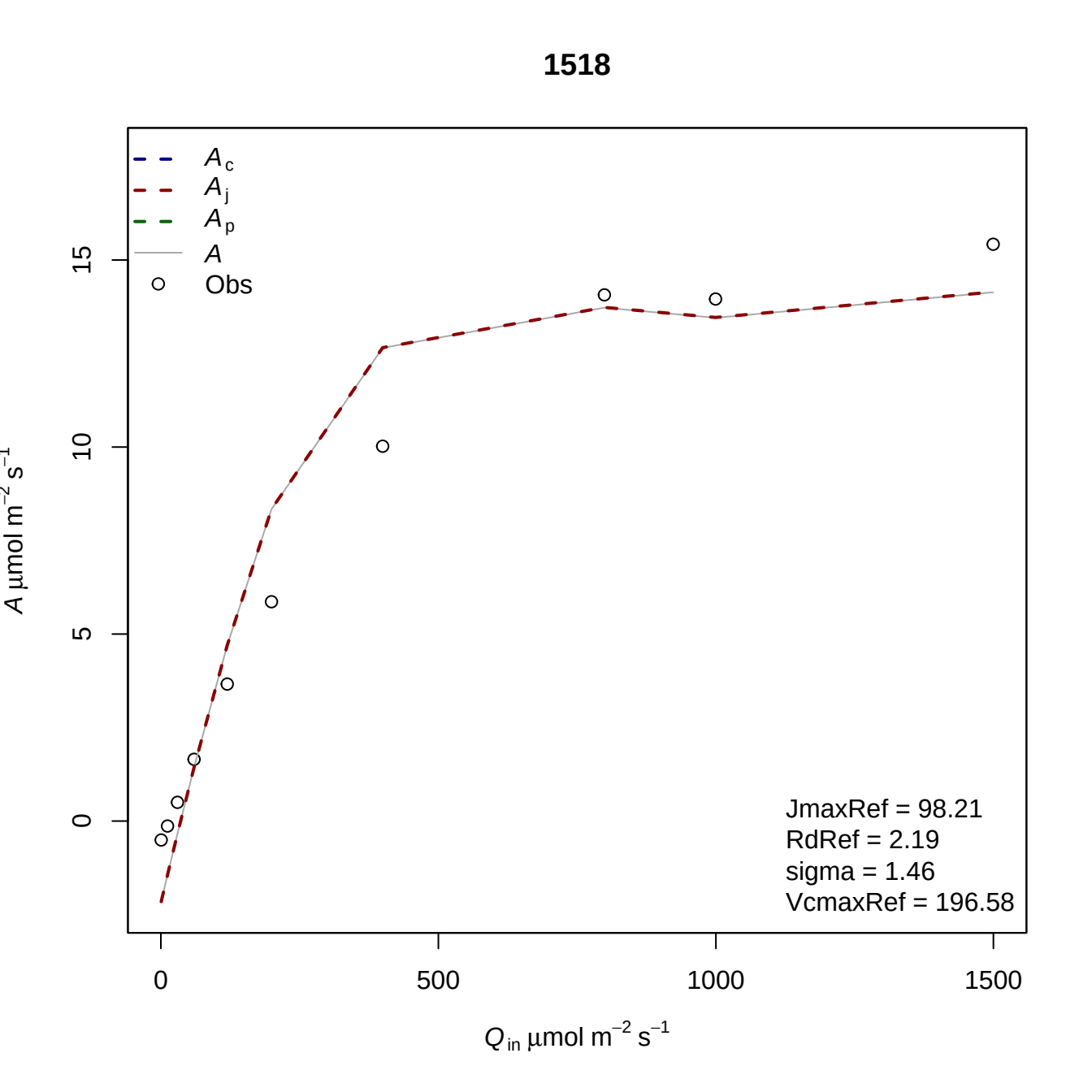

1513
A_c
A_j
A_p
A
Obs
JmaxRef = 34.58
RdRef = 2.39
sigma = 0.6
VcmaxRef = 98.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

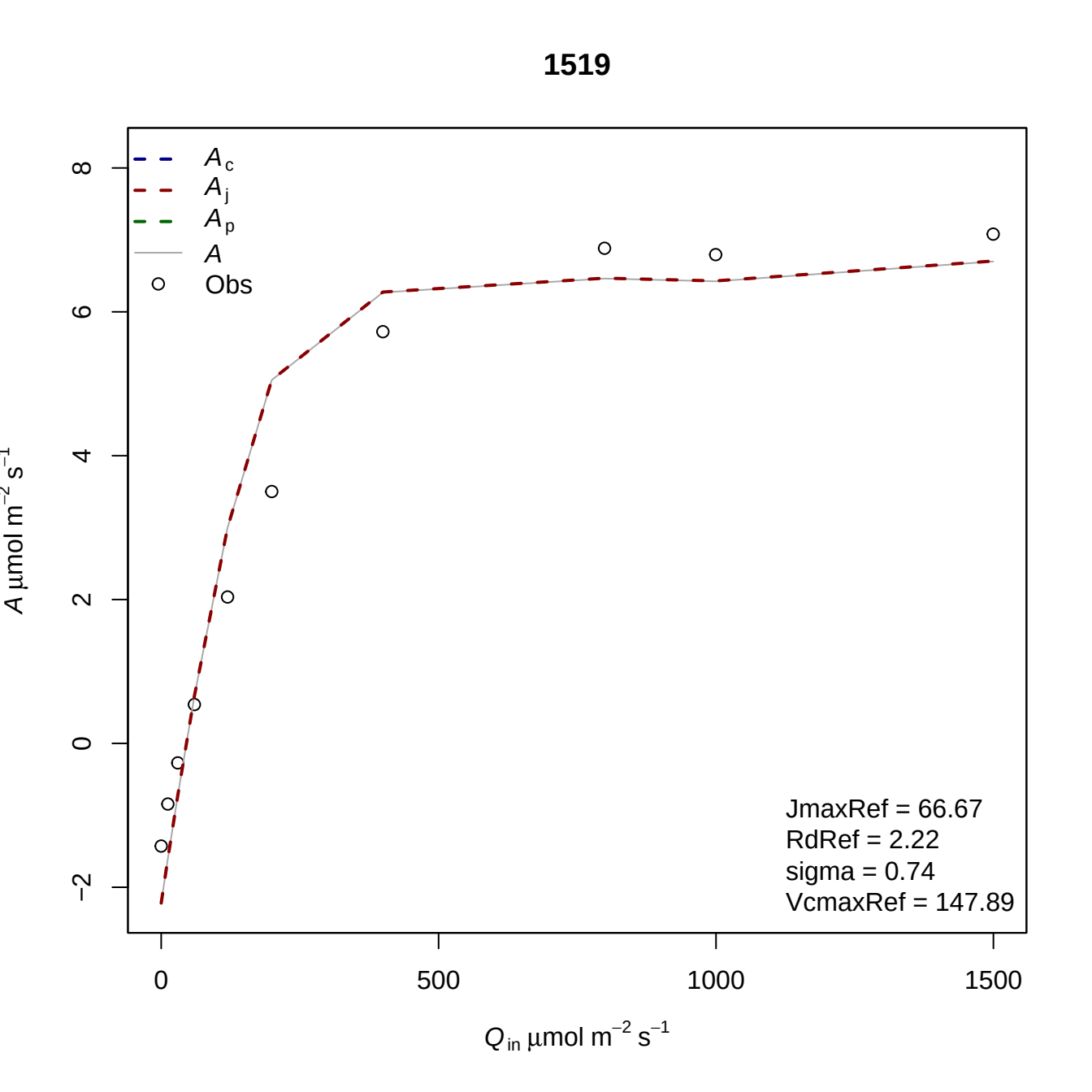

1514
A_c
A_j
A_p
A
Obs
JmaxRef = 80.85
RdRef = 2.46
sigma = 0.89
VcmaxRef = 122.99
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

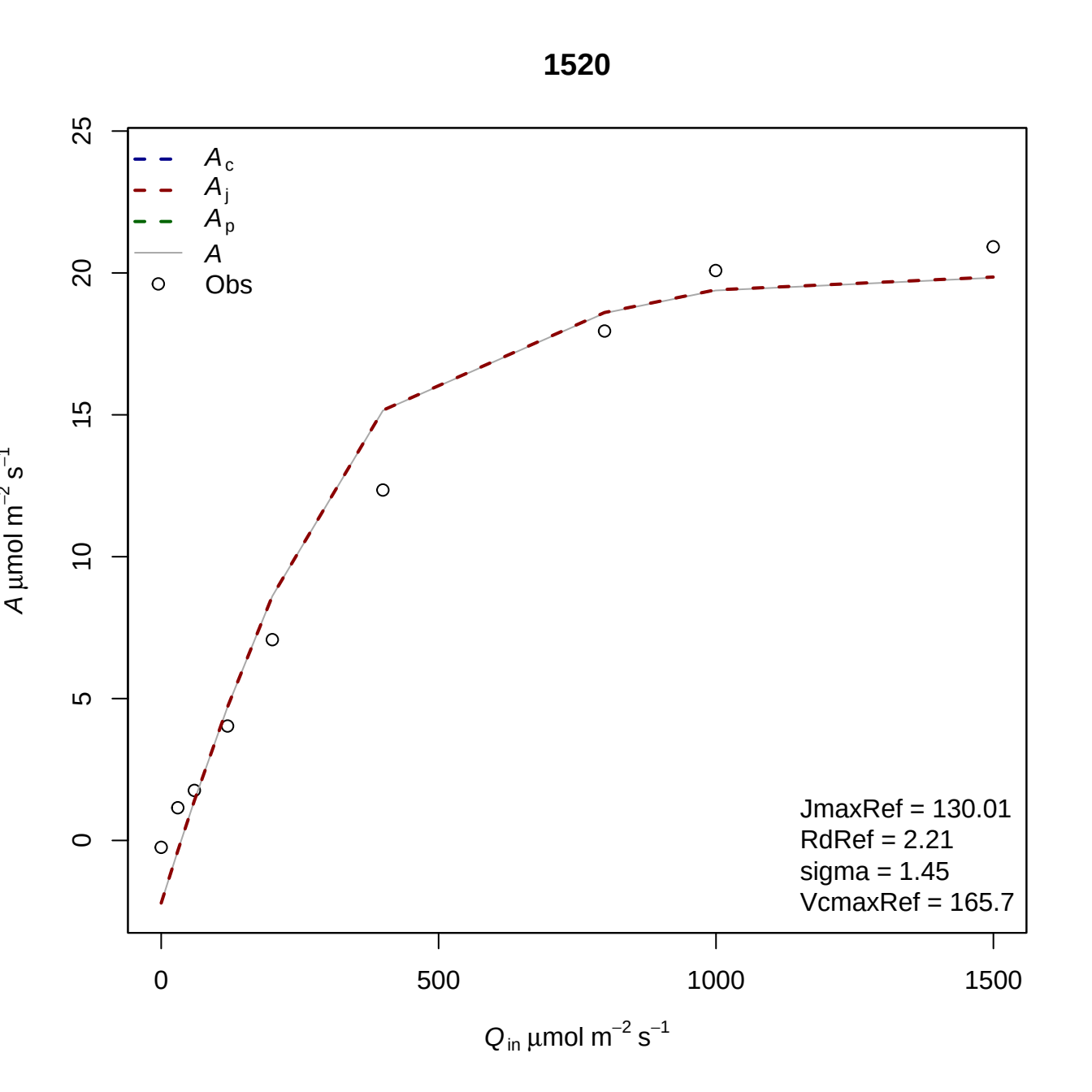

1515
A_c
A_j
A_p
A
Obs
JmaxRef = 35.16
RdRef = 1.56
sigma = 0.64
VcmaxRef = 74.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

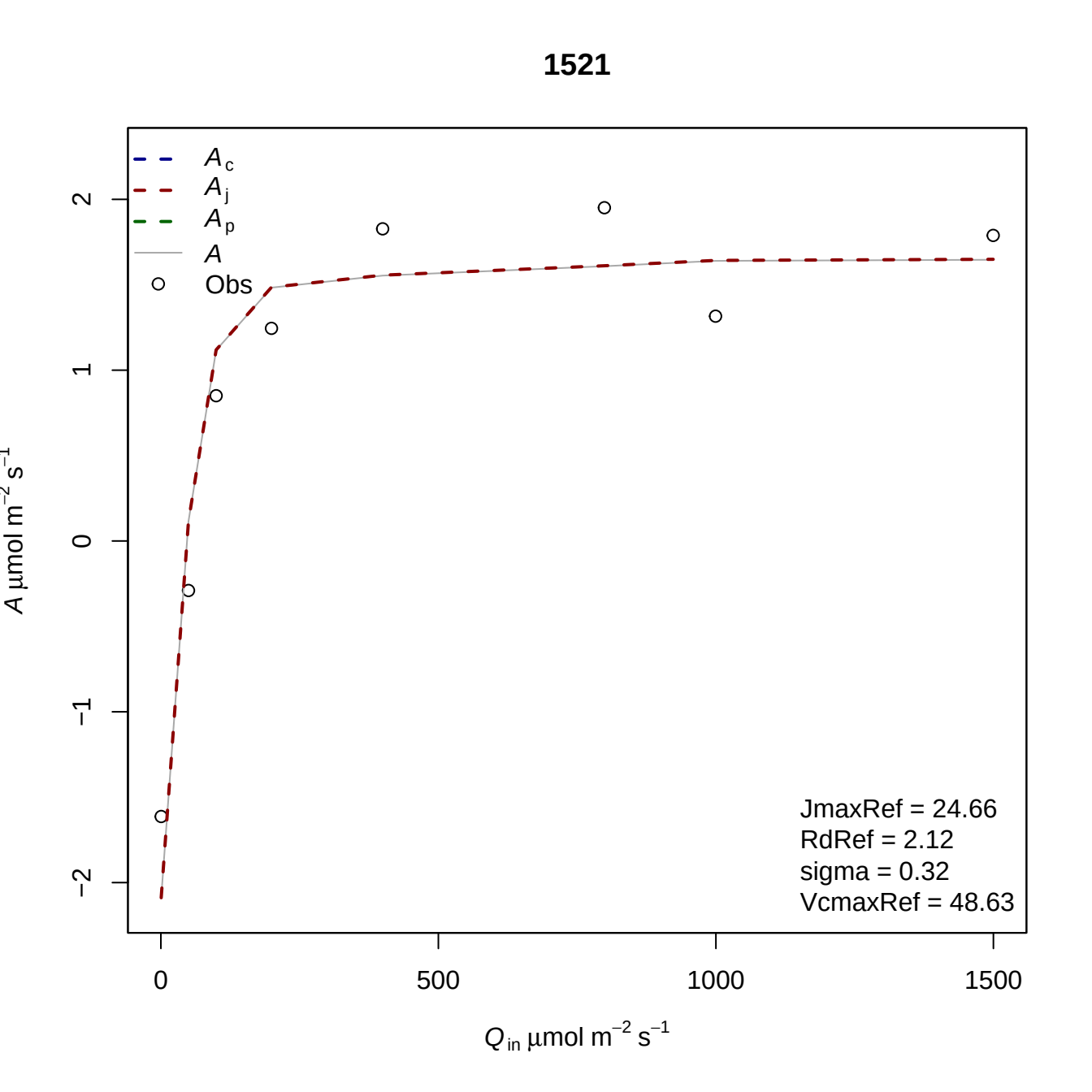

1516
A_c
A_j
A_p
A
Obs
JmaxRef = 91.37
RdRef = 1.78
sigma = 1.1
VcmaxRef = 125.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

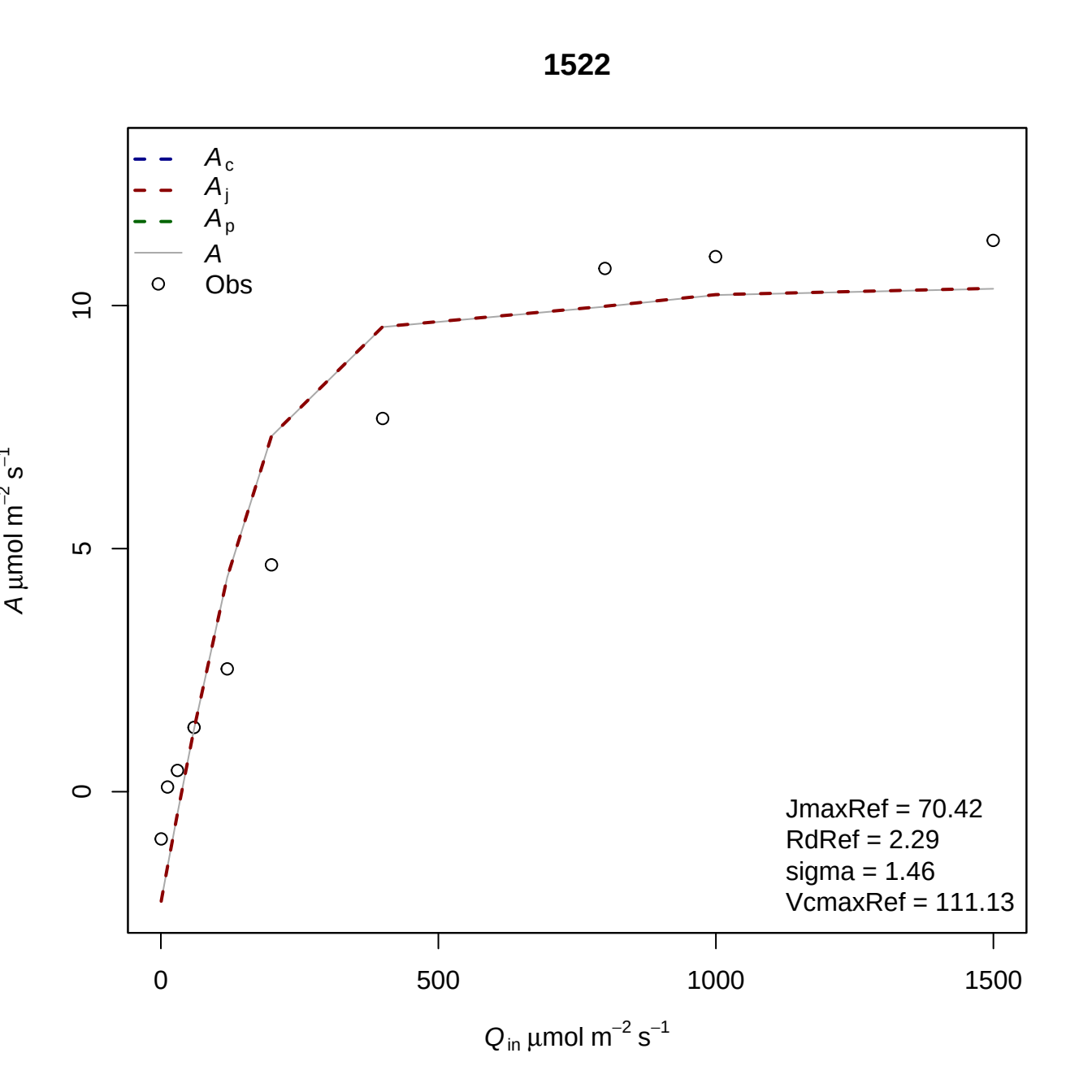

1517
A_c
A_j
A_p
A
Obs
JmaxRef = 81.24
RdRef = 1.89
sigma = 1.01
VcmaxRef = 188.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

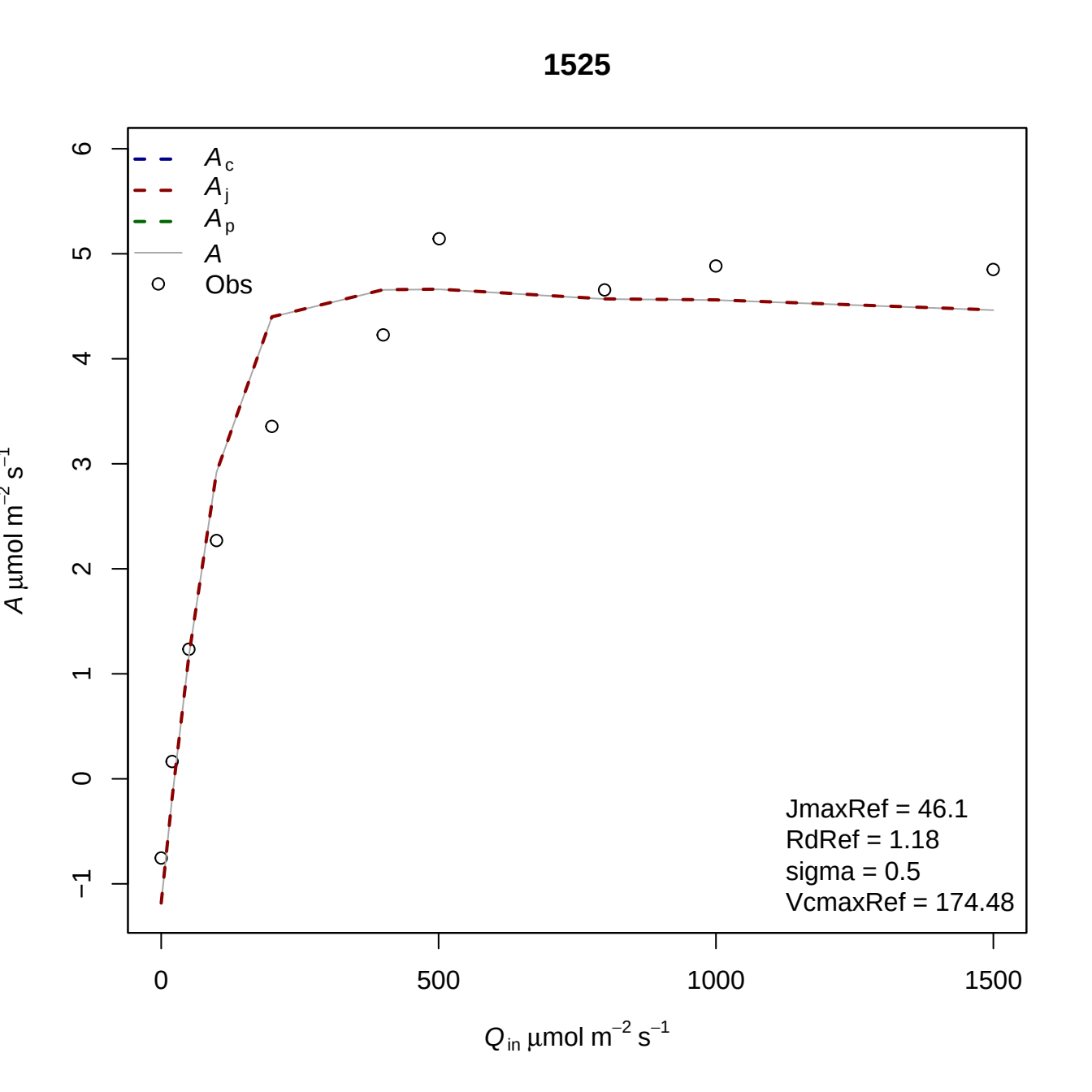

1518
A_c
A_j
A_p
A
Obs
JmaxRef = 98.21
RdRef = 2.19
sigma = 1.46
VcmaxRef = 196.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

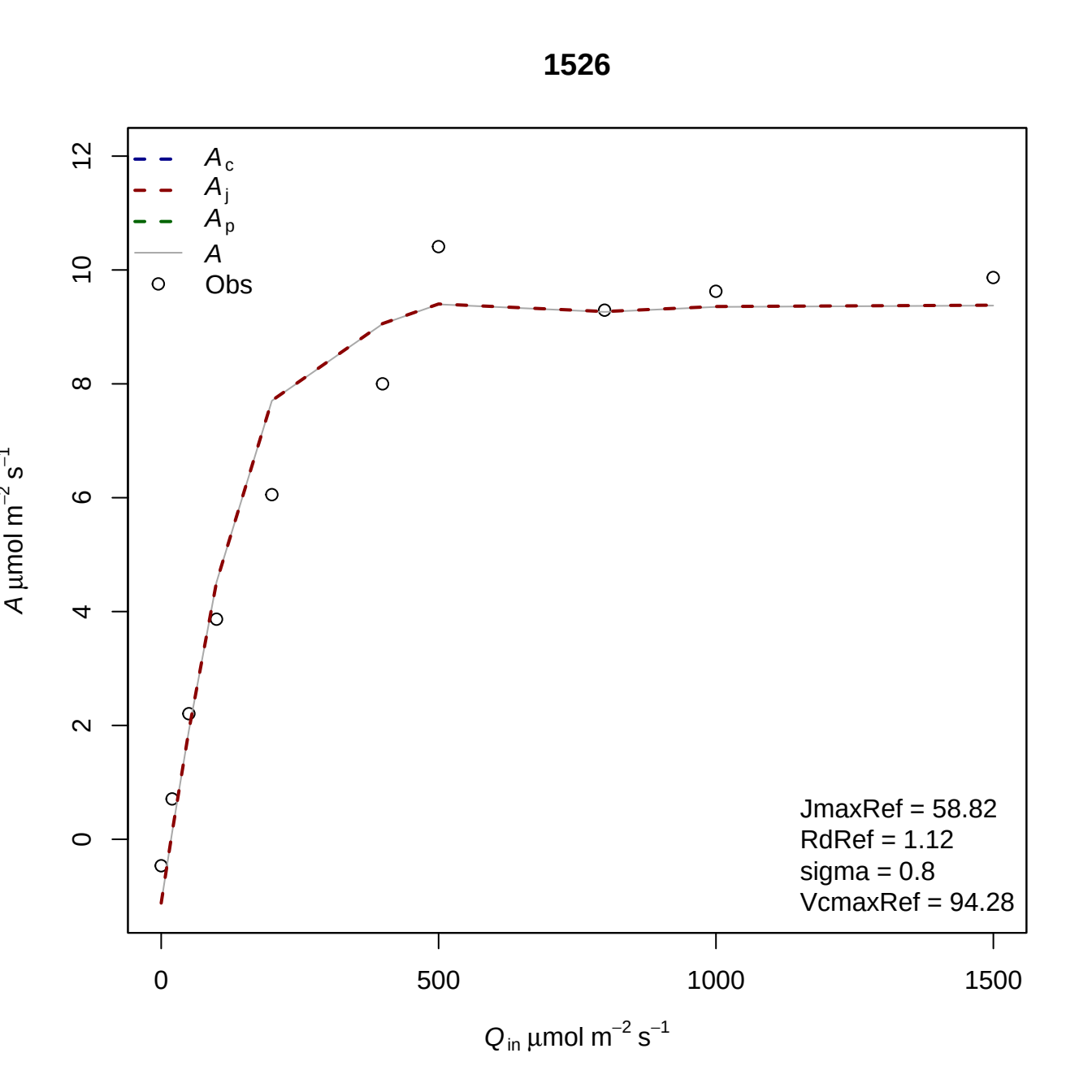

1519
A_c
A_j
A_p
A
Obs
JmaxRef = 66.67
RdRef = 2.22
sigma = 0.74
VcmaxRef = 147.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

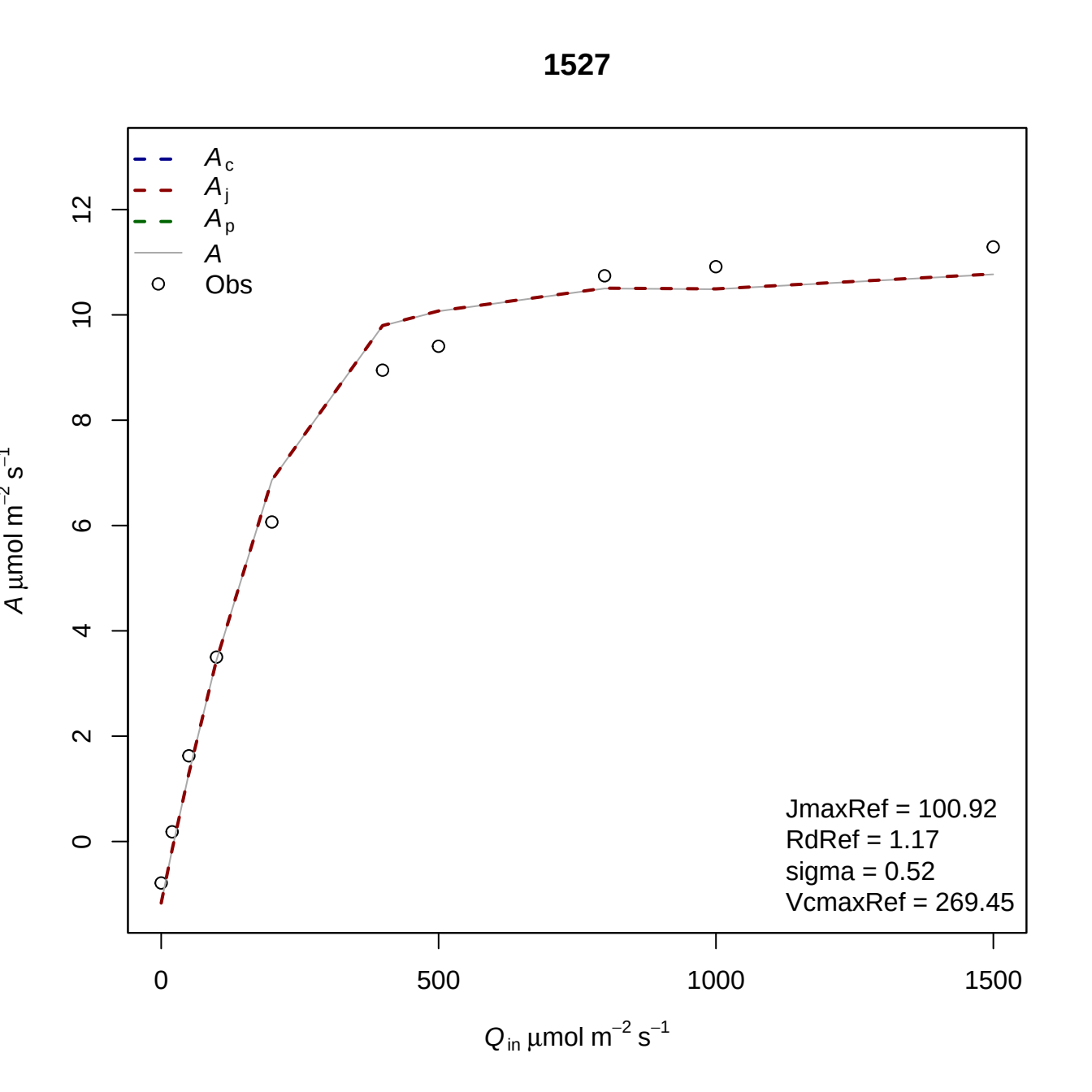

1520
A_c
A_j
A_p
A
Obs
JmaxRef = 130.01
RdRef = 2.21
sigma = 1.45
VcmaxRef = 165.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

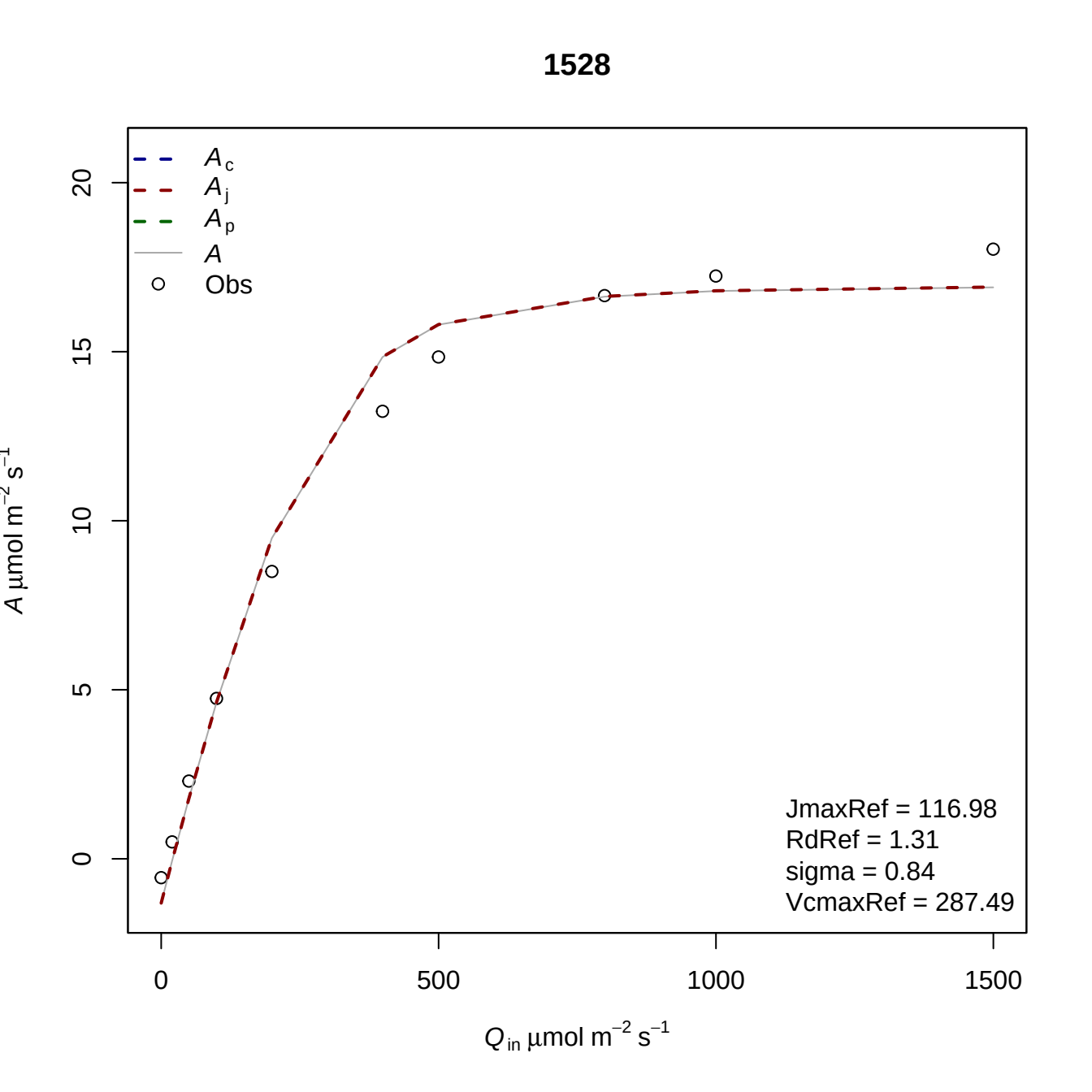

1521
A_c
A_j
A_p
A
Obs
JmaxRef = 24.66
RdRef = 2.12
sigma = 0.32
VcmaxRef = 48.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500

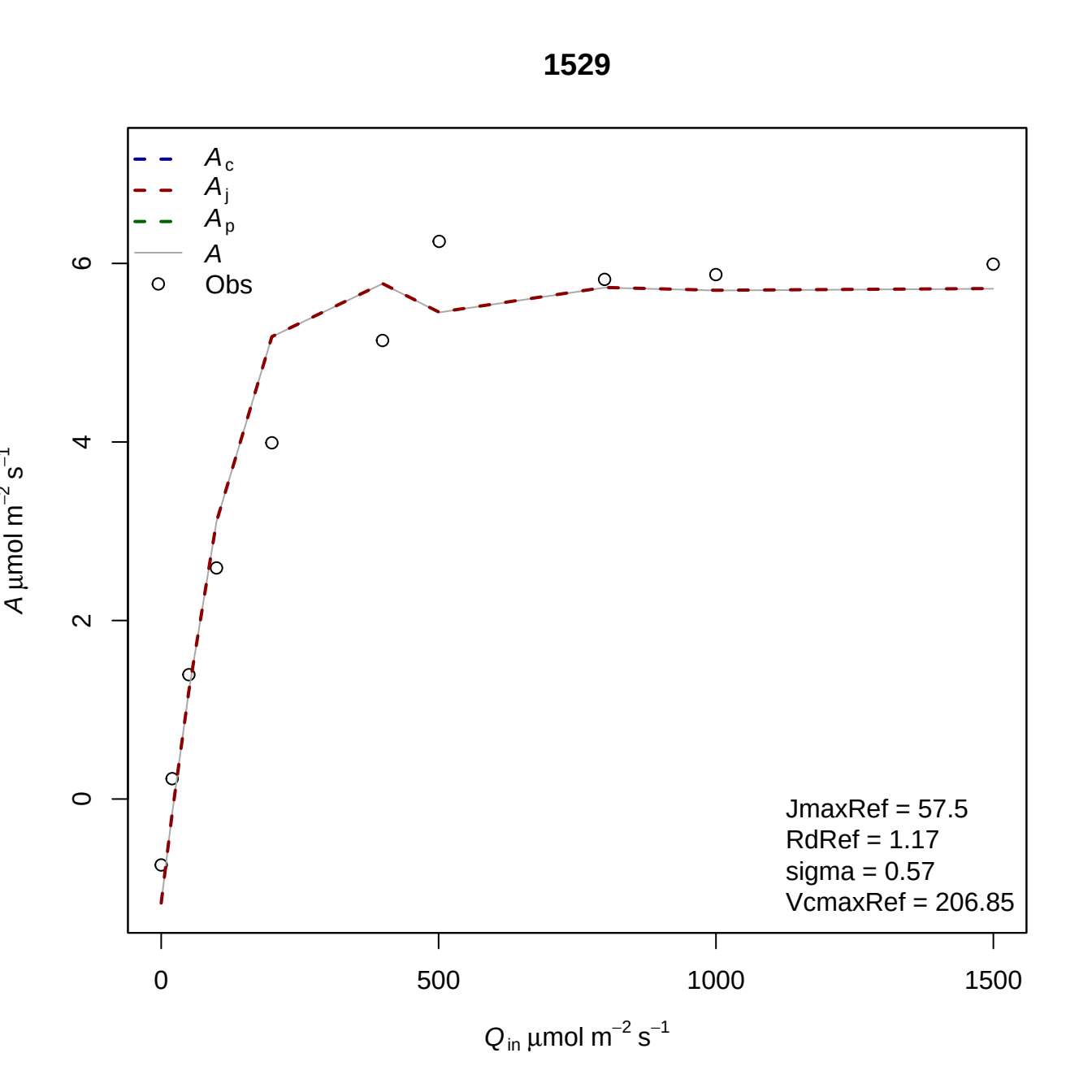

1522
A_c
A_j
A_p
A
Obs
JmaxRef = 70.42
RdRef = 2.29
sigma = 1.46
VcmaxRef = 111.13
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

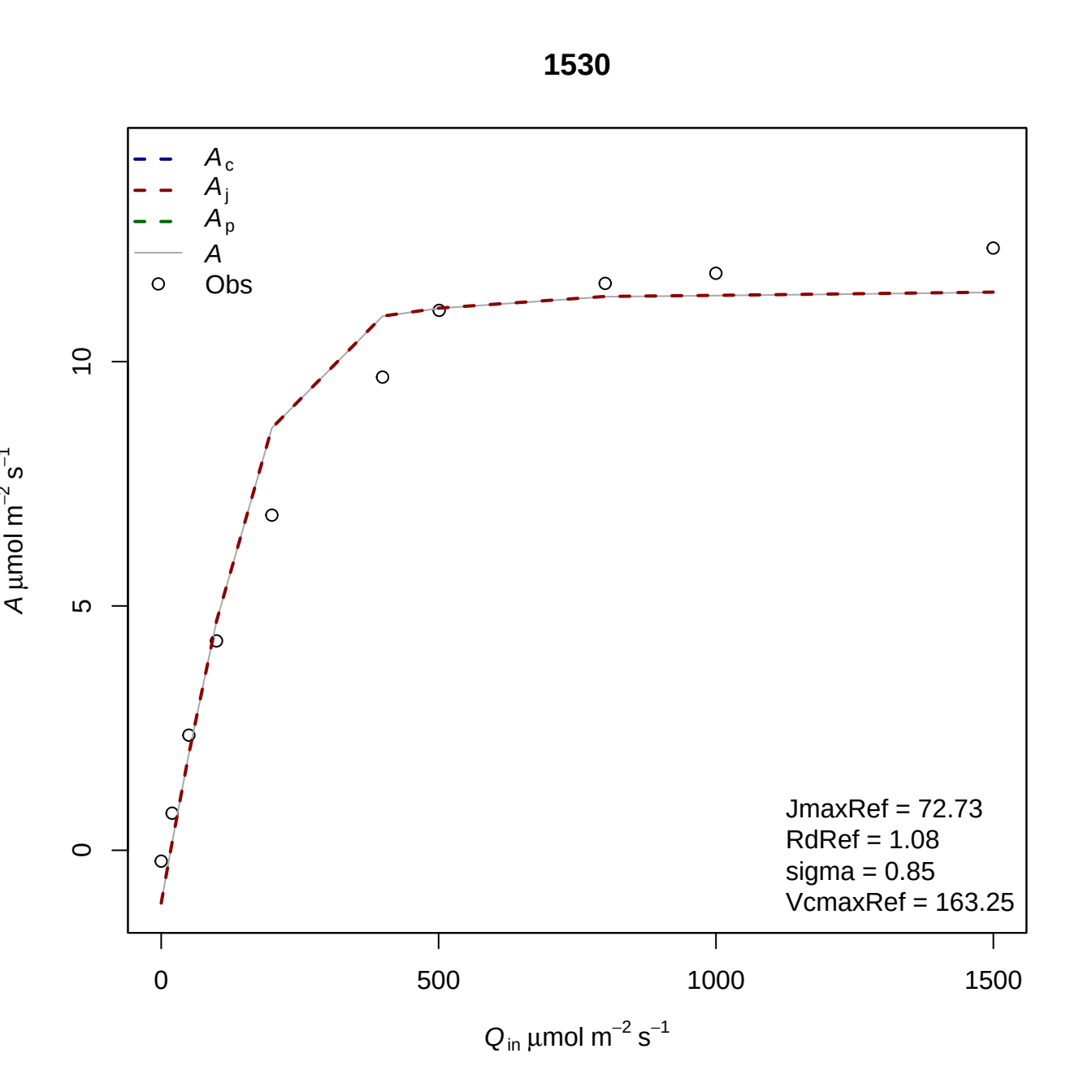

1525
A_c
A_j
A_p
A
Obs
JmaxRef = 46.1
RdRef = 1.18
sigma = 0.5
VcmaxRef = 174.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

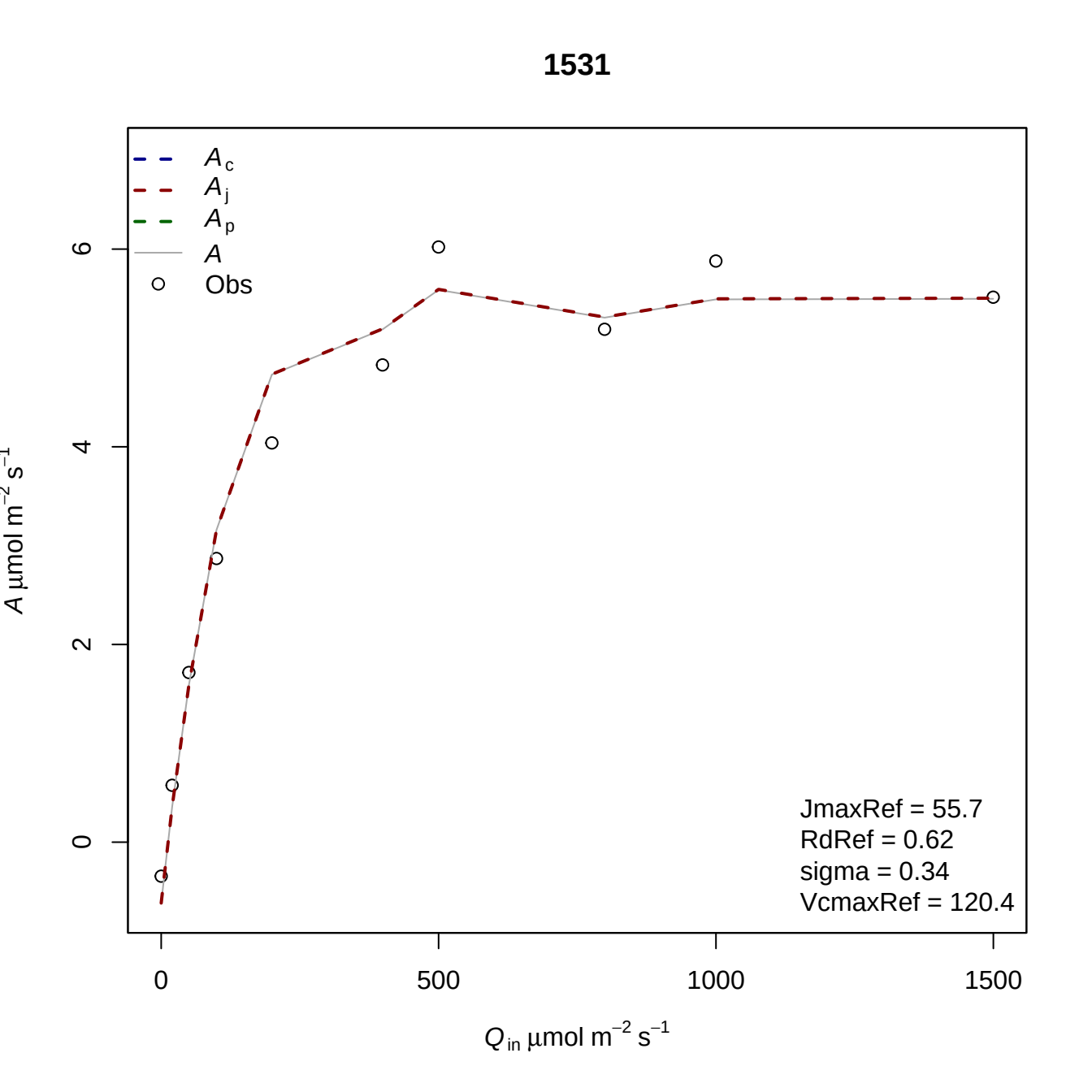

1526
A_c
A_j
A_p
A
Obs
JmaxRef = 58.82
RdRef = 1.12
sigma = 0.8
VcmaxRef = 94.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

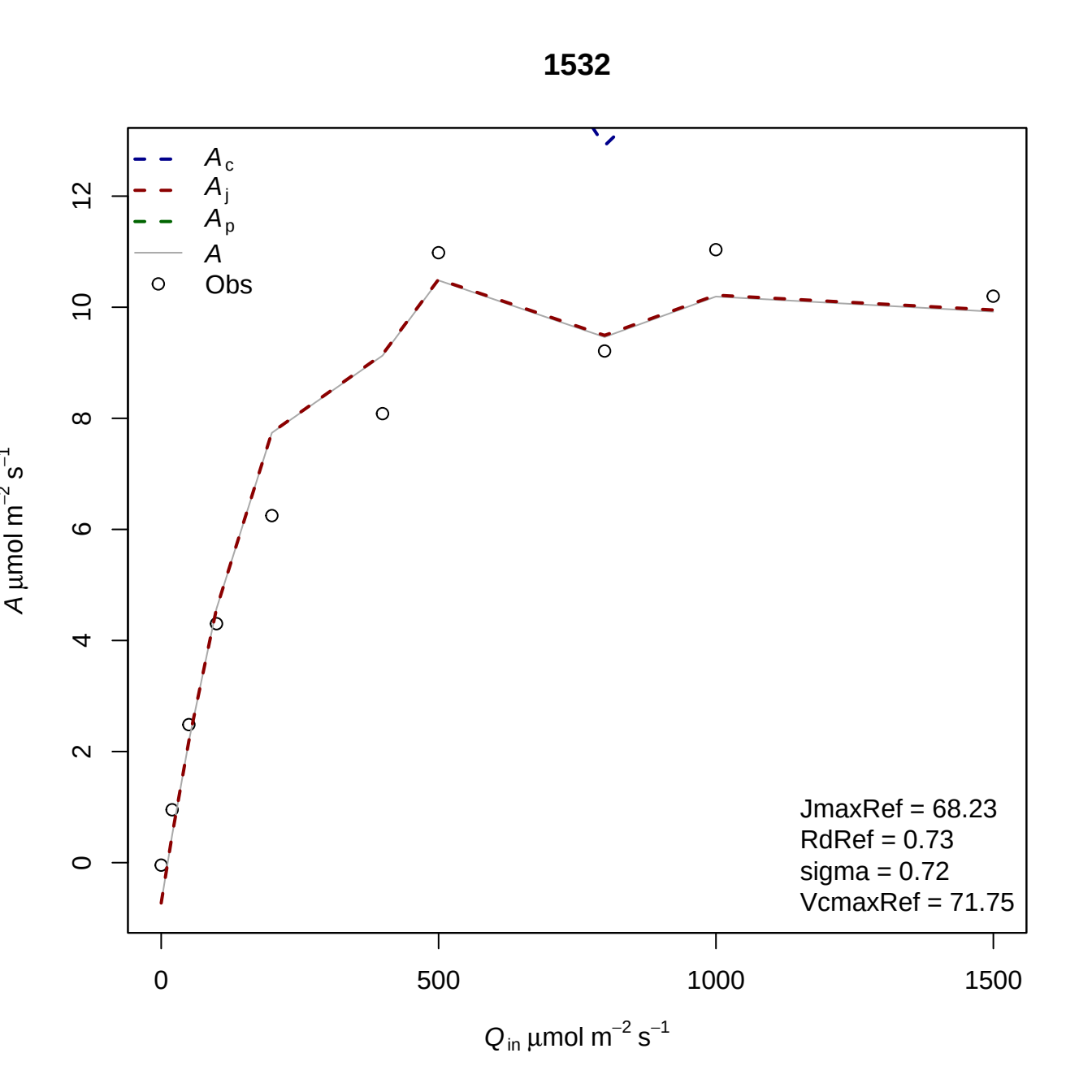

1527
A_c
A_j
A_p
A
Obs
JmaxRef = 100.92
RdRef = 1.17
sigma = 0.52
VcmaxRef = 269.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

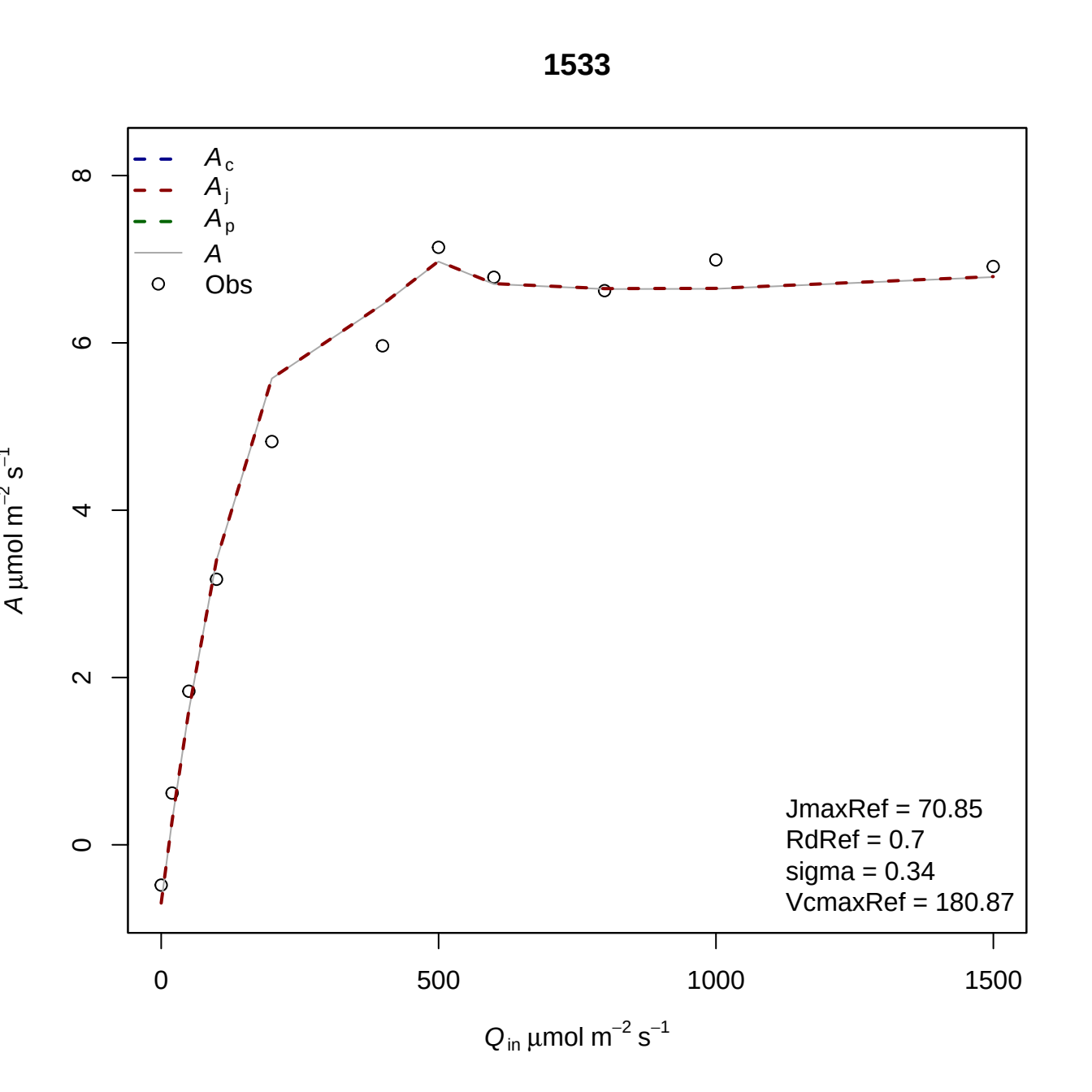

1528
A_c
A_j
A_p
A
Obs
JmaxRef = 116.98
RdRef = 1.31
sigma = 0.84
VcmaxRef = 287.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

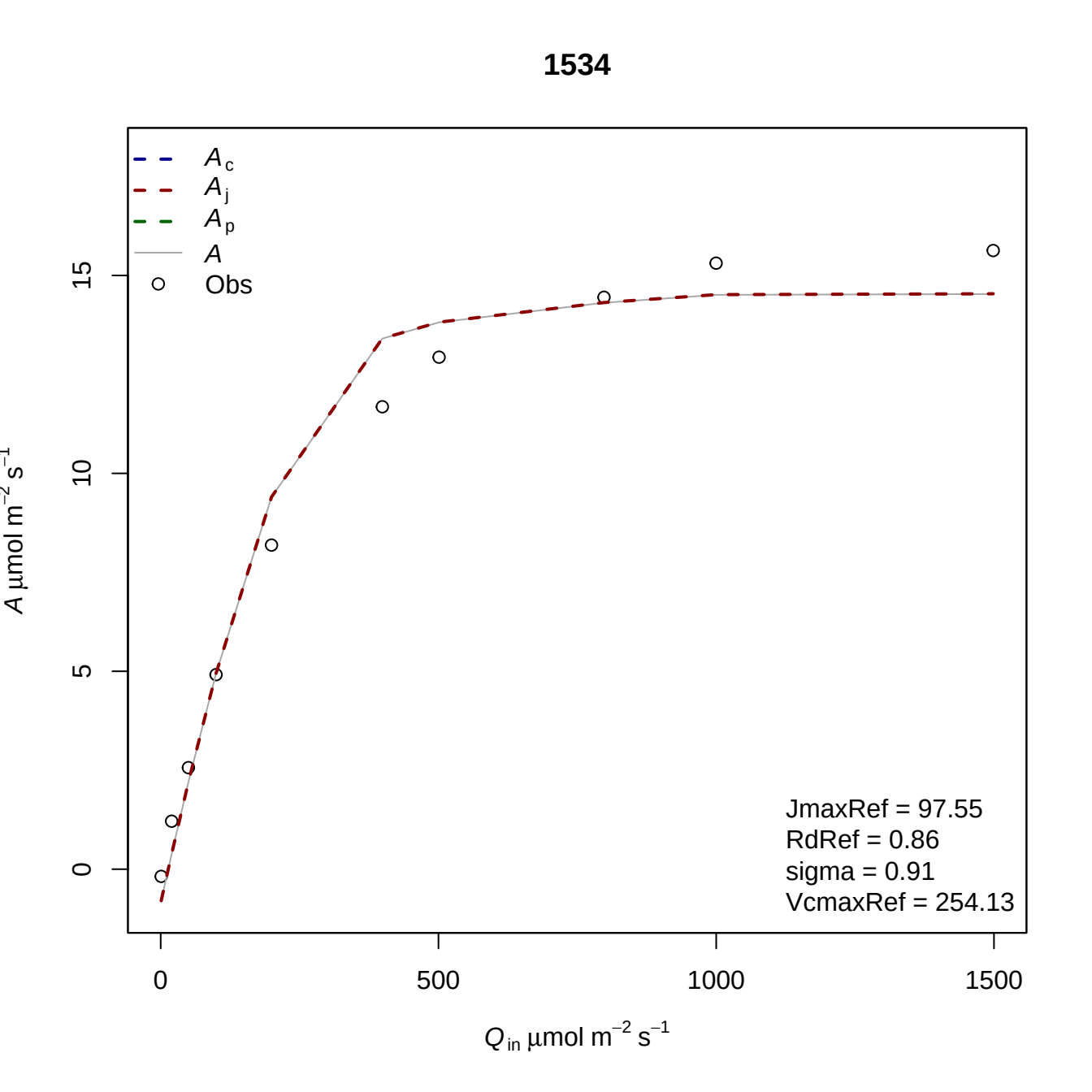

1529
A_c
A_j
A_p
A
Obs
JmaxRef = 57.5
RdRef = 1.17
sigma = 0.57
VcmaxRef = 206.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

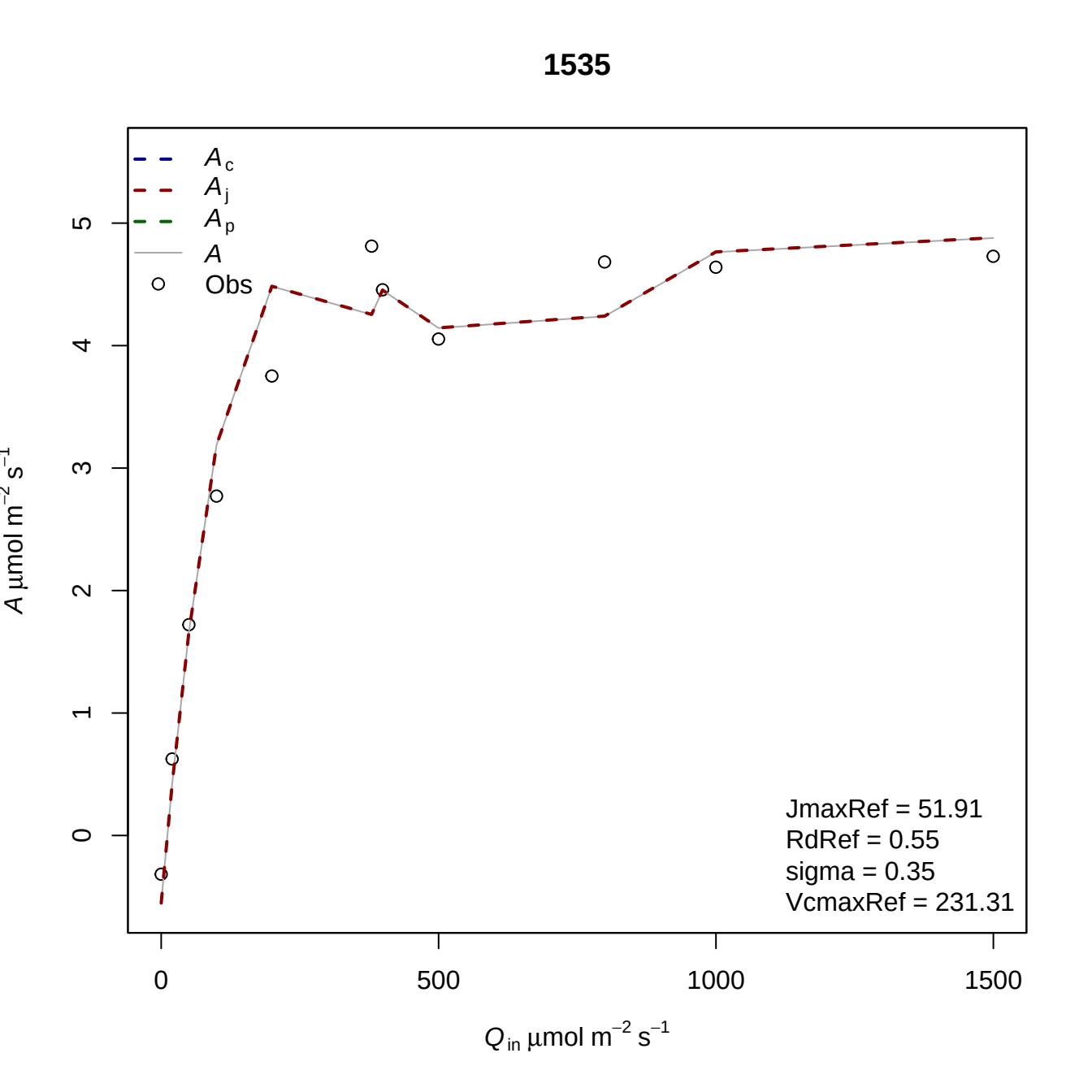

1530
A_c
A_j
A_p
A
Obs
JmaxRef = 72.73
RdRef = 1.08
sigma = 0.85
VcmaxRef = 163.25
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

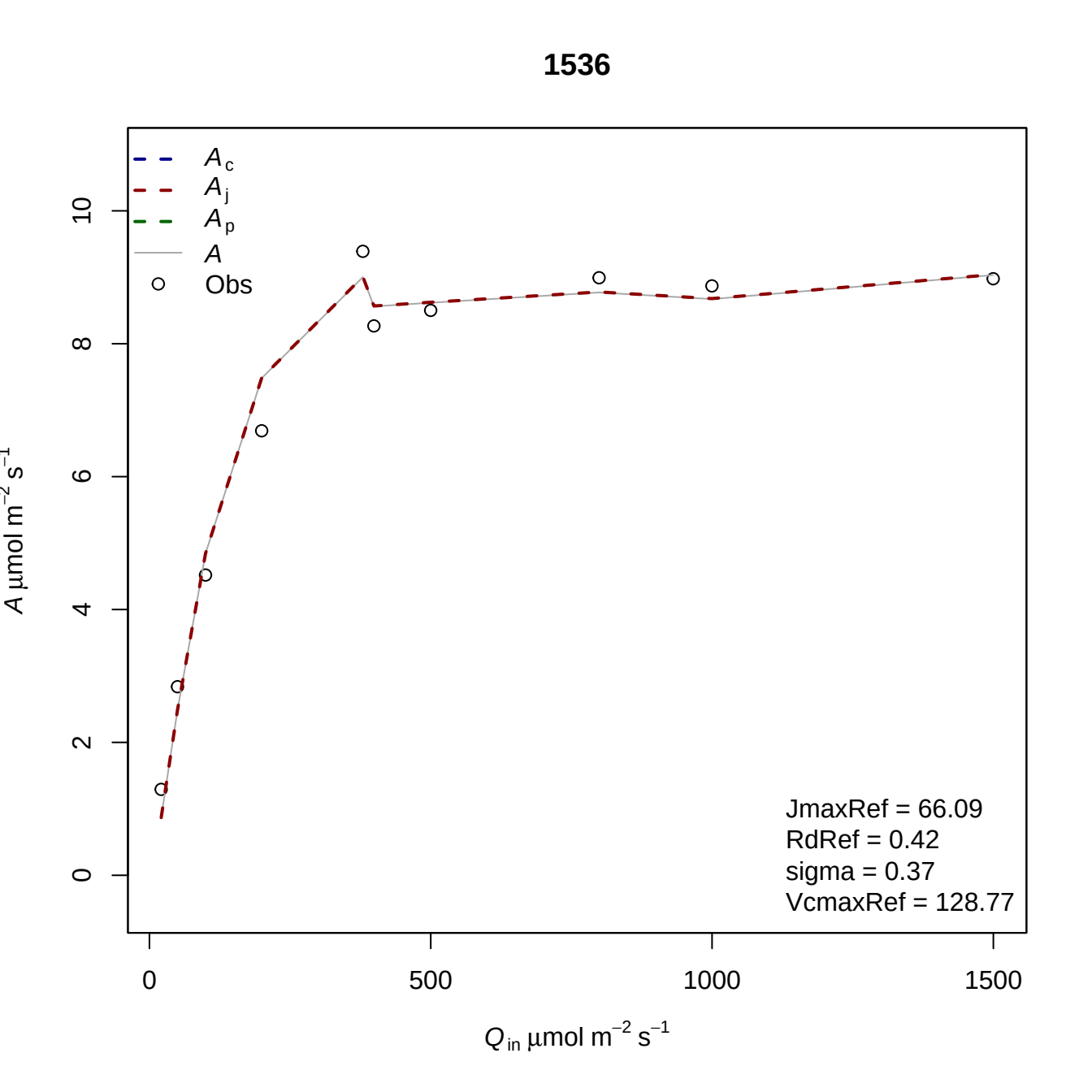

1531
A_c
A_j
A_p
A
Obs
JmaxRef = 55.7
RdRef = 0.62
sigma = 0.34
VcmaxRef = 120.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

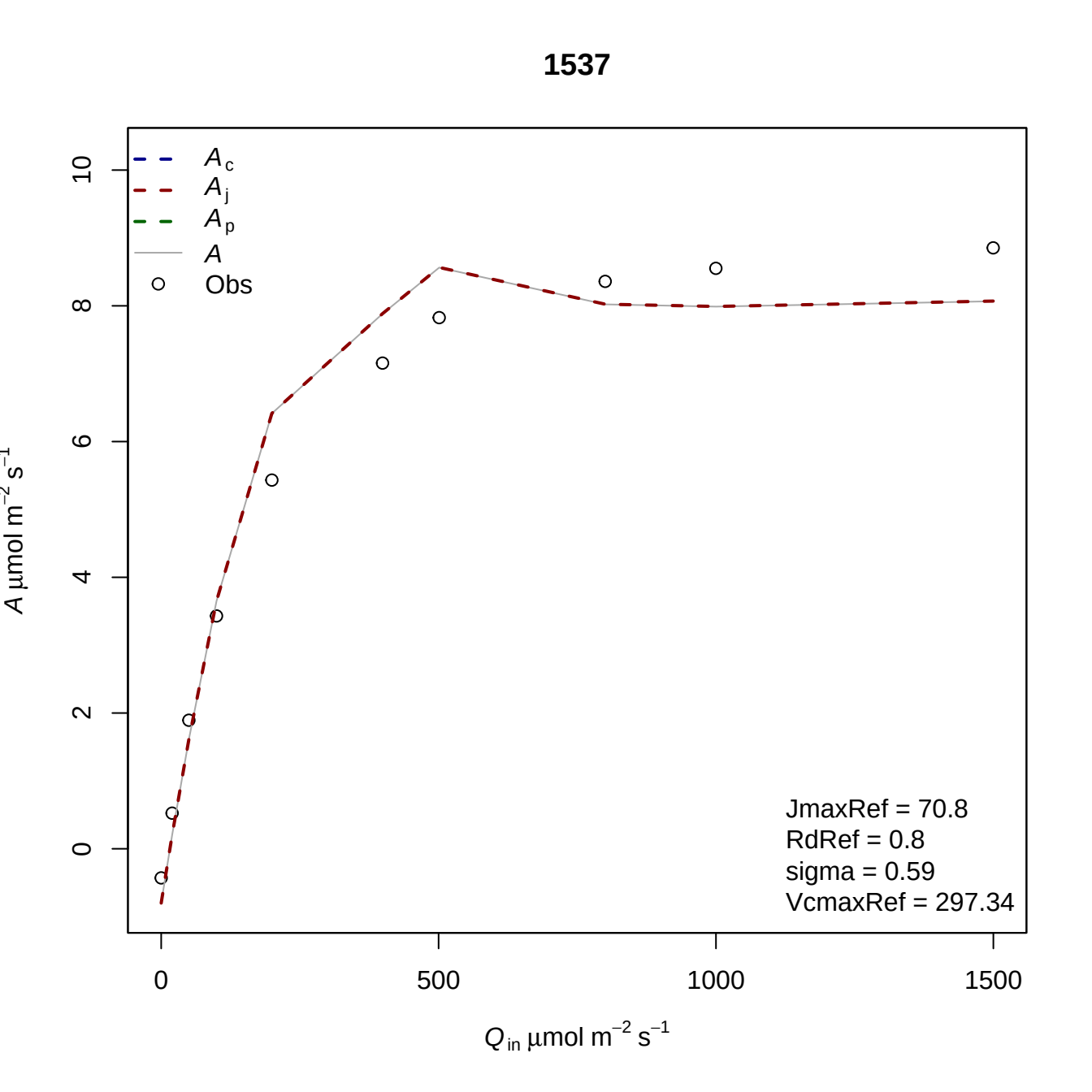

1532
A_c
A_j
A_p
A
Obs
JmaxRef = 68.23
RdRef = 0.73
sigma = 0.72
VcmaxRef = 71.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

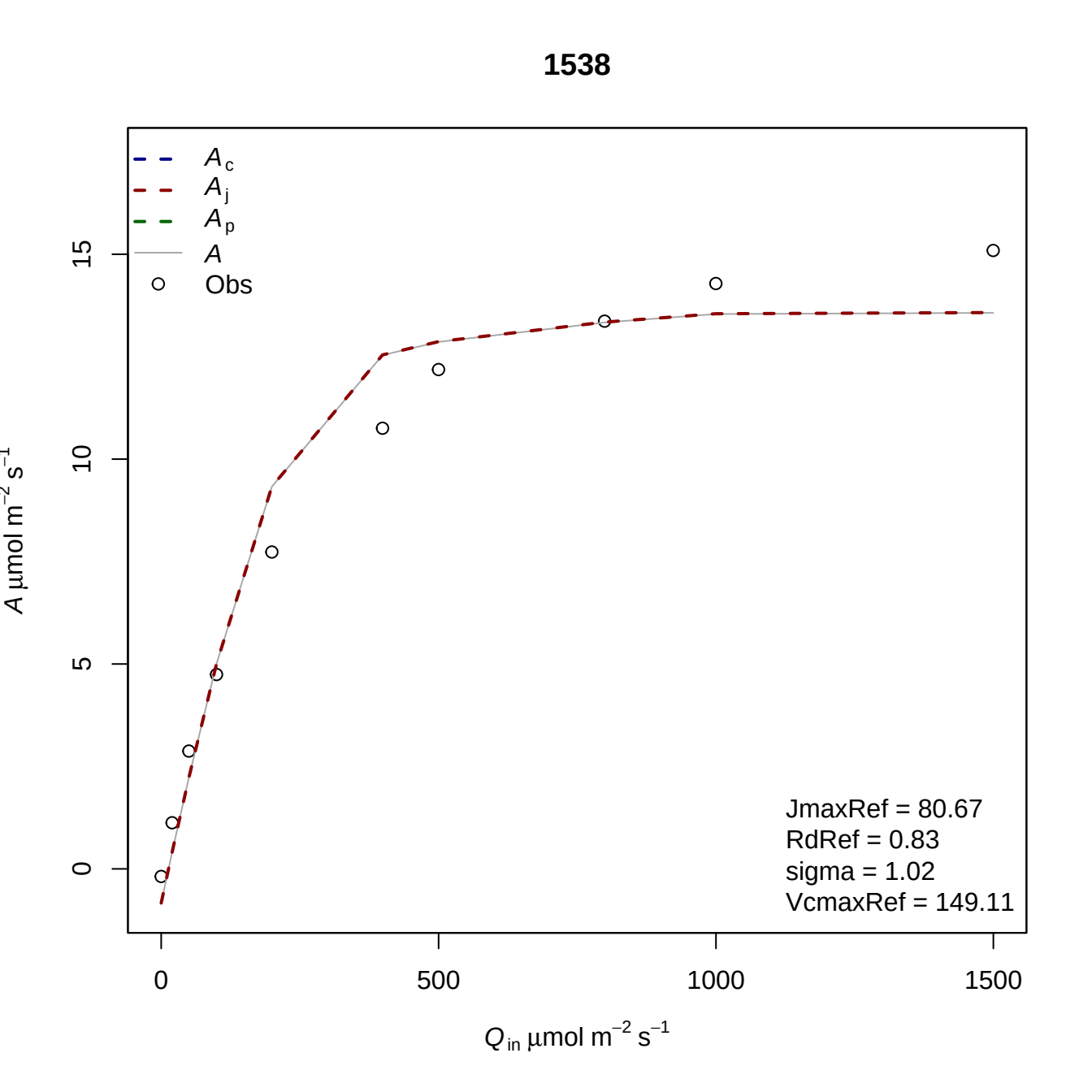

1533
A_c
A_j
A_p
A
Obs
JmaxRef = 70.85
RdRef = 0.7
sigma = 0.34
VcmaxRef = 180.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

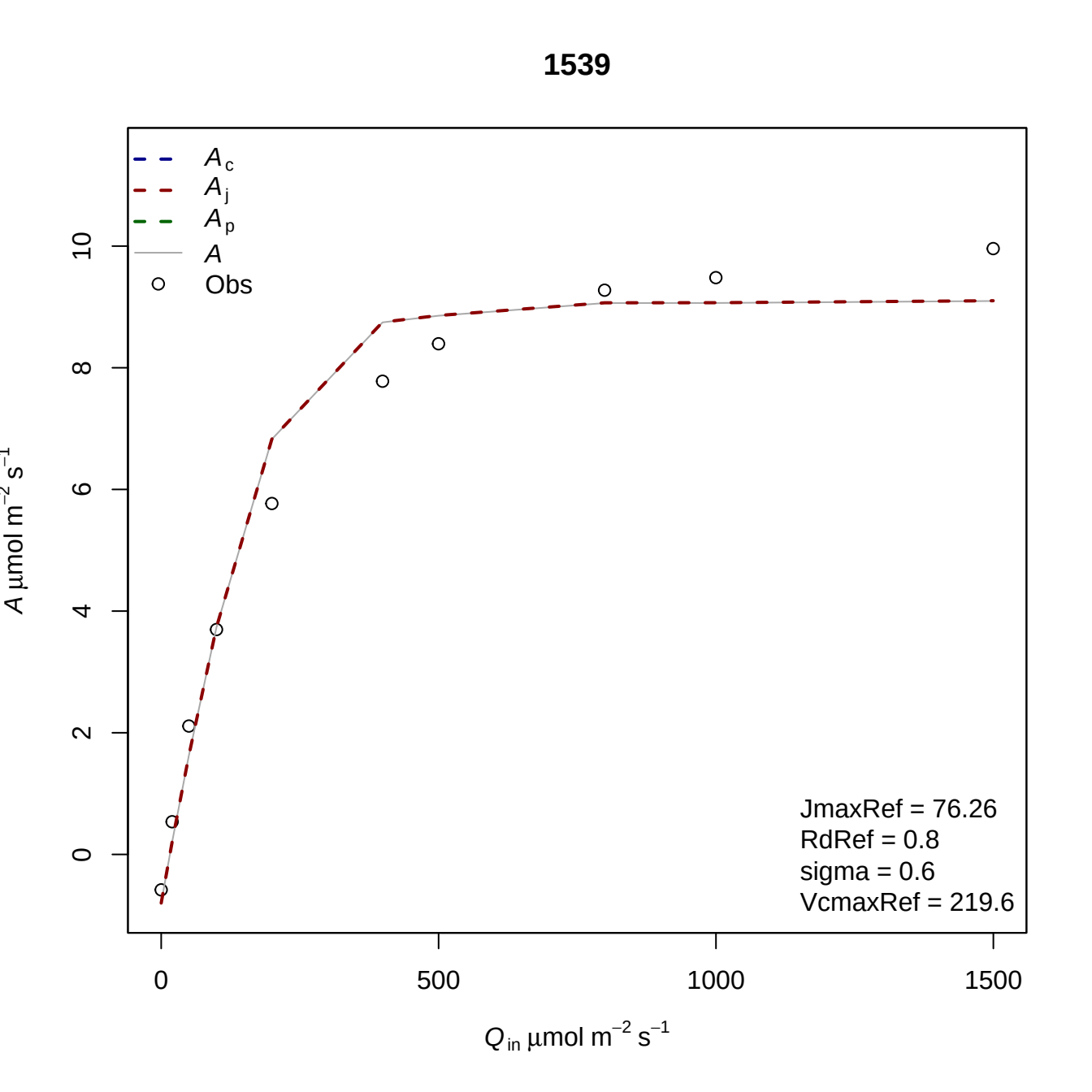

1534
A_c
A_j
A_p
A
Obs
JmaxRef = 97.55
RdRef = 0.86
sigma = 0.91
VcmaxRef = 254.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

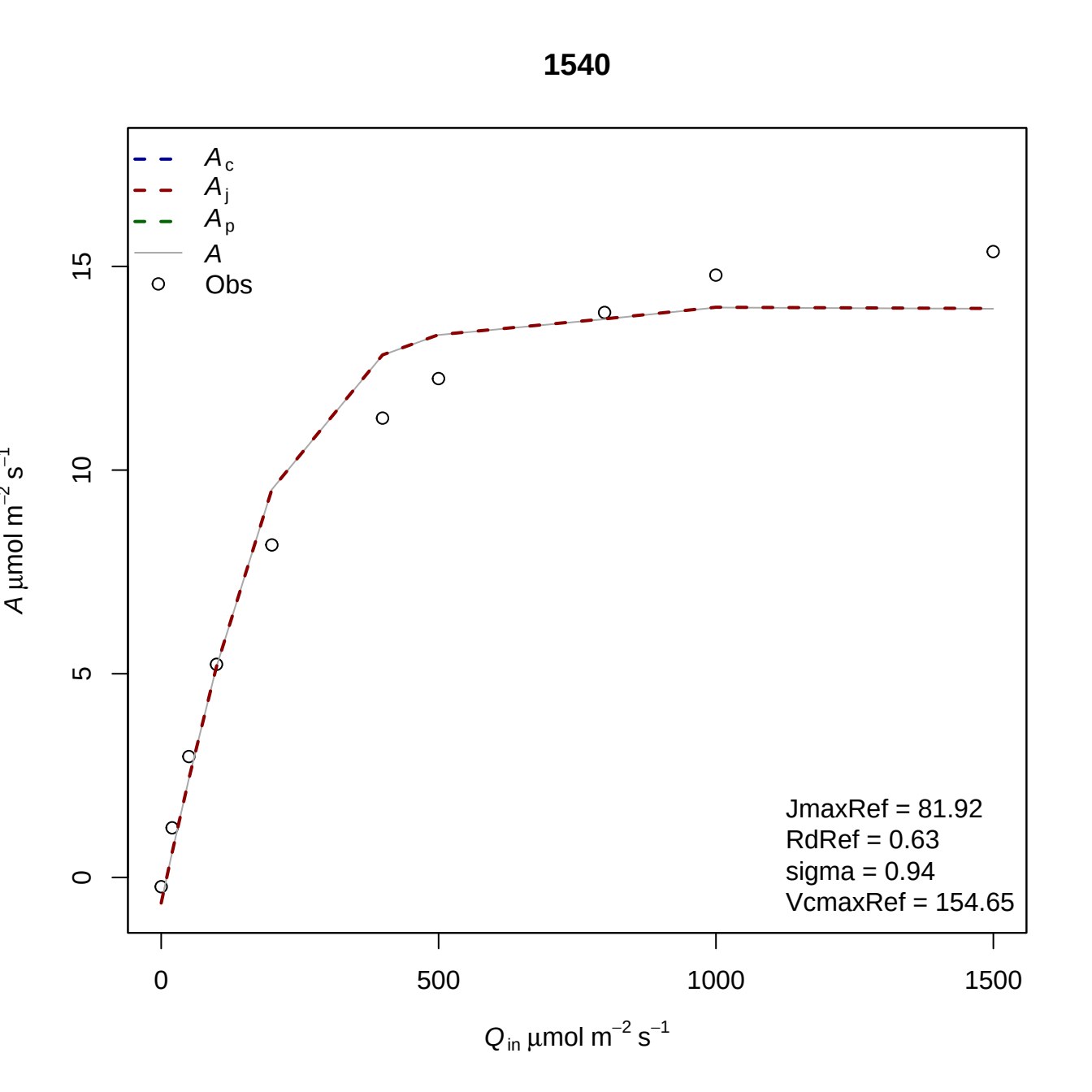

1535
A_c
A_j
A_p
A
Obs
JmaxRef = 51.91
RdRef = 0.55
sigma = 0.35
VcmaxRef = 231.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

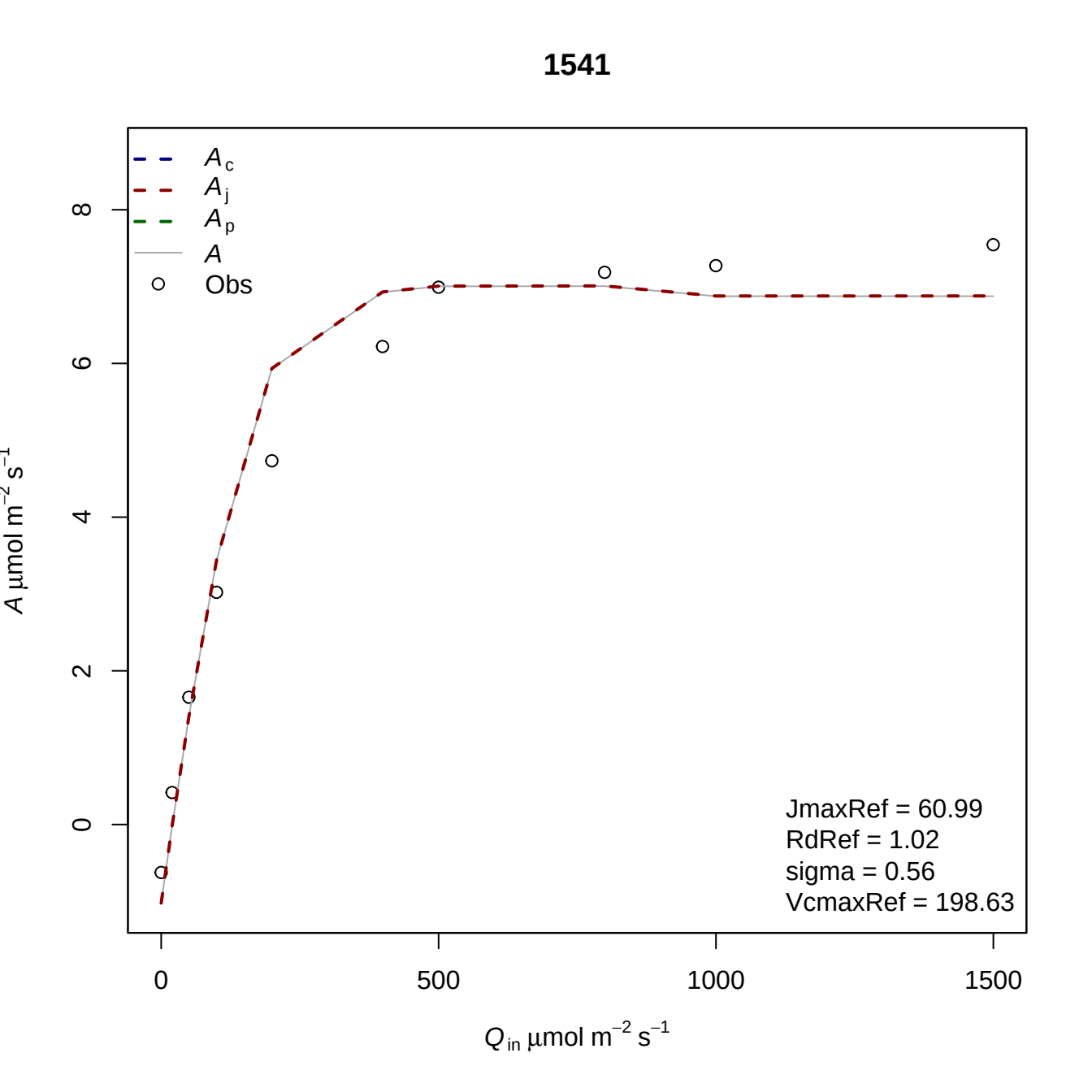

1536
A_c
A_j
A_p
A
Obs
JmaxRef = 66.09
RdRef = 0.42
sigma = 0.37
VcmaxRef = 128.77
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

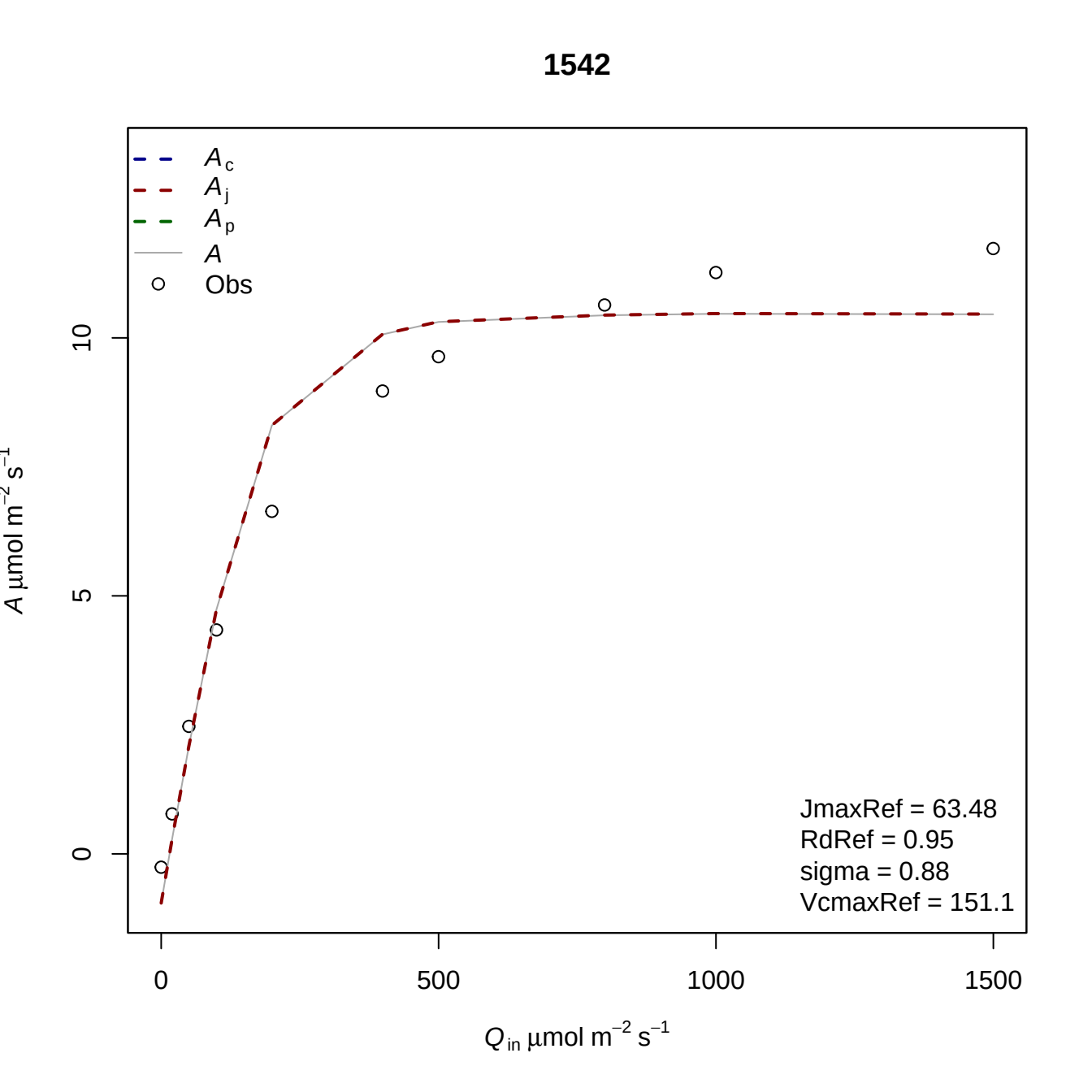

1537
A_c
A_j
A_p
A
Obs
JmaxRef = 70.8
RdRef = 0.8
sigma = 0.59
VcmaxRef = 297.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

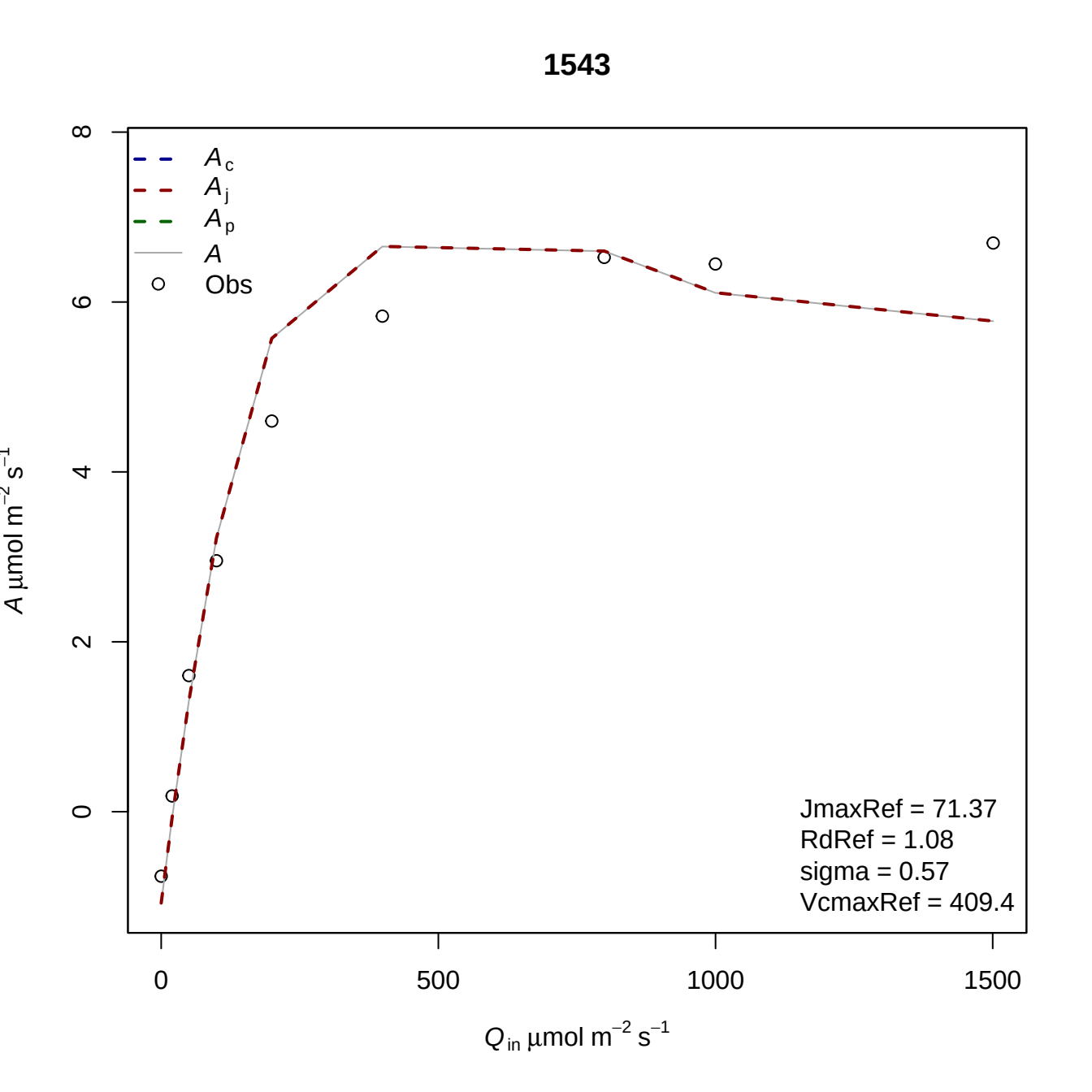

1538
A_c
A_j
A_p
A
Obs
JmaxRef = 80.67
RdRef = 0.83
sigma = 1.02
VcmaxRef = 149.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

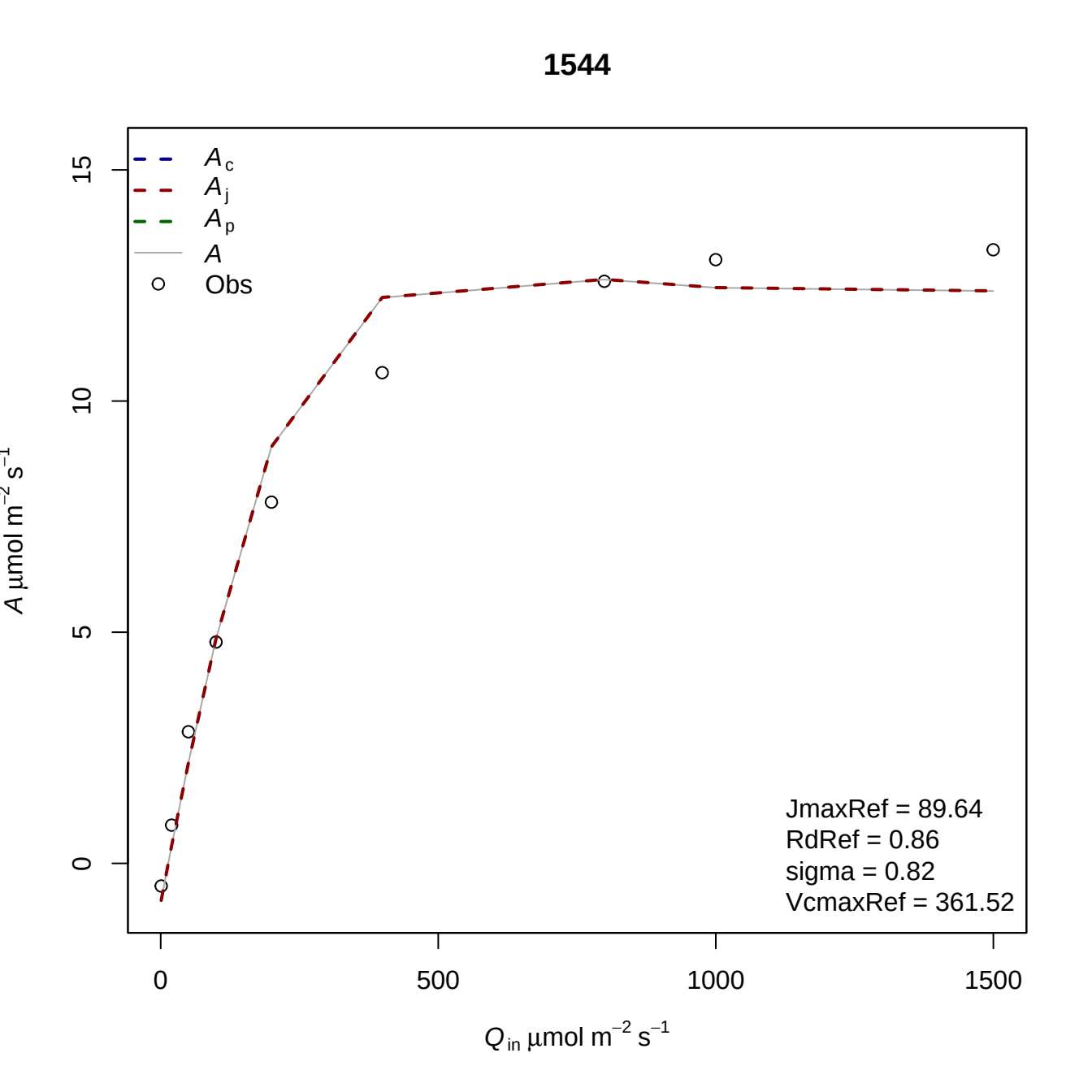

1539
A_c
A_j
A_p
A
Obs
JmaxRef = 76.26
RdRef = 0.8
sigma = 0.6
VcmaxRef = 219.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

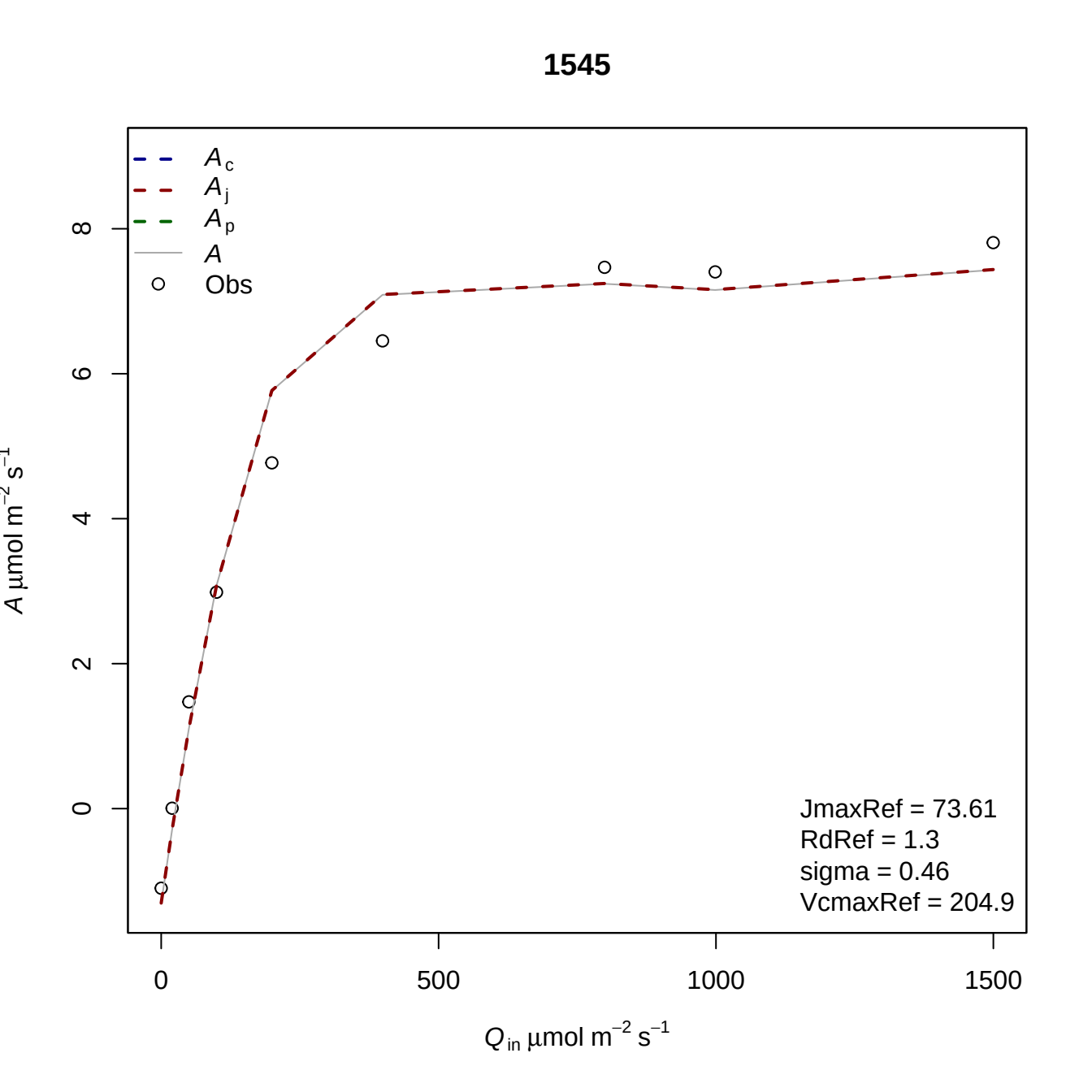

1540
A_c
A_j
A_p
A
Obs
JmaxRef = 81.92
RdRef = 0.63
sigma = 0.94
VcmaxRef = 154.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

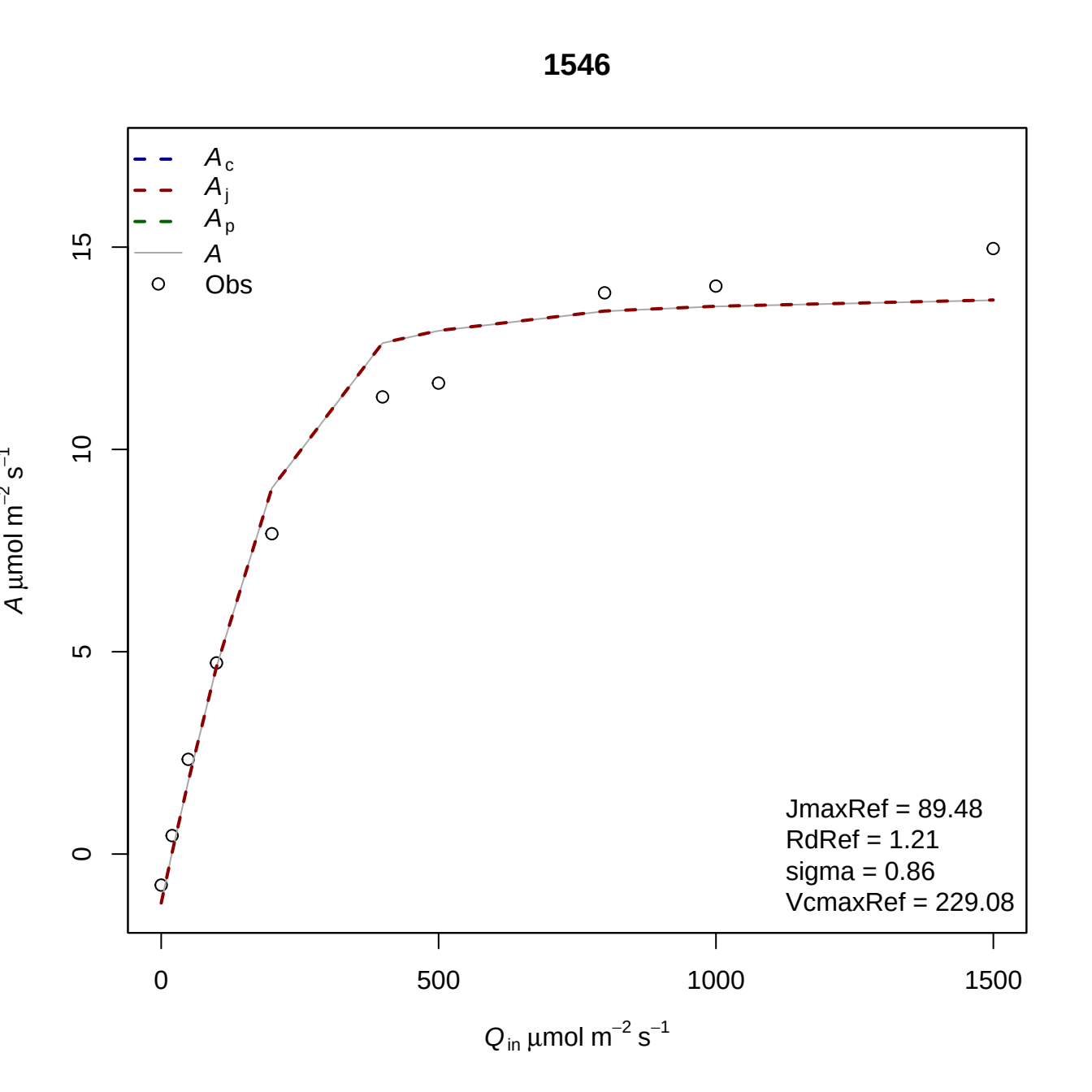

1541
A_c
A_j
A_p
A
Obs
JmaxRef = 60.99
RdRef = 1.02
sigma = 0.56
VcmaxRef = 198.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

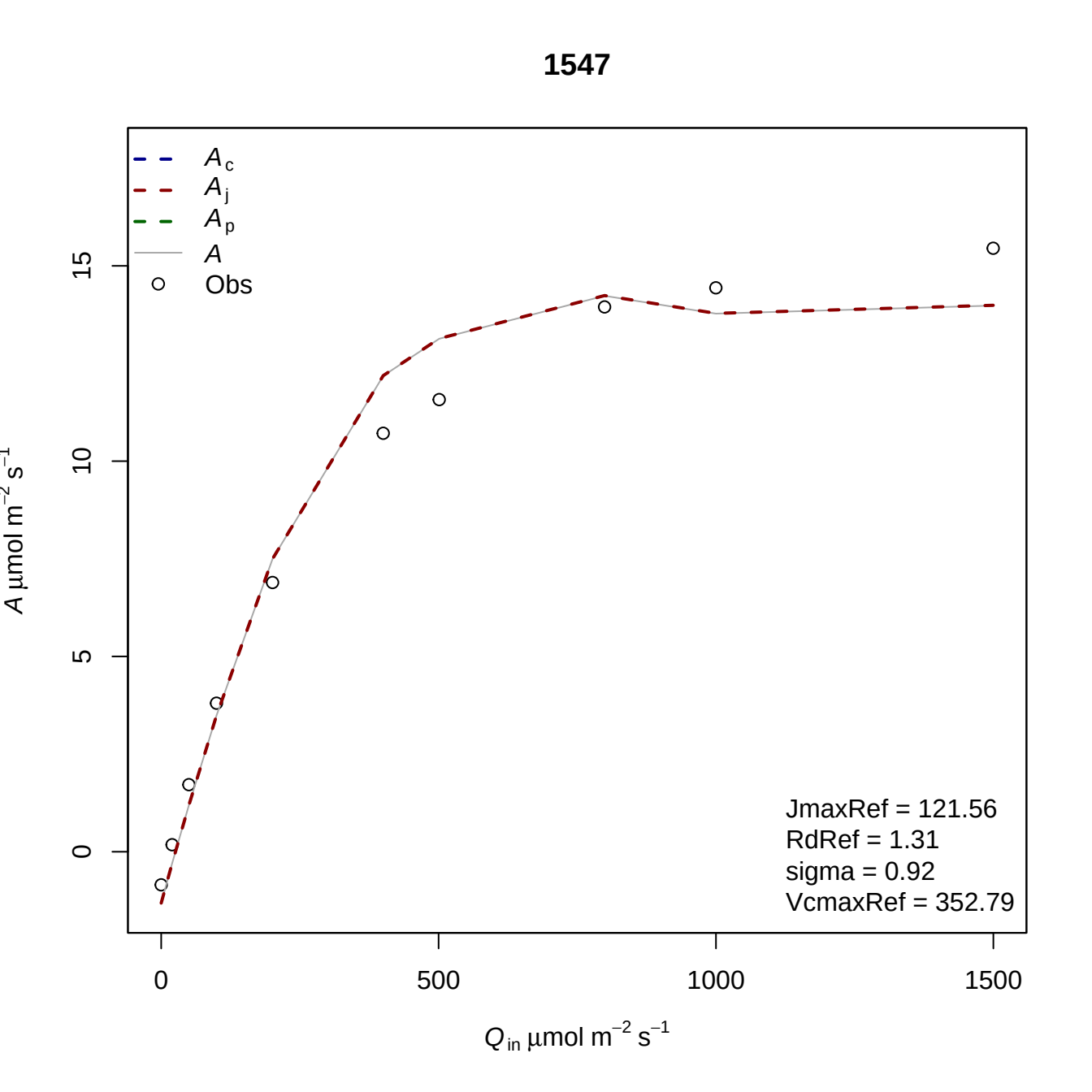

1542
A_c
A_j
A_p
A
Obs
JmaxRef = 63.48
RdRef = 0.95
sigma = 0.88
VcmaxRef = 151.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

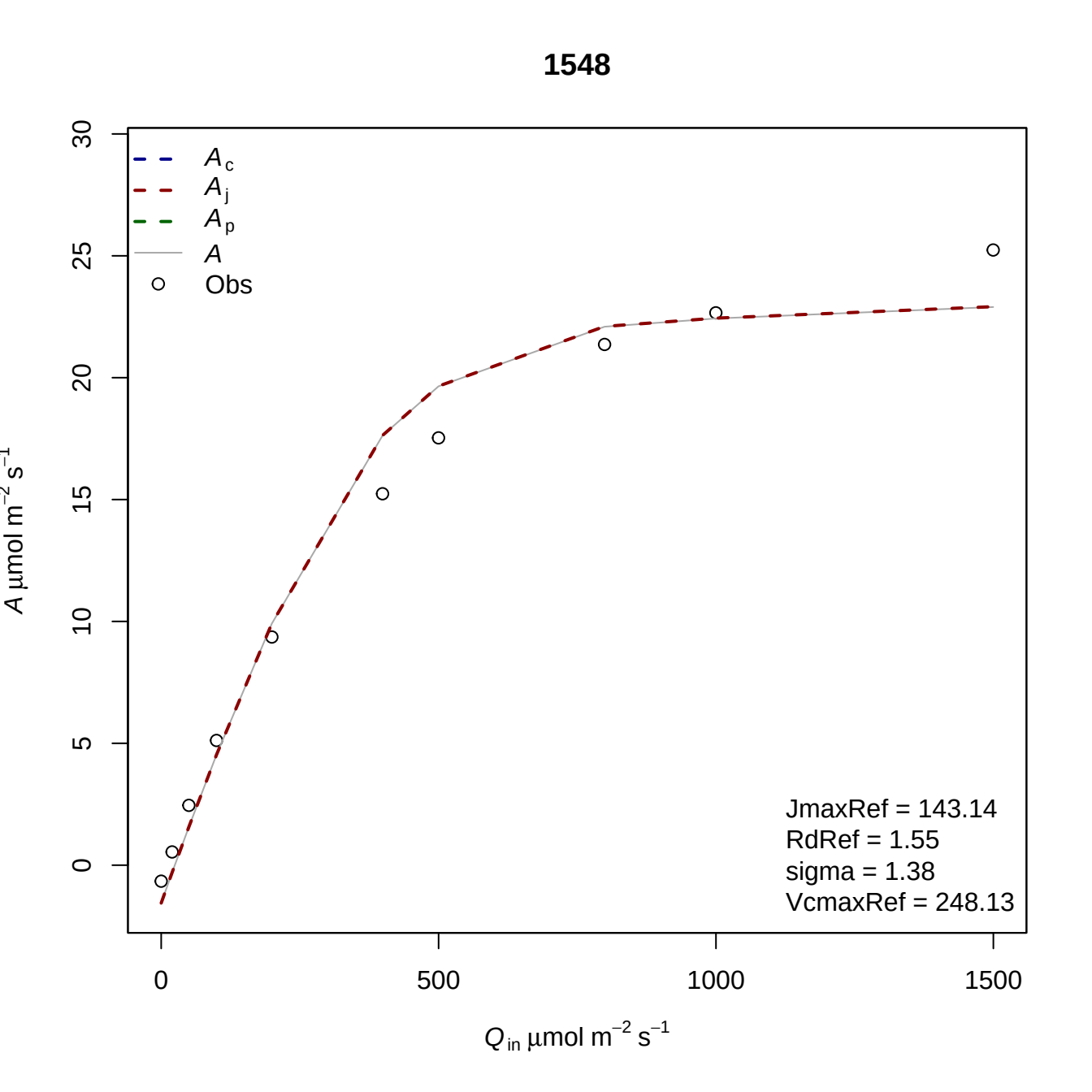

1543
A_c
A_j
A_p
A
Obs
JmaxRef = 71.37
RdRef = 1.08
sigma = 0.57
VcmaxRef = 409.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

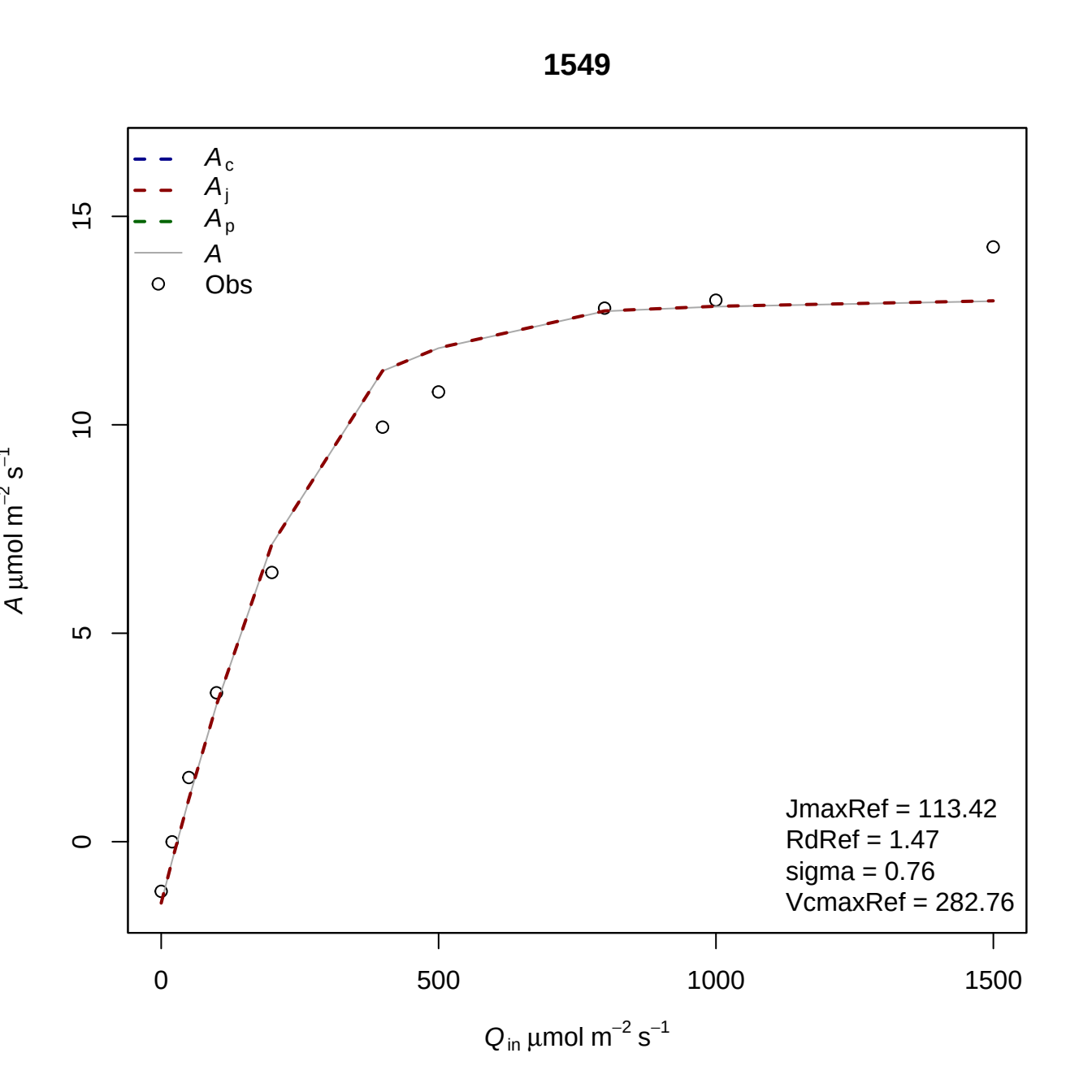

1544
A_c
A_j
A_p
A
Obs
JmaxRef = 89.64
RdRef = 0.86
sigma = 0.82
VcmaxRef = 361.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

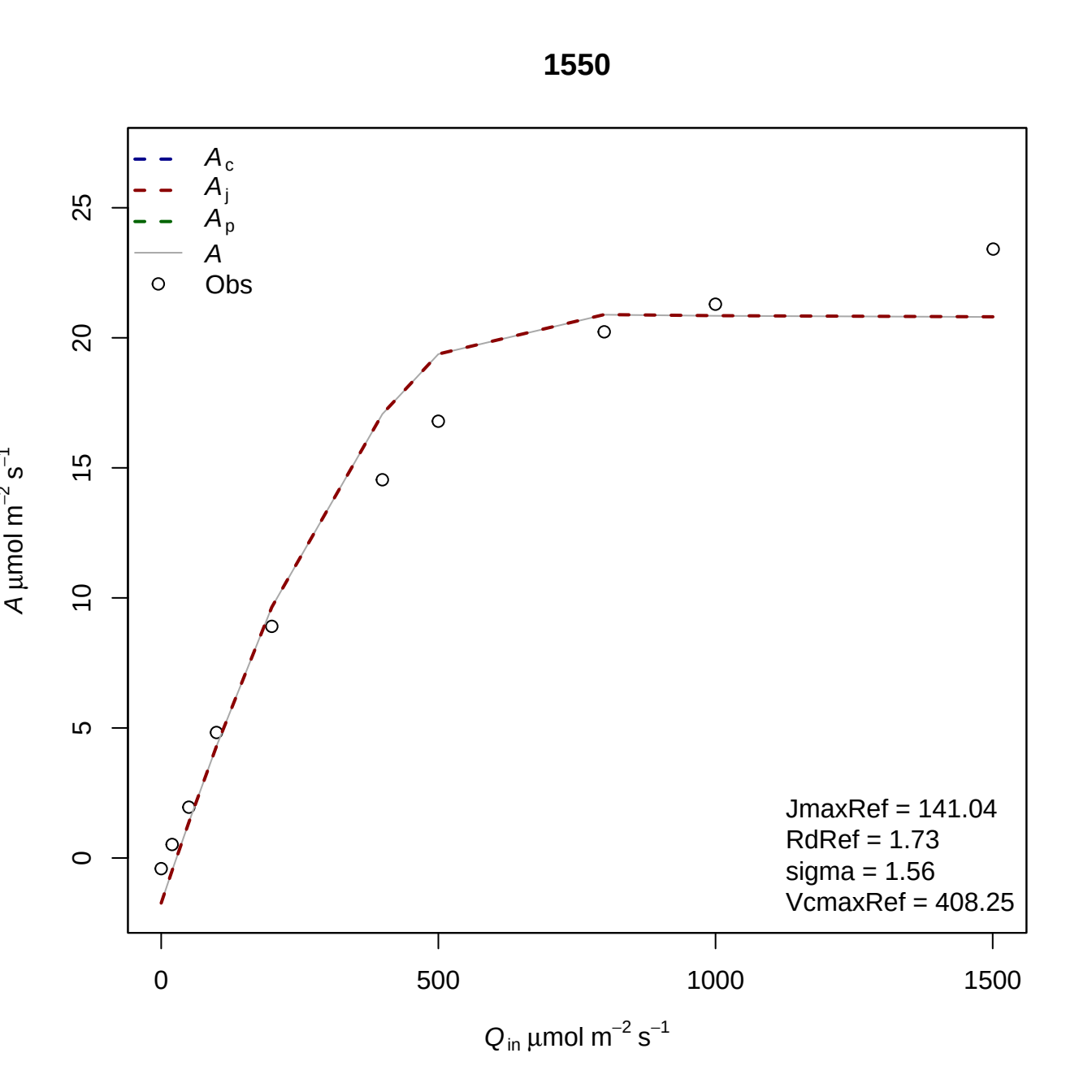

1545
A_c
A_j
A_p
A
Obs
JmaxRef = 73.61
RdRef = 1.3
sigma = 0.46
VcmaxRef = 204.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

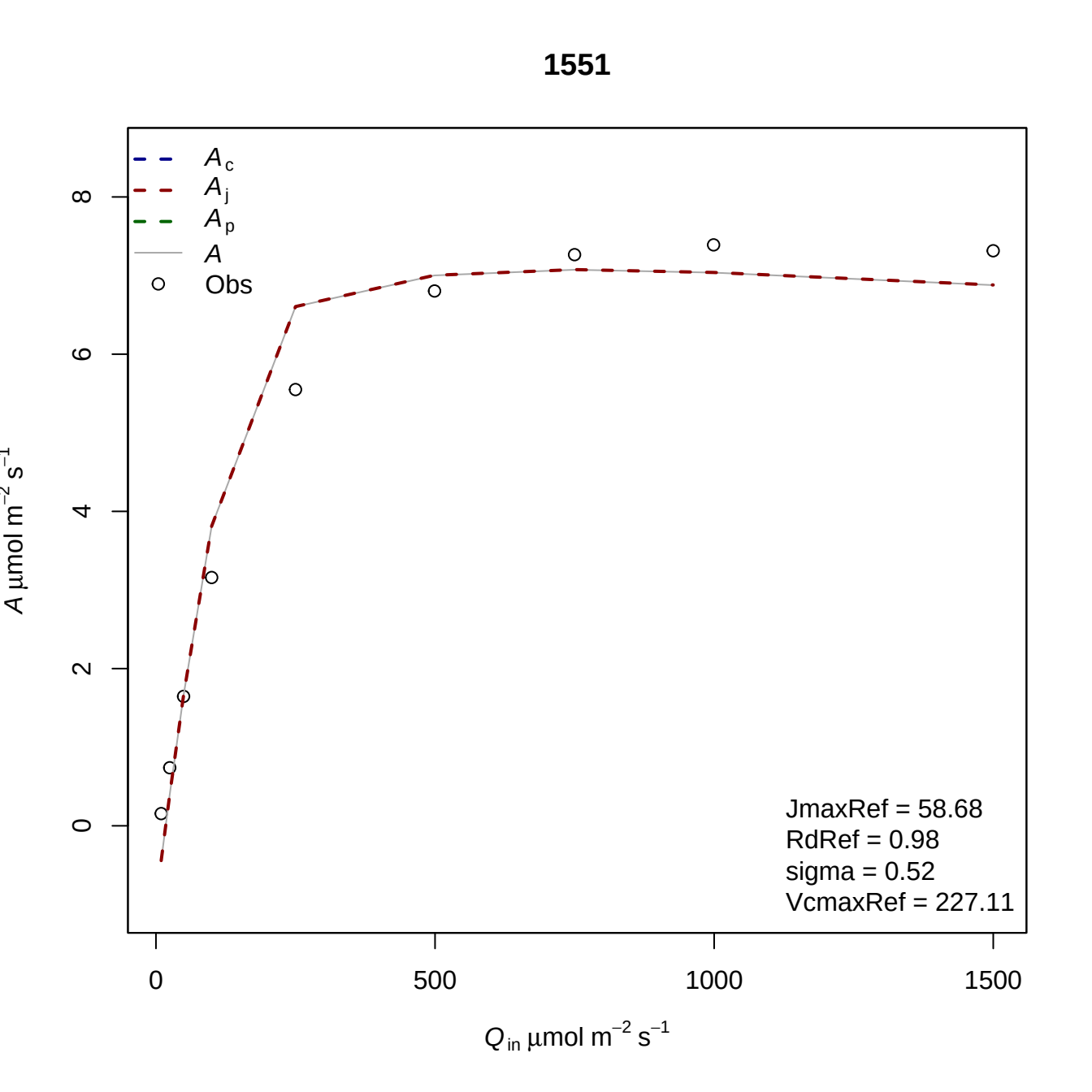

1546
A_c
A_j
A_p
A
Obs
JmaxRef = 89.48
RdRef = 1.21
sigma = 0.86
VcmaxRef = 229.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

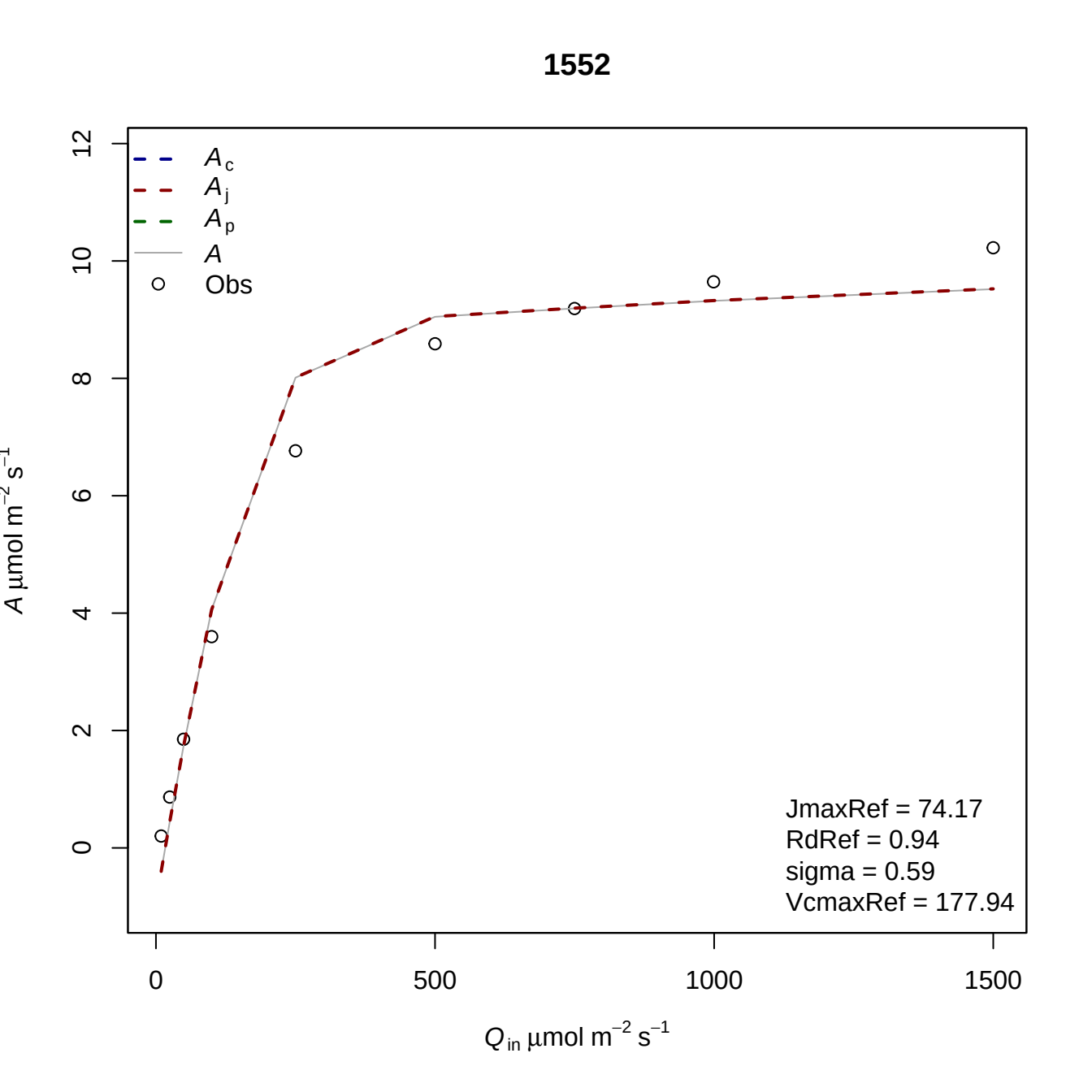

1547
A_c
A_j
A_p
A
Obs
JmaxRef = 121.56
RdRef = 1.31
sigma = 0.92
VcmaxRef = 352.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

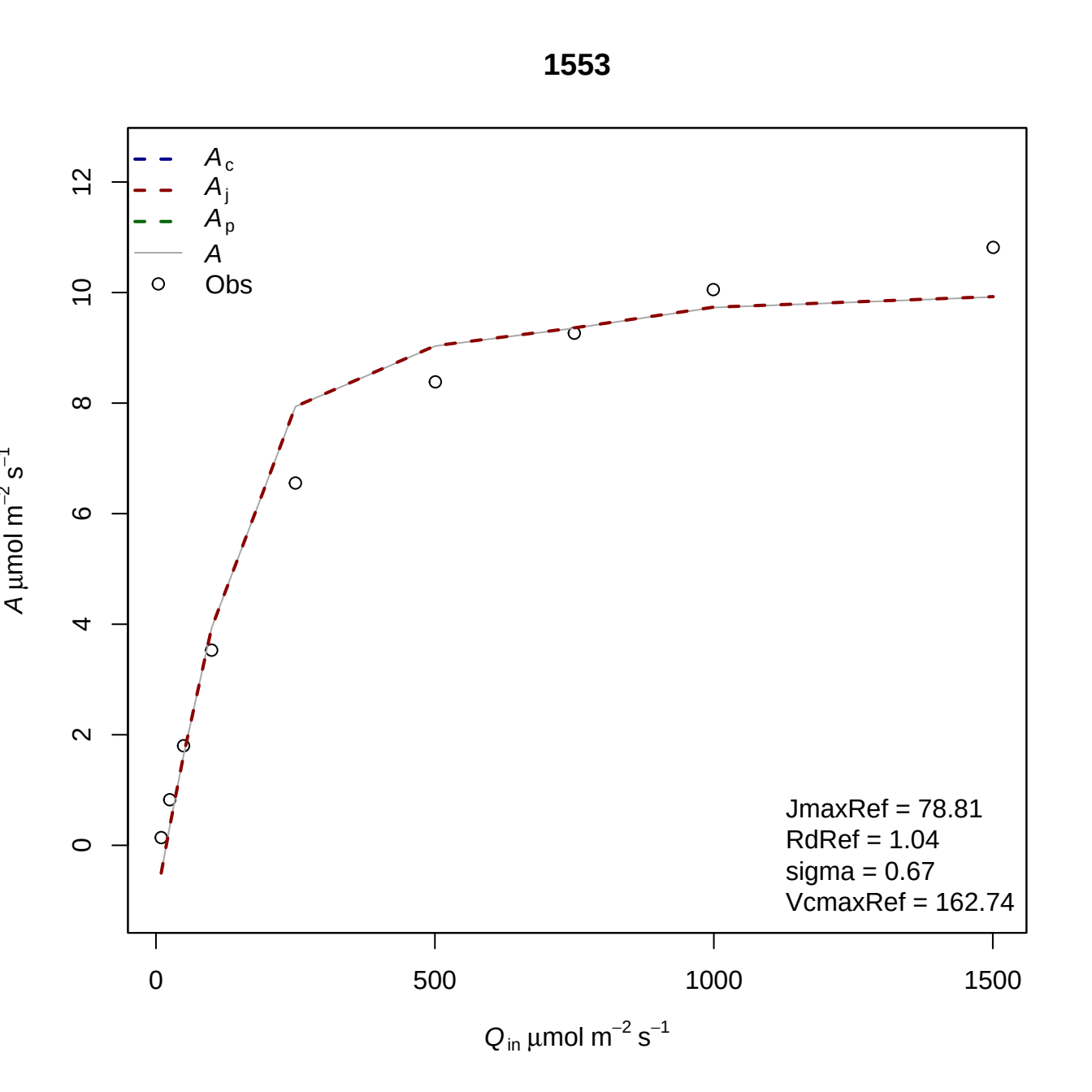

1548
A_c
A_j
A_p
A
Obs
JmaxRef = 143.14
RdRef = 1.55
sigma = 1.38
VcmaxRef = 248.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500

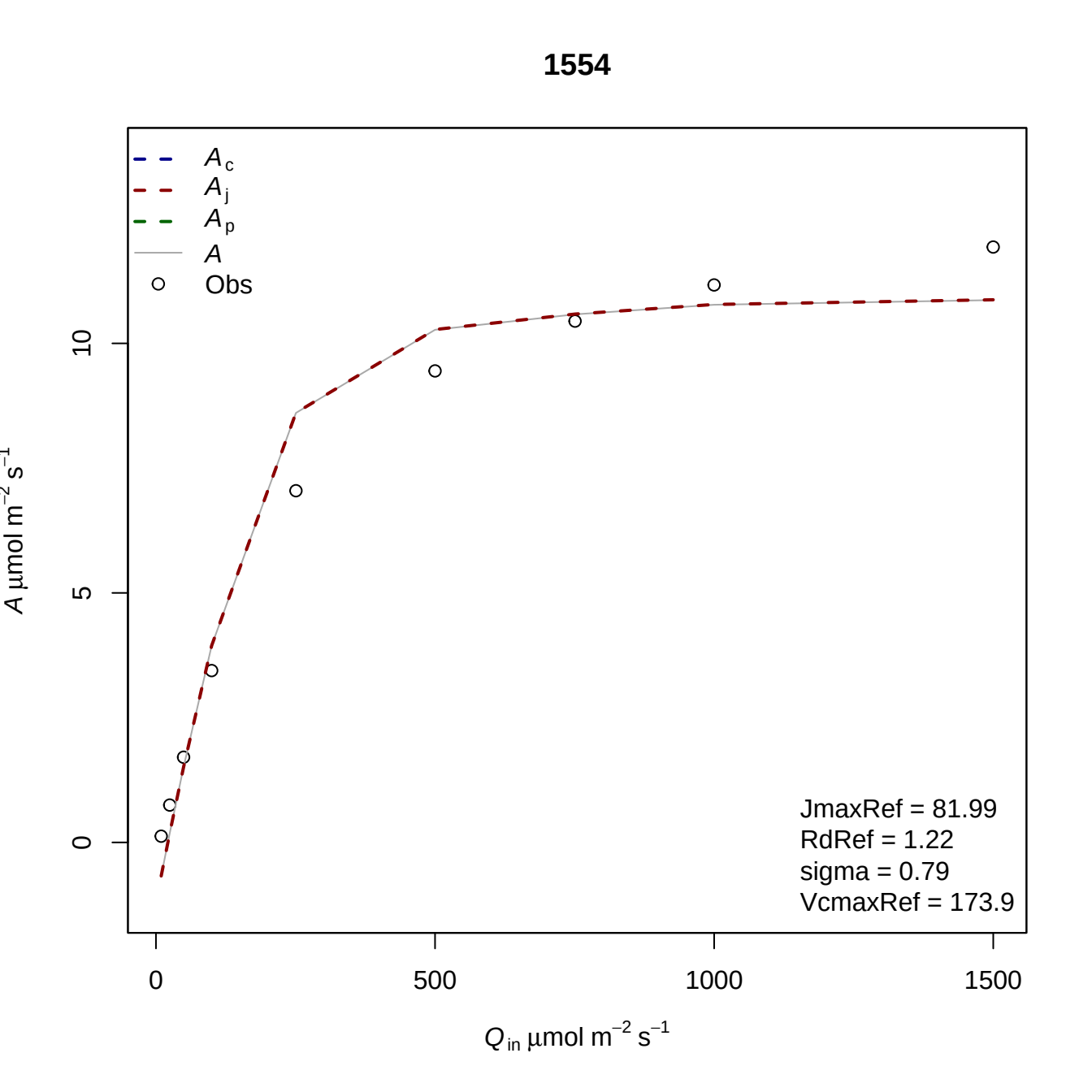

1549
A_c
A_j
A_p
A
Obs
JmaxRef = 113.42
RdRef = 1.47
sigma = 0.76
VcmaxRef = 282.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

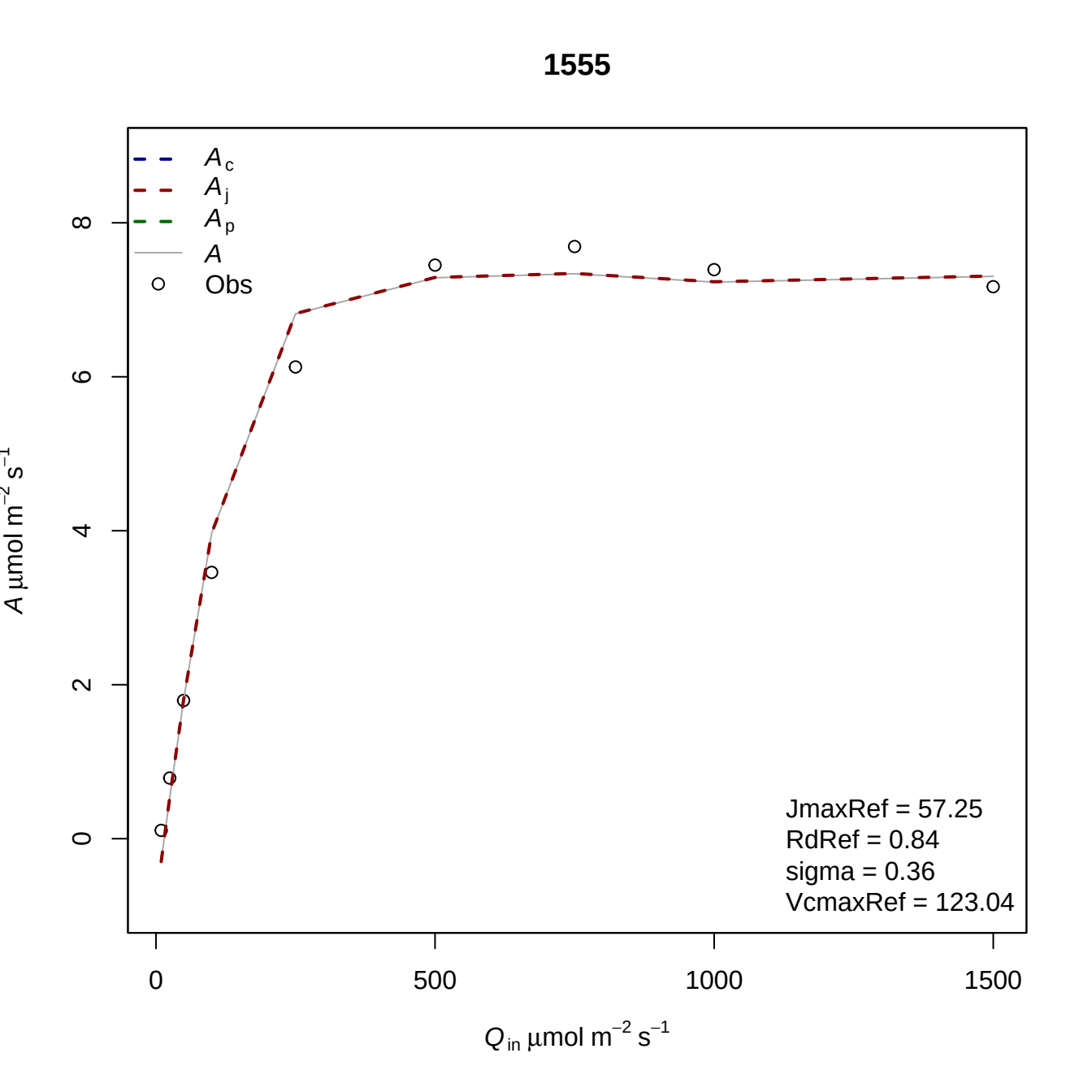

1550
A_c
A_j
A_p
A
Obs
JmaxRef = 141.04
RdRef = 1.73
sigma = 1.56
VcmaxRef = 408.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

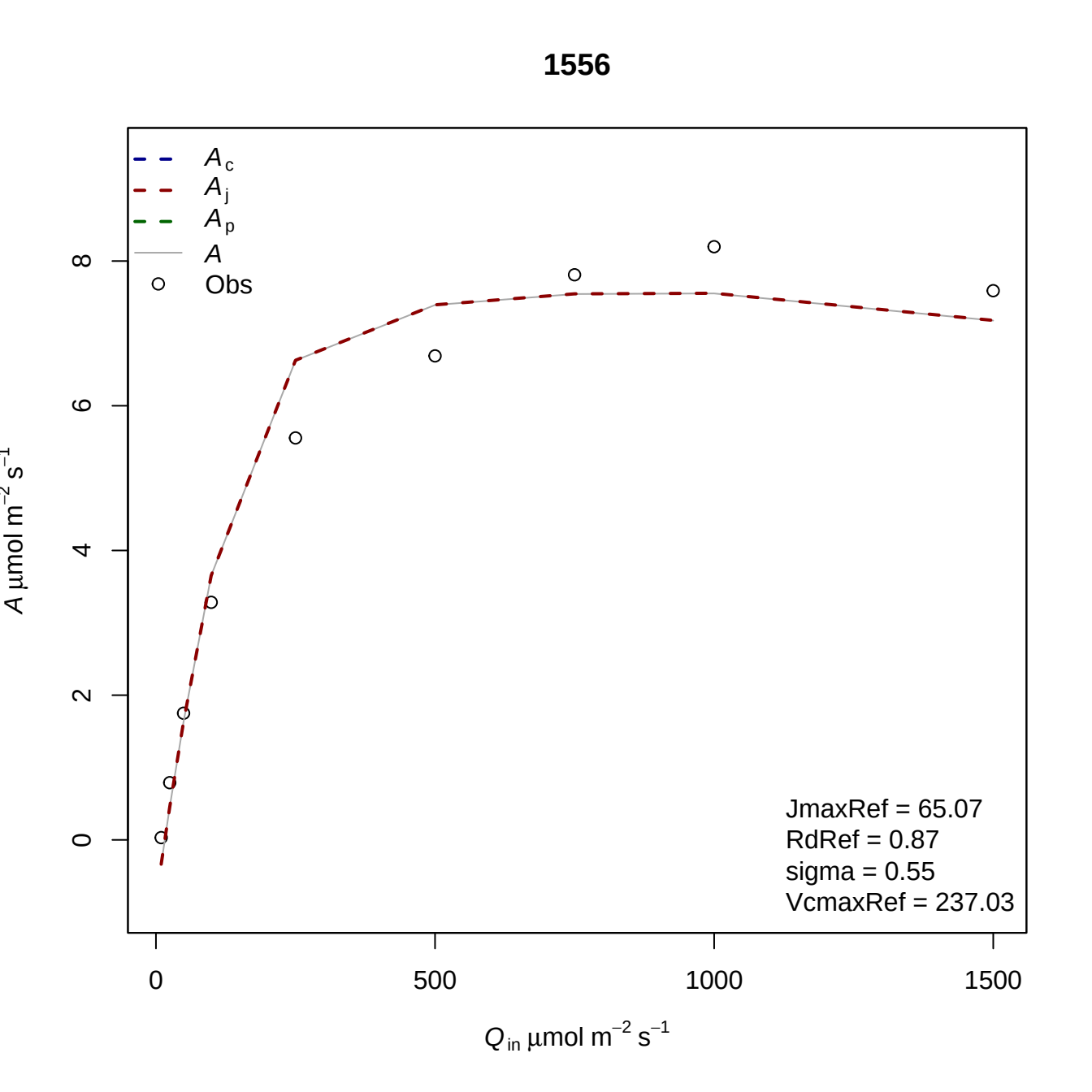

1551
A_c
A_j
A_p
A
Obs
JmaxRef = 58.68
RdRef = 0.98
sigma = 0.52
VcmaxRef = 227.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

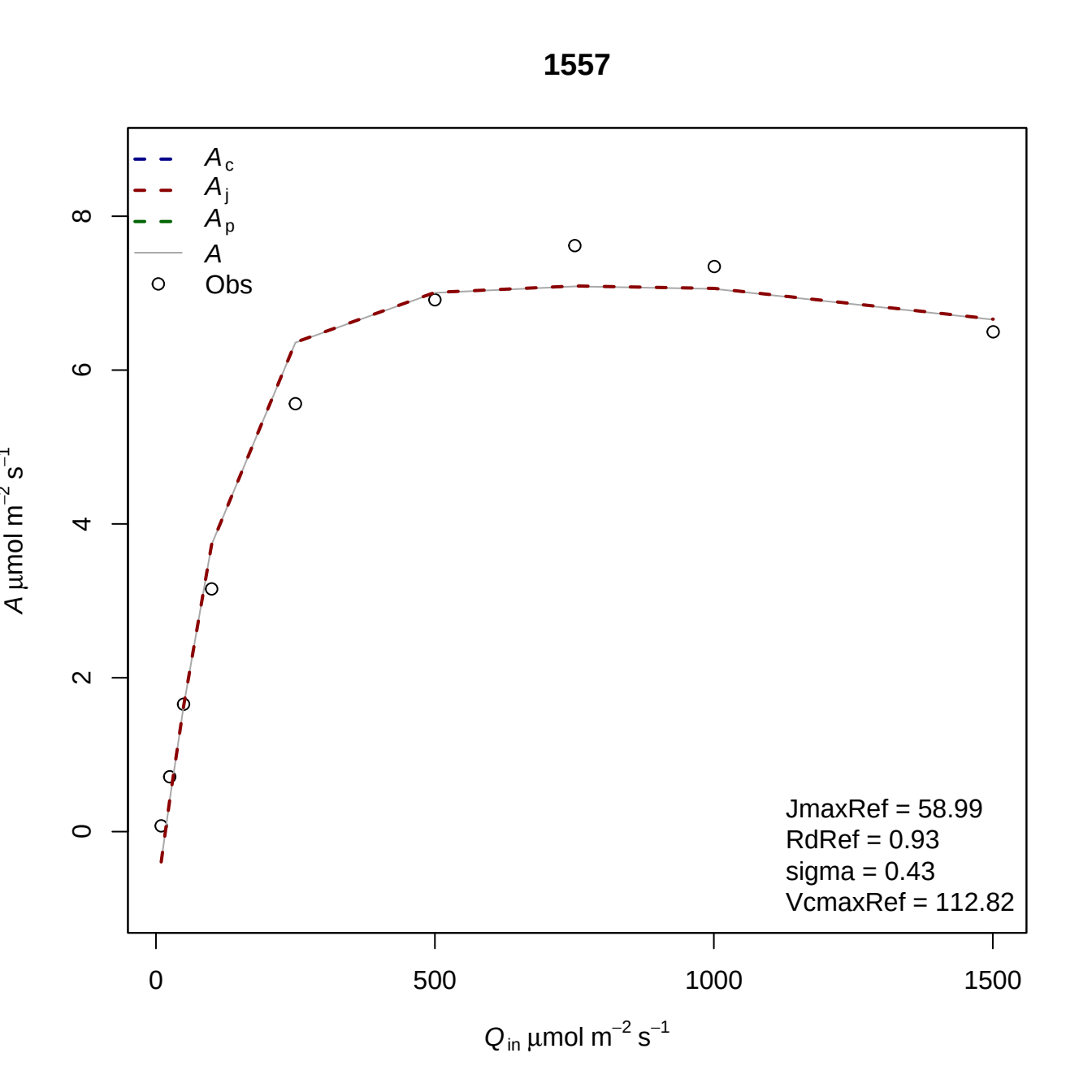

1552
A_c
A_j
A_p
A
Obs
JmaxRef = 74.17
RdRef = 0.94
sigma = 0.59
VcmaxRef = 177.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

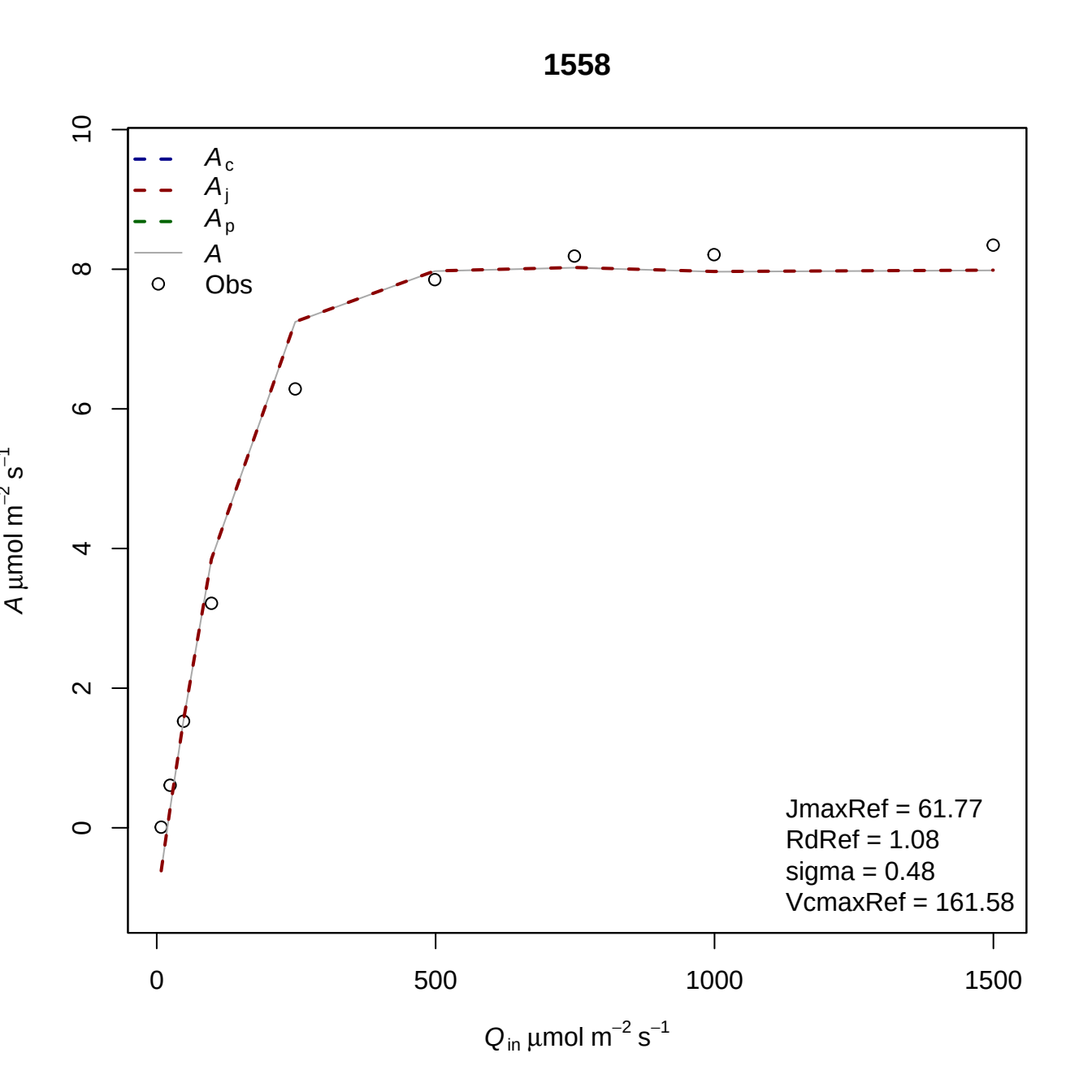

1553
A_c
A_j
A_p
A
Obs
JmaxRef = 78.81
RdRef = 1.04
sigma = 0.67
VcmaxRef = 162.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

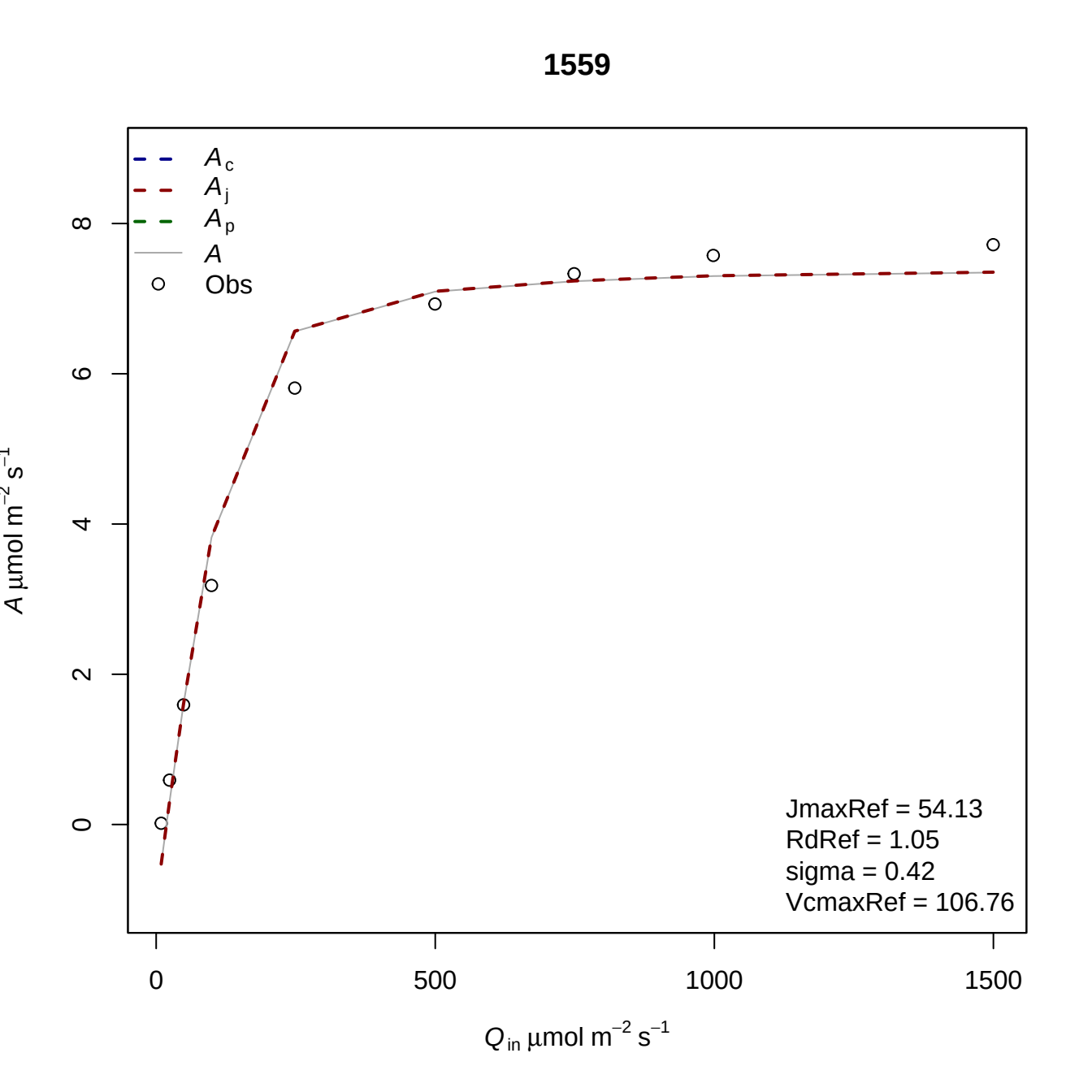

1554
A_c
A_j
A_p
A
Obs
JmaxRef = 81.99
RdRef = 1.22
sigma = 0.79
VcmaxRef = 173.9
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

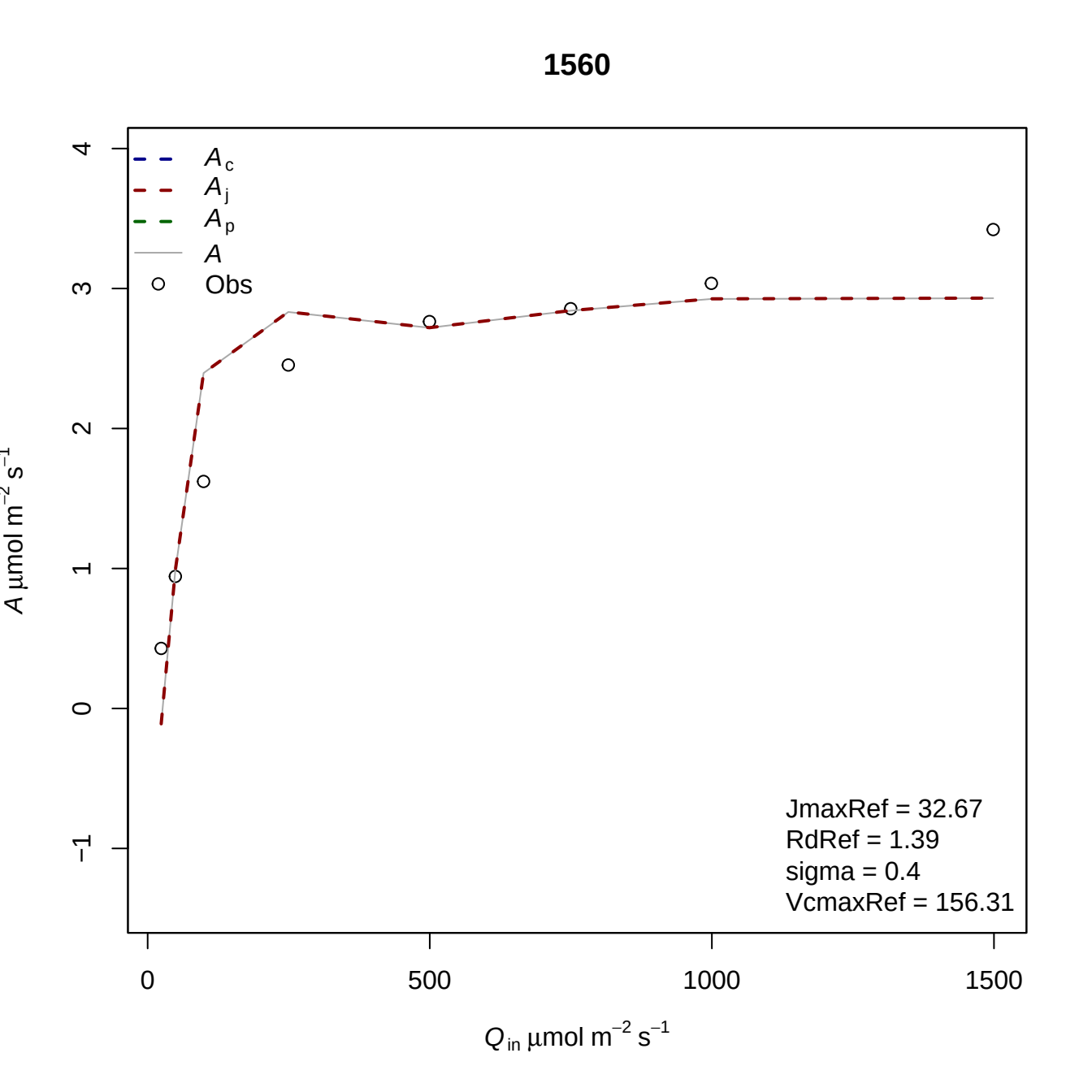

1555
A_c
A_j
A_p
A
Obs
JmaxRef = 57.25
RdRef = 0.84
sigma = 0.36
VcmaxRef = 123.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

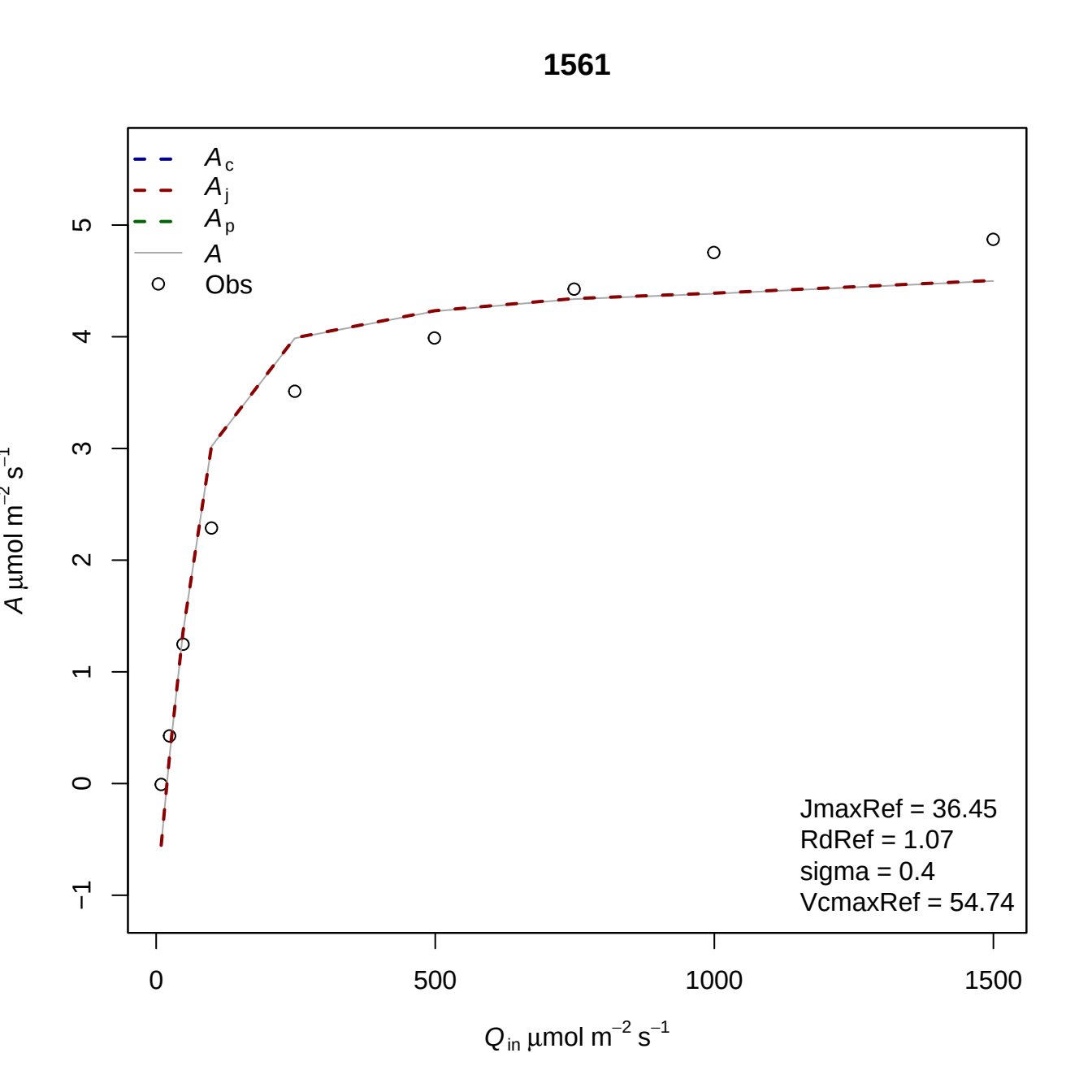

1556
A_c
A_j
A_p
A
Obs
JmaxRef = 65.07
RdRef = 0.87
sigma = 0.55
VcmaxRef = 237.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

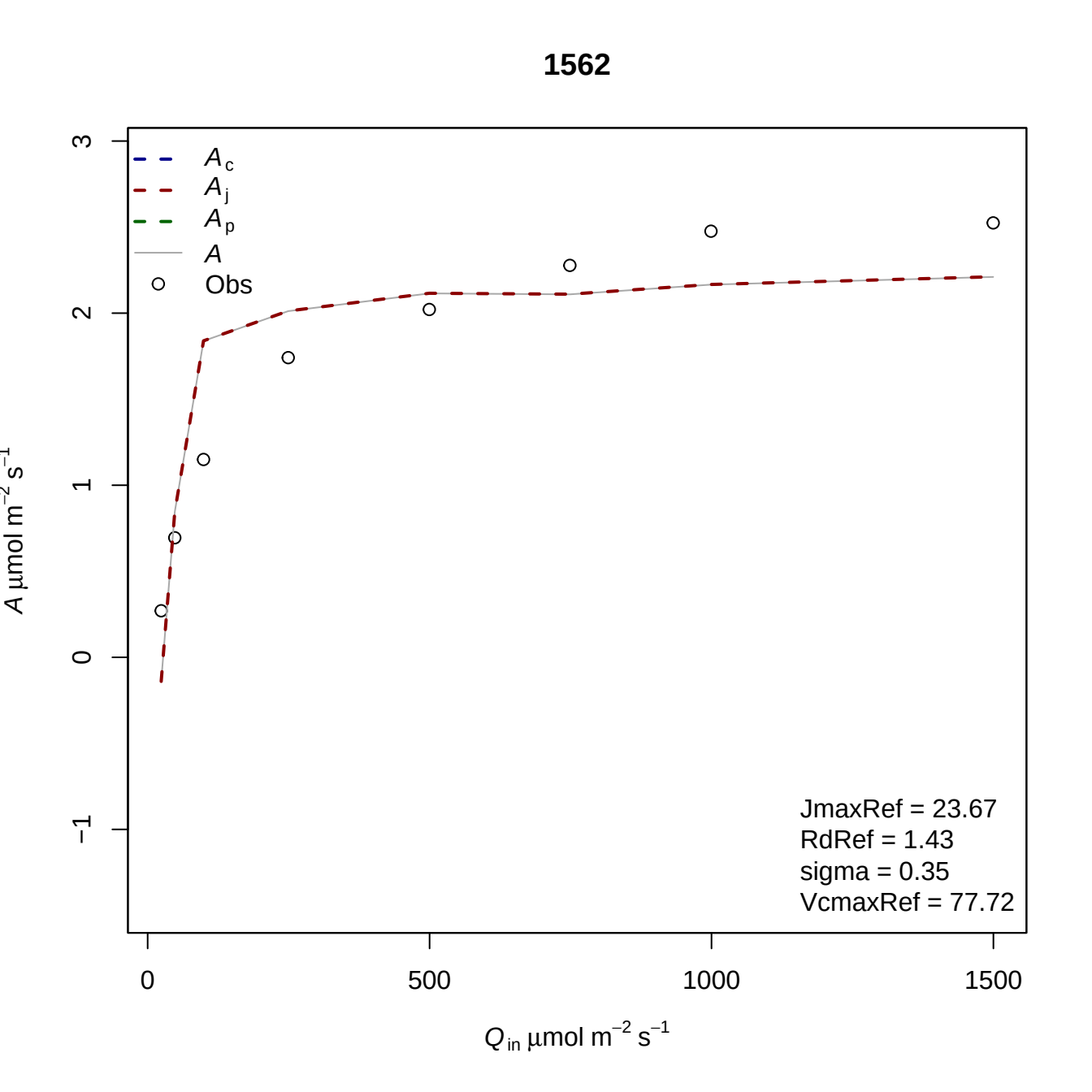

1557
A_c
A_j
A_p
A
Obs
JmaxRef = 58.99
RdRef = 0.93
sigma = 0.43
VcmaxRef = 112.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

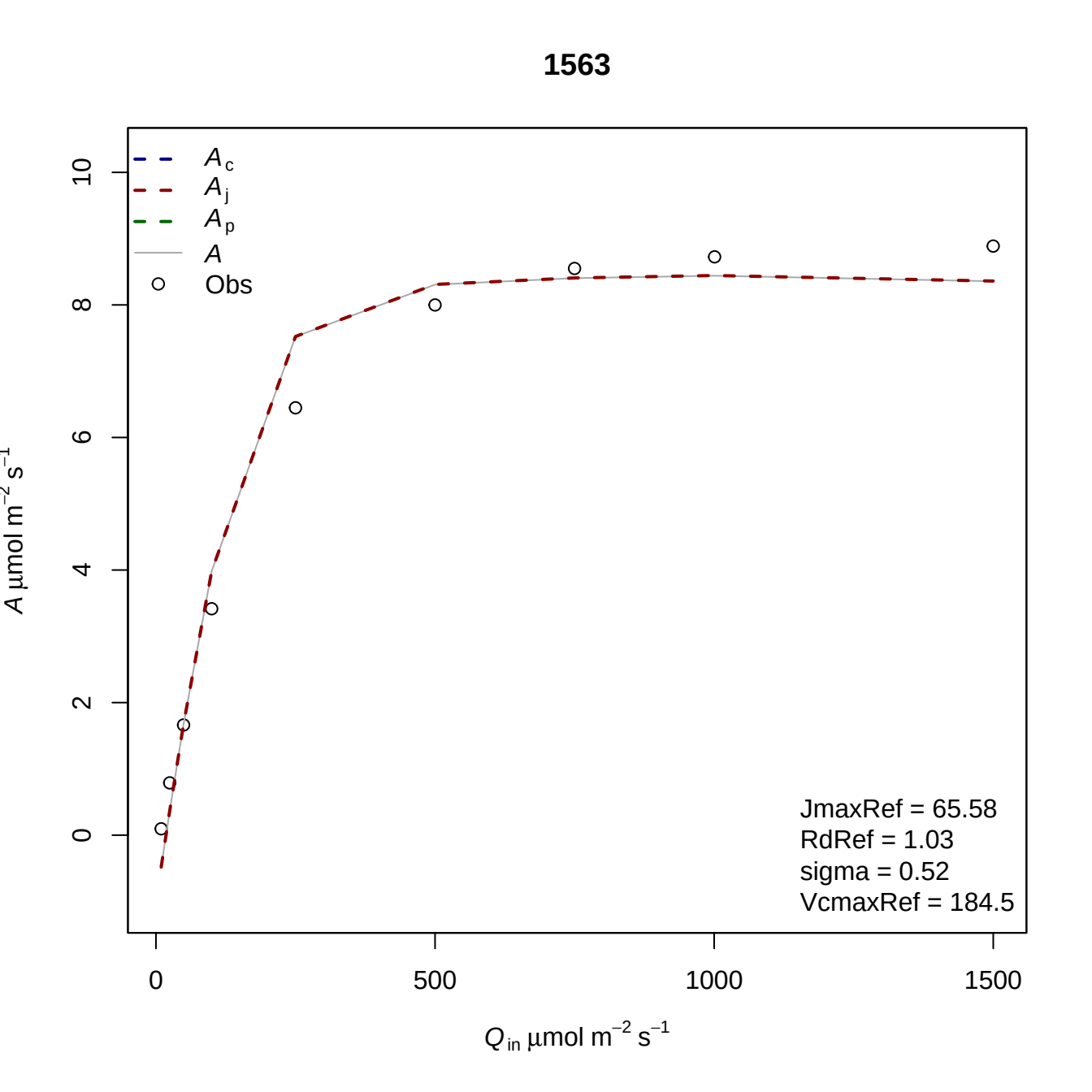

1558
A_c
A_j
A_p
A
Obs
JmaxRef = 61.77
RdRef = 1.08
sigma = 0.48
VcmaxRef = 161.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

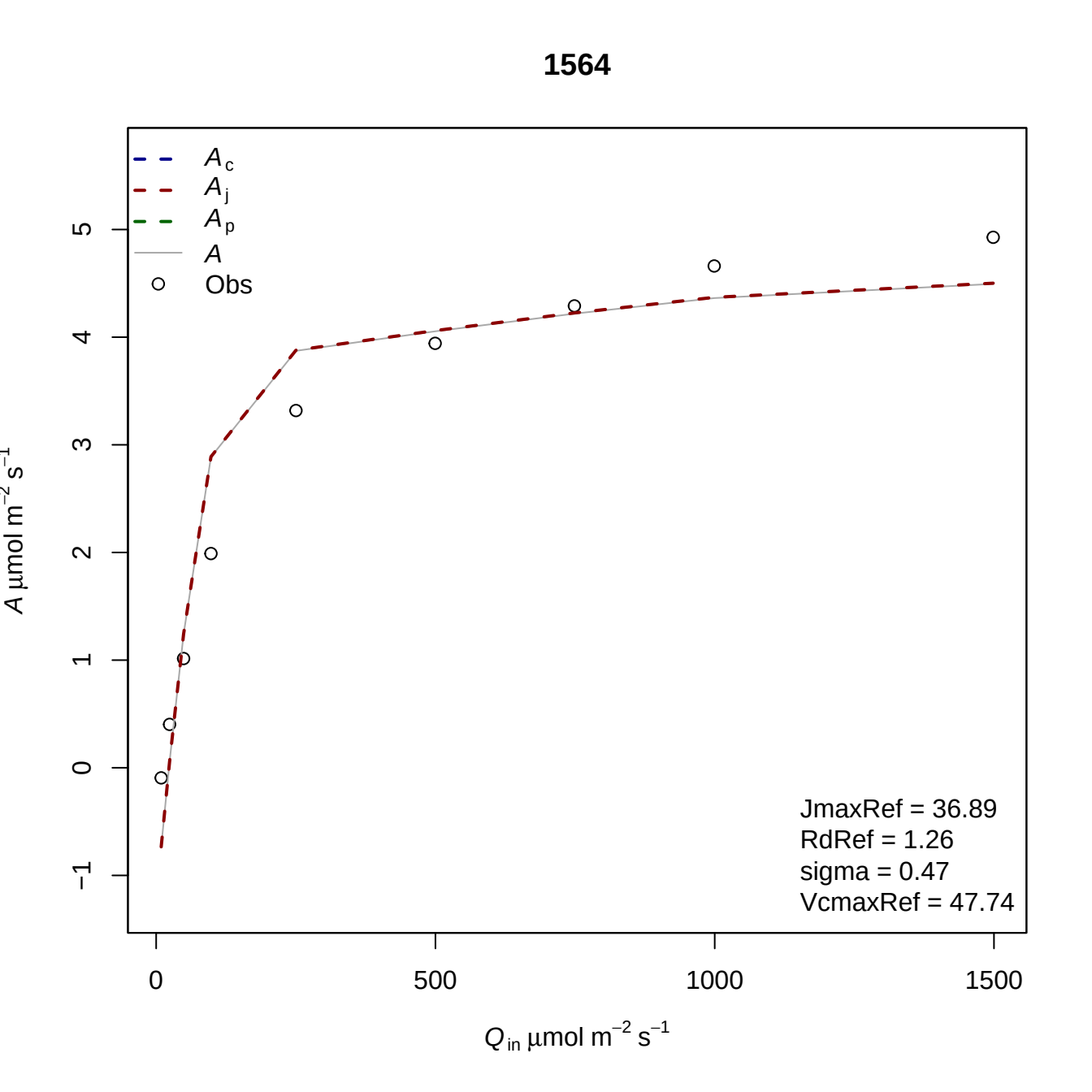

1559
A_c
A_j
A_p
A
Obs
JmaxRef = 54.13
RdRef = 1.05
sigma = 0.42
VcmaxRef = 106.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

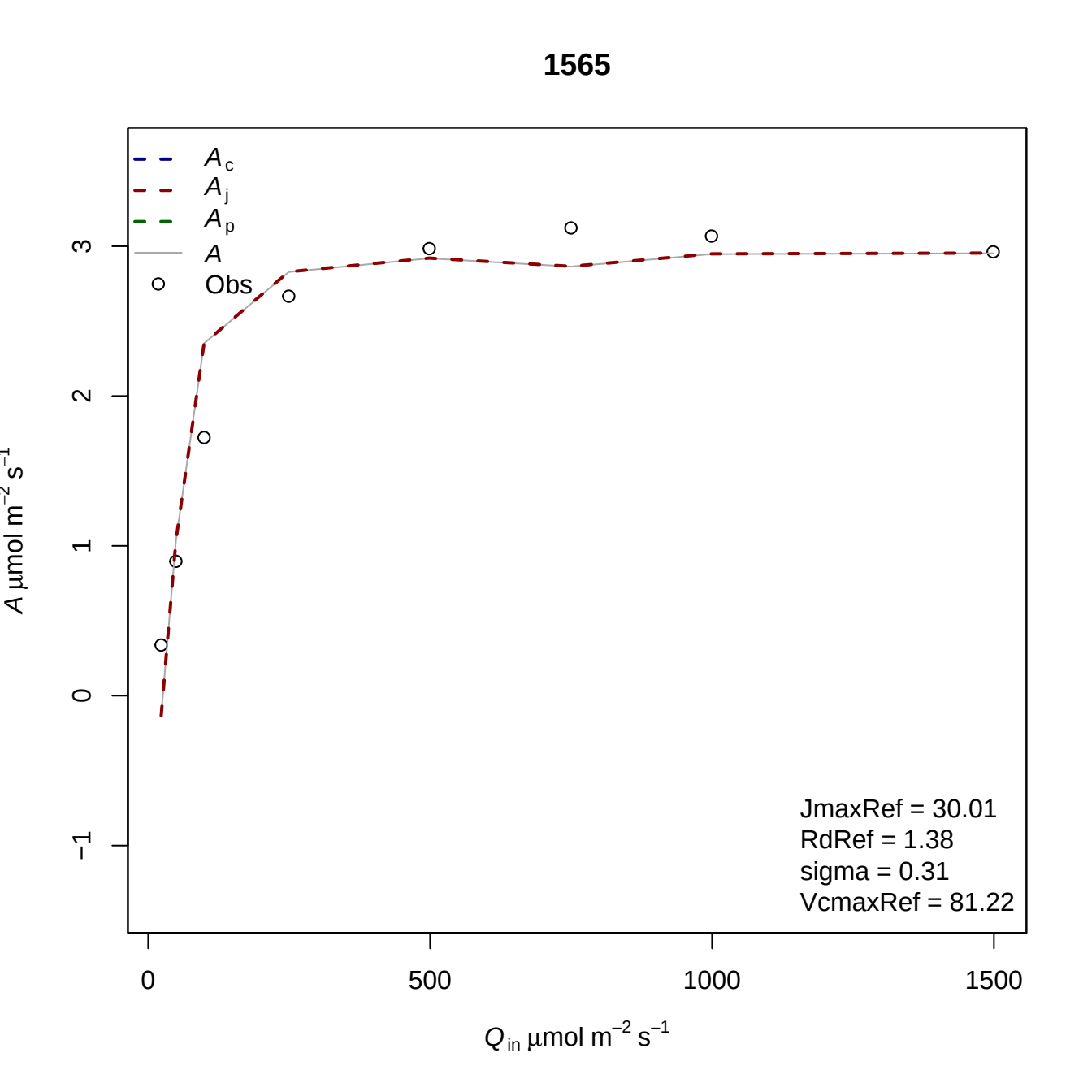

1560
A_c
A_j
A_p
A
Obs
JmaxRef = 32.67
RdRef = 1.39
sigma = 0.4
VcmaxRef = 156.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

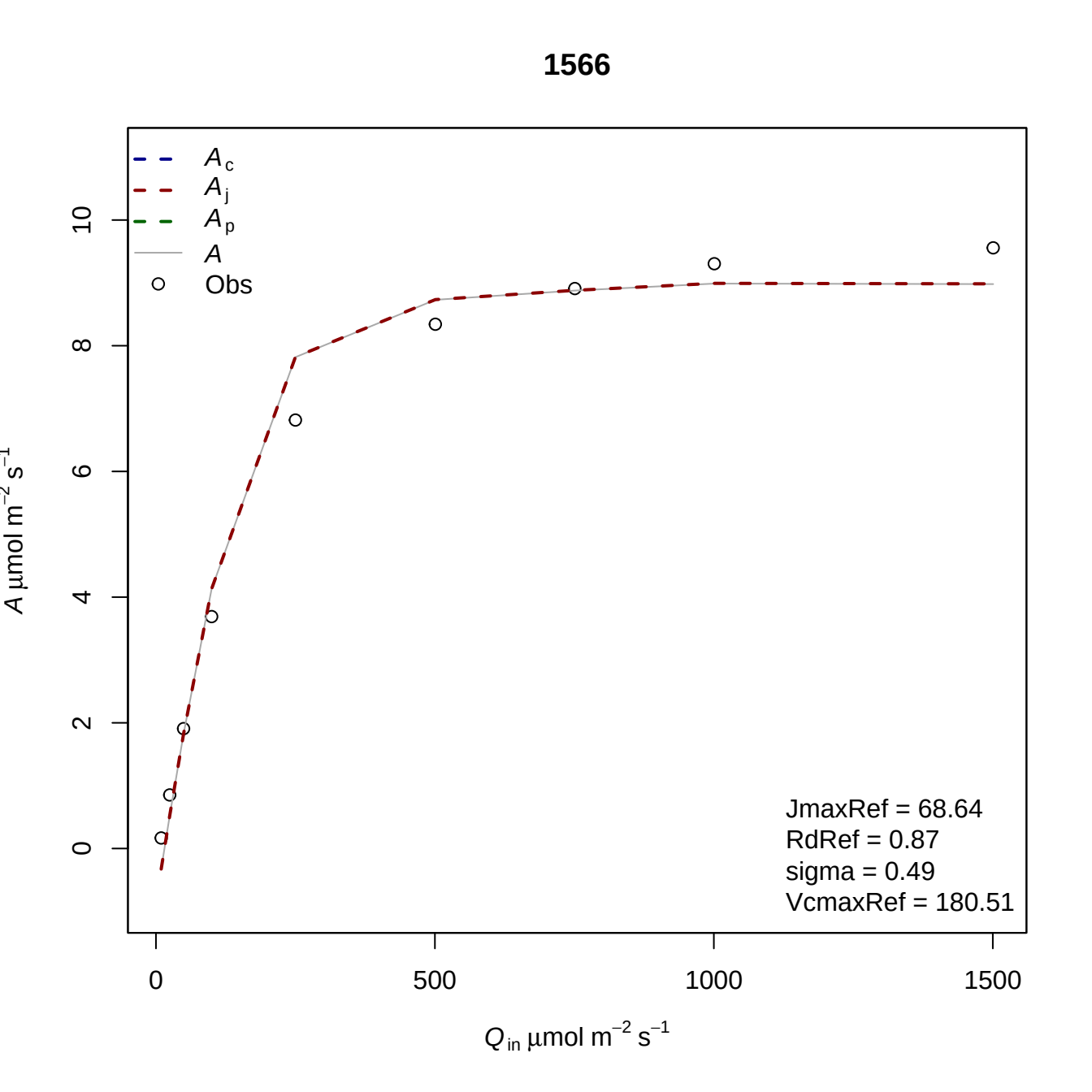

1561
A_c
A_j
A_p
A
Obs
JmaxRef = 36.45
RdRef = 1.07
sigma = 0.4
VcmaxRef = 54.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

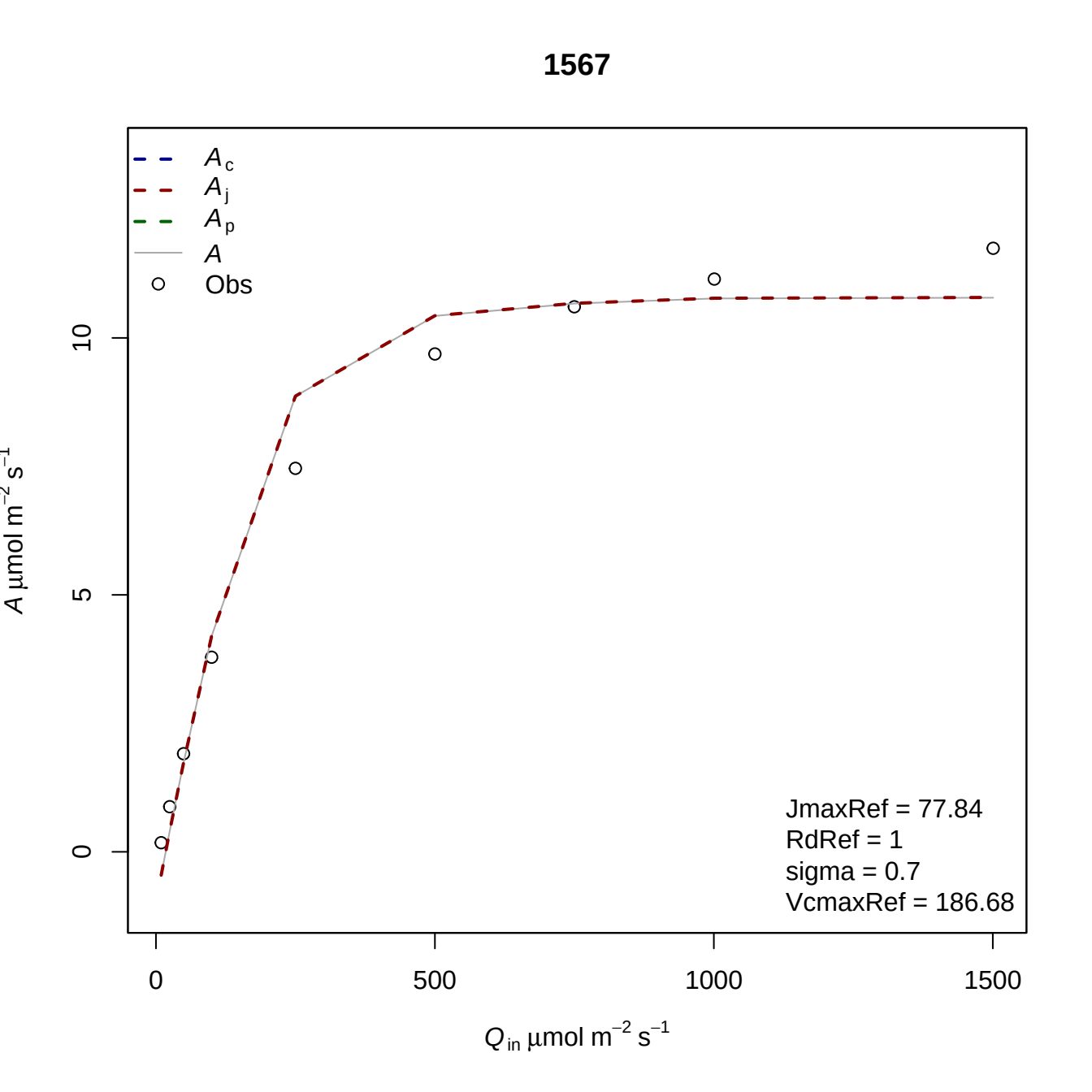

1562
A_c
A_j
A_p
A
Obs
JmaxRef = 23.67
RdRef = 1.43
sigma = 0.35
VcmaxRef = 77.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

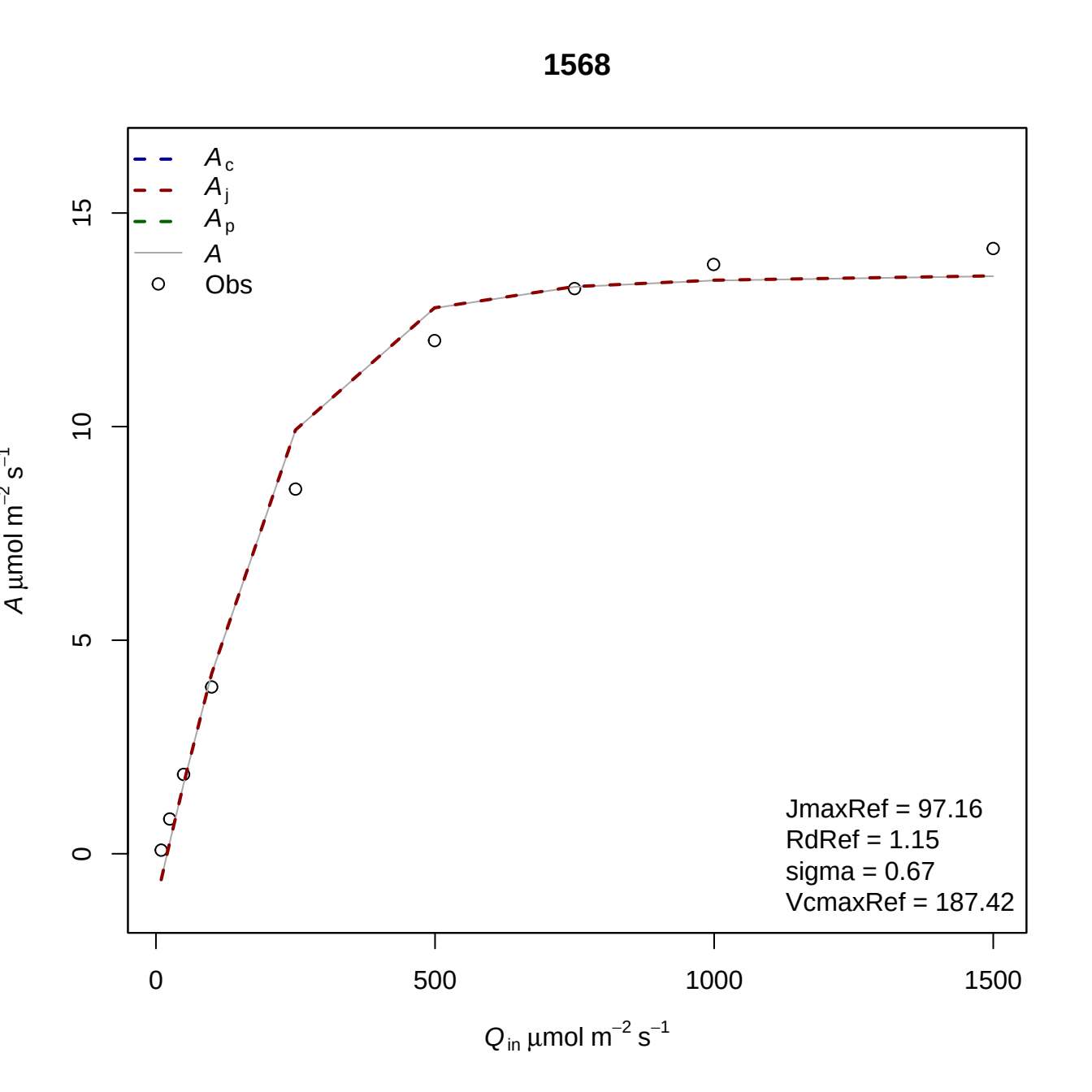

1563
A_c
A_j
A_p
A
Obs
JmaxRef = 65.58
RdRef = 1.03
sigma = 0.52
VcmaxRef = 184.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

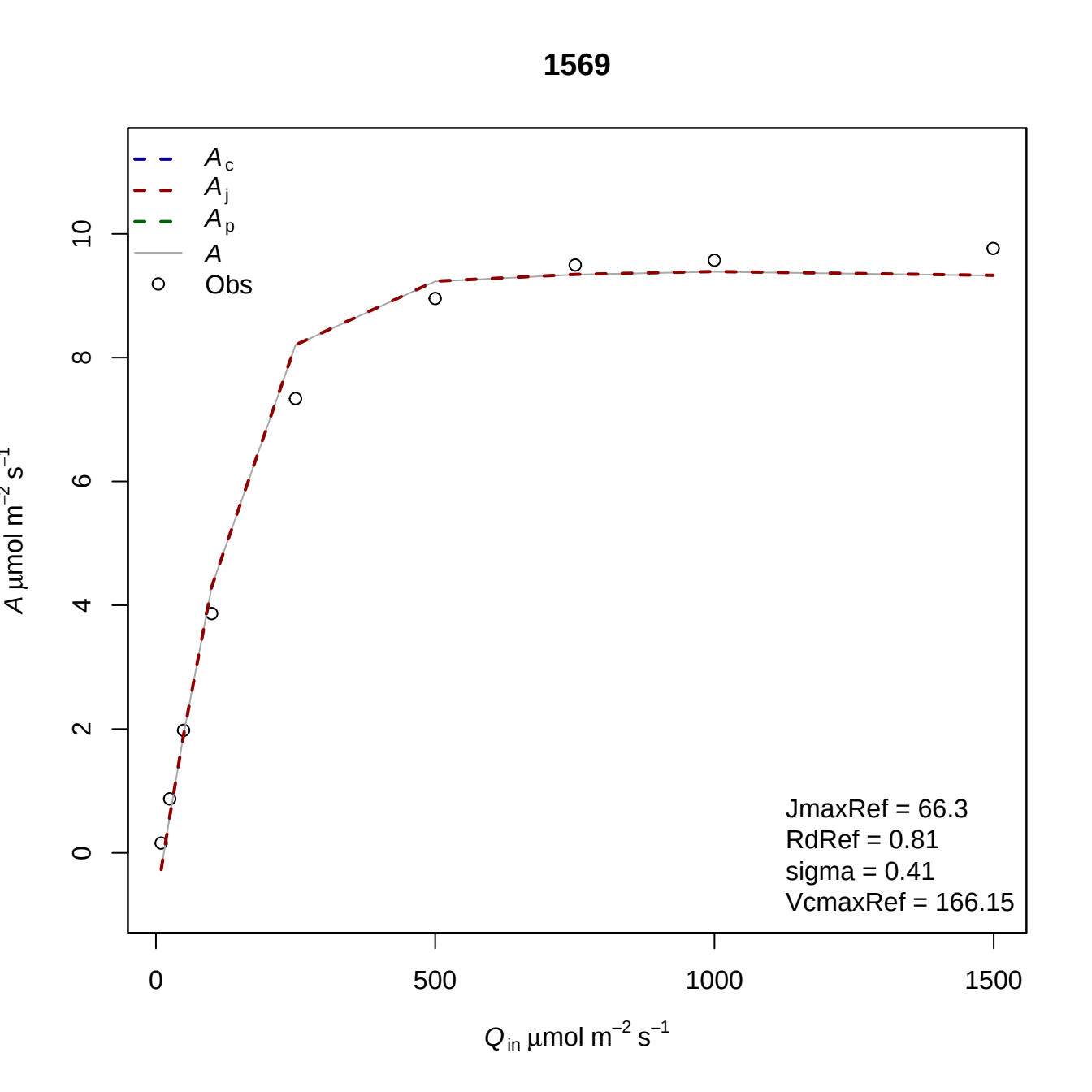

1564
A_c
A_j
A_p
A
Obs
JmaxRef = 36.89
RdRef = 1.26
sigma = 0.47
VcmaxRef = 47.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

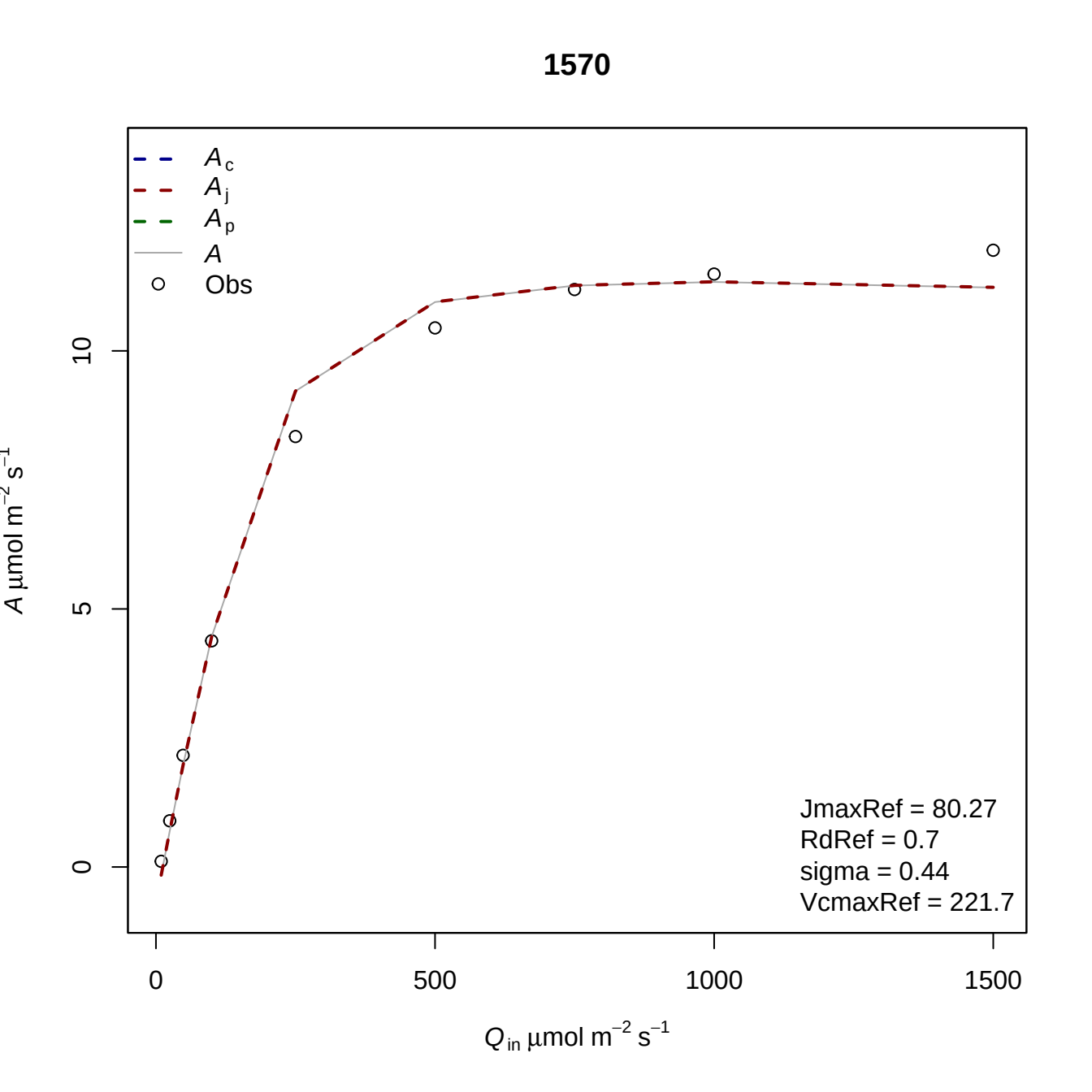

1565
A_c
A_j
A_p
A
Obs
JmaxRef = 30.01
RdRef = 1.38
sigma = 0.31
VcmaxRef = 81.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

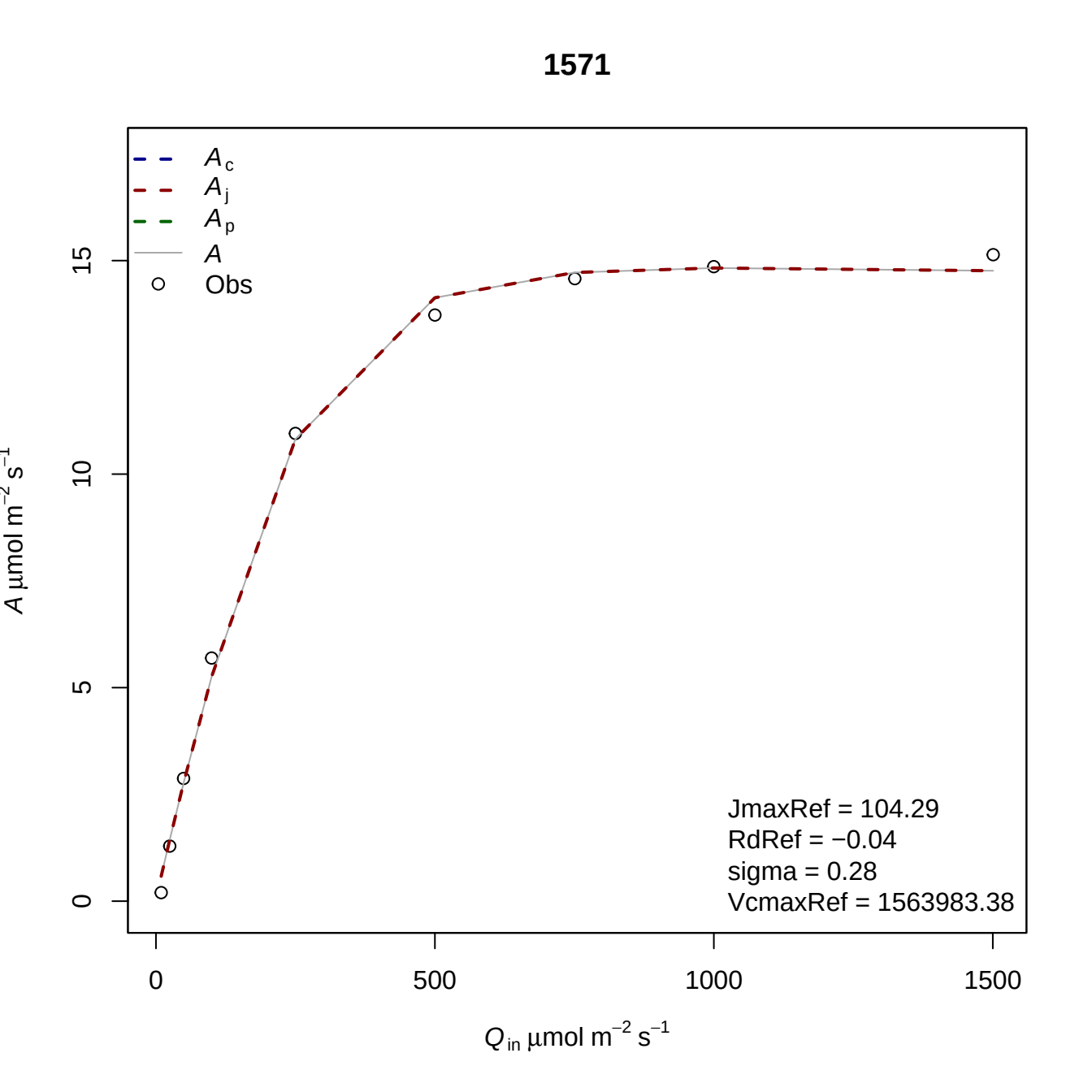

1566
A_c
A_j
A_p
A
Obs
JmaxRef = 68.64
RdRef = 0.87
sigma = 0.49
VcmaxRef = 180.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

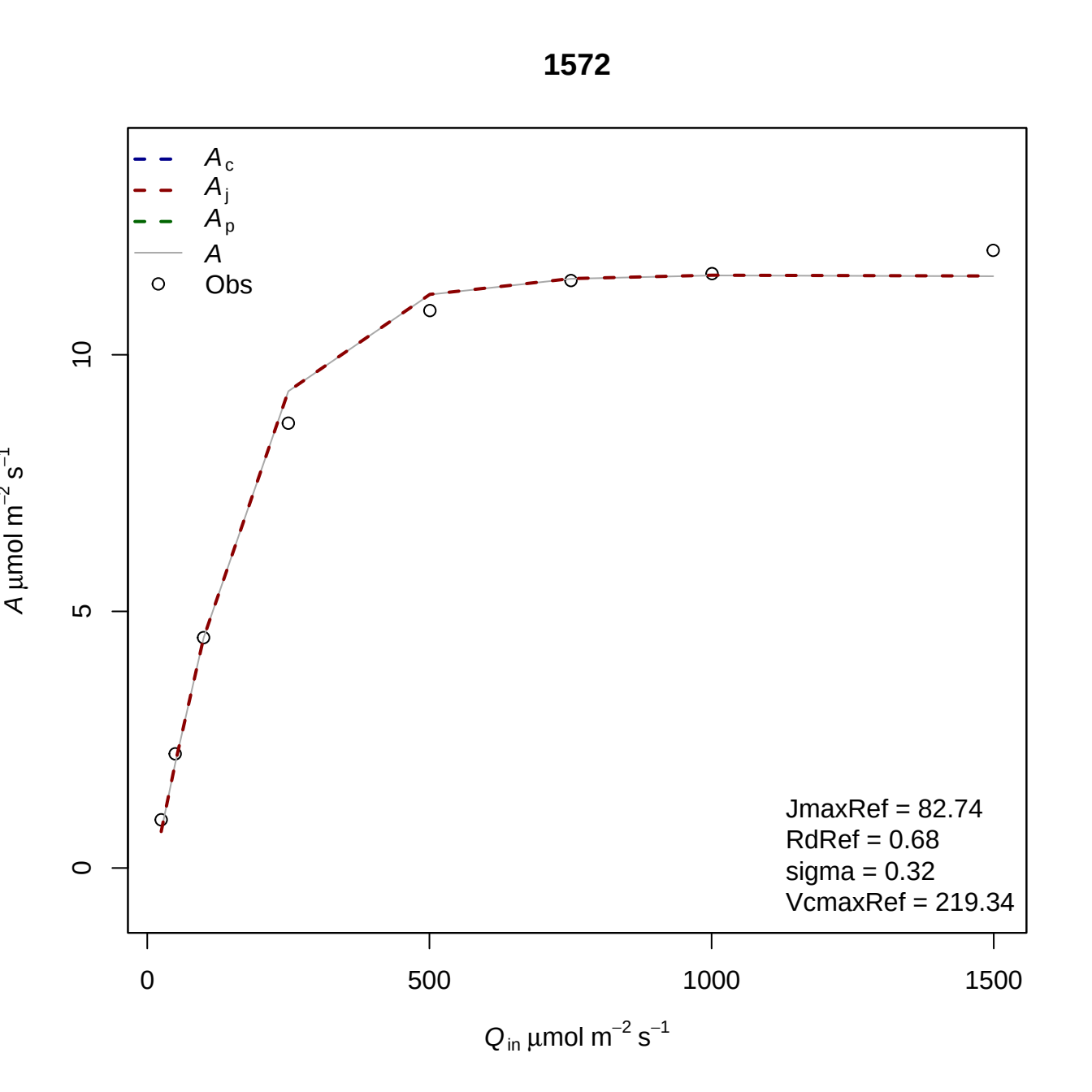

1567
A_c
A_j
A_p
A
Obs
JmaxRef = 77.84
RdRef = 1
sigma = 0.7
VcmaxRef = 186.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

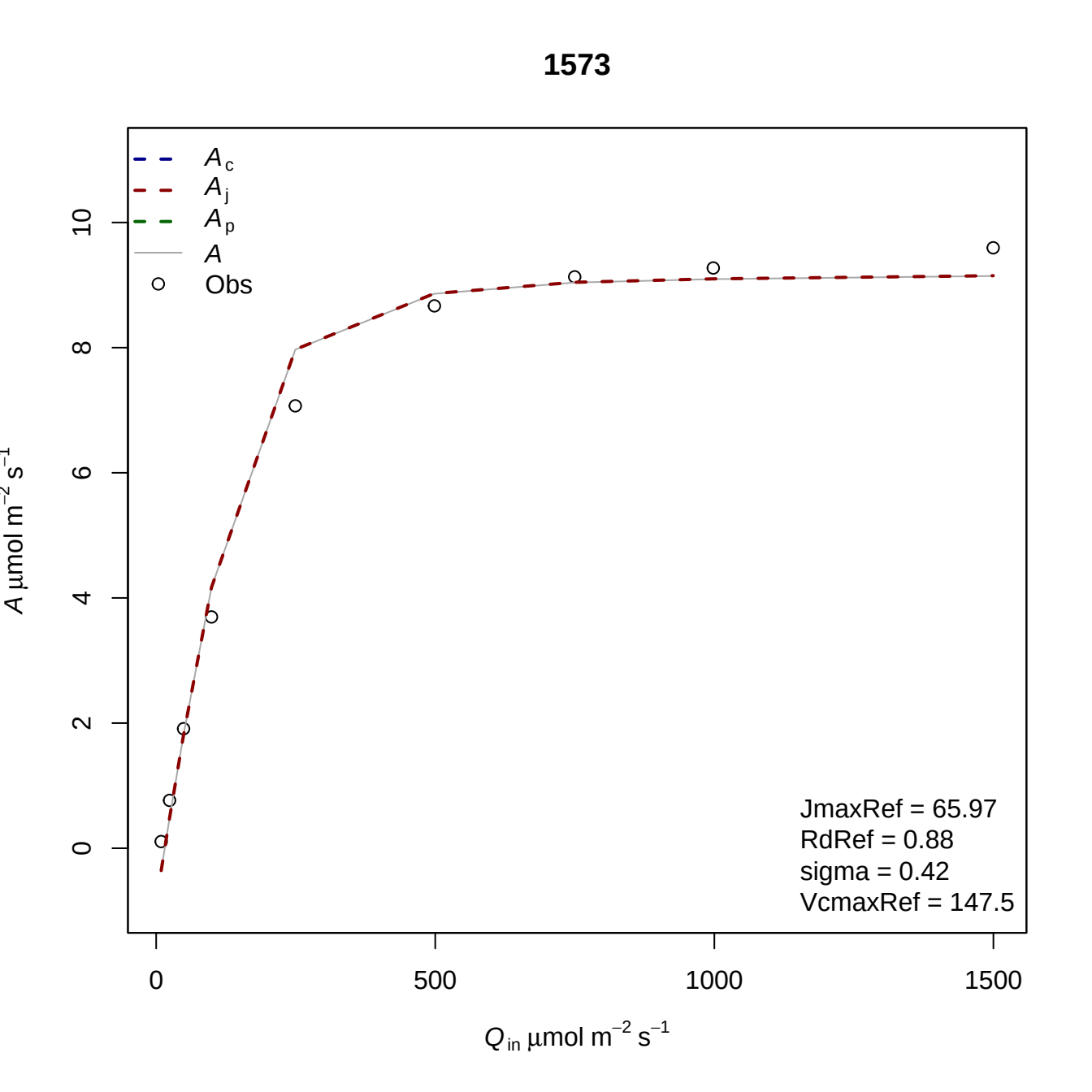

1568
A_c
A_j
A_p
A
Obs
JmaxRef = 97.16
RdRef = 1.15
sigma = 0.67
VcmaxRef = 187.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

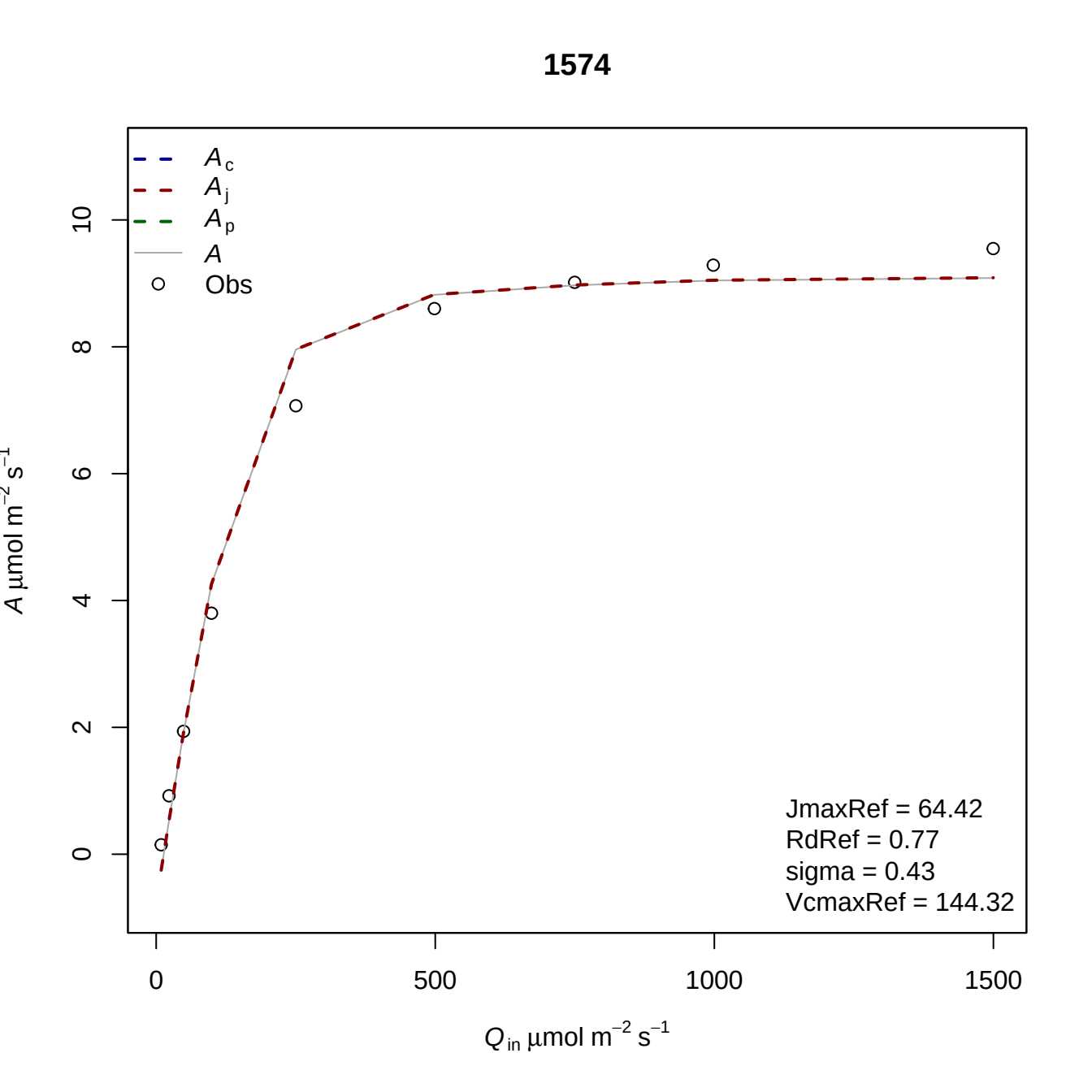

1569
A_c
A_j
A_p
A
Obs
JmaxRef = 66.3
RdRef = 0.81
sigma = 0.41
VcmaxRef = 166.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

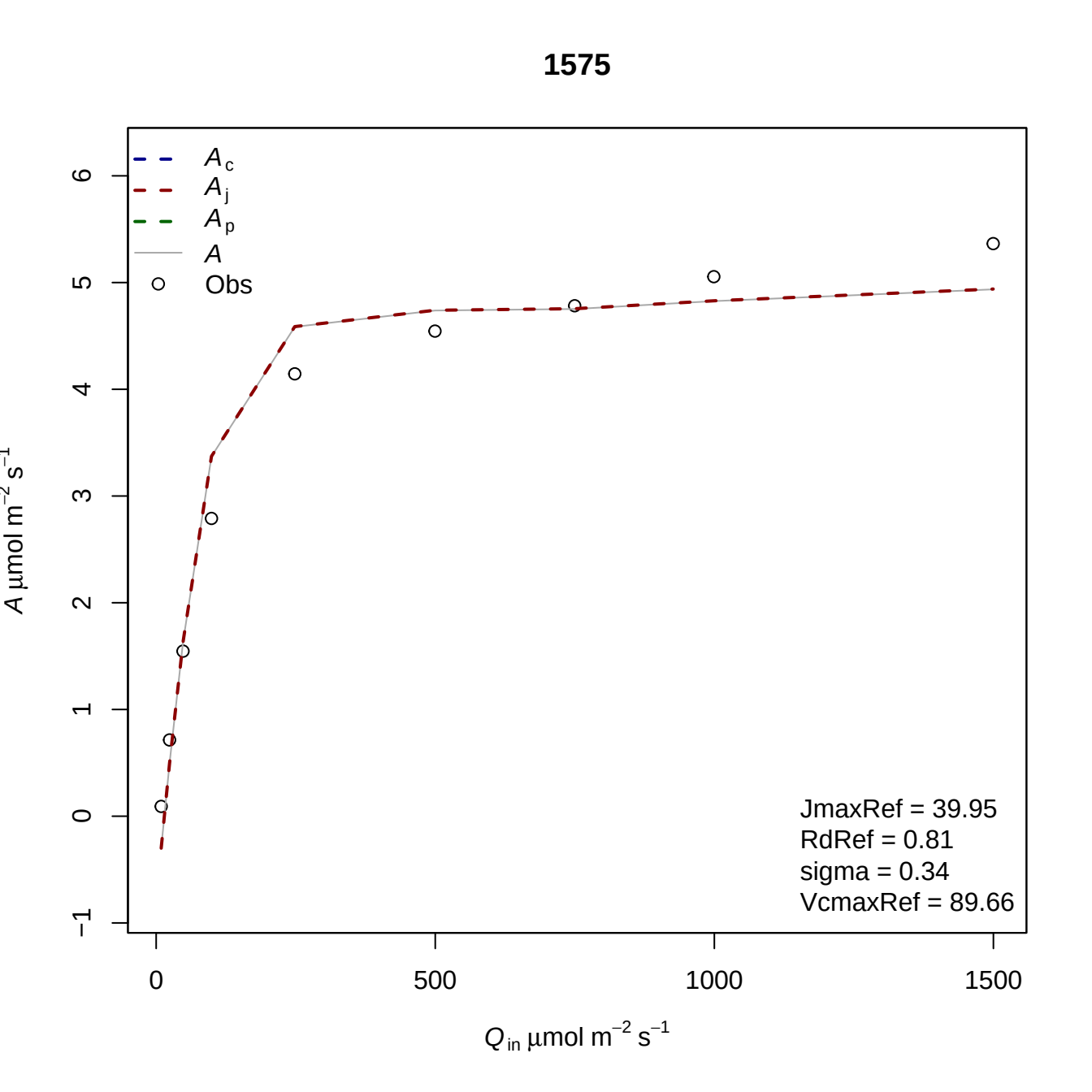

1570
A_c
A_j
A_p
A
Obs
JmaxRef = 80.27
RdRef = 0.7
sigma = 0.44
VcmaxRef = 221.7
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

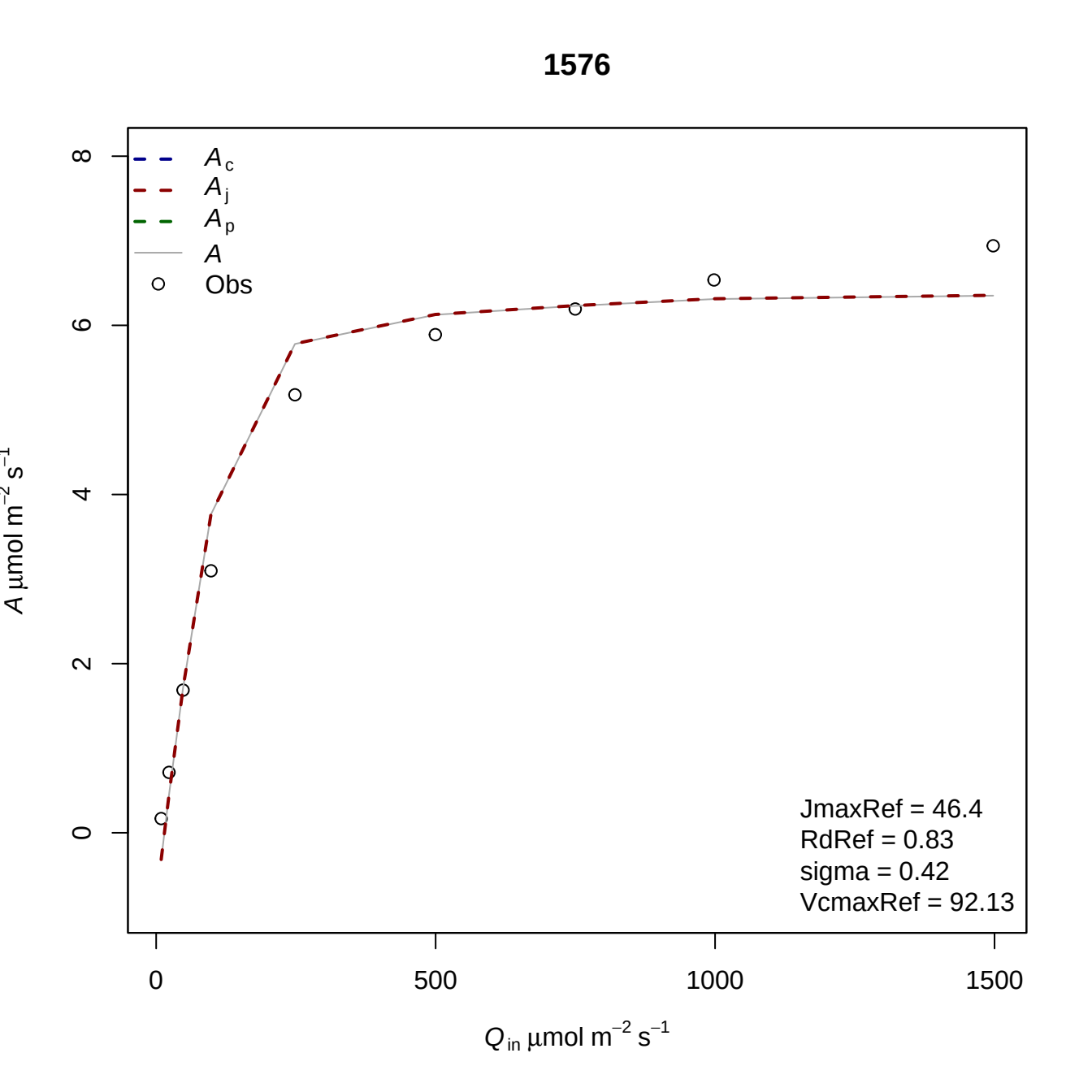

1571
A_c
A_j
A_p
A
Obs
JmaxRef = 104.29
RdRef = −0.04
sigma = 0.28
VcmaxRef = 1563983.38
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

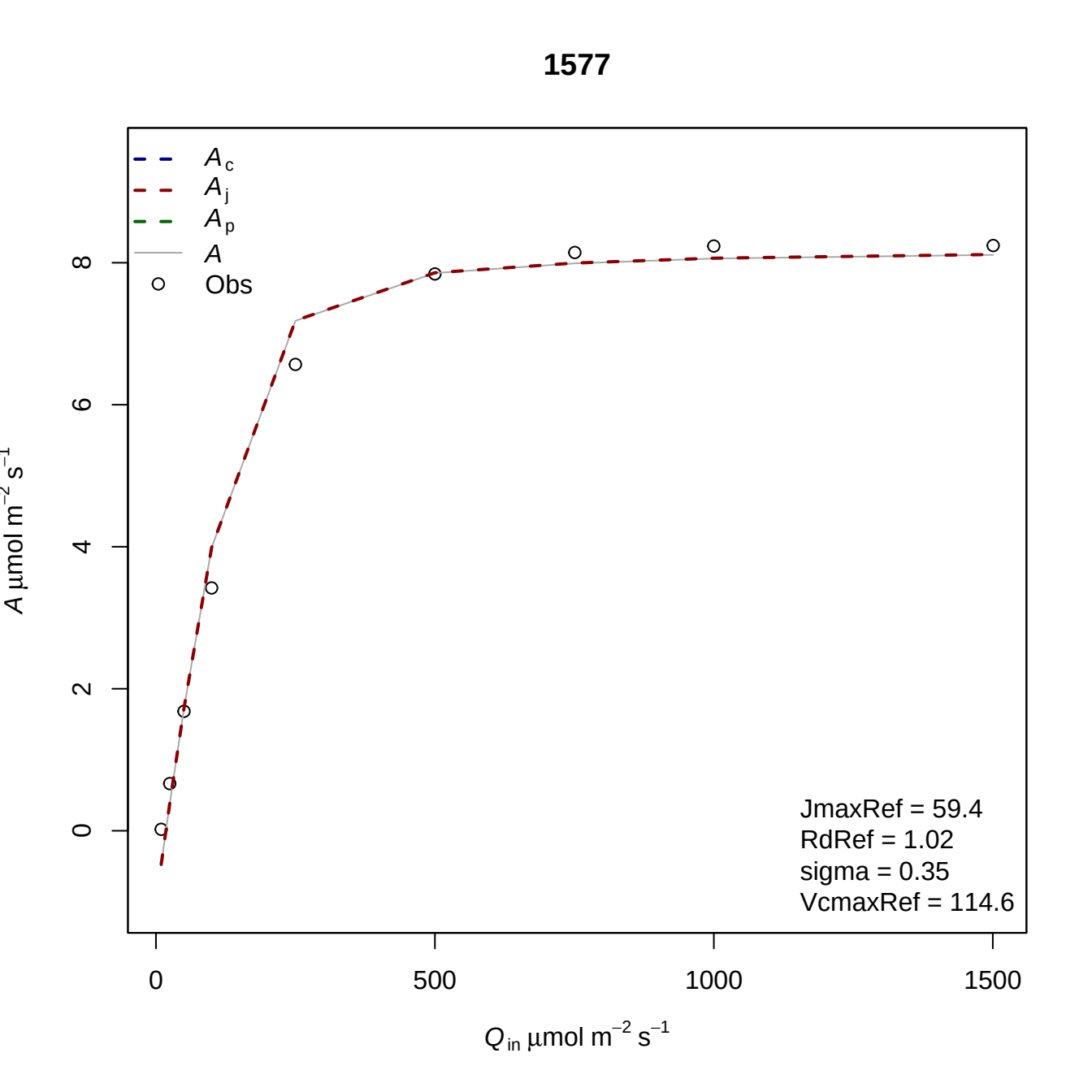

1572
A_c
A_j
A_p
A
Obs
JmaxRef = 82.74
RdRef = 0.68
sigma = 0.32
VcmaxRef = 219.34
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

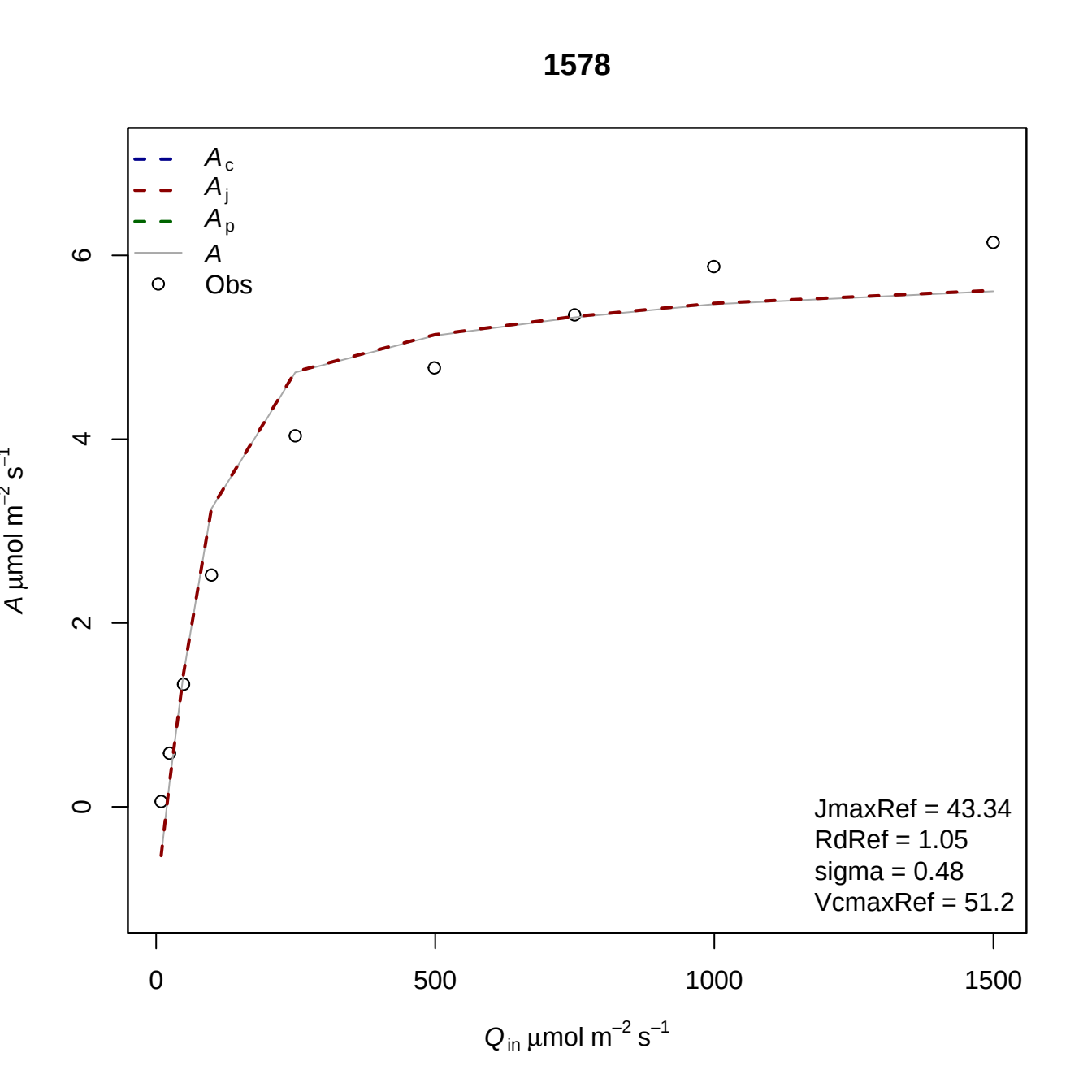

1573
A_c
A_j
A_p
A
Obs
JmaxRef = 65.97
RdRef = 0.88
sigma = 0.42
VcmaxRef = 147.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

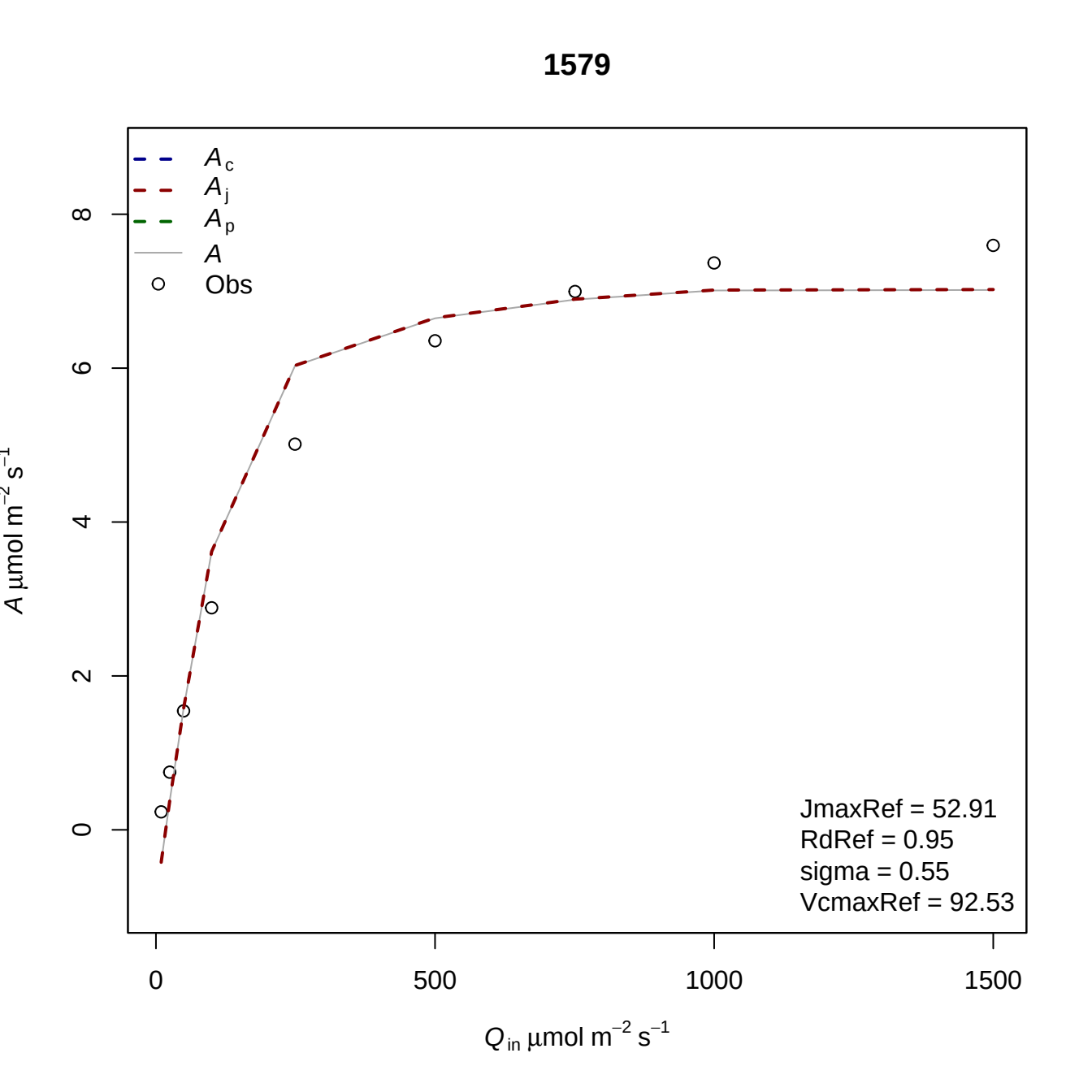

1574
A_c
A_j
A_p
A
Obs
JmaxRef = 64.42
RdRef = 0.77
sigma = 0.43
VcmaxRef = 144.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

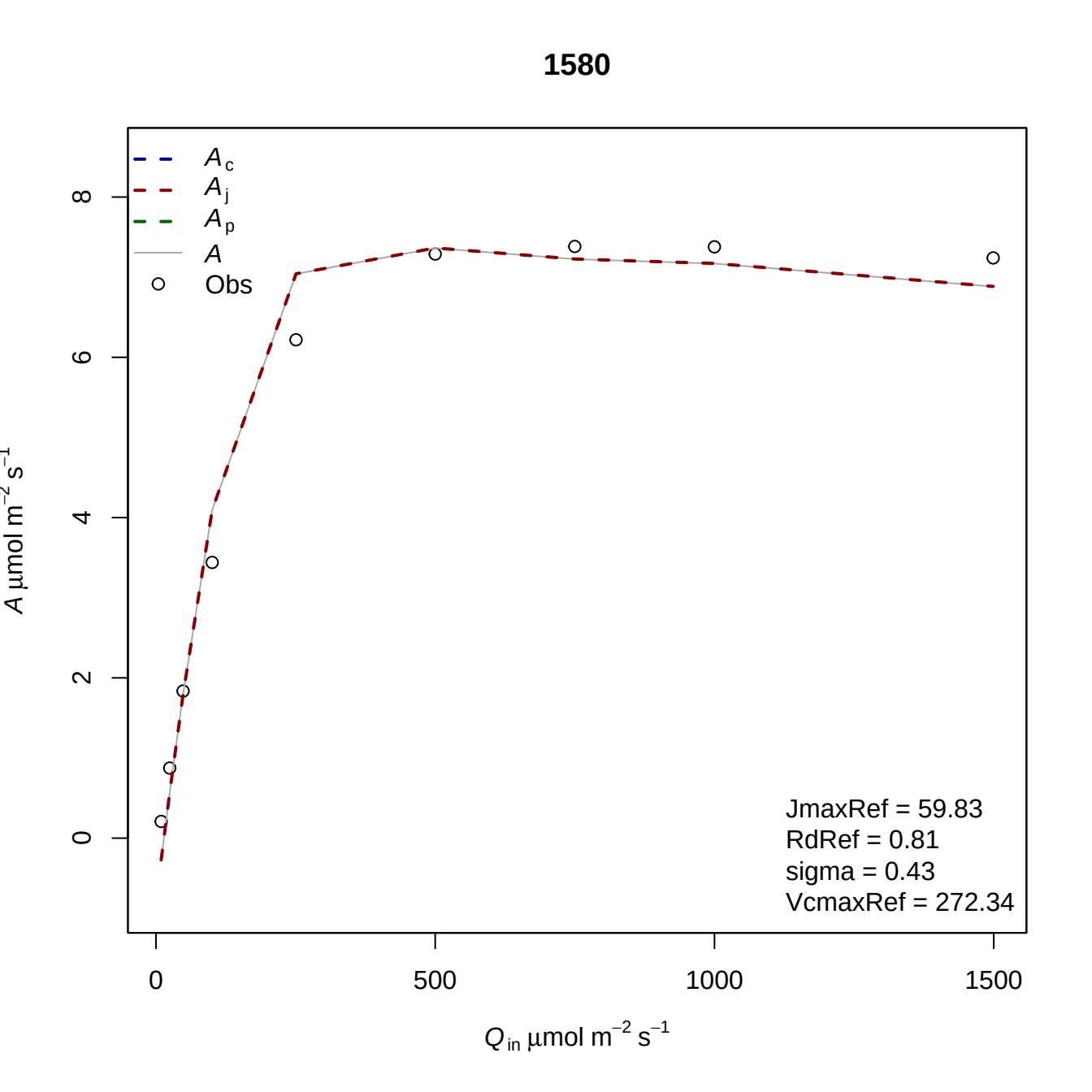

1575
A_c
A_j
A_p
A
Obs
JmaxRef = 39.95
RdRef = 0.81
sigma = 0.34
VcmaxRef = 89.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

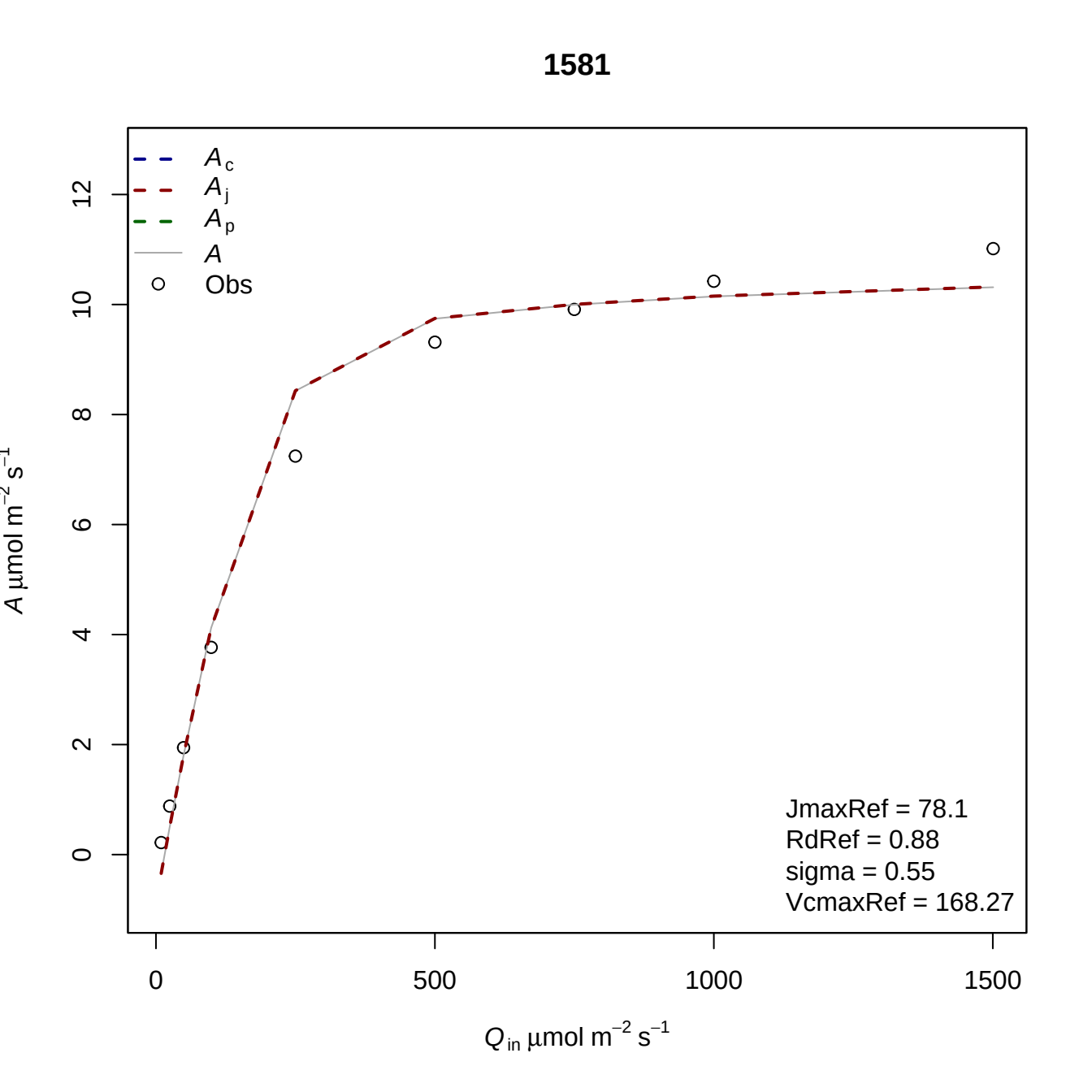

1576
A_c
A_j
A_p
A
Obs
JmaxRef = 46.4
RdRef = 0.83
sigma = 0.42
VcmaxRef = 92.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

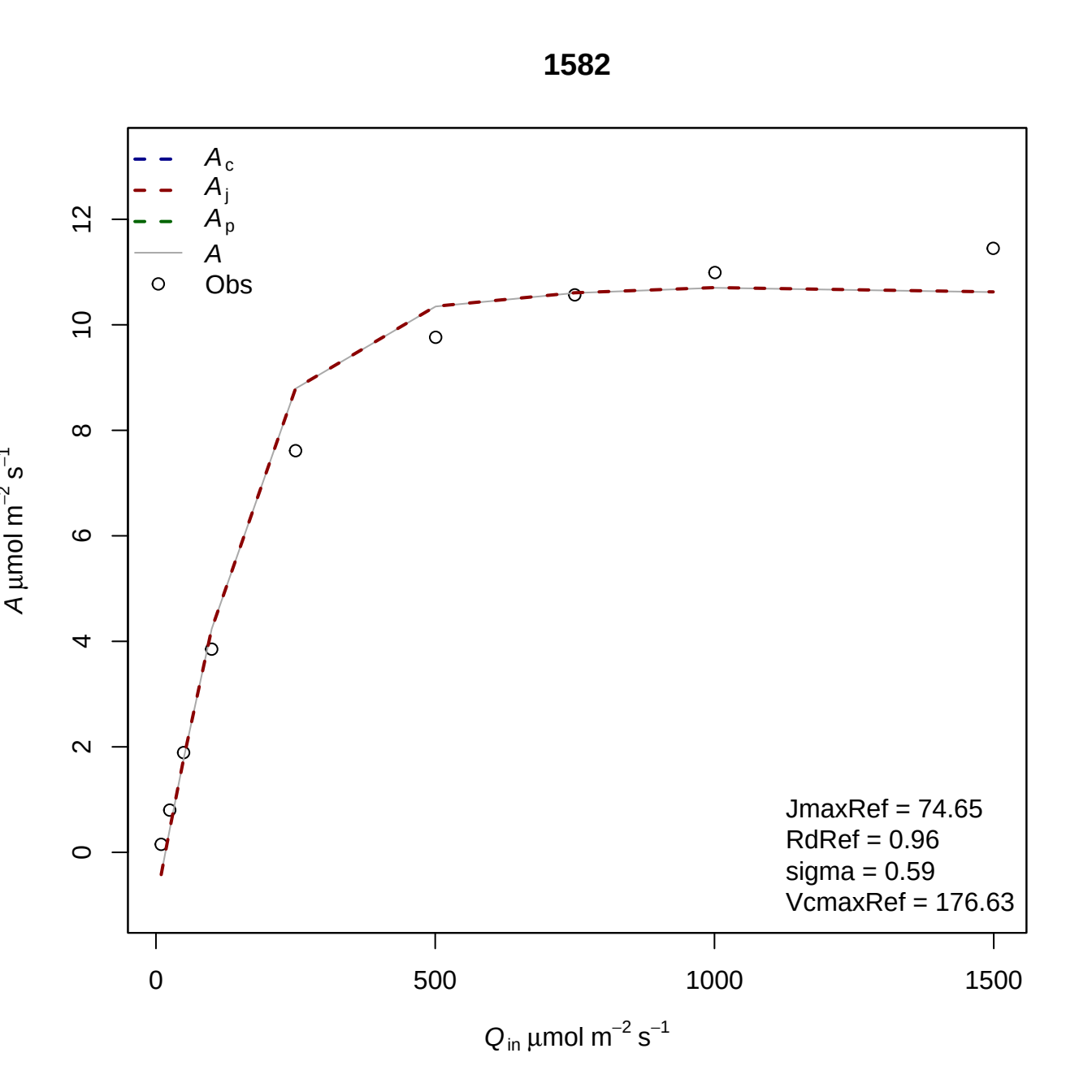

1577
A_c
A_j
A_p
A
Obs
JmaxRef = 59.4
RdRef = 1.02
sigma = 0.35
VcmaxRef = 114.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

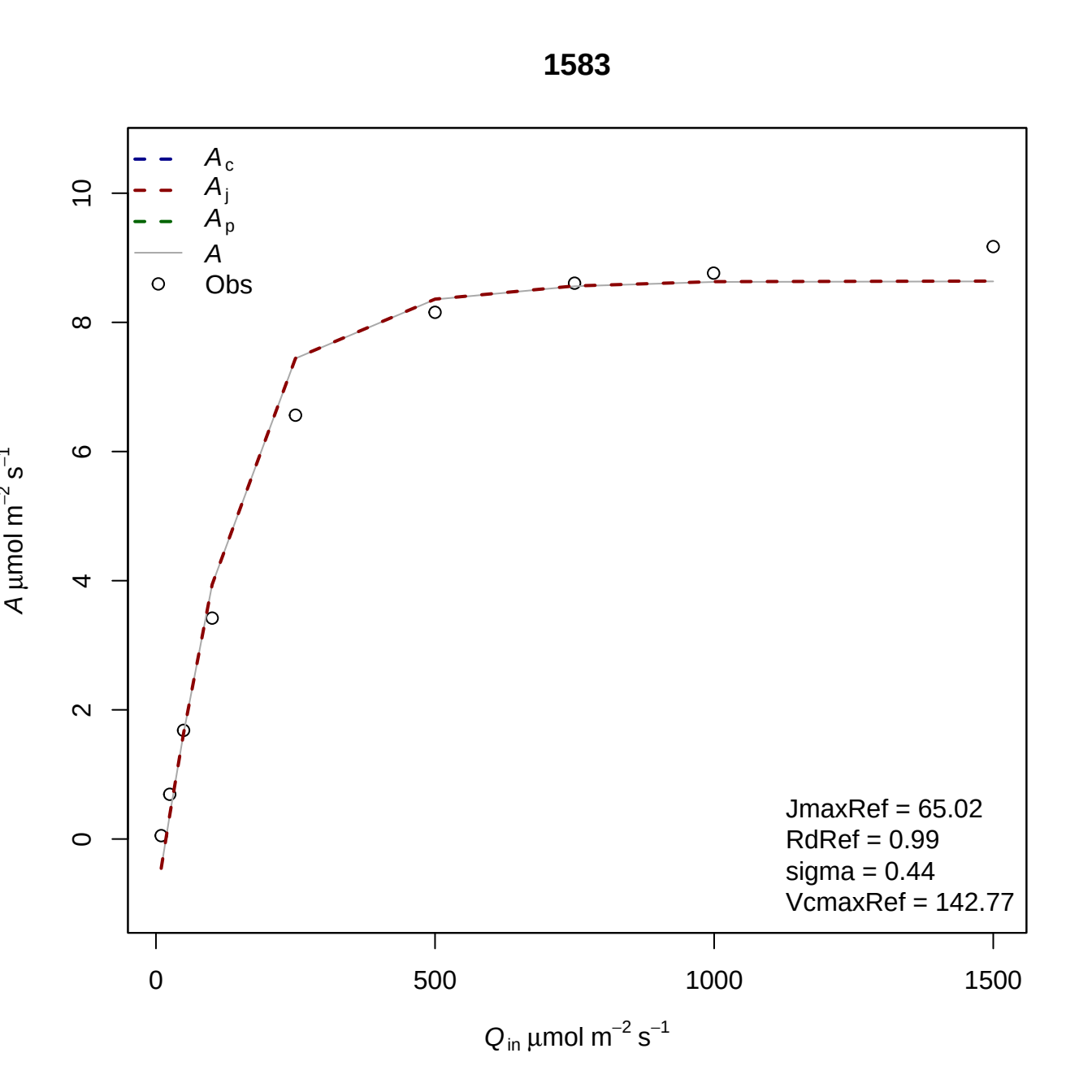

1578
A_c
A_j
A_p
A
Obs
JmaxRef = 43.34
RdRef = 1.05
sigma = 0.48
VcmaxRef = 51.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

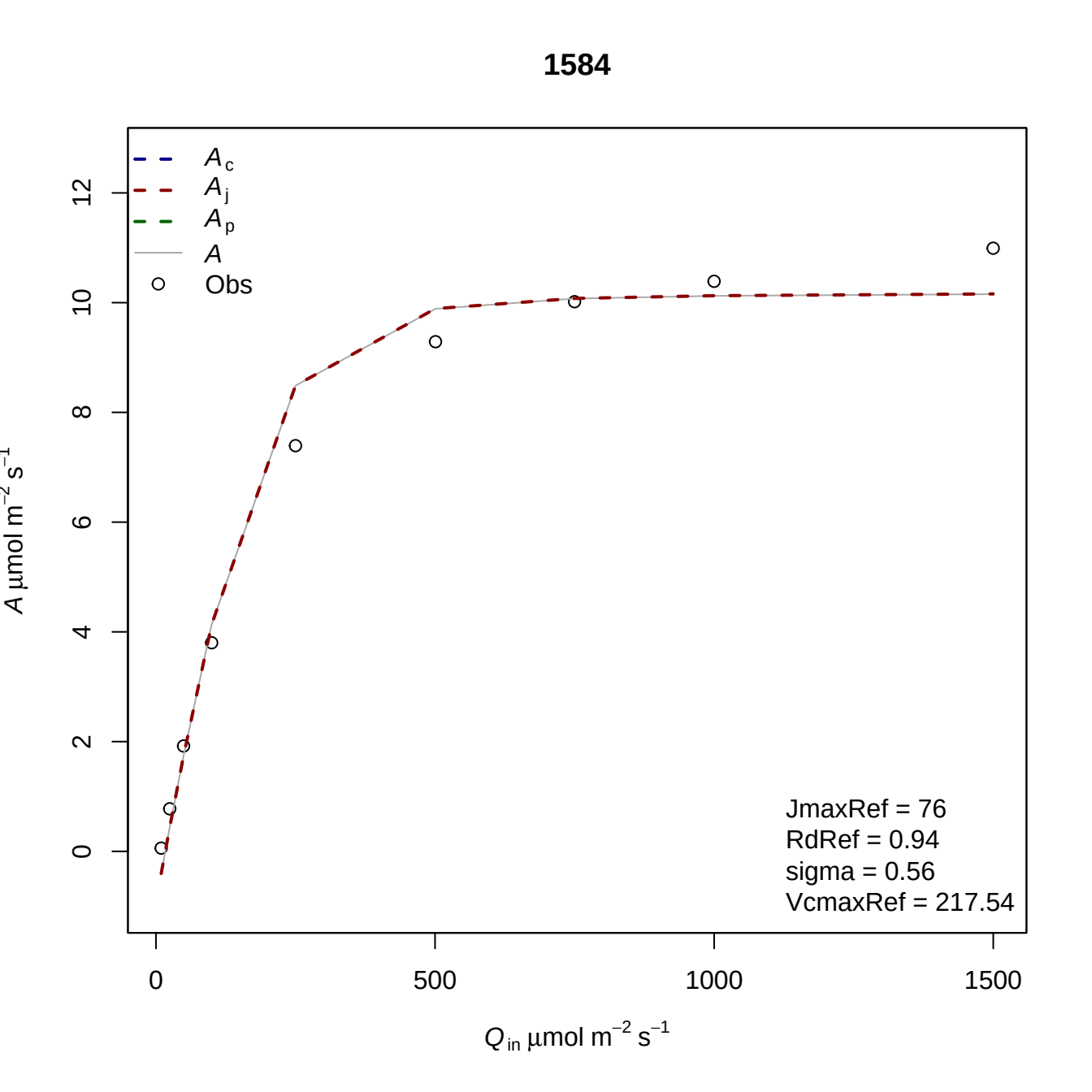

1579
A_c
A_j
A_p
A
Obs
JmaxRef = 52.91
RdRef = 0.95
sigma = 0.55
VcmaxRef = 92.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

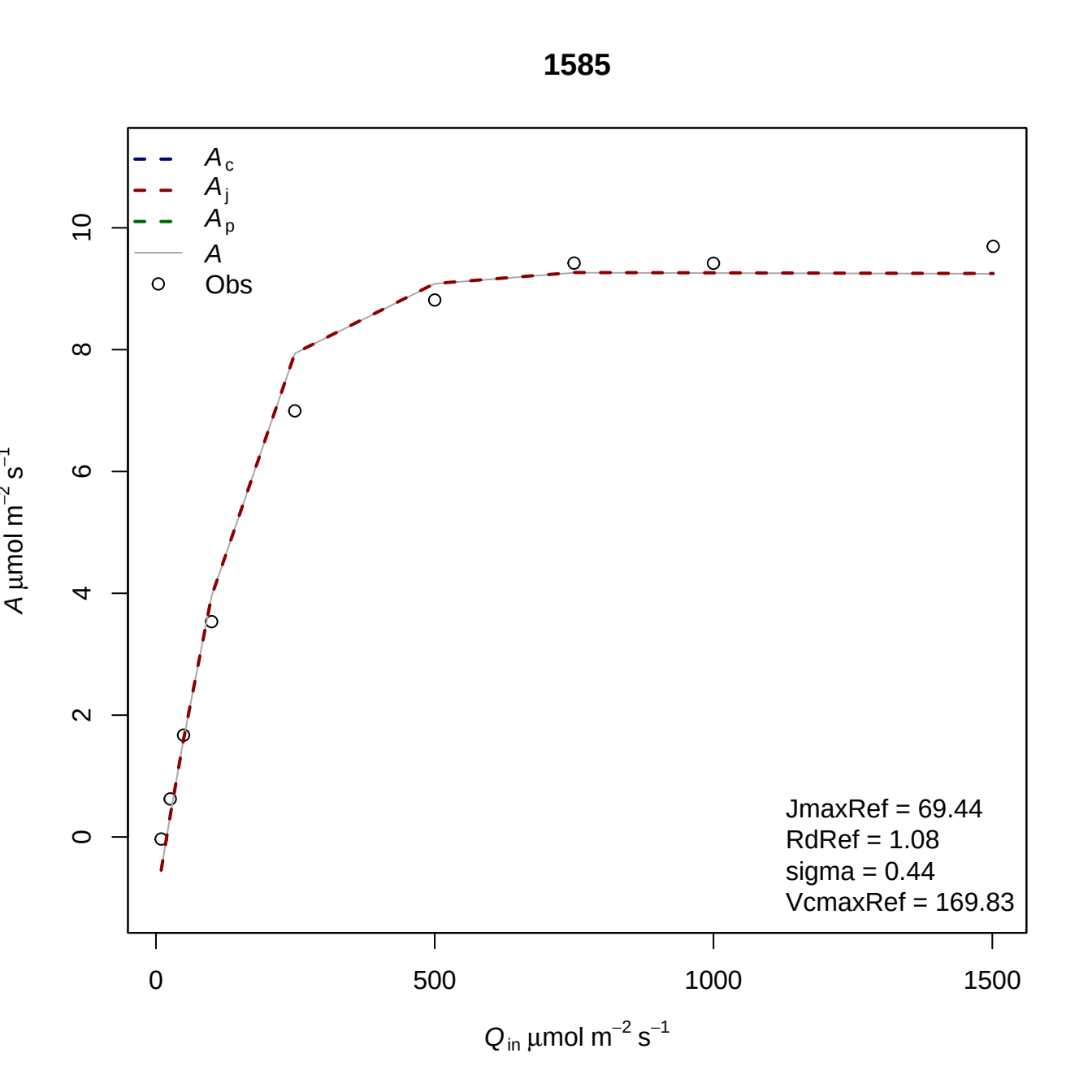

1580
A_c
A_j
A_p
A
Obs
JmaxRef = 59.83
RdRef = 0.81
sigma = 0.43
VcmaxRef = 272.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

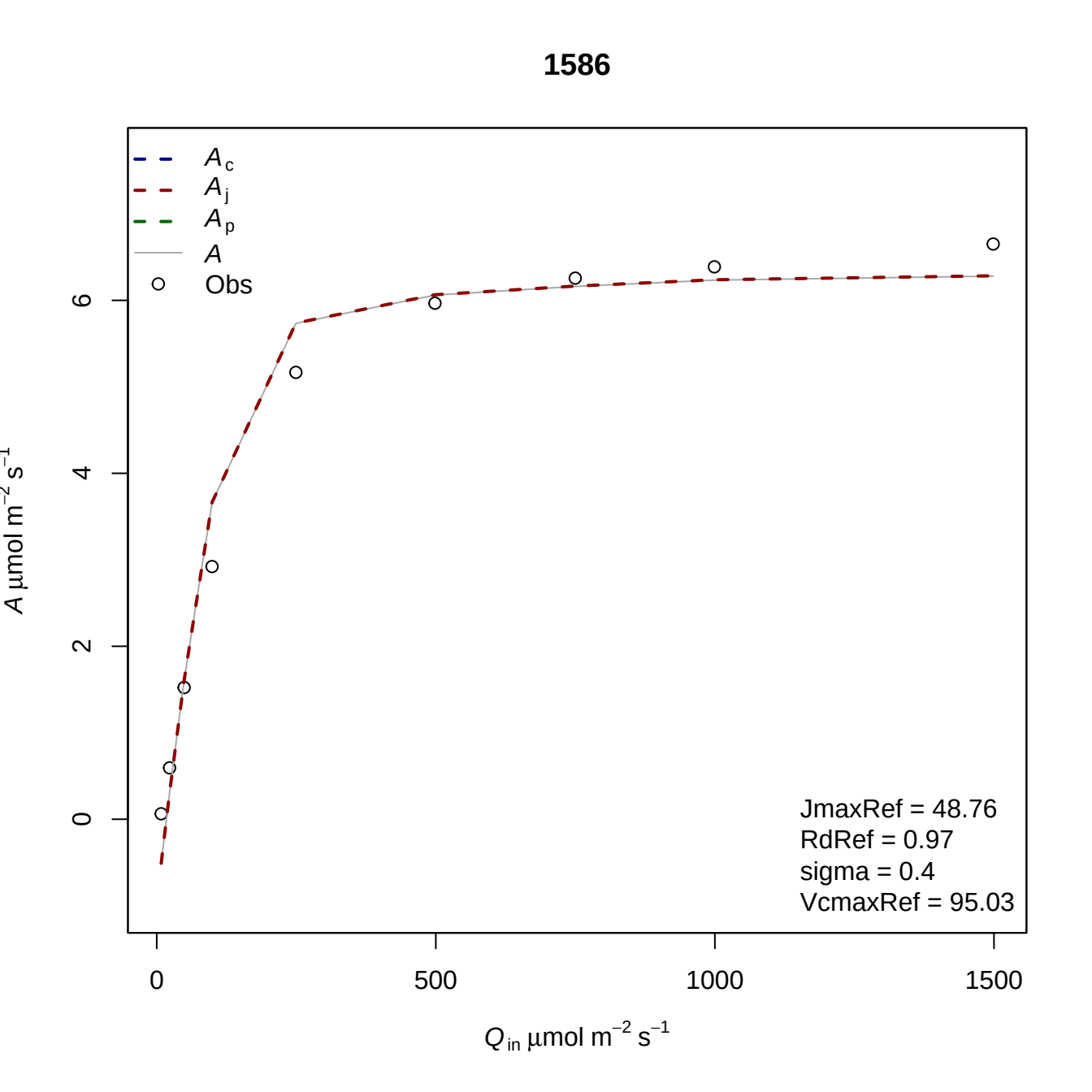

1581
A_c
A_j
A_p
A
Obs
JmaxRef = 78.1
RdRef = 0.88
sigma = 0.55
VcmaxRef = 168.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

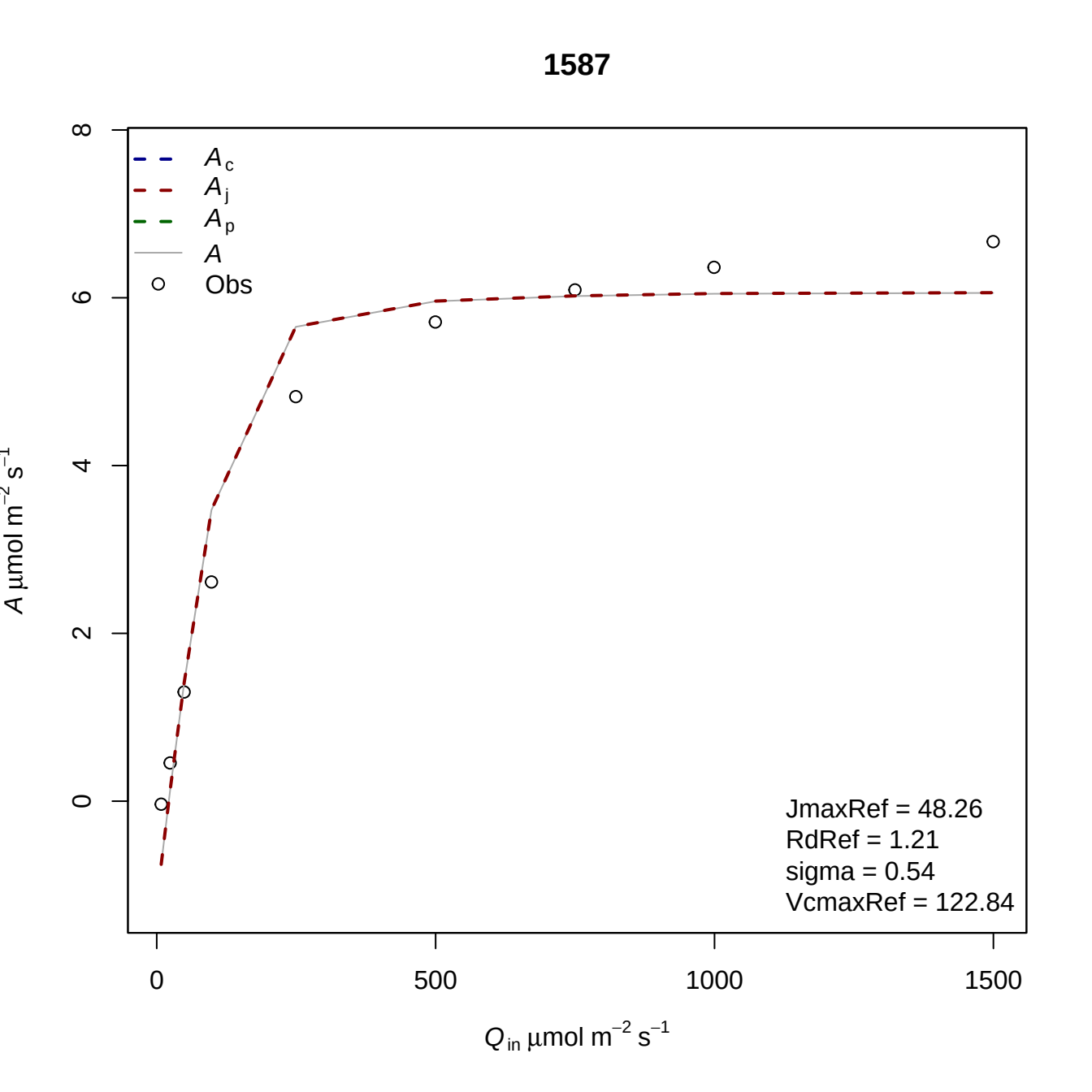

1582
A_c
A_j
A_p
A
Obs
JmaxRef = 74.65
RdRef = 0.96
sigma = 0.59
VcmaxRef = 176.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

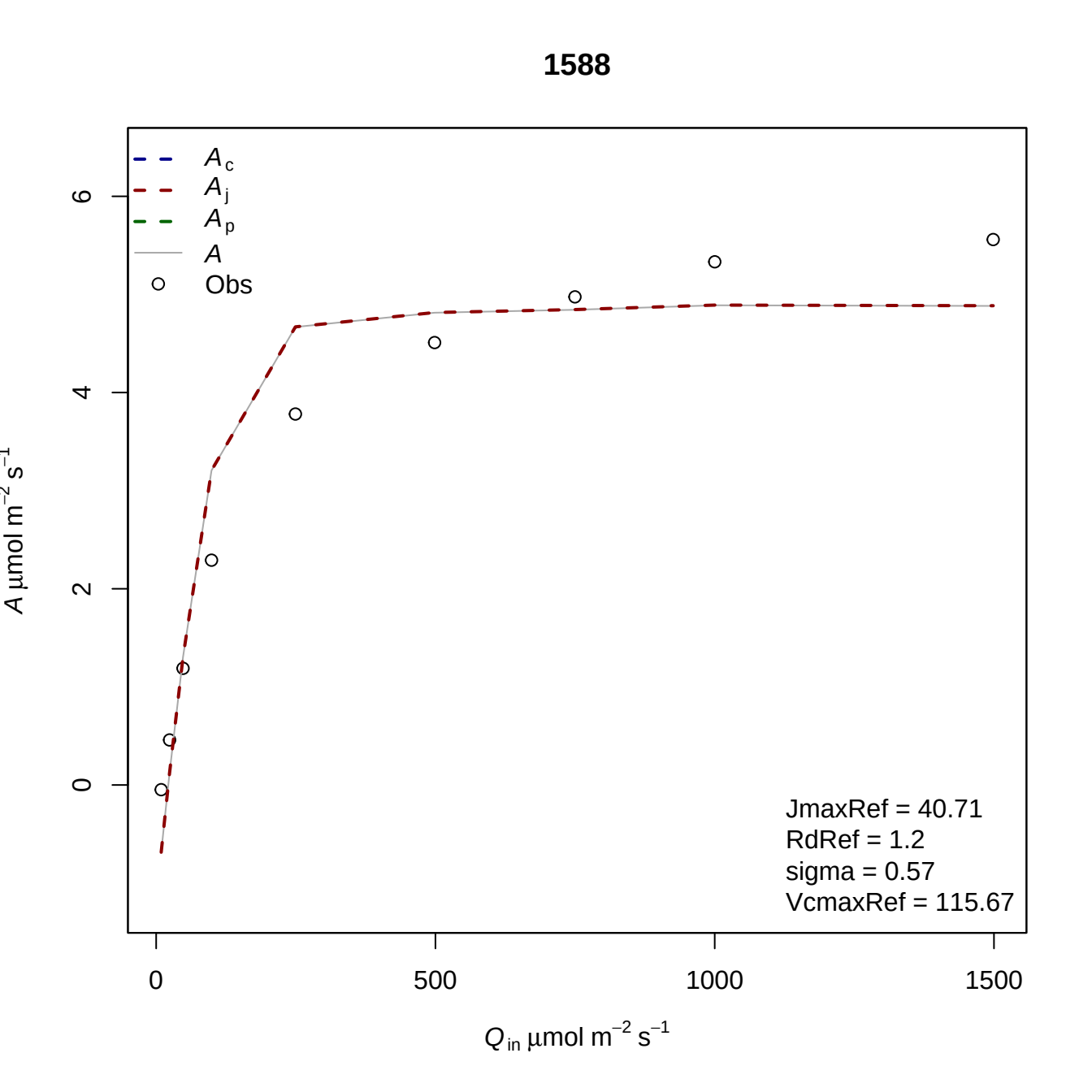

1583
A_c
A_j
A_p
A
Obs
JmaxRef = 65.02
RdRef = 0.99
sigma = 0.44
VcmaxRef = 142.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

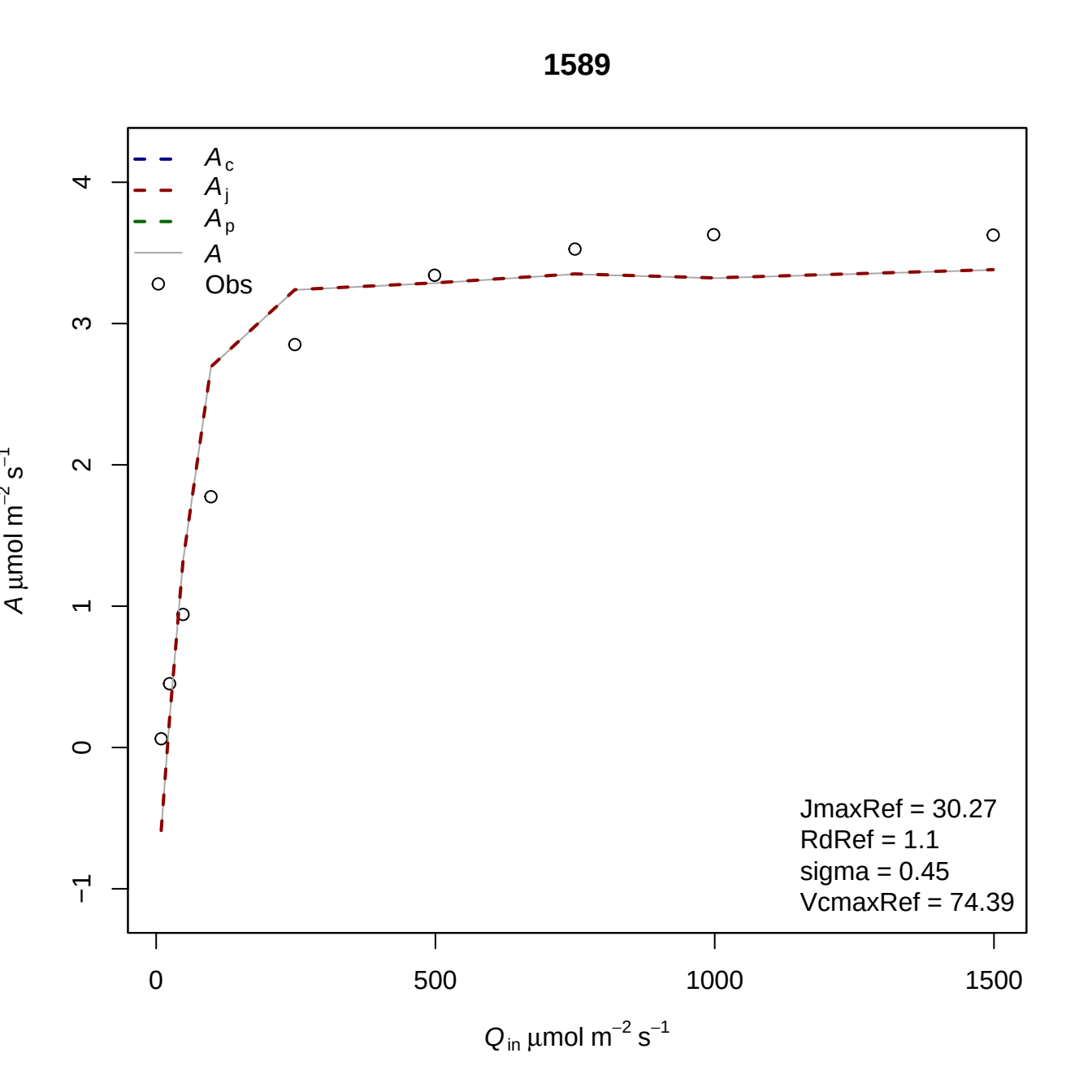

1584
A_c
A_j
A_p
A
Obs
JmaxRef = 76
RdRef = 0.94
sigma = 0.56
VcmaxRef = 217.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

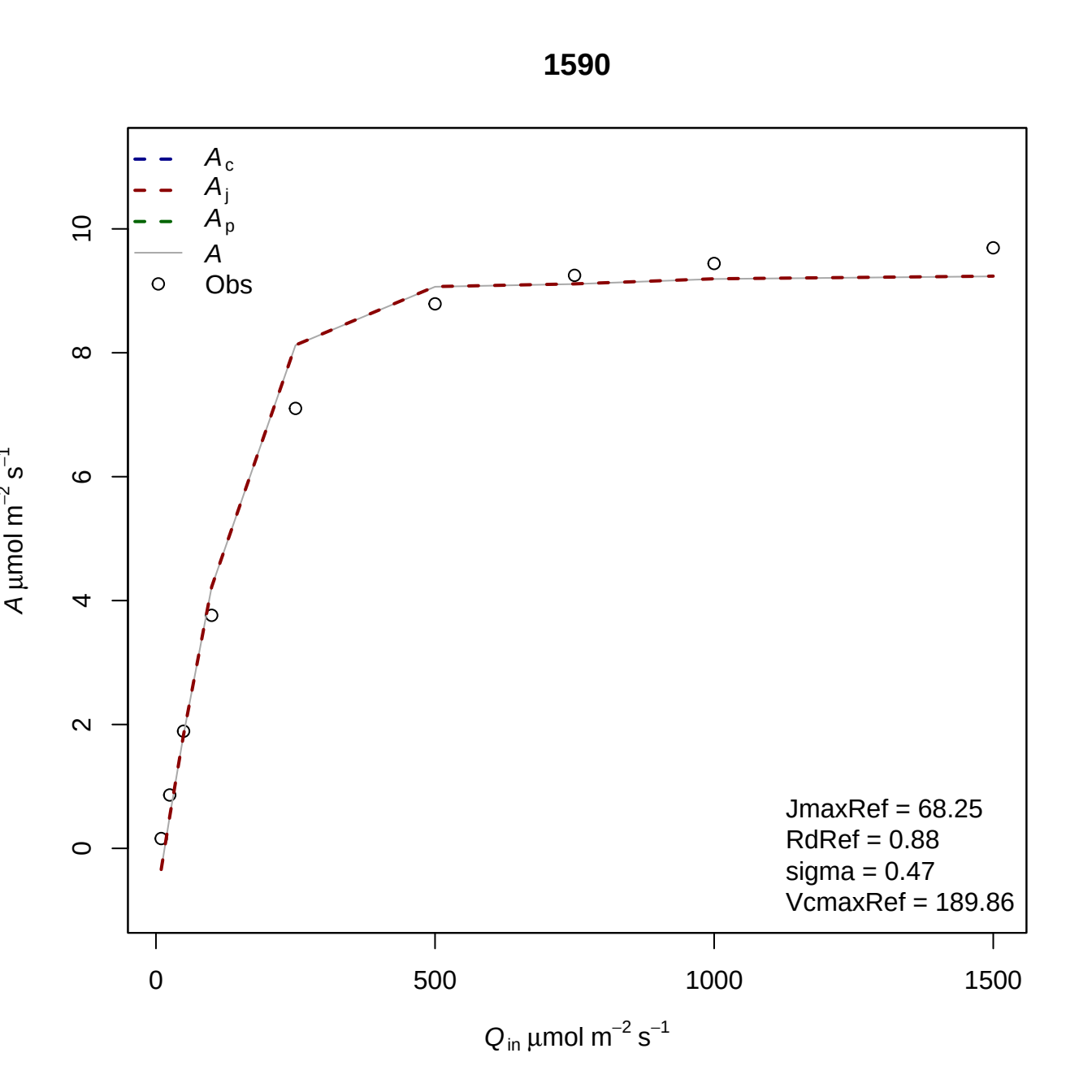

1585
A_c
A_j
A_p
A
Obs
JmaxRef = 69.44
RdRef = 1.08
sigma = 0.44
VcmaxRef = 169.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

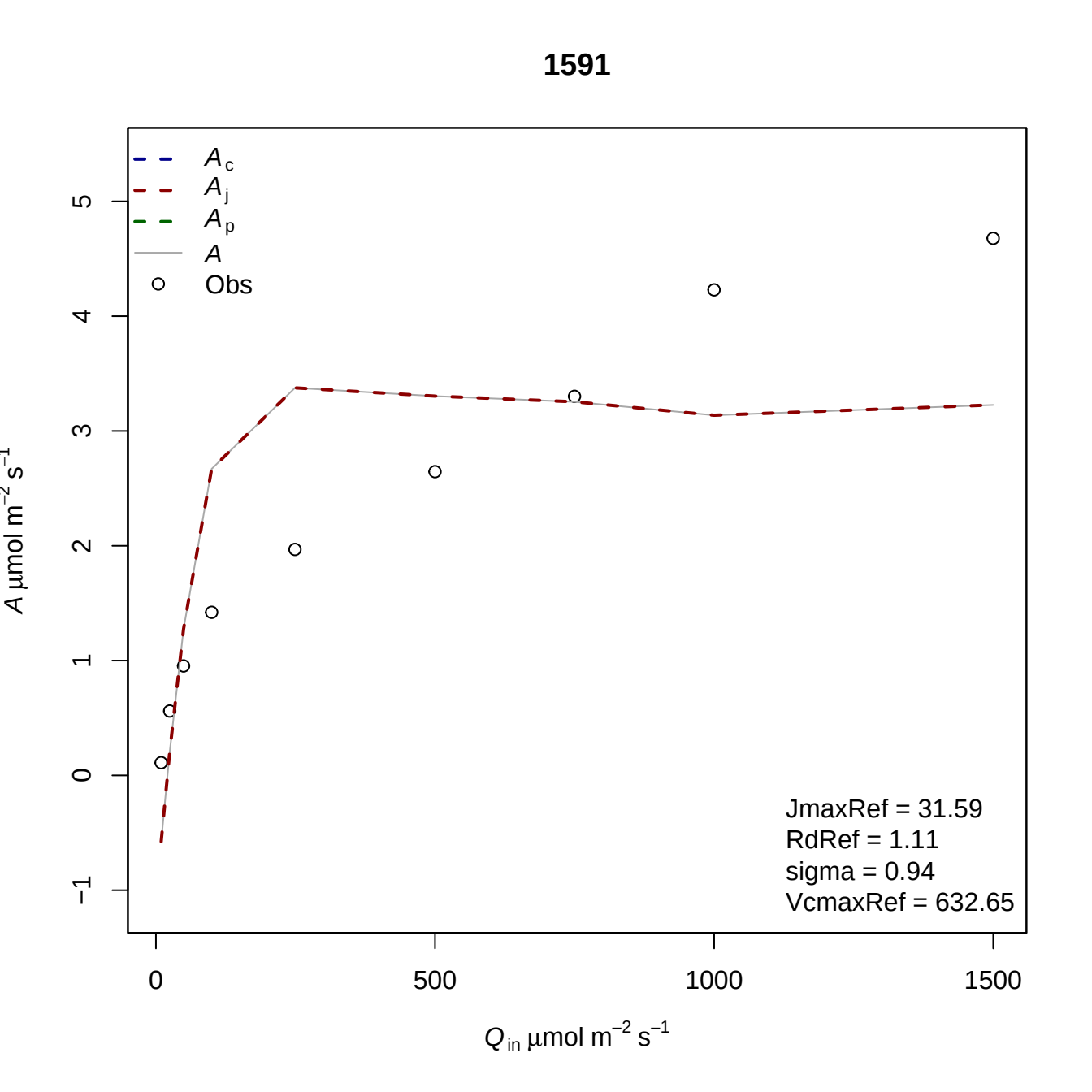

1586
A_c
A_j
A_p
A
Obs
JmaxRef = 48.76
RdRef = 0.97
sigma = 0.4
VcmaxRef = 95.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

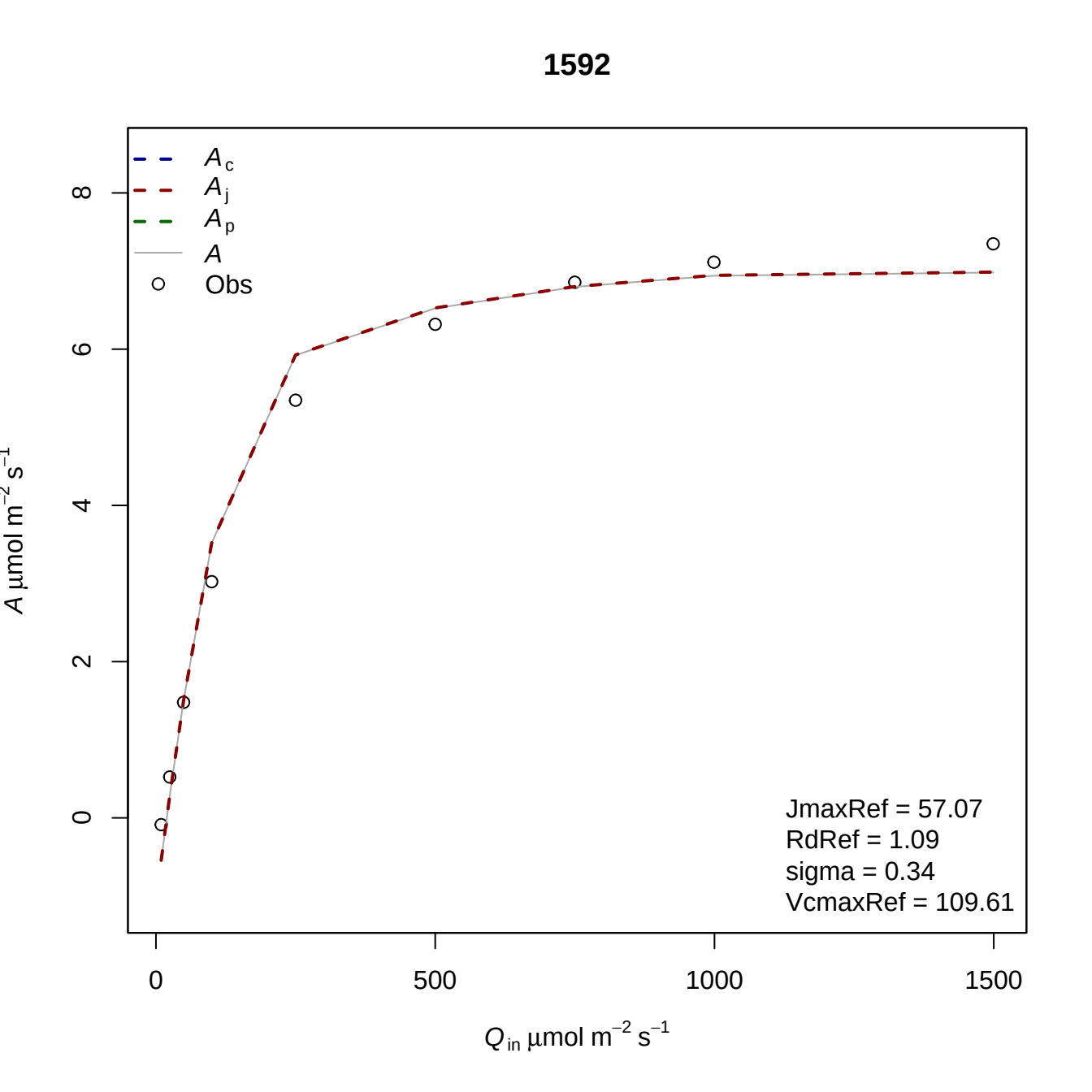

1587
A_c
A_j
A_p
A
Obs
JmaxRef = 48.26
RdRef = 1.21
sigma = 0.54
VcmaxRef = 122.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

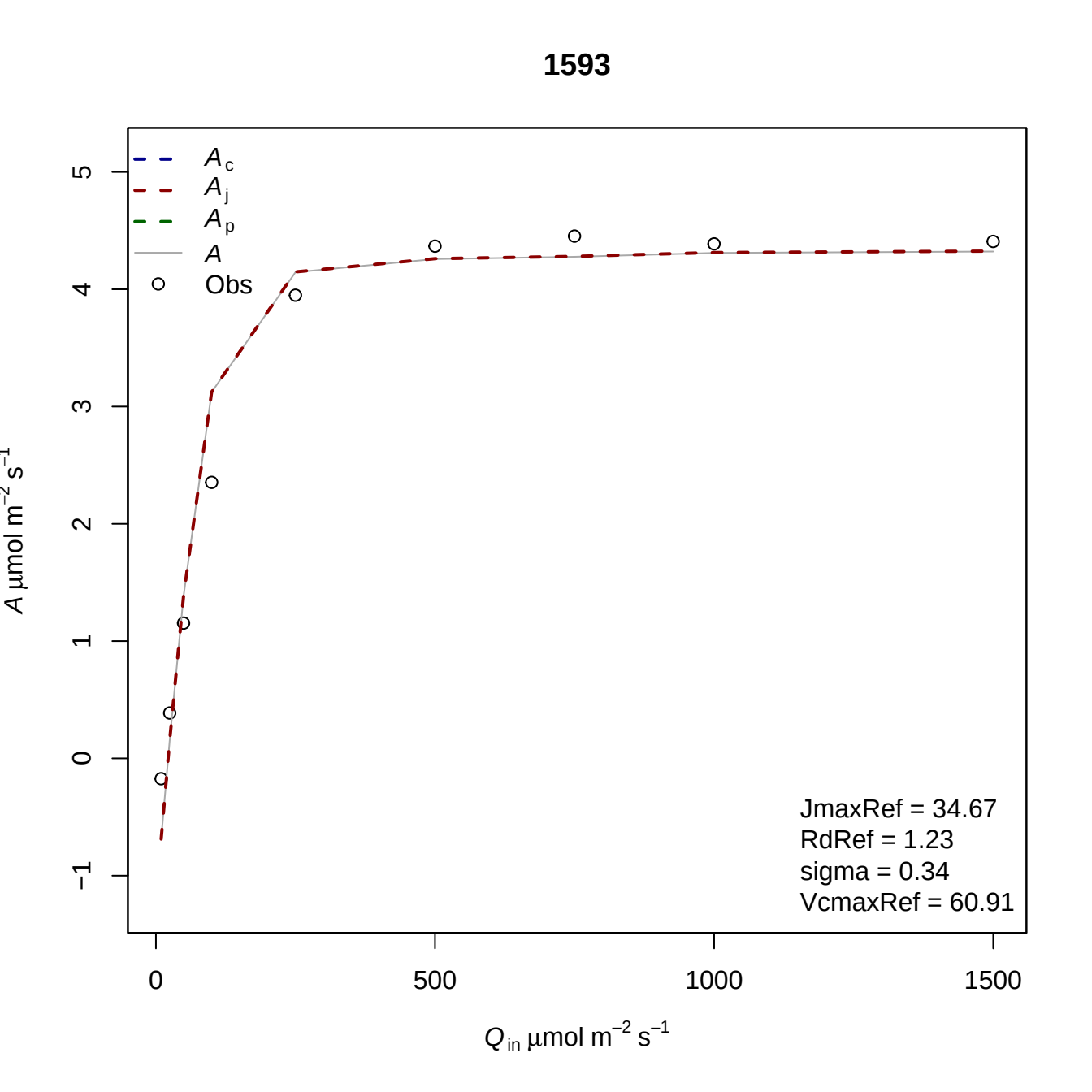

1588
A_c
A_j
A_p
A
Obs
JmaxRef = 40.71
RdRef = 1.2
sigma = 0.57
VcmaxRef = 115.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

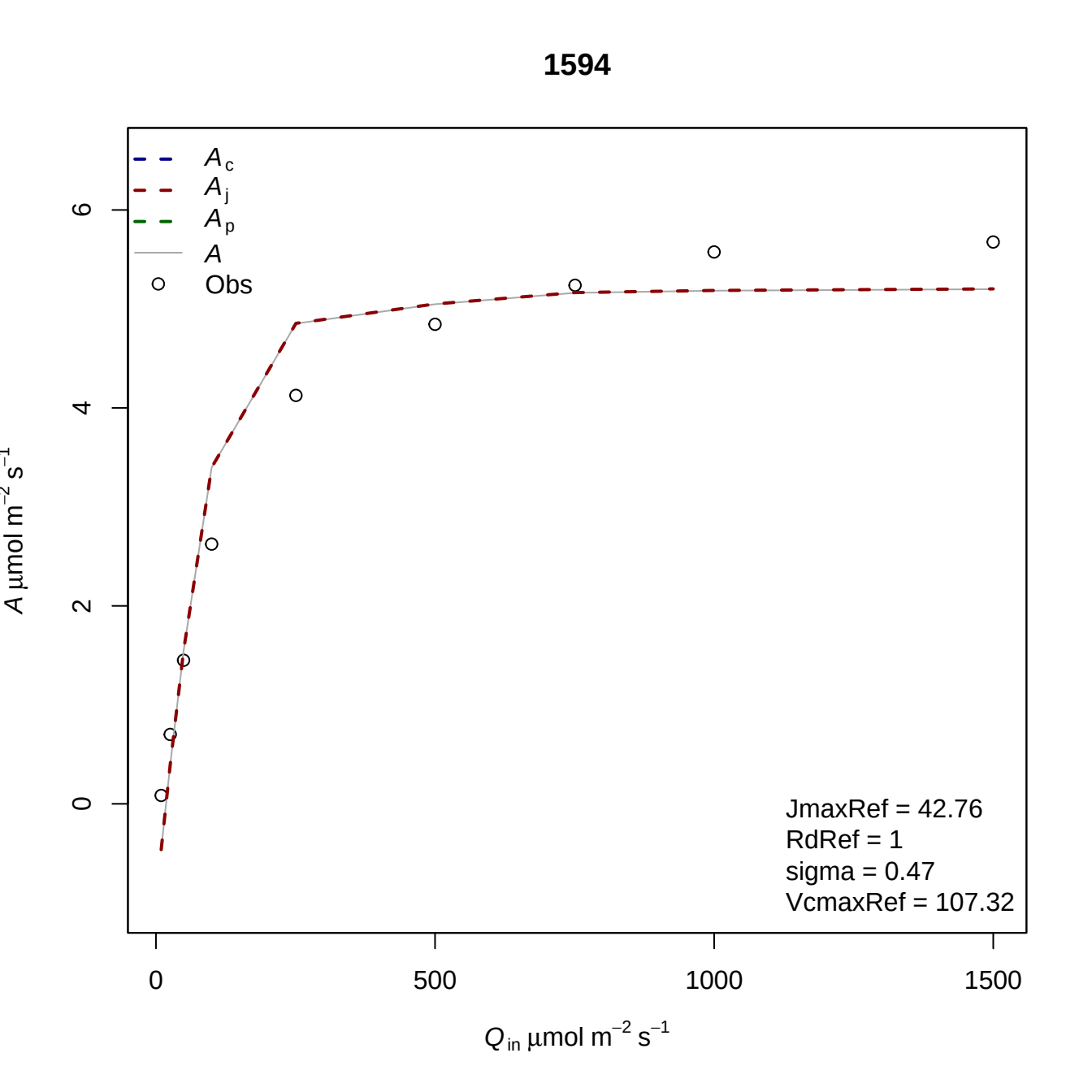

1589
A_c
A_j
A_p
A
Obs
JmaxRef = 30.27
RdRef = 1.1
sigma = 0.45
VcmaxRef = 74.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

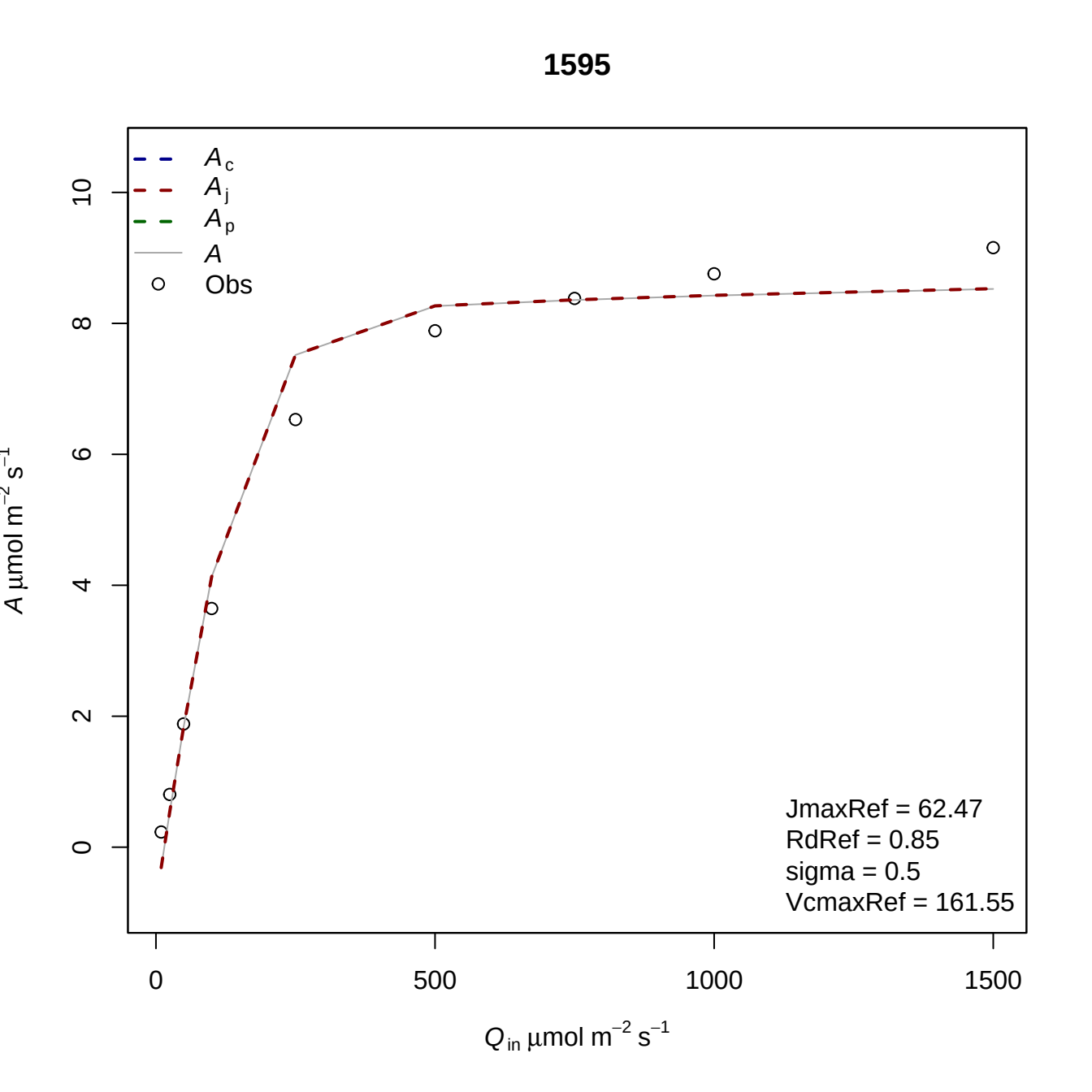

1590
A_c
A_j
A_p
A
Obs
JmaxRef = 68.25
RdRef = 0.88
sigma = 0.47
VcmaxRef = 189.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

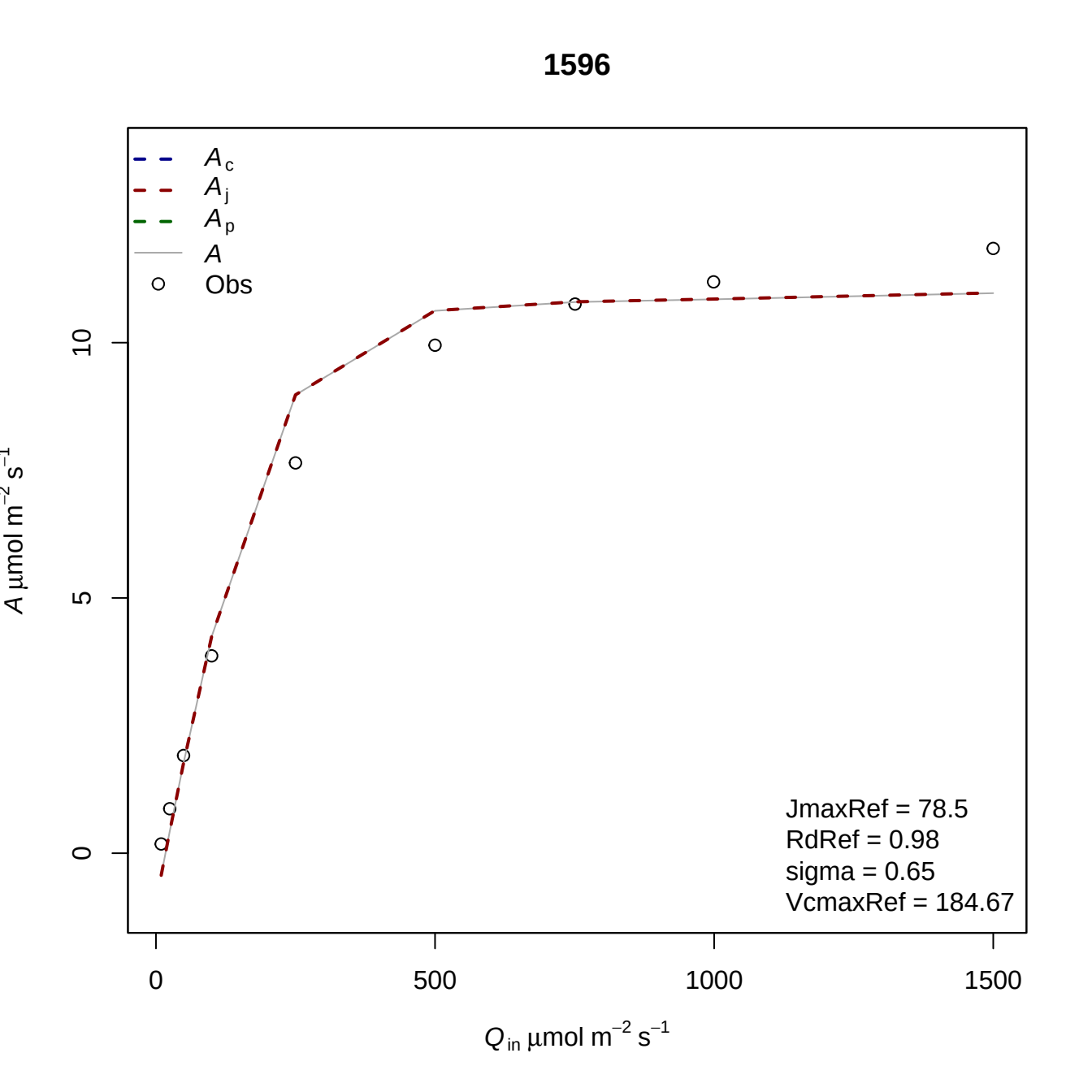

1591
A_c
A_j
A_p
A
Obs
JmaxRef = 31.59
RdRef = 1.11
sigma = 0.94
VcmaxRef = 632.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

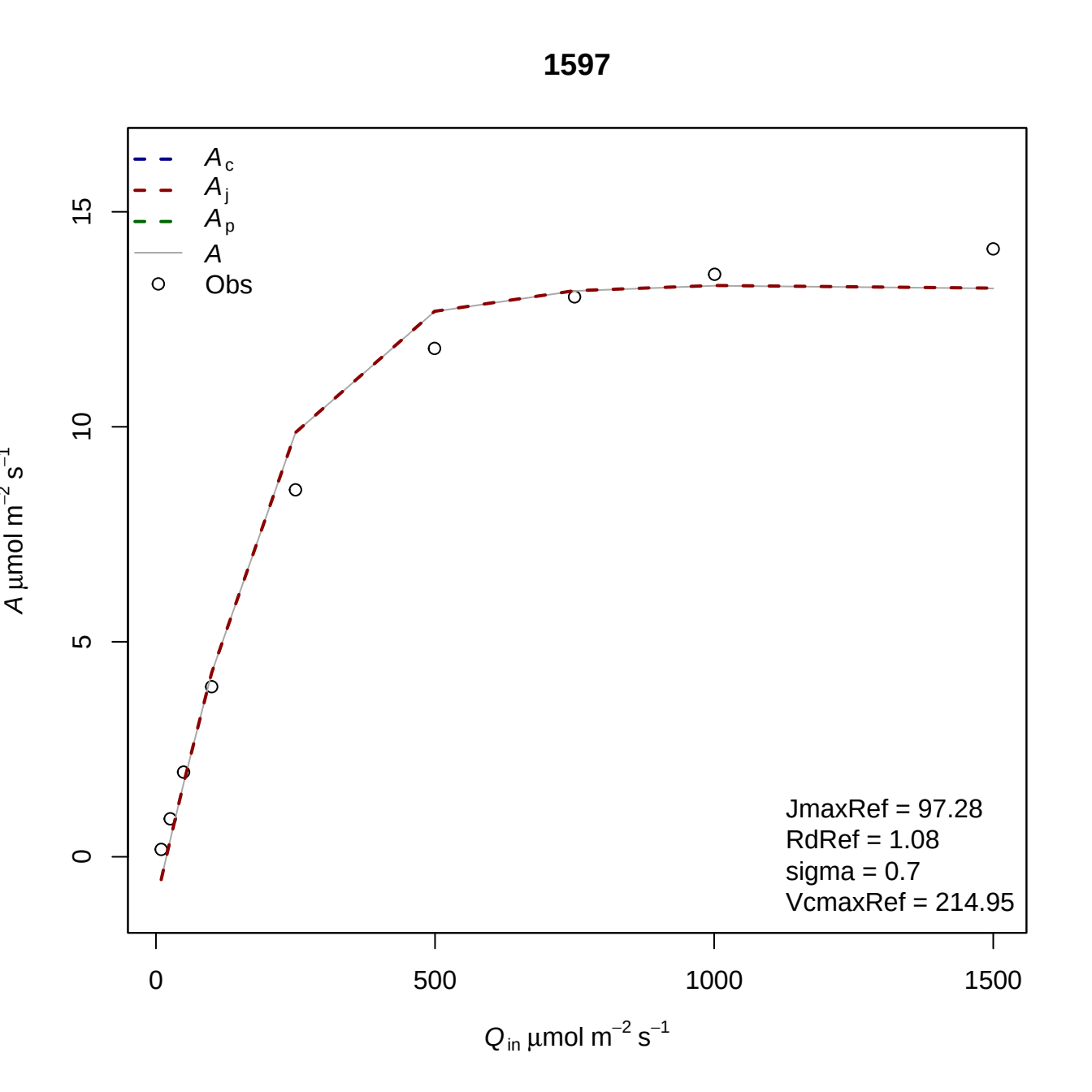

1592
A_c
A_j
A_p
A
Obs
JmaxRef = 57.07
RdRef = 1.09
sigma = 0.34
VcmaxRef = 109.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

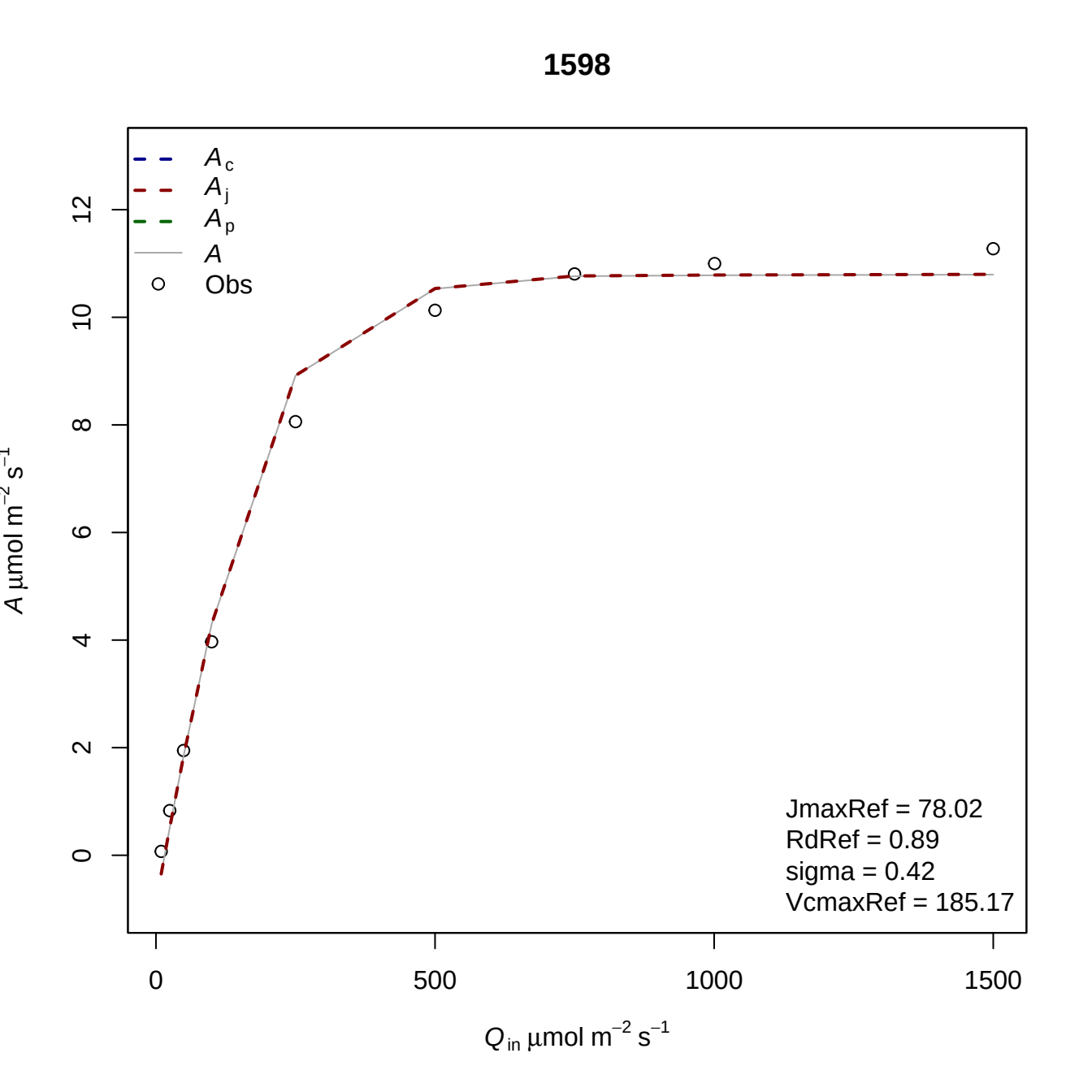

1593
A_c
A_j
A_p
A
Obs
JmaxRef = 34.67
RdRef = 1.23
sigma = 0.34
VcmaxRef = 60.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

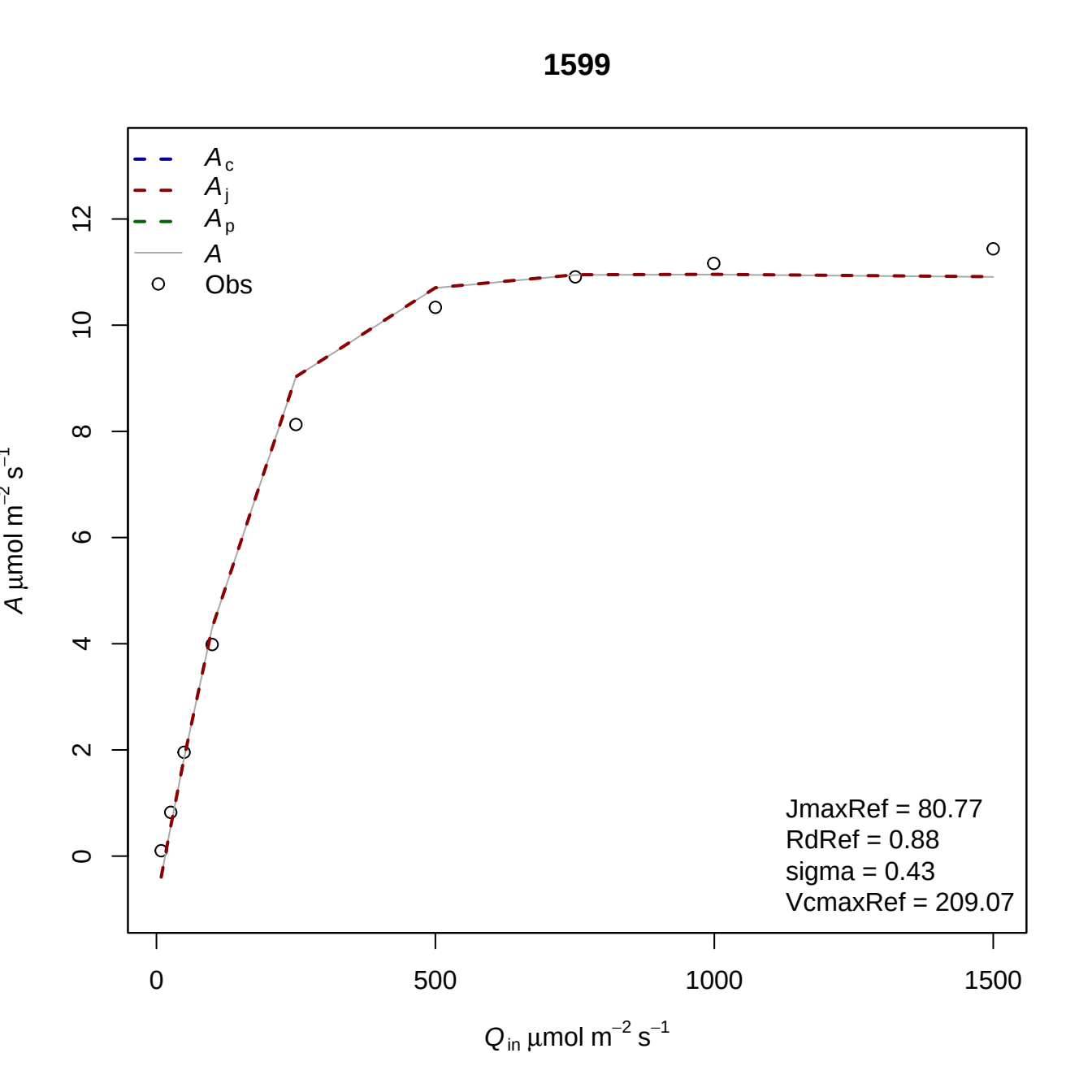

1594
A_c
A_j
A_p
A
Obs
JmaxRef = 42.76
RdRef = 1
sigma = 0.47
VcmaxRef = 107.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

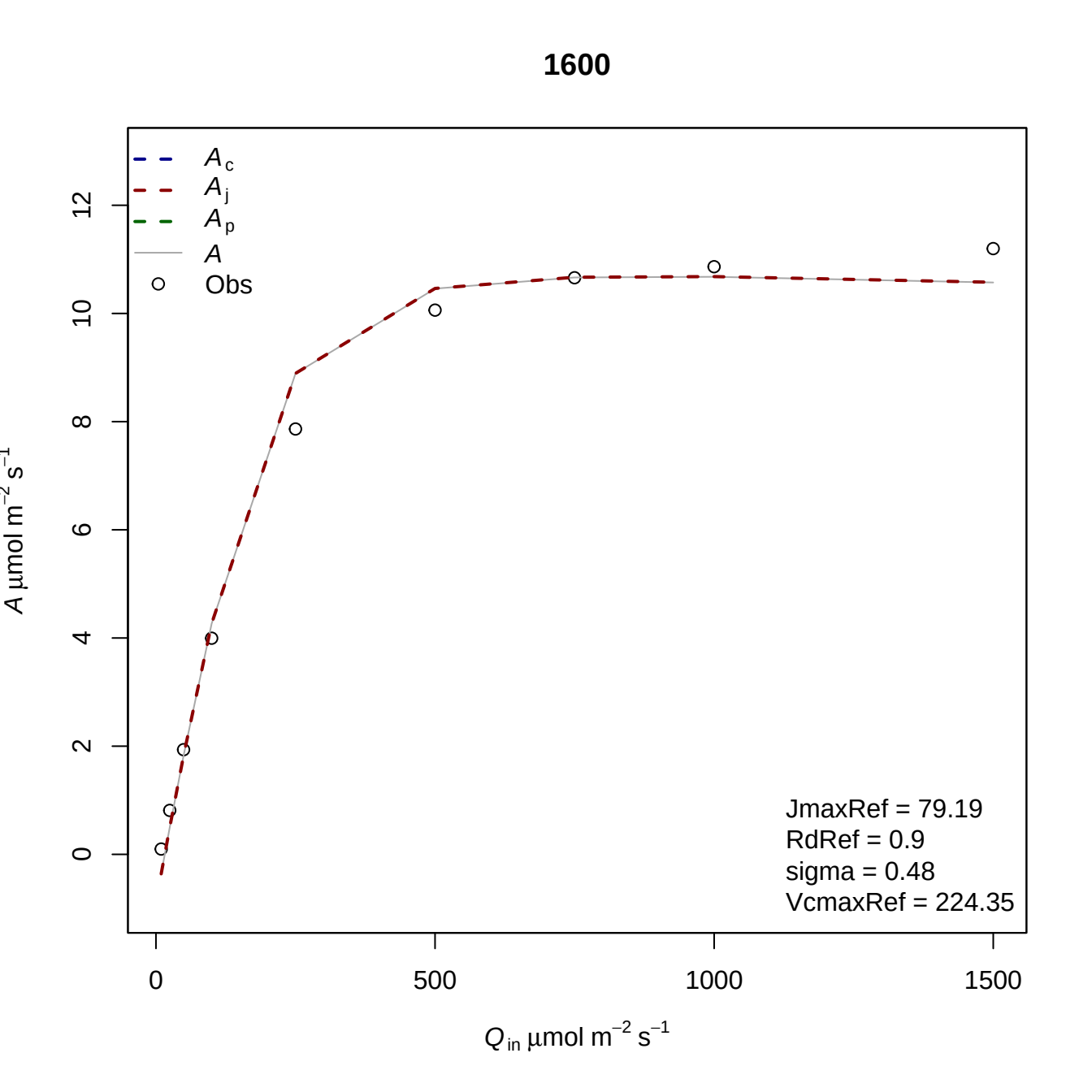

1595
A_c
A_j
A_p
A
Obs
JmaxRef = 62.47
RdRef = 0.85
sigma = 0.5
VcmaxRef = 161.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

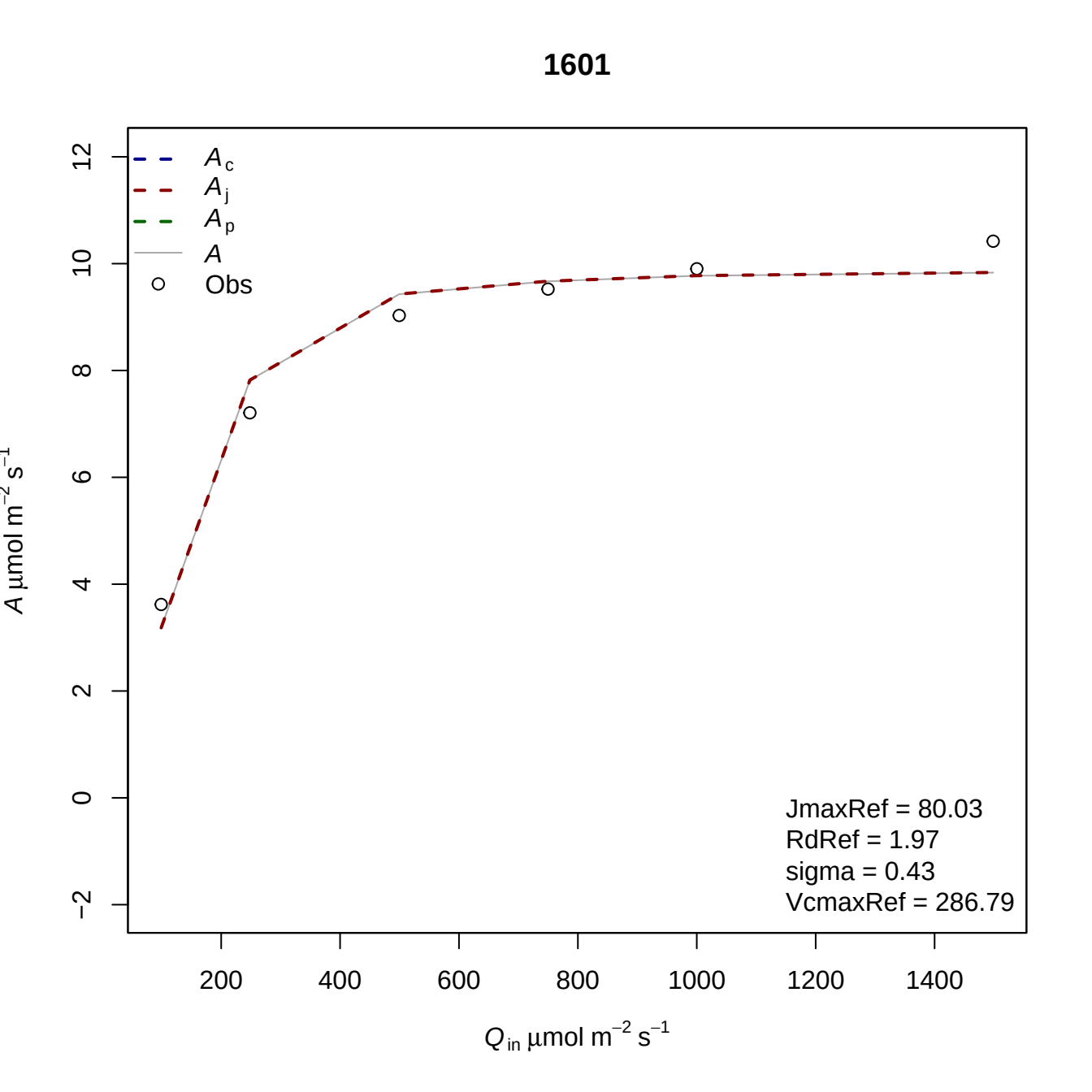

1596
A_c
A_j
A_p
A
Obs
JmaxRef = 78.5
RdRef = 0.98
sigma = 0.65
VcmaxRef = 184.67
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

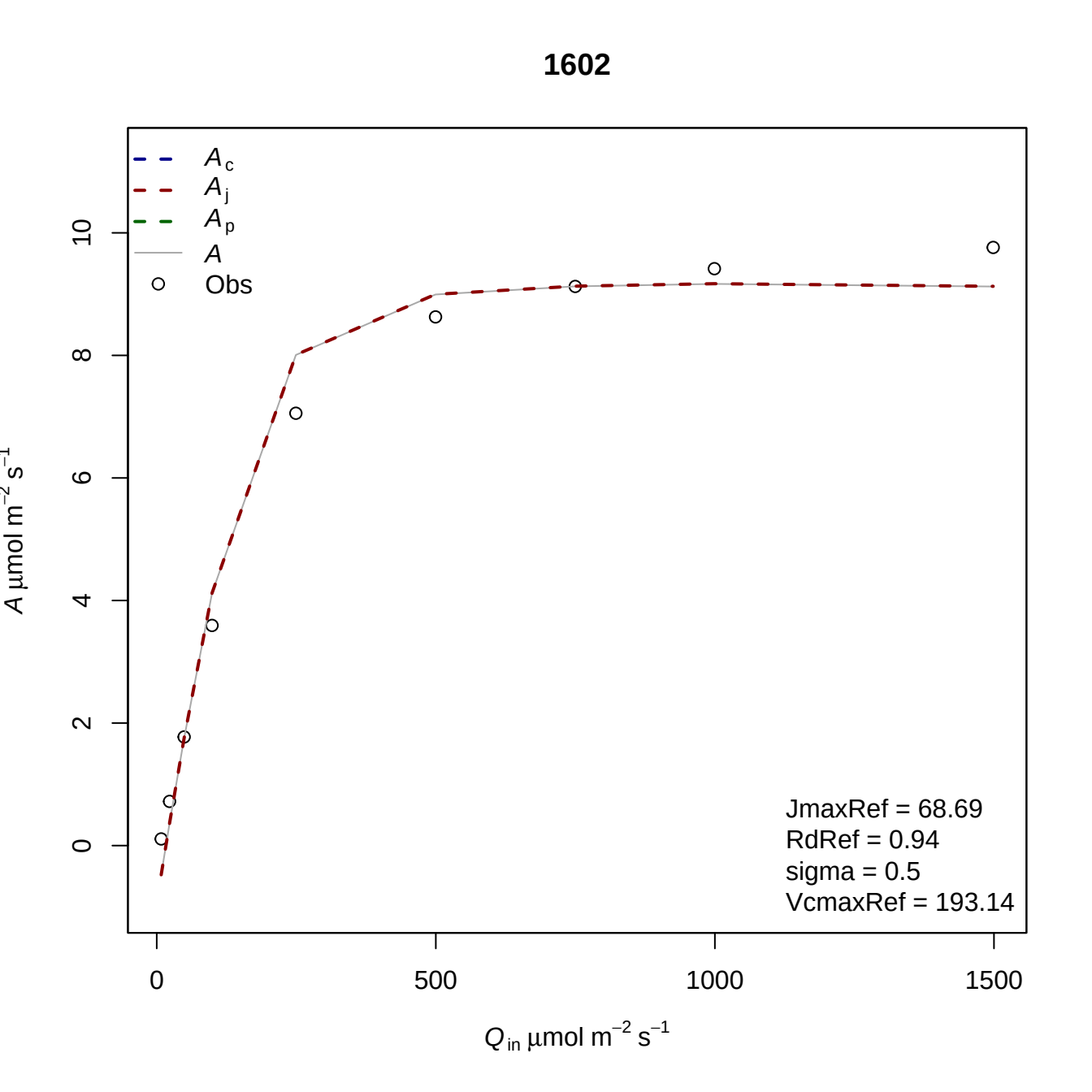

1597
A_c
A_j
A_p
A
Obs
JmaxRef = 97.28
RdRef = 1.08
sigma = 0.7
VcmaxRef = 214.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

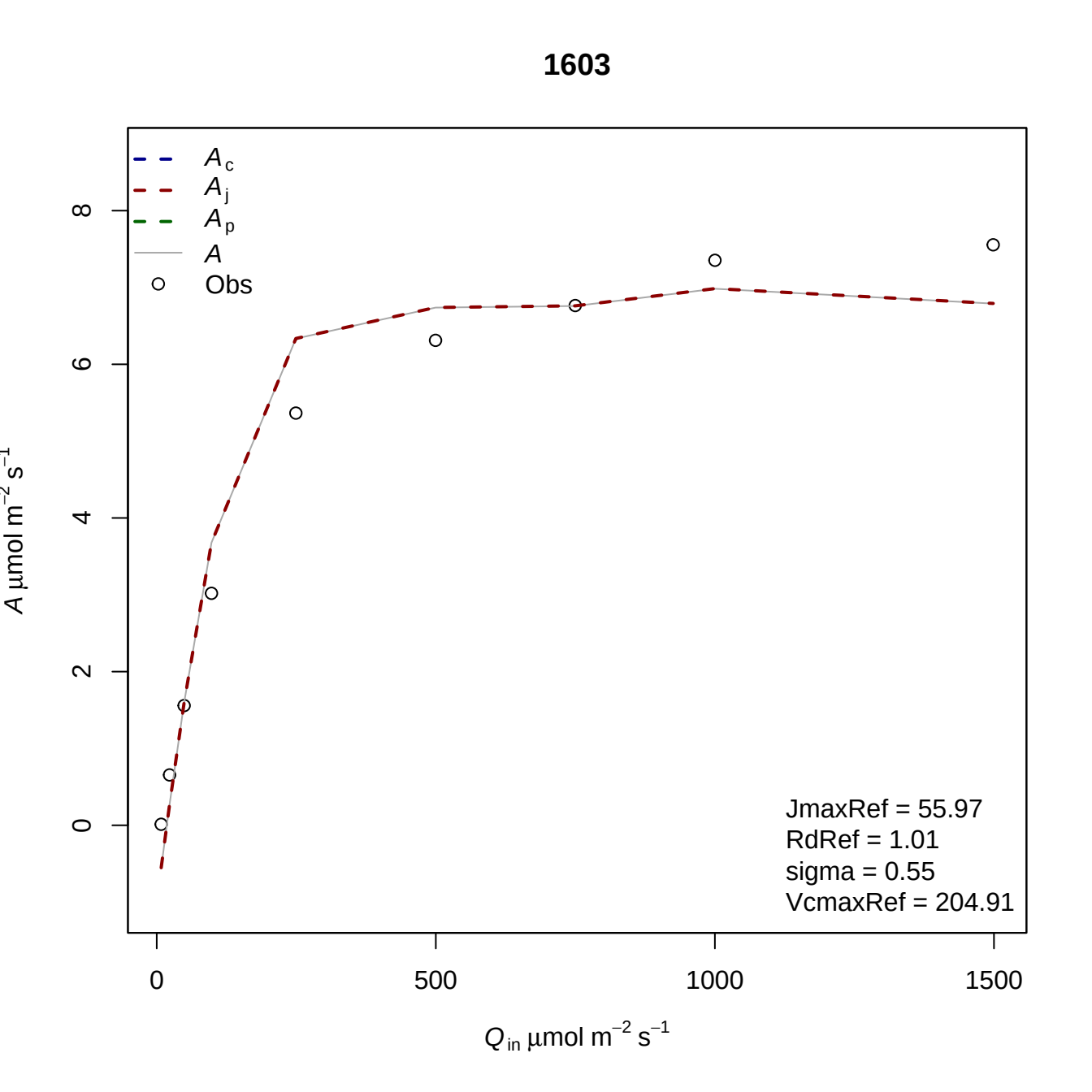

1598
A_c
A_j
A_p
A
Obs
JmaxRef = 78.02
RdRef = 0.89
sigma = 0.42
VcmaxRef = 185.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

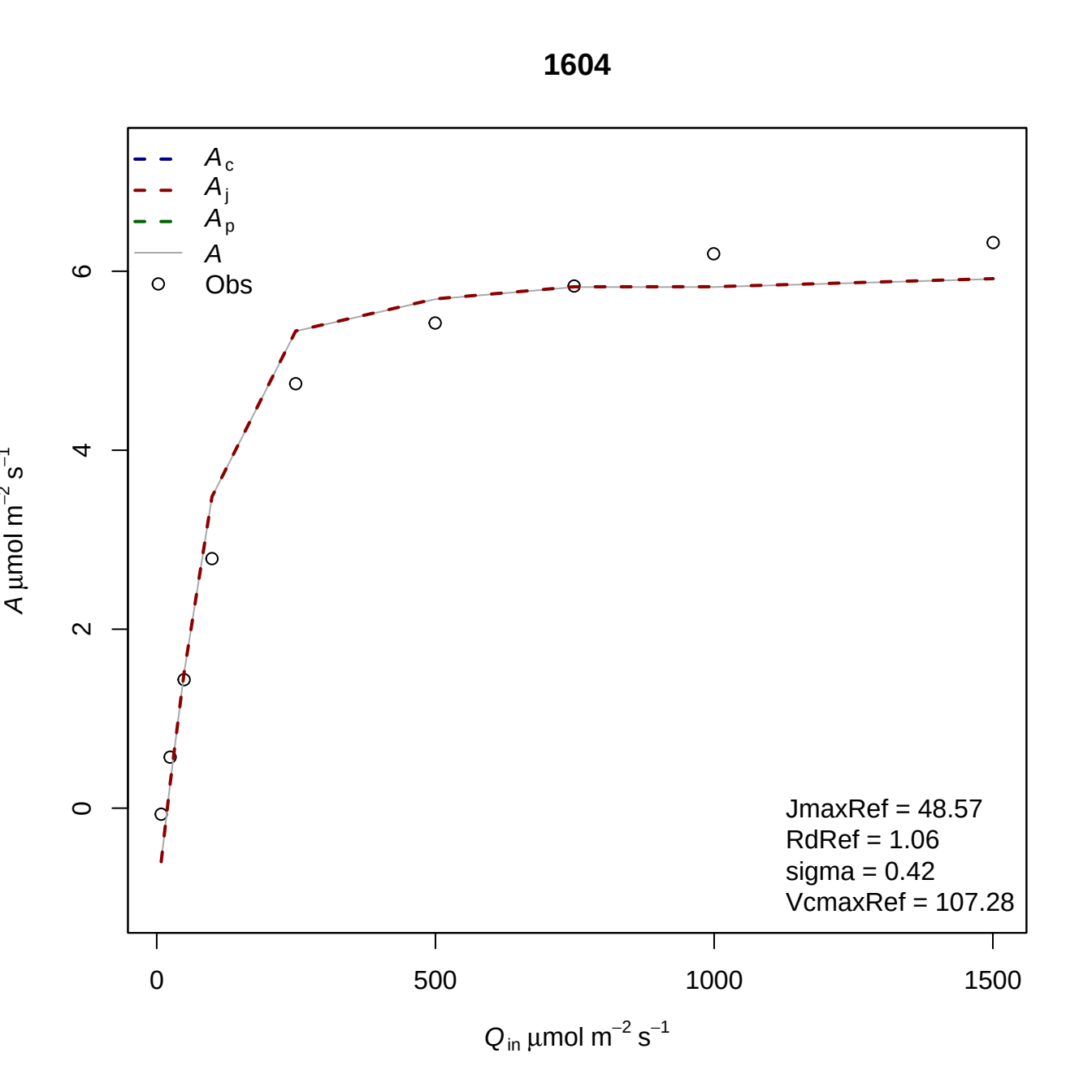

1599
A_c
A_j
A_p
A
Obs
JmaxRef = 80.77
RdRef = 0.88
sigma = 0.43
VcmaxRef = 209.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

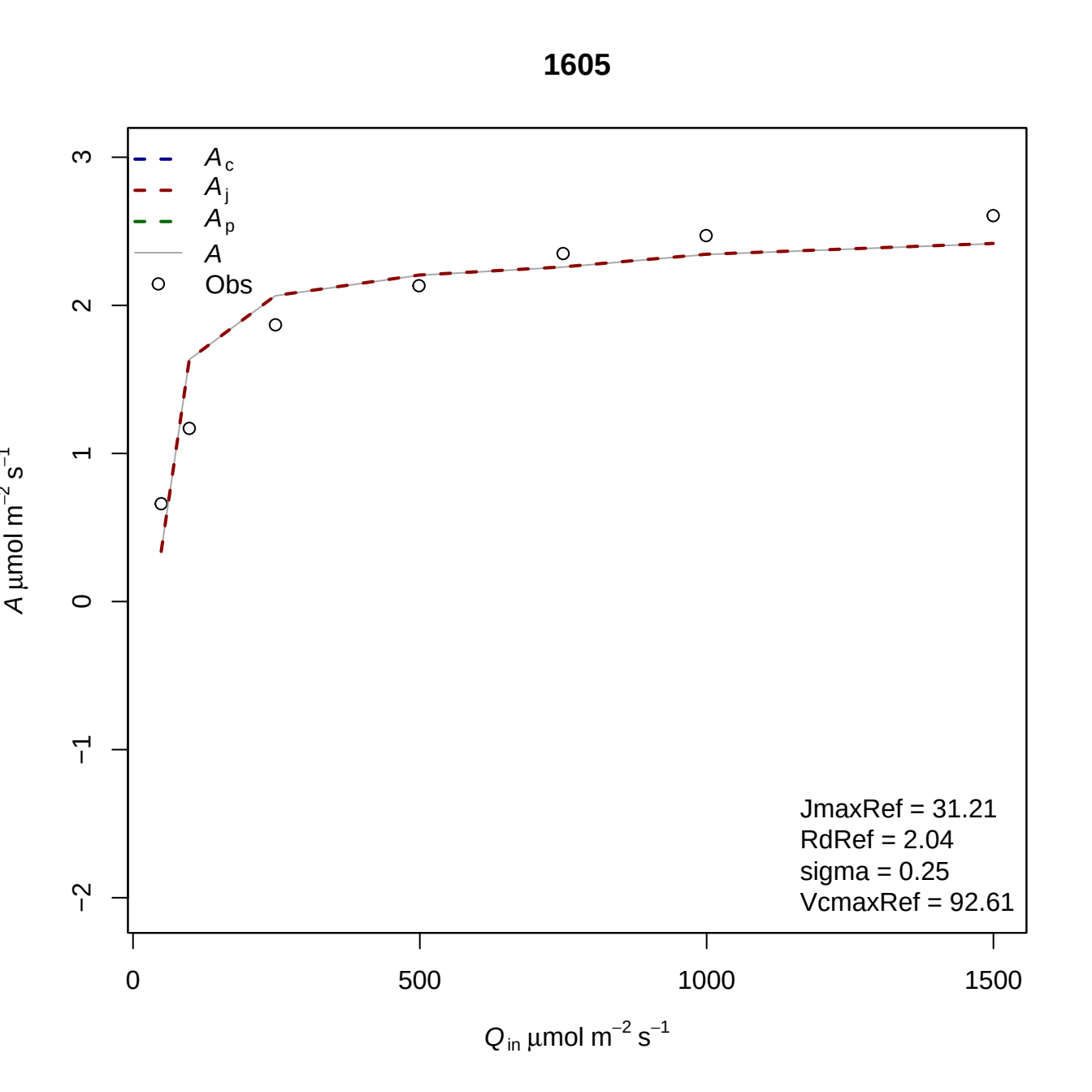

1600
A_c
A_j
A_p
A
Obs
JmaxRef = 79.19
RdRef = 0.9
sigma = 0.48
VcmaxRef = 224.35
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

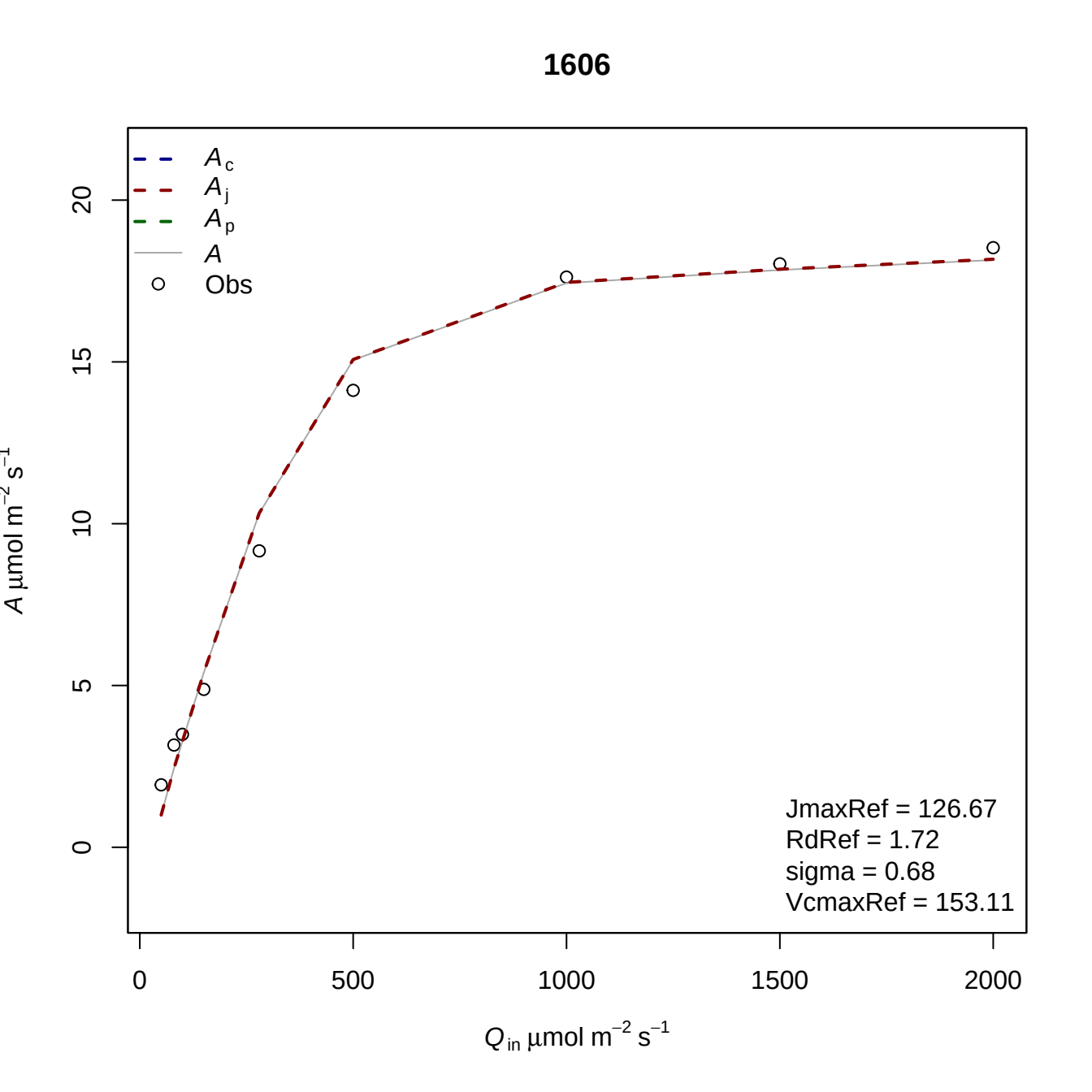

1601
A_c
A_j
A_p
A
Obs
JmaxRef = 80.03
RdRef = 1.97
sigma = 0.43
VcmaxRef = 286.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

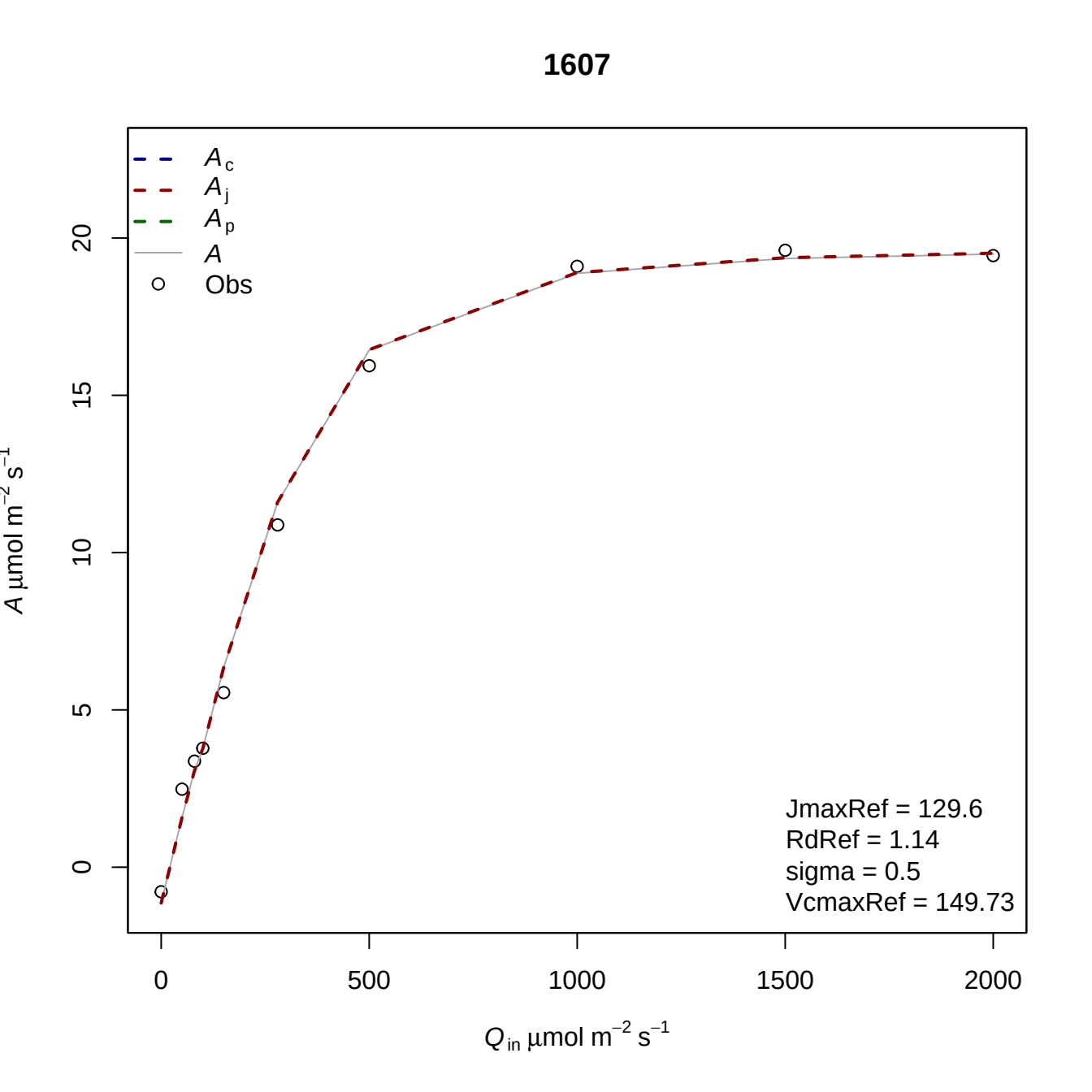

1602
A_c
A_j
A_p
A
Obs
JmaxRef = 68.69
RdRef = 0.94
sigma = 0.5
VcmaxRef = 193.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

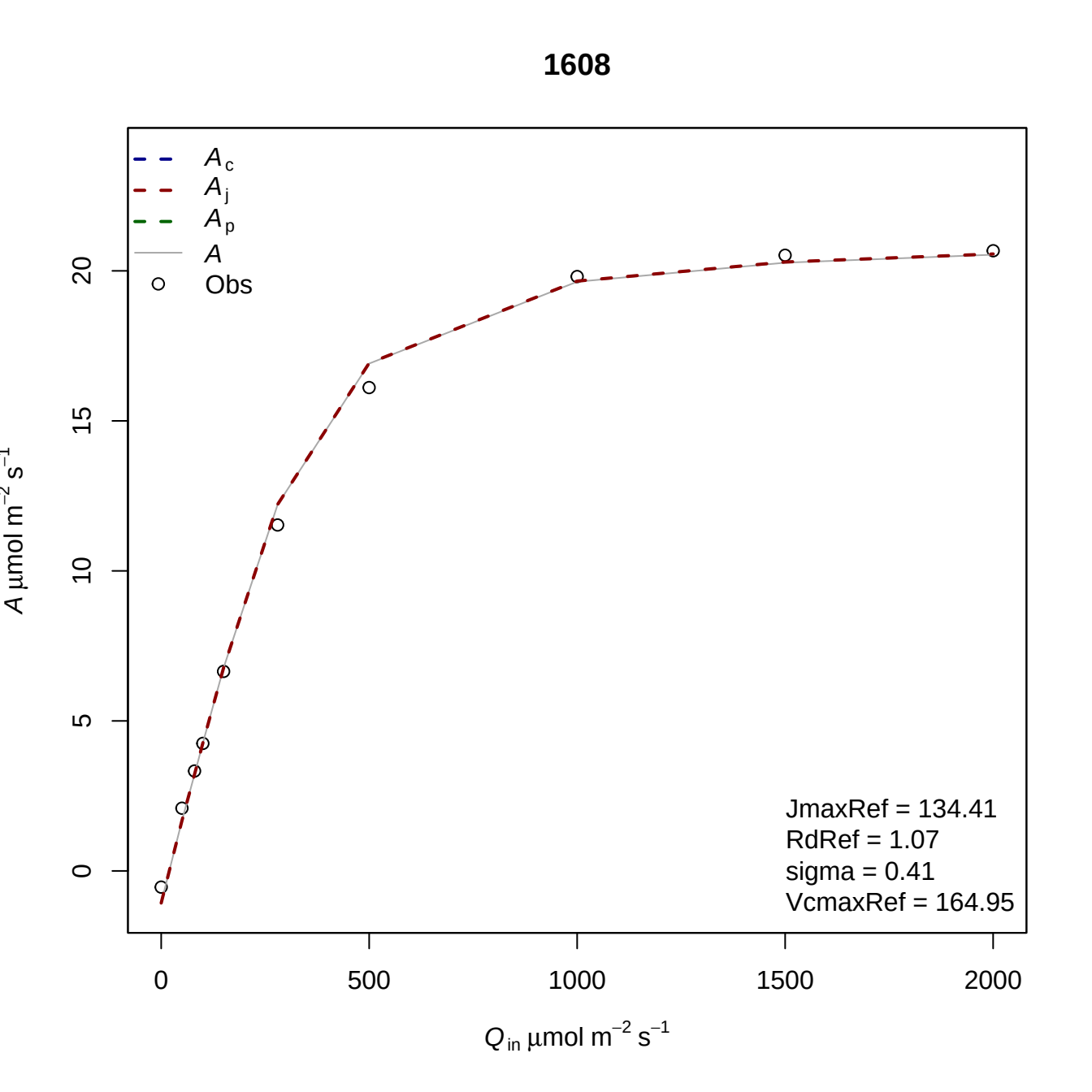

1603
A_c
A_j
A_p
A
Obs
JmaxRef = 55.97
RdRef = 1.01
sigma = 0.55
VcmaxRef = 204.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

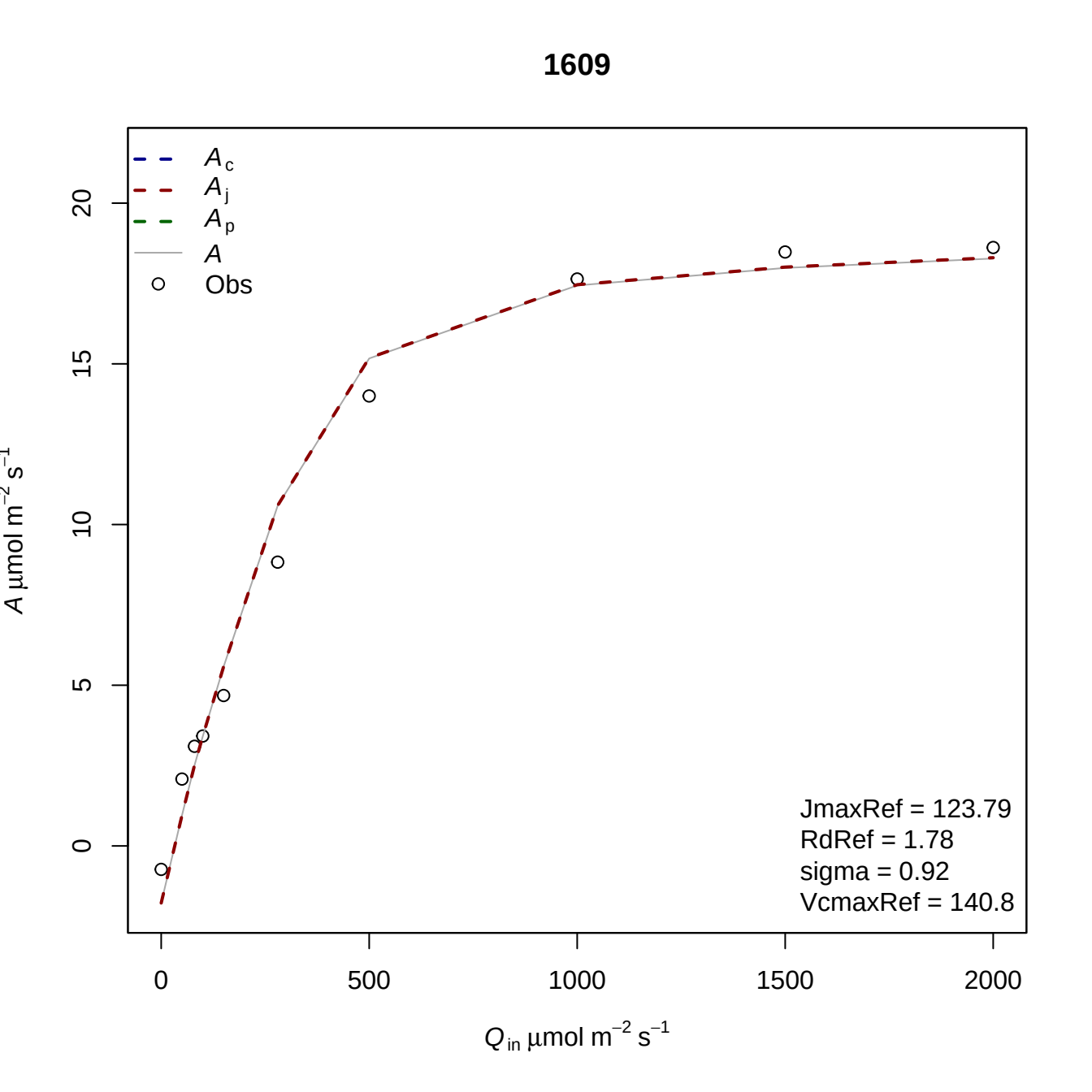

1604
A_c
A_j
A_p
A
Obs
JmaxRef = 48.57
RdRef = 1.06
sigma = 0.42
VcmaxRef = 107.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

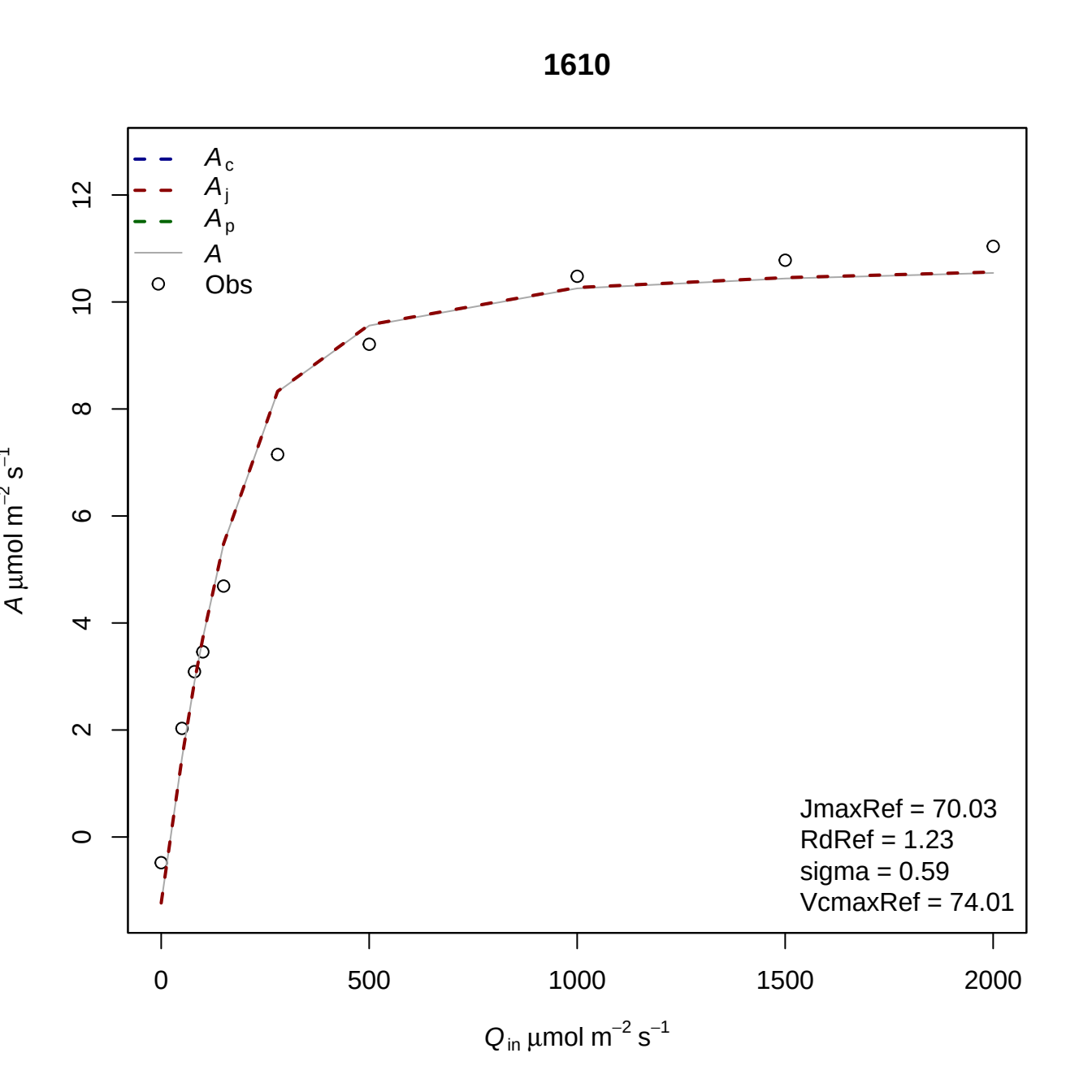

1605
A_c
A_j
A_p
A
Obs
JmaxRef = 31.21
RdRef = 2.04
sigma = 0.25
VcmaxRef = 92.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500

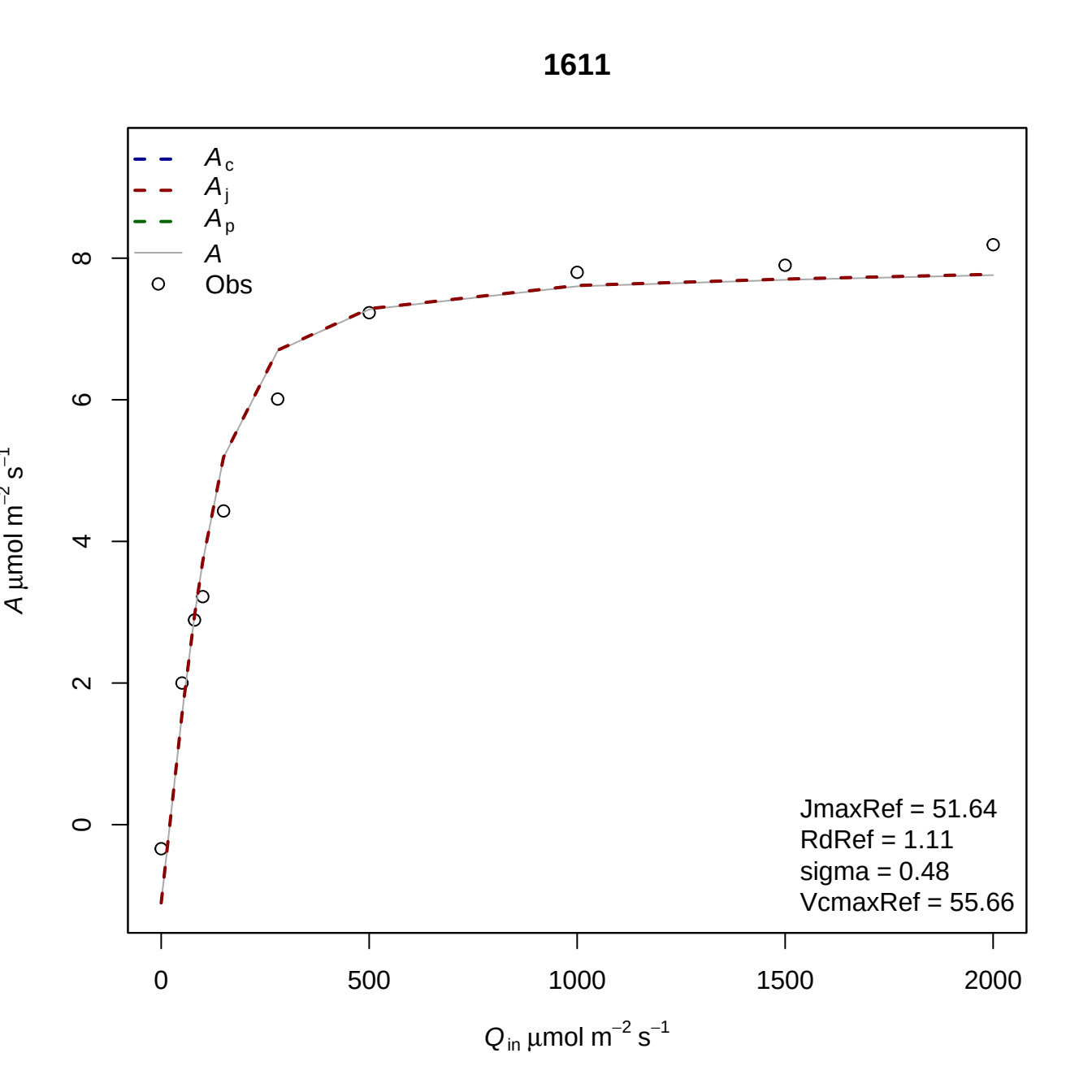

1606
A_c
A_j
A_p
A
Obs
JmaxRef = 126.67
RdRef = 1.72
sigma = 0.68
VcmaxRef = 153.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

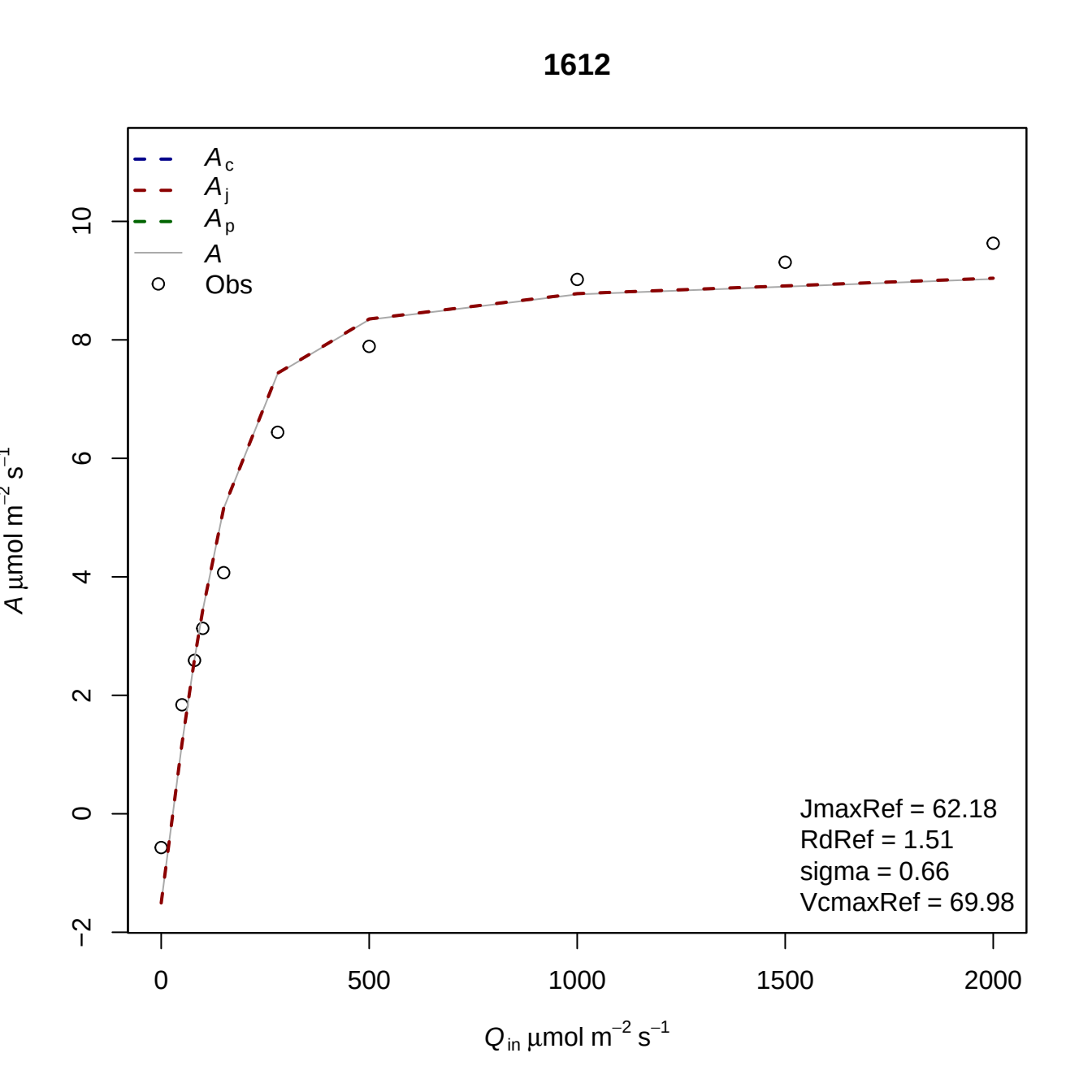

1607
A_c
A_j
A_p
A
Obs
JmaxRef = 129.6
RdRef = 1.14
sigma = 0.5
VcmaxRef = 149.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

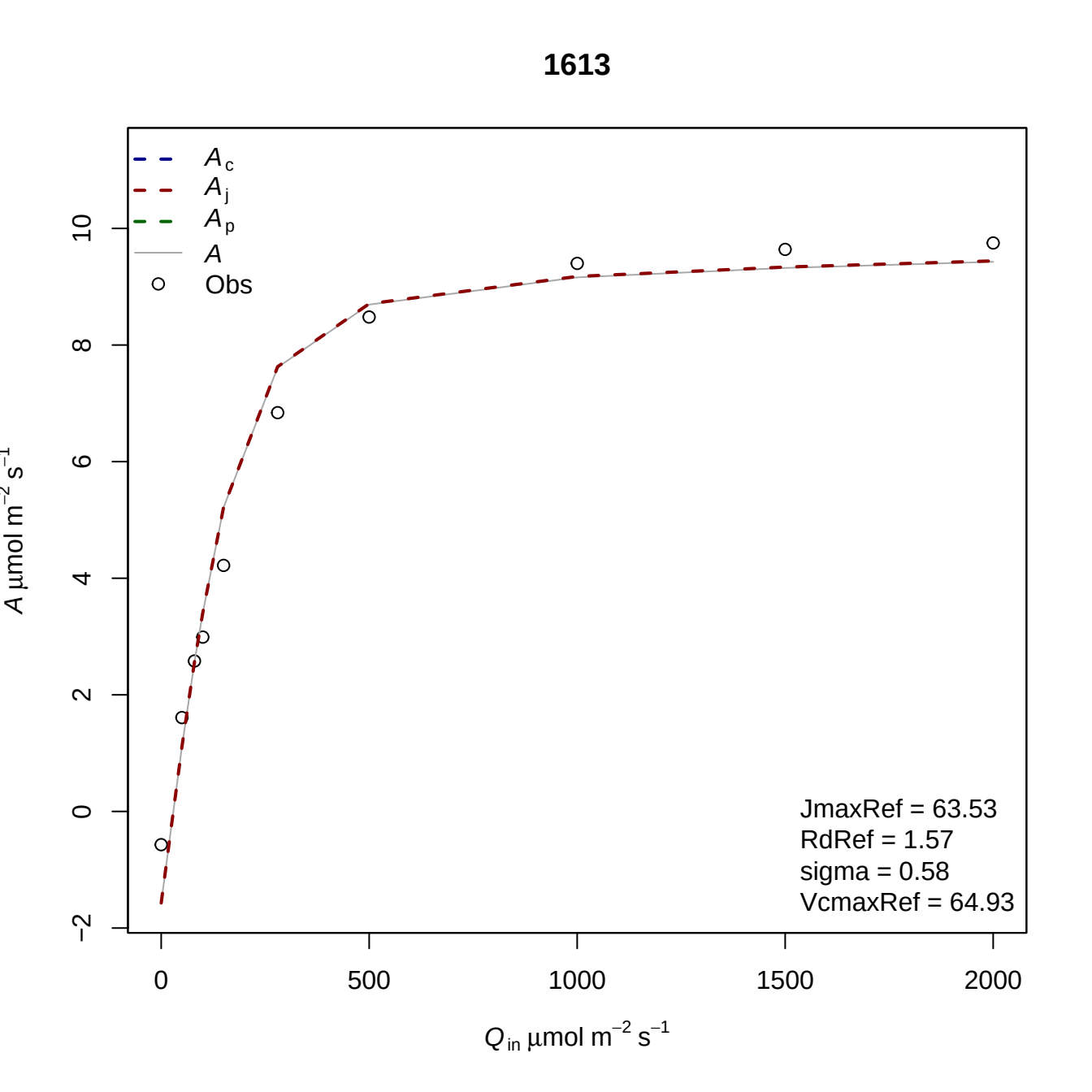

1608
A_c
A_j
A_p
A
Obs
JmaxRef = 134.41
RdRef = 1.07
sigma = 0.41
VcmaxRef = 164.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

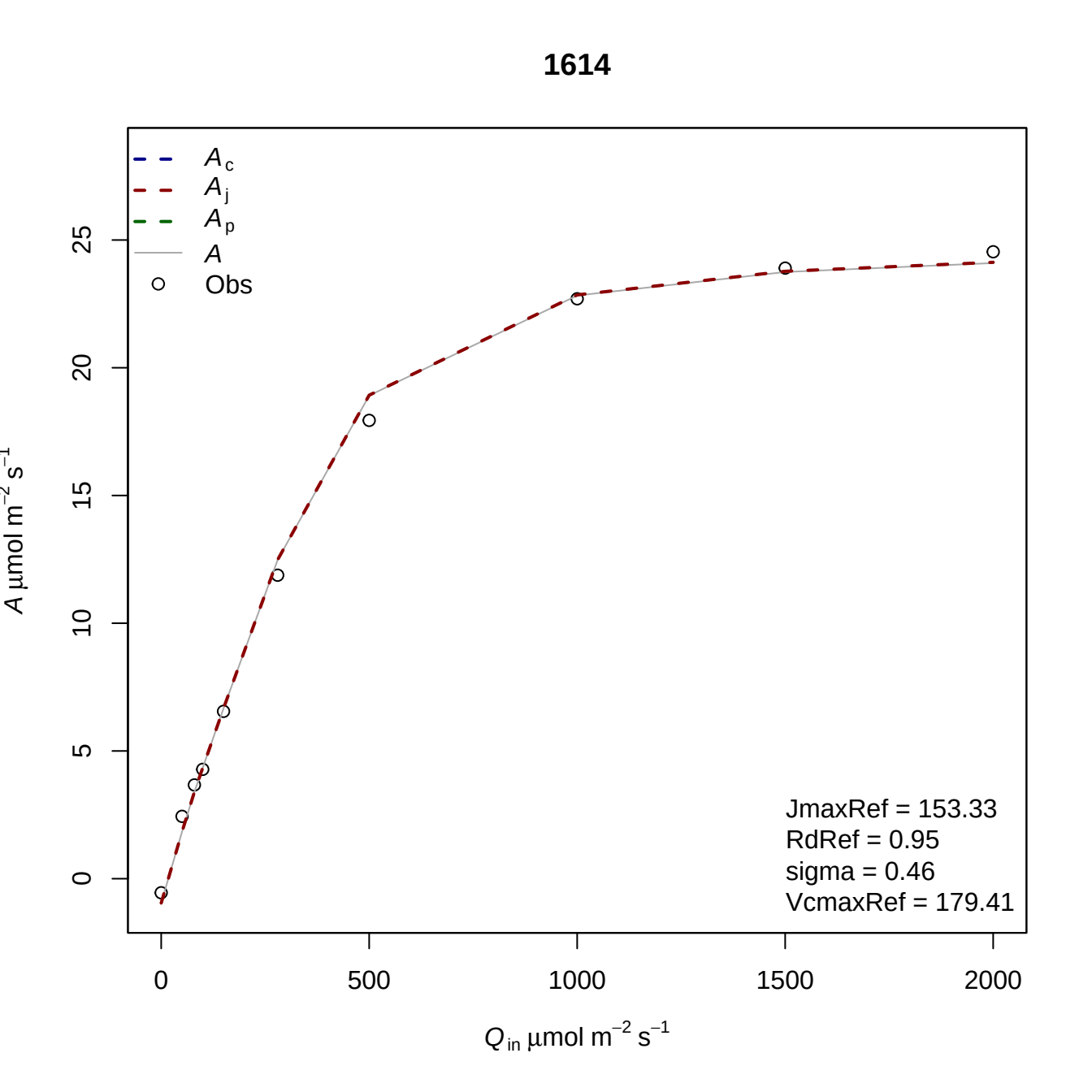

1609
A_c
A_j
A_p
A
Obs
JmaxRef = 123.79
RdRef = 1.78
sigma = 0.92
VcmaxRef = 140.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

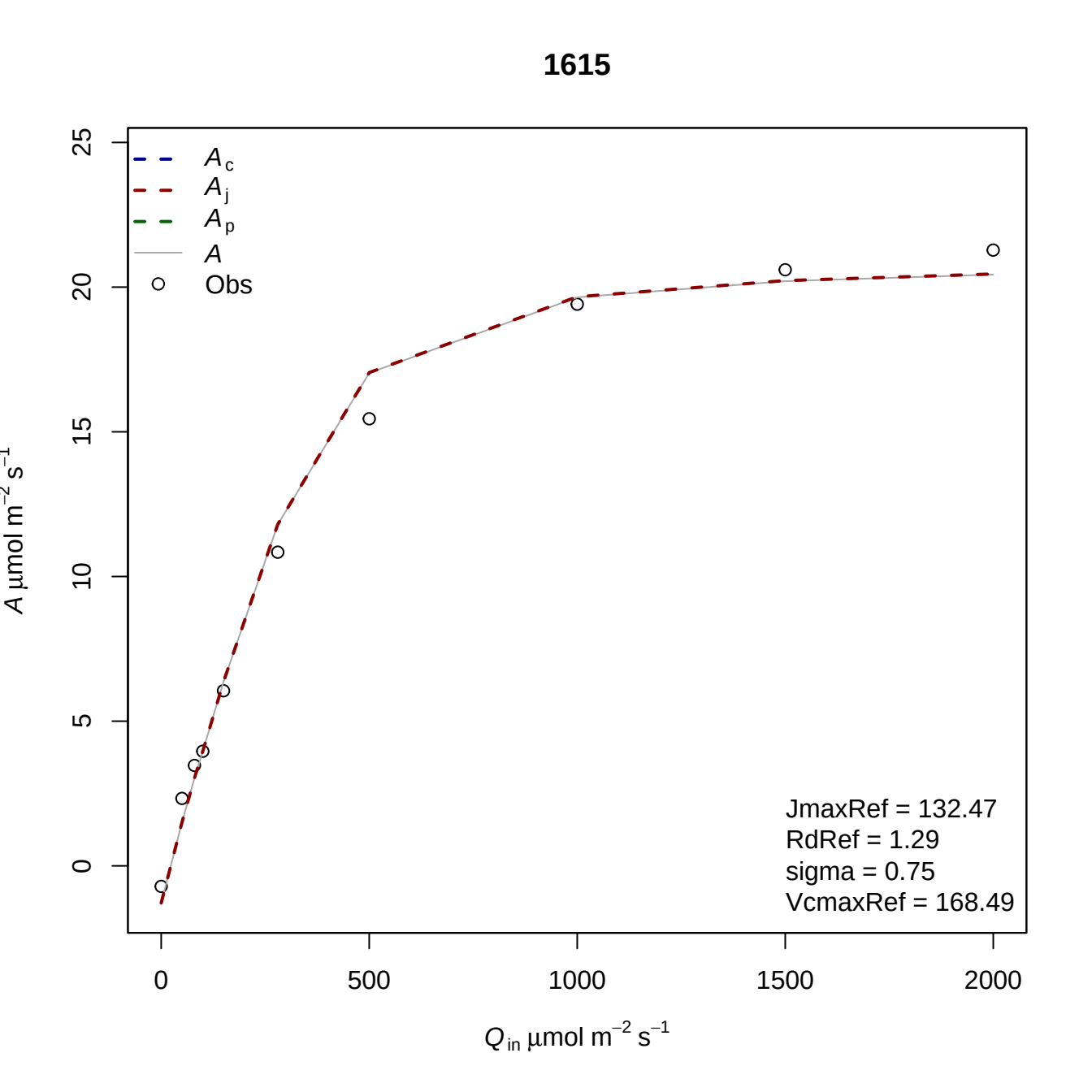

1610
A_c
A_j
A_p
A
Obs
JmaxRef = 70.03
RdRef = 1.23
sigma = 0.59
VcmaxRef = 74.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

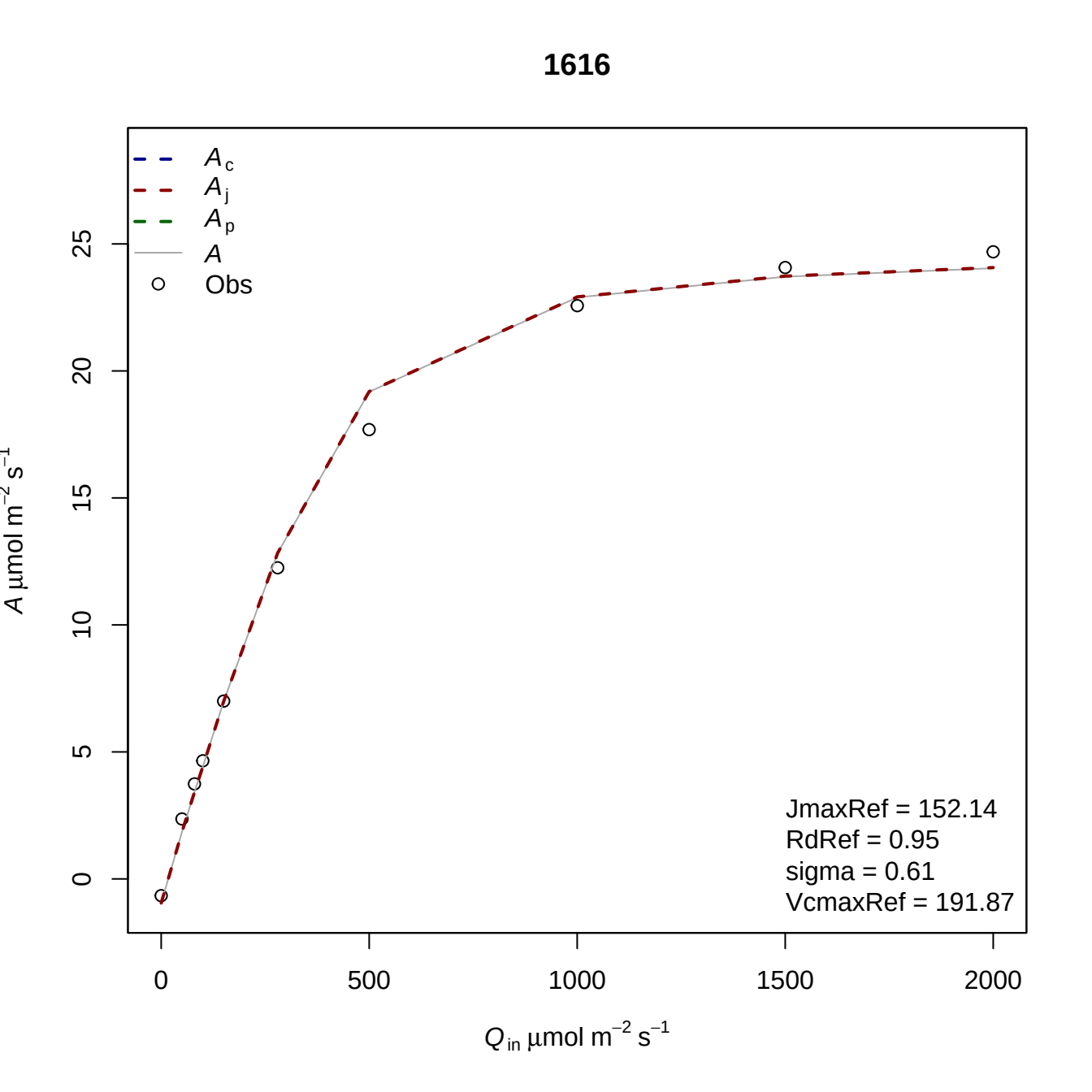

1611
A_c
A_j
A_p
A
Obs
JmaxRef = 51.64
RdRef = 1.11
sigma = 0.48
VcmaxRef = 55.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

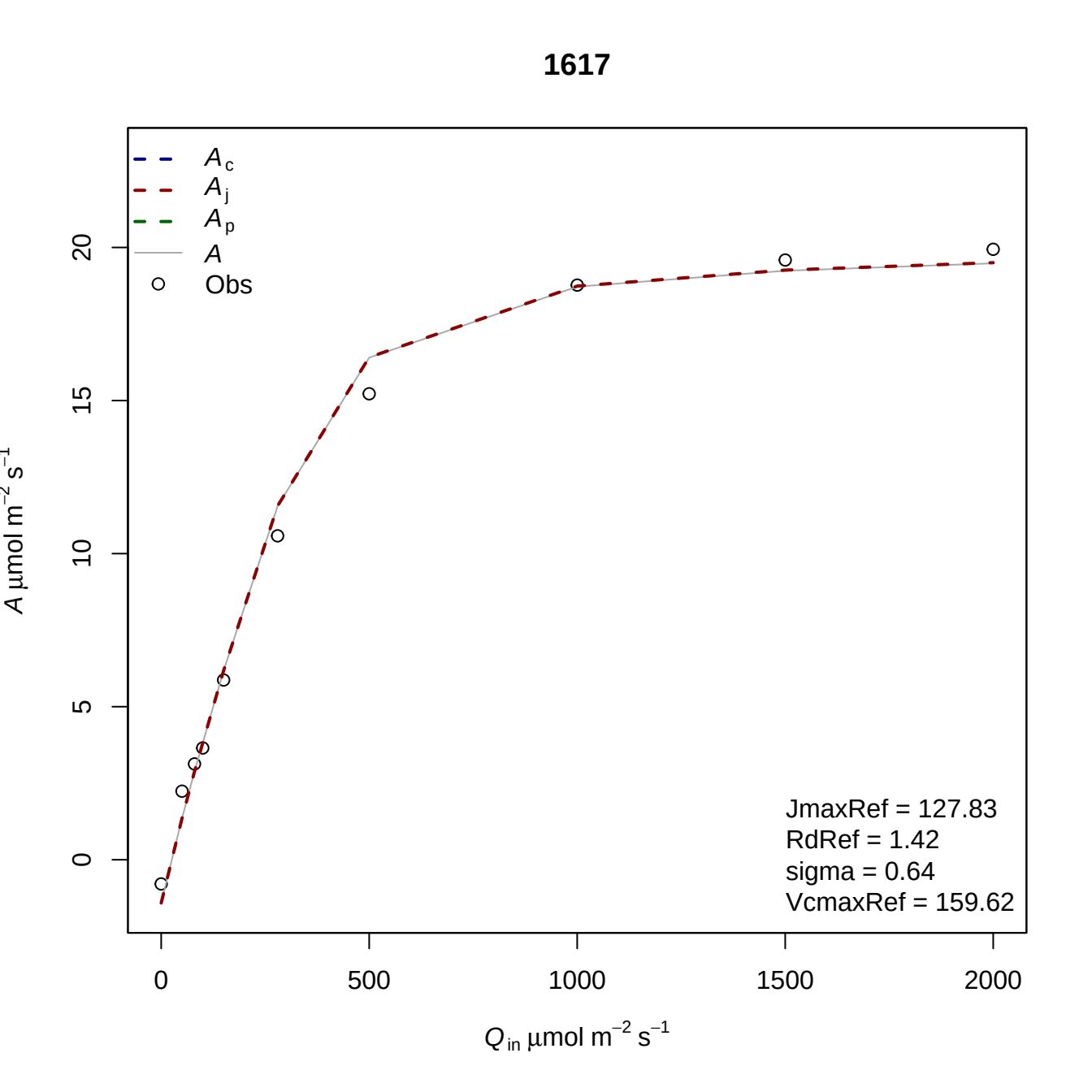

1612
A_c
A_j
A_p
A
Obs
JmaxRef = 62.18
RdRef = 1.51
sigma = 0.66
VcmaxRef = 69.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

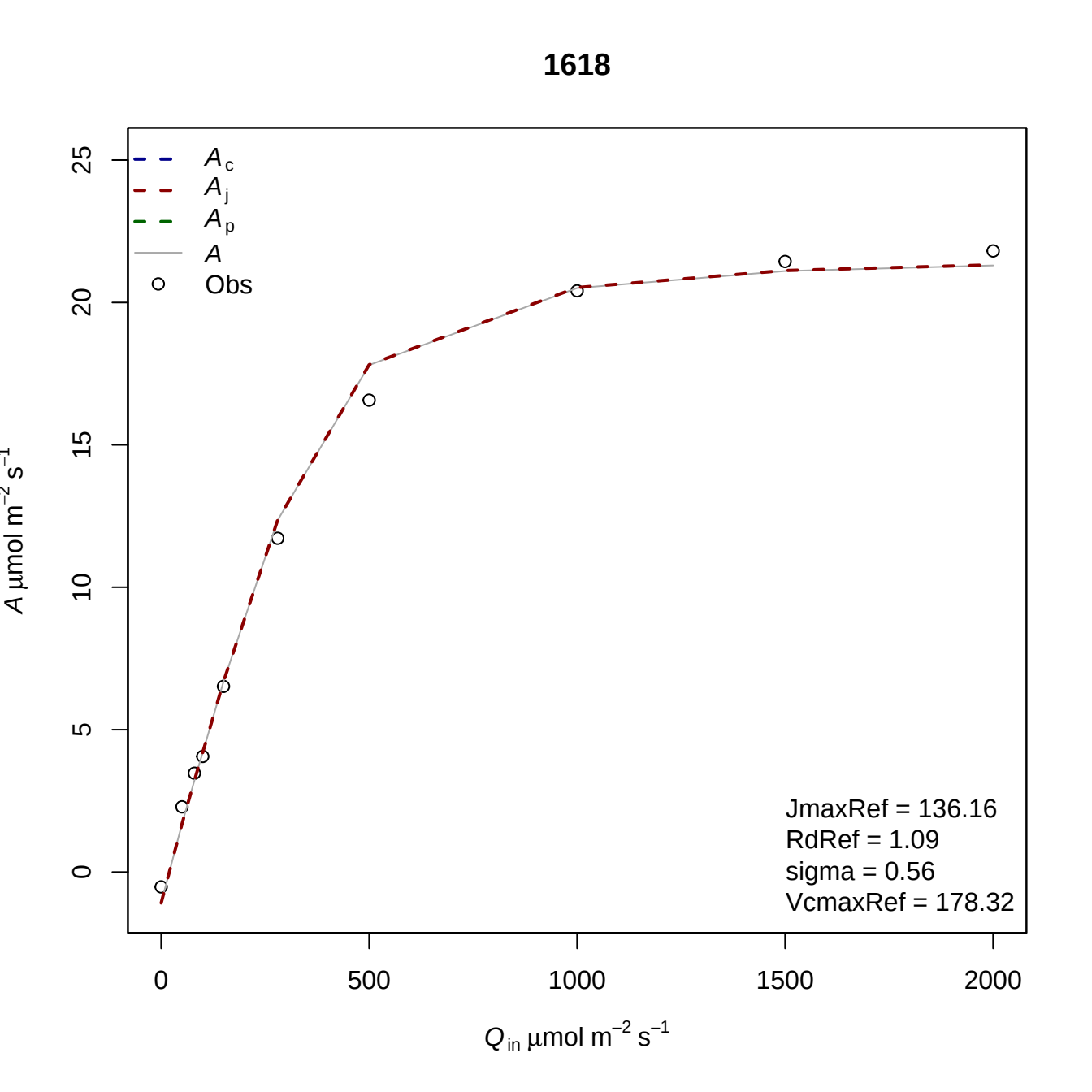

1613
A_c
A_j
A_p
A
Obs
JmaxRef = 63.53
RdRef = 1.57
sigma = 0.58
VcmaxRef = 64.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

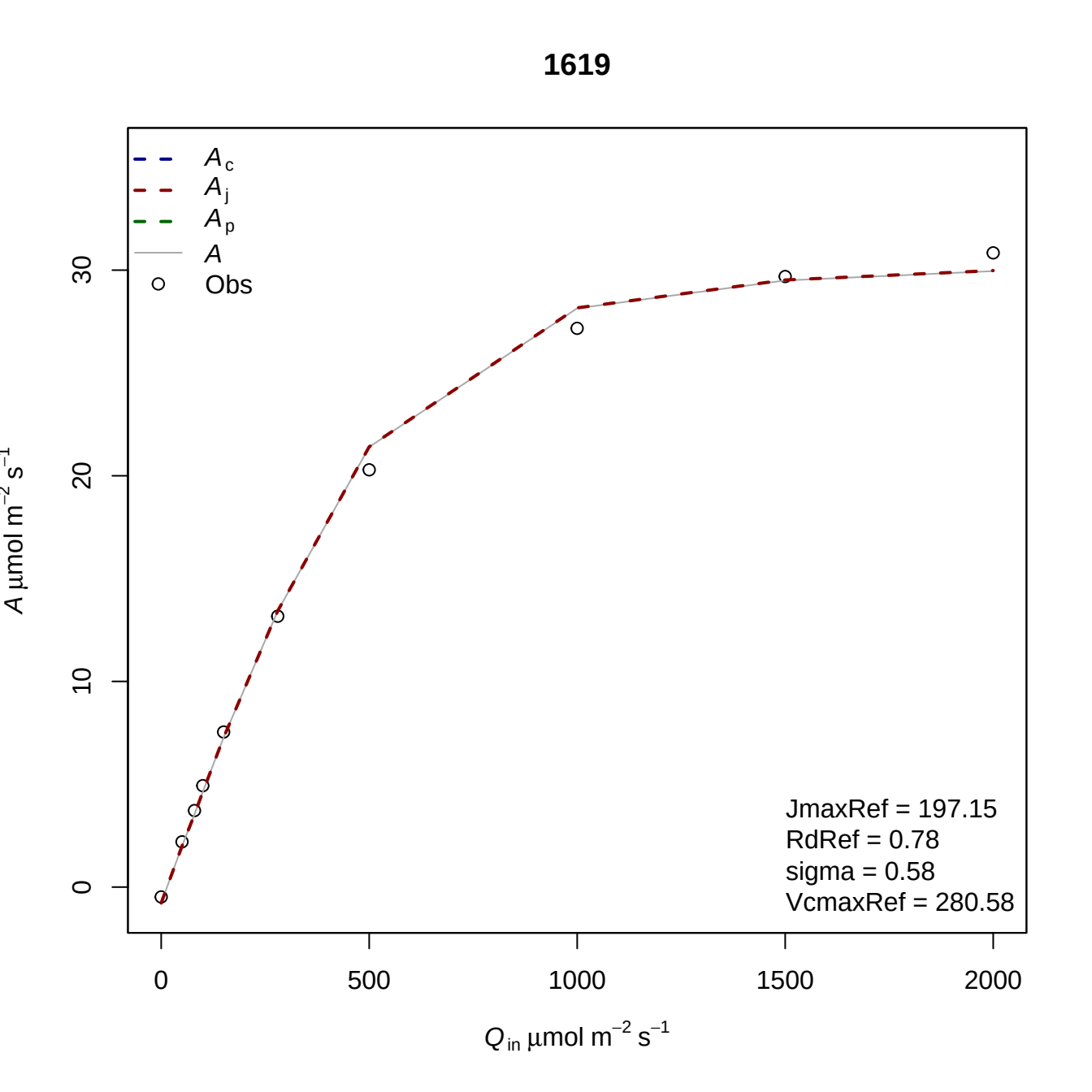

1614
A_c
A_j
A_p
A
Obs
JmaxRef = 153.33
RdRef = 0.95
sigma = 0.46
VcmaxRef = 179.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

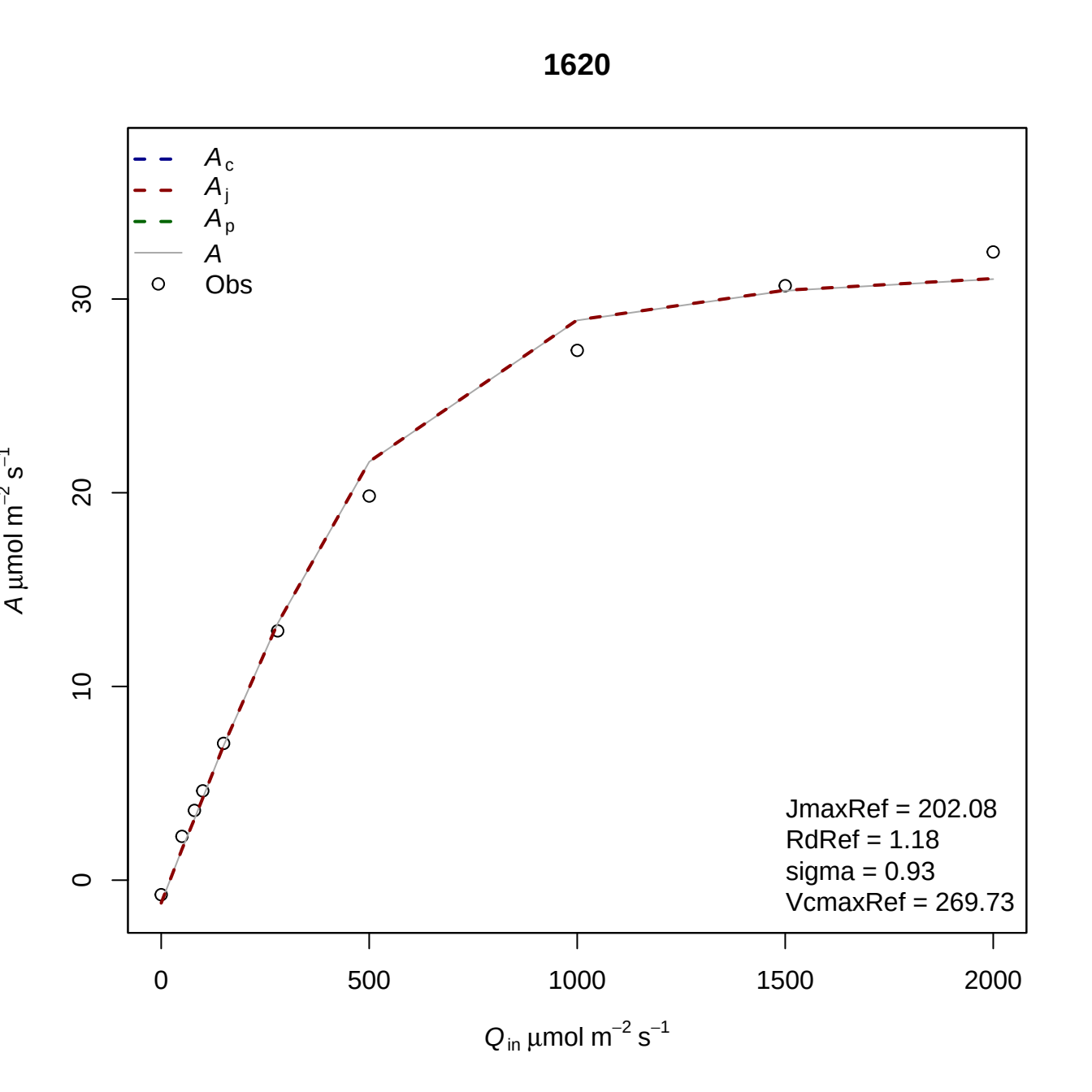

1615
A_c
A_j
A_p
A
Obs
JmaxRef = 132.47
RdRef = 1.29
sigma = 0.75
VcmaxRef = 168.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

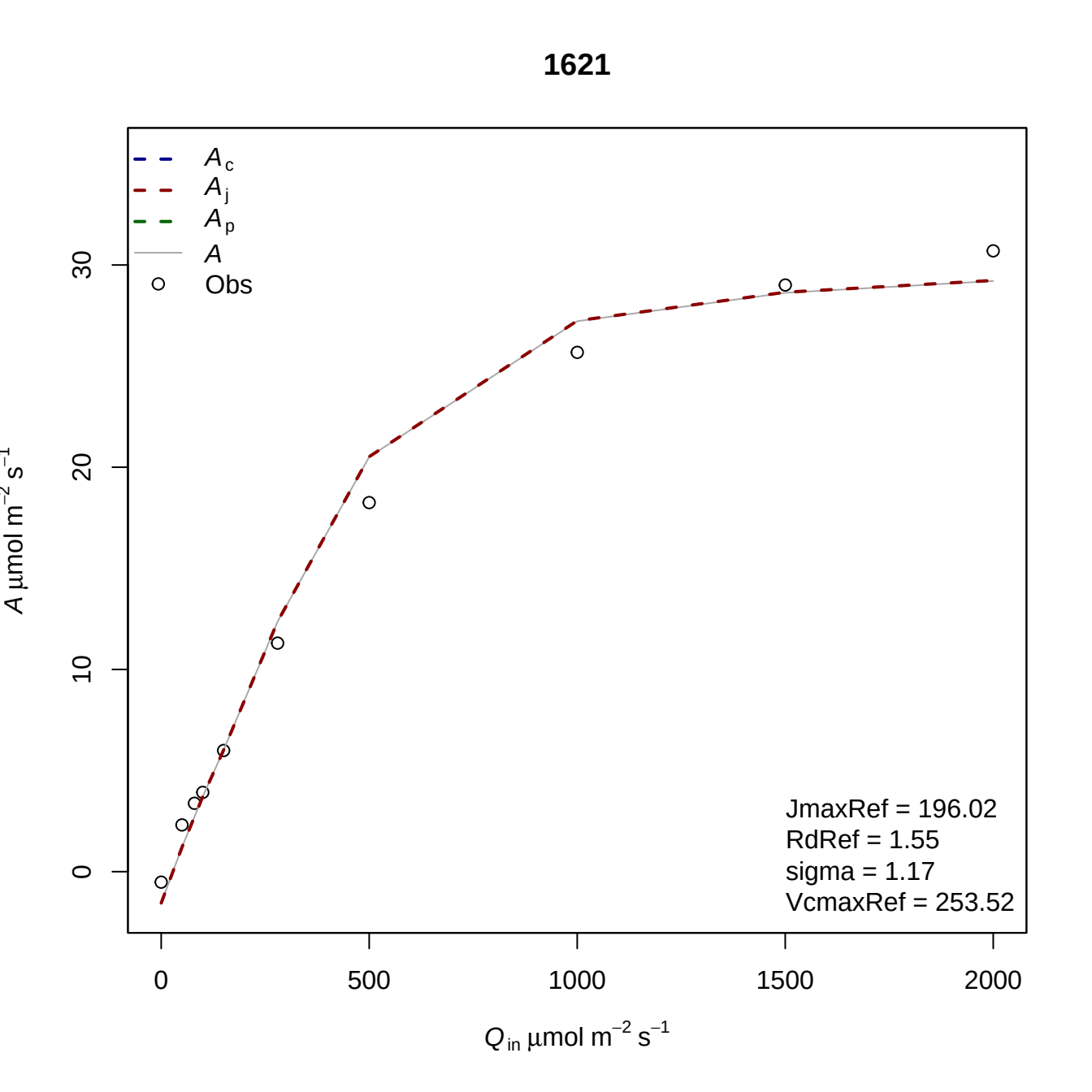

1616
A_c
A_j
A_p
A
Obs
JmaxRef = 152.14
RdRef = 0.95
sigma = 0.61
VcmaxRef = 191.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

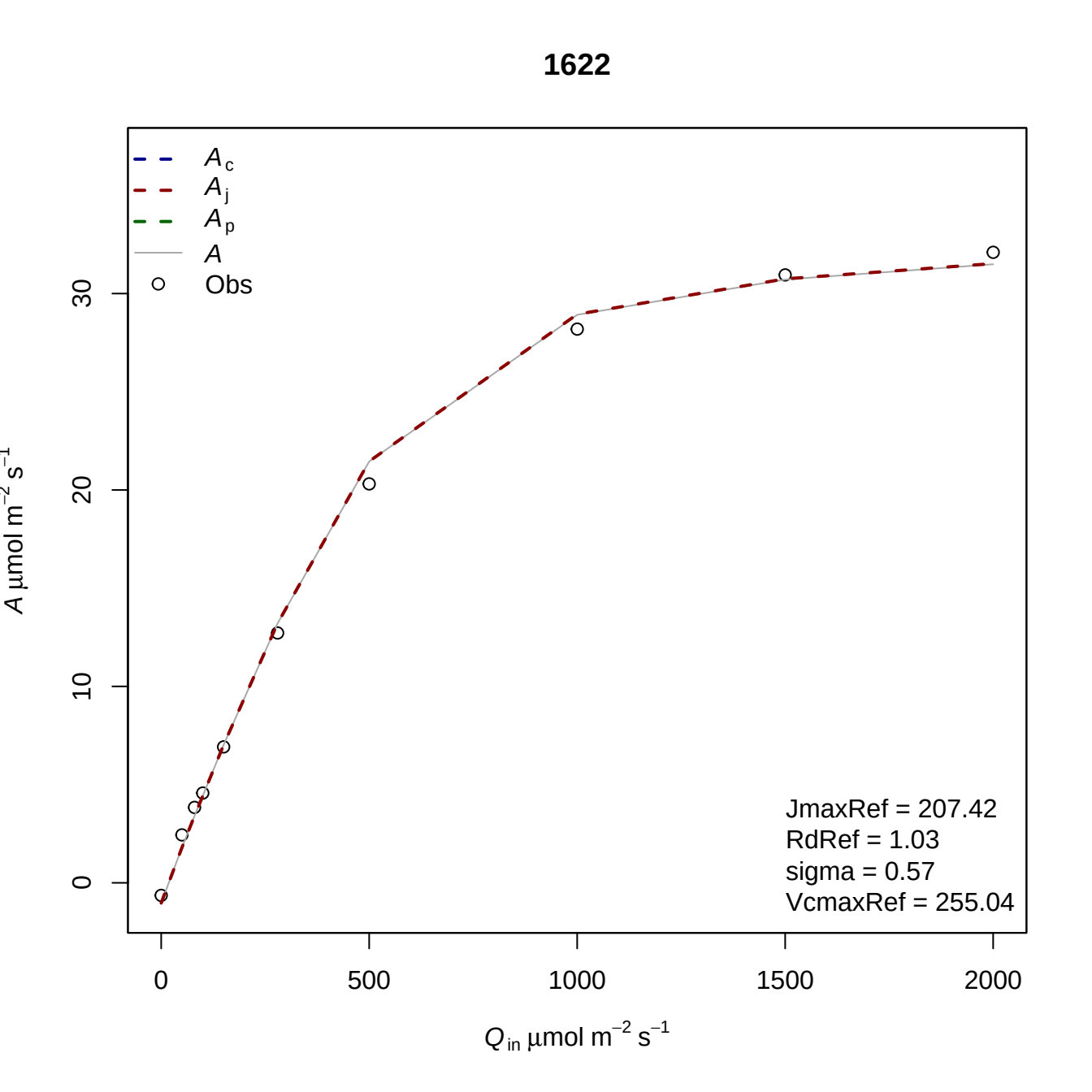

1617
A_c
A_j
A_p
A
Obs
JmaxRef = 127.83
RdRef = 1.42
sigma = 0.64
VcmaxRef = 159.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

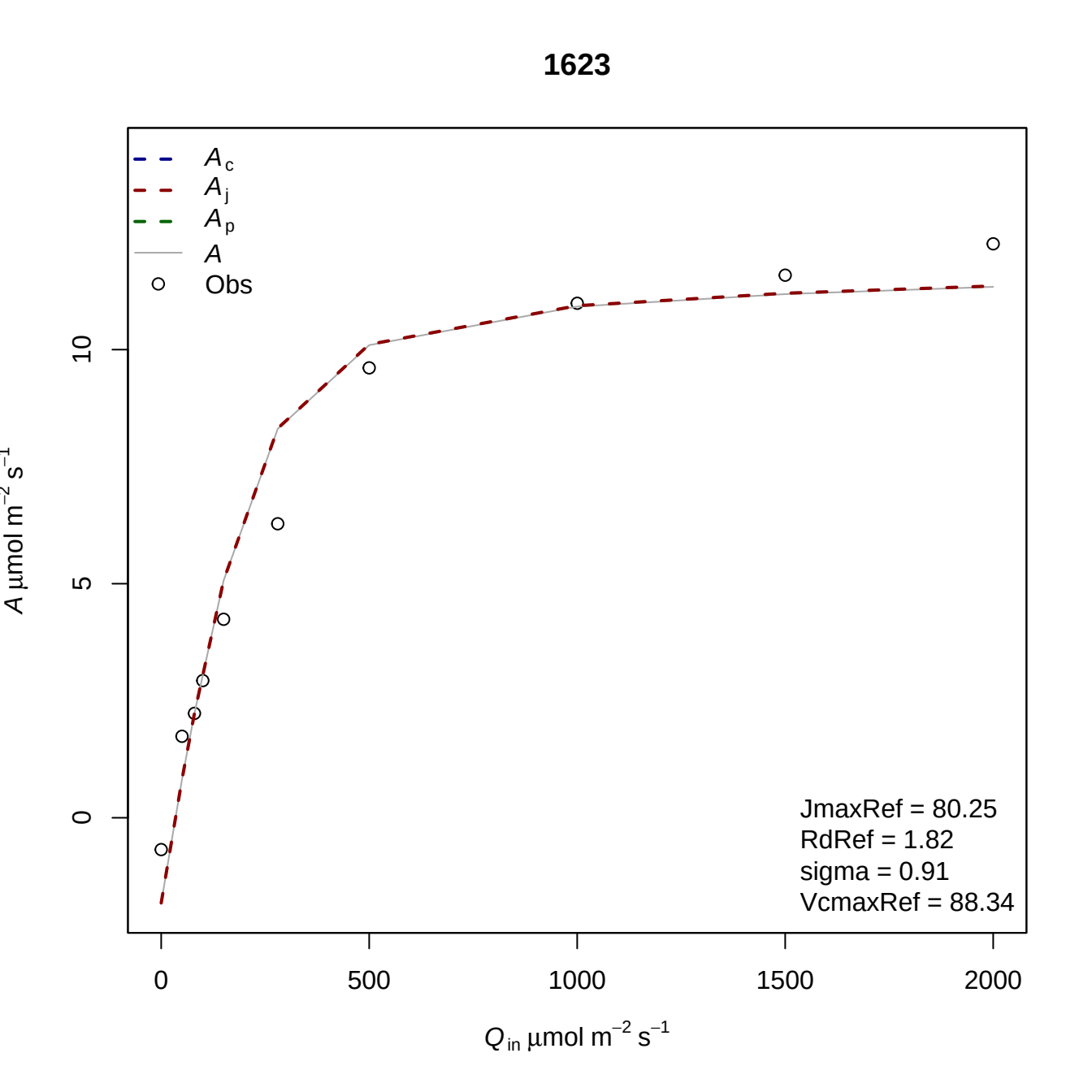

1618
A_c
A_j
A_p
A
Obs
JmaxRef = 136.16
RdRef = 1.09
sigma = 0.56
VcmaxRef = 178.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

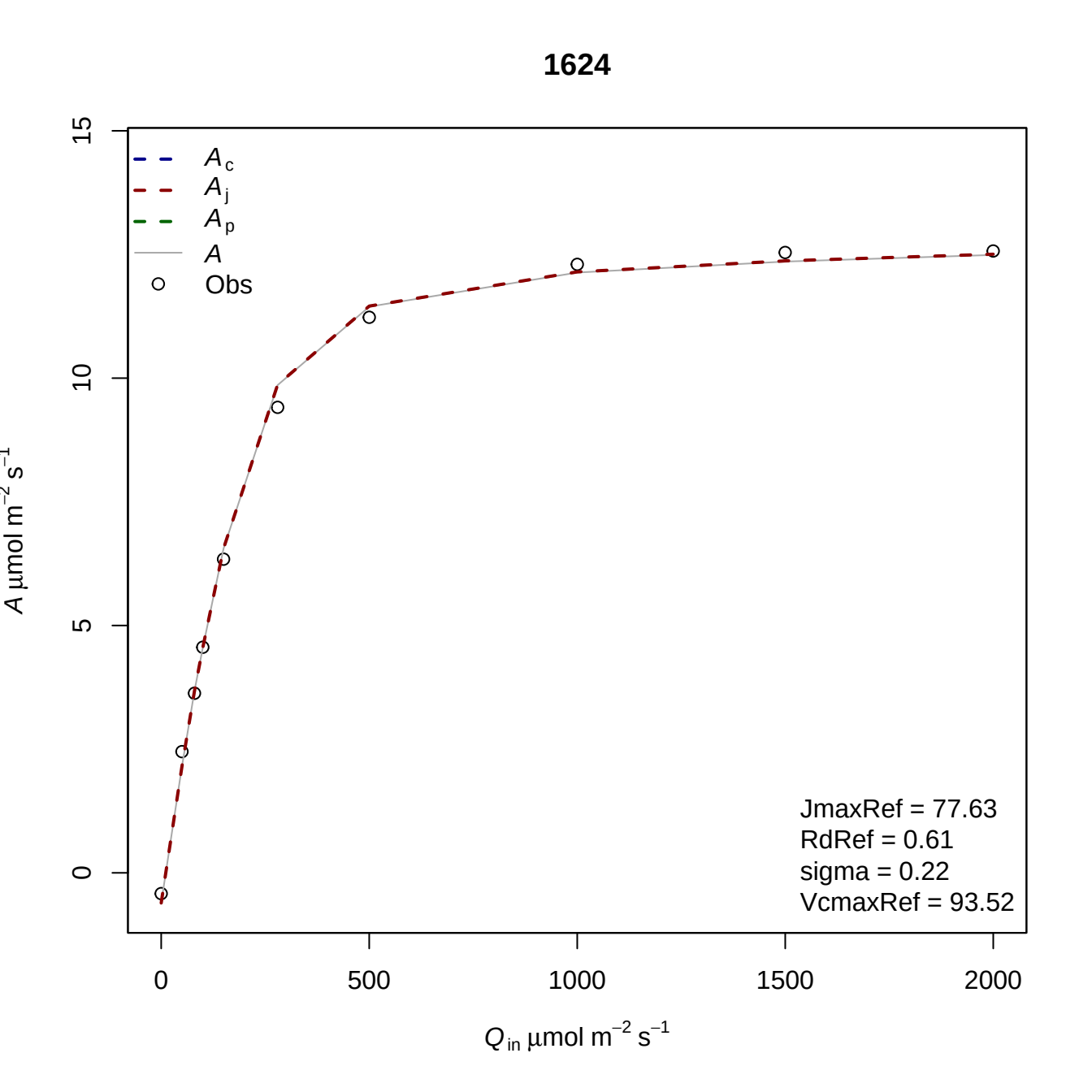

1619
A_c
A_j
A_p
A
Obs
JmaxRef = 197.15
RdRef = 0.78
sigma = 0.58
VcmaxRef = 280.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

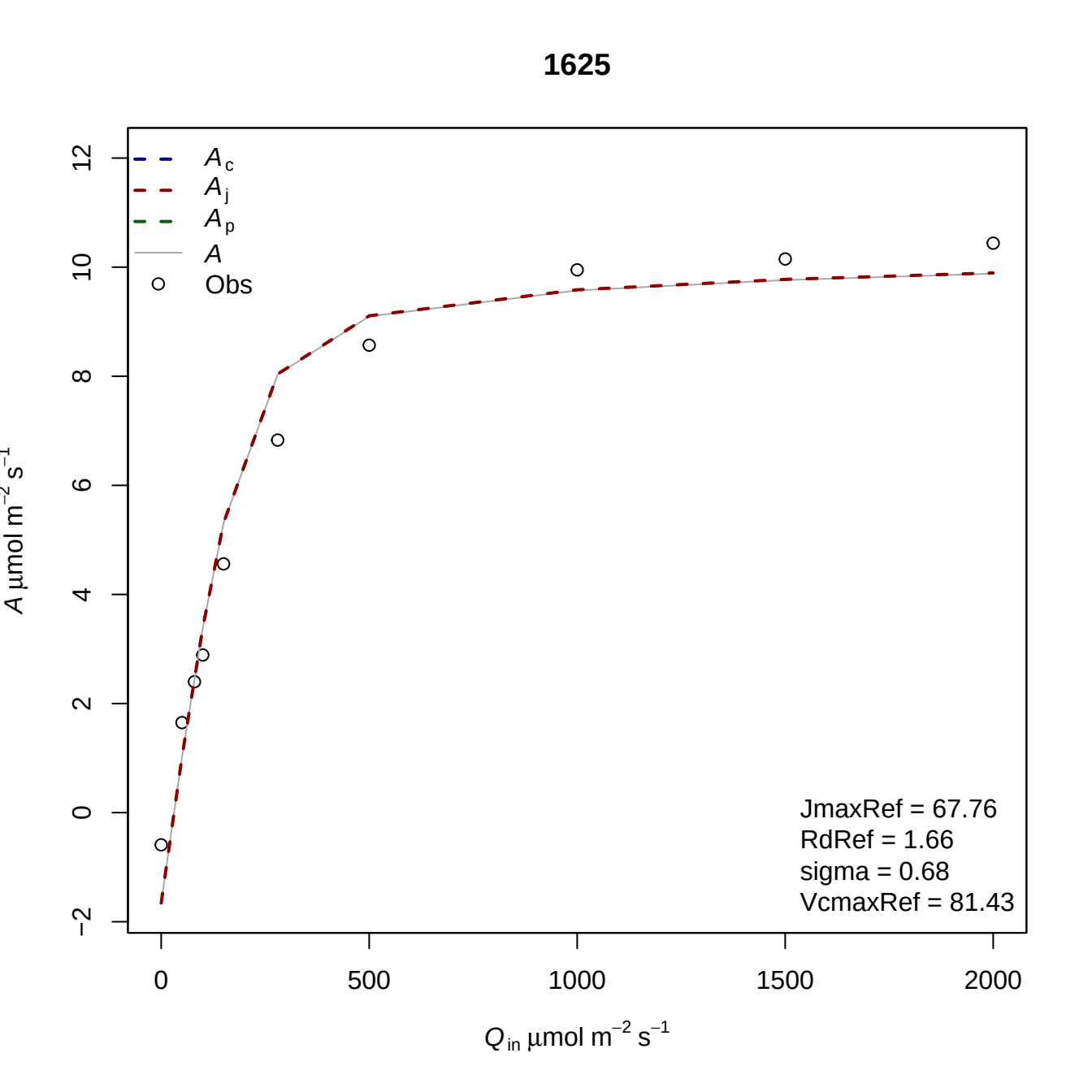

1620
A_c
A_j
A_p
A
Obs
JmaxRef = 202.08
RdRef = 1.18
sigma = 0.93
VcmaxRef = 269.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

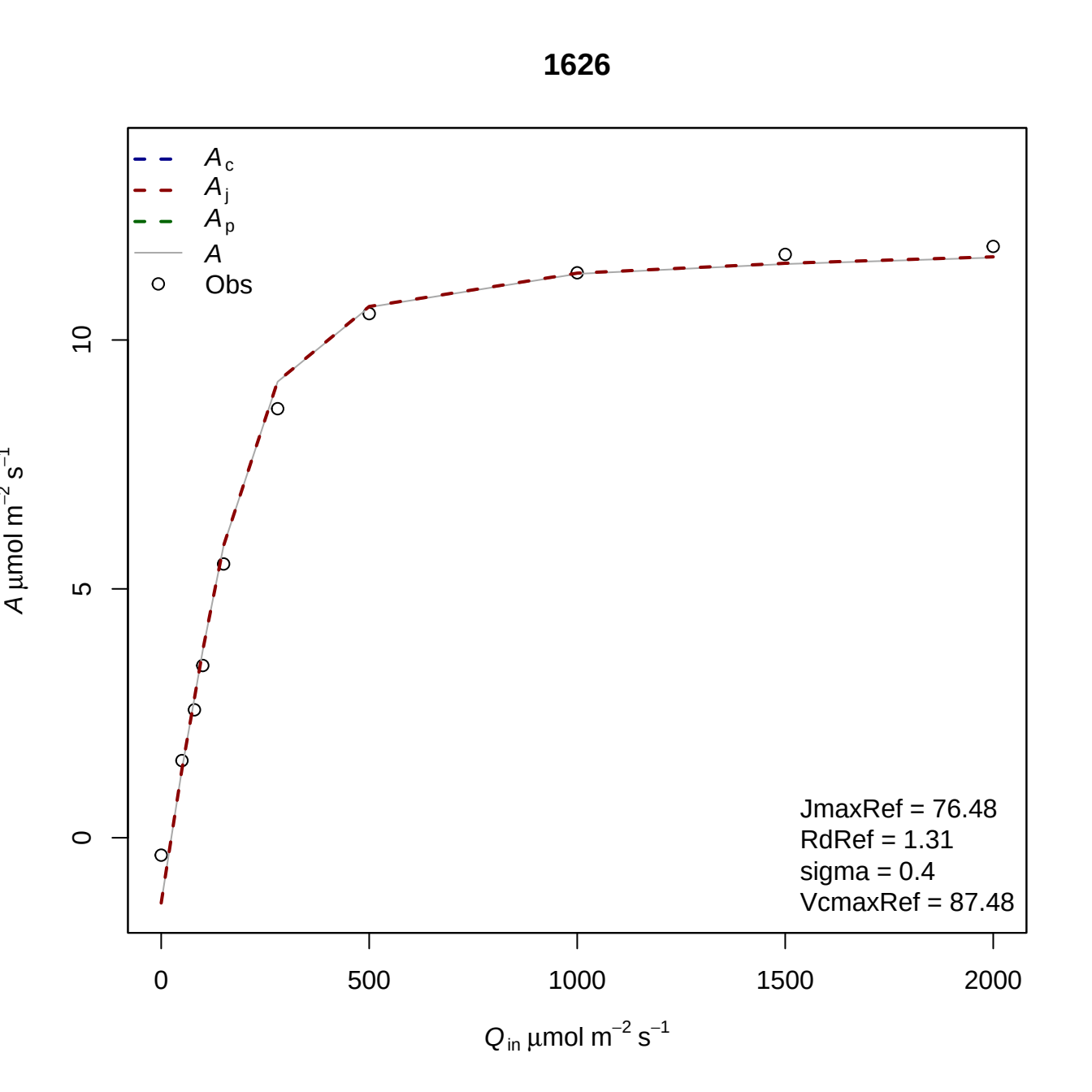

1621
A_c
A_j
A_p
A
Obs
JmaxRef = 196.02
RdRef = 1.55
sigma = 1.17
VcmaxRef = 253.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

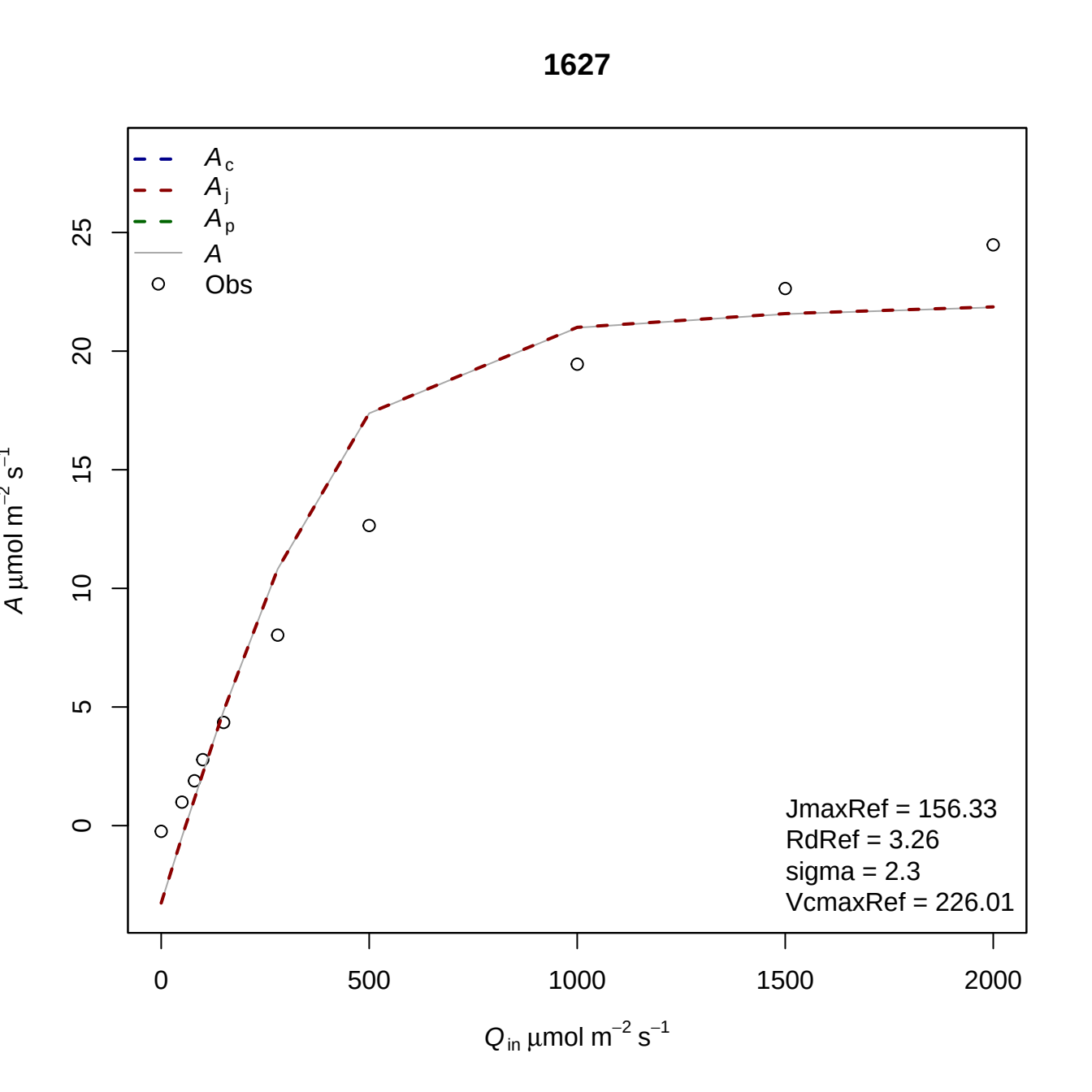

1622
A_c
A_j
A_p
A
Obs
JmaxRef = 207.42
RdRef = 1.03
sigma = 0.57
VcmaxRef = 255.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

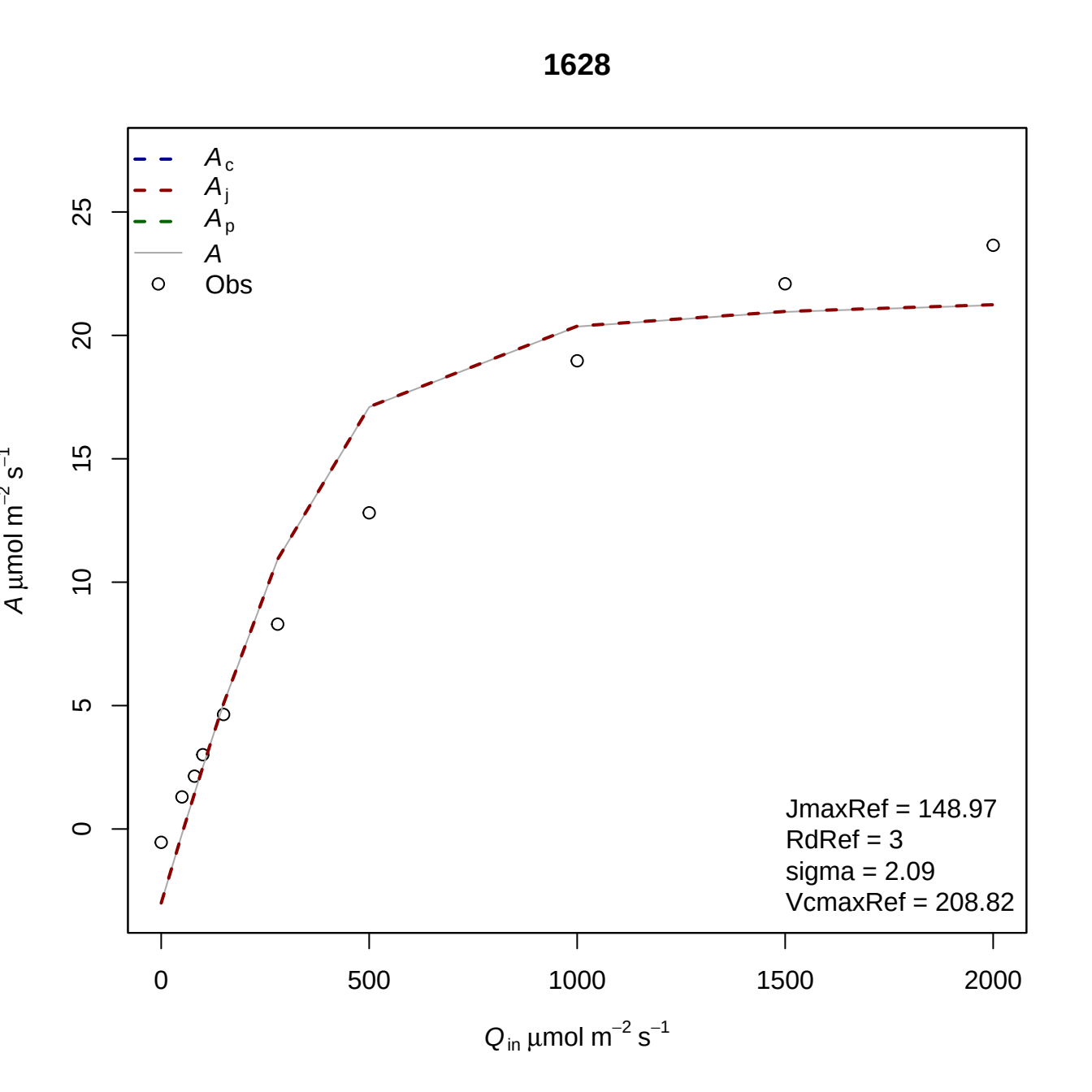

1623
A_c
A_j
A_p
A
Obs
JmaxRef = 80.25
RdRef = 1.82
sigma = 0.91
VcmaxRef = 88.34
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

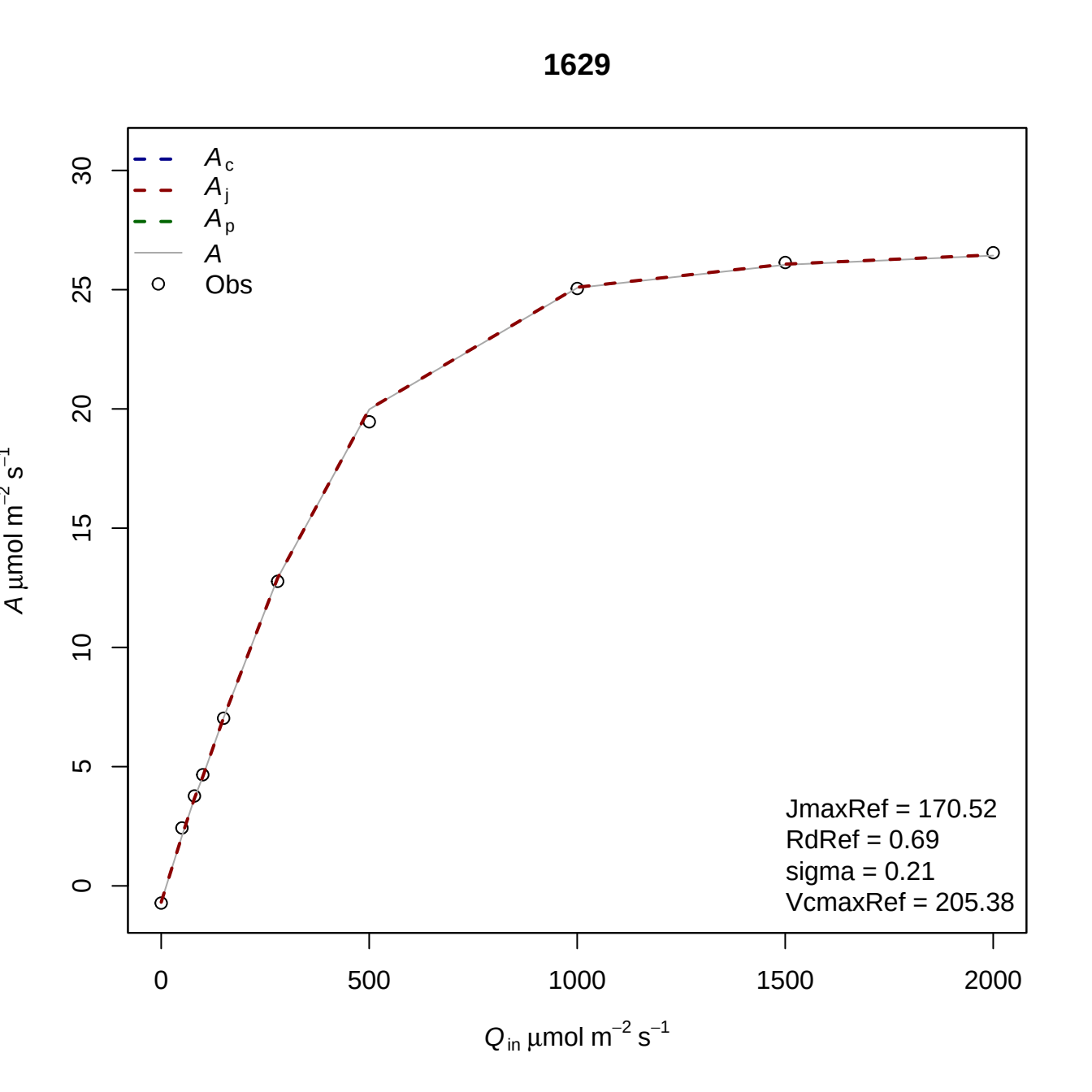

1624
A_c
A_j
A_p
A
Obs
JmaxRef = 77.63
RdRef = 0.61
sigma = 0.22
VcmaxRef = 93.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

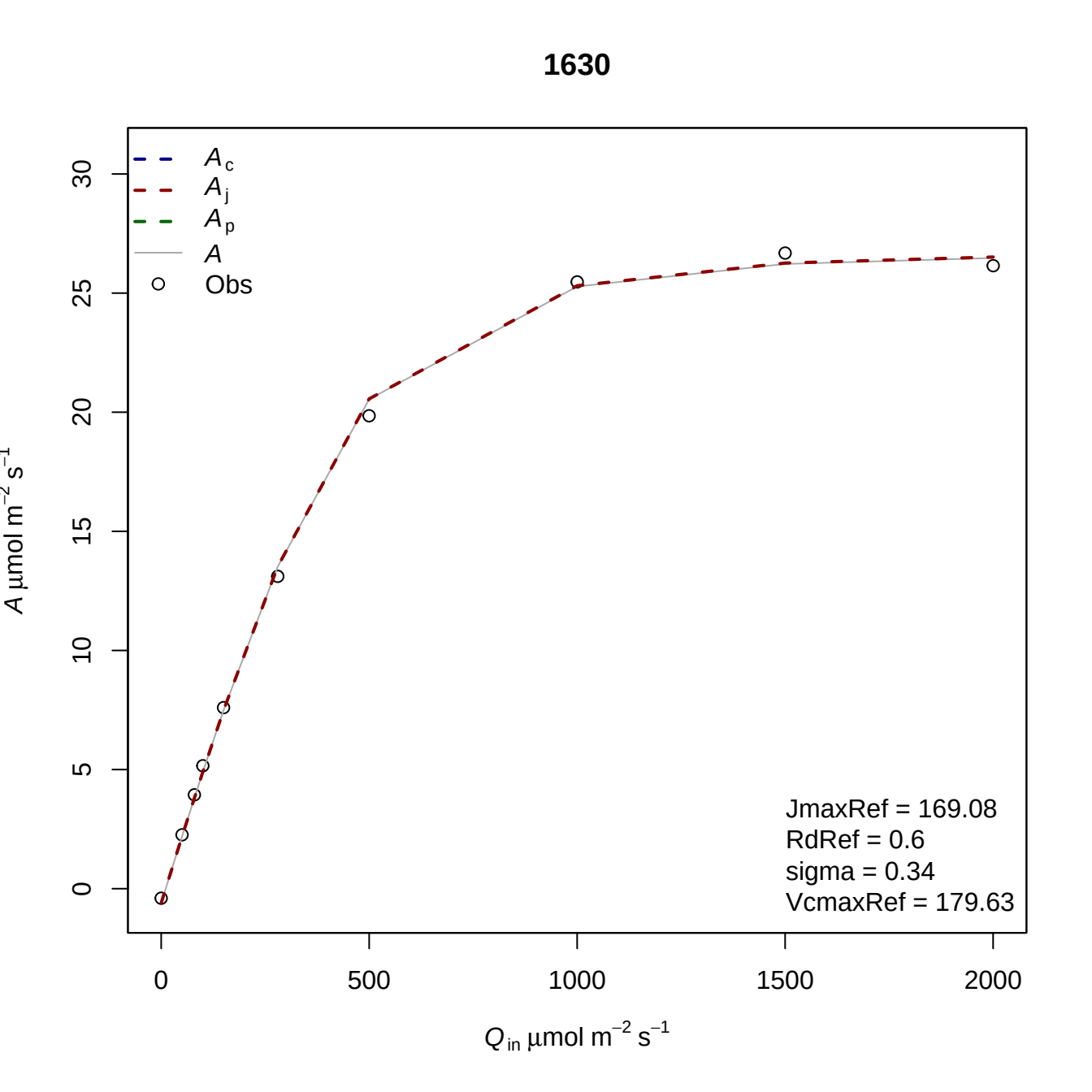

1625
A_c
A_j
A_p
A
Obs
JmaxRef = 67.76
RdRef = 1.66
sigma = 0.68
VcmaxRef = 81.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

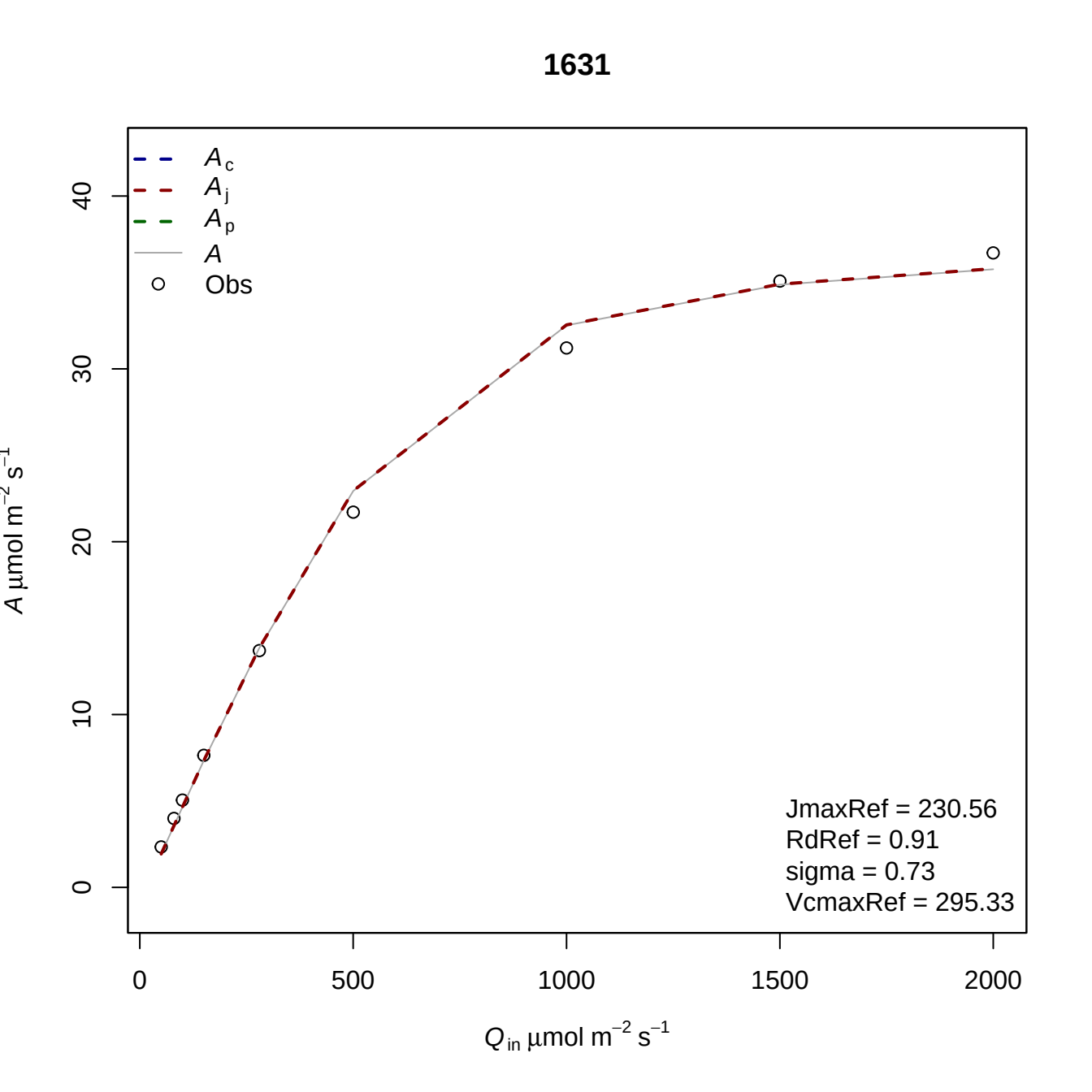

1626
A_c
A_j
A_p
A
Obs
JmaxRef = 76.48
RdRef = 1.31
sigma = 0.4
VcmaxRef = 87.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

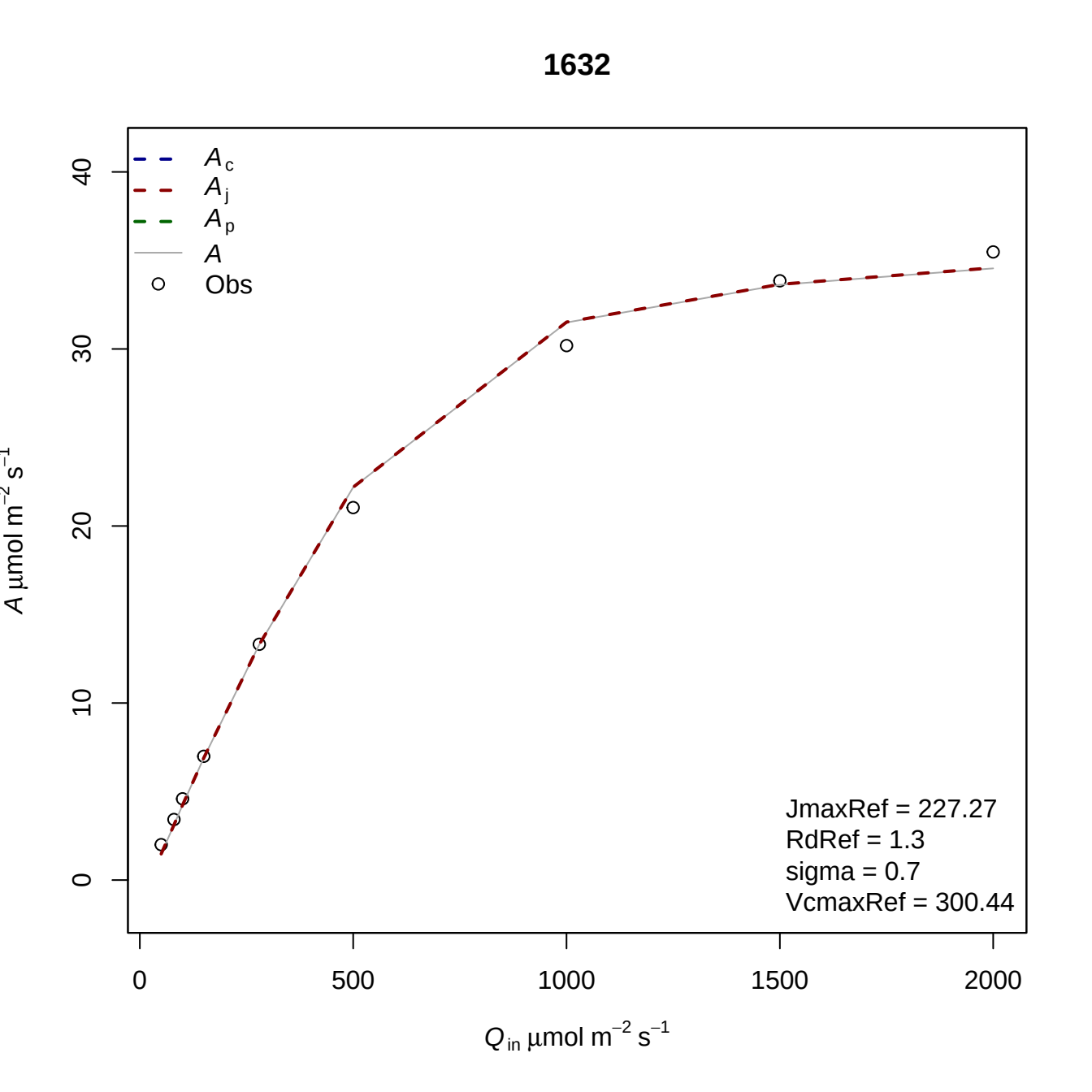

1627
A_c
A_j
A_p
A
Obs
JmaxRef = 156.33
RdRef = 3.26
sigma = 2.3
VcmaxRef = 226.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

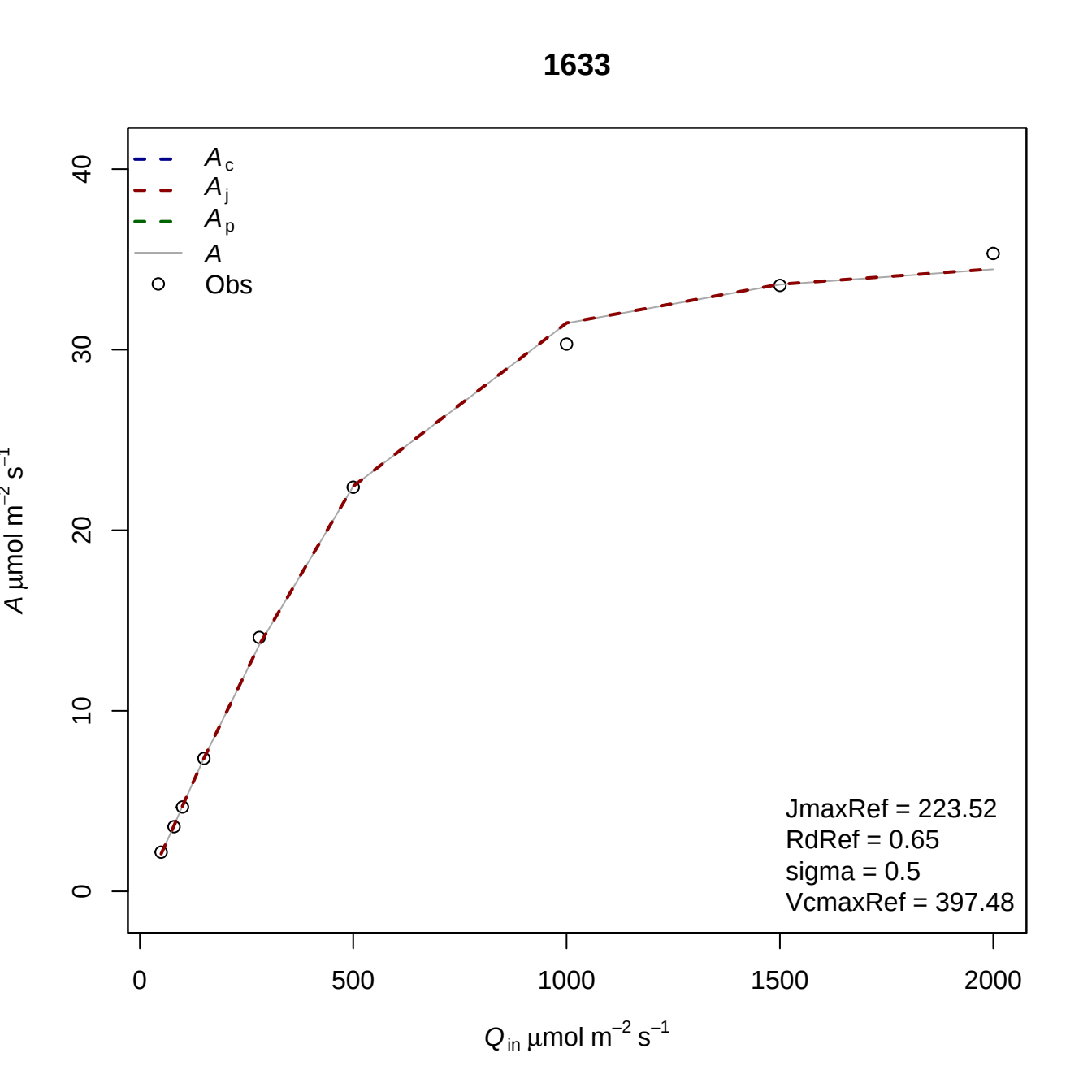

1628
A_c
A_j
A_p
A
Obs
JmaxRef = 148.97
RdRef = 3
sigma = 2.09
VcmaxRef = 208.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

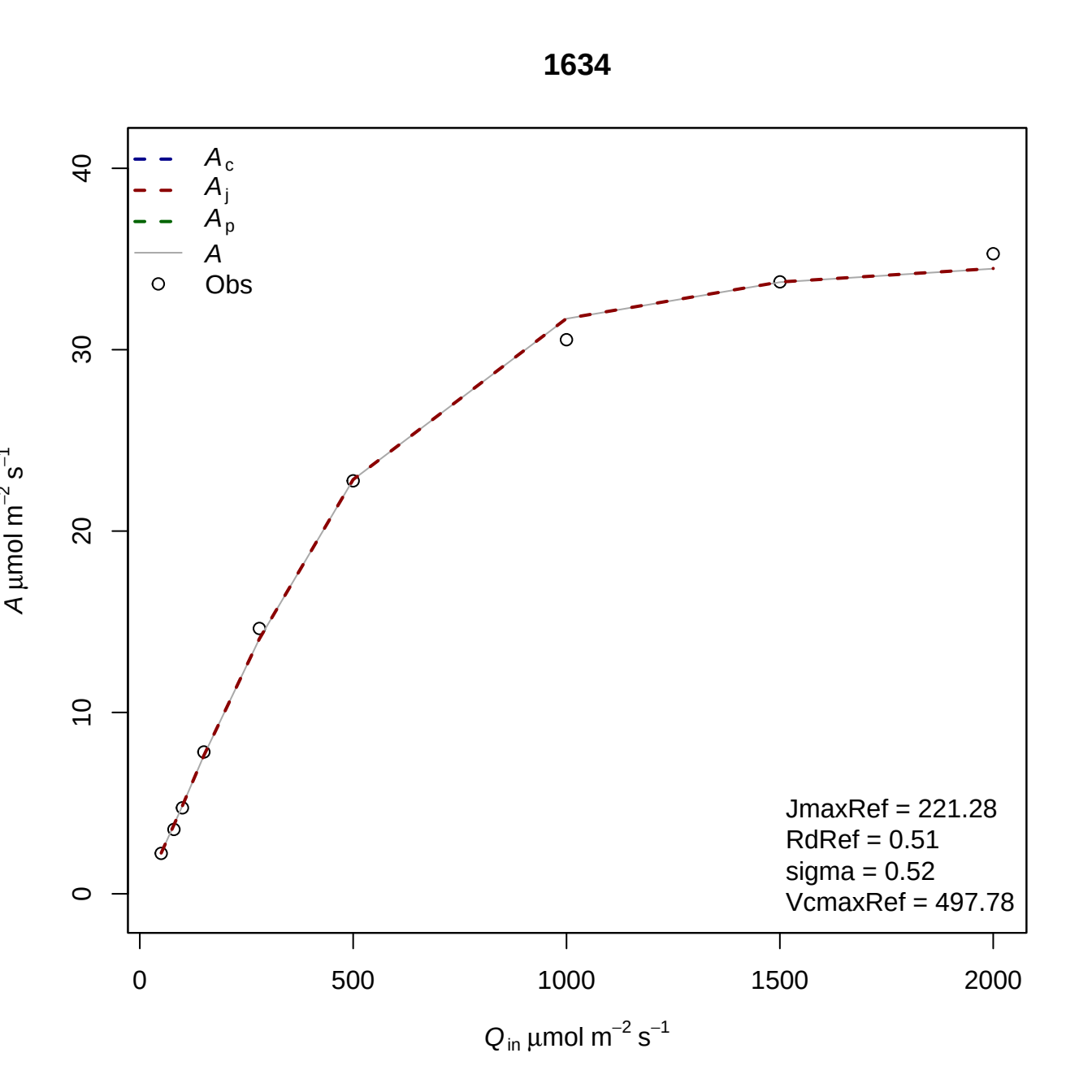

1629
A_c
A_j
A_p
A
Obs
JmaxRef = 170.52
RdRef = 0.69
sigma = 0.21
VcmaxRef = 205.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

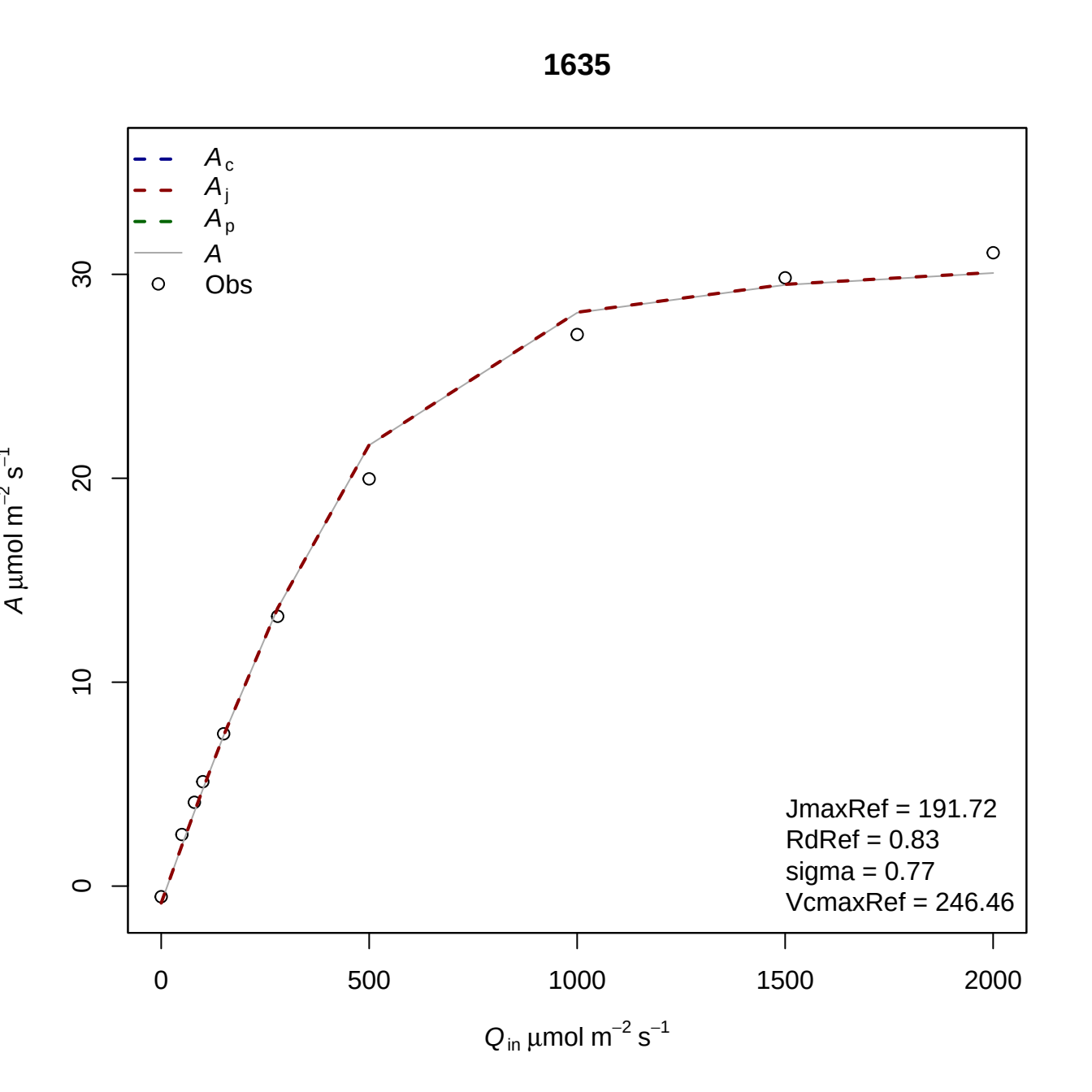

1630
A_c
A_j
A_p
A
Obs
JmaxRef = 169.08
RdRef = 0.6
sigma = 0.34
VcmaxRef = 179.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

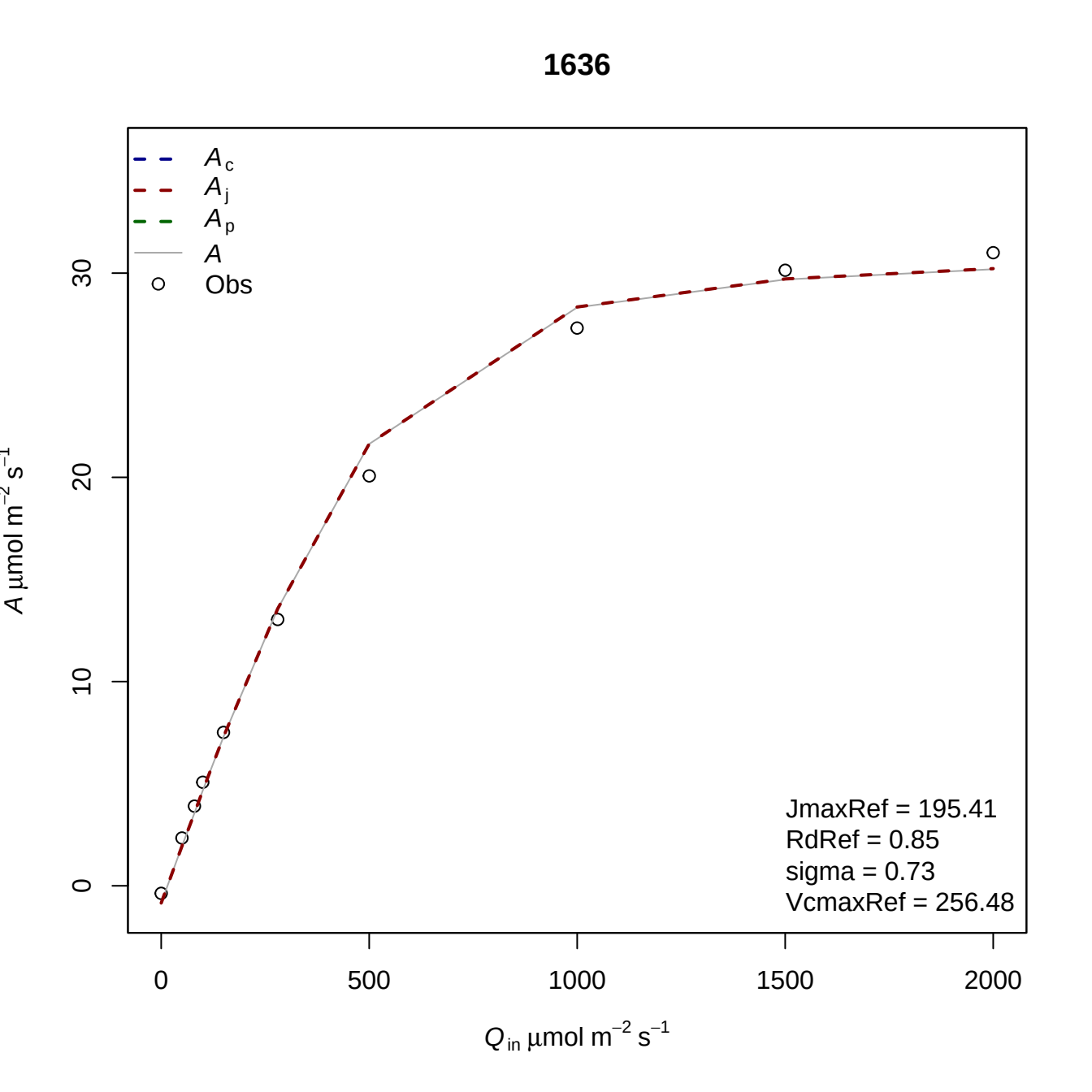

1631
A_c
A_j
A_p
A
Obs
JmaxRef = 230.56
RdRef = 0.91
sigma = 0.73
VcmaxRef = 295.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

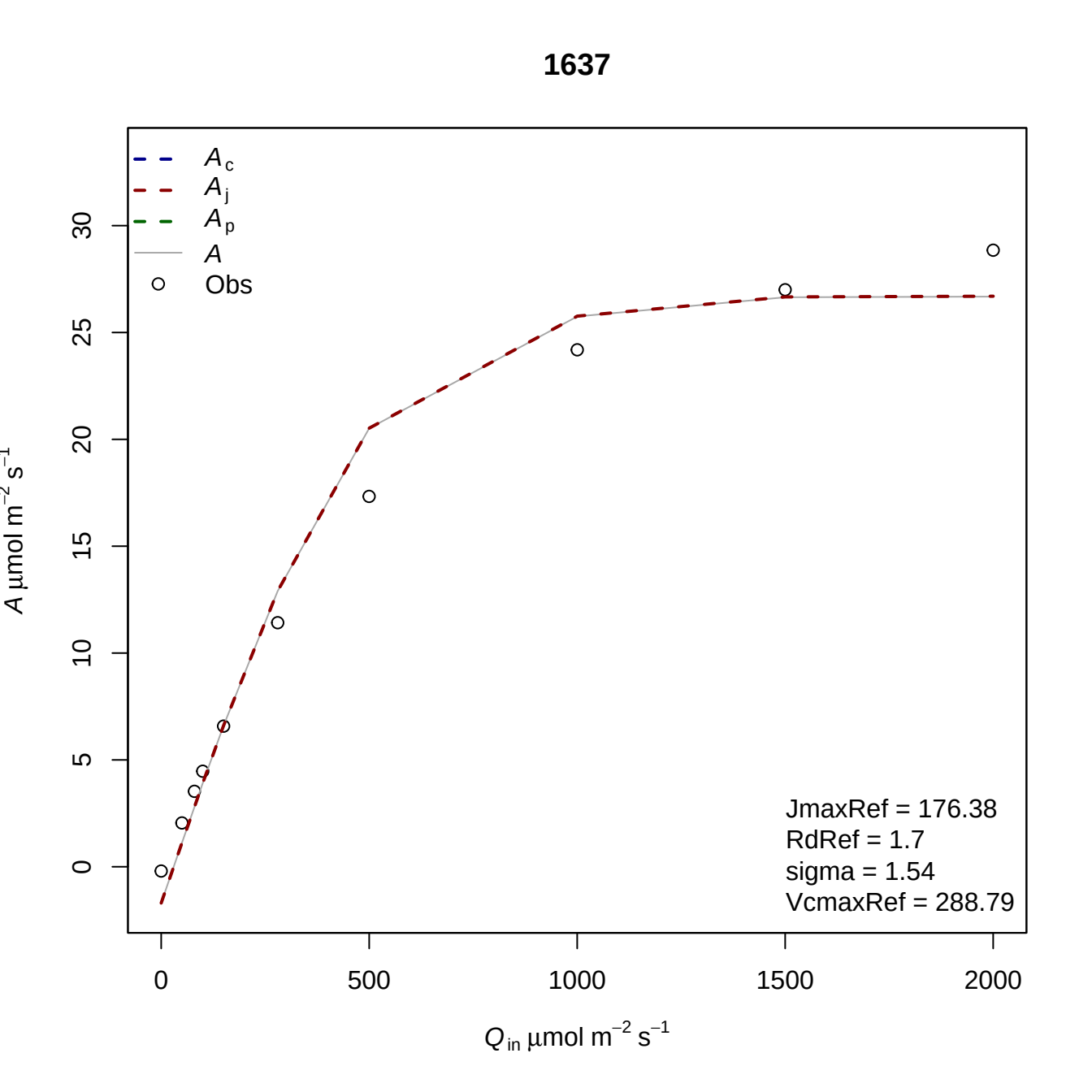

1632
A_c
A_j
A_p
A
Obs
JmaxRef = 227.27
RdRef = 1.3
sigma = 0.7
VcmaxRef = 300.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

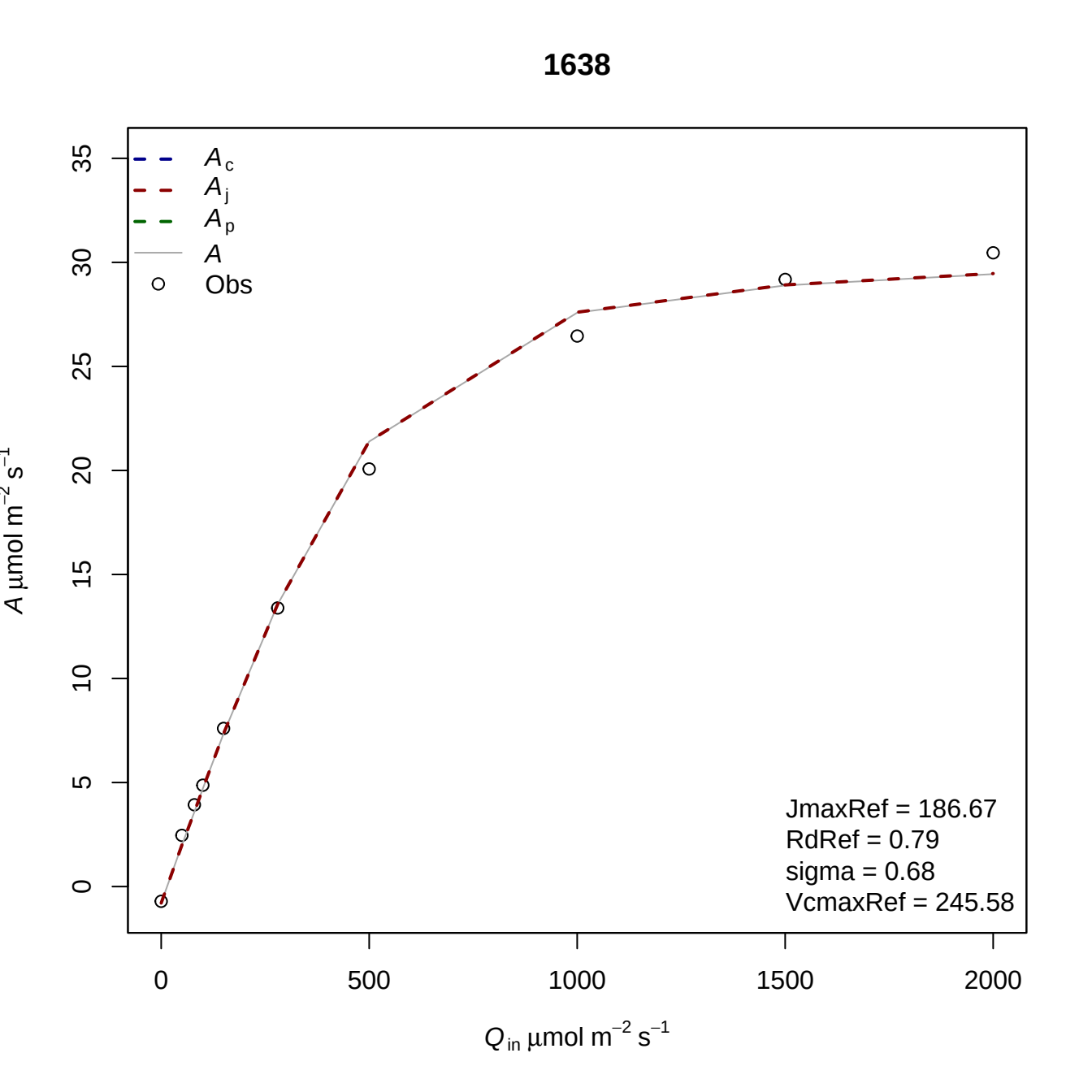

1633
A_c
A_j
A_p
A
Obs
JmaxRef = 223.52
RdRef = 0.65
sigma = 0.5
VcmaxRef = 397.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

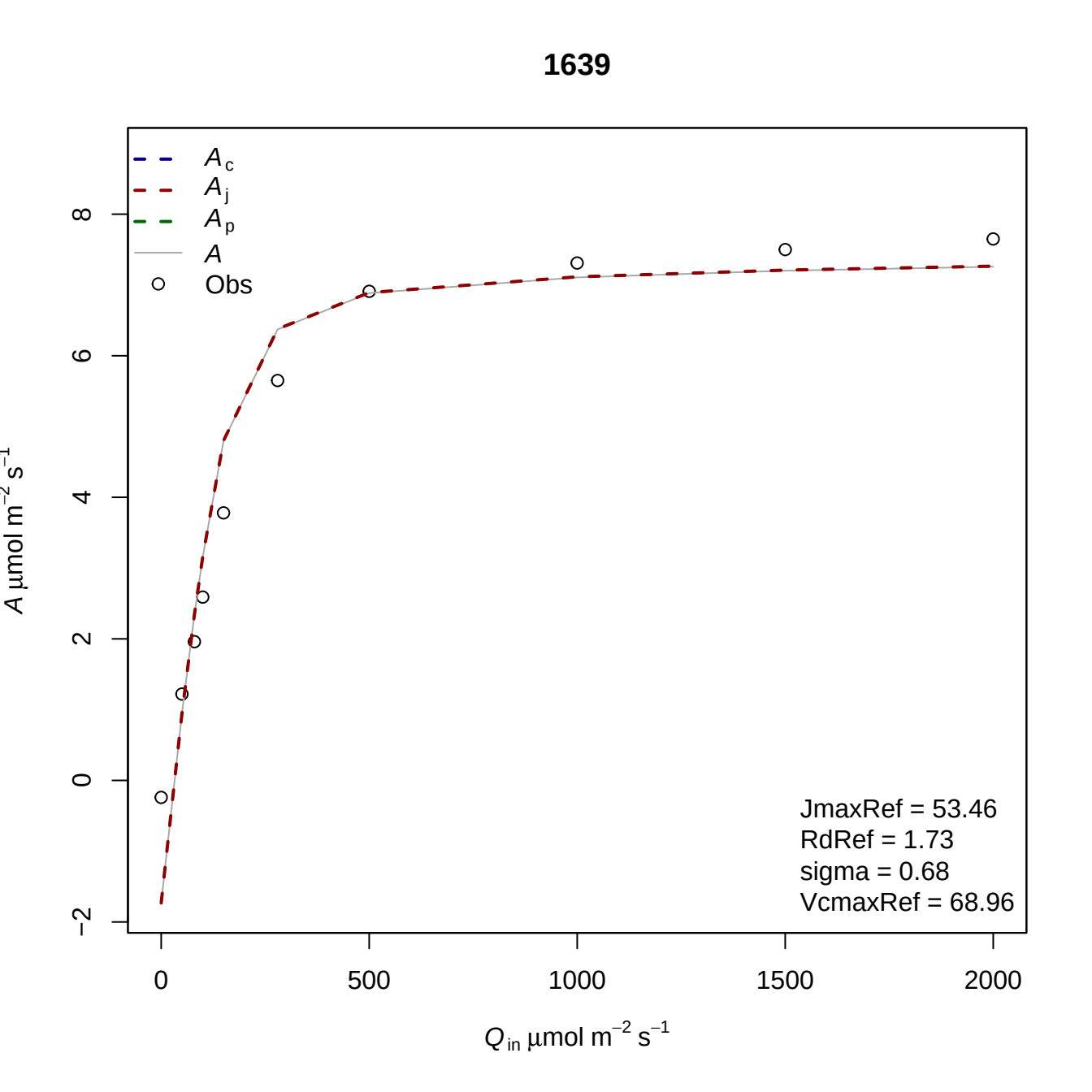

1634
A_c
A_j
A_p
A
Obs
JmaxRef = 221.28
RdRef = 0.51
sigma = 0.52
VcmaxRef = 497.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

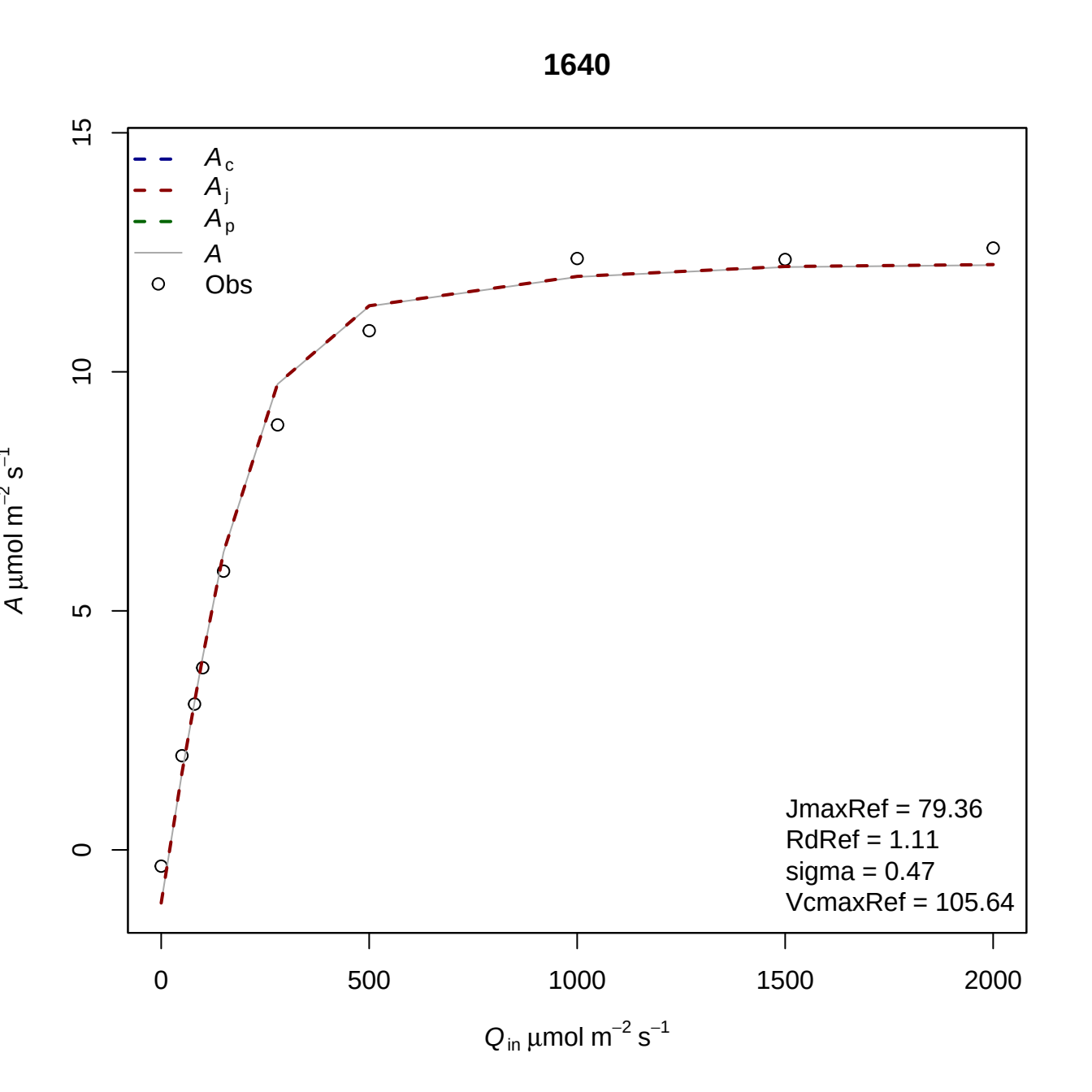

1635
A_c
A_j
A_p
A
Obs
JmaxRef = 191.72
RdRef = 0.83
sigma = 0.77
VcmaxRef = 246.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

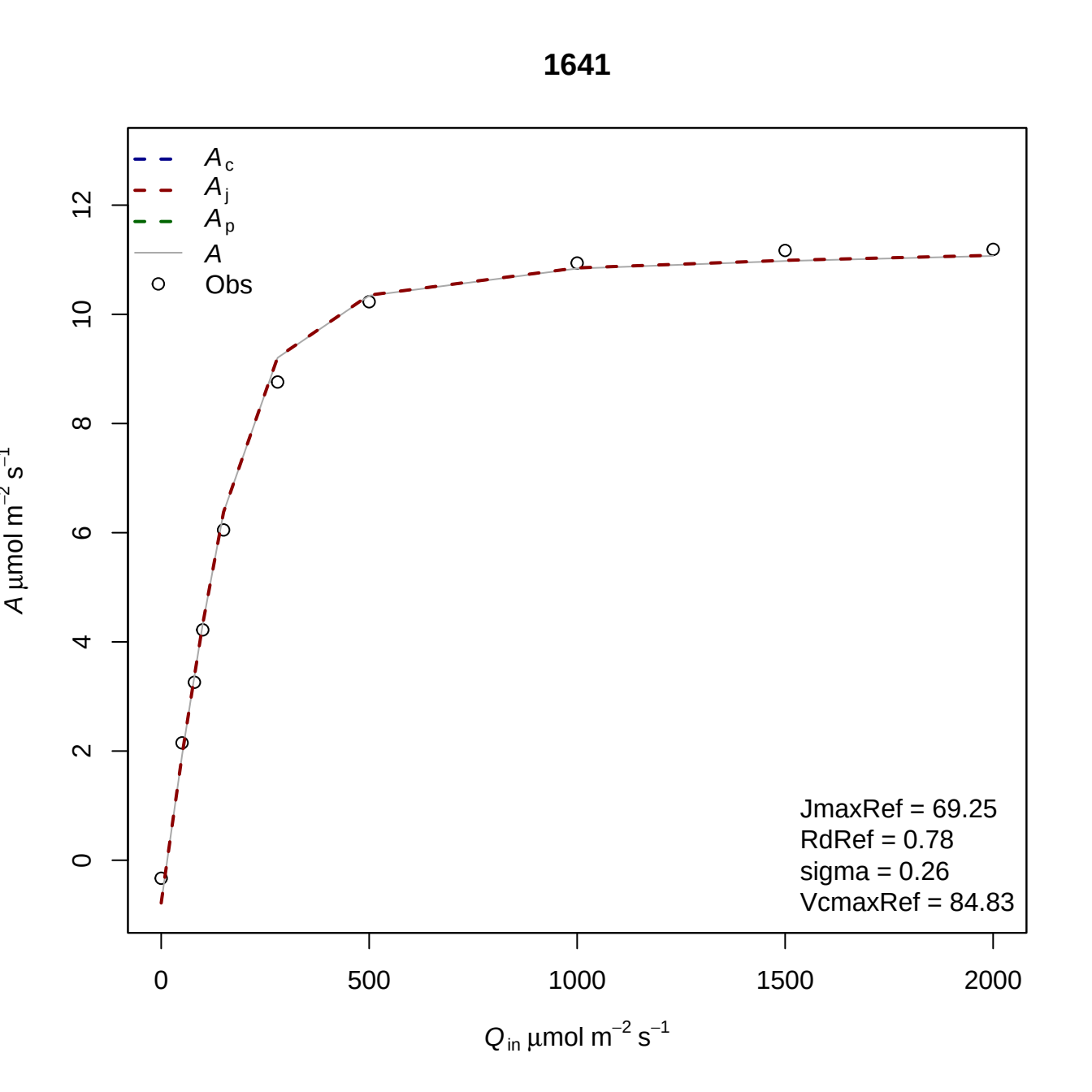

1636
A_c
A_j
A_p
A
Obs
JmaxRef = 195.41
RdRef = 0.85
sigma = 0.73
VcmaxRef = 256.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

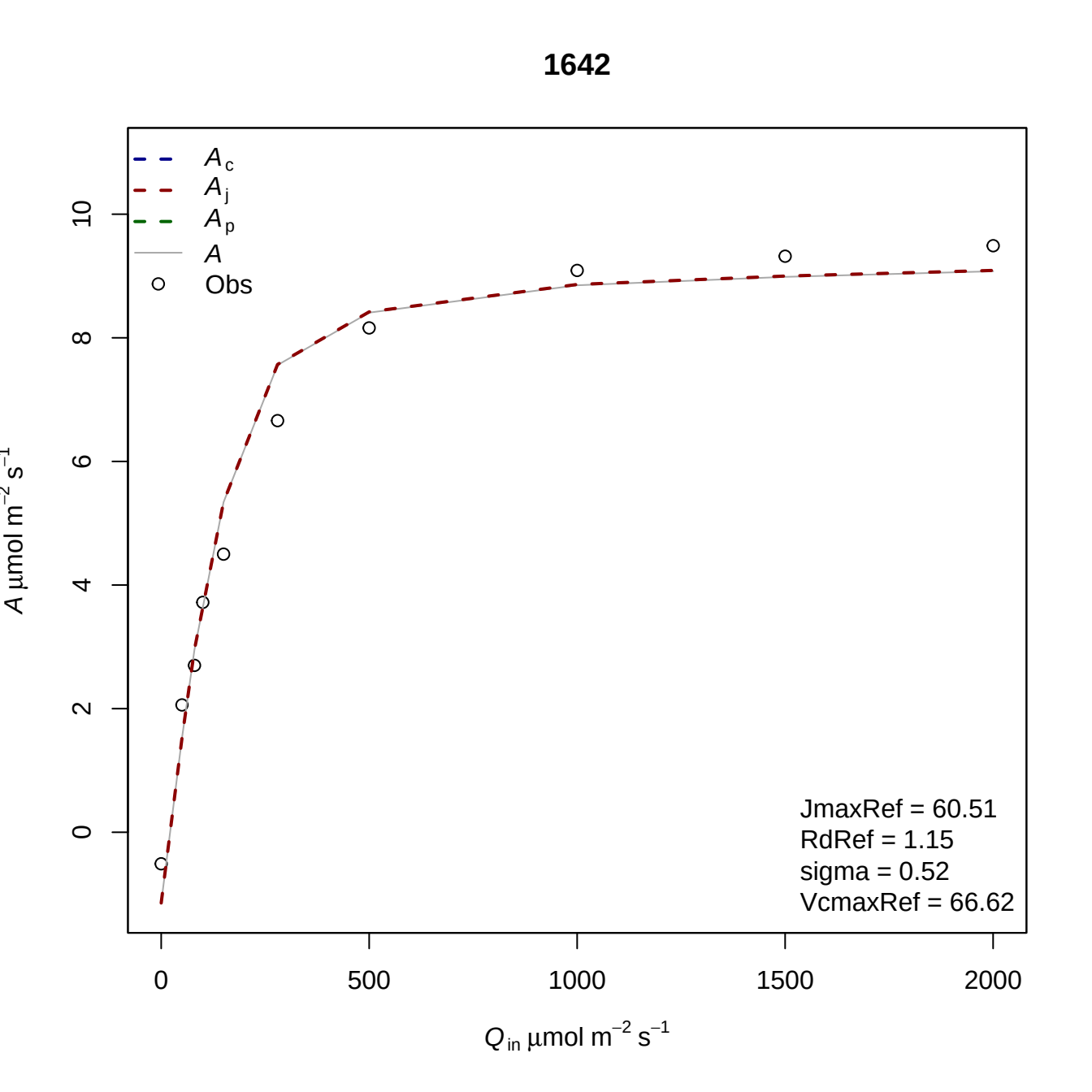

1637
A_c
A_j
A_p
A
Obs
JmaxRef = 176.38
RdRef = 1.7
sigma = 1.54
VcmaxRef = 288.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

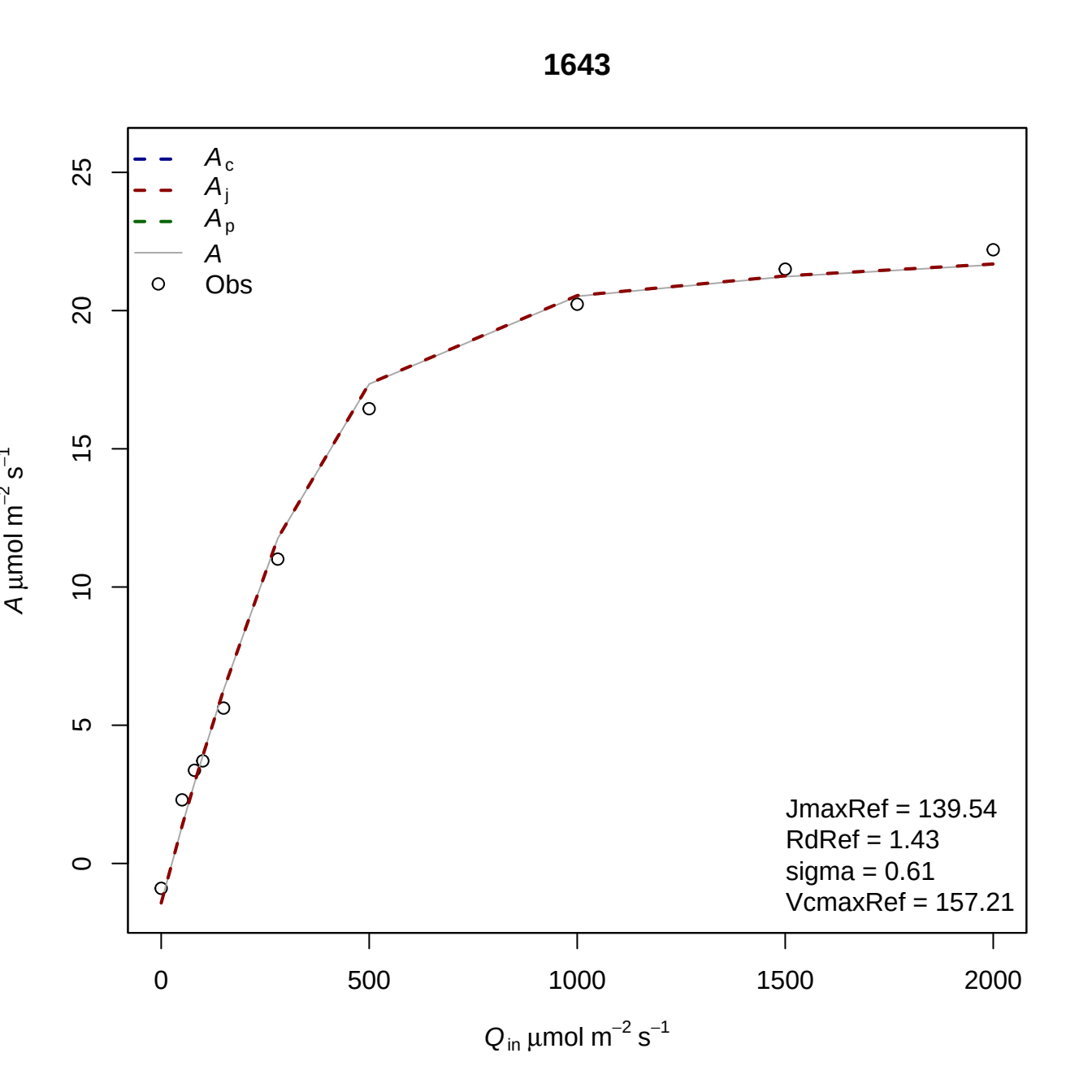

1638
A_c
A_j
A_p
A
Obs
JmaxRef = 186.67
RdRef = 0.79
sigma = 0.68
VcmaxRef = 245.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

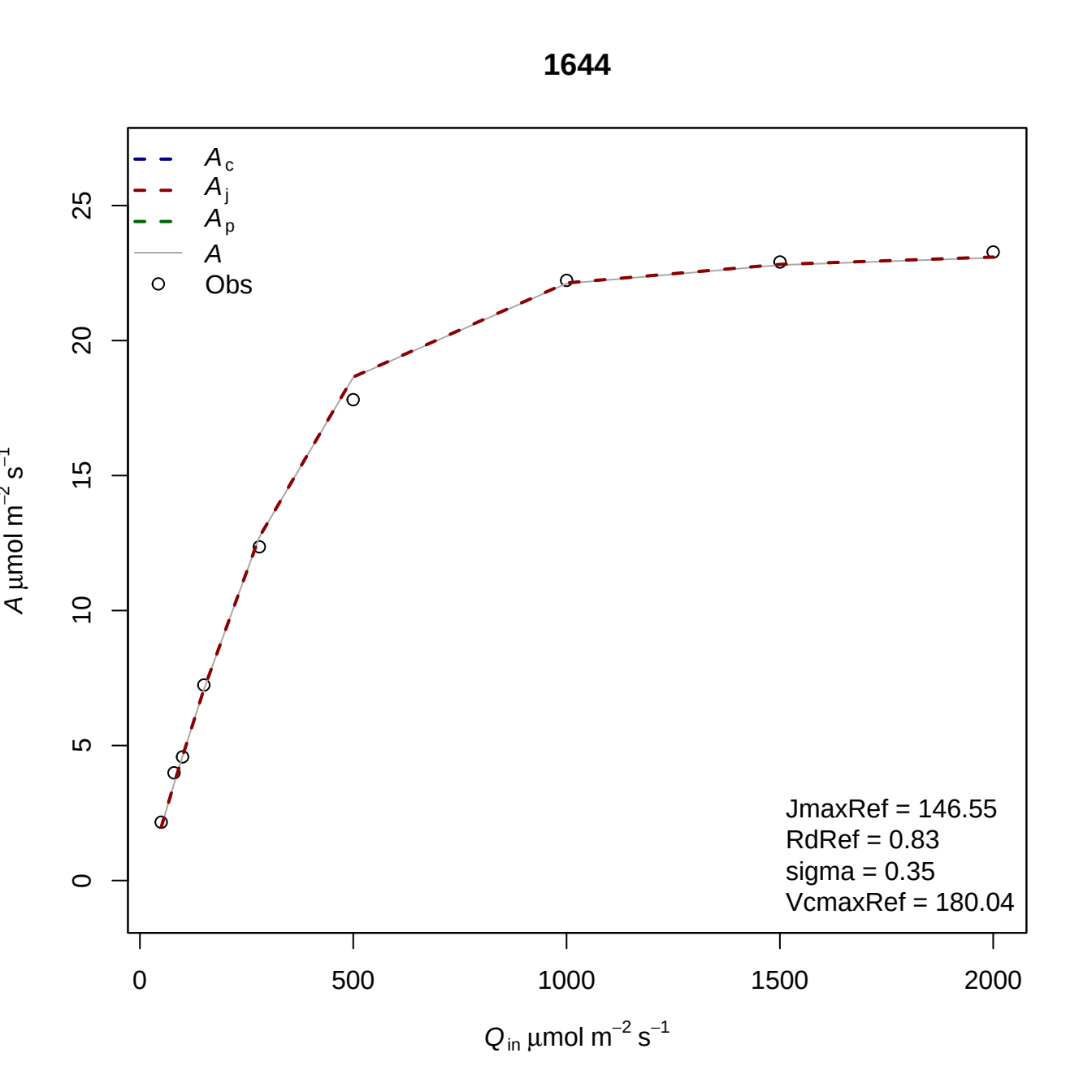

1639
A_c
A_j
A_p
A
Obs
JmaxRef = 53.46
RdRef = 1.73
sigma = 0.68
VcmaxRef = 68.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

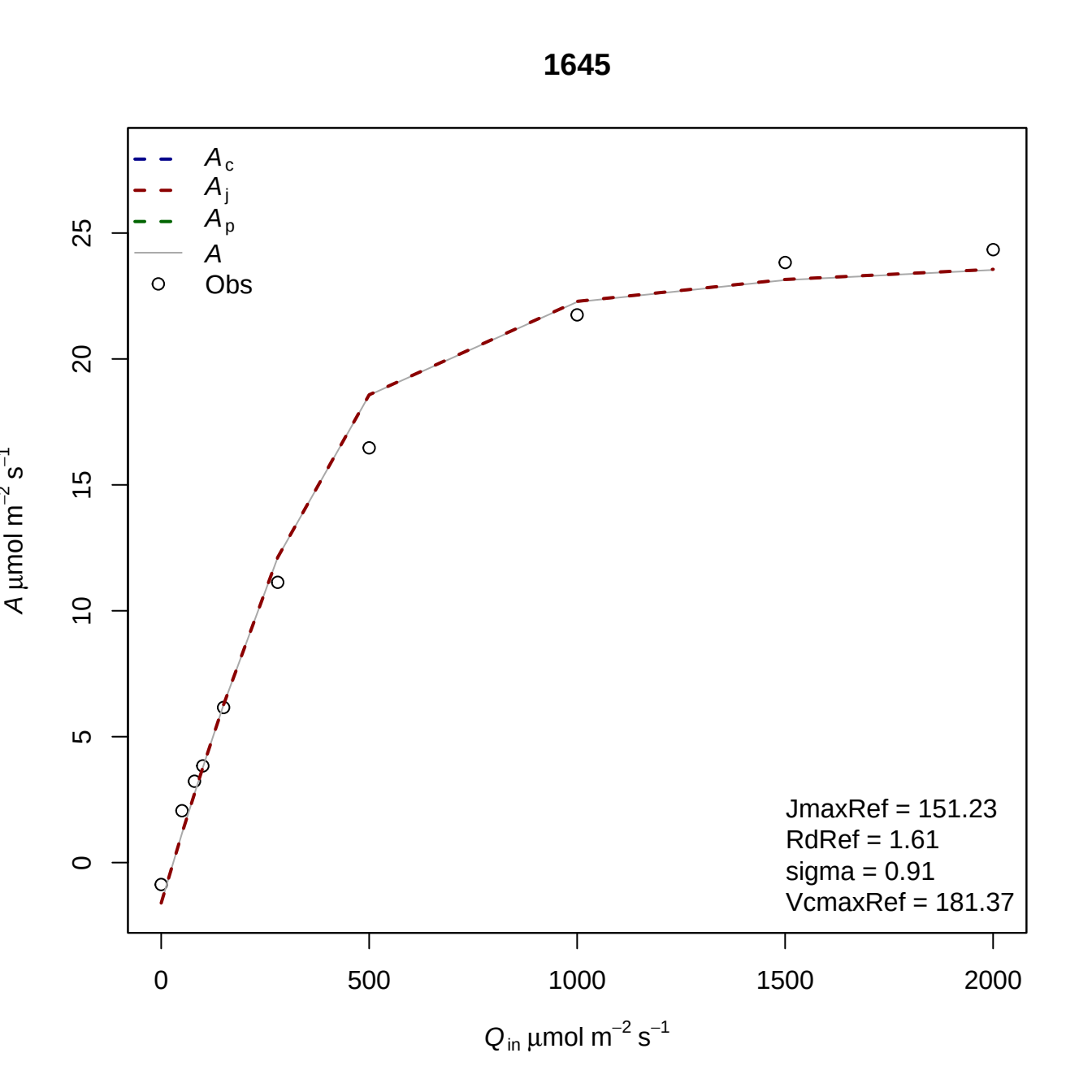

1640
A_c
A_j
A_p
A
Obs
JmaxRef = 79.36
RdRef = 1.11
sigma = 0.47
VcmaxRef = 105.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

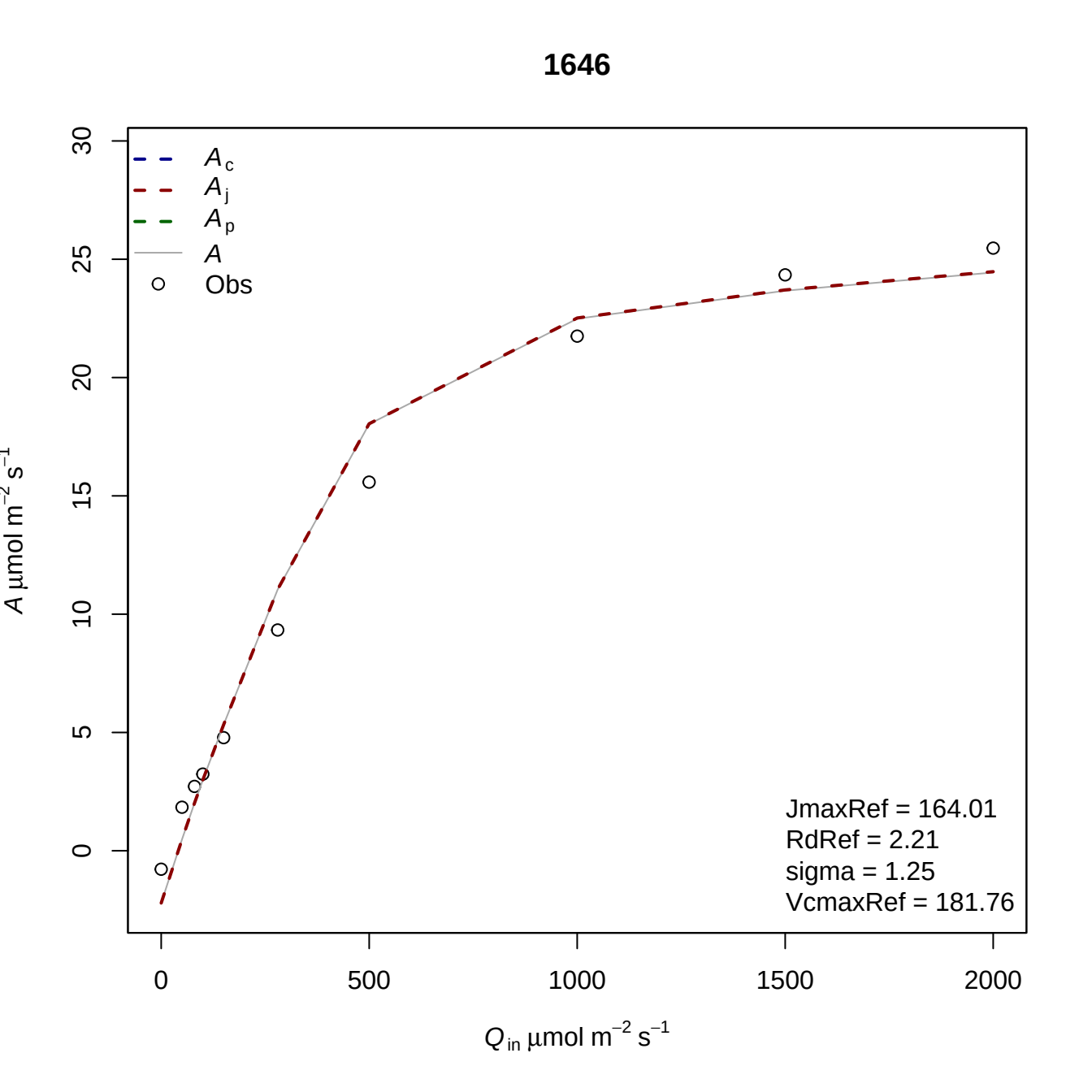

1641
A_c
A_j
A_p
A
Obs
JmaxRef = 69.25
RdRef = 0.78
sigma = 0.26
VcmaxRef = 84.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

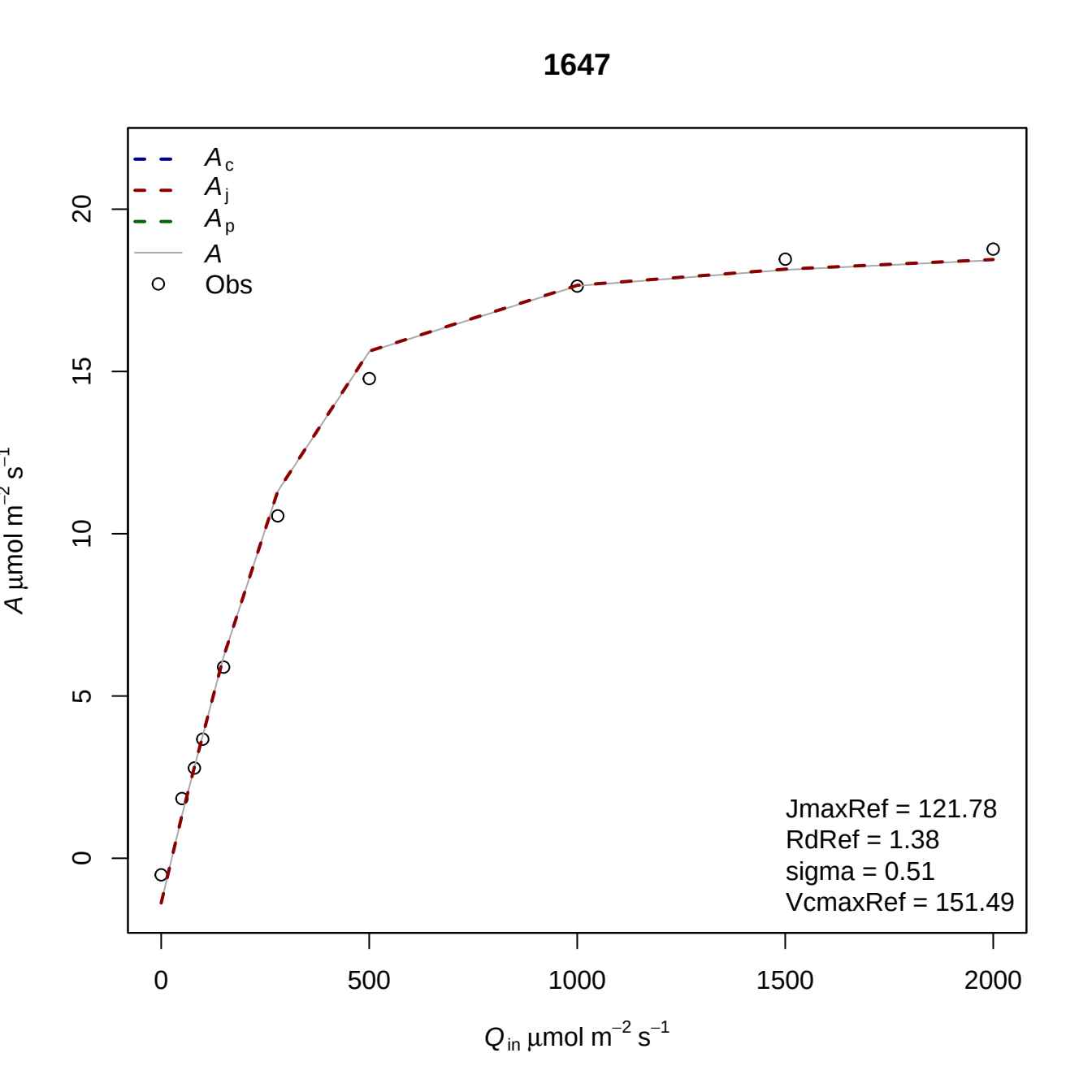

1642
A_c
A_j
A_p
A
Obs
JmaxRef = 60.51
RdRef = 1.15
sigma = 0.52
VcmaxRef = 66.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

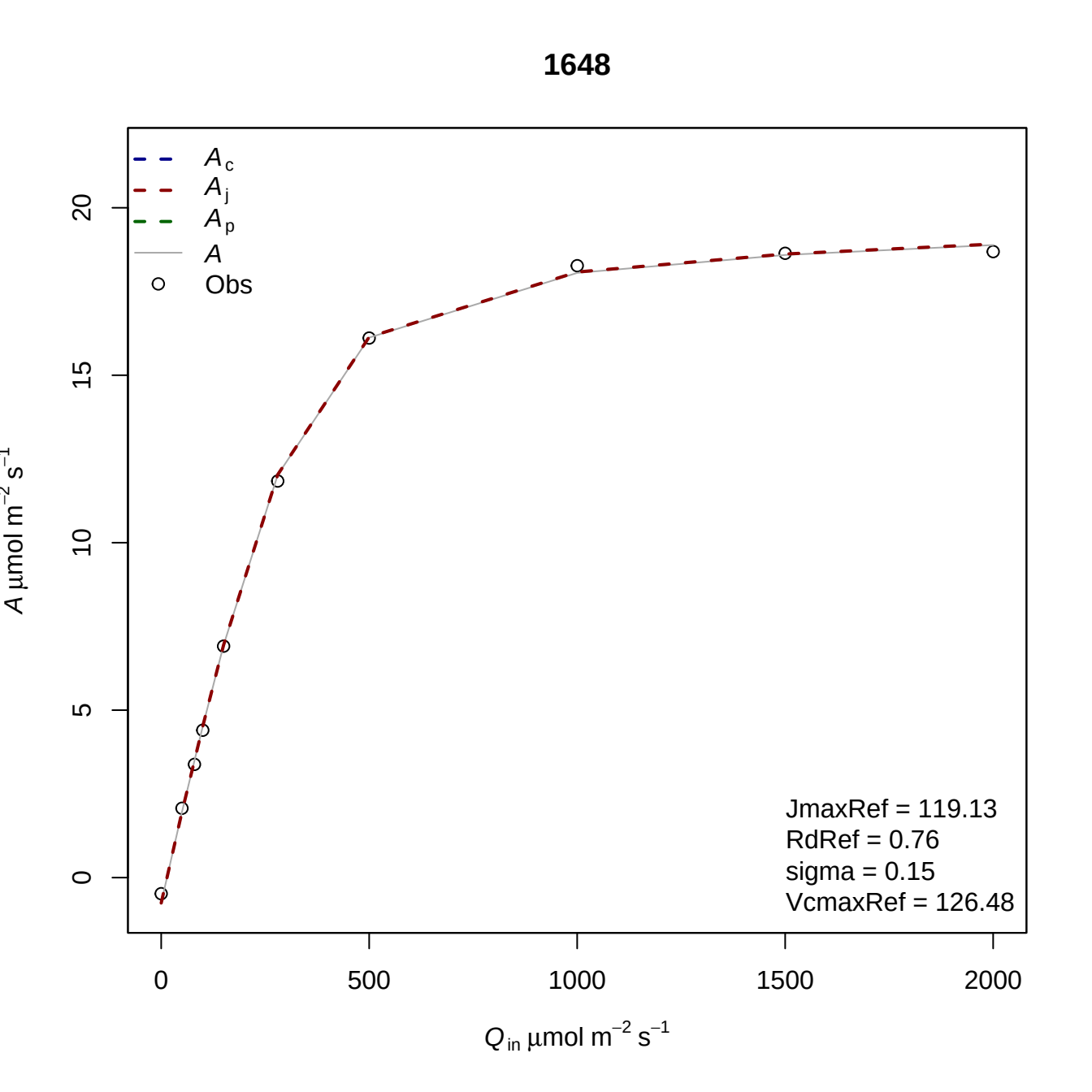

1643
A_c
A_j
A_p
A
Obs
JmaxRef = 139.54
RdRef = 1.43
sigma = 0.61
VcmaxRef = 157.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

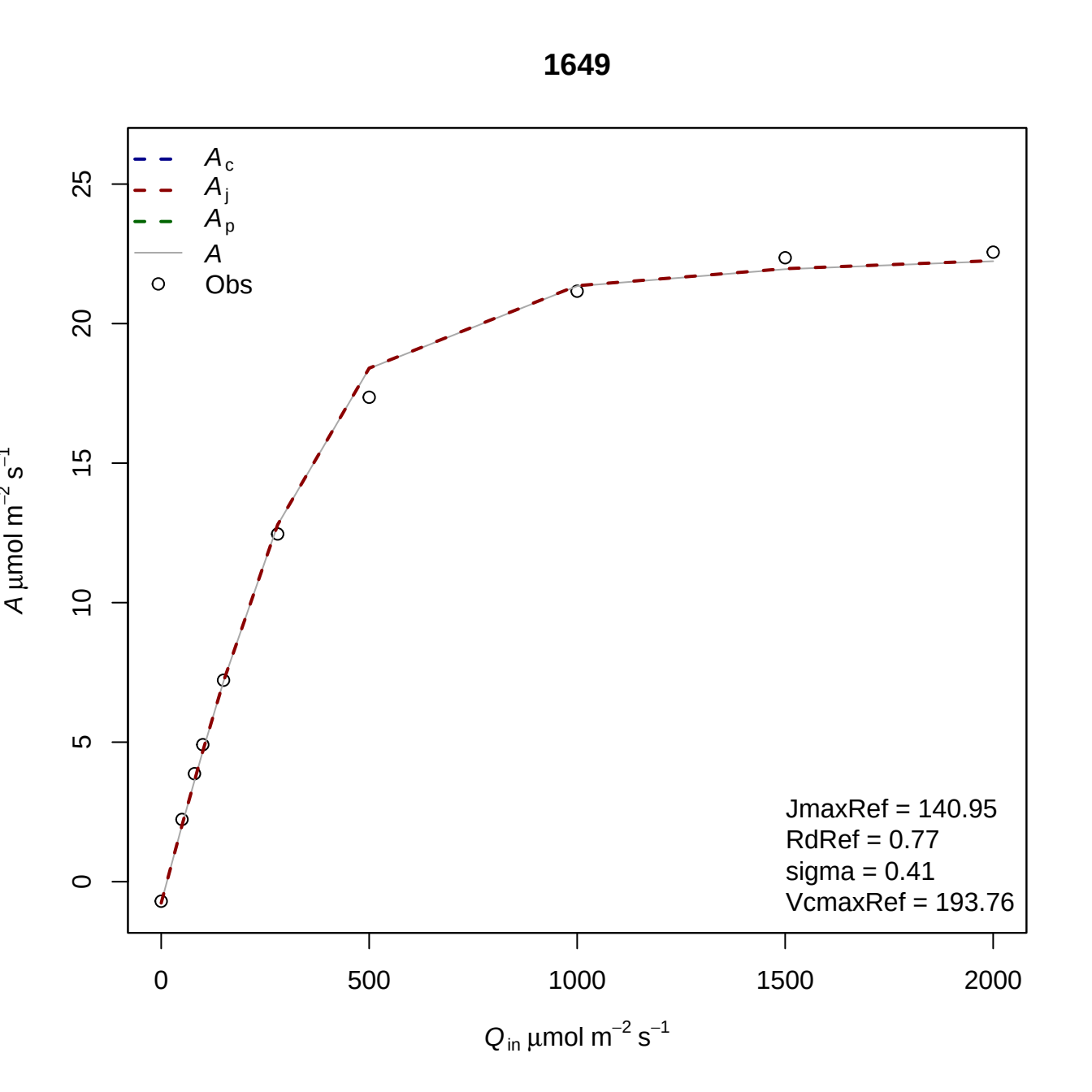

1644
A_c
A_j
A_p
A
Obs
JmaxRef = 146.55
RdRef = 0.83
sigma = 0.35
VcmaxRef = 180.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

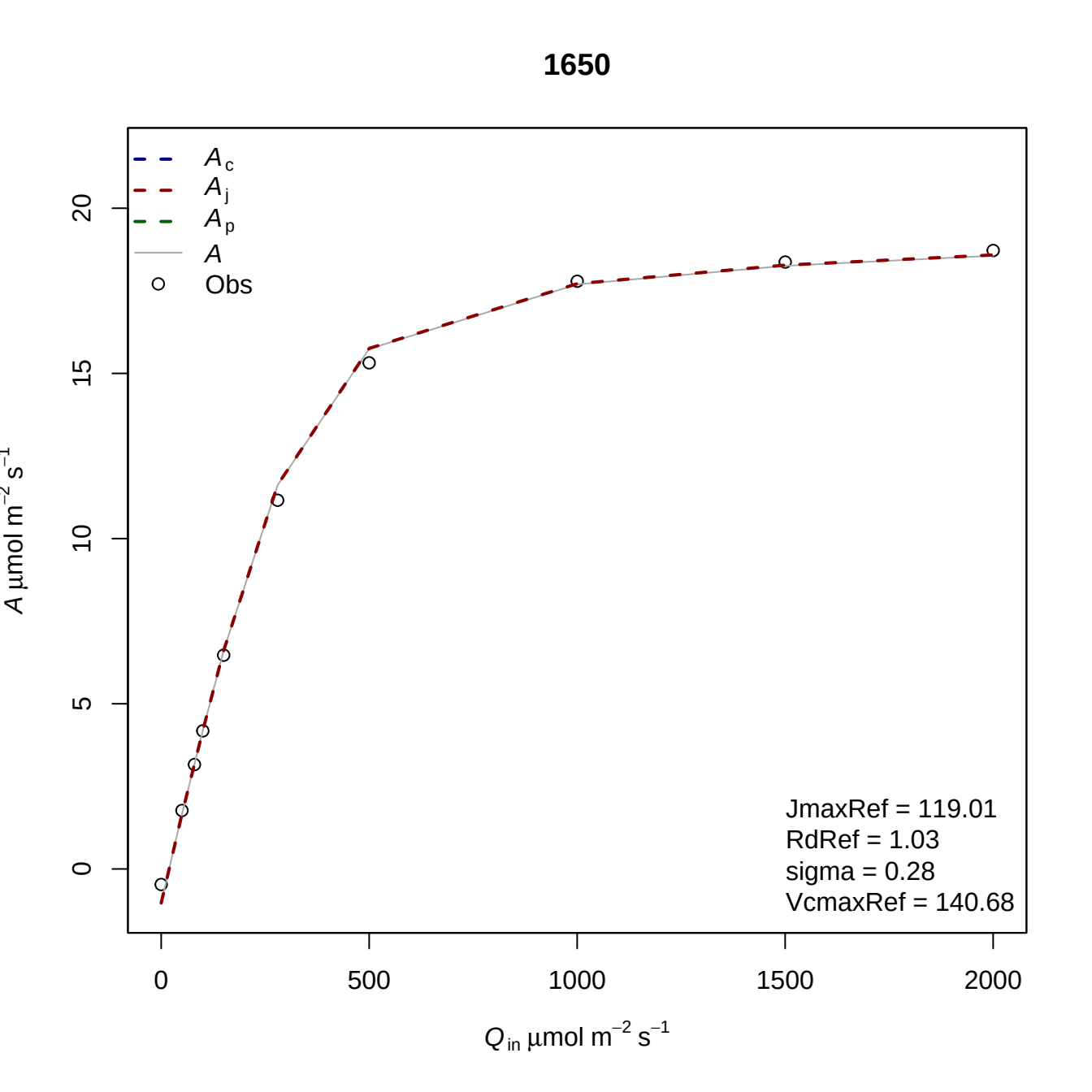

1645
A_c
A_j
A_p
A
Obs
JmaxRef = 151.23
RdRef = 1.61
sigma = 0.91
VcmaxRef = 181.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

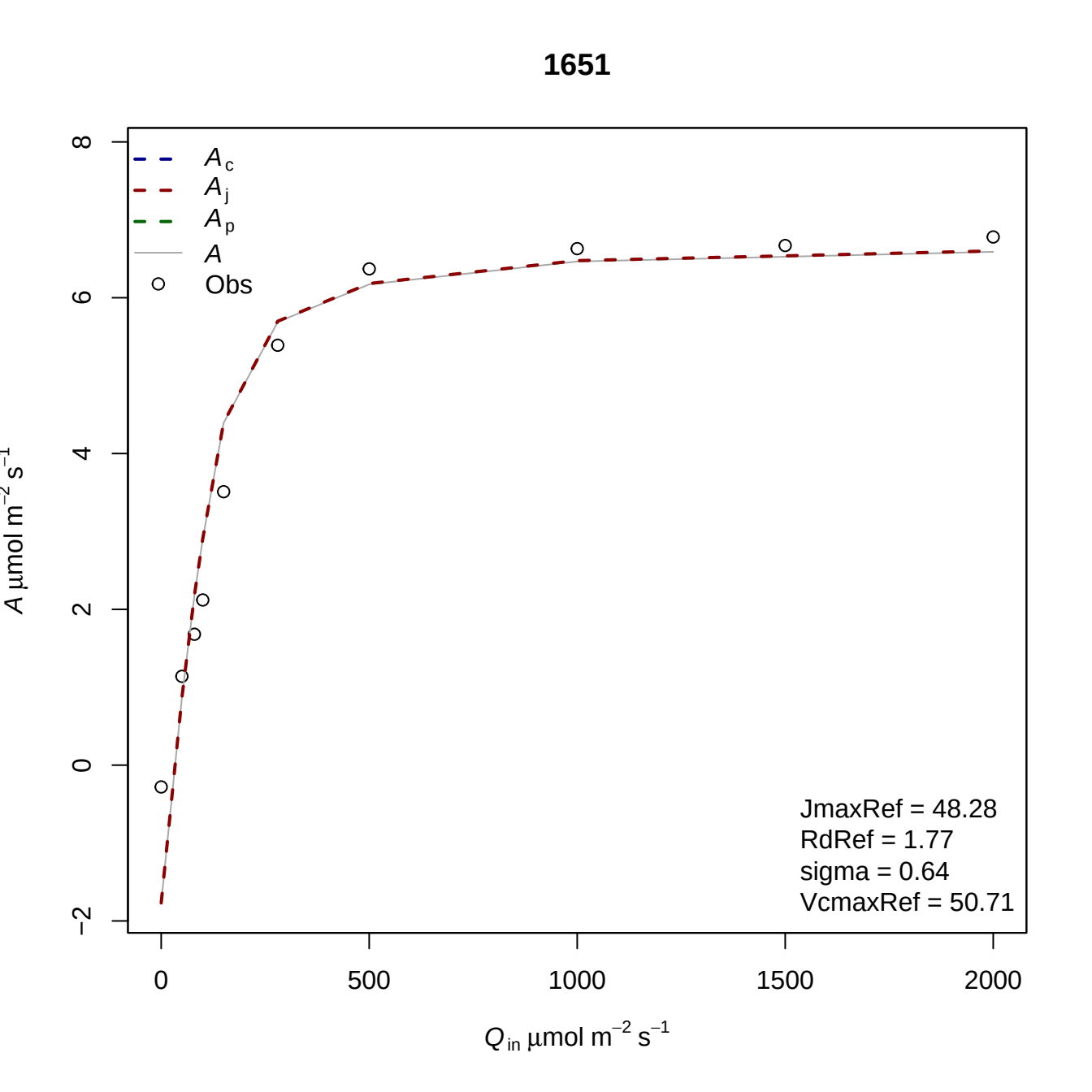

1646
A_c
A_j
A_p
A
Obs
JmaxRef = 164.01
RdRef = 2.21
sigma = 1.25
VcmaxRef = 181.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

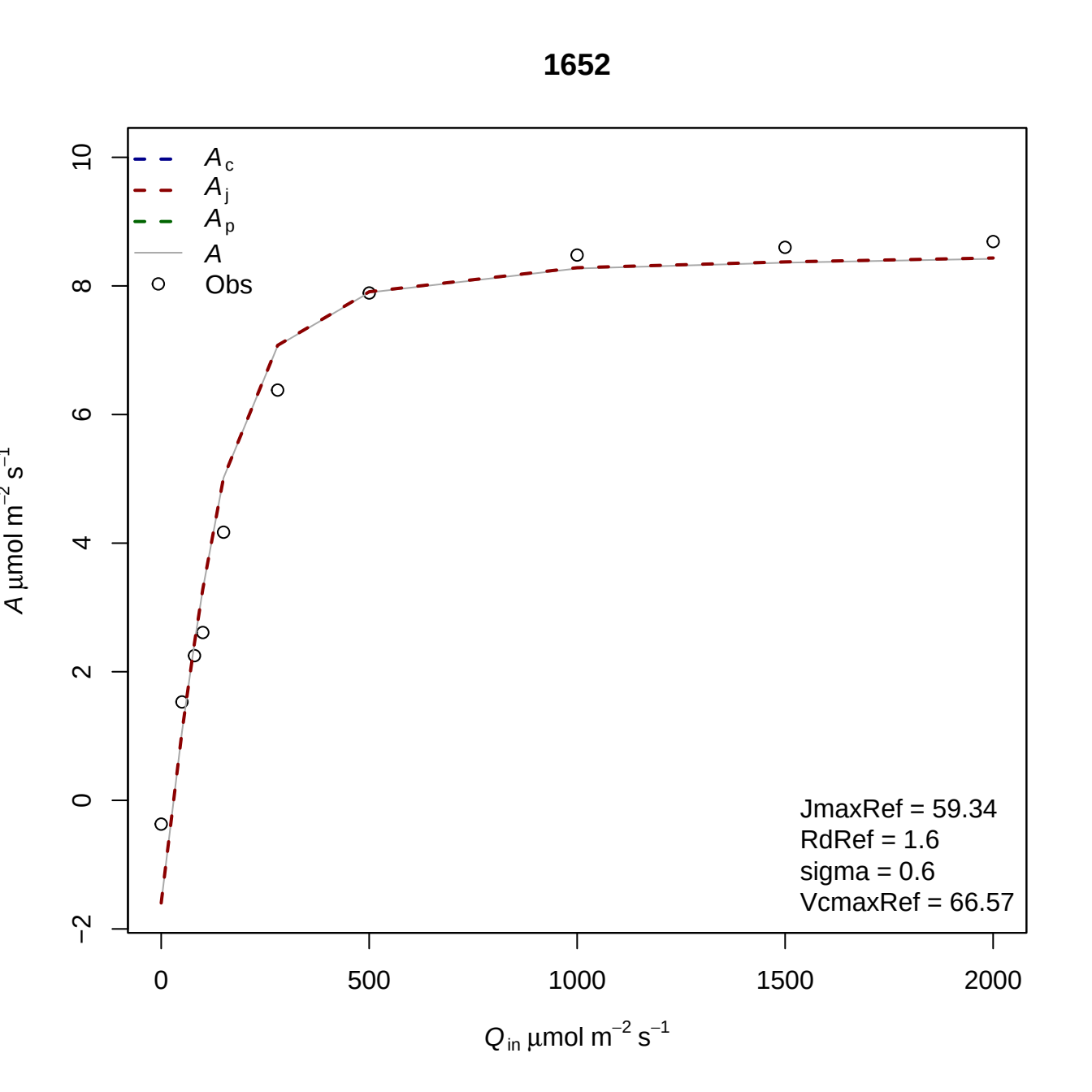

1647
A_c
A_j
A_p
A
Obs
JmaxRef = 121.78
RdRef = 1.38
sigma = 0.51
VcmaxRef = 151.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

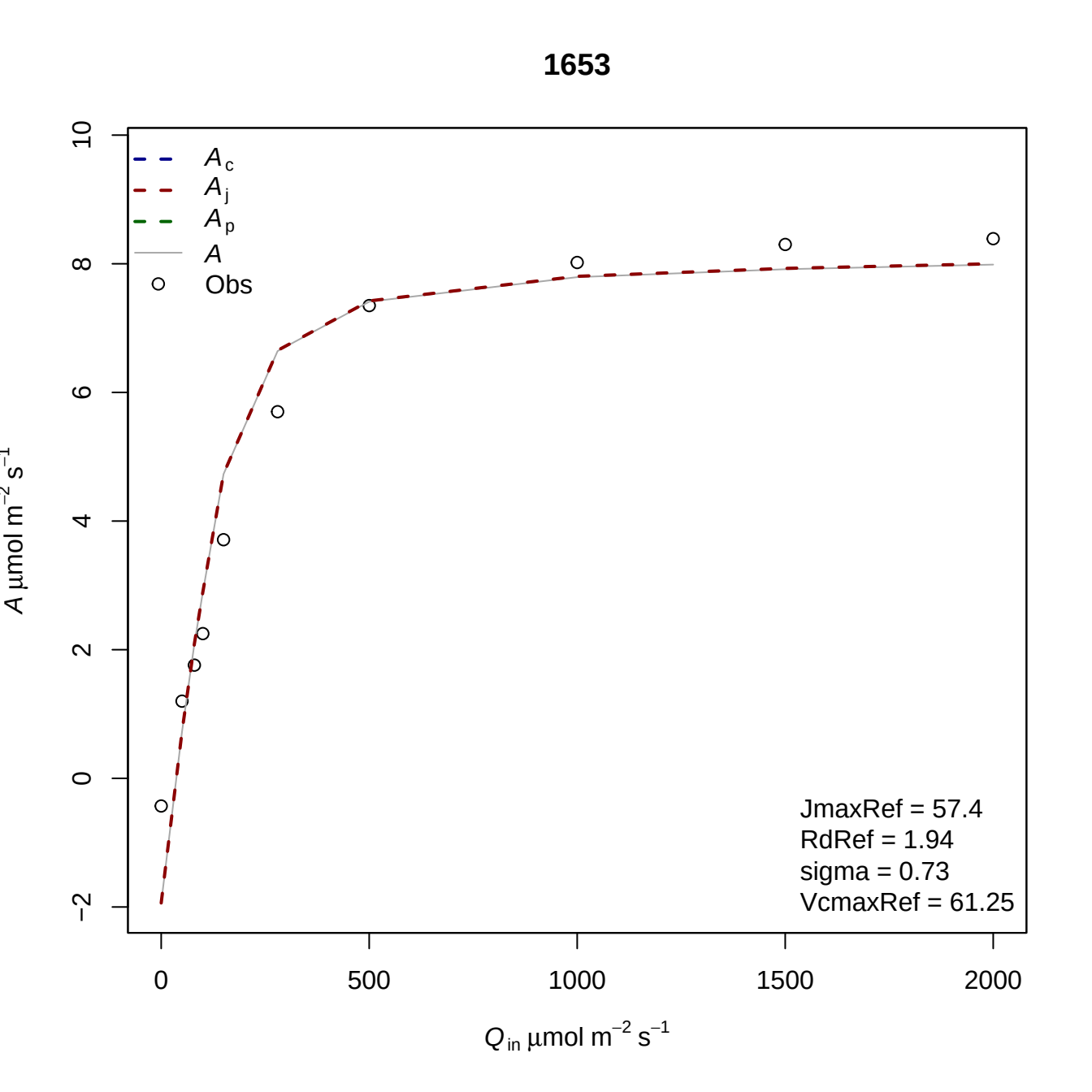

1648
A_c
A_j
A_p
A
Obs
JmaxRef = 119.13
RdRef = 0.76
sigma = 0.15
VcmaxRef = 126.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

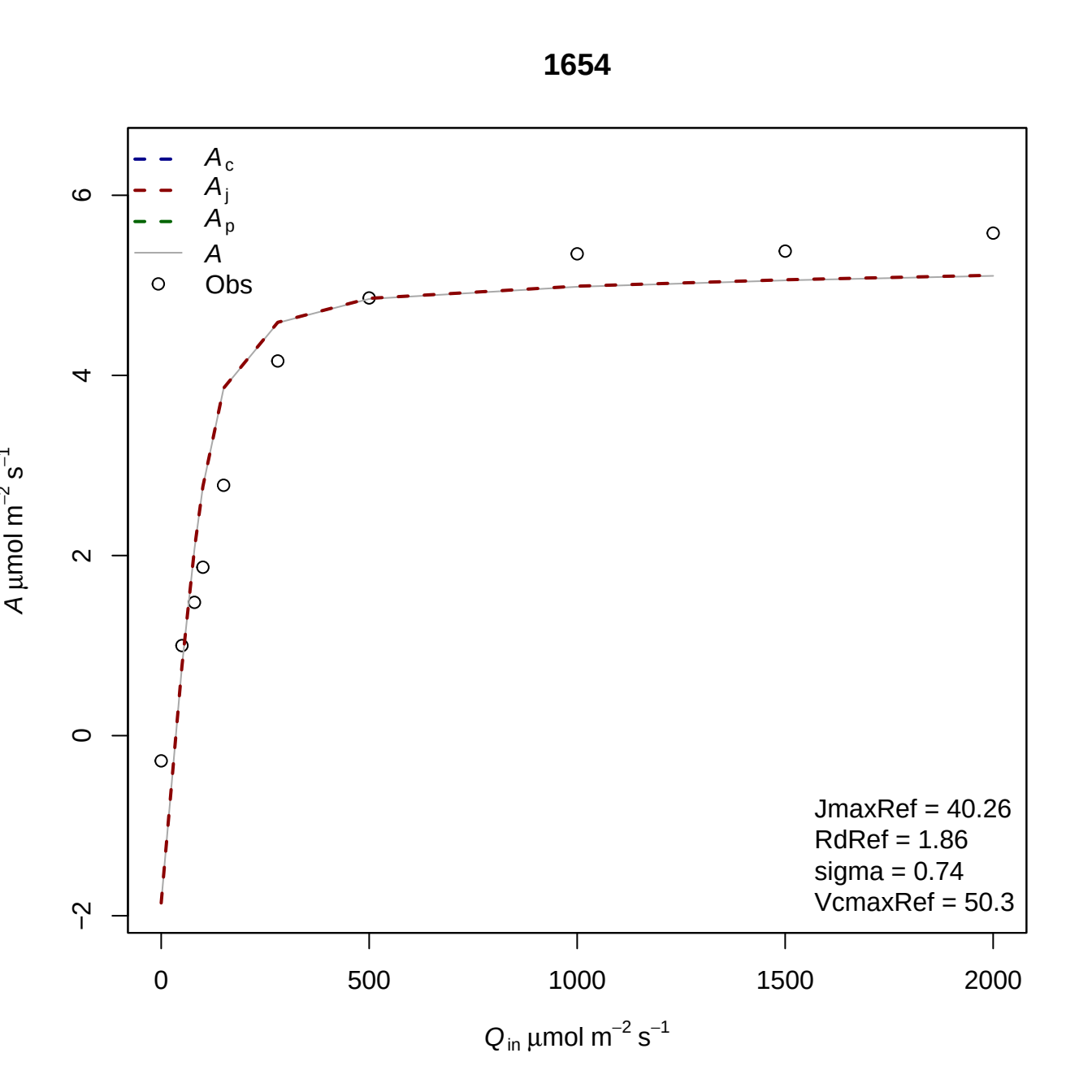

1649
A_c
A_j
A_p
A
Obs
JmaxRef = 140.95
RdRef = 0.77
sigma = 0.41
VcmaxRef = 193.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

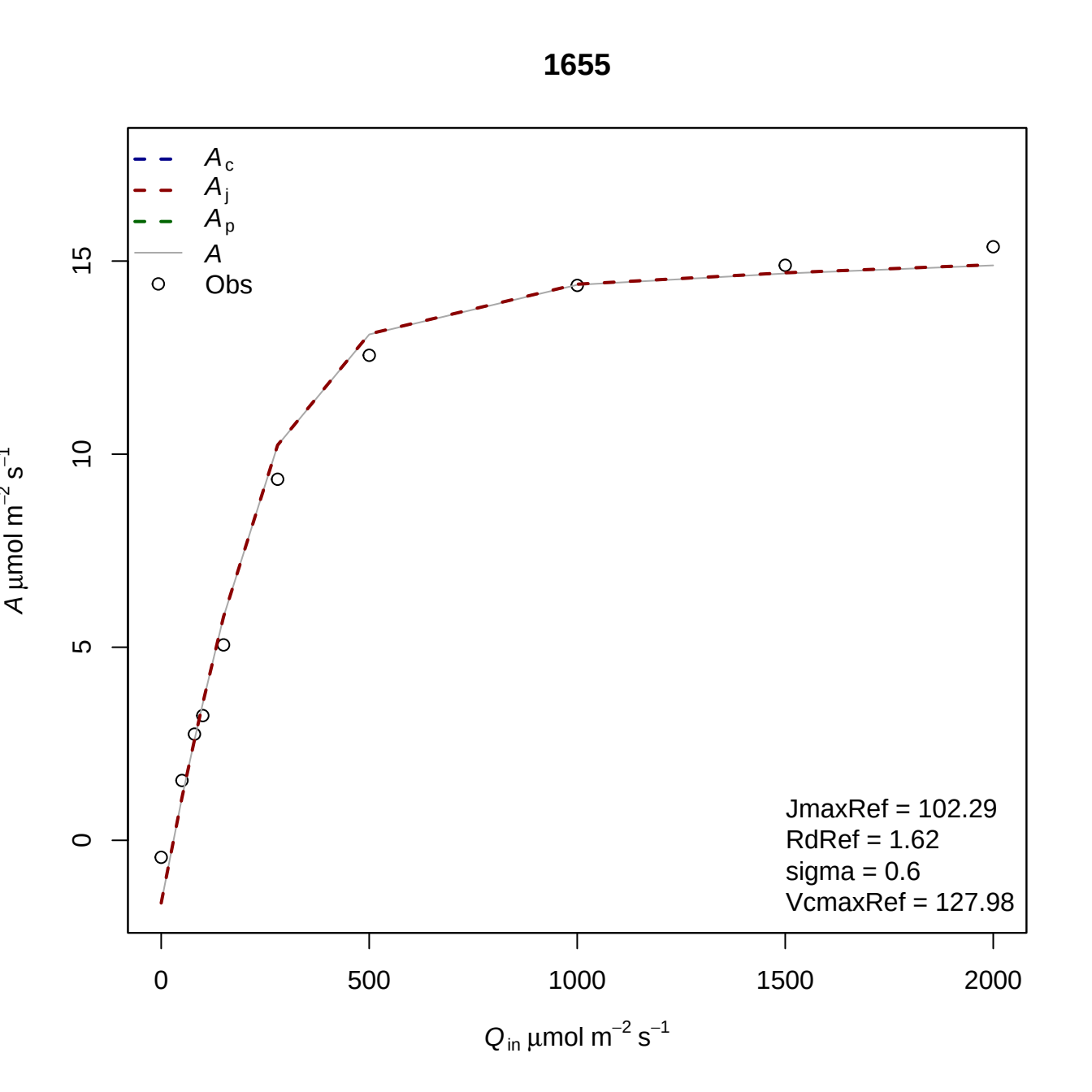

1650
A_c
A_j
A_p
A
Obs
JmaxRef = 119.01
RdRef = 1.03
sigma = 0.28
VcmaxRef = 140.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

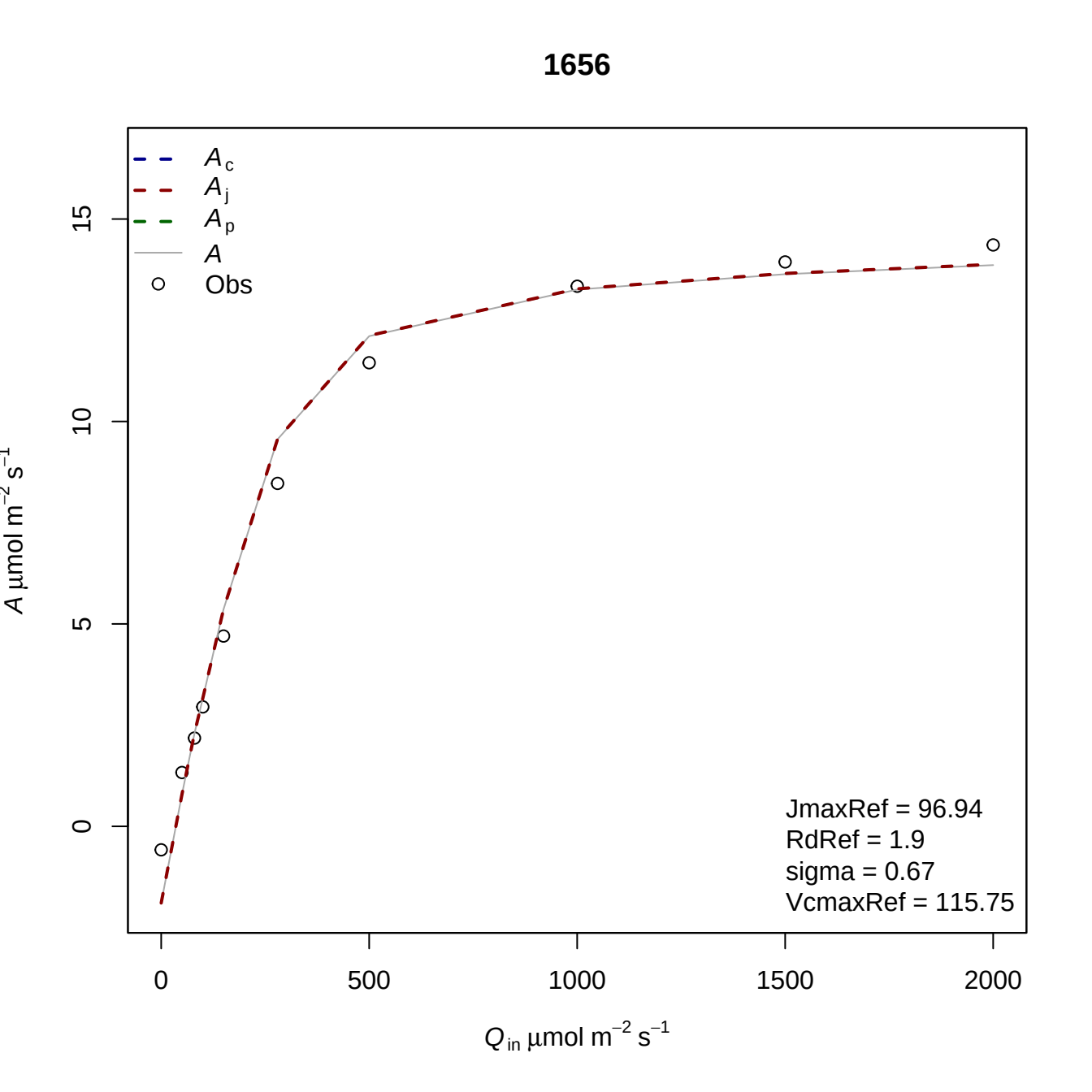

1651
A_c
A_j
A_p
A
Obs
JmaxRef = 48.28
RdRef = 1.77
sigma = 0.64
VcmaxRef = 50.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

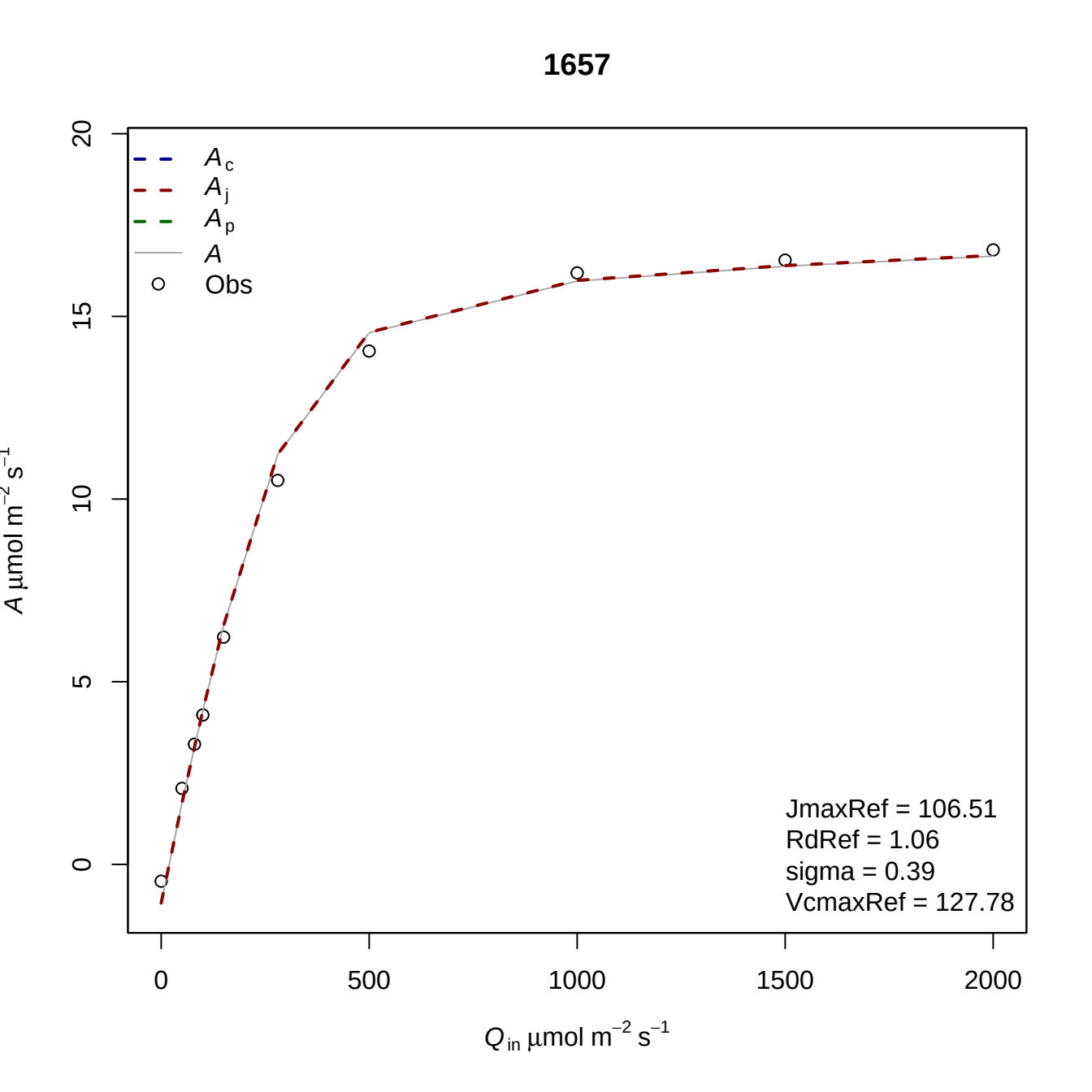

1652
A_c
A_j
A_p
A
Obs
JmaxRef = 59.34
RdRef = 1.6
sigma = 0.6
VcmaxRef = 66.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

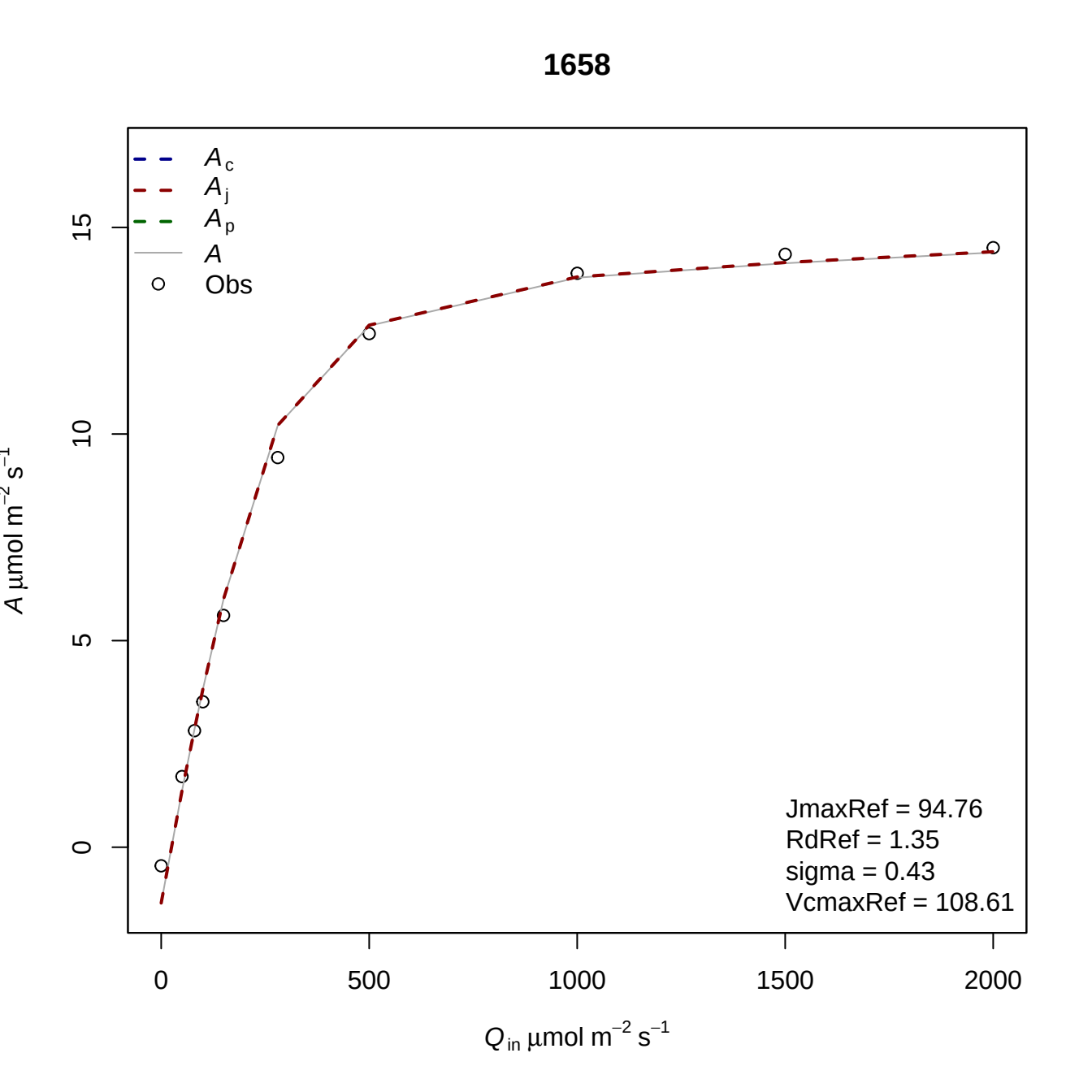

1653
A_c
A_j
A_p
A
Obs
JmaxRef = 57.4
RdRef = 1.94
sigma = 0.73
VcmaxRef = 61.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

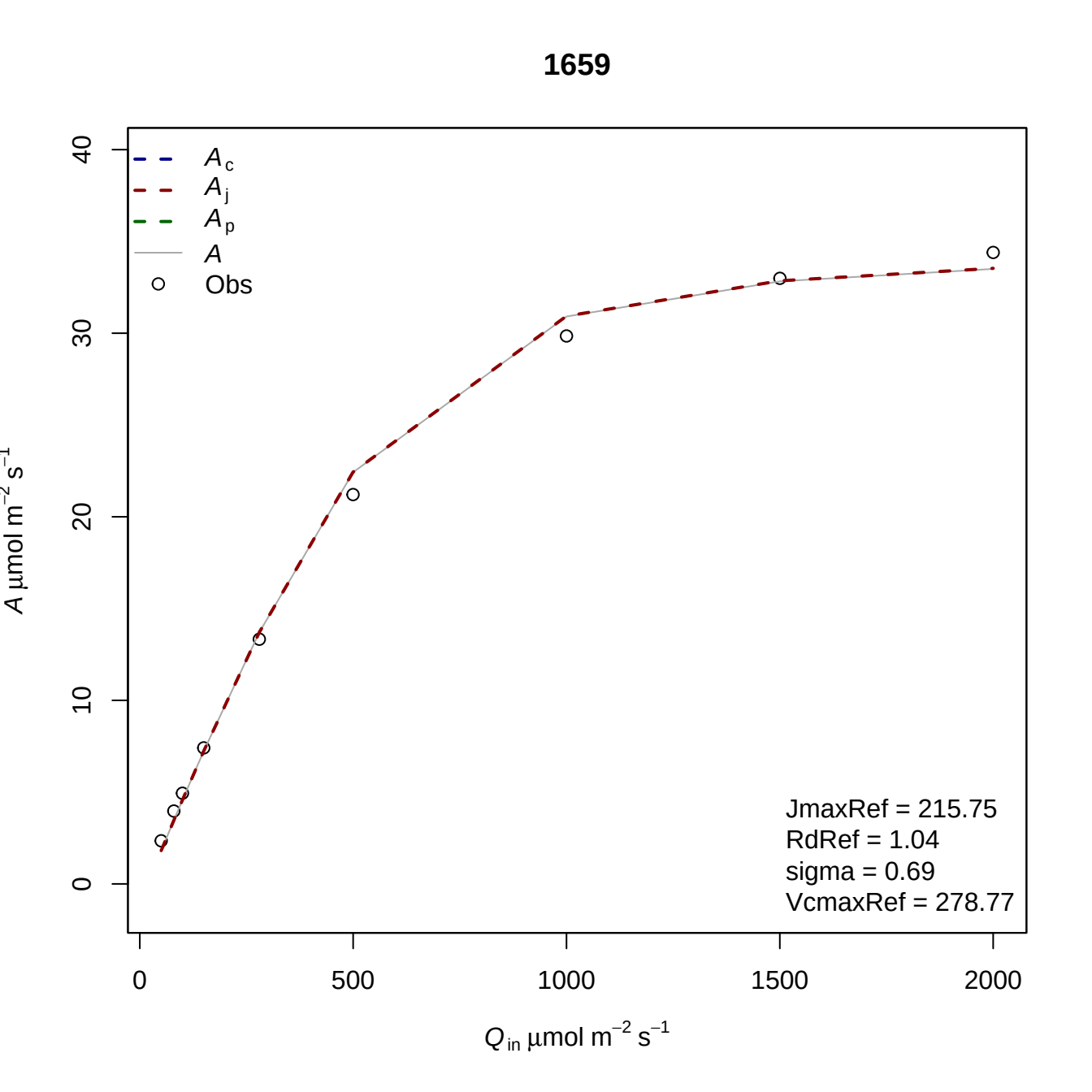

1654
A_c
A_j
A_p
A
Obs
JmaxRef = 40.26
RdRef = 1.86
sigma = 0.74
VcmaxRef = 50.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

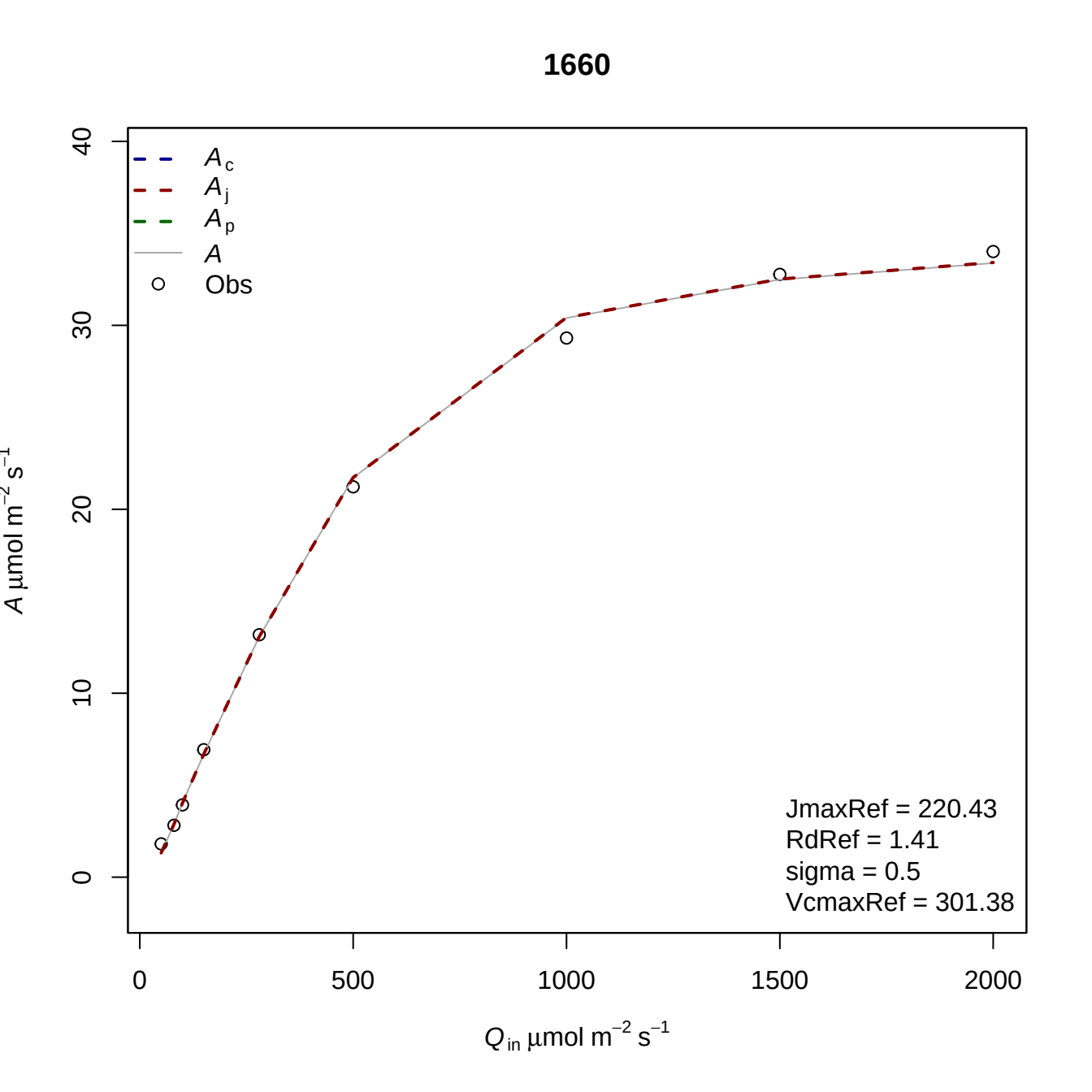

1655
A_c
A_j
A_p
A
Obs
JmaxRef = 102.29
RdRef = 1.62
sigma = 0.6
VcmaxRef = 127.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

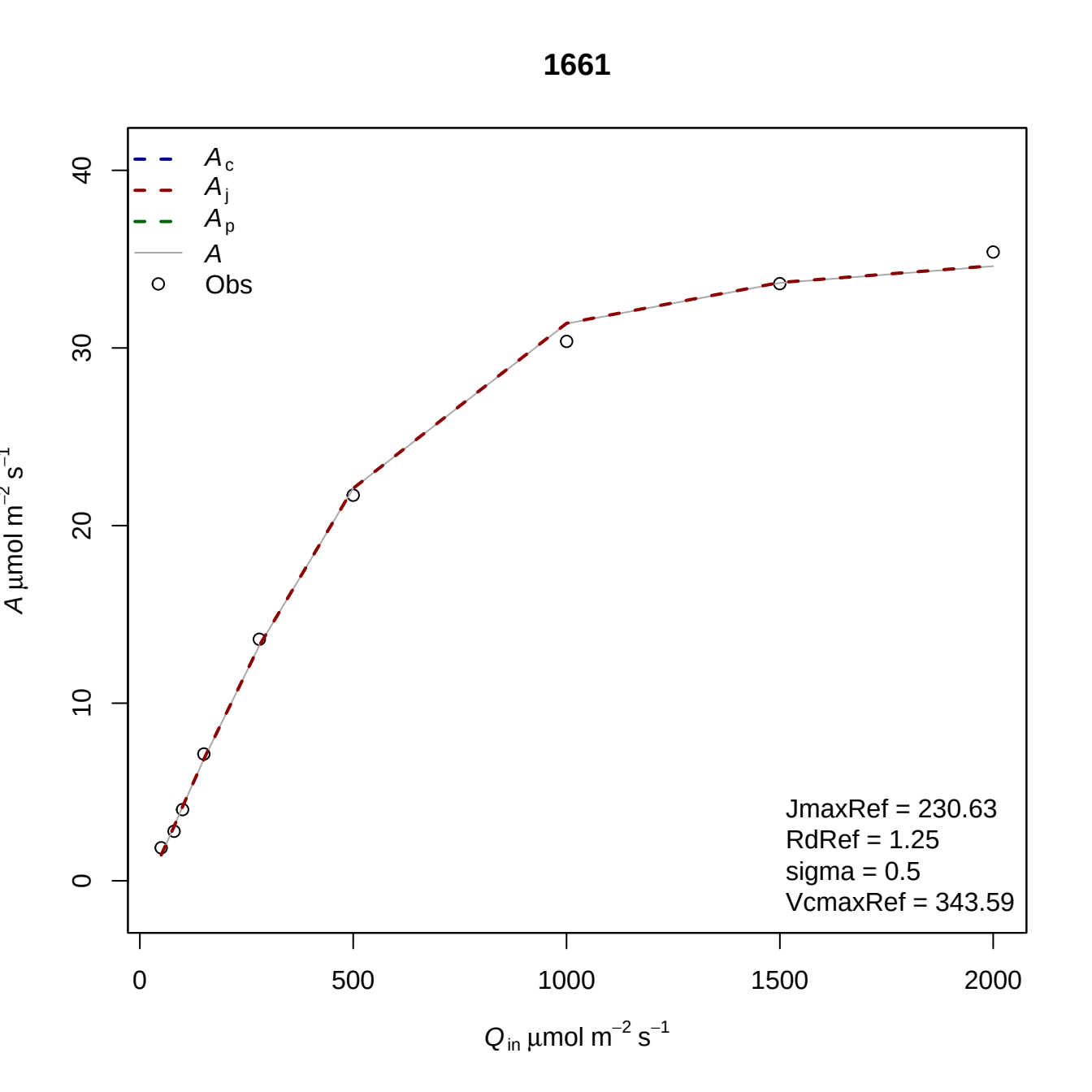

1656
A_c
A_j
A_p
A
Obs
JmaxRef = 96.94
RdRef = 1.9
sigma = 0.67
VcmaxRef = 115.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

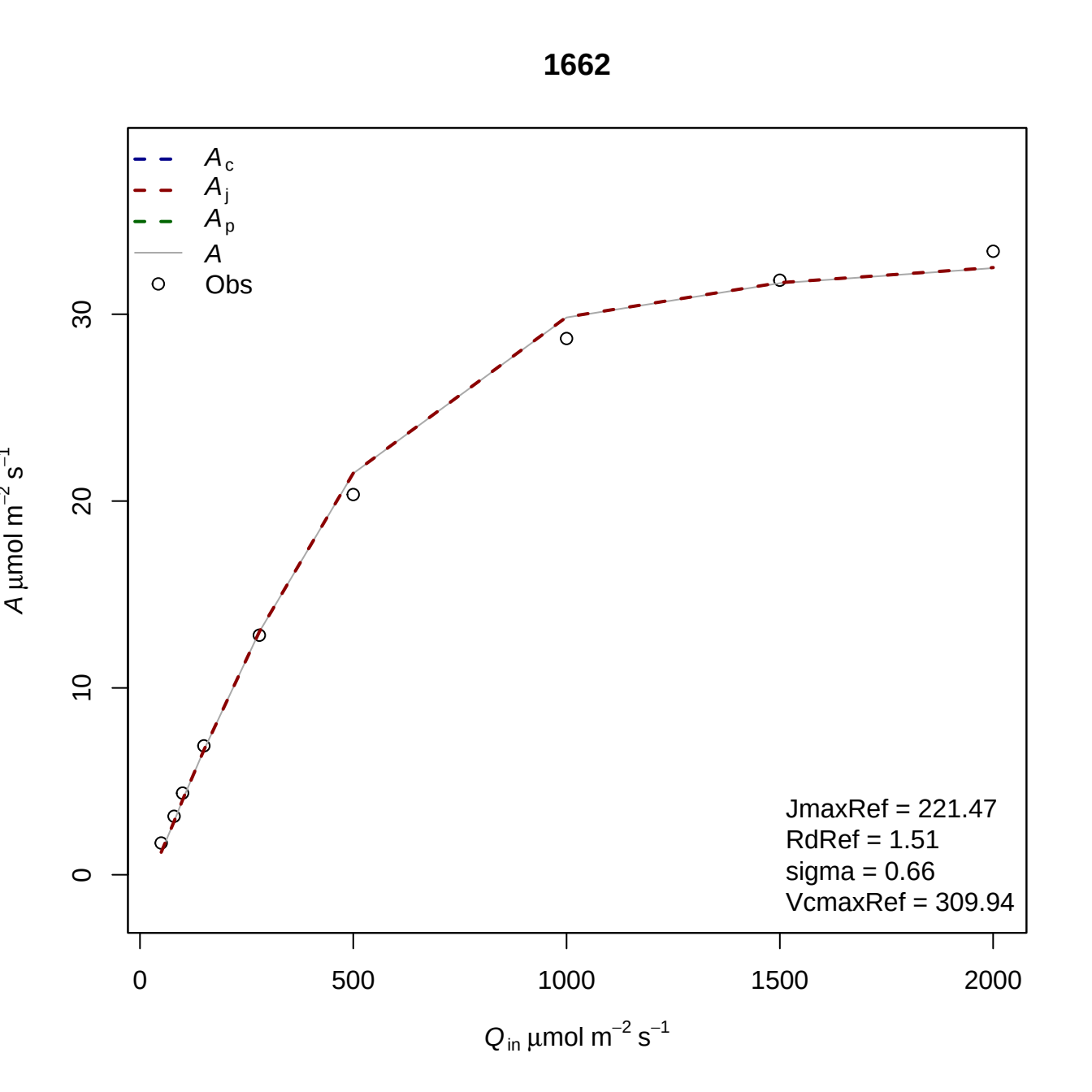

1657
A_c
A_j
A_p
A
Obs
JmaxRef = 106.51
RdRef = 1.06
sigma = 0.39
VcmaxRef = 127.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000


1658
A_c
A_j
A_p
A
Obs
JmaxRef = 94.76
RdRef = 1.35
sigma = 0.43
VcmaxRef = 108.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000


1659
A_c
A_j
A_p
A
Obs
JmaxRef = 215.75
RdRef = 1.04
sigma = 0.69
VcmaxRef = 278.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1660
A_c
A_j
A_p
A
Obs
JmaxRef = 220.43
RdRef = 1.41
sigma = 0.5
VcmaxRef = 301.38
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1661
A_c
A_j
A_p
A
Obs
JmaxRef = 230.63
RdRef = 1.25
sigma = 0.5
VcmaxRef = 343.59
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1662
A_c
A_j
A_p
A
Obs
JmaxRef = 221.47
RdRef = 1.51
sigma = 0.66
VcmaxRef = 309.94
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}